Hyperseed ontology:
formal presentation and theoretical development

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Abstract

Hyperseed is a concept network intended to describe mind, experience, and reality using a compact set of mutually interdefinable primitives (e.g. occasions of experience, distinction, repetition, variety, non-duality), together with successive layers of derivative notions (e.g. effort, simplicity, pattern, emergence, habit, morphic resonance, mind-world correspondence, reflective self, reality-systems, and social/cosmic structures).

Hyperseed has earlier been presented in a semi-formalized fashion. This document aims to rebuild Hyperseed as a mathematically grounded ontology. The organizing hypothesis is that a small formal "core" can generate most of Hyperseed by composition: (i) paraconsistent truth values ("p-bits") to represent partial and conflicting evidence without explosion; (ii) a commutative quantale of "weakness" to model simplicity/generalization as (weighted) failure to make distinctions; (iii) quantale-enriched (and, where needed, $(\infty, 1)$-categorical) structure to compose contexts, processes, and approximate morphisms; and (iv) a resonance construction derived from paraconsistent evaluation dynamics, giving a principled bridge from logical conflict/coherence to dynamical coupling.

The development proceeds in several phases:

- a dependency DAG of Hyperseed concepts and a minimal formal core together with basic "sanity theorems" that anchor the core.
- a toy universe with a finite set X and a $\mathsf{p\text{-}bit}$-valued quantale, including a minimal worked example that reaches the level of morphic resonance and anti-resonance.
- a systematic crosswalk, in which every concept in the original Hyperseed ontology is assigned to a section and mapped to explicit mathematical constructions in the core.
- a series of simple "helper theorems" articulating elementary formal properties of the Hyperseed concepts, and then a series of less trivial theorems relating the concepts to each other, intended to be helpful in applying them to particular cases
- a specific, sustained theoretical development aimed at using Hyperseed to explore and formalize the ethical and cognitive properties of hypothetical AGI systems (or other intelligences) with massive amounts of cognitive and energetic resources. *A conclusion of this development is that, if a series of self-improving AGI systems starts off with a compassionate orientation, it is reasonably likely to maintain such an orientation as it becomes increasingly resource-rich and powerful.*

A (much shorter) companion paper titled *Hyperseed: A Scrutability-Style Core Ontology and a Practical Inference-Control Scaffold for Probabilistic Logic Networks* summarizes the intended use of Hyperseed within AI commonsense reasoning systems, including the potential use of Hyperseed as an initial condition for automatic learning and adaptation of ontologies.

Note on AI Co-Authoring and Co-Utilization .

- This document was primarily written by LLMs, though in a process of repeated prompting, sweeps across the whole document by scripts running prompts, and so forth. There was also human editing here and there when it was an easier way to fix things than doing prompting. Every part of the document was human-reviewed at one or more stages in the document creation process, but it can't be claimed that every part has been carefully reviewed in the final form – there may still be glitches!

- This is not a product envisioned as the subject of frequent leisure reading. It does contain a significant number of ideas, insights and conclusions that will be of interest to humans oriented toward philosophical and conceptual foundations of science, AI, and so forth; however, these are not usually presented here in the most delightfully consumable fashion, although effort has been made to include nontechnical explanations along with formalism.

- A primary use-case for this document is envisioned as: Ingestion by AI systems, which may more fully formalized the material given here, and then leverage this more formalized version for practical guidance of automated uncertain reasoning in commonsense and scientific domains. The brief companion document *Hyperseed: A Scrutability-Style Core Ontology and a Practical Inference-Control Scaffold for Probabilistic Logic Networks* outlines some ideas on how this might be done, along with some associated theoretical work elaborating "what makes a good ontology," and some ideas on how AIs may evolve even better and more useful ontologies using Hyperseed as, well, a seed.

Contents

1 Introduction

Hyperseed is an ontology – a systems of concepts – intended to serve as a basic framework for the explanation and elaboration of other concepts of interest to human beings and other intelligent systems. In more ordinary language, the aim is to have a shared "map legend": a small set of ideas and relationships that can be reused when talking about very different domains (perception, learning, ethics, culture, scientific inference, etc.). It is not intended as a set of semantic primes in a reductive sense, but rather as a universal-ish guide and toolkit, the working hypothesis being that by expressing diverse concepts in terms of this ontological core, deep and effective reasoning about the concepts and their relations and applications can often be efficiently conducted. The emphasis on "toolkit" is practical: the ontology is meant to help one build models that can be checked, compared, and composed, rather than only described informally.

Hyperseed is written as a "semantic stack": it starts from phenomenological primitives (occasion of experience, distinction, repetition, variety, non-duality) and builds upward via effort and simplicity into patterns and processes, then into mind-level constructs (representation, attention, control, goals, emotion, consciousness), and finally into collective and cosmological constructs (society, culture, science, mindplex/global brain, multiverse/eurycosm). The informal picture is that higher-level notions should be intelligible in terms of lower-level ones, in the same way that a complex program is built from simpler components. The stack metaphor also suggests a direction of explanation: when a high-level phenomenon is unclear, one can ask which lower-level ingredients and composition rules would generate it.

The ontology aims to be simultaneously (1) ontological (speaking about what exists), (2) phenomenological (speaking about what is given in experience), and (3) computational (speaking about what bounded agents can represent and do). These three aspects are in tension in many philosophical projects; here they are treated as constraints on the same formal objects. For example, a notion like "distinction" should have a phenomenological reading (what it is like to separate figure from ground), an ontological reading (what structures in the world support stable separation), and a computational reading (what a limited system can actually compute or store).

The Appendix contains a relatively concise list of the concepts included the ontology. Beyond the core, this is a list with some degree of arbitrariness to it – concepts may be added in future, and some could be subtracted without huge harm. But we believe this is a reasonably useful set of choices. One reason to explicitly list concepts, rather than only gesture at them, is to make the project falsifiable in a modest sense: the reader can check whether later constructions actually cover the promised vocabulary.

Remark 1. *The phrase "semantic stack" is meant quite literally: each layer is supposed to constrain (and in some cases determine) the next layer, much as in a protocol stack each level provides an interface whose regularities enable the levels above it. In a philosophical register one may hear echoes of process ontology and triadic semiotics—experience as a primitive given, distinction as the act that carves it, and habit/law as what stabilizes it across repetitions—themes congenial to various historical philosophers such as Whitehead and Peirce [15, 14]. Hyperseed's distinctive move is to insist that these are not merely poetic analogies: they are intended to admit a formal reconstruction that preserves compositional structure, limits, and approximation. In particular, "preserves compositional structure" means that if a complex situation is built by combining simpler situations, then the semantics should allow one to compute properties of the whole from properties of the parts, up to explicitly tracked approximation.*

The goal here is not to settle metaphysical debates, but to provide a rigorous scaffold that makes the Hyperseed concept network mathematically precise enough to: (1) support theorem statements and proofs; (2) enable toy and mid-scale computational models; (3) make explicit where paraconsistency, approximation, and observer-relativity enter; (4) formally derive theorems about the relationship between the concepts and their utilization in various domains. This is intended to move discussions of otherwise slippery notions (e.g. emergence, attention, selfhood) toward a setting where one can ask concrete questions like "what follows from what?" and "what fails if we weaken this assumption?"

Remark 2. *We can view Hyperseed metaphorically as a scaffold for building diverse concepts; however, a modest but important philosophical point is hidden in the word "scaffold." A scaffold is not the building; it is what allows one to test whether the building could stand. Here the "building"* *includes contested notions (e.g. consciousness, morphic resonance, non-duality), and the wager is that one can isolate a minimal algebraic/categorical substrate on which these notions can be treated with the same discipline that we apply to, say, probabilistic inference or programming language semantics. Even if one ultimately rejects the ontology as metaphysics, the scaffold may still be valuable as mathematics for graded, resource-bounded, and non-classical reasoning (cf. the general approach to cognitive formalization in [19] and the pattern/emergence framing in [5]). Concretely, one can separate two questions: (i) whether a concept is metaphysically "real," and (ii) whether there is a coherent formal object that plays the role the concept is supposed to play in reasoning and modeling.*

Method. We proceed by identifying a minimal formal "core" that can support the entire ontology. The guiding idea is to start with a small number of mathematical structures that are expressive enough to encode the recurring moves the ontology needs: making a distinction, weakening it, composing processes, and coping with conflict. The core must be able to express: (i) pairwise distinctions and their loss (weakness); (ii) composition of contexts and processes; (iii) inconsistent and incomplete knowledge without trivialization; (iv) emergent pattern intensity; and (v) resonance/morphic resonance as structure-sensitive coupling. To do this we combine a p-bit semantics (paraconsistent evidence), quantale weakness (simplicity/generalization as undistinguished pairs), and enriched-categorical composition. The intent is that later concepts are not introduced ad hoc; rather, they arise as constructions on top of this core, so that their interactions are constrained by shared algebraic laws.

Remark 3. *Several technical terms appear here that will be defined carefully later, but it is helpful to state the intended reading now. The symbol* p-bit *denotes a paraconsistent evidence unit: rather than assigning each proposition a single truth value, we allow a* pair *(support, opposition), so that both can be nonzero simultaneously without the system collapsing into triviality. This is the formal counterpart of taking value conflict and ambiguous boundaries as first-class phenomena (Hyperseed-Concepts 198 and 165). Logically, this draws on the general tradition of non-explosive semantics (see also [23]) and is developed in a resonance-oriented direction in [24]. An informal example is a dataset or social situation where there is strong evidence for and against a claim (or for and against a boundary between "self" and "other") that does not disappear merely by collecting more of the same kind of evidence; the pair representation records both pressures explicitly.*

Remark 4. *"Quantale weakness" is the second pillar: a* quantale *is an algebraic structure that behaves like a lattice of grades together with an associative "multiplication" (a way to compose grades). In this paper, weakness measures (Hyperseed-Concept 202) will be aggregated and composed using such a structure, enabling the slogan "loss of distinctions composes." This line of thought is elaborated in [3], and the broader cognitive motivation for taking "weakness" as primitive (rather than error-free crispness) is consonant with the perspective in [2]. We will also connect this with patterns and emergence (Hyperseed-Concept* **??***) as treated in [5]. The key conceptual point is that blur is not just a nuisance: if two distinctions are each slightly unreliable, then the combined distinction produced by chaining them should be predictably (and quantifiably) weaker, rather than treated as either perfectly valid or completely broken.*

Remark 5. *Finally, "enriched-categorical composition" indicates the following: instead of treating contexts merely as labels, we treat translations between contexts as morphisms whose quality/strength is graded (by the same sort of quantale used for weakness). This provides a disciplined language for observer-/context-indexing (Hyperseed-Concept 86) and for composing approximate correspondences rather than exact equalities. The resulting structure is meant to serve as the backbone for later* *discussions of mind–world correspondence (Hyperseed-Concept 112) and transfer-like phenomena (Hyperseed-Concepts 192 and 193; see [7] for related mathematical motivations). Informally, this is a way to say: two observers (or two models) may not match perfectly, but there can still be a structured, composable notion of "translation" between them, and the translation can carry a measured degree of fidelity.*

Remark 6. *The mention of "resonance/morphic resonance" (Hyperseed-Concepts 159 and 115) signals a specific ambition: to model coupling not as a generic diffusion term, but as a structure-sensitive propagation whose strength depends on pattern-alignment. The phrase "morphic resonance" is historically associated with Sheldrake's speculative proposal [13]; here it is treated as a name for a mathematically specifiable operator that can be studied independently of any particular metaphysical interpretation. The modeling goal is to make room for the idea that some influences propagate preferentially along shared structure (similar organization, similar habits, similar representations), which is a common motif in cognitive science and machine learning even when the specific term "morphic resonance" is not used.*

What counts as success. A successful mathematical Hyperseed should make it easy to answer questions of the form:

- "Given an observer with limited representational capacity, what distinctions can it make?"
- "How do patterns emerge from compositional simplicity?"
- "When do habits stabilize, reverse, or transmit via morphic resonance?"
- "How do self-models and reflective will arise as fixed points or higher morphisms?"

Even if one disagrees with Hyperseed's philosophical stance, this reconstruction should still be valuable as a modular formal toolkit for paraconsistent, resource-sensitive modeling of cognition. For early-stage AGI learning, these are exactly the kinds of questions that recur: an agent must decide what to represent, how to compress experience into patterns, how to remain stable under self-modification, and how to coordinate or transfer knowledge across contexts.

Remark 7. *These questions already hint at the kinds of mathematics that will appear later. "Limited representational capacity" anticipates explicit capacity/constraint functionals and the associated notion of what an observer can stably discriminate (Hyperseed-Concepts 98, 82, 181). "Habits stabilize, reverse, or transmit" points toward dynamical notions (Hyperseed-Concepts 189, 188, and 115) and their interaction with paraconsistent evidence accumulation. And "fixed points or higher morphisms" foreshadows the use of order-theoretic fixed point theorems to express reflective stability conditions for goal systems and self-modification (Hyperseed-Concept 110; see [10] for a closely related application pattern). The informal moral is that "reflective agency" is treated as something one can characterize by invariance or stability properties, rather than only by narrative description.*

Remark 8. *The intended notion of "observer" here is not merely a human subject but any system that (i) makes distinctions, (ii) carries representations, and (iii) allocates attention and control under resource constraints (Hyperseed-Concepts 157, 60, 87). This explicitly permits both biological and engineered instances (Hyperseed-Concepts 68 and* **??***) and aligns with an engineering-oriented stance toward general intelligence [19]. In later sections, when consciousness is discussed more directly, one may also connect this observer-relativity with formal "locations" for individuals and collectives (Hyperseed-Concepts* **??** *and 75; see [4, 12] for related perspectives). This broad notion* *of observer is useful for AGI research because it lets one apply the same definitions to, say, a single model, a team of models, or a model interacting with tools and institutions, while keeping track of which capacities and contexts are doing the explanatory work.*

Organization of the paper. Sections 2–5 cover the "first sprint": concept DAG, formal core with sanity theorems, and a finite toy model (including morphic resonance). Sections 6–26 then map the full Hyperseed ontology into rigorous constructions. Appendix C contains a complete checklist of Hyperseed concepts and where they are treated. The division is meant to help the reader separate foundational commitments (the core machinery) from downstream elaborations (the larger vocabulary), so that one can reuse pieces even if one does not adopt the full stack.

Remark 9. *The phrase "first sprint" is meant to set expectations: early sections aim for a small set of definitions and compositional laws that are strong enough to support meaningful theorems, yet weak enough to remain applicable across many domains (cognition, culture, science, cosmology). In particular, the appearance of "sanity theorems" is meant to reassure the reader that the algebraic choices are not merely aesthetic: they enforce non-trivial constraints on how approximation and composition can interact. Later, when the discussion turns toward wu wei and least-effort style principles (Hyperseed-Concepts 206 and 207), it will be useful to have already fixed the underlying graded-compositional apparatus (see [21] for a related development). One can think of these sanity theorems as the analogue of type-safety or soundness results in programming languages: they prevent certain pathologies and clarify what constructions are legitimate.*

Scope notes. Throughout, "ontology" is used in a technical sense: a typed vocabulary with axioms, composition rules, and semantics. Many Hyperseed notions are observer-relative (e.g. distinction, simplicity, uncertainty) and some are intrinsically paraconsistent (e.g. value conflict, self/other boundaries). These are not bugs; they are treated as explicit design requirements. In particular, the goal is not to eliminate context dependence or conflict, but to represent them explicitly so that one can reason about them and measure their effects.

Remark 10. *In particular, observer-relativity is not taken to mean "arbitrary subjectivism." Rather, it is treated as the claim that many predicates are best understood as* indexed *(by observer, context, or both), and that the mathematics should track how such indices compose when observers communicate, aggregate, or form collectives (Hyperseed-Concepts 86,* **??***, 170, 91). This stance is compatible with a social-computational-probabilistic view of scientific practice [20], and it also sets the stage for later efficiency claims about prosocial coordination in multi-agent settings [6]. A simple motivating example is that two agents can disagree not only about facts but about which distinctions are salient; indexing helps separate "disagreeing about the world" from "using different coordinate systems for describing it."*

Remark 11. *Paraconsistency is likewise not a license for confusion; it is a discipline for representing situations where evidence genuinely supports incompatible conclusions (Hyperseed-Concept 198) without forcing an explosion of entailments. The practical motivation is clear in cognitive and ethical modeling: minds (and groups of minds) often must act under persistent conflict, not merely transient uncertainty. The formal aim is to make such conflict compositional, so that one can ask when it localizes, when it propagates, and when it is resolved by additional structure rather than by fiat. From an AGI-safety and alignment perspective, this matters because real agents must integrate competing objectives, inconsistent testimony, and shifting self-model boundaries without degenerating into either paralysis or arbitrary collapse.*

1.1 Use of Hyperseed for Guiding AI Reasoning

A (much shorter) companion paper titled *Hyperseed: A Scrutability-Style Core Ontology and a Practical Inference-Control Scaffold for Probabilistic Logic Networks* summarizes the intended use of Hyperseed within AI commonsense reasoning systems, including the potential use of Hyperseed as an initial condition for automatic learning and adaptation of ontologies.

2 Hyperseed at a glance: concept inventory and dependency DAG

2.1 Purpose and reading conventions

Hyperseed is presented (in its original form) as a tightly interlinked concept-network rather than as a linear theory [1]. Many concepts are intended to be mutually definable or at least mutually constraining, so there is no unique "correct" topological ordering. In particular, what counts as "basic" in Hyperseed is often a matter of which questions one wants the formalism to answer first: one can start from distinctions and build upward toward agency-like notions, or start from process and build toward stable patterns, or start from value-laden notions and work backward toward representational constraints. The reconstruction strategy adopted here is therefore pragmatic: we choose primitives and dependencies so that the resulting definitions are checkable (in the ordinary mathematical sense) and reusable across later sections without reintroducing implicit circularity.

Remark 12. *It is worth pausing over what it means, mathematically, for an ontology to be "interlinked." In an ordinary axiomatic theory one expects a small set of primitives and a long chain of definitions; but in a concept network one expects a kind of mutual illumination, where definitions are less like one-way reductions and more like constraints that cut down a space of possible meanings. Hyperseed leans into this latter style: concepts such as* distinction *(Hyperseed-Concept 98),* pattern *(Hyperseed-Concept 130), and* process *(Hyperseed-Concept 140) are not merely "introduced" and then used; they are repeatedly refined by their relations to effort, simplicity, emergence, and resonance. In technical terms, one may think of the "meaning" of a concept as the collection of constraints it imposes on admissible models: a formal model of Hyperseed is not merely a set-theoretic structure with extra operations, but a structure satisfying a web of compatibility conditions that jointly determine which interpretations are coherent. This way of proceeding is familiar in areas where definitions are stabilized by the role they play (e.g. universal properties, adjunctions, or fixed-point characterizations), even if Hyperseed's motivating vocabulary is not usually presented in that language.*

Nevertheless, a mathematically grounded reconstruction benefits from an explicit *expository orientation*: we choose a directed acyclic graph (DAG) whose edges represent "is used in the formal definition of" or "is a prerequisite for the cleanest formalization of". This DAG should be read as a *build order* rather than a metaphysical claim. Concretely, the DAG functions as a promise that each later object can be defined in terms of earlier ones in a way that can be verified line-by-line, even if (conceptually) Hyperseed intends later notions to feed back and refine earlier intuitions. The build order is thus an editorial device: it is a compromise between the networked character of the source material and the linear constraints of a written proof-oriented exposition.

Remark 13. *The phrase "build order" is doing real work. It signals that the DAG is a discipline of exposition: it tells the reader what must be* already fixed *so that later definitions are not silently circular. In particular, it separates definitional dependence (what you need in order to write down a definition) from causal or dynamical dependence (what influences what in time) and from explanatory dependence (what helps us understand what). Hyperseed will later return to feedback,* *heterarchy, and closure (Hyperseed-Concept* **??***, 168), but those will be treated as* structures within *the already-defined mathematical universe rather than as reasons to permit circular definitions at the foundational level. A useful analogy is the difference between (i) allowing mutually recursive* equations *inside a system once the ambient semantics is fixed (e.g. via least fixed points in a complete lattice), and (ii) allowing mutually recursive* definitions *without any semantic grounding. The former can be made precise and non-pathological; the latter typically produces ambiguity. The DAG is meant to keep us in the former regime: when Hyperseed-like closure is formalized, it will appear as a theorem or as a definable construction (often a fixed point, an invariant, or a limit), not as a tacit permission to define objects by appealing to themselves.*

Throughout the paper we will use two recurring indexing ideas: these are not merely notational conveniences, but design constraints on the formal core, since they force many predicates to be typed and parameterized rather than treated as absolute, global properties. The intent is that whenever an informal Hyperseed statement appears to be unconditional, the reader should ask whether it is actually conditional on (i) a representational point of view and (ii) a possibly inconsistent evidence state.

- **Observer-/context-indexing:** many Hyperseed predicates are implicitly of the form "*relative to observer O and context C*". Mathematically this becomes typed, indexed structure (e.g. relations valued in a quantale, enriched categories of contexts, etc.). This indexing is also a guardrail against category mistakes: without it, one is tempted to conflate statements about the world with statements about an agent's internal discrimination of the world, or to conflate stable properties with those that only hold under a particular conditioning regime. In the formalization, the same underlying entity may support multiple, incompatible partitions or similarity relations when viewed under different (O, C), and the theory aims to keep those variations explicit rather than hidden in prose.

  Remark 14. *Here "observer" and "context" should be read technically, not anthropomorphically: an observer is any system whose distinctions and updates we model, and a context is any conditioning structure that selects which distinctions and compositions are currently active (Hyperseed-Concept 86). One can think of O as fixing representational resources and update rules, while C fixes the slice of the world (or of the internal state) relative to which a judgment is made. Thus, a claim like "x and y are equivalent" is never purely absolute; it is an equivalence* relative to *which distinctions the observer can draw (Hyperseed-Concept* **??***, 98). From a modeling standpoint, this means that the formal objects in later sections should be read as living in families indexed by (O, C): for example, a "pattern" may be a pattern-for-O-in-C, and a "simplicity" measure may be a simplicity-relative-to-the-coding-scheme-available-to-O and the constraints activated by C. This does not weaken the theory; rather, it clarifies what kind of invariance would be meaningful. When we do obtain invariants (e.g. quantities or constructions insensitive to certain changes of context), that will be an explicit result rather than an implicit assumption.*

- **Paraconsistent indexing:** some Hyperseed judgments are intrinsically allowed to be both supported and opposed at the same time (e.g. value conflicts, ambiguous boundaries, "both random and predictable" limit constructions). Mathematically this becomes a non-explosive truth/evidence domain (the p-bit layer in Section 3). This indexing is meant to preserve information: it distinguishes "unknown" from "both-ways" and distinguishes "inconsistent evidence" from "lack of evidence." As the exposition progresses, this will permit us to combine multiple sources (subagents, sensors, heuristics, or evaluators) without first forcing agreement, while still supporting principled aggregation and comparison operations.

Remark 15. *Paraconsistency is not introduced here as a metaphysical indulgence but as a representational necessity: when evidence streams, subagents, or value sources disagree, we want to record the disagreement rather than force an artificial collapse into a single scalar certainty. This is particularly vivid for ethical and motivational predicates (Hyperseed-Concept 198, 197), but it also appears for boundary predicates such as self/other (Hyperseed-Concept 165) and for limit notions such as ineffability and unknowability (Hyperseed-Concept* **??***, 196). The formal strategy we adopt aligns with paraconsistent semantics in the spirit of multi-valued logics [23] and with later resonance-oriented applications [24]. Operationally, the point is that later definitions can be written so that they are monotone in evidence and robust under fusion: adding a new (possibly conflicting) source should refine the state of information rather than crash the logic. When this works well, it also clarifies which parts of Hyperseed are genuinely normative (depending on value inputs) and which parts are structural (depending only on the algebra of evidence combination). The* p-bit *layer is therefore both a technical tool and a conceptual separation principle: it lets us say, precisely, when a claim is "in tension" rather than merely "false."*

2.2 Layered view: the Hyperseed "semantic stack"

For navigation, it is helpful to group the ontology into a small number of layers. Table 1 is not meant to be exhaustive; it is a map of the main build-up route. In this section, "layer" should be understood as a *dependency stratum*: later terms are intended to presuppose (at least implicitly) earlier ones, so that definitions can be staged rather than mutually recursive. The goal is not to enforce a rigid hierarchy, but to provide an orienting spine that makes it easier to locate where a concept is *introduced*, what kinds of prior distinctions it assumes, and what kinds of constructions it subsequently supports. In particular, the layering is designed to align with the dependency DAG discussed in this section: edges should predominantly point "upward" (from prerequisites to derived notions), even if, in practice, some modeling workflows will traverse the layers in the opposite direction (starting from agency or values and then unpacking the primitives they rely on).

Layer	Core vocabulary (representative)	What it enables
Primitives	occasions of experience; difference/distinction; repetition/variety; non-duality; presentational immediacy; abstract/concrete; first/second/third/fourth	typed "raw materials" for observer-relative modeling
Order and becoming	after/before; proto-time; becoming; event/process calculi	history-sensitive structure, causal/pre-causal ordering
Effort and simplicity	effort; resistance/submission; opening/closing; simplicity; compositional simplicity; structural complexity	resource-sensitive compression and generalization
Patterns and structure	pattern; pattern intensity; property; emergence; blend; hierarchy/heterarchy; pattern web	reusable structure; compositional organization; multi-scale description
Dynamics and coupling	tendency to take/reverse habits; self-weaving webs; resonance/dissonance; morphic resonance/anti-resonance	stabilization, amplification/decay, cross-context pattern transmission
Mind and agency	mind; representation (icon/index/symbol); perception; prediction/attraction; cause; control; attention; self/other; consciousness; autonomy	closed-loop cognition; self-models; coherent action under constraints
Values, life, society, cosmos	emotion/compassion; values/goals; rationality/ethics; energy/genenergy; belief systems/science; societies/mindplex/global brain; reality-systems; eurycosm/multiverse/y-verse; wu-wei	motivation, culture, long-range structure of "reality" as system-relative

Table 1: A pragmatic layered view of Hyperseed. The full inventory is listed in Appendix C.

The table can also be read as a set of *design affordances*. Each row suggests a typical "unit of work" for formalization: first establish a minimal observational alphabet (primitives), then add directed structure for change (order/becoming), then introduce explicit boundedness constraints (effort/simplicity), then define reusable regularities (patterns/structure), then account for their stabilization and transfer (dynamics/coupling), then specify control-relevant internal organization (mind/agency), and finally situate agents within normative and collective contexts (values/life/society/cosmos). This makes it clearer which later claims are meant to be *model-relative* consequences of earlier commitments (e.g., a notion of "control" presupposing some notion of "before/after," and a notion of "ethics" presupposing agents capable of goal selection and trade-offs). Conversely, when a concept appears to "skip" layers in exposition, the intent is typically that the missing dependencies are implicit and can be expanded by following the DAG edges back down.

Remark 16. *The table should be read as a map of* constraints propagating upward. *The primitives (Hyperseed-Concept 124, 98, 121) supply the minimal types of "what can be talked about"; the order/becoming layer (Hyperseed-Concept 63, 52, 142, 62) supplies a way to speak about directed change; effort and simplicity (Hyperseed-Concept 100, 169) supply a way to treat cognition as resource-bounded; and so on. In particular, the transition from simplicity to pattern is meant to be sharp enough to support later formal definitions of emergence (Hyperseed-Concept* **??***) rather than leaving "pattern" as a merely aesthetic label [5]. A useful practical reading is that each layer introduces a new kind of* invariance claim *and therefore a new family of admissible abstractions. For example, once distinctions and repetitions are in place, the order/becoming layer constrains which sequences of occasions can be treated as coherent histories; once effort and simplicity are explicit, admissible abstractions are additionally constrained by resource budgets (what can be represented, learned, or inferred under bounded computation and attention). The patterns/structure layer then adds criteria for when compressed descriptions count as stable "things" (properties, blends, hierarchical modules) rather than ad hoc summaries; dynamics/coupling adds criteria for when such things persist, transmit, or synchronize across contexts (resonance, anti-resonance, habit formation and reversal); mind/agency adds criteria for when pattern webs close into self-referential loops capable of prediction and control; and the final layer adds criteria for how such loops are steered and coordinated by valuations, norms, and shared world-models. This upward constraint flow is also intended to reduce ambiguity in later sections: when a term such as "cause" or "goal" is used, it should be interpretable as a derived notion built from the earlier commitments about directed change, bounded effort, and pattern stability, rather than as an unanalyzed primitive. When these dependencies are respected, disagreements about higher-level notions can often be localized to specific upstream choices (e.g., what counts as a distinction, what counts as simplicity, or what kind of ordering is assumed), making the ontology easier to test, modify, and extend without collapsing into purely verbal disputes.*

2.3 Concept inventory by module

Appendix C contains a completeness checklist. Here we group the same vocabulary into modules that will become the main "subgraphs" of the dependency DAG.

Remark 17. *These modules are not claimed to be natural kinds. They are closer to what a working mathematician might call a "good set of coordinates": a choice that makes the subsequent constructions easier to state, even if other coordinate choices are possible. In later sections, some concepts will deliberately cut across module boundaries (for example,* information *and* uncertainty *constrain nearly everything; Hyperseed-Concept* **??***, 195).*

Module P: Phenomenological primitives.
Primitivest primitives; occasions of experience; difference; distinction; repetition; variety; non-duality; non-dual variety; first/second/third/fourth; number; quantity; grouping; presentational immediacy; abstract/concrete.

Module T: Order, time, and becoming.
after/before; the asymmetry constraint (e.g. $A < B$ implies not$(B < A)$); proto-time; becoming; event and process calculi.

Module E: Effort and simplicity.
effort; resistance/submission; opening/closing; simplicity; compositional simplicity; (generalized) Kolmogorov complexity (as an interpretive anchor); structural complexity; minimum representational effort.

Remark 18. *The mention of Kolmogorov complexity is meant as an* anchor*, not a reduction: it provides a familiar paradigm where "simple" means "short description" [16]. Hyperseed's effort-based simplicity (Hyperseed-Concept 100, 169) is broader: effort may represent energetic cost, attentional cost, time, or any other resource relevant to an observer. This is also where the notion of "weakness" enters later as the systematic encoding of collapsed distinctions (Hyperseed-Concept 202, 143) [2, 3].*

Module S: Patterns and combinational structure.
process; system; coherence; recognition process; equivalence relation; combination; inheritance; association; lifting from instances to groups; pattern; pattern intensity; property; specific entity; emergence; blend; combinatorial-categorical pattern; combination system; interpreted combination system; combinational dynamical causal model.

Module I: Information, truth, and limits of representation.
probability; entropy/information; uncertainty; ineffability; soggy predicates; logical/empirical truth; experiential truth; unknowability/infinity/infinitesimally (including "potentially infinite" / "potentially infinitesimal" conceptual roles).

Module D: Higher-order structure and dynamics.
hierarchy/systemic hierarchy; subpattern hierarchy; heterarchy; pattern web; tendency to take habits; tendency to reverse habits; self-weaving webs; autocatalytic sets (as a closure motif); resonance/dissonance; morphic resonance/anti-resonance; pattern flow networks; mind-world correspondence.

Remark 19. *This module is where Hyperseed's "network metaphysics" becomes mathematically demanding: once pattern webs allow loops (Hyperseed-Concept 132,* **??***), we must be careful to distinguish (a) harmless dynamical cycles from (b) forbidden definitional cycles. The subsequent use of resonance and morphic resonance (Hyperseed-Concept 159, 115) is intended to make "coupling across contexts" an operator with algebraic properties, not a metaphor [13].*

Module M: Mind and cognition.
mind; contexts and aspects; registration; sensory system; "X registers A"; perception; representation; icon; index; symbol; prediction and attraction; cause; extensional predictive attraction; control; control hierarchy; perceptual hierarchy (as needed); attention/attentional focus; varieties of knowledge; cognitive synergy; self/other; self-continuity; development; space.

Remark 20. *The appearance of "cognitive synergy" here is deliberate (Hyperseed-Concept 73). In the engineering literature on AGI, synergy names the phenomenon where multiple weak inference/control processes, each insufficient alone, combine into a more general competence [19]. In the present reconstruction, this will later be expressed in compositional terms: interactions between context-indexed processes that preserve enough structure to be nontrivially cumulative.*

Module V: Values, emotion, and ethics.
joy/woe; emotion; compassion; values/feelings/goals; reward; goals implicit/explicit; reward maximization/goal seeking; value paraconsistency; criteria for rational decision; rationality; pleasure/pain/open-ended intelligence; categorical imperative; rational ethical decision; evil; cultural morality; (plus auxiliary terms in Appendix C, e.g. humility/grandiosity/ambition/humbition).

Module L: Life and energy.

energy/genenergy; metabolism; reproduction (within species); alive/dead; lifeform (biological/synthetic/evolved); biological lifecycle strata (infant/young/mature/old/elderly).

Module K: Knowledge, belief systems, and science.

questions; answers; beliefs; belief system; axiom systems; question networks; knowledge; thinking; science; state-dependent science; open-ended intelligence/open-ended benefit (as community-level properties, when applicable).

Module A: Agency, society, tools, and embodiment.

intent; stimulate/inhibit; agent; autonomous agent; tasks/intellectuality/general intelligence/narrow intelligence (as assessment vocabulary); society; tribe; collective mind systems; mindplex; global brain; social roles (teacher/student/worker/colleague); communicative acts and systems; culture; tools; machines; computers and programs; engineered; physical realizations; physical reality/body; algebraically asymmetric physical reality.

Taken together, these terms fix the minimal interface between an individual cognitive process and the wider world it must act within. In particular, "intent" and "stimulate/inhibit" supply a compact language for goal-directedness and control: an agent is characterized not only by internal states, but by the kinds of interventions it can apply (stimulate) or withhold/suppress (inhibit) in itself or in its environment. The pair "agent / autonomous agent" is used to distinguish systems that merely participate in causal chains from systems that can maintain and revise policies (e.g., select actions under changing constraints) without being micromanaged by an external controller.

The assessment vocabulary "tasks/intellectuality/general intelligence/narrow intelligence" is listed here as a deliberately lightweight scaffold: it does not commit the paper to any single benchmark suite, but it enables later statements of the form "system X is more/less general than system Y relative to task family $\mathcal{T}$" without conflating capability with implementation. This also clarifies that "intellectuality" is treated as an evaluative label that can be made more precise (by specifying tasks, environments, or information constraints) rather than as a primitive substance.

The society terms are included because agency is rarely purely individual in practice: "tribe" and "society" point to social coordination problems, while "collective mind systems" is a neutral umbrella for cases where group-level dynamics exhibit memory, attention, and decision-making analogues. The role vocabulary (teacher/student/worker/colleague) is meant as a concrete handle on asymmetric interaction patterns: roles encode who is expected to provide feedback, who is expected to learn, who delegates, and who executes, and thus they support later formalization of learning channels and authority channels. Likewise, "communicative acts and systems" and "culture" serve as the medium through which norms, knowledge, and coordination strategies persist beyond any single individual, functioning as shared external state.

The "tools; machines; computers and programs" cluster isolates the engineered substrate through which agency is amplified and stabilized: tools are treated as persistently available transformations (often with standardized interfaces), and programs represent repeatable procedures whose behavior can be referenced, audited, and recomposed. "Engineered; physical realizations" indicates that the same abstract procedure can have multiple embodiments, and that the analysis will sometimes need to separate an algorithmic claim (what transformation is computed) from a physical claim (what constraints and costs arise in a particular realization).

Finally, "physical reality/body; algebraically asymmetric physical reality" flags that embodiment is not only a matter of being in the world, but of being in a world with directional constraints. "Algebraic asymmetry" is meant to capture the idea that the environment can impose non-reversible mappings (e.g., irreversible dissipation, one-way channels, or constraints that break naive symmetry assumptions), so that an agent's reachable states and control options form a structured set rather than an unconstrained space. This is relevant whenever the paper later compares what is possible "in principle" to what is possible under thermodynamic, causal, or communication constraints.

Remark 21. *The cluster "mindplex / global brain" is best read as a claim about* coherence at scale*: when communication, shared artifacts, and shared goals generate high mutual constraint, the collective begins to behave like a single cognitive system (Hyperseed-Concept 113,* **??***) [12]. This paper does not assume a single sociological mechanism; rather, it builds enough formal vocabulary to state what such coherence would mean in the same language used for individual minds.*

One practical implication of reading mindplex/global-brain as "coherence at scale" is that the relevant unit of analysis can shift with coupling strength: weakly coupled groups may be better modeled as interacting agents, while strongly coupled groups may be modeled as a single composite agent with internal substructure. This motivates keeping both individual-level and collective-level terms in the same inventory, since the paper will later need to state conditions under which the composite description is warranted (e.g., high-bandwidth communication, stable shared memory artifacts, or robust alignment of goal proxies) without presuming that those conditions always hold.

Module C: Communion, spirituality, aesthetics, cosmos.
intersubjective reality; intersubjective communion; spiritual and psychedelic communion; spirituality; entheogen; religion; aesthetics/art; archetypes; empirical reality/reality-systems; cosmos/eurycosm; near eurycosm; universe; multiverse; guidable multiverse; y-verse; anthropic principle; big bounce; wu-wei; wu-wei dynamics; fluid flow and optimal control.

This module groups vocabulary for (i) shared experiential structure, (ii) meaning-making and value formation, and (iii) large-scale cosmological hypothesis spaces that contextualize agency. "Intersubjective reality" indicates that some "realities" are maintained by shared interpretation, shared language, and shared practices, even when participants disagree on metaphysics; "intersubjective communion" then names the act or process of aligning experience and interpretation across persons. "Spiritual and psychedelic communion" is included because altered states and ritualized contexts can change the effective channel capacity and the salience landscape by which meaning and value are negotiated, and the framework aims to be able to talk about such shifts without either endorsing or dismissing any particular tradition.

The cluster "spirituality; entheogen; religion; aesthetics/art; archetypes" is not introduced as a separate domain from cognition, but as a set of commonly observed generators of shared symbols, shared narratives, and shared evaluative patterns. "Aesthetics/art" is included because aesthetic judgment can function as an implicit optimization signal (e.g., compressive preference for certain structures), and "archetypes" is treated as a shorthand for recurrent, culturally transmissible patterns that can organize both personal experience and collective behavior.

"Empirical reality/reality-systems" distinguishes (a) measurement-grounded constraint and (b) broader systems of practice and interpretation that embed those measurements into a usable world-model. The term "reality-systems" is intended to allow multiple coexisting model bundles (scientific, mythic, pragmatic, artistic) to be compared by their constraints, predictions, and coordination affordances, rather than being treated as mutually exclusive declarations. "Cosmos/eurycosm; near eurycosm" supplies a scale vocabulary: the eurycosm is the broadest context under discussion, while a near eurycosm is a restricted but still expansive cosmic neighborhood relevant to local agency, observation, or intervention.

"Universe; multiverse; guidable multiverse; y-verse" enumerates increasingly permissive hypothesis classes for what exists and what can be influenced. A "guidable multiverse" is listed to name a regime where selection or steering among branches/worlds is at least conceptually meaningful within the theory's causal structure, enabling discussions of strategy that range over cosmological-level degrees of freedom. The "y-verse" serves as a compact placeholder for a particular cosmological construction used elsewhere in the Hyperseed vocabulary, and is kept here to ensure that later references do not silently import assumptions about uniqueness of history or uniqueness of laws.

The "anthropic principle" and "big bounce" entries are included as examples of how observer-conditions and cosmological dynamics can constrain plausible world-models and thus constrain what "reasonable" priors look like for agents embedded in such worlds. They also function as named handles for arguments where the existence of observers (or cycles of expansion/contraction) changes what kinds of explanations are admissible.

Finally, "wu-wei; wu-wei dynamics; fluid flow and optimal control" marks a bridge between experiential/ethical vocabulary and technical control theory. "Wu-wei" is used as a label for policy regimes where effective action arises from minimal forcing and maximal exploitation of system dynamics; "wu-wei dynamics" then points to the idea that such regimes can be analyzed, at least in part, with the same mathematics used for flow and control. The mention of "fluid flow and optimal control" is meant to license later analogies and formal translations (e.g., between effortless action narratives and solutions to constrained optimization in dynamical systems) while keeping clear that the paper is not claiming a one-to-one identity between spiritual descriptions and differential-equation models.

2.4 Dependency DAG: definition and interpretation

We now define the intended meaning of the DAG used throughout the paper. In particular, we separate *definitional* dependence (what must be fixed in order to state a definition unambiguously) from other looser relations such as historical inspiration, pedagogical convenience, or empirical correlation. The goal is to make clear what kind of "depends on" is being asserted when we draw an arrow.

Definition 1 (Expository dependency DAG)**.** *Let $\mathcal{C}$ be the set of Hyperseed concepts (as listed in Appendix C). An* expository dependency DAG *is a directed acyclic graph $G = (\mathcal{C}, \to)$ such that for concepts $A, B \in \mathcal{C}$, an edge $A \to B$ means:*

> *"In this paper's formalization, the definition (or primary semantics) of B uses A as a primitive or as an already-defined construction."*

Here "uses" is meant in the strict sense relevant to formalization: writing down the core clauses that fix the meaning of B requires that the meaning of A has already been fixed (even if only up to *isomorphism or up to some declared equivalence notion). Equivalently, $A \to B$ asserts that A is among the prerequisites for stating what counts as an instance of B, what data B carries, or what laws B must satisfy, as formalized in this paper. If the original Hyperseed presentation treats A and B as mutually interdefinable, we may still orient one direction for exposition; such choices are explicitly recorded as "cuts". A cut is not a claim that one direction is* mathematically prior *in all possible reconstructions; it is a bookkeeping decision that pins down one acyclic presentation order for the purposes of this document.*

Remark 22. ***Notation and conventions.*** *Here $\mathcal{C}$ is literally a set whose elements are* concept names *(or, more precisely, the formal tokens by which we refer to concepts). The symbol "$\to$" is the edge relation of the graph G; it is* not *logical implication and should not be read as a claim that A entails B. "Directed acyclic" means that there is no sequence $A_0 \to A_1 \to \cdots \to A_n$ with $A_n = A_0$; so the DAG forbids definitional loops, even though later dynamical structures will happily contain feedback loops. In particular, acyclicity is a constraint on* presentational dependence*: it ensures that we can choose a topological ordering of $\mathcal{C}$ so that each concept is introduced only after its declared prerequisites. When we later speak informally of one concept "depending" on another, the intended meaning is always via this edge relation (or its transitive closure), not merely that the concepts tend to co-occur in narrative explanations.*

It is sometimes convenient to distinguish the edge relation $\to$ from its transitive closure $\to^*$. An edge $A \to B$ is a claim of direct *definitional use, while $A \to^* B$ expresses that A is an* indirect *prerequisite for B through some chain of intermediate concepts. The DAG representation deliberately does not insist that all indirect prerequisites be drawn as edges; doing so would add visual clutter without adding information, since those edges would be derivable from reachability in G.*

Remark 23. ***Intuition.*** *An edge $A \to B$ says: "to speak clearly about B, we first commit to a formal meaning for A." This is a modest claim, but it is surprisingly stabilizing: it turns a web of mutually suggestive ideas into a sequence of obligations. In philosophical terms, it is a way of keeping the freedom of a conceptual network while still making room for the stern virtues of definition and proof. One can also read the DAG as separating two questions: (i) what it is* useful *to mention when motivating a concept, versus (ii) what it is* necessary *to have already fixed in order to state the concept precisely. The dependency DAG only records (ii), even though our prose will often use (i) for intuition-building.*

Examples. *A typical edge is "Distinction $\to$ Equivalence Relation": we treat an equivalence relation as something induced by which pairs the observer does* not *distinguish (Hyperseed-Concept 98,* **??***). Another example is "Effort $\to$ Simplicity": once an observer-relative cost notion is fixed, simplicity can be defined as what costs less (Hyperseed-Concept 100, 169). At the module level, one may think of this DAG as a coarse "dependency tower" that orders which formal commitments are made first [8]. In both examples, the arrow is meant to be read as a constraint on the* form *of the definition: the clauses defining "Equivalence Relation" refer to "Distinction" (or to data structures already introduced to formalize distinction), and the clauses defining "Simplicity" quantify over or otherwise invoke the previously fixed cost/effort notion. Conversely, we do* not *mean that every informal discussion of effort must precede every discussion of simplicity; rather, we mean that the formal semantics of simplicity, as we pin it down, is parameterized by (or constructed from) the earlier effort notion.*

Usefulness. *The DAG is useful because it lets us locate disagreements. If two formalizations differ, it is often because they choose different incoming prerequisites for a concept, or because they place different "cuts" between mutually constraining notions. Recording cuts explicitly also prepares the ground for later tradeoff reasoning: sometimes one can take "route A" (define B from*

A) or "route B" (define A from B), and the two routes may not be equivalent under resource constraints [9]. In practice, this also means that the DAG can function as a checklist for hidden assumptions: when a definition of B appears to "reach backward" to an undeclared prerequisite A, we either revise the definition, add the missing edge, or acknowledge that we have made an implicit cut. Thus the DAG is simultaneously a representational artifact (a picture of dependence) and a consistency discipline (a way to prevent accidental circularity in the formal layer).

Remark 24. *A DAG is not the only reasonable representation: Hyperseed itself emphasizes heterarchy and feedback. The DAG is a* scaffold*: later sections (especially those on pattern webs, habits, and mind) reintroduce cycles as dynamical and/or higher-categorical structure, rather than as definitional circularity. One can think of this as a separation of levels: the DAG controls the order in which* meaning is fixed*, while later cyclic structures describe how, once meanings are fixed, the corresponding entities may interact, compose, and self-modify over time. In particular, permitting feedback in dynamics does not require permitting feedback in definitions; the former concerns trajectories and updates, whereas the latter concerns well-foundedness of semantic commitments.*

Remark 25. *One may view this as a gentle compromise between two temperaments. The first temperament wants everything to be founded on primitives; the second temperament suspects that primitives are often just temporarily frozen knots in a larger fabric of mutual dependence. The expository DAG admits the second suspicion while still satisfying the first temperament's demand: "show me where you started, and show me that you did not smuggle in what you later claim to derive." Put differently, we allow that many conceptual pairs can be made equivalent by adding enough structure (or by weakening the notion of equivalence), but we still insist on choosing and documenting a one-way street for the purpose of a linear exposition. This is especially important in a document that mixes formal definitions with interpretive commentary: the DAG is the device that keeps the interpretive layer from silently importing circular justifications into the formal layer.*

2.5 High-level DAG (module graph)

At the coarsest granularity, the build order used by this paper is:

```
(P) primitives
  -> (T) order/time/becoming
  -> (E) effort/simplicity
  -> (S) patterns/combination systems
  -> (D) webs/habits/resonance
  -> (M) mind/representation/control
  -> (A) agency/society/tools
  -> (C) reality-systems/cosmos/wu-wei

(I) information/truth/limits is a cross-cutting module:
  (P,T,E,S,D,M,V,A,C) -> (I) and (I) feeds back as constraints on all modules.

(V) values/emotion/ethics draws on:
  (P,T,E,S,D,M,I) -> (V), and then informs (M,A,C) via control and culture.

(L,K) life and knowledge/science sit between (M) and (A,C):
  (E,S,D,M,V) -> (L,K) -> (A,C).
```

Read the arrows as "uses as an input vocabulary" rather than as a claim that later modules are metaphysically prior: the intent is that each module introduces a handful of new definitions, operators, or invariants that become available to subsequent modules. In particular, the direction of the edges is chosen to support reconstruction: later modules should be expressible using the earlier ones with minimal back-references, so that proofs and toy implementations can proceed in a mostly one-way manner.

The labels are meant to be mnemonic anchors for a small number of recurrent themes. The primitives (P) are the minimal typed atoms and relations the rest of the paper is willing to assume. The order/time/becoming layer (T) introduces the infrastructure for sequencing, precedence, and change. Effort/simplicity (E) packages the bias toward compressive description and feasible action (so that "doing" and "describing" do not float free of resource constraints). Patterns/combination systems (S) then provide the combinators and closure principles needed to build reusable structures. Webs/habits/resonance (D) describe the way local patterns stabilize into networks, attractors, and learned regularities. Mind/representation/control (M) formalizes internal models, feedback, and policy-like constructs, after which agency/society/tools (A) treats multi-agent coordination and externalization into artifacts. Finally, reality-systems/cosmos/wu-wei (C) is where global "world"-level organization and the stance of non-forcing or alignment-with-dynamics can be stated using the preceding machinery.

The module graph is therefore not a claim that (say) society is "made of" mind in a reductionist sense; it is a claim about what the paper chooses to introduce first so that later constructions have a clear dependency footprint. This is also why the verbatim presentation resembles a build system: the goal is to make it easy for a reader (or an implementer) to locate which earlier notions are permitted as dependencies when a later notion is defined.

Remark 26. *The "feedback" from (I) back into all modules is not a violation of acyclicity because it is not a claim of definitional dependence in the strict sense; it is a claim of* constraint propagation*. Once one formalizes probability, entropy, uncertainty, or ineffability (Hyperseed-Concept 139,* **??***, 195,* **??***), one typically discovers that earlier definitions need to be indexed or weakened to remain coherent under limited representation. In other words, the mathematics of limits of representation is not merely another module; it is a set of lenses that refract the whole ontology.*

Concretely, this is the same phenomenon that occurs when a previously informal ontology is "typed": one may have introduced a clean notion of state, cause, or agent, and then later discover that any usable account must respect ambiguity, partial observability, finite code length, or statistical identification limits. The resulting revision is not that the earlier objects were ill-formed, but that their intended use requires additional side-conditions (e.g., measurability, identifiability, bounded description length, admissible inference rules). In this sense, (I) acts like a global coherence check: it does not supply new everyday entities, but it determines which formulations survive contact with compression, uncertainty, and representational finitude.

The placement of (V) similarly reflects a methodological choice. Values/emotion/ethics is downstream of (I) because any explicit treatment of preferences, norms, or affect that is meant to constrain control must confront uncertainty, aggregation, and the distinction between "what is true" and "what is endorsed." At the same time, (V) is upstream of key parts of (M) and (A): control without a value-like objective is underdetermined, and culture without norm-like stabilizers is structurally incomplete. The indicated influence of (V) on (M,A,C) should thus be read as the introduction of evaluative functionals that shape learning, feedback, and coordination, not as an attempt to reduce ethics to a single scalar.

The intermediate status of (L,K) is meant to capture that "life" and "knowledge/science" are simultaneously internal and external: they depend on representational and control notions (M), but they also generate the reliable regularities and instrumented practices that make large-scale agency and world-modeling possible. The edge $(E, S, D, M, V) \to (L, K)$ highlights that accounts of living systems and scientific inference typically require (i) resource-bounded optimization (E), (ii) compositional structure (S), (iii) networked stabilization and habit-formation (D), (iv) internal representation and feedback (M), and (v) evaluative constraints (V) such as survival, coherence, or epistemic norms. The subsequent $(L, K) \to (A, C)$ then reflects that collective agency and "cosmos"-level claims are best stated after one has a story about adaptive organization and disciplined knowledge production.

This orientation matches the intended mathematical reconstruction: Sections 3–5 build the formal substrate and an executable toy instance, and Sections 6–26 then walk the modules in roughly the above order. The ordering is not meant to be the only possible traversal; rather, it is a default path that minimizes forward references while keeping the reader's stack small. When later sections temporarily refer back to earlier modules, it is intended to be for reuse of established operators or for the kind of representational constraint updates described in the remark above, not for introducing new foundational primitives retroactively.

2.6 Key concept-level dependencies (selected adjacency list)

The module DAG is useful but too coarse to guide formalization choices. Below is a selected adjacency list for the most load-bearing concept dependencies. Each arrow indicates a likely definitional prerequisite in the systematic reconstruction. In particular, an arrow $A \to B$ should be read as: "to state B cleanly (and to make its intended invariances explicit), one typically needs a prior handle on A." This is weaker than logical implication and weaker than ontological priority; it is primarily a claim about what scaffolding is needed so that later definitions do not smuggle in unexamined structure.

Remark 27. *The adjacency list is intentionally phrased in informal English, because at this stage it is serving as a* planning document*. Later sections will replace many arrows by explicit definitions and theorems. Still, it is useful to read these lists as hypotheses about what must come first, and to notice where one might want alternative cuts. In practice, an "alternative cut" often amounts to changing which notions are taken as primitive (e.g. whether to treat effort or simplicity as basic), or to changing the ambient notion of "context" under which distinctions, groupings, and cost are evaluated. Where the list suggests a linear prerequisite chain, one should also watch for latent mutual dependence: some concepts may be most naturally characterized by a small package of interlocking definitions rather than a single strict ordering.*

Primitives to structure.

```
Occasions of experience
  -> (difference, presentational immediacy, abstract/concrete)

Difference
  -> distinction
  -> repetition, variety, non-duality, non-dual variety

Distinction
  -> grouping, equivalence relation (observer-relative)
  -> contexts/aspects (as "which distinctions are active")
  -> (proto-)entropy via distinction graphs (cross to Module I)
```

The intended reading here is that "occasions of experience" supply the minimal arena in which anything like comparison can occur; "difference" is then the first relational articulation within that arena, and "distinction" is difference stabilized into a usable form (e.g. something that can be applied repeatedly, carried across contexts, or used to define sameness-by-absence-of-separation). The parenthetical cluster "(difference, presentational immediacy, abstract/concrete)" is deliberately grouped because, in some reconstructions, one might treat these as co-emergent facets rather than as strictly sequential.

Remark 28. *Philosophically, this part of the graph is where "the given" is first permitted to articulate itself. A distinction (Hyperseed-Concept 98) is not merely a label; it is a commitment to treating some pair as different* within a context*. Grouping and equivalence then arise as the shadow cast by repeated non-distinctions: to say "these are the same" is to say "my active distinctions do not separate them" (Hyperseed-Concept* **??***,* **??***). The cross-edge to entropy anticipates later constructions where a distinction structure induces an information measure (Hyperseed-Concept* **??***), in the spirit of distinction-based accounts of information such as logical entropy [17]. Two technical cautions are implicit. First, the observer-relativity of equivalence is not meant to trivialize equivalence; rather, it highlights that an equivalence relation is indexed by a selected set of active distinctions (and, therefore, by the granularity at which the observer is operating). Second, the route from distinctions to entropy is not only "counting distinctions"; it also depends on what is taken to be the space of possible distinctions (or the space of distinguishable states) and on whether distinctions are weighted, probabilistic, or context-conditioned. These choices determine whether the induced information measure behaves more like a simple partition measure, a graph-based impurity, or a probability-weighted entropy analogue.*

Order, time, and process.

```
After/before (partial order; directional flow)
  -> proto-time
  -> becoming (process compatible with proto-time)
  -> event and process calculi (history-indexed predicates)
```

Here "after/before" is treated as an ordering primitive that can be applied wherever directed dependence is meaningful (causal, computational, inferential, or phenomenological). The phrase "partial order" matters: proto-time is allowed to have incomparable elements (e.g. independent subprocesses), which is essential if one later wants to accommodate concurrency, branching, or multi-scale organization without forcing an artificial total order.

Remark 29. *The asymmetry implicit in after/before (Hyperseed-Concept 52, 63) is not necessarily a claim about physical time; it is a way of encoding directed constraint. Proto-time (Hyperseed-Concept 142) is then the minimal structure needed to speak about "earlier" and "later" without yet committing to metric time. This is congenial to process-ontological perspectives (compare [15]) while remaining fully compatible with purely computational histories. One reason to keep this layer thin is that later formalisms (e.g. event calculi, process calculi, rewriting systems, or state-transition semantics) can impose additional structure as needed: discrete versus continuous time, branching versus linear histories, and explicit notions of simultaneity. By placing "becoming" after proto-time, the graph flags that becoming is not merely change, but change* tracked *along an ordering—a requirement that becomes important when defining persistence, novelty, and history-indexed predicates without circularity.*

Effort, simplicity, and pattern bootstrapping.

```
Effort (resource/cost, observer-relative)
  -> resistance/submission, opening/closing (effort gradients or regimes)
  -> simplicity (what costs less effort)
  -> minimum representational effort (principle)

Simplicity + combination (composition law)
  -> compositional simplicity
  -> pattern (representation-as-simpler)
  -> pattern intensity (quantitative degree of simplification)
```

This cluster is written to keep the reconstruction honest about where optimization enters. Treating effort as primitive allows one to talk about "cheaper" and "costlier" representations before committing to any particular currency (time, memory, description length, energy, attention, or risk), while still permitting later specialization. The "minimum representational effort" line should be read as a bridge principle: it does not claim that observers always achieve minima, but that there is explanatory pressure toward lower-effort representations when available.

Remark 30. *This is the hinge on which the formal reconstruction turns. If effort is primitive (Hyperseed-Concept 100), then simplicity becomes a derived notion, not an aesthetic preference (Hyperseed-Concept 169). The move from "simplicity + combination" to "pattern" is the point where structure becomes reusable: a pattern (Hyperseed-Concept 130) is, roughly, a way of representing many instances with less effort than enumerating them. This is closely aligned with the "weakness" viewpoint, where collapsed distinctions are not merely failures but also generalizations [3, 2]. The explicit mention of a "composition law" is meant to prevent ambiguity about how effort aggregates: does combining two representations add costs, take a maximum, introduce overhead, or allow shared substructure? Different choices yield different notions of compositional simplicity and, consequently, different notions of what counts as a "pattern." By treating pattern intensity as a quantitative degree of simplification, the graph anticipates later questions of scale: when does a small local gain in representational effort accumulate into a robust, system-level regularity, and when is it too marginal to matter?*

Patterns to properties, emergence, and blends.

```
Pattern intensity
  -> property attribution (patterns-as-properties)
  -> emergence (intensity not reducible to components)
  -> blend (compositional merge of pattern systems)

Combination systems / interpreted combination systems
  -> combinational dynamical causal models
  -> combinatorial-categorical patterns
```

The first block indicates a conceptual shift from "patterns as compressions" to "patterns as predicates we are willing to project." If a pattern robustly reduces effort across a family of instances, it becomes natural to reify it as a property attribution: we treat the pattern as something the instances *have*. The second block flags that once combination is formalized (syntactically and/or semantically), one can state more structured claims about dynamics and causality using the same underlying compositional machinery.

Remark 31. *Pattern intensity is meant to be the quantitative pivot: it allows one to say not merely that a pattern exists, but that it compresses (or coheres) more or less strongly (Hyperseed-Concept 82, 181). Once intensity exists, emergence becomes a testable claim: does the whole exhibit an intensity that cannot be accounted for by the intensities of the parts (Hyperseed-Concept* **??***)? These ideas connect directly to Hyperseed's broader account of pattern and hierarchical/heterarchical organization [5]. The dependency "pattern intensity $\to$ blend" is intended to guard against arbitrary mixing: without some notion of strength (or salience) of patterns, a "blend" risks being merely a disjunction or a bag-of-features. Intensity provides a handle for principled merge policies (e.g. prioritize high-intensity substructures, penalize incoherent overlaps, or preserve components that drive most of the simplification). Similarly, the route through interpreted combination systems emphasizes that "dynamical causal models" and "combinatorial-categorical patterns" are not separate add-ons: they are ways of making the same compositional substrate carry semantics (interpretation) and temporal dependence (dynamics).*

From patterns to webs and dynamics (habits/resonance).

```
Patterns
  -> subpattern hierarchy and systemic hierarchy
  -> heterarchy and pattern web (loops and multi-path dependence)

Pattern webs + proto-time
  -> pattern flow networks
  -> tendency to take habits / reverse habits (time-indexed reinforcement)

Habit dynamics + coupling across contexts
  -> morphic resonance / anti-resonance
  -> self-weaving webs (autocatalytic closure motif)
```

The intent of this final cluster is to mark where "pattern" stops being a static representational convenience and becomes an active participant in system evolution. A subpattern hierarchy suggests nested reuse (patterns built from patterns), whereas heterarchy and pattern webs allow mutual constraint and feedback (patterns supporting, inhibiting, or redefining one another in loops). Adding proto-time turns these webs into directed flow structures, so that one can speak about reinforcement, stabilization, and path-dependence without presupposing a particular physics.

Remark 32. *The step from "pattern web" to "pattern flow networks" is meant to introduce a graph-of-graphs viewpoint: nodes may themselves be patterns (or pattern systems), edges encode influence or transport of constraint, and temporal indexing (via proto-time) makes it meaningful to compare a web "now" to the same web "later." In that setting, a "habit" can be treated as a statistically or structurally stabilized transition tendency: repeated traversals of certain subwebs lower effective effort along those routes, which in turn increases their future likelihood, producing reinforcement. The explicit inclusion of "reverse habits" is a reminder that stabilization is not monotone: regimes can switch, gradients can invert, and previously reinforced pathways can be actively suppressed when contexts shift. Finally, "morphic resonance / anti-resonance" is placed downstream of coupling across contexts to emphasize that resonance is not magic correspondence; it is a claim about cross-context influence mediated by shared (or alignable) pattern structure. Anti-resonance, correspondingly, captures systematic interference: coupling that increases effort, destroys coherence, or forces incompatible distinctions to coexist. The "self-weaving webs" arrow indicates a* *closure motif: webs that, through their own habit dynamics, maintain and elaborate the very pattern infrastructure that enables further habit formation, yielding an autocatalytic style of organization.*

Remark 33. *The progression here is from static structure to time-sensitive reinforcement. Pattern webs (Hyperseed-Concept 132) allow the same local pattern to participate in multiple global organizations (heterarchy; Hyperseed-Concept* **??***). Adding proto-time makes reinforcement meaningful: habits are, mathematically, stable tendencies of update rules over histories (Hyperseed-Concept 189, 188). Morphic resonance is then a claim about how habit-strength can propagate across contexts without requiring literal identity of the underlying substrate (Hyperseed-Concept 115, 114) [13]. In other words, the dependency chain is not merely thematic: each step introduces an additional degree of freedom that the previous level cannot express cleanly. "Static structure" accounts for compositional organization, but cannot on its own distinguish a pattern that is merely present from a pattern that is* maintained *by repeated successful updating across episodes; proto-time supplies the ordering needed to define reinforcement as a property of trajectories rather than snapshots. Heterarchy matters here because reinforcement can stabilize multiple, partially incompatible global organizations simultaneously, so that "the same" local regularity can be reinforced in different roles depending on which context is active. On this reading, morphic resonance (and anti-resonance) is best interpreted as a constraint on cross-context generalization of such stabilized tendencies: what transfers is not a token pattern but a bias in the update dynamics that makes certain organizations easier (or harder) to reconstitute elsewhere.*

Mind, representation, control, and self.

```
Pattern web + contexts/aspects
  -> mind (as a system of interlocking representations and processes)

Registration/sensory system + contexts/aspects
  -> perception (registration causing representation updates)

Representation
  -> icon/index/symbol (structure-preservation constraints)
  -> prediction/attraction (probabilistic or inferential dependence)
  -> cause (attraction plus precedence plus simplicity bias)
  -> control and control hierarchy (action selection shaping future perception)

Attention/attentional focus
  -> cognitive synergy (interaction of multiple partial inference/control processes)

Self/other + self-continuity
  -> development (directed transformation of self-model patterns)
  -> consciousness/reflective consciousness (closure conditions on self-modeling)
  -> autonomy (degree of internally sourced causation/control)
```

Remark 34. *This list expresses a sober thesis: mind is not taken as a primitive substance, but as a certain kind of organized pattern web with context-sensitive update rules (Hyperseed-Concept 111, 132, 86). Representation then subdivides by semiotic constraint (icon/index/symbol; Hyperseed-Concept* **??***,* **??***, 184), while control and attention formalize the closed-loop character of cognition (Hyperseed-Concept 87, 60). The final cluster (self/other, consciousness, autonomy) names closure properties: the system not only predicts and acts, but also models itself as a locus of prediction* *and action (Hyperseed-Concept 165, 85, 119). A useful way to read the arrows is as "explanatory sufficiency" rather than strict logical implication: given a pattern web plus a disciplined notion of context/aspect, one can* construct *a notion of mind as the emergent regime in which representations and processes mutually constrain one another over time. Likewise, perception is not treated as a passive channel but as the special case where registration events are explicitly permitted to drive representation updates under contextual gating; this makes perception inherently fallible and revisable, rather than definitionally veridical. The icon/index/symbol subdivision then clarifies what is being preserved across updates (similarity structure, contiguity/trace, or convention-like mapping), which in turn constrains what kinds of prediction are cheap or robust. "Cause" appears here as an enriched notion of attraction: precedence supplies temporal direction, and a simplicity bias supplies a compressive preference that turns mere correlational pull into a causal-ascription heuristic within the system's inference machinery. Finally, the self/other and self-continuity cluster can be read as the point at which the representational/control loop becomes reflexive: development tracks systematic changes in the self-model, consciousness adds a closure requirement on how that self-model participates in updating, and autonomy measures the extent to which control signals are generated endogenously rather than merely relayed from external perturbations.*

Values, society, and cosmos.

```
Paraconsistent valuation (both-for and both-against)
  -> joy/woe, emotion, compassion
  -> values/feelings/goals, reward, rationality/ethics

Multiple minds + communication + shared artifacts
  -> society/tribe/culture, collective mind systems
  -> mindplex/global brain (high-coherence collectives)

Reality-systems (prediction-stabilized patterns)
  -> physical reality/body (special constrained reality-system)
  -> intersubjective reality (shared constraints)
  -> cosmos/eurycosm and (multi-, y-)verse constructions (observer- and signal-relative)
```

Remark 35. *The first arrow block emphasizes that valuation need not be a single number: it can be a structured, even conflicting evidence object (hence paraconsistency; Hyperseed-Concept 198) [23]. The second block emphasizes that collectives are not just sums of individuals; coherence can be a new mode of organization (Hyperseed-Concept 170, 91, 113). The third block makes explicit that "reality-system" is a technical phrase: it refers to those stabilized patterns that survive prediction and action loops, including intersubjective stabilization (Hyperseed-Concept 149,* **??***) and, at the widest scale, eurycosmic constructions (Hyperseed-Concept* **??***, 116) [18]. The inclusion of wu-wei (Hyperseed-Concept 206, 207) anticipates later discussion where "least-effort" dynamics are treated in a quantale-geometric style [21]. It is also important that paraconsistent valuation is presented upstream of emotion and ethics: on this dependency view, affective phenomena are not external add-ons to "rational" cognition, but are emergent summaries of multi-sided evaluative evidence that can legitimately contain tension (simultaneous approach/avoid, endorsement/repulsion) without forcing immediate collapse to a scalar utility. The bridge to reward and goals then becomes a question of how such structured valuations are operationalized into policy updates, including cases where different sub-processes endorse incompatible actions and attention/control must negotiate a workable compromise. For the social block, communication and shared artifacts are not merely channels for* *transmitting already-formed beliefs; they function as persistence mechanisms that stabilize representations across individuals and across time, thereby enabling collective memory and coordinated control policies that no isolated agent could maintain. The mindplex/global brain line can therefore be read as a further closure condition: when inter-agent coupling becomes sufficiently high-bandwidth and mutually predictive, the collective can exhibit its own attractors, norms, and self-maintaining dynamics. For the reality-system block, "prediction-stabilized" highlights that what counts as "real" in this framework is inseparable from the success of coupled prediction–action loops under constraints. The "physical reality/body" item is then not an ontological demotion but a special case in which constraints are unusually tight, regular, and widely shared, making certain pattern families exceptionally stable across observers and interventions. Intersubjective reality is explicitly listed to mark that many constraints are social (linguistic, institutional, technological) and hence partially negotiable, yet still objectively operative for agents embedded in them. Finally, the observer- and signal-relative phrasing for (multi-, y-)verse constructions is meant to keep the dependency honest: cosmological scope here is treated as an extrapolation of stabilization principles, not as a leap to unconstrained metaphysics; the later wu-wei discussion then adds a variational/geometric intuition for why some stabilized trajectories appear "effortless" relative to the agent's control costs.*

2.7 Mathematical load-bearing walls

The DAG above is large; only a few edges are truly load-bearing for the mathematical reconstruction. These are the edges that force the choice of formal core (Section 3) and that determine whether the rest of the ontology can be reduced cleanly. In practice, "load-bearing" here means: if one weakens or removes the edge, one typically loses either (i) compositionality (the ability to combine local descriptions into coherent global ones), (ii) gradedness (the ability to speak about degrees rather than only yes/no predicates), or (iii) robustness under partial inconsistency (the ability to reason without requiring total agreement across perspectives).

1. **Distinction → weakness/simplicity.** We need a compositional way to encode "which distinctions an observer fails to make" and how that composes. This is the primary motivation for quantale weakness.

   Remark 36. *The idea is that an observer's limitations can be treated as algebraic data: a structured record of which pairs are* not *separated by the observer's active distinctions (Hyperseed-Concept 98). Quantale weakness (Hyperseed-Concept 143) then aggregates these undistinguished pairs in a way that composes cleanly across contexts [3]. Philosophically, one may say: weakness is not merely error; it is the condition that makes generalization possible. A technical point worth making explicit is that the algebra is doing two jobs at once: it records indistinguishability data, and it supplies an internal notion of accumulation. When weakness is modeled in a quantale-like structure (complete lattice with a monoidal product), one can interpret joins as "taking the least upper bound of limitations" and the monoidal product as "combining limitations along a pipeline of observations." This is exactly the kind of compositional bookkeeping that later allows "simplicity" to be defined relative to what distinctions are even available to the observer in the first place.*

2. **Effort → observer-relative simplicity.** Hyperseed treats effort as primitive; simplicity is defined by effort cost. This commits the formalization to explicit resource indexing, not just abstract truth-valuation.

   Remark 37. *This is where the reconstruction refuses a common temptation: to identify simplicity with some absolute minimality. Instead, simplicity becomes relative to a system's costs* *and constraints (Hyperseed-Concept 100, 169). This is congenial both to computational realism and to the more philosophical point that "the same" world can be carved differently by different modes of access. Formally, the point of taking effort as primitive is that it provides a parameter against which many later notions can be measured: approximation quality, compressibility, action feasibility, and even which inferences are practically stable. In particular, if two representations have equal truth-conditions but different effort profiles, Hyperseed treats them as non-equivalent for an embodied agent; this is the step that prevents the framework from collapsing into purely extensional semantics.*

3. **Simplicity → patterns and pattern intensity.** Patterns are "representations as something simpler"; pattern intensity is the quantitative pivot that generates properties, emergence, and structure.

   Remark 38. *Pattern intensity is crucial because it turns "pattern" from a binary predicate into a graded quantity that can support inequalities, monotonicity, and stability claims (Hyperseed-Concept 130,* **??***) [5]. Without some intensity-like notion, emergence risks becoming merely verbal; with it, emergence can be articulated as a failure of additive decomposition, or as a fixed point phenomenon in dynamical settings. It is also the bridge from local comparisons ("this description is simpler than that one for this observer") to global structural claims ("these macroscopic regularities persist under perturbation"). Intensity gives a handle for statements of the form: under a given effort budget, a pattern remains identifiable (or predictive) across a family of contexts; this supports the later use of order-theoretic or enriched-categorical machinery, where morphisms can carry not only existence but strength.*

4. **Pattern webs → habits → morphic resonance.** To reach morphic resonance without handwaving, one needs (i) a dynamical reinforcement operator on webs, and (ii) a cross-context coupling notion that can transmit reinforcement without forcing consistency.

   Remark 39. *Here the demand for rigor is especially sharp: morphic resonance (Hyperseed-Concept 115) is only interesting if it is more than a poetic synonym for "things influence each other." The formal core must therefore supply operators that (a) update pattern webs over proto-time (habits; Hyperseed-Concept 189) and (b) couple distinct contexts in a way that can amplify or damp signals even under partial conflict [13, 24]. The load-bearing content is that "habit" must be representable as an* iterable *update rule on structured objects (pattern webs), rather than as an informal narrative of repetition. In other words, the formalism must make sense of the question "what is the cumulative effect of applying reinforcement n times?" and must allow comparison of the resulting webs (e.g., by a refinement order or graded similarity). Likewise, "resonance" must be a controlled interaction between contexts: strong enough to transfer reinforcement, but weak enough to avoid imposing a single global state that would erase the very contextuality the framework aims to preserve.*

5. **Value conflict → paraconsistent evaluation.** Ethics, motivation, and even stable agency require a semantics that can represent genuine conflict without collapsing everything into a single scalar utility and without explosive inference. This motivates p-bit-style paraconsistent valuation.

   Remark 40. *This is, in a sense, the ethical analogue of observer-relativity. Just as an observer may be unable to draw every distinction, an agent may be unable (or unwilling) to collapse every value into a total order (Hyperseed-Concept 197, 198). The formal core therefore treats conflict* *as first-class data, rather than as a bug to be eliminated by definition [23]. The mathematical pressure is that evaluation must remain compositional even when inputs disagree: combining reasons should not force immediate resolution, but it should still yield structured outputs that can guide action. A paraconsistent scheme can represent, for the same proposition or action, simultaneous support and opposition at varying strengths; this makes it possible to model ethical tradeoffs, motivational ambivalence, and multi-objective control without reducing everything to a single number that hides internal tensions.*

6. **Context and composition $\to$ enriched/categorical structure.** The ontology repeatedly composes processes, representations, and contexts. A quantale-enriched category (and in some places higher-categorical structure) is the natural container for these compositions.

   Remark 41. *The guiding thought is Russellian in spirit: to avoid confusion, one should make the types and modes of combination explicit. Categorical structure provides disciplined composition laws; quantale enrichment provides graded comparison and approximation. Together, they let us say not only that two contexts relate, but* to what degree *and in what compositional manner (Hyperseed-Concept 86, 141). Concretely, enrichment is what allows "maps between contexts" to carry nontrivial semantics: a hom-object can encode effort, approximation loss, weakness, or similarity, and composition can express how these quantities accumulate along a chain of translations. Where the ontology speaks about patterns-of-patterns, or transformations between transformation-schemes, higher-categorical structure becomes the disciplined way to avoid conflating levels (processes, meta-processes, and so on), while still permitting the "web" viewpoint in which multiple compositional routes can coexist and be compared.*

A final methodological note is that these edges are not merely a list of topics, but constraints on *what kinds of definitions are admissible*. For example, if weakness does not compose, then observer-relativity becomes inexpressible at scale; if effort does not index semantics, then "simplicity" becomes detached from agency; if conflict explodes inference, then any realistic value theory becomes trivial. The formal core is therefore selected less by aesthetic preference than by the requirement that these specific forms of composition, grading, and controlled inconsistency all coexist in one coherent mathematical envelope.

2.8 Where observer-relativity and paraconsistency enter

Two "non-classical" themes appear early and remain present throughout:

Observer-relativity. Distinctions, simplicity, probability, reality, and even "what counts as the same" are indexed by a system or context. Formally, this means we will not treat most predicates as absolute, but as living in a structure like "context C supports relation/predicate R with value v". One can read this as replacing a global truth-valuation $R(\cdot)$ with a family of context-indexed valuations $R_C(\cdot)$, or more generally with an evidential semantics in which C contributes both the available vocabulary (what can be expressed) and the operational constraints (what can be measured, computed, or distinguished). In particular, "indexed by a system or context" is meant literally: changing C is allowed to change not only numerical outputs (e.g. probabilities) but also which comparisons are meaningful at all (e.g. whether two states are distinguishable or whether a coarse-graining identifies them). This perspective also makes room for partiality: a context may return a value v that is not simply `true/false` but reflects limited resolution, missing resources, or an explicitly graded assessment.

A useful mental model is that C packages a viewpoint together with its interface: sensors, internal state, background theory, and cost constraints. Then the same underlying situation can induce different supported relations R in different contexts, not merely because of disagreement, but because the contexts genuinely implement different partitions of possibility space and therefore different notions of "sameness," "cause," or "relevance."

Remark 42. *The phrase "supports ... with value v" anticipates the algebra of graded evidence used later. The point is not merely that different observers disagree; it is that the ontology is built to record* how *they disagree, e.g. which distinctions are unavailable, which resources are scarce, or which contexts are operative (Hyperseed-Concept 98, 100, 86). In particular, the additional slot v is intended to store more than a Boolean; it can encode confidence, strength, cost-adjusted plausibility, or multi-component evidence, depending on the later choice of evidence domain. Thus, rather than saying "R is true," the framework aims to say something closer to "R is supported to degree v in context C," where the phrase "in context C" may itself include which measurements were admissible and which inferential moves were allowed.*

It is also important that observer-relativity is not an optional "interpretation layer" placed on top of an otherwise absolute theory. Instead, later constructions (e.g. combining aspects, projecting to coarser views, or comparing agents with different resources) will use the context-indexing as a primitive bookkeeping device: if two contexts cannot even express the same predicate in the same way, the formalism should make that mismatch explicit rather than silently identifying them.

Paraconsistency. Hyperseed allows (and sometimes requires) situations where evidence supports both a claim and its negation, or where boundaries are both present and absent depending on perspective. This shows up in at least: (i) conflicting evidence about equivalence/distinction, (ii) value conflicts in motivation and ethics, (iii) ambiguous self/other boundaries, and (iv) limit constructions where signals are "both random and predictable" (y-verse style motivation). The formal core therefore must support non-explosive reasoning with explicit conflict. Concretely, the target behavior is that from a contradiction $P \wedge \neg P$ one should *not* be able to infer arbitrary conclusions; instead, contradictions are treated as localized states of tension that can be carried, compared, and acted upon. A standard way to model this is to let the "evidence value" for a proposition contain separate components for support and refutation (so that both can be simultaneously nonzero), yielding the familiar four qualitative modes: supported-only, refuted-only, both (inconsistent), and neither (undetermined). In this reading, the phrase "supports both a claim and its negation" is not a rhetorical flourish but a structural allowance: the evidence algebra is expected to represent such pairs directly, rather than forcing an immediate collapse to classical inconsistency.

Paraconsistency also interacts with observer-relativity in two directions. First, conflicts can arise *across* contexts (different C give incompatible supported values for the same R), and the framework should be able to store and reason about those incompatibilities without erasing them. Second, conflicts can arise *within* a single context when the context itself aggregates multiple subsystems, time-slices, or heuristics that are each partially reliable; in that case the context-indexing does not remove contradiction, and paraconsistency is needed even before comparing observers.

Remark 43. *The intended stance is pragmatic: if a system is to reason and act under uncertainty and disagreement, it must be able to represent conflict without self-destruction. A paraconsistent evidence domain can encode "for" and "against" simultaneously and then permit later operators (attention, control, learning) to modulate which parts of the conflict matter in which contexts (Hyperseed-Concept 60, 87, 198) [23, 24]. This is also a way to keep "contradiction" distinct* *from "noise": the system may track that there are coherent reasons on both sides, and that resolving the conflict requires additional effort, new measurements, different abstractions, or a change in priorities rather than merely more samples. In the limit, classical reasoning can be recovered as a special case when the evidence domain happens to forbid simultaneous support and refutation (or when an operator projects inconsistent states down to a consistent subdomain), but the default theory keeps the richer state space available so that conflicts can be carried forward until some later stage makes them decision-relevant.*

2.9 How this DAG drives the rest of the paper

The remainder of the paper follows the DAG in three passes: the goal is that each pass tightens the link between informal motivation and formal commitments, while keeping the direction of dependence explicit. In particular, earlier material is intended to be sufficient to justify later constructions without silently importing extra structure, so that the reader can check (locally) what is doing the work at each step.

- Sections 3 and 4 instantiate the load-bearing walls as a minimal core. In practice, this means fixing the smallest set of primitives, typing/axiom schemes, and admissible inference moves that all downstream modules will reuse.
- Section 5 demonstrates that the core reaches nontrivial dynamics (including morphic resonance) in a finite universe. The point of placing this early is diagnostic: it serves as a concrete witness that the core is not merely consistent-by-design, but also expressive enough to generate structured behavior under explicit, fully checkable choices.
- Sections 6–26 systematically assign each Hyperseed concept (Appendix C) to a precise construction, definition, or reduction within the core. This is the "accounting" pass: wherever a term appears in the concept inventory, the reconstruction supplies an unambiguous formal handle (or a declared reduction) and makes clear which earlier nodes in the DAG it actually depends on.

Because the DAG is acyclic, this organization also enforces a reading discipline: if a later module seems to require a premise not yet introduced, that is a signal either that the dependency edge is missing from the diagram or that the module is in fact using additional assumptions that should be surfaced and labeled. Accordingly, forward references are meant to be informational rather than foundational: they may preview where a definition will be used, but the definition itself should only rely on what lies upstream.

Remark 44. *A useful way to read what follows is to treat the DAG as a promise: whenever a later section appeals to a concept A, the reader should be able to trace backwards along $\rightarrow$-edges to see what was already fixed, and therefore what is genuinely being assumed. If the reconstruction succeeds, this trace should feel like descending a well-built staircase rather than like falling through a trapdoor. Operationally, this means that apparent "intuitions" should be recoverable as consequences of earlier commitments, and that any genuinely new assumption should correspond to an explicit new node or edge rather than an unnoticed shift in meaning.*

Concept coverage in this section. This section explicitly inventories (by module) the full Hyperseed vocabulary and provides the master dependency DAG used to structure the formal reconstruction (including the load-bearing edges that determine the formal core). As a result, the reader can use this section as an index of where each term is anchored: the inventory locates the first point of formal responsibility for a concept, while the DAG clarifies which earlier notions must be in place for that anchoring to be legitimate.

3 Minimal formal core

Outline

- Specify the minimal "data types" (entities, contexts, valuations, relations) and the design requirements (observer-relativity, paraconsistency, compositionality).
- Define the `p-bit` truth-value domain and a small set of paraconsistent connectives and derived observables (support, opposition, conflict, indeterminacy).
- Define a commutative quantale V that can simultaneously serve as (i) an evidence domain and (ii) an aggregation domain for weakness and pattern measures.
- Define V-valued relations and their quantale-style composition, and interpret them as approximate correspondences.
- Define a V-enriched category (and/or quantaloid) of contexts/processes and the basic notion of approximate morphism used later for mind–world correspondence.
- Define resonance as an operator induced from `p-bit` evaluations (via a logic-to-complex embedding) and give the abstract form of resonance-driven propagation.

Summary and Hyperseed concepts covered

This section fixes a small algebraic/categorical substrate that will be used throughout the paper. The guiding idea is that Hyperseed's "soft" notions (distinction, simplicity, pattern, emergence, habit, resonance) are best treated as *graded* and often *paraconsistent*, and that they must compose cleanly across contexts [1].

Concretely, we build on: (i) `p-bit`-valued paraconsistent semantics for partial and conflicting evidence [23, 24]; (ii) a commutative quantale V for aggregating evidence, weakness, and intensities [3]; (iii) V-valued relations, modules, and V-enriched categories to express composition, approximation, and context translation [7]; and (iv) a resonance construction that turns paraconsistent evaluation structure into a coupling signal [24].

Hyperseed concepts covered.

- Distinction; indistinction/equivalence (observer-relative, graded). Hyperseed-Concept 98; Hyperseed-Concept **??**.
- Contexts/aspects; representation; approximate morphism / correspondence. Hyperseed-Concept 86; Hyperseed-Concept 157; Hyperseed-Concept 112.
- Effort-adjacent core proxies: weakness/simplicity as "failed distinctions" (full effort theory comes later). Hyperseed-Concept 202; Hyperseed-Concept 143; Hyperseed-Concept 169.
- Pattern and emergence (only the minimal scaffolding needed here; the main treatment is in Section 9). Hyperseed-Concept 130; Hyperseed-Concept **??**.
- Resonance/dissonance; the core coupling idea that later becomes morphic resonance/anti-resonance. Hyperseed-Concept 159; Hyperseed-Concept 97; Hyperseed-Concept 115; Hyperseed-Concept 114.

Purpose and "minimality" criterion. By "minimal formal core" we mean the smallest collection of mathematical primitives sufficient to (a) state the main constructions unambiguously, and (b) ensure that all later notions are *portable across contexts* via explicit composition laws. In particular, the formal layer here is not intended to be a full foundational reconstruction of Hyperseed; rather, it functions as a common interface between informal conceptual vocabulary and the later technical developments (patterns, emergence, translation, and resonance dynamics). A useful litmus test for whether a primitive belongs in this section is whether it is needed to define: (i) observer-relative evaluation, (ii) graded combination/propagation of evaluations, and (iii) approximate correspondence between structures.

Observer-relativity as an explicit parameter. The term "context" is treated as a first-class parameter rather than implicit background. Operationally, this means that any evaluation (truth, evidence, similarity, or correspondence) will be indexed by, or otherwise depend on, a context/aspect, so that the same entity may support different (even incompatible) attributions under different observational regimes. This explicit context dependence is what allows later constructions to represent translation, alignment, and weaving between viewpoints without collapsing all evaluations into a single globally consistent picture.

Why paraconsistency is structural rather than exceptional. Hyperseed-style modeling frequently involves partial evidence and mutually antagonistic constraints (e.g. an item that is simultaneously supported and opposed by different processes, or by the same process under different approximations). The point of introducing `p-bit` at the core is that "conflict" should be representable *without* forcing triviality: contradiction is recorded as a stable, graded state rather than an error condition. Accordingly, derived observables like support/opposition/conflict/indeterminacy are not merely diagnostic; they become quantitative signals that later participate in resonance, selection, and stabilization.

Compositionality as the non-negotiable design requirement. The repeated emphasis on relations, enriched categories, and quantale composition reflects a single requirement: evaluations must compose in a way that is associative and monotone (and hence usable for chaining approximations and propagating constraints). This is the formal analog of the informal claim that patterns and distinctions can be built out of simpler ones and transported through intermediate representational layers. In later sections, this compositionality is what permits multi-step mind–world correspondences (and inter-context translations) to be compared, aggregated, and iterated.

Interpreting the quantale V as a shared measurement substrate. The commutative quantale V is introduced to unify several graded notions that otherwise remain disconnected: strength of evidence, degree of weakness (as an effort-adjacent proxy), and pattern intensity or salience. The intent is that V provides both an ordering (to compare degrees) and a monoidal combination (to aggregate degrees across components, steps, or channels). Because the same algebraic structure supports both "evidence accumulation" and "cost/weakness aggregation," later definitions can express tradeoffs and co-variation (e.g. a correspondence that is strong but "expensive," or simple but weakly supported) within a single calculus.

V-valued relations as approximate correspondences. A V-valued relation may be read as assigning, to each candidate pair, a graded degree of correspondence rather than a Boolean yes/no match. Quantale-style composition then implements the idea that correspondences can be chained through intermediates, with the resulting degree determined by aggregating the degrees along the path and taking the appropriate supremum over alternatives. This recovers a standard intuition: an approximate match from A to C can be mediated by many different B-level representations, and the best available mediation determines the effective correspondence.

Enrichment and the "shape" of translation. The move from V-valued relations to V-enriched categories/quantaloids provides a language in which "processes" or "contexts" themselves can be organized compositionally. Enrichment is used here to avoid treating translation as a binary predicate; instead, morphisms carry degrees that can represent approximation quality, stability, or adequacy of representation. This becomes important when later sections compare alternative representational pipelines: two translations may exist between the same contexts but differ in graded adequacy, and those differences should themselves compose and propagate.

Resonance as an induced coupling signal. The resonance construction is included in the minimal core because it links logical evaluation to dynamical interaction: paraconsistent assessment is turned into a quantity that can drive propagation, reinforcement, or suppression. The mention of a logic-to-complex embedding indicates that resonance will be treated not only as magnitude but also as having a phase-like structure suitable for interference, alignment, and cancellation effects. Abstractly, resonance-driven propagation should be read as a schema: evaluations generate couplings; couplings modulate future evaluations; and the resulting feedback can stabilize patterns (resonance) or destabilize them (dissonance).

How the listed Hyperseed concepts attach to the formal layer. The items in the "Hyperseed concepts covered" list are meant to be grounded immediately by the above primitives: distinctions correspond to graded separations induced by valuations; equivalences are context-relative identifications supported by V-relations; weakness/simplicity are tracked by quantale aggregation over "failed distinctions"; patterns and emergence are prepared by having a compositional notion of correspondence; and resonance/dissonance are realized as operators acting on graded, possibly conflicting, evaluations. This anchoring is intentionally lightweight here, so that later sections can introduce richer versions (e.g. full effort theory and detailed pattern criteria) without changing the underlying compositional interface.

3.1 Design requirements and core data types

Hyperseed's informal presentation mixes phenomenology, ontology, and computation [1]. To make this mathematically tractable without destroying its intent, we treat the following features as *requirements* rather than optional add-ons:

1. **Observer-relativity.** Distinctions, simplicity, and uncertainty are indexed by a context/observer. Hyperseed-Concept 86.

2. **Paraconsistency.** Conflicting evidence ("both yes and no") must not entail triviality. Hyperseed-Concept 198.

3. **Gradation.** Most predicates (distinction, similarity, salience) are not crisp in realistic settings. Hyperseed-Concept 96; Hyperseed-Concept 195.

4. **Compositionality.** Combining processes or contexts should correspond to algebraic composition. Hyperseed-Concept 140; Hyperseed-Concept **??**.

5. **Approximation.** Many Hyperseed claims are about approximate morphisms (not exact isomorphisms). Hyperseed-Concept 112.

These requirements are intended to function together rather than independently. In particular, observer-relativity makes it unavoidable that distinct contexts can generate apparently incompatible verdicts about the same underlying situation, which motivates paraconsistency; gradation then refines this further by allowing "how incompatible" and "to what degree" to be part of the mathematical state rather than an informal afterthought. Compositionality ensures that once we admit multiple contexts and multiple processes, we can still build large models from smaller ones without losing interpretability, while approximation clarifies that the compositional laws we use should support robust *near-preservation* of structure (e.g., preservation up to tolerance, slack, or bounded distortion) rather than demanding brittle exactness.

Remark 45 (Why these requirements are not merely technical)**.** *Observer-relativity makes the mathematics mirror the epistemic situation: distinctions are not properties of the world in isolation, but of a world-as-grasped by some situated system. Paraconsistency acknowledges that real cognitive systems often carry simultaneous reasons for and against a claim; insisting on classical consistency would force premature collapse of lived ambiguity into brittle binaries. Gradation then supplies the missing continuum between "yes" and "no," allowing us to speak of weak, partial, and contested distinctions without erasing them.*

Compositionality and approximation are the two pillars that prevent this from being mere bookkeeping. Compositionality ensures that combining contexts or processes has a principled algebraic meaning, rather than being an ad hoc patchwork. Approximation recognizes that most correspondences in cognition and science are structural homologies, not perfect matches: a map can be useful precisely because it is good enough *while cheaper than exactness (a theme that will later connect to effort and simplicity; cf. Hyperseed-Concept 100 and Hyperseed-Concept 169).*

A further non-technical point is that these requirements jointly preserve a phenomenological reading: agents routinely operate with partially formed categories, revise them under pressure of evidence, and nevertheless act compositionally (they build larger plans from smaller habits) under approximate guidance (plans succeed when they are close enough). The formal core is therefore meant to support stable action under graded, context-indexed ambiguity*, rather than to eliminate ambiguity by stipulation.*

We therefore work with a minimal set of *core data types*:

Definition 2 (Universe of entities (local))**.** *A* local universe *is a set X whose elements are called* entities*. In applications, X may represent occasions of experience, perceived objects, internal representational states, or abstract pattern-tokens.*

Remark 46 (Intuition and examples for a local universe)**.** *This definition is deliberately modest: a "universe" here is not the cosmos but a chosen domain of discourse. One may think of it as the scope of what a particular model, observer, or subsystem is currently willing to count as distinct* things*. In Russell's sense, we begin with a type of objects, and only later attach predicates and relations; this postpones metaphysical commitments while keeping the mathematics clean.*

Examples: (1) In a toy perceptual model, X might be a finite set of perceived stimuli, say $\{a, b, c\}$. (2) In a cognitive architecture, X might be a set of internal memory items or attention tokens (Hyperseed-Concept **??***). The usefulness is that essentially all later structure (relations, valuations, composition) can be built on top of this minimal carrier set; we avoid prematurely committing to whether entities are "physical," "mental," or "abstract" (Hyperseed-Concept 136; Hyperseed-Concept 51; Hyperseed-Concept 83).*

It is also important that "local" allows X to be small, partial, or purpose-bound. For instance, an entity may be a coarse-grained token standing for a cluster of finer-grained sub-entities not presently tracked; conversely, what counts as a single entity in one modeling choice can be expanded into many entities in a refined choice. This flexibility is essential later when we talk about correspondence and approximation: a mapping between domains typically relates different *local universes (e.g., a perceptual token-space and a hypothesized external object-space), and the formalism should allow those universes to have different granularities while remaining comparable.*

Definition 3 (Contexts (observers/aspects))**.** *A* context *(or observer/aspect) C is a package of structure on a local universe X_C that includes:*

- *a* p-bit*-valued evaluation of propositions or relations (Section 3.2);*
- *a distinguished family of V-valued relations used to represent (in)distinction, similarity, and correspondence (Section 3.5);*
- *optional weights/saliences on entities (used by weakness and intensity functionals).*

Concretely, one may think of C as determining not only *which* entities are in play (via X_C), but also the admissible questions about them (via the chosen relations) and the permitted answer-values (via p-bit and V). Put differently: X_C supplies the "vocabulary" of entities, while the remaining components of the context supply a "semantics" that can tolerate conflict and degrees. This separation will later make it possible to speak precisely about translating information across contexts (e.g., carrying a graded similarity relation from one observer to another) and about composing contexts (e.g., forming joint standpoints that aggregate evidence with controlled loss).

Remark 47 (Contexts as perspectival packages)**.** *A context is not merely a "parameter"; it is an organized standpoint: it determines what counts as evidence, how evidence aggregates, and what sorts of comparisons can be composed. In Peircean terms [14], one may regard the context as providing the mediating habits (Thirdness) by which raw encounters (Secondness) are stabilized into usable generalities (Firstness). In Hyperseed's vocabulary, this is the formal shadow of* Contexts and Aspects *(Hyperseed-Concept 86).*

Example: Two contexts might share the same underlying set X of stimuli but disagree on which pairs are "the same for purpose P," because one context emphasizes color while another emphasizes shape. The point of packaging is practical: observer-relativity is not sprinkled across the theory as a warning label; it becomes explicit structure that can be translated, compared, and composed (later connecting to mind–world correspondence; Hyperseed-Concept 112, and to engineering concerns in cognitive systems [19]).

This packaging also clarifies what it means for two observers to "disagree." They need not disagree by assigning opposite classical truth-values; they may instead (i) assign incompatible paraconsistent values to the same proposition, (ii) agree on a proposition's evaluation but disagree on the induced similarities and distinctions, or (iii) differ in salience so that the same informational state leads to different downstream choices. Stated formally, contexts let us locate disagreement in a specific component (evaluation, relation-family, weighting) rather than treating it as an unanalyzable global mismatch.

The later ontology will add time, dynamics, effort, habit reinforcement, and self-reference on top of this; here we only build the minimal algebraic scaffolding.

3.2 Paraconsistent p-bit values

Definition 4 (p-bit domain)**.** *A* p-bit *truth value is a pair* $v = (v^+, v^-)$ *with* $v^+, v^- \in [0,1]$*, interpreted as degrees of* positive *and* negative *evidence. We write*

$$\mathbb{V} := [0,1]^2.$$

Remark 48 (Notation unpacking and philosophical motivation)**.** *The notation* $v = (v^+, v^-)$ *should be read literally as an* evidence record*: the superscripts* $+$ *and* $-$ *denote support-for and support-against, not "probability" and "one minus probability." The square* $[0,1]^2$ *means we allow any pair of real numbers between* 0 *and* 1*; thus* $\mathbb{V}$ *is the unit square, interpreted epistemically rather than geometrically. This choice parallels evidence-pair traditions in paraconsistent and bilattice-inspired semantics [23].*

Intuitively, paraconsistency enters because cognitive and scientific life routinely supplies mixed reasons. A system may have strong evidence for a claim in one subsystem and strong evidence against it in another, without immediately dissolving into incoherence. The p-bit *domain is a minimal way to encode this while remaining computationally cheap and algebraically tractable (and it aligns with later resonance constructions based on interference [24]).*

Remark 49 (Simple examples and why the domain is useful)**.** *Example 1:* $v = (1,0)$ *can be read as "maximally supported and not opposed," a classical truth-like corner. Example 2:* $v = (1,1)$ *is "both strongly supported and strongly opposed," representing genuine conflict rather than confusion. Example 3:* $v = (0,0)$ *represents lack of evidence either way (ignorance). Between these corners lie graded states such as* $(0.7, 0.2)$ *(mostly supported, weakly opposed).*

The utility is that we can carry conflicting or incomplete information forward through compositions and aggregations without forcing premature resolution. Later, when values and goals conflict (Hyperseed-Concept 198) or when boundaries between self and other are ambiguous (Hyperseed-Concept 165), this capacity becomes structurally necessary rather than merely convenient.

Remark 50 (Belnap corners as extremes)**.** *The four extreme points*

$$(1,0),\ (0,1),\ (1,1),\ (0,0)$$

correspond to the classical four-valued paraconsistent states: true, false, both, neither. The interior points represent graded mixtures.

Derived observables. From $v = (v^+, v^-)$ we will frequently use the following derived quantities:

$$\mathrm{supp}(v) := v^+, \qquad \mathrm{opp}(v) := v^-,$$

$$\mathrm{bias}(v) := v^+ - v^- \in [-1,1], \qquad \det(v) := \max(v^+, v^-) \in [0,1], \qquad \mathrm{conf}(v) := \min(v^+, v^-) \in [0,1].$$

Here $\mathrm{conf}(v)$ is a simple "contradiction/co-presence" score (both-sides present), while $1 - \det(v)$ is a simple "indeterminacy" score (neither-side strong).

Remark 51 (Why these observables are the right minimal statistics)**.** *The pair* (v^+, v^-) *contains two degrees of freedom; the derived quantities extract different "faces" of the same datum.* bias *tells us which way the evidence leans;* conf *tells us how much the evidence is* internally at war. det *measures whether there is any strong commitment (either pro or con), and* $1 - \det$ *measures a kind of uncertainty. These are not the only possible summaries, but they are monotone and inexpensive, and they harmonize with later constructions where we must compare many judgments quickly.*

A useful mental picture is to imagine a compass: bias *corresponds to east–west pull, while* conf *corresponds to a northward pull toward "both." This geometric metaphor will become literal once we embed* $\mathbb{V}$ *into* $\mathbb{C}$ *in Section 3.9.*

Definition 5 (Negation)**.** *Define the involution*

$$\neg(v^+, v^-) := (v^-, v^+).$$

Remark 52 (Intuition, example, and why involution matters)**.** *Negation swaps positive and negative evidence: what was counted as support is now counted as opposition and vice versa. For example, $\neg(1,0) = (0,1)$ and $\neg(0.8, 0.3) = (0.3, 0.8)$. This is an involution: applying it twice returns the original value, $\neg(\neg v) = v$.*

This operation is useful because it cleanly separates the representation of evidence from any particular choice of connectives. It also makes anti-resonance propagation (Definition 22) formally simple: one can propagate "the contrary" without inventing a new evidence calculus each time.

Definition 6 (Evidence-style conjunction and disjunction)**.** *Define*

$$v \wedge w := \big(\min(v^+, w^+),\ \max(v^-, w^-)\big), \qquad v \vee w := \big(\max(v^+, w^+),\ \min(v^-, w^-)\big).$$

Remark 53 (Intuition via constraints and two quick examples)**.** *The connective $v \wedge w$ behaves like an "and" for positive evidence and like an "or" for negative evidence: to support $A \wedge B$ you must support both (hence* min*), but to oppose $A \wedge B$ it suffices to oppose either (hence* max*). Dually, $v \vee w$ takes the easiest support (*max*) but the hardest opposition (*min*).*

Example: if $v = (0.9, 0.1)$ and $w = (0.4, 0.8)$ then

$$v \wedge w = (0.4, 0.8), \qquad v \vee w = (0.9, 0.1).$$

So the conjunction inherits the weaker support and the stronger opposition; the disjunction inherits the stronger support and the weaker opposition. This is useful because it respects the intuition that a single strong refutation can undermine a conjunction, while a single strong support can sustain a disjunction.

Remark 54 (Interpretation)**.** *Under these connectives, supporting $A \wedge B$ requires supporting both (hence* min *on positive evidence), while refuting $A \wedge B$ requires refuting either (hence* max *on negative evidence). Dually, supporting $A \vee B$ requires supporting either (*max*), while refuting $A \vee B$ requires refuting both (*min*). This is one standard "evidence pair" paraconsistent semantics; other monotone choices are possible.*

It may help to keep the intended reading explicit: a value $v = (v^+, v^-)$ records independent *degrees of positive and negative evidence, so v^+ and v^- need not sum to 1 and may simultaneously be large (a conflicted state), or simultaneously be small (an uninformed state). With that in mind, the $\wedge/\vee$ clauses above are just the order-theoretically simplest way to implement "need both" versus "need either" for support/refutation, using the lattice operations* min *and* max *on each evidence coordinate.*

In particular, if A is strongly supported but also strongly opposed, and similarly for B, then $A \wedge B$ remains strongly supported only insofar as both *supports remain high (hence the* min*), while it becomes strongly opposed as soon as* either *conjunct attracts strong opposition (hence the* max*). The dual intuition for $\vee$ is analogous: disjunction is easy to support (any disjunct suffices) but hard to refute (both must be refuted). These asymmetries are exactly what lets the semantics tolerate inconsistency without collapsing into triviality.*

Definition 7 (Implication (one convenient choice))**.** *Define implication by*

$$v \to w := (\neg v) \vee w.$$

Remark 55 (What this implication expresses and why we do not overcommit to it)**.** *The definition $v \to w := (\neg v) \vee w$ is the familiar material-implication pattern translated into the evidence-pair setting: "if v then w" is treated as "either not-v or w." For example, if $v = (1, 0)$ (strongly supported) and $w = (0, 1)$ (strongly opposed), then $v \to w$ becomes maximally conflicted, reflecting the idea that a supported antecedent with an opposed consequent is problematic.*

One should also read this connective operationally: increasing evidence against *v (i.e. increasing the negative coordinate of v, which becomes positive evidence for $\neg v$) makes $v \to w$ easier to support, while increasing evidence* for *w also makes $v \to w$ easier to support, since it is a disjunction. Conversely, to refute $v \to w$ one must refute* both *$\neg v$ and w, meaning (intuitively) one must have strong support for v together with strong opposition to w; this matches the classical idea that an implication fails precisely when the antecedent holds but the consequent does not, while still allowing the paraconsistent possibility that additional counterevidence coexists.*

We include implication mainly as a bridge to conventional logical reading. Much of the later theory leans more on monotone aggregation and composition (quantale joins/products and enriched-category inequalities) than on any particular internal logic. This restraint is intentional: Hyperseed aims to model cognition where inference is often heuristic, resource-bounded, and context-sensitive, so algebraic monotonicity often does more work than a single, universal implication connective [19, 20].

Accordingly, nothing in the sequel depends on material implication validating all familiar proof principles (e.g. unrestricted modus ponens under inconsistency), and alternative paraconsistent implications could be substituted if one wishes to enforce different inferential behavior. The point of Definition 7 is thus representational convenience: it supplies a recognizable "if–then" connective that interacts predictably with $\neg$ and $\vee$ in the evidence-pair algebra, without claiming to be the unique or cognitively privileged notion of implication.

For the toy model and much of the later ontology, we will rely more heavily on *monotone aggregation* (quantale joins and monoidal products) than on any particular choice of implication; the above is included mainly to provide a familiar logical vocabulary when needed.

3.3 Quantales as aggregation and composition domains

Definition 8 (Commutative quantale)**.** *A* commutative quantale *is a tuple $(V, \leq, \bigvee, \otimes, e)$ such that:*

1. *$(V, \leq)$ is a complete lattice with joins $\bigvee S$ for all $S \subseteq V$;*
2. *$(V, \otimes, e)$ is a commutative monoid;*
3. *$\otimes$ distributes over arbitrary joins:*

$$a \otimes \Big(\bigvee_i b_i\Big) = \bigvee_i (a \otimes b_i), \qquad \Big(\bigvee_i b_i\Big) \otimes a = \bigvee_i (b_i \otimes a).$$

Remark 56 (Basic consequences worth keeping in mind)**.** *Because $(V, \leq)$ is a complete lattice, it in particular has a bottom element $\bot := \bigvee \emptyset$ and a top element $\top := \bigvee V$, and it also has all meets $\bigwedge S$ (as joins in the opposite order). The distributivity axiom implies that, for each fixed $a \in V$, the map $(-) \mapsto a \otimes (-)$ preserves arbitrary joins; similarly $(-) \mapsto (-) \otimes a$ preserves arbitrary joins. In particular, $\otimes$ is monotone in each argument:*

$$b \leq c \implies a \otimes b \leq a \otimes c, \qquad b \leq c \implies b \otimes a \leq c \otimes a.$$

This monotonicity is the order-theoretic statement that "adding evidence cannot reduce a composed influence." It is the minimal property needed to make repeated update steps behave predictably when we later iterate propagation operators.

A useful special case of join-preservation is the interaction with $\perp$*:*

$$a \otimes \perp = a \otimes \left(\bigvee \emptyset\right) = \bigvee \emptyset = \perp, \qquad \perp \otimes a = \perp,$$

so $\perp$ *acts as an absorbing element for composition. Intuitively, "impossible/absent influence" remains impossible/absent under chaining.*

Remark 57 (Notation and intuition for quantales)**.** *The symbols are meant to be read in an order-theoretic way. The relation* $\leq$ *is an information/strength ordering;* $\bigvee$ *is the join (least upper bound), i.e. the operation of aggregating many pieces of evidence into the weakest upper bound that dominates them. The operation* $\otimes$ *is the monoidal product, interpreted as a form of sequential or conjunctive composition. The element* e *is the unit for* $\otimes$*.*

A quantale can be thought of as "logic with arithmetic": it allows you to accumulate *many influences (*$\bigvee$*) and also to* chain *influences (*$\otimes$*) in a way that interacts well with accumulation (distributivity). This is precisely what we need for Hyperseed's graded and compositional notions of distinction, weakness, and pattern propagation [3].*

The commutativity assumption is not strictly necessary in all quantale applications, but it matches the intended reading here where the combination of two influences is not sensitive to their left/right ordering at the level of scalar weights. Later, directionality will be represented in the relational structure (source/target of a link) rather than in the scalar itself, so a commutative $\otimes$ *keeps the scalar calculus simple while still supporting non-symmetric networks.*

Remark 58 (Simple examples and why quantales are useful here)**.** *Example 1: The Boolean quantale* $(\{0,1\}, \leq, \vee, \wedge, 1)$ *yields ordinary relational composition and crisp distinctions. Example 2: The unit interval with* $\bigvee = \max$ *and* $\otimes = \cdot$ *yields fuzzy relational composition. Our later choice* $V = [0,1]^2$ *is a paraconsistent generalization of this.*

To connect these examples to the way Hyperseed will use them: if one represents a pattern-propagation relation as a V*-valued adjacency matrix, then the "two-step influence" from* x *to* z *through intermediate nodes* y *is computed by composing along* y *using* $\otimes$ *and then aggregating over all* y *using* $\bigvee$*. In the Boolean case this recovers existential path composition; in the fuzzy case it recovers the standard max-product or sup-product style composition. The quantale axioms are exactly what ensures that this matrix-style multiplication is associative, so that multi-step propagation does not depend on arbitrary parenthesization.*

Quantales serve Hyperseed as a disciplined replacement for informal phrases like "combine these influences" or "propagate this pattern." They guarantee that iterative updates behave monotonically and that multi-step compositions can be reassociated without ambiguity (associativity of relational composition ultimately rests on the distributivity axiom). This becomes essential once we discuss resonance-driven propagation and habit reinforcement, where repeated composition is the rule rather than the exception (Hyperseed-Concept 189; Hyperseed-Concept 188).

The paraconsistent choice $V = [0,1]^2$ *can be read as tracking two graded channels (e.g. support and counter-support) simultaneously. In that setting,* $\leq$ *will typically be chosen so that moving "up" corresponds to having at least as much support and at least as much counter-support, and the join* $\bigvee$ *becomes a pointwise "take the strongest available components." What matters at this stage is that quantales give a single abstract interface: regardless of whether we are in a crisp, fuzzy, or paraconsistent regime, the same definitions of aggregation and chaining apply.*

Quantales are a natural "glue" structure for Hyperseed because they simultaneously support: (i) graded aggregation (joins); (ii) sequential or conjunctive composition (monoidal product); and (iii) order-theoretic monotonicity (needed for observer-relativity and refinement arguments).

Operationally, items (i)–(iii) are the minimum needed to speak about networks where many partial influences converge on a node (hence $\bigvee$), where influences can traverse multiple links (hence $\otimes$), and where strengthening local information cannot create a global weakening (hence monotonicity). This is why the quantale layer sits in the "minimal formal core": it pins down the algebraic laws that make later propagation operators well-defined, compositional, and stable under iteration.

3.4 A canonical p-bit quantale

Definition 9 (Canonical commutative p-bit quantale)**.** *Let $V := ([0,1]^2, \leq, \oplus, \otimes, e)$ where:*

- $(a^+, a^-) \leq (b^+, b^-)$ *iff* $a^+ \leq b^+$ *and* $a^- \leq b^-$ *(componentwise order),*
- $(a^+, a^-) \oplus (b^+, b^-) := (\max(a^+, b^+), \max(a^-, b^-))$ *(join),*
- $(a^+, a^-) \otimes (b^+, b^-) := (a^+b^+, a^-b^-)$ *(monoidal product),*
- $e := (1, 1)$ *(unit).*

Then V is a commutative quantale.

Remark 59 (Why this is a quantale (explicit properties))**.** *The statement that V is a commutative quantale unfolds into three routine facts: (i) $([0,1]^2, \leq)$ is a complete lattice; (ii) $(V, \otimes, e)$ is a commutative monoid; and (iii) $\otimes$ distributes over arbitrary joins in each argument.*

For (i), completeness is componentwise: given any family $\{(a_i^+, a_i^-)\}_{i \in I} \subseteq [0,1]^2$, its join is

$$\bigvee_{i \in I} (a_i^+, a_i^-) = \Big(\sup_{i \in I} a_i^+,\ \sup_{i \in I} a_i^-\Big),$$

and its meet is defined analogously with infima; these exist because $[0,1]$ is complete. For finite joins, this recovers $\oplus$ as the componentwise maximum.

For (ii), associativity and commutativity of $\otimes$ follow from associativity and commutativity of multiplication in $[0,1]$ applied componentwise, and $e = (1,1)$ is a two-sided unit since $a^\pm \cdot 1 = a^\pm$.

For (iii), fix (b^+, b^-) and a family $\{(a_i^+, a_i^-)\}_{i \in I}$. Then

$$\Big(\bigvee_{i \in I} (a_i^+, a_i^-)\Big) \otimes (b^+, b^-) = \Big((\sup_i a_i^+) b^+,\ (\sup_i a_i^-) b^-\Big) = \Big(\sup_i (a_i^+ b^+),\ \sup_i (a_i^- b^-)\Big) = \bigvee_{i \in I} \big((a_i^+, a_i^-) \otimes (b^+, b^-)\big),$$

using that multiplication by a fixed $b^\pm \in [0,1]$ preserves suprema in $[0,1]$ (it is monotone and Scott-continuous on the complete lattice $[0,1]$). The same argument applies in the other argument, hence $\otimes$ preserves arbitrary joins separately, as required for a quantale.

Remark 60 (How to read $\oplus$ and $\otimes$ in this specific quantale)**.** *The operation $\oplus$ takes componentwise maxima: it is an "evidence accumulation" operator. If one source gives evidence $(0.7, 0.2)$ and another gives $(0.5, 0.9)$, then the aggregate is $(0.7, 0.9)$: we keep the strongest support and the strongest opposition, because both may matter in a paraconsistent setting.*

This reading also highlights that $\oplus$ is idempotent and commutative: aggregating the same piece of evidence twice does not inflate it, and order of aggregation does not matter. Moreover, since $\oplus$ is the join induced by $\leq$, the inequality $(a^+, a^-) \leq (a^+, a^-) \oplus (b^+, b^-)$ expresses that aggregation is information-increasing in each coordinate.

The operation $\otimes$ multiplies components: it is a "co-occurrence" or "chaining" operator. If two independent stages each preserve positive evidence at rates a^+ and b^+, then the two-stage preservation is a^+b^+ (and similarly for negative evidence). This multiplicativity is computationally convenient and aligns with common probabilistic intuitions without reducing evidence-pairs to ordinary probabilities.

Algebraically, $\otimes$ is monotone in each argument with respect to $\leq$: if $(a^+, a^-) \leq (a'^+, a'^-)$ then $(a^+, a^-) \otimes (b^+, b^-) \leq (a'^+, a'^-) \otimes (b^+, b^-)$, because multiplication by $b^{\pm} \in [0,1]$ preserves order. Intuitively, strengthening either the positive or negative component cannot weaken the chained outcome.

Remark 61 (Bottom and top)**.** *The least element is $\bot = (0,0)$ and the greatest element is $\top = (1,1) = e$. Intuitively, $\bot$ represents "no evidence either way", while $\top$ represents "maximal evidence on both sides". This is acceptable (and sometimes desirable) in paraconsistent settings; if one wants $\top$ to mean "maximally true" instead, one may instead work with a different ordering (e.g. a truth order vs. a knowledge order). For the present paper, the componentwise order is treated as an* information aggregation order*.*

Concretely, with the information aggregation (knowledge) order, $(a^+, a^-) \leq (b^+, b^-)$ means that b contains at least as much supporting evidence and at least as much opposing evidence as a. Under this interpretation, increasing either coordinate represents adding information rather than resolving a truth-value; hence it is consistent for $\top$ to carry maximal values in both coordinates.

Remark 62 (A concrete example and why $e = (1,1)$ is not a paradox)**.** *It may look strange that the monoidal unit e carries* maximal *evidence in both components. But e is not intended to represent "the proposition that is perfectly true." It represents "no loss under composition": composing with e should not attenuate either positive or negative evidence. In that sense, e plays the role of a neutral element for information-flow, not a metaphysical truth.*

This choice is particularly appropriate when the lattice order is read as "having at least as much information" (knowledge order). Under that reading, $(1,1)$ is indeed the most informative point: it contains maximal positive and maximal negative evidence simultaneously. This echoes the bilattice tradition that separates truth-order from knowledge-order [23].

A small numerical check may help: for any (a^+, a^-) we have $(a^+, a^-) \otimes (1,1) = (a^+, a^-)$, whereas $(a^+, a^-) \oplus (1,1) = (1,1)$. Thus e is neutral for chaining ($\otimes$) but absorbing for aggregation ($\oplus$), exactly as one expects when $\oplus$ is a join operation and e happens to coincide with the top element in the chosen order.

Remark 63 (Residuals (optional but often useful))**.** *Many quantale-based constructions use the (right) residual $x \Rightarrow y := \bigvee\{z \mid x \otimes z \leq y\}$ when it exists; in a quantale it always exists by completeness. In the present $[0,1]^2$ instance, the residual is componentwise and can be written explicitly using the usual residuum of multiplication on $[0,1]$:*

$$(a^+, a^-) \Rightarrow (b^+, b^-) = \big(a^+ \Rightarrow_\times b^+,\ a^- \Rightarrow_\times b^-\big), \qquad a \Rightarrow_\times b := \begin{cases} 1, & a = 0, \\ \min\{1,\ b/a\}, & a > 0. \end{cases}$$

This observation is not required for the basic development, but it clarifies that the canonical choice supports implication-like operations (as adjoints to chaining) without leaving the p-bit *domain.*

Remark 64 (Other choices)**.** *The specific operations above are not mandatory. Any commutative quantale compatible with the intended monotonicities can be substituted. The canonical choice is used because it is simple, computable, and supports the toy model in Section 5.*

For example, one could replace componentwise multiplication by a different t-norm (and componentwise maximum by the corresponding join if one changes the underlying order), or keep the componentwise order but use an alternative monoidal product reflecting a different notion of sequential combination. The key constraints are that $\otimes$ be associative, commutative, monotone, and distribute over joins, so that the general quantale calculus remains available.

3.5 V-valued relations and quantale composition

Definition 10 (V-valued relations)**.** *Given a set X, a V-*valued relation *on X is a map*

$$R : X \times X \to V.$$

More generally, a V-valued relation from X to Y is a map $R : X \times Y \to V$.

Remark 65 (Basic special cases and why "relations" is the right word)**.** *When $V = \mathbf{2} = \{\bot, \top\}$ with $\oplus = \vee$ and $\otimes = \wedge$, a V-valued relation $R : X \times Y \to V$ is exactly an ordinary (crisp) relation: $R(x, y) = \top$ means "related" and $R(x, y) = \bot$ means "not related." When $V = [0, 1]$ with $\oplus = \sup$ and $\otimes$ a t-norm, one recovers standard fuzzy relations. The present setting keeps the same relational syntax $R(x, y)$ but allows richer evidence types, so that the same formalism can express classical, fuzzy, and paraconsistent/bi-graded correspondences as instances.*

Remark 66 (Intuition: relations as graded, possibly conflicting correspondences)**.** *A classical relation says only whether (x, y) is related (yes/no). A V-valued relation assigns to each pair (x, y) a value in V encoding graded evidence. When $V = [0, 1]^2$, this means we can simultaneously record degrees of "x matches y" and "x does not match y," without forcing a collapse. This directly supports Hyperseed's graded distinction/indistinction (Hyperseed-Concept 98).*

Example: If X is a set of perceived objects and Y is a set of internal symbols, then $R(x, y) = (0.8, 0.1)$ might mean the system mostly endorses interpreting object x as symbol y, with mild counterevidence. In a conflicting case, $R(x, y) = (0.8, 0.8)$ records a genuine ambiguity: strong reasons to map and strong reasons not to map. Such relations are a natural substrate for representation and correspondence (Hyperseed-Concept 157; Hyperseed-Concept 112) as discussed in cognitive-systems settings [19].

Remark 67 (Terminology: evidence values live in a poset, not necessarily numbers)**.** *It is important that V is used only through its order and its two operations. Even when one writes pairs such as $(0.8, 0.1)$, the formal reading is: $R(x, y)$ is an element of an ordered structure in which one can (i) combine pieces of evidence along a path using $\otimes$, and (ii) aggregate alternative paths using $\oplus$ (a join). Thus the same definitions apply when V is non-numeric (e.g. logical truth values, cost semirings, or other complete lattices equipped with a compatible "multiplication").*

Definition 11 (Quantale "matrix" composition)**.** *Given $R : X \times Y \to V$ and $S : Y \times Z \to V$, define their composite*

$$(S \circ R)(x, z) := \bigoplus_{y \in Y} R(x, y) \otimes S(y, z).$$

When V is complete and $\otimes$ distributes over joins, this composition is associative.

Remark 68 (Well-definedness for infinite Y and why completeness matters)**.** *For finite Y, the expression $\bigoplus_{y \in Y}$ is just a finite join. For general (possibly infinite) Y, the definition requires that arbitrary joins exist in V so that $\bigoplus_{y \in Y}(\cdots)$ is meaningful. This is exactly the "completeness" condition in the quantale assumption, and it is the reason quantales (rather than merely monoids or semirings) are the natural ambient algebra for relation-like composition over arbitrary sets.*

Remark 69 (Notation unpacking and a small example)**.** *The symbol $\bigoplus_{y\in Y}$ denotes the join (supremum) over all intermediates y. It plays the role of "there exists a good intermediary" but in a graded way: we aggregate the evidence contributed by each path $x \to y \to z$ using $\otimes$, and then take the best/most informative aggregate using $\oplus$.*

Example (finite Y): if $Y = \{y_1, y_2\}$ then

$$(S \circ R)(x,z) = \big(R(x,y_1) \otimes S(y_1,z)\big) \ \oplus \ \big(R(x,y_2) \otimes S(y_2,z)\big).$$

This is literally matrix multiplication with $(+,\cdot)$ replaced by $(\oplus,\otimes)$. The usefulness is that it gives a principled notion of chaining approximate correspondences, essential for modeling multi-step representation pipelines (e.g. perception $\to$ internal model $\to$ action), and it will later support pattern-web propagation rules (Hyperseed-Concept 132).

Remark 70 (Associativity: what "$\otimes$ distributes over joins" buys you)**.** *The associativity claim can be seen by expanding both sides pointwise. Given $R : X \times Y \to V$, $S : Y \times Z \to V$, and $T : Z \times W \to V$, one has*

$$\big(T \circ (S \circ R)\big)(x,w) = \bigoplus_{z\in Z}\Big(\bigoplus_{y\in Y} R(x,y) \otimes S(y,z)\Big) \otimes T(z,w),$$

and distributivity of $\otimes$ over joins allows one to rewrite this as

$$\bigoplus_{z\in Z}\bigoplus_{y\in Y} R(x,y) \otimes S(y,z) \otimes T(z,w).$$

Similarly,

$$\big((T \circ S) \circ R\big)(x,w) = \bigoplus_{y\in Y} R(x,y) \otimes \Big(\bigoplus_{z\in Z} S(y,z) \otimes T(z,w)\Big),$$

which also becomes

$$\bigoplus_{y\in Y}\bigoplus_{z\in Z} R(x,y) \otimes S(y,z) \otimes T(z,w).$$

Since joins are associative/commutative up to the ambient order (and $\otimes$ is associative), these two expressions coincide. Thus the "matrix" composition is not merely suggestive: it is genuinely functorial under the standard quantale axioms.

Remark 71 (Interpretation as approximate correspondence)**.** *A V-relation $R : X \times Y \to V$ can be read as a graded, possibly inconsistent correspondence: $R(x,y)$ is the amount of evidence that x in X "matches" y in Y (or is "the same" for some purpose), with both positive and negative components allowed. Composition then corresponds to chaining correspondences through an intermediate representation.*

Remark 72 (Reading $\oplus$ as aggregation of alternatives rather than averaging)**.** *The join $\oplus$ should be read as an order-theoretic "best supported" (or "most informative") combination of alternatives, not as an arithmetic average. In particular, if one alternative path yields strictly stronger evidence than another, the join simply retains the stronger one. This is one reason quantale composition is well-suited to propagation of constraints or supports in a pattern web: multiple routes can contribute, but the algebra specifies precisely how they compete or reinforce.*

Definition 12 (Identity V-relation)**.** *For a set X, define $\mathrm{id}_X : X \times X \to V$ by*

$$\mathrm{id}_X(x,y) := \begin{cases} e & \text{if } x = y, \\ \bot & \text{if } x \neq y. \end{cases}$$

Remark 73 (Why this is the correct identity and what $\perp$ does)**.** *The identity relation is meant to say: an element matches itself with full compositional strength, and matches distinct elements with none. Here $\perp$ is the least element of V (in the* p-bit *quantale, $\perp = (0,0)$), so it contributes no evidence under joins and attenuates completely under products.*

A quick check in the finite case confirms the intuition: composing any $R : X \times Y \to V$ with id_X *on the left leaves R unchanged because only the $x = y$ term contributes nontrivially. This identity is essential because it lets us treat V-relations as morphisms in a bona fide category (so that associativity and identity laws hold), which in turn is what makes later compositional arguments about contexts and translations well-typed rather than metaphorical.*

Remark 74 (Left and right unit laws pointwise)**.** *It is useful to keep in mind the explicit pointwise verification. For $R : X \times Y \to V$ and $x \in X$, $y \in Y$,*

$$(R \circ \mathrm{id}_X)(x,y) = \bigoplus_{x' \in X} \mathrm{id}_X(x,x') \otimes R(x',y) = e \otimes R(x,y) \;\oplus\; \bigoplus_{x' \neq x} \perp \otimes R(x',y) = R(x,y),$$

using $\perp \otimes v = \perp$ and $\perp \oplus v = v$. Similarly $\mathrm{id}_Y \circ R = R$. *These computations are routine but conceptually important: they show that* id_X *acts exactly like the diagonal matrix with e on the diagonal and $\perp$ elsewhere, matching the matrix analogy from Definition 11.*

With these definitions, sets and V-relations form a category (indeed, a standard quantaloid construction when one varies X).

3.6 V-enriched categories and approximate morphisms

Hyperseed frequently talks as if there are "morphisms" between internal and external pattern structures, but these morphisms are rarely exact. Quantale enrichment provides a uniform language for this [7].

A useful way to think about the role of the quantale $(V, \leq, \otimes, e)$ is that it supplies a *scale of comparison* (the order $\leq$) together with a *compositional aggregator* (the monoidal product $\otimes$) and a *baseline unit* (the element e). In many applications V is not merely a set of "weights": it is chosen so that arbitrary joins $\bigvee$ exist and $\otimes$ distributes over them, ensuring that "taking the best available evidence" and "composing evidence" interact coherently. Although we assume commutativity here for simplicity, many of the intended readings (e.g. asymmetric translation effort) can be modeled in noncommutative quantales as well; the thin axioms below are written so that the directional nature of $\mathcal{C}(A,B)$ is retained regardless.

Definition 13 (V-enriched category (thin form))**.** *Let $(V, \leq, \otimes, e)$ be a commutative quantale. A* V-enriched category $\mathcal{C}$ *consists of:*

- *a class of objects* $\mathrm{Ob}(\mathcal{C})$*;*
- *for each $A, B \in \mathrm{Ob}(\mathcal{C})$, a hom-value $\mathcal{C}(A,B) \in V$;*

such that for all A, B, C:

$$e \leq \mathcal{C}(A,A), \qquad \mathcal{C}(A,B) \otimes \mathcal{C}(B,C) \leq \mathcal{C}(A,C).$$

Remark 75 (Canonical choices of V)**.** *Different choices of quantale recover familiar "approximate" semantics. If $V = \mathbf{2} = \{\perp \leq \top\}$ with $\otimes = \wedge$ and $e = \top$, then a thin V-category is exactly a preorder: $\mathcal{C}(A,B) = \top$ means "A entails/reaches B" and the composition axiom is transitivity. If*

$V = [0,1]$ with $\leq$ the usual order, $e = 1$, and $\otimes$ a t-norm (often multiplication), then $\mathcal{C}(A, B)$ can be read as a graded confidence/support. If one instead wishes to model cost *or* distance*, a standard choice is the Lawvere quantale $V = ([0,\infty], \geq, +, 0)$ (note the reversed order), in which case the same inequality becomes the triangle inequality for extended pseudo-metrics: going via B cannot be cheaper than going directly. These examples underline that the "same" formalism can encode either support-like or cost-like readings by a systematic choice of order and tensor.*

Remark 76 (What "thin form" means and how to picture it)**.** *In an ordinary category, $\mathcal{C}(A, B)$ is a* set *of morphisms. In the thin enriched form used here, we collapse that set to a single graded value: $\mathcal{C}(A, B) \in V$ measures "how well A leads to B" (as support, feasibility, similarity, etc.). This is closer to a preorder than to a category with many arrows, but it is often exactly what is needed for resource-sensitive modeling: we care about* degree of accessibility *more than about enumerating all witnesses.*

Example: objects could be contexts, models, or descriptions; then $\mathcal{C}(A, B)$ might encode the degree to which B can be obtained from A by a translation or refinement step. The axioms then say: each object can reach itself at least as well as the unit e, and going from A to B and then B to C cannot yield better support than a direct A to C value (a quantitative triangle-like law). This is a convenient formal home for Hyperseed's many "approximate correspondence" claims (Hyperseed-Concept 112).

Equivalently, one can read a thin V-category as a V-valued preorder*: it records, for every ordered pair (A, B), a single "strength" of the statement "A is at most as informative as B" (or "A can be transformed into B") in a way that is reflexive and transitive in the enriched sense. This perspective is often helpful when translating informal monotonicity claims into lemmas, because one can appeal directly to enriched reflexivity/transitivity rather than to the existence of explicit witnessing arrows.*

Remark 77 (How to read the axioms)**.** *These axioms say: identity evidence is maximal (or at least not worse than e), and chaining two correspondences yields a correspondence that is (weakly) no stronger than the direct one. The direction of the inequalities depends on whether one models "cost" (where smaller is better) or "support" (where larger is better). Here we model graded support, so composition multiplies support and is bounded above by direct support.*

It is often useful to keep two derived intuitions in mind. First, the enriched composition law is a submultiplicativity *constraint: whatever proxy for "evidence" $\mathcal{C}(A, B)$ represents, indirect evidence obtained by composing steps cannot systematically dominate direct evidence. Second, the axiom is robust under coarse-graining: if one later refines the modeling to track multiple pathways explicitly, taking a join/supremum over pathways typically yields a thin hom-value that still satisfies the same inequality because $\otimes$ distributes over joins in a quantale.*

Definition 14 (V-functor (thin form))**.** *A* V-functor *$F : \mathcal{C} \to \mathcal{D}$ between V-enriched categories is a function on objects such that*

$$\mathcal{C}(A, B) \leq \mathcal{D}(F(A), F(B))$$

for all objects A, B.

Remark 78 (Intuition: structure preservation up to monotone improvement)**.** *A V-functor sends each object in $\mathcal{C}$ to an object in $\mathcal{D}$ and guarantees that any accessibility/support present in $\mathcal{C}$ is* not *lost under translation. In other words, if A supports reaching B to degree $\mathcal{C}(A, B)$, then $F(A)$ supports reaching $F(B)$ at least as strongly in $\mathcal{D}$.*

Example: $\mathcal{C}$ might be a category of world-states and $\mathcal{D}$ a category of internal representations. Then F models a representational mapping that does not reduce the system's ability to track certain *transitions. This is one rigorous way to capture "approximate morphism" talk in cognitive modeling [19], while retaining the flexibility that F need not be invertible or information-preserving in the classical sense.*

In practice, many mappings of interest are not strictly structure-preserving but are lax *or* approximate *in the sense that they preserve lower bounds on support rather than exact values. The thin V-functor inequality is exactly such a laxness condition: it asks only that the image correspondence in $\mathcal{D}$ be* at least as permissive *(in the $\leq$-order sense) as the original correspondence. This matches the modeling stance in which internal representations may "fill in" or "smooth over" distinctions present in the world while still enabling reliable downstream composition of inferences.*

Why enrichment is enough for "approximate morphisms." In later sections we will represent many Hyperseed correspondences as either: (i) V-functors (structure-preserving maps), or (ii) V-relations/modules (structure-preserving correspondences that need not be functional). Both are stable under composition and admit quantitative comparison.

Concretely, "admit quantitative comparison" can be read pointwise: given two candidate correspondences of the same type, one can often define an ordering by saying that one is no worse than the other exactly when it assigns everywhere-greater (support-like) hom-values (or everywhere-lower costs, in the dual convention). This makes it possible to phrase optimization or "best available translation" problems entirely inside the enrichment order, without prematurely committing to a particular algorithmic realization.

Remark 79 (Why this suffices for a large portion of Hyperseed's needs)**.** *One might worry that a "thin" enriched category cannot express all the nuance of internal representations, which often involve many distinct mechanisms. But much of Hyperseed's ontology, at least at the scaffold level, asks questions of the form: "to what degree does structure A constrain or entail structure B?" For this, a single graded hom-value is often the right abstraction: it forgets implementation details while preserving compositional inequality laws, which are the main invariants needed to prove monotonicity and stability results later.*

This choice is also methodologically Russellian: replace metaphysical speculation with a calculus of relations whose algebraic laws are explicit. When more expressivity is required (multiple arrows, higher morphisms, transformations between transformations), the enriched perspective remains compatible with refinement rather than needing replacement.

A further pragmatic benefit is that thin enrichment cleanly separates two concerns that are often conflated in informal discussions: (a) the existence *of a pathway from A to B, and (b) the* quality *or* reliability *of that pathway. In a thin V-category the former is encoded by whether $\mathcal{C}(A, B)$ is above a chosen threshold, while the latter is encoded by its precise value; proofs can then be parameterized by the threshold without changing the underlying formal statements.*

Remark 80 (Optional higher-categorical extension)**.** *Nothing in this section requires higher categories. However, some Hyperseed themes (e.g. perspective shifts, equivalence up to reparameterization, and self-reference) are naturally expressed using groupoids and $(\infty, 1)$-categorical equivalence. Where needed later, one may treat "entities" as objects of an $(\infty, 1)$-groupoid and treat observer translations as higher morphisms. The present section provides the 1-categorical "spine" on which that extension can be built.*

Even in the presence of higher structure, thin enrichment remains useful as a decategorified "shadow": one can map a higher groupoid of transformations to a V-valued accessibility relation by assigning to each pair (A, B) an aggregate strength of all higher paths from A to B (e.g. a supremum over witnesses, a probability, or a robust similarity score). Thus the enriched spine is not merely a preliminary simplification but also a stable interface between qualitative and quantitative layers.

3.7 Quantale weakness as "failed distinctions"

We now formalize the simplest quantitative proxy for Hyperseed's simplicity/generalization idea: an observer is "simpler" (or more general) when it collapses more distinctions [3, 2]. Hyperseed-Concept 202; Hyperseed-Concept 143; Hyperseed-Concept 169.

The key move in this subsection is to represent "what the observer does not (or chooses not to) discriminate" as a relation on X, and then to aggregate the "importance" of the collapsed pairs using the ambient quantale structure. This keeps the formal core small while still allowing different downstream readings (perceptual limitations, abstraction, compression, or deliberate coarse-graining).

Definition 15 (Salience/importance weights)**.** *Let X be a set and let $\mu : X \to V$ be a weight map. Intuitively, $\mu(x)$ encodes how much the observer cares about distinguishing x (or how much "mass" x has for aggregation).*

Remark 81 (Intuition and examples for μ)**.** *The weight map μ is a formal admission that not all entities matter equally. Two observers may agree on which pairs are indistinct yet differ on the consequences because one treats certain entities as salient and others as negligible. This aligns with Hyperseed's emphasis on attention and salience (Hyperseed-Concept 60).*

Example 1: In a perception task, $\mu(x)$ might reflect how often x occurs or how costly it is to misclassify it. Example 2: In an internal cognitive model, $\mu(x)$ might encode how emotionally charged or goal-relevant an item is, which will later connect to values and emotion (Hyperseed-Concept **??***; Hyperseed-Concept* **??***). The usefulness is that weakness and pattern measures become sensitive not only to* how many *distinctions are lost, but to* which *ones.*

Operationally, μ is the hook by which context enters the otherwise combinatorial object $H \subseteq X \times X$: changing attention, goals, or risk tolerance can be modeled simply by changing μ, without changing X itself. In particular, assigning $\mu(x) = \bot$ (or the least element of V) can be read as declaring x negligible for the present analysis, since such elements will not contribute to weakness after multiplication by $\otimes$ and aggregation by $\bigoplus$.

Definition 16 (Crisp failed-distinction set)**.** *A* failed-distinction set *is a subset $H \subseteq X \times X$ whose elements are the pairs an observer treats as indistinguishable (in a given context and for a given purpose).*

Remark 82 (Intuition: a formal shadow of "cannot tell apart")**.** *A failed-distinction set H records where the observer's discriminative power collapses: $(u, v) \in H$ means that, for the relevant purpose, u and v are treated as the same. If one were modeling classical equivalence, one might demand reflexivity, symmetry, and transitivity. Here we do* not *impose those axioms: in cognitive reality, indistinction can be asymmetric or context-fragmented, and transitivity can fail under noise or shifting attention (Hyperseed-Concept* **??***; Hyperseed-Concept 195).*

Example: If X consists of similar visual stimuli, H may contain (u, v) when the system confuses u for v. The usefulness of keeping H separate from a fully axiomatized equivalence relation is that it allows us to measure weakness and then later study what closure operations (e.g. compositional propagation) do to H as a form of emergence (Hyperseed-Concept **??***).*

Note also that H is allowed to include or exclude diagonal pairs (u, u). If $(u, u) \in H$ is permitted, it can be interpreted as "self-collapse" in the sense that the observer does not maintain a stable distinction of u from itself across time or internal state (e.g. due to temporal aliasing); if it is excluded, then weakness is driven purely by cross-entity confusions. The formalism itself does not force either convention; one can choose the modeling stance appropriate to the domain.

Definition 17 (Weakness of a failed-distinction set)**.** *Given (X, μ) and $H \subseteq X \times X$, define the* weakness *of H by*

$$w(H) := \bigoplus_{(u,v)\in H} \mu(u) \otimes \mu(v).$$

Remark 83 (Notation unpacking, intuition, and a tiny example)**.** *The symbol $\bigoplus$ denotes the join/supremum in the quantale V (in the canonical* p-bit *quantale, this is componentwise* max*). The product $\otimes$ is the quantale monoidal product (in the canonical case, componentwise multiplication). Thus each indistinct pair contributes a combined weight $\mu(u) \otimes \mu(v)$, and $w(H)$ aggregates all such contributions.*

Example (canonical p-bit *quantale): if $\mu(u) = (0.9, 0.9)$ and $\mu(v) = (0.5, 0.5)$ then $\mu(u) \otimes \mu(v) = (0.45, 0.45)$. If H contains several pairs, then $w(H)$ keeps the maximum positive component and maximum negative component among the pair-weights. This is a minimal but robust way to say: "weakness is at least as large as the most significant collapsed distinction." The definition is useful because it creates a monotone scalar-like measure (in V) that can be compared across observers/contexts and will later interface with pattern intensity and effort tradeoffs [3].*

A useful basic property, immediate from the definition and the order structure of V, is monotonicity in H: if $H \subseteq H'$ then every join-summand for H is also present for H', hence $w(H) \leq w(H')$. In words, declaring additional pairs as indistinct cannot reduce weakness. Likewise, if $\mu \leq \mu'$ pointwise (meaning $\mu(x) \leq \mu'(x)$ for all x), then $w_{\mu}(H) \leq w_{\mu'}(H)$, expressing that increasing salience can only increase (or preserve) measured weakness.

When V has a top element $\top$ and $\otimes$ is order-preserving, one can also read $w(H) \leq \bigoplus_{(u,v)\in H} \top = \top$ as a trivial upper bound, and $w(\varnothing) = \bot$ as the base case: an observer that collapses no tracked distinctions (in the current context) has minimal weakness.

Remark 84 (Fuzzy failed distinctions)**.** *One can generalize H from a crisp set to a V-valued relation $H : X \times X \to V$ and define*

$$w(H) := \bigoplus_{(u,v)\in X\times X} H(u, v) \otimes \mu(u) \otimes \mu(v),$$

which reduces to Definition 17 when H is $\{e, \bot\}$-valued. The crisp form is sufficient for the sanity theorem in Section 4 and for minimal exposition.

In this fuzzy reading, $H(u, v)$ can be interpreted as a graded degree of indistinguishability or confusion: $\bot$ means "fully distinguished" and larger values mean "more collapsed." The algebraic form then says that weakness is obtained by combining (via $\otimes$) three factors: how much the observer fails to distinguish the pair, and how salient each endpoint is. The crisp case corresponds to the indicator-like choice

$$H(u, v) = \begin{cases} e & (u, v) \in H, \\ \bot & (u, v) \notin H, \end{cases}$$

so that only the designated indistinct pairs contribute to the join.

Remark 85 (Why weakness formalizes "simplicity" rather than merely "error")**.** *It is tempting to read H as a set of mistakes, but in Hyperseed the collapse of distinctions is not always a failure; it is often a strategy of generalization. A system with limited resources may deliberately treat many cases as "the same" to gain compression and transfer. The weakness functional captures this tradeoff by measuring the extent of such collapse, without presuming it is bad. In later sections, effort and description-length considerations will supply the missing normative layer, connecting to algorithmic compression intuitions [16].*

This is also why the definition is phrased purely in terms of indistinction and salience, rather than in terms of an externally provided "ground truth": the same H may be advantageous or disastrous depending on which downstream tasks are prioritized (encoded, here, primarily through μ and later through explicit utility/effort structure).

Interpretive link to Hyperseed simplicity. Hyperseed treats simplicity as observer-relative and grounded in the distinctions an observer can afford to make. The weakness $w(H)$ is a direct algebraic instantiation: larger failed-distinction sets (or heavier collapsed pairs) produce larger weakness. Later, effort and description length will provide additional structure on top of this.

Because $w(H)$ lives in the same quantale V used elsewhere in the minimal formal core, it can be compared, combined, or bounded using the same operations as other quantities (e.g. pattern intensities). This is one reason to prefer a quantale-valued measure rather than a single real number: the ordering and aggregation structure needed for later composition is already present at the type level.

3.8 Interpretation map: from Hyperseed talk to core objects

The formal core is intended to be a "dictionary" rather than a full metaphysics [1]. In particular, the role of the mapping below is to make each piece of Hyperseed vocabulary point to a *typed* mathematical placeholder with a clear domain of definition (e.g. elements, relations, functionals) and with explicit rules for how it composes with the rest of the framework. This helps keep later constructions invariant under changes of philosophical reading: one may reinterpret the words, but the algebra of the surrogates stays fixed. A convenient minimal mapping is:

- **Entity / occasion token** $\rightsquigarrow$ element $x \in X_C$ in a context C. Hyperseed-Concept 124. Here X_C is not assumed to be a set of mind-independent "things"; it is simply the carrier of whatever items are *available to be related* inside the chosen standpoint C, so its intended meaning is operational rather than metaphysical.

- **Context / aspect / observer** $\rightsquigarrow$ package C with X_C and V-relations and valuations. Hyperseed-Concept 86. Concretely, C functions as the minimal bookkeeping device that fixes (i) what counts as an admissible token, (ii) what graded comparisons or constraints between tokens can be stated, and (iii) which valuation scheme interprets those graded statements in the chosen evidence structure V.

- **Distinction / indistinction** $\rightsquigarrow$ a designated V-relation $R_C : X_C \times X_C \to V$ (read as evidence-for/evidence-against "same for purpose P"), plus thresholded crisp projections if needed. Hyperseed-Concept 98; Hyperseed-Concept **??**. The point is that "distinguishability" is represented as *graded* rather than all-or-nothing: $R_C(x, y)$ records how strongly C supports treating x and y as the same (or as different, depending on the chosen polarity) relative to a purpose parameter P that may be implicit in C's choice of relation. When later arguments require a classical equivalence relation, one can recover it by selecting a threshold in V and projecting R_C to a crisp predicate, thereby making explicit exactly where discretization enters.

- **Simplicity / generalization** $\rightsquigarrow$ weakness $w(H)$ of an induced failed-distinction relation H. Hyperseed-Concept 169; Hyperseed-Concept 202; Hyperseed-Concept 143. Intuitively, H summarizes where a proposed distinction does *not* hold (or holds only weakly), and the functional $w(H)$ turns that summary into a scalar-like comparison value inside the ambient order of V (or an associated preorder). Reading "simplicity" as "ability to ignore differences without breaking too much," greater weakness corresponds to more aggressive generalization: the formalism makes that tradeoff measurable so that later optimization principles can compare different candidate abstractions.

- **Approximate morphism / correspondence** $\rightsquigarrow$ a V-relation (module) or V-functor between enriched structures. Hyperseed-Concept 112. This treats "correspondence" as something that may be many-to-many and degree-valued rather than a strict function: a module captures soft matching between carriers, while a V-functor captures structure preservation at the enriched level (so that similarity, entailment, or reachability relations are transported coherently). The choice between them is therefore not merely technical: it encodes whether the intended alignment is functional, relational, or compositional.

- **Pattern support** $\rightsquigarrow$ a constraint family on V-relations (or a subobject in an enriched category), ranked by weakness/intensity (developed later). Hyperseed-Concept 130; Hyperseed-Concept 132. In this dictionary, a "pattern" is not a primitive shape but a *condition* that admissible relations (or admissible objects) satisfy; support is then the degree to which the condition is met under the valuation scheme. The weakness/intensity ranking is included so that "pattern is present" becomes a comparative statement (which patterns are stronger, more stable, or more informative) rather than a binary label.

- **Resonance / coupling** $\rightsquigarrow$ a functional of paraconsistent valuations (Section 3.9) used to modulate cross-context propagation operators. Hyperseed-Concept 159; Hyperseed-Concept 115. The intent is that resonance is not itself another relation on X_C, but a higher-level aggregator acting on valuations (which may contain inconsistent or incomplete evidence) and returning a control parameter that gates how strongly information, constraints, or alignments are allowed to flow between contexts. Framed this way, "coupling" becomes something that can be tuned, composed, and compared, rather than a purely qualitative metaphor.

Remark 86 (Why a dictionary is philosophically honest)**.** *Hyperseed's ambition is ontological, but a mathematical reconstruction can remain neutral about ultimate metaphysics while still being precise about* inference rules *and* composition laws. *The dictionary approach acknowledges, in a Whiteheadian spirit [15], that much of what we call "reality" in practice is a stabilized network of relations and transformations between standpoints. What matters for theoremhood is not whether a term is metaphysically primitive, but whether its formal surrogate behaves coherently under the operations the theory demands.*

This mapping is also a safeguard: when later sections speak in richer language (pattern, habit, resonance, mindplex), we can always ask, "what is the corresponding object here, and what operations does it support?" The core thus acts as a semantic spine that keeps the later conceptual elaborations from drifting into purely metaphorical territory.

A further virtue is methodological: disagreements about interpretation can be localized. If two readings of "context" lead to the same algebraic interface (carriers, V-relations, valuations, and their induced operators), then the subsequent theorems apply to both; if they lead to different interfaces, the point of divergence is made explicit and can be analyzed as a change of axioms rather than as a purely verbal dispute.

This dictionary will be refined repeatedly in the systematic reconstruction part of the paper (Part II). In particular, later sections will add explicit assumptions on V (e.g. whether it is a quantale, bilattice, or other ordered algebra), clarify how valuations interact with composition of V-relations, and specify which additional structure on contexts is required to make cross-context propagation and comparison theorems well-posed.

3.9 Resonance derived from paraconsistency

Hyperseed uses "resonance" as a primitive-seeming word for coherent coupling. Hyperseed-Concept 159. To make it mathematically explicit while keeping the paraconsistent structure visible, we embed p-bit values into the complex plane [24]. This also sets the stage for later discussion of morphic resonance (Hyperseed-Concept 115) and related speculative inspirations [13].

Definition 18 (Logic-to-complex map)**.** *Define* $\sigma_C : \mathbb{V} \to \mathbb{C}$ *by*

$$\sigma_C(v^+, v^-) := (v^+ - v^-) \ + \ i\big((v^+ + v^-) - 1\big).$$

Remark 87 (Notation unpacking and intuition)**.** *Here* $\mathbb{C}$ *is the complex plane, and* i *is the imaginary unit. The map* σ_C *takes an evidence-pair* (v^+, v^-) *and produces a complex number whose real part is the evidence bias* $v^+ - v^-$ *and whose imaginary part is the total evidence* $(v^+ + v^-)$ *recentered so that* $v^+ + v^- = 1$ *maps to imaginary part* 0*.*

Intuitively, this is a way of turning paraconsistent structure into a phase-like *object: different kinds of certainty and conflict point in different directions. Once evidence lives in* $\mathbb{C}$*, we can use the geometry of angles and inner products to speak of alignment, opposition, and interference—concepts that mirror the phenomenology of resonance more closely than scalar similarities do.*

Remark 88 (Geometric meaning)**.** *The real part is the bias toward positive vs negative evidence. The imaginary part measures over- vs under-determination relative to the midpoint* $v^+ + v^- = 1$*:*

- $(1, 0) \mapsto 1 + 0i$ *("true" axis),*
- $(0, 1) \mapsto -1 + 0i$ *("false" axis),*
- $(1, 1) \mapsto 0 + 1i$ *("both" axis),*
- $(0, 0) \mapsto 0 - 1i$ *("neither" axis).*

Thus the four Belnap corners become the four cardinal directions in $\mathbb{C}$*.*

Remark 89 (Two simple examples and why complex geometry helps)**.** *Example 1:* $(0.9, 0.1)$ *maps to* $0.8 + i(0.0)$*, strongly positive and determinate. Example 2:* $(0.9, 0.9)$ *maps to* $0 + i(0.8)$*, indicating maximal conflict without bias. In the plane, these points are orthogonal directions: a bias-driven certainty is geometrically distinct from a conflict-driven certainty.*

This matters because many cognitive couplings are sensitive to kind *of certainty, not just amount. Two contexts can both be "highly committed" while one is committed via consensus and the other via internal contradiction; the embedding keeps these apart so that resonance can distinguish alignment from mere magnitude.*

Definition 19 (Local resonance kernel)**.** *For* $v, w \in \mathbb{V}$ *with* $\sigma_C(v) \neq 0$ *and* $\sigma_C(w) \neq 0$*, define the normalized alignment*

$$\mathrm{align}(v, w) := \frac{\Re\big(\sigma_C(v)\overline{\sigma_C(w)}\big)}{|\sigma_C(v)|\,|\sigma_C(w)|} \in [-1, 1].$$

If either $\sigma_C(v) = 0$ *or* $\sigma_C(w) = 0$*, set* $\mathrm{align}(v, w) := 0$*.*

Remark 90 (Notation unpacking: $\Re$, $\overline{(\cdot)}$, and $|\cdot|$)**.** *The symbol* $\Re(\cdot)$ *denotes the real part of a complex number. The bar* $\overline{z}$ *denotes complex conjugation, which reflects* z *across the real axis. The absolute value* $|z|$ *denotes the complex modulus (Euclidean norm). Thus* $\Re(\sigma_C(v)\overline{\sigma_C(w)})$ *is the* *standard real inner product on* $\mathbb{R}^2$ *written in complex notation, and the denominator normalizes by lengths to yield a cosine similarity.*

So $\mathrm{align}(v,w)$ *is literally the cosine of the angle between the two embedded evidence vectors. In geometric language: resonance is alignment of directions; anti-resonance is pointing in opposite directions; orthogonality is non-coupling.*

Remark 91 (Reading align)**.** $\mathrm{align}(v,w) = 1$ *indicates perfect alignment in the complex embedding (maximal "resonance"),* $\mathrm{align}(v,w) = -1$ *indicates perfect opposition (maximal "anti-resonance"), and values near* 0 *indicate weak coupling or orthogonality (e.g. true vs both).*

Remark 92 (Why define alignment locally before aggregating)**.** *Resonance in practice is rarely a single comparison; it is an emergent property of many coupled judgments across a network (Hyperseed-Concept 131; Hyperseed-Concept 132). The local kernel* $\mathrm{align}(v,w)$ *gives the primitive comparison from which larger interference measures can be built. By keeping the primitive symmetric and normalized, we avoid arbitrary scale artifacts when later summing many terms.*

In the paraconsistent resonance perspective [24], such kernels are the atoms of a broader interference calculus: coherent coupling corresponds to constructive interference of many phase-like contributions.

Many resonance phenomena involve aggregation over many coupled evaluations. The following interference functional is one simple way to obtain a scalar coupling score that increases under coherent alignment.

Definition 20 (Interference-based resonance for a family)**.** *Let* $z_1, \dots, z_n \in \mathbb{C}^d$ *and weights* $\alpha_1, \dots, \alpha_n \in \mathbb{R}_{\geq 0}$*. Define*

$$z_{\mathrm{tot}} := \sum_{k=1}^{n} \alpha_k z_k, \qquad I := \|z_{\mathrm{tot}}\|^2 - \sum_{k=1}^{n} \|\alpha_k z_k\|^2.$$

A bounded resonance score may be obtained by normalizing I *(e.g. by* $\sum_k \|\alpha_k z_k\|^2$ *when nonzero) and applying any monotone squashing map into* $[0,1]$.

Remark 93 (Intuition: constructive vs. destructive interference)**.** *The quantity* $\|z_{\mathrm{tot}}\|^2$ *measures the squared magnitude of the vector sum, while* $\sum_k \|\alpha_k z_k\|^2$ *measures the sum of squared magnitudes if there were no cross-terms. Their difference* I *is exactly the contribution of pairwise cross-terms: it is positive when vectors tend to align (constructive interference) and negative when they tend to cancel (destructive interference).*

More explicitly (for the usual inner product on $\mathbb{C}^d$*), expanding the square gives*

$$\left\|\sum_k \alpha_k z_k\right\|^2 = \sum_k \|\alpha_k z_k\|^2 \; + \; \sum_{i \neq j} \Re\big(\langle \alpha_i z_i, \alpha_j z_j \rangle\big),$$

so I *aggregates the (real parts of the) pairwise alignments* $\langle z_i, z_j \rangle$ *after scaling by* α_i, α_j*. In particular,* I *is sensitive not just to magnitudes but to relative directions/phases, which is exactly what one wants from a numerical proxy for "resonance" versus "anti-resonance."*

Example (scalar case $d = 1$*): if* $z_1 = z_2 = 1$ *and* $\alpha_1 = \alpha_2 = 1$*, then* $z_{\mathrm{tot}} = 2$ *so* $I = 4 - (1+1) = 2 > 0$*. If* $z_1 = 1$ *and* $z_2 = -1$*, then* $z_{\mathrm{tot}} = 0$ *so* $I = 0 - (1+1) = -2 < 0$*. This makes* I *a natural numerical surrogate for resonance/anti-resonance at the aggregate level.*

In intermediate situations I *can be close to* 0 *even when individual terms are strong, meaning that the aggregate is "balanced" by competing contributions; this is the quantitative sense in which interference captures the difference between mere presence of strong components and their ability to jointly amplify a shared direction.*

Remark 94 (Why we allow $z_k \in \mathbb{C}^d$)**.** *The extra dimension d allows us to embed multiple aspects of evaluation simultaneously—for example, multiple propositions, multiple relational edges, or multiple features of a pattern. This is a small but important generalization: resonance in rich systems is seldom one-dimensional. The formalism here is kept minimal, but it anticipates later constructions where patterns and contexts are multi-aspect objects (Hyperseed-Concept 86; Hyperseed-Concept 79).*

Allowing $\mathbb{C}$ (rather than restricting to $\mathbb{R}$) is useful even if one ultimately interprets relation-strengths as real-valued: complex phases provide a compact way to encode contextual "orientation" or timing/ordering effects, so that two strong components can either reinforce (similar phase) or suppress (opposite phase) without changing their magnitudes. In particular, the same magnitude data $\|z_k\|$ can yield different aggregate behavior depending on relative phases, which mirrors the idea that the same local evidence can support different global outcomes depending on context alignment.

Resonance as a driver of propagation. To connect resonance to later morphic-resonance dynamics, we introduce the abstract form of a resonance-driven update rule.

The point of isolating an abstract operator at this stage is that it factors "what is being propagated" (the V-relation) from "how propagation composes" (the quantale algebra), so later sections can swap in different concrete semantics for V without changing the propagation skeleton.

Definition 21 (Resonance-driven propagation operator (abstract form))**.** *Let $R_1, R_2 : X \times X \to V$ be V-relations in two contexts (or two subsystems). Let $K \in V$ be a coupling strength. Define*

$$\mathsf{Res}_K(R_1 \to R_2)(x,y) := R_2(x,y) \;\oplus\; \big(K \otimes R_1(x,y)\big).$$

Remark 95 (Intuition and a minimal example of an update)**.** *This update says: the new relation-value in context 2 is the old value, plus (in the quantale join sense) a contribution from context 1 scaled by coupling strength K. In the canonical* p-bit *quantale, $\oplus$ is componentwise* max*, so this means the update can only increase evidence components: resonance propagation accumulates rather than overwrites.*

Conceptually, K plays the role of a gain or transmissibility parameter: when K is low, even strong structure in R_1 has limited influence on R_2; when K is high, R_2 rapidly inherits (in the join sense) the strongest components present in R_1. Because the update is pointwise in (x,y), this is the minimal "local" propagation law; later dynamics can make K depend on (x,y) or on higher-order pattern statistics.

Example: if $R_2(x,y) = (0.2, 0.1)$, $K = (0.5, 0.5)$, and $R_1(x,y) = (0.9, 0.8)$, then

$$K \otimes R_1(x,y) = (0.45, 0.4),$$

so $\mathsf{Res}_K(R_1 \to R_2)(x,y) = (\max(0.2, 0.45), \max(0.1, 0.4)) = (0.45, 0.4)$*. This captures the intuition that strongly resonant structure in one context can "pull" another context toward similar evaluations, without requiring them to become consistent or to erase prior opposition (Hyperseed-Concept 159; Hyperseed-Concept 86).*

Note also that the form $R_2 \oplus (\cdots)$ makes the prior state of context 2 explicit: propagation is not a replacement map $R_2 \mapsto K \otimes R_1$, but a cumulative enrichment map. This is the algebraic analogue of treating contexts as memory-bearing substrates rather than as ephemeral registers.

Remark 96 (Why this is the right minimal form)**.** *The operator* Res_K *is:*

- *monotone in R_1, R_2, and K;*
- *explicitly paraconsistent: it can increase positive evidence without erasing negative evidence (and vice versa);*

- *compositional: repeated application composes via the quantale product.*

Section 5 instantiates this operator numerically and shows how it reaches a "morphic resonance" style update.

The compositionality point can be read operationally as a semigroup law for propagation strength: applying Res_{K_1} *and then* Res_{K_2} *yields an effect equivalent to a single step whose effective contribution from* R_1 *is scaled by a combination of* K_1 *and* K_2 *as dictated by* $\otimes$ *and* $\oplus$ *(with the precise normal form depending on distributivity properties available in the chosen* V*). This is one reason to phrase the update at the quantale level: the algebra already knows how to combine influences.*

Remark 97 (Why monotonicity is a feature, not a limitation)**.** *One might object that real learning sometimes decreases beliefs. That is correct; but the monotone operator here is intentionally the minimal* propagation *primitive. Decrease and revision will later enter via additional dynamics: decay, inhibition, competition between patterns, and anti-resonance mechanisms. In other words, we first isolate a clean "transmission" law and then layer conflict-resolution and forgetting on top, which is closer to how dynamical systems are often built in cognitive modeling [19].*

Technically, monotonicity is also what makes iterated updates well-behaved: one can form increasing chains $R_2 \preceq \mathsf{Res}_K(R_1 \to R_2) \preceq \mathsf{Res}_K(R_1 \to \mathsf{Res}_K(R_1 \to R_2)) \preceq \cdots$ *and then study their convergence or fixed points using standard order-theoretic tools when* V *supports them. This is helpful later when "habit" is modeled as a stabilized relation-value under repeated exposure.*

Definition 22 (Anti-resonance propagation (minimal choice))**.** *Define the anti-resonance operator by propagating negated evidence:*

$$\mathsf{AntiRes}_K(R_1 \to R_2)(x,y) := R_2(x,y) \ \oplus \ \big(K \otimes (\neg R_1(x,y))\big).$$

Remark 98 (Intuition: transmitting the contrary and modeling habit reversal)**.** *Anti-resonance propagation uses the same algebra as resonance but flips the incoming evidence via* $\neg$*. If* $R_1(x,y)$ *strongly supports a match, then* $\neg R_1(x,y)$ *strongly supports a mismatch, and vice versa. The resulting update tends to induce opposition or reversal in the target relation-values.*

Because $\neg$ *is applied before the same* $\otimes$*-scaling and* $\oplus$*-accumulation, this operator is not a different propagation mechanism but a different* polarity *of what is being transmitted. In paraconsistent settings (such as* $\mathsf{p\text{-}bit}$*), this is crucial: transmitting the contrary does not require removing the original support already present in* R_2*, so the target context can legitimately accumulate mixed evidence and thereby represent tension, ambiguity, or internal conflict rather than being forced into a prematurely consistent state.*

This gives a minimal formal hook for Hyperseed's idea that some couplings do not reinforce but rather destabilize or invert habits (Hyperseed-Concept 188; Hyperseed-Concept 114). The operator is intentionally simple; later sections can refine it by making K *depend on interference scores, context similarity, or resource constraints.*

In particular, one can anticipate variants where K *is large precisely when the interference indicator* I *is strongly negative (net cancellation), so that anti-resonant channels become more active when signals are out of phase; the present definition isolates the algebraic core needed for such couplings without committing to a specific functional dependence.*

Remark 99 (Toward "morphic" resonance)**.** *In Hyperseed terms, morphic resonance will arise when resonance-driven propagation interacts with (i) repeated exposure over proto-time, (ii) reinforcement/decay (habit-taking and habit-reversal operators), and (iii) a pattern web structure that makes some relation-values self-supporting. Those dynamical ingredients are added in Sections 5 and 12.*

From the perspective of the present minimal core, the key transition to "morphic" behavior is the emergence of feedback loops: once propagated relation-values in one context begin to act as sources for further propagation (possibly back into the original context, or into overlapping subcontexts), the system can amplify particular relational regularities and thereby exhibit the characteristic persistence associated with habits. The same feedback architecture can also sustain stable paraconsistent states (simultaneous support and opposition), which is one reason the update primitives above are designed to be compatible with non-classical evidence aggregation from the outset.

4 Core sanity theorems

Outline

- Prove monotonicity properties that justify interpreting $w(H)$ as "weakness."
- Prove basic boundedness/consistency properties for pattern intensity and resonance.
- State minimal categorical coherence properties needed later (functoriality, enrichment laws).

Summary and Hyperseed concepts covered

These theorems are intentionally simple. They exist to ensure that the chosen core definitions behave in the expected direction: more collapsed distinctions means greater weakness, and resonance behaves monotonically under coherent aggregation.

Hyperseed concepts covered.

- Distinction; equivalence; simplicity/weakness; pattern intensity (as a later derived quantity). (Hyperseed-Concepts 98, **??**, 169, 202, 143.)
- Resonance/dissonance (as interference monotonicity). (Hyperseed-Concepts 159, 97.)

Standing notation. Fix a commutative quantale $(V, \leq, \oplus, \otimes, e)$ and a set of entities X. Let $\mu : X \to V$ be an object-valuation. For a relation $H \subseteq X \times X$ interpreted as the set of *undistinguished pairs*, recall the (quantale) weakness functional

$$w(H) \;=\; \bigoplus_{(u,v)\in H} \mu(u) \otimes \mu(v).$$

(Here $\oplus$ is the quantale join/supremum.)

Remark 100. *A brief decoding of the symbols may help fix the intended reading. A* commutative quantale *is, in particular, a complete lattice $(V, \leq)$ equipped with a commutative, associative "multiplication" $\otimes$ with unit e, such that $\otimes$ distributes over arbitrary joins $\oplus$. Thus $\oplus$ should be read as a supremum (a generalized "maximum"), while $\otimes$ is the operation used to aggregate two pieces of graded evidence or intensity. This is precisely the algebraic setting in which one can speak coherently about combining local contributions and then taking their overall "best possible" aggregate; it is also the basic technical vehicle for quantale-style weakness in the sense of [3] (see Hyperseed-Concept 143).*

The set X is the local universe of entities (occasion-tokens, perceived items, or internal states), and $\mu : X \to V$ assigns each entity a weight/importance in the evidence domain V. The relation

$H \subseteq X \times X$ collects pairs that an observer/context fails to distinguish (Hyperseed-Concept 98). Then $w(H)$ takes every undistinguished pair (u, v), scores it by the combined weight $\mu(u) \otimes \mu(v)$, and aggregates all these scores via the join $\oplus$. Philosophically, this is a very Russellian move: one takes an apparently qualitative notion ("weakness" as failure-to-separate) and pins it to a monotone functional that can be reasoned about with lattice algebra, while leaving ample room for the phenomenological interpretation (Hyperseed-Concepts 202, 169).

It is also worth noting what information is, and is not, being retained. The functional $w(H)$ depends only on the set *of undistinguished pairs, not on any additional structure (e.g. transitivity, symmetry, or a metric-like notion of "how hard" a pair is to distinguish). Later sections may impose additional axioms on H (for instance, taking H to be an equivalence relation, or the kernel of an observation map), but the present sanity checks deliberately avoid such commitments: they isolate the order-theoretic behavior that should hold for* any *coarsening of distinctions.*

Finally, the commutativity of $\otimes$ ensures that $\mu(u) \otimes \mu(v)$ does not depend on the ordering of the pair. This matches the intended semantics when H is thought of as an undirected indistinction; if one later studies directed or asymmetric indistinction (e.g. attentionally one-way confusions), then one can still use the same formula, but should be explicit about whether H is closed under swapping coordinates.

Theorem 1 (Weakness is monotone under adding undistinguished pairs)**.** *Let $H, K \subseteq X \times X$ with $H \subseteq K$. Then $w(H) \leq w(K)$.*

Remark 101. *Intuitively, this theorem says: if you collapse* more *distinctions (i.e. you declare more pairs to be "undistinguished"), then you cannot become* less *weak. This is exactly the direction one expects if weakness is meant to proxy simplicity-as-coarsening: a coarser partition of experience (more identifications, fewer discriminations) carries less articulated structure, hence represents a stronger "failure to distinguish" (Hyperseed-Concepts 98, 202; see also [2, 3] for related motivations).*

This monotonicity is also a sanity check for later constructions. When patterns are modeled as constraints that reduce the space of distinctions in a controlled way (Hyperseed-Concepts 130, **??***), we will repeatedly compare "baseline" and "pattern-induced" indistinction relations. The present theorem guarantees that these comparisons can be made in the order-theoretic language of V without logical surprises.*

One can also read the result contrapositively: if some intervention (a new measurement, a refined conceptual scheme, an additional sensor) strictly decreases the computed weakness value, then it must have eliminated *at least one undistinguished pair of nontrivial weight. Thus w behaves like a coarse "certificate" that refinement has occurred, even when one does not track which distinctions were recovered.*

Proof. Since $H \subseteq K$, every term $\mu(u) \otimes \mu(v)$ included in the join for $w(H)$ also appears in the join for $w(K)$. Because $\oplus$ is a join (supremum) and hence monotone in each argument, adding more terms cannot decrease the result. ☐

Remark 102. Proof sketch. *The strategy is to reduce everything to the elementary fact that a supremum cannot go down when you enlarge the set you take the supremum over. Here the set being "enlarged" is the collection of pair-scores $\mu(u) \otimes \mu(v)$ indexed by $(u, v) \in H$ versus $(u, v) \in K$.*

Spelled out in lattice-theoretic terms, define two indexed families in V:

$$A = \{\mu(u) \otimes \mu(v) \mid (u, v) \in H\}, \qquad B = \{\mu(u) \otimes \mu(v) \mid (u, v) \in K\}.$$

Then $H \subseteq K$ implies $A \subseteq B$, and $w(H) = \bigoplus A$, $w(K) = \bigoplus B$. Since $\bigoplus$ is the least upper bound, $\bigoplus A \leq \bigoplus B$ follows from $A \subseteq B$ alone, with no need for any further algebraic properties of $\otimes$

beyond well-typedness. In particular, commutativity of $\otimes$ is conceptually natural but not used by this specific monotonicity argument.

This is one reason quantales are a convenient ambient structure: they provide (i) enough completeness so that arbitrary joins exist, and (ii) an order in which "more evidence aggregated" is reflected as "larger value." Even in the degenerate case $H = \varnothing$, the same reasoning applies: $\bigoplus_{(u,v)\in\varnothing}(\cdots)$ is the join of the empty subset of V, hence equals the bottom element of the lattice (often denoted $\perp$), so monotonicity includes the base case $\perp = w(\varnothing) \leq w(K)$ for all K.

A useful mental model is the Boolean quantale $V = \{0,1\}$ with $\oplus = \vee$ and $\otimes = \wedge$. If $\mu(x) = 1$ for all x, then $w(H) = 1$ iff H is nonempty, and the theorem becomes the trivial statement that if $H \subseteq K$ and H is nonempty then K is nonempty. More informative is the fuzzy/probabilistic-like setting $V = [0,1]$ with $\oplus = \max$ and (for instance) $\otimes = \cdot$; then $w(H)$ returns the maximum pairwise combined salience among the undistinguished pairs, so adding more undistinguished pairs can only maintain or increase that maximum.

In later applications, one often considers $K = H \cup H'$ where H' is the new set of collapsed distinctions introduced by an additional pattern or aggregation step. The theorem then reads $w(H) \leq w(H \cup H')$, making explicit that pattern-driven coarsening can be tracked purely by order comparison in V.

Proposition 1 (Weakness is monotone in each argument under unions)**.** *For any $H, K \subseteq X \times X$ one has*

$$w(H) \leq w(H \cup K) \qquad \text{and} \qquad w(K) \leq w(H \cup K).$$

Proof. Both inclusions $H \subseteq H \cup K$ and $K \subseteq H \cup K$ hold, so the claim follows immediately from Theorem 1. □

Remark 103. *This simple corollary is often the operational form of monotonicity: "adding an additional source of indistinction" is mathematically modeled by taking a union of relations. In contexts where multiple pattern constraints contribute independently to collapse of distinctions, the inequality above ensures that combining constraints by union can only increase (or leave unchanged) the overall weakness score.*

If one later introduces a directed *family of relations $(H_i)_{i\in I}$ with $H_i \subseteq H_j$ for $i \leq j$ (successive coarsenings), then repeated application yields a chain*

$$w(H_i) \leq w(H_j) \quad (i \leq j),$$

so the weakness values themselves form an increasing net in the complete lattice V. This observation is frequently used implicitly when taking limits of refinement/coarsening procedures, or when

The key step is not computational but conceptual: interpreting $w(H)$ literally as a join over a family makes monotonicity immediate. In particular, if $w(H) = \bigvee\{\,\phi(e) \mid e \in H\,\}$ for some assignment ϕ into V, then enlarging H simply enlarges the index set of the join, and completeness of the lattice ensures that taking the join over a larger family cannot decrease the value. One may visualize H as a set of edges in a graph on X (edges mark "undistinguished" pairs). Adding edges increases the pool of candidate weights, and a join/supremum can only stay the same or increase. Equivalently, $H \subseteq H'$ implies every upper bound of $\{\phi(e) \mid e \in H'\}$ is also an upper bound of $\{\phi(e) \mid e \in H\}$, so the least such upper bound (the supremum) satisfies $w(H) \leq w(H')$. This is exactly the sense in which "monotonicity is baked into the definition": no further algebraic identities are required beyond the order-theoretic behavior of joins.

Definition 23 (One possible pattern-intensity functional (placeholder))**.** ***[TODO: Replace with the final definition used in the paper.]*** *A* pattern intensity *functional is a map* $\mathrm{Int} : \mathcal{P} \to V$

from a space of candidate patterns $\mathcal{P}$ to the quantale V that increases when a pattern yields a more compressive (weaker/simpler) description of data relative to a baseline. Concretely, the intended reading is that V plays the role of a graded evidence/score domain: larger elements represent "more intensity" (however that is instantiated later), and the order $\leq$ is the sole structure used to compare intensities across patterns. The definition does not require $\mathcal{P}$ to carry additional structure (such as a lattice order), but later uses typically equip $\mathcal{P}$ with a refinement preorder so that monotonicity statements become meaningful.

Remark 104. *This definition is deliberately schematic: it identifies what a pattern-intensity* must do *(be a graded score in V) without committing to a single formula. Intuitively, $\mathcal{P}$ is the space of "possible regularities" one might impose on data (Hyperseed-Concept 130), and $\mathrm{Int}(P)$ measures how much* structural economy *the pattern P buys you compared to not using it (Hyperseed-Concepts 169, 82). In particular, "baseline" is meant to capture whatever reference situation corresponds to "no pattern" (or a default pattern family), so that intensity can be interpreted as a relative rather than absolute score. In compression language, one imagines that a stronger pattern allows a shorter description; in the weakness language, one imagines that a stronger pattern organizes indistinction in a way that can be exploited for prediction or representation (cf. the pattern/emergence viewpoint in [5]). The quantale viewpoint is that whatever concrete coding length or predictive advantage is used, it is ultimately funneled into an ordered structure where joins, tensors, and monotone maps preserve the comparisons one cares about.*

A simple toy example (consistent with later sections) is: let P specify that certain pairs in $X \times X$ must be treated as indistinct (or, conversely, must be distinguished), inducing a relation H_P of undistinguished pairs. Then $\mathrm{Int}(P)$ can be a monotone function of $w(H_P)$ relative to some baseline H_{base}. For instance, one may take $F(a,b) = b$ (pure weakness-as-intensity), or $F(a,b) = a \otimes b$ (combining baseline and pattern weakness multiplicatively/tensorially), or $F(a,b) = b \oplus a$ (a join-style aggregation); the theorem below is formulated to cover such choices as long as monotonicity is preserved. The point of keeping the definition abstract here is methodological: the later theory wants theorems that depend only on quantale-algebraic monotonicity, not on the idiosyncrasies of one specific intensity formula. In particular, by avoiding a fixed numeric encoding at this stage, one can later change the enrichment base V (e.g. to Boolean truth values, costs, probabilities, or degrees of evidence) without having to redo the monotonicity proofs from scratch.

Theorem 2 (Pattern intensity is bounded and respects refinement)**.** *Assume the intensity functional* Int *is defined in the form*

$$\mathrm{Int}(P) \;=\; F\big(w(H_{\mathrm{base}}),\, w(H_P)\big)$$

for some map $F : V \times V \to V$ that is monotone in each argument and is constructed using only the quantale operations $\oplus, \otimes$ and the order $\leq$. Then:

1. *(*Boundedness*) $\mathrm{Int}(P)$ lies between the least and greatest elements of V. Equivalently, writing $\perp$ and $\top$ for the bottom and top of the underlying complete lattice, one has $\perp \leq \mathrm{Int}(P) \leq \top$ for every $P \in \mathcal{P}$.*

2. *(*Refinement monotonicity*) If P refines Q in the sense that $H_P \subseteq H_Q$ (i.e. P leaves weakly fewer pairs undistinguished than Q), then $\mathrm{Int}(P) \leq \mathrm{Int}(Q)$. In words: moving to a finer pattern (fewer declared indistinguishabilities) cannot increase the weakness-based ingredient, so any monotone intensity aggregator cannot increase either.*

Remark 105. *In plain language, this theorem is saying two unsurprising but indispensable things. First, the intensity score never "escapes" the universe of discourse: it is always some element of the* *evidence lattice V, hence automatically confined between bottom and top. This is not a deep estimate but a type discipline: once the codomain is V and V is a complete lattice, every term built from V-operations has a well-defined place in the global order. Second, the ordering of patterns by* refinement *is respected by the intensity score: when one pattern P is "sharper" than another Q (it collapses fewer distinctions, i.e. it declares fewer pairs undistinguished), then the corresponding weakness-based ingredient $w(H_P)$ is smaller, and this order propagates through any monotone aggregator F. Note that the subset condition $H_P \subseteq H_Q$ is oriented so that refinement corresponds to* removing *edges from the "undistinguished graph": fewer edges means more distinctions are enforced.*

This connects directly back to Theorem 1: refinement monotonicity is not a new phenomenon but a consequence of the same lattice-theoretic discipline. In later sections, this is what allows one to treat "pattern selection" as a legitimate optimization or comparison problem, rather than as a collection of metaphors (Hyperseed-Concepts 130, **??***, 169; compare [5]). In particular, once refinement induces an order on patterns, one can speak coherently about maximizing intensity, computing Pareto frontiers under multiple scores, or proving existence of optima using completeness properties (when they apply), all without leaving the order-enriched setting.*

Proof. Because V is a complete lattice, it has a least element (bottom) and a greatest element (top). Any expression built from $\oplus$ and $\otimes$ on elements of V evaluates to an element of V, hence is automatically bounded between bottom and top. More explicitly, for any $v \in V$ one has $\perp \leq v \leq \top$ by definition of $\perp$ and $\top$, and therefore the same holds for $v = \mathrm{Int}(P)$ regardless of how F is parenthesized or composed, provided its values lie in V.

For refinement monotonicity: if $H_P \subseteq H_Q$, then by Theorem 1 we have $w(H_P) \leq w(H_Q)$. Monotonicity of F in its second argument (and the fact that $w(H_{\mathrm{base}})$ is fixed across P, Q) yields $\mathrm{Int}(P) \leq \mathrm{Int}(Q)$. If one also varies the baseline across comparisons, then monotonicity in the first argument would be needed as well; the present statement isolates the typical situation where the baseline is held constant and only the pattern-induced relation changes. □

Remark 106. Proof sketch. *The proof separates into two independent observations. Boundedness is immediate once one remembers that V is a complete lattice: there is simply nowhere else for* $\mathrm{Int}(P)$ *to land. Refinement monotonicity is a one-line reduction: $H_P \subseteq H_Q$ implies $w(H_P) \leq w(H_Q)$ by Theorem 1, and monotonicity of F propagates the inequality to* Int. *In many concrete instantiations, F will be built as a composition of monotone primitives (e.g. $F(a, b) = (a \otimes b) \oplus a$), and the statement "constructed using only $\oplus, \otimes$ and $\leq$" is precisely what guarantees monotonicity is preserved under such compositions.*

The key step is the use of structural monotonicity *rather than any specific numeric property. This is a recurring theme of the reconstruction: rather than rely on special arithmetic facts, we aim to work in the generality of order, join, and tensor, so that the same argument can later be transported to other enrichment bases via change-of-quantale results (cf. Proposition 2 below). One can also read this as a robustness claim: as long as the semantics of "combining evidence" is expressed by monotone quantale operations, refinement comparisons of patterns will behave predictably across models.*

Definition 24 (Interference-based resonance and dissonance (sanity model))**.** *Let I be a finite index set and let $a = (a_i)_{i \in I}$ be a finite family of complex-valued contributions $a_i \in \mathbb{C}$. Define the* resonance *and* dissonance *by*

$$\mathrm{Res}(a) \;=\; \left| \sum_{i \in I} a_i \right|, \qquad \mathrm{Dis}(a) \;=\; \sum_{i \in I} |a_i| \;-\; \mathrm{Res}(a).$$

Intuitively, $\mathrm{Res}(a)$ *measures coherent constructive interference, while* $\mathrm{Dis}(a)$ *measures the destructive-interference gap relative to perfect alignment.*

Remark 107. *A few immediate sanity checks are worth keeping in view. Because* $|\sum_i a_i| \le \sum_i |a_i|$ *by the triangle inequality,* $\mathrm{Dis}(a) \ge 0$ *always, so dissonance is a* nonnegative defect term *measuring the strictness of that inequality. At the extremes, if* $|I| = 1$ *then* $\mathrm{Res}(a) = |a_i|$ *and* $\mathrm{Dis}(a) = 0$, *while if* $\sum_i a_i = 0$ *then* $\mathrm{Res}(a) = 0$ *and* $\mathrm{Dis}(a) = \sum_i |a_i|$ *(complete cancellation). Both* Res *and* Dis *are also permutation-invariant in* I *(they depend only on the multiset of contributions), reflecting that the model is agnostic to any ordering of "inputs."*

It is also useful to note the scaling behavior: for any $c \in \mathbb{C}$ *one has*

$$\mathrm{Res}\big((ca_i)_i\big) = |c|\,\mathrm{Res}(a), \qquad \mathrm{Dis}\big((ca_i)_i\big) = |c|\,\mathrm{Dis}(a),$$

so the model separates a global strength factor $|c|$ *from the internal alignment structure of the phases.*

Remark 108. *Here the notation is meant to be read with the ordinary geometry of the complex plane in mind. The family* $a = (a_i)_{i\in I}$ *is a collection of "phasors" (vectors) in* $\mathbb{C}$, $|z|$ *denotes the complex modulus, and* $\sum_{i\in I} a_i$ *is their vector sum. Thus* $\mathrm{Res}(a)$ *is the length of the resultant vector, while* $\sum_i |a_i|$ *is the sum of the lengths. The quantity* $\mathrm{Dis}(a)$ *is exactly the shortfall between these two, i.e. how much cancellation occurs because the* a_i *point in different directions.*

In particular, $\mathrm{Dis}(a)$ *can be viewed as a quantitative proxy for "phase dispersion": holding the magnitudes* $|a_i|$ *fixed, any increase in disagreement among arguments* $\arg(a_i)$ *tends to reduce* $|\sum_i a_i|$ *and thereby increases* $\mathrm{Dis}(a)$*. For two contributions* $a_1, a_2 \neq 0$ *with relative angle* $\varphi = \arg(a_2) - \arg(a_1)$, *one has the explicit formula*

$$\mathrm{Res}(a) = \sqrt{|a_1|^2 + |a_2|^2 + 2|a_1||a_2|\cos\varphi}, \qquad \mathrm{Dis}(a) = |a_1| + |a_2| - \mathrm{Res}(a),$$

so dissonance increases monotonically as $\cos\varphi$ *decreases (i.e. as the vectors rotate away from one another toward opposition).*

This interference model is intentionally modest: it supplies a mathematically sharp proxy for the phenomenological language of resonance/dissonance (Hyperseed-Concepts 159, 97) without yet importing the full paraconsistent-to-complex embedding discussed elsewhere in the document. It is also aligned with the broader theme that "coherence" is a kind of constructive alignment and that "conflict" is measured by cancellation; related developments coupling paraconsistent structure to resonance-style propagation appear in [24].

Theorem 3 (Resonance is invariant under global phase and increases under coherent alignment)**.** *Let* $a = (a_i)_{i\in I}$ *be as in Definition 24.*

1. *(*Uniform phase rotation invariance*) For any real* θ,

$$\mathrm{Res}\big((e^{i\theta}a_i)_{i\in I}\big) \;=\; \mathrm{Res}(a), \qquad \mathrm{Dis}\big((e^{i\theta}a_i)_{i\in I}\big) \;=\; \mathrm{Dis}(a).$$

2. *(*Coherent-alignment monotonicity*) For any real* θ, *define the aligned family* $a^{(\theta)} = (|a_i|e^{i\theta})_{i\in I}$*. Then*

$$\mathrm{Res}(a) \;\le\; \mathrm{Res}\big(a^{(\theta)}\big) \;=\; \sum_{i\in I} |a_i|, \qquad 0 \;\le\; \mathrm{Dis}\big(a^{(\theta)}\big) \;\le\; \mathrm{Dis}(a).$$

   Moreover, $\mathrm{Res}(a) = \sum_i |a_i|$ *iff all nonzero* a_i *share a common phase.*

Remark 109. *A useful way to read the statement is that* Res *depends only on* relative *phases and magnitudes. Part (1) says that there is no distinguished "absolute reference direction" in the complex plane: rotating the entire configuration rigidly cannot affect either the resultant length or the cancellation defect. Part (2) identifies an extremal configuration at fixed magnitudes: among all families with the same* $|a_i|$*, the maximum possible resultant is achieved exactly by placing every vector on the same ray (common phase), and the dissonance is then forced to vanish.*

The inequalities also clarify an interpretive asymmetry: alignment can only help resonance (never hurt it), while misalignment can only increase dissonance (never decrease it), as long as magnitudes are held fixed. In later applications, this gives a clean monotonicity principle: any mechanism that reduces relative phase disagreement (or, more abstractly, reduces mutual contradiction among contributing terms) must move the model in the direction of larger Res *and smaller* Dis.

Remark 110. *This theorem formalizes two pieces of intuition that often remain implicit when one speaks of "resonance." First, resonance is not about absolute phase but about* relative *phase: rotating every contribution by the same angle changes nothing. Second, resonance is maximized when all contributions are aligned, in which case the resultant amplitude equals the sum of the individual amplitudes and dissonance vanishes.*

The result is important mainly because it validates a particular direction of explanation used later: if a coupling mechanism tends to align phases (or, more abstractly, tends to bring separate evaluative contributions into mutual agreement), then resonance increases and dissonance decreases. This is precisely the sort of monotone behavior one wants when resonance is used as a driver for cross-context pattern propagation (Hyperseed-Concepts 131, 159, 168) rather than as a merely poetic label.

Proof. (1) We have

$$\mathrm{Res}\big((e^{i\theta}a_i)_i\big) = \left|\sum_i e^{i\theta}a_i\right| = \left|e^{i\theta}\sum_i a_i\right| = \left|\sum_i a_i\right| = \mathrm{Res}(a).$$

Also $|e^{i\theta}a_i| = |a_i|$, so the sum of magnitudes is unchanged, hence Dis is unchanged as well. Equivalently, multiplying by $e^{i\theta}$ is an isometry of $\mathbb{C} \cong \mathbb{R}^2$, so it preserves all Euclidean lengths appearing in the definitions.

(2) By the triangle inequality,

$$\left|\sum_i a_i\right| \;\le\; \sum_i |a_i|.$$

But $\sum_i |a_i|e^{i\theta} = e^{i\theta}\sum_i |a_i|$, whose magnitude is exactly $\sum_i |a_i|$. Thus $\mathrm{Res}(a) \le \mathrm{Res}(a^{(\theta)}) = \sum_i |a_i|$. Since $\mathrm{Dis}(a) = \sum_i |a_i| - \mathrm{Res}(a)$, increasing Res (with the same $\sum_i |a_i|$) decreases Dis, and $\mathrm{Dis}(a^{(\theta)}) = 0$. One can also read this as: at fixed magnitudes, the maximal possible resultant is achieved by placing every term so that its contribution to the projection along the common direction is positive and maximal.

Equality in the triangle inequality holds iff all nonzero summands have the same argument, i.e. the a_i are all nonnegative real multiples of a common $e^{i\theta}$. □

Remark 111. Proof sketch. *Part (1) uses the fact that multiplying a complex number by* $e^{i\theta}$ *rotates it without changing its length, and that absolute value is invariant under such rotations. Part (2) is essentially the triangle inequality: the magnitude of a sum cannot exceed the sum of magnitudes, with equality exactly when all vectors point the same way.*

The key step is recognizing that "alignment" can be expressed by replacing each a_i with a vector of the same length but common phase. Geometrically, one may picture the a_i as arrows in the plane; the resonant case is when the arrows lie on top of each other, so their endpoint-to-endpoint addition is maximal, and the dissonant case is when they partially cancel. In this sense, $\mathrm{Dis}(a)$ *is not an additional primitive but a bookkeeping device that records precisely "how far" the configuration is from saturating the triangle inequality, which is why it behaves well under the sanity checks above (nonnegativity, vanishing in the fully aligned case, and maximality under total cancellation).*

Proposition 2 (Minimal categorical coherence for weakness: join-homomorphism and change of base)**.** *With $w(H) = \bigoplus_{(u,v)\in H} \mu(u) \otimes \mu(v)$ as above (where $\bigoplus$ denotes the* arbitrary *join in the complete lattice V, so the definition makes sense for finite or infinite H):*

1. *(*Complete join preservation*) For any family of relations $\{H_j\}_{j\in J}$,*

$$w\left(\bigcup_{j\in J} H_j\right) \;=\; \bigoplus_{j\in J} w(H_j), \qquad w(\emptyset) = \bot_V,$$

   where $\bot_V$ is the least element of the lattice V. In particular, overlaps among the H_j do not affect the result: if a given pair (u, v) appears in more than one H_j, it contributes the same element $\mu(u) \otimes \mu(v)$ multiple times, but because $\bigoplus$ is a join (hence idempotent), repeated contributions do not change the supremum.

2. *(*Functoriality under quantale homomorphisms*) If $\phi : V \to W$ is a (commutative) quantale homomorphism (preserving $\oplus, \otimes, e$), and if $\mu_W := \phi \circ \mu : X \to W$, then for every $H \subseteq X \times X$,*

$$\phi\big(w_V(H)\big) \;=\; w_W(H),$$

   i.e. computing weakness commutes with change of enrichment base. Equivalently, the assignment $H \mapsto w(H)$ is natural with respect to strict change-of-base maps ϕ that preserve the operations used in the definition of w (joins and tensor), so that the same relational evidence is merely re-expressed *in W rather than altered.*

Remark 112. *This proposition is the small categorical spine hidden inside an apparently elementary functional. Item (1) says that w behaves like a homomorphism from unions of relations to joins in V: the weakness of "either H_1 or H_2 or $\cdots$" is the join of their weaknesses. This is exactly what one wants if weakness is to be compatible with compositional constructions that build complex indistinction relations by taking unions (for example, when forming closures or aggregating evidence across subcontexts). Stated in slightly more algebraic terms, (1) says that $w : \mathcal{P}(X \times X) \to V$ is a* complete join-semilattice morphism *when $\mathcal{P}(X \times X)$ is ordered by inclusion and equipped with joins given by unions: it preserves arbitrary joins and therefore is automatically monotone (if $H \subseteq K$ then $w(H) \leq w(K)$), which matches the intuition that adding more indistinguishability pairs cannot decrease the overall weakness.*

Item (2) is a "change of base" principle: if one maps the quantale V into another quantale W in a structure-preserving way (via ϕ), then one can compute weakness either before or after applying ϕ and get the same answer. This is the order-enriched analogue of functoriality, and it is what allows later sections to shift between different evidence domains (e.g. probability-like versus possibilistic-like) while keeping the formal meaning of weakness stable (cf. the quantale-oriented development in [3]). Concretely, the point is that $w(H)$ is built from μ using only two constructors: taking tensor-products $\mu(u) \otimes \mu(v)$ and then taking their join; a quantale homomorphism preserves exactly these constructors, so it cannot "see" any difference between "compute-then-map" and "map-then-compute."

Proof. (1) Expand definitions and use that $\oplus$ is the join in V:

$$w\left(\bigcup_j H_j\right) = \bigoplus_{(u,v)\in\cup_j H_j} \mu(u)\otimes\mu(v) = \bigoplus_j \bigoplus_{(u,v)\in H_j} \mu(u)\otimes\mu(v) = \bigoplus_j w(H_j).$$

Also, the empty join is $\perp_V$, so $w(\emptyset) = \perp_V$. A slightly more explicit way to read the middle equality is that taking the join over the set $\bigcup_j H_j$ is the same as taking the join over the disjoint union of indexing data $\{(j,(u,v)) \mid (u,v)\in H_j\}$ and then observing that repetitions do not matter for suprema: if the same pair (u,v) is indexed multiple times, its contribution is duplicated but the join remains unchanged.

(2) Using preservation of joins and tensor by ϕ:

$$\phi\big(w_V(H)\big) = \phi\left(\bigoplus_{(u,v)\in H} \mu(u)\otimes\mu(v)\right) = \bigoplus_{(u,v)\in H} \phi(\mu(u)\otimes\mu(v)) = \bigoplus_{(u,v)\in H} \phi(\mu(u))\otimes\phi(\mu(v)) = w_W(H).$$

Here the second equality is exactly the "arbitrary-join preservation" part of being a quantale homomorphism, while the third equality is multiplicativity with respect to $\otimes$; together they ensure that ϕ commutes with the syntactic formation of w from μ. In particular, no special finiteness assumptions on H are needed: the argument applies uniformly to infinite joins because quantales are, by definition, complete join-semilattices with $\otimes$ distributing over arbitrary joins, and ϕ is assumed to respect this structure. ☐

Remark 113. Proof sketch. *For (1), the proof strategy is to "flatten" the indexing: taking the join over a union of sets is the same as taking a join over each set and then joining those results. This is the lattice-theoretic content of complete additivity with respect to unions. For (2), the strategy is to push ϕ through the defining expression for w using the fact that a quantale homomorphism preserves the operations used to build that expression. One can also view both statements as "no-surprises" coherence conditions: w is assembled from universal constructions (joins) and monoidal structure (tensor), so it must respect the canonical ways of rearranging those constructions.*

The crucial steps are exactly where the quantale axioms enter: preservation of arbitrary joins (to move ϕ past $\bigoplus$) and preservation of tensor (to rewrite $\phi(\mu(u)\otimes\mu(v))$ as $\phi(\mu(u))\otimes\phi(\mu(v))$). Visually, one can think of (2) as asserting that "measuring weakness" is invariant under a consistent change of measuring instrument: applying a monotone, structure-respecting regraduation of the evidence scale does not alter the computed weakness, aside from expressing it in the new units. For example, if V encodes evidence in one scale (say, additive costs) and W encodes it in another (say, a normalized or clipped cost domain), then a homomorphism ϕ that preserves the relevant joins and combinations ensures that the computed weakness is transported faithfully; what changes is only the codomain in which the resulting bound is reported.

5 Toy model: finite universe, p-bit quantale, and morphic resonance demo

Outline

- Fix a small finite universe X and two contexts C_1, C_2.
- Represent each context's "indistinction judgments" by a V-valued relation $R_i : X \times X \to V$.

- Compute weakness from R_i (as an "importance-weighted failure to distinguish").
- Represent patterns as finite constraint-sets on pairs in X, and define pattern support.
- Demonstrate "emergence" via quantale-compositional closure (transitive-like propagation of indistinction).
- Define morphic resonance and anti-resonance as cross-context coupling updates on relations.
- Add a tiny resonance score demo using the paraconsistent logic $\to$ complex mapping and interference.

Summary and Hyperseed concepts covered

This section instantiates the formal core of Section 3 in the smallest nontrivial setting. It provides an explicit computational picture of: (i) what it means to (fail to) make distinctions in a context, (ii) how simplicity/weakness aggregates, (iii) how a "pattern" can be a constraint that is supported to varying degrees, (iv) how pattern structure can *emerge* from composition/closure, and (v) how resonance can transmit pattern support across contexts, producing morphic resonance (and how anti-resonance can transmit opposition, producing habit reversal).

Hyperseed concepts covered.

- Distinction; equivalence relation; contexts/aspects.
- Weakness/simplicity; compositional simplicity; pattern; emergence.
- Resonance/dissonance; morphic resonance/anti-resonance; habits (as iterated updates).

Remark 114. *The point of this section is not to claim a faithful model of cognition or physics, but to give a* fully explicit *microcosm in which the core algebraic moves can be seen with one glance and checked with a calculator. In the language of Hyperseed's initial presentation [1], we are building a tiny "reality-system" where observer-relativity (Hyperseed-Concept 86), graded distinction (Hyperseed-Concept 98), weakness (Hyperseed-Concept 202), pattern (Hyperseed-Concept 130), and emergence (Hyperseed-Concept* **??***) appear as simple operations on a finite table.*

In particular, the "finite universe" assumption means that every object of interest can be concretely enumerated. Thus, once X is fixed, each context C_i can be treated as providing a complete *judgment table* $\big(R_i(x, y)\big)_{x,y\in X}$, i.e. a matrix whose entries live in a chosen truth-value/weight structure V. One should read $R_i(x, y)$ as: "*in context C_i, to what degree are x and y not distinguished?*" Depending on the choice of V, this degree may be probabilistic, fuzzy, possibilistic, or (as in this section) paraconsistent in the sense of allowing simultaneous support for and against indistinction.

The role of the p-bit quantale is to make three operations simultaneously available and computationally transparent: (i) an order $\leq$ expressing when one indistinction-value is *at most as strong* as another, (ii) a join (typically written $\vee$) expressing aggregation of evidence ("taking the better supported indistinction"), and (iii) a monoidal product (often written $\otimes$) that will be used to *compose* relations. Concretely, composition will take the familiar relational form

$$(R \circ S)(x, z) \;=\; \bigvee_{y\in X} \big(R(x, y) \otimes S(y, z)\big),$$

so that indistinction can propagate along intermediate y's; this is the precise sense in which a "transitive-like" closure can be computed and inspected entry-by-entry.

The weakness calculation is included here to make the informal phrase "failure to distinguish" operational. Because X is finite, weakness can be written as an explicit sum (or other finite aggregation) over pairs (x, y), with optional weights expressing that some confusions matter more than others in a given task or setting. In this way, weakness becomes a *scalar summary* extracted from a full table R_i, while still being sensitive to where (which pairs) the indistinction is concentrated. This also makes it clear how one can speak of "simplicity" as a property of a context: a context is "simple" (not because its world is objectively simple, but) because its distinction-table is coarse or permissive in a way that reduces discriminative burden.

Patterns enter in the most concrete possible way: as finite collections of constraints on pairs in X that a context may or may not support. A pattern can be read as a small "theory" about which objects should be treated as indistinct (or, depending on the constraint language chosen later in the section, which should be treated as distinct). Because R_i is graded, pattern satisfaction is graded as well: a pattern is not merely true/false in a context, but has a *support value* that can be computed by combining the relevant table entries with the operations of V. This gives an explicit bridge between the algebra (quantale-valued relations) and the intuitive notion of a recurring regularity (a pattern supported by the context's own judgments).

The point of demonstrating emergence via closure is that even when the pattern constraints are initially "local" (i.e. mention only a few pairs), compositional propagation can generate additional, *derived* indistinction facts. In the finite setting, this is visible as new nontrivial entries appearing (or strengthening) in the closure of R_i under $\circ$ (and, if taken, reflexive and/or symmetric completion), thereby turning an initially sparse set of judgments into a richer structure. This is the toy-model analogue of an "emergent pattern": not stipulated as a primitive constraint, but produced by the algebraic dynamics of the representational apparatus.

Finally, morphic resonance and anti-resonance are introduced here as explicit *cross-context update rules*. The guiding idea is that a context C_1 can influence a context C_2 by pushing R_2 toward (resonance) or away from (anti-resonance) the indistinction-structure present in R_1. Because everything is finite, such coupling can be written as an explicit formula (e.g. an update of the form $R_2 \leftarrow f(R_2, R_1)$ applied entrywise or via composition), iterated a few times, and numerically inspected. Interpreting repeated updates as "habit" formation is then not a metaphor but a literal iteration of a map on the space of V-valued relation tables.

The final bullet in the outline connects this to the paraconsistent logic aspect of p-bit by showing how a resonance (or interference) score can be computed even when evidence is inconsistent. Mapping the paraconsistent implication $\rightarrow$ into complex numbers is used in the demo as a compact way to encode phase-like reinforcement versus cancellation: agreement can add constructively, while certain mismatches can subtract, producing an explicit toy example of how "dissonance" can be more than mere absence of support. The purpose is not physical analogy per se, but a minimal, checkable instance where opposition is not collapsed into zero and can therefore drive anti-resonant change rather than being ignored.

5.1 Finite universe and two contexts

Let $X := \{a, b, c\}$. We consider two contexts C_1 and C_2 (think: two agents, or two spatial locations, or two subsystems) that each maintain a p-bit-valued relation

$$R_i : X \times X \to V, \qquad i \in \{1, 2\},$$

where $V = [0,1]^2$ is the p-bit quantale from Section 3.4. Concretely, each R_i can be pictured as a 3×3 table (a weighted adjacency matrix) whose entries are pairs in $[0,1]^2$. Keeping X finite makes it possible to compute closure, composition, and cross-context comparison explicitly and transparently: every "global" quantity is obtained by finitely many $\oplus$-aggregations of finitely many $\otimes$-products.

Remark 115 (Notation unpacking)**.** *Here $X \times X$ is the set of ordered pairs of entities, so $R_i(x,y) \in V$ is a* graded *statement about the pair (x,y) as judged in context C_i. The symbol $V = [0,1]^2$ means each value has two coordinates, typically written $R_i(x,y) = (R_i^+(x,y), R_i^-(x,y))$. The order on V is the componentwise order (so larger means "more evidence" in that channel), and the toy computations later implicitly use the standard* p-bit *quantale operations: $\oplus$ as componentwise maximum and $\otimes$ as componentwise multiplication, with unit $e = (1,1)$. This is the minimal paraconsistent evidence calculus needed for the rest of the examples. In particular, the choice of $\oplus = \max$ means that when multiple independent "routes" (or justifications) support a claim, the aggregated evidence keeps the strongest available support in each coordinate rather than averaging it away, while $\otimes = \cdot$ models attenuation along multi-step chains (products of confidences). The componentwise order makes the induced lattice structure transparent: $u \leq v$ iff $u^+ \leq v^+$ and $u^- \leq v^-$, so a value can increase in one channel without forcing any change in the other.*

Remark 116 (Two contexts as two ways of cutting the world)**.** *A context is, philosophically, a stance: a way an observer (or subsystem) decides what differences matter. Mathematically we encode that stance by a relation-valued table. This operationalizes observer-relativity: the same pair (x,y) can be judged "nearly identical" in C_1 and "clearly distinct" in C_2 without contradiction at the meta-level. This is one of the core motivations for paraconsistent semantics in the Hyperseed program (Hyperseed-Concept 86, 98). One can also view C_i as fixing a measurement protocol: which features are compared, what counts as noise, and which correlations are trusted. On this reading, changing contexts corresponds to changing the observational interface rather than changing the underlying set X.*

Remark 117 (Interpretation of $R_i(x,y)$)**.** *We interpret $R_i(x,y) = (R_i^+(x,y), R_i^-(x,y))$ as two-channel evidence about the proposition "x and y should be treated as the same (undistinguished) in context C_i":*

- *$R_i^+(x,y)$ is positive evidence supporting indistinction ($x \sim y$),*
- *$R_i^-(x,y)$ is negative evidence opposing indistinction (supporting distinction).*

Paraconsistency means both channels may be simultaneously high. It is useful to emphasize that this is not merely "probability of sameness" plus "probability of difference"; the two coordinates are not required to sum to 1 (and generally should not), because they represent potentially independent evidential sources. Thus $(1,1)$ represents maximal support and *maximal opposition, i.e. a fully conflicted or overdetermined state, whereas $(0,0)$ represents lack of information (neither support nor opposition). This separation is what allows later constructions to treat agreement and conflict as distinct, simultaneously trackable resources.*

Remark 118 (A simple concrete reading)**.** *If $R_i(x,y) = (0.9, 0.1)$ then context C_i has strong support for collapsing x and y and little opposition. If $R_i(x,y) = (0.9, 0.8)$ then C_i is in a conflicted state: it has strong reasons to identify and strong reasons to separate. And if $R_i(x,y) = (0.1, 0.8)$ then C_i is largely opposed to identifying x and y. This two-axis geometry is precisely what later allows "resonance" and "dissonance" to be treated as interference-like phenomena (Hyperseed-Concept*

159, 97). Geometrically, one can think of $R_i(x, y)$ as a point in the unit square, with qualitatively different regions: a "consensus-identify" region near $(1, 0)$, a "consensus-separate" region near $(0, 1)$, a "conflict" region near $(1, 1)$, and an "uninformative" region near $(0, 0)$. Later, when we compare contexts or compose relations along paths, these regions behave differently under $\oplus$ and $\otimes$, which is exactly what makes the toy model nontrivial even though the underlying set X is tiny.

For convenience we assume symmetry $R_i(x, y) = R_i(y, x)$ and set diagonal entries to the quantale unit $e = (1, 1)$ so that diagonal paths do not attenuate relation-composition (this is an algebraic convenience for the closure demo below). Equivalently, each R_i may be regarded as an undirected V-weighted graph with a designated self-loop of weight e at every node. This convention is especially convenient when we later form powers or closures of R_i: it ensures that length-k paths can always be extended to length $k + 1$ without changing their value by inserting a self-loop, so comparisons across path lengths are not artifacts of path padding.

Remark 119 (Why symmetry and why $R_i(x, x) = e$?)**.** *Symmetry is not logically necessary, but it matches the intended reading "x and y are (un)distinguished" as an (approximate) equivalence-like judgment (Hyperseed-Concept* **??***). Setting $R_i(x, x) = e = (1, 1)$ is likewise a modeling decision: it ensures that when we later compose relations, a path that "stays at x" contributes neutrally rather than shrinking values. In graph terms, we are giving each node a self-loop of maximal strength so that closure emphasizes multi-step connections between* distinct *nodes. Technically, this is the enriched-graph analogue of putting* 1 *on the diagonal of an adjacency matrix so that "no move" is always an available step. With $\otimes$ as multiplication, a diagonal value strictly below* 1 *would systematically damp all multi-step compositions, making it harder to interpret whether a decrease came from genuinely weak links between distinct nodes or merely from repeated diagonal attenuation. Note also that symmetry and reflexive diagonals together do* not *force transitivity: the later closure computation is precisely the mechanism that constructs the least transitive (or path-complete) strengthening consistent with the chosen $\oplus/\otimes$ aggregation.*

5.2 From relations to weakness: "failed distinctions"

Section 3.7 defined weakness for a crisp set $H \subseteq X \times X$ of undistinguished pairs. For the toy model it is useful to use the standard V-enriched generalization. In particular, passing from crisp H to a V-valued relation R lets us represent not only *whether* two items are treated as indistinct, but also *to what degree* that indistinction holds in each channel of the quantale, and then aggregate these degrees in a quantale-native way.

Remark 120. *Conceptually, weakness is meant to quantify "how much important structure is being collapsed": how costly it is, relative to what matters, to treat too many things as indistinct. In the Hyperseed literature this idea is developed as a quantale-native aggregation principle (Hyperseed-Concept 143; see also [3, 2]). Here we simply instantiate that principle on a finite set. One can view the weakness functional as a deliberately coarse proxy for "information lost under a compression": it does not attempt to reconstruct which distinctions were lost, but returns a single severity score that can be compared across candidate relations R.*

Definition 25 (Weakness of a V-valued relation)**.** *Fix an importance valuation $\mu : X \to V$. For a V-valued relation $R : X \times X \to V$, define*

$$w(R) \;:=\; \bigoplus_{(u,v) \in X \times X} \mu(u) \otimes R(u, v) \otimes \mu(v) \;\; \in V.$$

Remark 121. *The type of this definition is worth making explicit: since $\mu(u), R(u, v), \mu(v) \in V$ and $\otimes$ is the quantale product, each term $\mu(u) \otimes R(u, v) \otimes \mu(v)$ is again an element of V, and the (possibly large) family of such terms is then aggregated by the quantale join $\bigoplus$. Because X is finite in the toy model, the join is a finite join; in an infinite setting one would typically require the appropriate completeness so that the join exists.*

Remark 122 (Intuition)**.** *The expression for $w(R)$ is the same idea as the crisp case, but with two refinements. First, the relation $R(u, v)$ is graded, so "being undistinguished" is a matter of degree. Second, the valuation μ weights entities by importance, so collapsing a with something may count more than collapsing c with something. The quantale join $\bigoplus$ then aggregates all these weighted collapses into a single global score. In this toy quantale, the aggregation is a bottleneck (a max), so weakness is driven by the most severe important collapse; other quantales would yield sum-like or softmin-like notions of weakness. Equivalently, one can think of $w(R)$ as a "worst-case after weighting" criterion: among all pairs (u, v), it selects the pair whose weighted indistinction is most damaging, with the notion of "most damaging" determined by the join on V.*

Remark 123. *In many relational settings one may want to exclude diagonal pairs (u, u) from the aggregation, since an object being "undistinguished from itself" is not a failure of distinction. Here the definition sums over all of $X \times X$ for notational uniformity; in the worked toy instances one typically takes $R(u, u)$ to be the least element $(0, 0)$ (or otherwise fixed by convention) so that diagonal terms do not affect $w(R)$. If instead one sets $R(u, u) = e$, then the diagonal contributes the baseline importance $\mu(u) \otimes e \otimes \mu(u)$ and one should interpret the resulting weakness as including self-collapse as a fixed offset.*

Remark 124 (A tiny example)**.** *If $\mu(x) = (1, 1)$ for all x and R has only one substantial off-diagonal entry, say $R(a, b) = (0.9, 0.2)$ and all other off-diagonal entries are $(0, 0)$, then*

$$w(R) = (0.9, 0.2)$$

(up to symmetry/duplication conventions in the join). So in the bottleneck semantics, the global weakness simply mirrors the worst undistinguished edge. When multiple edges are present, $w(R)^+$ becomes the largest product-weighted positive indistinction, as noted below. In this example, if one also had $R(b, a) = (0.9, 0.2)$, the join would still return the same value, illustrating that duplicating the same worst edge does not change a max-type aggregator; by contrast, in an additive quantale, doubling could change the score and symmetry conventions would matter more.

Remark 125. *If $H \subseteq X \times X$ is crisp, embed it as a V-valued relation $\mathbf{1}_H$ via $\mathbf{1}_H(u, v) = e$ if $(u, v) \in H$ and $\mathbf{1}_H(u, v) = (0, 0)$ otherwise. Then $w(\mathbf{1}_H) = \bigoplus_{(u,v)\in H} \mu(u) \otimes \mu(v)$, recovering the earlier definition. This embedding is the standard "characteristic relation" construction: elements of H are assigned the multiplicative unit e so that the only contribution of such a pair is through the endpoint weights $\mu(u)$ and $\mu(v)$, while pairs not in H are assigned the least element so that they contribute nothing under $\otimes$ and $\bigoplus$.*

Definition 26 (A concrete importance valuation)**.** *In the worked example below we use*

$$\mu(a) = (1, 1), \qquad \mu(b) = (0.8, 1), \qquad \mu(c) = (0.6, 1).$$

Thus the positive channel weights importance (how costly it is to collapse distinctions involving an item), while the negative channel is left unweighted for simplicity. Note that these values impose an explicit ranking of "stakes": collapsing distinctions involving a is treated as most consequential, then b, then c, at least in the positive coordinate that drives the bottleneck score.

Remark 126 (Why weight only the positive channel here?)**.** *This is purely for pedagogical transparency. In many applications one would also want to weight negative evidence: e.g. "distinguishing a from b is more important than distinguishing c from b." The present choice keeps the numeric algebra minimal: the negative coordinate of $w(R)$ will be dominated by whatever negative evidence is largest, independent of μ. Nothing essential hinges on this choice; it merely fixes one explicit convention for the toy. Concretely, with $\mu(\cdot)^- = 1$ in this section, the negative coordinate behaves like an unweighted bottleneck over $R(u,v)^-$ (up to the way $\otimes$ combines negatives in the chosen quantale), so that the only place where "importance" visibly enters is the positive-coordinate term $\mu(u)^+\mu(v)^+R(u,v)^+$.*

Remark 127 (Reading $w(R)$ in this toy)**.** *With the quantale operations from Section 3.4, the first coordinate of $w(R)$ is*

$$w(R)^+ = \max_{u,v} \ \mu(u)^+\mu(v)^+R(u,v)^+,$$

i.e. a bottleneck-style "largest important collapsed link" score. This matches a minimax/bottleneck notion of simplicity: the strongest (most important) collapsed distinction dominates. Other aggregators (e.g. additive) are possible, but we keep the quantale-native choice here. For completeness, the second coordinate $w(R)^-$ is computed analogously using the $\otimes$-combination rule for the negative coordinate and the join $\bigoplus$ on that coordinate (which, for p-bit*, is again max-like); in the present toy convention $\mu(\cdot)^- = 1$, so the negative contribution is driven directly by the largest negative component appearing among the $\mu(u) \otimes R(u,v) \otimes \mu(v)$ terms.*

Remark 128. *The form $\mu(u)^+\mu(v)^+R(u,v)^+$ also makes it clear why $w(R)^+$ can be read as a "failed distinction" score rather than merely a raw strength score: even if $R(u,v)^+$ is moderately large, the weakness only becomes large when the endpoints are themselves important in the positive channel. Thus, collapsing two low-importance entities can remain cheap even if the relation strongly identifies them, while a weaker identification between high-importance entities can still dominate the global bottleneck.*

Remark 129 (Connection to "simplicity as failed distinction")**.** *It is easy to hear "simplicity" as a virtue and "failure to distinguish" as a vice. Hyperseed treats them as two faces of a single structural phenomenon: compression always forgets distinctions, and the question is* which *ones and at* what cost*. The quantale-functional $w(R)$ is precisely a knob for that trade: it quantifies the degree and importance of collapsed distinctions in a context (Hyperseed-Concept 169, 202). In particular, when used alongside a competing objective (e.g. a fit-to-data or coherence score), $w(R)$ provides the "simplicity pressure" by penalizing relations that identify too much of what the valuation μ declares significant.*

5.3 Patterns as finite constraints and pattern support

In this toy setting, a "pattern" will be a finite set of pair-constraints that should be jointly supported. This choice builds in two simplifying assumptions that are useful later: (i) patterns are *finitely checkable* objects (one can compute their support by a finite aggregation), and (ii) the only structure inside a pattern is *which pairs are demanded to cohere.* Nothing in this subsection requires that the constraints be consistent in the classical sense; instead, consistency becomes an emergent, graded property of the support values.

Definition 27 (Pattern as a constraint-set on pairs)**.** *A* pattern *is a finite set*

$$P \ \subseteq \ X \times X$$

(typically symmetric and excluding the diagonal) interpreted as a conjunction of constraints "for all $(u, v) \in P$, treat u and v as (approximately) undistinguished."

Remark 130 (Conventions: symmetry, diagonal, and multiplicity)**.** *Formally, nothing breaks if P includes ordered pairs or diagonal pairs; these are conventions chosen to keep the toy notion aligned with the intended reading. Symmetry corresponds to treating "u matches v" as interchangeable with "v matches u," and excluding (u, u) avoids constraints that are tautological in most similarity-like contexts. Because P is a* set*, not a multiset, repeating the same constraint does not artificially increase its influence; if one wanted repeated evidence, that would be modeled in R (as stronger support) or by a different pattern formalism.*

Remark 131 (Intuition and relation to the general notion of pattern)**.** *In a large theory, a "pattern" is often something like a model, a regularity, or a compression ([5, 16]). Here we take the smallest possible fragment of that idea: a pattern is just a finite list of pairwise identifications that are supposed to hold together. This is deliberately austere: it makes patternhood into a constraint satisfaction problem, and the only subtlety is that satisfaction is graded and paraconsistent. One can regard P as specifying a tiny "hypothesis" about which distinctions the system is willing to ignore (at least locally), and the role of R is to provide context-dependent evidence for and against each such local identification.*

Remark 132 (Simple examples)**.** *With $X = \{a, b, c\}$, the set $P = \{(a, b)\}$ is the pattern "a is identified with b." The set $P = \{(a, b), (b, c)\}$ is the pattern "$a \sim b$ and $b \sim c$." If one additionally includes (a, c) then P is a triangle/cluster constraint. In later sections, such constraint-sets will be replaced by richer pattern objects, but the logical spine is the same: a pattern is something that asks multiple local judgments to cohere (Hyperseed-Concept 130). It is worth noting that, at this toy level, $P = \{(a, b), (b, c)\}$ does* not *itself enforce transitivity; it merely requests two local identifications. Whether "a and c should also cohere" is a further constraint, represented by adding (a, c) explicitly (or by a separate closure principle introduced later, if desired).*

Definition 28 (Pattern support)**.** *Given a context relation R and a pattern P, define the (conjunctive) pattern support*

$$S_R(P) \;:=\; \bigotimes_{(u,v)\in P} R(u, v) \;\in V.$$

In particular,

$$S_R(P)^+ = \prod_{(u,v)\in P} R(u, v)^+, \qquad S_R(P)^- = \prod_{(u,v)\in P} R(u, v)^-.$$

Remark 133 (Empty pattern and the unit of conjunction)**.** *Since $\bigotimes$ is a multiplicative "big conjunction," the empty pattern $P = \varnothing$ evaluates to the multiplicative unit:*

$$S_R(\varnothing) = 1_V.$$

This matches the logical convention that an empty conjunction is trivially satisfied. In the two-channel setting, this should be read as "the empty set of demands contributes no constraint at all," rather than as a substantive claim about the world; nontrivial content only appears once at least one pair-constraint is included.

Remark 134 (Notation unpacking)**.** *The symbol $\bigotimes$ is the quantale analogue of a big conjunction: we multiply the evidence-values for each required pair. Because our $\otimes$ is componentwise multiplication, $S_R(P)^+$ is literally the product of the positive supports for each constraint, and similarly for*

$S_R(P)^-$. *Thus the more constraints you ask for, the harder it becomes to keep the support high—an algebraic reflection of the simple thought that conjunction is stricter than a single predicate. Equivalently, each additional constraint* (u, v) *acts as a multiplicative "gate" that can only decrease (or, in the best case, preserve) the overall degree whenever* $R(u, v)^{\pm} \leq 1$ *in the ambient scale.*

Remark 135 (Monotonicity in the pattern argument)**.** *Because* $\otimes$ *is monotone in each argument, enlarging a pattern can only make it harder to satisfy: if* $P \subseteq Q$*, then (componentwise in the* V*-order)*

$$S_R(Q) \;\leq\; S_R(P).$$

Concretely, $S_R(Q)^+$ *is* $S_R(P)^+$ *multiplied by the additional positive factors required by* $Q \setminus P$*, and likewise for the negative channel. This is the toy analogue of the general principle that adding conjuncts strengthens a theory.*

Remark 136 (A concrete computation)**.** *For intuition, suppose* $P = \{(a, b), (b, c)\}$ *and (in some context)* $R(a, b) = (0.9, 0.2)$ *and* $R(b, c) = (0.6, 0.8)$*. Then*

$$S_R(P) = (0.9 \cdot 0.6,\ 0.2 \cdot 0.8) = (0.54,\ 0.16).$$

In this example, the pattern is moderately supported (0.54*) and only weakly opposed (*0.16*), even though one constituent pair has high opposition in isolation; the multiplicative aggregation makes the overall opposition sensitive to* joint *opposition across all constraints rather than to a single contested link. This "jointness" is exactly the behavior expected of a conjunction-like operator.*

Remark 137 (Paraconsistent reading of $S_R(P)$)**.** *When* $S_R(P) = (s^+, s^-)$*, the pattern is:*

- *supported to degree* s^+*,*
- *opposed to degree* s^-*,*
- *and may be simultaneously both (if both are high).*

This is exactly why the two-channel semantics is useful: the theory can represent "the pattern is strongly present and strongly resisted" without collapsing into triviality. That tension is one of the seeds from which later notions of dissonance, reversal, and habit dynamics grow (Hyperseed-Concept 159, 97). In particular, a high s^+ *indicates that each demanded identification is locally well-supported, while a high* s^- *indicates that each demanded identification is also locally well-opposed; these can coexist without forcing an all-or-nothing verdict.*

Remark 138 (Why conjunctive aggregation is the right toy default)**.** $S_R(P)$ *is a deliberately minimal "all-of" aggregator (a toy conjunction). More elaborate pattern measures used later in the full paper (pattern intensity, emergence criteria, etc.) can be layered on top of this. For Section 5, the goal is simply to have a compact, checkable notion of "this pattern is supported / opposed / contradictory." The multiplicative form is especially convenient because it makes the dependence on individual constraints explicit and local: every pair in* P *contributes a visible factor, so changes in* R *(or edits to* P*) have transparent effects on the aggregate.*

5.4 Emergence via compositional closure

A key Hyperseed intuition is that patterns can emerge from composition rather than being explicitly present. In the relation-quantale, this is exactly what composition and closure do: if $R(a, b)$ and $R(b, c)$ are strong, then $R(a, c)$ becomes strong in the closure even if it started weak. In other words, the "emergent" link $a \to c$ is not stipulated as a primitive datum; it is forced by the algebra of chaining. This is the graded analogue of how (crisp) transitive closure turns a set of edges into all reachable edges, except that here reachability is replaced by a quantale-valued notion of support for a connection.

Remark 139. *This is a precise algebraic analogue of the informal slogan "structure implies more structure": a system that treats a as (approximately) the same as b, and b as the same as c, is under pressure to treat a as the same as c. In classical logic that pressure is transitivity of equivalence. In a graded setting the same pressure becomes a closure operator built from $\oplus$ and $\otimes$. This is one minimal formal doorway into Hyperseed's notion of emergence (Hyperseed-Concept* **??**; *see also [5] for broader discussion). One can also read this as a disciplined way of converting* local *judgments (pairwise links) into* global *judgments (consistency across chains) while remaining inside the same algebraic universe V. When V is the* p-bit *quantale used in this toy model, the same mechanism propagates not only "positive support" but also any built-in notion of "opposition" or "incompatibility" that $\otimes$ and $\oplus$ encode.*

Definition 29 (Quantale composition of V-valued relations)**.** *For V-valued relations $R, S : X \times X \to V$, define*

$$(R \circ S)(x, z) \;:=\; \bigoplus_{y \in X} R(x, y) \otimes S(y, z).$$

Remark 140 (What this composition means)**.** *The variable y plays the role of an intermediate "bridge" entity. For each bridge y, the quantity $R(x, y) \otimes S(y, z)$ measures how strongly x connects to z by going through y. Then the join $\bigoplus_{y \in X}$ takes the best such bridge in the quantale order. Because $\oplus$ is a max in this toy, composition chooses the strongest multi-step path. This is the same algebra that underlies weighted path problems in idempotent semirings, but here it is interpreted as propagation of indistinction judgments. Equivalently, if one thinks of R and S as $|X| \times |X|$ matrices with entries in V, then $\circ$ is the usual "matrix product" but with $(+, \times)$ replaced by $(\oplus, \otimes)$; the bridge index y is exactly the summed-over index. This perspective is often useful for concrete computations in the finite-universe toy setting, since repeated composition becomes iterated (quantale-)matrix multiplication.*

Definition 30 (Transitive-like closure)**.** *Define $R^{(1)} := R$ and $R^{(n+1)} := R^{(n)} \circ R$. Define the closure*

$$\mathrm{Cl}(R) \;:=\; \bigoplus_{n \geq 1} R^{(n)}.$$

Remark 141 (Intuition: closure as "all finite chains")**.** *The power $R^{(n)}(x, z)$ measures evidence for connecting x to z by a chain of n edges. Taking $\bigoplus_{n \geq 1}$ then means: allow* any *finite chain length, and keep whichever chain provides the strongest support (and strongest opposition) for the resulting link. In a finite universe, $\mathrm{Cl}(R)$ can often be computed by iterating composition until no entry improves. Note that the definition starts at $n \geq 1$, so it is a "transitive-like" closure rather than a reflexive-transitive closure: we do not automatically force a maximal self-link $R(x, x)$ via an $n = 0$ identity step. If one wanted the reflexive-transitive variant, one would typically adjoin the identity relation I (with $I(x, x) = \top$ and $I(x, z) = \bot$ for $x \neq z$ in the V-order) and take $\bigoplus_{n \geq 0} R^{(n)}$; the present choice isolates what is generated purely by chaining the given evidence in R.*

Remark 142 (Finite stabilization in the toy universe)**.** *Because X is finite and $\oplus$ is idempotent in this setting, the ascending sequence*

$$R^{(1)} \;\leq\; R^{(1)} \oplus R^{(2)} \;\leq\; \cdots \;\leq\; \bigoplus_{k=1}^{N} R^{(k)} \;\leq\; \cdots$$

often stabilizes after finitely many steps when computed entrywise: beyond some length, longer chains do not improve any value. Operationally, this is why one can implement $\mathrm{Cl}(R)$ *by repeated composition and pointwise* $\oplus$*-accumulation until a fixed point is reached, mirroring classical algorithms for transitive closure (e.g. Floyd–Warshall) but with the boolean semiring replaced by* $(V, \oplus, \otimes)$*. The theorem below identifies the mathematical reason that this fixed point is the "right" one: it is exactly the least self-composition-stable extension above* R*.*

Theorem 4 (Closure is transitive and is the least transitive extension above R)**.** *Let* $R : X \times X \to V$*. Then:*

1. *(*Extensive*)* $R \leq \mathrm{Cl}(R)$*.*

2. *(*Transitive*)* $\mathrm{Cl}(R) \circ \mathrm{Cl}(R) \leq \mathrm{Cl}(R)$*.*

3. *(*Least*) If* T *is any relation with* $R \leq T$ *and* $T \circ T \leq T$*, then* $\mathrm{Cl}(R) \leq T$*.*

Remark 143 (What the theorem is saying, in plain language)**.** *The theorem asserts that* $\mathrm{Cl}(R)$ *behaves exactly as a closure should. First, it does not forget what you started with: every direct judgment in R remains present (possibly strengthened). Second, once you have closed, composing closed relations does not generate anything genuinely new: the closure is stable under composition, which is the graded analogue of transitivity. Third, it is the* smallest *such stable extension: any other relation* T *that contains* R *and is closed under self-composition must also contain* $\mathrm{Cl}(R)$*. In particular, statement (2) says that after closure, every composite chain-of-chains can be "flattened" back into a single finite chain already accounted for in* $\mathrm{Cl}(R)$*, so no additional emergent links remain hidden behind further iteration.*

Remark 144 (Why it matters here)**.** *This result is a sanity check that the toy "emergence" mechanism is not ad hoc. If we claim that emergent pattern structure arises from iterating compositional implications, then* $\mathrm{Cl}(R)$ *is the canonical object that embodies "everything implied by finite compositional chaining." In later categorical language,* Cl *is a standard closure operator induced by enrichment. So this theorem connects the concrete numeric demo back to the abstract compositionality requirement in Section 3. It also clarifies the sense in which emergence here is* conservative*:* $\mathrm{Cl}(R)$ *can add new strong links, but it does so only when they are justified by existing links and the fixed rules* $(\oplus, \otimes)$*, i.e. by the chosen notion of compositional propagation encoded in the quantale.*

Proof. (1) The join defining $\mathrm{Cl}(R)$ includes $R^{(1)} = R$, hence $R \leq \mathrm{Cl}(R)$.

(2) Using distributivity of $\circ$ over joins (true in the relation quantale) and associativity of $\circ$:

$$\mathrm{Cl}(R) \circ \mathrm{Cl}(R) = \left(\bigoplus_{m \geq 1} R^{(m)} \right) \circ \left(\bigoplus_{n \geq 1} R^{(n)} \right) = \bigoplus_{m,n \geq 1} \left(R^{(m)} \circ R^{(n)} \right).$$

Here distributivity can be read entrywise: for each (x, z), the definition of $\circ$ and the fact that $\otimes$ distributes over $\oplus$ allow one to pull joins outside and regroup them into a single (possibly larger) join over pairs (m, n). But $R^{(m)} \circ R^{(n)} = R^{(m+n)}$ by associativity of composition, so

$$\mathrm{Cl}(R) \circ \mathrm{Cl}(R) = \bigoplus_{m,n \geq 1} R^{(m+n)} \leq \bigoplus_{k \geq 1} R^{(k)} = \mathrm{Cl}(R).$$

The final inequality holds because every index of the form $k = m + n$ with $m, n \geq 1$ satisfies $k \geq 2$, and the join over all $k \geq 1$ dominates the join over the subset $k \geq 2$ in the quantale order.

(3) If $R \le T$ and $T \circ T \le T$, then by induction $R^{(n)} \le T$ for all n. Taking joins yields $\mathrm{Cl}(R) = \bigoplus_{n \ge 1} R^{(n)} \le T$. Concretely, the induction step uses monotonicity of $\circ$: if $R^{(n)} \le T$, then

$$R^{(n+1)} \;=\; R^{(n)} \circ R \;\le\; T \circ T \;\le\; T,$$

where $R \le T$ is used in the middle inequality. □

Proof sketch. The strategy is to recognize $\mathrm{Cl}(R)$ as the join of all finite compositional powers of R. Extensiveness is immediate because R is one of those powers. Transitivity follows because composing an m-step chain with an n-step chain yields an $(m+n)$-step chain, and all such powers are already included in the join. Leastness follows because any transitive extension T that contains R must contain every power $R^{(n)}$ (by repeated composition), hence must contain their join. □

Remark 145 (Why the key steps work)**.** *The only real work is done by two structural facts about quantale-valued relations:*

- *composition $\circ$ is associative, so path concatenation is well-defined, and*
- *composition distributes over joins, so "take the best bridge" is compatible with "take the best path length."*

These are precisely the algebraic conditions that make composition a sensible model of propagation. Visually, one can imagine a weighted graph with two weights per edge; $\mathrm{Cl}(R)$ *is the result of allowing all paths and then assigning each pair (x, z) the best weight achievable by any path from x to z.*

Remark 146 (Hyperseed reading)**.** *The closure* $\mathrm{Cl}(R)$ *is an extremely literal toy formalization of "pattern emergence": new constraints become supported because they are* implied by compositions of already-supported constraints*.*

5.5 Miledorphic resonance and anti-resonance as cross-context coupling

We now model morphic resonance as the transport of pattern support between contexts.

Remark 147. *The phrase "morphic resonance" is historically associated with Sheldrake's speculative proposal [13]. Hyperseed uses the phrase in a more formal-and-operational spirit: it denotes a cross-context coupling that tends to make similar patterns more likely in different places, without requiring direct causal contact in the narrow sense. In this paper, the coupling is explicitly represented as an algebraic update rule on context relations (Hyperseed-Concept 115, 159).*

Definition 31 (Pointwise scaling of a relation)**.** *For $K \in V$ and a relation $R : X \times X \to V$, define $(K \otimes R) : X \times X \to V$ by*

$$(K \otimes R)(x, y) := K \otimes R(x, y).$$

Remark 148 (Intuition and example)**.** *Pointwise scaling means: take every pairwise judgment in R and attenuate (or amplify) it by the same coupling strength K. For instance, if $K = (0.9, 0.9)$ and $R(x, y) = (0.8, 0.2)$, then $(K \otimes R)(x, y) = (0.72, 0.18)$. In other words, we transmit the same "shape" of evidence but at reduced intensity.*

Definition 32 (Morphic resonance update operator)**.** *Fix a coupling strength $K \in V$. Define the resonance propagation from C_1 to C_2 by*

$$\mathsf{Res}_K(R_1 \to R_2) \;:=\; R_2 \;\oplus\; (K \otimes R_1),$$

where the join $\oplus$ and product $\otimes$ are taken pointwise on $X \times X$.

Remark 149 (Plain-language reading)**.** *The update says: "keep what C_2 already believes, but also import (a scaled version of) what C_1 believes, taking the better-supported evidence in each channel." Because $\oplus$ is a join, the update is conservative in the sense that it cannot decrease either the positive or negative evidence for any pair; it can only add support and/or add opposition. This is the algebraic core of paraconsistent transport: influence without forced consistency.*

Remark 150 (Why a join, rather than overwriting?)**.** *If one overwrote R_2 by $K \otimes R_1$, then resonance would erase local evidence and turn into domination. Using $\oplus$ instead makes resonance an inclusion-like process: new evidence accumulates. This matches the intended semantics of contexts as partially independent stances that can be coupled without being annihilated.*

Definition 33 (Anti-resonance update operator)**.** *Anti-resonance is modeled as preferential transport of opposition to the pattern. A minimal way to express this with the same quantale is to use a coupling element that acts mainly on the negative channel:*

$$K^- := (0, k^-) \in V.$$

Then define

$$\mathsf{ARes}_{k^-}(R_1 \to R_2) \ := \ R_2 \ \oplus \ (K^- \otimes R_1).$$

Concretely, $(K^- \otimes R_1)(x,y) = (0,\ k^- \, R_1^-(x,y))$, so only negative evidence is propagated.

Remark 151 (Anti-resonance as structured "no")**.** *The anti-resonance operator is not "the negation" of resonance; it is a directed transport of resistance. In terms of habit dynamics, it provides a minimal mechanism by which a context can inherit reasons* not *to instantiate a pattern that another context opposes. This aligns with the Hyperseed idea that habit reversal is not mere noise but can be driven by imported constraint pressure (Hyperseed-Concept 188, 114).*

Proposition 3 (Monotonicity of resonance and anti-resonance)**.** *For fixed K, the maps $R_2 \mapsto \mathsf{Res}_K(R_1 \to R_2)$ and $R_2 \mapsto \mathsf{ARes}_{k^-}(R_1 \to R_2)$ are monotone (order-preserving). They are also monotone in R_1 and in the coupling parameters.*

Remark 152 (What monotonicity means here)**.** *Monotonicity formalizes a simple but important ethical/epistemic constraint on the update rules: if R_2 already had* more *evidence (in either channel) than some R_2^*, then after resonance it should still have at least as much. In other words, adding evidence and then resonating should not perversely reduce what you had. This proposition says the update rules are "well-behaved" in precisely that minimal sense.*

Remark 153 (Connection to the rest of the document)**.** *This is the same monotonicity logic that underlies the sanity theorems in Section 4: weakness and pattern measures are intended to behave directionally under addition of structure. Here we ensure that the* dynamical *operators (resonance/anti-resonance) respect the same order-theoretic discipline, which becomes crucial later when iterating updates and taking fixed points.*

Proof. Each operator is built from pointwise $\oplus$ and $\otimes$, which are monotone in each argument with respect to the componentwise order on V. □

Proof sketch. The proof reduces the claim to a basic fact about the $\mathsf{p\text{-}bit}$ quantale: both $\oplus$ (join) and $\otimes$ (product) are order-preserving in each input. Since the update operators apply these operations pointwise on $X \times X$, the order-preservation lifts immediately from V to V-valued relations. □

Remark 154 (Why this matters conceptually)**.** *Order-preservation is the minimal algebraic substitute for the intuitive statement "resonance transports influence without making things worse by fiat." In paraconsistent settings, one often worries that allowing contradictory evidence will break basic reasoning. Monotonicity reassures us that the update dynamics at least respect the lattice structure that makes graded, nontrivial aggregation possible in the first place.*

Definition 34 (A simple habit-taking / habit-reversal statistic)**.** *To connect with the Hyperseed "tendency to take habits" definitions, we map a* p-bit *value* $v = (v^+, v^-)$ *to a single decision-biased probability via the truth-bias*

$$d(v) := v^+ - v^- \in [-1, 1], \qquad p_{\text{dec}}(v) \ := \ \frac{d(v) + 1}{2} \ \in [0, 1].$$

For a pattern P*, we define* $p_{\text{dec}}(P \mid R) := p_{\text{dec}}(S_R(P))$*.*

Remark 155 (Intuition and scope)**.** *This statistic is intentionally not presented as a universal "probability of truth." It is a* decision proxy*: a way of turning two-channel evidence into a single scalar that increases with net support and decreases with net opposition. One should read it as "if the system must act as if* P *holds, how disposed is it to do so?" This bridges the toy evidence algebra to the language of habit formation and reversal (Hyperseed-Concept 189, 188).*

Remark 156. *Morphic resonance tends to increase* $p_{\text{dec}}(P \mid R)$ *for patterns* P *supported in the source, while anti-resonance tends to decrease it by increasing opposition. This is the toy analogue of "tendency to take habits" versus "tendency to reverse habits".*

5.6 A concrete nontrivial instance reaching morphic resonance

We now specify concrete R_1 and R_2, compute weakness, show emergence via closure, and then show morphic resonance.

Example 1 (Initial relations in two contexts)**.** *Let* $X = \{a, b, c\}$ *and define symmetric relations* $R_1, R_2 : X \times X \to V$ *by setting the diagonal to* $e = (1, 1)$ *and*

$$R_1(a, b) = (0.90, 0.80), \quad R_1(b, c) = (0.85, 0.20), \quad R_1(a, c) = (0.10, 0.70),$$

$$R_2(a, b) = (0.20, 0.70), \quad R_2(b, c) = (0.30, 0.60), \quad R_2(a, c) = (0.10, 0.80).$$

Intuitively: C_1 *strongly supports* $a \sim b$ *(but with conflict), and supports* $b \sim c$*;* C_2 *is initially skeptical about all three pairwise identifications.*

Remark 157. *This is the smallest size where the closure phenomenon is nontrivial: with only two entities there is no room for a genuinely emergent third relation. With three entities, a two-edge chain can force an implied edge. So* $\{a, b, c\}$ *is the first stage where "emergence by composition" is visible.*

Example 2 (Weakness comparison)**.** *Use* μ *from Definition 26 and weakness from Definition 25. A short check shows:*

$$w(R_1) = (0.72, 0.80), \qquad w(R_2) = (0.16, 0.80).$$

Thus, in the bottleneck sense of Section 3.7, C_1 *is "weaker" (collapses a more important distinction) than* C_2*.*

Remark 158 (How to read the numbers)**.** *The positive weakness* 0.72 *for* C_1 *comes from combining (i) the strong positive indistinction* $R_1(a,b)^+ = 0.90$ *with (ii) the high importance weights* $\mu(a)^+ = 1$ *and* $\mu(b)^+ = 0.8$*. By contrast,* C_2 *never strongly supports an important collapse, so its positive weakness is small. The negative weakness being* 0.80 *in both contexts reflects that each contains at least one strongly opposed identification somewhere; since we used a bottleneck/max aggregator, a single strong opposition dominates.*

Example 3 (Emergence in C_1 via closure)**.** *Compute the closure* $\mathrm{Cl}(R_1) = R_1 \oplus (R_1 \circ R_1) \oplus \cdots$*. Because* X *has three elements, the first nontrivial emergent edge is* $a \to c$ *via* b*:*

$$(R_1 \circ R_1)(a,c) \;\geq\; R_1(a,b) \otimes R_1(b,c) = (0.90 \cdot 0.85,\; 0.80 \cdot 0.20) = (0.765,\; 0.16).$$

Joining with the direct edge $R_1(a,c) = (0.10, 0.70)$ *yields*

$$\mathrm{Cl}(R_1)(a,c) \;\geq\; R_1(a,c) \;\oplus\; \big(R_1(a,b) \otimes R_1(b,c)\big) = (0.765,\; 0.70).$$

So even though C_1 *initially had very low positive evidence for* $a \sim c$*, the compositional structure* $a \sim b$ *and* $b \sim c$ *causes* $a \sim c$ *to become strongly supported in the closure. This is a concrete toy instance of* emergence *as compositional implication.*

Remark 159 (Visual intuition)**.** *If one draws a triangle with vertices* a, b, c*, then* R_1 *begins with two thick edges (a–b and b–c) in the positive channel and a thin edge (a–c). Closure thickens the missing edge because the two-step walk* $a \to b \to c$ *provides an alternate route. In this toy quantale, the positive-channel strength of that route is the product of the two edge strengths.*

Example 4 (Morphic resonance transmission from C_1 to C_2)**.** *Let the coupling strength be*

$$K := (0.90, 0.90).$$

Apply the morphic resonance update (Definition 32):

$$R_2' := \mathsf{Res}_K(R_1 \to R_2) = R_2 \oplus (K \otimes R_1).$$

Compute, for example,

$$R_2'(a,b) = (0.20, 0.70) \;\oplus\; \big((0.90, 0.90) \otimes (0.90, 0.80)\big) = (0.20, 0.70) \;\oplus\; (0.81, 0.72) = (0.81, 0.72),$$

$$R_2'(b,c) = (0.30, 0.60) \;\oplus\; \big((0.90, 0.90) \otimes (0.85, 0.20)\big) = (0.30, 0.60) \;\oplus\; (0.765, 0.18) = (0.765, 0.60),$$

and

$$R_2'(a,c) = (0.10, 0.80) \;\oplus\; \big((0.90, 0.90) \otimes (0.10, 0.70)\big) = (0.10, 0.80) \;\oplus\; (0.09, 0.63) = (0.10, 0.80).$$

Thus C_2 *receives strong support for the* structure $a \sim b$ *and* $b \sim c$ *from* C_1 *even though the direct edge* $a \sim c$ *does not (yet) increase.*

Now close internally in C_2*:*

$$\mathrm{Cl}(R_2')(a,c) \;\geq\; R_2'(a,b) \otimes R_2'(b,c) = (0.81 \cdot 0.765,\; 0.72 \cdot 0.60) = (0.61965,\; 0.432).$$

Joining with the direct edge $(0.10, 0.80)$ *yields*

$$\mathrm{Cl}(R_2')(a,c) \;\geq\; (0.61965,\; 0.80).$$

So morphic resonance transmits the pattern skeleton *(the chain* $a \sim b$*,* $b \sim c$*), and then compositional closure produces the missing implied link* $a \sim c$*. This is a minimal toy model of "spatially separated habit taking":* C_2 *becomes disposed to instantiate the same cluster-pattern present in* C_1*.*

Remark 160 (What is being transmitted?)**.** *Notice that resonance did not directly increase* $R_2(a,c)^+$*, because the source itself had* $R_1(a,c)^+$ *small. What was transmitted was the* two-edge scaffold *that makes* $a \sim c$ *compositional. This is philosophically important: it models the transmission of a relational* form *rather than a single belief, which is closer to the intended "morphic" flavor (pattern-shape transfer) than naive copying of propositions.*

Example 5 (Pattern support and a habit-taking inequality)**.** *Let the pattern (constraint-set) be the chain*

$$P := \{(a,b),(b,c)\}.$$

Compute support using Definition 28. Intuitively, this pattern asks for a simultaneous "fit" of two links, $a \sim b$ *and* $b \sim c$*, so its support is a* p-bit*-valued conjunction of the two corresponding edge-values. In this toy* p-bit *quantale, the tensor* $\otimes$ *acts componentwise on* (t,f)*-pairs (so it behaves like a multiplicative "and" that propagates both positive and negative evidence along the chain).*

In C_1*:*

$$S_{R_1}(P) = R_1(a,b) \otimes R_1(b,c) = (0.765,\ 0.16).$$

In particular, the pattern-level positive support 0.765 *is the product of the two positive edge-scores in* C_1*, while the pattern-level opposition* 0.16 *is the product of the two edge-level opposition scores; hence a single weak link (small* t*) or a single strong objection (large* f*) can materially affect the combined pattern support.*

In C_2 *initially:*

$$S_{R_2}(P) = (0.20, 0.70) \otimes (0.30, 0.60) = (0.06,\ 0.42).$$

*Here the pre-update pattern has relatively low positive support (*0.06*) and comparatively high opposition (*0.42*), reflecting that both constituent constraints are, initially, not well-endorsed in* C_2*. It is sometimes helpful to look at the net inclination* $t - f$ *at the pattern level:*

$$(0.06 - 0.42) = -0.36,$$

which already indicates that the combined pattern is disfavored before any coupling.

After morphic resonance:

$$S_{R_2'}(P) = (0.81, 0.72) \otimes (0.765, 0.60) = (0.61965,\ 0.432).$$

So the resonance update has sharply increased the positive component of the pattern support (from 0.06 *to* 0.61965*) while leaving the negative component in the same general range (from* 0.42 *to* 0.432*). At the level of net inclination this changes the sign:*

$$(0.61965 - 0.432) = 0.18765,$$

so the same multi-constraint form moves from "overall opposed" to "overall favored" according to the simple difference $t - f$*.*

Using the decision probability from Definition 34,

$$p_{\mathrm{dec}}(P \mid R_2) = \frac{(0.06 - 0.42) + 1}{2} = 0.32, \qquad p_{\mathrm{dec}}(P \mid R_2') = \frac{(0.61965 - 0.432) + 1}{2} \approx 0.593825.$$

Recall that the affine rescaling $p_{\mathrm{dec}}(t,f) = \frac{(t-f)+1}{2}$ *maps the score* $t - f \in [-1,1]$ *to a number in* $[0,1]$*, so that equal support and opposition (*$t = f$*) corresponds to* $p_{\mathrm{dec}} = \frac{1}{2}$*, net support (*$t > f$*) yields* $p_{\mathrm{dec}} > \frac{1}{2}$*, and net opposition (*$t < f$*) yields* $p_{\mathrm{dec}} < \frac{1}{2}$*.*

Thus P becomes substantially more likely in C_2 after the resonance update. This is the toy analogue of Hyperseed's "tendency to take habits": the post-update probability is higher than the pre-update probability for a pattern supported elsewhere. In other words, a pattern that is already coherently instantiated in one context (C_1) can, under resonance, become easier to enact in a different context (C_2) even though C_2 initially opposed it at the multi-constraint level.

Remark 161. *In the language of graded habits, the inequality $p_{\mathrm{dec}}(P \mid R_2') > p_{\mathrm{dec}}(P \mid R_2)$ is the simplest possible signature of habit-taking: the system's disposition toward enacting a multi-constraint form increases after coupling. Note that, because the support is computed by chaining constraints with $\otimes$, the phenomenon is genuinely pattern-level rather than edge-local: one is not merely increasing confidence in a single relation, but increasing the likelihood that an entire small configuration is jointly satisfied. More sophisticated theories would track trajectories under repeated updates and study stability; this toy merely exhibits the direction of motion. One could also distinguish between (i) amplifying positive evidence for the pattern, (ii) reducing opposition to the pattern, and (iii) doing both; in the numbers above, the dominant contribution is (i), since the truth component jumps while the falsity component changes only mildly.*

5.7 Anti-resonance as habit reversal (tiny numeric demo)

Example 6 (Anti-resonance increases opposition and can reverse a habit)**.** *Let $k^- = 0.9$ and use ARes_{k^-} from Definition 33. Suppose C_2 has begun to accept $a \sim b$ with $R_2(a,b) = (0.60, 0.20)$, but a different context (or a new stream of evidence) provides strong opposition modeled as $R_1(a,b) =$* $(0.10, 0.90)$. *Conceptually, anti-resonance differs from resonance by selectively importing (scaled)* opposition *rather than support, so it acts like a controlled mechanism for "unlearning" or counter-conditioning without requiring that prior positive evidence be erased.*

Then

$$\mathsf{ARes}_{0.9}(R_1 \to R_2)(a,b) = (0.60, 0.20) \oplus \big((0, 0.9) \otimes (0.10, 0.90)\big) = (0.60, 0.20) \oplus (0, 0.81) = (0.60, 0.81).$$

Here $(0, 0.9) \otimes (0.10, 0.90) = (0, 0.81)$ illustrates the same componentwise multiplication as before: anti-resonance scales only the negative coordinate and passes no additional positive mass (the 0 in the first coordinate). Likewise, the join $\oplus$ acts componentwise as an aggregation of available evidence; in particular, the second coordinate increases from 0.20 to 0.81, capturing that the imported counter-evidence dominates the earlier, weaker opposition. Notice that the positive coordinate remains 0.60 throughout, so the update changes behavior by increasing conflict, not by deleting prior support.

The decision probability drops from

$$p_{\mathrm{dec}}(0.60, 0.20) = \frac{(0.40) + 1}{2} = 0.70 \quad \textit{to} \quad p_{\mathrm{dec}}(0.60, 0.81) = \frac{(-0.21) + 1}{2} = 0.395.$$

Equivalently, the net inclination shifts from $0.60 - 0.20 = 0.40$ (net support) to $0.60 - 0.81 = -0.21$ (net opposition), which is exactly the sense in which a "reversal" occurs in this graded setting: the sign of $t - f$ flips even though t itself does not decrease.

So anti-resonance can drive a reversal: a previously formed habit becomes less likely by importing opposition-to-the-pattern from elsewhere. In this way, ARes_{k^-} can be read as a minimal mathematical caricature of how external critique, adverse feedback, or a competing norm can suppress an emerging tendency by strengthening the "reasons against".

Remark 162 (A philosophical aside)**.** *In ordinary discourse, "reversal" sounds like negation: the mind flips from yes to no. In a paraconsistent setting, reversal is subtler: the system may keep its* *positive evidence while importing so much opposition that the net disposition changes sign. This resembles how real deliberation often feels: one does not forget the reasons* for*, but the reasons* against *become weightier. Formally, the point is that (t, f) is not collapsed to a single scalar before updating: the calculus allows t and f to vary semi-independently, so an agent can become less disposed to act even while retaining (and acknowledging) substantial supporting considerations.*

5.8 A tiny resonance score demo via paraconsistent interference

We now connect the above "transport of evidence" to a resonance score derived from the p-bit geometry.

Remark 163. *The goal of this subsubsection is to show that "resonance" can be read in two compatible ways:*

- *as an order-theoretic / quantale-theoretic propagation rule on evidence structures, and*
- *as a coherence or interference statistic after embedding evidence into a complex plane.*

This second reading echoes approaches that relate paraconsistent evaluation to oscillatory or resonant dynamics ([24]), but here it is presented only as a tiny computational demo.

Definition 35 (p-bit to complex mapping)**.** *Define $\sigma_{\mathbb{C}} : [0,1]^2 \to \mathbb{C}$ by*

$$\sigma_{\mathbb{C}}(w^+, w^-) := d + ic, \qquad d := w^+ - w^- \in [-1,1], \qquad c := (w^+ + w^-) - 1 \in [-1,1].$$

Here d is the truth-bias axis (net support vs net opposition) and c is the contradiction/ignorance axis.

Remark 164 (Geometric intuition)**.** *The map $\sigma_{\mathbb{C}}$ turns a* p-bit *value into a point in the complex plane whose real part d measures net inclination (support minus opposition), while the imaginary part c measures deviation from classical consistency. When $w^+ + w^- = 1$ (no "extra" evidence), we have $c = 0$ and the amplitude is purely real. When both channels are high, $c > 0$ and we move into the upper half-plane, marking contradiction. When both are low, $c < 0$ and we move into the lower half-plane, marking ignorance/indeterminacy. This gives a compact way to treat "truth" and "conflict" as orthogonal components.*

Definition 36 (Two-context interference (resonance) score)**.** *Given two complex amplitudes $z_1, z_2 \in \mathbb{C}$ and weights $w_1, w_2 \geq 0$ with $w_1 + w_2 = 1$, define*

$$\mathrm{Int}(z_1, z_2) := |w_1 z_1 + w_2 z_2|^2 - |w_1 z_1|^2 - |w_2 z_2|^2 .$$

Positive values indicate constructive interference (resonance), negative values indicate destructive interference (dissonance).

Remark 165 (What this formula measures)**.** *The quantity $|w_1 z_1 + w_2 z_2|^2$ is the squared magnitude of the combined amplitude. Subtracting $|w_1 z_1|^2 + |w_2 z_2|^2$ isolates the* cross term *that depends on relative phase. Thus $\mathrm{Int}(z_1, z_2)$ is zero when the two contributions are orthogonal in phase on average, positive when they align, and negative when they oppose. Interpreted back in* p-bit *terms: it measures whether two contexts "pull" in the same evidential direction after accounting for both bias and contradiction.*

Remark 166. *For $w_1 = w_2 = \frac{1}{2}$ one has*

$$\mathrm{Int}(z_1, z_2) = \frac{1}{2} \operatorname{Re}(z_1 \overline{z_2}).$$

So resonance is literally the (scaled) real part of the complex inner product, i.e. phase alignment.

Remark 167 (Why this is a reasonable toy resonance score)**.** *The interference score is not claimed as a final definition of resonance for the full theory. Its pedagogical value is that it is:*

- *symmetric in the two contexts,*
- *sensitive to both direction (truth-bias) and conflict (imaginary component),*
- *and continuous/graded.*

These are precisely the desiderata one wants when translating qualitative talk of resonance/dissonance into a computable diagnostic (Hyperseed-Concept 159, 97).

Example 7 (Resonance increases after morphic resonance update)**.** *Consider the single proposition "$a \sim b$" as evaluated in C_1 and C_2. Using Example 1 and Example 4:*

$$R_1(a,b) = (0.90, 0.80), \quad R_2(a,b) = (0.20, 0.70), \quad R_2'(a,b) = (0.81, 0.72).$$

Map to complex amplitudes via Definition 35:

$$z_1 = \sigma_{\mathbb{C}}(0.90, 0.80) = 0.10{+}0.70i, \qquad z_2 = \sigma_{\mathbb{C}}(0.20, 0.70) = -0.50{-}0.10i, \qquad z_2' = \sigma_{\mathbb{C}}(0.81, 0.72) = 0.09{+}0.53i.$$

With equal weights $w_1 = w_2 = \frac{1}{2}$:

$$\mathrm{Int}(z_1, z_2) = -0.06 \qquad \textit{(dissonant)},$$

while after the morphic resonance update,

$$\mathrm{Int}(z_1, z_2') = 0.19 \qquad \textit{(resonant)}.$$

Thus the same evidence-transport step that increases pattern support across contexts also increases an interference-derived resonance score.

One can likewise compute resonance for the pattern $P = \{(a,b),(b,c)\}$ *by using the pattern supports $S_{R_1}(P)$ and $S_{R_2}(P)$ from Example 5 and mapping those* p-bit *values through $\sigma_{\mathbb{C}}$. This again yields a negative interference score before morphic resonance and a positive one after, i.e. increased coherence at the level of the whole pattern-constraint.*

Remark 168 (What changed, structurally?)**.** *Before resonance, C_1 and C_2 assign almost opposite truth-bias to $a \sim b$: z_1 lies in the upper-right (net positive but conflicted) while z_2 lies in the lower-left (net negative and slightly ignorant). After resonance, z_2' rotates toward z_1 in both real and imaginary components. The increase in* Int *is thus a numeric witness that the contexts have become more phase-aligned on that proposition.*

Remark 169 (Toy moral)**.** *In this toy model, "morphic resonance" has two separable mathematical faces:*

1. *as a monotone, quantale-native evidence transport $R_2 \mapsto R_2 \oplus (K \otimes R_1)$ that increases pattern support across contexts without forcing consistency, and*

2. *as an increase in an interference-based coherence score after mapping* p-bit*s to complex amplitudes.*

This is exactly the bridge needed later in the paper: resonance is simultaneously a structural update in the ontology and a dynamical/energetic coherence signal.

Remark 170 (Where this is going)**.** *The toy computations show how a single algebraic move can produce: (i) transferred pattern scaffolds, (ii) emergent closure-completions, and (iii) increased coherence scores. Concretely, "transferred pattern scaffolds" can be read as the appearance of a reusable template: once a pattern is encoded as an algebraic element (or as a map between such elements), the same formal manipulation can be re-applied in a new location, effectively transporting structure along the available morphisms. In this sense the scaffold is not a new axiom but a consequence of how composition and order interact in the chosen algebra.*

Likewise, "emergent closure-completions" refers to the fact that passing to a closure (or completion) is often forced by the operations already present: an initially partial family of pattern-combinations can generate its own minimal fixed point under the closure operator, yielding a completed object in which previously missing joins/meets (or stable consequences) become available. Even when the starting data are intentionally finite and schematic, the closure step makes explicit the difference between "what is stipulated" and "what is entailed" by the algebraic rules.

Finally, "increased coherence scores" can be understood as the numerical or order-valued witness that the move has improved internal compatibility: after the transfer and closure, more constraints can be simultaneously satisfied, or the same constraints can be satisfied in a stronger way. In later sections, the same ideas are lifted to richer pattern spaces and iterated dynamics, where questions of stability, habit persistence, and controlled reversal become meaningful. Here "iterated dynamics" should be taken literally: repeating the move defines an update process, so one can ask whether repeated application converges to a fixed point, cycles among a small family of states, or bifurcates when parameters are changed. Correspondingly, "controlled reversal" anticipates the need for operations that can partially undo an update (or at least trace its provenance), so that one can distinguish irreversible closure effects from reversible transport effects. The present section should be read as the smallest "seed crystal" of that later structure. In particular, it isolates the minimal set of ingredients—a finite universe, an order-sensitive combination rule, and a coherence witness—so that later generalizations can be recognized as elaborations rather than as unrelated constructions.

Part I
Systematic reconstruction of Hyperseed concepts

6 Phenomenological primitives

Roadmap

This section turns the most primitive Hyperseed vocabulary into a small, typed interface. The guiding principle is that the primitives should be *structural* rather than verbal: the mathematics should remain invariant under renaming of symbols, choice of language, and (to a large extent) choice of representational format. In practice, "typed interface" means that each primitive comes with an explicit domain of application (what sort of experiential item it ranges over) and a codomain of evaluation (what sort of value it returns), so that later constructions can be checked for compositional well-formedness rather than defended only by prose. The intended effect is analogous to exposing an API: once the minimal signatures are fixed, alternative phenomenological stories can be layered on top without modifying the underlying formal objects. This is also why the early definitions are deliberately austere: the goal is not to capture every nuance of introspective report, but to isolate a stable core that is preserved across rephrasings, translations, and modeling choices.

Remark 171. *The philosophical stance here is deliberately Russellian in its demand for clarity: if two descriptions differ only by a renaming of symbols, then whatever is essential has not changed. But it is also Peircean in its tolerance for "signs" whose meaning depends on use: the same experiential content can be regimented differently in different contexts, without supposing that one regiment is the one true metaphysical grammar [14]. In the Hyperseed ontology [1], this structural emphasis is what permits a single formal core to underwrite multiple phenomenological and scientific readings. A useful way to reconcile these attitudes is to treat the formalism as fixing* invariants *(what must be preserved under admissible transformations), while leaving room for multiple* coordinatizations *(how those invariants are presented, narrated, or operationalized). Thus the text will repeatedly separate (i) the minimal algebraic or relational constraints that a primitive must satisfy from (ii) any particular interpretation in terms of language, imagery, embodiment, or cognitive architecture. The reader should therefore not expect the primitives to settle interpretive disputes; rather, they provide a common frame in which such disputes can be stated precisely as questions about which additional axioms, identifications, or equivalences are being assumed.*

We proceed in three steps. First we introduce *occasions of experience* as the basic carriers of primitive phenomenology, together with *contexts* that implement observer-relativity. Here "occasion" should be read neutrally as a minimal unit of phenomenal bearing (whatever ultimately plays the role of a token of experience in the theory), while "context" names the parameter that records the standpoint relative to which evaluations are made. The point of making context explicit is to prevent hidden shifts of reference: many apparent paradoxes about sameness or difference in experience dissolve once one distinguishes differences across occasions from differences induced by changing the evaluative frame. Second we define the core pairwise experiential relations (difference, distinction, repetition, variety, non-duality, non-dual variety) using p-bit-valued evaluations. The use of p-bit-valued outputs is meant to keep the interface flexible: it accommodates crisp/bivalent judgments as a special case, while also permitting graded, partial, or context-sensitive assessments without changing the type signature of the primitives. It also forces a discipline about what is being asserted: instead of saying "x and y are different" as an untyped slogan, one records *how* and *to what extent* a given context rates them as different, and one can later study how such ratings compose. Third we formalize the Peircean/Jungian numerical-metaphysical categories *First, Second, Third, Fourth* as an inductive ladder of relational constructions. The intended reading is that these categories are not introduced as mysterious primitives in their own right, but as progressively richer patterns that can be built from the earlier relational vocabulary (so that "Thirdness," for example, is expressed as a certain kind of mediated or triadic structure rather than a mere label). We conclude by formalizing *presentational immediacy* (intensity) and the *abstract/concrete* distinction. These closing notions serve two roles: they connect the relational core to phenomenological salience (why some distinctions matter more than others in lived experience), and they provide a handle for later passages where the theory must distinguish between structural descriptions and the more "felt" or "given" character of an occasion.

Hyperseed concepts covered

- Occasions of experience.
- Difference; distinction; repetition; variety; non-duality; non-dual variety.
- First; second; third; fourth.
- Presentational immediacy (intensity).
- Abstract/concrete.

The list above is meant as a contract with the reader: later sections will freely reuse these terms, but only after they have been pinned down here in a way that supports reparameterization by context and comparison across representational choices. In particular, each named relation will be introduced so that it can be applied uniformly to occasions across diverse modeling settings (introspective reports, cognitive architectures, or scientific proxies), with any domain-specific assumptions made explicit as additional structure rather than smuggled into the primitive itself.

6.1 Occasions and contexts

Hyperseed begins from *occasions of experience*: primitive experiential "atoms" that do not presuppose an external experiencer, except in the minimal sense that an occasion "is what it is" from its own inside. The mathematical reconstruction makes two moves that keep this notion precise without over-committing metaphysically:

1. We treat occasions as *objects* in a structure that only cares about identity *up to equivalence.*
2. We treat every act of distinction-making as *context-indexed* (observer-relative), implemented by explicit context data.

Definition 37 (Occasion space)**.** *An* occasion space *is a small* $(\infty, 1)$*-groupoid* $\mathcal{E}$*. Its objects* $\mathrm{Ob}(\mathcal{E})$ *are occasions. A 1-morphism* $f : o \to o'$ *represents an admissible "identification" or "transport" between occasions (e.g. recognizing the same thing under a change of viewpoint), and higher morphisms encode coherent equivalences between such identifications.*

In toy models one may replace $\mathcal{E}$ *by an ordinary groupoid, or even by a set* E *(regarding only equality), by applying the truncation* π_0*.*

Remark 172 (Intuition and examples for occasion spaces)**.** *This definition packages a simple idea: experience does not usually present us with an inflexible equality relation, but rather with a family of ways of* treating *two occasions as "the same enough." A groupoid is the algebraic prototype of this: objects are things, arrows are reversible identifications, and composition expresses chaining of identifications. The notation* $(\infty, 1)$ *indicates that we may also want to track identifications-between-identifications (and so on), because in many domains the* coherence *of how we identify things matters as much as the identification itself.*

A minimal example is a set E *of snapshots of sensory input, where the only identifications are equalities; this is the discrete groupoid case. A slightly richer example is a groupoid whose objects are images, and whose arrows are invertible transformations (e.g. rotations) that a perceptual system treats as preserving "the same object." The utility of the definition is that it gives us a disciplined place to put "sameness up to viewpoint" without prematurely choosing a metric, a probability model, or a logical theory. This is congenial to process views in the spirit of Whitehead, where the primitive units are occasions and their relations, not enduring substances [15].*

Remark 173 (Why $(\infty, 1)$?)**.** *Hyperseed uses language like "interchangeable in a context" and (later) "self-model recursion". These are naturally modeled using identity-up-to-equivalence rather than rigid equality. An $(\infty, 1)$-groupoid is the minimal formal object that (i) admits nontrivial identifications and (ii) keeps track of coherence of identifications. Nothing in the present section depends on advanced higher-categorical technology; we use the $(\infty, 1)$ language mainly to record that such coherence data* may matter *later.*

Remark 174 (Notation clarification)**.** *Here $\mathcal{E}$ denotes the occasion space;* $\mathrm{Ob}(\mathcal{E})$ *is the set (or class) of objects of $\mathcal{E}$; and a "1-morphism" $f : o \to o'$ should be read as a directed identification from o to o' (invertible up to higher structure in an $(\infty, 1)$-groupoid). The truncation π_0 is the operation that forgets higher identifications and keeps only connected components, i.e. "equivalence classes up to identification."*

Definition 38 (Contexts)**.** *A* context *(or viewpoint) is a tuple*

$$c := \big(X_c,\ \Pi_c,\ \delta_c,\ \iota_c\big)$$

where:

- X_c *is a set of* presented entities *(what the context treats as the "things" it can refer to);*
- $\Pi_c : \mathrm{Ob}(\mathcal{E}) \to X_c$ *is a* presentation map *(a coarse-graining of occasions into context-level entities);*
- $\delta_c : X_c \times X_c \to \mathbb{V}$ *is a* p-bit*-valued* distinctness evaluator*;*
- $\iota_c : X_c \times X_c \to \mathbb{V}$ *is a* p-bit*-valued* interchangeability evaluator.

We interpret $\delta_c(x, y)$ as evidence for/against the proposition "x and y are distinct (diverge)", and $\iota_c(x, y)$ as evidence for/against "x and y are interchangeable for the purposes relevant to c."

Remark 175 (Intuition, examples, and why contexts are necessary)**.** *A context c is the formal surrogate for "an observer-as-a-way-of-carving." The set X_c is what the context can even* name*; the map Π_c says which raw occasions are presented as which named entities; and the evaluators δ_c, ι_c say how the context judges difference and substitutability. This makes observer-relativity a matter of explicit structure rather than a vague proviso. It also makes room for paraconsistent judgments: a context may have simultaneously high evidence for "distinct" and high evidence for "not distinct," a point developed below using* p-bit *semantics (see also paraconsistent treatments such as [23, 24]).*

Remark 176 (Distinctness versus interchangeability)**.** *Although δ_c and ι_c are related in ordinary language, we do not assume that one is the logical negation of the other. In many practical situations a context can regard two entities as* distinct *(they differ in some tracked respect) while still treating them as* interchangeable *for the tasks at hand (the difference is irrelevant), and conversely it can regard two entities as* not distinct *in some coarse description while refusing to substitute one for the other (e.g. due to causal, temporal, or deontic constraints). Formally, allowing δ_c and ι_c to vary independently is what lets the framework represent "same object, different role" and "different object, same role" judgments without forcing a single equivalence relation to do all conceptual work.*

Remark 177 (Contexts as $\mathbb{V}$-valued relations)**.** *The evaluators δ_c, ι_c can be read as $\mathbb{V}$-valued binary relations on X_c. This viewpoint emphasizes that a context need not deliver crisp Boolean predicates but rather graded and potentially inconsistent evidence states, with $\mathbb{V}$ serving as the codomain of* *semantic values. In particular, one may later impose optional structural conditions (symmetry, reflexivity, transitivity up to evidence, etc.) when modeling a specific domain, but the present definition stays neutral: it merely provides a slot in which such constraints can be stated explicitly rather than smuggled in as informal assumptions.*

Definition 39 (Context-induced evaluations on occasions)**.** *Given a context $c = \big(X_c, \Pi_c, \delta_c, \iota_c\big)$, define the induced evaluators on occasions by*

$$\delta_c^{\mathcal{E}}(o, o') := \delta_c\big(\Pi_c(o), \Pi_c(o')\big), \qquad \iota_c^{\mathcal{E}}(o, o') := \iota_c\big(\Pi_c(o), \Pi_c(o')\big),$$

for $o, o' \in \mathrm{Ob}(\mathcal{E})$.

Remark 178 (What the presentation map is doing)**.** *The map $\Pi_c : \mathrm{Ob}(\mathcal{E}) \to X_c$ expresses the idea that a context does not have direct access to "raw occasions as such," but only to a quotient-like presentation into context-level entities. This is intentionally weaker than requiring that Π_c be invariant under all identifications in $\mathcal{E}$: an admissible identification $f : o \to o'$ in the occasion space need not be one that the context can or will recognize, and contexts may collapse distinctions that $\mathcal{E}$ retains (or refine them by remembering additional tags in X_c). In concrete models, one may nevertheless choose to impose compatibility conditions of the form "if o and o' are connected in $\mathcal{E}$, then $\Pi_c(o)$ and $\Pi_c(o')$ are judged highly interchangeable," but such requirements are part of a modeling choice rather than part of the primitive ontology.*

Remark 179 (Example patterns for contexts)**.** *Typical sources of context-dependence fit neatly into the tuple c: (a)* Resolution/attention: *X_c contains only the coarse entities available at a given resolution, with Π_c forgetting fine-grained differences between occasions; (b)* Coordinate/viewpoint choice: *Π_c factors out transformations the context treats as irrelevant, while δ_c may still register residual differences important for the task; (c)* Memory and identity-tracking: *X_c may include persistent labels ("the cup from earlier"), so that Π_c encodes a policy for binding new occasions to existing labels, and ι_c describes when substituting one label for another preserves predicted consequences. These examples illustrate why separating Π_c (what is presented) from δ_c, ι_c (how it is evaluated) is useful: the same coarse-graining can be paired with different evaluators depending on goals and background commitments.*

Remark 180. *A simple example is a vision context in which X_c is a set of tracked objects on a screen, Π_c maps pixel-level occasions to object IDs, δ_c scores how different two tracks are, and ι_c scores whether they can be swapped without changing the task outcome (e.g. two identical chess pawns in symmetric positions might be interchangeable for some purposes). Concretely, one can think of the underlying occasion space as a stream of low-level sensor events (frames, patches, keypoints, or other "glimpses"), while Π_c performs the role of a context-dependent binding/association rule that declares which of those low-level events count as evidence about which putative object. In that setting, δ_c is not required to be a metric in the strict mathematical sense; it is simply whatever evaluation the context treats as "difference-relevant," possibly combining appearance, trajectory, and role features into a paraconsistent value. Likewise, ι_c is best read as an invariance or symmetry score: it records whether exchanging two hypothesized entities preserves whatever objective or downstream action policy defines "the same outcome" for the current task. This allows the formalism to express both perceptual distinctions (two tracks are strongly non-identical) and practical equivalences (two tracks are distinct but functionally interchangeable), which need not coincide once observer goals are made explicit. The definition is useful because later notions (weakness/simplicity, patterns, resonance) require a stable place to anchor "who is distinguishing what, and under what rules" [5]. In particular, once later constructions quantify over repeated occasions, compressibility,* *or alignment across contexts, it becomes important that the act of "counting as the same thing" has been made an explicit, indexable datum rather than an implicit assumption. In Hyperseed terms, this is the structural backbone for Hyperseed-Concept 86 and Hyperseed-Concept 124.*

Remark 181 (Observer-relativity as explicit data)**.** *In this reconstruction, observer-relativity is not a philosophical afterthought; it is built in as explicit context-indexing. Different contexts can: (i) carve the same occasion space $\mathcal{E}$ into different entity sets X_c; (ii) disagree paraconsistently about whether two entities are distinct or interchangeable. The point of allowing (ii) is not merely to permit "uncertainty," but to permit structured coexistence of competing discriminations: a context may accumulate simultaneous evidence that two tracks are the same (e.g. perfect appearance match) and that they are not (e.g. mutually exclusive trajectories), without collapsing into triviality. This also clarifies that observer-relativity here includes both representational choices (what the observer treats as an entity at all) and normative choices (what differences are treated as relevant to a task, and what differences are ignored as symmetries). On this view, two contexts may share the same raw stream $\mathcal{E}$ yet implement different "grain sizes" by choosing different X_c and different binding maps Π_c, leading them to construct different object inventories from the same experiential substrate. This is the formal home for Hyperseed's claim that much of "ontology" is bookkeeping about which distinctions are made by which observers, and at what grain.*

Remark 182 (Notation clarification: $\mathbb{V}$ and p-bit-valued evaluations)**.** *The codomain $\mathbb{V}$ is the value space for paraconsistent evidence, intended to be the* p-bit *quantale (typically $[0,1]^2$ with a chosen order and monoidal product; cf. Section 3.4). When we write $\delta_c(x,y) \in \mathbb{V}$ we mean that $\delta_c(x,y)$ has (at least) two coordinates, often read as (positive-evidence, negative-evidence). Intuitively, this lets the model represent "evidence for difference" and "evidence against difference" (or, depending on the chosen polarity, "evidence for sameness" and "evidence against sameness") side by side, rather than forcing a single scalar that must trade one off against the other. Componentwise comparisons, multiplication-like combination $\otimes$, and join-like aggregation $\oplus$ then provide a disciplined calculus for combining experiential judgments. For example, if two independent cues both support distinctness, $\otimes$ can be used to accumulate that support; if two alternative cues provide competing routes to the same judgment, $\oplus$ can be used to aggregate them without prematurely discarding minority evidence. The need for this richer codomain shows up immediately when contexts are allowed to disagree or when binding is ambiguous: the system must be able to carry forward incompatible partial assessments until later constraints (task loss, additional observations, or higher-level regularities) resolve or compartmentalize them.*

6.2 Difference and distinction

Hyperseed distinguishes *difference* from *distinction*. Difference is directed: one occasion diverges from another. Distinction is symmetric: two occasions mutually diverge. In practice, the separation is useful because many cognitive and operational comparisons have an intrinsic "reference" (what is being compared *to*), even when ordinary language uses a symmetric phrasing. The context parameter c is intended to hold fixed the aspect, task, or perspective relative to which divergence is evaluated (e.g. perceptual modality, feature set, goal constraints, or a representational code), so that directedness is not conflated with merely changing evaluative criteria.

Definition 40 (Difference as a directed evaluation)**.** *Given a context c, define the* directed difference *evaluator*

$$\mathrm{Diff}_c(x \to y) := \delta_c(x,y) \in \mathbb{V}.$$

The positive channel $\mathrm{Diff}_c(x \to y)^+$ *is evidence that* x *diverges from* y*. The negative channel* $\mathrm{Diff}_c(x \to y)^-$ *is evidence that* x *does* not *diverge from* y *(e.g.* x *is absorbed into, explained by, or indistinguishable-from* y *along the relevant aspect).*

It is worth emphasizing that Diff_c is not assumed to be a metric, a distance, or even a preorder; it is simply an evaluator returning a p-bit-style evidential value. Accordingly, nothing in the definition forces $\mathrm{Diff}_c(x \to y)^+$ and $\mathrm{Diff}_c(x \to y)^-$ to be complements, and (in paraconsistent regimes) they may both be high, representing simultaneously strong reasons for divergence and strong reasons for non-divergence under different sub-aspects aggregated into the same context c. Conversely, they may both be low, representing insufficient evidence either way.

Remark 183 (Intuition and examples for directed difference)**.** *The directedness in* $\mathrm{Diff}_c(x \to y)$ *matters whenever "x differs from y" is not symmetric in the lived or operational sense. For instance, a coarse prototype* y *may "explain" a fine-grained observation* x *better than* x *explains* y*. In perception, a noisy instance may be well-absorbed into a clean template (low divergence of* x *from* y*) even though the template does not fit the instance in reverse (higher divergence of* y *from* x*) if the evaluators are aspect-sensitive.*

This definition is useful because it isolates the smallest unit in which "otherness" can appear: a directed contrast. It corresponds naturally to Hyperseed-Concept 96 and provides the raw material out of which symmetry (distinction), grouping (relations), and ultimately patterns are built [5]. A further way to read the direction $x \to y$ *is as "evaluate* x using y *as the comparator or explanatory basis": if* y *is a model, type, or hypothesis and* x *is a datum, then* $\mathrm{Diff}_c(x \to y)^+$ *expresses the degree to which the datum resists assimilation to the model, while* $\mathrm{Diff}_c(x \to y)^-$ *expresses the degree to which it is successfully covered by the model. This interpretation also clarifies why the negative channel is phrased as "does not diverge" rather than "is identical": absorption and explainability are often weaker (and more context-dependent) than identity.*

Remark 184 (Self-comparison and neutral cases)**.** *Although many applications will set* $\mathrm{Diff}_c(x \to x)^+$ *to be small and* $\mathrm{Diff}_c(x \to x)^-$ *to be large (reflecting the expectation that an occasion does not diverge from itself), the formalism does not require this. Allowing nontrivial self-difference can model temporal or perspectival drift (an "occasion" treated as a process with internal variation), sensor noise (a self-sample may not reproduce exactly), or representation mismatch (different encodings of "the same" occasion). When a context* c *is too coarse or too uncertain to decide,* $\mathrm{Diff}_c(x \to y)$ *may be near-indeterminate in both channels, which is precisely the situation later "weakness" notions are designed to register rather than forcibly resolve.*

Definition 41 (Distinction as mutual difference)**.** *Define the* (symmetric) distinction *evaluator by*

$$\mathrm{Dist}_c(x, y) := \mathrm{Diff}_c(x \to y) \otimes \mathrm{Diff}_c(y \to x),$$

where $\otimes$ *is the* p-bit *product from Section 3.4. Thus evidence that* x *and* y *are mutually distinct compounds multiplicatively (and likewise for evidence against mutual distinctness).*

Because $\otimes$ combines evidential values, Dist_c is best read as "joint support for two directed claims" rather than as an averaged similarity score. In particular, if either direction strongly fails (e.g. $\mathrm{Diff}_c(x \to y)^+$ is very low because x is thoroughly absorbed into y in context c), then the compounded support for mutual distinctness will typically be reduced. When $\otimes$ is commutative (as in standard p-bit constructions), $\mathrm{Dist}_c(x, y) = \mathrm{Dist}_c(y, x)$ holds automatically, aligning the formal object with the intended symmetric reading.

Remark 185 (Intuition and examples for distinction)**.** *The move from difference to distinction is the move from "x contrasts with y in this direction" to "x and y stand apart from each other." If one thinks geometrically,* $\mathrm{Diff}_c(x \to y)$ *is like a directed arrow recording mismatch; taking* $\mathrm{Dist}_c(x, y)$ *amounts to requiring mismatch in both directions and combining the two pieces of evidence via* $\otimes$*. In a crisp (non-paraconsistent) setting, one could imagine* $\mathrm{Diff}_c(x \to y)$ *as a Boolean; then* $\mathrm{Dist}_c(x, y)$ *reduces to conjunction.*

The definition is necessary because later measures of weakness and simplicity depend on how many distinctions a context fails to make. Distinction provides the symmetric "edge" concept needed for graphs, equivalence-like closures, and entropy-like measures such as graphtropy (cf. Hyperseed-Concept 98 and Hyperseed-Concept **??***). One can also view* Dist_c *as the minimal symmetry-enforcing operation applied to raw directed evaluations: rather than attempting to symmetrize by averaging or by taking a maximum,* $\otimes$ *encodes the idea that mutual distinctness is supported only to the extent that* both *directed divergences are supported together. This choice aligns with later graph-theoretic constructions, where an undirected edge typically represents a mutual relation that should not be present if either directed justification for separating the nodes is missing.*

Remark 186 (Paraconsistent readings of mutuality)**.** *Since values live in* $\mathbb{V}$*, mutuality can itself be conflicted: it is possible to have substantial evidence that x and y are distinct and, simultaneously, substantial evidence that they are not distinct (for example, two stimuli may differ in shape but be functionally interchangeable for the current task). Under* $\mathrm{Dist}_c(x, y) = \mathrm{Diff}_c(x \to y) \otimes \mathrm{Diff}_c(y \to x)$*, such conflicts can arise either from conflicts already present in one direction or from an interplay between directions (e.g. one direction supports divergence while the other direction supports non-divergence). This is not treated as a defect but as a faithful representation of contexts in which "making a distinction" is itself unstable, multi-aspect, or sensitive to what is taken as explanans versus explanandum.*

Remark 187 (Thresholding)**.** *To obtain a crisp "made distinction" relation (useful for counting, graphs, and later definitions of graphtropy), choose thresholds* $\tau_\delta, \tau_\iota \in (0, 1)$*. One may then declare:*

$$x \#_c y \iff \mathrm{Dist}_c(x, y)^+ \geq \tau_\delta.$$

Hyperseed frequently reasons at both levels: a fuzzy background of graded distinctions together with crisp distinctions induced by task needs. In applications, τ_δ *governs the willingness to treat weak positive evidence as sufficient to "separate" two occasions, and adjusting it changes graph density in the induced crisp distinction graph. The additional parameter* τ_ι *is naturally available for dual thresholding of the negative channel (e.g. to assert a crisp "indistinguishable" or "identified" relation when* $\mathrm{Dist}_c(x, y)^-$ *is large), allowing a context to support both separation and identification decisions with explicit, independent criteria when required by downstream constructions.*

Remark 188 (Notation clarification: channels and threshold parameters)**.** *When we write* $\mathrm{Dist}_c(x, y)^+$ *we mean the positive-evidence coordinate of the* p-bit *value* $\mathrm{Dist}_c(x, y) \in \mathbb{V}$*; similarly for* $^-$*. The thresholds* τ_δ, τ_ι *are real numbers in* $(0, 1)$ *used to turn graded evidence into crisp relations when needed. It is often helpful to keep in mind the typing discipline implicit here:* Diff_c *and* Dist_c *return objects of the same evidential type* $\mathbb{V}$*, while channel extraction* $(-)^+$ *and* $(-)^-$ *maps those objects to ordinary reals (or whatever ground scale* $\mathbb{V}$ *uses for each coordinate), which is why comparing to numerical thresholds is well-formed.*

6.3 Repetition, variety, non-duality, and non-dual variety

Beyond difference/distinction, Hyperseed elevates four additional pairwise experiential categories to the primitive level. In the present reconstruction, these are defined by combining (i) distinctness information and (ii) interchangeability information, both potentially paraconsistent. In particular, the intent is that a context may track not only "are these two presented as different?" but also "can one stand in for the other without loss for the currently salient purpose?"; the latter is a practical/operational relation rather than a purely perceptual one. Because both axes are evaluated with potentially incomplete and competing cues, it is important that the formalism can represent mixed evidence without forcing an immediate resolution.

Definition 42 (Soft interchangeability). *Given c, interpret $\iota_c(x,y) \in \mathbb{V}$ as a paraconsistent evaluation of interchangeability. We will often use its two channels as shorthand:*

$$\iota_c(x,y)^+ \text{ (evidence interchangeable)}, \qquad \iota_c(x,y)^- \text{ (evidence not interchangeable)}.$$

Remark 189 (Intuition and examples for interchangeability). *Interchangeability is not sameness. Two items can be distinct and yet substitutable for a purpose: two identical screws in a machine, two equivalent lemmas in a proof, or two tokens of a symbol-type. Conversely, two items can be nearly indistinguishable in raw perception and yet not interchangeable because of hidden constraints (one is a live wire, the other insulated). The paraconsistent reading allows both kinds of evidence to be present, which is often how real systems behave under partial information [23, 24].*

Formally, ι_c gives us a second axis beyond distinctness. This is useful because Hyperseed's later notions of pattern and emergence rely not only on detecting difference, but on discovering stable substitutions and invariances across contexts (Hyperseed-Concept 59 and Hyperseed-Concept 130).

Interchangeability should also be read as explicitly context-indexed: $\iota_c(x,y)$ can change when the operative goal, constraints, or "interface" changes. For example, two objects may be interchangeable with respect to mass distribution (balancing a scale) but not interchangeable with respect to provenance (authenticity, ownership, legal responsibility). Nothing in the primitive requires ι_c to be an equivalence relation: symmetry and transitivity may hold in some contexts (e.g. standardized parts) but can fail in others (e.g. substitutability under resource limits, one-way emulation, or asymmetric access rights). The paraconsistent two-channel representation is therefore meant to cover both well-behaved "mathematical" substitution and the more brittle substitutions encountered in empirical systems.

Definition 43 (Repetition and variety (threshold form)). *Fix thresholds $\tau_\delta, \tau_\iota \in (0,1)$. For $x \neq y$ in X_c define:*

$$\begin{aligned} \mathrm{Rep}_c(x,y) \quad &\iff \quad \mathrm{Dist}_c(x,y)^+ \geq \tau_\delta \text{ and } \iota_c(x,y)^+ \geq \tau_\iota, \\ \mathrm{Var}_c(x,y) \quad &\iff \quad \mathrm{Dist}_c(x,y)^+ \geq \tau_\delta \text{ and } \iota_c(x,y)^- \geq \tau_\iota. \end{aligned}$$

Intuitively, a repetition *is a pair that is experienced as distinct yet context-substitutable; a* variety *is distinct and non-substitutable.*

Remark 190 (Intuition and examples for repetition and variety). *A repetition is the lived form of "same pattern, different instance." Two distinct footsteps in sand may be distinct events (Dist_c^+ high) but interchangeable as tokens of one gait (ι_c^+ high). A variety is the contrary: the distinction matters. Two keys on a keyboard are distinct and not interchangeable if one is "Enter" and the other is "Escape" for the current task.*

These notions are useful because they separate mere multiplicity from structured multiplicity. Repetition is where compression begins: if many distinct things are interchangeable, a context can represent them with less descriptive effort, feeding directly into simplicity/weakness reasoning [3, 2]. Variety, by contrast, is where complexity is forced upon the system because substitutions break. This corresponds to Hyperseed-Concept 155 and Hyperseed-Concept 199.

It is worth emphasizing that the threshold presentation is a "crisping" of inherently graded judgments. The parameters τ_δ and τ_ι should be understood as knobs for when evidence becomes actionable for a context: in some regimes one may require near-certainty before treating two items as substitutable, while in others weak evidence may suffice. Because both Dist_c and ι_c have two channels, these thresholds can be used to carve out robust regions of decision even when the complementary channel is non-negligible; for instance, a system may classify $\mathrm{Rep}_c(x, y)$ based on $\iota_c(x, y)^+$ being high, while still retaining "residual" $\iota_c(x, y)^-$ as a warning that substitution is not unconditional. This separation between stored evidence (the p-bit valuation) and downstream categorization (thresholding) is part of what allows Hyperseed to represent pragmatic choice without losing contradictory or minority cues.

Definition 44 (Non-duality and non-dual variety (threshold form))**.** *Fix thresholds $\tau_\delta, \tau_\iota \in (0, 1)$. For $x \neq y$ in X_c define:*

$$\begin{aligned} \mathrm{NonDual}_c(x, y) \quad &\iff \quad \mathrm{Dist}_c(x, y)^+ \geq \tau_\delta \textit{ and } \mathrm{Dist}_c(x, y)^- \geq \tau_\delta, \\ \mathrm{NDVar}_c(x, y) \quad &\iff \quad \iota_c(x, y)^+ \geq \tau_\iota \textit{ and } \iota_c(x, y)^- \geq \tau_\iota. \end{aligned}$$

Thus non-duality *is "distinct and not-distinct" (paraconsistent distinctness), while* non-dual variety *is "interchangeable and not interchangeable" (paraconsistent substitutability).*

Remark 191 (Intuition and examples for non-duality and non-dual variety)**.** *Non-duality here is not a metaphysical decree that all things are identical; it is a disciplined way of representing a familiar phenomenological tension: the same pair can be simultaneously experienced as separate and as not-separate. In perceptual bistability (e.g. ambiguous figures), one may have strong evidence for two incompatible parsings. In social cognition, one may experience another person as "other" and yet as "part of me/us"; the formal point is that both channels can be high without forcing contradiction to trivialize the system [24].*

Non-dual variety likewise captures cases where substitution is both licensed and prohibited depending on which constraints are salient. For example, two roles in an organization may be interchangeable in a reporting chart but not interchangeable for legal responsibility. These definitions are useful because they prevent an overly classical ontology from erasing precisely the regimes where experience is richest: the zones where evidence does not settle into a single polarity (Hyperseed-Concept 121 and Hyperseed-Concept 120).

Non-dual variety can also arise in technical settings where multiple compatibility layers coexist. Two software modules may be interchangeable at the level of an interface (they implement the same signature, so ι_c^+ is high) yet not interchangeable at the level of non-functional constraints (performance, security policy, licensing, nondeterminism), yielding substantial ι_c^- as well. Similarly, in mathematics two proof steps might be interchangeable from the perspective of entailment, but not interchangeable from the perspective of constructive content, proof length, or allowed axioms; what counts as "salient" is again context-indexed, so it is natural that the evidence channels can both be high before a downstream agent commits to a choice.

Remark 192 (Non-duality as paraconsistency rather than collapse)**.** *A common mathematical temptation is to interpret "not-distinct" as strict equality. Hyperseed explicitly resists this: non-duality is not the claim that $x = y$ as symbols, but that experience supplies simultaneous evidence for and against the distinction. The* p-bit *semantics makes this precise: non-duality is simply a region of the square $[0, 1]^2$ where both channels are large. No logical explosion occurs because the logic is paraconsistent.*

The four primitives discussed in this subsection should not be read as mutually exclusive, nor as exhausting all possible evidence states. Because the underlying evidence is stored in two-channel form, it is possible (and often realistic) that $\mathrm{Rep}_c(x, y)$ holds while $\iota_c(x, y)^-$ is also nontrivial (a "normally safe" substitution with known caveats), or that $\mathrm{Var}_c(x, y)$ holds while $\iota_c(x, y)^+$ is also nontrivial (items are usually not substitutable, yet there are emergency procedures or partial substitutions). Likewise, non-duality in distinctness can co-occur with either repetition or variety judgments: one can have strong evidence for both distinction and non-distinction while also having a fairly settled stance on interchangeability (or vice versa). In this sense, $\mathrm{NonDual}_c$ and NDVar_c track tensions *within* a single evidence axis, whereas Rep_c and Var_c combine information *across* the distinctness and interchangeability axes.

A useful geometric picture is to treat each p-bit-valued predicate as placing the pair (x, y) at a point in $[0, 1]^2$ (one coordinate for positive evidence and one for negative evidence). Thresholding then selects regions: $\mathrm{NonDual}_c(x, y)$ corresponds to the "upper-right" region of the distinctness square, while $\mathrm{NDVar}_c(x, y)$ corresponds to the upper-right region of the interchangeability square. By contrast, $\mathrm{Rep}_c(x, y)$ and $\mathrm{Var}_c(x, y)$ are defined by combining one channel from the distinctness square (the positive channel) with one channel from the interchangeability square (positive or negative, respectively). This helps clarify why non-dual categories are not simply "in between" repetition and variety: they represent high-confidence conflict rather than low-confidence ambiguity.

Finally, there is a classical limit worth keeping in view. If, for a given context, the evidence is *nearly consistent* in the sense that $\mathrm{Dist}_c(x, y)^-$ is typically small whenever $\mathrm{Dist}_c(x, y)^+$ is large (and similarly for ι_c), then $\mathrm{NonDual}_c$ and NDVar_c will be rare, and the remaining categories behave more like classical dichotomies: distinct pairs separate into mostly interchangeable repetitions and mostly non-interchangeable varieties. Hyperseed retains the non-classical regimes not because they are always present, but because they become decisive precisely in settings involving partial observability, layered constraints, or competing interpretations—the same settings where pattern discovery and emergent structure are most informative.

6.4 First, Second, Third, Fourth

Hyperseed uses the numerical-metaphysical categories *First, Second, Third* (Peirce) and extends them with *Fourth* (synergy, Jungian "mandalic" structure). The key point for a rigorous ontology is that these are not mystical numerology; they are *structural levels* obtained by iterating a small set of constructions. Concretely, the intent is that each higher "number" is not an extra ingredient but a disciplined *closure step* on what came before: Firsts are carriers, Seconds are directed contrasts between carriers, Thirds are reusable organizations of such contrasts, and Fourths are higher-order organizations of multiple Thirds whose joint behavior is not reducible to separate consideration of each Third in isolation. This "iterated construction" view is what will allow later sections to treat phenomenological language as shorthand for precise mathematical objects and maps between them, rather than as a separate interpretive layer.

Remark 193. *Peirce's categories are often presented as metaphysics [14], but they are equally well read as an austere typology of how relations enter experience: firstness as suchness, secondness as reaction, thirdness as mediation/law. Hyperseed's maneuver is to make this typology operational: we identify canonical mathematical carriers for each level, so that later theorems can depend on them without importing any particular spiritual vocabulary [1]. A further benefit of making the carriers explicit is that one can compare different contexts c by comparing their associated carrier-sets X_c and the maps between them; in other words, the Peircean ladder becomes compatible with ordinary mathematical practices of translating structures across domains (e.g. from perceptual contexts to linguistic contexts) without presuming that the underlying "stuff" is the same.*

6.4.1 A minimal inductive ladder

The ladder below is "minimal" in the sense that each step adds only as much structure as needed to express the corresponding Peircean intuition. In particular, nothing forces Diff_c, V, or $\mathrm{Synergize}_c$ to have any special form beyond what is required for subsequent constructions; the framework is designed so that one may instantiate them with metrics, divergences, logical orderings, probabilistic scores, or other application-specific choices as long as they fit the declared types.

Definition 45 (Firsts)**.** *For a context c, a* First *is an entity $x \in X_c$ considered* without relating it *to anything else. Operationally, it is the unit of description before any distinctions are made.*

Remark 194 (Intuition and examples for Firsts)**.** *A First is the bare "this-such" prior to comparison: a tone as heard before labeling it as higher than another, a patch of red before classifying it as stop-sign, a number-symbol "17" before embedding it into arithmetic structure. In formal terms, it is just an element of X_c, but the point is methodological: we allow ourselves to speak of entities even when no relational judgments have yet been asserted about them.*

This is useful because it separates existence-in-a-context from structure-in-a-context. Many later constructions (weakness, abstraction, patterns) talk about how relations over X_c behave; identifying Firsts as the carriers of later relations keeps the levels clean (Hyperseed-Concept **??***). It is also helpful to read X_c as a* typed *carrier: the context c fixes what counts as an admissible "atom of discourse" at all (pixels, tokens, hypotheses, actions, etc.), and the subsequent ladder then describes increasingly structured ways of speaking about those admissible atoms, rather than silently changing the underlying domain midstream.*

Definition 46 (Seconds)**.** *For a context c, a* Second *is a directed difference event, i.e. an ordered pair $(x, y) \in X_c \times X_c$ together with its directed difference evaluation $\mathrm{Diff}_c(x \to y)$. Seconds are the smallest units at which "reaction" or "opposition" can be represented.*

Remark 195 (Intuition and examples for Seconds)**.** *A Second is what appears when a First meets resistance: "this is not that," "I cannot substitute this for that," "this pushes back." The ordered pair (x, y) records directionality: x can fail to fit y even if y fits x in some aspect-sensitive sense. In a predictive setting, one may read a Second as the primitive discrepancy between a model state and an observation.*

The definition is useful because it makes explicit the granularity at which "difference" lives: not as a global property of a system but as a local, typed datum that can later be aggregated into Thirds (relations) and Fourths (synergistic webs). This aligns with Hyperseed-Concept 163 and Hyperseed-Concept 163. One can think of $\mathrm{Diff}_c(x \to y)$ as an annotation on the arrow $x \to y$: depending on the application, it may be a real-valued cost, an element of an ordered evidence lattice, a probability of mismatch, or a logical witness of incompatibility. The only commitment at this stage is that the evaluation is part of the Second, so that later operations have access to both the fact of pairing *(x, y) and the* strength/character *of the directed contrast.*

Definition 47 (Thirds)**.** *For a context c, a* Third *is a* grouping of Seconds*. Concretely, we model a Third as a (possibly fuzzy) relation*

$$R : X_c \times X_c \to V,$$

where V is a commutative quantale as in Section 3.4. In the simplest crisp case, a Third is just a subset $R \subseteq X_c \times X_c$.

Remark 196 (Intuition and examples for Thirds). *If Seconds are individual dyadic events, then a Third is a policy, a rule, or a stable pattern of such events: "whenever x is of type A and y is of type B, treat them as distinct," or "these pairs are usually interchangeable." In the crisp case, a Third is literally a set of ordered pairs; in the V-valued case it is a graded and potentially paraconsistent assignment of evidence to each pair.*

The usefulness is twofold. First, Thirds are composable: relations can be combined, closed, propagated, and compared—all operations that become central in later discussions of patterns and emergence [5]. Second, Thirds are where weakness/simplicity naturally lives: weakness measures are aggregations over relations of which distinctions are left unmade [3, 2] (Hyperseed-Concept 190 and Hyperseed-Concept 132). It is worth emphasizing why the codomain V is taken to be a quantale: it supports a join operation (for pooling/merging evidence across sources) and a monoidal product (for chaining/propagating relational strength), so that one can later model both "either of these reasons supports the relation" and "these reasons compose along a path" in a uniform algebraic language.

Remark 197 (Relation-as-Third). *This choice is deliberately aligned with the observation that "a relation is a set of ordered pairs grouped together." The mathematical point is that the move from Second to Third is the move from isolated dyadic differences to a reusable* schema *of dyadic differences. This becomes the backbone for later constructions: patterns, rules, causal links, and interpretive frames all live at the level of Thirds. Equivalently, Seconds correspond to* tokens *of interaction, while Thirds correspond to* types *(or stable regularities) extracted from many such tokens; this token/type distinction is one of the simplest ways to see how "lawlike mediation" can be implemented without positing any additional metaphysical substance beyond the data of directed contrasts and their organization.*

Definition 48 (Fourths). *A* Fourth *is a higher-order grouping that joins multiple interconnected Thirds into a larger web in a way that can create emergent "whole is greater than the parts" behavior.*

Formally, let Thi_c denote a chosen class of Thirds (e.g. V-valued relations on X_c with finite support). A Fourth is specified by a finite diagram D in Thi_c (a network of relations together with chosen overlap maps), together with a chosen composite/colimit-like operation

$$\mathrm{Synergize}_c(D) \in \mathsf{Thi}_c.$$

The degree of Fourthness (synergy) of D is then measured by an emergence functional *(defined precisely in Section 9) comparing the "intensity" of* $\mathrm{Synergize}_c(D)$ *to the intensities of the individual relations in D.*

A diagram D should be read as more than a mere set of Thirds: the overlap maps encode how the component relations are intended to *share* variables, subdomains, or intermediate identifications, i.e. where one relation's "output" becomes another relation's "input" or constraint. In concrete applications, $\mathrm{Synergize}_c(D)$ may be instantiated as a closure operation, a pushout/pullback-like construction, an inference step that propagates constraints across the network, or any other principled mechanism that produces a new Third capturing what is jointly implied by the entire configuration. This is also the sense in which the Jungian "mandalic" image is meant: not a claim about symbolism, but a reminder that Fourths are characterized by *global coherence* arising from a structured interlocking of multiple mediations rather than from any single dyadic interaction or single law taken alone.

Remark 198 (Intuition and examples for Fourths). *A Fourth is where "mere aggregation" becomes "organization." Said differently, a Fourth is not just "many Thirds at once" but a* joint constraint system *in which the participating Thirds co-determine what counts as admissible structure. If each Third is a relational lens (a rulebook, constraint-set, or association pattern), then a Fourth is a structured conjunction of lenses whose interaction yields consequences none had in isolation. In particular, the interaction is not merely logical conjunction at the level of predicates; it includes a* chosen mode of interaction *(a way the lenses compose, interfere, reinforce, or propagate along each other).*

In a toy model, take two relations R_1 and R_2 and close them under a composition rule; the closure may entail new pairwise links, and those entailments are precisely the mathematical shadow of "emergence." Concretely, if R_1 and R_2 are viewed as permitted transitions, then closing under composition asserts that multi-step transitions are also permitted, producing new edges (x, z) from paths $(x, y) \in R_1$ and $(y, z) \in R_2$. Even when R_1 and R_2 are each individually sparse or locally meaningful, their combined closure can produce global connectivity patterns, equivalence classes, or invariants that were not readable from either relation alone.

This definition is useful because it is flexible while still principled: by making a Fourth a diagram plus a chosen universal-like composition, we ensure that Fourthness is not a vague adjective but a reproducible construction. The "diagram" component specifies which *Thirds are present and how they overlap or are glued; the "composition" component specifies* how *information is allowed to flow across that overlap. As a result, Fourthness can be tested by checking whether the composite object supports new entailments, new conserved quantities, or new effective rules that are stable under the chosen composition.*

It directly anticipates later pattern/emergence formalisms (Hyperseed-Concept **??** *and Hyperseed-Concept* **??***) and aligns with Hyperseed's emphasis on synergy as a core engine of intelligence [19]. In that framing, Fourthness is the minimal formal layer at which "synergy" becomes something we can compute, compare across contexts, and reason about via functorial or closure properties rather than by informal appeal to holism.*

Remark 199 (Why the definition is "diagram + composition")**.** *Hyperseed's description of Fourthness emphasizes* embedding *a local network of relations into a broader web. This emphasis is important: the same local Third can behave very differently when placed inside different ambient relational environments, because the ambient links provide additional paths of inference, constraint propagation, or coordination. In category theory, "a local network of composable pieces plus an embedding into a larger web" is naturally represented by a diagram and a universal construction (colimit, pushout, Kan extension, etc.). The diagram records the local pieces and their interface maps, while the universal construction enforces that the resulting composite is the* most general *object compatible with those interfaces (or, depending on direction, the* best approximation *living in a target category). This is precisely the categorical way to ensure that the composite object depends on the parts only through their specified relationships, rather than through arbitrary presentation artifacts.*

We keep the definition abstract here because different applications suggest different concrete choices of diagram shape and composition law. For example, when relations represent constraints, a colimit-like construction captures gluing constraints together; when relations represent observations, a Kan extension-like construction can express how local observations induce global predictions; and when relations represent rewrites or proofs, closure under a proof calculus implements the intended notion of inferential completion. The intended invariance is: if two diagrams are isomorphic (or equivalent up to the relevant notion of equivalence in context), their synergized outputs should be correspondingly equivalent, so that Fourthness behaves stably under re-indexing and re-description.

Remark 200 (Notation clarification: Thi_c and D)**.** *The symbol* Thi_c *is not a new axiom but a placeholder for "whatever collection of Thirds we are willing to manipulate in context c" (e.g. finite-*

support relations, measurable relations, etc.). The point of leaving Thi_c *schematic is that Fourthness should not depend on a single privileged substrate: one can instantiate Thirds as binary relations, weighted relations, stochastic kernels, logical theories, constraint graphs, or other relational objects, provided the context supplies a sensible notion of combination. In practice, the context c encodes what counts as an entity, what counts as a relation over entities, and what operations (composition, closure, saturation) are admissible or computable.*

A "finite diagram D in Thi_c*" means a finite indexing shape (a small finite category) together with an assignment of an object of* Thi_c *to each node and overlap maps along arrows. Here "finite" is primarily a technical convenience: it guarantees that the specified pattern of interaction is bounded and can be treated as a discrete schema, which is particularly relevant for computational implementations and for avoiding size issues. The "overlap maps" should be read as interface constraints: they specify how one Third is to be matched against another (e.g. shared variables, shared subrelations, restriction maps, or inclusion of a common substructure).*

The operation $\mathrm{Synergize}_c$ *is then a chosen way to combine the diagram into a single Third. It is useful to think of* $\mathrm{Synergize}_c$ *as a context-dependent "completion" or "fusion" operator: it takes multiple partial lenses and returns a single lens that reflects what is jointly implied once they are allowed to interact by the chosen rule. Depending on the application,* $\mathrm{Synergize}_c$ *may be idempotent (a closure operator), may be monotone with respect to refinement of diagrams, and may satisfy naturality properties under change of context, but these are additional structure one may impose later rather than assumptions built into the notation.*

6.4.2 A concrete set-theoretic model (optional)

Example 8 (Set-theoretic First–Fourth ladder)**.** *Fix a context c with entity set X_c. Define*

$$\begin{aligned}
\mathsf{First}_c &:= X_c,\\
\mathsf{Second}_c &:= X_c \times X_c,\\
\mathsf{Third}_c &:= \mathcal{P}(X_c \times X_c),\\
\mathsf{Fourth}_c &:= \mathcal{P}(\mathsf{Third}_c) \ \textit{(families of relations)}.
\end{aligned}$$

This ladder is intentionally coarse-grained: it treats a Third as a plain subset of pairs, so that the only built-in structure is membership. Even so, it already separates (i) entities, (ii) raw dyadic interactions, (iii) stabilized patterns of interaction (relations), and (iv) ensembles of such patterns.

A "synergy" operator might map a family of relations $\{R_i\}$ to their closure under a composition rule (e.g. relational composition, transitive closure, or closure under a proof calculus). For instance, one may define

$$\mathrm{Synergize}_c(\{R_i\}) \ := \ \mathrm{cl}\Big(\bigcup_i R_i\Big),$$

where cl *is a chosen closure operator (transitive closure, reflexive-transitive closure, or a domain-specific saturation procedure). The key point is that the closure is not merely union: it adds whatever the rulebook says must follow from jointly having those relations available.*

The resulting closure typically contains entailments that were not present in any single R_i. In graph language, new edges appear because multi-edge paths become single-step consequences under the closure rule; in logic language, new theorems appear because premises from different R_i can be combined in a single derivation. This is the simplest rigorous picture behind "the whole is greater than the parts." It also exhibits a minimal notion of "organization": the output relation can encode higher-level regularities (e.g. connectivity components, reachability, or induced equivalences) that effectively act like emergent macroscopic features of the system.

Remark 201 (Why include the set-theoretic model)**.** *The optional ladder is a sanity anchor: it shows that the definitions do not require higher category theory to be meaningful. It also makes explicit which step introduces which kind of structure: moving from* Second_c *to* Third_c *is the shift from* individual interactions *to* patterns *of interaction, while moving from* Third_c *to* Fourth_c *is the shift from* one pattern *to* interacting patterns*. Even in this elementary setting, the conceptual point survives: Fourthness corresponds to closure/interaction effects among multiple relations. Moreover, one can already ask meaningful questions such as: when does synergizing preserve symmetry, when does it generate equivalence relations, and when does it create long-range dependencies that were absent from the input family?*

This is also the simplest place to test computational toy models before lifting to enriched or higher-categorical settings [5]. For example, one can implement $\mathrm{Synergize}_c$ *as a saturation loop, measure the growth of the closure as a function of input family size, and compare different closure rules as different hypotheses about what kind of "synergy" a given domain supports. Such experiments help distinguish effects that are artifacts of representation from effects that persist under more abstract formulations.*

6.5 Presentational immediacy and intensity

Hyperseed treats presentational immediacy as a primitive correlate of "raw consciousness": some occasions are foreground, others background, and this distinction is fuzzy. We model this fuzziness as a context-indexed salience (intensity) field. In particular, the index c is meant to cover whatever factors jointly determine "what it is like right now": task demands, bodily state, sensory modality, memory activation, and ongoing control state. Nothing in this subsection presumes that the salience assignment is stable across time, agents, or tasks; it is explicitly permitted (and expected) that I_c changes as the context changes.

Definition 49 (Presentational immediacy)**.** *A* presentational immediacy *function in context c is a map*

$$I_c : X_c \to [0,1].$$

If $I_c(x)$ is near 1, x is in the foreground of experience in c. If $I_c(x)$ is near 0, x is in the background.

The choice of codomain $[0,1]$ is a normalization convention rather than an ontological claim that salience is intrinsically bounded. It allows comparisons within a context and makes later constructions (e.g., thresholds, resource budgets, and weighted graphs) notationally simple. Importantly, I_c is not intended to be a probability, and no additivity constraints are imposed; multiple entities may simultaneously have high immediacy. Similarly, we do not require $\max_{x\in X_c} I_c(x) = 1$ (although one may impose such a gauge-fixing in specific models), because contexts may differ in overall "vividness" or saturation.

Remark 202 (Intuition and examples for presentational immediacy)**.** *The function I_c is the formal shadow of the commonplace fact that experience has a figure/ground structure. While staring at a page, the letters are foreground; the pressure of one's feet is fringe; the hum of distant traffic may be nearly absent. The map $I_c : X_c \to [0,1]$ is intentionally modest: it asserts only that a context can assign graded salience to its presented entities.*

This definition is useful because it provides a primitive quantity that later theories can explain rather than assume. In later sections, intensity will be connected to effort, resource allocation, and attention dynamics (Hyperseed-Concept 138 and Hyperseed-Concept 60; see also [19] for related cognitive resource themes).

A further virtue of treating immediacy as a field over X_c is that it can, in principle, be applied uniformly across heterogeneous kinds of presented entities. For example, X_c may contain perceptual items (a letter-shape), interoceptive signals (heart rate sensations), affective tones (unease), action affordances ("turn the page"), and memory fragments (a recalled phrase). The same formalism covers all of these without committing to a single representational format. This is one reason the definition is kept at the level of a simple map rather than, say, a modality-specific vector or a hand-built attentional structure.

Remark 203 (Foreground/background is not (yet) attention)**.** *Later (Section 8 and onward) we will connect intensity to effort/energy expenditure and obtain attention as a dynamical/resource concept. Here we keep intensity primitive: it is an observational datum about what is "present" and with what degree.*

It is also useful to distinguish immediacy from related but nonidentical notions that often travel under the same name in informal discussions. For instance, an item may be highly immediate because it is perceptually intrusive (a loud bang), even if it is not selected for deliberate manipulation or report; conversely, an item may be the target of deliberate control (a rehearsed intention) while remaining phenomenologically "thin" compared to vivid sensory input. By separating (i) a phenomenological grading I_c from (ii) later dynamical and computational mechanisms, Hyperseed leaves room for models in which selection, report, working memory access, or executive control can dissociate from subjective vividness.

Definition 50 (Focus and fringe)**.** *Given a threshold* $\tau_I \in (0,1)$*, define the* focus set

$$\mathrm{Focus}_c(\tau_I) := \{x \in X_c : I_c(x) \geq \tau_I\},$$

and the fringe set

$$\mathrm{Fringe}_c(\tau_I) := \{x \in X_c : 0 < I_c(x) < \tau_I\}.$$

These are coarse operationalizations of the fuzzy foreground/fringe distinction.

Note that $\mathrm{Fringe}_c(\tau_I)$ explicitly excludes entities with $I_c(x) = 0$. This leaves open two common interpretations, both compatible with the present formalism: either (i) $I_c(x) = 0$ designates entities not presented at all in context c (outside the lived field), or (ii) it designates entities presented so weakly that they are functionally negligible for the purposes at hand. Which interpretation is appropriate can be fixed later when X_c and the mechanisms generating I_c are specified more concretely.

Remark 204 (Intuition and examples for focus/fringe thresholding)**.** *The threshold τ_I is a pragmatic bridge between graded phenomenology and discrete computation. For instance, a system may allocate symbolic reasoning only to elements in* $\mathrm{Focus}_c(\tau_I)$ *while allowing statistical/background processes to operate on* $\mathrm{Fringe}_c(\tau_I)$*. Different tasks warrant different thresholds; this is one reason Hyperseed keeps thresholding explicit rather than implicit.*

The definition is useful because many later constructions (graphs of salient entities, resource budgets, bounded planning) require finite or crisp sets even when the underlying salience is continuous. This is a standard move in cognitive architectures [19].

In practice, one may treat τ_I as a tunable interface parameter between phenomenology and computation. A high threshold yields a narrow "spotlight" that supports fast, low-branching deliberation; a lower threshold yields a broad set that supports integrative or exploratory processing at the cost of increased combinatorics. Because τ_I is explicit, the theory can later describe adaptive policies (e.g., lowering τ_I under uncertainty, raising it under time pressure) without changing the underlying meaning of I_c. One can also define multiple thresholds (e.g., $\tau_I^{(1)} > \tau_I^{(2)}$) to obtain layered bands of salience, but the single-threshold version already captures the minimal coarse-graining needed for many constructions.

Finally, although the present definitions are pointwise, they are compatible with temporal refinements. If contexts are indexed by time (or contain a time parameter), then c and c' at nearby moments may induce fields I_c and $I_{c'}$ whose changes describe shifts of lived salience. Later dynamical models can then constrain the rate or cost of such shifts (e.g., inertia in salience, hysteresis under fatigue) while preserving the basic primitive notion introduced here.

6.6 Abstract and concrete

Hyperseed distinguishes abstract experiences (e.g. the number 17) from concrete experiences (e.g. kicking a brick), while emphasizing that the same "content" may sometimes be experienced with more concreteness under unusual states. For a rigorous ontology, we need a notion of abstractness/concreteness that is: (i) context-relative, (ii) compatible with graded intensity, and (iii) compatible with the weakness/simplicity formalism. In particular, "abstract vs. concrete" here is not treated as a property of an isolated symbol or proposition, but as a property of how some collection of experienced tokens is organized *within a context* (what distinctions the context makes salient, which equivalences it tolerates, and which discriminations it refuses to carry out). This is why the definitions below are phrased in terms of maps out of X_c, rather than in terms of an intrinsic "abstractness score" assigned to a content in isolation. The graded aspect (ii) will be represented indirectly: one can move along chains in the abstraction preorder, or consider degrees of refinement/coarsening, rather than requiring a single scalar that must fit every use-case.

Remark 205. *This section's definitions deliberately echo a long-standing mathematical habit: abstraction is quotienting. To abstract is to identify; to concretize is to refuse identification unless constraints are satisfied. The point is not to reduce phenomenology to set theory, but to give a clean interface between lived categories (abstract/concrete) and later formal tools (weakness, effort, compression). See Hyperseed-Concept 51 and Hyperseed-Concept 83. A practical reading is: an "abstract experience" is one in which many token-level variations are treated as irrelevant (and therefore collapsed), whereas a "concrete experience" is one in which the system maintains, attends to, or is compelled by distinctions among tokens that might otherwise be quotiented away. On this view, unusual states can make the same nominal content feel more concrete by effectively refining the relevant context c (enlarging X_c and/or making finer discriminations available), so that fewer identifications are tolerated without experiential "error" or friction.*

6.6.1 Abstraction as quotienting of distinctions

Definition 51 (Abstraction as a quotient)**.** *Let c be a context with entity set X_c. An* abstraction *of c is a surjection*

$$q : X_c \twoheadrightarrow A$$

to a set of abstracta A. The abstraction collapses distinctions by identifying elements of the same fiber. Equivalently, q determines an equivalence relation

$$x \sim_q y \iff q(x) = q(y).$$

In this sense, the abstractum $a \in A$ can be read as the "type" corresponding to the entire equivalence class $q^{-1}(a) \subseteq X_c$ of concrete tokens that are treated as interchangeable for the purposes of the context.

Remark 206 (Intuition and examples for abstraction-as-quotient)**.** *The surjection $q : X_c \twoheadrightarrow A$ says: "many concrete tokens map to one abstract type." For example, take X_c as handwritten glyphs and let A be the set of letters; then q sends each glyph to the letter it instantiates, collapsing distinctions among different handwriting styles. Or let X_c be physical trajectories of balls and let A be a coarse state descriptor (position rounded to a grid), collapsing micro-differences.*

This definition is useful because it expresses "collapsing distinctions" in a way that composes: two abstractions can factor through one another, and the induced equivalence relations can be ordered. It connects directly to weakness: collapsing more distinctions increases undistinguished pairs, hence increases weakness in the sense used elsewhere [3, 2] (Hyperseed-Concept **??** *and Hyperseed-Concept 202). One can also read q as a formalization of an* invariance*: the map forgets differences along directions the context treats as irrelevant. In the glyph example, stylistic variation (stroke thickness, slant) becomes an "ignored degree of freedom," while in the trajectory example, micro-physical perturbations become ignorable, leaving only a coarse descriptor. This aligns with the phenomenological claim that abstraction often feels like "stability under variation": what is preserved by q is what the experience treats as the same.*

Definition 52 (Abstraction preorder)**.** *Given abstractions $q_i : X_c \twoheadrightarrow A_i$, define*

$$q_1 \preceq q_2 \quad \iff \quad \exists\, r : A_1 \to A_2 \text{ with } q_2 = r \circ q_1.$$

Thus q_2 is at least as abstract as q_1 when it factors through q_1 (i.e. collapses at least as many distinctions). Equivalently, $q_1 \preceq q_2$ means that every q_1-fiber is contained in a q_2-fiber, so the partition of X_c induced by q_2 is a coarsening of the partition induced by q_1.

Remark 207 (Intuition and examples for the abstraction preorder)**.** *The preorder $\preceq$ formalizes "is no less abstract than." If $q_2 = r \circ q_1$, then whatever identifications q_1 makes, q_2 makes them as well (and possibly more). For instance, in image recognition one might first quotient by viewpoint to get object identity (q_1), and then quotient object identities by category to get "animal vs. vehicle" (q_2). Then $q_1 \preceq q_2$ because category forgets more.*

This is useful because it gives a rigorous notion of abstraction order *without requiring a numeric scale. Later, when we attach cost/effort to abstraction, the preorder provides the qualitative backbone for those quantitative refinements. It is also useful that $\preceq$ is only a preorder: two different surjections can induce the same equivalence relation (e.g. via different codomains A_i that merely relabel classes), so antisymmetry need not hold at the level of functions even when the induced partitions are identical. If desired, one can pass to equivalence classes of abstractions under mutual factorization, yielding a partial order isomorphic to the lattice of partitions (or equivalently, of equivalence relations) on X_c. This observation is conceptually helpful later when "graded concreteness" is implemented by moving between finer and coarser partitions, rather than by changing the underlying set of tokens.*

Proposition 4 (Abstraction increases weakness)**.** *Assume a context c supplies an "undistinguished pairs" relation $H_c \subseteq X_c \times X_c$. If $q_1 \preceq q_2$ are abstractions and H_{q_i} denotes the induced relation that treats all $\sim_{q_i}$-equivalent pairs as undistinguished, then*

$$H_{q_1} \subseteq H_{q_2} \qquad \Longrightarrow \qquad w(H_{q_1}) \leq w(H_{q_2}).$$

Remark 208 (What the proposition says, and why it matters)**.** *This proposition states that "more abstraction" (in the sense of identifying more things) produces "more weakness" (in the sense of leaving more pairs undistinguished), provided the induced undistinguished-pairs sets are nested.*

Intuitively, if you agree to treat more pairs as the same, you have deliberately declined to make distinctions; weakness is designed to increase exactly in that situation [3, 2].

The result connects this section's abstract/concrete interface to Section 4, where monotonicity properties justify interpreting $w(H)$ as a coherent weakness measure. Here we see a concrete instantiation: quotienting creates inclusion of undistinguished-pair sets, and monotonicity of weakness carries the inclusion to an inequality. Phenomenologically, this captures a common pattern: as an experience becomes more abstract, it becomes "easier to satisfy" because fewer discriminations are demanded; conversely, concrete experience imposes more specific constraints (more opportunities for mismatch), which corresponds to fewer undistinguished pairs and hence lower weakness. Note that the antecedent $H_{q_1} \subseteq H_{q_2}$ is presented explicitly to keep the statement modular: in some formalisms H_{q_i} may include both the base relation H_c and the identifications induced by q_i, and the inclusion condition is then the clean hypothesis under which monotonicity of $w(\cdot)$ applies without committing to a particular construction of H_{q_i}.

Proof. The factorization condition $q_1 \preceq q_2$ implies that $\sim_{q_2}$ is coarser than $\sim_{q_1}$, hence any pair identified by q_1 is also identified by q_2, giving $H_{q_1} \subseteq H_{q_2}$. The conclusion then follows from Theorem 1. More explicitly: if $x \sim_{q_1} y$ then $q_1(x) = q_1(y)$, so $q_2(x) = r(q_1(x)) = r(q_1(y)) = q_2(y)$, hence $x \sim_{q_2} y$; therefore every $\sim_{q_1}$-identified pair is also $\sim_{q_2}$-identified, which is exactly the coarsening relationship needed to apply monotonicity of w to the induced undistinguished-pairs relations. $\square$

Remark 209 (Concretization as refinement (informal))**.** *Although the formal development here is phrased in terms of abstraction maps $q : X_c \twoheadrightarrow A$, the opposite motion—toward concreteness—can be read as moving to a* finer *quotient (i.e. reversing $\preceq$), or equivalently as refining the induced partition of X_c so that fewer tokens are treated as interchangeable. In this sense, "concretizing" does not require introducing a new primitive beyond the preorder: it can be modeled as selecting an abstraction q' with $q' \preceq q$ replaced by $q \preceq q'$ failing, i.e. by choosing q' that distinguishes more. This dovetails with graded intensity (ii): to become "more concrete" is not necessarily to jump categories, but to traverse a chain of refinements, each step adding discriminations that the context now treats as salient.*

Remark 210 (Proof sketch and commentary)**.** Proof sketch. *Factorization means that q_2 cannot distinguish anything that q_1 already identifies, so the equivalence classes (fibers) of q_2 are unions of fibers of q_1. Equivalently, every identification made by q_1 is* preserved *by q_2, because q_2 only has access to information that already passed through q_1. Hence the set of pairs declared equivalent by q_1 is contained in the set declared equivalent by q_2. In other words, $\sim_{q_2}$ is at least as coarse as $\sim_{q_1}$: it can collapse further, but it cannot "un-collapse" any pair that q_1 has already merged. Weakness is monotone under adding undistinguished pairs, so the inclusion yields the inequality. This monotonicity can be read operationally: if a quotient is allowed to forget* more *distinctions (i.e. it identifies more pairs), then it can only become weaker in the sense of losing discriminative or task-relevant structure.* $\square$

The key step is the passage from $q_2 = r \circ q_1$ to "$\sim_{q_2}$ is coarser than $\sim_{q_1}$." This is simply the observation that if $q_1(x) = q_1(y)$ then automatically $q_2(x) = r(q_1(x)) = r(q_1(y)) = q_2(y)$. It is sometimes helpful to rephrase this in terms of kernels: the relation $\sim_q$ is the kernel congruence of q, and composition enlarges kernels, i.e. $\ker(q_1) \subseteq \ker(r \circ q_1)$. Geometrically, quotienting can be pictured as merging points into blobs; factoring corresponds to merging blobs into larger blobs, so the set of merged pairs can only grow. One can also view the map r as defining a partition of the q_1-blobs, after which q_2 assigns the same label to all blobs in each part of that partition.

6.6.2 Concreteness as resistance to quotienting

The previous definitions make "more abstract" precise as "collapses more distinctions." In particular, abstraction is measured by how large a family of potentially different entities is treated as interchangeable once mapped through q. To model concreteness we add a complementary notion: some entities resist being safely quotiented away. Here "resist" is not metaphysical stubbornness but a contextual fact: in some practices, merging certain items produces errors, surprises, or loss of control. We represent this as a context-indexed constraint on admissible abstractions. The role of the context index c is to keep track of which invariances are acceptable: the same physical entity may be concrete for one task and abstract for another.

Definition 53 (Concreteness constraint)**.** *Let c be a context. A* concreteness constraint *is a predicate $\mathcal{K}_c$ on abstractions $q : X_c \twoheadrightarrow A$ specifying which quotientings count as "benign". The point of treating $\mathcal{K}_c$ as a predicate (rather than a single privileged quotient) is that there may be many acceptable ways to abstract, and concreteness is meant to be robust across that admissible family. An entity $x \in X_c$ is* concrete (relative to c) *if, for every admissible abstraction q in the sense of $\mathcal{K}_c$, the fiber $q^{-1}(q(x))$ is "small" (i.e. x is not merged with many other entities without violating $\mathcal{K}_c$). Here "small" is intentionally left schematic: depending on the application it may mean small cardinality, small measure, bounded diameter under a similarity metric, or small expected variability in downstream predictions. Conversely, x is* abstract *if there exist admissible abstractions under which x represents a large equivalence class. This existential clause captures the idea that an entity can function as a* representative *or* type *when the context licenses aggressive identification, even if other contexts would forbid it.*

Remark 211 (Intuition and examples for concreteness constraints)**.** *A concreteness constraint $\mathcal{K}_c$ encodes what a context will not allow itself to forget. Equivalently, $\mathcal{K}_c$ expresses which distinctions are treated as non-negotiable because collapsing them would damage some capability the context aims to preserve. For a robotic context, $\mathcal{K}_c$ might forbid quotienting that breaks sensorimotor prediction: two objects cannot be identified if pushing one produces different outcomes than pushing the other. More generally, any abstraction that would destroy controllability or introduce systematic prediction error would be deemed non-benign. For a mathematical context, $\mathcal{K}_c$ might be looser: many concrete instances can be safely quotiented into a single abstract structure if all proofs and computations of interest are invariant under that quotient. For example, different presentations of the same group may be quotiented together when the task concerns only group-theoretic properties, but not when the task concerns a specific representation or algorithmic cost.*

This definition is useful because it prevents a degenerate triviality: without constraints, maximal abstraction is always possible. Indeed, the quotient map that sends all of X_c to a single point is always available set-theoretically, but it is rarely benign relative to any nontrivial practice. By making admissibility explicit, we keep the abstract/concrete distinction tethered to practice and consequence. Later sections can propose specific quantitative forms of $\mathcal{K}_c$ based on effort or predictive utility, connecting to Hyperseed-Concept 100 and Hyperseed-Concept **??***. One can anticipate, for instance, that $\mathcal{K}_c(q)$ might require that some loss functional (prediction loss, control loss, or explanation loss) remain below a context-dependent threshold.*

Remark 212 (Why a constraint is necessary)**.** *If we allowed arbitrary quotient maps, every entity could be collapsed into a single point, making everything maximally abstract. This would trivialize any attempt to compare abstractions, because there would always exist a strictly coarser quotient that "wins" by collapsing more. Hyperseed's distinction between abstract and concrete presupposes that not all quotientings are experienced as legitimate. The predicate $\mathcal{K}_c$ stands in for whatever makes a quotient "valid" in a context: sensorimotor coupling, causal consequences, pragmatic stakes, or* *other constraints. In practice, $\mathcal{K}_c$ can be thought of as delimiting a feasible set of compressions: compressions outside this set are experienced as errors, category mistakes, or harmful oversimplifications. Later sections will propose quantitative candidates for $\mathcal{K}_c$ in terms of effort and predictive utility.*

6.7 Non-duality as a bridge between perspectives

Hyperseed treats non-duality as primitive and also as an organizing constraint relating (at least) first-person and third-person framings. We record one clean mathematical way to express "non-duality without collapse" using the language of context translation. In particular, the aim is to state a condition that is strong enough to prevent a translation from trivializing distinctions, while still weak enough to permit genuine coarse-graining and abstraction (as is unavoidable when moving between descriptive standpoints).

Remark 213. *In philosophical terms, this is a proposal to treat "bridging perspectives" as a problem in approximate structure-preservation rather than exact identity. One does not demand that first-person and third-person descriptions coincide; one asks instead for translations that preserve what must be preserved, blur what may be blurred, and make the extent of blurring legible as a parameter. This resonates with process-oriented and sign-oriented metaphysics (Whitehead and Peirce) in which relations and mediations are primary [15, 14]. It also anticipates later Hyperseed themes of mind–world correspondence and self/other boundaries (Hyperseed-Concept 112 and Hyperseed-Concept 165). A further motivation is methodological: if one treats "first-person" and "third-person" as two contexts with different observational capacities, then any realistic bridge must allow information loss in at least one direction, yet should still provide an auditable account of what was preserved and what was sacrificed.*

Definition 54 (Context translation and non-dual compatibility)**.** *Let c, d be contexts. A* translation *from c to d is a function $T : X_c \to X_d$. We say that T is* non-dually compatible *with c and d if for all $x, y \in X_c$ we have*

$$\delta_c(x, y) \le \delta_d\big(Tx, Ty\big) \quad \textit{and} \quad \delta_c(x, y) \le \delta_d\big(Ty, Tx\big)$$

(componentwise order on $\mathbb{V}$), i.e. d does not erase distinctions that are strong in c. Dually, a translation $S : X_d \to X_c$ is compatible if it does not create spurious distinctions.

A non-dual bridge *between c and d is a pair of translations (T, S) together with a witness that the composites $S \circ T$ and $T \circ S$ act like identities only* up to controlled weakness *(to be made precise using the quantale V and weakness ordering).*

One may read the compatibility inequalities as a minimal "no-forgetting" requirement: if $\delta_c(x, y)$ certifies that x and y are distinguishable in context c (in whichever multivalued sense $\mathbb{V}$ encodes), then translating to d should not make them *less* distinguishable. Because $\mathbb{V}$ is ordered componentwise, each component of δ (e.g. evidential strength, salience, modality-specific discriminability) is protected separately; thus a translation is allowed to blur some components only insofar as the order relation permits, and any such blurring is tracked at the level of $\mathbb{V}$ rather than hidden in informal interpretation.

Remark 214 (Intuition and examples for translations and bridges)**.** *A translation $T : X_c \to X_d$ is a recipe for re-expressing what c can talk about in the language of d. Non-dual compatibility demands that if c has strong evidence that x and y diverge, then d must not flatten that divergence after translation. In other words, T must not turn strong distinctions into weak ones. The symmetric-looking condition with (Tx, Ty) and (Ty, Tx) ensures this protection of distinction in both directions,* *matching the earlier emphasis on mutuality for distinctions. A useful way to think about this is that even when δ is not literally symmetric (e.g. if it encodes directional evidential flow, or asymmetric accessibility), the bridge should not depend on an arbitrary choice of ordering of the pair (x, y) when deciding whether a distinction has been preserved.*

A simple example is mapping a fine-grained introspective context (many subtle feelings) into a coarse behavioral context (few categories). A compatible translation cannot claim two feelings are indistinct if, in the introspective context, the evidence for their distinctness is high. The notion of a non-dual bridge then says: even if round-tripping $x \mapsto Tx \mapsto S(Tx)$ does not return exactly x, it returns something that is "the same up to controlled weakness," making the loss of detail explicit and measurable via weakness. This is the formal analogue of "non-duality without collapse" (Hyperseed-Concept 121 and Hyperseed-Concept 143; see also [3]).

A second example (closer to scientific practice) is translating from a richly parameterized dynamical model to a low-dimensional summary statistic used for prediction or control. The translation to the summary statistic is permitted to discard degrees of freedom, but it should not erase a distinction that the original model treats as robust. Conversely, the translation back from the statistic to a representative dynamical state should not hallucinate fine distinctions unsupported by the statistic, i.e. it should not "over-interpret" the coarse description as if it contained more information than it does.

In both examples, the "witness" for the bridge can be understood as a formal certificate of how far $S \circ T$ and $T \circ S$ deviate from identity in the weakness order. Concretely (at an intuitive level), one expects such a witness to amount to inequalities of the form "x is close to $S(Tx)$ up to an allowed weakening" and likewise "u is close to $T(Su)$ up to an allowed weakening," where "close" is expressed using δ and the allowed weakening is expressed using the quantale structure on V. Thus, the bridge does not pretend that translations are invertible; it instead makes non-invertibility quantitative and therefore discussable.

Remark 215. *The point of this definition is not to force an equivalence between first-person and third-person descriptions. Rather, it provides a formal knob: a bridge can preserve some distinctions while allowing others to blur, and non-dual states are those where the blurring is substantial but non-trivial structure remains. This prepares the ground for later sections where self/other boundaries, intentionality, and "mind–world correspondence" are modeled as approximate morphisms rather than strict identities. In particular, the framework makes room for the possibility that different contexts are related by many admissible bridges, not a unique privileged one; the choice of bridge can then be treated as part of the modeling assumptions, and the witness explicitly records the cost of that choice in terms of weakness. This is one way to reconcile the phenomenological insistence that perspectives are irreducible with the scientific insistence that perspectives can nonetheless constrain one another.*

7 Order, time, and becoming

Outline

- Formalize after/before as strict partial orders (and graded/paraconsistent refinements).
- Define proto-time as "time before physics-time": any experienced partial order.
- Define linear time axes as serializations/quotients/approximations of proto-time.
- Define becoming as boundary non-duality between successive occasions/entities.

- Give a paraconsistent, quantale-valued event calculus as a canonical "time reasoning" layer.
- Define event regularity predicates: persistent, continuous, increasing, decreasing, via time-context similarity.

Summary and Hyperseed concepts covered

Hyperseed treats time as built from ordering. The most primitive ingredient is an experienced after/before relation, i.e. a family of asymmetric distinctions whose transitive closure yields a (strict) partial order. Any such partial order is a *proto-time.* A linear time axis is a special case (or approximation) that serializes events; this can lose information when concurrency or branching is present.

The section then connects this ordering-first viewpoint to concrete temporal reasoning formalisms. We define a paraconsistent, quantale-valued event calculus in which predicates such as Happens and HoldsAt take values in the p-bit quantale V from the formal core. Finally, we formalize Hyperseed's qualitative temporal predicates (persistence, continuity, increasingness, decreasingness) using a notion of time-context and temporal similarity.

Hyperseed concepts covered.

- After / Before; $A < B$ implies NOT($B < A$); NOT($A < A$); transitivity.
- Proto-Time; linear time axis (serialization); "information loss under linearization".
- Becoming (boundary non-duality / non-dual variety).
- Event and process calculi; events; fluents; holds/initiate/terminate; "holds sometime in".
- Persistent / continuous / increasing / decreasing (events or fluents).
- Time context; temporal similarity.

Remark 216 (Orientation and sources)**.** *The present formalization is an intentionally austere extraction of the "ordering-first" stance on temporality used throughout the Hyperseed ontology [1]. The guiding thought is Russellian in its analytic economy: if we can explain "before/after" without presupposing a global time coordinate, we obtain a foundation that can accommodate branching, concurrency, subjective ordering, and inconsistent evidence. At the same time, the section keeps open a Whiteheadian reading in which temporal structure is a derived regularity of processes rather than a primitive container [15].*

The key methodological move is to treat "temporal talk" as downstream of a more basic relation of order. Concretely, an agent (or a theory) may register multiple local discriminations of the form "A is after B" without thereby committing to a single global clock or even to a total ordering. The strictness assumptions in the bullet list can be read as idealizations: irreflexivity NOT($A < A$) and asymmetry $A < B \Rightarrow$ NOT($B < A$) express that "after" is not meant as mere difference, while transitivity captures the compounding of experienced precedence. In practice, the motivating data may be fragmentary, cyclic, or in tension (e.g. multiple observers, noisy sensors, retrospective reconstruction), which is why the section explicitly anticipates graded/paraconsistent refinements rather than insisting on classical consistency from the start.

Calling any such partial order a *proto-time* is meant to separate two notions that are often conflated: (i) having an ordering at all, and (ii) having a privileged metric parameter with uniform units. Proto-time in this sense is compatible with incomparability (neither $A < B$ nor $B < A$), which operationally corresponds to concurrency, causal independence, or simply missing evidence. It is also compatible with branching structures, where a single event is before two mutually incomparable events, capturing divergent possible continuations or genuinely parallel processes. This emphasizes that proto-time is not merely "early physics" but rather "pre-metric order"—a notion that can appear in phenomenology, computation traces, causal graphs, and narrative accounts.

The phrase "linear time axis" is then treated as an additional construction on top of proto-time. A serialization can be understood as selecting (or synthesizing) a total order that is consistent with (some portion of) the partial order, for example by choosing a linear extension when one exists, or by imposing tie-breaking conventions when it does not. A quotient/approximation perspective is also useful: if many events are only distinguishable up to simultaneity or resolution limits, one may collapse equivalence classes and order the classes instead. Either way, "information loss under linearization" is not merely a philosophical slogan: incomparabilities and branching structure can be erased when represented on a single axis, so that distinct proto-times may project to the same linearized representation.

The concept of *becoming* is positioned as a bridge between ordering and ontology: if we speak of successive occasions/entities, the boundary between them is not treated as a hard cut but as a non-dual interface in which "earlier" and "later" partially interpenetrate. In this framing, becoming is not an extra event occurring *in* time; rather, it is the structural relation that makes a succession intelligible as a succession of occasions at all. This is why becoming is stated as "boundary non-duality": it names the continuity of inheritance/transition even when the order relation itself is strict.

Finally, the event-calculus layer is introduced as a disciplined way to reason with proto-time when evidence is inconsistent or graded. Interpreting $\mathrm{Happens}(e, t)$ and $\mathrm{HoldsAt}(f, t)$ as V-valued predicates allows the theory to track not only whether an event happens or a fluent holds, but *to what extent* and under what kind of logical status in p-bit. In particular, the quantale structure supports aggregating temporal evidence (via its join/meet structure) and sequencing/composition of supports (via its monoidal operation), which is essential when temporal claims are derived from multiple partial observations. On this basis, regularity notions such as persistence and continuity can be cast as constraints over families of contexts: a fluent is persistent when its V-valuation remains stable across sufficiently similar neighboring contexts, continuous when changes are small relative to a similarity threshold, and increasing/decreasing when there is a directional trend relative to the proto-time order. The explicit use of time-context and temporal similarity is meant to keep these predicates qualitative and robust, so that they remain meaningful even when no globally consistent linear clock is available.

7.1 After/before as strict partial order

Hyperseed's first move is to treat time as a special case of ordering. Mathematically, the minimal clean object capturing this is a *strict partial order*. This choice deliberately separates the *direction* of precedence from any further commitments about metric structure (how long something takes), topology (continuity), or global synchrony (a single shared clock). In particular, it keeps open whether time is discrete or continuous, whether it branches, and whether every pair of events admits comparison.

Definition 55 (Strict partial order)**.** *A* strict partial order *on a set T is a binary relation $<$ on T such that:*

1. (Irreflexive) for all $t \in T$, not$(t < t)$, *2. (Transitive) for all $t_1, t_2, t_3 \in T$, if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$.*

We say t_1 is before *t_2 if $t_1 < t_2$, and t_2 is* after *t_1.*

It is often convenient (and conceptually clarifying) to note that a strict order $<$ canonically induces a non-strict companion relation $\leq$ by defining $t_1 \leq t_2$ to mean "$t_1 < t_2$ or $t_1 = t_2$." This does not add structure; it simply packages the option of equality explicitly. Conversely, starting from a reflexive partial order $\leq$, one recovers a strict version by setting $t_1 < t_2$ iff $t_1 \leq t_2$ and $t_1 \neq t_2$. This standard conversion is useful later when one wants to talk about "no later than" versus "strictly earlier" without changing the underlying conceptual commitments.

Remark 217 (Intuition, examples, and why this matters)**.** *The relation $<$ is meant to capture only the irreducible directional fact "this precedes that," not necessarily measured duration. Irreflexivity forbids loops of the form "an event is before itself," and transitivity enforces that "before" composes: if t_1 is before t_2 and t_2 before t_3, then t_1 must be before t_3. Together these clauses prevent directed cycles, so the order can be read as an acyclic dependency structure (a "can-happen-before" relation).*

Two standard examples are: (i) $T = \mathbb{N}$ with the usual $<$; (ii) $T = \mathcal{P}(S)$ for a set S with $A < B$ defined as $A \subsetneq B$ (strict inclusion). In the temporal reading, the second example is suggestive: temporal precedence can behave like a refinement relation on information states. This strict-partial-order abstraction is useful because it is the weakest structure that still lets us speak rigorously about "past" and "future" without assuming that every pair of events is comparable. This aligns directly with the Hyperseed notions After and Before (Hyperseed-Concepts 52 and 63).

One can also visualize a strict partial order as a directed graph whose vertices are elements of T and whose edges indicate immediate precedence; transitivity then says that reachability agrees with $<$. In the finite case, a Hasse-style picture (showing only "cover" relations and omitting transitive edges) makes the distinction between a *chain* (a totally ordered subset) and an *antichain* (a set of pairwise incomparable elements) explicit. In temporal terms, a chain reads as a history-like thread of definite succession, while an antichain reads as a simultaneous (or mutually unrelated) slice at the resolution being modeled.

A further reason this matters is that many familiar time models implicitly assume *total* order, i.e. for all distinct t_1, t_2 one has either $t_1 < t_2$ or $t_2 < t_1$. A strict partial order drops exactly that assumption. Thus it can represent branching futures, merging causal lines, and observer-dependent coarse-grainings where the model refuses to invent an arbitrary order when the underlying situation does not warrant one.

Remark 218 (Hyperseed's asymmetry axiom)**.** *For strict partial orders, the Hyperseed clause "$A < B$ implies NOT$(B < A)$" follows from irreflexivity and transitivity: if $A < B$ and $B < A$ then transitivity yields $A < A$, contradicting irreflexivity.*

It is worth emphasizing that this derivation uses only the internal logic of the relation $<$: the contradiction is not empirical but formal. In particular, the step "$A < B$ and $B < A$ implies $A < A$" is exactly the closure property transitivity provides, so asymmetry is not an additional stipulation about time; it is a structural fact about any relation intended to behave like strict precedence.

Remark 219 (Intuition: why asymmetry is not a separate axiom)**.** *One sometimes writes "asymmetric" as a separate requirement for a strict order, but it is already latent in the two clauses given. Philosophically, this is a small instance of a broader pattern in foundational work: what appears at first as an additional metaphysical constraint can often be seen as a theorem of a simpler algebra of relations. Here, the prohibition of self-precedence and the compositionality of precedence suffice.*

Another way to say the same thing is that strict partial orders are precisely those relations whose "before" arrows cannot be followed around a directed cycle. If a theory of time wants to allow closed timelike curves or other forms of cyclic dependence, then it is not (in that region) a strict partial order; one would need a different primitive. Hyperseed's choice here therefore commits to acyclicity at the level of the basic After/Before relation, while leaving open many other possibilities (density, branching, finiteness, etc.).

Remark 220 (Incomparability and concurrency)**.** *If neither $t_1 < t_2$ nor $t_2 < t_1$ holds, then t_1 and t_2 are* incomparable. *In the time interpretation, incomparability represents concurrency, branching, or the absence of a well-defined ordering relation at the observer's grain.*

Given any $t \in T$, the strict order also induces the associated "past" and "future" sets

$$\mathrm{Past}(t) = \{s \in T : s < t\}, \qquad \mathrm{Future}(t) = \{u \in T : t < u\}.$$

These are purely order-theoretic constructions (no metric required). Incomparability between t_1 and t_2 can then be read as the failure of inclusion between their past/future profiles: neither event lies in the other's future, so the order records no precedence constraint between them. This gives a precise sense in which partial ordering supports talk of "past" and "future" locally, even when a global linear timeline is unavailable.

Remark 221 (Example: concurrency as a mathematical primitive)**.** *In a distributed system, two messages may be causally unrelated: neither is provably earlier. A strict partial order captures this directly by leaving the pair incomparable, rather than forcing a spurious choice. This is precisely the sort of "information-preserving humility" that is lost when one insists prematurely on linear time.*

In such settings, it is common to distinguish the *causal* order (the strict partial order that must be respected) from any *schedule* or *linearization* that totally orders events for execution or narration purposes. Mathematically, a schedule is a total order that *extends* the partial order: whenever $t_1 < t_2$ causally, the schedule also places t_1 before t_2. The key point is that many different extensions may exist, and their differences correspond exactly to choices made among incomparable events. Hyperseed's use of strict partial orders therefore cleanly separates what is forced by After/Before from what is merely conventional or perspective-dependent.

7.2 Proto-time as observer-relative ordering

Hyperseed's term *proto-time* refers to "every partial ordering considered as a sort of time". This can be taken literally. In particular, the proposal is that the minimal formal residue of "temporal" talk is not duration or a privileged coordinate, but an ordering relation that can support the distinctions "earlier than", "later than", and "not (yet) comparable at the present resolution". The emphasis on partial (rather than total) order is crucial: it allows the formalism to treat indeterminacy, concurrency, and limited access as first-class rather than as defects to be repaired.

Definition 56 (Proto-time)**.** *A* proto-time *is a pair $(T, <)$ where T is a set of temporal indices (e.g. occasions, events, or representations of them) and $<$ is a strict partial order on T. In other words, $<$ is irreflexive ($\neg(t < t)$ for all $t \in T$) and transitive (if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$); antisymmetry is automatic for a strict order in the sense that cycles are excluded by transitivity and irreflexivity.*

Remark 222 (Intuition and connection to experience). *Proto-time is time "before clocks": any experienced ordering of episodes, perceptions, or internal updates qualifies, regardless of whether it resembles a line. The set T is deliberately generic: it may be a set of occasions (Hyperseed-Concept 124), a set of events in an event calculus, or a set of internal representational states. The only essential ingredient is the precedence relation $<$, which records whatever the system takes to be "after" and "before" at its current resolution. The point of using a* strict *order is that it matches the phenomenological asymmetry of "before/after" without forcing an identity criterion for simultaneous or indistinguishable episodes; where equality or "same time" is needed, it can be supplied separately (e.g. via an equivalence relation or by quotienting), rather than being silently baked into the time structure. Likewise, incomparability ($\neg(t_1 < t_2)$ and $\neg(t_2 < t_1)$) need not mean "simultaneous" in any physical sense: it can mean "no ordering information available", "different narrative threads", or "the system declines to commit".*

A simple example is a causal DAG whose vertices are events and whose edges represent direct causal influence; the reachability relation induces a strict partial order. This illustrates how proto-time can be extracted from constraints rather than posited as an ambient parameter: causation (or influence) supplies the arrows, and time is the induced ordering. Another example is the "edit history" of a document with branching: commits form a partially ordered set under ancestor relation. These are proto-times in exactly the present sense, and they make vivid why a linear axis is often a lossy summary rather than the primitive. In both examples, a linear "timeline" can be recovered only by adding choices (e.g. selecting a linear extension, choosing a topological sort, or imposing additional tie-breaking conventions), and different choices may be equally compatible with the underlying proto-time. This definition operationalizes the Hyperseed notion Proto-Time (Hyperseed-Concept 142).

Remark 223 (Proto-time is local and aspectual). *Different contexts/aspects (in Hyperseed's sense) may carry different proto-times. For instance, the same collection of occasions may admit a fine-grained causal proto-time and a coarser narrative proto-time, or multiple incompatible orderings inferred from different evidence. This is one of the places where paraconsistency is not an afterthought but a design requirement. Concretely, a system may maintain one ordering for sensorimotor updating, another for autobiographical memory, and yet another for planning, even when these do not admit a single common refinement without introducing contradictions or arbitrarily discarding constraints. The locality claim can also be read computationally: proto-time is indexed by what is tracked and what is ignored, so changing resolution (e.g. by coarse-graining episodes into "chunks") can change T and thereby change which order-relations are expressible at all. In that sense, proto-time functions as an* aspectual interface*: it describes the temporal commitments of an aspect, not an absolute inventory of "all events".*

Remark 224 (Why "observer-relative" is not a concession but a method). *To say proto-time is context-indexed is not to deny a mind-independent world; it is to keep separate (i) the structure realized by whatever generates events and (ii) the structure inferred or imposed by a bounded system attempting to navigate them. Hyperseed repeatedly exploits this separation later when relating time to effort-minimization and "least-effort paths" in inference or control [21]. The methodological payoff is that one can ask two different questions without conflating them: "What ordering constraints does the world impose?" versus "Which ordering constraints does this agent (or aspect) represent, and how costly is it to refine them?". On this view, observer-relativity is not a relativism about reality, but an explicit accounting of representation: a proto-time is the agent's current ordering model, which may be revised as evidence arrives, resources change, or goals shift. It also clarifies why proto-time does not presuppose a metric, a topology, or a global "now": those are additional* *structures that may be constructed when useful, but the base concept remains the ordering relation that supports predictions, explanations, and control.*

7.3 Graded and paraconsistent temporal ordering

Hyperseed also gestures at *graded* orderings ("how long or fast the flow feels"). A simple rigorous way to do this within the formal core is to treat "before" as a p-bit-valued relation whose positive component records support for "t_1 is before t_2" and whose negative component records support against it. One can view this as replacing a single Boolean-valued edge in a temporal graph by a pair of weights: a pro-weight and a con-weight, both of which may be nonzero. The point is not merely to allow uncertainty (low support in both channels), but also to allow *tension* (simultaneously high support and high counter-support) without collapsing into triviality.

Definition 57 (p-bit-valued before-relation)**.** *Let $V = [0,1]^2$ be the* p-bit *quantale from Section 3.4. A* p-bit-valued temporal evidence relation *is a map*

$$B : T \times T \to V, \qquad B(t_1, t_2) = (B^+(t_1, t_2),\ B^-(t_1, t_2)).$$

Here $B^+(t_1, t_2)$ is positive evidence for $t_1 < t_2$, and $B^-(t_1, t_2)$ is negative evidence.

Remark 225 (Order and entailment on V)**.** *Since V is used as a codomain for evidential values, inequalities such as $B(t_1, t_3) \geq B(t_1, t_2) \otimes B(t_2, t_3)$ presuppose a comparison relation on pairs. In the basic* p-bit *setup, one typically orders $V = [0,1]^2$* componentwise*:*

$$(v^+, v^-) \leq (w^+, w^-) \quad \textit{iff} \quad v^+ \leq w^+ \textit{ and } v^- \leq w^-.$$

Under this reading, "$\geq$" means "has at least as much positive evidence and at least as much negative evidence." This is deliberately not *a reduction to a single net score: increasing v^- does not automatically decrease v^+, so the structure can represent both evidential accumulation and evidential conflict as first-class phenomena rather than as noise to be normalized away.*

Remark 226 (Notation unpacking)**.** *The symbol $V = [0,1]^2$ denotes ordered pairs (v^+, v^-) of reals in $[0,1]$. We read v^+ as "support for" and v^- as "support against" a proposition. Thus $B^+(t_1, t_2)$ and $B^-(t_1, t_2)$ are simply the two coordinate functions of B. This two-channel semantics is one concrete paraconsistent option; related four-valued approaches appear in constructible duality logics [23], and the use of such values as composable evidence is aligned with the general "paraconsistent resonance" perspective [24]. A convenient mnemonic is that $(1,0)$ behaves like "confidently yes," $(0,1)$ like "confidently no," $(0,0)$ like "no information," and $(1,1)$ like "inconsistent but fully activated," with intermediate points capturing graded mixtures of these modes.*

Remark 227 (Intuition, examples, and purpose)**.** *A crisp order says either "$t_1 < t_2$" or not; the* p-bit*-valued relation says how strongly the system leans in each direction, and allows it to record conflict explicitly. For example, a system may have sensory evidence suggesting that a stimulus s_1 preceded s_2 (high B^+), while a higher-level narrative model suggests the opposite (high B^-). A single classical truth value would force premature resolution; the* p-bit *pair preserves both pressures so that later inference (e.g. transitive closure, inertia reasoning) can integrate them. In particular, the representation supports at least three qualitatively different cases that a single probability-like scalar conflates: (i)* uncertain *(both channels near* 0*), (ii)* confident *(one channel high, the other low), and (iii)* contested *(both channels high).*

This graded form is useful because it connects time to the same algebra used for distinction, weakness, and compositional aggregation elsewhere in the document [3]. In other words, temporal *uncertainty is not treated as an exceptional problem requiring a separate probabilistic apparatus; it is folded into the same quantale-enriched calculus. From a modeling standpoint, this uniformity matters: if temporal judgments are produced by the same pipelines that produce other relational judgments (causal, spatial, conceptual), then having a common aggregation and comparison algebra makes it possible to share learning rules, regularizers, and compositional mechanisms across domains.*

Definition 58 (Coherence constraints for temporal evidence (optional))**.** *One may impose any of the following (model-dependent) constraints:*

1. *(Anti-symmetry as evidence) $B^+(t_1, t_2) \le B^-(t_2, t_1)$ for all t_1, t_2.*

2. *(Transitive support) $B(t_1, t_3) \ge B(t_1, t_2) \otimes B(t_2, t_3)$ where $\otimes$ is the quantale tensor (componentwise multiplication).*

3. *(Irreflexive support) $B^+(t, t) = 0$ for all t (while allowing $B^-(t, t)$ to encode strong evidence against $t < t$).*

We emphasize that these are constraints on evidence*, not metaphysical axioms.*

Remark 228 (Componentwise reading of the constraints)**.** *It may be helpful to unpack clause (2) in coordinates. If $\otimes$ is componentwise multiplication, then*

$$B(t_1, t_2) \otimes B(t_2, t_3) = \big(B^+(t_1, t_2)B^+(t_2, t_3),\ B^-(t_1, t_2)B^-(t_2, t_3)\big).$$

With the componentwise order on V, the inequality $B(t_1, t_3) \ge (\cdots)$ requires both

$$B^+(t_1, t_3) \ \ge\ B^+(t_1, t_2)B^+(t_2, t_3) \qquad \textit{and} \qquad B^-(t_1, t_3) \ \ge\ B^-(t_1, t_2)B^-(t_2, t_3).$$

Thus transitivity is treated as a statement about how evidence propagates along a chain in each channel, rather than as a demand that the sign *of the final verdict must be classically consistent. This makes it possible for a system to infer that a longer chain can inherit both support and counter-support, which is often exactly what happens when different subsystems contribute competing temporal cues.*

Remark 229 (Intuition and examples for the evidence constraints)**.** *Clause (1) says: evidence that t_1 is before t_2 should count as counter-evidence for t_2 being before t_1. It is a soft version of asymmetry, expressed in the vocabulary of evidence rather than in the vocabulary of facts. Clause (2) says: if we have support for $t_1 < t_2$ and for $t_2 < t_3$, then we should have at least the composed support for $t_1 < t_3$; the tensor $\otimes$ (componentwise multiplication in the* p-bit *quantale) plays the role of "and then," i.e. accumulation of evidential strength along a chain. Clause (3) enforces that the positive channel does not directly support self-precedence, while leaving room for the negative channel to record explicit rejection of it.*

These constraints are useful when B is learned or estimated: they provide regularizers that encourage temporal evidence to behave like an order without demanding perfect consistency. As a concrete numerical illustration, if $B^+(t_1, t_2) = 0.9$ and $B^+(t_2, t_3) = 0.8$, then clause (2) requires $B^+(t_1, t_3) \ge 0.72$; similarly, if both negative evidences are 0.6, it requires $B^-(t_1, t_3) \ge 0.36$. This matches an intuitive "attenuation along a path" reading: longer compositions retain only what survives repeated conjunction.

Remark 230 (Alternative tensors and modeling choices)**.** *The use of componentwise multiplication is a clean default because it is monotone and matches the usual "independent conjunction" intuition for graded evidence. However, nothing essential forces this particular t-norm: depending on the application, one might replace multiplication by* min *(for a bottleneck interpretation), by the Łukasiewicz t-norm (for a bounded-sum interpretation), or by a learned monotone aggregator, provided it remains compatible with the chosen order on V. The present subsection only needs that V is a quantale so that such aggregations compose associatively and distribute over suprema, enabling systematic propagation of temporal evidence in networks rather than ad hoc case-by-case combination.*

Remark 231 (Why keep a crisp proto-time at all?)**.** *A useful stance is: the world (or an internal simulation) may realize a crisp partial order $(T, <)$, while a bounded observer maintains only B. Alternatively, one can work entirely with B and treat "proto-time" as an emergent crisp skeleton (e.g. the set of pairs whose B^+ exceeds a threshold) when needed. This split between an (ideal) underlying order and a (bounded) evidential surrogate is also a way to separate ontological questions (what actually happened) from epistemic and computational questions (what can be inferred, stored, and composed under constraints), which is the main methodological role played here by paraconsistent grading.*

Remark 232 (A practical reading: skeletonization by threshold)**.** *In computation, one often picks a threshold θ and defines a derived crisp relation $t_1 <_\theta t_2$ iff $B^+(t_1, t_2) \geq \theta$ (or by a net score). This is not merely a convenience: it is an explicit model of the act of "making time definite" by expending effort to force a decision, a theme that reappears when simplicity and weakness are defined as observer-relative resource tradeoffs [2, 3]. One may also impose auxiliary tie-breaking rules at the skeletonization stage (for instance, requiring simultaneously that $B^-(t_1, t_2) \leq \varepsilon$ for some small ε) to avoid declaring an edge "present" precisely when the system is signaling strong internal disagreement.*

Remark 233 (From evidence to derived temporal structures)**.** *Once a crisp skeleton $<_\theta$ is extracted, standard constructions become available: one can take its transitive reduction (to obtain a Hasse-like diagram), compute its strongly connected components (to diagnose cycles created by thresholding), or topologically sort it when it happens to be acyclic. Conversely, one can remain in the enriched setting and compute a graded transitive closure by taking suprema over all intermediary chains, in direct analogy with path problems in weighted graphs, but with weights in V and composition given by $\otimes$. The purpose of these remarks is not to fix a single pipeline, but to make explicit that B is meant to be operational: it is a data structure on which one can run order-theoretic and graph-theoretic procedures, either before or after discretization, depending on the resource budget and the desired degree of commitment.*

7.4 Linear time axes as serializations and quotients

A linear time axis is a special case of proto-time: a total (linear) order. Hyperseed also suggests a specific *quotienting* move: group together events with the same "before-set" and "after-set". In the present section it is helpful to keep two distinct operations conceptually separate: *serialization* (choosing a linear order that respects a given proto-time) and *coarse-graining* (identifying events that the proto-time already fails to distinguish). The quotienting move is of the second kind: it does not "decide" any new comparisons, but only collapses redundancies in the order-theoretic description of temporal location.

Definition 59 (Linear time axis)**.** *A* linear time axis *is a proto-time* $(L, \prec)$ *where* $\prec$ *is a strict total order: for all distinct* $\ell_1, \ell_2 \in L$*, either* $\ell_1 \prec \ell_2$ *or* $\ell_2 \prec \ell_1$*. Equivalently,* $\prec$ *is irreflexive and transitive, and it satisfies totality (comparability) on distinct elements; in many texts this is packaged as a trichotomy principle for* $\prec$ *on* L*.*

Remark 234 (Intuition and examples)**.** *A strict total order is the familiar "single-file" time of classical narratives and many physics models: every two distinct times are comparable. For example,* $(\mathbb{Z}, <)$ *is a linear time axis; so is any finite list with its left-to-right order. Linear axes are computationally convenient because they allow iteration "from past to future" without branching cases. But they are also epistemically assertive: they force comparability even where proto-time may honestly have none (e.g. concurrent events). The later remark on information loss makes this precise. One may also distinguish discrete linear axes (like* $\mathbb{Z}$*) from dense ones (like* $(\mathbb{Q}, <)$*): both are linear time axes, but they support different intuitions about "next" moments and granularity. The present development does not assume discreteness; what matters is total comparability.*

Definition 60 (Past/future sets and quotient-by-indistinguishability)**.** *Let* $(T, <)$ *be a proto-time. Define for each* $t \in T$*:*

$$\mathrm{Past}(t) := \{u \in T : u < t\}, \qquad \mathrm{Future}(t) := \{u \in T : t < u\}.$$

Define an equivalence relation $\sim$ *on* T *by*

$$t \sim t' \iff \mathrm{Past}(t) = \mathrm{Past}(t') \ \textit{and} \ \mathrm{Future}(t) = \mathrm{Future}(t').$$

Let $[t]$ *denote the equivalence class of* t*, and let* $T/{\sim}$ *be the set of classes. Define a relation* $<$ *on* $T/{\sim}$ *by*

$$[t] < [t'] \iff t < t' \text{ for some (equivalently any) representatives.}$$

In order-theoretic language, $\mathrm{Past}(t)$ *is the strict down-set of* t *and* $\mathrm{Future}(t)$ *is the strict up-set of* t*. The pair* $(\mathrm{Past}(t), \mathrm{Future}(t))$ *can therefore be read as a complete "order signature" of how* t *sits inside* $(T, <)$*, ignoring any additional labels on events.*

Remark 235 (Intuition: quotienting as "time slicing")**.** *The pair* $(\mathrm{Past}(t), \mathrm{Future}(t))$ *describes the "causal address" of* t *inside the partial order: what lies strictly before it and what lies strictly after it. Two elements with the same past and future are indistinguishable with respect to the order-theoretic information available; quotienting identifies them. In temporal terms, this implements a rigorous version of "simultaneous at the current grain": if the order cannot distinguish* t *and* t'*, the quotient refuses to pretend otherwise.*

A simple example is a proto-time with two incomparable events x *and* y *occurring after an initial event* s *and before a final event* e*. Then* x *and* y *have the same past* $\{s\}$ *and the same future* $\{e\}$*, hence* $x \sim y$ *and they collapse to a single slice. This is useful as a canonical compression of proto-time prior to attempting linearization. It is worth noting that* $\sim$ *is generally finer than "incomparability": two incomparable events need not be equivalent, because they may have different sets of predecessors or successors. Thus the quotient is not merely "turn all antichains into points," but rather "merge only those nodes whose entire comparative context agrees."*

Lemma 1 (The quotient order is well-defined and strict)**.** *The relation* $<$ *on* $T/{\sim}$ *is well-defined and makes* $(T/{\sim}, <)$ *a strict partial order.*

Remark 236 (What the lemma asserts, in plain language)**.** *The definition of* $[t] < [t']$ *is only meaningful if it does not depend on which representatives* $t \in [t]$ *and* $t' \in [t']$ *we pick. The lemma* *says exactly this: the quotient construction does not introduce ambiguity, and it preserves the essential "no cycles" and "compositional precedence" properties of a strict order. This result is conceptually important because it licenses the quotient as an information-preserving coarse-graining of time: we are not smuggling in extra structure, merely collapsing what the order already cannot tell apart. In particular, after quotienting, distinct classes are automatically* order-distinguishable*: if $[t] \neq [t']$, then either their past sets differ or their future sets differ (by definition of $\sim$), so the quotient has removed precisely the "duplicate addresses" and nothing more.*

Proof. Well-definedness: suppose $t \sim s$ and $t' \sim s'$ with $t < t'$. If $s \not< s'$, then either (i) $s' = s$ (impossible since $t < t'$ and $t \sim s$ would force incompatible past/future sets), (ii) $s' < s$, which together with $t < t'$ and the equalities of past/future sets implies a contradiction with transitivity/irreflexivity in $(T, <)$, or (iii) s and s' incomparable, which again contradicts the equality of past/future sets across the $\sim$-classes once one representative is comparable. Strictness: irreflexivity and transitivity descend to the quotient because any cycle in the quotient would lift to a cycle in T. A more explicit way to see the "equivalently any representatives" clause is as follows. Assume $t < t'$ and $s \sim t$, $s' \sim t'$. Then $t \in \mathrm{Past}(t')$, hence $t \in \mathrm{Past}(s')$ because $\mathrm{Past}(t') = \mathrm{Past}(s')$. Since $\mathrm{Past}(t) = \mathrm{Past}(s)$ and $t < t'$, we also have that every predecessor of t is a predecessor of s, and conversely; in particular, the statement "t is before s'" transfers to "s is before s'" by comparing down-sets at s' and using transitivity in $(T, <)$ to rule out the alternative possibilities $s' = s$ and $s' < s$. Thus $s < s'$. This establishes that if one pair of representatives witnesses $[t] < [t']$, then all pairs do. For strictness, irreflexivity on classes follows because $[t] < [t]$ would require some representative $t < t$, impossible. Transitivity follows because if $[a] < [b]$ and $[b] < [c]$, pick representatives $a < b$ and $b' < c$ with $b' \sim b$; by well-definedness one may replace b' with b and obtain $a < b < c$, hence $a < c$ and therefore $[a] < [c]$. □

Remark 237 (Proof sketch and key ideas). *The strategy is to show that $\sim$-equivalent elements have exactly the same comparability profile. If $t \sim s$, then every element that is before t is before s, and every element after t is after s. Hence, if t is comparable to t', then s must be comparable to any $s' \sim t'$ in the same direction; otherwise the past/future sets would differ. Once well-definedness is secured, irreflexivity and transitivity are inherited because any violation in the quotient would correspond to a directed cycle among representatives in T, which cannot exist in a strict partial order.*

Visually, one may imagine the Hasse diagram of $(T, <)$: the quotient merges nodes that have identical "upward" and "downward" reachability sets. Such a merge cannot create a new directed cycle, because cycles are already forbidden in the original graph. Another way to phrase the same idea is that $\sim$ identifies vertices with identical strict principal ideals and filters; this is a standard "separation" step in order theory that removes redundant copies of the same abstract position in the poset.

Remark 238 (Why this quotient matters). *The quotient $T/\sim$ collapses "simultaneity at the observer's grain": events with the same causal/narrative placement become a single "time-slice". Even after quotienting, the resulting order need not be total; concurrency remains possible. What changes is that concurrency is no longer obscured by multiple indistinguishable nodes: the remaining incomparabilities in $T/\sim$ represent genuine branching (or genuine lack of information) that cannot be removed without making an additional modeling choice, such as selecting a linear extension.*

Theorem 5 (Finite proto-times admit linear extensions). *If $(T, <)$ is a finite proto-time, then there exists a strict total order $\prec$ on T such that $t_1 < t_2$ implies $t_1 \prec t_2$. Such a $\prec$ is called a* linear extension *(or topological sort) of $(T, <)$.*

A linear extension can be viewed as a systematic "tie-breaking" of incomparabilities: whenever the proto-time does not constrain the order of two events, the extension chooses one of the two possible orientations while keeping all required precedence relations intact. For finite T, such an extension may be constructed algorithmically by repeatedly selecting a $<$-minimal element (an event with no strict predecessors), placing it next on the linear axis, and removing it; strictness ensures no cycles, so this process terminates. The existence of multiple linear extensions is typical and is precisely one formal measure of how much concurrency or underdetermination a proto-time contains.

Remark 239 (Meaning and importance)**.** *This theorem says that for a finite proto-time we can always choose a "story order" that respects all the precedence constraints, even if the underlying structure is branching or concurrent. In practice, this is what allows one to render a partial order as a sequence for purposes of simulation, explanation, logging, or training a model that expects sequential input. The result is not that linear time is fundamental, but that linear time is always* available as a representation *for finite structures.*

A useful way to read this is: whenever the only temporal facts you insist on are of the form "t_1 must occur before t_2," you can always embed those facts into a single list, provided the system is finite and has no cycles. This is the mathematical justification behind routine engineering constructions such as topological sorting of a dependency graph, or choosing an execution order for independent tasks that still respects required prerequisites.

This connects directly to the earlier warning that linearization loses information: the theorem guarantees existence of a serialization, while the later remark clarifies that such a serialization is typically many-to-one and hence not invertible. In other words, the existence result is about compatibility *(there is at least one linear narrative that does not contradict the proto-time), whereas the information-loss warning is about* identifiability *(many different proto-times can collapse to the same linear narrative once concurrency is forced into an order).*

Proof. Induct on $|T|$. For $|T| = 0, 1$ the claim is trivial. Assume the claim holds for all smaller finite sets. Because $<$ is a strict partial order on a finite set, there exists a minimal element $m \in T$ (i.e. no $u \in T$ satisfies $u < m$). Remove m to obtain $T' := T \setminus \{m\}$, and restrict $<$ to T'. By induction, T' admits a linear extension $\prec'$. Define $\prec$ on T by placing m first and then following $\prec'$ on T'. Minimality of m ensures that no constraint $u < m$ is violated, and all constraints within T' are respected by construction.

For clarity on the existence of m: if every element had a predecessor, we could build a descending chain $t_0 > t_1 > t_2 > \cdots$ by repeatedly choosing a predecessor. In a finite set such a process must eventually repeat an element, producing a cycle, which contradicts the irreflexivity/transitivity of a strict partial order. Hence at least one element has no predecessor and is minimal. The same argument also highlights why finiteness is essential for this simple induction: in infinite posets a minimal element need not exist, so one cannot always "start" a linear narrative without adding extra choice principles or additional structure. $\square$

Remark 240 (Proof sketch, and a visual intuition)**.** *The proof is the standard "peel off a minimal element" argument: in any finite acyclic directed graph, there is at least one node with no incoming edges. Place such a node first; then repeat on the remaining graph. The inductive step works because removing a minimal element cannot destroy the strict-order property on what remains, and placing it first cannot violate any constraint since nothing was required to precede it.*

Geometrically, imagine repeatedly shaving off the "earliest" layer of events in the Hasse diagram. A linear extension is just the record of this shaving process. Different choices of minimal element at each step correspond to different linear narratives compatible with the same proto-time.

From an algorithmic perspective, this is exactly the idea behind a topological sort: at each step one selects an available event (one with no unmet prerequisites), outputs it, and deletes it from the dependency graph. The "visual intuition" is therefore not merely metaphorical; it directly encodes a constructive procedure for producing a serialization, and it makes explicit where nondeterminism enters (namely, whenever there is more than one currently-minimal event).

Remark 241 (Information loss under linearization)**.** *A linear extension preserves all constraints $t_1 < t_2$ but destroys explicit incomparability: if t_1 and t_2 were incomparable in $<$, the extension still forces either $t_1 \prec t_2$ or $t_2 \prec t_1$. Hence linearization is generally a many-to-one representation of proto-time.*

Equivalently, the linear order $\prec$ adds extra comparabilities that were not entailed by $<$. These additional pairwise decisions are exactly what a serialization must supply in order to become a total order, but they should be read as choices of presentation *rather than as new facts about precedence in the original proto-time. In applications, this is why one should treat the resulting sequence as an* annotated view *of the underlying partial order, not as a faithful encoding of all its structure.*

Remark 242 (Why "information loss" is philosophically substantive)**.** *The point is not merely technical. When a system commits to a linear axis, it is choosing one possible way of resolving concurrency and ambiguity; this choice can feed back into prediction, responsibility assignment, and causal explanation. Hyperseed treats such commitments as aspects of cognitive effort and design, not as inevitable reflections of reality [5].*

In particular, once incomparable events are forced into an order, downstream reasoning may begin to treat the imposed order as if it were evidentially supported (e.g. inferring causal direction from narrative order, or attributing agency based on who "acted first"). The philosophical weight comes from the fact that such inferences are not just errors in bookkeeping; they can systematically bias models of agency, credit, blame, or counterfactual dependence whenever concurrency is a real feature of the underlying situation.

Remark 243 (Serialization as a quotient viewpoint)**.** *It is often useful to describe the many-to-one aspect more structurally. Fixing a partial order $(T, <)$, each linear extension $\prec$ can be seen as a* choice of totalization*: it retains all required relations but adds enough additional comparisons to make every pair comparable. Conversely, if one starts from a total order $\prec$ on T and "forgets" which comparisons were merely chosen to resolve incomparability, one recovers only a coarser object—some partial order compatible with $\prec$—and many distinct partial orders may be compatible with the same $\prec$.*

From this perspective, passing from proto-time to a linear time axis behaves like quotienting by an equivalence relation that identifies structures that differ only in how they arrange events that were originally concurrent. The theorem guarantees that at least one representative exists on the linear side for every finite proto-time, while the information-loss remark explains why the projection to that quotient is not injective.

7.5 Becoming as boundary non-duality

Hyperseed defines *becoming* as a relation between temporally adjacent entities whose boundary is "non-dual": it is not crisply determined whether the two are distinct or identical near the transition.

To model this, we assume:

- a proto-time $(T, <)$,
- a set E of entities (or entity-representations),

- and a p-bit-valued *distinction* (or equivalence) relation whose value may vary by time.

Remark 244 (Proto-time versus physical time)**.** *The order* $(T, <)$ *is intentionally weak: it provides temporal precedence without committing to a metric, topology, or even continuity. This allows the same formalism to cover discrete update steps (e.g. iterations of a learning algorithm), event sequences (e.g. observations in an experiment), or continuous physical time as a special case. When additional structure exists (e.g. a topology or a metric), it can be used to make the notion of "neighborhood" in Definition 63 more explicit, but the core idea of becoming does not require it.*

Definition 61 (Time-indexed distinction relation)**.** *Let* E *be a set of entities and* T *a proto-time carrier. A* time-indexed distinction relation *is a map*

$$\mathrm{Dist} : T \times E \times E \to V,$$

where $\mathrm{Dist}(t; x, y) = (d^+, d^-)$ *records evidence at time* t *that* x *and* y *are* distinct *(positive) and evidence that they are* not distinct *(negative).*

Remark 245 (Notation and intended reading)**.** *The notation* $\mathrm{Dist}(t; x, y)$ *is a three-argument evaluation: first pick a time* $t \in T$*, then two entities* $x, y \in E$*, and obtain a* p-bit *value in* V*. The first coordinate* d^+ *supports the proposition "*x *and* y *are distinct at* t*," and the second coordinate* d^- *supports its opposition. This is an instance of the general Hyperseed notion Distinction (Hyperseed-Concept 98) made time-dependent, so that identity and difference can vary across a transition.*

Remark 246 (Optional structural constraints on Dist)**.** *The definition does not require* $\mathrm{Dist}(t; x, y) = \mathrm{Dist}(t; y, x)$*, nor any form of reflexivity such as* $\mathrm{Dist}(t; x, x)$ *being maximally non-distinct. This is deliberate: in some applications the distinction judgment is viewpoint-dependent (e.g. an asymmetric classifier, or a directed comparability relation where* x *is treated as a refinement of* y *but not conversely). When symmetry and reflexivity are appropriate, they can be imposed as modeling assumptions, for example:*

$$\mathrm{Dist}(t; x, y) = \mathrm{Dist}(t; y, x), \qquad \mathrm{Dist}(t; x, x) \approx (0, 1).$$

Likewise, some domains may benefit from a weak "triangle-type" constraint (distinctions compose), but becoming as boundary non-duality does not presuppose such coherence.

Remark 247 (Interpreting the value space V)**.** *The comparison* $\mathrm{Dist}(t; x, y) \geq (\theta, \theta)$ *presumes a coordinatewise preorder on* V *(e.g.* $(a^+, a^-) \geq (b^+, b^-)$ *iff* $a^+ \geq b^+$ *and* $a^- \geq b^-$*). Intuitively, larger* d^+ *means more support for "distinct," while larger* d^- *means more support for "not distinct." In this sense, non-duality is not the absence of information, but the presence of* competing *information. If* V *additionally supports operations such as negation, conjunction, or normalization (as many* p-bit*-style constructions do), those can later be used to define derived notions like "net distinctness" or confidence-weighted salience, but they are not required for the basic definitions here.*

Remark 248 (Intuition and examples)**.** *If* E *contains successive snapshots of a physical object,* $\mathrm{Dist}(t; x, y)$ *can be read as "how much the observer at time* t *treats the snapshot* x *as different from snapshot* y*." In cognition,* x *and* y *might be two concepts or two internal states; during learning, a system may oscillate between treating them as the same category and as different ones, yielding paraconsistent values.*

The usefulness of time-indexing is that it avoids baking identity into the entities themselves. Instead of asking "are x *and* y *the same?", we ask "under what temporal and contextual conditions does the system treat them as distinct?" This is a formal handle on the otherwise slippery boundary language used in becoming.*

Remark 249 (Context dependence and "who" is doing the distinguishing)**.** *In practice,* Dist *often aggregates multiple sources: sensory measurements, internal inference, linguistic labels, and social or institutional criteria for sameness. Time-indexing allows these sources to shift in influence: the same pair (x, y) can become more or less distinct as attention changes, as measurement resolution improves, or as the governing equivalence standard is revised. Thus, even when the underlying world-state changes smoothly, the distinction relation may change non-monotonically, which is one of the main reasons a paraconsistent representation is useful in boundary regimes.*

Definition 62 (Non-dual variety at time t)**.** *We say entities $x, y \in E$ display* non-dual variety at time t *if*

$$\mathrm{Dist}(t; x, y) \text{ is strongly paraconsistent, e.g. } \mathrm{Dist}(t; x, y) \geq (\theta, \theta)$$

for a chosen threshold $\theta \in (0, 1]$. Intuitively: there is simultaneously strong evidence for "distinct" and strong evidence for "not distinct".

Remark 250 (Alternative strong-paraconsistency criteria)**.** *The coordinatewise threshold condition is a simple sufficient test. Depending on how V is constructed, one might instead use a scalarization such as $\min(d^+, d^-) \geq \theta$, or require that both coordinates be high* relative to *total support (e.g. $d^+/(d^+ + d^-) \approx 1/2$ with large $d^+ + d^-$), distinguishing non-duality from mere uncertainty. The present definition keeps the criterion explicit and easily checked, while leaving room for domain-specific refinements.*

Remark 251 (Intuition: a boundary where both stories are true)**.** *Non-dual variety is the formal way of saying: at the boundary, the system has reasons to keep the two relata apart and reasons to identify them, and neither reason is negligible. A classic illustrative situation is a gradual morph: as a shape continuously changes from a circle to a square, there can be a region where classification as "circle" and as "not circle" both have strong support. Another is the Ship of Theseus: near certain thresholds, identity can be supported by continuity of function while non-identity is supported by replacement of parts.*

The threshold θ is a modeling choice: higher θ demands stronger simultaneous support. This definition instantiates the Hyperseed notion Non-Dual Variety (Hyperseed-Concept 120), and makes explicit why a paraconsistent truth space is convenient: it lets us represent such boundary states without collapsing into triviality.

Remark 252 (Non-dual variety versus indeterminacy)**.** *It is useful to separate* conflict *from* lack of evidence*. A boundary is "non-dual" in the intended sense when there is substantial support on both sides; this differs from a case where both d^+ and d^- are small (the system simply has not formed a judgment). In applications, this difference often corresponds to whether the system is actively maintaining competing models or interpretations (non-duality) versus being under-informed (ignorance).*

Definition 63 (Becoming along a proto-time)**.** *Fix a proto-time $(T, <)$ and a time-indexed distinction relation* Dist*. We say $x \in E$* becomes *$y \in E$ (along $(T, <)$) if there exist $t_1 < t_2$ and an interval (or neighborhood) $U \subseteq T$ with $t_1 \leq u \leq t_2$ for $u \in U$ such that:*

1. *(Temporal succession) x is salient/present up to (near) t_1 and y is salient/present from (near) t_2 onward (the salience notion can be instantiated later via pattern intensity or attention),*

2. *(Boundary non-duality) for all $u \in U$ near the transition, x and y display non-dual variety at u,*

3. *(Outside the boundary) away from U, x and y are (comparatively) more distinguishable (e.g.* $\mathrm{Dist}(t; x, y)$ *is not strongly paraconsistent).*

Remark 253 (Making "interval" and "near" precise)**.** *If T is equipped only with an order, "interval" can be read in the order-theoretic sense (e.g. a convex subset) and "near" can be treated informally as "close to the transition region in the ordering." If T has a topology or metric, one can instead take U to be an open neighborhood of a boundary time t_*, or a small window $[t_* - \varepsilon, t_* + \varepsilon]$, and interpret clause (2) as holding throughout that window. The definition is written to accommodate both discrete and continuous settings: in discrete time, U might be a short run of indices where judgments flip or overlap; in continuous time, it can represent a genuine boundary band.*

Remark 254 (Salience as a needed companion notion)**.** *Clause (1) separates* what exists in the representational universe *from* what is currently active *for the system. Without salience, x and y could both be permanently present in E (as representations) while never participating in a transition. Later formalizations can operationalize salience via attention weights, activation levels, pattern intensities, or selection functions; the present definition only requires that the model be able to say "x is what we are tracking before" and "y is what we are tracking after."*

Remark 255 (Multiple boundaries and staged becoming)**.** *The definition allows U to be any neighborhood capturing the transition; in many realistic cases, there may be several disjoint or partially overlapping boundary regions (e.g. repeated back-and-forth reclassifications during learning). One can model such cases either by taking U to include the full transition band, or by treating becoming as occurring in stages $x \to z \to y$ with successive boundary intervals. This flexibility matters in paraconsistent regimes, where the path of distinctions can be non-monotone even when the underlying process is directed.*

Remark 256 (Intuition and why this definition is structured as it is)**.** *The definition isolates three ingredients that are often conflated in informal talk about change: (i) there is a before/after placement ($t_1 < t_2$), (ii) there is a boundary region where identity is unsettled (paraconsistent non-duality), and (iii) outside that region the system regains the ability to treat the two as distinct. The third clause is crucial: without it, one might confuse becoming with permanent ambiguity or with simple synonymy.*

A simple example is phase change: "water becomes ice." Around the melting/freezing point, the system may simultaneously recognize crystalline structure (supporting distinction) and fluidity (supporting non-distinction with the earlier state), depending on measurement scale and noise. Another example is conceptual learning: a child may treat whales as fish, then as mammals; during the transition there can be a period of mixed judgments that is naturally modeled as non-dual variety.

This definition implements the Hyperseed notion Becoming (Hyperseed-Concept 62) in a way compatible with process ontology: becoming is not a primitive arrow glued onto time, but a pattern of changing distinction structure along proto-time [15].

Remark 257 (Why clause (3) matters for explanatory power)**.** *Clause (3) ensures that the boundary region functions as a* localized *site of tension rather than a global feature of the pair (x, y). In empirical terms, it supports counterfactual diagnostics: if we sample times well before and well after the transition, the model should usually predict that the system can again make a stable distinction (or stable identification) according to its standards. This makes becoming detectable from time series of judgments: one looks for a band of strong paraconsistency bracketed by regions of relative determinacy.*

Remark 258 (Sorites-style boundaries)**.** *The third clause encodes the intended "boundary" behavior: becoming is not permanent identity, but a transition region where identity/distinction is not* *crisply settled. Paraconsistent truth values let us represent such boundaries without forcing collapse to triviality.*

One can read this as a Sorites-friendly stance: across a chain of micro-variations, it is generally unreasonable to demand that each step preserve a classical predicate (e.g. "same object") while also insisting that the endpoints differ classically. The boundary region is precisely where the theory records that both continuist and discontinuist descriptions have non-negligible support. In particular, the role of a p-bit *value is not to "decide" the boundary but to keep track of how much the available constraints and observations support each side of the would-be dichotomy.*

Remark 259 (Aesthetic aside: boundary as a locus of form)**.** *There is an aesthetic dimension here: boundaries are where classification fails gracefully rather than catastrophically. In the present formalism, the boundary is not a logical defect but a region where the system explicitly represents competing commitments. This is mathematically tractable and, arguably, phenomenologically faithful (Hyperseed-Concept 53).*

More concretely: if a classification scheme is forced to return a single crisp label everywhere, then any perturbation near the boundary produces discontinuous "label flicker," which is both computationally brittle and experientially unlike how agents often cope with vague transitions. By contrast, allowing the boundary to carry structure—here, two-channel support/denial that can coexist—turns the boundary into a site where form can be articulated (e.g. by gradients of support, by competing explanations, or by alternative decompositions of a process into events). This makes the boundary a locus for further inference rather than an error condition.

7.6 A paraconsistent, quantale-valued event calculus

Hyperseed's "event and process calculi" discussion can be realized cleanly by taking the standard event calculus vocabulary and interpreting all core predicates in the p-bit quantale V. This subsection instantiates Event and Process Calculi (Hyperseed-Concept **??**) in a way that is algebraically continuous with the rest of the document.

The guiding idea is that "temporal reasoning" is not a separate module with its own ad hoc uncertainty mechanics; rather, it is another place where the same algebra of evidence is applied. In particular, we want the same operations that combine perceptual cues or conceptual constraints to also combine (i) evidence that an event occurs, (ii) evidence that it changes a fluent, and (iii) evidence that the fluent holds over an interval.

7.6.1 Signature and intended semantics

Definition 64 (Event calculus signature (paraconsistent, quantale-valued))**.** *Fix:*

- *a proto-time carrier* $(T, <)$*,*
- *a set* $\mathcal{A}$ *of actions/events,*
- *a set* $\mathcal{F}$ *of fluents (time-varying propositions).*

A p-bit*-valued event calculus assigns:*

$$\begin{aligned}
\mathrm{Happens}&: \ \mathcal{A} \times T \to V,\\
\mathrm{HoldsAt}&: \ \mathcal{F} \times T \to V,\\
\mathrm{Initiates}&: \ \mathcal{A} \times \mathcal{F} \times T \to V,\\
\mathrm{Terminates}&: \ \mathcal{A} \times \mathcal{F} \times T \to V.
\end{aligned}$$

Remark 260 (Time structure and modeling latitude). *The modest assumption "proto-time carrier $(T, <)$" is deliberately weak: in applications one may take T to be discrete (e.g. $\mathbb{N}$ for step-indexed simulation), continuous (e.g. $\mathbb{R}$ for physical time), partially ordered (for branching-time or concurrency models), or even an interval domain when the primary observations are temporally coarse. What matters for the present subsection is that $<$ provides enough structure to state propagation constraints from earlier to later times. The quantale-valued semantics then handles gradedness and inconsistency orthogonally to the choice of temporal granularity.*

Remark 261 (Notation unpacking: actions, fluents, and V). *Here $\mathcal{A}$ is the set of event-types (or action-tokens, depending on modeling choice), and $\mathcal{F}$ is the set of fluents, i.e. propositions whose holding status may vary over time. The codomain $V = [0,1]^2$ means each predicate returns a two-channel evidence value. Thus* $\text{Happens}(a,t) = (h^+, h^-)$ *is evidence for/against "event a happens at time t," and similarly for the other predicates.*

It is important that the two channels are not required to sum to 1 *(as they would in a probabilistic simplex), nor to be complements; this is what permits "boundary" and "conflict" states such as* $(0.8, 0.7)$*, which would otherwise be disallowed. Conversely, complete ignorance may be represented by something like* $(0,0)$*, where neither support nor denial is available.*

Remark 262 (Intuition, examples, and why this signature is useful). *In classical event calculus,* $\text{HoldsAt}(f,t)$ *is Boolean; here it is graded and paraconsistent. This matters precisely at boundaries: a fluent like "the door is open" may have conflicting evidence from different sensors; an initiation rule may be supported by a learned model but contradicted by a physics-based model. By putting all these predicates in the same quantale-valued space, we can aggregate evidence compositionally rather than by ad hoc priority rules.*

As a minimal example, let $\mathcal{F} = \{f\}$ and $\mathcal{A} = \{a\}$. If $\text{Happens}(a,t)$ *is high and* $\text{Initiates}(a,f,t)$ *is high, we can infer support for* $\text{HoldsAt}(f,t')$ *at later times, subject to clipping/termination evidence. This provides a concrete "time reasoning" layer compatible with Hyperseed's broader pattern-based cognitive architecture [19].*

A slightly richer example illustrates the boundary use-case directly. Suppose a is "push door" and f is "door open." A camera may support $\text{HoldsAt}(f,t)$ *while a latch sensor denies it; simultaneously, the dynamics model may support* $\text{Initiates}(a,f,t)$ *while a safety interlock model supports* $\text{Terminates}(a,f,t)$ *(e.g. the push triggers a mechanism that closes the door). In a Boolean setting this can force inconsistent axioms; here it yields a controlled state of tension that can be resolved (if at all) by additional evidence, temporal propagation, or explicit bias terms.*

Remark 263 (Why p-bit values here?). *In realistic temporal reasoning, the boundary of when an event happens, whether a fluent holds, and whether an action initiates/terminates a fluent are all uncertain and can be mutually conflicting. The* p-bit *representation makes these conflicts explicit and composable.*

Technically, the point is that the same algebraic operations used elsewhere in the document (e.g. $\otimes$ for conjunction-like composition and $\oplus$ for proof/evidence accumulation) can be reused here without special casing time. This uniformity is what makes the approach "algebraically continuous": temporal inference becomes another instance of quantale-enriched constraint propagation rather than an isolated logical subsystem.

7.6.2 Monotone rule evaluation over a quantale

A convenient semantics for Horn-style rules over a quantale is: conjunction is interpreted by the tensor $\otimes$, and alternative proofs are aggregated by join $\oplus$ (in V this is componentwise max). This yields a monotone "proof accumulation" logic.

Remark 264 (Connection to standard event calculus axioms). *Classical event calculus is often presented with axiom schemata for persistence (inertia), initiation, termination, and "clipping" (events that end a fluent's holding). The present subsection does not repeat the full axiom inventory; instead it isolates the evaluation principle that makes those axioms computationally well-behaved in the quantale-valued setting. When the familiar axiom schemata are added, each becomes a Horn-style rule (or a family of such rules) whose body combines pieces of evidence, and whose head accumulates whatever support those bodies provide.*

Definition 65 (Quantale-valued implication schema). *Let $P, Q_1, \ldots, Q_n$ be predicate symbols with appropriate arities. A* quantale-valued rule *has the form*

$$P(\vec{x}) \ \Leftarrow\ Q_1(\vec{x}_1) \ \wedge\ \cdots\ \wedge\ Q_n(\vec{x}_n),$$

and is evaluated pointwise by

$$[\![P(\vec{x})]\!] \ \geq\ [\![Q_1(\vec{x}_1)]\!] \otimes \cdots \otimes [\![Q_n(\vec{x}_n)]\!].$$

If multiple rules (or multiple instantiations) derive the same head, their contributions are joined using $\oplus$.

Remark 265 (Reading the inequality direction). *The use of "$\geq$" reflects that rules provide* lower bounds *on the evidence for the head: any derivation gives at least some support/denial profile for $P(\vec{x})$, and additional derivations can only increase that profile via joins. This matches the fixpoint view where one starts from minimal information and iteratively accumulates consequences until closure. In proof-theoretic terms, the model is the least assignment that satisfies all such inequalities.*

Remark 266 (Intuition: proofs as resources that accumulate). *The tensor $\otimes$ plays the role of "and": to support the head, we must support each body literal, and the combined support is the multiplicative accumulation of those supports. The join $\oplus$ plays the role of "or": multiple derivations contribute, and we keep the strongest support (componentwise maximum in the* p-bit *quantale). This is the algebraic analogue of how one informally reasons with multiple pieces of evidence: a chain is only as strong as its weakest link (multiplication attenuates), but parallel arguments compete and the best available argument dominates (maximum selects).*

The usefulness of this schema is that it yields a monotone immediate-consequence operator, enabling least-fixpoint semantics and iterative computation, a theme reused later when stability properties of self-modifying systems are studied via fixpoints [10].

In the p-bit *case, the same rule simultaneously propagates positive and negative information. For instance, if one body literal strongly denies a condition, then the negative channel can propagate through $\otimes$ in a controlled way. This is exactly the mechanism that lets "boundary" regions be stable under inference: the system can accumulate both pro and con evidence without exploding into triviality, while still allowing later evidence to tilt the balance.*

Remark 267 (Negation and non-monotonicity). *Classical event calculus uses negation-as-failure (e.g. "not clipped") and circumscription. In the present paper we keep the* algebraic core *monotone and represent negative information explicitly via the negative component of* p-bit *values. Non-monotone defaults (frame assumptions) are then expressed by an additional* weakness bias *(Section 8), rather than by embedding classical negation-as-failure into the core semantics.*

This separation is especially relevant for temporal reasoning because inertia is a default: in a purely monotone setting, one does not want the mere absence *of termination evidence to behave* *like a hard fact. Instead, "persistence" can be treated as a defeasible contribution whose weight is governed by effort/weakness tradeoffs, while explicit termination evidence contributes on the same algebraic footing as other sources of information.*

Remark 268 (Why monotonicity is a design choice)**.** *Monotonicity means: adding evidence cannot invalidate previous inferences, it can only enrich them. This is not how human default reasoning always behaves; however, it is how the present algebraic layer behaves so that computation and compositional semantics remain tractable. Departures from monotonicity are pushed to explicit bias terms and effort/weakness tradeoffs [3, 2], which makes the source of "non-monotone behavior" inspectable.*

From an implementation perspective, monotonicity also aligns with streaming and incremental updates: as new observations arrive, one can update Happens *or* HoldsAt *values and rerun (or incrementally maintain) the immediate-consequence operator without needing to retract inferences at the algebraic level. Any retraction-like behavior is then a matter of shifting bias, discounting stale evidence, or changing the governing rules—all of which are explicit modeling moves rather than hidden in the semantics of negation.*

7.6.3 Clipping and inertia (paraconsistent sketch)

We now give one clean paraconsistent way to express inertia. First define a derived predicate $\mathrm{Clipped}(f; t_1, t_2)$ meaning that between t_1 and t_2 there is evidence that some action terminates f.

Definition 66 (Clipped (derived; paraconsistent sketch))**.** *Define the* p-bit *value*

$$\mathrm{Clipped}(f; t_1, t_2) := \bigoplus_{\substack{a \in \mathcal{A} \\ u \in T: \\ t_1 < u < t_2}} \mathrm{Happens}(a, u) \otimes \mathrm{Terminates}(a, f, u).$$

When $\mathrm{Clipped}^+$ *is high there is strong evidence that f was terminated in* (t_1, t_2)*. When* $\mathrm{Clipped}^-$ *is high there is strong evidence against such termination.*

Remark 269 (Intuition and a simple example)**.** *This definition says: f is clipped on the interval if there exists some event a at some intermediate time u that both happens and terminates f; the degree of evidence is aggregated over all such candidates by $\oplus$ (max). If there are multiple possible terminating events, we keep the strongest combined evidence among them.*

For example, suppose two events might terminate a fluent, but only one is strongly supported by sensor data; then $\mathrm{Clipped}^+$ *will reflect the stronger candidate. Conversely, if we have explicit evidence that no terminating events occurred (perhaps from a reliable log), this can raise* $\mathrm{Clipped}^-$ *and thereby later support inertia.*

We then express inertia as: if f holds at t_1 and there is no (or weak) evidence that it was clipped before t_2, then f holds at t_2.

Definition 67 (Inertia update operator (paraconsistent sketch))**.** *Define an operator Φ on assignments to* HoldsAt *by*

$$(\Phi\,\mathrm{HoldsAt})(f, t_2) := \mathrm{HoldsAt}_0(f, t_2) \;\oplus\; \bigoplus_{\substack{t_1 \in T: \\ t_1 < t_2}} \mathrm{HoldsAt}(f, t_1) \otimes \neg\mathrm{Clipped}(f; t_1, t_2),$$

where $\mathrm{HoldsAt}_0$ *encodes initial conditions and $\neg$ is the* p-bit *involution* $\neg(v^+, v^-) = (v^-, v^+)$*.*

Remark 270 (Notation and meaning of $\neg$ in this setting)**.** *The involution $\neg(v^+, v^-) = (v^-, v^+)$ simply swaps positive and negative evidence channels. Thus $\neg\text{Clipped}(f; t_1, t_2)$ is evidence for/against "unclipped" obtained by swapping the channels of the clipped evidence. This is not classical negation; it is a bookkeeping operation appropriate to the* p-bit *evidence interpretation, and it is compatible with the monotone proof-accumulation semantics described above.*

Remark 271 (Intuition: inertia as propagation through gaps)**.** *The update says: f holds at t_2 either because it is an initial fact $\text{HoldsAt}_0(f, t_2)$, or because there exists an earlier time t_1 where f held and the interval (t_1, t_2) carries counter-evidence to termination. One may view this as a kind of temporal path propagation: holding status flows forward unless clipped, and the quantale operations determine how strongly it flows.*

This is the temporal analogue of closure constructions used elsewhere (e.g. composition/closure of relations in the toy model section): local constraints propagate globally by repeated application of a monotone operator until a fixpoint is reached [7].

Remark 272. *Using $\neg$Clipped here is not "absence of evidence"; it is the explicitly represented counter-evidence (or support for "unclipped"). This is consistent with the paraconsistent stance: one can simultaneously carry evidence that a termination occurred and evidence that it did not, and inertia will then yield a correspondingly paraconsistent holding status.*

Theorem 6 (Existence of least fixpoint for monotone temporal inference)**.** *Assume T is finite and all event-calculus rules are built using only $\oplus$, $\otimes$, and monotone operations on V. Then the induced immediate-consequence operator (e.g. Φ above) is monotone on the complete lattice of assignments $(V^{\mathcal{F}\times T}, \leq)$ and hence has a least fixpoint.*

Remark 273 (What the theorem says and why it matters)**.** *The theorem guarantees that the iterative "accumulate evidence until stable" procedure is not merely heuristic: there exists a least stable assignment of truth values consistent with the rules. This is crucial for making the event calculus a well-defined semantic object rather than just a collection of update equations. It also connects temporal reasoning to the same fixpoint machinery used elsewhere in Hyperseed-style analyses of stability and self-reference [10]: time inference becomes a special case of order-theoretic stabilization.*

Proof. $V^{\mathcal{F}\times T}$ is a complete lattice under pointwise order. Each clause contribution is monotone because $\otimes$ and $\oplus$ are monotone in each argument, and joins over index sets preserve monotonicity. Therefore the overall operator is monotone. By the Knaster–Tarski theorem, a monotone operator on a complete lattice has a least fixpoint. $\square$

Remark 274 (Proof sketch and intuition)**.** *The high-level strategy is: (i) identify the space of all candidate interpretations as an ordered set, (ii) show the update operator never reverses order (monotonicity), and then (iii) invoke the general fixpoint theorem. Pointwise order on $V^{\mathcal{F}\times T}$ means: one assignment is $\leq$ another if it assigns $\leq$-smaller* p-bit *values to every fluent-time pair. Monotonicity follows because $\otimes$ and $\oplus$ are monotone operations and because taking a join (supremum) over any family of monotone expressions preserves monotonicity. A visual intuition is to imagine starting from the bottom assignment (no evidence anywhere) and repeatedly applying Φ to "grow" the set of supported conclusions. Because the update can only add or strengthen evidence (never retract it), the sequence ascends in the lattice and must stabilize in finitely many steps when T is finite. The stabilized point is the least fixpoint. In more algebraic terms, this is the familiar picture behind Kleene iteration for monotone operators: starting at $\perp$ and iterating Φ yields an increasing chain whose supremum is a fixpoint, and when the underlying carrier is finite (or,* *more generally, when the lattice has finite height) the chain cannot strictly increase forever. The least-fixpoint characterization is important conceptually because it encodes a minimality principle: nothing is concluded unless it is forced by the update rules, and no extra evidence is introduced beyond what is generated by repeated rule application.*

7.7 Time contexts and temporal similarity

Hyperseed defines several important "event regularity" notions using the idea that each timepoint comes with a *time context* (what holds then), and that nearby/similar timepoints have similar time contexts. The key modeling move is to treat temporal structure as being (at least partly) induced by patterns in fluent valuations: rather than assuming a fixed external metric on T, we derive a notion of proximity from how the world is described at those times. This is especially natural when T is a proto-time carrier (e.g. a partially ordered or otherwise non-metric structure) and when different observers may coarse-grain or refine what distinctions matter.

Definition 68 (Time context)**.** *Fix an event calculus instance with fluent set $\mathcal{F}$ and proto-time carrier T. The* time context *at time $t \in T$ is the vector of fluent truth values*

$$\mathrm{TimeCtx}(t) \ := \ \big(\mathrm{HoldsAt}(f,t)\big)_{f\in\mathcal{F}} \in V^{\mathcal{F}}.$$

Remark 275 (Intuition and example)**.** $\mathrm{TimeCtx}(t)$ *is the full "state description" of what is (paraconsistently) true at time t, as far as the chosen fluent vocabulary can express. For example, if $\mathcal{F} = \{\mathsf{DoorOpen}, \mathsf{LightOn}\}$ then* $\mathrm{TimeCtx}(t)$ *is the ordered pair of* p-bit *values assigned to these two propositions at time t. This concept operationalizes the idea that a timepoint is not merely a coordinate but a bundle of conditions; it is an explicit bridge between ordering-based time and state-based dynamics. Note that* $\mathrm{TimeCtx}(t)$ *depends on the chosen fluent set $\mathcal{F}$: refining the vocabulary (splitting one fluent into several more specific ones) generally increases the dimension of $V^{\mathcal{F}}$ and can make previously indistinguishable timepoints separable, while a coarser vocabulary can collapse distinct situations into the same apparent context. In this sense, a "time context" is not a metaphysically complete state of the world but an observer-relative description at a particular granularity.*

Definition 69 (Truth-value projection and similarity)**.** *Choose any monotone scalar projection $\pi : V \to [0,1]$; two useful examples are*

$$\pi_{\mathrm{pos}}(v^+,v^-) = v^+, \qquad \pi_{\mathrm{net}}(v^+,v^-) = \frac{1}{2}\big(1 + (v^+ - v^-)\big).$$

Define a similarity on V by

$$\mathrm{sim}_V(v,w) \ := \ 1 - |\pi(v) - \pi(w)|.$$

Define time-context similarity for $t_1, t_2 \in T$ by

$$\mathrm{sim}_T(t_1,t_2) \ := \ \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}} \mathrm{sim}_V\big(\mathrm{HoldsAt}(f,t_1),\, \mathrm{HoldsAt}(f,t_2)\big).$$

Remark 276 (Notation and modeling choice)**.** *The map π compresses a two-channel* p-bit *value to a single scalar in $[0,1]$ so that we can use a simple absolute-difference similarity. π_{pos} measures similarity of positive support only; π_{net} measures a signed "net support" that discounts negative* *evidence. Different projections correspond to different stances about how to treat contradiction: one may focus on what is supported, or on the balance between support and opposition.*

The similarity $\mathrm{sim}_T(t_1, t_2)$ *is just the average similarity across all fluents; it is a computable proxy for "how similar the world looks at* t_1 *and* t_2 *in this vocabulary." Because* sim_V *takes values in* $[0, 1]$*, so does* sim_T*, and the endpoints have a clear reading:* $\mathrm{sim}_T(t_1, t_2) = 1$ *exactly when* $\pi(\mathrm{HoldsAt}(f, t_1)) = \pi(\mathrm{HoldsAt}(f, t_2))$ *for every* $f \in \mathcal{F}$*, while smaller values quantify the average magnitude of disagreement under the chosen projection. The monotonicity requirement on* π *(with respect to the lattice order on* V*) ensures that strengthening evidence in* V *cannot perversely decrease the projected scalar in a way that would invert comparisons; this makes similarity judgments stable under the "adding evidence" dynamics of the underlying paraconsistent calculus.*

Remark 277 (Why similarity is defined via projection rather than directly in V)**.** *One could define a similarity directly on* $V = [0, 1]^2$ *using a 2D metric, but the projection method makes explicit which aspect of the paraconsistent value is being compared. This is philosophically in the spirit of Hyperseed: similarity is not an absolute given but an observer-relative construction, and the choice of* π *is part of the observer model. Technically, the projection also makes it easy to swap in alternative scalar summaries without changing the surrounding definitions: for instance, one could use* $\pi(v^+, v^-) = \max(v^+, 1 - v^-)$ *to privilege "unopposed support," or* $\pi(v^+, v^-) = 1 - v^-$ *to track absence of counterevidence. Each such choice corresponds to a different operational meaning of "two truth values being close."*

Remark 278 (Basic properties of the induced similarities)**.** *For any fixed projection* π*,* sim_V *is symmetric and satisfies* $\mathrm{sim}_V(v, v) = 1$ *for all* $v \in V$*, hence* $\mathrm{sim}_T(t, t) = 1$ *for all* $t \in T$*. However,* sim_V *(and therefore* sim_T*) is generally not transitive when turned into a crisp relation via thresholding: it is possible to have* t_1 *similar to* t_2 *and* t_2 *similar to* t_3 *while* t_1 *is not similar to* t_3*, especially when similarity is defined by an average across fluents and disagreements are distributed across different coordinates. This non-transitivity is not a bug in the present setting: it reflects the familiar phenomenon that "looks similar" is typically a neighborhood notion rather than an equivalence relation, and it supports a flexible notion of continuity that does not require global clustering.*

Definition 70 (Temporal similarity)**.** *Fix a threshold* $\tau \in (0, 1]$*. Define*

$$\mathrm{TemporalSimilarity}(t_1, t_2) \textit{ to hold if } \mathrm{sim}_T(t_1, t_2) \geq \tau.$$

Remark 279 (Intuition and example)**.** $\mathrm{TemporalSimilarity}(t_1, t_2)$ *is a derived crisp predicate that says: the system treats the two timepoints as "close enough" in state-space. For instance, if* $\tau = 0.95$*, then only timepoints with very similar fluent valuations are considered temporally similar. This definition is useful because it allows qualitative temporal predicates (like "continuous") to be stated without assuming a numeric metric on* T*; similarity is derived from what holds, not from an external clock. In particular, one can speak about "local smoothness" of a trajectory* $t \mapsto \mathrm{TimeCtx}(t)$ *by requiring that successive or causally adjacent timepoints (as given by the proto-time structure) satisfy* $\mathrm{TemporalSimilarity}$*, even when there is no canonical notion of successor in* T*.*

Remark 280. *Other choices are possible and sometimes preferable: e.g. using a weighted average over fluents, or using quantale weakness (Section 8) to measure how many distinctions between time contexts are collapsed by a given observer. The present choice is intentionally simple and computable. A weighted average can encode domain knowledge (some fluents are more salient or more reliably sensed) by assigning larger weights to dimensions that should dominate similarity judgments, and it can also mitigate the effect of many irrelevant fluents washing out a small but* *important change. Likewise, replacing the absolute-difference form with a different scalar similarity (e.g. a kernel) can change how sharply similarity decays, which in turn affects which temporal patterns count as regularities.*

Remark 281 (Connection to weakness and observer-dependent coarse-graining). *If an observer collapses many differences between time contexts (Hyperseed-Concept 202), then many pairs* (t_1, t_2) *will become temporally similar under that observer's projection and aggregation scheme. This provides a direct route from effort/simlicity theory to the phenomenology of "time flowing smoothly" versus "time being jagged" [3]. Concretely, weakening can be understood as reducing effective resolution in* $V^{\mathcal{F}}$*: distinct vectors of* p-bit *values that would be separated by a high-effort observer may map to nearby (or identical) projected summaries under a low-effort observer, thereby increasing* sim_T *on average. Under this lens, temporal coarse-graining is not merely subsampling in* T*; it is a transformation of the state-description map* $t \mapsto \mathrm{TimeCtx}(t)$ *that changes which variations are treated as signal versus noise.*

7.8 Persistent, continuous, increasing, decreasing

Hyperseed lists several meta-predicates about events/fluents. We implement these as constraints on the trajectory $t \mapsto \mathrm{HoldsAt}(f, t)$ relative to the proto-time order and the temporal similarity relation. In this reading, these meta-predicates are not additional primitive symbols; rather, they are properties of an interpreted event-calculus model, checkable from the induced HoldsAt relation together with whatever auxiliary similarity/projection structure the modeler provides.

Definition 71 (Continuous fluent (relative to a proto-time)). *A fluent* $f \in \mathcal{F}$ *is* continuous *(relative to* $(T, <)$, TemporalSimilarity*) if for all* $t_1, t_2 \in T$,

$$\mathrm{TemporalSimilarity}(t_1, t_2) \;\to\; \mathrm{sim}_V\big(\mathrm{HoldsAt}(f, t_1),\, \mathrm{HoldsAt}(f, t_2)\big) \;\geq\; \eta$$

for a chosen threshold $\eta \in (0, 1]$.

It is often convenient to read $\mathrm{TemporalSimilarity}(t_1, t_2)$ as designating a class of "small" or "contextually close" temporal transitions, even when T is discrete or only partially organized by $<$. Similarly, sim_V can be instantiated as a kernel, a cosine similarity on embeddings, or $1 - \mathrm{dist}_V(\cdot, \cdot)$ for some value-space distance; the point is that continuity is asserted in the value space V rather than in T. Note also that η acts as a tunable tolerance: taking $\eta = 1$ forces invariance of HoldsAt on all temporally similar pairs, whereas smaller η permits controlled variation.

Remark 282 (Intuition and example). *Continuity here is not about infinitesimals; it is about stability under the chosen notion of temporal similarity. If* t_1 *and* t_2 *look similar in their overall time contexts, then* f *should not change much between them. For example, if "temperature" is represented as a fluent and the system regards two moments as similar in relevant conditions, then the temperature fluent should have similar values at those moments. This captures a qualitative sense of continuity appropriate to proto-time, where* T *may not carry a metric topology.*

The definition is useful because it separates "what counts as a small time change" (encoded by TemporalSimilarity*) from "what counts as a small fluent change" (encoded by* sim_V *and* η*). Both are observer-relative and can be tuned.*

A useful operational consequence is that continuity can be evaluated as a family of constraints, one per temporally-similar pair, and therefore can be implemented either as a hard logical axiom or as a soft penalty in a probabilistic/optimization layer. For instance, in a weighted setting one may penalize violations by a hinge loss $\max\{0, \eta - \mathrm{sim}_V(\mathrm{HoldsAt}(f, t_1), \mathrm{HoldsAt}(f, t_2))\}$ whenever $\mathrm{TemporalSimilarity}(t_1, t_2)$ holds, thereby allowing noisy observations while still encoding the intended smoothness bias.

Definition 72 (Persistent fluent (relative to a proto-time))**.** *A fluent $f \in \mathcal{F}$ is* persistent *if it is continuous and, additionally, its truth-value varies only weakly across the entire interval between initiation and termination, i.e. there exists $\epsilon \geq 0$ such that whenever $t_1 < t_2$ and f holds "throughout" $[t_1, t_2]$ (as determined by the event calculus instance), then*

$$|\pi(\mathrm{HoldsAt}(f, t_1)) - \pi(\mathrm{HoldsAt}(f, t_2))| \ \leq \ \epsilon.$$

The phrase "holds throughout" can be understood in the event-calculus sense: f is supported on the interval because it is initiated at or before t_1, not terminated before t_2, and inertia (or the chosen persistence axioms) propagate its holding status across intermediate times. When T is dense or when HoldsAt is only partially observed, this "throughout" condition is best viewed as a model-relative judgment rather than a syntactic universal quantification over all $t \in [t_1, t_2]$; the intended meaning is that the theory treats the interval as one of uninterrupted holding.

Remark 283 (Intuition and why persistence is stronger than continuity)**.** *Continuity constrains local change: nearby time contexts should not flip the fluent abruptly. Persistence adds a global bound: across any span where the fluent is considered to hold, its projected value should not drift by more than ϵ. For example, "the light is on" may be persistent if it remains stably supported from switch-on to switch-off, even if minor sensor noise causes small fluctuations. By contrast, a fluent like "the music is loud" might be continuous but not persistent if it rises and falls gradually.*

This definition is useful because it lets us distinguish steady conditions from smooth evolutions. The parameter ϵ makes explicit how much variation is tolerated before we stop calling the fluent persistent.

Two limiting cases help orient the parameter choices. If $\epsilon = 0$ and π is chosen so that $\pi(\mathrm{HoldsAt}(f, t))$ is Boolean-valued, then the persistence constraint forces literal invariance of the projected value on any "holding" span, which aligns with the classical intuition of a fluent remaining true until terminated. At the other extreme, if ϵ is large (e.g. $\epsilon = 1$ for $\pi : V \to [0, 1]$), the additional persistence clause becomes vacuous and persistence collapses back toward continuity; in that regime, the distinction between "steady" and "smoothly varying" is pushed back into the choice of η and TemporalSimilarity.

Definition 73 (Increasing/decreasing fluent)**.** *Fix a scalar projection $\pi : V \to [0, 1]$. A fluent f is* increasing *(relative to $(T, <)$) if for all $t_1 < t_2$,*

$$\pi(\mathrm{HoldsAt}(f, t_1)) \ \leq \ \pi(\mathrm{HoldsAt}(f, t_2)).$$

Similarly, f is decreasing *if for all $t_1 < t_2$,*

$$\pi(\mathrm{HoldsAt}(f, t_1)) \ \geq \ \pi(\mathrm{HoldsAt}(f, t_2)).$$

In proto-time, $<$ need not be a complete order; it may be a partial order encoding precedence. The above monotonicity clauses should then be read as applying to all comparable pairs $t_1 < t_2$. In particular, if t_1 and t_2 are incomparable (neither $t_1 < t_2$ nor $t_2 < t_1$), the definition does not constrain $\pi(\mathrm{HoldsAt}(f, t_1))$ versus $\pi(\mathrm{HoldsAt}(f, t_2))$, which matches the idea that monotone "change over time" is only meaningful along directed temporal reachability.

Remark 284 (Intuition and examples)**.** *Monotonicity is stated after projection to a scalar because "increasing" is inherently a scalar notion. For example, if f represents "confidence that a goal is achieved," one might take $\pi = \pi_{\text{pos}}$ and ask that positive support does not decrease over time. If f represents "amount of fuel remaining," one might impose decreasingness.*

This is useful as a compact way to express directed change, and it can be combined with the earlier event calculus machinery: one can infer monotone trends from initiation/termination patterns and inertia, then check them against observed HoldsAt *values as a consistency constraint.*

It is also useful to note the interaction between increasing and decreasing: if a fluent is both increasing and decreasing with respect to the same π and order $<$, then $\pi(\text{HoldsAt}(f, t))$ must be constant on each chain in $(T, <)$ (i.e. on each totally ordered subset). This captures the idea that "no upward drift" together with "no downward drift" yields stability, but only relative to the comparabilities present in proto-time.

Remark 285 (Non-persistent events)**.** *Hyperseed notes that continuous/increasing/decreasing are most relevant for non-persistent events. In our formalization, persistence is the strong constraint (low total variation over a span), while continuity and monotonicity constrain only local or directional change. Thus a non-persistent fluent may be continuous yet non-constant, or increasing yet not persistent.*

One can also mix these predicates to express common qualitative regimes. For example, "warming up" can be modeled as (i) continuous with respect to a chosen temporal similarity and (ii) increasing with respect to a temperature projection π, without being persistent; conversely, "being connected to WiFi" might be persistent (small ϵ) but neither increasing nor decreasing (no directed trend is intended, only stability when it holds).

Remark 286 (A conceptual bridge to emergence)**.** *When a fluent violates persistence but satisfies continuity, it suggests the presence of a smooth transformation rather than a stable condition. Such smooth transformations are often where new patterns become detectable (emergence as a change in compressibility), foreshadowing the pattern and emergence layer in Section 9 [5].*

In particular, continuity provides a baseline expectation of "no abrupt jumps" under the system's own notion of temporal neighborhood, so sustained deviations from persistence can be interpreted as structured drift rather than noise. This is precisely the regime in which higher-level descriptions (e.g. a slowly varying latent factor, a staged process, or a new macro-variable) can become more compressive than a frame-by-frame account, making the persistence/continuity distinction serve as an interface between low-level event calculus dynamics and later pattern-level analyses.

7.9 Process view: becoming as functorial evolution (bridge to process calculi)

Event calculus is one way to reason about change; a complementary way is to treat the world (or mind) as a *process indexed by proto-time.* In this second lens, "becoming" is not primarily a list of event-occurrences but a rule that assigns, to each experienced position in an order, a structured state together with coherent transition maps. The emphasis is on how successive situations hang together (compositionally), rather than on how a theory entails that a fluent holds.

Definition 74 (Becoming as a time-indexed process (functorial sketch))**.** *Let $(T, <)$ be a proto-time. Consider the thin category $\mathbf{T}$ whose objects are elements of T and with a unique morphism $t_1 \to t_2$ iff $t_1 < t_2$. A* process *over T is a functor*

$$S : \mathbf{T} \to \mathbf{Sys}$$

into a chosen category of system states **Sys** *(e.g. sets, measurable spaces, or V-enriched structures). In particular, the functoriality condition forces compatibility with the proto-temporal order: if $t_1 < t_2 < t_3$, then the induced maps satisfy $S(t_1 \to t_3) = S(t_2 \to t_3) \circ S(t_1 \to t_2)$, so that multi-step becoming factors into successive becomings without ambiguity.*

Remark 287 (Notation and categorical intuition)**.** *A* thin category *is a category with at most one morphism between any two objects; here the existence of a morphism $t_1 \to t_2$ encodes exactly the relation $t_1 < t_2$. A functor S assigns to each timepoint t an object $S(t)$ (a "state") in* **Sys***, and to each precedence $t_1 < t_2$ a morphism $S(t_1) \to S(t_2)$ (an "evolution map"). Thus, time becomes a parameter indexing structure-preserving transformation, rather than a numerical axis labeling states. Equivalently, one can regard* **T** *as the categorical packaging of the proto-time order and S as a representation of that order inside* **Sys***.*

This is a canonical formalization of Process (Hyperseed-Concept 140) and clarifies how proto-time supports process calculi: the order supplies composable arrows, and the functor turns them into composable state transitions. From this viewpoint, the "laws of motion" (broadly construed) are precisely the chosen morphisms in **Sys** *that witness admissible change; changing* **Sys** *changes what counts as a lawlike update (deterministic function, stochastic kernel, simulation relation, refinement map, etc.).*

Remark 288. *This functorial viewpoint is the categorical form of "becoming": morphisms represent allowed transformations along the experienced order. It also makes explicit what is often only implicit in informal talk about time: the content is not merely that there are states at moments, but that there are coherent translations between them that respect the proto-temporal composition. In later sections, this will interact with pattern webs and mind-world correspondence by replacing* **Sys** *with a category of pattern-flow structures and by admitting approximate morphisms. Concretely, "approximate" can be implemented by moving to an enriched or metric setting in which morphisms carry a notion of distortion (or error budget), so that becoming can be stable under coarse-graining, attention limits, and perceptual noise.*

Remark 289 (Why include both event calculus and functorial process views?)**.** *Event calculus emphasizes* logical entailment *about discrete events and fluents; the functorial view emphasizes* structured evolution *and compositionality. They are complementary lenses on the same phenomenon: one can often derive an event calculus from a process model by extracting predicates from states, and conversely one can build a process model from an event calculus by taking time contexts as states. Keeping both views available supports later synthesis with pattern-based cognition and with the more explicitly process-ontological themes that trace back to Peircean and Whiteheadian traditions [14, 15]. The bridge to process calculi is particularly transparent in the functorial picture: many calculi distinguish between (i) the space of configurations and (ii) the step relation or transition system, and the functor packages exactly this distinction while enforcing associativity of successive steps via categorical composition. When proto-time is taken as a partial order (rather than a total one), the same construction supports branching and concurrency: incomparable elements of T represent independent or unordered developments, while comparable elements represent precedence constraints, aligning the categorical skeleton with causal or dependency structure rather than with a single global clock.*

8 Effort, resistance, and simplicity

Outline

- Represent "effort" as a resource/cost structure suitable for compositional reasoning.

- Model resistance/submission and opening/closing as (directed) changes to a system's distinction policy.
- Define observer-relative simplicity in terms of effort minimization and (dually) weakness maximization.
- Define compositional simplicity as a least-fixed-point/dynamic-programming object over a combination system.
- Connect these notions to generalized Kolmogorov complexity and to structural (grammar-like) complexity.

To reduce ambiguity, the word "effort" will be treated as a *typed* magnitude rather than an informal proxy for time, energy, or difficulty. In particular, the goal is to support reasoning of the form: (a) a task can be achieved by alternative constructions, (b) each construction incurs effort, (c) effort accumulates along chains of sub-operations, and (d) "simplicity" is the selection of low-effort constructions relative to a given system/observer. This subsection is therefore preparatory: it fixes the algebraic shape of the quantity that later sections will propagate through patterns, habits, and predictive correspondence.

The five bullet points above should be read as a progression from *local* to *global* structure. First we choose a representation of effort that supports addition (doing things in sequence) and comparison/minimization (choosing among alternatives). Then we interpret certain phenomenological notions—opening/closing and resistance/submission—as structured transformations whose costs can be measured in the same effort space. Finally, we define two notions of simplicity: one that is observer-relative ("simple for this system") and one that is compositional ("stable under building wholes from parts"), and we relate both to established complexity measures.

Summary and Hyperseed concepts covered

Hyperseed treats *effort* as primitive and observer-relative: what is "simple" is what a particular system can do, represent, or predict with low effort (Hyperseed-Concept 100). To make this mathematically usable, we represent effort via a cost quantale and use it to define (i) simplicity as minimum representational effort (Hyperseed-Concept 169), and (ii) compositional simplicity as a stability property under building wholes from parts (Hyperseed-Concept 82). We then formalize *opening/closing* (Hyperseed-Concepts 125, 72) as relaxing/tightening distinction policies (Hyperseed-Concept 98) and *resistance/submission* (Hyperseed-Concepts 158, 183) as the effort gradient associated with such changes.

More concretely, a *cost quantale* is used because it simultaneously provides (i) an order expressing "no more effort than," (ii) an operation expressing sequential composition of efforts, and (iii) a notion of "taking the best available option" as an infimum/meet (or, dually, "taking the worst" as a supremum/join, depending on conventions). This avoids committing to effort being numeric; it could be vector-valued (time, risk, attention), partially ordered (incomparable trade-offs), or even extended with an "impossible" element representing infeasible constructions. The essential requirement is that effort can be *propagated* through a composite description and then *optimized* over the space of descriptions.

The phrase "minimum representational effort" should be read as a design principle that converts a descriptive problem into an optimization problem. Given a target object (a percept, prediction, explanation, or action plan), and a family of available representational strategies (codes, models, grammars, decompositions), the simplicity of the target *for the observer* is identified with the least effort among those strategies. This makes simplicity explicitly dependent on the observer's repertoire: an object can be simple for one system (because it has an efficient internal code) and complex for another (because it lacks the relevant decomposition).

The "distinction policy" viewpoint is intended to make opening/closing precise. A distinction policy is a rule (implicit or explicit) for which differences in the world are treated as relevant, and which are collapsed as irrelevant. Opening corresponds to a directed change that *admits more distinctions* (finer discrimination, increased sensitivity, more degrees of freedom in representation), while closing corresponds to a directed change that *forbids distinctions* (coarser discrimination, compression, or ignoring degrees of freedom). Because these changes alter what can be represented at all, they naturally induce changes in feasible effort: certain representations become available/unavailable, and the costs of existing representations may change.

Within this frame, resistance/submission is not introduced as a separate primitive but as the *effort landscape* associated with changing the policy. Resistance expresses the additional effort required to move in a certain direction in policy-space (for example, to open when a system is currently closed, or to close when it is currently open), while submission expresses the relative ease of moving with the local gradient (the direction that decreases effort or increases weakness, depending on the dual convention). Thus "resistance" is not merely a psychological label; it is a measurable asymmetry in how effort changes under directed transformations of the system's discrimination and control structure.

The dual phrase "weakness maximization" is included to emphasize that effort minimization can be equivalently presented as maximizing what a system cannot (or does not need to) do. On the effort-minimizing view, a simple representation is one that requires little work. On the weakness-maximizing dual view, a simple representation is one that *intentionally leaves many distinctions unresolved* while still being adequate for the task, thereby preserving slack and avoiding commitment. This duality will matter later when discussing robustness: sometimes a system becomes more capable by refusing to overfit distinctions, i.e. by choosing policies that remain weak in the right directions.

Finally, compositional simplicity is treated as a least-fixed-point/dynamic-programming object because "building wholes from parts" introduces recursion and shared substructure. If a system can construct complex objects by combining simpler ones, then the effort to represent an object is not just a property of the object in isolation but depends on the best decomposition into sub-objects, the costs of the combination operations, and the reuse of previously constructed components. A fixed-point characterization captures the stable assignment of minimal effort to each constructible object under these combination rules, and it aligns with standard dynamic-programming logic: compute the cost of a whole by minimizing over ways of assembling it from cheaper parts.

The link to generalized Kolmogorov complexity is then conceptual rather than merely analogical. In classical Kolmogorov complexity, simplicity is the length of the shortest program producing a string, relative to a universal machine. Here, "program length" is replaced by the observer's effort quantale, and "universal machine" is replaced by the observer's available representational operations and distinction policy. Structural (grammar-like) complexity enters when representations are not flat codes but compositional descriptions (grammars, rules, factor graphs, rewrite systems), in which case the least-effort description typically reflects reusable structure rather than mere shortest encoding.

Remark 290. *Philosophically, this section attempts to treat "effort" not as a metaphor but as a formally composable magnitude: something one can add along a chain of operations, minimize over alternatives, and propagate through a system as a constraint. This makes effort play a role analogous to "energy" in physics, but interpreted as an experiential-cognitive primitive rather than* *a physical conserved quantity. The intended application is that later layers (patterns, habits, mind-world correspondence) can talk about simplicity and resistance without silently switching between incompatible notions of cost.*

Remark 291. *The intended level of generality is deliberately higher than "effort equals time" or "effort equals number of steps." Some observers have parallelism, caching, or learned shortcuts; others face bottlenecks such as attention, working memory, or sensorimotor precision. A quantale-based treatment is meant to keep these possibilities open while still enforcing the minimum structure needed for compositional proofs: monotonicity (more demanding tasks should not cost less), compositionality (doing two things should cost at least doing each), and optimizability (alternatives can be compared). This is also what makes it possible to talk about "effort gradients" for resistance/submission without presupposing differentiability or even numeric effort.*

Hyperseed concepts covered.

- Effort.
- Resistance/submission; opening/closing.
- Simplicity; compositional simplicity.
- (Generalized) Kolmogorov complexity; structural complexity.
- Minimum representational effort (as a design principle).

8.1 Effort as a cost structure

Hyperseed begins from the experiential primitive "effort" and then builds a derived notion of simplicity. In a mathematical reconstruction, it is useful to represent effort using a compositional algebra that supports: (i) sequential composition of acts/inferences/operations, (ii) choice among alternatives, and (iii) taking minima over competing decompositions.

One may view (iii) as the formal expression of "search over plans": a single high-level act can often be realized by many different decompositions into subacts, and effort should select the best available decomposition by taking an infimum over a (possibly infinite) family of candidates. The framework below is deliberately permissive about cardinality: in many realistic settings (e.g. continuous control, approximate inference, or anytime algorithms) the relevant set of decompositions is naturally infinite, so the availability of arbitrary infima is not merely a technical luxury but the algebraic counterpart of "optimize over all available implementations."

A standard choice (also used implicitly in shortest-path and dynamic programming) is the *Lawvere* cost quantale.

Definition 75 (Effort quantale)**.** *Let*

$$\mathcal{E} := ([0,\infty], \geq, \inf, +, 0),$$

where the order is reversed (so that $a \leq_{\mathcal{E}} b$ means $a \geq b$ as real numbers), the join in the quantale order is inf *(ordinary minimum), and the monoidal product is $+$ (sequential composition of costs). Then $\mathcal{E}$ is a commutative quantale: $+$ distributes over arbitrary infima.*

It is sometimes helpful to keep in mind why $[0,\infty]$ (with ∞ included) is used rather than $[0,\infty)$. Allowing ∞ provides a canonical value for "impossible" or "unavailable" processes within a context, and it ensures that arbitrary infima always exist in the underlying complete lattice structure. This aligns the formalism with optimization practice: if an operation cannot be performed under the current constraints, it can be assigned effort ∞, and taking minima over alternatives then correctly ignores it whenever a finite alternative exists.

Remark 292. ***Notation unpacking.*** *The tuple* $([0,\infty], \geq, \inf, +, 0)$ *should be read as follows: the underlying set is the extended nonnegative reals* $[0,\infty]$*; the order relation used for the quantale is the* reversed *real order (so "smaller effort" becomes "larger element" in the quantale order); the quantale join (least upper bound in the quantale order) is* inf *in the usual real order; the monoidal operation is* $+$ *with unit* 0*. Saying "+ distributes over arbitrary infima" is exactly the familiar identity*

$$a + \inf_i b_i \;=\; \inf_i (a + b_i),$$

which is what makes dynamic programming algebraically natural.

One can also read the distributivity law operationally: if one has a fixed prefix act of cost a and then must choose an optimal continuation among many options with costs b_i, the optimal total cost is obtained by adding a once and then optimizing the remainder, rather than re-solving the whole problem from scratch for each candidate continuation. This is precisely the algebraic skeleton behind Bellman-style optimal substructure.

Remark 293. ***Intuition.*** *The reversed order is a small but important conceptual maneuver: we want "better" to mean "easier," i.e. lower cost. But many later constructions (especially those expressed in quantale language, including weakness; Hyperseed-Concepts 202, 143) are more naturally written as monotone statements in an order where "better" corresponds to "larger." So we flip the order once and then keep monotonicity statements in their natural direction; this is consonant with the quantale-based treatment of weakness in [3] (and also with the broader Hyperseed ontology presentation [1]).*

A related perspective (useful later, though not required here) is that $\mathcal{E}$ is the canonical "cost domain" used in Lawvere's approach to generalized metric spaces and enriched categories: addition plays the role of path concatenation, and infimum plays the role of taking the best path among many. This connection is one reason the same algebra reappears across shortest-path problems, program semantics, and compositional models of planning.

Remark 294. ***Examples.*** *If a system can solve a task by either: (1) doing method A costing* 3 *units, or (2) doing method B costing* 5 *units, then "choosing among alternatives" corresponds to* $\inf(3,5) = 3$*. If it must do two steps in sequence costing* 3 *and* 5*, then the sequential effort is* $3 + 5 = 8$*. Thus, even before one speaks of "minds," this algebra captures the basic grammar of optimization.*

It is also worth emphasizing that the "alternative set" over which inf is taken need not be limited to two options. For instance, if a process can be implemented by selecting an integer parameter k (say, the number of refinement iterations) and the effort scales as $f(k)$, then the best available implementation corresponds to $\inf_{k\in\mathbb{N}} f(k)$. If the infimum is not achieved by any finite k (a common situation in asymptotic approximations), the formalism still cleanly represents "arbitrarily close" improvements, which is often the right abstraction when one separates idealized capability from finite-resource constraints.

Remark 295. *We use the reversed order so that "better" (lower cost) corresponds to "larger" in the quantale order. This makes formulas with* inf *and* $+$ *align with the algebra of optimality:*

choosing among alternatives corresponds to taking inf*, and composing steps corresponds to adding costs.*

Definition 76 (Effort of a process)**.** *Let $\mathcal{P}$ be a set of processes available to an observer in a fixed context C. An* effort assignment *is a map*

$$\mathrm{Eff}_C : \mathcal{P} \to [0, \infty]$$

satisfying (as modeling assumptions):

- *(Sequential subadditivity)* $\mathrm{Eff}_C(p; q) \leq \mathrm{Eff}_C(p) + \mathrm{Eff}_C(q)$ *whenever $p; q$ denotes running p then q,*
- *(Choice) if p can be implemented by choosing between p_1 and p_2, then* $\mathrm{Eff}_C(p) = \min(\mathrm{Eff}_C(p_1), \mathrm{Eff}_C(p_2))$*.*

The definition intentionally separates two layers: the abstract quantale $\mathcal{E}$, which specifies the algebra of combining and comparing costs, and the concrete map Eff_C, which instantiates that algebra for a particular observer and context. In categorical language (not required for later sections, but conceptually clarifying), Eff_C behaves like a *lax* monoidal measure of sequential composition: it respects the monoidal operation $+$ up to an inequality rather than on-the-nose equality, reflecting that real implementations can reuse intermediate results, amortize setup costs, or exploit structure.

Remark 296. ***Intuition.*** *An effort assignment is an observer-indexed bookkeeping device: it says, within context C, how costly it is for the observer to execute certain operations. The inequalities are deliberately weak. Sequential subadditivity says that doing two things in sequence should not cost* more *than paying for each separately (because one can always execute them in the naive way). In real systems there may be* synergies *(the combined act costs less than the sum) or* frictions *(overheads not modeled here); the subadditivity axiom allows synergies and leaves room to represent frictions elsewhere (e.g. via t_C later).*

One can also interpret context dependence explicitly: the same "nominal" process p may have very different effort in different contexts C because the observer's available tools, cached state, permissions, training, or environmental affordances differ. This is part of why the effort assignment is not treated as an intrinsic property of p alone. For example, access to a GPU, a precomputed index, or a specialized library can change $\mathrm{Eff}_C(p)$ by orders of magnitude without changing the abstract specification of p.

Remark 297. ***Examples.*** *If p is "load a dataset" and q is "train a model," then $p; q$ is the sequential procedure. If training reuses cached preprocessing from loading, then $\mathrm{Eff}_C(p; q)$ may be strictly less than $\mathrm{Eff}_C(p) + \mathrm{Eff}_C(q)$. For "choice," if p is "sort a list" and p_1 is quicksort while p_2 is mergesort, then the observer may treat $\mathrm{Eff}_C(p)$ as the cheaper of the two implementations available in that context.*

A further example of frictions (not captured by subadditivity alone) is a fixed setup cost: suppose running any nontrivial process incurs a one-time initialization overhead (allocating memory, starting a service, warming a cache). Then one may have $\mathrm{Eff}_C(p; q) \approx \mathrm{Eff}_C(p) + \mathrm{Eff}_C(q) -$ (shared setup), which is synergy-like, but in other regimes one may see the opposite effect if the environment forces repeated setup (e.g. sandboxing, rate limits), motivating later context parameters that explicitly model such overheads.

Remark 298. ***Usefulness.*** *These axioms are the minimal interface needed to make later definitions (simplicity as an infimum; compositional simplicity as a least fixed point) behave the way one expects of optimization problems. They permit a clean separation between (i) the algebraic form of "effort reasoning" and (ii) the empirical or architectural details of what an observer can actually do, as emphasized in [19].*

In particular, because "choice" is modeled by a minimum and sequential composition is modeled (at the quantale level) by addition, later notions of simplicity can be defined as the cost of the cheapest description, explanation, or program that achieves a target. The present section therefore fixes the cost calculus that makes such infimum-based definitions well-typed and compositional: one can substitute equivalent subprocedures, refine a plan by expanding a step into substeps, and still reason algebraically about the resulting effort.

8.2 Distinctions as policies; opening and closing

Hyperseed's account of simplicity is inseparable from the idea that an observer may or may not make certain distinctions. We model this using a *distinction policy*, conveniently represented by an *indistinction* relation (the pairs the observer treats as the same for the task at hand). In particular, the policy is meant to be *task-relative*: the same underlying observer may adopt different policies for different contexts C, and even within a fixed C may move along a spectrum from very fine to very coarse discrimination.

Definition 77 (Indistinction policy)**.** *Fix a finite universe X (of entities, states, events, or features relevant in context C). An* indistinction policy *is a subset $H \subseteq X \times X$, interpreted as:*

$$(x, y) \in H \quad \text{means “the observer does not (currently) distinguish } x \text{ from } y \text{ in context } C\text{.”}$$

Write $\mathcal{H}_C \subseteq \mathcal{P}(X \times X)$ for the set of admissible policies for context C.

Remark 299. ***Scope and admissibility.*** *The set $\mathcal{H}_C$ is left abstract on purpose: in some applications, not every relation should be allowed. For example, an engineered sensor might be required to treat indistinction symmetrically, or a legal/organizational setting might impose that indistinction respects certain categories (e.g. cannot collapse "approved" with "unapproved"). Conversely, allowing $\mathcal{H}_C$ to include non-equivalence relations makes it possible to represent partial, asymmetric, or temporarily inconsistent discrimination patterns that occur under time pressure, noise, or conflicting heuristics.*

Remark 300. ***Intuition.*** *An indistinction policy is the observer's current answer to the question: "Which differences matter here?" (Hyperseed-Concept 96) and, more sharply, "Which distinctions will I actually enforce?" (Hyperseed-Concept 98). Representing it as a subset $H \subseteq X \times X$ emphasizes that the policy is a* relation*, not necessarily an equivalence relation. We are not assuming symmetry, transitivity, or reflexivity; an observer may be inconsistent or paraconsistent about what it merges, and later sections allow more graded/quantale-valued versions.*

Remark 301. ***Extremes and calibration.*** *Two boundary cases help calibrate the interpretation. If $H = \varnothing$, then no pairs are explicitly treated as indistinct, corresponding to an idealized stance of "distinguish everything from everything" (maximal discrimination, typically high effort). If $H = X \times X$, then every pair is treated as indistinct, corresponding to "everything is the same for this task" (maximal collapse, typically low effort but also low discriminatory power). Most realistic policies sit between these extremes, and the framework is designed to let later sections quantify the cost/benefit of moving within this lattice of relations.*

Remark 302. ***Examples.*** *If X is a set of colors and the observer cannot distinguish nearby shades, then H contains pairs of shades it treats as the same for the present task. If X is a set of world-states and the observer ignores microphysical details, then H contains pairs of microstates that are collapsed into one macrostate. In the most classical special case, H could be the graph of an equivalence relation corresponding to a partition (coarse-graining), matching the probabilistic coarse-graining discussion in Section 10.2.*

Remark 303. ***Relation to representation maps.*** *It is often helpful to think of a policy as being implemented by some representation or measurement map $\pi\colon X \to R$ into a smaller set of reports R, where $(x, y) \in H$ whenever $\pi(x) = \pi(y)$. This produces an equivalence relation (the kernel of π), but the present definition is broader: it also covers situations where the observer's "same/different" judgments depend on direction, history, or limited local comparisons (e.g. x is treated as indistinct from y for one subroutine but not vice versa). Keeping H as an arbitrary relation allows these cases to be discussed without forcing a premature commitment to partitions.*

Remark 304. ***Usefulness.*** *Making the policy explicit allows later constructions to talk about* changing *what is distinguished: opening and closing are then simply inclusion relations. This is a way to treat "attention," "granularity," and "resolution" as first-class mathematical objects rather than informal modifiers (cf. Hyperseed-Concept 60).*

Remark 305. ***Order-theoretic viewpoint.*** *Because policies are subsets of $X \times X$, they inherit the inclusion partial order. This gives a simple, compositional notion of "more indiscriminating" versus "more discriminating" without requiring a numeric measure. Later, numeric notions of effort or simplicity can be layered on top of this order (e.g. by assigning costs to relations or to their generators), but the inclusion order is the minimal structural scaffold needed to compare policies.*

Definition 78 (Opening and closing)**.** *Given two policies $H, H' \in \mathcal{H}_C$:*

- *The transition $H \to H'$ is an* opening *(relaxation of distinctions) if $H \subseteq H'$.*
- *The transition $H \to H'$ is a* closing *(tightening of distinctions) if $H' \subseteq H$.*

Remark 306. ***Intuition.*** *Opening means: "I will now treat* more *pairs as indistinct," i.e. I will collapse more distinctions (Hyperseed-Concept 125). Closing means: "I will now treat* fewer *pairs as indistinct," i.e. I will enforce more distinctions (Hyperseed-Concept 72). It is useful to note the direction: $H \subseteq H'$ is an opening because H' is a* bigger set of collapsed pairs*.*

Remark 307. ***Monotonic consequences.*** *Under any reasonable downstream use of H (e.g. defining which states are treated as interchangeable, or which features are ignored), opening is monotone in the sense that anything permissible under H remains permissible under H' when $H \subseteq H'$. Closing is the reverse move: it may invalidate previous simplifications by reintroducing distinctions that must now be tracked, measured, stored, or acted upon. This monotonicity is what makes openings/closings suitable as "policy moves" in later optimization and dynamical-control discussions.*

Remark 308. ***Examples.*** *If X is the set of pixel configurations and the observer switches from "full image" to "only average brightness," the new policy collapses many more pairs of images and is thus an opening. Conversely, switching from "only average brightness" to "average brightness plus edge map" is a closing: fewer pairs remain indistinct.*

Remark 309. ***Sequential changes.*** *Openings and closings compose in the expected way along inclusion chains. For instance, $H_0 \subseteq H_1 \subseteq H_2$ corresponds to successively coarsening discrimination, while $H_2 \supseteq H_1 \supseteq H_0$ corresponds to successively refining it. If a system alternates between opening and closing, the resulting path may not be monotone; such non-monotone trajectories are precisely what one expects when attention is reallocated over time (e.g. zooming out to gain tractability, then zooming in on a salient subproblem).*

Remark 310. ***Why needed.*** *The opening/closing distinction is the bridge from phenomenological vocabulary to optimization: it provides the partial order along which effort and weakness can be compared. In later sections, opening/closing also becomes a dynamical control lever: a system can spend effort to close (gain precision) or open (gain tractability), echoing the tradeoff themes emphasized in [9].*

Remark 311. *Opening increases the set of collapsed distinctions (more pairs treated as "the same"); closing decreases it. In Hyperseed terms: opening generally reduces representational effort (but may reduce control/predictive sharpness), whereas closing generally increases effort (but may increase control/predictive sharpness).*

Remark 312. ***Link to resistance and simplicity.*** *Within this section's theme (effort, resistance, and simplicity), the policy language allows a clean separation between (i) what the world presents (the complexity of X and its dynamics) and (ii) what the observer commits to tracking (the chosen H). Resistance can then be modeled as the cost of maintaining a closing against pressures to open (e.g. noise, limited bandwidth, competing tasks), while simplicity can be treated as the availability of effective openings that do not destroy task performance. This sets up later formalizations in which simplicity is not merely "small description length," but "small description length relative to a tolerated indistinction policy."*

8.3 Effort of distinctions; resistance and submission

To connect policies to effort, we assume the observer incurs effort to maintain or compute distinctions. This can be modeled in many ways; for this section we only require the *directionality* to match the intended interpretation. Concretely, the partial order by inclusion on $\mathcal{H}_C$ is read as an "at least as discriminating as" relation: $H \subseteq H'$ means that H keeps (at least) all the constraints/distinctions that H' keeps, so H' is a coarsening (an opening) of H. In many realizations one can think of H as a partition, a σ-algebra, a hypothesis class, or a set of admissible histories; in each case, enlarging H corresponds to allowing more states to be treated as equivalent, hence reducing the informational or computational burden of maintaining sharp boundaries.

Definition 79 (Policy effort and marginal effort)**.** *A* policy effort functional *for context C is a map*

$$\mathrm{Eff}_C : \mathcal{H}_C \to [0, \infty]$$

that is antitone *with respect to inclusion:*

$$H \subseteq H' \quad \Longrightarrow \quad \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H').$$

Define the marginal effort *of changing policy from H to H' as*

$$\Delta\mathrm{Eff}_C(H \to H') := \max\big(0,\ \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H)\big).$$

Remark 313. ***Domain/codomain choices.*** *Allowing Eff_C to take values in $[0, \infty]$ (rather than only $[0, \infty)$) makes it possible to represent policies that are, in context C, effectively unrealizable for the observer (e.g. a discrimination requiring unattainable sensor resolution or unbounded computation). No further regularity (continuity, convexity, additivity, etc.) is assumed here: the framework is designed so that later sections can specialize Eff_C according to the application.*

Remark 314. ***Notation and conventions.*** *The symbol Eff_C has been used above as an effort assignment on processes; here it is specialized to an effort assignment on* policies $H \in \mathcal{H}_C$. *This* *is intentional: the policy is itself something the observer must implement (by allocating computation, attention, measurement, memory, etc.). Also note that* $\max(0, \cdot)$ *makes* $\Delta\mathrm{Eff}_C$ *a one-sided "increase only" measure; it is a minimal operational proxy for resistance rather than a full signed gradient. A signed analogue* $\mathrm{Eff}_C(H') - \mathrm{Eff}_C(H)$ *could be introduced if one wanted to track both "relief" (negative changes) and "load" (positive changes), but for resistance the positive part is the invariant needed.*

Remark 315. ***Intuition.*** *Antitonicity expresses the central phenomenological claim that fine-grained discrimination is costly: if you collapse more distinctions (opening), you should not pay* more *effort. The model is deliberately permissive: it does not insist that opening strictly decreases effort, only that it does not increase it. Equivalently, if one reads* $\subseteq$ *as a refinement order, then* Eff_C *is monotone with respect to "refinement makes things harder": adding constraints cannot make the observer's implementation problem easier.*

Remark 316. ***Examples.*** *If* H *represents "distinguish 100 categories" and* H' *represents "distinguish only 10 categories," then* $H \subseteq H'$ *and* $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H')$ *says that maintaining 100 categories requires at least as much effort as maintaining 10. If closing from* H' *to* H *demands extra memory or sensory precision, then* $\Delta\mathrm{Eff}_C(H' \to H)$ *quantifies the added cost. A concrete toy family is* $\mathrm{Eff}_C(H) = \log|\Pi(H)|$ *when* H *induces a finite partition* $\Pi(H)$ *(or, more generally, a description-length/MDL-type functional); then coarsening reduces the number of cells and does not increase* Eff_C*. Another common choice in probabilistic settings is to let* $\mathrm{Eff}_C(H)$ *track the expected coding or inference cost of operating with policy* H *under a context-dependent distribution; antitonicity then expresses that enforcing additional distinctions cannot reduce expected operational load.*

Remark 317. ***Usefulness.*** *This is the smallest assumption needed to define "resistance" without psychologizing it. It also prepares the ground for more elaborate, history-dependent models of resistance (habits) where opening can become costly because the system has invested in a high-resolution policy and must now "unlearn" it. In particular, the present definition isolates an "instantaneous" effort requirement associated to a policy as such; later, additional state variables can encode sunk costs, commitments, or stabilizing feedbacks that make effort depend on the trajectory by which* H *was reached.*

Remark 318. ***Context dependence.*** *The index* C *is essential: the same abstract policy* H *can have different effort in different environments, embodiments, or task regimes (e.g. a classification with 100 categories may be easy with a fast lookup table but hard if the categories must be inferred from raw sensory streams). Formally, nothing forces* Eff_C *and* $\mathrm{Eff}_{C'}$ *to be comparable across contexts; the intended reading is that effort is observer-and-situation indexed, and comparisons are meaningful only within a fixed* C *unless an explicit identification is provided.*

Remark 319. *If* $H \to H'$ *is an opening (*$H \subseteq H'$*), antitonicity implies* $\mathrm{Eff}_C(H') \leq \mathrm{Eff}_C(H)$*, hence* $\Delta\mathrm{Eff}_C(H \to H') = 0$*: opening does not* require *additional effort in this minimal model (it releases constraints). If* $H \to H'$ *is a closing (*$H' \subseteq H$*), then* $\Delta\mathrm{Eff}_C(H \to H')$ *measures the added effort required to enforce more distinctions. A richer model (used later when habits are introduced in Section 12) adds "habit penalties" so that opening may also be resisted. Note that no path-additivity is assumed: even with a fixed* Eff_C*, one need not have* $\Delta\mathrm{Eff}_C(H \to H'') = \Delta\mathrm{Eff}_C(H \to H') + \Delta\mathrm{Eff}_C(H' \to H'')$*; this absence of a required triangle law is deliberate, because resistance will later be allowed to depend on intermediate reorganizations and nonlocal reparameterizations of policy.*

Definition 80 (Resistance and submission (minimal operationalization))**.** *For a directed policy change $H \to H'$ in context C:*

- Resistance *is identified with* $\Delta\mathrm{Eff}_C(H \to H')$.
- Submission *(in this minimal operationalization) occurs when the system follows a policy update that does not increase required effort, i.e. when* $\Delta\mathrm{Eff}_C(H \to H') = 0$.

Remark 320. ***Intuition.*** *Resistance (Hyperseed-Concept 158) is treated here as the "extra effort demanded" by a change, and submission (Hyperseed-Concept 183) as the special case where no extra effort is demanded. This minimal reading is intentionally behavioristic: it does not claim to capture the whole phenomenology of resistance, only an invariant that can be transported into later mathematics. In particular, it separates (i) the* direction *of a policy update in the lattice/order of distinctions from (ii) the* load *required to implement that update for a given observer in context C.*

Remark 321. ***Examples.*** *If a system is asked to refine its model (closing) and this requires additional computation, memory, or sensing, then $\Delta\mathrm{Eff}_C > 0$ and the change is resisted in this operational sense. If it is asked to simplify (opening) and this reduces constraints, then $\Delta\mathrm{Eff}_C = 0$ and the change is "submitted to" without additional load. In later dynamical models, opening may also incur cost because simplifying can break internal commitments (learned habits), leading to nonzero resistance even for openings. One can also interpret $\Delta\mathrm{Eff}_C(H \to H')$ as an* activation barrier*: the policy H' might be globally "simpler" than H in a static sense, yet transitioning to it can still be difficult if the model is extended to include switching/rewiring costs not captured by the static* Eff_C *alone.*

Remark 322. ***Why this definition matters.*** *One can view this as a precise analogue of a thermodynamic gradient: forcing a system into a more constrained configuration costs work, and that "work" is what resistance measures. This resonates with the "least-effort" orientation in [21] while staying purely formal and observer-indexed. The analogy is structural rather than literal:* Eff_C *is not assumed to be physical energy, but it plays the same mathematical role of a potential whose increases quantify the minimal additional "input" required to sustain tighter constraints.*

Remark 323. *Hyperseed's phenomenological language distinguishes several kinds of resistance/submission (somatic, cognitive, affective, spiritual). The formalization above is deliberately thin: it isolates a single load-bearing invariant that will matter throughout the ontology:* tightening distinctions is effort-consuming*. Later sections add dynamical structure to represent different "channels" of resistance. The point of starting thin is that the ontology can then refine (rather than revise) the concept: richer models may decompose* Eff_C *into multiple terms (e.g. sensing vs. inference vs. memory), but antitonicity and the resulting operational meaning of* $\Delta\mathrm{Eff}_C$ *remain stable.*

8.4 Simplicity as minimum representational effort

Hyperseed's simplicity is observer-relative and task-relative: an object is simple for an observer if it can be handled (represented, predicted, manipulated) with low effort. This framing treats "simplicity" not as an intrinsic geometric or syntactic attribute of x, but as a property of the best interface an observer can construct between x and the demands of context C. In particular, what counts as "handling" can vary: the same x may be easy to *store* but hard to *simulate*, easy to *predict* but hard to *control*, etc., and these differences are absorbed into the choice of admissible representations and the effort functional.

We therefore define simplicity as an *infimum over representations.* The use of an infimum (rather than, say, an arbitrary chosen representation) emphasizes that simplicity is an *optimal* notion: $s_C(x)$ records the best achievable effort among all admissible representational strategies in C. It also allows that the "best" representation may be approached only in the limit (e.g., by increasingly refined approximations), which is often the realistic case in continuous or resource-bounded settings.

Definition 81 (Representations and representational effort)**.** *Fix a context C and a domain of items $\mathcal{X}_C$ (entities, states, experiences, or patterns). Let $\mathcal{R}_C(x)$ denote the set of admissible representations of $x \in \mathcal{X}_C$. Assume an effort assignment* Eff_C *on representations. Define the* (subjective) simplicity *of x in context C by*

$$s_C(x) := \inf\{\mathrm{Eff}_C(r) : r \in \mathcal{R}_C(x)\}.$$

Remark 324. ***Notation unpacking.*** *Here $\mathcal{R}_C(x)$ is a set (possibly large) of alternative representations r that "stand for" x in context C. The functional* $\mathrm{Eff}_C(r)$ *measures the cost of using representation r (storage, computation, inference, sensing, etc.). The infimum* inf *is taken in the ordinary real order: $s_C(x)$ is the best (least costly) representation effort available. When $\mathcal{R}_C(x)$ is rich, it is helpful to read "$r \in \mathcal{R}_C(x)$" as encoding both an* encoding language *(what forms representations may take) and a set of* acceptability constraints *(what counts as representing x well enough for the purposes of C).*

Remark 325. ***Intuition.*** *This definition turns "simplicity" into a controlled form of compression: x is simple if there exists a cheap way to represent it. The philosophy is close in spirit to algorithmic information theory—the object is "simple" if it has a short description—but we have replaced "short" with a general observer-dependent effort measure. This observer-relativity is central in Hyperseed (Hyperseed-Concept 169) and aligns with the broader pattern/emergence approach in [5]. One can also view $s_C(x)$ as a generalized "minimum description length" objective in which description length, compute time, energy, bandwidth, or even required sensor acuity may all contribute to effort, depending on what the context regards as scarce.*

Remark 326. ***Examples.*** *A circle is simple for a context that has the concept "circle" and can store "center + radius" cheaply; it may be complex for a context that can only store point clouds. A large integer like 10^{100} is simple for a context that allows exponent notation, and less simple for a context that requires explicit decimal expansion. Similarly, a periodic signal may be simple for a context that admits Fourier representations (a few coefficients) but complex for a context restricted to raw time-series samples; conversely, a sharply localized transient may be cheaper as a sparse time-domain description than as a global frequency expansion.*

Remark 327. ***Why useful.*** *By making simplicity an infimum, we obtain a quantity that behaves well under optimization and can be compared across tasks once adequacy constraints are imposed. This is also the mathematical doorway through which patterns will enter in Section 9: a pattern is, roughly, a representation whose effort is much smaller than some baseline. In practice, this invites a two-part modeling stance: first choose what counts as an admissible representation (the hypothesis class, the interface, the "vocabulary"), then measure the best achievable effort within that class. Changing $\mathcal{R}_C(x)$ (e.g., by adding a new representational primitive) can strictly decrease $s_C(x)$, formalizing the idea that learning new concepts can make previously complex items simple.*

Remark 328. *Low $s_C(x)$ means "x is simple for C." Different contexts yield different simplicity orderings. This formalizes the Hyperseed claim that simplicity is not an absolute property of the world but is indexed by an observer's capacities and goals. In particular, if two contexts C and C' differ only by resource budgets (time, memory, precision), then $s_C(x)$ can be interpreted as a* *context-indexed "resource complexity" of x. Also note that, depending on conventions, it is natural to allow $s_C(x) = +\infty$ when no admissible representation exists (e.g., $\mathcal{R}_C(x) = \varnothing$ or all candidates have unbounded effort), which captures the possibility that some items are effectively unhandleable in a given context.*

8.5 Simplicity as weakness: the dual perspective

Section 3.7 defined a quantale-valued *weakness* $w(H)$ for an indistinction policy H: weakness increases as more (and/or more important) distinctions are collapsed (Hyperseed-Concepts 202, 143; see also [3, 2] for related motivation). This provides an algebraic dual to effort-based simplicity. In particular, it lets us speak about "being simpler" either in an operational sense (less work is required to maintain fine-grained distinctions) or in a semantic/informational sense (fewer distinctions are being asserted at all). The point of introducing w is that the latter can be aggregated and compared even when an explicit effort model is unavailable or context-dependent.

The key relation is monotonic: *opening decreases effort and increases weakness*, while *closing increases effort and decreases weakness*. Here "opening" is the order-theoretic operation of enlarging H by adding more pairs declared indistinct, i.e. moving upward in the inclusion order on $\mathcal{H}_C$; "closing" is the reverse movement. Because both w and Eff_C are defined as order-respecting maps (but in opposite directions), this monotonicity is not an extra axiom: it is the compatibility condition that makes the two notions of simplicity track the same qualitative phenomenon. We record the abstract monotonic facts as a sanity theorem, to be used later.

Theorem 7 (Effort–weakness monotonicity)**.** *Fix a context C with admissible indistinction policies $\mathcal{H}_C$. Assume:*

1. *$w : \mathcal{H}_C \to V$ is monotone with respect to inclusion (as in Theorem 1 specialized to C),*

2. *$\mathrm{Eff}_C : \mathcal{H}_C \to [0, \infty]$ is antitone with respect to inclusion (Definition 79).*

Then for any $H \subseteq H'$ in $\mathcal{H}_C$ we have

$$w(H) \leq w(H') \qquad \textit{and} \qquad \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H').$$

Thus opening simultaneously increases weakness and decreases effort. Equivalently, along any inclusion chain

$$H_0 \subseteq H_1 \subseteq \cdots \subseteq H_n,$$

the sequence $\big(w(H_i)\big)_{i=0}^n$ is nondecreasing in the quantale order, while $\big(\mathrm{Eff}_C(H_i)\big)_{i=0}^n$ is nonincreasing in the usual order on $[0, \infty]$.

Remark 329. ***What this says, in plain language.*** *If you merge more cases together (open the policy), then you become "weaker" in the precise quantale sense (you assert fewer distinctions), and you also do not have to spend more effort maintaining distinctions. This theorem is not deep; its value is that it makes the duality between two kinds of "simplicity talk" explicit and checkable. In particular, it rules out the pathological possibility that a policy could be simultaneously "more indiscriminate" (collapsing distinctions) and yet require strictly* more *effort to implement, or vice versa, at least at the level of the abstract order-theoretic interface.*

Remark 330. ***Why it matters.*** *Later sections will compose policies across subsystems and contexts. Weakness is designed to compose well (quantale aggregation), while effort is designed to support minimization. This theorem is the hinge connecting those two calculi: it guarantees that* *one is not optimizing a quantity that behaves perversely with respect to the other. A typical use is: if a construction (composition, relaxation, marginalization) is known to correspond to opening in $\mathcal{H}_C$, then one immediately gets a one-line comparison statement for both w and Eff_C without unpacking their internal definitions in that particular setting.*

Remark 331. ***Connection to the rest of the document.*** *In Section 9, pattern intensity is defined by comparing effort (or weakness) before and after a representational move. In that setting, one often wants to reason with whichever quantity composes more conveniently; the monotonic relation here lets one translate qualitative conclusions across the two viewpoints. For example, if a representational move is shown (by construction) to only ever open the operative policy, then it cannot increase effort; if instead it is shown to strictly close the policy, then it cannot decrease weakness. These translations are intentionally "direction-only": they preserve the sign of comparisons even when the magnitudes live in different codomains (V versus $[0,\infty]$).*

Proof. The weakness inequality is exactly monotonicity of w under adding undistinguished pairs. The effort inequality is the assumed antitonicity of Eff_C. One can also view the statement as the conjunction of two functoriality properties: w is order-preserving and Eff_C is order-reversing on the poset $(\mathcal{H}_C, \subseteq)$. $\square$

Proof sketch. The argument is purely definitional: "opening" means $H \subseteq H'$. Monotonicity of weakness turns inclusion into $w(H) \leq w(H')$, while antitonicity of effort turns the same inclusion into $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H')$. There is no hidden construction; the point is to record that the intended phenomenological directionality has been built into the formalism. A useful way to remember the variance is: weakness measures what you *give up* by collapsing distinctions, so it grows when you add collapses; effort measures what you *must supply* to maintain distinctions, so it shrinks when you stop maintaining them. $\square$

Remark 332. *A useful image is to imagine a knob controlling perceptual or conceptual resolution. Turning the knob toward coarser resolution corresponds to opening: more things look the same. Weakness increases (more indistinction), and effort decreases (less computation/attention). Turning the knob toward finer resolution corresponds to closing: more things must be told apart, which costs effort. At the extremes, the fully open end of the knob corresponds to a maximally indiscriminate policy (everything relevant is treated as the same), which should intuitively have high weakness and low effort; the fully closed end corresponds to a maximally discriminating policy (no additional identifications beyond those forced by admissibility), which should have low weakness and high effort. The theorem does not assume such extrema exist in $\mathcal{H}_C$, but whenever they do, the inequalities specialize to immediate upper/lower bounds.*

Remark 333. ***Order conventions and interpretability.*** *The inequality $w(H) \leq w(H')$ is taken in the intrinsic order of the quantale V. In many concrete quantales used for aggregation (e.g. $([0,\infty], \geq, +, 0)$ with the* reverse *order, or a powerset quantale ordered by inclusion), it is worth checking once which direction corresponds to "more severe" weakness. The present statement fixes that convention by insisting that opening corresponds to* increasing *weakness in the chosen order; all later comparisons inherit this convention. Similarly, the effort inequality uses the standard order on $[0,\infty]$, so "decreases effort" always means "numerically no larger."*

Corollary 1 (No free lunch in both directions)**.** *If $H \subseteq H'$ and $w(H') < w(H)$ (strictly, in the order of V), then such a pair cannot occur under the stated assumptions; likewise, if $\mathrm{Eff}_C(H') > \mathrm{Eff}_C(H)$, then $H \subseteq H'$ cannot hold. In other words, under opening one cannot simultaneously get strictly* less *weakness, nor strictly* more *effort.*

8.6 Minimum representational effort as constrained weakness maximization

In applications, an observer typically cannot open indefinitely: too much collapsing of distinctions destroys predictive power and control. Hyperseed's "minimum representational effort" principle is therefore a *constrained* minimization: among all policies/representations that are adequate for a task, prefer those requiring the least effort (cf. [2, 19] for related design principles). In particular, the "constraint" is not an incidental add-on: it is the mathematical device that prevents the trivial solution "represent nothing" (or "distinguish nothing") from being optimal.

Definition 82 (Task constraint and feasible policy set)**.** *Let a task constraint in context C be a predicate $\mathsf{Adeq}_C(H) \in \{\text{true}, \text{false}\}$ meaning: "policy H is adequate for the task (prediction/control objective)". Define the feasible set*

$$\mathcal{F}_C := \{H \in \mathcal{H}_C : \mathsf{Adeq}_C(H)\}.$$

Remark 334. ***Intuition.*** *The predicate $\mathsf{Adeq}_C(H)$ is where "what the observer is trying to do" enters. Without it, the cheapest policy would typically be the maximally open one (collapse everything), which is useless for prediction/control. Adequacy formalizes the idea that some distinctions are not optional because the task depends on them (Hyperseed-Concept 186).*

Remark 335. ***Examples.*** *If the task is to predict whether a system will fail, then collapsing distinctions that separate failure states from non-failure states makes the policy inadequate. If the task is image classification among a small set of labels, then distinctions within each label class may be safely collapsed without harming adequacy.*

Remark 336. ***Why needed.*** *This definition isolates a general optimization shape that will recur in more elaborate guises (Pareto frontiers, tradeoff theorems, multi-objective feasibility). It is the formal core of what the Hyperseed overview treats as "do not spend representational effort unless you must" [1].*

Remark 337. ***Well-posedness.*** *The feasible set $\mathcal{F}_C$ can be empty if the task is impossible under the observer's available policy class $\mathcal{H}_C$ (or if the adequacy predicate is too strict). In that case the principle in Definition 83 becomes a diagnosis: either enlarge $\mathcal{H}_C$ (more expressive policies), weaken/relax Adeq_C (approximate adequacy, probabilistic success, slack constraints), or change the task/context C. This mirrors practical learning systems where "no model fits" is handled by revising the hypothesis class or by moving from hard constraints to soft penalties.*

Definition 83 (Minimum representational effort principle)**.** *An observer in context C satisfies* minimum representational effort *if it preferentially selects*

$$H^* \in \arg\min_{H \in \mathcal{F}_C} \mathrm{Eff}_C(H).$$

Remark 338. ***Intuition.*** *Among all adequate policies, choose the one that costs the least effort to maintain/compute. This is a cognitive analogue of an Occam-style constraint, but framed in terms of the observer's actual expenditures rather than a language-invariant notion of description length.*

Remark 339. ***Example.*** *Suppose two policies are adequate for a classification task: one distinguishes 1,000 fine-grained categories and then maps them to labels, while the other distinguishes only the labels directly. The minimum representational effort principle picks the latter, provided both are adequate for the actual objective.*

Remark 340. ***Usefulness.*** *This principle provides a normative and (later) dynamical bias: it tells us which policies are preferred in equilibrium, and it suggests learning dynamics that drift toward lower effort subject to constraints. Such dynamics are central in cognitive architectures that explicitly manage resource tradeoffs [19].*

Remark 341. *This is a formalization of a central Hyperseed slogan: "simplicity is what costs less effort." Note that "adequacy" may itself be paraconsistent, fuzzy, or multi-objective; the definition above only assumes we can describe a feasible region.*

Remark 342. ***Effort as an internal cost functional.*** *The point of writing* $\mathrm{Eff}_C(H)$ *rather than a syntactic complexity measure is that effort can include costs that are not captured by descriptions alone: latency, memory footprint, sampling cost, training instability, energy expenditure, or (in embodied settings) sensor/actuator wear. Thus two extensionally equivalent policies (same input–output behavior) may differ in effort because they are implemented differently in the observer. This is also why the context subscript* C *matters: what is "cheap" depends on available hardware, priors, and interaction regime.*

Under mild closure assumptions, minimum-effort selection is equivalent to selecting the *maximally open* adequate policy, hence the *maximally weak* adequate policy. Here "mild" is primarily the assumption that adequacy is monotone with respect to opening, i.e., once the task can be done, discarding additional distinctions that the task does not use will not break adequacy.

Theorem 8 (Maximally open adequate policy under upward closure)**.** *Assume* $\mathcal{F}_C$ *is* upward closed *under opening:*

$$H \in \mathcal{F}_C \text{ and } H \subseteq H' \implies H' \in \mathcal{F}_C.$$

Let

$$\widehat{H} := \bigcup_{H \in \mathcal{F}_C} H.$$

Then:

1. $\widehat{H} \in \mathcal{F}_C$*,*

2. *for all* $H \in \mathcal{F}_C$*, we have* $H \subseteq \widehat{H}$*,*

3. *if* Eff_C *is antitone, then* $\widehat{H}$ *minimizes effort over* $\mathcal{F}_C$*,*

4. *if* w *is monotone, then* $\widehat{H}$ *maximizes weakness over* $\mathcal{F}_C$*.*

Remark 343. ***What this says, in plain language.*** *If adequacy is preserved when you open further (collapse extra distinctions that the task does not need), then there is a single "most open" adequate policy: take the union of all adequate policies. This most open adequate policy is automatically the least-effort one (because effort decreases under opening) and the most weak one (because weakness increases under opening).*

Remark 344. ***Why it is important.*** *The theorem turns an optimization problem (minimize effort subject to adequacy) into a purely order-theoretic construction (take a union). This is conceptually valuable: it says that, in the upward-closed regime, the hard part is not optimization but specifying adequacy correctly. It also foreshadows later fixed-point reasoning about stable policies and goal systems (cf. [10] for a related fixed-point style in a different layer).*

Remark 345. ***Connection to weakness.*** *This theorem makes precise the slogan "minimum effort equals maximum weakness subject to task constraints," at least when $\mathcal{F}_C$ is upward closed. This is one of the main reasons weakness is useful: it can be maximized compositionally across contexts, while effort is minimized.*

Remark 346. ***Proof sketch (unpacking the order-theoretic steps).*** *Item (2) holds by construction: every $H \in \mathcal{F}_C$ is a subset of the union of all such H. For item (1), pick any $H_0 \in \mathcal{F}_C$ (when $\mathcal{F}_C \neq \emptyset$); then $H_0 \subseteq \widehat{H}$, and by upward closure $\widehat{H} \in \mathcal{F}_C$. For item (3), antitonicity of Eff_C means $H \subseteq H'$ implies $\mathrm{Eff}_C(H') \leq \mathrm{Eff}_C(H)$, so combining $H \subseteq \widehat{H}$ with antitonicity gives $\mathrm{Eff}_C(\widehat{H}) \leq \mathrm{Eff}_C(H)$ for all feasible H. Item (4) is analogous: monotonicity of weakness means $H \subseteq H'$ implies $w(H) \leq w(H')$, hence $w(H) \leq w(\widehat{H})$ for all feasible H. The content is therefore not a delicate inequality, but the existence of a* greatest *element (a maximal opening) in the feasible set once it is upward closed.*

Remark 347. ***When upward closure is reasonable (and when it can fail).*** *Upward closure is natural for tasks whose adequacy depends only on retaining certain* necessary *distinctions: if a policy is adequate, then merging additional states that are irrelevant to the objective will keep it adequate. It can fail for tasks where the representation itself must satisfy side-constraints that are not monotone under opening (e.g., "must separately track two protected groups" or "must retain a diagnostic variable for auditing"), or where opening changes action selection in a way that violates a safety constraint even if prediction remains acceptable. In such cases, $\mathcal{F}_C$ need not have a single maximally open element, and minimum-effort selection may involve genuine optimization over incomparable feasible policies rather than the union construction.*

Remark 348. ***Interpretation in common representational formalisms.*** *If policies correspond to partitions (or σ-algebras) over a state space, "opening" corresponds to* coarsening *the partition (forgetting distinctions). The union $\widehat{H}$ then corresponds to the coarsest partition that is still adequate: it merges every pair of states that can be merged without breaking the task. This matches the informal idea of "keep exactly the distinctions the task forces you to keep" and discard the rest.*

Remark 349. ***Algorithmic reading.*** *Although Theorem 8 is stated extensionally (via a union over all feasible policies), it suggests an intensional procedure: repeatedly open (merge distinctions) while checking adequacy, stopping only when any further opening would violate Adeq_C. In learning settings, this corresponds to a bias toward representational compression with constraint checks (or constraint penalties), which is a common pattern in resource-bounded inference and control.*

Proof. (1) Upward closure implies that since each $H \in \mathcal{F}_C$ satisfies $H \subseteq \widehat{H}$, we have $\widehat{H} \in \mathcal{F}_C$. To make the inclusion $H \subseteq \widehat{H}$ explicit, recall that $\widehat{H}$ is defined as the union of the feasible policies (in particular, $\widehat{H} = \bigcup_{G \in \mathcal{F}_C} G$), so every element of a given $H \in \mathcal{F}_C$ is, by definition of union, an element of $\widehat{H}$. Thus $\widehat{H}$ is an upper bound for $\mathcal{F}_C$ in the inclusion order, and upward closure then upgrades this pointwise upper-boundedness into feasibility of the upper bound itself.

(2) Trivial from the definition of union. More concretely, if some relation/distinction/permission x is in $\widehat{H}$, then $x \in H$ for at least one feasible H, and conversely any $x \in H$ for a feasible H must lie in $\widehat{H}$; this is the sense in which $\widehat{H}$ aggregates all the "openings" present in any adequate policy.

(3) If $H \subseteq \widehat{H}$ and Eff_C is antitone, then $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(\widehat{H})$ for all $H \in \mathcal{F}_C$, so $\widehat{H}$ is an effort minimizer. Here antitone means that moving upward in $(\mathcal{H}_C, \subseteq)$ cannot increase effort: whenever $H \subseteq H'$, one has $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H')$. Since every feasible H is below $\widehat{H}$, the inequality specializes to show that $\widehat{H}$ achieves effort no larger than any other feasible candidate, i.e. it is a minimizer

over the feasible set. In particular, $\widehat{H}$ is not merely locally improving along a chain; it is globally extremal because it is a (feasible) top element among all feasible policies under inclusion.

(4) If w is monotone and $H \subseteq \widehat{H}$ then $w(H) \leq w(\widehat{H})$ for all feasible H, hence $\widehat{H}$ is a weakness maximizer. Monotonicity means weakness is aligned with openness: enlarging H (collapsing more distinctions) cannot decrease w. Thus, because $\widehat{H}$ contains every feasible H, it achieves weakness at least as large as any feasible policy. Equivalently, $\widehat{H}$ realizes the maximum of w over $\mathcal{F}_C$ provided $\mathcal{F}_C$ is nonempty, and the argument shows that this maximum is witnessed by the same union object that witnesses minimal effort under antitonicity. $\square$

Proof sketch. The strategy is to build a candidate policy that is "at least as open as any adequate one," namely the union $\widehat{H}$. Intuitively, $\widehat{H}$ is constructed by taking every permission/identification that appears in some adequate policy and allowing it simultaneously; it is therefore the least restrictive object that still stays within the closure assumptions. Upward closure guarantees adequacy is preserved when moving upward in the inclusion order, so $\widehat{H}$ remains feasible. This step is the only place where closure is used: without it, the union could overshoot the feasible region even if each constituent policy is feasible. Then antitonicity/monotonicity convert the inclusion comparisons $H \subseteq \widehat{H}$ into the desired extremal properties for effort and weakness. In other words, once $\widehat{H}$ is known to be a feasible upper bound, order-respecting objective functions automatically turn that upper bound into an optimizer: antitone objectives favor higher elements, while monotone objectives reward higher elements. $\square$

Remark 350. *A visual way to think of this is to picture the poset $(\mathcal{H}_C, \subseteq)$ as a landscape of policies. The feasible region $\mathcal{F}_C$ is a subset of this landscape. Upward closure means $\mathcal{F}_C$ is a "cone" pointing upward: once you are feasible, moving upward (opening) stays feasible. The union $\widehat{H}$ is then the apex of this cone. In order-theoretic terms, $\widehat{H}$ functions as a greatest element (or at least a canonical maximal element) of $\mathcal{F}_C$ with respect to $\subseteq$, because every feasible H lies below it. This is why the optimization statements become essentially immediate: if the objective is compatible with the order (antitone for effort, monotone for weakness), then an extremum is attained at such an apex.*

Remark 351. *The upward-closure assumption formalizes a natural regime: if a policy is already adequate, then collapsing* additional *distinctions that the task does not depend on remains adequate. This corresponds to the intuition that adequacy depends only on preserving distinctions that matter for C; any extra granularity is gratuitous and can be removed without harming performance. Later sections refine this using pattern-intensity and emergence: adequacy can fail under excessive opening because emergent patterns may disappear. In that refined regime, $\widehat{H}$ may cease to be feasible, so the existence of a single "most open adequate" policy is no longer guaranteed, and the extremal argument must be replaced by a constrained optimum that balances these competing effects.*

8.7 Compositional simplicity

Hyperseed insists that simplicity must be *compositional* to support a serious theory of pattern. We formalize compositional simplicity in a way that is compatible with the effort-quantale viewpoint and later categorical constructions.

Remark 352. ***Why "compositional" is a substantive constraint.*** *A non-compositional simplicity score can always be defined by fiat (e.g. "declare the objects we like to be simple"). Requiring a recursive upper bound in terms of parts forces simplicity to be* witnessable *by an explicit construction history, and thereby makes simplicity sensitive to the structure induced by the context C (available building blocks, allowable compositions, and their overheads).*

Definition 84 (Combination system and overhead)**.** *A* combination system *is a tuple* $(\mathcal{X}_C, *, e)$ *where:*

- $\mathcal{X}_C$ *is a set of items in context* C*,*
- $* : \mathcal{X}_C \times \mathcal{X}_C \to \mathcal{X}_C$ *is a (not necessarily associative) combination operation,*
- $e \in \mathcal{X}_C$ *is an identity-like element (when applicable).*

An overhead *is a function* $t_C : \mathcal{X}_C \times \mathcal{X}_C \to [0, \infty]$ *representing the extra effort needed to "glue"* y *and* z *into* $y * z$*.*

Remark 353. ***On non-associativity.*** *Allowing* $*$ *to be non-associative is not a technicality: in many realistic contexts,* $(y * z) * w$ *and* $y * (z * w)$ *represent different assembly orders with different intermediate artifacts and different integration burdens. The framework therefore treats a decomposition as implicitly* tree-shaped *rather than simply set- or multiset-based, and later "parse tree" viewpoints make this explicit.*

Remark 354. ***On the role of*** e**.** *The element* e *is included to support contexts where there is a natural "do nothing" or "empty" item (empty string, empty program module, empty diagram, etc.). When a true unit does not exist or is not meaningful, one can still include a distinguished baseline element for which* $s_C(e) = 0$ *expresses a normalization convention. This normalization prevents degenerate admissible solutions obtained by shifting all costs down by a constant.*

Remark 355. ***Intuition.*** *A combination system abstracts the idea that complex items can be built from simpler ones. The overhead* $t_C(y, z)$ *isolates the cost of composition itself: bookkeeping, interface matching, coordination, translation between formats, etc. In cognitive terms, even if you can represent* y *and* z *cheaply, combining them may require additional work (aligning contexts, resolving conflicts, finding a common schema).*

Remark 356. ***Examples.*** *If* $*$ *is concatenation of strings,* t_C *might be* 0 *(gluing is free). If* $*$ *is merging two knowledge graphs,* t_C *may be substantial because entity alignment is nontrivial. If* $*$ *is composing two programs,* t_C *can represent integration overhead such as type conversions or interface code.*

Remark 357. ***Usefulness.*** *By making overhead explicit, we can distinguish "intrinsic simplicity" of parts from "integration simplicity" of wholes. This becomes important when discussing emergence and pattern: sometimes a whole is simple not because its parts are simple, but because there is a low-overhead way to organize them.*

Remark 358. ***Overhead as a contextual parameter.*** *In practice,* t_C *is often where the context* C *enters most strongly. For example, the same pair of modules* (y, z) *may have small overhead in a shared ecosystem (common protocols, shared ontologies) and large overhead across ecosystems. Thus compositional simplicity is not intended to be an intrinsic property of an abstract object alone, but of an object* as situated *in a particular representational and operational environment.*

Definition 85 (Compositional simplicity functional)**.** *Let* $(\mathcal{X}_C, *, e)$ *be a combination system with overhead* t_C*. A function* $s_C : \mathcal{X}_C \to [0, \infty]$ *is* compositionally admissible *if:*

1. $s_C(e) = 0$,

2. for all $x \in \mathcal{X}_C$,

$$s_C(x) \le \inf_{\substack{y,z \in \mathcal{X}_C \\ y*z=x}} \big(s_C(y) + s_C(z) + t_C(y,z)\big),$$

with the understanding that $\inf \emptyset = \infty$*.*

Define the compositional simplicity s_C^* *to be the least (pointwise) function satisfying these constraints.*

Remark 359. ***Why an inequality (rather than equality).*** *The admissibility condition requires that $s_C(x)$ not exceed the best available two-part construction cost. This leaves room for x to be even simpler for reasons not captured by binary decomposition alone (e.g. if x is deemed primitive, or if x has a special direct representation not modeled as $*$-composition). Taking the* least *admissible solution s_C^* then eliminates arbitrary slack: any simplicity that cannot be justified by the recursive constraints is removed.*

Remark 360. ***The "no decomposition" case.*** *If x cannot be expressed as $y * z$ for any pair (y,z), then the right-hand side is* $\inf \emptyset = \infty$*, and the inequality becomes $s_C(x) \le \infty$, i.e. imposes no upper bound. This is deliberate: whether such an x should have finite simplicity is determined by the global least-solution requirement, not by forcing every item to decompose. Consequently, items can be treated as primitives in a given system (finite or infinite cost depending on what is consistent with the rest of the constraints).*

Remark 361. ***Notation unpacking.*** *The set under the infimum is the set of all pairs (y,z) that decompose x via the operation $*$. For each such decomposition, we assign a candidate cost $s_C(y)+s_C(z)+t_C(y,z)$. Then we take the best (minimum) cost among decompositions, via* $\inf$*. If x has no decomposition at all (empty set), we set* $\inf \emptyset = \infty$*, meaning "cannot be built compositionally (within this system)".*

Remark 362. ***Intuition.*** *This is a Bellman-style optimality condition: the simplicity of a whole should be no more than the best way to build it from parts, paying the simplicity of parts plus integration overhead. Taking the* least *solution s_C^* says we are not allowed to assign arbitrary low costs; we want the smallest costs consistent with the recursive constraints. In effect, s_C^* is "the cost of the cheapest parse" of x.*

Remark 363. ***Fixed-point viewpoint (effort-quantale compatibility).*** *Define an operator F on functions $s : \mathcal{X}_C \to [0,\infty]$ by setting $F(s)(e) = 0$ and, for $x \ne e$,*

$$F(s)(x) = \inf_{\substack{y,z \in \mathcal{X}_C \\ y*z=x}} \big(s(y) + s(z) + t_C(y,z)\big),$$

interpreting $\inf \emptyset = \infty$ *as above. Then compositionally admissible s are precisely (pointwise)* post-fixed points *of F (i.e. $s \le F(s)$), and s_C^* is the least post-fixed point. This framing matches the quantale intuition: $[0,\infty]$ with $+$ and $\le$ supports monotone recursion, and* inf *plays the role of taking best achievable effort among alternatives.*

Remark 364. ***On existence and computability.*** *Because $[0,\infty]$ is complete under arbitrary infima, one can construct the least admissible solution by taking the pointwise infimum over all admissible functions, or (under mild finiteness/well-foundedness assumptions on decomposition) by iterative improvement reminiscent of dynamic programming. When there are cycles of decompositions, s_C^* still exists as a least consistent assignment, but it may take the value ∞ on elements that cannot be finitely justified without circularity.*

Remark 365. ***Examples.*** *If items are arithmetic expressions and $*$ combines subexpressions, $s^*_C(x)$ measures the minimal cost of a derivation tree of x. If items are designs built from components, it measures the cheapest assembly plan accounting for component simplicity and assembly overhead.*

Remark 366. ***Further example (segmentation).*** *If $\mathcal{X}_C$ is a set of strings, $*$ is concatenation, and $t_C \equiv 0$, then s^*_C can be read as the best achievable cost under all binary segmentations of a string into substrings. Different binary parenthesizations correspond to different parse trees, but because $t_C = 0$ and concatenation is associative, the best cost is effectively governed by the best segmentation into primitive or previously-simplified pieces.*

Remark 367. ***Why needed.*** *Hyperseed's pattern theory requires not just that some objects are simple, but that simplicity behaves predictably under composition. Otherwise "pattern intensity" would not be stable when patterns are nested or combined. Compositional simplicity is one way to enforce this stability, and it aligns with the hierarchical/heterarchical pattern-web perspective in [5].*

Remark 368. *The recursion above is the effort-quantale analogue of the informal Hyperseed sketch (see the motivating discussion around compositional simplicity in the Hyperseed overview). The "least" solution corresponds to the best achievable decomposition costs, i.e. dynamic programming over all factorization trees.*

Remark 369. ***Binary vs. multiway composition.*** *Only binary composition is built into the definition, but multiway composition is implicitly covered by iterating $*$ along a tree. The overhead function t_C then determines how integration costs accumulate across intermediate stages, which is precisely why parenthesization can matter when $*$ is non-associative or when overheads depend strongly on the intermediate artifacts.*

Proposition 5 (Immediate subadditivity bound)**.** *For any $y, z \in \mathcal{X}_C$ we have*

$$s^*_C(y * z) \leq s^*_C(y) + s^*_C(z) + t_C(y, z).$$

Remark 370. ***What this says, intuitively.*** *Even if there might exist a cleverer way to build $y * z$ than "build y, build z, glue them," the naive plan is always available. Therefore the optimal cost of $y * z$ cannot exceed the cost of that naive plan. This is the compositional counterpart of the basic subadditivity property of effort.*

Remark 371. ***Connection.*** *This proposition is a one-step corollary of the defining inequality of compositional admissibility. It is also the local inequality that later becomes global when we interpret $s^*_C(x)$ as the minimum over parse trees in Theorem 9.*

Remark 372. ***When the bound is informative.*** *If $t_C(y, z)$ is small (or zero), the proposition says that simplicity is nearly subadditive under composition, so building larger structures from simple parts does not incur disproportionate penalty. If $t_C(y, z)$ is large, then even very simple parts can yield a complex whole, capturing the common phenomenon that "integration dominates implementation."*

Proof. In the definition of $s^*_C(y * z)$, the infimum is taken over all decompositions of $y * z$. One admissible decomposition is precisely (y, z), hence the infimum is bounded above by the corresponding cost. $\square$

Proof sketch. The proof simply points to one specific candidate decomposition inside the infimum defining $s^*_C(y * z)$. An infimum over many candidates is always less than or equal to any particular candidate. Concretely, the defining inequality for $s^*_C(x)$ (as the least solution of the compositional recursion) considers *all* pairs (y, z) with $y * z = x$; selecting the specific pair that actually witnesses the composition $y * z$ gives an admissible term in that infimum, hence an immediate upper bound. This is the same one-line argument that underlies subadditivity bounds for shortest-path or dynamic-programming value functions: "evaluate at a specific action" bounds the optimum. □

Remark 373. *A geometric metaphor is useful: think of all decompositions of x as offering different "routes" to construct x, each with a cost. The simplicity $s^*_C(x)$ is the cost of the cheapest route. The particular route "split into y and z" provides an upper bound on the cheapest route, because it is one of the routes under consideration. Equivalently, if one imagines a landscape of candidate constructions, then $s^*_C(x)$ is the lower envelope of their costs, and any explicitly described construction gives a point on that landscape. This perspective also clarifies why such bounds are often easy to produce: it is typically simpler to describe* one *feasible construction than to certify optimality among all constructions.*

Theorem 9 (Compositional simplicity equals minimum parse-tree cost (finite case))**.** *Assume $\mathcal{X}_C$ is finite, and for each $x \in \mathcal{X}_C$ the set of decompositions*

$$\mathrm{Dec}(x) := \{(y, z) \in \mathcal{X}_C \times \mathcal{X}_C : y * z = x\}$$

is finite. Assume also $t_C(y, z) \geq 0$ for all (y, z) and $s^*_C(e) = 0$. Define the cost of a full binary parse tree T evaluating to x by summing t_C over internal nodes and 0 on leaves labeled by e (or by primitive atoms assigned fixed base costs, if desired). Then $s^*_C(x)$ equals the minimum cost among all such parse trees for x. Moreover, in this finite setting the minimum is attained by at least one parse tree (so the statement is not only an "infimum equals infimum" claim, but an actual minimum over a finite search space of well-formed derivations).

Remark 374. ***Plain-language meaning.*** *In the finite case, $s^*_C(x)$ is exactly what you would compute by enumerating all ways of parenthesizing/decomposing x into binary combinations, adding up the overhead at each combination, and taking the cheapest total. So the abstract least-solution definition really does coincide with the familiar notion of "minimum derivation cost" from parsing and dynamic programming. In particular, the recursion for s^*_C can be read as the usual dynamic-programming recurrence: the best way to build x is to choose a last split (y, z) that yields x and pay the best costs for y and z plus the overhead $t_C(y, z)$.*

Remark 375. ***Why it matters.*** *This theorem justifies calling s^*_C "compositional simplicity" rather than an arbitrary fixed-point artifact. It says the definition matches an operational procedure: build x by a tree of compositions and pay overhead at each internal node. This is the form in which simplicity will interface with grammar-like structural complexity (Hyperseed-Concept 181) and, later, with pattern hierarchies. It also means that many standard intuitions about parsers carry over: sharing subcomputations, reusing repeated substructures, and excluding impossible decompositions correspond to pruning the parse-tree search and thus computing s^*_C efficiently when the finite assumptions hold.*

Remark 376. ***Connection to later sections.*** *Pattern recognition often constructs candidate explanations via hierarchical decompositions. The parse-tree interpretation implies that if patterns supply cheap decompositions of observations, then they also supply low compositional simplicity. This will be used implicitly when emergence is defined by comparing pattern intensities under combination in Section 9. A useful way to keep the linkage in mind is: patterns propose "good splits"* *(low t_C and/or low child costs), and the theorem formalizes that repeatedly choosing such splits corresponds exactly to building a low-cost derivation tree.*

Proof. Because $\mathcal{X}_C$ is finite and each $\mathrm{Dec}(x)$ is finite, the recursion defining s_C^* reduces to a finite system of Bellman-style optimality equations over a finite directed hypergraph. Standard dynamic-programming arguments imply that the least fixed point assigns each x the minimum cost among all finite derivations (parse trees) that construct x from e using $*$, with per-node cost t_C. One convenient way to make this precise is to define the Bellman operator B on functions $f : \mathcal{X}_C \to [0, \infty]$ by

$$(Bf)(x) = \inf_{(y,z)\in\mathrm{Dec}(x)} \big(f(y) + f(z) + t_C(y, z)\big),$$

together with the boundary condition $f(e) = 0$ (and any chosen base costs for other primitives, if the variant with primitives is used). In the finite nonnegative-cost setting, iterating B from the pointwise-zero (or pointwise-minimum admissible) function produces a monotone nondecreasing sequence that stabilizes at the least fixed point, and that limit coincides with the minimum over all finite parse trees because each application of B corresponds to adding one more internal-node layer to a partial derivation. □

Proof sketch. The high-level strategy is: (i) view the decomposition relation $y * z = x$ as a finite hypergraph of production rules; (ii) interpret the defining inequality for s_C^* as the Bellman optimality condition for that hypergraph; (iii) invoke the standard fact that, in finite nonnegative-cost settings, the least fixed point of the Bellman operator yields the minimum cost over all finite derivations, i.e. parse trees. If one wants a slightly more operational phrasing, step (iii) can be seen as: dynamic programming on a finite acyclic-or-nonnegative-cost graph cannot benefit from infinite unfolding, so optimal values are witnessed by finite trees, and the least solution is exactly the one computed by taking minima over all finite unfoldings. □

Remark 377. *The key step is the identification of "least solution of recursive inequalities" with "minimum over trees." Nonnegativity of t_C prevents pathological negative-cost cycles, and finiteness ensures the relevant minima are achieved among finitely many candidate derivations. The "directed hypergraph" phrasing is simply the combinatorial shadow of the binary constructor: each equation $x = y*z$ is a hyperedge from (y, z) to x. It may also help to note that the hypergraph viewpoint cleanly separates* syntax *(which decompositions are allowed, i.e. which hyperedges exist) from* weights *(the overheads t_C), and s_C^* is exactly the induced shortest-derivation-cost function in that weighted hypergraph.*

Remark 378. *A visual intuition is to imagine building x by repeatedly merging pieces. Each merge has an overhead cost. A parse tree records the history of merges. The theorem says: among all possible merge histories that end at x, the least fixed point $s_C^*(x)$ picks the cheapest. In this picture, internal nodes correspond to merge operations, leaves correspond to the simplest available building blocks (here normalized so that e has zero cost), and the total cost is additive along the merge history, mirroring how many compositional systems accumulate "effort" across steps.*

Remark 379. *The theorem is stated in a minimal form to emphasize the intended interpretation. In the full development one typically assigns base costs to a set of primitives and allows leaves to be any primitive, not only e; the same proof pattern applies. The only substantive change in that extension is that the base-cost function contributes an additional leaf term in the tree-cost sum, while the internal-node contributions remain exactly the overheads t_C.*

8.8 Generalized Kolmogorov complexity and structural complexity

Classical Kolmogorov complexity measures the length of the shortest program that outputs an object (Hyperseed-Concept **??**; see [16]). Hyperseed's observer-relative simplicity suggests a more general construct: replace "program length" by *effort* measured in whatever resources the observer actually expends.

Remark 380. ***A small but important shift.*** *In the classical setting, the observer fixes (implicitly) a reference machine and a bitlength measure; both are taken as background structure. Here, the context C makes that background structure explicit: the observer not only chooses a decoding procedure, but also chooses which objects count as admissible descriptions and which expenditures count as cost. This makes it possible to speak about complexity in settings where "programs" are not the natural currency (e.g. physical actions, laboratory protocols, or interactive strategies), without losing the essential "best available redescription" character of Kolmogorov-style definitions.*

Definition 86 (Generalized description systems)**.** *A generalized description system for context C consists of:*

- *a set of descriptions (programs, proofs, encodings) $\mathcal{D}_C$,*
- *a partial decoding map $\mathsf{Dec}_C : \mathcal{D}_C \rightharpoonup \mathcal{X}_C$,*
- *an effort assignment $\mathrm{Eff}_C : \mathcal{D}_C \to [0, \infty]$ satisfying $\mathrm{Eff}_C(d_1 d_2) \leq \mathrm{Eff}_C(d_1) + \mathrm{Eff}_C(d_2)$ for a suitable concatenation/combination operation on descriptions.*

Remark 381. ***Notation unpacking.*** *The arrow $\rightharpoonup$ denotes a partial function: not every description must decode successfully. The expression $d_1 d_2$ denotes a chosen notion of concatenation/combination of descriptions (for instance, string concatenation, program composition, proof concatenation). The inequality $\mathrm{Eff}_C(d_1 d_2) \leq \mathrm{Eff}_C(d_1) + \mathrm{Eff}_C(d_2)$ is the same sequential subadditivity idea as before, now applied to description objects.*

Remark 382. ***What "partial" buys you.*** *Allowing Dec_C to be partial captures several common situations: ill-formed strings, programs that do not halt, proofs that fail to check, circuits with undefined behavior under the chosen simulator, or sensory-motor policies that fail to reach a terminal state. Formally, partiality lets $\mathcal{D}_C$ include a broad space of candidates while keeping Dec_C as the filter that determines which candidates are meaningful for the observer. This mirrors the classical use of partial computable functions (universal machines) but does not restrict Dec_C to that regime.*

Remark 383. ***Effort as a resource preorder.*** *The codomain $[0, \infty]$ is chosen so that infeasible descriptions can be assigned infinite effort, and so that infima of effort sets always exist (possibly as ∞). The subadditivity condition is deliberately weak: it says only that performing two description-steps in sequence should not cost* more *than paying for each step separately. In many concrete models one expects an additional fixed overhead (e.g. for glue code, delimiters, or interface adaptation), and this can be incorporated either by modifying the combination operation or by absorbing the overhead into the definition of Eff_C (as is done later via t_C in compositional systems).*

Remark 384. ***Intuition.*** *A description system is a general "language + interpreter + cost" triple. Kolmogorov complexity chooses a specific universal programming language and uses bitlength as cost; we instead allow any description language that an observer in context C can actually use, and any cost model that tracks the observer's real expenditures. This is one way to reconcile the philosophical power of algorithmic information with the Hyperseed insistence on observer-relativity [1].*

Remark 385. ***Examples.*** *$\mathcal{D}_C$ could be: binary strings interpreted by a fixed machine; formulas decoded by a parser; circuits decoded by a simulator; or proofs decoded by a proof checker. Effort could count length, runtime, energy, memory, or a weighted blend (Hyperseed-Concept 100).*

Remark 386. ***Two clarifying example patterns.*** *If $\mathcal{D}_C$ is a set of plans (action sequences) and Dec_C maps each plan to its realized outcome in an environment model, then $K_C(x)$ becomes the least action-cost required to produce x. If $\mathcal{D}_C$ consists of probabilistic model specifications together with random seeds, and Dec_C is "run the sampler and return its sample," then effort can represent expected compute and $K_C(x)$ becomes a best-effort generative route to x (where the partiality can encode non-termination or rejection conditions). These are not classical program-length settings, but they still instantiate the same abstract "search for cheapest description that yields x" template.*

Remark 387. ***Why needed.*** *Later, when patterns are defined as effort-reducing redescriptions, we need the notion of "description" to be context-dependent: the patterns available to a symbolic reasoner differ from those available to a sensory-motor controller. Generalized description systems are the formal slot where that dependence lives.*

Remark 388. ***On comparability across contexts.*** *Because K_C depends on $\mathcal{D}_C$, Dec_C, and Eff_C, values $K_{C_1}(x)$ and $K_{C_2}(x)$ are not automatically commensurate. This is a feature rather than a bug: Hyperseed uses C precisely to record which tools, priors, and resource-accounting conventions the observer is using. When cross-context comparisons are needed, they must be mediated by explicit translations between description systems (e.g. an embedding of $\mathcal{D}_{C_1}$ into $\mathcal{D}_{C_2}$ with controlled overhead), which is the generalized analogue of the classical invariance discussion around universal machines [16].*

Definition 87 (Generalized Kolmogorov complexity)**.** *Given a generalized description system, define*

$$K_C(x) := \inf\{\mathrm{Eff}_C(d) : d \in \mathcal{D}_C,\ \mathsf{Dec}_C(d) = x\}.$$

Remark 389. ***Well-definedness and edge cases.*** *If there is no description d with $\mathsf{Dec}_C(d) = x$, then the set inside the infimum is empty and, by the usual convention, $K_C(x) = \infty$. If there are descriptions achieving arbitrarily low effort without attaining a minimum (e.g. due to a limiting process in Eff_C), the infimum still exists even though no optimal description exists. Thus K_C should be read as an "optimal achievable bound" rather than a guarantee of an attained optimum.*

Remark 390. ***Intuition.*** *This is the same infimum-over-descriptions pattern as the simplicity definition $s_C(x)$, but with an explicit decoding map. It is therefore best read not as a single universal complexity, but as an observer-indexed family of complexities K_C.*

Remark 391. ***Examples.*** *If x is an image and $\mathcal{D}_C$ contains generative-model parameters plus a decoder that renders images, then $K_C(x)$ measures the cheapest generative explanation of x available to C. If x is a theorem statement and $\mathcal{D}_C$ consists of proofs in a formal system, then $K_C(x)$ measures the cheapest proof effort to derive x (relative to that system).*

Remark 392. ***Interpreting "cheapest" in non-length models.*** *When Eff_C is time or energy, $K_C(x)$ is not a static measure of description* size *but of description* feasibility*. In such cases $K_C(x)$ is naturally paired with a specific execution model (the Dec_C dynamics) and may depend sensitively on implementation details (e.g. hardware assumptions, parallelism, or available caches). This dependence is exactly what is wanted in observer-relative analysis: two observers with different hardware, training, or libraries can legitimately assign different complexities to the same x.*

Remark 393. ***Usefulness.*** *This definition provides a rigorous bridge between the classical "shortest program" perspective [16] and the Hyperseed "least effort" perspective. It allows later claims about simplicity/patterns to be interpreted either as compression statements or as resource-allocation statements, depending on the chosen* Eff_C.

Remark 394. *If* $\mathcal{D}_C = \{0,1\}^*$, $\mathrm{Eff}_C(d) = |d|$ *is bitlength, and* Dec_C *is a universal Turing machine, then* K_C *reduces (up to an additive constant) to classical Kolmogorov complexity. If* Eff_C *counts time, energy, memory, proof steps, or a weighted blend, one obtains different observer-relative complexity notions.*

Remark 395. ***Relationship to classical invariance (informal).*** *The "additive constant" caveat in the classical case comes from comparing two universal machines via compiler overhead. In the generalized setting, analogous comparison statements require explicit assumptions: a translation* $\tau : \mathcal{D}_{C_1} \to \mathcal{D}_{C_2}$ *such that* $\mathsf{Dec}_{C_2}(\tau(d)) = \mathsf{Dec}_{C_1}(d)$ *and* $\mathrm{Eff}_{C_2}(\tau(d)) \leq \mathrm{Eff}_{C_1}(d) + \kappa$ *(or a multiplicative/affine variant). When such a translation exists, one can transport upper bounds on* $K_{C_1}(x)$ *into upper bounds on* $K_{C_2}(x)$ *with controlled overhead, which is the operational content of invariance in this observer-relative language.*

Proposition 6 (Structural complexity as compositional complexity)**.** *Suppose* $\mathcal{D}_C$ *is generated by a finite set of primitives and a binary constructor corresponding to* $*$ *(with overhead* t_C*) so that every description induces a parse tree whose yield is* $x \in \mathcal{X}_C$. *Then the generalized complexity* $K_C(x)$ *coincides with a compositional simplicity functional of the form* $s^*_C(x)$ *(up to fixed base costs for primitives).*

Remark 396. ***Plain-language meaning.*** *When your description language is essentially "build objects by repeatedly combining primitives," then the cheapest description is the cheapest parse tree. So structural complexity (Hyperseed-Concept 181) is not a separate mystery: it is compositional simplicity under a particular choice of description system.*

Remark 397. ***Proof idea (informal).*** *Under the hypothesis, any* $d \in \mathcal{D}_C$ *corresponds to a finite parse tree whose leaves are primitives and whose internal nodes are applications of the binary constructor. If* Eff_C *assigns each primitive a fixed base cost and each constructor application the overhead* t_C *(possibly plus child costs), then* $\mathrm{Eff}_C(d)$ *is (up to constants) the same as the recursively defined compositional cost of the induced tree. Taking the infimum over all* d *yielding* x *is therefore the same optimization as taking the infimum over all* $*$*-parse trees yielding* x, *which is exactly what* $s^*_C(x)$ *encodes. The "up to fixed base costs" clause accounts for the choice of primitive-cost normalization, which shifts all complexities by a bounded amount when the primitive set is finite.*

Remark 398. ***Why it matters.*** *This proposition explains why grammar-like notions of complexity appear naturally once one takes compositionality seriously. It also clarifies how the later pattern layer can speak simultaneously about "compression" and "structure": both are costs of derivations in appropriate description systems [5].*

Remark 399. ***Connection to grammars and circuits.*** *When primitives are terminals and the constructor corresponds to applying production rules, parse trees are exactly derivations in a grammar, and* $K_C(x)$ *becomes a least-derivation-cost measure. When primitives are gates and the constructor corresponds to wiring subcircuits, parse trees become circuit expressions, and* $K_C(x)$ *becomes a size/energy/time cost of realizing the circuit that yields* x. *These familiar "structural" complexity measures are therefore instances of the same generalized template, differing only in what counts as a primitive, what the constructor does, and how effort is tallied.*

Remark 400. ***Connection to Theorem 9.*** *The proposition reduces an infimum over descriptions to a minimum over parse trees, and Theorem 9 identifies that minimum with the least fixed point s^*_C. So the chain is: descriptions $\to$ parse trees $\to$ fixed point. More explicitly, the description language can be viewed as a syntactic presentation of the same compositional data encoded by a derivation/parse tree: the leaves specify which primitives are used, and each internal node records one application of the binary constructor (together with any side-information required by the combination system). The key point is that the objective being optimized (effort) is* structural*: it depends only on the multiset of leaves (base costs) and the pattern of internal nodes (overhead), so it is invariant under switching between these two representations.*

Proof. Each description corresponds to a finite parse tree in the combination system, whose effort is the sum of base costs and overhead costs. Taking the infimum over descriptions is therefore equivalent to taking the minimum over parse trees. By Theorem 9, this is exactly the compositional simplicity $s^*_C(x)$ (with the appropriate base-cost initialization). To make the equivalence precise, note that the decoding map from descriptions to objects factors through the induced tree: once a description is parsed, it determines (i) the primitive labels at the leaves and (ii) the recursive bracketing specifying how the binary constructor is applied. Conversely, given any finite labeled parse tree that evaluates to x under the combination rules, one can encode it as a description by writing down the corresponding parenthesized expression (together with the same primitive labels), so the two optimization domains are coextensive up to the chosen encoding/decoding conventions. Moreover, because effort is defined additively—a leaf contributes its base cost and each internal node contributes the constructor overhead for that particular combination step—the effort assigned to a description coincides with the effort assigned to its tree. Under this identification, optimizing over descriptions or over trees is the same optimization problem, and Theorem 9 supplies the value of that optimum as the least fixed point s^*_C with the stated initialization. □

Proof sketch. The proof strategy is a change of variables. Instead of optimizing over descriptions directly, we optimize over their induced parse trees. Because the description language is generated by primitives and a binary constructor, descriptions and parse trees carry the same information up to the chosen decoding. Effort additivity over constructors becomes summation of overhead along internal nodes, so the best description is the cheapest tree, which Theorem 9 already characterizes as s^*_C. In other words, the apparent "infimum over strings" is not an additional analytic complication here: the strings are merely a linearization of a finite tree-shaped derivation. Once we switch to the tree view, the cost decomposition aligns directly with the dynamic-programming/fixed-point viewpoint: the cost of a node is the overhead for combining its two children plus the costs already assigned to those children, so the global optimum is obtained by choosing, at each object, the cheapest available decomposition into subobjects. This is exactly the recursion captured by the operator whose least fixed point is s^*_C. □

Remark 401. *One can think of this as a Russellian move: replace an informal notion ("structure") by an explicit construction ("a derivation tree in a generating grammar") and then observe that the resulting measure of complexity is just the optimization problem already solved by the compositional simplicity fixed point. From this perspective, "structural complexity" is not an extra ingredient beyond the grammar/combination system: it is the minimal cost of a witness that x can be generated by the primitives via the constructor. The move is useful because it separates (a) the* representational *choices (how we write down derivations as descriptions) from (b) the* invariant *content (which derivations exist and what costs they accrue), and it is the latter that the fixed-point characterization computes.*

8.9 Discussion: how this feeds later Hyperseed layers

This section provides the minimal mathematical "landing zone" for several later constructs: In particular, the goal is to make the informal notions of "trying," "difficulty," and "having a good way to see something" available in a form that can be transported into later layers without reintroducing hidden anthropomorphic assumptions. The emphasis on *relative* effort (relative to a baseline, relative to an observer, relative to an available policy class) is deliberate: later Hyperseed layers will repeatedly compare systems that differ in representational resources, history, or control authority, and the present setup makes those comparisons explicit rather than tacit. One may also read this section as fixing a common currency for subsequent tradeoffs: whenever a later construct is said to be "simpler," "more natural," or "more learnable," the intended operational meaning is that some decomposition or control objective can be achieved with systematically reduced effort, possibly after a change of representation.

- **Patterns (Section 9).** A pattern will be a representation r of x that yields a significant reduction in effort relative to a baseline. Pattern intensity will compare baseline effort to achieved effort (or dually compare baseline weakness to achieved weakness). This links the present material to the theory of pattern and emergence in [5]. In this sense, a "pattern" is not merely a regularity in x, but a *useful* regularity: it is evidenced by an improved effort profile when the agent adopts (or is given) the representation r. The baseline is crucial: the same r may be intensely patterned for one observer (who lacks alternative efficient decompositions) and nearly trivial for another (who already possesses a better policy or representation class), which is exactly the observer-relativity that later emergence arguments rely upon. When later sections speak of the "strength" or "salience" of a pattern, the intent is that such terms can be cashed out in the effort/weakness comparison induced by substituting r into a fixed task family, rather than by appealing only to descriptive compression in the abstract.

- **Habits and morphic resonance (Section 12).** Habit formation will be modeled as repeated updates to a distinction policy (or to the representations supported by a policy) that progressively reduce effective effort for certain decompositions. Resistance/submission will become dynamical (history-dependent) rather than purely static. (For the broader ontological framing of habit and resonance, see [13] alongside the Hyperseed presentation [1].) The core addition in the habit layer is that "effort" is not treated as fixed by the current task alone: instead, the system carries a trajectory of learned biases, cached distinctions, and preferred decompositions that change what is easy to do next. Formally, this means that whatever object plays the role of a distinction policy is iteratively updated so that future instances of a decomposition problem are solvable at lower marginal effort, e.g. by reusing internal structure rather than reconstructing it from scratch. In this frame, "resistance" to a proposed decomposition can be read as an induced cost that persists across time (a kind of inertial term), while "submission" corresponds to a learned alignment in which the same decomposition pathway becomes increasingly low-effort. This is also where the present algebraic notions become temporally textured: weakness and effort will be evaluated not only at a single time slice, but along an updating process, making the relevant quantities explicitly history-dependent.

- **Mind-world correspondence (Section 14 and onward).** "Good models" are those that achieve task adequacy with low effort, i.e. they lie near the Pareto frontier of adequacy vs effort. Weakness provides an algebraic handle for composing such models across contexts, as emphasized in [3]. The Pareto framing is meant to prevent a false dichotomy between "accurate" and "simple": in later sections, adequacy (predictive, explanatory, or control success) and effort (cognitive/work cost under a policy) will be treated as jointly constrained objectives. On this view, a model that is "too complex" is not rejected for having many parts per se, but because its achieved adequacy does not justify its effort relative to available alternatives; conversely, a model that is "too simple" is one whose low effort comes at an avoidable adequacy loss. Weakness enters as a compositional bridge: when moving between contexts or composing submodels, one wants to track how far the composed construction is from an ideal of effortless adequacy, and weakness provides a way to express and bound that deviation without re-deriving everything from first principles each time. This is especially important when the later causality/control layer introduces interventions and control authority: the "effort" budget then includes not only internal computation but also the cost of acquiring leverage over the environment, and the same algebraic bookkeeping is intended to remain applicable.

In short: effort provides the primitive experiential "thermodynamics" of cognition, and simplicity is its observer-relative, compositional shadow. Read this as a methodological constraint as well as a slogan: whenever later layers introduce new entities (patterns, habits, control structures), they should be interpretable as mechanisms for reallocating, reducing, or regularizing effort across tasks and contexts. Correspondingly, "simplicity" is not an intrinsic label attached to an object x in isolation, but a relational statement about how an observer can interact with x under some representational and policy constraints, and about how such interactions compose when x is embedded into larger constructions.

9 Patterns, emergence, properties, and blending

Outline

- Fix a general "description-and-decoding" setup in a context C: entities, a combination operation, a description language, and a cost/effort functional.
- Define *pattern* as a compressive re-description (a representation-as-something-simpler), and define *pattern intensity* as normalized compression gain (optionally p-bit-valued to allow "strong but wrong" patterns).
- Define *properties* as fuzzy sets of patterns (membership = intensity), and define *specific entities* as temporally coherent bundles of patterns (identity-as-a-derived-pattern).
- Define *emergence* for a combination $A*B$ as patterns whose intensity in $A*B$ exceeds a weighted combination of their intensities in A and B.
- Formalize *combination*, *inheritance*, *association*, and *lifting from instances to groups* as operations on pattern/property structures (with quantale-style aggregation as a unifying tool).
- Define *blends* in property-set terms (large overlap of properties), and present a categorical blend construction (pushout/colimit with weakness/effort ranking).
- Define *combinatorial-categorical patterns* (patterns that appear after category-theoretic substitutions), and define *combination systems*, *interpreted combination systems*, and *combinational dynamical causal models*.

Summary and Hyperseed concepts covered

Hyperseed's "pattern layer" is the pivot from phenomenological primitives and effort/simplicity to structured cognition. The core move is operational: a *pattern* in x is a way to represent x as something simpler (lower effort / lower description cost) in a given observer/context. Once pattern intensity is available, one can define properties (as fuzzy sets of patterns), emergence (as patterns whose intensity is amplified by combination), and blending (as property overlap). This section also supplies formal counterparts of Hyperseed's auxiliary constructs that are needed to make patterns interact with computation and dynamics: combination systems, interpreters, substitution-lifted (combinatorial-categorical) patterns, and combination-preserving representations of dynamical causality.

The intended reading is that the "observer/context" C fixes not only what counts as a valid description, but also what resources are salient: the same underlying entity can exhibit different patterns under different description languages, different decoders, different noise models, or different effort functionals. In particular, the cost/effort functional should be understood broadly: it may be a code length, a runtime/space budget, an energy expenditure, a statistical risk, or any proxy for what the context treats as "simplicity." This is what makes patterns comparable and "intensity" meaningful: intensity is not an intrinsic property of the entity alone, but a context-indexed degree of compressive advantage.

The phrase "representation-as-something-simpler" is meant to cover both exact and lossy re-descriptions. An exact pattern corresponds to a shorter description that still decodes to the same target (up to the equivalence notion induced by the decoder), whereas a lossy pattern permits a controlled mismatch when the context does not distinguish the difference. Allowing p-bit-valued intensity is a way to explicitly represent the case where a description is highly compressive yet systematically incorrect: such a pattern can be strong as a proposal while being wrong as a reconstruction, which becomes important when modeling biased or partial observers, adversarial settings, or situations where the decoder is itself fallible. In that sense, intensity is not merely "how present a pattern is," but how much the pattern functions as an economical explanatory handle within the context.

Defining properties as fuzzy sets of patterns makes the ontology "pattern-first" rather than "predicate-first." A property is not an atomic label attached to entities; it is a distribution of pattern intensities, so that property membership is graded and decomposable. This supports a notion of similarity as overlap of property-sets (equivalently, similarity of pattern profiles), and it also supports compositional updates: changing context C changes which patterns are cheap, thereby changing property membership without requiring a separate re-axiomatization of predicates. The notion of a "collective-property-set" then captures the idea that a group or aggregate can carry properties that are not merely pointwise averages of its members' properties, but are properties arising from aggregation of pattern evidence under the chosen combination operator.

Similarly, defining specific entities as temporally coherent bundles of patterns makes identity a derived, dynamical construct. What persists through time is not a metaphysically primitive "thing," but a stable configuration of patterns whose intensities remain coherent under a temporal update rule (the temporal coherence algebra). This lets the framework treat tracking, object permanence, and re-identification as instances of pattern maintenance under resource constraints, rather than as separate primitives. It also clarifies why boundaries of entities can be context-sensitive: what counts as "the same" entity can depend on which patterns are cheap enough to enforce over time.

The emergence criterion for $A * B$ is intended to separate three cases: (i) mere inheritance of patterns already intense in A or B, (ii) dilution or interference where intensities decrease upon combination, and (iii) genuinely emergent amplification where the combined object supports a pattern more strongly than expected from its parts. The weighted comparison baseline makes this precise without forcing linearity: the weights can encode asymmetry of contributions, attention, or relevance, and the aggregation can be tuned to match the semantics of the combination operator $*$. In particular, emergence here is not a mysterious ontological surplus; it is a detectable departure in pattern intensity relative to a specified compositional expectation.

The operations listed (combination, inheritance, association, and lifting) are included to make pattern/property structures usable as algebraic objects rather than mere annotations. Quantale-style aggregation is singled out because it provides a unified way to combine graded evidence (intensities), compose relational links (associations), and transport structure between levels (from instances to groups) while preserving monotonicity and resource-sensitivity. This is the sense in which the pattern layer serves as infrastructure: it supplies the algebra needed to manipulate intensional content in a way that is stable under composition and aggregation.

Blending is presented in two complementary ways: as a high-overlap relation on property-sets and as a categorical construction. The property-overlap view captures an intuitively geometric notion of conceptual mixture: two entities (or two concepts) blend when they share a large portion of their salient properties, with "large" measured relative to the chosen intensity aggregation and any relevance weighting. The categorical pushout/colimit view then captures the constructive side: a blend is not only a comparison between two sources but an explicit amalgam built by identifying shared structure. The mention of weakness/effort ranking indicates that there may be multiple candidate pushouts or colimits, and the framework chooses among them by preferring lower-effort identifications, thereby keeping the construction aligned with the underlying description-cost semantics.

Finally, combinatorial-categorical patterns, interpreted combination systems, and combinational dynamical causal models are included to connect patterns to computation and dynamics. The key point is that new patterns can become visible only after a systematic change of representation (a substitution or functorial re-encoding), so the framework must account for patterns that are invariant or salient under such transformations. A combination system provides the abstract algebra of how entities compose; an interpreted combination system adds a semantics that links abstract compositions to concrete decoded entities and their costs; and a combinational dynamical causal model adds time and causality in a way that is compatible with combination. Together, these notions ensure that patterns are not merely descriptive snapshots but can be propagated, compared, and preserved under dynamical evolution and under the computational processes that implement decoding and substitution.

Hyperseed concepts covered.

- Pattern; pattern intensity.
- Emergence.
- Property; property-set; collective-property-set.
- Specific entity (as a derived pattern with a temporal coherence algebra).
- Blend.
- Combination; inheritance; association; lifting from instances to groups.
- Combinatorial-categorical pattern.
- Combination system; interpreted combination system.

- Combinational dynamical causal model.

9.1 Setup: entities, contexts, descriptions, and effort

We assume the effort/simplicity layer from Section 8 provides, for each context (or observer-relative aspect) C, a universe of entities $\mathcal{E}_C$ together with:

- a (possibly partial) *combination* operation

$$*_C : \mathcal{E}_C \times \mathcal{E}_C \rightharpoonup \mathcal{E}_C,$$

  intended to model "putting things together" in the sense relevant to C;

- a *description system* (syntax) $\mathcal{L}_C$ and an *interpreter/decoder*

$$\mathrm{Dec}_C : \mathcal{L}_C \to \mathcal{E}_C,$$

  intended to model how the observer encodes/constructs entities; and

- an *effort or description cost* functional

$$\mathrm{cost}_C : \mathcal{L}_C \to \mathbb{R}_{\geq 0} \cup \{\infty\}.$$

Here "context" should be read broadly: C may represent a sensory modality, a measurement apparatus, a scientific modeling stance, a downstream task, or a cognitive "view" that determines which distinctions are salient. Accordingly, $\mathcal{E}_C$ is not assumed to be a context-independent ontology; rather, it is the space of *entities as individuated by* C. For example, for a text-processing context C, $\mathcal{E}_C$ might be strings; for a vision context, $\mathcal{E}_C$ might be images up to some equivalence; and for an engineering context, $\mathcal{E}_C$ might be structured designs equipped with tolerances.

The partiality of $*_C$ is deliberate: some pairs of entities may be non-composable (e.g., their interfaces do not match, their physical constraints are incompatible, or the context has no rule for combining them). No global algebraic laws (associativity, commutativity, identities) are assumed unless explicitly stated later; different applications may impose such laws, or may instead treat them as approximate regularities that can themselves be learned or encoded by descriptions. Intuitively, $x *_C y$ captures the *context-specific* notion of composing x and y (e.g., concatenation of strings, overlaying images, wiring components, forming a conjunction of properties, or bundling features into a composite representation).

The role of $\mathcal{L}_C$ is to make explicit that "having an entity" is operationally mediated by representational resources. The map Dec_C then specifies how a description ℓ is interpreted as an entity. Although written as a total function, one can fold ill-formedness or decoding failure into this framework by allowing $\mathrm{cost}_C(\ell) = \infty$ for descriptions that are syntactically valid but semantically unusable in C, or by treating $\mathcal{L}_C$ as already restricted to well-formed, decodable descriptions. The cost functional cost_C can represent many things (bits, time, memory, energy, cognitive effort, or a weighted mixture), and it may incorporate both fixed overheads (e.g., calling a subroutine, invoking a concept) and variable costs (e.g., specifying parameters at a given precision). Allowing ∞ is important: it makes room for hard constraints (some descriptions are forbidden or unattainable), and it ensures the infimum below behaves sensibly even when some candidates are inadmissible.

The *(contextual) simplicity/complexity* of an entity is then the minimal effort required to describe it:

Definition 88 (Contextual description complexity)**.** *For $x \in \mathcal{E}_C$, define*

$$s_C(x) := \inf\{\mathrm{cost}_C(\ell) \ : \ \ell \in \mathcal{L}_C, \ \mathrm{Dec}_C(\ell) = x\}.$$

We interpret lower $s_C(x)$ as "simpler for C" (less representational effort).

The use of inf (rather than min) accommodates description languages with arbitrarily fine approximations or parameterizations: one may be able to describe x increasingly cheaply while approaching it in a limiting sense, or there may be no single cheapest description even though costs can be driven down to a greatest lower bound. In many discrete settings (finite strings, finite programs with integer-valued costs), the infimum is attained and $s_C(x)$ is a true minimum. If there is no ℓ with $\mathrm{Dec}_C(\ell) = x$, the set in the infimum is empty and one may take $s_C(x) = \infty$, expressing that x is indescribable (or not individuated) within the representational resources of context C.

It is also useful to keep in mind that s_C is *relative* in two ways: it depends on what counts as a description (the choice of $\mathcal{L}_C$) and on how descriptions are interpreted (the choice of Dec_C). Thus, changes in measurement conventions, available primitives, background knowledge, or permissible abstractions are all legitimately represented as changes in C (and hence changes in s_C).

Remark 402. *One can equivalently define s_C via compositional simplicity recursion (as in Hyperseed's informal definition) by choosing a grammar for building entities and taking $s_C(x)$ to be the least total effort of any parse tree yielding x, including glue/interaction overheads. The description-based definition above is the most portable because it separates* (i) *the space of possible representations from* (ii) *the evaluator/decoder.*

Concretely, the "parse tree" viewpoint treats a description as an explicit construction plan: internal nodes correspond to applications of combination rules (often modeled by $*_C$ or by a family of typed composition operators), while leaves correspond to primitives (atomic symbols, stored concepts, measurements, or retrieved exemplars). In that view, glue/interaction overheads capture the costs of making parts compatible: specifying alignment, resolving interfaces, setting boundary conditions, or adding correction terms needed for the composite to decode to the intended entity. The description-based formulation packages all of these choices into $\mathcal{L}_C$, Dec_C, and cost_C, which is why it can be transported across domains even when the natural "parts" and "composition" operations differ dramatically.

To connect to the formal core (Section 3), it is often useful to allow the system to store not only scalar costs but also p-bit-valued evidence about whether a putative description actually matches its target. This is optional, but it is exactly where paraconsistency becomes practically relevant: many cognitively useful "patterns" are compressive but partly wrong.

One motivation is that, in practice, an observer often deploys short descriptions that intentionally ignore exceptions (e.g., "birds fly", "objects are rigid", "noise is Gaussian"). Such descriptions can be excellent compression devices (low cost) even when they conflict with some data (nonzero mismatch). Separating "how cheap is the description?" from "how well does it fit?" allows us to model the common situation where a representation is preferred because it is simple *and* informative, not because it is perfectly accurate.

Definition 89 (p-bit-valued mismatch evidence)**.** *Fix a context C. A* mismatch evidence map *is any function*

$$\mathrm{mis}_C : \mathcal{L}_C \times \mathcal{E}_C \to [0,1]^2$$

where $\mathrm{mis}_C(\ell, x) = (m^+, m^-)$ is interpreted as (positive, negative) evidence that ℓ fails to match x in context C.

The two coordinates in $\mathrm{mis}_C(\ell, x) = (m^+, m^-)$ are intended to support paraconsistent situations in which evidence can accumulate on both sides: m^+ can be high because there are clear counterexamples to the description, while m^- can simultaneously be high because there are strong confirmations (or because the context provides reasons the mismatch diagnosis is unreliable). In particular, the framework does not require that "evidence for mismatch" and "evidence against mismatch" sum to 1; allowing both to be nontrivial is what lets the system represent partial, conflicting, or heterogeneous fit in a controlled way.

Operationally, one can use mis_C in several (compatible) ways without fixing a single choice here: as a feasibility filter (only accept descriptions with m^+ below a threshold), as an additive penalty term combined with cost_C (trading off simplicity and mismatch), or as a record used downstream to decide which aspects of a description are trustworthy. In applications where a single scalar is desired, one may derive a mismatch score from the pair (e.g., by a context-dependent aggregation), but keeping the pair explicit preserves information about conflict and uncertainty that would otherwise be collapsed.

9.2 Combination, inheritance, and association as pattern-level notions

Hyperseed distinguishes *dynamic* combination (co-becoming in time) from *static* combination (property union/blend). The present section focuses on the static side (the dynamic side is reconstructed via proto-time and event/process calculi in Section 7). In particular, "static" should be read as "evaluated at a fixed descriptive moment": we intentionally ignore the micro-history of how a compound came about, and only keep the resulting blend of relevant traits. This separation is methodological: the same pair of entities may admit both a temporally detailed story of co-formation (dynamic combination) and a temporally collapsed account in which only the resulting pattern profile matters (static combination).

Definition 90 (Static combination in a context)**.** *A* static combination system *in context C is a pair $(\mathcal{E}_C, *_C)$ where $*_C$ is a partial binary operation on $\mathcal{E}_C$. We write $A *_C B$ when the combination is defined.*

The explicit dependence on C is essential: which combinations are meaningful, admissible, or even nameable depends on the background ontology and the representational resources assumed in that context. Thus $\mathcal{E}_C$ should be read as "the entity-kinds currently in play," and $*_C$ as the rule by which the context permits us to form composites/blends at the static (property-level) description scale. The partiality of $*_C$ is not merely a technical convenience: it encodes the idea that some pairs do not have a well-formed blend in the chosen descriptive scheme, or that a purported blend would require introducing new entity-types not available in $\mathcal{E}_C$. No algebraic laws (such as commutativity or associativity) are assumed a priori; when such laws hold in a given application, they should be stated as additional structure on $(\mathcal{E}_C, *_C)$ rather than taken for granted.

Remark 403. *Many important cases are naturally typed or multi-sorted: $*_C$ is only defined on compatible pairs. This is not a defect; it is exactly what makes combination systems suitable as ontologies.*

Concretely, "compatible" may mean sharing an interface (e.g. two modules with matching ports), belonging to the same sort (e.g. two color swatches that can be mixed), or satisfying a constraint (e.g. two roles that can co-occupy a narrative position without contradiction). In such cases one can think of $*_C$ as implicitly carrying a typing judgment: only when the judgment succeeds do we get a combined entity in $\mathcal{E}_C$. When combination is undefined, that is informative: it marks a boundary of what the context treats as a coherent object of discourse, rather than forcing an artificial "junk" element to stand for incoherence.

Inheritance and association are also most naturally defined using patterns and properties, but we give early "pre-definitions" now (and tighten them after properties are defined). The intent is that these notions operate at the level of pattern-roles (how an entity behaves in typical constructions) rather than at the level of internal constitution: inheritance concerns what can stand in for what, while association concerns what tends to co-appear or co-fit within the same descriptive frame.

Definition 91 (Inheritance as substitutability (preorder form))**.** *Fix a context C. A relation $\preceq_C$ on $\mathcal{E}_C$ is an* inheritance preorder *if $A \preceq_C B$ means "A can substitute for B in many C-relevant contexts without increasing representational effort or breaking too many patterns."*

The phrase "preorder form" signals that, in typical instantiations, $\preceq_C$ is intended to be at least reflexive and transitive (while not necessarily antisymmetric): distinct entities may be mutually substitutable at the granularity that C cares about, yielding equivalence classes of "effectively the same role". The qualifier "many C-relevant contexts" is also deliberate: substitutability is rarely absolute, so $\preceq_C$ should be understood as summarizing a stability of use across a sufficiently rich family of pattern-instances judged important in C. Likewise, "representational effort" is meant to capture the cost of accommodating the substitute (e.g. extra clauses, exceptions, or ad hoc repairs), so inheritance becomes a notion tied to the economy of pattern-description rather than mere set inclusion.

Remark 404. *Later we instantiate $\preceq_C$ by comparing property-sets (Definition 98) or by requiring existence of low-effort rewrite maps in a combination system.*

In property-set terms, the guiding picture is that $A \preceq_C B$ holds when A carries (most of) the properties that make B usable in the patterns of interest, possibly together with additional properties that do not interfere too often. In rewrite-map terms, the idea is that there is a systematic way to translate occurrences of B into occurrences of A with small overhead, so that pattern-instances remain largely valid after substitution. Both readings emphasize that inheritance is not simply a static taxonomic claim, but a claim about preservation of pattern-function under controlled replacement.

Definition 92 (Association (static version))**.** *A (static)* association score *in context C is a symmetric map*

$$\mathrm{assoc}_C : \mathcal{E}_C \times \mathcal{E}_C \to [0, 1]$$

intended to measure "how much A and B tend to occur together" or "how many properties/patterns they share."

The symmetry requirement reflects the intended reading as co-occurrence or overlap rather than directional implication. No further axioms are imposed here: depending on the application, one might want $\mathrm{assoc}_C(A, A) = 1$, or monotonicity with respect to inheritance, or triangle-type constraints, but these should be added only when justified by the chosen construction. The codomain $[0, 1]$ should be interpreted as a normalization convention: association is meant to be comparable across pairs in the same context, even when raw counts or unnormalized overlaps would be scale-dependent. When association is later implemented via property-sets, typical examples include normalized overlap coefficients (such as Jaccard similarity) or weighted variants in which some properties contribute more heavily because they anchor more central patterns in C.

Dynamic notions of association (via short chains of becoming, or temporal co-focusing) are treated in Sections 7 and 12. Here we will implement association in terms of overlap between property-sets (Definition 101). This implementation is meant to make explicit the bridge between "co-occurs" and "co-describes": if two entities share many of the properties that matter to the context, then they will tend to appear together in pattern-instances and will also be easy to group, compare, or blend without introducing exceptional-case bookkeeping.

9.3 Patterns and pattern intensity

We now formalize Hyperseed's key operational definition:

> A pattern is a representation as something simpler.

There are two complementary ways to say this rigorously: *(i)* the description-theoretic way (a code ℓ for x with lower effort than baseline), and *(ii)* the compositional way (a factorization of x into simpler components with bounded glue cost). In practice, these two views are often intertranslatable: a compositional factorization can be encoded as a structured description (e.g. a parse tree), while a good description typically implies some latent factorization (e.g. reuse of subroutines, shared parameters, or repeated motifs). The point of stating both is that different applications make different primitives more natural: for a string, a description is immediate; for a physical or semantic object, an explicit "parts + interaction" picture can be the clearer witness.

9.3.1 Patterns as compressive descriptions

Definition 93 (Pattern witness (description form))**.** *Fix a context C. A* pattern witness *for $x \in \mathcal{E}_C$ is any $\ell \in \mathcal{L}_C$ with* $\mathrm{Dec}_C(\ell) = x$*. Its* description cost *is* $\mathrm{cost}_C(\ell)$*.*

The role of $\mathcal{L}_C$, Dec_C, and cost_C is to make explicit what counts as an admissible representation and what "effort" means for the observer in context C. For example, cost_C might be literal bit-length, runtime, description length under a grammar, MDL-style codelength (data plus model), or an experimentally grounded cognitive effort proxy. Allowing such flexibility is essential later when the same underlying entity is judged by different observers or under different resource constraints.

In this raw form, every entity has infinitely many "patterns" (by trivial re-encoding). The nontrivial content comes from *relative simplicity.* Concretely, without a reference point, one can always define a baroque coding language in which a particular x has a one-token name, thereby making it "simple" by fiat. The baseline mechanism below is what prevents such vacuity: it forces simplicity to be measured relative to a fixed default scheme rather than an adversarially chosen code.

Definition 94 (Baseline description and compression gain)**.** *Fix a context C. Choose a baseline description scheme for entities, represented by a function* $\mathrm{base}_C : \mathcal{E}_C \to \mathcal{L}_C$ *with* $\mathrm{Dec}_C(\mathrm{base}_C(x)) = x$*. Define the baseline cost*

$$s_C^{\mathrm{base}}(x) := \mathrm{cost}_C(\mathrm{base}_C(x)).$$

Given any witness ℓ of x, define the (raw) compression gain

$$\Delta_C(\ell; x) := s_C^{\mathrm{base}}(x) - \mathrm{cost}_C(\ell).$$

The quantity $\Delta_C(\ell; x)$ is positive exactly when ℓ improves on the baseline, and negative when ℓ is a worse-than-default way to represent x. Keeping Δ_C in "absolute units" (whatever units cost_C uses) is convenient for aggregating costs across compositions; later, the normalized intensity I_C will rescale this gain into a unit interval.

Remark 405. *The baseline is part of the observer model: different observers have different "default" ways of representing the same thing, so they will see different patterns. This is not optional; it is Hyperseed's observer-relativity made explicit. A useful way to read $s_C^{\mathrm{base}}(x)$ is "how hard x is* before *I notice any structure"; then a pattern witness is precisely a certificate that the observer can exploit some structure to do better.*

It is often helpful to distinguish the baseline cost $s_C^{\mathrm{base}}(x)$ from any other intrinsic or task-dependent simplicity measure $s_C(x)$ used elsewhere in the paper. Here, the baseline is deliberately operational: it is defined by a concrete encoding choice base_C, so that "compression" is not merely metaphorical but literally an improvement over what the observer would otherwise do by default.

9.3.2 Patterns as compositional factorization

To mirror Hyperseed's (y, z)-pattern picture, we represent patterns as explicit decompositions. Intuitively, $x = y *_C z$ asserts that x can be assembled from parts y and z using the context's composition operator, while t accounts for the additional coordination or interaction cost not already paid for by describing y and z separately. This makes the notion of a "pattern" align with common scientific and engineering practice: a system is understood by specifying components plus a (hopefully small) interaction law.

Definition 95 (Compositional pattern witness)**.** *Fix a context C. A* compositional pattern witness *for $x \in \mathcal{E}_C$ is a tuple*

$$P = (y, z, t)$$

such that $y, z \in \mathcal{E}_C$, $x = y *_C z$, and $t \in \mathbb{R}_{\geq 0}$ is an explicit glue/interaction overhead *intended to upper bound the extra effort needed to assemble the composite from its parts in context C.*

Define the composite cost

$$h_C(P) := s_C(y) + s_C(z) + t.$$

The requirement $t \geq 0$ encodes the idea that composition is never "free" in the accounting: even when the parts are known, there is at least the bookkeeping cost of specifying how they are combined. At the same time, by allowing t to vary by witness, this framework can represent both clean modular decompositions (small t) and strongly entangled composites (large t), which will later be relevant for distinguishing mere aggregation from genuine emergent interaction.

Remark 406. *If one has a direct syntactic representation ℓ for the composite (e.g. a parse tree), t can be derived from the overhead terms in the interpreter; in algorithmic settings t is often a "model class" penalty (e.g. number of parameters, constraint complexity). In MDL terms, one may think of $s_C(y) + s_C(z)$ as the cost of stating the components, and t as the cost of stating the linking hypothesis (alignment, coupling constants, wiring diagram, interface spec, etc.).*

Note that $h_C(P)$ is written using $s_C(y)$ and $s_C(z)$ rather than $s_C^{\mathrm{base}}(y)$ and $s_C^{\mathrm{base}}(z)$. This is intentional: a compositional witness is meant to exploit whatever simplicity notion s_C the context provides for parts, while still being judged against the baseline for the *whole* x when we compute intensity. This matches the informal picture "a pattern is worthwhile if it makes the whole cheaper than my default way of describing the whole," even if the parts themselves are described using specialized (non-baseline) codes.

9.3.3 Pattern intensity

Hyperseed defines pattern intensity as normalized improvement over baseline (in the compositional case, improvement of h_C over the baseline $s_C(x)$). We adopt that, with two small refinements: *(i)* we clamp at 0 so that "non-patterns" have intensity 0, and *(ii)* we optionally add a p-bit negative component to record mismatch evidence.

The normalization by $s_C^{\text{base}}(x)$ makes intensity comparable across entities of different absolute sizes: saving 10 units of cost is a bigger deal when the baseline is 20 than when it is 2000. Clamping at 0 ensures that "actively worse" representations do not create misleading negative intensities; instead, they simply fail to count as patterns under the intensity measure, while their costs can still be tracked separately if needed.

Definition 96 (Scalar pattern intensity)**.** *Fix C and $x \in \mathcal{E}_C$ with $s_C^{\text{base}}(x) > 0$. For any description witness ℓ of x, define the scalar intensity*

$$I_C(\ell; x) := \max\left\{0,\ \frac{s_C^{\text{base}}(x) - \text{cost}_C(\ell)}{s_C^{\text{base}}(x)}\right\} \in [0,1].$$

For any compositional witness $P = (y, z, t)$ of x, define

$$I_C(P; x) := \max\left\{0,\ \frac{s_C^{\text{base}}(x) - h_C(P)}{s_C^{\text{base}}(x)}\right\} \in [0,1].$$

If $s_C^{\text{base}}(x) = 0$, set $I_C(\cdot; x) = 0$ by convention.

As a quick calibration: if $\text{cost}_C(\ell) = s_C^{\text{base}}(x)$, then $I_C(\ell; x) = 0$ (no improvement); if $\text{cost}_C(\ell) = \frac{1}{2} s_C^{\text{base}}(x)$, then $I_C(\ell; x) = \frac{1}{2}$; and if $\text{cost}_C(\ell) \geq s_C^{\text{base}}(x)$, the clamping forces $I_C(\ell; x) = 0$ rather than a negative value. Thus, intensities near 1 correspond to dramatic simplifications relative to baseline, and intensities near 0 correspond to either negligible simplification or none at all.

Proposition 7 (Basic sanity properties of intensity)**.** *Fix C and x.*

1. *$0 \leq I_C(P; x) \leq 1$ for all witnesses P of x.*
2. *$I_C(P; x) > 0$ if and only if $h_C(P) < s_C^{\text{base}}(x)$ (strict compression).*
3. *If P and Q are witnesses of x with $h_C(P) \leq h_C(Q)$, then $I_C(P; x) \geq I_C(Q; x)$.*

Proof. All items follow directly from Definition 96: the ratio lies in $(-\infty, 1]$, clamping yields $[0, 1]$, and the map $u \mapsto \max\{0, (s - u)/s\}$ is monotone decreasing in u for fixed $s > 0$. □

One additional consequence (used implicitly later) is that intensity is invariant under uniform rescaling of cost units within a fixed context: if all costs are multiplied by a positive constant, the ratio $\left(s_C^{\text{base}}(x) - \text{cost}_C(\ell)\right)/s_C^{\text{base}}(x)$ is unchanged. What *does* change intensity is any shift in what the observer treats as baseline, which is exactly as intended: intensity measures improvement over what the observer would have done by default.

Now incorporate paraconsistency: the positive component measures compressive strength, while the negative component records evidence of mismatch ("strong but wrong"). This lets us represent the common situation in which a hypothesis yields a compact description but only approximately fits the data (or fits in one regime while failing in another), without forcing the entire representational apparatus to collapse into triviality.

Definition 97 (p-bit pattern intensity). *Fix C and assume a mismatch map* mis_C *(Definition 89). For any description witness ℓ of x, define a* p-bit *intensity*

$$\mathrm{Int}_C(\ell; x) := \big(I_C(\ell; x),\ \mathrm{mis}_C(\ell, x)^+\big) \in [0,1]^2,$$

where $\mathrm{mis}_C(\ell, x)^+$ *is the positive-evidence component that ℓ mismatches x. (Alternative conventions are reasonable; the only requirement for later sections is monotonicity in compression-gain and mismatch evidence.)*

This two-coordinate view separates two axes that are often conflated: how *useful* the pattern is as a compressor versus how *trustworthy* it is as an account of x. For instance, a coarse model might yield high $I_C(\ell; x)$ by exploiting a strong regularity, yet still incur non-negligible mismatch evidence because it ignores boundary effects or rare events; conversely, an exact but cumbersome encoding may have low intensity while having essentially zero mismatch evidence.

Remark 407. *This construction is intentionally permissive: it lets a system retain a compressive pattern even when it is partially false, without forcing triviality. That is precisely the role of paraconsistent logic in the ontology. Operationally, later selection rules can prefer patterns that are simultaneously high-intensity and low-mismatch, but the representation itself does not have to discard a high-intensity idea merely because it is not perfectly accurate.*

9.4 Properties as fuzzy sets of patterns

Hyperseed's property notion is derived: properties are patterns, collected and graded by intensity. Concretely, one begins with whatever family of pattern-detectors $\mathcal{P}_C$ is deemed relevant to a context C (logical predicates, learned features, dynamical motifs, compression primitives, etc.), and then treats an object's "having a property" as nothing more (and nothing less) than exhibiting the corresponding pattern with some graded strength. In this sense, the property vocabulary is not primitive: it is induced by the pattern vocabulary, and any change in $\mathcal{P}_C$ (or in the intensity functional I_C) changes the resulting property semantics. It is also important that C is explicit: the same entity may support very different salient patterns in different contexts, so $\mathrm{Prop}_C(x)$ should be read as a context-indexed description rather than an absolute inventory of intrinsic attributes.

Definition 98 (Property-set). *Fix a context C and a space of candidate patterns $\mathcal{P}_C$. The* property-set *of $x \in \mathcal{E}_C$ is the fuzzy set*

$$\mathrm{Prop}_C(x) : \mathcal{P}_C \to [0,1], \qquad \mathrm{Prop}_C(x)(P) := I_C(P; x).$$

If one uses p-bit *intensity (Definition 97), then* $\mathrm{Prop}_C(x)$ *is* p-bit*-valued instead.*

This is an explicit "feature map" induced by patterns: it sends an entity x to a membership function on $\mathcal{P}_C$. If $\mathcal{P}_C$ is finite, then $\mathrm{Prop}_C(x)$ can be viewed as a vector in $[0,1]^{|\mathcal{P}_C|}$; if $\mathcal{P}_C$ is countable, as a bounded sequence; and if one generalizes further (e.g. measurable pattern families), as a bounded function suitable for integration against weights. The fuzzy-set reading is deliberate: a pattern P is not simply "present" or "absent," but can be present to a degree, matching the idea that real-world regularities are often partial, noisy, or scale-dependent.

When p-bit-valued intensities are used, the codomain changes but the role does not: $\mathrm{Prop}_C(x)(P)$ still represents the strength of evidence (in the particular p-bit sense) that x supports P. One can always recover an $[0,1]$-valued proxy (for instance, by mapping a p-bit value to an expected degree, a probability of detection, or another application-specific scalarization), but keeping p-bit values can be useful when one wants intensities that compose more naturally under operations introduced elsewhere.

Definition 99 (Mass of a property-set (one simple choice))**.** *Assume $\mathcal{P}_C$ is finite or countable. Define the* mass *of the property-set as*

$$\|\mathrm{Prop}_C(x)\|_1 := \sum_{P\in\mathcal{P}_C} \mathrm{Prop}_C(x)(P).$$

This L^1-style mass is the simplest aggregate notion of "how many patterns are present, counting multiplicity by intensity." In particular, it distinguishes the case where x weakly supports many patterns from the case where it strongly supports only a few, while still remaining additive over the pattern index. For a finite $\mathcal{P}_C$, the mass is automatically finite and lies in $[0, |\mathcal{P}_C|]$; for a countably infinite $\mathcal{P}_C$, finiteness becomes a substantive condition and can be enforced by choosing $\mathcal{P}_C$ so that only finitely or summably many patterns have non-negligible intensity on typical entities, or by introducing weights $w(P) \geq 0$ and considering $\sum_P w(P)\mathrm{Prop}_C(x)(P)$ when different patterns carry different saliencies or costs.

Remark 408. *This mass is a pattern-theoretic analogue of "how much structured regularity" is present. More sophisticated variants can correct for overlap between patterns (see the "structural complexity" discussion in Section 8).*

A useful way to interpret the overlap issue is that two distinct patterns $P, Q \in \mathcal{P}_C$ may be systematically co-triggered by the same underlying regularity, so summing their intensities can double-count evidence. Correcting for this can be done by (i) discounting redundancy via similarity/implication relations among patterns, (ii) selecting approximately independent subsets of patterns, or (iii) passing from a raw pattern-family to a factorized or minimal basis (when such a notion is available). The present definition intentionally stays minimal: it supplies a baseline quantity that is easy to compute and easy to replace.

9.4.1 Inheritance and association via properties

We can now instantiate inheritance and association in a canonical way. The guiding idea is that once entities are represented by their pattern-intensity profiles, standard set-theoretic relations and similarities can be lifted to the fuzzy setting, yielding orderings (for "is-a"-like relations) and symmetric scores (for "related-to"-like relations) directly from the same data.

Definition 100 (Inheritance via property inclusion)**.** *Fix C. Define an inheritance preorder on $\mathcal{E}_C$ by*

$$A \preceq_C B \iff \mathrm{Prop}_C(A)(P) \geq \mathrm{Prop}_C(B)(P) \quad \textit{for all } P \in \mathcal{P}_C.$$

Because the condition is pointwise in P, $\preceq_C$ is automatically reflexive and transitive, hence a preorder (antisymmetry may fail if two distinct entities have identical property-sets on $\mathcal{P}_C$). Equivalently, $A \preceq_C B$ means that $\mathrm{Prop}_C(B)$ is a fuzzy subset of $\mathrm{Prop}_C(A)$ under the usual product order on $[0,1]^{\mathcal{P}_C}$. This matches the intuition that A can stand in for B with respect to all pattern-tests in $\mathcal{P}_C$: any pattern that manifests in B manifests no less in A.

Remark 409. *This reads: "A inherits from B" when every pattern/property that characterizes B is present (at least as strongly) in A. This is a direct mathematical rendering of Hyperseed's "substitutability in contexts." In applications, one usually weakens the $\forall P$ requirement to a weighted or thresholded version.*

Two common weakenings, both compatible with the same conceptual picture, are: (i) a slackened inclusion condition such as $\mathrm{Prop}_C(A)(P) \geq \mathrm{Prop}_C(B)(P) - \varepsilon(P)$ for pattern-dependent tolerances $\varepsilon(P) \geq 0$ (useful when intensities are noisy), and (ii) a weighted aggregate condition such as

$$\sum_{P \in \mathcal{P}_C} w(P) \left(\mathrm{Prop}_C(B)(P) - \mathrm{Prop}_C(A)(P)\right)_+ \leq \delta,$$

where $(t)_+ := \max\{t, 0\}$ measures shortfall of A relative to B and $w(P)$ encodes salience. These variants preserve the idea that inheritance is driven by systematic dominance of property intensity, while allowing controlled exceptions or trading off minor deficits against major matches.

Definition 101 (Association from property overlap)**.** *Fix C. Define the (symmetric) association score*

$$\mathrm{assoc}_C(A, B) := \frac{\sum_{P \in \mathcal{P}_C} \min\{\mathrm{Prop}_C(A)(P), \mathrm{Prop}_C(B)(P)\}}{\sum_{P \in \mathcal{P}_C} \max\{\mathrm{Prop}_C(A)(P), \mathrm{Prop}_C(B)(P)\}}$$

with the convention $0/0 := 0$.

This ratio lies in $[0, 1]$ whenever the denominator is nonzero. It equals 1 precisely when the two property-sets agree pointwise (on all patterns that appear in at least one of them), and it approaches 0 when their supports (in the fuzzy sense) are largely disjoint. The $0/0 := 0$ convention corresponds to treating two entities with identically zero intensity on every candidate pattern (i.e. indistinguishable "blank" descriptions relative to $\mathcal{P}_C$) as having no meaningful association signal under this particular score; alternative conventions are possible, but the present one keeps the score conservative in the absence of evidence.

Remark 410. *This is a fuzzy Jaccard similarity. Other similarity measures (inner product, Hellinger distance, Hutchinson metric on induced measures) fit the same role; the later "pattern web" construction (Section 11) can be built from any reasonable choice.*

The choice of similarity is partly a modeling decision about what "overlap" should mean. Jaccard emphasizes shared intensity relative to total exhibited intensity, hence it is naturally scale-sensitive to the union of properties; inner-product-based scores emphasize co-activation but can be dominated by overall mass; and distributional distances (e.g. Hellinger) become particularly natural if one normalizes $\mathrm{Prop}_C(x)$ to a probability distribution over patterns when $\|\mathrm{Prop}_C(x)\|_1 > 0$. All of these remain instances of the same general recipe: represent entities by graded pattern profiles, then apply a similarity/divergence on those profiles to obtain an association primitive suitable for building higher level relational structure.

9.5 Emergence

Hyperseed defines emergence of (A, B) (relative to a combination operation $*$) as the collection of patterns that become more intense in $A * B$ than is explained by A and B separately. Intuitively, the comparison is made pattern-by-pattern: a candidate pattern P is deemed emergent when its measured intensity in the composite exceeds an explicit baseline prediction formed from the intensities of P in each part. The role of the baseline is to distinguish genuinely "interaction-driven" structure from structure that is already present in A or in B alone and would therefore be expected to appear in the composite even without additional organization.

Definition 102 (Weighted baseline for emergence)**.** *Fix C and $A, B \in \mathcal{E}_C$ with $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) > 0$. Define weights*

$$w_C(A) := \frac{s_C^{\mathrm{base}}(A)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}, \qquad w_C(B) := \frac{s_C^{\mathrm{base}}(B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

Remark 411. *Definition 102 produces a convex weighting in the sense that $w_C(A), w_C(B) \geq 0$ and $w_C(A) + w_C(B) = 1$. The condition $s_C^{\text{base}}(A) + s_C^{\text{base}}(B) > 0$ rules out the degenerate case in which neither entity contributes any baseline "mass" for constructing a meaningful prediction. When $s_C^{\text{base}}(A) = 0$ and $s_C^{\text{base}}(B) > 0$ (or vice versa), the baseline reduces to the nonzero contributor, so emergence is measured relative to the part that carries the entire baseline weight in context C.*

Definition 103 (Emergent intensity and emergent set)**.** *Fix C and assume $A *_C B$ is defined. For a pattern candidate $P \in \mathcal{P}_C$, define the* emergent surplus

$$\mathrm{Sur}_C(P; A, B) := I_C(P; A *_C B) - \big(w_C(A)\, I_C(P; A) + w_C(B)\, I_C(P; B)\big).$$

Define the emergent intensity

$$\mathrm{Emer}_C(P; A, B) := \max\{0, \mathrm{Sur}_C(P; A, B)\}.$$

The emergence of (A, B) *is the fuzzy set $P \mapsto \mathrm{Emer}_C(P; A, B)$, or the crisp set $\{P : \mathrm{Emer}_C(P; A, B) > 0\}$ if desired.*

Remark 412. *The surplus $\mathrm{Sur}_C(P; A, B)$ is permitted to be negative: this corresponds to a pattern candidate whose intensity is* suppressed *in the composite relative to the weighted baseline. Definition 103 focuses the notion of emergence on "positive novelty" by clamping negative surplus to 0 via $\max\{0, \cdot\}$. In settings where suppression is also of interest, one can still recover it directly from $\mathrm{Sur}_C(P; A, B)$ without altering the present definition.*

Remark 413. *Because the baseline term is a convex combination of $I_C(P; A)$ and $I_C(P; B)$, the quantity $\mathrm{Emer}_C(P; A, B)$ measures a* relative gain *over what would be expected if the composite carried forward only a weighted mixture of the parts' pattern intensities. In particular, if $I_C(P; \cdot)$ is normalized (for example, $I_C(P; X) \in [0, 1]$ for all $X \in \mathcal{E}_C$), then $\mathrm{Emer}_C(P; A, B)$ inherits corresponding interpretability as a bounded, nonnegative degree of emergence. The definition itself does not require a particular normalization, but it does require that $I_C(P; X)$ be comparable across $X \in \{A, B, A *_C B\}$ within the same context C.*

Proposition 8 (Basic properties of emergent intensity)**.** *Fix C and assume $A *_C B$ is defined. Then for every $P \in \mathcal{P}_C$:*

1. $\mathrm{Emer}_C(P; A, B) \geq 0$.
2. *If $I_C(P; X) \geq 0$ for all $X \in \{A, B, A *_C B\}$, then*
$$\mathrm{Emer}_C(P; A, B) \leq I_C(P; A *_C B).$$
3. $\mathrm{Emer}_C(P; A, B) = 0$ *whenever* $I_C(P; A *_C B) \leq w_C(A)\, I_C(P; A) + w_C(B)\, I_C(P; B)$.

Proof. (1) holds by definition since Emer_C is the maximum of 0 and a real number. For (2), the baseline term $w_C(A)\, I_C(P; A) + w_C(B)\, I_C(P; B)$ is nonnegative under the stated assumption, so

$$\mathrm{Sur}_C(P; A, B) = I_C(P; A *_C B) - (\text{nonnegative}) \leq I_C(P; A *_C B),$$

and clamping cannot increase the quantity beyond $I_C(P; A *_C B)$. Statement (3) is immediate from the definition of Sur_C and the clamping in Emer_C. $\square$

Proposition 9 (Symmetry of emergence under commutative combination)**.** *Fix C and assume $*_C$ is commutative and $A *_C B$ is defined. Then for every $P \in \mathcal{P}_C$,*

$$\mathrm{Emer}_C(P; A, B) = \mathrm{Emer}_C(P; B, A).$$

Proof. If $*_C$ is commutative, then $A *_C B = B *_C A$, hence $I_C(P; A *_C B) = I_C(P; B *_C A)$. The weights in Definition 102 satisfy $w_C(A)$ and $w_C(B)$ swap under exchanging A and B, so the baseline term swaps accordingly. Therefore the surplus and its clamped version agree. □

Remark 414. *Emergence is the formal bridge from "patterns in parts" to "integration". A high density of high-intensity emergent patterns spanning a composite corresponds to the Hyperseed intuition of a system whose organization is not reducible to its components. The weighting by $s_C^{\mathrm{base}}(\cdot)$ makes this bridge sensitive to the relative "amount" of each component in the context: in many applications s_C^{base} can be chosen to track size, mass, duration, or any other baseline contribution that would make a simple additive expectation reasonable before accounting for interactions.*

Definition 104 (Collective-property-set)**.** *Fix C. Given an entity $A \in \mathcal{E}_C$ and a reference class $\mathcal{B} \subseteq \mathcal{E}_C$ of "other entities" relevant to A, define the* collective-property-set *of A relative to $\mathcal{B}$ as*

$$\mathrm{Prop}_C^{\mathrm{coll}}(A)(P) := \sup_{B \in \mathcal{B}} \mathrm{Emer}_C(P; A, B).$$

Remark 415. *Definition 104 turns emergence into a property assignment for A by asking: across all relevant interaction partners $B \in \mathcal{B}$, how strongly can A support the emergence of pattern P? The use of* sup *(rather than* max*) allows $\mathcal{B}$ to be infinite or to admit limiting sequences of partners for which the emergent intensity approaches, but does not necessarily attain, its least upper bound. When $\mathcal{B}$ is finite (or compact under suitable regularity assumptions on Emer_C), the supremum can often be replaced by a maximum.*

Proposition 10 (Monotonicity in the reference class)**.** *Fix C and $A \in \mathcal{E}_C$. If $\mathcal{B}_1 \subseteq \mathcal{B}_2 \subseteq \mathcal{E}_C$, then for all $P \in \mathcal{P}_C$,*

$$\mathrm{Prop}_C^{\mathrm{coll}}(A; \mathcal{B}_1)(P) \le \mathrm{Prop}_C^{\mathrm{coll}}(A; \mathcal{B}_2)(P),$$

where $\mathrm{Prop}_C^{\mathrm{coll}}(A; \mathcal{B})(P)$ denotes the right-hand side of Definition 104 with the chosen reference class $\mathcal{B}$.

Proof. If $\mathcal{B}_1 \subseteq \mathcal{B}_2$, then the set of values $\{\mathrm{Emer}_C(P; A, B) : B \in \mathcal{B}_1\}$ is a subset of $\{\mathrm{Emer}_C(P; A, B) : B \in \mathcal{B}_2\}$, hence its supremum cannot exceed the supremum over the larger set. □

9.6 Blends

Hyperseed's blend concept is property-theoretic: C is a blend of A and B when a significant fraction of C's properties are also properties of A or B. Here "properties" are understood in the same weighted sense used throughout the property semantics: $\mathrm{Prop}_C(X)(P)$ assigns (possibly graded) salience, strength, or evidential mass to the claim that X has property P in context C, so overlap can be partial rather than all-or-nothing.

Definition 105 (Blend score and blend predicate)**.** *Fix C and choose a threshold $\theta \in (0, 1]$. Define the* property overlap mass *of C with (A, B) as*

$$\mathrm{ov}_C(C; A, B) := \sum_{P \in \mathcal{P}_C} \min\Big\{\mathrm{Prop}_C(C)(P),\ \max\big(\mathrm{Prop}_C(A)(P), \mathrm{Prop}_C(B)(P)\big)\Big\}.$$

Define the blend score

$$\mathrm{blendscore}_{\mathcal{C}}(C;A,B) := \begin{cases} \mathrm{ov}_{\mathcal{C}}(C;A,B)/\|\mathrm{Prop}_{\mathcal{C}}(C)\|_1, & \|\mathrm{Prop}_{\mathcal{C}}(C)\|_1 > 0, \\ 0, & \|\mathrm{Prop}_{\mathcal{C}}(C)\|_1 = 0. \end{cases}$$

We say C is a blend of A and B (in context $\mathcal{C}$) *if* $\mathrm{blendscore}_{\mathcal{C}}(C;A,B) \geq \theta$.

The use of $\max(\mathrm{Prop}_{\mathcal{C}}(A)(P), \mathrm{Prop}_{\mathcal{C}}(B)(P))$ treats the sources A and B as an "either-parent" supplier of property mass for P, while the outer $\min\{\cdot,\cdot\}$ enforces that only as much of C's mass on P can be credited as is supported by at least one parent. Thus $\mathrm{ov}_{\mathcal{C}}$ is a fuzzy-set analogue of $|C \cap (A \cup B)|$ when properties are crisp indicators, and $\mathrm{blendscore}_{\mathcal{C}}$ becomes a normalized coverage ratio for how much of C can be explained as inherited from A or B at the property level. In particular, whenever $\|\mathrm{Prop}_{\mathcal{C}}(C)\|_1 > 0$ one has $0 \leq \mathrm{blendscore}_{\mathcal{C}}(C;A,B) \leq 1$, with $\mathrm{blendscore}_{\mathcal{C}}(C;A,B) = 1$ precisely when, for every $P \in \mathcal{P}_{\mathcal{C}}$, $\mathrm{Prop}_{\mathcal{C}}(C)(P) \leq \max(\mathrm{Prop}_{\mathcal{C}}(A)(P), \mathrm{Prop}_{\mathcal{C}}(B)(P))$.

Because the normalization uses $\|\mathrm{Prop}_{\mathcal{C}}(C)\|_1$, the score measures the fraction of C's own asserted property mass that is supported by A or B; properties that are salient for A or B but not salient for C do not inflate the score. This keeps the predicate focused on explaining C rather than on measuring generic similarity between A and B. The special case $\|\mathrm{Prop}_{\mathcal{C}}(C)\|_1 = 0$ corresponds to a context where C has no active property mass (e.g. missing data, a fully uncommitted representation, or an explicitly "null" property profile), in which case the conservative convention sets the score to 0.

Remark 416. *This definition is intentionally parametric: the threshold θ and the mass functional can be chosen based on application. In cognitive modeling, blends are typically context-sensitive; the present definition makes that explicit. A higher θ demands that most of what is asserted about C be attributable to at least one parent, whereas a lower θ allows C to introduce more novel or emergent properties not present in A or B. Likewise, replacing* max *by another pooling operator (e.g. a t-conorm, a softmax, or an additive rule with a cap) changes whether joint support from both parents can strengthen the accounting for a property.*

For intuition, consider the crisp/unweighted limit where $\mathrm{Prop}_{\mathcal{C}}(X)(P) \in \{0,1\}$ and $\|\mathrm{Prop}_{\mathcal{C}}(C)\|_1 = |\{P : \mathrm{Prop}_{\mathcal{C}}(C)(P) = 1\}|$. Then $\mathrm{ov}_{\mathcal{C}}(C;A,B)$ counts the number of properties that C has and that at least one of A or B also has, and $\mathrm{blendscore}_{\mathcal{C}}(C;A,B)$ is the fraction of C's properties covered by $A \cup B$. In the graded case, the same picture holds but with fractional contributions: if C partially satisfies P to degree 0.6 while the stronger parent support is 0.4, then only 0.4 counts as overlap and the remaining 0.2 of C's mass on P is treated as not inherited from either parent.

9.6.1 Categorical blend construction (pushout intuition)

A more structural notion of blend is useful when entities are themselves compositional objects (e.g. graphs, terms, enriched presheaves). In that case, a blend is naturally modeled by a colimit that glues two objects along a shared substructure, with a weakness/effort bias to prefer minimal glue. This captures the common cognitive intuition that blending reuses a matched subpattern and then merges the remaining structure around it, rather than arbitrarily combining the entirety of A and B.

Definition 106 (Blend as colimit with weak-glue preference (sketch))**.** *Fix a category (or enriched category) $\mathcal{C}$ of entity-structures appropriate to the domain. Given $A, B \in \mathrm{Ob}(\mathcal{C})$, a* blend diagram *is a span*

$$S \xrightarrow{i_A} A, \qquad S \xrightarrow{i_B} B$$

where S is an extracted "shared subpattern." A categorical blend *is a choice of pushout (or more generally colimit)*

$$A \xleftarrow{i_A} S \xrightarrow{i_B} B \quad \rightsquigarrow \quad A \to \mathrm{Blend}(A, B; S) \leftarrow B,$$

together with a ranking functional (effort/weakness) that prefers spans S yielding lower glue overhead.

Operationally, S represents the alignment hypothesis: it specifies which parts of A and B are to be treated as "the same" for the purpose of blending, and the pushout performs the minimal identification consistent with that hypothesis. The effort/weakness preference then plays a role analogous to regularization: among many possible spans S (and many possible ways to extract them from data), one favors those that explain the perceived commonality while introducing as few identifications, coercions, or ad hoc repairs as possible. In concrete categories (e.g. graphs), the glue overhead can be understood as the size or complexity of the interface S relative to A and B, the number of forced identifications, or the amount of additional structure required to make the span commute in the chosen modeling formalism.

Remark 417. *In practice, one rarely needs literal pushouts; approximate colimits in a quantale-enriched setting often match cognition better (multiple partial shared subpatterns, conflicting identifications, etc.). Paraconsistency enters here in a controlled way: one can glue while retaining evidence against some identifications. From the property-theoretic perspective, such "approximate gluing" corresponds to allowing the blend to inherit some properties strongly while keeping other candidate inheritances tentative or contested, which then manifests as intermediate masses in* $\mathrm{Prop}_C(\mathrm{Blend}(\cdot))$ *rather than hard yes/no attributions.*

9.7 Lifting from instances to groups

Hyperseed notes that once a relation is defined for instances, there are multiple sensible ways to lift it to groupings of instances. We formalize three common lifts (average-like, $\forall\exists$-like, and $\exists\forall$-like) in a form that generalizes well.

Definition 107 (Three lifts of an instance relation)**.** *Let $C(\cdot, \cdot) : X \times Y \to [0, 1]$ be any graded instance-level relation. For finite nonempty $A \subseteq X$ and $B \subseteq Y$, define:*

$$\begin{aligned} C_{\mathrm{avg}}(A, B) &:= \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} C(a, b), \\ C_{\forall\exists}(A, B) &:= \min_{a \in A} \max_{b \in B} C(a, b), \\ C_{\exists\forall}(A, B) &:= \max_{b \in B} \min_{a \in A} C(a, b). \end{aligned}$$

Remark 418. *These correspond (respectively) to "on average", "for most a there exists some b", and "there exists some b that works for most a", in a graded sense. The same pattern works in quantale-valued settings by replacing* $(\min, \max)$ *with meet/join and the average with a chosen aggregation functional.*

Remark 419. *The nonemptiness of A, B ensures each lift is well-defined without additional conventions. In applications where empty groups may arise (e.g., sparse clustering outputs), one can extend the definitions by selecting neutral elements (e.g.,* 1 *for "vacuously true" lifts,* 0 *for "no evidence") or by explicitly carrying an "undefined" value; the present formalization keeps the core algebra clean.*

Remark 420. *When C is crisp, i.e. $C(a,b) \in \{0,1\}$, the second and third lifts reduce to familiar quantifier patterns:*

$$C_{\forall\exists}(A,B) = 1 \iff \forall a \in A\, \exists b \in B : \; C(a,b) = 1, \qquad C_{\exists\forall}(A,B) = 1 \iff \exists b \in B\, \forall a \in A : \; C(a,b) = 1.$$

Thus, in the Boolean case, $C_{\forall\exists}$ behaves like an "every a can be matched" score, while $C_{\exists\forall}$ behaves like a "single prototype b covers all a" score.

Remark 421. *Basic bounds and order-theoretic behavior are immediate from the definitions. Writing*

$$C_{\min}(A,B) := \min_{a\in A,\, b\in B} C(a,b), \qquad C_{\max}(A,B) := \max_{a\in A,\, b\in B} C(a,b),$$

one always has $C_{\min}(A,B) \le C_{\forall\exists}(A,B) \le C_{\max}(A,B)$ and $C_{\min}(A,B) \le C_{\exists\forall}(A,B) \le C_{\max}(A,B)$, and the minimax inequality gives

$$C_{\exists\forall}(A,B) \;\le\; C_{\forall\exists}(A,B).$$

In particular, $C_{\exists\forall}$ is the more "demanding" lift: it asks for one element of B that uniformly scores well against all of A, whereas $C_{\forall\exists}$ allows the best witness in B to depend on a.

Remark 422. *If C is interpreted as a similarity or compatibility, then C_{avg} measures overall affinity between two clouds, $C_{\forall\exists}$ measures whether each point in A is well-covered by some point in B (a directed "coverage" score), and $C_{\exists\forall}$ measures whether B contains a single representative that fits all of A (a directed "prototype" score). These readings often align with empirical choices in clustering, retrieval, and exemplar-based modeling.*

This is the generic mechanism behind lifting reference, association, and even emergence-like relations from episodes to categories.

9.8 Combinatorial-categorical patterns

Hyperseed introduces "combinatorial-categorical patterns" as patterns that become visible only after running a category-theoretic substitution/relabeling process. We formalize this as: apply a rule system that assigns category labels to instances, producing a derived structure in which ordinary patterns can be mined.

Definition 108 (Category domain and membership state)**.** *Fix a finite set of categories (types) W. A* membership state *on a set of instances S is a map*

$$m : S \to 2^W.$$

Remark 423. *Equivalently, a membership state can be seen as a binary relation $R_m \subseteq S \times W$ via $(s,w) \in R_m \Leftrightarrow w \in m(s)$, or as a $|S| \times |W|$ incidence matrix. This viewpoint makes the subsequent closure operation look like a monotone update of a relation by rules.*

Definition 109 (Combinatorial-categorical theory)**.** *Fix a set of instances S and a category domain W. A* combinatorial-categorical theory *T is a finite set of rules of the form*

$$(x \in A) \;\wedge\; (y \in B) \implies (z \in C),$$

where $x, y, z \in S$ and $A, B, C \in W$.

Remark 424. *Syntactically, these are Horn-style implication rules over the atomic predicates "$s \in w$" (read: the instance s has category label w). The restriction to conjunctions in the antecedent ensures that the associated operator on membership states is monotone under set inclusion, which is what makes least-fixed-point semantics natural and robust.*

Definition 110 (Substitution output / closure)**.** *Given an initial membership state $m_0 : S \to 2^W$ and a theory T, define the closure m_T as the least fixed point (under set inclusion) obtained by repeatedly applying the rules of T: start from m_0 and whenever a rule's antecedent holds under the current m, add C to $m(z)$. Let*

$$\mathrm{sub}(T, S, m_0) := (S, m_T)$$

denote the resulting labeled structure.

Remark 425. *Because S and W are finite and rules only add labels, the iterative forward-chaining process must terminate after at most $|S| \cdot |W|$ successful additions, independent of rule order. Moreover, the least-fixed-point characterization implies confluence: any fair rule-application schedule yields the same m_T, so* $\mathrm{sub}(T, S, m_0)$ *is well-defined without committing to an operational order.*

Remark 426. *It is often useful to name the associated closure operator explicitly: define F_T on membership states by adding, for each rule $(x \in A) \wedge (y \in B) \Rightarrow (z \in C)$, the label C to z whenever $A \in m(x)$ and $B \in m(y)$. Then m_T is the least fixed point of F_T above m_0 (i.e. the least $m \supseteq m_0$ with $F_T(m) = m$). This makes clear that* sub *is monotone in m_0 and in T (adding initial labels or rules can only add, never remove, derived labels).*

Definition 111 (Combinatorial-categorical pattern)**.** *Fix a context C and assume a pattern mining scheme has been defined on labeled structures. A pattern candidate P is a* combinatorial-categorical pattern *in S relative to T and m_0 if P is a (standard) pattern in the derived structure* $\mathrm{sub}(T, S, m_0)$.

Remark 427. *Concretely, one can take the "labeled structure" to be the multiset $\{m_T(s) : s \in S\}$ and apply any off-the-shelf pattern formalism (frequent itemsets, association rules, MDL-based codes, subgraph motifs after encoding m_T as a bipartite graph, etc.). The key is that patterns are not searched for in the raw instance space, but in the feature space induced by closure under T.*

Remark 428. *This construction cleanly separates two kinds of inductive bias. The theory T determines which composite categories are even expressible (a kind of "feature generation"), while the mining scheme determines which regularities count as salient (frequency, compressibility, predictive utility,* etc.*). In this sense, "combinatorial-categorical patterns" are ordinary patterns, but in a theory-generated representation.*

Remark 429. *This is a clean formal docking port to "pattern theory" constructions: T generates a derived feature space (categories as features), after which one mines compressive structures (patterns) over that space.*

9.9 Combination systems, interpreters, and combinational dynamical causal models

We now formalize the computational and dynamical notions Hyperseed attaches to combinations. The intent of this subsection is to keep the algebraic structure minimal (a set equipped with a not-necessarily-total binary operation), while still being expressive enough to cover common cases such as term concatenation, multiset union, function composition, event interaction, and process coupling. In particular, the partiality of the operator is used to encode compatibility constraints: some pairs of objects simply do not combine (or are not allowed to combine) under the rules of a given model.

Definition 112 (Combination system)**.** *A* combination system *is a pair* $(S, *)$ *where* S *is a set and* $* : S \times S \rightharpoonup S$ *is a partial binary operation.*

The use of a partial operation $*\colon S \times S \rightharpoonup S$ means that there may exist $x, y \in S$ for which $x * y$ is undefined. This can be read either operationally (the system refuses to execute that composition) or semantically (there is no meaningful composite object in the chosen ontology). No associativity, commutativity, identity, or inverses are assumed unless explicitly added later; this allows the same template to describe both highly structured algebraic settings and ad hoc gluing rules that arise in modeling and representation.

Definition 113 (Interpreted combination system)**.** *An* interpreted combination system *is a triple* $(S, *, \mathrm{Dec})$ *where* $(S, *)$ *is a combination system,* $\mathcal{L}$ *is a syntactic language of expressions, and* $\mathrm{Dec} : \mathcal{L} \to S$ *is an interpreter such that whenever* $\ell_1 \cdot \ell_2$ *denotes a syntactic composition corresponding to* $*$*, one has*

$$\mathrm{Dec}(\ell_1 \cdot \ell_2) = \mathrm{Dec}(\ell_1) * \mathrm{Dec}(\ell_2)$$

whenever the right-hand side is defined.

Here $\mathcal{L}$ may be thought of as a grammar-generated set of expressions, while $\cdot$ is the syntactic constructor for putting two expressions together in whatever way is appropriate (concatenation, application, pairing, sequencing, etc.). The condition

$$\mathrm{Dec}(\ell_1 \cdot \ell_2) = \mathrm{Dec}(\ell_1) * \mathrm{Dec}(\ell_2)$$

is a compositionality requirement: decoding a composite expression agrees with composing the decoded parts, whenever that semantic composition exists. Equivalently, Dec behaves like a homomorphism from the syntactic combination (where defined as a constructor) into the semantic combination system (where defined as a partial operation), and undefinedness on the semantic side signals that the syntactic composite has no coherent meaning under the intended interpretation.

Remark 430. *This is the formal core of "a computational model is an interpreted combination system": syntax plus an evaluation map into semantic objects equipped with composition.*

One can also view $(S, *)$ as the model's semantic workspace, and Dec as the mechanism that embeds symbolic expressions into that workspace. The definition deliberately does not constrain $\mathcal{L}$ to be closed under $\cdot$ for all pairs: even at the syntactic level there may be formation rules, and at the semantic level there may be compatibility conditions. When $\mathcal{L}$ *is* closed under $\cdot$ but $*$ is partial, then some well-formed syntactic composites may decode to undefined semantic composites, which captures the common situation where an expression is grammatical but ill-typed or semantically incoherent.

Next, a combination-preserving representational mapping allows a system to talk about a dynamical domain using a symbolic combination system. The central idea is that the same combinational structure appears on two sides: in the world-domain of processes/events and in the representational-domain of symbols. A map that preserves combination is the minimal requirement for symbols to track how interactions in the domain assemble into larger composite processes.

Definition 114 (Combination-preserving representational mapping)**.** *Let* $(D, **)$ *be a domain of processes or events with a (possibly partial) interaction/combination operator* $**$*. Let* $(S, *)$ *be a combination system of symbols. A map* $f : D \to S$ *is* combination-preserving *if*

$$f(a * *b) = f(a) * f(b)$$

whenever $a * *b$ *and* $f(a) * f(b)$ *are defined.*

This condition can be read as a consistency constraint between what happens in the dynamical domain and how we name it in the symbol system. Because both $**$ and $*$ may be partial, the definition only enforces equality on the intersection of their defined regions: if an interaction $a ** b$ exists and the corresponding symbolic combination $f(a) * f(b)$ is also meaningful, then the symbol for the interaction must agree with the combination of the symbols. In modeling terms, this is precisely the requirement that representation respect the compositional granularity of the domain. When f is not injective, it intentionally collapses distinct domain elements into the same symbol; the later validity condition for causal attribution is designed to ensure that causal claims are invariant under such representational collapse.

Finally, Hyperseed's "combinational dynamical causal model" is a graded causal attribution for triples. The triple (a, b, c) is intended to capture a two-input interaction resulting in (or contributing to) an outcome. The range $[0, 1]$ supports soft or probabilistic attributions, and it is compatible with both deterministic and statistical readings depending on how Γ is instantiated.

Definition 115 (Combinational dynamical causal model)**.** *Let $(D, **)$ be a dynamical process domain. A* combinational dynamical causal model *is a function*

$$\Gamma : D \times D \times D \to [0, 1]$$

where $\Gamma(a, b, c)$ is interpreted as the degree to which "interaction of a and b caused c."

Given a combination-preserving representational mapping $f : D \to S$ into a symbol system $(S, *)$, we say Γ is valid relative to f *if*

$$f(a) = f(a'),\ f(b) = f(b'),\ f(c) = f(c') \quad \Longrightarrow \quad \Gamma(a, b, c) = \Gamma(a', b', c')$$

for all $a, b, c, a', b', c' \in D$.

The valid relative to f condition can be understood as invariance under the equivalence relation induced by the representation. If we define $x \sim_f y$ to mean $f(x) = f(y)$, then validity requires Γ to be constant on $\sim_f$-equivalence classes in each argument simultaneously. Equivalently, Γ factors through the quotient induced by f: there exists a well-defined function $\widetilde{\Gamma}$ on the level of symbols/classes such that $\Gamma(a, b, c)$ depends only on $(f(a), f(b), f(c))$. This makes explicit that the causal scores are not absolute facts about D alone, but are constrained to be compatible with the descriptive alphabet provided by f.

Remark 431. *Validity means the causal model "respects the symbols": it does not distinguish processes that the representation identifies. This is the causal analogue of observer-relativity: causality is computed relative to a representational alphabet (and thus relative to an ontology).*

In particular, if f merges multiple micro-level processes into a single macro-symbol (e.g., many distinct trajectories mapped to one coarse label), then validity forces Γ to assign the same causal degree to any two triples that are indistinguishable at that macro level. This ensures that causal attributions are stable under re-description: once an ontology has decided what counts as the same process, the causal model cannot re-introduce distinctions that the ontology has explicitly discarded. Conversely, if a proposed Γ violates validity, then either the representation f is too coarse for the causal distinctions being made, or the causal model is implicitly using additional latent features not present in $(S, *)$.

9.10 Specific entities as temporally coherent patterns

Hyperseed treats "specific entities" (ordinary objects, persons, institutions) as complex derived constructions rather than primitives. The key observation is that specific-entity identity behaves like a pattern with a temporal algebra. In particular, what is being modeled is not an additional metaphysical glue beyond the entities already available in $\mathcal{E}_C$, but a rule-governed way of *tracking* a coherent thread through time-slices: the stream provides the candidate trajectory, while the coherence rule $\mathcal{R}$ fixes which kinds of variation count as "the same" within the intended resolution of the context C. This makes explicit that persistence is always relative to (i) the granularity of proto-time, (ii) the measurement/description vocabulary encoded by the properties used in assoc_C, and (iii) the tolerance encoded by $\mathcal{R}$.

Definition 116 (Occurrence stream)**.** *Fix a context C with proto-time ordering (Section 7). An* occurrence stream *is a family $\{x_t\}_{t\in T}$ of entities $x_t \in \mathcal{E}_C$ indexed by times (or time-slices) t in a partially ordered set $(T, \leq)$.*

The choice of $(T, \leq)$ as a partially ordered set, rather than assuming a total order, is intentional: it allows proto-time to represent branching, concurrency, or causal precedence structures (e.g. distributed records, parallel sub-processes, or partially ordered observations), without forcing a single linear history. In this setting, an occurrence stream may be *partial*: for some $t \in T$ there need not be an x_t at all, reflecting missing observations, gaps in memory, or times at which the entity is simply not represented in $\mathcal{E}_C$. The phrase "if … both occur" in the coherence conditions is meant to accommodate such partiality.

Definition 117 (Specific entity (pattern-theoretic criterion))**.** *A* specific entity *in context C is specified by an occurrence stream $\{x_t\}_{t\in T}$ together with a coherence rule $\mathcal{R}$ such that:*

1. *(Same-time identity) If $t = t'$ and both x_t and $x_{t'}$ occur, then they are identified: $x_t = x_{t'}$.*

2. *(Near-time similarity) If t and t' are close in proto-time, then the association $\mathrm{assoc}_C(x_t, x_{t'})$ is high (Definition 101), or equivalently the property-sets overlap strongly.*

The strength of "being the same specific entity" is graded by how well the stream satisfies these rules.

Condition (1) should be read as a determinacy constraint internal to the representation: within a fixed time-slice, the stream does not bifurcate into multiple candidates that are *simultaneously* treated as distinct occurrences of the same tracked entity. This is a minimal consistency requirement for a tracking convention, and it separates questions of "co-reference at a time" from questions of persistence across time. Condition (2) provides the persistence signal: if proto-time steps are small, then large discontinuities in properties are penalized (low assoc_C), while small variations are tolerated. The coherence rule $\mathcal{R}$ can be taken to parameterize what counts as "close" in proto-time (e.g. adjacent steps, bounded distance in the order, or neighborhood relations induced by the context) and what counts as "high" association (e.g. a threshold, a soft score, or a weighted overlap depending on which properties are privileged in C). Thus, even when the same underlying object is being discussed informally, different contexts C (legal, biological, physical, social) can yield different specific-entity criteria because they select different property vocabularies and tolerances.

The graded reading in the final sentence is crucial: rather than producing a binary verdict, the framework supports *degrees of identity* driven by the quality of temporal coherence. For example, a heavily repaired artifact (a "Ship of Theseus" trajectory) can yield a stream where local similarity remains high even though long-range similarity between distant times is low; the framework then naturally supports the common phenomenon that identity feels secure locally but disputable globally. Likewise, institutional entities (companies, governments) are often stable under large personnel changes, so $\mathcal{R}$ in such contexts would weight organizational roles, charters, or registries more than physical continuity; by contrast, for biological organisms $\mathcal{R}$ would privilege different properties (e.g. bodily continuity, metabolic functioning).

Remark 432. *This captures the intended Hyperseed idea: "specific entity" is a pattern (a temporally stable bundle of properties) with conventional default rules, but those rules can be violated in exotic scenarios (yielding graded/ambiguous identity rather than forcing contradiction).*

A further consequence is that identity questions can be reframed as optimization or inference problems: given a set of candidate occurrences in $\mathcal{E}_C$ at various times, one seeks streams that maximize coherence under $\mathcal{R}$ (high near-time association, minimal violations of same-time determinacy), possibly subject to additional constraints coming from the context. This aligns the informal practice of "tracking an object/person" with a precise, context-sensitive criterion: the best-supported stream is the one that forms the most temporally coherent pattern. In cases with competing high-coherence streams (e.g. fission, duplication, imperfect records), the model predicts *plural* partially supported identifications rather than a forced all-or-nothing answer.

9.11 Worked micro-example (optional but useful)

Example 9 (A minimal pattern and emergent pattern)**.** *Let C be a context with a combination operator $*$ and baseline costs s_C^{base}. Suppose $x = y * z$ and $s_C^{\text{base}}(x) = 10$, $s_C(y) = 3$, $s_C(z) = 4$, and glue overhead $t = 1$. Then $h_C(P) = 3 + 4 + 1 = 8$ so*

$$I_C(P; x) = \max\left\{0, \frac{10 - 8}{10}\right\} = 0.2.$$

Thus (y, z, t) is a pattern in x of intensity 0.2. Here $h_C(P)$ is the hypothesized cost of "explaining" or "reconstructing" x via the structured description provided by the pattern P (namely, reuse y and z and pay the additional glue overhead t to combine them in the appropriate way). The baseline $s_C^{\text{base}}(x)$ is the cost of representing x without exploiting any special structure, so the numerator $10-8$ measures the absolute savings attributable to recognizing P inside x. Dividing by 10 normalizes this savings, making the resulting intensity scale-free relative to the baseline size of x in context C. The outer $\max\{0, \cdot\}$ enforces that an alleged pattern cannot have negative intensity: if the hypothesized description is no cheaper than baseline (i.e., if $h_C(P) \geq s_C^{\text{base}}(x)$), then the correct conclusion is simply that P provides no compression advantage in x (intensity 0) rather than treating it as "anti-pattern" evidence.

Now suppose also that $A *_C B$ exists and a pattern P has intensities $I_C(P; A) = 0.1$, $I_C(P; B) = 0.1$, but $I_C(P; A *_C B) = 0.5$. If $s_C^{\text{base}}(A) = s_C^{\text{base}}(B)$ then $w_C(A) = w_C(B) = 1/2$ and

$$\mathrm{Sur}_C(P; A, B) = 0.5 - (0.5 \cdot 0.1 + 0.5 \cdot 0.1) = 0.4,$$

so P is emergent in $A *_C B$ with emergent intensity 0.4. The point of this second computation is that emergence is assessed relative to what one would "expect" from the parts when combined, where that expectation is a weighted average of part-wise intensities. In this symmetric case, equal baseline sizes lead to equal weights, so the predicted intensity absent interaction effects is just the simple average of 0.1 and 0.1, namely 0.1. The surplus $\mathrm{Sur}_C(P; A, B) = 0.4$ quantifies the genuinely interaction-dependent gain: the pattern becomes much more detectable (or much more cost-saving) in the composite than one would anticipate from its weak presence in each component *alone. Equivalently, this example isolates a situation where the combined object $A *_C B$ supports a strong structural shortcut described by P that is not available, or not salient enough to be valuable, when considering A and B in isolation.*

Remark 433. *This example is numerically trivial, but it is exactly the formal shape of the Hyperseed definitions. In later sections, the toy resonance model (Section 5) and the habit dynamics (Section 12) will turn these static notions into time-evolving structures. One should read the numbers here as placeholders for whatever cost model is supplied by the context C: for instance, costs may encode description length, energy, inference effort, or any other resource that the agent tracks. Likewise, the "glue overhead" t is the minimal place where contextual idiosyncrasy can enter: even if y and z are individually cheap, composing them may require additional bookkeeping, alignment, interface matching, or constraints to be satisfied, all of which are abstracted into t. The emergent calculation is also intentionally schematic: it highlights that emergence is not merely "having high intensity," but rather "having higher intensity than predicted from the parts under the chosen weighting scheme." Consequently, changing s_C^{base} (and hence the weights w_C) can change the quantitative emergent intensity, reflecting the fact that what counts as "surprising" depends on how the context measures and compares the relative contribution of the components.*

10 Information, uncertainty, and ineffability

10.1 Motivation: information as observer-indexed bookkeeping

Hyperseed treats *distinction* as primitive and observer-relative (Hyperseed-Concept 98). As soon as we take this stance seriously, "information" and "uncertainty" can no longer be treated as absolute scalars attached to a world-state; they must be indexed by a *context* (or observer) (Hyperseed-Concept **??**, Hyperseed-Concept 195; see also [1]). This section provides a small formal layer that lets us: (i) speak about probabilistic belief (Hyperseed-Concept 139); (ii) relate classical Shannon information to distinction-based measures (logical entropy and graphtropy) (Hyperseed-Concept 105, Hyperseed-Concept **??**; [17]); (iii) make "ineffability" precise as a representational limitation (Hyperseed-Concept **??**); and (iv) clarify what "potential infinity" and "infinitesimals" can mean in a resource-sensitive ontology (Hyperseed-Concept **??**, Hyperseed-Concept **??**).

Throughout, let X denote a (typically finite) set of "world items" (entities, events, situations) and let O denote an observer/context. The basic data carried by O are: (a) a family of distinctions it can make about X; and (b) a belief state over those distinctions.

In this framing, X should be read as a domain of discourse relative to a modeling task: it might be a set of perceptual episodes, candidate hypotheses, experimental outcomes, or coarse-grained situations that are treated "as if" they were atomic for current purposes. The parenthetical "typically finite" is not merely a convenience: in Hyperseed the finiteness (or boundedness) of X often stands in for resource limits on attention, memory, time, and representational capacity, while still allowing limits to be taken when discussing "potential" infinities.

Item (a) can be made concrete in multiple equivalent ways, depending on which mathematical lens is most appropriate. For example, a distinction-making capacity can be encoded as a partition of X into equivalence classes (items that O cannot tell apart), or as a set of binary questions of the form "does x have property P?", or (more generally) as a σ-algebra of subsets of X specifying which events are *measurable* for O. Each representation emphasizes a different intuition: partitions emphasize *indistinguishability*, question-sets emphasize *interrogability*, and σ-algebras emphasize closure properties needed for consistent probability assignments. In particular, once distinctions are taken as primary, it becomes natural to treat two items $x, y \in X$ as observationally equivalent for O when no available distinction separates them; the "world-state" relative to O is then more faithfully represented by the cell of a partition (or, dually, by the truth-values of all questions in the algebra) than by a bare element of X.

Item (b) is then the observer's weighting over the distinguishable alternatives. Operationally, one may think of O as assigning probabilities to the events it can express (e.g., to the cells of a partition, or to measurable subsets), which is why a probability measure naturally appears alongside the distinction structure. This is intentionally broad: it includes Bayesian degrees of belief, empirical frequencies within a data stream, and hybrid or imprecise credal assignments, so long as they respect the representational constraints imposed by (a). The key point is that probabilities are not assigned to an unstructured "reality" but to *recognizable* possibilities—hence even a numerically identical probability vector may correspond to different informational situations if the underlying distinctions differ.

Remark 434. *The philosophical point is modest but consequential: an "information measure" is never purely a property of* the world*; it is a property of a world* as carved *by some scheme of recognition, and as weighted by some state of belief. Russell would say that the definitions below are not metaphysical proclamations about Being, but carefully chosen bookkeeping devices; Peirce would add that this bookkeeping is itself a habit of interpretation, hence belongs as much to the interpreter as to the interpreted. Formally, the bookkeeping will appear as (i) a σ-algebra encoding what can be asked, and (ii) a probability measure encoding how strongly the observer leans toward different answers.*

A useful way to read the remark is as a constraint on what it could even *mean* to compare information across observers. If two contexts O_1, O_2 carve X differently, then an expression like "O_1 has more information than O_2" is not automatically well-defined until one specifies a translation between their distinction structures (e.g., refinement/coarsening maps between partitions, or homomorphisms between σ-algebras). When such a translation exists, many familiar monotonicity intuitions reappear (finer distinctions can support lower uncertainty, higher potential information gain), but these become theorems about relationships between algebras/partitions rather than metaphysical claims about a single observer-independent quantity.

This observer-indexed bookkeeping also motivates why distinction-based measures (logical entropy, graphtropy) are not merely alternative "formulas" for the same idea. Shannon entropy presumes a fixed alphabet of outcomes and measures expected code length relative to that alphabet; logical entropy directly counts (in expectation) how often two draws fall into *different* distinguishability classes, making the role of distinctions explicit. Graphtropy generalizes the same intuition from partitions to graphs of indistinguishability or comparability relations on X, which is often closer to how real cognitive and computational systems represent "can/cannot tell apart" in a graded or relational manner. Thus, the formal layer introduced here is meant to keep the distinction structure visible, so that later statements about uncertainty reduction, learning, and ineffability remain tethered to what the observer can actually represent.

Finally, the promised link to ineffability can be previewed already at this introductory stage: something is "ineffable for O" not because it is mystical, but because it cannot be stably encoded within O's distinction system and the associated algebra of questions. On this view, ineffability is an internal, structural limitation: there may be facts about X that exist in some richer meta-description, yet are not measurable (or not finitely describable) in the σ-algebra available to O, or cannot be assigned coherent probabilities without extending O's representational resources. This is also where resource-sensitive readings of infinity and infinitesimals enter: what looks like an "infinitary" refinement may be treated as a limit of finite refinements that an observer can only ever approximate, and "infinitesimal" differences may correspond to distinctions that are in principle definable but in practice unresolvable at the observer's current granularity.

10.2 Observer-indexed probability spaces

Definition 118 (Observable events)**.** *Fix an observer/context O. An* event observable to O *is a predicate on X whose truth value is determined at the resolution of O. Formally, we model the collection of O-observable events as a sigma-algebra $\Sigma_O \subseteq 2^X$. A* probability model for O *is a probability measure*

$$P_O : \Sigma_O \to [0, 1].$$

The triple (X, Σ_O, P_O) is the observer-indexed probability space.

Remark 435. ***Notation/conventions.*** *Here 2^X denotes the power set of X (all subsets of X); $\Sigma_O \subseteq 2^X$ is a σ-algebra, meaning it is closed under complements and countable unions (in the finite case, this reduces to closure under complements and finite unions). The probability measure P_O assigns to each observable event $A \in \Sigma_O$ a number in $[0, 1]$ with $P_O(X) = 1$ and countable additivity on disjoint unions. It is also standard that $P_O(\varnothing) = 0$ and that (by additivity) $P_O(A^c) = 1 - P_O(A)$ whenever $A \in \Sigma_O$; these identities emphasize that P_O is defined only on the observer's event-language Σ_O rather than on arbitrary subsets of X. When X is large or continuous, the restriction to Σ_O is not only philosophical but technically necessary: many subsets of X are not measurably well-behaved, and σ-algebras are precisely the minimal structure needed to make probability assignments coherent.*

Intuition. *Σ_O is the mathematical shadow of the observer's* question-set*: it lists precisely which yes/no questions about X are meaningful at O's resolution. If $A \notin \Sigma_O$, it is not that A is false; it is that A is* not a well-formed observable event *for O. Once the observer has a question-set, P_O encodes the observer's uncertainty over answers. In this sense, an "observer" O can be read broadly: a physical sensor suite, an agent with finite memory and categories, a modeling choice about which variables are tracked, or even a scientific community that has stabilized a repertoire of admissible experimental distinctions.*

Example. *Let X be the set of microstates of a gas. A coarse observer might only measure temperature and pressure, so Σ_O is generated by macroscopic equivalence classes (microstates with the same macro-observables). A finer observer might distinguish many more subsets, having a much larger Σ_O. This is the formal version of saying: different observers inhabit different informational worlds even when they speak about the same underlying X. Equivalently, one can think of the coarse observer as having access only to a map $f_O : X \to Y_O$ of microstates into macrostates, so that Σ_O is the pullback of the power set (or Borel σ-algebra) on Y_O; in that view, P_O is the pushforward of whatever uncertainty the observer implicitly has over X into the observer's accessible variables.*

Remark 436 (Coarse-graining as "loss of distinction")**.** *If Σ_O is generated by a partition π_O of X, then the observer does not represent individual elements of X but only the block in π_O that contains them. The move from 2^X to Σ_O is a formal encoding of "failure to make distinctions." In particular, any two microstates $x, x' \in X$ that lie in the same block of π_O are observationally interchangeable for O: no event in Σ_O can separate them. This makes explicit that "coarse-graining" is not merely a lossy summary of a distribution, but a restriction on the very predicates that can be formed and evaluated.*

Remark 437. *A partition-induced σ-algebra is the simplest case, and it corresponds to crisp in-distinction: O treats two items as the same iff they fall in the same block. Later we generalize* *this to graded and even asymmetric indistinction (via indistinction graphs), which is closer to how cognition behaves in practice (cf. the pattern-based picture in [5]). In the Hyperseed vocabulary this is an instance of treating* distinctions *as primary structure (Hyperseed-Concept 98). It is useful to note that "finer" and "coarser" observers can be compared directly at the level of σ-algebras: if $\Sigma_{O_1} \subseteq \Sigma_{O_2}$, then O_2 can express every question O_1 can (and possibly more), so O_2 is informationally at least as fine as O_1. This inclusion relation is a precise, order-theoretic version of "one observer refines another," and it will later support principled ways to relate models across levels (e.g. restricting a finer model to a coarser question-set, or lifting a coarse belief to a family of compatible fine beliefs).*

Definition 119 (Paraconsistent evidence annotations)**.** *To connect to the paraconsistent core (Section 3), we may annotate events with a* p-bit evidence value

$$E_O(A) = (E_O^+(A), E_O^-(A)) \in [0,1]^2 \qquad (A \in \Sigma_O),$$

where $E_O^+(A)$ is degree of positive evidence and $E_O^-(A)$ is degree of negative evidence. No consistency constraint is imposed: it is allowed that $E_O^+(A) + E_O^-(A) > 1$. In particular, E_O is not assumed to be a probability measure, nor is it assumed that $E_O^-(A)$ equals $E_O^+(A^c)$; the point is to retain two channels of support that need not be reducible to a single additive calculus.

Remark 438. ***Intuition.*** *Ordinary probability tries to compress all epistemic nuance into one scalar $P_O(A)$. Paraconsistent annotation instead records two partially independent intensities: reasons* for *A and reasons* against *A. Allowing $E_O^+(A)+E_O^-(A) > 1$ is not a bug but a formal admission that real epistemic agents can be pulled strongly in incompatible directions at once (e.g. contradictory testimonies, model mismatch, sensor conflict). This connects directly to paraconsistent logics such as Constructible Duality Logic [23] and to resonance-style dynamics studied in [24]. One can read $E_O^{\pm}$ operationally as aggregating heterogeneous sources that are not forced to "cancel" into a net score; the annotation thereby separates* amount of support *from* degree of resolution*, and it allows an agent to represent "I have strong reasons on both sides" without collapsing that state into a near-*1/2 *probability. At a technical level, this is a way of preserving informational structure that would otherwise be lost under a single-number summary.*

Examples. *If $E_O(A) = (0.9, 0.1)$ the observer has strong support for A with little counterpressure. If $E_O(A) = (0.9, 0.9)$ the observer has a deep conflict: A is simultaneously strongly supported and strongly opposed, reflecting a state closer to dissonance than ignorance. If $E_O(A) = (0.1, 0.1)$, by contrast, the agent has little evidence either way; this can be interpreted as genuine informational poverty rather than a balanced clash of considerations.*

Usefulness. *This annotation is not meant to replace probability, but to complement it when the observer's representational regime allows mutually incompatible local inferences. Later, when habit dynamics and resonance/dissonance are modeled as update rules on graded supports (Section 12), the possibility of conflict becomes structurally central. In particular, keeping P_O alongside E_O allows one to distinguish (i) the agent's betting-odds or action-guiding summary from (ii) the internal tension profile that may drive future reorganization of distinctions, attention, or model class.*

Remark 439 (Two kinds of "uncertainty")**.** *The pair (P_O, E_O) supports two distinct phenomena:*

- Ignorance*: probabilities near $1/2$ and/or low total evidence.*
- Conflict*: simultaneous strong positive and strong negative evidence.*

In Hyperseed terms, ignorance is "not enough distinction-making power or data," whereas conflict is "incompatible distinction-making pressures that are both locally supported." Later (Section 12) we will connect conflict to resonance/dissonance dynamics. It is often useful to treat these as empirically and behaviorally different regimes: ignorance tends to invite exploration or refinement of sensing (seek more data or sharper distinctions), whereas conflict tends to invite adjudication (seek error sources, reconcile models, or split contexts so that incompatible pressures are not forced into the same representational frame).

Remark 440. *When comparing observers O_1 and O_2, two further distinctions are helpful. First, if $\Sigma_{O_1} \subseteq \Sigma_{O_2}$ and P_{O_2} is given, then P_{O_2} restricts to a probability measure on Σ_{O_1}; this corresponds to "forgetting" distinctions while keeping coherent odds on the coarser questions. Second, even when $\Sigma_{O_1} = \Sigma_{O_2}$, different P_O (or different E_O) can represent differences in priors, training histories, or sensor reliabilities: the observer-indexing is therefore not only about* what can be asked, *but also about* how uncertainty is distributed *over what can be asked. These two axes (expressivity via Σ_O and credence via P_O and E_O) will matter later when we separate changes in "world-model resolution" from changes in "belief within a fixed resolution."*

10.3 Shannon information, coding, and divergence

Assume X is finite and $\Sigma_O = 2^X$ (or that we have reduced to the finite set of blocks of a partition). Write $p_O(x) := P_O(\{x\})$.

Remark 441. ***Notation/conventions.*** *Because X is finite, a probability measure P_O is equivalent to its point-mass function $p_O : X \to [0, 1]$ with $\sum_{x \in X} p_O(x) = 1$. The expression $p_O(x) := P_O(\{x\})$ is this identification. When we write $H(X)$ later, we mean the Shannon entropy of the distribution of the random variable X, i.e. the entropy of $p(x)$ on its finite support.*

Definition 120 (Entropy and surprise)**.** *The* surprisal *of an outcome x is*

$$s_O(x) := -\log p_O(x),$$

where log *denotes the natural logarithm. The* Shannon entropy *of P_O (in nats) is*

$$H(P_O) := \sum_{x \in X} p_O(x)\, s_O(x) \;=\; -\sum_{x \in X} p_O(x) \log p_O(x).$$

Define also the entropy in bits as

$$H_{\text{bits}}(P_O) := \frac{H(P_O)}{\log 2}.$$

Remark 442. ***Intuition.*** *Surprisal $s_O(x)$ is high when x was deemed unlikely by O; entropy $H(P_O)$ is the average surprisal under the observer's own expectations. One may read this as the expected amount of* conceptual work *needed to accommodate an observation, provided that "work" is measured in logarithmic units.*

Examples. *If p_O is uniform on $|X| = N$, then $H(P_O) = \log N$ (maximal uncertainty). If p_O concentrates on one point, entropy is 0 (no uncertainty). A biased coin with $p(heads) = 0.9$ has lower entropy than a fair coin, because the observer is rarely surprised.*

Usefulness in this document. *Entropy is a bridge from belief to description length: if an observer believes outcomes are distributed according to p, then optimal codes have expected length about $H_{\text{bits}}(p)$. This will later be tied to effort-based simplicity (Section 8) and pattern intensity (Section 9), and it parallels the algorithmic viewpoint on description length emphasized in [16].*

Definition 121 (Cross-entropy and KL divergence)**.** *Given two distributions p and q on X with $q(x) > 0$ whenever $p(x) > 0$, the* cross-entropy *and* Kullback–Leibler divergence *are*

$$H(p,q) := -\sum_{x\in X} p(x)\log q(x), \qquad D_{\mathrm{KL}}(p\|q) := \sum_{x\in X} p(x)\log\frac{p(x)}{q(x)}.$$

Remark 443. ***Notation/conventions.*** *The condition $q(x) > 0$ whenever $p(x) > 0$ ensures $\log\frac{p(x)}{q(x)}$ is well-defined wherever it is weighted by $p(x)$. $D_{\mathrm{KL}}(p\|q)$ is not symmetric in (p,q) and is not a metric; it is an information-theoretic divergence.*

Intuition. *Cross-entropy $H(p,q)$ is the expected code length (in nats) if the world is distributed as p but you encode using a code optimized for q. KL divergence is the penalty for that mismatch: $D_{\mathrm{KL}}(p\|q) = H(p,q) - H(p)$.*

Usefulness. *Later, when we define fidelity and effability, we effectively compare a target item x with a reconstructed distribution $q = \mathrm{dec}_O(d)$. KL divergence is a canonical way to measure such mismatches; we do not enforce it as the unique notion of error here, but the reader should recognize it as a standard candidate.*

Proposition 11 (Entropy as optimal expected code length)**.** *Let p be a distribution on a finite alphabet X. Among all prefix-free binary codes with codeword lengths $\ell(x) \in \mathbb{N}$, the optimal expected length*

$$L^*(p) := \inf_{\ell} \sum_{x\in X} p(x)\,\ell(x)$$

obeys

$$H_{\mathrm{bits}}(p) \le L^*(p) < H_{\mathrm{bits}}(p) + 1.$$

Remark 444. ***What the proposition says (plainly).*** *If an observer believes outcomes follow p, then there exists an essentially optimal way to encode outcomes in binary so that the* average *number of bits used is about the entropy of p. One cannot beat $H_{\mathrm{bits}}(p)$ on average (lower bound), but one can get within 1 bit of it with an explicit construction (upper bound).*

Why it matters. *This is the classical theorem that makes entropy operational: it is not merely a numerical ornament on a probability distribution; it is the best achievable average compression rate under an ideal coding scheme. In the Hyperseed framing, this is one key bridge from* belief *to* effort*: shorter codes mean less representational work, and longer codes mean more (cf. the effort/simplicity lens in Section 8 and the coding/description lens in [16]).*

Connection to later sections. *Patterns (Section 9) are defined by compressive re-description; the inequality above is one of the classical reasons compression and probability models are inseparable. When we later speak of observer-relative "simplicity" or "weakness" as an optimization principle, this proposition is a basic exemplar of how such minima arise.*

Proof. (Lower bound.) For any prefix-free code, Kraft's inequality gives $\sum_x 2^{-\ell(x)} \le 1$. Let $K := \sum_x 2^{-\ell(x)}$ and define $q(x) := 2^{-\ell(x)}/K$. Then q is a probability distribution, and

$$0 \le D_{\mathrm{KL}}(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)}.$$

Rearranging yields

$$-\sum_x p(x)\log p(x) \le -\sum_x p(x)\log q(x) = \log K + (\log 2)\sum_x p(x)\ell(x).$$

Since $K \leq 1$ we have $\log K \leq 0$, hence $H(p) \leq (\log 2) \sum_x p(x)\ell(x)$. Dividing by $\log 2$ gives $H_{\text{bits}}(p) \leq \sum_x p(x)\ell(x)$. Taking an infimum over codes proves the lower bound.

(Upper bound.) Define candidate lengths by

$$\ell(x) := \lceil -\log_2 p(x) \rceil \in \mathbb{N}.$$

Then $2^{-\ell(x)} \leq 2^{\log_2 p(x)} = p(x)$, hence $\sum_x 2^{-\ell(x)} \leq \sum_x p(x) = 1$. By Kraft's inequality, there exists a prefix-free binary code whose codeword lengths are exactly these $\ell(x)$ (for instance, via a Shannon–Fano construction, or by building a binary tree whose leaf depths match ℓ).

For this code,

$$\sum_x p(x)\ell(x) \;\leq\; \sum_x p(x)\bigl(-\log_2 p(x) + 1\bigr) \;=\; H_{\text{bits}}(p) + 1.$$

Taking an infimum over all codes yields $L^*(p) \leq H_{\text{bits}}(p) + 1$, and since the bound is strict at the level of $L^*(p)$ (because an integer-length constraint cannot in general achieve the real-valued optimum exactly), we obtain $L^*(p) < H_{\text{bits}}(p) + 1$ as stated. ☐

Remark 445. ***On the*** $+1$ ***slack.*** *The additive* 1 *bit arises from the integrality of* $\ell(x)$ *and the ceiling operation. If one allows idealized non-integer code lengths (or, operationally, block coding over long sequences and arithmetic coding), the achievable average length can be made arbitrarily close to* $H_{\text{bits}}(p)$*. Equivalently: the entropy is the fundamental rate, while the* $+1$ *term is a single-symbol overhead due to discretizing lengths.*

KL divergence as a coding penalty. *The proof of the lower bound already contains a more general statement: if a prefix-free code induces an implicit distribution* $q(x) \propto 2^{-\ell(x)}$*, then the expected length satisfies*

$$\sum_x p(x)\ell(x) \;=\; H_{\text{bits}}(p) + \frac{1}{\log 2} D_{\text{KL}}(p\|q) - \frac{1}{\log 2} \log K,$$

with $K = \sum_x 2^{-\ell(x)} \leq 1$*. Thus, mismatch between the "true"* p *and the code-implied* q *shows up as a nonnegative penalty term (the KL divergence), mirroring the earlier identity* $H(p,q) = H(p) + D_{\text{KL}}(p\|q)$ *in coding form.*

Observer-relativity. *In the present document, it is often* p_O *(the observer's model) rather than an objective distribution that matters. The same theorem applies with* p *replaced by* p_O*: if the observer's beliefs are wrong, the code that is optimal under* p_O *need not be optimal under the world's distribution, but it is still optimal relative to the observer's own representational commitments.*

Remark 446 (Why this matters for Hyperseed)**.** *In Hyperseed, "simplicity" is grounded in representational effort. The proposition formalizes one standard bridge: a probability model induces an optimal expected description length. Later sections connect description length to weakness and to pattern intensity.*

Additional detail on the "one checks" step. The feasibility claim for $\ell(x) = \lceil -\log_2 p(x) \rceil$ is the concrete place where discretization (integer lengths) meets the continuous ideal $-\log_2 p(x)$. Since $\ell(x) \geq -\log_2 p(x)$, we have

$$2^{-\ell(x)} \leq 2^{-(-\log_2 p(x))} = p(x),$$

and therefore $\sum_x 2^{-\ell(x)} \leq \sum_x p(x) = 1$. This is exactly Kraft's inequality for a binary prefix code, so the length assignment is not merely formal: it is realizable by an actual prefix-free set of codewords (equivalently, by a pruned binary tree whose leaves occur at depths $\ell(x)$).

Where the strict inequality comes from. For any real number r, the ceiling satisfies $\lceil r \rceil < r+1$. Applying this pointwise to $r = -\log_2 p(x)$ gives $\ell(x) < -\log_2 p(x) + 1$, hence

$$\sum_x p(x)\ell(x) < \sum_x p(x)\big(-\log_2 p(x) + 1\big) = \sum_x p(x)\big(-\log_2 p(x)\big) + \sum_x p(x) = H_{\text{bits}}(p) + 1,$$

making explicit that the sole source of slack is integer rounding. In particular, when all probabilities are dyadic (i.e. $p(x) = 2^{-k_x}$ for integers k_x), the ideal lengths are already integers and the rounding loss vanishes, so the upper bound can meet $H_{\text{bits}}(p)$ exactly.

More explicit link between coding and D_{KL}. Given any prefix lengths $\ell(x)$ satisfying Kraft, the unnormalized weights $2^{-\ell(x)}$ form a sub-probability mass function; normalizing by $Z := \sum_x 2^{-\ell(x)} \leq 1$ yields

$$q(x) = \frac{2^{-\ell(x)}}{Z}.$$

Then $\ell(x) = -\log_2 q(x) - \log_2 Z$, so taking expectation under p gives

$$\mathbb{E}_p[\ell(X)] = \mathbb{E}_p[-\log_2 q(X)] - \log_2 Z.$$

Because $Z \leq 1$, we have $-\log_2 Z \geq 0$, so $\mathbb{E}_p[\ell(X)] \geq \mathbb{E}_p[-\log_2 q(X)]$. Finally,

$$\mathbb{E}_p[-\log_2 q(X)] - H_{\text{bits}}(p) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)} = D_{\text{KL}}(p\|q) \quad \text{(in bits)},$$

so $D_{\text{KL}}(p\|q) \geq 0$ implies $\mathbb{E}_p[\ell(X)] \geq H_{\text{bits}}(p)$. This makes concrete the interpretation that a code induces a model q, and the penalty for using the "wrong" model is precisely a KL divergence term.

Units and interpretation. The notation $H_{\text{bits}}(p)$ emphasizes that the logarithm base is 2, so entropy is measured in bits and code lengths are measured in binary digits. If one instead uses natural logarithms, the same statements hold with lengths measured in nats and with the conversion factor $\log_2 t = (\log t)/(\log 2)$ accounting for the change of units.

Tightness and algorithmic realizations. The $+1$ in the upper bound is a worst-case guarantee for single-symbol (one-shot) coding with integer lengths. In practice, Huffman coding achieves an expected length within 1 bit of the entropy and is optimal among prefix codes for a fixed alphabet. Moreover, by coding blocks of n symbols from the product distribution $p^{\otimes n}$, the per-symbol overhead can be driven to 0 as $n \to \infty$, reflecting that the discreteness penalty is not fundamental but a finite-length artifact.

Further connection to the Hyperseed framing. In settings where a system maintains an internal predictive distribution q while the world follows p, the expected description length under q decomposes into $H_{\text{bits}}(p)$ plus a mismatch cost $D_{\text{KL}}(p\|q)$. Thus, "simplicity" (short expected code length) can be read simultaneously as (i) compressibility of observations under the true law and (ii) adequacy of the internal model. This dual view is often useful when later notions (e.g. weakness, pattern intensity) depend not only on how much structure is present in p but also on how well an agent-like representation aligns with it.

10.4 Mutual and interaction information

Let (X, Y) be a pair of random variables on finite sets with joint distribution $p(x, y)$. In this finite setting all entropies and divergences below are well-defined (possibly taking value $+\infty$ only if one allows zero-probability events in ways that make the KL divergence diverge), and $I(X;Y)$ can be read as an *average* quantity over outcomes rather than a pointwise guarantee about any particular sample.

Definition 122 (Mutual information)**.** *The* mutual information *between X and Y is*

$$I(X;Y) := H(X) + H(Y) - H(X,Y).$$

Equivalently, $I(X;Y) = D_{\mathrm{KL}}(p(x,y)\|p(x)p(y))$. In particular, $I(X;Y) \geq 0$, with equality if and only if $p(x,y) = p(x)p(y)$ (independence).

Remark 447. ***Intuition.*** *Mutual information measures how much knowing X reduces uncertainty about Y (and symmetrically, knowing Y reduces uncertainty about X). The KL expression says: it is the divergence between the true joint distribution and the distribution you would get if you falsely assumed X and Y were independent. Equivalently, it is the expected log-likelihood ratio*

$$I(X;Y) = \mathbb{E}_{(X,Y)\sim p}\left[\log \frac{p(X,Y)}{p(X)p(Y)}\right],$$

so it quantifies the average extent to which joint observations look "surprising" relative to an independence model. A useful identity that often clarifies the "uncertainty reduction" reading is

$$I(X;Y) = H(Y) - H(Y \mid X) = H(X) - H(X \mid Y),$$

so it is literally the drop in entropy of one variable after conditioning on the other. From this, one sees immediately the bounds $0 \leq I(X;Y) \leq \min\{H(X), H(Y)\}$: no channel can convey more information about Y than the total uncertainty that was present in Y to begin with.

Example. *If $Y = X$ (perfect copying), mutual information equals $H(X)$: observing Y tells you everything about X. If X and Y are independent, then $I(X;Y) = 0$. More generally, if Y is a noisy copy of X through some stochastic channel, then $I(X;Y)$ decreases as noise increases; in this sense it is a quantitative measure of "fidelity" that is independent of any particular decoding rule.*

Usefulness. *In a Hyperseed setting, $I(X;Y)$ is a clean numerical proxy for "how many distinctions about Y are made available by access to X," i.e. how one representational channel supports another. This will reappear when discussing pattern webs and dependency-like relations between representational units (Sections 11 and 12). A further practical reason it is favored is its invariance under bijective reparameterizations of X or Y: relabeling symbols does not change $I(X;Y)$, so it depends on the structure of dependence rather than the names of states.*

Chain rules and conditioning. *For later use it is convenient to recall the chain rule*

$$I(X;Y,Z) = I(X;Y) + I(X;Z \mid Y),$$

which formalizes the idea that information provided by a compound representation (Y,Z) decomposes into the part already provided by Y plus the additional part provided by Z once Y is known. Here $I(X;Z \mid Y)$ is the conditional mutual information*, defined by*

$$I(X;Z \mid Y) = H(X \mid Y) - H(X \mid Y,Z) = H(Z \mid Y) - H(Z \mid X,Y),$$

and it is always nonnegative as well.

Definition 123 (Interaction information)**.** *For three random variables (X,Y,Z), the* interaction information *is*

$$I(X;Y;Z) := I(X;Y) - I(X;Y \mid Z).$$

This quantity may be positive or negative, reflecting synergy or redundancy. It can also be written symmetrically in entropies as

$$I(X;Y;Z) = H(X) + H(Y) + H(Z) - H(X,Y) - H(X,Z) - H(Y,Z) + H(X,Y,Z),$$

so it is invariant under permutation of X, Y, Z despite being defined using $I(X;Y \mid Z)$.

Remark 448. ***Intuition.*** *Interaction information asks whether the relationship between X and Y becomes* more *or* less *informative once Z is known. If Z explains away the dependence between X and Y, the quantity tends to be negative (redundancy). If Z unlocks a dependence that is not visible marginally, it may be positive (synergy). One way to read the sign is:*

$$I(X;Y;Z) > 0 \iff I(X;Y \mid Z) < I(X;Y),$$

so conditioning on Z reduces the apparent X–Y association (suggesting that what X and Y "share" is, in part, already present in Z), whereas

$$I(X;Y;Z) < 0 \iff I(X;Y \mid Z) > I(X;Y),$$

so learning Z reveals dependence between X and Y that was not apparent marginally. The possibility of negative values is not a paradox: it does not *contradict nonnegativity of mutual information, because interaction information is a* difference *of two nonnegative quantities.*

Example. *In the classical XOR example, X and Y are independent fair bits and $Z = X \oplus Y$. Then $I(X;Y) = 0$ but conditioning on Z makes X and Y perfectly dependent, leading to a positive interaction information: information is present only in the* triad. *Conversely, if Z is a common cause that copies into both X and Y (e.g. $X = Z$ and $Y = Z$), then $I(X;Y)$ is large but $I(X;Y \mid Z) = 0$, yielding negative interaction information: the apparent X–Y dependence is explained by the shared variable. These two extremes motivate the informal language of "synergy" versus "redundancy," while also hinting that any single scalar can only be a coarse summary of multi-way dependence.*

Usefulness. *Hyperseed frequently emphasizes that cognition is not purely additive: wholes can carry relations not present in parts. Interaction information is one standard quantitative hint of such non-additivity, and it resonates with emergence-style notions introduced later in the pattern section (Section 9). It is also a warning label against naive "Venn diagram" interpretations of information: although the symmetric entropy formula resembles inclusion–exclusion, the sign changes show that multivariate dependence does not behave like ordinary set overlap.*

Remark 449 (Observer-relative "second law" reading)**.** *Entropy-like quantities depend on which variables are being tracked. If O changes its representational partition (coarse-grains or refines), then the induced random variables change and so do H and I. Thus, statements that resemble a "second law" (monotone entropy increase under typical dynamics) should always be read as:* for the chosen coarse variables, under the chosen effective dynamics, typical trajectories move toward larger macro-uncertainty. *This matches Hyperseed's perspectivism: "the same world" can have different entropic arrows at different resolutions. A concrete way to see this is that coarse-graining typically reduces the maximum achievable mutual information between a microscopic state and a macro-description: when distinctions are merged, there is (by design) less that can be learned from the coarse variable about fine structure. Dually, refining a partition can create apparent information flows that were previously invisible, simply because the representational vocabulary has changed.*

Remark 450. *This remark is a formal caution against turning a contextual regularity into a metaphysical absolute. The second law (Hyperseed-Concept 164) is robust in physics, but its informational analogs are always indexed by (i) what we count as a macrostate and (ii) what we treat as the effective dynamics. Hyperseed's insistence on observer-indexing is, in this sense, a reminder that entropy is a measure of* description *as much as a measure of* disorder*. In particular, when one says "entropy increases," what is usually meant is that the distribution over the tracked variables spreads out under the chosen dynamics and constraints; but changing the tracked variables* *changes the sample space, the constraint set, and the induced notion of "spread." The same underlying micro-dynamics can therefore support multiple, seemingly competing, informational narratives depending on which distinctions are treated as salient.*

10.5 Logical entropy and graphtropy from distinction structure

Logical entropy (also called "Gini impurity" in some applied contexts) measures the probability that two independent draws fall into different "distinction classes." In Hyperseed terms, it measures the fraction of distinctions that are actually made.

Definition 124 (Logical entropy of a partition)**.** *Let $\pi = \{B_1, \ldots, B_k\}$ be a partition of X. Given a distribution p on X, define the block masses*

$$p_i := \sum_{x \in B_i} p(x).$$

The logical entropy *of (π, p) is*

$$h(\pi, p) := 1 - \sum_{i=1}^{k} p_i^2.$$

Remark 451. ***Intuition.*** *Where Shannon entropy measures expected code length under optimal compression, logical entropy measures expected* distinguishability *under the partition π. It does not ask: "How many bits to encode?" but rather: "How often do two random observations fall into different blocks?" This makes it an especially direct companion to Hyperseed's primitive of distinction (Hyperseed-Concept 98) and aligns with Ellerman's development of logical entropy [17].*

Example. *If π is the discrete partition into singletons, then $p_i = p(x)$ and $h(\pi, p) = 1 - \sum_x p(x)^2$. If π is the trivial one-block partition, then $h(\pi, p) = 0$ (no distinctions are made).*

Why useful. *Logical entropy is quadratic in p, hence behaves differently from Shannon entropy under fine-graining; in many cognitive settings, quadratic forms correspond naturally to "pairwise comparison" processes (e.g. sampling two items and checking whether they are treated as the same). Graphtropy will generalize this beyond partitions, preserving this pairwise-comparison intuition.*

Remark 452. *The quantity $\sum_i p_i^2$ is the probability that two iid draws fall in the* same *block. Hence $h(\pi, p)$ is the probability that two iid draws fall in* different *blocks, i.e. that the partition makes a distinction between them.*

Definition 125 (Distinction graphs and graphtropy)**.** *Let G be a (possibly directed) graph on vertex set X. We interpret an edge $x \to y$ as "O does* not *distinguish x from y" (an* indistinction*). Let $\iota_G(x, y) \in [0, 1]$ be a weight encoding degree of indistinction, with $\iota_G(x, y) = 1$ meaning "fully indistinct" and $\iota_G(x, y) = 0$ meaning "fully distinct." Given a probability distribution p on X, define the* graphtropy *as*

$$\mathrm{GT}(G, p) := 1 - \sum_{x \in X} \sum_{y \in X} p(x)p(y)\, \iota_G(x, y).$$

Remark 453. ***Notation/conventions.*** *$\iota_G : X \times X \to [0, 1]$ is an* indistinction weight *rather than an adjacency matrix in the usual graph-theoretic sense. The double sum $\sum_x \sum_y p(x)p(y)\, \iota_G(x, y)$ is an expectation over two independent draws $X_1, X_2 \sim p$ of the indistinction degree $\iota_G(X_1, X_2)$.*

Intuition. *Graphtropy asks: if I sample two items according to p, how indistinguishable are they on average? Then it flips the answer by taking $1-$, so that higher graphtropy means "more* *expected distinction". This makes graphtropy a direct numerical shadow of an observer's distinction structure, even when that structure is graded, context-dependent, and not transitive.*

Examples. *If* $\iota_G(x,y) = \mathbf{1}[x = y]$ *(only identical items are treated as indistinct), then* $\mathrm{GT}(G,p) = 1 - \sum_x p(x)^2$. *If* $\iota_G(x,y) = 1$ *for all pairs (everything indistinct), then* $\mathrm{GT}(G,p) = 0$.

Why useful. *Partitions (equivalence relations) are often too rigid for cognitive modeling. A mind can treat* x *as indistinct from* y *in one direction but not the other (asymmetry), and can treat* $x \approx y$ *and* $y \approx z$ *without treating* $x \approx z$ *(non-transitivity). Graphtropy keeps the core idea—distinction measured by pairwise sampling—without demanding equivalence-relation structure (Hyperseed-Concept* **??***).*

Remark 454 (Fuzzy and asymmetric cases)**.** *If* ι_G *is crisp and symmetric and transitive (an equivalence relation), then* G *encodes a partition and graphtropy reduces to logical entropy. If* ι_G *is fuzzy and/or asymmetric, then graphtropy measures "expected distinguishability" without requiring a partition. This extra generality is important for modeling minds, which often treat "x indistinguishable from y" as context-sensitive, graded, and non-transitive.*

Proposition 12 (Reduction to logical entropy)**.** *Let* π *be a partition of* X*, and define* $\iota_\pi(x,y) = 1$ *iff* x *and* y *lie in the same block, and* 0 *otherwise. Let* G_π *be the induced indistinction graph. Then*

$$\mathrm{GT}(G_\pi, p) = h(\pi, p).$$

Remark 455. ***What the proposition says.*** *Graphtropy is not a competing definition but a strict generalization: in the special case where indistinction really is a partition (crisp equivalence classes), graphtropy collapses exactly to logical entropy.*

Why it matters. *This justifies graphtropy as a conservative extension. We gain the ability to represent graded/non-transitive indistinction without losing compatibility with classical partition-based measures such as logical entropy [17].*

Connection to the rest of the document. *Earlier we treated distinctions as policies (Section 8); graphtropy is a canonical way to turn such a policy into a scalar measure of how "finely" the observer is carving* X *under a distribution* p*. This scalar will later be useful when comparing distinction regimes in the definition of effability and in discussions of resolution changes.*

Proof. Two iid draws (X_1, X_2) from p fall in the same block with probability $\sum_i p_i^2$, which equals $\sum_{x,y} p(x)p(y)\,\iota_\pi(x,y)$. Here $\iota_\pi(x,y)$ is the (0–1) indistinction indicator for the partition π, so the right-hand side is simply the probability (under independent sampling) of the event "X_1 and X_2 land in the same block": independence gives $\Pr[X_1 = x, X_2 = y] = p(x)p(y)$, and the indicator $\iota_\pi(x,y)$ selects exactly those pairs (x,y) that are in a common block. Equivalently, one may write

$$\sum_{x,y} p(x)p(y)\,\iota_\pi(x,y) = \sum_i \sum_{x\in B_i} \sum_{y\in B_i} p(x)p(y) = \sum_i \Big(\sum_{x\in B_i} p(x)\Big)\Big(\sum_{y\in B_i} p(y)\Big) = \sum_i p_i^2,$$

which makes explicit how the pairwise sum "collapses" onto block masses. Taking $1-$ of both sides yields the claim. ☐

Remark 456. ***Proof sketch.*** *Compute the same probability in two ways: (i) by summing block masses squared, and (ii) by summing over pairs* (x,y) *with an indicator that they lie in the same block. Since graphtropy is defined as* $1-$ *that pairwise indistinction probability, it matches logical entropy.*

Key step. *The identity* $\sum_{x,y} p(x)p(y)\,\iota_\pi(x,y) = \sum_i p_i^2$ *is simply the law of total probability applied to blocks: it groups the pairwise sum by which block both elements fall into. Another way* *to say this is that the indicator ι_π is block-constant in the sense that it is* 1 *exactly on the union of the Cartesian products $B_i \times B_i$, and* 0 *off that union; summing against $p(x)p(y)$ therefore sums the total $p \otimes p$-mass of those diagonal block rectangles.*

Visual intuition. *Imagine drawing two balls from an urn labeled by blocks. The chance that both draws land in the same block is the sum over blocks of (chance of landing in that block)^2. If one instead thinks in terms of ordered pairs, the same probability is obtained by summing the probability of each ordered pair (x, y) and then discarding (via ι_π) all pairs that straddle two different blocks; graphtropy/logical entropy is precisely the complement of this "pair-collision" event.*

Proposition 13 (Monotonicity under loss of distinction)**.** *Let G and G' be two indistinction graphs on X with weights satisfying $\iota_G(x, y) \le \iota_{G'}(x, y)$ for all x, y. Then for every distribution p,*

$$\mathrm{GT}(G, p) \ge \mathrm{GT}(G', p).$$

In particular, adding more indistinction edges (or strengthening indistinction weights) cannot increase graphtropy.

Remark 457. ***What the proposition says.*** *If an observer becomes* less discriminating *(treats more pairs as indistinct, or treats them as more strongly indistinct), then the expected distinguishability cannot go up.*

Why it matters. *This is the formal monotonicity property one expects from any sensible "distinction content" measure. It lets us reason order-theoretically about resolution changes: coarsening is an order move, and graphtropy moves monotonically with it. This will be echoed later when effability is shown to be monotone in the indistinction regime. Concretely, if G' is obtained from G by declaring additional pairs (x, y) to be indistinct (changing $\iota_G(x, y)$ from* 0 *toward* 1*) or by increasing their indistinction weight in a fuzzy model, then G' represents a genuine loss of discriminative power, and* GT *registers that loss by decreasing (or staying equal) for every p.*

Connection. *Notice the parallel between this proposition and Proposition 14 below: both say that making fewer demands on distinctions expands what is achievable/expressible. In both cases, the underlying mechanism is order preservation: increasing indistinction enlarges the set of "identified" pairs, which increases the expected indistinction and thereby reduces the complement quantity (graphtropy) that counts distinctions.*

Proof. Immediate from the definition of $\mathrm{GT}(G, p)$ as $1-$ a weighted average of ι_G. More explicitly, if $\mathrm{GT}(G, p) = 1 - \sum_{x,y} p(x)p(y)\iota_G(x, y)$ (or the corresponding normalized variant used in the surrounding text), then pointwise $\iota_G \le \iota_{G'}$ implies

$$\sum_{x,y} p(x)p(y)\iota_G(x, y) \;\le\; \sum_{x,y} p(x)p(y)\iota_{G'}(x, y),$$

and subtracting from 1 reverses nothing (it is an order-preserving affine transformation), yielding $\mathrm{GT}(G, p) \ge \mathrm{GT}(G', p)$. □

Remark 458. ***Proof sketch.*** *The quantity subtracted from* 1 *is an expectation of $\iota_G(X_1, X_2)$ under independent sampling from p. If $\iota_G \le \iota_{G'}$ pointwise, then the expectation under G is no larger than under G', hence $1-$ that expectation is no smaller. Said differently,* $\mathrm{GT}(G, p)$ *is the expectation of the complementary "distinction indicator" $1 - \iota_G(X_1, X_2)$, and if ι_G increases pointwise then $1 - \iota_G$ decreases pointwise, so its expectation cannot increase.*

Geometric intuition. *Think of ι_G as a "blur kernel" over $X \times X$. Increasing blur reduces contrast; graphtropy measures contrast, so it decreases. In the crisp (0–1) case, adding indistinction* *edges literally increases the region in $X \times X$ on which the kernel equals 1, so the $p \otimes p$-mass of indistinct pairs grows and the complementary mass of distinct pairs shrinks.*

Theorem 10 (Shannon entropy lower-bounds quadratic/logical distinguishability)**.** *Let p be a distribution on a finite set X and define*

$$S_2(p) := \sum_{x \in X} p(x)^2, \qquad H_2(p) := -\log S_2(p).$$

Then

$$H(p) \geq H_2(p) = -\log S_2(p).$$

If π is a partition of X and p_i are the block masses, then

$$H(\pi) := -\sum_i p_i \log p_i \geq -\log \sum_i p_i^2 = -\log(1 - h(\pi, p)).$$

Remark 459. ***What the theorem says (plainly).*** *Shannon entropy is always at least the order-2 Rényi entropy H_2 (sometimes called "collision entropy" because it depends on $\sum p(x)^2$, the collision probability). So if an observer's belief distribution is highly uncertain in the Shannon (coding) sense, then it is also uncertain in this quadratic (pairwise-collision) sense.*

Why this is important here. *Logical entropy and graphtropy are built from quadratic pairwise expressions. This theorem links the coding-theoretic world (Shannon) to the distinction/pairwise world (logical entropy/graphtropy). It is, in effect, a compatibility guarantee: the different notions of "uncertainty" discussed in this section are not unrelated numbers but are constrained by inequalities. In particular, whenever one uses $S_2(p)$ (or its complement $1 - S_2(p)$) as a proxy for uncertainty/heterogeneity, the theorem bounds how far that proxy can drift from Shannon's coding interpretation.*

Connection to earlier/later material. *Earlier we motivated information as observer-indexed; this theorem remains valid under that indexing, because it is a statement about any finite distribution p_O the observer might hold. Later, in pattern and habit sections, we will move between coding-like and distinction-like quantities; having such monotone bridges makes those translations principled rather than merely metaphorical. One can also view this as a special case of a more general monotonicity phenomenon for Rényi entropies: as the order α increases, H_α decreases, so $H = H_1$ sits above H_2; the present proof gives a direct inequality tailored to the quadratic quantity that appears in graphtropy.*

Proof. Define $r(x) := p(x)^2/S_2(p)$, which is a probability distribution on X. Nonnegativity of KL divergence gives

$$0 \leq D_{\mathrm{KL}}(p\|r) = \sum_x p(x) \log \frac{p(x)}{r(x)}.$$

Substituting $r(x) = p(x)^2/S_2(p)$ yields

$$D_{\mathrm{KL}}(p\|r) = \sum_x p(x) \log \frac{p(x)}{p(x)^2/S_2(p)} = \sum_x p(x) \log \frac{S_2(p)}{p(x)}.$$

Hence

$$0 \leq \log S_2(p) - \sum_x p(x) \log p(x) = \log S_2(p) + H(p),$$

which rearranges to $H(p) \geq -\log S_2(p) = H_2(p)$. The partition statement follows by applying the same argument to the block distribution (p_i). Note that no special cases are needed for $p(x) = 0$ (or $p_i = 0$): the conventional interpretation $0 \log 0 := 0$ makes all displayed sums well-defined, and KL divergence remains nonnegative as long as $r(x) > 0$ whenever $p(x) > 0$, which holds here because $r(x) \propto p(x)^2$. $\square$

Remark 460. ***Proof sketch.*** *Construct an auxiliary distribution r proportional to p^2, then apply the universal inequality $D_{\mathrm{KL}}(p\|r) \geq 0$ to derive an inequality between $\sum p \log p$ and $\log \sum p^2$.*

Key step. *The choice $r(x) \propto p(x)^2$ is not arbitrary: it is exactly the distribution that turns $\log S_2(p)$ into a normalization constant inside the KL divergence. Once that substitution is made, the inequality becomes algebraic. One can also read the algebra as comparing the average of $-\log p(x)$ under p (which is $H(p)$) to the log of an L^2-type quantity $S_2(p)$; the KL step is what guarantees the comparison has the correct direction for all p (not just for nearly uniform distributions).*

Visual intuition. *Shannon entropy is sensitive to the entire distribution shape, while $S_2(p)$ is dominated by larger probabilities (since it squares them). The inequality $H \geq H_2$ expresses that if a distribution is "peaky," its S_2 becomes large (so H_2 becomes small), and Shannon entropy cannot be larger than that quadratic constraint allows. In the common "collision" interpretation, $S_2(p) = \sum_x p(x)^2$ is the probability that two independent draws from p coincide; thus $H_2 = -\log S_2$ measures how difficult it is (in bits) to force a collision. Under this reading, $H \geq H_2$ says that the coding-theoretic uncertainty H cannot undercut the collision-based notion of spread captured by H_2, and it becomes tight (up to support effects) when the distribution is close to uniform over its effective support.*

Remark 461 (Hyperseed reading)**.** *The theorem says: if an observer's distribution is spread out enough to have large Shannon entropy, then it must also have a large "quadratic spread," hence a large logical entropy/graphtropy. In other words, uncertainty in the coding sense lower-bounds the expected rate at which distinctions are encountered. Equivalently, when the observer is forced (by high Shannon uncertainty) to distribute probability mass broadly, the probability that two independent samples fall into the* same *indistinction class drops, so the complement quantity (logical entropy / graphtropy) rises. This is one way to connect the "uncertainty" modality (coding/inference spread) to the "distinction" modality (how often the observer can separate states): the former constrains the latter because concentrated mass simultaneously increases collision probability and reduces the expected number of effective distinctions available to the observer.*

10.6 Uncertainty and ineffability as observer-relative modalities

Hyperseed uses "ineffable" in a practical sense: something may exist and matter to a mind, yet remain resistant to being stably and communicably represented. We formalize this by combining (i) a cost model for representations and (ii) a fidelity predicate that depends on the observer's distinctions. The key point is that both ingredients are explicitly indexed by the observer/context O, so "ineffability" is not an absolute metaphysical label but a relative statement about what can be encoded *given* a representational interface and a limited budget.

Definition 126 (Observer representational resources)**.** *An observer/context O is equipped with:*

- *a description space (language) $\mathcal{D}_O$;*
- *a cost/effort function $\sigma_O : \mathcal{D}_O \to [0, \infty)$;*

- *a decoding map* $\mathrm{dec}_O : \mathcal{D}_O \to \Delta(X)$*, where* $\Delta(X)$ *is the set of probability distributions on* X*.*

Given $d \in \mathcal{D}_O$*, the distribution* $\mathrm{dec}_O(d)$ *is interpreted as the observer's best reconstruction of the target in* X *from description* d*.*

Remark 462. ***Notation/conventions.*** $\mathcal{D}_O$ *is a set of descriptions (strings, programs, structured messages, etc.).* $\sigma_O(d)$ *is the effort/cost of using description* d*; it plays the same formal role as description length or resource use, but we do not insist it be literal bit-length.* $\Delta(X)$ *denotes the simplex of probability distributions on* X*.*

Intuition. *A description* d *is not required to point to a single* $x \in X$*; instead it may decode to a distribution, reflecting that a finite description can determine only a region of possibilities. This is a deliberately observer-friendly model: what the observer can* say *need not uniquely identify what the world* is*.*

Example. *If* X *is a set of images and* $\mathcal{D}_O$ *is a natural-language vocabulary, then* $\mathrm{dec}_O(d)$ *is the distribution over images compatible with the phrase* d *(e.g. "a red apple on a table"). If* X *is microstates and* $\mathcal{D}_O$ *is macroscopic parameters, then* $\mathrm{dec}_O(d)$ *is a macrocanonical distribution over microstates.*

Why useful. *This definition provides the scaffold needed to define effability/ineffability as a resource-bounded property, aligning with Hyperseed's emphasis that what can be expressed depends on effort and available representational moves (Hyperseed-Concept 100, Hyperseed-Concept* **??***).*

Further clarification. *The map* dec_O *can be read in several compatible ways: (a) as a conventional semantics map (a description denotes a set/region of states, represented here by a distribution), (b) as Bayesian posterior inference given a message* d*, or (c) as a lossy decompressor in a coding system. Allowing* $\mathrm{dec}_O(d)$ *to be a distribution rather than a point makes room for ambiguity, underspecification, and context-dependence, which are precisely the regimes where ineffability is most natural to discuss. Similarly, the cost* σ_O *may measure time, energy, cognitive effort, social negotiation cost, or any other constrained resource, so that "saying more precisely" is not free.*

Definition 127 (Fidelity relative to a distinction structure)**.** *Fix an indistinction weight function* ι_G *on* X*. Given a target* $x \in X$ *and a reconstructed distribution* $q \in \Delta(X)$*, define the* distinction error

$$\mathrm{Err}_G(x, q) := \sum_{y \in X} q(y)\,(1 - \iota_G(x, y)).$$

Define the corresponding distinction fidelity

$$\mathrm{Fid}_G(x, q) := 1 - \mathrm{Err}_G(x, q) = \sum_{y \in X} q(y)\,\iota_G(x, y).$$

Remark 463. ***Intuition.*** $\mathrm{Fid}_G(x, q)$ *is the expected indistinction between the true target* x *and a random draw* $y \sim q$*. So fidelity is high when* q *puts most of its mass on states* y *that the observer would treat as indistinct from* x*. This makes fidelity explicitly dependent on the observer's distinction policy.*

Example. *If* ι_G *is induced by a partition into categories (say, "cat" vs. "dog"), then fidelity measures whether your reconstruction distribution stays within the correct category. A description might be "good enough" even if it cannot identify the exact microstate, as long as it lands in the right coarse class.*

Why useful. *This is the technical move that makes ineffability meaningful rather than mystical: we can say precisely which aspects of* x *the observer fails to capture, because the failure is* *measured by the observer's own indistinction structure. It also anticipates the later use of graded relations/quantales to represent cognition (Sections 11 and 12).*

Basic properties. *When $\iota_G(x,y) \in [0,1]$ for all x,y, one has $\mathrm{Err}_G(x,q) \in [0,1]$ and $\mathrm{Fid}_G(x,q) \in [0,1]$. Moreover, $\mathrm{Fid}_G(x,q)$ is linear in q, so mixing reconstructions mixes fidelities in the obvious way. At the extremes: if $\iota_G(x,y) = \mathbf{1}_{x=y}$ (perfect distinction of microstates), then $\mathrm{Fid}_G(x,q) = q(x)$ is just the probability that q hits the correct state exactly; if $\iota_G(x,y) \equiv 1$ (no distinctions at all), then $\mathrm{Fid}_G(x,q) \equiv 1$ and everything is trivially "perfectly faithful" in this coarse sense.*

Remark 464. *If ι_G encodes a partition, then $\mathrm{Fid}_G(x,q)$ is exactly the probability mass that q assigns to the block of x. Thus Fid_G measures whether the reconstruction lands in the "right distinction class." For fuzzy ι_G, it becomes a graded correctness criterion. In particular, when ι_G is derived from a weighted graph (or similarity kernel), $\mathrm{Fid}_G(x,q)$ can be read as the expected similarity between x and a q-sample, so the same formalism covers both crisp categorization and graded perceptual similarity.*

Definition 128 ((ϵ, K)-effability and ineffability)**.** *Fix thresholds $\epsilon \in (0,1)$ and $K \in (0,\infty)$. An element $x \in X$ is* (ϵ, K)-effable for O (relative to G) *if there exists a description $d \in \mathcal{D}_O$ such that*

$$\sigma_O(d) \le K \quad \textit{and} \quad \mathrm{Fid}_G\big(x, \mathrm{dec}_O(d)\big) \ge 1-\epsilon.$$

Otherwise x is (ϵ, K)-ineffable for O (relative to G). *We write $\mathrm{Eff}_O(\epsilon, K; G) \subseteq X$ and $\mathrm{Ineff}_O(\epsilon, K; G) := X \setminus \mathrm{Eff}_O(\epsilon, K; G)$.*

Remark 465. *Intuition.* *Effability is a two-threshold notion:*

- *K limits how much representational effort the observer may spend;*
- *ϵ limits how much distinction error the observer is willing to tolerate.*

An item is ineffable if every description cheap enough fails the fidelity test.

Examples. *A detailed inner sensation might be ineffable for an observer with a coarse language (small $\mathcal{D}_O$) or small budget K. A high-dimensional physical state may be ineffable relative to a coarse distinction structure G if the observer demands fine fidelity (tiny ϵ) but lacks resources.*

Monotonicity and limiting cases. *Holding O and G fixed, $\mathrm{Eff}_O(\epsilon, K; G)$ is monotone increasing in both parameters: if $K' \ge K$, then any description feasible under K is feasible under K', and if $\epsilon' \ge \epsilon$, then any reconstruction meeting the stricter fidelity target $1-\epsilon$ also meets the weaker target $1-\epsilon'$. Thus ineffability is most stringent at small K and small ϵ, matching the informal idea that "hard to say" can mean either "too expensive to say precisely" or "no available description hits the demanded granularity." At the opposite extreme, if $\mathcal{D}_O$ contains arbitrarily long descriptions and dec_O can represent point masses δ_x (perfectly specific reconstructions), then every x becomes (ϵ, K)-effable for sufficiently large K and sufficiently small ϵ; the interesting regime is where either the language cannot express such pinpoint reconstructions or the budget K forbids them.*

Relation to lossy coding. *Formally, the condition $\mathrm{Fid}_G(x, \mathrm{dec}_O(d)) \ge 1-\epsilon$ parallels a distortion constraint in rate–distortion theory, with $(1-\iota_G)$ acting as an observer-relative distortion measure. The novelty here is that distortion is not imposed externally but induced by the observer's distinction structure, so "what counts as an acceptable lossy description" is itself observer-relative.*

Remark 466. *Why this is useful.* *This definition separates three sources of "ineffability" that are often conflated: (i) lack of resources (small K), (ii) lack of expressive language (restricted $\mathcal{D}_O$*

and dec_O*), and (iii) demanding too fine a notion of correctness (too discriminating an* ι_G *or too small* ϵ*). In Hyperseed terms, it makes ineffability a modality indexed by* effort *and* distinction regime*, rather than an absolute metaphysical boundary (Hyperseed-Concept* **??***). Equivalently, the question "is x ineffable?" is replaced by the operational question "for which budgets* K*, tolerances* ϵ*, and regimes* G *does* x *fall outside the reachable set of reconstructions?". This shift matters because it lets one diagnose* which knob *(capacity, vocabulary/decoding, or resolution) is responsible for the failure, rather than treating all failures as the same phenomenon. It also clarifies that "ineffability" can arise even in a fully deterministic world: it is about limits of mediation and evaluation, not about indeterminacy in the target.*

Proposition 14 (Monotonicity in resources and resolution)**.** *If* $K' \geq K$ *then* $\mathrm{Eff}_O(\epsilon, K; G) \subseteq \mathrm{Eff}_O(\epsilon, K'; G)$*. If* G' *is coarser than* G *in the sense that* $\iota_G(x,y) \leq \iota_{G'}(x,y)$ *for all* x, y *(more indistinction), then*

$$\mathrm{Eff}_O(\epsilon, K; G) \subseteq \mathrm{Eff}_O(\epsilon, K; G').$$

Remark 467. ***What the proposition says.*** *Two monotonicities hold:*

- *More resource budget* ($K' \geq K$) *cannot make previously sayable things unsayable.*
- *Coarser resolution (more indistinction) cannot make previously acceptable reconstructions unacceptable.*

Here "resource budget" is whatever complexity/cost measure is encoded in the definition of $\mathrm{Eff}_O(\epsilon, K; G)$ *(e.g., description length, time, energy, sample size, or any composite bound), so the first monotonicity is a statement about feasibility under constraint relaxation. Likewise, "coarser" is not an informal metaphor but the pointwise order* $\iota_G \leq \iota_{G'}$*, which guarantees that every pair* (x, y) *is judged at least as "similar" (or at least as "indistinct") in* G' *as it was in* G*.*

Why it matters. *This is the minimal sanity condition for an effability notion: adding capacity should not reduce what you can express, and relaxing your standards of distinction should not reduce what you can count as successfully expressed. It also gives a clean formal handle on developmental/cultural claims: as a community gains representational resources or shifts its distinction regime, the boundary between effable and ineffable moves monotonically. In particular, if one thinks of* ϵ *as a tolerance parameter for "good enough" reconstruction, then this proposition isolates two other monotone directions (in* K *and in* G*) that move the boundary without changing* ϵ*. This is useful when comparing observers: two agents can agree on* ϵ *but still differ radically in effability because they differ in budget* K *or in the similarity structure* ι_G *they implicitly apply.*

Connection. *Compare with Proposition 13: graphtropy decreases under coarsening, while effability increases under coarsening. Together they express a trade: fewer distinctions encountered means more targets count as representable. One can read this as a duality between (i) the "complexity of the world as cut" (how many distinctions are treated as real) and (ii) the "complexity of a message needed to succeed" (how hard it is to meet the fidelity test). When* G *is made coarser, the observer is, in effect, permitted to ignore more microstructure, so many different underlying states can share a single acceptable description.*

Proof. The first inclusion is immediate: any description of cost at most K is also of cost at most K'. Concretely, if $d \in \mathcal{D}_O$ is feasible under K (i.e., it belongs to the allowed-cost subset of descriptions), then it remains feasible under any larger bound K', and the corresponding decoded reconstruction $\mathrm{dec}_O(d)$ does not change; only the admissible set of d expands. For the second, if $\iota_G \leq \iota_{G'}$ then for any $q \in \Delta(X)$ we have

$$\mathrm{Fid}_{G'}(x, q) = \sum_y q(y)\, \iota_{G'}(x, y) \geq \sum_y q(y)\, \iota_G(x, y) = \mathrm{Fid}_G(x, q).$$

A coarser G' makes the fidelity test easier to satisfy, so the effable set can only expand. Equivalently, for each fixed x and each candidate reconstruction q, the set of regimes under which q is ϵ-acceptable is upward closed in the pointwise order on ι; taking an existential quantifier over descriptions preserves this monotonicity. □

Remark 468. ***Proof sketch.*** *The first part is pure set inclusion under a relaxed constraint: enlarging the allowed-cost set enlarges the feasible descriptions. The second part is an inequality under a pointwise domination $\iota_G \leq \iota_{G'}$, which transfers directly through the expectation defining fidelity. What is doing the work is that $\mathrm{Fid}_G(x, q)$ depends on G only through $\iota_G(x, \cdot)$, so comparing regimes reduces to comparing two functions pointwise and then averaging.*

Key step. *Fidelity is linear in ι_G (it is an average of $\iota_G(x, y)$ under q), so pointwise ordering of ι implies ordering of fidelities. In particular, no additional regularity assumptions (convexity, continuity, etc.) are needed: the argument is purely order-theoretic.*

Intuition. *If you blur your distinctions, you become easier to satisfy: many reconstructions that were previously "wrong in the details" become "close enough" in the new regime. Conversely, sharpening distinctions (moving from G' to a finer G) can render previously adequate summaries inadequate, not because the agent "lost" expressive power, but because the scoring rule became stricter.*

Remark 469 (Ineffability versus unknowability)**.** *Ineffability is not the same as (absolute) unknowability. Even if x is ineffable to O at some (ϵ, K), it may become effable after: (i) increasing resources (larger K); (ii) changing the representational language $\mathcal{D}_O$; or (iii) changing the distinction regime (a different G). Hyperseed's point is that such changes are common in development and culture: a concept can become sayable once a community learns to carve the world differently. Said differently, ineffability here is indexed by an* interface*: a particular coding/decoding practice together with a particular way of scoring similarity. Under that interpretation, an "ineffable" x is often better described as "not currently compressible into the observer's available description forms without exceeding the tolerated error". This also prevents a common equivocation: a fact may be perfectly learnable in principle (there exists some richer observer O' or some larger K for which it is effable) while being practically unsayable for a given agent in a given context.*

Remark 470. *In Peircean terms, ineffability is not a statement about the absence of a Third (law/representation), but about the current poverty of mediating habits available to the agent or community. In Whiteheadian terms, it is a limitation in the available forms of "prehension" and their integration, not a denial that the occasion exists (cf. [14, 15] for the broader metaphysical framing). The present formalism simply makes those philosophical diagnoses testable against parameters: one can ask whether the obstacle is primarily one of habit/mediation (modeled by $\mathcal{D}_O$ and dec_O), one of effort (modeled by K), or one of discriminatory stance (modeled by G and ϵ). On that reading, a "new concept" in a community can be seen as simultaneously (a) an expansion of $\mathcal{D}_O$ (new symbols, new grammars, new practices of description) and (b) a reconfiguration of ι_G (new similarities taken as salient), either of which can move items from the ineffable to the effable region without changing the underlying target space.*

10.7 Potential infinity and infinitesimals as "limit objects"

Hyperseed frequently speaks about "potentially infinite" or "potentially infinitesimal" quantities. Rather than invoke full nonstandard analysis, it is often enough to work with a constructive surrogate: *germs of sequences at infinity.* In this subsection, "limit object" should be read in an informal operational sense: we take a potentially unbounded or potentially refining process (a sequence), and then identify processes that differ only by a finite amount of "startup behavior." What remains is a stable tail-pattern that functions like an idealized object for asymptotic reasoning.

Definition 129 (Eventual equality and germs)**.** *Let $\mathbb{R}^{\mathbb{N}}$ be the set of real sequences. Define an equivalence relation $\sim$ on $\mathbb{R}^{\mathbb{N}}$ by*

$$a \sim b \quad \iff \quad \exists N \in \mathbb{N}\, \forall n \geq N: \; a_n = b_n.$$

The quotient

$$\mathbb{R}_\infty := \mathbb{R}^{\mathbb{N}}/\sim$$

is called the set of germs at infinity*. We write $[a]$ for the germ of the sequence a. Addition and multiplication are defined pointwise on representatives.*

Remark 471. ***Well-definedness and algebra.*** *Because $\sim$ is compatible with pointwise addition and multiplication (if two pairs of sequences agree eventually, then so do their pointwise sums and products), the operations "defined pointwise on representatives" do not depend on which representative of a germ one chooses. Thus $\mathbb{R}_\infty$ is naturally a commutative ring. It is not a field in general: a germ can fail to have a multiplicative inverse, e.g. if a representative has infinitely many zero terms (so that no pointwise reciprocal sequence exists beyond any finite stage).*

Embedding of standard reals. *There is a canonical map $\mathbb{R} \hookrightarrow \mathbb{R}_\infty$ sending a real number c to the germ of the constant sequence $(c, c, c, \dots)$. This makes it precise to compare a germ to an ordinary real "benchmark" without leaving the germ language. In what follows, inequalities such as $\varepsilon <_\infty c$ implicitly use this embedding.*

Notation/conventions. *$\mathbb{R}^{\mathbb{N}}$ is the set of sequences $(a_0, a_1, a_2, \dots)$ with real entries. The relation $a \sim b$ means "a and b are eventually the same," i.e. they may differ at finitely many initial indices but coincide from some point onward. $[a]$ denotes the equivalence class (germ) of a.*

Intuition. *A germ forgets finite beginnings and remembers only tail behavior. This captures the idea of "potential" processes: what matters is not having an actually completed infinity, but having a rule whose behavior stabilizes beyond any finite horizon. Equivalently, a germ is a bookkeeping device for statements of the form "for all sufficiently large n," which will recur whenever one reasons about limits, asymptotics, or tolerances.*

Examples. *The sequences $(0, 1, 2, 3, \dots)$ and $(5, 1, 2, 3, \dots)$ define the same germ: they differ only at the first entry. The sequences $(1, 1, 1, \dots)$ and $(1, 1, 1, 0, 1, 1, \dots)$ do* not *define the same germ if they differ infinitely often. Similarly, the sequences (n) and $(n+1000)$ define the same germ difference as a constant shift: $[n + 1000] = [n] + [1000]$, so the "potential infinity" is insensitive to finite offsets in starting point.*

Why this is useful. *Resource-sensitive ontology often wants to say "can be made arbitrarily large" or "arbitrarily small" without committing to completed infinities. Germs provide a minimal algebraic language for such talk, compatible with an operational view of approximation. They also separate two kinds of information that often get conflated: the finite transient behavior of a process (which can be idiosyncratic or context-specific) and its eventual regime (which is what asymptotic comparison and "limit" talk typically tracks).*

Definition 130 (Eventual order)**.** *For germs $[a], [b] \in \mathbb{R}_\infty$ define*

$$[a] \leq_\infty [b] \quad \iff \quad \exists N \in \mathbb{N}\, \forall n \geq N: \; a_n \leq b_n.$$

This is a partial order compatible with addition and multiplication by nonnegative germs.

Remark 472. ***Basic properties.*** *The relation $\leq_\infty$ is reflexive and transitive, and it is antisymmetric in the sense appropriate for a quotient: if $[a] \leq_\infty [b]$ and $[b] \leq_\infty [a]$, then $a_n = b_n$ eventually, hence $[a] = [b]$. It is generally not a total order: there may be incomparable germs whose representatives cross infinitely often.*

Intuition. *$[a] \leq_\infty [b]$ means: after some finite stage, a_n never exceeds b_n. So the order compares asymptotic tail behavior and ignores finitely many initial anomalies. This is exactly the quantifier pattern used in many "limit" arguments: the tail is constrained uniformly beyond some cutoff.*

Example. *If $a_n = n$ and $b_n = n^2$, then $[a] \leq_\infty [b]$ since eventually $n \leq n^2$. But $[a] \leq_\infty [c]$ and $[c] \leq_\infty [a]$ can both fail if the sequences oscillate forever. For instance, if $c_n = (-1)^n n$, then neither $[c] \leq_\infty [0]$ nor $[0] \leq_\infty [c]$ holds: the sign keeps flipping, so no single tail bound of one by the other stabilizes.*

Why useful. *The eventual order lets us define "infinitely large" and "infinitesimal" elements as order-theoretic comparisons with constants, matching Hyperseed's use of potential infinity/infinitesimals (Hyperseed-Concept **??**, Hyperseed-Concept **??**). Concretely, a germ x is "(potentially) infinite" if it eventually exceeds every constant, and it is "(potentially) infinitesimal" if it is eventually positive yet eventually below every fixed positive constant.*

Example 10 (A potentially infinite and a potentially infinitesimal element)**.** *Let $\omega := [n]$ where $n \mapsto n$. Then ω dominates every standard constant: for any $c \in \mathbb{R}$, eventually $n > c$. Let $\varepsilon := [1/n]$. Then ε is positive but smaller than any fixed positive constant in the eventual order.*

Remark 473. *This example is the operational core: ω stands for an unbounded growth process, while ε stands for a decay process that can be made smaller than any fixed tolerance by going far enough along the sequence. Neither requires a completed infinite magnitude; both require only the ability to continue the process as needed. In particular, the germ formalism explicitly tolerates the fact that real computations have "initial conditions": changing finitely many initial values of n or $1/n$ does not change the potential object.*

A useful consistency check is that these two limit objects interact as expected under multiplication:

$$\omega \cdot \varepsilon \;=\; [n] \cdot [1/n] \;=\; [\, n \cdot (1/n)\,] \;=\; [1],$$

since $n \cdot (1/n) = 1$ for all $n \geq 1$, i.e. the product is eventually the constant 1. This illustrates how the germ quotient encodes the idea that "after a finite warm-up," scaling up and scaling down can cancel in a stable way.

Lemma 2 (Infinitesimal property in the germ order)**.** *Let $\varepsilon = [1/n] \in \mathbb{R}_\infty$. Then $0 <_\infty \varepsilon$ and for every standard $c > 0$ we have $\varepsilon <_\infty c$.*

Remark 474. ***What the lemma says.*** *The germ of $1/n$ behaves exactly as one informally expects of an infinitesimal: it is eventually positive, yet eventually smaller than any fixed positive real constant.*

Relation to "limit" language. *In ordinary analysis one writes $\lim_{n\to\infty} 1/n = 0$. Here, instead of collapsing* $1/n$ to *the value* 0, *we keep the whole tail-behavior as an object ε that remains distinguishable from 0 but is dominated by every positive constant. This is precisely the kind of "nonzero but negligible" quantity that potential-infinitesimal talk is aiming at.*

Why it matters. *This provides a precise, lightweight sense in which we can talk about "arbitrarily weak couplings" or "arbitrarily small errors" inside a framework that treats infinity as potential rather than completed. Such language is later useful when describing diminishing influences* *across increasingly remote contexts (e.g. in resonance-style stories), where one wants "nonzero but negligible" rather than exactly zero. In particular, the lemma justifies replacing a family of tolerance statements "for every $c > 0$ there exists a stage after which the error is $< c$" with a single order-theoretic comparison against all constants.*

Proof. For $n \geq 1$, $1/n > 0$, so $\varepsilon >_\infty 0$. Given $c > 0$, choose $N > 1/c$. Then for all $n \geq N$, $1/n < c$, so $\varepsilon <_\infty c$. □

Remark 475. ***Proof sketch.*** *Both claims follow by choosing a sufficiently large index N. Positivity is immediate because all terms $1/n$ are positive; smallness follows because $1/n$ eventually falls below any fixed threshold c.*

Key step. *The eventual order bakes "there exists an N such that for all $n \geq N$" into the definition, so asymptotic estimates translate directly into order relations. This is the same quantifier structure that appears in ε–δ reasoning; germs package it into algebraic objects so that one can reason about tails without constantly reintroducing quantifiers.*

Visual intuition. *Plot $1/n$ against n; the curve descends toward 0. The lemma is simply the statement that any horizontal line at height $c > 0$ is eventually above the curve.*

Interpretive note. *Although ε is "smaller than any fixed constant," it is not* equal *to 0 in $\mathbb{R}_\infty$: the sequence $1/n$ is never eventually the zero sequence. This distinction is exactly what makes ε a useful "limit object" rather than merely a limit value.*

Remark 476 (Conceptual role)**.** *Germs provide a lightweight way to talk about "approaching infinity" or "approaching zero" without assuming that a mind must represent completed infinite objects. In this sense, a germ functions as a stable* direction of refinement*: it records what remains invariant as one continues to sharpen a description (for example, by increasing resolution, adding measurement time, or enlarging a comparison class), while refusing to reify the unattainable endpoint as a finished entity. This is the intended "limit object" reading: not an actually-given limit, but a disciplined placeholder for an indefinitely extensible process of approximation.*

In the ontology, this supports statements like: "the distinction granularity can be refined without bound" (potential infinity) and "a coupling can be arbitrarily weak" (potential infinitesimal). Here "without bound" should be read as a modal claim about available refinements (there is always a further distinction regime available in principle), rather than as a claim that an infinite set of distinctions is ever simultaneously present to an observer. Likewise, "arbitrarily weak" emphasizes a comparative ordering of couplings under refinement: for any operational threshold fixed by an observer, one can consider contexts in which the coupling falls below that threshold, even if no absolute "zero-coupling" state is ever asserted.

When we later discuss morphic resonance across increasingly remote contexts, it can be useful to treat resonance strength as an infinitesimal germ rather than as exactly zero. This avoids a sharp discontinuity between "no influence" and "some influence" that would otherwise be an artifact of representational coarseness: the germ formalizes the idea that influence may persist in a way that is systematically negligible for any fixed finite resource bound, yet not logically excluded. Equivalently, the infinitesimal germ lets us separate two questions that are often conflated: whether a coupling is detectable *at a given resource level, and whether it is* structurally present *in the relational organization being modeled.*

10.8 Section wrap-up

This section supplied three bridges that will be used repeatedly later:

- Shannon information provides the standard coding-theoretic notion of uncertainty. In particular, it supplies a quantitative baseline for how many bits are required, on average, to resolve a choice among alternatives given an assumed probability model, and it supports the usual trade-offs between compression, prediction, and error.
- Logical entropy and graphtropy connect uncertainty to the Hyperseed primitive of distinction. This makes uncertainty explicitly about *separations* (which pairs are distinguished and which are not), so that information measures can be read directly as properties of a distinction graph or partition structure, rather than only as properties of symbol sequences.
- Ineffability becomes a monotone property relative to observer resources and distinction regimes. That is, an object or pattern may be perfectly effable for one observer (or one modeling interface) while remaining ineffable for another, and increasing resources or refining distinctions can only decrease ineffability in the sense that more becomes stably nameable and communicable.

With these in place, we can treat hierarchy and pattern webs (Section 11) and habit dynamics (Section 12) as information-processing structures whose meaning depends explicitly on what is, and is not, distinguished. This dependence is not merely epistemic but structural: changing a distinction regime can change what counts as an entity, a relation, or a causal regularity in the model, thereby shifting which compressions are valid and which predictions are well-posed. Conversely, by keeping the distinction regime explicit, later sections can state claims about organization and dynamics in a way that remains robust under changes of scale, granularity, or observational interface.

11 Hierarchy, heterarchy, and pattern webs

11.1 Motivation: why Hyperseed needs both hierarchies and webs

Patterns (Section 9) do not occur in isolation. Once an observer can represent "P is a pattern in x" with some intensity, it can also represent relationships *between* patterns and *between* patterned entities. This is not an optional embellishment: as soon as patterns are treated as reusable cognitive objects, the observer must be able to compare them, compose them, and propagate information from one to another. In particular, even a minimal pattern-recognizer tends to (a) reuse the same patterns across many inputs and (b) relate those reused patterns via co-occurrence, entailment, explanation, control, abstraction, and part–whole decomposition. Two structural motifs appear immediately:

- *Hierarchy:* some entities are treated as "higher" than others (more valuable, more general, more controlling, more explanatory) and this induces an order-like structure.
- *Heterarchy (web):* most entities have multiple different paths of relationship to other entities, forming loops and multi-path dependence networks rather than trees.

The emphasis on "order-like" versus "network-like" is deliberate: in practice, the same pair of patterns may be related in multiple ways at once (e.g. one pattern may be more general, while the other is more predictive in a restricted domain), and the cognitive model must be able to represent both the comparative directionality (who outranks whom, or who constrains whom) and the lateral connectivity (who interacts with whom, possibly reciprocally). In this section, the word "hierarchy" will be used broadly enough to include not only strict trees but also preorders and partial orders, i.e. structures where many elements may be incomparable, and where "being higher" can mean "being at least as good as" rather than "being the unique parent of."

Remark 477. *The philosophical pressure behind this bifurcation is simple: a mind (or any modeling system) must both (i) compress the world into* comparative priorities *(a ranking of what matters, what explains, what controls), and (ii) keep track of the fact that the world is not a ladder but a tangle, where explanation and influence typically move along many partially redundant and mutually constraining routes. Hierarchy and heterarchy are thus not competing metaphors; they are complementary projections of the same cognitive act, seen from the angles of order and connectivity (Hyperseed-Concepts* **??** *and* **??***).*

A useful way to read this remark is that hierarchy expresses a kind of *compression* (reduce many degrees of freedom to a comparative relation), while heterarchy expresses a kind of *closure under interaction* (allow many routes of influence and evidential support, including feedback). A cognitive system that only ranks without interconnection becomes brittle (it cannot represent mutual constraint, circular causation, or context sensitivity), while a cognitive system that only connects without ranking becomes diffuse (it cannot decide what to attend to, what to optimize, or what to treat as explanatory leverage). Thus, the two motifs show up not merely as descriptive metaphors but as recurring representational necessities.

Hyperseed uses these motifs in several places: subpattern hierarchy (patterns within patterns), control/perceptual hierarchies, self-hierarchies, and the "pattern web" picture of cognition. Here "subpattern hierarchy" points to part–whole and compositional structure (a pattern may be built from constituent patterns), whereas "control/perceptual hierarchies" point to layered organization in action and inference (e.g. high-level goals constraining lower-level controllers, and higher-level interpretations organizing lower-level percepts). "Self-hierarchies" similarly reflect that an agent can maintain multiple self-models with differing levels of generality, authority, or temporal scale. Meanwhile, the "pattern web" picture emphasizes that these hierarchical decompositions are embedded in a larger network: patterns participate in many overlapping decompositions and predictive/explanatory links, and any given pattern typically inherits constraints from many sources at once.

The mathematical point of this section is that all of these can be treated as special cases of *quantale-enriched structure* (Section 3): the same formal machinery that represents "how related" two entities are (possibly with degrees, costs, or truth-values) can specialize to order-theoretic hierarchy, metric-like connectivity, and more general graded relations. Concretely, enrichment replaces the crisp statement "there is a relation from a to b" with a value that can encode intensity, confidence, cost, or compatibility, and it equips these values with a notion of composition. When applied to patterns, this means we can model not only whether one pattern subsumes or controls another, but also *to what extent* and *through which chains of intermediate patterns* that relationship holds.

- A crisp hierarchy is enrichment in the Boolean quantale $\{0, 1\}$.
- A metric heterarchy is enrichment in the Lawvere quantale $([0, \infty], \geq, +, 0)$.
- A paraconsistent, graded hierarchy is enrichment in a p-bit quantale (Section 3.4).

To unpack the intuition: Boolean enrichment yields relations valued in $\{0, 1\}$, so composition reduces to logical conjunction and the structure behaves like a preorder (and, with suitable antisymmetry, a partial order), matching the usual mathematical abstraction of "hierarchy." Lawvere enrichment replaces "true/false relatedness" with extended nonnegative "distance" or "cost," where composing paths adds costs; this naturally captures web-like multi-path structure because indirect connections compete with direct ones via triangle-inequality-style constraints. Finally, p-bit enrichment allows relationship-values that can represent graded support and graded conflict simultaneously (paraconsistency), which is important when the same pair of patterns can be connected by evidence that points in different directions, or when an agent maintains multiple partially incompatible models without immediate collapse into triviality.

Remark 478. *The "pattern web" framing is central in Hyperseed's earlier presentations and related work on patterns, emergence, and cognitive network structure [1, 5]. The present section is deliberately modest: it does not add new metaphysics; it supplies a single mathematical dialect in which these recurrent motifs can be spoken without ambiguity.*

In this sense, the goal is methodological: once hierarchies and webs are both expressed as enriched structures, one can translate intuitions and results across domains. For example, "subpattern-of" can be treated as an order-like relation, while "influences" or "predicts" can be treated as graded, directional, and compositional; and in both cases, the formalism makes explicit what it means for indirect relationships to accumulate through intermediate patterns.

We will define the Hyperseed terms (hierarchy, systemic hierarchy, subpattern hierarchy, heterarchy, pattern web) and then show how they fit into the enriched-categorical core.

11.2 Hierarchy as observer-assessed order

Hyperseed treats "hierarchy" as inherently observer-relative: an observer O assesses some entities as higher than others, *in some respect* (importance, generality, desirability, control, explanatory priority). The simplest mathematical representation of "higher than" is a partial order, but we will allow graded and paraconsistent structure. In particular, allowing "graded" comparisons accommodates the common situation where O can compare u and v only weakly or provisionally, while allowing "paraconsistent" comparisons accommodates the situation where O simultaneously carries evidence both for and against the claim that u outranks v without collapsing into triviality. Note also that even in the ungraded case, a hierarchy need not be total: many pairs of entities may remain incomparable, reflecting underspecification, competing criteria, or simply irrelevance under the current respect-of-comparison.

Definition 131 (Order-like relations)**.** *Let U be a set of entities (objects, patterns, processes, roles).*

- *A* preorder *on U is a relation $\preceq \subseteq U \times U$ that is reflexive and transitive.*
- *A* partial order *is a preorder that is also antisymmetric.*
- *A* graded preorder *(relative to a commutative quantale V) is a map*

$$h : U \times U \to V$$

with reflexivity $e \leq h(u, u)$ *and* transitivity

$$h(u, v) \otimes h(v, w) \leq h(u, w) \qquad (u, v, w \in U),$$

where $(V, \leq, \oplus, \otimes, e)$ is as in Section 3.

It is worth stressing what is *not* required by these definitions. A preorder permits ties (distinct $u \neq v$ with $u \preceq v$ and $v \preceq u$), which can be interpreted as observational or pragmatic equivalence under the chosen respect-of-comparison. Antisymmetry upgrades this to a partial order, ruling out such ties except when $u = v$; thus, moving from preorder to partial order amounts to deciding that "mutual outranking" should be treated as identity rather than as a legitimate equivalence class. Conversely, neither preorders nor partial orders require comparability: it can happen that neither $u \preceq v$ nor $v \preceq u$ holds, which is often the appropriate formalization of "O has no basis to rank them (yet)".

Remark 479 (Notation unpacking: what the symbols mean). *The data of a graded preorder live in a commutative quantale $(V, \leq, \oplus, \otimes, e)$ (Section 3). Here:*

- *V is the space of "grades" (truth-strengths, costs, evidences, similarities, etc.).*
- *$\leq$ is the order on grades ("no stronger than").*
- *$\oplus$ is the join/supremum operation (aggregating alternative supports).*
- *$\otimes$ is the monoidal product (aggregating sequential supports, as in composition along a path).*
- *e is the unit grade (the neutral element for $\otimes$), playing the role of identity.*

Thus $h(u, v) \in V$ should be read as "the degree (in V) to which u is at least as high as v" in the hierarchy. The transitivity inequality says: if u is above v and v is above w, then (by composing these two claims) u is above w to at least the composed degree.

One additional interpretive point is that $\oplus$ and $\otimes$ encode two qualitatively different ways of accumulating support for a comparison. Typically, $\otimes$ is used when comparisons are chained (e.g., "u outranks v" and "v outranks w" together provide a route from u to w), while $\oplus$ is used when comparisons arise through multiple alternative routes or arguments (e.g., two different explanatory pathways both suggest that u outranks v). This separation becomes especially important when the same domain U simultaneously participates in multiple interacting structures: a "hierarchy" viewed as an order-like relation can later be combined with other relational webs using the same quantale operations, so that path-composition and route-aggregation are not redefined ad hoc for each application.

Remark 480 (Intuition, examples, and why graded preorders matter). *A crisp preorder answers a yes/no question: "is u above v?" But in cognition and modeling we usually have degrees: we may have strong reasons, weak reasons, or even simultaneous reasons for and against a comparison. A graded preorder is exactly a disciplined way to represent such comparative gradations (Hyperseed-Concept* **??***).*

Two simple examples:

- *If $V = \{0, 1\}$ with $\otimes = \wedge$ and $\oplus = \vee$, then $h(u, v) = 1$ means "$u \preceq v$" in the ordinary preorder sense. Transitivity becomes $(u \preceq v \ \wedge \ v \preceq w) \Rightarrow u \preceq w$.*
- *If $V = [0, 1]$ with $\otimes$ as multiplication and $\oplus$ as maximum, then $h(u, v)$ can be read as "confidence" in the comparison $u \preceq v$. A longer chain of comparisons generally decreases confidence by multiplication, while multiple independent routes can increase it by the maximum.*

The usefulness is structural: once comparisons are V-valued, one can reuse the same algebra both for "hierarchy" and for "web" propagation later in the section, rather than maintaining separate formalisms. This is also where Hyperseed's notion of weakness naturally enters, because weakness is precisely an order-theoretic measure of what distinctions one has collapsed [3, 2] (Hyperseed-Concept 202).

To connect these examples to more applied readings: when $V = [0, 1]$ is interpreted as confidence, the inequality $h(u, v) \otimes h(v, w) \leq h(u, w)$ enforces the idea that indirect support for $u \preceq w$ via v should never exceed what the model assigns directly. If the model later learns a stronger direct comparison for $u \preceq w$, this can be seen as adding a new "route" whose contribution can be combined (via $\oplus$) with the routed support. If instead V represents a cost or penalty scale, then

$\leq$ may be interpreted as "no worse than," and $\otimes$ can be interpreted as accumulating costs along a chain; in that case the same axiom becomes a disciplined form of "taking paths seriously" while still preferring cheaper direct comparisons when available. These shifts of interpretation are exactly why the definition is parameterized by a quantale rather than hard-coding a particular numeric scheme.

Remark 481 (Thin V-categories)**.** *A graded preorder is exactly the same data as a* thin V*-enriched* *category: there is at most one hom-value between any two objects, and composition is encoded by the quantale transitivity inequality. Thus, "hierarchy" is naturally treated as a special case of the enriched-categorical core.*

This thinness perspective also clarifies the difference between "having a hierarchy" and "having a richer relational structure." A thin V-category records only the best available grade $h(u, v)$ of the comparison, not the internal structure of *why* u outranks v (e.g., which intermediate arguments, subcriteria, or evidential sources support it). Later, when we discuss pattern webs, those internal justifications can be represented explicitly as additional nodes and edges, while the induced graded preorder can be recovered as an "overall reachability/strength" summary through the same $(\oplus, \otimes)$ machinery.

Definition 132 (Observer-assessed hierarchy)**.** *Fix an observer/context O. A* hierarchy for O on *a domain U is a graded preorder*

$$h_O : U \times U \to V_O$$

for some quantale V_O chosen to match the observer's representational regime. In the p-bit *case,* $h_O(u, v) = (h^+_O(u, v), h^-_O(u, v))$ *encodes positive and negative evidence for "u is above v."*

The freedom to choose V_O is not merely technical: it expresses that different observers may treat the same underlying signals as probabilities, costs, plausibilities, bipolar evidences, or even multi-criteria vectors. Even for a fixed observer, different respects-of-comparison can justify different codomains: "priority under time pressure" might be naturally cost-like, while "explanatory generality" might be naturally evidence-like. In the p-bit (positive/negative bit) case, allowing both h^+_O and h^-_O to be simultaneously large permits the model to represent genuine tension (e.g., competing heuristics) without forcing an immediate resolution; the graded preorder axioms then constrain how such tensions propagate along chains of comparisons.

Remark 482 (Intuition and simple examples)**.** *This definition bakes in a central Hyperseed stance: hierarchy is not "in the world" as a bare relation, but in the triad (world, observer, respect-of-comparison). Formally, h_O is indexed by O, and even the codomain V_O is chosen to suit O's mode of representation (Hyperseed-Concept 86).*

A simple example is a task-based priority ordering in an intelligent system: U might be a set of goals or subgoals, and $h_O(u, v)$ encodes the degree to which u has priority over v given the current task context (cf. the broader AGI motivation for context-relative prioritization in [19]). Another example is an explanatory hierarchy: U might be hypotheses, and $h_O(u, v)$ measures how strongly u is judged a "higher-level" explanation than v in the sense of compressing more data with fewer assumptions [5]. The utility of this observer-indexed view is that it lets us discuss multiple coexisting hierarchies without contradiction: distinct observers (or the same observer at different times) may impose distinct but internally coherent V_O-orders.

A further consequence of indexing by O is that changes in context can be modeled as changes in h_O itself rather than as inconsistencies in a single absolute hierarchy. For instance, if O shifts from "short-term reward" to "long-term safety" as the respect-of-comparison, the domain U may remain the same while the ranking relation changes substantially; this is naturally expressed by moving to a different h_O (or, equivalently, treating the observer as updated). Similarly, disagreements between observers can be tracked at the level of their chosen quantales and aggregation rules: two observers might share the same underlying comparisons but combine them differently (different $\otimes$), or they might treat alternative supports differently (different $\oplus$), yielding different induced hierarchies even when looking at the same evidence. In this sense, "observer-assessed hierarchy" is a portability layer: it isolates the order-like constraints (reflexivity, transitivity) from the representational conventions by which O encodes and composes comparative judgments.

Remark 483 (Values and "higherness")**.** *Hyperseed's gloss is that entities higher in a hierarchy are "more valuable in some sense." Formally, this can be captured in several equivalent ways:*

- Order-first: *define h_O directly as an order of priority or importance.*
- Value-first: *define a valuation $\nu_O : U \to V$ (or to $\mathbb{R}$) and set $h_O(u, v)$ to be evidence that $\nu_O(u) \geq \nu_O(v)$.*
- Constraint-first: *define h_O as the order induced by a model class (e.g. the simplest causal explanations), so that "higher" means "more explanatory" or "more compressive."*

In later sections, these perspectives will be linked: in a resource-sensitive ontology, "valuable" typically means "valuable for prediction/control under bounded effort," which often correlates with explanatory compression. In particular, the "value-first" stance can be read as a representation theorem in miniature: whenever one has a (possibly graded) comparative judgment of "at least as important as," it is often convenient to embed that judgment into a numerical or lattice-valued score ν_O that makes comparisons mechanically checkable. Conversely, the "order-first" stance emphasizes that the comparison itself is primary: one may know that u outranks v even when no single scalar score captures all the reasons, motivating valuations into richer codomains V (e.g. vectors, intervals, or evidence pairs). The "constraint-first" stance makes explicit that "higher" can be tied to a background standard of adequacy: if two descriptions fit observed behavior equally well, then the one requiring fewer degrees of freedom, fewer exceptions, or fewer ad hoc auxiliary assumptions can be treated as "above" the other because it supports more robust generalization under perturbation or distribution shift.

Remark 484. *The equivalence of these perspectives is not merely a convenience; it expresses a deeper pragmatic unity. In a Whiteheadian idiom, "value" is a mode of selection among processes, while "explanation" is a mode of organization among descriptions; the point of a formalism like a graded preorder is precisely that it allows these to be treated as two views of the same comparative act [15]. One can read the graded preorder as recording a* practice *of preference rather than a discovery of a metaphysical ladder: what counts as "higher" is the result of repeated comparative acts (selection, attention, justification) stabilized into a relation. This also clarifies why "equivalence" here should be taken up to the observer's operational needs: different encodings (order-first vs. value-first) may disagree on fine structure while remaining interchangeable with respect to the tasks that motivated the hierarchy in the first place.*

11.3 Systemic hierarchy via models of a dynamical system

Hyperseed introduces "systemic hierarchy" to emphasize that a hierarchy is often not a primitive of the system, but a structure induced by a *model*. The intended case is: a dynamical system D is mapped into a symbolic domain S equipped with a partial order. In this framing, "systemic" is meant to contrast with purely stipulative taxonomies: the hierarchy is anchored in regularities of D as *captured* by ϕ, even though it is not assumed to be uniquely determined by D without an observer, a language, and a purpose.

Definition 133 (Systemic hierarchy (model-induced))**.** *Let D be a dynamical system (in any sense suitable for the application). Let $\mathrm{Proc}(D)$ denote a set of processes (or process-types) associated with D. A* hierarchical model *of D consists of:*

- *a symbol domain S;*
- *a (possibly partial) order relation $\prec_S$ on (part of) S;*
- *an interpretation map (a model) $\phi : \mathrm{Proc}(D) \to S$.*

We say that D manifests a systemic hierarchy relative to *$(S, \prec_S, \phi)$ if ϕ is sufficiently faithful for the observer's purposes and $\prec_S$ is used to guide prediction, explanation, or control. The qualifier "for the observer's purposes" is doing substantive work: faithfulness may mean predictive adequacy, counterfactual reliability, or control-relevant abstraction rather than isomorphism to an underlying "true" state space. Likewise, allowing $\prec_S$ to be defined only on part of S permits models in which only some symbols participate in the hierarchical comparison (e.g. only those corresponding to actionable options, stable patterns, or salient macrostates).*

Remark 485 (Intuition and examples)**.** *The definition is a reminder that hierarchy often appears* after *representation. The system D unfolds in time; the observer selects a set of processes $\mathrm{Proc}(D)$ to pay attention to; and only then does an ordering $\prec_S$ become meaningful as a cognitive or engineering tool (Hyperseed-Concept 133).*

Example: in control, one may model a robot's dynamics by a set of nested controllers and treat high-level policies as "above" low-level motor primitives. Here S is a symbolic space of controller-states, and $\prec_S$ encodes dominance/override relationships among control modules, used to guide action selection [19]. Example: in explanation, one may map physical micro-processes into a symbolic domain of macro-regularities; then "higher" means "more compressive" or "more predictive" relative to some descriptive cost (echoing the pattern-based compression stance [5]). The usefulness of the definition is that it legitimizes plural hierarchies as plural models: different ϕ and $\prec_S$ can both be "right" insofar as they serve distinct purposes. A helpful way to read ϕ is as a choice of coarse-graining*: many distinct trajectories or micro-events in D may be sent to the same symbol in S, and the induced hierarchy then ranks the* symbols *(i.e. the equivalence classes the observer cares about), not the underlying microstates directly. In the control example, "dominance" may itself be context-sensitive (e.g. safety overrides goal pursuit), so $\prec_S$ can encode normative or design constraints (what* should *happen) as much as empirical regularities (what* does *happen). In the explanation example, "compressive" can be understood relative to an explicit description length, a model class capacity, or an algorithmic resource budget; thus the same D can yield different "higher" macropatterns when the observer changes the allowable explanatory vocabulary.*

Remark 486 (Fuzzy and paraconsistent systemic hierarchy)**.** *In realistic cognitive settings, the order $\prec_S$ may be incomplete (only some pairs are comparable) and may be graded or disputed. This is naturally modeled by taking $\prec_S$ to be a graded preorder $h_S : S \times S \to V$. If $V = [0,1]^2$, then $h_S(s,t) = (h_S^+(s,t), h_S^-(s,t))$ can encode both evidence and counterevidence for "s precedes t" (or "s dominates t"). Such a representation is useful when different data sources, different evaluators, or different time-slices of the same evaluator support incompatible rankings: rather than forcing a premature linearization, one records the support structure and postpones commitment to a single* *crisp order. Operationally, graded comparisons can be used to implement decision procedures such as "prefer s to t when $h_S^+(s,t)$ exceeds a threshold and $h_S^-(s,t)$ remains below a tolerance," making explicit how hierarchy-guided action depends on risk appetite and error costs.*

Remark 487 (Why the model view matters)**.** *The same underlying system D may admit many different hierarchical models depending on:*

- *the observer's values and tasks;*
- *the level of abstraction (which distinctions are ignored);*
- *which variables are treated as controls versus observations;*
- *which explanatory language is chosen.*

Hyperseed treats this plurality as normal. Thus, "systemic hierarchy" is best read as: there exists a useful hierarchy in some modeling space that guides successful interaction with the system*. This perspective also clarifies what it means for two hierarchies to "disagree": they may be ranking different symbolizations of the same processes, or the same symbolization under different loss functions and intervention capabilities. For example, an engineer optimizing reliability may place fault-monitoring processes "above" throughput-optimizing processes, while an economist modeling the same organization may rank activities by revenue contribution; both hierarchies can be faithful in their respective senses, because faithfulness is indexed to what is being predicted, explained, or controlled.*

Remark 488. *In Peircean terms, one may say that a systemic hierarchy is a particular way in which Thirdness (law-like mediation) is projected onto a flux of Seconds (actual interactions) and a field of Firsts (qualities felt as salient) [14]. The formal point remains: the hierarchy is carried by $(S, \prec_S)$ and by the interpretive act ϕ, not by D alone. Equivalently, the hierarchy is a property of a semiotic triad (world, signs, interpreter) rather than a property of the world taken in isolation: change the signs (S), the comparative practice ($\prec_S$), or the interpretive mapping (ϕ), and the resulting "systemic hierarchy" may change even when D does not. This does not trivialize hierarchy; it locates its objectivity in reproducible modeling and successful coordination: if many observers with similar access and aims converge on similar $(S, \prec_S, \phi)$, then the induced hierarchy can be treated as stable for those purposes.*

11.4 Subpattern hierarchy: patterns within patterns

Hyperseed defines a subpattern hierarchy by declaring $y \le x$ when y is a *compositional subpattern* of x. The informal criterion is: y can be combined with something else to form a pattern in x. We formalize this using the combination operation from Section 9.

Definition 134 (Compositional subpattern)**.** *Fix a domain of entities X and a combination operation $* : X \times X \to X$ (not assumed commutative). Let $\mathcal{P}$ be a set of pattern-representers (in simple settings one may take $\mathcal{P} = X$, but we keep it separate). Assume we have a (possibly graded) predicate*

$$\mathrm{Pat}(p; x) \in [0,1]$$

interpreted as "degree to which p is a pattern in x" (e.g. pattern intensity).

For $p, q \in \mathcal{P}$, we say that p is a compositional subpattern of q*, written*

$$p \preceq_{\mathrm{sub}} q,$$

if there exists some $r \in X$ such that $\mathrm{Pat}(p * r; q)$ *is nontrivial (e.g. strictly positive, or above a chosen threshold). In a graded version, define the subpattern strength*

$$h_{\mathrm{sub}}(p, q) := \sup_{r \in X} \mathrm{Pat}(p * r; q).$$

Remark 489 (Intuition, examples, and usefulness)**.** *The guiding intuition is almost embarrassingly concrete: p is a subpattern of q if p can serve as a reusable "piece" in constructing a pattern that fits q (Hyperseed-Concept 182). The quantifier "there exists r" is the formal way to say "p can be completed."*

Example 1 (strings): let X be strings and $*$ be concatenation. Let $\mathrm{Pat}(p; x)$ *be high when p is a compressive description of x (Section 9, cf. [5]). Then $p \preceq_{\mathrm{sub}} q$ says: there is some string r such that the concatenation $p*r$ is itself a strong pattern for q. Example 2 (feature composition): let X be structured feature-bundles, and let $*$ combine feature-sets. Then $p \preceq_{\mathrm{sub}} q$ says: the feature-pattern p can be augmented to yield a pattern adequate for q.*

This definition is useful because it turns the informal "patterns within patterns" idea into an algebraic relation that can be composed and closed (later yielding hierarchy-like structure and, via path closure, a web-like structure).

Remark 490 (Two readings)**.** *The relation $p \preceq_{\mathrm{sub}} q$ can be read in (at least) two complementary ways:*

- Constructive: *q can be built (in part) from p by composition.*
- Explanatory: *p is a reusable regularity that helps compress or predict q.*

The constructive reading matches "patterns within patterns." The explanatory reading matches the idea that higher-level patterns summarize or bundle lower-level ones. Both will be used later (especially in Sections 12 and 13).

Proposition 15 (Subpattern is a preorder under associativity)**.** *Assume $*$ is associative and has an identity element $e \in X$. Assume also that for each $p \in \mathcal{P}$ there is an embedding of p into X (so that $p * e$ is defined) and that* $\mathrm{Pat}(p * e; p)$ *is nontrivial. Then $\preceq_{\mathrm{sub}}$ is a preorder on $\mathcal{P}$.*

Remark 491 (What the proposition says, and why it matters)**.** *The statement is: once "combination" behaves like a monoid operation (associative with identity), the subpattern relation behaves like a genuine order-like comparison: every pattern is a subpattern of itself (reflexivity), and subpatternhood chains (transitivity). This is important because it shows that an apparently heuristic notion ("p is a piece of q") inherits the same formal stability properties as more familiar hierarchies. It also connects this section back to the enriched-order viewpoint above: preorders are precisely the crisp case of V-enriched thin categories.*

Proof. (Reflexivity.) For $p \in \mathcal{P}$, choose $r = e$. By assumption $\mathrm{Pat}(p*e; p)$ is nontrivial, so $p \preceq_{\mathrm{sub}} p$.

(Transitivity.) Assume $p \preceq_{\mathrm{sub}} q$ and $q \preceq_{\mathrm{sub}} s$. Then there exist $r_1, r_2 \in X$ such that $\mathrm{Pat}(p*r_1; q)$ and $\mathrm{Pat}(q*r_2; s)$ are nontrivial. Using associativity, $(p*r_1)*r_2 = p*(r_1*r_2)$. Under the intended compositional semantics of patterns (Section 9), a pattern used to form q can be re-used inside a composite pattern used to form s. Thus $\mathrm{Pat}(p*(r_1*r_2); s)$ is nontrivial (possibly with reduced degree), implying $p \preceq_{\mathrm{sub}} s$. □

Remark 492 (Proof sketch and intuition)**.** Proof sketch. *Reflexivity uses the identity element e: completing p by "doing nothing" yields p again, so p is trivially a component of itself. Transitivity uses associativity: if p can be completed to something that patterns q, and q can be completed to* *something that patterns s, then concatenating the completions yields a completion witnessing that p can be completed to pattern s.* $\square$

The key step is the reassociation $(p * r_1) * r_2 = p * (r_1 * r_2)$, which licenses the informal move "reuse the subpattern inside the larger composition." Geometrically, one can picture p as a reusable module, and r_1, r_2 as successive "adapters"; associativity says that the order of bracketing the assembly does not change the assembled whole, so the module relationship persists through multi-stage construction.

Remark 493 (Antisymmetry and "pattern identity")**.** *To obtain a partial order one needs a notion of when two patterns are "the same." In practice, pattern identity is observer-relative: two patterns may be distinct at one resolution and equivalent at a coarser resolution. Thus, it is often best to treat the subpattern hierarchy as a preorder and then quotient by an equivalence relation appropriate to the observer. This aligns with the general Hyperseed stance that equivalence is a re-description of which distinctions are being ignored (Hyperseed-Concept* **??***). In particular, antisymmetry fails precisely when the observational interface cannot (or chooses not to) distinguish two patterns that are mutually subpatterns of one another: one may have $p \preceq q$ and $q \preceq p$ without wanting to conclude $p = q$ as syntactic objects. The quotient construction makes this explicit by replacing "pattern identity" with an equivalence class $[p]$ of mutually inter-subpattern patterns, so that the resulting relation on classes becomes antisymmetric. Concretely, if $\preceq$ is a preorder on patterns and one defines $p \sim q \iff (p \preceq q \ \wedge \ q \preceq p)$, then $\sim$ captures "indistinguishable for this observer, at this granularity"; different observers (or the same observer under a different coarse-graining map) may choose different $\sim$. One can also read the observer-relative choice of $\sim$ as specifying which internal degrees of freedom are treated as gauge: two patterns are identified when they differ only by transformations that the observer treats as non-informative.*

11.5 Heterarchy: multi-path structure and loops

Hyperseed uses "heterarchy" as the opposite of "tree thinking": multiple distinct paths connect entities, and the network contains loops. In cognition and society this is typical: concepts mutually constrain each other, roles interlock, values conflict, and control is distributed. Formally, the point is not merely that there are many nodes and edges, but that relational structure does not factor through a single spine: there is generally no privileged unique lineage of explanation, delegation, or causation. Accordingly, heterarchy is also a warning against forcing a directed acyclic graph (DAG) model too early: cycles often encode precisely the feedback that makes an organization, a mind, or a culture self-maintaining.

Definition 135 (Heterarchy)**.** *A* heterarchy *on a set of nodes U is a directed graph (or weighted digraph)*

$$G = (U, E, w)$$

with the property that for many pairs (u, v) there exist multiple distinct directed paths from u to v, and G contains directed cycles. If weights live in a quantale V, we interpret $w(u, v) \in V$ as a graded link strength. In this definition "for many pairs" is intentionally informal: heterarchy is meant as a structural regime rather than a sharp graph property, and the relevant notion of "many" is domain- and scale-dependent (e.g. dense conceptual association versus sparse but consequential institutional channels). When V is a quantale, the intended semantics is that path composition aggregates along a route via the monoidal product $\otimes$, while competing routes can be compared or pooled using the join structure (often denoted $\oplus$ or $\bigvee$); the enriched viewpoint below makes this explicit.

Remark 494 (Intuition, examples, and why the definition is phrased this way). *A heterarchy is a formal acknowledgement that influence is rarely a single chain. In a conceptual network, there may be several different reasons why u suggests v; in a social network, several institutional routes by which one role constrains another; in a cognitive architecture, several routes by which perception and action co-determine each other [19]. Such multiplicity is not noise; it is the medium by which robustness and conflict coexist (Hyperseed-Concept* **??**).

Example: let U be concepts, and put an edge $u \to v$ when u reliably predicts v in a dataset; then loops encode mutually reinforcing concept-clusters. Example: let U be agents/roles in an organization, and put an edge $u \to v$ when u can request actions from v; loops encode delegation and feedback. The definition is phrased in terms of multiple paths and cycles because these are precisely the features that distinguish a web from a tree: a tree has unique simple paths and no directed cycles. A further reason to phrase the definition this way is methodological: in most empirical settings, it is easier to detect (i) the existence of distinct routes between two points and (ii) the presence of feedback loops than it is to justify any single total ranking or layered decomposition. Equivalently, cycles and multi-path connectivity are "witnesses" that no single linear explanation (or command chain) can be globally valid without loss. In applications one often refines this intuition by analyzing strongly connected components (SCCs): an SCC is a maximal region of mutual reachability, and the condensation graph of SCCs is a DAG. From this angle, a heterarchy can be seen as a system whose dynamics and semantics live primarily inside (possibly large) SCCs, rather than being well-described by the acyclic skeleton alone.

Remark 495 (Heterarchy as a non-thin enriched category). *A hierarchy is "thin": between any two nodes there is (at most) one order value. A heterarchy is better modeled as a general (possibly non-thin) enriched category, where there may be multiple morphisms between u and v representing different explanatory or causal routes. If one wants to remain in the thin setting, one may collapse multiple routes by taking a join (supremum) over path strengths, yielding a single composite strength. Both options appear in Hyperseed practice. In the enriched reading, a distinct directed path is not merely redundant evidence: it can correspond to a different* type *of entailment (e.g. statistical association versus mechanistic constraint), a different mediator (institutional channel A versus channel B), or a different timescale. Keeping multiple morphisms allows one to track these differences explicitly, while composition encodes how routes concatenate. By contrast, the thin collapse (taking a supremum over routes) yields a convenient "best available" or "dominant" linkage, which is often what one wants for scoring, visualization, or approximate reasoning, but it intentionally forgets the internal pluralism that generated the linkage in the first place.*

Definition 136 (Cycle mass (one simple heterarchy index)). *Let $V = [0, 1]$ with $\otimes$ as multiplication and $\oplus$ as maximum, and let $G = (U, E, w)$ be a weighted digraph. Define the* cycle mass *of G by*

$$\mathrm{Cyc}(G) := \max \Big\{ \prod_{i=1}^{k} w(u_i, u_{i+1}) : (u_1, \ldots, u_k) \text{ is a directed cycle} \Big\},$$

with $u_{k+1} = u_1$. Then $\mathrm{Cyc}(G) > 0$ iff G contains a directed cycle with nonzero weight. If G has no directed cycles, the displayed set is empty; in that case one may take $\mathrm{Cyc}(G) := 0$, consistent with the stated iff condition and with the intuition that an acyclic digraph has no "returning signal" at any positive strength.

Remark 496 (Intuition and a toy example). *If each edge-weight is read as a multiplicative "reinforcement factor," then the product around a cycle measures how strongly a signal can return to its starting point after traversing that loop. For a simple 2-cycle $u \to v \to u$ with weights a and b, the* *cycle mass includes the product ab. If either $a = 0$ or $b = 0$, the loop is broken; if both are near 1, the loop is strong. The usefulness of this (admittedly toy) index is that it anticipates the later dynamical claim: habit-taking is autocatalytic and tends to strengthen existing loops (Section 12), so quantifying "loop strength" matters. One can also view $\mathrm{Cyc}(G)$ as a "best feedback gain" under multiplicative composition: it selects the most self-reinforcing directed cycle, not the average cycle. This is appropriate when a single dominant feedback loop can govern qualitative behavior (e.g. a stable institutional feedback channel, or a self-supporting belief cluster), even if many weaker loops also exist. A mild variant replaces the maximum by another aggregator (e.g. a softmax or a sum over cycles) when one wants a notion of distributed cyclicity rather than dominance.*

Remark 497. *The particular definition of $\mathrm{Cyc}(G)$ is not canonical; it is only a toy measure. In later sections, cycles in pattern webs will matter because habit dynamics (Section 12) are autocatalytic: they preferentially reinforce existing loops. It is also worth noting that even this toy measure depends on modeling choices: using multiplication privileges long cycles by accumulating factors, whereas other semiring/quantale choices (e.g. max–min, or additive log-weights) emphasize bottlenecks or total gain differently. Thus, $\mathrm{Cyc}(G)$ should be read as one concrete instantiation of a broader idea: heterarchy invites quantitative indices that are sensitive to feedback, and different indices correspond to different interpretations of what it means for a loop to be "strong" in the domain at hand.*

11.6 Pattern webs from pattern profiles and pattern-based distances

Hyperseed introduces a "pattern web" as a heterarchical structure derived from patterns. One natural construction is:

1. assign to each entity $x \in X$ its fuzzy set of patterns (a *pattern profile*);
2. define a distance between entities in terms of distances between their pattern profiles;
3. convert the distance into a weighted graph (the pattern web).

We give a general construction that can be instantiated in several ways. In particular, the construction deliberately factors the modeling choices into (i) a pattern vocabulary $\mathcal{P}$, (ii) an intensity assignment I, (iii) a notion of profile distance D, and (iv) a graph-building rule. This separation is useful because each component can be varied independently (e.g. refining $\mathcal{P}$ without changing the graph construction rule, or changing D while keeping I fixed), and because it makes explicit where observer- and task-dependence enters.

Definition 137 (Pattern profile)**.** *Let X be a set of entities and $\mathcal{P}$ a set of patterns. Assume we have a pattern-intensity map*

$$I : \mathcal{P} \times X \to [0,1], \qquad I(p,x) = \text{"intensity of pattern } p \text{ in } x\text{"}.$$

For each $x \in X$ define its pattern profile *(a fuzzy set of patterns)*

$$F_x : \mathcal{P} \to [0,1], \qquad F_x(p) := I(p,x).$$

It is often convenient (though not required) to regard $I(p,x)$ as implicitly relative to a context, observation process, or modeling interface, so that one could write $I_C(p,x)$ when emphasizing contextual dependence. In this reading, two analysts with different contexts or sensors may induce different pattern profiles for the "same" underlying x, and hence different pattern webs, without any contradiction: the web is a structure over the modeled entities as distinguished by the chosen pattern interface.

Remark 498 (Intuition, example, and why the definition is useful). *A pattern profile is simply the "fingerprint" of an entity in the pattern vocabulary: it tells you which patterns occur in x and with what strength (Hyperseed-Concept 130). If $\mathcal{P}$ is a finite set of candidate regularities, then F_x is a vector in $[0,1]^{\mathcal{P}}$.*

Example: suppose $\mathcal{P} = \{p_{\text{round}}, p_{\text{red}}, p_{\text{edible}}\}$ and x is an apple (in some context C). Then $F_x(p_{\text{round}}) \approx 0.8$, $F_x(p_{\text{red}}) \approx 0.6$ (green apples exist), $F_x(p_{\text{edible}}) \approx 0.9$. The utility is that once entities are mapped to profiles, we can speak about similarity, clustering, neighborhood, and network propagation without committing to the raw internal structure of x. This is the formal bridge from "pattern" to "pattern web" emphasized in [5].

A further practical advantage is that profiles support partial comparability: two entities can be compared even when their "internal" representations are heterogeneous, so long as they admit pattern intensities against a shared $\mathcal{P}$. Conversely, if two domains cannot share a useful $\mathcal{P}$ (or cannot be mapped into it by a meaningful I), then the construction makes clear that the limitation lies in the pattern interface rather than in the downstream graph machinery.

Remark 499 (Connection to Hyperseed properties). *In Hyperseed, the* property-set *of x is essentially the fuzzy set of patterns in x, with membership degree equal to pattern intensity. Thus F_x can also be read as a property profile. We keep the term "pattern profile" here to emphasize its role in constructing higher-order structure.*

In applications, one may additionally include pattern weights to encode salience or reliability of patterns (e.g. downweighting noisy detectors, or upweighting rare but informative patterns). Such weights can be folded either into the intensity map I (by rescaling intensities) or into the subsequent choice of D (by using a weighted distance), and the present definitions accommodate both approaches.

Definition 138 (Pattern-based pseudo-metric). *Fix a metric (or pseudo-metric) D on the space of fuzzy subsets $[0,1]^{\mathcal{P}}$. Define*

$$d_P(x, y) := D(F_x, F_y).$$

We call d_P the pattern-based pseudo-metric *on X.*

When $\mathcal{P}$ is infinite (e.g. a very large or even uncountable pattern family), specifying D typically entails additional structure (such as a measure on $\mathcal{P}$, a summability condition, or a choice of feature map into a normed space). For instance, one may replace finite sums by integrals, or restrict attention to patterns with non-negligible intensity, yielding a well-defined comparison even when $\mathcal{P}$ is conceptually open-ended.

Remark 500 (Intuition and conventions). *A pseudo-metric is like a metric except it may assign distance 0 to distinct points; this matches the idea that two distinct entities might be indistinguishable at the pattern-resolution induced by $\mathcal{P}$. The symbol D denotes any chosen distance on profile space; d_P is then the induced distance on entities. This is a clean instance of Hyperseed's broader theme: geometry is observer-relative because the chosen distinctions (here, the chosen pattern vocabulary and intensity map) determine which entities are near or far (Hyperseed-Concept 98).*

From a modeling standpoint, allowing $d_P(x, y) = 0$ for $x \neq y$ is often desirable: it expresses that the pattern vocabulary may be coarse-grained, or that the available evidence cannot separate the entities. One may then treat the induced equivalence relation $x \sim y \iff d_P(x, y) = 0$ as an informational quotient, and view the pattern web as operating on equivalence classes at the chosen pattern-resolution.

Example 11 (Finite-pattern ℓ^1 distance). *If $\mathcal{P}$ is finite, we may define*

$$D(F,G) := \sum_{p\in\mathcal{P}} \big|F(p) - G(p)\big|.$$

Then $d_P(x,y)$ is the total absolute difference in pattern intensities across the pattern vocabulary.

A common variant is a weighted ℓ^1 distance,

$$D_w(F,G) := \sum_{p\in\mathcal{P}} w(p)\,\big|F(p) - G(p)\big|,$$

where $w(p) \geq 0$ encodes the relative importance, confidence, or discriminative value of pattern p. This makes explicit that the induced geometry can prioritize certain pattern dimensions without altering the underlying profile representation.

Remark 501 (A second simple example). *Another common choice (when $\mathcal{P}$ is finite) is the ℓ^2 distance $D(F,G) := \Big(\sum_{p\in\mathcal{P}}(F(p) - G(p))^2\Big)^{1/2}$, which penalizes large deviations more strongly. The formalism does not privilege either; it only insists that whatever notion of similarity one uses should be made explicit as a D.*

Depending on the application, one might prefer distances that compare profiles up to overall "mass" or scale. For example, cosine distance (derived from cosine similarity) emphasizes angular agreement between vectors and can be useful when absolute intensity magnitudes are less meaningful than relative pattern mixtures. Likewise, one can normalize profiles (e.g. by ℓ^1 or ℓ^2 norm when appropriate) prior to computing D, thereby distinguishing "composition" from "quantity" of pattern evidence.

Remark 502 (Hutchinson-style choices). *Hyperseed mentions using standard metrics (such as the Hutchinson metric) to compare fuzzy pattern-sets. The present formalism isolates the key idea: any reasonable choice of D yields a pattern-based geometry on X. The specific choice of metric is application-dependent.*

In the same spirit, if patterns carry an internal geometry or ground space (e.g. patterns are distributions over some base domain, or are themselves structured objects), then D can be chosen to respect that structure (e.g. optimal-transport-type distances, kernel-induced distances, or other task-aligned divergences). The point is not to mandate a particular metric family, but to make the dependency of the resulting web on the comparison principle explicit and inspectable.

Definition 139 (From distance to a weighted pattern web). *Fix a monotone decreasing kernel $\kappa : [0,\infty) \to [0,1]$. Define a similarity weight*

$$s(x,y) := \kappa\big(d_P(x,y)\big).$$

A pattern web *on X is a weighted (undirected or directed) graph*

$$W = (X, E_W, w_W)$$

constructed from $s(x,y)$, for example by:

- threshold graph: *include edge (x,y) iff $s(x,y) \geq \tau$;*

- k-NN graph: *connect each x to its k nearest neighbors under d_P;*
- full graph: *take $E_W = X \times X$ with weights $w_W(x, y) = s(x, y)$.*

If a directed web is desired, one can obtain directionality either by using an inherently asymmetric comparison (e.g. a divergence-like D on profiles, or an inclusion/entailment score between fuzzy sets) or by using an asymmetric graph construction rule (e.g. k-NN edges $x \to y$ when y is among the k nearest to x). Even when D is symmetric, k-NN graphs are typically directed unless symmetrized (e.g. by mutual k-NN), and this choice affects the reachability and propagation properties of the resulting pattern web.

Remark 503 (Intuition, examples, and necessity)**.** *This definition performs a familiar maneuver: convert distance ("far means unrelated") into similarity ("near means connected"). The kernel κ is monotone decreasing so that larger distances yield smaller similarities. Example: $\kappa(t) = e^{-\lambda t}$ (a Gaussian-like choice) makes similarity decay exponentially with distance.*

Why is this useful? Because a graph is the natural substrate for heterarchical reasoning: once we have edges, we can talk about multi-step paths, loops, and propagation of support (Hyperseed-Concept 132). In cognitive terms, the pattern web is the externalized form of the judgment "these things share patterns and thus should inform each other," as in earlier Hyperseed discussions [1, 5].

The kernel scale (e.g. λ above, or an analogous bandwidth) plays the role of a resolution parameter: small changes in bandwidth can turn a sparse web into a dense one, changing the balance between local neighborhoods and long-range connections. Similarly, the choice between thresholding, k-NN, and full graphs reflects a trade-off between interpretability and computational cost: thresholding yields explicit "strong ties," k-NN yields bounded degree (often beneficial for scalability), and the full graph preserves all graded similarities but may be impractical for large X. These are not merely engineering details; they influence which heterarchical pathways exist for multi-step association and which loops can form in the induced structure.

Remark 504 (Pattern web as an enriched space)**.** *There are two closely related enriched-category readings:*

- Metric enrichment: *(X, d_P) is a Lawvere metric space, i.e. a category enriched in the quantale $([0, \infty], \geq, +, 0)$.*
- Similarity enrichment: *(X, w_W) is a $[0, 1]$-enriched (or V-enriched) graph/category, where composition along paths multiplies similarities and joins over alternative paths.*

Both are useful. Metric enrichment emphasizes geometry and neighborhood structure; similarity enrichment emphasizes propagation of support along multiple routes. The formal core (Section 3) was designed to support either viewpoint.

11.7 Pattern webs as free V-categories

Given a weighted directed graph of patterns or entities, there is a standard way to "close" it under compositional reasoning. This is exactly the enriched-category analogue of taking the transitive closure of a relation.

Definition 140 (Path composition in a V-graph)**.** *Let V be a commutative quantale. A* V-graph *is a set U with a weight function*

$$g : U \times U \to V.$$

For a directed path $\pi = (u_0 \to u_1 \to \cdots \to u_n)$ *define its* path weight

$$w(\pi) := g(u_0, u_1) \otimes g(u_1, u_2) \otimes \cdots \otimes g(u_{n-1}, u_n).$$

Define the path-closure hom

$$\widehat{g}(u, v) := \bigoplus_{\pi : u \rightsquigarrow v} w(\pi),$$

where the join ranges over all finite directed paths from u *to* v *(including the empty path).*

Remark 505 (Notation unpacking and intuition)**.** *The symbol* $\pi : u \rightsquigarrow v$ *denotes a directed path starting at* u *and ending at* v*. The "empty path" is the path of length* 0 *from* u *to itself; by convention its weight is* e *(the unit of* $\otimes$*). Thus* $\widehat{g}(u, v)$ *aggregates evidence/support from* all *routes from* u *to* v *by:*

- *composing along a fixed route using* $\otimes$ *(sequential aggregation), and*
- *taking the join* $\oplus$ *over alternative routes (parallel aggregation).*

In the Boolean case this is exactly transitive closure: $\widehat{g}(u, v) = 1$ *iff there exists a path from* u *to* v*. In a graded case, it becomes "best-path" (or "best-aggregate") closure, depending on the quantale operations.*

Remark 506 (A tiny example)**.** *Let* $U = \{a, b, c\}$ *and suppose* $g(a, b) = 0.9$*,* $g(b, c) = 0.8$*,* $g(a, c) = 0.1$ *in* $V = [0, 1]$ *with* $\otimes = \times$ *and* $\oplus = \max$*. Then the path* $a \to b \to c$ *has weight* 0.72*, so* $\widehat{g}(a, c) = \max(0.1, 0.72) = 0.72$*. This captures the heterarchical idea that an indirect route can dominate a direct (but weak) link.*

Theorem 11 (Free V-category on a V-graph)**.** *Let* V *be a commutative quantale and* (U, g) *a* V*-graph. Define* $\widehat{g}$ *by path-closure as above, taking the empty path to have weight* e*. Then* $\widehat{g}$ *defines a* V*-enriched category on object set* U*. In particular:*

- *(identity)* $e \leq \widehat{g}(u, u)$ *for all* u*;*
- *(composition)* $\widehat{g}(u, v) \otimes \widehat{g}(v, w) \leq \widehat{g}(u, w)$ *for all* u, v, w*.*

Remark 507 (What the theorem says in plain language)**.** *Starting from a weighted directed graph, the theorem constructs the "best available" composite connection between any two nodes by looking at all finite paths and aggregating their weights. It then asserts that these composite connections automatically satisfy the axioms of a* V*-enriched category: there are identities (the empty paths), and composing a* $u \to v$ *connection with a* $v \to w$ *connection yields a valid* $u \to w$ *connection. So, path closure is not an ad hoc trick; it is the canonical completion of a graph into a compositional reasoning structure.*

Remark 508 (Why it matters here)**.** *This result is the formal hinge between "web" as a static picture and "web" as an inferential substrate. In a pattern web, one typically wants to propagate support (or inhibition, or relevance) along multi-step routes. The theorem guarantees that once we pick a quantale* V *describing how supports combine, the propagation algebra is coherent by construction. This coherence is exactly what later habit dynamics exploit: repeated updating tends to amplify already-available composite routes (Section 12).*

Proof. Identity holds because the empty path from u to u is included and has weight e, hence $e \le \widehat{g}(u,u)$.

For composition, fix u, v, w. A path from u to w can be formed by concatenating a path $\pi_1 : u \rightsquigarrow v$ and a path $\pi_2 : v \rightsquigarrow w$. The weight of the concatenated path is $w(\pi_1) \otimes w(\pi_2)$ by associativity of $\otimes$. Thus, for each pair of paths (π_1, π_2) we have

$$w(\pi_1) \otimes w(\pi_2) \le \widehat{g}(u,w),$$

because $\widehat{g}(u,w)$ is the join over *all* $u \rightsquigarrow w$ paths. Taking joins over all π_1 and π_2 and using distributivity of $\otimes$ over joins gives

$$\Big(\bigoplus_{\pi_1 : u \rightsquigarrow v} w(\pi_1) \Big) \otimes \Big(\bigoplus_{\pi_2 : v \rightsquigarrow w} w(\pi_2) \Big) \le \widehat{g}(u,w).$$

The left-hand side is $\widehat{g}(u,v) \otimes \widehat{g}(v,w)$. □

Remark 509 (Proof sketch and key-step commentary)**.** Proof sketch. *The strategy is to show that the empty path supplies identities, and that concatenating paths supplies composition. Since $\widehat{g}(u,w)$ is defined as a join over* all *$u \rightsquigarrow w$ paths, every particular concatenated path contributes a lower bound, and distributivity lets us lift from particular paths to joins over all paths.* □

The heart of the argument is the monotonicity encoded by joins: if a value is among the terms joined to make $\widehat{g}(u,w)$, then it is $\le \widehat{g}(u,w)$. Concatenation produces the relevant term, and distributivity is what permits "join of left paths" $\otimes$ "join of right paths" to be bounded by the "join of all concatenations." Visually, one can imagine all paths as threads in a fabric: $\widehat{g}(u,v)$ summarizes all threads from u to v, and composition corresponds to splicing threads at v to create longer threads from u to w.

Remark 510 (Interpretation)**.** *If $g(u,v)$ encodes "u supports/predicts/constructs v", then $\widehat{g}(u,v)$ encodes overall support aggregated across all pathways. This is the mathematical backbone of the "web" picture: multi-path reinforcement is represented by the join over paths. If V is* p-bit*-valued, then the same construction supports paraconsistent webs in which multiple routes can provide both support and counter-support. In particular, when V is a join-semilattice (or more generally a quantale), the hat-construction $\widehat{g}$ can be read as a kind of transitive closure: it aggregates not only direct edges $u \to v$ but also indirect chains $u \to \cdots \to v$ by composing along the chain and then taking the join over all chains. Thus $\widehat{g}(u,v)$ measures the best (or most permissive, depending on V) composite evidence that v is reachable from u through the pattern web. When $V =$* p-bit*, the aggregation by joins can simultaneously retain distinct "positive" and "negative" contributions: different paths can witness different polarities without forcing collapse to a single consistent truth value. This is precisely the sense in which the same path-join formalism can model webs that are tolerant of contradiction: the global value $\widehat{g}(u,v)$ records what the web as a whole supports, rather than demanding that all routes agree.*

11.8 Multi-resolution transforms and weakness

Hierarchies are often produced by *abstraction*: collapsing distinctions and bundling lower-level details into higher-level summaries. Pattern webs, likewise, are usually considered at multiple resolutions. We express this using maps (functors) between pattern spaces. At an informal level, the point is that the same underlying system can be described by many different "alphabets": a fine alphabet with many entity- and pattern-types, and a coarse alphabet in which many fine types are identified. The usefulness of a multi-resolution formalism is that it allows one to compare statements made at different descriptive granularities while keeping track of which claims survive abstraction and which are artifacts of fine detail.

Definition 141 (Resolution maps on entities and patterns)**.** *Let X be an entity set and $\mathcal{P}$ a pattern vocabulary. A* resolution change *from a fine description to a coarse one consists of:*

- *a surjective map $\alpha : X \to X'$ (coarse-graining of entities);*
- *a map $\beta : \mathcal{P} \to \mathcal{P}'$ (coarse-graining of patterns);*
- *a rule for pushing forward intensities $I(p, x)$ to $I'(p', x')$.*

A simple pushforward rule is

$$I'(p', x') := \sup\{I(p, x) : \beta(p) = p',\ \alpha(x) = x'\}.$$

Remark 511 (Intuition and example)**.** *Coarse-graining is the formal act of deciding that many differences do not matter for the present purpose. The maps α and β implement this decision on entities and patterns, respectively. Example: X might be images, X' might be object-categories; α maps an image to its category label. Similarly, $\mathcal{P}$ might contain fine-grained visual motifs while $\mathcal{P}'$ contains coarser semantic patterns.*

The pushforward rule via sup *says: a coarse pattern p' holds of a coarse entity x' if* any *of its fine refinements held strongly in* any *representative fine entity mapping to x'. This matches one common cognitive stance: a coarse label is licensed by the strongest supporting instance. The definition matters because multi-resolution comparison is ubiquitous in transfer and abstraction workflows; it is also a bridge to later frameworks emphasizing structured transfer and abstraction [7]. One can also read the* sup *rule as a left Kan extension–style "best upper approximation" of the fine intensity profile by a coarse one: the coarse description is the least committal (largest) assignment consistent with the existence of a supporting witness downstairs. In settings where intensities live in an ordered domain (e.g. $[0, 1]$, a Boolean algebra, or* p-bit*), the use of* sup *ensures that I' is monotone with respect to adding more fine-grained evidence: enlarging the set of fine representatives for a given (p', x') can only increase (never decrease) the pushed-forward intensity. Surjectivity of α is a convenient way to insist that every coarse entity $x' \in X'$ corresponds to at least one fine representative; if α were not surjective, then some coarse entities would be "empty bins," and one would need a convention for the* sup *over an empty set (often the bottom element of V).*

Remark 512. *The pushforward rule above is "optimistic": it preserves any strong evidence that a coarse pattern holds of a coarse entity. Other rules (averaging, expectation, evidence-weighted pooling) may be more appropriate in specific domains. The key point is that coarse-graining can be treated as a monotone operator on pattern profiles. In particular, if $I \le J$ pointwise (every fine intensity in I is bounded by the corresponding one in J), then the induced coarse profiles satisfy $I' \le J'$ under the* sup *rule, since* sup *is monotone in its arguments. This monotonicity is what makes multi-resolution moves compatible with order-enriched viewpoints: resolution change is not merely a lossy projection, but a structured map that respects the underlying preorder of evidential strength.*

Proposition 16 (Coarse-graining is monotone and increases weakness)**.** *Assume an observer's indistinction relation $H \subseteq X \times X$ is pushed forward along α to an indistinction relation $H' \subseteq X' \times X'$ by declaring*

$$(\alpha(x), \alpha(y)) \in H' \text{ whenever } (x, y) \in H.$$

Then H' collapses at least as many distinctions as H. In particular, for any monotone weakness functional $w(\cdot)$ (Section 3.7),

$$w(H) \leq w(H').$$

Remark 513 (Plain-English meaning and connection to earlier results)**.** *The proposition says: when you merge fine entities into coarse ones, you cannot create new distinctions; you only lose them. Therefore any sensible measure of "weakness" (in Hyperseed's technical sense: degree of collapsed distinction-making power) can only increase. This connects directly to the monotonicity theorem for weakness (Theorem 1) and provides a small formal justification for the intuitive idea that abstraction trades detail for simplicity [3, 2] (Hyperseed-Concept 202). Equivalently, passing from H to H' is an order-preserving move in the lattice of indistinction relations: H' contains at least the images of the indistinguishabilities already present in H, and so represents an observer who is (at best) as discriminating as before. If H were an equivalence relation capturing exact perceptual indistinguishability, then the pushforward along α can be seen as inducing a coarser equivalence on X'; the proposition is then the familiar idea that quotienting (or identifying states) cannot increase information.*

Proof. Every indistinction asserted by H induces an indistinction asserted by H'. Thus the set of undistinguished (coarse) pairs contains the image of the set of undistinguished (fine) pairs. Monotonicity of weakness under adding undistinguished pairs (Theorem 1) yields $w(H) \leq w(H')$. More explicitly: if $(x, y) \in H$, then by definition $(\alpha(x), \alpha(y)) \in H'$, so $\{(\alpha(x), \alpha(y)) : (x, y) \in H\} \subseteq H'$. Hence H' is at least as permissive an indistinction relation on the space of coarse entities as the one induced from H, and any weakness functional that is monotone under enlarging the indistinction set must satisfy the stated inequality. □

Remark 514 (Proof sketch and intuition)**.** Proof sketch. *Pushforward along α maps each "confused pair" in X to a "confused pair" in X'. So the coarse observer cannot distinguish at least the pairs the fine observer could not. Then apply monotonicity of w.* □

The key step is the containment claim: H's indistinction assertions survive (as images) in H'. Geometrically, one can think of α as collapsing points into blobs; any pair of points that were already indistinguishable lie in blobs that are indistinguishable in the quotient. The inequality $w(H) \leq w(H')$ is then the numeric shadow of this collapsing. Another way to visualize the same argument is via fibers of α: each coarse point $x' \in X'$ represents a fiber $\alpha^{-1}(x') \subseteq X$. Indistinguishability at the fine level propagates to indistinguishability between fibers at the coarse level, so the coarse description inherits all the "blind spots" of the fine one and may add new ones due to fiber-merging.

Remark 515 (Why this belongs in the hierarchy section)**.** *A hierarchy is often the* result *of a multi-resolution move: the "higher" nodes are coarse summaries that ignore fine distinctions. The proposition formalizes the directionality: moving upward in an abstraction hierarchy typically increases weakness (simplicity) while reducing detail. Later, wu-wei style dynamics (Section 26) will treat "acting with minimal forcing" as moving along flows that are biased toward such weak (simple) representations [21]. In this sense, the same formal move (pushing forward along α) explains two commonplace observations about hierarchies: (i) higher levels are cheaper to reason about because they track fewer distinctions, and (ii) higher levels may be more robust because they do not depend on unstable fine-grained differences. The weakness increase inequality gives a compact way to state that these benefits come with an intrinsic limitation: any decision rule that relies on fine discriminations must be expected to degrade under repeated coarse-graining.*

11.9 Section wrap-up

This section reconstructed four Hyperseed notions in a uniform mathematical language:

- *Hierarchy/systemic hierarchy* as observer-indexed (graded) order, often model-induced. In particular, the "grade" can be read as an explicit resource parameter (e.g. resolution, budget, attention, or tolerance) that determines which distinctions are available, so that what counts as "above" or "below" is not absolute but arises from a chosen standpoint. The phrase "model-induced" emphasizes that the ordering is frequently an artifact of a representation scheme: change the features, the abstraction map, or the inference rule, and the resulting order can refine, coarsen, or even reorder comparisons.

- *Subpattern hierarchy* as the preorder generated by compositional subpattern relationships. Here "generated" signals a closure process: one starts from primitive subpattern links (e.g. "appears as a component of," "factors through," "can be assembled into"), and then takes the smallest reflexive and transitive relation containing them. The fact that the result is a preorder (rather than necessarily a partial order) is conceptually important: different construction paths can yield mutual reachability without forcing equality, capturing the idea that two patterns may embed into one another under composition while remaining distinct as descriptions or mechanisms.

- *Heterarchy* as multi-path, loop-rich network structure. The emphasis on multiple paths encodes the coexistence of alternative explanations, realizations, or routes of influence; the emphasis on loops encodes feedback, mutual reinforcement, and circular dependence. In this view, "non-tree-like" is not merely a combinatorial complication but a signal that comparative relations are context-sensitive and can be revisited: traversing a loop can accumulate costs, change effective granularity, or expose tensions between competing constraints.

- *Pattern web* as a heterarchy derived from the geometry of pattern profiles. Geometric language is meant literally: patterns are represented by profiles (e.g. vectors, distributions, or enriched morphisms) whose relative position determines adjacency, similarity, and feasible transitions under bounded resources. The resulting web is therefore not an arbitrary graph but one induced by a metric-, order-, or quantale-enriched notion of distance/effort, so edges and path weights reflect how distinctions deform under abstraction, noise, or compression.

Read together, these four reconstructions explain how seemingly different vocabularies—rank, inclusion, feedback, and profile geometry—can be treated as different presentations of the same underlying compositional machinery. In particular, the shift from "hierarchy" to "web" is not a change of topic but a change of emphasis: from single-chain comparability to the coexistence of many compatible (and incompatible) comparison pathways, with explicit accounting of the resources required to traverse them.

Remark 516. *The unifying thread is compositionality: in hierarchies, comparisons compose; in webs, paths compose; in multi-resolution transforms, abstraction composes with indistinction and thereby reshapes weakness. Once these are all placed within the same quantale-enriched idiom, it becomes easier to see why Hyperseed can move fluidly between "order" talk and "network" talk without changing subject: both are presentations of structured distinction-making under resource bounds. Concretely, enrichment supplies a common semantics for "how much" (cost, effort, or uncertainty) accompanies a comparison or transition: composing comparisons adds/aggregates that quantity, while taking meets/joins expresses the presence of multiple competing routes. In this sense, hierarchy is the special case where the web of comparisons is sufficiently rigid (or sufficiently* *observed through a single grading) that it behaves like an ordered scaffold, whereas heterarchy is the generic case in which composition exposes parallelism, feedback, and nontrivial trade-offs between pathways.*

The next section (Section 12) adds time and dynamics: it studies how webs reinforce themselves into habits, how they sometimes reverse, and how morphic resonance couples habit dynamics across contexts. From the perspective of the present section, this amounts to endowing the pattern web with update rules that change edge strengths, accessibility, or effective grades as interactions repeat. The same compositional principles then reappear in temporal form: repeated traversals compose into long-run tendencies, loops become attractors or oscillations, and the resource bounds that shaped static comparison now shape learning rates, persistence, and the conditions under which a previously weak distinction can become stable.

12 Habits, self-weaving webs, and morphic resonance

This section reconstructs the Hyperseed dynamical cluster *tendency to take habits* (Hyperseed-Concept 189), *tendency to reverse habits* (Hyperseed-Concept 188), *self-weaving webs* (Hyperseed-Concept 168), and *morphic resonance/anti-resonance* (Hyperseed-Concept 115, Hyperseed-Concept 114) using the minimal formal core (Section 3) and the pattern-web machinery (Sections 11–9). For the broader Hyperseed framing of these notions, see [1].

The key modeling move is to treat a *pattern web* (Hyperseed-Concept 132) as a V-valued directed graph (equivalently, a V-relation) whose edges encode reinforcement or "flow" of pattern support, and to treat habit formation as a family of order-theoretic update operators on V-valued pattern-support states. Morphic resonance is then a cross-context coupling of these same dynamics, in a way philosophically reminiscent of "habit" as a cosmological tendency in Peirce and process-like propagation in Whitehead [14, 15]. In more concrete terms, one may think of a pattern web as a weighted adjacency structure on a set of pattern-indices, where the weight on an arrow $p \to q$ summarizes "how much" support for p tends to induce, propagate, or transform into support for q. The choice of a quantale V allows these weights and supports to be more expressive than ordinary real-valued probabilities: in particular, V can encode graded truth, graded evidence, multi-aspect valuation, or even paraconsistent combinations of support and counter-support, while still providing a disciplined algebra (join, multiplication, order) for composing influences along paths and aggregating multiple sources of reinforcement.

A central theme in what follows is that the above informal notions can be separated into two interacting ingredients: (i) a *structural* ingredient, namely the pattern web itself as a V-relation capturing potential channels of influence; and (ii) a *dynamical* ingredient, namely the operators that update pattern-support states in response to that structure (and to possible external inputs). The "tendency to take habits" will be modeled by update maps that, in a precise order-theoretic sense, consolidate and amplify frequently traversed channels of support, so that repeated activation makes certain state configurations easier to re-enter. The "tendency to reverse habits" will be modeled by companion updates that weaken, invert, or otherwise reorganize these consolidations under suitable triggers, so that entrenched trajectories can become unstable and alternative trajectories can become accessible. The concept of "self-weaving webs" will be treated as the case where the web is not fixed in advance but is itself an evolving object influenced by the very states whose propagation it shapes, yielding a feedback loop between representation (edges) and activity (support values).

Remark 517 (Notation and reading conventions for this section)**.** *We will repeatedly use function spaces of the form V^P, meaning the set of all functions $s : P \to V$ (pattern-support states). When* *we write inequalities such as $s \leq s'$ for $s, s' \in V^P$, this is always the* pointwise *order: $s \leq s'$ iff $s(p) \leq s'(p)$ for all $p \in P$. Similarly, when we write $\oplus$ between two states (or two inputs) we mean the pointwise join induced by the quantale join in V. It is often useful to keep in mind the two complementary readings of the pointwise order: $s \leq s'$ can mean that s' contains at least as much support as s for every pattern, but it can also be read operationally as "s' is reachable from s by adding support and never removing it" (relative to the chosen order on V). This second reading is especially convenient when we treat habit-formation operators as inflationary or extensive maps on V^P, since then $s \leq H(s)$ expresses that an update H can only accrete (or at least not diminish) support, thereby formalizing a minimal sense in which repeated application stabilizes configurations.*

Throughout this section:

- V denotes a commutative quantale. For the concrete toy instantiations we use the p-bit quantale from Section 3.4. In particular, the commutativity assumption ensures that the "combination" of independent support flows (encoded via the quantale multiplication) does not depend on an arbitrary ordering of factors, which matches the intuition that multiple reinforcing sources can be fused without privileging a sequence.
- A *pattern class* is a set P of pattern-indices (abstract "pattern types"). In applied settings P can be a finite or countable subset of the space of candidate patterns from Section 9 (see also the pattern-web discussion in [5]). The abstraction here is deliberate: we only require that elements of P can be used as coordinates for states and as nodes in a directed V-graph, without committing to whether a "pattern" is a perceptual feature, a symbolic structure, a program, a hypothesis, a dynamical attractor, or some hybrid object.
- A *pattern-support state* is a function $s : P \to V$ assigning each pattern a graded (possibly paraconsistent) support value. In this formalism, a single state s may simultaneously encode multiple interacting kinds of information (e.g. evidence-for and evidence-against, or different modalities of support) as long as these are represented in the chosen quantale; the ensuing update dynamics then become a principled way of turning the "static" valuations in s into a time-evolving propagation process along the edges of the web.

The rest of the section will repeatedly move between three closely related objects: (i) a V-relation on P (the web), (ii) a state $s \in V^P$ (the current distribution of support over patterns), and (iii) a state transformer $H : V^P \to V^P$ (a habit operator). One reason for emphasizing this triad is that each of the Hyperseed notions under discussion can be cast as a constraint on, or a construction from, one of these objects: habit-taking concerns the algebraic and order-theoretic properties of H; self-weaving concerns rules for updating the underlying V-relation using s (and possibly also H); and morphic resonance concerns how multiple such triples (P, V, web) interact via coupling maps that transfer support across contexts. When we later speak of "reinforcement" or "anti-reinforcement," the formal interpretation will be that the relevant update either increases support along certain channels (via joins and multiplicative composition in V) or decreases it relative to other channels (e.g. by redirecting flow, introducing inhibitory edges, or applying an antagonistic coupling), always in a way that can be stated in the language of monotone maps, adjunctions, closure-like operators, or their duals.

12.1 From Hyperseed's probability phrasing to V-valued dynamics

Hyperseed's informal definitions of habit-taking and habit-reversal are stated in terms of probabilities of patterns occurring before versus after times T. To connect this to the paraconsistent core we separate two issues:

1. how to represent "pattern occurs" in a paraconsistent way; and

2. how to extract a scalar "occurrence propensity" from that representation when needed.

Definition 142 (p-bit occurrence evidence and scalarization)**.** *Assume the* p-bit *semantics of Section 3.2. A pattern-support value is* $s(p) = (s^+(p), s^-(p)) \in [0,1]^2$*, where* $s^+(p)$ *is positive evidence for "p occurs" and* $s^-(p)$ *is negative evidence.*

A scalarization *is a function* $\pi : V \to [0,1]$ *used when we want a single number comparable across time. For the* p-bit *quantale we will use one of:*

$$\pi_{\text{pos}}(v^+, v^-) = v^+, \qquad \pi_{\text{net}}(v^+, v^-) = \max(0,\, v^+ - v^-).$$

Remark 518 (Intuition: two-channel evidence and why scalarization is optional)**.** *A* p-bit *value* (v^+, v^-) *should be read as a compact record of two distinct pressures: how much the context is being pushed to affirm occurrence, and how much it is being pushed to deny it. This is the paraconsistent posture: we do not demand that the world (or the observer's record of it) collapse into a single consistent bit. One may view this as a formal version of the practical epistemic condition "I have reasons for and against," without forcing premature reconciliation (cf. the paraconsistent perspective developed in Section 3 and related work such as [23, 24]).*

Scalarization π *is then a* choice of lens*. For example, if* $s(p) = (0.7, 0.6)$ *then* $\pi_{\text{pos}}(s(p)) = 0.7$ *declares "there is strong positive pull," whereas* $\pi_{\text{net}}(s(p)) = 0.1$ *declares "net inclination is weak because conflict is strong." Neither is "the truth"; each is a deliberately chosen compression of a richer state, useful for connecting to probability-style statements about averages over time.*

Remark 519. π_{pos} *matches the simplest reading of Hyperseed's "probability of occurrence."* π_{net} *treats strong contradictory evidence as reducing effective occurrence propensity, which is often convenient when modeling anti-resonance or habit reversal as "net suppression."*

Definition 143 (Observer-indexed occurrence process)**.** *Fix a context/observer* C *and a pattern class* P*. A* (discrete-time) occurrence process *in* C *is a sequence*

$$s_0, s_1, s_2, \dots \quad \textit{with} \quad s_t : P \to V.$$

Given a scalarization π*, define the scalar occurrence propensity*

$$p_t(p) := \pi(s_t(p)) \in [0,1].$$

Remark 520 (Intuition and a minimal example)**.** *The sequence* $(s_t)_{t\geq 0}$ *is simply "how supported is each pattern, at each time, in this context?" For* $P = \{p\}$ *a singleton, this reduces to a one-dimensional time series in* V*; in the* p-bit *case, it is two coupled scalar time series* $(s_t^+(p), s_t^-(p))$*. For instance, repeated exposure might drive* $(s_t^+(p), s_t^-(p))$ *upward in the first coordinate while leaving the second roughly constant, reflecting growing positive evidence without resolving doubt.*

The value of formalizing this as a function $s_t : P \to V$ *is that we can later propagate support through a web (via a* V*-relation) and reason about global properties like monotonicity and fixed points using order theory, rather than relying on ad hoc verbal metaphors.*

Definition 144 (Tendency to take habits and tendency to reverse habits)**.** *Fix a distribution* ν *over a class* P *of patterns, a distribution* τ *over times* $T \in \{0, 1, 2, \dots\}$*, and a scalarization* π*. Define, for each* $p \in P$ *and time* T*,*

$$\mathrm{Pre}_T(p) := \mathbb{E}[\, p_t(p) \mid t < T\,], \qquad \mathrm{Post}_T(p) := \mathbb{E}[\, p_t(p) \mid t > T\,],$$

whenever these conditional expectations are defined.

The context C manifests the tendency to take habits *(relative to (P, ν, τ, π)) if*

$$\mathbb{E}_{p\sim\nu,\, T\sim\tau}\big[\mathrm{Post}_T(p) - \mathrm{Pre}_T(p)\big] > 0.$$

It manifests the tendency to reverse habits *if the above expectation is < 0.*

Remark 521 (Intuition and why the quantifiers matter)**.** *The quantities $\mathrm{Pre}_T(p)$ and $\mathrm{Post}_T(p)$ compare average propensities before and after a sampled "cut" time T. The outer expectation over $p \sim \nu$ and $T \sim \tau$ makes the definition intentionally meta-statistical: one is not declaring that* every *pattern increases, but that there is a systematic bias in the time direction on average, relative to a chosen ensemble of patterns and time windows.*

A simple example is the following. Suppose P is finite and for each $p \in P$ the scalar propensity is $p_t(p) = \min(1, a_p + bt)$ with $b > 0$. Then for any reasonable τ the average after T will exceed the average before T, yielding habit-taking. Conversely, exponential forgetting $p_t(p) = c_p\lambda^t$ with $0 < \lambda < 1$ yields habit-reversal.

Remark 522. *Definition 144 is intentionally "meta-statistical": it measures a bias in time-directional change averaged over a chosen pattern class and time sampling scheme. This matches Hyperseed's wording ("on average over patterns P in class C, and times T") while making the indexing explicit.*

Lemma 3 (Monotone time series imply habit-taking; antitone imply reversal)**.** *Assume $p_t(p)$ is nondecreasing in t for every $p \in P$. Then for every T and p for which the conditional expectations exist,*

$$\mathrm{Post}_T(p) \geq \mathrm{Pre}_T(p).$$

Consequently, the tendency-to-take-habits expectation in Definition 144 is ≥ 0.

If instead $p_t(p)$ is nonincreasing in t for every $p \in P$, then $\mathrm{Post}_T(p) \leq \mathrm{Pre}_T(p)$ and the tendency-to-reverse-habits expectation is ≤ 0.

Remark 523 (What the lemma is saying, in plain terms)**.** *If every pattern's scalar propensity is drifting upward over time, then the "after" average cannot be below the "before" average: the future is at least as supportive as the past, in this coarse statistical sense. Likewise, if everything drifts downward, then the future is (on average) less supportive. The lemma is not deep; its importance is methodological: it reduces a philosophical notion (habit-taking as a time-asymmetric tendency) to a verifiable monotonicity property of the time-indexed propensities.*

Remark 524 (How this connects forward)**.** *Later in this section we will define concrete update operators $s_{t+1} = \mathsf{Upd}(s_t)$. Proposition 17 will show these operators are monotone (order-preserving), and in inflationary regimes the induced scalarized series $p_t(p)$ often becomes monotone as well, giving an immediate bridge from dynamics to the habit-taking criterion of Definition 144.*

Remark 525 (On the role of conditional expectations)**.** *The lemma is phrased pointwise in (T, p): whenever $\mathrm{Pre}_T(p)$ and $\mathrm{Post}_T(p)$ are defined (as conditional expectations over the "before" and "after" portions of the time axis, conditioned on the cut time T and the pattern p), the stated inequality holds. In particular, no distributional assumptions are needed beyond existence: neither stationarity nor independence of the time series is invoked, and the inequality is not an "in expectation" statement but an inequality between two conditional averages computed from the same underlying monotone sequence. This is why the conclusion survives the subsequent outer averaging over (T, p) in Definition 144: the sign is already determined at the conditional level.*

Remark 526 (Finite-window intuition as a special case)**.** *Although* $\mathrm{Pre}_T(p)$ *and* $\mathrm{Post}_T(p)$ *may be defined via semi-infinite averages (or other conditional averages, depending on the model), the lemma is already evident in the finite-window case: if one defines, for integers* $m, n \geq 1$*,*

$$\mathrm{Pre}_{T,m}(p) = \frac{1}{m}\sum_{t=T-m}^{T-1} p_t(p), \qquad \mathrm{Post}_{T,n}(p) = \frac{1}{n}\sum_{t=T}^{T+n-1} p_t(p),$$

then monotonicity implies every term in the right block dominates every term in the left block, hence $\mathrm{Post}_{T,n}(p) \geq \mathrm{Pre}_{T,m}(p)$*. Any limiting or conditional-expectation definition of* $\mathrm{Pre}_T(p)$ *and* $\mathrm{Post}_T(p)$ *that is compatible with such windowed approximations inherits the same inequality whenever the limits exist.*

Proof. If a sequence is nondecreasing, then the average of its terms after T is at least the average of its terms before T (whenever these averages exist). Apply this pointwise to each pattern p, then average over p and T. The nonincreasing case is analogous.

For completeness, one can spell out the elementary comparison in the common case of block averages: if $a_0 \leq a_1 \leq a_2 \leq \cdots$, then for any indices $i < j$ we have $a_i \leq a_j$. Hence if I is any finite set of indices all $< T$ and J is any finite set of indices all $\geq T$, then

$$\frac{1}{|I|}\sum_{t\in I} a_t \;\leq\; \frac{1}{|J|}\sum_{t\in J} a_t,$$

since every term in the left sum is bounded above by every term in the right sum. Taking $a_t := p_t(p)$ gives Pre $\leq$ Post for these approximating averages; passing to the chosen conditional-expectation notion yields $\mathrm{Pre}_T(p) \leq \mathrm{Post}_T(p)$ whenever the conditional expectations exist. □

Proof sketch. The proof uses only the elementary fact that if $a_0 \leq a_1 \leq a_2 \leq \cdots$, then any average of later terms is at least any average of earlier terms, because later terms dominate earlier ones pointwise. We apply this to each pattern's propensity series $p_t(p)$ and then use linearity of expectation to lift the inequality through the outer averaging over p and T.

Equivalently, one may view $\mathrm{Pre}_T(p)$ and $\mathrm{Post}_T(p)$ as two Cesàro-type functionals applied to the left and right tails of the same monotone sequence. Monotonicity forces the right tail to lie above the left tail, and any averaging functional that is monotone with respect to pointwise order preserves this comparison. □

Remark 527 (Commentary and intuition)**.** *One may picture the propensity sequence for a fixed* p *as a curve on the unit interval. If the curve never goes down, then any horizontal "cut" at time* T *separates smaller values (to the left) from larger values (to the right), so the right-side mean dominates the left-side mean. The lemma is thus a formal version of a simple geometric picture: monotone curves have a built-in temporal bias.*

Remark 528 (When the inequality is strict (and when it is not))**.** *The lemma only asserts a weak inequality because constant stretches carry no directional information. If* $p_t(p)$ *is strictly increasing on a set of times with nonzero weight in the definition of* $\mathrm{Pre}_T(p)$ *and* $\mathrm{Post}_T(p)$ *(for example, if there exist indices* $t < T \leq t'$ *with* $p_t(p) < p_{t'}(p)$ *that are sampled with positive probability), then typically one obtains* $\mathrm{Post}_T(p) > \mathrm{Pre}_T(p)$*. Conversely, if the series is eventually constant (or if* T *is chosen so that both "before" and "after" averages sample only a flat region), then equality is expected and corresponds to a neutral "no-habit-signal" situation under the tendency statistic.*

Lemma 3 reduces the Hyperseed habit notions to verifiable monotonicity properties of a dynamical update rule. We now provide such update rules.

12.2 Pattern webs as V-relations and habit update operators

Definition 145 (V-valued reinforcement relation (pattern-flow kernel)). *Let P be a pattern class. A* reinforcement relation *(or* pattern-flow kernel*) on P is a V-valued relation*

$$A : P \times P \to V.$$

Intuitively, $A(p, q)$ measures how strongly the presence of pattern p tends to increase the support for pattern q at the next time step, within a given context.

Remark 529 (Intuition: a weighted directed graph of support flow). *Think of P as nodes in a directed graph. Then $A(p, q)$ is the weight on the arrow $p \to q$, but with weights living in a quantale V rather than in $\mathbb{R}$. In the* p-bit *toy case, $A(p, q) = (0.9, 0.1)$ might be read as: "when p is present, it provides strong positive pressure toward q, and slight negative pressure against q as well." That is: causal/associative support is itself permitted to be ambivalent.*

This is useful because it lets us treat pattern webs (Hyperseed-Concept 132) as algebraic objects whose propagation can be computed and iterated, connecting the informal "web" metaphor to explicit operators (compare the broader pattern-network picture in [5]).

Remark 530 (Categorical reading: A as a V-relation in $\mathbf{Rel}_V$). *It is often helpful to view A as a morphism in the quantale-enriched category of V-relations: objects are sets (or classes) like P, and a morphism $P \to P$ is exactly a function $P \times P \to V$. Under this reading, the operator $s : P \to V$ is a "V-valued predicate" (a V-subset) on P, and $A \star s$ is the standard action of a V-relation on a V-predicate (an enriched direct image). This perspective makes later constructions feel less ad hoc: one is simply using the native algebra of V-enrichment to push support forward along edges.*

Definition 146 (Relational image operator). *Given $A : P \times P \to V$ and a state $s : P \to V$, define the* image *(or one-step propagation)*

$$(A \star s)(q) := \bigoplus_{p \in P} A(p, q) \otimes s(p).$$

Here $\oplus$ and $\otimes$ are the quantale join and product; the join is taken in V.

Remark 531 (Well-definedness when P is large). *The expression $\bigoplus_{p \in P}$ uses arbitrary joins in V. Thus, when P is infinite, the definition still makes sense provided the quantale is complete (as assumed) and we are working at a set-theoretic level where the join over P is legitimate. In applications one often has either (i) P finite, (ii) s with finite support, or (iii) a sparsity/decay condition on A ensuring only "effectively many" p contribute materially to each q.*

Remark 532 (Notation unpacking: what $\star$, $\oplus$, and $\otimes$ do). *The symbol $\star$ is not new structure; it is simply a name for the standard "relational image" construction induced by the quantale operations. For each target node q, we look at all sources p, combine the edge strength $A(p, q)$ with the source support $s(p)$ using $\otimes$, and then aggregate all these contributions using $\bigoplus$ (the join in V). When $V = [0, 1]^2$ with componentwise max as $\oplus$ and componentwise multiplication as $\otimes$, this says: "q receives the strongest (componentwise) multiplicative contribution from any predecessor."*

In a finite example with $P = \{p_1, p_2\}$, we have

$$(A \star s)(q) = \big(A(p_1, q) \otimes s(p_1)\big) \oplus \big(A(p_2, q) \otimes s(p_2)\big),$$

which is a quantale-analogue of summing influences, except that we use the join operation appropriate to the chosen notion of graded truth/evidence.

Remark 533. ***Special cases: Boolean and*** $[0,1]$***-valued kernels*** *If* $V = \{\bot, \top\}$ *with* $\oplus = \vee$ *and* $\otimes = \wedge$*, then* A *is an ordinary binary relation on* P*,* s *is a subset indicator, and* $(A \star s)(q) = \top$ *exactly when there exists some* $p \in P$ *with* $s(p) = \top$ *and* $A(p,q) = \top$*. In other words,* $A \star s$ *becomes one-step reachability along directed edges. If* $V = [0,1]$ *with* $\oplus = \max$ *and* $\otimes = \cdot$*, then* $A \star s$ *computes a "max-product" propagation common in fuzzy relation equations and certain log-domain message-passing schemes; the present setup can be read as a multi-channel generalization of that idea.*

Remark 534. *If* P *is finite,* A *can be viewed as a* $|P| \times |P|$ *matrix over a quantale semiring, and* $A \star s$ *is the usual matrix-vector multiplication with* $\oplus$ *as addition and* $\otimes$ *as multiplication (Section 3.4).*

Remark 535 (Monotonicity: propagation respects the V-order)**.** *Because* $\otimes$ *is monotone in each argument and* $\bigoplus$ *is a join, the map* $s \mapsto A \star s$ *is monotone with respect to the pointwise order on* V^P*. Concretely, if* $s \leq t$ *pointwise, then* $(A \star s) \leq (A \star t)$ *pointwise. This is one of the reasons quantales are a convenient ambient algebra: order-theoretic properties needed for iterating updates (e.g. existence of least fixed points) come for free under mild additional assumptions.*

To allow both habit formation and habit decay (reversal), we include an explicit "persistence/forgetting" factor.

Definition 147 (Persistence/decay scalar)**.** *Let* $d \in V$*. Given a state* $s : P \to V$*, define the pointwise scaled state*

$$(d \odot s)(p) := d \otimes s(p).$$

For the p-bit *quantale* $V = [0,1]^2$ *with* $\otimes$ *given by componentwise multiplication, choosing* $d = (d^+, d^-)$ *with* $d^+, d^- \in [0,1]$ *scales down both positive and negative evidence.*

Remark 536 (Intuition: persistence as inertia, decay as forgetting)**.** *The role of* d *is to encode what remains of the old support if nothing else happens. If* $d = e$ *(the quantale unit), then* $d \odot s = s$ *and the past persists perfectly. If* d *is strictly smaller (in an appropriate scalar sense), then the past is discounted, capturing forgetting or habit reversal by attrition.*

In the p-bit *toy case with componentwise multiplication, taking* $d = (0.95, 0.95)$ *is literal exponential decay applied symmetrically to both evidence channels. One can also imagine asymmetric decay, e.g.* $(0.95, 0.7)$*, representing a context that retains positive evidence longer than negative evidence (or vice versa).*

Remark 537 (On choosing d: timescales and stability heuristics)**.** *Heuristically,* d *sets a memory timescale: values closer to the unit* e *correspond to long persistence, while values farther from* e *enforce faster turnover. In many concrete instantiations (e.g.* $V = [0,1]$ *or* $V = [0,1]^2$ *with multiplicative* $\otimes$*), choosing* d *strictly below* e *also prevents indefinite accumulation from persistence alone and makes it easier for new input* u_t *to reshape the state, which matches the intended "habit can change" reading.*

Definition 148 (Habit dynamics operator)**.** *Fix a reinforcement relation* $A : P \times P \to V$ *and a persistence/decay scalar* $d \in V$*. Let* $u_t : P \to V$ *be an external input sequence.*

Define the update operator $\mathsf{Upd}_{A,d,u_t} : V^P \to V^P$ *by*

$$\mathsf{Upd}_{A,d,u_t}(s) := u_t \oplus (d \odot s) \oplus (A \star s),$$

where $\oplus$ *on the right-hand side is applied pointwise in* V^P*.*

A habit dynamics *in a context is any process satisfying*

$$s_{t+1} = \mathsf{Upd}_{A,d,u_t}(s_t).$$

Remark 538 (Interpretation: superposition as join, not arithmetic addition)**.** *The use of $\oplus$ (join) rather than a numeric sum is deliberate: the update aggregates contributions according to the evidence-combination logic encoded by V. In max-like quantales, this corresponds to a "winner-take-strongest" or "most-supported" aggregation; in other quantales, $\oplus$ can encode alternative combination rules (e.g. logical disjunction, cost minimization, or other idempotent semiring behaviors). Thus, although the formula resembles a linear recurrence, it is best read as a generic evidence propagation rule parametrized by the choice of quantale semantics.*

Remark 539 (Iterating the operator: trajectories and equilibria)**.** *Given an initial state s_0, the dynamics defines a trajectory $s_0, s_1, s_2, \dots$ by repeated application of Upd_{A,d,u_t}. When $u_t = u$ is time-independent, one can ask for a (contextual) equilibrium state s^* satisfying*

$$s^* = u \oplus (d \odot s^*) \oplus (A \star s^*),$$

i.e. a fixed point of the update. Because the update is built from monotone operations, fixed-point methods (least/greatest fixed points, Kleene iteration) are natural tools when additional conditions ensuring convergence are available in the chosen V and in the scale of P.

Remark 540 (Intuition: three sources of support)**.** *The update is the join of three contributions: (i) u_t, the new stimulation (what the world, body, or other subsystems inject at time t); (ii) $(d \odot s)$, persistence (what the system carries forward from its own past); and (iii) $(A \star s)$, internal propagation through the pattern web. This is deliberately schematic: it abstracts the idea that habit is neither purely memory nor purely current stimulus, but a synthesis in which the web's structure re-expresses and amplifies what is already present.*

A minimal worked toy example: let $P = \{p, q\}$ and suppose u_t stimulates only p. If $A(p, q)$ is large, then even with $u_t(q) = \mathbf{0}$ the pattern q can grow over time because it is downstream of p. Thus "habit-taking" can appear as a derived phenomenon: repeated stimulation of one pattern recruits others via the web.

Remark 541 (Reading $A \star s$ as self-weaving reinforcement)**.** *The term $A \star s$ formalizes the "self-weaving" aspect of a pattern web: whenever a pattern becomes supported, it can in turn become a source of support for other patterns according to the context-specific kernel A. In this sense, A is not merely a static adjacency structure; it is a learned or given bias about which co-activations tend to follow which other activations, and the update equation makes that bias operational as a propagating field over P.*

Remark 542 (About $\mathbf{0}$ and "no input")**.** *When we write $\mathbf{0}$ for a relation or state, we mean the bottom element in the appropriate function space: for a state $s : P \to V$, $\mathbf{0}$ is the constant function $p \mapsto \bot_V$ (the bottom of V); for a relation $A : P \times P \to V$, $\mathbf{0}$ is the constant relation $(p, q) \mapsto \bot_V$. In the* p-bit *toy quantale, $\bot_V = (0, 0)$.*

It is worth emphasizing the semantic distinction between absence of evidence *and* negative evidence*: writing $\mathbf{0}$ encodes the former, i.e. that no support is present (or no coupling is present) in the sense of the order of V. This is not the same as assigning a special "inhibitory" value unless such inhibition is explicitly represented by elements of V and their order (in which case the bottom element would still be the least inhibitory/least supportive option, not an additional kind of signal). In particular, "no input" means that the exogenous drive term contributes the least possible value everywhere, not that the node is removed from P or that the dynamics is undefined.*

Concretely, if the update rule combines terms by $\oplus$ (a join), then using $\mathbf{0}$ behaves as an identity for "adding nothing" in the lattice-theoretic sense: $x \oplus \bot_V = x$. Likewise, using $A = \mathbf{0}$ means no propagation through the web, since every path contribution is uniformly bottom and hence cannot raise any join.

Remark 543 (Special cases)**.** • Pure reinforcement (no forgetting)*: $d = e$ and $u_t \equiv \mathbf{0}$ (bottom element), so $s_{t+1} = s_t \oplus (A \star s_t)$. This is inflationary and models cumulative habit formation. In this case, any node can only keep or increase its current level of support, and the only source of further increase is the propagation term $A \star s_t$. If one iterates this update from a given initial seed, one obtains an increasing chain of states (with respect to the pointwise order) whose growth is controlled entirely by the connectivity pattern encoded in A.*

- Pure decay (no reinforcement)*: $A = \mathbf{0}$ and $u_t \equiv \mathbf{0}$, so $s_{t+1} = d \odot s_t$. If d is strictly below the unit in an appropriate scalar sense, this models habit reversal via forgetting. Here the dynamics factorizes across nodes: each $s_t(p)$ evolves independently by repeated scaling, so any nontrivial long-range structure disappears and the web plays no role. This clarifies why forgetting alone cannot create new tendencies: it can only attenuate what is already present.*

- Driven dynamics*: nontrivial u_t models repeated external stimulation; the long-run effect depends on how A propagates and amplifies the input. In particular, even if u_t is localized in P (e.g. non-bottom on a small set of nodes), repeated application of the update can spread that support along edges where A is non-bottom, creating a broader pattern. Depending on whether decay dominates reinforcement, the result may settle to a stable pattern, oscillate in response to time-varying u_t, or keep accumulating if the update is inflationary.*

Proposition 17 (Monotonicity of the habit update)**.** *For fixed A and d, the map $s \mapsto \mathsf{Upd}_{A,d,u}(s)$ is monotone in s (order-preserving). It is also monotone in u and in A (pointwise order on relations).*

Remark 544 (Meaning and why it matters)**.** *This proposition says: if you start with more support, you cannot get less support after one update; likewise, stronger input and stronger couplings cannot decrease the next state. It is a minimal coherence condition for a reinforcement-style model. Without it, one could not reliably connect structural properties of A and u to the emergent time-bias criteria of Definition 144, nor could one safely apply standard fixed-point tools later in the section.*

Monotonicity is also the bridge to closure constructions: order-preserving inflationary operators are precisely the terrain in which Knaster–Tarski/Kleene style least-fixed-point reasoning becomes available, a theme that will recur in other Hyperseed stability analyses [10].

A further reason monotonicity matters is interpretability under parameter changes. If $A \leq A'$ pointwise, then any explanation phrased as "increasing a coupling" becomes unambiguous: the model guarantees that such a change cannot reduce *downstream support in the next step, so comparative statics can be done at the level of the order on V. Similarly, if $u \leq u'$, then increasing the external drive is guaranteed not to suppress any node, which matches the intended "reinforcement" reading of $\oplus$ as accumulation rather than competition.*

Proof. Each component of $\mathsf{Upd}_{A,d,u}(s)$ is built from $\oplus$ and $\otimes$ and a join over $p \in P$. In a quantale, $\otimes$ is monotone in each argument and distributes over arbitrary joins, and $\oplus$ is a join hence monotone. Therefore increasing s, u, or A can only increase the result.

More explicitly, the relevant order is the pointwise order on functions $P \to V$ and $P \times P \to V$: we write $s \leq s'$ iff $s(p) \leq s'(p)$ for all $p \in P$, and $A \leq A'$ iff $A(p,q) \leq A'(p,q)$ for all $p, q \in P$. Fix $p_0 \in P$. The value $\mathsf{Upd}_{A,d,u}(s)(p_0)$ is obtained by combining (via $\oplus$) contributions such as $(d \odot s)(p_0)$, $u(p_0)$, and the propagated term $(A \star s)(p_0) = \bigoplus_{p \in P} A(p_0, p) \otimes s(p)$. If $s \leq s'$, then for each p we have $A(p_0,p) \otimes s(p) \leq A(p_0,p) \otimes s'(p)$ by monotonicity of $\otimes$ in its second argument, hence taking the join over p preserves the inequality, yielding $(A \star s)(p_0) \leq (A \star s')(p_0)$. The same pointwise reasoning applies to $u \leq u'$ (monotonicity of $\oplus$ in the corresponding argument) and to $A \leq A'$ (monotonicity of $\otimes$ in its first argument), establishing monotonicity in all stated parameters. □

Proof sketch. The update operator is assembled from monotone building blocks. Pointwise, it is a join of three terms. Joins are monotone; $\otimes$ is monotone in each argument; and the $\bigoplus_{p\in P}$ aggregation preserves monotonicity because it is itself a join. Therefore any pointwise increase in s, u, or A propagates through the formula without creating decreases.

Equivalently, one can view $\mathsf{Upd}_{A,d,u}$ as a composition of monotone maps between complete lattices: first apply scalar/pointwise transformations (such as $s \mapsto d \odot s$), then apply the linear-propagation map $s \mapsto A \star s$, and finally merge all contributions by $\oplus$. Since the composition of monotone maps is monotone, the whole update is monotone. $\square$

Remark 545 (Visual intuition)**.** *If one imagines the web as channels that can only transmit or accumulate support (in the sense encoded by V), then "adding" an extra amount of support at any node cannot rationally lead to less support downstream if the update consists only of propagation and accumulation. The proposition is the algebraic reflection of this picture.*

The same picture applies to the parameters: increasing an edge weight in A corresponds to widening a channel, and increasing u corresponds to injecting additional support at some nodes. If the dynamics had subtractive or normalizing steps, such monotonicity could fail; the present formulation avoids that by restricting to the order-theoretic primitives $\oplus$, $\otimes$, and joins.

Monotonicity (Proposition 17) is a minimal sanity property: stronger present support or stronger couplings cannot lead to weaker next-step support. In particular, it ensures that iterating the update from $\mathbf{0}$ yields a well-defined increasing approximation process whenever the update is also inflationary, setting up the later use of least fixed points as canonical "habit closures" of a seed state.

12.3 Self-weaving webs and autocatalytic closure

Hyperseed's "self-weaving web" notion is meant to capture a pattern network whose own internal structure reinforces and recreates itself. In our setting, this becomes a fixed-point and closure story for the propagation operator.

Definition 149 (One-step closure operator)**.** *Fix $A : P \times P \to V$. Define the inflationary operator*

$$\mathsf{Cl}_A(s) := s \oplus (A \star s).$$

Remark 546 (Intuition: adding what the web can infer/propagate in one step)**.** $\mathsf{Cl}_A(s)$ *takes a support state and enriches it by one round of internal propagation. If s is a seed, then $(A \star s)$ is what the web can "grow" from that seed in one step, and the join $\oplus$ records that we keep the original seed support as well. This is the basic algebraic gesture behind "weaving": we do not replace the old pattern; we overlay it with its consequences.*

Definition 150 (Iterated closure and Kleene star)**.** *Define* $\mathsf{Cl}_A^{(0)}(s) := s$ *and* $\mathsf{Cl}_A^{(n+1)}(s) := \mathsf{Cl}_A(\mathsf{Cl}_A^{(n)}(s))$.

When V^P is a complete lattice (e.g. P finite and V complete), define the limit (join of the ascending chain)

$$\mathsf{Cl}_A^{(\omega)}(s) := \bigoplus_{n\geq 0} \mathsf{Cl}_A^{(n)}(s).$$

Also define powers of A by relational composition: $A^0 := I$ (identity relation) and $A^{n+1} := A^n \circ A$. Then define the Kleene star

$$A^* := \bigoplus_{n\geq 0} A^n.$$

Remark 547 (Notation unpacking: A^n and A^* as "all finite paths")**.** *The composition A^n is the n-step reinforcement relation: $A^2(p,r)$ aggregates reinforcement along all two-edge paths $p \to q \to r$, and so on. Thus A^* is the join of all A^n, i.e. the aggregate relation capturing reinforcement along* any finite chain*. This is the same algebraic pattern behind Kleene star in automata theory and Kleene algebras, here expressed in quantale language.*

Accordingly, $\mathsf{Cl}_A^{(\omega)}(s)$ is the seed s closed under repeated propagation: what you get by allowing the web to "echo" its own implications arbitrarily many times, without invoking any infinitary paths beyond the join already present in the lattice structure.

Theorem 12 (Closure as least fixed point; explicit A^* form)**.** *Assume P is finite and V is a commutative quantale. Then for every seed state $s : P \to V$:*

1. *The chain $\mathsf{Cl}_A^{(0)}(s) \le \mathsf{Cl}_A^{(1)}(s) \le \cdots$ is ascending and has a join $\mathsf{Cl}_A^{(\omega)}(s)$.*
2. *$\mathsf{Cl}_A^{(\omega)}(s)$ is a fixed point of Cl_A, and is the least fixed point of Cl_A above s.*
3. *The fixed point can be written as*
$$\mathsf{Cl}_A^{(\omega)}(s) = A^* \star s.$$

Remark 548 (Plain-English statement and why it is central here)**.** *Start with some initial support s. Apply one-step closure, then close again, and so on. Because closure only adds support (never subtracts), this produces an ascending sequence. The theorem says there is a well-defined "limit" state obtained by joining all these stages, and that this limit is exactly the smallest self-consistent state containing the seed: once you reach it, closing again adds nothing new.*

This is important because it gives a rigorous meaning to "self-weaving" as a fixed-point phenomenon. Instead of narrating that the web "recreates itself," we can point to a precise object $\mathsf{Cl}_A^{(\omega)}(s)$ and identify it both as a least fixed point and as an explicit propagation construction $A^* \star s$.

Remark 549 (Connection to the rest of the document)**.** *Fixed-point characterizations of "stable organization" recur in later discussions of agency and goal stability. The present theorem is a local, pattern-web-scale instance of that general motif, akin in spirit to other fixed-point applications in Hyperseed analyses [10].*

Proof. (1) Cl_A is inflationary because $\mathsf{Cl}_A(s) = s \oplus (A \star s) \ge s$, so iterating yields an ascending chain. Since V^P is complete (finite product of complete lattices), the join exists.

(2) Monotonicity of Cl_A and completeness of V^P imply the set of fixed points is nonempty and forms a complete lattice (Knaster–Tarski). Because Cl_A is inflationary, the join of the iterates from s is a post-fixed point; a standard argument (Kleene fixed-point theorem in complete lattices for such algebraic operators) gives it is the least fixed point above s.

(3) Unfolding the recurrence shows $\mathsf{Cl}_A^{(n)}(s) = \bigoplus_{k=0}^{n} A^k \star s$. Taking $\bigoplus_{n\ge 0}$ and using distributivity of $\otimes$ over joins yields $A^* \star s$. □

Proof sketch. The operator Cl_A is inflationary and monotone, so iterating it from a seed produces an ascending chain. In a complete lattice, the join of this chain exists. Standard fixed-point theory says that for such operators, this join is the least fixed point above the seed. Finally, expanding $\mathsf{Cl}_A^{(n)}(s)$ shows it collects exactly the contributions along paths of length at most n, hence the ω-join corresponds to summing over all finite path lengths, which is precisely $A^* \star s$. □

Remark 550 (Key steps: why they work)**.** *The essential technical facts are (i) completeness of V^P when P is finite, (ii) monotonicity and inflationarity of Cl_A, and (iii) distributivity of*

$\otimes$ over joins (the quantale axiom). Inflationarity ensures we have an increasing approximation sequence; distributivity ensures we can commute propagation with taking joins, so that "propagate after joining" matches "join after propagating." This is why the explicit formula $A^* \star s$ appears: it is the algebraic normal form of "all finite chains of reinforcement."

Concretely, when P is finite and V is complete, the pointwise order makes V^P a complete lattice, so the iteration $s \le \mathsf{Cl}_A(s) \le \mathsf{Cl}_A^2(s) \le \cdots$ has a well-defined join $\bigoplus_{n\ge 0} \mathsf{Cl}_A^n(s)$. Monotonicity is what guarantees that these iterates are comparable and that joins of approximants behave as expected under further applications of Cl_A; in standard fixed-point terms, this is exactly the hypothesis one needs for a least fixed point construction in a complete lattice. In typical quantale semantics one can also unpack A^* as a Kleene-style star, $A^* = \bigoplus_{n\ge 0} A^{\otimes n}$ (with $A^{\otimes 0}$ the identity relation), so that the closed-form expression $A^* \star s$ is literally the join of all n-step propagations $A^{\otimes n} \star s$. The distributivity axiom is what lets this decomposition be pushed through the $\star$-action: it ensures that accumulating alternative routes of support (joins) is compatible with composing successive reinforcements (tensor products), so that the closure captures "all ways of getting there" rather than privileging a particular bracketing of compositions.

Remark 551 (Hyperseed reading)**.** *$A^* \star s$ is the total "downstream closure" of an initial seed s under the reinforcement relation A: it aggregates the support transmitted along all finite reinforcement chains. This is a precise sense in which repeated co-occurrence and associative structure can "weave" a larger support pattern from a smaller seed.*

Equivalently, if one defines $s_0 := s$ and $s_{n+1} := s_n \oplus (A \star s_n)$, then the increasing family $(s_n)_{n\ge 0}$ is an explicit approximation to the closure, and $A^* \star s$ is the join of all stages of this self-amplifying propagation. This makes clear that no single application of A is privileged: what matters is the accumulation of support across any finite number of reinforcement steps, including the possibility of revisiting intermediate patterns and thereby compounding evidence. In particular, if s concentrates mass on a small set of generators, then $A^* \star s$ describes the full footprint of what those generators can collectively activate, modulated by the weights in A and the way $\otimes$ combines successive transmissions.

Remark 552 (Geometric/graph intuition)**.** *In graph terms, $A^* \star s$ assigns to each node q the aggregate influence of all directed paths starting at any seed node, with path contributions combined multiplicatively along edges (via $\otimes$) and aggregated across alternative paths (via $\oplus$). If one pictures each edge as a conduit that attenuates and possibly distorts evidence, then A^* is the total field of influence generated by repeatedly sending signals through the network.*

One can read A^* as the weighted reachability relation: $(A^*)(q,p)$ is the total support by which p can be reached from q through any finite directed walk, with composition of edges evaluated by $\otimes$ and competition/branching between walks evaluated by $\oplus$. On this reading, $A^* \star s$ is a path-sum (or walk-sum) semantics: it is the analogue of taking a transitive closure in the crisp case, but with a quantitative accumulation law determined by the chosen quantale. This is also why cycles matter: feedback loops correspond to families of paths that revisit nodes, and the star construction is the algebraic device that folds all such finite unfoldings into a single closed expression.

Definition 151 (Autocatalytic (self-weaving) support)**.** *Fix a reinforcement relation $A : P \times P \to V$ and a threshold $\theta \in V$.*

*A nonempty subset $S \subseteq P$ is called θ-*autocatalytic* if there exists a state $s : P \to V$ such that:*

1. *(support on S) for all $p \in S$, $s(p) \ge \theta$; and*

2. *(self-production) for all $p \in S$, $(A \star s)(p) \ge \theta$.*

A pattern web (P, A) *is a* self-weaving web *(at threshold* θ*) if it contains a* θ*-autocatalytic subset.*

The two clauses separate "having enough of the patterns present" from "being able to reproduce that presence using the web's own reinforcement dynamics." The existential quantifier over s *is important: it says that there is at least one coherent way of assigning support levels so that every element of* S *is simultaneously above threshold and is simultaneously regenerated above threshold by the influence coming from* S *(and possibly from outside* S *if* s *has additional mass elsewhere). In applications one often considers* s *that is concentrated on* S *(e.g. an indicator-like state in crisp cases, or a uniform* θ*-level assignment on* S *in graded cases), but the definition is stated to allow any state witnessing the closure property.*

Remark 553 (Intuition and a simple example)**.** *This definition formalizes the "autocatalytic set" motif (Hyperseed-Concept 61): a set* S *of patterns is self-sustaining if, once* S *is supported above threshold, the web dynamics generated by* S *re-produces support above threshold for every member of* S*. In the simplest crisp case where* $V = \{0, 1\}$ *and* $\theta = 1$*, the condition says: each node in* S *has at least one incoming edge from some node in* S *(support can be regenerated from within the set).*

The usefulness is conceptual: it provides a minimal criterion for "the web can keep itself going" independent of external input. In later discussions of minds and systems, this is the pattern-web analogue of closure under production dependencies.

In graded settings (e.g. $V = [0, 1]$ *with a t-norm for* $\otimes$*), the same clauses express that each member of* S *receives enough aggregate reinforcement from the current support configuration to stay above* θ*. The condition does not force a unique witness state* s*; rather, it identifies sets* S *for which there exists at least one support configuration making* S *self-maintaining, and this flexibility is often what one wants when modeling multiple possible "operating points" of a system. Graphically, the crisp characterization can be strengthened in familiar ways (e.g. insisting on a directed cycle through each node, or on strong connectivity), and the present definition should be read as the quantitative analogue of such internal-regeneration conditions, expressed at the level of the* $\star$*-propagation operation.*

Remark 554. *Definition 151 is intentionally minimal and matches the "autocatalytic set" motif: each pattern in* S *is supported by the reinforcement generated from patterns in* S *itself. Stronger definitions can require minimality of* S*, robustness under perturbations, or explicit compositional generation mechanisms (Section 9).*

For example, one may ask that S *be inclusion-minimal among* θ*-autocatalytic sets (no proper subset has the same property), or that the inequalities hold with a margin above* θ *to ensure persistence under small decreases of support. One may also require that the witness state* s *be supported* only *on* S *(a stricter internal-closure reading), or that self-production be witnessed not just in one step* $(A \star s)$ *but as a post-fixpoint condition* $s \leq \mathsf{Cl}_A(s)$*, making the closure operator itself the object of stability. These refinements become relevant when one wants to distinguish mere possibility of self-support from dynamical attractiveness or from modular compositionality of the generating mechanisms.*

12.4 Morphic resonance and morphic anti-resonance as cross-context coupling

Hyperseed defines morphic resonance as habit-taking among spatially separated patterns, and morphic anti-resonance as the reverse. We model "spatial separation" abstractly by indexing multiple contexts (locations, subsystems, or reality-systems) and adding coupling kernels between their pattern webs. This formalization is mathematically orthodox (coupled dynamical systems) while leaving room for the broader interpretive reading of morphic resonance as a nonlocal "field of habit" in the sense associated with Sheldrake [13]. Concretely, "coupled dynamical systems" here means that each context has its own local state-update rule, but the rule is allowed to depend (via an explicit input term) on the current states of other contexts; the only "distance" notion needed is membership in distinct indices $\ell \in \mathcal{L}$. The intent is that the coupling term is written in exactly the same algebra as internal reinforcement, so that cross-context influence can be treated as another instance of the same habit-amplifying mechanism, rather than as an ad hoc external disturbance.

Definition 152 (Family of contexts and internal habit kernels)**.** *Let $\mathcal{L}$ be an index set of contexts (e.g. locations). For each $\ell \in \mathcal{L}$, fix:*

- *a pattern class P_ℓ;*
- *an internal reinforcement relation $A_\ell : P_\ell \times P_\ell \to V$;*
- *a decay scalar $d_\ell \in V$; and*
- *an external input process $u_{\ell,t} : P_\ell \to V$.*

A global state at time t is a tuple $s_t = (s_{\ell,t})_{\ell \in \mathcal{L}}$ with $s_{\ell,t} : P_\ell \to V$.

Remark 555 (Intuition: many webs at once)**.** *One can think of each $\ell \in \mathcal{L}$ as its own local pattern web with its own internal reinforcement. The global state s_t is simply the collection of all local support states at time t. This is the minimal formal move needed to discuss "at a distance": we do not need literal Euclidean distance, only a multiplicity of contexts and a notion of cross-context influence.*

Remark 556 (What is shared vs. what is local)**.** *Each context ℓ has its own pattern vocabulary P_ℓ and its own reinforcement operator A_ℓ, so contexts can differ in what they can represent and how quickly they self-reinforce. At the same time, all contexts share the same value object V (and thus the same $\oplus, \otimes, \odot$ operations), so that internal and external influences can be aggregated in a uniform way. This "shared algebra, distinct vocabularies" split is what makes it possible to discuss cross-context influence even when P_ℓ and $P_{\ell'}$ are not identical: the coupling kernels will serve as the translation layer.*

Definition 153 (Morphic coupling kernel)**.** *For $\ell \neq \ell'$, a* morphic coupling kernel *from ℓ' to ℓ is a V-relation*

$$K_{\ell' \to \ell} : P_{\ell'} \times P_\ell \to V.$$

Its image on a state $s_{\ell',t}$ is the induced input on P_ℓ:

$$(K_{\ell' \to \ell} \star s_{\ell',t})(q) := \bigoplus_{p \in P_{\ell'}} K_{\ell' \to \ell}(p, q) \otimes s_{\ell',t}(p).$$

Remark 557 (Intuition and examples of what $K_{\ell' \to \ell}$ can encode)**.** *The kernel $K_{\ell' \to \ell}$ plays the same mathematical role as A, but across contexts rather than within one context. In applications, $K_{\ell' \to \ell}(p, q)$ could encode: (i) similarity or "translation" between pattern vocabularies $P_{\ell'}$ and P_ℓ; (ii) a learned association that when p is active in one context, q tends to become active elsewhere; or (iii) a hypothesized nonlocal coupling. In the simplest same-pattern case where $P_{\ell'} = P_\ell = P$ and $K_{\ell' \to \ell}(p, q)$ is nonbottom only when $p = q$, the coupling just transmits support for a pattern to the same pattern-index in the other context.*

Remark 558 (Algebraic reading of the $\star$-action). *The expression $K_{\ell' \to \ell} \star s_{\ell',t}$ is formally the same type of construction as applying a weighted adjacency matrix to a state vector, except that the weights and aggregations are taken in the semiring-like structure V. Intuitively, the induced support for $q \in P_\ell$ is an $\oplus$-sum over all source patterns $p \in P_{\ell'}$ of "(strength of link from p to q) $\otimes$ (current support of p)." This makes it explicit that cross-context influence is graded and compositional: multiple distinct source patterns can contribute to the same target pattern, and their contributions combine by the same aggregator used everywhere else in the habit dynamics.*

Remark 559 (Directionality, reciprocity, and sparsity). *Nothing in Definition 153 requires $K_{\ell' \to \ell}$ to equal $K_{\ell \to \ell'}$ (even when $P_\ell = P_{\ell'}$), so the model allows directed coupling (influence from ℓ' to ℓ without a matching reverse channel). In many applications one may impose additional structure, e.g. reciprocity ($K_{\ell' \to \ell}$ related to $K_{\ell \to \ell'}$), boundedness (to prevent unrealistically large induced inputs), or sparsity (only a small subset of inter-context pattern pairs interact). These are modeling choices rather than requirements of the formalism.*

Definition 154 (Morphic resonance dynamics). *Given internal dynamics (A_ℓ, d_ℓ) and couplings $(K_{\ell' \to \ell})$, define the coupled update for each ℓ by*

$$s_{\ell,t+1} = u_{\ell,t} \oplus (d_\ell \odot s_{\ell,t}) \oplus (A_\ell \star s_{\ell,t}) \oplus \bigoplus_{\ell' \neq \ell} (K_{\ell' \to \ell} \star s_{\ell',t}).$$

We call this a morphic resonance dynamics *when the couplings $K_{\ell' \to \ell}$ are intended to encode cross-context propagation of habit bias, rather than ordinary local causal influence.*

Remark 560 (Intuition: one equation, two interpretations). *The equation is formally just "local update plus cross-input from others." The interpretive distinction is in what we take the coupling term to mean. If the coupling is ordinary causal influence, we are modeling a standard interacting multi-system process. If the coupling is "morphic," we are treating cross-context similarity itself as a conduit for habit, in the sense that a stabilized pattern in one place can seed stabilization elsewhere without a mediated causal chain in the usual physical sense. The latter reading is the one Hyperseed associates to morphic resonance (Hyperseed-Concept 115) and that is historically linked to morphic field narratives [13].*

Remark 561 (Time-variation and learning of couplings). *Although the definition treats $K_{\ell' \to \ell}$ as fixed, nothing prevents using time-indexed couplings $K_{\ell' \to \ell, t}$ to represent learned or adaptive cross-context links (e.g. strengthening of translation correspondences between vocabularies). Under such a reading, morphic resonance can be modeled either as a static "field" (fixed K) or as a co-evolving structure where repeated co-activation across contexts modifies the coupling itself, producing a second-order habit: a habit of coupling.*

Remark 562 (Why this captures "habit-taking at a distance"). *If a pattern p becomes increasingly supported in context ℓ' due to its internal habit dynamics, then the induced term $K_{\ell' \to \ell} \star s_{\ell',t}$ becomes increasingly supported (by Proposition 17), providing an increasing drive for corresponding patterns in ℓ. Thus the coupled system can realize the same before/after bias of Definition 144 across separated contexts.*

Remark 563 (A minimal two-context picture). *For $\mathcal{L} = \{\ell_1, \ell_2\}$ the update for ℓ_1 contains exactly one cross term $K_{\ell_2 \to \ell_1} \star s_{\ell_2,t}$ (and symmetrically for ℓ_2). In this simplest setting, one can interpret morphic resonance as the claim that "whatever becomes easier to realize in ℓ_2 becomes, via the kernel, easier to realize in ℓ_1", with the particular mapping from "what" to "what" specified by the nonbottom entries of $K_{\ell_2 \to \ell_1}$. This makes explicit that the phenomenon is not a single scalar effect but a structured transport of bias over a pattern vocabulary.*

To model morphic *anti*-resonance, we need a way to transport "opposition" or "reversal pressure" across contexts. In the p-bit setting, negation swaps positive and negative evidence, so we can define an "oppositional channel" by precomposing with negation.

Definition 155 (Anti-resonance via negated-source coupling (p-bit case))**.** *Assume the* p-bit *negation* $\neg(v^+, v^-) = (v^-, v^+)$*. Define an anti-resonant coupling contribution by*

$$\mathsf{AntiRes}_{\ell' \to \ell}(s_{\ell',t}) := K^-_{\ell' \to \ell} \star (\neg \circ s_{\ell',t}),$$

where $K^-_{\ell' \to \ell} : P_{\ell'} \times P_\ell \to V$ *is a* V*-valued kernel. Here* $(\neg \circ s)(p)$ *means applying* $\neg$ *to the* p-bit *value* $s(p)$*.*

Remark 564 (How anti-resonance enters the coupled update)**.** *In direct analogy with Definition 154, one can incorporate anti-resonant influence by adding* $\bigoplus_{\ell' \neq \ell} \mathsf{AntiRes}_{\ell' \to \ell}(s_{\ell',t})$ *as an additional input term in the update for* $s_{\ell,t+1}$*. Operationally, this means that strong evidence for* p *in context* ℓ' *can become strong evidence* against *related* q *in context* ℓ*, with the mapping again controlled by the kernel* $K^-_{\ell' \to \ell}$*. This clarifies that "anti-resonance" is not merely weaker resonance, but a qualitatively different channel that reverses polarity at the level of evidence.*

Remark 565 (Resonance and anti-resonance as distinct channels)**.** *The formalism permits both kinds of coupling to coexist: one may have a positive (resonant) kernel* $K_{\ell' \to \ell}$ *and a negative (anti-resonant) kernel* $K^-_{\ell' \to \ell}$ *simultaneously, representing systems where some cross-context correspondences reinforce while others suppress. In the* p-bit *interpretation, this supports mixed phenomena such as: "pattern* p *in* ℓ' *makes* q *more plausible in* ℓ*, but makes* r *less plausible," with different targets and strengths specified by the respective kernels.*

Remark 566 (Intuition: transmitting "the other side" of evidence)**.** *If the source context* ℓ' *contains strong negative evidence against a pattern* p*, then under* p-bit *negation this becomes strong positive evidence for* $\neg p$ *(or, more cautiously, strong positive evidence in the "negative channel" for* p*). Feeding* $\neg \circ s_{\ell',t}$ *through a kernel and adding it to the target dynamics allows the target to be pushed away from the same patterns that the source is rejecting. This is an abstract formalization of morphic anti-resonance (Hyperseed-Concept 114): not merely absence of resonance, but active propagation of reversal pressure. Concretely, if one thinks of* $s_{\ell,t}(p)$ *as a two-channel evidential state (e.g.* $s_{\ell,t}(p) = (v^+_{\ell,t}(p), v^-_{\ell,t}(p))$*), then* p-bit *negation can be read as an operation that swaps and/or reinterprets these channels so that "evidence against* p*" becomes usable input for biasing the evolution of* p*-related coordinates elsewhere. In this view, the kernel is not merely a similarity map but a transport mechanism for* constraints *or* inhibitions*: the transported signal acts like an inhibitory coupling that suppresses the target's tendency to reinstantiate the rejected pattern. The qualifier "more cautiously" is important: in a paraconsistent setting, strong* v^- *for* p *need not entail weak* v^+ *for* p*, so what is reliably transmitted is the intensity of the negative channel (or the intensity of support for* $\neg p$*), rather than a forced flip to an exclusive alternative.*

The use of paraconsistent negation here also connects to later resonance/dissonance themes (Hyperseed-Concept 97): conflict can be transported and can seed further conflict elsewhere, rather than being suppressed by consistency constraints (cf. paraconsistent dynamics perspectives such as [24]). In particular, anti-resonant coupling can propagate a tension pattern *even when the target context contains its own positive evidence for* p*: the resulting superposition is precisely the kind of structured incompatibility that dissonance concepts are meant to track. This clarifies why anti-resonance is not simply "negative similarity": it is a mechanism by which one context exports a local veto-like signal, and the receiving context integrates it without requiring global agreement or resolution.*

Remark 567. *There are multiple reasonable choices for anti-resonance. The above is the minimal one that mirrors the toy computation in Section 5: it allows negative evidence in one context to be transmitted as negative (or even as positive-for-not) evidence in another, without enforcing consistency. This minimality is useful pedagogically because it makes the "sign flip" intuition explicit: first apply* $\neg$ *at the level of evidential states, then transport via the same (or an analogous) kernel used for resonance, and finally aggregate into the target update as an extra driving term. However, the choice of whether to transport* $\neg \circ s_{\ell',t}$ *or to transport* $s_{\ell',t}$ *and negate after transport can matter if the kernel is nonlinear, state-dependent, or channel-mixing; in such cases, the ordering encodes a modeling decision about whether "reversal" is a property of the source signal itself or a property of how the target interprets incoming information. More sophisticated anti-resonance operators can be built using two-channel kernels that mix* (v^+, v^-) *components asymmetrically. For instance, one may wish to model a setting where negative evidence is more portable than positive evidence (or vice versa), or where negative evidence produces a broad, diffuse repulsion in the target while positive evidence produces a narrow, pattern-specific attraction. One can also introduce saturation, thresholds, or gating so that weak negative evidence does not create spurious repulsion, while strong negative evidence yields a robust anti-resonant push.*

Remark 568 (Further modeling notes)**.** *Anti-resonance can also be interpreted as transporting a* constraint *rather than transporting a* hypothesis*. If the source learns that* p *is systematically incompatible with its own stable trajectories, then exporting* $\neg p$ *(or exporting a strong* v^- *for* p*) amounts to telling the target: "do not allocate attractor mass to* p *unless you have compensating evidence." This suggests a natural stability heuristic: anti-resonant couplings should often be scaled (e.g. by kernel normalization or context-dependent confidence weights) to avoid globally destabilizing the dynamics when many sources simultaneously export negative pressure. In the two-channel perspective, such scaling can be applied differently to the* v^+*-transport and* v^-*-transport pathways, yielding a controlled way to represent that a system may be more conservative about spreading rejections than about spreading endorsements (or the reverse), depending on the application.*

12.5 Stability, amplification, and decay

The Hyperseed story needs three qualitative regimes:

1. *amplification*: repeated exposure yields increasing pattern support (habit-taking);

2. *decay*: support fades away (habit-reversal);

3. *stabilization*: a nontrivial persistent pattern-support structure emerges (habits as stable attractors).

We give basic sufficient conditions for these regimes in a finite P setting.

Remark 569 (A philosophical aside: stability as "thirdness")**.** *In Peircean terms, a habit is a kind of lawfulness: not a single event (secondness) nor a mere quality (firstness), but a stabilizing generality across time [14]. Mathematically, this is precisely what fixed points and attractors encode: an organization that, once present, reproduces itself under the dynamics. The following estimates and fixed-point statements are small technical shadows of this broader idea.*

One can also read "thirdness" here as the appearance of constraints *that are not explicitly imposed at each time step, but are nevertheless respected by the evolution once they have formed. In dynamical terms, this corresponds to invariant sets, absorbing regions, or attractor basins: after transients, the system repeatedly returns to (or remains inside) a family of states characterized by the same structured regularities.*

Definition 156 (A scalar upper bound for the p-bit quantale)**.** *Assume $V = [0,1]^2$ with componentwise order. For $v = (v^+, v^-)$ define $\|v\|_\infty := \max(v^+, v^-)$. For a state $s : P \to V$ define*

$$\|s\|_\infty := \max_{p \in P} \|s(p)\|_\infty.$$

For a relation $A : P \times P \to V$ define

$$\|A\|_\infty := \max_{p,q \in P} \|A(p,q)\|_\infty.$$

Remark 570 (Intuition: measuring the largest evidence magnitude)**.** *The quantity $\|v\|_\infty$ ignores the distinction between positive and negative channels and simply records the larger of the two. This is a conservative measure: if either channel is large, then $\|v\|_\infty$ is large. For states and relations, taking the max over all patterns (or all edges) gives a simple global bound suitable for elementary contraction-style estimates.*

This norm-like bound is useful here not because it is subtle, but because it is easy: it lets us give a clean sufficient condition for forgetting/decay in the toy p-bit *model without developing a full spectral theory over quantales.*

It is also worth noting what is being discarded by this bound. Because $\oplus$ is max, there is no cancellation between positive and negative evidence at the level of the semiring/quantale operations, so an ℓ_∞-style "largest-entry" estimate is often remarkably faithful. However, the bound still throws away all distributional *information (how many entries are large, whether large entries are isolated or form reinforcing motifs, etc.), and therefore cannot distinguish between many qualitatively different webs that share the same single largest edge weight.*

Remark 571 (Operator-level intuition for the update)**.** *The homogeneous update*

$$s_{t+1} = (d \odot s_t) \oplus (A \star s_t)$$

has two contributions. The term $d \odot s_t$ is a pointwise persistence/discounting term: it keeps each pattern's current evidence but shrinks it by a per-pattern factor d. The term $A \star s_t$ is a propagation/reinforcement term: evidence at pattern p is transmitted along the edge (p,q) to contribute to pattern q, and the join $\oplus$ aggregates all such incoming contributions at q by taking a componentwise maximum. In this toy p-bit *semantics, the state at time* t+1 *is therefore "whatever survives locally"* or *"whatever is newly supported by at least one incoming source," channelwise.*

Theorem 13 (A simple decay criterion in the p-bit toy quantale)**.** *Assume P is finite, $V = [0,1]^2$ with $\oplus$ as componentwise max and $\otimes$ as componentwise multiplication. Consider the homogeneous (no input) single-context update*

$$s_{t+1} = (d \odot s_t) \oplus (A \star s_t), \qquad u_t \equiv \mathbf{0}.$$

Let $\rho := \max(\|d\|_\infty, \|A\|_\infty)$. Then

$$\|s_{t+1}\|_\infty \le \rho \, \|s_t\|_\infty.$$

In particular, if $\rho < 1$ then $\|s_t\|_\infty \to 0$ as $t \to \infty$, so scalar propensities $p_t(p) = \pi(s_t(p))$ decay to 0 *for both π_{pos} and π_{net}.*

Remark 572 (A quantitative consequence: geometric forgetting)**.** *The inequality $\|s_{t+1}\|_\infty \le \rho \, \|s_t\|_\infty$ iterates immediately to*

$$\|s_t\|_\infty \le \rho^t \|s_0\|_\infty.$$

Thus when $\rho < 1$ the decay is not only eventual but geometric, with a characteristic "forgetting timescale" controlled by ρ. For example, to guarantee $\|s_t\|_\infty \leq \varepsilon$, it suffices that $t \geq \frac{\log(\varepsilon/\|s_0\|_\infty)}{\log(\rho)}$, where $\log(\rho) < 0$. This makes explicit how stronger persistence/reinforcement (larger ρ) slows forgetting even in the still-decaying regime.

Remark 573 (Plain-English meaning)**.** *If every edge weight in the web and the persistence factor d are bounded by some $\rho < 1$ (in the conservative $\|\cdot\|_\infty$ sense), then the system cannot sustain evidence: each update step scales the maximum evidence down by at least a factor ρ. Thus, in the absence of ongoing input, whatever support the system had is eventually forgotten. This gives a clean sufficient condition for habit-reversal-by-decay.*

The key point is that the model has no mechanism for generating evidence "from nothing" under $u_t \equiv \mathbf{0}$: all evidence at time $t+1$ is obtained by multiplying previously existing evidence by quantities in $[0,1]$ and then taking maxima. When all multiplicative factors are uniformly < 1, even the most favorable reinforcement path cannot avoid overall shrinkage.

Remark 574 (Why this is connected to earlier habit definitions)**.** *Under $\rho < 1$, the scalar propensities $p_t(p)$ decrease to 0 regardless of scalarization choice among π_{pos} and π_{net}. Hence, for sufficiently large T, "after T" averages become smaller than "before T" averages, aligning with the habit-reversal direction in Definition 144.*

More explicitly, because both $\pi_{\text{pos}}(v)$ and $\pi_{\text{net}}(v)$ are bounded above by $\|v\|_\infty$ (up to the trivial clipping to $[0,1]$ in the net case), one has a uniform implication of the form

$$\|s_t\|_\infty \to 0 \quad \Longrightarrow \quad \pi(s_t(p)) \to 0 \text{ for each fixed } p \in P,$$

so any average-of-propensities criterion based on time windows will eventually detect a decline.

Proof. Fix $q \in P$. Using the definitions and that $\oplus$ is max while $\otimes$ is multiplication (componentwise), we have

$$\|(d \odot s_t)(q)\|_\infty \leq \|d\|_\infty \, \|s_t\|_\infty,$$

and also

$$\|(A \star s_t)(q)\|_\infty = \left\| \bigoplus_{p \in P} A(p,q) \otimes s_t(p) \right\|_\infty \leq \max_{p \in P} \|A(p,q)\|_\infty \, \|s_t(p)\|_\infty \leq \|A\|_\infty \, \|s_t\|_\infty.$$

Taking the max of these two bounds yields $\|s_{t+1}(q)\|_\infty \leq \rho \, \|s_t\|_\infty$. Finally maximize over $q \in P$. $\square$

Remark 575 (Where finiteness enters)**.** *The assumption that P is finite ensures that the maxima in Definition 156 are attained and that the propagation bound*

$$\left\| \bigoplus_{p \in P} A(p,q) \otimes s_t(p) \right\|_\infty \leq \max_{p \in P} \|A(p,q)\|_\infty \, \|s_t(p)\|_\infty$$

is literally a bound by a finite maximum. Conceptually, the same proof pattern extends to infinite P if one replaces maxima by sups and assumes the relevant sups exist in $[0,1]$. The present finite setting keeps the estimate elementary and avoids measure-theoretic or completeness issues that are orthogonal to the qualitative message about decay.

Proof sketch. Bound each component of the next state by separately bounding the persistence term and the propagation term. Because the join $\oplus$ is componentwise max, the magnitude of the join is at most the maximum of the magnitudes. Because $\otimes$ is componentwise multiplication, magnitudes multiply. Taking maxima over p (for propagation) and then over q (for the whole state) yields the stated inequality, which iterates to show convergence to 0 when $\rho < 1$. $\square$

Remark 576 (Commentary)**.** *The proof is a straightforward "largest-entry" estimate: it ignores cancellations (there are none, since $\oplus$ is max) and tracks only the largest evidence component present anywhere. This is why the conclusion is a* sufficient *condition: many webs will still decay even when $\rho \geq 1$, but $\rho < 1$ guarantees decay by a simple contraction argument.*

One can compare this to classical matrix norm bounds: in linear systems, a condition like $\|A\| < 1$ implies contraction and hence decay, while $\|A\| \geq 1$ is inconclusive because the true stability threshold is governed by more refined structure (eigenvalues/spectral radius). Here the estimate plays the same role: it is intentionally crude but robust, and it emphasizes the logical shape of the argument ("uniform shrinkage everywhere implies global forgetting") rather than the sharp boundary between stability and instability.

Remark 577 (Interpretation)**.** *The condition $\rho < 1$ says that neither persistence nor reinforcement is strong enough to maintain evidence; the web forgets. This is a clean sufficient condition for habit-reversal in the absence of ongoing input.*

Equivalently, every path by which evidence might persist—either by staying at a node via d or moving along an edge via A—incurs at most a factor $\rho < 1$ per step in the channelwise magnitude. Even if the system continually selects the "best" incoming support at each step (because $\oplus$ is max), it still cannot overcome the uniform multiplicative discounting.

Remark 578 (A minimal illustrative example)**.** *Consider $P = \{1, 2\}$, with a single nonzero edge $A(1, 2) = (a^+, a^-)$ and persistence $d(1) = (d_1^+, d_1^-)$, $d(2) = (d_2^+, d_2^-)$. If $\max(d_1^+, d_1^-, d_2^+, d_2^-, a^+, a^-) < 1$, then regardless of where evidence starts (at node 1 or node 2), each time step multiplies every available contribution by a factor < 1 before taking maxima, so the overall magnitude must shrink. This makes concrete that the condition does not depend on detailed topology: even a "chain" that can move evidence forward cannot preserve it without at least one unit-sized multiplicative channel somewhere.*

Amplification and stabilization typically require persistent input and/or autocatalytic structure.

Proposition 18 (Driven closure yields a stable attractor in the inflationary case)**.** *Assume $d = e$ (no decay) and $u_t \equiv u$ is constant. Then the update*

$$s_{t+1} = u \oplus s_t \oplus (A \star s_t)$$

produces an ascending chain $s_0 \leq s_1 \leq \cdots$ whose join $s_\infty := \bigoplus_{t \geq 0} s_t$ exists and satisfies

$$s_\infty = u \oplus (A \star s_\infty).$$

If $u \neq \mathbf{0}$ and A connects u to many patterns, then s_∞ exhibits habit-taking (Lemma 3). Moreover, the statement is order-theoretic: the inequality $s_0 \leq s_1 \leq \cdots$ is interpreted pointwise in V^P (i.e., for each pattern $p \in P$ we have $s_t(p) \leq s_{t+1}(p)$ in V), and the join $\bigoplus_{t \geq 0} s_t$ is taken in the complete lattice V^P (again pointwise).

Remark 579 (What this proposition says)**.** *With no forgetting ($d = e$) and a constant source of stimulation u, the system's support can only increase over time. Thus it converges (in the order-theoretic sense of taking the join of an ascending chain) to a stable state s_∞. The fixed-point* *equation $s_\infty = u \oplus (A \star s_\infty)$ says: at equilibrium, the support is exactly what is injected plus what the web can propagate from that equilibrium.*

This is the simplest formal incarnation of "habits as stable attractors" in this framework. It is not a metric convergence statement; it is the order-theoretic statement appropriate to quantale-valued, possibly paraconsistent support. In particular, nothing here presupposes that supports are real numbers with a topology, nor that successive states become close in any metric; instead, the chain stabilizes in the sense that it reaches the least upper bound of all iterates. Equivalently, s_∞ is the least element (with respect to $\le$) that simultaneously dominates the initial iterates and is closed under the update rule. Because the update includes the explicit persistence term s_t, the map $F(s) := u \oplus s \oplus (A \star s)$ is inflationary ($s \le F(s)$), and s_∞ can be viewed as the closure obtained by repeatedly applying F starting from s_0.

Proof. Inflationarity is immediate because $s_{t+1} \ge s_t$ pointwise. Completeness of V^P gives existence of s_∞. Monotonicity plus distributivity of $\otimes$ over joins yields the fixed-point equation at the join. More explicitly, define $F(s) := u \oplus s \oplus (A \star s)$. The update is $s_{t+1} = F(s_t)$, and since F is monotone and inflationary, the sequence $(s_t)_{t\ge0}$ is an ascending chain. By completeness of V^P, the join $s_\infty = \bigoplus_{t\ge0} s_t$ exists. Then, using monotonicity of F and that $\oplus$ computes joins, one has

$$F(s_\infty) = u \oplus s_\infty \oplus (A \star s_\infty) = u \oplus \Big(\bigoplus_{t\ge0} s_t\Big) \oplus \Big(A \star \bigoplus_{t\ge0} s_t\Big),$$

and the distributivity/continuity property for $\star$ (spelled out in Remark 580 below) yields $A \star (\bigoplus_{t\ge0} s_t) = \bigoplus_{t\ge0}(A \star s_t)$, so

$$F(s_\infty) = u \oplus \bigoplus_{t\ge0} s_t \oplus \bigoplus_{t\ge0}(A \star s_t) = \bigoplus_{t\ge0}\big(u \oplus s_t \oplus (A \star s_t)\big) = \bigoplus_{t\ge0} s_{t+1} = s_\infty,$$

where the last equality uses that the join of the tail $\{s_{t+1} : t \ge 0\}$ is the same as the join of the whole chain $\{s_t : t \ge 0\}$. Finally, since $s_\infty = s_\infty \oplus u \oplus (A \star s_\infty)$ implies $s_\infty \ge u \oplus (A \star s_\infty)$ and also $s_\infty \le u \oplus (A \star s_\infty) \oplus s_\infty$, the displayed fixed-point identity $s_\infty = u \oplus (A \star s_\infty)$ follows because $\oplus$ is idempotent and $s_\infty \oplus s_\infty = s_\infty$. □

Proof sketch. The update map is inflationary, so iterating it yields an ascending chain. In a complete lattice, the join of this chain exists. Taking the join on both sides of $s_{t+1} = u \oplus s_t \oplus (A \star s_t)$ and using that $\otimes$ distributes over joins lets us pass the join through the propagation term, producing the fixed-point equation for s_∞. One can read this as a standard "iterate-and-take-the-supremum" construction: s_∞ collects everything that can be reached from the seed s_0 by any finite number of applications of the forcing u and the propagation $A \star (\cdot)$, with persistence preventing any loss. This is the order-theoretic analogue of reaching equilibrium by repeatedly applying an expanding operator until no genuinely new support can be added. □

Remark 580 (Why the distributivity step is the hinge)**.** *The subtlety (such as it is) lies in justifying that $A \star (\bigoplus_t s_t) = \bigoplus_t (A \star s_t)$. This is exactly where the quantale axioms matter: $\otimes$ distributes over arbitrary joins, so the propagation operator commutes with the limit-by-join. Without this property, the closure and fixed-point story would lose its clean algebraic form. Concretely, writing $(A \star s)(p) = \bigoplus_{q\in P} A(p,q) \otimes s(q)$ (the usual quantale-enriched matrix–vector product), we compute*

$$(A\star\bigoplus_t s_t)(p) = \bigoplus_{q\in P} A(p,q)\otimes\Big(\bigoplus_t s_t(q)\Big) = \bigoplus_{q\in P}\bigoplus_t\big(A(p,q)\otimes s_t(q)\big) = \bigoplus_t\bigoplus_{q\in P}\big(A(p,q)\otimes s_t(q)\big) = \bigoplus_t (A\star s_t)(p),$$

where the interchange of joins is justified by completeness and the distributivity of $\otimes$ over arbitrary joins. Thus, $\star$ is Scott-continuous with respect to the order structure induced by $\oplus$, and the fixed-point argument is a direct application of that continuity.

Remark 581 (Least fixed point and "driven closure")**.** *The equality $s_\infty = u \oplus (A \star s_\infty)$ exhibits s_∞ as a fixed point of the monotone map*

$$G(s) := u \oplus (A \star s).$$

Because the actual update uses $F(s) = s \oplus G(s)$, iteration from any s_0 produces a post-fixed chain $s_t \leq s_{t+1}$ that approaches the least fixed point above s_0 in the order. In particular, if one starts from the empty state $s_0 = \mathbf{0}$, then s_∞ is the least solution of $s = G(s)$, i.e., the smallest stable support pattern compatible with both the constant drive u and the propagation web A. This is the sense in which the attractor is a "closure": it is the minimal closed (stable) element containing the forced content.

12.6 A worked micro-example reaching self-weaving and morphic resonance

We now give a small numerical instance (compatible with the toy model style in Section 5) that shows:

- morphic resonance creating a paraconsistent "both-sides" state; and
- internal reinforcement weaving a larger support pattern from that seed.

The example is deliberately minimal (two contexts and two patterns) so that the mechanisms can be read off from the update equations themselves: the "morphic" step is an *external* injection of evidence into a target context, while the "self-weaving" step is an *internal* propagation of evidence along a small directed web of pattern-to-pattern reinforcement links.

Example 12 (Morphic resonance seeds a self-weaving two-cycle)**.** *Work in the* p-bit *quantale $V = [0,1]^2$ with $\oplus$ as componentwise max and $\otimes$ as componentwise multiplication (Section 3.4). Here a value $(x^+, x^-) \in [0,1]^2$ is read as a* pair *of evidential degrees: x^+ measures positive support and x^- measures negative support. The key choice is that $\oplus =$* max accumulates *evidence (it never retracts prior evidence in either coordinate), while $\otimes$* gates *or* attenuates *evidence multiplicatively (so a weak coupling or weak premise yields a weak contribution). This combination is what makes it easy to exhibit paraconsistent states: the negative coordinate can remain high even as the positive coordinate is increased by new inputs.*

Consider two contexts (locations) C_1 and C_2 that share a small pattern class $P = \{p, q\}$. Interpret p as a "seed" pattern and q as a downstream pattern that can participate in a reinforcing loop with p inside C_2. The intended reading is that C_1 already "knows" (or has stabilized) a habit supporting p, while C_2 has not stabilized it and may even have reasons to reject it; nevertheless, C_2 can be nudged by a morphic coupling that carries only partial and possibly one-sided influence.

State in C_1 (source). *Assume C_1 strongly supports p:*

$$s^{(1)}(p) = (0.9, 0.1), \qquad s^{(1)}(q) = (0.1, 0.1).$$

Thus, in the source context C_1, p has high positive evidence and only a small amount of negative evidence, while q is essentially neutral/weak in both directions. One can think of this as p being an established habit in C_1 and q not yet being a meaningful downstream consequence there.

Initial state in C_2 (target). *Assume C_2 initially leans against p:*

$$s_0^{(2)}(p) = (0.2, 0.7), \qquad s_0^{(2)}(q) = (0.1, 0.1).$$

So before any cross-context influence, C_2 has relatively low positive support for p but relatively strong negative evidence against it. This is the setup in which a purely monotone, non-retractive update rule is most visibly paraconsistent: later increases in the positive coordinate for p need not (and in this model, will not) remove or "explain away" the pre-existing negative coordinate.

Morphic coupling $C_1 \to C_2$. *Let the morphic field strength be $K = (0.6, 0.2)$, and apply a direct same-pattern coupling to p:*

$$s_1^{(2)}(p) := s_0^{(2)}(p) \oplus (K \otimes s^{(1)}(p)).$$

This update should be read as: the new state of p in the target is obtained by taking the old state and then adding *(via $\oplus$) a coupled contribution obtained by attenuating the source evidence by K (via $\otimes$). Because $\otimes$ is componentwise multiplication, K plays the role of a pair of channel strengths: it transmits positive evidence with weight 0.6 and negative evidence with weight 0.2. Compute*

$$K \otimes s^{(1)}(p) = (0.6 \cdot 0.9,\ 0.2 \cdot 0.1) = (0.54, 0.02),$$

so

$$s_1^{(2)}(p) = (0.2, 0.7) \oplus (0.54, 0.02) = (0.54, 0.7).$$

Thus C_2 receives positive support for p without erasing its negative support: it enters a paraconsistent "both positive and negative" regime. Concretely, after coupling we have simultaneously *(i) relatively strong positive evidence for p (raised from 0.2 to 0.54) and (ii) the original strong negative evidence against p unchanged at 0.7. The point is not that this is "inconsistent" in a classical sense, but that the state now encodes a live tension: p is pushed forward as a candidate habit while remaining counter-supported. In a classical single-score model, one would typically be forced to collapse this to a single compromise value; in the* p-bit *model the two-sidedness is explicit and stable under $\oplus$.*

Internal reinforcement in C_2. *Now define an internal reinforcement relation in C_2 by*

$$A_2(p, q) = (0.9, 0.1), \qquad A_2(q, p) = (0.8, 0.2),$$

and take all other pairs to have value $(0, 0)$. These two nonzero entries specify a tiny directed web: p pushes evidence toward q, and q pushes evidence back toward p, forming a two-cycle. The values also allow the web to transmit both positive and negative evidence (albeit with different strengths), so that downstream patterns can inherit not only support but also potential counter-support. In particular, $A_2(p, q) = (0.9, 0.1)$ says that if p is present, it tends to generate q with a strong positive channel and a weak negative channel; $A_2(q, p) = (0.8, 0.2)$ similarly gives a feedback link.

Update q once by

$$s_1^{(2)}(q) := s_0^{(2)}(q) \oplus (A_2(p, q) \otimes s_1^{(2)}(p)).$$

This is the local "weaving" step: the newly morphically-amplified state of p in C_2 is used as an input that propagates along the internal edge $p \to q$. Because $s_1^{(2)}(p)$ already contains a paraconsistent tension, the propagated contribution can, in principle, carry forward both aspects of that tension. Compute

$$A_2(p, q) \otimes s_1^{(2)}(p) = (0.9 \cdot 0.54,\ 0.1 \cdot 0.7) = (0.486, 0.07),$$

hence

$$s_1^{(2)}(q) = (0.1, 0.1) \oplus (0.486, 0.07) = (0.486, 0.1).$$

Here the positive coordinate of q jumps substantially (from 0.1 *to* 0.486*), while the negative coordinate of q does not increase because the pre-existing negative evidence* 0.1 *dominates the newly imported* 0.07 *under* max*. This illustrates a general feature of $\oplus = \max$: once a coordinate is already at some level, additional weaker contributions in that coordinate do not change it, whereas stronger contributions do. In other words, the web transmits* new *evidence when it exceeds what is already locally present.*

Finally, propagate back from q to p (one step of self-weaving closure):

$$s_2^{(2)}(p) := s_1^{(2)}(p) \oplus (A_2(q,p) \otimes s_1^{(2)}(q)).$$

This feedback step is the smallest nontrivial instance of "closure": evidence that was triggered by p and flowed to q is now allowed to flow back and potentially further reinforce p. In larger webs, iterating such steps (or taking a least fixed point) is what builds up an extended habit-supporting substructure from a small initiating seed. Compute

$$A_2(q,p) \otimes s_1^{(2)}(q) = (0.8 \cdot 0.486,\ 0.2 \cdot 0.1) = (0.3888, 0.02),$$

so

$$s_2^{(2)}(p) = (0.54, 0.7) \oplus (0.3888, 0.02) = (0.54, 0.7).$$

In this specific numeric instance, the back-propagation does not further increase p because p is already saturated in positive evidence and remains high in negative evidence; but it does demonstrate the closure mechanism and shows how a morphically seeded p produces downstream q within C_2. Said differently: the loop $p \to q \to p$ is present and active, but the idempotent nature of $\oplus = \max$ means that once the dominant positive and negative coordinates for p have been reached (here, positive 0.54 *and negative* 0.7*), additional weaker returns do not alter the state. This is a feature rather than a bug: it models a kind of "stability after saturation," whereby a context can become locked into a conflicted yet persistent stance.*

Hyperseed interpretation. *Relative to scalarization π_{pos}, q in C_2 has increased from* 0.1 *to* 0.486 *after the morphic seeding of p, which is a direct instance of "habit-taking at a distance" (morphic resonance) followed by local habit closure on the internal web. The scalarization π_{pos} is used here only to make the growth easy to read numerically; the underlying state remains two-dimensional, and in other scenarios one might equally track π_{neg} or a combined measure. Because negative evidence for p persisted (it stayed at* 0.7*), the same example also illustrates how morphic resonance can create ambivalent or conflicted habit structures without logical explosion. In particular, the model permits the following qualitative narrative: a remote context can make a pattern feel "compelling" (increasing positive evidence) while local countervailing constraints, memories, or inhibitions remain in place (maintaining negative evidence). The internal web then turns this unstable co-presence into additional structured consequences (here, support for q), which is the minimal sense in which a seed becomes a weave.*

Remark 582 (What the example is meant to teach)**.** *The arithmetic is not the point; the point is the qualitative possibility. A coupling term can inject new positive evidence into a context without removing existing negative evidence, creating a stable paraconsistent tension. Then, internal reinforcement can transmit that tension downstream, producing new supported patterns (here q) as a consequence of the web structure. This is precisely the sense in which morphic resonance can be* *a* seed *and the web's closure can be the* weave*. One may also view the example as separating two kinds of "cause": an external cause (the morphic field term) that changes what is locally available to the system, and an internal cause (the reinforcement web) that determines what the system can build from whatever is available. The two-step structure matters: without the morphic injection, p would remain low in positive evidence in C_2, and the edge $p \to q$ would not generate a substantial increase in q; without the internal web, the morphic injection would remain a local change to p without generating additional downstream organization.*

12.7 Notes and bridges to later sections

Where attention and effort enter. The update operators above are "structural." In Hyperseed, attention (Hyperseed-Concept 60) and effort (Hyperseed-Concept 100) modulate which pattern classes are tracked and which reinforcement links are strengthened. Formally, attention can be implemented as a time-varying reweighting of A and u_t (and even of the pattern class P itself) driven by resource constraints from Section 8. Concretely, one may treat attention as a gating field w_t that rescales the effective adjacency and reinforcement updates (e.g., $A \mapsto w_t \odot A$ and $u_t \mapsto w_t \odot u_t$ with appropriate normalization), so that limited processing budget is expressed directly as selective amplification/suppression of candidate pattern links. On this view, "effort" can be modeled not only as a scalar budget but also as a constraint on the allowable complexity of the currently tracked subweb (e.g., limiting the number of active nodes/edges, the depth of multi-step propagation, or the entropy of the attention weights), thereby coupling computational cost to the dynamics of habit-formation. This makes explicit how the same structural operator can yield markedly different learning trajectories under different resource regimes: under scarcity, updates become sparse and conservative; under abundance, they become denser and can support rapid structural reconfiguration. This perspective aligns with the general cognitive-systems view that attention is a resource-allocation operator acting on inference/learning dynamics (see, e.g., [19]).

Relation to resonance/dissonance (Section 3.9). Section 3.9 sketches a logic-to-complex mapping and an interference-based resonance score. In the present section, morphic resonance is implemented as a coupling kernel $K_{\ell' \to \ell}$ that transports pattern support across contexts. One can combine these views by: (i) using paraconsistent interference scores to adaptively learn K; and/or (ii) interpreting high interference-coherence as a signal that two contexts share a near-morphism between their pattern webs, warranting stronger morphic coupling. In more operational terms, $K_{\ell' \to \ell}$ may be taken as a (row-)stochastic or otherwise mass-preserving operator so that "transport" can be read literally as redistribution of support, with constraints (e.g., positivity, bounded norm, or decay with contextual distance) preventing spurious amplification. Likewise, when interference is used as a learning signal, it can be used either to increase coupling for coherent overlaps (resonance) or to decrease coupling when overlaps are systematically contradictory (dissonance), yielding a unified knob that can implement both attraction and repulsion between webs depending on the sign/structure of the paraconsistent evidence. The near-morphism interpretation can be made explicit by asking whether there exists a structure-respecting map that approximately preserves key relational signatures (connectivity motifs, implication-like edges, or compositional paths) from one web to the other; high coherence then indicates that such a map is available with low distortion, making cross-context transfer reliable rather than merely associative. This is one natural meeting point between paraconsistent evidence dynamics and resonance-style measures [24].

Reality-systems and beyond. In later sections (reality-systems, intersubjective reality, eurycosm), the same coupled-web formalism can be re-used with $\mathcal{L}$ indexing different reality-systems and K encoding cross-system transduction pathways. This provides a mathematically explicit "port" for discussing morphic resonance in settings where "spatial separation" is generalized to separation across representational, social, or cosmological contexts (Hyperseed-Concept 149, Hyperseed-Concept **??**, Hyperseed-Concept **??**; see also [18]). Here it is helpful to read "transduction" broadly: K can stand for translation between symbol systems, alignment between agents internal models, propagation through communication networks, or coupling across scales (e.g., individual $\to$ group $\to$ institution) when $\mathcal{L}$ is organized hierarchically. In such cases, constraints on K can encode empirical or normative limitations (bandwidth limits, trust/credibility filters, institutional friction), making the same mathematics serve both descriptive and prescriptive roles: describing how habits and meanings spread, and prescribing how to shape coupling to improve coordination or reduce pathological contagion. This framing also anticipates later uses where one distinguishes "within-system" stabilization (habit consolidation inside a reality-system) from "between-system" transfer (morphic resonance across reality-systems), since the two can be tuned independently by the intra-web reinforcement rules versus the inter-web kernel K.

13 Mind, representation, and perception

This section introduces the representational layer of Hyperseed [1]. The preceding sections provide (i) an observer-relative notion of distinction/weakness, (ii) patterns as simplicity-improving representations, and (iii) dynamical operators (habits and morphic resonance) that move pattern support across contexts. Here we focus on how a system becomes a *mind* by supporting *registration* of patterns, *representation* of what is registered, and the beginnings of *closed-loop organization* (with control treated formally in Section 14).

Remark 583. *Conceptually, this section is the hinge between the "static" side of the formalism (distinctions, weakness, and patterns) and the "active" side (prediction, control, and attention). A mind, in the Hyperseed sense, is not merely a container of data; it is a system that can* stabilize *some internal stand-ins for what it encounters, and then use these stand-ins to guide future behavior. This framing aligns with the emphasis in [19] on internal representational dynamics as the substrate for flexible cognition.*

13.1 Systems, minds, and recognition processes

Hyperseed uses "system" in the autopoietic sense: not every arbitrary distinguished collection counts as a system; a system must have internal coherence and internal production. For this paper, it is useful to model "system" at two complementary levels: *(i) a process level* (how state evolves) and *(ii) a pattern level* (what compressive regularities are stably present).

Definition 157 (Process and system)**.** *A* process *is a triple* (T, Σ, δ) *where:*

- T *is a proto-time index (a poset, interpreted as a partial order of "before/after"),*
- Σ *is a (possibly structured) state space,*
- $\delta : T \to (\Sigma \to \Sigma)$ *assigns to each* $t \in T$ *a state-update map.*

A system *is a finite directed multigraph* $\mathcal{S} = (P, E)$ *whose nodes* $p \in P$ *are processes and whose edges encode* production dependencies*:* $(p \to q) \in E$ *means that the evolving state of* p *supplies* *part of the input (or constraint) for the evolution of q. We say $\mathcal{S}$ is* inter-producing *if every node lies on a directed cycle.*

Remark 584. *A few notational clarifications help anchor this definition. The symbol T denotes a partially ordered set (poset): one should read $t \preceq t'$ as "t is no later than t'" in the system's proto-time (Hyperseed-Concept 142). The state space Σ is the mathematical home of instantaneous configurations, and $\delta(t) : \Sigma \to \Sigma$ is the time-indexed update map (so the state at t advances according to $\delta(t)$). Thus, the composite object (T, Σ, δ) is a minimalist process in the style of dynamical systems theory, but allowing partial time order fits well with later branching-time and context machinery.*

The shift from a single process to a system $\mathcal{S} = (P, E)$ is a shift from "one evolving thing" to "many evolving things with dependencies." A directed multigraph *allows multiple edges between the same pair of processes, which is useful when a module can depend on another in several distinct ways. The inter-producing condition (every node lies on a directed cycle) is a concise way of insisting on mutuality: nothing in the system is purely a passive input; every component is also (perhaps indirectly) sustained by others. This resonates with process-ontological intuitions in the background of Hyperseed (compare [15]) and with Hyperseed's emphasis on self-maintaining webs (Hyperseed-Concept 168).*

Remark 585. *The "inter-producing" condition is a minimal formal proxy for the informal Hyperseed idea that systems are coherent sets of inter-producing processes. In Section 12 we strengthen this idea using autocatalytic closure properties expressed on pattern webs.*

Definition 158 (Mind)**.** *A* mind *is a system $\mathcal{M} = (P, E)$ equipped with additional structure:*

1. *A distinguished* registration interface *(a "sensory" subgraph) $\mathcal{S}_{\text{in}} \subseteq \mathcal{M}$.*
2. *A* representation space *$\mathcal{R}$ (objects, tokens, or states that can stand in for other entities).*
3. *A* reference relation $\text{Ref}_C \subseteq \mathcal{R} \times \mathcal{U}$ *for each context/aspect C, where $\mathcal{U}$ denotes the ambient universe of entities under discussion.*
4. *An* update rule *that turns registrations into representational changes (e.g. $r_{t+1} = \mathsf{Update}(r_t, \text{Reg}_t)$ for some internal state $r_t \in \mathcal{R}$).*

The remaining sections refine these components and give semantics in terms of patterns, weakness, and paraconsistent evidence.

Remark 586. *This definition should be read as a deliberately spare "constitution" rather than a detailed psychology. A mind is, first, a system in the sense above: an interdependent complex of processes. But it becomes a mind by having (i) a privileged gateway where the world impresses itself upon the system (registration; Hyperseed-Concept 153), (ii) an internal realm of stand-ins (representation; Hyperseed-Concept 157), and (iii) a relation* Ref_C *that ties stand-ins to what they are about (reference; Hyperseed-Concept 152). The context index C matters philosophically: it encodes the Russellian insistence that meaning is not floating in a void; it is always meaning* under some way of taking *the world, i.e. under some partition of distinctions (Hyperseed-Concept 86).*

A simple example: let $\mathcal{U}$ contain physical objects, and let $\mathcal{R}$ contain internal feature vectors or symbolic tokens. A camera-like subsystem can serve as $\mathcal{S}_{\text{in}}$. Then $\text{Ref}_C(r, u)$ *may mean "token r currently refers to object u as seen in visual context C." The update rule is the bridge from mere causal impact to sustained internal content. This is why Hyperseed treats registration as weaker than perception: the update rule can fail, be blocked, or be too noisy to stabilize representation. The general program is consonant with the cognitive-systems stance of [19], while remaining compatible with the pattern/emergence framing in [5].*

Definition 159 (Pattern recognition process)**.** *Fix a context/aspect C and a set of candidate patterns $\mathcal{P}_C$. Assume each system-state S at time t determines a (possibly fuzzy) intensity map $\mathrm{Int}_{S,t} : \mathcal{P}_C \to V$ into the quantale V from Section 3. A process Y is a* pattern recognition process aimed at *an entity X (in aspect C) if there exists a nonempty set $\mathcal{P}_{X,C} \subseteq \mathcal{P}_C$ such that for each $P \in \mathcal{P}_{X,C}$:*

1. *(alignment) $\mathrm{Int}_{Y,t}(P) \leq \mathrm{Int}_{X,t}(P)$ for all relevant t, and*
2. *(learning) $\mathrm{Int}_{Y,t}(P)$ is nondecreasing in t (w.r.t. the order on V), with strict increase for at least one P at least once.*

Remark 587. *Here $\mathcal{P}_C$ is the repertoire of patterns that the context C makes available (Hyperseed-Concept 130). The quantale V supplies a graded scale of "how much a pattern is present": the order $\leq$ on V is the comparison of intensities, and the monoidal structure (introduced in the formal core) is what ultimately makes these intensities compositional. The notation $\mathrm{Int}_{S,t}(P)$ is the intensity of pattern P in the system-state S at time t; one may picture it as the system's current "degree of embodiment" of P.*

Intuitively, Y recognizes X when Y comes to mirror (some of) X's patterns, and to do so ever more stably. Condition (alignment) prevents us from calling arbitrary growth in Y a recognition of X; it must be growth in patterns that are genuinely patterns of X. Condition (learning) demands a temporal direction: recognition is not a static coincidence but a historical acquisition. A toy example: if X is a melody and P ranges over rhythmic/melodic motifs, then a listener-process Y recognizes X as it internalizes these motifs (its $\mathrm{Int}_{Y,t}(P)$ rises), while remaining within what is actually present in the melody (alignment).

This concept is useful because it makes recognition expressible without presupposing crisp symbols or propositional truth: recognition is framed in the same graded pattern vocabulary as the rest of the theory. That uniformity becomes crucial once we allow paraconsistent evidence, habit dynamics, and cross-context resonance (Hyperseed-Concept 151).

A further point of the quantale-valued formulation is that "presence" need not be probabilistic or frequency-like: depending on the chosen V, intensities can encode salience, constraint-satisfaction, degree of activation, similarity, or any other orderable evidence measure that supports the required compositional operations. In particular, allowing $\mathrm{Int}_{S,t}$ to be fuzzy means that Y can partially recognize X—e.g. strongly for some motifs and weakly for others—without forcing a binary recognized/not-recognized cut. The set $\mathcal{P}_{X,C}$ can be read as the (aspect-relative) collection of patterns through which X is "trackable" by Y in the given context; different choices of C can therefore yield different recognition relations even between the same X and Y.

The clause "for all relevant t" is intentionally flexible: in applications one may take t to range over a training episode, over a window in which X is presented, or over a period of interaction in a shared environment. Nothing in the definition requires t to be discrete; the monotonicity condition in (learning) can be interpreted for continuous time (as an order-preserving map along the temporal order) as readily as for stepwise updates. Likewise, (learning) allows plateaus: $\mathrm{Int}_{Y,t}(P)$ may stabilize once Y has acquired the pattern, but the requirement of at least one strict increase rules out the degenerate case where Y merely happens to match X without any acquisition at all.

Remark 588. *Definition 159 formalizes a core Hyperseed intuition: recognition means that, through interaction, the recognizer comes to contain more of the same patterns that are present in what it recognizes. When combined with habit dynamics (Section 12), this becomes an explicit model of perceptual learning. In that combined picture, the monotone rise of $\mathrm{Int}_{Y,t}(P)$ can be understood as the trace left by repeated successful engagements with P—a memory-like accumulation that need* *not be explicit or symbolic. The alignment condition then plays the role of an external constraint: it ties the direction of learning to the target entity X rather than to arbitrary self-amplification within Y, and it permits a clean distinction between "becoming more patterned" and "becoming more patterned* in the way that matches *X." The nonemptiness of $\mathcal{P}_{X,C}$ similarly enforces that recognition is about at least one concrete pattern channel in the specified aspect, making explicit that recognition is always selective and aspect-relative rather than an undifferentiated global relation.*

Remark 589. *It can be helpful to view $\mathcal{P}_{X,C}$ as encoding what counts as "evidence about X" for the recognizer in context C: if $P \notin \mathcal{P}_{X,C}$, then even large values of $\mathrm{Int}_{Y,t}(P)$ do not contribute to recognition* of X in that aspect*. This separation makes room for familiar phenomena such as confounds and misattribution: Y may increase intensity on patterns that are predictive in its environment but are not actually patterns of X, in which case (alignment) fails and the process is better described as learning a surrogate. Conversely, the definition allows recognition to be partial and distributed: Y need not match X on all patterns in $\mathcal{P}_C$, only on some nonempty subset, and only to the extent measured in V. In settings where one wants to model over-shooting or hallucination (i.e. $\mathrm{Int}_{Y,t}(P) > \mathrm{Int}_{X,t}(P)$ for some P), one can treat such departures as a distinct kind of process rather than folding them into recognition; the present definition deliberately marks that boundary by requiring alignment.*

13.2 Contexts and aspects

Hyperseed treats most predicates as context-dependent. For representation, this context dependence is not optional: what a token represents is always "a representation of B *in some aspect C*."

Definition 160 (Context and aspect extraction)**.** *Let* **Ctx** *be a (small) poset of contexts with order $C \preceq D$ meaning "D refines C" (makes at least as many distinctions). An* aspect extraction family *is a family of maps*

$$\mathrm{asp}_C : \mathcal{U} \to \mathcal{U}_C \qquad (C \in \mathbf{Ctx})$$

with the property that if $C \preceq D$ then there is a canonical "forgetful" map $\pi_{D\to C} : \mathcal{U}_D \to \mathcal{U}_C$ satisfying $\mathrm{asp}_C = \pi_{D\to C} \circ \mathrm{asp}_D$. We write $X|C := \mathrm{asp}_C(X)$ for the aspect of X in context C.

Remark 590. *The notation is doing quiet but important work. The poset* **Ctx** *is a bookkeeping device for "ways of carving reality": $C \preceq D$ means that D is at least as discriminating as C (Hyperseed-Concept 86). The map asp_C is a projection that forgets everything about X that C is not sensitive to. The codomain $\mathcal{U}_C$ is the universe as viewed through C: one may think of it as equivalence classes, coarse descriptors, or structured observations, depending on the application.*

A simple example is geometric: let $\mathcal{U}$ be the set of 3D objects, and let C be "silhouette shape" while D is "full 3D mesh." Then D refines C, and $\pi_{D\to C}$ maps a 3D mesh to its silhouette descriptor; extracting the silhouette directly from the object or extracting the mesh and then forgetting down to silhouette yield the same result. Another example is linguistic: C might keep only "part of speech," while D keeps full syntactic parse; again, D refines C.

This definition is necessary because reference and representation will be indexed by C, and without a disciplined notion of aspect we risk equivocation: a token can be a faithful representation in one aspect (e.g. location) while being wildly unfaithful in another (e.g. color). Hyperseed's insistence that meaning is aspect-relative makes this explicit and prevents category mistakes.

Remark 591. *The condition $\mathrm{asp}_C = \pi_{D\to C} \circ \mathrm{asp}_D$ says that refining context and then forgetting back down is equivalent to extracting the coarse aspect directly. This formalizes the idea that "aspects" are stable observer-relative projections.*

Remark 592. *It is helpful to view the family* $\{\mathcal{U}_C\}_{C\in\mathbf{Ctx}}$ *as a system of compatible "views" of* $\mathcal{U}$*, where refinement introduces additional structure but never contradicts coarser views. Concretely, if* $C \preceq D \preceq E$*, then one typically expects the forgetful maps to compose coherently, i.e.* $\pi_{E\to C} = \pi_{D\to C}\circ\pi_{E\to D}$*, so that there is a unique way to forget from a very fine context down to a very coarse one. This kind of coherence is often implicit when one says the map* $\pi_{D\to C}$ *is "canonical," and it mirrors the informal idea that there is not more than one principled way to discard distinctions.*

Remark 593. *The adjective "small" for* **Ctx** *is primarily a set-theoretic convenience: it ensures that the indexing of contexts does not introduce size issues when later constructions quantify over contexts, form products over families of contexts, or speak of "all" contexts relevant to an agent. Intuitively, it corresponds to the modest claim that the agent's repertoire of ways-of-carving is enumerable within the theory, rather than a proper class of all imaginable measurements.*

Remark 594. *In many applications, each* asp_C *may be understood as inducing an equivalence relation on* $\mathcal{U}$*: two entities* $X, Y \in \mathcal{U}$ *are* C*-equivalent if* $X|C = Y|C$*. Under this reading,* $\mathcal{U}_C$ *can be taken (up to isomorphism) as the quotient of* $\mathcal{U}$ *by* C*-equivalence, and* $\pi_{D\to C}$ *expresses that* D*-equivalence refines* C*-equivalence when* $C \preceq D$*. This can clarify the slogan that a context "forgets distinctions": it identifies many fine-grained states as the same coarse state.*

Definition 161 (Contextualized groupings)**.** *For a grouping (collection)* $G \subseteq \mathcal{U}$ *and context* C*, define the* contextualization *of* G *to* C *by*

$$G|C := \{\, x|C : x \in G \,\} \subseteq \mathcal{U}_C.$$

Remark 595. *This is the natural lifting of aspect extraction from individuals to sets (Hyperseed-Concept 103). Intuitively, if* G *is "the set of all cups on the table," then* $G|C$ *is that same set as seen through context* C*—for example, perhaps as a set of silhouettes, or as a set of rough positions, depending on what distinctions* C *preserves.*

The usefulness is pragmatic: later constructions (pattern webs, attention allocation, prediction) often operate on collections rather than single entities. By contextualizing the collection, we can compare group-level phenomena across contexts without pretending that the contexts share the same granularity.

Remark 596. *Note that the definition uses an ordinary set image, so distinct elements of* G *may collapse to the same element of* $\mathcal{U}_C$ *when context* C *is coarse. This is not a bug: it records the fact that, relative to* C*, those distinct individuals are indistinguishable. When multiplicity matters (e.g. "two identical-looking cups" vs. "one cup"), one may later choose to work with multisets, measures, or probability distributions on* $\mathcal{U}_C$ *rather than plain subsets, but the present definition captures the minimal invariant needed for many aspect-relative comparisons.*

Remark 597. *Contextualization interacts with basic set operations in the expected monotone way. If* $G \subseteq H$ *then* $G|C \subseteq H|C$*, and if* $\{G_i\}$ *is any family of groupings then*

$$\Big(\bigcup_i G_i\Big)|C = \bigcup_i (G_i|C).$$

In general, contextualization does not commute with intersections: $(G\cap H)|C$ *may be strictly smaller than* $(G|C) \cap (H|C)$*, because two different elements—one from* G *and one from* H*—may become* C*-indistinguishable even if the original sets were disjoint. This is another manifestation of how a context can erase distinctions, and it will be relevant whenever later arguments reason about "overlap" between groupings in an aspect-relative way.*

13.3 Registration, sensory systems, and perception

Registration is Hyperseed's primitive "information intake" notion. It is weaker than perception: perception requires that registration triggers an internal representation. In particular, registration is meant to capture the minimal sense in which the system's dynamics come to *depend* on some aspect of the world, even when that dependence is too coarse, too unstable, or too weakly integrated to count as content-bearing cognition. This makes registration suitable as a common substrate for later distinctions: for instance, one can speak of registration that is accurate but nonconceptual, registration that is noisy or ambiguous, and registration that is driven by internal couplings (as in dreams) rather than exogenous causes (as in ordinary sensing).

Definition 162 (Registration)**.** *Fix a context C and an equivalence relation $\sim_C$ on $\mathcal{U}_C$ (interpreted as "indistinguishable in context C" for the registering system). A system X* registers *an entity A at time t (in context C) if there exists $A' \in \mathcal{U}$ such that $A'|C \sim_C A|C$ and $A'|C$ becomes a pattern in X by time t. Concretely, this can be witnessed by a pattern $P \in \mathcal{P}_C$ such that*

$$\mathrm{Int}_{X,t}(P) \not\leq \theta \qquad \text{and} \qquad \mathrm{Ref}_C(P, A|C) \text{ is nontrivial},$$

for some threshold $\theta \in V$.

Remark 598. *Registration (Hyperseed-Concept 153) is intentionally defined so that it can occur without any rich internal model. The equivalence relation $\sim_C$ encodes what the system cannot tell apart in context C (Hyperseed-Concept* **??***). Thus, registration is compatible with vagueness: to register A is not necessarily to pin down A as a unique individual, but to have one's internal dynamics come to embody a pattern that is at least compatible with A under the system's current discriminations.*

A minimal example: let C be a low-resolution visual context in which many distinct objects share the same coarse "blob" descriptor. Then $\sim_C$ identifies objects with the same blob. The system registers a particular object A when a blob-pattern becomes salient in the system ($\mathrm{Int}_{X,t}(P)$ *crosses threshold) and is linked by reference to the extracted aspect $A|C$. Here the notation* $\mathrm{Int}_{X,t}(P) \not\leq \theta$ *means "the intensity is above threshold" in the quantale order; in a numeric quantale this would read as the usual inequality "greater than θ," but the order-theoretic phrasing accommodates nonstandard evidence domains.*

This definition is useful because it separates (i) being impacted by the world *from (ii)* constructing an internal token that stands for the world*. That separation is needed for later distinctions among mere sensation, perception, hallucination, and reflective awareness.*

Two further clarifications help fix the intended reading. First, the role of A' is to accommodate the fact that the system may only ever interact with (or encode) a surrogate *that is equivalent to A under the coarse-graining imposed by C and $\sim_C$; one should not read the definition as requiring access to A "as it is in itself," but only to some entity that is indistinguishable from A at the resolution that matters for the registering system. Second, the "becomes a pattern in X by time t" clause is temporal in a permissive way: it allows registration to be an event that takes time to settle (e.g., integration over a short window, the closing of a feedback loop, or the accumulation of evidence), rather than an instantaneous snapshot. In settings where t is discrete and updates are synchronous, this can be read as "at or before the t-th update."*

It is also useful to distinguish nontrivial reference *from* maximally specific reference*. The condition* $\mathrm{Ref}_C(P, A|C)$ *nontrivial demands that the pattern is not merely active, but active* as about *(or at least systematically linked to) the contextual aspect $A|C$. However, the definition does not force the pattern to determine a unique referent beyond the equivalence class fixed by $\sim_C$: the same P may be compatible with many distinct A's in $\mathcal{U}$ so long as their C-aspects are identified.*

This is the intended locus of perceptual ambiguity at the level of registration (e.g., "something moved" or "a sound occurred" without a stable object assignment).

Definition 163 (Sensory system)**.** *Let X be a system and let $\mathcal{U}^{\text{ext}} \subseteq \mathcal{U}$ denote those entities considered "external" to X in a given modeling decomposition. A* sensory system *for X is a coherent subsystem $S \subseteq X$ such that:*

1. *most registrations produced by S have referents in $\mathcal{U}^{\text{ext}}$, and*

2. *the update dynamics of S are driven primarily by coupling to $\mathcal{U}^{\text{ext}}$ (rather than by coupling internal to X).*

Remark 599. *The point is not that the sensory system is metaphysically "outside-facing," but that, under a chosen decomposition into "system" and "environment," its causal drivers are predominantly exogenous. In robotics this might literally be sensors; in social cognition it might be language interfaces; in introspective cognition it might even be sub-systems that "sense" other parts of the mind. The definition thus accommodates both the ordinary case and the more Peircean/Whiteheadian view that there are many layers of mutual prehension, some treated as external only by pragmatic abstraction [14, 15].*

As an example, in a neural network model of an agent, an early vision stack could serve as S if its inputs are mostly pixels from the environment and its learned features are mostly driven by those pixels. The utility of naming S is that perception, attention, and control will later be analyzed as transformations that begin at such interfaces and propagate inward.

The qualifier "most" in item (1) is deliberate: many realistic sensory subsystems also generate registrations whose immediate referents are internal variables (e.g., gain control states, prediction error channels, or learned latent features) while still being primarily shaped by exogenous coupling. Similarly, item (2) is intended to be robust under modeling choices: one may refine or coarsen the system–environment boundary, and the definition only asks that, under the chosen decomposition, the dominant causal influence on S comes from $\mathcal{U}^{\text{ext}}$. This keeps the notion compatible with settings in which sensorimotor loops blur the boundary (e.g., active sensing where X moves to create informative inputs) while still allowing one to identify subsystems whose input channels *are anchored in the external world.*

It is also helpful to note that coherence of S is functional rather than anatomical: a sensory system may be distributed (e.g., multiple modalities or parallel transducers) so long as it forms a relatively unified interface from the standpoint of downstream processing (for instance, by supporting stable patterns $P \in \mathcal{P}_C$ that later stages can treat as evidence).

Definition 164 (Perception)**.** *A mind M* perceives *an entity A at time t (in context C) if:*

1. *M registers A at time t (Definition 162), and*

2. *this registration causally triggers the emergence of a representation token $R \in \mathcal{R}$ with $\mathrm{Ref}_C(R, A|C)$ nontrivial.*

In slogan form: perception is registration that causes representation.

Remark 600. *Perception (Hyperseed-Concept 134) is the first place where the theory demands a causal bridge from world-impact to internal content. Item (2) is what blocks a trivialization: mere registration can be fleeting, local, and unintegrated, whereas perception is registration that organizes the mind's representational economy by producing a token R that can participate in later inference, memory, and action.*

A simple example: a thermostat may "register" a temperature (its state changes in response to heat), but unless it produces an internal token that stands for "it is hot" in a way that can be recombined with other tokens, we would hesitate to call it perception in this framework. Conversely, an agent that forms an explicit map entry or object-file when stimulated by A meets the condition. This is why perception is the natural bridge to the prediction/control machinery of Section 14.

The "causally triggers" clause is meant to exclude cases where a representation happens to co-occur with a registration without being appropriately dependent on it. For instance, if R is generated by a purely endogenous process and merely coincides with the time at which M registers A, then (2) fails even if $\mathrm{Ref}_C(R, A|C)$ is nontrivial. Conversely, if the registration initiates a cascade (possibly mediated by attention, gating, or working memory) that yields R, then the perception condition is satisfied even when the representation is delayed relative to the initial registration, so long as the causal dependence is preserved.

It is also important that (2) requires a representation token *rather than merely a strengthened pattern. Intuitively, a token $R \in \mathcal{R}$ is something the mind can reuse, store, compare, and bind into larger structures; it is the kind of internal item that can later be evaluated for accuracy, recalled, or used as a premise in reasoning. Thus, perception is not merely "strong sensation": it is the emergence of a content-bearing element whose reference to $A|C$ is stabilized enough to enter the mind's representational repertoire.*

Proposition 19 (Perception implies registration)**.** *If a mind M perceives A at time t in context C, then M registers A at time t in context C.*

Proof. Immediate from Definition 164, item (1), which requires that M registers A at time t in context C as a precondition for perception. □

Remark 601. *Intuitively, this proposition says: if you genuinely perceived something (in the present formal sense), then you must at least have been affected by it in the weaker registration sense. The result is not surprising, but it is structurally important: it guarantees that perception is a strengthening of registration rather than a separate notion that could drift away. This monotonic relationship will be quietly used later when we model attention as selectively modulating the pipeline from registration to representation.*

Proof. Immediate from Definition 164, item (1). □

Proof sketch. Unpack the definition of perception: registration is literally a required conjunct. □

Remark 602. *The key step "item (1)" is doing all the work because perception was defined as a two-part condition, with registration as the first part. One may think of this as a design choice: the formalism encodes a conceptual hierarchy of notions (registration $\Rightarrow$ perception) directly into the logical form of the definitions.*

13.4 Representation as reference plus pattern inheritance

Hyperseed defines representation as "re-presentation": there must be a *reference* relation to what is represented, and the representation must inherit (some of) the patterns of the represented entity. We now formalize both components.

Definition 165 (Pattern signature in an aspect)**.** *Fix a context C and a finite pattern basis $\mathcal{B}_C \subseteq \mathcal{P}_C$. The* pattern signature *of an entity X in aspect C is the map*

$$\mathrm{Sig}_C(X) : \mathcal{B}_C \to V, \qquad \mathrm{Sig}_C(X)(P) := \mathrm{Int}_X(P),$$

where $\mathrm{Int}_X(P)$ *denotes the intensity of pattern* P *in* X *as defined in Section 9 (or its later refinement).*

Remark 603. *The pattern signature is an "aspect-relative fingerprint" of an entity in terms of a chosen basis* $\mathcal{B}_C$ *(Hyperseed-Concept 141). Concretely,* $\mathcal{B}_C$ *is a finite list of patterns we have decided are salient for the present discussion; the signature* $\mathrm{Sig}_C(X)$ *then assigns each basis pattern* P *an intensity value in* V *measuring how strongly* P *is present in* X*.*

A simple example: if C *is a sensory context and* $\mathcal{B}_C$ *is a set of visual features (edges, colors, textures), then* $\mathrm{Sig}_C(X)$ *is analogous to a feature vector. If* C *is a relational context and* $\mathcal{B}_C$ *is a set of graph motifs, then* $\mathrm{Sig}_C(X)$ *is a profile of how much* X *exhibits those motifs. The definition is useful because representation will be defined by comparing signatures: to represent is, in part, to share patterns in a controlled (typically reduced) way. This directly reflects Hyperseed's notion of patterns as compressive regularities [5].*

Definition 166 (Strong pattern inheritance)**.** *Let* $A, B \in \mathcal{U}$ *and context* C *be given. We say that* A inherits patterns from B *in aspect* C *if*

$$\mathrm{Sig}_C(A|C)(P) \leq \mathrm{Sig}_C(B|C)(P) \qquad \forall P \in \mathcal{B}_C.$$

Equivalently, $\mathrm{Sig}_C(A|C) \leq \mathrm{Sig}_C(B|C)$ *pointwise.*

Remark 604. *The inequality is intentionally one-sided: inheritance means "*A *has no more of each basis-pattern than* B *does," so* A *is (at most) as pattern-rich as* B *in the monitored directions. In a numeric setting, if* Sig_C *were feature activations in* $[0,1]$*, then inheritance would say each activation in* A *is bounded above by the corresponding activation in* B*.*

A toy example is projection: if B *is a high-resolution image and* A *is a compressed thumbnail, then the thumbnail inherits many coarse patterns (large-scale edges) but not the fine ones; in a basis that monitors only coarse patterns, one expects* $\mathrm{Sig}_C(A) \leq \mathrm{Sig}_C(B)$*. The definition is useful because it captures, in a single monotone condition, the intuition that a representation should be a simplification: it may discard detail, but should not invent detail in the same aspect without further explanation.*

Definition 167 (Representation in an aspect)**.** *Let* $A, B \in \mathcal{U}$ *and context* C *be given. We say that* A represents B *in aspect* C *if:*

1. *(reference)* $\mathrm{Ref}_C(A, B|C)$ *is nontrivial, and*

2. *(inheritance)* $A|C$ *inherits patterns from* $B|C$ *in the sense of Definition 166.*

Remark 605. *This definition makes precise Hyperseed's slogan that representation is "re-presentation":* A *presents again, in a cheaper form, something that is already present in* B*. The reference condition ensures "aboutness" (Hyperseed-Concept 152), while inheritance ensures a structured similarity in the chosen aspect (Hyperseed-Concept 157). Either condition alone is insufficient: mere similarity without reference yields accidental resemblance, and mere reference without inherited patterns yields a bare pointer with no internal content.*

As a simple example, a written name "Paris" may refer to Paris (reference), but in a sensory context it inherits almost none of Paris's sensory patterns; thus it is not an icon. In a relational context, however, it can inherit relational patterns (linked to "France," "capital-of," etc.), hence it can count as a symbol. The usefulness of the definition is that it lets us treat iconic, indexical, and symbolic representation uniformly and then specialize by aspect.

Remark 606. *The inheritance condition is deliberately "one-way": representations are allowed (and usually expected) to drop patterns. Dropping patterns is what makes a representation simpler and cheaper to manipulate.*

Proposition 20 (Transitivity of representation under compositional reference)**.** *Assume the reference relation composes in the following weak sense: if* $\mathrm{Ref}_C(A, B|C)$ *and* $\mathrm{Ref}_C(B, C|C)$ *are nontrivial, then* $\mathrm{Ref}_C(A, C|C)$ *is nontrivial. Then representation is transitive: if* A *represents* B *in aspect* C *and* B *represents* C *in aspect* C*, then* A *represents* C *in aspect* C*.*

Remark 607. *In plain terms: if* A *stands for* B*, and* B *stands for* C*, then (under a mild assumption on how reference chains behave)* A *stands for* C*. This is the familiar idea that representations can be layered: a mental image can represent a drawing which represents an object; a symbol can represent a definition which represents a concept; and so on. In particular, the "mild assumption" is doing the standard compositional work one expects of any usable notion of reference: when one token is used* via *another to pick out a target, the mediated link is still a bona fide instance of reference rather than a mere accidental correlation. The point is not that* every *associative chain composes (spurious co-occurrences should not), but that the chains that satisfy the definition's reference clause are closed under concatenation.*

What may be slightly surprising is that we also get the inheritance part "for free" by order transitivity: if each layer only drops patterns, then the composite also only drops patterns. Put another way, if passing from C *to* B *forgets (at most) some structure and passing from* B *to* A *forgets (at most) some additional structure, then passing from* C *to* A *cannot have introduced any new structure not already present in* C*; it can only have forgotten what the intermediate steps forgot. This is the precise sense in which "pattern inheritance" behaves like information monotonicity along a pipeline.*

This proposition connects the present section to later ones in two ways. First, prediction and control will often operate on multi-step representational pipelines, so transitivity ensures that the end-to-end token still counts as representing the target. Here one can think of a learned model whose internal states represent latent variables that in turn represent environment-level quantities: even if the agent only ever directly manipulates internal states, the claim guarantees that such manipulation is still, in the defined sense, manipulation of a representation of *the distal target. Second, attention allocation may be described as choosing which representational chains to extend; transitivity guarantees that extending a chain does not break representation, provided reference composes. This matters because attention policies often introduce new intermediate codes (additional "layers"), and we want a criterion under which adding such codes preserves representational status rather than requiring a fresh justification from scratch at each extension.*

Proof. By assumption, nontrivial reference composes. Here "nontrivial" rules out degenerate cases in which everything is taken to refer to everything (or reference is satisfied vacuously); the intended assumption is that whenever (A, B) and (B, C) meet the reference clause of Definition 167, then so does (A, C). This isolates the reference-conjunct from any additional constraints coming from inheritance and makes clear that the compositional step is a closure property of the reference relation itself.

For inheritance, if $\mathrm{Sig}_C(A|C) \leq \mathrm{Sig}_C(B|C)$ and $\mathrm{Sig}_C(B|C) \leq \mathrm{Sig}_C(C|C)$ pointwise, then by transitivity of $\leq$ we have $\mathrm{Sig}_C(A|C) \leq \mathrm{Sig}_C(C|C)$ pointwise. The word "pointwise" is essential: it means that for each coordinate (each pattern-dimension tracked by the signature), the inequality holds separately, so inequalities can be chained coordinate by coordinate without any need for aggregation or weighting. In particular, nothing like convexity, linearity, or an averaging argument is required; the conclusion follows purely from the order structure on the codomain of $\mathrm{Sig}_C(\cdot \mid C)$.

Thus both the reference and inheritance conditions of Definition 167 hold for (A, C). Equivalently, the composite map from C-relevant patterns to those preserved in A factors through B but does not depend on which factorization is chosen: any intermediate layer that merely deletes patterns cannot enable A to exceed C in signature strength. □

Proof sketch. The proof separates the two conjuncts in the definition of representation. Reference transitivity is assumed directly. Inheritance transitivity is obtained by chaining pointwise inequalities of pattern signatures. One can regard this as a "two-channel" argument: the semantic channel (reference) composes by hypothesis, while the structural channel (inheritance) composes because it is expressed as an order constraint, and orders are designed to support such chaining. □

Remark 608. *The key move is the pointwise ordering on signatures: it turns "inherits patterns" into an ordinary order-theoretic statement. Once that translation is made, the proof becomes an instance of the simplest Russellian principle: if $a \leq b$ and $b \leq c$ then $a \leq c$. Visually, one can picture signatures as vectors in a positive cone and inheritance as coordinatewise domination; then domination is clearly transitive. In more concrete terms, if each coordinate tracks the salience, presence, or strength of a given pattern relative to C, then "X inherits patterns from Y" means X does not score higher than Y on any such coordinate, i.e. X never claims to preserve a pattern more strongly than its source. The positive-cone picture is helpful because it emphasizes that we are not comparing vectors by length or angle (which could behave non-monotonically under composition), but by the product order, for which transitivity is immediate.*

It is also worth noting what this does not *require. The argument does not assume that any layer is lossless, only that loss is monotone in the sense captured by $\leq$. Nor does it require that signatures be complete descriptions of internal states; they only need to be the features relevant to the inheritance clause. As long as inheritance is expressed as coordinatewise non-increase of these features, any number of intermediate representational formats can be inserted without endangering the inheritance relation from the initial target to the final token.*

13.5 Semiotic modes: icon, index, symbol

Hyperseed adopts a simple Peircean triad [14]. All three modes are special cases of representation, distinguished by which aspect C is being preserved. In this framing, the triad is not a claim that there are three disjoint kinds of things "in the world," but that there are three recurring emphases in *how* a token functions as a representation: it may primarily preserve sensory patterning, spatiotemporal placement, or relational constraints. Which mode is appropriate can therefore depend on the analytic choice of aspect C (and thus on what the agent is treating as relevant structure) rather than on the token alone.

Definition 168 (Sensory, spatiotemporal, and relational aspects)**.** *Assume the context poset* $\mathbf{Ctx}$ *contains (at least) three marked sub-posets:*

- $\mathbf{Ctx}_{\mathrm{sens}}$ *(sensory domains),*
- $\mathbf{Ctx}_{\mathrm{st}}$ *(spatiotemporal positioning domains),*
- $\mathbf{Ctx}_{\mathrm{rel}}$ *(relational/network domains).*

These need not be disjoint; a single context may mix multiple concerns.

Remark 609. *This definition is a convenience: it declares three families of aspects that correspond, roughly, to "how it looks/feels," "where/when it is," and "how it relates." The sub-posets may* *overlap because real representational systems rarely separate these cleanly: a perceptual token may encode both sensory features and spatial location, and a symbol may have sensory typography. What matters is that we can talk about emphasizing one family of patterns over another when classifying a representational mode. This prepares the ground for the icon/index/symbol definitions (Hyperseed-Concept* **??**, **??**, *184).*

The role of the marked sub-posets is not to constrain what representations can *exist, but to provide a vocabulary for which invariances are being appealed to when we say "A represents B." For instance, a single token may preserve a sensory pattern (e.g. a silhouette) while also locating an event (e.g. a timestamped recording); the present taxonomy says that these are different representational* aspects*, even if realized by the same artifact. In later applications, this separation allows one to ask whether an inference step is justified by sensory similarity, by spatiotemporal anchoring, or by relational constraints in a model.*

Definition 169 (Icon)**.** *An entity A is an* icon *for an entity B if there exists a sensory context $C \in \mathbf{Ctx}_{\mathrm{sens}}$ such that A represents B in aspect C.*

Remark 610. *An icon is, in effect, a representation that preserves sensory-pattern structure. A photograph is the standard example: in an appropriate sensory context it inherits many of the visual patterns of its referent (though often with distortions). Another example is a melody hummed from memory: it can be iconic of the original tune insofar as auditory patterns are preserved.*

The definition is intentionally broad about what counts as "sensory." Many scientific and technical representations function iconically once a suitable sensory context is fixed: a spectrogram can be an icon of an acoustic event in a visual sensory context; a plot of a signal over time can be iconically related to the underlying process insofar as shape, periodicity, and relative magnitude are preserved. Even schematic diagrams (e.g. circuit diagrams) can be partially iconic if the relevant sensory aspect is taken to be geometric adjacency or shape-based regularities, rather than photorealism.

The usefulness of the definition is that it reduces "iconicity" to the general representation notion plus a restriction on aspect. This makes it possible to treat icons within the same mathematical machinery used for symbols, rather than as an unrelated primitive. In particular, the present viewpoint allows degrees and failures of iconicity to be stated as failures to preserve specific sensory patterns (e.g. loss of resolution, occlusion, aliasing), rather than as a vague departure from "resemblance."

Definition 170 (Index)**.** *An entity A is an* index *for an entity B if there exists a spatiotemporal context $C \in \mathbf{Ctx}_{\mathrm{st}}$ such that A represents B in aspect C.*

Remark 611. *An index preserves spatiotemporal positioning patterns: it is a pointer, a trace, or a coordinate-like token. A deictic gesture "that" is indexical; a memory address in a computer is indexical; smoke can be indexical of fire when treated in a spatiotemporal/causal-trace context.*

The spatiotemporal aspect may be literal (a GPS coordinate, a clock time, a bounding box in an image) or implemented via a proxy that is treated as preserving location-like structure (e.g. a database key that is stable under operations interpreted as "moving through" a record space). What matters for the present classification is that the representation is used primarily to pick out *an occurrence or region by its placement within some ordering of space, time, or traceable causal propagation, rather than to encode its detailed sensory qualities.*

Indices are theoretically useful because they allow reference with minimal inherited content. In many control and prediction settings, what you need is not a rich description but a stable handle to the relevant portion of the world. Indexicality supplies this handle in a form that can later be elaborated into richer representation by adding further aspects. This also clarifies why indexical *tokens often appear in pipelines: an agent may first secure an index to "the object over there" and only subsequently attach iconic features (appearance) or symbolic labels (category, name) once additional processing becomes available.*

Definition 171 (Symbol)**.** *An entity A is a* symbol *for an entity B if there exists a relational context $C \in \mathbf{Ctx}_{\mathrm{rel}}$ such that A represents B in aspect C. In addition, symbolic representations are assumed to participate in a* combinational system *(Section 9): symbols can be composed and recombined, and their referents are constrained by their relationships to other symbols.*

Remark 612. *A symbol is a representation whose salient preserved structure is relational, not sensory. The word "electron" does not look like an electron; it inherits its meaning from the web of relations it bears to other symbols (laws, measurement terms, mathematical operators). The additional requirement of participation in a combinational system captures this: symbols are not isolated labels; they live in a grammar, calculus, or graph of constraints (Hyperseed-Concept 77).*

On this view, "conventionality" or "arbitrariness" is not a separate axiom but a common consequence: if what is preserved is primarily relational structure, then many different tokens could in principle occupy the same node in the relational web without changing the role they play. Conversely, if the relational neighborhood changes (e.g. the inferential rules governing a term), then the symbol's meaning shifts even if the token's sensory form remains identical. This matches everyday cases in which a symbol retains its typography while acquiring new use through altered definitions, laws, or embedding discourse.

This definition is valuable because it connects semiotics to computation. Once symbols are combinable, they support systematic inference and counterfactual reasoning; hence symbolic representation is a key substrate for the later sections on prediction, causality, and control. The move also echoes Peirce's emphasis on the triadic, law-like mediation characteristic of symbols [14] (compare also Hyperseed-Concept 190). A further consequence is that symbolic systems typically admit explicit notions of well-formedness and equivalence (syntactic validity, derivability, rewriting), which can be treated as additional relational constraints internal to $\mathbf{Ctx}_{\mathrm{rel}}$.

Remark 613. *These definitions deliberately leave room for hybrid tokens. For instance, a labeled photograph may be both an icon (sensory similarity) and a symbol (networked relationship to other labeled items).*

Hybridity is not an exception but a common mechanism of grounding and coordination: an index may anchor a symbolic description to a particular event, while an icon may provide a perceptual interface for symbolic manipulation (e.g. diagrammatic reasoning where visual similarity guides steps that are nonetheless licensed by relational rules). In practice, agents often move between modes by changing which aspect C they treat as primary: the same token can be read iconically (attend to resemblance), indexically (attend to location or provenance), or symbolically (attend to its role in a combinational system).

13.6 Soggy predicates and paraconsistent truth

Hyperseed's "soggy predicates" are intended as a bridge between: (i) fuzzy truth values attached to predicates, and (ii) probabilistic semantics grounded in (possibly hypothetical) collections of observations. We now define soggy predicates in a way that is compatible with the $[0,1]^2$ (paraconsistent) evidence domain [23, 24].

Definition 172 (p-bit-valued predicate)**.** *Let X be a universe of discourse. A* p-bit-valued predicate *on X is a function $F : X \to [0,1]^2$. We interpret $F(x) = (F^+(x), F^-(x))$ as degrees of positive and negative evidence for the statement "x has property F."*

Remark 614. *A* p-bit*-valued predicate replaces a single truth degree with a pair of evidential degrees: how much evidence supports the predicate and how much evidence opposes it. The notation* $F^+(x)$ *and* $F^-(x)$ *simply refers to the first and second components of* $F(x)$*, respectively. This is a natural fit for paraconsistent reasoning: one can simultaneously have high positive and high negative evidence without collapsing into triviality (Hyperseed-Concept 198).*

A simple example: let X *be a set of patients and* $F(x)$ *encode evidence that patient* x *has a disease. A lab test might contribute positive evidence while a conflicting scan contributes negative evidence. Then* $F(x)$ *can legitimately be* $(0.8, 0.7)$*, reflecting serious conflict rather than forcing an artificial resolution. This is useful later because habit dynamics, prediction, and control will often propagate and transform such evidential pairs rather than crisp truths.*

Remark 615. *It is helpful to keep distinct (i)* evidence *and (ii)* truth *in the classical sense. In a* p-bit*-valued predicate, the pair* $(F^+(x), F^-(x))$ *is not required to satisfy* $F^-(x) = 1 - F^+(x)$*, and therefore it is not a probability assignment or a fuzzy membership degree by default. Instead, it records two potentially independent aggregates: "how much pushes toward acceptance" and "how much pushes toward rejection." This independence is precisely what allows explicit representation of conflict, as well as "informational gaps" where both coordinates are small.*

A common derived reading (not assumed, but often useful as intuition) is to view

$$\mathit{conflict}(x) \ := \ \min(F^+(x), F^-(x)), \qquad \mathit{ignorance}(x) \ := \ 1 - \max(F^+(x), F^-(x)),$$

so that $(1, 0)$ *indicates maximal support with no opposition,* $(0, 1)$ *indicates maximal opposition with no support,* $(1, 1)$ *indicates maximal conflict, and* $(0, 0)$ *indicates maximal lack of evidence either way. The framework itself only commits to the evidential pair; any downstream "decision rule" (accept/reject/undetermined) can be layered on later, depending on the application.*

Definition 173 (Soggy predicate)**.** *Fix a finite observation set* $\mathcal{O}$ *and weights* $w_o \geq 0$ *with* $\sum_{o \in \mathcal{O}} w_o = 1$*. Assume each observation* $o \in \mathcal{O}$ *yields an evidential evaluation* $e_o : X \to [0, 1]^2$*. Define the* soggy predicate *induced by* $(\mathcal{O}, w, e)$ *as*

$$F(x) := \sum_{o \in \mathcal{O}} w_o \, e_o(x),$$

where the sum is taken componentwise in $[0, 1]^2$*.*

Remark 616. *A soggy predicate (Hyperseed-Concept 172) is, mathematically, just an average of evidential evaluations across an ensemble. The normalization* $\sum_o w_o = 1$ *ensures* $F(x)$ *remains in* $[0, 1]^2$ *when each* $e_o(x)$ *does. Componentwise summation means:*

$$F^+(x) = \sum_o w_o e_o^+(x), \qquad F^-(x) = \sum_o w_o e_o^-(x).$$

Intuitively, each $o \in \mathcal{O}$ *is a possible "way of looking" (an observation, measurement, or test), and* w_o *measures its importance or frequency.*

A simple example: let $\mathcal{O}$ *be a set of diagnostic tests. Each test* o *yields a pair* $e_o(x)$ *of support/opposition for "*x *has condition* F*." Then* $F(x)$ *aggregates them. This is useful for two reasons. First, it grounds fuzzy/paraconsistent truth in an observation model, which is important for the later philosophy-of-science stance. Second, it provides a canonical way to combine heterogeneous evidence without prematurely forcing complementarity; this aligns with the paraconsistent motivation emphasized in [23, 24].*

Remark 617. *Because the weights $(w_o)_{o\in\mathcal{O}}$ form a simplex (nonnegative and summing to* 1*), the map $F(x) = \sum_o w_o e_o(x)$ is a* convex combination *of the points $\{e_o(x)\}_{o\in\mathcal{O}} \subseteq [0,1]^2$. Thus $F(x)$ lies in the convex hull of the per-observation evidential evaluations, and in particular cannot introduce out-of-range evidence values. This convexity perspective is often the right mental model: "sogginess" amounts to pooling an ensemble of perspectives without forcing them to agree.*

In the intended applications, $\mathcal{O}$ can be read either extensionally (a literal finite set of measurements) or intensionally (a finite set of measurement types *or contexts). Correspondingly, the weights w_o can be read as empirical frequencies, reliabilities, priors over contexts, or (in engineered settings) design choices that allocate attention among sensors or sub-models.*

Remark 618. *Soggy predicates are "simple observation grounded" in the sense that $F(x)$ is the (weighted) average evidential assessment over an ensemble of observations. The ensemble $\mathcal{O}$ need not be the mind's actual observations; it may encode hypothesized observations, cultural priors, or counterfactual experiments.*

Remark 619. *The finiteness of $\mathcal{O}$ is a convenience rather than a deep restriction: it ensures the sum is elementary and avoids measure-theoretic overhead. Conceptually, one can regard $(\mathcal{O}, w)$ as a finite probability space on "observation modes" and $e_o(x)$ as a random evidential evaluation; then $F(x)$ is the expectation of that random evaluation. This probabilistic reading is deliberately weak: it treats weights as a bookkeeping device for aggregation, without requiring that the evidential coordinates themselves be probabilities.*

Proposition 21 (Probabilistic semantics as a special case)**.** *Suppose that for every observation o and entity x the evidence is complementary, meaning $e_o^-(x) = 1 - e_o^+(x)$. Then the induced soggy predicate F satisfies $F^-(x) = 1 - F^+(x)$ for all x. Thus $F^+(x)$ can be read as an ordinary fuzzy membership degree in $[0,1]$ and, if $e_o^+(x)$ are interpreted as per-observation probabilities, then $F^+(x)$ is the expected probability of satisfaction.*

Remark 620. *This proposition says that the paraconsistent framework strictly generalizes the familiar probabilistic (or fuzzy) one. If every observation insists that "negative evidence is exactly one minus positive evidence," then the two-coordinate representation collapses back down to a single number, and soggy predicates become ordinary expectations. This is important because it reassures us that we have not abandoned classical semantics; we have merely stopped assuming that the world (or the observer) always provides perfectly complementary information.*

The result connects backward to the earlier probability-flavored phrasing of habits (Section 12.1) and forward to prediction and control (Section 14), where one may sometimes choose to work in the complementary subcase for convenience.

Remark 621. *The complementarity assumption $e_o^-(x) = 1 - e_o^+(x)$ can be understood as embedding a single scalar evaluation into the square $[0,1]^2$ via the diagonal map $p \mapsto (p, 1-p)$. Under this embedding, the soggy construction commutes with averaging: averaging on the diagonal in $[0,1]^2$ is the same as averaging in $[0,1]$ and then re-embedding. In this sense, soggy predicates are "conservative" over ordinary fuzzy/probabilistic semantics: the extra coordinate does not change anything when the data already lives on the diagonal, but it becomes available when the data departs from complementarity.*

It is also worth noting what complementarity rules out: if two observations deliver $e_{o_1}(x) = (0.9, 0.1)$ and $e_{o_2}(x) = (0.1, 0.9)$, then each observation is individually complementary, yet the disagreement between observations is not represented as within-observation conflict; rather, it manifests as uncertainty in the aggregate when averaged. By contrast, genuinely paraconsistent evidence allows an individual observation mode (or model) to carry its own internal tension, e.g. $(0.9, 0.8)$*, which is a different phenomenon than mere inter-observation disagreement.*

Proof. Compute componentwise:

$$F^-(x) = \sum_o w_o e_o^-(x) = \sum_o w_o(1 - e_o^+(x)) = 1 - \sum_o w_o e_o^+(x) = 1 - F^+(x).$$

□

Proof sketch. Use the complementarity assumption $e_o^- = 1 - e_o^+$ inside the weighted sum, then pull out the constant 1 using $\sum_o w_o = 1$. □

Remark 622. *The decisive step is the normalization of weights. Without $\sum_o w_o = 1$, one would obtain $F^-(x) = (\sum_o w_o) - F^+(x)$ instead. Geometrically, complementarity restricts evidence pairs to the diagonal line $v^- = 1 - v^+$ in the unit square; averaging preserves this line, so the soggy predicate stays on the diagonal and can be identified with its first coordinate.*

Remark 623. *In later sections, it is sometimes useful to treat F as defining a family of classical predicates by thresholding one or both coordinates, e.g. "accept F at level α" meaning $F^+(x) \geq \alpha$ (support-based), or "reject F at level β" meaning $F^-(x) \geq \beta$ (opposition-based), or a three-way policy that separates accept/reject/undecided by simultaneously requiring high support and low opposition. The point of the soggy construction is that such policies can be formulated* after *aggregation, once the evidential landscape is explicit, rather than being silently forced by an assumption of complementarity at the outset.*

Moreover, because F is an affine combination of the e_o, any downstream affine transformation of evidence (for example, a linear calibration map applied to each observation mode) can be moved inside the sum when convenient. This algebraic flexibility is part of why soggy predicates behave well as building blocks for habit-updates and for compositional models of perception.

Remark 624. *The point of the $[0, 1]^2$ form is that complementary evidence is not required. A mind may have strong positive and strong negative evidence simultaneously for the same predicate (conflict), or may have neither (lack of evidence). Soggy predicates simply average such evidence profiles over an observation ensemble. In particular, the two coordinates can be read as* independently accumulated *support-for and support-against signals, rather than as probabilities that must sum to one. This makes room for familiar cognitive situations: perceptual ambiguity (both channels low), illusion or misinformation (high negative evidence against a claim that is nevertheless asserted), and genuine inconsistency in a knowledge base (both channels high due to incompatible but salient cues). Averaging over an ensemble should be understood as smoothing over time, viewpoints, or sub-systems (e.g. different sensory modalities), so that a "stable" soggy predicate can coexist with momentary spikes of conflict without forcing an immediate collapse to a single bivalent verdict.*

13.7 Micro-example: registering and representing a pattern

We illustrate the chain

$$\text{external pattern} \to \text{registration} \to \text{representation}$$

on a minimal finite instance, aligned with the toy universe style of Section 5. The intent is to separate (i) what is present in the external entity, (ii) what the sensory apparatus can extract in a particular situation, and (iii) what the cognitive system subsequently stabilizes as a manipulable internal item. Even in this minimal setting, the example shows that "representation" is not a single relation but a family of context-indexed relations, each preserving different structural constraints.

Example 13 (Icon, index, and symbol for one referent)**.** *Let $\mathcal{U}$ contain an external entity B. Assume a mind M has a sensory subsystem S that registers B in a sensory context $C_{\text{sens}} \in \mathbf{Ctx}_{\text{sens}}$. Registration produces a token $r_{\text{icon}} \in \mathcal{R}$ whose sensory signature matches a subset of B's sensory signature, i.e. $\text{Sig}_{C_{\text{sens}}}(r_{\text{icon}}) \leq \text{Sig}_{C_{\text{sens}}}(B)$. Here the order $\leq$ can be read as "no more detailed than" or "contained within" the external signature as constrained by the context (e.g. limited resolution, occlusion, viewpoint, or modality). Thus, even when B has a rich sensory profile, the registered icon may preserve only a small pattern family (edges but not texture; pitch but not timbre; silhouette but not color), and the context index specifies* which *family is being compared. If $\text{Ref}_{C_{\text{sens}}}(r_{\text{icon}}, B|C_{\text{sens}})$ is nontrivial, then r_{icon} is an icon for B. Nontriviality can be taken to rule out purely accidental resemblance or degenerate cases where the token fails to track B at all in that context (e.g. noise that happens to match a feature).*

Now suppose the same registration also yields a spatiotemporal pointer token $r_{\text{index}} \in \mathcal{R}$ (e.g. "the thing at location ℓ at time t"), with nontrivial reference in some $C_{\text{st}} \in \mathbf{Ctx}_{\text{st}}$. Then r_{index} is an index for B. The index need not share sensory features with B; its fidelity is instead to the coordinate structure of the spatiotemporal context (tracking, co-location constraints, persistence conditions, and so on). This makes explicit how a mind can keep track of "that thing there" even when the iconic channel is weak or corrupted, and conversely how an icon may be vivid without being anchored to a stable pointer.

Finally, suppose M maintains a relational knowledge graph in a context $C_{\text{rel}} \in \mathbf{Ctx}_{\text{rel}}$. A new node token $r_{\text{sym}} \in \mathcal{R}$ is introduced into this graph and linked via learned relations (e.g. "is-a", "part-of", "causes"). If r_{sym} represents B in C_{rel} and participates in the graph's combinational operations (composition/inference rules), then r_{sym} is a symbol for B. The emphasis on participation is crucial: the symbol is not merely a label, but a handle that supports rule-governed recombination (e.g. inheritance along "is-a", constraint propagation along "part-of", or abductive links along "causes"). In this sense the symbol preserves a pattern family that is primarily relational *rather than sensory or spatiotemporal, and its adequacy is tested by the success of downstream inferences rather than by immediate resemblance.*

In this way, one referent B can simultaneously have iconic, indexical, and symbolic representations within the same mind, each preserving different pattern families. The three tokens may be created together by a single episode, but nothing in the setup requires a one-to-one correspondence: a single referent can spawn many icons (different viewpoints), many indices (different tracking frames), and many symbols (different conceptual roles), and conversely a single token can sometimes serve multiple roles when contexts align.

Remark 625. *This example highlights why the aspect index C is not decorative. The same referent B can generate three tokens with three different "fidelities," each faithful in its own slice of structure. The icon inherits sensory patterns; the index inherits spatiotemporal patterns; the symbol inherits relational patterns and gains power from combinational constraints. In realistic cognitive architectures, these modes interact: symbols can cue imagery, indices can anchor symbolic facts, and icons can be annotated symbolically. The present formalism supports that interaction precisely because all modes are instances of the same representation schema. More formally, the context index indicates (i) which signature map is in play, (ii) which reference conditions count as nontrivial tracking, and (iii) which transformations are licensed as "structure-preserving" within that mode. Interactions across modes can then be modeled as context-bridging operations (e.g. an index-to-symbol update that binds a pointer to a graph node, or a symbol-to-icon query that retrieves a stored prototype), rather than as an informal mixing of incomparable representational types. This also clarifies how conflict can arise and be managed: an icon may strongly support a property while the symbol-graph entails its negation, yielding the kind of simultaneous positive and negative* *evidence that motivates the soggy/paraconsistent treatment above, without forcing an immediate collapse of either representational channel.*

13.8 What this buys us for later sections

This section sets up the representational vocabulary needed for later parts of the paper: in particular, it provides a disciplined way to move between (a) causal interaction with the world, (b) internal state that carries information across time, and (c) the system's own graded assertions about what is the case. The payoff is that later sections can rely on these distinctions without having to re-argue them each time, and can state claims in a way that is stable under changes of implementation details (e.g. the chosen evidence calculus).

- In Section 14, prediction and control can be defined over representations, with sensory registration providing the perception channel. This means that what gets predicted (and what gets steered) is not raw stimulation as such, but the system's internal stand-ins that persist and can be compared, composed, and acted upon; registration then functions as the principled interface by which the environment constrains and corrects those stand-ins.
- In Section 15, attention can be modeled as selective updating of subsets of the representation/predicate network. On this framing, "attention" is not a mysterious extra faculty but a policy over which representational tokens and which predicate-attachments are revised given new registrations; this makes it natural to treat attentional bottlenecks and synergy effects as properties of update locality, bandwidth, and coupling between subnets.
- In Sections 16 and 17, self models and reflective content will be formalized as representational fixed points and higher-order registrations (e.g. "M registers that M registers A"). The fixed-point language is intended to capture the way a stable "self" representation can be maintained even as particular subordinate representations change, while higher-order registration provides a uniform syntax for reflexive and meta-level content (for example, when the system not only registers A but also registers something about its own registering of A).

Remark 626. *In philosophical terms, we have separated three layers that are often conflated: (i) the world's impact on a system (registration), (ii) the system's stabilized internal stand-ins (representation tokens), and (iii) the evidential grading of assertions about entities (soggy predicates). This separation is what permits later sections to talk rigorously about agency: prediction and control operate on representations, attention shapes which representational subnets are updated, and reflective consciousness can be treated as iterated registration/representation operating on the system itself (Hyperseed-Concept 166, 85). A useful way to read (i)–(iii) is as a progression from causal contact, to internal re-identification, to doxastic stance: registration is the "impression" or incoming constraint, representation tokens are the stabilized carriers that allow re-use and cross-context comparison, and soggy predicates express not just whether a claim is stated but how strongly it is supported given the system's currently available registrations and representational structure. In particular, soggy predicates make room for the possibility that the same representation token can be maintained while its evidential status shifts (e.g. strengthened, weakened, or put into tension by new registrations), which will matter later when we discuss conflict, revision, and partial observability.*

Mathematically, the benefit is modularity: one may change the evidence domain (e.g. Boolean, probabilistic, p-bit*-paraconsistent) or change the context poset, while keeping the same definition-shape of mind, perception, and representation. This is a recurring theme in the Hyperseed program [1]. Concretely, the definitions are arranged so that (a) representation-management operations (e.g. prediction, control, and selective update) depend primarily on the abstract interfaces provided by* *the chosen evidence domain and context structure, and (b) the perception channel can be treated as a morphism-like linkage from registrations into representational updates. As a result, later claims can be phrased at a level where one can swap out, for instance, a crisp two-valued evidential semantics for a graded or paraconsistent one, and still interpret the same formal roles for "perception," "belief-like grading," and "self-referential registration" without rewriting the surrounding conceptual architecture.*

14 Prediction, attraction, causality, and control

14.1 From patterns to forecasts and interventions

Sections 9–13 built a scaffold in which an observer/context O (a mind, or a mind-plus-environment slice) (i) makes distinctions, (ii) registers patterns, (iii) organizes those patterns into hierarchies/webs, and (iv) maintains representations whose content is indexed by contexts and aspects.

This scaffold is intentionally descriptive: it explains how a structured "world of patterns" can be available *to* an observer, without yet committing to any particular stance on time, action, or counterfactual dependence. In particular, the indexing-by-context machinery implies that what counts as "the same" pattern may vary with aspect (what features are being attended to) and with the surrounding representational commitments (what background regularities are currently assumed). Forecasting and intervention add further structure: they require that the observer's stored patterns be usable not only as compressions of what has been registered, but also as guides to what is likely, what is avoidable, and what is achievable under feasible actions.

Remark 627. *In the Hyperseed framing, the passage from "patterns" to "agency" is not a metaphysical jump but a change of grammatical tense: we move from describing what is coherently present to describing what is* expected *to become present, and what may be* made *to become present by action. This shift is precisely what brings prediction, control, and therefore causality into focus, and it connects the pattern-oriented representational layer to the later resource-bounded attentional layer (Section 15). See [1, 5] for the motivating ontology-level narrative.*

One can read this "tense shift" as adding a minimal temporal semantics to the representational architecture: patterns are no longer merely relations among currently co-present distinctions, but also constraints on transitions between contexts. In that sense, the same representational web that supports recognition ("this looks like that") is repurposed into a transition model ("from here, that tends to follow"). Likewise, the same distinction-making that supports categorization is repurposed into a space of candidate interventions: an "action" is, at minimum, an internally available distinction whose enactment changes which subsequent contexts become accessible.

This section adds the "forward arrow" needed for agency: *prediction* (how likely are future patterns/events?) and *control* (how can actions steer future patterns/events?), as well as the intermediate notion of *causality* (Hyperseed-Concept 70).

A useful way to separate these three is by their roles, not their vocabulary. Prediction concerns inference under uncertainty: given present context/aspects and available evidence, what distribution over future distinctions is warranted? Control concerns selection under constraints: given possible actions and limited resources, which action policies best steer the distribution of future contexts toward desired regions? Causality, in this framing, is the bridge that makes control non-magical: it is the claim that some distinctions (candidate actions, conditions, or structural changes) make a *systematic difference* to future patterns beyond mere co-occurrence, thereby licensing intervention planning rather than passive extrapolation.

Hyperseed's key move here is to make the dependence notions explicitly *differential*. For decision making, what matters is rarely $P(B \mid A)$ by itself, but rather how different $P(B \mid A)$ is from $P(B \mid \neg A)$. This is packaged in the notions of *attraction*, *predictive implication*, and *predictive attraction* (Hyperseed-Concept **??**).

Concretely, a non-differential statement like "B is likely" can be true in a way that is irrelevant to agency: B may be likely regardless of what the agent does, because of background regularities that dominate the dynamics. Differential dependence, by contrast, asks whether A changes the likelihood of B *relative to a baseline*, where the baseline is typically "not A" or "the default policy" in the given context. This is why notions like attraction naturally support ranking and prioritization: if two candidate actions both yield high $P(B \mid A)$, but one of them yields only a tiny improvement over $P(B \mid \neg A)$, then that action carries less leverage and is typically less valuable for control.

It is also where the representational indexing by context matters operationally. The quantities $P(B \mid A)$ and $P(B \mid \neg A)$ are only meaningful relative to a specified evidential context, including which aspects are held fixed, which are allowed to vary, and which background conditions are treated as stable. Thus, the "difference" being computed is not merely a numerical subtraction but a context-sensitive comparison between two nearby worlds (or two nearby policy regimes) as represented by O. When later sections introduce resource-bounded attention, this contextuality becomes even more pronounced: an agent may only be able to approximate such differences using compressed, aspect-limited summaries of the evidence.

From the standpoint of interventions, it is often helpful to distinguish evidential conditioning from interventional forcing: $P(B \mid A)$ can be high because A is a symptom of hidden causes, even if setting A would not increase B. Hyperseed's "control" reading therefore implicitly invites a distinction between "observing A" and "doing A"—the latter corresponding to an intervention that breaks certain background dependencies. Even when the formalism is written in ordinary conditional-probability terms, the intended use is counterfactual: the agent is comparing what happens under a contemplated action to what would have happened without it, within the same contextual slice.

Remark 628. *The differential stance is philosophically modest but practically decisive: it acknowledges that many future outcomes are overdetermined by background regularities, and that an agent's action matters only insofar as it* tilts *the odds relative to what would have happened otherwise. This is also where ordinary probability talk (Hyperseed-Concept 139) becomes insufficient on its own: in borderline, vague, or internally conflicted contexts, we often possess evidence both for and against an event, and the* difference *between conditional propensities must be computed from that evidence rather than assumed crisp at the outset. Paraconsistent treatments of evidence streams of this sort are developed in various logical settings; see, e.g., [23, 24].*

A further pragmatic consequence is that differential notions can be estimated and improved incrementally. An observer may begin with crude qualitative comparisons ("A seems to help B") and refine them into more stable graded relations ("A attracts B more strongly than A' does") as additional evidence accumulates, aspects are reweighted, or confounders are represented explicitly. In this sense, attraction-like quantities function as "handles" for learning and control: they are directly tied to what changes when the agent changes something, and thus to the empirical signature of agency within a pattern-based world-model.

14.2 Event types, timelines, and paraconsistent occurrence data

We assume the temporal scaffolding from Section 7: a (possibly branching, partially ordered) timeline $(T, \preceq)$ for a given context/observer, where $t \preceq t'$ means "t is not later than t' as experienced/represented by O."

Remark 629. *The notation $(T, \preceq)$ should be read as follows: T is a set of time indices, and $\preceq$ is a partial order encoding the observer's "before/after" structure (Hyperseed-Concept 142). The possibility that $\preceq$ is not total allows branching and incomparability, which is useful when the observer's model has multiple consistent continuations or when retrospective reconstruction leaves some order relations undecided.*

Remark 630. *It is helpful to keep three distinct readings of $t \in T$ in mind, depending on the application: (i) t may be a physical clock reading (as in sensor timestamps), (ii) t may be an internal computational step or belief-state index (as in an agent's recurrent update cycle), or (iii) t may be a reconstructed "scene time" used to order remembered or narrated events. The partial order $\preceq$ is deliberately flexible enough to cover all three cases while remaining agnostic about whether time is continuous, discrete, or hybrid.*

Remark 631. *When $\preceq$ is partial, one can still speak about* histories *(or* branches*) as chains $H \subseteq T$ such that any two elements of H are $\preceq$-comparable. Many constructions later can be understood as first restricting attention to a particular chain H (a single "possible continuation" as tracked by O), and only then applying sequence-like reasoning. This is one way to connect branching time intuitions with sequence-based prediction rules.*

Definition 174 (Event types and event tokens)**.** *Let $\mathcal{E}$ be a set of* event types *(also called "event predicates"), such as `kiss(Ben)` or `get(ball)`. An* event token *is a pair (A, t) with $A \in \mathcal{E}$ and $t \in T$. We write $A@t$ for "event type A occurs at time t." We assume the event language is closed under a formal negation operator $A \mapsto \neg A$, read as "not A". Because boundaries may be vague or inconsistent, it is allowed that both $A@t$ and $(\neg A)@t$ carry substantial evidence.*

Remark 632. *Intuitively, an* event type *is a reusable description (a predicate schema), while an* event token *is a particular alleged occurrence located at a particular time index. The notation $A@t$ is a compact "addressing" convention: it keeps the type A and the time t visible so we can talk about temporal relations between occurrences.*

Remark 633. *The operator $\neg$ is treated here as* syntactic *negation: it simply forms a new event type $\neg A$ from A. In particular, we do not assume that $\neg A$ exhausts all alternatives to A, or that it behaves like a classical complement with respect to evidence. This matters in practice: for many perceptual or social predicates, "not A" is not a well-delimited natural kind, and the observer may have independent reasons to support both A and $\neg A$ (e.g., due to incompatible measurement channels or shifting criteria of application).*

Remark 634. *A simple example is: let $A =$`get(ball)` and let t be the moment a camera frame shows a hand grasping a ball. Then $A@t$ is the token claim "the ball is gotten at time t." If the camera view is occluded, one may simultaneously have evidence for $A@t$ (a motion consistent with grabbing) and for $(\neg A)@t$ (no clear visual confirmation). Allowing both is not a flaw here; it is a way of keeping track of the epistemic situation without prematurely forcing a crisp boundary between "occurs" and "does not occur." This connects back to Hyperseed's emphasis on distinction-making as observer-relative (Sections 3 and 13).*

Remark 635. *In examples like* `kiss(Ben)` *or* `get(ball)`*, the event type may implicitly carry arguments (participants, objects, locations). One can regard $\mathcal{E}$ as a set of fully instantiated predicates (ground atoms), or as a set of predicate schemas paired with a domain of entities. Nothing below depends on this choice, as long as the notation $A@t$ picks out a well-formed token claim that can be supported or opposed by the observer's evidence.*

Remark 636. *This definition is useful because it cleanly separates (i) the* syntax *of events (what we are talking about), (ii) the* temporal indexing *(when we are talking about it), and (iii) the* evidence semantics *(how strongly we support or deny the token). That separation will let us define prediction and control without assuming an underlying classical truth assignment.*

Hyperseed is explicitly comfortable with borderline, context-dependent, and even inconsistent boundaries. We therefore allow paraconsistent evidence about event occurrences.

Definition 175 (Paraconsistent occurrence evidence)**.** *Fix an observer/context O. A* paraconsistent occurrence assignment *is a map*

$$E_O : \mathcal{E} \times T \to [0,1]^2, \qquad E_O(A@t) = (E_O^+(A@t), E_O^-(A@t)),$$

where E_O^+ is the degree of positive evidence for occurrence and E_O^- is the degree of negative evidence for occurrence. No consistency constraint is imposed; it may happen that $E_O^+(A@t) + E_O^-(A@t) > 1$.

Remark 637. *The pair $(E_O^+(A@t), E_O^-(A@t))$ is the* p-bit*-style encoding of evidence: the first coordinate is support for occurrence, the second is support for non-occurrence. The subscript O emphasizes that this is* observer-indexed *evidence, not an observer-independent fact (Hyperseed-Concept* **??** *is treated elsewhere).*

Remark 638. *A useful way to read $(p, q) \in [0,1]^2$ is as answering two distinct questions that need not be complements: "How much do I have reasons for $A@t$?" (measured by p) and "How much do I have reasons against $A@t$?" (measured by q). Paraconsistency enters precisely by refusing the classical identification of "reasons against" with "lack of reasons for," thereby allowing the agent to represent conflict, ambiguity, or model mismatch without forcing an immediate resolution.*

Remark 639. *As a toy example, suppose a microphone picks up a sound that is somewhat like a door slam. An audio classifier might assign $E_O^+ = 0.7$ for $A =$*`door-slam` *at time t, while a concurrent vision module that sees no door movement assigns $E_O^- = 0.6$. Then $E_O^+ + E_O^- = 1.3 > 1$, which records a genuinely conflicted evidential state. This is especially useful when modeling agents that must act under contradictory cues rather than wait for idealized consistency.*

Definition 176 (Derived evidence diagnostics (consistency, conflict, and indeterminacy))**.** *Given a token $A@t$ with evidence $(p, q) = E_O(A@t)$, define:*

$$\mathrm{conf}_O(A@t) := \min(p, q), \qquad \mathrm{gap}_O(A@t) := 1 - \max(p, q), \qquad \mathrm{bal}_O(A@t) := |p - q|.$$

Remark 640. *The quantity $\mathrm{conf}_O(A@t)$ measures* conflict*: it is large only when both occurrence and non-occurrence are substantially supported. The quantity $\mathrm{gap}_O(A@t)$ measures* indeterminacy *(or evidential "silence"): it is large when neither side is strongly supported. Finally, $\mathrm{bal}_O(A@t)$ measures how* decisive *the net evidence is, regardless of whether the decisiveness arises from consistent support (large p with small q) or consistent refutation (small p with large q). These diagnostics are optional but often clarify which kind of uncertainty is present: "both" (conflict) versus "neither" (gap).*

Remark 641. *This definition is necessary for later sections where "self/other" boundaries and other high-level partitions can be simultaneously supported and resisted (Section 16). It also aligns with the broader paraconsistent stance that contradiction need not entail collapse of inference, but may instead be treated as structured information [23, 24].*

To speak about attraction, we want a scalar "plausibility" derived from a p-bit. Many choices are possible; the following is algebraically convenient and reduces to ordinary probabilities in the consistent case.

Definition 177 (Plausibility projection)**.** *Define* $\pi : [0,1]^2 \to [0,1]$ *by*

$$\pi(p,q) := \frac{1+p-q}{2}.$$

We call $\pi(E_O(A@t))$ *the* plausibility *of* $A@t$ *for* O*.*

Remark 642. *The symbol* π *here is a* projection *from two-dimensional evidence to a single scalar in* $[0,1]$*. If* p *is positive evidence and* q *is negative evidence, then* $p-q$ *is a signed net tendency toward occurrence; the affine transformation* $(1+(p-q))/2$ *merely rescales this net tendency so that* 0 *corresponds to "strongly implausible,"* 1 *corresponds to "strongly plausible," and* $1/2$ *corresponds to a neutral balance.*

Remark 643. *The mapping* π *has a few immediate properties that motivate its later use as an "attraction proxy." First, it is monotone in the intended directions: increasing* p *(holding* q *fixed) increases plausibility, and increasing* q *decreases it. Second, it is symmetric around the neutral point: swapping* p *and* q *sends* $\pi(p,q)$ *to* $1-\pi(p,q)$*, expressing that "more reasons against" mirrors "more reasons for." Third, in a consistent probabilistic special case where the observer's evidence is such that* $q = 1-p$*, one obtains* $\pi(p, 1-p) = p$*, so* π *reduces to the familiar single probability parameter.*

Remark 644. *It is also worth emphasizing what* π *does* not *do: it does not distinguish between high-conflict and low-information states when they have the same net difference* $p-q$*. For example,* $(p,q) = (1,0)$ *and* $(p,q) = (0.6,0)$ *both yield high plausibility, while* $(0.9, 0.9)$ *and* $(0.1, 0.1)$ *both yield* $\pi = 1/2$*. This is intentional at this stage: attraction-oriented dynamics later will often be driven by a one-dimensional "direction" (toward/away), while diagnostics such as* $\mathrm{conf}_O(A@t)$ *and* $\mathrm{gap}_O(A@t)$ *can be carried alongside* π *when it matters to distinguish "conflicted" from "uninformative" neutrality.*

Remark 645. *Alternative scalarizations are possible (e.g., using* p *alone, using* $p/(p+q)$ *when* $p+q > 0$*, or using a nonlinear squashing of* $p-q$*). The choice of* π *is a minimal linear map that preserves the qualitative ordering induced by net evidence while keeping values in* $[0,1]$*, which makes it convenient for coupling to later definitions that resemble probabilistic prediction or utility-like attraction scores without presupposing classical semantics.*

Remark 646. *For example,* $\pi(1,0) = 1$*,* $\pi(0,1) = 0$*, and* $\pi(0.7, 0.6) = \frac{1+0.1}{2} = 0.55$*, reflecting that the token is only slightly more plausible than not. This scalarization is useful because attraction is defined as a difference of conditional plausibilities, and differences are most straightforwardly taken in a totally ordered set like* $[0,1]$*.*

Remark 647 (Consistent case)**.** *If the evidence is consistent in the sense that* $E_O^-(A@t) = 1 - E_O^+(A@t)$*, then* $\pi(E_O(A@t)) = E_O^+(A@t)$*. Thus the projection* π *exactly recovers the "usual" probability-like degree.*

Remark 648. *The consistent case remark shows that we are not abandoning classical probabilistic intuition (Hyperseed-Concept 139); rather, we are extending it so that classical probability appears as the special case where positive and negative streams are perfectly complementary. This is a common pattern in foundational work: one seeks a conservative extension whose additional degrees of freedom encode epistemic subtleties rather than metaphysical extravagance.*

14.3 Prediction and attraction

14.3.1 Conditionals

To define attraction we need conditionals. We deliberately stay lightweight and treat conditionals as something computed by the observer's modeling machinery (statistical estimation, simulation, logical inference, or combinations).

Definition 178 (Predictive conditional operator)**.** *Fix a lag window Δ (e.g. a set of allowed lags, or an interval $[\delta_{\min}, \delta_{\max}]$). A* predictive conditional operator *for O is a map*

$$\mathrm{Cond}_{O,\Delta} : \mathcal{E} \times \mathcal{E} \to [0,1]^2, \qquad (A, B) \mapsto \mathrm{Cond}_{O,\Delta}(B \mid A),$$

where $\mathrm{Cond}_{O,\Delta}(B \mid A)$ is intended to represent evidence about "B occurs at some time t' with $t \prec t'$ and $t' - t \in \Delta$, given that A occurs at t."

Remark 649. *Notation unpacking: $\mathrm{Cond}_{O,\Delta}(B \mid A)$ is a* p-bit*-valued object in $[0,1]^2$, not a single probability. The vertical bar "$|$" is read "given," and Δ specifies what counts as an acceptable temporal separation between the antecedent token time t and the consequent token time t'. The strict order $t \prec t'$ (derived from $\preceq$) encodes temporal direction.*

Remark 650. *Intuitively, $\mathrm{Cond}_{O,\Delta}$ is the observer's* forecasting interface*: it answers the question, "if A happens, how much evidence do I have that B will happen soon after (within the window Δ)?" A simple example is: $A =$`press(button)`, $B =$`light(on)`, and $\Delta = \{1\}$ in discrete time. Then $\mathrm{Cond}_{O,\Delta}(B \mid A)$ captures the learned (or inferred) reliability of the button-to-light connection.*

Remark 651. *This definition is useful because it remains neutral about* how *the observer computes conditionals: from frequencies, from simulation, from mechanistic modeling, or from logical rules. This neutrality matters for Hyperseed because minds may contain multiple inference processes whose outputs must be fused (Section 15 discusses such multi-process regimes; see also [19] for a broader cognitive-systems perspective).*

Remark 652 (Empirical instance)**.** *In a finite dataset of event tokens, one may estimate $\mathrm{Cond}_{O,\Delta}(B \mid A)$ from relative frequencies of co-occurrence of $A@t$ and $B@t'$ with $t' - t \in \Delta$, separately for positive and negative evidence streams. We use the abstract operator $\mathrm{Cond}_{O,\Delta}$ so the formalism can also cover more structured inference.*

14.3.2 Attraction

Definition 179 (Attraction)**.** *For event types $A, B \in \mathcal{E}$ and lag window Δ, define the* attraction

$$\mathrm{Att}_{O,\Delta}(A, B) := \pi\big(\mathrm{Cond}_{O,\Delta}(B \mid A)\big) - \pi\big(\mathrm{Cond}_{O,\Delta}(B \mid \neg A)\big).$$

This is a real number in $[-1, 1]$.

Remark 653. *Attraction is the simplest formal embodiment of the "difference that makes a difference." If* $\mathrm{Att}_{O,\Delta}(A, B)$ *is positive, then (in the observer's model) A makes B* more *plausible than it would be under $\neg A$; if it is negative, A suppresses B relative to the $\neg A$ baseline. This is the formal heart of Hyperseed's* PredictiveAttraction *notion (Hyperseed-Concept* **??***).*

Remark 654. *A concrete example: let $A =$*`water(plant)` *and $B =$*`plant(healthy)` *within a two-week lag window Δ. If the model estimates $\pi(\mathrm{Cond}(B \mid A)) = 0.8$ but $\pi(\mathrm{Cond}(B \mid \neg A)) = 0.5$, then attraction is 0.3: watering provides a substantial differential lift. Conversely, if a healthy plant is likely anyway due to rain (0.75 without watering), then even a high absolute conditional (0.8) yields low attraction (0.05), signaling weak control leverage.*

Remark 655. *The definition is necessary because control and planning should be built on* leverage, *not on mere correlation: if B happens regardless, then A is not a good handle on the world. Attraction is the low-level quantity that will later be mixed with temporal precedence and simplicity bias to yield a causal notion (Section 14.5).*

Proposition 22 (Basic properties of attraction)**.** *For all A, B and Δ:*

(a) (Boundedness.) $\mathrm{Att}_{O,\Delta}(A, B) \in [-1, 1]$.

(b) (Negation antisymmetry.) $\mathrm{Att}_{O,\Delta}(\neg A, B) = -\mathrm{Att}_{O,\Delta}(A, B)$.

(c) (Independence gives zero.) If $\pi(\mathrm{Cond}_{O,\Delta}(B \mid A)) = \pi(\mathrm{Cond}_{O,\Delta}(B \mid \neg A))$ *then* $\mathrm{Att}_{O,\Delta}(A, B) = 0$.

Remark 656. *This proposition says that attraction behaves like a signed, normalized measure of differential predictive influence: it cannot exceed 1 in magnitude, flipping the antecedent to its negation flips the sign, and if the antecedent makes no difference to the consequent (in the projected conditional plausibility), attraction vanishes. These are not deep theorems; rather, they are sanity checks that the formal definition matches the intended conceptual role.*

Remark 657. *The connection to later material is direct: control strength will be defined using causal implication, and causal implication will incorporate predictive attraction. Thus, verifying simple algebraic properties here prevents later definitions from inheriting unintended asymmetries or scale problems.*

Proof. (a) Each term is in $[0, 1]$, hence their difference lies in $[-1, 1]$. (b) Swap A and $\neg A$ in the definition. (c) Immediate. □

Proof sketch. All three items follow by reading the definition literally. The key observation is that the projection π maps into $[0, 1]$, so subtraction yields a quantity in $[-1, 1]$, and exchanging A with $\neg A$ simply swaps the minuend and subtrahend, reversing the sign. In slightly more detail: for any conditional object $\mathrm{Cond}(\cdot \mid \cdot)$ in the definition, the scalarization $\pi(\mathrm{Cond}(\cdot \mid \cdot))$ is by construction a real number between 0 and 1. Therefore, any expression of the form $\pi(\mathrm{Cond}(B \mid A)) - \pi(\mathrm{Cond}(B \mid \neg A))$ is automatically bounded below by $0 - 1 = -1$ and above by $1 - 0 = 1$. Moreover, the "antisymmetry under negation" is not a separate theorem but a one-line algebraic check:

$$\pi(\mathrm{Cond}(B \mid \neg A)) - \pi(\mathrm{Cond}(B \mid \neg\neg A)) = -\big(\pi(\mathrm{Cond}(B \mid A)) - \pi(\mathrm{Cond}(B \mid \neg A))\big),$$

using only $\neg\neg A = A$ and commutativity of subtraction in the sense of swapping operands. □

Remark 658. *The proof works because the definition of attraction is intentionally built from very rigid pieces: a bounded scalarization followed by subtraction. The "negation antisymmetry" is not a mysterious logical fact; it is an algebraic symmetry that we engineered into the definition so that "doing A" and "not doing A" are treated as complementary alternatives. Note that this design also forces a clear interpretation of the sign: positive attraction means that, after projection, B is more plausible conditional on A than conditional on $\neg A$; negative attraction means the opposite; and attraction near* 0 *means that A carries little* differential *information about B compared to its negation, even if both one-sided conditionals are individually high.*

Remark 659 (Why attraction matters)**.** *Suppose B is a goal condition and A is an action. Even if $\pi(\mathrm{Cond}(B \mid A))$ is large, taking action A is not very useful if $\pi(\mathrm{Cond}(B \mid \neg A))$ is equally large. Attraction measures the* differential *predictive value of A for B. This is the basic reason attraction is closer to the intuitive notion of a "handle" or "lever": it discounts baseline prevalence of B that would be present regardless of whether A occurs. In particular, if B is already near-certain in the background circumstances (so both conditionals project close to* 1*), then attraction correctly reports that A adds little, whereas a raw conditional may misleadingly suggest that A is strongly associated with B.*

14.3.3 Predictive implication and predictive attraction

Hyperseed distinguishes ordinary implication from *predictive implication* (implication plus a temporal precedence constraint). In the present lightweight setting, we simply make this temporal aspect explicit via Δ. Operationally, Δ should be read as a fixed prediction horizon: we are not merely asking whether B is compatible with A, but whether B is expected to occur *after* A within a specified window, so that the resulting quantity is tied to forecasting and (later) to action evaluation.

Definition 180 (Predictive implication and predictive attraction)**.** *Define the* predictive implication strength *as*

$$\mathrm{PImp}_{O,\Delta}(A, B) := \pi\big(\mathrm{Cond}_{O,\Delta}(B \mid A)\big) \in [0, 1].$$

Define the predictive attraction *as*

$$\mathrm{PAtt}_{O,\Delta}(A, B) := \mathrm{PImp}_{O,\Delta}(A, B) - \mathrm{PImp}_{O,\Delta}(\neg A, B).$$

Remark 660. *Here $\mathrm{PImp}_{O,\Delta}(A, B)$ is the one-sided, probability-like strength of the forward conditional "A predicts B within Δ" (after projection). It is intentionally* not *yet causal: it does not require that A precedes B typically in the stronger sense, nor that there be any intensional connection. Predictive attraction* PAtt *then restores the differential stance by contrasting A against $\neg A$. One can also view* PImp *as a score assigned to the ordered pair (A, B), whereas* PAtt *is a score assigned to the* contrast *between two rival predictors, A and $\neg A$, holding B fixed. This contrastive structure is what makes* PAtt *more stable as a decision signal in settings where the background rate of B fluctuates across contexts.*

Remark 661. *Example: if a smoke detector rings (A) then a person wakes up (B) with high conditional plausibility, say* $\mathrm{PImp} = 0.9$*. But if the person would wake anyway because of a loud storm even without the detector (*$\mathrm{PImp}(\neg A, B) = 0.85$*), then* $\mathrm{PAtt} = 0.05$*. The model is saying: the detector is a good predictor but a weak* lever*. This distinction becomes crucial once we interpret A as an action or intervention candidate in control. The same arithmetic also clarifies the limiting cases: if* $\mathrm{PImp}(\neg A, B) \approx 0$*, then* $\mathrm{PAtt} \approx \mathrm{PImp}(A, B)$ *and the action-like reading of A becomes more credible; if* $\mathrm{PImp}(A, B) \approx \mathrm{PImp}(\neg A, B)$*, then* $\mathrm{PAtt} \approx 0$ *and A provides little incremental predictive advantage even if the absolute prediction is strong.*

Remark 662 (Relationship to attraction)**.** *With the above definitions,* $\mathrm{PAtt}_{O,\Delta}(A,B) = \mathrm{Att}_{O,\Delta}(A,B)$*. We keep both notations because (i) the ontology uses both; and (ii) in richer formalisms one may separate "mere conditional dependence" from "dependence with a specific temporal semantics." In particular, future refinements may encode the* Δ*-constraint in the conditional object itself (e.g., as an event construction) rather than as an external parameter, at which point it can become useful to reserve* Att *for an atemporal notion and* PAtt *for the explicitly forward-looking notion used in prediction and control.*

Remark 663. *This duplication of notation is philosophically telling: in the minimal formalism, the concepts coincide; in richer settings, they may come apart. It is often fruitful in foundational work to keep two names for what is presently the same quantity, if one expects later refinements to split the notion along different conceptual joints (as happens below when we separate extensional from intensional predictive attraction). Concretely, one expects such a split once the formalism distinguishes (a) extensional regularities in observed sequences and (b) intensional linkages that support counterfactual and interventional reasoning. The present section keeps the arithmetic simple, but the duplicated naming convention acts as a reminder that the surrounding interpretation can change even when the underlying number happens to be computed by the same formula at this stage.*

Remark 664 (Range and interpretive calibration)**.** *Because* $\mathrm{PImp}_{O,\Delta}(A,B) \in [0,1]$ *by definition, it follows immediately that* $\mathrm{PAtt}_{O,\Delta}(A,B) \in [-1,1]$*. The endpoints have a clear meaning:* $\mathrm{PAtt} = 1$ *occurs only when* $\mathrm{PImp}(A,B) = 1$ *and* $\mathrm{PImp}(\neg A,B) = 0$*, i.e.,* A *perfectly predicts* B *within* Δ *and* $\neg A$ *perfectly predicts* $\neg B$ *within the same horizon (after projection);* $\mathrm{PAtt} = -1$ *reverses these roles. Intermediate values can be read as a signed "margin" of predictive advantage, and the sign convention is fixed so that* $\mathrm{PAtt} > 0$ *favors* A *over* $\neg A$ *as a predictor for* B*.*

Remark 665 (Dependence on the projection π)**.** *The role of* π *is to force a common numeric scale for comparison, independent of the internal representation of* $\mathrm{Cond}_{O,\Delta}(B \mid A)$*. This makes the subtraction in* PAtt *well-typed and ensures boundedness, but it also means that the qualitative behavior of* PImp *and* PAtt *is sensitive to how* π *treats uncertainty, indeterminacy, or graded evidence. Accordingly, later sections can change the empirical behavior of predictive attraction either by changing the conditional object* Cond *or by changing the projection* π*, without changing the surrounding algebraic facts used above.*

14.4 Sequential and parallel temporal composition

Hyperseed emphasizes that predictive dependence concepts should compose along time. In the original ontology this is expressed using operators such as `SequentialAND`, `SimultaneousAND`, and `SimultaneousOR`. Here we sketch one clean way to realize this using the quantale and enriched-category machinery from Section 3.4.

Remark 666. *The guiding intuition is that a mind rarely reasons about isolated one-step predictions. It reasons about* chains *(if I do this, then that happens, then something else becomes possible) and about* bundles *(multiple conditions jointly supporting a later outcome). Quantales provide a compact algebraic language for this kind of composition: a tensor* $\otimes$ *aggregates along a sequence, and a join* $\oplus$ *aggregates across alternatives. This style is consonant with the pattern-web machinery used earlier (Sections 9 and 12) and foreshadows later planning/optimal-control principles (Section 26; see [21]).*

14.4.1 Sequential composition as quantale multiplication

Let $\mathcal{E}$ be viewed as a set of "states" or "event-conditions" and suppose the observer has a p-bit-valued predictive relation

$$R_{O,\Delta} : \mathcal{E} \times \mathcal{E} \to [0,1]^2, \qquad R_{O,\Delta}(A,B) := \mathrm{Cond}_{O,\Delta}(B \mid A).$$

If we interpret $R_{O,\Delta}(A,B)$ as an "edge weight" from A to B, then sequential composition of two predictive steps corresponds to relational composition, with the quantale tensor as the multiplicative aggregator.

Remark 667. *Notation unpacking: $R_{O,\Delta}$ is a V-valued relation (here $V = [0,1]^2$ with* p-bit *operations), so $R_{O,\Delta}(A,B)$ is the edge weight from A to B. The symbol $\otimes$ is the quantale tensor (multiplicative aggregation along a path), and $\oplus$ is the quantale join (additive aggregation across alternative intermediate nodes). This is the same algebraic pattern used in enriched-category composition, but here read operationally as "compose predictions across time."*

Definition 181 (Two-step predictive composition)**.** *Define the composite relation*

$$(R_{O,\Delta} \circ R_{O,\Delta'})(A,C) := \bigoplus_{B \in \mathcal{E}} R_{O,\Delta}(A,B) \otimes R_{O,\Delta'}(B,C),$$

where $\otimes$ and $\oplus$ are the tensor and join of the p-bit *quantale (Section 3.4).*

Remark 668. *This is the familiar "sum over intermediates, multiply along edges" pattern, generalized from classical probability or weighted automata to the* p-bit *quantale setting. One may visualize $\mathcal{E}$ as nodes in a directed graph of event-types; then $(R \circ R')(A,C)$ aggregates the strength of all two-step paths $A \to B \to C$.*

Remark 669. *A simple example: if A predicts B strongly and B predicts C strongly, then the tensor $R(A,B) \otimes R'(B,C)$ yields a strong composite for that B. If there are multiple plausible bridges B, the join $\oplus$ allows them to compete/cooperate depending on the quantale's join. This definition is useful because it makes explicit what is often left informal in planning discussions: the semantics of chaining predictions is not automatic; it is a chosen algebra.*

Remark 670 (Reading)**.** *The join $\oplus$ aggregates over intermediate events B ("there exists a reasonably good bridge"), while the tensor $\otimes$ aggregates evidence across a temporal chain ("both steps work"). This is the algebraic content behind the informal operator `SequentialAND`.*

14.4.2 A tiny worked schematic

Consider an action sequence (cf. the ontology's example):

`Teacher says fetch` $\to$ `I get the ball` $\to$ `I bring the ball` $\to$ `I get a reward`.

If each arrow has a predictive implication strength (in $[0,1]$ after projection π), then the overall sequence can be given a composite strength by multiplying along the chain (tensor) and joining over alternative intermediate paths. Predictive attraction then compares the "do the sequence" case with the "do not do the sequence" case, exactly as in the single-step definition.

Remark 671. *Even this schematic illustrates an important philosophical point: "the plan works" is not a primitive fact but a compositional judgment assembled from local predictive regularities. The Hyperseed approach tries to keep such judgments* graded *and* evidence-sensitive*, so that a plan may be adopted with appropriate humility rather than with binary certainty.*

14.5 Causality as attraction plus temporal precedence plus simplicity bias

The ontology proposes that causality is not captured by extensional dependence alone. Rather, causal judgments combine: (i) a temporal precedence constraint; (ii) a surprising extensional dependence (predictive attraction); (iii) an intensional (property-level) connection; and (iv) a simplicity/weakness bias.

These four ingredients are meant to be read as jointly necessary for the ordinary, practice-guided use of "cause" in modeling and intervention. Temporal precedence rules out purely symmetric associations as candidates for causal direction; predictive attraction encodes the idea that causes provide leverage beyond background expectations; intensional linkage encodes the idea that the dependence should be mediated by comparatively stable properties rather than arbitrary labels; and the simplicity/weakness bias resolves underdetermination by preferring causal stories that introduce as little representational "force" as possible.

Remark 672. *Temporal precedence here is a constraint on admissible directionality rather than a full dynamical model: it says that if A is judged a cause of B at resolution Δ, then the dependence must be expressible with A occurring "before" B in the observer's discretization. This keeps the causal notion compatible with the predictive setup (which is indexed by Δ) while allowing that different observers—or the same observer at different time scales—may legitimately report different causal relations when the ordering becomes ambiguous at coarse granularity.*

Remark 673. *The simplicity/weakness bias is not merely aesthetic. It functions as a regularizer on causal attribution: among multiple hypotheses that fit extensional predictive facts equally well, prefer the one that commits to weaker predicates, fewer special cases, and less brittle boundary drawing. In the Hyperseed framing, this is the same pressure that, in other contexts, selects simpler patterns as better compressive descriptions, but here it is applied to causal predicates and their associated intervention rules.*

Remark 674. *This cluster deliberately echoes several long-standing tensions in philosophy of science: Humean regularity (extensional dependence) is not enough for the lived concept of cause; yet pure metaphysical necessity is too heavy a tool for scientific practice. Hyperseed's compromise is operational: define causality as a* structured *kind of predictive leverage, constrained by time and guided by an Occam-like preference for weak distinctions (Hyperseed-Concept 202, and in the quantale form, Hyperseed-Concept 143). For related discussions of weakness as a central organizing principle, see [2, 3].*

14.5.1 Extensional and intensional predictive attraction

Definition 182 (Extensional predictive attraction)**.** *The* extensional predictive attraction *is the predictive attraction computed on the level of occurrence of event types:*

$$\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B) := \mathrm{PAtt}_{O,\Delta}(A,B).$$

Remark 675. *Calling this "extensional" emphasizes that it looks only at the surface fact of whether A and B occur, not at what internal structure A and B may have as patterns or bundles of properties. In many empirical settings, extensional attraction is the easiest to estimate directly from time series data, and so it often serves as the first approximation to causal influence.*

Remark 676. *One can also read $\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B)$ as a statistic of* type-level *tokens: it aggregates over the observer's instances of A and asks whether B is enriched in the future of those instances relative to an appropriate baseline. In this sense it is closer to population-level causal evidence than* *to a single mechanistic explanation; it answers "does A carry predictive leverage about B?" rather than "through which internal channel does A bring about B?"*

Remark 677. *Example: if we take $A =$`take(aspirin)` and $B =$`headache(reduces)`, extensional attraction asks only whether the aspirin-token tends to be followed by a reduction-token, compared to the baseline where aspirin is not taken.*

Remark 678. *The same example also illustrates why extensional attraction alone is not the full causal story: the observed dependence might be mediated by confounders represented (or not represented) by O, by selection effects in when aspirin is taken, or by hidden structure within the event types (dose, timing, severity). The role of the intensional layer below is to make explicit some of the within-type structure that can stabilize causal attribution across contexts.*

To define the intensional counterpart we use the property machinery from Section 9. Recall that the *property set* of an entity/event is the fuzzy set of patterns in it, weighted by pattern intensity (Hyperseed-Concept 130).

Definition 183 (Property events)**.** *Fix a finite "property basis" $\mathcal{P}$ (patterns-as-properties) and a map $\mathrm{Prop}_O : \mathcal{E} \to [0,1]^{\mathcal{P}}$. For $p \in \mathcal{P}$ we write $p \in_O A$ for the derived "property event type" "pattern/property p is present in A" and set*

$$\pi(p \in_O A) := \mathrm{Prop}_O(A)(p) \in [0,1].$$

Remark 679. *Notation unpacking: $\mathcal{P}$ is a set of properties (implemented as patterns), and $\mathrm{Prop}_O(A)$ is a function $\mathcal{P} \to [0,1]$ assigning each property p an intensity for event type A. The expression $p \in_O A$ is not literal set membership but a* derived event type *whose plausibility is defined to be $\mathrm{Prop}_O(A)(p)$. The subscript O again signals observer-relative representation.*

Remark 680. *The requirement that $\mathcal{P}$ be finite is a pragmatic restriction that matches how an observer/model typically operates: causal claims are made in a bounded vocabulary of features, measurements, or patterns. Technically, finiteness ensures that the averaging constructions below are unproblematic, while conceptually it emphasizes that intensionality is always relative to a chosen representational basis (which may be expanded when the observer learns new patterns).*

Remark 681. *Intuitively, this turns "A has property p" into something that can be fed back into the same predictive machinery used for ordinary events. For example, if A is* `fire(burning)` *and p is the pattern/property* `hot`*, then $\pi(p \in_O A)$ is the degree to which the observer's representation of the event-type includes hotness. This is useful because it lets us ask not only whether A predicts B, but whether the* properties *characteristic of A predict the properties characteristic of B.*

Remark 682. *In particular, property events let the same observer represent two different "internal decompositions" of the same extensional event type. For instance, the event type* `take(aspirin)` *may include properties such as* `dose(large)` *or* `dose(small)` *with different intensities in different subpopulations or contexts. If those properties have different downstream predictive signatures, the intensional analysis can distinguish them even when the extensional event label is the same.*

Remark 683. *The purpose of introducing property events is theoretical: it provides a bridge between the "pattern as compressive description" view (Section 9) and causal judgments, by allowing causal dependence to be sensitive to intensional structure rather than only to extensional co-occurrence. This is one way to formalize the thought that causation should reflect not merely what happens, but something about* why *it happens, insofar as "why" is accessible to the observer's representational vocabulary.*

Remark 684. *A further benefit is robustness under redescriptions: if two event types are extentionally similar but differ in property profile, then extensional attraction may be unstable across shifts in data collection or coding, while intensional attraction can remain informative provided $\mathcal{P}$ captures the stable aspects of the phenomenon. Conversely, if $\mathcal{P}$ omits the relevant mechanistic properties, then the intensional score will collapse back toward an extensional proxy, which is appropriate: the observer cannot legitimately claim a structured causal story without the representational capacity to express it.*

Definition 184 (Intensional predictive attraction)**.** *Define the* intensional predictive attraction *by averaging predictive attraction over properties:*

$$\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B) := \sum_{p\in\mathcal{P}}\sum_{q\in\mathcal{P}} w_{p,q}\,\mathrm{PAtt}_{O,\Delta}(p \in_O A,\, q \in_O B),$$

where $w_{p,q} \geq 0$ are weights that may be chosen, for example, proportional to $\mathrm{Prop}_O(A)(p)\,\mathrm{Prop}_O(B)(q)$ and normalized to sum to 1*.*

Remark 685. *The weights $w_{p,q}$ implement a choice about what it means for a property-level linkage to be "about" A and B. Taking $w_{p,q} \propto \mathrm{Prop}_O(A)(p)\mathrm{Prop}_O(B)(q)$ focuses the average on properties that are actually present in the two event types, while still allowing rare but highly diagnostic properties to matter if their associated* PAtt *terms are large. Other weighting schemes are possible (e.g. emphasizing only properties of A, or penalizing very common q to reduce trivial predictability), and can be understood as different observer policies for extracting mechanistically meaningful signal from the same raw dependence statistics.*

Remark 686. *This definition says: look at all property pairs (p,q), measure how much the presence of p in A predicts the presence of q in B (differentially, via* PAtt*), and then average these numbers with weights $w_{p,q}$ emphasizing the properties actually salient in A and B. The result is a single scalar capturing structured, property-level predictive linkage.*

Remark 687. *When $\mathcal{P}$ contains a very coarse property corresponding to the event's own occurrence (or a near-indicator pattern that tracks it), the intensional construction can recover the extensional one as a special case: the average places most of its mass on the indicator-like property pair and $\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B)$ approaches $\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B)$. This provides a consistency check: intensional attraction refines extensional attraction rather than replacing it with an unrelated quantity.*

Remark 688. *A simple schematic example: let A be `strike(match)` and B be `fire(ignites)`. If properties like `friction` and `heat` are strongly present in A, and properties like `heat` and `light` are present in B, then property-level attractions such as $\mathrm{PAtt}(\texttt{friction} \in_O A, \texttt{heat} \in_O B)$ and $\mathrm{PAtt}(\texttt{heat} \in_O A, \texttt{light} \in_O B)$ should be positive, raising $\mathrm{PAtt}^{\mathrm{int}}(A,B)$.*

Remark 689. *In that match example, the intensional score is closer to what one informally calls a mechanism: it does not merely say that the labels `strike(match)` and `fire(ignites)` are statistically associated, but that specific properties characteristic of striking (e.g. friction) are linked to specific properties characteristic of igniting (e.g. heat). This is also where the simplicity/weakness bias interacts with intensionality: if two candidate causal explanations fit the extensional data, the preferred one is the one whose property linkages can be expressed with fewer and weaker distinguishing properties (for example, a small, stable set of features rather than a highly conjunctive, brittle property list).*

Remark 690. *This notion is useful because it can distinguish between (i) brute co-occurrence and (ii) co-occurrence mediated by shared structure in the observer's vocabulary of patterns-as-properties. In complex systems, many correlations are accidental; intensional attraction is one principled way to bias causal inference toward correlations that align with stable internal descriptions (cf. [5] on hierarchical/ heterarchical pattern structure).*

Remark 691. *A further intuition is that "shared structure" should be understood as invariance under reasonable changes of presentation: if the observer refines, factorizes, or re-encodes a situation using roughly equivalent internal predicates, genuinely causal linkages should remain expressible with similar property-level dependencies. By contrast, accidental correlations often disappear under such re-descriptions because they do not correspond to a stable pattern-level relation in the observer's representational scheme.*

Remark 692 (Interpretation)**.** $\mathrm{PAtt}^{\mathrm{int}}$ *is high when the properties of A (as patterns) tend to predict the properties of B. This captures the intuitive idea that a causal connection is not merely a statistical regularity, but reflects a structured relationship between what A* is like *and what B* is like.

Remark 693. *In particular,* $\mathrm{PAtt}^{\mathrm{int}}$ *is meant to be sensitive to "mechanism-shaped" features: even if A and B co-occur frequently, intensional attraction should increase only when the mapping from salient attributes of A to salient attributes of B is compressible in the observer's property language (e.g. the same few properties of A explain many properties of B). This helps separate mere proxy variables from variables that participate in a coherent descriptive pipeline.*

14.5.2 Causal implication and simple causal implication

Definition 185 (Causal implication)**.** *Fix a lag window Δ and a mixing parameter $\lambda \in [0,1]$. Define the* causal implication strength

$$\mathrm{CImp}_{O,\Delta}(A,B) := \big(\lambda\,\sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B)) + (1-\lambda)\,\sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B))\big)\cdot \mathbf{1}[A \prec B],$$

where $\mathbf{1}[A \prec B]$ is an indicator that A typically precedes B in the relevant context, and $\sigma : [-1,1] \to [0,1]$ is the affine rescaling $\sigma(x) = (1+x)/2$.

Define also the strong *(product) variant*

$$\mathrm{CImp}^{\times}_{O,\Delta}(A,B) := \sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B))\;\sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B))\;\mathbf{1}[A \prec B].$$

Remark 694. *One can read λ not only as "trust in statistics" versus "trust in structure," but also as a tuning between two kinds of robustness: extensional attraction can be robust to vocabulary drift (it depends less on how properties are named) but brittle under distribution shift, whereas intensional attraction can be robust across contexts where the same mechanism holds but brittle under changes in the observer's property set. The mixture is therefore an explicitly observer-relative compromise.*

Remark 695. *Notation unpacking: σ converts a signed attraction in $[-1,1]$ into a $[0,1]$ score, so that negative attraction maps below $1/2$ and positive attraction maps above $1/2$. The indicator $\mathbf{1}[A \prec B]$ enforces temporal direction: even a strong predictive linkage does not count as causal implication if A does not typically precede B in the observer's time-ordering.*

Remark 696. *The affine rescaling is chosen so that a "neutral" attraction of 0 maps to $1/2$, which functions as a convenient baseline when combining multiple sources of evidence. In particular, the product variant then has the interpretation that a neutral score in either extensional or intensional attraction yields a mid-level contribution, while strongly negative attraction in either channel pushes the overall score downward even when the other channel is strong.*

Remark 697. *Intuitively,* $\mathrm{CImp}_{O,\Delta}(A, B)$ *is a compromise between two ideas of cause: "A makes B happen (extensional leverage)" and "A is connected to B through properties the observer recognizes (intensional structure)." The parameter* λ *expresses how much the observer (or modeling choice) trusts extensional statistics versus property-level structure.*

Remark 698. *The two variants also correspond to different stances on "defeaters." In the sum form, a strongly mechanistic-looking intensional linkage can partially rescue a weak extensional signal (e.g. when the available samples are sparse), and conversely strong extensional predictability can partially rescue an immature vocabulary. In the product form, either channel can act as a defeater: if the observer cannot express a stable property-level relation, or if extensional evidence is strongly negative, then causal implication is suppressed.*

Remark 699. *A simple example: in a world where barometers correlate with storms, extensional attraction* $\mathrm{PAtt}^{\mathrm{ext}}(\texttt{barometer(low)}, \texttt{storm})$ *may be high. But the intensional attraction might be low if the properties of* `barometer(low)` *do not structurally connect to the properties of storms in the observer's mechanistic vocabulary. The mixing/product choices then control whether barometer readings are treated as causes or merely predictors.*

Remark 700. *The same distinction shows up in "common cause" settings: a hidden atmospheric-pressure variable may generate both* `barometer(low)` *and* `storm`*, producing high extensional attraction, while a mechanistic vocabulary that explicitly represents pressure as an intermediate property could raise intensional attraction for* $\texttt{pressure(low)}\to\texttt{storm}$ *and reduce it for* $\texttt{barometer(low)}\to\texttt{storm}$*. Thus, intensional attraction can be interpreted as a bias toward models that expose plausible intermediates in the observer's descriptive hierarchy.*

Remark 701 (Temporal precedence as a statistic)**.** *One may implement* $A \prec B$ *in many ways, e.g. as a thresholded statistic computed from the same occurrence data used for* $\mathrm{Cond}_{O,\Delta}$*: conditional on co-occurrence of* A *and* B *within a broad window, the median (or mean) lag* $\mathrm{lag}(A, B)$ *is positive. The ontology treats precedence as context-relative (observer-relative time and event boundaries), so this indicator should be understood as relative to* O*.*

Remark 702. *The role of the lag window* Δ *is to encode a causal timescale hypothesis: if* Δ *is too small, genuine delayed effects will be missed; if it is too large, unrelated events may be treated as eligible for precedence tests, washing out directionality. In practice, one can treat* Δ *as a hyperparameter and evaluate sensitivity of* $\mathrm{CImp}_{O,\Delta}$ *to its choice, which is itself a kind of "simplicity" consideration (prefer time-scale assumptions that do not require fine-tuning).*

Remark 703 (Sum vs product)**.** *The "sum" form aligns with the ontology's* `SimpleCausalImplication` *intuition (a weighted average of extensional and intensional aspects), while the "product" form aligns with the ontology's* `CausalImplication` *intuition (both aspects must be simultaneously strong).*

Remark 704. *In a decision-theoretic reading, the sum form behaves more like an expected-evidence aggregator (useful for ranking many candidate causes when evidence is heterogeneous), while the product form behaves more like a conjunction of tests (useful when one wants high precision and is willing to sacrifice recall). This difference matters when causal implication is later used for control: permissive attribution can increase exploratory interventions, whereas severe attribution can reduce spurious control attempts.*

Remark 705. *The product form is, in a sense, more severe: it demands that extensional and intensional evidence cohere. The sum form is more permissive: it allows a cause to be inferred from strong extensional leverage even if the property vocabulary is impoverished, or from strong* *intensional coherence even if the raw statistics are noisy. This mirrors a practical tradeoff in scientific reasoning between mechanistic understanding and purely statistical prediction (cf. [20] for a related philosophy of science stance).*

Remark 706. *Edge cases help clarify interpretation: if $\mathbf{1}[A \prec B] = 0$ then both scores are 0, making the temporal ordering a hard gate rather than a soft preference. If $\lambda = 1$ then $\mathrm{CImp}_{O,\Delta}$ reduces to a purely extensional, precedence-gated score; if $\lambda = 0$ it reduces to a purely intensional, precedence-gated score. Thus the definition contains, as limiting cases, both a Granger-like "prediction with time-order" notion and a more structural "mechanism in the observer's vocabulary" notion.*

14.5.3 Weakness-biased causal model selection

Causal attribution is also biased by simplicity. In the Hyperseed formal core, simplicity is tracked by weakness (Section 3.7): all else equal, prefer explanations that make fewer special-case distinctions (Hyperseed-Concept 169, Hyperseed-Concept 202).

Remark 707. *This is an explicitly Russellian move: when two descriptions fit the appearances equally well, prefer the one that posits fewer independent discriminations. In Hyperseed, this preference is not only a methodological maxim but is formalized via quantale weakness measures [3, 2]. The result is that causal inference becomes a kind of model selection under a simplicity prior, rather than a mere reading-off of correlations.*

Remark 708. *Here "fewer special-case distinctions" can be read operationally as: fewer conditional branches, fewer context-specific parameters, fewer ad hoc predicates, or fewer exception clauses required to maintain predictive adequacy across the contexts recognized by O. Importantly, weakness is not merely "small cardinality" of S; it concerns the complexity of the* description *of the explanation, so that a slightly larger parent set may still be weaker if it enables a more uniform rule.*

Definition 186 (Weak causal parent selection (schematic))**.** *Fix a target event type B. Let $\mathcal{C}$ be a finite set of candidate causes. A* causal parent set *is a subset $S \subseteq \mathcal{C}$. Define a score*

$$\mathrm{Score}(S \to B) := \underbrace{\mathrm{Fit}(S \to B)}_{\textit{predictive adequacy}} \qquad +\eta \underbrace{\mathrm{Weak}(S \to B)}_{\textit{simplicity/weakness}},$$

where $\eta > 0$ is a tradeoff parameter. A weak causal explanation *for B is a parent set S^* maximizing this score.*

Remark 709. *The separation into* Fit *and* Weak *is intended to parallel standard regularization:* Fit *rewards predictive performance (e.g. likelihood, reduction in uncertainty, or improvement in $\mathrm{Cond}_{O,\Delta}$-style predictability when conditioned on S), while* Weak *acts like a simplicity prior (e.g. a penalty for representational complexity, or a reward for compressibility). The hyperparameter η therefore controls how readily the observer trades marginal predictive gains for a simpler causal story.*

Remark 710. *Although presented schematically, this parent-selection view connects back to $\mathrm{CImp}_{O,\Delta}$: one can treat* CImp *(or $\mathrm{CImp}^{\times}$) as a primitive pairwise signal used to generate $\mathcal{C}$ (screening candidates), and then use weakness-biased selection to choose a multivariate explanation that resolves redundancy among candidates (e.g. when several correlated A_i compete as apparent causes of the same B). In that role, weakness functions as a bias toward selecting upstream or more "explanatorily central" parents when multiple nearly equivalent predictors exist.*

Remark 711. *This definition is deliberately schematic: it isolates the* form *of the causal selection rule without committing to a particular* Fit *or* Weak*. Conceptually, it says that causality is not a single number but an optimization: we search for a set of antecedents that both predict B and do so without excessive complexity. In particular, the definition is meant to be read as a regularized model-selection principle: among many candidate antecedent-sets S that could be placed "before" B in the relevant sequence, we prefer those that score well under* Fit *while paying an explicit penalty (or softness constraint) measured by* Weak*. This keeps the notion of "cause" distinct from mere association, since high association achieved only by ad hoc, highly brittle conditions is treated as less causal under the same observational data. The temporal-precedence component is encoded by restricting attention to candidates that occur in the antecedent portion of* $\mathrm{Seq}(S)$ *relative to B, so that the optimization is not over arbitrary correlates but over prospective parents in the temporal (or otherwise directed) ordering.*

Remark 712. *A simple example (in outline): suppose B is `alarm(rings)`, and the candidate causes $\mathtt{C}$ include `smoke`, `steam`, and `dust`. A high-fit but overly specific rule might include a long conjunction of rare conditions; a weaker (simpler) rule might treat `smoke` as the primary parent because it captures most of the predictive signal with fewer special cases. The parameter η governs how aggressively the observer prefers such simplicity. One can view this as preferring a small, stable explanatory "core" over an arbitrarily detailed exception list: e.g. a contrived rule like `(smoke AND weekday AND sensor_A AND NOT window_open AND ...)` may fit a limited dataset extremely well while generalizing poorly, whereas `smoke` alone may capture most of the predictive mass across contexts. In this toy scenario, `steam` might be a frequent correlate in kitchens (thus sometimes predictive) but also a common benign source of false alarms, and `dust` might correlate with sensor noise in certain rooms; the weakness term is intended to suppress treating such context-fragile correlates as "the cause" unless the added distinctions genuinely buy enough predictive performance to justify the added complexity. The role of η is then analogous to a regularization strength: small η permits richly contextual causes when the fit gain is large, while large η pushes toward more parsimonious parent-sets even at some cost in fit.*

Remark 713. *This is useful for the broader theory because it harmonizes with later "Wu Wei" optimization: the same weakness bias that prefers simple causal stories can also prefer minimally contrived control policies (Section 26; Hyperseed-Concept 206, 207; see [21]). It also connects to general tradeoff-theorem thinking about balancing dependency, fit, and complexity [9]. In other words, the same mathematical knob that discourages baroque explanatory structure also discourages baroque interventions: both "overfitted explanations" and "overengineered controls" can be seen as solutions that achieve a target outcome only by relying on an excessive number of finely tuned distinctions. When the observer (or agent) is constrained to prefer weaker descriptions, it tends to select causal parents that are robust across contexts, and likewise tends to select policies that succeed without requiring a long list of special-case contingencies. This is precisely the sense in which a shared simplicity/weakness bias can provide a unifying regularity principle across prediction (which antecedents matter?), causality (which antecedents count as parents?), and control (which actions achieve outcomes without contrivance?), making the later Wu Wei criteria appear less as a separate doctrine and more as the control-theoretic face of the same selection rule.*

Remark 714 (What "Fit" and "Weak" may be)**.** *There are many concrete instantiations. For example,* Fit *may be the projected predictive implication* $\pi(\mathrm{Cond}(B \mid \mathrm{Seq}(S)))$ *or a negative log-loss; and* Weak *may be a quantale weakness of the distinctions needed to state the conditional "B depends on S" (e.g. a description-length proxy or a partition-based weakness of contexts). The point is the structure: causality is not "just statistics" but statistics plus an Occam-like bias. Concretely,* Fit *can be read as any score that rewards accurate forecasting of B from the antecedent* *sequence, including likelihood-based criteria, proper scoring rules, or information-theoretic dependence measures so long as they increase when S makes B more predictable in the intended sense. Likewise,* Weak *is intentionally permissive: it may be instantiated as an MDL-style codelength of the rule, a complexity of the partition of contexts required to make the rule true, a penalty on the number of conjuncts/clauses, or a more semantic notion of "how many distinctions the observer must track" to use the conditional reliably. The quantale phrasing is meant to allow weakness to compose and compare across heterogeneous kinds of distinctions (e.g. temporal, contextual, and representational), rather than restricting it to a single numeric proxy. Under any such instantiation, the selection principle can be read as: choose parents that are predictive* and *that can be stated and applied with a small, reusable inventory of distinctions, thereby building temporal precedence and simplicity directly into the operational meaning of "cause."*

14.6 Control, control hierarchies, and perceptual hierarchies

Hyperseed's slogan is: *control is systematic causation.* We formalize this by treating control as causal implication where the antecedent is an intervention/action variable (Hyperseed-Concept 87). In this framing, "systematic" emphasizes repeatability across relevant contexts: a controller is not defined by a single lucky coincidence but by a stable mapping from *doable* antecedents to downstream consequences. The use of causal implication rather than mere co-occurrence is meant to exclude cases where A and G happen to correlate due to a common cause, and to favor cases where actively setting or instantiating A would (in the model) make G more likely. The context parameter O and lag window Δ are therefore not incidental bookkeeping: they encode what background conditions are held fixed and what temporal scope is treated as the "effect horizon" of action.

Definition 187 (Actions, goals, and control strength)**.** *Let $\mathcal{A} \subseteq \mathcal{E}$ be a designated set of* action event types. *Let $G \in \mathcal{E}$ be a goal condition (or a conjunction/disjunction of such). For $A \in \mathcal{A}$ define the* control strength *of A over G as*

$$\mathrm{Ctrl}_{O,\Delta}(A \to G) := \mathrm{CImp}_{O,\Delta}(A, G).$$

Remark 715. *The formal move is simple: we do not invent a new scalar for control; we reuse causal implication, restricted to antecedents interpreted as actions or interventions. Philosophically, this encodes the idea that agency is not an extra substance added to the world, but an organizational stance toward certain event types as* doable *(actions) and certain event types as* desirable *(goals). A technical advantage of this reuse is that any refinements already built into causal implication (e.g. sensitivity to temporal precedence, background context O, and windowing by Δ) carry over automatically to control: what counts as "the same action" and what counts as "achieving the goal" are evaluated inside the same event-type and context machinery.*

Remark 716. *Example: if $A =$`turn(key)` and $G =$`engine(starts)`, then* $\mathrm{Ctrl}(A \to G)$ *is high precisely when turning the key differentially raises the plausibility of starting the engine, does so with temporal precedence, and (optionally) aligns with intensional property connections. This is useful because it makes control a graded quantity, allowing partial control and contested control (e.g. conflicting evidence about whether the key mechanism is functional). A further benefit of grading is comparative evaluation: if $\{A_i\}$ are alternative actions, then control strengths* $\mathrm{Ctrl}(A_i \to G)$ *provide a principled way to rank actions by their expected goal-directed causal leverage under the same context and lag assumptions. In that sense, the definition is compatible with (but does not itself impose) decision-theoretic selection rules such as choosing an A that maximizes control subject to costs or constraints.*

Remark 717. *Although the definition is written for a single antecedent A, many control situations are sequential: a policy is a temporally extended pattern of actions. One can model such cases by letting A denote a composite event type (e.g. a short action script within Δ) or by applying the definition to intermediate subgoals $G_1, G_2, \dots$ that are chained together. This keeps the basic notion of control strength intact while allowing hierarchical decomposition of complex tasks into locally controllable steps.*

Definition 188 (System-level control)**.** *Let $\mathcal{U}$ and $\mathcal{V}$ be groupings of event types (subsystems, modules, or coarse variables). We say $\mathcal{U}$* controls *$\mathcal{V}$ (in context O with lag window Δ) if there exist $A \in \mathcal{U} \cap \mathcal{A}$ and $B \in \mathcal{V}$ such that $\mathrm{Ctrl}_{O,\Delta}(A \to B)$ exceeds a chosen threshold.*

Remark 718. *This lifts control from individual event types to coarse subsystems: "$\mathcal{U}$ controls $\mathcal{V}$" means that something in $\mathcal{U}$, construed as an action, exerts sufficiently strong causal implication toward something in $\mathcal{V}$. The threshold is a modeling choice, reflecting how strict we wish to be in calling a relation "control" rather than "influence." Because the definition uses an existential quantifier, it captures the minimal claim that* some *action-like degree of freedom in $\mathcal{U}$ can steer* some *event type in $\mathcal{V}$; stronger notions (sometimes useful in engineering) would replace "there exists" with requirements like "for many $B \in \mathcal{V}$" or would aggregate via $\max_{A \in \mathcal{U} \cap \mathcal{A}} \max_{B \in \mathcal{V}} \mathrm{Ctrl}(A \to B)$ to define a graded* inter-module *control strength.*

Remark 719. *Example: let $\mathcal{U}$ be motor-command events and $\mathcal{V}$ be arm-position events. Then system-level control holds if there exists a motor command that reliably steers the arm. In more abstract cognitive systems, $\mathcal{U}$ might be an executive module and $\mathcal{V}$ a memory module; the definition then captures the idea that executive actions can gate or modulate memory states. This kind of modular control analysis is widely used in cognitive architectures (cf. [19]). In both examples, the window Δ implicitly distinguishes fast, direct effects (a command changes position within tens of milliseconds) from slower, mediated effects (a command changes posture after a sequence of reflex loops). Choosing Δ therefore selects what counts as "the controlled variable" at the scale of interest.*

Remark 720 (Asymmetry and bidirectionality)**.** *Nothing forces control to be symmetric. One may have $\mathcal{U}$ controlling $\mathcal{V}$ without $\mathcal{V}$ controlling $\mathcal{U}$. Bidirectional control corresponds to feedback coupling. In feedback-coupled systems, bidirectionality is common: $\mathcal{U}$ may act on $\mathcal{V}$ while $\mathcal{V}$ supplies observations or error signals that modulate subsequent actions in $\mathcal{U}$. The present definition is deliberately agnostic about stability and optimality: it identifies the presence of causal leverage, not whether the resulting closed-loop dynamics converge, oscillate, or diverge. Those dynamical questions can be layered on top by analyzing control strengths across multiple lags or across recurrent cycles.*

Definition 189 (Control hierarchy)**.** *A* control hierarchy *is a partial order $(\mathcal{M}, \succcurlyeq)$ on a family of modules $\mathcal{M}$ such that $M_1 \succcurlyeq M_2$ is aligned with "M_1 has (potentially asymmetric) control over M_2."*

Remark 721. *The term "hierarchy" is used in the order-theoretic sense: $\succcurlyeq$ is a partial order, so it need not compare every pair of modules. This accommodates heterarchical organization (Hyperseed-Concept* **??***) where some modules are incomparable or form local loops, while still capturing a global tendency for some modules to sit "above" others in the sense of exerting more control (Hyperseed-Concept 88). Operationally, one can imagine estimating pairwise system-level control relations (possibly at different thresholds or different Δ) and then taking the transitive closure to obtain an ordering over modules. When mutual control is pervasive, it is often useful to treat strongly mutually controlling clusters as equivalence classes, so that the partial order applies between clusters rather than insisting on antisymmetry between tightly coupled components.*

Remark 722. *A simple example is a two-level robot controller: a high-level planner module M_{plan} chooses goals and sequences, and a low-level motor module M_{motor} executes them. Then $M_{\text{plan}} \succcurlyeq M_{\text{motor}}$. In biological or cognitive settings, such a clean order may not exist globally; using a partial order rather than a total order respects this. One also frequently encounters multi-level cases where M_{plan} controls not the motor plant directly but the* parameters *of lower controllers (gains, setpoints, task constraints), while lower controllers control moment-to-moment actuator outputs. The partial-order notion is intended to cover both "direct command" and "parameter-setting" as varieties of control, so long as they show up as strong causal implication from action-like events in the higher module to state changes in the lower one.*

Remark 723. *This definition is useful because it gives a formal handle on a pervasive explanatory motif: many systems are organized so that some processes constrain or parameterize others. Later sections revisit this motif under attention allocation and cognitive synergy, where control of resource flow is as important as control of external actions. A further motif is* abstraction*: higher modules often act over coarser variables and longer time horizons, while lower modules act over finer variables and shorter time horizons. The parameters (O, Δ) can be chosen to reflect this scale separation, so that "higher controls lower" is not merely a metaphor but corresponds to measurable differences in what kinds of event types are treated as actions, what kinds of event types are treated as goals, and over what temporal windows the causal implications are evaluated.*

Definition 190 (Perceptual hierarchy)**.** *A* perceptual hierarchy *is a control hierarchy in which the relevant control events are instances of pattern recognition (Section 13): controlling a module consists in selecting, stabilizing, or transforming the patterns it registers.*

Remark 724. *This definition treats perception as a special case of control: higher levels "control" lower levels by constraining what patterns are recognized, amplified, or suppressed. It thereby links representation (Section 13) with action: to perceive is to enact a selection among possible organizations of the incoming stream (Hyperseed-Concept 133). Concretely, "control" here can be realized by mechanisms such as attentional gating, adaptive thresholding, top-down priming, or predictive templates that bias which hypotheses the lower module considers. On this view, a higher perceptual level does not merely* read *a fixed set of features; it can actively shape the feature space and the salience landscape, thereby changing what the lower level will count as a match on subsequent inputs. This is especially clear in ambiguous stimuli, where stabilizing one interpretation over another is itself a form of goal-directed regulation of internal state, even if no overt motor action is taken.*

Remark 725. *Example: in vision, a high-level hypothesis like `face-present` can bias lower-level feature detectors to stabilize edge/texture patterns consistent with a face; conversely, ambiguous low-level features may fail to stabilize any high-level pattern. In the present formalism, such biasing is represented abstractly as control relations among modules whose event types include recognition events. This is consonant with active-perception and predictive-processing intuitions, while not committing to a particular neural implementation.*

The point of phrasing this as a control relation *is that the "top–down" influence is treated as a directional constraint on which downstream events are made more (or less) likely to occur in a stable way, rather than as a mere correlation. In particular, a recognition event at a higher layer functions as a context-setting signal: it changes which lower-level event sequences are admissible (or evidentially supported) as the system settles into an interpretation. Conversely, persistent instability at the lower layer can be read as a failure to supply sufficient evidential support for any higher-level recognition event to become dominant, which explains why the overall percept remains indeterminate in ambiguous stimuli.*

Remark 726 (Perception as action). *In this view, perception is not passive "taking in data" but an active control process that modulates which patterns become salient and how they are organized. This aligns naturally with predictive-processing and active-inference perspectives, but remains agnostic about particular neuroscience.*

More concretely, the "action" here need not be overt motor behavior; it can also be internal action in the space of hypotheses, such as adjusting attention, gain, precision, or other modulatory knobs that determine which prediction errors are amplified and which are attenuated. The formal upshot is that perceptual outcomes are treated as controlled *equilibria: what is perceived is whatever interpretation the system can stably maintain under its current control policy and evidential constraints, rather than a direct readout of raw sensory input.*

14.7 Planning and optimal control: integration hooks

14.7.1 Event calculus hook

The Event Calculus represents temporally extended domains using predicates such as $\text{Happens}(a,t)$, $\text{Initiates}(a,f,t)$, $\text{Terminates}(a,f,t)$, and $\text{HoldsAt}(f,t)$. In a Hyperseed-style uncertain setting, one can attach p-bit evidence values to these predicates and treat planning as search for action sequences whose composed predictive implication toward the goal is high. The sequential composition operator from Section 14.4 is designed to match the temporal compositionality required by planning.

The intended reading is that each of these predicates can be treated as an *evidential claim* rather than a Boolean fact: an agent may have partial support that an action occurred, mixed support that it initiates a fluent, or even contradictory support due to noisy sensing, conflicting reports, or model mismatch. Planning then becomes an evidential optimization problem: instead of searching for a single logically consistent narrative, one searches for trajectories whose chained, time-respecting implications make the goal fluent more strongly supported at the relevant horizon. The sequential operator is the mechanism that turns per-step evidential implications into an aggregate multi-step assessment while respecting temporal order (i.e. earlier actions can only influence later holds).

Remark 727. *This paragraph is an interface point rather than a full development: it says that one may take a classical event/process calculus (Hyperseed-Concept* **??***) and replace its crisp truth values with* p-bit *evidence values, thereby allowing plans to be assessed under contradictory and incomplete information. The quantale-composition rule then supplies an algebra for aggregating multi-step plan reliability without leaving the evidential domain.*

In particular, the "interface" claim is that existing symbolic planning infrastructure can be reused at the level of syntax and temporal structure (events, fluents, and their persistence), while the semantics of entailment is softened into evidential support. This makes it possible to speak meaningfully about plans that are good enough *even when no plan can be proven correct in the classical sense, and it clarifies how plan refinement can proceed: one improves a plan either by strengthening the evidential links between steps (better predictive implication) or by inserting actions whose main role is epistemic, i.e. to reduce contradiction and increase the stability of later inferences about* $\text{HoldsAt}(f,t)$.

14.7.2 Wu Wei preview

Section 26 introduces Wu Wei dynamics: a bias toward minimal contrivance, i.e. achieving goals via the weakest (least special-case) control policies that still work. In the present terms, Wu Wei is a global optimization principle layered on top of the local notions of attraction/causality/control. Roughly: among plans that achieve high projected predictive implication toward the goal, prefer those that maximize weakness (simplicity) and minimize experienced effort (Section 8) (Hyperseed-Concept 206, Hyperseed-Concept 100; see [21]).

The integration hook to optimal control is that "weakness" plays a role analogous to regularization or complexity penalties: it biases the solution set toward policies that are robust under model error and that do not rely on brittle coincidences in the environment. Meanwhile "effort" functions like a cost functional over control signals, including both overt energy expenditure and internal computational or attentional load. The resulting preference ordering is thus not merely about whether a goal can be reached, but about whether it can be reached in a way that remains stable under the agent's limited resources and imperfect knowledge.

Remark 728. *The philosophical tenor here is that of "least action," but translated from physics to agency: do not seek maximal domination of the world, but maximal alignment with the world's own tendencies, so that small interventions suffice. Formally, this becomes a multi-objective optimization over (i) plan success (predictive implication), (ii) simplicity/weakness, and (iii) effort. This is also where control theory (in the engineering sense) and weakness theory (in the Hyperseed sense) meet.*

One can also read this as a constraint on control hierarchies*: higher layers should prefer to set broad, low-commitment goals or biases that allow lower layers to exploit existing attractors in the environment, rather than micromanaging every degree of freedom. In that sense, Wu Wei is not a rejection of control but a discipline of control: intervene at the level where a small informational or policy change yields a large shift in downstream trajectories, and avoid interventions that must be continually "propped up" against the system's natural dynamics.*

Hyperseed concepts covered

- Attraction, PredictiveImplication, PredictiveAttraction; sequential temporal operators (SequentialAND, SimultaneousAND, SimultaneousOR) at a formal level.
- Cause, CausalImplication, SimpleCausalImplication; extensional vs intensional predictive attraction; temporal precedence; weakness/simplicity bias.
- Control, asymmetric/symmetric control; Control Hierarchy; Perceptual Hierarchy.
- Planning hook (Event Calculus) and connection to later Wu Wei optimal control.

15 Attention and cognitive synergy

15.1 From control to attention

Section 14 introduced prediction, attraction, and control. For bounded systems, however, "control" is always exercised through a bottleneck: only a small fraction of the system's internal processes can be actively updated at any moment. This bottleneck can be read quite literally as a limit on concurrently maintainable state, updatable parameters, searchable hypotheses, or executable action plans—and, more abstractly, as the simple fact that any physical mechanism has finite throughput. Hyperseed calls the structured management of this bottleneck *attention.* In other words, attention is not an optional add-on to control; it is the mechanism by which control becomes realizable under finite capacity.

Informally, attention is not (only) a spotlight on sensory data. It is the system's allocation of its limited resources (time, energy, computation, bandwidth) across the recognition, inference, simulation, and action processes that operate on patterns. This includes not only *which* patterns are processed, but also *how deeply* they are processed: e.g., whether a hypothesis is merely queued, weakly checked, or iteratively refined until it becomes a stable, reusable chunk in the system's pattern inventory. When the allocation becomes highly concentrated, the system has an *attentional focus*. When the system's self-model lies inside the focus, it has *self-reflective attention*, a key ingredient in Hyperseed's later notion of reflective consciousness (Section 17). Equivalently, self-reflective attention can be viewed as the case where the system treats parts of its own modeling-and-control loop as objects of ongoing inference and revision, rather than as fixed machinery.

A useful way to read the above is that attention defines an *attentional policy* over internal processes: it determines which inference chains get additional compute, which simulations get extended time horizons, which sensors are sampled at higher fidelity, and which candidate actions are evaluated with higher resolution. Since these choices affect what the system learns and what it can do next, attention couples short-term performance to long-term capability: the system that repeatedly allocates effort to the same limited set of patterns may become very efficient there, but risks neglecting alternative explanations or actions. This trade-off between exploitation (deepening a current focus) and exploration (shifting focus to new patterns) will reappear later when we discuss synergy across processes and the conditions under which collections of patterns become mutually supportive rather than mutually distracting.

Remark 729. *The guiding idea is that agency is never merely about possessing a good model; it is also about* deciding which parts of the model to refine now*. This is a resource-theoretic thought in the style of bounded rationality, but expressed in the native Hyperseed vocabulary of patterns and genenergy. In this sense, attention is the operative bridge between the "can in principle" of prediction/control and the "can right now" of a finite mechanism [19, 1]. One concrete implication is that even an accurate generative model may fail to yield effective control if the system cannot reliably route effort to the model components that matter for the present context (e.g., the causal variables relevant to a current goal, or the uncertainty sources that dominate expected loss).*

Remark 730. *When we later refer to the Hyperseed core-concept set, the notions of attention and focus here should be read alongside Hyperseed-Concept 60 and the more general resource/effort notions (Hyperseed-Concept 100) that implicitly underwrite the formal budget constraints below. In particular, "effort" can be interpreted as the accounting unit that allows different internal activities (updating beliefs, running simulations, searching memories, selecting actions) to be compared and traded off within a single bounded budget, even when their concrete computational substrates differ.*

Remark 731. *It is also helpful to distinguish the* target *of attention (which patterns or processes are selected) from the* mode *of attention (how selection is enforced). In engineered systems this mode might look like explicit scheduling, priority queues, or dynamic resource allocation; in biological systems it may appear as gating, inhibition, or neuromodulatory gain control. Hyperseed abstracts away from these implementations and treats attention primarily as the functional role of reliably concentrating limited genenergy on the pattern-transformations most relevant to the current trajectory of inference and action.*

15.2 Genenergy budgets and attention measures

Hyperseed measures attention in terms of "genenergy": any fungible resource that gates what the system can do (metabolic energy, compute cycles, working memory bandwidth, time-on-task, etc.). Section 19 gives a broader treatment of energy/genenergy; here we only need a minimal accounting formalism.

Definition 191 (System, modules, and genenergy flow)**.** *A* system S *is assumed to have:*

(a) a finite set of modules $\mathcal{M} = \{1, \dots, n\}$ *(subsystems/process bundles);*

(b) a finite (or countable) set of patterns $\mathcal{P}$ *(Section 9);*

(c) an association map $\mathrm{Assoc} : \mathcal{M} \to 2^{\mathcal{P}}$ *indicating which patterns each module is currently "about" (recognizes, maintains, manipulates);*

(d) a time index set T (Section 7);

(e) a genenergy expenditure *function*

$$g : T \times \mathcal{M} \to [0, \infty), \qquad (t, i) \mapsto g_t(i),$$

where $g_t(i)$ is the amount of genenergy spent by module i during the interval around t.

The total genenergy spent at time t is

$$G_t := \sum_{i \in \mathcal{M}} g_t(i).$$

Remark 732. *Intuitively, the definition separates* what the system is *(a collection of modules that can spend resources) from* what the system is about *(the patterns those modules are currently tracking or manipulating). The association map* Assoc *is intentionally coarse: it is not a semantic truth-condition, only a bookkeeping relation indicating which patterns are in the working ambit of which processes (Hyperseed-Concept 59; Hyperseed-Concept 130).*

Remark 733. *A simple example is a toy architecture with three modules: vision ($i = 1$), planning ($i = 2$), and motor control ($i = 3$). At time t, we might have $g_t(1) = 5$, $g_t(2) = 2$, $g_t(3) = 1$, so $G_t = 8$. Even in this minimal case, attention is already present as the uneven allocation of genenergy across modules (Hyperseed-Concept* **??***).*

Remark 734. *The notation $g_t(i)$ follows a standard convention: g is the underlying function, while the subscript t indicates we freeze time and view g_t as a function of the module index i. Similarly, G_t is the scalar total expenditure at time t. This small notational discipline matters later, when we push forward module-level allocations to pattern-level distributions.*

Remark 735. *Nothing in the definitions requires genenergy to be conserved or fixed across time. If $G_t = 0$ we interpret S as effectively inactive at time t.*

15.2.1 Attention to a pattern class

Hyperseed defines attention to a class of patterns as genenergy spent on processes associated with those patterns. To avoid double-counting when modules touch multiple patterns, we distribute each module's expenditure over its associated pattern set.

Definition 192 (Pattern-level attention)**.** *Assume each module i comes with a nonnegative weighting*

$$w_i : \mathcal{P} \to [0, 1], \qquad \sum_{p \in \mathcal{P}} w_i(p) = 1,$$

with $w_i(p) = 0$ unless $p \in \mathrm{Assoc}(i)$. Define the attention mass *on pattern p at time t by*

$$\mathrm{Attn}_t(p) := \sum_{i \in \mathcal{M}} g_t(i)\, w_i(p).$$

For a class of patterns $X \subseteq \mathcal{P}$ define

$$\mathrm{Attn}_t(X) := \sum_{p \in X} \mathrm{Attn}_t(p).$$

When $G_t > 0$, define the normalized attention distribution

$$\alpha_t(p) := \frac{\mathrm{Attn}_t(p)}{G_t}, \qquad \sum_{p \in \mathcal{P}} \alpha_t(p) = 1.$$

Remark 736. *The intent is to get a clean quantitative proxy for the intuitive claim "pattern p is in the spotlight." Each module spends genenergy, and each module spreads its expenditure over the patterns it is currently engaging. The weights $w_i(p)$ can be read as a soft assignment: how much of module i's work is effectively "about" p rather than something else.*

Remark 737. *A concrete example: suppose a planning module i is simultaneously considering two candidate subgoals, represented by patterns p_1 and p_2. One might set $w_i(p_1) = 0.7$, $w_i(p_2) = 0.3$. Then if $g_t(i) = 10$, this module contributes 7 units of attention mass to p_1 and 3 to p_2. Summing across modules yields a global pattern-level allocation.*

Remark 738. *The distribution $\alpha_t \in \Delta(\mathcal{P})$ is the normalized version of attention mass. Here $\Delta(\mathcal{P})$ denotes the probability simplex on $\mathcal{P}$: the set of functions $\alpha : \mathcal{P} \to [0,1]$ summing to 1. We use probabilistic notation not because attention is always stochastic, but because normalization turns "share of resources" into a form that can be compared across times and systems, and connected to entropy-like concentration measures.*

Remark 739 (Module-level vs pattern-level)**.** *One can also work directly with the module allocation vector $\hat{\alpha}_t(i) := g_t(i)/G_t$. The pattern-level distribution α_t is a pushforward of $\hat{\alpha}_t$ along the weights w_i.*

Remark 740. *The pushforward comment can be read categorically: $\hat{\alpha}_t$ lives on modules, while α_t lives on patterns; the weights w_i implement a (time-dependent) stochastic map from modules to patterns. In the philosophical idiom, this is how the system's internal economy of effort becomes a distribution over what is* present to its cognition *(Hyperseed-Concept **??**).*

15.2.2 Attentional focus and self-reflective attention

Hyperseed's notion of attentional focus is a *concentration* condition: a relatively small subset of the system absorbs a relatively large fraction of genenergy. In this section we treat the attention weights $\alpha_t(p)$ as an abstract accounting device for how much of the system's limited update capacity is effectively routed through pattern p at time t. In particular, one can read α_t as a probability distribution over $\mathcal{P}$ (hence $\sum_{p \in \mathcal{P}} \alpha_t(p) = 1$), but the intended interpretation is budgetary: the mass is a normalized proxy for "genenergy share" rather than a claim about any specific implementation.

Definition 193 (Attentional focus)**.** *Fix parameters $\theta \in (0,1)$ (mass threshold) and $k \in \mathbb{N}$ (size bound). We say S has an* attentional focus *at time t (at level (θ, k)) if there exists a subset $F_t \subseteq \mathcal{P}$ with $|F_t| \le k$ such that*

$$\alpha_t(F_t) := \sum_{p \in F_t} \alpha_t(p) \ge \theta.$$

Equivalently, if $\alpha_t^{\downarrow}$ is the nonincreasing rearrangement of the weights $(\alpha_t(p))_{p \in \mathcal{P}}$, then focus holds iff $\sum_{j=1}^{k} \alpha_t^{\downarrow}(j) \ge \theta$.

Remark 741. *The definition formalizes an old introspective fact: at any moment, a mind is usually "mostly doing one thing," perhaps with a few side computations. The parameters (θ, k) let us tune what we mean by "mostly" and "few." The nonincreasing rearrangement $\alpha_t^{\downarrow}$ is simply the list of attention weights sorted from largest to smallest; the criterion then says: the top k patterns together carry at least θ fraction of the attention mass. It is also useful to keep in mind that the set F_t is typically* not *unique: whenever there are ties among the largest weights, multiple choices of F_t witness the same level of focus. Nothing in later arguments depends on a canonical choice; only the existence of some small set with large mass matters.*

Remark 742. *A simple example: if $\mathcal{P}$ has* 100 *patterns and the top $k = 5$ patterns carry* 85% *of the mass, then the system has focus at $(\theta, k) = (0.85, 5)$. Conversely, if attention is spread nearly uniformly, then no small subset can reach a high threshold and focus fails. This aligns with the intuition that "diffuse attention" is the absence of a narrow bottleneck. Note that this is sensitive to* how *nonuniformity is distributed: e.g. two medium-sized clusters can each be salient without either cluster alone clearing a large θ for small k, which matches the everyday phenomenon of "split attention" (two active tasks) as distinct from either single-task focus or fully uniform diffusion.*

Remark 743. *The usefulness of this definition is pragmatic: it lets later sections talk about focus-dependent phenomena (e.g. workspace access, self-reference, and stable control loops) without committing to a specific psychological or neural mechanism. It is thus a minimal mathematical hook for Hyperseed-Concept 60. A further pragmatic benefit is that the condition is scale-free: only the* relative *genenergy shares matter. This is helpful when comparing subsystems or agents whose absolute compute budgets differ, since attentional bottlenecks are often about allocation rather than total capacity.*

Remark 744 (Basic monotonicities)**.** *The definition behaves monotonically in the expected way: if focus holds at (θ, k), then it holds at (θ', k') for any $\theta' \leq \theta$ and any $k' \geq k$. Conversely, increasing θ or decreasing k makes the test strictly harder. This lets one speak of "strong" focus (high θ with small k) versus "weak" focus (lower θ and/or larger k) without changing the underlying concept.*

Remark 745 (Canonical witnessing set via top-k)**.** *Given α_t, one can always witness the maximal attainable mass under a size bound k by choosing F_t to be any set of k patterns achieving the k largest weights (ties arbitrary). Then $\alpha_t(F_t) = \sum_{j=1}^{k} \alpha_t^{\downarrow}(j)$, which shows that the "existence of F_t" phrasing and the rearrangement criterion are not merely equivalent but operationally the same test: compute the top-k share and compare to θ.*

Definition 194 (Self-model and self-reflective attention)**.** *Let $\mathcal{P}_{\mathrm{self}} \subseteq \mathcal{P}$ be the set of patterns that constitute the system's current self-model (Section 16). We say S has* self-reflective attention *at time t if*

$$\alpha_t(\mathcal{P}_{\mathrm{self}}) \geq \theta_{\mathrm{self}}$$

for some chosen threshold $\theta_{\mathrm{self}} \in (0, 1)$. More strongly, S has focused self-reflective attention *at time t if F_t can be chosen so that $F_t \cap \mathcal{P}_{\mathrm{self}} \neq \emptyset$.*

Remark 746. *Here the point is not metaphysical privilege but routing: self-model patterns are those whose update affects how the system predicts, controls, and narrates itself. When a significant share of attention mass is allocated to $\mathcal{P}_{\mathrm{self}}$, the system is spending real resources on maintaining or revising its self-description (Hyperseed-Concept* **??***; Hyperseed-Concept 165). The two thresholds θ (general focus) and θ_{self} (self allocation) need not coincide: one may want to detect self-involvement even when overall focus is weak (e.g. rumination spread across many self-related fragments), or conversely require that self-related patterns occupy a significant fraction of a very tight focus set.*

Remark 747. *A typical example is a learning agent that allocates attention to error-monitoring patterns such as "I am failing" or "my model is uncertain." These are self-model patterns in the present sense, and when they enter the focus set F_t, one gets a crisp formal correlate of the everyday report "I noticed that I was confused." This is a small but essential ingredient for the later consciousness interface (Section 17) and resonates with empirical discussions of self-related processing (Hyperseed-Concept 85). It is worth emphasizing that the stronger condition (intersection with F_t) captures a difference between (i) dedicating some background mass to self-maintenance and (ii) having self-model content be among the currently dominant patterns. The latter is the mathematically clean version of the phenomenological distinction between "the self is in the foreground" and "the self is merely implicitly present."*

Remark 748 (Focus is graded)**.** *The parameters (θ, k) provide a simple family of "focus tests." Other concentration measures can be used (entropy, Gini coefficient, top-k share), but the above definition is adequate for connecting to later sections. In practice, one may also care about* temporal *aspects (how long a focus persists): a brief spike where $\sum_{j=1}^{k} \alpha_t^{\downarrow}(j) \geq \theta$ may have different functional consequences than a sustained interval on which the inequality holds. Later when we discuss stable control loops, it will be natural to consider not only instantaneous focus but also its persistence under small perturbations of α_t.*

15.3 Attention as a bounded control policy

Attention is itself an action: it chooses which parts of the internal model are updated, refined, or queried. Formally, we treat attention as a policy that selects a distribution over internal update actions. This framing makes "what to think about next" a first-class control variable, rather than a side-effect of perception or representation: the system can actively steer its own computational trajectory by deciding which internal transformation to apply next.

Definition 195 (Internal update actions)**.** *Let $\mathcal{U}$ be a set of* internal update actions*. An element $u \in \mathcal{U}$ may represent:*

(a) refining a distinction (Section 3);

(b) updating a pattern intensity estimate (Section 9);

(c) querying memory or a submodule (Section 20);

(d) simulating a causal consequence (Section 14);

(e) reweighting or re-indexing representations (Section 13).

Assume each update action u has a genenergy cost $c(u) \geq 0$.

Remark 749. *The philosophical move here is to treat attention not as a passive "light" but as an internal intervention: an update action u changes the system's internal state, and thus changes what future actions are possible. This matches the control-theoretic reading of attention used in engineered AGI systems, where "choosing what to compute next" is itself a crucial policy decision [19].*

Remark 750. *It is useful to emphasize that $\mathcal{U}$ can mix heterogeneous operations that live at different algorithmic levels. For example, "refine a distinction" may change the representational vocabulary, while "retrieve one memory" may only change the active context, and "simulate a consequence" may instantiate a temporary forward model. Treating all of these as elements of a single action set is a deliberate unification: the attention mechanism need not commit in advance to a single computational paradigm, and can instead arbitrate among many possible internal moves.*

Remark 751. *Examples of $u \in \mathcal{U}$ include: running one step of a planner; performing a memory retrieval; updating a Bayesian posterior; or running a short roll-out simulation. The cost $c(u)$ is deliberately abstract so it can stand in for time, energy, FLOPs, bandwidth, or any conserved-or-not resource (Hyperseed-Concept 87; Hyperseed-Concept 100).*

Remark 752. *The cost function $c(u)$ can also be interpreted as a* capacity consumption *measure: two update actions may both be fast, but one may monopolize scarce working memory or interfere with concurrent processes. In that sense, bounded attention is not only a time-budget story; it can represent any constraint that forces internal computation to be selectively scheduled rather than globally applied.*

Definition 196 (Attention policy with budget)**.** *Fix a budget $B_t \geq 0$ for time t. An* attention policy *is a map*

$$\pi_t : \Omega_t \to \Delta(\mathcal{U}),$$

where Ω_t is the system's internal state space at time t and $\Delta(\mathcal{U})$ is the set of probability distributions over $\mathcal{U}$. A draw $u \sim \pi_t(\omega)$ is feasible if $c(u) \leq B_t$.

Remark 753. *One can equivalently define, for each (t, ω), a feasible subset*

$$\mathcal{U}_t(\omega) := \{u \in \mathcal{U} \ : \ c(u) \leq B_t\},$$

and view $\pi_t(\omega)$ as a distribution supported on $\mathcal{U}_t(\omega)$. This highlights that the budget may be context-dependent in practice: the same nominal update action might be cheap in one internal configuration (e.g., when a relevant cache is warm) and expensive in another (e.g., when a submodule must be re-initialized).

Remark 754. *The notation $\pi_t : \Omega_t \to \Delta(\mathcal{U})$ is standard in decision theory: given an internal state $\omega \in \Omega_t$, the policy π_t returns a distribution over actions. The only twist is that actions here are internal computations or updates rather than external motor moves. The feasibility condition $c(u) \leq B_t$ makes the resource constraint explicit, so that attention is a bounded-control problem rather than a purely epistemic one.*

Remark 755. *In this language, "attention failure" can be described without invoking any special pathology: it is simply the selection (perhaps repeatedly) of update actions that are locally feasible but globally unhelpful, e.g., over-investing in cheap but low-value computations, or repeatedly applying an update rule that no longer improves internal predictions. Conversely, "skilled attention" is a policy π_t that reliably selects update actions whose downstream effects improve later decision quality under budget constraints.*

Remark 756. *A simple example: if B_t is small, the policy may assign high probability only to cheap actions like "retrieve one memory" rather than expensive ones like "run a long simulation." This gives a clean way to formalize attentional phenomena such as shallow vs. deep reasoning under time pressure, without altering the underlying pattern semantics.*

Remark 757. *The budgeted-policy view also clarifies how attentional allocation can couple to external action. If an external decision must be made imminently, B_t effectively shrinks, shifting π_t toward quick diagnostic updates; if the system is in a deliberative regime, B_t can expand, permitting costly counterfactual simulation and deeper representational refinement. In both cases, the same internal machinery can be used; only the feasible set and the policy's trade-offs change.*

The definitions above are intentionally abstract: Hyperseed is compatible with many concrete attention mechanisms. One common form is "soft" allocation by expected value-of-information or expected control benefit. In particular, we can separate (i) a mechanism that scores candidate foci (e.g., patterns) for their relevance to control, from (ii) a mechanism that turns those scores into a bounded allocation that respects the need to diversify across multiple competing opportunities for improvement.

Definition 197 (Priority scores from predictive attraction)**.** *Let $\mathcal{G}$ be a set of goal conditions (Section 18). For each pattern $p \in \mathcal{P}$ assume there is an associated event type A_p meaning "pattern p is currently active/salient" (Section 14). Define a* priority score *at time t by*

$$\mathrm{Prio}_t(p) := \max_{G \in \mathcal{G}} \left|\mathrm{Att}_{O,\Delta}(A_p, G)\right|,$$

where $\mathrm{Att}_{O,\Delta}$ is the attraction operator from Section 14.3.

Remark 758. *This definition is the promised "differential" idea in miniature: a pattern is prioritized if its presence makes a significant difference to what the system expects about its goals. The absolute value allows both positive and negative attraction to matter: either "this pattern tends to bring about the goal" or "this pattern tends to prevent the goal" can warrant attention, because both can guide effective control.*

Remark 759. *The use of $\max_{G \in \mathcal{G}}$ encodes a conservative, goal-sensitive criterion: a pattern is high-priority if it substantially affects at least one goal condition, even if it is irrelevant to others. Alternative aggregations are possible (e.g., expectation under a goal-importance distribution, or a weighted sum), but the max operator cleanly captures the idea of "whatever is most at stake right now" without committing to a particular utility encoding beyond the existence of $\mathcal{G}$.*

Remark 760. *One may read $\mathrm{Prio}_t(p)$ as a bridge between prediction/control (Section 14) and resource allocation: it converts the attraction relation into a scalar that can be traded off against entropy-regularization or other concentration penalties. This is a typical move in architectures emphasizing cognitive synergy and integrated attention allocation [19].*

Remark 761. *To connect the pattern-level priority score back to the update-action view, one may imagine that many update actions $u \in \mathcal{U}$ are* pattern-indexed *(e.g., "update intensity of p", "refine distinction involving p", "retrieve memories associated with p"). In that case, $\mathrm{Prio}_t(p)$ induces a natural family of heuristics for scoring update actions: an action is valuable insofar as it is expected to reduce uncertainty, sharpen distinctions, or improve control predictions for high-priority patterns. This is one concrete route from attraction-based semantics to implementable attention scheduling.*

Proposition 23 (Entropy-regularized attention allocation)**.** *Fix $\beta > 0$ and suppose $\mathcal{P}$ is finite. Among all distributions $\alpha \in \Delta(\mathcal{P})$, the maximizers of*

$$J(\alpha) := \sum_{p \in \mathcal{P}} \alpha(p)\, \mathrm{Prio}_t(p) \;-\; \frac{1}{\beta} \sum_{p \in \mathcal{P}} \alpha(p) \log \alpha(p)$$

are exactly the softmax distributions

$$\alpha^*(p) = \frac{\exp(\beta\, \mathrm{Prio}_t(p))}{\sum_{q \in \mathcal{P}} \exp(\beta\, \mathrm{Prio}_t(q))}.$$

Remark 762. *A standard proof uses Lagrange multipliers: optimize $J(\alpha)$ subject to $\sum_p \alpha(p) = 1$ and $\alpha(p) \geq 0$. Differentiating yields*

$$\mathrm{Prio}_t(p) - \frac{1}{\beta}(1 + \log \alpha(p)) + \lambda = 0,$$

so $\alpha(p) \propto \exp(\beta \, \mathrm{Prio}_t(p))$, and normalization gives the stated form. The entropy term is strictly concave in α (on the simplex interior), so the optimizer is unique when all $\alpha(p) > 0$.

Remark 763. *This proposition says: if you want to allocate attention to maximize expected priority, but you also penalize excessively peaky allocations using Shannon entropy, then the optimal allocation has the familiar exponential form. In other words, "soft attention" is not an arbitrary heuristic; it is the exact optimizer of a simple and interpretable objective.*

Remark 764. *The parameter β plays the role of an inverse temperature. As $\beta \to \infty$, the entropy penalty vanishes and α^* concentrates on patterns with maximal $\mathrm{Prio}_t(p)$, yielding a nearly greedy attentional focus; as $\beta \to 0^+$, the entropy term dominates and α^* tends toward a uniform allocation over $\mathcal{P}$, representing highly diffuse or exploratory attention. Thus, β provides a compact handle on the sharpness of attentional selection, independent of the underlying semantics of priority.*

Remark 765. *The result matters here because it gives a canonical mathematical mechanism for turning attraction scores into a distribution α_t that can be compared to the earlier focus definition. It therefore makes contact between the qualitative Hyperseed narrative about attention and a quantitative optimization principle common in both cognitive modeling and machine learning. In particular, it turns an* ordinal *ranking of patterns (what is more or less attractive) into a* cardinal *allocation rule (how much attention each pattern receives), which is necessary if attention is later treated as a measurable bottleneck. The same mechanism also clarifies what it means for attention to be "bounded": the entropy term encodes a soft resource constraint that penalizes overly concentrated allocations unless the priorities justify them.*

Remark 766. *The connection to the rest of the document is direct: predictive attraction (Section 14.3) feeds Prio_t, which feeds an attention distribution α_t, which then gates which patterns and knowledge types can participate in cognitive synergy (Section 15.6). This makes α_t a control-like interface between evaluation and participation: it is computed from internal scores, but it has externally visible consequences for which inference steps, memory retrievals, and action proposals are even available to be composed. Seen this way, Prio_t plays the role of an energy landscape over patterns, while α_t is the induced sampling or allocation policy used by downstream cognitive processes.*

Proof. This is the standard Lagrange-multiplier calculation for maximizing a linear functional with an entropy regularizer under the simplex constraint $\sum_p \alpha(p) = 1$. Differentiating the Lagrangian yields $\mathrm{Prio}_t(p) - \frac{1}{\beta}(1 + \log \alpha(p)) = \lambda$ for all p, which implies $\alpha(p) \propto \exp(\beta \, \mathrm{Prio}_t(p))$. To make the dependence explicit, one may start from an objective of the form $J(\alpha) = \sum_p \alpha(p) \, \mathrm{Prio}_t(p) + \frac{1}{\beta} H(\alpha)$ with $H(\alpha) = -\sum_p \alpha(p) \log \alpha(p)$, and impose $\alpha(p) \geq 0$ with $\sum_p \alpha(p) = 1$. The stationarity condition above holds on the interior (where $\alpha(p) > 0$); at the optimum one indeed gets strictly positive weights whenever β is finite and the priorities are finite, so the interior calculation is self-consistent. Solving the first-order condition gives $\alpha(p) = \exp(\beta(\mathrm{Prio}_t(p) - \lambda) - 1)$, and the scalar λ is fixed by the normalization constraint, yielding the familiar normalized softmax form. □

Remark 767. *At a high level, the proof works because the objective $J(\alpha)$ is strictly concave in α on the interior of the simplex: a linear term plus (negative) entropy. Strict concavity guarantees a* *unique optimum (up to degeneracies when priorities tie), and the first-order condition characterizes it. Equivalently, one can view the optimization as minimizing a Kullback–Leibler divergence to a Gibbs distribution: maximizing* $\sum_p \alpha(p)\mathrm{Prio}_t(p) + \frac{1}{\beta}H(\alpha)$ *is the same as minimizing* $\mathrm{KL}(\alpha \,\|\, \pi_\beta)$ *where* $\pi_\beta(p) \propto \exp(\beta\,\mathrm{Prio}_t(p))$*. This perspective makes clear why the solution has full support for finite* β *(no pattern is given exactly zero attention unless forced by infinite negative priority), and why ties in* Prio_t *can yield symmetry-induced degeneracies only in the limiting cases where the objective effectively becomes linear.*

Proof sketch. Introduce a Lagrange multiplier λ to enforce $\sum_p \alpha(p) = 1$. Taking derivatives with respect to each coordinate $\alpha(p)$ yields an equation of the form $\log \alpha(p) = \beta(\mathrm{Prio}_t(p) - \lambda) - 1$. Exponentiating gives an unnormalized exponential distribution, and the normalization constant is the partition function $\sum_q \exp(\beta\,\mathrm{Prio}_t(q))$. In other words, the multiplier λ is nothing but (a shifted version of) the log-partition function that ensures the simplex constraint is satisfied, and it couples all coordinates of α through a single global normalization. The same algebra also shows the invariance under shifting priorities by a constant: replacing $\mathrm{Prio}_t(p)$ by $\mathrm{Prio}_t(p) + c$ multiplies all unnormalized weights by $e^{\beta c}$, which cancels after normalization, so only relative priority differences matter. □

Remark 768. *Geometrically, one may picture the simplex* $\Delta(\mathcal{P})$ *as a polytope and the entropy term as a barrier that discourages sliding onto its faces. As* $\beta \to \infty$*, the barrier weakens and the optimum approaches a hard argmax (full focus on the best pattern). As* $\beta \to 0$*, the barrier dominates and the optimum approaches uniform attention. Thus* β *is a temperature-like knob interpolating between exploration and exploitation, but now phrased as an internal allocation of genenergy. A complementary interpretation is that* β *sets the* sensitivity *of allocation to priority gaps: for moderate* β*, small differences in* Prio_t *produce gently varying weights, while for large* β *the distribution becomes sharply peaked and effectively implements winner-take-most selection. This matters for cognitive synergy because it controls how many distinct pattern types can remain active enough to interact; overly large* β *risks premature single-pattern lock-in, whereas overly small* β *disperses genenergy so widely that no pattern is sufficiently amplified to contribute strongly.*

Remark 769. *Proposition 23 does* not *claim Hyperseed requires softmax attention. It only shows one clean way to connect the earlier notion of predictive attraction to a concrete bounded-resource allocation rule. More broadly, the proposition functions as an existence proof that "attraction* $\to$ *gated participation" can be implemented by a principled optimization criterion, rather than by an ad hoc mapping. Alternative mechanisms (e.g., sparsemax/entmax variants, thresholded mixtures, or constraint-based allocations) could be substituted if they better match the desired phenomenology, but the softmax case provides a simple baseline with a clear control parameter and well-understood limiting behavior.*

15.4 Varieties of knowledge as typed pattern structures

Hyperseed treats "knowledge" as a family resemblance notion: multiple distinct types of stored content play different roles in prediction and control. The ontology offers a pragmatic classification for human-like minds; we formalize it as a family of typed stores. In this framing, a "store" is not merely a container of propositions, but a structured substrate whose items behave like *typed pattern constraints*: each type comes with characteristic generalization behavior, characteristic forms of evidence, and characteristic ways it can be brought to bear on ongoing perception/action loops.

Definition 198 (Knowledge type)**.** *A* knowledge type T *for a system* S *consists of:*

(a) a carrier set (or space) K_T of knowledge items;

(b) a semantics map $\mathrm{Sem}_T : K_T \to \mathcal{S}$ that interprets items as constraints on patterns, processes, or goals;

(c) an update operator (or family of operators) Upd_T specifying how items are learned, revised, or forgotten;

(d) an interface specifying which other types it can exchange information with.

Remark 770. *One can make the "update operator" more explicit as a rule family of the form*

$$\mathrm{Upd}_T : K_T \times \mathcal{E}_T \to K_T,$$

where $\mathcal{E}_T$ denotes the space of admissible evidential inputs for type T (e.g. sensory episodes, counterexamples, rewards, demonstrations, proofs, or internal audit signals). What matters for the present section is that different knowledge types have different $\mathcal{E}_T$ and different invariants under update (for example, exemplar memories may be capacity-limited while declarative stores may be consistency-limited only by attention and time).

Remark 771. *The point of calling these "types" is not to reify a metaphysical taxonomy, but to mark distinct* update dynamics *and distinct* failure modes*. A knowledge item is not only something that can be true or false; it is something that can be learned, revised, forgotten, communicated, and used for control. These verbs differ across types, and this difference is precisely what makes cross-type synergy possible [19].*

Remark 772. *The "interface" component (d) is where cognitive synergy becomes operational: it specifies what queries a store can answer, what messages it can accept, and what it can emit to other stores. Concretely, an interface may include (i) a set of accessors (e.g. retrieve-by-key, retrieve-by-similarity, retrieve-by-goal), (ii) a set of translators into other representational formats, and (iii) gating conditions (e.g. only export high-confidence items, or only import items when attention flags a relevant context boundary). In practice, interfaces are also where resource constraints show up: two knowledge types may be semantically compatible but interface-incompatible because translation is too expensive under current genenergy/attention budgets.*

Remark 773. *One can view Sem_T as a disciplined way of saying "what does this store constrain?" For declarative knowledge it constrains beliefs about events; for procedural knowledge it constrains the space of policies; for attentional knowledge it constrains how the system searches its own computations. In each case the semantics map ties a symbolic or stored item back to patterns and processes (Hyperseed-Concept 157; Hyperseed-Concept* **??***). A useful way to read Sem_T is as a* bridge *from a compact stored token to a (possibly large) set of permitted pattern instantiations: the token is small, but its denotation is a constraint over the system's pattern space.*

15.4.1 Declarative knowledge

Definition 199 (Declarative knowledge)**.** *Declarative knowledge is a store of* pattern beliefs*, often (but not necessarily) linguistically expressible. Formally, let $\mathcal{L}$ be a language for talking about events/patterns. A declarative state is a valuation*

$$K_{\mathrm{dec}} : \mathcal{L} \to \mathbb{V} = [0, 1]^2,$$

assigning each formula a p-bit *degree of positive/negative evidence (Section 3.2).*

Remark 774. *Although $\mathcal{L}$ can be taken as a linguistic/propositional language, nothing in the definition requires natural language; $\mathcal{L}$ may equally be a structured pattern description language (e.g. relational templates over perceived entities, temporal claims, or causal sketches). The essential ingredient is that formulas in $\mathcal{L}$ are the kind of objects that can be* queried *by other subsystems (e.g. planning, explanation, counterfactual reasoning), and that their evidence values can be updated by a mix of perception, testimony, and internal consistency checks.*

Remark 775. *Intuitively, $K_{\mathrm{dec}}(\varphi)$ is the system's present evidential stance toward the claim φ: not merely "how likely," but "how supported" and "how refuted," separately. This is a natural fit for observer-relative and potentially inconsistent boundaries, where one can have both substantial evidence for and against the same proposition without collapse (Hyperseed-Concept 65; see also paraconsistent motivations in [23, 24]).*

Remark 776. *Update for declarative knowledge typically consists of accumulating or discounting evidence while preserving the separation between positive and negative support. For example, a single observation may increase the first coordinate without decreasing the second, while a debunking argument may increase the second coordinate without forcing the first to vanish. This is relevant to attention because conflict (high support and high refutation) is a natural trigger for targeted information gathering: attention can treat such formulas as "open loops" whose resolution is valuable but not always urgent.*

Remark 777. *A simple example is a vague predicate like "the room is quiet." A system might store $K_{\mathrm{dec}}(quiet) = (0.7, 0.4)$, reflecting positive evidence from a low sound meter and negative evidence from intermittent noise. The usefulness is that later inference can proceed without forcing premature consistency; attention can then decide whether to spend genenergy resolving the conflict or to act under it.*

15.4.2 Procedural knowledge

Definition 200 (Procedural knowledge)**.** *Procedural knowledge consists of "if in context C, doing procedure π tends to achieve goal G." Let $\mathcal{C}$ be a set of contexts and $\mathcal{G}$ a set of goals. Let Π be a set of procedures (policies, programs, controllers). A procedural knowledge item is a triple (C, π, G) together with an evidence value $\mathrm{ProcEvd}(C, \pi, G) \in \mathbb{V}$.*

Remark 778. *Here "context" C should be read broadly: it may denote an external situation (terrain type, social setting), an internal state (fatigue, low battery), or a conjunction of both, and it may be crisp or fuzzy depending on the system's perceptual and conceptual granularity. In particular, procedural knowledge can be indexed by* attentionally constructed *contexts: the system may only carve out a context boundary (and thus create a stable key C) once attention has repeatedly found that a certain cluster of features predicts procedure success/failure.*

Remark 779. *Procedural knowledge is "knowing how" rather than "knowing that." The evidence value $\mathrm{ProcEvd}(C, \pi, G)$ plays the same paraconsistent role as in declarative knowledge, but now attached to a claim about the efficacy of a procedure. This lets a system represent, for instance, that a policy often works but has known failure modes, without reducing everything to a single success probability (Hyperseed-Concept **??**; Hyperseed-Concept **??**).*

Remark 780. *Procedural update is typically driven by rollouts, trial-and-error, imitation, and counterfactual evaluation: a single attempted execution of π under an estimated C contributes evidence to $\mathrm{ProcEvd}(C, \pi, G)$, while diagnosis of why an attempt failed can refine the context key* *(the system may split C into subcontexts where the procedure behaves differently). This is one of the primary points of interaction with attention: attention determines which trials to run, which failures to analyze deeply, and which context splits are worth the representational complexity.*

Remark 781. *As an example, C might be "slippery ground," π might be "walk slowly," and G might be "reach destination without falling." The system may have positive evidence from past successes and negative evidence from rare but salient failures. Such items feed directly into control selection (Section 14) and can be prioritized by attention when a context boundary becomes active.*

15.4.3 Sensory knowledge

Definition 201 (Sensory knowledge)**.** *Sensory knowledge is memory enabling discrimination: after sensing an item s, the system can later distinguish a copy/recurrence of s from similar non-copies. Formally, let $\mathcal{S}$ be a stimulus space and let d be a similarity (pseudo)metric. A sensory memory state is a finite set of stored exemplars $\{s_j\}$ together with an embedding $\phi : \mathcal{S} \to Y$ such that discrimination is performed by thresholding distances $d_Y(\phi(s), \phi(s_j))$.*

Remark 782. *This definition abstracts over many implementations (template matching, nearest-neighbor retrieval, metric learning, associative memories) by highlighting the functional role: a* discrimination operator *induced by storing exemplars and comparing them in an internal space. The finiteness of $\{s_j\}$ is not merely technical: it encodes the idea that sensory memory is usually capacity-limited, so forgetting (or prototype compression) is a normal and often desirable update dynamic.*

Remark 783. *This definition captures the bare bones of recognition by similarity: store a few exemplars, map stimuli into an internal space Y, and then declare "match" when the embedded distance is small. The pseudo-metric d allows the usual flexibility: different stimuli may be deemed zero-distance by the system because it cannot or does not care to distinguish them at the current resolution.*

Remark 784. *The embedding ϕ can be fixed, learned, or attention-modulated. In particular, attentional control can act by temporarily changing the effective metric (or the threshold) so that, for the current task, some distinctions are sharpened while others are collapsed. This gives one route to cognitive synergy between sensory and declarative/procedural stores: declarative hypotheses can cue which sensory dimensions should matter, and procedural demands can cue which discriminations are worth allocating representational capacity to.*

Remark 785. *A simple example is face recognition: $\mathcal{S}$ is the space of face images, Y is an embedding space learned by a neural network, and the exemplars $\{s_j\}$ are stored reference faces. The utility of formalizing sensory knowledge this way is that it makes explicit where genenergy is spent: improving ϕ, storing more exemplars, or changing thresholds are all internal update actions (Section 15.3).*

Remark 786. *The same schematic separation also clarifies what can be held fixed while something else is adapted. For instance, if ϕ is treated as stable (a "frozen" encoder), then sensory improvement must occur via exemplar management (adding, pruning, reweighting, or clustering $\{s_j\}$) and via decision rules (thresholds, rejection options, abstention policies). Conversely, if exemplar memory is constrained, genenergy can be redirected toward learning a more invariant ϕ so that fewer exemplars suffice. Making these tradeoffs explicit is useful when diagnosing failure: poor recognition might come from an inadequate embedding, from stale exemplars, or from a miscalibrated acceptance criterion, and these correspond to distinct internal actions and costs.*

15.4.4 Attentional knowledge

Hyperseed highlights a specific meta-knowledge type: learned regularities of attention shift.

Definition 202 (Attentional knowledge)**.** *Let $\mathcal{X}$ be a set of "attention targets" (pattern classes, module clusters, or questions). Attentional knowledge consists of regularities of the form: "when attention is on X, it often shifts to Y, and this shift is associated with goal success." Formally, an attentional knowledge state may be represented as a family of kernels*

$$K_{\mathrm{att}}(Y \mid X, G) \in \mathbb{V}, \qquad X, Y \in \mathcal{X},\ G \in \mathcal{G},$$

encoding (paraconsistent) evidence that shifting focus from X to Y tends to support G.

Remark 787. *This is knowledge about the system's own search procedure: it stores which transitions of attentional focus have historically been fruitful. The notation $K_{\mathrm{att}}(Y \mid X, G)$ is deliberately reminiscent of a conditional probability kernel, but valued in $\mathbb{V} = [0,1]^2$ to permit mixed evidence. The targets $\mathcal{X}$ can be pattern sets, modules, or even explicit questions (Hyperseed-Concept* **??***).*

Remark 788. *Interpreting K_{att} as a kernel emphasizes that it can be many-to-many and contextually gated by goals: from the same X there may be several productive next foci Y, and the ranking can depend strongly on G. The paraconsistent valuation $\mathbb{V} = [0,1]^2$ can be read as separately tracking evidence-for and evidence-against a shift, so that the system may represent both (i) that a transition is often helpful in some regimes, and (ii) that it is also often harmful or distracting in others, without collapsing this tension into a single scalar. This matters for attention control, since overly aggressive "always shift to Y" habits are a common pathology in bounded systems.*

Remark 789. *One can also regard K_{att} as an internal, learned "operator selection" map: X names a current representational neighborhood (e.g., a memory store, a perception stream, a planning module), and Y names which neighborhood to query next. In this reading, attentional targets are not only semantic objects but also* interfaces *to computational resources, so the kernel summarizes regularities about which interface-switches pay off under which goals. This makes attentional knowledge a natural site for genenergy expenditure: updating K_{att} changes how future computation is allocated even when the underlying declarative beliefs remain unchanged.*

Remark 790. *A simple example: when debugging code (target X), shifting attention to "construct a minimal reproducible example" (target Y) often supports the goal G of fixing the bug. The system may not be able to justify why this works declaratively, yet it can learn the attentional transition as a useful habit. This is one of the clearest loci where cognitive synergy manifests in practice: the system uses one knowledge type (attentional) to orchestrate updates in others.*

Remark 791. *In that debugging example, the shift $X \to Y$ is valuable partly because it* changes the data distribution *seen by other subsystems: a minimal reproducible example compresses the search space, reduces confounds, and yields cleaner counterfactual tests. Thus attentional knowledge can be viewed as knowledge about which transformations of the agent's own input streams make downstream learning or inference cheaper. This perspective also clarifies why attentional habits can remain useful even when they are not explicitly articulated: the relevant regularity may be procedural and distributed across modules rather than available as a compact declarative rule.*

15.4.5 Intentional knowledge

Definition 203 (Intentional knowledge)**.** *Intentional knowledge captures goal specialization: "when pursuing goal G in context C, it is appropriate to switch to a specialized goal G_1." Formally, represent intentional knowledge as a relation*

$$K_{\mathrm{int}}(G_1 \mid G, C) \in \mathbb{V}, \qquad G, G_1 \in \mathcal{G},\ C \in \mathcal{C}.$$

Remark 792. *This is knowledge about the* internal goal topology *of the agent: how abstract goals refine into more concrete subgoals when contexts shift. It is not merely planning, but a learned mapping of which refinements tend to be appropriate. In Hyperseed terms, it is a pattern of goal-development (Hyperseed-Concept 95; Hyperseed-Concept* **??***).*

Remark 793. *The emphasis on topology is meant literally: the agent's goal system can be treated as a graph (or hypergraph) where edges correspond to refinement, decomposition, or constraint addition. Then $K_{\text{int}}(G_1 \mid G, C)$ plays a role analogous to a context-sensitive adjacency relation, encoding which outgoing refinements are supported or discouraged by experience. Because it is $\mathbb{V}$-valued, the representation can capture that a refinement is plausible yet risky, or that it is appropriate only when further conditions (not yet represented in C) hold. This allows the agent to maintain multiple candidate specializations without premature commitment.*

Remark 794. *An example: for a general goal G = "be healthy," and context C = "winter," a specialized goal might be G_1 = "get vitamin D." The system may have mixed evidence about the appropriateness of this specialization depending on other latent factors, hence the* p-bit*-valued relation. Formalizing K_{int} is useful because it provides a precise target for attentional update actions: one can spend genenergy revising not only beliefs and policies but also the mapping from goals to subgoals.*

Remark 795. *The "appropriate to switch" phrasing is intentionally compatible with several control regimes. In a strict regime, K_{int} may be used to* replace *G by G_1 (commitment); in a softer regime, it may* spawn *G_1 as a concurrent subgoal (branching); and in a reflective regime, it may be treated as a prompt for information gathering (e.g., attend to whether latent conditions hold before specializing). These distinctions matter operationally because they correspond to different resource commitments and different ways of recovering from a poor specialization.*

Remark 796 (Why multiple knowledge types)**.** *Different types have different update dynamics and different failure modes. Procedural knowledge can be robust without being explainable; declarative knowledge can be communicable without being actionable; attentional knowledge can guide search even when the system cannot articulate why the guidance works. This plurality is one of Hyperseed's motivations for cognitive synergy.*

Remark 797. *The failure-mode differences are not merely philosophical; they induce different diagnostics and different interventions. When procedural knowledge fails, the symptom may be brittle performance under distribution shift; when declarative knowledge fails, the symptom may be contradiction or confabulation under questioning; when attentional knowledge fails, the symptom may be wasted compute on unproductive loci even if the necessary facts are present somewhere in memory. Treating these as distinct knowledge types supports targeted repair: e.g., revise a kernel like K_{att} to change where the system looks, rather than trying to directly patch the content it will eventually retrieve or infer.*

Remark 798. *In the Hyperseed framing, the plurality is not a defect but a structural response to finitude: because a bounded system cannot maintain one perfect, unified representational format, it maintains several partially overlapping ones. Synergy is then the disciplined practice of translating between them, so that the weaknesses of one type are compensated by the strengths of another (Hyperseed-Concept* **??***; [19]).*

Remark 799. *On this view, cognitive synergy is implemented by concrete cross-type couplings: attentional knowledge decides which declarative store to consult; declarative knowledge can propose new attention targets (e.g., a hypothesis suggests a test); procedural knowledge can supply fast* *heuristics that seed deliberate reasoning; and intentional knowledge can reshape the very criteria by which attention shifts are judged successful. The point of explicitly typing these structures (kernels over $\mathcal{X}$, relations over $\mathcal{G}\times\mathcal{C}$, etc.) is to make such couplings engineerable: each type exposes specific "handles" for internal update actions, and genenergy can be budgeted among them rather than being spent implicitly and opaquely.*

15.5 Cognitive synergy as cross-type weakness improvement

Hyperseed's term *cognitive synergy* names a ubiquitous architectural fact: complex cognition typically requires frequent exchange of intermediate products between distinct knowledge processes (not merely "final answers"). We now give a weakness-based formalization. In particular, the guiding hypothesis is that many of the practical benefits of "integrated cognition" can be expressed as *improved explanation quality at lower contrivance* when multiple knowledge types are allowed to interact, i.e. when one representational mode can supply constraints, priors, candidate structures, or abstractions to another.

Remark 800. *The motivating intuition is that "the whole is more than the sum of its parts" becomes operational when translations between representational modes yield new compressions. This is the same spirit in which pattern discovery is a compression phenomenon (Section 9; [5]), but now lifted to an architectural level: the compression is enabled by cross-type interaction rather than by a single homogeneous inference engine (Hyperseed-Concept 73). One may read "translation" broadly: it can be an explicit encoder/decoder, a shared latent space, a symbolic labeling of subsymbolic states, a compilation of rules into a policy prior, or any other mechanism by which structure found in one knowledge store reduces the search burden in another.*

15.5.1 Usefulness of a learning algorithm for a knowledge type

Definition 204 (Learning algorithms indexed by knowledge type)**.** *For each knowledge type T, let $\mathsf{Alg}(T)$ denote a family of learning/update algorithms appropriate to T (e.g. logical inference for declarative, reinforcement learning for procedural, metric learning for sensory). An algorithm $A \in \mathsf{Alg}(T)$ induces an update map $\mathrm{Upd}_{T,A} : K_T \to K_T$.*

Remark 801. *This definition is a modest piece of type discipline: it says that not every learning method is appropriate for every store. One does not typically update a sensory embedding ϕ by syllogistic inference, nor update a logical theory by gradient descent—though hybrid methods exist. By making $\mathsf{Alg}(T)$ explicit, we can later speak precisely about cross-type communication: outputs of one type become inputs to algorithms of another. A convenient side effect is that it lets us talk about* where *an intervention occurs: an update that changes K_{dec} but leaves K_{proc} fixed is a different kind of cognitive act than one that changes both, even if both ultimately serve the same downstream goal.*

Remark 802. *A simple example is: $T = \mathrm{proc}$, $\mathsf{Alg}(T)$ includes Q-learning, and $\mathrm{Upd}_{T,A}$ takes a table of action values and updates it after an episode. For $T = \mathrm{dec}$, $\mathsf{Alg}(T)$ might include a paraconsistent inference procedure (Section 3.2). The usefulness of writing $\mathrm{Upd}_{T,A}$ is that it lets us treat learning steps as internal update actions in the sense of Section 15.3. It also makes iteration explicit: repeated application $\mathrm{Upd}^{(n)}_{T,A} := \underbrace{\mathrm{Upd}_{T,A} \circ \cdots \circ \mathrm{Upd}_{T,A}}_{n\ times}$ is the formal proxy for "training for n steps" or "performing n rounds of inference," and so properties like convergence, brittleness, or sensitivity to initialization can be attached to the map rather than left implicit.*

Definition 205 (Usefulness)**.** *Fix a goal set $\mathcal{G}$ and a success predicate* $\mathrm{Succ}(G)$*. An algorithm* $A \in \mathsf{Alg}(T)$ *is* useful for knowledge type T *if, across the contexts of interest, repeatedly applying* $\mathrm{Upd}_{T,A}$ *tends (in expectation) to increase the probability of achieving some goals in $\mathcal{G}$. Equivalently, "invoking A to learn T-knowledge" forms a valuable pattern in the system's activity (Section 9).*

Remark 803. *The definition says: a learning algorithm is not judged by elegance but by teleology—by whether it improves goal achievement in the contexts that matter. This is a deliberately pragmatic stance in the spirit of Hyperseed's observer-relative epistemology (Hyperseed-Concept* **??***; Hyperseed-Concept 160). Here "contexts of interest" can be understood as a (possibly implicit) distribution over task situations, sensory histories, internal states, and resource constraints; the expectation is taken with respect to that distribution together with any stochasticity in the algorithm or environment. This framing also makes room for bounded rationality: an algorithm may be "useful" because it improves* timely *success under limited compute, even if an idealized unbounded method would eventually do better.*

Remark 804. *As an example, a memory consolidation method might be "useful" if it increases the probability of later successful recognition or planning, even if it occasionally strengthens some false declarative beliefs. The paraconsistent setting makes room for such tradeoffs: usefulness is an expected-value criterion rather than a demand for global consistency. In the same vein, a heuristic that increases success on a target regime while slightly degrading performance on rare corner cases can still qualify as useful—and such regime-dependence will later be important when we discuss synergy, since cross-type information flow often acts as a mechanism for shifting the system toward the regimes where its individual methods are strongest.*

15.5.2 Explanations, weakness, and contrivance cost

We measure the "simplicity" or "generality" of an explanatory construct by a scalar weakness. This is a specialization of the quantale weakness ideas from Section 3.7. Intuitively, "weak" explanations are permissive (they rule out fewer possibilities), while "strong" ones are more committal (they encode many fine distinctions); contrivance cost is then the currency in which committal structure is paid for.

Definition 206 (Explanations and weakness)**.** *Fix a behavior (or dataset) b. Let $\mathcal{E}(b)$ be a set of candidate* explanations *of b (models, hypotheses, plans, proofs, causal graphs, etc.). Assume each explanation $E \in \mathcal{E}(b)$ comes with:*

(a) an adequacy score $\mathrm{Fit}(E) \in [0,1]$ *(how well it predicts/explains/controls b);*

(b) a weakness $\mathrm{Weak}(E) \in (0,1]$ *(how few distinctions it makes; larger is weaker).*

Define the contrivance cost *(description-length proxy)*

$$\mathrm{Con}(E) := -\log \mathrm{Weak}(E) \in [0,\infty).$$

Remark 805. *Weakness is a formal proxy for a familiar epistemic virtue: an explanation that makes fewer distinctions is, other things equal, more general and less brittle. In the Hyperseed development, weakness is not mere vagueness; it is a structured notion built to live in quantale settings where composition and partial order matter (Hyperseed-Concept 202; Hyperseed-Concept* **??***; [3, 2]). At a purely scalar level, one can read* $\mathrm{Weak}(E)$ *as measuring "how large a set of worlds" (or latent configurations) remain compatible with E; then larger* $\mathrm{Weak}(E)$ *corresponds to a coarser partition of possibilities, hence fewer encoded distinctions.*

Remark 806. *A simple example: suppose b is a dataset generated by a linear relation plus noise. A linear model may have high* $\mathrm{Weak}(E)$ *(few degrees of freedom, fewer distinctions) and decent* $\mathrm{Fit}(E)$*, while a high-degree polynomial can achieve higher fit but lower weakness (more contrivance). The cost* $\mathrm{Con}(E) = -\log \mathrm{Weak}(E)$ *then behaves like a description length in the spirit of algorithmic information theory [16]. One may also view this as a minimal-description-length style tradeoff: among explanations with comparable adequacy, those with larger weakness (smaller* Con*) are preferred because they are less likely to be overfit to incidental structure in b.*

Remark 807. *The usefulness of* Con *is that it makes the Russellian demand for clarity operational: to explain is to compress, and to compress is to pay for distinctions. The logarithm converts multiplicative weakness comparisons into additive costs, which is exactly the currency in which tradeoffs and budgets are most naturally expressed. Because* $\mathrm{Weak}(E) \in (0, 1]$*, the sign convention ensures* $\mathrm{Con}(E) \geq 0$ *and assigns infinite penalty only to the limiting case of* $\mathrm{Weak}(E) \to 0$*, i.e. to explanations so highly contrived that they are maximally distinction-heavy. The choice of logarithm base merely selects units (e.g. natural nats versus bits with* $\log_2$*), and the additive form is what matters for comparing composite explanations built from multiple sub-structures.*

Remark 808. *It is often helpful to interpret the pair* $(\mathrm{Fit}(E), \mathrm{Con}(E))$ *as defining a Pareto-style frontier: improving fit typically requires paying additional contrivance, while reducing contrivance may reduce fit. In that light, "weakness improvement" refers to moving to an explanation E' with comparable (or higher) adequacy but lower cost, e.g.* $\mathrm{Fit}(E') \geq \mathrm{Fit}(E)$ *and* $\mathrm{Con}(E') < \mathrm{Con}(E)$*. This is the axis along which cross-type interaction will be cast as synergistic: a translation from one representational mode to another can supply constraints or latent variables that let the second mode reach the same adequacy with fewer ad hoc distinctions.*

Remark 809. *The use of* $-\log$ *is optional but convenient: multiplicative weakness becomes additive cost. Any monotone transform would serve similarly.*

Definition 207 (Best achievable contrivance under a fit threshold)**.** *Fix $\tau \in (0, 1]$. For a collection of knowledge types $\mathcal{T}$ (e.g. a subset of the full architecture), let $\mathcal{E}_{\mathcal{T}}(b)$ be the explanations constructible using only those types. Define*

$$\mathrm{Con}^*(\mathcal{T}; \tau) := \inf\{\mathrm{Con}(E) : E \in \mathcal{E}_{\mathcal{T}}(b),\ \mathrm{Fit}(E) \geq \tau\}.$$

Equivalently, $\mathrm{Con}^*(\mathcal{T}; \tau)$ *is the minimal contrivance needed to reach fit τ using the knowledge in $\mathcal{T}$.*

Remark 810. *The quantity* $\mathrm{Con}^*(\mathcal{T}; \tau)$ *is a performance frontier: it asks for the least contrived (most weak/general) explanation that still meets an adequacy demand τ. The infimum allows that the best tradeoff might be approached by a sequence of explanations even if not attained exactly, which is important when the space of explanations is large or continuous.*

Remark 811. *In practice, one may think of τ as a required success rate, prediction accuracy, or control competence. Then* $\mathrm{Con}^*(\mathcal{T}; \tau)$ *is the minimal "model complexity" that the architecture fragment $\mathcal{T}$ must deploy to achieve that competence. This creates a clean interface to later "intelligence under budget" notions (Section 21).*

15.5.3 Synergy as strict subadditivity of contrivance

Definition 208 (Cognitive synergy between knowledge sets)**.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be two disjoint sets of knowledge types and fix $\tau \in (0, 1]$. We say there is* cognitive synergy at level τ *between $\mathcal{T}_1$ and $\mathcal{T}_2$ if*

$$\mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau) \ < \ \mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau).$$

The synergy gain *is the positive part*

$$\Sigma(\mathcal{T}_1, \mathcal{T}_2; \tau) := \Big(\mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau) - \mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau)\Big)_+.$$

Remark 812. *In plain terms: synergy means that letting the two knowledge sets interact yields an explanation meeting the same adequacy threshold with strictly less contrivance than one could achieve by using each set in isolation and then "adding" the results. The positive-part operator $(\cdot)_+$ ensures Σ is always nonnegative, so it behaves like a measurable gain rather than a signed difference.*

Remark 813. *A simple example is a system where declarative knowledge supplies a compact causal rule that lets procedural learning drastically reduce exploration. Separately, each subsystem must encode many exceptions; together, one subsystem supplies a compression that the other can exploit, lowering the overall contrivance needed for fit. This is precisely the kind of architectural phenomenon discussed under the name "cognitive synergy" in [19] (Hyperseed-Concept 73).*

Remark 814. *The definition captures Hyperseed's informal criterion: exchanging intermediate representations is useful when it enables a* less contrived *(weaker, more general) explanation to achieve the same level of adequacy.*

15.5.4 A sufficient condition: monoidal compositionality of weakness

The simplest theorem states that if weakness is compositional across independent subsystems, then combining subsystems cannot be worse (in contrivance cost) than treating them separately.

Definition 209 (Multiplicative weakness model)**.** *Assume that when two explanations E_1, E_2 are combined (e.g. run in parallel, or glued along an interface) to form $E_1 \otimes E_2$, their weakness satisfies*

$$\mathrm{Weak}(E_1 \otimes E_2) \geq \mathrm{Weak}(E_1)\,\mathrm{Weak}(E_2).$$

Remark 815. *This condition says that composing explanations does not force additional distinctions beyond those already made by the parts. Since weakness is larger when fewer distinctions are made, the inequality* $\mathrm{Weak}(E_1 \otimes E_2) \geq \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$ *asserts that the composed system is at least as "weak" as the multiplicative baseline. This is a natural monoidal-style assumption consistent with the quantale flavor of weakness developed earlier [3] (Hyperseed-Concept* **??***).*

Remark 816. *A concrete example is when E_1 and E_2 explain disjoint aspects of the data b and are glued without sharing parameters. Then the combined description length is roughly additive, corresponding to multiplicative weakness. The reason to formalize this is not because independence always holds, but because it gives a clean baseline against which emergent (strict) synergy can later be measured.*

Proposition 24 (Subadditivity of contrivance under multiplicative weakness)**.** *If* $\mathrm{Weak}(E_1 \otimes E_2) \geq \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$ *for all composable E_1, E_2, then for all disjoint knowledge sets $\mathcal{T}_1, \mathcal{T}_2$ and all τ,*

$$\mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau) \;\leq\; \mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau).$$

In particular, the synergy gain $\Sigma(\mathcal{T}_1, \mathcal{T}_2; \tau)$ is well-defined and nonnegative.

Remark 817. *The proposition says that, under the multiplicative weakness model, there is never a* penalty *(in minimal contrivance) for allowing two disjoint knowledge sets to be used together. Even if they do not yield strict synergy, they can at least be composed without worsening the best achievable tradeoff.*

Remark 818. *This result is important mainly as a sanity check: it ensures that the synergy gain defined earlier behaves as a nonnegative quantity under a natural compositionality assumption. It thereby connects the architectural idea of cross-type interaction to the algebraic idea that weakness behaves well under composition (a theme running through the weakness/pattern layers of the document). In particular, the inequality can be read as saying that "having access to more explanation types" cannot make the best achievable contrivance frontier worse, provided that the architecture admits a way of juxtaposing explanations without destroying their adequacy. This is the minimal monotonicity one expects from any notion of joint modeling capacity: if it were violated, then the definitions of* Weak, Con, *and the admissible explanation classes would be misaligned with the intended semantics of "adding representational degrees of freedom."*

Remark 819. *The link to later sections is that attention (as a gating mechanism) determines whether compositions like $E_1 \otimes E_2$ are actually realized in time. The inequality here is about representational possibility; Section 15.6 is about resource-mediated actuality. Said differently, Proposition 24 is "offline": it assumes the agent is permitted to form the composite explanation and then measures what cost is achievable in principle. Attention introduces an "online" constraint: even if E_1 and E_2 exist and can be composed, the system may fail to instantiate $E_1 \otimes E_2$ because the gating policy does not co-activate the relevant pathways, or because partial activation yields an effectively degraded fit that falls below τ.*

Proof. Let $\varepsilon > 0$. Choose explanations $E_1 \in \mathcal{E}_{\mathcal{T}_1}(b)$ and $E_2 \in \mathcal{E}_{\mathcal{T}_2}(b)$ with $\mathrm{Fit}(E_i) \geq \tau$ and $\mathrm{Con}(E_i) \leq \mathrm{Con}^*(\mathcal{T}_i;\tau) + \varepsilon$. (Here we use that $\mathrm{Con}^*(\mathcal{T}_i;\tau)$ is an infimum: for any $\varepsilon > 0$ there exists an admissible explanation whose contrivance lies within ε of the optimum.) Combine them to form $E := E_1 \otimes E_2$. Assuming the combination preserves fit at level τ (e.g. the two parts address disjoint aspects of b), we have $E \in \mathcal{E}_{\mathcal{T}_1\cup\mathcal{T}_2}(b)$. The "disjoint aspects" condition is one sufficient way to guarantee that the joint explanation does not introduce interference; more generally, any architectural condition ensuring that the composite retains $\mathrm{Fit} \geq \tau$ is enough for the argument. Then

$$\begin{aligned} \mathrm{Con}(E) &= -\log \mathrm{Weak}(E) \leq -\log(\mathrm{Weak}(E_1)\mathrm{Weak}(E_2)) \\ &= \mathrm{Con}(E_1) + \mathrm{Con}(E_2) \leq \mathrm{Con}^*(\mathcal{T}_1;\tau) + \mathrm{Con}^*(\mathcal{T}_2;\tau) + 2\varepsilon. \end{aligned}$$

The first inequality is exactly where the multiplicative weakness hypothesis enters; it is also where one implicitly uses that the relevant weakness quantities lie in $(0,1]$ so that $-\log$ is well-defined and order-reversing in the expected way. Taking the infimum over admissible E and then letting $\varepsilon \to 0$ yields the inequality. Equivalently, the construction shows that for each ε one can exhibit a feasible joint explanation whose cost is within 2ε of the sum of the separate optima, which forces the joint optimum not to exceed that sum in the limit. $\square$

Remark 820. *The proof is a standard ε-approximation argument: one takes nearly optimal explanations from each side, composes them, and then uses the multiplicative weakness hypothesis to show the cost of the composite is no more than the sum of the costs. The passage to the infimum and the limit $\varepsilon \to 0$ then yields the stated inequality for the optimal frontiers. Conceptually, this is the same pattern as "subadditivity" arguments in coding and complexity: once a candidate composite description is written down, optimality can only improve upon it, never worsen relative to that explicit construction.*

Proof sketch. Pick E_1 and E_2 within ε of their respective optima. Form the composite $E = E_1 \otimes E_2$. Multiplicative weakness turns into additive cost because of the $-\log$. This gives an upper bound on $\mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2;\tau)$, and letting $\varepsilon \to 0$ removes slack. One can interpret the "slack" as coming solely from the choice of near-optimal witnesses rather than exact minimizers; no special convexity or continuity assumptions are required beyond the basic ability to approximate an infimum. □

Remark 821. *The key step is exactly why the logarithm was chosen: it converts the structural hypothesis about weakness under composition into the familiar subadditivity inequality for costs. Visually, if one thinks of weakness as a kind of "volume" of indistinction, then composing independent explanations multiplies volumes; contrivance cost is then the negative log-volume, hence additive. This is also the sense in which "independence" (or sufficiently weak interaction) is the baseline case: when the two explanations do not share compressive structure, one expects multiplication of volumes and hence mere additivity of costs, leaving no room for strict gains.*

15.5.5 When does synergy become *strict*?

Proposition 24 only guarantees "no penalty" for combining knowledge types. Hyperseed's intended phenomenon is stronger: *strict* improvement, driven by cross-type emergent patterns (Section 9). The strict case corresponds to the presence of shared structure that is representable once in the joint space but would have to be redundantly encoded (or simulated) in each type-specific space if one enforced modularity.

Definition 210 (Cross-type emergent compression)**.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be disjoint knowledge sets. We say there is a* cross-type emergent compression pattern *for b if there exists an explanation $E \in \mathcal{E}_{\mathcal{T}_1 \cup \mathcal{T}_2}(b)$ with $\mathrm{Fit}(E) \geq \tau$ such that for every decomposition $E \approx E_1 \otimes E_2$ with $E_i \in \mathcal{E}_{\mathcal{T}_i}(b)$ and $\mathrm{Fit}(E_i) \geq \tau$, one has*

$$\mathrm{Con}(E) < \mathrm{Con}(E_1) + \mathrm{Con}(E_2).$$

In this formulation, the strict inequality is uniform over all admissible factorizations: it is not enough that some *particular split is inefficient; rather,* every *attempt to realize the same explanatory content via separate type-restricted components incurs additional contrivance.*

Remark 822. *This definition formalizes the intuitive notion of emergence as* irreducible compression*: the joint explanation achieves a description-length advantage that cannot be recovered by any attempt to factor it into separate explanations for the two subsystems. The approximate equality $E \approx E_1 \otimes E_2$ is intentionally informal: it stands for any decomposition notion appropriate to the architecture (e.g. separable parameterizations, modular pipelines, or independent subgraphs). The intent is that $E \approx E_1 \otimes E_2$ should range over all "reasonable" ways of enforcing a two-part modular interpretation; the strict inequality then says that the best such modular account still fails to match the compression achieved by a genuinely cross-type representation.*

Remark 823. *A simple example is when a shared latent variable simultaneously explains sensory regularities and guides procedural choices; representing it once in a cross-type structure is less contrived than duplicating its effect separately in each module. In pattern language, the latent variable is an emergent pattern spanning multiple representational strata [5] (Hyperseed-Concept* **??***). One can also view this as a coordination advantage: a single latent factor couples prediction and control constraints, so that the same degrees of freedom satisfy both simultaneously, whereas separate modules must "rediscover" compatible structure through additional parameters or ad hoc interface conventions.*

Remark 824. *The usefulness of naming emergent compression explicitly is that it makes "synergy" empirically testable in principle: one can compare minimal contrivance frontiers under enforced modularization versus allowed interaction, and ask whether a strict gap appears. Operationally, this suggests experimental protocols where the architecture is trained under different constraints* *(e.g. blocked cross-attention, frozen communication channels, or enforced factorizations) and the resulting best-achievable* Con* *curves are compared at matched fit thresholds.*

Proposition 25 (Emergent compression implies positive synergy)**.** *If there exists a cross-type emergent compression pattern for b at level τ, then $\Sigma(\mathcal{T}_1, \mathcal{T}_2; \tau) > 0$.*

Remark 825. *The proposition says that the emergent compression condition is not merely poetic: it entails a strict inequality at the level of the optimal contrivance frontiers. Thus, emergence (as defined) is a sufficient condition for positive synergy gain. In other words, an explicit witness E whose cost beats every separable decomposition forces the joint optimum to beat the sum of separate optima by at least some nonzero margin (potentially small, but strictly positive in the sense of the inequality).*

Remark 826. *This connects tightly to the definition of Σ: the synergy gain is exactly the amount by which the joint optimum beats the sum of the separate optima. Emergent compression provides a witness E that beats* every *separable decomposition, hence it beats the optimal separable costs in particular. Concretely, if one denotes by C_{sep} the infimum of* $\mathrm{Con}(E_1)+\mathrm{Con}(E_2)$ *over all admissible (E_1, E_2) with* Fit $\geq \tau$*, then emergent compression asserts* $\mathrm{Con}(E) < C_{\text{sep}}$*, while Proposition 24 ensures the joint optimum is at most C_{sep}; together these force a strict improvement and hence $\Sigma > 0$.*

Proof. Let E witness emergent compression. By definition, $\mathrm{Con}(E) < \mathrm{Con}(E_1) + \mathrm{Con}(E_2)$ for any admissible decomposition. Taking infima over admissible E_1 and E_2 implies $\mathrm{Con}(E) < \mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau)$. Since $\mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau) \leq \mathrm{Con}(E)$, we obtain strict subadditivity and hence a positive synergy gain.

To unpack the two inequality steps: the emergent witness E is itself a jointly-constructed explanation drawn from the hypothesis class available when $\mathcal{T}_1$ and $\mathcal{T}_2$ are allowed to interact, so it is a feasible candidate in the joint optimization defining $\mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau)$. Meanwhile, each admissible decomposition corresponds to a pair of feasible candidates for the separate optimizations defining $\mathrm{Con}^*(\mathcal{T}_1; \tau)$ and $\mathrm{Con}^*(\mathcal{T}_2; \tau)$, so their infima provide a lower envelope over all separable ways of reaching fit threshold τ. The strictness is crucial: because $\mathrm{Con}(E)$ is strictly below *every* decomposed cost, it is strictly below the best decomposed cost as well. ☐

Remark 827. *The proof is a direct "witness to strictness" argument. The emergent compression hypothesis is already quantified over all decompositions; taking infima simply extracts the best possible decomposed costs and preserves strict inequality. The final step uses that the joint optimum is no worse than any particular joint explanation.*

Equivalently, one can view the argument as comparing two feasible regions: (i) the joint feasible set in which hypotheses may couple degrees of freedom across $\mathcal{T}_1$ and $\mathcal{T}_2$, and (ii) the restricted feasible set in which hypotheses must factor into components sourced separately. Emergent compression asserts that the restricted feasible set cannot realize the low-contrivance point achieved by E. The infimum operation then formalizes "best possible under the restriction" without changing the direction (or strictness) of the comparison.

Proof sketch. Use the emergent E to show it beats every decomposable pair (E_1, E_2). Therefore it beats the best decomposable pair, i.e. the sum of the separate optima. Since the joint optimum is at most the cost of E, the joint optimum is strictly less than the sum, giving $\Sigma > 0$.

In other words, E is a certificate that "interaction helps": once you exhibit a single joint construction whose contrivance lies below the entire separable baseline, you immediately obtain a nonzero gap between the joint optimum and the best separable performance. ☐

Remark 828. *Visually, if one plots the best-achievable contrivance as a function of enforced modularization constraints, emergent compression says the unconstrained curve dips below the sum of the constrained minima. This is a precise sense in which interaction creates a new "path of least contrivance" through hypothesis space.*

The same picture can be read as a "regularization relief" effect: the constraint that hypotheses must split across representational types can force each side to pay separate description length for structure that, in a coupled representation, is shared or reused. The dip corresponds to a regime where shared latent structure is expressible only when cross-type coupling is allowed.

15.5.6 A criterion for "synergy increases effective intelligence"

This paper later defines intelligence in terms of tasks (Section 21). For present purposes we state a local criterion that connects synergy to solvable task sets under resource limits.

Definition 211 (Budgeted solvability criterion)**.** *Fix a fit threshold τ and a contrivance budget B. A task/dataset b is* solvable by knowledge set $\mathcal{T}$ at budget B *if* $\mathrm{Con}^*(\mathcal{T};\tau) \leq B$.

Note that this definition is intentionally one-sided: it declares solvability whenever there exists some *explanation meeting the fit threshold with contrivance at most B, i.e. it uses the minimal contrivance* Con^* *as a summary of the most efficient available strategy within $\mathcal{T}$.*

Remark 829. *This is the simplest possible "resource-bounded competence" definition: competence is meeting fit τ, and resource use is contrivance cost. The budget B should be understood as an abstract proxy for the system's available representational and computational resources, mirroring the genenergy budgeting earlier in the section (Hyperseed-Concept* **??***).*

In particular, B can be interpreted as fixing a "feasible complexity band" of hypotheses: if the system's architecture or training regime can only realize hypotheses up to contrivance B, then any task whose best achievable contrivance exceeds B is effectively out of reach, even if it is learnable with unbounded resources.

Remark 830. *A concrete reading is: if B is small, only very weak (highly general, low-description-length) explanations are permitted. If the task requires a very contrived explanation, then it is not solvable at that budget even if solvable in principle. This provides a clean interface between architectural synergy and "effective intelligence" under scarcity [9].*

This criterion also makes the "capabilities under scaling" story explicit: increasing B expands the solvable set monotonically, but architectural synergy can shift the required budget for a given task downward by reducing Con^**. In that sense, synergy acts like an efficiency gain in how the system allocates limited representational capacity to meet the same fit level.*

Proposition 26 (Synergy expands the solvable set relative to non-synergistic partitions)**.** *Let $\mathcal{T}_1, \mathcal{T}_2$ be disjoint knowledge sets and suppose $\Sigma(\mathcal{T}_1, \mathcal{T}_2; \tau) > 0$ for a given task/dataset b. Define the "partition-limited" contrivance level*

$$\mathrm{Con}^{\mathrm{part}}(\mathcal{T}_1, \mathcal{T}_2; \tau) := \mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau),$$

which corresponds to architectures in which the two knowledge sets may each build an explanation, but only in a separable (non-emergent) way. Then there exists a budget B such that b is solvable by $\mathcal{T}_1 \cup \mathcal{T}_2$ at budget B, but is not solvable by any separable partition-limited architecture at budget B.

The disjointness assumption is meant to rule out double-counting the same representational resources: if $\mathcal{T}_1$ and $\mathcal{T}_2$ overlapped heavily, then the sum $\mathrm{Con}^*(\mathcal{T}_1; \tau) + \mathrm{Con}^*(\mathcal{T}_2; \tau)$ could overstate the true cost of a separable design by charging twice for shared structure. Under disjointness, the comparison to a strictly synergistic joint solution is cleanly attributable to cross-type interaction rather than to bookkeeping artifacts.

Remark 831. *The statement translates synergy into an operational advantage: there is a range of budgets for which the synergistic architecture succeeds while any architecture forced to keep the two knowledge sets separable must fail. This is exactly the kind of "capability gap" one expects when intermediate results can be exchanged across representational types rather than only combined at the end.*

Concretely, a separable architecture must "pay" separately for whatever structure is required to reach fit τ from within each knowledge set, whereas a synergistic one may express a single joint hypothesis whose parts co-adapt and share explanatory burden. The proposition isolates this advantage as a strict separation in feasible budgets, not merely as a mild reduction in cost at high budgets.

Remark 832. *The result connects back to the earlier subadditivity propositions: if synergy is strict, then the joint contrivance frontier lies below the separable one. The proposition simply chooses a budget in the strict gap. This is a small but principled instance of a more general tradeoff theme in the Hyperseed line of work: architectural constraints translate into provable capability limits under budgets [9].*

One can also read the result as a robustness statement about evaluation under resource limits: if a benchmark is run at a fixed budget B lying in the gap, then observed success/failure becomes diagnostic of whether the system is exploiting cross-type coupling, rather than merely having a larger raw budget.

Proof. Let $C_{12} := \mathrm{Con}^*(\mathcal{T}_1 \cup \mathcal{T}_2; \tau)$ and $C_{\text{part}} := \mathrm{Con}^{\text{part}}(\mathcal{T}_1, \mathcal{T}_2; \tau)$. Positive synergy means $C_{12} < C_{\text{part}}$. Choose any budget B satisfying $C_{12} \le B < C_{\text{part}}$. Then b is solvable by $\mathcal{T}_1 \cup \mathcal{T}_2$ at budget B by definition. On the other hand, any architecture restricted to separable explanations built independently from $\mathcal{T}_1$ and $\mathcal{T}_2$ requires contrivance at least C_{part}, hence cannot meet the same fit threshold within budget B.

Existence of such a B follows immediately from strict inequality: there is at least one real value between C_{12} and C_{part} (for example, $B = \frac{1}{2}(C_{12} + C_{\text{part}})$ when both quantities are finite). If Con^* is defined via an infimum rather than a minimum, the use of "$\le B$" in the solvability criterion is important: even if the optimal contrivance is not attained, any B above the infimum still permits arbitrarily close feasible solutions, while the strict lower bound $B < C_{\text{part}}$ continues to exclude all separable architectures at that budget. ☐

Remark 833. *The proof is essentially interval arithmetic: strict inequality $C_{12} < C_{\text{part}}$ implies there exists a nonempty open interval of budgets between them. Picking any B in that interval creates a regime where joint explanations are affordable but separable ones are not.*

This also clarifies why the result is framed as an existence *statement: it does not claim that every budget benefits from synergy, only that whenever there is a strict cost gap, one can select a budget that lands inside that gap and thereby forces an observable difference in solvability.*

Proof sketch. Let the synergistic optimum cost be C_{12} and the best separable cost be C_{part}. Since $C_{12} < C_{\text{part}}$, choose B between them. Then the synergistic system meets the budget while every separable system exceeds it.

In the language of decision thresholds, B is chosen so that it accepts the joint model class but rejects the separable model class, turning a quantitative synergy gain into a qualitative difference in whether the task can be solved under the imposed resource constraint. ☐

Remark 834. *Geometrically, one can picture two thresholds on the number line: synergy moves the required budget leftward. One threshold corresponds to the minimum budget at which the task becomes solvable when types are allowed to interact emergently, while the other corresponds to the* *minimum budget when the types are forced to contribute in isolation. In this picture, "leftward" does not mean that the task becomes easier in an absolute sense; it means the architecture makes better use of the same scarce computational currency by enabling multi-type regularities to be captured earlier. The "intelligence increase" is then not mystical; it is the existence of tasks that cross from impossible to possible when the architecture permits cross-type emergent compression patterns (Hyperseed-Concept* **??** *provides a related intuition about gains from moving between representational levels). In particular, the shift can be interpreted as a reduction in contrivance: fewer ad hoc special cases are required because structure discovered in one type can constrain search or learning in another.*

Remark 835. *Proposition 26 isolates a simple operational implication: synergy can make a task solvable under a budget that would be insufficient for any architecture that forces the two knowledge sets to contribute only in a separable (non-emergent) way. This is one clean sense in which synergy can increase effective intelligence. Equivalently, if one thinks of a system designer choosing an architecture under a fixed resource envelope, synergy expands the set of tasks that can be completed without increasing the envelope. The "separable" constraint is important here: it rules out the possibility that partial results in one type reshape the hypothesis space, loss landscape, or action space explored by the other type, and thus rules out precisely the kind of cross-type compression that yields strict subadditivity.*

15.6 Attention as the gating mechanism for synergy

Cognitive synergy requires frequent exchange of intermediate results. In bounded systems, such exchange competes for attention. Thus attention is the architectural "gate" that decides whether potential synergy is realized. One can read this as an opportunity-cost claim: every unit of attention devoted to translation and cross-type integration is a unit not devoted to within-type refinement, and so the system must continuously trade off short-horizon progress against longer-horizon integrative gains. Conversely, if attention never "pays" for communication, then even a richly multimodal or multi-algorithmic architecture behaves like a set of loosely coupled modules with minimal mutual constraint.

Definition 212 (Inter-type communication graph)**.** *Let $\mathcal{T}$ be a set of knowledge types. Define a directed graph Γ on $\mathcal{T}$ where an edge $T \to T'$ indicates that items in K_T can be transformed into inputs for $\mathsf{Alg}(T')$. Assign each edge a time-dependent bandwidth allocation $b_t(T \to T') \geq 0$ with $\sum_{T,T'} b_t(T \to T') \leq B_t$. We interpret b_t as the fraction of the attention budget devoted to cross-type communication. In general b_t may be sparse (concentrated on a few edges) or diffuse (spread over many edges), and its temporal variation encodes when the system chooses to integrate versus when it chooses to specialize. If desired, one can also interpret Γ as including only genuinely cross-type edges (within-type processing handled separately), though allowing $T = T'$ edges can be convenient when one wants a uniform accounting of all attentional expenditures in a single graph formalism.*

Remark 836. *The graph Γ is a wiring diagram for cognitive synergy: it says which translations are available. The scalar $b_t(T \to T')$ then says which translations are being actively exercised at time t, under the same budget logic as earlier attention policies. The constraint $\sum_{T,T'} b_t(T \to T') \leq B_t$ treats cross-type communication as something that consumes the same scarce currency as within-type updates. It also makes explicit that "having a translator" is distinct from "running the translator often enough": the former is a structural capability (an edge exists), while the latter is an attentional commitment (a nontrivial b_t over time). In particular, a system can possess many potential cross-type edges but still behave effectively unimodally if the policy allocates negligible bandwidth to them.*

Remark 837. *A simple example is a pipeline* $T = \mathrm{sens} \to T' = \mathrm{dec}$*, where sensory embeddings are converted into declarative propositions, or* $T = \mathrm{dec} \to T' = \mathrm{proc}$*, where declarative constraints shape action selection. If the corresponding* b_t *values are near zero, the translations exist only as dormant possibilities; if they are substantial, the system is actively weaving together types into joint explanations [19, 7]. In realistic cases, the most synergy-rich subgraphs are often not simple pipelines but feedback loops, e.g.* $\mathrm{sens} \to \mathrm{dec} \to \mathrm{proc} \to \mathrm{sens}$*, where actions select new observations, observations revise beliefs, and revised beliefs reshape policies. The directed nature of* Γ *matters here: translation can be easier in one direction than the reverse (for example, procedural skill may be difficult to verbalize), so high synergy may require asymmetric allocations* $b_t(T \to T')$ *that reflect these representational asymmetries.*

Remark 838. *This makes precise the intuitive claim that synergy is not merely a structural property of an architecture, but a dynamical property of an architecture under attentional governance. A system may have the representational machinery for cross-type emergence, yet fail to realize it because attention is monopolized by short-term demands (Hyperseed-Concept* **??***). A complementary failure mode is oscillation without integration: attention repeatedly switches types but allocates insufficient bandwidth for any translation to become reliable, yielding overhead without commensurate cross-type constraint. On the other hand, sustained attention to translation can bootstrap itself: once intermediate results become easier to exchange (e.g. via learned interfaces, shared latent variables, or cached abstractions), the effective cost of future communication drops, making synergy more likely to persist.*

Remark 839 (Two limiting regimes)**.** *If the cross-type bandwidths are near zero, then even if synergy is latent (in principle), it will not be expressed. If the bandwidths are substantial, the system can form and exploit cross-type emergent compression patterns, supporting the strict inequalities in Section 15.5. Between these extremes lies a practically important intermediate regime: modest bandwidth can be sufficient if the translated artifacts are high-leverage (e.g. a small number of declarative constraints that sharply prune procedural search), while in other domains the same modest bandwidth may be ineffective because useful intermediate products are numerous, noisy, or require iterative back-and-forth refinement. In this sense, the "gate" metaphor should be read quantitatively: the question is not only whether the gate is open, but how wide it is and for how long.*

Hyperseed concepts covered

- Attention; attentional focus; genenergy-based attention measure; self-reflective attention (Hyperseed-Concept 60; Hyperseed-Concept **??**; Hyperseed-Concept **??**).
- Varieties of knowledge: declarative, procedural, sensory, attentional, intentional (Hyperseed-Concept **??**; Hyperseed-Concept 65; Hyperseed-Concept 157).
- Cognitive synergy: usefulness of learning algorithms by knowledge type; synergy as strict subadditivity of contrivance/weakness; attention as the gating mechanism for synergy (Hyperseed-Concept 73; Hyperseed-Concept 202; Hyperseed-Concept **??**).

16 Self, continuity, development, and mind-world correspondence

This section formalizes a cluster of Hyperseed notions that are often treated informally in cognitive science and philosophy of mind: *self/other boundaries*, *diachronic self-continuity*, *development*, and *mind–world correspondence* (world-model quality). Hyperseed's distinctive move is to define these notions in terms of *pattern structure and its persistence*, rather than in terms of a privileged substance or an observer-independent identity relation [1, 5]. In the present reconstruction, "self" is thus not a metaphysical atom but a repeatedly reconstituted cut through a web of relations, and continuity is not an all-or-nothing identity but a degree of structural persistence. This cluster of ideas connects directly to the Hyperseed core-concept set: Self vs. Other (Hyperseed-Concept 165), Self-Continuity (Hyperseed-Concept 166), Development (Hyperseed-Concept 95), and Mind-World Correspondence (Hyperseed-Concept 112).

Throughout, fix an observer/context O and a commutative quantale $(\mathbb{V}, \leq, \otimes, \bigvee, \mathbf{1})$ as in Section 3.

Remark 840 (Notation and conventions used throughout)**.** *Here $\mathbb{V}$ is the domain of "degrees" (truth-like, evidence-like, or strength-like values), $\leq$ is its order (so $u \leq v$ means "u is no stronger than v"), $\otimes$ is the monoidal product (interpretable as conjunction/combination of constraints), and $\bigvee$ is the join (supremum; interpretable as "best possible" or "least upper bound"). We will also use $\bigwedge$ (the infimum/meet) when defining degrees via "for all" constraints. The element $\mathbf{1}$ is the unit for $\otimes$ and typically plays the role of a maximal (or perfectly adequate) degree. Quantales (and their "weakness"-flavored interpretations) are discussed more broadly in [3, 2].*

When we want explicit paraconsistency we take $\mathbb{V}$ to be the p-bit-valued quantale used in the toy model (Section 5). Time is treated as an observer-indexed proto-time ordering (Section 7); we write I, J for time intervals, and we treat a "time slice" as data aggregated over such an interval.

The key representational objects are pattern webs and pattern-flow networks (Sections 9 and 12). In this section we focus on morphisms between such networks: maps that preserve pattern-relational structure up to controlled degradation. This emphasis on morphism and transport is aligned with Hyperseed's broader stance that cognition is best modeled by what can be carried along transformations, rather than by what is statically possessed [7]. Pattern and Pattern Flow Networks are treated as core notions (Hyperseed-Concepts 130 and 131).

To situate the above commitments: fixing an observer/context O is not merely a notational convenience, but a way to make explicit that the relevant pattern decompositions, degrees of fit, and even the effective "time slices" are all relative to an information-gathering and information-organizing standpoint. In particular, what counts as the "same" pattern across two intervals I and J will typically be mediated by representational choices (feature sets, granularity, segmentation) that are part of O; the formalism does not attempt to erase this dependence, but instead makes it explicit and therefore comparable across contexts.

Likewise, the quantale of degrees $(\mathbb{V}, \leq, \otimes, \bigvee, \mathbf{1})$ is meant to carry the minimal algebra needed to talk about *approximate preservation* and *graded failure*. In concrete applications one may instantiate $\mathbb{V}$ as a unit interval $[0, 1]$ with a t-norm, as a Boolean algebra, or as a more information-structured object (e.g. the paraconsistent p-bit choice mentioned above). The point is that the same definitions can be read as (i) graded truth of relational constraints, (ii) graded evidence for relational constraints, or (iii) graded strength of pattern-couplings, provided that $\otimes$ plays the role of "putting constraints together" and that joins/meets provide the appropriate extremal aggregations.

The phrase "controlled degradation" will be made precise by requiring that morphisms between pattern-flow networks preserve relations not necessarily *exactly* but to a specified degree in $\mathbb{V}$. Informally, such morphisms can be read as transports that (a) identify which parts of a network correspond, (b) quantify how much relational structure survives the transport, and (c) localize where structure is lost, distorted, or newly introduced. This is the technical hinge on which the section turns: self/other boundaries, self-continuity, development, and mind–world correspondence will each be cast as particular patterns of existence (or non-existence) of such structure-respecting transports.

With this in mind, the four headline notions can be previewed as follows (with full definitions deferred to the subsequent subsections). A *self/other boundary* will be treated as a context-relative partitioning of a pattern web into subnetworks whose internal transports are comparatively strong and whose cross-boundary transports are comparatively weak (all measured in $\mathbb{V}$). *Diachronic self-continuity* will then be a degree of transportability from the "self"-designated subnetwork at interval I to the corresponding subnetwork at interval J, where the degree tracks which relational constraints remain satisfiable (or approximately so) over time. *Development* will be expressed not as mere accumulation of states but as a structured sequence of network morphisms whose typical effect is to expand or reorganize the space of patterns and flows available to O (for example, by adding new stable motifs or by reweighting existing habit-like transitions). Finally, *mind–world correspondence* will be expressed as the existence and quality of morphisms (or adjoint-like pairs of morphisms) that align an internal pattern-flow network (the "model") with an external or action-embedded pattern-flow network (the "world as encountered"), again with degradation explicitly represented rather than idealized away.

The use of paraconsistent degrees (via p-bit) is especially relevant here because both self-models and world-models are typically inconsistent in the weak sense: they contain partial, conflicting, and context-shifting constraints that remain practically useful. In such cases, mind–world correspondence is not well represented by a single global consistency requirement, but rather by a pattern of locally strong correspondences coexisting with tolerated contradictions—a situation the quantale semantics is designed to encode.

16.1 Self/other boundary as a paraconsistent, observer-relative cut

Hyperseed treats "self" as something *constructed* by a mind: a self is a distinguished subsystem (often hierarchical) carved out of a broader coherence field. A crucial point is that the self/other boundary is not always crisp; it can be fuzzy, context-dependent, and even genuinely inconsistent (one may simultaneously experience something as "me" and "not-me"). This is the formal face of Self vs. Other (Hyperseed-Concept 165) and of Non-Duality motifs (Hyperseed-Concept 121) in an otherwise highly analytic framework; compare the paraconsistent stance motivating such coexistence of opposites [23, 24]. In particular, "observer-relative" here should be read literally: different observers (or the same observer under different modeling stances) can induce different partitions of the same underlying dynamics, because the cut is made in representational space rather than assumed to be a pre-given ontological seam in the world. The time-interval parameter I likewise matters: self/other boundaries are treated as *temporally situated* practices, so that drift, context-switching, and rapid oscillation can be expressed as changes in σ_I across I.

We model this by attaching to each representational item a p-bit degree of self-membership. The use of a two-component degree is intended to capture not only vagueness (weak evidence in either direction) but also conflict (strong evidence in both directions), which is precisely the configuration that classical single-valued membership functions cannot represent without forcing a premature resolution.

Definition 213 (Self-evidence and self-groupings)**.** *Fix a time interval I. Let X_I be the set of representational tokens (entities, situations, pattern-nodes) used by the observer/model O to describe the system under study on I. A* self-evidence function *is a map*

$$\sigma_I : X_I \to [0,1]^2, \qquad \sigma_I(x) = (\sigma_I^+(x), \sigma_I^-(x)),$$

where $\sigma_I^+(x)$ is the degree of evidence that "x is part of the self" and $\sigma_I^-(x)$ is the degree of evidence that "x is not part of the self." No consistency constraint is imposed. Equivalently, σ_I may be read as a graded, paraconsistent labeling of tokens by the observer O: the two coordinates are not complements, and their joint values are informative precisely when they fail to sum to 1.

Given thresholds $\theta^+, \theta^- \in [0,1]$, the corresponding crisp self-grouping *(induced by σ_I) is*

$$\mathrm{Self}_\theta(I) := \{x \in X_I : \sigma_I^+(x) \geq \theta^+ \text{ and } \sigma_I^-(x) \leq \theta^-\},$$

where $\theta = (\theta^+, \theta^-)$. The complementary crisp other-grouping *is $\mathrm{Other}_\theta(I) := X_I \setminus \mathrm{Self}_\theta(I)$. The particular choice of θ is best understood as a downstream modeling decision (e.g. needed to define an agent boundary for credit assignment, control, or responsibility), rather than as an intrinsic parameter of the phenomenon.*

Remark 841 (Threshold monotonicity and decision-sensitivity)**.** *For fixed σ_I, the map $\theta \mapsto \mathrm{Self}_\theta(I)$ behaves monotonically in the expected directions: increasing θ^+ (demanding stronger positive evidence) can only shrink $\mathrm{Self}_\theta(I)$, while decreasing θ^- (demanding weaker negative evidence) can only shrink $\mathrm{Self}_\theta(I)$. Thus, varying θ interpolates between permissive and conservative self-ascriptions without altering the underlying evidence profile encoded by σ_I. This makes explicit a methodological separation between (i) evidence accumulation about self-relevance and (ii) the practical need to make a binary cut for a particular analytic task.*

Remark 842 (Intuition and examples for self-evidence)**.** *Intuitively, $\sigma_I(x)$ is not a statement of fact about an object x but a statement about O's* boundary-making practice *on interval I. The pair $(\sigma_I^+(x), \sigma_I^-(x))$ records that evidence can point both ways at once: this is the mathematical counterpart of the phenomenological observation that some contents are experienced as "mine" and "not mine" in different respects (or even simultaneously). One can also view $(\sigma_I^+(x), \sigma_I^-(x))$ as distinguishing four qualitative regimes in a graded way: predominantly-self (high $+$, low $-$), predominantly-other (low $+$, high $-$), mixed or conflicted (high $+$, high $-$), and unassigned/irrelevant (low $+$, low $-$). The last regime is important in practice because many tokens in X_I may simply not participate in the self-model at all, even though they are present in the observer's overall description of the situation.*

A simple example is bodily sensation: let $x =$ "pain in the arm." One may have high $\sigma_I^+(x)$ because the pain is experienced as happening "to me," while also having nontrivial $\sigma_I^-(x)$ because the pain is experienced as an intrusive, alien event. Another example is a tool in skilled use (e.g. a tennis racket): during fluent action one may assign it substantial positive self-evidence, while still retaining negative evidence because it is not literally part of the body. The usefulness of this definition is that it turns boundary vagueness into a manipulable object: thresholds θ yield crisp sets when needed for downstream constructions, without erasing the underlying ambiguity. A further class of examples arises in social and extended-cognition settings: a group "we" token, a conversational role, or an institutional identity can carry high σ_I^+ during coordination (the group-agent is treated as self-like) while simultaneously carrying high σ_I^- because the same observer recognizes non-identity at the individual level; the model records this tension rather than forcing an all-or-nothing verdict.

Remark 843 (Boundary and non-duality)**.** *The* boundary region *between self and other is naturally modeled as*

$$\partial\mathrm{Self}_\theta(I) := \{x \in X_I : \sigma_I^+(x) \geq \theta^+ \text{ and } \sigma_I^-(x) \geq \theta^-\}.$$

Elements in $\partial\mathrm{Self}_\theta(I)$ carry substantial evidence in both directions. This is a simple formal representation of the Hyperseed motif that experienced categories can be simultaneously distinct and *not-distinct. Note that $\partial\mathrm{Self}_\theta(I)$ is not a topological boundary but a decision-relative "zone of conflict": it depends on θ, and it expresses the coexistence of incompatible self-ascriptions at the level of representation rather than an ambiguity about where matter ends and begins.*

Remark 844 (Further intuition: why model a boundary region?)**.** *The set $\partial\mathrm{Self}_\theta(I)$ is where explanatory work tends to be needed: it marks tokens whose role in agency, responsibility, and prediction is inherently unstable. In Russellian terms, it allows us to speak precisely about cases where the "class"* Self *fails to be sharply defined, without thereby abandoning logical discipline. In Peircean terms, it acknowledges that the sign-process by which the self is constituted may place the same token under multiple interpretants (Hyperseed-Concepts 190 and 157); and in Whiteheadian terms it treats the self-boundary as a process-dependent abstraction rather than a substance boundary [14, 15]. Operationally, boundary-region tokens are also the ones for which small shifts in attention, affect, or narrative framing can lead to large shifts in downstream attributions (e.g. "I did that" versus "it happened to me"), which is precisely the kind of sensitivity that motivates keeping σ_I^+ and σ_I^- separate rather than forcing a single net score.*

Remark 845 (Multiple selves and hierarchical self-models)**.** *In many systems it is natural to consider multiple nested "selves" (e.g. bodily self, actional self, narrative self, social self). Formally, one may use a family of self-evidence functions $\sigma_I^{(k)} : X_I \to [0,1]^2$ indexed by a level k (or by a context label). Nestedness can be encoded either by pointwise inequalities $\sigma_I^{(k+1),+}(x) \le \sigma_I^{(k),+}(x)$ (stricter self at higher k) or by explicit inclusion of crisp groupings $\mathrm{Self}_\theta^{(k+1)}(I) \subseteq \mathrm{Self}_\theta^{(k)}(I)$. Conceptually, this accommodates the possibility that "what counts as me" is not a single flat set but a stratified construction: some tokens are self-like only at certain descriptive levels, and conflicts can occur not only within a level (high $+$ and high $-$) but also between levels (e.g. bodily self-evidence positive while narrative self-evidence negative).*

Remark 846 (Example: nested selves)**.** *As a concrete illustration, let $k = 0$ denote "bodily self" and $k = 1$ denote "narrative self." A heartbeat token may have high $\sigma_I^{(0),+}$ (it is bodily-me) but low $\sigma_I^{(1),+}$ (it is not typically part of the narrative-me), while a biographical memory may show the reverse. The nestedness conditions formalize the idea that higher-level selves are often more selective: they include fewer tokens, but possibly with greater stability across time. This multi-level framing is also compatible with phenomenological taxonomies that sort experience into "locations" or "modes" of selfhood [4, 12]. One can also treat social roles (e.g. "teacher in the classroom") as a higher-level self that activates only on certain intervals I, making $\sigma_I^{(k)}$ an explicitly context-gated construction; in such cases, the same underlying token (say, a verbal utterance) may alternate between being ascribed to the role-self and being experienced as alien or forced, naturally landing in $\partial\mathrm{Self}_\theta(I)$ during periods of role strain.*

Where does σ_I come from? Hyperseed motivates selfhood via *persistence of patterns over time* and via *coherence* (a mind often draws a boundary separating a mutually coherent group from other entities). In this paper we treat σ_I as derived data, typically computed from: (i) persistence statistics across time (Section 16.3); (ii) coherence/interaction weights within the pattern web (Section 9); and (iii) control/agency signals (Sections 14 and 15). The formal point is that none of these drivers need be consistent.

Remark 847 (Why self-evidence is placed "upstream" of identity)**.** *A tempting move in philosophy is to posit an identity relation and then to explain experience by reference to that identity. Hyperseed inverts this: boundary-making and persistence are primary, and "identity" is (at best) an* *emergent, context-sensitive convenience (Hyperseed-Concepts* **??** *and* **??***). This inversion matters in engineering terms as well: in adaptive systems, the representational vocabulary and the modeled objects can both change, so insisting on fixed identity would force one either into brittle bookkeeping or into ignoring precisely the developmental dynamics one wants to describe [19, 5].*

16.2 Approximate morphisms of quantale-weighted networks

Self-continuity and mind–world correspondence are both defined by Hyperseed using "approximate morphisms" between *pattern-flow networks*. We now make this precise in a form that is compatible with enriched-category theory.

Definition 214 (Quantale-weighted directed graphs)**.** *A* $\mathbb{V}$-weighted directed graph *(briefly, a* $\mathbb{V}$-graph*) is a pair* $G = (X, G)$ *where* X *is a set of nodes and*

$$G : X \times X \to \mathbb{V}$$

assigns a weight to each ordered pair of nodes. We write $G(x, y)$ *for the weight on the directed edge* $x \to y$*.*

Remark 848 (Intuition and examples for $\mathbb{V}$-graphs)**.** *A* $\mathbb{V}$*-graph is the simplest object that can carry graded relational structure. If* $\mathbb{V} = [0, 1]$ *with* $\leq$ *the usual order and* $\otimes$ *interpreted as multiplication or* min*, then* $G(x, y)$ *can be read as the strength of an influence or a similarity from* x *to* y*. If* $\mathbb{V} = [0, 1]^2$ *(a* p-bit *domain), then* $G(x, y)$ *can record positive and negative evidence for that influence simultaneously, echoing the earlier paraconsistent attitude [23, 24].*

This definition is useful because both "mind" and "world" can be represented as such graphs: nodes stand for tokens in the representational scheme, and edges for pattern-conditioned flow, predictive linkage, or habitual propagation (Hyperseed-Concepts 132 and 131). The key point is that we do not assume the node sets match across time or across domains; instead we will compare graphs by maps between node sets.

Definition 215 (Residuation)**.** *Because* $\mathbb{V}$ *is a quantale, the map* $u \otimes (-)$ *preserves arbitrary joins and hence has a right adjoint. Define the* residuum *(implication) by*

$$u \to v := \bigvee \{w \in \mathbb{V} : u \otimes w \leq v\}.$$

Equivalently, for all u, v, w *one has*

$$u \otimes w \leq v \quad \textit{iff} \quad w \leq (u \to v).$$

Remark 849 (Notation and intuition for the residuum)**.** *The symbol* $\to$ *here is not material implication from classical logic; it is the* residuated implication *determined by* $\otimes$*. The defining equivalence*

$$u \otimes w \leq v \iff w \leq (u \to v)$$

says that $(u \to v)$ *is the* largest *degree* w *such that combining* u *with* w *does not exceed* v*. One may read it as: "given that* u *holds to some degree, how much slack is available to reach* v*?" In quantitative settings it behaves like a best-possible conditional bound; in paraconsistent settings it functions as a controlled, non-explosive form of implication, which is precisely why residuation is valuable in the Hyperseed stack [3].*

Remark 850 (Example computations). *If $\mathbb{V} = [0,1]$ and $\otimes = \min$, then*

$$u \to v = \begin{cases} 1 & \text{if } u \le v, \\ v & \text{if } u > v, \end{cases}$$

so the implication is "perfect" when u is already no stronger than v, and otherwise degrades to v. If instead $\otimes$ is multiplication, then $u \to v$ behaves like a truncated ratio (v/u capped at 1), again matching the idea of a maximal compensating factor. These examples illustrate why residuation is the right tool for measuring how well a map preserves edges: it compares source edge-strength with target edge-strength in a way consistent with the quantale's native combination operator.

Definition 216 (Degree of a structure-preserving map). *Let $G = (X, G)$ and $H = (Y, H)$ be $\mathbb{V}$-graphs and let $f : X \to Y$ be a function. Define the* preservation degree *of f by*

$$\deg(f; G, H) := \bigwedge_{x,x' \in X} \big(G(x,x') \to H(f(x), f(x'))\big).$$

Equivalently, $q \le \deg(f; G, H)$ if and only if

$$G(x,x') \otimes q \le H(f(x), f(x')) \qquad \text{for all } x, x' \in X.$$

*We say that f is a q-*approximate morphism *$G \to H$ if $q \le \deg(f; G, H)$.*

Remark 851 (Intuition for $\deg(f; G, H)$ and why the meet appears). *The degree $\deg(f; G, H)$ measures the worst-case adequacy of f over all edges in G. The meet $\bigwedge_{x,x'}$ enforces a universal constraint: the map is only as good as its weakest-preserved relational link. The equivalent condition*

$$G(x,x') \otimes q \le H(f(x), f(x'))$$

should be read as: if an edge in G has strength $G(x,x')$, then after discounting by q it must still fit under the corresponding edge in H. Thus q quantifies a uniform level of tolerated degradation.

A simple example: if G and H encode similarity graphs, and f maps internal concepts to external referents, then q bounds how much similarity structure is lost under interpretation. This is useful because it yields a single composable number capturing the fidelity of a whole map, which will allow us to speak rigorously about continuity and correspondence without assuming any hard identity of tokens.

Remark 852 (Relation to enriched functors). *If G and H are actually* $\mathbb{V}$-categories *(reflexive and transitive in the $\mathbb{V}$-enriched sense), then q-approximate morphisms with $q = \mathbf{1}$ are precisely $\mathbb{V}$-enriched functors. The definition above is intentionally more general: it applies to raw weighted graphs and can be combined with a closure operation (e.g. reflexive-transitive closure) when one wants a genuine enriched categorical semantics.*

Remark 853 (Why we avoid requiring categorical axioms here). *Requiring reflexivity/transitivity at the outset would build in a strong coherence assumption that many empirical pattern webs do not satisfy before closure or smoothing. Hyperseed repeatedly emphasizes that cognition begins in rough, partially contradictory data and only later acquires stable structure via closure processes (compare Hyperseed-Concepts 72 and* **??***). The present choice therefore keeps the interface maximally general while remaining compatible with enriched-category intuitions when the additional axioms hold.*

More concretely: reflexivity and transitivity are precisely the axioms that would let a $\mathbb{V}$-graph be regarded (after suitable interpretation) as the hom-object assignment of a $\mathbb{V}$-enriched category, where *identities and composition are already "baked into" the structure. By postponing these axioms, we allow the weights $G(x, x')$ to be read as raw affinities, similarities, causal supports, or co-occurrence strengths, without presuming that self-links are maximally strong or that indirect support must be mediated by $\otimes$ in a globally consistent way. The closure/smoothing steps can then be understood as the point at which the empirical web is forced into (or approximated by) a more coherent, category-like regime, rather than assuming coherence from the beginning.*

Theorem 14 (Composition law for approximate morphisms)**.** *Let $G = (X, G)$, $H = (Y, H)$, and $K = (Z, K)$ be $\mathbb{V}$-graphs. For functions $f : X \to Y$ and $g : Y \to Z$,*

$$\deg(g \circ f; G, K) \;\geq\; \deg(f; G, H) \otimes \deg(g; H, K).$$

In particular, if f is a q_1-approximate morphism and g is a q_2-approximate morphism, then $g \circ f$ is a $(q_1 \otimes q_2)$-approximate morphism.

It is worth stressing what the direction of the inequality encodes: the composite cannot be worse *than the tensor-combination of the two degradations, but it may be* better *in special cases (for instance, if g collapses distinctions that were responsible for f's worst violations, or if the feasibility constraints become looser after passing through H). Thus the theorem provides a guaranteed lower bound on composite fidelity rather than an exact formula.*

Remark 854 (What the composition law says, and why it matters)**.** *In plain terms: if f preserves the structure of G inside H up to degradation q_1, and g preserves the structure of H inside K up to degradation q_2, then the composite $g \circ f$ preserves the structure of G inside K up to the combined degradation $q_1 \otimes q_2$. This is the algebraic statement that "approximate structure preservation composes."*

The result is crucial for everything that follows: self-continuity will be defined by chaining together approximate morphisms across successive time-slices, and mind–world correspondence will be used inside control and prediction loops (Section 14). Without a composition law, these degrees would not behave like a stable currency of fidelity across multiple steps.

Two limiting cases help anchor the interpretation. First, when $\deg(f; G, H)$ is the monoidal unit (of $\mathbb{V}$) so that f is "exact" in the sense of not requiring any weakening, the theorem reduces to the familiar closure of exact structure preservation under composition. Second, in common numerical quantales (e.g. $\mathbb{V} = [0, 1]$ with $\otimes$ as multiplication or a t-norm), the combination $q_1 \otimes q_2$ matches the intuition that independent degradations multiply, so long chains decay in a controlled, algebraically predictable way rather than in an ad hoc manner.

Proof. Let $q_1 := \deg(f; G, H)$ and $q_2 := \deg(g; H, K)$. By definition of deg, for all $x, x' \in X$,

$$G(x, x') \otimes q_1 \leq H(f(x), f(x'))$$

and for all $y, y' \in Y$,

$$H(y, y') \otimes q_2 \leq K(g(y), g(y')).$$

Apply the second inequality with $y = f(x)$ and $y' = f(x')$ and compose:

$$G(x, x') \otimes (q_1 \otimes q_2) = (G(x, x') \otimes q_1) \otimes q_2 \leq H(f(x), f(x')) \otimes q_2 \leq K(g(f(x)), g(f(x'))).$$

Thus $(q_1 \otimes q_2)$ satisfies the defining feasibility condition for $\deg(g \circ f; G, K)$, so $\deg(g \circ f; G, K) \geq q_1 \otimes q_2$.

Note that the proof uses only the basic quantale/enrichment primitives already in play: the monotonicity of $\otimes$ in each argument (so inequalities can be tensored on the right), and associativity (so "weakening twice" is unambiguous). No reflexivity or transitivity assumptions on G, H, K are needed, which is consistent with the preceding remark about avoiding categorical axioms at this stage. □

Remark 855 (Proof sketch and geometric intuition). Proof sketch. *Use the equivalent characterization of* deg*: a lower bound q is feasible exactly when every edge in the source, weakened by q, fits under the corresponding edge in the target. Apply this twice (first for f, then for g) and associate the tensor product to conclude feasibility for $g \circ f$ with weakening $q_1 \otimes q_2$.* □

The key step is substituting $y = f(x)$ and $y' = f(x')$ so that the second inequality speaks about exactly the images of the nodes used in the first inequality. Associativity of $\otimes$ then ensures the degradation factors combine cleanly. Visually: think of each morphism as allowing you to "shrink" all edge weights by a factor and still embed them into the next graph; doing this twice multiplies (or otherwise $\otimes$-combines) the shrinkage.

Order-theoretically, the same picture can be read as follows: the set of feasible degradations for a map is downward closed in $\mathbb{V}$ (if q works, any weaker $q' \leq q$ works), and deg *is the best achievable bound among them. The theorem exhibits an explicit feasible element for the composite—namely $q_1 \otimes q_2$—so the degree of the composite must lie* above *that element in the $\mathbb{V}$-order. This viewpoint clarifies why the statement is a $\geq$ bound on degrees even though it arises from chaining $\leq$ constraints on weakened edges.*

16.3 Self-continuity as approximate self-morphism across time

We now instantiate the preceding definitions on time-indexed pattern-flow networks.

Definition 217 (Pattern-flow network). *Fix a system S (as modeled by observer O) and a time interval I. A* pattern-flow network *for S on I is a $\mathbb{V}$-graph*

$$F_{S,I} : X_{S,I} \times X_{S,I} \to \mathbb{V},$$

where $X_{S,I}$ is a set of nodes representing specific entities or pattern-tokens, and where $F_{S,I}(a, b)$ quantifies the extent to which pattern that "first appears" in a subsequently appears in b over the interval I. The precise estimator used to compute $F_{S,I}(a, b)$ is observer-dependent and may include sub-interval weighting (temporal discounting).

Remark 856 (Intuition and examples for pattern-flow networks). *A pattern-flow network is a time-slice summary of "how patterns propagate" inside the modeled system. If nodes are tokens like* `belief_X`, `memory_Y`, `action_schema_Z`, *then $F_{S,I}(a, b)$ measures how reliably activity or content associated with a predicts or feeds into activity/content associated with b during I. This packages the Hyperseed emphasis on Pattern and Process (Hyperseed-Concepts 130 and 140) in a way that is directly comparable across times and across domains.*

For a simple example, let a = "see_cup" and b = "reach_cup". Then a high value $F_{S,I}(a, b)$ indicates that within the interval I the perceptual token "see cup" tends to be followed by (or to support) the action token "reach cup." If $\mathbb{V}$ is paraconsistent, then $F_{S,I}(a, b)$ may simultaneously record evidence for and against the flow, capturing the fact that behavior can be context-fragmented or internally conflicted.

Remark 857 (Why observer-dependence is not a defect). *The estimator for $F_{S,I}$ is allowed to be observer-dependent because "pattern" is already observer-relative in Hyperseed: it depends on the distinctions the observer can draw and the resources it allocates (Sections 9 and 15). Rather than pretending away this relativity, we keep it explicit and then build invariants (like morphism degrees) that remain meaningful under representational change [5, 19].*

Definition 218 (Pattern-flow-network self-continuity)**.** *Let I and J be successive time intervals. The* pattern-flow-network self-continuity *of S from I to J is*

$$\mathrm{SC}_S(I \to J) := \bigvee_{f: X_{S,I} \to X_{S,J}} \deg\big(f; F_{S,I}, F_{S,J}\big).$$

We say that S has θ-self-continuity *from I to J if* $\mathrm{SC}_S(I \to J) \geq \theta$.

Remark 858 (Intuition: continuity as best possible relabeling)**.** *This definition says: among all possible ways of matching (or "translating") the system's internal tokens at time-slice I into tokens at time-slice J, pick the one that best preserves the flow structure; that best achievable degree is $\mathrm{SC}_S(I \to J)$. So continuity is not "are the tokens identical?" but "can we find a renaming under which the organizational dynamics remain largely the same?" This is exactly the kind of continuity one expects in developing or learning systems, where representational vocabulary shifts while deeper regularities persist (Hyperseed-Concept 166; compare [5]).*

As a toy example, suppose a child first uses a token `dog` *and later refines it into* `dog` *and* `wolf`*. The node sets differ across time, but there may be a map f that sends earlier* `dog` *into later* `dog` *and preserves many flow relations (e.g. "seeing dog" $\to$ "fear"), yielding high self-continuity despite vocabulary refinement.*

Remark 859 (Interpretation)**.** $\mathrm{SC}_S(I \to J)$ *is the best achievable degree of structure preservation between the system's pattern-flow network on I and on the next interval J. The definition allows the system to "rename" internal nodes across time via the map f. This is essential when the representational vocabulary itself evolves.*

Remark 860 (Why use a join over maps?)**.** *The join $\bigvee_f$ plays the role of an existential quantifier in the quantale: it says "there exists a correspondence between times that works this well." This existential flavor is appropriate because the system need not commit to a single privileged identity map; different contexts and downstream tasks may induce different best correspondences. This flexibility will matter again when we discuss mind–world correspondence, where multiple incompatible interpretations of the world may coexist without explosion [23, 24].*

Corollary 2 (Longer-horizon self-continuity along a realized chain)**.** *Suppose we have a chain of successive intervals $I_0 \to I_1 \to \cdots \to I_n$. Then*

$$\mathrm{SC}_S(I_0 \to I_n) \;\geq\; \mathrm{SC}_S(I_0 \to I_1) \otimes \mathrm{SC}_S(I_1 \to I_2) \otimes \cdots \otimes \mathrm{SC}_S(I_{n-1} \to I_n).$$

Remark 861 (What the chain inequality means)**.** *If the system is fairly continuous from one slice to the next, then it is at least that continuous over multiple steps, with the degree decaying (at worst) by repeated combination via $\otimes$. This is the formal statement that "continuity compounds" rather than resets. It connects directly back to Theorem 14: the corollary is essentially the repeated application of compositionality, but lifted through the join-over-maps definition of* SC_S.

Proof. Choose (for each k) a map $f_k : X_{S,I_k} \to X_{S,I_{k+1}}$ whose degree approximates $\mathrm{SC}_S(I_k \to I_{k+1})$. More explicitly, for any tolerance parameter $\delta \succ 0$ (in the order of $\mathbb{V}$, or after scalarization if one prefers a numerical proxy), one may pick f_k so that its degree is within δ of the join defining $\mathrm{SC}_S(I_k \to I_{k+1})$; this is the standard "$\varepsilon$-optimizer" maneuver when the supremum is not attained. Then Theorem 14 shows that the composition $f_{n-1} \circ \cdots \circ f_0$ has degree at least the stated tensor product. Here the tensor product encodes the cumulative degradation under composition: each step contributes a factor in $\mathbb{V}$, and the quantale multiplication (or its chosen monoidal operation) aggregates these factors in the same order as the composites. Taking joins over choices of the f_k

yields the inequality. In other words, since $\mathrm{SC}_S(I_0 \to I_n)$ is defined as the join of degrees over *all* candidate maps $X_{S,I_0} \to X_{S,I_n}$, it must dominate the degree of any particular composite built from near-optimal adjacent witnesses. □

Remark 862 (Proof sketch and intuition)**.** Proof sketch. *Pick near-optimal matching maps between each adjacent pair of slices, compose them into a single map from I_0 to I_n, use the composition law to bound the degree of this composite map, and then observe that $\mathrm{SC}_S(I_0 \to I_n)$ is the join over* all *such maps, so it must be at least as large as the degree of this particular composite.* □

The only subtlety is the use of "approximates": since the join defining SC_S may not be attained by a single maximizer, one chooses maps close enough to the supremum for the intended application. Conceptually, the statement says that if you can translate your internal vocabulary forward slice-by-slice, you can translate it forward over the whole horizon by composing those translations, with predictable loss. A useful way to read the quantale-valued inequality is as a generalized "triangle inequality" (or, depending on $\otimes$, a multiplicative analogue): stepwise continuity lower-bounds long-range continuity, but cannot in general prevent decay as n grows. This also clarifies why the result is framed as a lower *bound: composing specific witnesses constructs one admissible long-range witness, and the join defining $\mathrm{SC}_S(I_0 \to I_n)$ can only be larger when additional direct correspondences exist that are not decomposable into the chosen stepwise maps.*

Relative-information self-continuity. Hyperseed distinguishes pattern-flow-network self-continuity from a more "information geometric" notion: the mind at I and the mind at J should share a large amount of relative information. To express this in a quantale-friendly way, we use a monotone scalarization of $\mathbb{V}$. This shift is deliberate: the morphism-based notion is invariant under relabelings (and thus tracks *structural* persistence), whereas an information-style comparison presupposes a shared coordinate system (a fixed node set) and then measures *numerical* drift within that coordinate system.

Remark 863 (Connection to information theory)**.** *This "relative-information" variant resonates with classical measures of model drift and state distance. When $\mathbb{V}$ is ultimately derived from probabilistic quantities, one may connect the divergence to familiar information-theoretic functionals [16, 17]. Hyperseed's point, however, is not to privilege any single divergence, but to keep the interface open while preserving monotonicity and composability. In particular, the monotonicity requirement ensures that "more disagreement in $\mathbb{V}$" (in the ambient order) cannot be mapped to a smaller scalar penalty, so that downstream bounds and thresholds behave predictably when one refines or coarsens the underlying edge annotations.*

Definition 219 (Scalarization and graph divergence)**.** *A* scalarization *is a monotone map $\|\cdot\| : \mathbb{V} \to [0, \infty]$. Given two $\mathbb{V}$-graphs $G = (X, G)$ and $H = (X, H)$ on the same node set, define the* edgewise divergence

$$D(G, H) := \sum_{(x,x') \in X \times X} \big\|G(x, x') - H(x, x')\big\|,$$

where subtraction is interpreted componentwise when $\mathbb{V}$ is a product domain (e.g. a p-bit *quantale), and otherwise D is taken as a schematic placeholder for any reasonable scalar divergence induced by $\|\cdot\|$. We say that G and H have* ε-relative-information proximity *if $D(G, H) \leq \varepsilon$. When X is infinite (or merely large), the same definition can be read as selecting a summation convention (e.g. absolute convergence, truncation, or a weighted sum), depending on the modeling choice; the present formulation isolates the conceptual ingredient, namely an edgewise comparison mediated by a monotone scalarization.*

Remark 864 (Intuition and examples for scalarization). *A scalarization $\|\cdot\|$ is simply a way of extracting a nonnegative "size" or "cost" from a $\mathbb{V}$-value. For $\mathbb{V} = [0,1]$ one might take $\|u\| = 1-u$ (turning similarity into distance) or $\|u\| = u$ (treating u itself as a magnitude). For $\mathbb{V} = [0,1]^2$ one might take $\|(p,q)\| = p+q$ or $\|(p,q)\| = \max(p,q)$, depending on whether one wants to penalize total evidence mass or only the strongest polarity.*

The divergence $D(G,H)$ then sums edgewise differences. This is useful when the node set is held fixed (e.g. a stable vocabulary) and one wants to quantify how much the network changed in a numerical sense. It complements the morphism-based continuity, which is structurally invariant under relabeling and thus better suited to developmental regimes where the vocabulary itself evolves. One can also view $D(G,H)$ as a "Hamming-like" comparison generalized to weighted, typed edges: rather than counting mismatched symbols, it accumulates per-edge disagreement costs determined by $\|\cdot\|$. If desired, one may normalize by $|X|^2$ (when finite) to obtain an average per-edge drift; the unnormalized form is often preferable when absolute drift magnitude, rather than density, is the salient quantity (e.g. when small localized changes matter less than widespread global rewiring).

Remark 865 (Two complementary views). *Pattern-flow self-continuity (existence of a high-weight approximate morphism) is a* structural *condition. Relative-information proximity is a* numerical *condition. They are related but not identical: self-continuity allows nontrivial relabelings and can hold even when many edge weights drift, provided the drift is coherent under the relabeling. For example, if a system consistently renames an internal symbol (or splits one concept into two with a systematic bookkeeping map), then a high-degree morphism may still exist even though a naive edgewise comparison on a fixed node set would report substantial divergence. Conversely, a small $D(G,H)$ can occur even when no good relabeling exists (e.g. many small but incoherent edge perturbations), so the two notions probe different failure modes.*

Remark 866 (Why keep both notions). *In applications, one often wants both: structural continuity to justify that a system is still "the same agent" in the operational sense, and numerical proximity to detect subtle instability within a fixed representational frame. Hyperseed treats these as complementary lenses rather than competing definitions, reflecting a broader methodological pluralism: different invariants answer different questions about persistence (Hyperseed-Concept 200). In particular, structural continuity is naturally adapted to scenarios where the system's representational basis is itself part of the dynamical state (so comparing raw coordinates is ill-posed), while numerical proximity is adapted to monitoring and validation tasks in a fixed interface (so that deviations can be alarmed on a stable scale).*

Reflective self-continuity. Hyperseed further distinguishes *reflective self-continuity*: the system is explicitly aware (in its own representational language) of its continuity. Formally this requires a notion of internal self-models and declarative content, developed in Section 17. In the present section we only note the minimal scaffold: reflective self-continuity arises when a self-model contains an explicit representation of (approximate) morphisms witnessing $\mathrm{SC}_S(I \to J)$, and this representation is used in attention and control (Sections 15 and 14). One can think of this as internalizing not only the fact that "there exists a correspondence," but also storing (possibly schematically) the correspondence itself as an object that can be queried, composed, revised, and used as a premise in planning.

Remark 867 (Why reflectivity is an extra requirement). *A system may be continuous without having a representational handle on that fact, just as a river may persist as a process without representing its own persistence. Reflective self-continuity adds a meta-level constraint: the continuity witness must appear as content in the system's representational economy and must be* actionable

(able to steer attention or control). This anticipates the closure phenomena discussed later in the consciousness layer (Section 17) and aligns with fixed-point flavored analyses of self-model stability [10]. Operationally, reflectivity is what allows the agent to treat identity-over-time as a manipulable hypothesis: it can allocate resources to maintaining a correspondence, detect when the correspondence degrades, and initiate repair (e.g. by learning a new alignment map) rather than merely exhibiting continuity as an unexamined byproduct of dynamics.

16.4 Development as continuity plus expanding capability

Hyperseed defines *development* as the conjunction of self-continuity with the acquisition of fundamentally new capabilities over time, typically in a way that extends and enriches rather than obsoletes earlier capabilities. To formalize this we separate the "identity" condition (self-continuity) from a capability preorder. This targets Development as a core notion (Hyperseed-Concept 95) and also interfaces naturally with task-based views of intelligence [19]. In particular, the separation makes explicit that (i) continuity is about *who* or *what* persists across change, whereas (ii) capability is about *what can be done* and how well, which may increase, plateau, or even partially regress while still remaining "the same system" in the sense of Section 16. This also anticipates later discussions where apparent "improvement" in one regime can be explained by changes in attention, habit structure, or observation/interface layers rather than by a simple accumulation of skills.

Definition 220 (Capability profile and capability preorder)**.** *Fix a family $\mathcal{G}$ of goals (or tasks) that may be relevant to the system. For each interval I, define a* capability profile *as a function*

$$\mathrm{Cap}_S(I) : \mathcal{G} \to \mathbb{V},$$

where $\mathrm{Cap}_S(I)(g)$ quantifies the system's ability to achieve goal g (as modeled by O) when operating in the regime represented by I. The pointwise order on $\mathbb{V}^{\mathcal{G}}$ defines a preorder:

$$\mathrm{Cap}_S(I) \preceq \mathrm{Cap}_S(J) \quad \textit{iff} \quad \mathrm{Cap}_S(I)(g) \leq \mathrm{Cap}_S(J)(g) \textit{ for all } g \in \mathcal{G}.$$

Remark 868 (Regimes, evaluation protocol, and dependence on O)**.** *The phrase "regime represented by I" is intended to cover not only internal state (weights, memory, habits) but also evaluation conditions: the distribution of inputs encountered, the available tools, and the interface by which success on g is measured. Formally, this dependence is pushed into the observation/evaluation machinery O, so that $\mathrm{Cap}_S(I)(g)$ is not an abstract Platonic competence but an* operational *quantity: it is always relative to the channel through which performance is read out and aggregated over I. This matters in later sections on mind–world correspondence: an apparent gain in capability can sometimes be a gain in alignment between internal representation and the evaluation context, without a commensurate increase in underlying generality.*

Remark 869 (Intuition and examples for capability profiles)**.** *A capability profile is a many-goal, graded generalization of the idea of "competence." For each $g \in \mathcal{G}$ (say, "solve linear equations," "cooperate with peers," or "navigate a room"), the value $\mathrm{Cap}_S(I)(g)$ records how well the system can achieve g during interval I, in the same quantale-valued language used elsewhere.*

For example, if $\mathbb{V} = [0, 1]$, then $\mathrm{Cap}_S(I)(g)$ can be interpreted as success probability or normalized performance. If $\mathbb{V}$ is p-bit*-valued, then $\mathrm{Cap}_S(I)(g)$ can represent simultaneous evidence for and against competence (e.g. the system sometimes succeeds and sometimes fails in incompatible contexts). The preorder $\preceq$ then formalizes the natural notion that "J is at least as capable as I" when it is no worse on any goal in the family.*

Remark 870 (Scope of $\mathcal{G}$ and "fundamentally new" capabilities)**.** *The family $\mathcal{G}$ should be understood as a modeling choice: it may be narrow (a benchmark suite), broad (a library of tasks parameterized by context), or even implicitly defined (e.g. by a goal generator used in training). In practice, "fundamentally new capabilities" correspond to strict improvements on goals that were previously low or undefined in the relevant evaluation scheme, or to the appearance of new goals that become salient as the system interacts with richer environments. The formalism can accommodate either view: one may keep $\mathcal{G}$ fixed and track new abilities as movement from low to high values on previously included goals, or allow $\mathcal{G}$ to expand over time and interpret development as the ability to maintain performance on incumbent goals while raising performance on newly introduced ones.*

Remark 871 (Why a preorder rather than a total order)**.** *Capabilities across diverse tasks are rarely totally comparable: one system may improve at g_1 while regressing at g_2. The preorder keeps the formalism honest about partial comparability and dovetails with later value-paraconsistent and multi-objective reasoning (Section 18). In engineering terms, this is also the right level of abstraction for discussing tradeoffs and dependency constraints across skill acquisition [9, 8].*

Remark 872 (Strict improvement and ties)**.** *Because $\preceq$ is only a preorder, it may identify distinct profiles as mutually comparable in both directions (i.e. ties up to the induced equivalence). Accordingly, $\mathrm{Cap}_S(I) \prec \mathrm{Cap}_S(J)$ in Definition 221 should be read as "$\mathrm{Cap}_S(I) \preceq \mathrm{Cap}_S(J)$ and not $\mathrm{Cap}_S(J) \preceq \mathrm{Cap}_S(I)$," i.e. a strict increase on at least one goal without any decrease on any goal. This notion is intentionally strong; later variants can weaken it by allowing bounded losses on a small set of goals, which is often necessary for resource-bounded agents.*

Definition 221 (Development path)**.** *A sequence of successive intervals $I_0 \to I_1 \to \cdots$ is a* development path *for S if there exists a threshold $\theta \in \mathbb{V}$ such that:*

(a) (continuity) $\mathrm{SC}_S(I_k \to I_{k+1}) \geq \theta$ for all k;

(b) (nontrivial growth) there exist infinitely many k such that $\mathrm{Cap}_S(I_k) \prec \mathrm{Cap}_S(I_{k+1})$;

(c) (enrichment bias) for typical goals g with high previous capability, $\mathrm{Cap}_S(I_k)(g)$ does not sharply decrease when new goals are added.

Condition (c) is qualitative; in concrete models it is replaced by a quantitative constraint (e.g. bounded loss on a weighted set of incumbent goals).

Remark 873 (On the continuity threshold θ)**.** *The single threshold θ serves as a tunable notion of "identity granularity": higher θ demands a stronger form of persistence across steps, while lower θ tolerates more drift. One can equivalently think of θ as picking out a subcategory of transitions regarded as identity-preserving, or as a robustness margin that separates development from replacement. In empirical settings, θ can be chosen to reflect how stable an external observer (or the system itself, via self-modeling) must judge the system to be in order to treat its learning trajectory as belonging to one continuing agent.*

Remark 874 (Quantitative surrogates for enrichment bias)**.** *A common quantitative replacement for (c) is to fix, at each step k, an "incumbent" subset $\mathcal{G}_k^{\mathrm{inc}} \subseteq \mathcal{G}$ of goals deemed already mastered, together with weights $w_k(g)$ emphasizing what must not be forgotten, and then require a bounded drop such as*

$$\inf_{g \in \mathcal{G}_k^{\mathrm{inc}}} \big(\mathrm{Cap}_S(I_{k+1})(g) - \mathrm{Cap}_S(I_k)(g)\big) \geq -\varepsilon \quad \textit{or} \quad \sum_{g \in \mathcal{G}_k^{\mathrm{inc}}} w_k(g)\, \mathrm{Cap}_S(I_{k+1})(g) \geq \sum_{g \in \mathcal{G}_k^{\mathrm{inc}}} w_k(g)\, \mathrm{Cap}_S(I_k)(g) - \varepsilon,$$

interpreting subtraction and summation in whatever enrichment of $\mathbb{V}$ is being used. When $\mathbb{V}$ is not numeric, the same idea can be phrased order-theoretically as a constraint that the restriction of $\mathrm{Cap}_S(I_{k+1})$ *to* $\mathcal{G}_k^{\mathrm{inc}}$ *remains above a designated lower envelope. This connects directly to catastrophic forgetting in continual learning, where development requires not merely acquiring new tasks but retaining competence on earlier ones under distribution shift.*

Remark 875 (Intuition: development is not mere change). *Condition (a) insists that the system remains recognizably continuous: development is not a replacement of one system by another unrelated system. Condition (b) insists that something genuinely new keeps appearing: without it, a perfectly stable adult-like agent would count as "developing" just by persisting. Condition (c) is the "enrich rather than erase" clause: it encodes the idea that development, in the Hyperseed sense, is not simply optimization for a moving target but a bias toward retaining earlier competences while acquiring new ones.*

A simple example is language learning: early competence at phoneme discrimination should not be destroyed by later acquisition of syntax; rather, later skills should build on earlier ones. This is also precisely where resource-bounded systems encounter tradeoffs, and why Hyperseed connects development to attention allocation and dependency management [19, 9].

Remark 876 (Local regressions and realistic trajectories). *The definition allows that between some consecutive intervals,* $\mathrm{Cap}_S(I_{k+1})$ *may fail to dominate* $\mathrm{Cap}_S(I_k)$*: development paths can include plateaus and periods of reorganization. The requirement is only that strict improvements occur infinitely often, capturing the idea of an open-ended trajectory rather than a one-off training phase. In practice, one may also consider softened variants of (b) that measure improvement relative to a curriculum, or that require strict growth on a sequence of increasingly complex goal families, which better captures development under shifting environments and expanding affordances.*

Remark 877 (Self-transcendence as a special case). *Hyperseed treats "self-transcendence" as a form of development. In the present framework, a mathematically simple proxy is: self-transcendence corresponds to development in which the self/other boundary function σ_I changes in a way that reduces conflict on the boundary (smaller $\partial\mathrm{Self}_\theta(I)$) while preserving continuity. This is only a proxy; richer formalizations depend on the resonance machinery (Sections 12 and 17).*

Remark 878 (Why the proxy focuses on the boundary region). *The boundary region $\partial\mathrm{Self}_\theta(I)$ concentrates the system's "ontological tension" about what counts as self. A reduction in this region (at fixed continuity) formalizes, in one clean variable, the intuition that the self becomes less conflicted about its own scope. In experiential terms this can correspond to a shift toward non-dual integration (Hyperseed-Concept 121); in control terms it can reduce internal contention for attentional resources (Section 15); and in resonance terms it can reflect an increase in coherence across subsystems (compare [5, 13] for broader resonance motifs).*

Remark 879 (Boundary shrinkage versus capability expansion). *The proxy for self-transcendence is intentionally orthogonal to the capability preorder: a system may become less conflicted about self/other while gaining, losing, or redistributing task capabilities. Nevertheless, in many practical settings boundary clarification can indirectly support capability growth by reducing internal interference (fewer competing self-models) and by improving the mind–world interface through which goals are perceived and pursued. This provides a conceptual bridge between "development" in the skill-acquisition sense and "development" in the integration/non-duality sense: both can be viewed as trajectories that increase coherence while maintaining sufficient self-continuity to count as the same ongoing agent.*

16.5 Mind–world correspondence via pattern-flow morphisms

Hyperseed's "mind–world correspondence" is a generalized notion of world-model quality. It is defined in terms of approximate morphisms between two pattern-flow networks: one describing relevant entities inside the system (mind/brain organization), and one describing relevant entities in the environment. This corresponds to Mind-World Correspondence as a core notion (Hyperseed-Concept 112) and is aligned with the engineering view that a model is good insofar as it supports successful prediction and control [19, 20]. A key point is that "quality" here is not identified with any particular representational format (e.g. propositional symbols, neural features, latent-state vectors): it is identified with the existence of a structure-preserving translation between whatever internal units the system uses and whatever external regularities are relevant to its task-context.

Definition 222 (Internal and external pattern-flow networks)**.** *Fix a system S and interval I. Let $M_{S,I}$ be a pattern-flow network on a set $X^{\mathrm{in}}_{S,I}$ representing internal entities (states, representations, memory items, action schemas). Let $E_{S,I}$ be a pattern-flow network on a set $X^{\mathrm{out}}_{S,I}$ representing relevant external entities (objects, events, social partners, affordances) in the environment. Both are $\mathbb{V}$-graphs.*

Remark 880 (Further intuition on what $M_{S,I}$ and $E_{S,I}$ are tracking)**.** *The interval parameter I is included to allow both networks to be context- and history-sensitive: the system's internal organization (what distinctions it makes salient, which memory items are active, which policies are available) may change over time, and the relevant external "world" being tracked may likewise shift (lighting changes, new interlocutors appear, different tools become available). Treating $M_{S,I}$ and $E_{S,I}$ as $\mathbb{V}$-graphs emphasizes that "edges" encode graded strengths of pattern-flow rather than binary relations: the same pair of nodes can have weak, moderate, or strong flow depending on the circumstances and on the semantics of $\mathbb{V}$.*

Remark 881 (Intuition: why two networks?)**.** *The internal network $M_{S,I}$ describes how patterns propagate within the system's own representational economy: which internal tokens lead to which others. The external network $E_{S,I}$ describes how patterns propagate in the environment as the system experiences or models it: which external events lead to which others, which affordances follow which contexts, and so on. By keeping these separate, Hyperseed avoids the simplistic assumption that the system's internal vocabulary is already "the same as" the world's vocabulary.*

A simple example is a robot with an internal node `heat_warning` *and an external node* `high_temperature`*. The system may need to learn a correspondence between these nodes even if their internal causal neighborhoods differ in detail. The value of this definition is that it sets up the exact mathematical situation in which a correspondence morphism can be sought.*

Remark 882 (Many-to-one, partiality, and re-encoding)**.** *Nothing in the setup requires f to be injective: distinct internal states can legitimately map to the same external entity when the system uses multiple internal encodings (e.g. a fast heuristic feature and a slower deliberative concept) that nevertheless pick out the same worldly regularity. Conversely, a single internal node may correspond only coarsely to a cluster of external nodes; in practice this is handled either by allowing $X^{\mathrm{out}}_{S,I}$ to contain suitably coarse-grained external entities (equivalence classes, macro-objects) or by viewing f as selecting a task-relevant proxy in the environment graph. If one wants to model representational "gaps" explicitly, one can also interpret the supremum over f as ranging over maps defined on a chosen subset of $X^{\mathrm{in}}_{S,I}$ (with the subset-selection encoded into the domain), but the present definition keeps the interface simple and delegates such choices to goal-conditioning below.*

Definition 223 (Mind–world correspondence score)**.** *The* mind–world correspondence score *of S on interval I is*

$$\mathrm{MWC}_S(I) := \bigvee_{f : X^{\mathrm{in}}_{S,I} \to X^{\mathrm{out}}_{S,I}} \deg(f; M_{S,I}, E_{S,I}).$$

*We say that S has θ-*mind–world correspondence *on I if* $\mathrm{MWC}_S(I) \geq \theta$*.*

Remark 883 (About the supremum $\bigvee$ and what is being optimized)**.** *The operator $\bigvee$ is the join/supremum in the order on $\mathbb{V}$ (or, when $\mathbb{V}$ is numerical, simply the least upper bound). Conceptually, $\mathrm{MWC}_S(I)$ is an "optimal achievable" alignment score: it asks for the best interpretive translation f available, rather than committing a priori to a particular hand-chosen decoding of internal states. When the set of candidate maps is finite (e.g. finite node sets), $\bigvee$ reduces to a maximum; in more general settings, $\bigvee$ expresses that correspondence is defined by an optimization principle even when a maximizer may not be unique (or may only be approached). This choice matches the intended role of $\mathrm{MWC}_S(I)$ as a capability-like quantity: a system should not be penalized for having multiple equally good internal codings of the same external structure.*

Remark 884 (How $\deg(f; M_{S,I}, E_{S,I})$ should be read)**.** *The degree $\deg(f; M_{S,I}, E_{S,I})$ measures how well the translation f preserves pattern-flow constraints from the internal graph to the external graph. Operationally, it can be read as the largest $q \in \mathbb{V}$ such that f is a q-approximate morphism (in the sense used throughout the pattern-flow formalism): higher q means that whenever there is strong internal flow from x to x', there is correspondingly strong external flow from $f(x)$ to $f(x')$. This is why $\mathrm{MWC}_S(I)$ is naturally interpreted as "world-model quality": it evaluates whether internal dynamics can be soundly reinterpreted as statements about the environment, rather than merely being internally coherent.*

Remark 885 (Intuition and examples for $\mathrm{MWC}_S(I)$)**.** *The score $\mathrm{MWC}_S(I)$ is the best achievable degree to which internal pattern-flow can be interpreted as external pattern-flow. It asks: is there a translation f from internal tokens to external tokens such that internal edge-structure (flow regularities) survives under interpretation?*

For example, if internally the system has a rule-like flow `see_red` $\to$ `stop` *and externally there is a flow* `red_light` $\to$ `cars_stop`*, then a map sending* `see_red` $\mapsto$ `red_light` *and* `stop` $\mapsto$ `cars_stop` *will score highly when these flows align. The usefulness is that correspondence becomes a measurable, composable quantity that can feed into capability profiles and control policies, without requiring perfect isomorphism between mind and world. A further advantage is that the criterion is inherently relational: it does not demand that individual nodes be "correctly named" or that internal tokens have any privileged semantics in isolation; it only demands that the transition/implication structure among tokens can be matched to a transition/implication structure among worldly entities.*

Remark 886 (Goal-conditioned correspondence)**.** *Hyperseed notes that mind–world morphisms are often strongest on the subgraphs most relevant to the system's active goals. A convenient formalization is to use a goal-dependent node-filter $U_{g,I} \subseteq X^{\mathrm{in}}_{S,I}$ and $V_{g,I} \subseteq X^{\mathrm{out}}_{S,I}$ representing the internal/external entities relevant to goal g on I. Define*

$$\mathrm{MWC}_S(I; g) := \bigvee_{f : U_{g,I} \to V_{g,I}} \deg\bigl(f; M_{S,I}|_{U_{g,I}}, E_{S,I}|_{V_{g,I}}\bigr).$$

The family $g \mapsto \mathrm{MWC}_S(I; g)$ can be used as one component of the capability profile $\mathrm{Cap}_S(I)$. In many systems, the filters $U_{g,I}$ and $V_{g,I}$ can be understood as formal counterparts of attention and state-abstraction: they specify which internal distinctions the system is actually deploying, and which external distinctions matter for success, for the particular goal and time window under consideration.

Remark 887 (Why goal-conditioning is philosophically natural)**.** *What counts as a "relevant" portion of the world is not fixed; it is indexed by purpose, attention, and action (Hyperseed-Concepts* **??***, 60, 87). Goal-conditioned correspondence makes this indexicality explicit, preventing the notion of world-model quality from silently smuggling in an observer-independent ontology. In Russell's idiom, it avoids treating "the world" as a single completed totality; it treats it as a structured field carved by inquiry and action. Put differently, it treats correspondence as a relation between an agent's operative semantics and a task-salient slice of environmental structure, rather than as a once-and-for-all alignment between two complete descriptions.*

Lemma 4 (Transport of predicted flow under correspondence)**.** *Assume $\mathbb{V}$ is ordered so that larger values mean "stronger flow". Let $f : X^{\text{in}}_{S,I} \to X^{\text{out}}_{S,I}$ be a q-approximate morphism from $M_{S,I}$ to $E_{S,I}$. Then for all internal nodes x, x',*

$$M_{S,I}(x, x') \otimes q \leq E_{S,I}(f(x), f(x')).$$

In particular, if the system uses $M_{S,I}$ as a predictive model of external flow (under the interpretation f), then the correspondence degree q provides an explicit lower bound on the predicted external flow strength.

Remark 888 (What this lemma says and why it is important)**.** *The lemma is the operational heart of mind–world correspondence: it says that an approximate morphism is exactly the kind of object that allows the system to* transport *predictions from internal dynamics to external dynamics. If q is high, then strong internal flow $M_{S,I}(x, x')$ guarantees strong external flow between the interpreted nodes; if q is low, the guarantee weakens accordingly. This makes correspondence actionable: the score is not merely a descriptive similarity measure, but a quantitative certificate about which inferences made in the internal model remain valid (up to degree q) when read as claims about the environment. It also clarifies why $\otimes$ appears: the same conjunction-like operator that combines evidence or flow within the $\mathbb{V}$-semantics is used to discount internal strength by the quality of the interpretation.*

Remark 889 (Practical reading: robustness, calibration, and failure modes)**.** *When $\mathrm{MWC}_S(I)$ is low, the system may still exhibit rich internal dynamics, but those dynamics do not reliably "latch onto" the environment: internal transitions may correspond to spurious correlations, hallucinated causal links, or purely self-referential loops. When $\mathrm{MWC}_S(I)$ is high only for certain goals g, the system may be well-calibrated in domains it actively trains on or attends to, while remaining poorly grounded elsewhere; this matches the empirical observation that competent behavior can be highly domain-local. In design terms, raising correspondence can be pursued either by improving $M_{S,I}$ (learning better internal predictive structure), improving $E_{S,I}$ (using a better external abstraction of the environment), or improving the map f (learning a better grounding/decoder); the definition is agnostic about which of these is the appropriate intervention.*

Remark 890. *This connects back to Section 14: prediction and control require not just a model, but a way to interpret model-variables as world-variables. Here that interpretive act is formalized as f, and its reliability is summarized by q. In particular, f plays the role of an "observational semantics" that turns internal distinctions (nodes and edges in the internal pattern-flow network) into externally meaningful distinctions (nodes and edges in the world pattern-flow network), so that claims like "x tends to lead to x'" can be transported to claims like "$f(x)$ tends to lead to $f(x')$." The scalar (or truth-degree) q is then a single, uniform lower bound on how well this transportation preserves edge-constraints across* all *internal pairs, thereby separating the questions "what is the agent internally representing?" (encoded by $M_{S,I}$) and "how accurate is that representation when interpreted externally?" (encoded by q and f together).*

Proof. Immediate from the definition of q-approximate morphism. More explicitly: the statement of the lemma is exactly the defining inequality (or its equivalent reformulation via the degree $\deg(\cdot;\cdot,\cdot)$), so no additional construction is required beyond instantiating the definition with the particular internal and external edge-structures under discussion. □

Remark 891 (Proof sketch and intuition)**.** Proof sketch. *Unpack the definition: $q \le \deg(f; M_{S,I}, E_{S,I})$ is equivalent to $M_{S,I}(x, x') \otimes q \le E_{S,I}(f(x), f(x'))$ for all x, x'. Here $\otimes$ is the monoidal product of the underlying truth-value/weight structure (e.g. a t-norm in a fuzzy setting, multiplication in a probabilistic-weight setting, or $\wedge$ in an order-theoretic setting), and $\le$ is the corresponding entailment/order on edge-weights; thus the inequality states that "internal support, discounted by q, is never stronger than the external support after interpretation."* □

There is no further trick: the lemma simply names, in predictive language, the inequality that constitutes "being an approximate morphism." Geometrically, q is the uniform amount by which internal edge strengths may be discounted so that the interpreted edges fit inside the world graph. Equivalently, q can be read as a global safety margin*: if $q = 1$ then f is fully structure-preserving at the level of edge-weights (an exact morphism in the chosen enrichment), while smaller q quantifies how much one must "shrink" internal claims to guarantee external validity. This is useful when internal edges encode strong expectations or control-relevant dependencies that are only partially realized in the world: the same f may remain meaningful even as conditions change, by lowering q to reflect reduced reliability rather than abandoning the interpretive mapping entirely.*

Remark 892 (Why this is a useful notion of "world model")**.** *A classical world model in AI typically consists of: (i) a representational vocabulary for external states, and (ii) a transition/prediction structure on that vocabulary. Mind–world correspondence adds two important generalizations: (1) it does not assume the internal vocabulary matches the external one; and (2) it measures fidelity by the existence of an approximate structure-preserving map, not by equality of probability distributions. This is well-suited to observer-relative modeling and to cases where multiple incompatible models coexist (paraconsistency). One consequence is that the internal model may be* coarser *(many-to-one) or* refined *(one-to-many, via auxiliary internal distinctions) relative to the world description, and the framework still assigns a clear quantitative notion of adequacy: the question becomes whether there exists an f that makes the internal transition claims conservatively true about the world, up to the uniform slack q. In contrast, equality (or close match) of full distributions typically presupposes a shared event space and a shared semantics of state variables; here those assumptions are replaced by an explicit correspondence map. Moreover, the approximate-morphism criterion aligns with the practical role of a model in control: for action selection, one often needs* sound *(non-overconfident) predicted constraints more than perfectly calibrated probabilities, and the inequality $M_{S,I}(x, x') \otimes q \le E_{S,I}(f(x), f(x'))$ is exactly such a soundness condition after interpretation. When the agent maintains multiple internal descriptions that disagree, paraconsistency becomes manageable because each description can be paired with its own (f, q) witness of partial correspondence, rather than forcing premature reconciliation into a single globally consistent probabilistic account.*

Remark 893 (Relation to transfer and abstraction)**.** *The use of a morphism f as an interpretation map parallels the role of representation mappings in transfer learning: one succeeds by finding a structure-preserving translation between domains. Hyperseed's stance is that "modeling the world" is itself a transfer problem between internal and external pattern-flow networks [7]. This also clarifies why mind–world correspondence is a natural ingredient in general intelligence: it is the quantitative form of "having the right abstractions" (Hyperseed-Concept 51). Concretely, if an agent has learned an internal pattern-flow network in a latent space (shaped by its sensors, embodiment, or prior tasks), then applying that competence to a new environment requires a map f that aligns latent* *nodes and transitions with externally meaningful ones; the parameter q then measures how much of the learned relational structure survives that alignment. In this sense, abstraction is not merely discarding detail, but creating internal variables whose induced transition structure admits a high-q morphism into a broad class of world-structures. This viewpoint also supports compositionality: when correspondences can be chained (e.g. internal $\to$ intermediate $\to$ external), one can interpret successive abstractions as composing morphisms, where the effective reliability degrades in a controlled way (typically through the monoidal product of the respective q values). Such compositional degradation formalizes an intuitive fact about transfer: each additional representational "gap" may introduce some loss, but good abstractions are exactly those for which the loss remains bounded and predictable across contexts.*

16.6 Space as a structure on which correspondences are defined

Hyperseed uses "space" in a broad sense: not only physical space, but also conceptual/semantic spaces on which similarity, neighborhood, and correspondences can be formulated. A pattern web already induces a notion of distance/similarity between entities. Here we package the minimal structure needed for correspondence-based reasoning. This connects to Space as a core notion (Hyperseed-Concept 173) and to the general idea that similarity is not merely geometric but can be representational and process-relative.

In particular, the relevant "space" may be tied to a task, a sensorimotor loop, or a model class: two entities can count as "near" when they are hard to distinguish by the agent's available measurements, when they afford similar actions, or when they play the same role in a recurrent process (even if they are far apart geometrically). This makes the choice of space part of the agent's theory of what distinctions matter, and it is precisely these distinctions that later support self/world separation and the persistence of identity across time.

Definition 224 (Enriched spaces)**.** *A* $\mathbb{V}$-space *is a* $\mathbb{V}$*-category* $\mathcal{S}$ *(objects with* $\mathbb{V}$*-valued hom/similarity) together with a chosen scalarization* $\|\cdot\| : \mathbb{V} \to [0,\infty]$*. The induced extended pseudo-metric on objects is*

$$d_{\mathcal{S}}(a,b) := \big\|\mathcal{S}(a,b)\big\|.$$

Depending on $\mathbb{V}$ *and* $\|\cdot\|$*, this may behave like a similarity, a cost, or a distance.*

A useful way to read Definition 224 is as a separation between (i) a compositional notion of comparison, encoded by enrichment (how comparisons compose along paths), and (ii) a numerical projection used for heuristics, thresholds, and optimization. The enrichment supplies the analogue of a "triangle inequality" at the $\mathbb{V}$-level (via composition in the $\mathbb{V}$-category), while the scalarization supplies a computable summary. In applications, the scalarization is often chosen to be monotone and to respect the intended direction of comparison (e.g. "larger in $\mathbb{V}$ means more similar" versus "larger means more costly"), but the definition is deliberately permissive: the agent may carry multiple scalarizations for different downstream uses (planning, clustering, anomaly detection), all over the same underlying enriched structure.

Remark 894 (Intuition and examples for $\mathbb{V}$-spaces)**.** *A* $\mathbb{V}$*-space is an enriched generalization of a metric space: instead of real-valued distances, we have* $\mathbb{V}$*-valued comparisons* $\mathcal{S}(a,b)$*. If* $\mathbb{V} = [0,\infty]$ *ordered oppositely and* $\otimes$ *is addition, then one recovers the Lawvere view of generalized metric spaces; if* $\mathbb{V} = [0,1]$ *with* $\otimes = \min$ *one obtains a fuzzy similarity space.*

The scalarization $\|\cdot\|$ *then turns enriched similarities into an ordinary extended pseudo-metric* $d_{\mathcal{S}}$ *that can be used for numerical reasoning (e.g. nearest neighbors), while preserving the option to work directly in* $\mathbb{V}$ *when composability is more important. This is useful because correspondence* *reasoning often needs both: categorical structure for compositional semantics and scalar distances for optimization.*

One practical intuition is that $\mathcal{S}(a, b)$ can encode *typed* or *structured* evidence of relatedness that would be lost by prematurely collapsing to a number. For instance, $\mathbb{V}$ could record not only a magnitude but also the "mode" of similarity (perceptual, functional, linguistic), or it could be a product quantale that combines independent channels of evidence. Scalarization then becomes an explicit design choice: when projecting to $[0, \infty]$, one decides how to trade off these channels, which in turn affects what the agent treats as stable identity versus incidental variation. This makes it easier to state, within the same formalism, both (a) hard correspondences induced by strong and consistent evidence, and (b) soft correspondences that are provisional and revisable.

It is also important that $d_{\mathcal{S}}$ is only required to be an *extended pseudo*-metric: distinct objects may have zero distance (indistinguishability at the agent's resolution), and distances may be infinite (no meaningful comparison available). Both behaviors occur naturally in developmental settings, where early representations may collapse many distinct world states, and later refinement splits them as new sensors, concepts, or actions become available.

Definition 225 (Correspondence space)**.** *Let $M = (X^{\text{in}}, M)$ and $E = (X^{\text{out}}, E)$ be $\mathbb{V}$-graphs. A* correspondence space *for (M, E) is a $\mathbb{V}$-space $\mathcal{S}$ together with maps $i : X^{\text{in}} \to \mathrm{Ob}(\mathcal{S})$ and $e : X^{\text{out}} \to \mathrm{Ob}(\mathcal{S})$. The pair (i, e) induces a $\mathbb{V}$-valued correspondence relation*

$$R_{\mathcal{S}}(x, y) := \mathcal{S}(i(x), e(y)) \qquad (x \in X^{\text{in}},\ y \in X^{\text{out}}).$$

This definition is intentionally weak: i and e need not be injective (multiple tokens can be embedded to the same point of $\mathcal{S}$, representing aliasing or abstraction), and they need not preserve any structure beyond whatever regularities are captured by the induced relation $R_{\mathcal{S}}$. In particular, i and e can be read as *representation functions*: they say how the agent chooses to represent internal items and external items in a common comparison medium. When these representation functions change over time (learning, sensor drift, concept acquisition), the correspondence space view still applies by treating i and e as time-indexed families, so that development becomes movement (or re-embedding) within a stable $\mathcal{S}$, or else co-development of $\mathcal{S}$ itself.

Remark 895 (Intuition: an interlingua for comparison)**.** *The correspondence space $\mathcal{S}$ plays the role of a shared medium in which internal and external entities can be compared. The maps i and e embed internal tokens and external tokens into this medium, and $R_{\mathcal{S}}(x, y)$ then measures how well internal token x matches external token y in the space.*

For example, if $\mathcal{S}$ is ordinary physical space, then i may map internal body-schema tokens to physical locations and e may map external objects to their locations, making correspondence spatial. If $\mathcal{S}$ is a semantic embedding space, then i and e may map tokens to vectors, making correspondence conceptual. The utility of this definition is that it supports a third mode of correspondence reasoning: not only by direct morphisms f, but also by metric/neighbor structure in $\mathcal{S}$ (Hyperseed-Concept 59).

The "interlingua" reading is especially apt when internal and external tokens have different native structures (different feature vocabularies, different temporal granularities, different modalities). Rather than forcing a direct alignment $X^{\text{in}} \leftrightarrow X^{\text{out}}$, one chooses $\mathcal{S}$ so that the agent can ask meaningful comparative questions: Which external candidates lie in the neighborhood of an internal hypothesis? Which internal states cluster around the same external referent? Which correspondences remain stable under small perturbations of sensors or attention? These are neighborhood questions, and they can be answered even when no single crisp morphism is yet justified.

Remark 896 (Interpretation). *A correspondence space provides an "interlingua" in which internal and external entities can be compared. When $\mathcal{S}$ is physical space (or spacetime), i and e may represent embodied localization. When $\mathcal{S}$ is a conceptual space, i and e encode semantic embeddings. In either case, approximate morphisms $M \to E$ can be seen as arising from (or inducing) regularities in $R_{\mathcal{S}}$.*

One can also regard $R_{\mathcal{S}}$ as a *field of constraints* that guides the search for morphisms. A high-quality morphism f typically selects, for each $x \in X^{\text{in}}$, an output $f(x)$ lying in a high-similarity region of the slice $y \mapsto R_{\mathcal{S}}(x, y)$, while also respecting relational constraints coming from the edges/flows of M and E. Conversely, if repeated successful morphisms are found over time, they provide empirical pressure to adjust i, e, or $\mathcal{S}$ so that the corresponding pairs become nearer, i.e. so that the interlingua becomes better adapted to the agent's world.

Remark 897 (Why introduce correspondence spaces if we already have morphisms?). *The morphism view (f with degree q) is crisp and compositional, but it chooses a single matching. A correspondence space supports softer, many-to-many relations: one can represent ambiguity and polysemy as neighborhoods rather than as a hard assignment. This is especially important in paraconsistent settings, where it may be rational for a system to retain multiple incompatible correspondences until further evidence or action resolves them [24, 23].*

In addition, the correspondence-space view makes explicit room for *partial* alignment. When the agent only knows how to compare some internal tokens to some external tokens (because other tokens are newly formed, occluded, or not currently observable), a crisp morphism may be forced to guess, whereas $R_{\mathcal{S}}$ can simply remain diffuse or uninformative on those pairs. This matters for continuity of self-models: the agent can maintain a persistent "core" alignment (e.g. body schema) while leaving peripheral correspondences unresolved, thereby avoiding the need to collapse uncertainty into premature commitments.

Example 14 (A minimal embodied correspondence sketch). *Consider a toy agent with internal nodes $X^{\text{in}} = \{\texttt{body}, \texttt{pain}, \texttt{reach}\}$ and external nodes $X^{\text{out}} = \{\texttt{arm}, \texttt{heat}, \texttt{cup}\}$. Suppose M encodes the internal regularity "pain follows heat-on-body" and "reach tends to precede cup-in-hand," while E encodes the external regularity "heat tends to follow contact between arm and heat source" and "cup-in-hand tends to follow reaching." A correspondence map f sending* `body` $\mapsto$ `arm`*,* `pain` $\mapsto$ `heat`*, and* `reach` $\mapsto$ `reach` *(if a matching external token exists) will have high degree precisely when these flow regularities align. The same data can be represented by a correspondence space $\mathcal{S}$ in which* `body` *and* `arm` *are close,* `pain` *and* `heat` *are close, and so on.*

In such examples, the self/other boundary (Section 16.1) can also be grounded: `body` *tends to be a persistent, highly self-evidenced node, while* `cup` *is typically other-evidenced, and* `pain` *may live on the paraconsistent boundary (both "me" and "not-me").*

To connect this sketch to continuity and development: if the agent experiences repeated episodes in which `reach` is followed by `cup`-contact and then by proprioceptive changes in `arm`, the neighborhood structure in $\mathcal{S}$ can support a stable triangulation between intended action, observed outcome, and body state. On this reading, "self" corresponds not to a single node but to a region (or subspace) of $\mathcal{S}$ that is repeatedly implicated by high-confidence internal–external correspondences across time, while "other" corresponds to regions that are only indirectly predictable or only sporadically coupled. This also clarifies how paraconsistency can be localized: incompatible evidence can be retained as multiple nearby candidates in $\mathcal{S}$ (a multi-modal neighborhood) until interaction reduces the ambiguity.

Remark 898 (Reading the example as a miniature theory of embodiment)**.** *The example shows how embodiment can be described without reducing "mind–world matching" to direct identity of variables. The correspondence f is justified not because* `pain` *and* `heat` *are the same thing, but because their* relational roles *in their respective networks align. In particular, what matters is that the incoming and outgoing constraints of the nodes (how they co-vary with other nodes, what they modulate, and what they are modulated by) are preserved up to the tolerance encoded by the enrichment: the match is a claim about structural position in a web of dependencies, not about shared intrinsic labels. This is precisely the structuralist spirit behind the morphism definitions. At the same time, the mention of the boundary illustrates how "self" is not merely the set of internal nodes: it is the subset that the system treats as self-evidenced, potentially with a nontrivial boundary region, echoing Self vs. Other (Hyperseed-Concept 165). One can read the "boundary region" as the locus where evidence is mixed or contested (e.g. partially self-generated and partially externally driven), so that self/other is not a crisp partition but a graded or even paraconsistent classification that can still support stable control policies. On this reading, embodiment is the practical situation in which the system must maintain a workable correspondence despite such mixed provenance, and the morphism formalism supplies a way to state when the correspondence is coherent enough to guide action.*

Summary of Hyperseed concepts handled in this section. The constructions above provide rigorous handles for:

- **Self/other boundary:** a (possibly paraconsistent) self-evidence function and induced self/other groupings. Here the function operationalizes "what counts as mine" as a rule for tagging states or flows, while allowing inconsistent or context-sensitive tagging without collapse.
- **Self-continuity:** existence (and degree) of high-weight approximate morphisms between successive internal pattern-flow networks. The weight (or enrichment value) makes continuity a matter of degree rather than an all-or-nothing identity, which matches the phenomenology of gradual change, interruption, and recovery.
- **Relative-information self-continuity:** proximity of time-slice networks under a scalar divergence. This provides a complementary summary statistic that can be compared across systems or scales, even when explicit node-to-node matching is underdetermined.
- **Reflective self-continuity:** flagged here, and developed further in Section 17 using self-model recursion. The key additional requirement is that some of the preserved structure is explicitly *about* the system's own continuity claims (i.e. meta-representations that constrain future updates).
- **Development:** continuity plus expanding capability profile. In the present language, capabilities correspond to families of morphisms or controlled transformations that become available, so development can be tracked as an enlarging repertoire together with sufficient diachronic coherence.
- **Pattern flow networks and mind–world correspondence:** approximate morphisms between internal and external flow networks. This treats "aboutness" as a map between relational structures (the dynamics of internal patterns and the dynamics attributed to the world), rather than as a static lookup table of symbols.

- **Space:** an enriched correspondence space on which mind/world embeddings can be compared. Concretely, this is the arena in which different candidate correspondences (or embeddings) can be measured against one another, so that "spatial" comparison is recovered as comparison of mappings by their induced distortions, costs, or constraint-violations.

Remark 899 (How these pieces compose with the rest of the document)**.** *Self/other boundary feeds into attention and control by determining what is treated as "inside" the locus of action (Sections 15 and 14). This matters because downstream optimization and resource allocation typically presuppose a boundary: credit assignment, risk attribution, and intervention selection all depend on which variables are treated as under the agent's remit versus merely observed. Self-continuity and development provide the diachronic scaffold needed for later discussions of reflective will and consciousness (Section 17), where closure and self-reference become explicit. In particular, without a notion of "same system over time" (even only approximately), recursive self-modeling would have no stable target, and reflective endorsement would be ill-posed. Mind–world correspondence is the bridge from internal pattern dynamics to external prediction and intervention, providing a quantale-valued measure of model adequacy that remains meaningful under vocabulary drift and paraconsistent evidence [19, 20]. The quantale-valued aspect is what allows adequacy to be aggregated and compared even when evidence is multi-graded or partially ordered (e.g. different kinds of constraint satisfaction that do not collapse to a single probability), and the paraconsistent tolerance is what prevents isolated contradictions from trivializing the evaluation. In this sense, the same enriched structure that supports "space" as a comparison domain also supports robustness of correspondence across re-parameterizations, relabelings, and incremental revision of the internal vocabulary.*

17 Consciousness, reflective will, and autonomy

17.1 Orientation: what gets formalized here

By this point in the reconstruction we have: (i) paraconsistent p-bit evidence values and a p-bit-quantale substrate (Section 3, especially Sections 3.2–3.4); (ii) pattern webs and habit dynamics (Sections 11–12); (iii) mind/representation/perception and semiotic modes (Section 13); (iv) prediction/control and temporal composition (Section 14); (v) attention and synergy as resource-allocation over competing processes (Section **??**); and (vi) self/continuity as a context-dependent boundary and a constraint on self-model evolution (Section 16).

Hyperseed's consciousness layer is not introduced as a new primitive, but as a *closure phenomenon* involving:

- *global accessibility* of some representational content ("what is present to experience"),
- *integration/coherence* across multiple subsystems (often mediated by resonance),
- *self-reference* in the weak (non-explosive) sense appropriate to paraconsistent settings,
- and *control leverage*: conscious contents can steer attention and action.

The goal of this section is to pin down a small mathematical interface for these ideas that is compatible with: (1) observer-relativity; (2) paraconsistency; and (3) resource sensitivity.

What is meant by "small interface" is that we are not trying to axiomatize everything that could be called conscious life, but to isolate the minimal structural commitments that let the earlier pieces talk to each other in a recognizably "conscious" regime. Concretely, the earlier machinery already provides (a) a semantics of graded, possibly inconsistent evidence; (b) a dynamics in which patterns stabilize into habits; and (c) an attentional scheduler that allocates limited computational and behavioral bandwidth. The present task is to specify when these ingredients collectively implement a *globally available, integrated, self-involving* representation that also has downstream causal consequences for control. In other words, consciousness will be treated as a particular kind of *functional closure* in the agent's ongoing inference–attention–action loop, rather than as an extra ontological ingredient.

It is important that all four bullets above are intended to be read in a way that respects paraconsistency. "Global accessibility" does not mean that all modules agree, or that a single classical "belief state" is broadcast without contradiction; rather, it means that certain representational items become accessible *as shared constraints* across multiple processes even when those processes maintain divergent or locally inconsistent p-bit valuations. Likewise, "integration/coherence" is not synonymous with full logical consistency: it is closer to the notion that, at the level relevant for control and report, the system achieves a stable working alignment between competing evidence streams, typically via resonance or other synergy mechanisms introduced earlier. A system can therefore exhibit conscious access to a content while still containing unresolved tensions (e.g., competing interpretations of the same stimulus), and the formalism must allow such tensions to be represented without collapse into triviality.

The emphasis on "control leverage" also ties this section to the forthcoming themes of reflective will and autonomy. A content is not merely conscious because it is present in some inner arena; it is conscious, in the present reconstruction, to the extent that it participates in the causal economy that selects policies, re-allocates attention, and reshapes habits. This point matters because the step from consciousness to autonomy is not made by adding a new faculty, but by showing how globally accessible and self-referential contents can *modulate* the very mechanisms that generate future attention distributions, action tendencies, and self-model updates. In this sense, reflective will is anticipated here as a special case of control leverage: the system can represent not only world-directed contents, but also the structure of its own decision situation (including conflicts, constraints, and higher-order preferences), and thereby change what it will do next under resource limitations.

Observer-relativity enters at two levels. First, "what is present to experience" is not treated as an absolute, system-independent predicate, but as indexed to a modeling context: different observers (or different internal meta-models) may disagree about which contents are globally accessible, or even about the granularity at which "contents" are individuated. Second, the closure phenomenon itself is evaluated relative to a chosen decomposition into subsystems and channels (perception, memory, valuation, motor control, etc.), so that the same underlying dynamics can be described at multiple scales. This is aligned with the earlier treatment of self/continuity as context-dependent: the boundary of "the experiencing system" and the boundary of "the conscious workspace" need not coincide, and both are subject to pragmatic constraints from modeling and control.

Resource sensitivity is similarly not an afterthought but a defining constraint. Global accessibility cannot be taken to mean unlimited broadcast; it must be implemented as a selective, capacity-limited mechanism that trades off precision, breadth, and speed. Integration is likewise bounded: coherence must be measured relative to limited attention and finite synergy budgets, rather than as an ideal limit of perfect mutual consistency. In paraconsistent terms, this means that the system may carry contradictions forward when the cost of resolving them exceeds the value, and that consciousness can co-exist with such tolerated inconsistency as long as the resulting closure remains stable enough to support coordinated action.

Remark 900. *Philosophically, the stance here is deliberately anti-mysterian: "consciousness" (Hyperseed-Concept 85) is treated neither as an ineffable substance nor as a mere synonym for behavior, but as a particular* organizational closure *in the dynamics of evidence, attention (Hyperseed-Concept 60), and control. The formalism is a kind of austere Russellian bookkeeping: we do not pretend to capture the whole phenomenology, but we insist that whatever consciousness is, it must show up as a stable pattern in the inferential and control economy of a bounded system [1, 19].*

17.2 A minimal interface: evidence states, workspace closure, and access

17.2.1 Languages and evidence states

Fix an observer/context Ob (a mind, or mind-plus-reality slice). Let $\mathcal{L}_{\mathrm{Ob}}$ be a finite (or at least well-founded) set of propositions the observer can represent at the current level of modeling granularity. Elements of $\mathcal{L}_{\mathrm{Ob}}$ may include:

- world propositions, e.g. "there is an apple on the table";
- bodily propositions, e.g. "the left arm is moving";
- internal propositions, e.g. "I feel tension" (self-report tokens in the sense of Section 13);
- relational propositions, e.g. "A tends to lead to B" (predictive implication from Section 14).

Definition 226 (p-bit evidence state)**.** *An* evidence state *for* Ob *is a map*

$$E : \mathcal{L}_{\mathrm{Ob}} \to \mathbb{V} = [0,1]^2, \qquad \varphi \mapsto E(\varphi) = (E^+(\varphi), E^-(\varphi)).$$

We order evidence states pointwise by the componentwise order on $\mathbb{V}$.

Remark 901. *Intuitively, an evidence state is the observer's current "epistemic posture" toward each representable proposition: $E^+(\varphi)$ is how much support there is for φ, while $E^-(\varphi)$ is how much support there is for "not-φ". The crucial choice is that we store* both *numbers explicitly rather than forcing them to sum to* 1 *(as in ordinary probability) or to be mutually exclusive (as in classical truth values). This is the* p-bit *move from earlier sections, aligned with constructive/paraconsistent semantics such as those explored in paraconsistent logics and their computational interpretations [23, 24].*

A simple example is a tiny language $\mathcal{L}_{\mathrm{Ob}} = \{\varphi, \psi\}$ where $E(\varphi) = (0.9, 0.1)$ (strongly supported) while $E(\psi) = (0.7, 0.8)$ (both supported and opposed). The latter represents a situation of genuine tension: the system has substantial evidence on both sides. The pointwise order means that $E \leq E'$ precisely when every proposition has no less *positive evidence and* no less *negative evidence in E' than in E; i.e. E' is, in this formal sense, "more evidenced" than E. This definition is useful because it makes the space of epistemic states into an ordered structure well-suited to monotone "closure" operators and fixed-point reasoning later in the section.*

Remark 902 (Paraconsistency is a feature)**.** *Allowing $(E^+(\varphi), E^-(\varphi))$ with $E^+(\varphi) + E^-(\varphi) > 1$ is essential to modeling: (1) borderline predicates ("soggy predicates" in Hyperseed's terminology; Hyperseed-Concept 172); (2) perceptual conflict (ambiguous or adversarial signals); (3) value conflict (next section; Hyperseed-Concept 198); and (4) self-referential modeling without explosion.*

Remark 903 (Granularity and well-foundedness)**.** *The "finite (or at least well-founded)" condition on $\mathcal{L}_{\mathrm{Ob}}$ is not intended to deny that agents can in principle form arbitrarily complex thoughts, but to delimit a* current *workspace-level interface at a given moment of modeling. In practice, $\mathcal{L}_{\mathrm{Ob}}$*

can be taken as the set of propositions currently expressible in the agent's active representational scheme (including compressed, schematic, or indexical propositions), while the well-foundedness assumption is what makes iterative closure constructions converge rather than chasing an infinite regress of ever-finer re-representations. This also matches the intended use of evidence states as interfaces*: they summarize what is currently at stake for the agent, not an exhaustive ontology of the world.*

17.2.2 Subsystem reports and attention-weighted integration

Let $\mathcal{M}$ be a finite index set of concurrently running subsystems/modules for Ob (perception streams, memory retrieval, imagination, verbal thought, somatic monitoring, etc.). Each module produces its own evidence state.

Definition 227 (Module evidence and attention weights)**.** *For each $m \in \mathcal{M}$, let*

$$E_m : \mathcal{L}_{\mathrm{Ob}} \to \mathbb{V}$$

*be the evidence state generated by module m. Let $a : \mathcal{M} \to [0,1]$ be an attention allocation, with $a(m)$ interpreted as the fraction of available cognitive resource dedicated to m (cf. Section **??**).*

Remark 904. *This definition separates two things that are often conflated in informal discussion: (i) what a module computes (its evidence state E_m), and (ii) how much the system is currently "listening" to that module (the allocation $a(m)$). A classical cognitive-science picture might call this a distinction between local processing and global broadcast; Hyperseed frames it in terms of genenergy bottlenecks and attentional focus (Hyperseed-Concept 60) [19].*

As a small example, one can imagine $\mathcal{M} = \{vision, memory\}$ and a proposition $\varphi =$ "there is a cat". Vision might output $E_{vision}(\varphi) = (0.8, 0.2)$ while memory, primed by a recent cat discussion, outputs $E_{memory}(\varphi) = (0.6, 0.4)$. If attention is mostly on vision, say $a(vision) = 0.9$, then vision dominates the integrated story; if attention shifts to memory, the integrated state changes without any change in the modules' intrinsic computations. This is useful because it gives a precise handle on the idea that "conscious content depends on attention" without reducing either to the other.

To combine module reports we use the p-bit-quantale operations. We need a way to treat a real attention weight as a quantale element.

Definition 228 (Scalar embedding into p-bit values)**.** *For $\alpha \in [0,1]$ define its* diagonal embedding *into $\mathbb{V}$ by*

$$\overline{\alpha} := (\alpha, \alpha) \in [0,1]^2.$$

Remark 905. *The diagonal embedding is the simplest way to regard a scalar "strength" α as affecting both positive and negative evidence magnitudes uniformly. It is not the only possibility, but it is the most conservative: α is treated as a resource-like multiplier rather than as a truth-value bias. For instance, $\overline{0.5} \otimes E_m(\varphi)$ will (given the* p-bit*-quantale tensor from Section 3.4) act like scaling down the module's contribution in a symmetric way.*

Remark 906 (Why uniform scaling is the right default)**.** *Uniform scaling is appropriate when attention is interpreted as* bandwidth or gain *rather than as* valence*. In that reading, allocating less attention to a module reduces the magnitude of whatever it would have contributed, regardless of whether that contribution is for or against a given φ. This matches the intended role of attention in a minimal interface: it gates access to the workspace rather than directly changing the module's internal epistemic stance. More expressive embeddings (e.g. scaling E^+ and E^- differently, or mixing them) can be treated as additional modeling assumptions about bias, framing, or motivated cognition, but are not required for the basic global-workspace story developed here.*

Definition 229 (Attention-weighted integration (pre-closure)). *Given module evidence states $\{E_m\}_{m\in\mathcal{M}}$ and attention weights a, define the* integrated pre-closure evidence state E_{pre} *by*

$$E_{\mathrm{pre}}(\varphi) \;:=\; \bigoplus_{m\in\mathcal{M}} \overline{a(m)} \otimes E_m(\varphi), \qquad \varphi \in \mathcal{L}_{\mathrm{Ob}},$$

where $\oplus$ and $\otimes$ are the p-bit*-quantale join and tensor (Section 3.4), applied pointwise over $\mathcal{L}_{\mathrm{Ob}}$.*

Remark 907. *The choice of $\oplus$ in the integration step implements a minimal "accumulation of support" picture: the workspace-facing state records whatever evidence is currently contributed by any attended module, without forcing premature reconciliation of conflicts. Because $\oplus$ is monotone and associative, increasing attention to a module or strengthening a module's own evidence state can only move E_{pre} upward in the pointwise order (i.e. toward "more evidenced"). This monotonicity is one of the reasons for setting the interface up in an order-theoretic way: it ensures that later closure steps (which will also be monotone) can be studied using fixed-point and convergence arguments rather than ad hoc case analyses.*

Note also that no normalization condition such as $\sum_{m\in\mathcal{M}} a(m) = 1$ is required for the definition to make sense: $a(m)$ is interpreted as a resource fraction relative to a maximal per-module gain, and the quantale operations already enforce the codomain constraint to $[0,1]^2$. If one does impose a normalization constraint, it can be viewed as an additional model of limited global resource, but it is not logically necessary for defining E_{pre}.

Remark 908 (Integration does not resolve contradiction). *Even after integration, it is possible (and often expected) that $E_{\mathrm{pre}}(\varphi)$ has both large positive and large negative components. This is not a bug: on the present view, "what is globally available" is not identical to "what is already consistent". The workspace can contain tensions (e.g. perceptual ambiguity, competing predictions, or value conflict) and the next modeling step—workspace closure—is where inferential propagation and constraint satisfaction are represented. Keeping contradiction visible at the interface level is precisely what prevents the system from spuriously collapsing into a single forced narrative when the underlying evidence is genuinely mixed.*

Remark 909. *This is useful because attention is meant to be a* capacity allocation *(genenergy budget) more than a semantic commitment: turning down attention should reduce how strongly a module's results enter the global workspace, regardless of whether those results are pro-φ or anti-φ. In later sections, one can generalize this to non-diagonal embeddings to model systematic biases (e.g. anxiety amplifying negative evidence), but the diagonal case provides a clean baseline. In particular, the diagonal case isolates the purely "gain control" role of attention: it scales a module's contribution without reinterpreting its internal polarity structure. This matters when one wants the algebra to distinguish* what *is represented (the evidence values themselves) from* how strongly *it competes for global access (the weight), so that attenuation can be understood as lowering broadcast priority rather than as a change of mind about φ.*

Definition 230 (Attention-weighted aggregation operator). *Define the integrated evidence state $\Gamma_a(\{E_m\}_{m\in\mathcal{M}})$ by*

$$(\Gamma_a E_\bullet)(\varphi) := \bigoplus_{m\in\mathcal{M}} \overline{a(m)} \otimes E_m(\varphi),$$

where $\oplus, \otimes$ are the join and tensor of the p-bit *quantale (Section 3.4), applied pointwise in φ.*

Remark 910. *Notation check: $E_\bullet$ is a convenient "bundle" notation for the family $\{E_m\}_{m\in\mathcal{M}}$, and $(\Gamma_a E_\bullet)(\varphi)$ denotes the $\mathbb{V}$-value assigned to proposition φ after aggregating all module contributions* *with weights $a(m)$. The symbol $\bigoplus$ is the quantale join over the finite index set $\mathcal{M}$, and the tensor $\otimes$ is the quantale's multiplicative operation; both act in $\mathbb{V} = [0,1]^2$ and here are applied separately for each proposition φ ("pointwise in φ").*

It is helpful to keep in view the ambient order: evidence states are compared pointwise *in φ, and values in $\mathbb{V} = [0,1]^2$ are compared componentwise (so "more evidence" means no less positive support and no less negative support, as determined by the order specified in Section 3.4). With this convention, both $\oplus$ and $\otimes$ are monotone in each argument, which ensures that increasing a module's evidence or increasing its attention weight cannot decrease the integrated result. This monotonicity is one of the reasons for using quantale operations rather than an ad hoc averaging rule.*

The overline in $\overline{a(m)}$ is doing conceptual work: it is the embedding that turns an attention scalar (a resource or gain parameter) into an element of the evidence-value algebra so that it can interact with $E_m(\varphi)$ via $\otimes$. In the diagonal baseline, this embedding treats positive and negative channels symmetrically, so that "listening less" to a module downweights its entire *evidential profile rather than selectively distorting one channel.*

Intuitively, Γ_a is a formal version of "what you get when you listen to multiple voices according to how much attention you grant them." A minimal example with two modules $m = 1, 2$ gives

$$(\Gamma_a E_\bullet)(\varphi) = \overline{a(1)} \otimes E_1(\varphi) \ \oplus \ \overline{a(2)} \otimes E_2(\varphi).$$

In this two-module case one can already see the separation between within-module *structure (whatever constraints relate $E_m(\varphi)$ across different φ) and* between-module *arbitration (implemented by $\oplus$ after attention-weighting). The join $\oplus$ should be read as an "integration" or "pooling" operation internal to the chosen semantics: it may behave like a max-like merge, a bounded sum, or another quantale join, but in all cases it provides a principled way of expressing that multiple sources can contribute to the global evidential standing of the same proposition.*

The utility of this definition is compositionality: once module outputs and attention weights live in the same algebraic setting, later "workspace closure" can be written as a map on evidence states without having to mention the modules again. Equally importantly, Γ_a makes explicit what parts of the story are architectural *(the choice of $\mathcal{M}$, the attention allocation a, and the integration rule) and what parts are* semantic *(the evidence values that the modules compute). This distinction will matter when comparing different hypotheses about deficits of access: e.g. whether a failure is due to low attention (small $a(m)$), weak local evidence (small $E_m(\varphi)$), or the downstream effects of workspace dynamics (the behavior of W below).*

Remark 911 (Interpretation)**.** *This makes explicit a Hyperseed-compatible picture:* conscious-accessible content is a product of representation plus attention*. Low-attention modules may still compute rich content, but it will not appear strongly in the integrated state unless it is "amplified" by attention. One can also read this as a minimal formal analogue of the common distinction between* availability *(a module has computed something) and* access *(that content is poised to influence report, planning, or control): Γ_a converts a distributed availability profile $\{E_m\}$ into a single access-oriented evidential profile by explicitly modulating each module with its current attentional gain. On this view, attention does not create representational content; it changes which already-computed contents become decision-relevant in the global aggregate.*

17.2.3 Global accessibility and workspace closure

Hyperseed's "conscious/unconscious" contrast can be modeled as a distinction between:

- content that is globally accessible for downstream reasoning/control/report; and

- content that exists in local modules but is not globally broadcast.

We model this using a *workspace closure* operator. The intended separation is thus not between two different ontological kinds of content, but between two regimes of *functional connectivity*: some information remains encapsulated (local), whereas some is placed into a format that can be consumed by many downstream processes (global).

Definition 231 (Workspace operator)**.** *A* workspace operator *is a monotone map*

$$W : \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}} \to \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}}$$

intended to represent the effect of "global broadcasting" and "cross-module stabilization": given an integrated evidence state E, the workspace produces a refined state $W(E)$ in which globally accessible propositions may be mutually reinforced, disambiguated, or inhibited.

Remark 912. *Notation check: $\mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}}$ denotes the set of all functions $E : \mathcal{L}_{\mathrm{Ob}} \to \mathbb{V}$, i.e. all evidence states over the language. Thus W is an* operator on evidence states*. Monotonicity means: if $E \leq E'$ pointwise (every proposition has no less evidence in E'), then $W(E) \leq W(E')$.*

Intuitively, W stands in for whatever mechanisms make some contents "globally present": recurrent stabilization, cross-checking, binding, report preparation, etc. A very simple example is a workspace rule that enforces a single implication "φ implies ψ": it could map a state E to a new state $W(E)$ that increases $E^{+}(\psi)$ whenever $E^{+}(\varphi)$ is high. More generally, one can think of W as encoding a family of (possibly soft) constraints among propositions in $\mathcal{L}_{\mathrm{Ob}}$ that only become effective once information is broadcast into a shared arena: consistency pressures, binding constraints (e.g. linking a feature to an object), winner-take-most competitions, or normalization steps that reduce ambiguity by distributing support across mutually exclusive hypotheses.

The definition is useful because it packages all such cross-module interactions into one map, so that consciousness can be analyzed as a property of the map's fixed points rather than as a mysterious extra ingredient. In particular, the map-based view allows one to ask structural questions about the workspace independently of any specific module architecture: e.g. whether W tends to sharpen evidence into stable patterns, whether it admits multiple distinct stable states (supporting a notion of multistability), and how sensitive stable states are to small changes in the integrated input.

It is also useful to note that this is deliberately stated at the level of a single-step *operator. In applications, one may imagine iterating W (or iterating W interleaved with renewed integration) until changes become negligible, and then treating the resulting stabilized state as the workspace-resolved outcome. The fixed-point formulation below captures the equilibrium notion without committing to any particular implementation of the iterative dynamics.*

Remark 913 (What monotonicity means here)**.** *Monotonicity is the minimal sanity constraint: if all modules provide at least as much positive/negative evidence for every proposition, then the workspace should not reduce evidence. More sophisticated models may introduce inhibitory competition via additional state variables, but the present reconstruction stays deliberately lightweight. In this lightweight setting, inhibition is modeled (when needed) not by literally decreasing evidence when inputs increase, but by how the workspace may preferentially* increase *some propositions more than others in response to the same increment of input, thereby changing comparative salience while preserving the basic "more in, no less out" ordering. This keeps the formalism compatible with the idea that the workspace is a stabilizing broadcast mechanism rather than an adversarial filter that destroys information.*

Definition 232 (Conscious content as a fixed point). *An evidence state E is* workspace-stable *if $W(E) = E$. We define a* conscious evidence state *to be a workspace-stable state that is reachable from module integration, i.e. of the form*

$$E = W\big(\Gamma_a(E_\bullet)\big) \quad \textit{for some module collection } E_\bullet \textit{ and attention } a.$$

Remark 914. *The fixed-point condition $W(E) = E$ is the formal avatar of "closure": once the content is broadcast and mutually adjusted, it does not immediately change under the same broadcasting dynamics. In other words, a conscious evidence state is not merely a momentary aggregate, but a* self-consistent global posture *of the system's representational economy. This aligns with the Hyperseed emphasis that consciousness is stabilized accessibility, not a raw data dump. The additional reachability condition (being of the form $W(\Gamma_a(E_\bullet))$ for some inputs) prevents the definition from treating* every *abstract fixed point of W as conscious: it ties workspace-stable states to those that can actually be produced from the system's modules under some attention allocation. Conceptually, this separates "mathematical equilibria" from "attainable equilibria," mirroring the architectural idea that conscious contents arise from the interaction between what modules deliver and what the workspace can stabilize and broadcast.*

Finally, note that the definition leaves open whether conscious states are unique: different attentional settings a, or different module bundles $E_\bullet$ (e.g. reflecting different sensory inputs or internal memory retrievals), may lead to different workspace-stable outcomes. This provides a clean place to later discuss phenomena like context dependence, priming, and multistable perception as differences in which reachable fixed point (or approximate fixed point) is selected.

Remark 915. *As an example, suppose W enforces that "if pain is present then I am aware of pain" by boosting a proposition I(pain) whenever pain is supported (we implement this more explicitly below with introspection). One can think of this as a minimal "linking rule" from first-order content to a matching higher-order access claim: W does not create pain evidence ex nihilo, but it ensures that, once pain has become salient enough to enter the workspace dynamics, the corresponding access marker is also made available for downstream use (e.g., for verbal report, control, or memory routing). Then a workspace-stable state cannot have strong pain evidence but zero evidence for awareness of pain; closure forces a consistency relation at the level of access. In this sense, workspace closure behaves like an access-level coherence condition: it suppresses configurations in which the system would be poised to act as if pain is present while simultaneously lacking any globally available representation that it is aware of pain. This is intentionally weaker than an identity claim between pain and awareness-of-pain; it is merely a constraint on which evidence profiles can be stable once W has done its organizing work. The definition is useful because it lets us separate two questions: (i) what modules compute and attention selects (Γ_a), and (ii) what becomes globally organized and reportable (W). Equivalently, Γ_a concerns the formation and prioritization of candidate contents, whereas W concerns the system-level integration that turns some of those contents into accessible, mutually supporting commitments that can guide centralized decision and communication.*

Remark 916 (Conscious vs unconscious). *Given a workspace-stable state E, one can define (for a chosen threshold $\theta \in (0,1)$) the* conscious proposition set

$$\mathrm{Con}_\theta(E) := \{\varphi \in \mathcal{L}_{\mathrm{Ob}} : E^+(\varphi) \geq \theta\}$$

and treat the complement as "unconscious" at that moment. Formally, $\mathrm{Con}_\theta(E)$ is monotone in the expected direction: if $\theta' > \theta$ then $\mathrm{Con}_{\theta'}(E) \subseteq \mathrm{Con}_\theta(E)$, so raising the bar yields a stricter notion of what counts as conscious. This allows one to model "degrees" of access without changing the *underlying evidence state: the same E can support a thin conscious set (high θ) or a thick one (low θ), depending on how demanding we want the operational criterion to be. This is not meant as a metaphysical claim; it is a modeling move that operationalizes the conscious/unconscious boundary in observer-relative terms. In particular, the threshold is a tool for linking the graded internal quantities $E^+(\varphi)$ to binary behavioral predicates (e.g., "reportable now" or "available for deliberate choice") that often appear in experiments and in ordinary descriptions.*

Remark 917. *The thresholding move is deliberately pragmatic: it turns graded evidence into a crisp set that can be used to define reports, action conditions, or "what is currently in mind". It also makes explicit that the model is not committed to a sharp boundary in the mechanism itself: the sharpness is introduced by the analyst as a way to interface a continuous internal description with discrete explanatory targets such as yes/no report, categorical memory encoding, or rule-based action selection. Different observers or modeling contexts may choose different θ, reflecting different granularities of introspective access and different criteria for "present to experience." For example, a coarse-grained behavioral readout may justify a higher θ (only very strong evidence counts as conscious), while a fine-grained phenomenological or neural readout may justify a lower θ (allowing weak but systematic availability to count). This observer-relativity is not a bug: it is consistent with the broader Hyperseed stance that many ontological boundaries are context-indexed constructions rather than absolute partitions. In that spirit, the model can also accommodate intermediate categories (e.g., "fringe" or "preconscious" contents) by considering multiple thresholds or by tracking how close $E^+(\varphi)$ is to θ, without changing the underlying definition of workspace stability.*

17.3 Reflective consciousness as higher-order representational closure

Hyperseed distinguishes ordinary consciousness from *reflective consciousness*. The simplest way to capture the difference is:

- ordinary consciousness: workspace-stable integration of world/body/internal content;
- reflective consciousness: workspace-stable integration that includes *meta-content* about the agent's own representations and attention.

Remark 918. *This contrast parallels a familiar philosophical distinction between merely having experiences and also* taking oneself *to be having them. But the present reconstruction refuses to treat this as an unanalysable "light" added on top of cognition. Instead, reflective consciousness is characterized by adding a further layer of representational tokens and requiring that the global workspace stabilize* including *those tokens. In Hyperseed terms, reflective consciousness is a particular kind of stable self-reference enabled by paraconsistency rather than forbidden by it (compare Hyperseed-Concept 84 and 166).*

17.3.1 Adding a reflective layer

Introduce a syntactic operator $\mathsf{I}(\cdot)$ producing "introspective" propositions. For each $\varphi \in \mathcal{L}_{\mathrm{Ob}}$, $\mathsf{I}(\varphi)$ is read as "Ob is aware of φ" (or "φ is present to experience").

Definition 233 (Reflective language extension)**.** *Define the* reflective extension *of the language by*

$$\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}} := \mathcal{L}_{\mathrm{Ob}} \ \cup \ \{\mathsf{I}(\varphi) : \varphi \in \mathcal{L}_{\mathrm{Ob}}\}.$$

An evidence state on $\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}$ is a map $E : \mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}} \to \mathbb{V}$.

Remark 919. *Intuitively, $\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}$ doubles the vocabulary: for every base proposition φ we include a corresponding meta-proposition saying "φ is in awareness". This is not claimed to be psychologically exhaustive; it is a minimal representational hook that lets the model distinguish between (i) a proposition being supported somewhere in the system and (ii) that proposition being* accessible as experienced content*. In Russellian terms, we introduce a new set of propositions that talk about the* status *of propositions, thus allowing the theory to make "aboutness of aboutness" explicit.*

A simple example: if φ is "there is an apple" then $\mathsf{I}(\varphi)$ is "I am aware that there is an apple" or "the apple is present to experience." The definition is useful because it makes reflective consciousness an ordinary fixed-point/stability question in an enlarged evidence space, rather than a sui generis phenomenon.

Definition 234 (Introspection operator)**.** *An* introspection operator *is a monotone map*

$$\mathsf{Int} : \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}} \to \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}}$$

that assigns evidence to introspective propositions based on base evidence. A minimal choice is:

$$\big(\mathsf{Int}(E)\big)\big(\mathsf{I}(\varphi)\big) := \big(E^{+}(\varphi),\ 1 - E^{+}(\varphi)\big),$$

*but more realistic choices incorporate attention (Section **??**) and coherence (resonance, Section 3.9).*

Remark 920. *The minimal choice says: "awareness of φ tracks positive support for φ." It is intentionally crude, but it has two virtues: it is monotone, and it makes introspective support computable from the same evidence quantities already present. More elaborate* Int *can (and in realistic modeling should) depend on attention and workspace organization: one may have high $E^{+}(\varphi)$ in a local module but low evidence for $\mathsf{I}(\varphi)$ if attention is elsewhere.*

For example, let φ be "background music is playing." A perception module may support φ strongly, yet if the system is deeply focused on a math problem, attention to that stream is low and introspective awareness of the music may be absent. The usefulness of Int *is that it gives a formal bridge from first-order content to meta-content, enabling a precise statement of reflective closure and self-reference in a paraconsistent setting [23, 24].*

Remark 921. *It is also important that $\mathsf{I}(\cdot)$ is treated here as a* representational *operator rather than a factive epistemic one: $\mathsf{I}(\varphi)$ is not read as "φ is true and known to be true," but as "φ is tokened in a way that is globally available as experience." This matches the intended use in global-workspace-style models, where what matters for consciousness is accessibility and stabilization, not guaranteed veridicality. Accordingly, $\mathsf{I}(\varphi)$ can be present even when φ is false, and (conversely) φ can be well-supported in a peripheral subsystem without $\mathsf{I}(\varphi)$ being supported.*

Remark 922 (Paraconsistent self-reference)**.** *Nothing prevents the model from supporting evidence for both $\mathsf{I}(\varphi)$ and $\mathsf{I}(\neg\varphi)$ simultaneously. This is a direct formal analog of "I am aware of conflicting impressions" without collapse into triviality.*

17.3.2 Reflective workspace closure and a fixed-point existence theorem

Let W^{ref} be a workspace operator on the reflective language:

$$W^{\mathrm{ref}} : \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}} \to \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}}.$$

Combine module integration, introspection, and workspace closure into a single update map.

Definition 235 (Reflective update map)**.** *Fix attention a and module evidence $E_\bullet$ on $\mathcal{L}_{\mathrm{Ob}}$. Define the* reflective update *map*

$$F := W^{\mathrm{ref}} \circ \mathsf{Int} \circ \Gamma_a,$$

with codomain $\mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$.

Remark 923. *The intended reading of the composition is operational: Γ_a performs attention-weighted integration of first-order contents into a candidate workspace state;* Int *then generates (or updates) the corresponding introspective tokens $\mathsf{I}(\varphi)$; finally, W^{ref} imposes the global-workspace constraint* on the extended state *in which first-order and meta-level tokens co-determine stability. On this picture, reflective consciousness is not a separate process added after ordinary integration, but a closure condition for a larger state space in which awareness-claims are among the stabilized representations.*

Definition 236 (Reflective closure / reflective consciousness state)**.** *An evidence state $E^* \in \mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$ is* reflectively closed *(with respect to F) if it is a fixed point:*

$$F(E^*) \;=\; E^*.$$

When such a fixed point is reached (or approximately reached under iterative dynamics), the model treats the agent as being reflectively conscious in the sense that the workspace has stabilized over both object-level content and the corresponding introspective/meta-content.

Remark 924. *This definition makes the higher-order aspect precise: what is stabilized is not only φ-tokens about the world/body/internal state, but also $\mathsf{I}(\varphi)$-tokens about what is* present. *In particular, the model distinguishes (a) stable first-order representations without stable $\mathsf{I}(\cdot)$ support (a formal analog of "processing without reportable experience") from (b) stable joint support for φ and $\mathsf{I}(\varphi)$ (a formal analog of reportable, globally accessible experience). Because the workspace operates over $\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}$, the availability of meta-content can feed back into which first-order propositions remain stable, capturing the intuitive idea that "noticing" something can itself reorganize what is experienced.*

Theorem 15 (Fixed-point existence under monotonicity)**.** *Assume Γ_a,* Int*, and W^{ref} are monotone with respect to the pointwise order on evidence states (inherited from the order on $\mathbb{V}$). Then the reflective update map*

$$F = W^{\mathrm{ref}} \circ \mathsf{Int} \circ \Gamma_a$$

is monotone, hence (on the complete lattice $\mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$) admits at least one fixed point. Moreover, F has a least fixed point and a greatest fixed point.

Remark 925. *The theorem expresses the "closure" idea in a mathematically standard way: if the dynamics are order-preserving, then reflective stabilization is guaranteed in the sense of fixed-point existence (even when the contents stabilized may be paraconsistent). Interpretively, the least fixed point can be read as the minimal reflectively closed workspace compatible with the update rules, while larger fixed points represent richer stabilized profiles of both object-level and introspective tokens. In applications, one typically studies the iteration $E_{t+1} := F(E_t)$ and asks when (and in what sense) it converges, using the fixed points as attractors or equilibrium candidates for reflectively conscious episodes.*

Remark 926. *This is the formal "assembly line" of reflective consciousness in one equation. Starting from distributed module reports, Γ_a builds an integrated base evidence state; then* Int *adds meta-evidence about what is present to awareness; then W^{ref} applies the global stabilization/broadcasting* *dynamics in the enlarged language. The composition order matters: it reflects a modeling judgment that introspective tokens are built from base content (and its attention-mediated integration) and then are themselves subject to workspace-level reinforcement, inhibition, and coherence constraints.*

A minimal example is: a sensory module yields evidence for φ; attention is high; introspection adds $\mathsf{I}(\varphi)$; the reflective workspace then may propagate this to further tokens like "I am attending to φ" (if included in the language) or may suppress contradictions. The usefulness of the definition is that it makes reflective consciousness amenable to standard order-theoretic tools: we can ask when F (or a variant) has fixed points and how they are selected. Moreover, reading the pipeline left-to-right helps separate three distinct roles that are often run together in informal discussions: (i) integration *of first-order contents (modeled by Γ_a), (ii)* re-representation *of those contents as available-to-awareness (modeled by* Int*), and (iii)* global constraint satisfaction and propagation *over the resulting mixed vocabulary (modeled by W^{ref}). In particular, the model does not assume that introspection is infallible or "transparent": since we work in an evidence lattice,* Int *can add graded support and/or graded counterevidence for reflective propositions, allowing misintrospection, uncertainty about one's own states, or even simultaneous partial evidence for and against $\mathsf{I}(\varphi)$ without trivialization.*

To speak about fixed points we consider the complete lattice $\mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$ with pointwise order. Concretely, an element of $\mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$ assigns to each sentence $\psi \in \mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}$ a pair of values in $\mathbb{V} = [0,1]^2$ (e.g. degrees of support and counter-support), and the pointwise order compares such assignments proposition-by-proposition. This is the natural setting for treating reflective updates as global operators on entire evidence *profiles*, rather than as isolated updates to single tokens.

Theorem 16 (Existence of reflective conscious fixed points)**.** *Assume W^{ref} and* Int *are monotone and $\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}$ is finite (or more generally that $\mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$ is a complete lattice). Then the operator*

$$\widehat{W} := W^{\mathrm{ref}} \circ \widehat{\mathsf{Int}} \quad \textit{on} \quad \mathbb{V}^{\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}}$$

admits at least one fixed point, where $\widehat{\mathsf{Int}}$ is any monotone extension of introspection that can read reflective propositions as input (e.g. by ignoring them or by treating them as additional signals). In particular, $\widehat{W}$ has a least fixed point and a greatest fixed point.

Remark 927. *In plain language, the theorem says: if your reflective workspace update rule is monotone, then "reflectively stable" global states are guaranteed to exist. This is important because self-reference is often thought to invite paradox; in classical logic, naive self-reference can indeed be explosive. Here, by working in an ordered paraconsistent evidence lattice and using monotone closure, we obtain an existence guarantee rather than a contradiction. This theorem thus connects directly to the Hyperseed theme that "weak" (non-explosive) forms of self-reference are not only possible but structurally natural in bounded cognitive systems [2, 3]. A useful way to read the monotonicity requirement is as a "no retraction of evidence by fiat" constraint: when the input state is informationally larger (more support and/or more counterevidence, in the lattice order), the output is not allowed to become informationally smaller. This captures, at an abstract level, the idea that the workspace dynamics are a kind of closure or propagation process rather than a capricious rewrite rule; it is precisely this closure-like behavior that makes fixed-point existence the default expectation rather than a delicate special case.*

The result also connects forward: later, when we talk about reflective will and autonomy as fixed points of meta-control maps, we again rely on the idea that monotone self-modification dynamics can have stable regimes; compare the use of fixed-point theorems to analyze stability of self-modifying goal systems [10]. In that later application, the presence of multiple fixed points should be interpreted *as a formal analogue of multiple admissible "self-stabilizing" agent regimes (e.g. different coherent reflective stances), with least/greatest fixed points providing extremal anchors among them.*

Proof. The set $\mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}}$ is a complete lattice under pointwise order because $\mathbb{V} = [0,1]^2$ is a complete lattice under componentwise order and products of complete lattices are complete. By monotonicity, Tarski's fixed point theorem applies, yielding existence of least and greatest fixed points of $\widehat{W}$. One may also note that the "finite language" hypothesis is a convenient sufficient condition: if $\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}$ is finite, then the product lattice is automatically complete, so no additional set-theoretic assumptions are needed to ensure existence of arbitrary joins and meets. □

Remark 928. Proof sketch. *The strategy is purely order-theoretic: build a big ordered space of possible reflective evidence states, check that it has enough completeness (all joins/meets exist), and then invoke the general fixed-point principle that monotone maps on complete lattices must have fixed points. No special properties of the particular* p-bit *operations are needed beyond completeness and monotonicity. This robustness is part of the point: the existence claim is intended to survive a wide range of choices for how evidence is aggregated, so long as those choices respect the underlying order.*

The key step is recognizing $\mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}}$ as a product lattice: each proposition contributes a copy of $[0,1]^2$, and the product of complete lattices is complete. Once this is in place, Tarski's theorem does the heavy lifting. If one wants a visual intuition, imagine iterating $\widehat{W}$ from the "least evidence" state and watching evidence accumulate monotonically until it can no longer increase; this limiting state is the least fixed point. Dually, iterating downward from the top element yields the greatest fixed point. In many cognitive readings, the least fixed point can be interpreted as the minimally committed reflective stabilization compatible with the dynamics (only what is forced by closure), whereas the greatest fixed point represents a maximally saturated stabilization (everything that can be maintained without violating the order constraints). Nothing in the theorem requires that the fixed point be unique; rather, the lattice-theoretic framework anticipates the possibility of multiple stable reflective equilibria, which can be selected among by additional constraints (e.g. cost, simplicity, or policy-level preferences) not encoded in monotonicity alone.

Remark 929 (Interpretation)**.** *Theorem 16 is a small but useful "sanity anchor": it ensures that if one models reflective consciousness as a closure process on a paraconsistent evidence lattice, then reflective conscious states exist as fixed points. This matches the Hyperseed intuition that reflective awareness is a* stabilized *regime of self-referential modeling rather than a one-shot predicate. It also clarifies what kind of mathematical object a "reflective conscious state" is in this setup: not a single privileged sentence, but a whole valuation over an enriched language that includes both world-directed claims and (possibly fallible) claims about what is present to awareness. Finally, the extremal fixed points provided by Tarski offer canonical reference states even when empirically plausible dynamics admit many equilibria; they supply baseline targets for subsequent questions about selection, robustness under perturbation, and the effect of adding further reflective operators to $\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}$.*

17.4 Resonance and coherence as stability selectors

In Hyperseed, resonance functions both as: (i) a dynamical amplifier (habit reinforcement, morphic resonance; Section 12); and (ii) an experiential signature of coherence vs conflict (motivational resonance; see also the core construction in Section 3.9).

Here we use resonance as a *selector* among the fixed points guaranteed by Theorem 16. Concretely, the existence of multiple fixed points can be read as the formal analogue of multiple internally stable "interpretations" or "self-consistent closures" of the same evidential situation; resonance is then the additional principle that ranks (or biases trajectories toward) some closures rather than others. In other words, the closure theorem supplies *possibility* (stable self-maps exist), while resonance supplies *preference* (some stable self-maps are lived as coherent, others as fragmented or tense).

Remark 930. *The existence theorem above is generous: a monotone closure map may have many fixed points. Phenomenologically, however, consciousness is not an arbitrary fixed point; it has a felt signature of coherence, dissonance, tension, and release (Hyperseed-Concept 159 and 97). The role of resonance here is to provide a principled mechanism for preferring some stable closures over others, echoing the broader Hyperseed account of pattern emergence and reinforcement (Hyperseed-Concept* **??**, *130, and 115) [5, 13]. A useful way to think about this is as a two-stage constraint: (a) a workspace dynamics imposes logical and evidential closure, producing a set of fixed points compatible with monotonic updating; (b) a resonance criterion then acts like a Lyapunov-style or energy-style selector that makes some fixed points attractors (or makes them more robust against perturbation), while others remain merely mathematically admissible. This distinction mirrors the everyday fact that many "interpretations" of oneself and the world are compatible with the available evidence, yet only some are experienced as settling, integrated, or action-guiding.*

17.4.1 Coherence scores from paraconsistent evidence

Assume we have a logic-to-complex mapping σ_C as in Section 3.9. For an evidence state E and a chosen feature map Φ from propositions to vectors (e.g. by stacking $\sigma_C(E(\varphi))$ for a designated list of propositions), define a coherence score. The key modeling move is that σ_C allows one to treat paraconsistent structure as a signal in a continuous space, so that "coherence" can be computed, compared, optimized, and coupled to dynamics, rather than remaining a purely qualitative label.

Definition 237 (Coherence functional)**.** *Let $\mathcal{K} \subseteq \mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}$ be a finite set of propositions regarded as the "currently relevant workspace contents" (this may depend on attention and task). Define*

$$z_E \in \mathbb{C}^{|\mathcal{K}|}, \qquad (z_E)_\varphi := \sigma_C\big(E(\varphi)\big),$$

and define the coherence functional

$$\mathrm{Coh}(E) := \frac{1}{|\mathcal{K}|} \sum_{\varphi \in \mathcal{K}} |(z_E)_\varphi|^2.$$

Remark 931. *Notation check: $\mathcal{K}$ is a finite index set of propositions; $|\mathcal{K}|$ is its cardinality. The vector z_E has one complex coordinate for each $\varphi \in \mathcal{K}$, obtained by applying σ_C to the* p-bit *value $E(\varphi)$. The modulus $|(z_E)_\varphi|$ is the usual complex absolute value, and $\mathrm{Coh}(E)$ is the mean squared magnitude over the chosen proposition set.*

Intuitively, this turns a structured paraconsistent evidence configuration into a single scalar that can play the role of a "stability" or "coherence" signal. A simple example: if σ_C maps high net support to large real magnitude, then a workspace state with many strongly supported propositions yields high $\mathrm{Coh}(E)$. If instead σ_C maps tension to imaginary magnitude, then a state full of conflicts also yields high $\mathrm{Coh}(E)$ (but in a different complex direction), allowing the modeler to decide whether to favor or penalize tension, as in paraconsistent resonance constructions [24]. This is useful because it provides an explicit numerical criterion to prefer one fixed point over another, rather than appealing to an informal "coherence" notion.

One can also unpack the contribution of each proposition: if $(z_E)_\varphi = a_\varphi + ib_\varphi$, then $|(z_E)_\varphi|^2 = a_\varphi^2 + b_\varphi^2$. Thus the chosen σ_C implicitly defines which aspects of evidential state count as "large": pure support-dominant regimes correspond to a_φ dominating, while conflict/energy-dominant regimes correspond to b_φ dominating. The squared magnitude is convenient because it is additive across coordinates and insensitive to global phase conventions, but it is not the only possible choice; it is simply a minimal functional that is (i) nonnegative, (ii) decomposable across propositions, and (iii) compatible with treating σ_C as a feature embedding.

Remark 932 (Reading). *Large $|(z_E)_\varphi|$ corresponds (by design of σ_C) to either strong net support (real part) or strong paraconsistent tension/energy (imaginary part). Depending on modeling goals, one can prefer net support, prefer "energized tension", or penalize tension. The point is not the specific formula but the existence of a principled bridge from* p-bit *evidence patterns to a continuous stability/interaction signal. A further modeling degree of freedom lies in the choice of $\mathcal{K}$: taking $\mathcal{K}$ to be attention-weighted makes coherence explicitly state-dependent, while taking $\mathcal{K}$ to include self-referential and meta-level propositions makes coherence sensitive to reflective consistency rather than only object-level consistency. In this sense,* $\mathrm{Coh}(E)$ *can be read as a task- and attention-conditioned "felt integrity" measure: it is not a global property of an agent's total belief state, but of the currently active workspace slice that is available for report, control, and reflective endorsement.*

17.4.2 Reflective consciousness as coherence-weighted closure

A simple way to model "states of consciousness" is to treat the workspace operator W^{ref} as depending on parameters controlling:

- external coupling (how strongly sensory evidence anchors the workspace),
- internal coupling (how strongly association/imagination fills in content),
- reflective gain (how strongly introspection amplifies meta-content),
- and coherence preference (how strongly the workspace selects coherent states).

The last parameter is the locus at which resonance enters as a stability selector: it does not create new logical consequences, but changes which closures become salient, persistent, or self-reinforcing under iteration.

Definition 238 (Parametric reflective workspace family). *A* parametric reflective workspace family *is a family*

$$W_\lambda^{\mathrm{ref}} : \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}} \to \mathbb{V}^{\mathcal{L}_{\mathrm{Ob}}^{\mathrm{ref}}}$$

indexed by parameters $\lambda = (\lambda_{\mathrm{ext}}, \lambda_{\mathrm{int}}, \lambda_{\mathrm{ref}}, \lambda_{\mathrm{coh}})$, where each W_λ^{ref} is monotone and can be interpreted as: external evidence anchoring (λ_{ext}), internal completion (λ_{int}), introspection amplification (λ_{ref}), and coherence selection (λ_{coh}).

Remark 933. *Intuitively, λ is a "control panel" for the style of conscious closure. If λ_{ext} is high, the workspace behaves like a sober accountant tethered to registration (Hyperseed-Concept 153); if λ_{int} is high, it behaves more like a novelist weaving completions and associations; if λ_{ref} is high, it tends to generate and amplify meta-content; if λ_{coh} is high, it prefers states that score well according to the chosen coherence criterion. This way of speaking is intentionally suggestive rather than neuroscientific: it says that many qualitative shifts in "state" can be treated as parameter changes, not as changes of metaphysical kind (Hyperseed-Concept 178). Moreover, the parameters* *can be read as implementing different* selection pressures *on the set of fixed points: varying* λ_{ext} *and* λ_{int} *changes which propositions are likely to enter and persist in the workspace, varying* λ_{ref} *changes how strongly the system closes over meta-level descriptions of its own state, and varying* λ_{coh} *changes whether the dynamics tolerates patchwork tension or instead drives toward integrative resolutions. In this sense, "reflective will" can be modeled as the agent's capacity to modulate* λ *(in particular* λ_{coh} *and* λ_{ref}*) so that the eventual fixed point is not only reachable but also endorseable: the closure is selected not merely because it is stable, but because it resonates with the agent's preferred mode of integration.*

Remark 934 (Selector interpretation of fixed points)**.** *Given* λ*, Theorem 16 yields at least one fixed point* $E^* \in \mathrm{Fix}(W_\lambda^{\text{ref}})$*. When there are many, one can impose a resonance-based preference by ranking fixed points via* $\mathrm{Coh}(E^*)$ *(or via a variant that explicitly rewards support and penalizes tension). Formally, one can treat the "selected" conscious closure as an element of*

$$\arg\max \left\{ \mathrm{Coh}(E^*) \ : \ E^* \in \mathrm{Fix}(W_\lambda^{\text{ref}}) \right\},$$

when this set is nonempty and the modeling context licenses a maximization principle. Alternatively (and more dynamically), one can view λ_{coh} *as shaping the iteration trajectory so that high-coherence fixed points have larger basins of attraction, thereby capturing the phenomenological idea that resonant configurations are easier to maintain and return to after perturbation, while dissonant configurations are fragile or effortful to sustain.*

Remark 935 (On the choice of normalization and workspace size)**.** *The factor* $\frac{1}{|\mathcal{K}|}$ *makes* $\mathrm{Coh}(E)$ *comparable across different workspace sizes, which is useful if* $\mathcal{K}$ *is attention-dependent and can expand or contract. If one instead wants larger workspaces to have systematically larger coherence signals (e.g. to model the felt "intensity" of a richly populated conscious scene), one can drop the normalization or replace it with a softer scaling. Likewise, one can introduce proposition weights* w_φ *to model salience:*

$$\mathrm{Coh}_w(E) := \frac{1}{\sum_{\varphi \in \mathcal{K}} w_\varphi} \sum_{\varphi \in \mathcal{K}} w_\varphi |(z_E)_\varphi|^2,$$

without changing the basic role of coherence as a selector; such weights provide a direct handle on how attention and motivational relevance modulate resonance.

Remark 936. *As a concrete example, one could define* W_λ^{ref} *to take a convex combination (in the quantale sense) of "sensory anchoring" constraints and "associative completion" rules, with weights controlled by* λ_{ext} *and* λ_{int}*. In particular, one may read* λ_{ext} *as strengthening the degree to which external error-correction, cross-modal consistency, and stimulus-locked constraints dominate the closure, and* λ_{int} *as strengthening the degree to which internally generated completions, priors, and memory-driven predictions dominate it. The quantale-convex phrasing is meant to emphasize that the combination need not be linear in a vector-space sense: the "mixture" can be expressed through the underlying order and monoidal structure that governs how constraints compose and how entailments accumulate. The definition is useful because it formalizes the claim that waking, dreaming, hallucination, etc., can be understood as regimes of the same computational architecture, tuned differently. On this view, what varies across regimes is not the presence or absence of a workspace/closure mechanism, but which families of constraints are allowed to dominate the selection of stable fixed points of the induced operator.*

This provides a clean formal slot for the Hyperseed claim that many distinct "states of consciousness" are *parameter regimes* rather than separate ontological kinds. It also makes room for intermediate and mixed cases (e.g. lucid dreaming, absorption, meditation, or drug-induced states) as regions in a continuous parameter landscape rather than as discrete labels, since small changes in λ_{ext} and λ_{int} can shift which patterns become coherent enough to persist.

17.5 Reflective will as meta-control on self-model dynamics

Hyperseed uses "(reflective) will" to denote more than mere action selection: it is the capacity to intentionally reshape the mind's own internal dynamics (especially attention and self-model updates). In particular, the relevant "internal dynamics" include not only which contents enter a global workspace, but also which self-descriptions (e.g. "I am safe," "I am failing," "I am curious") are repeatedly re-instantiated by the closure process, thereby becoming behaviorally and phenomenologically dominant.

Remark 937. *In ordinary engineering terms, "will" here is closer to* meta-control *than to control: it is the selection of how control itself is carried out, including how attention is allocated and how self-representations are stabilized. One can think of it as operating on the parameters and operators that determine which internal signals are amplified, which are suppressed, and which are treated as "authoritative" inputs to decision-making. This aligns with Hyperseed's broader interest in self-modifying agents and stability of internal goal/attention structures [10]. It also connects to the concept of Weakness (Hyperseed-Concept 202) in the sense that the system must manage self-reference and self-modification without collapsing into inconsistency or paralysis [2, 3]. In more concrete terms, meta-control must include safeguards (implicit or explicit) that prevent runaway self-editing (e.g. compulsive policy rewriting) while still allowing sufficient plasticity for learning and recovery.*

17.5.1 Will as choice of an intervention on the update operator

Let $\mathcal{S}_{\text{Ob}}$ be a state space of the observer (internal memory, body state, etc.). For example, $\mathcal{S}_{\text{Ob}}$ may include latent variables corresponding to working memory contents, affective tone, estimated uncertainty, and the current self-model, insofar as these are available to the system as inputs to further updating. Let U be a set of interventions that can affect: (i) attention allocation a (Section **??**), (ii) the workspace parameters λ (Section 17.4), and/or (iii) the control policy used to act in the world (Section 14). It is important that U ranges over *internal* as well as external-facing interventions: the notion of "intervening" here is not restricted to motor outputs, but includes changes to the internal routing and gain-control by which information becomes influential.

Definition 239 (Will operator)**.** *A* will operator *is a map*

$$\mathsf{Will} : \mathcal{S}_{\text{Ob}} \to U$$

that selects an intervention $u \in U$ based on the current internal state. Applying u induces an update of the reflective closure operator, i.e. changes $F = W^{\text{ref}} \circ \mathsf{Int} \circ \Gamma_a$ to a new operator F_u. One may view u as selecting (or modulating) one or more components of the composite map—for instance, by altering Γ_a through attention shifts, by altering Int through changes in how sensory evidence is incorporated, or by altering W^{ref} through changes in reflective stabilization strength.

Remark 938. *Intuitively,* Will *is the system's "choice of how to proceed," not only in the world but also in its own cognitive configuration. An example of $u \in U$ might be: "shift attention from rumination to sensory grounding," which changes a and potentially increases λ_{ext}. Another example is: "engage deliberative mode," which might increase the weight of explicit reasoning processes and* *adjust workspace inhibition rules. A further example (in the same spirit) is: "reduce precision on intrusive imagery," which would down-weight certain internally generated candidates before they are stabilized by closure, thereby reducing the chance that they become fixed-point contents.*

This definition is useful because it makes "will" something that has a clear causal footprint in the formalism: it changes the operator whose fixed points are the conscious states. That is, will is not a ghostly add-on; it is a map that selects an intervention that changes the dynamics of closure, thereby changing what is stably present to experience and action. Equivalently, it lets us speak precisely about cases where two agents (or two moments of the same agent) have similar sensory inputs but systematically different stable experiences because the meta-selected operator F_u differs.

Remark 939 (Why this is "will" rather than "control")**.** *Ordinary control (Section 14) chooses actions in the world. Here the intervention may also choose:* what the agent attends to*,* how reflective closure is computed*, and* how self-representations are stabilized*. This corresponds closely to Hyperseed's use of "will" as control over inner dynamics. In particular, it distinguishes between (a) selecting an action under a fixed cognitive pipeline and (b) selecting the pipeline (or its parameters) under which subsequent actions and perceptions will be generated.*

17.5.2 Reflective will as meta-selection

Reflective will occurs when the system can represent and modify the will operator itself. We capture this by allowing interventions that change the policy class. This is the point at which the agent is no longer only choosing within a given meta-control scheme, but is able to restructure the scheme that generates such choices, thereby altering its own future degrees of freedom.

Definition 240 (Reflective will)**.** *Let Π be a space of possible will operators $\pi : \mathcal{S}_{\mathrm{Ob}} \to U$. A* reflective will state *is a pair (π, s) with $\pi \in \Pi$ and $s \in \mathcal{S}_{\mathrm{Ob}}$. A* reflective will update *is a map*

$$\mathsf{RWill} : \Pi \times \mathcal{S}_{\mathrm{Ob}} \to \Pi \times \mathcal{S}_{\mathrm{Ob}},$$

which may change both the internal state and the will operator. In effect, RWill *can be read as a learning (or self-programming) dynamic on Π coupled to an ordinary state-update dynamic on $\mathcal{S}_{\mathrm{Ob}}$.*

Remark 940. *In plain language: a non-reflective will chooses what to do; a reflective will can also choose* how it will choose *in the future. For example, an agent might adopt a new attentional policy "when anxious, ground attention in breath"; this is a modification to the will operator itself, not merely a single action. The pair (π, s) represents "the current chooser" together with the current state it is choosing from. One can also interpret π as encoding commitments or self-binding constraints (e.g. "do not deliberate about X after midnight," or "when uncertain, consult external evidence"), which then shape subsequent trajectories of F_u by restricting or biasing which interventions are selected.*

This definition is necessary if we want to model long-term autonomy and development: an agent that cannot revise its own will policy is confined to a fixed meta-level, whereas reflective agents can enter regimes of self-training, self-regulation, and ethical self-binding. Order-theoretic and dynamical tools then become applicable to questions of stability and drift, linking naturally to fixed-point approaches to self-modifying systems [10]. It also makes it possible to distinguish benign adaptation (slow improvement of π with respect to some internal viability criterion) from destabilizing meta-dynamics (rapid churn in π that prevents coherent stabilization of goals, attention, or self-model contents).

Remark 941 (Fixed point picture)**.** *A stable reflective will regime corresponds to a fixed point (or slowly drifting orbit) of* RWill*. This is one of the cleanest mathematical ways to interpret the*

Hyperseed idea that "reflective will" is a stabilized meta-level capability rather than a single act. In such a regime, the agent's meta-policy π is itself resilient under the pressures of short-term perturbations in s (fatigue, affect, noise), and this resilience can be studied with the same stability intuitions used elsewhere in the framework (e.g. as a selector for coherent, self-sustaining patterns).

17.6 Autonomy and the reflective self as fixed points and higher morphisms

Hyperseed's "natural autonomy" is not maximal independence from world and society; it is the capacity to maintain coherent self-directed dynamics while remaining coupled to larger systems. The earlier sections already provide most of the needed machinery: self as boundary in a pattern web (Section 16), control as attraction-sensitive intervention (Section 14), and habits/self-weaving dynamics (Section 12).

Here we package these ideas into a minimal autonomy criterion.

Remark 942. *This notion of autonomy (Hyperseed-Concept 119) is intentionally non-romantic: it is not "freedom from causation" but the ability to keep one's dynamics within a viable region by selecting interventions, including internal interventions. It is also explicitly relational: the viable region depends on coupling to environment and society, and the self/other boundary (Hyperseed-Concept 165) is itself observer-relative and potentially paraconsistent (Section 16).*

17.6.1 Autonomy as viability under self-directed closure

We treat autonomy as *viability* of self-maintaining closure.

Definition 241 (Viable set and autonomy)**.** *Let $D_u : \mathcal{S}_{\mathrm{Ob}} \to \mathcal{S}_{\mathrm{Ob}}$ denote the state update induced by applying intervention $u \in U$ (this includes both world actions and internal interventions). Let $\mathcal{V} \subseteq \mathcal{S}_{\mathrm{Ob}}$ be a set of "viable" states (states where the mind remains functional, coherent enough, and continuous enough; this is observer- and model-dependent). We say* Ob *has* natural autonomy relative to $\mathcal{V}$ *if for every $s \in \mathcal{V}$ there exists an intervention $u \in U$ such that $D_u(s) \in \mathcal{V}$.*

Remark 943. *Intuitively, $\mathcal{V}$ is the set of states in which the agent "still counts as itself and still works". The condition says: whenever the agent is in a viable state, it has at least one available move that keeps it viable. This is a minimal viability-style autonomy notion, echoing how "life" and "agency" are often made precise via invariance or controlled invariance of a viability region (compare Section 19.2).*

A simple example: if $\mathcal{S}_{\mathrm{Ob}}$ includes a resource variable (energy, attention capacity, etc.) and $\mathcal{V}$ is defined by "resource above threshold," then autonomy means the agent can always choose some intervention that does not immediately exhaust the resource. The definition is useful because it separates autonomy from omnipotence: autonomy is not "I can do anything," but "I can keep going in a coherent way," which is closer to the lived phenomenon and to Hyperseed's emphasis on self-maintaining pattern-web dynamics.

Remark 944 (Connection to self-weaving)**.** *If $\mathcal{V}$ is defined in terms of persistence of a self-weaving subweb (Section 12) plus a continuity constraint (Section 16), then "autonomy" is precisely the ability to choose actions (internal and external) that keep the self-weaving process alive without losing identity.*

17.6.2 Reflective self as a stabilized self-representation

Hyperseed's "reflective self" is not merely a self-model; it is a self-model that becomes a stable reference point for reflective consciousness and reflective will.

Definition 242 (Self-model object and reflective self condition)**.** *Assume $\mathcal{L}^{\mathrm{ref}}_{\mathrm{Ob}}$ includes special propositions of the form $\mathsf{Self}(\psi)$ meaning "ψ is a property of me" (e.g. a pattern/property token as in Section 9 applied to the self-boundary of Section 16). A workspace-stable evidence state E is said to exhibit a* reflective self *if:*

(a) (Self-presence) for a designated set $\mathcal{P}_{\mathrm{self}}$ of self-properties, $E^{+}(\mathsf{Self}(p))$ is above threshold for many $p \in \mathcal{P}_{\mathrm{self}}$;

(b) (Reflective access) for these p, the introspective propositions $\mathsf{I}(\mathsf{Self}(p))$ are also supported above threshold;

(c) (Stability) these supports persist under small perturbations of module evidence and attention.

Remark 945. *This definition makes "having a self" (as a representational fact) different from "having a reflective self" (as a stable, accessible, meta-accessible representational fact). Condition (a) says that many self-attributions are present; condition (b) says they are not merely implicit but are themselves available to awareness; condition (c) says the whole configuration is robust under small changes. In informal terms: it is not enough that the system encodes a self-model somewhere; it must be able to* stand by it *in the workspace and to know (in the formal introspective sense) that it is standing by it.*

A simple example is a bodily self-property p = "this arm is mine." In a reflective self regime, one expects both $\mathsf{Self}(p)$ and $\mathsf{I}(\mathsf{Self}(p))$ to be supported, and not to vanish under minor sensory noise. This definition is useful because it ties together three threads from earlier sections: self/other boundary (Hyperseed-Concept 165), self-continuity (Hyperseed-Concept 166), and attention-mediated global accessibility (Hyperseed-Concept 60). It also provides a place to connect with individual and collective "consciousness location" ideas in the broader Hyperseed program [4, 12].

Remark 946 (Higher morphism reading)**.** *In the enriched-categorical viewpoint, a stable reflective self can be treated as a higher morphism (or a higher cell) relating:*

$$\text{(self-boundary as subobject of a pattern web)} \quad \to \quad \text{(self-representation in the reflective workspace)} \quad \to \quad \text{(meta}$$

We avoid heavy $(\infty, 1)$-categorical machinery here, but the conceptual alignment is direct.

17.7 States of consciousness as parameter regimes

Hyperseed lists a number of "states of consciousness" (ordinary waking consciousness, dreaming, hallucinating, being, oceanic consciousness, creative inspiration, enlightenment, Vedantic strata). In the present reconstruction, these can be modeled as regimes of the parameter vector $\lambda = (\lambda_{\mathrm{ext}}, \lambda_{\mathrm{int}}, \lambda_{\mathrm{ref}}, \lambda_{\mathrm{coh}})$ together with the habit/resonance coupling strengths (Sections 12 and 3.9).

Remark 947. *The modeling posture here is pluralist but disciplined: we do not deny the qualitative diversity of states of consciousness (Hyperseed-Concept 178); rather, we claim that much of this diversity can be represented as changes in coupling strengths and closure preferences inside a single architecture. This is reminiscent of how, in physics, different phases of matter are not different substances but different regimes of the same underlying constituents; here the "constituents" are attention, evidence, workspace closure, and resonance.*

Definition 243 (State-of-consciousness parameters)**.** *A* state-of-consciousness parameter tuple *is*

$$\Lambda := (\lambda_{\mathrm{ext}}, \lambda_{\mathrm{int}}, \lambda_{\mathrm{ref}}, \lambda_{\mathrm{coh}}, \rho_{\mathrm{hab}}, \rho_{\mathrm{res}}),$$

where:

- λ_{ext} *controls anchoring to current sensory registration (Section 13);*
- λ_{int} *controls internal completion/association (pattern web propagation; Section 11);*
- λ_{ref} *controls introspection gain (Section 17.3);*
- λ_{coh} *controls coherence preference (Section 17.4);*
- ρ_{hab} *controls habit reinforcement strength (Section 12);*
- ρ_{res} *controls resonance coupling strength (Sections 3.9 and 12).*

Remark 948. *Intuitively, Λ extends the earlier workspace parameters λ with two further knobs: habit reinforcement and resonance coupling. If ρ_{hab} is high, recently active patterns are rapidly entrenched (Hyperseed-Concept 155); if ρ_{res} is high, distant parts of the pattern web can synchronize and amplify one another (Hyperseed-Concept 159). Thus Λ acts as a compact descriptor of how "tight" or "loose" the system is in binding together contents and in reinforcing repeated flows.*

As an example, a dreaming-like regime might be described by low λ_{ext}, high λ_{int}, and moderate-to-high ρ_{res} (strong internal coupling) with weaker external anchoring. The usefulness of introducing Λ is that it allows the narrative taxonomy of states to be replaced by a parameterized family of operators, which is mathematically tractable and composable with the rest of the framework.

17.7.1 Canonical regimes

The following table is not intended as neuroscience; it is a compact map from Hyperseed vocabulary to a minimal mathematical control panel.

- **Ordinary waking consciousness.** High λ_{ext}; moderate λ_{int}; moderate λ_{ref}; λ_{coh} tuned to prefer sensory-consistent closure. Habits and resonance are active but constrained by sensory feedback.
- **Dreaming.** Low λ_{ext}; high λ_{int}; variable λ_{ref}. Coherence preference may be lower, allowing more internally generated blends and narrative jumps. Habit dynamics may replay or reweight pattern-flow motifs (Section 12) without external error correction.
- **Hallucinating.** Low λ_{ext} but strong internal propagation and high self-consistency pressure: high λ_{int} and high λ_{coh}, with closure dominated by internally generated evidence. Paraconsistency becomes prominent: the system may carry simultaneous support for incompatible world propositions without collapse, while still experiencing strong presence.
- **Being (non-conceptual presence).** Moderate-to-low λ_{int} (less elaborative completion), moderate λ_{ext} (or also low, depending on context), and high λ_{coh} with coherence defined more by low tension than by high net propositional content. In this regime the workspace fixed point contains fewer explicit propositions but higher stability.
- **Oceanic consciousness.** This is naturally modeled by a *reduction of self/other boundary sharpness* (Section 16) together with high global coherence. Formally, this corresponds to: (i) fewer strongly supported self-boundary propositions; (ii) higher coupling between self-typed and world-typed pattern nodes; and (iii) a coherence preference that does not penalize loss of boundary distinctions. This connects back to Hyperseed's non-duality motif (Section 6; Hyperseed-Concept 121).

- **Creative inspiration.** A transient regime with increased resonance coupling ρ_{res} between normally distant subwebs, enabling new blends (Section 9). Here ρ_{res} can be read as a control parameter governing the effective "conductance" of associative links across otherwise weakly communicating regions of the cognitive network, so that activation can percolate between contexts that are usually conditionally independent. One can model this as a temporary increase in cross-context coupling in the habit/resonance kernel, leading to rapid emergence of high-intensity composite patterns [5]. Concretely, if the kernel K weights transitions between subweb states, then "inspiration" is a short-lived deformation $K \mapsto K + \Delta K$ with ΔK concentrated on off-diagonal (cross-context) blocks, so that previously subthreshold combinations become reachable within a few update steps. Phenomenologically, this corresponds to the felt "suddenness" of a novel synthesis: the composite pattern is not built by slow local refinement alone, but by a burst of long-range binding that stabilizes once coherence constraints are satisfied.

- **Enlightenment.** Modeled here as a stable, low-effort, high-coherence regime in which: (i) reflective closure remains stable under perturbation; (ii) will/intervention choices tend to preserve viability with minimal contrivance (foreshadowing Wu Wei; Section 26; Hyperseed-Concept 206 and 207) [21]; and (iii) self/other boundary is flexible without inducing disorganized conflict. Item (i) can be understood as a robustness property: small changes in incoming evidence, affect, or bodily state do not cause reflective propositions to oscillate or fragment, i.e. the closure operator returns to (or remains near) the same fixed point after bounded disturbance. Item (ii) emphasizes that "control" is not absent but becomes economical: interventions align with system dynamics so that fewer corrective moves are required to maintain constraints (e.g. viability, social coordination, or goal satisfaction), resembling a descent along a shallow energy landscape rather than repeated forcing against steep gradients. Item (iii) indicates that increased permeability of self/other modeling does not collapse into indiscriminate identification; instead, boundary flexibility is paired with sufficient coherence to avoid runaway contagion of competing action policies. In quantale terms: low "tension" components across the reflective proposition set while retaining functional coupling for control. More explicitly, if reflective propositions are combined via a monoidal product that aggregates commitments, then "tension" informally tracks the degree to which jointly held propositions require active suppression of incompatibilities; the enlightened regime corresponds to configurations where incompatibilities are either resolved (by re-framing) or rendered non-disruptive (by appropriate contextualization), while the coupling needed for action selection remains intact.

- **Vedantic strata.** A simple formal analogue is a hierarchy of reflective closures: $\mathcal{L}_{\text{Ob}} \subset \mathcal{L}_{\text{Ob}}^{\text{ref}} \subset \mathcal{L}_{\text{Ob}}^{\text{ref},2} \subset \cdots$, where $\mathcal{L}_{\text{Ob}}^{\text{ref},k}$ adds propositions about awareness-of-awareness iterated k times. This mirrors the idea that certain reports distinguish (a) first-order contents, (b) awareness of those contents, and (c) awareness that one is aware, etc., with each level introducing new constraints on what counts as a stable self-description. Different strata correspond to stable fixed points at different k (or different gains λ_{ref}), with the paraconsistent setting allowing such iteration without trivial paradox. Operationally, λ_{ref} can be interpreted as a weighting (or gain) on reflective update terms relative to first-order dynamics: higher λ_{ref} makes meta-level propositions more influential in the closure computation, which can stabilize higher-order self-modeling when the network can sustain it. The reference to paraconsistency matters because awareness-of-awareness claims often generate self-referential structures; allowing controlled inconsistency prevents the entire reflective lattice from collapsing into triviality while still permitting meaningful fixed points that correspond to distinct phenomenological "strata".

Remark 949. *The point of these regime descriptions is not to legislate phenomenology but to show* *that the formalism has enough expressive degrees of freedom to represent qualitative shifts without adding new primitives. In this sense the regimes are parameterized views of one and the same machinery: varying coupling, gain, noise, and boundary-regularization terms can move the system between recognizable macrostates without changing the underlying ontology of operators, propositions, or update rules. In particular, the transition from waking to dreaming becomes intelligible as a movement along the axis from externally anchored closure to internally generated completion; the more "mystical" regimes become movements along the axes of self/other boundary softness and coherence preference, echoing Hyperseed's process-oriented stance in which identity and boundary are dynamically constructed (compare [15, 14] for metaphysical inspirations). One can also view these axes as coordinates on a stability landscape: "waking" corresponds to closures strongly constrained by sensory-anchored evidence and prediction-error correction, whereas "dreaming" corresponds to closures where generative priors and internal completion dominate, with external correction weakened. Likewise, changes in boundary softness can be modeled as shifts in how strongly the system penalizes cross-agent (or cross-model) identification errors, and changes in coherence preference can be modeled as shifts in how aggressively the system resolves or tolerates local inconsistency in the reflective set.*

17.8 Intuition and reason as dual inference channels

Hyperseed explicitly distinguishes intuition and reason, and treats both as essential. A minimal way to integrate this into the present framework is to posit two update channels:

- *Intuitive update*: fast, low-cost propagation in pattern webs, strongly driven by resonance.
- *Reasoned update*: slower, higher-cost composition of explicit predictive/causal relations.

Remark 950. *This distinction corresponds to two complementary modes of cognition: intuition (Hyperseed-Concept* **??***) as quick pattern completion and resonance-driven association, and reason (Hyperseed-Concept 150) as explicit manipulation of articulated relations (implications, causal arrows, plans). The framework does not declare one mode superior; rather, it provides a mathematical place for their interaction, consistent with the Hyperseed emphasis on cognitive synergy (Hyperseed-Concept 73) [19].*

Definition 244 (Two-channel update sketch)**.** *Let E be the current reflective evidence state. Define:*

$$E_{\text{int}} := U_{\text{int}}(E), \qquad E_{\text{rea}} := U_{\text{rea}}(E),$$

where U_{int} is a monotone operator that spreads support along pattern-web morphisms (Section 11) and resonance couplings (Section 17.4), and U_{rea} is a monotone operator that propagates support along explicitly represented implications and attractions (Section 14), with explicit composition. The combined update may be taken as

$$U(E) := W^{\text{ref}}\big(E \oplus \overline{\beta} \otimes E_{\text{int}} \oplus \overline{\gamma} \otimes E_{\text{rea}}\big),$$

with channel weights $\beta, \gamma \in [0, 1]$.

Remark 951. *Notation check: U_{int} and U_{rea} are endomaps on the space of reflective evidence states (they take an evidence state and output another). The scalars β, γ are embedded into $\mathbb{V}$ via $\overline{\beta}, \overline{\gamma}$, and then used to weight the contributions of the two channels using the quantale tensor $\otimes$. The join $\oplus$ combines the baseline state with the two channel-updated states, and finally W^{ref} re-applies reflective workspace stabilization.*

Intuitively, this says: take the current state, add in a resonance-driven intuitive propagation and a relation-driven reasoned propagation, each with its own strength, then stabilize globally. A concrete toy example is: intuition quickly spreads support from "dark clouds" to "it will rain" via a learned association, while reason slowly constructs support for "bring an umbrella" by chaining explicit implications. The definition is useful because it gives a formal grammar for how two inference modes can cooperate without requiring them to be reducible to one another.

Remark 952. *The two-channel form also makes room for an important asymmetry often observed in practice: the intuitive channel is typically* broad *(high fan-out in the pattern web) but* shallow *(limited explicit depth of justification), whereas the reasoned channel is typically* narrower *but* deeper *(longer chains of explicit composition). In the present notation, this difference is not encoded by changing the carrier space of E, but by the structure of U_{int} versus U_{rea}: the former is dominated by resonance couplings and morphism-local propagation steps, while the latter is dominated by explicit relational composition (e.g. repeated application of implications, causal links, or plan operators). This helps clarify why "slower" does not merely mean "smaller γ": it can also mean that U_{rea} itself is implemented as a multi-step computation whose intermediate results may enter the workspace only after partial stabilization by W^{ref}.*

Remark 953. *Although the definition presents a single combined update $U(E)$, the same ingredients can be used to model alternating or interleaved schedules, which better match many cognitive architectures. For example, one may consider a two-phase cycle*

$$E^{(t+\frac{1}{2})} := W^{\text{ref}}\big(E^{(t)} \oplus \overline{\beta} \otimes U_{\text{int}}(E^{(t)})\big), \qquad E^{(t+1)} := W^{\text{ref}}\big(E^{(t+\frac{1}{2})} \oplus \overline{\gamma} \otimes U_{\text{rea}}(E^{(t+\frac{1}{2})})\big),$$

so that intuition rapidly proposes candidate completions that are then subjected to explicit reasoning on the next phase. This interleaving is still monotone at each step (assuming monotonicity of the constituents), and it highlights that "interaction" can mean not only additive combination via $\oplus$ but also sequential coupling through the shared workspace.

Remark 954. *From the perspective of reflective will and autonomy, the weights β, γ can be understood as coarse global controls over where computational effort is spent. A high-β regime privileges fast associative completion (useful for time pressure and exploration), while a high-γ regime privileges explicit justification and longer-horizon planning (useful for high-stakes deliberation). Nothing in the formalism requires β and γ to be fixed constants; they may be taken as functions of the current state (or of a separate control state) to represent attentional gating. For instance, one may write $\beta(E), \gamma(E)$ to indicate that the system shifts toward reason when the workspace contains unresolved conflicts, or shifts toward intuition when the workspace is sparse and requires rapid hypothesis generation.*

Remark 955. *A further use of the two-channel grammar is to separate* proposal *from* justification *without forcing either to be infallible. Intuition may propose a high-support candidate that is later weakened by reason when explicit causal structure does not sustain it; conversely, reason may construct a formally valid chain that remains low-impact in practice if it fails to resonate with the currently active pattern web. In the combined update, these cases correspond to situations where $\overline{\beta} \otimes E_{\text{int}}$ dominates the join but is then attenuated by stabilization, or where $\overline{\gamma} \otimes E_{\text{rea}}$ introduces structured support that nonetheless competes with other attractors already present in E. This is one place where cognitive synergy becomes operational: the channels are allowed to disagree, and the workspace provides the arena in which their contributions are reconciled into an actionable state.*

Remark 956 (Why paraconsistency matters for intuition/reason)**.** *Intuition often produces "both-leaning" states: simultaneous attraction toward incompatible interpretations. In a classical logic* *semantics this would be immediately problematic; in the* p-bit *setting it is a first-class representational state that can later be refined by reason, by attention shifts, or by additional evidence.*

Remark 957. *The preceding point can be sharpened by noting that the intuitive channel may increase support for multiple mutually incompatible hypotheses* at once *because resonance can be triggered by shared features (e.g. similar surface cues) even when deeper causal models differ. Paraconsistent representation allows such "multi-attractor" activation to remain expressible in E long enough for U_{rea} to apply discriminating structure (explicit constraints, explanations, counterfactual checks, or plan feasibility tests). In this sense, paraconsistency is not merely a tolerance of inconsistency but a functional resource: it provides a stable intermediate regime in which the system can entertain competing candidates without collapsing into triviality or prematurely committing to a single narrative.*

17.9 Closing remarks and link to values/ethics

This section has treated consciousness and reflective will as closure and meta-control phenomena. In that framing, "closure" is the emergence of a globally available, self-consistent-enough (in the paraconsistent sense) workspace configuration, while "meta-control" is the system's capacity to act on its own internal dynamics (e.g., to sustain, redirect, inhibit, or reconfigure processes that would otherwise run automatically). Two crucial ingredients have been intentionally bracketed:

- *valuation*: what the system cares about (joy/woe, goals, values);
- *ethical structure*: how value conflicts and social constraints are represented and resolved.

Here, "valuation" is meant broadly: it includes affective salience (what feels good or bad), instrumental priorities (what advances goals), and more reflective endorsements (what the agent takes itself to have reason to pursue), all of which can bias which content is amplified into the workspace and which is ignored or suppressed. Likewise, "ethical structure" is not restricted to explicit moral rules; it includes any internal representation of constraints that have a normative character for the system (e.g., prohibitions, commitments, fairness-like considerations, role obligations, or coordination requirements in multi-agent settings), as well as the mechanisms by which such constraints interact with other goals under conflict. In Hyperseed, these are not add-ons: they shape attention, shape which fixed points are selected, and shape whether reflective will stabilizes into autonomy or degenerates into fragmentation. More concretely, the same capacity for reflective iteration that enables autonomy can also enable rumination, oscillation among incompatible commitments, or opportunistic switching, unless there are stable evaluative gradients and constraint-handling policies that give reflective control a direction and a stopping condition. Accordingly, Section 18 extends the same paraconsistent and resonance-enabled machinery to evaluation, motivation, and rational/ethical decision. This extension matters because evaluation is not merely a label attached after the fact; it enters into the dynamical loop that determines which representations couple resonantly, which attractors are reachable, and which equilibria are robust under perturbation, learning, and social feedback.

Remark 958. *One way to read the logical arc is: this section supplies the* stage *(workspace closure and its reflective iteration), while the next supplies the* drama *(what is attractive, what is aversive, which conflicts matter, and which resolutions count as rational or ethical). The metaphor is intentional: closure and reflective will specify a capacity for integration and self-modification, but they do not by themselves specify* why *the system should integrate in one direction rather than another, nor* which *internal tensions are to be treated as errors versus acceptable trade-offs. Without values, a* *consciousness modeled as "just" a fixed point is underdetermined: many fixed points may exist, and the system needs selection pressures to inhabit some rather than others. This underdetermination appears both at the level of content (which interpretations, plans, or self-models can close) and at the level of policy (which meta-control actions are deemed improvements), so valuation and norm-like constraints function as discriminators over a landscape of possible closures. Hyperseed's claim is that affect and value are among the deepest such selectors, co-evolving with attention, habit, and resonance [5, 24]. On this view, autonomy is not merely the persistence of a reflective fixed point, but the stability of a reflective process under evaluative and normative pressures: it is the ability to maintain coherent commitments over time while still revising them in response to evidence, conflict, and social constraint.*

18 Emotion, values, goals, and rationality

18.1 Orientation: why "emotion" belongs in the formal stack

Hyperseed treats *effort* and *distinction* as primitive and observer-relative. Once effort is taken seriously as a first-class quantity (Sections 8–9), it becomes natural to treat many affective phenomena as *evaluative signals about (i) expected effort/resistance, (ii) expected coherence/resonance, and (iii) expected alignment with values/goals.* In this section we give a minimal, mathematically explicit vocabulary for:

- **joy/woe and pleasure/pain** as scalar projections of paraconsistent evaluative evidence;
- **emotion** as structured (vector-valued, context-dependent) evaluation fields rather than a single utility number;
- **values, feelings, and goals** as stable patterns and constraints in these evaluation fields;
- **implicit vs explicit goals** as a representational distinction;
- **value paraconsistency** as an intended feature (conflict without explosion);
- **rationality** as coherence between belief, value, and action under resource constraints;
- **ethics** (categorical imperative, cultural morality, evil) as constraints on goal selection and action selection that can be stated in the same formal language.

A key motivation for placing "emotion" inside the formal stack is that, in real agents, evaluation is not an optional afterthought appended to belief: it is part of the control loop that determines which distinctions are worth making, which inferences are worth paying for, and which actions are worth attempting under limited time/energy. If the framework already assigns a formal status to *effort* (as an accounting of cost, resistance, and compression difficulty), then it is consistent to treat affective variables as the agent's internal, dynamically updated estimates of that effort landscape, coupled to its goal constraints. On this view, emotion is not primarily a noise source to be "filtered out" from rationality, but a low-latency summary of how current perception and prediction interact with constraints on action selection.

The phrase "observer-relative" is doing additional work here: what counts as "effortful" or "coherent" depends on the representational resources and habits of the observer (or subsystem) doing the evaluation. Thus, the same external situation can induce different affective profiles in different agents, or even in the same agent across time, without requiring that emotion be arbitrary;

it is lawful relative to the agent's current distinctions, priors, and available control policies. This also clarifies why a single scalar utility is often too coarse: distinct appraisal channels can simultaneously register "high resistance," "high resonance," and "high value alignment," yielding mixed feelings that are not well-captured by total ordering.

The three parenthesized components—expected effort/resistance, expected coherence/resonance, and expected alignment with values/goals—should be read as separable but interacting coordinates of evaluation. Expected effort/resistance tracks anticipated cost-to-execute (physical, computational, social) and cost-to-maintain (attention, working memory, ongoing commitment). Expected coherence/resonance tracks how well a candidate interpretation/action integrates with existing patterns (including narrative self-models, predictive models, and social models), and therefore how stable it is under perturbation. Expected alignment tracks the degree to which a candidate is compatible with constraints that the system treats as non-negotiable or priority-weighted (explicit commitments, learned norms, or implicit drives). Later formalisms will allow these coordinates to be aggregated, contrasted, or kept separate depending on the control regime.

The formalisms here are intentionally lightweight: we aim for a scaffold that composes cleanly with the formal core (Section 3), with the control/prediction machinery (Section 14), and with the resonance construction (Section 3.9). In particular, we want definitions that remain meaningful when the agent's beliefs are incomplete, when its goals are partially articulated, and when its evaluative evidence is internally conflicted. This motivates the use of paraconsistent structures: an agent can have simultaneously supported and opposed evaluations of the same option (e.g., "approach" and "avoid") without collapsing into triviality, and without requiring that the conflict be immediately resolved before action is possible.

Concretely, the intended picture is that an agent maintains a collection of context-indexed evaluation fields over its state/action space (or over structured descriptions of situations), and emotions are identifiable patterns in these fields as they change over time. Scalar quantities such as pleasure/pain are then treated as projections or summaries extracted from a richer object, typically for fast control decisions. This retains the practical role of simple affective labels while keeping the underlying representation expressive enough to capture ambivalence, tradeoffs, and norm conflict.

Remark 959. *At the level of philosophical posture, this section takes seriously the idea that "value" is not a decorative gloss atop cognition, but a structural constraint on what a system can coherently do. In Peircean terms, values function as a species of* Thirdness*—habits, norms, and generalities shaping inference and action—while emotions are the time-local manifestations of how that Thirdness collides with the recalcitrance of the world (Secondness) and the immediacy of felt quality (Firstness) [14]. In a Whiteheadian accent, emotions are not opaque atoms but events in a process: they are patterns in the ongoing concrescence of appraisal and action [15]. Put differently, the "formal stack" is not meant to exclude phenomenology, but to provide a disciplined way to talk about the constraints and transformations that make phenomenology actionable for an agent. A value, in this posture, behaves like a general rule that can be instantiated across contexts; an emotion behaves like the local error-signal (or opportunity-signal) generated when that general rule meets a particular world-situation under resource bounds.*

Remark 960. *This section should be read alongside the Hyperseed core-concepts: Effort (Hyperseed-Concept 100), Distinction (Hyperseed-Concept 98), Emotion (Hyperseed-Concept* **??***), Values (Ethical) (Hyperseed-Concept 197), Goals: Explicit/Implicit (Hyperseed-Concepts* **??***,* **??***), Value Paraconsistency (Hyperseed-Concept 198), Rationality (Hyperseed-Concept 148), and Compassion/Evil (Hyperseed-Concepts 81,* **??***). One way to keep the reading coherent is to treat "emotion" (as used here) as the interface layer between (i) descriptive models of the world (belief/prediction), (ii) prescriptive constraints (values/goals), and (iii) the cost model (effort). In that sense, the section is not* *proposing a separate psychology module, but a unifying vocabulary that lets the same mathematics speak about appraisal, commitment, conflict, and action choice.*

18.2 Paraconsistent evaluative evidence and affective projections

18.2.1 Evaluation fields

Let S denote a set of (observer-relative) *situations* or *states* and A a set of *actions*. We also fix a finite set $\mathcal{D}$ of *value dimensions*. Typical elements of $\mathcal{D}$ include:

comfort, safety, truth, beauty, competence, belonging, compassion, individuation, self_transcendence.

Nothing in the math forces these labels; the point is that Hyperseed does not assume values are reducible to one scalar.

Remark 961 (Notation and conventions)**.** *Here $s \in S$ is a modeled situation (which may already bundle a time-slice, an internal state-estimate, and relevant context), and $a \in A$ is a candidate intervention. The finite index set $\mathcal{D}$ is not assumed to be "the true list of human values"; it is the observer's current basis of evaluative distinctions. The choice to make $\mathcal{D}$ finite is pragmatic: it lets us treat value as a vector of coordinates while still allowing the coordinates themselves to evolve later (Section 18.6).*

Remark 962 (Why $S \times A$ rather than S alone)**.** *The dependence of evaluation on (s, a) (rather than on s alone) is essential: many values are primarily* interventional *rather than merely descriptive. For example, a situation s may be high in* safety *as a snapshot, while a contemplated action a may nonetheless be evaluated as strongly anti-*safety *because it increases risk; conversely, s may be low in* comfort *but a may have high pro-*comfort *evidence due to expected relief. The field E_O is thus best read as attaching evaluation to* counterfactuals *("what would happen if I do a in s") rather than to world-states in isolation.*

Definition 245 (Paraconsistent evaluative field)**.** *Fix an observer/context O. A* paraconsistent evaluative field *is a map*

$$E_O : S \times A \times \mathcal{D} \to [0, 1]^2,$$

where

$$E_O(s, a; d) = \left(E_O^+(s, a; d),\, E_O^-(s, a; d)\right)$$

encodes degree of evidence that action a in situation s promotes (resp. opposes) the value dimension d. No consistency constraint is imposed; it is allowed that $E_O^+(s, a; d) + E_O^-(s, a; d) > 1$.

Remark 963 (Interpretation of the two channels)**.** *The pair $\left(E_O^+, E_O^-\right)$ should be read as two separately supported appraisals, not as a single net valence split into positive/negative parts. Concretely, E_O^+ may be driven by cues of opportunity, growth, alignment, or anticipated benefit, while E_O^- may be driven by cues of threat, violation, loss, or anticipated harm. Allowing both to be simultaneously high captures the common case where an option is "meaningful but dangerous," "honest but socially risky," or "safe but stifling." In this sense, the paraconsistent format is not merely tolerant of conflict; it makes conflict representable without forcing an immediate resolution step.*

Remark 964. *Intuitively, E_O is a kind of "moral/affective sensorium" that maps each contemplated option to a two-sided appraisal. The two coordinates are not probabilities; they are independent evidence channels, so one may have strong reasons both for and against. This is the evaluative analogue of the paraconsistent evidence states used earlier for beliefs, and it aligns with the general Hyperseed stance that conflict is often data rather than error (Hyperseed-Concept 198).*

Remark 965 (Sources and observer-relativity)**.** *The subscript O emphasizes that evaluative evidence is indexed to an observer/context, which can include embodiment, developmental history, culture, role, and current commitments. In particular, E_O can legitimately encode information that is not "in the world" as a neutral fact (e.g., attachment dynamics, trauma-conditioned threat detection, social-norm priors), because these factors causally shape both experienced affect and action guidance. This is not a concession to arbitrariness: it is an explicit modeling choice that makes room for calibration, learning, negotiation among observers, and reflective updates of O itself.*

Remark 966. *A simple example is the choice "tell a difficult truth" in a social situation. Relative to $d =$ truth one may have high positive evidence and low negative evidence; relative to $d =$ belonging one may have both substantial positive evidence (authenticity builds trust) and substantial negative evidence (risk of social rupture). The formalism records this without prematurely collapsing it into a scalar utility. This is useful because later decision rules can treat some dimensions as constraints, others as objectives, and can invoke resonance to measure coordination among them rather than forcing a false unanimity.*

Remark 967 (Value paraconsistency is the default, not an exception)**.** *Hyperseed explicitly expects that real agents have value conflicts, self/other boundary ambiguities, and mixed evidence about what is good or bad. The p-bit representation is a direct formalization of this stance: one can carry "pro-d" and "anti-d" evidence simultaneously without collapsing the system into triviality.*

Remark 968 (Incompleteness and neutrality)**.** *Paraconsistency should be distinguished from mere uncertainty: it is possible (and common) that both channels are low, e.g. $E_O(s, a; d) \approx (0, 0)$, representing a kind of evaluative silence or irrelevance with respect to d. This matters because "no strong evidence either way" is not equivalent to "moderate net endorsement": downstream rules can treat low-information dimensions differently (e.g. defer, seek more data, or discount) rather than mistaking neutrality for agreement.*

Remark 969. *This stance resonates with the broader paraconsistent program used in Constructible Duality Logic and related approaches: inconsistency is handled by design rather than by denial [23]. In the Hyperseed context, this also connects to paraconsistent resonance constructions, where structured interference is treated as informative rather than merely pathological [24].*

18.2.2 Joy/woe and pleasure/pain

To connect to the phenomenological words, we need scalar projections of the p-bit.

Definition 246 (Joy/woe intensities as projections)**.** *Given a p-bit $v = (v^+, v^-) \in [0, 1]^2$, define:*

$$J(v) := v^+ \qquad \textit{(joy intensity)}, \qquad W(v) := v^- \qquad \textit{(woe intensity)}.$$

Given an evaluative field E_O, define:

$$J_O(s, a; d) := J\big(E_O(s, a; d)\big), \qquad W_O(s, a; d) := W\big(E_O(s, a; d)\big).$$

Remark 970 (Why take projections rather than a single valence)**.** *The choice to define $J(v)$ and $W(v)$ as direct coordinate projections (rather than, say, $v^+ - v^-$ or $v^+/(v^+ + v^-)$) is deliberate: it preserves the possibility that joy and woe co-occur without being forced into a one-dimensional tradeoff. A net-valence reduction would erase the distinction between $(0.9, 0.9)$ and $(0.9, 0.1)$ even though they plausibly correspond to very different phenomenology and very different action-guidance (the former signaling "high stakes" and the latter signaling "clear endorsement").*

Remark 971. *The intuition is almost austere: joy and woe are taken here as* readouts *of the positive and negative channels of evidence. This does not claim that joy is merely "belief in goodness"; rather, it asserts that whatever else joy and woe are, they can be operationally summarized (for decision purposes) by two monotone coordinates that can coexist. This corresponds to the Hyperseed core-concepts Joy and Woe (Hyperseed-Concepts* **??***, 204).*

Remark 972 (Joy/woe as action-guiding signals)**.** *Under this projection, $J_O(s, a; d)$ can be read as the strength of "pull" toward a along dimension d, while $W_O(s, a; d)$ is the strength of "push-back" or inhibition along the same dimension. Importantly, neither signal by itself dictates choice: a high J_O can be overridden by high W_O (e.g. morally tempting but dangerous actions), and a high W_O can coexist with a high J_O when the agent faces a costly duty or a meaningful sacrifice. This sets up later mechanisms in which regulation, prioritization, or resonance can decide how such tensions are handled rather than assuming they must be eliminated at representation time.*

Remark 973. *As a concrete example, take $v = (0.8, 0.7)$. Then $J(v) = 0.8$ and $W(v) = 0.7$: the system has strong pull and strong repulsion simultaneously. Phenomenologically this resembles ambivalence, dread-excitement, or "I want it and I fear it." The usefulness is that such a state need not be treated as an error to be eliminated; it becomes a stable input to downstream coherence and choice mechanisms.*

Remark 974 (Pleasure/pain)**.** *We treat* pleasure *as joy-dominant affect and* pain *as woe-dominant affect, but we do not identify them with mere sensation. Instead, pleasure/pain are taken to be* evaluative *and can attach to internal or external processes (physical, mental, social, spiritual). Formally, pleasure and pain are scalar summaries of (J_O, W_O) after aggregating across value dimensions and time horizons.*

Remark 975 (Aggregation across dimensions and horizons)**.** *The last sentence is intentionally underspecified: different tasks call for different aggregators. One natural family is a weighted combination across dimensions,*

$$\mathsf{Joy}(s, a) = \sum_{d \in \mathcal{D}} w_d \, J_O(s, a; d), \qquad \mathsf{Woe}(s, a) = \sum_{d \in \mathcal{D}} w_d \, W_O(s, a; d),$$

with context-dependent weights $w_d \geq 0$ (possibly normalized, possibly not), together with a temporal pooling rule that can emphasize short-term impact, long-term impact, or risk-sensitive tails. The key point is that whatever aggregation is used, it is applied after *representing the two-channel evidence per dimension, so that mixed or conflicting evidence remains visible until the stage where explicit policy choices (e.g. prioritizing* safety *as a constraint) are permitted to matter.*

Remark 976 (Why "pleasure/pain" are not identical to any single $d \in \mathcal{D}$)**.** *Although* comfort *is an obvious contributor, pleasure and pain as used here are not confined to the comfort axis: a person can experience pleasure in truth-telling, competence, belonging, or self-transcendence, and can experience pain in betrayal, shame, incoherence, or moral injury even in the absence of physical discomfort. Treating pleasure/pain as aggregated evaluative summaries is therefore a way to model how multiple value dimensions can jointly produce global affective tone, while still allowing finer-grained analysis at the level of $(s, a; d)$ when needed.*

Remark 977. *This remark aligns Pleasure and Pain (Hyperseed-Concepts 137, 129) with the idea that affect is not just peripheral sensation but a control-relevant appraisal. One simple toy aggregation is: average $J_O(s, a; d)$ and $W_O(s, a; d)$ over a chosen subset of d and over a short temporal window, and then compare the two. Yet the formalism deliberately leaves the aggregation operator* *open, because in real agents pleasure/pain is plastic, trained, and context-dependent. In particular, the same underlying stream of two-sided evaluative evidence can be re-weighted by attention, habituation, framing, or learned predictors of downstream outcomes, so that an "affective summary" is best treated as an agent-relative readout rather than a fixed physical observable. Formally, one can regard an aggregation operator as any map from a collection of per-dimension, time-indexed p-bits to a single p-bit (or small tuple of p-bits) that is subsequently used for control; leaving this open allows the theory to represent both hard-coded reflex-like aggregation and slowly learned, meta-cognitively editable aggregation.*

18.2.3 Aggregating across value dimensions without collapsing conflicts

Hyperseed wants aggregation, but not at the cost of erasing conflict. The simplest approach is: keep the vector structure for decision constraints, and use a scalar only as a *tie-breaker*. Concretely, the vector is used to define feasibility or dominance (so that certain tradeoffs are forbidden or at least not silently normalized away), while any scalarization is invoked only after the space of options has already been filtered to those that are acceptable under the explicit multi-dimensional constraints. This separation mirrors common practical decision procedures: first rule out options that violate non-negotiables, then optimize within what remains.

Definition 247 (Value-vector for an option)**.** *For fixed (s,a) define the* value-vector*:*

$$\mathbf{E}_O(s,a) := \big(E_O(s,a;d)\big)_{d\in\mathcal{D}} \in ([0,1]^2)^{\mathcal{D}}.$$

Remark 978. *This is the precise mathematical expression of "values are plural." Each coordinate is itself paraconsistent, so $\mathbf{E}_O(s,a)$ is a structured object: a vector of two-sided appraisals. A simple example is $\mathcal{D} = \{\mathsf{truth}, \mathsf{comfort}\}$, in which case $\mathbf{E}_O(s,a)$ is just a pair of p-bits. The usefulness is that we can express constraints like "must not strongly violate comfort" while still optimizing truth among feasible options. More generally, the formalism supports heterogeneous dimensions whose evidential sources differ: e.g.* truth *may be driven by predictive accuracy and calibration signals, whereas* comfort *may be driven by affective forecasting and physiological proxies; keeping separate coordinates prevents premature averaging from hiding which subsystem is "objecting." This also makes room for asymmetric policy rules, such as treating some dimensions as veto constraints (e.g. "never exceed woe threshold on safety") and others as objectives (e.g. "maximize joy on curiosity"), without forcing all of them into a single commensurate scale.*

Definition 248 (Preference order on p-bits)**.** *For evaluative purposes, define a partial order $\preceq_{\text{pref}}$ on $[0,1]^2$ by*

$$(v^+,v^-) \preceq_{\text{pref}} (u^+,u^-) \quad\iff\quad v^+ \le u^+ \text{ and } v^- \ge u^-.$$

Thus "better" means more positive evidence and less negative evidence.

Remark 979. *The order $\preceq_{\text{pref}}$ is the simplest monotone notion of "at least as good" compatible with two-sided evidence: you can improve by increasing the positive channel and/or decreasing the negative channel. It is partial rather than total because many pairs are incomparable (one has more joy but also more woe). This incomparability is not a technical nuisance; it is the formal reflection of conflicted evaluation (Hyperseed-Concept 198). Note that $\preceq_{\text{pref}}$ makes explicit that "net" affect is not the primitive object: a p-bit with high (v^+,v^-) may represent a tempting but risky option, whereas a p-bit with lower (u^+,u^-) may represent a dull but safe option; neither dominates the other without an additional principle. In this sense, a later tie-breaker scalar (if used at all) is properly interpreted as an* extra *policy commitment about how to resolve incomparabilities, not as part of the evidence itself.*

Example 15. *Let* $v = (0.7, 0.2)$ *and* $u = (0.8, 0.2)$*. Then* $v \preceq_{\text{pref}} u$ *because* u *has at least as much positive evidence and no more negative evidence. But* $v' = (0.9, 0.7)$ *and* $u' = (0.8, 0.2)$ *are incomparable: one is higher joy and higher woe, the other lower joy and lower woe. This is a minimal formal model of ambivalence versus calm satisfaction. One can also view* $(0.9, 0.7)$ *as "high activation conflict" (strong pull and strong push), whereas* $(0.8, 0.2)$ *is "cleanly positive"; treating them as incomparable preserves the psychologically natural distinction between intensity and valence-mixing, which would be lost under a single net score.*

Definition 249 (Pareto dominance for value-vectors)**.** *For two options* (s, a) *and* (s, a') *define:*

$$\mathbf{E}_O(s, a) \preceq_{\text{Pareto}} \mathbf{E}_O(s, a') \iff E_O(s, a; d) \preceq_{\text{pref}} E_O(s, a'; d) \quad \forall d \in \mathcal{D}.$$

An option is Pareto-undominated *if there is no other option that strictly improves at least one dimension and is no worse in all others (under* $\preceq_{\text{pref}}$*).*

Remark 980. *Pareto dominance is the canonical way to aggregate multiple objectives without assigning tradeoff weights: it marks an option as "unambiguously worse" only when it is worse (or equal) in every dimension and strictly worse in at least one. In a value setting, this has a moral flavor: if one option is no worse on every value and strictly better on at least one, then choosing the dominated option looks like pure irrationality or error. This connects directly to the rationality conditions later (Hyperseed-Concept 148). Importantly, using Pareto-undominated options as a first-stage filter does not require the agent to decide how to trade* truth *against* comfort*; it only requires the weaker commitment to avoid options that are jointly worse by the agent's own lights in every tracked dimension. When the undominated set is large (as is typical in high-dimensional, conflict-laden settings), a tie-breaker can be applied* afterwards*—for example, by a context-conditioned scalar functional* $T(\mathbf{E}_O(s, a))$ *that is allowed to depend on situational features, learned norms, or risk sensitivity—but such a functional is then explicitly a second-stage policy choice rather than a covert averaging that would hide the underlying conflict structure.*

This keeps value paraconsistency explicit: conflicts remain visible as non-comparabilities. It also makes room for principled "refusal" or "deferral" behaviors: if all available actions are Pareto-undominated yet each carries substantial woe on some dimension, the formalism can represent the state as one of genuine normative tension rather than forcing a spurious numeric optimum. Conversely, when a clear dominance relation exists, the model predicts stable agreement across a wide class of tie-breakers, since any reasonable scalarization that is monotone in the sense of $\preceq_{\text{pref}}$ will also avoid selecting Pareto-dominated actions.

18.3 Emotion as structured evaluation and resistance signals

Hyperseed's notion of emotion is not merely an add-on to cognition; it is a *control-relevant summary* of value and effort structure.

Remark 981. *The guiding idea is that emotion compresses a large evaluative landscape into a small set of control knobs: what to approach, what to avoid, what to endure, what to postpone. Formally, we will keep the compression mild: rather than forcing a scalar utility, we package (i) predicted value-evidence, (ii) predicted resistance/effort, and (iii) a coherence indicator. This should be read as a mathematical scaffold for the Hyperseed concept Emotion (Hyperseed-Concept* **??***) rather than as an attempt to exhaust phenomenology.*

Definition 250 (Effort/resistance estimator)**.** *Let* $c_O(s, a) \in V$ *denote an observer-relative effort cost (weakness/effort quantale value; cf. Section 3.7 and Section 8). Let* $\rho : V \to [0, \infty)$ *be any* *monotone "readout" mapping from quantale cost to a nonnegative scalar (e.g. ρ could be a chosen norm or a projection). Define the* scalar resistance estimate*:*

$$R_O(s,a) := \rho\big(c_O(s,a)\big).$$

Remark 982. *Here V is the underlying quantale (or cost domain) used earlier to encode weakness/effort; $c_O(s,a)$ is the cost object in that domain; and ρ is a chosen way to read it out as a single nonnegative number so we can compare options by "how hard." The monotonicity requirement on ρ is the only structural assumption: more cost in the quantale sense should not turn into less scalar resistance. This ties the present section back to the weakness/effort program [3, 2] and to the core notion Quantale Weakness (Hyperseed-Concept 143).*

Remark 983. *A simple example is when $V = [0,\infty]$ with the usual order, and $c_O(s,a)$ is already a scalar time/energy estimate; then ρ can be the identity. A more Hyperseed-flavored example is when V encodes multi-resource cost (compute, attention, social risk) and ρ is a chosen projection that fits the observer's current strategic posture.*

Definition 251 (Emotion as an evaluation-resistance pattern)**.** *Fix a time horizon H and let $\widehat{E}_O(s,a;d)$ denote the predicted (future) value-evidence for d over horizon H if action a is taken at s. An* emotion token *for O at (s,a) is a tuple*

$$\mathrm{Em}_O(s,a) := \Big(\ \widehat{\mathbf{E}}_O(s,a),\ R_O(s,a),\ \kappa_O(s,a)\ \Big),$$

where $\kappa_O(s,a)$ is an optional coherence/resonance signal (defined below) indicating how aligned the value evidence is across dimensions and timescales.

Remark 984. *Intuitively, $\mathrm{Em}_O(s,a)$ is what an agent "feels about" doing a from s, in a way that is already poised to influence control: it includes (i) the forecasted evaluative consequences, (ii) the forecasted effort/resistance, and (iii) whether the forecasted consequences harmonize or fight each other. A simple example is "approach a difficult conversation": high predicted truth and belonging benefits, but also high resistance; and depending on the internal model, high or low coherence between dimensions.*

Remark 985. *The usefulness is compositional. The prediction/control layer can be used to compute $\widehat{E}_O$; the weakness/effort layer yields R_O; and the resonance layer yields κ_O. Emotion then becomes a derived interface object that is compatible with the rest of the stack, rather than a new primitive.*

Remark 986 (Why this matches the Hyperseed intent)**.** • *Joy/woe components live in $\widehat{\mathbf{E}}_O$.*

- *"Tension" and "relief" can be modeled as changes in R_O and/or changes in the coherence signal κ_O.*
- *Complex emotions correspond to characteristic patterns in the vector-valued evidence, rather than to single scalar utilities.*

18.3.1 Compassion

Compassion is treated as a value dimension whose evaluation depends on modeled others.

Definition 252 (Compassion as other-referenced evaluation)**.** *Let $\mathcal{A}$ be a set of agents, with distinguished self agent* self $\in \mathcal{A}$*. For each $j \in \mathcal{A}$ assume O carries an estimate of that agent's joy/woe* *for outcomes, encoded as p-bits* $E_{O\to j}(s,a;\textsf{joy})$ *and* $E_{O\to j}(s,a;\textsf{woe})$*. Define a compassion score (as a p-bit) by*

$$E_O(s,a;\textsf{compassion}) := \bigoplus_{j\in\mathcal{A}\setminus\{\mathrm{self}\}} \Big(E_{O\to j}(s,a;\textsf{joy}) \ \otimes\ \neg E_{O\to j}(s,a;\textsf{woe})\Big),$$

where $\neg$ *is p-bit negation (swap evidence components) and* $\oplus,\otimes$ *are the aggregators fixed in the core.*

Remark 987. *This definition treats compassion (Hyperseed-Concept 81) as intrinsically* model-based*: it depends on how* O *represents other agents' prospective joy and woe. In that sense it links value theory to social cognition and to the self/other boundary formalism earlier (Hyperseed-Concept 165). The algebraic structure—combine a model of others' joy with the negation of others' woe, then aggregate over others—makes explicit what is otherwise left as moral rhetoric.*

Remark 988. *A simple example: suppose there is one other agent* j*, and* $E_{O\to j}(s,a;\textsf{joy}) = (0.6, 0.2)$ *while* $E_{O\to j}(s,a;\textsf{woe}) = (0.4, 0.7)$*. Then* $\neg E_{O\to j}(s,a;\textsf{woe}) = (0.7, 0.4)$*, and the combined term is* $E_{O\to j}(\textsf{joy}) \otimes \neg E_{O\to j}(\textsf{woe})$*. The exact numeric result depends on the quantale operations, but the intended qualitative reading is: compassion rises when we foresee others' joy and do not foresee their woe.*

Remark 989. *This definition is only a template: it says compassion is evaluated by modeling others' joy/woe and aggregating. Different observers may use different agent sets, weights, and aggregators. The key point is structural: compassion is not a mysterious extra primitive; it is a particular kind of evaluative dependence on other-modeled affect.*

Remark 990. *In multi-agent contexts, formal compassion also connects to arguments that prosocial constraints can increase group-level efficiency and coordination capacity, rather than merely adding moral overhead [6]. In Hyperseed terms, compassion can be seen as an ingredient in building high-resonance cultural patterns that reduce collective resistance.*

18.4 Values, feelings, goals, and reward

Hyperseed distinguishes between transient *feelings* and persistent *values.* Goals are commitments that bind action selection to values.

18.4.1 Values as stable patterns; feelings as local evaluations

Definition 253 (Value as a stable evaluative pattern)**.** *A value dimension* $d \in \mathcal{D}$ *is said to be a* stable value for observer O *over an interval of proto-time* T *if the induced evaluation function*

$$(s,a) \mapsto E_O(s,a;d)$$

is (approximately) invariant under the observer's typical context changes in that interval (up to a tolerance set by O*'s weakness/effort limits).*

Remark 991. *The intuition is that a value is not a single episodic feeling; it is a* pattern of appraisal *that persists across changing circumstances. "Truth matters" means: across many contexts, actions predicted to support truth tend to receive stable positive evidence relative to that dimension. This explicitly ties values to the broader Hyperseed idea that ontology is reconstructed in terms of stable patterns (Hyperseed-Concept 130; cf. [5]).*

Remark 992. *A simple example: if $d = \textsf{compassion}$ is stable for O over T, then even as the specific people and situations vary, the mapping $(s, a) \mapsto E_O(s, a; d)$ continues to privilege actions that reduce others' woe. The usefulness of this definition is that it gives a crisp criterion for when we are justified in treating some evaluative coordinate as a relatively fixed constraint in planning, rather than as noise.*

Definition 254 (Feeling as time-local affective readout)**.** *A* feeling token *for O at time t is any scalar or low-dimensional readout*

$$F_O(t) = \Phi\Big(\mathrm{Em}_O(s_t, a_t)\Big),$$

where Φ is a bounded map (e.g. combining joy/woe across a small subset of dimensions, plus resistance).

Remark 993. *Here Φ formalizes the act of compression: the mind cannot (and need not) represent the entire evaluative tensor at every moment, so it reads out a small summary. In common terms, this includes things like "anxiety level," "relief," or "contentment." The formal definition is deliberately permissive: many different physiological and cognitive architectures may implement different Φ. This corresponds to Feelings (Hyperseed-Concept* **??***) as time-local indicators rather than enduring commitments.*

Remark 994. *A simple example is $\Phi(\widehat{\mathbf{E}}, R, \kappa) = \sum_{d \in D_0} \lambda_d (J - W) - \beta R$ for a small subset $D_0 \subseteq \mathcal{D}$ active in the current attentional focus. The usefulness is that the theory can talk about feelings without reifying them as primitives: they are observer-dependent readouts of deeper evaluative structure.*

This makes the common-sense relationship precise: feelings are local readouts; values are stable patterns in evaluative fields.

18.4.2 Goals: explicit and implicit

Definition 255 (Explicit goal)**.** *An* explicit goal *for O is a representable constraint G on trajectories or outcomes:*

$$G \subseteq \Omega$$

where Ω is a space of outcomes or histories (as formalized in the prediction/control layer). Explicitness means O can name G in its representational language (Section 13).

Remark 995. *The intuition is that an explicit goal is* declarative*: it lives in the agent's representational economy as something like "finish the proof," "do not lie," or "keep the system within a safety envelope." This corresponds to Goals: Explicit (Hyperseed-Concept* **??***). The set-theoretic form $G \subseteq \Omega$ is standard in control theory: it lets us treat goals as constraints on whole trajectories rather than one-step rewards.*

Remark 996. *A simple example is Ω being the set of finite-length action-observation histories up to horizon H, and G being those histories in which a safety predicate never becomes false. The usefulness is that such a goal can be used as a hard feasibility constraint, independent of how one trades off other values within the safe region.*

Definition 256 (Implicit goal)**.** *An* implicit goal *for O is any stable regularity in action selection induced by E_O and c_O that* need not be representable as a named constraint *in O's language. Equivalently: an implicit goal is a goal implemented by policy/habit rather than by declarative representation.*

Remark 997. *Implicit goals (Hyperseed-Concept **??**) live in the dynamics rather than in the discourse: the agent behaves* as if *it were optimizing or constraining something, even if it cannot articulate it. This matches both everyday psychology (habits, avoidance patterns) and machine learning practice (policies trained end-to-end without symbolic specification). In Hyperseed terms, this is also a bridge from the habit dynamics layer to the goal layer.*

Remark 998. *A simple example is "avoid embarrassment" in a person who never explicitly adopts that phrase as a goal, yet repeatedly chooses actions that lower predicted social woe. Another example is a learned controller that repeatedly chooses low-resistance actions even when no explicit "minimize effort" rule is represented. The usefulness is that it lets the formalism describe goal-directedness as a spectrum rather than a binary property.*

Remark 999 (Hyperseed intent)**.** *This matches the everyday distinction: a person may have an implicit goal of avoiding embarrassment without ever having stated it as a goal. Similarly, many learned systems optimize a reward function (implicit goal) without explicit symbolic commitments.*

18.4.3 Reward as a derived (optional) scalar

Hyperseed often speaks in terms of goals/values rather than rewards. Still, rewards are convenient for algorithms, so we define them as *derived*. In particular, the intent is that the underlying evaluative object remains the structured, potentially conflicting evidence (and any associated constraints), while the scalar is an interface layer for decision procedures that require a single number. This also clarifies a methodological point in this section: "reward" is not treated as metaphysically primitive, but as a deliberately lossy projection that can be turned on or off depending on the reasoning context.

Definition 257 (Paraconsistent reward from evaluative evidence)**.** *Fix weights $\lambda_d \geq 0$ for $d \in \mathcal{D}$ and a resistance weight $\beta \geq 0$. Define a scalar reward proxy:*

$$r_O(s,a) := \sum_{d \in \mathcal{D}} \lambda_d \Big(J_O(s,a;d) - W_O(s,a;d) \Big) \; - \; \beta \, R_O(s,a).$$

It is sometimes helpful to read the expression as a sequence of modeling choices: (i) treat each dimension d as producing two-sided evaluative evidence, (ii) convert that two-sided evidence into a signed "net" quantity via $(J - W)$, (iii) trade off dimensions by nonnegative weights, and (iv) subtract an additional term capturing friction, cost, or internal opposition via R_O. Nothing in the definition requires that the units of $(J - W)$ match those of R_O; rather, the choice of β (and the scale of the λ_d) is what makes the overall scalar comparable across actions. In applications, λ_d and β can be treated as hyperparameters to be elicited, learned, or tuned, with the important caveat that tuning changes the induced trade-offs and can therefore change which conflicts get "smoothed over" by the scalarization.

Remark 1000. *The intuition is utilitarian in form but not in metaphysics: r_O is a* computational convenience*, a single number extracted from a richer evaluative structure. The difference $(J - W)$ is a simple way to turn two-sided evidence into a signed quantity, while βR_O penalizes resistance. This relates to Reward and Reward Maximization (Hyperseed-Concepts 160, 161), while preserving the prior commitment that the vector structure is primary.*

A further implication of calling r_O "derived" is that two agents (or two subsystems) may share the same underlying (J, W, R) structure yet adopt different scalarizations for different tasks. For example, a fast planner might use a coarse scalar proxy for search, while a slower reflective process revisits the same candidate actions using the unsummed vector evidence to surface tensions and failure modes that scalarization can conceal. This is one way to reconcile the convenience of scalar interfaces with the broader aim of keeping value conflict legible.

Remark 1001. *A simple example: if $\mathcal{D} = \{\mathsf{truth}, \mathsf{comfort}\}$ with $\lambda_{\mathsf{truth}} = \lambda_{\mathsf{comfort}} = 1$ and $\beta = 0.5$, then $r_O(s, a)$ rewards actions with net positive evidence on each dimension while also preferring lower-resistance options. The usefulness is that this makes contact with standard planning and reinforcement learning interfaces (cf. broader AGI algorithmic discussions in [19]), without forcing the entire theory to be reward-centric.*

In that example, note that the scalar can increase either by raising J (more supporting evidence), lowering W (less opposing evidence), or lowering R (less resistance). These are not interchangeable changes in the underlying representation: increasing J is not the same as decreasing W, even if both increase $(J - W)$, and the distinction remains available to any procedure that inspects the full evidence structure rather than only the scalar. This is precisely the sense in which the scalar is optional: it can be used for action selection while leaving the evidence geometry intact for explanation, auditing, and conflict management.

Remark 1002 (Reward maximization vs goal seeking)**.** *Reward maximization uses the scalar r_O. Goal seeking treats some constraints as hard or lexicographic (e.g. "do not violate compassion below a threshold") and uses r_O only within the feasible set. Hyperseed's conceptual preference is typically the latter: keep conflicts visible, and use scalars carefully.*

A useful way to formalize the remark is to view goal seeking as a two-stage procedure: first restrict to a feasible region defined by constraints on (some functions of) (J, W, R), and only then optimize a scalar proxy on that region. This helps separate "non-negotiables" (hard constraints, thresholds, or lexicographic priorities) from "trade-off" dimensions where scalarization is acceptable. It also clarifies how resistance can be treated: in some contexts R_O is a soft cost (hence the βR_O term), while in other contexts it can itself induce feasibility constraints (e.g. "avoid actions that trigger extreme internal conflict"), again illustrating that r_O is not the sole decision object.

Proposition 27 (Scalarization implies Pareto optimality when used monotonically)**.** *Fix a finite set of actions A at state s and suppose we define a score*

$$\mathrm{Score}(a) := \sum_{d \in \mathcal{D}} \lambda_d \Big(J_O(s, a; d) - W_O(s, a; d) \Big) - \beta\, R_O(s, a),$$

with $\lambda_d > 0$ for all d and $\beta \geq 0$. If an action $a^\star$ maximizes Score*, then $a^\star$ is Pareto-undominated with respect to the componentwise order*

$$\Big(J_O(s, a; d),\ -W_O(s, a; d) \Big)_{d \in \mathcal{D}} \quad \textit{and} \quad - R_O(s, a).$$

Remark 1003 (Intuition and role in the section)**.** *The proposition says: if you insist on collapsing multiple value dimensions (and resistance) into one scalar by a strictly monotone linear combination, then any scalar maximizer cannot be* obviously worse *than another available action across all dimensions at once. In other words, scalarization cannot accidentally pick an option that is dominated on every criterion that the scalar score rewards.*

Remark 1004. *This result matters because Hyperseed repeatedly warns against premature scalar collapse, yet also acknowledges the practical convenience of scalar rewards. The proposition is a small sanity bridge between these stances: it shows that a monotone scalar proxy at least respects the minimal Pareto boundary of the richer evaluative geometry.*

A small but practically relevant nuance is why the proposition assumes a finite action set. With infinite (or continuous) action spaces, one typically needs additional regularity to guarantee that a maximizer exists (e.g. compactness of the feasible action set and continuity of Score, or else replacing "maximizes" with "is optimal" in the sense of a supremum). The Pareto-undominance conclusion is fundamentally about *any* maximizer that does exist: once an argmax is well-defined, the same dominance contradiction goes through. Thus the finiteness assumption is best read as a convenient sufficient condition for existence rather than a conceptual limitation of the monotonicity idea.

Proof. Assume for contradiction that $a^\star$ is dominated by some a such that for every d, $J_O(s,a;d) \geq J_O(s,a^\star;d)$ and $W_O(s,a;d) \leq W_O(s,a^\star;d)$, with strict inequality for at least one dimension or strictly smaller resistance. Then each term $\lambda_d(J-W)$ is at least as large for a as for $a^\star$, and strictly larger for at least one d. Also $-\beta R_O(s,a)$ is at least as large as $-\beta R_O(s,a^\star)$ (and strictly larger if resistance strictly decreases and $\beta > 0$). Hence $\text{Score}(a) > \text{Score}(a^\star)$, contradicting maximality of $a^\star$. $\square$

Proof sketch. The proof is a direct monotonicity argument. Dominance means every coordinate that contributes positively to Score is at least as favorable for a as for $a^\star$, and at least one is strictly more favorable. Because the weights λ_d are strictly positive (and $\beta \geq 0$), the strict improvement survives summation, contradicting the assumption that $a^\star$ maximizes the scalar score. $\square$

Remark 1005. *The key step is that dominance is expressed in the same direction as the scalar score: increasing J and decreasing W weakly increases each weighted term, and decreasing resistance weakly increases the $-\beta R$ term. This is why strict positivity of λ_d is important: it prevents an improved dimension from being "invisible" to the scalar. Geometrically, the argument says that a linear functional with positive coefficients cannot be maximized in the interior of a region that is strictly dominated in every outward direction.*

One can also read the strict positivity requirement as a warning about "dropping" dimensions during scalarization. If some $\lambda_d = 0$, then improvements in that dimension do not affect the scalar score at all, and the scalar maximizer can be dominated with respect to the full vector order that includes that dimension. In that sense, $\lambda_d > 0$ encodes a minimal respect condition: every dimension declared relevant to the Pareto comparison must have at least some influence on the scalar proxy. Similarly, if $\beta = 0$, then resistance is ignored by the scalar; the proposition still holds for the reduced vector order that omits $-R_O$, but it no longer guarantees undominance once resistance is added back as an additional coordinate.

Remark 1006. *The proposition does not claim scalar rewards are sufficient; it only states a sanity property: if you do collapse to a scalar in a strictly monotone way, you do not accidentally pick a dominated option. The paraconsistent structure still matters because the (J, W) values can encode conflict and incomplete evidence.*

A further limitation is that Pareto-undominance is only a minimal rationality property. Many Pareto-undominated actions can still be mutually incomparable, and the scalarization picks one according to a particular trade-off surface determined by (λ, β). Hyperseed's emphasis on keeping conflicts visible is partly a reminder that, when the underlying evidence is paraconsistent or incomplete, the set of Pareto-undominated actions can be large, and the choice among them may require additional deliberative structure (e.g. constraints, lexicographic rules, or context-sensitive priorities) rather than a single fixed scalar.

18.5 Resonance as coordination signal for value conflict

A core Hyperseed motivation is that action can be "coherent" even when values conflict. We formalize coherence using the resonance mapping from Section 3.9. In this subsection, "coherent" is meant in the dynamical/behavioral sense: the agent can stably select and execute a policy whose internal motivational components are sufficiently aligned to sustain follow-through, even if the agent cannot produce a single propositional description that renders all value-claims jointly satisfiable.

Remark 1007. *The philosophical point is subtle: coherence is not the same as consistency. An agent may coherently act under conflict by aligning many partial tendencies into a single trajectory, even if those tendencies cannot be made mutually noncontradictory at the representational level. Resonance (Hyperseed-Concept 159) is a formal proxy for this kind of alignment; dissonance (Hyperseed-Concept 97) is a proxy for cancellation and internal struggle.*

One practical reason to separate coherence from consistency is that real agents often face incommensurable desiderata (e.g., loyalty vs. fairness, exploration vs. safety, short-term relief vs. long-term health). In such settings, insisting on representational consistency can force an artificial collapse of plural values into a single scalar, whereas the resonance framing aims to preserve plurality while still yielding actionable structure.

Definition 258 (Complex value signal for a dimension)**.** *For each dimension d and option (s, a), define the complex signal*

$$z_O(s, a; d) := \sigma_C\big(E_O(s, a; d)\big) \in \mathbb{C},$$

where σ_C is the logic-to-complex map introduced in the core.

Remark 1008. *Here σ_C is the previously defined embedding of p-bit evidence into the complex plane, used to make "interference" and "phase alignment" algebraically explicit (Section 3.9). The intuition is that each value dimension contributes a complex phasor, and the overall alignment of these phasors measures how coordinated the value system is under the contemplated option.*

It is helpful to keep two roles of $z_O(s, a; d)$ conceptually distinct: its *magnitude* corresponds to the strength or confidence of the dimension-specific evaluation under (s, a), while its *argument* (phase) corresponds to the "direction" of the evaluation (broadly, pro vs. con, or endorsement vs. veto, depending on how σ_C was set up). This separation is exactly what makes it possible for strong but opposing evaluations to cancel, rather than being forced into an arbitrary compromise.

Remark 1009. *A toy example: if two dimensions yield complex signals of similar argument (phase), their sum has larger magnitude; if they yield signals with opposite phase, they partially cancel. This gives a clean mathematical image of "my values pull in the same direction" versus "my values fight each other."*

Note that cancellation can occur even when each individual dimension is "certain" (large magnitude): high-confidence opposition yields large destructive interference. Conversely, weak or ambiguous evaluations (small magnitude phasors) contribute little to either alignment or conflict, which matches the intuition that faint preferences should not dominate the global coordination signal.

Definition 259 (Weighted resonance magnitude)**.** *Fix weights $w_d \geq 0$ and define the total complex signal*

$$Z_O(s, a) := \sum_{d \in \mathcal{D}} w_d \, z_O(s, a; d).$$

Define the resonance magnitude

$$\kappa_O(s,a) := |Z_O(s,a)|.$$

Optionally define an interference term (as in Section 3.9) by

$$I_O(s,a) := |Z_O(s,a)|^2 - \sum_{d\in\mathcal{D}} |w_d z_O(s,a;d)|^2.$$

The nonnegativity constraint $w_d \geq 0$ ensures that weights encode relative salience or priority without inverting a dimension's meaning; sign flips are instead represented through the phase of z_O. In applications, w_d can be used to encode long-run commitments (stable importance weights) while allowing the evidence-to-phase map σ_C to reflect context-sensitive endorsement or rejection.

Remark 1010. *The magnitude $\kappa_O(s,a)$ is a scalar summary of alignment; it is large when many weighted dimensions agree in "direction" (as encoded by the complex phases induced by σ_C). The interference term $I_O(s,a)$ measures whether the whole is more or less than the sum of squared parts; negative values correspond to destructive interference, a particularly crisp signature of internal conflict.*

Because I_O is defined using squared magnitudes, it admits a familiar geometric reading: expanding $|Z_O|^2$ yields cross-terms that are proportional to pairwise phase agreement between dimensions. Thus I_O can be interpreted as aggregating pairwise "coordination" contributions, making it useful when one wants to diagnose which sets of dimensions are jointly supportive versus mutually cancelling (though such diagnosis requires looking beyond the scalar and into the underlying phasors).

Remark 1011 (Interpretation)**.** • *Large κ_O indicates that the dimension-signals line up in phase (coherent "yes" or coherent "no" structure).*

- *Negative interference $I_O < 0$ corresponds to cancellation between dimensions: this is one clean formal proxy for internal value conflict.*

  This gives a mathematically explicit handle on Hyperseed's qualitative talk of resonance/dissonance.

Importantly, κ_O alone does not claim that the option is "good" in a single-dimensional sense; rather, it measures the *degree of coordinated support* among the dimensions as encoded. An option can have high resonance because many dimensions coherently reject it (a coordinated "no"), which is still useful information for an agent that needs to avoid self-undermining action: coordination can be coordination toward refusal as well as toward pursuit.

Definition 260 (Resonant feasible choice rule)**.** *Let $A_{\text{Pareto}}(s) \subseteq A$ be the Pareto-undominated set at s (under $\preceq_{\text{pref}}$ across $\mathcal{D}$). Define a* resonant choice *as any maximizer of*

$$a^\star \in \arg\max_{a\in A_{\text{Pareto}}(s)} \Big(\kappa_O(s,a) - \beta R_O(s,a)\Big),$$

with $\beta \geq 0$.

The Pareto filter plays a normative role: it prevents resonance from favoring an option that is strictly worse on all dimensions (which could otherwise happen if the mapping σ_C induces phases that accidentally align even for uniformly dominated actions). In this sense, resonance is used as a *coordination tie-breaker* among options that are already defensible from a pluralistic standpoint.

Remark 1012. *The rule has a clear two-stage meaning. First, eliminate actions that are unambiguously worse in the Pareto sense. Second, among the survivors, prefer options whose value dimensions mutually reinforce (high resonance) and do not demand excessive resistance. The usefulness is that it preserves pluralism of values while still producing a practical decision rule. It also mirrors a theme that recurs in Hyperseed: do not resolve conflict by denial; resolve it by coordination.*

The parameter β controls the tradeoff between "internal alignment" and "effortfulness" (as measured by R_O). At $\beta = 0$, the rule reduces to selecting the most internally coordinated option among the Pareto-undominated set; as β increases, the agent increasingly favors options that are easier to execute or that require less overcoming of aversion, fatigue, or constraint. This makes the rule suitable both for idealized deliberation (small β) and for bounded, resource-aware choice (larger β).

This rule is intentionally conservative: it does not erase value conflicts; it selects among already-undominated options using coherence and effort. In particular, it treats conflict as a first-class signal: destructive interference lowers κ_O (and can render I_O negative), thereby disfavoring options that would likely produce vacillation, paralysis, or regret due to internal cancellation, even when no single dimension can decisively veto the action.

18.6 Individuation, self-transcendence, and open-ended intelligence

Hyperseed treats *individuation* and *self-transcendence* as key drivers of open-ended growth. Here we formalize them as meta-values: dimensions in $\mathcal{D}$ that regulate how the value system itself evolves. Concretely, calling these dimensions "meta-values" means that they do not merely score external world-states, but also score *candidate updates* to the agent's self-model and evaluative vocabulary: a change to $\mathcal{D}$, to its internal organization, or to the policies that consult it can itself be treated as an object of evaluation in $E_{O,t}$.

Definition 261 (Individuation and self-transcendence as meta-values)**.** *Assume* $\mathsf{individuation}, \mathsf{self\_transcendence} \in \mathcal{D}$.

- *High evaluation on* $\mathsf{individuation}$ *favors options that preserve coherence and continuity of the self-model (Section* **??***) and reduce destructive self-fragmentation.*
- *High evaluation on* $\mathsf{self\_transcendence}$ *favors options that expand the representational and motivational space, including revising or enlarging* $\mathcal{D}$ *itself.*

Remark 1013. *A useful operational reading is that* $\mathsf{individuation}$ *implicitly penalizes updates that increase internal inconsistency, unresolvable goal-conflict, or identity discontinuities across prototime, while* $\mathsf{self\_transcendence}$ *implicitly rewards updates that increase model class capacity, perspective-taking, and the ability to represent previously inexpressible goals. In this sense, these two dimensions can be understood as providing a principled "regularizer" and "expander" for value learning: one controls brittleness and fragmentation, the other controls stagnation and premature closure.*

Remark 1014. *These two dimensions capture a constructive tension. Individuation (Hyperseed-Concept* **??***) protects the integrity of the self as a temporally extended pattern (Hyperseed-Concept 166); self-transcendence (Hyperseed-Concept 167) prevents that integrity from ossifying into a closed world. In Russellian terms, individuation provides logical discipline, while self-transcendence supplies the imaginative expansion that keeps the logical system from mistaking its axioms for the cosmos.*

Remark 1015. *One way to see the tension is to treat the self-model as a hypothesis that must remain predictive enough to coordinate action over time, while also remaining revisable enough to accommodate new evidence and new possibilities. Overemphasizing individuation risks "identity lock-in" (a self-model that refuses to update even when it systematically mispredicts), whereas overemphasizing self-transcendence risks "identity diffusion" (a self-model that updates so freely that long-horizon commitments and responsibility become unstable). The intended regime is neither rigidity nor dissolution but a dynamic equilibrium in which continuity is maintained through intelligible, narratively compressible change.*

Remark 1016. *A simple example: in creative work, individuation may favor maintaining a coherent long-term project identity, while self-transcendence may favor learning a radically new tool or exploring a new domain that reshapes the project. The usefulness of making these explicit dimensions is that "growth" can be formalized as a controlled revision of the value basis rather than as an unstructured drift.*

Remark 1017. *The same structure applies outside creative work. In moral development, individuation can favor keeping faith with prior commitments and relationships (so that change does not become betrayal-by-forgetting), while self-transcendence can favor encountering unfamiliar perspectives that force a refinement of what those commitments mean. In epistemic terms, individuation supports calibration and internal consistency; self-transcendence supports the discovery of new latent variables, new abstractions, and new ethical patient-classes that were previously absent from* $\mathcal{D}$.

Definition 262 (Open-ended intelligence as stable-yet-revisable value dynamics)**.** *Let* $\mathcal{D}_t$ *be the set of active value dimensions at proto-time* t *and* $E_{O,t}$ *the corresponding evaluative field. An observer exhibits* open-ended intelligence *on an interval if:*

1. *(*stability*) there exist nontrivial dimensions in* $\mathcal{D}_t$ *that remain stable for substantial sub-intervals (values do not dissolve into noise);*

2. *(*revisability*) there exist times* $t < t'$ *such that* $\mathcal{D}_{t'} \neq \mathcal{D}_t$ *(the value vocabulary can grow or reorganize);*

3. *(*coherence under revision*) revisions do not make action selection trivial; i.e. the agent retains a non-degenerate Pareto set and nontrivial resonance structure.*

Remark 1018. *Condition (1) can be read as requiring that some evaluative commitments persist long enough to support planning, learning, and accountability, while condition (2) requires that the system can escape local optima in* value space *rather than merely optimizing within a fixed objective. Condition (3) then rules out a degenerate "anything goes" regime: if revisions create a situation where almost all options are mutually incomparable (or all options become equally optimal), then the value system no longer provides actionable guidance. The non-degenerate Pareto requirement is thus not an aesthetic preference but a functional criterion ensuring that evaluation continues to constrain behavior in informative ways.*

Remark 1019. *It is also helpful to distinguish revisability at three granularities: (i)* parameter *revision (reweighting dimensions or changing trade-off curves), (ii)* structural *revision (splitting/merging dimensions, introducing new latent factors, changing the geometry of* $\mathcal{D}$*), and (iii)* semantic *revision (changing what a dimension tracks via new world-model concepts). The definition above permits all three, provided that stability and non-degeneracy are maintained somewhere in the system so that the agent remains a coherent decision-maker throughout the revision process.*

Definition 263 (Value-basis revision operator (one formalization))**.** *Let* Rev *be a (possibly stochastic) operator such that*

$$(\mathcal{D}_{t+1}, E_{O,t+1}) \;=\; \mathsf{Rev}(\mathcal{D}_t, E_{O,t}, \mathcal{M}_t, \mathcal{H}_t),$$

where $\mathcal{M}_t$ denotes the agent's current world/self-model state and $\mathcal{H}_t$ denotes relevant history (experience, reflection, social input). We say Rev *is* individuation-respecting *if it places high weight on preserving identity-continuity constraints (e.g. bounded discontinuity of self-representations across steps), and* self-transcendence-capable *if it assigns nonzero probability (or admissibility) to expansions and reorganizations of $\mathcal{D}_t$ when such changes improve expressive adequacy or reduce systematic blind spots.*

Remark 1020. *This operator view makes explicit that "meta-values" can be implemented as preferences over* updates *rather than only over* outcomes*. For example, individuation-respecting revision can be modeled by continuity penalties (e.g. bounding divergences between successive self-model encodings), while self-transcendence-capable revision can be modeled by allowing hypothesis-class expansion steps (e.g. adding a new dimension when prediction error or moral uncertainty remains persistently high). Importantly, these are not merely engineering conveniences: they correspond to the normative intuition that growth should be both intelligible (continuous enough to remain "me") and exploratory (open enough to become more than prior limitations).*

Remark 1021. *This definition makes "open-ended intelligence" (Hyperseed-Concept 127) a property of* value dynamics*, not only of problem-solving performance. The agent must be stable enough to be itself, yet plastic enough to become otherwise; and the plasticity must not collapse decision-making into incoherence. This connects to the broader AGI view that intelligence is partly the ability to manage changing task and value regimes without losing functional unity [19].*

Remark 1022. *The "stable-yet-revisable" requirement is closely related to the exploration–exploitation balance, but lifted to the level of* normative representation*. Exploitation corresponds to acting coherently under a relatively stable $\mathcal{D}_t$; exploration corresponds not only to trying new actions but to considering new* criteria *for action. On this reading, self-transcendence is a driver of normative exploration (search over value bases), while individuation is a driver of normative exploitation (consolidation and integration so that the expanded basis yields a unified self).*

Remark 1023. *A simple example is an agent that begins with $\mathcal{D}_t = \{\mathsf{safety}, \mathsf{comfort}\}$, later adds* truth*, and later refines* belonging *into separate dimensions for "local community" and "global commons." Revisability is not mere addition: it can involve splitting, merging, or reweighting dimensions. The usefulness of the non-degeneracy condition is that it blocks a pathological case where everything becomes incomparable and the Pareto set becomes uninformative.*

Remark 1024. *A complementary pathological case is the opposite extreme: revisions that collapse the evaluative field so that almost everything becomes tied (e.g. by flattening all distinctions or by introducing an overriding dimension that trivializes the rest). Both extremes destroy guidance: pervasive incomparability yields paralysis, while pervasive ties yield arbitrariness. The point of requiring a nontrivial resonance structure is to ensure that the system retains patterned sensitivities—some options systematically "fit" the current integrated value basis better than others, in a way that can be learned, explained, and used for planning.*

Remark 1025 (Open-ended benefit)**.** Open-ended benefit *is the same idea applied normatively: value evolution is not only open-ended, but tends to expand the feasible set of futures in which multiple agents can realize joy and reduce woe. Formally, this is captured by including compassion and non-destructive coordination constraints in the revision rule for $\mathcal{D}_t$.*

Remark 1026. *The phrase "expand the feasible set of futures" can be read in a multi-objective sense: revisions should (in expectation and subject to uncertainty) increase the set of attainable trajectories that score well across a plurality of agents' evaluative fields, while avoiding expansions that are achieved by exploitation, coercion, or irreversible harm. This is stronger than simple "innovation" or "growth" because it adds constraints on the* means *of expansion (non-destructive coordination) and on the distribution of its benefits (compassion).*

Remark 1027. *Open-ended benefit (Hyperseed-Concept 126) adds a social and ethical tilt: it asks that the expansion of possibilities not be purchased by narrowing others' possibilities. In multi-agent settings, this is naturally connected to compassion and to collective coordination effects, echoing broader Hyperseed arguments about prosocial efficiency [6].*

Remark 1028. *In practice, "compassion and non-destructive coordination constraints" can be interpreted as admissibility conditions on* Rev*: certain updates to* $\mathcal{D}_t$ *(or to the weights and semantics of its dimensions) are disallowed if they systematically increase incentives for domination, deception, or zero-sum lock-in. This makes open-ended benefit a criterion not only on chosen actions but on the long-run* shape *of the evolving value system: it favors meta-stable attractors in which agents remain mutually legible and able to negotiate, rather than drifting toward value regimes that externalize costs onto others or render cooperation impossible.*

18.7 Rationality: coherence between belief, value, and action under constraints

Hyperseed uses "rationality" broadly: not merely logical consistency, but action selection that respects evidence, values, and resource limits. In particular, the term is intended to cover both (i) *epistemic discipline* (how strongly different outcomes are supported by the belief model) and (ii) *practical discipline* (how choices are filtered and ranked when multiple value-criteria and costs compete). This usage intentionally departs from the common identification of rationality with either deductive closure or with unconstrained expected-utility maximization, since the present framework makes room for conflict, incompleteness, and bounded computation as first-class structural features.

Remark 1029. *In this setting, rationality (Hyperseed-Concept 148) is not the classical ideal of a perfectly consistent belief set with a total utility function. It is instead a form of* coherence under limitation*: coherence across paraconsistent belief evidence, paraconsistent value evidence, and the resistances imposed by finite resources. This is consonant with the general Hyperseed ethos that weakness is fundamental rather than accidental [3, 2]. A useful way to read "coherence" here is as a family resemblance between internal components: beliefs constrain which futures are treated as live, values constrain which futures are treated as desirable, and resistances constrain which futures are treated as* reachable without collapse*. The relevant notion of "consistency" is therefore not global consistency of all attitudes at once, but local compatibility sufficient to support stable action selection.*

Definition 264 (Rational decision criterion (minimal version))**.** *Fix state* s*. A choice rule* $\pi_O(\cdot \mid s)$ *is* (Hyperseed-minimally) rational *if it satisfies:*

1. ***Non-explosion under value conflict:*** *if two actions have incomparable value-vectors (due to paraconsistency), the rule still returns a nonempty admissible set (or distribution) rather than degenerating into arbitrary choice over all actions.*

2. ***Dominance respect:*** *if* a *Pareto-dominates* a' *(under* $\preceq_{\text{pref}}$ *across all* d *and has no greater resistance), then* π_O *does not prefer* a' *over* a*.*

3. ***Resource sensitivity:*** *ceteris paribus, if two actions have equal value evidence, the rule prefers smaller resistance $R_O(s,\cdot)$.*

4. ***Belief/value/action coherence:*** *if the observer's belief model assigns negligible plausibility to an outcome, the value attached to that outcome does not dominate action selection unless explicitly designated as a faith-commitment (Section 20).*

The four clauses are intentionally framed as *failure-avoidance* constraints: they do not dictate a unique optimizer, but they rule out characteristic ways a decision procedure can become pathological when paraconsistency and bounded resources are treated as normal. They can also be read as "guardrails" for many different concrete implementations (e.g., rule-based filters, scalarizations, satisficing schemes, or distributions over an admissible set), provided those implementations preserve the stated invariants.

Remark 1030. *These criteria separate three kinds of failure. The first is* explosion *(everything becomes permissible) under conflict—the evaluative analogue of logical explosion. The second is* dominated choice *(choosing something plainly worse). The third is* resource blindness *(preferring an equally good but harder path). The fourth is* wishful incoherence *(acting as if an implausible outcome were assured). Together, they provide a minimal floor, not a full theory of rationality. It is worth emphasizing that "plainly worse" in the second item is defined relative to the agent's own current value-ordering and resistance structure, not relative to any external metric. In this way, the minimal criterion stays neutral about substantive ethics while still forbidding internal self-undermining moves such as paying strictly more resistance for strictly less-supported value outcomes.*

Remark 1031. *A simple example of criterion (4): an agent may value "winning the lottery," but if its belief model makes the probability effectively negligible, then "buy lottery tickets and quit my job" should not be selected unless the agent elevates that goal to a special status (faith-commitment). This links the present section to the epistemic layer (Section 20), where belief systems and commitments are treated as structured and resource-sensitive. More generally, this criterion blocks a common failure mode in multi-criteria settings: a high value attached to a near-impossible outcome can otherwise swamp moderate values attached to plausible outcomes, producing policies that are "optimistic" in a purely formal sense but behaviorally brittle. The "faith-commitment" escape hatch is included to represent deliberate, explicitly endorsed departures from evidential proportionality (e.g., vows, sacred values, or identity-defining goals), rather than allowing such departures to enter implicitly through numerical instabilities or unchecked aggregation.*

Remark 1032. *A simple example of criterion (2): suppose a and a' are two actions and, for every value-dimension d, the evidence-weighted preference ranks a at least as good as a' (and strictly better for at least one d), while also satisfying $R_O(s,a) \leq R_O(s,a')$. Then preferring a' amounts to accepting strictly less of what one already endorses while paying no less resistance. The minimal criterion does not require choosing a uniquely (there may be ties or incomparable alternatives), but it disallows elevating a' above a when a' is dominated in this strong sense.*

Remark 1033. *Criterion (1) can be viewed as the decision-theoretic analogue of paraconsistent non-triviality: a system that tolerates conflicting value evidence must still avoid the collapse in which* every *action becomes equally admissible simply because some values disagree. Practically, this often amounts to requiring that incomparability yields a* frontier *(a set of candidates) rather than a uniform indifference over all actions. In implementations that return a distribution, "non-explosion" can be operationalized by ensuring the support of the distribution remains within an admissible subset rather than spreading mass arbitrarily across clearly inferior or irrelevant options.*

Remark 1034. *Criterion (3) is deliberately stated as ceteris paribus because resistance is not an overriding value in the framework; it is a constraint-sensitive bias. When value evidence does not distinguish two actions (or distinguishes them only within a tolerance or uncertainty band), lower resistance functions as a tie-breaker that preserves operability and avoids gratuitous expenditure of time, energy, risk, or attention. This also connects to bounded rationality: limited resources are not merely external constraints but part of the internal structure that makes planning stable over time.*

Theorem 17 (Nonemptiness of resonant feasible choice in the finite case)**.** *Assume A is finite. Then the Pareto-undominated set $A_{\text{Pareto}}(s)$ is nonempty, and the resonant feasible choice rule admits at least one maximizer.*

Remark 1035 (Intuition and connection)**.** *The theorem says that once we refuse to collapse everything into a single scalar, we do not thereby doom ourselves to indecision—at least not in the finite-action case. There is always at least one undominated option, and once we restrict to the undominated set, any real-valued selection criterion (here $\kappa_O - \beta R_O$) attains a maximum. This provides a minimal existence guarantee for the decision scaffolds introduced above. One can also read this as a structural robustness statement: pluralism (many dimensions) and paraconsistency (incomplete or conflicting comparisons) can enlarge the set of admissible actions, but they do not force the admissible set to be empty. The nonemptiness of $A_{\text{Pareto}}(s)$ ensures that "refusing to scalarize" is not the same thing as "refusing to decide."*

Remark 1036. *This result is not surprising to order theorists, but it is conceptually important in the present document because it reassures us that paraconsistency and pluralism do not automatically destroy operability. It complements Proposition 27: scalarization respects Pareto boundaries, and Pareto boundaries exist (in the finite case), so one can move between scalar and vector viewpoints without falling into emptiness. In particular, the pairing of (i) an undominated-set filter and (ii) a subsequent scalar tie-breaker is a common pattern in practice: the first stage protects against dominated choices and some forms of value-conflict explosion, while the second stage yields a concrete selection when the frontier contains multiple elements.*

Proof. In any finite partially ordered set, there exists at least one minimal element. Apply this to the strict dominance relation (or equivalently to the complement of being dominated) to obtain a Pareto-undominated action. Thus $A_{\text{Pareto}}(s) \neq \emptyset$. Because $A_{\text{Pareto}}(s)$ is finite, the function $a \mapsto \kappa_O(s,a) - \beta R_O(s,a)$ achieves a maximum on it, so a maximizer exists. A slightly more explicit phrasing of the first step is: if every action were dominated by some other action, one could iterate the dominance relation indefinitely, constructing an infinite chain in a finite set, which is impossible; hence at least one action is not dominated. □

Proof sketch. The strategy is purely finitary. First, in a finite set you cannot have an infinite descending chain of "strictly dominated by" relations, so at least one action is not strictly dominated by any other (Pareto-undominated). Second, any real-valued function on a finite set attains a maximum, so the resonance-minus-resistance criterion yields at least one maximizer once restricted to the undominated set. The same schematic argument recurs throughout bounded frameworks: finiteness (or compactness, in continuous variants) is what turns "there are candidates" into "there is a best candidate" for a chosen scoring functional. □

Remark 1037. *The key step is recognizing Pareto-undominated actions as minimal elements in an appropriate order (or, dually, as maximal elements under the reverse order). Geometrically, if you plot all actions as points in a multi-criteria space, the Pareto set is the "frontier" that is not strictly* *overshadowed by any other point. Finite sets always have such a frontier. In paraconsistent settings, the geometry should be interpreted with care: incomparability can arise not only from tradeoffs between dimensions but also from genuinely unresolved or conflicting evidence about ordering within a dimension. Nevertheless, the "frontier" metaphor continues to serve as an intuition pump: the decision procedure first avoids points that are clearly worse by the agent's own lights, and then expends additional structure (e.g.,* $\kappa_O - \beta R_O$*) only within the remaining boundary.*

Remark 1038. *This theorem is deliberately simple, but it encodes a practical sanity check: once we refuse to collapse values into a single scalar, we must still guarantee that action selection does not become undefined. In later sections where action spaces may be large, continuous, or generated on the fly, this finitary sanity check can be treated as a baseline to be approximated: one seeks either finite candidate sets (via search and pruning) or conditions such as compactness and upper-semicontinuity that play the same role as finiteness in ensuring the existence of undominated elements and maximizers.*

18.8 Ethics: categorical imperative, cultural morality, and evil

Hyperseed treats ethics as a real phenomenon (not a mere epiphenomenal story) but also as observer-relative and socially embedded. We therefore model ethics as constraints and evaluation dimensions that are (i) representable, (ii) revisable, and (iii) partly shared across agents.

Remark 1039. *This subsection treats ethical notions as part of the same formal fabric as prediction, value, and action. The intent is not to settle meta-ethics, but to show that ethical constraints can be made computationally explicit without pretending that they are infallible. In Hyperseed terms, ethics is a species of socially stabilized evaluative pattern (Hyperseed-Concepts 197, 170, 91).*

18.8.1 Maxims and universalization

Definition 265 (Maxim)**.** *A* maxim *is a rule* m *mapping a representable context description to an action choice:*

$$m : \mathcal{C}_O \to A,$$

where $\mathcal{C}_O$ *is the set of contexts that* O *can represent (Section 13).*

Remark 1040. *A maxim is a compact representation of policy, but at the level of reasons rather than reflex. It says: "in contexts of type* c*, do action* a*." This corresponds to the Hyperseed emphasis that ethics must be expressible in the same representational medium as other knowledge, if it is to be debated, taught, revised, or enforced (Hyperseed-Concept 71).*

Remark 1041. *A simple example is: "When I borrow something, I return it promptly." Here* $\mathcal{C}_O$ *includes the representable condition "I borrowed item* x *from person* y*," and* m *selects the corresponding return action. The usefulness is that maxims can be fed into universalization tests and compared across agents as explicit social artifacts.*

Definition 266 (Universalization operator)**.** *Let* $\mathcal{W}$ *denote the modeled world dynamics (as in Section 14), including other agents. Define* $\mathrm{Univ}(m)$ *as the counterfactual world in which* all relevant agents *adopt maxim* m *whenever* m *applies. This induces a predicted outcome distribution over histories, denoted* $\mathbb{P}_O(\cdot \mid \mathrm{Univ}(m))$.

Remark 1042. *Universalization turns a first-person rule into a third-person dynamical hypothesis: what happens if everyone follows it? Technically,* $\mathrm{Univ}(m)$ *is not a single history but a modified policy regime inside the world model* $\mathcal{W}$*, which then induces a distribution* $\mathbb{P}_O(\cdot \mid \mathrm{Univ}(m))$ *over possible histories (because the world may be stochastic or partially known).*

Remark 1043. *This is useful because it connects ethical reasoning to the prediction machinery of the earlier sections. Ethics becomes a kind of counterfactual forecasting problem: evaluate the world you would get if the maxim were generalized. The paraconsistent machinery allows that the forecast can contain mixed evidence (e.g. universal lying might increase short-term safety but destroy long-term trust), rather than forcing a single verdict.*

Definition 267 (Categorical imperative test (paraconsistent))**.** *Fix a set $\mathcal{D}_{\mathrm{eth}} \subseteq \mathcal{D}$ of ethical dimensions (typically including* compassion *and possibly* truth*,* nonviolence*, etc.). Define the universalized ethical evaluation*

$$E_O^{\mathrm{univ}}(m; d) := \mathbb{E}_{\omega \sim \mathbb{P}_O(\cdot|\mathrm{Univ}(m))}\big[\, E_O(\omega; d)\,\big],$$

where $E_O(\omega; d)$ is the p-bit evidence that the realized history ω promotes/opposes d (obtained by aggregating stepwise evidence across the history using $\oplus, \otimes$). We say m passes a threshold categorical imperative *if*

$$J\big(E_O^{\mathrm{univ}}(m; d)\big) \geq \tau_d \quad \textit{and} \quad W\big(E_O^{\mathrm{univ}}(m; d)\big) \leq \eta_d \qquad \forall d \in \mathcal{D}_{\mathrm{eth}},$$

for fixed thresholds $\tau_d, \eta_d \in [0, 1]$.

Remark 1044. *This test is a direct formalization of the Kantian structure: evaluate a maxim by universalizing it. The novelty here is not the moral slogan but the computational interface: the output is a paraconsistent p-bit expectation for each ethical dimension, and the pass/fail decision is thresholded. This aligns with Categorical Imperative (Hyperseed-Concept 71) and also with the Hyperseed idea that ethical rules are operational constraints in a world-model, not metaphysical axioms.*

Remark 1045. *A simple example: suppose m is "lie when it is convenient." In a sufficiently realistic model $\mathcal{W}$,* Univ(m) *may predict that communication reliability collapses, undermining coordination and producing large expected woe on* belonging *and* truth*. The usefulness of the formalism is that such effects can be represented as predicted histories and aggregated evidence, rather than asserted by fiat.*

Remark 1046. *This is a formal analogue of "act only on maxims you can will as universal law": universalization is explicit, and paraconsistency is allowed (there can be both-for and both-against evidence). The test does not claim metaphysical finality; it is a computationally meaningful constraint an agent can actually evaluate approximately.*

18.8.2 Cultural morality

Definition 268 (Cultural morality as shared value field)**.** *Let G be a group/community. A* cultural morality *for G is a family of value dimension weights and thresholds*

$$\big(\lambda_d^{(G)}, \tau_d^{(G)}, \eta_d^{(G)}\big)_{d \in \mathcal{D}}$$

together with shared maxims $\mathcal{M}^{(G)}$ such that:

- *members of G tend to evaluate actions using these weights/thresholds (perhaps implicitly);*
- *maxims in $\mathcal{M}^{(G)}$ have high social resonance (Section 12 and Section 18.5), meaning they are reinforced by communication and imitation.*

Remark 1047. *This definition makes cultural morality (Hyperseed-Concept 92) into a partially shared* parameterization *of evaluation: the community supplies default weights and thresholds, plus a repertoire of endorsed maxims. The emphasis on resonance connects morality to habit formation and social reinforcement: moral rules are stabilized because they are repeated, communicated, and emotionally rewarded, not merely because they are "known." In particular, the $\lambda_d^{(G)}$ encode which dimensions are treated as salient or overriding ("what matters most"), while thresholds such as $\tau_d^{(G)}$ and $\eta_d^{(G)}$ encode where the community draws lines between acceptable and unacceptable (or between normal and exceptional) trade-offs within those dimensions.*

Remark 1048. *One should not read "shared" as "perfectly uniform." A cultural morality can be realized as a distribution over individuals clustered around community defaults, with disagreement arising from variance around (λ, τ, η) as well as from different interpretations of which real-world situations instantiate a maxim. Formally, this means that even if $(\lambda_d^{(G)}, \tau_d^{(G)}, \eta_d^{(G)})$ are treated as community-level parameters, the modeling stance can still accommodate subgroups, role-based moral specializations (e.g. professional roles), and context-conditional parameter shifts without abandoning the notion of a cultural field.*

Remark 1049. *A simple example is a professional culture (e.g. scientific practice) where* truth *has high weight and where maxims like "report negative results" have high resonance. The usefulness is that one can model moral disagreement as parameter mismatch (different λ, τ, η or different $\mathcal{M}^{(G)}$), and one can study cultural evolution as dynamics on these parameters. In such a setting, changes in institutional incentives or communication channels can be represented as interventions that alter resonance (which maxims are repeated and rewarded) even when explicit endorsement remains stable, thereby yielding divergence between "stated morality" and the effective evaluative field.*

Remark 1050. *Cultural morality is not assumed to be correct; it is modeled as a socially stabilized pattern in evaluative fields. This matches Hyperseed's stance that morality is both real (behaviorally and experientially) and historically contingent. The point of the definition is therefore descriptive and explanatory: it provides state variables with which to represent moral learning, acculturation, conformity pressures, and conflict between communities, rather than presupposing a final criterion of ethical truth.*

18.8.3 Evil

Definition 269 (Evil as a persistent anti-compassion pattern)**.** *Fix an observer O modeling a multi-agent setting. An action policy π is* evil relative to $(O, \mathcal{D}_{\mathrm{eth}})$ *if, over a substantial interval, it exhibits:*

1. *persistent high expected woe imposed on others, i.e. large W in the* compassion *and related ethical dimensions under universalized or repeated application; and*
2. *a tendency to maintain or increase the resistance/constraint load of others (increasing their R in typical contexts), thereby reducing their feasible action sets.*

Remark 1051. *The definition treats evil (Hyperseed-Concept* **??***) as neither a single act nor a metaphysical substance, but as a* stable pattern *across time: repeated production of others' woe, coupled with constriction of others' agency. The second clause is important because it captures a structural feature of domination: not only harm, but the engineering of a world in which alternatives are hard or impossible for the victims. The qualifier "over a substantial interval" is meant to exclude* *transient accidents and short-lived coordination failures, and to emphasize that the pattern is robust under feedback (e.g. the policy persists even after the harms are visible to the agent implementing π or to the institutions sustaining it).*

Remark 1052. *The reference to "universalized or repeated application" connects the criterion to a categorical-imperative-style stress test: if the policy were adopted as a general practice (or is enacted repeatedly as a standing policy), does it systematically generate woe in compassion-relevant dimensions? This does not require that O endorse Kantian ethics; rather, it uses universalization as an operational probe for whether the harm is structurally tied to the policy rather than to an unusual one-off context. In this sense, the formalism isolates a pattern-level property that is detectable even when agents rationalize or localize responsibility.*

Remark 1053. *A simple example is a policy that repeatedly extracts resources from others while imposing surveillance and dependency so that resistance becomes costly. In the formalism, this shows up as persistently high woe-evidence in compassion-like dimensions and rising R for others' typical actions. The usefulness is that it separates "accidental harm" from a self-maintaining harm pattern, which better matches the ordinary (and political) notion of evil as systemically reproducing injury. A further diagnostic implication is that even if the immediate W is episodically lowered (e.g. through superficial welfare concessions), a policy can remain evil if it continues to ratchet constraints and lock victims into narrowing feasible sets, thereby preserving the long-run capacity to impose woe.*

Remark 1054. *This definition intentionally mixes affect (woe), ethics (compassion), and agency (feasible sets / resistance). This mirrors the Hyperseed intuition: "evil" is not merely violating a rule; it is creating a stable pattern of harm and constriction. It also clarifies why purely outcome-based snapshots can be misleading: a momentary reduction in visible harm does not negate the presence of an agency-constricting mechanism that predictably regenerates harm under ordinary dynamics of compliance, enforcement, and retaliation.*

18.9 Humility, grandiosity, ambition, and "humbition"

Hyperseed highlights a family of self-attitudes that matter for rational agency: humility, grandiosity, ambition, and the proposed synthesis "humbition" (ambition with humility).

Remark 1055. *These notions are included because self-calibration is a hidden hinge between evaluation and action. If an agent systematically overestimates its capability, it will select actions with catastrophic resistance; if it underestimates itself, it will forgo feasible value. Hyperseed treats these not merely as personality adjectives, but as parameters shaping rational decision under uncertainty (Hyperseed-Concepts 148, 195).*

Remark 1056. *In particular, "self-attitude" is meant in a decision-theoretic sense: it determines how internal self-model evidence is translated into action selection, planning depth, risk appetite, and information-seeking. Two agents with identical outward goals and identical external observations can nevertheless diverge sharply in behavior if they map capability evidence to commitment differently.*

18.9.1 Capability p-bits and self-calibration

Definition 270 (Capability evidence)**.** *Let* cap *be a proposition meaning "I (or we) can successfully execute plan* Π *in context* c*." Represent evidence for/against this proposition by a p-bit*

$$C_O(\Pi, c) = \big(C_O^+(\Pi, c),\, C_O^-(\Pi, c)\big) \in [0,1]^2.$$

Remark 1057. *This is the capability analogue of the belief evidence states from Section 17.2: it stores both pro- and anti-evidence for success. A simple example is* Π *being "lift a heavy object" and* c *encoding fatigue state; one may have positive evidence from past successes and negative evidence from current weakness signals. The usefulness is that humility/grandiosity can be expressed as how one aggregates these two channels, rather than as a vague disposition.*

Remark 1058. *Nothing here requires* C_O^+ *and* C_O^- *to be complementary or to sum to* 1*; the representation is explicitly compatible with mixed, conflicting, or incomplete evidence. This matters because many real capability judgments are paraconsistent in practice: e.g. an agent may have strong reasons to think a plan is doable (training, past performance) while simultaneously having strong reasons to think it will fail (new constraints, time pressure, degraded resources). The role of calibration is then not to pretend the conflict is absent, but to decide how conflict is action-relevant.*

Remark 1059. *The context variable* c *is intended to include both external situation features (environmental difficulty, adversarial load) and internal state features (attention, fatigue, coordination quality). This makes capability evidence a function of state rather than a static trait, which is important for distinguishing "I can do this in general" from "I can do this now, under these constraints."*

Definition 271 (Humility and grandiosity parameters)**.** *Fix a calibration parameter* $h \in [0, 1]$ *(humility). Define a calibrated confidence scalar*

$$\mathrm{Conf}_h(\Pi, c) := C_O^+(\Pi, c) \cdot (1 - h) \ + \ C_O^+(\Pi, c) \cdot h \cdot \big(1 - C_O^-(\Pi, c)\big).$$

Interpretation:

- *Large* h *means negative evidence meaningfully suppresses confidence (humility).*
- *Small* h *means confidence tracks positive evidence while ignoring negative evidence (grandiosity tendency).*

Remark 1060. *The formula can be read as a convex interpolation between two attitudes toward negative evidence. When* $h = 0$*, confidence is simply* C_O^+*: "if I see reasons to think I can, I proceed." When* $h = 1$*, confidence becomes* $C_O^+(1 - C_O^-)$*: positive evidence is discounted when substantial negative evidence is present. The usefulness is that the moral-psychological idea of humility becomes a tunable parameter in a decision pipeline.*

Remark 1061. *Algebraically, one may note (without changing the intended interpretation) that*

$$\mathrm{Conf}_h(\Pi, c) = C_O^+(\Pi, c)\big(1 - h \cdot C_O^-(\Pi, c)\big),$$

so humility h *functions as a multiplicative "penalty sensitivity" to counterevidence. This makes several monotonicity properties transparent: for fixed* h*,* Conf_h *increases with* C_O^+*; for fixed* C_O^+*, it decreases with* C_O^- *at a rate proportional to* h*; and for fixed* (C_O^+, C_O^-) *with* $C_O^- > 0$*, it decreases as* h *increases. In other words,* h *is precisely the parameter controlling how costly warning signs are to one's felt feasibility.*

Remark 1062. *One motivation for keeping the two-channel form (rather than collapsing to a single probability) is that it allows "high confidence under dispute":* C_O^+ *can remain high even when* C_O^- *is also high, and humility then determines how the dispute is resolved for purposes of commitment. This provides a simple handle on phenomena like brittle overcommitment (low* h *despite substantial* C_O^-*) versus cautious non-commitment (high* h *even when* C_O^+ *is strong).*

Remark 1063. *A simple numerical example: if* $C_O^+ = 0.9$ *and* $C_O^- = 0.6$*, then* $\mathrm{Conf}_0 = 0.9$ *(grandiose) while* $\mathrm{Conf}_1 = 0.9 \cdot 0.4 = 0.36$ *(humble). This is an intentionally blunt model, but it captures the qualitative effect: humility treats warning signs as real constraints.*

Remark 1064. *The same computation also highlights a behavioral distinction: a low-*h *agent will tend to treat negative evidence as "noise to be overcome" and proceed as if near-certain, while a high-*h *agent is pushed toward either (i) selecting a modified plan* Π' *with lower* $C_O^-(\Pi', c)$*, (ii) performing information-gathering actions that can reduce* $C_O^-(\Pi, c)$ *(diagnostics, rehearsal, obtaining resources), or (iii) deferring the attempt until the context* c *changes. In this sense humility is operationally linked to strategy selection, not merely to a private attitude.*

Definition 272 (Ambition and humbition)**.** *Let* $a \in [0, 1]$ *be an ambition parameter controlling how high a target threshold is set for a goal dimension (e.g. competence, impact, or open-ended benefit). We call the pair* (a, h) *a* humbition profile*.*

- Ambition *corresponds to high* a*.*
- Humility *corresponds to high* h*.*
- Humbition *corresponds to high* a *and high* h *simultaneously: aiming high while not denying uncertainty, resistance, or conflict evidence.*

Remark 1065. *Ambition here is modeled not as sheer intensity but as* threshold-setting*: how demanding are the agent's constraints on success. Humbition (Hyperseed-Concept* **??***) is then the conjunction of demanding thresholds with careful calibration of capability evidence. The usefulness is that it predicts a specific behavioral profile: pursue high-value trajectories, but choose them in a way that is sensitive to resistance and negative evidence rather than propelled by denial.*

Remark 1066. *A useful way to read the interaction is:* a *determines what counts as "good enough to bother" among available objectives, while* h *determines what counts as "feasible enough to commit" among available plans. High ambition without humility tends toward selecting goals whose achievement requires fragile assumptions; high humility without ambition tends toward selecting only low-stakes, easily-verified goals; humbition aims to select high-value targets while actively managing feasibility constraints (e.g. by staging, redundancy, and incremental validation).*

Remark 1067. *Although* a *is introduced as a target threshold on a goal dimension, it can be coupled to capability confidence in concrete decision rules. For example, an agent might require that a plan meets both (i) a value/impact threshold (increasing in* a*) and (ii) a feasibility threshold expressed in terms of* $\mathrm{Conf}_h(\Pi, c)$*. This makes explicit how "aiming high" and "being careful" can jointly shape the frontier of actions the agent considers admissible.*

Remark 1068 (Why these are value-relevant)**.** *Humility and grandiosity change how evidence shapes action; ambition changes which goals are pursued. Thus these are not merely "personality traits"; they are parameters of rational decision making under uncertainty and paraconsistency.*

Remark 1069. *The value-relevance also appears in multi-agent and institutional settings: a "we can do it" judgment often aggregates across people, tools, and coordination protocols. In such cases, grandiosity can manifest as ignoring organizational failure modes (high* C_O^- *from known bottlenecks), while humility can manifest as appropriately discounting rosy projections in the presence of correlated risks. The humbition profile* (a, h) *then becomes a compact way to discuss when an agent (or team) should pursue ambitious projects while remaining sensitive to real constraints and failure evidence.*

18.10 Closing synthesis for this section

We can now summarize Hyperseed's motivation stack in a single sentence: *values are stable patterns in paraconsistent evaluative fields; emotions are time-local readouts of value and resistance structure; goals are commitments that constrain action selection; and rationality is coherent action selection under value conflict and resource limits, optionally constrained by ethical universalization tests.*

To make the sentence operationally interpretable: calling values "stable patterns" emphasizes that they are not momentary appetites but recurrent, learned regularities in evaluation—regularities that can be detected, reinforced, revised, and composed. Calling the field *paraconsistent* is not rhetorical flourish: it signals that the system can simultaneously carry partially incompatible evaluations (e.g., valuing honesty and valuing loyalty in a way that yields genuine tension), without collapsing into triviality or requiring immediate resolution by enforcing global consistency. In this view, value conflict is not an anomaly to be eliminated at all costs; it is a structural feature of realistic minds embedded in complex social and ecological environments.

Likewise, describing emotions as "time-local readouts" is meant to place them between raw sensation and deliberative judgment: an emotion summarizes, at a moment, what the value-system is registering about the current trajectory of events and the current *resistance structure* (friction, cost, threat, uncertainty, or constraint) encountered in pursuing or maintaining valued patterns. This makes emotions informative even when they are not themselves endorsable as final verdicts: they are fast, lossy projections of a richer evaluative landscape, and their intensity can be interpreted as a proxy for urgency or constraint rather than as a proof of correctness.

Goals, in this stack, are not merely desires but *commitments* that change the decision landscape: once adopted, they restrict and shape action selection by introducing persistence, coordination across time, and explicit or implicit policies for allocating attention and resources. This provides a natural bridge between implicit goals (which may be embedded in default policies, habits, or learned heuristics) and explicit goals (which are linguistically or symbolically represented and can be negotiated, justified, or revised). It also clarifies why "goal" is not synonymous with "reward": reward is an episodic signal that can train or reinforce policies, while a goal is a longer-horizon constraint that can remain in force even when local reward is absent or noisy.

Finally, the definition of rationality given here is intentionally bounded: it is *coherent* action selection under value conflict and resource limits, not omniscient optimization. Coherence is the thread that connects the parts: it asks that chosen actions make sense relative to the system's currently operative values and commitments, while acknowledging that the system may be navigating incompatible pulls, incomplete information, and finite compute, time, and energy. Ethical universalization tests (e.g., categorical-imperative-style constraints) then appear as optional but principled filters on the space of coherent plans: they do not remove the need for bounded rational tradeoffs, but they can rule out classes of strategies whose general adoption would undermine the very social-pattern substrate that sustains trust, coordination, and long-run benefit.

Remark 1070. *A final connective thread: the entire construction is meant to compose with the earlier pattern-web view of mind [5] and with the broader Hyperseed ontology program [1]. Values are patterns; emotions are local projections; ethics is socially stabilized pattern constraint; and rationality is the art of steering through this web without demanding an impossible purity of consistency.*

One way to read this connective thread is as a compatibility claim across levels of description: the "pattern" language is meant to let us speak about neurocomputational regularities, cognitive-symbolic commitments, and socio-cultural norms using a common conceptual currency, without reducing one level into another. On this reading, ethical norms are neither arbitrary overlays nor purely private preferences; they are constraints that become stable insofar as they are reinforced by collective learning, institutional scaffolding, and repeated interaction, and they remain revisable insofar as the underlying pattern ecology changes. The emphasis on "impossible purity" is also methodological: it licenses approximate, context-sensitive resolutions of conflict (including compromise, prioritization, deferral, and exploratory action) while warning against the pathological demand that every tension must be globally resolved before any action is permissible.

This section has covered the Hyperseed concept bundle: *joy/woe, emotion, compassion; values/feelings/goals/reward; implicit/explicit goals; value paraconsistency; individuation/self-transcendence/open-ended intelligence/open-ended benefit; rationality criteria; categorical imperative and ethical decision; humility/grandiosity/ambition/humbition; evil and cultural morality.*

Taken together, this bundle is intended to be used as a vocabulary for diagnosing motivational dynamics in agents that learn and self-modify: it distinguishes what is being valued (stable evaluative patterns) from how valuation is being expressed moment-to-moment (emotional readouts), and from what is being actively maintained across time (goal commitments). It also highlights where failures can occur: paraconsistent value structure can be navigated skillfully (via rational coherence under constraints) or unskillfully (via denial, compulsive simplification, or culturally amplified rationalization). In that sense, terms like humility, grandiosity, ambition, and "humbition" are not decorative; they name characteristic stances toward uncertainty and self-assessment that can either support or sabotage coherent action under conflict, especially when universalization-style ethical constraints are invoked to arbitrate between short-term advantage and long-run pattern sustainability.

19 Life, energy, and genenergy

19.1 Why "life" belongs in a Hyperseed ontology

Hyperseed treats *mind* as a richly structured special case of a broader phenomenon: persistent, self-maintaining pattern-web dynamics occurring in a context/environment. This section isolates the "life layer" in between the generic pattern-process layer (Sections 9–12) and the mind/agency layers (Sections 13–21). Concretely, the "life layer" is the point at which pattern persistence becomes *normatively loaded*: the persistence is no longer merely a fact about which patterns tend to recur, but is constrained by a viability notion that distinguishes "acceptable" from "unacceptable" trajectories for the system. This is also where the ontology begins to require explicit bookkeeping of limited resources, because without such bookkeeping the difference between "maintaining itself" and "just happening to persist" is hard to formulate in a way that remains robust across contexts.

The key thesis is that "life" can be usefully reconstructed as a conjunction of:

- a *viability* requirement (homeostasis: staying within a viable range of states);
- a *resource* requirement (having and replenishing capacity-for-action);
- a *self-weaving* requirement (patterns that help sustain the web tend to be reinforced);
- and, in many cases, a *trans-generational persistence* requirement (reproduction within a species).

Each clause here functions as a "handle" that lets us talk about life without assuming any particular substrate (biological, artificial, social, or hybrid). The viability clause isolates a set of constraints that are *stateful* (they refer to where the system is), the resource clause isolates constraints that are *capacitive* (they refer to what the system can do next), the self-weaving clause isolates constraints that are *structural* (they refer to which patterns tend to amplify), and the trans-generational clause isolates constraints that are *lineage-based* (they refer to persistence across copies). Hyperseed does not treat these as all-or-nothing; rather, they provide dimensions along which a system can more or less instantiate what observers ordinarily call "life."

Hyperseed's term *genenergy* is used here for a context-indexed, observer-relative generalization of "energy" as capacity-for-action. The intent is not to replace physical energy, but to provide a single variable that can stand in for whichever "currency" of action is salient in the modeled context: metabolic free energy, battery charge, attention, time, money, permissions, social credit, error budget, spare parts, or other bottleneck resources. In particular, genenergy is meant to track (i) the *availability* of action, (ii) the *cost* of actions and repairs, and (iii) the *coupling* between system behavior and the replenishment/depletion of that availability. This makes it possible to state viability claims in a way that remains meaningful even when the system boundaries, units of measure, or the observer's coarse-graining change.

19.2 Complex interactive systems and viability

We use a deliberately minimal dynamical scaffold. For concreteness (and to keep the formalism aligned with the finite toy-model ethos of Section 5), we present definitions in discrete time with finite or countable spaces. Nothing essential depends on this choice. The discrete-time presentation should be read as a modeling convenience: it can represent sampled continuous-time dynamics, event-driven updates, or coarse-grained episodes of interaction. Likewise, finiteness/countability here is a stand-in for "bounded representational resolution"—a reminder that the observer/context O ultimately determines what distinctions are made and what variables are tracked.

Definition 273 (System, environment, and interaction history)**.** *A* system-in-environment *is specified by:*

- *a system-state space $\mathcal{S}$ and environment-state space $\mathcal{E}$;*
- *an action space $\mathcal{A}$ (the set of interventions available to the system);*
- *a (possibly stochastic) transition kernel*

$$P\big((s_{t+1}, e_{t+1}) \mid (s_t, e_t), a_t\big);$$

- *an initial distribution $P(s_0, e_0)$.*

A length-T interaction history *is the sequence*

$$h_{0:T} := (s_0, e_0, a_0, s_1, e_1, a_1, \ldots, s_T, e_T).$$

The factorization into (s_t, e_t) is not intended to assert a sharp ontological boundary; it is a modeling move that lets us speak about "internal" versus "external" variables at the granularity relevant to O. In many applications, e_t can be understood to include other agents, spatial context, exogenous shocks, or a reservoir from which resources can be drawn, while s_t includes the degrees of freedom that are treated as belonging to the system. Similarly, the action variable a_t is intentionally broad: it may represent physical acts, control signals, information-gathering, communications, or internal reconfiguration steps, provided these are the kinds of interventions the model attributes to the system.

Hyperseed's notion of a *complex interactive system* is intentionally broad and observer-indexed: it is about the structural richness of the interaction between system and environment. The point is to exclude degenerate cases in which apparent "life-like" persistence is merely an artifact of trivial dynamics (e.g., an absorbing state with no meaningful coupling), while still allowing many kinds of complexity measures depending on the observer's available descriptions and interests.

Definition 274 (Complex interactive system (structural complexity criterion))**.** *Fix an observer/context O and a representational scheme that assigns a description length (or, more generally, a weakness/clarity score) to histories. Let $\mathrm{SC}_O(h_{0:T})$ denote the structural complexity of an interaction history as assessed by O (see Sections 8 and 9 for weakness-based and compression-based complexity measures). A system-in-environment is called a* complex interactive system (for O) *if $\mathbb{E}[\mathrm{SC}_O(h_{0:T})]$ exceeds a context-dependent threshold over relevant time horizons T.*

The expectation $\mathbb{E}[\mathrm{SC}_O(h_{0:T})]$ is taken with respect to the distribution over histories induced by the initial distribution, the kernel, and (when present) the policy governing actions. The "relevant time horizons" clause allows complexity to be scale-sensitive: some systems are simple at short horizons but complex over long horizons (e.g., due to slow variables or rare events), and others are the reverse (locally complicated but globally repetitive). The threshold is context-dependent because complexity here is not posited as an absolute property of the world, but as an observer-relative measure of how much structured interaction must be tracked to predict, compress, or explain the history to an acceptable degree.

The first life-specific ingredient is a viability/homeostasis notion. This is where "mere persistence" becomes "persistence under constraints": a system may continue to exist in a literal sense while no longer counting as viable under the observer's criteria (e.g., a dead organism persists as matter, but not as a viable living process).

Definition 275 (Viability set and homeostatic constraint)**.** *Fix an observer/context O. A* viability set *is a subset*

$$\mathcal{V}_O \subseteq \mathcal{S} \times \mathbb{R}_{\geq 0}$$

whose elements (s, g) are interpreted as "system state s together with resource level g is viable." A system is said to satisfy a homeostatic constraint *on a time interval $\{0, \ldots, T\}$ if*

$$(s_t, g_t) \in \mathcal{V}_O \quad \textit{for most } t \in \{0, \ldots, T\}$$

in a sense made precise by a probability threshold (below) or by a paraconsistent evidence threshold (Section 3.2).

The inclusion of g directly in the viability set makes explicit that viability is not solely about "being in the right configuration," but also about possessing enough capacity to correct deviations, repair damage, and respond to disturbances. This prevents a common failure mode in purely state-based viability criteria: declaring a system viable even when it is in a brittle corner of state space where any small perturbation leads to collapse. In practice, $\mathcal{V}_O$ can encode multiple tolerances at once (e.g., temperature bounds, integrity constraints, and a minimum resource reserve), and can be chosen to reflect the observer's modeling goals: a biologist and an engineer may use different $\mathcal{V}_O$ for the same physical system.

Definition 276 (Homeostatic policy)**.** *Let g_t be a resource variable (to be defined in detail in Section 19.3). A (possibly stochastic)* policy *is a map π that assigns a distribution over actions given the currently available information (at minimum the current state, and possibly also history):*

$$\pi(\cdot \mid s_t, e_t, h_{0:t}) \in \Delta(\mathcal{A}).$$

Fix a horizon T and a target confidence level $1 - \delta$. A policy π is homeostatic for $(\mathcal{V}_O, T, \delta)$ *if*

$$P_\pi\Big((s_t, g_t) \in \mathcal{V}_O \textit{ for all } t = 0, \ldots, T\Big) \geq 1 - \delta.$$

The dependence on $h_{0:t}$ allows policies with memory, learned internal state, or belief-state tracking, without committing to any particular representation of that internal information. The confidence parameter δ expresses a robustness requirement: the smaller δ is, the more stringent the notion of "homeostasis" becomes, ruling out policies that succeed only under lucky environmental sequences. Also note that this definition is deliberately conservative ("for all t"): later sections may consider relaxed variants (e.g., allowing brief excursions, or optimizing expected time-in-viability), but the strict form makes clear what it means for a system to *reliably* stay within bounds.

Remark 1071 (Observer-relativity and fuzzy boundaries)**.** *Hyperseed permits that what counts as "the same system" and what counts as "viable" can be fuzzy. In the present mathematical layer, this is expressed by: (i) allowing $\mathcal{V}_O$ to depend on O (which distinctions and patterns define identity), and (ii) later (Section 19.6) replacing crisp predicates like "alive" with* p-bit*-valued evidence.*

A further consequence of observer-relativity is that even the choice of $\mathcal{S}$ and $\mathcal{E}$ can be contextually renegotiated: an observer may "move the boundary" (e.g., include a tool, a symbiont, or a niche-structure inside the system description) if doing so yields a simpler or more stable account of viability and resource flows. This is one reason Hyperseed emphasizes interaction histories and structural complexity rather than fixed ontological categories: the same physical process can be modeled as a single life-like system, a coalition of sub-systems, or a component of a larger organism-like process, depending on which decomposition best preserves explanatory and predictive power for the questions at hand.

19.3 Energy and genenergy as capacity-for-action

Hyperseed distinguishes (a) the folk notion of energy as "resources you can deploy to get things done," (b) the physics notion of energy as a specific conserved quantity in particular dynamical theories, and (c) an intended generalization, *genenergy*, designed to capture capability-for-action across diverse contexts (physical, cognitive, social, spiritual) in a single formal motif.

We formalize this by combining:

- an *effort cost* functional (Section 8), and
- an *effect/capacity* functional grounded in causal influence (Section 14) and information flow (Section 10).

19.3.1 Effort budgets and energy

Definition 277 (Effort cost and energy budget)**.** *Fix an observer/context O. An* effort cost *is a function $c_O : \mathcal{A} \to \mathbb{R}_{\geq 0}$. An* energy budget process *is a sequence $(g_t)_{t \geq 0}$ with $g_t \in \mathbb{R}_{\geq 0}$ such that actions must satisfy the feasibility constraint*

$$c_O(a_t) \leq g_t$$

(or more generally $\mathbb{E}[c_O(A_t)] \leq g_t$ under a stochastic policy).

Remark 1072 (Energy as the dual of effort)**.** *In this reconstruction, effort is an observer-relative cost of performing a process or representation, whereas energy is the available budget to pay for effort. This makes "energy" inherently context-indexed, which aligns with Hyperseed's insistence that many resource notions are observer-relative rather than absolute.*

19.3.2 Genenergy via control-capacity

To link "capability for action" to something mathematically crisp, we model the system as having a controllable influence channel from actions to future environment observations. The maximal information that can be transmitted through this channel (subject to constraints) is a natural quantitative proxy for causal impact.

Definition 278 (Observer-coarse-grained environment observation)**.** *Fix an observer/context O. Let $\mathcal{Y}_O$ be a finite set of environment* observations *available to O, and let $\phi_O : \mathcal{E} \to \mathcal{Y}_O$ be the coarse-graining map. Define the observed environment variable $Y_t := \phi_O(e_t)$.*

Definition 279 (Action-to-observation channel at time t)**.** *Fix time t and a realized history prefix $h_{0:t}$. The induced channel from action to next observation is*

$$K_{O,t,h}(y \mid a) \;:=\; P\big(Y_{t+1} = y \,\big|\, h_{0:t},\, A_t = a\big), \qquad y \in \mathcal{Y}_O,\; a \in \mathcal{A}.$$

Definition 280 (Single-step genenergy as constrained channel capacity)**.** *Fix $(O, t, h_{0:t})$ and an effort cost c_O. For a budget $g \geq 0$, define the* constrained capacity

$$\mathrm{Cap}_{O,t,h}(g) \;:=\; \max_{p \in \Delta(\mathcal{A})} \Big\{ I(A; Y_{t+1} \mid h_{0:t}) \;:\; \mathbb{E}_{A \sim p}[c_O(A)] \leq g \Big\},$$

where $I(\cdot\,;\cdot)$ is Shannon mutual information (Section 10.4). We call $\mathrm{Cap}_{O,t,h}(g)$ *the* genenergy available at $(t, h_{0:t})$ with budget g.

Remark 1073 (Interpretation)**.** $\mathrm{Cap}_{O,t,h}(g)$ *measures how strongly the system can steer what O will observe next, given a budget g to pay for effort. It is explicitly: (i) observer-relative (depends on ϕ_O), (ii) context-relative (depends on $h_{0:t}$), and (iii) resource-relative (depends on the constraint g).*

We can aggregate single-step genenergy over time/histories.

Definition 281 (Realizable genenergy of a system over a horizon)**.** *Fix a horizon T. Given a policy π that induces a joint distribution over histories and actions, define*

$$G_O(\pi; T) \;:=\; \mathbb{E}_{h_{0:T-1} \sim P_\pi} \left[\frac{1}{T} \sum_{t=0}^{T-1} \mathrm{Cap}_{O,t,h}(g_t) \right].$$

Define the realizable genenergy *of the system over horizon T as*

$$G_O^{\mathrm{real}}(T) \;:=\; \sup_{\pi} G_O(\pi; T),$$

where the supremum ranges over policies satisfying the system's feasibility constraints (including energy-budget dynamics and any structural constraints on what actions are possible).

Remark 1074 (Potential vs realizable genenergy)**.** *Hyperseed distinguishes potential genenergy (what could be achieved by some system built from the entity-set) from realizable genenergy (what is achievable given contextual constraints). Formally, one may treat "potential" as a supremum taken over a larger family of policies and/or contexts, and "realizable" as the supremum constrained by the system's actual embedding.*

19.3.3 Basic properties

Proposition 28 (Zero genenergy means no controllable influence at the given resolution)**.** *Fix* $(O, t, h_{0:t})$*. For any budget* $g \geq 0$*,*

$$\mathrm{Cap}_{O,t,h}(g) = 0 \quad \Longleftrightarrow \quad Y_{t+1} \perp A_t \mid h_{0:t} \text{ for all feasible action-distributions under budget } g.$$

Proof. For any feasible distribution $p(a)$, mutual information satisfies $I(A; Y_{t+1} \mid h_{0:t}) \geq 0$. Moreover, $I(A; Y_{t+1} \mid h_{0:t}) = 0$ if and only if A and Y_{t+1} are conditionally independent given $h_{0:t}$ under the induced joint distribution. Taking the maximum over feasible p yields: the maximum is zero exactly when every feasible p yields zero mutual information, i.e. when the channel output distribution does not depend on the action at the resolution of O within the feasible set. □

Proposition 29 (Monotonicity and concavity in the budget)**.** *For fixed* $(O, t, h_{0:t})$*, the map* $g \mapsto \mathrm{Cap}_{O,t,h}(g)$ *is nondecreasing and concave on* $\mathbb{R}_{\geq 0}$*.*

Proof. Monotonicity is immediate: if $g_2 \geq g_1$, then the feasible set $\{p : \mathbb{E}[c_O(A)] \leq g_1\}$ is contained in $\{p : \mathbb{E}[c_O(A)] \leq g_2\}$, so the maximum cannot decrease. Equivalently, enlarging the budget can only add admissible input distributions and never removes any previously admissible one, so the supremum over mutual informations is taken over a weakly larger domain.

For concavity, use time-sharing. Let $g_1, g_2 \geq 0$ and $\lambda \in [0, 1]$. Choose (approximately) optimal feasible distributions p_1, p_2 for budgets g_1, g_2. Concretely, for any $\varepsilon > 0$ pick p_1, p_2 such that $\mathbb{E}_{p_i}[c_O(A)] \leq g_i$ and $I_{p_i}(A; Y_{t+1} \mid h_{0:t}) \geq \mathrm{Cap}_{O,t,h}(g_i) - \varepsilon$ for $i \in \{1, 2\}$, so that the argument goes through even if the supremum is not attained. Define a mixture policy: first sample $B \sim \mathrm{Bernoulli}(\lambda)$, then sample $A \sim p_1$ if $B = 1$ and $A \sim p_2$ if $B = 0$. This mixture has expected cost $\mathbb{E}[c_O(A)] = \lambda \mathbb{E}_{p_1}[c_O(A)] + (1 - \lambda)\mathbb{E}_{p_2}[c_O(A)] \leq \lambda g_1 + (1 - \lambda) g_2$, so it is feasible for budget $\lambda g_1 + (1 - \lambda) g_2$. (Here we use linearity of expectation in the mixture; this is the operational sense in which the agent can "spend" a fraction λ of the time behaving according to policy p_1 and the remaining fraction according to p_2 while meeting the averaged cost constraint.) Mutual information is concave in the input distribution for a fixed channel, hence

$$\mathrm{Cap}_{O,t,h}(\lambda g_1 + (1 - \lambda) g_2) \;\geq\; I(A; Y_{t+1} \mid h_{0:t}) \;\geq\; \lambda I_{p_1}(A; Y_{t+1} \mid h_{0:t}) + (1 - \lambda) I_{p_2}(A; Y_{t+1} \mid h_{0:t}).$$

The first inequality holds because $\mathrm{Cap}_{O,t,h}(\cdot)$ is defined as a supremum over all feasible input distributions, and the constructed mixture distribution is one such feasible candidate at budget $\lambda g_1 + (1 - \lambda) g_2$. The second inequality is the standard concavity property of $p \mapsto I_p(A; Y_{t+1} \mid h_{0:t})$ when the conditional law of Y_{t+1} given A and $h_{0:t}$ is fixed; intuitively, randomizing between two input strategies cannot yield less information than the corresponding convex combination because mutual information can be written as a KL divergence that is jointly convex in the pair of distributions and becomes concave in the input when the channel is held fixed. Taking p_1, p_2 arbitrarily close to optimal yields concavity, i.e.,

$$\mathrm{Cap}_{O,t,h}(\lambda g_1 + (1 - \lambda) g_2) \;\geq\; \lambda \mathrm{Cap}_{O,t,h}(g_1) + (1 - \lambda) \mathrm{Cap}_{O,t,h}(g_2),$$

after letting $\varepsilon \to 0$. □

19.3.4 Entity-sets and monotone genenergy

Hyperseed speaks about genenergy not only for a particular instantiated system, but also for an *entity-set* E as "what systems buildable from E can do." We formalize this with a family-of-systems construction. The point of this shift is that the same component library E may admit many distinct system designs and couplings, and genenergy at the level of E should reflect the best achievable design consistent with the available entities.

Definition 282 (Systems buildable from an entity-set)**.** *Let* Ent *be a universe of entities (Section 6). For a finite entity-set* $E \subseteq \mathsf{Ent}$*, let* $\mathsf{Sys}(E)$ *denote the class of all systems-in-environment whose internal components are drawn from (a subset of)* E*, together with any admissible couplings to an environment. (Equivalently:* $\mathsf{Sys}(E)$ *is the set of implementable system models whose "parts list" is contained in* E*.) In particular, the definition permits using only a strict subset of* E *(unused components may simply be absent), and it separates the intrinsic availability of components from the extrinsic choice of environment/interface, which can matter for realized genenergy.*

Definition 283 (Potential genenergy of an entity-set)**.** *Fix an observer/context* O *and horizon* T*. Define the* potential genenergy *of* E *by*

$$g_O(E;T) \;:=\; \sup_{S \in \mathsf{Sys}(E)} G^{\mathrm{real}}_{O,S}(T),$$

where $G^{\mathrm{real}}_{O,S}(T)$ *denotes the realizable genenergy of system* S *over horizon* T *according to the definitions above. This supremum formalizes the idea of "capacity of the library": even if most systems assembled from* E *are ineffective,* $g_O(E;T)$ *records the best attainable realizable genenergy among all admissible assemblies and couplings. The explicit dependence on* O *emphasizes that what counts as realizable genenergy is observer-relative (via* ϕ_O*, costs, and the chosen functional), and the dependence on* T *encodes that some entity-sets may only exhibit their advantages over longer horizons (e.g., via learning or accumulation of state).*

Proposition 30 (Monotone genenergy)**.** *If* $E \subseteq E'$ *then* $g_O(E;T) \leq g_O(E';T)$*.*

Proof. If $E \subseteq E'$, then any system buildable from E is also buildable from E', so $\mathsf{Sys}(E) \subseteq \mathsf{Sys}(E')$. Taking suprema over nested sets yields $g_O(E;T) \leq g_O(E';T)$. Intuitively, enlarging the parts list cannot make previously feasible designs unavailable; at worst, the additional entities are ignored, and at best they enable new constructions whose realizable genenergy can exceed what was possible before. $\square$

Remark 1075 (Conservation and omniscient observers)**.** *At a fine-grained physical level, many dynamical theories admit a conserved energy. Hyperseed's "conserved genenergy" intuition can be rendered as a conditional statement: for a closed set of interacting systems, an observer whose coarse-graining does not lose information (so that entropy does not increase under observation) may see a conserved total genenergy-like quantity, while coarse-graining observers may see apparent creation/destruction of genenergy due to information loss. One way to read this is that, when the observer tracks a sufficiently rich microstate, changes in "capacity-for-action" may be explainable purely as transfers between subsystems (or between degrees of freedom) rather than as net creation; whereas under a many-to-one observation map, hidden correlations and inaccessible microstate detail can make the observer's effective description non-conservative. We treat this as a modeling principle rather than a theorem, because it depends on the chosen dynamics, the chosen observation map* ϕ_O*, and the chosen genenergy functional (capacity, divergence, etc.). In particular, even in deterministic Hamiltonian dynamics, conservation at the micro-level does not automatically imply conservation of any observer-defined genenergy unless that functional is invariant under the dynamics and the observation retains the relevant invariants; conversely, in open systems with external driving or dissipation, one should expect genuine non-conservation relative to the subsystem boundary, regardless of observer acuity.*

19.4 Metabolism as genenergy conversion

Genenergy as "capacity for action" becomes life-relevant only when it can be replenished. Metabolism is the archetypal replenishment mechanism: it converts environmental resources into usable genenergy, typically while exporting waste. In this sense, metabolism is not merely a source term in a budget equation; it is a structured interface between a system and its environment that (i) identifies what counts as "resource" at a chosen level of description, (ii) specifies the internal transformations that turn that resource into actionable capacity, and (iii) manages byproducts so that the same interface remains usable over time.

Definition 284 (Metabolic accounting model). *Fix an observer/context O. Let $r_t \in \mathbb{R}_{\geq 0}$ denote the amount of usable resource ingested from the environment at time t (at O's level of description), and let $w_t \in \mathbb{R}_{\geq 0}$ denote waste expelled. Let $\Gamma_O : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be a conversion function mapping ingested resource to genenergy gain. Let $\lambda \geq 0$ be a leakage/decay rate. A simple budget dynamics is:*

$$g_{t+1} \;=\; (1-\lambda)g_t \;+\; \Gamma_O(r_t) \;-\; c_O(a_t).$$

The observer O matters twice in this model: first, through what is counted as an ingestible resource (the measurement of r_t), and second through what is counted as usable genenergy (the units and operational meaning of g_t). The conversion function Γ_O can be taken to encode both physical efficiency and descriptive coarseness: at a coarse level, many distinct micro-resources may be aggregated into a single scalar r_t, with Γ_O absorbing the aggregation and any saturation effects. In many applications one expects diminishing returns (e.g. bottlenecks or limited conversion capacity), so Γ_O may be concave, but the definition does not require this. Likewise, $c_O(a_t)$ plays the role of an action-dependent expenditure (maintenance, locomotion, sensing, planning, repair, etc.), and can be used to represent not only immediate energy draw but also wear or opportunity cost insofar as O treats those as budget-relevant.

Although w_t does not appear explicitly in the budget update, it is included because expelling waste is a constitutive part of metabolic coupling: waste handling can be modeled either by folding its costs into $c_O(a_t)$ (e.g. actions that perform filtration, excretion, garbage collection, heat export), by treating high w_t as required to keep λ low (failure to export waste increases effective decay), or by adding further state variables/constraints when the environment or internal compartments can saturate. The present scalar dynamics should be read as the simplest bookkeeping core around which richer constraints can be layered.

Definition 285 (Metabolism). *A system is said to carry out* metabolism *over a horizon T (for observer O) if, with nontrivial frequency over $t \in \{0, \dots, T-1\}$, it executes actions that:*

- *ingest resources (increase r_t relative to a baseline),*
- *expel waste (increase w_t relative to a baseline), and*
- *raise or sustain its genenergy budget via internal transformation, i.e. yield $g_{t+1} \geq g_t$ on average when maintenance is included.*

Here "baseline" can be understood as the counterfactual intake/expulsion rate that would occur without the system's organized activity at the same descriptive level (for example, passive diffusion, background inflow/outflow, or default operating mode). The requirement of "nontrivial frequency" is meant to rule out one-off charging events or purely accidental uptake: metabolism is a repeated, policy-shaped pattern of interaction rather than an isolated occurrence. The phrase "on average when maintenance is included" is also essential: a system may exhibit brief metabolic gains while still being non-metabolic in the relevant sense if those gains do not cover ordinary upkeep, or if they require unsustainable drawdown elsewhere (which, under O, would typically manifest either as an increase in λ or as rising costs $c_O(a_t)$).

Remark 1076 (Metabolism is a pattern-level notion)**.** *Metabolism need not be biochemical. Any system that reliably converts one form of resource into usable capacity-for-action counts as metabolic at the corresponding descriptive level. In a cognitive system, "metabolic resource" might be compute, time, attention budget, or social capital; in a physical organism, it might be chemical free energy; in an engineered robot, it might be battery charge and maintenance cycles.*

The pattern-level emphasis is important because it allows the same underlying substrate to support multiple metabolic descriptions depending on O. For instance, in an organism one may treat ATP production as the metabolic conversion (fine-grained O), or treat feeding and resting cycles as the conversion (coarser O); both can legitimately instantiate Γ_O, c_O, and λ, but they attribute different actions to intake versus expenditure. Similarly, for a social collective, "resource" may be funding or legitimacy, and "waste" may be churn, conflict, or reputational damage; the conversion function then summarizes the internal processes that turn inputs into coordinated capacity-for-action.

Proposition 31 (A minimal viability condition in the budget model)**.** *Assume there exists $\underline{g} > 0$ such that $(s, g) \in \mathcal{V}_O$ implies $g \geq \underline{g}$, and assume that there exists a policy producing a stationary resource-intake regime with*

$$\mathbb{E}[\Gamma_O(r_t)] \;\geq\; \mathbb{E}[c_O(a_t)] + \lambda \underline{g}$$

whenever $g_t \geq \underline{g}$. Then the expected budget $\mathbb{E}[g_t]$ can be maintained at or above $\underline{g}$ over arbitrarily long horizons (in expectation).

This condition has the natural interpretation that expected metabolic gains must cover (i) expected action costs and (ii) expected leakage at the minimal viable budget level. The "stationary regime" phrasing is doing conceptual work: it isolates a long-run, repeatable intake pattern rather than a transient windfall, and it permits stochasticity in both intake and cost while still requiring a nonnegative drift at the threshold $\underline{g}$. Note also that the statement is intentionally modest: it asserts invariance of a lower bound in expectation, not almost-sure non-ruin, and it does so within the scalar bookkeeping coordinate rather than the full state space.

Proof. Taking expectations in $g_{t+1} = (1 - \lambda)g_t + \Gamma_O(r_t) - c_O(a_t)$ gives

$$\mathbb{E}[g_{t+1}] \;=\; (1 - \lambda)\mathbb{E}[g_t] + \mathbb{E}[\Gamma_O(r_t)] - \mathbb{E}[c_O(a_t)].$$

Under the stated inequality, when $\mathbb{E}[g_t] \geq \underline{g}$ we have

$$\mathbb{E}[g_{t+1}] \;\geq\; (1 - \lambda)\mathbb{E}[g_t] + \lambda \underline{g} \;\geq\; (1 - \lambda)\underline{g} + \lambda \underline{g} \;=\; \underline{g}.$$

Thus $\underline{g}$ is an invariant lower bound in expectation. □

The final step can be read as a one-step inductive invariant: if $\mathbb{E}[g_t] \geq \underline{g}$ at some time t and the policy maintains the stated inequality whenever the system is above threshold, then the next expected budget also lies above threshold, and iterating yields the claimed "arbitrarily long horizons" result. When $\lambda = 0$, the condition reduces to the intuitive balance $\mathbb{E}[\Gamma_O(r_t)] \geq \mathbb{E}[c_O(a_t)]$; when $\lambda > 0$, it demands an additional margin proportional to the minimally viable stock $\underline{g}$.

Remark 1077 (From energy viability to pattern viability)**.** *Proposition 31 only protects the resource coordinate. A full "life" account also needs viability of the pattern-web structure itself (Section 12): metabolism must be coupled to repair, boundary maintenance, and habit stabilization so that the system-state s_t also remains in the viable region.*

In particular, keeping g_t above $\underline{g}$ does not prevent the system from drifting into states where conversion becomes impossible (e.g. loss of catalytic structure, broken sensors/actuators, degraded interfaces, or destabilized routines), which would correspond to changes in Γ_O, c_O, λ, or in the very mapping between micro-dynamics and the variables tracked by O. This is one reason "waste" and "maintenance" should be understood broadly: if waste accumulates and damages conversion machinery, the effective long-run condition for viability is not merely a budget inequality, but a coupled requirement that the mechanisms implementing Γ_O and controlling λ remain themselves within a self-sustaining pattern.

19.5 Reproduction, species, and lifeforms

Metabolism sustains an individual system; reproduction sustains a *type* of system over time. Hyperseed's definitions emphasize causal role and within-species reproduction. This distinction matters because many systems can maintain themselves for long periods (robust metabolism) without thereby generating any continuing lineage, while other systems can generate copies but only via external scaffolding that does not count as the system itself being the primary causal source. In the present stack, "type over time" is tracked by the persistence of structural/functional regularities in the pattern-web sense, rather than by any single material substrate.

19.5.1 Reproduction as a causal construction of a similar system

To define reproduction within our formal stack, we need (i) a similarity notion for systems and (ii) a causal notion of "plays a primary role in creating." Both are intentionally observer-relative: the same underlying physical process may or may not count as reproduction depending on which organizational features O treats as salient (e.g. genotype-level, phenotype-level, algorithm-level, institution-level), and depending on which causal granularity is used to attribute responsibility.

Similarity is naturally phrased using the pattern-web machinery: two systems are similar if their pattern webs are close under the pattern-web metric or under an approximate morphism (Section 11 and Section 12). This allows similarity to be defined at multiple descriptive levels: one may demand near-isomorphism of fine-grained internal mechanisms, or merely preservation of higher-level roles and interfaces. It also accommodates cases where there is no natural one-to-one component correspondence, as long as there exists an approximate structure-preserving map between organizational patterns.

Definition 286 (Pattern-web state of a system)**.** *Let $W_O(S,t)$ denote an observer-indexed summary of the system's internal organization at time t, e.g. a weighted graph/category whose nodes are salient patterns (Section 9) and whose edges represent compositional/probabilistic/causal relations (Section 11). We call $W_O(S,t)$ the* pattern-web state *of S at time t (relative to O).*

The point of $W_O(S,t)$ is not to insist that an agent (or scientist) literally constructs the entire graph, but that the ontology admits such a summary object as a target of distance/morphism comparisons. In practice, W_O may be realized implicitly by a learned representation, a mechanistic model, or a set of tests that jointly define what counts as "the same organization" for the observer/context at hand.

Definition 287 (Within-species similarity relation). *Fix an observer/context O. Let $d_O(\cdot,\cdot)$ be a distance (or dissimilarity) between pattern webs, for example a graph edit distance weighted by pattern intensities, or a metric derived from a Wasserstein-type distance between fuzzy pattern-sets. For $\epsilon > 0$, define*

$$S \approx_{O,\epsilon} S' \quad \Longleftrightarrow \quad \exists\, t, t' \text{ such that } d_O(W_O(S,t), W_O(S',t')) \leq \epsilon.$$

Allowing $\exists\, t, t'$ rather than fixing the same time index captures the idea that reproduction often relates different *life stages*: a parent at one phase and an offspring at another may be organizationally most comparable after development, bootstrapping, training, or maturation. The tolerance ϵ plays the role of a "species bandwidth": smaller ϵ yields narrower equivalence classes that require closer organizational match, while larger ϵ allows more drift, mutation, or implementation variation (e.g. different embodiments of the same algorithmic agent). Note also that $\approx_{O,\epsilon}$ need not be transitive for arbitrary d_O and ϵ; this is acceptable because the role of similarity here is to constrain reproduction events locally, while the species concept below supplies the lineage-closure structure.

Now define a reproduction event.

Definition 288 (Reproduction event). *Fix an observer/context O. A* reproduction event *from S to S' over a time window $[t_0, t_1]$ is an event in which:*

- *the existence/state of S and its actions in $[t_0, t_1]$ have a strong causal influence on the existence of S' at time t_1 (in the sense of predictive attraction / causal implication from Section 14), and*
- *$S \approx_{O,\epsilon} S'$ for a species-tolerance parameter ϵ.*

The time window formulation permits reproduction to be extended rather than instantaneous: copying may require a sequence of coordinated actions (construction, teaching, compilation, gestation, infection, fabrication) whose causal effect accumulates over $[t_0, t_1]$. The dependence on S's *actions* is important: it distinguishes reproduction from mere common-cause correlation (two similar systems produced by the same factory process) and from passive resemblance (unrelated convergent structures), by requiring that interventions on S (or on its relevant outputs) would substantially change the probability of S' existing as it does.

Remark 1078 (Primary causal role). *"Primary" can be formalized in multiple ways. One option is: among a set of candidate causal explanations for the existence of S', those that include S achieve strictly lower weakness (higher explanatory clarity). Another option is an information-flow criterion: conditional mutual information between S's actions and S''s formation is high after conditioning on other known factors. The ontology allows multiple instantiations; what matters is that reproduction is grounded in a quantitative causal contribution notion, not merely correlation.*

A useful intuition is that "primary" is meant to screen off background enabling conditions that are present in many contexts (ambient physics, general availability of matter/energy, a common environment), while retaining the particular causal pathway by which S helps bring about S'. In biological reproduction, the parent is primary even though the environment provides nutrients; in software reproduction, a running process may be primary even though the OS provides system calls. Conversely, if a system is merely a template stored in an archive and an external operator performs all the essential steps without dependence on the system's own organized activity, then the archive may fail to qualify as the primary causal source under many reasonable formalisms.

19.5.2 Species as a reproduction-closed set

Definition 289 (Species). *Fix an observer/context O and tolerance $\epsilon > 0$. Let $\to_O$ be the directed relation on systems given by*

$$S \to_O S' \iff \text{there exists a reproduction event from } S \text{ to } S' \text{ (within tolerance } \epsilon\text{)}.$$

A nonempty set $\mathcal{C}$ of systems is called a species (for O) *if:*

- *(closure) whenever $S \in \mathcal{C}$ and $S \to_O S'$, then $S' \in \mathcal{C}$; and*
- *(regeneration) for each $S' \in \mathcal{C}$, there exists $S \in \mathcal{C}$ with $S \to_O S'$ (except possibly for a designated "initial" subset, if one wishes to allow an origin event).*

This makes a species a lineage-stable class under the chosen reproduction relation: closure says that offspring remain in the class, while regeneration says that membership is not accidental but supported by incoming reproduction links from within the class (up to any explicitly granted origin boundary). The directedness of $\to_O$ allows asymmetric parent–offspring relations and accommodates cases where S can produce S' but not conversely (e.g. a mature organism producing a juvenile, or a trained model producing a distilled student). The definition is compatible with branching genealogies and with multiple parents, since $\to_O$ need not be functional. It also allows observer-dependent "species" notions such as: species by genetic organization, by morphology, by ecological role, or by algorithmic architecture, provided the similarity and causal criteria are specified.

One can view $\mathcal{C}$ as (approximately) a strongly connected component of the reproduction graph after contracting the optional initial subset: regeneration ensures incoming edges, while closure ensures outgoing edges stay inside. If the observer prefers a stricter biological notion, additional constraints (e.g. typical interbreeding, shared gene pool, or reproductive isolation) can be layered on top as refinements, without changing the core causal-closure idea.

Remark 1079 (Groups can reproduce too). *The ontology allows reproduction to be carried out by groups. Formally, one can replace S by a coalition object $\{S_1, \dots, S_k\}$ (or a composite system) and define a relation $\{S_1, \dots, S_k\} \to_O S'$. Nothing essential changes; it simply moves the reproduction arrow from individuals to composites.*

This group-level option covers cases like eusocial colonies, symbiotic consortia, cultural/institutional reproduction (where a coordinated group founds a new branch), and multi-agent systems that replicate via distributed protocols. It also supports "scaffolded" reproduction where no single component has enough causal influence alone, but the coalition does, matching many real-world cases where reproduction is inherently collaborative.

19.5.3 Lifeforms

We now combine the pieces into a lifeform notion. The intent is to separate (i) mere complex dynamics, (ii) self-maintenance (metabolism), and (iii) lineage continuation (reproduction), while keeping all three definable in a substrate-neutral way.

Definition 290 (Lifeform). *Fix an observer/context O. A system S is a* lifeform (for O) *if:*

- *S is a complex interactive system (Definition above),*
- *S routinely carries out metabolism (Section 19.4), and*

- *S routinely participates in reproduction events within some species $\mathcal{C}$.*

The species quantifier "within some species $\mathcal{C}$" is meant to ensure that reproduction is not a one-off coincidence but integrated into a reproduction-closed lineage class. Participation in reproduction events can be as a direct parent, as part of a coalition parent, or (in some contexts) as a necessary role in a structured reproductive process, depending on how O specifies the relevant causal attribution. This makes the lifeform concept robust across levels: an organism, a colony, or an engineered agent can qualify if it meets the metabolism and reproduction criteria at the chosen descriptive level.

Remark 1080 (Routine versus occasional)**.** *"Routinely" is intentionally graded. A crisp version would require reproduction and metabolism to occur with at least some frequency over long horizons; a fuzzy/paraconsistent version (below) treats routine-ness as evidence. This flexibility is essential for boundary cases (hibernation, spores, sterilized organisms, viruses).*

Graded routine-ness also handles time-scale mismatch: reproduction may be rare relative to metabolism (e.g. long-lived trees) but still a stable disposition, while metabolism may be intermittent relative to reproductive capability (e.g. dormant seeds). For sterilized organisms, the framework allows a nuanced classification: an individual may fail the "routinely participates" clause even if it is similar to members of a reproducing species, while the species itself remains well-defined via other members' reproduction links. For viruses and other borderline replicators, the causal-primacy condition can be tuned to decide whether the virus, the host, or the host–virus composite is the reproducing system, reflecting different observer commitments about what organization is doing the primary causal work.

19.5.4 Biological, synthetic, and evolved lifeforms

Hyperseed distinguishes lifeforms by substrate and origin. We encode this by adding labels/metadata on the realization of S. These labels are intended to be orthogonal to the core lifeform criteria: they classify *how* the organization is realized and *how* it arose, not whether it meets the metabolism/reproduction conditions.

Definition 291 (Lifeform subtypes: biological, synthetic, evolved)**.** *A lifeform S is:*

- biological *if its dominant implementation substrate is organic chemistry (at the relevant descriptive level);*
- synthetic *if its design/assembly was primarily guided by an intelligent engineering process (as opposed to arising via unguided population-level dynamics);*
- evolved *if it arose via evolutionary and self-organization processes without intentional design.*

These labels are not mutually exclusive at all descriptive levels (e.g. bio-engineered organisms), so we allow them to be multi-valued or p-bit*-valued when needed.*

At different descriptive levels, a single system may receive different subtype labels: a bio-engineered organism can be biological in substrate, synthetic in origin for particular modules, and evolved in the sense that its lineage later adapts via selection. Similarly, an artificial agent running on silicon may be synthetic in origin, but its later variants may be evolved if produced by an unguided evolutionary search or by an open-ended selection process. Allowing p-bit-valued labels supports intermediate or disputed cases where evidence is partial (e.g. uncertain origin), where the system contains a mix of components with different provenance, or where the observer treats "guidance" as a graded variable rather than a binary predicate.

19.6 Alive and dead as time-indexed (possibly paraconsistent) predicates

The ontology treats "alive" and "dead" as time-indexed properties and, crucially, as *boundary notions*: there are borderline cases, context dependence, and situations in which different evidential streams conflict. In particular, both predicates are intended to track *states of a system as reported through an observer/model interface*, not merely intrinsic microphysical facts: the same underlying system S at the same time t can receive different status assignments from distinct observers O because they use different sensors, different background theories, different coarse-grainings, or different practical decision thresholds.

We therefore define alive/dead predicates as p-bit-valued evidence fields (Section 3.2). Concretely, for any predicate P, a value $P(S,t) \in [0,1]^2$ is interpreted as a pair

$$P(S,t) = (P^+(S,t), P^-(S,t)),$$

where P^+ is the degree of *supporting* evidence and P^- is the degree of *counter*-evidence. The key point is that the framework does not require P^+ and P^- to sum to 1 (or to be otherwise complementary): simultaneously high P^+ and P^- represents an explicitly modeled evidential conflict, rather than an inconsistency that collapses the model.

Definition 292 (p-bit evidence for metabolism and reproduction)**.** *Fix an observer/context O. Let $\mathrm{Met}_O(S,t) \in [0,1]^2$ denote the* p-bit *evidence that S is metabolically active near time t, and let $\mathrm{Rep}_O(S,t) \in [0,1]^2$ denote the* p-bit *evidence that S is reproductively active (or participates causally in reproduction events) near time t.*

Both Met_O and Rep_O are meant to be operationally grounded: they may be produced by direct measurement (e.g. calorimetry, observed growth, molecular turnover), by inference from proxies, or by model-based attribution (e.g. participation of a viral genome in a host replication cycle). The phrase "near time t" allows the observer to use a small temporal neighborhood (e.g. a sampling interval), which is often necessary because metabolism and reproduction are rarely instantaneous at the resolution of observation.

Definition 293 (Alive predicate (paraconsistent))**.** *Fix weights $\alpha, \beta \geq 0$ with $\alpha + \beta > 0$. Define the alive evidence at time t by the monotone aggregation*

$$\mathrm{Alive}_O(S,t) \ := \ \Big(\min\{1, \tfrac{1}{\alpha+\beta}\big(\alpha\,\mathrm{Met}^+_O(S,t) + \beta\,\mathrm{Rep}^+_O(S,t)\big)\},\ \min\{1, \tfrac{1}{\alpha+\beta}\big(\alpha\,\mathrm{Met}^-_O(S,t) + \beta\,\mathrm{Rep}^-_O(S,t)\big)\}\Big).$$

We say S is alive for O at time t *if $\mathrm{Alive}^+_O(S,t)$ exceeds a chosen threshold; we say S is* not alive *if $\mathrm{Alive}^-_O(S,t)$ exceeds a chosen threshold. Both may occur simultaneously (paraconsistency).*

The normalization by $(\alpha+\beta)^{-1}$ makes the aggregation comparable across different weight choices, and the outer $\min\{1,\cdot\}$ ensures that the result remains within $[0,1]$ even if the underlying evidence sources are themselves already saturated. Allowing $\alpha = 0$ or $\beta = 0$ permits contexts in which one stream is judged irrelevant or unavailable; for example, an observer may treat reproduction as inapplicable to a particular entity type, or may lack reproductive data while still observing metabolic activity. Because the aggregation is componentwise monotone in each input coordinate, additional supporting evidence for metabolism or reproduction can never decrease Alive^+_O, and additional counter-evidence can never decrease Alive^-_O; this is important for ensuring that refinement of measurement (in the direction of more evidence) does not produce pathological reversals.

Thresholds may be chosen differently for the positive and negative components, and may depend on downstream use. For example, a safety-critical context might employ a low threshold for "not alive" (erring on the side of declaring non-viability), whereas a conservation context might use a low threshold for "alive" (erring on the side of continued protection). Nothing in the formalism forces a unique threshold policy; the intent is to separate the evidential field $\mathrm{Alive}_O(S,t)$ from the eventual binary (or multiway) decision rule applied to it.

Definition 294 (Dead predicate (time-indexed))**.** *Fix a lookback window length $\tau \in \mathbb{N}$. Define the "was-alive" evidence by*

$$\mathrm{WasAlive}_O(S,t) \;:=\; \bigoplus_{u\in\{t-\tau,\ldots,t-1\}} \mathrm{Alive}_O(S,u),$$

where $\oplus$ is the p-bit *join (componentwise max) from Section 3.4. Define*

$$\mathrm{Dead}_O(S,t) \;:=\; \mathrm{WasAlive}_O(S,t) \;\otimes\; \neg\mathrm{Alive}_O(S,t),$$

using the p-bit *negation and monoidal product from the core.*

The use of a lookback window encodes the idea that "dead" presupposes a recent history of life-like functioning, and thereby avoids classifying as dead entities for which there was never meaningful evidence of being alive (relative to O). The join $\bigoplus$ implements an existential-style temporal summary ("alive at some time in the window") rather than an averaging summary; this matches ordinary usage where a single sufficiently strong period of life in the recent past is enough to make subsequent non-life plausibly "death." The choice of τ is a modeling parameter that can be tuned to the timescales relevant to the organism or system class: for fast-lived microorganisms it may be short, while for long-lived organisms or slow metabolic processes it may be longer.

The monoidal product $\otimes$ in $\mathrm{Dead}_O(S,t)$ enforces a conjunctive structure: strong dead-ness requires both strong "was alive" evidence and strong present "not alive" evidence (expressed via $\neg\mathrm{Alive}_O(S,t)$). Because the underlying logic is paraconsistent, it is possible for $\mathrm{Dead}_O(S,t)$ to coexist with nontrivial $\mathrm{Alive}_O(S,t)$ when evidence streams disagree about current status; this is a feature rather than a bug in contexts such as partial observability, sensor failure, or disputes about what counts as metabolic activity.

Remark 1081 (Interpretation of dead-ness)**.** *$\mathrm{Dead}_O(S,t)$ is high when there is strong evidence that S was alive recently and strong evidence that it is not alive now. This captures the intended asymmetry: "dead" is not merely "not alive"; it is "not alive now, but was alive before."*

One can also read the construction as imposing a temporal directionality that is commonly implicit in practice: evidence of a prior living state changes the semantic category of subsequent inactivity. For example, a rock is ordinarily classified as "not alive" rather than "dead" because $\mathrm{WasAlive}_O$ remains low, while a recently functioning organism with a sudden cessation of activity yields high $\mathrm{WasAlive}_O$ and can therefore support high Dead_O if present non-life evidence is also high.

Example 16 (Boundary cases: viruses, spores, and cryopreservation)**.** *A virus may have high Rep^+_O (it participates in reproduction events) but low Met^+_O (it lacks autonomous metabolism). Depending on α, β and the observer's modeling choices, one may obtain: (i) borderline alive (moderate positive and negative evidence), or (ii) paraconsistent alive/not-alive (strong but conflicting evidence). Similarly, spores or cryopreserved organisms may have low current metabolic evidence but strong evidence for persistence of the underlying pattern web and strong evidence of future metabolic resumption; this pushes naturally toward a graded, time-indexed alive predicate rather than a crisp one.*

A further class of boundary cases arises from observational granularity and delayed reporting. For instance, an intensive-care monitor may show no detectable respiration (pushing Met_O^- upward) while other tests indicate ongoing cellular activity (pushing Met_O^+ upward), yielding simultaneous support for alive and not-alive at the same time index. In such cases, the time-indexing prevents the model from forcing an irreversible commitment: later evidence can be incorporated without retroactively rewriting earlier evidential states, while the paraconsistent representation preserves the fact that the earlier state was genuinely conflicted rather than merely unknown.

19.7 Life as self-weaving under resource constraints

We can now restate the intended Hyperseed synthesis succinctly: The aim of the synthesis here is not to propose an additional "vital" ingredient, but to make explicit how the already-introduced pattern-level dynamics acquire a concrete directionality once we acknowledge that maintaining any organized web requires ongoing throughput. In other words, the same self-weaving mechanism that can in principle elaborate arbitrarily rich structure becomes sharply selective when it must continually pay the costs of continuing to exist.

- The *self-weaving webs* of Section 12 provide the pattern-level mechanism: patterns that contribute to the persistence of the web tend to be reinforced (habit-taking), and those that undermine it tend to be suppressed (habit-reversal), potentially including cross-context coupling via morphic resonance. This can be read as an abstract selection process operating *within* an individual or system over time: reinforcement is not merely "learning" in the cognitive sense, but any dynamical bias that causes certain configurations to recur and stabilize. Likewise, suppression is not merely forgetting, but an active reduction in the probability that destabilizing configurations will be re-instantiated. The possible role of cross-context coupling highlights that stabilization may draw upon correlations that are not localized to a single immediate environment, so that persistence can be influenced by regularities that span situations, timescales, or nested sub-webs.
- The present section adds the *resource bottleneck*: self-weaving requires the continual availability of genenergy, replenished by metabolism and allocated via policies that manage effort. The bottleneck framing matters because it converts "could persist" into "must compete for limited continuation capacity": even if many patterns are in principle compatible with persistence, only those compatible with the available genenergy budget can actually be maintained. By "metabolism" we mean any process that converts environmental gradients, stored reserves, or external inputs into a usable internal currency, so that the web can repeatedly pay the costs of sensing, acting, repairing, and coordinating. The allocation "policies" can be explicit (deliberate choice, executive control) or implicit (homeostatic set-points, reflexive gating, priority dynamics); in either case they implement a distribution of effort across maintenance, exploration, exploitation, and reconfiguration. Under this view, life is not only a web that reinforces itself, but a web that must continually decide what to *fund*, what to *defer*, and what to *terminate* as conditions fluctuate.
- Reproduction provides a route for pattern-web persistence that exceeds the lifespan of a single instance, yielding species-level continuity. This adds an inter-instance channel for stabilization: patterns that cannot be sustained indefinitely within one bounded embodiment may nonetheless persist by being re-instantiated in successors. Importantly, reproduction need not be construed narrowly as biological replication; it can be treated more generally as any mechanism that transfers enough of the web's organization (or its generative constraints) to seed a new instance that can resume self-weaving. Species-level continuity then reflects the persistence of a *family* of webs

that share a core set of self-maintaining motifs while still permitting variation, adaptation, and drift across generations.

This prepares the ground for the next layers: knowledge and belief systems (Section 20) can be seen as specialized pattern-web regimes that manage genenergy allocation by compressing experience into reusable models, and intelligence/agency (Section 21) can be seen as the discovery and exploitation of high-transfer, high-capacity genenergy pathways across contexts. To "compress experience" here is to reduce the ongoing costs of prediction and control by replacing repeated costly inference with amortized structure (schemas, categories, causal expectations) that can be cheaply invoked. Such regimes therefore act as budgeting tools: they steer attention, simplify decision spaces, and pre-commit effort toward interventions that have historically yielded good persistence returns. In the same spirit, "high-transfer" pathways are those whose benefits generalize beyond the narrow situation in which they were learned (e.g., representations or skills that remain useful across tasks), while "high-capacity" pathways are those that can reliably sustain large flows of genenergy (e.g., robust metabolic routes, stable social coordination, or scalable tool-use) without collapsing under noise, scarcity, or complexity. The overall picture is that resource constraints do not merely limit life; they help define what counts as viable organization by making persistence an ongoing, budgeted achievement rather than a passive property.

20 Knowledge, belief systems, and science

20.1 Why an epistemic layer is needed

Sections 9–19 treated minds as pattern-using, energy-limited systems: observers make (and fail to make) distinctions, quantify simplicity via weakness/effort, and organize patterns into webs whose dynamics include attention and habit formation. In that framing, pattern webs already carry a great deal of structure, but they do not yet distinguish between (i) merely having an internal regularity that happens to correlate with the world and (ii) having a *normatively usable* internal regularity that can be recruited as an answer, criticized as an error, or improved via deliberate inquiry. The epistemic layer is introduced to make that distinction explicit: it adds mechanisms for *evaluation* (when is a pattern treated as a commitment?), *interrogation* (what question is a pattern answering?), and *revision* (how are commitments updated under limited time, compute, and evidence).

Hyperseed then takes an additional step: it treats *knowing* and *thinking* not as mysterious extras but as *structured dynamics on pattern webs.* The key move is operational:

- A system *knows* something to the extent that it can *answer a question* in a way that reliably reduces uncertainty and supports effective action.
- A system *thinks* to the extent that it can *generate, select, and refine* questions, answers, and the beliefs that connect them.

Here "question" and "answer" are not treated as purely linguistic objects. They are taken to be templates and constraints that can be instantiated in perception, planning, explanation, or social exchange. Reliability is likewise operationalized: an "answer" counts as reliable insofar as it remains usable across a relevant range of contexts (or can signal its own limits), rather than merely fitting a single episode. This connects directly to the earlier emphasis on attention and habit formation: attention provides a way to allocate limited resources across competing questions, while habits provide cached answer-patterns whose reuse is efficient but whose brittleness must be monitored.

Remark 1082. *Philosophically, this treats epistemology as a species of control theory: the point of "knowing" is not to mirror the world in some metaphysically immaculate way, but to compress and stabilize action-relevant distinctions in the face of uncertainty and limited resources. On this view, "uncertainty reduction" is not an abstract goal but a means of improving control: it is valuable insofar as it sharpens the system's ability to choose interventions, anticipate outcomes, and recover from error. This also implies that a system may rationally tolerate local inconsistency when the cost of immediate global reconciliation outweighs the benefit, or when conflicting evidence supports different actions in different regimes. This stance aligns with Hyperseed's broader pattern-centric view of cognition [5, 1] and with explicitly social and computational pictures of scientific practice [20]. In the Hyperseed core-concept set, this section primarily elaborates Knowing and Thinking (Hyperseed-Concept* **??***), Questions (Hyperseed-Concept 144), and Answers (Hyperseed-Concept 56).*

In this section we formalize:

(a) template patterns, questions, and answers;

(b) beliefs and axiom systems as paraconsistent, resource-sensitive commitments;

(c) question networks as (possibly cyclic) dependency structures;

(d) belief systems as coherent epistemic sub-webs within a community; and

(e) science and state-dependent science as belief-system dynamics with explicit empirical coupling.

The ordering reflects a build-up from local representational machinery (templates, questions, answers) to increasingly global organization (axioms, dependency networks, and community-level stabilization). In particular, allowing question networks to be cyclic is not an incidental complication: real inquiry often contains feedback loops (e.g., a hypothesis suggests a measurement protocol, whose results reshape the hypothesis space, which in turn changes what counts as a relevant measurement). The epistemic layer aims to model such loops without requiring idealized acyclic rational reconstructions.

Throughout, we reuse the paraconsistent p-bit domain and the quantale operations from Section 3. This reuse is intended to keep epistemic dynamics continuous with the earlier "weakness/effort" theme: epistemic commitments are treated as patterns with strengths that can be composed, competed, and updated, rather than as all-or-nothing propositions that instantly force global consistency.

Remark 1083 (Notation reminder: p-bit, $\mathbb{V}$, $\oplus$, $\otimes$)**.** *We write* $\mathbb{V} = [0,1]^2$ *for the paraconsistent evidence domain (a "p-bit"), where an element* (v^+, v^-) *is interpreted as* positive *and* negative *evidence held simultaneously (Hyperseed-Concept 65 and Hyperseed-Concept 198). The binary operation* $\oplus$ *denotes the componentwise join (so it* aggregates *evidence), and* $\otimes$ *denotes the quantale product used to* compose *strengths along a rule/premise chain. These choices are aligned with Hyperseed's broader quantale-weakness emphasis [3, 2] and with paraconsistent logics designed to avoid explosion [23, 24]. Interpreting beliefs in* $\mathbb{V}$ *makes it straightforward to represent common epistemic states that are awkward in purely consistent formalisms, such as "there are good reasons for and against," "the data are mixed," or "I have strong supporting evidence but also a salient defeater." In later formalizations, this will let question-answer dynamics treat conflict as a first-class signal (something to be routed, localized, or investigated) rather than as an immediate catastrophe that trivializes inference.*

20.2 Template patterns, questions, and answers

Hyperseed defines questions and answers in terms of *template patterns*. We will model a template pattern as a pattern with a fixed "frame" plus variable "slots".

Definition 295 (Template pattern)**.** *Fix a universe of discourse X. Let S be a finite set of* slots *and let $F \subseteq S$ be the subset of* fixed slots*. A* template *is a map*

$$\tau : S \to X \cup \{*\},$$

where $\tau(s) \in X$ for $s \in F$ and $\tau(s) = *$ (a placeholder) for $s \in S \setminus F$. An instantiation *of τ is a map $\iota : S \to X$ such that $\iota(s) = \tau(s)$ for all $s \in F$. We write $\mathrm{Inst}(\tau) \subseteq X^S$ for the set of instantiations.*

Remark 1084. *In practice, S corresponds to argument positions of a predicate or roles in a schema. The set $\mathrm{Inst}(\tau)$ is the space of candidates the system is trying to narrow down.*

Remark 1085. *It is often useful to view τ as specifying an* equivalence class *of fully specified patterns: two instantiations $\iota, \iota' \in X^S$ are equivalent with respect to the current frame if they agree on the fixed slots F. From this perspective, the placeholder symbol $*$ does not denote an element of X; it denotes* underdetermination *relative to a task or context. This makes explicit why templates are naturally context-sensitive: changing which slots are treated as fixed (i.e., changing F) changes which distinctions among candidates are currently meaningful.*

Remark 1086 (Intuition and examples)**.** *A template pattern is a disciplined way of saying: "I recognize the* form *of what I am looking for, but not all of its content." The slots S are the places where content may vary; the fixed set F pins down what is already determinate, and the symbol $*$ marks what is still open. The set X^S denotes all assignments of elements of X to each slot in S, and $\mathrm{Inst}(\tau) \subseteq X^S$ carves out those assignments that agree with the fixed frame.*

A simple example: let X be a set of people, $S = \{1, 2\}$ represent the two roles in a binary relation, and let $F = \{1\}$ with $\tau(1) = \mathsf{Alice}$ and $\tau(2) = *$. Then $\mathrm{Inst}(\tau)$ is the set of pairs (Alice, x) with $x \in X$. In a more schematic setting, S might index semantic roles like agent, $\mathsf{patient}$, $\mathsf{instrument}$, etc., matching Hyperseed's emphasis on patterns as re-usable structures (Hyperseed-Concept 130; see also [5]).

This definition is useful because it makes questions and answers mathematically comparable: a question will be about restricting $\mathrm{Inst}(\tau)$, and an answer will be anything that narrows, reweights, or otherwise simplifies the candidate space. Thus, the epistemic layer is expressed in the same language as the pattern layer: distinctions made among *instantiations are the concrete content of "learning."*

Remark 1087. *This template formalism also connects directly to familiar computational mechanisms such as pattern matching and unification. A fixed frame corresponds to a partially instantiated structure (e.g., a term with variables), and an instantiation corresponds to a substitution assigning concrete values to the remaining slots. In this light, $\mathrm{Inst}(\tau)$ can be read as the solution set of a constraint satisfaction problem induced by the fixed slots, with later evidence adding additional constraints or weights.*

Definition 296 (Question)**.** *A* question *for a system S is an action (or action policy) whose intent is to obtain information about $\mathrm{Inst}(\tau)$ for some template τ. Formally, we model a question as a pair (τ, Exp) where*

- *τ is a template pattern, and*

- Exp *is an* experiment/interaction *channel that produces data d which can be used to update beliefs about* $\mathrm{Inst}(\tau)$.

Remark 1088 (Intuition and examples)**.** *A question is not, in this formalism, merely a string of words; it is an* intervention into the world (or into ones own cognitive dynamics) *chosen to discriminate among candidate instantiations. The experiment channel* Exp *embodies the pragmatic fact that information is obtained through interaction: one measures, queries, probes, introspects, simulates, or otherwise acts to produce data.*

Example: if X is a set of possible diagnoses, a template might fix the patient identity and leave the diagnosis slot open; then Exp *might be a lab test producing an outcome that shifts the candidate set. Example: if X is a set of physical parameter values,* Exp *might be a measurement protocol. By treating questions as actions/policies, this section aligns the epistemic layer with the control machinery of Section 14 and with Hyperseed's general view that cognition is closed-loop (cf. [19]). In the Hyperseed core-concept set, this is precisely Questions as information-seeking actions (Hyperseed-Concept 144) interacting with Contexts and Aspects (Hyperseed-Concept 86).*

Remark 1089. *The pair* (τ, Exp) *separates two logically distinct components that are often conflated in informal discourse: (i)* what *is being asked (the target template τ), and (ii)* how *the system proposes to find out (the interaction channel* Exp*). Different experiments can address the same template (e.g., multiple tests for the same diagnosis), and the same experiment can serve multiple templates depending on which slots are being treated as open. This separation also makes room for internal questions:* Exp *may be an introspective or computational procedure (e.g., running a simulation, retrieving from memory, or checking a proof) rather than an external sensorimotor action.*

Definition 297 (Answer)**.** *An* answer *to a question* (τ, Exp) *is any artifact that reduces uncertainty about* $\mathrm{Inst}(\tau)$*. Concretely, an answer may be:*

- *a particular instantiation* $\iota \in \mathrm{Inst}(\tau)$*;*
- *a constraint* $C \subseteq \mathrm{Inst}(\tau)$ *that prunes candidates;*
- *a distribution (or* p-bit*-valued scoring) over* $\mathrm{Inst}(\tau)$*; or*
- *any representation that yields an entropy reduction or computational simplification relative to baseline inference.*

Remark 1090 (Intuition, examples, and usefulness)**.** *An answer is defined by its* function *rather than its format: it is whatever makes the space of candidates smaller, sharper, or cheaper to search. If the question is "which ι is correct?", then a complete answer is a single* $\iota \in \mathrm{Inst}(\tau)$*. But many answers are partial: a constraint C might say "the unknown is in this region" or "the unknown satisfies these invariants," and a distribution might say "these candidates are plausible, these are not."*

Remark 1091. *The four bulleted forms above intentionally mix* set-based *and* probabilistic/graded *answers. The set-based view corresponds to elimination and deduction (hard constraints), while the graded view corresponds to abduction and induction (soft evidence). In particular, the* p-bit*-valued scoring option is meant to cover situations in which the system uses bounded or many-valued belief representations rather than full real-valued probabilities, while still supporting the same conceptual role: ranking or weighting elements of* $\mathrm{Inst}(\tau)$ *so that resources are allocated toward more promising candidates.*

Remark 1092. *The phrase "entropy reduction or computational simplification" is meant broadly. In the probabilistic case, one can formalize the effect of an answer as moving from a prior $P(\iota)$ on* $\mathrm{Inst}(\tau)$ *to a posterior $P(\iota \mid d)$ after observing data d from* Exp, *with uncertainty reduction measured (for example) by $H(P)-H(P(\cdot \mid d))$ or by mutual information between ι and d. In the computational case, an answer can be a* compiled *representation—a rule, model, proof sketch, cached sufficient statistic, or program fragment—that makes subsequent inference over* $\mathrm{Inst}(\tau)$ *cheaper even if it does not drastically reduce the candidate set outright. This matches the operational notion that "knowing" often means being able to answer quickly or with fewer resources, not merely being able to exclude possibilities in principle.*

Remark 1093. *For example, if* $\mathrm{Inst}(\tau)$ *is large, then a constraint C may be vastly more valuable than a single guessed instantiation: it changes the computational geometry of the problem by pruning whole subspaces. In other words, a constraint acts like a* structural *answer: instead of selecting one element of* $\mathrm{Inst}(\tau)$*, it alters the shape of the remaining search landscape so that many candidate instantiations become unreachable or irrelevant under C, often turning an intractable exploration into a manageable one. The parenthetical reference to "entropy reduction" can be understood in several compatible ways: Shannon-style uncertainty reduction, logical-entropy reduction [17], or algorithmic description-length reduction [16]. On a Shannon view, the answer increases concentration of probability mass on fewer instantiations; on a logical-entropy view, it increases the rate at which pairs of distinct instantiations are distinguished by the system's predicates; on an algorithmic view, it permits a shorter effective description of the candidate set by exploiting regularities introduced by C or by the refined model. Hyperseeds point is not to legislate a single measure, but to assert that "answerhood" is tightly bound to* distinction-making *(Hyperseed-Concept 98) and* effort *(Hyperseed-Concept 100). Put informally, what makes an answer an answer is that it induces new discriminations the system can actually use, and that these discriminations are achieved at some nontrivial computational or experimental cost that must be justified by the resulting pruning or re-weighting of possibilities.*

Remark 1094 (Answer quality as weakness reduction)**.** *Let* Π *be the system's current "candidate set" representation for* $\mathrm{Inst}(\tau)$*. If an answer refines this representation (by removing candidates or sharpening scores), then the system has effectively* added distinctions *among instantiations. Here "representation" is meant broadly:* Π *could be an explicit set, a constraint-satisfaction encoding, a posterior distribution, a scoring function over candidates, or any other data structure that supports selection and elimination, and refinement can include either hard exclusion or merely making some candidates decisively less plausible. In the weakness formalism (Section 3.7), adding distinctions corresponds to* reducing *the set of undistinguished pairs, hence decreasing weakness. Equivalently, when fewer candidate pairs remain interchangeable with respect to the system's current tests, predicates, and internal symmetries, the system is less "weak" because it has more leverage to act differently depending on which instantiation is true. Thus, in a minimal formal sense, answering a question is "weakness reduction on a candidate space." This framing also clarifies why merely producing a token output is insufficient: an output counts as a higher-quality answer only insofar as it produces a measurable refinement of* Π *relative to the system's prior state, ideally in a way that is stable under re-encoding of the same candidate space.*

Remark 1095. *This remark is one of the key bridges in the document: it identifies epistemic progress with the same primitive currency used earlier (weakness/effort) [3, 2]. In particular, it treats "learning" and "problem-solving" as variations of a single underlying operation: investing effort in order to produce distinctions that make the environment (or the task space) less confusable to the system. Seen in this light, a "good question" is one that admits actions likely to yield* *high weakness reduction per unit effort, and a "good answer" is one that actually delivers such reduction. This makes explicit the normative direction implicit in the earlier sections: questions and answers are evaluated not only by truth-apt content, but by their capacity to reconfigure the system's candidate space so that subsequent decisions can be made with fewer costly ambiguities. This also echoes Hyperseed's general pattern-emergence framing [5]. Concretely, asking the right question can be understood as selecting a partition of* $\mathrm{Inst}(\tau)$ *that aligns with available tests and interventions, so that the system is nudged toward discovering compressive patterns (regularities) rather than merely accumulating isolated facts.*

20.3 Beliefs as paraconsistent valuations

We treat a belief state as a p-bit-valued valuation of a language of claims. The language can be instantiated in many ways (logical formulas, pattern claims, type assertions, causal rules, etc.). For this section it suffices to treat it abstractly. In particular, we do not assume that $\mathcal{L}$ comes equipped with a fixed proof system or semantics; the purpose of the abstraction is to separate (i) what the agent *takes itself* to have evidence for or against from (ii) whatever background logic one might later use to relate claims to each other. This separation is useful when comparing ordinary reasoning, scientific inference, and socially stabilized belief systems, since these often share *claims* while disagreeing about the sanctioned *derivations* between them.

Definition 298 (Claim language and belief state)**.** *Let $\mathcal{L}$ be a set of* claims*. A* belief state *is a function*

$$\beta : \mathcal{L} \to \mathbb{V} = [0,1]^2, \qquad \beta(\varphi) = (\beta^+(\varphi), \beta^-(\varphi)).$$

$\beta^+(\varphi)$ is positive evidence for φ and $\beta^-(\varphi)$ is negative evidence. No consistency constraint is imposed.

A useful way to read $\beta(\varphi) \in [0,1]^2$ is as recording two *independently accumulating* quantities: one channel aggregates support and one channel aggregates counterevidence. The absence of a constraint such as $\beta^+(\varphi) + \beta^-(\varphi) \le 1$ is deliberate: it allows the model to represent both partial information (both small) and conflicted information (both large) without forcing an artificial reconciliation step. When later reasoning requires a single scalar "credence-like" quantity, one can introduce derived summaries (e.g. a net-support score $\beta^+(\varphi) - \beta^-(\varphi)$ or an information-mass score $\beta^+(\varphi) + \beta^-(\varphi)$), but these are *projections* of the underlying state rather than the state itself.

Remark 1096 (Intuition and examples)**.** *This definition says: a minds epistemic posture is not a single probability distribution nor a set of crisp truths, but a paraconsistent field of* evidence *(Hyperseed-Concept 65). The same claim can be supported and opposed at once; this matches ordinary cognition, where we can have strong reasons both for and against a proposition without immediately dissolving into nonsense. Formally, allowing $\beta^+(\varphi)$ and $\beta^-(\varphi)$ to both be large blocks the classical principle that inconsistency entails triviality, aligning with paraconsistent traditions [23, 24].*

A simple example: let φ be "this medicine is safe." One may have clinical-trial support (large β^+) and also anecdotal reports of adverse events (nontrivial β^-). The usefulness of the definition is compositional: once belief is a function into $\mathbb{V}$, we can apply the same quantale operations used elsewhere to aggregate testimony, compose rules, and propagate constraints through networks of claims.

It is also helpful to connect the p-bit representation to familiar logical and epistemic categories. At the crisp extremes, $(1,0)$ behaves like "accepted," $(0,1)$ like "rejected," $(0,0)$ like "no information," and $(1, 1)$ like "inconsistent but maximally evidenced on both sides." Intermediate values allow graded versions of these four corners, so the model can represent "weakly supported," "mostly disconfirmed," or "strongly contested" claims without deciding prematurely which side must win. This graded, paraconsistent stance is especially important in scientific contexts where evidence streams arrive asynchronously, instruments disagree, or different experimental setups pull in different directions.

Definition 299 (Direct and indirect belief (operational form))**.** *Fix a temporal scaffold* $(T, \preceq)$ *as in Section 7. Let* $\mathcal{P}$ *be a set of pattern-claims of the form "pattern* P *holds at time* t*".*

(a) A direct belief *is a belief about* persistence*: a claim that an observed or inferred pattern will continue to hold at later times along some timeline relative to some world/context.*

(b) An indirect belief *is a belief about* belief production*: a claim that some cognitive content (rule, schema, procedure) will be used by the system's inference to generate (or modify) future direct beliefs.*

Operationally, one can think of direct beliefs as assigning p-bit-valuations to temporally indexed claims in $\mathcal{P}$ (e.g. $\beta(P@t)$), while indirect beliefs assign p-bit-valuations to higher-order claims about transitions, updates, and rule applications (e.g. that certain kinds of observations will be treated as reliable, or that certain inference steps will be applied when premises are present). The definition does not require that indirect beliefs be infallible or even consistent with one another: a system can simultaneously treat a source as "generally reliable" while also treating it as "often biased" in specific contexts, and this tension can be represented as nontrivial β^+ and β^- on the corresponding indirect claim.

Remark 1097 (Intuition and why the distinction matters)**.** *Direct beliefs are about how the world is expected to continue: they are, so to speak, commitments about the* trajectory *of pattern (Hyperseed-Concept 130) through time. Indirect beliefs are about the minds own machinery: which rules, sources, and internal procedures will shape those commitments. This split is philosophically significant: the mind not only predicts the world; it predicts itself as an epistemic actor, and it does so under resource constraints.*

Example: "the sun will rise tomorrow" is a direct belief about persistence. "my measurement instrument is reliable" or "modus ponens is a valid inference move" are indirect beliefs: they govern how new evidence becomes new direct belief. In Hyperseed terms, indirect beliefs are often stabilized socially as part of a shared practice, foreshadowing belief systems and science (Hyperseed-Concept 64; see also [20, 19]).

The direct/indirect split also clarifies how disagreement can persist even when agents share much of the same first-order data. Two agents may have similar direct beliefs about many immediate patterns while differing sharply in indirect beliefs about which models are admissible, which measurements are trusted, or which inference rules are appropriate under uncertainty. In such cases the conflict is not (only) over φ but over the mechanisms that generate future $\beta(\varphi)$, so resolution requires addressing the indirect layer (e.g. by calibration, methodological critique, replication practices, or negotiating shared standards).

Remark 1098. *In practice, indirect beliefs correspond to the "rules of the game" a mind uses in thinking: trusted inference patterns, trusted sources, and stable modeling biases. In a community, many indirect beliefs are socially maintained.*

20.4 Axiom systems and hypothetical reasoning

Hyperseed emphasizes that an axiom system, *as held by an agent*, is psychological/pragmatic: axioms are interrelated indirect beliefs. We model this as a designated subset of claims with high commitment. In particular, what makes a claim "axiomatic" in this sense is not a metaphysical property of the sentence, but the agent's stance that it will be treated as a stable inferential anchor while the agent reasons, explains, and plans. This framing is meant to cover both explicit axiom lists (as in formal mathematics) and tacit, practice-level commitments (as in scientific methodology or everyday reasoning).

Definition 300 (Axiom system)**.** *An* axiom system *is a pair* (A, θ) *where* $A \subseteq \mathcal{L}$ *is a set of claims and* $\theta \in (0, 1]$ *is a commitment threshold. A belief state* β treats (A, θ) *as axiomatic if*

$$\beta^+(\varphi) \geq \theta \quad \textit{for all } \varphi \in A.$$

It is important that this definition is intentionally one-sided: it constrains the positive commitment $\beta^+(\varphi)$ but does not require that $\beta^-(\varphi)$ be small. This is consistent with the paraconsistent stance of the framework: an agent may, for pragmatic reasons, proceed *as if* some φ is fixed for the purposes of a project, while still registering countervailing evidence in the background (e.g., unresolved anomalies or known edge cases). Likewise, A is not required to be deductively closed, consistent, or minimal; it is merely the set the agent has earmarked as "treated as fixed" at the given threshold. In practice, A may include redundant items, mutually supporting principles, and even explicitly recognized tensions (for example, idealizations in a model that are acknowledged not to be literally true).

Remark 1099 (Intuition and examples)**.** *An axiom system here is not a claim about eternal certainty; it is a policy-like stance: "within this context, I will treat these claims as fixed points around which inference turns." The threshold* θ *makes this explicit: axiomatic status is a* degree *of commitment, not an ungraded categorical mark. This matches the everyday phenomenon that even mathematical axioms are, psychologically, held with a certain kind of firmness that is stronger than ordinary conjectures, yet still situated within a practice.*

Example: in Euclidean geometry, A might contain the parallel postulate; in a physical theory, A might contain symmetry principles or conservation laws; in a social epistemology, A might contain methodological commitments like "randomized trials matter." The definition is useful because it gives a clear handle on how fixed assumptions enter a paraconsistent inference system without requiring global consistency, and it allows axiom change to be modeled as a continuous shift in evidence rather than an abrupt collapse.

A useful way to read θ is as a floor on "how hard it is to dislodge" an axiom within the relevant context: the closer θ is to 1, the more nearly the axiom behaves like a hard constraint for the agent's positive stance, whereas smaller θ corresponds to a looser, working-commitment notion of axiomaticity. Because $\theta \in (0, 1]$ rather than $\theta = 1$, the model can represent cases where an agent is highly committed but not maximally committed—for example, when the agent treats a principle as "good enough" for engineering design or for a particular line of argument, while conceding that future evidence could still rationally lower the commitment. Conversely, the definition also separates two roles often conflated in informal talk: (i) a claim's importance as an inferential premise, and (ii) the agent's total evidential situation about that claim, including recorded objections.

Definition 301 (Entertaining a hypothesis)**.** *Fix a belief state* β *and a claim* $h \in \mathcal{L}$*. To* entertain the hypothesis h at strength $s \in \mathbb{V}$ *is to form the hypothetically augmented state*

$$\beta \uparrow_s h \;:=\; \beta \oplus \delta_{h,s},$$

where $\delta_{h,s}(h) = s$ and $\delta_{h,s}(\varphi) = (0,0)$ for $\varphi \neq h$, and $\oplus$ is the componentwise join from Section 3.4. One then runs an inference/closure operator (defined below) to see what follows.

Here $s \in \mathbb{V}$ may itself be mixed, e.g. $s = (s^+, s^-)$ with both coordinates nonzero, so "entertaining" can represent not only a pure pro-assumption (boosting h) but also a controlled stress-test in which one injects both supportive and adverse weight and then observes which downstream commitments are robust. This is particularly relevant in domains where hypotheses are rarely adopted as crisp binaries, and instead come in graded forms (e.g. "h is approximately true," "h holds in the regime of interest," or "h is a useful idealization"). The fact that $\delta_{h,s}$ is concentrated at h is also deliberate: it isolates the direct hypothetical move from any additional modeling choices about how the rest of the network should be updated, leaving that propagation to the closure operator and the background inferential structure.

Remark 1100 (Notation unpacking and intuition)**.** *The map $\delta_{h,s}$ is an "evidence spike" concentrated on the single claim h: it assigns the chosen strength $s = (s^+, s^-)$ to h and contributes no evidence elsewhere. The operation $\beta \oplus \delta_{h,s}$ then means: add this hypothetical evidence to the current belief state, componentwise, without deleting prior evidence. This is the paraconsistent analogue of the classical act "assume h and derive consequences," but it is deliberately less brittle: one can entertain h while still retaining negative evidence for h in the background.*

Example: one might entertain h = "a new particle exists" at strength $(0.6, 0.0)$ to explore downstream predictions, even if current data yields some negative evidence. This definition is useful because it turns counterfactual reasoning into a standard operator on belief states, enabling systematic exploration of question networks (Section 20.6) and also clarifying how cycles can be temporarily "cut" by hypotheses.

A further point is that $\uparrow_s$ is designed to be compositional: in later use, one can entertain several hypotheses by successive applications (or by joining several spikes) and then compare the resulting closures. This makes the operation suitable not only for "what if h?" reasoning but also for controlled scenario analysis: one can vary s and watch which conclusions change continuously and which change only when certain thresholds are crossed. In that sense, $\uparrow_s$ plays a role analogous to parameter sweeps in modeling: it supplies a uniform interface for probing the sensitivity of an inferential system to hypothetical adjustments.

Remark 1101. *This implements the Hyperseed description: to assume an axiom is to explore what would be implied if a truth value were very different than currently believed. The paraconsistent setting allows the exploration to coexist with contradictory evidence rather than forcing an all-or-nothing toggle.*

One can also view this as separating two questions that are often run together: (i) what the agent currently believes, and (ii) what the agent would be prepared to infer under a stipulated supposition. By treating the supposition as an explicit, additive modification of the evidence state, the model makes room for disciplined hypothetical reasoning even when the agent's baseline state is already nonclassical (e.g. contains tensions, exceptions, or domain-limited rules). This becomes especially important when axioms are treated as context-sensitive commitments: the same agent may legitimately run different inferential closures from the same underlying β by entertaining different axiom-like hypotheses at different strengths, corresponding to different projects, explanatory aims, or modeling idealizations.

20.5 Weighted inference and closure

To connect beliefs to question answering we need a minimal notion of inference. We use a general "weighted rule" scheme compatible with the quantale operations. In particular, we treat inference as an evidence-transforming operation on belief states, so that the same algebraic primitives used for composing partial evidence elsewhere also govern derivations.

Definition 302 (Weighted inference rules)**.** *A* weighted inference rule *is a triple $r = (\Gamma, \psi, \lambda_r)$ where $\Gamma \subseteq \mathcal{L}$ is a finite set of premises, $\psi \in \mathcal{L}$ is a conclusion, and $\lambda_r \in \mathbb{V}$ is a rule strength. Given a belief state β, define the rule's contribution to ψ by*

$$\mathrm{contrib}_r(\beta) \;:=\; \lambda_r \otimes \bigotimes_{\varphi \in \Gamma} \beta(\varphi),$$

using the p-bit *quantale product $\otimes$ from Section 3.4.*

It is often helpful to read Γ as a conjunction-like bundle of requirements: all premises must be supported to some extent for the rule to fire with nontrivial strength, and the degree to which the premises jointly hold is computed by the iterated product. When $\Gamma = \{\varphi_1, \ldots, \varphi_k\}$, the expression $\bigotimes_{\varphi \in \Gamma} \beta(\varphi)$ is shorthand for $\beta(\varphi_1) \otimes \cdots \otimes \beta(\varphi_k)$; the finiteness of Γ ensures this is well-defined without additional limiting assumptions. In the special case $\Gamma = \emptyset$, the iterated product is understood as the $\otimes$-unit, so an "axiom rule" $(\emptyset, \psi, \lambda_r)$ contributes λ_r directly to ψ.

Remark 1102 (Intuition and examples)**.** *This definition formalizes inference as "strength flows through a rule." The premises Γ are a finite subset of claims; ψ is the claim supported by the rule; and λ_r is the intrinsic reliability (or methodological authority) of the inference pattern. The iterated product $\bigotimes_{\varphi \in \Gamma} \beta(\varphi)$ composes the evidence values for all premises, and then $\lambda_r \otimes (\cdots)$ scales that combined support by the rule strength.*

Example: a probabilistic-logic-inspired rule "from φ infer ψ" might use a large λ_r if it is a trusted implication, while a heuristic analogy rule might use a smaller λ_r. The usefulness is that all inference becomes a special case of the same algebra used for composing patterns and influences elsewhere: rules are patterns with weights, and applying a rule is pattern-composition [5, 3].

A further point is that λ_r can be used to encode more than "truth-preservation." It may reflect experimental reproducibility, the historical reliability of an instrument, the domain-of-applicability of an approximation, or even a social/organizational trust score. Because $\mathbb{V}$ is paraconsistent, the framework does not require that high support for ψ forces low support for $\neg\psi$; rather, it records how multiple strands of reasoning contribute (possibly in tension) to the evolving state. Thus, the notion of "rule strength" is closer to evidential warrant than to a classical entailment guarantee.

Definition 303 (One-step inference operator)**.** *Let $\mathcal{R}$ be a set of weighted inference rules. Define $F_{\mathcal{R}} : \mathbb{V}^{\mathcal{L}} \to \mathbb{V}^{\mathcal{L}}$ by*

$$F_{\mathcal{R}}(\beta)(\psi) := \beta(\psi) \oplus \bigoplus_{(\Gamma,\chi,\lambda)\in\mathcal{R}:\ \chi=\psi} \Big(\lambda \otimes \bigotimes_{\varphi\in\Gamma} \beta(\varphi)\Big).$$

By construction, $F_{\mathcal{R}}$ performs a form of weighted forward chaining: each rule computes a candidate increment for its conclusion, and then all such increments are aggregated by $\oplus$. The definition allows $\mathcal{R}$ to be large (even infinite) as long as the relevant joins exist in $\mathbb{V}$; this is exactly where the completeness assumptions on the quantale become operationally meaningful.

Remark 1103 (Notation unpacking and intuition)**.** *Here $\mathbb{V}^{\mathcal{L}}$ denotes the set of all belief states (all functions $\mathcal{L} \to \mathbb{V}$). For each conclusion $\psi \in \mathcal{L}$, the operator $F_{\mathcal{R}}$ keeps the current evidence $\beta(\psi)$ and joins it (via $\oplus$) with the evidence contributed by* every *rule whose conclusion is ψ. The symbol $\bigoplus$ denotes iterated join over that (possibly finite) set of rules.*

Intuitively: one inference step means "apply all rules once in parallel and add their conclusions to what you already believe." This is a deliberately extensional view of inference, compatible with different internal implementations (symbolic proof search, neural approximations, probabilistic programming), so long as their net effect can be summarized as evidence aggregation.

Two additional structural properties are worth keeping in mind when interpreting $F_{\mathcal{R}}$. First, it is *inflationary* (or extensive): for all β and ψ one has $\beta(\psi) \leq F_{\mathcal{R}}(\beta)(\psi)$ because $F_{\mathcal{R}}(\beta)(\psi)$ is defined as $\beta(\psi) \oplus (\cdots)$. This matches the intended reading that one-step inference accumulates consequences rather than retracting beliefs. Second, because different rules for the same conclusion are combined by $\oplus$, the operator treats rules as independent sources of support whose effects are merged in a commutative, associative way (as governed by the join structure of the quantale). In practice, this "all-at-once" aggregation is a semantic idealization; algorithmically one may schedule rules in any order, and the theorem below justifies that repeated application still converges to an appropriate closure under mild conditions.

Theorem 18 (Existence of inference closure)**.** *The operator $F_{\mathcal{R}}$ is monotone on the complete lattice $\mathbb{V}^{\mathcal{L}}$. Hence it has a least fixed point above any initial state β_0, given by*

$$\mathrm{Cl}_{\mathcal{R}}(\beta_0) := \bigoplus_{n \geq 0} F_{\mathcal{R}}^{n}(\beta_0),$$

where F^0 is the identity and $F^{n+1} = F \circ F^n$. If, additionally, all values that can arise belong to a finite subquantale $W \subseteq \mathbb{V}$ (e.g. a finite rational grid closed under $\oplus$ and $\otimes$), then for finite $\mathcal{L}$ the ascending chain stabilizes after finitely many steps.

The expression "least fixed point above β_0" should be read as: among all belief states $\beta \in \mathbb{V}^{\mathcal{L}}$ such that $\beta_0 \leq \beta$ and $F_{\mathcal{R}}(\beta) = \beta$, there is a smallest one in the pointwise order. This is the natural notion of closure for an accumulating (inflationary) inference step: closure must preserve the initial evidence and must be stable under further rule application. In settings where $\mathcal{L}$ or the effective range of β is finite (as in the theorem's second clause), the iterative characterization as a finite-stage saturation is especially literal: one can view $F_{\mathcal{R}}^{n}(\beta_0)$ as "beliefs derivable in at most n rounds," and the stabilization index N as a bound on the depth of rule-chaining needed to reach closure.

Remark 1104 (What the theorem is saying, in plain language)**.** *The theorem asserts that if one repeatedly applies the one-step inference operator $F_{\mathcal{R}}$—adding in all consequences, then consequences of consequences, and so on—this process has a well-defined "limit" belief state: the least state that contains the initial evidence and is closed under the rules. Mathematically, it is the least fixed point of $F_{\mathcal{R}}$ above β_0.*

This result is important because it guarantees that "run inference to closure" is not merely a metaphor but a legitimate construction in the paraconsistent quantale setting. It connects the epistemic layer to the document's broader use of fixed-point constructions for stability and self-consistency (compare Knaster–Tarski usage in other contexts, e.g. [10]). It also complements the earlier weakness-based view: closure is one way of operationalizing how indirect beliefs generate direct beliefs through repeated pattern application [5].

An additional intuition is that $\mathrm{Cl}_{\mathcal{R}}(\beta_0)$ is the smallest "saturated" belief state compatible with the rules, analogous to the least Herbrand model in logic programming, but with truth values in $\mathbb{V}$ rather than $\{0,1\}$. Because joins aggregate rather than choose, closure can be seen as a conservative way to incorporate *all* derivable support without committing to a single proof path. This is especially relevant in paraconsistent settings: if there are rules deriving both ψ and $\neg\psi$, closure records both streams of support, instead of forcing explosion or global inconsistency.

Proof. Monotonicity: $\oplus$ and $\otimes$ are monotone in each argument (componentwise on $[0,1]^2$). Therefore each $\mathrm{contrib}_r(\beta)$ is monotone in β, and the join over rules preserves monotonicity, so $F_{\mathcal{R}}$ is monotone.

The poset $\mathbb{V}^{\mathcal{L}}$ is a complete lattice under pointwise order, so by Knaster–Tarski, $F_{\mathcal{R}}$ has fixed points and the least fixed point above β_0 is the join of the iterates. For the finiteness claim, assume $\mathcal{L}$ is finite and values are restricted to a finite subquantale $W \subseteq \mathbb{V}$. Then $W^{\mathcal{L}}$ is a finite poset, and the ascending chain $\beta_0 \le F_{\mathcal{R}}(\beta_0) \le F_{\mathcal{R}}^2(\beta_0) \le \cdots$ cannot strictly increase forever. Hence it stabilizes at some finite stage N, yielding a fixed point. $\square$

It is also worth observing that the chain displayed in the proof is ascending not only because $F_{\mathcal{R}}$ is monotone, but because $F_{\mathcal{R}}$ is inflationary: $\beta \le F_{\mathcal{R}}(\beta)$ holds pointwise by definition. Thus the iterative process never loses evidence already present in β_0; it only accumulates additional support as rules fire on newly strengthened premises. In computational terms, when $\mathcal{L}$ and the effective value set are finite, one can implement closure as a standard saturation loop: repeatedly update β by $F_{\mathcal{R}}(\beta)$ until no coordinate changes.

Remark 1105 (Proof sketch and intuition)**.** Proof sketch. *The strategy is standard: show $F_{\mathcal{R}}$ is order-preserving, then invoke the fixed-point theorem for monotone maps on complete lattices. In the present setting, the relevant complete lattice is the space of valuations $\beta\colon \mathcal{L} \to [0,1]^2$ ordered pointwise by the product order on $[0,1]^2$, so that $\beta \le \beta'$ means $\beta(\varphi)^+ \le \beta'(\varphi)^+$ and $\beta(\varphi)^- \le \beta'(\varphi)^-$ for all $\varphi \in \mathcal{L}$. Joins in this lattice are computed componentwise (pointwise suprema), which is exactly what makes the iterative construction below well-behaved.*

The closure $\mathrm{Cl}_{\mathcal{R}}(\beta_0)$ is obtained by iterating $F_{\mathcal{R}}$ starting from β_0 and taking the join of all iterates; monotonicity ensures this is an ascending chain. Equivalently, one may view $\mathrm{Cl}_{\mathcal{R}}(\beta_0)$ as the least fixed point of $F_{\mathcal{R}}$ above β_0*: it is the smallest valuation extending the initial evidence that is stable under all rule applications. The usual fixed-point theorem guarantees existence of such fixed points and characterizes the least one as the join of the iterates (or, more abstractly, as the join of all post-fixed points above β_0).*

If the value domain and language are finite, the ascending chain must stabilize in finitely many steps. More concretely, finiteness implies there are only finitely many distinct valuations reachable by repeatedly applying $F_{\mathcal{R}}$ starting from β_0, and since each step can only increase evidence in the product order, no valuation can repeat without having stabilized; hence convergence occurs after at most that finite number of strict increases. (When the domain is infinite, the same monotone-iteration construction still makes sense, but the stabilization may require a limit stage and is then captured by taking the supremum of the chain.) $\square$

The key step is that $\oplus$ and $\otimes$ are monotone componentwise on $[0,1]^2$; thus any expression built from them by composition and joins is monotone. Here "componentwise monotone" means that increasing either the positive coordinate or the negative coordinate of any input cannot decrease the corresponding coordinate of the output, so rule-aggregation never retracts previously accumulated support. Since the rule operator $F_{\mathcal{R}}$ is assembled from these primitives by applying them pointwise across $\mathcal{L}$ (and then taking joins over applicable rules), the order-preservation of $F_{\mathcal{R}}$ follows by straightforward structural induction on the way rule conclusions are computed.

Geometrically, one may imagine each claim $\varphi \in \mathcal{L}$ carrying a point in the unit square (the (β^+, β^-) plane), and an inference step moving these points "upward" in the product order (never decreasing either component). The pointwise nature of the order is important: different formulas may move at different rates, and closure is reached only when every *formula has become stable under further applications of the rules. In particular, the "upward" motion is not along a single axis but in a partial order, so there can be incomparable intermediate states; nevertheless, the iteration from β_0 produces a chain because each iterate is obtained from the previous one by a uniform application of the same monotone operator.*

Closure is then the least stable configuration under repeated application of these monotone transformations: a fixed point reached by taking the supremum of all successive improvements. This least fixed point represents the minimal amount of evidence that must be assigned, given β_0 and $\mathcal{R}$, to satisfy the closure condition "all rule conclusions are already accounted for." In this sense, $\mathrm{Cl}_{\mathcal{R}}(\beta_0)$ functions as a weighted analogue of deductive closure: it is not merely a set of consequences, but a saturation of the initial valuation by all rule-supported increases in (β^+, β^-).

Remark 1106 (Non-explosion by design)**.** *The scheme above does not include any special rule of the form "from φ and $\neg\varphi$ conclude anything." Thus inconsistency does not automatically produce arbitrary conclusions: a claim receives evidence only if there is a rule path that contributes to it. This is the minimal operational sense in which the framework is* paraconsistent*. Put differently, even if both φ and $\neg\varphi$ carry high positive evidence, this fact by itself does not act as a universal trigger; only explicitly provided rules can propagate information from those premises to further conclusions. One may still choose to include rules whose premises mention both φ and $\neg\varphi$ (for example, to model specific forms of diagnostic reasoning), but any such propagation is then local, controlled, and weight-sensitive rather than automatic and unbounded.*

Remark 1107. *This remark highlights a design choice: paraconsistency is not an add-on but a consequence of what is* omitted*. In particular, since inference is mediated by explicit weighted rules, contradictions can persist locally without turning the entire lattice of beliefs into maximal evidence everywhere. The absence of explosion means that the closure operator does not collapse all formulas to a top element merely because some formula has both kinds of support; instead, the fixed-point computation preserves the distinction between "having conflicting evidence about φ" and "having evidence for an unrelated ψ." This aligns with the paraconsistent motivations developed in [23, 24]. It also clarifies how inconsistency interacts with graded evidence: tension between β^+ and β^- for a given claim is represented explicitly in the valuation, and closure propagates only what the rule base $\mathcal{R}$ licenses, thereby preventing a single inconsistency from saturating the entire language unless the rules themselves encode such saturation.*

20.6 Question networks and the dynamics of thinking

Hyperseed allows question networks whose answers are further questions, potentially without a well-founded base. We model this as a directed graph (or category) of question dependencies. In this framing, the "shape" of an inquiry matters: two agents may share the same local inference rules yet differ radically in what they can reach because their networks present different routes for evidence to enter and propagate.

Definition 304 (Question network)**.** *A* question network *is a directed graph $\mathsf{Q} = (V, E)$ where nodes $v \in V$ are questions and an edge $v \to w$ means "answering v requires (or strongly benefits from) answering w." Cycles are permitted.*

It is useful to keep in mind that the edge label "requires (or strongly benefits from)" can cover multiple epistemic relations. For instance, $v \to w$ may mean (i) w supplies a missing variable needed to even state a candidate answer to v, (ii) w supplies calibration/validation needed to treat v's data as trustworthy, or (iii) w supplies a model-selection choice without which v is ill-posed. In practice, one can regard the graph as a compressed summary of a richer structure (e.g., typed questions, dependence modalities, or weighted edges expressing how strong the benefit is), but the directed graph already captures the core phenomenon: the nontrivial ordering constraints induced by epistemic dependency.

Remark 1108 (Intuition and examples)**.** *A question network is an explicit representation of the structure of inquiry. Edges express epistemic dependency: to answer one question, you may need answers to others (or at least to reduce them sufficiently). Permitting cycles is essential: real inquiry often involves mutually supporting questions, like "what counts as a reliable instrument?" depending on "what theory of measurement do we accept?" and vice versa. This is the formal shadow of a familiar philosophical fact: inquiry is not always well-founded on indubitable primitives; it is often stabilized by a coherent web.*

Example: in a research program, one node may be "estimate parameter θ," another "validate the measurement protocol," another "choose a model class," and dependencies may loop. In the Hyperseed core-concept set this corresponds directly to Question Networks (Hyperseed-Concept 145) and also to Dependency (Hyperseed-Concept 93).

A further perspective is graph-theoretic: the strongly connected components (SCCs) of Q identify clusters of questions that mutually depend on one another. In a well-founded tree-like inquiry, SCCs are trivial; in realistic inquiry, SCCs often correspond to methodological "packages" (measurement standards, modeling assumptions, interpretive norms) that are revised together. This helps separate two issues: (i) what is locally answerable given current methods, and (ii) what is globally stabilized by coherence constraints across a loop.

One may also read the parenthetical "(or category)" literally. If we interpret each edge $v \to w$ as a generating morphism $v \to w$, then composites correspond to indirect dependence paths: if answering v benefits from w and w benefits from u, then there is a composite dependence of v on u. On this view, reachability in the underlying graph expresses the transitive closure of "benefits from," and SCCs correspond to objects mutually connected by morphisms in both directions. Nothing in what follows requires category theory, but the categorical reading aligns with the idea that inquiry is a process of composing partial reductions rather than merely accumulating facts.

Definition 305 (Thinking as controlled closure)**.** *Fix a question network Q, a belief state β, and an inference rule set $\mathcal{R}$. A* thinking episode *is a finite sequence of operations of the form:*

(a) choose a question node v (possibly based on goals/values and attention);

(b) obtain data via the question's experiment channel, producing an evidence increment e_v;

(c) update beliefs (e.g. $\beta \leftarrow \beta \oplus e_v$) and propagate by closure $\beta \leftarrow \mathrm{Cl}_{\mathcal{R}}(\beta)$; and

(d) optionally entertain hypotheses to break cycles or explore counterfactual branches.

The finiteness of the episode is important as an idealization of bounded agency: even if $\mathrm{Cl}_{\mathcal{R}}$ is defined as a least fixed point of rule application, an actual agent typically approximates it under time/attention constraints, and the "controlled" part of the loop determines where approximation error accumulates. Thus, a thinking episode is not assumed to be complete inference, but rather a bounded sequence of actions that can be repeated and interleaved with further data collection.

It is also helpful to make explicit a minimal set of closure-operator intuitions that motivate $\mathrm{Cl}_{\mathcal{R}}$. In many logical and computational settings, closure operators are (at least approximately) extensive, monotone, and idempotent:

$$\beta \preceq \mathrm{Cl}_{\mathcal{R}}(\beta), \qquad \beta \preceq \beta' \Rightarrow \mathrm{Cl}_{\mathcal{R}}(\beta) \preceq \mathrm{Cl}_{\mathcal{R}}(\beta'), \qquad \mathrm{Cl}_{\mathcal{R}}(\mathrm{Cl}_{\mathcal{R}}(\beta)) = \mathrm{Cl}_{\mathcal{R}}(\beta),$$

where $\preceq$ is an information ordering ("contains no more commitments than"). Hyperseeds paraconsistent stance suggests that $\preceq$ should be read as an ordering of evidence or commitments that need not collapse under inconsistency, so "extensive" means "adds what the rules license" rather than "forces global consistency."

Remark 1109 (Intuition and connections)**.** *This definition casts thinking as a controlled alternation between* active inquiry *(choose and perform a question) and* passive propagation *(apply closure to distribute the consequences through the belief web). The "controlled" aspect is crucial: the system does not apply all questions, but selects them under constraints of attention and value (Sections 18 and 14), echoing the cognitive-synergy and resource-allocation motifs in [19].*

Conceptually: inquiry supplies new distinctions (new evidence increments e_v), and closure turns these into a revised global stance by pushing them through indirect beliefs (rules, trusted sources, schemas). Thus thinking is neither pure perception nor pure deduction; it is the iterative weaving together of questions and inference.

One can read step (a) as a *policy* over nodes: a mapping from the current epistemic/valuational state to a choice of which question to invest in next. This makes explicit why networks matter: two networks with the same set of nodes but different edge structure can lead the same policy to very different trajectories of belief change, because different dependencies alter which questions become bottlenecks. In scientific practice, this corresponds to the familiar dynamic where an entire research program can be stalled by a single unresolved methodological node (e.g., an instrument calibration or a missing dataset), even when many downstream theoretical questions are already articulated.

Step (b) also deserves a slightly richer reading: the "experiment channel" of a question need not be a physical experiment. It may be an observation procedure, a simulation, a database query, an expert consultation, or even an introspective probe, so long as it yields an evidence increment e_v that the agent treats as relevant to updating β. This makes the notion of question nodes compatible with both empirical and conceptual inquiry: some nodes are primarily settled by measurement, others by argument, and many by a mixture of both.

Remark 1110 (Bottoming out cyclic dependencies)**.** *If Q has cycles (non-well-founded "questions about questions"), then an agent can still operate by (i) choosing a cut set of hypotheses that temporarily resolves the cycle, (ii) exploring the induced consequences, and (iii) revising the hypotheses or the network itself. This is a natural reading of Hyperseed's remark that one can "force bottoming out" by entertaining a hypothesis.*

A useful way to see this operationally is to treat the chosen hypotheses as provisional "anchors" that permit local progress. For example, if "is the instrument reliable?" depends on "does the theory predict stable readings?" and that depends on "are the readings trustworthy?", then one can temporarily posit a calibration prior or a simplified theory to obtain provisional predictions, then use discrepancies to revise the anchors. In other words, the cycle is not eliminated in principle; rather, the agent chooses a temporary direction of fit to obtain traction.

Remark 1111. *The "cut set" intuition is graph-theoretic: one chooses a set of nodes whose provisional assignment breaks the cycle, analogous to choosing boundary conditions for a differential* *equation. The paraconsistent mechanism makes this psychologically plausible: entertaining a hypothesis need not annihilate contrary evidence; it simply injects additional evidence that can later be retracted (or counterbalanced) without global collapse.*

In graph terms, the cut set can be compared to a feedback vertex set: removing (or temporarily "pinning") those nodes makes the remaining dependency structure acyclic, enabling a staged evaluation from upstream to downstream. In epistemic terms, the pinned nodes function like methodological stipulations or default assumptions, which can later be relaxed. This also clarifies why paraconsistency is not merely a technical convenience: if hypotheses can be introduced and later revised without forcing explosive inference, then the agent can explore multiple competing "boundary conditions" in parallel, retaining incompatible lines of investigation long enough to see which ones pay off.

Finally, the possibility of revising "the network itself" is not an afterthought: inquiry often changes what counts as a question, what counts as an experiment channel, and what counts as an acceptable dependency. For instance, learning that a measurement process is systematically biased can introduce a new node ("estimate bias parameters") and rewire edges so that previously "downstream" theoretical nodes now depend on bias correction. Thus, the dynamics of thinking are not only dynamics on β but also dynamics on Q and on the effective rule set $\mathcal{R}$ that governs closure.

20.7 Belief systems and productive belief systems

Hyperseed defines a belief system as a coherent network of beliefs held within a community of minds, with a characteristic "mutual implication" property. We capture this by combining (i) community aggregation, (ii) inference closure, and (iii) a complexity-sensitive notion of "surprising strength." In this subsection, "belief" is treated as a graded semantic attitude toward sentences of a shared language $\mathcal{L}$, and a "system" is not merely a set but a pattern of mutual evidential support as mediated by an explicit rule set $\mathcal{R}$.

Definition 306 (Community belief aggregation)**.** *Let M be a finite set of agents, each with a belief state $\beta_i : \mathcal{L} \to \mathbb{V}$. Define the* community aggregate *belief state by*

$$\beta_{\mathsf{M}}(\varphi) := \bigoplus_{i \in \mathsf{M}} w_i \otimes \beta_i(\varphi),$$

where weights $w_i \in \mathbb{V}$ model trust/authority/attention and the operations are those of the p-bit *quantale. In particular, $\otimes$ should be read as "evidence scaling" (how much an agent's attitude counts after weighting), while $\bigoplus$ is the quantale join that pools the contributions into a single community-level valuation.*

Remark 1112 (Intuition and examples)**.** *This is a minimal formalization of intersubjective knowledge: we take individual belief states and combine them into a community stance by (i) scaling each agents contribution by a weight w_i and (ii) joining the results. If w_i is high (in positive evidence), agent i has more epistemic influence; if w_i has a negative component, one can even represent distrust or systematic adversariality in a graded way.*

Example: in a lab group, a senior experimentalist might have higher weight on measurement-related claims, while a theoretician might have higher weight on model-selection rules. The usefulness of this definition is that it makes social epistemology compositional: a community is not a mysterious supra-mind but an aggregation operator on individual minds' evidence valuations (Hyperseed-Concept **??** *and Hyperseed-Concept 170).*

Two further points are worth making explicit. First, the aggregation is performed pointwise in φ: the community may defer to different agents on different claims, depending on how $\beta_i(\varphi)$ interacts with w_i under $\otimes$. Second, the weights w_i need not be normalized in the probabilistic sense; rather, their meaning is fixed by the algebra of $\mathbb{V}$ and by the intended operational semantics (e.g. "whose judgment is heeded" vs. "how much independent evidence is added"). This leaves room for modeling phenomena such as epistemic division of labor, institutional gatekeeping, and domains of recognized expertise within a single uniform operator.

Definition 307 (Coherence score)**.** *Fix β and a rule set $\mathcal{R}$. For $\varphi \in \mathcal{L}$ define the "leave-one-out" state*

$$\beta^{\setminus\varphi} := \text{the state equal to } \beta \text{ except with the value at } \varphi \text{ replaced by } (0,0).$$

Define the coherence score *of B (relative to β and $\mathcal{R}$) as*

$$\mathrm{Coh}_{\mathcal{R}}(B;\beta) := \frac{1}{|B|}\sum_{\varphi\in B}\pi\big(\mathrm{Cl}_{\mathcal{R}}(\beta^{\setminus\varphi})(\varphi)\big),$$

where $\pi(p,q) = (1+p-q)/2$ is the plausibility projection (cf. Section 14). When B is understood from context, $\mathrm{Coh}_{\mathcal{R}}(B;\beta)$ can be read as a single-number summary of how "self-explanatory" the set B is under the inferential practices encoded by $\mathcal{R}$.

Remark 1113 (Notation unpacking and intuition)**.** *The leave-one-out state $\beta^{\setminus\varphi}$ deletes the community's direct evidence for φ by setting it to $(0,0)$ while keeping all other claims unchanged. Then $\mathrm{Cl}_{\mathcal{R}}(\beta^{\setminus\varphi})(\varphi)$ asks:* how much support for φ can be reconstructed from the rest of the belief web, via inference rules? *Finally, $\pi(p,q) = (1+p-q)/2$ collapses a* p-bit *into a single plausibility scalar, increasing with positive evidence p and decreasing with negative evidence q.*

Thus $\mathrm{Coh}_{\mathcal{R}}(B;\beta)$ measures, on average over $\varphi \in B$, how well each belief is supported by the others*. This makes "mutual implication" precise without demanding classical consistency: coherence is about reconstructability under closure, not about absence of conflict.*

Operationally, this is a counterfactual test: if we "blank out" the community's explicit stance on φ while leaving its other stances intact, does the remaining web (together with $\mathcal{R}$) recover φ anyway? A belief set with high coherence is therefore one in which many beliefs are not epistemically isolated; they are downstream of shared principles, background assumptions, or explanatory constraints captured by the closure operator. Conversely, if many $\varphi \in B$ cannot be reconstructed once removed, then B behaves more like a mere list of stipulations than an integrated system.

Note that averaging over $\varphi \in B$ makes coherence sensitive to the "typical" member of the set, rather than to a single highly implied cornerstone claim; one can also consider variants such as a minimum-over-φ (worst-case) coherence or a distributional profile of the individual summands, but the mean is the simplest global proxy for networked support.

Remark 1114. Coh *is high when each belief in B is well supported by the others via the community's inference rules. This is a formalization of "on average, each belief is strongly implied by the other beliefs." Because we use $\pi \circ \mathrm{Cl}$, coherence is allowed to coexist with contradiction: a belief may be well supported and also well opposed.*

It is also worth emphasizing what coherence is not*. A high score does not imply that B is true, empirically adequate, or even stable under future evidence; it only indicates that, relative to $\mathcal{R}$ and β, the beliefs in B form an inferentially entangled cluster. This is one reason coherence is useful in a pluralistic setting: competing communities can each exhibit high internal coherence while disagreeing, and the framework can then ask additional questions (e.g. about predictive performance, causal control, or openness to revision) to distinguish productive belief systems from merely self-reinforcing ones.*

Definition 308 (Complexity cost and "surprise-adjusted" implication)**.** *Let* $\mathrm{Comp} : \mathcal{L} \to [0, \infty)$ *be a complexity cost (e.g. description length, or an order reversal of weakness). Fix* $\lambda > 0$ *and define a simplicity-biased prior weight*

$$\mathrm{Prior}(\varphi) := e^{-\lambda\,\mathrm{Comp}(\varphi)}.$$

For $\varphi \in \mathcal{L}$ *define the* surprise-adjusted implication score

$$\mathrm{SAI}(\varphi; \beta, \mathcal{R}) := \frac{\pi\big(\mathrm{Cl}_{\mathcal{R}}(\beta^{\setminus\varphi})(\varphi)\big)}{\mathrm{Prior}(\varphi)}.$$

Remark 1115 (Intuition, and relation to simplicity/weakness)**.** *The complexity cost* $\mathrm{Comp}(\varphi)$ *quantifies how "expensive" a claim is to represent or to search for: it could be a description length in the sense of algorithmic information theory [16], or a transformation of weakness/effort as developed earlier [3]. The parameter* $\lambda > 0$ *controls how strongly the community prefers simplicity.*

The score SAI *then formalizes a familiar epistemic phenomenon: a complex statement, if strongly implied by the rest of what we accept, is more noteworthy than a simple one. Dividing by* $\mathrm{Prior}(\varphi)$ *ensures that a claim with low prior weight must be especially well supported by the network to achieve a high* SAI*. In Hyperseed-concept terms, this is the interaction of Simplicity (Hyperseed-Concept 169), Weakness (Hyperseed-Concept 202), and Uncertainty (Hyperseed-Concept 195).*

Interpreting SAI *as a "productivity"-leaning signal is natural: the numerator rewards inferential integration (via closure), while the denominator penalizes gratuitous complexity. Thus, when two claims are equally implied by the rest of the system, the one with smaller* $\mathrm{Comp}(\varphi)$ *is treated as less epistemically surprising; conversely, a high-complexity claim must earn its keep by being tightly constrained by the rest of the web. The exponential form of* Prior *is convenient because it turns additive complexity into multiplicative penalties; however, the framework is compatible with other monotone decreasing mappings from complexity to prior weight if a different calibration is desired.*

Remark 1116 (Matching the Hyperseed phrasing)**.** *Hyperseed describes a belief as something "held true to a degree that is surprising given the complexity of the belief." The prior* $\mathrm{Prior}(\varphi)$ *makes complex claims a-priori unlikely. A large* SAI *means the community implies* φ *strongly despite its low prior.*

Equivalently, SAI *distinguishes between "easy" implications (those that follow from many simple, high-prior patterns) and "hard" implications (those that emerge as nontrivial consequences of the system). In this sense,* SAI *operationalizes the idea that an integrated belief system is not merely a heap of assertions: it can generate relatively unexpected consequences, and those consequences can be ranked by how much inferential pressure the rest of the system exerts on them relative to their complexity.*

Definition 309 (Belief system and productive belief system)**.** *A* belief system *(for a community* M*) is a pair* $(B, \mathcal{R})$ *where* $B \subseteq \mathcal{L}$ *is a focal belief set and* $\mathcal{R}$ *is a rule set, such that* $\mathrm{Coh}_{\mathcal{R}}(B; \beta_{\mathsf{M}})$ *exceeds a chosen threshold. Here "focal" is meant to emphasize that* B *is not the entire doxastic state of* M*, but a distinguished subfamily of commitments that are tracked as a unit (e.g. a theory, paradigm, doctrine, platform, or research program) and whose internal relations are evaluated using* $\mathcal{R}$*. Likewise,* $\mathcal{R}$ *is not only a list of inference schemata, but can also encode allowed explanatory moves, defeasible reasoning patterns, methodological norms, and admissible forms of updating; it is part of what makes* B *into a system rather than a mere set. The threshold requirement is intended to exclude arbitrarily assembled collections of propositions: a community may hold many beliefs, but only some coherent sub-webs are stable enough to function as objects of collective inquiry, pedagogy, and transmission.*

A belief system is productive *if, in addition:*

(a) (Individuation) there exists a "core" subset $B_{\text{core}} \subseteq B$ *that remains stable under typical evidence updates (high persistence of key commitments), and*

*(b) (Self-transcendence) the system regularly generates new high-*SAI *beliefs or new questions whose answers lead to novel patterns, without collapsing coherence.*

The two additional clauses should be read as constraints on the dynamics *of the pair* $(B, \mathcal{R})$ *as the community encounters new data, arguments, and social pressures. In particular, "typical evidence updates" can be understood as those updates that occur with non-negligible frequency in the epistemic environment of* M *(the ordinary course of observation, experiment, testimony, and criticism); the individuation requirement then says that some subset functions as a robust identity condition for the system across such ordinary perturbations. Similarly, the "without collapsing coherence" proviso rules out a degenerate kind of novelty in which the system generates ever more claims only by loosening* $\mathcal{R}$ *or fragmenting* B *so that* $\mathrm{Coh}_{\mathcal{R}}(B; \beta_{\mathsf{M}})$ *no longer remains above threshold.*

Remark 1117 (Intuition and why "productive" needs two constraints)**.** *A belief system is, on this definition, a coherent epistemic sub-web in a community (Hyperseed-Concept 64). But a merely coherent web can be sterile: it might resist novelty so strongly that it never learns, or it might be so plastic that it dissolves under every perturbation. The definition of "productive" therefore combines two tensions that recur throughout Hyperseed:* individuation *(a stable identity across time) and* self-transcendence *(ongoing creation of new structure).*

The individuation clause says there is a persistent core (Hyperseed-Concept **??***) that survives ordinary evidence fluctuations. One way to think of* B_{core} *is as the subset of commitments that play a high-"load-bearing" role in the community's reasoning: they constrain many other beliefs, are invoked across many contexts, and are not easily abandoned without a substantial reconfiguration of* B*. This does not require infallibility; it requires that, at the time-scale of ordinary inquiry, the system has enough internal continuity to accumulate results, train newcomers, and coordinate collective action. Conversely, if* no *such* B_{core} *exists, then the "system" behaves more like a shifting coalition of claims than a persisting object, and it becomes difficult to interpret its apparent successes (or failures) as attributable to anything stable.*

The self-transcendence clause says inquiry continues to generate surprising, complexity-defying implications (Hyperseed-Concept 167) and new Questions (Hyperseed-Concept 144) leading to Emergence (Hyperseed-Concept **??***). Here "generates" is meant in the constructive sense: the system, by applying* $\mathcal{R}$ *to* B *and to incoming evidence, yields new commitments, conjectures, models, predictions, or problem-statements that were not trivially present at the outset. The reference to high-*SAI *is intended to separate mere proliferation of claims from epistemically meaningful novelty: a productive system does not simply add ad hoc patches, but tends to add beliefs (or questions) that integrate with and reorganize the web in a way that increases understanding, explanatory reach, or compressive power while still maintaining coherence. In philosophical terms, one might hear an echo of process views in which stability is a maintained pattern rather than a static substance [15]. In the intended application to science, the pair of constraints approximates a familiar phenomenon: research traditions preserve enough continuity to be identifiable across time, yet are also compelled to generate new results, new measurement regimes, and new conceptual distinctions in response to anomalies and opportunities.*

A useful way to see why both constraints are needed is to consider two failure modes. If a system has individuation without self-transcendence, it may become a well-defended "museum" of doctrine: highly coherent and highly stable, but systematically unresponsive to the kinds of questions that would extend or revise it. If a system has self-transcendence without individuation, it may behave like a perpetual brainstorming session: many novel outputs, but little cumulative structure, making *it hard to distinguish progress from drift. The productive regime is the narrow corridor in which novelty is* assimilated *rather than merely* accumulated *or* repelled*.*

Remark 1118. *The individuation/self-transcendence pair mirrors Hyperseed's use of those terms in other contexts. The present definition is intentionally schematic; later sections can specialize "typical evidence" and "regularly generates" to concrete learning dynamics and novelty measures. In particular, one can operationalize "typical evidence" by specifying a distribution over evidence-events relevant to* M *(experiments, observations, counterarguments, replications, testimony streams), and then measuring persistence of candidate cores under the corresponding update rule. Likewise, "regularly generates" can be rendered as a rate condition (e.g. expected production of high-*SAI *outputs per unit time, per unit research effort, or per cycle of deliberation), together with a constraint that* $\mathrm{Coh}_{\mathcal{R}}(B;\beta_{\mathsf{M}})$ *does not fall below threshold after incorporation. On this reading, productivity becomes a property that can in principle be compared across communities or across historical phases of a single community: the same belief system may be productive during one era (when it opens fertile question-spaces) and less productive in another (when it mainly stabilizes and defends its inherited core).*

It is also worth noting that the definition does not presuppose that productive systems are always "true" in any final sense; rather, it characterizes a kind of epistemic fitness *for generating structured, coherent novelty. This allows the framework to distinguish (i) coherent but stagnant systems, (ii) inventive but unstable systems, and (iii) systems that exhibit cumulative inquiry. In later applications, one can ask which additional constraints (e.g. calibration to external feedback, robustness across subcommunities, or sensitivity to adversarial testing) connect productivity more tightly to truth-tracking or scientific reliability.*

20.8 Science as empirically coupled belief-system dynamics

Hyperseed defines science as a community belief system that generates predictive and causal models about observation sets accepted broadly by members of the community. We model this as a special case of the thinking loop where questions are experiments and answers are shared observations. This framing makes explicit that "scientific knowledge" is not a single agent's internal state but a coupled multi-agent process: the community stabilizes a repertoire of experimental questions, a shared interface for what counts as an answer, and a rule-governed method for turning those answers into durable belief updates. In particular, the "predictive and causal" emphasis is meant to distinguish scientific models from purely interpretive narratives: the central outputs of the belief system are claims that constrain what should be observed under specified conditions, and (when available) how observations would change under interventions.

Definition 310 (Observation set)**.** *Fix a community* M *and a context class of experiments* E*. An* observation set *is a set* O *together with an observation map*

$$\mathsf{Obs} : E \to \mathsf{O},$$

possibly stochastic and observer-indexed. An observation set is accepted broadly *by* M *if, for a large fraction of agents and a large fraction of eligible experiments, the induced observations are mutually consistent up to the community's tolerance thresholds. The phrase "context class of experiments" is intended to include not only the abstract experimental design but also the admissible range of protocols, instrument configurations, and background conditions that the community treats as "the same kind" of experiment for purposes of comparison. Likewise, "tolerance thresholds" should be read as including both quantitative tolerances (e.g., error bars, confidence intervals, calibration* *margins) and qualitative tolerances (e.g., coding reliability or procedural compliance in observational sciences).*

Remark 1119 (Intuition and examples). *An observation set is a formalization of what a community can "jointly see". The map* $\mathsf{Obs} : E \to \mathsf{O}$ *abstracts the full measurement pipeline into an input-output channel: an experiment* $e \in E$ *yields an observation in* O*, perhaps with stochasticity due to noise, differing instruments, or contextual effects. Broad acceptance is not metaphysical certainty; it is a social-epistemic stability condition: most relevant observers running the relevant protocols obtain compatible outcomes. Put differently,* Obs *is not assumed to be infallible, but it is assumed to be sufficiently well-behaved (under the community's standards) that disagreement is attributable to identifiable sources such as sampling variability, instrument drift, experimenter degrees of freedom, or genuine context sensitivity, rather than to arbitrary observer idiosyncrasy. The "observer-indexed" allowance captures cases where the observation pipeline includes human judgment (e.g., labeling, clinical assessment, ethnographic interpretation) while still permitting intersubjective convergence through training, rubrics, and reliability checks.*

Example: in physics, E *could be a class of experimental setups and* O *numerical readouts. In psychology,* O *might be survey responses or behavioral measures. In biology,* O *could be sequencing reads or microscopy counts; in astronomy,* O *could be processed spectra; in economics,* O *could be aggregated indicators computed from raw data. This makes explicit the dependence of "empirical truth" on intersubjective agreement and protocol, linking to Logical/Empirical Truth (Hyperseed-Concept 104) and Intersubjective Reality (Hyperseed-Concept* **??***), and aligning with social-computational accounts of science [20]. The point is not to reduce truth to consensus, but to specify the operational substrate by which a community can treat some claims as empirically constrained: without broadly accepted observation sets, "evidence" cannot enter the communal belief dynamics in a stable, reproducible way.*

Definition 311 (Scientific model class and predictive claims). *A* scientific model class *is a set* $\mathcal{M}$ *of candidate models. Each model* $m \in \mathcal{M}$ *induces:*

- *a family of predictive claims* $\mathrm{Pred}_m(e)$ *about observations* $\mathsf{Obs}(e)$*, and*
- *optionally a family of causal claims* Cause_m *about how interventions change predictions.*

The community treats these as elements of the claim language $\mathcal{L}$*. Here "predictive claims" should be read broadly to include point predictions, distributional predictions (e.g., likelihoods over* O*), conditional predictions given auxiliary variables, and robust qualitative constraints (e.g., monotonicity, invariances, sign predictions). Similarly, "causal claims" range from fully specified structural equations to weaker claims about directionality or invariance under a class of interventions; the definition leaves room for multiple grades of causal expressiveness within* $\mathcal{L}$*.*

Remark 1120 (Intuition and connection to earlier machinery). *A model class* $\mathcal{M}$ *is the space of representational hypotheses the community is willing to consider. The induced claims* $\mathrm{Pred}_m(e)$ *and* Cause_m *are then just ordinary claims in* $\mathcal{L}$*, so they can receive* p-bit *evidence and participate in closure with the same rule machinery as any other belief. This makes model comparison and theory choice instances of general belief management: models do not sit outside the belief state but are represented through the claims they entail and the support they accumulate via the update mechanisms. In particular, competing models may share some induced claims while differing on others; closure then propagates evidence to all logically or methodologically connected claims, which is how localized experimental outcomes can have global effects on a theory network.*

This is important structurally: science is not a separate kind of reasoning; it is a particular configuration of question/answer dynamics where questions are experiments and answers are shared *observations. Causal claims connect directly to Section 14's intervention viewpoint, and the pluralistic model-class stance aligns with Hyperseed's broader engineering orientation toward families of methods rather than single privileged formalisms [19]. The explicit model-class framing also clarifies why scientific communities invest in tool-building: new instruments, new statistical methods, and new representational languages expand E, enrich O, and enlarge (or restructure) $\mathcal{M}$, thereby changing what questions can be asked and what answers can count as decisive.*

Definition 312 (Empirical update operator)**.** *Let $e \in E$ be an experiment and $o = \mathsf{Obs}(e)$ its (shared) observation. An* empirical evidence increment *is a map $\epsilon_{e,o} : \mathcal{L} \to \mathbb{V}$ that assigns positive/negative evidence to those predictive/causal claims that agree or disagree with (e, o). A scientific update step is*

$$\beta \leftarrow \mathrm{Cl}_{\mathcal{R}}\big(\beta \oplus \epsilon_{e,o}\big),$$

where $\mathcal{R}$ includes both domain rules and methodological rules (statistics, measurement models, etc.). The definition is intentionally agnostic about the internal structure of $\mathbb{V}$ and the operator $\oplus$: different scientific subcultures can implement "evidence addition" by log-likelihood accumulation, score-based updates, hypothesis-test-like penalties, or other evidential bookkeeping, so long as the result can be represented as a belief-state update followed by rule closure. The phrase "agree or disagree" should be interpreted through $\mathcal{R}$: agreement may mean a high probability assigned to o under $\mathrm{Pred}_m(e)$, or satisfaction of a qualitative constraint, or being within a prespecified tolerance band after correction for known measurement error.

Remark 1121 (Intuition: why this captures "empirical coupling")**.** *Empirical coupling is represented by the fact that observations enter as an evidence increment $\epsilon_{e,o}$, which is then integrated into the belief state and propagated by closure. The methodological rule set $\mathcal{R}$ is where the community encodes its indirect beliefs about how to translate raw observations into support for claims (measurement error models, statistical tests, calibration conventions, etc.). Thus the empirical loop is: act (choose e), observe (o), translate to evidence, then infer. This highlights that "data" are not self-interpreting: the same (e, o) can produce different $\epsilon_{e,o}$ under different methodological commitments (e.g., different priors, different preprocessing pipelines, different correction procedures), and scientific disagreement can therefore be located either in the observation interface* Obs *or in the rule system $\mathcal{R}$.*

In Hyperseed terms, this is a controlled flow from Answers (Hyperseed-Concept 56) into Beliefs (Hyperseed-Concept 65), guided by Indirect Beliefs and stabilized socially as a Belief System (Hyperseed-Concept 64). The "controlled" aspect is precisely what distinguishes scientific coupling from mere exposure to experience: the community engineers reproducible questions (standardized experiments), shareable answers (broadly accepted observation sets), and disciplined update rules (methodological norms) so that belief change is not arbitrarily sensitive to individual perspective.

Proposition 32 (Occam bias from simplicity-weighted priors)**.** *Fix $\lambda > 0$ and a complexity cost* $\mathrm{Comp} : \mathcal{M} \to [0, \infty)$*. Define* $\mathrm{Prior}(m) = e^{-\lambda\,\mathrm{Comp}(m)}$*. Suppose two models $m_1, m_2 \in \mathcal{M}$ receive equal empirical support in the sense that the community assigns equal plausibility to their predictive adequacy. If* $\mathrm{Comp}(m_1) < \mathrm{Comp}(m_2)$ *then* $\mathrm{Prior}(m_1) > \mathrm{Prior}(m_2)$*, so any posterior weight proportional to (support) $\times$ (prior) will prefer m_1.*

Remark 1122 (What the proposition is saying and why it matters)**.** *The proposition states a minimal Occam principle: if two models fit the data equally well, the simplicity-biased prior assigns higher weight to the simpler model, so any Bayesian-like update that multiplies support by prior will prefer it. Nothing subtle is being claimed about the world; the point is that* a preference for simplicity is already present in the choice of prior*. In practice,* Comp *can proxy for many different* *simplicity notions: description length, number of parameters, circuit size, degrees of freedom, or norm-based regularity; different choices encode different methodological values, and communities can (and do) argue about which complexity measure best captures "simplicity" for their domain. The proposition also clarifies a common interpretive confusion: the Occam effect here is not an extra rule tacked onto inference after seeing data, but a consequence of committing (in advance) to a simplicity-penalizing representation of model plausibility. From the belief-system viewpoint, such priors are part of the community's methodological rule set: they are indirect beliefs about which hypotheses are worth taking seriously before evidence arrives, and they influence how quickly a community converges or fragments when empirical support is ambiguous.*

Remark 1123. *This connects directly to the earlier weakness preference: if* Comp *is taken as a form of description length or weakness-derived effort, then scientific model selection becomes a special case of the general Hyperseed bias toward lower effort representations [3, 2]. It also resonates with classic algorithmic justifications of Occam-style priors [16]. In particular, when* $\mathrm{Comp}(m)$ *is identified (up to an additive constant) with a code length for describing* m*, the weight* $e^{-\lambda \mathrm{Comp}(m)}$ *can be read as a softness-controlled analogue of assigning higher prior mass to shorter descriptions;* λ *then calibrates how aggressively the agent trades empirical fit against representational economy.*

Proof. Immediate from monotonicity of the exponential: $\mathrm{Comp}(m_1) < \mathrm{Comp}(m_2)$ implies $-\lambda\,\mathrm{Comp}(m_1) > -\lambda\,\mathrm{Comp}(m_2)$, hence $e^{-\lambda\,\mathrm{Comp}(m_1)} > e^{-\lambda\,\mathrm{Comp}(m_2)}$. Equivalently, taking logs shows $\log w(m) = -\lambda\,\mathrm{Comp}(m)$ is strictly increasing as $\mathrm{Comp}(m)$ decreases, so the ordering of weights is exactly the reverse ordering of complexities. □

Remark 1124 (Proof sketch and intuition)**.** Proof sketch. *The proof is a one-line order argument: since* $x \mapsto e^{-\lambda x}$ *is strictly decreasing for* $\lambda > 0$*, smaller complexity yields larger prior weight. Then equal empirical support implies the posterior preference is determined by the prior. More explicitly, if two models* m_1, m_2 *have the same likelihood for the observed data* D*, i.e.* $p(D \mid m_1) = p(D \mid m_2)$*, then Bayes' rule gives*

$$\frac{p(m_1 \mid D)}{p(m_2 \mid D)} = \frac{p(D \mid m_1)}{p(D \mid m_2)} \cdot \frac{p(m_1)}{p(m_2)} = \frac{p(m_1)}{p(m_2)},$$

so any posterior preference comes solely from the prior ratio $p(m_1)/p(m_2) = e^{-\lambda(\mathrm{Comp}(m_1)-\mathrm{Comp}(m_2))}$*.*
□

Geometrically, one may view $\mathrm{Comp}(m)$ *as a coordinate measuring where a model lies along a "simplicity axis." The prior* $e^{-\lambda \mathrm{Comp}(m)}$ *is then a decaying density along that axis; the proposition just states that this density is higher at smaller coordinates. One can also view* λ *as setting the "length scale" along this axis: for large* λ *the density drops sharply (strong simplicity pressure), whereas for small* λ *it decays slowly (weak simplicity pressure), so that empirical likelihood differences must be correspondingly larger to override the simplicity bias.*

Remark 1125 (Hyperseed reading)**.** *This is the epistemic analogue of weakness preference: among models consistent with observations, prefer the weaker/simpler one. When combined with paraconsistent evidence handling, "anomalies" do not force immediate abandonment; they become negative evidence that can coexist with positive evidence until a better model emerges. In this reading, the exponential prior plays the role of a continuously valued "default conservatism": it rewards models that require fewer special-case commitments, fewer auxiliary hypotheses, or less representational machinery to maintain internal coherence with what has already been learned. Thus, scientific change is not modeled as a brittle, all-or-nothing refutation step, but as a gradual reweighting in which anomalies reduce support without necessarily driving the posterior to zero in the presence of coexisting, partially conflicting evidential constraints.*

Remark 1126 (Additional clarification: coupling to empirical adequacy)**.** *The above preference statement is most informative when made relative to empirical coupling. In standard Bayesian terms, model selection compares posteriors $p(m \mid D) \propto p(D \mid m)\, p(m)$, so that the simplicity term $p(m) \propto e^{-\lambda \mathrm{Comp}(m)}$ acts as a regularizer rather than a substitute for data fit. Accordingly, if m_2 fits the data substantially better than m_1, then a higher* $\mathrm{Comp}(m_2)$ *can be compensated by a sufficiently larger likelihood $p(D \mid m_2)$; the present proposition isolates the limiting case where the likelihoods are equal (or, more generally, when the Bayes factor is near* 1*), so that the complexity penalty becomes the deciding factor. On the "belief-system dynamics" interpretation, this corresponds to cases where multiple explanatory stories accommodate the same observational constraints, and the update rule resolves the tie by favoring the representation that is cheaper to maintain, communicate, and integrate with the agent's existing web of commitments.*

20.9 State-dependent science

Hyperseed proposes that some observation sets are accessible (and intersubjectively stabilizable) only when the community is in particular states of consciousness. We model this by indexing the observation channel by a state parameter. Concretely, the role of the state parameter is to treat "the observer" (and the social/ritual/technical context that sustains the observer) as part of the experimental apparatus: changing the sustained cognitive–affective–attentional configuration can change which distinctions can be enacted, and hence which experimental questions are even well-posed. This is not merely an appeal to private experience; it is an attempt to express, in the same formal slot as an instrument model, the idea that observer-training and observer-context can be constitutive of measurement.

Definition 313 (State-indexed observation)**.** *Let* S *be a set of community-accessible states (meditative, "I–Thou", oceanic, etc.). A* state-indexed observation set *consists of* O *and maps*

$$\mathsf{Obs}_s : E_s \to \mathsf{O}, \qquad s \in \mathsf{S},$$

where E_s is the set of experiments that are performable/meaningful in state s.

In this definition, E_s plays a logically important role: it allows that the relevant difference between states is not only that they yield different outcomes for the *same* experimental description, but that some experimental descriptions may fail to be executable, intelligible, or stablely interpretable outside a given state. Equivalently, the state index can change both the "domain" (what can be asked) and the "channel" (how answers are produced). One can also package the same idea as a single partial map $\mathsf{Obs} : \bigcup_{s \in \mathsf{S}} (\{s\} \times E_s) \to \mathsf{O}$ with $\mathsf{Obs}(s, e) = \mathsf{Obs}_s(e)$; the present notation simply emphasizes that the observation channel itself is conditioned on state.

Remark 1127 (Intuition and conceptual placement)**.** *The new ingredient is the index $s \in \mathsf{S}$: what counts as an admissible experiment E_s and what observations result via Obs_s may depend on the cognitive/affective/attentional state of the agents performing the protocol. This operationalizes the idea that certain distinctions may be* state-gated*: one must become a certain kind of observer to stably enact certain observational couplings.*

Examples range from mundane to speculative: altered attentional regimes can change perceptual discrimination thresholds; disciplined contemplative practices may yield reproducible reports within a trained community; more conjectural proposals (e.g. morphic resonance) would also, if made protocol-stable, fit this formal slot [13]. In the Hyperseed core-concept set, this subsection touches States of Consciousness (Hyperseed-Concept 178) and Oceanic Consciousness (Hyperseed-Concept 123), and it is consonant with state/location taxonomies for individuals and collectives [4, 12].

A practical reading is that the "state" variable is itself something a community can (to some degree) prepare, verify, and maintain. In the same way that many laboratory instruments require calibration and controlled environmental conditions, a state-dependent protocol may require training regimes, selection criteria, social norms, and environmental supports that keep the community within an operational envelope corresponding to s. This suggests separating three layers: (i) a *state-preparation* procedure (how s is reached), (ii) a *state-verification* procedure (how membership in s is checked or at least bounded), and (iii) the *in-state* experimental protocol $e \in E_s$ (what is done once in s). The formalism above directly models (iii) and implicitly absorbs (i)–(ii) into the background conditions defining the domain E_s and the effective channel Obs_s.

Definition 314 (State-dependent science)**.** *A* state-dependent science *is a family of belief-system dynamics* $(B_s, \mathcal{R}_s, \mathsf{Obs}_s)$ *indexed by* $s \in \mathsf{S}$*, such that within each state:*

- *observations are broadly accepted by the in-state community;*
- *predictive/causal models are updated by the empirical loop of Section 20.8; and*
- *the resulting beliefs are stable under small perturbations of protocols within that state.*

The phrase "belief-system dynamics" here is intended to be read in the same sense as earlier sections: B_s is the in-state belief-state (or set of admissible belief-states), $\mathcal{R}_s$ is the in-state revision/update dynamic, and Obs_s is the in-state observation channel. Nothing in the definition requires that the beliefs be *true* in a metaphysical sense; the requirement is that they be generated and maintained by an empirically coupled, self-correcting loop relative to the state-conditioned access channel. In particular, one can allow that $\mathcal{R}_s$ includes explicit procedures for handling disagreement, uncertainty, and inter-observer variability, provided these procedures themselves are part of the stable in-state practice.

Remark 1128 (Intuition and why this remains "science" in the formal sense)**.** *The definition says: for each s, there is a coherent, empirically coupled belief-system dynamic. Thus state-dependent science is not defined by abandoning empirical discipline, but by acknowledging a further contextual variable that affects what can be observed and how. From a process perspective, the observer is not an inert point but a temporally extended pattern of capacities; changing that pattern changes the observation channel [15].*

The stability-under-perturbation clause is the analogue of experimental robustness: within a given state s, small variations of protocol should not wildly change the observation map. This is what blocks the notion from degenerating into "anything goes" subjectivism: reproducibility is enforced, but only relative to the state-indexed context.

To make the robustness clause more concrete, one may think of perturbations as spanning (at least) (i) "technical" perturbations (minor differences in timing, phrasing, apparatus configuration), (ii) "social" perturbations (minor differences in group composition or facilitation), and (iii) "state" perturbations that remain within the tolerance region of s (e.g. small fluctuations in attention, affect, or absorption that do not constitute leaving s). The intent is not to demand zero variability, but to demand that the in-state observation channel have a stable core: outcomes cluster, calibration is possible, and error models can be learned. This is also the point at which standard concerns about expectancy effects and demand characteristics must be treated as first-class methodological issues: if such effects dominate, then the purported Obs_s is not stable in the required sense, or else the channel being measured is actually "suggestibility under protocol" rather than the target phenomenon.

Remark 1129 (Why this is not automatically "unscientific"'). *The formal difference from ordinary science is not the absence of empirical coupling, but the presence of an additional context variable* s*. If a community can reliably enter a state* s *and reproduce observation protocols within* s*, then* $(\mathsf{Obs}_s, \mathcal{R}_s)$ *can support a genuine (state-indexed) empirical discipline. Cross-state translation and public accessibility become separate questions, not prerequisites.*

Public accessibility can then be treated as a spectrum rather than a binary. Some states may be widely reachable with modest training, making the resulting discipline closer to ordinary science in sociological character; other states may require long training horizons or unusual life conditions, making the resulting discipline closer to an expert craft (but still potentially empirical within that expert community). In either case, the formalism invites a "meta-scientific" layer: communities can study which state-preparation procedures are reliable, what the failure modes are, how long it takes to stabilize observers, and how intersubjective agreement scales with training. In that sense, even when s is hard to reach, claims about s can remain tethered to evidence about reachability, reliability, and robustness.

Remark 1130 (Normative feedback). *Hyperseed also suggests a feedback loop: if a state-indexed science pertains to states of extraordinary well-being, then the discipline can serve as a community-level control policy that helps shift the community toward those states. Within our formalism this is naturally expressed by letting states be reachable nodes in a policy-controlled Markov process, with "scientific practice" as an action that increases transition probabilities toward desirable states.*

The Markov framing also makes explicit that "doing the science" can have dual roles: it both (i) produces observations (epistemic output) and (ii) shifts the distribution over states (pragmatic/ethical output). This dual role is already present in many ordinary scientific institutions (e.g. training, credentialing, lab culture), but here it becomes central because the observation channel itself is state-conditioned. Accordingly, evaluation criteria can be layered: one may assess a practice by the coherence and predictive power of its in-state models, and separately assess it by the value profile of the states it tends to induce, while still keeping these assessments analytically distinct.

Remark 1131. *This remark connects epistemology back to values and control: inquiry is not merely descriptive but can become prescriptive insofar as a practice reliably induces states with preferred value profiles (Section 18). In Hyperseed language, this is an instance of Control (Hyperseed-Concept 87) operating on States of Consciousness (Hyperseed-Concept 178) via socially stabilized practice, which also relates naturally to community-level notions of culture and collective mind systems developed later (Section 22).*

20.10 Micro-example: paraconsistent transitivity in a question network

We conclude with a tiny instance showing how questions, axioms, and paraconsistent inference fit together. In this setting, it is helpful to read each claim $p \in \mathcal{L}$ as a node in a small question/answer graph, and to read the rule set as edges that transport evidential support between nodes. The purpose of the example is not to argue that "same for the current task" is literally transitive in every domain, but to show how a familiar schema can be represented and evaluated even when the inputs are partially contradictory.

Example 17 (Three entities and a transitivity rule). *Let* $X = \{a, b, c\}$*. Consider the claim language*

$$\mathcal{L} = \{p_{ab}, p_{bc}, p_{ac}\},$$

where p_{uv} abbreviates "u and v should be treated as the same (for the current task)." In many applications, one might imagine u, v as records, hypotheses, categories, or agents; the notation is intentionally minimal so that the only moving parts are (i) a rule schema and (ii) a paraconsistent treatment of evidence.

Let the community's rule set contain a weighted "transitivity" rule

$$r: \{p_{ab}, p_{bc}\} \to p_{ac} \quad \textit{with strength } \lambda_r = (0.8, 0.0).$$

Here the strength has two components, meant to encode (respectively) how strongly the community treats the rule as supporting its conclusion and how strongly it treats the rule as opposing *its conclusion. Thus, $\lambda_r = (0.8, 0.0)$ says "we have substantial positive license to apply this rule, and no explicit negative license attached to the rule itself."*

Suppose an agent has evidence

$$\beta(p_{ab}) = (0.9, 0.4), \qquad \beta(p_{bc}) = (0.7, 0.1), \qquad \beta(p_{ac}) = (0.0, 0.0).$$

It is useful to read $\beta(p) = (\beta^+(p), \beta^-(p))$ as a pair of nonnegative weights, where $\beta^+(p)$ measures degree of support for p and $\beta^-(p)$ measures degree of support for $\neg p$ (or, more generally, for "reasons against accepting p"). The evidence for p_{ab} is explicitly conflicted: both positive and negative support are present. In particular, nothing in the representation forces $\beta^+(p_{ab}) + \beta^-(p_{ab}) \leq 1$, so the framework can record "mixed signals" without first reconciling them into a single probability-like quantity.

Compute the one-step inference contribution:

$$\mathrm{contrib}_r(\beta) = (0.8, 0.0) \otimes (0.9, 0.4) \otimes (0.7, 0.1) = (0.8 \cdot 0.9 \cdot 0.7,\ 0.0 \cdot 0.4 \cdot 0.1) = (0.504, 0.0).$$

This multiplication-style $\otimes$ can be read as an "and-like" aggregation: a rule application is strong only to the extent that each ingredient (the rule license and each premise) is strong, and the particular choice here treats the positive and negative channels independently by multiplying like with like. In other words, for this illustrative $\otimes$, the positive component of the contribution depends only on the positive components of the inputs, and the negative component depends only on the negative components. Thus

$$F_{\mathcal{R}}(\beta)(p_{ac}) = \beta(p_{ac}) \oplus (0.504, 0.0) = (0.504, 0.0).$$

The $\oplus$ operation is acting as the accumulator for multiple sources of evidence (for instance, multiple rules or observations targeting the same claim); in this toy step there is only one contributing rule, so the update is transparent. The agent now has positive evidence for p_{ac}, derived from the transitivity schema, while the conflict in p_{ab} did not "explode" into arbitrary conclusions. Concretely, the presence of $\beta^-(p_{ab}) = 0.4$ does not force $\mathrm{contrib}_r(\beta)$ *to become negative, nor does it automatically block the positive pathway; it is simply recorded as a coexisting counter-weight on the premise node.*

If the agent wishes to "bottom out" a cycle in a larger question network, it may entertain a hypothesis such as p_{ac} at high strength, propagate by closure, and compare the resulting conflicts to decide whether the hypothesis is coherent with other commitments. In that broader usage, one can think of the hypothesis injection as an explicit, inspectable intervention: rather than silently assuming a fixed point exists and is unique, the agent tests candidate stabilizers and then evaluates how much inconsistency they induce elsewhere in the network.

Remark 1132 (What to notice in the computation)**.** *The crucial feature is that negative evidence in a premise does not automatically negate the ability to infer something; nor does it entail everything. This is the operational point of the paraconsistent setup: local inconsistency is permitted as a first-class datum, and inference is engineered so that inconsistency does not trivialize the entire theory.*

The rule strength $\lambda_r = (0.8, 0.0)$ carries only positive support, and (in the product used in this example) the negative component of the contribution becomes 0 because the rules negative component is 0. Equivalently, with this $\otimes$, a rule application can contribute negative evidence to its conclusion only if the rule itself (or, in other designs, some premise) carries negative evidence in the channel that is wired to generate negative output. This makes the "audit trail" of negativity explicit: one can point to exactly which pieces of the system are responsible for introducing counter-support for a claim.

Thus we obtain a derived positive claim without any requirement that the premises be consistent. In a network with multiple rules, an agent might simultaneously accumulate positive and negative evidence for the same conclusion (e.g., $F_{\mathcal{R}}(\beta)(p_{ac}) = (\alpha^+, \alpha^-)$ with both components nonzero), in which case the system records that tension rather than forcing an immediate collapse to either acceptance or rejection.

As a toy illustration, this mirrors the intended "conflict without explosion" behavior discussed throughout the paraconsistent parts of the document [23, 24]. The key methodological point is that the ability to proceed in the presence of conflict is not a purely informal principle; it is enforced by the algebra of $\otimes, \oplus$ and by the way rules are allowed to contribute to conclusions.

In a fuller implementation, alternative $\otimes$ choices could allow negative evidence to propagate in more nuanced ways; the point here is that the algebraic interface makes such choices explicit and auditable. For example, one could choose an $\otimes$ in which the negative component of a premise partially attenuates (rather than nullifies) the positive contribution, or in which a strongly negative premise injects explicit negative mass into the conclusion; the present micro-example simply exhibits the cleanest "separation of channels" behavior so that the mechanics can be read off directly from the numbers.

Hyperseed concepts covered

- Knowing and Thinking as structured dynamics on question/answer graphs (Hyperseed-Concept **??**).
- Template Patterns; Questions as information-seeking actions about instantiations; Answers as instantiations or other uncertainty-reducing artifacts (Hyperseed-Concept 130, Hyperseed-Concept 144, Hyperseed-Concept 56, Hyperseed-Concept 195).
- Beliefs (direct and indirect); Axiom Systems; "entertaining a hypothesis" as hypothetical belief injection followed by closure (Hyperseed-Concept 65, Hyperseed-Concept 64).
- Question Networks, including cyclic/non-well-founded dependencies and "bottoming out" by hypothesis (Hyperseed-Concept 145, Hyperseed-Concept 93).
- Belief System as a coherent network of beliefs held in a community; "productive" belief systems via individuation and self-transcendence constraints (Hyperseed-Concept 64, Hyperseed-Concept **??**, Hyperseed-Concept 167).
- Science as empirically coupled belief-system dynamics; State-Dependent Science as a state-indexed family of observation sets and update loops (Hyperseed-Concept 104, Hyperseed-Concept 178).

21 Intelligence, tasks, and agency

21.1 Motivation: from mind concepts to task competence

Sections **??**–20 built a picture of minds as pattern-using, resource-limited systems: they represent (some) distinctions, update paraconsistent beliefs, allocate attention, and exert control to shape future perception while negotiating value conflicts. In this framing, "mind" names an internal organization: a way of building and reusing structured regularities under bounded computation, bounded time, and bounded sensing. What remains missing at this stage is an *operational* connector between internal organization and externally checkable performance, i.e., a way to say not only what a system *is like* internally but what it can *reliably accomplish* when coupled to an environment.

Hyperseed's "intelligence" layer adds two commitments:

- *Tasks* provide an operational bridge from internal pattern webs to externally checkable competence.
- *Agency* is persistent closed-loop control; *autonomy* is the capacity to generate, sustain, and revise goals from within (rather than merely executing imposed objectives).

The first commitment is deliberately methodological: it treats performance as something that can be posed, repeated, and audited. In particular, a task statement is meant to pin down (i) what information is available to the system, (ii) what interventions are permitted, (iii) what counts as success or failure, and (iv) what resource budgets and time horizons apply. This makes it possible to distinguish "the system sometimes produces a desired outcome" from "the system is competent at a specified interaction pattern," and to separate one-off luck from stable capability. It also forces clarity about *interfaces*: what is observed, what is acted upon, and how feedback is delivered.

The second commitment clarifies a persistent ambiguity in informal discussions of intelligent behavior. Many systems can execute a goal when the goal is supplied from outside, yet do not select, maintain, or renegotiate goals under changing conditions. Hyperseed therefore separates (a) being an agent in the minimal cybernetic sense of maintaining a control loop over time, from (b) being autonomous in the stronger sense of owning a goal-management process. The distinction matters for evaluation: two systems might solve the same benchmark task while differing radically in how brittle their goal pursuit is when the task is perturbed, reframed, or partially underspecified.

This section formalizes tasks (and families of tasks), defines intelligence measures as transfer and generalization across task sets, and makes "agent," "autonomous agent," "intent," and the action primitives "stimulate"/"inhibit" mathematically explicit. We also introduce a compact notion of "engineered" as an ontological tag on artifacts produced by intelligent action. A further motivation for these formalizations is comparability: once tasks are explicit objects, one can speak precisely about when two seemingly different problems share a structure (hence belong to a common family), when a solution method transfers, and when improvement reflects genuine generalization rather than memorization of a narrow interaction script.

In particular, the move to *families* of tasks is meant to encode variation: different initial conditions, different noise realizations, different objective weights, different environments drawn from a distribution, or even different representational encodings of the same underlying problem. This is the natural setting in which "generalization" and "transfer" are not slogans but statements about performance stability under controlled shifts. A system that performs well on a single fixed instance may be competent at that instance; a system that performs well across a family demonstrates robustness to variation. Likewise, a system that learns within one family and then reliably improves performance on a related family exhibits transfer in a sense that can be measured and compared across architectures.

The emphasis on action primitives such as "stimulate"/"inhibit" is also explanatory rather than decorative: many agent models tacitly assume a rich action alphabet. By naming minimal primitives, Hyperseed aims to make clear what sort of causal interface is being assumed between an agent and its world. In this view, complex actions can be built compositionally from simpler primitives, while the core semantics remains grounded in how interventions modulate downstream dynamics (amplifying, suppressing, enabling, blocking). This keeps later notions of "intent" tied to concrete control structure: intent is not merely a post-hoc interpretation, but a commitment expressed through persistent selection of actions that shape trajectories toward goal conditions under feedback.

Finally, the "engineered" tag is introduced to mark a distinction that will matter later when discussing artifacts, tools, and constructed sub-agents. The tag is not meant to imply a specific material substrate or manufacturing process; rather, it indicates provenance: the object exists in its present functional organization *because* intelligent action brought it about (possibly via long chains of intermediate artifacts and delegation). This provenance marker becomes useful when reasoning about ecosystems that mix naturally occurring structures with designed ones, or when tracking how competence can be externalized into tools that then feed back into later task performance.

Remark 1133. *Philosophically, the move from "mind" to "intelligence" in Hyperseed is a move from* being a web of patterns *to* being assessable by what one can reliably do*. A task is not merely an external test: it is a formalized interface between an agent's internal organization and an environment that can answer back. This is why tasks belong to the Hyperseed core-concept set as an explicit bridge concept (Hyperseed-Concept 186), and why task families become the natural stage on which transfer learning is defined (Hyperseed-Concept 192; see also [19, 7]). The key word in "environment that can answer back" is* counterfactual responsiveness*: when the agent changes what it does, the world changes what it returns. This prevents collapsing "task" into a static dataset query and keeps the emphasis on interaction, feedback, and control. Reliability then refers not only to average success but to stability under repeated trials, bounded resources, and specified perturbations—a notion aligned with competence rather than isolated performance.*

21.2 Tasks as interaction specifications

We work with an intentionally lightweight task formalism: a task is an interaction channel plus an evaluation functional. This is compatible with classical decision theory (MDPs/POMDPs) but keeps the paraconsistent value layer explicit. In particular, we separate (i) the *interface* that generates observation streams in response to actions from (ii) the *evidence-bearing evaluation* attached to transitions, and we do so without assuming that the agent has access to any privileged "true state" variable beyond what the interaction itself provides.

Definition 315 (Histories)**.** *Fix finite sets O (observations) and A (actions). For $t \geq 0$ define the set of length-t histories by*

$$H_t := O \times (A \times O)^t = \{(o_0, a_0, o_1, \dots, a_{t-1}, o_t)\}.$$

Let $H := \bigcup_{t \geq 0} H_t$ be the set of all finite histories.

Remark 1134. *The notation here is deliberately explicit: O is the alphabet of percepts the task presents, and A is the alphabet of interventions the agent may select. A history in H_t is thus an alternating record*

$$(o_0)\ (a_0, o_1)\ (a_1, o_2)\ \cdots\ (a_{t-1}, o_t),$$

and H is simply the set of all such finite records of arbitrary length. This is the minimal object needed to define policies without assuming Markovian state access: a policy may depend on the entire past, not merely on the current observation.

As a simple example, if $O = \{0,1\}$ and $A = \{\mathsf{L}, \mathsf{R}\}$, then an element of H_2 is a quintuple $(o_0, a_0, o_1, a_1, o_2)$ such as $(1, \mathsf{L}, 0, \mathsf{R}, 0)$. Such strings are the "raw material" out of which an agent constructs patterns and habits (Hyperseed-Concept 130, 155; cf. [5]).

Two small technical conventions are worth keeping in mind for later sections. First, the set H includes the "time 0" history $(o_0) \in H_0$, so an agent may condition its very first action on an initial percept. Second, H is a disjoint union of H_t over t, so "history length" is always well-defined: a history is not merely a sequence but a sequence together with its time index.

Definition 316 (Policies)**.** *A (possibly stochastic)* policy *is a map*

$$\pi : H \to \Delta(A),$$

where $\Delta(A)$ is the probability simplex on A. A deterministic policy is the special case $\pi : H \to A$.

Remark 1135. *The symbol $\Delta(A)$ denotes the set of all probability distributions on the finite set A. Thus $\pi(h)$ is not an action but a distribution over actions, from which an action may be sampled. This is the standard way to model randomized decision-making; it is also the cleanest way to model bounded or exploratory control without changing the rest of the mathematics.*

For a concrete illustration, if $A = \{\mathsf{L}, \mathsf{R}\}$, then $\Delta(A)$ can be identified with $[0,1]$ by $p \mapsto (p, 1-p)$, and a policy returns a coin-bias conditioned on the history. In many later constructions, one can restrict attention to deterministic policies without changing the conceptual point; allowing stochasticity simply keeps the formalism honest about uncertainty.

It is also useful to note the modeling stance implicit in $\pi : H \to \Delta(A)$: the agent is not assumed to have a compact internal "state" that summarizes the past. Of course, real agents often compute summaries (belief states, feature maps, recurrent memories), but those are implementations *of a history-dependent policy rather than prerequisites of the task definition.*

Definition 317 (Task)**.** *A* task *is a tuple*

$$T = (O, A, P, r, \gamma),$$

where O and A are finite, $P : O \times A \to \Delta(O)$ is a transition kernel over observations, $r : O \times A \times O \to \mathbb{V}$ is a p-bit*-valued immediate evaluation (reward/penalty evidence), and $\gamma \in (0,1)$ is a discount factor.*

Remark 1136. *This definition should be read as a deliberately "interface-only" cousin of the standard MDP/POMDP framework: instead of positing a hidden state space, we treat the observation process itself as the primary object. The transition kernel $P(o,a)$ is a distribution over next observations o' given current observation o and action a. The immediate evaluation $r(o,a,o')$ is not a single real number but a value in $\mathbb{V} = [0,1]^2$, i.e. a* p-bit *of positive and negative evidence.*

A simple example is a bandit-like task where P ignores o and a and always returns the same distribution on O, while r assigns $(1,0)$ to "good" outcomes and $(0,1)$ to "bad" ones. More generally, this setup can encode tasks whose feedback is noisy, socially contested, or internally conflicted (Hyperseed-Concept 198), which is one reason Hyperseed avoids reducing value to a single scalar too early (cf. [19]).

Two clarifications help relate this "observation-level" formalism to more familiar ones. First, a classical MDP $(S, A, \tilde{P}, \tilde{r}, \gamma)$ induces a task of the present form by taking $O = S$, letting P be $\tilde{P}$

(viewed as a kernel on observations), and letting r be an evidence-valued lift of the scalar reward $\tilde{r}$ (for instance by mapping $\tilde{r} > 0$ to positive evidence and $\tilde{r} < 0$ to negative evidence, if one wishes). Second, a POMDP with latent state S and observation kernel Z can also be viewed through this lens by taking O to be the observation alphabet and absorbing the latent state into the induced observation-to-observation dynamics; in general this produces a non-Markov observation process, so the present Markov kernel $P : O \times A \to \Delta(O)$ is best seen as a simplifying interface choice rather than a claim that observations are always Markov in reality.

If one wishes to drop even this Markov-in-O simplification, the minimal generalization is to allow a history-dependent environment kernel $P : H \times A \to \Delta(O)$, i.e. $P(h, a)$ is a distribution over the next observation given the entire past. We do not need that generality for the constructions in this section, but it is conceptually aligned with the earlier point that the agent's policy may be fully history-dependent.

Remark 1137 (Why p-bit-valued evaluation)**.** *The evaluation r is allowed to be conflicted: it may assign both positive and negative evidence to a transition. This matches Hyperseed's value paraconsistency (Section 18) and supports tasks whose feedback is ambiguous, socially contested, or internally conflicted. In particular, allowing $r(o, a, o') = (p, n)$ with both $p > 0$ and $n > 0$ makes room for "mixed" events (e.g. actions that succeed on one criterion while failing on another, or actions that are praised by one evaluator while condemned by another) without forcing an early reconciliation into a single numerical tradeoff.*

Definition 318 (Trajectory value via discounted salience join)**.** *Write an infinite trajectory as $\omega = (o_0, a_0, o_1, a_1, \ldots)$. Define the* discount factor p-bit $\delta_t := (\gamma^t, \gamma^t) \in \mathbb{V}$ *and the trajectory value*

$$\mathrm{Val}_T(\omega) := \bigoplus_{t \geq 0} \delta_t \otimes r(o_t, a_t, o_{t+1}),$$

using the p-bit *quantale operations $\oplus$ and $\otimes$ from Section 3.4.*

Remark 1138. *A few symbols deserve a first-reading gloss. The infinite sequence ω is the full unfolding of interaction: observation, action, observation, action, and so on. The discount $\gamma \in (0, 1)$ yields weights γ^t that shrink with time, and the pair $\delta_t = (\gamma^t, \gamma^t)$ discounts positive and negative evidence symmetrically in $\mathbb{V}$. The operation $\otimes$ plays the role of "scaling" evidence by salience (here, discount), while $\oplus$ plays the role of an "accumulation" operator, which in the* p-bit *quantale is a componentwise maximum (Section 3.4; Hyperseed-Concept 143, 202; see [3, 2] for broader discussion of weakness).*

As an example, suppose along a trajectory there is one highly salient positive event early on with $r = (1, 0)$ and all later events have smaller discounted magnitude; then $\mathrm{Val}_T(\omega)$ will retain that early strong evidence. Conversely, a single striking failure early on will dominate the negative component. This reflects a bounded-attention reading: what is remembered and acted upon is often the most salient evidence, not the sum of all evidence.

The choice of $\bigoplus_{t \geq 0}$ (a join/max-like accumulation) also sidesteps the usual convergence questions that arise with infinite-horizon summation: because $\oplus$ is idempotent and bounded in $\mathbb{V}$, the expression is well-defined as soon as each discounted term $\delta_t \otimes r(o_t, a_t, o_{t+1})$ is defined. Intuitively, the value records the strongest *discounted positive evidence encountered and the* strongest *discounted negative evidence encountered, rather than aggregating all evidence into a single scalar.*

For readers who prefer to see the mechanics in a finite prefix, consider the first two transitions of ω with evidence $r_0 := r(o_0, a_0, o_1)$ and $r_1 := r(o_1, a_1, o_2)$. Then the two-step salience-joined evidence is

$$(\delta_0 \otimes r_0) \;\oplus\; (\delta_1 \otimes r_1),$$

so whichever transition carries more salient positive evidence will set the positive coordinate of the result, and likewise for the negative coordinate. This makes explicit the sense in which the formalism models "what stands out" rather than "what adds up" over time.

Remark 1139. *Because $\oplus$ is a componentwise maximum,* Val_T *emphasizes the most salient discounted success/failure signals along the trajectory. This matches an attention-like reading: a bounded agent may be driven primarily by the strongest evidence-laden events rather than by a fully additive cumulative utility. Other aggregators can be substituted if desired; the present choice keeps the link to the core quantale explicit. In particular, since $\oplus$ acts like a (discounted) supremum,* Val_T *behaves more like a* "peak evidence" *semantics than a* "sum of evidence" *semantics: once a decisive success signal has occurred, additional smaller successes do not further increase the positive component, and analogously for failures. This makes the evaluation robust to long stretches of low-level noise, while remaining sensitive to rare but high-confidence events (e.g. a single catastrophic failure state, or a single decisive achievement of a goal condition). It also clarifies why discounting interacts naturally with $\oplus$: discounting controls how much salience is granted to evidence as it recedes in time, while $\oplus$ controls how evidence is pooled across time once discounted.*

Definition 319 (Policy value and plausibility scalarization)**.** *Let $\mathbb{E}_\pi[\cdot]$ denote expectation over trajectories induced by P and π. Define the* p-bit*-valued policy value*

$$J_T(\pi) := \mathbb{E}_\pi\big[\mathrm{Val}_T(\omega)\big],$$

where expectation is taken componentwise in $[0,1]^2$*. Define the* plausibility projection $\mathrm{pl} : \mathbb{V} \to [0,1]$ *by*

$$\mathrm{pl}(v^+, v^-) := \frac{1 + v^+ - v^-}{2}.$$

The scalar score $\mathrm{pl}(J_T(\pi))$ *is interpreted as "net plausibility of success." In this reading, v^+ can be viewed as graded support for success, v^- as graded support for failure, and the affine centering at $\frac{1}{2}$ treats "no salient evidence either way" as epistemically neutral rather than as either success or failure.*

Remark 1140. *Componentwise expectation means: if* $\mathrm{Val}_T(\omega) = (V^+(\omega), V^-(\omega))$*, then* $J_T(\pi) = (\mathbb{E}_\pi[V^+], \mathbb{E}_\pi[V^-])$*. The projection* pl *then converts the evidence-pair into a single number in* $[0,1]$ *by treating positive evidence as increasing plausibility and negative evidence as decreasing it. The affine form $\frac{1+v^+-v^-}{2}$ is chosen so that $(1,0) \mapsto 1$, $(0,1) \mapsto 0$, and $(0,0) \mapsto \frac{1}{2}$ as a neutral baseline.*

One should not mistake pl *for a metaphysical collapse of value into one scalar; it is a pragmatic device for defining competence comparisons and breadth measures. The underlying* p-bit *value $J_T(\pi) \in \mathbb{V}$ retains the information about conflict (Hyperseed-Concept 198) even when one temporarily projects down to a real score. It is also useful to note some basic structural properties of* pl*: it is monotone nondecreasing in v^+, monotone nonincreasing in v^-, and* 1*-Lipschitz with respect to the ℓ_1-distance on* $[0,1]^2$ *up to a factor of $\frac{1}{2}$, i.e. changing (v^+, v^-) by (Δ^+, Δ^-) changes* pl *by at most $\frac{1}{2}(|\Delta^+| + |\Delta^-|)$. Thus, while* pl *is a lossy map, it behaves continuously and predictably under small changes in evidential components. Moreover,* $\mathrm{pl}(v^+, v^-) = \frac{1}{2} + \frac{1}{2}(v^+ - v^-)$ *makes explicit that the scalarization is equivalent to taking a signed "evidence balance" $v^+ - v^- \in [-1,1]$ and rescaling it into* $[0,1]$*; in particular,* pl *treats equal positive and negative evidence as neutral regardless of the absolute magnitudes, reflecting the intended paraconsistent stance that conflict should not be silently resolved without a modeling decision.*

Definition 320 (Task sets)**.** *A* task set *is any collection $\mathcal{T}$ of tasks (often equipped with a distribution* D *for sampling tasks). We will refer to* $(\mathcal{T}, \mathsf{D})$ *as a* task environment*. In the typical* *case where* D *is present, one can regard* D *as encoding both prevalence (which tasks occur) and importance (which tasks matter more) within the ecology.*

Remark 1141. *A task set* $\mathcal{T}$ *plays the role of an "ecology of challenges." A single task can be solved by memorization or brittle specialization; a distribution over tasks forces the question of generalization and transfer (Hyperseed-Concept* **??**, *118, 192; cf. [19, 7]).*

As a simple example, $\mathcal{T}$ *might be the set of all two-armed bandits with varying payoff probabilities, and* D *a prior over those probabilities. Then an "intelligent" agent is one that does not merely excel on one bandit instance, but does well on typical draws from the family. This example also highlights a distinction between* in-task *uncertainty (stochastic rewards within a fixed bandit instance) and* across-task *uncertainty (not knowing which instance will be drawn); task environments make the latter explicit and thereby separate adaptivity/learning from mere execution. In settings where tasks share structure (e.g. common latent parameters), good performance on* D*-typical tasks can be interpreted as evidence that the policy (or agent) has captured that structure rather than overfitting to idiosyncrasies of a single task. Finally, note that* $\mathcal{T}$ *need not be "small" or even countable: it may be a parametrized family (e.g. all MDPs in a class), and* D *may be a continuous distribution; the definition is intentionally agnostic about representation so that later breadth notions can range from finite benchmarks to open-ended task generators.*

21.3 Task morphisms and compositional transfer

Hyperseed treats intelligence as pattern discovery that *transfers* across contexts. A minimal formalization of transfer is: one task can be reduced to another by translating observations/actions while preserving evaluation.

Definition 321 (Task reduction / simulation morphism)**.** *Let* $T = (O, A, P, r, \gamma)$ *and* $T' = (O', A', P', r', \gamma)$ *share the same discount* γ*. A* reduction *(or* simulation morphism*)* $F : T \to T'$ *consists of maps*

$$f_O : O \to O', \qquad f_A : O \times A' \to A,$$

such that for all $o \in O$ *and* $a' \in A'$ *with* $a = f_A(o, a')$*:*

*(a) (*Dynamics compatibility*) The pushforward of* $P(o, a)$ *under* f_O *equals* $P'(f_O(o), a')$*, i.e.*

$$(f_O)_*\big(P(o, a)\big) = P'(f_O(o), a') \in \Delta(O').$$

*(b) (*Evaluation compatibility*) For all* $o_+ \in O$*,*

$$r(o, a, o_+) = r'\big(f_O(o), a', f_O(o_+)\big).$$

Remark 1142. *The notation* $(f_O)_*$ *denotes pushforward of a distribution along a function: if* $\nu \in \Delta(O)$ *and* $f_O : O \to O'$*, then* $(f_O)_*(\nu) \in \Delta(O')$ *is the distribution of* $f_O(X)$ *when* $X \sim \nu$*. Thus condition (a) says: if one observes* o *only through* $f_O(o)$ *and acts via an abstract action* a' *which is decoded into a concrete action* a*, then the next abstract observation has exactly the distribution prescribed by* T'*.*

The decoder $f_A : O \times A' \to A$ *is allowed to depend on the concrete observation* o*. This models a common phenomenon in representation transfer: the same abstract action token may be implemented differently depending on the finer-grained concrete context. In transfer-learning language,* F *is an exact semantics-preserving interface between tasks (Hyperseed-Concept 192, 157; see [7] for a related framework emphasizing compositional transfer).*

Remark 1143. *A reduction says: by observing o only through $f_O(o)$ and choosing an abstract action a' (which is then decoded into a concrete action a), the agent experiences a task equivalent to T'. This is a precise encoding of "reusing a solution by changing representation."*

Proposition 33 (Reductions compose)**.** *If $F : T \to T'$ and $G : T' \to T''$ are reductions (in the sense of Definition 321), then there is a composite reduction $G \circ F : T \to T''$.*

Remark 1144. *Intuitively, this proposition says that "doing one representation change after another" is itself a representation change of the same kind. This matters because transfer in realistic settings is rarely a single leap; it is often a chain of abstractions: raw perception to mid-level features to task-specific summaries. The proposition is a small but essential step toward treating transfer as a compositional algebra rather than a bag of tricks (cf. [7]).*

Proof. Write $F = (f_O, f_A)$ and $G = (g_O, g_A)$. Define $(g \circ f)_O := g_O \circ f_O$ and define the composite action decoder by

$$(g \circ f)_A(o, a'') := f_A\big(o,\ g_A(f_O(o), a'')\big),$$

so that an abstract a'' is first decoded to an a' using g_A (at the abstract observation $f_O(o)$), then decoded to a concrete a using f_A (at the concrete observation o). Dynamics compatibility follows from functoriality of pushforward of measures and the two given compatibility conditions; evaluation compatibility follows by substitution. □

Proof sketch. The strategy is to define the only composite maps one could reasonably mean: observations are re-encoded by composing the encoders, and actions are decoded in two stages. The two compatibility axioms are then checked by "plugging in" these definitions. Pushforward is functorial, so the dynamics condition composes automatically; reward preservation is a direct substitution. □

Remark 1145. *The key step is recognizing that actions must be decoded with awareness of the correct observation granularity at each stage: g_A expects an O'-observation (namely $f_O(o)$), while f_A expects the original O-observation o. Geometrically, one can picture a commutative diagram of stochastic kernels: the composite reduction ensures that the "abstract" observation process induced from T matches the observation process of T'' exactly.*

Corollary 3 (Category of tasks)**.** *There is a (small) category* Task *whose objects are tasks and whose morphisms are task reductions; identities are given by $f_O = \mathrm{id}_O$ and $f_A(o, a) = a$.*

Remark 1146. *This corollary says that tasks and exact transfer maps between them form a mathematical world in which one can speak of equivalence, composition, and invariants. The philosophical gain is that "transfer" becomes something one can reason about structurally, rather than narratively: an agent that discovers a morphism has discovered a re-description that preserves what matters for evaluation.*

In later developments (outside the strict setting here), one often relaxes exact equality to approximation and moves to enriched or higher-categorical variants; the point of beginning with an ordinary category is to make clear which parts of transfer are purely structural (cf. [7]).

Proposition 34 (Transfer by reduction)**.** *Let $F : T \to T'$ be a reduction. Given any policy $\pi' : H' \to \Delta(A')$ for T', there is an induced policy π for T such that*

$$J_T(\pi) = J_{T'}(\pi').$$

Remark 1147. *In plain terms: if you know how to act in the abstract task T', and you have an exact interface F telling you how to* view *T as T', then you can act in T with exactly the same expected value. This is the formal skeleton of "solving a new problem by translating it into a known one". It connects directly to the later breadth and intellectuality definitions: a large part of what makes an agent appear generally intelligent is not raw skill on each task, but the ability to locate such interfaces quickly (Hyperseed-Concept 192, **??**). In particular, the interface supplies both an* observation coding *(what the agent is allowed to treat as the "same situation") and an* action decoding *(how an abstract decision is realized as a concrete intervention). When such a translation is exact, there is no room for performance loss: the agent is not merely approximating T by T' but inhabiting T' as an isomorphic informational perspective on T.*

Proof. Define the encoded history map $\widehat{(\cdot)} : H \to H'$ by applying f_O to each observation in the history and leaving action slots as abstract placeholders. More explicitly, if a history in T has the form $h = (o_0, a_0, o_1, a_1, \ldots, o_t)$, then $\widehat{h} = (f_O(o_0),\ \square_0,\ f_O(o_1),\ \square_1,\ \ldots,\ f_O(o_t))$, where each $\square_i$ denotes an "action position" to be filled by an abstract action from A' when π' is executed. This separation emphasizes that π' is evaluated only on the information it expects in T', namely its own past abstract actions and the coded observations. At history $h \in H$ with current observation o, sample $a' \sim \pi'(\widehat{h})$ and execute $a = f_A(o, a')$ in T. This defines a policy π on T by the composite procedure $h \mapsto \widehat{h} \mapsto a' \mapsto a$; under the usual measurability assumptions on f_O, f_A, this induced π is a valid stochastic policy. By construction, the resulting encoded observation process matches the T' dynamics, and the step evaluation matches exactly; therefore the induced trajectory value distribution matches, hence the expected p-bit value matches. Concretely, one can couple the two rollouts so that at every time t the coded observation in the T-run equals the observation in the T'-run, and the abstract action sampled by π' in T' is the same abstract action used (via f_A) to generate the concrete action in T. The reduction axioms are precisely what make this coupling consistent across time: the transition law for $f_O(o_{t+1})$ given the past agrees with the T' transition law given $(\widehat{h}, a')$, and the evaluation functional applied to (o_t, a_t, o_{t+1}) agrees with the evaluation in T' applied to the corresponding coded transition. Under this coupling, the per-trajectory p-bit values coincide almost surely, so taking expectations over the common randomness yields equality of expected p-bit. ☐

Proof sketch. One defines π by "running π' on the encoded history" and decoding its chosen abstract action back into a concrete one. The reduction axioms guarantee that (i) the encoded observation stream has the same law as in T', and (ii) each step's evaluation agrees under encoding. Therefore trajectory values, and hence their expectations, coincide. Equivalently, the map F behaves like a semantics-preserving compiler from T-interaction traces to T'-interaction traces: the agent computes in the T'-space while the environment evolves in T, and the axioms ensure these two views remain synchronized. ☐

Remark 1148. *The proof works because the reduction conditions are exactly the two places where a mismatch could enter: in the stochastic evolution of observations or in the evaluation of transitions. Visually, one can imagine an agent living in T while wearing "glasses" that apply f_O to every observation; the world seen through the glasses behaves like T', and the decoded actions ensure the world is acted upon in the correct way to maintain the illusion. Said differently, the agent's internal state need not represent T directly: it can represent the coded process, plan there, and rely on the decoder to realize those plans faithfully in T. If either condition failed, transfer could break in a sharply identifiable way: the agent might see a process that no longer follows the assumed T' dynamics (model mismatch), or it might be optimizing the wrong objective because coded transitions are evaluated differently (reward/value mismatch).*

Remark 1149 (Where groupoids enter)**.** *If two tasks reduce to each other, they are equivalent objects in* Task*; the invertible reductions form a groupoid of task equivalences. In later sections, more flexible (approximate) morphisms can be handled by V-enrichment and higher-categorical structure, but the strict case above already captures the core transfer idea. Here "groupoid" means that every morphism regarded as an equivalence has an inverse and that these equivalences compose: if $T \to T'$ and $T' \to T''$ are invertible reductions, then their composite is an invertible reduction $T \to T''$, with identities given by the trivial "do nothing" interface. From the agent's perspective, equivalence identifies tasks that differ only by a change of representation: any optimal (or more generally any) policy can be transported across the equivalence without changing its induced value distribution. This categorical viewpoint separates questions of* capability *(what can be solved up to equivalence) from questions of* search *(how quickly an agent can find the relevant reduction).*

21.4 Competence, intellectuality, and intelligence measures

We now define task competence and several derived quantities that capture Hyperseed's distinction between narrow and general intelligence.

Definition 322 (Agent as a policy-producing process)**.** *Fix an interface (O, A). An* agent Ag *is any procedure that, when coupled to a task T on (O, A), produces a policy $\pi_{\mathsf{Ag},T} : H \to \Delta(A)$. (We make the internal state explicit in Section 21.5.)*

Remark 1150. *This definition intentionally abstracts away implementation details: it treats an agent as a* policy factory *whose output depends on the task it is placed in. The point is to allow many realizations (neural, symbolic, hybrid, social) under a single rubric (Hyperseed-Concept 111, 106, 80). Only later do we constrain the internal structure to express persistent agency.*

As an example, a table-lookup procedure that maps each history to a fixed action is an agent in this sense (though not an interesting one); a reinforcement learning system that updates its policy as it interacts is also an agent.

Remark 1151. *A few clarifications help avoid category errors later. First, the dependence $\pi_{\mathsf{Ag},T}$ on T is meant to include whatever coupling information is available when the agent is "placed in" the task: the task's observation and action semantics, any provided training corpus or simulator access, the interaction protocol, and (when relevant) the agent's own randomization. Second, writing $\pi_{\mathsf{Ag},T} : H \to \Delta(A)$ keeps open whether Ag is deterministic or stochastic: a deterministic agent yields point-mass distributions in $\Delta(A)$, whereas an exploratory learner yields non-degenerate distributions. Finally, the interface (O, A) matters: the same underlying decision-maker may induce very different policies depending on how information is represented in O or which interventions are available in A; later comparisons should therefore be understood as being made within a fixed interface unless explicitly stated otherwise.*

Definition 323 (Task competence)**.** *Given agent Ag and task T, define the competence*

$$\mathrm{Comp}(\mathsf{Ag}; T) := \mathrm{pl}\big(J_T(\pi_{\mathsf{Ag},T})\big) \in [0, 1].$$

Remark 1152. *Competence is the scalarized "how well did it do" number, obtained by first computing the* $\mathsf{p\text{-}bit}$ *policy value $J_T(\pi)$ and then projecting to $[0, 1]$ by* pl*. It is thus a derived quantity, suitable for aggregation and comparison, not the foundational notion of value itself.*

In the simplest non-paraconsistent case where r always has the form $(u, 1-u)$ for a real $u \in [0, 1]$, the projection pl *recovers u (up to an affine reparameterization). But in the general case, competence is merely a pragmatic summary of conflicted evidence.*

Remark 1153. *Because competence is defined via a projection* pl*, it should be read as an* ordinally *meaningful score unless further structure on* pl *is imposed: the same underlying* p-bit *value can map to different numerical scales depending on how cautious or optimistic* pl *is about inconsistency. In particular, if two tasks T and T' have very different reward-generating processes (different noise models, different degrees of inconsistency, or different ranges before scalarization), then* $\mathrm{Comp}(\mathsf{Ag}; T)$ *and* $\mathrm{Comp}(\mathsf{Ag}; T')$ *are comparable only to the extent that J_T and $J_{T'}$ have been normalized in a commensurate way. This is one reason the present section treats competence as a* measurement layer *on top of the task semantics rather than as a fundamental axiom of preference.*

Definition 324 (Capability profile)**.** *For a task set $\mathcal{T}$, the* capability profile *of* Ag *is the function*

$$\kappa_{\mathsf{Ag}} : \mathcal{T} \to [0, 1], \qquad \kappa_{\mathsf{Ag}}(T) := \mathrm{Comp}(\mathsf{Ag}; T).$$

Remark 1154. *The capability profile κ_{Ag} is a map from the ecology of tasks to measured performance. It is conceptually useful because it reframes intelligence as a* landscape *rather than a point: an agent may be strong in one region of $\mathcal{T}$ and weak in another, and questions about generalization become questions about the geometry of this function (Hyperseed-Concept* **??***, 118).*

For instance, a chess engine may have κ near 1 *on tasks involving legal chess positions and near* 0 *on tasks involving language understanding.*

Remark 1155. *The capability profile viewpoint also makes explicit what is sometimes implicit in informal discussions of "intelligence tests": any scalar score is necessarily an aggregation over (explicit or latent) sub-tasks. In this paper's terms, an "intelligence measure" corresponds to (i) a chosen task set $\mathcal{T}$, (ii) a way of sampling or weighting tasks (often a distribution* D*), and (iii) a rule for collapsing the resulting function κ_{Ag} (or learning curves derived from it) to one or a few summary numbers. Put differently, κ_{Ag} is the object that carries the most information; the various aggregates introduced below intentionally discard information in order to gain interpretability and comparability.*

Definition 325 (Breadth at tolerance ε)**.** *Let* D *be a probability distribution on $\mathcal{T}$. For $\varepsilon \in [0, 1]$, define the* breadth *(at tolerance ε) by*

$$\mathrm{Br}_\varepsilon(\mathsf{Ag}; \mathcal{T}, \mathsf{D}) := \mathsf{D}\big(\{T \in \mathcal{T} : \kappa_{\mathsf{Ag}}(T) \geq 1 - \varepsilon\}\big).$$

Remark 1156. *Breadth* Br_ε *is the probability (under* D*) that the agent performs within ε of the top score* 1*. Thus, as ε increases, the set inside* $\mathsf{D}(\cdot)$ *expands, and breadth increases. This parameter lets one tune how strict a notion of "competent" is.*

In the language of Hyperseed concepts, breadth formalizes one axis of General Intelligence (Hyperseed-Concept **??***) by explicitly quantifying how much of a task environment lies within reach at a given tolerance.*

Remark 1157. *Note that breadth is not purely a property of the agent: it is a property of the triple* $(\mathsf{Ag}, \mathcal{T}, \mathsf{D})$*. Changing* D *can change breadth dramatically, even when $\mathcal{T}$ is held fixed: a distribution that concentrates mass on a familiar subdomain can make a narrow specialist appear broad, while a distribution that emphasizes rare corner cases can make a generally capable system look brittle. For this reason,* D *should be interpreted as encoding an intended deployment ecology (what tasks are likely to matter), not merely as a mathematical convenience. One can also view ε as absorbing some of the normalization uncertainty of* pl*: if scalarization is slightly miscalibrated, reporting* Br_ε *across a range of ε can be more robust than selecting a single hard threshold.*

Remark 1158 (General vs narrow intelligence)**.** *A "general" agent has large breadth over a large/diverse task environment; a "narrow" agent may have near-perfect competence on a small region of $\mathcal{T}$ but low breadth overall. This makes the general/narrow distinction precise without committing to a single canonical task set.*

Remark 1159. *The breadth-based distinction also makes visible an important tradeoff: because breadth counts the mass of tasks above a competence threshold, it does not reward being* superlative *on a narrow slice beyond crossing that threshold. In settings where extreme performance on a small class of tasks is valuable (e.g. competitive games, highly standardized pipelines), a narrow agent can be preferable even if it is not "general" by this definition. Conversely, in open-ended environments, breadth aligns with the intuition that robustness across heterogeneous situations is central. Thus "general" and "narrow" here are descriptive labels relative to an environment, not moral judgments about which system is better in all contexts.*

Definition 326 (Intellectuality as rapid transfer)**.** *To make "intellectuality" capture* transfer speed*, assume the agent has a notion of interaction budget $n \in \mathbb{N}$ and produces a budget-indexed policy $\pi^{(n)}_{\mathsf{Ag},T}$ (e.g. after n episodes, n samples, or n inference steps). Define the learning-curve competence*

$$\mathrm{Comp}^{(n)}(\mathsf{Ag};T) := \mathrm{pl}\big(J_T(\pi^{(n)}_{\mathsf{Ag},T})\big).$$

Given a task distribution D, define the intellectuality curve

$$\mathrm{Int}^{(n)}(\mathsf{Ag};\mathcal{T},\mathsf{D}) := \mathbb{E}_{T\sim\mathsf{D}}\big[\mathrm{Comp}^{(n)}(\mathsf{Ag};T)\big].$$

An agent is "more intellectual" (relative to $(\mathcal{T},\mathsf{D})$) if it achieves high $\mathrm{Int}^{(n)}$ for smaller n.

Remark 1160. *This definition aims to isolate something like "grasp" rather than "polish." Two agents may eventually reach similar competence given enough interaction, but the more intellectual one climbs its learning curves quickly across typical tasks in the environment. This is closely aligned with the transfer-learning emphasis in AGI literature (Hyperseed-Concept* **??***, 192; cf. [19, 7]).*

Remark 1161. *Several modeling choices are bundled into the budget parameter n. In an episodic RL setting, n might count episodes of online interaction; in a supervised or self-supervised setting, it might count labeled examples or tokens; in a planning setting, it might count forward-model calls or wall-clock compute devoted to inference. The abstraction is intentional: the core idea is to compare* rates *of improvement, but one must take care to compare like with like when interpreting $\mathrm{Int}^{(n)}$ empirically. In particular, if one agent is allowed to amortize learning across tasks (meta-learning) while another is reset between tasks, then $\pi^{(n)}_{\mathsf{Ag},T}$ corresponds to different experimental protocols, and the resulting intellectuality curves answer different questions about transfer. Relatedly, one can treat $\mathrm{Int}^{(n)}$ as a curve-valued statistic (rather than a single number): comparisons such as "dominates for all n" or "achieves a given competence level with smaller n" often retain more information than collapsing the curve to, say, its area under the curve, even though such collapses can be useful in applications.*

Remark 1162. *A simple example: if $\mathcal{T}$ is a family of maze-navigation tasks differing only by a relabeling of wall textures, then an agent that quickly discovers the invariant structure (geometry of free space) will have a much higher $\mathrm{Int}^{(n)}$ for small n than an agent that relearns each maze from scratch. In particular, the texture relabeling changes superficial perceptual features while preserving the transition-relevant latent variables; high small-n performance therefore witnesses an ability to extract task symmetries and reuse a compact representation across instances rather than amortizing a separate model per instance.*

Remark 1163 (Connection to reductions)**.** *Reductions in Section 21.3 provide a concrete mechanism for high intellectuality: if the agent can identify (or learn) a reduction from a new task to an already-solved one, it can lift an existing policy with little additional data, improving small-n performance. Concretely, a reduction specifies how to translate observations, actions, and rewards of the new task into those of an old task; when such a mapping is available, the agent's "effective sample size" needed to achieve competent behavior can drop sharply because the remaining uncertainty is limited to estimating the reduction rather than learning the entire control problem end-to-end.*

21.5 Agents as persistent perception-action loops

We now pin down the core Hyperseed meaning of "agent": persistent closed-loop control. The key point is that the formal object called an agent is defined independently of any particular task; tasks enter only when we *couple* the agent to a task dynamics, producing an overall stochastic process.

Definition 327 (Agent as a controlled transducer)**.** *Fix (O, A). An* agent *is a tuple*

$$\mathsf{Ag} = (S, s_0, U, G),$$

where S is a (finite or measurable) internal state space, $s_0 \in S$ is an initial state, $U : S \times O \to S$ is an update map, and $G : S \to \Delta(A)$ is an action map. Here $\Delta(A)$ denotes the set of probability measures over A (or probability mass functions when A is finite), so G allows randomized policies as a first-class case. Given a history $h_t = (o_0, a_0, \ldots, o_t)$, the agent state is generated by

$$s_{t+1} := U(s_t, o_t), \qquad a_t \sim G(s_t).$$

Thus, the dependence of actions on the full history h_t is mediated entirely through the sufficient internal state s_t, which is why S can be read as the agent's "memory" or "belief" variable.

Remark 1164. *This is the mathematical core of agency as* process *rather than as a static mapping. The internal state S is what allows persistence: the agent at time t is not merely a response to o_t but the continuation of an internal trajectory $s_0, s_1, \ldots$. The update U represents perception and internal processing; the map G represents action selection (Hyperseed-Concept 87, 140, 151). In this view, "learning" can be modeled either by enlarging S to include parameters and letting U update them online, or by treating U and G as already-optimized maps obtained offline; the definition itself is agnostic to where the maps come from.*

A trivial example is a reactive agent with $S = O$ and $U(s, o) = o$, so the internal state simply caches the latest observation. In that case the action distribution at time t is conditionally independent of the earlier past given o_t, matching the usual notion of a memoryless policy. A more interesting example is a Bayesian filter in which S is a space of belief states, U is a belief update, and G selects actions based on expected value or plausibility; this matches the integration of epistemic dynamics and control emphasized earlier (Sections 20 and 14). In partially observed settings, s_t can be interpreted as an information state that renders optimal control Markovian in s_t even when the raw observation process is not Markov.

Remark 1165. *Definition 327 is deliberately minimal: it is a controlled Mealy-machine style model. Belief states (Section 20) and attention policies (Section 15) can be realized by taking S large enough to include those variables. Likewise, internal randomness (e.g. exploration noise, randomized tie-breaking, or stochastic computation) is captured by the fact that $G(s)$ is a distribution rather than a single action, and one may also represent randomized internal computation by adding a private random seed to S. When $G(s)$ is a Dirac measure for each s, the agent is deterministic at the action-selection level, and all stochasticity in the closed-loop system comes only from the task kernel P (and from any stochasticity one later chooses to place into U).*

Definition 328 (Closed-loop coupling to a task)**.** *Coupling* $\mathsf{Ag} = (S, s_0, U, G)$ *to a task* $T = (O, A, P, r, \gamma)$ *yields a Markov process on* $S \times O$ *with transition kernel*

$$\mathbb{P}\big((s,o) \to (s',o')\big) = \sum_{a \in A} G(s)(a)\ \mathbf{1}[s' = U(s,o)]\ P(o,a)(o').$$

Equivalently, given the current pair (s_t, o_t)*, we sample* $a_t \sim G(s_t)$*, set* $s_{t+1} = U(s_t, o_t)$*, and then sample* $o_{t+1} \sim P(o_t, a_t)$*; the displayed kernel is simply the one-step marginal of this generative description.*

Remark 1166. *The indicator* $\mathbf{1}[s' = U(s,o)]$ *enforces the deterministic state update: among all* $s' \in S$*, only the one equal to* $U(s,o)$ *is possible. The sum over* $a \in A$ *averages over the randomized action choice* $a \sim G(s)$*. Finally,* $P(o,a)(o')$ *is the probability that the task emits observation* o' *after seeing* (o,a)*. Because the resulting chain is Markov on* $S \times O$*, standard tools (occupancy measures, stationary distributions when they exist, mixing and recurrence conditions, etc.) become available for analyzing long-run performance and the effect of memory capacity encoded by* S*.*

This kernel makes concrete the idea that "agent + environment" is a single dynamical system. It is also the place where one can later insert richer causal or mechanistic structure: e.g. if U *is stochastic, the indicator would be replaced by an internal transition kernel. Similarly, if the task is written in a more latent-state form (e.g. an MDP on hidden states with an observation channel), the above coupling remains valid after folding the latent state into an augmented observation variable, or by extending the product space beyond* $S \times O$*.*

Remark 1167 (Agency vs mere dynamics)**.** *Any physical system has dynamics, but an "agent" in Hyperseed is a system for which (i) actions are computationally selected as a function of internal state, and (ii) internal state is updated as a function of perception, yielding an ongoing feedback loop. This is the minimal mathematical form of persistent agency. The requirement that action is selected* as a function of *internal state is what distinguishes closed-loop control from open-loop replay of a fixed action sequence; conversely, the requirement that internal state is updated* as a function of perception *distinguishes adaptive interaction from a self-contained dynamical system that never conditionally responds to its input stream.*

21.6 Autonomy, intent, and stimulate/inhibit primitives

Hyperseed distinguishes agency from autonomy, and uses "intent" plus the action polarities "stimulate"/"inhibit" as conceptual primitives for action semantics.

In this framing, *agency* is primarily about the presence of an action-selection loop (the system chooses actions as a function of its state and incoming information), while *autonomy* concerns where the system's effective objectives come from and how they change over time. The definitions below make this separation explicit by introducing intent as an explicit state component, and then defining autonomy in terms of whether that intent can be endogenously formed and updated rather than being fully imposed by the task interface.

Definition 329 (Intent variable)**.** *Let* $\mathcal{G}$ *be a set of goal-tokens (which may encode explicit goals, subgoals, or self-generated questions). An* intent-augmented agent *is an agent* Ag *whose internal state factors as* $S = \widetilde{S} \times \mathcal{G}$ *and whose action map depends on the current intent:*

$$G(\tilde{s}, g) \in \Delta(A).$$

We interpret $g_t \in \mathcal{G}$ *as the agent's* intent *at time* t*.*

Remark 1168. *The factorization $S = \widetilde{S} \times \mathcal{G}$ makes explicit a distinction that is often only implicit in control models: there is a "how am I doing" state $\tilde{s}$ and a "what am I trying to do" state g. Here $\mathcal{G}$ is intentionally token-level: it may represent explicit symbolic goals, implicit drives, active questions, or socially supplied objectives (Hyperseed-Concept* **??**, **??**, *144).*

As a toy example, $\mathcal{G} = \{\mathsf{eat}, \mathsf{hide}\}$ could encode two competing intents, and $G(\tilde{s}, g)$ may put high probability on different actions depending on whether the agent is in "eat" mode or "hide" mode, even if the observation stream is the same.

Remark 1169. *The object $G(\tilde{s}, g) \in \Delta(A)$ can be read as a policy conditioned on an explicit "goal-token" variable: it returns a distribution over actions rather than a single action so that both deliberate stochasticity (e.g. exploration) and unresolved internal conflict can be modeled without adding extra structure. When $\mathcal{G}$ is finite and relatively small, this resembles a mixture-of-policies view; when $\mathcal{G}$ is large (e.g. language-like), g functions more like a latent program pointer or a compact description of the currently active objective.*

Although we call g an "intent," nothing in the definition requires g to be linguistically reportable or consciously accessible. The main requirement is functional: g is a state component that mediates action choice in a way that is not reducible to $\tilde{s}$ alone.

Definition 330 (Autonomous agent (operational))**.** *An intent-augmented agent is* autonomous *if it contains an internal goal-formation mechanism: there exists a (possibly stochastic) update rule*

$$\Phi : \widetilde{S} \times O \to \Delta(\mathcal{G})$$

such that the intent is updated by sampling $g_{t+1} \sim \Phi(\tilde{s}_t, o_t)$, and such that Φ is not a fixed external script of the task alone (i.e. it depends nontrivially on the agent's internal state and value structure; cf. Section 18).

Remark 1170. *Operationally, autonomy is the refusal of the agent's goal dynamics to be reducible to the task's input alone. The update $\Phi(\tilde{s}, o)$ is permitted to depend on the observation o—autonomy does not mean blindness to circumstance—but it must also depend on internal organization and value, so that the agent's intent trajectory is genuinely* self-involving *(Hyperseed-Concept 119, 110; cf. [10] for a related use of fixed-point ideas to analyze self-modifying goal systems).*

A simple non-autonomous case is $\Phi(\tilde{s}, o) = \delta_{\psi(o)}$ for some externally imposed map ψ, forcing intent to be a deterministic function of observation. A more autonomous case is when $\tilde{s}$ contains long-term commitments or identity-like variables, and Φ expresses how those commitments bias goal selection.

Remark 1171. *The clause "not a fixed external script of the task alone" is meant to rule out trivial relabelings where the environment or supervisor effectively chooses g step-by-step, leaving the agent to merely execute a conditioned controller. In particular, even if Φ is learned from data, it is still non-autonomous in this sense if, at run time, its dependence on $\tilde{s}$ is vacuous or inert and all variation in g_{t+1} is driven by externally provided signals. Conversely, Φ may be highly reactive to o and still count as autonomous if that reactivity is mediated by persistent internal structure (e.g. personal constraints, commitments, or preferences encoded in $\tilde{s}$) rather than by a one-to-one task script.*

It is often useful to view Φ as a controller over controllers*: G maps $(\tilde{s}, g)$ to actions, while Φ maps $(\tilde{s}, o)$ to an updated distribution over goal-tokens. This makes room for hierarchical organization in which g selects a mode, skill, or subpolicy, and Φ implements deliberation, self-querying, or goal arbitration.*

Remark 1172 (Degree of autonomy). *Definition 330 is binary, but one can define a degree of autonomy by measuring how sensitive g_{t+1} is to internal variables vs external task signals (e.g. mutual information or causal influence measures). Hyperseed's "self-determined goal formation" can thus be treated quantitatively.*

Remark 1173. *One concrete quantitative proxy is to compare (i) the influence of $\tilde{s}_t$ on g_{t+1} holding o_t fixed versus (ii) the influence of o_t on g_{t+1} holding $\tilde{s}_t$ fixed, using either information-theoretic terms (e.g. $I(g_{t+1}; \tilde{s}_t \mid o_t)$) or interventionist ones (e.g. average causal effect under do-interventions on components of $\tilde{s}_t$). Such measures do not replace the conceptual definition, but they make the intended contrast operational in empirical or simulated settings where one can perturb internal memory, value parameters, or identity variables and observe resulting changes in the intent trajectory.*

Definition 331 (Action effects and stimulate/inhibit). *Fix a claim set $\mathcal{L}$ (Section 20) and let β be the agent's belief state. Assume that executing an action a in observation context o produces an* effect increment

$$\epsilon_{a,o} : \mathcal{L} \to \mathbb{V},$$

so that a belief update step has the form $\beta \leftarrow \beta \oplus \epsilon_{a,o}$ (possibly followed by closure). For a claim $\varphi \in \mathcal{L}$:

- *a* stimulates *φ at o if $\epsilon^+_{a,o}(\varphi) > \epsilon^-_{a,o}(\varphi)$;*
- *a* inhibits *φ at o if $\epsilon^-_{a,o}(\varphi) > \epsilon^+_{a,o}(\varphi)$;*
- *a is* neutral *on φ at o if the two are equal.*

Remark 1174. *Here $\epsilon_{a,o}(\varphi) = (\epsilon^+_{a,o}(\varphi), \epsilon^-_{a,o}(\varphi))$ is the evidence increment, contributed by doing a in context o, toward affirming or denying φ. The update $\beta \leftarrow \beta \oplus \epsilon_{a,o}$ uses $\oplus$ as evidence accumulation, which matches earlier paraconsistent belief dynamics (Section 20).*

The stimulate/inhibit distinction (Hyperseed-Concept 179, **??***) should be read as an* action semantics primitive*: actions are meaningful insofar as they tilt the evidential balance of claims. For example, if φ is "I am safe," then taking cover may stimulate φ (more positive than negative increment), while stepping into traffic may inhibit it.*

Remark 1175. *Because $\epsilon_{a,o}$ is indexed by both a and o, stimulate/inhibit is inherently context-dependent: the same motor primitive can stimulate φ in one situation and inhibit it in another. This matches ordinary action descriptions ("running" can stimulate "I am safe" when escaping danger, but inhibit it when running onto a freeway). It also highlights that stimulate and inhibit are not properties of actions in isolation, but of action-context pairs relative to a claim language $\mathcal{L}$ and an evidence scale $\mathbb{V}$.*

The strict inequalities in Definition 331 encode a simple polarity test. In applications where $\mathbb{V}$ is noisy or coarse, one may also introduce a tolerance (e.g. treat differences below a threshold as neutral), but the conceptual role remains the same: to classify action effects by which side of a claim they support.

Remark 1176. *This formalizes stimulate/inhibit as primitive polarities of action meaning. The same definition applies if φ is a pattern-claim, a goal-claim, a self-model claim, or a social claim. In later sections, $\epsilon_{a,o}$ can be derived from mechanistic models (neural, symbolic, or hybrid) or from learned causal predictors (Section 14).*

Remark 1177. *The evidence-increment view also makes room for multi-claim tradeoffs: a single action a can simultaneously stimulate some claims and inhibit others. For instance, in a social setting, speaking up might stimulate a claim like "my preference is expressed" while inhibiting "the group perceives me as agreeable." In this sense, stimulate/inhibit is not itself a utility theory; it is a semantics layer that describes how actions push on a structured set of propositions, leaving later machinery (e.g. value aggregation, intent selection, or constraint handling) to determine which claims are prioritized under a given g.*

Finally, note that nothing requires $\epsilon_{a,o}$ to be purely epistemic ("evidence about what is true") as opposed to dispositional ("evidence about what will be true"). When φ is a future-oriented claim (e.g. "I will be safe in the next minute"), $\epsilon_{a,o}$ can be interpreted as action-conditioned predictive support, which aligns stimulate/inhibit with causal control models while preserving the common representational interface of claim-level updates.

21.7 Engineered artifacts as externalized agency

Hyperseed uses "engineered" as an ontological category: a pattern (tool, machine, institution, protocol, etc.) brought into existence by intelligent action. In this sense an engineered artifact can be read as *externalized agency*: a stabilized structure in the world that continues to implement (or make easier to implement) an agent's goals even when the agent is not actively intervening at every moment. The point of treating "engineered" as a category is to capture not only material objects but also non-material but causally efficacious patterns (e.g., conventions, standards, and procedures) whose existence is reflected in reliable regularities and constraints on future trajectories.

Definition 332 (Engineered pattern / artifact)**.** *Let $\varphi_Y \in \mathcal{L}$ be the claim "artifact/pattern Y exists (in the relevant sense)." We say Y is* engineered by agent Ag *over an interval $[t_0, t_1]$ if:*

*(a) (*Emergence*) $\beta^+_{t_0}(\varphi_Y)$ is low and $\beta^+_{t_1}(\varphi_Y)$ is high;*

*(b) (*Intent alignment*) there exists an intent token g active for a nontrivial fraction of times in $[t_0, t_1]$ such that g entails (via the agent's internal rules) a positive commitment to φ_Y (Section 20);*

*(c) (*Causal dependence*) actions taken by Ag in $[t_0, t_1]$ are causally implicated in the increase of $\beta^+(\varphi_Y)$ in the sense of the causal/control notions of Section 14.*

If there exists some interval $[t_0, t_1]$ for which these hold, we call Y engineered.

The formulation intentionally uses the same representational substrate $\mathcal{L}$ and the same evidential/credal quantity $\beta^+_t(\cdot)$ that appear elsewhere, so that "coming into existence" is not treated as a primitive metaphysical event but as a transition that can be tracked within an agent/world-model interface. The parenthetical "in the relevant sense" is doing work: for a physical artifact it may mean spatiotemporal existence in an environment model, whereas for an institution or protocol it may mean existence as a socially instantiated pattern (e.g., a stable equilibrium of expectations) that the model can detect and reliably predict around.

Remark 1178. *The three conditions separate three ideas that are often conflated. Condition (a) is an epistemic/emergence criterion (Hyperseed-Concept* **??***): the world-model moves from "Y is not (evidentially) there" to "Y is there." Condition (b) asserts that the emergence is not accidental relative to the agent: it sits within the agent's intent dynamics (Hyperseed-Concept* **??***). Condition (c) requires that the agent's actions actually matter, so that "engineering" is not mere wishful thinking.*

A simple example: if Y is "a bridge across the river," then engineering requires that the bridge comes to exist, that building-a-bridge was an active intent, and that the agent's actions contributed causally to the bridge's existence. This definition is designed to interface smoothly with the later refinement into tools, machines, and programs (Section 23; Hyperseed-Concept 191, 106, 80). In particular, (b) is what distinguishes engineering from mere lucky correlation (e.g., the bridge appearing because another party built it while Ag *merely hoped for it), while (c) is what distinguishes engineering from mere endorsement (e.g., sincerely wanting a bridge and announcing that desire, but having no causal handle on the construction process).*

Remark 1179. *Definition 332 makes "engineered" neither purely physical nor purely intentional: it requires both an intentional commitment and a causal pathway from action to existence. Section 23 will refine this into tools/machines/programs and their realization semantics.*

Remark 1180. *Several boundary cases are worth making explicit because they motivate the particular choice of criteria. First, the "low"/"high" requirements in (a) are deliberately qualitative: in applications one can pick thresholds, but the conceptual role is to demand a* robust *shift in support for φ_Y, not a transient blip; this helps exclude cases where Y is merely hypothesized and then discarded. Second, (b) requires an intent token g to be active for a nontrivial fraction of $[t_0, t_1]$ so that momentary whim or post-hoc rationalization does not automatically count as engineering; the intent must play a sustained role in the policy dynamics. Third, the "causally implicated" phrasing in (c) is meant to cover indirect and distributed control: engineering a protocol or institution may proceed by writing documents, persuading others, or building coordination scaffolding, none of which is a direct physical construction step, yet each can provide the relevant control pathway under the notions of Section 14.*

Remark 1181. *The definition also accommodates multi-agent and social artifacts by allowing* Ag *to be read as an individual or a suitably defined collective agent. For example, for Y = "a ratified constitution" the emergence criterion (a) tracks the transition from nonexistence to existence of the constitutional regime in the relevant model; (b) corresponds to the presence of an active intent (e.g., "establish a constitutional order") in the collective decision process; and (c) is satisfied when the coalition's actions (drafting, negotiating, voting, enforcement) are causally responsible for the increase in support for φ_Y. Conversely, the definition is designed to exclude cases of mere discovery: learning that a natural resource deposit exists may raise $\beta^+(\varphi_Y)$ dramatically, but unless Y is defined as a newly instantiated pattern (e.g., "an operating mine" rather than "a mineral deposit"), (b) and (c) will not align in the right way for engineering.*

21.8 Intelligence, weakness minimization, and transfer by invariance

Hyperseed repeatedly links intelligence and simplicity: the intelligent system finds the weakest representation that preserves what matters for control and value. We now formalize one clean version of this principle: when a task is invariant under some coarse-graining, an agent loses nothing by failing to distinguish within the coarse-graining. Said differently, when a task "factors through" a representation map q, then planning can be performed entirely at the level of Q without sacrificing achievable return.

Definition 333 (Task quotient / abstraction)**.** *Let $T = (O, A, P, r, \gamma)$ be a task and let $q : O \to Q$ be a surjection onto a finite set Q. Define the pushforward kernel $q_* : \Delta(O) \to \Delta(Q)$ by $(q_*\nu)(B) = \nu(q^{-1}(B))$. We say q is a* task abstraction *if for all $o_1, o_2 \in O$ with $q(o_1) = q(o_2)$ and all $a \in A$:*

*(a) (*Lumpable dynamics*) $q_*(P(o_1, a)) = q_*(P(o_2, a))$;*

*(b) (*Evaluation invariance*) for all $o_+ \in O$,*

$$r(o_1, a, o_+) = r(o_2, a, o_+).$$

The pushforward q_* is the standard way to express "what distribution on abstract states is induced by a distribution on concrete observations": sampling $o_+ \sim \nu$ and then mapping to $q(o_+)$ yields a random variable on Q whose law is exactly $q_*\nu$. The surjectivity requirement means every abstract label corresponds to at least one concrete observation (no "dead" abstract states), and the finiteness of Q keeps measurability issues out of the foreground; conceptually the same definition extends to measurable Q by replacing subsets $B \subseteq Q$ with measurable sets.

Remark 1182. *The map $q : O \to Q$ is a coarse-graining: it groups fine observations into abstract classes. Condition (a) is a standard "lumpability" requirement: from the perspective of the abstract label $q(o)$, it does not matter which representative o in the fiber was actually seen, because the distribution of the next abstract label is the same. Condition (b) says the immediate evaluation likewise ignores the within-fiber distinctions.*

Intuitively, a task abstraction is an invariance: the task simply does not depend on certain details. This is the formal place where Hyperseed's simplicity/weakness thesis becomes visible: ignoring irrelevant distinctions can be free *(Hyperseed-Concept 202, 143, 169; cf. [3, 2]). One may also read this as an operational cousin of compression ideas: if the environment is invariant under identifying certain observations, then a shorter description (in the sense of algorithmic information) can suffice for control (cf. [16]).*

A useful way to internalize the two conditions is via "what the agent can affect and what the agent is scored on." Lumpability says that, after taking action a, the distribution of *future abstract observations* depends only on the *current abstract observation.* Evaluation invariance says that the instantaneous score attached to a transition (o, a, o_+) depends on o only through its abstract class; in particular, if two observations are identified by q, then they are interchangeable as far as one-step evaluation is concerned. Together these conditions ensure that the entire value computation (not just one-step statistics) can be performed on Q.

Theorem 19 (Invariant tasks admit quotient-optimal policies)**.** *Let $q : O \to Q$ be a task abstraction for $T = (O, A, P, r, \gamma)$. Then:*

(a) There exists a reduced task $T_Q = (Q, A, P_Q, r_Q, \gamma)$ such that q induces a reduction $T \to T_Q$ in the sense of Definition 321.

(b) Any policy π_Q for T_Q lifts to a policy π for T with identical p-bit *value: $J_T(\pi) = J_{T_Q}(\pi_Q)$.*

Consequently, an agent that represents observations only up to q (i.e. fails to distinguish within fibers of q) is not disadvantaged on task T.

Remark 1183. *In ordinary language, the theorem says: if the world does not care about some distinctions, then you do not need to care about them either—at least not for the sake of maximizing task value. The surprise (if any) is not that simplification can help, but that under exact invariance it can be made* rigorously lossless*: there exists an abstracted task whose optimal behavior is exactly as good as optimal behavior in the original task.*

This result connects the earlier categorical transfer machinery with the weakness/simplicity theme: the abstraction q becomes a specific reduction morphism $T \to T_Q$, so the value-preservation of lifted policies is an instance of Proposition 34. Thus, "simplicity helps generalization" here is not a slogan but a factorization through a task morphism (Hyperseed-Concept 192, 202).

One concrete way to read part (b) is that a policy for the quotient task can be executed on the original task by composing with q: the agent first maps its observation o to an abstract label $q(o)$ and then samples an action from $\pi_Q(\cdot \mid q(o))$. The equality $J_T(\pi) = J_{T_Q}(\pi_Q)$ asserts that this "act only on the abstraction" strategy is exactly value-preserving, not merely approximately good. In particular, the optimal value functions coincide under the same identification: the best return achievable by any history-independent policy on O is the same as the best return achievable by a history-independent policy on Q, because any optimal $\pi_Q^\star$ can be lifted back to O without loss.

Proof. Define $P_Q : Q \times A \to \Delta(Q)$ by $P_Q(q(o), a) := q_*(P(o, a))$. This is well-defined by lumpable dynamics. Define $r_Q(q(o), a, q(o_+)) := r(o, a, o_+)$; this is well-defined by evaluation invariance. Then (q, id_A) yields a reduction $T \to T_Q$ (take $f_O = q$ and $f_A(o, a) = a$). The lifting statement is a special case of Proposition 34. ☐

To make the "well-definedness" point completely explicit: if $q(o_1) = q(o_2)$, then condition (a) implies $q_*(P(o_1, a)) = q_*(P(o_2, a))$, so the expression $P_Q(q(o), a)$ depends only on $q(o)$ and not on the chosen representative o. Likewise, if $q(o_{+,1}) = q(o_{+,2})$, then by condition (b) (applied with fixed o and a) the value $r(o, a, o_{+,1})$ equals $r(o, a, o_{+,2})$, so writing $r_Q(q(o), a, q(o_+))$ is unambiguous as a function of abstract next-observation. Under these definitions, the lifted policy referred to in (b) can be taken concretely as $\pi(\cdot \mid o) := \pi_Q(\cdot \mid q(o))$, which matches the idea that the agent "forgets" all within-fiber distinctions before acting.

Proof sketch. Construct the quotient task by pushing the original dynamics forward along q and defining the quotient reward by the invariance condition. The abstraction axioms guarantee these definitions do not depend on which representative o of an abstract state $q(o)$ is chosen. Once this quotient task is built, the map q is immediately a reduction, so value-preserving policy lifting follows from the general transfer-by-reduction proposition. ☐

Remark 1184. *The decisive steps are the two "well-definedness" checks. Lumpability ensures that P_Q depends only on the abstract class $q(o)$, and evaluation invariance ensures r_Q is consistent across a fiber. Geometrically, one can imagine collapsing the observation space O by gluing together points in the same fiber; the theorem states that, when the task's dynamics and evaluation are constant along these fibers, the glued space supports an induced task with the same achievable values.*

This quotient construction is closely related to familiar notions of MDP homomorphism and bisimulation-style aggregation: the abstract state $q(o)$ is sufficient for predicting the next abstract state distribution and the reward. In those literatures, one often states the condition as "the abstract process is Markov" plus "rewards respect the partition," which is exactly what (a) and (b) ensure here. The present framing emphasizes the transfer consequence: invariance is not only a modeling nicety but a certificate that a weaker representation is guaranteed to be adequate for control.

Remark 1185 (Weakness reading)**.** *If an observer uses the abstraction q, it collapses distinctions among o_1, o_2 with $q(o_1) = q(o_2)$. This increases weakness (Section 3.7) relative to a fully discriminating observer. The theorem states that this "failure to distinguish" can be* free *for control on tasks that do not depend on the finer distinctions. This is a precise sense in which simplicity supports generalization and transfer.*

21.9 Micro-example: a task family with transfer by coarse-grained structure

Example 18 (Two-cluster task family on a four-element universe)**.** *Let $O = \{1, 2, 3, 4\}$ and $A = \{\mathsf{L}, \mathsf{R}\}$. Let $q : O \to Q = \{\alpha, \beta\}$ be the partition map with $q(1) = q(2) = \alpha$ and $q(3) = q(4) =$*

β. Equivalently, q identifies two equivalence classes (fibers) $\{1, 2\}$ and $\{3, 4\}$, so that any two observations in the same fiber are treated as the same "kind" of input at the abstract level.

Consider a family of tasks $\{T_\theta\}$ parameterized by a label $\theta \in \{0, 1\}$ meaning "which cluster should go left." In T_0, the intended policy is "choose L on α and R on β." In T_1, swap the actions. Thus, for each θ, there is a natural optimal policy $\pi_\theta : Q \to A$ at the abstract level, and any concrete policy $\pi : O \to A$ that factors through q (i.e., $\pi = \pi_\theta \circ q$) realizes the same behavior on all observations within a cluster.

Define dynamics so that the observation is i.i.d. uniform on O each time-step (no state dependence), and define a one-step evaluation

$$r_\theta(o, a, o_+) = \begin{cases} (1, 0) & \text{if } a \text{ matches the cluster-action label of } q(o) \text{ under } \theta, \\ (0, 1) & \text{otherwise.} \end{cases}$$

The two-component reward can be read as a minimal "success/failure" signal (e.g., first coordinate for "correct" and second for "incorrect"), and could be converted to a scalar reward by any fixed linear scalarization without changing the structural point of the example. Then q is a task abstraction for each T_θ: reward depends only on $q(o)$ and the observation law is constant on fibers. Concretely, if $q(o) = q(o')$ then (i) the conditional distribution of observations (here simply uniform, hence identical for all o) does not distinguish o from o', and (ii) the criterion for whether an action is correct is identical for o and o' under the same θ.

An agent that learns the abstraction q once (a coarse pattern: "there are two kinds of observations") can solve both *tasks with only one bit of additional information (which θ holds), achieving rapid transfer. In other words, after committing to the representation $Q = \{\alpha, \beta\}$, the remaining adaptation problem across the family $\{T_\theta\}$ is just to determine whether $\alpha \mapsto \mathsf{L}$ or $\alpha \mapsto \mathsf{R}$, which is a binary choice. An agent that insists on distinguishing all four observations has a larger hypothesis space and can require more data to discover the same transferable structure. For example, without using the quotient, a learner may treat the four mappings $1 \mapsto \mathsf{L}/\mathsf{R}$, $2 \mapsto \mathsf{L}/\mathsf{R}$, $3 \mapsto \mathsf{L}/\mathsf{R}$, $4 \mapsto \mathsf{L}/\mathsf{R}$ as independent degrees of freedom, even though the task family only varies along the single bit θ once the clustering is recognized.*

In Hyperseed language: the transferable pattern is the two-cluster schema; intellectuality appears as the speed with which the agent compresses experience into that schema and reuses it across task contexts. From this perspective, "transfer" is precisely the reuse of the learned factorization $O \xrightarrow{q} Q \to A$ across distinct values of θ, rather than relearning a fresh mapping $O \to A$ each time.

Remark 1186. *This micro-example is intentionally austere, but it displays a genuine phenomenon: the learnable invariant is not the particular mapping from $\{1, 2, 3, 4\}$ to actions, but the* partition *into two clusters (Hyperseed-Concept **??**, 103). Put differently, the family $\{T_\theta\}$ shares the same quotient structure q even though it disagrees about which abstract label should be assigned which action. Once the agent has the right abstraction, the remaining uncertainty (the bit θ) is small and can be resolved quickly, yielding high small-n intellectuality. This is a simple instance where "most" of what must be learned is representational (discovering the right grouping), while the task-specific learning reduces to a low-dimensional parameter fit.*

One can view this as a toy instance of a broader moral: discovering the right quotient is often the true work of intelligence, while selecting the right action within the quotient is then routine. Here the quotient can be seen explicitly as collapsing O by the equivalence relation $o \sim o'$ iff $q(o) = q(o')$, and the induced decision problem on Q is strictly smaller than the original one on O. This is also a point of contact with compression-based views of learning, where discovering a shorter description of the environment can reduce sample complexity (cf. [16]). In particular, once the agent encodes *the data using the two-symbol alphabet $\{\alpha, \beta\}$ rather than four unrelated symbols, fewer examples are needed to justify a stable generalization across the full observation set, because each new data point supports a whole cluster rather than a single element.*

21.10 Ensembles, attention, and cognitive synergy as intelligence multipliers

Finally, we relate intelligence to attention and synergy (Section 15) in a minimal theorem: combining monotone cognitive processes cannot reduce capability, and can strictly improve it when different processes capture complementary transferable patterns. In this subsection, "cognitive processes" can be read broadly: any subsystem that produces a belief state over a shared claim language (e.g. a perceptual channel, a model-based predictor, a memory retriever, a theorem-prover, or an attentional module that re-weights which claims are salient). The key structural move is that combining such processes is modeled as an order-theoretic join on beliefs, while the downstream control layer (action selection) is required to respect that order. This makes the non-degradation statement essentially a monotonicity fact: more (ordered) evidence should not force fewer (good) options.

Definition 334 (Monotone action selection from beliefs)**.** *Let β range over belief states $\mathcal{L} \to \mathbb{V}$ with pointwise order. An action-selection map $\mathcal{A}(\beta) \in \Delta(A)$ is* monotone *if $\beta_1 \leq \beta_2$ implies $\mathcal{A}(\beta_1)$ is no more restrictive than $\mathcal{A}(\beta_2)$ (e.g. the set of actions above a fixed value threshold only grows). In particular, one can interpret "no more restrictive" in several compatible ways depending on how $\Delta(A)$ is used: as inclusion of supports (actions assigned nonzero probability), as inclusion of an "admissible" subset of A underlying a stochastic choice, or as monotonic expansion of the set of actions exceeding a choice criterion computed from β. The requirement is intentionally permissive: it does not force a unique policy, only that improving beliefs (in the $\leq$ sense) cannot invalidate actions that were previously permissible under weaker belief.*

Remark 1187. *Pointwise order on belief states means $\beta_1 \leq \beta_2$ iff for every claim φ, $\beta_1(\varphi) \leq \beta_2(\varphi)$ in the* p-bit *order. Monotonicity then encodes a rational constraint: adding evidence should not arbitrarily forbid actions that were previously considered acceptable. This is a mild structural assumption, compatible with many decision rules, including thresholding, softmax-like schemes, and dominance filters. A useful way to read this is: if β_2 strengthens β_1 claim-by-claim, then any action that was defensible given β_1 should remain defensible given β_2 (even if it is no longer the most preferred). In practice, one can enforce such monotonicity by separating (i) a monotone feasibility or admissibility stage from (ii) a tie-breaking or exploration stage; only the former needs to be monotone for the proposition below to go through.*

Conceptually, monotonicity is a thin formal expression of a cognitive-synergy desideratum: when additional cognitive processes contribute additional distinctions or evidence, the control layer should be able to exploit them rather than being confused by them (Hyperseed-Concept 73, 60). This also clarifies the role of attention: attentional focus can be viewed as a resource-allocation choice about which claims are evaluated or refined, and the monotonicity requirement says that refining attention (thereby increasing certain belief values) should not itself destroy previously available behavioral options.

Proposition 35 (Parallel join of evidence cannot hurt under monotone control)**.** *Suppose two cognitive subsystems produce belief states $\beta^{(1)}$ and $\beta^{(2)}$ over the same claim language, and the agent combines them by $\beta := \beta^{(1)} \oplus \beta^{(2)}$ (pointwise join). If the downstream action-selection mechanism is monotone in the belief state, then for any task T the combined system can achieve competence at least as high as the better subsystem acting alone:*

$$\mathrm{Comp}(\mathsf{Ag}_{\oplus}; T) \;\geq\; \max\{\mathrm{Comp}(\mathsf{Ag}_1; T), \mathrm{Comp}(\mathsf{Ag}_2; T)\}.$$

The implicit intuition is that $\mathsf{Ag}_\oplus$ *has access to* at least *the evidential basis of each subsystem, so it can always choose to ignore whichever parts of the combined belief are unhelpful for a given decision and reproduce the choices of the stronger subsystem. When* Comp *is evaluated via expected task performance under the policy induced by* $\mathcal{A}(\beta)$*, the inequality should be read as an existence-style guarantee: there exists a valid way for the combined agent to act (consistent with monotone control) that attains at least the better subsystem's competence, even if a particular learning or optimization procedure may fail to find it.*

Remark 1188. *The proposition says, in effect, that "having two eyes cannot make you see worse," provided your action-selection respects the ordering of evidence. It is not claiming that naive ensembling always yields large gains; it is claiming a non-degradation property: if one subsystem was already good, joining in additional evidence does not force you to throw away the good option. This is one of the simplest rigorous footholds for the broader Hyperseed theme that cognitive synergy can multiply intelligence (Hyperseed-Concept 73; cf. [19]). A further operational reading is that the combined system can implement a* gating *or* fallback *strategy: if subsystem 1 is reliable on a task family while subsystem 2 contributes sporadically, then the joined belief state can preserve the reliable action set while occasionally unlocking additional high-value actions when subsystem 2 provides confirming evidence. This highlights why monotonicity is the right minimal assumption: without it, a control rule could react pathologically to extra information by collapsing to a worse action set, destroying the intended benefit of parallel cognition.*

Proof. Pointwise, $\beta^{(i)} \leq \beta^{(1)} \oplus \beta^{(2)}$ for $i = 1, 2$. By monotonicity of action selection, any action deemed feasible/good under $\beta^{(i)}$ remains feasible under the joined belief state. Therefore the combined agent can simulate whichever subsystem attains the higher competence on T. To make the simulation idea concrete, fix $i \in \{1, 2\}$. If under $\beta^{(i)}$ the control layer would select from some admissible set (or distributional support) $S_i \subseteq A$, monotonicity ensures that under β the admissible set S satisfies $S_i \subseteq S$ in the relevant "no more restrictive" sense. Thus the combined agent is permitted to choose exactly the same action (or sample from exactly the same distribution) that Ag_i would, yielding the same task performance; taking the better of the two establishes the stated lower bound. □

Proof sketch. The join $\oplus$ is an upper bound, so each subsystem's belief state is below the combined belief state. Monotonicity ensures that moving upward in belief cannot eliminate the actions available before. Hence the combined agent can always choose to behave exactly like the better subsystem, ensuring competence at least as large as the maximum of the two. Equivalently, the combined system has a "safe policy" option: default to the better-performing subsystem's behavior, and only deviate when the joined evidence expands (rather than shrinks) the set of admissible high-quality actions. □

Remark 1189. *The proof is short because the algebra is doing the work:* $\oplus$ *is a join, so it is literally the least upper bound, and monotonicity is exactly the condition needed to ensure upward movement in belief does not remove options. If one wants a visual intuition, imagine two partial maps of a landscape; taking their join is like overlaying them so that any marked safe path remains marked safe. Strict improvement becomes possible when the joined belief state reveals an action that neither subsystem, in isolation, had enough evidence to select. One simple sufficient condition for strictness (in threshold-based controllers) is: there exists an action* $a^\star$ *such that under each* $\beta^{(i)}$ *its score lies just below the selection threshold, but under* $\beta^{(1)} \oplus \beta^{(2)}$ *it crosses the threshold due to complementary evidence. This captures the intended "complementary transferable patterns" case: each subsystem supplies a different piece of reusable structure, and only their combination yields a decisive signal at the control interface.*

Remark 1190 (Why this matters for Hyperseed). *This proposition is a minimal mathematical expression of "cognitive synergy" as an intelligence multiplier: multiple partial processes, each limited in what distinctions it can track, can be combined without loss, and with potential gain when their pattern abstractions and transfer morphisms complement one another. Quantale weakness then provides the bookkeeping for which distinctions are being collapsed and where new distinctions are worth paying for. In particular, if weakness parameters encode which claims are represented coarsely (or ignored) by a subsystem, then different subsystems may be weak in different places; the join can partially undo those independent collapses without requiring either subsystem to bear the full representational cost alone. This clarifies a concrete route by which attention-like mechanisms (allocating representational precision to a subset of claims) and ensemble-like mechanisms (aggregating multiple such allocations) interact to yield a net capability increase.*

21.11 GTGI-inspired quantitative measures of general intelligence

This subsection introduces a family of quantitative functionals intended as concrete instantiations of *General Intelligence* (Hyperseed-Concept **??**) in the style of GTGI. These functionals are phrased in a policy–environment interaction formalism, but can be read as special cases of Hyperseed's broader "task environment" formalism (cf. the task-set/breadth machinery used to sharpen the general/narrow distinction; see Hyperseed-Concept 118).

Basic interaction model. Fix countable sets of actions $\mathcal{A}$ and observations $\mathcal{O}$, and a reward set $\mathcal{R} \subseteq [0,1] \cap \mathbb{Q}$. A (stochastic) policy is a map

$$\pi : (\mathcal{A} \times \mathcal{O} \times \mathcal{R})^* \to \Delta(\mathcal{A}),$$

and an (AIXI-style) environment is a map

$$\mu : (\mathcal{A} \times \mathcal{O} \times \mathcal{R})^* \times \mathcal{A} \to \Delta(\mathcal{O} \times \mathcal{R}),$$

where $\Delta(S)$ denotes the set of probability distributions on S. Writing $\mathbb{E}_{\mu,\pi}[\cdot]$ for expectation over trajectories induced by the interaction of π with μ, define the (undiscounted) expected total reward

$$V_\mu^\pi \;:=\; \mathbb{E}_{\mu,\pi}\left[\sum_{t=1}^{\infty} r_t\right].$$

Definition 335 (Reward-summable environment). *An environment μ is* reward-summable *iff for every policy π,*

$$V_\mu^\pi \;=\; \mathbb{E}_{\mu,\pi}\left[\sum_{t=1}^{\infty} r_t\right] \;\le\; 1.$$

(Glossary: Hyperseed-Concept 210.)

Definition 336 (Universal intelligence). *Let $\mathcal{E}$ be a chosen set of reward-summable environments. The* universal intelligence *of a policy π is*

$$\Upsilon(\pi) \;:=\; \sum_{\mu \in \mathcal{E}} 2^{-K(\mu)}\, V_\mu^\pi,$$

*where $K(\mu)$ is Kolmogorov complexity (Hyperseed-Concept **??**) with respect to a fixed description/decoding scheme. (Glossary: Hyperseed-Concept 211.)*

Definition 337 (GTGI goal function)**.** *A* GTGI goal function *is a map*

$$g : (\mathcal{A} \times \mathcal{O})^* \to \mathcal{R}.$$

Given a trajectory prefix $x_{1:t} := (a_1, o_1, \ldots, a_t, o_t) \in (\mathcal{A} \times \mathcal{O})^*$*, the goal-induced reward at time* t *is*

$$r_t \;:=\; g(x_{1:t}).$$

(Glossary: Hyperseed-Concept 212.)

Definition 338 (Natural timescale indicator)**.** *Given a goal function* g *and environment* μ*, a* natural timescale indicator *is a map*

$$\tau_{g,\mu} : \mathbb{N}_{>0} \to \{0, 1\},$$

where $\tau_{g,\mu}(n) = 1$ *indicates that evaluating performance on* g *in* μ *over* n *steps is considered meaningful, and* $\tau_{g,\mu}(n) = 0$ *indicates it is not. (Glossary: Hyperseed-Concept 213.)*

Definition 339 (GTGI context)**.** *A* GTGI context *is a triple*

$$c \;=\; (\mu, g, T),$$

where μ *is an environment,* g *is a goal function, and* $T = \{t_1, \ldots, t_2\}$ *is a finite time-interval of discrete steps. Its length is* $|T| := t_2 - t_1 + 1$*.*

Given a policy π*, the* expected goal-achievement *of* π *in context* (μ, g, T) *is*

$$V^{\pi}_{\mu,g,T} \;:=\; \mathbb{E}_{\mu,\pi}\left[\sum_{t=t_1}^{t_2} r_t\right], \qquad r_t := g(x_{1:t}).$$

(Glossary: Hyperseed-Concept 214.)

Definition 340 (Pragmatic general intelligence)**.** *Let* $\mathcal{E}$ *be a chosen class of environments,* $\mathcal{G}$ *a chosen class of goal functions, and* $\mathcal{T}$ *a chosen class of finite time-intervals. Let* ν *be a probability distribution on* $\mathcal{E}$*, and let* $\gamma(\cdot, \mu)$ *be a conditional probability distribution on* $\mathcal{G}$ *for each fixed* $\mu \in \mathcal{E}$*. The* pragmatic general intelligence *of a policy* π *(relative to* ν *and* γ*) is*

$$\Pi(\pi) \;:=\; \sum_{\mu \in \mathcal{E}} \sum_{g \in \mathcal{G}} \sum_{T \in \mathcal{T}} \nu(\mu)\, \gamma(g, \mu)\, \tau_{g,\mu}(|T|)\, V^{\pi}_{\mu,g,T},$$

whenever the sum is convergent. (Glossary: Hyperseed-Concept 215.)

Definition 341 (Resource-consumption distribution)**.** *Fix a discrete resource scale* $\mathcal{Q} \subseteq \mathbb{R}_{>0}$ *(e.g. time-steps, FLOPs, bytes). For each context* (μ, g, T)*, a* resource-consumption distribution *is a probability distribution*

$$\eta_{\mu,g,T} : \mathcal{Q} \to [0, 1], \qquad \sum_{Q \in \mathcal{Q}} \eta_{\mu,g,T}(Q) = 1,$$

where $\eta_{\mu,g,T}(Q)$ *is the probability that* Q *units of computational resources are consumed by the agent/policy while pursuing* g *in* μ *over* T*. (Glossary: Hyperseed-Concept 216.)*

Definition 342 (Efficient pragmatic general intelligence)**.** *With the same $\mathcal{E}, \mathcal{G}, \mathcal{T}, \nu, \gamma, \tau$ as in Definition 340, and with resource distributions $\eta_{\mu,g,T}$ as in Definition 341, the* efficient pragmatic general intelligence *of π is*

$$\Pi^{\mathrm{eff}}(\pi) \;:=\; \sum_{\mu\in\mathcal{E}} \sum_{g\in\mathcal{G}} \sum_{T\in\mathcal{T}} \sum_{Q\in\mathcal{Q}} \nu(\mu)\,\gamma(g,\mu)\,\tau_{g,\mu}(|T|)\,\eta_{\mu,g,T}(Q)\,\frac{V^{\pi}_{\mu,g,T}}{Q},$$

whenever the sum is convergent. (Glossary: Hyperseed-Concept 217.)

Definition 343 (Intellectual breadth)**.** *Define the* context-competence weight *of π on (μ, g, T) by*

$$\chi_{\mathrm{Con}_\pi}(\mu, g, T) \;:=\; \sum_{Q\in\mathcal{Q}} \eta_{\mu,g,T}(Q)\,\frac{V^{\pi}_{\mu,g,T}}{Q}.$$

Define a normalized distribution over contexts by

$$\chi^{P}_{\mathrm{Con}_\pi}(\mu, g, T) \;:=\; \frac{\nu(\mu)\,\gamma(g,\mu)\,\tau_{g,\mu}(|T|)\,\chi_{\mathrm{Con}_\pi}(\mu, g, T)}{\sum_{\mu',g',T'} \nu(\mu')\,\gamma(g',\mu')\,\tau_{g',\mu'}(|T'|)\,\chi_{\mathrm{Con}_\pi}(\mu', g', T')},$$

assuming the denominator is finite and nonzero. The intellectual breadth *of π is the Shannon entropy of this distribution:*

$$B_{\mathrm{int}}(\pi) \;:=\; -\sum_{\mu,g,T} \chi^{P}_{\mathrm{Con}_\pi}(\mu, g, T)\,\log\big(\chi^{P}_{\mathrm{Con}_\pi}(\mu, g, T)\big),$$

with a fixed choice of log base (e.g. base 2). (Glossary: Hyperseed-Concept 218.)

Intuitively, $\chi_{\mathrm{Con}_\pi}(\mu, g, T)$ aggregates performance $V^{\pi}_{\mu,g,T}$ across an explicit "difficulty" or "complexity" index $Q \in \mathcal{Q}$, weighted by $\eta_{\mu,g,T}(Q)$. The factor $1/Q$ implements a conventional complexity penalty: holding value fixed, competence concentrated on lower-complexity (easier) contexts contributes more weight, while competence only at higher-complexity contexts is discounted. This makes the induced distribution $\chi^{P}_{\mathrm{Con}_\pi}$ sensitive not merely to raw reward, but to where (across context families and difficulty) that reward is being obtained.

The normalized quantity $\chi^{P}_{\mathrm{Con}_\pi}(\mu, g, T)$ can be read as a posterior-like mass function over contexts, proportional to (i) prior plausibility of environments $\nu(\mu)$, (ii) a coupling term $\gamma(g, \mu)$ indicating which goals g are meaningfully posed in which environments μ, (iii) a length- or horizon-weighting $\tau_{g,\mu}(|T|)$ over task sets of size $|T|$, and (iv) the context-competence weight itself. Under this interpretation, $B_{\mathrm{int}}(\pi)$ measures how widely the policy's competence is spread across the space of contexts after *accounting for these priors/weights: high entropy corresponds to competence that is distributed across many distinct (μ, g, T) rather than concentrated on a small subset.*

The finiteness/nonzero assumption on the denominator is not only technical but conceptual: it enforces that the weighted competence mass is normalizable, i.e. the choice of $(\nu, \gamma, \tau, \eta)$ yields a well-posed comparison across policies. In particular, if $\mathcal{Q}$ or the context index set is infinite, then the decay of $\eta_{\mu,g,T}(Q)$ (and the interaction with $1/Q$) governs whether χ_{Con_π} is finite; similarly, the tails of $\nu(\mu)$ and $\tau_{g,\mu}(|T|)$ govern whether the global sum is finite.

Finally, the choice of log base only rescales $B_{\mathrm{int}}(\pi)$ (e.g. bits for base 2, nats for base e), so comparisons of breadth across policies are invariant up to a constant factor. In applications, one often pairs this entropy with a separate level term (e.g. average competence) to distinguish "broad but weak" from "broad and strong" policies, although the present definition isolates breadth as dispersion of competence mass.

Definition 344 (Multi-criterion driven general intelligence)**.** *Fix a feasible policy class* Π_{adm} *and a vector of objective functionals*

$$\mathbf{J}(\pi) := (J_1(\pi), \dots, J_k(\pi)) \in \mathbb{R}^k, \qquad \pi \in \Pi_{\mathrm{adm}},$$

intended to represent multiple desiderata (e.g. efficiency Π^{eff}*, breadth* B_{int}*, and additional Hyperseed-relevant criteria such as joy/growth/choice). Define Pareto dominance by* $\pi' \succ \pi$ *iff*

$$(\forall i \in \{1, \dots, k\})\ J_i(\pi') \geq J_i(\pi) \quad \textit{and} \quad (\exists j)\ J_j(\pi') > J_j(\pi).$$

The multi-criterion general intelligence set *(Pareto front) is*

$$\mathrm{Pareto}(\mathbf{J}) \ := \ \{\pi \in \Pi_{\mathrm{adm}} : \nexists \pi' \in \Pi_{\mathrm{adm}} \textit{ with } \pi' \succ \pi\}.$$

(Glossary: Hyperseed-Concept 219.)

This definition makes explicit that "general intelligence" (as operationalized here) is not a single scalar score but a family *of trade-offs among multiple desiderata. Concretely, the feasible class* Π_{adm} *encodes admissibility constraints (e.g. computational budgets, safety constraints, embodiment limits, permitted interventions, or epistemic limitations), so that the Pareto set is taken over policies that are actually realizable under the system's design and deployment assumptions.*

Each coordinate $J_i(\pi)$ *can be interpreted as a measurement functional induced by some benchmark family: for instance, an efficiency-like functional might quantify reward per unit time/compute; a breadth-like functional can be instantiated by* $B_{\mathrm{int}}(\pi)$*; and additional coordinates can encode desiderata that are not reducible to reward maximization (e.g. robustness, corrigibility, preference learning quality, or more explicitly Hyperseed-aligned constructs such as joy/growth/choice). The Pareto-front framing prevents premature scalarization: if two policies differ such that neither is uniformly better across all criteria, both remain "optimal" in the multi-criterion sense.*

The strict inequality requirement in at least one coordinate ensures that Pareto dominance captures genuine improvement rather than equality. In practice, one may also consider ε*-dominance (to model measurement noise) or impose partial orders induced by constraints (e.g. treat some* J_i *as hard constraints and others as objectives). These are refinements of the same core idea: multi-criterion general intelligence is identified with non-dominated policies relative to a declared objective vector, rather than with a unique maximizer of a single aggregated score.*

Because $\mathrm{Pareto}(\mathbf{J})$ *is a set, it naturally supports subsequent selection principles (e.g. choose among Pareto-optimal policies using a domain-specific preference over trade-offs, or choose policies that remain Pareto-optimal under shifts in* ν, γ, τ *defining breadth). This separation of* measurement *(the* J_i*) from* selection *(how one picks a point on the front) is often crucial when aligning agent behavior to pluralistic or evolving desiderata.*

Hyperseed concepts covered

- Tasks; task sets; intellectuality as rapid transfer/generalization across tasks.
- General intelligence vs narrow intelligence (breadth and learning-curve measures).
- Agent as persistent perception-action loop; closed-loop coupling to a task.
- Autonomous agent via internal goal/intent formation and revision.
- Intent as an explicit variable shaping policy; autonomy as nontrivial internal intent dynamics.
- Stimulate/inhibit as action primitives defined by p-bit-valued effect increments on claims.

- Engineered as an ontological category for artifacts/patterns brought into existence by intent-aligned, causally efficacious intelligent action.
- Various formal criteria for measuring degres of general intelligence (and related quantities)

The two definitions above provide a quantitative bridge from task-centric competence to agency-level evaluation. The "intellectual breadth" measure turns competence across environments, goals, and task-sets into an information-theoretic dispersion score, making it possible to distinguish a policy that is competent in many context families from one that is competent in only a few. The "multi-criterion" construction then places such breadth alongside other desiderata and treats general intelligence as residing in the set of non-dominated feasible policies, rather than collapsing all considerations into a single number.

22 Society, culture, and collective mind systems

22.1 From individual agents to social pattern webs

Previous sections treated a single observer/context as the primary bearer of distinctions, patterns, beliefs, and action (Sections 9–21). Hyperseed then scales this picture to groups: societies, tribes, cultures, and collective minds. The key methodological move is conservative:

- Keep the same core mathematics (paraconsistent `p-bit` evidence values and quantale-style composition), but
- allow *multiple* agents to be coupled by communication and shared artifacts, so that *patterns can span agents.*

Concretely, "scaling to groups" is not treated as changing the kind of object under study, but as changing the *boundary* of the system whose pattern web we track. The individual case studies a pattern web internal to one agent's perceptual-cognitive-action loop; the social case studies a pattern web distributed across many such loops, with additional coupling edges corresponding to message passing, imitation, norm enforcement, and environment-mediated coordination. The key consequence of keeping the same evidence and composition machinery is that all the usual operations for combining, propagating, and constraining patterns (including accumulation of partial evidence, composition of subroutines, and coexistence of incompatible commitments) remain valid, but they now apply to a larger graph whose nodes may be multi-agent processes rather than intra-agent ones.

The phrase "patterns can span agents" should be read literally: some patterns are not located in any single head, but are only well-defined at the level of *interaction.* For example, a question–answer routine, a bargaining protocol, a classroom lecture, or a market price formation process can be treated as a pattern whose realizations require at least two roles (and typically multiple agents), with regularities that persist even as individual participants change. In the same spirit, a shared artifact (a text, law, map, software repository, or ritual object) can be treated as a persistent substructure in the overall pattern web, providing stable constraints and coordination points that outlast particular communicative episodes.

Remark 1191. *The philosophical point is that "sociality" is not introduced as an extra substance. Rather, it is introduced as an additional layer of* composition*: we keep the individual-level formalism, but we allow morphisms (influence, communication, artifact-mediated constraint) to connect individual pattern webs into a larger web. This aligns with Hyperseed's general stance that much* *of mind is a matter of structured pattern dynamics rather than an ineffable residue; see also the general Hyperseed ontology framing in [1].*

This compositional framing also makes it natural to treat social phenomena as having multiple simultaneous "carriers": (a) transient signals (spoken utterances, gestures, posts), (b) relatively stable media (documents, institutions, codebases), and (c) embodied habits (skills, practices, and dispositions). In the formalism, these carriers correspond to different kinds of coupling edges and different persistence timescales, but they can still be integrated by the same evidence calculus: signals supply short-lived evidence pulses, institutions provide long-lived constraints, and habits implement default transitions that bias future pattern activation.

In Hyperseed-Concept terms, what is being formalized here is chiefly Society *(Hyperseed-Concept 170),* Culture *(Hyperseed-Concept 91), and the move from individual* Pattern Web *dynamics to group-scale* Pattern Web *dynamics (Hyperseed-Concept 132), with explicit allowance for contradiction via* Value Paraconsistency *(Hyperseed-Concept 198).*

The explicit role of paraconsistency becomes especially salient at the social scale: groups commonly exhibit stable patterns of behavior while holding mutually inconsistent narratives, norms, or self-models. Rather than forcing premature consistency (which would erase important empirical structure), `p-bit`*-valued evidence allows a society-level pattern web to encode, for example, that a norm is publicly endorsed, privately doubted, and selectively enforced, without collapsing into triviality. This supports modeling phenomena such as pluralism, taboo, propaganda, institutional hypocrisy, and multi-layer norm systems as structured configurations of evidence and composition rather than as exceptions to rational modeling.*

In this section we formalize: (i) societies and tribes as coupled multi-agent pattern webs; (ii) communicative acts and communicative systems (speech acts and their physical realizations); (iii) social roles as interaction templates (teacher, student, worker, colleague); (iv) culture as stabilized intersubjective pattern web; and (v) collective mind systems, including mindplex and global brain.

Item (i) focuses on the graph-theoretic and algebraic structure of coupling: agents are treated as pattern-web-bearing subsystems, while social ties become morphisms that transmit constraints, evidence, and action affordances. Importantly, coupling can be asymmetric (authority, expertise, dependence), delayed (archives, reputation), or mediated (institutions, markets), and these variants correspond to different composition pathways and update schedules in the distributed web.

Item (ii) separates the *illocutionary* aspect of communication (what act is performed: asking, promising, warning, asserting) from its *physical realization* (sound waves, ink, bits). This distinction matters because the same physical channel can realize different speech acts depending on context, and the same speech act can be realized across many channels; the formalism therefore treats communicative acts as patterns that can be instantiated in multiple substrates, with evidence accumulating both from content and from channel-dependent credibility cues.

Item (iii) treats roles as reusable interface specifications for multi-agent patterns: a role constrains what inputs are expected, what outputs are permissible, and what transitions are typical. Roles thus act as higher-level templates that compress social complexity: instead of modeling each dyad from scratch, one models role-structured interactions and then binds agents to roles in particular contexts, allowing rapid composition of large-scale social dynamics.

Item (iv) characterizes culture as what remains invariant (or changes only slowly) when individuals enter and leave: a stabilized, intersubjective pattern web whose persistence is supported by enculturation, shared artifacts, and institutional reinforcement. On this view, cultural "beliefs" need not be located as propositional attitudes inside individuals; they can be seen as constraints on what patterns are easy to activate, acceptable to express, or rewarded to enact in the social web.

Item (v) extends the same apparatus to collective mind systems in which coordination and shared artifact layers become sufficiently rich that group-level cognitive functions emerge: distributed attention (what the group monitors), distributed memory (archives and traditions), distributed planning (organizations and governance), and distributed self-modeling (public narratives and metrics). The terms "mindplex" and "global brain" are treated here as endpoints on a spectrum of integration: a mindplex emphasizes tightly coupled subcommunities (e.g., institutions or networks) whose interaction yields coherent higher-level dynamics, while the global brain emphasizes planetwide information flow and coordination patterns spanning many such subsystems, without presuming a single central controller.

22.2 Minimal social data: agents, coupling, and shared artifacts

We assume the general agent notion from Section 21: agents are persistent perception–action loops interacting with an environment, and autonomous agents are those for which a significant fraction of action has simplest-causal explanations internal to the agent.

For social modeling we need, minimally, (a) a set of agents, (b) an interaction structure, and (c) a way to represent communication and shared artifacts.

Remark 1192. *The qualifier "minimally" is important: richer models often include internal state spaces, explicit message contents, norms, payoffs, or learning rules. Here we separate what is* structural *(who can affect whom, and what durable objects mediate coordination) from what is* dynamical *(how agents update). The point of this subsection is to provide just enough scaffolding that later sections can state fixed-point, stabilization, and propagation claims without committing to a specific cognitive architecture.*

Definition 345 (Interaction graph and coupling weights)**.** *Let A be a (finite or infinite) set of agents. An* interaction graph *on A is a directed graph*

$$G = (A, E), \qquad E \subseteq A \times A,$$

where $(i, j) \in E$ means agent i can directly influence agent j (by communication, direct action, or mediated action that is "social" from the modeling perspective).

A coupling weight *is an assignment*

$$K : E \to [0, 1] \quad or \quad K : E \to \mathbb{V} = [0, 1]^2,$$

depending on whether we want a scalar coupling or a paraconsistent (two-axis) coupling. We interpret larger $K(i, j)$ as higher bandwidth, trust, salience, or effectiveness of i's influence on j.

Remark 1193. *When A is infinite, the interaction graph is best thought of as a* relation *rather than a finite data structure. In applications one may impose additional regularity (e.g. bounded out-degree, locality constraints, or measurability assumptions) so that aggregate quantities (such as total incoming coupling to an agent) are well-defined. Nothing in the present definition forces such assumptions, but later analytic arguments may add them as needed.*

Remark 1194. *A directed edge (i, j) is not meant to imply a moral direction, but an* operational *direction: there exists some channel (speech, coercion, imitation, infrastructural control, shared tools, etc.) through which i can nontrivially alter j's future internal state or external behavior. The weight $K(i, j)$ then compresses many concrete facts (bandwidth, reliability, attention, authority) into a single scalar or* p-bit*-pair.*

As a simple example, in a small lab group one might take $K(i, j) \approx 0.8$ for supervisor-to-student instruction, but $K(j, i) \approx 0.3$ for the reverse direction. In a peer group, one might have $K(i, j) \approx K(j, i) \approx 0.6$ for many pairs. These are not "true" numbers; they are a modeling device for the later fixed-point and stabilization arguments.

Remark 1195. *In practice, $K(i, j)$ may summarize multiple distinct channels (e.g. direct conversation, group chat, shared work products, or observation of third-party interactions). One can interpret $K(i, j)$ as an* effective *coupling after coarse-graining across channels and timescales. This is useful when the model aims to capture macroscopic phenomena (consensus, polarization, institutional persistence) rather than the microstructure of individual conversations.*

Remark 1196 (Why allow $\mathbb{V}$-valued coupling?)**.** *If coupling is scalar, it can only say "more" or "less" influence. If coupling is* p-bit*-valued, it can also represent* ambivalent *links: e.g. high positive coupling (strong influence) together with high negative coupling (active resistance, distrust, or adversarial framing). This matters in social settings where attention and belief updates can be both attracted and repelled.*

Remark 1197. *Allowing $\mathbb{V}$-valued coupling is also a mathematical way to treat "mixed valence" social relations without forcing them into a single real axis. In many real societies, a source may be simultaneously salient and distrusted; the recipient may attend closely in order to rebut, parody, or guard against manipulation. Paraconsistency thus enters not only at the level of* beliefs *but at the level of* channels *themselves, echoing Hyperseed's broader stance that contradiction is often a stable feature of complex adaptive systems rather than an immediate catastrophe; compare [3] for related uses of resource-sensitive algebraic structure.*

Remark 1198. *One convenient reading of a* p-bit*-valued edge $K(i, j) = (k^+(i, j), k^-(i, j))$ is that k^+ parameterizes how strongly j is pulled toward adopting or integrating i's outputs, while k^- parameterizes how strongly j is pushed toward counter-signaling, discounting, or producing anti-correlated outputs in response. This interpretation allows a single graph to represent settings in which opposition is coupled (and thus structured) rather than merely absent interaction.*

Remark 1199. *Nothing requires the coupling to be static across time. One may generalize to $K_t(i, j)$ (discrete time) or $K(i, j; t)$ (continuous time) to represent attention cycles, institutional reconfigurations, or learning about trustworthiness. In such cases the interaction graph itself can be time-varying, e.g. $E_t \subseteq A \times A$, with edges appearing or disappearing as agents enter or leave a community, or as communication channels become available or are blocked.*

Definition 346 (Shared artifacts)**.** *Let $\mathcal{A}$ be a set of* artifacts *(documents, tools, rituals, institutions, machines, shared environments, shared memory stores, etc.). An* access relation *is a relation*

$$\mathrm{Acc} \subseteq A \times \mathcal{A},$$

where $(i, a) \in \mathrm{Acc}$ means agent i can read/use/modify artifact a (in the sense relevant to the model). Optionally, we enrich access with weights $\kappa(i, a) \in [0, 1]$ or $\kappa(i, a) \in \mathbb{V}$ to represent accessibility, trust, and friction.

Remark 1200. *The access relation is intentionally broad: "read" may mean perception (seeing a public sign), "use" may mean executing a procedure embodied in a tool, and "modify" may mean editing a document, changing an institutional policy, or physically altering an environment. In many cases, access is asymmetric: some agents can modify an artifact that others can only observe, which can be represented either by directional refinement (separate read/write relations) or by encoding this asymmetry in $\kappa(i, a)$.*

Remark 1201. *One may also view $(A, \mathcal{A}, \mathrm{Acc})$ as a bipartite graph and then "project" it to obtain an induced agent–agent coupling via shared artifacts: two agents become effectively coupled if they co-access or co-modify the same artifact. This makes explicit how environments and institutions can generate social structure even when direct communication is sparse.*

Remark 1202. *Artifacts are the simplest route from ephemeral interaction to durable structure. An utterance fades, but a written protocol, a code repository, a ritual, or an institution can persist and thereby "store" patterns that many agents can later re-import into their own webs. In Hyperseed-Concept terms, this is a core mechanism by which* Culture *(Hyperseed-Concept 91) acquires memory beyond individual lifetimes.*

As an example, a shared calendar is an artifact: it constrains coordination by offering a persistent representation of commitments. A library is an artifact: it stores a vast set of template patterns and inference habits that can be repeatedly instantiated by different minds. Even a physical tool can be an artifact in this sense; Section 23 treats this more explicitly via physical realizations of abstract processes.

Remark 1203. *The artifact view also clarifies how "the same" cultural object can be instantiated in many media. A norm may exist as an explicit written rule, as a repeated ritual practice, and as a set of implicit expectations encoded in many agents' policies; modeling it as an artifact (or a small constellation of artifacts) provides a common handle for these heterogeneous realizations. This will matter later when distinguishing transient alignment (momentary coordination) from persistent alignment (coordination maintained by externalized memory and enforcement).*

Artifacts are crucial in Hyperseed because they create durable cross-agent coupling: a belief, pattern, or procedure can persist outside any one individual and propagate through time.

Remark 1204. *This durability can be treated as a timescale separation: interactions among agents may be fast and noisy, while artifact dynamics are slower and more inertial. Modeling artifacts explicitly therefore supports explanations of social stability (and of abrupt regime shifts) that do not rely on assuming stable individual psychology. In particular, once artifacts are present, a society can exhibit path-dependence: early contingencies become "locked in" by persistent representations that shape later learning and coordination.*

22.3 Society as a coupled multi-agent system

Hyperseed's informal definition emphasizes sustained close interaction and dependence. Our formal definition keeps the spirit but stays minimal. In particular, the aim is to isolate a small set of *coupling primitives* that can later be instantiated in many ways: communication, imitation, exchange, instruction, coercion, care, and other recurrent channels of mutual influence. The guiding idea is that "being a society" is less about any one mechanism (e.g. kinship, markets, or law) than about the *existence of persistent feedback loops* among agents, mediated both directly (agent–agent interaction) and indirectly (through durable shared structures).

Definition 347 (Society)**.** *A* society *is a tuple*

$$\mathsf{Soc} := (A, G, K, \mathcal{A}, \mathrm{Acc}, \mathcal{D}),$$

where:

- *A is a set of agents;*
- *$G = (A, E)$ is an interaction graph;*
- *K is a coupling weight on E;*
- *$\mathcal{A}$ is a set of shared artifacts;*

- $\mathrm{Acc} \subseteq A \times \mathcal{A}$ *is an access relation; and*
- $\mathcal{D}$ *is (optional) additional domain structure capturing the resource-flow and reproduction dependencies relevant to the society (metabolism, exchange, child rearing, maintenance of shared infrastructure, etc.).*

Remark 1205. *The tuple* Soc *is intentionally austere: it says that a society is (i) a population of agents, (ii) a directed coupling structure among them, and (iii) a set of external structures that agents can jointly read and write. In this sense it is closer to a "minimal ontology of social coupling" than to a sociological theory. The formalism is designed to support later questions such as: which patterns stabilize; which beliefs become shared; which roles emerge; and when the group behaves in a mind-like fashion. One should read* G *and* K *as describing* effective influence *rather than any single physical channel: an edge* $(i, j) \in E$ *indicates that, in the relevant timescale, agent* i *can affect agent* j*'s state or action selection in a repeatable way, while the weight* $K(i, j)$ *parameterizes the strength, rate, or reliability of that effect (depending on the modeling choice used later). The "shared artifacts"* $\mathcal{A}$ *play the role of an external memory and coordination substrate: they include objects like documents, codebases, signage, norms-as-inscriptions, ledgers, shared tools, or institutions understood as* stateful *entities that persist beyond any single interaction. The access relation* Acc *makes explicit that not all agents necessarily have symmetric visibility or control over each artifact, allowing the same formalism to represent, for example, private vs. public channels, credentialed systems, censorship, and asymmetric power to modify common resources.*

A simple example is a two-agent society with $A = \{1, 2\}$*,* $E = \{(1, 2), (2, 1)\}$*, a symmetric* $K(1, 2) = K(2, 1) = 0.7$*, and* $\mathcal{A}$ *containing a shared notebook. Even this toy case already permits nontrivial cultural stabilization: repeated reinforcement through the notebook can induce long-lived habits that neither agent would sustain alone. Here the notebook functions as a minimal "public world": each agent can offload intermediate results, commitments, and cues, and then later re-import them as if they were environmental facts. This kind of indirect coupling is important because it can create* path dependence*: once certain entries exist, they bias future behavior even if the immediate interpersonal influence* K *is weakened or intermittent.*

Remark 1206. *This definition is also a convenient "socket" into which more specialized theories can be plugged. For instance, if one wants to reason about prosocial coordination as efficiency gain, one can enrich* $\mathcal{D}$ *with explicit resource-flow and bargaining constraints, connecting with arguments like [6]. But the present section keeps* $\mathcal{D}$ *optional so the mathematics does not presuppose any particular economics or evolutionary story. Concretely,* $\mathcal{D}$ *can be used to add whatever additional state variables and constraints are needed to make "dependence" precise in a given application: endowments, production functions, task graphs, consumption needs, enforcement mechanisms, spatial structure, or demographic turnover. Leaving* $\mathcal{D}$ *optional is also a way to separate two questions that are often conflated: (a) what makes interaction* socially coupled *at the level of information and control (handled by* $G, K, \mathcal{A}, \mathrm{Acc}$*), and (b) what makes that coupling* necessary or stable *given material constraints (often captured in* $\mathcal{D}$*).*

Remark 1207 (Where "metabolism and reproduction" enter)**.** *If one wants to explicitly encode Hyperseed's biological emphasis,* $\mathcal{D}$ *can include a resource-flow network (who supplies what to whom), plus reproduction/maintenance processes. For many cognitive and cultural applications, it is enough to treat* $\mathcal{D}$ *as implicit and to work directly with* G, K *and the shared artifact layer. One can view this as a modeling choice about the boundary between "social cognition" and "social ecology": if the focus is on belief dynamics, norm propagation, institutional memory, or collective decision-making, then the informational coupling via* G, K *and the persistence provided by* $\mathcal{A}$ *may be the dominant* *explanatory variables. By contrast, when studying collapse, resilience, demographic transitions, or long-run selection pressures, making $\mathcal{D}$ explicit becomes important because it specifies the feasibility constraints under which cultural patterns must reproduce.*

Remark 1208 (Interpretation notes (noncommittal but operational))**.** *The components of* Soc *are intentionally typed only loosely at this stage. For example, K may be taken as a function $K : E \to \mathbb{R}_{\geq 0}$, a stochastic matrix on A, a time-dependent weight K_t, or even a vector of modality-specific couplings (communication, observation, sanction, etc.); later sections can choose the level of structure appropriate to theorems being proved. Similarly, $\mathcal{A}$ may be instantiated as a set of discrete objects, as a shared state space, or as a collection of writable variables with an update semantics; the only essential requirement is that artifacts provide a medium through which effects can persist and propagate beyond a single dyadic exchange.*

22.4 Tribes as tightly intertwined sub-societies

Hyperseed treats a tribe as a society that is "very tightly intertwined" and typically small enough for many direct interactions among members. We formalize this as a high-coupling, high-connectivity regime.

Definition 348 (Strong-tie threshold graph)**.** *Fix a threshold $\tau \in (0, 1]$. Given a society* $\mathsf{Soc} = (A, G, K, \ldots)$ *with scalar couplings $K : E \to [0, 1]$, define the* strong-tie graph $G_\tau = (A, E_\tau)$ *by*

$$E_\tau := \{(i, j) \in E : K(i, j) \geq \tau\}.$$

If K is $\mathbb{V}$-valued, one may threshold on a projection, e.g. the plausibility map $\pi(p, q) = (1+p-q)/2$ (Section 14) or simply on the positive axis.

Remark 1209. *Intuitively, G_τ discards the weak, occasional, or negligible interactions and keeps only the links strong enough to function as reliable conduits for habit, language, norm, and belief transfer. The threshold τ is not a metaphysical constant; it is a modeling knob that selects a time-scale and a notion of "meaningful influence."*

As a small example, if in a workplace $K(i, j)$ measures average weekly effective communication, then a high τ might isolate a close-knit team (a tribe-like substructure), while a low τ might recover the whole organizational chart. In Hyperseed-Concept terms, this is a way to carve out a Tribe *(Hyperseed-Concept 194) as a high-coupling region inside a* Society *(Hyperseed-Concept 170).*

Remark 1210. *Depending on the modeling choice for G and K, the strong-tie graph G_τ may be treated as undirected (mutual strong ties) or directed (asymmetric influence or attention). In the directed case, "(strongly) connected" in later definitions can be interpreted as strong connectivity when influence is genuinely directional, whereas in the undirected case it reduces to ordinary graph connectivity. When $K(i, j)$ is not symmetric, one can also consider symmetrizations such as $\min\{K(i, j), K(j, i)\}$ (for mutuality) or $\max\{K(i, j), K(j, i)\}$ (for one-way influence) before thresholding, depending on whether tribal cohesion is intended to require reciprocal contact.*

Remark 1211. *Varying τ induces a natural* filtration *of graphs: if $\tau_1 \leq \tau_2$ then $E_{\tau_2} \subseteq E_{\tau_1}$. Thus, as τ increases, strong-tie components can only split (or disappear), never merge. This monotonicity supports a multi-scale reading of "tribal structure" in which higher τ reveals the most robust, repeatedly reinforced cores, while lower τ reveals looser coalitions that may still be socially meaningful at a longer time-scale.*

Definition 349 (Tribe)**.** *A* tribe *(relative to τ) is a subset $T \subseteq A$ such that:*

(a) the induced strong-tie subgraph $G_\tau[T]$ is (strongly) connected; and

(b) T satisfies a boundedness condition reflecting finite memory/interaction bandwidth, modeled either as an explicit size bound $|T| \leq N$ or as a density condition such as

$$\frac{1}{|T|(|T|-1)} \sum_{i \neq j \in T} \mathbf{1}[(i,j) \in E_\tau] \geq \delta,$$

for some δ close to 1.

Remark 1212. *Condition (a) enforces that strong ties form a single influence-component rather than a scattering of dyads. Condition (b) encodes the idea that tribes are not merely connected but* thickly *connected (or at least bounded in size so that connectivity is experientially meaningful). The two modeling options for (b) reflect two ways of cashing out bounded cognitive bandwidth: either limit the number of close ties, or require that close ties be dense among the members.*

A canonical example is an extended family or small religious community where most members interact directly and repeatedly. By contrast, an online fandom might be connected but not dense; in that case it may be a society with many tribes rather than a tribe itself.

Remark 1213. *The density expression in (b) is written in a directed-edge style (summing over ordered pairs $i \neq j$). If one instead models G_τ as undirected, the analogous normalization uses $|T|(|T|-1)/2$ and the indicator for $\{i,j\} \in E_\tau$; the conceptual role of the condition is the same: it approximates the claim that "nearly everyone in T is in frequent contact with nearly everyone else." In empirical settings, one may also replace density by an average-degree constraint (e.g. $\frac{1}{|T|}\sum_{i\in T} \deg_{G_\tau[T]}(i) \geq d$) when ties are not expected to be fully meshed but are still expected to be richly redundant.*

Remark 1214. *The size-bound option $|T| \leq N$ can be read as a formal proxy for conversational and attentional limits (often discussed informally via "Dunbar-like" constraints), while the density option can be read as a proxy for repeated joint context (shared routines, shared reference frames, and repeated mutual updating). These are not competing metaphysical theses: they are alternative handles on the same modeling intent, and they can also be combined when one wants "small and dense" as the operational definition of tribal cohesion.*

Remark 1215. *In many real societies, tribes overlap: an agent may simultaneously belong to a family unit, a work cell, and a religious circle. The present definition treats a tribe as a subset $T \subseteq A$ and does not require a partition of A; consequently, different tribes may intersect, and the same agent may serve as a high-coupling "bridge" across multiple tribal contexts. When overlap is common, one can interpret G_τ as supporting a* cover *of A by connected, bounded, high-density sets, rather than a disjoint decomposition.*

Remark 1216 (Why tribes matter for later sections)**.** *Tribal substructure is where morphic resonance and cultural stabilization become most visible: high coupling and repeated interaction create strong reinforcement dynamics. In the later "mindplex" discussion, tribes are a natural scale at which notable coherence can arise.*

Remark 1217. *One practical reason to isolate tribes is that many dynamical claims (diffusion of habits, norm formation, convergence of belief-like variables, stabilization of shared symbols) have qualitatively different behavior on sparse-but-connected graphs versus dense, redundant graphs. Dense strong-tie subgraphs support multiple independent paths of reinforcement, making them less* *fragile to occasional edge failures, episodic disengagement, or the removal of a single highly connected individual. In this sense, the density/size boundedness condition can be read as a robustness surrogate: it aims to rule out "connected only through a single corridor" structures that behave more like chains of acquaintance than like tightly intertwined sub-societies.*

Remark 1218. *Hyperseed's mention of "morphic resonance" gestures at reinforcement across time and across instances; the present section uses it only as an intuition, not as an extra axiom. Where it is used as an explanatory frame, it is natural to connect it to [13] and to the formal habit and resonance machinery earlier in the document. In Hyperseed-Concept terms, see* Morphic Resonance *(Hyperseed-Concept 115) and* Autocatalytic Sets *(Hyperseed-Concept 61) for related self-reinforcement imagery.*

Remark 1219. *Finally, the thresholding perspective makes contact with standard community-detection intuitions without committing to a particular algorithm: tribes can be approximated by connected components of G_τ (when boundedness is enforced by N) or by searching for dense connected subgraphs inside G_τ (when boundedness is enforced by δ). In applications, one may choose τ by a measurement convention (e.g. "at least once per day"), by calibration to observed cohesion (e.g. self-reported trust), or by stability across nearby thresholds (selecting tribes that persist over a range of τ values as especially robust sub-societies).*

22.5 Communicative acts and communicative systems

Hyperseed treats communication as an action by one complex interactive system aimed at effecting a communicative act on another system, and emphasizes that communicative acts come in multiple logical forms (speech acts) and multiple physical forms. In particular, the same abstract message can be instantiated as spoken language, written text, gestures, signals embedded in artifacts, or machine-to-machine protocol events; the formal layer below is deliberately indifferent to these carrier modalities, treating them as different physical realizations of a shared state-transformer schema.

Our minimal formalization is built on the epistemic layer (Section 20). Fix a claim language $\mathcal{L}$ and recall the $\mathsf{p\text{-}bit}$ domain $\mathbb{V} = [0,1]^2$. The purpose of reusing $\mathbb{V}$ here is to make "communication" compatible with the same evidence geometry already used for learning, inference, and aggregation, so that receiving testimony is treated as one more evidence-producing interaction rather than as a separate logical primitive.

Remark 1220. *Here $\mathcal{L}$ is simply a set of well-formed "claims" (propositions, statements, hypotheses) about which agents can store and exchange evidence. We do not assume $\mathcal{L}$ is complete, consistent, or even closed under all classical connectives; the epistemic layer already allows paraconsistent inference and resource-sensitive closure. In communicative settings, this flexibility is essential: natural-language discourse rarely presents itself as a tidy Boolean algebra, and social actors routinely trade in partially formalizable claims (e.g. vague predicates, defeasible generalizations, or context-indexed statements) whose logical closure properties are not globally stable.*

Definition 350 (Messages as paraconsistent evidence bundles)**.** *A* message *is a finitely supported function*

$$m : \mathcal{L} \to \mathbb{V},$$

interpreted as "the sender is transmitting this bundle of positive/negative evidence about claims." Let $\mathsf{Msg}(\mathcal{L})$ denote the set of messages.

Remark 1221. *"Finitely supported" means that $m(\varphi) \neq (0,0)$ for only finitely many claims φ. This reflects the mundane constraint that each communicative act carries only bounded explicit content, even if it triggers further inferences in the recipient. The codomain $\mathbb{V} = [0,1]^2$ means that each claim receives two independent coordinates: positive evidence and negative evidence, as in the* p-bit *framework used throughout the epistemic and value layers. One can also regard a message as the "public" part of a richer communicative event: prosody, timing, source identity, and shared situational context are not encoded in m itself, but can be incorporated indirectly through how the recipient chooses the coupling k (or through auxiliary state variables in more detailed models).*

As a toy example, if $\mathcal{L}$ contains claims {"it will rain", "the bus is late"}, a message might assign $(0.8, 0.1)$ to "it will rain" and $(0.0, 0.0)$ to everything else, thereby asserting rain with mild doubt, and staying silent about the bus.

Definition 351 (Belief states and message assimilation)**.** *Let $\beta_j : \mathcal{L} \to \mathbb{V}$ be agent j's belief state. Given a coupling value $k \in [0,1]$ (or $k \in \mathbb{V}$), define the* assimilated message *$k \odot m$ by*

$$(k \odot m)(\varphi) := \begin{cases} (k,k) \otimes m(\varphi) & \text{if } k \in [0,1], \\ k \otimes m(\varphi) & \text{if } k \in \mathbb{V}, \end{cases}$$

where $\otimes$ is the quantale product on $\mathbb{V}$ from Section 3. Define the belief update

$$\beta_j \leftarrow \beta_j \oplus (k \odot m),$$

where $\oplus$ is the componentwise join on $\mathbb{V}$. Optionally, after receiving a message the agent applies an inference closure (Section 20.5) to propagate implications.

Remark 1222. *The operator $\odot$ here is not a new primitive; it is just notation for "scale the message by the coupling strength." If $k \in [0,1]$, we embed it as $(k,k) \in \mathbb{V}$ so that the same $\otimes$ operation can be used uniformly. The update $\beta_j \leftarrow \beta_j \oplus (k \odot m)$ then says: the recipient retains their existing evidence and also joins in whatever evidence arrives through the channel, discounted by coupling. Operationally, k can be read as a compressed stand-in for many social and cognitive factors: trust in the sender, perceived expertise, attention allocation, channel noise, and incentives. Allowing $k \in \mathbb{V}$ further permits asymmetric coupling—e.g. a recipient might treat the sender as reliable for positive testimony but unreliable for denials, or conversely discount affirmations while taking warnings seriously.*

A simple numerical example: if $k = 0.5$ and $m(p) = (0.8, 0.2)$, then $(k,k) \otimes m(p)$ yields a down-weighted evidence pair for p (depending on the exact $\otimes$ defined in the p-bit *quantale). Joining this into $\beta_j(p)$ accumulates evidence without requiring consistency. In richer settings one may also consider claim-dependent coupling $k(\varphi)$, or context-dependent coupling chosen as a function of the act type α and conversational state; the present formalization keeps k uniform to isolate the core mechanism.*

Remark 1223 (Paraconsistency is the default at social scale)**.** *Even if each individual is internally "mostly consistent" (in some approximate sense), the join of many individuals' evidence streams quickly yields conflicting evidence on many claims. Hyperseed's paraconsistent stance therefore becomes more natural, not less, as we scale up. The point is not merely that different agents disagree, but that a single recipient typically integrates heterogeneous sources whose evidential standards and error modes are not aligned, making some degree of contradiction an ordinary equilibrium condition rather than a pathology.*

Remark 1224. *Conceptually, this is the same reason classical logic is a poor description of large bodies of testimony: the world is large, observers are limited, and social incentives are mixed. Hyperseed treats this not as an anomaly but as a structural property of any large coupled epistemic system. The mathematical role of the* p-bit *domain is precisely to let contradiction be represented without allowing it to collapse the entire inferential apparatus; compare the paraconsistent framing in [24] for a different (but thematically related) use of non-classical structure. In communicative terms, this means that "he said, she said" does not force the system into triviality; instead it yields a state in which both positive and negative evidence coordinates can be simultaneously high, and downstream decisions can explicitly condition on that mixed evidential profile.*

Definition 352 (Communicative act types)**.** *A* communicative act type *is an intended state-transformer that uses a message as a vehicle. We use the following Hyperseed-aligned types:*

(a) Question*: aims to elicit an answer (a message or action) or to provoke inquiry in the recipient.*

(b) Response*: aims to answer a question by transmitting evidence or constraints.*

(c) Command*: aims to cause the recipient to carry out some particular action.*

(d) Statement*: aims to cause the recipient to possess a certain belief (evidence update).*

(e) Interjection*: aims to cause the recipient to possess a belief about the sender's emotional state or attitude (a value/affect update; Section 18).*

Formally, an act is a triple (i, j, α, m) *where* i *is sender,* j *is recipient,* α *is act type, and* m *is message content. The act type determines which internal variables of* j *are targeted (beliefs, goals, attention, affect), and how* m *is interpreted. In particular, the same evidence bundle may be processed differently depending on whether it is treated as an answer to a standing question, a freestanding assertion, or a socially binding directive; act types therefore function as a routing layer that selects the relevant update operators and any associated normative or procedural constraints.*

Remark 1225. *This definition separates* content *(the evidence bundle* m*) from* illocutionary force *(the act type* α*). That separation is useful because the same propositional content can be used as a statement, a hint, a threat, or a question depending on context; conversely, the same act type (question, command) can be realized with many different contents. In addition, it clarifies how communicative success and communicative uptake can diverge: the sender may intend a command while the recipient treats it as mere information, or the sender may pose a question whose primary function is to shift attention rather than to obtain new evidence. This gap between intended and realized state-transformation is one locus where social power, shared norms, and institutional roles enter the model (e.g. by modulating* k*, by constraining admissible responses, or by altering which state variables are allowed to be targeted by particular senders).*

Definition 353 (Communicative systems as coupled message dynamics)**.** *A* communicative system *over a population of agents* A *is given by (i) belief states* $\{\beta_i\}_{i \in A}$*, (ii) a set of admissible act types (as above), and (iii) coupling parameters* $\{k_{ij}\}_{i,j \in A}$ *(with* $k_{ij} \in [0,1]$ *or* $k_{ij} \in \mathbb{V}$*) specifying how strongly* j *assimilates messages from* i*. A communicative episode can then be modeled as a finite sequence of acts*

$$(i_1, j_1, \alpha_1, m_1), \ldots, (i_T, j_T, \alpha_T, m_T),$$

with each act inducing an update to the targeted state of the recipient (e.g. belief update via $\beta_{j_t} \leftarrow \beta_{j_t} \oplus (k_{i_t j_t} \odot m_t)$ *in the case of statements).*

Remark 1226. *This system-level view makes explicit that "communication" is not only a dyadic event but a networked process: repeated updates, feedback, and multi-party routing (broadcast, gossip, institutional reporting) can be represented by composing these elementary acts. It also highlights that coupling is typically directional ($k_{ij} \neq k_{ji}$) and role-sensitive, and that stability phenomena (e.g. convergence to shared high-positive evidence, polarization into incompatible evidence profiles, or persistence of mixed evidence) are properties of the induced dynamics rather than of any single message in isolation.*

Remark 1227. *As an example, "Close the window" is a command. "Could you close the window?" is syntactically a question, but pragmatically often functions as a command; in a richer model, that pragmatic mapping would live in the recipient-side update rules associated with α and the context encoded in K and attention variables. In particular, the same surface form can map to different illocutionary forces depending on relational context (e.g. power asymmetries, politeness conventions, shared projects), local attentional state, and the agent's estimate of channel costs (e.g. whether direct imperatives are socially penalized). Formally, this means that the decoding and update stages may depend not only on the message token but also on latent variables internal to the receiver (and, in multi-turn settings, on dialogue state), so that the "meaning" relevant for state change is a function of $(m, \beta_j, attention_j, K(i,j), \dots)$ rather than of m alone.*

Definition 354 (Communicative system)**.** *A* communicative system *on a society is a choice, for each ordered pair $(i,j) \in E$, of:*

- *a message alphabet/space Msg_{ij} (often a subset of $\mathsf{Msg}(\mathcal{L})$);*
- *encoding/decoding maps (possibly stochastic) that implement the physical channel; and*
- *update rules on the recipient (belief update, goal update, attention update, etc.).*

We write this data collectively as $\mathsf{Com} = (\mathsf{Msg}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Upd})$. It is important that the choice is made for each ordered *pair (i,j): communication can be asymmetric in both capacity and interpretation (e.g. agent i can reliably address j but not vice versa; or j treats i as authoritative while i discounts j). The dependence on (i,j) also lets the model represent institutionally mediated channels (moderation, bureaucracy, publication) as special cases of edges with distinctive $\mathsf{Enc}/\mathsf{Dec}$ and with characteristic filtering or amplification in Upd. Finally, allowing Msg_{ij} to differ across edges captures the fact that not all agents share the same vocabulary, protocol, or representational format, even when they participate in the same society-wide network.*

Remark 1228. *The communicative system Com is where "language" and "media" enter the formalism, but in a way that keeps us honest: communication is not magic, it is a channel with an encoding, a decoding, and a state-update. This is compatible with Shannon-style abstraction, but the state being updated is paraconsistent and value-sensitive rather than a single probability distribution. Put differently, Shannon theory idealizes the transmission of symbols; here we additionally specify what receiving a symbol* does *to an agent's cognitive state, and we do not assume that this effect can be reduced to a single scalar belief. The separation into Enc, Dec, and Upd also forces an explicit distinction between (i) channel errors (noise, loss, distortion), (ii) interpretive divergence (the same token mapped to different internal content), and (iii) normative or strategic response (the receiver may understand but decline to update, or may update attention/values without accepting propositional content).*

As a simple example, Enc could be "serialize evidence pairs into a JSON packet" and Dec could be "parse the packet," while Upd could be the join-update $\beta_j \leftarrow \beta_j \oplus (k \odot m)$. In face-to-face speech, Enc/Dec *are biophysical and socially trained processes, and* Upd *includes pragmatic inference and attention modulation (Section* **??***; see also the broader cognitive-systems framing in [19]). Note that even in the "JSON packet" case, the decoding stage typically includes a schema agreement (what fields mean) and a trust calibration (how much weight to place on the packet), both of which can be modeled either as part of* Dec *(mapping bytes to structured internal objects) or as part of* Upd *(mapping structured objects to state change). Likewise, the weighting factor k can be interpreted as folding into a single scalar multiple heterogeneous features such as channel reliability, source credibility, alignment of incentives, and current cognitive load; making these components explicit yields richer update operators without changing the overall architecture.*

A further consequence is that communication protocols (turn-taking rules, quoting, forwarding, cryptographic signing, moderation queues) can be represented by enriching Enc/Dec *and by letting* Upd *depend on message metadata (timestamps, signatures, provenance, thread position). This matters in collective settings because many social phenomena (rumor cascades, institutional trust, polarization) are driven at least as much by these protocol-level variables as by the propositional content transmitted.*

Physical realizations. Hyperseed emphasizes that the same logical act types can be physically realized in different ways. We therefore treat physical forms as implementations of Enc/Dec: the same act type α (e.g. asserting p, requesting an action, signaling approval) may be instantiated as a spoken utterance, a gesture, a written note, or a machine-readable command, with potentially different error profiles and with different degrees of persistence and publicness. This perspective also makes it natural to treat "translation" (between languages or modalities) as composition of encoding/decoding maps, where translation quality becomes part of the effective noise model and can be learned or institutionalized.

- gesture (iconic/indexical signaling),
- "Hmmmmm" (holistic, manipulative, multi-modal, musical, mimetic signaling),
- spoken language,
- written language,
- social language (used inside a society),
- natural language (evolved by use),
- constructed language (engineered, e.g. Lojban),
- programming language (communication with a machine; see Section 23).

We will not formalize each form separately; instead we treat them as different channel families with different noise models, bandwidth, grounding constraints, and habit formation dynamics. Here "bandwidth" includes not only raw symbol rate but also effective expressive capacity under shared conventions; "grounding constraints" include the degree to which signals are anchored to shared perception/action loops (e.g. pointing in a shared environment versus reference in a text-only forum); and "habit formation dynamics" covers how repeated use of a channel changes both availability (ease of production) and interpretation (default pragmatic enrichments). This is also where latency and synchronization enter: some channels support rapid repair (immediate clarification) while others require asynchronous clarification, which alters the feasible structure of multi-step coordination.

Remark 1229. *This list is not merely anthropological; it matters for the mathematics because different physical realizations induce different effective couplings $K(i, j)$ and different update operators* Upd*. Written language increases persistence (artifact-layer memory), programming languages couple agents to machines (Section 23), and holistic/affective signaling can primarily update value and attention variables rather than propositional belief states. In Hyperseed-Concept terms, this is one place where* Representation *(Hyperseed-Concept 157) and* Attention / Attentional Focus *(Hyperseed-Concept 60) become inseparable from "purely epistemic" content. Persistence changes the temporal structure of interaction: stored messages can be revisited, quoted, and aggregated, which effectively increases the time horizon over which updates compound and can create path dependence in β_j (e.g. via selective retrieval). Publicness changes the audience model: a message in a broadcast medium may induce different updates because the recipient infers not only what the sender intends them to know, but also what others are likely to know, affecting coordination, reputational incentives, and second-order beliefs.*

Moreover, modalities differ in how much they privilege different state components: a terse textual assertion may mostly target propositional belief, while tone, rhythm, facial expression, or musical mimesis may more directly target attention allocation and value salience. Mathematically, this can be reflected by allowing Upd *to be anisotropic in the space of state variables (large effects on attention/value coordinates, smaller effects on propositional coordinates), and by allowing $K(i, j)$ to represent not just "how strongly i influences j" but which* channels of influence *(belief, goal, affect, attention) are most strongly coupled for that edge.*

Finally, treating "social language," "natural language," and "constructed language" as distinct realizations highlights that Com *is partly a learned, collectively maintained technology. A constructed language may reduce syntactic ambiguity (altering* Dec*) but increase adoption cost (altering effective K via reduced usage); a natural language may maximize ease and coverage at the expense of ambiguity, pushing disambiguation into pragmatic components of* Upd*. These tradeoffs are central when modeling how communicative conventions co-evolve with institutions and with collective cognitive constraints.*

22.6 Universal Grammar as a weakest interface theory

Hyperseed treats *distinction* as primitive and observer-relative (Hyperseed-Concept 98, 86), and treats *weakness* as the controlled collapsing of distinctions not needed for a given purpose (Hyperseed-Concept 202, 143, **??**). The UG manuscript can be read as an instance of this program: "Universal Grammar" is modeled as the *weakest* operational congruence on grammar derivations that preserves a chosen *interface observation layer* (LF/PF-style invariants), up to (i) swaps of independent actions and (ii) declared symmetries (parametric equivalences).

In this section we introduce a Hyperseed-aligned formal package for this idea: Trace-based Universal Grammar (TUG) (Hyperseed-Concept 221). The core ingredients are: an observation interface (Hyperseed-Concept 222), an independence relation on visible grammatical actions (Hyperseed-Concept 225), and a symmetry group on visible labels (Hyperseed-Concept 226).

22.6.1 Grammar dynamics as a process on representations

Grammars are treated here as *engineered* cultural artifacts (Hyperseed-Concept **??**, 91), but their mathematical core is a process on representations (Hyperseed-Concept 157, 184).

Definition 355 (Grammar state space: pattern plus canonical store)**.** *Let $\mathcal{T}$ be a set of syntactic term-trees ("patterns") and let Σ be a set of finite "interface atoms" representing commitments* *relevant to semantic and/or phonological interpretation. A* grammar state space *is*

$$\mathrm{St} := \mathcal{T} \times \Sigma.$$

Elements $s = (t, \sigma) \in \mathrm{St}$ *are viewed as a representation token of syntactic structure* t *together with a canonicalized store* σ *that retains the interface-relevant residue of derivational choices (Hyperseed-Concept 223).*

Definition 356 (Grammar labeled transition system (LTS))**.** *A* grammar LTS *is a triple*

$$G = (\mathrm{St}, \mathrm{Lab}, \to)$$

where St *is as in Definition 355,* Lab *is a set of labels, and* $\to \subseteq \mathrm{St} \times \mathrm{Lab} \times \mathrm{St}$ *is a transition relation. We write* $s \xrightarrow{\ell} s'$ *for* $(s, \ell, s') \in \to$*. A* derivation *is a finite path*

$$\pi : \quad s_0 \xrightarrow{\ell_1} s_1 \xrightarrow{\ell_2} \cdots \xrightarrow{\ell_n} s_n.$$

22.6.2 Observation interfaces (LF/PF) as context-selected distinctions

The UG manuscript treats "what the grammar must preserve" as an *observation interface* (LF-only vs LF+PF, or other invariants). In Hyperseed terms, this is a context/aspect choice that selects which distinctions count as live (Hyperseed-Concept 86, 98).

Definition 357 (Observation interface)**.** *An* observation interface *is a set* $\mathcal{O}$ *together with a map*

$$\mathrm{obs}_{\mathcal{O}} : \mathrm{St} \to \mathcal{O}.$$

Two standard examples are:

- *LF-only:* $\mathrm{obs}_{\mathrm{LF}}(t, \sigma) := \sigma_{\mathrm{LF}}$.
- *LF+PF:* $\mathrm{obs}_{\mathrm{LF+PF}}(t, \sigma) := (\sigma_{\mathrm{LF}}, \sigma_{\mathrm{PF}})$.

Given a fixed start state s_0*, two derivations* π, π' *from* s_0 *are* $\mathcal{O}$-observationally equivalent*, written* $\pi \equiv_{\mathcal{O}} \pi'$*, if their endpoints have the same observation:* $\mathrm{obs}_{\mathcal{O}}(s_n) = \mathrm{obs}_{\mathcal{O}}(s'_m)$*.*

22.6.3 Visible vs internal actions; resources and independence

Following standard process-algebra practice, we distinguish administrative normalization steps (internal, τ-like) from meaning-bearing choices (visible). Independence is the formal locus of "can commute without changing the interface."

Definition 358 (Visible vs internal labels)**.** *Assume a disjoint partition*

$$\mathrm{Lab} = \mathrm{Lab}_V \uplus \mathrm{Lab}_\tau,$$

where Lab_τ *are internal (administrative) labels and* Lab_V *are visible (meaning-bearing) labels. The* visible trace *of a derivation* π *is the word* $\mathrm{tr}_V(\pi) \in \mathrm{Lab}_V^*$ *obtained by deleting* Lab_τ*-labels.*

Definition 359 (Positions, resources, and rely footprints)**.** *Let* $\mathrm{Pos} := \mathbb{N}^*$ *be the set of finite sequences of naturals (tree positions), with length* $|p|$*. Let* Res *be a finite set of abstract resources. Assume maps*

$$\mathrm{pos} : \mathrm{Lab}_V \to \mathrm{Pos}, \qquad \mathrm{rely} : \mathrm{Lab}_V \to 2^{\mathrm{Res}}.$$

Intuitively, $\mathrm{rely}(\ell)$ *is the set of semantic/interface "slots" touched by* ℓ *(Hyperseed-Concept 224).*

Definition 360 (Disjoint positions). *For $p, q \in \mathrm{Pos}$, write* $\mathrm{disjointPos}(p, q)$ *if neither is a prefix of the other.*

Definition 361 (Grammatical independence). *Define a binary relation $\perp$ on Lab_V by*

$$\ell \perp \ell' \quad :\Longleftrightarrow \quad \mathrm{disjointPos}(\mathrm{pos}(\ell), \mathrm{pos}(\ell')) \;\; \textit{and} \;\; \mathrm{rely}(\ell) \cap \mathrm{rely}(\ell') = \emptyset.$$

When $\ell \perp \ell'$, the intended operational reading is that the two visible actions act on disjoint subtrees and disjoint interface resources, so their order is interface-irrelevant.

22.6.4 Symmetries and trace quotienting

Parametric variation can be modeled as *declared symmetries* on visible labels: differences that (by stipulation) do not matter to the chosen interface. This is an explicit instance of Hyperseed's equivalence-via-indistinction stance (Hyperseed-Concept **??**, 98).

Definition 362 (Label symmetry group). *A* label symmetry group *is a group Γ acting on Lab_V. We require that symmetries preserve observation at the interface level in the following sense: for any $\gamma \in \Gamma$ and any derivation step $s \xrightarrow{\ell} s'$ with $\ell \in \mathrm{Lab}_V$, there exists a step $s \xrightarrow{\gamma \cdot \ell} s''$ such that* $\mathrm{obs}_{\mathcal{O}}(s') = \mathrm{obs}_{\mathcal{O}}(s'')$.

Definition 363 (Trace quotient and TUG congruence). *Let $\approx_\perp$ be the least congruence on Lab_V^* generated by swaps of adjacent independent labels:*

$$\alpha\, \ell\, \ell'\, \beta \;\approx_\perp\; \alpha\, \ell'\, \ell\, \beta \quad \textit{whenever} \quad \ell \perp \ell'.$$

Let $\approx_\Gamma$ be the least congruence generated by pointwise symmetry replacements:

$$\alpha\, \ell\, \beta \;\approx_\Gamma\; \alpha\, (\gamma \cdot \ell)\, \beta \quad \textit{for any} \quad \gamma \in \Gamma.$$

Define the TUG congruence $\approx_{\perp,\Gamma}$ *as the least congruence containing both $\approx_\perp$ and $\approx_\Gamma$. The resulting quotient monoid $\mathrm{Lab}_V^*/\approx_{\perp,\Gamma}$ is the* trace quotient *induced by $(\perp, \Gamma)$ (Hyperseed-Concept 227).*

Definition 364 (Trace-based Universal Grammar (TUG)). *Fix a grammar LTS $G = (\mathrm{St}, \mathrm{Lab}, \rightarrow)$ (Definition 356), an observation interface $(\mathcal{O}, \mathrm{obs}_{\mathcal{O}})$ (Definition 357), an independence relation $\perp$ (Definition 361), and a label symmetry group Γ (Definition 362). The* Trace-based Universal Grammar *induced by these choices is the package*

$$\mathrm{TUG} \;:=\; (\mathcal{O}, \perp, \Gamma)$$

together with the induced equivalence on derivations from a start state s_0:

$$\pi \equiv_{\perp,\Gamma} \pi' \quad :\Longleftrightarrow \quad \mathrm{tr}_V(\pi) \approx_{\perp,\Gamma} \mathrm{tr}_V(\pi').$$

This is a Hyperseed-style "weak" theory: it identifies derivations that differ only by reordering independent actions and/or applying interface-preserving symmetries. We call $\equiv_{\perp,\Gamma}$ the UG core congruence.

Remark 1230 (Principles and parameters, Hyperseed reading). *In this formalization, "parameters" can be represented by (i) changing what is observable at the interface (changing $\mathcal{O}$), and/or (ii) changing the symmetry group Γ (declaring more or fewer label-identifications). "Principles" are those constraints that survive these changes, i.e., invariants of the dependence/independence topology and the resulting trace quotient. This is a concrete instantiation of Hyperseed's theme that what looks like a "law" is often an invariance forced by a choice of active distinctions (Hyperseed-Concept 86, 202, 169).*

22.6.5 Helper definitions for economy and locality (used later)

Universal Grammar theory derives economy and locality consequences from the above substrate. We record two definitions that will be referenced by later theorems.

Definition 365 (Movement labels and structural distance)**.** *A* movement label *has the form* $\mathrm{Move}(x, p_{\mathrm{from}}, p_{\mathrm{to}})$*, where x is the moved element and* $p_{\mathrm{from}}, p_{\mathrm{to}} \in \mathrm{Pos}$ *are tree positions. Define the* structural distance

$$d(\ell) \;:=\; |p_{\mathrm{to}}| - |p_{\mathrm{from}}| \quad \textit{for} \quad \ell = \mathrm{Move}(x, p_{\mathrm{from}}, p_{\mathrm{to}}).$$

(Interpretation: depth difference, or more generally the number of phase-like domain boundaries crossed, if such resources are included in Res*.)*

Definition 366 (Redundant visible step)**.** *Fix an observation interface* $(\mathcal{O}, \mathrm{obs}_{\mathcal{O}})$*. A visible label* $\ell \in \mathrm{Lab}_V$ *occurring in a derivation* π *is* redundant *if deleting that occurrence from the visible trace yields a derivation* π' *with the same observed endpoint:*

$$\pi \equiv_{\mathcal{O}} \pi'.$$

Equivalently: ℓ contributes no distinction at the chosen interface (Hyperseed-Concept 202, 222).

22.7 Social roles as behavioral templates

Hyperseed lists concrete (teacher, student, worker, colleague) and treats them as patterns of activity within a society. We formalize roles as *interaction templates* that can be matched against observed interaction subgraphs or sequences. In particular, the intent is that a role is neither merely a verbal label nor merely a statistical cluster, but a structured hypothesis about which participants tend to occupy which positions and which kinds of communicative or artifact-mediated acts tend to connect them.

Definition 367 (Role templates as small labeled graphs)**.** *Fix a set of act types* $\mathcal{T}$ *(question/response/command/sta and a set of artifact operations (read, write, modify, use). A* role template *is a finite labeled directed graph*

$$R = (V_R, E_R, \lambda),$$

where V_R *are abstract participant slots (e.g. "teacher", "student") and* $\lambda : E_R \to \mathcal{T} \cup \{\textit{artifact ops}\}$ *labels edges by the act type (or artifact operation) that should occur between the corresponding slots.*

Remark 1231. *The definition is intentionally minimal: it captures only who interacts with whom and what kind of act (or artifact operation) mediates the interaction. More elaborate variants can be obtained by enriching* λ *to include parameters such as (i) expected frequency ranges, (ii) permissible time lags, (iii) polarity or valence (support vs. opposition), or (iv) context guards specifying when an edge is considered "active." These enrichments are not required for the basic template-matching story, but they are often helpful when roles are temporally structured (e.g. meetings, lessons, reviews) rather than purely relational.*

Remark 1232. *The separation between act types* $\mathcal{T}$ *and artifact operations is meant to keep visible the common empirical fact that many social roles are stabilized by recurring interactions with shared objects: documents, tools, codebases, instruments, checklists, ledgers, and so on. In this sense, the artifact edges act as a coarse proxy for "institutional memory" and coordination infrastructure: the template can require not only talk but also patterned reading/writing/modifying behavior that persists beyond any single utterance.*

Remark 1233. *The point of the template R is to represent a role as something like a small "program" of interaction: who speaks to whom, in what mode, and with what recurring artifact usage. This is deliberately compatible with earlier template-pattern machinery (Section 20.2), now applied not to objects in a universe X but to social interaction slots and labeled edges.*

For example, a minimal "teacher–student" template might have two slots and edges labeled (question from student to teacher) and (statement/response from teacher to student), plus an artifact edge (student reads textbook). A "colleague" template might have symmetric coordination edges and shared artifact writes. In Hyperseed-Concept terms, this is precisely the formalization of Social Roles *(Hyperseed-Concept 171).*

Remark 1234. *When we say "subgraph or sequence," we are implicitly allowing two observational regimes: (i) interaction episodes that are best treated as a time-ordered trace (messages/events with timestamps), and (ii) episodes that are best treated as an aggregated graph (counts or existence of certain act types over a window). The same template can be matched in either regime, but the notion of matching may respect order in the sequential case and ignore order in the aggregated case; this distinction becomes important for roles like "interviewer–candidate" where the order of question/response is part of the role itself.*

Definition 368 (Role instantiation and fit score)**.** *Let $\mathsf{Soc} = (A, G, K, \mathcal{A}, \mathrm{Acc}, \mathcal{D})$ be a society and let R be a role template. An* instantiation *of R is a map $\rho : V_R \to A$ assigning each slot to an agent.*

Given an observed interaction episode Ep (a finite subgraph or sequence of acts in the society), define a fit score

$$\mathrm{Fit}(R, \rho; \mathsf{Ep}) \in [0, 1]$$

measuring how well the labeled edges of R match edges/acts in Ep under the assignment ρ, weighted by couplings K and (optionally) attention weights from Section **??***. A role is considered* present *in* Ep *if* Fit *exceeds a chosen threshold.*

Remark 1235. *Although* Fit *is left abstract, it is useful to keep in mind a few canonical constructions. For graph-episodes, one can define* Fit *via a normalized maximum-weight subgraph matching score between the required labeled edges of R and the observed labeled edges in Ep under ρ. For sequence-episodes, one can define* Fit *via an edit-distance-style alignment between the template's expected act-types and the observed act-types, allowing skips and substitutions with penalties. In either case, couplings K can weight edges by social importance (e.g. trust, authority, or bandwidth), and attention weights can downweight interactions that are present but not actually processed or remembered.*

Remark 1236. *The threshold is not meant to be universal: different analytical tasks motivate different operating points. For detection and monitoring, a lower threshold may be appropriate to avoid false negatives and to support early warnings ("a mentoring relationship might be forming"). For attribution or policy decisions, a higher threshold may be appropriate to avoid false positives ("this episode genuinely counts as instruction/command/coordination"). In learning settings, the threshold can itself be tuned to optimize downstream predictive accuracy of role-labeled episodes.*

Remark 1237. *The map ρ is the social analogue of a template instantiation: it plugs concrete agents into abstract slots. The fit score then recognizes that social reality is noisy and approximate: roles are rarely perfectly instantiated, yet they can be real in the sense of being stable, predictive patterns.*

For instance, if Alice asks Bob many questions and Bob provides many answers that change Alice's skills, the teacher–student template has high fit for $\rho(\text{teacher}) = \text{Bob}$, $\rho(\text{student}) = \text{Alice}$. If *the interaction is symmetric, the colleague template may have higher fit. The thresholding step is a way of turning graded pattern matching into a crisp judgment when needed.*

Remark 1238. *This formalism also allows role ambiguity and role drift to be represented quantitatively: one can track* $\mathrm{Fit}(R, \rho; \mathsf{Ep}_t)$ *over time windows* t *and observe when a relationship changes category (e.g. student becomes collaborator) or oscillates between competing descriptions depending on topic. In this sense, roles become time-varying hypotheses about interaction structure rather than static labels, which aligns with the document's emphasis on adaptive agents and evolving societies.*

Remark 1239 (Approximate morphism viewpoint)**.** *The fit score is a lightweight proxy for a more categorical statement: "an episode implements the role template up to approximation." If one models episodes as morphisms in an enriched category of processes, then role instantiation becomes an approximate morphism (Sections* **??** *and 12).*

Remark 1240. *This viewpoint is useful because it makes roles compositional: a role template can be nested inside a larger template, and episodes can partially implement multiple roles simultaneously (often with conflict). That, again, is paraconsistency in social form: one can be simultaneously "teacher" and "student" in different subtopics, or simultaneously "colleague" and "competitor."*

Remark 1241. *Composition also highlights that some roles are inherently multi-slot and cannot be reduced to pairwise dyads without loss. For example, "manager" and "team member" are often mediated by shared artifact operations (planning documents, task boards) and by broadcast-style communication patterns (one-to-many statements, many-to-one reports). Such patterns can be captured by including additional slots (e.g. multiple workers) or by allowing the same template to match repeatedly with different* ρ *assignments to represent a group-level role configuration.*

We now specialize to the roles Hyperseed highlights.

Definition 369 (Teacher and student (operational form))**.** *Fix a claim language* $\mathcal{L}$ *and an update scheme for messages as in Section 22.5. An agent* i *plays the role of* teacher *for an agent* j *over an episode if:*

- i *repeatedly performs statement/response acts toward* j *with messages that (when assimilated) reduce* j*'s weakness/uncertainty on some task-relevant question family (Sections 20 and 21); and*
- j *performs question acts toward* i *and updates its beliefs/skills so that its downstream performance on relevant tasks improves (at fixed or reduced effort).*

An agent j *plays the role of* student *relative to* i *if it exhibits the complementary pattern: question/attention allocation plus sustained assimilation leading to capability gain.*

Remark 1242. *The definition intentionally allows many concrete pedagogical modalities: direct explanation, worked examples, corrective feedback, Socratic questioning (where the teacher's "statements" are themselves structured prompts), and even purely artifact-mediated instruction (e.g.* i *curates a document or tooling that* j *reads/uses). The key requirement is not the surface form of the interaction but the causal effect on* j*'s measurable competence and uncertainty proxies under the assumed update scheme.*

Remark 1243. *This is an operational definition: it identifies "teacher" not by credential but by measurable effect on the student's epistemic and task competence variables. The phrase "reduce weakness/uncertainty" connects directly to the document's resource-sensitive simplicity and weakness machinery (see also [2, 3] for related motivations for treating weakness as a primary quantity).*

It also ties pedagogy to control: teaching is a way of shaping another agent's future capability landscape.

As a simple example, if j is learning to solve a class of tasks and i supplies messages that let j prune the hypothesis space (Section 20), then i is a teacher in precisely the sense that matters for later cultural stabilization: patterns of explanation are being transmitted and absorbed.

Remark 1244. *The emphasis on downstream performance "at fixed or reduced effort" is meant to rule out a purely illusory improvement (e.g. short-term compliance without internalization) and to distinguish teaching from mere delegation. In practice, one can operationalize this via before/after evaluation on a task distribution, or via a reduction in the number of steps/queries j requires to achieve a comparable success rate. This keeps the role definition aligned with the broader document theme that competence is a resource-bounded quantity rather than a purely declarative one.*

Definition 370 (Worker and colleague (operational form))**.** *A* worker *is an agent that repeatedly executes commands or participates in a coordinated plan issued by some coordinating structure (a boss, institution, or collective policy), in exchange for resources or other value signals (Section 18).*

Two agents are colleagues *(in a given interval/episode) if they are co-participants in a collaborative project: a sustained multi-agent pattern of activity aimed at a shared goal, with repeated coordination acts and sharing of intermediate results via communication or artifacts.*

Remark 1245. *The worker definition emphasizes asymmetry of command and exchange of resources/value, whereas the colleague definition emphasizes symmetry of participation in a shared project. Neither is inherently ethical or unethical; they are structural roles describing coordination topology and value flow.*

A concrete example: in a software team, an engineer may be a worker relative to an organizational planner (commands via tickets), but also a colleague relative to peers (shared design discussions and co-authored artifacts). In Hyperseed-Concept terms, Worker *(Hyperseed-Concept 205) and* Colleague *(Hyperseed-Concept 74) are best seen as recurring interaction templates rather than as intrinsic properties of persons.*

22.8 Culture as a stabilized intersubjective pattern web

Hyperseed defines culture as a system of patterns of external and cognitive action associated with a society. Mathematically, we treat culture as a population-level stabilization of certain patterns and procedures, maintained by communication, artifacts, and habit dynamics.

Remark 1246. *This subsection makes precise a familiar but subtle thought: culture is neither merely "inside" minds nor merely "outside" in artifacts, but a stabilized coupling between the two. In more process-oriented language (in the spirit of [15]), culture is a pattern of ongoing reproduction: a process that continually re-instantiates itself through communication, imitation, and institutional constraint.*

Population-level pattern intensity. Let $\mathcal{P}$ be a set of patterns (Section 9), which may include: behavioral scripts, norms, language constructions, rituals, institutions, and shared explanatory models. Assume each agent $i \in A$ carries a (possibly fuzzy) pattern intensity function

$$I_i : \mathcal{P} \to [0,1] \quad \text{or} \quad I_i : \mathcal{P} \to \mathbb{V}.$$

(Concrete constructions were given earlier via compositional simplicity and property-sets.)

Remark 1247. *Here $I_i(P)$ is best read as: "to what degree does pattern P actually occur in, guide, or constrain agent i?" The scalar case $[0,1]$ treats endorsement and opposition as one dimension; the $\mathbb{V}$ case allows a society to carry patterns that are both enacted and resisted, a common phenomenon in norms and ideologies.*

For example, P might be a greeting ritual. Then $I_i(P)$ can be high if i usually performs it and expects it. Or P might be an explanatory model ("germs cause disease"); then $I_i(P)$ measures the extent to which the model shapes inference and action. This connects back to the pattern-web framing of [5].

Definition 371 (Cultural state)**.** *A* cultural state *for a society is a population-level intensity assignment*

$$C : \mathcal{P} \to [0,1] \quad or \quad C : \mathcal{P} \to \mathbb{V},$$

defined as an aggregation of individual intensities and artifact-embedded intensities, e.g.

$$C(P) := \Big(\bigoplus_{i \in A} w_i \odot I_i(P)\Big) \oplus \Big(\bigoplus_{a \in \mathcal{A}} u_a \odot I_a(P)\Big),$$

where w_i, u_a are weights (importance, population fractions, accessibility), and $\odot$ is scalar or p-bit *scaling as in Section 22.5. A* culture *at threshold θ is the set*

$$\mathrm{Cult}_\theta := \{P \in \mathcal{P} : \pi(C(P)) \geq \theta\},$$

where π is a chosen projection $\mathbb{V} \to [0,1]$ if needed.

Remark 1248. *The aggregation formula has two terms: one from agents and one from artifacts. This matters because some cultural patterns live primarily as habits in minds, while others live primarily as procedures encoded in institutions or tools (e.g. bureaucratic forms, legal codes, software). The join $\oplus$ indicates that culture, in this minimal model, records the presence of strong instances anywhere in the population or artifact layer, rather than averaging them away.*

As a concrete example, consider a society where only a small minority actively practices a ritual, but the ritual is also embedded in a calendar artifact and regularly displayed in public spaces. Then the artifact term can keep $C(P)$ high even if the mean individual intensity is low.

Remark 1249 (Paraconsistent cultures)**.** *If $C(P) \in \mathbb{V}$, culture can include patterns that are simultaneously strongly supported and strongly opposed in the same society. This matches the empirical fact that many cultures carry internal contradictions (norm conflicts, contested narratives, ambivalent values) without collapsing into triviality.*

Remark 1250. *The usefulness of defining* Cult_θ *is that it gives a crisp object on which later mathematics can act: we can ask about attractors, perturbations, and transitions as θ changes, or as coupling weights K change. In Hyperseed-Concept terms, this is one formal route to the* Culture *concept (Hyperseed-Concept 91) without presupposing any specific anthropological taxonomy.*

Stabilization via social habit dynamics. To model cultural stabilization, we adopt the simplest monotone reinforcement dynamics that mirrors the habit and morphic resonance constructions (Section 12).

Let $\mathcal{P}$ be finite for the theorem below (or restrict attention to a finite relevant subset).

Definition 372 (Social reinforcement operator)**.** *Fix a society with couplings $K : E \to [0,1]$ (scalar case for simplicity) and let $I \in [0,1]^{A \times \mathcal{P}}$ be the matrix of agent pattern intensities.*

Define $F : [0,1]^{A\times\mathcal{P}} \to [0,1]^{A\times\mathcal{P}}$ *by*

$$F(I)_{i,P} := I_{i,P} \ \vee \bigvee_{(j,i)\in E} \big(K(j,i) \wedge I_{j,P}\big),$$

where $\vee$ *and* $\wedge$ *are max/min on* $[0,1]$. *(Equivalently, one can use the* $\mathbb{V}$ *quantale operations from Section 3; we state the scalar version to keep notation light.)*

Remark 1251. *The operator* F *says: agent* i *keeps whatever intensity it already has for pattern* P*, and it also acquires (up to min with coupling strength) whatever intensity for* P *is present in its incoming neighbors. In other words, the strongest pattern instance among trusted neighbors "lifts"* i*'s intensity upward, but never beyond the channel capacity* $K(j,i)$*. A toy example: if* i *has intensity* 0.2 *for a slang term* P*, and a neighbor* j *has* $I_{j,P} = 0.9$ *with coupling* $K(j,i) = 0.6$*, then* j *can pull* i *up to at least* $\min(0.6, 0.9) = 0.6$ *(and then the* $\vee$ *with the old* 0.2 *yields* 0.6*). Repeating this across the network yields cultural spread. One can read* $\min(K(j,i), I_{j,P})$ *as a "bottlenecked transmission": even if* j *uses the term very strongly, the influence on* i *is capped by the strength of their social channel, and even if the channel is strong, nothing stronger than* j*'s own intensity can be conveyed. The outer* $\vee$ *(max) then implements a pure reinforcement assumption: the updated intensity at* i *is whatever is larger between what* i *already had and what the neighborhood can support.*

Theorem 20 (Existence of stabilized cultural states as fixed points)**.** *Assume* A *and* $\mathcal{P}$ *are finite. Then:*

(a) The operator F *is monotone on the complete lattice* $[0,1]^{A\times\mathcal{P}}$ *under pointwise order.*

(b) Hence F *has fixed points; the least fixed point above any initial intensity matrix* $I^{(0)}$ *is given by*

$$I^{(\infty)} := \bigvee_{n\geq 0} F^n(I^{(0)}),$$

where the supremum is taken pointwise.

(c) If intensities are restricted to a finite grid (e.g. values in $\{0, 1/N, \ldots, 1\}$*), then the ascending chain stabilizes after finitely many steps, yielding a computable stabilized state.*

Remark 1252. *Intuitively, the theorem says that if cultural transmission is "only reinforcing" (never decreasing intensity) and if we have only finitely many agents and finitely many patterns under consideration, then iterating social influence must converge to a stable regime. There is nothing mystical here: it is the general logic of monotone dynamics on a complete lattice. Yet it is important because it shows that "cultural fixed points" are not an extra hypothesis; they are what monotone reinforcement* must *produce.*

A helpful way to parse the statement is to separate three layers of content. First, monotonicity formalizes the idea that strengthening any agent's commitment to any pattern cannot (by the model's rules) make anyone else less *committed after updating. Second, fixed points correspond to "culturally self-consistent" states: applying the transmission rule* F *does not change the intensity matrix, so every agent already reflects all reinforcement available from its neighbors. Third, the expression* $I^{(\infty)} = \bigvee_{n\geq 0} F^n(I^{(0)})$ *shows that the stabilized state is obtained by accumulating all reinforcement that can propagate through the network over arbitrarily many steps, including indirect influence (friends-of-friends, etc.) as represented by repeated iteration.*

This connects to other parts of the document in a direct way. Earlier sections introduced habits and morphic resonance as reinforcement-like dynamics; here we show that, at social scale, the *same mathematical shape yields stable cultural states. Fixed-point reasoning also reappears later in goal and self-modification analysis; compare [10] for a related application of fixed-point theorems to stability.*

It is also worth noting what is not *assumed. We do not require any particular geometry of the social graph beyond the existence of the couplings $K(j,i)$, and we do not assume symmetry ($K(j,i)$ may differ from $K(i,j)$). The result is therefore robust under directed influence, highly heterogeneous connectivity, and strong clustering; these features affect* which *fixed point is reached from $I^{(0)}$, not the existence of some stabilized state under monotone updating.*

Proof. Monotonicity follows because max/min are monotone in each argument and the pointwise join preserves monotonicity. The space $[0,1]^{A\times\mathcal{P}}$ is a complete lattice under pointwise order, so by Knaster–Tarski, F has fixed points and the least fixed point above $I^{(0)}$ is the supremum of the ascending Kleene chain $F^n(I^{(0)})$.

Spelling out the first sentence more explicitly: if $I \le I'$ pointwise, then for each agent–pattern pair (i,P) and each neighbor j, we have $I_{j,P} \le I'_{j,P}$, hence $\min(K(j,i), I_{j,P}) \le \min(K(j,i), I'_{j,P})$. Taking the supremum (or maximum, since finiteness of neighborhoods is typical in network settings) over all j preserves the inequality, and finally taking the max with the old value $I_{i,P}$ preserves it again. Therefore $F(I) \le F(I')$ pointwise.

If the value set is finite, the ascending chain cannot strictly increase forever (there are finitely many possible matrices), so it stabilizes in finitely many steps at a fixed point. In this discretized setting one can also bound the runtime crudely by the number of possible states: $(N+1)^{|A||\mathcal{P}|}$ is an absolute upper bound on the number of distinct matrices on the grid $\{0, 1/N, \ldots, 1\}$, so stabilization occurs no later than that many iterations (though in practice one typically stabilizes much sooner). The substantive point for the present section is not computational efficiency but the existence of a well-defined "end state" obtained by repeated reinforcement. □

Proof sketch. The strategy is to recognize $[0,1]^{A\times\mathcal{P}}$ (with pointwise $\le$) as a complete lattice and to notice that F is built entirely from monotone operations (max, min, and sup over neighbors), hence is monotone. Monotone endomaps on complete lattices have fixed points by Knaster–Tarski, and the least fixed point above a starting state is obtained by iterating F and taking the supremum of the resulting ascending chain (Kleene iteration).

The key step is monotonicity: if we increase any entry of I, then every term of the form $K(j,i) \wedge I_{j,P}$ can only increase, and taking $\vee$ over these terms preserves the increase. Geometrically, one can picture each $I_{i,P}$ as a "fluid level" that can only rise, with incoming neighbors pushing it up subject to channel caps $K(j,i)$; the fixed point is the state where no further pushes can raise any level.

Another useful mental model is reachability under attenuation: repeated application of F allows an intensity at some source agent to propagate along directed edges, but at each step it is clipped by the coupling on that edge. The eventual fixed point therefore encodes, for each (i,P), the best support for P that can arrive at i through any chain of influences starting from the initial configuration, combined with i's own initial holding (since reinforcement never erases the past in this toy setup). This is one reason the least fixed point above $I^{(0)}$ is the natural stabilized state to associate with the dynamics: it represents the minimal commitments consistent with all possible reinforcements generated from $I^{(0)}$. □

Remark 1253 (Interpretation)**.** *Theorem 20 is not meant as a realistic sociological model. Its role is structural: it shows that once we treat social transmission as a monotone reinforcement process (habit-taking across agents), stable cultural regimes arise automatically as fixed points. More detailed models (with decay, anti-resonance, bounded attention, and competing patterns) can* *be built by modifying F while preserving monotonicity or by switching to stochastic dynamics with attractors.*

In particular, adding decay typically breaks monotonicity (since an update may reduce intensities), and then one must use different tools (e.g. contraction mappings, Lyapunov functions, or Markov-chain ergodicity) to recover convergence guarantees. Conversely, one can often keep a monotone backbone by separating reinforcement from decay into alternating steps, or by enlarging the state to include resources/attention so that the extended update is again monotone in an appropriate order. The point here is simply that the present theorem isolates a clean mathematical mechanism for stabilization that can be reused as a component inside more elaborate models.

Remark 1254. *From a philosophical point of view, this is one of the places where the difference between "explanation" and "description" becomes vivid. The fixed-point theorem does not tell us which culture we will get; it tells us that if the underlying transmission has a certain abstract form, then stabilization is a structural necessity. In Peircean terms [14], this is a move toward Thirdness: a law-like regularity extracted from the flux of particular dyadic interactions.*

One can also view this as a limited but genuine explanatory gain: even without forecasting the content of the stabilized regime, we can predict that certain kinds of interventions (those that increase some intensities or couplings) cannot lead, under the model assumptions, to a net decrease in eventual adoption levels. In that sense, the theorem supports counterfactual reasoning about qualitative constraints: it tells us which directions of change are ruled out by the abstract structure of monotone reinforcement, independent of the detailed sociology of any specific pattern.

22.9 Collective mind systems

Hyperseed includes "collective mind systems" as a headline concept: groups can function as minds insofar as they carry distributed patterns of perception, inference, and action. Our formalization treats a collective mind system as the society endowed with a derived pattern web and (optional) derived epistemic/agentive variables. In this subsection, "derived" means that these society-level objects are constructed (perhaps with modeling choices) from agent-internal state, artifact-embedded state, and the structured interaction channels among them; they are not assumed to exist as additional primitive entities.

Definition 373 (Collective pattern web)**.** *Let W_i denote agent i's internal pattern web (Section **??**), and let W_a denote the pattern web embedded in artifact $a \in \mathcal{A}$ (external memory, procedures, institutions).*

The collective pattern web *of a society is the (weighted) union*

$$W_{\text{col}} := \Big(\bigsqcup_{i \in A} W_i \Big) \sqcup \Big(\bigsqcup_{a \in \mathcal{A}} W_a \Big) \sqcup W_{\text{comm}},$$

*where W_{comm} adds edges corresponding to communication and coordination links. Weights on W_{comm} are derived from K and attention/effort constraints (Sections **??** and 8).*

Remark 1255. *The symbol $\bigsqcup$ is used informally here to indicate a disjoint/typed union of webs: we keep track of which subweb came from which agent or artifact, but we allow edges to be added that connect them. The additional web W_{comm} is where cross-agent influence becomes explicit: it is the connective tissue that turns many local pattern webs into a single larger web.*

As an example, in a research community, W_i contains each individual's concepts and inference habits, W_a contains textbooks and codebases, and W_{comm} contains seminar talks, peer review, and informal discussion channels. The collective web is then the natural object on which to ask whether *the community is collectively "smart" in a way not reducible to isolated members; see [19] for compatible systems-level motivations.*

Remark 1256. *It is often useful to think of W_{comm} as itself decomposing into multiple typed layers (even if we keep a single symbol): (a)* broadcast *edges (one-to-many dissemination such as announcements, published papers, mass media), (b)* dialogue *edges (bidirectional conversational links such as collaboration or deliberation), (c)* protocol *edges (standardized interfaces such as submission systems, forms, APIs, or legal procedures), and (d)* control/authority *edges (routing of decisions or permissions such as managerial reporting lines, access control lists). The weights on these edges can be interpreted as effective channel capacities: they summarize reliability, latency, interpretability, and willingness to transmit/receive, all of which are constrained by attention and effort budgets. In particular, high K but low attention can still yield low effective weight, because agents may be capable of understanding each other but not able to allocate the time to do so.*

Remark 1257. *The artifact term $\bigsqcup_{a\in\mathcal{A}} W_a$ is intended to cover more than passive storage. Many artifacts implement* procedural *patterns (workflows, algorithms, norms, bureaucratic scripts) that shape inference and action. For instance, a clinical guideline, a unit test suite, or a legal code can be modeled as a subweb that connects observational patterns to action patterns through standardized intermediate representations. This is one reason artifacts matter for emergence: they can stabilize and transmit patterns across time even when individual membership changes, thereby giving the collective web a degree of continuity and path dependence not present in purely conversational groups.*

Definition 374 (Collective belief state (one simple construction))**.** *Fix a claim language $\mathcal{L}$ and individual belief states $\beta_i : \mathcal{L} \to \mathbb{V}$. Define the* collective belief state *by pointwise join:*

$$\beta_{\text{col}}(\varphi) := \bigoplus_{i\in A} \beta_i(\varphi).$$

This is the minimal aggregation that preserves all evidence held by any agent.

Remark 1258. *Pointwise join is "maximally inclusive": if any agent holds strong positive evidence for φ, the collective holds it; if any agent holds strong negative evidence, the collective holds that too. This is appropriate if one wants the collective belief state to represent the society's total available evidential resources, rather than an official consensus.*

A different choice (e.g. averaging, weighted pooling, deliberation dynamics) could represent different social epistemologies. Hyperseed keeps the join construction prominent because it meshes cleanly with paraconsistent inference: contradictions remain visible rather than being suppressed by premature aggregation.

Remark 1259. *The "minimality" claim can be understood in the lattice-theoretic sense: $\beta_{\text{col}}(\varphi)$ is the least upper bound (with respect to the informational order implicit in $\oplus$) among all aggregate values that dominate every $\beta_i(\varphi)$. Equivalently, any aggregate belief state $\tilde{\beta}$ that satisfies $\tilde{\beta}(\varphi) \sqsupseteq \beta_i(\varphi)$ for all i must also satisfy $\tilde{\beta}(\varphi) \sqsupseteq \beta_{\text{col}}(\varphi)$. This makes β_{col} a natural representation of "everything the society could in principle cite" before filtering, bargaining, or forming a public stance.*

Example 19 (Collective belief is naturally paraconsistent)**.** *Suppose two agents disagree strongly about a claim p:*

$$\beta_1(p) = (0.9, 0.1), \qquad \beta_2(p) = (0.2, 0.8).$$

Then the collective join yields

$$\beta_{\mathrm{col}}(p) = (0.9, 0.8),$$

which represents high positive and high negative evidence simultaneously. This is not a bug: it formalizes the fact that societies often "contain" incompatible sub-beliefs.

Remark 1260. *In ordinary language, one might say: "the society has both a strong case for* p *and a strong case against* p *in circulation." This does not entail that any individual is irrational; it entails that the society, being larger than any one viewpoint, contains incompatible lines of testimony, theory, and incentive. The* p-bit *formalism makes that simultaneously explicit and computationally tractable.*

Remark 1261. *This paraconsistent stance is also useful for modeling institutional settings where mutually inconsistent records coexist for extended periods: for example, a large organization may have conflicting datasets, divergent process documents, and incompatible local norms that nonetheless all remain "live" because they are supported by different subcommunities and artifacts. In such cases, suppressing inconsistency at the collective level can be misleading; it can erase the very structure one needs to model how conflicts are detected, escalated, and eventually resolved (or entrenched) through coordination dynamics.*

Definition 375 (Collective mind system)**.** *A* collective mind system *is a society* Soc *together with chosen derived structures (e.g.* W_{col}, β_{col}*, a collective goal/attention policy, and a cultural state* C*) such that:*

- *the derived structures support persistent, society-level question answering and action (Section 20); and*
- *the society exhibits nontrivial emergence: some patterns or capabilities exist at the collective level that are not well-modeled as independent sums of individual patterns/capabilities (Section 9).*

Remark 1262. *The first condition says: the collective must actually* do *epistemic and agentive work over time (e.g. science, governance, coordinated building). The second condition says: we do not want to label every crowd as a mind; the collective must manifest emergent capabilities that depend on coupling and artifact-mediated memory.*

A standard example is a modern scientific community: no individual contains all the relevant evidence, techniques, and instruments, yet the community can answer questions and build artifacts that no isolated member could. In Hyperseed-Concept terms, this is the core of Collective Consciousness Location *discussions (Hyperseed-Concept 75), though the present section remains agnostic about phenomenology and focuses on functional structure. For a broader framing of collective mind categories and mindplex notions, see [12].*

Remark 1263. *The phrase "persistent, society-level question answering and action" is intended to rule in cases where the collective maintains long-horizon state and can revisit problems across time. Concretely, persistence typically requires (i) some form of durable external memory (archives, repositories, precedents), (ii) role specialization and handoff (so that tasks survive individual turnover), and (iii) a mechanism that selects which questions receive attention and which actions are executed (agenda setting, budgeting, prioritization, or market allocation). These ingredients can be modeled within the same derived-structure framework: artifacts support (i), communication/control edges support (ii), and an explicit or implicit policy supports (iii).*

Remark 1264. *The "collective goal/attention policy" component can be interpreted narrowly (a formal governance procedure) or broadly (any stable regularity that routes attention and effort). For example, in a firm, it may be a managerial planning cycle and incentive scheme; in an open-source project, it may be issue triage norms and maintainer review practices; in a market, it may be the price system coupled with institutional constraints. In each case, the policy shapes W_{comm} weights over time by selectively amplifying some channels (e.g. urgent incident response) and attenuating others (e.g. low-priority requests), thereby changing which subwebs effectively influence collective inference and action.*

Remark 1265. *The cultural state C is included to represent slow-moving, population-level regularities that are not conveniently reducible to a single agent or artifact: shared norms, taboos, aesthetic standards, interpretive frames, and background ontologies. Operationally, C can be viewed as a constraint or prior over pattern formation and communication: it affects which hypotheses are considered, which testimony is trusted, which institutional scripts are legitimate, and which coordination equilibria are stable. Thus C can modulate both the internal composition of W_i (via social learning) and the structure/weights of W_{comm} (via norm-governed interaction).*

Remark 1266. *The emergence condition is intentionally stated qualitatively, but it can be probed in several complementary ways. One approach is* capability gap*: identify tasks that the collective reliably performs while typical individuals cannot (e.g. maintaining a global logistics network, sustaining a legal system, producing a large-scale scientific theory). Another is* counterfactual dependence*: if one removes key coupling mechanisms (communication edges, shared artifacts, standardized protocols), does the capability collapse even when individuals remain intact? A third is* compression/description*: if modeling the collective requires representing new macroscopic variables (institutions, norms, reputations) that are not simply sums of individual variables, this is evidence that society-level state has explanatory autonomy. These diagnostics align with the general patterns discussion in Section 9 while remaining compatible with multiple philosophical views.*

Remark 1267. *Finally, the framework allows "collective mind systems" at multiple scales and with partial integration. A small team with shared documents and frequent dialogue may form a tight collective web with high effective W_{comm} weights and rapid feedback. A large society may instead consist of loosely coupled sub-minds (communities, institutions, platforms) whose interactions are mediated by slower, lower-bandwidth channels and standardized artifacts. In such cases, it can be useful to study not only W_{col} as a whole, but also its community structure, bottlenecks, and boundary objects, since these often determine which contradictions persist in β_{col} and which become resolved through coordination.*

22.10 Mindplex, notable coherence, and the global brain

Hyperseed defines: (i) coherence as systemicity (replaceability under removal), (ii) a notably-coherent system as one whose coherence is greater than slightly smaller subsets and slightly greater supersets, (iii) a mindplex as a notably-coherent mind composed of components that are themselves notably-coherent minds, and (iv) the global brain as a mindplex consisting of people and devices and other organisms on the planet.

We now give a mathematical proxy for these ideas using the earlier notion of approximate morphism and the observer-indexed distances underlying pattern webs.

Definition 376 (Replaceability and coherence (observer-indexed proxy))**.** *Fix an observing context O that provides: (i) a representation of system behavior and (ii) a distance d_O on behaviors (or on induced pattern webs).*

Let S be a finite set of components (agents and/or artifacts) inside a larger system. Write $\mathrm{Beh}_O(S)$ *for the behavior/pattern-flow summary of S as modeled by O. Define the* replaceability error *of a component $x \in S$ by*

$$\mathrm{Err}_O(x; S) := \inf_f d_O\big(\mathrm{Beh}_O(\{x\}),\ f(\mathrm{Beh}_O(S \setminus \{x\}))\big),$$

where the infimum ranges over the class of admissible approximators f (predictors/simulators) available to O. Define the coherence *of S (relative to O) by*

$$\mathrm{Coh}_O(S) := 1 - \max_{x \in S} \mathrm{Err}_O(x; S),$$

after normalizing d_O so that $\mathrm{Err}_O \in [0, 1]$.

Remark 1268. *The definition is deliberately observer-indexed. Coherence is not an intrinsic scalar floating in the void; it is measured relative to what the observer can model and predict. The quantity* $\mathrm{Err}_O(x; S)$ *asks: if we remove x, how well can the observer approximate the effect of x using the rest of the system? If every component is, in that sense, reconstructible from the rest, then the system exhibits strong redundancy and integration, and* $\mathrm{Coh}_O(S)$ *is high.*

As an example, in a tightly synchronized team with overlapping skills and shared documentation, the removal of one individual may be partially compensable by others plus artifacts, yielding low error and thus higher coherence. In a team where one person is the sole holder of a critical secret, their removal yields high error and thus low coherence.

Remark 1269. *Several modeling choices are bundled into* Beh_O *and d_O, and different choices capture different notions of "replaceability." For example,* $\mathrm{Beh}_O(S)$ *might encode time series of actions, message flows, task throughput, or induced pattern-web structure; correspondingly, d_O might be a metric on distributions (e.g. Wasserstein, total variation), a trajectory distance (e.g. dynamic time warping), or a distance between graphs/pattern webs (e.g. via spectral embeddings). The normalization of d_O to* $[0, 1]$ *can be done by fixing a reference scale (e.g. maximal expected distance under a null model, or by clipping via* $d'_O := \min\{1, d_O/\tau\}$ *for a chosen tolerance τ), making the "1−" transformation interpretable as a bounded coherence score.*

Remark 1270. *The choice of* $\max_{x \in S}$ *makes* $\mathrm{Coh}_O(S)$ *a "weakest-link" notion: a set is coherent only insofar as* every *component is individually replaceable (to the observer) by the remainder. In some applications, an average- or quantile-based aggregation (e.g.* $1 - \frac{1}{|S|} \sum_x \mathrm{Err}_O(x; S)$, *or* $1 - \mathrm{Quantile}_p(\{\mathrm{Err}_O(x; S)\})$*) may better match intuitions about partial cohesion in large populations. The max-based proxy is useful here because it corresponds closely to the Hyperseed phrasing about removal of* a *component and emphasizes crisp boundary detection when searching for notably coherent units.*

Remark 1271 (Interpretation)**.** *High* $\mathrm{Coh}_O(S)$ *means: for each component, the rest of the system can (in O's best model) closely approximate what that component contributes. This captures Hyperseed's "if removed, would be replaced by the other portions by some close approximation" in a way compatible with observer-relativity and approximate morphisms.*

Remark 1272. *The infimum over approximators f is a compact way to fold in both (i) the representational limits of O and (ii) the computational/inferential limits of O (what predictors/simulators are admissible). If O is taken to be an idealized Bayesian observer with unbounded compute, f may be extremely rich; if O is a bounded observer (e.g. a human analyst, or a particular monitoring system), the admissible class may be quite small, and coherence becomes correspondingly harder to* *"see." This makes it possible for the same underlying system to be notably coherent for one observer and not for another, which is often the right outcome when discussing societies, institutions, and multi-agent infrastructures.*

Definition 377 (Notably coherent)**.** *A subset S of components is* notably coherent *(relative to O) if it is a local maximum of* Coh_O *in the subset lattice:*

$$\mathrm{Coh}_O(S) > \mathrm{Coh}_O(S \setminus \{x\}) \text{ for all } x \in S,$$

and

$$\mathrm{Coh}_O(S) > \mathrm{Coh}_O(S \cup \{y\}) \text{ for all } y \notin S \text{ with } S \cup \{y\} \text{ admissible as a unit.}$$

(Variants with non-strict inequalities and neighborhood restrictions are also reasonable.)

Remark 1273. *The idea of "notable coherence" is that a coherent system should have a certain boundary: remove a part and coherence drops; add an unrelated part and coherence drops. This is a pragmatic way to detect integrated subsystems inside a larger network, akin to finding "communities" but with a semantics tied to functional replaceability rather than purely edge density.*

As a simple example, within a large company, a particular product team plus its internal tools may be notably coherent: adding random employees outside the team decreases coherence, and removing a key team member decreases coherence. This is one mathematical proxy for the intuitive claim that some subsystems are more mind-like units than others.

Remark 1274. *The "admissible as a unit" qualifier matters in practice: the observer O may only be able to form* $\mathrm{Beh}_O(S)$ *for subsets satisfying constraints (e.g. spatial contiguity, organizational boundaries, data-access permissions, or bandwidth limits). Likewise, the "neighborhood restrictions" alluded to in the definition can be understood as restricting attention to "small perturbations" of S (e.g. add/remove one component, or add/remove from a specified candidate pool), which avoids pathologies in very large systems and aligns with the Hyperseed intuition of "slightly smaller" and "slightly greater" sets. Computationally, searching the subset lattice is generally intractable, so practical detection of notably coherent systems would typically use greedy heuristics, multiscale clustering, or optimization relaxations guided by the observer's representation.*

Definition 378 (Mindplex and society of mind)**.** *A* society of mind *is a collective mind system whose components are (to a reasonable degree) autonomous agents (Section 21).*

A mindplex *(relative to O) is a society of mind S such that:*

(a) S is notably coherent (as above), and

(b) S can be decomposed into components $S_1, \ldots, S_n$ (agents or subgroups) each of which is itself a notably coherent mind (relative to O).

Remark 1275. *This definition bakes in a multi-scale picture: a mindplex is not merely a coherent group, but a coherent group composed of coherent subminds. This is close in spirit to hierarchical/heterarchical pattern-web pictures (cf. [5]) and to Hyperseed's emphasis on nested organization. In Hyperseed-Concept terms, this is the formal proxy for* Mindplex *(Hyperseed-Concept 113), and it is meant to be compatible with the graded collective categories discussed in [12].*

As a concrete example, one might model a research institute as a mindplex: each lab is notably coherent, and the institute as a whole is notably coherent due to shared infrastructure, shared norms, and high-bandwidth communication among labs.

Remark 1276. *The decomposition $S = \bigcup_i S_i$ is not assumed unique, and in many real systems it is more realistic to allow overlaps or fuzzy membership (e.g. a person participates in multiple teams, or an artifact is shared infrastructure). One can treat the present definition as a crisp proxy for a more general multiresolution picture in which notable coherence is evaluated across scales and partitions. In particular, a mindplex is intended to capture the case where there are "units within units": coherent subminds that remain meaningful even as the surrounding collective exhibits its own boundary and persistence. This multi-scale constraint helps distinguish (for instance) a crowd that is briefly coordinated by a single broadcast signal (which may exhibit some coherence) from an organization with enduring internal structure, specialized subgroups, and stable interfaces among them.*

Definition 379 (Global brain)**.** *Let $A_\oplus$ be the population of persons (and other relevant organisms) on a planet, and let $\mathcal{A}_\oplus$ include major communication and computation infrastructures (devices, networks, archives, institutions). A* global brain *is a mindplex constructed on*

$$(A_\oplus, G_\oplus, K_\oplus, \mathcal{A}_\oplus, \mathrm{Acc}_\oplus, \mathcal{D}_\oplus),$$

when the coupling/communication regime is strong enough that large-scale notable coherence emerges.

Remark 1277. *The clause "strong enough that large-scale notable coherence emerges" is doing conceptual work: it signals that merely having many agents and devices is insufficient; what matters is whether the planet-scale ensemble forms an integrated unit* relative to some observer *and representation. In the present proxy, this would mean that for a substantial set S spanning large fractions of $(A_\oplus \cup \mathcal{A}_\oplus)$, the removal of typical components yields relatively low replaceability error because the remainder can approximate their contributions via redundancy, routing, institutional memory, and adaptive reconfiguration. Conversely, if large portions are irreplaceable single points of failure (or if the observer cannot model cross-domain coupling),* Coh_O *would remain low and no planet-scale local maximum would appear. One may also take O to be an internal observer (e.g. an institution attempting to model global behavior), in which case "global brain" becomes an emergent, partially self-modeling phenomenon rather than a purely external description.*

Remark 1278. *The global brain concept (Hyperseed-Concept* **??***) is intentionally conditional: it is not asserted that the planet already forms a mindplex, only that under sufficiently strong, structured coupling it may. The tuple displayed is simply the society tuple at planetary scale, with artifacts now including digital networks, archives, and large institutions. This conditionality matters because "planetary scale" changes the relevant timescales and bottlenecks: communication latencies, institutional decision cycles, and archival retrieval dynamics can become as important as raw connectivity. In particular, "structured coupling" is meant to exclude mere density of contacts and instead emphasize patterned, reliable, and semantically aligned interactions (e.g., stable protocols, shared ontologies, and recurrent coordination pathways) that can support error-correcting circuits of collective action and belief update.*

A key reason to separate definition from existence is methodological humility: large-scale coherence is an empirical and dynamical question, not something settled by terminology. The present formalism provides variables one could, in principle, estimate (couplings, access, replaceability errors) and thus turn the question into a scientific one in Hyperseed's sense [20]. Concretely, one could operationalize coupling strength via observed information flow, coordination success rates, or causal influence estimates across layers; access via differential reachability of archives, institutions, or communication channels; and replaceability errors via counterfactual robustness tests (e.g., whether removing a platform, institution, or key actor leads to graceful degradation or systemic collapse). On this view, "global brain" becomes a hypothesis about a regime: a phase in which *the planetary tuple supports sustained integration and collective memory, rather than a rhetorical label.*

Remark 1279 (Where morphic resonance enters)**.** *The mindplex/global brain notions are not only graph-theoretic. Their distinctive Hyperseed flavor is dynamical: habits and morphic resonance (Section 12) can operate across agents via the communication and artifact layers, amplifying certain collective patterns and stabilizing them across time. This amplification is intended to cover both fast channels (imitation, social contagion, coordinated attention) and slow channels (institutionalization, codification in texts and software, and the training of new agents on existing corpora), so that patterns can persist even as individual participants change. In this sense, "resonance" functions as a shorthand for mechanisms that bias future trajectories toward previously realized configurations, including reinforcement learning loops, reputational feedback, and path-dependent standards.*

In this sense, a mindplex is not just "a network" but a regime of high mutual reinforcement and cross-component pattern re-use. The regime language is meant to be taken literally: the same underlying nodes and edges may or may not constitute a mindplex depending on parameters such as signal-to-noise, incentives, interpretability, and the availability of shared external memory. Cross-component pattern re-use includes the re-deployment of solutions, norms, and representations across domains (e.g., legal templates applied to new technologies, or computational schemas migrating into organizational design), which increases compression and coordination capacity at the collective level.

Remark 1280. *One can visualize this as a multi-layer dynamical system: the interaction graph supplies channels, artifacts supply long-term memory, and habit dynamics supply reinforcement. When these align, the collective web begins to exhibit "self-maintaining" patterns, not unlike autocatalytic sets but in a socio-cognitive substrate (Hyperseed-Concept 61). A useful way to read the analogy is functional rather than biochemical: artifacts and institutions can play the role of catalysts by lowering the cost of re-instantiating practices, while communication channels play the role of transport that connects otherwise isolated sub-processes. Alignment across layers can be partial and still significant, yielding localized mindplex behavior (e.g., within scientific communities, technical ecosystems, or governance networks) even if the whole planet does not exhibit uniform coherence.*

Whether one wishes to describe this in the evocative language of resonance [13] or in the austere language of fixed points and attractors, the mathematical role is the same: to describe persistence of structure across time. Here "persistence" can be understood as invariance (patterns that remain approximately unchanged), recurrence (patterns that reappear after perturbation), or robustness (patterns that survive turnover in agents or artifacts). In dynamical terms, this corresponds to attractors or metastable sets in the state space of collective variables, with "noise" supplied by shocks, innovation, and demographic churn; the empirical question is then whether observed trajectories concentrate near such sets, and on what timescale transitions occur.

22.11 Summary: Hyperseed concepts covered in this section

This section provided a rigorous scaffold for group-level Hyperseed concepts: society and tribe as multi-agent pattern webs; communicative acts and systems; social roles (teacher, student, worker, colleague) as interaction templates; culture as stabilized intersubjective pattern web; and collective mind systems culminating in mindplex and global brain. The intent of this scaffold is to make "group-level" talk precise without reducing it to mere metaphor: a society or tribe is modeled as a recurrently coupled ensemble whose stable patterns are identifiable across time, membership changes, and context shifts. In this framing, the same analysis that applies to an individual agent's patterns (formation, reinforcement, decay, and reconfiguration) is extended to coordinated patterns spanning many agents and their shared artifacts. The section also emphasized that many apparently "soft" social phenomena (norms, conventions, institutions) can be described operationally as constraints on interaction that shape which patterns are likely to be reproduced.

Remark 1281. *The unifying theme is that "collective mind" is treated as a question of* structure and dynamics *rather than as a special ontological category. When couplings, artifacts, and reinforcement are arranged so that group-level patterns can persist, propagate, and answer questions, then the society begins to behave mind-like in the operational sense developed in Sections 20 and 21. In Hyperseed-Concept terms, this is the bridge between* Society *(Hyperseed-Concept 170),* Culture *(Hyperseed-Concept 91),* Collective Consciousness Location *(Hyperseed-Concept 75), and* Mindplex *(Hyperseed-Concept 113). In particular, "answer questions" should be read in the same pragmatic sense as for individual cognition: the system exhibits reliable input–output mappings, can integrate distributed information, and can select actions (or policies) that improve its performance relative to some goal or constraint. This does not require positing a single inner observer; it requires that the interaction network implements functional loops of sensing, memory, inference, and control at the group level. The bridge is therefore not a leap from individuals to an entirely new kind of entity, but an account of how coordination mechanisms (communication protocols, role structures, shared symbols, and institutional feedback) can yield emergent, yet analyzable, cognitive regimes.*

In later sections, tools/machines (Section 23) and intersubjective communion (Section **??**) will add additional coupling modes that can dramatically increase coherence and stabilization, thereby changing when mindplex-like regimes become plausible. These later coupling modes matter because they can alter the bandwidth, fidelity, and persistence of coordination: tools and machines can externalize memory and computation into durable substrates, while communion-like mechanisms can tighten alignment by increasing shared context and mutual predictability. As a result, the boundary between loosely coordinated social webs and more tightly integrated mindplex dynamics becomes contingent on concrete engineering and cultural practices, not merely on group size. This also clarifies why "global brain" claims should be evaluated via measurable properties of coupling (latency, robustness, error correction, incentive compatibility) rather than by rhetoric alone.

23 Tools, machines, computation, and embodiment

23.1 Why engineered artifacts belong in a rigorous Hyperseed

Hyperseed treats minds as pattern webs embedded in (and coupled to) reality-systems. Within that picture, *tools*, *machines*, and *computers* are not ontological afterthoughts: they are stable, engineered pattern-systems that let an agent extend its perception-action loop across additional substrates in physical reality. In particular, "engineered" here signals that the artifact is intentionally shaped to yield repeatable causal regularities under a specified range of conditions (tolerances, loads, noise), so it is naturally described as part of the agent–world coupling rather than as a passive backdrop. This is also why the tool/machine/computer triad is not a mere list: it spans a spectrum from direct mechanical mediation (tools), to self-sustaining coordinated mediation (machines), to symbolically mediated mediation (computers), all of which can be treated uniformly as realizers of process structure.

Remark 1282. *At an intuitive level, the point is that cognition is never merely "in the head" once one admits closed-loop agency: an agent's competent behavior is a whole causal circuit that includes body, environment, and (often) engineered artifacts. A tool is therefore not merely an inert object;*

relative to an observer/context, it is a reliable bridge *between action tokens and effect tokens, so it belongs naturally beside the formal notions of agency and task competence (Sections 21 and 14). This corresponds directly to Hyperseed-Concept 191 and Hyperseed-Concept 106 as elements of a reality-system that restructure what an agent can do.*

From the standpoint of the overall reconstruction, engineered artifacts are where abstract process structure (morphisms, compositionality) meets physical constraint (effort, friction, reliability). That meeting is one of the main arenas in which the quantale notions of evidence and weakness become empirically consequential [1, 3, 2]. A further point (useful later when we talk about embodiment) is that the "bridge" can run both ways: tools not only amplify outward action but also reshape inward perception by changing what is sensed, how it is filtered, and what degrees of freedom become salient. A microscope, a telescope, an fMRI scanner, or even a simple ruler can be treated as a reparameterization of the agent's observation channel, thereby changing the evidence available for downstream inference; conversely a wrench, a lathe, or a numerical optimizer can be treated as a reparameterization of the agent's actuation channel, thereby changing the effort required to reliably reach desired effects. On this view, engineering is a way of sculpting the agent's effective sensorimotor manifold.

Mathematically, the clean way to express this is:

- an *abstract process* is a morphism in a category of processes (or a V-enriched category as in Section 3);
- a *physical process* is a morphism in a category describing transformations available within a physical reality-system (Section 25);
- a *physical realization* is a structured relationship (encoding/decoding + dynamics) connecting an abstract process to a physical process; and
- a *tool* is a physical entity whose coupling to an agent yields additional realized processes (typically with reduced effort, increased reliability, or both).

To make the above list more than a slogan, it is helpful to keep two distinctions explicit. First, the abstract/physical split is not a mind/body dualism; it is a modeling move that separates (1) the compositional structure we wish to preserve ("what computation or procedure is being performed?") from (2) the constraints that determine whether a real system can stably enact that structure (energy, noise, wear, latency, finite precision, and so on). Second, "realization" is not mere association: it is a constraint-respecting correspondence that must commute with the relevant compositions, at least approximately, if we want the engineered artifact to be predictable and reusable as a component. This is where interfaces matter: an artifact is engineered so that the conditions under which its input-output behavior is well-approximated are themselves describable, testable, and (often) composable.

A particularly important special case is *computation*. An algorithm (or program semantics) is naturally modeled as an abstract process, while a concrete run on hardware is a physical process. The "encoding/decoding + dynamics" clause then packages familiar notions such as representation, compilation/interpretation, signal encoding, error correction, and timing discipline. For example, a logical gate network and its transistor-level implementation can be treated as two process levels linked by a realization relation; the engineering content is precisely the set of design choices that make the correspondence robust against temperature variation, component drift, and noise. In the Hyperseed setting, such robustness is not merely practical: it is part of the explanation of why certain morphisms are available as stable resources to an agent in a given reality-system.

Remark 1283. *The phrase "morphism in a category" is doing philosophical work here: it encourages us to treat processes as first-class citizens and to regard composition as primitive. This resonates with process-oriented views (in the broad Whiteheadian spirit [15]) while keeping the mathematics explicitly compositional: if a complex artifact is built by wiring together sub-artifacts, the formal semantics should reflect this wiring by categorical composition. One can add that engineering practice tacitly relies on exactly this idea: subcomponents are designed to satisfy contracts (interfaces, tolerances, input/output ranges), and complex devices are built by composing these contracts. Categorically, this motivates not only ordinary composition but also monoidal structure (parallel composition), products/coproducts (bundling/switching of channels), and enrichment (where hom-objects can encode graded notions such as cost, effort, risk, latency, or fidelity). In that enriched setting, "reduced effort" and "increased reliability" in the tool bullet above can be read literally as improvements in the value assigned to a morphism by the enrichment: a tool changes which morphisms are accessible at what grade.*

This section makes these ideas precise enough to support later discussions of embodiment, implementation semantics, and the "algebraic asymmetry" between physical and mental realities. Concretely, embodiment enters because the agent's body is itself an engineered (or evolved) artifact from the standpoint of control: it defines the baseline action/observation morphisms available to the mind-pattern. External tools then act as additional composable modules that alter the reachable set of world-states and the evidential structure of observations. Implementation semantics enters because a realization relation must say what it means for an abstract procedure to be enacted by a physical device in a way that is stable under the kinds of perturbations the reality-system induces. Finally, the "algebraic asymmetry" is foreshadowed by the fact that physical realization is graded and fragile (bounded by conservation laws, noise floors, and finite resources), whereas mental or symbolic composition is often treated as frictionless; the theory needs a principled way to relate these two modes of composition without collapsing them.

23.2 Tools as externalized affordance morphisms

Hyperseed's informal definition can be summarized as: a tool is something a user manipulates purposefully toward an objective, where the objective is not primarily about the tool's own state of consciousness.

To express this without importing unnecessary machinery, we model a tool in terms of what it *affords* when coupled to an agent. One can read "coupled" here broadly: the agent may hold the artifact, wear it, operate it at a distance, or interact through an interface layer (buttons, APIs, protocols). The intent is to treat the tool as part of the agent–environment control loop, while still isolating a crisp interface between what the agent can *do* (its action tokens) and what changes in the world are made more (or less) attainable (its effect tokens). In particular, the same physical object may induce different affordances under different couplings (e.g. bare hands versus gloved hands; direct use versus remote teleoperation), which is why the formalization keeps the observer/context explicit.

Definition 380 (Affordance profile)**.** *Fix an observer/context O. Let $\mathcal{A}$ be a set of* action tokens *available to an agent (motor commands, control signals, or higher-level actions), and let $\mathcal{E}$ be a set of* effect tokens *in some environmental slice of O's reality-system (events, state changes, or achieved constraints).*

An affordance profile *is a map*

$$\mathrm{Aff} : \mathcal{A} \times \mathcal{E} \to \mathbb{V} = [0, 1]^2$$

whose value $\mathrm{Aff}(a, e)$ *represents (paraconsistent) evidence that performing* a *using the artifact tends to produce effect* e *within the relevant context.*

Remark 1284. *The use of* tokens *emphasizes that* $\mathcal{A}$ *and* $\mathcal{E}$ *are modeling choices rather than metaphysical primitives. An action token may be low-level (a torque command), mid-level (a gesture), or high-level ("compile and run test suite"); similarly an effect token may be a raw state transition, a perceptual event, or a constraint satisfaction statement ("object is aligned"). The appropriate granularity depends on* O *and on the explanatory task: for fine motor control one might refine* $\mathcal{A}$ *to continuous commands, whereas for social or institutional tools one might coarsen* $\mathcal{A}$ *to normative or procedural actions ("submit form") and let* $\mathcal{E}$ *encode institutional state changes ("request accepted"). This flexibility is also what lets the same mathematical object cover both embodied tools (hammers) and informational tools (calculators, software libraries) without forcing a single privileged ontology of actions and effects.*

Remark 1285. *Intuitively,* $\mathrm{Aff}(a, e)$ *is an "action-to-effect" interface description: it says what the artifact makes easy, difficult, reliable, unreliable, encouraged, or resisted. The codomain* $\mathbb{V} = [0, 1]^2$ *is the* p-bit *evidence space used throughout Hyperseed: the first coordinate is positive evidence and the second is negative evidence. Thus an affordance can be simultaneously supported and undermined by experience, reflecting the common situation that a tool can work well in some contexts and fail (or be dangerous) in others. This is one instance of Hyperseed-Concept 86 interacting with Hyperseed-Concept 191.*

A simple example is a lever used to lift a weight: an action token might be "apply downward force at distance d *from fulcrum," and an effect token might be "weight rises by* Δh*." One expects strong positive evidence for that action-effect pair in ordinary mechanical contexts, but one may also have nontrivial negative evidence (slippage, breakage, user error). Another example is a password manager: an action token might be "request credentials for site* s*," and an effect token might be "credentials filled into browser form"; again, positive evidence may coexist with negative evidence when the environment is hostile or the tool is misconfigured. Formally, the point of* Aff *is that it allows us to treat a tool as a* morphism-like *object mediating between an action vocabulary and an effect vocabulary, which is exactly what later sections will need when tools become participants in social and intersubjective systems.*

Remark 1286. *The "morphism-like" phrasing is meant operationally:* Aff *behaves like an interface that can be compared, composed, and substituted. For example, if one tool* T_1 *maps actions to an effect token that is itself an action token for another tool* T_2 *(e.g. "produce a tightened screw" as a precondition for "mount a circuit board"), then one can model multi-tool workflows by composing their affordance profiles at the level of shared intermediate tokens. Even when such strict token matching is unavailable, coarse-graining and refinement mappings between token sets can be used to relate affordance profiles across levels of description (e.g. translating a family of low-level controller actions into a higher-level macro-action).*

Remark 1287. *The paraconsistent encoding in* $\mathbb{V} = [0, 1]^2$ *can also be read as separating two kinds of observations: evidence* for *the claim "*a *tends to produce* e*" and evidence* against *it. This separation is useful when the world is nonstationary or heterogeneous: repeated success in one subcontext does not erase repeated failure in another, and both should remain available for context-sensitive inference. In practice, one can interpret* $\mathrm{Aff}(a, e)$ *as a summary of experience under* O *(counts, weighted frequencies, model-based estimates), and later decision procedures can condition on additional context variables or aspects to resolve when the positive versus negative component should dominate.*

Remark 1288. *The affordance profile is a compact interface between the (agent-relative) action vocabulary and the (context-relative) effect vocabulary. In practice,* Aff *can be learned from interaction data, derived from physics models, or constructed compositionally from sub-artifacts.*

Remark 1289. *Learning* Aff *from interaction data can be understood as estimating a structured action–effect relation under partial observability. For instance, given logs of attempted actions and observed outcomes, one may update positive evidence when* e *is observed after* a*, and update negative evidence when* e *fails to occur or when an incompatible effect occurs. When the relevant effects occur with delay, one can either expand* $\mathcal{E}$ *to include temporally extended tokens (e.g. "achieved within* τ*") or treat* Aff *as summarizing an induced policy-conditional transition model over an appropriate time window. This keeps the definition agnostic about the details of time, memory, and credit assignment, while still letting later sections talk about tools in a uniform vocabulary.*

Definition 381 (Tool)**.** *A* tool *(relative to an agent class and an observer/context* O*) is an entity* T *in a reality-system together with an affordance profile* Aff_T *that is:*

(a) usable*: there exists a nontrivial subset* $\mathcal{A}_T \subseteq \mathcal{A}$ *and effects* $\mathcal{E}_T \subseteq \mathcal{E}$ *such that* $\mathrm{Aff}_T(a, e)$ *has substantial positive evidence for some* $(a, e) \in \mathcal{A}_T \times \mathcal{E}_T$*; and*

(b) purpose-supporting*: for some objective functional (Section 18), the availability of* T *strictly increases expected objective achievement for typical users in the class, via the induced action-effect couplings.*

Remark 1290. *The "nontrivial" qualifier in the usability clause is intended to rule out degenerate cases where* $\mathcal{A}_T$ *or* $\mathcal{E}_T$ *contains only vacuous tokens (e.g. an action that is never available to the agent, or an effect that is always true regardless of action). Intuitively, a tool must open up at least some meaningful region of action–effect space that was previously absent or less reliable. Depending on the application, "substantial positive evidence" can be instantiated as a threshold on the positive coordinate of* $\mathrm{Aff}_T(a, e)$*, a gap between positive and negative evidence, or a downstream decision-theoretic criterion tied to improved control.*

Remark 1291. *The purpose-supporting clause is deliberately phrased in terms of* expected *objective achievement to allow tools that are beneficial in aggregate despite occasional failures, and to allow tools that shift risk profiles rather than deterministically guaranteeing outcomes. The reference to "typical users" acknowledges that toolhood is relative not only to* O *but also to a distribution over user skills, training, and embodiment: a lathe may be purpose-supporting for a trained machinist but not for a novice; a user interface may be purpose-supporting for sighted users and not for blind users unless paired with assistive technology. This connects directly to the motivating theme that tools are part of* embodied *computation: their effective affordances depend on bodily and cognitive properties of the agent class.*

Remark 1292. *The usability clause prevents "tools" from including arbitrary inert objects whose action-effect links are too weak to matter; the purpose-supporting clause prevents us from calling every causal mediator a tool (e.g. a random obstacle) unless it helps with some objective. In plainer language: a tool is something that* makes some intended outcomes easier or more reliably achievable *for some class of agents, relative to an observer/context.*

Examples: a hammer is a tool because it supports the objective "join two pieces of wood" by increasing the probability of the effect "nail is driven" for actions in a natural user action set. A microscope is a tool because it supports objectives that require fine perceptual distinctions by reliably producing effects like "image reveals small-scale structure," thus interacting with Hyperseed-Concept 134 and Hyperseed-Concept 98. The usefulness of this definition is that it makes "toolhood" *a property that can be discussed in the same formal vocabulary as tasks and values: it becomes possible to compare tools by their affordance maps and by the induced changes in expected objective achievement.*

Remark 1293. *This framing also clarifies why "tool" is not synonymous with "technology" or "artifact." An entity may be technologically sophisticated yet fail the purpose-supporting clause for a given objective functional (e.g. a complicated gadget that distracts rather than helps), and conversely a simple artifact (a wedge, a string) may be a powerful tool relative to an agent class and objective. Moreover, a single physical object can implement multiple tools depending on which affordance profile is salient: a smartphone can be a camera, a map, a payment instrument, or a distraction source, each corresponding to different subsets of $\mathcal{A}$ and $\mathcal{E}$ and different induced changes in objective achievement.*

Remark 1294 (Tools as patterns that extend agency)**.** *Earlier sections treated agency as closed-loop control (Section 14) and tasks as externally specified objective-achievement problems (Section 21). A tool is then an* engineered pattern-system *that enlarges the agent's effective action set (or reduces the effort of actions) by adding new reliable couplings from actions to effects.*

Remark 1295. *In this sense, tools can be seen as externalized "affordance morphisms" that restructure the agent's reachable set in state space: they do not merely add raw power, but reparameterize control. A steering wheel converts small hand rotations into large changes in vehicle trajectory; a search engine converts short text queries into high-probability access to relevant information; a musical instrument converts constrained gestures into complex acoustic effects. In each case, what matters for agency is the induced mapping from a feasible action alphabet to a task-relevant effect alphabet, not the internal mechanism by which the artifact accomplishes the transformation.*

Remark 1296. *Note the deliberate observer-relativity: "tool" is not asserted as an intrinsic essence, but as a role an entity plays in a coupled system of agent, objective functional, and context. This is consistent with Hyperseed's general stance that many ontological tags are relative to an observer/context and a modeling purpose [1].*

Remark 1297. *Observer-relativity also helps separate two questions that can otherwise be conflated: (i) whether an entity* can *mediate certain action–effect couplings (a descriptive claim about the reality-system under O), and (ii) whether those couplings are* counted *as the entity's affordances in a given modeling task (a choice of $\mathcal{A}$, $\mathcal{E}$, and the relevant aspect of context). For example, a rock affords "smash" in a coarse-grained model of improvised tools, but in a fine-grained safety-critical model its negative evidence for controlled, repeatable effects may dominate, changing whether it is treated as a tool for the objective at hand. This explicit separation is one reason the affordance profile representation remains useful when the discussion later shifts to social tools (protocols, norms, institutions) whose "actions" and "effects" are themselves observer- and context-mediated.*

23.3 Machines as internally patterned tool-systems

Hyperseed distinguishes machines from generic tools by emphasizing that the causal interactions among parts of a machine form a pattern in the machine's overall behavior. Put differently, a machine is not merely *used* by an agent to produce an effect; it is also a system whose *internal* use-relations (parts acting on parts) are engineered so that the whole has a robust, repeatable mode of operation.

Definition 382 (Machine)**.** *A* machine *is an* engineered *physical system M (Definition 383) that can be decomposed into components $(M_i)_{i\in I}$ such that:*

(a) each component M_i is (typically) a tool for acting on other components; and

(b) the internal causal interaction structure among the components induces a stable input-output behavior of the whole M that exhibits nontrivial patterns (Section 9) with high intensity relative to the decomposition baseline.

Remark 1298. *In ordinary speech, a machine is often "a tool with moving parts." The formal definition refines this: a machine is a* network of tools acting on each other *in such a way that the composite behavior is stable and patterned. The phrase "patterns with high intensity" appeals to the earlier formalization of pattern and emergence [5]; the idea is that the overall behavior is not merely a bag of componentwise behaviors, but a structured regularity that survives perturbation and supports prediction and control (Hyperseed-Concept 130 and Hyperseed-Concept* **??***). In particular, "relative to the decomposition baseline" indicates that the relevant comparison is not against an arbitrary null model, but against what one would expect if the components were only weakly coordinated (e.g., if they acted largely independently, or if their couplings were randomly rewired while keeping the same parts). On this view, a system qualifies as a machine not merely because it is complicated, but because its couplings* constrain *the accessible behaviors into a smaller, more regular family: the whole exhibits a characteristic repertoire of operation that can be recognized, reproduced, and relied upon.*

Examples: a mechanical clock decomposes into gears, springs, and escapement components that constrain each other; the high-intensity pattern is periodic motion and reliable timekeeping. A modern CPU decomposes into logic gates and subsystems; the high-intensity pattern is reliable execution of instruction semantics under a clocked discipline. One may also view a bicycle as a machine in this sense: its components (frame, chain, sprockets, bearings, brakes) are individually tools that act on each other to yield a stable input-output relation between rider-applied forces and predictable motion/steering/braking, with the "pattern" being the constrained dynamics that make riding learnable and controllable rather than chaotic. These examples underscore that the "input" and "output" of a machine need not be purely informational; they may be forces, motions, material transformations, or other embodied variables, so long as the system as a whole implements a stable mapping (possibly stochastic, but with reproducible statistics) from relevant operating conditions to outcomes.

This definition is useful because it lets us connect machines to compositional semantics: if the components are tools and their couplings are stable, then one can treat the machine as a higher-level tool whose affordance profile is itself constructed from lower-level affordances. The same idea supports hierarchical abstraction: once the higher-level tool behavior is stable, it can be "black-boxed" for many purposes, allowing agents to plan using the machine's macroscopic affordances without explicitly simulating all internal component interactions.

Remark 1299. *Clause (a) uses "typically" to allow that some components may function primarily as supports, constraints, or media (e.g., a chassis, casing, lubrication, or shielding) rather than as direct actuators on other parts. Nevertheless, even these components often participate causally by setting boundary conditions, maintaining tolerances, or channeling energy and information, so that their presence is part of what makes the overall pattern stable. In this sense, the decomposition $(M_i)_{i\in I}$ is not required to be unique: different granularities (coarser modules vs. finer subparts) may each reveal patterned interactions, and the definition only requires that there exists some decomposition at which the machine-like character is made explicit.*

Remark 1300 (Machines induce internal "interaction languages")**.** *When a family of machines shares standardized components and couplings, one gets an internal "language" of parts and compositions (gears and shafts, logic gates, instruction sets, network protocols). Formally, this can be* *treated as a subcategory of realizable processes whose morphisms are precisely the allowed component compositions. The "language" metaphor is apt because the standards function like a syntax (which parts may connect, in what orientations, under what tolerances) together with an operational semantics (what behaviors result when they are connected in permitted ways). In mature machine families, this interaction language is often explicitly documented as interface specifications (e.g., bolt patterns and torque specs, voltage levels and timing constraints, ISA and ABI conventions), which makes the induced patterns portable across instances: one can reason about the behavior of a class of compositions without re-deriving everything from first principles.*

Remark 1301. *This "interaction language" viewpoint is also a bridge to social coordination: once a machine family stabilizes, communities can build training, documentation, and division of labor around it. In that sense, machines become social artifacts that mediate collective competence, anticipating the culture-level constructions of Section 22 (Hyperseed-Concept 91 and Hyperseed-Concept 171). The point is not merely that people* use *machines, but that the stabilized internal language of a machine family reshapes how labor is decomposed: roles emerge that correspond to modular boundaries (designer vs. fabricator vs. operator vs. maintainer), and competence becomes partly a matter of fluency in the relevant interaction language. This also clarifies why machines are closely linked to computation and embodiment: they provide reliable, repeatable causal structure that can be embedded into workflows, plans, and protocols, so that abstract procedures (including symbolic ones) can be physically realized with predictable behavior across time, contexts, and operators.*

23.4 Engineered artifacts

Hyperseed uses "engineered" as a broad ontological tag: created via purposeful, goal-oriented activity of one or more intelligent minds. In particular, the term is meant to track *origin and causal history* (how a thing came to be in the modeled reality-system), rather than merely its present-day use or its aesthetic resemblance to human-made objects. The tag is intentionally permissive: it covers cases where design is explicit and centralized (a blueprint and a factory), as well as cases where design is implicit, distributed, or only partially articulated (e.g., incremental improvement across generations of makers, or selection among variants produced by a mixed intentional/accidental process).

Definition 383 (Engineered)**.** *An entity E is* engineered *(relative to an observer/context O) if there exists:*

- *an agent (or agent collective) A as in Section 21, with goal/valuation structure as in Section 18; and*
- *a nontrivial interval of time (Section 7)*

such that E arises as a consequence of a goal-oriented design-and-construction process within O's modeled reality-system. In this observer-relative framing, the requirement that the process occur "within O's modeled reality-system" is doing substantive work: it excludes hypothetical external designers that the observer does not model, while including indirect and mediated causal routes (e.g., a committee designs a standard, which reshapes incentives, which in turn leads many independent actors to build compatible artifacts). The "nontrivial interval" clause is also intended to rule out degenerate cases where E would be treated as engineered only by stipulating an instantaneous act of creation; engineering here is tied to temporally extended activity such as iteration, refinement, procurement, testing, coordination, or maintenance, any of which may be the bottleneck that makes the outcome non-accidental relative to the goals of A.

Remark 1302. *Intuitively, "engineered" names a causal pedigree: the artifact exists because some mind (or coalition of minds) pursued a goal and successfully constrained reality to instantiate a pattern. This is not restricted to industrial engineering; it includes language artifacts (alphabets, grammars), institutional artifacts (contracts, courts), and computational artifacts (code and protocols), depending on what the observer/context counts as part of the modeled reality-system. This is Hyperseed-Concept* **??** *stated in the same observer-relative style as the rest of the ontology [1]. A useful consequence of this broad scope is that "engineered" can apply to abstracta and semi-abstracta insofar as they have realized implementations and stabilizing causal roles in the world the observer models (e.g., a legal procedure instantiated by documents, officials, and sanction mechanisms; or a protocol instantiated by software, hardware, and operator practices). Conversely, mere* use *does not suffice: a naturally occurring stone used as a hammer is not, by that fact alone, engineered; but a stone intentionally shaped or selected and curated through an organized process for hammering may be engineered, depending on the causal history the observer ascribes.*

A simple example is a screwdriver: it is engineered because it arises from an iterative process of designing and manufacturing an object that reliably couples twisting actions to screw-rotation effects. A more subtle example is a musical notation system: it is engineered (in this sense) because it is a designed symbol technology enabling new patterns of reproduction and coordination across time. The usefulness of explicitly labeling engineered entities is that later arguments about society, culture, and collective minds can treat engineered artifacts as stabilizers of cross-agent coupling, rather than as incidental background [19]. Note that these examples also illustrate that engineering can be multi-layered*: a screwdriver depends on engineered metallurgical practices and measurement conventions, while notation depends on engineered inscriptions, pedagogy, and social norms. In such cases, engineering is not a single event but a scaffold of interlocking engineered subsystems whose combined effect is to make certain actions easy, reliable, and interoperable across agents.*

Remark 1303. *In borderline cases, it can be helpful to distinguish engineered entities from entities that are merely* selected *or* filtered *by agents without being constructed in detail. For example, breeding and domestication involve goal-oriented selection operating through natural reproduction; whether a domesticated organism counts as engineered relative to O depends on whether O models the selective process as an intentional design-and-construction process (often distributed over time and across many agents). Similarly, artifacts produced by autonomous systems (e.g., code emitted by a learned model) can be engineered relative to O if O models the upstream training, prompting, evaluation, and deployment pipeline as a goal-oriented process attributable to an agent collective. These clarifications preserve the intent that "engineered" is about goal-directed constraint of outcomes, even when the fine details of the resulting pattern are not explicitly specified by any single mind at any single moment.*

Remark 1304. *In the quantale-weakness framing, one can treat engineering as constrained search over candidate artifacts whose realized affordances reduce effort (Section 8) while meeting goal constraints. We develop this viewpoint explicitly in Section 26. One can also interpret this as a shift in the feasible action-set available to agents: an engineered artifact is not merely an object, but a transformation of the coupling between intentions and outcomes, often by making some transitions in the agent–environment system more reliable, cheaper, or more reversible. On this view, the "constraints" include not only physical laws and resource limits, but also institutional compatibility conditions, coordination costs, and cognitive limits of the agents that must adopt the artifact for it to function as intended.*

Remark 1305. *The "constrained search" phrasing foreshadows the wu-wei formalism: one may view successful engineering as finding low-effort geodesic-like trajectories in a space of designs,* *where effort is not merely physical exertion but also cognitive and organizational cost [21]. This links engineered artifacts to Hyperseed-Concept 100 and Hyperseed-Concept 207. In practical terms, this emphasizes that many engineered artifacts are engineered* interfaces*: they compress complicated causal structure into a small set of stable handles (buttons, APIs, rituals, forms) that let agents achieve goals without re-solving the underlying problem each time. Accordingly, "engineering" in this sense includes the creation of standards and abstractions whose primary function is to reduce coordination effort by making agent expectations align.*

23.5 Computers and programs as emulation machines

Hyperseed defines a computer (relative to a group of minds) as a machine that can, with modest effort, be made to emulate a large variety of other machines; and defines a program as a set of signals telling the computer what other machine to emulate.

We formalize this by separating syntax, semantics, and realization.

Remark 1306. *The phrase "relative to a group of minds" is meant to foreground that programmability is partly an interface notion: what counts as "modest effort" depends on a user class's tools, training, and available infrastructure, and what counts as a "large variety" depends on which behaviors are distinguishable and valuable to that class. For example, to an embedded-systems engineer, changing a firmware image may be modest; to a lay user, even installing a compiler toolchain may not be. Likewise, two machines might be physically capable of the same reconfigurations, but differ greatly in the effort required to exploit them (documentation, debugging support, safety constraints, access to I/O), which is why the definition is intentionally not purely extensional.*

Definition 384 (Abstract process semantics)**.** *Let* $\mathbf{Abs}$ *be a category (or* V*-enriched category) of abstract processes. Objects are abstract state spaces, types, or contexts; morphisms are abstract transitions/processes. We write* $f : A \to B$ *for an abstract process from* A *to* B*.*

Remark 1307. *The notation* $\mathbf{Abs}$ *indicates a category: its objects are "types of states" and its morphisms are the processes that carry states of one type to states of another. One may keep in mind standard examples: (i) sets and functions, (ii) measurable spaces and stochastic kernels, (iii) state machines and their transition relations, or (iv) typed terms modulo equivalence in a lambda calculus. The abstractness here is methodological: we want to speak about a process* f *independently of any particular physical substrate that might implement it. This is a mathematical expression of Hyperseed-Concept 51 and Hyperseed-Concept 140 (and, more specifically, Hyperseed-Concept 141).*

The definition is useful because it gives a precise target for implementation: when we later say a physical system realizes f*, the claim is that the physical dynamics, suitably encoded/decoded, approximate this morphism in* $\mathbf{Abs}$*.*

The parenthetical "(or V*-enriched category)" is included to accommodate quantitative structure that is common in engineering uses of computation. For instance, one may want hom-objects to carry probabilities, costs, resource bounds, or metrics of approximation error. In such settings, the statement that a physical system "approximately realizes" a morphism can be made internal to the enrichment (e.g. an order capturing refinement, or a metric capturing simulation distance), rather than treated as an informal afterthought.*

Definition 385 (Program language and semantics)**.** *A* programming language *is a syntactic category* $\mathbf{Syn}$ *together with a semantics functor*

$$[\![\cdot]\!] : \mathbf{Syn} \to \mathbf{Abs}.$$

A program *is a morphism* $p : A \to B$ *in* $\mathbf{Syn}$*; its intended behavior is the abstract process* $[\![p]\!] : [\![A]\!] \to [\![B]\!]$*.*

Remark 1308. *The brackets* $[\![\cdot]\!]$ *denote the interpretation map from syntax to semantics: given a piece of code (a morphism in* **Syn***), it produces an abstract process (a morphism in* **Abs***). Calling* $[\![\cdot]\!]$ *a* functor *is the formal way of insisting that semantics respects composition: interpreting "do p then q" should equal "interpret p, interpret q, then compose." This is the semantic counterpart of the engineering fact that wiring subroutines together yields a composite behavior.*

As examples: **Syn** *could be a category generated by flowcharts or by typed lambda terms;* **Abs** *could be a category of partial functions or of probabilistic state transformers. This definition is useful because it cleanly separates three levels that are often conflated: the* symbolic *artifact (program text), the* mathematical *object it denotes (abstract process), and the* physical *evolution that enacts it (realization below). This separation is also congenial to algorithmic-information viewpoints in which programs function as descriptions of processes [16], while remaining compatible with richer semantic theories.*

It is often helpful (though not required here) to think of **Syn** *as providing the* compositional interface *available to a programmer: the primitives and composition rules that are exposed for assembling behaviors. Dually,* **Abs** *provides the space of meanings in which one can state correctness claims. On this view, different semantics functors* $[\![\cdot]\!]$ *correspond to different intended interpretations of the same surface syntax (e.g. operational vs denotational semantics, or a resource-aware semantics), and the functoriality condition is what makes correctness proofs scale from primitives to composites.*

Definition 386 (Computer as an emulation machine)**.** *A* computer *is a machine* C *equipped with:*

(a) a set (or category) of admissible programs $\mathbf{Syn}_C$ *(the "program signals" the user can provide); and*

(b) an execution mechanism that, for many $p \in \mathbf{Syn}_C$*, yields a physical realization of* $[\![p]\!]$ *(Definition 388 below);*

such that the family $\{[\![p]\!] : p \in \mathbf{Syn}_C\}$ *spans (approximately) a large variety of machine behaviors within the relevant engineering domain, and reconfiguration (from* p *to* p'*) has modest effort cost for the relevant user class.*

Remark 1309. *The clause "set (or category) of admissible programs" allows* $\mathbf{Syn}_C$ *to carry additional structure beyond mere membership. For instance, one may want to talk about translations/compilers between sublanguages, program equivalences, or program composition as an internal operation, in which case it is natural to model programs as morphisms in a category rather than as elements of a set. Likewise, the phrase "program signals" is intentionally broad: for different machines these may be voltages on pins, bitstrings in memory, punched cards, network packets, or even physical rewiring that selects among a finite library of behaviors, so long as the resulting reconfiguration is systematic and cheap relative to rebuilding.*

Remark 1310. *In plain language: a computer is a machine whose behavior can be widely varied by supplying different symbolically representable "recipes," and doing so is cheap compared to rebuilding the machine. This captures the practical heart of programmability: the user moves in a space of behaviors by manipulating signs rather than metal. In Hyperseed terms, the computer is a tool (indeed, a machine) whose affordance profile is unusually broad and compositional, and whose use is mediated by a stable syntax/semantics interface (Hyperseed-Concept 80).*

A simple example is a microcontroller that can emulate many small control circuits by changing firmware. A more expansive example is a general-purpose CPU that can approximate a huge range of behaviors by interpreting different programs as different abstract processes. The definition is *useful because it avoids privileging any single formal model of computation: what matters is not a particular axiomatization but an empirically grounded emulation capacity relative to a user class and engineering domain, in the same spirit as Hyperseed's operational approach to intelligence [19].*

The qualifier "(approximately)" is doing real work: physical computers are constrained by finite precision, noise, timing jitter, component tolerances, and resource limits, so their realizations typically match $[\![p]\!]$ only up to an error model and within an operating regime. This is why emulation here should be read in an engineering sense: the machine is a reliable stand-in *for the target behavior under specified assumptions (e.g. clock bounds, numeric ranges, memory availability), not a metaphysical identity. In particular, the same abstract process may admit many realizations (different chips, different microarchitectures, different compilation strategies), and the point of the syntax/semantics interface is to make these interchangeable for the relevant users.*

It is also worth distinguishing the present use of "emulate" from everyday "simulate." A simulation may reproduce aspects of a target system for purposes of prediction while remaining embedded in a larger computational context (e.g. a weather model run inside a program that also handles I/O and visualization). By contrast, the emulation perspective emphasizes behavioral substitution*: supplying p configures C so that, from the perspective of a designated interface, the resulting physical behavior can play the same role as the emulated machine's behavior.*

Remark 1311 (Curry–Howard and typed semantics (optional))**.** *If one chooses* **Syn** *to be a typed lambda calculus (or related proof calculus), then $[\![\cdot]\!]$ expresses a Curry–Howard style correspondence between proofs and programs. In that setting, "computer programs are physical realizations of mathematical constructions" becomes a literal statement: proof terms (syntactic objects) are realized as physical signal-configurations that drive a machine to enact the corresponding semantic morphisms.*

This optional perspective is also a reminder that the choice of **Abs** *can encode more than input–output behavior: it can encode invariants, specifications, or logical propositions that the program is meant to witness. Then correctness can be phrased as a claim that execution realizes not only a function-like mapping but also a structured piece of meaning (e.g. a proof-relevant morphism), which is particularly useful when connecting "program" to "procedure" and "tool use" in embodied contexts.*

Remark 1312. *The Curry–Howard remark emphasizes a broader methodological moral: the boundary between "mathematics" and "mechanism" is in part a boundary between description languages and the physical realizations that make those descriptions causally effective. This resonates with Hyperseed-Concept 107 and Hyperseed-Concept 136, without requiring any particular foundational stance.*

23.6 Physical realizations of abstract processes

Hyperseed describes a physical realization as a case where a physical process P and an abstract process A have strong *intensional similarity*, but one is physical and the other abstract. We express this as a realizability structure: encodings/decodings plus dynamics.

Definition 387 (Physical process category)**.** *Fix a physical reality-system R_{phys} for an embodied mind (Section 25). Let* **Phys** *be a category whose objects are physical substrates (bodies, tools, machines, laboratory setups) and whose morphisms are physically realizable processes (possibly stochastic, possibly context-dependent).*

Remark 1313. *The category* **Phys** *is deliberately broad: it is not restricted to deterministic Newtonian mechanics, but may include stochasticity, noise, measurement back-action, and context-dependence. This is important for Hyperseed because the observer/context O is always present in* *the modeling: what counts as a morphism is what the observer regards as a realizable transformation in the reality-system [1]. The point is not to deny physics, but to represent physics as a structured family of allowed transformations inside a reality-system (Hyperseed-Concept 135).*

A concrete example: an object of **Phys** *might be a particular circuit board; a morphism might be "apply voltage profile $u(t)$ for $t \in [0, T]$," resulting in a physical state transition. This definition is useful because it provides the codomain in which implementations live: abstract morphisms are realized by (structured) physical morphisms.*

Definition 388 (Physical realization / implementation)**.** *Let $f : A \to B$ be an abstract process in* **Abs***. A* physical realization *of f is a tuple*

$$\mathcal{R} = (P, \mathrm{enc}, \mathrm{dec}, \phi)$$

where:

- *P is a physical substrate (an object of* **Phys***),*
- $\mathrm{enc} : A \to \mathrm{State}(P)$ *is an encoding of abstract inputs into physical states,*
- $\mathrm{dec} : \mathrm{State}(P) \to B$ *is a decoding of physical states into abstract outputs, and*
- $\phi : \mathrm{State}(P) \to \mathrm{State}(P)$ *is (a model of) the relevant physical dynamics of P induced by the realization setting (including any control signals regarded as part of the realization),*

such that $\mathrm{dec} \circ \phi \circ \mathrm{enc}$ *approximates f in the intended sense.*

Remark 1314. *The notation* $\mathrm{State}(P)$ *is used informally for the state space associated to the substrate P (for instance, a set of microstates, a manifold of configurations, or a space of probability distributions over configurations, depending on modeling choices). The functions* enc *and* dec *are the formal representation of what engineers call "representation" or "interface": how abstract inputs are written into the physical medium, and how physical outcomes are read back as abstract outputs (Hyperseed-Concept 157 and Hyperseed-Concept 136). The map ϕ is the effective dynamics of the substrate during execution; it may itself depend on context and control, but that dependence is treated as part of the realization setting.*

A simple example is an adder circuit realizing the abstract function $f : \{0,1\}^2 \to \{0,1\}$ given by XOR. Here P is a gate-level circuit, enc *maps bits to voltage levels, ϕ is the electrical evolution over a short time window, and* dec *maps the output voltage back to a bit. The definition is useful because it makes explicit where approximation can enter: errors can come from imperfect encodings, noisy dynamics, or lossy decoding, and these can be measured and managed.*

Remark 1315 (Quantale- / p-bit-valued correctness scores)**.** *In realistic settings, realization quality is empirical and observer-relative. A convenient representation is a score*

$$\mathrm{Score}(\mathcal{R}, f) \in \mathbb{V} = [0,1]^2,$$

where the positive component measures evidence that the realized behavior matches f and the negative component measures evidence of mismatch. This score can be obtained by tests, formal verification, physical modeling, or mixtures.

Remark 1316. *The* p-bit*-valued score again encodes the possibility of conflict: one may simultaneously possess strong evidence of correctness on a test suite and strong evidence of failure in edge cases. This is not a defect but a faithful representation of engineering knowledge, which is often paraconsistent in practice even when not acknowledged as such. It also connects implementation to the broader Hyperseed value layer: correctness, safety, and reliability are themselves evaluated through evidence-bearing processes rather than assumed as absolute, context-free facts.*

One of the main reasons to state physical realizations explicitly is to guarantee that "wiring tools together" corresponds to composition of abstract processes.

Proposition 36 (Compositionality of realizations (lax functoriality))**.** *Let $f : A \to B$ and $g : B \to C$ be abstract processes. Suppose $\mathcal{R}_f = (P_f, \mathrm{enc}_f, \mathrm{dec}_f, \phi_f)$ realizes f and $\mathcal{R}_g = (P_g, \mathrm{enc}_g, \mathrm{dec}_g, \phi_g)$ realizes g. Assume the interface is compatible in the sense that the output of $\mathcal{R}_f$ can be fed as input to $\mathcal{R}_g$ (e.g.* enc_g *accepts the decoded outputs of $\mathcal{R}_f$, or there is a fixed adaptor with negligible additional error). Then there exists a composite realization $\mathcal{R}_{g\circ f}$ of $g \circ f$ on the composite substrate $P_f \triangleright P_g$ ("run P_f then P_g") such that*

$$\mathrm{Score}(\mathcal{R}_{g\circ f}, g \circ f) \;\; \geq \;\; \mathrm{Score}(\mathcal{R}_g, g) \, \otimes \, \mathrm{Score}(\mathcal{R}_f, f),$$

where $\otimes$ is the chosen monoidal product on $\mathbb{V}$ (Section 3.4). In particular, the direction of the inequality emphasizes that we are claiming only a guaranteed lower bound on end-to-end correctness evidence: even if the composite system has additional sources of error, the combined stage-level evidence still yields a conservative certificate in the order on $\mathbb{V}$.

Remark 1317. *Informally, the proposition says: if you have a physical implementation of f and a physical implementation of g, and you can connect them so that the output of the first is a valid input to the second, then you can implement the composite computation $g \circ f$ by running the first and then the second. Moreover, a conservative lower bound on the composite correctness evidence is given by combining the stage-level correctness evidence using the same evidence-aggregation operation $\otimes$ used elsewhere in the* p-bit *quantale.*

This matters because much of engineering (and much of cognition) proceeds by composition: we build systems out of subsystems. Hyperseed needs a formulation of this compositionality that is compatible with paraconsistent, evidence-based correctness rather than requiring idealized perfect correctness. This proposition is also the implementation-analogue of the categorical principle that semantics should respect composition (as in the functor $[\![\cdot]\!]$ above). One can read $P_f \triangleright P_g$ as an abstract model of a pipeline: a temporally ordered coupling (not necessarily a mere product of substrates) in which the physical degrees of freedom used to represent the intermediate value are transferred, re-encoded, or otherwise adapted so that the second stage can consume them. The point is not that the intermediate representation must be identical across stages, but that the realization data includes whatever interface mechanism makes it meaningfully the "same value" for the purposes of the overall abstract computation.

Proof. Define the composite substrate to execute ϕ_f on P_f and then ϕ_g on P_g. Define the composite encoding as enc_f, and the composite decoding as dec_g, with the intermediate adaptor given by the compatibility assumption. Concretely, the composite realization packages (i) the first-stage preparation of a physical state from an abstract input via enc_f, (ii) evolution by ϕ_f on P_f, (iii) an interface map that turns the relevant output degrees of freedom of P_f into admissible input degrees of freedom for P_g (which may include copying, buffering, re-timing, transduction, or re-encoding), and finally (iv) evolution by ϕ_g on P_g followed by readout via dec_g. This makes explicit that the intermediate "wire" is itself part of the physical story: the claim only goes through when the adaptor is treated as realization structure rather than an implicit idealization.

Under the interpretation of $\otimes$ as multiplicative aggregation of evidence, independent positive evidence for correctness multiplies across stages, and (at minimum) the resulting composite score is bounded below by the product of the stage scores. This is the standard "pipeline" principle of implementation: correctness does not improve by composition, but a reliable stage chained after a reliable stage remains reliably correct in the aggregated-evidence sense. Formally, what is used here is that $\otimes$ is monotone in each argument with respect to the order on $\mathbb{V}$: if a realization of the composite can be analyzed as first satisfying f to some degree and then satisfying g to some degree (with compatibility ensuring that the second analysis is applicable to the first stage's outputs), then combining the two pieces of evidence via $\otimes$ yields an evidence value that is *no stronger than* any additional, possibly more complex end-to-end analysis, hence the inequality as a lower bound. The "independence" language should be read in the conservative sense appropriate to paraconsistent evidence: we are not assuming that errors are uncorrelated in a probabilistic model, only that the evidence calculus represented by $\otimes$ is designed to aggregate stage-level support without introducing unjustified strengthening. In practical terms, even when the stages share failure modes (for example, both depend on a common power supply, clock, or calibration procedure), the proposition still serves as a template for what one may safely claim given only the two stage scores and an interface condition. □

Proof sketch. The strategy is to build the composite realization by literally running the first physical process and then the second, using the interface-compatibility assumption to connect the two runs. The inequality then follows from the intended semantics of $\otimes$ as an evidence aggregator: stage-level evidence is combined to yield a conservative bound on end-to-end evidence. Intuitively, one can think of $\mathcal{R}_{g\circ f}$ as a "program" for the laboratory or machine shop: prepare according to enc_f, run the first device, translate its output into the input format of the second device, run the second device, and finally read out using dec_g. □

Remark 1318. *The key step is conceptual rather than technical: we treat encodings/decodings as part of the realization structure, so the "wiring" between stages is not a handwave but an explicit adaptor condition. Once that is granted, the remaining content is the monotonicity and aggregative meaning of $\otimes$ in $\mathbb{V}$. Geometrically one may picture each stage as a noisy channel between abstract spaces; composing two channels yields another noisy channel whose reliability is (at best) the product-like combination of the reliabilities of the components. A helpful concrete picture is a sensing-and-control loop: f might be a sensor pipeline mapping a physical quantity to a digital estimate, and g a control law mapping that estimate to an actuator command; the composite $g \circ f$ is the overall perception-to-action transform. Even if the sensor and controller are each only partially correct in the paraconsistent evidence sense, chaining them yields an overall system with an evidence score that is at least the $\otimes$-combination of the two, provided the estimate produced by the sensing stage is indeed in the admissible input class for the control stage (units, ranges, timing, and representation all count as part of this admissibility).*

Remark 1319. *Proposition 36 is intentionally weak: it asserts only a monotone lower bound on composite correctness evidence. This matches engineering reality: composition can introduce new failure modes, but one can still obtain a conservative guarantee by multiplying (or otherwise aggregating) stage-level guarantees. The weakness is also a feature for Hyperseed's intended use: it allows modular reasoning in the presence of inconsistency, where one may have strong evidence for some aspects of each stage and counterevidence for others, and still wish to propagate whatever support is available without collapsing into triviality. In particular, the proposition does not claim that* $\mathrm{Score}(\mathcal{R}_{g\circ f}, g \circ f)$ *equals* $\mathrm{Score}(\mathcal{R}_g, g) \otimes \mathrm{Score}(\mathcal{R}_f, f)$*, nor that the bound is tight; it merely guarantees that the evidential calculus respects the basic operational fact that running two verified-enough stages in sequence yields a verified-enough composite, in the sense captured by the order on* $\mathbb{V}$*.*

23.7 Embodiment and the body-channel

Hyperseed defines a physical reality (for an embodied mind) as a reality-system containing the body where the mind's causal impact on that reality is mostly mediated through the body-channel. We represent this using factorization constraints on causal influence. This framing is deliberately structural: rather than stipulating what "matter" or "the physical" is in itself, it specifies which parts of a mind–world interaction graph are treated as primary for modeling, prediction, and intervention.

Definition 389 (Body as a coupled pattern-system). *Let M be a mind-level pattern web (Sections 13–17). A* body *for M is a physical substrate B such that:*

(a) there is strong mutual causal coupling between M and B (modeled as bidirectional channels for perception and action), and

(b) B is largely contained in a designated physical reality-system R_{phys}.

Remark 1320. *Intuitively, this definition says that a body is not merely "owned" by a mind, but* coupled *to it: the mind changes because the body is sensed, and the body changes because the mind acts. The bidirectionality clause is what distinguishes a body from a mere instrument occasionally used: the coupling is persistent and forms a significant part of the mind's ongoing dynamics. This is the formal doorway into Hyperseed-Concept 135 and Hyperseed-Concept* **??** *(the latter being implicit in the ontology even when not singled out as a separate primitive).*

Examples: for a human mind-model M, B may be the organism's nervous system and musculoskeletal system as represented in a physical reality-system. For a robot controller M, B may be the sensor-actuator suite plus onboard compute substrate. The usefulness of the definition is that it makes "having a body" a structural property that can be used in later causal and cosmological definitions: a physical reality-system is then the one in which this body is situated and through which most causal influence flows.

Two clarifications are often useful in applications. First, "strong mutual causal coupling" is not intended as a binary predicate but as a dominance claim: relative to the observer's modeling granularity, there are sustained causal pathways $B \to M$ (perception, interoception) and $M \to B$ (action, regulation) that explain a large fraction of M's state transitions and a large fraction of B's state transitions on the timescales of interest. Second, the boundary of B may be context-dependent: prostheses, implants, or tightly integrated tools can be treated either as part of B or as external artifacts depending on whether their coupling to M is persistent, high-bandwidth, and governed by the same closed-loop dynamics that ordinarily characterize sensorimotor control.

Definition 390 (Body-channel mediation (factorization form)). *Fix a physical reality-system R_{phys} and an embodied mind (M, B). We say that M's causal impact on R_{phys} is* mostly mediated through the body *if, for most effects e in R_{phys} that are attributable to M (relative to observer O), the corresponding influence morphism factors as*

$$M \xrightarrow{\text{Act}} B \xrightarrow{\text{Phys}} R_{\text{phys}},$$

up to the chosen approximation notion in the process formalism (Section 3). In particular, "most" and "attributable" are to be read as observer-indexed judgments about explanatory and predictive relevance: O selects a causal model, a notion of intervention/counterfactual dependence, and a tolerance under which small residual channels may be ignored without materially changing forecasts or control policies.

Remark 1321. *The factorization diagram is the mathematical version of the everyday claim that, in physical reality, minds act through bodies. The arrow labeled* Act *stands for the mind-to-body channel (motor control, endocrine control, etc.), and the arrow labeled* Phys *stands for body-to-world causal propagation. The qualifier "mostly" is essential: it permits exceptions (e.g. indirect influence via other agents or artifacts) while asserting that the body-channel carries most of the causal mass relevant for prediction and control.*

This is useful because it provides a criterion for when a reality-system should be called "physical" relative to a mind: if there are large, reliable mind-to-world influence channels that do not factor through the body, then the modeling stance changes (as the next remark states). In short, physicality is treated operationally via channel structure rather than metaphysically.

Although the definition is stated in terms of impact *(an $M \to R_{\mathrm{phys}}$ direction), the same modeling posture typically comes with a complementary perception-side regularity: for most observations o in M that are about R_{phys}, the relevant information flow factors as*

$$R_{\mathrm{phys}} \xrightarrow{\mathrm{Sens}} B \xrightarrow{\mathrm{Per}} M,$$

again up to approximation. This makes explicit why B functions as an interface: it is simultaneously the dominant actuator path outward and the dominant sensor path inward, yielding a closed loop whose stability and bandwidth strongly constrain the kinds of policies and representations that are feasible for M in R_{phys}.

Remark 1322. *If a mind can reliably influence a reality-system without using the body-channel, then (in the Hyperseed sense) that reality-system is less "physical" relative to that mind. The point here is not metaphysical; it is a modeling decision about which channels carry most causal mass for the observer's predictions. Concretely, if the observer must posit persistent high-bandwidth morphisms $M \to R$ that bypass B in order to explain regularities (e.g. dependable "direct" effects of intention on distant states), then it becomes misleading to treat R as a standard physical reality-system for that mind, because the dominant control affordances would not be organized around embodied action.*

Remark 1323. *This factorization stance also prepares later social and communion discussions: many causal influences in society do not run directly through one's own body but through artifacts, institutions, and other agents. Separating "body-channel" influence from "artifact-mediated" influence lets one speak precisely about extended agency and collective mind systems without losing the embodied anchor. In particular, it allows a clean distinction between (i) amplification of embodied action (where* Act *still initiates the relevant causal chain, even if subsequent propagation is mediated by machines and institutions) and (ii) putative non-embodied channels (which would require a different ontological placement in the causal web). This becomes important when describing delegation, coordination, and the emergence of higher-level agents whose effective "bodies" may be distributed across many human bodies and technical substrates.*

23.8 Algebraic asymmetry of physical reality

Hyperseed emphasizes that physical reality tends to have a relatively rigid, low-dimensional (vector-space-like) structure, while mental and intersubjective realities tend to have weaker algebraic structure (often best treated topologically), and that the latter can only approximately be projected into the former. This "rigidity" is meant in the sense that physical descriptions typically come with many simultaneously interacting constraints (e.g. compositional laws, conservation principles, and dynamical compatibility conditions), whereas mental and social descriptions often preserve only those aspects that remain stable under coarse changes of implementation (e.g. continuity rather than linearity, reachability rather than exact trajectories, or preference/ordering rather than precise quantities).

A minimal formalization uses a "forgetful" map from stronger structure to weaker structure. Concretely, one can treat "physical structure" and "mental/intersubjective structure" as different *signatures of operations and relations* together with axioms, where the mental signature is obtained from the physical signature by dropping some operations, equations, or distinguished predicates. In that common situation, the corresponding forgetful functor is not an additional assumption but a standard construction: every model of the richer theory canonically determines a model of the poorer theory by restriction of structure.

Definition 391 (Algebraic asymmetry (structure implication form))**.** *Let* $\mathsf{Alg}_{\mathrm{phys}}$ *be an algebraic theory capturing salient physical structure (e.g. vector-space operations, linear dynamics, conservation laws) and let* $\mathsf{Alg}_{\mathrm{mind}}$ *be a weaker theory capturing salient mental/intersubjective structure (e.g. topology, partial order, or merely the quantale-enriched relational structure).*

We say that a community's physical and mental realities exhibit algebraic asymmetry *if:*

(a) there is a canonical forgetful functor

$$U : \mathrm{Alg}(\mathsf{Alg}_{\mathrm{phys}}) \to \mathrm{Alg}(\mathsf{Alg}_{\mathrm{mind}})$$

that maps each physically structured system to its underlying mental/intersubjective structure; but

(b) there is no functorial inverse that assigns to each mental/intersubjective system a unique physically structured system realizing it (at best, one has approximations, constraints, or choices).

Remark 1324. *The definition says, in effect, that physical structure is (for the relevant community) more constraining than mental/intersubjective structure. There is a systematic way to* forget *some structure (go from richer algebra to poorer algebra), but there is no systematic way to* reconstruct *the richer structure uniquely from the poorer. This is Hyperseed-Concept 55 expressed in standard mathematical language (forgetful functors and non-invertibility).*

A guiding example is that many different vector spaces can share the same underlying set, and many different norms can induce the same coarse topology; the "mental" structure (say, topology or order) typically underdetermines the "physical" structure (say, linear operations and conserved quantities). The usefulness of the definition is that it provides a clean formal lever for later claims about implementation and projection: mental contents may be realized physically only by selecting among many physically distinct instantiations, and that selection is constrained by effort, resources, and context. In particular, the "no inverse" clause should be read as a non-uniqueness *claim: even when a realization exists, it generally does not exist* canonically *(i.e. in a way that is natural with respect to morphisms in* $\mathrm{Alg}(\mathsf{Alg}_{\mathrm{mind}})$*), and so any reconstruction procedure necessarily imports additional conventions, priors, or external constraints.*

Remark 1325. *It is worth separating three notions that can otherwise be conflated: (i) existence of some right-inverse* on objects *(each mental structure is realizable by at least one physical structure), (ii) existence of a* section S *with* $U \circ S = \mathrm{id}$ *(a consistent choice of a realization for each mental object), and (iii) existence of a* functorial *section (the chosen realizations respect structure-preserving maps between mental systems). The definition rules out (iii), which is the categorical form of "no systematic unique reconstruction." In many settings (especially when mental descriptions ignore metrics, coordinates, or conserved quantities), one may have (i) for broad classes of cases, occasionally (ii) after imposing extra choices, but typically not (iii) without adding additional structure* *to* $\mathsf{Alg}_{\text{mind}}$ *that effectively re-introduces the missing physical constraints. This captures the sense in which embodiment and implementation require* extra degrees of freedom *beyond what is present in the mental description alone.*

Remark 1326 (The simplest motivating example)**.** *A real vector space canonically determines a topological space (via the metric induced by a norm), but an arbitrary topological space does not determine a vector space structure. This is an archetype for Hyperseed's claim: physical structure typically implies (and constrains) mental/intersubjective structure, whereas mental structure only partially constrains physical structure and often requires approximation to be physically instantiated. One can sharpen the same point with even simpler forgetful functors: a group canonically determines a set, but a set does not canonically determine a group; a smooth manifold canonically determines a topological space, but a topological space does not canonically determine a smooth structure. These examples emphasize that "forgetting" is often a many-to-one collapse of possibilities: the fibers of* U *over a given mental object can be large, reflecting many distinct physical implementations compatible with the same mental/intersubjective description.*

Remark 1327. *The community-relative phrasing is material: what counts as "salient physical structure" depends on the community's instrumentation, embodiment, and practical invariants. A community that can only measure coarse macroscopic variables may treat distinct microphysical configurations as physically irrelevant, thereby shifting the boundary between* $\mathsf{Alg}_{\text{phys}}$ *and* $\mathsf{Alg}_{\text{mind}}$*. In that sense, algebraic asymmetry is compatible with multiple nested layers of description: one may have a chain of forgetful functors from microphysics to macrophysics to phenomenology, each step discarding structure and enlarging the space of admissible realizations.*

Proposition 37 (Asymmetry and weakness)**.** *If mental/intersubjective structure is modeled as a coarse-graining (a controlled collapse of distinctions) of physical structure, then the passage from physical to mental reality increases quantale weakness (Section 3.7) relative to the finest-grained physical observer.*

Remark 1328. *In plain terms, the proposition says: if we go from a fine-grained physical description to a coarser mental/intersubjective description by identifying distinctions, then we become "weaker" in the sense that more states are treated as undistinguished. This is not a pejorative claim; it is a structural one. It links the philosophical idea of coarse-grained experience to the formal order-theoretic notion of weakness developed earlier [3, 2]. The connection to the preceding definition is that the forgetful passage* U *is a structured way of* discarding *discriminations: when operations/observables are forgotten, the induced semantics can merge previously separable possibilities, thereby enlarging the indistinguishability relation that defines weakness.*

Proof. Coarse-graining identifies (or partially identifies) physical states that were previously distinct. In the weakness formalism, this corresponds to adding undistinguished pairs. By Theorem 1, adding undistinguished pairs monotonically increases weakness. More explicitly, if the finest-grained physical observer induces an undistinguished relation $\sim_{\text{phys}}$ on a state space, and the coarse-grained mental/intersubjective observer induces $\sim_{\text{mind}}$, then coarse-graining means $\sim_{\text{phys}} \subseteq \sim_{\text{mind}}$ (at minimum, every pair that was already indistinguishable remains so, while some formerly distinguishable pairs become indistinguishable). The monotonicity theorem applies to this inclusion. □

Proof sketch. The proof reduces the statement to monotonicity: coarse-graining is exactly the act of declaring previously distinct states to be undistinguished, and weakness was defined so that enlarging the undistinguished relation increases weakness. Equivalently, coarse-graining is an order-preserving map in the direction that collapses distinctions, and weakness is ordered so that more collapse corresponds to greater weakness. □

Remark 1329. *The key step is the identification of "coarse-graining" with "adding undistinguished pairs." Once this is made explicit, the result is essentially a corollary of how weakness is ordered. A visual intuition is to imagine a high-resolution image being pixelated: multiple microstates collapse into one macrostate, and the observer loses discriminative power. In Hyperseed terms, this loss is not merely epistemic; it shapes what patterns can be represented and what control can be exerted, because both depend on distinctions. A further intuition, aligned with the functorial language above, is that the collapse induced by U can be understood as pushing forward along an "observation" that identifies states whenever the remaining (mental/intersubjective) invariants agree; since fewer invariants are tracked, more pairs agree, and the indistinguishability relation expands.*

23.9 Two tiny examples

23.9.1 A tool that reduces effort via mechanical advantage

Let a task be "raise a weight" within a physical reality-system. Model the relevant effort scalar as $E \in [0, 1]$ where larger means more effort. Let $E_{\text{bare}} = 0.9$ be the effort without a tool, and let a lever tool T yield $E_T = 0.3$ for the same achieved effect in the same context. One can read the choice of $[0, 1]$ here as a normalization convention: it fixes units only up to monotone rescaling, but it is enough to express ordinal comparisons (this action is easier than that action) and threshold claims (an agent can afford at most $E \leq \epsilon$ in a given episode). The phrase "same achieved effect" is also doing work: the comparison is not between different goals, but between different *means* to the same end, so that the effort change can be attributed to the tool rather than to a change in objective. In many environments the effort would be context-dependent (terrain, grip, fatigue, precision requirements), and the toy scalar E should be read as already incorporating those background conditions into a single budget-like quantity.

Remark 1330. *This is the archetypal case where the affordance profile changes the mapping from action to effect by changing the required resource expenditure: the lever increases the space of achievable effects for a bounded agent because actions that were previously too costly become feasible. This illustrates why Hyperseed keeps effort (Section 8) close to the ontology of tools: tools are not merely causal mediators but* resource transformers *(Hyperseed-Concept 100 and Hyperseed-Concept 191).*

In Hyperseed terms: the lever is a tool because it is purposefully manipulable toward the objective, and it extends agency by lowering effort for the same outcome. In the quantale-weakness perspective, lowering effort typically enables the agent to maintain finer distinctions elsewhere (attention, memory, control), indirectly reducing global weakness. Equally importantly, the reduction $E_{\text{bare}} \mapsto E_T$ can be understood as a change in the *effective* action space: even if the bare body can in principle generate the required force, the lever may move the task from "barely possible" to "reliably repeatable" under noise, limited precision, and finite time. This motivates treating tools not just as add-ons but as parts of an extended sensorimotor loop, where the tool shifts which policies are stable under perturbations and which plans remain within budget. On this reading, mechanical advantage is just one especially transparent instance of a more general pattern: an external artifact can reshape the effort landscape so that the same target state becomes reachable by a broader set of trajectories.

23.9.2 A program as a physical realization of an abstract construction

Let **Syn** contain a program p whose semantics $[\![p]\!]$ is an abstract process $f : A \to B$ (e.g. a parsing algorithm, a planning operator, or a proof normalization map). Running p on a computer C yields a physical realization $\mathcal{R} = (P, \mathrm{enc}, \mathrm{dec}, \phi)$ of f in the sense of Definition 388. Here P can be read as the relevant physical state space (register contents, memory configurations, I/O buffers, and possibly environmental degrees of freedom that the computation couples to), enc and dec specify how abstract inputs and outputs are represented at that physical level, and ϕ names the concrete evolution (often a time-indexed dynamics induced by the machine executing the compiled form of p). Even in this simple packaging, the role of representation becomes explicit: a program does not act on A and B directly, but on their encodings, and the "same" abstract function can be realized by different choices of code, data layout, and hardware, yielding different resource profiles.

Remark 1331. *The example highlights the three-layer separation:* program *(a syntactic object),* meaning *(an abstract morphism), and* execution *(a physical evolution plus an interface). In practice, many confusions about computation come from sliding between these layers without noticing. The formalism forces us to keep track of where a claim lives: is it about denotation ($[\![p]\!]$), or about implementation quality (*$\mathrm{Score}(\mathcal{R}, f)$*), or about resource usage (effort, time, energy) of the physical substrate? This is precisely why Hyperseed-Concept 80 is placed alongside Hyperseed-Concept 136.*

This makes precise (at a minimal level) the slogan:

> Computer programs are physical realizations of mathematical constructions.

The slogan becomes more structured if **Syn** is taken to be a proof calculus (Curry–Howard), but the basic point already holds at the level of "syntax $\to$ semantics $\to$ physical realization." The additional structure in $\mathcal{R}$ is what allows one to distinguish correctness from mere intention: $[\![p]\!] = f$ is an abstract statement about denotation, while the realized process may approximate, fail, or only probabilistically track f because ϕ is embedded in a noisy, finite-precision substrate. Moreover, two realizations can agree extensionally on f yet differ significantly in embodied properties (latency, energy draw, heat dissipation, fault tolerance), which are invisible if one collapses everything into the abstract morphism alone. In Hyperseed terms, this is exactly the point at which computation becomes part of embodiment: the interface maps abstract types into physical carriers, and the carrier dynamics determines which abstract constructions are feasible under an agent's resource constraints. This also clarifies why "running the same program" can be an underspecified claim: without fixing enc, dec, and the operational context of ϕ, one has not fixed the realized process that the world actually instantiates.

23.10 What this section contributed to the overall reconstruction

This section mapped the Hyperseed nodes *tools*, *machines*, *engineered*, *computers and programs*, *physical realizations*, and *physical reality/body with algebraic asymmetry* into the paper's core mathematical vocabulary. In particular, it translated what can look like informal engineering talk (devices, implementations, programs, bodies) into a small number of reusable structural moves: coupling, emulation, factorization through channels, and order-theoretic coarse-graining. This matters because it makes later claims about "shared worlds" and "scaling" provably sensitive to the constraints and degrees of freedom that artifacts and bodies impose, rather than treating them as externalities.

- Tools and machines become physical pattern-systems whose affordances can be represented as p-bit-valued couplings from actions to effects. Concretely, one can treat an action-set (or action-algebra) A and an effect-set E as interfaces, and model the tool as a constrained relation or channel $A \rightsquigarrow E$ whose entries live in p-bit, so that "what the tool lets you do" is captured by a structured map rather than a verbal description. On this reading, a tool is not merely an object but an input–output regularity stabilized in the world, and the coupling encodes both possibilities and impossibilities as first-class data.

- Computers and programs become emulation machines plus syntax/semantics layers, with physical realization connecting semantics to physics. Here the central explanatory move is to separate (i) the physical machine as an emulator (a pattern-system capable of reliably realizing a family of state transitions), from (ii) a syntactic layer (descriptions, code, symbols) and (iii) a semantic layer (what those symbols are taken to denote or implement), and then to insist that a *realization* map is what makes the semantics causally efficacious. This is precisely what lets later sections speak about "the same program" across different substrates without erasing embodiment: sameness is routed through emulation and realization, not assumed as a primitive identity.

- Embodiment becomes a factorization constraint on causal impact through a body-channel. Instead of treating an agent's influence as a direct action on the external world, the reconstruction forces a decomposition in which the agent's internal degrees of freedom can only affect distal variables via a mediating interface (the body), i.e., via a channel that both enables and limits control. This makes constraints such as bandwidth, locality, actuation delay, and sensor noise part of the formal story: what an agent can bring about is restricted to what can pass through that body-channel and be amplified by the surrounding physics.

- Algebraic asymmetry becomes a formal "strong structure forgets to weak structure" relation, with coarse-graining captured by monotone increases in weakness. The point of phrasing this as an asymmetry is that there is typically an easy, structure-forgetting map from a richer description to a poorer one, while reconstruction in the opposite direction is underdetermined or impossible without extra information. By encoding this as a monotone order ("more weak" = "less structured"), the framework gains a disciplined way to talk about abstraction layers (e.g., physical $\to$ computational $\to$ semantic) and about why certain equivalences are stable under forgetting while others are not.

Remark 1332. *From a broader perspective, these constructions place "engineering" inside the same conceptual frame as "pattern" and "agency": engineered artifacts are patterns stabilized by goal-directed processes, and computers are machines whose stable patterns include a semantics-respecting reconfiguration mechanism. Put differently, an engineered object is a pattern whose persistence is not merely accidental but maintained (historically and/or presently) by optimization, selection, or design, and this maintenance can be tracked using the same coupling-and-channel vocabulary used elsewhere in the reconstruction. Moreover, emphasizing semantics-respecting reconfiguration highlights what distinguishes general-purpose computation from mere dynamics: the machine is organized so that changes in symbolic descriptions (programs) systematically correspond to changes in realized behavior, with the realization map doing the work of tying symbols to physical transitions. This is why these notions are not merely ancillary but central to the later discussion of intersubjective communion and cosmological scaling: communities build shared realities partly by building shared artifacts [1, 19]. The shared artifact, on this view, is a coordination device precisely because it induces shared couplings: multiple agents can reliably predict how actions propagate to effects through the artifact, and this predictability is a precondition for stable joint practice.*

These constructions are used in the next sections to formalize intersubjective communion and cosmological scaling in a way that keeps the engineering/implementation dimension explicit. Intersubjective communion can then be treated not only as alignment of beliefs or symbols, but also as alignment of accessible channels and realizations (e.g., agreeing on what a sign, tool, or protocol *does* because the underlying couplings are shared and repeatedly testable). Cosmological scaling, likewise, can be discussed in terms of how such couplings, emulators, and body-channels compose and persist across increasing spatial, temporal, and energetic scales, rather than being idealized away as if agency operated on an unbounded, implementation-free substrate.

24 Intersubjectivity, Communion, and Aesthetics

24.1 Hyperseed concepts handled in this section

This section gives a mathematically grounded treatment of the following Hyperseed concepts:

- intersubjective reality;
- intersubjective communion and explicit intersubjective communion;
- spiritual experience as communion with a broader-scope mind; psychedelics/entheogens;
- emotion and compassion;
- archetypes (as cross-context pattern templates);
- aesthetics/art/beauty as "surprising fulfillment of expectations" and the role of paraconsistent logic;
- religion and ritual as persistent social technologies for inducing and stabilizing communion;
- (briefly) state-dependent science as a community-level epistemic practice conditioned on states of consciousness.

To avoid ambiguity, the intent of the list above is not merely thematic but structural: each item will be assigned an explicit representational role inside the same formal apparatus (patterns, carriers, links, and state/observer parameters), so that claims about communion, archetypes, or beauty can be expressed as statements about how patterns distribute and couple across minds and artifacts. In particular, "intersubjective reality" is treated here as an *object of analysis* rather than a vague backdrop: it is the specific pattern-structure that a community can jointly sustain, predict within, and coordinate action around. Likewise, "communion" is treated as a *dynamical regime* of that structure (a change in the balance between within-mind and cross-mind pattern dominance), and "explicit communion" as the additional condition that some agents in the community represent (possibly imperfectly) that regime within their self- and world-models.

The inclusion of psychedelics/entheogens and "state-dependent science" signals that the relevant variables are not only social (who is connected to whom) but also *stateful*: the same community graph can instantiate different intersubjective realities under different distributions of attention, affect, and altered states, because those states modulate which patterns become salient, which links are strengthened, and which contradictions can be tolerated without collapse. Emotion and compassion are included not as informal add-ons but because they function as pattern-selection and coupling mechanisms: they bias which patterns are amplified, which others are suppressed, and how quickly patterns propagate across interpersonal channels.

24.2 Motivation and placement in the formal development

Hyperseed treats minds as pattern-bearing systems, and treats communities of minds as giving rise to *intersubjective realities*: reality-systems constituted by patterns distributed across multiple minds, together with causal and predictive couplings between these minds.

A key motivation for formalizing intersubjective realities is that many phenomena of interest (language, norms, shared meanings, institutions, artistic canons, collective emotions, religious experiences, and scientific paradigms) are neither well-captured by purely individualist models nor by undifferentiated "group mind" metaphors. The proposed framework aims to model precisely what is shared (patterns and their carriers), how it is shared (links with weights and directions), and how it can remain stable even when it is not fully consistent from any single point of view. In this sense, the formal objects introduced here are intended to be reusable building blocks for later sections: once intersubjective reality is expressed as a pattern-structured object, one can compare communities, track phase transitions in coordination, and study how artifacts and rituals act as external memory and alignment mechanisms.

The goal here is not to reduce intersubjectivity, communion, or beauty to a single scalar. Rather, the goal is to place these notions inside a rigorous ontology in a way that is: (i) compositional (community structure matters), (ii) explicitly state- and observer-relative (different communities and states of consciousness yield different regimes), and (iii) compatible with non-classical reasoning, since Hyperseed's core claims explicitly allow tensions and contradictions (e.g. the "surprising yet fulfilling" character of beauty).

Compositionality here means that the theory should distinguish, for example, a tightly connected triad from a loosely coupled crowd, even if both have the same number of participants: higher-order structure (clusters, bridges, hubs, and shared artifacts) changes what patterns can be sustained and how quickly they synchronize. State- and observer-relativity means that the same underlying environment can support multiple concurrently valid intersubjective realities, depending on which patterns are accessible in a given state of consciousness and which measurements/queries are being posed by which observers. Compatibility with non-classical reasoning is required because many of the target phenomena are characterized by stable co-presence of apparently incompatible descriptions: rituals can be "symbolic and real," archetypes can be "abstract and concrete," and aesthetic experiences can be "unexpected yet exactly right." A paraconsistent setting allows such co-presence to be represented without trivializing inference, which is crucial if one wants a mathematics that can model the phenomenology rather than explain it away.

We proceed in three steps:

1. Define an intersubjective reality for a community as a weighted pattern hypergraph with cross-mind predictive/causal links.

2. Define communion as a dominance of cross-mind pattern intensity over within-mind pattern intensity, and define *explicit communion* via self-model recognition using paraconsistent truth values.

3. Define aesthetic experience and beauty in a way that directly formalizes "surprising fulfillment" and explains why a paraconsistent logic is a natural fit; then relate aesthetics to communion via shared pattern carriers (artifacts, rituals) and archetypes.

Step (1) is meant to make "shared reality" operational: the hypergraph representation allows patterns to be carried not only by individual minds but also by multi-agent configurations and by external artifacts (texts, images, songs, places, symbols), with hyperedges encoding that certain patterns are only instantiated or only predictive when several carriers jointly participate. The weights on links and hyperedges will be interpreted as strengths of coupling, reliability of prediction, and/or causal influence, so that the formalism can express both weak cultural resonance and strong, high-fidelity synchrony. The "predictive/causal" language is intentionally inclusive: in some contexts the relevant link is a learned model of another agent, while in other contexts it is a direct causal pathway (speech, gaze-following, imitation, coordinated movement), and the framework aims to support both without forcing an artificial reduction.

Step (2) isolates communion as a regime change: instead of treating communion as an ineffable binary property, it is treated as a measurable shift in where pattern-intensity resides (primarily within individuals versus primarily across the interpersonal links and shared carriers). The "explicit" qualifier is then reserved for cases where at least some agents track, represent, or can report the communion regime in their self-models, even if that representation is partial, conflicted, or simultaneously affirmed and denied. The invocation of paraconsistent truth values at this step reflects a common empirical feature of communion reports: agents may sincerely endorse both "I remained myself" and "I dissolved into the group" without experiencing this as a failure of cognition. A logic that can register such dual attribution without collapse is thus not a decorative choice but a way to keep the ontology faithful to the data the theory is supposed to cover.

Step (3) ties aesthetics to inference and prediction: "surprising fulfillment of expectations" will be made precise by distinguishing (a) a departure from locally expected patterns (surprise) and (b) a higher-level coherence or resolution that nonetheless satisfies deeper constraints (fulfillment). The paraconsistent connection arises because many aesthetic resolutions operate by sustaining tension (two readings, two moods, two interpretations) while still yielding an integrated experience; the formalism should therefore permit a controlled coexistence of incompatible pattern-attributions that remains informative rather than degenerating into arbitrariness. Finally, the link back to communion is not merely sociological: artworks, rituals, and archetypal narratives act as shared pattern carriers that can entrain attention and expectation across multiple minds, thereby increasing cross-mind pattern intensity and stabilizing communal regimes over time. This is also where "religion and ritual" and the brief discussion of "state-dependent science" are placed: both can be analyzed as repeatable protocols for shifting state variables (attention, affect, interpretive priors) in a coordinated population, thereby selecting which intersubjective realities become accessible, credible, and action-guiding under those conditions.

24.3 Intersubjective reality as a community pattern system

We assume (as in earlier sections) that "pattern" is an observer-relative regularity associated with compression/effort, and that pattern *intensity* is a nonnegative quantity reflecting salience, coherence, frequency, and/or causal relevance within a time interval. In particular, the same underlying events may support different patterns for different observers, and the same nominal "kind" of pattern (e.g. a linguistic convention) may register with different intensities depending on how stable it is, how often it is instantiated, how much predictive power it contributes, or how costly it would be (in description length or cognitive effort) to ignore it. The time interval I is included explicitly so that both short-lived synchronizations (e.g. a fleeting shared glance) and longer-lived regularities (e.g. a persistent norm) can be treated within one scheme by varying the temporal window over which intensities are evaluated.

Definition 392 (Intensity quantale)**.** *Let $V_I = ([0,\infty], \leq, \oplus, 0)$ where $\oplus$ is addition with $x \oplus y := x + y$ and $[0,\infty]$ is ordered by the usual $\leq$. We will use V_I as the codomain of pattern-intensity maps and as an aggregation monoid.*

Remark 1333. *V_I is a commutative quantale if equipped with the complete lattice structure given* *by arbitrary suprema, and with monoidal product + distributing over suprema. We will not need the full quantale structure here, but it is convenient to keep the "quantale habit" so intensity and weakness can be treated uniformly later. Concretely, the choice of + reflects the idea that (when appropriate independence assumptions hold) the combined salience of distinct supported patterns can be accumulated, while the order $\leq$ supports the comparison of alternative descriptions by dominance of intensity. Allowing ∞ is a technical convenience: it can represent "effectively unignorable" regularities within I (for instance, hard constraints induced by the environment or by institutional enforcement), and it guarantees the existence of suprema for arbitrarily large families of intensities when such joins are useful for analysis.*

Definition 393 (Minds, communities, and supported patterns)**.** *Let* Mind *be a set (or groupoid) of minds. Fix a (finite) community* $C \subseteq \mathsf{Mind}$ *and a time interval* I*. For each nonempty subset* $U \subseteq C$*, let* $\mathsf{Pat}_U(I)$ *denote the set of patterns whose realization is supported across exactly the minds in* U *(e.g. shared attention, synchronized appraisal cycles, coordinated linguistic constructions, mutually constrained expectations, shared affective dynamics). We write* $\mathsf{Pat}_{\{m\}}(I)$ *as* $\mathsf{Pat}_m(I)$ *for singleton supports.*

The phrase "supported across exactly the minds in U" is intended to separate (i) patterns that are purely idiosyncratic to a single mind from (ii) patterns that are genuinely sustained by interaction, mutual modeling, or common constraints within a subcommunity. Here "exactly" should be read as a modeling choice rather than a metaphysical claim: in practice one may treat $\mathsf{Pat}_U(I)$ as consisting of patterns for which U is the *minimal* support set in C (with the understanding that a pattern instantiated by all of U may also be compatible with, or embedded in, larger-U' patterns). Nothing in what follows depends on having a unique minimal support; the only essential point is that we can distinguish patterns by how broadly in the community they are realized.

When Mind is taken as a groupoid rather than a set, the intention is that there may be structure-preserving identifications between minds (e.g. treating two roles as equivalent in a certain analysis, or quotienting by a notion of observational indistinguishability), and that pattern support can be considered up to such identifications. This flexibility is useful later when we want to discuss institutional roles, replaceability, and anonymity in large-scale intersubjective systems.

Definition 394 (Community intensity data)**.** *A* community intensity datum *on* (C, I) *is a family of functions*

$$\mathbf{w} = \{\, w_U : \mathsf{Pat}_U(I) \to V_I \mid \emptyset \neq U \subseteq C \,\}$$

assigning an intensity to each supported pattern.

Intuitively, $\mathbf{w}$ encodes not only what patterns occur but how strongly they matter to the community-level dynamics on I. One may read $w_U(p)$ as a community-relative measure of how much predictive compression is gained by tracking the pattern p as a regularity jointly carried by the minds in U (as opposed to treating the constituents of p as unrelated fluctuations). Because V_I is additive, it is natural—when appropriate—to consider aggregated quantities such as the total intensity of all patterns supported on a given U,

$$W(U) \;:=\; \bigoplus_{p \in \mathsf{Pat}_U(I)} w_U(p),$$

whenever the relevant sum is well-defined (or replaced by a supremum, depending on the modeling choice). Such aggregated views will later make it easy to state conditions like "most of the action is dyadic" or "the community has a strong global mode," without committing to a particular enumeration of patterns.

Definition 395 (Intersubjective reality-system). *Fix (C, I) and a community intensity datum* $\mathbf{w}$*. An* intersubjective reality-system *for C on I is a triple*

$$\mathcal{R}_{C,I} = \big(\mathcal{E}_{C,I},\ \mathcal{L}_{C,I},\ \mathbf{w}\big)$$

where:

- $\mathcal{E}_{C,I}$ *is a set of* entities-as-modeled-in-community *(tokens for persons, objects, shared symbols, joint goals, norms, roles, etc., as represented across the community);*
- $\mathcal{L}_{C,I}$ *is a set of weighted links encoding* predictive *and* causal *relationships between entities and between minds' representations of entities. Formally, one may take $\mathcal{L}_{C,I}$ to be a V_I-weighted directed hypergraph or simplicial set on $\mathcal{E}_{C,I}$;*
- $\mathbf{w}$ *assigns intensities to patterns supported on subcommunities $U \subseteq C$.*

We say $\mathcal{R}_{C,I}$ is intersubjective *(rather than merely a disjoint union of individual realities) if $\mathcal{L}_{C,I}$ contains nontrivial cross-mind links of significant weight and* $\mathbf{w}$ *assigns non-negligible total intensity to patterns supported on $|U| \geq 2$.*

The three components play complementary roles. The entity set $\mathcal{E}_{C,I}$ is not meant to be a list of mind-independent objects, but rather the community's *working ontology* on I: the handles that appear in joint descriptions, coordination, and expectation management. Thus "the same" external object may correspond to different community-entities across different communities (or time intervals), and conversely a single community-entity (e.g. a role like "referee" or a symbol like a brand) may be realized by many different individuals or physical tokens while remaining stable at the level of communal modeling.

The link structure $\mathcal{L}_{C,I}$ is where predictive and causal organization is recorded. Treating $\mathcal{L}_{C,I}$ as a weighted directed hypergraph emphasizes that relations in social cognition are often many-to-one or many-to-many (e.g. several cues jointly predict an interpretation; multiple people jointly enforce a norm; a shared goal shapes several downstream actions). The V_I-weight on a link can be interpreted as the strength, reliability, or relevance of the predictive/causal dependency during I, in the same broad sense of salience used for pattern intensity. Links may connect not only "external" entities but also representations (e.g. "Alice believes Bob intends X" linking an Alice-representation node to a Bob-intention node), which is why cross-mind links are a natural place to represent higher-order social inference and mutual modeling.

The condition for being genuinely intersubjective is deliberately qualitative ("significant weight", "non-negligible total intensity") because the intended use is comparative and context-sensitive. In some settings, a small number of high-weight cross-mind links (e.g. a rigid command structure) can dominate; in others, a diffuse field of moderate-weight mutual expectations (e.g. conversational turn-taking or politeness norms) can be the primary stabilizer. The formalism permits either extreme, and the role of $\mathbf{w}$ is precisely to let the analyst say where, within C, the realized regularities actually live.

Remark 1334. *This definition is intentionally permissive: "intersubjective reality" is not a single kind of object. It ranges from dyadic conversations to large-scale institutions, markets, scientific communities, and cultures. What unifies these is the presence of significant cross-mind prediction/causation and significant cross-mind patterns.*

A useful way to read the permissiveness is that $\mathcal{R}_{C,I}$ is a *pattern system indexed by community support* rather than a single monolithic graph: it can represent simultaneously (i) individual-level regularities ($|U| = 1$), (ii) small-group synchronizations ($|U| = 2, 3$), and (iii) whole-community stabilizers ($U = C$). For example, in a conversation between two participants $a, b \in C$, one expects substantial intensity on patterns in $\mathsf{Pat}_{\{a,b\}}(I)$ such as turn-taking rhythms, accommodation of vocabulary, and mutual repair strategies, together with cross-mind links in $\mathcal{L}_{C,I}$ encoding rapid reciprocal prediction (e.g. "a's utterance predicts b's next move"). In an institution, by contrast, one expects prominent entities in $\mathcal{E}_{C,I}$ corresponding to roles and norms, and high-intensity patterns supported on larger subcommunities (sometimes approaching $U = C$), such as shared compliance expectations or shared symbol grounding.

Finally, note that nothing requires $\mathsf{Pat}_U(I)$ to be explicitly enumerated in applications. One can treat it as an abstract carrier for whatever pattern-extraction procedure is used (statistical, ethnographic, mechanistic, or hybrid), with $\mathbf{w}$ serving as the interface between such extraction and the higher-level statements made about intersubjective structure.

24.4 Intersubjective communion as cross-mind intensity dominance

Hyperseed's definition of *intersubjective communion* can be treated as a comparative statement: cross-mind patterns become more intense than within-mind patterns. In this framing, communion is not a primitive "togetherness" predicate but a measurable dominance relation: the community-level phenomenology is taken to be explained by whether jointly-supported structure outweighs the structure that remains private to each individual mind. The intent is that W_{cross} and W_{in} serve as coarse but tractable aggregates of "how much pattern-intensity is being carried" by intersubjective versus merely intrasubjective organization, relative to a fixed community C, an index set I (e.g. times, contexts, or interaction windows), and a chosen intensity datum $\mathbf{w}$.

Definition 396 (Within-mind and cross-mind total intensity)**.** *Fix (C, I) and a community intensity datum $\mathbf{w}$. Define the* within-mind total intensity *and* cross-mind total intensity *by:*

$$W_{\text{in}}(C, I; \mathbf{w}) := \sum_{m \in C} \sum_{p \in \mathsf{Pat}_m(I)} w_{\{m\}}(p),$$
$$W_{\text{cross}}(C, I; \mathbf{w}) := \sum_{\substack{U \subseteq C \\ |U| \geq 2}} \sum_{p \in \mathsf{Pat}_U(I)} w_U(p).$$

(We assume these sums are finite; if not, replace sums by suitable suprema or integrals.)

It is useful to keep in mind the intended separation of roles: W_{in} aggregates intensities of patterns whose supporting set is a singleton $\{m\}$ (so they are, by construction, "owned" by one mind), whereas W_{cross} aggregates patterns supported by some coalition U of size at least 2. The indexing by U permits graded forms of intersubjectivity: a pattern supported by a dyad, a triad, or the whole community can be assigned intensities at the corresponding support level(s), depending on how $\mathsf{Pat}_U(I)$ and w_U are specified. In applications, one often ensures that patterns are *support-tagged* (so that the same abstract content appearing with different support sets is treated as distinct pattern tokens), which makes the above sums unambiguous even when there is conceptual overlap between "the same" pattern instantiated at different levels of support.

The parenthetical finiteness clause also carries conceptual content: the theory only needs enough regularity to compare within-mind and cross-mind aggregates. If the pattern collections are infinite (for example, if $\mathsf{Pat}_U(I)$ is generated by a continuous-time process), replacing sums with integrals makes the comparison sensitive to the choice of measure; replacing sums with suprema yields a "peak-dominance" reading rather than a "total-mass" reading. The present section uses the summation presentation because it most directly matches the interpretation of W_{in} and W_{cross} as total intensity budgets.

Definition 397 (Communion predicate and communion margin)**.** *We say the community C exhibits* intersubjective communion *on I (relative to $\mathbf{w}$) if*

$$\mathsf{Comm}(C, I; \mathbf{w}) \quad \text{holds, i.e.} \quad W_{\mathrm{cross}}(C, I; \mathbf{w}) > W_{\mathrm{in}}(C, I; \mathbf{w}).$$

Define the communion margin *by*

$$\Delta_{\mathsf{Comm}}(C, I; \mathbf{w}) := W_{\mathrm{cross}}(C, I; \mathbf{w}) - W_{\mathrm{in}}(C, I; \mathbf{w}).$$

The strict inequality in Comm is intended to encode a genuine dominance rather than a tie: if $W_{\mathrm{cross}} = W_{\mathrm{in}}$, then the community is at a boundary where neither intrasubjective nor intersubjective structure clearly governs the overall intensity accounting. In many empirical or computational settings, one may introduce a tolerance parameter $\tau > 0$ and require $W_{\mathrm{cross}} \geq W_{\mathrm{in}} + \tau$ to obtain robustness to estimation noise; the margin Δ_{Comm} already provides the quantity one would threshold in such a variant.

It is also sometimes convenient to separate W_{cross} by coalition size:

$$W_{\mathrm{cross}}(C, I; \mathbf{w}) = \sum_{k=2}^{|C|} \sum_{\substack{U \subseteq C \\ |U| = k}} \sum_{p \in \mathsf{Pat}_U(I)} w_U(p),$$

so that one can distinguish dyadic resonance from broadly shared patterns. On this view, Δ_{Comm} measures not only whether intersubjectivity dominates, but also (when decomposed) *how* it dominates—e.g. by many small coalitions or by a smaller number of large, community-wide supports.

Remark 1335. *One can normalize Δ_{Comm} to obtain a bounded score, e.g. $\Delta_{\mathsf{Comm}}/(W_{\mathrm{cross}}+W_{\mathrm{in}}+\varepsilon)$. We keep the unnormalized margin because (a) it composes additively in the simplest way, and (b) it keeps clear the ontology-level meaning: a literal dominance of cross-supported pattern intensity.*

A further reason the unnormalized margin can be informative is that it retains sensitivity to the *scale* of the interaction: two communities might have the same normalized score while differing drastically in absolute cross-mind intensity, which may matter if one aims to connect communion to downstream effects (e.g. behavioral coordination, communication bandwidth, or aesthetic salience). At the same time, the reader should note an elementary invariance: if one scales all intensities by a common factor $\lambda > 0$ (i.e. replaces $\mathbf{w}$ by $\lambda\mathbf{w}$ pointwise), then both W_{in} and W_{cross} scale by λ, and hence $\mathsf{Comm}(C, I; \mathbf{w})$ is unchanged while Δ_{Comm} scales by λ. Thus the predicate captures a comparative dominance, while the margin captures a dominance magnitude.

For intuition, consider the smallest nontrivial case $C = \{m_1, m_2\}$. Then

$$W_{\mathrm{in}} = \sum_{p \in \mathsf{Pat}_{m_1}(I)} w_{\{m_1\}}(p) + \sum_{p \in \mathsf{Pat}_{m_2}(I)} w_{\{m_2\}}(p), \qquad W_{\mathrm{cross}} = \sum_{p \in \mathsf{Pat}_{\{m_1,m_2\}}(I)} w_{\{m_1,m_2\}}(p),$$

so communion says that the dyad's jointly-supported intensity outweighs the sum of their private intensities. This makes explicit the conceptual reading: the dyad is "more constituted" by what is mutually supported than by what remains individually supported.

Proposition 38 (Monotonicity of communion under adding cross-supported patterns)**.** *Fix (C, I) and a community intensity datum $\mathbf{w}$. If $\mathbf{w}'$ differs from $\mathbf{w}$ only by increasing some cross-mind intensities (i.e. $w'_U(p) \geq w_U(p)$ for all $|U| \geq 2$, and $w'_{\{m\}} = w_{\{m\}}$ for singletons), then $\Delta_{\mathsf{Comm}}(C, I; \mathbf{w}') \geq \Delta_{\mathsf{Comm}}(C, I; \mathbf{w})$. In particular, if $\mathsf{Comm}(C, I; \mathbf{w})$ holds, then $\mathsf{Comm}(C, I; \mathbf{w}')$ holds.*

Proof. Immediate from Definition 396 and Definition 397: only W_{cross} can increase, while W_{in} is unchanged. □

This proposition formalizes a minimal adequacy condition for the notion of communion: strengthening jointly-supported pattern intensity (without changing private pattern intensity) cannot make the community *less* communal under the dominance criterion. Equivalently, the only way to decrease Δ_{Comm} is to decrease cross-mind intensity or to increase within-mind intensity; the definition thereby cleanly separates "what adds to togetherness" from "what adds to privateness" at the level of intensity bookkeeping. When Δ_{Comm} is used as a target quantity for optimization or learning, Proposition 38 ensures that any update rule that monotonically increases cross-supported weights (holding singleton weights fixed) is guaranteed to be non-worsening with respect to the communion margin.

24.5 Explicit intersubjective communion via self-model recognition

Hyperseed distinguishes implicit communion (cross-mind patterns dominate) from *explicit* communion, where the participants' self-models recognize the communion as such. To formalize "recognize" we use a simple paraconsistent truth-value space. This keeps the ontology compatible with simultaneously holding (or partially holding) seemingly incompatible self-reports, commitments, and interpretations. In particular, "recognition" here is treated as a kind of *endorsement state* internal to each participant: one can endorse that communion is occurring while also endorsing (to some extent) that it is *not* occurring, or that it is unsafe, or that one is uncertain what to call the experience. This is common in real reports (e.g. "I feel connected, but I also feel guarded"), and it is precisely the kind of case where a classical truth assignment would force premature resolution.

Definition 398 (Uncertain paraconsistent truth values)**.** *Let*

$$\mathbb{B} := [0,1] \times [0,1].$$

An element $b = (t, f) \in \mathbb{B}$ is read as "degree of truth" t and "degree of falsity" f. Define negation, conjunction, and disjunction by

$$\begin{aligned} \neg(t,f) &:= (f,t), \\ (t_1,f_1) \wedge (t_2,f_2) &:= (\min\{t_1,t_2\},\ \max\{f_1,f_2\}), \\ (t_1,f_1) \vee (t_2,f_2) &:= (\max\{t_1,t_2\},\ \min\{f_1,f_2\}). \end{aligned}$$

Fix a designation threshold $\tau \in (0,1]$. We say $b = (t,f)$ is designated *(accepted/endorsed) if $t \geq \tau$.*

The intended reading is that t tracks the strength with which the self-model supports a proposition, while f tracks the strength with which the self-model supports its denial. Nothing in the definition enforces $t + f = 1$, so the framework permits both *incompleteness* (low t and low f) and *inconsistency* (high t and high f). The connective definitions make this interpretation operational: conjunction preserves the weaker truth-support (via min) while accumulating falsity-support (via max), whereas disjunction preserves the stronger truth-support (via max) while reducing falsity-support (via min). Negation simply swaps the two degrees, reflecting that support for $\neg\varphi$ is measured by support-for-falsity of φ in the same self-model.

The designation threshold τ isolates the specific notion of "counts as endorsed" needed for explicit communion. For instance, taking $\tau = 1$ yields a strict reading: only full truth-support counts as recognition. Taking $\tau < 1$ permits graded recognition: partial endorsement suffices, which can be appropriate when self-model access is noisy, introspective reports are hesitant, or participants are linguistically misaligned but still clearly signaling a "yes" in practice. Importantly, designation depends only on t; thus a proposition may be designated even when f is simultaneously large, capturing the case "I endorse that communion is occurring, even though part of me also rejects that description."

Remark 1336. *This is a continuous generalization of the familiar four-valued Belnap-Dunn bilattice. The key property for our purposes is: contradictions do not entail triviality.*

Concretely, the familiar four corners embed as $(1, 0)$ (true only), $(0, 1)$ (false only), $(1, 1)$ (both true and false), and $(0, 0)$ (neither), with the continuous interior representing intermediate mixtures. The paraconsistent motivation is practical rather than purely philosophical: if a participant's self-model includes both "we are in communion" and "we are not in communion" (or "this is safe" and "this is unsafe") at nonzero strength, then we do not want the formalism to collapse into a state where every proposition becomes equally endorsable. Instead, we want to preserve the informational content of ambivalence and keep explicit-communion criteria well-defined.

Definition 399 (Self-model valuation)**.** *Fix a mind $m \in C$. Let $\mathcal{L}_m$ be the (informal) language of propositions expressible in m's self-model on I (e.g. "we are in communion now", "I feel connected", "I feel threatened", etc.). A* self-model valuation *is a map*

$$v_m : \mathcal{L}_m \to \mathbb{B}.$$

The point of allowing $\mathcal{L}_m$ to be informal is to avoid overcommitting to a particular cognitive architecture. Different minds may carve the space of self-descriptions differently, may have different granularity of introspective access, and may attach different connotations to the same surface sentence. What matters for the present definition is only that each mind can represent (in some way) a proposition playing the role of "the community is currently in communion on I," and that the valuation records the mind's endorsement and rejection degrees for such propositions. In settings where the self-model is partly implicit or sub-symbolic, v_m can be viewed as an idealization of an accessible report channel (e.g. a confidence score, a felt-sense rating, or a post-hoc linguistic report) rather than a literal internal data structure.

Definition 400 (Explicit communion)**.** *Fix (C, I), intensity datum $\mathbf{w}$, and self-model valuations v_m for $m \in C$. Assume there is a proposition symbol $\mathsf{CommNow}_{C,I}$ available in each $\mathcal{L}_m$ meaning "the community C is currently in communion on I" (as represented by m). We say* explicit intersubjective communion *holds if:*

1. $\mathsf{Comm}(C, I; \mathbf{w})$ *holds (implicit communion), and*
2. *for every $m \in C$, the value $v_m(\mathsf{CommNow}_{C,I})$ is designated.*

We denote this by $\mathsf{ExplComm}(C, I; \mathbf{w}, \{v_m\})$.

Clause (2) is intentionally phrased as a universal condition over participants: explicit communion is not merely that "someone notices," nor that "on average the group notices," but that each member's self-model contains sufficient endorsement of the communion proposition. Because designation ignores the falsity coordinate, this allows explicit communion even when some members are conflicted, defensive, or simultaneously carrying counter-interpretations, provided the endorsement of $\mathsf{CommNow}_{C,I}$ clears the agreed threshold τ. For example, if $\tau = 0.7$, then a participant with

$$v_m(\mathsf{CommNow}_{C,I}) = (0.8, 0.8)$$

still counts as explicitly recognizing communion (designated), even though they are also strongly pulled toward denying it; this models "yes, and no" states without forcing a resolution. Conversely, a participant with

$$v_m(\mathsf{CommNow}_{C,I}) = (0.6, 0.0)$$

would not count as recognizing it at $\tau = 0.7$, even though they are not actively rejecting it; this models hesitant or unclear acknowledgement.

The explicitness criterion can also be read as a coordination condition on *meta-representation*: implicit communion concerns the dominance of cross-supported patterns in the system dynamics, whereas explicit communion adds that each self-model has tracked and endorsed that very fact. In that sense, the additional structure in $\mathsf{ExplComm}$ is not merely epistemic; it is part of the phenomenon being defined, since Hyperseed distinguishes communion that is lived but unrecognized from communion that is jointly identified and thus available for deliberate action, narrative stabilization, and norm formation.

Remark 1337. *Explicit communion is a* joint *condition: it is possible that* Comm *holds but some participants do not endorse (or do not have access to) the proposition* $\mathsf{CommNow}_{C,I}$ *in their self-models. Conversely, a group can self-report communion while cross-supported patterns do not dominate, in which case* $\mathsf{ExplComm}$ *fails due to the first clause.*

A further subtlety is that "do not have access" can occur even when a participant is behaviorally aligned with the communion: they may enact the pattern without categorizing it as communion, or they may lack the linguistic/conceptual resources to form $\mathsf{CommNow}_{C,I}$ as a stable self-model proposition. The definition treats this as a failure of explicitness (by assumption, the symbol exists, but the valuation may fail to designate it), which matches the intended distinction between tacit participation and self-conscious recognition. At the same time, the paraconsistent setting prevents such cases from forcing a binary verdict about the participant's entire self-model: the model can represent partial endorsement, partial denial, and partial indeterminacy simultaneously, and explicit communion depends only on whether the endorsement passes the chosen threshold.

24.6 Spiritual experience as communion with a broader-scope mind

Hyperseed treats spiritual experience as a special case of communion: communion between a mind M and another mind that is experienced as broader in scope. In this framing, "spiritual" does not name a separate ontological substance; it names a particular *relational configuration* within the general communion machinery, namely one in which the partner mind is comparatively broader (in a sense made checkable below) over the interval of interest.

Definition 401 (Scope of a mind on an interval)**.** *Fix a mind m and an interval I. Assume we have: (i) a multiset of patterns $\mathsf{Pat}_m(I)$, (ii) intensities $w_{\{m\}}$ on these patterns, and (iii) a complexity/cost function $\kappa : \mathsf{Pat}_m(I) \to [0, \infty)$ measuring pattern depth (e.g. description length in some fixed representational language, or effort to represent/maintain the pattern). Define the* scope *as*

$$\mathsf{Scope}(m, I) := \alpha \cdot H_m(I) + (1 - \alpha) \cdot D_m(I), \quad \alpha \in [0, 1],$$

where

$$H_m(I) := -\sum_{p \in \mathsf{Pat}_m(I)} \pi_m(p) \log(\pi_m(p) + \varepsilon), \quad \pi_m(p) := \frac{w_{\{m\}}(p)}{\sum_{q \in \mathsf{Pat}_m(I)} w_{\{m\}}(q) + \varepsilon},$$

and

$$D_m(I) := \sum_{p \in \mathsf{Pat}_m(I)} \pi_m(p)\, \kappa(p).$$

Here $\varepsilon > 0$ is a small numerical stabilizer.

Remark 1338. *H_m measures diversity of salient patterns (breadth), while D_m measures average depth. Any other combination of diversity and depth consistent with the intended semantics could be used. The ontology only requires that "broader scope" be a mathematically checkable comparative.*

Remark 1339. *The normalization π_m makes H_m and D_m depend on* relative *saliencies rather than the absolute scale of $w_{\{m\}}$; in particular, if $w_{\{m\}}$ is multiplied by a positive constant, π_m is unchanged (up to the ε term), so scope is driven by how attention/importance is distributed across patterns. The multiset language allows multiple tokenings of the "same" schematic pattern to be treated as distinct when the representational system of m distinguishes them (e.g. repeated motifs, recurring affective complexes), while still permitting coarse-graining by choosing $\mathsf{Pat}_m(I)$ appropriately.*

Remark 1340. *The parameter α interpolates between two limiting readings: when $\alpha = 1$, scope reduces to diversity of salient patterns (a purely breadth-like notion), and when $\alpha = 0$, scope reduces to average depth under κ (a purely depth-like notion). The choice of log base in H_m only rescales the contribution of the entropy term and can be absorbed into α (or into a rescaling of κ), so the operational content is the comparative ordering of scopes across minds/intervals rather than any absolute unit. The stabilizer ε can be taken small enough that it does not affect comparisons except in degenerate cases (e.g. extremely sparse or near-zero intensity mass), where it serves to keep the expressions well-defined.*

Definition 402 (Spiritual communion)**.** *Let m, n be minds and consider $C = \{m, n\}$. We say m has a* spiritual experience with *n on I if:*

1. $\mathsf{Comm}(\{m, n\}, I; \mathbf{w})$ *holds, and*
2. $\mathsf{Scope}(n, I) \geq \mathsf{Scope}(m, I) + \delta$ *for some fixed $\delta > 0$,*

where the second condition expresses that n is broader-scope than m.

Remark 1341. *The threshold δ encodes that "broader" is intended as a* robust *rather than hairline comparative: it rules out labeling as spiritual those cases where m and n have essentially the same breadth/depth profile on I up to negligible fluctuations. In applications, δ may be treated as a context-dependent parameter (e.g. varying by domain, by the granularity of Pat, or by the scaling of κ), but the logical role remains the same: it implements a margin that makes the broader-scope relation stable under small modeling perturbations (choice of pattern vocabulary, noise in w, etc.).*

Remark 1342. *The definition is deliberately stated in terms of $\mathsf{Scope}(n, I)$, not merely m's* belief *about it. This yields a structural notion that can, in principle, be assessed from a third-person model of both pattern systems. At the same time, many episodes described as spiritual involve an* appearance *of vastness or encompassment that may or may not match the external comparative; if desired, one can represent this by adding an epistemic layer in which m maintains an estimate $\widehat{\mathsf{Scope}}_m(n, I)$ and the phenomenology tracks $\widehat{\mathsf{Scope}}_m(n, I) - \widehat{\mathsf{Scope}}_m(m, I)$, while the present definition tracks the underlying relational configuration.*

Remark 1343. *The "broader-scope mind" n may be (i) a collective mind of a group including m, (ii) a mind of another individual, animal, ecosystem-scale process, etc., or (iii) an unusual mind-model accessed in altered states. The ontology does not decide which of these is metaphysically correct; it only supplies a structural criterion for when the experience is of the relevant kind.*

Remark 1344. *In case (i), the scope inequality can be read as a formal version of a familiar intuition: groups can sustain a richer or deeper joint pattern repertoire than any single member on the same interval (e.g. via division of cognitive labor, distributed memory, or complementary perspectives). In case (ii), the inequality can represent the phenomenology of encountering an agent whose lived world is experienced as "larger" (more varied or more structurally articulated). In case (iii), the scope inequality can model episodes where m relates to an internally generated partner n (a god-form, archetype, "nature," "the universe") that carries a broader pattern economy than m's ordinary self-model, even if the ontological status of n is left open by design.*

Definition 403 (Entheogen as a communion-shifting intervention)**.** *Fix an embodied mind m with body B_m and suppose interventions on B_m can change m's pattern system. An* entheogen *(or more generally an entheogenic procedure) is an intervention E on B_m such that, for a relevant class of intervals I and relevant classes of partner minds n, the induced change in intensity data $\mathbf{w} \mapsto \mathbf{w}^E$ tends to increase*

$$\Delta_{\mathsf{Comm}}(\{m,n\}, I; \mathbf{w}^E) \quad \textit{and/or} \quad v_m(\mathsf{CommNow}_{\{m,n\},I})$$

toward designation.

Remark 1345. *This definition intentionally avoids committing to a single mechanism. Entheogens may act by changing attention dynamics, relaxing self-model rigidity, increasing cross-subsystem coupling, altering predictive priors, etc. All of these show up as changes in $\mathbf{w}$ and/or in the valuation maps v_m.*

Remark 1346. *The "and/or" is important: some interventions primarily change the* structural *conditions for communion (as measured by Δ_{Comm}), while others primarily change whether m counts an already-present relation as salient, valuable, or real (as measured by $v_m(\mathsf{CommNow}_{\{m,n\},I})$). The phrase "toward designation" is meant to cover thresholded decision rules commonly used elsewhere in the ontology: even if Δ_{Comm} is continuous, communion-as-designated may require crossing a context-specific cutoff, and entheogens are characterized here by their tendency to push the system across that boundary for a nontrivial class of (I, n) rather than by guaranteeing communion in every instance.*

24.7 Emotion and compassion as mind-body-and-community patterns

Hyperseed treats emotions as dynamical patterns spanning large portions of mind and body over time. Compassion is treated as an emotion whose content and dynamics are essentially intersubjective. In this framing, an "emotion" is not primarily a propositional judgment, nor merely an isolated bodily sensation, but a coordinated regime that recruits perception, appraisal, action-readiness, autonomic regulation, and attention into a temporally extended configuration. Likewise, "intersubjective" does not mean that an emotion must be publicly expressed or linguistically reported; it means that some of the constitutive structure of the episode depends on, tracks, or anticipates patterns that are jointly realized across more than one mind (e.g. co-regulation, mutual attention, joint action, or reciprocity constraints).

Definition 404 (Emotion episode)**.** *Fix an embodied mind m with body B_m and interval I. Let $S_m(t)$ denote the (modeled) mind-state at time $t \in I$ and $S_{B_m}(t)$ the body-state. An* emotion episode *is a pattern $e \in \mathsf{Pat}_m(I)$ such that:*

1. *e is supported on both mind and body degrees of freedom (not purely cognitive, not purely somatic),*

2. *e has nontrivial temporal extent (persists across a subinterval $J \subseteq I$),*

3. *e organizes (constrains) a large fraction of the system's active patterns during J.*

Concretely, condition (1) is meant to exclude both "cold" belief updates that do not recruit bodily regulation and also purely reflexive physiological events that do not enter the mind-level organization of attention and action; the episode must be realized as a coupled mind–body pattern. Condition (2) distinguishes an episode from instantaneous perturbations (e.g. a momentary startle spike) by requiring persistence on a subinterval with internal temporal structure, allowing onset, maintenance, and decay phases. Condition (3) captures the familiar phenomenology that emotions are global organizers: they bias perception, reshape salience, modulate memory access, and gate policy selection, rather than remaining local to a single representational module. The membership $e \in \mathsf{Pat}_m(I)$ is intended to emphasize that episodes are identified as patterns at the level of the whole embodied system across time, not as single-state labels; in particular, two distinct micro-realizations can count as the "same" episode-type if they share the relevant organizing constraints at the pattern level.

Definition 405 (Compassion episode)**.** *Fix a community C containing m and another mind $m' \neq m$. A* compassion episode in m toward m' on I *is an emotion episode $e \in \mathsf{Pat}_m(I)$ such that:*

1. *e is causally and predictively coupled to cross-mind patterns supported on $\{m, m'\}$ (i.e. intersubjective content),*

2. *e carries positive valence in m's valuation scheme and includes patterns representing m' as included in m's concern-set ("love inclusive of the other mind").*

The coupling requirement in (1) is intended to be stronger than mere causal dependence on a perceptual stimulus: the compassion episode must participate in a dynamical loop in which m tracks and forecasts relevant aspects of m' (e.g. affect, needs, intentions, vulnerability), and where that tracking is itself stabilized or corrected by cross-mind interaction. In particular, the "predictive" clause allows compassion to be constituted partly by anticipatory organization (e.g. preparing supportive actions, maintaining readiness to coordinate, sustaining attention to m'), rather than requiring that all relevant information about m' be presently observed. The concern-set clause in (2) is meant to mark the difference between benevolent attention and neutral monitoring: m' is represented as a beneficiary of m's protective, affiliative, or helping dispositions, so that action selection is biased toward outcomes favorable to m' (subject to m's broader constraints). Nothing in the definition requires that m' reciprocate, that m' be aware of m's state, or that the episode be socially successful; it only requires that the episode in m be constituted by intersubjective coupling and inclusive valuation directed toward m'.

Remark 1347. *The ontology does not insist on a single scalar "valence". All that is required is that there is a valuation component (possibly multi-dimensional) that can distinguish benevolent inclusion from aversive exclusion, and that compassion corresponds to inclusion along that axis.*

This remark allows the model to accommodate cases where affective tone, motivational direction, and normative appraisal dissociate (for example, compassion that is experienced as painful or heavy while remaining prosocial and inclusive). It also permits valuation schemes in which "positive" is not hedonically pleasant but instead reflects a structured preference for relationship-preserving, other-regarding, or harm-avoiding outcomes. Accordingly, the "axis" of inclusion can be realized by a family of coordinated valuation components (e.g. attachment, affiliation, care, fairness) so long as they jointly implement the inclusion/exclusion distinction relevant to compassion.

Proposition 39 (Communion-bias of compassion)**.** *Suppose a mind m has a compassion episode toward m' on I as in Definition 405. Then, for the dyadic community $\{m, m'\}$, the cross-supported intensity W_{cross} is bounded below by the total intensity of those cross-supported patterns coupled to the compassion episode. In particular, compassion in this sense cannot occur in a purely individual reality-system with $W_{\text{cross}} = 0$.*

Intuitively, the proposition formalizes a "communion bias": if compassion is constitutively intersubjective, then it necessarily allocates nonzero pattern-intensity to structures that are not realizable within a single isolated mind. The bound is stated in terms of cross-supported patterns *coupled* to the episode to emphasize that not all interpersonal structure in $\{m, m'\}$ need be relevant, only that the compassion episode cannot be constituted without some positive share of cross-mind organization. This is compatible with compassion also recruiting many purely intra-mind patterns in m (e.g. autobiographical memory, private imagery, internal deliberation), while still requiring that a nontrivial component of the episode's dynamics is anchored in patterns supported on the dyad.

Proof. By Definition 405, the compassion episode is causally and predictively coupled to cross-supported patterns on $\{m, m'\}$, hence at least one such pattern has positive intensity, so $W_{\text{cross}} > 0$ and is bounded below by the sum of intensities of these coupled patterns. $\square$

The proof uses only the minimal content of the definition: coupling entails the existence of at least one cross-supported pattern participating in the episode's causal/predictive structure, and the intensity formalism assigns a nonzero contribution to any such participating pattern. The inequality is therefore a bookkeeping statement about how the total cross-supported intensity cannot fall below the portion that is actually recruited by the compassion episode. Under a strict isolation assumption (no cross-supported patterns at all, i.e. $W_{\text{cross}} = 0$), the definitional coupling condition cannot be satisfied, so compassion-as-defined is ruled out in that regime.

24.8 Aesthetics and beauty as paraconsistent "surprising fulfillment"

Hyperseed proposes: (i) aesthetic experience is the experience of beauty, (ii) art is whatever has significant aesthetic effect for a class of experiencers, and (iii) beauty involves an experience that is both fulfilling-of-expectations and surprising, and whose internal representation behaves like an open-endedly intelligent mental subnetwork.

We now encode these claims in a form compatible with the earlier quantale/pattern formalism.

24.8.1 Fulfillment and surprise as distinct evaluators

Definition 406 (Predictive fulfillment and compression surprise)**.** *Fix a mind m and an encountered stimulus/artifact/event B over interval I. Assume m maintains:*

- *a predictive model Π_m assigning likelihoods to observations of B,*
- *an internal coding/compression scheme with code-length functional $L_m(\cdot)$ on representations of B.*

Define:

$$\mathsf{Ful}_m(B;I) := \exp\bigl(-\lambda \cdot \mathbb{E}[-\log \Pi_m(obs(B))]\bigr) \in (0,1],$$
$$\mathsf{Sur}_m(B;I) := \max\{0,\ L_m^{\mathrm{old}}(B) - L_m^{\mathrm{new}}(B)\} \in [0,\infty),$$

where $\lambda > 0$ is a scaling parameter and "old/new" refer to before/after updating m's internal codes after the encounter with B.

Remark 1348. *With these definitions, fulfillment and surprise are not forced to be complements. A stimulus can be broadly predictable (high* Ful*) while still yielding nontrivial compression gain (high* Sur*), which is a precise mathematical reading of "surprising fulfillment".*

24.8.2 Beauty as a paraconsistent conjunction

Hyperseed explicitly notes a tension (even a contradiction) between fulfillment and surprise. A convenient way to keep that tension in the formalism without collapsing reasoning is to use a paraconsistent logic.

Definition 407 (Beauty proposition and valuation)**.** *Fix a mind m and stimulus B on I. Let* $\mathsf{Ful}(B)$ *and* $\mathsf{Sur}(B)$ *be proposition symbols in m's self-model language $\mathcal{L}_m$. Define a valuation v_m on these by mapping to $\mathbb{B}$ (Definition 398) as:*

$$v_m(\mathsf{Ful}(B)) := (\min\{1, \mathsf{Ful}_m(B;I)\},\ 1 - \min\{1, \mathsf{Ful}_m(B;I)\}),$$
$$v_m(\mathsf{Sur}(B)) := \Bigl(\min\{1, \tfrac{\mathsf{Sur}_m(B;I)}{\mathsf{Sur}_m(B;I)+1}\},\ 1 - \min\{1, \tfrac{\mathsf{Sur}_m(B;I)}{\mathsf{Sur}_m(B;I)+1}\}\Bigr).$$

Define the beauty proposition *by*

$$\mathsf{Beaut}(B) \ := \ \mathsf{Ful}(B) \wedge \mathsf{Sur}(B),$$

with $\wedge$ interpreted as paraconsistent conjunction on $\mathbb{B}$.

Theorem 21 (Non-explosion for "surprising fulfillment")**.** *In the paraconsistent semantics of Definition 398, there exist valuations v and propositions P, Q such that P and $\neg P$ are both designated, but Q is not designated. Hence contradictions do not entail triviality.*

Proof. Let $\tau = 1$ (designation means $t = 1$). Choose $v(P) = (1,1)$ and $v(Q) = (0,0)$. Then $v(\neg P) = \neg(1,1) = (1,1)$, so both P and $\neg P$ are designated. But $v(Q)$ has truth degree 0, so Q is not designated. □

Remark 1349. *This formalizes the intended use: one can explicitly represent the tension between "fulfills expectations" and "is surprising" (including by adding additional rules that may generate contradictions), yet still reason nontrivially about other matters.*

24.8.3 Beauty and open-endedly intelligent subnetworks

Hyperseed's definition of beauty includes that the internal representation of the beautiful object "behaves as an open-endedly intelligent mental subnetwork." We capture this with a minimal, ontology-compatible criterion: the representation should contribute to both individuation and self-transcendence while maintaining self-continuity.

Definition 408 (Individuation gain, transcendence gain, self-continuity)**.** *Fix a mind m and an encounter with B on I. Assume we have numeric functionals:*

$$G_I(m, B; I) \in [0, \infty), \quad G_T(m, B; I) \in [0, \infty), \quad \mathsf{Self}C(m; I) \in [0, 1],$$

interpreted respectively as individuation gain, self-transcendence gain, and self-continuity during I. Assume monotonicity constraints consistent with Hyperseed's intent:

1. *G_I is nondecreasing in $\mathsf{Ful}_m(B; I)$ (fulfillment supports individuation),*

2. *G_T is nondecreasing in $\mathsf{Sur}_m(B; I)$ and in $\mathsf{Self}C(m; I)$ (surprise with continuity supports self-transcendence).*

Definition 409 (Aesthetic subnetwork is OEI-active)**.** *Let $\mathsf{Rep}_m(B; I)$ denote the induced pattern subweb/subnetwork in m representing B during I. We say $\mathsf{Rep}_m(B; I)$ is* OEI-active *if both $G_I(m, B; I) > 0$ and $G_T(m, B; I) > 0$.*

Theorem 22 (Beauty implies OEI-activity under mild thresholds)**.** *Fix thresholds $\theta_F \in (0, 1]$, $\theta_S > 0$, and $\theta_C \in (0, 1]$. Assume:*

1. *$\mathsf{Ful}_m(B; I) \geq \theta_F$ implies $G_I(m, B; I) > 0$,*

2. *$\mathsf{Sur}_m(B; I) \geq \theta_S$ and $\mathsf{Self}C(m; I) \geq \theta_C$ imply $G_T(m, B; I) > 0$.*

Then if m undergoes a beauty episode with B on I in the sense that $\mathsf{Beaut}(B)$ is designated and $\mathsf{Self}C(m; I) \geq \theta_C$, it follows that $\mathsf{Rep}_m(B; I)$ is OEI-active.

Proof. If $\mathsf{Beaut}(B) = \mathsf{Ful}(B) \wedge \mathsf{Sur}(B)$ is designated, then both $\mathsf{Ful}(B)$ and $\mathsf{Sur}(B)$ have designated truth components, hence $\mathsf{Ful}_m(B; I) \geq \theta_F$ and (after rescaling) $\mathsf{Sur}_m(B; I) \geq \theta_S$ for suitable corresponding numeric thresholds. By assumptions (1) and (2) and $\mathsf{Self}C(m; I) \geq \theta_C$, we get $G_I > 0$ and $G_T > 0$. Thus $\mathsf{Rep}_m(B; I)$ is OEI-active by Definition 409. In other words, the designated status of $\mathsf{Beaut}(B)$ forces *both* constituents—fulfillment and surprise—to be present to a model-dependent but nontrivial degree (captured by θ_F, θ_S), and the OEI-activity condition is triggered precisely because the growth terms G_I, G_T are required to be strictly positive rather than merely nonnegative. The "after rescaling" clause is included to emphasize that $\mathsf{Sur}_m(\cdot)$ may be measured on a different natural scale than $\mathsf{Ful}_m(\cdot)$ (e.g. entropic novelty versus reward-like satisfaction), yet the designatedness constraint can still be pushed down to a single inequality by choosing a monotone calibration map and an appropriate θ_S. □

Remark 1350. *This theorem is intentionally a "sanity theorem": it shows that the ontology can formalize Hyperseed's claim that beauty is tied to open-ended intelligence (individuation + self-transcendence) without requiring a single reductionist scalar. Stronger results can be obtained once specific models of $G_I, G_T, \mathsf{Self}C$ are chosen. In particular, the point is not that* Beaut equals *any one quantitative score, but that the logical structure of beauty-as-"surprising fulfillment" can be made to entail (via thresholding) the activation of a representation subnetwork with both individuative and transcendent growth directions. Accordingly, the theorem isolates the minimal commitments needed for the implication $\mathsf{Beaut}(B) \Rightarrow$ "OEI-active representation": designatedness of the conjunction, and positivity of the growth terms under adequate self-coherence.*

24.9 Archetypes, art, and religion as communion technologies

Hyperseed treats archetypes as patterns that occur surprisingly often across a collection of systems but with diverse instantiations, and treats art as something with strong aesthetic effect. A natural ontology-level synthesis is: *artifacts and rituals are pattern carriers that can induce cross-mind patterns, thus facilitating communion.* Here "pattern carrier" is intended broadly: it includes not only static objects (images, texts, buildings) but also repeatable procedures (chants, dances, liturgies) whose execution can synchronize attention, affect, and interpretation across a group. The central technical idea is that such carriers can be analyzed in the same language as other representational substrates, by tracking how they change cross-supported intensities and thereby shift communion margins.

Definition 410 (Archetype as a cross-context pattern template)**.** *Fix a class of systems (minds, texts, cultures, etc.) indexed by J. An* archetype *is a template τ together with an instantiation map sending each $j \in J$ to a (possibly empty) set of instantiations $\mathsf{Inst}_j(\tau)$, such that:*

1. *τ has high support across J (many j have $\mathsf{Inst}_j(\tau) \neq \emptyset$), and*
2. *the typical instantiations are diverse (not near-copies), and*
3. *the support is "surprising" relative to the description complexity of τ (e.g. high occurrence despite nontrivial $\kappa(\tau)$).*

Intuitively, (1) captures the "recurrence" aspect of archetypes, (2) rules out trivial replication effects (e.g. a single meme copied verbatim), and (3) is the anti-tautology constraint ensuring that τ is neither so vague that it matches anything nor so simple that high frequency is automatically expected. Depending on application, "diverse" can be operationalized via a distance on instantiations (semantic, structural, stylistic), and "surprising" can be operationalized via deviation from a null model on J in which occurrences are generated independently of τ's internal structure.

Remark 1351. *A useful way to read Definition 410 is as a three-way tension: increasing $\kappa(\tau)$ typically decreases baseline occurrence under simple generative priors, so an archetype is precisely a template whose empirical support remains high despite that penalty. This aligns the informal "archetypal" notion (recurring forms that feel deeper than coincidence) with a quantitative perspective in which recurrence is evaluated against compressibility and expected frequency.*

Definition 411 (Art as an intersubjective pattern carrier)**.** *Fix a class of experiencers $\mathcal{X} \subseteq \mathsf{Mind}$. An* art object *$A$ (relative to $\mathcal{X}$) is any entity such that, for many $m \in \mathcal{X}$, encountering A induces a designated beauty proposition $\mathsf{Beaut}(A)$ in m's valuation. Equivalently, it induces a high rate of beauty episodes in the population $\mathcal{X}$. Because $\mathsf{Beaut}(\cdot)$ lives in a paraconsistent setting, this criterion allows that some experiencers may also (simultaneously) assign non-designated or even conflicting evaluations to A without preventing A from counting as an art object relative to $\mathcal{X}$, so long as the designated-beauty response is sufficiently widespread. The explicit relativization to $\mathcal{X}$ is meant to capture the familiar phenomenon that what functions as art for one community may not function as art for another, not merely due to preference but due to differences in learned perceptual and interpretive repertoires.*

Definition 412 (Communion capacity of an artifact)**.** *Fix a community C and interval I, and let A be an artifact jointly attended to by the community on I. Define the* communion capacity *of A for (C, I) as the expected change in communion margin:*

$$\Gamma_{\mathsf{Comm}}(A; C, I) := \mathbb{E}\Big[\Delta_{\mathsf{Comm}}(C, I; \mathbf{w}^A) - \Delta_{\mathsf{Comm}}(C, I; \mathbf{w})\Big],$$

where $\mathbf{w}$ *is the baseline intensity datum and* $\mathbf{w}^A$ *is the datum after (or during) joint participation in* A*, and the expectation is over relevant internal stochasticity and contextual variation. The quantity* Γ_{Comm} *is therefore an* intervention effect*: it isolates the marginal contribution of adding joint participation in* A *to the community's dynamics, holding fixed (in expectation) other sources of fluctuation. In particular,* $\Gamma_{\mathsf{Comm}}(A; C, I)$ *may be negative for artifacts that fragment interpretation or intensify within-mind idiosyncrasy more than cross-mind alignment, so the sign of* Γ_{Comm} *functions as a formal separator between communion-promoting and communion-eroding interventions.*

Proposition 40 (Broadly beautiful artifacts tend to have positive communion capacity)**.** *Assume:*

1. *Joint attention to an artifact* A *creates at least one cross-supported pattern (a shared representation, shared affective response, shared narrative token, etc.) with intensity increasing in the number of participants who undergo a beauty episode with* A*.*

2. *Within-mind intensities are not increased by joint attention to* A *faster than cross-mind intensities (i.e. the dominant new patterns created by joint participation are cross-supported).*

Then for communities drawn from a population in which A *is an art object (Definition 411), we have* $\Gamma_{\mathsf{Comm}}(A; C, I) > 0$ *for a substantial measure of* (C, I)*. The "substantial measure" clause is meant to exclude degenerate cases (e.g. communities of size one, or intervals too short for any shared pattern to form) while not requiring universality across all possible* C *and* I*; it is enough that the positive effect persists across a non-negligible region of typical group compositions and engagement durations.*

Proof. Under assumption (1), the probability that multiple members of C undergo beauty episodes with A implies creation of cross-supported patterns with positive intensity, increasing W_{cross}. Assumption (2) ensures the within-mind total does not outgrow the cross-supported total under this intervention. Therefore Δ_{Comm} increases in expectation, giving $\Gamma_{\mathsf{Comm}}(A; C, I) > 0$. More explicitly, when A is an art object for the ambient population, draws of (C, I) from that population have a nontrivial chance that several members of C will instantiate designated $\mathsf{Beaut}(A)$ during I; by (1) this increases at least one cross-supported component of the intensity datum, so $\mathbf{w}^A$ differs from $\mathbf{w}$ by adding (or amplifying) cross-linked structure. Condition (2) prevents this amplification from being offset by a larger (or faster) growth in purely within-mind intensities, so the net effect on the communion margin remains positive on average under the stated expectation. □

Remark 1352. *This proposition is the formal core of a qualitative Hyperseed theme: art and ritual can function as reliable methods for inducing communion, hence as social technologies for shaping intersubjective realities. In this framing, the reliability is not mystical but statistical: if an artifact robustly elicits designated beauty episodes across a class of experiencers, then joint exposure predictably increases the weight of cross-supported patterns (shared interpretations, shared affect, shared narrative anchors), which is exactly the structural move required for communion as measured by* Δ_{Comm}*. This also clarifies why the same artifact can fail to produce communion in some settings: if the induced increments concentrate in within-mind idiosyncratic elaborations rather than in shared tokens, assumption (2) can fail and* Γ_{Comm} *can be small or negative.*

24.10 Religion, ritual, and state-dependent science

Hyperseed treats religion as a system of cultural beliefs and social roles associated with spirituality within a society, and treats "state-dependent science" as science relative to observation-sets accepted broadly within a community while they are in particular states of consciousness. In this framing, "religion" is not reduced to propositional doctrine alone; it also includes institutional roles, pedagogies, calendars, and shared interpretive practices that shape how spiritual episodes are recognized, narrated, and transmitted. Likewise, "state-dependent" is meant in a precise epistemic sense: it marks a restriction of the evidential interface (what counts as an observation) and of the community's background constraints (what counts as a permissible inference) to those intervals in which the relevant state predicate is satisfied. The intent is not to grant automatic authority to any such state, but to make explicit that many communities already operate with tacit state-conditions (e.g. prayer, meditation, fasting, liturgical participation) that filter both experience and evaluation.

The ontology-level reading is that:

- Religions are persistent, socially stabilized pattern systems that (among other things) scaffold repeated access to spiritual communion (Definition 402), making such episodes more frequent and more explicit (Definition 400). This includes stabilizing a repertoire of symbols, narratives, and practices that lower the activation threshold for communion-relevant patterns and provide socially shared "handles" for interpreting them. In particular, the persistence of a religion can be understood as a kind of cultural memory that preserves the conditions under which communion has been historically attainable, even when individuals' private access is intermittent.

- Rituals are repeated procedures that reliably increase cross-supported pattern intensity, thereby tending to push a community toward Comm and sometimes $\mathsf{ExplComm}$. Here "procedure" is meant broadly: bodily synchrony, chanting, call-and-response, incense, architectural acoustics, and coordinated attention can all function as controllable parameters that increase mutual predictability and shared salience. On the present ontology, a ritual's efficacy is therefore not primarily a matter of private belief, but of interlocking supports that raise the intensity of patterns that are jointly tracked, jointly enacted, and jointly reinforced.

- State-dependent science is an epistemic practice internal to an intersubjective reality-system that is conditioned on particular community states; formally, it is ordinary science carried out on a restricted subdomain of intervals I satisfying a state predicate $\Sigma(I)$. The restriction to $\Sigma(I)$ should be read as a domain restriction rather than a suspension of ordinary epistemic norms: within the selected intervals one still compares models, checks predictiveness, and negotiates intersubjective agreement, but one does so using the observation-sets and inferential affordances that are actually available in those states. In this sense, the proposal parallels familiar cases where instruments or training open new observation channels, except that the "instrument" is partly constituted by communal state and practice.

Definition 413 (State-dependent science schema)**.** *Fix a community C and a state predicate Σ selecting intervals I (e.g. "oceanic state", "I-Thou state", "high compassion resonance"). A* state-dependent science *for (C, Σ) is a family of model-building procedures that, restricted to intervals I with $\Sigma(I)$, produces predictive/causal models of observation-sets that are accepted broadly by members of C in those states.*

The schema is intentionally permissive about what counts as a "model-building procedure." In some cases the procedures may look like familiar experimental cycles (induction, controlled variation, replication, and error analysis), while in others they may be closer to disciplined phenomenological reporting with structured elicitation protocols and inter-rater calibration. The key constraints are (i) the restriction to Σ-intervals is explicit, (ii) the outputs are not merely idiosyncratic but admit communal uptake, and (iii) there is some articulated sense in which the resulting models succeed or fail relative to the observation-sets available under Σ. In particular, the acceptance clause is meant to exclude purely private revelation from counting as "science" in this narrow technical sense, unless it is embedded in practices that render its contents intersubjectively assessable within C.

Remark 1353. *This definition is intentionally schematic; it exists to locate the concept inside the ontology. Filling it out requires specifying: the observation algebra, the acceptance criterion, and how altered states modify the available observation channels and inference biases.*

A useful way to read "observation algebra" in this setting is as the community's formal or informal account of which experiential discriminations are admissible as stable observables under Σ, and which combinations, comparisons, and transformations of those observables preserve meaning. For example, a tradition might treat certain affective contours, attentional unifications, or relational gestalts as reliably reportable only after training; that training then functions analogously to calibration in laboratory practice. Similarly, altered-state inference biases need not be understood only as sources of error; they may also function as structured priors that are appropriate to the restricted domain (though they can mislead when exported outside it). On the present view, the main methodological risk is equivocation between domains: treating claims stabilized under Σ as if they were validated over all intervals, or conversely dismissing Σ-restricted regularities simply because they do not appear in the complement domain.

24.11 Summary and forward links

We have given:

- a community pattern-system definition of intersubjective reality (Definition 395);
- a quantitative criterion for intersubjective communion via cross-supported pattern dominance (Definition 397);
- a formalization of explicit communion via self-model recognition using uncertain paraconsistent truth values (Definition 400);
- a scope-based definition of spiritual communion and an intervention-based definition of entheogen (Definitions 401–403);
- emotion and compassion as mind-body-and-community patterns (Definitions 404–405);
- beauty as paraconsistent conjunction of fulfillment and surprise and a non-explosion theorem explaining why this is coherent (Theorem 21);
- a minimal theorem linking beauty episodes to OEI-activity of the representing subnetwork (Theorem 22);
- archetypes, art, and ritual as cross-mind pattern carriers with positive communion capacity (Definitions 410–412, Proposition 40);
- a schema locating religion and state-dependent science as persistent community-level pattern practices (Definition 413).

Taken together, these items supply a single "ladder" from ontology to practice. At the base, intersubjective reality is treated as neither a private mental construction nor an observer-independent substrate, but as what is stabilized when multiple agents participate in and reinforce a shared pattern system. The communion criteria then provide a way to distinguish mere agreement (e.g., coincidental similarity of reports) from structural coupling in which patterns become mutually sustaining across minds and media.

The progression from intersubjective communion to explicit communion clarifies the role of reflexivity: explicit communion is not only that agents co-enact compatible patterns, but that their self-models (and their models of others) register this co-enactment under truth values that may be incomplete or even locally inconsistent. The use of uncertain paraconsistent valuations is thus not a stylistic choice but a technical mechanism for allowing realistic cognitive states—where agents can partially endorse and partially reject the same proposition—without trivializing the overall inferential system.

The scope-based definition of spiritual communion and the intervention-based definition of entheogen can be read as a controlled extension of the communion framework to cases where (i) the relevant pattern web extends beyond ordinary interpersonal exchange (scope enlargement), and (ii) there exist interventions that systematically alter the salience, accessibility, or binding of patterns in $\mathbf{w}$ and in the agent valuations. This framing makes it possible to compare "spiritual" episodes to other kinds of high-scope communal episodes (scientific, artistic, political) using the same underlying vocabulary while still marking where special interventions or altered state dynamics are doing explanatory work.

Emotion and compassion enter as patterns precisely because they mediate between internal valuation updates and outwardly observable coordination. In this setting, emotions are not merely "inner feelings" but structured, learnable, and socially conditioned pattern complexes that affect attention, memory, and action selection; compassion is then a special case with characteristic coupling properties (e.g., tendencies to align the welfare-related evaluations of self and other), which makes it directly relevant to communion as a stabilizing or destabilizing force depending on context.

The aesthetic items connect evaluation to coherence under contradiction. Defining beauty as a paraconsistent conjunction of fulfillment and surprise formalizes an everyday phenomenology—that the beautiful often both "fits" and "exceeds" expectation—while the non-explosion theorem guarantees that allowing such conjunctions does not collapse the logic into triviality. The OEI link then supplies a minimal dynamical bridge: beauty episodes are not only definitional artifacts of a valuation scheme, but correlate with identifiable activity of the representing subnetwork, which is the sort of claim needed for later empirical contact (e.g., computational or neurocognitive modeling).

Archetypes, art, and ritual are positioned as portable pattern carriers that can propagate and synchronize across individuals, timescales, and media. Their "communion capacity" is the key functional notion: it identifies when a carrier does not merely transmit content but tends to increase cross-support among participants' patterns (including affective and normative patterns), thereby enhancing the conditions under which communion predicates are satisfied. Finally, the schema for religion and state-dependent science frames both as persistent community-level pattern practices: each involves stabilized methods of generating, selecting, legitimating, and teaching patterns, and each can be analyzed in terms of the same underlying machinery while differing in their typical interventions, scopes, and validation dynamics.

In later sections, these constructions can be specialized by: (i) choosing concrete pattern languages and complexity measures (linking to weakness/MDL formalisms), (ii) choosing explicit dynamical models for how artifacts and rituals modify $\mathbf{w}$ and the valuations v_m, and (iii) connecting repeated communal practice to habit formation (morphic resonance) as a long-run stabilizer of intersubjective pattern webs.

Concretely, item (i) determines what counts as a "pattern" in the first place and how competing descriptions are penalized or preferred; this is where the abstract pattern-system talk acquires operational bite, since the same empirical stream can admit multiple encodings with different complexity profiles. The MDL/weakness link is especially important here because it yields a principled way to trade off expressivity and generalization, making intersubjective stabilization a matter of converging on pattern sets that are jointly compressive under shared constraints.

Item (ii) then fixes the time-evolution story: the definitions above specify relational and logical conditions, but dynamical models specify mechanisms by which those conditions arise, persist, or fail. In particular, modeling how artifacts and rituals modify $\mathbf{w}$ and the valuations v_m makes it possible to distinguish (a) mere informational transmission from (b) systematic reweighting of saliences, priorities, and inferential pathways. This also clarifies how the same ritual form can have different communion outcomes depending on parameter regimes (e.g., strength of coupling, noise levels, or the degree of paraconsistent tolerance in participants' valuation dynamics).

Item (iii) provides the long-horizon glue: repeated practice does not only reinforce explicit beliefs but also trains priors, attentional habits, and default interpretive moves, thereby shaping the background conditions under which future communion is more or less likely. Casting this as habit formation (morphic resonance) supplies a way to model persistence at the community timescale without requiring that every episode of communion be re-constructed from scratch; instead, stabilized pattern webs can be treated as attractors that new participants enter and that existing participants revisit, with the possibility of drift, bifurcation, or sudden reconfiguration under strong interventions.

25 Reality-systems, cosmos, multiverse, and eurycosm

This section scales Hyperseed's observer-relative ontology outward. Earlier sections developed the formal core (paraconsistent evidence, quantale weakness, and compositional model/process structure) and used it to formalize patterns, habits, and resonance (cf. [1, 5, 3]). Here we use the same primitives to define:

(i) *empirical reality* as predictive usefulness to a mind over an interval;
(ii) *reality-systems* as stabilized networks of mutually predictive entities/patterns;
(iii) *physical reality and body* as a particular reality-system with a privileged sensorimotor interface;
(iv) *cosmos/universe* as an observer- and signal-class-indexed reachability notion;
(v) *multiverse* and *guidable multiverse* as (controlled) stochastic evolution on universes;
(vi) the *Yverse* as a recursive limit of multiverse towers (best treated paraconsistently);
(vii) the *eurycosm* and *near eurycosm* as the "wider world" beyond any single well-defined mind or universe; and
(viii) optional speculative anchors: the anthropic principle and big-bounce cosmology.

To motivate the ordering of (i)–(viii), it is helpful to view the constructs as successive closures of the same operational loop: a mind conditions on evidence, makes predictions, and adjusts its internal models/processes; "reality" is then whichever parts of this loop remain stable enough to support compression and counterfactual reliability across time. In this sense, the section does not introduce new metaphysical primitives so much as new *scales* at which the same inferential machinery is applied: from short-horizon usefulness (empirical reality) to long-horizon mutual stabilization (reality-systems), then to reachability under constraints (cosmos/universe), then to stochastic families of such reachability structures (multiverse towers), and finally to limiting constructions intended to capture what remains when no single observer-indexed universe is taken as final (Yverse, eurycosm).

A practical reading is that each definition below will be paired with an "operational test" of the form: what evidence is admissible, what predictions are counted, what closure operator is being used, and what is treated as a fixed point or attractor. This makes the terminology functional: for instance, whether some entity belongs to a reality-system is not a binary metaphysical decree, but a question about whether it participates in sufficiently strong mutual predictability with other entities (relative to the mind, the time interval, and the permitted evidence transformations).

Remark 1354. *Conceptually, the aim is to keep the metaphysical vocabulary honest by tethering it to explicit operations: conditioning, prediction, closure, reachability, and fixed points. One may read this as a Russell-style attempt to replace vague nouns ("the real," "the universe") by precise relations between a mind and its prospective experience, without thereby denying that there can be a world that exceeds any one mind. In Hyperseed's idiom, the exceeding is captured not by a single absolute set, but by a family of observer-indexed constructions and their limits.*

The emphasis on reachability and fixed points is not accidental: many of the objects named in this section can be presented as solutions to recursive equations (e.g., "the smallest closed structure containing what the mind can stably track," or "the maximal set of states reachable given a class of signals and updates"). When the recursion is well-founded and convergent, classical set-based reasoning can suffice; when the recursion is open-ended (as in multiverse towers or the Yverse), paraconsistent treatment is advocated so that one can work with partially specified limits without trivializing the theory in the presence of inevitable self-reference and boundary ambiguities.

Hyperseed concepts covered here. Empirical reality / reality-systems (**??**, 149); physical reality and body (135); algebraically asymmetric physical reality (55); cosmos/eurycosmos (89, **??**); universe, multiverse, guidable multiverse, Yverse (116, 209); eurycosm and near -eurycosm (**??**); anthropic principle (57); big bounce (66).

For orientation, "signal-class-indexed" in (iv) anticipates that a mind's accessible universe may depend not only on its internal inferential capacities but also on which channels of interaction are counted as legitimate interventions or observations (e.g., a restricted sensor suite versus an augmented one, or a narrower versus richer class of admissible measurements). Likewise, the "privileged sensorimotor interface" in (iii) should be read as a structural asymmetry: the body is not merely another predicted object, but the particular interface through which prediction errors arrive and actions are issued, thereby shaping the update dynamics that define empirical reality and reality-systems in the first place.

Finally, the "optional speculative anchors" in (viii) are included as interpretive touchpoints rather than as load-bearing axioms: they indicate how Hyperseed-style definitions might interface with familiar cosmological and selection-effect discussions without presupposing that any single cosmological narrative exhausts the eurycosm. In particular, the near eurycosm is intended to name those aspects of the "wider world" that remain robust across a broad range of observer-indexed constructions, even when the full eurycosm is treated only via partial, limit-like characterizations.

25.1 Empirical reality as predictive usefulness

Hyperseed characterizes "empirically real" in explicitly operational terms: roughly, something is empirically real to a mind over a time interval insofar as perceiving it improves that mind's predictions about its future. To formalize this, we separate (a) *entities/patterns* that may be perceived and (b) the *prediction task* induced by the mind's ongoing coupling to its environment. This framing treats "reality" as a role played inside an inferential control-loop: what matters is whether incorporating a candidate distinction measurably improves forward-looking performance, not whether the distinction is metaphysically primitive or phenomenologically forceful.

Definition 414 (Entities and perceptions)**.** *Fix a mind (or observer) M and an interval of (proto-)time I. Let $\mathcal{U}$ be a set of entities, situations, or patterns that M may treat as objects of perception. We represent "perceiving $x \in \mathcal{U}$" over the interval I by an observation token $\mathsf{obs}_I(x)$.*

Remark 1355. *The notation here is deliberately spare. The set $\mathcal{U}$ is not assumed to be "the set of all things that exist"; it is merely the repertoire of candidate distinctions available to the mind M in the context at hand. Likewise $\mathsf{obs}_I(x)$ is a token standing for the event "M has (somehow) taken in x during I," not a claim that x is thereby known with certainty. The proto-time interval I is the minimal temporal scaffolding needed to speak of "future observations"; it matches the Hyperseed emphasis that time may be mind- and process-relative (see Hyperseed-Concept 142).*

Remark 1356. *A useful way to read $\mathsf{obs}_I(x)$ is as an* interface variable *rather than a raw datum: it stands for whatever internal representation M actually makes available to its predictor. Thus, two minds may share the same external stimulus while having different x-tokens (different feature sets, categorizations, or attention policies), and consequently different empirical realness scores for "the same" ostensible thing. This observer-indexing is not a bug in the framework; it is the intended mechanism by which Hyperseed makes "empirical reality" a property of mind–world coupling rather than a view-from-nowhere.*

Remark 1357. *A simple example is to let $\mathcal{U}$ contain items like "the traffic light is red," "the stove is hot," "my colleague looks annoyed," or "this symbol-string has the form of a proof." Then $\mathsf{obs}_I(x)$ is an abstract placeholder for whichever perceptual/interpretive process produces that classification. The definition is useful because it lets us discuss reality in terms of what changes in the mind's predictive situation when such a token is available, without taking a stand on whether the token came from a retina, a dream, a telescope, or a theorem-prover.*

Definition 415 (Observer-indexed prediction loss)**.** *Let $\mathcal{F}_I$ denote a space of future observation streams relevant to M on the next interval (e.g. I^+). A* predictor *for M is a map Π_M that assigns to each current informational state a predictive distribution over $\mathcal{F}_I$. Let L be a bounded loss function comparing predicted to realized futures. Define the expected prediction error of Π_M on I by*

$$\mathsf{Err}_I(\Pi_M) := \mathbb{E}\big[L(\Pi_M, \mathsf{future})\big]. \tag{1}$$

If M conditions on perceiving x, write $\Pi_M \mid \mathsf{obs}_I(x)$ and $\mathsf{Err}_I(\Pi_M \mid \mathsf{obs}_I(x))$.

Remark 1358. *The symbol $\mathcal{F}_I$ is a space of possible "next-step" futures as M encodes them: e.g., strings of sensor readings, sequences of internal thoughts, or higher-level events. The predictor Π_M is left abstract on purpose: it may be Bayesian, neural, symbolic, hybrid, or something else. The expectation $\mathbb{E}[\cdot]$ is taken with respect to the distribution over futures that actually governs M's coupling to the world (whatever that means in the model). Boundedness of L is a technical convenience ensuring the expectation exists and that normalizations below behave well.*

Remark 1359. *The phrase "current informational state" can be understood broadly to include: memory traces, latent beliefs, currently attended features, and any background context that persists through I. In this sense, conditioning on $\mathsf{obs}_I(x)$ does not assert that x is the* only *input to prediction; it isolates the marginal contribution of having x available on top of whatever else M already carries. One can therefore interpret $\mathsf{Err}_I(\Pi_M) - \mathsf{Err}_I(\Pi_M \mid \mathsf{obs}_I(x))$ as an operational proxy for "how much x helps, given the rest of M."*

Remark 1360. *As an example, suppose L is log loss and Π_M outputs a probability distribution over the next sensory frame; then $\mathsf{Err}_I(\Pi_M)$ is a cross-entropy-like quantity measuring how surprised M*

is, on average, by what happens next. Conditioning on $\mathsf{obs}_I(x)$ corresponds to giving the predictor access to a feature x ("I see smoke"), and the resulting change in error formalizes the pragmatic sense in which perceiving x makes the world more navigable. This is precisely the Hyperseed-Concept **??** *cast in a predictive mold.*

Remark 1361. *In the special (idealized) case where Π_M is the Bayes-optimal predictor for M's true generative coupling and L is log loss, the expected reduction in loss from conditioning on x admits an information-theoretic reading: it coincides with an expected log-likelihood ratio and is closely related to the mutual information between $\mathsf{obs}_I(x)$ and the next-step future in $\mathcal{F}_I$. This highlights an intended interpretation of "empirically real" as "carries compressive, future-relevant information for this mind," while still allowing non-log-loss settings where other operational desiderata (calibration, squared error, asymmetric costs, safety margins) matter more than pure information gain.*

Definition 416 (Empirical realness score)**.** *Fix a small constant $\delta > 0$. The* empirical realness *of $x \in \mathcal{U}$ to mind M on interval I (relative to predictor Π_M and loss L) is*

$$\rho_{M,I}(x) := \frac{\mathsf{Err}_I(\Pi_M) - \mathsf{Err}_I(\Pi_M \mid \mathsf{obs}_I(x))}{\mathsf{Err}_I(\Pi_M) + \delta}. \tag{2}$$

Thus $\rho_{M,I}(x)$ is high when conditioning on x yields a large relative reduction in prediction loss.

Remark 1362. *The fraction defining $\rho_{M,I}(x)$ is a normalized error reduction. The numerator is the improvement in expected predictive performance when x is perceived; the denominator $\mathsf{Err}_I(\Pi_M)+\delta$ prevents division by zero and makes the score scale-comparable across regimes where overall prediction is easy or hard. In particular, $\rho_{M,I}(x)$ can be negative: an observation token that systematically misleads the predictor (say, due to illusion, biased interpretation, or adversarial manipulation) will increase error rather than decrease it.*

Remark 1363. *Because L is bounded, $\rho_{M,I}(x)$ is also controlled in magnitude (though not necessarily confined to $[-1,1]$ under this particular normalization). Concretely, if $0 \le L \le L_{\max}$, then $0 \le \mathsf{Err}_I(\Pi_M) \le L_{\max}$ and the numerator lies in $[-L_{\max}, L_{\max}]$, giving the crude bound*

$$-\frac{L_{\max}}{\mathsf{Err}_I(\Pi_M) + \delta} \;\le\; \rho_{M,I}(x) \;\le\; \frac{L_{\max}}{\mathsf{Err}_I(\Pi_M) + \delta}. \tag{3}$$

This makes explicit the role of δ as more than a division-by-zero patch: it also prevents extreme blow-ups when $\mathsf{Err}_I(\Pi_M)$ is very small (i.e., when the mind is already in an easy-to-predict regime). In applications, δ may be chosen to reflect a minimum meaningful scale of predictive error (a noise floor) below which additional "improvement" should not count as large empirical realness.

Remark 1364. *The score $\rho_{M,I}(x)$ is explicitly relative to a predictor Π_M. This is intentional: if M uses a systematically mis-specified predictor, then even perfectly reliable cues may fail to improve its forecasts, yielding low measured realness in practice. In that sense, empirical realness here is not only about the environment; it is also about the competence and inductive biases of the mind. This matches the Hyperseed stance that "world" and "knower" cannot be fully separated when reality is cashed out operationally.*

Remark 1365. *Two instructive examples:*

(a) If x is "the kettle is whistling" and the future includes "steam will appear," then conditioning on x can sharply reduce error, giving $\rho_{M,I}(x) > 0$.

(b) *If x is "I am certain this random sequence has a hidden message" but this belief leads to consistently wrong anticipations, then $\rho_{M,I}(x)$ may be < 0 even though the experience feels compelling.*

This illustrates why Hyperseed refuses to identify the real with the merely vivid: empirical realness is tied to the predictive economy of the mind-world coupling, not to a phenomenological intensity taken in isolation (cf. Hyperseed-Concept **??***).*

Remark 1366. *The time-indexing by I matters. Some tokens x can be locally predictive (high $\rho_{M,I}(x)$) while degrading longer-horizon performance if they incentivize brittle policies or overfitting to short-lived regularities. Conversely, some distinctions may appear useless on a single step but become empirically real over extended horizons (e.g., slow variables, latent traits, stable causal mechanisms). A natural extension is therefore to consider multi-interval aggregates such as $\rho_{M,[I_1:I_T]}(x)$ defined by replacing Err_I with an averaged or discounted error across a sequence of proto-time steps, which better captures the pragmatic persistence often associated with "real objects."*

Remark 1367. *Finally, nothing in the setup restricts x to being atomic. In many cognitive and scientific settings, empirical traction comes from* bundles *of cues (joint features, models, or hypotheses) that are only predictive together. One may therefore treat x as a structured object (e.g., a tuple, a submodel, or a learned representation) or generalize the conditioning event to a set $S \subseteq \mathcal{U}$ by writing $\Pi_M \mid \mathsf{obs}_I(S)$ for "M has access to all tokens in S on I," enabling a notion of incremental realness (marginal contribution) and synergy (non-additive gains) within the same operational framework.*

Remark 1368 (Paraconsistent evidence)**.** *The scalar $\rho_{M,I}(x)$ measures predictive utility, not logical certainty. If one wishes, one may lift ρ to a* p-bit *value by tracking (i) evidence that conditioning on x improves prediction and (ii) evidence that conditioning on x worsens or destabilizes prediction. This is compatible with Hyperseed's tolerance for borderline and inconsistent boundaries, and connects naturally to paraconsistent valuation frameworks such as Constructible Duality Logic [23] and resonance-based paraconsistent semantics [24].*

25.2 Reality-systems as stabilized mutually predictive pattern complexes

Hyperseed motivates "reality" as a property of *systems of entities* rather than isolated objects. The circularity in the slogan "x is real if perceiving it helps predict real things" is not a bug; it indicates a fixed-point structure. We now package this via a closure operator on subsets of $\mathcal{U}$.

Definition 417 (Predictive adjacency at threshold)**.** *Fix M, I and a threshold $\theta \in (0,1)$. Define a directed relation $\to_\theta$ on $\mathcal{U}$ by setting*

$$x \to_\theta y \quad \textit{iff} \quad \rho_{M,I}(x \to y) \geq \theta, \tag{4}$$

where $\rho_{M,I}(x \to y)$ is a chosen score measuring how much conditioning on $\mathsf{obs}_I(x)$ improves prediction of observation tokens associated with y. (Concrete choices include conditional loss reduction restricted to features belonging to y.)

Remark 1369. *The symbol $\to_\theta$ is a thresholded "predicts" relation: $x \to_\theta y$ means that, at the selected confidence level θ, access to x makes y more predictable. The auxiliary score $\rho_{M,I}(x \to y)$ is intentionally underspecified; it should be read as a* directed *version of the earlier $\rho_{M,I}(x)$ that focuses on the part of the future stream attributable to y. In a feature-based setting, one may take y to denote a subset of coordinates of the future observation stream and compute loss reduction on those coordinates only.*

Remark 1370. *A simple example: let $\mathcal{U}$ include weather-relevant tokens, and suppose x is "dark clouds" and y is "rain within 10 minutes." Then $x \to_\theta y$ holds for a fairly high θ in many climates. By contrast, if x is "I had a particular dream" and y is "rain within 10 minutes," the relation may fail for typical observers, even if some observers attribute meaning to the dream. The definition is useful because it turns "seems to go with" into a directed graph that can be closed and iterated, preparing the ground for fixed-point constructions of reality-systems (Hyperseed-Concept 149).*

Definition 418 (Forward/backward predictive closure)**.** *For $R \subseteq \mathcal{U}$, define*

$$\mathsf{Pred}^{\to}(R) := \{y \in \mathcal{U} : \exists x \in R \text{ s.t. } x \to_\theta y\}, \tag{5}$$

$$\mathsf{Pred}^{\leftarrow}(R) := \{y \in \mathcal{U} : \exists x \in R \text{ s.t. } y \to_\theta x\}. \tag{6}$$

Define the mutual predictive closure operator

$$F(R) := \mathsf{Pred}^{\to}(R) \cap \mathsf{Pred}^{\leftarrow}(R). \tag{7}$$

Remark 1371. *The operators $\mathsf{Pred}^{\to}$ and $\mathsf{Pred}^{\leftarrow}$ are the one-step out- and in-neighborhood operators in the directed graph given by $\to_\theta$. Thus:*

- $\mathsf{Pred}^{\to}(R)$ *collects what R predicts;*
- $\mathsf{Pred}^{\leftarrow}(R)$ *collects what predicts R; and*
- $F(R)$ *keeps only those nodes that stand in both relations to R.*

The intersection is where the circularity enters: a candidate real set should not merely emit predictions, nor merely be predicted; it should participate in a mutually supporting web.

Remark 1372. *As a toy example, suppose $\mathcal{U} = \{a, b, c, d\}$ with edges $a \to b$, $b \to a$, $b \to c$, $c \to b$, and d isolated. Then the mutually predictive closure of $\{a\}$ yields $\{a, b\}$ (and may further expand to include c depending on the thresholded edges), while d never enters any nontrivial fixed point. This makes vivid the intended metaphysical moral: isolated tokens that do not participate in stable mutual prediction are, by this operational criterion, not part of a well-formed reality-system.*

Lemma 5 (Monotonicity)**.** *The operator $F : 2^{\mathcal{U}} \to 2^{\mathcal{U}}$ is monotone: if $R \subseteq R'$ then $F(R) \subseteq F(R')$.*

Remark 1373. *Intuitively, monotonicity says: if you start with more candidate "real" items (a larger seed set), then the set of things that are mutually predictively adjacent to it cannot shrink. This is a minimal sanity condition for any closure-like operator and is precisely what we need to invoke general fixed-point theorems later. In the logic of the development, this lemma is the hinge connecting the empirical/predictive notions above to lattice-theoretic existence results below.*

Proof. If $R \subseteq R'$, then any witness $x \in R$ for membership in $\mathsf{Pred}^{\to}(R)$ or $\mathsf{Pred}^{\leftarrow}(R)$ is also in R'. Thus $\mathsf{Pred}^{\to}(R) \subseteq \mathsf{Pred}^{\to}(R')$ and $\mathsf{Pred}^{\leftarrow}(R) \subseteq \mathsf{Pred}^{\leftarrow}(R')$, hence their intersection is monotone. □

Remark 1374. *At a high level, the proof uses only the existential nature of the definitions: membership in $\mathsf{Pred}^{\to}(R)$ is witnessed by the existence of some $x \in R$ with an edge $x \to_\theta y$. If R grows to R', every old witness remains available, so the image can only expand. Geometrically, enlarging a set of nodes in a directed graph cannot reduce its one-step neighborhoods.*

Proof sketch. Unpack the definitions: both $\mathsf{Pred}^{\to}$ and $\mathsf{Pred}^{\leftarrow}$ are built from an $\exists x \in R$ condition. Replacing R by a superset R' preserves all witnesses, so each of the two neighborhood operators is monotone, hence so is their intersection. □

Definition 419 (Reality-system (fixed-point form)). *A* reality-system *for mind M on interval I (at threshold θ) is a nonempty set $R \subseteq \mathcal{U}$ such that*

$$F(R) = R. \tag{8}$$

Elements of R are then "real relative to R" in the sense that they are both predicted by and predictive of the rest of R (at least along some edges at the chosen threshold).

Remark 1375. *The definition is intentionally parameterized by M, I, and θ to make explicit that "what counts as mutually predictive" can depend on the observer, the temporal or contextual window, and the granularity at which predictive links are deemed strong enough to count. In particular, changing θ changes the edge set of $\to_\theta$ and therefore changes the fixed points of F; one can view θ as controlling the tradeoff between permissive "reality-systems" (low θ, many weak links) and conservative ones (high θ, only robust links survive).*

Remark 1376. *The fixed-point condition can also be read as a closure principle: R contains exactly those tokens that survive the operator F when applied to R itself. Equivalently, if one starts from R and applies the "mutual predictability filter" encoded by F, nothing is gained and nothing is lost. This makes $F(R) = R$ a formal way to say that R is self-maintaining under the inferential norms induced by $\to_\theta$ (for the chosen mind and interval).*

Remark 1377. *The fixed-point equation $F(R) = R$ is the formal expression of the earlier circular slogan. Rather than attempting to define "the real" by a one-directional reduction (from world to mind or from mind to world), we define it as a* stabilized relation *between tokens inside a predictive web. The requirement that R be nonempty prevents the trivial solution in which nothing is real because nothing predicts anything.*

Remark 1378. *Nonemptiness also forces "reality" to pick out at least one informationally anchored token, which matters when $\to_\theta$ is sparse or when the mind is in a regime where predictive links degrade (e.g. noise, novelty, or adversarial inputs). Without the nonemptiness clause, the empty set would always be a fixed point for many natural choices of F, and the theory would risk collapsing into the claim that vacuity is always an available "reality-system". The clause therefore functions like a minimal coherence constraint: there must be at least some token(s) that participate in the mutual-support structure.*

Remark 1379. *A practical example: in ordinary perception, the set R might include "solid objects," "my hand," "gravity," "sound sources," and so on, because these tokens support a dense network of mutual predictability (touch predicts proprioception; object permanence predicts future visual input; etc.). In a scientific sub-community, R might also include theoretical entities (fields, genes, curvature) insofar as conditioning on them improves predictions of experimental outcomes. This is a conceptual bridge to Hyperseed's view of science as observer- and community-relative model selection [20].*

Remark 1380. *This example can be sharpened by observing that the same underlying sensory stream can support different stabilized sets depending on which auxiliary tokens the community treats as eligible members of $\mathcal{U}$ (measurement conventions, inferential schemas, instrument readouts, and mathematical constructs). On this view, disagreements about "what exists" often reduce to disagreements about which tokens belong in the shared candidate universe and which predictive relations clear the threshold θ under the community's modeling practices. The fixed-point framing makes those dependencies explicit without implying that reality is arbitrary: fixed points are still constrained by the actual pattern of predictive relations.*

Theorem 23 (Existence of minimal and maximal reality-systems)**.** *For any mind M and interval I, the monotone operator F has a complete lattice of fixed points (ordered by inclusion). In particular there exist:*

(a) a least fixed point $\mathsf{lfp}(F)$ (possibly empty),

(b) a greatest fixed point $\mathsf{gfp}(F)$ (possibly all of $\mathcal{U}$), and

(c) for any seed set $R_0 \subseteq \mathcal{U}$, a least fixed point containing R_0 given by iterating F from R_0 until convergence.

Remark 1381. *The "least" and "greatest" qualifiers are with respect to inclusion, so $\mathsf{lfp}(F)$ is the smallest reality-system (if any) that satisfies the fixed-point constraint, while $\mathsf{gfp}(F)$ is the largest stabilized predictive web compatible with F. Interpreting fixed points as possible "worlds-as-modeled," the theorem guarantees both an extremal minimal commitment (take as real only what must be in any fixed point) and an extremal maximal commitment (take as real everything that can be stably included without breaking mutual predictive closure). This is useful later when comparing conservative and expansive ontologies inside the same formal framework.*

Remark 1382. *Part (c) can be read as a procedure for "ontological bootstrapping": begin with some tokens treated as nonnegotiable (e.g. immediate experiences, measurement outcomes, or postulated primitives), then repeatedly close under the mutual predictability operator until the process stops changing the set. The resulting fixed point is the smallest stabilized reality-system that honors the initial commitments R_0. This provides a clean way to formalize how different starting assumptions can lead to different self-consistent stabilized webs even when F is held fixed.*

Remark 1383. *In plain terms, the theorem says that once we define "mutual predictive closure" in a monotone way, reality-systems exist automatically: there is always a smallest self-consistent predictive web and a largest one, and any initial guess can be iteratively refined to a stabilized web. This is important because it shows that the fixed-point approach is not merely poetic; it is supported by general order-theoretic machinery (Knaster–Tarski) rather than by ad hoc assumptions. It also connects directly with later uses of fixed points in Hyperseed to study self-stabilizing systems, including goal systems of self-modifying agents [10].*

Remark 1384. *The lattice-of-fixed-points perspective also highlights that reality-systems need not be unique: there can be many distinct fixed points, partially ordered by inclusion. When multiple fixed points exist, the formalism separates two questions: (i) which stabilized webs are* possible *under the predictive relations encoded by F, and (ii) which one(s) are* selected *by a particular mind or community, perhaps via pragmatic criteria (compression, robustness, intervention success) not contained in the bare fixed-point equation. Thus, the existence theorem supplies the space of candidates, while selection principles (if any) determine where within that lattice an actual observer tends to land.*

Proof. The powerset $2^{\mathcal{U}}$ is a complete lattice under inclusion. By Lemma 5, F is monotone, so the Knaster–Tarski theorem implies that the set of fixed points of F is a complete lattice and that least/greatest fixed points exist. If $\mathcal{U}$ is finite, then the ascending chain $R_0 \subseteq F(R_0) \subseteq F^2(R_0) \subseteq \cdots$ stabilizes in at most $|\mathcal{U}|$ steps at the least fixed point above R_0. In the infinite case, one may use transfinite iteration or (under mild continuity assumptions) Scott-continuous fixed-point theory. $\square$

Remark 1385. *The proof is a direct application of a general theorem: monotone endomaps on complete lattices have fixed points, and in fact their fixed points themselves form a complete lattice.*

The finite case adds a computational intuition: iterating F from a seed set is like repeatedly expanding to the mutually predictive neighborhood until no new nodes appear. Visually, one is "growing" a subgraph until it becomes closed under the relevant in- and out-neighborhood operators.

Remark 1386. *In the finite case, the bound of $|\mathcal{U}|$ steps is a worst-case guarantee; in typical sparse predictive graphs convergence can be much faster because each application of F only adds tokens supported by the existing predictive structure. Conversely, if one begins from a large R_0, iteration may also* remove *tokens depending on the exact definition of F (e.g. if F enforces mutuality rather than one-way reachability), so the intuitive picture is not only "expansion" but "pruning toward stability." This clarifies why the fixed-point condition is aptly described as stabilization: it is the point at which neither expansion nor pruning changes the set.*

Remark 1387. *The reference to Scott-continuity is not essential for the existence of fixed points (which is already guaranteed by Knaster–Tarski) but becomes relevant if one wants the iterative procedure to converge by taking countable limits rather than transfinite stages. In applications where $\mathcal{U}$ is effectively enumerable and F has suitable continuity properties, this can justify practical approximation schemes: compute $F^n(R_0)$ for increasing n and interpret the limit (when it stabilizes) as the induced reality-system.*

Proof sketch. Show the ambient space $2^{\mathcal{U}}$ is a complete lattice. Verify F is monotone (Lemma 5). Invoke Knaster–Tarski to obtain existence of $\mathsf{lfp}(F)$ and $\mathsf{gfp}(F)$ and completeness of the fixed-point set. For finite $\mathcal{U}$, use the fact that a strictly increasing chain of subsets cannot exceed length $|\mathcal{U}|$, so iteration stabilizes. □

Remark 1388. *The sketch can be operationalized as follows: one chooses an explicit representation for $\mathcal{U}$ (tokens), an explicit rule for the predictive relation $\to_\theta$ at the chosen threshold, and then defines F as the induced closure (or mutuality) operator. At that point, the abstract order-theoretic argument guarantees that the resulting computation is not chasing a mirage: there really are stabilized sets to be found, even if one later adds additional criteria for preferring one fixed point over another.*

Remark 1389 ("Stabilized pattern complex")**.** *Interpreting $\to_\theta$ as a weighted directed graph, a reality-system is a subgraph that is closed under taking one-step predictive in- and out-neighborhoods. If we refine F to use multi-step reachability rather than one-step edges, fixed points coincide more closely with unions of strongly connected components. Either way, the fixed-point condition captures the intended circularity: reality is attributed to collections of entities that mutually support predictive coherence.*

Remark 1390. *The phrase "pattern complex" is meant to stress that membership in R is not an intrinsic label attached to a token in isolation, but a property of a* configuration *of tokens linked by reliable predictive constraints. Graph-theoretically, the closure requirement ensures that once a token is admitted, the local predictive dependencies that sustain it (incoming support) and that it sustains (outgoing support) are also represented within the same subgraph, at least to the degree enforced by F. This turns "being real relative to R" into a coherence notion: reality is the stable core of a mutual-modeling relation rather than a primitive unary predicate.*

Remark 1391. *When F is strengthened from one-step to multi-step conditions, the strong connectivity analogy helps explain why reality-systems tend to behave like "modules" of inference: within a strongly connected predictive module, information can circulate and reinforce itself, supporting robust counterfactual and temporal predictions. At the same time, the threshold θ controls whether weak long-range links suffice to merge modules into larger fixed points or whether they remain separate, yielding multiple coexisting stabilized pattern complexes inside the same $\mathcal{U}$.*

25.3 Physical reality and body as an interface-mediated reality-system

Hyperseed defines *physical reality* (relative to an embodied mind) as a reality-system containing the body, with the additional property that the mind's causal impact on that reality is mostly mediated through the body-channel. We formalize this by adding a factorization condition on causal/process morphisms.

Definition 420 (Embodied mind and body)**.** *Let M be a mind modeled as a dynamical subsystem with internal states* Mind*, percept tokens* $\mathsf{Percepts}$*, and action tokens* Acts*. A* body *for M is a pattern-system B together with a distinguished interface map*

$$\iota : (\mathsf{Percepts}, \mathsf{Acts}) \longrightarrow B \tag{9}$$

such that there is a highly significant mutual causal relationship between M and B. (In enriched categorical terms: the coupling morphisms $M \leftrightarrow B$ have large weight and are stable under refinement.)

Remark 1392. *The typing of ι encodes a simple but crucial idea: percepts and actions are not free-floating; they are anchored in some structured subsystem B that mediates between inner and outer. Here* Mind*,* $\mathsf{Percepts}$*, and* Acts *are names for state/token sets internal to the model of M; B is a pattern-system (in the sense developed earlier, cf. [5]) that carries stable regularities over time. The parenthetical enriched-categorical phrasing points to the earlier "weights" (weakness/residuals) that quantify coupling strength (Hyperseed-Concept 202, 143; see [3, 2]).*

Remark 1393. *A concrete example is an animal body:* $\mathsf{Percepts}$ *may include retinotopic arrays and proprioceptive signals, while* Acts *includes muscle activations. Then ι maps these tokens to bodily subsystems (eyes, joints, muscles) whose dynamics couple to the environment. For a software agent, B could be a robot platform, a server interface, or even a stable toolchain. The definition is necessary because later notions of "physical reality" depend on which causal paths are privileged, and those privileges are expressed by factorization through B.*

Definition 421 (Physical reality)**.** *A* physical reality *for an embodied mind M is a reality-system R such that:*

(a) the body B is largely contained in R (i.e. $B \subseteq R$ as a pattern-subsystem);

(b) most percepts and actions of M are perceptions/actions of entities in R;

(c) ***body-channel mediation:*** *for any action-effect morphism $f : M \to R$ there exist morphisms $g : M \to B$ and $h : B \to R$ with*

$$f \approx h \circ g, \tag{10}$$

where "$\approx$" denotes approximation in the enrichment (e.g. bounded residual weakness or bounded prediction-error increase when replacing f by $h \circ g$).

Remark 1394. *Clause (c) is the formal core: the mind's effective influence on R must (up to controlled error) factor through the body. In symbols, $f : M \to R$ is any causal/process morphism capturing an action-to-effect link; then $g : M \to B$ is the "issue a bodily action" component, and $h : B \to R$ is the downstream physical influence of the body on the larger reality-system. The approximation relation $\approx$ is where the enrichment matters: one may measure the deviation between f and $h \circ g$ by weakness residuals, by increased prediction loss, or by any compatible metric-like structure supplied by the formal core.*

Remark 1395. *An example: if M is you, B includes your vocal tract and hands, and R includes your room, then speaking to move air molecules and thereby alter another person's beliefs is still mediated through B. By contrast, if one posits a mind that can directly alter far-away objects without any intervening bodily channel, then (in this framework) its "physicalness" relative to that R would be lower. This provides a mathematically expressible sense in which "physical reality and body" (Hyperseed-Concept 135) are intertwined.*

Remark 1396 (Degree of "physicalness")**.** *One can define a graded physicalness score by measuring how well action-effect morphisms factor through B. For example, if the enrichment provides a distance d, define*

$$\mathsf{phys}(M, R; B) := 1 - \sup_{f:M\to R} \inf_{g,h} d\big(f, h \circ g\big). \tag{11}$$

If M can significantly impact R without using the body-channel, this score decreases.

Remark 1397 (Laws of physics)**.** *Within this ontology, the "laws of physics" relative to a physical reality R are simply a system of effective hypotheses (models) for predicting and explaining what happens in R. Formally, this can be cast as selecting a model class $\mathcal{H}_R$ that minimizes prediction error subject to simplicity/weakness constraints (a Minimum Description Length or weakness-regularized choice).*

Remark 1398. *The last parenthetical points to a convergence of ideas: Minimum Description Length is a way of expressing that models should be predictive yet parsimonious, often formalized using description lengths related to algorithmic information [16]. Hyperseed's weakness-regularized variants reinterpret this parsimony as a representational/effort constraint (Hyperseed-Concept 82, 202), anticipating the Wu Wei control formalism developed later [21].*

25.4 Algebraically asymmetric physical reality

Hyperseed stresses an *algebraic asymmetry* between physical reality (often low-dimensional and "vector-space-like") and mental/intersubjective realities (typically weaker, more topological, and only approximately projectable into low-dimensional coordinates). We can express this without committing to a particular physical theory by treating "algebraic structure" as additional operations/axioms on the state space. In particular, the asymmetry is not meant to assert that mental reality is "vague" in every respect, but that the kinds of structure that make physical systems rigid (e.g. smoothness, metrics, conservation laws, differential constraints, compatibility conditions) are not, in general, fully present in intersubjective representations. Thus "algebraic weakness" here is a statement about *recoverability of structure*: the mental/intersubjective level typically underdetermines many physically distinct realizations.

Definition 422 (Structure strength and forgetful maps)**.** *Let $\mathcal{R}_{\text{phys}}$ be a category (or enriched category) of models of physical reality, whose objects carry some strong algebraic structure (e.g. smooth manifold, metric space, vector bundle, etc.). Let $\mathcal{R}_{\text{ment}}$ be a category of models of mental and intersubjective reality (e.g. topological or relational pattern spaces). An* algebraic asymmetry *is the presence of a structure-forgetting functor*

$$U : \mathcal{R}_{\text{phys}} \longrightarrow \mathcal{R}_{\text{ment}} \tag{12}$$

that is not (even approximately) invertible on the images relevant to a given community of minds. In this case we say that mental reality is algebraically weaker *than physical reality.*

Remark 1399. *The phrase "category (or enriched category)" allows one to treat morphisms as carrying graded costs, probabilities, or fidelities when needed. For example, in an enriched setting one can measure how well a mental representation matches an underlying physical state, rather than treating representation as all-or-nothing. This is useful for articulating the "approximately projectable" clause: minds may track some physical degrees of freedom with high fidelity while collapsing or ignoring others.*

Remark 1400. *The functor U is called "forgetful" because it discards structure: it takes an object with rich operations/axioms (say, a differentiable manifold with metric) and remembers only whatever mental/intersubjective representation retains (say, adjacency relations, coarse regions, or pattern links). The non-invertibility condition encodes the asymmetry: from the weaker description alone one generally cannot reconstruct the full physical structure. This is the categorical way to articulate the Hyperseed-Concept 55.*

Remark 1401. *One can make the non-invertibility intuition more concrete by considering the* fibers *of U. For an object $m \in \mathrm{Ob}(\mathcal{R}_{\mathrm{ment}})$, the preimage (informally) $U^{-1}(m)$ consists of all physically structured objects that "look like" m under the forgetting process. Algebraic asymmetry asserts that these fibers are typically large: many distinct physical configurations collapse to the same mental/intersubjective pattern. The inability to choose a canonical element of the fiber (even approximately, for the relevant class of observers) is exactly the sense in which U fails to be invertible where it matters.*

Remark 1402. *A simple example is the ordinary map/territory gap: the same topological graph of "places connected by roads" can be embedded in many different geometries with different distances and curvatures. If mental reality records mostly graph-like relations, then lifting it to a physical geometry requires extra choices. This is useful because it explains, without mysticism, why physical reality feels "hard" or "constraining": the constraint is precisely the missing information needed to select one physical refinement among many mental descriptions that collapse under U.*

Remark 1403. *The same pattern appears in familiar mathematical forgetful functors. For instance, forgetting a metric yields a topological space; forgetting a topology yields a set; forgetting a smooth structure yields a topological manifold; forgetting phase information yields an intensity pattern. In each case, the weaker object retains certain invariants (e.g. continuity properties, adjacency relations, cardinality, qualitative connectivity), but it does not determine the stronger object without additional data. The point of stating this categorically is that one can swap in different notions of "physical" and "mental" models while keeping the same asymmetry schema.*

Remark 1404 (Why asymmetry "feels like constraint")**.** *If U discards operations/constraints present in physical reality, then many distinct physical states become indistinguishable in the weaker mental description. From the mind's perspective, trying to realize a mentally-specified pattern physically requires choosing (or discovering) additional structure to lift it back into $\mathcal{R}_{\mathrm{phys}}$. This lifting typically incurs effort and may be impossible, producing the phenomenology that physical reality "constrains" mind.*

Remark 1405. *The "may be impossible" clause has two distinct readings that can coexist. First, there may be no object in $\mathcal{R}_{\mathrm{phys}}$ that maps (even approximately) to the desired m under U, meaning the mental pattern is not physically instantiable in that reality-system. Second, there may be many such objects, but the agent may lack access to the additional structure needed to select and stabilize one. In either case, the mental specification underdetermines the physical implementation, and that underdetermination shows up phenomenologically as friction, failed attempts, or the need for iterative refinement.*

Proposition 41 (Free lift as a source of effort)**.** *Suppose U has a left adjoint F (a "free" structure-adding functor), so $F \dashv U$. Then any mental description $m \in \mathrm{Ob}(\mathcal{R}_{\mathrm{ment}})$ has a canonical lift $F(m)$ into physical-structured form. If a mind's goal pattern is specified in $\mathcal{R}_{\mathrm{ment}}$, then realizing it physically requires at least the "free" lift plus additional constraint satisfaction (selecting a morphism out of $F(m)$ compatible with the actual physical reality-system). This separates* conceptual feasibility *from* physical realizability*.*

Remark 1406. *Intuitively, an adjunction $F \dashv U$ formalizes the idea of "adding structure in the cheapest possible way." The proposition says that even the cheapest lift $F(m)$ already introduces commitments not present in m, and that making the lifted structure* actually occur *in a given physical reality-system is an additional step. This is important because it lets us speak rigorously about effort: effort is not only about searching a space of possibilities, but about bridging a categorical gap between levels of structure.*

Remark 1407. *The word "free" should be read in the technical sense, not as implying physical ease. In many settings, $F(m)$ is a large object: it can represent the space of all minimally constrained implementations of m consistent with the algebraic axioms of $\mathcal{R}_{\mathrm{phys}}$. As a result, the agent's labor is not removed by having F; rather, the adjunction clarifies* what kind *of labor is required: selecting, constraining, and stabilizing a specific physical realization within the degrees of freedom left open by $F(m)$.*

Proof. An adjunction $F \dashv U$ provides, for each m, a unit map $\eta_m : m \to U(F(m))$ that is universal among maps from m into U-images. Thus $F(m)$ is the least additional structure needed to represent m in the physical category. Any further realization must factor through $F(m)$ and so must satisfy additional physical constraints not present in the mental description. $\square$

Remark 1408. *The key step is the universality of the unit η_m. It ensures that every attempt to interpret the mental description m as something physically structured must pass through $F(m)$, so the free lift acts as a bottleneck: it is the common interface between the weak and strong worlds. Geometrically, one may picture $F(m)$ as the "bundle of all minimal physical implementations" of a mental specification, and "physical realizability" as the further act of selecting a section compatible with the actual physics. The proof works because adjunctions are precisely the machinery that captures "best approximation" across categories.*

Remark 1409. *There is also a useful dynamical reading. The composite $T := U \circ F$ is a monad on $\mathcal{R}_{\mathrm{ment}}$, meaning it is an endofunctor that takes a mental description and returns its "freely physically-coherent" enrichment as seen back in the mental category. In this language, iterating T can be interpreted as repeatedly applying a coherence-imposing closure: the agent refines a pattern until it becomes stable under the constraints implicitly imported from $\mathcal{R}_{\mathrm{phys}}$. This does not add a new assumption beyond $F \dashv U$; it only spells out a standard categorical consequence that often matches how agents learn to respect physical constraints through repeated feedback.*

Proof sketch. Use the unit $\eta_m : m \to U(F(m))$ provided by $F \dashv U$. By the adjunction's universal property, any map from m into an underlying physical object factors uniquely through η_m, showing that $F(m)$ is the canonical minimal-structure lift. Realizing m physically then necessarily involves (at least) choosing and satisfying additional constraints beyond those encoded in m. $\square$

25.5 Signals and observer-indexed universes

Hyperseed defines a "universe" relative to a system and a signal class. Intuitively: what counts as "in the universe" depends on what kinds of signal exchanges the system believes are possible.

This usage is intentionally operational: it tracks what is *reachable for interaction* (in the sense of potential bidirectional coordination, observation, or control), rather than what is "out there" in a mind-independent inventory.

Definition 423 (Signals and signal classes)**.** *Let $\mathcal{S}$ be a category (or enriched category) whose objects are systems/entities. A* signal *from A to B is a pair of patterns $(p_{\rm tx}, p_{\rm rx})$ (transmission, receipt) together with a causal interpretation that the transmission pattern is causal for the receipt pattern. A* signal class *C is a specified family of signals (morphisms) in $\mathcal{S}$. Because these are observer-indexed, we allow a mind M to assign a* p-bit *evidence value $E_M(s) \in [0,1]^2$ to the proposition "signal s is usable."*

Remark 1410. *The category $\mathcal{S}$ supplies the ambient "systems and interactions" universe in which we talk about signaling. A signal $s : A \to B$ is not merely a physical disturbance; it is a disturbance together with an interpretation that it transmits information in a usable way. The pair $(p_{\rm tx}, p_{\rm rx})$ emphasizes that the same underlying process may be encoded one way at the transmitter and decoded another way at the receiver; this matches the earlier emphasis that "pattern" is observer-relative [5]. Allowing $E_M(s) \in [0,1]^2$ makes the epistemic status of "usability" paraconsistent: a mind can have simultaneous evidence for and against the usability of a purported channel [23, 24].*

Remark 1411. *It is helpful to separate three layers that are bundled in ordinary language: (i) a dynamical coupling in the substrate (some causal influence from A toward B), (ii) a* coding/decoding *convention that makes that coupling function as information transfer, and (iii) a* pragmatic *constraint that the transfer is reliable enough (or controllable enough) to count as usable for the mind's current purposes. The definition of "signal" packages (i) and (ii) into a morphism-like object, while the evidence value $E_M(s)$ records (iii) in a way that can accommodate ambiguity, adversarial environments, deception, and internal uncertainty.*

Remark 1412. *When $\mathcal{S}$ is taken as an enriched category, the enrichment can encode graded notions of interaction (e.g. costs, capacities, fidelities, probabilities, or resource bounds). In that case, "a specified family of signals" may mean "morphisms satisfying a constraint in the enriching structure" (for example, only channels above a minimum capacity, or only interactions within a given energy budget). The present development does not require fixing an enrichment, but it is useful to keep in mind that many physically relevant restrictions on communication are naturally expressed in enriched-categorical terms.*

Remark 1413. *Example: in an embodied physical cosmos, C might be electromagnetic signals detectable by the body's sensors (light, radio) or mechanical signals (sound, touch). In a social/intersubjective cosmos, C might include linguistic utterances and symbolic artifacts. The definition is useful because it makes "what exists for me" depend on the reachability structure induced by which signals are taken seriously, which is exactly the sort of observer-relativity Hyperseed intends.*

Remark 1414. *The same underlying interaction may fall into different signal classes depending on the mind's model. For instance, the same acoustic wave can be treated as "noise" (excluded from C), as a generic mechanical disturbance (included in a broad C), or as a linguistic utterance with semantic content (included in a narrower but more structured C that presupposes shared conventions). In this way, choosing C captures not only physical modality but also interpretive commitment.*

Definition 424 (Observable universe relative to a signal class)**.** *Fix a system S (typically including a mind and its current reality-system model) and a signal class C. For a threshold $\eta \in (0,1)$, define the* observable universe *of S relative to C as*

$$\mathsf{Univ}_{C,\eta}(S) := \{X \in \mathrm{Ob}(\mathcal{S}) : \exists \text{ a finite } C\text{-signal path } S \rightsquigarrow X \text{ with } E_M^+ \geq \eta\}. \tag{13}$$

In words: X is in the universe if S believes it can (now or eventually) send/receive a C-class signal to/from X.

Remark 1415. *Here* $\mathrm{Ob}(\mathcal{S})$ *denotes the objects of the category* $\mathcal{S}$*. The notation* $S \rightsquigarrow X$ *indicates a finite composable chain of signals in the class* C *from* S *to* X*. The condition "with* $E_M^+ \geq \eta$*" means that along the relevant path(s), the mind's* positive *evidence component for usability exceeds the chosen threshold. (One may similarly incorporate negative evidence, for instance by requiring high positive and low negative evidence, but the present definition keeps the reachability notion uncluttered.)*

Remark 1416. *Concretely, if* $E_M(s) = (E_M^+(s), E_M^-(s)) \in [0,1]^2$*, then* E_M^+ *refers to the first coordinate, interpreted as evidence-for-usability. The definition leaves implicit how evidence is aggregated across a multi-step path; the intended minimal reading is existential: there exists at least one finite path* $s_1, \ldots, s_n$ *of* C*-signals for which each step is usable to degree* $\geq \eta$ *(or, more generally, for which the mind judges the* path as a whole *to clear the threshold under its preferred aggregation rule). This keeps the reachability predicate flexible enough to model different epistemic policies (e.g. "weakest-link" aggregation via* $\min_i E_M^+(s_i)$ *versus Bayesian-like compounding), without changing the surrounding categorical picture.*

Remark 1417. *The phrase "now or eventually" is meant to include mediated reachability:* S *may not directly signal to* X*, but may be able to do so via relays, intermediaries, or composed protocols. The use of a* finite *path matches the operational idea that interaction proceeds through finitely many steps that the mind can in principle represent as a procedure, even if the intermediate systems are not themselves salient in ordinary description.*

Remark 1418. *As an example, if* C *is "radio transmission" and* S *is an Earth-based civilization, then* $\mathsf{Univ}_{C,\eta}(S)$ *includes objects that are radio-reachable in principle (satellites, nearby spacecraft), but excludes galaxies beyond any plausible ability to send/receive radio signals at the required strength. Changing* C *changes the universe: if* C *expands to include hypothetical faster-than-light signals, the universe expands accordingly. This formalizes Hyperseed's distinction between cosmos/universe talk and absolute ontology (Hyperseed-Concept 89).*

Remark 1419. *The threshold* η *plays a normative role: it encodes how demanding the system is about what counts as a usable channel. A cautious scientific instrument may require a high* η *(only high-confidence channels are admitted), whereas an exploratory or imaginative cognition may use a lower* η *(allowing tenuous, speculative, or weakly supported channels to count as potential links). Thus* $\mathsf{Univ}_{C,\eta}(S)$ *is sensitive not only to physics and infrastructure, but also to epistemic temperament and context.*

Proposition 42 (Monotonicity in signal class)**.** *If* $C \subseteq C'$ *then* $\mathsf{Univ}_{C,\eta}(S) \subseteq \mathsf{Univ}_{C',\eta}(S)$*.*

Remark 1420. *This proposition states the obvious but important structural fact: allowing more kinds of signals cannot reduce what you can reach. It mirrors Lemma 5 at a different level: both are monotonicity principles ensuring that subsequent closure/fixed-point constructions behave well.*

Proof. Any C-signal path is also a C'-signal path. Thus reachability using C implies reachability using C'. □

Remark 1421. *The proof is immediate because the definition of universe membership is existential: it suffices that there* exists *a path of allowed signals. Enlarging the allowed set preserves existing paths. Visually, if one adds more edge-types to a graph, the set of vertices reachable from a fixed start vertex can only increase.*

Remark 1422. *Operationally, this monotonicity is a sanity check for model revision: if a mind updates its reality-system model by admitting a previously disallowed channel type into C (e.g. it learns a new protocol, discovers a new sensor modality, or accepts testimony it previously discounted), then previously reachable entities remain reachable, while additional entities may become reachable. This aligns the formalism with ordinary learning dynamics.*

Proof sketch. Universe membership is witnessed by a finite path using signals in C. If $C \subseteq C'$, the same path is valid in C', so reachability (and hence membership) is preserved. □

Proposition 43 (Monotonicity in threshold)**.** *If $\eta \leq \eta'$ then $\mathsf{Univ}_{C,\eta'}(S) \subseteq \mathsf{Univ}_{C,\eta}(S)$.*

Proof. If $X \in \mathsf{Univ}_{C,\eta'}(S)$ then there exists a finite C-signal path $S \rightsquigarrow X$ whose positive evidence satisfies $E_M^+ \geq \eta'$. Since $\eta \leq \eta'$, the same path also satisfies $E_M^+ \geq \eta$, hence $X \in \mathsf{Univ}_{C,\eta}(S)$. □

Remark 1423. *This captures the dual way in which the "observer-indexed universe" can grow: either by broadening C (new modalities of interaction) or by lowering η (relaxing standards of usability). Conversely, a stricter evidential policy (raising η) can shrink the operational universe even if the underlying substrate and signal class remain unchanged.*

Remark 1424 (Cosmos vs universe)**.** *We use* cosmos *informally for a privileged or stabilized universe notion, often induced by a canonical signal class (e.g. "physical" signals mediated through the body-channel for embodied minds). Different minds or different coupling regimes may induce different cosmos notions.*

Remark 1425. *In this terminology, a "cosmos" is typically what remains after a process of convergence: shared measurement conventions, stable instrumentation, and negotiated intersubjective standards produce a relatively fixed C and an implicit threshold policy η. By contrast, "universe" here is deliberately parameterized, so that shifts in embodiment, interface, language-game, or trust policy are represented as explicit parameter changes rather than as contradictions about what exists.*

25.6 Multiverse and guidable multiverse

In Hyperseed, a multiverse is not introduced as a purely metaphysical postulate. Rather, it is an observer-relative construct: a multiverse is a class of possible universes such that (from the observer's perspective) stochastic events determine which universe the observer will inhabit after the event. In this sense, "universe" should be read as a structured world-hypothesis or world-state at the level of description available to the observer's modeling interface, rather than as a commitment to a specific cosmological ontology. The same underlying reality-system may admit many such universe-descriptions depending on which signal classes are available and which variables the observer can stably track.

Definition 425 (Stochastic events relative to an observer)**.** *Fix an observer O. An event e is* stochastic relative to *O if O cannot predict its outcome. An* event class *is stochastic relative to O if O cannot predict the outcome of any particular event in the class, but can predict some features of the outcome distribution over many instances.*

Remark 1426. *This definition is epistemic rather than metaphysical: "stochastic" means "unpredictable to the observer," not "uncaused." One may read this in a Peircean spirit: chance is a name for a limit of habit, a region where regularities have not yet been formed or recognized (cf. the emphasis on habit and Thirdness in [14]). In Hyperseed's operational language, the mind lacks a model that reduces prediction error for the outcomes of e. A practical consequence is that* *"stochastic" depends on the observer's representational capacity, data access, and computational limits; it can therefore change when the observer learns, upgrades instrumentation, or acquires a new signal class.*

Remark 1427. *Example: a fair coin flip is stochastic relative to a typical human observer. But the same physical process may be non-stochastic relative to an idealized observer with complete microstate information and unlimited computation (if one adopts a deterministic underlying model). The definition is useful because it lets multiverse talk arise naturally from limits of prediction and control, rather than requiring an ontological proclamation that "all possibilities exist." More generally, what matters is not whether the microphysics is deterministic, but whether the observer can compress the relevant causal details into a predictive habit at the scale of interaction; when the required compression is unavailable, the observer is forced into a distributional (rather than point) forecast.*

Definition 426 (Multiverse (controlled Markov form))**.** *Fix an observer O and a base universe U_0 (itself observer- and signal-class-indexed). A* multiverse *associated with (U_0, O) is:*

(a) a set (or measurable space) $\mathcal{U}$ of candidate universes consistent with O's model class;

(b) a transition kernel $P(\cdot \mid U, e)$ giving a distribution over next universes after a stochastic event e occurring while in universe U.

Thus, the observer's universe evolves as a stochastic process on $\mathcal{U}$.

Remark 1428. *The kernel $P(\cdot \mid U, e)$ is a Markov-style update rule on the space of universes: given that the observer is currently in universe U and event e occurs, the observer assigns a distribution over which universe comes next. The measurable-space option is included because $\mathcal{U}$ may be infinite or continuous in realistic modeling. The point is not that universes are literally "jumped between" in a physical sense, but that the observer's* self-location *among competing world-hypotheses changes stochastically as unpredictable events unfold. This formalizes Hyperseed-Concept 116 in a way compatible with decision/control extensions. One can also read U as an information-state (e.g., a hypothesis index, a parameter setting, or a latent regime label), in which case $P(\cdot \mid U, e)$ expresses how surprises and updates move the observer among regimes.*

Remark 1429. *A simple example is Bayesian model selection over a hypothesis class of "which underlying dynamics governs my environment." After a surprising observation, probability mass shifts among hypotheses, which can be read as transitioning among "universes" in $\mathcal{U}$. The definition is useful because it makes multiverse structure available wherever there is irreducible uncertainty and competing world-models, including mundane settings like partial observability and nonstationarity. In this Bayesian reading, the "state" of the multiverse can equivalently be taken as a belief distribution over $\mathcal{U}$; the present definition instead treats the realized (self-located) universe as the stochastic variable and keeps the belief implicit in O's kernel assignment. Both viewpoints are compatible: the former emphasizes inference dynamics, while the latter emphasizes experienced branch-selection.*

Remark 1430. *The "event" label e may be instantiated at different granularities. At one extreme, e can be a single measurement or observation; at the other, e can stand for an extended episode whose internal microstructure is not modeled by O. The Markov form should then be read as a modeling commitment that whatever details of the past matter for predicting the next universe are encoded in the current U (as represented by O), which is exactly the usual sufficiency intuition behind state variables in stochastic process modeling.*

Definition 427 (Guidable multiverse). *Let S_1 and S_2 be two signal classes and let O be an observer. A* guidable multiverse *(relative to S_1, S_2, O) is a multiverse $\mathcal{U}$ defined using S_1-reachability such that, in universe U as modeled by O, signals in class S_2 can systematically influence which element of $\mathcal{U}$ the observer will be in after a transition. Formally, this means there is an action set A (implementing S_2 signals) and a controlled transition kernel*

$$P(\cdot \mid U, a) \tag{14}$$

on $\mathcal{U}$.

Remark 1431. *The only new ingredient is control: $a \in A$ indexes interventions available to the observer (or to an agent coupled to the observer) and the kernel $P(\cdot \mid U, a)$ replaces the purely event-driven kernel. This is the same structural move that turns an uncontrolled Markov chain into a Markov decision process. The definition is useful because it makes room for the Hyperseed idea that broader or subtler signal classes could enlarge not only what is reachable, but what is steerable. In particular, if S_2 is weak or noisy, then distinct actions may induce nearly identical kernels $P(\cdot \mid U, a)$, making the multiverse effectively unguidable at the level of $\mathcal{U}$ even if many universes are in principle possible.*

Remark 1432. *Example: in ordinary life, S_1 might be "sensory signals that define my everyday physical universe," while S_2 might be "actions I can take using tools and institutions." Then $P(\cdot \mid U, a)$ encodes how my choices alter which future regimes ("universes" in the hypothesis/experience space) become actual for me. This connects to Hyperseed's general emphasis on action as selection among futures, later formalized variationally in Wu Wei dynamics [21]. In a more explicitly decision-theoretic register, one can introduce a policy $\pi(a \mid U)$ and consider the induced (uncontrolled) kernel $P_\pi(\cdot \mid U) = \int_A P(\cdot \mid U, a)\, \pi(da \mid U)$, which makes clear that guidance is realized through a choice rule that couples the observer's current universe-description to action selection.*

Remark 1433 (Interpretation). *A guidable multiverse is simply a multiverse equipped with controllable degrees of freedom. In conventional physical modeling, "guidance" is limited by the causal structure of the physical reality-system. Hyperseed allows (as a conceptual possibility) that broader signal classes may exist, which would make the multiverse "more guidable" relative to a mind. One concrete way to interpret "more guidable" (still at the level of modeling, not metaphysics) is that different actions a induce more distinguishable outcome distributions over $\mathcal{U}$ (e.g., larger separations between $P(\cdot \mid U, a)$ across a), thereby increasing the set of achievable transitions and decreasing the entropy of the next-universe distribution under well-chosen interventions.*

Remark 1434. *The guidable multiverse definition deliberately remains neutral about whether the observer has direct access to U (full observability) or only to signals generated by U (partial observability). In the latter case, U may be treated as latent and the effective control problem lives on a belief state over $\mathcal{U}$, but the intuitive claim remains: certain signal classes (S_2) can act as handles that reshape which world-hypothesis becomes experientially realized next. This preserves the Hyperseed motif that expansion of signal classes is simultaneously an expansion of both epistemic discrimination and practical steering capacity.*

25.7 The Yverse as a recursive limit of multiverse towers

Hyperseed introduces the *Yverse* as the limit of iterating the "multi" construction:

$$\texttt{universe} \to \texttt{multiverse} \to \texttt{multimultiverse} \to \cdots$$

In this view, signals that appear random at one level may become predictable at a higher level, and in the limit a signal may be paraconsistently both random and predictable. The name "Yverse" references the Y-combinator from combinatory logic, which makes its argument recursive. One can read the informal "$\to \cdots$" not merely as an unending sequence, but as a prompt to replace indefinite iteration with a stabilization principle: rather than asking for the tower at stage n, one asks for a description that is already closed under the "make it multi" operation. This is the sense in which the Yverse is a *recursive limit*: it is characterized by an equation rather than by an external indexing set of stages.

Remark 1435. *The "random vs. predictable" slogan can be understood in a coarse-graining sense: at a given level of description, a signal may be effectively unpredictable because relevant hidden variables are not represented. A higher-level description may add latent structure (e.g. by turning a single-world description into a distribution over refinements), making the same signal conditionally predictable relative to the expanded information state. In the limit, if the description is self-referentially closed under such expansions, it becomes coherent (in a paraconsistent sense) to treat the signal as both "random" (under a projection or forgetting map back down to a lower level) and "predictable" (under the enriched viewpoint that retains the higher-order structure).*

Definition 428 (Abstract multi-operator)**.** *Let $\mathcal{D}$ be a complete lattice of "world-descriptions" (these may encode universes, multiverses, and higher-level constructions). A* multi-operator *is a monotone map*

$$\mathsf{Multi} : \mathcal{D} \to \mathcal{D} \tag{15}$$

that sends a description to the corresponding "multi" description.

Remark 1436. *The lattice $\mathcal{D}$ is an abstract container for levels of description: an element might be a single universe model, a probability distribution over universe models, a distribution over such distributions, and so on. The order structure (making $\mathcal{D}$ a complete lattice) is what permits fixed-point reasoning: it provides suprema/infima needed by general theorems. Monotonicity of* Multi *means that refining a description cannot yield a strictly coarser multi-level description; it is the analog, at the "tower of worlds" level, of the earlier monotonicity lemmas. It is often helpful to read the order $\leq$ on $\mathcal{D}$ epistemically: $d_1 \leq d_2$ may mean "d_2 contains at least as much information/constraint as d_1," or "d_2 is at least as strong a theory as d_1." Under such readings, monotonicity of* Multi *expresses the idea that adding constraints at the base level should not reduce the constraints (or information) available after passing to the multi-level wrapper.*

Remark 1437. *A concrete mental picture is to let $\mathcal{D}$ be a powerset lattice of admissible meta-descriptions, or a lattice of theories ordered by entailment/strength. Then* Multi *wraps a description into a "distribution over refinements" object. The definition is useful because it isolates the recursion in a single operator, making the Yverse a fixed point rather than an indefinitely postponed iteration. Another concrete picture (compatible with the same formalism) is to treat elements of $\mathcal{D}$ as* state spaces with semantics*: a description might specify both (i) a space of possible world-histories and (ii) an interpretation map telling us which signals are observable. Then* Multi *can be understood as lifting a description to a meta-space in which the previous space appears as one component among many alternatives, together with a weighting or selection mechanism. The formal abstraction deliberately does not commit to whether this lifting is probabilistic, possibilistic, or logical; it only commits to the order-theoretic property needed for fixed-point existence.*

Definition 429 (Yverse as a fixed point)**.** *A* Yverse *is a fixed point $Y \in \mathcal{D}$ of* Multi*:*

$$Y = \mathsf{Multi}(Y). \tag{16}$$

When multiple fixed points exist, one may select $Y := \mathsf{lfp}(\mathsf{Multi})$ (the least fixed point) or $Y := \mathsf{gfp}(\mathsf{Multi})$ (the greatest fixed point) depending on the intended semantics.

Remark 1438. *The equation $Y = \mathsf{Multi}(Y)$ is the formal signature of self-reference: the Yverse is a world-description that, when "multiversed," yields itself. Philosophically, this is the point where classical insistence on well-founded constructions begins to strain: the object is defined by its own image. Hyperseed treats this not as a defect but as a clue that paraconsistent and/or non-well-founded semantics may be appropriate when discussing ultimate "wider world" notions (Hyperseed-Concept 209). In analogy with the Y-combinator, the role of Multi is that of a functional F for which one seeks a fixed point satisfying $Y = F(Y)$. The combinatory-logic analogy is not meant to collapse set-theoretic and computational fixed points, but to highlight a shared structural theme: self-application is not an anomaly to be eliminated, but a generative mechanism that produces objects characterized by closure or invariance under a transformation.*

Remark 1439. *One may think of $\mathsf{lfp}(\mathsf{Multi})$ as the most conservative fixed point: the smallest self-consistent Yverse description compatible with the multi-operator. Conversely, the greatest fixed point can be read as the most expansive closure under "multi." The availability of both mirrors familiar least/greatest fixed point choices in semantics (induction vs. coinduction), which in turn mirrors Hyperseed's willingness to keep multiple, potentially tensioned, but useful perspectives in play. In semantic terms, least fixed points are typically associated with "generated" structures (what must be true if one starts from nothing and repeatedly closes under the operator), whereas greatest fixed points are associated with "allowed" structures (what can be true if one permits any consistent completion that remains closed under the operator). Under a "tower of worlds" interpretation, these correspond to two different attitudes: building up the multi-tower from a minimal base versus admitting any sufficiently rich background and requiring only that it be stable under further multiversing.*

Remark 1440. *When $\mathcal{D}$ has a least element $\perp$ and greatest element $\top$ (as every complete lattice does), one may form the ascending chain*

$$\perp \;\le\; \mathsf{Multi}(\perp) \;\le\; \mathsf{Multi}^2(\perp) \;\le\; \cdots \tag{17}$$

and the descending chain

$$\top \;\ge\; \mathsf{Multi}(\top) \;\ge\; \mathsf{Multi}^2(\top) \;\ge\; \cdots, \tag{18}$$

where Multi^n denotes n-fold iteration. In many familiar settings these chains stabilize at $\mathsf{lfp}(\mathsf{Multi})$ and $\mathsf{gfp}(\mathsf{Multi})$ respectively; in full generality, one can require transfinite iteration and take suprema/infima at limit ordinals. This makes precise the earlier informal idea of "iterating the multi construction and taking a limit," with the lattice operations providing the needed notion of limit.

Proposition 44 (Existence of Yverse fixed points)**.** *If Multi is monotone on a complete lattice $\mathcal{D}$, then fixed points exist and the set of fixed points forms a complete lattice.*

Remark 1441. *This proposition says that the Yverse is not a merely aspirational infinity: under the same minimal hypotheses used earlier (complete lattice + monotone operator), fixed points exist by purely mathematical necessity. In the narrative of the paper, this repeats the structural motif of Section 25.2: wherever Hyperseed defines something by a stabilization condition, it also provides the monotonicity needed to guarantee existence. This is a recurring theme in the broader Hyperseed program [1]. A further useful consequence (often taken as part of the Knaster–Tarski package) is that least and greatest fixed points admit order-theoretic characterizations:*

$$\mathsf{lfp}(\mathsf{Multi}) \;=\; \bigwedge\{d \in \mathcal{D} : \mathsf{Multi}(d) \le d\}, \qquad \mathsf{gfp}(\mathsf{Multi}) \;=\; \bigvee\{d \in \mathcal{D} : d \le \mathsf{Multi}(d)\}. \tag{19}$$

Here the meet ranges over pre-fixed points *and the join ranges over* post-fixed points*. This connects the fixed-point selection question to a semantics question: choosing the least fixed point amounts to intersecting all descriptions that are "closed downward" under multiversing, whereas choosing the greatest fixed point amounts to uniting all descriptions that can be extended in a way preserved by multiversing.*

Proof. This is again Knaster–Tarski. ☐

Remark 1442. *The proof deliberately does not add detail because it is identical in structure to Theorem 23: Knaster–Tarski applies to any monotone operator on a complete lattice. The visual intuition is again closure under iteration, except that the objects being iterated are now themselves "world towers." One may imagine repeatedly applying* Multi *and taking a limit in the lattice order; fixed points are precisely where the process stabilizes. It is also worth noting what is* not *assumed: no continuity, no contractiveness, and no metric or topological structure is required. The cost of this generality is that the fixed point need not be reached in finitely many steps or even in countably many steps by naive iteration; the benefit is that the existence of a stabilized "recursive limit" does not depend on a particular implementation of multiversing, only on its monotone behavior with respect to refinement/strength.*

Proof sketch. Invoke Knaster–Tarski on the complete lattice $\mathcal{D}$ and the monotone endomap Multi. Conclude existence of least and greatest fixed points and completeness of the fixed-point set. ☐

Remark 1443 (Why paraconsistent logic helps)**.** *If the "multi" tower diverges or leads to self-referential definitions, classical well-founded reasoning may fail. Paraconsistent logic and non-well-founded set-like constructions (used elsewhere in Hyperseed to model self-reference) provide principled semantics in which such recursive limits can be discussed without trivialization [23, 24]. Concretely, if one tries to force a strict stratification in which level* $n+1$ *must be built only from level* n*, then the equation* $Y = \mathsf{Multi}(Y)$ *appears illicit, because the right-hand side mentions* Y *"from above." Paraconsistent and non-well-founded approaches relax the demand that definitions be grounded in a well-founded hierarchy, allowing circularity while controlling explosion (the principle that from a contradiction everything follows). This aligns with the intended reading of the Yverse as a stable closure of the multiverse-building act itself, rather than as a final object obtained only after completing an impossible traversal of all stages.*

25.8 Eurycosm and near eurycosm

Hyperseed's *eurycosm* ("wider world") is the scope of distinctions, entities, and experiences beyond any single well-defined mind or physical universe. The intent is to capture, in one term, the idea that what is "out there" may not be exhausted by any one internally coherent physics, ontology, or observer-interface; different observers and different admissible channels of interaction can carve the wider world into different operational "worlds" that may partially overlap, partially disagree, or be mutually incommensurable. The *near eurycosm* relative to a given mind is the portion of the eurycosm that the mind can at least partially comprehend. In particular, "comprehend" here does not require full simulation or perfect prediction; it can mean any stable representational handle, such as a compressive model, a useful analogy, or a reliable decision-relevant summary. Universe, multiverse, and Yverse are then different ways of conceptualizing aspects of the eurycosm. One can think of these as different lenses: "universe" emphasizes a single consistent context, "multiverse" emphasizes families of related contexts, and "Yverse" emphasizes structures that remain meaningful across diverse contexts (e.g. cross-context invariants and correspondences).

Definition 430 (Eurycosm)**.** *Let* Mind *be a class of minds/observers and let* $\mathsf{SigClass}$ *be a class of signal classes. The* eurycosm *is the (typically large) union of observer-indexed universes across signal classes:*

$$\mathsf{Eury} := \bigcup_{M \in \mathsf{Mind}} \bigcup_{C \in \mathsf{SigClass}} \mathsf{Univ}_{C,\eta}(M). \tag{20}$$

Here $\mathsf{Univ}_{C,\eta}(M)$ *should be understood as the "universe" induced for observer* M *when* M *treats signals of class* C *as admissible and reliable up to the tolerance parameter* η *(e.g. a coarse-graining or error bar that determines which distinctions count as stable). This makes the union sensitive not only to* what *channels are allowed but also to* how finely *the mind is able or willing to discriminate structure within those channels. This definition is schematic; formally one may treat* Eury *as a large category of contexts with morphisms given by admissible signals and cross-context pattern correspondences. In that categorical picture, objects are contexts (observer-relative worlds), and morphisms represent structured ways of carrying information, models, or identifications from one context to another without insisting on literal identity of ontology.*

Remark 1444. *The formula for* Eury *should be read with a logician's caution: it is not claiming that all such unions form a small set. Rather, it expresses an* organizing principle*: the wider world is what you get when you range over minds and over the classes of signals they treat as real. The suggestion that* Eury *may be treated as a large category emphasizes that relations between contexts (translations, analogies, correspondences of patterns) matter at least as much as the contexts themselves. This aligns with the euryphysics framing [18]. Concretely, depending on ones foundational stance,* Eury *may be a proper class, or it may be internalized by working inside a Grothendieck universe or a type-theoretic hierarchy; the mathematical bookkeeping is secondary to the intended invariant: do not privilege one observer-context as* the *unique container of reality. The categorical emphasis also helps separate two kinds of questions: questions about the internal laws of a context (object-level structure) and questions about how patterns recognized in one context can be transported or matched to patterns in another (morphism-level structure).*

Remark 1445. *A helpful example is to consider* $\mathsf{SigClass}$ *ranging from ordinary physical channels to social-symbolic channels and to any hypothetical channels contemplated by a community. Then* Eury *contains, in a single schematic envelope, the diverse "universes" induced by those different signal regimes. The definition is necessary because Hyperseed wishes to speak about "beyond any one universe" while retaining operational content: the beyond is not a mysterious elsewhere, but a quantified range over observer-indexed reachability structures (Hyperseed-Concept* **??***). In practice,* $\mathsf{SigClass}$ *can also include formal channels such as mathematical proof systems, computational experiment, and mediated instrumentation; these are not "mere language" in this framing, but structured ways of making distinctions and establishing constraints that can stabilize parts of the eurycosm for a community of minds. On this view, "physical" and "social-symbolic" are not competing substances but different signal-mediated interfaces, each with its own notions of invariance, noise, and admissible intervention.*

Definition 431 (Near eurycosm)**.** *Fix a mind* M*. Let* $\mathsf{Comp}_M(X) \in [0, \infty]$ *be a complexity/effort cost for* M *to represent or comprehend* X *(e.g. a weakness-regularized description length). For a budget* B*, define the* near eurycosm *of* M *at budget* B *by*

$$\mathsf{NearEury}_B(M) := \{X \in \mathsf{Eury} : \mathsf{Comp}_M(X) \le B\}. \tag{21}$$

The parameter B *is intentionally abstract: it may stand for compute, time, memory, training data, attention, or any mixture of resources that constrain the minds modeling capacity. One* *natural consistency condition (not required but often convenient) is monotonicity: if $B_1 \leq B_2$ then $\mathsf{NearEury}_{B_1}(M) \subseteq \mathsf{NearEury}_{B_2}(M)$, expressing that additional resources weakly expand what the mind can stably grasp.*

Remark 1446. *The function Comp_M is where resource-boundedness enters explicitly. It may be defined via description length (algorithmic complexity [16]), via weakness in the sense of quantale-enriched representational limitation [3], or by any other effort-like cost compatible with the formal core. The budget B turns that cost into a boundary: what the mind can grasp without exceeding its representational and computational means. Depending on the application, Comp_M may include (i) the cost to* learn *a representation from data, (ii) the cost to* use *the representation for inference or action, and (iii) the cost to* communicate *the representation to other minds; these can diverge sharply even when the underlying object X is the same. It is also useful to allow $\mathsf{Comp}_M(X) = \infty$ for objects that are, for that mind, effectively unrepresentable (for example, because no stable interface exists within its available signal classes).*

Remark 1447. *Example: for a human mind M, parts of Eury corresponding to high-dimensional microscopic dynamics may be outside $\mathsf{NearEury}_B(M)$ for any realistic B, while coarse statistical laws fall inside. For a rich-resource mind (Hyperseed-Concept 162) with the ability to simulate vast portions of the cosmos (as discussed in [11]), the boundary may be dramatically farther out. The definition is useful because it makes "the wider world that is in principle there" distinct from "the portion of it that this mind can meaningfully engage." This distinction also clarifies why scientific progress can look like an outward motion of the near eurycosm: new instruments, mathematical languages, and training regimes can decrease $\mathsf{Comp}_M(X)$ for previously inaccessible X, thereby pulling additional structure into $\mathsf{NearEury}_B(M)$ without changing Eury itself. Conversely, the same underlying region of Eury may be "near" for one mind and "far" for another because Comp_M is observer-relative: what is a direct percept for one agent can be an exotic hypothesis for another.*

Remark 1448 (Operational meaning)**.** *The near eurycosm depends on the mind's representational resources. This makes "beyond our universe" precise in the same observer-relative style as the rest of Hyperseed: it means "outside the portion of the wider world that the mind can currently model with bounded effort." Operationally, a claim that some phenomenon lies outside $\mathsf{NearEury}_B(M)$ can be read as a prediction about failure modes: the mind will lack stable compressions, will be forced into brittle ad hoc models, or will be unable to maintain cross-context correspondences robustly under noise and limited data. Likewise, bridging work (new signal channels, new representational formalisms, or better translations between contexts) can be understood as constructing morphisms that effectively lower Comp_M for relevant targets, thereby making parts of the wider world newly actionable.*

25.9 Anthropic principle as self-locating evidence

Hyperseed treats the anthropic principle as a straightforward statistical point: one's own existence is data, and should be conditioned on when reasoning about the universe. In this sense, "anthropic" updates are not a special-purpose loophole in inference, but an ordinary application of Bayesian reasoning in the presence of indexical (self-locating) information: not only do we learn facts about the world, we also learn that we are located in a world-history compatible with our being here to observe it.

Definition 432 (Anthropic conditioning)**.** *Let Θ be a parameter space of universe models (physical constants, laws, initial conditions, etc.). Let $\pi(\theta)$ be a prior distribution over Θ. Let A be an* *"anthropic" event such as "observers of type $\mathcal{O}$ exist." The anthropic posterior is*

$$\pi(\theta \mid A) \propto \pi(\theta)\,\mathbb{P}(A \mid \theta). \tag{22}$$

Remark 1449. *The notation is standard Bayesian conditioning: $\pi(\theta)$ is a prior over models $\theta \in \Theta$, and $\mathbb{P}(A \mid \theta)$ is the probability that observers exist under model θ. The proportionality indicates normalization by the marginal $\mathbb{P}(A)$. The definition is useful because it demystifies anthropic arguments: they are, at their core, conditioning on self-location information.*

Remark 1450. *In many applications, $\mathbb{P}(A \mid \theta)$ is understood as an "observer-weight" or "selection" factor: models in which observers are common receive greater posterior support after conditioning on the fact that an observer is making the inference. More explicitly, the omitted normalization constant is*

$$\mathbb{P}(A) = \int_{\Theta} \pi(\theta)\,\mathbb{P}(A \mid \theta)\,d\theta, \tag{23}$$

so that the posterior is a reweighted version of the prior in which the weight is precisely the model-dependent chance (or frequency, or measure) of producing observers of the relevant kind.

Remark 1451. *The event A can be defined at different levels of granularity depending on the question. A coarse-grained choice is "there exists at least one observer somewhere in the universe," while a more fine-grained choice might be "there exists an observer with my evidence-state" (including, for example, being in a galaxy of roughly the Milky Way's age, observing a low-entropy past, and so on). Hyperseed's usage emphasizes that such choices are not arbitrary decorations: they encode what information is actually being conditioned on, and therefore what hypotheses are being compared.*

Remark 1452. *A simple example: if some models predict sterile universes with essentially zero probability of observers, then conditioning on A will suppress those models regardless of their prior weight. This aligns with Hyperseed's insistence that epistemology is observer-indexed: what is rational to believe depends on what kind of observer you are and that you are one at all (Hyperseed-Concept 57).*

Remark 1453. *It is also useful to distinguish "possibility" from "probability" in $\mathbb{P}(A \mid \theta)$. A model might permit observers in principle but only in an exceedingly small region of its parameter-realization space (or along an exceedingly narrow subset of histories), making $\mathbb{P}(A \mid \theta)$ tiny. In that case, anthropic conditioning treats the model as strongly disfavored, not because it is logically inconsistent with observers, but because observers are atypical under it. This captures the intuition that "compatible with our existence" is weaker than "makes our existence probable."*

Remark 1454 (Selection effects are not "explanations")**.** *Anthropic conditioning does not explain why a parameter choice is "likely" in some absolute sense. Rather, it formalizes a constraint on what an observer should expect to see given that the observer exists. This is fully compatible with Hyperseed's emphasis on observer-relativity.*

Remark 1455. *In particular, anthropic conditioning separates two questions that are often conflated: (i) what the universe is like (encoded by θ), and (ii) what an observer should expect to see given that they are sampling from the subset of world-histories containing observers. The latter is a statement about conditional expectation, not about a teleological "purpose" of the universe. On this reading, anthropic arguments do not compete with dynamical or microphysical explanations; they specify how observational evidence is filtered by the fact that only certain worlds contain observers capable of gathering that evidence.*

Remark 1456. *Finally, the anthropic event A is a minimal form of self-locating evidence, but it is not the only one. One may also condition on additional indexical data such as "I am an observer at a cosmic time when heavy elements are abundant" or "I find myself in a region with a particular coarse-grained environment." In Bayesian terms, these are further events E to be conditioned on, leading to $\pi(\theta \mid A, E) \propto \pi(\theta)\mathbb{P}(A, E \mid \theta)$, which makes explicit that anthropic reasoning is continuous with ordinary scientific updating: the only novelty is that some components of E describe the observer's location within the model rather than only the model itself.*

25.10 Big bounce as two-ended boundary coupling (speculative)

Hyperseed suggests a "big bounce" picture as an optional speculative anchor: if the large-scale cosmos is modeled by a least-contrivance (Wu Wei) principle between boundary conditions, then one may ask where those boundary conditions come from. A big-bounce hypothesis replaces a strict "beginning" and "end" with a coupled pair of boundaries, potentially allowing certain patterns to "pass through" from a previous cycle.

A further motivation for writing the hypothesis in a two-ended form is that, in many inferential and variational settings, specifying *both* endpoints is a compact way to encode constraints that would otherwise appear as time-dependent "forces" or finely tuned initial microstates. In cosmological language, this also echoes the idea that macroscopic regularities (including low-entropy constraints or special smoothness conditions) may be more naturally stated as boundary information than as ad hoc midstream interventions. In this register, the "bounce" is not only a geometric picture (contraction followed by expansion) but also an informational coupling between terminal and subsequent initial data.

Definition 433 (Two-ended cosmic boundary model)**.** *Let $\mathcal{X}$ be a space of coarse cosmic states (or state distributions), and let $T > 0$ be a cosmic timescale. A two-ended model specifies boundary data (μ_0, μ_T) and selects a history distribution $\{\mu_t\}_{t\in[0,T]}$ by minimizing a contrivance functional (e.g. a KL divergence relative to a reference dynamics), subject to these boundary constraints. This is the same mathematical idiom used for Schrödinger bridges and Wu Wei dynamics.*

Remark 1457. *The objects μ_0 and μ_T are boundary distributions on $\mathcal{X}$, and $\{\mu_t\}$ is a path (in distribution space) connecting them. The "contrivance functional" is meant to quantify how much the selected history deviates from a reference, "natural" dynamics; KL divergence is a canonical choice in Schrödinger bridge theory. Hyperseed uses this idiom to formalize Wu Wei as least forcing [21]; the present definition merely transposes the same mathematics to a cosmic scale (Hyperseed-Concept 66, 207).*

Remark 1458. *For readers who prefer a slightly more explicit bridge to standard formulations: one common setup places a reference path measure P_0 on trajectories (induced, for instance, by a Markov kernel or stochastic differential equation regarded as "baseline physics" at the chosen coarse-graining), and then selects a constrained path measure P whose time-marginals are $\{\mu_t\}$. The contrivance cost is then written as $\mathsf{KL}(P\|P_0)$, and the boundary conditions appear as marginal constraints $P(X_0 \in \cdot) = \mu_0$ and $P(X_T \in \cdot) = \mu_T$. In that language, the family $\{\mu_t\}$ is not merely an arbitrary interpolation but the set of marginals of the least-contrived path ensemble consistent with the endpoint information. This is one reason the two-ended specification is methodologically attractive: it turns endpoint information into a well-posed selection principle over entire histories.*

Remark 1459. *A simple intuition is to imagine that instead of specifying an initial condition and letting dynamics run, one specifies both an "initial-like" and "final-like" macroscopic condition and asks for the least contrived interpolation. This resembles variational principles in physics and* *control, and in Hyperseed's philosophical register it resonates with the idea that "history" is a selection among possible trajectories subject to constraints, rather than an inert given.*

Remark 1460. *At the level of interpretation, a two-ended formulation also helps separate three notions that are often conflated: (i) microscopic time-reversal symmetry (a property of candidate baseline dynamics), (ii) macroscopic arrow-of-time (a property of typical solutions under special boundary conditions), and (iii) epistemic updating (how an observer infers histories given partial data). In the present subsection, Hyperseed is not committing to a specific microphysical model of the bounce; it is instead emphasizing that, whatever the microphysics, a coupled-boundary representation can make explicit which part of the "arrow" is being attributed to boundary data rather than to asymmetric dynamical laws.*

Definition 434 (Bounce operator and cross-cycle morphic resonance)**.** *A* bounce operator *is a map B sending the terminal boundary to the next-cycle initial boundary:*

$$\mu_0^{\text{next}} = B(\mu_T). \tag{24}$$

A simple way to express "two-sided morphic resonance" across the bounce is to let B depend on pattern content:

$$B(\mu_T) := \arg\min_{\nu} \Big(\mathsf{KL}(\nu \,\|\, \mu_\star) + \lambda\, \mathsf{Res}(\nu, \mu_T) \Big), \tag{25}$$

where $\mu_\star$ is a reference prior for initial conditions and Res *is a (bounded) resonance functional favoring re-instantiation of patterns present in μ_T.*

Remark 1461. *In the displayed formula, $\arg\min_\nu(\cdots)$ selects the distribution ν (candidate next initial condition) that trades off proximity to a baseline prior $\mu_\star$ (via KL divergence) against "resonant" similarity to the previous terminal condition μ_T (via* Res*). The scalar λ tunes the tradeoff. This is intended as a formal placeholder showing that, if one wants to talk about cross-cycle pattern carryover, one can do so in the same optimization and resonance language used elsewhere in Hyperseed. The mention of "morphic resonance" connects to speculative proposals in the broader literature [13] as well as Hyperseed's own resonance formalism [5].*

Remark 1462. *The requirement that* Res *be bounded is not merely cosmetic: it is one simple way to avoid trivializing the optimization by letting resonance dominate all other terms. More generally, one can view* Res *as encoding a* pattern statistic *extracted from μ_T and compared to a candidate ν. For instance, if patterns are represented by a feature map $\phi : \mathcal{X} \to \mathbb{R}^d$ (coarse observables, topological signatures, long-range mode amplitudes, algorithmic motifs, etc.), then a resonance penalty/reward could depend on moment matching such as $\|\mathbb{E}_\nu[\phi] - \mathbb{E}_{\mu_T}[\phi]\|^2$, or on a kernel similarity between distributions. Writing B in this way makes explicit where the hypothesis lives: in the choice of pattern representation and in the form/weight of the coupling.*

Remark 1463. *One should also distinguish the "bounce operator" B from the within-cycle reference dynamics used in the two-ended model. In the present idiom, B is not a time-step of ordinary evolution on $\mathcal{X}$; it is an inter-cycle* boundary map *that may include coarse-graining, information loss, or projection onto a family of admissible next-cycle initial macrostates. Put differently, even if the within-cycle history is selected by least-contrivance relative to a baseline dynamics, the cross-cycle linkage can still be nontrivial because it acts on boundary distributions rather than on instantaneous microstates.*

Remark 1464. *One should notice the methodological point: even when Hyperseed permits speculative anchors, it seeks to express them as operators and functionals that could, in principle,* *be constrained by evidence. This is the Dyson/Sagan virtue in a metaphysical setting: keep the conjecture legible to calculation, so that disagreement can become quantitative rather than merely rhetorical.*

Remark 1465. *In the same methodological spirit, the bounce-operator parametrization suggests at least three distinct kinds of constraints one might imagine (even if only in principle): (i) constraints on the "memory strength" λ (how much cross-cycle similarity is permitted without contradicting observed decoherence or cosmological mixing), (ii) constraints on which observables can plausibly remain informative across a bounce (encoded by the choice of ϕ or of the admissible class of* Res*), and (iii) consistency constraints between within-cycle least-contrivance histories and cross-cycle boundary maps (e.g. fixed-point or cycle-consistency relations such as $\mu_0 \approx B(\mu_T)$ for a stationary cycling regime). These are not claims that such data are currently available; they clarify what, in this formalization, would count as sharpening or refuting the hypothesis.*

Remark 1466 (Status)**.** *This "big bounce" construction is included here as a formal placeholder: it shows how to talk about cross-cycle pattern carryover using the same operators already developed for resonance and least-effort path selection. Hyperseed explicitly notes that quantitative aspects of such a phenomenon are unclear.*

Taken together, these constructions place Hyperseed's cosmological vocabulary inside the same mathematical toolbox as its cognitive vocabulary. The next section (Wu Wei dynamics) uses the variational viewpoint more directly, treating action and history selection as geodesic flow in a weakness-shaped geometry [21].

26 Wu Wei dynamics and optimal control

Hyperseed uses the Taoist term *wu wei* as an ontological-computational slogan: *effective action with minimal forcing.* The informal imagery is "like water" or "like a plant bending in the wind": adaptive, responsive, and skillful, without unnecessary struggle. Earlier sections of this paper formalize (i) representational limits via weakness and (ii) conflicting valuation via paraconsistent evidence and resonance. This section adds a third piece: *control as variational flow in a geometry induced by weakness.*

Remark 1467. *The philosophical gesture behind wu wei in Hyperseed is that* agency need not be identified with strain. *One may act, and act effectively, while remaining close to a "passive" unfolding of events—provided that the passive unfolding already embodies a rich store of learned regularities and habits. In this sense, wu wei is an engineering principle as much as a contemplative one: it says that a good controller should look, mathematically, like a* gentle reweighting *of what would have happened anyway, rather than a continual override of the world and the agent's own default tendencies; compare [21].*

Remark 1468. *Conceptually, this section is primarily about Hyperseed-Concept 206 and Hyperseed-Concept 207, and it also relies on Hyperseed-Concept* **??***, Hyperseed-Concept 100, Hyperseed-Concept 158, and Hyperseed-Concept 202. The formal development below is meant to show how these concepts can be grounded in standard (but remarkably elegant) variational principles, while keeping faith with Hyperseed's emphasis that "minimal forcing" is inseparable from "minimal representational effort" [3, 2].*

The key move is to treat "effort" as having (at least) two components:

- **Effort to move**: changing internal (or external) state has a kinetic/transport cost.
- **Effort to resist**: maintaining a trajectory that diverges from a reference ("natural") dynamics has an informational/forcing cost.

Wu wei then means: among all ways of achieving the intended result, follow the path that minimizes these costs, subject to constraints.

Hyperseed concepts covered here. Wu wei; wu wei dynamics; fluid flow and optimal control; minimum representational effort as the bias shaping the flow.

To make the phrase "geometry induced by weakness" concrete, it is useful to imagine that the agent's state (or belief-state) lives on a manifold (or simply a high-dimensional space) $x \in \mathcal{X}$, and that "weakness" supplies a *local notion of difficulty* for representing, predicting, or enacting changes around x. Mathematically, this can be encoded as a position-dependent metric (or, more generally, a convex local norm) $g_x(\cdot, \cdot)$ that determines which directions are "easy" and which are "hard" given the agent's limited descriptive resources. In such a setting, the effort-to-move term naturally takes the form of a kinetic energy measured in this metric,

$$\mathcal{E}_{\text{move}}[x(\cdot)] \;\equiv\; \int_0^T \frac{1}{2} \langle \dot{x}(t), \dot{x}(t) \rangle_{g_{x(t)}} \, dt,$$

so that a geodesic under g is precisely a path that changes state in the least effortful way *relative to the agent's representational constraints*. This is the sense in which wu wei is not "doing nothing," but rather "moving along directions that are already cheap," i.e., moving in the grain of what the system can express without strain.

The effort-to-resist term can be framed in parallel by introducing a reference dynamics (the "passive unfolding") and measuring how strongly a controlled trajectory must depart from it. For example, one may posit a baseline dynamics $\dot{x} = f(x)$ (or a stochastic baseline, if noise is essential to the model), and represent deliberate intervention via an additive control u,

$$\dot{x}(t) \;=\; f(x(t)) + B(x(t))\, u(t).$$

In this case, "minimal forcing" means not merely that u is small in magnitude, but that it is small *in the right units*: the units are again dictated by weakness, via a cost or norm that penalizes interventions that are informationally expensive for the agent to specify or sustain. A canonical form is

$$\mathcal{E}_{\text{resist}}[u(\cdot), x(\cdot)] \;\equiv\; \int_0^T \frac{1}{2} \langle u(t), u(t) \rangle_{R_{x(t)}} \, dt,$$

where R_x can be interpreted as an inverse "control authority" or, equivalently, as a local resistance tensor (so large R_x means forcing is costly). In probabilistic terms, and especially when one works with stochastic reference dynamics, this same idea can be expressed as an informational divergence between the controlled and uncontrolled *path measures*; then "gentle reweighting" can be taken literally as changing the likelihood of trajectories without inventing fundamentally alien ones, which aligns with the remark above that effective agency can appear as a modest tilt of what would have happened anyway.

Finally, the phrase "subject to constraints" should be read broadly: constraints may encode reaching a target set, matching terminal conditions, satisfying safety bounds, or (in the valuation language of earlier sections) tracking a potential-like objective derived from resonance-weighted evidence. In the variational picture, these constraints enter as boundary conditions or as Lagrange-multiplier terms that shape the Euler–Lagrange equations (or, in control language, the Hamilton–Jacobi–Bellman conditions). The resulting wu wei trajectory is thus the one that (i) respects what the agent *can* represent and execute with low complexity (weakness), (ii) remains close to the reference unfolding unless deviation is genuinely needed (minimal forcing), and (iii) still attains the intended outcome via the least costly reconfiguration of the flow.

26.1 Wu wei as minimal-forcing control relative to a passive dynamics

We start with a discrete-time, finite-state formulation that makes the "minimal forcing" idea explicit. In this setting, "control" does not mean selecting an arbitrary action in a separate action set; instead, the agent's decision is directly a choice of *next-state distribution* $P(\cdot \mid x)$, and the central question becomes how much this chosen distribution departs from a baseline $P_0(\cdot \mid x)$.

Definition 435 (Passive and controlled dynamics)**.** *Let X be a finite state space. A* passive *(or* unforced*) dynamics is a Markov kernel*

$$P_0(\cdot \mid x) \in \Delta(X) \qquad (x \in X),$$

where $\Delta(X)$ is the probability simplex on X. A controlled *dynamics is a kernel $P(\cdot \mid x) \in \Delta(X)$ that the agent may choose. A (stationary)* policy *is a map π that assigns to each state x a controlled kernel $P_\pi(\cdot \mid x)$.*

Remark 1469. *Intuitively, $P_0(\cdot \mid x)$ is "what the world (and agent) tend to do" when currently at x, before we impose any task-specific steering. Saying that $P_0(\cdot \mid x) \in \Delta(X)$ simply means it is a probability distribution over next states: for each fixed x, the numbers $P_0(x' \mid x)$ are nonnegative and sum to 1 as x' ranges over X. A controlled kernel $P(\cdot \mid x)$ is another such distribution, representing the agent's choice of how to bias the next step from x.*

Remark 1470. *It is worth emphasizing what is* not *assumed here: we do not posit an explicit action space A with transition law $P(\cdot \mid x, a)$. One may recover that standard viewpoint by taking $P(\cdot \mid x)$ to be the marginal transition kernel induced by choosing an action (or a randomized action) at x. However, the present formulation is more direct for wu wei because it foregrounds the comparison between an "effortless" flow P_0 and any "steered" flow P.*

Remark 1471. *A simple example is a random walk on a small graph whose nodes are X. The passive kernel $P_0(x' \mid x)$ could put equal probability on each neighbor of x (a "default exploration"). A controlled kernel $P(\cdot \mid x)$ might, at some states, assign higher probability to neighbors leading toward a goal region. The definition is useful because it isolates the central wu wei contrast: instead of choosing arbitrary dynamics from scratch, we measure* how far the controlled choice departs from the passive baseline*.*

Remark 1472. *In graph terms, P_0 encodes the geometry and friction of the environment: which moves are "easy" (high passive probability) and which are "unnatural" (low or zero passive probability). This becomes important later because the KL penalty makes it extremely expensive (indeed impossible at finite cost) to assign positive probability to transitions that are impossible under the passive flow, so the passive dynamics functions as a feasibility scaffold as well as an effort prior.*

Definition 436 (Forcing cost as divergence from the passive flow)**.** *Fix a scale parameter $\lambda > 0$. The* forcing cost *at state x for using a controlled kernel $P(\cdot \mid x)$ is*

$$\mathsf{Force}_\lambda(P \,\|\, P_0)(x) := \lambda\, \mathsf{KL}\big(P(\cdot \mid x) \,\|\, P_0(\cdot \mid x)\big). \tag{26}$$

We interpret this as resistance/effort*: it is zero when we "go with the flow" ($P = P_0$) and grows as we push the dynamics away from its natural tendencies.*

Remark 1473. *Here* $\mathsf{KL}(\cdot\|\cdot)$ *denotes the Kullback–Leibler divergence: for two distributions* $P, Q \in \Delta(X)$ *with* P *absolutely continuous with respect to* Q*,*

$$\mathsf{KL}(P\|Q) = \sum_{x' \in X} P(x') \log \frac{P(x')}{Q(x')}.$$

It is always ≥ 0 *and is* 0 *exactly when* $P = Q$*. The parameter* λ *is a unit/scale parameter that determines how expensive it is to deviate from the passive flow: large* λ *means "forcing is costly," so wu wei strongly prefers near-passive behavior; small* λ *makes deviation cheap. This KL-based penalty is a standard information-geometric way to quantify effort-to-resist [16, 21].*

Remark 1474. *Two additional facts about* $\mathsf{KL}(P\|Q)$ *matter for the wu wei reading. First, it is* asymmetric*: steering away from* P_0 *is penalized according to how surprising the controlled flow would look to an observer who expects* P_0*. Second, the absolute continuity condition (*P *must not put mass outside the support of* P_0*) formalizes a kind of "non-violent" constraint: one cannot, at finite forcing cost, demand transitions that the passive dynamics regards as impossible. In many physical settings this corresponds to respecting hard constraints (walls, forbidden moves), while still allowing soft steering among feasible moves.*

Remark 1475. *As a concrete example, suppose from a state* x *the passive kernel assigns* $P_0(\cdot \mid x) = (1/2, 1/2)$ *on two successors. If the controller changes this to* $(3/4, 1/4)$*, the forcing cost is positive but moderate; if it changes to* $(1, 0)$*, the forcing cost becomes larger (and in fact infinite if* P_0 *assigns zero mass where* P *assigns positive mass). This captures the wu wei intuition that "hard commitment" is a kind of strain when it contradicts what the passive dynamics naturally supports.*

Remark 1476. *The dependence on* λ *can also be read in this example: the same change from* $(1/2, 1/2)$ *to* $(3/4, 1/4)$ *is interpreted as "mild effort" when* λ *is small and as "significant effort" when* λ *is large. Thus* λ *mediates a tradeoff between task achievement and non-interference: it acts like a temperature parameter, where high temperature (small* λ*) permits sharper tilting of probabilities, while low temperature (large* λ*) keeps the controlled kernel close to the passive one.*

Definition 437 (Task cost and total objective)**.** *Let* $q : X \to \mathsf{R}$ *be a bounded* state cost *(negative reward). Given initial state distribution* $\rho_0 \in \Delta(X)$ *and horizon* $T \in \mathbb{N}$*, define*

$$J_T(\pi) := E\Big[\sum_{t=0}^{T-1} \big(q(x_t) + \mathsf{Force}_\lambda(P_\pi \parallel P_0)(x_t)\big) + q_T(x_T)\Big], \tag{27}$$

where $x_0 \sim \rho_0$*,* $x_{t+1} \sim P_\pi(\cdot \mid x_t)$*, and* q_T *is a terminal cost. A* wu wei policy *for this problem is any minimizer*

$$\pi^* \in \arg\min_{\pi} J_T(\pi).$$

Remark 1477. *The assumption that* q *is bounded ensures, in particular, that the task part of the objective does not by itself create divergences in the finite-horizon expectation. Any remaining possibility of infinite cost typically comes from the forcing term (via support mismatch), which is conceptually appropriate: if one tries to "force" transitions not permitted by* P_0*, the penalty becomes prohibitive. The terminal cost* q_T *is included to model goals that are evaluated at the end of the horizon (e.g., "be in a target set at time* T*"), and it also supports dynamic-programming formulations where boundary conditions are naturally expressed at* $t = T$*.*

Remark 1478. *The objective $J_T(\pi)$ combines two desiderata in a single expectation $E[\cdot]$ over trajectories: (i) we want to avoid high-cost states (small q if thinking in rewards, or small negative reward if thinking in costs), and (ii) we want to avoid "fighting" the passive dynamics, quantified by Force_λ. The random variables $x_0, x_1, \ldots, x_T$ are generated by first drawing x_0 from the initial distribution ρ_0 and then propagating forward according to the controlled kernel selected by π at each visited state.*

Remark 1479. *Although the policy is defined as stationary (state-dependent but not time-dependent), the finite-horizon criterion still makes sense: one may either restrict attention to stationary policies for conceptual simplicity, or later allow nonstationary π_t when emphasizing optimality for each remaining time-to-go. In linearly-solvable and KL-regularized control, the stationary restriction is often adopted because the optimal kernels can be expressed as statewise tilts of $P_0(\cdot \mid x)$ by functions of successor-state "desirability," and these tilts have a consistent form across time in the time-homogeneous case.*

Remark 1480. *A simple example is navigation with stochastic drift: $q(x)$ is large in dangerous regions and small near the target; P_0 models wind/current drift; P_π models the effect of steering. A wu wei policy is then one that reaches the target while (in expectation) expending minimal forcing against the drift. This definition is necessary for the later claims about "exponential tilting" and "linear solvability": the KL-regularized form is exactly what makes the optimization admit closed-form local solutions and linear recursions.*

Remark 1481. *It is also useful to notice that the forcing cost is* state-local *while the task cost is* trajectory-coupled *through the state evolution. This is precisely the structure that allows one to write Bellman-style recursions: the immediate cost at x_t is $q(x_t) + \lambda\, \mathsf{KL}(P_\pi(\cdot \mid x_t) \| P_0(\cdot \mid x_t))$, and the future cost depends on the chosen next-state distribution. When one minimizes over $P(\cdot \mid x)$, the KL term acts as a convex regularizer, leading (under standard assumptions) to a unique minimizing controlled kernel at each state given an appropriate value-to-go function.*

Remark 1482 (Where weakness enters)**.** *The forcing term is an information-geometric penalty. To connect it to the paper's weakness theme, interpret P_0 as the dynamics induced by a* weak model *(a maximally general representation that makes few distinctions), while P_π is the refined, task-specific dynamics requiring extra distinctions. Minimizing $\mathsf{KL}(P_\pi \| P_0)$ is then a precise form of* minimum representational effort*: do not add distinctions unless they buy real task value.*

Remark 1483. *This connection can be made even more explicit by reading a controlled kernel $P_\pi(\cdot \mid x)$ as an "annotated" version of the passive kernel: it reallocates probability mass among successor states, thereby encoding which distinctions among successors are behaviorally relevant. If the weak model P_0 already concentrates on a small set of plausible next states, then steering within that set can be cheap, while insisting on a qualitatively different move (outside the passive support) corresponds to introducing a distinction the weak model does not even represent. In this sense, wu wei control implements a disciplined refinement of the weak baseline: it intervenes only to the extent that intervention is justified by the task cost.*

Remark 1484. *This remark is pointing at Hyperseed-Concept 143 and Hyperseed-Concept 202. In the weakness-centered view [3, 2], a "strong" model draws sharper boundaries and therefore supports more specialized, more brittle interventions. The wu wei objective says: sharpen boundaries only where the task truly compels it, and otherwise remain close to the weak/default dynamics. Thus what looks like an external control regularizer can also be read as an internal epistemic ethic: avoid gratuitous distinctions. One can also read this as a warning about* self-induced overfitting of agency*: if the controller commits to distinctions not demanded by the objective, it pays an avoidable "maintenance cost" in the form of higher sensitivity to mismatch between model and world. In that sense, the KL penalty is not merely a numerical convenience but a formal way to encode the preference that intervention should be* locally justified *by task-relevant gradients rather than by the availability of controllable degrees of freedom.*

26.2 A one-step wu wei principle: exponential tilting

The forcing regularizer has a canonical optimization consequence: optimal control is achieved by *tilting* the passive dynamics by an exponential of negative cost. Equivalently, the controller chooses the least-informative (closest-to-passive) distribution that still achieves an improved expected cost, where "least informative" is measured in the KL geometry induced by the passive kernel. This is the same variational template that appears in maximum-entropy inference: an energy term (expected cost) plus an entropic or divergence-based regularizer yields a Gibbs/Boltzmann form.

Proposition 45 (Gibbs form for minimal forcing)**.** *Fix a state $x \in X$, a passive distribution $P_0(\cdot \mid x)$ with full support, and an immediate cost function $c : X \to \mathsf{R}$. Consider the one-step optimization problem*

$$\min_{P(\cdot|x)\in\Delta(X)} \Big\{ E_{x'\sim P(\cdot|x)}[c(x')] + \lambda\,\mathsf{KL}\big(P(\cdot \mid x)\|P_0(\cdot \mid x)\big) \Big\}. \tag{28}$$

Then the unique minimizer is

$$P^*(x' \mid x) = \frac{P_0(x' \mid x)\,\exp\big(-c(x')/\lambda\big)}{Z(x)} \qquad \text{with} \qquad Z(x) = \sum_{y\in X} P_0(y \mid x)\exp\big(-c(y)/\lambda\big). \tag{29}$$

Remark 1485. *In plain terms, the proposition says: if you pay a cost both for "going to expensive next states" and for "departing from the default distribution," then the optimal way to choose your next-state distribution is to take the default $P_0(\cdot \mid x)$ and reweight it by an exponential preference for low cost. The normalization constant $Z(x)$ (often called a* partition function*) is just what makes the reweighted numbers sum to 1 again. It is useful to notice that the exponential weight is* relative*: what matters is not the absolute scale of c but its scale compared to λ. For example, adding a constant to $c(\cdot)$ does not change the minimizer, because it factors out of the numerator and is cancelled by the same factor in $Z(x)$; only cost* differences *across next states affect the tilt.*

Remark 1486. *A further quantitative reading is that $-\lambda \log Z(x)$ is a "soft minimum" of c under the passive distribution. Indeed, $Z(x)$ is a log-sum-exp moment of $-c/\lambda$ under $P_0(\cdot \mid x)$, so $\log Z(x)$ plays the role of a cumulant-generating function and interpolates between averaging and minimizing as λ varies. This is one of the reasons λ is often called a* temperature*: high temperature smooths preferences and low temperature sharpens them. In control terms, this makes explicit how the wu wei regularizer controls the curvature (and therefore brittleness) of the effective objective landscape over actions/policies.*

Remark 1487. *This is important because it is one of the rare places in control theory where a constrained optimization over the whole simplex yields an explicit, closed-form answer. It connects directly to later results in this section: the multi-step "linear desirability recursion" is essentially this one-step tilt applied repeatedly inside dynamic programming. It also connects to the broader Hyperseed theme that "soft" (quantitative, graded) commitments are often the right formalization of non-forcing behavior. A second reason it matters is methodological: having an explicit optimizer* *makes it possible to reason about limits ($\lambda \to 0$ and $\lambda \to \infty$), sensitivity (how P^* changes as c changes), and compositionality (how local tilts combine across time) without appealing to generic existence theorems or black-box numerical solvers.*

Remark 1488. *One can also explicitly write the optimal objective value by substituting P^* back into the functional. A standard calculation yields*

$$\min_{P(\cdot|x)\in\Delta(X)} \Big\{ E_P[c] + \lambda\, \mathsf{KL}(P\|P_0) \Big\} \;=\; -\lambda \log Z(x),$$

so the partition function is not merely a normalization device: it is the quantity that summarizes the best achievable cost–forcing tradeoff at state x. This identity is the one-step version of the "free energy" variational principle, and it is the algebraic reason that log-partition functions recur when one passes from one-step control to multi-step value functions.

Proof. The objective is strictly convex in P because $\mathsf{KL}(\cdot\|P_0)$ is strictly convex on the simplex when P_0 has full support. Introduce a Lagrange multiplier α for the constraint $\sum_{x'} P(x' \mid x) = 1$. Differentiating the Lagrangian with respect to $P(x' \mid x)$ and setting to zero yields

$$c(x') + \lambda\big(\log(P(x' \mid x)/P_0(x' \mid x)) + 1\big) + \alpha = 0.$$

Solving gives $P(x' \mid x) \propto P_0(x' \mid x)\exp(-c(x')/\lambda)$, with the proportionality constant fixed by normalization. Strict convexity implies uniqueness. Because $P_0(\cdot \mid x)$ has full support, the stationarity condition is valid at an interior point of the simplex and the optimizer does not need to place exact zeros in order to be feasible; instead, it can trade off small probabilities continuously against the KL penalty. This interiority is what makes the Euler–Lagrange (first-order) condition sufficient here, rather than merely necessary. □

Remark 1489. *A useful way to read the proof is: KL divergence is the "right" strictly convex barrier on the simplex that makes the calculus work cleanly. The Lagrange multiplier step encodes the single essential constraint (probabilities sum to 1), and the derivative calculation produces a logarithm, which is why the solution comes out exponential. The full-support assumption on P_0 ensures the logarithm is well-defined and enforces uniqueness. Another way to say the same thing is that $\mathsf{KL}(P\|P_0)$ supplies a Legendre-type duality between log-partition functions and moment constraints: exponential families are the natural coordinates in which the tradeoff between expected cost and informational deviation becomes linear. This is why the same mathematical move appears in both "soft" control and in maximum-entropy inference: the optimizer lives in an exponential family anchored at the reference measure $P_0(\cdot \mid x)$.*

Proof sketch. The strategy is to minimize a strictly convex functional on a compact convex set (the simplex), so there is a unique minimizer characterized by first-order optimality. One writes the Lagrangian with a normalization constraint and solves the resulting stationarity condition, which yields an exponential reweighting of the passive distribution. In particular, the stationarity condition forces $\log P$ to differ from $\log P_0$ by an affine function of c, and exponentiating converts that affine relation into multiplicative reweighting. □

Remark 1490. *Geometrically (in information geometry), this minimizer can be viewed as the closest distribution to P_0 (in KL sense) among those that achieve the desired reduction in expected cost. The "tilt" is not an arbitrary trick: it is the canonical way to trade off expectation constraints against divergence penalties, and it recurs throughout variational formulations of inference and control [21]. More precisely, the level sets of expected cost are affine slices of the simplex, while* *the KL divergence supplies a strictly convex "distance-like" function; the optimizer is therefore the unique point where a KL ball around P_0 first becomes tangent to a cost level set. This tangency picture also makes the role of λ intuitive: increasing λ inflates the effective penalty on moving away from P_0, so the tangency occurs closer to P_0.*

Remark 1491 (Interpretation)**.** *The passive kernel supplies a* default motion*. The exponential tilt supplies a* soft bias *toward lower-cost next states. As $\lambda \to \infty$ the policy approaches the passive dynamics (high acceptance / low forcing). As $\lambda \to 0$ the policy concentrates on argmin states (high forcing, potentially brittle). Wu wei is not "do nothing" but rather "bias gently unless strongly necessary." A further operational reading is that λ governs how much evidence (in the form of expected-cost improvement) is required to justify deviating from the default. Large λ demands strong, persistent cost advantages before it will significantly reallocate probability mass, while small λ makes the controller "decisive" in the sense of rapidly collapsing onto whichever next state currently looks best. In applications, this means that wu wei regularization can be used to tune not only average performance but also* mode of failure*: softer tilts tend to fail gracefully (by remaining near baseline behavior under uncertainty), whereas harder tilts can fail catastrophically if the cost model is misspecified.*

26.3 Multi-step wu wei control and linear solvability

The same structure persists over many steps and yields a particularly clean dynamic programming recursion.

Definition 438 (Finite-horizon value function)**.** *Define the optimal cost-to-go*

$$V_t(x) := \inf_{\pi} \; E\Big[\sum_{s=t}^{T-1} \big(q(x_s) + \mathsf{Force}_\lambda(P_\pi \parallel P_0)(x_s)\big) + q_T(x_T) \;\Big|\; x_t = x\Big]. \tag{30}$$

Set $V_T(x) := q_T(x)$.

Remark 1492. *The value function $V_t(x)$ is the standard dynamic programming object: it compresses all future consequences of optimal behavior from time t onward, given that we are currently at state x. The conditioning "$\big|\; x_t = x$" means we evaluate the expected future cost assuming the current state is fixed to x. This definition is useful because it allows the global trajectory optimization problem to be solved recursively (Bellman's principle), and it is precisely this recursion that becomes linear after a change of variables.*

Remark 1493. *As an example, if $T = 1$ then $V_0(x)$ is exactly the one-step objective minimized in Proposition 45 (up to the addition of the immediate $q(x)$ term and terminal cost). For larger horizons, V_t stitches together many such one-step optimizations, with the crucial twist that the "immediate cost" in the tilt becomes the* next-step value V_{t+1}.

Theorem 24 (Linear recursion in desirability variables)**.** *Assume $P_0(\cdot \mid x)$ has full support for all x. Define the* desirability *variables*

$$z_t(x) := \exp\big(-V_t(x)/\lambda\big) \in (0, \infty). \tag{31}$$

Then the Bellman optimality equations are equivalent to the linear *recursion*

$$z_t(x) = \exp\big(-q(x)/\lambda\big) \sum_{x' \in X} P_0(x' \mid x)\, z_{t+1}(x'). \tag{32}$$

Moreover, the optimal controlled kernel at time t is

$$P_t^*(x' \mid x) = \frac{P_0(x' \mid x)\, z_{t+1}(x')}{\sum_{y \in X} P_0(y \mid x)\, z_{t+1}(y)}. \tag{33}$$

Remark 1494. *The theorem says that, for this particular KL-regularized control problem, dynamic programming becomes almost suspiciously simple: if one exponentiates (with scale λ) the negative value function, the nonlinear Bellman minimization turns into a linear update of the transformed quantity z_t. In other words, the "difficulty" of optimal control has been moved into a change of variables, after which the recursion resembles a passive expectation under P_0.*

Remark 1495. *This result matters because linear recursions are algorithmically and conceptually easier than nonlinear ones: they admit superposition, can be estimated by sampling, and connect directly to spectral methods. Within this section, it is the multi-step counterpart of Proposition 45; within the larger Hyperseed development, it supports the claim that "minimal forcing" yields a kind of* computational grace*: one does not merely suffer less, one often computes more cleanly. This "linear solvability" viewpoint is also part of the motivation for connecting wu wei to "geodesics" and "fluid flow" later [21].*

Proof. Start from the Bellman equation

$$V_t(x) = q(x) + \min_{P(\cdot|x)} \Big\{ \lambda\, \mathsf{KL}(P \| P_0)(x) + E_{x' \sim P}[V_{t+1}(x')] \Big\}$$

Apply Proposition 45 with $c(x') := V_{t+1}(x')$. The minimizing kernel satisfies $P_t^*(x' \mid x) \propto P_0(x' \mid x) \exp(-V_{t+1}(x')/\lambda) = P_0(x' \mid x)\, z_{t+1}(x')$. The minimal value of the inner optimization is

$$-\lambda \log\Big(\sum_{x'} P_0(x' \mid x) \exp(-V_{t+1}(x')/\lambda)\Big) = -\lambda \log\Big(\sum_{x'} P_0(x' \mid x)\, z_{t+1}(x')\Big).$$

Thus

$$V_t(x) = q(x) - \lambda \log\Big(\sum_{x'} P_0(x' \mid x)\, z_{t+1}(x')\Big).$$

Exponentiating $-V_t/\lambda$ gives the stated linear recursion. □

Remark 1496. *The proof is essentially "plug Proposition 45 into Bellman recursion." The key move is recognizing that the Bellman minimization over $P(\cdot \mid x)$ has exactly the same KL-regularized form as the one-step problem, with V_{t+1} playing the role of the one-step cost. Once this is seen, everything becomes algebra: the minimizer is an exponential tilt, the minimized value is a log-partition term, and exponentiating cancels the logarithm and produces linearity.*

Proof sketch. One first writes Bellman's equation for V_t and notices that the inner minimization is the one-step KL-regularized optimization from Proposition 45. The minimizer is therefore a Gibbs tilt by $\exp(-V_{t+1}/\lambda)$, and the minimized objective becomes a negative log of a passive expectation. Finally, defining $z_t = \exp(-V_t/\lambda)$ converts the log-sum-exp into a linear sum. □

Remark 1497. *A visual intuition is to imagine a "field" over states whose height is $V_t(x)$; exponentiating turns steep cliffs into near-zero desirabilities. Then the update for z_t says: propagate desirability backward by averaging future desirability under the passive dynamics, while discounting by immediate state cost. This is the sense in which wu wei control can look like "letting passive trajectories vote," rather than issuing a single deterministic command.*

Remark 1498 (Wu wei as "path integral control")**.** *The recursion for z_t expresses desirability as a passive expectation of exponentiated negative cost. This makes explicit a wu wei reading: instead of forcing a single hard plan, we compute a soft "desirability field" and let the passive dynamics flow through it. Algorithmically, this often yields efficient estimation by Monte Carlo sampling.*

26.4 Continuous-time limit: fluid flow, optimal transport, and Schrodinger bridges

Hyperseed also links wu wei to the intuition that *fluid flow behaves like holistic decision*: not as independent particles choosing, but as a coordinated transformation of the entire state. A standard mathematical way to express such "holistic transformation" is to view the state as a probability density (or mass density) $\rho(\cdot, t)$ and the action as a velocity field $v(\cdot, t)$.

Remark 1499. *This subsection is best read as a continuous analogue of the discrete KL-control story. Instead of individual sample paths $x_0, x_1, \dots,$ we work with an evolving distribution of mass $\rho(\cdot, t)$ on a domain $\Omega \subseteq \mathrm{R}^d$. The control variable becomes a velocity field $v(\cdot, t)$, which "pushes" the density around. The continuity equation $\partial_t \rho + \nabla \cdot (\rho v) = 0$ is simply the formal statement that mass is conserved as it flows; it is the PDE counterpart of probability normalization in discrete time. This is a central instance of Hyperseed-Concept* **??** *[21].*

Remark 1500. *At an intuitive level, the pair (ρ, v) is the continuum analogue of a controlled Markov evolution: $\rho(\cdot, t)$ plays the role of the time-marginal distribution, while $v(\cdot, t)$ summarizes the "instantaneous drift" that re-shapes the entire ensemble. This is one precise sense in which the decision is "holistic": rather than selecting separate actions for separate particles, one selects a field that coherently transports mass everywhere in Ω.*

Definition 439 (Transport form of representational effort)**.** *Let $\Omega \subseteq \mathrm{R}^d$ be a domain and let $t \in [t_0, t_1]$. Let $\rho(x, t)$ be a density on Ω and let $v(x, t)$ be a velocity field. Assume the continuity equation*

$$\partial_t \rho + \nabla \cdot (\rho v) = 0 \tag{34}$$

in the weak sense. Let $p(x, t)$ be a reference density (the passive flow). Define the Schrodinger-bridge effort functional

$$\mathsf{WSB}[\rho, v] := \int_{t_0}^{t_1} \int_\Omega \rho(x, t) \Big(\tfrac{1}{2} \|v(x, t)\|^2 + \epsilon \log \frac{\rho(x, t)}{p(x, t)} \Big) \, dx \, dt. \tag{35}$$

Remark 1501. *The weak-sense continuity equation means that for every smooth test function ψ compactly supported in space-time, one has the integral identity*

$$\int_{t_0}^{t_1} \int_\Omega \big(\partial_t \psi(x, t) \, \rho(x, t) + \nabla \psi(x, t) \cdot (\rho(x, t) v(x, t)) \big) \, dx \, dt + \int_\Omega \psi(x, t_0) \rho(x, t_0) \, dx - \int_\Omega \psi(x, t_1) \rho(x, t_1) \, dx \ = \ 0.$$

This formulation is important because, in transport problems, minimizers often exist in classes where ρ is not classically differentiable in t and v is defined only ρ-a.e.; the weak form is the correct notion of "mass conservation" for such low-regularity flows.

Remark 1502. *The notation here is classical in PDE/optimal-transport formalisms. $\|v(x, t)\|$ is the Euclidean norm of the velocity vector at point x and time t. The symbol $\epsilon > 0$ plays a role analogous to λ in the discrete formulation: it sets the scale of the entropic/regularization term. The logarithmic term $\log(\rho/p)$ is the pointwise log-likelihood ratio; integrating $\rho \log(\rho/p)$ over space yields a relative-entropy-type quantity, making this a continuous cousin of a KL penalty.*

Remark 1503. *One can also view $\rho \log(\rho/p)$ as a* local *(in time) mismatch penalty: at each time slice t, it penalizes the divergence between the controlled marginal $\rho(\cdot, t)$ and the passive marginal $p(\cdot, t)$. In many Schrödinger-bridge settings, the passive p itself arises as the time-marginal of an uncontrolled diffusion (e.g. a Fokker–Planck flow); in that case, the entropic term measures how much the controlled evolution "leans away" from what the environment would do without intervention.*

Remark 1504. *A simple example is to take $\Omega = \mathrm{R}$ and imagine $\rho(\cdot, t)$ as a "cloud" of probability mass that begins concentrated near one location at time t_0 and must end concentrated near another at time t_1. The passive reference $p(\cdot, t)$ might represent diffusion or drift (what the cloud would do naturally). The functional* WSB *then measures: how much kinetic energy it takes to transport the cloud (the $\|v\|^2$ term) plus how much informational strain it takes to keep the cloud's distribution from simply behaving like p (the $\log(\rho/p)$ term). This makes the definition useful as a precise continuous formalization of Hyperseed-Concept 100 and Hyperseed-Concept 158.*

Remark 1505 (Two kinds of effort, again)**.** *The kinetic term $\int \rho \, \frac{1}{2}\|v\|^2$ penalizes rapid rearrangement (effort to move). The entropic term $\int \rho \, \log(\rho/p)$ penalizes deviation from the passive reference (effort to resist). Thus* WSB *is a continuous analogue of the discrete forcing-regularized objective.*

Remark 1506. *Two limiting cases are especially informative. If the entropic weight ϵ is set to* 0 *(formally), the functional reduces to the Benamou–Brenier kinetic energy for dynamic optimal transport, and the minimizer becomes a displacement interpolation (a Wasserstein geodesic) between ρ_0 and ρ_1. Conversely, when ϵ is large, the minimization increasingly prefers trajectories whose time-marginals remain close to $p(\cdot, t)$, so the controlled evolution behaves more like a "softly steered" version of the passive flow rather than a purely distance-minimizing rearrangement.*

Definition 440 (Wu wei geodesic)**.** *Fix boundary densities $\rho(\cdot, t_0) = \rho_0$ and $\rho(\cdot, t_1) = \rho_1$. A* wu wei trajectory *between ρ_0 and ρ_1 (relative to passive prior p) is any minimizer (ρ^*, v^*) of* $\mathsf{WSB}[\rho, v]$ *subject to the continuity constraint and boundary conditions.*

Remark 1507. *In practice, the phrase "subject to the continuity constraint" means the admissible set consists of pairs (ρ, v) with $\rho(\cdot, t) \geq 0$, $\int_\Omega \rho(x, t)\, dx = 1$ for each t (when ρ is a probability density), finite kinetic energy $\int \rho \|v\|^2 < \infty$, and with $\partial_t \rho + \nabla \cdot (\rho v) = 0$ holding in the weak sense. These conditions ensure that the velocity field genuinely transports the mass of ρ and that* $\mathsf{WSB}[\rho, v]$ *is well-defined (possibly $+\infty$ if ρ is not absolutely continuous with respect to p at some times).*

Remark 1508. *Calling the minimizer a "geodesic" is not merely metaphor. One is endowing the space of densities with a geometry whose path-length-like functional is* WSB*, and then selecting shortest paths subject to endpoints. The "wu wei" content is that the geometry is not purely kinetic (as in classical least-action transport) but also measures closeness to a passive reference p, so the shortest path is the one that accomplishes the endpoint change with minimal combined motion-and-resistance. This is exactly Hyperseed-Concept 207 [21].*

Remark 1509. *This "geodesic relative to p" viewpoint also clarifies why the reference is called "passive": it is not merely a prior over endpoints, but a preferred* background evolution *that defines what counts as effortless. In the same way that a physical medium can have a prevailing current, the density $p(\cdot, t)$ encodes a baseline flow; wu wei is then the principle that one should exploit that baseline unless there is compensating value in deviating from it.*

Remark 1510. *As a toy case, if $\rho_0 = \rho_1$ and p is stationary, then the trivial path $\rho(x, t) = \rho_0(x)$ and $v \equiv 0$ achieves zero kinetic cost and (typically) minimal entropic cost, embodying the "do nothing because nothing is required" corner of wu wei. At the opposite extreme, if ϵ is tiny and the endpoints are far apart, the minimizer will behave like an almost-deterministic transport map: wu wei does not forbid decisive movement; it penalizes* unnecessary *deviation from the passive flow.*

Remark 1511. *A further subtlety is that "do nothing" depends on what p encodes. If $p(\cdot, t)$ already drifts substantially (for instance, due to a strong ambient drift field), then matching endpoints may still require nontrivial motion in the laboratory frame. In that case, wu wei is naturally interpreted in the co-moving frame defined by the passive dynamics: the minimizer prefers to "ride" the baseline drift and only apply control where the endpoint constraints force a correction.*

Theorem 25 (Euler–Lagrange structure (informal)). *Under standard regularity and positivity assumptions (smooth p, strictly positive ρ_0, ρ_1), there exists a unique minimizer (ρ^*, v^*) of* WSB. *Moreover v^* is a gradient field $v^* = \nabla\phi$ and (ρ^*, ϕ) satisfy a coupled system (the Schrodinger system / entropic optimal transport equations).*

Remark 1512. *The condition that v^* is a gradient field is the continuum analogue of optimal policies being expressible via a value function or potential: it says that, at optimum, the flow is driven by a scalar "pressure" or "cost-to-go" ϕ rather than by an arbitrary solenoidal component. In geometric terms, one is selecting the steepest (most efficient) direction for changing the density subject to the endpoint constraints, which is consistent with the "minimal interference" reading of wu wei.*

Remark 1513. *While the theorem statement above deliberately suppresses formulas, a common way to write the coupled optimality conditions is as a pair consisting of (i) the continuity equation with $v^* = \nabla\phi$, and (ii) a Hamilton–Jacobi-type equation for ϕ with an additional term encoding the entropic preference relative to p. In the classical Schrödinger bridge formulation (with a diffusion-based passive process), the same content is often expressed through two space-time potentials (sometimes called Schrödinger factors) whose product, together with the passive kernel, reconstructs the optimal time-marginals $\rho^*(\cdot, t)$. These equivalent representations make precise the idea that the optimal interpolation is simultaneously a least-effort transport and a least-deviation-from-prior inference problem.*

Remark 1514. *Terminology note: the literature often spells this as "Schrödinger bridge" and "Schrödinger system" (with an umlaut), but the present spelling "Schrodinger" refers to the same object. The essential point is that the minimizer can be characterized either dynamically (via (ρ^*, v^*)) or probabilistically (via an entropy minimization on path space relative to a passive process), and these viewpoints coincide under standard assumptions.*

Remark 1515. *In ordinary language: not only does a best (least-effort) flow exist, but it is essentially unique, and it has a special "potential flow" form. The claim that $v^* = \nabla\phi$ says the optimal velocity field has no rotational component (no "vorticity") in the idealized setting; it is driven by a scalar potential ϕ. This is important because potential flows are mathematically tractable and geometrically interpretable: the system moves "downhill" along a potential landscape rather than stirring itself in complicated loops. In particular, once ϕ is known (or characterized), the dynamics are reduced to solving for a single scalar field coupled to the continuity equation, which is substantially simpler than working with a general vector field. Equivalently, the optimality conditions enforce that any divergence-free "circulation" component would only add kinetic cost without helping satisfy the endpoint constraints, and is therefore ruled out at the minimizer.*

Remark 1516. *This theorem connects to the discrete results above via a shared variational structure: both the Gibbs tilt (Proposition 45) and the linear desirability recursion (Theorem 24) can be seen as discrete shadows of entropic optimal transport/Schrodinger bridge formulations. Within Hyperseed, this provides one route to treat wu wei as a literal "least effort path" principle, consistent with the broader claim that cognition may be organized around minimal representational forcing [21, 3]. A useful way to read this connection is that "tilting" a reference process by exponentiated costs is the discrete analogue of regularizing a transport problem by relative entropy with respect to a reference path measure, so the same object appears either as a reweighted transition law (discrete) or as an entropy-penalized flow (continuous). In this sense, the bridge viewpoint explains why linear recursions and log-transform tricks arise: they are the computational footprint of an underlying convex optimization problem in probability space.*

Proof sketch. The functional WSB is convex in $(\rho, \rho v)$ and strictly convex in appropriate function spaces once boundary conditions are fixed. Existence follows from lower semicontinuity and compactness (via standard calculus of variations arguments). Uniqueness follows from strict convexity. The gradient-field form for v^* arises from introducing a Lagrange multiplier for the continuity constraint and taking first variations; this yields that the optimal momentum ρv is the gradient of a potential. More explicitly, writing the Lagrangian with a multiplier ϕ for $\partial_t \rho + \nabla \cdot (\rho v) = 0$ and differentiating with respect to v yields an optimality condition of the form $v = \nabla \phi$ (up to the precise sign convention), while differentiating with respect to ρ produces a complementary equation that can be recognized as a Hamilton–Jacobi–Bellman-type condition coupled to the continuity/Fokker–Planck side. This pair of first-order conditions is the continuous-time analogue of the "desirability" linearization: the potential ϕ plays the role of a log-transform of a value-like function. □

Remark 1517. *At a high level, the proof follows the classical pattern of the direct method in the calculus of variations: (1) show the functional is bounded below, (2) take a minimizing sequence, (3) use compactness to extract a convergent subsequence, and (4) use lower semicontinuity to pass the limit through the functional. Strict convexity eliminates the possibility of two distinct minimizers. The Euler–Lagrange condition comes from enforcing the continuity equation constraint with a multiplier and computing first variations, which forces the optimal momentum to be a gradient. In addition, the endpoint constraints $\rho(\cdot, 0) = \rho_0$ and $\rho(\cdot, 1) = \rho_1$ remove the usual gauge freedom in ϕ (up to an additive constant in space), which is why the potential is determined essentially uniquely at optimality. From a control perspective, these steps formalize the intuition that any deviation from the gradient structure would represent wasted "sideways" motion that cannot improve the endpoint fit but does increase action.*

Proof sketch. View $(\rho, \rho v)$ as the primary variables so that the constraint becomes linear and the objective becomes convex. Apply existence/uniqueness results for strictly convex constrained variational problems with fixed endpoints, and then compute first variations to obtain the potential-flow condition $v = \nabla \phi$. Concretely, the strict convexity is in the momentum variable ρv (quadratic kinetic term), while the entropic regularizer provides coercivity/regularity that helps rule out oscillatory minimizing sequences; together these properties support the standard compactness arguments needed by the direct method. □

Remark 1518. *A geometric intuition is to picture the density $\rho(\cdot, t)$ as an incompressible-looking "mist" that must morph from shape ρ_0 to shape ρ_1. The potential ϕ acts like a time-dependent height function whose gradient tells the mist how to flow. The entropic term (scaled by ϵ) is like a preference to remain near the reference mist $p(\cdot, t)$, preventing the solution from collapsing into overly sharp, brittle transport when such sharpness is not demanded by the endpoints. Equivalently, ϵ controls how strongly the bridge is allowed to "deviate" from the reference dynamics: small ϵ permits decisive, high-curvature rearrangements, whereas larger ϵ favors smoother, more diffusive, reference-aligned evolution. This also clarifies why the solution can be interpreted as a compromise between matching the boundary marginals and staying statistically typical under the prior flow.*

Remark 1519 (Optimal transport and "fluid optimal control")**.** *When $\epsilon \to 0$ the entropic term vanishes and* WSB *approaches the Benamou–Brenier formulation of Wasserstein-2 optimal transport. Thus wu wei trajectories interpolate between: (i)* pure transport geodesics *(strong commitment, low noise), and (ii)* entropically regularized bridges *(soft, exploratory flow). This is one precise route by which "fluid flow and optimal control" become the same mathematics. In the opposite regime (moderate to large ϵ), the minimizer can be read as the most likely evolution connecting ρ_0 to ρ_1 under a stochastic reference process, with the control v acting as the minimal intervention* *needed to steer the law of the process to the desired endpoint. This makes the "least effort" interpretation literal: the action measures the smallest kinetic forcing (in a quadratic-energy sense) compatible with the imposed marginal constraints, with entropic regularization preventing degenerate, measure-concentrating solutions.*

26.5 Exploration, turbulence, and a cognitive Reynolds number

Hyperseed suggests reading the Reynolds number as a knob for "creative exploration" versus "smooth exploitation." We can capture the same qualitative point with the regularization/noise scale. The classical fluid Reynolds number compares inertial transport to viscous smoothing, and the same structural comparison appears here: a transport scale ("how hard you are trying to move mass/attention through state space") versus a smoothing scale ("how much the dynamics is softened by noise/regularization"). In this sense, the Reynolds analogy is less about literal Navier–Stokes dynamics and more about a dimensionless *ratio of mechanisms* that compete in the objective.

Definition 441 (A simple cognitive Reynolds number)**.** *In the transport setting, define a dimensionless quantity*

$$\mathrm{Re}_c := \frac{\|v\|_{\mathrm{typ}}\, L}{\epsilon}, \tag{36}$$

where L is a characteristic length scale and $\|v\|_{\mathrm{typ}}$ is a characteristic velocity. In discrete-time KL control, an analogous knob is $1/\lambda$. One can think of L as encoding the "task horizon" in state space (how far the distribution must be rearranged), while $\|v\|_{\mathrm{typ}}$ encodes the typical rate of rearrangement demanded by the task (how quickly the rearrangement must occur). The parameter ϵ plays the role of the regularization/noise amplitude, so that increasing ϵ increases smoothing (analogous to increasing viscosity), and decreasing ϵ allows sharper, more "ballistic" rearrangements.

Remark 1520. *This is an analogy, not a claim of literal fluid identity: the point is that ϵ (and similarly λ) behaves like an effective viscosity/regularization. When ϵ is large, the entropic term dominates and the flow prefers to remain close to p, exploring broadly; when ϵ is small, the kinetic term dominates and the flow can "commit" to sharp rearrangements. The definition is useful because it provides a single dimensionless control parameter that summarizes how "strained" the system is allowed to become in pursuit of goals, linking directly to Hyperseed-Concept 100 and Hyperseed-Concept* **??**. *A helpful way to read the ratio is: $\|v\|_{\mathrm{typ}}\, L$ sets a scale for "transport intensity" (distance $\times$ speed), while ϵ sets a scale for "smoothing capacity." Thus Re_c increases either because the task demands faster/farther motion (larger $\|v\|_{\mathrm{typ}}$ or L) or because the system is permitted to regularize less (smaller ϵ). In many entropic transport/Schrödinger-bridge formulations, ϵ is directly tied to diffusion strength, so increasing ϵ literally increases stochastic mixing along paths; here we only need the qualitative implication that larger ϵ widens the effective exploration kernel around passive dynamics. Similarly, in KL control, λ is often interpretable as an inverse-temperature-like parameter in a soft-min/soft-max: larger λ makes control more "cost-aware" and conservative (more weight on staying close to the passive prior), while smaller λ makes the policy more decisive and more willing to pay KL to reshape trajectories.*

Remark 1521. *As a very simple mental picture: take L to be the diameter of the region in state space you must traverse and $\|v\|_{\mathrm{typ}}$ to be the typical speed needed to do so in time. Then Re_c is large when the problem demands fast movement relative to the smoothing scale ϵ. In cognitive terms, this corresponds to contexts where sharp decisions are forced (high "closing") rather than gently sampled (high "opening"). Equivalently, if one fixes a time budget T, a common rough choice is $\|v\|_{\mathrm{typ}} \approx L/T$, yielding $\mathrm{Re}_c \approx L^2/(\epsilon T)$, which makes explicit that shortening the horizon T or* *increasing the spatial span L pushes the system toward higher "cognitive Reynolds" demand. This highlights why deadlines, urgency, or high-stakes constraints tend to shift behavior from exploratory sampling to more committed, lower-entropy action selection: they effectively increase the transport-to-smoothing ratio even if the underlying noise level is unchanged.*

Remark 1522. *Small* Re_c *(large* ϵ *or large* λ*) corresponds to high regularization: the flow stays close to the passive prior and remains smooth (exploration dominates). Large* Re_c *corresponds to low regularization: the system can commit sharply, but may also exhibit brittle forcing and complex transient dynamics. This is a formal handle on the exploration–exploitation tradeoff. In particular, the "turbulence" metaphor can be read as a statement about* sensitivity *and* structure*: at large* Re_c*, small changes in goals or boundary conditions can induce large rearrangements of the optimal transport plan/policy, and intermediate-time behavior can become highly non-uniform (e.g., temporarily concentrating probability mass in narrow channels of state space before spreading again). Conversely, at small* Re_c*, regularization enforces a kind of global coherence: mass moves along many paths with moderate weights rather than a few extreme paths, which in cognitive terms resembles maintaining multiple hypotheses or action options with non-negligible probability. It is also useful to note that* Re_c *is meant as a* scaling guide*, not a precisely measurable constant: different reasonable choices of "typical" speed or length yield different numerical values, but they preserve the core monotonicity that increasing task intensity or decreasing smoothing pushes the system toward sharper, more exploitative behavior.*

26.6 Opening/closing, resistance/acceptance, and resonance-modulated control

Wu wei is not only about external dynamics; it also has a phenomenological signature: low felt resistance, low grasping, and high "openness" to the unfolding situation. Within Hyperseed's formalism, these can be treated as properties of the control objective.

Definition 442 (Resistance and acceptance)**.** *Fix passive dynamics* P_0*. For a policy* π*, define its* instantaneous resistance *at state* x *as*

$$\mathsf{Resist}_\lambda(\pi; x) := \lambda \mathsf{KL}\big(P_\pi(\cdot \mid x) \| P_0(\cdot \mid x)\big).$$

Define its acceptance *at* x *as the complementary score*

$$\mathsf{Accept}_\lambda(\pi; x) := \exp\big(-\mathsf{Resist}_\lambda(\pi; x)/\lambda\big) = \exp\big(-\mathsf{KL}(P_\pi \| P_0)(x)\big).$$

Thus acceptance is high exactly when resistance is low.

Remark 1523. *This definition makes Hyperseed-Concept 158 operational: resistance is not a vague feeling but a measurable divergence from a baseline. The acceptance score is simply a monotone transformation of the same quantity, chosen so that it lies in* $(0, 1]$ *with value* 1 *at perfect non-resistance (*$P_\pi = P_0$*). In this way, "acceptance" becomes mathematically the degree to which action remains compatible with the passive unfolding.*

Remark 1524. *A small example clarifies the scale: if* $\mathsf{KL}(P_\pi \| P_0)(x) = 0.1$ *then acceptance is* $e^{-0.1} \approx 0.905$*; if* $\mathsf{KL} = 2$*, acceptance is* $e^{-2} \approx 0.135$*. Thus acceptance sharply penalizes large divergences. This is useful later when resonance is introduced: resonance can bias action without necessarily producing large KL divergence, so it can yield decisiveness together with high acceptance.*

Definition 443 (Opening and closing as entropy of action)**.** *Suppose the agent chooses an action distribution $\pi(\cdot \mid x)$ over a finite action set A. Define its* opening level *at x by the Shannon entropy*

$$\mathsf{Open}(\pi; x) := -\sum_{a \in A} \pi(a \mid x) \log \pi(a \mid x).$$

High opening means the system keeps many options live; low opening means it commits strongly.

Remark 1525. *This definition links Hyperseed-Concept 125 and Hyperseed-Concept 72 to a standard information measure. If $\pi(\cdot \mid x)$ is uniform on A, opening is maximal; if π concentrates almost all mass on a single action, opening is small. The entropy here is Shannon entropy; one may also compare with alternative measures such as logical entropy (Hyperseed-Concept 105) in contexts where distinctions are the primitive [17].*

Remark 1526. *For example, if $A = \{a_1, a_2\}$ and $\pi(a_1 \mid x) = \pi(a_2 \mid x) = 1/2$, then $\mathsf{Open}(\pi; x) = \log 2$. If instead $\pi(a_1 \mid x) = 0.99$ and $\pi(a_2 \mid x) = 0.01$, then $\mathsf{Open}(\pi; x) \approx 0.056$. The point is not that "high opening is always good," but that wu wei often manifests as avoiding unnecessary premature closing: one stays open until reality (task constraints) demands commitment.*

Remark 1527. *In KL-regularized control, the hyperparameter λ simultaneously: (i) penalizes resistance (forcing), and (ii) prevents premature closing by keeping action distributions soft. Thus "opening" can be viewed as a controlled consequence of maintaining wu wei.*

Adding resonance. Earlier sections derive resonance from paraconsistent evaluation by mapping evidence values to complex amplitudes and measuring constructive vs destructive interference. We can plug a resonance term directly into the minimal-forcing control principle.

Definition 444 (Paraconsistent evaluation amplitudes and resonance reward)**.** *Let a set of internal evaluators (subsystems, value-aspects) be indexed by $k \in \{1, \ldots, K\}$. Given an action $a \in A$ in state x, evaluator k assigns a* p-bit *evidence value $E_k(a \mid x) = (E_k^+(a \mid x), E_k^-(a \mid x)) \in [0,1]^2$. Define a complex amplitude*

$$z_k(a \mid x) := (2E_k^+(a \mid x) - 1) + i\,(2E_k^-(a \mid x) - 1) \in \mathrm{C}. \tag{37}$$

Given nonnegative weights w_k, define the total amplitude

$$z_{\mathrm{tot}}(a \mid x) := \sum_{k=1}^{K} w_k\, z_k(a \mid x) \tag{38}$$

and the (bounded-below) resonance reward

$$r_{\mathrm{res}}(a \mid x) := \left| z_{\mathrm{tot}}(a \mid x) \right|^2. \tag{39}$$

Remark 1528. *The notation is doing something simple but philosophically pointed. Each evaluator supplies* two *evidence channels, E_k^+ and E_k^-, as is natural in paraconsistent settings where positive and negative support need not sum to 1 and need not exclude each other. Mapping (E_k^+, E_k^-) to a complex number z_k treats the two channels as orthogonal components of an "opinion vector". The shift-and-scale $(2E - 1)$ simply recenters $[0,1]$ to $[-1,1]$, so that "neutral" evidence sits near 0 in each component. This connects to Hyperseed-Concept 159 and the paraconsistent-logical machinery motivating it [24, 23].*

Remark 1529. *As a small example, if an evaluator says "strongly positive and weakly negative"* $(E^+, E^-) = (1, 0)$*, then* $z = (1) + i(-1)$*. If it says "strongly positive and strongly negative"* $(1, 1)$*, then* $z = (1) + i(1)$*, which is a different phase: paraconsistency is not erased but encoded. Summing amplitudes* $z_{\text{tot}} = \sum_k w_k z_k$ *causes agreement to add coherently and disagreement to partially cancel, and then* $|z_{\text{tot}}|^2$ *turns that net coherence into a scalar reward. This is useful because it lets "internal harmony" become a control-relevant quantity, without requiring the system to collapse conflicting evidence into a single classical truth value.*

Remark 1530 (What this encodes)**.** *If evaluators agree (phases align), then* $|z_{\text{tot}}|^2$ *is large: constructive interference. If evaluators conflict (phases oppose), then* $|z_{\text{tot}}|^2$ *is smaller: destructive interference. Thus* r_{res} *rewards internal coherence. One can also read this as a "coherence-sensitive gain" term: it is insensitive to which evaluator is "right" in isolation, and instead tracks whether the ensemble forms a mutually reinforcing signal. In particular, if the individual complex votes have comparable magnitudes, then alignment produces a superlinear increase in* $|z_{\text{tot}}|^2$*, whereas cancellation keeps the total near the scale of an individual contribution. This is the sense in which resonance operationalizes the phenomenological idea that an action can feel "easy" when many sub-evaluations point the same way, even if no single evaluation dominates.*

Proposition 46 (Resonant wu wei action selection)**.** *Fix a state* x*, a baseline action distribution* $\pi_0(\cdot \mid x)$ *with full support, a bounded "task cost"* $c(a \mid x)$*, a resonance weight* $\beta \geq 0$*, and a forcing scale* $\lambda > 0$*. Consider*

$$\max_{\pi(\cdot|x)\in\Delta(A)} \Big\{ E_{a\sim\pi}[\beta\, r_{\text{res}}(a \mid x) - c(a \mid x)] - \lambda\, \mathsf{KL}\big(\pi(\cdot \mid x) \| \pi_0(\cdot \mid x)\big) \Big\}. \tag{40}$$

Then the unique maximizer is the tilted distribution

$$\pi^*(a \mid x) = \frac{\pi_0(a \mid x)\, \exp\big((\beta\, r_{\text{res}}(a \mid x) - c(a \mid x))/\lambda\big)}{\sum_{b\in A} \pi_0(b \mid x)\, \exp\big((\beta\, r_{\text{res}}(b \mid x) - c(b \mid x))/\lambda\big)}. \tag{41}$$

Moreover, the denominator is a (finite) partition function for the state x *because* $c(\cdot \mid x)$ *is bounded and* $\beta r_{\text{res}}(\cdot \mid x)$ *is typically bounded by construction of the evaluator ensemble; this guarantees the normalization is well-defined for all* $\lambda > 0$*. In the limiting cases, one recovers familiar behaviors: if* $\beta = 0$ *then* π^* *reduces to the standard cost-tilted baseline (no resonance influence), while for large* λ *the distribution remains close to* π_0 *(strong resistance to change), and for small* λ *the mass concentrates on actions maximizing* $\beta r_{\text{res}} - c$ *(nearly deterministic choice).*

Remark 1531. *This proposition says that adding resonance does not break wu wei's computational simplicity. One still gets a Gibbs/softmax form: start from a baseline* π_0 *(the agent's default habit), and reweight actions by the exponential of (resonance reward minus task cost), scaled by* λ*. Thus resonance enters as a* gentle biasing potential*, not as a hard constraint. In Hyperseed terms, this is a concrete mechanism by which internal coherence can guide action without necessarily increasing resistance (KL divergence) from baseline behavior. It is also worth emphasizing the role of* λ *as a "temperature" or "compliance" parameter: smaller* λ *means the agent is willing to depart more sharply from its default policy in response to the combined score* $\beta r_{\text{res}} - c$*, whereas larger* λ *means it will only adjust mildly even if resonance is high. From the perspective of optimal control with information constraints, the KL term quantifies control effort measured in bits relative to the default controller, so resonance can be interpreted as providing additional internal "reward signal" that justifies spending some of that informational/control budget.*

Remark 1532. *The connection to earlier parts of the document is direct: paraconsistent evaluators can disagree, yet their disagreement does not force the agent into paralysis or into brittle resolution. Instead, the system can behave as if it is "listening" to the interference pattern produced by its own competing value-aspects [24]. This is, mathematically, the same kind of "softness" that makes the minimal-forcing principle compatible with bounded rationality and weakness-based representation [3]. In particular, disagreement need not be eliminated by an explicit tie-breaking rule; it is transduced into a graded modulation of the effective utility via r_{res}, and the KL term ensures that any resulting policy change is smooth and bounded. This makes room for action under unresolved internal tension: the agent can still sample from π^*, with the sampling itself representing a form of non-brittle commitment.*

Proof. This is the standard Gibbs variational principle. Equivalently, minimize the convex functional $\lambda \mathsf{KL}(\pi \| \pi_0) - E_\pi[\beta r_{\text{res}} - c]$ over the simplex. The first-order optimality condition is

$$\lambda \log \frac{\pi(a \mid x)}{\pi_0(a \mid x)} = \beta r_{\text{res}}(a \mid x) - c(a \mid x) + \alpha,$$

where α enforces normalization, yielding the stated tilt. Strict convexity of $\mathsf{KL}(\cdot \| \pi_0)$ implies uniqueness. Concretely, exponentiating the stationarity condition gives $\pi(a \mid x) = \pi_0(a \mid x) \exp((\beta r_{\text{res}}(a \mid x) - c(a \mid x) + \alpha)/\lambda)$, and α is then determined by the requirement $\sum_{a \in A} \pi(a \mid x) = 1$, so $-\alpha/\lambda$ is precisely the log of the partition function in the denominator of the displayed formula. Full support of $\pi_0(\cdot \mid x)$ ensures that the logarithm and the ratio π/π_0 are well-defined at the optimum and that the KL divergence is strictly convex on the feasible set. ☐

Remark 1533. *The proof is structurally identical to Proposition 45, with two substitutions: the state distribution $P(\cdot \mid x)$ is replaced by an action distribution $\pi(\cdot \mid x)$, and the immediate cost c is replaced by the combined score $c - \beta r_{\text{res}}$ (up to sign conventions). The logarithm again produces an exponential form, and full support of π_0 ensures the KL term remains well-defined and strictly convex. One can also view the result as the action-analog of entropic regularization in control: the KL penalty selects, among all policies with similar expected score, the one that stays closest to the baseline in relative-entropy distance. In that reading, resonance does not introduce a new optimization primitive; it only modifies the effective reward shaping term inside the same variational template.*

Proof sketch. Optimize a strictly concave objective (linear utility minus KL penalty) over the simplex. Use a Lagrange multiplier for normalization, solve the stationarity equation for π, and normalize; strict convexity of KL gives uniqueness. Because the objective is strictly concave in π (linear term plus negative KL), any stationary point is automatically the global maximizer, so the softmax form is not merely a convenient solution but the unique optimum compatible with the information-cost geometry induced by $\mathsf{KL}(\cdot \| \pi_0)$. ☐

Remark 1534. *A helpful intuition is that π_0 defines what the agent would do "by habit" (or by weak/default modeling), while the exponential term defines a smooth preference landscape over actions. Resonance affects that landscape by increasing the utility of actions that align internal evaluators. Thus coherence acts like an internal "tailwind": it can make one action stand out without requiring a violent departure from baseline. Equivalently, r_{res} changes the relative odds between actions by a multiplicative factor $\exp(\beta(r_{\text{res}}(a \mid x) - r_{\text{res}}(b \mid x))/\lambda)$, so coherence differences matter primarily through their contrasts, and their influence is softened when λ is large. This clarifies why the mechanism is robust: even if resonance is noisy or imperfectly estimated, the KL-regularized tilt prevents it from producing discontinuous policy jumps.*

Remark 1535 (How this matches the phenomenology)**.** *When internal evaluations are coherent, the resonance reward creates a strong but smooth bias, allowing a decisive choice without large divergence from baseline behavior. When evaluations are incoherent (low resonance), the distribution stays closer to π_0 unless the external task cost strongly forces a choice. Thus "acting without forcing" becomes: let coherence guide action whenever possible; otherwise, minimize resistance while resolving the task. In experiential terms, this predicts a graded transition between "effortless" and "effortful" action: high resonance increases decisiveness at fixed λ, while low resonance shifts behavioral control back toward habit and toward externally imposed costs. It also makes precise how acceptance/allowing can be action-relevant: acceptance corresponds to tolerating evaluator conflict (not artificially collapsing it), while still permitting the induced interference pattern to steer action probabilistically rather than via rigid, overconfident commitment.*

26.7 Closing remarks: wu wei as an implementable principle

The section's message is that wu wei is not merely poetic; it corresponds to a standard and useful mathematical design pattern:

- choose a reference (passive) flow P_0 induced by weak modeling and learned habits,
- express goals as costs on trajectories or endpoints,
- penalize divergence from the reference flow (resistance/forcing), and
- optionally add a resonance term that rewards paraconsistent coherence among value-aspects.

Here "flow" can be read concretely as a path measure (a probability distribution over state–action trajectories) or, in continuous time, as a controlled diffusion law; in either case, P_0 supplies the baseline dynamics that would occur absent deliberate intervention. The second step fixes what is to be achieved, while the third step fixes *how* it is to be achieved: not by arbitrary steering, but by minimal deviation from what is already natural or well-learned. In this language, the "forcing" cost is not an aesthetic add-on but a quantitative proxy for intervention effort, model-mismatch, or cognitive strain, depending on the chosen interpretation of the control channel and state representation. The optional resonance term can be understood as an additional structured preference that is *not* reducible to a single scalar reward without loss, but can still be implemented by coupling multiple value-aspects into a single optimization objective.

The resulting dynamics can be viewed equivalently as: (i) KL-regularized optimal control, (ii) entropic optimal transport / Schrodinger bridge interpolation between boundary conditions, or (iii) "fluid decision" in which the whole probability mass distribution moves along minimal representational effort geodesics. A compact way to phrase the shared structure is that one selects a path measure P by trading off task cost against relative entropy to the baseline,

$$P^\star \in \arg\min_P \Big(\mathbb{E}_P[\,\text{task-cost}\,] + \lambda\,\mathrm{KL}(P\|P_0) - \eta\,(\text{resonance})\Big),$$

with weights λ, η chosen to set the strength of "non-forcing" and the strength of coherence-seeking, respectively. In discrete-time Markov settings this objective specializes to the familiar "soft" control laws in which optimal policies are reweighted versions of passive dynamics, and in continuous-time diffusion settings it yields drift adjustments that are small in the sense of relative entropy rate. The equivalence to Schrödinger bridges emphasizes that, when one constrains initial and terminal marginals (or more general boundary conditions), the KL penalty produces the most likely interpolation under P_0, i.e., the least-informative change consistent with the constraints; this is a precise sense in which wu wei implements "getting there without forcing."

Remark 1536. *From the Hyperseed point of view, these equivalences are not merely technical: they suggest a unifying way to speak about behavior, inference, and phenomenology. What looks like "control" in one vocabulary appears as "inference under a passive prior" in another, and as "least-effort geometry" in a third. The wu wei slogan, then, is not an appeal to mystification; it is a reminder that the same formal object can be read as action, as belief update, or as the unfolding of a field. In particular, wu wei becomes implementable precisely because the KL/entropy regularizers yield tractable Gibbs forms and (in special cases) linear recursions [21]. Concretely, the Gibbs form implies that optimal choices reweight baseline tendencies by exponentiated (negative) cost, so that "trying" corresponds to* tilting P_0 *rather than replacing it by an unrelated plan. This is also the technical reason that many such models admit efficient dynamic programming variants: after an exponential transform, the Bellman optimality conditions become linear in the transformed value, so the computation aligns with standard inference machinery (message passing, path integrals, or linear solvers), depending on the domain. On the phenomenological reading, the same mathematics says that deliberate action can feel like "allowing" a trajectory when the objective is achieved by small informational departures from an already coherent baseline.*

Remark 1537. *Finally, the weakness connection is not optional ornamentation. If P_0 is learned from a weak representational stance (few distinctions, broad generalization), then staying close to P_0 is simultaneously (a) low forcing in the environment and (b) low forcing in the agent's own representational apparatus. This recovers a single mathematical mechanism underlying Hyperseed-Concept 206 and Hyperseed-Concept 202: act effectively, but let effectiveness arise from well-chosen priors and coherent internal structure rather than continual override [3, 2]. More explicitly, a weak stance implies that the hypothesis class used to learn P_0 prefers simple, stable regularities; the KL penalty then discourages policies that demand fine-grained distinctions the agent does not robustly maintain. In that sense, the "cost of forcing" doubles as a cost of representational overreach: if one must continually inject detailed corrections to compensate for brittle modeling, then* $\mathrm{KL}(P\|P_0)$ *grows, signaling a departure from both environmental and cognitive economy. Conversely, when the baseline already captures broad invariances, the same KL-regularized update yields behavior that is both low-intervention and resilient under uncertainty, because the solution inherits the generalization properties of P_0 while still permitting targeted deviations where the task-cost strongly demands them.*

27 Discussion, limitations, and next steps

This section is intentionally reflective rather than deductive: it does not add new primitives, but asks what has (and has not) been achieved by the formalization so far. In Russell's spirit, one may say that the gain of formal work is a certain disciplined clarity about *what follows from what*; but the price is that every formal system must declare (sometimes tacitly) which distinctions it is willing to draw, and which it will treat as blurred, incomplete, or even contradictory. Hyperseed leans into this price by treating inconsistency and vagueness as first-class citizens, via paraconsistent evidence and weakness-based compositional structure (cf. Hyperseed-Concept 198, 143, 202 and [3, 2]).

Remark 1538 (How to read this section)**.** *This section is a map of plausible continuations, not a list of settled conclusions. The formal core (paraconsistent evidence, quantale weakness, compositional structure) was built to be extended; the items below indicate* where *extensions may be required for particular applications (cognitive architectures, philosophy of science, or speculative cosmology). When the text mentions a Hyperseed notion (e.g. morphic resonance), it should be read as an interface point: a place where one might connect the ontology to either mathematical or empirical constraints.*

Outline

- Discuss alternative formalizations (topos semantics, other paraconsistent logics, other quantales).
- Discuss what parts of Hyperseed remain underspecified and how to tighten them.
- Discuss computational realizations (knowledge graphs, proof assistants, agent architectures).
- List open problems and potential empirical tests (especially around morphic resonance).

Remark 1539 (Why these four bullets are the natural "pressure points")**.** *The four bullets correspond to four distinct questions one can ask of any formal ontology: (i)* Semantics: *what mathematical universes can interpret the axioms (toposes, Kripke models, algebraic semantics)? (ii)* Completeness of specification: *which parts of the intended meaning are still carried by informal prose rather than axioms, and which extra axioms would capture them without overcommitting? (iii)* Implementability: *what data structures and algorithms support reasoning with the primitives at scale? (iv)* Falsifiability or at least empirical constraint: *which claims are intended to be empirically tethered (even weakly), and what would it mean for the tether to fail? In Hyperseed, these questions are especially salient because the theory explicitly treats* context, evidence*, and* weakness *as observer-relative (Hyperseed-Concept 86,* **??** *is not in the core-concept list, but paraconsistent evidence relates directly to 198 and 104).*

Summary and Hyperseed concepts covered

This section is explicitly meta: it reviews the extent to which the formal core captures the original ontology, where additional axioms are needed, and how to implement the ontology in software.

Hyperseed concepts covered.

- Cross-cutting: all concepts, with emphasis on morphic resonance, paraconsistent valuation, and weakness-based pattern theory.

Remark 1540 (Pointers to the concept index for the emphasized notions)**.** *For quick navigation, the three emphasized themes correspond most directly to: morphic resonance (Hyperseed-Concept 115), paraconsistent valuation (Hyperseed-Concept 198 and also the broader idea of logical vs. empirical truth, Hyperseed-Concept 104), and weakness-based pattern theory (Hyperseed-Concept 202, 143, 130,* **??***). Background sources that motivated these emphases include [5, 3, 2, 24].*

Alternative formalizations: semantics and logical choices

Remark 1541 (Topos semantics as a principled way to internalize "context")**.** *A recurring implicit structure in Hyperseed is that what counts as a "fact" depends on a context of observation, inference, and action. One classical mathematical way to express context-dependence without collapsing into mere relativism is to work in a topos: truth values live in an internal logic whose semantics varies with the chosen topos, often realized as sheaves over a space of contexts. In such a setting, the move from absolute propositions to observer-indexed propositions can be rendered as the move from Set-based semantics to sheaf semantics. This resonates with the paper's general stance that "reality" is (at least partly) stabilized predictive structure relative to observers (cf. Hyperseed-Concept* **??** *and the philosophy-of-science perspective in [20]).*

Remark 1542 (Other paraconsistent logics: keeping inconsistency local)**.** *Hyperseed uses paraconsistent evidence to prevent contradiction from entailing triviality. The specific choice of paraconsistent machinery can vary. For example, 4-valued paraconsistent logics (including constructive variants) provide a truth-functional account of "both true and false" and "neither true nor false"; these align naturally with the idea of maintaining positive and negative evidence channels. In such settings, the bookkeeping distinction is not merely semantic but operational: one can treat "positive support" and "negative support" as independently accruing resources, and then interpret the four truth-values as different coarse summaries of that resource state (supported, refuted, both, or neither). Constructible Duality Logic [23] is one relevant anchor when one wants a typed, constructive discipline around such multi-valued reasoning. The appeal of a typed, constructive discipline is that it supports a more explicit separation between (1) judgments that are computationally witnessed (proof-relevant) and (2) judgments that are merely consistent with current information, which matters when Hyperseed is instantiated as an actual system that must decide when it is permitted to act versus when it must suspend judgment. In other directions, one could emphasize dynamic or modal paraconsistent logics, to better encode temporal updating of evidence (connecting to the later helper-theorem sections on time and becoming). Here the modal/dynamic operators can be read as tracking how the evidence state evolves under actions, observations, or interventions, so that "inconsistency local" means not only "non-explosive" but also "confined to the regions of the state space that the update rules actually touch." The key desideratum is preserved: contradictions should be* registered *and* reasoned with*, not suppressed, and not allowed to explode. One practical consequence is that the logic should permit selective inferences that remain stable under the addition of further evidence, so that local inconsistencies do not globally invalidate downstream planning or learning. This is consonant with the use of resonance-like mechanisms linking inconsistency with dynamical effects [24].*

Remark 1543 (Other quantales: what changes when the "evidence algebra" changes)**.** *Quantales serve as a flexible algebraic substrate for combining graded evidence and measuring weakness/effort (Hyperseed-Concept 143, 202). The canonical choice $V = [0,1]^2$ with componentwise order and product tensor is mathematically convenient, but not obligatory. One can view this choice as fixing a particular notion of "how independent pieces of support compose" and a particular notion of "how costs/weaknesses accumulate," and different applications may require different compositional semantics. If one changes the tensor (e.g. from multiplication to* min *or to a Lukasiewicz t-norm), one changes the interpretation of "independent accumulation" versus "bottleneck conjunction." Concretely:*

- *Using multiplication makes repeated moderate evidence decay (or amplify) multiplicatively, matching an "independent likelihood" intuition. This is especially natural when one expects approximate conditional independence between evidence sources, or when one wants long chains of weak support to gradually attenuate rather than saturate.*
- *Using* min *makes conjunction governed by the weakest link, matching a "necessary condition" intuition. This is appropriate when a claim is only as credible as its most fragile prerequisite (e.g. safety constraints, brittle sensor requirements, or proof obligations where any missing lemma blocks the result).*

The philosophical point is that such a choice is not just technical: it declares what sort of composition the ontology is willing to treat as basic. In particular, it fixes whether "more evidence" behaves like accumulating mass, like satisfying a set of constraints, or like consuming a limited budget of slack. The paper's emphasis on weakness suggests that these choices should be evaluated by *how they support minimum-distinction representation and compositional control [3, 2]. From an engineering perspective, a useful additional test is whether the chosen quantale admits well-behaved residuals/implications (when present), since these determine how one can pose and solve inverse questions of the form "how much additional evidence/effort is needed to reach a target threshold?"*

What remains underspecified, and how to tighten it

This subsection is best read as a set of explicit "degrees of freedom" that separate the core ontology from any particular instantiation. The goal is not to postpone precision indefinitely, but to make clear which precisifications are domain commitments (and therefore revisable) versus which are constitutive of the framework.

Remark 1544 (Underspecification is not a defect, but it creates obligations)**.** *Any broad ontology that aims to serve both metaphysical and computational ends must initially leave some degrees of freedom. However, leaving degrees of freedom creates obligations of two kinds: (i) to state clearly* what is fixed *(axioms) and* what is variable *(model class), and (ii) to articulate criteria by which a variable choice becomes justified in a given application. A useful way to think about (ii) is that justification can be empirical (predictive performance), computational (tractability and robustness), or interpretive (faithfulness to the intended metaphysical reading), and in practice a system design will trade these against one another rather than optimize a single axis. Hyperseed's core is deliberately permissive about representational granularity and about which distinctions are "worth making"; this permissiveness is exactly what weakness is meant to formalize (Hyperseed-Concept 202 and [3, 2]). In particular, weakness is meant to make it possible to say not merely that two models differ, but that the* cost of maintaining the difference *(in attention, memory, proof burden, or control complexity) may dominate its benefits, thereby providing a principled reason to coarsen a representation.*

Remark 1545 (Examples of places one may want extra axioms)**.** *The following are typical points where a research program must decide whether to add structure:*

- ***Threshold choices and stability.*** *Many definitions use thresholds (for distinctions, focus/fringe, etc.). To "tighten" the theory one may specify how thresholds vary across contexts, or impose robustness constraints: e.g. require that small perturbations of thresholds do not flip key classifications too often. One may also require monotonicity constraints (e.g. as resources increase, thresholds do not become harder to satisfy) or hysteresis-like constraints (to prevent oscillation when evidence hovers near a boundary), depending on whether the intended system is more like a classifier or more like a controller.*
- ***Coherence constraints on evidence updating.*** *Paraconsistency permits inconsistency, but does not determine how evidence should be updated in time or merged across agents. One may impose additional coherence axioms (probabilistic, order-theoretic, or dynamical) when building concrete systems, at the cost of narrowing the model class. For instance, one might demand commutativity/associativity of certain merge operations (to model order-insensitive aggregation) or explicitly reject them (to model path-dependence and learning trajectories), and either choice should be treated as a substantive modeling commitment rather than an implementation detail.*
- ***Pattern/evidence interfaces.*** *The ontology of patterns, emergence, and pattern webs is deliberately broad (Hyperseed-Concept 130,* **??***, 132; see also [5]). A tighter theory might specify a privileged family of pattern formalisms (graphical models, programs, logical theories, dynamical systems) together with translations between them. Such translations may themselves carry*

weakness/effort costs (e.g. compilation blowups, loss of fine structure, or the introduction of approximation error), and spelling out these costs is one way to connect the abstract ontology of patterns to concrete computational budgets.

These tightenings do not change the spirit of Hyperseed; they determine how the spirit becomes an engineering discipline. In other words, the point is not to eliminate pluralism, but to ensure that pluralism is controlled by explicit principles so that different instantiations remain comparable.

Remark 1546 (Dependency structure as a recurring hidden variable)**.** *A subtle but pervasive underspecification concerns* dependency*: which assertions depend on which representational commitments, resources, or background regularities (Hyperseed-Concept 93, 94). Formal work on dependency towers and tradeoff theorems provides one route to make this explicit [8, 9]. In practice, making dependency explicit can mean recording not only* that *a claim holds but* under which supports it was derived *(data sources, inference rules, time windows, or modeling assumptions), so that revisions can be localized when some supports are withdrawn or contradicted. The philosophical benefit is a more candid separation between what is claimed "in the abstract" and what is claimed "given the scaffold" (Hyperseed-Concept 51, 83). This also clarifies how paraconsistency interacts with engineering: if contradiction is confined to a particular scaffold component, then one can degrade gracefully by revising or isolating that component rather than treating the entire knowledge state as unusable.*

Computational realizations

Remark 1547 (Knowledge graphs with paraconsistent annotations)**.** *A straightforward implementation path is to treat entities/patterns as nodes in a knowledge graph, and attach to each edge or assertion a p-bit-like evidence annotation capturing positive and negative support. Concretely, one can view each annotation as a small record (e.g.* $(\mathrm{e}^+, \mathrm{e}^-)$*) whose components are updated independently as new sources arrive, so that contradictory sources need not collapse the representation into triviality. This aligns with the intent of paraconsistent valuation (Hyperseed-Concept 198) and makes explicit the difference between "absence of evidence" and "evidence of absence." In practice, the former corresponds to leaving* e^+ *and/or* e^- *near a designated neutral element, while the latter corresponds to actively increasing the negative channel; the distinction is especially important in sparse graphs where missing edges are common. One then implements inference as propagation and aggregation of these annotations, respecting the chosen quantale operations (Hyperseed-Concept 143). Operationally, propagation specifies how evidence moves along paths (e.g. composing along relations), while aggregation specifies how multiple paths, witnesses, or sources combine at a node, and quantale structure provides a principled way to make these combinations associative and order-respecting. Such an approach is compatible with the broader engineering perspective in [19]. It also suggests a natural interface to existing KG toolchains: the paraconsistent layer can often be implemented as an annotation algebra on top of otherwise standard storage, indexing, and query machinery, with the main novelty living in the update rules and the inference semiring/quantale.*

Remark 1548 (Proof assistants and typed internalization)**.** *A different realization strategy is to internalize fragments of Hyperseed in a proof assistant. Here the goal is not merely to compute, but to ensure that complex chains of reasoning about contexts, translations, and compositional structure remain checkable. This shifts the emphasis from runtime performance to auditability: proofs become artifacts that record which context-shifts were permitted and which coherence conditions were used. Dependently typed settings are particularly apt for expressing context-indexed notions without losing rigor. For example, one can index claims by explicit context parameters and require that any "change of context" be witnessed by a term that states the translation/transport* *map, preventing accidental equivocation between similar-looking but non-identical situations. Work on program structure and type-theoretic constructions (e.g. McBride-style derivations in type theory) offers inspiration for making "change of context" and "differentiation of structure" first-class [22]. A practical deliverable along these lines would be a small library of primitives (contexts, morphisms/translation witnesses, and composition laws) together with a collection of derived rules encoding common Hyperseed patterns, so that later mechanized arguments can reuse them rather than re-proving basic bookkeeping lemmas.*

Remark 1549 (Agent architectures, cognitive synergy, and transfer)**.** *For agent architectures, the central design question is how to allocate attention and computation among many partial models and heuristics. In concrete systems, this becomes a scheduling and resource-allocation problem: which models get run, at what fidelity, for how long, and with what sharing of intermediate results. Hyperseed's emphasis on weakness suggests policies that prefer coarse, low-effort representations until task pressure forces refinement. One can interpret "task pressure" as any signal of mismatch (loss, surprise, failure rate, or utility gradient) that justifies paying the cost of a more discriminating model, and interpret "refinement" as moving along a ladder of increasingly structured representations rather than committing to maximal structure from the start. This rhymes with the cognitive-synergy engineering stance in [19] and with transfer-learning frameworks aiming to reuse structure across tasks (Hyperseed-Concept 192, 193; see [7]). From an implementation viewpoint, transfer then becomes the disciplined reapplication of previously validated partial morphisms: components that were weak-but-useful in one context are tested as candidate scaffolding in a new one, with explicit bookkeeping of where the fit succeeds and fails. In multi-agent settings, one also needs a theory of when cooperation improves group-level effectiveness; the pro-social efficiency argument in [6] provides one candidate mathematical lens (touching Hyperseed-Concept 170, 81). Here, "improves" should be read operationally: cooperation must pay for its coordination costs, and thus the relevant predictions concern regimes where communication bandwidth, trust calibration, and division of labor create net gains in collective inference or exploration.*

Remark 1550 (Complexity and information measures as operational proxies for weakness)**.** *When implementing weakness, one often needs computable proxies. In particular, many theoretical notions of "minimal structure" or "least commitment" are defined abstractly, whereas engineering requires scalars or efficiently estimated quantities that can guide search, compression, or model selection. Algorithmic information theory (Kolmogorov complexity) provides one principled family of proxies (Hyperseed-Concept* **??***; [16]). Even when true Kolmogorov complexity is uncomputable, practical approximations (compression-based scores, MDL-style criteria, or restricted model classes) can still serve as consistent heuristics for preferring weaker descriptions that nonetheless fit the data. Logical entropy offers an alternative, partition-centric lens that can be more natural when the primitive notion is "distinction" rather than "code length" (Hyperseed-Concept 105; [17]). This viewpoint is often convenient in settings where the agent's first act is to carve the world into equivalence classes (what is treated as the "same") and only later to assign probabilistic weights or codes within each class. These measures do not replace the ontology; they offer handles by which the ontology can be made computationally testable. They also provide a bridge to empirical evaluation: by tying weakness to measurable surrogates, one can compare alternative implementations by checking whether the proxies track improved generalization, transfer, or robustness under perturbations of context.*

Open problems and potential empirical tests

Remark 1551 (Morphic resonance: what it would mean to test a speculative coupling)**.** *Morphic resonance (Hyperseed-Concept 115) is the most empirically charged and controversial notion named in the outline. Sheldrake's proposal [13] is not used here as a settled fact, but as an invitation to state clearly what sort of empirical signature Hyperseed would treat as evidence for a nonlocal, pattern-mediated coupling. That is, the role of the concept is methodological: it forces the framework to spell out what would count as a residual effect after ordinary causal pathways have been accounted for, rather than leaving the notion vague or purely rhetorical. In Hyperseed terms, a "morphic" effect would look like a reproducible improvement in prediction, learning speed, or pattern stabilization that cannot be accounted for by the explicit causal and informational channels included in the modeled context. A useful sharpening is to demand not merely statistical significance, but robustness across modest changes of experimental realization: if the effect depends delicately on uncontrolled details, it is more plausibly an unmodeled ordinary channel than a genuine context-transcending coupling. The practical challenge is to design experiments where such a residual cannot be too easily explained away by hidden ordinary channels, while also not demanding an unrealistic level of isolation. In other words, the experiments must balance two failure modes: being so open that mundane leakage dominates, and being so closed that the study becomes infeasible or so artificial that external validity is lost.*

Remark 1552 (Paraconsistent experimental methodology)**.** *If one expects borderline effects (small, context-sensitive, partially inconsistent), a classical binary hypothesis-testing posture may be epistemically mismatched. In such regimes, it is common for replications to produce mixed outcomes, and the key scientific question often becomes whether the pattern of success/failure aligns with specific contextual moderators rather than whether a single global null can be rejected. A paraconsistent stance instead records both supporting and disconfirming evidence channels and asks how stable the* net *pattern is under changes of context and analysis pipeline. Operationally, this means tracking which preprocessing choices, inclusion criteria, modeling assumptions, and instrumentation changes strengthen or weaken the effect, and representing the result as an evidence profile rather than a single scalar verdict. This is a natural extension of the social-computational-probabilistic philosophy of science view in [20], where science is treated as a dynamical process of community-level belief revision rather than a one-shot adjudication. The methodological implication is that publishing "negative" and "positive" channels together is not a concession but a requirement, because downstream reasoning needs access to the full inconsistency structure in order to update responsibly.*

Remark 1553 (Stability of self-modifying goals as an additional testbed)**.** *Another empirical-leaning direction is to test formal predictions about stability of goal systems in self-modifying agents (Hyperseed-Concept 110). Here "stability" can be understood in several related senses (e.g. invariance of a goal under allowed self-modifications, bounded drift under noise, or convergence to an attractor set of policies), and different formalisms may target different senses. Fixed-point methods applied to goal dynamics provide one route to generate concrete mathematical criteria for stability/instability that can be simulated and, in limited domains, experimentally checked [10]. One advantage of this testbed is that it permits closed-loop experiments: the agent's own updates change the object being studied, so one can measure not only performance but also the geometry of goal evolution under explicit intervention. This connects back to the theme that "realness" and "reality-systems" are stabilized predictive structures, now applied to the internal economy of goals rather than external perception. In this light, a goal system counts as "real" for the agent insofar as it persists as a coherent predictive constraint through the agent's own attempts to optimize, compress, or re-express it.*

Remark 1554 (Resource-rich minds and shifts in cognitive regime)**.** *Some open questions become visible only when the agent's resources become large. For example: if a mind can afford extensive simulation, model search, or self-replication, then its "natural" weakness profile and attention-allocation strategies may qualitatively change (Hyperseed-Concept 162; [11]). One way to read this is as a warning against extrapolating human-default priors: scarcity shapes which approximations feel "natural," and abundance may make previously unaffordable forms of meta-reasoning (e.g. maintaining many candidate contexts in parallel) the dominant strategy. This matters because the ontology aims to be scale-robust: it should describe not only humans and current machines, but also potential successors. Accordingly, a next step is to identify which claims are invariant under resource scaling (true structural constraints) versus which are artifacts of a particular computational regime, so that future empirical work can separate principled predictions from contingent ones.*

Remark 1555 (Philosophical anchoring: Peirce and Whitehead as interpretive constraints)**.** *Finally, it is worth noting that Hyperseed's stance is not only technical but metaphysical: it treats process, mediation, and relational structure as primary. Peirce's categories of Firstness, Secondness, and Thirdness [14] offer an interpretive schema for understanding why the formalism insists on both raw qualitative givenness (First), brute interaction and resistance (Second), and lawful mediation/pattern (Third) (Hyperseed-Concept* **??***, 163, 190). In this reading, the point is not to "reduce" the mathematical objects to Peircean terms, but to treat the triad as a constraint on what a satisfactory account must be able to represent without smuggling in extra assumptions: a place for immediacy as such, a place for constraint as such, and a place for the mediating structures that render sequences of constraints intelligible as patterns. This is also a way of articulating why purely extensional descriptions (e.g., sets of outcomes) are often inadequate on their own: they may list what happened, yet fail to encode the modalities of how it was encountered (First), what resisted (Second), and what stabilized as a rule-like linkage (Third). Whitehead's process ontology [15] likewise provides a philosophical context in which "entities" are stabilized occasions and relations rather than timeless substances (Hyperseed-Concept 124, 140). On that view, persistence is something achieved—a repeated compatibility of successive occasions—and the formalism's emphasis on relational consistency conditions can be read as a technical analogue of that achievement, rather than as an a priori commitment to fixed, globally valid objecthood. It also clarifies a limitation: if one imports substance-style intuitions (fixed identity across all contexts), one may misread observer-relative or paraconsistent constructions as defects rather than as deliberate representations of partial stabilization in the presence of incompatible constraints. These references do not prove the mathematics; they help one see what the mathematics is trying to be faithful to. They also function as a guardrail against overinterpreting convenience: when a representation is chosen for tractability, the philosophical anchoring prompts the question of which aspects of process and mediation have been idealized away, and whether that idealization is appropriate for the intended domain of application.*

Remark 1556 (A note on euryphysics and cosmological ambition)**.** *The earlier sections that extend toward cosmos/multiverse/eurycosm aim at a general formal vocabulary for talking about "the wider world" beyond any one stabilized observer-bound reality (Hyperseed-Concept* **??***, 89; see [18]). One motivation for introducing such vocabulary is pragmatic: without a disciplined way to talk about "external" structure, it is easy to oscillate between naive realism (treating one observer's stabilized world as the world simpliciter) and instrumentalism (treating all structure as mere bookkeeping with no cross-context constraint). The caution is that cosmological reach can outrun empirical grip. In particular, once the formalism is used to compare or relate multiple observer-relative stabilizations, it becomes tempting to read purely formal degrees of freedom as ontological commitments; resisting that temptation requires an explicit account of what would count as evidence for, or constraint on,* *those degrees of freedom. The opportunity is that a paraconsistent, observer-relative formalism can at least let us state the limits of our grip precisely, rather than masking them as false certainty. This "precision about limits" is itself a methodological next step: to classify which claims are invariant across observer-stabilizations, which are only locally stable, and which are artifacts of representational choices, thereby turning philosophical caution into concrete criteria for model comparison. A further limitation, closely related, is communicability: the more general the euryphysical language becomes, the more work is required to show how it reduces to familiar empirical practices in special cases (e.g., when observers agree, when inconsistencies are negligible, or when a classical limit is appropriate). Conversely, a near-term opportunity is to develop explicit translation principles that connect eurycosmic statements to testable predictions in restricted regimes, making cosmological ambition continuous with ordinary scientific inference rather than a separate, unconstrained layer of speculation.*

Part II

Helper theorems for reasoning about Hyperseed concepts

28 Helper theorems for phenomenological primitives (Section 6)

Standing assumptions and notation. Fix a context $c = (X_c, \Pi_c, \delta_c, \iota_c)$ and a (commutative) p-bit quantale V with componentwise order $\leq$ and monotone commutative product $\otimes$. Write $v = (v^+, v^-) \in V$ for positive/negative evidence channels. Fix thresholds $\tau_\delta, \tau_\iota, \tau_I \in (0, 1)$. Recall:

$$\mathrm{Diff}_c(x \to y) := \delta_c(x, y), \qquad \mathrm{Dist}_c(x, y) := \mathrm{Diff}_c(x \to y) \otimes \mathrm{Diff}_c(y \to x),$$

and the crisp distinction $x \#_c y$ means $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$. Repetition/variety/non-duality/non-dual-variety are the threshold predicates of Defs. 42–43. Intensity is $I_c : X_c \to [0, 1]$, with $\mathrm{Focus}_c(\tau_I)$ and $\mathrm{Fringe}_c(\tau_I)$ as in Def. 49. An abstraction is a surjection $q : X_c \twoheadrightarrow A$ with preorder $q_1 \preceq q_2$ iff $\exists r : A_1 \to A_2$ such that $q_2 = r \circ q_1$ (Def. 51). A context translation is a function $T : X_c \to X_d$, and T is non-dually compatible if it does not erase strong δ-distinctions (Def. 53).

It is useful to keep in mind the qualitative role of the three thresholds. The parameter τ_δ governs when directed evidential differences "count" as a stable, symmetric separation at the level of Dist_c; it is the place where the evidence-valued notion is intentionally collapsed to a crisp predicate $x \#_c y$ only for the purpose of stating combinatorial claims. Similarly, τ_ι controls when the ι_c-based predicates (Defs. 42–43) transition from graded evidence to crisp repetition/variety conditions, while τ_I is the intensity cutoff that turns the scalar field I_c into the two regions $\mathrm{Focus}_c(\tau_I)$ and $\mathrm{Fringe}_c(\tau_I)$. In later lemmas, monotonicity properties typically show that if a statement holds at some threshold, then it continues to hold at weaker thresholds (e.g. lowering τ_δ makes $x \#_c y$ easier to satisfy), whereas strengthening a threshold expresses a more demanding notion of "strong evidence".

Also note that Dist_c is built to enforce a "two-way" requirement: even when δ_c is highly directed, $\mathrm{Dist}_c(x, y)$ aggregates the evidence for x differing from y together with the evidence for y differing from x. Thus Dist_c should be read as a symmetry-enforcing composite, while Diff_c is the primitive directed quantity. The helper results in this section repeatedly exploit exactly this separation: asymmetry is allowed at the Diff level, but many of the later predicates are deliberately phrased using Dist in order to avoid baking in any a priori directionality.

Remark 1557. *This subsection of helper theorems is a kind of "elementary geometry" for the phenomenological primitives: it records the algebraic consequences of how* Difference *(Hyperseed-Concept 96),* Distinction *(Hyperseed-Concept 98),* Repetition *(Hyperseed-Concept 155),* Variety *(Hyperseed-Concept 199),* Non-Duality *(Hyperseed-Concept 121), and* Non-Dual Variety *(Hyperseed-Concept 120) were defined earlier. The guiding idea is that these are not treated as crisp, classical predicates from the outset; rather, they are built from evidence-bearing quantities taking values in a p-bit quantale, i.e. a two-channel structure that can simultaneously support evidence* for *and evidence* against *the same claim. This is one clean way to keep faith with paraconsistent intuitions without abandoning formal discipline (see, e.g., constructive/paraconsistent semantics as discussed in [23] and the resonance-motivated uses in [24]; for the broader Hyperseed framing see [1]).*

A practical consequence of this "two-channel" setup is that later statements can be formulated so that they only depend on the +-channel (as with $x \#_c y$), or depend on both channels when non-duality is at stake. In particular, one should not read the presence of a large v^- component as a contradiction in the classical sense; rather, it represents structured counterevidence that is retained rather than forced into a binary truth value. This is why many helper lemmas are phrased as monotonicity or preservation results: they guarantee that the constructions used to define phenomenological predicates do not behave erratically when evidence is strengthened in either channel.

Remark 1558. *A few notational conventions are worth making explicit the first time they are used in this helper section.*

- *The arrow in* $\mathrm{Diff}_c(x \to y)$ *is* not *a function type; it is a suggestive way of marking that δ_c is generally* directed*: $\delta_c(x, y)$ need not equal $\delta_c(y, x)$.*
- *An element $v \in V$ is written as $v = (v^+, v^-)$, where v^+ and v^- are the positive and negative evidence coordinates. When we write inequalities like $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$, we mean: take the +-component of the p-bit value and compare it to the scalar threshold.*
- *"Componentwise order" means: $(a^+, a^-) \leq (b^+, b^-)$ iff $a^+ \leq b^+$ and $a^- \leq b^-$ (in the underlying order on each component). This matters because it guarantees that threshold comparisons behave monotonically under $\leq$.*
- *"Monotone commutative product" means: $u \leq u'$ and $v \leq v'$ implies $u \otimes v \leq u' \otimes v'$, and also $u \otimes v = v \otimes u$. The commutativity will be the only property used in Lemma 6; monotonicity will be the key engine in Lemma 18.*

For a concrete mental model, one may keep in mind the canonical example $V = [0, 1]^2$ with componentwise order and a componentwise multiplicative $\otimes$, though the lemmas below intentionally avoid committing to that special case.

Two further interpretive points will help when reading the proofs that follow. First, in the canonical picture $V = [0, 1]^2$, the multiplication-like behavior of $\otimes$ makes $\mathrm{Dist}_c(x, y)$ act like an "AND"-style aggregator: both directions $x \to y$ and $y \to x$ must carry substantial positive evidence in order for $\mathrm{Dist}_c(x, y)^+$ to be large. This is why the crisp predicate $x \#_c y$ is naturally phrased in terms of Dist_c rather than Diff_c: it encodes the idea that a distinction is not merely a one-way detectability but a mutual separation.

Second, the preorder on abstractions (Def. 51) is best read as "q_2 is at least as coarse as q_1". Indeed, if $q_2 = r \circ q_1$, then any two points identified by q_1 are necessarily identified by q_2 after applying r, so q_2 forgets at least as much as q_1. Several helper lemmas later on implicitly use this coarsening intuition when relating repetition/variety properties across different quotients of the same underlying X_c.

Finally, the notion of non-dual compatibility for translations (Def. 53) can be viewed as a minimal invariance requirement: a translation T is allowed to change intensities, reorder elements, or even merge weakly distinguished items, but it should not collapse pairs that were already separated by strong positive evidence at the δ-level. In that sense, Lemma 18 will function as a "functoriality" statement for the geometry induced by δ_c, ensuring that the basic combinatorics of strong distinctions survive passage between contexts whenever T is intended to respect phenomenological structure.

28.1 Difference, distinction, repetition, variety, and non-duality

Remark 1559. *These first lemmas are "sanity checks" in the best sense: they verify that the formal primitives behave as their names suggest. The theme is that while δ_c is directed (so difference can be asymmetric), the derived quantity Dist_c is engineered to be symmetric by combining both directions. This separation mirrors a philosophical caution: one may experience x as differing-from y in a way that is not reciprocated, yet the* mutual *relation of distinction ought to reflect a bilateral comparison. In the background, the thresholds (e.g. τ_δ and τ_ι) serve as explicit "decision levels" that turn graded, possibly multi-channel evidence into crisp assertions; the lemmas below confirm that these decision levels interact with the primitives in the intended minimal ways.*

Lemma 6 (Symmetry of mutual distinction)**.** *For all $x, y \in X_c$, $\mathrm{Dist}_c(x, y) = \mathrm{Dist}_c(y, x)$. Hence the induced crisp relation $\#_c$ is symmetric.*

Remark 1560. *Intuitively, the lemma says: even if "x points away from y" differs in strength from "y points away from x," once we define* mutual *distinction by multiplying the two directed evidences, the order of the two factors no longer matters. This is important because $\#_c$ is used as a crisp proxy for "the context actually makes a distinction here": crisp distinctions should not depend on which element is named first. A related point is that this symmetry is achieved without requiring δ_c itself to be symmetric; instead, it is the* construction *of Dist_c that enforces reciprocity at the level where later relational predicates will operate.*

Sketch. By definition, $\mathrm{Dist}_c(x, y) = \delta_c(x, y) \otimes \delta_c(y, x)$. Since $\otimes$ is commutative, this equals $\delta_c(y, x) \otimes \delta_c(x, y) = \mathrm{Dist}_c(y, x)$. For crisp symmetry: $\mathrm{Dist}_c(x, y)^+ = \mathrm{Dist}_c(y, x)^+$, so $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$ iff $\mathrm{Dist}_c(y, x)^+ \geq \tau_\delta$. ☐

Remark 1561. Proof sketch and intuition. *The strategy is to unfold the definition of Dist_c and use only one algebraic axiom: commutativity of $\otimes$. Geometrically (in the heuristic $[0, 1]^2$ picture), one can imagine $\delta_c(x, y)$ and $\delta_c(y, x)$ as two "directed arrows" with lengths in each evidence channel; taking $\otimes$ is a way of insisting that mutual distinction is strong only when both arrows support it. Swapping the endpoints swaps the arrows, but the combined strength stays the same. From the standpoint of later constructions, this is also what makes $\#_c$ a well-behaved undirected edge relation on X_c: it can be used to define graphs, neighborhoods, and partitions without having to keep track of an orientation that would otherwise be arbitrary at the level of crisp distinguishability.*

Lemma 7 (Repetition/variety imply a made distinction)**.** *For all $x \neq y$:*

$$\mathrm{Rep}_c(x, y) \;\to\; x \#_c y, \qquad \mathrm{Var}_c(x, y) \;\to\; x \#_c y.$$

Remark 1562. *This lemma expresses a design constraint baked into the definitions:* repetition *and* variety *are only meaningful when there is at least some robust distinction being made between*

x and y in the first place. Said differently: the ontology does not allow "repetition" to collapse into mere identity, nor "variety" into unstructured difference; both are modes of relation under distinction*. This will matter later when non-duality is allowed to create simultaneous support for seemingly opposing relational predicates. One can also read the condition $x \neq y$ as a guardrail against trivial self-comparisons: even if a formal system could assign nonzero directed difference to (x, x) in degenerate cases, the intended phenomenological reading treats repetition/variety as relations between two items presented as distinct positions in experience.*

Sketch. Both $\mathrm{Rep}_c(x, y)$ and $\mathrm{Var}_c(x, y)$ include the conjunct $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$ by definition; that conjunct is exactly $x \#_c y$. □

Remark 1563. Proof sketch and intuition. *The proof is purely definitional: it observes that $\#_c$ is literally a named subcondition of* Rep_c *and* Var_c*. The conceptual picture is that repetition/variety are "decorations" on top of a basal fact of distinguishability: without the basal fact, the decoration is not even syntactically permitted. This definitional dependence is intentional: it ensures that later arguments can freely use $x \#_c y$ whenever either* $\mathrm{Rep}_c(x, y)$ *or* $\mathrm{Var}_c(x, y)$ *is assumed, without separately re-checking a distinction threshold.*

Lemma 8 (Non-dual variety yields simultaneous repetition and variety under distinction)**.** *If $x \neq y$ and* $\mathrm{NDVar}_c(x, y)$ *and* $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$*, then* $\mathrm{Rep}_c(x, y)$ *and* $\mathrm{Var}_c(x, y)$ *both hold.*

Remark 1564. *This lemma makes precise a subtle but central phenomenological possibility: a pair (x, y) can be unmistakably distinguished and yet also carry strong evidence for both "repetition" and "variety." In ordinary classical logic, one is tempted to treat these as mutually exclusive labels; the present framework refuses that temptation by allowing a controlled, thresholded form of co-presence. One may read* $\mathrm{NDVar}_c(x, y)$ *as asserting that the indistinction signal $\iota_c(x, y)$ is* bi-valent*: it is simultaneously high in the positive and negative evidence channels, and therefore can support both readings when combined with a baseline distinction. In particular, the lemma emphasizes that non-dual variety does not erase distinction: the hypothesis explicitly retains* $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$*, so the "both at once" reading is not a collapse into undifferentiated identity, but rather an instance of jointly sustained, potentially tension-bearing relational structure.*

Sketch. $\mathrm{NDVar}_c(x, y)$ means $\iota_c(x, y)^+ \geq \tau_\iota$ and $\iota_c(x, y)^- \geq \tau_\iota$. Together with $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$, these are exactly the defining conjuncts of both $\mathrm{Rep}_c(x, y)$ (uses ι^+) and $\mathrm{Var}_c(x, y)$ (uses ι^-). □

Remark 1565. Proof sketch and intuition. *Again the proof proceeds by unpacking conjunctive definitions. The key step is noticing that* Rep_c *and* Var_c *differ only in which channel of ι_c they consult. Thus, when* NDVar_c *asserts that* both *channels are above threshold, it automatically supplies the missing half needed for each predicate. A useful mental image is a two-axis meter for ι: the point lies in the "upper-right quadrant," so both one-sided tests pass. One may also view this as a formal version of a familiar experiential report: two moments can be clearly told apart (high* Dist_c^+*) while still being "the same again" in one respect (captured by ι^+ crossing threshold) and "genuinely different" in another respect (captured by ι^- crossing threshold), with the framework explicitly allowing these to be simultaneously licensed rather than forcing a premature choice.*

Lemma 9 (Consistency condition makes repetition and variety disjoint)**.** *Assume $x \neq y$ and* $\neg\mathrm{NDVar}_c(x, y)$*. Then* $\neg(\mathrm{Rep}_c(x, y) \wedge \mathrm{Var}_c(x, y))$*.*

Remark 1566. *This lemma articulates the complementary regime to Lemma 8: if non-dual variety is* not *present, then the framework behaves more like a classical either/or with respect to the pair of predicates* Rep_c *and* Var_c*. Formally,* $\neg\mathrm{NDVar}_c(x, y)$ *means that it is* not *the case that both* *evidence channels of $\iota_c(x, y)$ simultaneously clear the threshold τ_ι. Consequently, one cannot meet the definitional requirements for repetition (which needs $\iota_c(x, y)^+ \geq \tau_\iota$) and for variety (which needs $\iota_c(x, y)^- \geq \tau_\iota$) at the same time. The role of $x \neq y$ is again to keep the statement aligned with the intended domain of application of these predicates.*

Sketch. Assume for contradiction that $\mathrm{Rep}_c(x, y) \wedge \mathrm{Var}_c(x, y)$ holds. Unpacking definitions, $\mathrm{Rep}_c(x, y)$ supplies $\iota_c(x, y)^+ \geq \tau_\iota$ (and the shared distinction conjunct), while $\mathrm{Var}_c(x, y)$ supplies $\iota_c(x, y)^- \geq \tau_\iota$ (and the shared distinction conjunct). Therefore both $\iota_c(x, y)^+ \geq \tau_\iota$ and $\iota_c(x, y)^- \geq \tau_\iota$ hold, i.e. $\mathrm{NDVar}_c(x, y)$ holds, contradicting the assumption $\neg\mathrm{NDVar}_c(x, y)$. Hence $\neg(\mathrm{Rep}_c(x, y) \wedge \mathrm{Var}_c(x, y))$. □

Remark 1567. Proof sketch and intuition. *The argument is a straightforward contrapositive-style use of the definitions: if both repetition and variety were simultaneously true, then the positive and negative channels of ι_c would both have to be above threshold, which is precisely what* NDVar_c *asserts. Thus* $\neg\mathrm{NDVar}_c$ *functions as a* consistency constraint *preventing the co-presence of the two predicates. Conceptually, the lemma clarifies that non-duality is not an automatic background feature of the system; rather, it is an explicit, checkable condition. When it fails, the framework reverts to a more exclusive reading in which the same pair cannot be certified as both "repetition" and "variety" at once.*

Remark 1568. *Where Lemma 8 shows how coexistence is possible, the present lemma shows how coexistence is* controlled. *If* $\mathrm{NDVar}_c(x, y)$ *fails, then at least one of the two evidence channels of* $\iota_c(x, y)$ *is not strong enough, and therefore the theory prevents both* Rep_c *and* Var_c *from holding simultaneously. In this way, the system distinguishes (i) genuine non-dual superposition (both channels strong) from (ii) ordinary situations where one channel dominates. In particular, the control mechanism is not a separate axiom but is enforced by the same thresholding convention used throughout: "coexistence" is permitted only when* each *requisite inequality clears the stipulated bar (here τ_ι), so a single weak channel is sufficient to block the simultaneous attribution. This provides an operational reading of "dominance": it is precisely the regime in which only one of $\iota_c(x, y)^+$ and $\iota_c(x, y)^-$ is above threshold, so the model records an asymmetric evidential profile rather than a balanced one.*

Sketch. If $\mathrm{Rep}_c(x, y) \wedge \mathrm{Var}_c(x, y)$ held, then by definitions we would have $\iota_c(x, y)^+ \geq \tau_\iota$ and $\iota_c(x, y)^- \geq \tau_\iota$, which is exactly $\mathrm{NDVar}_c(x, y)$, contradicting the assumption. Equivalently, one can view the argument as unpacking the conjunctive structure of the predicates: the conjunction $\mathrm{Rep}_c(x, y) \wedge \mathrm{Var}_c(x, y)$ forces *both* of the requisite lower bounds on the two coordinates of $\iota_c(x, y)$, and that joint lower-boundedness is precisely the definitional content of $\mathrm{NDVar}_c(x, y)$. □

Remark 1569. Proof sketch and intuition. *This is a standard contrapositive pattern: assume both predicates hold and derive the defining condition for* NDVar_c. *Conceptually, it says that the only route to "both repetition and variety" is the explicitly named non-dual-variety regime. Thus paraconsistency is not an ambient fog; it is localized and testable via thresholds. A further conceptual payoff is that the lemma makes the interaction between predicates diagnostically reversible: if an application ever yields credible evidence for both* Rep_c *and* Var_c*, then the framework commits you to the stronger claim that both evidence channels of ι_c are simultaneously high, rather than allowing an unprincipled mixture of partial conditions. This is exactly the sense in which the coexistence is "controlled": the model does not merely tolerate tension, it specifies the unique evidential pattern under which tension is licensed.*

Lemma 10 (Non-duality entails paraconsistent distinctness, but does not force equality). *If* $\mathrm{NonDual}_c(x, y)$ *then* $\mathrm{Dist}_c(x, y)^+ \geq \tau_\delta$ *and* $\mathrm{Dist}_c(x, y)^- \geq \tau_\delta$ *simultaneously (hence* $x \#_c y$*), i.e. the system carries strong evidence for and against the distinction. Moreover, the conclusion is explicitly two-sided: it asserts the co-presence of high positive and high negative evidence about the same distinction claim, not an averaged or reconciled value. In particular, the lemma is compatible with* x *and* y *remaining syntactically separate items in the language and, semantically, with later updates that may lower one channel without forcing any retroactive identification.*

Remark 1570. *This lemma states, in a deliberately careful way, what non-duality is* not. *It is not the collapse of two items into literal identity; it is the simultaneous presence of strong evidence on both sides of a distinction. In paraconsistent terms, the distinction is both supported and opposed at the evidence level; classical explosion is avoided because we do not equate "evidence for* $x = y$*" with the syntactic rewrite rule "replace* x *by* y*" (compare the motivations for paraconsistent semantics in [23] and the resonance-oriented perspective in [24]). Stated differently, the framework treats "distinctness" as a graded, two-channel report rather than as a classical bivalent predicate:* $\mathrm{Dist}_c(x, y)^+$ *and* $\mathrm{Dist}_c(x, y)^-$ *are tracked independently, and non-duality corresponds to both being large enough to pass* τ_δ*. The notation* $x \#_c y$ *is therefore best read as "paraconsistently distinguished in context* c*" rather than as a commitment to either* $x \neq y$ *or* $x = y$ *in the classical sense, and the lemma highlights that this is a stable middle position rather than a temporary inconsistency to be immediately resolved.*

Sketch. Immediate from the defining conjuncts of $\mathrm{NonDual}_c(x, y)$. The extra point for implementation is that this is evidence-level paraconsistency, not syntactic equality: one should not rewrite $x = y$ from $\mathrm{NonDual}_c(x, y)$. Concretely, if $\mathrm{NonDual}_c(x, y)$ is defined by the conjunction of the two threshold conditions on $\mathrm{Dist}_c(x, y)^+$ and $\mathrm{Dist}_c(x, y)^-$ (possibly together with auxiliary side conditions such as $x \neq y$), then the conclusion follows by direct projection onto those conjuncts, without any additional inference principles. □

Remark 1571. Proof sketch and intuition. *Formally, nothing more is happening than reading off the conjuncts in the definition of* $\mathrm{NonDual}_c$*. The interpretive force lies in the last sentence: the algebra remembers two channels of evidence, so it can represent the felt "both/and" without collapsing the computational universe by identification. A helpful visual is to imagine two dials, one for* Dist^+ *and one for* Dist^-*; non-duality means both dials are high, not that the two entities have been merged. One can also read the lemma as a guardrail on downstream reasoning: any subsequent rule that requires a* univocal *distinction should check for a one-sided condition (e.g. only* Dist^+ *being high, or only* Dist^- *being low), whereas rules that are explicitly designed for non-dual regimes may safely assume the simultaneous high-high configuration. In this way the lemma clarifies which inferences are legitimate in a non-dual context and why the framework refrains from identifying items even when the phenomenology suggests a strong "togetherness".*

Lemma 11 (Threshold monotonicity of the primitive predicates). *Let* $\tau'_\delta \leq \tau_\delta$ *and* $\tau'_\iota \leq \tau_\iota$*. Then for all* $x \neq y$*,*

$$\mathrm{Rep}_c^{(\tau_\delta, \tau_\iota)}(x, y) \to \mathrm{Rep}_c^{(\tau'_\delta, \tau'_\iota)}(x, y),$$

and similarly for Var_c*,* $\mathrm{NonDual}_c$*,* NDVar_c*. In other words, the parameterization by thresholds is monotone in the expected direction: weakening the acceptance criteria (by lowering thresholds) can only expand, never shrink, the extension of each predicate. The restriction* $x \neq y$ *ensures that the claim is about the intended "two-item" use of these predicates, and prevents any degenerate boundary case for which the predicates might have special conventions when* $x = y$*.*

Remark 1572. *This lemma confirms that thresholds behave the way one expects: lowering the bar cannot invalidate a previously valid claim. It is not deep, but it is structurally important because it lets one compare contexts or agents that choose different strictness levels. In particular, any later theorem that is proved at stringent thresholds automatically holds (in the same direction) for more permissive thresholds, which is often the right notion of robustness for phenomenological predicates. A common use is "calibration transfer": if one analyst uses conservative settings* $(\tau_\delta, \tau_\iota)$ *and another uses more permissive settings* $(\tau'_\delta, \tau'_\iota)$*, then any instance certified by the conservative analyst is automatically certified by the permissive one. The lemma thus supports principled comparisons across experimental regimes, interpretive stances, or modeling choices without requiring any further assumptions about the underlying evidence values beyond their order structure.*

Sketch. All four predicates are conjunctions of lower bounds of the form $u \geq \tau$. If a value exceeds a larger threshold, it exceeds any smaller threshold. More explicitly, suppose $\mathrm{Rep}_c^{(\tau_\delta,\tau_\iota)}(x, y)$ holds. Unpacking its definition yields a finite list of inequalities, each of which compares some evidence coordinate (e.g. a component of Dist_c or ι_c) to either τ_δ or τ_ι. Replacing τ_δ by the smaller τ'_δ and τ_ι by the smaller τ'_ι preserves truth of each inequality, hence of their conjunction, giving $\mathrm{Rep}_c^{(\tau'_\delta,\tau'_\iota)}(x, y)$. The other predicates follow identically. ☐

Remark 1573. Proof sketch and intuition. *The argument uses only the transitivity of* $\geq$ *on* $[0, 1]$ *(or whatever ordered set the components live in). The key step is recognizing that each predicate is assembled from statements of the form "some evidence coordinate is at least a threshold," so monotonicity in thresholds is inherited componentwise. It is worth noting that no continuity, additivity, or probabilistic interpretation is required: the lemma is purely order-theoretic. Consequently, it remains valid under many possible implementations of evidence coordinates (e.g. scores, normalized measures, or bounded utilities), provided only that the comparison relation is a partial or total order with the usual transitivity property.*

28.2 Presentational immediacy: focus and fringe

Remark 1574. *We now move from relational predicates to an internal scalar field* $I_c : X_c \to [0, 1]$ *encoding* intensity *in a context. The induced sets* $\mathrm{Focus}_c(\tau_I)$ *and* $\mathrm{Fringe}_c(\tau_I)$ *formalize a simple attentional phenomenology: some items are in the bright center of awareness, others are faintly present at the periphery. This connects directly to* Presentational Immediacy *(Hyperseed-Concept 138) and* Attention / Attentional Focus *(Hyperseed-Concept 60).*

Remark 1575. *The only structure assumed here is that each context c comes with a carrier set* X_c *and a numerical intensity assignment* I_c*. In particular, no topology, metric, similarity relation, or ordering on* X_c *is required in order to state (and later reuse) the basic focus/fringe facts. All comparisons are pushed into the codomain* $[0, 1]$*, so the subsequent lemmas are essentially consequences of elementary order properties of real numbers.*

Remark 1576. *The threshold parameter* τ_I *should be read as a* convention for discretizing *a graded phenomenon. Different choices of* τ_I *correspond to different attentional "grain sizes": low thresholds treat many items as focused, while high thresholds reserve focus for only the most intense items. This explicit dependence on* τ_I *is useful later when one wants to model shifts of attention without changing the underlying intensity field* I_c *itself.*

Lemma 12 (Focus/fringe are disjoint)**.** $\mathrm{Focus}_c(\tau_I) \cap \mathrm{Fringe}_c(\tau_I) = \emptyset$.

Remark 1577. *The lemma says there is no ambiguous "half-in, half-out" element: at a fixed threshold τ_I, an item cannot be both above threshold (focus) and strictly below it while still positive (fringe). This is important because later constructions may treat focus and fringe as distinct computational roles (e.g. a focus-like set receiving expensive processing while the fringe is handled heuristically).*

Remark 1578. *Note also that the disjointness hinges on the* asymmetric boundary convention*: focus is defined using $\geq \tau_I$ while fringe uses $< \tau_I$ (together with > 0). Thus an item with $I_c(x) = \tau_I$ is classified as focused rather than fringed, which matches the idea that "meeting the criterion" counts as being in the attended set. Other conventions (e.g. making both sets closed or both open at τ_I) would require separate handling of the boundary case and can introduce unwanted ambiguity.*

Sketch. If x were in both, we would have $I_c(x) \geq \tau_I$ and $0 < I_c(x) < \tau_I$, contradiction. ☐

Remark 1579. Proof sketch and intuition. *The proof is a direct contradiction: it combines the defining inequalities for membership in each set. Visually, if one plots $I_c(x)$ on the unit interval,* Focus *is the right-closed ray $[\tau_I, 1]$ while* Fringe *is the open interval $(0, \tau_I)$; these do not overlap.*

Remark 1580. *An additional way to see the same point is to observe that, for each fixed τ_I, the predicates*

$$x \in \mathrm{Focus}_c(\tau_I) \iff I_c(x) \in [\tau_I, 1], \qquad x \in \mathrm{Fringe}_c(\tau_I) \iff I_c(x) \in (0, \tau_I)$$

pull back two disjoint subsets of $[0,1]$ along the function I_c. No properties of I_c beyond being single-valued are needed for this pullback-disjointness argument.

Lemma 13 (Focus/fringe partition the positive-intensity region)**.** *Let $P_c := \{x \in X_c : I_c(x) > 0\}$. Then $P_c = \mathrm{Focus}_c(\tau_I) \,\dot\cup\, \mathrm{Fringe}_c(\tau_I)$ (disjoint union).*

Remark 1581. *This lemma states that, among whatever is present at all (positive intensity), the threshold τ_I induces an exhaustive dichotomy: everything present is either salient enough to be in focus or weakly present enough to be in fringe. The restriction to P_c matters: elements with $I_c(x) = 0$ are treated as absent rather than as "infinitely fringe."*

Remark 1582. *Operationally, one can read P_c as the set of "currently activated" or "presented" items, with focus and fringe giving a two-tier stratification of that active set. This avoids forcing a trichotomy in which $I_c(x) = 0$ would have to be assigned to one of the two attentional tiers; instead, $I_c(x) = 0$ is reserved for non-presentation. In many applications, this makes later aggregation steps cleaner: sums, averages, or normalizations over "present" items can be taken over P_c without inadvertently counting absences.*

Sketch. If $I_c(x) > 0$ then either $I_c(x) \geq \tau_I$ (so $x \in$ Focus) or $I_c(x) < \tau_I$ (so $x \in$ Fringe). Disjointness is Lemma 12. ☐

Remark 1583. Proof sketch and intuition. *The proof is a case split on whether a real number is at least τ_I or below it. One can think of τ_I as a moving "attentional horizon": everything above the horizon is foreground, everything below (but not zero) is background.*

Remark 1584. *At the extremes, the statement behaves as expected (when the definitions are instantiated): if $\tau_I = 0$, then* $\mathrm{Focus}_c(\tau_I)$ *coincides with P_c and* $\mathrm{Fringe}_c(\tau_I)$ *is empty, corresponding to an "everything present is focused" idealization. If $\tau_I = 1$, then* $\mathrm{Focus}_c(\tau_I)$ *consists precisely of items at maximal intensity (those with $I_c(x) = 1$), and all other present items lie in the fringe. These boundary cases can be useful as sanity checks when calibrating τ_I in examples.*

Lemma 14 (Monotonicity in the focus threshold). *If $\tau_1 \leq \tau_2$ then*

$$\mathrm{Focus}_c(\tau_2) \subseteq \mathrm{Focus}_c(\tau_1), \qquad \mathrm{Fringe}_c(\tau_1) \subseteq \mathrm{Fringe}_c(\tau_2).$$

Remark 1585. *Raising the focus threshold makes focus smaller (harder to qualify) and fringe larger (easier to qualify), exactly as one would expect. This monotonicity is useful when comparing different "modes" of the same agent or different agents: an agent with a stricter focus criterion is literally focusing on fewer items, in the precise sense of set inclusion.*

Remark 1586. *Formally, Lemma 14 says that the assignment $\tau \mapsto \mathrm{Focus}_c(\tau)$ is* antitone *(order-reversing) in τ, while $\tau \mapsto \mathrm{Fringe}_c(\tau)$ is* monotone *(order-preserving). Thus, as τ varies, the focus sets form a nested family (a decreasing "filtration" by thresholds), and the fringe sets form a nested family in the opposite direction. This nestedness is often what one needs when proving stability statements under changes of attentional resolution.*

Sketch. If $I_c(x) \geq \tau_2$ and $\tau_1 \leq \tau_2$ then $I_c(x) \geq \tau_1$. If $0 < I_c(x) < \tau_1$ and $\tau_1 \leq \tau_2$ then $0 < I_c(x) < \tau_2$. □

Remark 1587. Proof sketch and intuition. *The argument is again order-theoretic on* $[0, 1]$. *On a number line, moving the threshold right shrinks the $\geq$-region and expands the $<$-region; the inclusions are just that picture translated into set language.*

Remark 1588. *In applications, one sometimes varies τ_I dynamically (e.g. to model fatigue, time pressure, or deliberate concentration). Lemma 14 then provides immediate "directional" guarantees: any item that remains focused under a stricter threshold must also have been focused before the shift, and any item that was in the fringe under a looser threshold will remain in the fringe when the criterion is tightened. These implications can be used to justify reusing previously computed information for items that are guaranteed not to have changed attentional tier.*

28.3 Abstraction and concreteness

Remark 1589. *Abstraction here is treated as a controlled* forgetting*: a surjection $q : X_c \twoheadrightarrow A$ partitions X_c into fibers, and elements in the same fiber are "identified" at the abstract level. The preorder $q_1 \preceq q_2$ says: q_2 is at least as abstract as q_1 (it can be obtained by further collapsing the outputs of q_1). This is a standard idea in mathematics (factorization through quotients), but in Hyperseed it is also a disciplined way of discussing* Abstract *vs.* Concrete *(Hyperseed-Concepts 51 and 83) within a fixed context (Hyperseed-Concept 86).*

Remark 1590. *It is sometimes helpful to keep in mind the quotient-set picture: a surjection $q : X_c \twoheadrightarrow A$ is (up to bijection of codomains) the same data as an equivalence relation $\sim$ on X_c, where $x \sim y$ iff $q(x) = q(y)$, together with a choice of* names *for the equivalence classes. From this viewpoint, "controlled forgetting" means replacing X_c by the set of its $\sim$-classes, and the order relation $\preceq$ compares how much information is retained by comparing the induced partitions.*

Lemma 15 (The abstraction preorder is a preorder). *The relation $\preceq$ on abstractions $q : X_c \twoheadrightarrow A$ is reflexive and transitive.*

Remark 1591. *The point is modest but foundational: we can reason about "more abstract than" in the usual order-theoretic way, without worrying that the relation fails basic coherence conditions. Once this is established, one may meaningfully talk about chains of abstractions, maximal elements under admissibility constraints, and so on.*

Remark 1592. *Note that a preorder need not be antisymmetric: it can happen that $q_1 \preceq q_2$ and $q_2 \preceq q_1$ without $q_1 = q_2$ as functions, for example when q_1 and q_2 induce the same partition of X_c but use different codomains or different labels for the same fibers. Thus, the correct "equality notion" implicit in $\preceq$ is often equality of induced equivalence relations (or isomorphism of quotients) rather than literal equality of maps.*

Sketch. Reflexive: $q = \mathrm{id}_A \circ q$. Transitive: if $q_2 = r_{12} \circ q_1$ and $q_3 = r_{23} \circ q_2$, then $q_3 = (r_{23} \circ r_{12}) \circ q_1$. □

Remark 1593. Proof sketch and intuition. *Reflexivity uses the trivial observation that composing with an identity changes nothing. Transitivity uses associativity of function composition: if q_1 factors to q_2 and q_2 factors to q_3, then q_1 factors directly to q_3. In terms of partitions, refining (collapsing) and then refining again is the same as refining once by the composite collapse.*

Lemma 16 (Factorization iff inclusion of induced equivalence relations)**.** *Let $q_i : X_c \twoheadrightarrow A_i$ and define $x \sim_{q_i} y \iff q_i(x) = q_i(y)$. Then*

$$q_1 \preceq q_2 \quad \iff \quad (\sim_{q_1} \subseteq \sim_{q_2}).$$

Equivalently: $q_1 \preceq q_2$ iff for all x, y, $q_1(x) = q_1(y) \to q_2(x) = q_2(y)$.

Remark 1594. *This lemma gives an extremely useful reformulation: instead of explicitly producing a map r with $q_2 = r \circ q_1$, we can check an inclusion between the equivalence relations (partitions) induced by q_1 and q_2. Intuitively, "q_2 is coarser" means: whenever q_1 fails to distinguish two points (puts them in the same fiber), q_2 must also fail to distinguish them. Thus, the lemma translates a statement about existence of a factor map into a purely relational statement about fibers.*

Remark 1595. *Two small technical clarifications are often useful when applying Lemma 16. First, surjectivity of q_1 is exactly what guarantees that the definition of $r : A_1 \to A_2$ by "pick an x with $q_1(x) = a$" is total. Second, the factor map r is then forced (and hence unique) by the requirement $q_2 = r \circ q_1$: for each $a \in A_1$, $r(a)$ must be the common value of q_2 on the fiber $q_1^{-1}(a)$ (well-definedness is the only real issue).*

Sketch. ($\to$) If $q_2 = r \circ q_1$ and $q_1(x) = q_1(y)$ then $q_2(x) = r(q_1(x)) = r(q_1(y)) = q_2(y)$. ($\Leftarrow$) Assume q_2 is constant on fibers of q_1. Define $r : A_1 \to A_2$ by choosing any x with $q_1(x) = a$ (exists since q_1 is surjective) and setting $r(a) := q_2(x)$. Well-definedness follows from the fiber-constancy assumption. Then $q_2 = r \circ q_1$. □

Remark 1596. Proof sketch and intuition. *The forward direction is immediate: composing with r cannot split a fiber that q_1 has already collapsed. The reverse direction is the more instructive step: if q_2 does not distinguish within any fiber of q_1, then each fiber-label $a \in A_1$ can be sent to the common value of q_2 on that fiber, producing the required r. The only delicate point is well-definedness, and it is secured precisely by the assumption that q_2 is constant on fibers.*

Corollary 4 (Fiber inclusion along $\preceq$)**.** *If $q_1 \preceq q_2$, then for all $x \in X_c$,*

$$q_1^{-1}(q_1(x)) \ \subseteq \ q_2^{-1}(q_2(x)).$$

Remark 1597. *This corollary states the same "coarsening" idea at the level of individual fibers: the equivalence class of x under the finer abstraction is contained in the equivalence class of x under the coarser abstraction. This is often the most directly used form in arguments about concreteness constraints, because many such constraints are phrased in terms of fiber sizes.*

Remark 1598. *In particular, whenever a size notion is monotone under inclusion (e.g. cardinality, measure, diameter bounds stated as "at most", or any predicate* Small *that is downward closed as in Lemma 17), Corollary 4 immediately transports such statements from a coarser fiber to a finer one, and contrapositively transports "not small" statements from a finer fiber to a coarser one. This is the basic mechanism behind many "abstractness persists under further abstraction" arguments.*

Sketch. If $y \in q_1^{-1}(q_1(x))$, then $q_1(y) = q_1(x)$. By Lemma 16, this implies $q_2(y) = q_2(x)$, hence $y \in q_2^{-1}(q_2(x))$. $\square$

Remark 1599. Proof sketch and intuition. *Pick any point y identified with x by q_1. Since q_2 is coarser, it must also identify y with x. In partition language: every block of the fine partition sits inside a block of the coarse partition.*

Lemma 17 (Abstractness propagates to coarser admissible abstractions (mild assumption))**.** *Fix a concreteness constraint K_c and a predicate* $\mathrm{Small}(\cdot)$ *on subsets of X_c. Assume* Small *is downward closed: $A \subseteq B$ and* $\mathrm{Small}(B)$ *implies* $\mathrm{Small}(A)$*. Let $x \in X_c$ be abstract in the sense of Def. 52, witnessed by some K_c-admissible q_1 such that $\neg\mathrm{Small}(q_1^{-1}(q_1(x)))$. If $q_1 \preceq q_2$ and q_2 is also K_c-admissible, then x is also witnessed abstract by q_2.*

Remark 1600. *The "mild assumption" is the downward-closure of* Small*, which is exactly the property needed to ensure that "not small" is upward closed with respect to inclusion: if $A \subseteq B$ and A is not small, then B cannot be small. This matches the intended reading in typical cases: for instance, if* $\mathrm{Small}(S)$ *means "S has at most N elements" or "S has diameter at most ε", then any superset of a non-small set remains non-small.*

Sketch. By Corollary 4, from $q_1 \preceq q_2$ we have

$$q_1^{-1}(q_1(x)) \subseteq q_2^{-1}(q_2(x)).$$

Suppose for contradiction that $\mathrm{Small}(q_2^{-1}(q_2(x)))$ holds. Since Small is downward closed, the inclusion would imply $\mathrm{Small}(q_1^{-1}(q_1(x)))$, contradicting the witnessing condition $\neg\mathrm{Small}(q_1^{-1}(q_1(x)))$. Hence $\neg\mathrm{Small}(q_2^{-1}(q_2(x)))$. Together with the assumed K_c-admissibility of q_2, this makes q_2 a valid witness (in the sense of Def. 52) that x is abstract. $\square$

Remark 1601. Proof sketch and intuition. *Coarsening an abstraction can only make fibers larger (never smaller), so any "largeness" phenomenon exhibited by the q_1-fiber of x persists for the corresponding q_2-fiber. The admissibility hypotheses are logically separate: they ensure that both q_1 and q_2 are permitted abstractions under the same concreteness constraint K_c, so that the witness can legitimately be replaced by the coarser abstraction without leaving the intended model class.*

Remark 1602. *The content is: if an element is already "too coarse" (its fiber is not small) under some admissible abstraction, then making the abstraction even coarser cannot restore concreteness. This matches an intuitive irreversibility: once you have thrown away distinctions, you cannot recover them by throwing away even more. The only extra hypothesis is a weak regularity condition on what counts as "small": downward closure says that if a big set is small (which is unusual), then all its subsets are small; contrapositively, if a subset is not small then no superset can be small. This is exactly what the proof uses. Equivalently, the statement can be read as a monotonicity principle: "being abstract" is upward closed along coarsenings of the abstraction, because coarsening only ever enlarges the equivalence class of x (its fiber), and enlarging a set cannot turn a non-small set into a small one when smallness is downward closed. In this sense, the remark isolates the precise point where regularity of the size predicate enters: nothing about cardinality is needed, only the order-theoretic behavior of the predicate* $\mathrm{Small}(\cdot)$ *with respect to $\subseteq$.*

Sketch. By Corollary 4, $q_1^{-1}(q_1(x)) \subseteq q_2^{-1}(q_2(x))$. If $\mathrm{Small}(q_2^{-1}(q_2(x)))$ held, downward closure would imply $\mathrm{Small}(q_1^{-1}(q_1(x)))$, contradicting the witness. So $\neg\mathrm{Small}(q_2^{-1}(q_2(x)))$; hence q_2 also witnesses abstractness of x. In slightly more explicit contrapositive form: since $q_1^{-1}(q_1(x))$ is *not* small and is contained in $q_2^{-1}(q_2(x))$, the closure condition forces $q_2^{-1}(q_2(x))$ to be not small as well, because otherwise the smaller set would inherit smallness. The admissibility condition is used only to ensure that q_2 is among the abstractions one is allowed to consider as potential witnesses. □

Remark 1603. Proof sketch and intuition. *The proof is a simple order argument on sets. Step (1) uses the previously proved fiber inclusion: coarsening enlarges fibers. Step (2) uses downward closure in contrapositive form: "not small" propagates upward to supersets. Thus, the witness of abstractness persists along $\preceq$ so long as admissibility (K_c) is preserved. One can also view this as a stability property of counterexamples: once x has a fiber that is "too large to be concrete" under some q_1, any q_2 that forgets at least as much as q_1 must produce a fiber that contains that same counterexample, hence cannot pass the smallness test. This makes precise the informal slogan that loss of distinctions is one-way: refinement may restore concreteness, but coarsening cannot.*

28.4 Non-duality as a bridge between perspectives (context translations)

Remark 1604. *A context translation $T : X_c \to X_d$ formalizes the act of viewing the "same" situation through a different lens. The compatibility condition is deliberately phrased negatively: it forbids erasing strong distinctions. In cognitive terms, this is a minimal requirement for* transfer*: if a salient distinction in one context becomes invisible in the translated context, then predicates like repetition/variety/non-duality will not be stably comparable across perspectives. This resonates with the general transfer-learning motivation of Hyperseed (Hyperseed-Concepts 192 and 193) and the associated mathematical program (see [7] for broader context). At the level of interpretation, this means T is not required to preserve* all *structure (it may merge some states, rename features, or otherwise reinterpret them), but it is required to preserve whatever the theory treats as evidence of distinction. The point of stating the condition in terms of δ-inequalities is that it directly controls the quantities that feed into later definitions (e.g. mutual distinction, crisp thresholds, and the paraconsistent pairing that underlies non-duality), rather than imposing an external notion of similarity.*

Lemma 18 (Non-dually compatible translations preserve distinction evidence)**.** *Let $T : X_c \to X_d$ be non-dually compatible (Def. 53). Then for all $x, y \in X_c$,*

$$\mathrm{Dist}_c(x, y) \;\leq\; \mathrm{Dist}_d(Tx, Ty).$$

In particular, if $x \#_c y$ then $Tx \#_d Ty$ (using the same τ_δ). Moreover, $\mathrm{NonDual}_c(x, y)$ *implies* $\mathrm{NonDual}_d(Tx, Ty)$*. In other words, T is monotone with respect to the derived evidence ordering on mutual distinction: it cannot reduce the "two-way" support for distinguishing x from y. The threshold clause emphasizes that the result is not merely qualitative: it preserves whatever fixed operational cutoff τ_δ is used to declare a distinction to be crisp in the source context.*

Remark 1605. *In plain language: a non-dually compatible translation cannot make two things* less *mutually distinct (in the evidence order), so any crisp distinction is preserved. The second clause says that even the more delicate paraconsistent situation of non-duality is preserved: strong evidence both for and against distinction survives translation. This is a rigorous expression of a phenomenological aspiration: shifting perspective may reinterpret an experience, but it should not arbitrarily delete its salient tensions. Put differently, a translation may add nuance (e.g. by increasing evidence for distinction in one or both directions), but it may not "flatten" a distinction* *that was already present at or above the salient level. The non-duality preservation is especially important because it says the translation respects not only clean separations but also ambiguous or tension-laden pairings where both poles of evidence are simultaneously significant.*

Sketch. Non-dual compatibility gives (for all x, y) inequalities of the form $\delta_c(x, y) \leq \delta_d(Tx, Ty)$ and, by swapping x, y, also $\delta_c(y, x) \leq \delta_d(Ty, Tx)$. Monotonicity of $\otimes$ yields $\delta_c(x, y) \otimes \delta_c(y, x) \leq \delta_d(Tx, Ty) \otimes \delta_d(Ty, Tx)$, i.e. $\mathrm{Dist}_c(x, y) \leq \mathrm{Dist}_d(Tx, Ty)$. Componentwise order then preserves threshold exceedance in both channels, giving the crisp and non-dual consequences. Concretely, for the crisp clause: if $\mathrm{Dist}_c(x, y) \geq \tau_\delta$ (in the relevant ordering), then the inequality forces $\mathrm{Dist}_d(Tx, Ty) \geq \tau_\delta$ as well, because increasing a p-bit value cannot make it fall below a fixed threshold. For the non-dual clause, the same componentwise monotonicity applies simultaneously to the "distinguish" and "do-not-distinguish" components that define NonDual, so having both components above their respective thresholds in context c implies the same in context d. □

Remark 1606. Proof sketch and intuition. *The proof composes three monotonicities: (1) compatibility gives two directed inequalities for δ (one for each direction); (2) monotonicity of $\otimes$ lifts directed inequalities to an inequality about mutual distinction* Dist*; (3) componentwise order ensures that if a p-bit value increases, then any threshold condition that held before still holds after. One may picture T as an embedding that may stretch distances but is not allowed to collapse them below salient levels. A useful mental model is that $\delta(\cdot, \cdot)$ supplies directed "pushes apart" evidence; $\otimes$ aggregates the two directed pushes into a single mutual signal. Non-dual compatibility says each directed push is not weakened by translation; therefore the aggregate mutual signal is not weakened either. The preservation of non-duality then follows because non-duality is formulated as a conjunction of threshold conditions on the same underlying evidence channels, and conjunction of monotone properties remains monotone under such translations.*

Lemma 19 (Closure under composition; identity translation)**.** *If $T : X_c \to X_d$ and $U : X_d \to X_e$ are non-dually compatible, then $U \circ T : X_c \to X_e$ is non-dually compatible. The identity map id_{X_c} is non-dually compatible. This shows that "non-dually compatible translation" behaves like a structural constraint rather than an ad hoc property of a single map: it is stable under chaining of perspective shifts, and it is satisfied by the trivial shift that changes no perspective at all.*

Remark 1607. *This lemma says that non-dually compatible translations form a small but robust algebra: they compose and include identities. As a result, contexts and their admissible translations can be treated categorically (at least informally): one can build multi-step perspective shifts without losing the guarantee that distinctions are not erased. In particular, once a downstream analysis is stated in a target context X_e, one may safely precompose it with any sequence of such translations from a source context without worrying that crisp distinctions (or non-dual tensions) were artifacts of a particular viewpoint and disappear under the change of lens. The lemma is thus a technical enabler for discussing families of contexts connected by permissible reinterpretations.*

Sketch. For all x, y, $\delta_c(x, y) \leq \delta_d(Tx, Ty) \leq \delta_e(U(Tx), U(Ty))$ by transitivity of $\leq$, and similarly for the swapped-argument inequality. This is exactly Def. 53 for $U \circ T$. For the identity, both inequalities reduce to $\delta_c(x, y) \leq \delta_c(x, y)$. No further properties of δ are needed beyond the order structure already assumed, which is why the argument is purely diagrammatic: it is just the fact that an order-preserving lower bound remains a lower bound after applying another order-preserving lower bound. □

Remark 1608. Proof sketch and intuition. *Composition uses nothing mysterious: it is simply the transitivity of the underlying order $\leq$. If T does not reduce distinction evidence and U does not* *reduce distinction evidence, then doing T and then U does not reduce it either. The identity case is the degenerate translation that changes nothing, so it trivially preserves every distinction. From the perspective-bridging viewpoint, this means that "respecting salient distinctions" is a property that can be required locally at each step of a translation pipeline, while still guaranteeing the global property for the entire pipeline. This aligns with modularity: one can design or learn translations in stages, checking the same simple inequality condition at each stage, and then compose them without re-verifying everything from scratch.*

29 Helper theorems for order, time, and becoming (Section 7)

Standing assumptions and notation. A *proto-time* is a pair $(T, <)$ with $<$ a strict partial order (Defs. 54–55). Where needed, fix the canonical p-bit quantale $V = [0, 1]^2$ with componentwise order $\leq$, join $\oplus$ as componentwise max, and tensor $\otimes$ as componentwise multiplication. A p-bit temporal evidence relation is $B : T \times T \to V$ (Def. 56). Optional coherence constraints are those in Def. 57. For linearization/quotients, Past/Future sets and $\sim$ are as in Def. 59. For becoming, $Dist : T \times E \times E \to V$ and NDVar/Becoming are as in Defs. 60–62. For event calculus, predicates and operators are as in Defs. 63–66. TimeCtx/similarity/TemporalSimilarity are as in Defs. 67–69. Regularity predicates are as in Defs. 70–72. The functorial process view is Def. 73.

Remark 1609. *The organizing thought here is that "time" is treated in two layers. First there is a thin, structural layer: a set T of "time-like" indices equipped with a strict partial order $<$, yielding what the core calls a proto-time (Hyperseed-Concept 142). This is intentionally weaker than a linear timeline: it permits branching and incomparability, which is often what one has before imposing a narrative or a measurement protocol.*

Second there is a graded, paraconsistent evidential layer, in which comparisons $t_1 < t_2$ may be supported or disputed by data. Concretely, the canonical p-bit quantale $V = [0, 1]^2$ encodes a pair $v = (v^+, v^-)$ of positive *and* negative *evidence channels (in the sense of paraconsistent valuation; see also [23, 24]). Componentwise order means $(a^+, a^-) \leq (b^+, b^-)$ iff $a^+ \leq b^+$ and $a^- \leq b^-$. The join $v \oplus w$ is componentwise* max*, representing "take the stronger of two arguments"; the tensor $v \otimes w$ is componentwise multiplication, representing "compose pieces of support" along a chain. A temporal evidence map $B : T \times T \to V$ thus assigns to each ordered pair (t_1, t_2) a degree of support $B^+(t_1, t_2)$ and a degree of counter-support $B^-(t_1, t_2)$ for the claim "t_1 is before t_2" (Hyperseed-Concepts 63 and 52). This style of quantale-based graded reasoning connects to the broader weakness/quantale program [3] and to the original Hyperseed presentation [1].*

How the two layers interact. Although $(T, <)$ and B are introduced separately, the intended use is that the structural order provides a baseline constraint (e.g. what is allowed in a model), while the evidential map provides a flexible mechanism for learning, aggregating, and revising order-claims from heterogeneous sources. In particular, one should read $B(t_1, t_2)$ as *evidence about* the ordering claim, not as the ordering claim itself: the strict partial order $<$ can be taken as a conservative "skeleton" that may later be refined, quotiented, or linearized (Def. 59) using the evidential layer. This separation is what allows the later helper results to state clean monotonicity and closure properties for the constructions built from B without requiring that the underlying proto-time be total, well-founded, or otherwise timeline-like.

Interpreting p-bits and inconsistency. The p-bit value $v = (v^+, v^-) \in [0, 1]^2$ can be read as allowing independent degrees of affirmation and denial. Thus, values near $(1, 0)$ represent strong support with little objection, values near $(0, 1)$ represent strong counter-evidence, and values with both components large represent a state of *inconsistent* information (in the paraconsistent sense) rather than a collapse of the framework. The componentwise operations are chosen so that common evidential patterns have direct algebraic expressions: $\oplus$ merges alternative arguments by retaining the strongest available pro and con components, while $\otimes$ models the idea that support (and counter-support) can be propagated along compositional patterns such as chains of "before" comparisons. Optional coherence constraints (Def. 57) can then be seen as enforcing minimal compatibility conditions between direct and composed evidence (for example, requiring that direct evidence for $t_1 < t_3$ not be weaker than what is obtained by composing evidence for $t_1 < t_2$ and $t_2 < t_3$), while still permitting the presence of both support and objection in the same comparison.

Why the remaining definitions are grouped here. The later notions referenced above (Past/Future sets and $\sim$, becoming via *Dist* and NDVar/Becoming, the event-calculus operators, TimeCtx and TemporalSimilarity, regularity predicates, and the functorial process view) are all ways of turning proto-time plus graded evidence into derived structures that behave like temporal semantics. The point of this section is not to reintroduce those constructions, but to collect helper theorems that show they respect the underlying order/evidence architecture: closure properties under $\oplus$ and $\otimes$, invariance under quotienting by $\sim$, and the stability of becoming- and event-level predicates under the similarity/regularity constraints. Establishing these routine facts once lets later sections focus on modeling choices rather than re-proving basic algebraic and order-theoretic properties.

29.1 Strict proto-time order: basic derived rules

Lemma 20 (Asymmetry from strictness)**.** *If $(T, <)$ is a strict partial order, then for all $t_1, t_2 \in T$,*

$$t_1 < t_2 \;\to\; \neg(t_2 < t_1).$$

Remark 1610. *Intuitively, this lemma says that if "t_1 is strictly before t_2," then t_2 cannot be strictly before t_1. In ordinary temporal language this is almost tautological, but it is worth isolating because it is exactly where the* strict *nature of $<$ matters: asymmetry is derived from irreflexivity plus transitivity.*

More explicitly, recall that a strict partial order *is, by definition, a relation that is (i)* irreflexive *($\neg(t < t)$ for all t) and (ii)* transitive *(if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$). The point of the lemma is that one does not need to add asymmetry as an independent axiom: it is already forced by (i)–(ii). This matters later because we will treat $<$ as the base "proto" structure and layer additional semantic/evidential notions on top; keeping the base assumptions minimal prevents accidental circularity when richer principles are introduced.*

This result functions as the minimal sanity check on proto-time (Hyperseed-Concept 142): it rules out the simplest temporal contradiction inside the underlying order, even before we add graded/paraconsistent evidence. Later results on cycles and linear extensions reuse the same idea in increasingly global forms.

A useful reformulation is that the binary relation $<$ defines an orientation on pairs: once $t_1 < t_2$ holds, the opposite orientation is ruled out. In particular, even if $<$ is not total (so that some pairs are incomparable), whenever comparability does occur it is one-way only; this is the precise sense in which strictness provides a minimal directional constraint.

Sketch. Assume $t_1 < t_2$ and $t_2 < t_1$. By transitivity, $t_1 < t_1$, contradicting irreflexivity. $\square$

Remark 1611. Proof sketch (strategy). *Suppose both directions hold; then compose them to obtain a self-loop, which strictness forbids.*

Key step. *The only real move is applying transitivity to the pair of inequalities; irreflexivity then closes the door. One may picture $t_1 < t_2$ as an arrow $t_1 \to t_2$ in a directed graph: having also $t_2 \to t_1$ forces a 2-cycle, hence a loop.*

Geometric intuition. *In a strict order, time cannot "bend back" onto itself at a point; if it did, the point would be both before and after itself, collapsing the meaning of "before" (Hyperseed-Concept 63).*

It is also worth noting what the lemma does not *assert: it does not say that either $t_1 < t_2$ or $t_2 < t_1$ must hold. Thus, the absence of symmetry is compatible with branching or partially ordered temporal structures, where two moments may be incomparable (neither earlier nor later in the proto-time sense), even though any asserted ordering must be coherent.*

Lemma 21 (No directed cycles)**.** *If $(T, <)$ is a strict partial order, there is no finite cycle $t_0 < t_1 < \cdots < t_{n-1} < t_0$ for any $n \geq 1$.*

Remark 1612. *This lemma globalizes Lemma 20: not only are 2-cycles forbidden, but* any *finite directed cycle is. In graph terms, the directed graph of the relation $<$ is a DAG.*

*Two boundary cases are worth keeping in view. When $n = 1$, the "cycle" would read $t_0 < t_0$, which is excluded exactly by irreflexivity; so Lemma 21 may be seen as extending irreflexivity from length-*1 *loops to all finite loops. When $n = 2$, the cycle is precisely the configuration ruled out by Lemma 20. For $n \geq 3$, the lemma says that one cannot return to a starting point by stepping strictly forward finitely many times.*

The importance is practical as well as conceptual. Many constructions—linear extensions, topological sorts, inductive proofs over time—silently presuppose acyclicity. Here we see that acyclicity is not an extra axiom but a theorem of strictness, which will be exploited immediately in Lemma 22 and Theorem 27.

From the standpoint of later "becoming"-style talk, acyclicity is the minimal condition under which iterative "advance" operations are guaranteed not to revisit a prior state in finitely many steps. This provides a clean separation: any apparent temporal looping must come from additional structure (e.g. evidential layers) rather than from the underlying proto-time relation itself.

Sketch. By repeated transitivity, the cycle implies $t_0 < t_0$, contradicting irreflexivity. □

Remark 1613. Proof sketch (strategy). *Collapse the whole cycle into a single inequality $t_0 < t_0$ by composing along the chain.*

Key step. *"Repeated transitivity" is simply iterated application of transitivity to the successive links $t_0 < t_1$, $t_1 < t_2$, …, $t_{n-1} < t_0$.*

Visual intuition. *Any directed cycle is a closed directed path; transitivity lets us treat a path as a single arrow. But a closed path becomes a self-arrow, which strictness excludes.*

For readers who prefer a fully explicit symbolic form: apply transitivity to $t_0 < t_1$ and $t_1 < t_2$ to get $t_0 < t_2$; then combine this with $t_2 < t_3$ to get $t_0 < t_3$; continue until reaching $t_0 < t_{n-1}$, and finally combine with $t_{n-1} < t_0$ to conclude $t_0 < t_0$. The proof uses no finiteness of T, only the finiteness of the alleged cycle.

Lemma 22 (Minimal and maximal elements exist in the finite case)**.** *If T is finite and $(T, <)$ is a strict partial order, then T has at least one minimal element (no predecessor) and at least one maximal element (no successor).*

Remark 1614. *In a finite proto-time, there must be at least one "earliest" point (a minimal element) and at least one "latest" point (a maximal element), though there may be many of each.*

This does not *mean time is linear; it means only that in any finite acyclic directed graph there is a source and a sink.*

Concretely, "minimal" here means: $t \in T$ such that there is no $s \in T$ with $s < t$; "maximal" means: there is no $u \in T$ with $t < u$. In the directed-graph picture (with an arrow $s \to t$ whenever $s < t$), a minimal element is a vertex of indegree 0 *and a maximal element is a vertex of outdegree* 0*. The lemma therefore states a familiar graph-theoretic fact about finite DAGs, restated in order-theoretic language suited to proto-time.*

This lemma is the engine behind linearization (Theorem 27): the standard topological sorting argument repeatedly removes a minimal element and continues. Philosophically, it exhibits a mild form of "temporal directionality" without imposing a single global present: direction emerges from the impossibility of cycles.

It is also useful to compare this finite result with the stronger (and non-equivalent) notion of well-foundedness *in the infinite case. Finiteness guarantees that one cannot keep moving to predecessors forever; in infinite structures, the existence of minimal elements can fail unless one assumes additional well-foundedness conditions. The present lemma thus isolates exactly what finiteness buys us, without building in stronger assumptions than needed.*

Sketch. If there were no minimal element, start at any t_0 and choose $t_1 < t_0$, then $t_2 < t_1$, etc. Finiteness forces a repeated element, yielding a cycle, contradicting Lemma 21. The maximal case is dual. □

Remark 1615. Proof sketch (strategy). *Assume the opposite and build an infinite descending (or ascending) chain by always stepping to a predecessor (or successor); finiteness forces repetition, hence a cycle.*

Key step. *The pigeonhole principle turns an infinite walk in a finite set into a repeated vertex; strictness turns the repeat into a contradiction via Lemma 21.*

Geometric intuition. *Think of repeatedly walking "backward in time." If you can always take one more step, then in a finite universe you eventually revisit a moment, implying time has looped.*

The "dual" maximal-element argument can be read either by reversing all inequalities or, equivalently, by applying the same reasoning to the opposite relation $>$ (which is also a strict partial order whenever $<$ is). This symmetry clarifies why the finite DAG picture yields both a source and a sink rather than only one of them.

29.2 Graded/paraconsistent temporal evidence: useful inequalities

Lemma 23 (Path lower bound under transitive support)**.** *Assume the optional* transitive support *constraint (Def. 57.2): $B(t_0, t_2) \geq B(t_0, t_1) \otimes B(t_1, t_2)$. Then for any chain $t_0, t_1, \ldots, t_n \in T$,*

$$B(t_0, t_n) \;\geq\; \bigotimes_{i=1}^{n} B(t_{i-1}, t_i).$$

In particular, on the positive channel, $B^+(t_0, t_n) \geq \prod_{i=1}^{n} B^+(t_{i-1}, t_i)$ in the canonical $[0,1]^2$ quantale.

Remark 1616. *This lemma states a very general principle: if evidence for "before" composes transitively, then the evidence for a long-range ordering is at least as strong as the cumulative composition of the short-range evidences along any chosen chain. In the canonical quantale, "composition" on the positive channel is multiplication, so support decays (unless values are* 1*).*

The usefulness is algorithmic and epistemic. Algorithmically, it provides a guaranteed lower bound when one infers distant temporal relations from local ones. Epistemically, it mirrors a sober

Russellian caution: chained inferences should not become more certain merely because they are longer. In paraconsistent settings [23, 24], one may carry positive and negative channels in parallel; this lemma is about the monotone accumulation of support (not the resolution of conflict).

Sketch. Induct on n. The case $n = 2$ is Def. 57.2. The inductive step composes $B(t_0, t_n) \geq B(t_0, t_{n-1}) \otimes B(t_{n-1}, t_n)$ and applies the hypothesis to $B(t_0, t_{n-1})$. $\square$

Remark 1617. Proof sketch (strategy). *Reduce an n-step chain to an $(n-1)$-step chain plus one final step, then apply induction.*

Key step. *The transitive-support axiom (Def. 57.2) is exactly the $n = 2$ case; everything else is bookkeeping of repeated composition via associativity of $\otimes$ (implicit in the quantale structure).*

Visual intuition. *Picture a path $t_0 \to t_1 \to \cdots \to t_n$ with each arrow labeled by a strength. Transitive support says the direct arrow $t_0 \to t_n$ must be at least as strong as the composed label of the path.*

Lemma 24 (Two-step threshold propagation)**.** *Assume transitive support (Def. 57.2) and the canonical $[0,1]^2$ tensor. If $B^+(t_1, t_2) \geq \alpha$ and $B^+(t_2, t_3) \geq \beta$, then*

$$B^+(t_1, t_3) \geq \alpha\beta.$$

Equivalently, if $B^+(t_1, t_2) \geq \tau$ and $B^+(t_2, t_3) \geq \tau$, then $B^+(t_1, t_3) \geq \tau^2$.

Remark 1618. *This is the concrete numerical form of Lemma 23 for a chain of length 2. The statement is almost pedagogical: if each of two links has at least some reliability, the inferred two-step relation has at least the product reliability.*

Its conceptual role is to make explicit how thresholds behave under inference. If one chooses a single threshold τ for "strong evidence," then chaining two strong evidences yields a weaker lower bound τ^2. This warns us that thresholding is not invariant under composition: in long chains, one may need renormalization, alternative t-norms, or different proof policies.

Sketch. Take the positive component of Lemma 23 for $n = 2$. $\square$

Remark 1619. Proof sketch (strategy). *Specialize the general chain bound to $n = 2$ and read off the positive coordinate.*

Key step. *In the canonical $[0,1]^2$ quantale, $\otimes$ is componentwise multiplication, so the composed positive evidence is $\alpha\beta$.*

Intuition. *Each step is a filter on certainty; two filters in series multiply their pass-through.*

Lemma 25 (No positive-evidence 2-cycles under transitive+irreflexive support)**.** *Assume transitive support (Def. 57.2),* irreflexive support $B^+(t,t) = 0$ *(Def. 57.3), and the canonical $[0,1]^2$ tensor. Then for all $t_1, t_2 \in T$,*

$$B^+(t_1, t_2) \cdot B^+(t_2, t_1) = 0.$$

So it is impossible to have strictly positive evidence for both directions simultaneously.

Remark 1620. *This lemma translates the strict-order intuition "no two-way arrows" into the evidence layer. Under transitive support, a 2-cycle would yield evidence for a self-loop; irreflexive support forbids any positive self-loop, so at least one direction of the 2-cycle must have zero positive evidence.*

The significance is methodological: it separates two notions of inconsistency. One may have paraconsistent *evidence (positive and negative simultaneously) without having* positive *evidence for both temporal directions. The latter would be a particularly strong kind of incoherence, and this lemma shows how to preclude it with simple constraints.*

Sketch. Apply transitive support to the length-2 loop $t_1 \to t_2 \to t_1$: $B^+(t_1, t_1) \geq B^+(t_1, t_2) \cdot B^+(t_2, t_1)$. But $B^+(t_1, t_1) = 0$, hence the product must be 0. $\square$

Remark 1621. Proof sketch (strategy). *Turn the two-way evidence into evidence for a self-loop via transitivity, then use irreflexivity to force the product to vanish.*

Key step. *The inequality $B^+(t_1, t_1) \geq B^+(t_1, t_2) \cdot B^+(t_2, t_1)$ is the transitive-support axiom applied to $t_0 = t_1$, $t_1 = t_2$, $t_2 = t_1$.*

Graph intuition. *If both arrows $t_1 \to t_2$ and $t_2 \to t_1$ had positive weight, then the round trip $t_1 \to t_1$ would inherit positive weight; irreflexive support declares that no round trip is allowed to count as "forward in time."*

Lemma 26 (Conflict detection from anti-symmetry-as-evidence)**.** *Assume the optional constraint (Def. 57.1): $B^+(t_1, t_2) \leq B^-(t_2, t_1)$. Then any strong positive evidence for $t_1 < t_2$ forces at least as-strong negative evidence for $t_2 < t_1$. If additionally $B^+(t_2, t_1)$ is also strong, then (t_1, t_2) is* paraconsistently ordered*: both directions have strong positive and strong negative evidence.*

Remark 1622. *The axiom $B^+(t_1, t_2) \leq B^-(t_2, t_1)$ can be read as a disciplined bookkeeping rule: support for one direction counts as (counter-)evidence against the reverse direction. Under this discipline, strong belief that t_1 is before t_2 automatically implies strong disbelief that t_2 is before t_1.*

The second sentence isolates a characteristic paraconsistent phenomenon (Hyperseed-Concept 198 in the general style, and paraconsistent valuation as in [23, 24]): one may still end up with strong $B^+(t_1, t_2)$ and strong $B^+(t_2, t_1)$ simultaneously, in which case the bookkeeping forces strong negative evidence in both directions too. This is not "resolved" here; rather, it is detected and made explicit, so later mechanisms can decide how to act on it.

Sketch. Immediate from Def. 57.1 and symmetry of the statement under swapping indices. $\square$

Remark 1623. Proof sketch (strategy). *Unpack the constraint and apply it directly.*

Key step. *The condition is already pointwise: for each ordered pair it gives an inequality relating $+$ in one direction to $-$ in the reverse.*

Intuition. *The lemma is less a "derivation" than a clarifying rephrasing: it tells you what Def. 57.1 means operationally.*

Remark 1624. *To make the phrase "strong" fully concrete, one may read it relative to any chosen strength threshold θ (often with θ near the top of the scale in use). In that case the first conclusion can be restated as the monotonic implication*

$$B^+(t_1, t_2) \geq \theta \implies B^-(t_2, t_1) \geq \theta,$$

which is exactly what the inequality $B^+(t_1, t_2) \leq B^-(t_2, t_1)$ guarantees. No additional algebra is needed: the inequality transfers lower bounds on $B^+(t_1, t_2)$ into the same lower bounds on $B^-(t_2, t_1)$.

In the same thresholded reading, the "paraconsistently ordered" situation described in the lemma can be written as the joint satisfaction of four strong-evidence conditions,

$$B^+(t_1, t_2) \geq \theta, \qquad B^-(t_1, t_2) \geq \theta, \qquad B^+(t_2, t_1) \geq \theta, \qquad B^-(t_2, t_1) \geq \theta,$$

where the two negative terms arise from the two positive terms by applying Def. 57.1 once to (t_1, t_2) and once to (t_2, t_1). Thus the bookkeeping constraint functions as a conflict amplifier *in the following precise sense: if the data source or aggregation procedure supplies strong support in both directions, then the constraint necessarily yields strong opposition in both directions as well, making the inconsistency explicit rather than latent.*

Remark 1625. *The name "anti-symmetry-as-evidence" is intended to highlight the distinction between (i) enforcing classical antisymmetry of an order, which would prohibit $t_1 < t_2$ and $t_2 < t_1$ from both holding, and (ii) enforcing a relation between* evidence values *for opposite directions. Def. 57.1 is of type (ii): it does not directly forbid having strong $B^+(t_1, t_2)$ and strong $B^+(t_2, t_1)$ at the same time, but it ensures that such a situation cannot occur without simultaneously producing correspondingly strong B^- values. In other words, the constraint does not remove paraconsistency; it makes its presence measurable in the $B^\pm$ fields, which is the relevant output for later decision policies.*

Further detail (optional). Fix t_1, t_2 and assume Def. 57.1, i.e. $B^+(t_1, t_2) \leq B^-(t_2, t_1)$. If $B^+(t_1, t_2)$ is strong (e.g. $\geq \theta$ for a chosen θ), then by transitivity of $\leq$ we have $\theta \leq B^+(t_1, t_2) \leq B^-(t_2, t_1)$, hence $B^-(t_2, t_1)$ is strong. If, in addition, $B^+(t_2, t_1)$ is strong, then applying the same constraint with indices swapped gives $B^+(t_2, t_1) \leq B^-(t_1, t_2)$, so $B^-(t_1, t_2)$ is also strong. This yields the four strong-evidence statements described in the lemma's final sentence. □

29.3 Linearization and quotient-by-indistinguishability

Lemma 27 ($\sim$ is an equivalence relation)**.** *Let $\sim$ be defined by equality of Past/Future sets (Def. 59). Then $\sim$ is reflexive, symmetric, and transitive.*

Remark 1626. *The relation $\sim$ collapses time points that are observationally indistinguishable with respect to the order structure: two points are equivalent if they have the same Past-set and the same Future-set. This is a structural instance of an equivalence relation (Hyperseed-Concept* **??***) arising from an invariance principle: if nothing in the order-theoretic neighborhood distinguishes t from s, we treat them as the same "abstract time."*

This quotienting move is philosophically suggestive: it echoes the idea that time is partially constituted by relations (Whitehead's process-relational stance [15]), and technically it prepares the ground for a clean strict order on the quotient (Theorem 26).

Sketch. Reflexive/symmetric are immediate from set equality. For transitivity, if Past(t)=Past(s) and Past(s)=Past(r), then Past(t)=Past(r), and similarly for Future. □

Remark 1627. Proof sketch (strategy). *Use the standard fact that equality is an equivalence relation, applied separately to Past and Future.*

Key step. *Transitivity is inherited from transitivity of equality: if two sets are equal to a third, they are equal to each other.*

Intuition. *This is the formal stamp that "indistinguishable" behaves the way one expects: it partitions T into classes.*

Lemma 28 (Indistinguishable points are incomparable)**.** *If $t \sim s$ and $t \neq s$, then neither $t < s$ nor $s < t$.*

Remark 1628. *This lemma asserts that if two distinct points have identical Past/Future profiles, then the order cannot place one before the other. If it did, that very fact would change the Past/Future profile and hence destroy indistinguishability.*

The result matters because it ensures the quotient $T/\sim$ is not hiding genuine order information: when we collapse t and s, we are not collapsing a real before/after distinction (Hyperseed-Concept 98); rather, we are collapsing redundancy in the representation of proto-time (Hyperseed-Concept 142).

Sketch. If $t < s$, then $t \in \mathit{Past}(s)$ but $t \notin \mathit{Past}(t)$ (irreflexivity), contradicting Past(t)=Past(s). The case $s < t$ is symmetric. □

Remark 1629. Proof sketch (strategy). *Show that comparability forces an asymmetry in the Past sets.*

Key step. *Irreflexivity is crucial:* $t \notin \mathit{Past}(t)$, *but if* $t < s$ *then* $t \in \mathit{Past}(s)$. *Thus* $\mathit{Past}(t) \neq \mathit{Past}(s)$.

Visual intuition. *If t were before s, then t would cast a "shadow into the past" of s that is absent from the past of itself.*

Lemma 29 (Quotient map is order-preserving and order-reflecting)**.** *Let* $q : T \to T/\sim$ *be* $q(t) = [t]$ *and let the quotient order be as in Def. 59. Then for all* $t_1, t_2 \in T$,

$$t_1 < t_2 \iff [t_1] < [t_2].$$

Remark 1630. *This lemma says the quotient is faithful to the original strict order: it neither invents new strict order relations nor loses old ones. Order-preserving* ($t_1 < t_2 \Rightarrow [t_1] < [t_2]$) *is expected; order-reflecting is the deeper assurance that if two equivalence classes are ordered, then their representatives really were ordered.*

This is the technical heart of treating $T/\sim$ *as an "abstracted" proto-time. In Hyperseed language, it is a disciplined abstraction (Hyperseed-Concept 51) that respects temporal structure (Hyperseed-Concept 142), a theme that also appears elsewhere in the system as abstraction-by-quotient.*

Sketch. ($\rightarrow$) If $t_1 < t_2$, then by definition $[t_1] < [t_2]$. ($\Leftarrow$) If $[t_1] < [t_2]$, there exist representatives $s_1 \sim t_1$ and $s_2 \sim t_2$ with $s_1 < s_2$. If t_1 were not $< t_2$, Past/Future equalities plus the existence of a comparable representative force a contradiction (same kind of reasoning used to show well-definedness in Lemma 1). □

Remark 1631. Proof sketch (strategy). *The forward direction is definitional. The reverse direction argues that comparability is a class property: if some representatives are comparable, then all representatives must align, because* $\sim$ *fixes Past/Future structure.*

Key step. *The Past/Future invariants prevent a situation in which* $s_1 < s_2$ *but* t_1 *and* t_2 *are not ordered: if* s_1 *is in the Past of* s_2*, then (via equality of Past/Future sets within each class) the corresponding membership relations must transfer to* t_1 *and* t_2.

Intuition. *Equivalence classes are meant to be "structural positions" in the order. If one representative sits before another, the whole class sits before the other class; otherwise, the classes would not be positions but merely bags of points.*

Theorem 26 (Quotient order is strict (helper restatement of Lemma 1))**.** *The relation* $<$ *on* $T/\sim$ *defined in Def. 59 is well-defined and makes* $(T/\sim, <)$ *a strict partial order.*

Remark 1632. *This theorem assures that the quotient construction yields a genuine proto-time again: it is not merely a set of classes, but a set of classes equipped with a strict partial order. "Well-defined" is the subtle point: we must not let the ordering depend on which representative of a class one picks. It connects directly to Lemmas 27–29: the equivalence relation gives the partition, incomparability protects against collapsing real order, and order-reflection preserves the semantics of before/after. Taken together, they justify the quotient as an abstraction step in the sense of Hyperseed (Hyperseed-Concept 51). In particular, the quotient does not "invent" new temporal distinctions: it merely re-expresses the original proto-time in a coarser vocabulary where indistinguishable points are treated as a single abstract point, while all genuine* $<$*-constraints remain detectable at the level of equivalence classes.*

Sketch. Well-definedness: comparability of one representative is incompatible with incomparability of another when Past/Future sets are equal; strictness descends because any cycle in the quotient would lift to a cycle in T. (This is the proof idea of Lemma 1.) More explicitly, if the quotient order is defined by $[t_1] < [t_2]$ iff there exist representatives $t_1' \in [t_1]$, $t_2' \in [t_2]$ with $t_1' < t_2'$, then Past/Future equality is used to show that whenever one such witnessing pair exists, every choice of representatives remains compatible with $<$ in the sense that the opposite inequality cannot be witnessed without contradicting incomparability preservation. For strictness, if $[t_0] < [t_1] < \cdots < [t_k] = [t_0]$ were a directed cycle among classes, choosing witnesses for each step produces a directed cycle in T, contradicting Lemma 21. □

Remark 1633. Proof sketch (strategy). *For well-definedness, show that if* $[t_1] < [t_2]$ *holds via some representatives, it cannot fail via others. For strictness, transfer irreflexivity/transitivity (or equivalently, acyclicity) from* T *to the quotient.*

Key steps. *(i) Past/Future equality makes "being before" a class invariant. (ii) Any directed cycle among classes would choose representatives to yield a directed cycle in* T*, contradicting Lemma 21. Concretely, (i) is used in the contrapositive form: if two elements have identical Past/Future sets, then there is no third point that can distinguish them by being before one but not the other; this blocks the possibility that swapping representatives could flip a strict inequality into a non-inequality. Step (ii) is the familiar "lift a cycle" maneuver: since each edge in the quotient is witnessed by at least one edge in* T*, a cyclic chain of class-edges forces a cyclic chain of element-edges.*

Intuition. *Quotienting is safe here because the equivalence relation was designed from the order itself; it forgets only redundancy, not structure. Put differently: proto-time is the primary structure, and the quotient is a controlled compression of that structure, justified exactly because the defining indistinguishability criterion is stated in terms of observable order-theoretic behavior (Past/Future), rather than in terms of an external labeling.*

Theorem 27 (Finite proto-times admit linear extensions (helper restatement of Theorem 5))**.** *If* $(T, <)$ *is a finite proto-time, there exists a strict total order* $\prec$ *on* T *such that* $t_1 < t_2 \to t_1 \prec t_2$*.*

Remark 1634. *A linear extension is a way of telling a complete story from a partial order: it imposes a total order* $\prec$ *that respects all the constraints of* $<$*. When proto-time represents a web of "must-be-before" constraints, a linear extension is a choice of one consistent timeline compatible with them.*

This is important because many algorithms (simulation, scheduling, event calculus iteration) are most naturally run over a linear order. The theorem says that for finite proto-times this is always possible: one may add arbitrary tie-breaks among incomparable points without creating contradictions. The proof relies on Lemma 22 and is the standard topological sorting argument. Equivalently, one may view $(T, <)$ *as a finite acyclic directed graph and produce* $\prec$ *by repeatedly selecting a node with no incoming edges; the finiteness assumption ensures the process terminates, while acyclicity ensures one can always continue. Note that the construction generally produces many possible* $\prec$*, reflecting the underdetermination present in the original proto-time.*

Sketch. Induct on $|T|$ using existence of a minimal element (Lemma 22), place it first, and extend the remainder inductively (standard topological sort proof). More explicitly, choose a $<$-minimal $m \in T$, define $\prec$ to start with m, restrict $<$ to $T \setminus \{m\}$ (which remains a finite proto-time), apply the induction hypothesis to obtain a linear extension $\prec'$ on the remainder, and then concatenate m with $\prec'$. Minimality of m ensures no constraint $t < m$ is violated, and the induction hypothesis ensures all constraints internal to $T \setminus \{m\}$ are respected. □

Remark 1635. Proof sketch (strategy). *Repeatedly peel off a minimal element and put it next in the total order.*

Key step. *Lemma 22 guarantees there is always at least one minimal element to peel off until the set is empty. Induction then assembles these choices into a strict total order. The strictness of $\prec$ is automatic because the construction never places two distinct elements in the same position, and transitivity follows from the fact that each stage appends a well-ordered remainder.*

Intuition. *In a finite acyclic dependency graph, there is always something you can do first; do it, remove it, and repeat. When there are several minimal choices at some stage, each choice corresponds to a different but equally valid "completion" of the partial temporal information encoded by $<$.*

Lemma 30 (Soundness of linearization; incomparability is not preserved)**.** *Let $\prec$ be a linear extension of $(T, <)$. Then:*

$$t_1 < t_2 \to t_1 \prec t_2 \quad \textit{(soundness)},$$

but in general,

$$t_1 \prec t_2 \not\to t_1 < t_2 \quad \textit{(not complete)}.$$

In particular, if t_1 and t_2 are incomparable in $<$, $\prec$ still forces exactly one of $t_1 \prec t_2$ or $t_2 \prec t_1$.

Remark 1636. *This lemma is a cautionary counterpart to Theorem 27. A linear extension is* sound *with respect to $<$: it never violates the original constraints. But it is not* complete*: it introduces additional comparabilities that were not present in the proto-time. Incomparable events become artificially ordered.*

The point is conceptual as much as technical. Linearization is an interpretive act: it selects a narrative from a partial structure. Hyperseeds stance is to keep proto-time primary and treat any $\prec$ as a derived convenience, not as ontology itself (Hyperseed-Concept 142). This distinction prevents one from mistaking a bookkeeping choice for a metaphysical fact. A simple example is a proto-time with two events a, b such that neither $a < b$ nor $b < a$; any linear extension must pick either $a \prec b$ or $b \prec a$, but that choice encodes no additional constraint in the underlying model. Accordingly, any subsequent reasoning that depends on the newly introduced order (e.g., treating $a \prec b$ as evidence that a "really" precedes b in proto-time) is methodologically unsound unless it is explicitly stated as a convention or an extra assumption.

Sketch. Soundness is the definition of linear extension. Non-completeness holds because a total order must compare incomparable pairs, introducing an arbitrary choice that does not imply $<$. Formally, if t_1 and t_2 are incomparable under $<$, then $t_1 < t_2$ is false and $t_2 < t_1$ is false, but totality of $\prec$ forces exactly one of $t_1 \prec t_2$ or $t_2 \prec t_1$; hence either way one obtains an instance of $t_1 \prec t_2$ (or its reverse) that does not correspond to a $<$-fact. This shows that $\prec$ may strictly extend $<$ by adding relations not entailed by the proto-time. $\square$

Remark 1637. Proof sketch (strategy). *The first implication is immediate from the definition; the second is forced by the nature of total orders.*

Key step. *Totality means every pair is comparable under $\prec$, so $\prec$ contains at least one of (t_1, t_2) or (t_2, t_1) even when $<$ contains neither. Thus, while $<$ embeds into $\prec$ (soundness), the embedding is typically proper: $\prec$ contains additional ordering information not justified by proto-time alone.*

Intuition. *Linearization is like choosing an ordering of chapters in a book built from partially ordered notes: it respects the dependency constraints but still chooses an order where none was mandated. The extra order is best understood as a convenient serialization for execution or exposition, not as a discovery of hidden temporal structure.*

29.4 Becoming as boundary non-duality: operational lemmas

Lemma 31 (Necessary paraconsistent boundary condition)**.** *If x becomes y along $(T,<)$ (Def. 62), then there exists a neighborhood $U \subseteq T$ near the transition such that for all $u \in U$,*

$$Dist(u;x,y) \geq (\theta,\theta)$$

for the chosen NDVar threshold θ (Def. 61).

Remark 1638. *The core idea of* becoming *(Hyperseed-Concept 62) here is neither simple replacement nor mere succession. Rather, it is characterized by a boundary region in which x and y are held together in a kind of structured ambiguity: strong evidence both for and against their distinction, formalized via NDVar (Hyperseed-Concept 120) and ultimately via a paraconsistent Dist valuation.*

This lemma extracts the operational test that any implementation must satisfy: if the theory declares "x becomes y," then there must exist some time neighborhood where Dist is simultaneously high in both channels. This is the minimal trace of Whiteheadian transition (process and becoming [15]) rendered in quantale-valued evidence.

Sketch. Immediate: boundary non-duality (Def. 62.2) is NDVar at each u near the transition, and NDVar is $Dist(u;x,y) \geq (\theta,\theta)$ by Def. 61. □

Remark 1639. Proof sketch (strategy). *Unpack the definitions: Def. 62 reduces becoming to boundary NDVar; Def. 61 reduces NDVar to a threshold on Dist.*

Key step. *The inequality $Dist(u;x,y) \geq (\theta,\theta)$ is simply the bundled form of "both positive and negative components are at least θ."*

Intuition. *Becoming must leave a signature: a zone where the system is compelled to treat x and y as both distinct and not-distinct (Hyperseed-Concept 121), rather than switching instantaneously.*

Remark 1640. Further operational reading. *The neighborhood U in Lemma 31 should be understood as an* interval of indeterminacy *around the transition locus, not necessarily symmetric and not necessarily unique. In applications where T carries additional structure (e.g. a topology or a metric), "near the transition" can be instantiated as a small open interval, a finite window in discrete time, or any other admissible neighborhood notion used by Def. 62. The lemma itself only needs the minimal neighborhood concept already assumed in the definition of becoming.*

Why the inequality is two-channel. *Since $Dist(t;x,y)$ is paraconsistent, the condition $Dist(u;x,y) \geq (\theta,\theta)$ does not mean "maximal confusion"; it means that* both *(i) evidence for distinction and (ii) evidence against distinction cross the same operational floor. This is exactly the boundary-style signal: the system is pulled in incompatible directions strongly enough that a classical crisp predicate (distinct vs. identical) cannot be consistently applied in that region.*

Threshold dependence. *For fixed Dist, the choice of θ tunes how demanding the test is. Increasing θ makes becoming harder to certify, shrinking the set of times counted as boundary; decreasing θ enlarges it. Thus the lemma is best read as a* relative *statement: given the θ that defines NDVar (Def. 61), becoming forces a θ-robust paraconsistent band to exist around the transition.*

Remark 1641. Implementation note (minimal check). *In a computational or empirical setting, Lemma 31 can be used as a necessary diagnostic: sample or estimate $Dist(t;x,y)$ on a candidate transition window and verify that there exists a contiguous set of times on which both components exceed θ. Failure of this diagnostic rules out becoming* regardless *of any higher-level narrative about x and y.*

Lemma 32 (Non-becoming tests)**.** *Fix $(T,<)$ and Dist.*

1. *If there is no $t \in T$ with $Dist(t;x,y) \geq (\theta,\theta)$, then x does* not *become y.*
2. *If $Dist(t;x,y) \geq (\theta,\theta)$ for* all *$t \in T$, then x does* not *become y (because Def. 62.3 cannot be satisfied).*

Remark 1642. *This lemma gives two complementary failure modes for becoming. The first is the obvious one: if x and y are never in the required non-dual-variety relation, then there is no boundary region and hence no becoming. The second is subtler: if x and y are in NDVar* everywhere*, then the boundary ceases to be a boundary—there is no "outside" where the system recovers ordinary distinguishability.*

Operationally, these tests matter because they prevent over-ascription. Without them, one might call everything a becoming, either by never checking for the boundary signature or by allowing a perpetual blur to count as transition. The formalism insists that becoming is a localized phenomenon: neither absent everywhere nor present everywhere.

Sketch. (1) Contrapositive of Lemma 31. (2) Def. 62 requires a boundary region U where NDVar holds, but also that away from U the pair is more distinguishable (not strongly paraconsistent). If NDVar holds everywhere, the "outside" clause fails. ☐

Remark 1643. Proof sketch (strategy). *Part (1) is direct contrapositive reasoning; part (2) checks the "boundary vs. outside" clause in the definition of becoming.*

Key step. *The distinction between "there exists a neighborhood U" and "outside U something different happens" is what blocks the degenerate case where NDVar is ubiquitous.*

Intuition. *A boundary is meaningful only if it separates regions with different character; otherwise it is just uniform fog.*

Remark 1644. Clarifying the two failure modes. *Items (1) and (2) exclude opposite extremes:*

- *In (1), Dist never reaches the paraconsistent threshold in both channels simultaneously, so there is no candidate region in which x and y can be held in the structured ambiguity required by boundary non-duality. Even if Dist occasionally makes x and y look* similar *(e.g. one channel high), the defining feature of NDVar is the joint lower bound.*
- *In (2), the condition holds everywhere, so NDVar becomes the background state rather than a transition signature. The intended reading of Def. 62.3 is that becoming is witnessed not only by a boundary band but also by a recovery of "ordinary" discriminability away from that band. If there is no such recovery, then the situation is better described as persistent non-duality (a standing ambiguity) rather than becoming (a bounded passage).*

These tests therefore delineate a middle regime*: becoming is compatible with NDVar holding for* some *times (enough to form a neighborhood) but not for* all *times.*

Remark 1645. Discrete vs. continuous time. *When T is discrete (e.g. $T = \mathbb{Z}$ or a finite sequence of frames), a "neighborhood" U can be instantiated as a finite block of indices around a transition index. Lemma 32(1) then becomes a simple scan for any index with $Dist(t;x,y) \geq (\theta,\theta)$, while Lemma 32(2) becomes a check for the degenerate case where* every *index satisfies the NDVar inequality. When T is continuous, the same logic applies, but the existence of U is naturally read as the presence of an interval (possibly small) on which the inequality persists.*

Remark 1646. Robustness under strengthening the outside clause. *The exclusion in Lemma 32(2) is intentionally weak: it does not require specifying* how much *more distinguishable x and y must become outside U, only that Def. 62.3 fails if NDVar holds everywhere. In settings where one strengthens Def. 62.3 (for example, by requiring $Dist(t; x, y) \ngeq (\theta, \theta)$ on a nontrivial set outside U, or by requiring a quantitative drop in at least one component), the same failure mode remains: ubiquitous NDVar leaves no room for any strengthened "outside" behavior either.*

29.5 Quantale-valued event calculus: monotonicity and inertia rules

Lemma 33 (Rule contributions are lower bounds; multiple proofs join)**.** *For a quantale-valued Horn rule (Def. 64),*

$$P(\bar{x}) \Leftarrow Q_1(\bar{x}_1) \wedge \cdots \wedge Q_n(\bar{x}_n),$$

the semantics enforces the pointwise lower bound

$$[\![P(\bar{x})]\!] \;\geq\; [\![Q_1(\bar{x}_1)]\!] \otimes \cdots \otimes [\![Q_n(\bar{x}_n)]\!].$$

If multiple rules/instantiations derive the same head, the resulting head value is at least the $\oplus$-join (componentwise max in $[0,1]^2$) of all contributions.

Remark 1647. *This lemma clarifies how quantale-valued inference behaves: each rule instance contributes a* lower bound *on the heads truth-value. Conjunction in the body corresponds to $\otimes$ (so evidence composes), and multiple independent derivations aggregate via $\oplus$ (so the strongest derivation in each channel dominates, in the canonical model).*

The importance is that it makes explicit the proof-theoretic meaning of the quantale connectives. One can read $\oplus$ as "alternative reasons" and $\otimes$ as "joint reasons," which aligns with the broader goal of implementing a graded, potentially paraconsistent event calculus (Hyperseed-Concept **??***) using algebraic semantics [23, 24].*

Sketch. This is exactly Def. 64: conjunction maps to $\otimes$ and alternative derivations aggregate by $\oplus$. □

Remark 1648. Proof sketch (strategy). *This is definitional: interpret the rule under the chosen semantics.*

Key step. *The only "work" is remembering which connective maps to which quantale operation.*

Intuition. *A rule is not a claim that the head equals the bodys value; rather, it guarantees at least that much support. Additional rules can only add more support (via $\oplus$), never subtract it.*

Remark 1649. *A useful way to read the displayed inequality is as a semantic analogue of modus ponens for graded proofs: whenever the body holds to some degree (in the quantale order), the head must hold to at least the corresponding combined degree. In particular, if $V = [0,1]^2$ with the pointwise order, the inequality means that* each coordinate *of $[\![P(\bar{x})]\!]$ is bounded below by the corresponding coordinate of the $\otimes$-product of the body coordinates. This coordinatewise reading is what makes the "componentwise max" description of $\oplus$ operationally transparent: independent derivations can improve one channel without harming the other.*

Note also that the lemma is phrased as a lower bound rather than an equality because the canonical model is designed to be proof-accumulative: the semantics must admit additional support for $P(\bar{x})$ coming from other rules, from facts, or from later iterations in a fixpoint construction. Thus the inequality is the correct algebraic statement of "sound rule application" in a setting where multiple sources of evidence may coexist.

Lemma 34 (Monotonicity of proof accumulation)**.** *Assume all rules are built using only $\oplus$, $\otimes$, and monotone operations on V (as in Theorem 6). Then increasing any premise value (in the pointwise order on assignments) cannot decrease any derived head value.*

Remark 1650. *Monotonicity is the quiet structural condition that makes fixpoint semantics possible. It says: if you strengthen the premises (in either evidence channel), you cannot end up with weaker conclusions. This is the algebraic analogue of the ordinary Horn-clause property that adding facts does not invalidate derivations.*

This lemma is the hinge connecting local rule evaluation to global least-fixpoint constructions (Theorem 28). Without monotonicity, iterative semantics may oscillate or depend on evaluation order. With it, one may safely "accumulate proofs" as a growing approximation.

Sketch. $\oplus$ and $\otimes$ are monotone in each argument; joins over index sets preserve monotonicity. Thus each rule contribution is monotone, and the aggregate is monotone. □

Remark 1651. Proof sketch (strategy). *Reduce the claim to the monotonicity of the constructors used to build rule bodies and aggregation.*

Key step. *The pointwise order on $V^{(\cdot)}$ means monotonicity is checked coordinatewise at each ground atom/time; $\oplus$ and $\otimes$ preserve inequalities coordinatewise.*

Intuition. *If inference is built only from operations that never reverse inequalities, then inference itself never reverses them.*

Remark 1652. *Concretely, if $\nu \leq \nu'$ are two assignments of values in V to ground atoms (so $\nu(A) \leq \nu'(A)$ for every atom A), then for any fixed rule instance, the body evaluation under ν is $\leq$ the body evaluation under ν' because it is computed by composing monotone operations (iterated $\otimes$ and any other admissible monotone connectives). Applying the lemma repeatedly over all rule instances shows that a single "round" of consequence generation defines a monotone operator on the complete lattice of assignments, which is the exact hypothesis needed to justify least fixpoints (and hence canonical minimal models) by standard order-theoretic arguments.*

This also explains why non-monotone connectives (for example, a classical negation interpreted as order-reversing) are excluded from the rule building blocks in this fragment: they would break the monotone-operator property and thereby undermine the intended "accumulate and join" reading of proofs.

Lemma 35 (Clipped lower bound and monotonicity)**.** *Define Clipped as in Def. 65:*

$$Clipped(f;t_1,t_2) := \bigoplus_{a\in A,\ u\in T:\ t_1<u<t_2} Happens(a,u) \otimes Terminates(a,f,u).$$

Then for any particular witness (a,u) with $t_1 < u < t_2$,

$$Clipped(f;t_1,t_2) \ \geq\ Happens(a,u) \otimes Terminates(a,f,u).$$

Moreover, Clipped is monotone in each of $Happens$ and $Terminates$.

Remark 1653. *The operator Clipped expresses the idea that a fluent f might fail to persist from t_1 to t_2 because some terminating action happened in between. Quantale-valuedly, we aggregate* all *such potential terminating witnesses by a big $\oplus$. The lemma says two things: any particular terminating witness contributes a guaranteed lower bound, and strengthening either the evidence for actions happening or for their terminating effects can only increase the clippedness.*

This is a typical pattern in the algebraic event calculus (Hyperseed-Concept **??***): existential search over possible witnesses becomes a join over an index set, and the order-theoretic properties (lower bounds, monotonicity) become immediate consequences of the quantale operations.*

Sketch. The witness inequality holds because *Clipped* is a $\oplus$-join over a set that includes that term. Monotonicity follows from monotonicity of $\otimes$ and $\oplus$. $\square$

Remark 1654. Proof sketch (strategy). *Use the universal property of joins: a join is above each joined element.*

Key step. *The specific pair (a, u) is among the indices being joined, so its term is bounded above by the join.*

Intuition. *"Clipped" means "there exists evidence of a terminator"; $\oplus$ is the algebraic stand-in for that existential.*

Remark 1655. *It is often helpful to separate two layers in the definition of Clipped: (i) for each candidate witness (a, u), the conjunction "$Happens(a, u)$ and $Terminates(a, f, u)$" is evaluated by $\otimes$, producing a single combined piece of evidence that f is terminated in the open interval (t_1, t_2); and (ii) these candidate pieces of evidence are pooled by $\oplus$ across all intermediate times and actions. The first layer corresponds to requiring both an occurrence and an appropriate effect; the second corresponds to allowing any intervening terminator to suffice.*

*In particular, in the $[0, 1]^2$ setting, if one coordinate tracks positive support and the other tracks negative support (or, more generally, two independent evidential channels), then Clipped can increase in either coordinate as soon as there is a single witness whose combined $\otimes$-value increases in that coordinate. This coordinatewise behavior is precisely what is needed later when inertia is formulated as a rule with a condition like "*not clipped*": the clippedness score provides a graded obstruction to persistence that can be compared and combined monotonically in the subsequent fixpoint computation.*

Lemma 36 (One-step inertia propagation bound)**.** *Let Φ be the inertia update operator (Def. 66):*

$$(\Phi HoldsAt)(f, t_2) := HoldsAt_0(f, t_2) \;\oplus\; \bigoplus_{t_1 < t_2} HoldsAt(f, t_1) \otimes \neg Clipped(f; t_1, t_2).$$

Then for any $t_1 < t_2$,

$$(\Phi HoldsAt)(f, t_2) \;\geq\; HoldsAt(f, t_1) \otimes \neg Clipped(f; t_1, t_2).$$

Remark 1656. *This lemma isolates a single contribution to inertia: if f held at some earlier time t_1, and there is strong evidence that it was* not *clipped between t_1 and t_2, then the update Φ ensures that f holds at t_2 with at least the composed strength. It is the quantale-valued analogue of the familiar persistence axiom.*

The reason to spell this out is practical: in implementations, one often wants interpretable justifications for why a fluent holds. This inequality tells you exactly which term in the big join witnesses that justification.

Remark 1657. *Two order-theoretic facts are being used implicitly here and will reappear throughout: (i) in any join-semilattice, a join dominates each of its summands, i.e. $\bigoplus_i x_i \geq x_j$ for every j; and (ii) $\oplus$ itself is a join-like operation, so for any a, b one has $a \oplus b \geq b$ (and also $a \oplus b \geq a$). In particular, even if $HoldsAt_0(f, t_2)$ is weak (or $\perp$), the joined inertia support still lower-bounds the total update.*

Note also that the term $HoldsAt(f, t_1) \otimes \neg Clipped(f; t_1, t_2)$ is a single causal path *from past to present: it composes (i) the degree to which f held at t_1 with (ii) the degree to which the interval $[t_1, t_2]$ is free of terminating events for f. The lemma says that Φ never discards such a candidate path; it can only keep it or be strengthened by other paths via $\oplus$.*

Sketch. Immediate: the right-hand term is one of the joined summands in Def. 66. □

Remark 1658. Proof sketch (strategy). *Again use the fact that a join dominates each summand.*

Key step. *The index t_1 is included in the join over all $t_1 < t_2$, so its corresponding term provides a lower bound.*

Intuition. *Inertia is modeled as "take the best surviving support from the past"; any particular past-support path is a candidate, and the operator keeps at least the best one via $\oplus$.*

Remark 1659. *For readers who want the inequality written as an explicit chain, one may unpack Def. 66 as follows. Let*

$$S := \bigoplus_{t<t_2} HoldsAt(f,t) \otimes \neg Clipped(f;t,t_2).$$

Then $(\Phi HoldsAt)(f,t_2) = HoldsAt_0(f,t_2) \oplus S \geq S$ by join-domination, and $S \geq HoldsAt(f,t_1) \otimes \neg Clipped(f;t_1,t_2)$ because t_1 is one of the indices in the join defining S. Chaining these inequalities yields the claimed bound.

Operationally, the lemma can be read as a local certificate*: to justify a lower bound on $(\Phi HoldsAt)(f,t_2)$, it suffices to exhibit one earlier witness time t_1 together with the corresponding unclipped evidence, without needing to inspect the other summands.*

Theorem 28 (Least fixpoint semantics for monotone temporal inference (helper restatement of Theorem 6))**.** *Assume T is finite and all event-calculus rules use only $\oplus$, $\otimes$, and monotone operations on V. Then the induced immediate-consequence operator (e.g. Φ) is monotone on the complete lattice $(V^{F\times T}, \leq)$ and hence has a least fixpoint.*

Remark 1660. *This theorem provides the semantic foundation for the quantale-valued event calculus: the meaning of the theory is given by the* least *stable assignment of truth-values to all fluent-time pairs that satisfies the rules. That least fixpoint corresponds to "nothing is believed unless forced by the rules," generalized to graded/paraconsistent belief.*

The result is important because it replaces ad hoc procedural evaluation with an order-theoretic guarantee: iterating the immediate-consequence operator from the bottom converges (in finitely many steps when T is finite) to a canonical model. This style of reasoning via fixed points is a recurring tool in analyzing self-referential or self-updating systems (see also [10] for related fixed-point analyses in agent goal dynamics), and it sits naturally within the process view (Hyperseed-Concept 140) where "what holds" is stabilized by iterated update.

Remark 1661. *A useful way to read the statement is to separate (a) the* domain *and (b) the* operator*. The domain $V^{F\times T}$ is the space of all functions assigning each fluent-time pair (f,t) a value in V; the order is pointwise: $X \leq Y$ iff $X(f,t) \leq Y(f,t)$ for all (f,t). Completeness then follows from the fact that arbitrary joins and meets can be taken pointwise, e.g.*

$$\Big(\bigoplus_{i\in I} X_i\Big)(f,t) = \bigoplus_{i\in I} X_i(f,t), \qquad \Big(\bigwedge_{i\in I} X_i\Big)(f,t) = \bigwedge_{i\in I} X_i(f,t),$$

so the Knaster–Tarski theorem applies as soon as monotonicity of the immediate-consequence map is established.

The finiteness of T is used for the computational reading (finite convergence of iteration). Existence of lfp *itself does not require T finite; it is guaranteed by Knaster–Tarski on any complete lattice. What finiteness buys is that the ascending chain produced by Kleene iteration from $\perp$ cannot increase indefinitely through distinct time-slices without eventually stabilizing in the finite product.*

Sketch. $V^{F\times T}$ is a complete lattice under pointwise order. Each rule contribution is monotone (Lemma 34), hence the overall operator is monotone. Apply Knaster–Tarski to obtain a least fixpoint. $\square$

Remark 1662. Proof sketch (strategy). *Prove the operator is monotone; then invoke the Knaster–Tarski theorem, which guarantees least and greatest fixpoints for monotone maps on complete lattices.*

Key steps. *(i) Completeness of $V^{F\times T}$ comes from taking joins/meets pointwise. (ii) Monotonicity is inherited from the rule constructors by Lemma 34. (iii) Knaster–Tarski supplies existence of* lfp.

Intuition. *The lattice is the space of all possible temporal valuations; the operator is "one round of reasoning." A least fixpoint is the smallest valuation that survives another round unchanged: the minimal stable story consistent with the rules.*

Remark 1663. *For later use, it is often convenient to recall the equivalent characterization of the least fixpoint:*

$$\mathrm{lfp}(\Phi) = \bigwedge\{X \in V^{F\times T} \mid \Phi(X) \le X\},$$

i.e. the meet of all pre-fixpoints*. This emphasizes why the semantics is conservative: any valuation that is stable (or over-approximates stability) must lie above* $\mathrm{lfp}(\Phi)$*. Dually,* $\mathrm{gfp}(\Phi) = \bigoplus\{X \mid X \le \Phi(X)\}$ *exists as well, but the event-calculus reading typically privileges* lfp *as the "minimal commitment" model.*

When implementing the iteration from bottom, one typically starts from the pointwise least element $\bot \in V^{F\times T}$ (assigning $\bot$ everywhere) and forms the chain $\bot \le \Phi(\bot) \le \Phi^2(\bot) \le \cdots$. Monotonicity ensures this is ascending, and any stage in the sequence provides an under-approximation to $\mathrm{lfp}(\Phi)$*, which is useful when one wants progressively refined bounds on $HoldsAt(f,t)$.*

29.6 Time contexts, similarity, and regularity predicates

Lemma 37 (Basic properties of similarity and TemporalSimilarity)**.** *Let $\pi : V \to [0,1]$ be monotone and define $sim_V(v,w) = 1 - |\pi(v) - \pi(w)|$ (Def. 68). Let sim_T be the average over fluents (Def. 68) and TemporalSimilarity be the threshold predicate (Def. 69). Then:*

1. *$sim_V(v,w) = sim_V(w,v)$ and $0 \le sim_V(v,w) \le 1$, with $sim_V(v,v) = 1$.*
2. *$sim_T(t_1,t_2) = sim_T(t_2,t_1)$ and $0 \le sim_T(t_1,t_2) \le 1$, with $sim_T(t,t) = 1$.*
3. *TemporalSimilarity is symmetric, and is reflexive whenever the threshold $\tau \le 1$.*
4. *If $\tau_1 \le \tau_2$ then $\mathit{TemporalSimilarity}_{\tau_2}(t_1,t_2) \to \mathit{TemporalSimilarity}_{\tau_1}(t_1,t_2)$.*

Remark 1664. *The map $\pi : V \to [0,1]$ is a chosen* projection *from p-bit values to a single scalar—for instance, one might take $\pi(v) = v^+$ to measure "support" alone, or something more nuanced. Similarity $sim_V(v,w)$ is then the simplest possible metric-like notion: values are similar when their projected scalars are close. The time similarity sim_T averages this across all fluents, producing a single number in* $[0,1]$ *that summarizes how alike two timepoints look in the given time context (Def. 67).*

The lemma collects basic sanity properties: symmetry, boundedness, reflexivity, and monotonicity in the threshold. These are not deep, but they are the kinds of properties one needs later when building higher-level reasoning about regularity (Hyperseed-Concept **??***) and when interpreting temporal clustering or compression as abstraction.*

Sketch. (1) Absolute value is symmetric and bounded; (2) averages preserve symmetry and bounds; (3) follows from (1–2); (4) is threshold monotonicity. □

Remark 1665. Proof sketch (strategy). *Reduce everything to elementary properties of* $|\cdot|$ *and of averaging.*

Key steps. *(i)* $|\pi(v) - \pi(w)| = |\pi(w) - \pi(v)|$ *gives symmetry. (ii) Since* $\pi(v) \in [0, 1]$*, the difference lies in* $[-1, 1]$*, so* $1 - |\cdot|$ *lies in* $[0, 1]$*. (iii) Averaging preserves inequalities and range. (iv) Threshold predicates are monotone by definition.*

Intuition. *Similarity is engineered to behave like a sane resemblance score; the lemma verifies it does so.*

Lemma 38 (Continuity gives a projection-difference bound)**.** *If a fluent* f *is continuous (Def. 70) with parameter* $\eta \in (0, 1]$*, then for any* t_1, t_2*,*

$$TemporalSimilarity(t_1, t_2) \to |\pi(HoldsAt(f, t_1)) - \pi(HoldsAt(f, t_2))| \leq 1 - \eta.$$

Remark 1666. *Here "continuity" is a regularity condition on a fluent: whenever two times are deemed similar enough (TemporalSimilarity holds), the fluents value cannot change too much (after projection by* π*). The lemma simply rewrites this in the more familiar form of an upper bound on a difference.*

This matters because it turns a similarity-based condition into an inequality that can be composed with other numerical reasoning. In effect, it lets one treat TemporalSimilarity as a kind of control on variation, linking the qualitative predicate to quantitative bounds.

Sketch. Def. 70 gives $sim_V(HoldsAt(f, t_1), HoldsAt(f, t_2)) \geq \eta$. By Def. 68, $sim_V = 1 - |\pi(\cdot) - \pi(\cdot)|$, so $1 - |\Delta| \geq \eta$, i.e. $|\Delta| \leq 1 - \eta$. □

Remark 1667. Proof sketch (strategy). *Substitute the definition of* sim_V *and rearrange.*

Key step. *The transformation from* $1 - |\Delta| \geq \eta$ *to* $|\Delta| \leq 1 - \eta$ *is purely algebraic, but it is the step that makes the condition operational.*

Intuition. *"High similarity" means "small distance," because similarity was defined as* $1 -$ *distance.*

Lemma 39 (Increasing and decreasing imply constancy)**.** *Fix* $(T, <)$ *and projection* π *(Def. 72). If* f *is both increasing and decreasing, then for all* $t_1 < t_2$*,*

$$\pi(HoldsAt(f, t_1)) = \pi(HoldsAt(f, t_2)).$$

So $\pi(HoldsAt(f, t))$ *is constant along each* $<$*-chain.*

Remark 1668. *This lemma is an order-theoretic truism with a useful interpretive punch: a quantity that never goes down and never goes up is constant. Here the quantity is the projected fluent value along the proto-time order.*

Its role is to show how qualitative monotonicity constraints collapse degrees of freedom. In a system where fluents represent properties evolving through time, imposing both monotone directions forces stasis (at least in π*), which can be used either as a diagnostic (your constraints are too strong) or as a characterization (the fluent is invariant).*

Sketch. Increasing gives $\pi(t_1) \leq \pi(t_2)$, decreasing gives $\pi(t_1) \geq \pi(t_2)$; hence equality. □

Remark 1669. Proof sketch (strategy). *Squeeze the value between itself: if $a \leq b$ and $a \geq b$, then $a = b$.*

Key step. *The lemma relies only on antisymmetry of $\leq$ on $[0, 1]$, not on any special property of proto-time.*

Intuition. *Think of a curve that is everywhere nondecreasing and everywhere nonincreasing; the only such curve is flat.*

29.7 Functorial process view: composition along proto-time

Lemma 40 (Functorial composition along proto-time)**.** *Let T be the thin category generated by $(T, <)$ (Def. 73), and let $S : T \to Sys$ be a process (functor). If $t_1 < t_2 < t_3$, then the induced transition morphisms satisfy*

$$S(t_1 \to t_3) = S(t_2 \to t_3) \circ S(t_1 \to t_2).$$

Remark 1670. *This lemma expresses, in categorical language, the minimal coherence one expects of "process" (Hyperseed-Concept 140): going from t_1 to t_3 in one step must agree with going from t_1 to t_2 and then from t_2 to t_3. The thin category associated to a strict partial order has at most one arrow between any two objects, so the only question is whether a process respects this compositional structure.*

Its importance is conceptual unity. It is one of the places where the formalism aligns with a process ontology in the Whiteheadian spirit [15]: what persists across time is not a static substance but a web of transitions, and consistency means that these transitions compose. It also complements the event-calculus fixpoint view: both are ways of ensuring that temporal evolution is not arbitrary but structure-preserving.

Sketch. In the thin category, $(t_1 \to t_3)$ is the composite of $(t_1 \to t_2)$ and $(t_2 \to t_3)$. A functor preserves identity and composition, hence the equality. □

Remark 1671. Proof sketch (strategy). *Use the defining property of a functor: it preserves composition.*

Key step. *In a thin category generated by an order, the composite arrow $(t_1 \to t_2) \circ (t_2 \to t_3)$ is* by construction *the arrow $(t_1 \to t_3)$. Functoriality transfers this identity into Sys.*

Intuition. *If a process is a lawful passage, it cannot depend on whether you regard an interval as one stretch or two consecutive stretches; the "meaning" of the passage must be compositional.*

Remark 1672. *A useful way to read the statement is that S assigns to each comparable pair $t < t'$ a morphism in Sys that can be interpreted as an* evolution map *(or transition) from the system-at-t to the system-at-t'. The lemma then says that these evolution maps are compatible with* refinement of a proto-time interval*: if one refines the passage $[t_1, t_3]$ by inserting an intermediate moment t_2, the "overall" transition is unchanged, because it is determined by composing the refined transitions.*

This is exactly the categorical notion of path-independence *in a poset-shaped index category: there may be multiple ways to witness $t_1 < t_3$ by a chain of inequalities, but in a thin category all such chains represent the same unique arrow $t_1 \to t_3$. A functor cannot distinguish between these witnesses; it must send the unique arrow to a unique morphism in Sys, forcing all corresponding composites in Sys to agree. In this sense, the lemma is not an extra axiom on top of functoriality; rather, it is an unpacking of what functoriality means when the domain is generated by an order relation.*

Remark 1673. *The identity part of functoriality, though not written in Lemma 40, is often interpreted as the "no time passes" constraint: for each t there is an identity arrow $\mathrm{id}_t : t \to t$ in T, and functoriality forces $S(\mathrm{id}_t) = \mathrm{id}_{S(t)}$. Together with composition, this says that "doing nothing" is represented by an identity morphism and that successive passages concatenate without ambiguity. In applications, these two properties typically underwrite invariance principles (what is preserved by identity) and concatenation principles (how evolutions combine).*

Corollary 5 (Iterated composition along a chain)**.** *Let T and $S : T \to Sys$ be as in Lemma 40. For any finite chain $t_0 < t_1 < \cdots < t_n$, one has*

$$S(t_0 \to t_n) = S(t_{n-1} \to t_n) \circ \cdots \circ S(t_0 \to t_1).$$

Sketch. Apply Lemma 40 repeatedly (or argue by induction on n) using associativity of composition in Sys. □

Remark 1674. *Corollary 5 makes explicit the "independence from discretization" intuition: if one approximates a passage from t_0 to t_n by inserting more intermediate proto-times, the composite of the finer-grained transition morphisms must still agree with the coarse-grained transition. This is one categorical expression of a Markovian or semigroup-like intuition, but without requiring any additional analytic structure: it is a purely structural compatibility between (i) the order-theoretic shape of proto-time and (ii) the composition law of system morphisms in Sys.*

30 Helper theorems for effort, resistance, and simplicity (Section 8)

Standing assumptions and notation. Fix a context C. Let $E := ([0,\infty], \leq_E, \bigvee_E, \otimes_E, 0)$ be the effort quantale of Def. 74, where $\leq_E$ is the reversed numerical order, $\bigvee_E = \inf$ (numerical minimum), and $\otimes_E = +$ (sequential composition of costs). In particular, $0 \in [0,\infty]$ functions as the tensor unit for $\otimes_E$, so that composing a process with a zero-effort process leaves the effort unchanged, and ∞ serves as the absorbing/impossible cost that dominates sequential composition when present. The choice of $[0,\infty]$ (rather than $[0,\infty)$) ensures that infeasible or undefined operations can be represented internally without leaving the quantale.

Let P be a set of processes with effort assignment $\mathrm{Eff}_C : P \to [0,\infty]$ (Def. 75). One should read $\mathrm{Eff}_C(p)$ as a context-dependent cost model (time, energy, data requirements, or any aggregate resource) rather than as an intrinsic property of p in isolation; changing C may change what counts as "expensive" even for the same underlying algorithmic step. The separation between the abstract process set P and the numerical valuation Eff_C is used repeatedly in later statements to phrase claims as inequalities in the quantale rather than as ad hoc estimates.

Let X be a finite universe and $\mathcal{H}_C \subseteq \mathcal{P}(X \times X)$ the admissible indistinction policies (Def. 76). For policies $H, H' \in \mathcal{H}_C$, opening/closing are as in Def. 77. Policy effort $\mathrm{Eff}_C : \mathcal{H}_C \to [0,\infty]$ is antitone w.r.t. inclusion and marginal effort is $\Delta\mathrm{Eff}_C(H \to H') := \max(0, \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H))$ (Def. 78). Resistance is identified with $\Delta\mathrm{Eff}_C$ and submission is the case $\Delta\mathrm{Eff}_C(H \to H') = 0$ (Def. 79). The antitonicity convention encodes that allowing *more* indistinction (larger H) should not require *more* effort from the observer: once one relaxes discriminations, one can always "ignore" the additional distinctions for free, so the required effort is weakly smaller. The positive-part definition of $\Delta\mathrm{Eff}_C$ is important for directional comparisons: it distinguishes a genuine increase in required effort from a change that merely makes the task easier, and it makes "submission" behave like a zero-resistance edge in the policy graph (so that sequences of submissions remain submissions under addition of marginal efforts).

Simplicity $s_C(x)$ is the infimum of representation effort over admissible representations of x (Def. 80). Weakness $w(H)$ is monotone w.r.t. inclusion and satisfies the monotonic sanity theorem with effort (Thm. 7). Task adequacy $\mathrm{Adeq}_C(H)$, feasible set F_C, and MRE are as in Defs. 81–82; upward closure and the maximally open feasible policy H_b are as in Thm. 8. The infimum in the definition of $s_C(x)$ should be read operationally as "best available description under the rules of C": it permits multiple admissible representation schemes while still producing a single scalar that can be compared across objects. Monotonicity of weakness $w(H)$ with respect to inclusion matches the reading that opening a policy (adding more indistinction pairs) can only increase the space of confusions available to an adversary or learner, hence cannot decrease weakness; Thm. 7 then links this monotonicity to the effort ordering so that "easier" policies are not accidentally judged "stronger". For feasibility, the upward-closure property emphasizes that if a policy H is feasible (adequate at tolerable effort), then any further opening $H' \supseteq H$ remains feasible because it only relaxes constraints; the maximally open feasible policy H_b packages this closure into a canonical boundary object that can be used as a single representative of the feasible region.

Compositional simplicity s^*_C for a combination system $(X_C, *, e)$ with overhead t_C is Def. 84, with Prop. 4 and Thm. 9. Generalized description systems and generalized Kolmogorov complexity $K_C(x)$ are Defs. 85–86 with structural/compositional equivalence Prop. 5. The role of the overhead t_C is to isolate the "glue cost" of composition (parsing, interfacing, protocol negotiation, or other coordination burdens) from the base costs of the parts, so that s^*_C measures a compositional tradeoff rather than merely restating s_C. Prop. 4 and Thm. 9 are used later to justify that, under suitable regularity, compositional simplicity behaves like a controlled relaxation of plain simplicity: building an object from simpler parts can be cheaper, but not arbitrarily cheaper unless the overhead model permits it. Finally, the generalized description-system viewpoint provides a bridge to more classical complexity measures: $K_C(x)$ is a context-indexed analogue of description length, and Prop. 5 ensures that (within the permitted structural hypotheses) the "program-like" and "composition-like" formulations coincide up to the equivalence notion fixed there.

Remark 1675 (Intuition on the standing notation and conventions)**.** *The effort quantale E is a precise way to say: "smaller numerical cost means better / easier." This is why $\le_E$ is defined as the* reversed *numerical order: when we write $a \le_E b$ we are declaring that a is* no more effortful *than b in the sense relevant for the quantale. Concretely, if a process costs* 3 *units and another costs* 5 *units numerically, then the former should count as "larger" in the algebra of feasibility/ease; reversing the order makes the quantale joins and monotonicity statements align with that intuition (Hyperseed-Concepts 100, 158). Equivalently, one can keep the usual numerical order but then every monotonicity statement about "increasing effort" would have to be rewritten with order reversal; the present convention avoids that constant translation and lets inequalities read in the intended direction. This becomes especially convenient when effort is combined with other ordered quantities (e.g., weakness) via monotone maps: the categorical/quantale apparatus then directly reflects "harder implies no better guarantee"-type reasoning without sign changes.*

The symbol $\bigvee_E$ denotes the quantale join. Here $\bigvee_E =$ inf *(numerical minimum), reflecting the idea that when one may choose among multiple ways to achieve a result, the effective effort is the least effort among the options. The tensor $\otimes_E$ is sequential composition; here it is simply $+$, reflecting that doing p and then q generally costs the sum of the costs. These conventions place effort in the same algebraic family as the classic "tropical" or "min-plus" arithmetic often used for shortest paths; the present development uses it specifically to support the notions of resistance/submission and the effort–weakness tradeoff [3, 2] (Hyperseed-Concept 143). From this perspective, a statement such as "the effort of a composite plan is the tensor of the efforts" becomes an instance of a general algebraic law (associativity, unitality, and distributivity over joins), rather than a special-*

case arithmetic fact. Likewise, the interpretation of $\bigvee_E$ as "choose the best available option" is what allows later lemmas to take infima over families of candidate procedures without leaving the formal setting.

For policies $H \subseteq X \times X$, inclusion $H \subseteq H'$ should be read as "H' permits at least as much indistinction/coarse-graining as H." Thus an opening *is movement to a superset (loosening the observer's discriminations), while a* closing *is movement to a subset (tightening distinctions). The marginal effort $\Delta\mathrm{Eff}_C(H \to H') = \max(0, \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H))$ is the positive-part operator applied to an effort difference; it formalizes the idea that only* additional *required effort counts as resistance. In particular, if $H \subseteq H'$ is an opening, then antitonicity suggests $\mathrm{Eff}_C(H') \leq \mathrm{Eff}_C(H)$ numerically, so $\Delta\mathrm{Eff}_C(H \to H')$ is typically 0; this is the formal counterpart of "relaxing constraints should not be resisted." Conversely, for a closing $H' \subseteq H$ one expects (numerically) $\mathrm{Eff}_C(H') \geq \mathrm{Eff}_C(H)$, and then $\Delta\mathrm{Eff}_C(H \to H')$ records the extra work needed to regain discriminatory power. This directional asymmetry is why resistance is defined on ordered pairs $(H \to H')$ rather than as a symmetric distance: opening and closing have different operational meanings, and the cost of moving between policies depends on the direction of change.*

30.1 Effort quantale and effort of processes

Lemma 41 (Distributivity law (cost quantale))**.** *For any $a \in [0, \infty]$ and any set $S \subseteq [0, \infty]$,*

$$a + \inf S \;=\; \inf_{s \in S}(a + s), \qquad \inf S + a \;=\; \inf_{s \in S}(s + a).$$

Remark 1676 (What the lemma says, and why it matters)**.** *This lemma states that adding a fixed cost a "commutes" with taking the best available option (the infimum) among a family of costs. In plain terms: if you must pay a no matter what, then choosing the cheapest option from S and then adding a yields the same total as adding a to each option first and then choosing the cheapest.*

This is important because it is exactly the algebraic backbone that makes the min-plus effort structure behave like a well-formed quantale: sequential composition $(+)$ distributes over choice $(\inf)$. Later arguments about infima over representations or parse trees tacitly rely on being able to "push" fixed overhead through minima.

Remark 1677. ***Edge cases and the role of*** $[0, \infty]$ *Two minor points are worth keeping in view because they occur implicitly later. First, the lemma is stated in the extended nonnegative reals $[0, \infty]$, so $\inf S$ is always defined (possibly 0 or ∞), and addition is interpreted with the usual convention $a + \infty = \infty$ for all $a \in [0, \infty]$. In particular, if $S = \varnothing$, then $\inf S = \infty$ and both sides reduce to $a + \infty = \infty$ (respectively $\infty + a = \infty$), so the identities remain valid even for empty families of options.*

Second, the statement is really about the order structure: it uses only that the map $x \mapsto a + x$ is monotone and preserves arbitrary infima. This is why analogous distributivity identities continue to hold for many other "cost" structures used in semantics, provided they are complete lattices with a compatible monoidal action.

Sketch. This is exactly the statement that $+$ distributes over arbitrary infima in Def. 74. It can also be checked directly from basic properties of infimum in $[0, \infty]$. $\square$

Remark 1678 (Proof sketch and intuition)**.** *The strategy is to use the defining property of $\inf S$: it is the greatest lower bound of S. Adding a is order-preserving in the usual numerical order, so it sends lower bounds of S to lower bounds of $\{a + s : s \in S\}$, and likewise it sends the greatest lower bound to the greatest lower bound. Geometrically, if you picture the set S as points on the number* *line, adding a simply shifts the entire set to the right by a; the leftmost point (or limiting leftmost point) shifts by the same amount.*

Remark 1679 (A slightly more explicit inequality argument)**.** *For readers who prefer to see the two inequalities separately: let $m = \inf S$. Since $m \le s$ for all $s \in S$, monotonicity gives $a + m \le a + s$ for all $s \in S$, hence $a+m$ is a lower bound of $\{a+s : s \in S\}$, so $a+m \le \inf_{s\in S}(a+s)$. Conversely, if b is any lower bound of $\{a+s : s \in S\}$, then $b \le a+s$ for all $s \in S$, so $b-a \le s$ for all $s \in S$ (in the extended-real sense), hence $b - a \le \inf S = m$, so $b \le a + m$. Taking $b = \inf_{s\in S}(a + s)$ yields $\inf_{s\in S}(a+s) \le a+m$, giving equality. The right-hand identity $\inf S + a = \inf_{s\in S}(s+a)$ is the same argument, using commutativity of $+$ in $[0,\infty]$ (or simply re-running the monotonicity argument for $x \mapsto x + a$).*

Lemma 42 (Multi-step sequential subadditivity)**.** *Assume the sequential subadditivity axiom of Def. 75: $\mathrm{Eff}_C(p; q) \le \mathrm{Eff}_C(p) + \mathrm{Eff}_C(q)$. Then for any composable sequence $p_1; \cdots ; p_n$,*

$$\mathrm{Eff}_C(p_1; \cdots ; p_n) \ \le \ \sum_{i=1}^{n} \mathrm{Eff}_C(p_i).$$

Remark 1680 (What the lemma says, and why it matters)**.** *The lemma extends a two-step bound to an n-step bound: if doing p then q never costs more than doing them separately and summing, then the same is true for any finite chain. Intuitively, "any efficiencies of composition" are allowed (the inequality points in the permissive direction), but there can be no mysterious superlinear penalty that appears only after many steps.*

This kind of inequality is the minimal tool needed to compare a complex routine to the sum of its parts, which is precisely what later compositional simplicity and description-length arguments do: they break constructions into smaller pieces and add overheads.

Remark 1681 (Associativity and parenthesization conventions)**.** *The expression $p_1; \cdots ; p_n$ implicitly assumes that sequential composition is associative up to the intended notion of equality (or at least that Eff_C is insensitive to the choice of parenthesization). Concretely, one can read $p_1; \cdots ; p_n$ as the left-associated composite $((p_1; p_2); \cdots); p_n$, which is the form used in the induction proof. This is the only place where a bookkeeping convention is needed: the inequality then follows by repeated application of the two-step axiom, regardless of whether there may be other equivalent bracketings.*

Sketch. Induct on n. The case $n = 2$ is Def. 75. For $n + 1$, apply Def. 75 to $(p_1; \cdots ; p_n); p_{n+1}$ and then apply the inductive hypothesis. □

Remark 1682 (Proof sketch and intuition)**.** *The proof is the familiar "parenthesize-and-repeat" technique. One first groups the first n steps into a single macro-process $(p_1; \cdots ; p_n)$ and then applies the given two-step inequality to append p_{n+1}. The inductive hypothesis supplies the needed bound on the macro-process. Visually, one may imagine a chain whose total length is at most the sum of segment lengths; adding one more segment preserves the bound.*

Remark 1683 (Related consequence: bounded overhead accumulation)**.** *A useful way to rephrase the lemma is that any fixed per-stage upper bound propagates linearly: if $\mathrm{Eff}_C(p_i) \le M$ for all i, then $\mathrm{Eff}_C(p_1; \cdots ; p_n) \le nM$. Later, when "effort" is used to upper-bound the resource expenditure of a composite construction, this is the step that converts local (componentwise) estimates into a global estimate for a whole pipeline.*

Lemma 43 (Nested choice collapses to a global minimum)**.** *Assume the choice axiom of Def. 75: if p is implemented by choosing between p_1 and p_2, then $\mathrm{Eff}_C(p) = \min(\mathrm{Eff}_C(p_1), \mathrm{Eff}_C(p_2))$. If p can be implemented by choosing among a finite set $\{p_i\}_{i=1}^k$ via a binary-choice tree, then*

$$\mathrm{Eff}_C(p) = \min_{1 \le i \le k} \mathrm{Eff}_C(p_i).$$

Remark 1684 (What the lemma says, and why it matters)**.** *This lemma says that if "choice" is evaluated by taking the cheaper branch, then it does not matter how one nests binary choices: after all the nesting, the effort is simply the minimum among all leaf options. A simple example is a decision procedure that chooses between algorithms A and B, and then, if it chose A, chooses between A_1 and A_2; the overall effort is just* $\min(\mathrm{Eff}(A_1), \mathrm{Eff}(A_2), \mathrm{Eff}(B))$.

The point is conceptual as much as technical: it justifies treating a finite menu of alternatives as a single "min" operation, which is the algebraic dual of saying that opening up more options cannot increase the minimal required effort. This resonates with the later policy-union construction H_b and with compositional search over parse trees.

Remark 1685 (Ties, duplicates, and "do-nothing" options)**.** *Because the evaluation is by* min*, ties between branches are harmless: if two leaves have the same effort, the overall effort is that common value, and nothing depends on which tie-breaking rule an implementation might use. Likewise, duplicates among the p_i (or syntactically different branches with the same effort) do not change the result. This matches the intended interpretation: adding redundant alternatives cannot make the best achievable effort worse, and adding an explicitly available "do-nothing" or "fallback" branch simply introduces one more candidate in the global minimum.*

Sketch. Induct on the size of the choice tree: each binary decision replaces a node effort by the minimum of its children, so the root evaluates to the minimum over all leaves. □

Remark 1686 (Proof sketch and intuition)**.** *The high-level strategy is to observe that a binary-choice tree is evaluated bottom-up: every internal node applies the operation* min *to its two children. Since* min *is associative and commutative over finite sets (in the sense that repeated pairwise minima yield the overall minimum), the final value at the root must be the minimum over all leaves. One can picture water flowing downhill in a branching channel: it will always end up at the lowest reachable point, regardless of the branching pattern.*

Remark 1687 (Connection to finite infima)**.** *Since* $\min_{1 \le i \le k}$ *is the infimum over a finite set, this lemma can also be read as the finite version of replacing nested binary infima by a single infimum over all leaves. This is exactly the form used when one enumerates a finite family of candidate representations (or parse trees) and then defines the effort of the overall procedure to be the least effort among them. In that sense, the lemma is the operational counterpart of the algebraic identity that finite infima are computed by iterated binary infima.*

30.2 Policies, opening/closing, and resistance/submission

Lemma 44 (Compositionality of opening and closing)**.** *Let $H_0, H_1, H_2 \in \mathcal{H}_C$.*

1. *If $H_0 \subseteq H_1 \subseteq H_2$, then $H_0 \to H_2$ is an opening.*
2. *If $H_2 \subseteq H_1 \subseteq H_0$, then $H_0 \to H_2$ is a closing.*

Remark 1688 (What the lemma says, and why it matters). *An "opening" is defined purely by inclusion $H \subseteq H'$, and a "closing" by reverse inclusion. This lemma observes that these notions are stable under multi-step motion: if you open in stages, you have opened overall; if you close in stages, you have closed overall. The content is logically elementary, but it matters because later we will compare marginal efforts along multi-step paths, and we must know that the qualitative direction (opening vs. closing) is well-defined for the whole path.*

Conceptually, this is the set-theoretic shadow of the more phenomenological idea that one can relax distinctions gradually without ever reversing the direction of relaxation (Hyperseed-Concepts 125, 72).

Sketch. Both are immediate from Def. 77 and transitivity of $\subseteq$. □

Remark 1689 (Proof sketch and intuition). *The proof is a one-line use of transitivity: inclusion composes. If H_0 is contained in H_1 and H_1 is contained in H_2, then H_0 is contained in H_2. Visually, if each step enlarges your permissible indistinction relation, then the endpoint is an enlargement of the starting point.*

Lemma 45 (Marginal effort for opening vs closing). *Assume policy effort is antitone (Def. 78): $H \subseteq H' \to \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H')$. Then:*

1. *If $H \to H'$ is an opening, then $\Delta\mathrm{Eff}_C(H \to H') = 0$.*
2. *If $H \to H'$ is a closing, then $\Delta\mathrm{Eff}_C(H \to H') = \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H) \geq 0$.*

Remark 1690 (What the lemma says, and why it matters). *This lemma makes precise a guiding intuition:* opening should not cost extra effort *in the minimal model, whereas* closing may. *Under antitonicity, larger policies (more indistinction permitted) are easier/cheaper, so moving to a superset cannot increase effort. Therefore the positive part* $\max(0, \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H))$ *vanishes.*

The asymmetry between opening and closing is philosophically suggestive: it encodes a directionality reminiscent of thermodynamic coarse-graining (though here it is purely definitional and order-theoretic). In the later MRE result, this is what makes the "most open feasible" policy H_b into an effort minimizer (Hyperseed-Concepts 100, 158).

Sketch. (1) Opening means $H \subseteq H'$, so antitonicity gives $\mathrm{Eff}_C(H') \leq \mathrm{Eff}_C(H)$. Thus $\max(0, \mathrm{Eff}_C(H') - \mathrm{Eff}_C(H)) = 0$. (2) Closing means $H' \subseteq H$, so antitonicity gives $\mathrm{Eff}_C(H') \geq \mathrm{Eff}_C(H)$, so the max is attained by the difference. □

Remark 1691 (Proof sketch and intuition). *The proof is a direct case analysis on the sign of the difference* $\mathrm{Eff}_C(H') - \mathrm{Eff}_C(H)$. *Antitonicity tells you the sign once you know the inclusion direction. The operator* $\max(0, \cdot)$ *then acts as a diode: negative differences are clipped to* 0 *(no resistance), positive differences pass through unchanged (resistance equals the increase). One may picture opening as "releasing a brake" and closing as "tightening a brake" (Hyperseed-Concept 183).*

Corollary 6 (All openings are submissions in the minimal model). *Under the assumptions of Lemma 45, every opening $H \to H'$ satisfies submission: $\Delta\mathrm{Eff}_C(H \to H') = 0$.*

Remark 1692 (What the corollary says, and why it matters). *This corollary identifies a clean conceptual alignment: in the minimal model, opening is not merely "non-resistant" but is literally the definition of submission (zero marginal effort). This clarifies how resistance/submission are not separate phenomenological primitives here, but arise from the effort calculus.*

This is also a warning label: if one wants a model where opening can be difficult (e.g. because letting go of distinctions can require cognitive work), then one must weaken the antitonicity assumption or refine the effort definition. The present result is therefore a baseline, not a metaphysical necessity (Hyperseed-Concepts 125, 183).

Sketch. Immediate from Lemma 45(1) and Def. 79. □

Remark 1693 (Proof sketch and intuition)**.** *The proof simply matches two definitions: Lemma 45(1) gives the numeric condition $\Delta\mathrm{Eff}_C = 0$ for openings, and Def. 79 names that condition "submission." Conceptually, the corollary says that in this model, submission is nothing over and above the absence of additional cost.*

Lemma 46 (Triangle inequality for resistance (path subadditivity))**.** *For any real numbers a, b, c,*

$$\max(0, c-a) \;\le\; \max(0, b-a) + \max(0, c-b).$$

Hence for any policies H_0, H_1, H_2,

$$\Delta\mathrm{Eff}_C(H_0 \to H_2) \;\le\; \Delta\mathrm{Eff}_C(H_0 \to H_1) \;+\; \Delta\mathrm{Eff}_C(H_1 \to H_2).$$

Remark 1694 (What the lemma says, and why it matters)**.** *The first inequality is a purely numerical statement about the "positive part" function $x \mapsto \max(0, x)$: the positive part of the total change from a to c is bounded by the sum of the positive parts of the two changes $a \to b$ and $b \to c$. Interpreted for policies, it says resistance is* subadditive along a path*: going from H_0 to H_2 directly cannot require more marginal effort than taking a detour through H_1 and paying marginal effort at each step.*

This matters because it gives resistance a weak "metric-like" behavior (though it need not be symmetric, nor satisfy identity of indiscernibles). It also connects naturally to the idea that resistance is about where effort increases*, and increases can be accounted for incrementally (Hyperseed-Concept 158).*

A useful way to read the inequality is: any decrease along the path is "free" from the perspective of resistance (since the positive part ignores negative changes), and so the only contribution comes from the segments where effort goes up. In particular, if $a \le b \le c$, then all changes are nonnegative and the inequality becomes an equality $(c-a) = (b-a) + (c-b)$; whereas if one segment goes down (say $a < b$ but $c < b$), then the total increase from a to c may be smaller than the sum of segmentwise increases because the intermediate rise can be "undone" by a later drop. The lemma asserts that even in such mixed cases, counting rises separately along the broken line $a \to b \to c$ can only overestimate (never underestimate) the rise seen by jumping directly from a to c.

Equivalently, the statement can be viewed as a triangle-inequality analogue for the one-sided "distance" $d_+(u, v) := \max(0, v-u)$ on $\mathbb{R}$: it always holds that $d_+(a, c) \le d_+(a, b) + d_+(b, c)$. When resistance is defined via d_+ applied to effort levels, this immediately yields the corresponding path subadditivity for resistance values.

Sketch. Let $x = b - a$ and $y = c - b$, so $c - a = x + y$. Note $\max(0, x) \ge x$ and $\max(0, y) \ge y$, hence $\max(0, x) + \max(0, y) \ge x + y = c - a$. Also $\max(0, x) + \max(0, y) \ge 0$, so it is at least $\max(0, c-a)$. Substitute $a = \mathrm{Eff}_C(H_0)$, $b = \mathrm{Eff}_C(H_1)$, $c = \mathrm{Eff}_C(H_2)$.

For completeness, one can also see the inequality directly by a short case split on the signs of x and y. If $x \ge 0$ and $y \ge 0$, then $\max(0, x) + \max(0, y) = x + y = \max(0, x + y)$. If $x \ge 0$ and $y < 0$, then $\max(0, x) + \max(0, y) = x$, while $\max(0, x + y) \le \max(0, x) = x$ because adding a negative amount cannot increase the positive part beyond x. The case $x < 0$ and $y \ge 0$ is symmetric, yielding $\max(0,x)+\max(0,y)=y\geq\max(0,x+y)$. Finally, if $x<0$ and $y<0$, then both sides are 0 since $x+y<0$. This is the same numerical fact expressed in the more algebraic inequalities above, but it makes explicit that the only nontrivial situation is when one increment is positive and the other is negative. $\square$

Remark 1695 (Proof sketch and intuition)**.** *The key move is to rewrite the total difference as a sum of incremental differences:* $c-a=(b-a)+(c-b)$. *The positive-part function is convex enough that* $\max(0,x)+\max(0,y)$ *always dominates both* $x+y$ *and* 0, *hence dominates* $\max(0,x+y)$. *Visually, if you graph* $f(u)=\max(0,u)$, *the inequality expresses that the "upward jump" from* a *to* c *is no bigger than the sum of the two upward jumps along the broken line* $a\to b\to c$.

In policy terms, if $\mathrm{Eff}_C(H)$ *is interpreted as the effort level required by the agent (under constraint/context* C*) to implement or sustain policy* H*, then the resistance of changing from* H_i *to* H_j *is computed by applying the positive part to the effort difference. The lemma then says that if one contemplates a transition that passes through an intermediate policy* H_1*, the total "paid" resistance is the sum of the positive parts at each step, and this can only be larger than the resistance of moving directly from* H_0 *to* H_2*. Intuitively: intermediate overexertion is still overexertion, even if the endpoint ends up being easier than the intermediate.*

A further consequence that is often used implicitly is the extension to longer chains: by iterating the same inequality, for any sequence $H_0,H_1,\dots,H_n$ *one has*

$$\max\bigl(0,\mathrm{Eff}_C(H_n)-\mathrm{Eff}_C(H_0)\bigr)\leq\sum_{k=1}^{n}\max\bigl(0,\mathrm{Eff}_C(H_k)-\mathrm{Eff}_C(H_{k-1})\bigr),$$

so resistance accumulates at most additively when one decomposes a change into multiple stages. This is exactly the sense in which resistance can be "accounted for incrementally": the model never misses an increase by splitting into smaller steps, but it may double-count increases that are later reversed by decreases, which is appropriate if increases themselves are what generate resistance.

30.3 Simplicity as minimum representational effort and weakness duality

Lemma 47 (Representation upper bound)**.** *For any* $x\in X_C$ *and any admissible representation* $r\in R_C(x)$,

$$s_C(x)\ \leq\ \mathrm{Eff}_C(r).$$

Remark 1696 (What the lemma says, and why it matters)**.** *By definition, simplicity* $s_C(x)$ *is the least effort among all admissible ways to represent* x*. This lemma merely states the obvious consequence: any particular representation cannot beat the optimum. A simple example: if* x *has two encodings costing* 10 *and* 12*, then* $s_C(x)=10$*, and indeed* $10\leq 12$ *for the second encoding.*

The usefulness is methodological: whenever one constructs a representation explicitly (a code, a proof, a parse tree), this lemma turns that construction into an upper bound on simplicity. In complexity-theoretic language, it is the basic "exhibit a program" technique (Hyperseed-Concept 169; cf. [16]).

Remark 1697 (A small but useful refinement: infimum vs. minimum)**.** *Because Def. 80 uses an infimum, one should not silently assume that an optimal representation achieving* $s_C(x)$ *necessarily exists. For example, it is consistent with the definition that there is a sequence of admissible representations* (r_n) *with* $\mathrm{Eff}_C(r_n)\downarrow s_C(x)$*, while no single* r *attains* $\mathrm{Eff}_C(r)=s_C(x)$*. The inequality in Lemma 47 therefore has the correct logical strength: it gives an upper bound from* any *exhibited* r*, without requiring a minimizer.*

In particular, the lemma supports a common style of argument: to show that $s_C(x)$ is "small", it suffices to provide one concrete representation whose effort is small, even if the true optimum is approached only asymptotically in the representation space. Conversely, when lower bounds are sought, one must reason about all *admissible $r \in R_C(x)$, since no single worst-case witness suffices.*

Sketch. By Def. 80, $s_C(x) = \inf\{\mathrm{Eff}_C(r') : r' \in R_C(x)\}$, and an infimum is bounded above by any element of the set. $\square$

Remark 1698 (Proof sketch and intuition)**.** *The proof is an instance of a general order fact: the infimum of a set is below every element of the set. Intuitively, $s_C(x)$ is what you get when you "shop around" among all representations; any one receipt r provides a concrete price tag, and the best possible price cannot be higher than that.*

Remark 1699 (Connection to "two-part codes" and model selection heuristics)**.** *Interpreting $\mathrm{Eff}_C(r)$ as a description-length-like quantity makes Lemma 47 the abstract form of the familiar guarantee that any candidate code (or hypothesis description) provides a valid upper bound on the unknown optimal code length. In MDL-style reasoning, one often decomposes a description into "model" + "data given model"; in the present notation, this corresponds to constructing a composite representation r and then immediately obtaining $s_C(x) \leq \mathrm{Eff}_C(r)$. Even when the effort functional is not literally a bit-length, the lemma justifies the same workflow: propose a representation strategy, compute its effort, and treat that number as a certified (though possibly loose) simplicity bound.*

Lemma 48 (Monotonicity in the representation set)**.** *Let $R_C(x) \subseteq R'_C(x)$ and use the same effort assignment on representations. Let s_C and s'_C be defined by Def. 80 using R_C and R'_C respectively. Then $s'_C(x) \leq s_C(x)$.*

Remark 1700 (What the lemma says, and why it matters)**.** *If you allow* more *candidate representations, the minimal effort cannot go up: more options cannot worsen the best option. For instance, if you initially only permit handwritten descriptions but later also permit a compressed digital format, the simplest description can only become simpler (or remain the same).*

This monotonicity is crucial whenever we compare different modeling choices: enlarging a representation language, adding constructors, or relaxing admissibility constraints should never artificially inflate simplicity. It also foreshadows the analogous monotonicity in policy sets and in parse-tree decompositions.

Remark 1701 (When do we get strict improvement?)**.** *Lemma 48 only asserts $\leq$, and equality is common: if every newly admitted representation in $R'_C(x) \setminus R_C(x)$ has effort at least $s_C(x)$, then nothing changes. Strict inequality $s'_C(x) < s_C(x)$ occurs precisely when the enlargement introduces at least one new admissible representation whose effort is strictly below the previous infimum.*

This observation can be operationally important: it separates representational "expressivity" (having more syntactic forms available) from representational "efficiency" (actually lowering minimal effort for the specific x under study). It also cautions that claims like "language L' is more expressive than L" do not automatically imply that all objects become simpler; monotonicity guarantees non-increase, but improvement is instance-dependent.

Sketch. The infimum over a larger set cannot be larger: $\inf_{r \in R'_C(x)} \mathrm{Eff}_C(r) \leq \inf_{r \in R_C(x)} \mathrm{Eff}_C(r)$. $\square$

Remark 1702 (Proof sketch and intuition)**.** *The proof uses the elementary fact that if $A \subseteq B$ then* $\inf B \leq \inf A$*. Geometrically, if you plot all allowed effort values on the number line, adding more points can only move the leftmost point leftward (or keep it where it is).*

Remark 1703 (Relation to admissibility constraints and "free lunches")**.** *In applications, $R_C(x)$ often depends on constraints that encode what counts as a valid explanation, derivation, or encoding (e.g., a grammar, a proof system, a resource bound, or a notion of observational access). Lemma 48 ensures that relaxing such constraints cannot perversely make an object harder to describe. However, the lemma also highlights the hidden cost of relaxation: if one changes the admissibility notion, one may also implicitly change what is being measured. The monotonicity statement therefore supports careful bookkeeping: one can enlarge $R_C(x)$ to gain smaller $s_C(x)$, but one should record that this reflects a more permissive representational regime rather than a property of x alone.*

Theorem 29 (Effort-weakness monotone tradeoff along opening/closing)**.** *Assume weakness w is monotone under inclusion and policy effort is antitone (Thm. 7 assumptions). Then for any $H \subseteq H'$ in $\mathcal{H}_C$:*

$$w(H) \leq w(H'), \qquad \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H').$$

So opening (moving to a superset) simultaneously increases weakness and decreases effort, while closing does the opposite.

Remark 1704 (What the theorem says, and why it matters)**.** *The theorem articulates a clean tradeoff: if H' allows more indistinction than H (so $H \subseteq H'$), then two things happen in lockstep. First, weakness increases: $w(H) \leq w(H')$. Second, effort decreases: $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H')$. Thus, in this formalism, "to be weaker" is literally to be "more open" in policy space, and openness is easier in the effort sense (Hyperseed-Concepts 202, 143).*

This is important because it turns what might otherwise feel like a metaphor (weakness vs. effort) into an inference rule. It is also a conceptual hinge connecting the phenomenological primitives (effort/resistance) to the policy/observer apparatus: a monotone weakness functional is a way of measuring how much the observer permits indistinction, and the antitone effort makes that permission cheap. This monotone relationship is central in the broader "weakness is all you need" perspective [3, 2].

Remark 1705 (A boundary-case sanity check)**.** *At the extremes, the theorem behaves as expected. If H is already maximally open among the policies under consideration, there is no strict superset H', and thus no further reduction in effort via opening in that restricted policy class. If H is very closed (highly discriminating), moving to a slightly more open H' can be interpreted as allowing additional identifications (equivalences) among states or observations; the theorem then guarantees that this relaxation cannot cost* more *effort under an antitone effort functional.*

In that sense, the pair of inequalities can be read as a consistency condition on the primitives: if one adopts the normative stance that "being permissive about indistinction is easier", then antitonicity of Eff_C is the formal expression of that stance, and the theorem is the immediate payoff.

Sketch. This is exactly Thm. 7, restated as an inference rule. □

Remark 1706 (Proof sketch and intuition)**.** *No new argument is required: the proof simply invokes the previously established theorem. The underlying intuition is that the policy lattice is ordered by inclusion, while effort is contravariant and weakness is covariant with that order. One may picture policy space as a cone opening outward: moving outward (opening) broadens permissible indistinction, which by design counts as weaker but costs less.*

Remark 1707 (How to use the tradeoff as a design principle)**.** *In practice, Theorem 29 can be used in either direction. If one has a target effort budget, opening the policy (moving to a larger H') is a principled way to reduce $\mathrm{Eff}_C(H)$ at the price of increased weakness. Dually, if one needs* *a stricter notion of distinction (smaller weakness), closing the policy (moving to a smaller H) is guaranteed to raise the required effort.*

This is also the point where the "duality" language becomes operational: the partial order simultaneously organizes two quantities with opposite variance. As a result, inclusion chains $H_0 \subseteq H_1 \subseteq \cdots$ *automatically produce monotone sequences* $w(H_0) \leq w(H_1) \leq \cdots$ *and* $\mathrm{Eff}_C(H_0) \geq \mathrm{Eff}_C(H_1) \geq \cdots$*, which can be interpreted as Pareto-front trajectories in the* (w, Eff) *plane.*

30.4 Minimum representational effort (MRE) as maximal opening

Lemma 49 (Maximally open feasible policy is a greatest element)**.** *Assume the feasible set* $F_C := \{H \in \mathcal{H}_C : \mathrm{Adeq}_C(H)\}$ *is upward closed (Thm. 8 assumption):* $H \in F_C$ *and* $H \subseteq H' \to H' \in F_C$*. Define* $H_b := \bigcup_{H \in F_C} H$*. Then* $H_b \in F_C$ *and for all* $H \in F_C$*,* $H \subseteq H_b$*. Hence* H_b *is a greatest element of* $(F_C, \subseteq)$*.*

Remark 1708 (What the lemma says, and why it matters)**.** *This lemma says that if feasibility is stable under opening (upward closed), then there exists a single "most open" feasible policy* H_b*, obtained by unioning all feasible policies. In concrete terms: if every time a policy is adequate, any policy that forgets* even more *distinctions is also adequate, then the policy that forgets all distinctions that are forgettable while staying adequate is simply the union of all adequate forgettings.*

This is the order-theoretic heart of the MRE principle in this setting: it identifies a canonical extremal point of the feasible region, so optimization over F_C *reduces to selecting this extremal policy. The construction is intentionally simple: it uses the lattice structure of* $\mathcal{P}(X \times X)$ *under* $\subseteq$*.*

It is also worth making explicit how "opening" is encoded by $\subseteq$*. In this framework, a policy* $H \subseteq X \times X$ *is a set of allowed identifications (or "indistinguishabilities"); enlarging* H *means declaring* more *pairs indistinguishable, i.e., representing the world with fewer distinctions. Thus the inclusion order is aligned with an "openness" or "forgetfulness" order:* $H \subseteq H'$ *reads as "*H' *is at least as open as* H*."*

Finally, one minor edge case is that if $F_C = \varnothing$ *then the union* $\bigcup_{H \in F_C} H$ *is the empty set by convention, and the conclusion* $H_b \in F_C$ *is not automatic. In typical applications of Thm. 8 one implicitly has at least one adequate baseline policy (e.g., a maximally discriminating or otherwise trivially adequate policy), so* F_C *is nonempty and the lemma applies in its intended (nonvacuous) form.*

Sketch. This is Thm. 8(1–2): upward closure makes the union feasible, and union contains each member. □

Remark 1709 (Proof sketch and intuition)**.** *The proof uses two facts about unions. First,* $\bigcup_{H \in F_C} H$ *contains every* $H \in F_C$*, hence is above them in the inclusion order. Second, upward closure means that anything above a feasible element is feasible; since* H_b *is above every feasible element, it is feasible in particular. Visually, imagine stacking translucent sheets (each sheet a feasible policy); their overlay is the union, and upward closure says the overlay remains feasible.*

From an order-theoretic standpoint, what is being used is that $(\mathcal{P}(X \times X), \subseteq)$ *is a complete lattice, so any subset has a join (supremum), given concretely by union. The lemma is precisely the statement that when* F_C *is an* upper set *(i.e., upward closed), its join is not merely an upper bound but remains inside* F_C*, thereby becoming a greatest element.*

Corollary 7 (MRE reduces to choosing H_b)**.** *Under the assumptions of Lemma 49 and antitonicity of* Eff_C*,* H_b *is an effort minimizer over* F_C *(Thm. 8(3)). Moreover, if weakness* w *is monotone,* H_b *is also a weakness maximizer over* F_C *(Thm. 8(4)).*

Remark 1710 (What the corollary says, and why it matters)**.** *Once H_b is known to be the greatest feasible policy, the optimization becomes almost tautological: if effort is antitone, then greater policies have lower effort, hence the greatest feasible policy has the least effort. If weakness is monotone, then greater policies have greater weakness, hence the same H_b maximizes weakness among feasible policies.*

This is an instance of a general pattern: when an objective is monotone and the feasible set has an extremal element, optimization is solved by selecting that extremal element. The corollary therefore ties together feasibility (adequacy), effort (minimization), and weakness (maximization) in a single stroke [9, 3] (Hyperseed-Concepts 202, 100).

Intuitively, the antitonicity of Eff_C *formalizes the idea that representing* more *distinctions costs* more *representation: if $H \subseteq H'$ then H enforces at least as many distinctions as H', and therefore demands at least as much representational effort. The adequacy constraint* Adeq_C *then restricts how far one can open while still meeting the context C; the lemma identifies the farthest open point that remains feasible, and the corollary states that any monotone objective must take its optimum there.*

In particular, this "monotone objective + extremal feasible element" structure is what makes the MRE conclusion robust to many possible choices of effort functional: one does not need a detailed analytic form of Eff_C*, only its order-compatibility with opening. This is why the result reads as a structural theorem about the poset $(F_C, \subseteq)$ rather than as a computation.*

Sketch. If $H \subseteq H_b$ and Eff_C is antitone, then $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H_b)$. Similarly monotonicity of w gives $w(H) \leq w(H_b)$. ☐

Remark 1711 (Proof sketch and intuition)**.** *The proof is a direct application of monotonicity/antitonicity along the inclusion order. The key step is recognizing that Lemma 49 provides $H \subseteq H_b$ for every feasible H; the rest is just pushing inequalities through monotone functions. One may picture F_C as a region in the poset with a "top" element; any monotone quantity reaches an extreme at the top (or bottom, depending on variance).*

A helpful way to read the inequalities is as a comparison principle: any feasible policy H is dominated (in openness) by H_b, and so any antitone cost is dominated (in value) by its value at H_b. In the same way, any monotone "capacity to fail gracefully" or "weakness" quantity is maximized at the same dominating policy. No additional combinatorics about F_C is needed beyond the existence of this top element.

Lemma 50 (Uniqueness under strict antitonicity)**.** *Assume strict antitonicity: if $H \subset H'$ then* $\mathrm{Eff}_C(H) > \mathrm{Eff}_C(H')$*. Then under the assumptions of Lemma 49, H_b is the unique effort minimizer in F_C.*

Remark 1712 (What the lemma says, and why it matters)**.** *The previous corollary identified H_b as an effort minimizer, but it allowed ties. This lemma says that if opening* strictly *reduces effort (not just weakly), then there can be no tie: only the most open feasible policy achieves the minimum.*

This uniqueness is useful when one wants determinacy in the theory: it means that "the" minimum representational effort policy is well-defined, rather than being a choice among several incomparable minimizers. It also clarifies which assumption is doing the work: uniqueness does not come from feasibility but from strictness of the effort decrease.

Conceptually, strict antitonicity rules out "flat directions" in the effort landscape: whenever one can strictly open a policy (move to a strict superset), one must strictly decrease effort. In contexts where effort is derived from, e.g., description length, memory usage, or constraint count, strictness corresponds to the claim that every additional allowed identification eliminates some representational burden rather than merely leaving it unchanged.

Sketch. Let $H \in F_C$ with $H \neq H_b$. Then $H \subset H_b$ (since H_b is greatest), so strict antitonicity implies $\mathrm{Eff}_C(H) > \mathrm{Eff}_C(H_b)$. Thus no other H can tie the minimum. $\square$

Remark 1713 (Proof sketch and intuition). *The proof is a standard extremal-element argument. Any feasible $H \neq H_b$ must lie strictly below the greatest element H_b, and strict antitonicity turns "strictly below" into "strictly higher effort." Visually, there is a strictly descending slope of effort as one moves upward in the policy poset, so only the top point can be lowest in effort.*

Equivalently, strict antitonicity makes Eff_C an order-embedding (up to reversal) on comparable elements: comparability $H \subset H_b$ is transported to a strict inequality of effort values. Since Lemma 49 guarantees that every feasible H is comparable to H_b (indeed, below it), the strict comparison applies to all competitors, yielding uniqueness without needing any additional structure on F_C (such as convexity, finiteness, or chain conditions).

30.5 Compositional simplicity: parse trees, monotonicity, and optimal substructure

Lemma 51 (Parse-tree upper bound). *Assume the setting of Defs. 83–84 and nonnegative overhead $t_C \geq 0$. For any full binary parse tree T evaluating to x with internal-node costs given by t_C,*

$$s_C^*(x) \;\leq\; \mathrm{cost}(T).$$

Remark 1714 (What the lemma says, and why it matters). *Compositional simplicity $s_C^*(x)$ is defined as the least effort required to build x from primitives by binary composition, paying an overhead t_C at each composition step (Hyperseed-Concept 82). A parse tree T is a concrete witness of one particular way to build x; this lemma says that the optimal cost is no more than the cost of that witness.*

As a simple example, if $x = (a * b) * c$ and you have a tree that first combines a, b (paying $t_C(a,b)$) and then combines the result with c (paying $t_C(a*b,c)$), then $s_C^*(x)$ is bounded above by the sum of the leaf costs (here 0 for the unit e in the simplest base case) plus those overheads. This lemma is the core "certificate gives an upper bound" principle that makes dynamic programming and optimal-substructure reasoning possible; it is the same structural idea used in dependency tower analyses [8].

Sketch. Proceed by structural induction on T. If T is a leaf labeled by e, then $x = e$ and $s_C^*(e) = 0 = \mathrm{cost}(T)$ (Def. 84). If T has root children T_1, T_2 evaluating to y, z with $y * z = x$, then Prop. 4 gives $s_C^*(x) \leq s_C^*(y) + s_C^*(z) + t_C(y,z)$. Apply the induction hypothesis to bound $s_C^*(y) \leq \mathrm{cost}(T_1)$ and $s_C^*(z) \leq \mathrm{cost}(T_2)$, and note $\mathrm{cost}(T) = \mathrm{cost}(T_1) + \mathrm{cost}(T_2) + t_C(y,z)$. $\square$

Remark 1715 (Proof sketch and intuition). *The strategy is structural induction, which mirrors the recursive nature of parse trees. At each internal node, one uses the compositional inequality from Prop. 4 to relate the simplicity of the parent to the simplicities of the children plus overhead. The induction hypothesis then converts those child simplicities into the concrete subtree costs. A helpful mental picture is that cost accumulates locally at each internal node; the global bound is obtained by summing those local contributions along the tree.*

Corollary 8 (Minimum parse tree characterization (finite case)). *In the finite case of Thm. 9, $s_C^*(x)$ equals the minimum $\mathrm{cost}(T)$ over all parse trees T for x.*

Remark 1716 (What the corollary says, and why it matters). *The previous lemma provides an inequality $s_C^*(x) \leq \mathrm{cost}(T)$ for every tree T for x. This corollary says that, in the finite case, this* *inequality is tight:* $s^*_C(x)$ *is not merely bounded by tree costs, it is* exactly *the minimal tree cost. Thus compositional simplicity can be computed (in principle) by searching over parse trees.*

This is a pivotal bridge between a fixed-point/least-solution definition (Def. 84, Thm. 9) and a combinatorial optimization view (minimization over trees). The latter view enables algorithmic intuition: optimal substructure, pruning, and monotonicity in overhead.

Sketch. Lemma 51 shows $s^*_C(x)$ is at most any tree cost. Thm. 9 states that the least solution of Def. 84 achieves the minimum parse-tree cost, i.e. the bound is tight. □

Remark 1717 (Proof sketch and intuition)**.** *The proof is a two-sided squeeze: (i) every tree gives an upper bound on* $s^*_C(x)$*, so* $s^*_C(x)$ *is at most the minimum tree cost; (ii) Thm. 9 ensures that the recursive definition does not leave a gap: the least fixed point corresponds to actually attaining that minimum. One may think of Thm. 9 as ensuring that the recursive "Bellman-style" equations faithfully encode the global optimization.*

Lemma 52 (Optimal substructure)**.** *Assume the finite case of Thm. 9. Let* T *be a minimum-cost parse tree for* x *with root split* $x = y * z$ *and subtrees* T_1, T_2 *for* y, z*. Then* T_1 *is minimum-cost among parse trees for* y*, and* T_2 *is minimum-cost among parse trees for* z*.*

Remark 1718 (What the lemma says, and why it matters)**.** *Optimal substructure is the classic principle behind dynamic programming: an optimal whole is built from optimal parts. Here it says: if* T *is the cheapest way to build* x*, and the root decomposes* x *into* y *and* z*, then the chosen subtree for* y *must be the cheapest way to build* y *(and similarly for* z*). Otherwise, we could swap in a cheaper subtree and improve* T*.*

This lemma matters because it legitimizes bottom-up computation of s^*_C*: one can compute best costs for smaller components and reuse them, rather than enumerating all global trees naively. In the broader program, this resonates with how complex systems reuse optimal subassemblies, and how dependency structures enforce local optimality constraints [8] (Hyperseed-Concept 94).*

Sketch. If T_1 were not optimal for y, replace it by a cheaper tree T_1' for y. This strictly decreases the cost of the full tree for x, contradicting optimality of T. Similarly for T_2. □

Remark 1719 (Proof sketch and intuition)**.** *The proof is by contradiction via local improvement: assume a subtree is not optimal, then perform a local surgery (replace* T_1 *by a cheaper* T_1'*) that leaves the root combination structure intact but strictly lowers total cost, contradicting minimality. Geometrically, one can think of the full tree cost as the sum of independent contributions from disjoint subtrees plus the root overhead; lowering one disjoint contribution lowers the whole.*

One way to formalize the "local surgery" is: fix a parse tree T *for* x *and identify a node whose associated subexpression evaluates to some intermediate object* u*. The subtree below that node is itself a parse tree for* u*, contributing a well-defined additive amount to the total cost of* T *(the sum of the overheads on the subtree's internal nodes, plus whatever base/leaf costs are part of the model). If there exists another parse tree* T_u' *for the same* u *with strictly smaller cost, then substituting* T_u' *in place of the old subtree produces a new parse tree for* x *with the same outer shape above that node and strictly smaller total cost. The only requirement for this substitution argument is that the cost functional be additive across disjoint subtrees together with the overhead at the parent node—i.e., that the cost of a full tree decomposes as "root overhead + cost(left subtree) + cost(right subtree)" (and similarly recursively). This is the standard "optimal substructure" property familiar from dynamic programming: global optimality forces local optimality of every component.*

It is also worth noting why strict improvement is available: if the subtree is not optimal, then by definition there is some alternative with strictly smaller cost, so the replacement decreases the *total by at least the gap between the two subtree costs. If there are ties (multiple distinct subtrees with the same minimal cost), the argument still goes through with "$\leq$" but yields non-uniqueness rather than a contradiction; the contradiction requires the assumption that one has chosen a globally minimal tree and then found a strict local improvement.*

Lemma 53 (Monotonicity in overhead and in available decompositions)**.** *Assume the finite case of Thm. 9.*

1. *If $t'_C(y,z) \leq t_C(y,z)$ for all (y,z), then the corresponding compositional simplicities satisfy $s^*_{C,t'}(x) \leq s^*_{C,t}(x)$ for all x.*
2. *If the decomposition set is enlarged (more admissible pairs (y,z) with $y * z = x$), then $s^*_C(x)$ cannot increase.*

Remark 1720 (What the lemma says, and why it matters)**.** *Part (1) says: if you make composition cheaper everywhere, then the optimal compositional cost cannot go up. Part (2) says: if you allow more ways to decompose/build things, you may find cheaper constructions, but you cannot be forced into a more expensive optimum. Both are sanity checks that the formalism respects the everyday logic of optimization.*

These monotonicity principles are also philosophically important: they express that simplicity is not an intrinsic property of x alone, but of x relative to a constructive regime (allowed decompositions, overhead schedule). Changing the regime changes simplicity in a directionally predictable way (Hyperseed-Concept 82).

A useful way to read (1) is pointwise monotonicity of the objective: each parse tree T induces a real number $\mathrm{cost}_t(T)$, and switching from t to t' moves every one of these real numbers downward (or leaves it unchanged). Since the optimum is an infimum/ minimum over the same set of trees, it cannot move upward. In particular, if $t'_C < t_C$ on at least one overhead actually used by every optimal tree under t, then one expects a strict decrease in s^*; but the lemma correctly states only the weak inequality because it may happen that the only decreases occur on decompositions that no optimal construction ever uses.

Likewise, (2) is monotonicity in the feasible set: enlarging the admissible decomposition relation enlarges the set of realizable parse trees. The cost function itself does not become worse; there are simply more candidate trees to compare. Thus the optimal value can only stay the same or drop. This captures an important modeling intuition: granting the agent an extra construction rule cannot rationally make its best achievable simplicity worse, even if the agent chooses to ignore the new rule. (The statement is about the optimal *cost, not about what a particular algorithm might output if it were constrained to use the new rule.)*

Finally, the finiteness assumption in Thm. 9 matters only to justify "minimum" rather than merely "infimum": in an infinite setting one would typically restate the lemma with inf *over all parse trees. The same order-theoretic logic would apply, but one must be slightly more careful about whether the optimal value is attained by an actual tree.*

Sketch. Both are immediate from the minimum-parse-tree characterization: decreasing per-node costs or adding new candidate trees can only decrease (or preserve) the minimum.

More explicitly, fix x and let $\mathcal{T}(x)$ denote the set of admissible parse trees for x under the given regime. For (1), the hypothesis $t'_C \leq t_C$ implies $\mathrm{cost}_{t'}(T) \leq \mathrm{cost}_t(T)$ for every $T \in \mathcal{T}(x)$, because the total cost is a sum of node overheads (and other nonnegative components, if present) evaluated at the same node labels and child pairs. Taking minima over the same set $\mathcal{T}(x)$ yields $\min_{T\in\mathcal{T}(x)} \mathrm{cost}_{t'}(T) \leq \min_{T\in\mathcal{T}(x)} \mathrm{cost}_t(T)$, i.e., $s^*_{C,t'}(x) \leq s^*_{C,t}(x)$. For (2), if $\mathcal{T}(x) \subseteq \mathcal{T}'(x)$ after enlarging the decomposition set, then $\min_{T\in\mathcal{T}'(x)} \mathrm{cost}(T) \leq \min_{T\in\mathcal{T}(x)} \mathrm{cost}(T)$ by the elementary fact that the minimum over a superset cannot exceed the minimum over a subset. $\square$

Remark 1721 (Proof sketch and intuition)**.** *The proof reduces everything to minimization over a set of real numbers: the set of parse-tree costs. In (1), every tree's cost weakly decreases when node overheads decrease, hence the minimum weakly decreases. In (2), adding decompositions enlarges the set of available trees, and the minimum over a larger set cannot be larger. This is the same order fact used earlier for representation sets (Lemma 48).*

It can be helpful to keep separate the two "directions" in which monotonicity may act:

- *Changing t_C changes the* cost assigned to each fixed tree *(the feasible set stays the same).*
- *Changing the admissible decompositions changes the* feasible set of trees *(the cost assignment stays the same).*

Both are instances of the same basic principle: if you make some options cheaper, or you provide additional options, an optimum defined as a minimum cannot worsen. The lemma thus functions as a consistency check on the definition of s^*_C: it behaves like an honest optimization objective rather than an ad hoc complexity score.

As a concrete mental model, consider building x from two parts y and z with overhead $t_C(y, z)$. If one decreases all such overheads by a constant $\delta \geq 0$, then every *internal node becomes cheaper by at most δ, so the total cost of any fixed tree decreases by (at most) δ times the number of internal nodes. The optimal tree may change when costs change, but the optimum value cannot increase. Similarly, if one adds a new admissible decomposition $x = y * z$ that was previously forbidden, then one has strictly more candidate trees for x; if the added decomposition is never beneficial, the optimum stays the same, and if it is beneficial, the optimum drops.*

In this sense, Lemma 53 is the compositional analogue of the monotonicity properties one expects from shortest paths: lowering edge weights cannot increase the shortest-path distance, and adding edges cannot increase it. Here, parse trees play the role of paths, and node overheads play the role of local weights.

30.6 Generalized Kolmogorov complexity and structural complexity

Lemma 54 (Description upper bound)**.** *For any generalized description system (Def. 85) and any $d \in D_C$ with $\mathrm{Dec}_C(d) = x$,*

$$K_C(x) \;\leq\; \mathrm{Eff}_C(d).$$

Remark 1722 (What the lemma says, and why it matters)**.** *This is the direct analogue of Lemma 47, now in the setting of generalized Kolmogorov complexity: $K_C(x)$ is defined as the infimum effort among descriptions that decode to x, so any particular description gives an upper bound. In the classical setting of Kolmogorov complexity, the slogan is: "any program that prints x shows $K(x)$ is at most the program's length" [16] (Hyperseed-Concept* **??***).*

The lemma is useful whenever we design a description language: it allows one to bound complexity by constructing explicit descriptions, and it provides the basic comparison principle used in the compositional bound below.

Concretely, one can read the statement as separating two tasks: (i) exhibit a description d that works *(i.e., decodes to the target object x), and (ii) evaluate or upper bound the effort* $\mathrm{Eff}_C(d)$ *in the chosen effort model. The lemma then turns (ii) into an immediate bound on $K_C(x)$. This is typically how upper bounds are proved in practice: instead of reasoning directly about the infimum over all possible descriptions, one presents a structured family of candidate descriptions and analyzes their effort.*

Note that nothing in the lemma requires optimality, uniqueness, or even computability of the infimum: it is purely an order-theoretic consequence of defining $K_C(x)$ as an infimum over the set

$\{ \mathrm{Eff}_C(d) : d \in D_C,\ \mathrm{Dec}_C(d) = x \}$. *In particular, if there are many distinct descriptions of x (e.g., many programs that output the same string), each yields a possibly different upper bound, and the best such upper bound is precisely $K_C(x)$.*

Sketch. Immediate from Def. 86: $K_C(x)$ is the infimum of $\mathrm{Eff}_C(d)$ over all descriptions decoding to x. ☐

Remark 1723 (Proof sketch and intuition)**.** *As before, the proof is the general order fact that an infimum is below every element of its set. Intuitively: the "best possible" description cannot be worse than any particular description you already have in hand.*

It is sometimes helpful to spell out the quantifiers explicitly: Def. 86 says

$$K_C(x) = \inf\{\mathrm{Eff}_C(d) : d \in D_C,\ \mathrm{Dec}_C(d) = x\}.$$

Fixing any particular d with $\mathrm{Dec}_C(d) = x$ simply means $\mathrm{Eff}_C(d)$ is one element of this set, so the infimum cannot exceed it. This remains true even when the infimum is not attained (i.e., even if there is no single description whose effort equals $K_C(x)$).

Lemma 55 (Compositional upper bound under a homomorphism assumption)**.** *Assume there is a binary constructor/concatenation $\cdot$ on descriptions such that $\mathrm{Eff}_C(d_1 \cdot d_2) \le \mathrm{Eff}_C(d_1) + \mathrm{Eff}_C(d_2)$ (Def. 85), and additionally assume a decoder compatibility (homomorphism) condition:*

$$\mathrm{Dec}_C(d_1) = x,\ \mathrm{Dec}_C(d_2) = y \ \rightarrow\ \mathrm{Dec}_C(d_1 \cdot d_2) = x * y.$$

Then

$$K_C(x * y) \ \le\ K_C(x) + K_C(y).$$

Remark 1724 (What the lemma says, and why it matters)**.** *This lemma says that complexity is subadditive under composition: if you can describe x and describe y, then you can describe their composite $x * y$ with effort at most the sum of efforts (up to the constructor overhead encoded in the inequality for $\cdot$). The extra assumption $\mathrm{Dec}_C(d_1 \cdot d_2) = x * y$ is a homomorphism condition ensuring that "concatenate the descriptions" really corresponds to "compose the objects."*

This is a generalized version of a familiar fact from algorithmic information theory: $K(x, y) \le K(x) + K(y) + O(1)$, where the $O(1)$ accounts for pairing/concatenation overhead [16]. Here the overhead is absorbed into the effort inequality for $\cdot$. The lemma is also a conceptual precursor to the equivalence theorem below: it suggests that description complexity and parse-tree (compositional) complexity obey the same algebra.

The compatibility assumption can be viewed as a correctness constraint on the description language: the syntax-level operation $\cdot$ must implement the intended semantic operation $*$ after decoding. Without such a constraint, one could still have a subadditive effort inequality on raw descriptions, but it would not translate into any bound on the complexity of semantic composites $x * y$. In other words, the lemma isolates the two minimal ingredients needed to inherit subadditivity at the level of decoded objects: a low-overhead way to combine descriptions, and a decoder that respects that combination.

Depending on the application, the effort inequality $\mathrm{Eff}_C(d_1 \cdot d_2) \le \mathrm{Eff}_C(d_1) + \mathrm{Eff}_C(d_2)$ can be interpreted in different concrete ways. If Eff_C measures bit-length, it says concatenation does not increase total length beyond the sum (often up to a fixed delimiter cost). If Eff_C measures time or energy, it says the work to execute the combined description is no worse than running the two parts sequentially (again, up to the stipulated overhead encoded in Def. 85).

Remark 1725 (How it connects to earlier results). *Compare the inequality $K_C(x * y) \le K_C(x) + K_C(y)$ with the compositional recursion for s^*_C and with Lemma 42: all are manifestations of the same principle that sequential/constructive composition should not create superlinear effort by fiat. In that sense, this lemma is the "description-language" counterpart of compositional simplicity.*

From the perspective of dynamic programming or optimal substructure, the lemma can also be read as stating that if an object is formed by a binary composition, then a near-optimal description of the whole can be obtained by stitching together near-optimal descriptions of the parts, provided the description system is designed so that such stitching is valid and does not introduce disproportionate overhead. This is exactly the kind of closure property one wants when relating "how objects are built" to "how objects are described."

Sketch. Fix $\epsilon > 0$. Choose d_1, d_2 with $\mathrm{Dec}_C(d_1) = x$, $\mathrm{Dec}_C(d_2) = y$ and $\mathrm{Eff}_C(d_1) \le K_C(x) + \epsilon$, $\mathrm{Eff}_C(d_2) \le K_C(y) + \epsilon$ (approximate infimum). Then $\mathrm{Dec}_C(d_1 \cdot d_2) = x * y$ and $\mathrm{Eff}_C(d_1 \cdot d_2) \le \mathrm{Eff}_C(d_1) + \mathrm{Eff}_C(d_2) \le K_C(x) + K_C(y) + 2\epsilon$. Apply Lemma 54 to $d_1 \cdot d_2$ and let $\epsilon \to 0$. $\square$

Remark 1726 (Proof sketch and intuition). *The strategy is the standard "ϵ-approximate minimizer" technique: because $K_C(x)$ is an infimum, we may not have a literal minimizer, but we can get arbitrarily close. We then compose two near-optimal descriptions and use the assumed subadditivity of effort under $\cdot$ to bound the composite's effort. Finally, we remove the slack by sending $\epsilon \to 0$.*

A useful picture is that of two funnels: one funnel collects all descriptions of x and narrows down to $K_C(x)$; similarly for y. By choosing points close to each funnel's narrowest part and then combining them, we land close to the narrowest part of the funnel for $x * y$.

Formally, the only subtlety relative to a minimum-attaining setting is that one cannot necessarily pick d_1 with $\mathrm{Eff}_C(d_1) = K_C(x)$ exactly. The ϵ-argument is the standard replacement: for every tolerance $\epsilon > 0$ there exists a witness within ϵ of the infimum. The final limit step uses that the bound holds for all $\epsilon > 0$, hence also in the limit, yielding the inequality without slack.

Theorem 30 (Structural/compositional equivalence as an inference bridge). *Under the generation assumptions of Prop. 5 (finite primitives + binary constructor corresponding to $*$ with overhead),*

$$K_C(x) \text{ coincides with } s^*_C(x) \text{ up to fixed base costs.}$$

Hence helper theorems for s^*_C (parse-tree bounds, monotonicity, optimal substructure) can be re-used for K_C.

Remark 1727 (What the theorem says, and why it matters). *This theorem states that, under the specified generation assumptions, the two notions of complexity in this section are essentially the same quantity computed in two different languages. On one side, $s^*_C(x)$ measures the cheapest way to assemble x via binary composition (a "construction history" represented by a parse tree). On the other, $K_C(x)$ measures the cheapest description that decodes to x. The theorem says these coincide up to fixed base costs, meaning that differences are absorbed into constant offsets attributable to the choice of primitives and representation conventions.*

The phrase "fixed base costs" is meant to parallel the ubiquitous $O(1)$ terms in classical Kolmogorov complexity: changing the universal machine changes $K(x)$ by at most an additive constant independent of x. Here, once the set of primitives and the constructor overhead are fixed (as in Prop. 5), the conversion between "a parse tree that constructs x" and "a description that decodes to x" incurs only constant-size scaffolding: delimiters, tags indicating which constructor node is being used, and a uniform decoding convention. These costs do not scale with the size or internal complexity of x; they depend only on the agreed-upon representation scheme.

Operationally, the inference bridge works in both directions. A parse-tree upper bound on $s^*_C(x)$ yields an explicit construction plan, which can be compiled into a description d with $\mathrm{Dec}_C(d) = x$, giving an upper bound on $K_C(x)$ via Lemma 54. Conversely, if one has a short/low-effort description of x in the generalized Kolmogorov sense, the generation assumptions ensure that the description can be interpreted (perhaps after a constant-cost normalization) as specifying a finite composition tree over the primitives, thereby bounding $s^*_C(x)$ by essentially the same value.

In particular, once this equivalence is in place, any structural lemma proved for s^*_C can be transported to K_C without repeating the full argument at the level of descriptions. This is the main methodological payoff: one may reason in the more convenient domain (parse trees or descriptions) while obtaining conclusions about the other domain "for free," up to the stipulated constant offsets.

Remark 1728 (Why an equivalence theorem is plausible under Prop. 5)**.** *Prop. 5 posits a finite generating basis together with a binary constructor aligned with $*$, and it controls overhead at each composition step. Under these assumptions, any construction of x determines a finite rooted binary tree whose leaves are primitives and whose internal nodes are applications of $*$. Encoding such a tree as a description is a matter of choosing a traversal and adding a fixed amount of syntax to indicate branching structure; conversely, any description that decodes via repeated application of the constructor can be unfolded into such a tree. The effort model then assigns comparable total cost to both representations because both are built from the same local ingredients: primitive costs at leaves plus per-node overhead at internal constructor applications.*

Remark 1729. *This is important because it allows a transfer of results: any optimization or monotonicity statement proved for parse trees becomes a statement about description complexity. Philosophically, it supports a Russell-style moral about definitions: apparently different descriptions of "simplicity" can converge when they are forced to respect the same structural constraints [16]. In the broader Hyperseed frame, it links compositional construction (Hyperseed-Concept 82) with structural complexity (Hyperseed-Concept 181) through a rigorous equivalence. More concretely, it justifies treating a shortest description (in the generalized Kolmogorov sense) and a cheapest construction (in the parse-tree sense) as two presentations of the same minimization problem, differing at most by the bookkeeping conventions used to account for composition. In particular, once a fixed encoding of constructors and arguments is chosen, the "overhead" of describing how parts are combined can be made uniform across descriptions, so that the remaining variability is exactly the structural choice of how to decompose x. This is the point at which the usual invariance intuition from Kolmogorov complexity becomes relevant: changes of description language affect costs only up to a controlled translation penalty, while the structural constraints determine the leading behavior of the minimum [16].*

Remark 1730 (How it connects to the surrounding theory)**.** *This equivalence is the technical reason the document can speak fluidly about "simplicity" both as minimal effort of assembly and as minimal effort of description, without constantly reproving parallel lemmas. It is also aligned with the perspective that dependency structures (parse trees, construction graphs) are not mere bookkeeping, but the geometry of complexity itself [8]. Operationally, it means that any lemma stated for the effort functional on trees (e.g., that refining a decomposition cannot reduce cost under a monotone overhead model, or that certain normal forms are optimal) can be reinterpreted as a lemma about description length for a decoder that implements the same constructors. Conversely, lower bounds stated in terms of unavoidable description length can be viewed as constraints on the minimal achievable assembly effort. This is especially helpful when later sections mix viewpoints: one can argue in the tree language when exploiting compositionality and in the description language when appealing to information-theoretic intuitions, while relying on this equivalence to ensure the two arguments are genuinely about the same quantity.*

Sketch. Prop. 5: each description induces a parse tree whose effort is the sum of base and overhead costs, and taking the infimum over descriptions equals taking the minimum over parse trees; then apply Thm. 9. More explicitly, fix a decoding convention in which a description is parsed into (i) literals or base symbols and (ii) constructor calls that combine previously produced subobjects. Reading such a description left-to-right (or via its own syntactic parse) yields a tree whose leaves correspond to base symbols and whose internal nodes correspond to constructor applications; by design, the description cost splits into the base costs of the leaves plus a per-node overhead for each constructor application. Minimizing description cost therefore reduces to choosing the best tree shape together with the best leaf labels, which is exactly the tree minimization problem. The passage from "infimum" to "minimum" is justified in the finite/compact setting assumed by Prop. 5 (e.g., finitely many admissible constructors at each step and discrete costs), while in more general settings one keeps the infimum and obtains equality up to the same limiting argument. Thm. 9 then identifies the resulting tree minimum with the corresponding fixed-point/minimum characterization required for the statement. □

Remark 1731 (Proof sketch and intuition)**.** *The proof strategy is to build a two-way correspondence between descriptions and parse trees. One direction maps a description to a parse tree witnessing how the decoder constructs x; the effort of the description decomposes into base costs plus per-composition overhead, matching the tree cost. The other direction maps a parse tree to a composed description built by repeated application of the binary constructor. Once these mappings preserve (or tightly relate) costs, taking an infimum over descriptions becomes equivalent to taking a minimum over trees (in the finite case), and Thm. 9 supplies the fixed-point/minimum identification needed to conclude. A useful way to view the "tightness" condition here is that the encoding of tree structure (which node combines which subresults, and in what order) is required to have a predictable cost: it is either counted explicitly as overhead at each internal node or absorbed into a uniform additive constant determined by the chosen description formalism. In this sense, the equivalence is not merely existential; it depends on the modeling choice that descriptions must respect the same admissible composition rules as the parse trees they are being compared to.*

The geometric intuition is that both descriptions and parse trees are coordinate systems on the same underlying constructive space: they slice the same object x into parts and record the gluing operations. The theorem says that, once the slicing rules are aligned, the measured "length" of the shortest route to x is independent of which coordinate system you use. One can also read this as a statement about where complexity "lives": if two descriptions differ only by superficial syntactic rearrangements but induce the same dependency graph, then they have essentially the same effort, whereas genuinely different decompositions correspond to different routes through the constructive space and can change the minimum. This explains why later arguments can treat the parse tree as the primary carrier of structure and treat description length as a derived measurement attached to that structure, rather than as an unrelated notion of succinctness.

31 Helper theorems for patterns, properties, emergence, and blending (Section 9)

Standing assumptions and notation. Fix a context C with: (i) entities E_C, (ii) a (partial) static combination $*_C$ on E_C, (iii) a description language L_C and decoder $\mathrm{Dec}_C : L_C \to E_C$, (iv) a cost functional $\mathrm{cost}_C : L_C \to \mathbb{R}_{\geq 0} \cup \{\infty\}$, (v) a baseline encoder $\mathrm{base}_C : E_C \to L_C$ with $\mathrm{Dec}_C(\mathrm{base}_C(x)) = x$.

The role of the context C is to package the "model of description" (the pair (L_C, Dec_C)), the "notion of resource" used to score descriptions (the functional cost_C), and the admissible way of composing entities (the partial operation $*_C$). The operation $*_C$ is allowed to be partial to accommodate domains where not every pair of entities can be meaningfully combined, and the qualifier "static" emphasizes that $*_C$ is treated as a fixed operation of the context (rather than something optimized over within a theorem).

Let $s_C(x) := \inf\{\mathrm{cost}_C(\ell) : \ell \in L_C,\ \mathrm{Dec}_C(\ell) = x\}$ (Def. 87), and $s_C^{\mathrm{base}}(x) := \mathrm{cost}_C(\mathrm{base}_C(x))$ (Def. 93).

Thus $s_C(x)$ is the best achievable (infimal) cost of any valid description of x in C, whereas $s_C^{\mathrm{base}}(x)$ is a distinguished reference cost obtained from the baseline encoder. In particular, by the defining property $\mathrm{Dec}_C(\mathrm{base}_C(x)) = x$, the set over which the infimum defining $s_C(x)$ is taken is nonempty whenever $\mathrm{base}_C(x)$ is defined, and one always has the basic upper bound $s_C(x) \leq s_C^{\mathrm{base}}(x)$ (possibly with both sides infinite). The use of inf rather than min permits contexts where optimal descriptions may not exist but can be approached arbitrarily well.

A compositional witness for x is $P = (y, z, t)$ with $x = y *_C z$ and overhead $t \geq 0$ (Def. 94), with composite cost $h_C(P) := s_C(y) + s_C(z) + t$ (Def. 94).

Intuitively, a witness P certifies that x can be obtained by combining constituents y and z at an additional nonnegative overhead t, where t is intended to account for glue code, alignment/typing constraints, coordination information, or any other extra description needed to make the composition well-formed in the given context. The quantity $h_C(P)$ should be read as the total cost of describing x via the route "describe y, describe z, and then pay t to combine them," with the understanding that the constituents are scored using their intrinsic description costs $s_C(y)$ and $s_C(z)$ rather than their baseline costs. (In later arguments, this separation allows one to compare direct baselines against potentially more efficient compositional encodings.)

Scalar intensity (Def. 95) is

$$I_C(\ell; x) = \max\left(0, \frac{s_C^{\mathrm{base}}(x) - \mathrm{cost}_C(\ell)}{s_C^{\mathrm{base}}(x)}\right) \in [0,1], \quad I_C(P; x) = \max\left(0, \frac{s_C^{\mathrm{base}}(x) - h_C(P)}{s_C^{\mathrm{base}}(x)}\right) \in [0,1].$$

This normalization makes intensity a relative improvement over the baseline: $I_C(\ell; x)$ (respectively $I_C(P; x)$) is positive exactly when the candidate description ℓ (respectively the compositional explanation P) beats the baseline cost for x, and it saturates at 1 in the extreme case where the candidate cost is 0. The outer $\max(0, \cdot)$ enforces that "no improvement" is represented as intensity 0 rather than a negative number, so the scale is compatible with viewing intensities as fuzzy membership values. When $s_C^{\mathrm{base}}(x)$ is finite and positive, the formula is literal; in degenerate cases (e.g. $s_C^{\mathrm{base}}(x) = 0$ or $s_C^{\mathrm{base}}(x) = \infty$), one typically treats the displayed expression as a conventionally normalized score and restricts attention to settings where the baseline is informative enough to serve as a denominator.

A property-set is $\mathrm{Prop}_C(x) : P_C \to [0,1]$ given by $\mathrm{Prop}_C(x)(Q) := I_C(Q; x)$ for a chosen pattern vocabulary P_C (Def. 97), with mass $\|\mathrm{Prop}_C(x)\|_1 := \sum_{Q \in P_C} \mathrm{Prop}_C(x)(Q)$ (Def. 98). Here the pattern vocabulary P_C is an externally fixed collection of "candidate explanations" or "recognizable regularities" (e.g. descriptions ℓ, compositional witnesses P, or other structured patterns Q) whose intensities are evaluated against the target entity x. The mass $\|\mathrm{Prop}_C(x)\|_1$ is the total amount of recognized structure from P_C present in x according to this scoring; in applications one often chooses P_C so that this sum is finite (for example by restricting to a finite vocabulary, to patterns below a cost threshold, or to patterns supported by a finite dataset), ensuring that the ℓ^1-quantity is well-defined and comparable across entities.

Inheritance is the preorder

$$A \preceq_C B \quad :\Longleftrightarrow \quad \mathrm{Prop}_C(A)(Q) \geq \mathrm{Prop}_C(B)(Q) \ \forall Q \in P_C \qquad \text{(Def. 99).}$$

This order reads "A inherits B" (or A is at least as pattern-rich as B relative to P_C) in the sense that every pattern in the vocabulary is present in A with intensity at least as large as it is present in B. Because the definition is pointwise over $Q \in P_C$, it is immediate that $\preceq_C$ is reflexive and transitive (hence a preorder), and it becomes a partial order after quotienting by the induced equivalence relation of mutual inheritance.

Association is fuzzy Jaccard (Def. 100):

$$\operatorname{assoc}_C(A, B) := \frac{\sum_{Q \in P_C} \min(\operatorname{Prop}_C(A)(Q), \operatorname{Prop}_C(B)(Q))}{\sum_{Q \in P_C} \max(\operatorname{Prop}_C(A)(Q), \operatorname{Prop}_C(B)(Q))} \quad \text{with } 0/0 := 0.$$

This is the standard Jaccard similarity generalized from crisp sets to fuzzy membership weights, where min plays the role of intersection and max plays the role of union. The convention $0/0 := 0$ corresponds to declaring two entities to have association 0 when neither exhibits any pattern from the vocabulary (so both numerator and denominator vanish). When the denominator is nonzero, $\operatorname{assoc}_C(A, B) \in [0, 1]$, achieving 1 exactly when the property profiles coincide pointwise on P_C and approaching 0 when their overlap is negligible compared to their union.

Emergence is defined by weights $w_C(A), w_C(B)$ from baseline costs (Def. 101), surplus $\operatorname{Sur}_C(Q; A, B)$ and emergent intensity $\operatorname{Emer}_C(Q; A, B) := \max(0, \operatorname{Sur}_C(Q; A, B))$ (Def. 102). The conceptual intent is that $w_C(A)$ and $w_C(B)$ calibrate how much the constituents A and B should contribute to the baseline expectation for a combined object (for instance, to avoid double-counting improvements that are already explained by one constituent alone). A positive surplus $\operatorname{Sur}_C(Q; A, B)$ indicates that the pattern Q is exhibited by the joint entity more strongly than can be accounted for by the weighted contributions of A and B separately, and the truncation to $\operatorname{Emer}_C(Q; A, B)$ ensures that only genuinely new (nonnegative) gains are counted as emergence. In later helper theorems, these definitions are used to bound how pattern masses, inheritances, and associations behave under composition and to isolate conditions under which emergent structure must appear.

31.1 Pattern intensity algebra

Lemma 56 (Intensity as truncated relative savings)**.** *Assume $s_C^{\text{base}}(x) > 0$. Then for any witness with cost-value $u \in [0, \infty]$ (either $u = \operatorname{cost}_C(\ell)$ or $u = h_C(P)$),*

$$I_C(\cdot; x) = \max\left(0, 1 - \frac{u}{s_C^{\text{base}}(x)}\right).$$

In particular:

$$I_C(\cdot; x) > 0 \iff u < s_C^{\text{base}}(x), \qquad I_C(\cdot; x) = 0 \iff u \geq s_C^{\text{base}}(x).$$

Sketch. Immediate algebra from Def. 95 and the max-clamp. More explicitly, Def. 95 defines intensity as a normalized savings fraction relative to the baseline scale $s_C^{\text{base}}(x)$, with the outer $\max(0, \cdot)$ ensuring nonnegativity even when a proposed witness is no better than baseline. When $u = \infty$ (i.e., an unusable or undefined witness cost), the expression inside the clamp is $-\infty$, hence $I_C(\cdot; x) = 0$ as expected. □

Lemma 57 (Recovering cost from positive intensity)**.** *Assume $s_C^{\text{base}}(x) > 0$ and $I_C(\cdot; x) > 0$. Then the witness cost is exactly*

$$u = s_C^{\text{base}}(x) \cdot (1 - I_C(\cdot; x)).$$

Sketch. If $I > 0$, the clamp is inactive, so $I = 1 - u/s_{\text{base}}$ and rearrange. Note that the assumption $I_C(\cdot; x) > 0$ is essential: at intensity 0 the clamp discards information about how large u is beyond the threshold $s_C^{\text{base}}(x)$, so u cannot be uniquely recovered from $I = 0$ alone. □

Lemma 58 (Monotonicity: cheaper witnesses yield higher intensity)**.** *Assume $s_C^{\mathrm{base}}(x) > 0$. If two witnesses of x have costs $u \le v$, then $I(u;x) \ge I(v;x)$.*

Sketch. The map $u \mapsto \max(0, 1 - u/s)$ is monotone decreasing in u for fixed $s > 0$. Equivalently, before truncation the expression is affine with slope $-1/s_C^{\mathrm{base}}(x)$, and the truncation at 0 preserves the order because it is itself monotone. □

Lemma 59 (Lipschitz control: intensity changes at most linearly with cost)**.** *Assume $s_C^{\mathrm{base}}(x) > 0$. Then for any two witness costs $u, v \in [0, \infty]$,*

$$\big|I(u;x) - I(v;x)\big| \;\le\; \frac{|u-v|}{s_C^{\mathrm{base}}(x)},$$

with the convention that the right-hand side is ∞ if one of u, v is ∞.

Sketch. On the region where the clamp is inactive, $I(u;x) = 1 - u/s_C^{\mathrm{base}}(x)$ is affine with slope magnitude $1/s_C^{\mathrm{base}}(x)$. Applying $\max(0, \cdot)$ cannot increase distances (it is 1-Lipschitz on $\mathbb{R}$), so the same bound holds globally after truncation. □

Lemma 60 (Overhead sensitivity bound (compositional witnesses))**.** *Assume $s_C^{\mathrm{base}}(x) > 0$ and let $P = (y, z, t)$ witness x. For $\Delta \ge 0$, define $P' := (y, z, t + \Delta)$ (same parts, larger glue bound). Then*

$$I_C(P';x) \;\ge\; I_C(P;x) - \frac{\Delta}{s_C^{\mathrm{base}}(x)}.$$

Sketch. $h_C(P') = h_C(P) + \Delta$. Apply Lemma 56 and use that truncation at 0 can only increase the left-hand side. Concretely, before truncation the intensity drops by exactly $\Delta/s_C^{\mathrm{base}}(x)$, and if this pushes the value below 0 the clamp sets it back to 0, so the realized decrease can only be smaller in magnitude than the linear prediction. □

Lemma 61 (Baseline dominates optimal complexity)**.** *For all $x \in E_C$,*

$$s_C(x) \le s_C^{\mathrm{base}}(x).$$

Consequently, if $s_C^{\mathrm{base}}(x) > 0$ then the best achievable *normalized compression fraction is at least*

$$I_C^\star(x) := \max\!\left(0, \frac{s_C^{\mathrm{base}}(x) - s_C(x)}{s_C^{\mathrm{base}}(x)}\right) \in [0,1],$$

and every witness intensity is bounded by $I_C(\cdot;x) \le I_C^\star(x)$.

Sketch. $\mathrm{base}_C(x)$ is a witness for x, so the infimum defining $s_C(x)$ is $\le \mathrm{cost}_C(\mathrm{base}_C(x))$. For the bound, note $u \ge s_C(x)$ for any witness cost u, then apply Lemma 56. In particular, if $s_C(x) = s_C^{\mathrm{base}}(x)$ then $I_C^\star(x) = 0$ and no witness can achieve positive intensity (the baseline is already optimal under the cost model), whereas if $s_C(x) = 0$ then $I_C^\star(x) = 1$, matching the interpretation that perfect compression relative to a positive baseline would yield unit intensity. □

31.2 Properties and inheritance

Theorem 31 (Inheritance via properties is a preorder)**.** *The relation $\preceq_C$ from Def. 99 is reflexive and transitive on E_C.*

Sketch. Reflexive: $\mathrm{Prop}_C(A)(Q) \geq \mathrm{Prop}_C(A)(Q)$ pointwise. Transitive: if $\mathrm{Prop}_C(A) \geq \mathrm{Prop}_C(B)$ and $\mathrm{Prop}_C(B) \geq \mathrm{Prop}_C(D)$ pointwise, then $\mathrm{Prop}_C(A) \geq \mathrm{Prop}_C(D)$ pointwise. In other words, $\preceq_C$ is precisely the product (pointwise) order on property-vectors $\mathrm{Prop}_C(\cdot) \in [0,1]^{P_C}$ pulled back along the map $\mathrm{Prop}_C \colon E_C \to [0,1]^{P_C}$, so reflexivity and transitivity reduce to the corresponding facts about $\geq$ on $[0,1]$ evaluated coordinatewise. Note that antisymmetry need not hold on E_C: it is possible that $A \preceq_C B$ and $B \preceq_C A$ while $A \neq B$ as entities, provided they induce the same property function (cf. Lemma 66 for an equality criterion at the level of Prop_C). □

Lemma 62 (Threshold inheritance rule)**.** *If $A \preceq_C B$ and $\mathrm{Prop}_C(B)(Q) \geq \tau$, then $\mathrm{Prop}_C(A)(Q) \geq \tau$. Equivalently, any property holding strongly of B holds at least as strongly of A.*

Sketch. Immediate from the pointwise inequality in Def. 99. Concretely, $A \preceq_C B$ means $\mathrm{Prop}_C(A)(Q) \geq \mathrm{Prop}_C(B)(Q)$ for each fixed $Q \in P_C$; chaining this with $\mathrm{Prop}_C(B)(Q) \geq \tau$ yields $\mathrm{Prop}_C(A)(Q) \geq \tau$. This is the basic mechanism by which property-level constraints propagate "downward" along $\preceq_C$ at any chosen cutoff $\tau \in [0,1]$. □

Lemma 63 (Mass monotonicity under inheritance)**.** *Assume P_C is finite or countable so the ℓ^1 mass is defined. If $A \preceq_C B$, then $\|\mathrm{Prop}_C(A)\|_1 \geq \|\mathrm{Prop}_C(B)\|_1$.*

Sketch. Sum the pointwise inequalities $\mathrm{Prop}_C(A)(Q) \geq \mathrm{Prop}_C(B)(Q)$ over $Q \in P_C$. Because all terms are nonnegative, the (finite or countable) sum preserves inequalities, so the aggregate "amount of property evidence" (total mass) cannot decrease when moving from B to an inheritor A in the sense of $\preceq_C$. In particular, if $\|\mathrm{Prop}_C(B)\|_1$ is finite then $\|\mathrm{Prop}_C(A)\|_1$ is also finite and bounded below by it. □

31.3 Association (fuzzy Jaccard) calculus

The quantity $\mathrm{assoc}_C(A,B)$ used below is the standard fuzzy Jaccard similarity of the property profiles $\mathrm{Prop}_C(A)$ and $\mathrm{Prop}_C(B)$: it measures overlap via min (soft intersection) normalized by max (soft union). The $0/0 := 0$ convention ensures the definition is total even when both entities carry zero total property mass, i.e. when $\mathrm{Prop}_C(A)$ and $\mathrm{Prop}_C(B)$ are both identically 0 on P_C.

Lemma 64 (Association bounds and symmetry)**.** *For all $A, B \in E_C$,*

$$0 \leq \mathrm{assoc}_C(A,B) \leq 1, \qquad \mathrm{assoc}_C(A,B) = \mathrm{assoc}_C(B,A).$$

Sketch. Numerator and denominator are sums of nonnegative terms, and $\min(u,v) \leq \max(u,v)$ pointwise, giving the $[0,1]$ bound. More explicitly, for each Q one has $0 \leq \min(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)) \leq \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$, hence after summing over Q the numerator is between 0 and the denominator. If the denominator is 0 then both numerator and denominator are 0 (since $\min \leq \max$ and both are nonnegative), and the convention $0/0 := 0$ yields the stated bounds as well. Symmetry holds because min and max are symmetric in their arguments, so both numerator and denominator are unchanged under swapping A and B. □

Lemma 65 (Intersection/union rewrite)**.** *Define the* intersection mass

$$M_\cap(A,B) := \sum_{Q \in P_C} \min(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)).$$

Then

$$\sum_{Q\in P_C} \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)) = \|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1 - M_\cap(A,B),$$

and if the denominator is nonzero,

$$\mathrm{assoc}_C(A,B) = \frac{M_\cap(A,B)}{\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1 - M_\cap(A,B)}.$$

Sketch. Use the pointwise identity $\max(u,v) = u + v - \min(u,v)$ and sum over Q. It is sometimes convenient to name the denominator as a "union mass" $M_\cup(A,B) := \sum_{Q\in P_C} \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$, in which case the displayed equality reads $M_\cup(A,B) = \|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1 - M_\cap(A,B)$ and $\mathrm{assoc}_C(A,B) = M_\cap(A,B)/M_\cup(A,B)$ whenever $M_\cup(A,B) > 0$. □

Lemma 66 (When is association 1?)**.** *If* $\|\mathrm{Prop}_C(A)\|_1 > 0$ *or* $\|\mathrm{Prop}_C(B)\|_1 > 0$*, then*

$$\mathrm{assoc}_C(A,B) = 1 \iff \mathrm{Prop}_C(A) = \mathrm{Prop}_C(B) \textit{ pointwise.}$$

If both masses are 0, then $\mathrm{assoc}_C(A,B) = 0$ *by the* $0/0 := 0$ *convention.*

Sketch. If assoc = 1 and denominator is nonzero, then numerator equals denominator, so min = max termwise, hence the two property functions agree pointwise. To justify the termwise conclusion: if $a_Q := \min(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$ and $b_Q := \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$, then $0 \le a_Q \le b_Q$ for all Q and $\sum_Q a_Q = \sum_Q b_Q$ forces $a_Q = b_Q$ for each Q (otherwise some strict inequality would remain after summing nonnegative differences). Conversely, if they agree pointwise and have nonzero mass, then min = max termwise and the ratio is 1. The separate zero-mass case rules out the degenerate situation where both numerator and denominator vanish; under the stated convention this yields similarity 0 rather than 1, which aligns with the interpretation that there is no positive evidence of shared properties. □

Lemma 67 (Association lower-bounds shared property mass)**.** *Assume the denominator is nonzero and let* $\alpha := \mathrm{assoc}_C(A,B) \in (0,1]$*. Then the intersection mass satisfies*

$$M_\cap(A,B) = \frac{\alpha}{1+\alpha}\Big(\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1\Big).$$

In particular, if $\mathrm{assoc}_C(A,B) \ge \tau$ *then*

$$M_\cap(A,B) \;\ge\; \frac{\tau}{1+\tau}\Big(\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1\Big).$$

Sketch. Solve the equation in Lemma 65 for $M_\cap$. Indeed, from $\alpha = M_\cap/(\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1 - M_\cap)$ one obtains $\alpha(\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1 - M_\cap) = M_\cap$, hence $\alpha(\|\mathrm{Prop}_C(A)\|_1 + \|\mathrm{Prop}_C(B)\|_1) = (1+\alpha)M_\cap$, yielding the displayed formula. The monotone implication for $\mathrm{assoc}_C(A,B) \ge \tau$ follows since the map $\alpha \mapsto \alpha/(1+\alpha)$ is increasing on $[0,\infty)$. □

Lemma 68 (Finite-vocabulary witness of shared patterns)**.** *Assume* P_C *is finite with* $|P_C| = N$*. If* $M_\cap(A,B) \ge \epsilon > 0$*, then there exists some pattern* $Q \in P_C$ *such that*

$$\min(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)) \;\ge\; \epsilon/N.$$

Sketch. If every term were $< \epsilon/N$, the sum of N terms would be $< \epsilon$. Equivalently, this is a pigeonhole/averaging argument: the average value of the N nonnegative numbers $\min(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$ is $M_\cap(A,B)/N \ge \epsilon/N$, so at least one coordinate attains at least the average. Thus positive intersection mass in a finite vocabulary necessarily yields a concrete "witness" pattern shared to a quantitatively bounded degree. □

31.4 Emergence calculus

This subsection collects elementary algebraic facts about the emergence quantities introduced earlier, with an emphasis on inequalities and simplifications that can be used as plug-in steps inside longer arguments. Throughout, C denotes the ambient context, P_C the property space, E_C the space of eligible entities, and $I_C(Q;\cdot)$ the (normalized) degree to which an entity instantiates the property Q. The definitions of Sur_C and Emer_C are arranged so that Sur_C measures "excess property instantiation" of the composite relative to a baseline mixture of the parts, while $\mathrm{Emer}_C = \max(0, \mathrm{Sur}_C)$ retains only the positive (i.e., genuinely emergent) component. In particular, the weights $w_C(A), w_C(B)$ encode how strongly each part contributes to the baseline, so the baseline is a convex combination whenever the total baseline mass is nonzero.

Lemma 69 (Emergence weights form a convex pair)**.** *Assume $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) > 0$. Then the weights from Def. 101 satisfy*

$$w_C(A) \geq 0, \quad w_C(B) \geq 0, \quad w_C(A) + w_C(B) = 1.$$

Sketch. Immediate from Def. 101 as ratios of nonnegative numbers with common positive denominator. □

The condition $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) > 0$ is precisely what ensures that the normalization used to define $(w_C(A), w_C(B))$ is well-posed. When this hypothesis holds, $w_C(A)$ and $w_C(B)$ can be read as barycentric coordinates: any weighted baseline term of the form $w_C(A)\,x + w_C(B)\,y$ lies between $\min\{x, y\}$ and $\max\{x, y\}$, a fact that is repeatedly used implicitly when comparing Sur_C to $I_C(Q; A *_C B)$.

Lemma 70 (Basic emergence bounds)**.** *For any $Q \in P_C$ and $A *_C B$ defined:*

$$0 \leq \mathrm{Emer}_C(Q; A, B) \leq I_C(Q; A *_C B) \leq 1.$$

Sketch. By definition $\mathrm{Emer} = \max(0, \mathrm{Sur}) \geq 0$. Also $\mathrm{Emer} \leq \max(0, I_C(Q; A * B)) \leq I_C(Q; A * B)$ since the baseline term is nonnegative. Finally $I_C(\cdot;\cdot) \in [0, 1]$ by Def. 95 / Prop. 6. □

To make the middle inequality explicit, recall that $\mathrm{Sur}_C(Q; A, B)$ subtracts a baseline mixture of $I_C(Q; A)$ and $I_C(Q; B)$ (with weights from Lemma 69) from $I_C(Q; A *_C B)$. Since the baseline mixture is pointwise ≥ 0 whenever $I_C(\cdot;\cdot) \geq 0$, one has

$$\mathrm{Sur}_C(Q; A, B) \leq I_C(Q; A *_C B),$$

and therefore $\mathrm{Emer}_C(Q; A, B) = \max(0, \mathrm{Sur}_C(Q; A, B)) \leq \max(0, I_C(Q; A *_C B)) = I_C(Q; A *_C B)$ using $I_C(Q; A *_C B) \geq 0$. The upper bound by 1 is the normalization built into I_C, and it propagates to emergence because Emer_C never exceeds the composite score.

Lemma 71 (Equal-part simplification)**.** *Assume $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) > 0$ and $I_C(Q; A) = I_C(Q; B) = q$. Then*

$$\mathrm{Sur}_C(Q; A, B) = I_C(Q; A *_C B) - q, \qquad \mathrm{Emer}_C(Q; A, B) = \max(0, I_C(Q; A *_C B) - q).$$

Sketch. Use Lemma 69 to replace the weighted average by $w_C(A)q + w_C(B)q = q$. □

This simplification is often the fastest route to a usable estimate: whenever the two parts are indistinguishable with respect to the queried property Q, the entire baseline collapses to that common value q, independent of how the baseline weights split between A and B. In particular, in the equal-part regime, positivity of emergence for Q is equivalent to the single inequality $I_C(Q; A *_C B) > q$, i.e., the composite must score strictly above either part (since both parts share the same score).

Theorem 32 (Symmetry under commutative combination)**.** *If $*_C$ is commutative and $A *_C B$ is defined, then*

$$\mathrm{Emer}_C(Q; A, B) = \mathrm{Emer}_C(Q; B, A) \quad \textit{for all } Q \in P_C.$$

Sketch. This is Proposition 7 in Section 9: commutativity implies $A * B = B * A$ and the weights swap, leaving the surplus unchanged. □

Concretely, the Sur_C expression depends on (A, B) through (i) the composite $A *_C B$ and (ii) a weighted baseline that is affine in the pair of numbers $\big(I_C(Q; A), I_C(Q; B)\big)$. Under commutativity, the composite term is invariant, while exchanging (A, B) simply permutes the two weight–score products in the baseline, so the baseline sum is unchanged. Since Emer_C is obtained from Sur_C by applying $\max(0, \cdot)$ pointwise, the same invariance transfers immediately from surplus to emergence.

Lemma 72 (Collective-property-set monotonicity)**.** *Let $\mathcal{B} \subseteq E_C$ be a reference class and define*

$$\mathrm{Prop}_C^{\mathrm{coll}}(A)(Q) := \sup_{B \in \mathcal{B}} \mathrm{Emer}_C(Q; A, B) \qquad \textit{(Def. 103).}$$

Then:

1. $\mathrm{Prop}_C^{\mathrm{coll}}(A)(Q) \geq \mathrm{Emer}_C(Q; A, B)$ *for all $B \in \mathcal{B}$.*
2. *If $\mathcal{B} \subseteq \mathcal{B}'$, then $\mathrm{Prop}_{C,\mathcal{B}}^{\mathrm{coll}}(A)(Q) \leq \mathrm{Prop}_{C,\mathcal{B}'}^{\mathrm{coll}}(A)(Q)$.*

Sketch. (1) Supremum dominates each element. (2) Supremum over a larger set cannot decrease. □

The use of sup (rather than max) allows $\mathcal{B}$ to be infinite and also covers the common case where the "best" partner B is only approached by an improving sequence of candidates but not attained by any single $B \in \mathcal{B}$. Part (1) states that $\mathrm{Prop}_C^{\mathrm{coll}}(A)(Q)$ is an upper envelope of the family of functions $B \mapsto \mathrm{Emer}_C(Q; A, B)$ indexed by the reference class; part (2) is the corresponding isotonicity with respect to enlarging the admissible comparison set. Together these properties justify interpreting $\mathrm{Prop}_C^{\mathrm{coll}}(A)$ as a reference-class-dependent "potential for emergence" of A across possible blends with elements of $\mathcal{B}$, in the sense that giving oneself more admissible partners cannot reduce the measured potential.

31.5 Blend score calculus

Lemma 73 (Blend score bounds and symmetry)**.** *Let $\mathrm{blendscore}_C(C; A, B)$ be as in Def. 104. Then $0 \leq \mathrm{blendscore}_C(C; A, B) \leq 1$ and it is symmetric in (A, B).*

Sketch. The overlap term uses $\min(\mathrm{Prop}_C(C), \max(\mathrm{Prop}_C(A), \mathrm{Prop}_C(B))) \leq \mathrm{Prop}_C(C)$ pointwise, so $\mathrm{ov}_C(C; A, B) \leq \|\mathrm{Prop}_C(C)\|_1$, giving the ≤ 1 bound when mass is nonzero. Nonnegativity is immediate. Symmetry holds because $\max(\mathrm{Prop}(A), \mathrm{Prop}(B))$ is symmetric.

More explicitly, for each $Q \in P_C$ we have

$$0 \;\le\; \min\Big(\mathrm{Prop}_C(C)(Q),\, \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))\Big) \;\le\; \mathrm{Prop}_C(C)(Q),$$

so summing over Q (or integrating, depending on the ambient model underlying $\|\cdot\|_1$) yields

$$0 \;\le\; \mathrm{ov}_C(C; A, B) \;\le\; \|\mathrm{Prop}_C(C)\|_1.$$

Dividing by $\|\mathrm{Prop}_C(C)\|_1$ (under the standing nonzero-mass condition in Def. 104) gives $0 \le \mathrm{blendscore}_C(C; A, B) \le 1$. The symmetry statement follows pointwise from

$$\max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)) = \max(\mathrm{Prop}_C(B)(Q), \mathrm{Prop}_C(A)(Q)),$$

hence the entire integrand (and therefore the overlap mass and the score) is unchanged by swapping A and B. □

Lemma 74 (Trivial blends (full property coverage))**.** *Assume* $\|\mathrm{Prop}_C(C)\|_1 > 0$*. If for all* $Q \in P_C$*,*

$$\mathrm{Prop}_C(C)(Q) \le \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)),$$

then $\mathrm{blendscore}_C(C; A, B) = 1$ *(hence C is a blend at every threshold $\theta \le 1$). In particular, if $C = A$ or $C = B$ and* $\|\mathrm{Prop}_C(C)\|_1 > 0$*, then* $\mathrm{blendscore}_C(C; A, B) = 1$.

Sketch. Under the assumption, the min in Def. 104 equals $\mathrm{Prop}_C(C)(Q)$ for every Q, so $\mathrm{ov}_C = \|\mathrm{Prop}_C(C)\|_1$.

To spell out the pointwise step: the hypothesis says that for each Q,

$$\mathrm{Prop}_C(C)(Q) \;\le\; \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)),$$

so the smaller of the two quantities $\mathrm{Prop}_C(C)(Q)$ and $\max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))$ is necessarily $\mathrm{Prop}_C(C)(Q)$, i.e.

$$\min\Big(\mathrm{Prop}_C(C)(Q),\, \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q))\Big) = \mathrm{Prop}_C(C)(Q).$$

Summing over $Q \in P_C$ gives $\mathrm{ov}_C(C; A, B) = \|\mathrm{Prop}_C(C)\|_1$, hence the normalized score in Def. 104 equals 1. The special case $C = A$ (or $C = B$) is immediate because then $\max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)) \ge \mathrm{Prop}_C(A)(Q) = \mathrm{Prop}_C(C)(Q)$ pointwise, so the coverage condition holds without further assumptions beyond $\|\mathrm{Prop}_C(C)\|_1 > 0$. □

Lemma 75 (Blend score monotonicity under stronger sources)**.** *Assume* $\|\mathrm{Prop}_C(C)\|_1 > 0$*. If* $A' \preceq_C A$ *and* $B' \preceq_C B$*, then*

$$\mathrm{blendscore}_C(C; A', B') \;\ge\; \mathrm{blendscore}_C(C; A, B).$$

Sketch. $A' \preceq_C A$ means $\mathrm{Prop}_C(A') \ge \mathrm{Prop}_C(A)$ pointwise, similarly for B'. Therefore $\max(\mathrm{Prop}(A'), \mathrm{Prop}(B')) \ge \max(\mathrm{Prop}(A), \mathrm{Prop}(B))$ pointwise, so the overlap mass in Def. 104 cannot decrease, while the denominator $\|\mathrm{Prop}(C)\|_1$ is unchanged.

Concretely, fix any $Q \in P_C$. From $A' \preceq_C A$ and $B' \preceq_C B$ we have

$$\mathrm{Prop}_C(A')(Q) \ge \mathrm{Prop}_C(A)(Q), \qquad \mathrm{Prop}_C(B')(Q) \ge \mathrm{Prop}_C(B)(Q),$$

hence

$$\max(\mathrm{Prop}_C(A')(Q), \mathrm{Prop}_C(B')(Q)) \;\ge\; \max(\mathrm{Prop}_C(A)(Q), \mathrm{Prop}_C(B)(Q)).$$

Applying $\min(\mathrm{Prop}_C(C)(Q), \cdot)$ to both sides preserves the inequality, so the integrand defining $\mathrm{ov}_C(C; \cdot, \cdot)$ in Def. 104 is pointwise larger (or equal) for (A', B') than for (A, B). Summation then yields

$$\mathrm{ov}_C(C; A', B') \;\geq\; \mathrm{ov}_C(C; A, B).$$

Dividing by the common positive normalization $\|\mathrm{Prop}_C(C)\|_1$ gives the claimed monotonicity of $\mathrm{blendscore}_C$. $\square$

It is sometimes helpful to read Lem. 75 as a kind of "antitonicity" in the preorder symbols: moving to A', B' that dominate A, B in the $\preceq_C$ sense (i.e. that have pointwise *larger* Prop_C values) can only increase the achievable overlap with the fixed target C. In particular, whenever one source is strengthened while the other is held fixed, the blend score cannot go down.

31.6 Lifting instance relations to groups

Lemma 76 (Singleton consistency)**.** *For an instance relation $C(\cdot, \cdot) : X \times Y \to [0, 1]$ and singletons $A = \{a\}$, $B = \{b\}$, the three lifts from Def. 106 satisfy*

$$C_{\mathrm{avg}}(A, B) = C_{\forall\exists}(A, B) = C_{\exists\forall}(A, B) = C(a, b).$$

Sketch. Each definition collapses to the unique entry $C(a, b)$ when $|A| = |B| = 1$. More explicitly, every aggregation involved in the three lifts is taken over an index set of size one: any average over $A \times B = \{(a, b)\}$ equals the single summand, any $\max_{b' \in B}$ equals evaluation at $b' = b$, and any $\min_{a' \in A}$ equals evaluation at $a' = a$. Thus no information is lost in lifting from instances to singleton groups: all three group-level scores coincide with the original instance score. $\square$

Lemma 77 (Minimax inequality for lifted relations)**.** *For finite nonempty $A \subseteq X$, $B \subseteq Y$:*

$$C_{\exists\forall}(A, B) \;\leq\; C_{\forall\exists}(A, B).$$

Sketch. This is the standard minimax inequality: $\max_b \min_a C(a, b) \leq \min_a \max_b C(a, b)$ for real matrices. To see the inequality directly, write $m(b) := \min_{a \in A} C(a, b)$ and $M(a) := \max_{b \in B} C(a, b)$. For any fixed pair $(a, b) \in A \times B$, we have $m(b) \leq C(a, b) \leq M(a)$, because $m(b)$ is the minimum in column b and $M(a)$ is the maximum in row a. Taking $\max_{b \in B}$ of the left inequality yields $\max_{b \in B} m(b) \leq C(a, b)$ for all a, b, and taking $\min_{a \in A}$ of the right inequality yields $C(a, b) \leq \min_{a \in A} M(a)$ for all a, b. Combining these bounds gives $\max_{b \in B} \min_{a \in A} C(a, b) \leq \min_{a \in A} \max_{b \in B} C(a, b)$, which is exactly the stated claim after matching to the definitions of $C_{\exists\forall}$ and $C_{\forall\exists}$. The finiteness and nonemptiness assumptions ensure these minima and maxima are attained and well-defined without additional conventions. $\square$

Lemma 78 (Monotonicity in group inclusion for the quantifier-like lifts)**.** *Let $A \subseteq A' \subseteq X$ and $B \subseteq B' \subseteq Y$ be finite nonempty. Then:*

$$C_{\forall\exists}(A', B) \leq C_{\forall\exists}(A, B), \qquad C_{\forall\exists}(A, B) \leq C_{\forall\exists}(A, B'),$$

and similarly

$$C_{\exists\forall}(A', B) \leq C_{\exists\forall}(A, B), \qquad C_{\exists\forall}(A, B) \leq C_{\exists\forall}(A, B').$$

Sketch. For $\forall\exists$: enlarging A adds more terms to a minimum (can only decrease), enlarging B adds more options to a maximum (can only increase). For $\exists\forall$: enlarging A decreases each inner minimum, enlarging B increases the outer maximum. Spelling out one representative inequality, for $C_{\forall\exists}$ we have

$$C_{\forall\exists}(A,B) = \min_{a\in A}\ \max_{b\in B} C(a,b), \qquad C_{\forall\exists}(A',B) = \min_{a\in A'}\ \max_{b\in B} C(a,b).$$

Since $A \subseteq A'$, the set of candidate values $\{\max_{b\in B} C(a,b) : a \in A'\}$ contains (at least) the set $\{\max_{b\in B} C(a,b) : a \in A\}$, so taking a minimum over the larger index set A' cannot produce a larger value; hence $C_{\forall\exists}(A',B) \le C_{\forall\exists}(A,B)$. For monotonicity in B, note that if $B \subseteq B'$ then for each fixed a,

$$\max_{b\in B} C(a,b) \ \le\ \max_{b\in B'} C(a,b),$$

and therefore taking $\min_{a\in A}$ on both sides yields $C_{\forall\exists}(A,B) \le C_{\forall\exists}(A,B')$. The corresponding statements for $C_{\exists\forall}$ follow by the same two elementary monotonicity facts for extrema: (i) enlarging the index set of a minimum can only decrease (or leave unchanged) the minimum, and (ii) enlarging the index set of a maximum can only increase (or leave unchanged) the maximum; applied in the correct order to

$$C_{\exists\forall}(A,B) = \max_{b\in B}\ \min_{a\in A} C(a,b).$$

Intuitively, $C_{\forall\exists}$ behaves like a "for all a there exists a good b" score, so adding more required a's makes the constraint stricter, while adding more available b's makes it easier; $C_{\exists\forall}$ behaves like "there exists a b that works for all a," so enlarging A makes each candidate b harder to satisfy and enlarging B offers more candidates b to choose from. □

31.7 Combinatorial-categorical closure (rule-based substitution)

Setup. Fix finite instances S and finite category domain W (Def. 107). A membership state is $m : S \to 2^W$. Given a theory T of rules $(x \in A) \wedge (y \in B) \to (z \in C)$ (Def. 108), define an immediate-consequence operator Φ_T on states by: for each $z \in S$,

$$\Phi_T(m)(z) := m(z) \cup \{\, C \in W : \exists \text{ rule } (x \in A)\wedge(y \in B) \to (z \in C) \in T \text{ with } A \in m(x), B \in m(y) \,\}.$$

Then the closure state m_T in Def. 109 is the least fixed point above m_0.

In more explicit lattice-theoretic terms, the space of all states is the finite product

$$\mathcal{M} := (2^W)^S,$$

ordered pointwise by $m \le m'$ iff $m(s) \subseteq m'(s)$ for all $s \in S$. With this order, $\mathcal{M}$ is a finite complete lattice: joins and meets are computed componentwise, e.g.,

$$\Big(\bigvee_i m_i\Big)(s) = \bigcup_i m_i(s), \qquad \Big(\bigwedge_i m_i\Big)(s) = \bigcap_i m_i(s).$$

The operator $\Phi_T : \mathcal{M} \to \mathcal{M}$ is precisely the usual "one-step consequence" operator familiar from forward-chaining: it leaves all existing labels in place and adds exactly those consequent labels C whose antecedents A, B are already present at the specified source instances x, y. Because S and W are finite, each state can be identified with a finite set of labeled facts $\{(s,\omega) : \omega \in m(s)\} \subseteq S \times W$, and Φ_T can be viewed as repeatedly saturating this set under the rules in T.

Lemma 79 (Monotonicity and extensivity of Φ_T)**.** *For all states m, m':*

$$m \le m' \implies \Phi_T(m) \le \Phi_T(m'), \qquad m \le \Phi_T(m),$$

where $m \le m'$ means $m(s) \subseteq m'(s)$ for all $s \in S$.

Sketch. If $m \leq m'$, any rule antecedent satisfied under m is also satisfied under m', so all added labels under m are also added under m'. Extensivity holds because $\Phi_T(m)(z)$ includes $m(z)$ by definition. $\square$

It is sometimes useful to spell out the monotonicity check componentwise: fix $z \in S$. If $C \in \Phi_T(m)(z)$, then either $C \in m(z)$ (hence $C \in m'(z) \subseteq \Phi_T(m')(z)$), or else C is justified by some rule $(x \in A) \wedge (y \in B) \to (z \in C)$ with $A \in m(x)$ and $B \in m(y)$. But $m \leq m'$ implies $A \in m'(x)$ and $B \in m'(y)$, so the same rule witnesses $C \in \Phi_T(m')(z)$. Thus $\Phi_T(m)(z) \subseteq \Phi_T(m')(z)$ for all z, i.e. $\Phi_T(m) \leq \Phi_T(m')$.

Lemma 80 (Finite convergence bound)**.** *Because S and W are finite and Φ_T is monotone and extensive, the iterative sequence $m_{n+1} := \Phi_T(m_n)$ starting from m_0 stabilizes after at most $|S| \cdot |W|$ strict label-additions. The limit is the least fixed point m_T (Def. 109).*

Sketch. Each strict step adds at least one new pair (s, ω) with $\omega \in m(s)$, and there are only $|S| \cdot |W|$ such pairs. By monotone iteration on a finite complete lattice, the limit is the least fixed point above m_0. $\square$

Concretely, extensivity implies a chain

$$m_0 \leq m_1 \leq m_2 \leq \cdots, \qquad \text{where } m_{n+1} = \Phi_T(m_n),$$

so no label ever disappears. Writing $E_n := \{(s, \omega) : \omega \in m_n(s)\} \subseteq S \times W$, each strict step satisfies $E_n \subsetneq E_{n+1}$ and therefore increases $|E_n|$ by at least 1. Since $|S \times W| = |S| \cdot |W|$, there can be at most $|S| \cdot |W|$ such strict increases before stabilization. At a stabilized stage N we have $m_{N+1} = m_N$, hence m_N is a fixed point of Φ_T.

Moreover, in this finite setting one can equivalently describe the resulting closure as the union of all iterates:

$$m_T = \bigvee_{n \geq 0} m_n, \qquad \text{that is, } m_T(s) = \bigcup_{n \geq 0} m_n(s) \text{ for each } s \in S,$$

and the union is attained at some finite N by the stabilization argument above. This is the usual Kleene construction for least fixed points specialized to finite lattices.

Lemma 81 (Monotonicity of closure in data and theory)**.** *If $m_0 \leq m'_0$ then $m_T \leq m'_T$. If $T \subseteq T'$ then $m_T \leq m_{T'}$ (same S, W, m_0).*

Sketch. Both follow from Lemma 79 and standard monotone-iteration / least-fixpoint arguments. $\square$

For the first claim, one can compare the two Kleene sequences $m_{n+1} = \Phi_T(m_n)$ and $m'_{n+1} = \Phi_T(m'_n)$ with $m_0 \leq m'_0$: Lemma 79 gives by induction that $m_n \leq m'_n$ for all n, hence also $\bigvee_n m_n \leq \bigvee_n m'_n$, i.e. $m_T \leq m'_T$. For the second claim, define Φ_T and $\Phi_{T'}$ from the two rule sets; then $T \subseteq T'$ implies $\Phi_T(m) \leq \Phi_{T'}(m)$ for every state m (since T' can only add more rule-justified consequents at each step). Iterating from the same m_0 preserves the inequality at each stage, yielding $m_T \leq m_{T'}$ at the stabilized limits.

Lemma 82 (Idempotence of closure)**.** *Let m_T be the closure of (T, S, m_0) (Def. 109). Then applying closure again does nothing:*

$$\Phi_T(m_T) = m_T.$$

Sketch. By definition, m_T is a fixed point of rule-application and is obtained as the least fixed point reached by iteration. $\square$

In particular, Lemma 82 says that m_T is already "saturated" under T: every consequent label that is derivable from the labels present in m_T is already included in m_T itself. Equivalently, if one views closure as the endofunction $\mathrm{cl}_T : \mathcal{M} \to \mathcal{M}$ defined by $\mathrm{cl}_T(m_0) := m_T$, then the three properties

$$m \leq \mathrm{cl}_T(m), \qquad m \leq m' \Rightarrow \mathrm{cl}_T(m) \leq \mathrm{cl}_T(m'), \qquad \mathrm{cl}_T(\mathrm{cl}_T(m)) = \mathrm{cl}_T(m),$$

are exactly extensivity, monotonicity, and idempotence of a closure operator on the lattice $\mathcal{M}$. This makes the construction compatible with standard notions of closure in algebra and logic: the closed states are precisely the fixed points of Φ_T, and m_T is the smallest closed state extending the initial data m_0.

Finally, because the rules are purely additive (no negation, no deletion), the computation of m_T can be organized as forward chaining: repeatedly scan for rules whose antecedents are satisfied and insert the missing consequent labels, halting when a full pass adds nothing. Lemma 80 ensures termination in the finite setting, and Lemma 81 ensures robustness under adding initial labels or enlarging the theory.

31.8 Combination systems, interpreters, and causal validity

Lemma 83 (Interpreter homomorphism extends to all parse trees)**.** *Let $(S, *, \mathrm{Dec})$ be an interpreted combination system (Def. 112) with syntax-composition $\cdot$. For any well-formed expression e built from $\cdot$ and atomic symbols, the value $\mathrm{Dec}(e)$ equals the result of evaluating the corresponding binary parse tree by $*$ (when defined). Moreover, when $*$ is a partial operation, the statement is to be read with the usual convention that only those parse-tree evaluations in which every internal node operation is defined are considered, and in that case the evaluation yields a unique element of S independent of any notational bracketing of e consistent with the given binary parse tree.*

Sketch. Structural induction on expressions: base case atoms is trivial; inductive step uses Def. 112: $\mathrm{Dec}(e_1 \cdot e_2) = \mathrm{Dec}(e_1) * \mathrm{Dec}(e_2)$. Concretely, for the induction hypothesis assume that for each immediate subexpression e_i the value $\mathrm{Dec}(e_i)$ agrees with the $*$-evaluation of the parse subtree rooted at e_i (when that subtree evaluation is defined). The parse tree for $e = e_1 \cdot e_2$ has root corresponding to $\cdot$ with left and right subtrees for e_1 and e_2, so evaluating the full tree is exactly applying $*$ to the two subtree values. Def. 112 provides precisely this compatibility between Dec and the syntactic constructor $\cdot$, and thus the root value produced by tree evaluation matches $\mathrm{Dec}(e)$. If $*$ is partial, then definedness of the full-tree evaluation is equivalent to definedness of the two subtree evaluations together with definedness of $\mathrm{Dec}(e_1) * \mathrm{Dec}(e_2)$, matching the recursive clause above. $\square$

Lemma 84 (Combination-preserving maps extend to composites)**.** *Let $f : D \to S$ be combination-preserving (Def. 113) between $(D, **)$ and $(S, *)$. Then for any composite term E built from elements of D using $**$ (when defined), $f(E)$ equals the corresponding composite of the f-images in $(S, *)$. Equivalently, if E is represented by a binary tree whose internal nodes are labeled by $**$ and whose leaves are labeled by elements of D, then applying f after evaluating the D-tree yields the same result as first mapping all leaves by f and then evaluating the resulting S-tree using $*$, provided each intermediate composition required is defined.*

Sketch. Induction on the composite structure using Def. 113 as the induction step. In the base case where E is a single element $d \in D$, the claim is $f(d) = f(d)$. For the inductive step, write $E = E_1 ** E_2$ with E_1, E_2 smaller composites and assume the claim holds for E_1 and E_2. Then

$$f(E) = f(E_1 ** E_2) = f(E_1) * f(E_2),$$

where the last equality is exactly the combination-preservation clause of Def. 113 (under the relevant definedness assumptions). Substituting the induction hypotheses for $f(E_1)$ and $f(E_2)$ identifies the right-hand side with the composite of the f-images, yielding the desired extension to all composite terms. $\square$

Theorem 33 (Validity implies factorization through the representation quotient)**.** *Let $\Gamma : D^3 \to [0,1]$ be a combinational dynamical causal model (Def. 114), and let $f : D \to S$ be a combination-preserving representational mapping (Def. 113). Assume Γ is valid relative to f (Def. 114). Define an equivalence relation $a \sim a'$ iff $f(a) = f(a')$. Then there exists a unique function $\overline{\Gamma} : (f(D))^3 \to [0,1]$ such that*

$$\Gamma(a,b,c) = \overline{\Gamma}(f(a), f(b), f(c)) \quad \textit{for all } a,b,c \in D.$$

In particular, Γ depends on (a,b,c) only through their representational images $(f(a), f(b), f(c))$, so any two triples (a,b,c) and (a',b',c') with the same f-images necessarily receive the same causal score.

Sketch. Define $\overline{\Gamma}(s_1, s_2, s_3) := \Gamma(a,b,c)$ for any a,b,c with $f(a) = s_1, f(b) = s_2, f(c) = s_3$. Validity ensures this is well-defined (independent of chosen representatives). Uniqueness is immediate. To spell out the well-definedness argument: suppose (a,b,c) and (a',b',c') are two choices with $f(a) = f(a') = s_1$, $f(b) = f(b') = s_2$, and $f(c) = f(c') = s_3$. By validity relative to f (Def. 114), replacing any argument of Γ by another element with the same f-image does not change the value, so one obtains $\Gamma(a,b,c) = \Gamma(a',b,c) = \Gamma(a',b',c) = \Gamma(a',b',c')$. Hence the definition of $\overline{\Gamma}(s_1,s_2,s_3)$ does not depend on the representatives chosen from each equivalence class. For uniqueness, if $\widetilde{\Gamma} : (f(D))^3 \to [0,1]$ also satisfies $\Gamma(a,b,c) = \widetilde{\Gamma}(f(a), f(b), f(c))$ for all a,b,c, then for any $(s_1,s_2,s_3) \in (f(D))^3$ choose representatives a,b,c with those images and conclude $\widetilde{\Gamma}(s_1,s_2,s_3) = \Gamma(a,b,c) = \overline{\Gamma}(s_1,s_2,s_3)$. $\square$

31.9 Specific entities as temporally coherent pattern bundles

Lemma 85 (Association threshold implies shared property mass)**.** *Let $x_t, x_{t'} \in E_C$ be two time-slices in an occurrence stream (Def. 115) and suppose the association is computed by Def. 100. If $\mathrm{assoc}_C(x_t, x_{t'}) \geq \tau > 0$ and the denominator is nonzero, then*

$$M_\cap(x_t, x_{t'}) \;\geq\; \frac{\tau}{1+\tau}\Big(\|\mathrm{Prop}_C(x_t)\|_1 + \|\mathrm{Prop}_C(x_{t'})\|_1\Big),$$

so "near-time similarity" (Def. 116) entails a quantitative lower bound on shared property overlap. In particular, the right-hand side scales linearly with the total property mass of the two slices (as measured by the ℓ_1-norm), so a fixed association threshold τ rules out the degenerate situation where x_t and $x_{t'}$ have large total property mass but negligible overlap. The condition that the denominator is nonzero is the standard well-definedness requirement in ratio-style similarity scores: it excludes the case where both slices carry no property mass at all (so that both $\|\mathrm{Prop}_C(x_t)\|_1$ and $\|\mathrm{Prop}_C(x_{t'})\|_1$ vanish and association becomes undefined or vacuous, depending on the chosen convention). Equivalently, for any fixed $\tau > 0$, association above τ forces $M_\cap(x_t, x_{t'})$ to be a nontrivial fraction $\tau/(1+\tau)$ of the combined mass $\|\mathrm{Prop}_C(x_t)\|_1 + \|\mathrm{Prop}_C(x_{t'})\|_1$, making the notion of "temporal coherence" operationally testable via overlap computations.

Sketch. Apply Lemma 67 with $A = x_t$ and $B = x_{t'}$. The inequality is the same statement specialized to two slices of a single occurrence stream; the proof obligation is therefore only to check that the association score used here is exactly the one referenced in Lemma 67 (Def. 100) and that the nonzero-denominator hypothesis matches the hypotheses of that lemma. $\square$

Lemma 86 (Finite-vocabulary: near-time similarity yields at least one shared strong pattern)**.** *Assume P_C is finite with $|P_C| = N$, and that $M_\cap(x_t, x_{t'}) \ge \epsilon > 0$. Then there exists some pattern $Q \in P_C$ such that Q is present in both slices with*

$$\mathrm{Prop}_C(x_t)(Q) \ge \epsilon/N \quad \textit{and} \quad \mathrm{Prop}_C(x_{t'})(Q) \ge \epsilon/N.$$

The content is a simple but important discretization principle: if the total *shared mass across a finite list of patterns is at least ϵ, then some* individual *pattern must contribute at least an ϵ/N share on both sides. This converts a global overlap guarantee (a single scalar lower bound on $M_\cap$) into the existence of a concrete witness pattern Q that can be used downstream as an explicit "identity carrier" (e.g., a named feature that persists across t and t') in logic-engine implementations.*

Sketch. Combine Lemma 68 with the definition of $M_\cap$ as a sum of mins. Concretely, writing $M_\cap(x_t, x_{t'}) = \sum_{R \in P_C} \min(\mathrm{Prop}_C(x_t)(R), \mathrm{Prop}_C(x_{t'})(R))$, the assumption $M_\cap(x_t, x_{t'}) \ge \epsilon$ implies that at least one summand satisfies $\min(\mathrm{Prop}_C(x_t)(Q), \mathrm{Prop}_C(x_{t'})(Q)) \ge \epsilon/N$; otherwise all N summands would be $< \epsilon/N$ and the total would be $< \epsilon$, contradicting the hypothesis. Unpacking the minimum then yields both inequalities $\mathrm{Prop}_C(x_t)(Q) \ge \epsilon/N$ and $\mathrm{Prop}_C(x_{t'})(Q) \ge \epsilon/N$. $\square$

Engineering note (logic-engine friendliness). Defs. 115–116 describe specific-entity identity as graded satisfaction of coherence rules. Lemmas 85–86 provide direct inference rules from an assoc threshold to guaranteed shared pattern overlap, which is often the most useful computable proxy for "temporal coherence" in automated reasoning. In practice, these lemmas support a two-stage strategy: first, use $\mathrm{assoc}_C(x_t, x_{t'}) \ge \tau$ as a fast filter to certify that overlap must be present at scale (Lemma 85); second, when P_C is finite (as is typical in engineered feature vocabularies), extract a concrete witness pattern Q that is provably shared above a known quantitative floor (Lemma 86). This is particularly helpful when subsequent rules require an explicit shared predicate rather than a real-valued similarity score: the witness Q can be reified as the common handle anchoring the two slices to a single temporally coherent bundle. Moreover, the bounds are monotone in the thresholds: increasing τ (or ϵ) strengthens the guaranteed overlap and thus reduces the risk that "identity links" are supported only by diffuse low-level noise spread across many patterns. .

32 Helper theorems for Section 10 (Information, uncertainty, and ineffability)

Standing conventions. Throughout, X is finite (or has been reduced to a finite partition of blocks), log is the natural logarithm, and we use the standard convention $0 \log 0 := 0$. An "indistinction weight" is a function $\iota(x, y) \in [0, 1]$; a reconstruction is a distribution $q \in \Delta(X)$. When log is taken to be natural, entropies and divergence-like quantities are measured in nats; changing the logarithm base merely rescales these quantities by a fixed constant, so statements that compare such values (e.g. inequalities or minimization principles) are invariant under this choice. The phrase "reduced to a finite partition of blocks" should be read literally: in many arguments the only structure that matters is the σ-algebra generated by an observer's coarse partition, and a finite partition ensures that all relevant objects may be represented by finite vectors and matrices (for example, by passing from microstates to block-labels). In particular, if X is replaced by a set of blocks $\{B_1, \ldots, B_k\}$, then a distribution on the reduced space is equivalently a distribution that is constant on each block of the original space, making explicit the idea that the observer cannot (or will not) resolve distinctions within a block.

Remark 1732. *The finiteness assumption on X is not merely a technical convenience: it ensures that all sums are finite and that σ-algebras generated by partitions behave in the simplest possible way. In the main text, X often stands for a world of possibilities available to an observer or modeling system; reducing to a finite partition amounts to a deliberate coarse-graining, i.e. a choice to treat many microstates as indistinguishable. This is one clean mathematical avatar of* uncertainty *(Hyperseed-Concept 195) and of the way* information *(Hyperseed-Concept* **??***) depends on what distinctions an observer is able or willing to draw. Equally importantly, finiteness avoids measure-theoretic pathologies that can obscure the intended interpretation: conditional probabilities and entropies on finite spaces can be handled by explicit finite sums, and statements about refinement/coarsening of partitions reduce to elementary combinatorics on blocks. When X is infinite but replaced by a finite partition, the loss of resolution is not an afterthought but part of the modeling stance: two microstates that lie in the same block are, by stipulation, observationally identical for the purposes of the theory being developed in Section 10. Thus, when later notions depend on what an agent can "say" or encode, the finitary reduction makes the dependence on representational granularity explicit rather than implicit.*

Remark 1733. *The convention $0 \log 0 := 0$ is a disciplined way of expressing the idea that events assigned probability 0 contribute no uncertainty: the limit $\lim_{t \downarrow 0} t \log t = 0$ makes the formula for Shannon entropy continuous at the boundary of the simplex. The symbol $\Delta(X)$ denotes the probability simplex on X, i.e. the set of functions $q : X \to [0,1]$ with $\sum_{x \in X} q(x) = 1$ (Hyperseed-Concept 139). An "indistinction weight" $\iota(x,y) \in [0,1]$ may be read as a graded notion of "x could be taken for y" (Hyperseed-Concept 98); it is the soft, quantitative cousin of the crisp equivalence relation behind a partition (Hyperseed-Concept* **??***). Such weights are the raw material from which we later define fidelity and ineffability (Hyperseed-Concept* **??***) in observer-relative terms. It is useful to keep in mind that, unless explicitly assumed later, ι need not be symmetric and need not satisfy transitivity: an agent might reliably mistake x for y without equally mistaking y for x (asymmetric confusability), and "could be taken for" may fail to propagate cleanly along chains of resemblance. Requiring only that $\iota(x,y) \in [0,1]$ keeps the framework broad enough to encompass both hard partitions (recoverable by setting ι to be 1 on a block and 0 across blocks) and more graded perceptual or representational limitations. Similarly, calling $q \in \Delta(X)$ a reconstruction emphasizes an intended role: q is not merely any distribution but a candidate posterior, guess, or synthesized representation of what the world could be, given limited access to distinctions. In later helper theorems, q typically appears as a variable over which some functional is minimized or maximized, formalizing the idea that an agent chooses (or is forced into) the "best available" representation under its constraints.*

Remark 1734. *For orientation, one may view many forthcoming quantities as functionals on $\Delta(X)$ assembled from finite sums of elementary terms such as $q(x) \log q(x)$ or $p(x) \log\big(p(x)/q(x)\big)$, possibly modified by an indistinction kernel $\iota(x,y)$ that blurs the contribution of specific states. The standing conventions above guarantee that all such expressions are well-defined without additional measurability hypotheses and that boundary cases (where some probabilities vanish) can be handled uniformly. This uniformity is especially helpful when formal claims about "ineffability" are meant to be robust under coarse-graining: replacing X by a partition should change the values in an interpretable way, rather than introducing technical exceptions.*

32.1 Observer-indexed coarse-graining facts

Lemma 87 (Partition-generated observables are exactly unions of blocks)**.** *Suppose $\Sigma_O \subseteq 2^X$ is generated by a partition $\pi_O = \{B_1, \ldots, B_k\}$. Then an event $A \subseteq X$ is O-observable (i.e. $A \in \Sigma_O$) iff A is a union of blocks of π_O. Equivalently, if $\alpha : X \to \{1, \ldots, k\}$ maps $x \mapsto i$ when $x \in B_i$, then $A \in \Sigma_O$ iff $A = \alpha^{-1}(S)$ for some $S \subseteq \{1, \ldots, k\}$. Moreover, in this finite-partition setting, Σ_O contains exactly 2^k events (one for each choice of subcollection of blocks), and Σ_O may be identified with the full power set $2^{\{1,\ldots,k\}}$ via $S \mapsto \alpha^{-1}(S)$.*

Remark 1735. *Intuitively, a partition π_O specifies exactly which distinctions the observer O can make. If O can only tell which block B_i the true state lies in, then the only well-posed yes/no questions are those whose answers depend only on the block-label, not on the fine-grained element within the block. In this sense, Σ_O is the algebra of "questions that make sense to O" given their perceptual or representational granularity (Hyperseed-Concept 134, and more broadly Hyperseed-Concept 86). One can also read this as a statement about invariance: A is O-observable exactly when it is constant on each block (i.e. whenever x, y lie in the same B_i, either both are in A or both are not in A). In that sense, π_O induces an "indistinguishability" relation on X, and O-observables are precisely those events that respect that relation.*

Remark 1736. *This lemma is a small but essential bridge between qualitative epistemology and quantitative information theory: once observables are characterized as unions of blocks, one can push probability distributions forward along α and compute entropies on the coarse space $\{1, \ldots, k\}$. This is one of the cleanest ways to see how information is* observer-indexed *rather than absolute. Concretely, if $(X, \Sigma, \mathbb{P})$ is a probability space with $\Sigma_O \subseteq \Sigma$, then α becomes a random variable taking values in $\{1, \ldots, k\}$, and $\mathbb{P}(\alpha = i) = \mathbb{P}(B_i)$. Any O-observable event A corresponds to a subset $S \subseteq \{1, \ldots, k\}$, so that $\mathbb{P}(A) = \mathbb{P}(\alpha \in S) = \sum_{i \in S} \mathbb{P}(B_i)$. Thus probabilities of O-events depend only on the block-masses, not on any finer distribution inside each B_i, which is exactly what it means for the observer to be insensitive to within-block structure.*

Remark 1737. *Before the proof, note the role of the symbols. The set 2^X is the power set of X (all events). A σ-algebra Σ_O is a collection of events closed under complements and countable unions; in the finite case, "countable unions" reduces to finite unions. The map $\alpha : X \to \{1, \ldots, k\}$ is simply the "block ID" map, and $\alpha^{-1}(S)$ is the event that the block ID lies in S. It is often helpful to note that Σ_O can also be written as $\sigma(\alpha)$, the σ-algebra generated by the random variable (measurable map) α; equivalently, $\Sigma_O = \{\alpha^{-1}(S) : S \subseteq \{1, \ldots, k\}\}$ when the codomain carries the discrete σ-algebra (which, for a finite set, is just its power set). This repackages the lemma into a standard measurability fact: an event is measurable with respect to α exactly when it is a preimage of an event in the codomain.*

Sketch. The σ-algebra generated by a partition is, by definition, the smallest σ-algebra containing each block; closure under complements and countable unions forces inclusion of all unions of blocks, and nothing else is needed (finite case). More explicitly, let

$$\mathcal{U} \;=\; \Big\{ \bigcup_{i \in S} B_i \;:\; S \subseteq \{1, \ldots, k\} \Big\}.$$

Then $\mathcal{U}$ contains each B_i (take $S = \{i\}$), contains $\emptyset$ (take $S = \emptyset$), and contains X (take $S = \{1, \ldots, k\}$). If $A = \bigcup_{i \in S} B_i \in \mathcal{U}$, then its complement is

$$X \setminus A \;=\; \bigcup_{i \notin S} B_i \in \mathcal{U},$$

so $\mathcal{U}$ is closed under complements. If $A_n = \bigcup_{i \in S_n} B_i \in \mathcal{U}$ is any sequence, then

$$\bigcup_{n=1}^{\infty} A_n \;=\; \bigcup_{n=1}^{\infty} \bigcup_{i \in S_n} B_i \;=\; \bigcup_{i \in \cup_n S_n} B_i \in \mathcal{U},$$

so $\mathcal{U}$ is closed under countable unions as well (and in the present finite setting, this closure is equivalent to closure under finite unions). Hence $\mathcal{U}$ is a σ-algebra containing the blocks, so by minimality of the generated σ-algebra we have $\Sigma_O \subseteq \mathcal{U}$. Conversely, since Σ_O contains each block and is closed under unions, it must contain every union of blocks, so $\mathcal{U} \subseteq \Sigma_O$. Therefore $\Sigma_O = \mathcal{U}$, proving the "union of blocks" characterization. The equivalence with $A = \alpha^{-1}(S)$ follows by observing that $\alpha^{-1}(S) = \bigcup_{i \in S} B_i$ by definition of α and the partition. □

Remark 1738. Proof sketch. *The strategy is purely structural: show that the family of all unions of blocks is itself a σ-algebra containing each block, and then use minimality of the generated σ-algebra to conclude equality.* □

The key step is that closure under complements turns a union of blocks into a union of the remaining blocks, while closure under unions composes unions-of-blocks into unions-of-blocks. Visually, one may imagine the blocks as "pixels"; the only shapes expressible to O are those made by selecting whole pixels, never by cutting a pixel in half. An equivalent visualization is via quotienting: the partition induces a quotient set $X/\!\sim$, where $x \sim y$ iff they lie in the same block. The map α is (up to relabeling) the quotient map, and Σ_O consists precisely of the events that are inverse images of subsets downstairs. So an O-observable event is one that is well-defined on equivalence classes, i.e. it makes sense to ask it without specifying which representative inside the class is realized.

Remark 1739. *Two small corollaries are often useful in later information-theoretic manipulations. First, every O-observable event A is determined by the* index set $S_A = \{i : B_i \subseteq A\}$, *and conversely $A = \bigcup_{i \in S_A} B_i$. Second, if one works with indicator functions, then $\mathbf{1}_A$ is Σ_O-measurable iff $\mathbf{1}_A = f \circ \alpha$ for some $f : \{1, \ldots, k\} \to \{0, 1\}$, i.e. iff the truth value of A is a function of the coarse label alone. This provides a convenient functional form of the same idea that will align with later uses of random variables and coarse-grained statistics.*

32.2 Shannon identities that are useful as rewrite rules

Lemma 88 (Cross-entropy decomposition)**.** *For distributions p, q on X with $q(x) > 0$ whenever $p(x) > 0$,*

$$H(p, q) \;=\; H(p) \;+\; D_{\mathrm{KL}}(p \| q).$$

In particular, $H(p, q) \geq H(p)$ with equality iff $p = q$.

Remark 1740. *This identity says: the expected code-length of using the "wrong" model q when the truth is p equals the intrinsic uncertainty of p plus a nonnegative penalty measuring how far q deviates from p. In philosophical terms, $H(p)$ is the ineliminable opacity in the source itself, whereas $D_{\mathrm{KL}}(p \| q)$ is the price of an epistemic misfit between representation and reality (Hyperseed-Concept* **??**, *Hyperseed-Concept* **??***). This decomposition is standard in information theory and underlies many minimum-cross-entropy and maximum-likelihood principles; it also resonates with algorithmic perspectives on description and model mismatch [16]. One can also read the identity as a "rewrite rule" for optimizing models: minimizing cross-entropy $H(p, q)$ over q is equivalent to minimizing $D_{\mathrm{KL}}(p \| q)$, since $H(p)$ does not depend on q. In this way, KL divergence measures the* excess *description length beyond what is unavoidable given the source p itself. Finally, the numerical value depends on the base of the logarithm: base 2 gives bits, base e gives nats, and the identity holds verbatim in any base as long as it is used consistently in all three quantities.*

Remark 1741. *A simple example: let $X = \{0, 1\}$ and $p(1) = 1/2$ (a fair coin), but you insist on using $q(1) = 0.9$. Then $H(p)$ is $\log 2$, while $H(p, q)$ is larger because your model predicts 1 too often; the gap is exactly $D_{\mathrm{KL}}(p\|q) > 0$. Thus, even when the world is maximally uncertain (fair coin), you can still do strictly worse by holding an incorrect belief. If we take base-2 logarithms to make the code-length interpretation literal (in bits), then*

$$H(p, q) = -\tfrac{1}{2}\log_2(0.9) - \tfrac{1}{2}\log_2(0.1) = \tfrac{1}{2}\log_2\left(\tfrac{1}{0.09}\right) = \tfrac{1}{2}\log_2(11.\overline{1}) \approx 1.736,$$

whereas $H(p) = 1$. Hence $D_{\mathrm{KL}}(p\|q) = H(p, q) - H(p) \approx 0.736$ bits: on average, using the wrong model costs roughly three quarters of a bit per outcome relative to the optimal code matched to the fair coin. This makes vivid that the penalty term is not merely qualitative; it is an explicit and operationally meaningful loss.

Remark 1742. *Before the proof, recall the meaning of the notation: $H(p)$ is Shannon entropy $-\sum_x p(x)\log p(x)$; $H(p, q)$ is cross-entropy $-\sum_x p(x)\log q(x)$; and $D_{\mathrm{KL}}(p\|q)$ is the Kullback–Leibler divergence $\sum_x p(x)\log\frac{p(x)}{q(x)}$. The support condition $q(x) > 0$ whenever $p(x) > 0$ ensures that $\log\frac{p(x)}{q(x)}$ never evaluates to $\log\frac{\text{positive}}{0}$. Equivalently, it ensures that $H(p, q)$ is finite: if some event has $p(x) > 0$ but $q(x) = 0$, then the term $-p(x)\log q(x)$ contributes $+\infty$, reflecting that a code optimized for q would assign infinite length to an event that actually occurs with positive probability under p. In this sense, the support condition is not an artificial technicality: it encodes the minimal requirement that the model q does not declare genuinely possible outcomes to be impossible. For completeness, one typically adopts the convention $0\log 0 := 0$ (as a limit) in Shannon entropy, so terms with $p(x) = 0$ do not cause difficulties in $H(p)$; the real obstruction is only when $q(x) = 0$ on the support of p.*

Sketch. Expand definitions: $D_{\mathrm{KL}}(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)} = \sum_x p(x)\log p(x) - \sum_x p(x)\log q(x) = -H(p) + H(p, q)$. Rearrange. The inequality follows from $D_{\mathrm{KL}} \geq 0$. To connect this directly to the intended "rewrite" usage: the displayed computation is precisely the algebraic step that allows one to replace an expression involving $H(p, q)$ by one involving $H(p)$ and $D_{\mathrm{KL}}(p\|q)$ (or vice versa), whichever is more convenient for subsequent bounds or conceptual interpretation. ☐

Remark 1743. Proof sketch. *Compute $D_{\mathrm{KL}}(p\|q)$ and observe it is exactly the difference between cross-entropy and entropy; then appeal to the general fact $D_{\mathrm{KL}} \geq 0$.* ☐

The algebra works because $\log\frac{p}{q} = \log p - \log q$, so the sum splits into two sums, one being $-H(p)$ and the other being $H(p, q)$. Geometrically, D_{KL} is a kind of directed "separation" between distributions, so it is natural that it measures the excess cost of coding under the wrong model. It is worth stressing the "directed" aspect: typically $D_{\mathrm{KL}}(p\|q) \neq D_{\mathrm{KL}}(q\|p)$, so the penalty depends on which distribution is the data-generating truth and which distribution is the assumed model. This asymmetry matches the operational story: data are drawn from p, but the coder chooses q, so the expectation is taken with respect to p. As a practical rewrite rule, the identity is often used in the form

$$H(p, q) - H(p) = D_{\mathrm{KL}}(p\|q),$$

i.e. "excess average codelength equals divergence." In arguments about uncertainty and ineffability, this can be read as: some portion of opacity is irreducible ($H(p)$), while the rest is an avoidable artifact of misrepresentation ($D_{\mathrm{KL}}(p\|q)$).

Lemma 89 (Entropy bounds via uniform comparison)**.** *Let $|X| = n$ and let u be uniform on X. Then $0 \leq H(p) \leq \log n$. Moreover, $H(p) = 0$ iff p is a point-mass, and $H(p) = \log n$ iff $p = u$.*

Remark 1744. *This lemma formalizes two extremes of uncertainty (Hyperseed-Concept 195). At one end, a point-mass distribution corresponds to complete certainty: there is no surprise, no freedom of outcome, hence entropy* 0*. At the other end, the uniform distribution spreads belief evenly across* n *possibilities, maximizing ignorance subject to the finite support constraint, hence entropy* $\log n$*. The result is also a moral: entropy is not "disorder" in the abstract; it is a measure of how evenly probability mass is distributed across the distinguishable alternatives (Hyperseed-Concept* **??***). A further useful perspective is that the upper bound* $\log n$ *is not merely a numerical ceiling but the entropy of the maximally symmetric state on a set of* n *labeled alternatives. Thus, whenever an argument bounds some complicated distribution by comparing it to uniform, it is implicitly appealing to symmetry as the criterion for maximal uncertainty.*

Remark 1745. *A concrete example: with* $n = 4$ *equiprobable outcomes,* $H = \log 4 = 2\log 2$*. If instead all mass is on one outcome,* $H = 0$*. If mass is split* $(1/2, 1/2, 0, 0)$*, then* $H = \log 2$*; one may read this as "only one bit of uncertainty remains" after coarse elimination of two possibilities. One can also see the concavity of entropy numerically here: averaging two point-masses (certainty in one of two outcomes) yields the mixed distribution* $(1/2, 1/2, 0, 0)$*, whose entropy* $\log 2$ *is larger than the average entropy* 0 *of the extremes. This is the same mechanism, in miniature, behind why the uniform distribution is the unique maximizer on the simplex.*

Remark 1746. *Before the proof, note that comparing to uniform is a standard technique:* $D_{\mathrm{KL}}(p\|u) \geq 0$ *is equivalent to* $H(p) \leq H(u) = \log n$*. This is one place where the cross-entropy decomposition of Lemma 88 functions as a rewrite rule: by choosing a particular* q*, one obtains inequalities for* $H(p)$*. In fact, specializing Lemma 88 to* $q = u$ *yields an identity (not only an inequality):*

$$H(p) = H(u) - D_{\mathrm{KL}}(p\|u) = \log n - D_{\mathrm{KL}}(p\|u),$$

which makes the upper bound transparent: entropy falls short of $\log n$ *by exactly the amount of non-uniformity measured by* $D_{\mathrm{KL}}(p\|u)$*. This form is often useful when one wants a quantitative stability statement: if* p *is close to uniform in KL divergence, then* $H(p)$ *is close to its maximum.*

Sketch. Nonnegativity: $H(p) = -\sum_x p(x)\log p(x)$ and $\log p(x) \leq 0$. Upper bound: $D_{\mathrm{KL}}(p\|u) \geq 0$ gives $\sum_x p(x)\log\frac{p(x)}{1/n} \geq 0$, i.e. $\sum_x p(x)\log p(x) + \log n \geq 0$, so $H(p) \leq \log n$. Equality conditions are the standard equality cases of Gibbs' inequality. For the lower bound, one may also note explicitly that each term satisfies $-p(x)\log p(x) \geq 0$ for $p(x) \in [0, 1]$ (with $0\log 0 := 0$), so the sum cannot be negative. For the upper bound, the equality case $D_{\mathrm{KL}}(p\|u) = 0$ occurs exactly when $p = u$, yielding $H(p) = \log n$, whereas entropy 0 occurs exactly at the vertices of the simplex, i.e. point masses. ☐

Remark 1747. Proof sketch. *Show* $H(p) \geq 0$ *termwise, and show* $H(p) \leq \log n$ *by comparing* p *to the uniform distribution using KL divergence.* ☐

The key mechanism is convexity (hidden inside Gibbs' inequality): uniform is the maximizer of entropy because it is the unique distribution that does not privilege any outcome. One may visualize the simplex $\Delta(X)$*: entropy is a concave function on this simplex, achieving its maximum at the center (uniform) and minima at the vertices (point masses). Another standard route to the same conclusion (consistent with the "rewrite rule" framing) is to maximize* $H(p)$ *under the constraint* $\sum_x p(x) = 1$ *via Lagrange multipliers: the stationarity condition forces* $p(x)$ *to be constant in* x*, hence uniform. The KL-divergence method is often preferred because it yields both the bound and a canonical measure of the gap to optimality:* $\log n - H(p) = D_{\mathrm{KL}}(p\|u)$*. In later arguments, it is frequently helpful to remember that these inequalities are tight and structural: they are not artifacts of a loose estimate, but rather reflect the geometry of the probability simplex together with the strict concavity of the entropy functional.*

32.3 Mutual and interaction information: algebraic normal forms

Lemma 90 (Mutual information: nonnegativity and independence test)**.** *For finite random variables (X, Y) with joint $p(x, y)$,*

$$I(X;Y) \;=\; D_{\mathrm{KL}}(p(x,y)\,\|\,p(x)p(y)) \;\geq\; 0,$$

and $I(X;Y) = 0$ iff $p(x,y) = p(x)p(y)$ (i.e. X and Y are independent).

Remark 1748. *It is often useful to keep in mind the equivalent entropy identities (all for finite variables)*

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(X) - H(X|Y) = H(Y) - H(Y|X),$$

which make explicit the interpretation of mutual information as a "reduction in uncertainty." These equalities are compatible with the KL-divergence definition because, expanding $D_{\mathrm{KL}}(p(x,y)\|p(x)p(y))$ as a sum $\sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$ and collecting terms, one recovers precisely the entropy combination above. In particular, the unit of $I(X;Y)$ (bits vs. nats) is determined by the base of the logarithm used in $H(\cdot)$ and $D_{\mathrm{KL}}(\cdot\|\cdot)$.

Remark 1749. *Mutual information measures the extent to which knowing X reduces uncertainty about Y (and symmetrically). It is the quantitative shadow of the intuitive idea of* constraint *or* coupling *between variables: if the joint distribution factors, then the variables do not constrain one another, and the mutual information vanishes. If the joint cannot be factored, the mutual information is strictly positive, encoding precisely how far the world is from being decomposable into independent parts (Hyperseed-Concept 117).*

Remark 1750. *A simple example: if $Y = X$ for a fair bit, then $I(X;Y) = H(X) = \log 2$; all uncertainty about Y is resolved by observing X. If X and Y are independent fair bits, then $I(X;Y) = 0$. Thus mutual information behaves like a measure of "shared structure" between two descriptions of reality, and it is foundational both in classical information theory and in algorithmic analogues [16].*

Remark 1751. *Before the proof, note that $p(x)p(y)$ denotes the product of the marginals, i.e. the joint distribution one would have if X and Y were independent. The identity in the lemma therefore states that mutual information is precisely the KL divergence between the true joint and the closest "independent-world" model obtained by fixing marginals.*

Remark 1752. *To avoid any ambiguity about domains: since we assumed (X, Y) are finite, all sums below are over finite supports. As usual in information theory, terms with $p(x,y) = 0$ contribute 0 to expressions of the form $p(x,y)\log(\cdot)$, while $D_{\mathrm{KL}}(p\|q)$ is defined to be $+\infty$ if there exists an atom with $p > 0$ but $q = 0$. In the present lemma, $q(x,y) = p(x)p(y)$, so this pathology cannot occur unless one allows marginals with zeros that exclude points where $p(x,y) > 0$; for ordinary discrete random variables, $p(x,y) > 0$ implies $p(x) > 0$ and $p(y) > 0$, hence $q(x,y) > 0$.*

Sketch. Immediate from the KL expression and $D_{\mathrm{KL}} \geq 0$, with equality iff the two distributions coincide. □

Remark 1753. Proof sketch. *Rewrite $I(X;Y)$ as a KL divergence, then use the general theorem that KL divergence is always nonnegative and is zero exactly when its arguments match.* □

The decisive step is conceptual: once mutual information is recognized as a KL divergence, its basic order-theoretic behavior (nonnegativity and equality case) becomes automatic. The geometry is that the set of independent distributions $p(x)p(y)$ forms a curved submanifold inside the simplex of all joint distributions, and $I(X;Y)$ quantifies departure from that submanifold.

Remark 1754. *If one wants a more explicit inequality behind $D_{\mathrm{KL}} \geq 0$, one can appeal to the log-sum inequality or to Jensen's inequality applied to the convex function $t \mapsto t \log t$. Concretely, writing $D_{\mathrm{KL}}(p\|q) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{q(x,y)}$ and using $\log u \leq u - 1$ (equivalently $-\log u \geq 1 - u$) with $u = q/p$ yields*

$$D_{\mathrm{KL}}(p\|q) \;=\; \sum_{x,y} p(x,y) \log \frac{p(x,y)}{q(x,y)} \;\geq\; \sum_{x,y} \big(p(x,y) - q(x,y)\big) = 0,$$

and the equality condition forces $p(x,y) = q(x,y)$ for all (x,y) with $p(x,y) > 0$, i.e. precisely the independence condition in this case.

Lemma 91 (Mutual information upper bounds)**.** *For finite (X,Y),*

$$I(X;Y) \leq H(X) \qquad \textit{and} \qquad I(X;Y) \leq H(Y).$$

Remark 1755. *This lemma states an epistemic sanity constraint: the information X can provide about Y cannot exceed the total uncertainty in X itself (and vice versa). One cannot extract more "shared information" than there is information to share. In contexts where X is interpreted as a message or measurement, the bound reads as: the channel cannot transmit more than the entropy of its input.*

Remark 1756. *A slightly different but equivalent viewpoint is obtained from the identity $I(X;Y) = H(X) - H(X|Y)$. Since conditional entropy is nonnegative, $H(X|Y) \geq 0$, one immediately gets $I(X;Y) \leq H(X)$, with equality precisely when $H(X|Y) = 0$, i.e. when X is a deterministic function of Y (almost surely). Symmetrically, $I(X;Y) \leq H(Y)$, with equality when Y is (a.s.) a function of X. This makes clear that the upper bounds are tight and characterize the "noiseless decoding" extremes.*

Remark 1757. *Before the proof, recall that the chain rule expresses $H(X,Y)$ as $H(X)+H(Y|X)$ and also as $H(Y)+H(X|Y)$, where conditional entropies are always ≥ 0. The inequality $H(X,Y) \geq H(X)$ is precisely the statement that joint uncertainty cannot be smaller than uncertainty in a marginal.*

Sketch. Use the standard chain-rule identity $H(X,Y) = H(X) + H(Y|X) \geq H(X)$ and also $H(X,Y) \geq H(Y)$. Then $I(X;Y) = H(X)+H(Y)-H(X,Y) \leq H(X)$ and symmetrically $\leq H(Y)$. (Any preferred proof of $H(X,Y) \geq H(X)$ can be used, e.g. via log-sum inequality.) ☐

Remark 1758. Proof sketch. *Use the chain rule to lower bound $H(X,Y)$ by $H(X)$ and $H(Y)$; then substitute into $I(X;Y) = H(X) + H(Y) - H(X,Y)$.* ☐

The key move is recognizing conditional entropy as a nonnegative remainder term. Visually, if one imagines entropy as "volume" of uncertainty, then $H(X,Y)$ is a volume in the product space, which must dominate the projection volumes $H(X)$ and $H(Y)$.

Remark 1759. *The upper bounds can also be read as a constraint on correlation structure: if $H(X)$ is small (for instance, X is nearly deterministic), then regardless of how complicated Y is, the shared information cannot exceed that small budget. This is one reason mutual information is often more robust than correlation coefficients when comparing variables with different alphabets or different degrees of intrinsic randomness: the entropy caps automatically normalize what is even possible.*

Lemma 92 (Interaction information as an alternating sum of entropies)**.** *Assuming the standard conditional-mutual-information identity $I(X;Y|Z) = H(X|Z)+H(Y|Z)-H(X,Y|Z)$ (equivalently $I(X;Y|Z) = H(X,Z) + H(Y,Z) - H(Z) - H(X,Y,Z)$), the interaction information satisfies the symmetric expansion*

$$I(X;Y;Z) = H(X) + H(Y) + H(Z) - H(X,Y) - H(X,Z) - H(Y,Z) + H(X,Y,Z).$$

In particular, $I(X;Y;Z)$ is invariant under permutation of (X,Y,Z).

Remark 1760. *One common definition (equivalent to the expansion above, given the stated identities) is*

$$I(X;Y;Z) \;=\; I(X;Y) - I(X;Y|Z),$$

which makes explicit that interaction information compares the X–Y dependence "before" and "after" revealing Z. From this viewpoint, the alternating-sum normal form is obtained simply by substituting $I(X;Y) = H(X)+H(Y)-H(X,Y)$ and the given expression for $I(X;Y|Z)$, and then collecting like entropy terms. The permutation invariance is then a consequence of the fact that the resulting polynomial in joint entropies is manifestly symmetric.

Remark 1761. *Interaction information $I(X;Y;Z)$ is a higher-order quantity: it measures how the dependence between X and Y changes when one conditions on Z. Unlike ordinary mutual information, it can be positive or negative, reflecting the subtlety of three-way constraint: Z can either explain away a correlation (yielding negative interaction information in some conventions) or create synergy where the pair (X,Y) together carries information about Z not present in either alone (Hyperseed-Concept* **??***).*

Remark 1762. *The sign can be sanity-checked on canonical examples. If Z is a common cause that renders X and Y conditionally independent (so $I(X;Y|Z) = 0$ but $I(X;Y) > 0$), then $I(X;Y;Z) = I(X;Y) > 0$ under the convention $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$. Conversely, if X and Y are marginally independent but become dependent after conditioning on Z (a collider pattern), then $I(X;Y) = 0$ while $I(X;Y|Z) > 0$, producing $I(X;Y;Z) < 0$. These examples illustrate why interaction information is not a simple "amount of dependence" but rather an interference term measuring how dependencies compose.*

Remark 1763. *The alternating-sum normal form is important because it allows one to compute and transform interaction information using only entropies of various marginals. This makes it a convenient "algebraic normal form" for proofs and for deriving identities: once expanded, symmetry under permutation becomes visible by inspection rather than by argument.*

Remark 1764. *A practical advantage of the alternating-sum form is that it makes cancellations transparent in proofs that involve adding and subtracting information quantities. For instance, when comparing two interaction-information terms that share marginals (e.g. replacing Z by (Z,W)), many entropy terms cancel immediately, leaving a small remainder that can often be recognized as a conditional mutual information. This "algebraic bookkeeping" role is analogous to how inclusion–exclusion turns set-cardinality calculations into an alternating sum of intersections.*

Remark 1765. *Before the proof, observe the bookkeeping: the formula is an inclusion–exclusion pattern in which single-variable entropies appear with $+$, pairwise entropies with $-$, and the triple entropy with $+$. This is a signature that interaction information is, in a precise sense, a third-order correction term measuring what is* not *captured by all lower-order marginals.*

Sketch. Start from the definition $I(X;Y;Z) = I(X;Y) - I(X;Y|Z)$. Replace $I(X;Y)$ by $H(X) + H(Y) - H(X,Y)$, and replace $I(X;Y|Z)$ by the entropy form above, then simplify by cancellation. The final expression is manifestly symmetric. □

Remark 1766. Proof sketch. *Expand the definition in terms of $I(X;Y)$ and $I(X;Y|Z)$, rewrite each in terms of entropies, and cancel like terms to obtain an inclusion–exclusion expression.* □

The cancellation works because conditioning on Z replaces each entropy term by a corresponding term that includes Z (e.g. $H(X|Z) = H(X,Z) - H(Z)$). The geometric intuition is that the alternating sum is analogous to subtracting overlaps between sets: pairwise overlaps are removed, but then the triple overlap must be added back.

32.4 Logical entropy, graphtropy, and their links to fidelity

Lemma 93 (Logical entropy monotone under refinement)**.** *Let π' refine π (every block of π is a union of blocks of π'). Then for any distribution p on X,*

$$h(\pi, p) \;\le\; h(\pi', p).$$

Remark 1767. *Logical entropy $h(\pi, p)$ (Hyperseed-Concept 105) is the probability that two independent draws from p fall into* different *blocks of the partition π; it is, in Ellerman's terminology, a measure of distinctions rather than surprises [17]. A refinement π' makes* more *distinctions, so it is natural that it can only increase the chance that two samples are separated.*

Remark 1768. *As an example, if π has two blocks $\{a, b\}$ and $\{c, d\}$ under the uniform distribution, then $h(\pi, p) = 1 - (1/2)^2 - (1/2)^2 = 1/2$. If one refines $\{a, b\}$ into $\{a\}$ and $\{b\}$ while leaving $\{c, d\}$ intact, then the block masses become $(1/4, 1/4, 1/2)$ and $h(\pi', p) = 1 - (1/4)^2 - (1/4)^2 - (1/2)^2 = 5/8 > 1/2$.*

Remark 1769. *Before the proof, note the refinement relation: writing $\pi = \{B_i\}$ and $\pi' = \{B'_j\}$, refinement means each B_i is a disjoint union of some subcollection of the B'_j. Thus the coarse block mass p_i is the sum of the fine block masses p'_j over those j in the subcollection.*

Sketch. Let coarse block masses be $p_i = \sum_{j \in J_i} p'_j$ where p'_j are fine masses. Then by convexity of $t \mapsto t^2$, $(\sum_{j \in J_i} p'_j)^2 \ge \sum_{j \in J_i} (p'_j)^2$. Summing over i gives $\sum_i p_i^2 \ge \sum_j (p'_j)^2$. Since $h(\cdot, p) = 1 - \sum (\cdot)^2$, the inequality reverses as stated. □

Remark 1770. Proof sketch. *Express coarse masses as sums of fine masses and apply convexity of the square function to show $\sum p_i^2$ decreases under refinement; since $h = 1 - \sum p_i^2$, h increases.* □

The key step is that squaring punishes concentration: merging probabilities (coarsening) increases the sum of squares, while splitting probabilities (refinement) decreases it. Visually, refinement spreads mass across more bins, which increases the chance that two draws land in different bins.

Lemma 94 (Logical entropy maximum on k blocks)**.** *If π has k blocks with block masses $(p_1, \dots, p_k)$, then*

$$0 \le h(\pi, p) = 1 - \sum_{i=1}^{k} p_i^2 \le 1 - \frac{1}{k},$$

with equality $h(\pi, p) = 1 - \frac{1}{k}$ iff $p_i = \frac{1}{k}$ for all i.

Remark 1771. *This is the logical-entropy analogue of the Shannon statement "entropy is maximized by uniform". Here the maximum distinction-rate is achieved when the observer's coarse blocks are equiprobable: then two random samples are most likely to fall into different blocks. The bound $1-1/k$ is also intuitively satisfying: with k equally likely blocks, the chance of matching blocks is $1/k$, so the chance of a distinction is $1-1/k$.*

Remark 1772. *Before the proof, notice that the constraint $\sum_i p_i = 1$ places (p_i) on the standard simplex, and $\sum_i p_i^2$ is minimized (hence h maximized) at the barycenter. Cauchy–Schwarz provides a one-line certificate of that minimization.*

Sketch. By Cauchy–Schwarz, $\sum_i p_i^2 \geq \frac{(\sum_i p_i)^2}{k} = \frac{1}{k}$. Thus $h = 1 - \sum_i p_i^2 \leq 1 - \frac{1}{k}$. Equality in Cauchy–Schwarz gives p_i all equal. $\square$

Remark 1773. Proof sketch. *Use Cauchy–Schwarz to lower bound $\sum_i p_i^2$ by $1/k$, then translate to an upper bound on $h = 1 - \sum_i p_i^2$.* $\square$

The proof works because Cauchy–Schwarz captures the idea that, among nonnegative numbers with fixed sum, the sum of squares is minimized when the numbers are equal. One may picture h as the amount of "separation" created by the partition under p; equal block masses are the configuration that makes the partition maximally separating in expectation.

Lemma 95 (Graphtropy is expected distinction error under the "prior" reconstruction)**.** *Let G be an indistinction structure with weights $\iota_G(x,y) \in [0,1]$ and let $p \in \Delta(X)$. Define the distinction error* $\mathrm{Err}_G(x,q) = \sum_y q(y)\,(1-\iota_G(x,y))$ *and fidelity* $\mathrm{Fid}_G(x,q) = \sum_y q(y)\,\iota_G(x,y)$*. Then*

$$GT(G,p) \;=\; \mathbb{E}_{x\sim p}\big[\mathrm{Err}_G(x,p)\big] \qquad \textit{and} \qquad 1 - GT(G,p) \;=\; \mathbb{E}_{x\sim p}\big[\mathrm{Fid}_G(x,p)\big].$$

Remark 1774. *Graphtropy (Hyperseed-Concept **??**) generalizes logical entropy from crisp partitions to graded indistinction graphs. The lemma tells us that graphtropy is literally an expected* distinction error *when one uses the "prior" p as the reconstruction of a state drawn from p. In other words, it quantifies how much indistinction structure G fails to support faithful self-reconstruction, when no additional evidence beyond the prior is used. This connects the combinatorial notion of "distinctions" directly to an epistemic notion of representation quality (Hyperseed-Concept 157).*

Remark 1775. *A simple sanity check: if $\iota_G(x,y) = 1$ for all (x,y) (everything is indistinguishable), then $\mathrm{Err}_G(x,q) = 0$ and $GT(G,p) = 0$; perfect indistinction yields no distinction errors precisely because the scoring rule never penalizes confusion. At the other extreme, if $\iota_G(x,y) = \mathbf{1}[x=y]$ (the crisp identity relation), then $\mathrm{Err}_G(x,p) = 1 - p(x)$ and $\mathbb{E}_{x\sim p}[1-p(x)] = 1 - \sum_x p(x)^2$, recovering logical entropy as expected mismatch. This illustrates how graphtropy interpolates between classical partitions and graded similarity structures, as discussed in broader pattern-theoretic settings [5]. In particular, one can read ι_G as a (possibly asymmetric) "confusability profile": the closer $\iota_G(x,y)$ is to 1, the less costly it is to confuse x with y under Err_G. Thus, as ι_G is made sharper (mass concentrated nearer the diagonal $x=y$), the same distribution p yields larger expected error and hence larger graphtropy, reflecting that the underlying notion of distinction has become more discriminating.*

Remark 1776. *Before the proof, unpack the notation: $\mathbb{E}_{x\sim p}[\cdot]$ denotes expectation when x is drawn according to p. The functions Err_G and Fid_G are linear in q and sum the (in)distinction weights against a reconstruction distribution. The identity $\mathrm{Fid}_G(x,q) = 1 - \mathrm{Err}_G(x,q)$ is immediate from $\iota_G(x,y) \in [0,1]$ and $(1-\iota_G(x,y))$ being complementary at each pair (x,y). Equivalently, for fixed x the quantity $\sum_y q(y)\iota_G(x,y)$ is an average similarity of the reconstruction $y \sim q$ to the* *target x, and $\sum_y q(y)(1-\iota_G(x,y))$ is the corresponding average dissimilarity; the complementarity holds pointwise before taking any expectation over x. This also highlights that the construction does not assume $q=p$: one can interpret $\mathrm{Err}_G(x,q)$ as the expected "distinction loss" when x is encoded/approximated by a distribution q, and the main identity specializes this to the self-consistent case $q=p$.*

Sketch. Expand: $\mathbb{E}_{x\sim p}[\mathrm{Err}_G(x,p)] = \sum_x p(x)\sum_y p(y)(1-\iota_G(x,y)) = 1-\sum_{x,y} p(x)p(y)\iota_G(x,y) = GT(G,p)$. The fidelity identity is the complement. For avoidance of doubt, the step from $\sum_x p(x)\sum_y p(y)(1-\iota_G(x,y))$ to the double sum $\sum_{x,y} p(x)p(y)(1-\iota_G(x,y))$ is just Fubini/Tonelli for finite (or absolutely summable) expressions, and it makes explicit that (x,y) are being sampled independently from p. □

Remark 1777. Proof sketch. *Write the expectation as a double sum over (x,y) drawn independently from p, then recognize the definition of $GT(G,p)$ (and its complement) inside the algebra.* □

The decisive maneuver is exchanging the order of summation: expectation over x of a q-average over y is just the expectation over independent (x,y). Geometrically, this is averaging a "blurring kernel" ι_G over the product measure $p\otimes p$, so graphtropy is a global property of how p sits inside the indistinction geometry induced by G. In particular, if p concentrates on a region where $\iota_G(x,y)$ is typically large (many pairs are mutually indistinct), then $\sum_{x,y} p(x)p(y)\iota_G(x,y)$ is large and $GT(G,p) = 1-\sum_{x,y} p(x)p(y)\iota_G(x,y)$ is small; conversely, if p spreads mass across mutually distinct regions, then the expected indistinction drops and graphtropy rises. This mirrors the usual partition picture: "spread out over many blocks" increases the probability that two independent draws fall in different blocks.

Corollary 9 (Entropy–distinction conversion bounds)**.** *Let π be a partition with block masses (p_i) and logical entropy $h(\pi,p) = 1-\sum_i p_i^2$. Then the quadratic entropy $H_2 := -\log\sum_i p_i^2$ satisfies*

$$H(\pi) \geq H_2 = -\log(1-h(\pi,p)).$$

Equivalently, any lower bound $h(\pi,p)\geq h_0$ implies $H(\pi)\geq -\log(1-h_0)$. In this partition setting, $\sum_i p_i^2$ is also the collision probability (the probability that two independent draws land in the same block), so the bound can be read as: if collisions are sufficiently unlikely, then Shannon entropy must be large. The monotonicity of $x\mapsto -\log x$ on $(0,1]$ explains why a lower bound on h (i.e., an upper bound on $\sum_i p_i^2$) turns directly into a lower bound on H_2, and then on H.

Remark 1778. *This corollary provides a translation dictionary between two ways of speaking about "how much is distinguished." Logical entropy measures distinctions directly as a probability of inequality of blocks, while Shannon entropy measures average surprise. The inequality says that a sufficiently high distinction-rate forces a nontrivial Shannon entropy lower bound. Thus, even though H and h encode different intuitions, they constrain one another, and one may use whichever is easier to bound in a given argument [17]. One practical point is that $h(\pi,p)$ is quadratic and often admits direct estimation or bounding by Cauchy–Schwarz-type arguments, while $H(\pi)$ involves a logarithm and can be harder to handle algebraically. The corollary lets one turn such quadratic control into an information-theoretic lower bound without committing to a particular coding interpretation beyond the standard meaning of Shannon entropy.*

Remark 1779. *Before the proof, note that H_2 is the order-2 Rényi ("quadratic") entropy, and it is often algebraically simpler because it depends on $\sum_i p_i^2$ rather than $\sum_i p_i\log p_i$. The identity $\sum_i p_i^2 = 1-h(\pi,p)$ makes the conversion particularly transparent. Also recall that for a fixed* *distribution, Rényi entropies are nonincreasing in the order parameter, so $H(\pi) \geq H_2$ is consistent with the general fact $H = H_1 \geq H_2$; here we are simply extracting the explicit dependence on $h(\pi, p)$ via $\sum_i p_i^2$.*

Sketch. This is the partition specialization of the Shannon-vs-quadratic bound $H \geq H_2$ (derived via KL nonnegativity by comparing p to $p^2/\sum p^2$), followed by the identity $\sum_i p_i^2 = 1 - h(\pi, p)$. Concretely, letting $r_i := p_i^2/\sum_j p_j^2$ and using $D_{\mathrm{KL}}(p\|r) \geq 0$ yields $\sum_i p_i \log p_i \geq \sum_i p_i \log r_i = \sum_i p_i(2 \log p_i - \log \sum_j p_j^2)$, which rearranges to $-\sum_i p_i \log p_i \geq -\log \sum_j p_j^2$. □

Remark 1780. Proof sketch. *Use a known inequality $H \geq H_2$ (provable via KL divergence), then substitute $\sum_i p_i^2 = 1 - h(\pi, p)$ to express the quadratic entropy bound in terms of logical entropy.* □

The proof works because KL nonnegativity is a versatile engine: one compares p to a carefully chosen distribution built from p itself (here proportional to p^2), extracting an inequality between two functionals. The logarithm then turns multiplicative concentration (captured by $\sum p_i^2$) into an additive lower bound on H. It is worth emphasizing that the comparison distribution $p^2/\sum p^2$ biases toward heavier blocks, so $D_{\mathrm{KL}}(p\|p^2/\sum p^2) \geq 0$ is precisely measuring the impossibility of p being "more spread out" than its own squared reweighting; the resulting inequality formalizes the intuition that if p has low collision probability then it must also have high average surprise.

32.5 Effability / ineffability: monotonicity and optimization forms

Lemma 96 (Fidelity is a bounded linear functional)**.** *For fixed x and G, the map $q \mapsto \mathrm{Fid}_G(x, q) = \sum_y q(y)\iota_G(x, y)$ is linear in q, and satisfies $0 \leq \mathrm{Fid}_G(x, q) \leq 1$.*

Remark 1781. *This lemma expresses a comforting fact: fidelity is an expectation value. Given x, the number $\iota_G(x, y)$ is the degree to which y counts as an acceptable reconstruction of x under the indistinction structure G. If one randomizes reconstructions according to q, fidelity is simply the average acceptability. Linearity means that mixing strategies mixes fidelities with no hidden nonlinearity, and boundedness means fidelity is a genuine "score" in $[0, 1]$. This is the basic quantitative substrate for talking about effability and ineffability (Hyperseed-Concept **??**) as optimization problems over descriptions.*

Remark 1782. *Example: if $\iota_G(x, y) = \mathbf{1}[x = y]$, then $\mathrm{Fid}_G(x, q) = q(x)$, the probability of reconstructing x exactly. If instead $\iota_G(x, y)$ gives partial credit to near-misses, then $\mathrm{Fid}_G(x, q)$ is expected partial credit, still linear in q.*

Remark 1783. *Before the proof, note that linearity is with respect to the vector space structure on functions $q : X \to \mathbb{R}$; restricting to $\Delta(X)$ makes $\mathrm{Fid}_G(x, q)$ an affine functional on the simplex. The bound $[0, 1]$ comes from the fact that q is a convex combination and $\iota_G(x, \cdot)$ takes values in $[0, 1]$.*

Sketch. Linearity is immediate from the sum. Since $\iota_G(x, y) \in [0, 1]$ and q is a probability distribution, $\mathrm{Fid}_G(x, q)$ is a convex combination of numbers in $[0, 1]$. □

Remark 1784. Proof sketch. *View $\mathrm{Fid}_G(x, q)$ as an inner product of q with the vector $\iota_G(x, \cdot)$, and use that convex combinations preserve bounds.* □

The key step is recognizing "probability-weighted average" as the archetype of a bounded linear functional. Geometrically, for fixed x, $\mathrm{Fid}_G(x, \cdot)$ is a hyperplane functional on the simplex, so its extrema over $\Delta(X)$ occur at vertices (pure reconstructions), a useful fact for optimization intuition even when the formal optimization is constrained by description budgets.

Remark 1785. *A small strengthening that is often implicitly used in later arguments is the following: because* $\mathrm{Fid}_G(x,\cdot)$ *is linear on the ambient vector space and* $\Delta(X)$ *is convex, the image* $\mathrm{Fid}_G(x,\Delta(X))$ *is an interval in* $\mathbb{R}$*. In particular, once one knows the maximum achievable fidelity over a feasible set of posteriors/reconstructions, all smaller fidelities are automatically achievable by mixing with a low-fidelity reconstruction. This "interval property" is what makes threshold-style definitions of effability natural: there is a single frontier value, and any tolerance* ε *simply asks whether that frontier lies above* $1-\varepsilon$*.*

Remark 1786. *In finite* X*, one can make the "extrema occur at vertices" remark fully explicit: writing* $q = \sum_y q(y)\,\delta_y$*, linearity gives* $\mathrm{Fid}_G(x,q) = \sum_y q(y)\mathrm{Fid}_G(x,\delta_y)$*, so the maximum over all* $q \in \Delta(X)$ *equals* $\max_y \iota_G(x,y)$*. This is the special case where the reconstruction distribution* q *itself is unconstrained; later, when* q *must arise as* $\mathrm{dec}_O(d)$ *from a description* d *under a budget* $\sigma_O(d) \le K$*, the feasible set is typically nontrivial, and the maximizer need not be a delta distribution even though the objective remains linear.*

Lemma 97 (Effability is monotone in tolerance ε)**.** *Fix* O,K,G*. If* $0<\varepsilon\le\varepsilon'<1$*, then*

$$\mathrm{Eff}_O(\varepsilon,K;G) \subseteq \mathrm{Eff}_O(\varepsilon',K;G).$$

Remark 1787. *This lemma says that if you relax your standards, more things become expressible. Effability is an observer-relative notion of what can be said (or encoded) about* x *given a limited expressive budget* K*; increasing* ε *corresponds to tolerating lower fidelity. The monotonicity is thus conceptually inevitable: allowing more error cannot reduce the set of acceptable descriptions. Here ineffability (Hyperseed-Concept* **??***) is not mystical but operational: it is the complement of an attainable fidelity region under constraints.*

Remark 1788. *Before the proof, keep track of inequalities:* $\varepsilon \le \varepsilon'$ *implies* $1-\varepsilon' \le 1-\varepsilon$*. Thus any description meeting the stricter fidelity threshold* $1-\varepsilon$ *automatically meets the weaker one* $1-\varepsilon'$*.*

Sketch. If x has a description d with $\sigma_O(d) \le K$ and $\mathrm{Fid}_G(x,\mathrm{dec}_O(d)) \ge 1-\varepsilon$, then also $\mathrm{Fid}_G(x,\mathrm{dec}_O(d)) \ge 1-\varepsilon'$ because $1-\varepsilon' \le 1-\varepsilon$. □

Remark 1789. Proof sketch. *Reuse the same witnessing description* d*: the only thing that changes is the threshold, and the threshold becomes easier to satisfy.* □

The proof is a single inequality chase. The visual intuition is that for fixed budget K*, each* x *has an achievable fidelity interval* $[0,\mathrm{Fid}^\star(x;K)]$*; increasing* ε *lowers the required fidelity and therefore expands the feasible set.*

Remark 1790. *One can also read Lemma 97 contrapositively: if* x *is ineffable at a more permissive tolerance* ε'*, then it is ineffable at every stricter tolerance* $\varepsilon \le \varepsilon'$*. This emphasizes that "ineffability" is stable under tightening requirements: once the available descriptions fail even a lenient target, there is no hope of meeting a more demanding one without changing the resource bound* K *or the indistinction structure* G*.*

Lemma 98 (Effability as a best-achievable-fidelity threshold)**.** *Fix* O,G *and define the best achievable fidelity at budget* K *by*

$$\mathrm{Fid}^\star_{O,G}(x;K) \;:=\; \sup\{\mathrm{Fid}_G(x,\mathrm{dec}_O(d)) :\; d\in D_O,\; \sigma_O(d)\le K\}.$$

Then x *is* (ε,K)*-effable (relative to* G*) iff*

$$\mathrm{Fid}^\star_{O,G}(x;K) \;\ge\; 1-\varepsilon.$$

Moreover, $\mathrm{Fid}^\star_{O,G}(x;K)$ *is nondecreasing in* K*, and nondecreasing under coarse-graining of* G *(in the sense* $\iota_G \le \iota_{G'}$ *pointwise).*

Remark 1791. *This lemma repackages an existential definition ("there exists a description meeting a threshold") into an optimization statement ("the best possible score exceeds the threshold"). This is more than cosmetic: it lets one bring to bear the tools and intuitions of optimization, duality, and monotone resource tradeoffs. In the language of cognition, K is a resource bound on representation, and* $\mathrm{Fid}^\star$ *is the best attainable alignment between what is and what can be said about it (Hyperseed-Concept 157, Hyperseed-Concept* **??***).*

Remark 1792. *The definition of* $\mathrm{Fid}^\star_{O,G}(x;K)$ *is intentionally phrased as a supremum rather than a maximum: depending on the structure of the description space D_O and the decoder* dec_O*, the set of achievable fidelities need not contain its top element (for instance, if increasingly long descriptions approximate a limit behavior). In many discrete settings (finite D_O under the constraint $\sigma_O(d) \le K$, or compactness/continuity hypotheses), the supremum is attained and one may replace* sup *by* max *without changing the value. Even when it is not attained, the threshold characterization remains operational:* $\mathrm{Fid}^\star_{O,G}(x;K) \ge 1-\varepsilon$ *means that for every $\delta > 0$ there exists a description achieving fidelity at least $1-\varepsilon-\delta$.*

Sketch. For the threshold equivalence, unwind the definitions. By definition, x is (ε, K)-effable (relative to G) iff there exists some description $d \in D_O$ with $\sigma_O(d) \le K$ such that $\mathrm{Fid}_G(x, \mathrm{dec}_O(d)) \ge 1-\varepsilon$. This holds iff the supremum over all such descriptions is at least $1-\varepsilon$, i.e. iff $\mathrm{Fid}^\star_{O,G}(x;K) \ge 1-\varepsilon$.

For monotonicity in K, note that if $K \le K'$ then the feasible set $\{d : \sigma_O(d) \le K\}$ is contained in $\{d : \sigma_O(d) \le K'\}$, so taking a supremum over the larger set cannot decrease the value: $\mathrm{Fid}^\star_{O,G}(x;K) \le \mathrm{Fid}^\star_{O,G}(x;K')$.

For monotonicity under coarse-graining, assume $\iota_G \le \iota_{G'}$ pointwise. Then for every reconstruction distribution q,

$$\mathrm{Fid}_G(x,q) = \sum_y q(y)\iota_G(x,y) \;\le\; \sum_y q(y)\iota_{G'}(x,y) = \mathrm{Fid}_{G'}(x,q),$$

and in particular for every description d we have $\mathrm{Fid}_G(x, \mathrm{dec}_O(d)) \le \mathrm{Fid}_{G'}(x, \mathrm{dec}_O(d))$. Taking suprema over the same feasible set $\{d : \sigma_O(d) \le K\}$ yields $\mathrm{Fid}^\star_{O,G}(x;K) \le \mathrm{Fid}^\star_{O,G'}(x;K)$. ☐

Remark 1793. *The two monotonicities in Lemma 98 encode distinct "knobs." The resource knob K enlarges the admissible description set, so it can only improve best-case fidelity. The granularity knob G changes what counts as "close enough"; replacing G by a coarser indistinction structure G' (larger ι pointwise) weakens the success criterion, so any fixed decoding strategy scores at least as well. This clean separation is useful when analyzing tradeoffs: one can ask whether increasing K (better descriptions) or coarse-graining G (lower-resolution standards) is the more plausible route to effability for a given class of objects.*

Remark 1794. *A convenient way to visualize the threshold statement is to treat* $\mathrm{Fid}^\star_{O,G}(x;K)$ *as a "frontier function" of the resource K. Then the effability predicate $x \in \mathrm{Eff}_O(\varepsilon, K; G)$ is simply the superlevel condition* $\mathrm{Fid}^\star_{O,G}(x;K) \ge 1-\varepsilon$*. As K increases, the frontier moves upward (never downward), and as ε increases, the horizontal cutoff $1-\varepsilon$ moves downward. This is the basic geometry behind later statements that effability regions are nested in both parameters.*

Remark 1795. *The two monotonicity statements encode two different philosophical levers. Increasing K expands the expressive repertoire of the observer O, so attainable fidelity cannot decrease. Coarse-graining G (increasing indistinction weights so that more outcomes count as acceptable reconstructions) also cannot decrease fidelity. The second monotonicity captures a form of "forgiving* *semantics": if you decide to treat more things as the same, you make it easier to speak truthfully within your new, blurred ontology. In more operational terms, monotonicity in K is a claim about* resources *(how long/complex a description may be), whereas monotonicity in G is a claim about* evaluation *(how strictly reconstructions are judged). Both are order-theoretic statements: one is induced by inclusion of feasible-description sets, and the other by pointwise enlargement of the kernel used to score reconstructions.*

Remark 1796. *Before the proof, interpret the symbols: D_O is the description space available to observer O, $\sigma_O(d)$ is the description cost/size, and $\mathrm{dec}_O(d)$ is the reconstruction distribution induced by decoding d. The supremum is used rather than maximum because the feasible set may not contain an optimizer (e.g. if description costs are real-valued and the fidelity approaches a limit). Concretely, if $\mathrm{Fid}_G(x, \mathrm{dec}_O(d_n)) \uparrow L$ along a sequence of descriptions (d_n) with $\sigma_O(d_n) \le K$, it can happen that no single description attains L even though L is the best achievable value in the limit; the optimization problem is then naturally expressed using* sup. *One can also view σ_O as generating nested feasible sets*

$$\mathcal{F}_O(K) \;:=\; \{d \in D_O : \sigma_O(d) \le K\},$$

so that the quantity of interest is the value function $K \mapsto \sup_{d\in\mathcal{F}_O(K)} \mathrm{Fid}_G(x, \mathrm{dec}_O(d))$, which is automatically monotone under $\mathcal{F}_O(K) \subseteq \mathcal{F}_O(K')$ for $K \le K'$.

Sketch. The equivalence is just unrolling the $\exists d$ in the definition of effability into a supremum comparison. Monotonicity in K holds because the feasible-description set grows with K. Monotonicity in G follows from pointwise $\iota_G \le \iota_{G'}$ implying $\mathrm{Fid}_{G'}(x, q) \ge \mathrm{Fid}_G(x, q)$ for every q. More explicitly, the "$\exists d$" form says: there exists some d with $\sigma_O(d) \le K$ such that $\mathrm{Fid}_G(x, \mathrm{dec}_O(d)) \ge 1 - \varepsilon$. This holds if and only if the supremum of $\mathrm{Fid}_G(x, \mathrm{dec}_O(d))$ over all d with $\sigma_O(d) \le K$ is at least $1 - \varepsilon$ (because a supremum is $\ge 1 - \varepsilon$ exactly when some feasible point attains $\ge 1 - \varepsilon$, or at least can be approximated arbitrarily well from below when the target threshold is strict). For monotonicity in K, if $K \le K'$ then $\{d : \sigma_O(d) \le K\} \subseteq \{d : \sigma_O(d) \le K'\}$, hence

$$\sup_{\sigma_O(d)\le K} \mathrm{Fid}_G(x, \mathrm{dec}_O(d)) \;\le\; \sup_{\sigma_O(d)\le K'} \mathrm{Fid}_G(x, \mathrm{dec}_O(d)).$$

For monotonicity in G, the condition $\iota_G \le \iota_{G'}$ is a pointwise comparison of the scoring kernels, so the induced fidelity functional (which is linear in the reconstruction distribution q) increases pointwise: evaluating the same q under a more permissive G' cannot lower the score. □

Remark 1797. Proof sketch. *Translate "there exists a feasible description with fidelity $\ge 1 - \varepsilon$" into "the supremum of fidelities over feasible descriptions is $\ge 1 - \varepsilon$", and then note that enlarging the feasible set (via K) or enlarging the fidelity kernel (via G') cannot reduce a supremum.* □

The key steps are: (i) supremum-as-relaxed-existential, (ii) monotonicity of supremum under set inclusion, and (iii) pointwise domination of kernels implying domination of the resulting linear functionals in Lemma 96. A helpful picture is that of a Pareto frontier: $\mathrm{Fid}^\star(x; K)$ *is the upper envelope of achievable fidelity as a function of resources. One can make step (iii) concrete by writing fidelity in the typical bilinear form*

$$\mathrm{Fid}_G(x, q) \;=\; \sum_y \iota_G(x, y)\, q(y)$$

(or the corresponding integral), so that $\iota_G \le \iota_{G'}$ implies $\sum_y \iota_G(x, y)q(y) \le \sum_y \iota_{G'}(x, y)q(y)$ for every distribution q, since $q(y) \ge 0$ and $\sum_y q(y) = 1$. In this view, coarse-graining is literally an upward shift of the payoff vector $\iota_G(x, \cdot)$, and the decoder is choosing a distribution q inside

a constraint set determined by descriptions; the supremum simply picks the best payoff achievable under those constraints. The Pareto-frontier intuition can also be read as a tradeoff curve between description cost and semantic accuracy: allowing larger K moves one along the frontier, whereas replacing G by a coarser G' changes the scoring rule so that the same point in description space is evaluated more leniently.

Remark 1798. *It is sometimes useful to restate the same content as a value function and threshold test: define* $\mathrm{Fid}^{\star}_{O,G}(x;K) := \sup\{\mathrm{Fid}_G(x, \mathrm{dec}_O(d)) : d \in D_O,\ \sigma_O(d) \le K\}$*. Then "$x$ is (O, K, G, ε)-effable" is equivalent to* $\mathrm{Fid}^{\star}_{O,G}(x;K) \ge 1 - \varepsilon$*. This makes clear that ε is merely a reporting threshold on the monotone function $K \mapsto \mathrm{Fid}^{\star}_{O,G}(x;K)$, and that different choices of G correspond to different monotone families $K \mapsto \mathrm{Fid}^{\star}_{O,G}(x;K)$ ordered by kernel domination.*

32.6 Potential infinity and infinitesimals: algebra of germs

Lemma 99 (Well-defined ring structure on germs)**.** *Let $\mathbb{R}^{\mathbb{N}}/\sim$ be germs under eventual equality. If $a \sim a'$ and $b \sim b'$, then $a + b \sim a' + b'$ and $ab \sim a'b'$. Hence addition and multiplication on equivalence classes are well-defined.*

Remark 1799. *This subsection formalizes a mild version of "potential infinity" (Hyperseed-Concept* **??***): we reason about sequences, but we identify two sequences if they agree eventually, i.e. if their differences occur only at finitely many indices. The resulting objects are* germs*, capturing the idea that only the tail behavior matters. This is a pragmatic mathematical way to speak about infinite processes while ignoring finite initial transients—a move common in asymptotics, dynamics, and learning theory. In particular, the quotient $\mathbb{R}^{\mathbb{N}}/\sim$ can be read as a formalization of the slogan "only what happens after some finite time matters," with the understanding that the relevant "finite time" is allowed to depend on the statement being proved. This is exactly the pattern used when one writes $a_n = b_n$ "for n large enough," or when one argues that a learning algorithm eventually enters a stable regime and the warm-up phase can be ignored for asymptotic comparisons.*

Remark 1800. *The lemma is necessary because we want to do algebra on germs. If we define $[a]+[b] := [a+b]$, this must not depend on the chosen representatives. The statement is the standard compatibility condition that turns an equivalence relation into a quotient algebra. Equivalently, one may say that "eventual equality" is a congruence relation on the ring $\mathbb{R}^{\mathbb{N}}$ (with pointwise operations): it is not merely an equivalence relation on the underlying set, but one that is respected by addition and multiplication. This congruence viewpoint is what guarantees that $\mathbb{R}^{\mathbb{N}}/\sim$ inherits the structure of a commutative ring in a canonical way.*

Remark 1801. *Before the proof, the symbols mean: $\mathbb{R}^{\mathbb{N}}$ is the set of real sequences $a = (a_1, a_2, \dots)$, and $a \sim a'$ means $\exists N$ such that $a_n = a'_n$ for all $n \ge N$ ("eventual equality"). The equivalence class of a is written $[a]$. It is helpful to keep in mind that $\sim$ ignores a* finite *prefix of the sequence: two representatives may differ arbitrarily on indices $n < N$ without changing the germ. In particular, any statement about a germ that is well-posed should be invariant under such finite modifications. This is why proofs in this setting often take the form "choose a horizon N large enough so that all the desired tail properties hold simultaneously."*

Sketch. If $a_n = a'_n$ and $b_n = b'_n$ for all $n \ge N$, then $(a_n + b_n) = (a'_n + b'_n)$ and $(a_n b_n) = (a'_n b'_n)$ for all $n \ge N$ as well. □

Remark 1802. Proof sketch. *Take the index N after which both equalities hold, and check that addition/multiplication preserve pointwise equality for all $n \ge N$.* □

The key step is that ring operations are defined pointwise on sequences, so any property that holds for all sufficiently large n is preserved by these operations. One may visualize germs as "functions near infinity": equal near infinity implies sums and products are equal near infinity. A further useful intuition is that the set of "eventually zero" sequences forms an ideal in $\mathbb{R}^{\mathbb{N}}$, and $\mathbb{R}^{\mathbb{N}}/\sim$ is exactly the quotient by this ideal: two sequences define the same germ precisely when their difference is eventually 0. This makes the well-definedness checks feel analogous to the familiar construction of $\mathbb{Z}/m\mathbb{Z}$, except that the ideal is determined by tail-vanishing rather than divisibility.

Lemma 100 (Eventual order calculus)**.** *Let $\leq_\infty$ be eventual order on germs. If $[a] \leq_\infty [b]$ and $[c] \leq_\infty [d]$, then $[a] + [c] \leq_\infty [b] + [d]$. If also $[0] \leq_\infty [c]$, then $[a][c] \leq_\infty [b][c]$.*

Remark 1803. *Eventual order $\leq_\infty$ is the order-theoretic analogue of eventual equality: it compares sequences by what happens beyond some finite horizon. This is the natural order for asymptotic analysis and for speaking about infinitesimals as "eventually smaller than every positive constant" (Hyperseed-Concept* **??***). The lemma states that this asymptotic order behaves as one expects under addition, and under multiplication by an eventually nonnegative germ. In applications, this lets one manipulate tail bounds exactly as one manipulates ordinary inequalities, as long as one tracks the (usually irrelevant) fact that the inequalities may fail at finitely many early indices. For instance, if one has eventual error bounds for two learning subroutines, the lemma formalizes that their combined error bound is also valid eventually, without needing to synchronize the two subroutines at a single fixed start time in advance.*

Remark 1804. *Before the proof, recall that $[a] \leq_\infty [b]$ means $\exists N$ such that $a_n \leq b_n$ for all $n \geq N$. The condition $[0] \leq_\infty [c]$ means $c_n \geq 0$ eventually, which is exactly what is needed to preserve inequalities under multiplication. One subtlety worth keeping in view is that $\leq_\infty$ is naturally a preorder on representatives: different germs can be "eventually mutually $\leq$" without being literally equal as sequences, but on the quotient it behaves in the intended asymptotic sense. Concretely, the reason multiplication requires $c_n \geq 0$ eventually is the same as in ordinary algebra: without a sign restriction, the direction of an inequality can flip at indices where $c_n < 0$, and an "eventual" statement must rule out infinitely many such flips.*

Sketch. Unroll the definition: each inequality holds for all $n \geq N$ for some N, then combine pointwise for $n \geq \max N$. For multiplication, preserve order by multiplying by a nonnegative term. $\square$

Remark 1805. Proof sketch. *Choose horizons N_1, N_2 for the two eventual inequalities, then work beyond $\max(N_1, N_2)$ where both hold simultaneously. For the product inequality, also work beyond a horizon where $c_n \geq 0$.* $\square$

The proof works because eventual properties are stable under finite conjunction: "eventually P" and "eventually Q" implies "eventually (P and Q)". The geometric intuition is that we are doing order theory not on points but on tails of sequences. A useful operational takeaway is that one should freely introduce a single "master horizon" N_* that is larger than all the finitely many horizons appearing in the argument. After passing to $n \geq N_*$, all the needed inequalities become ordinary pointwise inequalities, and the familiar algebraic manipulations apply without further changes.

Lemma 101 ($\omega \cdot \varepsilon = 1$ and "infinite times infinitesimal" normalization)**.** *Let $\omega = [n]$ and $\varepsilon = [1/n]$. Then $\omega \cdot \varepsilon = [1]$ (the constant germ 1).*

Remark 1806. *This lemma gives a precise meaning to an often poetic slogan: "an infinite quantity times an infinitesimal can be finite." Here $\omega = [n]$ is a germ that grows without bound, and*

*$\varepsilon = [1/n]$ is a germ that tends to 0. Their product is nevertheless the constant germ [1]. While this is not full nonstandard analysis, it captures a workable fragment of the intuition needed for manipulating asymptotic scales in a rigorous quotient structure (Hyperseed-Concept **??**, Hyperseed-Concept **??**). In the language of asymptotics, ω and ε are reciprocal scales: multiplying by ε renormalizes quantities measured in "units of n" back to a stable, n-independent unit. This kind of normalization underlies many "dimensionless" reparameterizations, where a diverging or vanishing factor is absorbed into a change of scale so that the remaining expression has a finite, robust limit behavior.*

Remark 1807. *Before the proof, note that* [1] *denotes the equivalence class of the constant sequence* $(1, 1, 1, \ldots)$*. The statement* $\omega \cdot \varepsilon = [1]$ *means that the sequence* $(n) \cdot (1/n)$ *is eventually equal to the constant* 1*, i.e. equal for all sufficiently large indices. Also note that the representative* $(1/n)$ *may be treated as starting at* $n = 1$ *(or any other fixed positive index) to avoid the irrelevant issue of division by zero at* $n = 0$ *if one were to index sequences that way. Because germs ignore finitely many initial indices, any such convention changes at most a finite prefix and therefore does not affect the germ.*

Sketch. Pointwise, $(n) \cdot (1/n) = 1$ for all $n \geq 1$, hence the product sequence is eventually constant 1. □

Remark 1808. Proof sketch. *Compute the product pointwise and observe it equals* 1 *from the first index onward, hence certainly eventually.* □

The key step is that eventual equality ignores finitely many initial indices, so any identity that holds for all $n \geq 1$ (or all $n \geq N$) becomes an equality of germs. One may visualize ω and ε as two reciprocal asymptotic scales whose interaction renormalizes to a stable unit. In particular, this is a canonical example of how the quotient construction turns a "limit-style" intuition (growth versus decay) into an exact algebraic identity inside the germ ring: after passing to equivalence classes, the phrase "for all sufficiently large n" is promoted to literal equality of objects.

33 Helper Theorems for Section 11: Hierarchy, Heterarchy, and Pattern Webs

33.1 Standing notation

Fix a commutative quantale $(V, \leq, \oplus, \otimes, e)$. A (thin) V-hierarchy on a set U is a map $h : U \times U \to V$ satisfying

$$e \leq h(u, u), \qquad h(u, v) \otimes h(v, w) \leq h(u, w).$$

A V-graph is a weight function $g : U \times U \to V$ (not necessarily reflexive/transitive). For a finite directed path $\pi = (u_0 \to u_1 \to \cdots \to u_n)$, define the path weight

$$w(\pi) := g(u_0, u_1) \otimes g(u_1, u_2) \otimes \cdots \otimes g(u_{n-1}, u_n).$$

Define the path-closure $\widehat{g} : U \times U \to V$ by

$$\widehat{g}(u, v) := \bigoplus_{\pi : u \rightsquigarrow v} w(\pi),$$

where the join ranges over all finite directed paths from u to v, including the empty path (with weight e).

Remark 1809. *The quantale $(V, \leq, \oplus, \otimes, e)$ should be read as a unified "value domain" for graded entailment, graded reachability, or graded evidential support. Concretely: $\leq$ is the order of informativeness/strength; $\oplus$ is the (possibly infinite) join or "aggregation over alternatives"; $\otimes$ is the monoidal product that composes strength along a chain; and e is the unit strength. This is the standard setting in which many notions of hierarchy and connectivity become instances of a single enriched-order idea; in the Hyperseed core-concept set, this is closely allied to Quantale Weakness (Hyperseed-Concept 143) and Weakness (Hyperseed-Concept 202), as developed in [3].*

Remark 1810. *For later use, it is helpful to keep in mind that "quantale" here includes the completeness needed to interpret $\bigoplus$ over* arbitrary *(possibly infinite) families of path-weights. In particular, the join in the definition of $\widehat{g}(u, v)$ may range over infinitely many distinct finite paths when U is infinite or when cycles are present. The commutativity assumption on $\otimes$ is not strictly needed to* define *$w(\pi)$, but it ensures that many algebraic manipulations behave symmetrically and that the underlying enrichment has fewer left/right distinctions; this is consistent with the interpretation of $\otimes$ as a generic "combination of evidential support" rather than a direction-sensitive update.*

Remark 1811. *A "thin" V-hierarchy h is best viewed as an enriched preorder: the inequality $e \leq h(u, u)$ is reflexivity (each entity is at least as-related to itself as the unit allows), while*

$$h(u, v) \otimes h(v, w) \leq h(u, w)$$

is transitivity in graded form (composing the support from u to v and from v to w cannot exceed the direct support from u to w). If one specializes to $V = \{0, 1\}$ with $\oplus = \vee$, $\otimes = \wedge$, and $e = 1$, then $h(u, v) = 1$ simply means "u is below v" and the axioms reduce to the ordinary preorder axioms. If one specializes to $V = [0, 1]$ with $\otimes$ as multiplication and $\oplus$ as $\max$, then $h(u, v)$ can be read as the strength of a hierarchical entailment, and transitivity says that the strength along a two-step chain lower-bounds (in the order $\leq$) the direct strength. This language is used to speak about Hierarchy (Hyperseed-Concept **??***) and also its complementing phenomenon Heterarchy (Hyperseed-Concept* **??***) in the pattern-theoretic setting of [5].*

Remark 1812. *The order convention is worth keeping explicit: the inequalities are written with $\leq$ as the "increasing strength/informativeness" direction. In some applied settings one instead orders by "cost" or "distance" (so that smaller numbers are stronger), in which case the quantale is typically presented with the opposite order and with $\oplus$ behaving like an infimum (e.g. $\min$) rather than a supremum. The present notation fixes one consistent convention; any alternative convention can be accommodated by passing to the opposite order (or, equivalently, by re-reading $\leq$ as the intended strength comparison).*

Remark 1813. *A V-graph g drops the preorder axioms: it is merely a weighted directed graph. The path weight $w(\pi)$ multiplies (via $\otimes$) the edge weights along the path; then the closure $\widehat{g}(u, v)$ aggregates (via $\oplus$) over* all *finite paths from u to v. Including the empty path with weight e ensures that $\widehat{g}(u, u) \geq e$, i.e. reflexivity is built into the closure even if g itself has no self-loops. Informally, $\widehat{g}$ is "what g implies once one allows multi-step inference." In pattern-web language (Hyperseed-Concept 132) this is the formal analogue of saying: even if we only record immediate pattern-to-pattern links, we may still care about the total indirect influence obtainable by chaining links, a viewpoint emphasized in [5].*

Remark 1814. *The notation $\pi : u \rightsquigarrow v$ means that π is a directed path whose source is u and whose target is v. The empty path is the unique path of length* 0 *from u to itself, and its weight is* *declared to be e so that $w(\pi)$ behaves as expected with respect to concatenation: concatenating any path with an empty path does not change its weight. With this convention, $\widehat{g}(u,v)$ can be read as the total strength of* all *chains of immediate links from u to v, where "total" is taken in the $\oplus$-sense appropriate to the value domain (e.g. maximum strength, logical disjunction, or aggregation of alternative evidential routes).*

Remark 1815. *Although not yet used as a theorem here, it is useful intuition that $\widehat{g}$ is designed to be the canonical "reflexive–transitive closure" of a weighted digraph in the enriched sense: it is built to account for multi-step composition (via $\otimes$) and alternative routes (via $\oplus$). In many common quantales, this closure is also the least (thin) V-hierarchy extending g (i.e. the least reflexive/transitive weight function above g pointwise), mirroring the classical fact that the reachability relation is the least preorder containing an edge relation.*

33.2 Hierarchies and systemic hierarchies

Theorem 34 (Chain lower bound in a graded hierarchy)**.** *Let h be a V-hierarchy on U. For any finite chain $u_0, u_1, \dots, u_n \in U$,*

$$h(u_0,u_n) \;\geq\; h(u_0,u_1) \otimes h(u_1,u_2) \otimes \cdots \otimes h(u_{n-1},u_n).$$

Remark 1816. *This theorem says that in a graded hierarchy,* every explicit chain of intermediate steps provides a certified lower bound on the direct relation*. Philosophically, it is the formal expression of a very ordinary idea: if u_0 supports u_1 and u_1 supports u_2 and so on, then u_0 supports the end-point u_n at least as strongly as the composed support along the chain. The point is not that the chain is the only route; rather, the hierarchy axioms guarantee that chaining never fabricates illegitimate strength.*

Remark 1817. *A useful way to read the inequality is:* the composite along the chain is a lower estimate that is justified solely by transitivity*. In particular, if V is thought of as a space of "evidence values" or "confidence values," then the right-hand side is the confidence obtainable by* only *traversing the displayed edges. The theorem asserts that the direct assessment $h(u_0,u_n)$ must be at least that large, since otherwise the transitivity axiom would be violated at some stage of composition.*

Remark 1818. *In common special cases, the statement reduces to familiar laws:*

- *If $V = \{\perp, \top\}$ with $\otimes = \wedge$ (Boolean preorder case), then the theorem becomes: if $u_0 \leq u_1 \leq \cdots \leq u_n$ then $u_0 \leq u_n$.*
- *If $V = [0,1]$ with $\otimes = \cdot$ and the usual order, then the chain bound reads $h(u_0,u_n) \geq \prod_{i=0}^{n-1} h(u_i,u_{i+1})$, i.e. strengths multiply along an explicit path.*
- *If V is a tropical-type quantale where $\otimes$ is $+$ and the order is reversed (typical for "cost" semantics), then the displayed inequality corresponds to the usual triangle-inequality form: the direct cost cannot exceed the cost of a multi-step route.*

These are not new assumptions; they are just interpretations of the same abstract inequality in different enrichment semantics.

Remark 1819. *The relevance to Section 11 is that many constructions of hierarchy/heterarchy begin with local edges ("immediate influences") and then reason about multi-step consequences. Theorem 34 is the atomic rewrite rule enabling this multi-step reasoning inside an already-transitive structure, complementing the later closure results where transitivity is* added *freely.*

Remark 1820. *One small technical point worth making explicit is that the right-hand side is unambiguous because $\otimes$ is associative (as part of the standard quantale/monoidal hypotheses underlying a V-hierarchy). Thus one may parenthesize the product as*

$$\big(\cdots((h(u_0,u_1)\otimes h(u_1,u_2))\otimes h(u_2,u_3))\otimes\cdots\big)\otimes h(u_{n-1},u_n),$$

and the resulting element of V is independent of the parenthesization. The proof by induction effectively chooses one particular parenthesization (left-associated) and then applies transitivity repeatedly.

Sketch. Induct on n. The case $n=1$ is trivial. For the step, apply the hierarchy inequality $h(u_0,u_{n-1})\otimes h(u_{n-1},u_n)\le h(u_0,u_n)$ and use the induction hypothesis. □

Remark 1821. *If one wishes to see the induction step with the inequalities written out, it is:*

$$h(u_0,u_1)\otimes\cdots\otimes h(u_{n-2},u_{n-1}) \;\le\; h(u_0,u_{n-1})$$

by the induction hypothesis, hence by monotonicity of $\otimes$ in its first argument,

$$\big(h(u_0,u_1)\otimes\cdots\otimes h(u_{n-2},u_{n-1})\big)\otimes h(u_{n-1},u_n) \;\le\; h(u_0,u_{n-1})\otimes h(u_{n-1},u_n).$$

Applying transitivity to the right-hand side yields the desired bound by $h(u_0,u_n)$. (Here the monotonicity of $\otimes$ is part of the ambient ordered-monoid structure on V typically assumed when speaking of graded relations.)

Remark 1822. Proof sketch (high-level strategy). *One proves the statement for a chain of length 1, and then extends it by appending one more edge and using transitivity once. This is the same logical form as proving that an ordinary preorder respects finite composition of $\le$.*

Remark 1823. Why the key step works. *The inequality*

$$h(u_0,u_{n-1})\otimes h(u_{n-1},u_n)\le h(u_0,u_n)$$

is precisely the transitivity axiom of a V-hierarchy; it is invoked after the induction hypothesis has packaged the first $n-1$ edges into a single composite bound on $h(u_0,u_{n-1})$. Geometrically (in the common max-product example), one may picture a chain as multiplying edge strengths; the theorem asserts that the direct relation cannot be weaker than the chain-product when measured in the order of V.

Remark 1824. *It is also helpful to note what the theorem does* not *assert. It does not claim that $h(u_0,u_n)$ is* equal *to the best chain value, nor that the displayed chain is optimal among all chains; it only asserts that any specific chain yields a valid lower bound. In applications, one often considers many possible chains and takes their join/supremum (when V admits it) to obtain the strongest bound derivable from multi-step evidence, but that further "closure" operation is conceptually separate from the basic transitivity reasoning encapsulated here.*

Theorem 35 (Pullback of hierarchy along a model map)**.** *Let $\phi : A\to S$ be any function and let $h_S : S\times S\to V$ be a V-hierarchy. Define a relation on A by*

$$h_A(a,b) := h_S(\phi(a),\phi(b)).$$

Then h_A is a V-hierarchy on A.

Remark 1825. *This theorem says that a hierarchy on a space of "models" or "summaries" can be transported back to the underlying things being modeled. If ϕ extracts some representation of each $a \in A$ inside S, then h_A declares a to be below b exactly to the extent that $\phi(a)$ is below $\phi(b)$. In plainer language:* hierarchy is stable under re-description. *This is important because systemic hierarchies often live at a representational level (types, roles, abstractions) while we must reason about concrete processes that realize them.*

Remark 1826. *A practical interpretation is that S may be a quotient, coarse-graining, feature map, or embedding space in which comparisons are easier or more meaningful. The pullback h_A then inherits any "ordering logic" already validated on S without requiring one to directly define transitivity on A from scratch. In particular, if ϕ forgets details (many a mapping to one $\phi(a)$), then h_A may identify distinct objects as having identical hierarchical profiles; this is expected and corresponds to the modeling choice encoded by ϕ.*

Remark 1827. *In the idiom of Section 11, ϕ is a modeling map that turns processes into nodes of a systemic diagram; the theorem guarantees that any graded hierarchical structure imposed on the systemic level induces a graded structure on the process level. This is conceptually aligned with the broader pattern-and-system viewpoint of [5] and with the technical use of model maps in dependency-tower analyses [8].*

Sketch. Reflexivity: $e \leq h_S(\phi(a), \phi(a)) = h_A(a, a)$. Transitivity: $h_A(a, b) \otimes h_A(b, c) = h_S(\phi(a), \phi(b)) \otimes h_S(\phi(b), \phi(c)) \leq h_S(\phi(a), \phi(c)) = h_A(a, c)$. □

Remark 1828. *The proof is "one-line" because h_A is defined by direct substitution into h_S. No additional structure on ϕ is required: it need not be injective, surjective, order-preserving, or compatible with $\otimes$ in any special way, since h_A is not defined by transporting $\otimes$ across ϕ but rather by evaluating h_S after applying ϕ to each argument. This is exactly the sense in which V-hierarchy axioms are stable under re-indexing of the underlying set.*

Remark 1829. Proof sketch (high-level strategy). *One checks that the defining axioms of a V-hierarchy are purely equational/inequational, hence preserved by substitution. The function ϕ performs the substitution: wherever h_A appears, it is merely h_S evaluated at images under ϕ.*

Remark 1830. Key step and intuition. *The transitivity step works because $\phi(b)$ is literally the same object in the middle term of both factors $h_S(\phi(a), \phi(b))$ and $h_S(\phi(b), \phi(c))$, so the transitivity axiom of h_S applies directly. Visually, one can imagine a commuting triangle: edges in A are "shadowed" by edges in S, and the inequality in S pulls back along that shadowing.*

Remark 1831. *There is also a useful "functorial" intuition: if $A \xrightarrow{\phi} S \xrightarrow{\psi} T$ are functions and h_T is a V-hierarchy on T, then pulling back along $\psi \circ \phi$ produces the same V-hierarchy on A as pulling back along ψ to get a hierarchy on S and then pulling back along ϕ. This is immediate from the definition*

$$h_A(a, b) = h_T(\psi(\phi(a)), \psi(\phi(b))),$$

and it clarifies why pullback is the correct direction for transporting hierarchical structure along modeling maps (the same direction as inverse images of predicates).

Corollary 10 (Model-induced (systemic) hierarchy on processes)**.** *In the setting of a systemic hierarchy $(S, \prec_S, \phi)$, any graded preorder structure on S pulls back along ϕ to a graded preorder on* $\mathrm{Proc}(D)$*.*

Remark 1832. *The corollary is the application one actually uses: once one has chosen a systemic level S (often a compressed or conceptual representation of a domain), any hierarchical assessment made at that level immediately yields a hierarchical assessment on the processes in* $\mathrm{Proc}(D)$ *that are mapped into it. This makes precise how a theory of hierarchy can be simultaneously abstract (living on S) and operational (speaking about concrete process instances).*

Remark 1833. *Concretely, if $\phi : \mathrm{Proc}(D) \to S$ assigns to each process a type, role, macro-state, or pattern-label, then any graded ordering on those labels (e.g. "is a refinement of," "is more general than," "is more causally upstream than," depending on the intended semantics) becomes a derived graded ordering on the processes themselves by comparing their assigned labels. This is the formal step that lets one reason about process-level organization using system-level abstractions without conflating the two levels.*

Sketch. Apply Theorem 35 with $A = \mathrm{Proc}(D)$. $\square$

Remark 1834. Proof sketch. *There is no additional mathematical content beyond Theorem 35; the corollary is an instantiation. The conceptual content, however, is significant: it formalizes that "systemic" ordering principles become genuine order-like constraints on the dynamics that instantiate the system.*

33.3 Subpattern hierarchy

Theorem 36 (Subpattern preorder (crisp))**.** *Assume $*$ is associative with identity e, and assume $Pat(p * e; p)$ is nontrivial for each pattern-representer p (so $p * e$ is defined and recognized as a pattern in p). Define $p \preceq_{\mathrm{sub}} q$ iff there exists r such that $Pat(p * r; q)$ is nontrivial. Then $\preceq_{\mathrm{sub}}$ is a preorder.*

Remark 1835. *This theorem constructs a preorder out of a compositional pattern notion: $p \preceq_{\mathrm{sub}} q$ means, informally, that p can be embedded into q by adjoining some remainder r and then recognizing $p * r$ as (nontrivially) present in q. The requirement that $Pat(p * e; p)$ be nontrivial ensures reflexivity: every pattern contains itself, up to composing with a neutral "do nothing" remainder. This is the crisp (two-valued) backbone of what later becomes graded subpattern strength.*

Remark 1836. *Two simple mental examples help. First, if $*$ is string concatenation and $Pat(a; b)$ means "a occurs as a substring of b," then $p \preceq_{\mathrm{sub}} q$ says: there exists a string r such that pr occurs in q; for many reasonable Pat this recovers familiar containment-like notions. Second, if $*$ is multiset union and $Pat(A; B)$ means "A is a submultiset of B," then taking $r = B \setminus A$ witnesses $A \preceq_{\mathrm{sub}} B$. In Section 11 terms, this gives a formal entry point for Subpattern Hierarchy (Hyperseed-Concept 182) within Pattern (Hyperseed-Concept 130) and Pattern Web (Hyperseed-Concept 132) structures, as in [5].*

Remark 1837. *The importance of the result is not that it is difficult, but that it legitimizes a common move: to treat "is a subpattern of" as a genuine hierarchy relation (reflexive and transitive) rather than merely a heuristic similarity. Once it is a preorder, one can quotient by mutual subpatterning (to obtain a partial order) or enrich it (as in Theorem 37).*

Sketch. Reflexivity: choose $r = e$. Transitivity: if $Pat(p * r_1; q)$ and $Pat(q * r_2; s)$ are nontrivial, associativity gives $(p*r_1)*r_2 = p*(r_1*r_2)$; under the intended compositional semantics, this yields nontrivial $Pat(p * (r_1 * r_2); s)$. $\square$

Remark 1838. Proof sketch (high-level strategy). *Reflexivity is forced by the existence of an identity element for composition. Transitivity is forced by associativity: two successive embeddings can be merged into one by bundling the two remainders into a single remainder* $r_1 * r_2$*.*

Remark 1839. Key step and intuition. *The crucial move is rewriting*

$$(p * r_1) * r_2 = p * (r_1 * r_2),$$

which is exactly associativity. Conceptually, this is the idea that "glue" terms can be composed: if p *needs some context* r_1 *to live inside* q*, and* q *needs some context* r_2 *to live inside* s*, then* p *needs the combined context* $r_1 * r_2$ *to live inside* s*. One may picture this as nesting of contexts that can be flattened.*

Theorem 37 (Subpattern strength composes under a compositionality axiom)**.** *Let* $h_{\text{sub}}(p, q) := \sup_r Pat(p * r; q) \in [0, 1]$*. Assume there exists a t-norm* $\odot$ *on* $[0, 1]$ *such that for all* a, b, c*,*

$$Pat(a; q) \odot Pat(q; c) \ \leq \ Pat(a * c; s)$$

whenever the compositions are well-typed (this is a generic "patterns compose" axiom). Then for all p, q, s*,*

$$h_{\text{sub}}(p, q) \odot h_{\text{sub}}(q, s) \ \leq \ h_{\text{sub}}(p, s).$$

Remark 1840. *This theorem is the graded analogue of transitivity for subpatterning. The function* $h_{\text{sub}}(p, q)$ *takes the* best *(supremal) way to witness* p *inside* q *by allowing an arbitrary remainder* r*. The conclusion says: if* p *is strongly a subpattern of* q*, and* q *is strongly a subpattern of* s*, then* p *is a subpattern of* s *with at least the composed strength, where composition of strengths is mediated by a t-norm* $\odot$ *(a standard way to model "and" in* $[0, 1]$*).*

Remark 1841. *Why this matters in Section 11 is that it provides a principled route from* pattern recognition scores *to* hierarchical structure*: one can define a* $[0, 1]$*-valued hierarchy whose transitivity is guaranteed not by ad hoc thresholding, but by a compositional axiom about how pattern evidence behaves under composition. This directly supports building Pattern Web structures (Hyperseed-Concept 132) with meaningful multi-hop semantics, as advocated in [5].*

Sketch. Expand both h_{sub} as sups over witnesses r_1, r_2; apply the axiom to $Pat(p * r_1; q)$ and $Pat(q * r_2; s)$; use associativity to rewrite $(p * r_1) * r_2 = p * (r_1 * r_2)$; then take sups. □

Remark 1842. Proof sketch (high-level strategy). *One proves the inequality pointwise for arbitrary witnesses* r_1, r_2*, and then upgrades it to the supremal strengths by using the monotonicity of* sup *and of* $\odot$ *in each argument.*

Remark 1843. Key steps and intuition. *The compositionality axiom is the substantive ingredient: it asserts that evidence for* a *in* q *and evidence for* q *in* c *can be combined (via* $\odot$*) into evidence for the composed object. Associativity then ensures that the two-step remainder can be treated as a single remainder, so the final witness is of the allowed form for* $h_{\text{sub}}(p, s)$*. Visually, one may imagine two pattern-matches that share a middle interface* q*; the theorem says that the interface can be eliminated, leaving a direct match with a predictable weakening governed by* $\odot$*. In the language of "pattern webs," this is exactly the move of* collapsing *an intermediate node: whenever two links* $a \rightsquigarrow q$ *and* $q \rightsquigarrow c$ *are supported, the network can be summarized by a derived link* $a \rightsquigarrow c$ *together with an explicit, algebraically controlled loss term. This kind of step is what later allows us to discuss higher-level organization (hierarchical or heterarchical) without needing to keep every intermediate interface visible in the representation.*

33.4 Heterarchy and cycles

Theorem 38 (Cycle mass bounds and monotonicity in the max-product case)**.** *Let $V = [0,1]$ with $\oplus = \max$ and $\otimes = \cdot$. For a weighted digraph $G = (U, E, w)$, the cycle mass*

$$\mathrm{Cyc}(G) := \max\Big\{\prod_{i=1}^{k} w(u_i, u_{i+1}) : (u_1, \ldots, u_k) \text{ is a directed cycle}\Big\}$$

satisfies $0 \le \mathrm{Cyc}(G) \le 1$. Moreover, if $w' \ge w$ pointwise (or edges are added with nonzero weights), then $\mathrm{Cyc}(G') \ge \mathrm{Cyc}(G)$.

Remark 1844. *To avoid any ambiguity in degenerate cases, it is convenient to adopt the convention that if G has no directed cycles (i.e. G is acyclic), then the set in the definition is empty and $\mathrm{Cyc}(G)$ is taken to be 0. This aligns with the intended interpretation that "no feedback" corresponds to the minimal heterarchy score. When U is finite, the maximum is actually attained by at least one cycle; more generally, one can read the definition as a supremum over all directed cycles, and the stated inequalities remain valid.*

Remark 1845. *The theorem formalizes a basic quantitative proxy for heterarchy: the presence and strength of directed cycles. In a strict hierarchy there are no cycles; in a heterarchy (Hyperseed-Concept **??**) cycles are not anomalies but structure. Here $\mathrm{Cyc}(G)$ measures the strongest cycle under multiplicative aggregation of edge strengths and* max *selection of the best cycle. The bound $0 \le \mathrm{Cyc}(G) \le 1$ is sanity: multiplying numbers in $[0,1]$ stays in $[0,1]$. One may also read $\prod_{i=1}^{k} w(u_i, u_{i+1})$ as a "reliability of returning to the start along that loop": if any link is weak, the whole loop is weakened, and if a link has weight 0 the loop contributes nothing.*

Remark 1846. *The monotonicity clause is conceptually important: strengthening edges, or adding new nonzero edges, cannot make the best available cycle weaker. This provides a clean mathematical underpinning for the intuition that additional feedback links increase (or at least do not decrease) the degree of heterarchical interdependence in a network, a theme that appears repeatedly in pattern-web discussions [5]. In particular, even if a newly added edge does not participate in the previously strongest cycle, it may create a* new *cycle (possibly shorter, or using higher-weight links) whose product exceeds the former maximum, which is exactly the sense in which additional connectivity can create qualitatively new feedback structure.*

Sketch. Each edge weight lies in $[0,1]$, so any cycle product lies in $[0,1]$, hence so does the max. If weights increase, every cycle product weakly increases, so the max increases. More explicitly: for any fixed directed cycle $(u_1, \ldots, u_k)$ we have

$$0 \le \prod_{i=1}^{k} w(u_i, u_{i+1}) \le \prod_{i=1}^{k} 1 = 1,$$

and if $w'(e) \ge w(e)$ for all edges e, then termwise

$$\prod_{i=1}^{k} w'(u_i, u_{i+1}) \;\ge\; \prod_{i=1}^{k} w(u_i, u_{i+1}).$$

Taking the maximum over all cycles gives the stated monotonicity; if new edges are added, the collection of cycles can only grow, so the maximum cannot decrease. $\square$

Remark 1847. Proof sketch. *The argument is pointwise: establish the interval bound for each cycle product, then pass to the maximum. For monotonicity, compare corresponding cycle products under w and w' and then again pass to the maximum. One can regard this as an instance of a general principle: for monotone aggregations (here, product in each cycle and maximum across cycles), increasing the underlying weights cannot decrease the aggregated value.*

Remark 1848. Visual intuition. *One can view each directed cycle as a "feedback loop" that returns influence to its origin. In max-product semantics, the strength of a loop is the product of its link reliabilities; the strongest loop is the one with the best balance of shortness and high edge weights. Increasing any edge reliability cannot degrade any loop that uses it, and cannot affect loops that do not use it, hence the best loop can only improve. Equivalently, applying the transformation $\ell(e) := -\log w(e)$ (with the usual convention that $w(e) = 0$ corresponds to $\ell(e) = +\infty$) turns products into sums:*

$$-\log\Big(\prod_{i=1}^{k} w(u_i, u_{i+1})\Big) = \sum_{i=1}^{k} \ell(u_i, u_{i+1}),$$

so "maximizing cycle product" corresponds to "minimizing cycle log-cost." Under this view, strengthening an edge decreases its cost, so the best (minimum-cost) cycle cannot become worse, which is another way to see the same monotonicity.

33.5 Pattern profiles, distances, and webs

Theorem 39 (Pattern-based pseudo-metric inherits metric axioms)**.** *Let $F_x \in [0,1]^P$ be the pattern profile of $x \in X$, and let D be a (pseudo-)metric on $[0,1]^P$. Define $d_P(x,y) := D(F_x, F_y)$. Then d_P is a (pseudo-)metric on X.*

Remark 1849. *This theorem is a simple but powerful lifting principle: if each entity x is represented by a feature vector F_x indexed by patterns (a "pattern profile"), then* any *(pseudo-)metric on the profile space induces a (pseudo-)metric on the original space. In the idiom of Pattern Webs (Hyperseed-Concept 132), it tells us that once patterns have become coordinates, geometry follows: similarity and neighborhood structure can be imported wholesale from the chosen distance D.*

Remark 1850. *One practical consequence is that many modeling choices are cleanly separated: the map $x \mapsto F_x$ encodes* what *is being measured (which patterns, which normalizations, which aggregation), while the choice of D encodes* how *difference is scored (e.g. sparse vs. dense sensitivity, robustness to outliers, emphasis on maximal deviations vs. cumulative deviations). In particular, the same pattern profile map can support multiple downstream geometries without changing the representation.*

Remark 1851. *Concrete examples of admissible D include L^1, L^2, and L^∞ distances on $[0,1]^P$ (when P is finite, or more generally when restricted to suitable subspaces), as well as weighted variants such as*

$$D_w(F,G) = \sum_{p \in P} w(p)\,|F(p) - G(p)| \quad \textit{with} \quad w(p) \geq 0,$$

which reweights coordinates to reflect pattern importance or reliability. The theorem applies verbatim as long as the chosen D satisfies the metric axioms on the profile space under consideration.

Remark 1852. *The result is also a reminder of a philosophical subtlety: d_P will typically be only a* pseudo-*metric because distinct entities can share the same profile (representation is many-to-one). This is not a defect but a statement about Representation (Hyperseed-Concept 157): the geometry reflects the representational lens, not necessarily the metaphysical identity of objects.*

Remark 1853. *A standard way to turn this pseudo-metric into an honest metric is to pass to the quotient by representational indistinguishability. Define an equivalence relation $x \sim y$ iff $F_x = F_y$ (equivalently, iff $d_P(x, y) = 0$). Then d_P descends to a well-defined metric on the quotient set $X/\sim$ by*

$$\bar{d}_P([x], [y]) := d_P(x, y),$$

because any alternative representatives have the same profile and hence the same distance. This construction makes explicit that the induced geometry is really the geometry of profiles, with X observed through the pattern coordinate map.

Sketch. Nonnegativity, symmetry, and the triangle inequality are inherited pointwise because $d_P(x, y)$ is defined by applying D to the pair (F_x, F_y). Identity-of-indiscernibles holds iff it holds for D and $F_x = F_y \to x = y$ (which need not be true), hence "pseudo" is the generally correct claim. □

Remark 1854. Proof sketch. *This is a direct transport of structure: each metric axiom is an inequality (or equality) about D evaluated on profile vectors, and d_P is just D applied to the corresponding profiles.*

Remark 1855. Geometric intuition. *Think of $x \mapsto F_x$ as embedding X into a high-dimensional cube $[0, 1]^P$. The distance d_P is simply the ambient distance after embedding. If the embedding collapses distinct points, then the induced geometry cannot separate them: hence pseudo-metric rather than metric.*

Remark 1856. *When P is large (or conceptually infinite), it is often useful to interpret F_x as a point in a function space: $F_x : P \to [0, 1]$. The same lifting principle applies whenever D is a (pseudo-)metric on the chosen function space (for example, an L^1-type metric with respect to a measure on P, or a supremum metric when appropriate). In this view, the pattern web construction can be read as importing familiar functional-analytic geometries into the space of entities via their pattern responses.*

Theorem 40 (Finite L^1 bounds). *If P is finite and $D(F, G) = \sum_{p \in P} |F(p) - G(p)|$, then*

$$0 \ \le\ d_P(x, y) \ \le\ |P|.$$

Remark 1857. *This theorem gives an immediate scale calibration for one of the most common choices of D, namely the L^1 distance on pattern profiles. It tells us that distances are automatically bounded by the number of pattern coordinates. This matters operationally: bounded distances are convenient for turning distances into similarities (as in Theorem 41) and for comparing distances across datasets when $|P|$ is fixed.*

Remark 1858. *A frequently used normalization is the* average L^1 *disagreement,*

$$\tilde{d}_P(x, y) := \frac{1}{|P|}\, d_P(x, y),$$

which always lies in $[0, 1]$ under the assumptions of Theorem 40. This normalized distance is often easier to interpret across different choices of P (e.g. when comparing a coarse pattern vocabulary to a refined one), because it factors out the raw number of coordinates and emphasizes per-pattern disagreement.

Sketch. Each summand is in $[0, 1]$ because $F(p), G(p) \in [0, 1]$. Sum over $|P|$ terms. □

Remark 1859. Proof sketch. *The inequality is a sum of coordinate-wise inequalities: each coordinate differs by at most* 1*, and there are* $|P|$ *coordinates.*

Remark 1860. Intuition. *In the* L^1 *geometry, distance is "total coordinate-wise disagreement." If each coordinate is a dial between* 0 *and* 1*, then the largest possible disagreement per dial is* 1*, so the largest total disagreement is the number of dials.*

Remark 1861. *The bound also highlights a modeling choice: if many patterns are nearly always* 0 *(sparse profiles), then* L^1 *distance tends to count the number and magnitude of coordinate changes and is often more sensitive to sparse differences than* L^2*. Conversely, if one wants to emphasize a few large discrepancies rather than many small ones, one may prefer* L^∞ *or a suitably weighted distance. Theorem 39 ensures that any such choice, once made on profile space, yields a valid pseudo-metric on* X*.*

Theorem 41 (Kernel similarity is order-reversing in distance)**.** *Let* $\kappa : [0, \infty) \to [0, 1]$ *be monotone decreasing and define* $s(x, y) := \kappa(d_P(x, y))$*. If* $d_P(x, y) \leq d_P(x', y')$*, then* $s(x, y) \geq s(x', y')$*. Also* $s(x, x) = \kappa(0)$ *(since* $d_P(x, x) = 0$ *for any pseudo-metric).*

Remark 1862. *This theorem states the obvious but essential compatibility between two common ways of talking: distances and similarities. A monotone decreasing function* κ *converts "small distance" into "large similarity." This is the formal justification for constructing edges in a pattern web by thresholding a similarity kernel rather than working directly with a distance.*

Remark 1863. *In applications,* $\kappa(t) = \exp(-\lambda t)$ *or* $\kappa(t) = 1/(1 + t)$ *are typical; then the theorem says that the induced similarity ordering respects the distance ordering. Such kernels are often used to convert geometric structure into graph structure, hence to move from Pattern Profiles to Pattern Webs (Hyperseed-Concept 132).*

Remark 1864. *If* κ *is* strictly *decreasing, then order is not merely reversed but also reflected:* $d_P(x, y) < d_P(x', y')$ *implies* $s(x, y) > s(x', y')$*, and* $s(x, y) = s(x', y')$ *forces* $d_P(x, y) = d_P(x', y')$*. This can matter when one ranks candidate neighbors: strict monotonicity ensures that sorting by similarity is exactly the same as sorting by distance, just in reverse.*

Remark 1865. *For pattern webs built by thresholding, one common scheme is to include an edge* $x \sim y$ *whenever* $s(x, y) \geq \tau$*, equivalently whenever* $d_P(x, y) \leq \kappa^{-1}(\tau)$ *when* κ *is invertible on its range. Thus, distance thresholds and similarity thresholds are interchangeable representations of the same neighborhood rule, and the theorem guarantees that both definitions produce consistent edge orderings.*

Sketch. Immediate from monotonicity of κ and the defining property $d_P(x, x) = 0$. $\square$

Remark 1866. Proof sketch. *Apply* κ *to both sides of an inequality and use that* κ *reverses order because it is decreasing; the diagonal value follows from* $d_P(x, x) = 0$*.*

Remark 1867. *Finally, it is worth separating two notions that are sometimes conflated in practice: (i) a* similarity score s *that is merely order-reversing in a distance (as here), and (ii) a* positive definite kernel *in the sense of kernel methods. Theorem 41 only uses monotonicity and therefore applies broadly; additional conditions are required if one wants* s *to define an inner-product feature map or to guarantee that kernel matrices are positive semidefinite.*

33.6 Free V-category closure and practical corollaries

Theorem 42 (Edge domination and monotonicity of path-closure)**.** *For any V-graph g on U and its closure $\widehat{g}$:*

1. *(Edge domination) $g(u,v) \leq \widehat{g}(u,v)$ for all u,v.*
2. *(Monotonicity) If $g \leq g'$ pointwise then $\widehat{g} \leq \widehat{g'}$ pointwise.*
3. *(Idempotence) $\widehat{(\widehat{g})} = \widehat{g}$.*

Remark 1868. *This theorem collects three structural facts about the closure $g \mapsto \widehat{g}$ that one expects of any sensible "add all indirect consequences" operator. (1) says no direct edge is forgotten. (2) says strengthening the base graph cannot weaken its closure. (3) says closing twice does nothing new: once all indirect paths have been accounted for, repeating the operation is redundant. These properties make $\widehat{g}$ a canonical bridge from local relations to global, multi-step relations.*

Remark 1869. *In the language of Section 11, one can begin with a heterarchical or partially hierarchical pattern web (a V-graph) and then pass to a derived structure in which every multi-hop inference is represented explicitly (a V-hierarchy/V-category). This is one clean mathematical way to relate local Pattern Web edges to emergent global Hierarchy and Heterarchy phenomena [5].*

Sketch. (1) The length-1 path $u \to v$ is included in the join defining $\widehat{g}(u,v)$. (2) Every path weight is monotone in each edge weight, and joins preserve order. (3) Once all path information is already included in $\widehat{g}$, re-closing under paths adds nothing. Formally, paths in $\widehat{g}$ correspond to concatenations of paths in g, which are already represented in the original closure. □

Remark 1870. Proof sketch (high-level strategy). *For (1) exhibit a specific path (the single edge). For (2) compare corresponding path weights under g and g' and then take joins. For (3) show that any path in $\widehat{g}$ is built from pieces that were already joins of g-paths, so nothing strictly new can appear.*

Remark 1871. Intuition. *The closure is like taking a graph and adding an edge for every path, with the edge weight computed as the aggregated strength of that path. Doing this twice is like re-adding edges that already exist. In max-product semantics, for instance, $\widehat{g}(u,v)$ is the strength of the best path from u to v; taking "best paths of best paths" cannot improve on already having taken best paths once.*

Theorem 43 (Minimality (universal property) of the free V-category)**.** *Let $h : U \times U \to V$ be any V-category (reflexive/transitive in the enriched sense) such that $g(u,v) \leq h(u,v)$ for all u,v. Then $\widehat{g}(u,v) \leq h(u,v)$ for all u,v. Equivalently, $\widehat{g}$ is the least V-category majorizing g.*

Remark 1872. *This theorem is the universal property that justifies calling $\widehat{g}$ the "free" V-category generated by the V-graph g. It says: among all enriched-preorder structures h that dominate the given edges, $\widehat{g}$ is the smallest one. In plain terms, $\widehat{g}$ adds exactly the transitive consequences forced by g, and nothing extra. This is the precise sense in which $\widehat{g}$ is a principled completion of a mere web into a hierarchy-like structure.*

Remark 1873. *Within Section 11, this minimality is what allows one to move between heterarchical webs and hierarchical summaries without smuggling in unintended assumptions: any additional strength beyond $\widehat{g}$ must come from extra axioms or extra data, not from the closure construction itself.*

Sketch. Fix a path $\pi = (u = u_0 \to \cdots \to u_n = v)$. Since $g(u_i, u_{i+1}) \le h(u_i, u_{i+1})$ and h is a V-category, repeated use of transitivity yields

$$w(\pi) = \bigotimes_i g(u_i, u_{i+1}) \;\le\; \bigotimes_i h(u_i, u_{i+1}) \;\le\; h(u, v).$$

Taking the join over all paths π gives $\widehat{g}(u, v) \le h(u, v)$. $\square$

Remark 1874. Proof sketch. *Prove the claim first for a single path: h dominates each edge, hence dominates the $\otimes$-product along the path by monotonicity and transitivity. Then take the join over all paths; since $h(u, v)$ is an upper bound for each path weight, it is also an upper bound for their join.*

Remark 1875. Why the steps work. *The inequality $g \le h$ is used locally on edges; the V-category transitivity of h is used globally to compress an n-step chain into a single comparison with $h(u, v)$. The structure mirrors the classical fact that the reflexive-transitive closure of a relation is the smallest preorder containing it; here everything is graded and the operations are $\otimes$ and $\oplus$ rather than $\wedge$ and $\vee$.*

Theorem 44 (Max-product closure equals shortest paths in log-space)**.** *Let $V = [0, 1]$ with $\oplus = \max$ and $\otimes = \cdot$, and let $g : U \times U \to [0, 1]$. Define edge costs $c(u, v) \in [0, \infty]$ by*

$$c(u, v) := \begin{cases} -\log g(u, v) & \text{if } g(u, v) > 0, \\ +\infty & \text{if } g(u, v) = 0. \end{cases}$$

Let $d^*(u, v)$ be the shortest-path cost from u to v under additive costs c (min over all paths of sum of edge costs), with the convention that the empty path has cost 0. Then

$$\widehat{g}(u, v) \;=\; \exp(-d^*(u, v)).$$

Remark 1876. *This theorem provides a computational and conceptual translation: max-product path aggregation is equivalent to min-sum shortest paths after a logarithmic change of variables. Intuitively, multiplying probabilities along a path and then taking the best path (maximum product) is the same as adding "surprisal costs" $-\log g$ along a path and then choosing the cheapest path (minimum sum). Thus a seemingly specialized enriched-closure operation reduces to a classical shortest-path problem. In particular, the logarithmic transform is not merely a heuristic: it is an isomorphism between two standard algebraic "path aggregation" calculi, namely the multiplicative–max calculus on $(0, 1]$ and the additive–min (tropical) calculus on $[0, \infty)$. Any choice of logarithm base yields the same minimizing path, since changing bases scales all costs by a positive constant factor, leaving argmin structure invariant.*

Remark 1877. *In the setting of pattern webs, this equivalence allows one to interpret strong indirect pattern connections as "geodesics" in a cost geometry induced by $-\log$ weights. It also makes available the entire toolbox of shortest-path algorithms for computing $\widehat{g}$. When quantales are used to reason about least-effort trajectories more broadly, one meets similar log/energy dualities; see [21] for a thematically related quantale-based least-effort perspective. Concretely, if the underlying directed graph is finite and the costs $c(e) = -\log g(e)$ are nonnegative (as they are when $g(e) \in (0, 1]$), then standard algorithms such as Dijkstra's algorithm apply directly to compute $d^*(u, v)$, and then $\widehat{g}(u, v)$ is obtained by a single exponentiation. From a modeling perspective, this means one can reason about indirect pattern reinforcement either multiplicatively (via reliability/affinity propagation) or additively (via effort/energy accumulation), switching viewpoints depending on which is more interpretable or computationally convenient.*

Sketch. For any path π, $w(\pi) = \prod_i g(e_i)$ so $-\log w(\pi) = \sum_i(-\log g(e_i)) = \sum_i c(e_i)$. Taking max over $w(\pi)$ is equivalent to taking min over $-\log w(\pi)$, hence $-\log \widehat{g}(u,v) = d^*(u,v)$ and exponentiating gives the claim. Here one uses that $-\log$ is strictly decreasing on $(0,1]$, so for any two candidate path weights $w_1, w_2 \in (0,1]$ we have $w_1 \geq w_2 \iff -\log w_1 \leq -\log w_2$. In the common extension where some edges may have $g(e) = 0$, one may set $c(e) = +\infty$ so that any path containing such an edge has infinite cost and therefore cannot be minimizing, matching the fact that such a path has product weight 0 and therefore cannot be maximizing unless all paths have weight 0. □

Remark 1878. Proof sketch. *The proof is a change of coordinates:* $-\log$ *turns products into sums and* max *into* min *because* $-\log$ *reverses order on* $(0,1]$*. After this change,* $\widehat{g}$ *becomes the exponential of minus the shortest-path distance. Equivalently, one may view the construction as transporting the closure operation along the monotone anti-isomorphism* $(0,1], \leq \to [0,\infty), \geq$ *given by* $x \mapsto -\log x$*. This makes explicit why the closure respects composition: multiplicative composition of affinities corresponds exactly to additive composition of efforts, and the "best" composite (maximal affinity) corresponds to the "cheapest" composite (minimal effort).*

Remark 1879. Intuition. *One may think of* $g(u,v)$ *as a "reliability" or "affinity" and of* $c(u,v) = -\log g(u,v)$ *as the corresponding "effort." A path with high product reliability is exactly a path with low total effort. The closure* $\widehat{g}(u,v)$ *therefore reports the reliability of the least-effort route from* u *to* v*. This perspective also clarifies limiting cases: edges with* g *close to* 1 *correspond to very small effort (nearly "free" transitions), while edges with* g *close to* 0 *correspond to very large effort (nearly "forbidden" transitions). Moreover, since* $-\log$ *is additive, repeated moderate reliabilities compound into a linear accumulation of effort, which is often easier to compare across paths than products of many terms; the closure* $\widehat{g}$ *can therefore be read as a global measure of indirect support that is sensitive to both path length and edge strength in a principled way.*

33.7 Multi-resolution transforms and weakness

Theorem 45 (Sup-pushforward dominates original intensities)**.** *Let* $\alpha : X \to X'$ *be surjective,* $\beta : P \to P'$ *any map, and define the pushed-forward intensity*

$$I'(p', x') := \sup\{I(p,x) : \beta(p) = p',\ \alpha(x) = x'\}.$$

Then for all $p \in P$ *and* $x \in X$*,*

$$I'(\beta(p), \alpha(x)) \ \geq\ I(p,x).$$

Remark 1880. *This theorem addresses what happens to pattern intensities under coarse-graining (Hyperseed-Concept 86) or resolution change: multiple fine patterns* p *may map to one coarse pattern* p'*, and multiple fine entities* x *may map to one coarse entity* x'*. The sup-pushforward defines the coarse intensity as the best intensity among all fine representatives consistent with* (p', x')*. The claim is a monotonicity guarantee:* coarse-graining via supremum cannot lose intensity for any specific fine instance*, because that fine instance remains among the candidates.*

Remark 1881. *This "sup semantics" is a natural choice in transfer and abstraction pipelines (Transfer Learning: Hyperseed-Concept 192), where coarse concepts are intended to summarize whatever fine evidence is most salient. It is also aligned with the Transweave idea of transporting learned structure across representational changes [7] (Transweave: Hyperseed-Concept 193).*

Sketch. The set inside the supremum defining $I'(\beta(p), \alpha(x))$ includes the particular witness (p,x), so the supremum is at least $I(p,x)$. □

Remark 1882. Proof sketch. *This is the defining property of* sup*: it is an upper bound for every element of the set being aggregated. Since (p, x) is in that set after mapping, its intensity is bounded above by the supremum, yielding the inequality.*

Theorem 46 (Compositionality of sup-pushforward under successive coarse-graining)**.** *Suppose $X \xrightarrow{\alpha_1} X' \xrightarrow{\alpha_2} X''$ with both maps surjective, and $P \xrightarrow{\beta_1} P' \xrightarrow{\beta_2} P''$. Let I_1 be the sup-pushforward of I along (α_1, β_1), and I_2 the sup-pushforward of I_1 along (α_2, β_2). Let I_{12} be the sup-pushforward of I along $(\alpha_2 \circ \alpha_1,\ \beta_2 \circ \beta_1)$. Then $I_2 = I_{12}$ pointwise.*

Remark 1883. *This theorem says that sup-based coarse-graining is* associative*: performing abstraction in two stages yields exactly the same result as abstracting in one stage. This is crucial if one wants a consistent multi-resolution theory rather than a collection of ad hoc summaries. In epistemic terms, it means the order in which one forgets detail (first from X to X', then to X'') does not affect what intensities remain at the coarsest level, provided one uses the supremum aggregator.*

Remark 1884. *The result connects directly to Section 11's interest in multi-level pattern webs and hierarchical representations: it ensures that when one constructs pattern-web summaries at different levels of granularity, the induced intensities behave coherently across levels. This kind of consistency condition is central in transfer-learning formalisms [7].*

Sketch. Both sides evaluate to the supremum of $I(p, x)$ over the same constraint set: $(\beta_2 \circ \beta_1)(p) = p''$ and $(\alpha_2 \circ \alpha_1)(x) = x''$. The intermediate sup does not change the overall sup because $\sup_{x'} \sup_{x \in \alpha_1^{-1}(x')}$ equals $\sup_{x \in (\alpha_2 \circ \alpha_1)^{-1}(x'')}$, and similarly for patterns. $\square$

Remark 1885. Proof sketch. *Unfold both definitions until each side is written as a supremum over the original fine pairs (p, x) subject to the same composite constraints. Then observe that nested suprema over fibers of α_1 inside fibers of α_2 collapse to a single supremum over the composite fiber.*

Remark 1886. Intuition. *The supremum operator is indifferent to how you partition the candidate set before taking the maximum: taking the best element in each bin and then the best across bins yields the same as taking the best element overall. That invariance is exactly what fails for many other aggregators, hence the special utility of* sup *in a multi-resolution setting.*

Theorem 47 (Iterated coarse-graining monotonically increases weakness)**.** *Let $H \subseteq X \times X$ be an indistinction relation and let $\alpha : X \to X'$ be a coarse-graining map. Let H' be the pushed-forward indistinction relation defined by $(\alpha(x), \alpha(y)) \in H'$ whenever $(x, y) \in H$. For any weakness functional $w(\cdot)$ monotone under adding undistinguished pairs,*

$$w(H) \ \leq \ w(H').$$

Moreover, for a chain of coarse-grainings $X \to X' \to X'' \to \cdots$, the resulting weaknesses form a nondecreasing sequence.

Remark 1887. *This theorem expresses a fundamental directionality: coarser resolution produces more indistinction, hence (for any reasonable monotone measure) greater weakness. Here H encodes which pairs are treated as "not distinguished" (Indistinction is the flip-side of Distinction: Hyperseed-Concept 98). Pushing H forward along α can only identify more pairs, because multiple fine points may collapse to one coarse point, and any prior indistinction survives the collapse.*

Remark 1888. *The formal statement is intentionally generic about $w(\cdot)$: it may be any weakness functional in the sense of [3], or the more general cognitive-architectural notion of weakness discussed in [2]. The moral is that weakness is not merely an accident of a particular observer; it* *is structurally induced by coarse-graining operations. This provides a clean mathematical backbone for Weakness (Hyperseed-Concept 202) as a monotone companion to abstraction. In particular, by leaving w abstract one can accommodate both "relational" weakness (where w is defined directly from an indistinguishability relation H) and "informational" weakness (where H is first converted into an information-loss proxy such as a quotient structure, a partition, or an entropy-like statistic). The point of the lemma is then not the detailed formula for w, but the order property it respects: whenever a process enlarges the set of indistinguishable pairs (or, equivalently, reduces the set of discriminations), any such w must move monotonically in the direction of greater weakness. This makes explicit the sense in which abstraction is not neutral: even with an idealized observer, the coarse-graining map α can force formerly separable states to become inseparable, thereby raising weakness in a way that is induced by the representation change itself.*

Remark 1889. *The "moreover" clause states that repeated coarse-graining yields a monotone time-series of weakness values. In a multi-level pattern-web construction, this can be read as saying that as one moves upward through levels of abstraction, the space becomes progressively less discriminating, and any monotone measure of that loss of discrimination accumulates rather than oscillates. Equivalently, if one indexes abstraction levels by $t = 0, 1, 2, \dots$ and writes H_t for the induced indistinguishability relation at level t, then $H_t \subseteq H_{t+1}$ (in the natural inclusion order on relations), so the trajectory $w(H_t)$ is a nondecreasing sequence. This rules out "free" recovery of discriminative power under further coarse-graining: any later increase in discrimination would require an operation other than coarse-graining (e.g. refinement, additional sensors/features, or an explicit memory of distinctions lost at earlier levels). In pattern-web language, the higher-level nodes may still support useful inference, but the granularity of admissible distinctions is constrained monotonically by the abstraction pipeline.*

Sketch. The pushed-forward relation H' contains (at least) the image of H under $\alpha \times \alpha$, so it collapses at least as many distinctions. Apply monotonicity of $w(\cdot)$ once, and iterate for a chain. More concretely, if $H \subseteq X \times X$ encodes indistinguishability on the fine space X and $\alpha : X \to X'$ is the coarse-graining map, one standard pushed-forward choice is the least relation $H' \subseteq X' \times X'$ such that $(x, y) \in H \Rightarrow (\alpha(x), \alpha(y)) \in H'$. For indistinguishability, H' typically also contains all pairs that are forced to coincide by the map itself, i.e. whenever $\alpha(x) = \alpha(y)$ one has $(\alpha(x), \alpha(y)) \in H'$ even if $(x, y) \notin H$; thus H' is at least as large as the direct image $(\alpha \times \alpha)[H]$ and may be strictly larger due to many-to-one identification. If the weakness axiom is phrased as monotonicity under relation inclusion (or under any equivalent preorder capturing "having fewer distinctions"), then $H \preceq H'$ implies $w(H) \leq w(H')$, and repeating the same comparison along $\alpha_0, \alpha_1, \dots$ yields the monotone chain inequality. ☐

Remark 1890. Proof sketch. *The proof is order-theoretic: show that H' is "larger" than H in the sense relevant to the monotonicity axiom of w, and then conclude $w(H) \leq w(H')$. Iterating the one-step argument gives a nondecreasing sequence along a chain. Here "larger" can be read literally as set inclusion $H \subseteq H'$ when H and H' live on a common carrier, or more generally as the existence of a comparison map that embeds the fine indistinguishability information into the coarse one. In many presentations, indistinguishability relations (or partitions/equivalence relations derived from them) form a lattice ordered by refinement, and coarse-graining moves upward in that lattice (toward coarser partitions with bigger equivalence classes). Any weakness functional that is monotone with respect to that order—whether defined via counting indistinguishable pairs, measuring the size distribution of equivalence classes, or evaluating a cost of identification—inherits the same one-step monotonicity, and therefore the same monotone behavior under iteration.*

Remark 1891. Intuition. *Think of H as the set of pairs your observer cannot tell apart. Coarse-graining merges states, so if two fine states were already indistinct, their coarse images are certainly indistinct; and in addition, distinct fine states may map to the same coarse state and thus become indistinct* even if they were formerly distinguished. *Any weakness measure that increases when you add indistinct pairs must therefore increase (or stay constant) under coarse-graining. A useful mental model is to imagine X as high-resolution sensory microstates and X' as the corresponding macro-descriptions (features, categories, buckets). The map α forgets some coordinates; forgetting cannot create new evidence that separates two cases, but it can destroy separating evidence that used to exist. Thus the induced indistinguishability relation on X' necessarily contains at least the transported old indistinguishabilities and often strictly more, because collisions $\alpha(x) = \alpha(y)$ force a new indistinguishability regardless of what held at the fine level. This is exactly the sense in which weakness is structurally induced: once you decide to work at the coarse level, the loss of discriminations is not a contingent failure of attention but a consequence of the representational collapse encoded by α.*

34 Helper theorems for habits, self-weaving webs, and morphic resonance (Section 12)

Standing assumptions and notation. Fix a commutative quantale $(V, \leq, \oplus, \otimes, e)$ and a (finite, when needed) pattern class P. A pattern-support state is a function $s : P \to V$ (Section 12 preamble). Given a V-relation (kernel) $A : P \times P \to V$, define the relational image (Def. 145):

$$(A?s)(q) := \bigoplus_{p \in P} A(p, q) \otimes s(p).$$

Given $d \in V$, define scaling (Def. 146):

$$(d \odot s)(p) := d \otimes s(p).$$

Given an external input $u_t : P \to V$, define the habit update operator (Def. 147):

$$\mathrm{Upd}_{A,d,u_t}(s) := u_t \oplus (d \odot s) \oplus (A?s),$$

where $\oplus$ on the right is pointwise in V^P.

For scalar habit statistics (Defs. 141–143), fix a scalarization $\pi : V \to [0, 1]$ and define $p_t(p) := \pi(s_t(p))$.

Remark 1892. *The quantale $(V, \leq, \oplus, \otimes, e)$ plays two roles at once: it is an algebra of "intensities" (how much support a pattern has), and it is an algebra of "composition" (how supports propagate and combine). The order $\leq$ is the notion of "no more than" or "no stronger than." The operation $\oplus$ is a join-like combination (often interpreted as taking the stronger of two independent supports, or the least upper bound), while $\otimes$ is a multiplicative-like combination (often interpreted as chaining supports along a path, or conjoining evidence). The unit e is the neutral element for chaining, so that $e \otimes v = v$; this matches the intuition that an "empty chain" should not change a support value. Quantales (and in particular their use as graded logics/evidence algebras) connect to the general notion of* Quantale Weakness *in the Hyperseed core-concept set (Hyperseed-Concept 143) and to broader discussions of weakness-as-structure in adaptive systems [3, 2].*

Remark 1893. *The notation* $(A?s)$ *is the standard "relational image" (a.k.a. matrix-vector product, but in the quantale rather than over* $\mathbb{R}$*): for each target pattern* $q \in P$*, we look at all sources* $p \in P$*, combine the strength of the edge* $A(p,q)$ *with the current support* $s(p)$ *using* $\otimes$*, and then aggregate over all sources using* $\oplus$*. Concretely, if* $V = [0,1]$ *with* $\oplus = \max$ *and* $\otimes = \cdot$*, then* $(A?s)(q) = \max_p A(p,q)\, s(p)$*, i.e. the strongest incoming reinforcement chain into* q*. If instead* $V = \{0,1\}$ *with* $\oplus = \vee$ *and* $\otimes = \wedge$*, then* $(A?s)(q) = 1$ *iff there exists some* p *with both* $A(p,q) = 1$ *and* $s(p) = 1$*, i.e. ordinary reachability of active patterns. This relational viewpoint is the algebraic spine behind many "pattern web" and reinforcement intuitions in [5], and it links naturally to the Hyperseed notion of* Pattern *(Hyperseed-Concept 130).*

Remark 1894. *The update operator* Upd_{A,d,u_t} *says: next support is the pointwise join of (i) the exogenous drive* u_t*, (ii) the decayed previous support* $d \odot s$*, and (iii) the internally propagated support* $A?s$*. Philosophically, this is a minimal dynamical synthesis of three sources of "becoming": the outside world, the persistence of what already is, and the internal causal/associative web. One may hear here a faint echo of Peirce's triadic categories [14] (the immediate given, the brute secondness of constraint, and the mediating thirdness of habit), and of Whitehead's insistence that process must be understood through how it inherits and transmutes prior activity [15].*

Remark 1895. *The scalarization* $\pi : V \to [0,1]$ *is what allows one to compute scalar habit statistics from structured/graded evidence values. In the simplest case* $V = [0,1]$ *one may take* $\pi = \mathrm{id}$*; in the p-bit case* $V = [0,1]^2$ *later in this section,* π *may pick out a "positive" component or a "net" bias. This scalarization is a modeling choice: it specifies what an observer is willing to count as "more of the habit," and thus belongs as much to observer-indexed semantics as to pure algebra.*

Remark 1896. *It is often helpful to make explicit the ambient order on the function space* V^P*: we use the pointwise order (also written* $\leq$ *by abuse of notation), namely* $s \leq s'$ *iff* $s(p) \leq s'(p)$ *for all* $p \in P$*. With this convention, pointwise* $\oplus$ *is exactly the join operation in* V^P*, and the update* Upd_{A,d,u_t} *is naturally read as an order-theoretic combination of three monotone influences. In particular, whenever* $\otimes$ *is monotone in each argument (as it is in any quantale), both* $s \mapsto d \odot s$ *and* $s \mapsto A?s$ *are monotone maps* $V^P \to V^P$*, and hence* Upd_{A,d,u_t} *is monotone as well. This monotonicity is one of the basic "helper" properties used repeatedly when discussing convergence, fixed points, and the accumulation of habit under repeated updates.*

Remark 1897. *The clause "finite, when needed" for the pattern class* P *is mainly about how one wants to interpret the* $\bigoplus_{p\in P}$ *in the relational image. In a general quantale,* $\oplus$ *is defined for arbitrary joins, so* $\bigoplus_{p\in P}$ *is meaningful even for infinite* P *as the join of the set* $\{A(p,q) \otimes s(p) \mid p \in P\}$*. However, in applications one may restrict to finite* P *(or effectively finite support) to ensure that relational images can be computed concretely and to avoid measure-theoretic issues when* P *carries additional structure. Thus "finite when needed" is not a conceptual limitation of the formalism so much as a pragmatic stipulation about which helper theorems are intended to be computationally actionable.*

Remark 1898. *The kernel* $A : P \times P \to V$ *can be read as a weighted directed adjacency relation on the pattern class:* $A(p,q)$ *expresses "how much" activation of* p *can (directly) reinforce* q *in one step. Under iteration, the quantale product* $\otimes$ *governs how multi-step paths compose (chains of reinforcement), while* $\oplus$ *governs how alternative paths compete or accumulate. This is the sense in which the algebra is a "self-weaving web": even if* u_t *is sparse or intermittent, the term* $A?s$ *can sustain and spread support across the web in a way that depends only on the internal connectivity structure encoded by* A*.*

Remark 1899. *The decay/scaling element $d \in V$ is best thought of as a persistence parameter that regulates how much of the prior state is carried forward independently of web propagation. In many concrete instances one assumes $d \leq e$ (so that $d \otimes v \leq e \otimes v = v$), which makes $d \odot s$ a literal "decay" or "forgetting" term. Nonetheless, the formal definition allows more general choices (including context-dependent or pattern-dependent decay, by replacing d with a function $d : P \to V$), and later helper lemmas typically isolate exactly which inequalities on d are required for boundedness, stability, or contraction-like behavior under iteration.*

Remark 1900. *The time index in u_t and s_t is intentionally minimal: the operator Upd_{A,d,u_t} defines a one-step transition, and one may study either non-autonomous dynamics (time-varying input u_t) or autonomous dynamics (constant u). In the autonomous case, fixed points of $s \mapsto \mathrm{Upd}_{A,d,u}(s)$ provide a natural notion of "habit attractor" or "stable support profile" compatible with a steady environment; in the non-autonomous case, helper estimates typically compare trajectories $t \mapsto s_t$ against such fixed points or against each other to quantify the degree of resonance, drift, or entrainment caused by the changing u_t.*

Remark 1901. *Although this section foregrounds algebraic definitions, the motivating interpretations (habits, morphic resonance, and self-weaving webs) depend on iterating the update and examining how repeated exposure and internal propagation reshape the support landscape. Concretely, when u_t intermittently excites a small subset of patterns, the term $A?s$ can amplify recurrent subgraphs of A (feedback motifs), and the term $d \odot s$ can preserve partial activation between exposures. The helper theorems that follow are meant to make these informal statements precise in order-theoretic language (e.g. monotone iteration, inflationary behavior under suitable assumptions, and comparisons between kernels), while remaining agnostic about which particular quantale V is chosen in a given modeling instance.*

34.1 Habit-taking / habit-reversal as monotone time-series properties

Lemma 102 (Strict version of Lemma 3)**.** *Assume for every $p \in P$ the scalar propensity $p_t(p)$ is nondecreasing in t. Then for every p, T for which the conditional expectations exist, $\mathrm{Post}_T(p) \geq \mathrm{Pre}_T(p)$ (Lemma 3 in the main text).*

Moreover, if there is a set of pairs (p, T) of positive $(\nu \times \tau)$-measure such that $\mathrm{Post}_T(p) > \mathrm{Pre}_T(p)$, then the habit-taking statistic

$$\mathbb{E}_{p\sim\nu, T\sim\tau}\big[\mathrm{Post}_T(p) - \mathrm{Pre}_T(p)\big]$$

is strictly > 0 (hence the context manifests tendency-to-take-habits in Def. 143).

Remark 1902. *Intuitively, this lemma formalizes the idea that if each individual propensity time-series is monotone upward, then "after" averages must dominate "before" averages. The second clause adds a modest but important strengthening: if* some *non-negligible fraction of (p, T) pairs show a strict improvement, then the* average *improvement is strictly positive. This is exactly what one wants if one is trying to detect an intrinsic tendency-to-take-habits (Hyperseed-Concept 189) from data: a merely nonnegative statistic is consistent with stasis, but a strictly positive statistic witnesses genuine directional drift.*

Remark 1903. *The phrase "for which the conditional expectations exist" is doing real work: it rules out pathologies where the time-window averages defining $\mathrm{Pre}_T(p)$ and $\mathrm{Post}_T(p)$ are undefined (e.g. because of non-integrability). In the typical empirical setting one assumes bounded propensities $p_t(p) \in [0, 1]$, in which case these conditional expectations exist automatically. Under such boundedness, the lemma can be read as a statement purely about order: monotone sample paths force monotone conditional means.*

Remark 1904. *The result is not surprising once stated, but it is structurally important: it lets us replace a potentially messy definition involving conditional expectations with a simple sufficient condition (monotonicity). In later subsections, we will prove monotonicity of the underlying state dynamics under natural assumptions; this lemma is the bridge that transports those order-theoretic facts into the probabilistic/empirical habit statistics of Section 12.*

Sketch. The weak inequality is exactly Lemma 3: averages of a nondecreasing sequence after T dominate averages before T. For strictness, the expectation of a random variable is strictly positive if it is positive on a set of positive measure and never negative. □

Remark 1905. *One way to make the strictness clause completely explicit is to consider the random variable*

$$X(p,T) \;:=\; \mathrm{Post}_T(p) - \mathrm{Pre}_T(p).$$

By the first part of the lemma (or Lemma 3), $X(p,T) \geq 0$ for all (p,T) for which the conditional expectations exist. If, in addition, $X(p,T) > 0$ on a set of positive $(\nu \times \tau)$-measure, then the integral of X against $(\nu \times \tau)$ must be strictly positive. This is the standard "nonnegative and not almost everywhere zero" criterion for strict positivity of an expectation, and it shows that strict habit-taking is an identifiable *phenomenon at the level of averaged statistics (as opposed to being an artifact of a few measure-zero exceptional points).*

Remark 1906. *The same logic also isolates the only way strictness can fail: even if each trajectory $t \mapsto p_t(p)$ is nondecreasing, it may be* flat *on almost all relevant windows, so that $\mathrm{Post}_T(p) = \mathrm{Pre}_T(p)$ except on a negligible set. In that case the statistic is 0, correctly reflecting that (in the sense of the chosen sampling measures ν, τ) there is no detectable directional drift. Thus the strictness clause should be read as a joint condition: it requires not merely monotonicity but also that strict increases occur with non-negligible frequency under the sampling scheme.*

Remark 1907. Proof sketch. *The strategy is purely order-theoretic: if a sequence never goes down, then any average taken over later indices cannot be smaller than an average taken over earlier indices. The strictness part is a basic measure-theoretic observation: a nonnegative random variable with positive probability of being strictly positive has strictly positive expectation.* □

What makes the lemma useful is that it isolates the only place where time enters: once monotonicity in t is established, the before/after statistic becomes almost tautological. One can picture the graph of $t \mapsto p_t(p)$ as a nondecreasing curve; the "Post" average samples the curve to the right of T and the "Pre" average samples to the left, so the inequality is the statement that the center-of-mass of the right-hand sample cannot lie below the left-hand sample.

Remark 1908. *Because this subsection also targets habit-*reversal *(Hyperseed-Concept 188), it is worth spelling out the symmetric interpretation: if each $t \mapsto p_t(p)$ is nonincreasing, then the same averaging argument yields $\mathrm{Post}_T(p) \leq \mathrm{Pre}_T(p)$, and strict decrease on a set of positive measure yields a strictly negative mean $\mathbb{E}[\mathrm{Post} - \mathrm{Pre}] < 0$. In other words, the sign of the same statistic distinguishes "habit-taking" (upward drift) from "habit-reversal" (downward drift) once the appropriate monotonicity direction is established.*

Lemma 103 (Monotone scalarizations transport order to propensities)**.** *Assume $\pi : V \to [0,1]$ is monotone (order-preserving). If $s \leq s'$ in V^P (pointwise), then for all $p \in P$,*

$$\pi(s(p)) \leq \pi(s'(p)).$$

Remark 1909. *This lemma says that an order-preserving "readout" π cannot invert comparisons: if a state s' has at least as much support as s for every pattern, then the scalarized propensities must also be at least as large. It is a small fact, but it is the essential hinge between the abstract quantale-valued dynamics and any concrete scalar statistic we might compute. Without it, one could have a paradoxical situation where increasing evidence yields decreasing measured propensity merely because the observer chose a perverse scalarization.*

Remark 1910. *The pointwise order on V^P is crucial here: $s \leq s'$ means* every *pattern-coordinate has weakly increased. This is exactly the setting in which a monotone π can be applied coordinatewise without losing the comparison. In applications, π is often chosen to be a "forgetful" or "compression" map (e.g. projecting a multidimensional evidence record to a single confidence score); the lemma guarantees that such compression does not introduce spurious inversions when the underlying evidence order is respected.*

Remark 1911. *A simple example: if $V = [0,1]^2$ is a p-bit evidence domain, then $\pi_{\text{pos}}(v^+, v^-) = v^+$ is monotone, whereas $\pi(v^+, v^-) = v^- - v^+$ is* anti-monotone *in the first coordinate and so would not satisfy the lemma's assumption. The point is not that anti-monotone maps are illegitimate, but that they measure a different attitude: they interpret "more positive evidence" as "less" in the output scale.*

Remark 1912. *More generally, any map of the form $\pi(v) = \sum_i w_i v_i$ on $V = [0,1]^n$ with weights $w_i \geq 0$ is monotone in the product order, and so it respects increases in any coordinate. This kind of example is typical when V packages multiple "facets" of support (frequency, recency, coherence, etc.) and the observer reports a single propensity by an order-respecting aggregation.*

Sketch. Immediate from pointwise order and monotonicity of π. $\square$

Remark 1913. Proof sketch. *Unpack the definitions: $s \leq s'$ means $s(p) \leq s'(p)$ for each p, and monotonicity of π means $v \leq w \Rightarrow \pi(v) \leq \pi(w)$. Compose these two implications.* $\square$

The lemma is the precise way to say that π is an "observer-compatible" projection from structured support to a single dial reading: it respects the direction of increase encoded in $\leq$.

Corollary 11 (Inflationary dynamics imply habit-taking under monotone scalarization)**.** *Assume π is monotone and the state sequence satisfies $s_{t+1} \geq s_t$ pointwise for all t. Then each $p_t(p) = \pi(s_t(p))$ is nondecreasing in t, hence the habit-taking statistic in Def. 143 is ≥ 0 (and > 0 under the strictness condition of Lemma 102).*

Remark 1914. *This is the promised bridge: if the* state *evolves inflationarily (never losing support), and if the observer's measurement map π is monotone, then the measured propensities are monotone in time, so the tendency-to-take-habits statistic (Hyperseed-Concept 189) is nonnegative. In other words, an order-theoretic property of the underlying dynamics implies an empirically testable statistical signature.*

Remark 1915. *For strict positivity in the corollary, it is enough (though not necessary) that there exist a set of times t and patterns p of positive sampling measure for which $s_{t+1}(p) > s_t(p)$ in V and for which π is strictly order-reflecting at those points in the weak sense that $v < w$ implies $\pi(v) < \pi(w)$ along the realized comparisons. The lemma itself only assumes monotonicity, so strictness must ultimately come from the empirical fact that the dynamics sometimes make genuine progress (not merely weak progress) in directions that the scalarization can detect.*

Remark 1916. *This corollary is also a methodological statement: it tells us that in order to establish habit-taking it may suffice to prove inflationarity of* Upd *(or of a simplified update such as* Cl_A*) rather than to analyze the full before/after expectations directly.*

Sketch. Apply Lemma 103 to $s_t \leq s_{t+1}$ for each t, then invoke Lemma 3. □

Remark 1917. Proof sketch. *First transport monotonicity from* s_t *to* p_t *using the monotone scalarization lemma; then apply the already-established "monotone time-series implies Post* $\geq$ *Pre" fact.* □

Conceptually, the proof composes two monotonicities: (i) dynamical monotonicity in the state space V^P*, and (ii) observational monotonicity in passing from* V*-values to scalars. This composition is one of the recurring themes of the section: order structure is preserved through the whole pipeline.*

34.2 Update operator monotonicity and comparative statics

Lemma 104 (Monotonicity in all arguments)**.** *The operator* Upd_{A,d,u_t} *is monotone in each of* s*,* u_t*,* A*, and* d*: if* $s \leq s'$*,* $u_t \leq u'_t$*,* $A \leq A'$*, and* $d \leq d'$ *(pointwise order for relations), then*

$$\mathrm{Upd}_{A,d,u_t}(s) \leq \mathrm{Upd}_{A',d',u'_t}(s').$$

Remark 1918. *The lemma asserts a strong form of* comparative statics*: if you increase any of the ingredients of the update—start from a larger state, feed in a larger input, strengthen the internal kernel, or reduce decay by taking a larger* d *in the quantale order—then the resulting next state cannot decrease. For dynamical systems meant to model reinforcement and habit formation, this is the algebraic analogue of the intuition that "more cause cannot yield less effect," provided our algebra* $(V, \leq, \oplus, \otimes)$ *is chosen so that* $\leq$ *truly means "no stronger than."*

Remark 1919. *This monotonicity is a central reason for working in a quantale rather than in an arbitrary semiring: completeness and residuation-like order behavior make it possible to reason about dynamics with lattice-theoretic tools. In the background lies the same philosophy that makes fixed-point theorems practical for self-modifying systems and goal dynamics [10]: if the update is monotone on a complete lattice, then least/greatest fixed points and convergence-by-iteration become accessible.*

Sketch. Prop. 14 in the main text gives monotonicity in s, u_t, and A. Monotonicity in d follows similarly because $(d \odot s)(p) = d \otimes s(p)$ and $\otimes$ is monotone in each argument in a quantale. Join $\oplus$ is monotone as it is the least upper bound. □

Remark 1920. Proof sketch. *Prove each argument is monotone separately, then combine them:* Upd *is built by composing and joining monotone maps, and such constructions preserve monotonicity.* □

The key step is recognizing that the only "nontrivial" operations are $\otimes$ *and* $\oplus$*, and both are monotone in a quantale. Visually, one can think of* Upd *as three pipes (input, persistence, propagation) feeding into a single "max/join" combiner; increasing any pipe cannot make the combined output smaller.*

Corollary 12 (Trajectory dominance)**.** *Fix sequences* (A, d, u_t) *and* (A', d', u'_t) *with* $A \leq A'$*,* $d \leq d'$*, and* $u_t \leq u'_t$ *for all* t*. If* $s_0 \leq s'_0$*, and* $s_{t+1} = \mathrm{Upd}_{A,d,u_t}(s_t)$ *while* $s'_{t+1} = \mathrm{Upd}_{A',d',u'_t}(s'_t)$*, then* $s_t \leq s'_t$ *for all* t*.*

Remark 1921. *This corollary is the dynamical version of the monotonicity lemma: if one system is "everywhere stronger" in parameters and initialization than another, then it stays stronger at all future times. In practice, it lets us bound complicated trajectories by simpler ones (e.g. with larger kernels or inputs), and it guarantees that improving a learning environment cannot perversely reduce learned support, as long as the improvements correspond to increases in the quantale order.*

Sketch. Induction on t using Lemma 104. □

Remark 1922. Proof sketch. *Assume $s_t \leq s'_t$; apply the monotonicity lemma at time t to obtain $s_{t+1} \leq s'_{t+1}$; conclude by induction from the base case $s_0 \leq s'_0$.* □

The proof works because the inequality is preserved by one update step, and time is nothing but iteration of that step. Geometrically (in the poset V^P), the update map carries order intervals to order intervals.

Lemma 105 (Inflationarity in the no-decay case)**.** *Assume $d = e$ (no decay). Then for all s and t,*

$$s \leq \mathrm{Upd}_{A,e,u_t}(s).$$

In particular, if $u_t \equiv u$ is constant, the trajectory is pointwise ascending: $s_0 \leq s_1 \leq s_2 \leq \cdots$.

Remark 1923. *This lemma isolates a particularly transparent regime: when there is no decay, the system never forgets. Then the update always includes the previous state as a join-summand, so it must be inflationary. This is the simplest abstract situation in which "habit-taking" is almost built into the algebra: the dynamics is literally cumulative. One may see this as the most extreme formalization of a system with perfect retention.*

Sketch. If $d = e$ then $(d \odot s)(p) = e \otimes s(p) = s(p)$, hence $\mathrm{Upd}_{A,e,u_t}(s) = u_t \oplus s \oplus (A?s) \geq s$. □

Remark 1924. Proof sketch. *Substitute $d = e$ so that the decay term becomes exactly s, and then note that joining with s cannot yield something below s.* □

The proof is a one-line algebraic expression of an intuitive picture: the next state is the old state plus possible additions; if nothing is ever subtracted, monotone accumulation follows automatically.

34.3 Closure, Kleene star, and downstream support

Recall (Defs. 148–149). Define the one-step closure operator

$$\mathrm{Cl}_A(s) := s \oplus (A?s),$$

and define iterates $\mathrm{Cl}_A^{(0)}(s) := s$, $\mathrm{Cl}_A^{(n+1)}(s) := \mathrm{Cl}_A(\mathrm{Cl}_A^{(n)}(s))$. When V^P is a complete lattice (e.g. P finite, V complete), define the limit

$$\mathrm{Cl}_A^{(\omega)}(s) := \bigoplus_{n \geq 0} \mathrm{Cl}_A^{(n)}(s).$$

Also define $A^0 := I$, $A^{n+1} := A^n \circ A$ and $A^* := \bigoplus_{n \geq 0} A^n$ (Def. 149).

Remark 1925. *The operator Cl_A is the "pure reinforcement" update: keep whatever support you already have, and add what can be transmitted to you in one step through A. Iterating Cl_A therefore accumulates reinforcement through chains of length $0, 1, 2, \ldots$. The ω-iterate $\mathrm{Cl}_A^{(\omega)}(s)$ is the result of allowing* all finite *reinforcement chains to contribute. The definition uses $\bigoplus_{n \geq 0}$, so completeness of V^P ensures that this infinite join exists; in the finite P case this is automatic because joins reduce to finite ones after stabilization in many concrete quantales.*

Remark 1926. *For readers tracking the mechanics: the symbol "?" denotes the (left) action of a V-kernel $A : P \times P \to V$ on a state $s \in V^P$. Concretely, one typically has a formula of the form*

$$(A?s)(q) = \bigoplus_{p \in P} A(p,q) \otimes s(p),$$

so $A?s$ is an "incoming aggregation" into q from all sources p, weighted by $A(p,q)$ and gated by the current support $s(p)$. Under this reading, $\mathrm{Cl}_A(s) = s \oplus (A?s)$ literally means: preserve the current support at each node, and also add in any support that can flow to the node in one A-step. The quantale axioms (associativity of $\otimes$ and distributivity over $\oplus$) are exactly what make iterative propagation agree with path-composition via A^n later on.

Remark 1927. *The Kleene star A^* is the familiar transitive-closure construction, generalized from boolean adjacency matrices to V-weighted kernels: A^n aggregates all paths of length n, and A^* aggregates all finite lengths. In max-product semantics, $A^*(p,q)$ is the weight of the best (maximal) path from p to q; in boolean semantics it is reachability. This is why the next theorem can be read as saying that closure equals "propagate along all finite paths," an idea closely aligned with the "pattern web" viewpoint [5].*

Remark 1928. *It is also useful to keep in mind the familiar algebraic identities for the Kleene star (when the relevant joins exist): one has*

$$A^* = I \oplus (A \circ A^*) = I \oplus (A^* \circ A),$$

expressing the idea that a finite path is either empty (the I contribution) or a first step followed by a finite path (the $A \circ A^*$ contribution), equivalently a finite path followed by a last step (the $A^* \circ A$ contribution). These identities are a compact way to see why A^* behaves like a "saturation" or "closure" of A under finite chaining, and they parallel the idempotence phenomenon for $\mathrm{Cl}_A^{(\omega)}$ discussed below.

Theorem 48 ($\mathrm{Cl}^{(\omega)}$ is a closure operator)**.** *Assume P is finite and V is a commutative quantale. Then $\mathrm{Cl}_A^{(\omega)} : V^P \to V^P$ is:*

1. *inflationary: $s \le \mathrm{Cl}_A^{(\omega)}(s)$,*
2. *monotone: $s \le s' \to \mathrm{Cl}_A^{(\omega)}(s) \le \mathrm{Cl}_A^{(\omega)}(s')$,*
3. *idempotent: $\mathrm{Cl}_A^{(\omega)}(\mathrm{Cl}_A^{(\omega)}(s)) = \mathrm{Cl}_A^{(\omega)}(s)$,*
4. *a least-fixed-point operator: $\mathrm{Cl}_A^{(\omega)}(s)$ is the least fixed point of Cl_A above s (Thm. 12 in the main text).*

Remark 1929. *In plain language, the theorem says that if you repeatedly apply "add whatever A can transmit in one step" and then take the limit, you obtain a genuine closure operation: it only adds (inflationary), respects ordering (monotone), stabilizes after being applied once (idempotent), and in fact produces the* smallest *stable state that contains the original seed s. This is the formal underpinning of the intuition that a habit web has an "eventual downstream envelope" determined by reinforcement paths.*

Remark 1930. *One can also read these clauses operationally as guaranteeing a well-behaved notion of "downstream support." Inflationary says no support is ever lost by closure; monotonicity says that adding more initial support cannot reduce what is eventually supported downstream; idempotence says that once a state has been fully saturated with all its downstream consequences, re-saturating does not change it. In many applications, these are the exact properties needed for closure to serve as a semantic abstraction: it defines an equivalence relation* $s \sim s'$ *iff* $\mathrm{Cl}_A^{(\omega)}(s) = \mathrm{Cl}_A^{(\omega)}(s')$*, i.e. two seeds are behaviorally indistinguishable with respect to all downstream reinforcement consequences.*

Remark 1931. *The least-fixed-point characterization is the most conceptually potent clause: it says that* $\mathrm{Cl}_A^{(\omega)}(s)$ *is not merely some limit, but the* canonical *limit picked out by order theory (Kleene iteration / Knaster–Tarski). This connects the present dynamics to the same mathematical machinery used elsewhere in the document to discuss stability of self-referential systems [10].*

Remark 1932. *The condition "least fixed point above s" should be read with care: it is a* relative *least fixed point (a least post-fixed state containing s). Equivalently, among all states t such that* $s \leq t$ *and* $\mathrm{Cl}_A(t) = t$*, the element* $\mathrm{Cl}_A^{(\omega)}(s)$ *is the smallest. This is precisely the sense in which the closure is "minimal but sufficient": it includes everything forced by repeated reinforcement, and nothing beyond that forced content.*

Sketch. Inflationary and monotone follow because Cl_A is inflationary and monotone, so the iterates form an ascending chain; the join dominates all iterates. Idempotence and least-fixed-point characterization are exactly the Kleene/Knaster–Tarski content of Thm. 12. □

Remark 1933. Proof sketch. *The proof proceeds in two layers. First, show that one-step closure* Cl_A *is inflationary and monotone; therefore its iterates form an increasing chain. Second, take the join of this chain; completeness ensures it exists, and the Knaster–Tarski/Kleene theorem ensures this join is the least fixed point above the seed.* □

The reason idempotence holds is subtle but standard: once you have included the contribution of all *finite reinforcement chains, applying "one more step" cannot reveal anything genuinely new, because any would-be new contribution would already correspond to a finite chain (namely the previous chain plus one edge). A helpful picture is the reachability closure in a directed graph: once all reachable nodes are marked, repeating the marking rule does nothing.*

Remark 1934. *If one wants to see the monotonicity/inflationary claims at the level of a single line, they are inherited from the lattice and action structure:* $s \leq s \oplus x$ *always, hence* $s \leq \mathrm{Cl}_A(s)$*; and if* $s \leq s'$ *then typically* $A?s \leq A?s'$ *(monotonicity of the action), hence* $s \oplus (A?s) \leq s' \oplus (A?s')$*. This is the basic reason the Kleene iteration* $\mathrm{Cl}_A^{(0)}(s) \leq \mathrm{Cl}_A^{(1)}(s) \leq \cdots$ *forms an ascending chain whose join can be taken.*

Theorem 49 (Explicit path-sum form (Thm. 12 restated))**.** *Under the assumptions above,*

$$\mathrm{Cl}_A^{(\omega)}(s) = A^*?s.$$

Equivalently, for each $q \in P$*,*

$$\mathrm{Cl}_A^{(\omega)}(s)(q) = \bigoplus_{n \geq 0} (A^n?s)(q),$$

so closure aggregates support transmitted along all finite reinforcement chains.

Remark 1935. *A convenient intermediate identity (often proved by a short induction) is that finite iteration of* Cl_A *corresponds to truncating the star at finite depth:*

$$\mathrm{Cl}_A^{(n)}(s) = \bigoplus_{k=0}^{n}(A^k?s).$$

The base case $n = 0$ *reads* $\mathrm{Cl}_A^{(0)}(s) = A^0?s = s$*, while the inductive step uses distributivity of the action over joins together with the associativity implicit in* $A^{k+1} = A^k \circ A$*. Taking* $\bigoplus_{n\geq 0}$ *of both sides then yields exactly the stated formula* $\mathrm{Cl}_A^{(\omega)}(s) = \bigoplus_{k\geq 0}(A^k?s) = A^*?s$*, making explicit how "closure as a limit of iterates" coincides with "closure as a path-sum."*

Remark 1936. *This theorem gives the computational/semantic reading of* $\mathrm{Cl}_A^{(\omega)}$*: it is exactly "apply A along any finite number of steps, and aggregate." The expression* $A^*?s$ *is thus a compact way to say "everything that can be downstream of the seed, with weights given by chaining through A." When* V *is max-product, one can read* $A^n?s$ *as "best path of length* n *from any seeded pattern into* q*," and then the outer join as "best over all lengths."*

Remark 1937. *In particular, the formula* $\mathrm{Cl}_A^{(\omega)}(s)(q) = \bigoplus_{n\geq 0}(A^n?s)(q)$ *makes clear what "downstream" means: node* q *receives support not only from immediate predecessors (the* $n = 1$ *term), but also via predecessors-of-predecessors, and so on. The join over* n *expresses that a node can be supported in multiple ways (different path lengths and different intermediate nodes), and the quantale structure specifies how such alternatives are aggregated (e.g. maximum, sum, logical disjunction). Thus, while the notation resembles classical transitive closure, it simultaneously encodes both* reachability *and* strength of support *in a single object.*

Remark 1938. *The result connects the fixed-point view (closure as least stable state) with the path-combinatorics view (closure as sum over chains). This duality is one reason the Kleene star is so persistent in computer science and logic: it lets one translate between recursion and enumeration without changing meaning.*

Sketch. This is item (3) of Thm. 12: unfold the recurrence, observe $\mathrm{Cl}_A^{(n)}(s) = \bigoplus_{k=0}^{n} A^k?s$, then take $n \to \omega$ and use distributivity. More explicitly, the iteration defining $\mathrm{Cl}_A^{(n)}(s)$ can be read as a repeated "one-step propagate and then join" process: starting from the seed s, at each stage you add all contributions that can be obtained by applying A one more time and then joining with what has already been accumulated. This is precisely why the nth iterate collects all path contributions of length at most n, and why the closed form $\bigoplus_{k=0}^{n} A^k?s$ matches the operational unfolding. In the limit step $n \to \omega$, one uses that the ambient join-semilattice (or quantale/module structure, depending on the setup of Thm. 12) admits countable joins and that the action $A \mapsto (A?s)$ and the finite compositions A^k interact well with joins. The distributivity invoked here is the algebraic property that allows one to commute the join over iteration depth with the joins implicit in composition/action, so that the ω-iterate agrees with the Kleene-star form. □

Remark 1939. Proof sketch. *Inductively expand* Cl_A *to see that after* n *iterations you have accumulated contributions from paths of length up to* n*. Then take the join over all* n *and recognize the Kleene star.* □

The key step is the algebraic distributivity that permits rearranging "join over times" with "join over paths." Intuitively, each iteration adds one more hop of propagation, so the bookkeeping by iteration count coincides with bookkeeping by path length. In slightly more detail: the "time" index n *counts how many rounds of applying the propagation kernel* A *you have performed, while a "path"* *of length k corresponds to k consecutive uses of A, i.e. to the composite A^k. Thus, iterating Cl_A is equivalent to accumulating, stage by stage, the images $A^k?s$ for increasing k. Recognizing the Kleene star is then exactly recognizing that $\bigoplus_{k\geq 0} A^k$ is A^*, and hence that the eventual closure is the star-action $A^*?s$. This is also the point where it matters that joins are taken in the correct order: one wants to be allowed to first join over all ways of taking k steps (which is already encoded in A^k as a join over intermediate states), and then join over k, without changing the outcome. Distributivity guarantees that this "flattening" of a two-level join is legitimate.*

Lemma 106 (Monotonicity of A^* and of downstream closure)**.** *If $A \leq B$ (pointwise as V-relations), then $A^* \leq B^*$, and hence for all states s,*

$$A^*?s \leq B^*?s.$$

Also, if $s \leq s'$, then $A^*?s \leq A^*?s'$.

Remark 1940. *This lemma states that strengthening the kernel can only strengthen its long-run transitive closure, and strengthening the seed can only strengthen the propagated result. It formalizes an intuitive monotonicity of "downstream influence." In applied terms: if you make some reinforcement links stronger (or add new ones), you cannot end up with* less *eventual downstream habit support. Equivalently, the assignment $A \mapsto A^*$ is an order-preserving endomorphism on the poset of V-relations, and the downstream-closure operator $s \mapsto A^*?s$ is order-preserving both in the kernel and in the initial condition. This matters when comparing interventions or learning updates: any update that is monotone at the level of local edges (relations) remains monotone after taking "all finite chains" of influence. One can also read the second monotonicity statement as a form of positive homogeneity of propagation with respect to evidence/activation: starting from a larger seed s' (in the order of the underlying module of states) cannot lead to a smaller propagated state after allowing arbitrary finite downstream traversals.*

Sketch. $A \leq B$ implies $A^n \leq B^n$ for all n by induction using monotonicity of relational composition; joining over n yields $A^* \leq B^*$. Monotonicity of $(\cdot?\cdot)$ in each argument gives the image inequalities. To spell out the induction: for $n = 0$ one typically has $A^0 = I \leq I = B^0$ (identity relation), and for the step $n \mapsto n+1$ one uses that composition is monotone in both arguments, so $A^{n+1} = A^n \circ A \leq B^n \circ B = B^{n+1}$ whenever $A^n \leq B^n$ and $A \leq B$. Taking $\bigoplus_{n\geq 0}$ on both sides (the join defining the Kleene star) preserves the inequality because joins are least upper bounds. Finally, since $A^* \leq B^*$ and the action/image operator ? is assumed monotone in its relation argument, it follows that $A^*?s \leq B^*?s$ for each s; similarly, $s \leq s'$ and monotonicity in the state argument gives $A^*?s \leq A^*?s'$. ☐

Remark 1941. Proof sketch. *Use induction on n to transport $A \leq B$ through repeated composition, then take joins to get $A^* \leq B^*$, and finally apply monotonicity of the image operator in each argument.* ☐

A visual intuition: A^* is built from all finite concatenations of edges from A. If every edge in A is no stronger than the corresponding edge in B, then every path in A is no stronger than the corresponding path in B, and taking the best/combined over paths preserves this ordering. In more operational terms, A^* aggregates the effect of zero steps (the identity), one step (A), two steps (A^2), and so on. The inequality $A \leq B$ ensures that at each fixed path length n, the n-step influence computed using A is bounded above by the n-step influence computed using B. The Kleene star then simply collects these bounds uniformly over all finite lengths, so the global "allow any finite number of hops" influence remains monotone. This is the same kind of reasoning used in reachability and dynamic programming: if you increase the weight/availability of every edge, then every finite walk *can only become at least as good, and optimizing/aggregating over all walks respects that ordering. The lemma packages this intuition into a clean algebraic statement that can be reused whenever downstream support is compared across kernels or across initial activations.*

34.4 Self-weaving webs and autocatalytic support

Recall (Def. 150). Fix a threshold $\theta \in V$. A nonempty $S \subseteq P$ is θ-autocatalytic if there exists a state $s : P \to V$ such that: (i) $s(p) \geq \theta$ for all $p \in S$, and (ii) $(A?s)(p) \geq \theta$ for all $p \in S$.

Remark 1942. *It is useful to read the two conditions as a* static *and a* dynamic *requirement. Condition (i) says that every pattern in S is already "present" (or supported) at least at level θ in the seed state s. Condition (ii) says that if one applies a single round of propagation through the relational web A (via the image operator $A?$), the support fed back into each $p \in S$ is still at least θ. In particular, (ii) does not require that $A?s$ equals s on S, only that the propagated support does not drop below threshold on S; this is what makes the definition robust to the presence of additional incoming support from outside S or additional mass in s outside S.*

Remark 1943. *Autocatalysis is the idea that a set of patterns can collectively sustain itself: each member already has at least threshold support, and moreover the internal propagation $A?s$ reinforces each member back up to threshold. This matches the intuitive meaning of* Autocatalytic Sets *(Hyperseed-Concept 61): the set is not merely present, but self-producing via the web of relations A. The term "self-weaving web" (Hyperseed-Concept 168) suggests the same image: the fabric of patterns is woven by the very threads it contains, a theme emphasized in process-oriented accounts of organization [15] and in pattern-web treatments [5]. In graph language (when P is viewed as nodes and A as weighted edges), a θ-autocatalytic set can be thought of as a subgraph whose internal connectivity is strong enough that one step of message-passing keeps every node in the subgraph above the activation level θ.*

Remark 1944. *A simple example in the boolean quantale: let $P = \{a, b\}$ and $A(a, b) = A(b, a) = 1$, with $s(a) = s(b) = 1$. Then $A?s = s$ and $S = \{a, b\}$ is autocatalytic for $\theta = 1$. In max-product $[0, 1]$, take $A(a, b) = A(b, a) = 0.9$ and $s(a) = s(b) = 1$; then $(A?s)(a) = (A?s)(b) = 0.9$, so S is autocatalytic at any threshold $\theta \leq 0.9$. Note that in the $[0, 1]$ case the set is* not *autocatalytic at $\theta = 1$, even though the seed state itself places both elements at value 1; what fails is that propagation through A attenuates support. This highlights that autocatalysis depends on the interaction between the seed s and the coupling strengths encoded by A, not merely on the raw initial levels.*

Lemma 107 (Autocatalysis persists under closure)**.** *Assume S is θ-autocatalytic with witness state s. Let $s^\infty := \mathrm{Cl}_A^{(\omega)}(s)$. Then for all $p \in S$:*

$$s^\infty(p) \geq \theta \quad \textit{and} \quad (A?s^\infty)(p) \geq \theta.$$

Remark 1945. *The lemma says that once a set is autocatalytic at threshold θ, taking the full downstream closure cannot destroy that property. Closure can only add support, never remove it, so the autocatalytic set remains above threshold, and its internally propagated reinforcement remains above threshold as well. This is the formal reason self-weaving is stable under "letting the reinforcement dynamics run." In particular, even if the closure process introduces new patterns outside S (possibly strongly connected to S), the original members of S do not lose their guaranteed lower bound θ because all operators involved are order-preserving.*

Sketch. By inflationarity, $s \leq s^\infty$, hence $s^\infty(p) \geq s(p) \geq \theta$ on S. Monotonicity of the image operator gives $A?s^\infty \geq A?s$, hence $(A?s^\infty)(p) \geq \theta$ on S. $\square$

Remark 1946. *One can also view the argument as a two-step domination chain on S:*

$$s^\infty \;\geq\; s \;\geq\; \theta \quad\Rightarrow\quad A?s^\infty \;\geq\; A?s \;\geq\; \theta,$$

where the first implication uses inflationarity of $\mathrm{Cl}_A^{(\omega)}$ and the second uses monotonicity of $A?$. Nothing in the proof requires special algebraic properties beyond these order-theoretic ones, which is why the result transfers cleanly across different choices of quantale-like value domains V.

Remark 1947. Proof sketch. *Use two monotonicities: s^∞ dominates s because closure is inflationary, and $A?$ is monotone so applying it to a larger state yields a larger image. Combine these with the witness inequalities for s.* □

Geometrically, closure moves you upward in the lattice V^P, and the operator $A?$ is an order-preserving transformation; therefore once a coordinate is above threshold, it stays above threshold under these upward moves. If one imagines each coordinate $p \in P$ as a "height" of support, then Cl_A is an update rule that can only raise heights (never lower them), so any already-high region (the autocatalytic subset) remains high throughout the closure ascent.

Theorem 50 (Self-weaving implies a nontrivial stable attractor in pure reinforcement)**.** *Assume P is finite and V complete. If (P, A) is self-weaving at threshold θ (i.e. it contains a θ-autocatalytic subset), then there exists a non-bottom fixed point s^* of Cl_A with $s^*(p) \geq \theta$ on that subset. Moreover, in pure reinforcement dynamics $s_{t+1} = \mathrm{Cl}_A(s_t)$ started from the witness seed s, the trajectory converges (as a join) to $s^* = \mathrm{Cl}_A^{(\omega)}(s)$.*

Remark 1948. *This theorem asserts the existence of an "attractor" purely from the presence of an autocatalytic seed: if some subset can reinforce itself above threshold, then iterating pure closure produces a stable fixed point that is nontrivial (not bottom) and that maintains at least threshold support on that subset. The second sentence explains the dynamics: starting from the witness state, repeatedly applying Cl_A accumulates all downstream consequences and converges (in the order-theoretic sense of taking a join) to the fixed point. The "non-bottom" conclusion can be read concretely as: there is at least one pattern (indeed, every pattern in the autocatalytic subset) whose eventual stabilized support is not the minimal value of V, so the attractor represents a genuinely present structure rather than the null state.*

Remark 1949. *The finiteness of P helps connect the order-theoretic ω-iteration to an intuitive finite-time stabilization: ascending chains in V^P often stabilize after finitely many steps when the underlying combinatorics is finite and the closure operator is Scott-continuous (or otherwise well-behaved). Even when one does not assume a specific continuity condition, the ω-join formulation still provides a canonical "limit" object s^* that captures everything that can be generated by repeated reinforcement from the seed.*

Remark 1950. *The importance is that we get stability without metric arguments, Lyapunov functions, or contraction estimates: it is a* poset-based *stability result, hinging on monotonicity and completeness. This is one of the deep reasons monotone dynamical systems are so tractable: the "shape" of the order replaces much of the analytic complexity. In particular, rather than proving that trajectories get closer in some distance, one proves that trajectories are* nested *(ascending) and therefore have a least upper bound that is automatically forward-invariant under the closure operator.*

Sketch. Take $s^* := \mathrm{Cl}_A^{(\omega)}(s)$. Thm. 12 gives that s^* is a fixed point of Cl_A, and Lemma 107 gives $s^* \geq \theta$ on the autocatalytic subset, hence nontrivial. The convergence statement is the definition of $\mathrm{Cl}^{(\omega)}$ as the join of iterates. □

Remark 1951. *The proof can be mentally decomposed into three conceptual steps: (1) define the candidate limit state by iterating reinforcement transfinitely up to ω (i.e. taking the join of all finite iterations), (2) invoke the fixed-point characterization that ω-closure yields a Cl_A-stable state, and (3) use the persistence lemma to transfer the original threshold guarantee from the seed to the limit. In this way the existence of an attractor is not an additional assumption but a consequence of the internal self-support conditions already present in the autocatalytic seed.*

Remark 1952. Proof sketch. *Define the candidate attractor as the ω-closure of the witness seed. Fixed-point theorems (as packaged in Thm. 12) guarantee it is stable under Cl_A, and the persistence lemma guarantees it retains threshold support on the autocatalytic subset.* ☐

The "convergence" here is not convergence in a metric but in the lattice sense: the sequence $s, \mathrm{Cl}_A(s), \mathrm{Cl}_A^2(s), \ldots$ is ascending, so its least upper bound is a natural limit. Visually, one can picture repeatedly flooding a directed weighted graph with support starting from s; each iteration allows support to travel one more step, and the flooded region (with weights) stabilizes at s^*. In this picture, the fixed-point condition $\mathrm{Cl}_A(s^*) = s^*$ says that once the flooding has reached s^*, another round of propagation does not expand or intensify anything further: the web has finished weaving itself.

34.5 Morphic resonance and anti-resonance as cross-context coupling

Recall (Defs. 151–153). Let L index contexts. Each $\ell \in L$ has pattern class P_ℓ, internal kernel A_ℓ, decay d_ℓ, and external input $u_{\ell,t}$. For $\ell_0 \neq \ell$, a coupling kernel is a V-relation $K_{\ell_0 \to \ell} : P_{\ell_0} \times P_\ell \to V$, with induced input $(K_{\ell_0 \to \ell} ? s_{\ell_0,t})(q)$ defined by the same image formula. The coupled update is

$$s_{\ell,t+1} = u_{\ell,t} \oplus (d_\ell \odot s_{\ell,t}) \oplus (A_\ell ? s_{\ell,t}) \oplus \bigoplus_{\ell_0 \neq \ell} (K_{\ell_0 \to \ell} ? s_{\ell_0,t}).$$

Remark 1953. *This is the formal mechanism for* morphic resonance *(Hyperseed-Concept 115) in the present algebraic setting: each context ℓ has its own internal reinforcement dynamics, but it also receives inputs induced from other contexts via coupling kernels $K_{\ell_0 \to \ell}$. The term "morphic resonance" evokes the hypothesis that patterns formed in one domain can bias pattern-formation in another [13]; here it is treated operationally as cross-context propagation of support values through a kernel, without committing to any particular physical substrate.*

Remark 1954. *Notation check: $s_{\ell,t} \in V^{P_\ell}$ is the state in context ℓ at time t; $A_\ell : P_\ell \times P_\ell \to V$ is the within-context kernel; $K_{\ell_0 \to \ell} : P_{\ell_0} \times P_\ell \to V$ is the between-context kernel, and $(K_{\ell_0 \to \ell} ? s_{\ell_0,t}) \in V^{P_\ell}$ is an induced input into the target context. The outer $\bigoplus_{\ell_0 \neq \ell}$ is a join in the state space V^{P_ℓ}, i.e. taken pointwise over patterns in P_ℓ.*

Lemma 108 (Coupling images are monotone)**.** *Fix $\ell_0 \neq \ell$. If $s \leq s'$ in $V^{P_{\ell_0}}$ and $K \leq K'$ pointwise, then*

$$K ? s \leq K' ? s' \quad \text{in } V^{P_\ell}.$$

Remark 1955. *This lemma is the cross-context analogue of monotonicity for internal propagation: if the source context has more support and/or the coupling kernel is stronger, then the induced drive into the target cannot be weaker. It is the minimal algebraic sanity condition needed for interpreting $K_{\ell_0 \to \ell}$ as a "channel" of resonance rather than as an adversarial transformer.*

Sketch. Immediate from monotonicity of $\otimes$ and of joins in the definition $(K?s)(q) = \bigoplus_p K(p,q) \otimes s(p)$. ☐

Remark 1956. Proof sketch. *For each q, compare each summand $K(p, q) \otimes s(p)$ to $K'(p, q) \otimes s'(p)$ using monotonicity of $\otimes$, then compare the joins using monotonicity of $\oplus$.* □

The proof works coordinatewise: the state space order is pointwise, so one checks the inequality separately for each target pattern $q \in P_\ell$.

Theorem 51 (Global monotonicity of morphic resonance dynamics)**.** *Define the global state space $S := \prod_{\ell \in L} V^{P_\ell}$ with pointwise order. For each t, the global update map $F_t : S \to S$ induced by Def. 153 is monotone in the global state and monotone in each parameter family $(u_{\ell,t})$, (A_ℓ), (d_ℓ), and $(K_{\ell_0 \to \ell})$.*

Remark 1957. *The theorem says that the entire coupled multi-context system preserves order: if you start from a globally larger configuration (every context has at least as much support for every pattern), then after one update step you remain globally larger. Likewise, strengthening any of the parameters (inputs, internal kernels, decays, couplings) cannot decrease the next global state. This is crucial because it lets one reason about morphic resonance by bounding and comparison, rather than by solving coupled nonlinear recurrences directly.*

Remark 1958. *It also shows that "morphic resonance" in this formal sense is compatible with the same lattice-theoretic stability toolkit used for single-context habit dynamics: the price of coupling is not the loss of monotonic structure. This is one reason quantale-based models scale naturally from individual habit loops to networks-of-contexts.*

Sketch. Each component update is a pointwise join of monotone terms: $u_{\ell,t}$, $(d_\ell \odot s_{\ell,t})$, $(A_\ell ? s_{\ell,t})$, and the coupling images. Apply Lemma 104 plus Lemma 108 and closure of monotonicity under finite products. □

Remark 1959. Proof sketch. *Prove monotonicity for each context component separately by expressing it as a join of monotone maps; then assemble these componentwise monotone maps into a product map on S, which is again monotone.* □

The key observation is compositionality: monotonicity is stable under (i) forming joins, (ii) composing monotone functions, and (iii) taking product maps. This mirrors the architectural idea that complex cognitive/behavioral dynamics can be built from individually sensible modules without forfeiting global order properties [19].

Corollary 13 (Cross-context habit-taking under monotone scalarization)**.** *Assume a monotone scalarization π and suppose a source context ℓ_0 has $s_{\ell_0,t}$ pointwise nondecreasing in t. Then for any target ℓ, each induced coupling drive $t \mapsto \pi((K_{\ell_0 \to \ell} ? s_{\ell_0,t})(q))$ is nondecreasing. Thus coupling can create the same "before/after" bias of Def. 143 across contexts, provided the target dynamics is not strongly contractive.*

Remark 1960. *This corollary says: monotone growth in a source context propagates as monotone growth of the* input signal *into other contexts, after applying a monotone scalarization. In conceptual terms, once one domain starts "taking a habit," other domains coupled to it can inherit a habit-taking bias through resonance (Hyperseed-Concept 192 is a nearby conceptual cousin, though the present coupling is algebraic rather than task-reduction based). One way to read the statement is that it is purely a claim about the* ordering *of drives: if, over time, the source state $s_{\ell_0,t}$ becomes "no smaller" in the relevant partial order, then every downstream context sees a drive that is likewise "no smaller" once we (i) push the source state through the coupling operator and (ii) read it out through a monotone π. In particular, nothing in the claim depends on a specific parametric form of the dynamics internal to the target context; it is a structural property of the cross-context* *map. Equivalently, if one thinks of $K_{\ell_0 \to \ell}?(\cdot)$ as a kind of cross-context feature transport (kernel smoothing, message passing, or aggregation), the corollary isolates the simple but practically important fact that* monotone in $\Rightarrow$ monotone out *survives both the aggregation and the final scalar readout, provided both are order-preserving.*

Sketch. Monotonicity of $s_{\ell_0,t}$ implies monotonicity of $K_{\ell_0 \to \ell}?s_{\ell_0,t}$ by Lemma 108, then apply Lemma 103. Intuitively, the first step is the order-preservation of the image operator: if $s_{\ell_0,t} \leq s_{\ell_0,t'}$ (pointwise, or in the ambient lattice order) for $t \leq t'$, then applying the same kernel cannot reverse that inequality. The second step is that a monotone scalarization π cannot reverse inequalities either, so composing the two maps preserves the time-order of the resulting scalar signal. $\square$

Remark 1961. Proof sketch. *Monotone source state $\Rightarrow$ monotone induced image (by monotonicity of the image operator) $\Rightarrow$ monotone scalar readout (by monotonicity of π).* $\square$

The final sentence of the corollary flags the obvious caveat: even if an input signal increases, a sufficiently contractive target dynamics could still damp it. The corollary isolates what is guaranteed at the level of the induced drive itself. In more operational terms, the corollary is best viewed as separating signal formation *from* signal response*. The induced drive (produced by resonance and read out by π) is guaranteed to be monotone under the stated hypotheses, but the subsequent state trajectory in context ℓ depends on the update rule in ℓ (e.g., whether it saturates, forgets quickly, thresholds sharply, or exhibits strong negative feedback). Thus the result is a clean input-side monotonicity statement, not a stability or convergence theorem for the coupled system. A useful mental model is: resonance controls what gets "broadcast" across contexts, while the internal dynamics controls what gets "stored" or "amplified." The corollary formalizes that the broadcast itself respects growth, even when storage does not.*

Anti-resonance (Def. 154, p-bit case). Assume $V = [0,1]^2$ and p-bit negation is $\neg(v^+, v^-) = (v^-, v^+)$. Define $\mathrm{AntiRes}_{\ell_0 \to \ell}(s) := K^-_{\ell_0 \to \ell}?(\neg \circ s)$. Here $(\neg \circ s)$ is the pointwise (or componentwise) negation of the source state, so that any site (or feature coordinate) carrying evidence-for/evidence-against information has those roles swapped before being transported. The notation $K^-_{\ell_0 \to \ell}$ emphasizes that the anti-resonant channel can be parametrically distinct from the resonant one, allowing, for example, different receptive fields, different strengths, or different sparsity patterns for inhibitory versus excitatory transfer. In applications, it is often helpful to regard Res and AntiRes as two separately tunable conduits: one that imports support and one that imports counter-support. This makes it possible to model cross-context influence that is not merely "the negative" of resonance, but an independently shaped form of coupling that can preferentially transmit contradiction, skepticism, or inhibitory evidence.

Remark 1962. *Anti-resonance is the formal dual of resonance: instead of importing the "positive" support of patterns from a source context, one imports a negated version (swapping evidence-for and evidence-against) and then transmits it through a (typically different) kernel K^-. This aligns with the Hyperseed notion of* Morphic Anti-Resonance *(Hyperseed-Concept 114). The choice $V = [0,1]^2$ makes explicit a paraconsistent/evidence-pair semantics, where support and counter-support can coexist rather than annihilate each other, echoing themes in paraconsistent logics [23, 24]. The evidence-pair view is especially relevant for cross-context coupling because different contexts may legitimately disagree: a pattern might be strongly supported by one context while also strongly opposed by another. Encoding both coordinates allows the model to represent such tension explicitly rather than forcing an early collapse to a single signed scalar. Anti-resonance then becomes a principled way to import "the other side" of a pattern: what was treated as supporting evidence* *upstream becomes opposing evidence downstream (and vice versa), which can be used to model inhibitory transfer, adversarial correlation, or deliberate counter-conditioning between contexts. Note also that the duality here is algebraic rather than logical negation in the classical sense: $\neg$ is an involution on $[0,1]^2$ (applying it twice returns the original pair), and it preserves the total mass $v^+ + v^-$ while swapping the allocation between pro and con. This makes it well-suited for settings where one wants to conserve "amount of evidence" but reverse its polarity.*

Lemma 109 (Negation effects on common scalarizations)**.** *For $v = (v^+, v^-) \in [0,1]^2$:*

$$\pi_{\text{pos}}(\neg v) = v^-, \qquad \pi_{\text{net}}(\neg v) = \max(0,\, v^- - v^+).$$

Hence if $v^+ > v^-$ then $\pi_{\text{net}}(\neg v) = 0$ (net suppression).

Remark 1963. *This lemma simply computes what happens to two natural scalar readouts under evidence-pair negation. The first readout π_{pos} treats "positive evidence" as the scalar propensity, so negation swaps it to the negative coordinate. The second readout π_{net} treats only net-positive bias as salient; after negation, a pattern that was net-supported becomes net-suppressed (clipped to* 0*), capturing the idea that anti-resonance can selectively cancel rather than merely invert. It is worth emphasizing the behavioral difference between these two choices. Under π_{pos}, anti-resonance does not necessarily* decrease *the scalar signal; it reinterprets which coordinate is considered "positive" and therefore may yield a large scalar output whenever the source carried large counter-support. Under π_{net}, by contrast, the scalarization is explicitly one-sided: only* net *support survives, so swapping coordinates tends to turn net support into net opposition, which is then clipped away. Concretely, if $v = (0.9, 0.2)$ (strongly supported, weakly opposed), then $\neg v = (0.2, 0.9)$ and $\pi_{\text{net}}(\neg v) = \max(0, 0.9 - 0.2) = 0.7$ while $\pi_{\text{net}}(v) = \max(0, 0.9 - 0.2) = 0.7$ would not be correct since $\pi_{\text{net}}(v) = \max(0, 0.9 - 0.2) = 0.7$ and after negation becomes* $\max(0, 0.2 - 0.9) = 0$ *if the coordinates are interpreted as (v^+, v^-); this illustrates exactly why the swap matters: anti-resonance can extinguish what resonance would amplify, depending on which coordinate is treated as "for" versus "against" at readout time. From a modeling standpoint, this offers two distinct inhibitory mechanisms: (i) re-routing activation into a channel that the downstream system treats as negative evidence (via π_{pos}), or (ii) enforcing a strict "no drive unless net-positive" policy (via π_{net}), which yields genuine suppression in the presence of dominant counter-evidence.*

Sketch. Immediate from Def. 141 and the definition of p-bit negation (swap components). More explicitly: $\neg(v^+, v^-) = (v^-, v^+)$, so $\pi_{\text{pos}}(\neg v)$ returns the first coordinate of (v^-, v^+), namely v^-. Likewise, $\pi_{\text{net}}(\neg v) = \max(0, (v^-) - (v^+))$ by direct substitution into the net readout. The "net suppression" condition is then the observation that if $v^+ > v^-$, the quantity $(v^- - v^+)$ is negative and is clipped to 0 by the outer max. □

Remark 1964. Proof sketch. *Compute $\pi_{\text{pos}}(v^-, v^+) = v^-$ and $\pi_{\text{net}}(v^-, v^+) = \max(0, v^- - v^+)$ directly.* □

The "net suppression" clause is a useful interpretive guide: if a pattern is more supported than opposed, then anti-resonance followed by π_{net} yields no positive drive at all. In applications, this corresponds to a channel that can inhibit without generating spurious positive activation. A further way to see the same point is that π_{net} is intentionally asymmetric: it treats opposition as something that can only remove *drive, not create negative drive. Thus, after anti-resonance, any configuration in which the original v^+ dominated v^- becomes one in which the swapped pair has dominant second coordinate, producing a negative net that is discarded. This implements an "inhibit-only" effect at the scalar level, which is often desirable when downstream modules interpret scalar drives as nonnegative activation intensities rather than signed quantities. This also clarifies* *why anti-resonance need not be viewed as merely producing an "opposite habit." With π_{net}, anti-resonance can act as a* gate *that blocks transfer of strongly supported patterns while still allowing transfer of patterns that were originally contested (e.g., with $v^- \geq v^+$, where anti-resonance yields a nonzero net-positive signal). In that sense, the anti-resonant channel can be tuned to preferentially propagate cautionary or adversarial evidence across contexts.*

34.6 Stability, amplification, and decay in the p-bit toy quantale

Recall (Def. 155, Thm. 13). Assume $V = [0,1]^2$ with $\oplus = \max$ and $\otimes = \cdot$ componentwise. Define $|v|_\infty := \max(v^+, v^-)$ and extend to states/relations by maxima over entries. For homogeneous, no-input single-context updates $s_{t+1} = (d \odot s_t) \oplus (A?s_t)$, let $\rho := \max(|d|_\infty, |A|_\infty)$.

Remark 1965. *This "toy quantale" is deliberately concrete: $\oplus = \max$ means supports accumulate by taking the strongest available contributor, while $\otimes = \cdot$ means chaining along a link multiplies strengths. The ∞-norm $|v|_\infty$ measures the largest component of the evidence pair; extending it by maxima over entries yields a coarse but convenient global magnitude measure for the entire state or kernel. The parameter ρ is thus a crude upper bound on the strongest single-step amplification available either from persistence (d) or from propagation (A).*

Corollary 14 (Exponential decay bound (unrolling Thm. 13))**.** *Under the assumptions of Thm. 13, for all $t \geq 0$,*

$$|s_t|_\infty \leq \rho^t \, |s_0|_\infty.$$

In particular, if $\rho < 1$ then $|s_t|_\infty \to 0$ exponentially fast.

Remark 1966. *This corollary says that when the system has no external input, and the strongest possible single-step amplification is strictly less than 1, then all support must eventually vanish: forgetting dominates reinforcement. The result is the discrete-time analogue of linear stability under a spectral radius bound, but it is proved here purely with max-product inequalities rather than linear algebra. Its role in the overall narrative is to show that habit formation requires either sufficiently strong internal loops ($\rho \geq 1$) or sustained external/coupled inputs; otherwise decay wins.*

Sketch. Thm. 13 gives $|s_{t+1}|_\infty \leq \rho \, |s_t|_\infty$. Iterate the inequality. □

Remark 1967. Proof sketch. *Apply the one-step contraction estimate repeatedly: each step multiplies the bound by at most ρ, so after t steps the bound is ρ^t times the initial magnitude.* □

The geometric picture is a shrinking envelope: $|s_t|_\infty$ lies inside a ball whose radius decays like ρ^t. Even though the underlying dynamics uses max *and products rather than sums, the inequality behaves like the familiar "multiply by ρ each step" recurrence.*

Lemma 110 (Input-bounded contraction estimate)**.** *In the same p-bit setting, for the inhomogeneous update $s_{t+1} = u_t \oplus (d \odot s_t) \oplus (A?s_t)$, one has*

$$|s_{t+1}|_\infty \leq \max\big(|u_t|_\infty,\ \rho \, |s_t|_\infty\big).$$

If $\rho < 1$ and $|u_t|_\infty \leq U$ for all t, then

$$|s_t|_\infty \leq \max\big(\rho^t |s_0|_\infty,\ U\big),$$

so the long-run magnitude is controlled by the sustained input scale.

Remark 1968. *This lemma makes explicit the tradeoff between contraction and drive. If the internal dynamics is contractive ($\rho < 1$), then past support fades unless new support is injected via u_t. The inequality says precisely that the system's magnitude is bounded by whichever is larger: the current input magnitude or the contracted previous magnitude. Thus, even a forgetful system can maintain nontrivial long-run support if it is continually stimulated.*

Sketch. Use the same bounding steps as in the proof of Thm. 13 on the two terms $(d \odot s_t)$ and $(A?s_t)$, and note that $\oplus$ is componentwise max so the ∞-norm of a join is the max of the norms. Then unroll the recurrence by induction. □

Remark 1969. Proof sketch. *Bound the contribution of each join-summand to $|s_{t+1}|_\infty$, observe that max-join turns into an outer max on norms, and then iterate the resulting scalar recurrence.* □

The essential step is that in the max-product setting, norms behave well with respect to both $\oplus$ and $\otimes$: max does not create larger magnitudes than its largest input, and componentwise multiplication scales magnitudes multiplicatively. This makes the inequality almost inevitable once the norm is chosen.

Theorem 52 (Driven closure attractor as a Kleene-star closure)**.** *Assume $d = e$ and $u_t \equiv u$ is constant (Prop. 15 main text), and assume P finite and V complete. Let $s_{t+1} = u \oplus s_t \oplus (A?s_t)$ and define $s_\infty := \bigoplus_{t \geq 0} s_t$. Then*

$$s_\infty = A^*?(s_0 \oplus u) = \mathrm{Cl}_A^{(\omega)}(s_0 \oplus u).$$

In particular, if $s_0 \leq u$ then $s_\infty = A^*?u$.

Remark 1970. *This theorem identifies the long-run accumulated state under constant driving and no decay: it is exactly the closure of the combined seed $(s_0 \oplus u)$ under the reinforcement kernel A, equivalently the Kleene-star propagation $A^*?(s_0 \oplus u)$. The conceptual message is that constant input u behaves like an ever-present set of "roots" from which reinforcement chains can grow. In the long run, everything reachable (in the graded sense determined by V) from these roots will be activated to the extent permitted by A^*.*

Remark 1971. *The special case $s_0 \leq u$ is philosophically neat: if the system's initial support is already contained in what the environment continuously supplies, then the eventual attractor depends only on u and the internal kernel A. In that regime, "who the system was" matters less than "what it is being fed" plus "how it internally propagates." This resonates with the broader theme that sustained conditions can sculpt stable habit structure.*

Sketch. Since $d = e$, the update is inflationary (Lemma 105), so the join s_∞ exists. In particular, for the recurrence $s_{t+1} = F(s_t)$ given in this toy quantale update, inflationarity means $s_t \leq F(s_t) = s_{t+1}$ for every t, i.e. $(s_t)_{t \geq 0}$ is an ascending chain. Because we are working in a join-semilattice (indeed, the ambient setting provides the relevant completeness for the construction), the supremum of this chain exists, and we may write

$$s_\infty = \bigvee_{t \geq 0} s_t,$$

which is the least upper bound of all iterates.

After one step, $s_t \geq u$ for all $t \geq 1$, so the dynamics is pure closure starting from $s_0 \oplus u$: indeed $s_1 = F(s_0)$ contains the joined-in input u, and since the update always joins u again, once $u \leq s_1$

we have $u \leq s_t$ for all $t \geq 1$ by monotonicity and the ascending-chain property. Consequently, for $t \geq 1$ we may simplify the driven update by absorbing the constant drive into the state:

$$s_t \oplus u = s_t \quad (\text{because } u \leq s_t),$$

so the recurrence reduces to iterating the internal closure operator alone, with the input acting only through the choice of initial seed.

iterating yields $s_\infty = \mathrm{Cl}_A^{(\omega)}(s_0 \oplus u)$. Here $\mathrm{Cl}_A^{(\omega)}$ denotes the ω-iteration (countable iteration) of the one-step closure induced by A, i.e. the least fixed point above the seed $s_0 \oplus u$ obtained by repeatedly applying Cl_A and taking the join of all finite stages. Equivalently, s_∞ is characterized as the least element s such that $s_0 \oplus u \leq s$ and $\mathrm{Cl}_A(s) \leq s$, which matches the usual closure-as-least-fixed-point intuition in this quantale setting.

Apply Thm. 49. Concretely, this theorem supplies the explicit "Kleene star" description of the ω-closure in terms of the star A^* (built from the joins of finite powers of A), so that the limiting state can be written in the closed form provided there, with the dependence on the initial condition appearing only through the seed $s_0 \oplus u$. □

Remark 1972. Proof sketch. *Show the sequence is ascending (no decay, join with s_t), so the join limit exists. Observe that constant input can be absorbed into the initial seed after one step, reducing the driven recurrence to repeated closure. Then use the explicit A^* representation of* $\mathrm{Cl}_A^{(\omega)}$*.*
□

Show the sequence is ascending (no decay, join with s_t), so the join limit exists. Observe that constant input can be absorbed into the initial seed after one step, reducing the driven recurrence to repeated closure. Then use the explicit A^* representation of $\mathrm{Cl}_A^{(\omega)}$*. In slightly more detail: inflationarity gives $s_0 \leq s_1 \leq s_2 \leq \cdots$, hence the limit is computed by a join over all finite times. The absorption step is the observation that the "drive" u is not merely added once, but re-added at every step; after the first update it is therefore redundant to keep writing it explicitly, since joining it again does not change the state once u is already below the current iterate. This is what turns a driven system into an autonomous closure iteration without changing the eventual supremum.*

The key step is the absorption of u: because u is joined in at every step, once it has appeared it never disappears, and thus it may be treated as part of the seed for the closure dynamics. A graph picture helps: imagine adding a permanent set of "source nodes" that are always active at level u; the eventual activation pattern is exactly what you get by taking the transitive closure of influence from those sources through the internal edges A. Equivalently, one can regard A as encoding one-step propagation rules, so that applying Cl_A *once adds all consequences reachable in one propagation layer, applying it twice adds all consequences reachable in two layers, and so on; the ω-iteration collects all finite propagation depths. The A^* formula packages this iterative expansion into a single expression (a join of A^n over all $n \geq 0$ acting on the seed), making the roles of "amplification" (new reachability through repeated application of A) and "stability" (eventual fixed-point behavior under closure) manifest.*

35 Helper theorems for mind, representation, and perception (Section 13)

Standing notation. Let U be the ambient universe of entities. Let $(\mathrm{Ctx}, \preceq)$ be the poset of contexts (Def. 159), with $C \preceq D$ meaning "D refines C". For each $C \in \mathrm{Ctx}$, let $\mathrm{asp}_C : U \to U_C$ be the aspect extraction map, and for $C \preceq D$ let $\pi_{D \to C} : U_D \to U_C$ satisfy

$$\mathrm{asp}_C = \pi_{D \to C} \circ \mathrm{asp}_D.$$

Write $x|_C := \mathrm{asp}_C(x)$. Let V be the quantale/evidence domain used for intensities (Section 3), hence $(V, \leq)$ is a complete lattice.

Fix a context C and a finite pattern basis $B_C \subseteq P_C$. The pattern signature of an entity X in aspect C is

$$\mathrm{Sig}_C(X) : B_C \to V, \quad \mathrm{Sig}_C(X)(P) := \mathrm{Int}_X(P)$$

(Def. 164), and strong inheritance in aspect C is $\mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(B|_C)$ pointwise (Def. 165).

Remark 1973. *It is worth pausing over the meaning of the context machinery, because it is easy to be hypnotized by the notation. A context $C \in \mathrm{Ctx}$ should be read as a disciplined way of asking* which aspects of an entity we are currently attending to *(Hyperseed-Concept 86). The map $\mathrm{asp}_C : U \to U_C$ is then a kind of "lens" or "projection": given an entity $x \in U$, the symbol $x|_C$ means "x as seen through context C". The order $C \preceq D$ says that D is a* refinement *of C (more detailed, less coarse), and the map $\pi_{D\to C} : U_D \to U_C$ is the disciplined act of* forgetting detail *when passing from D back to C. The coherence equation $\mathrm{asp}_C = \pi_{D\to C} \circ \mathrm{asp}_D$ is precisely the statement that "extracting the coarse aspect directly" equals "extracting a finer aspect and then forgetting". This style of contextualization is central in the Hyperseed ontology presentation [1].*

Remark 1974. *Two further pieces of notation carry much of the weight in later lemmas. First, V is not merely an ordered set but a complete lattice: the phrase "joins exist" will repeatedly mean that for* any *family of intensities $\{v_i\} \subseteq V$ there is a least upper bound $\bigvee_i v_i \in V$. Second, $\mathrm{Sig}_C(X) : B_C \to V$ is best read as a finite "feature vector" (indexed by the basis patterns B_C), recording how intensely each basis pattern is present in X in aspect C (Hyperseed-Concept 130). Strong inheritance $\mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(B|_C)$ is then simply* coordinatewise domination*: every basis pattern appears in A no more strongly than it appears in B (Hyperseed-Concept* **??***). The quantale-based treatment of graded structure and evidence connects to the broader discussion of quantalic semantics in [3].*

Remark 1975. *A useful way to keep the arrows straight is to read U_C as an "observation space" associated to C. If one informally thinks of an entity $x \in U$ as having many potentially relevant degrees of freedom, then $x|_C \in U_C$ packages exactly those degrees of freedom that are expressible (or admissible) under C. Refinement $C \preceq D$ can then be interpreted as saying that D supports* more discriminations *than C: distinct elements of U_D may become identified after applying $\pi_{D\to C}$, because the coarser context is unable (or unwilling) to represent the difference. In many applications, one additionally expects the projection maps to compose as expected, i.e. for $C \preceq D \preceq E$ one has $\pi_{E\to C} = \pi_{D\to C} \circ \pi_{E\to D}$; when available, this expresses the idea that "forgetting in stages" is equivalent to "forgetting at once". Even when such compositionality is taken as background structure rather than foreground notation, it explains why the poset language is natural: contexts behave like levels of description ordered by informational content.*

Remark 1976. *The finiteness of the basis B_C is conceptually important even when it is not technically essential for a given lemma. It means that $\mathrm{Sig}_C(X)$ can be manipulated like an honest finite vector of evidence values, so pointwise comparisons, finite joins, and coordinatewise bounds are all concrete operations rather than schematic ones. In particular, the pointwise order on signatures is itself a complete lattice order inherited from V: for signatures $s, t : B_C \to V$ one has $s \leq t$ iff $s(P) \leq t(P)$ for all $P \in B_C$, and joins/meets are computed coordinatewise. This makes strong inheritance behave like the ordinary preorder on feature vectors used in many representation-learning settings, with the difference that the codomain V is allowed to carry non-Boolean and non-numeric evidence structure (as provided by the quantale apparatus). Viewed this way, strong inheritance is immediately reflexive and transitive at each fixed context C, since it is just the inherited preorder from the product order on V^{B_C}.*

Remark 1977. *Although the definition of $\mathrm{Sig}_C(X)$ is stated at a fixed context C, the refinement maps suggest a natural way to relate representations across contexts: if $C \preceq D$, then $x|_D$ contains at least as much descriptive material as $x|_C$, and $\pi_{D\to C}(x|_D) = x|_C$ ensures that any statement formulated purely in the coarser aspect is invariant under passage to the finer one. Later helper theorems will implicitly exploit this invariance pattern: one proves an inequality or entailment at a refined level D and then uses $\pi_{D\to C}$ (together with the coherence equation) to conclude a corresponding statement at the coarser level C. This is one of the main technical reasons to insist on explicit aspect maps rather than treating "context" as informal prose.*

35.1 System graphs and inter-production

Lemma 111 (No dead-end processes in inter-producing systems)**.** *Let $S = (P, E)$ be a system graph (Def. 156). If a node $p \in P$ lies on a directed cycle, then p has at least one incoming edge and at least one outgoing edge. Consequently, if S is inter-producing (every node lies on a directed cycle), then every node has in-degree ≥ 1 and out-degree ≥ 1.*

Remark 1978. *This lemma says that "being part of a genuine feedback loop" excludes isolation. A directed cycle is the graph-theoretic shadow of a process of mutual enabling: if p participates in a cycle, something leads into it and something proceeds out of it, so p cannot be a terminal dead-end nor a pure source. In inter-producing systems (every node is on some cycle),* every *process is both supported by others and supports others—a minimal formal trace of autocatalytic organization (Hyperseed-Concept 61). In the broader narrative about pattern webs and self-sustaining process networks, this is one of the simplest invariants one can prove; compare the general pattern-web viewpoint in [5].*

A small but useful clarification is that the statement is purely graph-theoretic*: it does not assume any additional algebraic structure on processes, nor any notion of quantitative flow. The conclusion only uses the existence of a cycle edge immediately "before" and "after" p along some directed cycle. In particular, even if p has many other incident edges (extra inputs/outputs), cycle membership already forces at least one of each direction. This is why the conclusion is phrased as a lower bound on in-degree and out-degree rather than an exact characterization.*

Note also that the lemma accommodates degenerate cycles when the underlying definition allows them. For instance, if there is a self-loop edge $(p, p) \in E$, then p lies on a directed cycle of length 1, and the same edge simultaneously contributes an incoming and an outgoing edge at p (so the conclusion still holds). Likewise, if the formalism allows multiple edges, then "at least one" should be read in the multigraph sense: there exists at least one directed edge with head p and at least one directed edge with tail p.

Conceptually, this lemma is the minimum consistency requirement for the slogan "inter-producing": if every node lies on a directed cycle, then there are no sinks (out-degree 0) and no sources (in-degree 0). This is weaker than strong connectivity of the whole graph (the graph may decompose into multiple cyclic components), but it is stronger than merely requiring that each node is reachable from itself by some trivial path: it enforces a nontrivial *return via directed edges.*

Sketch. On a directed cycle containing p, there is an edge entering p from the previous node and an edge leaving p to the next node. More explicitly, if the cycle is written as a sequence of vertices

$$p_0 \to p_1 \to \cdots \to p_{k-1} \to p_k = p_0,$$

and if $p = p_i$ for some i, then the edge $p_{i-1} \to p_i$ (indices modulo k) witnesses an incoming edge to p and the edge $p_i \to p_{i+1}$ witnesses an outgoing edge from p. Thus $\deg^-(p) \geq 1$ and $\deg^+(p) \geq 1$.

The "consequently" clause follows immediately by applying the same argument to each node when every node lies on some directed cycle. □

Remark 1979. Proof sketch. *The strategy is to unpack the definition of "directed cycle": it is literally a closed chain of directed edges. Any node on such a chain has a predecessor (giving an incoming edge) and a successor (giving an outgoing edge). The "consequently" part then just applies this fact to every node when the system is inter-producing.* □

The key step is that cycle membership is local*: to know degrees at p you only need the two adjacent edges of the cycle at p. Geometrically, picture tracing the arrows around the loop: you must enter p and you must leave p in order to keep moving and return.*

One can also rephrase the lemma in terms of the absence of one-way "boundary" processes. If we interpret edges as immediate enabling relations between processes, then a sink process would "consume" enabling without enabling anything in return, and a source process would "enable" without being enabled. The lemma shows that inter-production forbids both extremes, guaranteeing that each process is embedded in at least one immediate two-sided dependency pattern (incoming and outgoing). This does not yet guarantee global resilience or closure under perturbations, but it is the first necessary graph condition one checks before asking stronger questions (e.g. about strongly connected components, minimal feedback vertex sets, or the existence of larger autocatalytic subsets in the sense of 61).

Finally, the lemma is often used contrapositive-style: if a node has in-degree 0 or out-degree 0, then it cannot lie on any directed cycle, hence the system cannot be inter-producing. In practice, this gives a quick obstruction test when analyzing candidate system graphs.

35.2 Contexts, aspects, and contextualization

Lemma 112 (Aspect coherence under refinement)**.** *If $C \preceq D$, then for any $x \in U$,*

$$\pi_{D\to C}(x|_D) = x|_C.$$

Equivalently, extracting a refined aspect and then forgetting is the same as extracting the coarse aspect.

Remark 1980. *This is the formal expression of a commonplace methodological constraint: "if you look at something in high resolution and then blur it, you should get what you would have seen if you had looked in low resolution from the start." The equation $\pi_{D\to C}(x|_D) = x|_C$ is exactly the commutativity that makes refinement coherent rather than arbitrary. It is also what allows one to compare observations made in different contexts without treating them as incommensurable (Hyperseed-Concept 86). In particular, the lemma should be read as a* compatibility condition *between two kinds of operations: (i) changing the observational granularity (moving from C to D) and (ii) forgetting detail (moving back from D to C). If this compatibility failed, then the very meaning of "the aspect of x in context C" would depend on which refined context one passed through on the way to C, and the poset structure on contexts would not behave like a genuine refinement relation.*

Sketch. Immediate from $\mathrm{asp}_C = \pi_{D\to C} \circ \mathrm{asp}_D$ (Def. 159). Concretely, evaluate both sides at $x \in U$ and then unfold the notation $x|_C := \mathrm{asp}_C(x)$ and $x|_D := \mathrm{asp}_D(x)$. □

Remark 1981. Proof sketch. *One simply substitutes the definition $x|_D = \mathrm{asp}_D(x)$ and then applies the coherence axiom for* asp *and π.* □

The proof works because Def. 159 already encodes the intended semantics: $\pi_{D\to C}$ is defined *to be the map that makes the aspect extractions commute. Visually, one has a commutative triangle $U \xrightarrow{\mathrm{asp}_D} U_D \xrightarrow{\pi_{D\to C}} U_C$ and $U \xrightarrow{\mathrm{asp}_C} U_C$, and the lemma is merely the pointwise commutativity of that triangle. An equivalent way to read the same triangle is as a "change of coordinates" principle: U_D may carry strictly more information than U_C, but $\pi_{D\to C}$ provides the canonical comparison map, so that any statement about $x|_C$ can be checked either directly in U_C or indirectly via U_D without ambiguity. This is the minimal categorical content needed later when one composes several refinements $C \preceq D \preceq E$ and wants the resulting forgetful map $\pi_{E\to C}$ to behave consistently with $\pi_{E\to D}$ and $\pi_{D\to C}$.*

Lemma 113 (Contextualized groupings commute with forgetting)**.** *Let $G \subseteq U$ be a grouping (Def. 160). If $C \preceq D$, then*

$$\pi_{D\to C}(G|_D) = G|_C,$$

where $\pi_{D\to C}$ is applied elementwise to the set $G|_D \subseteq U_D$.

Remark 1982. *A grouping G is just a set of entities, but contextualization turns it into a set of* aspects *(Hyperseed-Concept* **??***). The lemma says that, just as individual aspect extraction commutes with forgetting (Lemma 112), the same remains true for a whole set: "contextualize in the refined context and then forget" equals "contextualize directly in the coarse context". This matters because many constructions in perception and representation concern sets (clusters, categories, collections of candidate referents) rather than single entities. Note that the statement is not merely about inclusions but about equality of the resulting contextualized collections: no element appears or disappears depending on whether one contextualizes before or after forgetting. At the level of methodology, this expresses that one can form a hypothesis class G at the "world" level U and then pass to any context without worrying that the bookkeeping depends on an arbitrary choice of intermediate refinements.*

Sketch. By definition $G|_D = \{x|_D : x \in G\}$ and $G|_C = \{x|_C : x \in G\}$. Apply Lemma 112 elementwise. Equivalently, show mutual inclusion by taking an arbitrary member of one side, rewriting it as $\pi_{D\to C}(x|_D)$ for some $x \in G$, and then identifying it with $x|_C$. □

Remark 1983. Proof sketch. *Rewrite both sides as sets defined by comprehension, then use Lemma 112 for each element $x \in G$ to identify the images.* □

The key step is that $\pi_{D\to C}$ is being applied elementwise, so set equality reduces to pointwise equality of members. Geometrically, imagine a cloud of points G in U; projecting the cloud into U_D and then further into U_C yields the same cloud in U_C as projecting directly. One can also view the lemma as the statement that the operation $(-)|_C$ behaves functorially on inclusions of contexts: it respects the refinement maps in the sense that the diagram of sets

$$G \subseteq U, \qquad G|_D \subseteq U_D, \qquad G|_C \subseteq U_C$$

commutes with the corresponding aspect-extraction and forgetting maps. This seemingly small observation is what later permits one to define context-dependent predicates or measures on groupings (e.g. "similarity" or "coherence" computed in U_C) without having to distinguish the route taken to arrive at U_C.

Lemma 114 (Monotonicity of contextualization)**.** *If $G \subseteq H \subseteq U$, then for all contexts C,*

$$G|_C \subseteq H|_C.$$

Remark 1984. *This monotonicity lemma is a sanity condition: if one collection is contained in another before contextualization, it remains contained after contextualization. It is formally trivial but conceptually important, because it means contextualization does not create spurious "new members"; it only changes the* view *of existing ones. In later reasoning, it permits order-theoretic arguments (inclusions, refinements) to pass cleanly through* $(-)|_C$ *(Hyperseed-Concept 86 and* **??***). A useful way to internalize the statement is: contextualization is an* order-preserving *map from* $\mathcal{P}(U)$ *to* $\mathcal{P}(U_C)$*. Thus, any reasoning step of the form "restrict to a smaller candidate set" remains valid after passing to any context* C*.*

Sketch. Apply asp_C elementwise to the inclusion $G \subseteq H$. More explicitly, if $y \in G|_C$ then $y = \mathrm{asp}_C(x)$ for some $x \in G$; since $x \in H$ as well, it follows that $y \in H|_C$. ☐

Remark 1985. Proof sketch. *Take any* $y \in G|_C$*. Then* $y = x|_C$ *for some* $x \in G$*. Since* $G \subseteq H$*, we also have* $x \in H$*, hence* $y \in H|_C$*.* ☐

The proof works because $G|_C$ *is defined as an image of* G *under* asp_C*. Images of subsets under a function are always subsets of the corresponding images. It is worth noting what is* not *claimed: one does not generally have* $G|_C = H|_C$ *when* $G \subseteq H$*, since distinct entities in* H *may collapse to the same aspect in context* C *(many-to-one* asp_C*), but the inclusion remains valid regardless of such collapses. This is precisely the kind of monotonicity one needs when later taking unions, intersections, or iterative refinements of candidate sets and then transporting them across contexts.*

35.3 Recognition, registration, and perception

Lemma 115 (Supremum of learned intensities exists)**.** *Fix a context/aspect* C *and a pattern recognition process* Y *aimed at* X *in aspect* C *(Def. 158), witnessed by a nonempty* $P_{X,C} \subseteq P_C$*. For each* $P \in P_{X,C}$*, the set* $\{\mathrm{Int}_{Y,t}(P) : t \in T\}$ *has a join (supremum) in* V*:*

$$\mathrm{Int}_Y^{\infty}(P) := \bigvee_{t \in T} \mathrm{Int}_{Y,t}(P) \ \in V.$$

Moreover, by alignment, for all relevant t*,*

$$\mathrm{Int}_Y^{\infty}(P) \ \leq \ \bigvee_{t \in T} \mathrm{Int}_{X,t}(P),$$

and if $\mathrm{Int}_{X,t}(P)$ *is constant in* t *then* $\mathrm{Int}_Y^{\infty}(P) \leq \mathrm{Int}_X(P)$*.*

Remark 1986. *A recognition process* Y *(Hyperseed-Concept 151) produces a time-indexed family of intensities* $\mathrm{Int}_{Y,t}(P)$ *measuring how strongly* Y *has come to "have" pattern* P *by time* t*. The first part of the lemma guarantees that we can speak coherently about a limiting or saturated intensity* $\mathrm{Int}_Y^{\infty}(P)$ *obtained by taking the join over time. This is not a probabilistic limit but an order-theoretic one: it is simply the least upper bound of everything* Y *ever reaches. The only real hypothesis needed is that* V *is a complete lattice. In particular, no assumption is made that* $t \mapsto \mathrm{Int}_{Y,t}(P)$ *is monotone, convergent, or even continuous in any analytic sense:* $\bigvee_{t \in T}$ *exists regardless, and is meaningful precisely because it records the best attainment of* Y *across the entire history* T*. If one prefers a concrete picture, in the common case* $T = \mathbb{N}$ *and* $V = [0,1]$ *with the usual order,* $\mathrm{Int}_Y^{\infty}(P)$ *is just* $\sup_{t \in \mathbb{N}} \mathrm{Int}_{Y,t}(P)$*, i.e. the maximum recognition strength achieved in the limit (whether or not it stabilizes). The requirement that* $P_{X,C}$ *be nonempty is part of the usual "aimed at" condition: the claim is not that* Y *learns arbitrary* $P \in P_C$*, but that for every pattern* P *that witnesses the aiming relationship, a well-defined "eventual" intensity can be formed.*

Remark 1987. *The second part says that learning cannot outrun its target: alignment bounds Y's intensity by X's intensity, pointwise in time. Thus the join over t for Y is bounded by the join over t for X. In the special case where X's intensity is time-invariant, we recover the intuitive statement that "the learner's eventual recognition strength cannot exceed the source's actual strength". This sort of pattern-centered view of learning and recognition is consonant with the pattern/emergence framework of [5]. One can also view the inequality as a minimal "soundness" condition for recognition: whatever internal degrees of pattern-presence Y develops must remain compatible with the external (or target) degrees of pattern-presence attributed to X in the same context. Even when $\mathrm{Int}_{X,t}(P)$ varies with t (e.g. due to changing illumination, occlusion, or context drift), the bound by $\bigvee_t \mathrm{Int}_{X,t}(P)$ says that Y's best-ever performance is still constrained by the best-ever availability of the pattern in X.*

Sketch. V is a complete lattice, so joins of subsets exist. Alignment gives $\mathrm{Int}_{Y,t}(P) \leq \mathrm{Int}_{X,t}(P)$ pointwise in t; taking joins over t preserves $\leq$. In more detail: $\mathrm{Int}_Y^\infty(P)$ is, by definition, an upper bound of $\{\mathrm{Int}_{Y,t}(P)\}_t$, and it is the *least* such upper bound; since each $\mathrm{Int}_{X,t}(P)$ is an upper bound for the corresponding $\mathrm{Int}_{Y,t}(P)$, the join over t of the X-intensities is an upper bound for the entire Y-family, and thus dominates the least upper bound $\mathrm{Int}_Y^\infty(P)$. □

Remark 1988. Proof sketch. *Existence of $\mathrm{Int}_Y^\infty(P)$ is pure lattice theory: completeness of $(V, \leq)$ provides $\bigvee_{t\in T}$. The inequality is proved by monotonicity of joins: if every term of one family is $\leq$ the corresponding term of another, then the join of the first family is $\leq$ the join of the second.* □

The key step is the passage "pointwise in t" $\Rightarrow$ "after taking $\bigvee_t$". This is exactly the order-theoretic analogue of taking suprema of bounded functions. If one pictures intensities as heights, alignment says Y's graph lies below X's graph; taking the supremum simply picks the highest point reached, and a curve lying below another cannot have a higher supremum. A useful corollary intuition is that $\mathrm{Int}_Y^\infty(P)$ is insensitive to transient lapses: if Y briefly attains a high recognition of P and later forgets, the join still remembers that achievement. Conversely, if one wants a notion of "stable" recognition, one would replace $\bigvee_t$ by a different temporal aggregate (e.g. an infimum over a tail, or a meet over a window); the present lemma isolates the simplest saturation operator compatible with the lattice structure.

Lemma 116 (Recognition plus threshold crossing yields registration)**.** *Fix a context C and an equivalence relation $\sim_C$ on U_C as in Def. 161. Assume Y is a pattern recognition process aimed at A in aspect C (Def. 158), and suppose there exists a time t and a pattern $P \in P_C$ such that:*

$$\mathrm{Int}_{Y,t}(P) \not\leq \theta \quad \textit{and} \quad \mathrm{Ref}_C(P, A|_C) \textit{ is nontrivial.}$$

Then Y registers A at time t in context C (Def. 161).

Remark 1989. *Registration (Hyperseed-Concept 153) is the moment where recognition ceases to be merely internal fluctuation and becomes an "event" in the ontology of the mind: something has been recognized* as *something, in a way stable enough to pass a threshold. The lemma states that two ingredients suffice: (i) some pattern intensity crosses the required threshold, and (ii) that pattern actually refers to (an aspect of) A. The equivalence relation $\sim_C$ (Hyperseed-Concept* **??***) encodes the idea that, within a context, registration is typically defined only up to some appropriate sameness criterion. It is worth emphasizing that the threshold clause $\mathrm{Int}_{Y,t}(P) \not\leq \theta$ is phrased in the ambient order on V, so it covers both numerical thresholds (e.g. $\mathrm{Int}_{Y,t}(P) > \theta$ when V is linearly ordered) and more general partially ordered notions (e.g. multiple independent resources or evidential dimensions). Likewise, "nontrivial" reference is the minimal semantic anchor needed to rule out degenerate patterns that do not actually pick out $A|_C$ in the referential scheme of the context.*

Remark 1990. *One can also read the statement as a bridge principle between three layers: recognition-strength (*Int*), semantic anchoring (*Ref_C*), and the discrete predicate "registers". In semiotic terms, it tells us that intensity without reference is blind, and reference without intensity is idle. This fits naturally with the representation-centric approach to cognition emphasized in [19]. The role of Def. 161 is to package this bridge into an existential condition, so the lemma is intentionally "low-tech": it does not require uniqueness of P, agreement among multiple patterns, or any temporal persistence beyond the single witnessing time t. Such stronger notions (e.g. robust registration, consensus registration, or temporally sustained registration) can be layered on top by strengthening the hypotheses, while the present lemma isolates the minimal trigger condition that turns graded recognition into an ontological registration event.*

Sketch. Take $A' = A$ in Def. 161; since $\sim_C$ is an equivalence relation it is reflexive, so $A|_C \sim_C A|_C$. The given P witnesses that $A|_C$ "becomes a pattern in Y by time t" with nontrivial reference. Concretely, Def. 161 asks for (a) an aspect-representative A' equivalent to A in context C, and (b) some pattern whose intensity exceeds the threshold and whose reference connects it to $A'|_C$; both are supplied immediately by setting $A' = A$ and using the hypothesized P, t. $\square$

Remark 1991. Proof sketch. *The proof is a direct match to the definition of registration: choose the trivial representative $A' = A$ of the equivalence class, use reflexivity of $\sim_C$ to satisfy the "same-aspect" clause, and use the given pattern P to satisfy the threshold-and-reference clause.* $\square$

The essential move is recognizing that registration is defined existentially: it suffices to provide one witnessing pattern P and one time t. Conceptually, the reflexivity of $\sim_C$ guarantees that "registering A" is always allowed to mean "registering something equivalent to A"; choosing A itself is then the simplest witness. This also highlights why the lemma does not require any global relationship between P and the entire witness-set $P_{A,C}$ from Def. 158: registration can occur via any pattern in the ambient repertoire P_C so long as it both (i) achieves sufficient intensity in Y at some moment and (ii) carries nontrivial reference to the relevant aspect. In applications, P might be a low-level feature (edge, tone, token) or a high-level gestalt, and the equivalence relation $\sim_C$ allows the registration event to ignore inessential variation (e.g. viewpoint changes) while still counting as "the same" registered target within C.

Lemma 117 (Perception yields both registration and a referential token)**.** *If a mind M perceives A at time t in context C (Def. 163), then:*

1. *M registers A at time t in context C (Prop. 16), and*
2. *there exists a representation token $R \in \mathcal{R}$ such that $\mathrm{Ref}_C(R, A|_C)$ is nontrivial (Def. 163, item (2)).*

In other words, the act of perception simultaneously yields an "internal trace" (a concrete token in the representational economy) and an "external tie" (the referential relation to the contextualized target $A|_C$). Note that the lemma only asserts existence of such a token R; it does not require that the token be unique, stable across time, or the only token produced by M while perceiving A.

Remark 1992. *Perception (Hyperseed-Concept 134) is stronger than mere registration: it asserts not only that the mind has crossed some recognitional threshold, but also that it has produced a* token *that refers to what was perceived. This lemma simply unpacks that dual nature. It is philosophically important because it makes perception inherently two-sided: an outward pole (anchoring to $A|_C$ via* Ref_C*) and an inward pole (a concrete representational act inside M) (Hyperseed-Concept 111 and 157).*

The context index is essential here: the lemma is not claiming that R refers to A "simpliciter," but rather to A as it is taken under *the contextual restrictions encoded by C, i.e. to $A|_C$. This matches ordinary cases in which the very same underlying object or event can be perceived under different contexts (lighting, task, background assumptions), yielding different referential successes or failures even when registration occurs.*

Sketch. Item (1) is Prop. 16. Item (2) is the second clause of Def. 163.

To make the dependency explicit: the hypothesis "M perceives A at time t in context C" activates the framework's perception conditions. Prop. 16 handles the registration component (ensuring that perception is not merely representational activity detached from recognition), while Def. 163 contributes the existence of at least one token $R \in \mathcal{R}$ whose contextual reference to $A|_C$ is nontrivial (ensuring that perception is not merely registration without any representational anchoring). $\square$

Remark 1993. Proof sketch. *No new argument is required: (1) appeals to an already-proved proposition connecting perception to registration, while (2) is literally one of the defining conjuncts of perception.* $\square$

The key "why" is definitional: the framework is designed so that perception is registration plus representational anchoring*. Visually, one can imagine perception as drawing an arrow from an internal token R to an external (contextualized) referent $A|_C$, while also recording that this arrow was formed at time t by the mind M.*

The modifier "nontrivial" in $\mathrm{Ref}_C(R, A|_C)$ *matters: it rules out degenerate or empty "reference" relations that would not genuinely connect the token to its target in context. Thus, the lemma can be read as providing a minimal witness to perceptual directedness: whenever perception occurs, there is at least one concrete representational item R for which the referential link to $A|_C$ is substantive rather than vacuous.*

Finally, observe what the lemma does not *claim. It does not assert that R is accurate, veridical, or exhaustive with respect to $A|_C$; it only extracts the structural consequence that perception, as defined, has both (i) a recognition/registration aspect and (ii) a token-based referential aspect. This is useful downstream because later arguments can quantify over such a witness R whenever they assume perception, without having to re-unpack the definition each time.*

35.4 Representation, inheritance, and semiotic modes

Lemma 118 (Inheritance is a preorder)**.** *Fix a context C. The strong pattern inheritance relation*

$$A \sqsubseteq_C B \quad :\Longleftrightarrow \quad \mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(B|_C)$$

is reflexive and transitive on U (Def. 165).

Remark 1994. *Here the role of the fixed context C is to select (i) the restriction operation $(-)|_C$ and (ii) the associated signature map Sig_C that lands in a space of V-valued feature assignments over the basis B_C. Thus $\sqsubseteq_C$ is not a global inheritance claim about A and B* simpliciter*, but a comparison of their signatures* as seen through *the aspect C. In particular, two entities may be incomparable in one context (because different features are salient or ordered differently) while being comparable in another; the lemma only asserts that, once C is fixed, the induced relation has the minimal stability properties expected of an order-like comparison.*

Remark 1995. *The point of this lemma is not that it is difficult, but that it stabilizes the logical status of inheritance: it behaves like a "$\leq$" relation. Reflexivity says every entity inherits from* *itself; transitivity says inheritance chains compose. Once we know inheritance is a preorder, we are justified in importing the familiar style of order-theoretic reasoning: one can speak of "more inherited" and "less inherited" in a context without demanding antisymmetry. This is exactly the right level of generality for pattern signatures, where distinct entities may share identical signatures in a fixed basis. (Hyperseed-Concept* **??** *and 130.)*

Sketch. Reflexive because $\leq$ is reflexive. Transitive because $\leq$ is transitive pointwise. More explicitly: reflexivity follows from $\mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(A|_C)$ in the codomain order. For transitivity, if $A \sqsubseteq_C B$ and $B \sqsubseteq_C D$, then $\mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(B|_C) \leq \mathrm{Sig}_C(D|_C)$, hence $\mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(D|_C)$. □

Remark 1996. Proof sketch. *Inheritance is defined by pointwise $\leq$ between functions $B_C \to V$. Pointwise order is always a preorder because the codomain order is.* □

The key step is that order on signatures is inherited *from order on V in the strict logical sense: once we have a preorder on values, the induced preorder on V-valued feature vectors is automatic. Geometrically, one may picture signatures as points in a V-valued hyper-rectangle; $\leq$ is then the standard product order. In particular, the reason antisymmetry is not required is that two distinct entities $A \neq B$ may satisfy $\mathrm{Sig}_C(A|_C) = \mathrm{Sig}_C(B|_C)$, yielding both $A \sqsubseteq_C B$ and $B \sqsubseteq_C A$ without forcing identity. This suggests the familiar construction of the induced equivalence relation $A \sim_C B$ iff $A \sqsubseteq_C B$ and $B \sqsubseteq_C A$, whose quotient is a genuine partial order on $U/{\sim_C}$ whenever such identification is useful.*

Theorem 53 (Representation becomes a preorder under mild reference axioms)**.** *Fix a context C and define representation-in-C as in Def. 166.*

Assume:

1. *(Reference reflexivity) $\mathrm{Ref}_C(A, A|_C)$ is nontrivial for all A, and*
2. *(Reference compositionality) if $\mathrm{Ref}_C(A, B|_C)$ and $\mathrm{Ref}_C(B, C|_C)$ are nontrivial, then $\mathrm{Ref}_C(A, C|_C)$ is nontrivial (the assumption used in Prop. 17).*

Then "A represents B in aspect C" is reflexive and transitive, hence a preorder.

Remark 1997. *The nontriviality language is intended to rule out degenerate reference relations that hold only vacuously (e.g. a bottom/zero witness that would make every putative reference arrow informationally empty). Under Def. 166, representation-in-C is a conjunction of an "internal" constraint (inheritance of signatures in C) and an "external" constraint (the existence of a suitable Ref_C witness). The theorem isolates exactly the two closure conditions needed for the reference component to match the closure conditions that inheritance already enjoys by Lemma 118. Without (1), even self-representation can fail because reference may not supply an identity-like arrow; without (2), chaining can fail because reference may not survive passage through intermediate carriers.*

Remark 1998. *The theorem says that representation, when defined via (i) pattern-signature inheritance and (ii) a nontrivial reference relation, has the same compositional stability as inheritance: it becomes a preorder. In plain terms, if representational tokens can always refer to themselves (reflexivity), and if reference arrows can be composed (compositionality), then "represents" is closed under identity and chaining. This is important because it allows one to build multi-step representational pipelines without losing well-formedness: if A represents B and B represents D, then A represents D. Such chaining is a minimal formal analogue of how complex cognitive systems stabilize symbol grounding across intermediate encodings [19]. One may also read this as an abstract guarantee that a representational system can be layered: intermediate stages may alter format or medium, but (under the axioms) cannot break representational reachability.*

Remark 1999. *The connection to Lemma 118 is structural: representation-in-C contains inheritance-in-C as one of its conjunctive ingredients (Def. 166), so the preorder behavior of inheritance is half the story. The other half is reference composition, which is the semantic counterpart of pattern-composition: it is the requirement that meaning does not evaporate when passing through intermediaries (Hyperseed-Concept 157 and 152). In categorical terms (purely as intuition, not as added structure), inheritance supplies an order-enrichment on "carriers" while* Ref_C *supplies the arrows that must compose; the theorem states that the combined relation inherits the same closure properties when both components admit identity and composition.*

Sketch. Reflexive: by (1) and Lemma 118 (inheritance reflexivity). Transitive: Prop. 17 gives transitivity provided reference composes as in (2), and inheritance composes by transitivity of $\leq$. Concretely, for reflexivity fix A and note $A \sqsubseteq_C A$ holds by Lemma 118, while $\mathrm{Ref}_C(A, A|_C)$ is nontrivial by (1); together these satisfy Def. 166. For transitivity, assume A represents B in C and B represents D in C; then $A \sqsubseteq_C B$ and $B \sqsubseteq_C D$ imply $A \sqsubseteq_C D$, and the corresponding nontrivial reference witnesses compose by (2) (as packaged in Prop. 17), yielding that A represents D in C. □

Remark 2000. Proof sketch. *To prove reflexivity, show each entity represents itself: inheritance is reflexive and reference is assumed reflexive. To prove transitivity, use Prop. 17 for the reference part (composition of* Ref_C*) and use transitivity of pointwise* $\leq$ *for the signature part.* □

The key steps mirror the two components of representation: the "intensional" ordering given by signatures and the "extensional" anchoring given by reference. A helpful picture is to imagine two parallel structures that must both compose: a ladder of pattern-strength comparisons and a chain of reference arrows. Only when both ladders can be climbed consistently does representation become preorder-like. Equivalently, representation-in-C fails to be transitive precisely when either the signature comparison breaks (which cannot happen because $\leq$ *is transitive) or the reference chain breaks (which is why axiom (2) is the essential additional assumption beyond Lemma 118).*

Corollary 15 (Semiotic modes inherit representation rules)**.** *For any* C*:*

1. *If* A *is an icon for* B*, then* A *represents* B *in some* $C \in \mathrm{Ctx}_{\mathrm{sens}}$*.*

2. *If* A *is an index for* B*, then* A *represents* B *in some* $C \in \mathrm{Ctx}_{\mathrm{st}}$*.*

3. *If* A *is a symbol for* B*, then* A *represents* B *in some* $C \in \mathrm{Ctx}_{\mathrm{rel}}$*, and* A *participates in a combinational system in that relational aspect (Def. 170).*

In particular, any inference rule proved for representation-in-C (e.g. transitivity under reference composition) specializes to icons/indices/symbols when the witnessing context is held fixed.

Remark 2001. *This corollary is best read as a transport principle: once icon/index/symbolhood is defined by* exhibiting *representation in some appropriate class of contexts, any property of representation that is stated* uniformly in the chosen C *immediately yields a corresponding property for the semiotic mode. The clauses* $C \in \mathrm{Ctx}_{\mathrm{sens}}, \mathrm{Ctx}_{\mathrm{st}}, \mathrm{Ctx}_{\mathrm{rel}}$ *can be understood as restricting which features and reference mechanisms are allowed to witness the representational link: icons privilege sensory similarity structure, indices privilege spatiotemporal or causal linkage, and symbols privilege relational/combinational structure. Accordingly, when a proof about representation uses only preorder reasoning (reflexivity/transitivity) and not any special facts about the particular* C*, the same proof schema applies verbatim after selecting the witnessing context supplied by the mode definition.*

Remark 2002. *This corollary places the classic Peircean triad—icon, index, symbol—into the present context-sensitive formalism (Hyperseed-Concept* **??**, **??**, *184; see also [14]). Each semiotic mode is defined by the existence of a context of an appropriate kind (sensory, spatiotemporal, relational) in which representation holds. The payoff is that general logical facts about representation can be "inherited" by each mode: once a mode is recognized as a specialization of representation, theorems about representation become reusable lemmas about icons, indices, and symbols. In particular, the mode labels do not add new inferential machinery on top of representation; they constrain* where *(i.e., in what kind of C) the representing relation is witnessed. Thus, any proposition of the form "for all contexts C of type τ, if Represents$_C(\cdot,\cdot)$ then ..." can be immediately re-read as a proposition about the corresponding semiotic mode by supplying $\tau \in \{\mathrm{sens}, \mathrm{st}, \mathrm{rel}\}$. Conversely, when one proves an icon/index/symbol claim, one should expect to uncover an underlying witness-context proof obligation, since the semantics is deliberately packaged as an existential over contexts.*

Sketch. Unfold Defs. 168–170: each mode is defined as existence of a context of the appropriate kind such that representation holds. More explicitly, the argument is a definitional expansion: $\mathrm{Icon}(A, B)$ means $\exists C \in \mathrm{Ctx}_{\mathrm{sens}}$ with $\mathrm{Represents}_C(A, B)$, $\mathrm{Index}(A, B)$ means $\exists C \in \mathrm{Ctx}_{\mathrm{st}}$ with $\mathrm{Represents}_C(A, B)$, and $\mathrm{Symbol}(A, B)$ means $\exists C \in \mathrm{Ctx}_{\mathrm{rel}}$ with $\mathrm{Represents}_C(A, B)$ together with whatever additional participation condition Def. 170 imposes. Once these abbreviations are expanded, any statement relating the semiotic predicates reduces to a statement about $\mathrm{Represents}_C$ and the type membership of the witnessing context. □

Remark 2003. Proof sketch. *Each item is immediate by definition: "icon" means "represents in some sensory context", etc. The final sentence then follows by fixing the witnessing context and reusing any rule about Represents$_C$.* □

The key idea is definitional reduction: the semiotic predicates are not new primitive relations here; they are existential wrappers around representation. This makes the theory modular: one proves algebraic properties once (for representation) and then applies them across semiotic categories. A useful way to read this is that the icon/index/symbol predicates function as typed interfaces to the same core relation: theorems about composition, monotonicity, invariance under appropriate renamings, or compatibility with other operators can be imported provided they are stated at the level of Represents$_C$. At the same time, because the wrapper is existential, there is a subtle but important methodological point: existence of some *witnessing context generally does not support chaining across steps unless one controls (or reuses) the witness; this motivates later context-fixed variants of familiar representation properties.*

Lemma 119 (Context-fixed transitivity for icon/index/symbol)**.** *Fix a context C and suppose reference composes in C as in Prop. 17. If $C \in \mathrm{Ctx}_{\mathrm{sens}}$ and A represents B in C and B represents D in C, then A is an icon for D. Analogous statements hold for $C \in \mathrm{Ctx}_{\mathrm{st}}$ (index) and $C \in \mathrm{Ctx}_{\mathrm{rel}}$ (symbol), with the additional symbolic combinational participation requirement inherited from Def. 170. In the symbolic case, the point is that one does not merely need a relational context in which $\mathrm{Represents}_C(A, D)$ holds, but also that A participates in the relevant combinational/structural constraints (as stipulated in Def. 170)* in that same context *C; the lemma asserts that this extra requirement is preserved when the premises already satisfy it within C.*

Remark 2004. *Where Corollary 15 moves from semiotic modes to representation, this lemma goes back in the other direction, but crucially* keeps the witnessing context fixed. *It says: if the chain of representation happens entirely within a sensory context C, then the composite relation is still sensory and therefore is an icon relation. This is precisely how one avoids the ambiguity of "A is an icon for B in some context" when reasoning over multiple steps: one chooses a context and remains* *within it. This context-fixing is the mathematical counterpart of a disciplined interpretive stance: do not silently change what you mean by "similar" or "connected" mid-argument. Technically, the lemma isolates the inference pattern that is* not *licensed by the bare existential wrappers: from Icon(A, B) and Icon(B, D) one cannot in general infer Icon(A, D), since the two icon premises may be witnessed by different sensory contexts. By contrast, if both premises are witnessed by the* same *C (as the lemma assumes), then Prop. 17 allows one to compose the underlying Represents$_C$ facts, after which the same C can be reused as witness for the semiotic conclusion. This is also why the lemma is stated with an explicit hypothesis about composition in C: it cleanly separates (i) general properties of Represents$_C$ as an internal relation from (ii) the external typing constraint $C \in \mathrm{Ctx}_{\mathrm{sens}}/\mathrm{Ctx}_{\mathrm{st}}/\mathrm{Ctx}_{\mathrm{rel}}$ that turns a plain representation into a specific semiotic mode.*

Sketch. Prop. 17 yields that A represents D in C. Then apply the corresponding semiotic definition (Def. 168/169/170) using the same witnessing C. Concretely: from $\mathrm{Represents}_C(A, B)$ and $\mathrm{Represents}_C(B, D)$, Prop. 17 gives $\mathrm{Represents}_C(A, D)$, and since C is already assumed to lie in the appropriate context class (sensory, spatiotemporal, or relational), one may discharge the existential in the corresponding definition by choosing this very C. In the symbolic case, one also checks (by the "inherited" clause from Def. 170) that whatever combinational participation condition is required is satisfied in C for the composed representation; this is why the lemma explicitly notes that the additional condition is part of what is transported through the same fixed witness. □

Remark 2005. Proof sketch. *First compose representations using Prop. 17 to get Represents$_C(A, D)$. Then, since C is of the appropriate type (sens/st/rel), the corresponding semiotic predicate holds with witness C.* □

The key step is that the semiotic definitions are existential: to prove Icon(A, D) it suffices to exhibit one *sensory context where representation holds. Here the context is already given, so the entire argument is a two-line "closure under composition" result. It is worth emphasizing that the closure here is* conditional *on staying inside a fixed C: the proof does not attempt to combine different witnessing contexts, nor does it require any global closure property of* $\mathrm{Ctx}_{\mathrm{sens}}$ *under some operation. In this sense the lemma is a controlled way of obtaining transitivity-like reasoning for semiotic predicates without strengthening them into universally quantified notions (which would change their intended meaning). The same pattern reappears whenever one wants to transport other representation-theoretic properties (e.g., invariance under isomorphisms of contexts, or compatibility with additional structure on C): one proves the property for Represents$_C$ and then rewraps it by reusing the same witness context.*

35.5 Soggy predicates as algebraic evidence aggregators

Setup (Defs. 171–172). Let X be a universe of discourse. Each observation $o \in O$ supplies an evidence map $e_o : X \to [0, 1]^2$, and weights $w_o \geq 0$ satisfy $\sum_{o \in O} w_o = 1$. Define the soggy predicate $F : X \to [0, 1]^2$ by $F(x) := \sum_o w_o e_o(x)$ componentwise.

Remark 2006. *The word "soggy" is a reminder that, unlike a crisp predicate, we allow evidence to have two components: support and counter-support. Thus $e_o(x) = (e_o^+(x), e_o^-(x)) \in [0, 1]^2$ can represent simultaneous partial reasons for and against x satisfying the predicate, as in paraconsistent and related multi-valued logics (Hyperseed-Concept 172; see [23, 24]). The aggregation $F(x) = \sum_o w_o e_o(x)$ is then a deliberately plain operation: a convex combination done componentwise. This simplicity is a virtue: it lets us prove robust inequalities about what aggregation can and cannot do without committing to a more baroque fusion rule.*

Remark 2007. *Notation check: the sum* $\sum_o w_o e_o(x)$ *is a vector sum in* $\mathbb{R}^2$ *(here constrained to* $[0,1]^2$*), so explicitly* $F^+(x) = \sum_o w_o e_o^+(x)$ *and* $F^-(x) = \sum_o w_o e_o^-(x)$*. The weights are nonnegative and sum to* 1*, making each component of* $F(x)$ *a convex combination of the corresponding components of* $e_o(x)$*.*

Lemma 120 (Convex bounds)**.** *For every* $x \in X$ *and each component* $\pm$*,*

$$\min_{o\in O} e_o^{\pm}(x) \;\leq\; F^{\pm}(x) \;\leq\; \max_{o\in O} e_o^{\pm}(x).$$

Remark 2008. *This lemma asserts that aggregation by weighted averaging cannot create component-values outside the observed range: the aggregated support* $F^+(x)$ *lies between the smallest observed support and the largest observed support, and likewise for the negative component. This is the sense in which the aggregation is "conservative": it does not invent extreme evidence; it interpolates among what the observations already provide.*

Sketch. Each component $F^{\pm}(x)$ is a convex combination of the scalars $e_o^{\pm}(x)$. □

Remark 2009. Proof sketch. *Convex combinations lie in the convex hull of their inputs; in one dimension this hull is exactly the interval* [min, max]. □

The key reason is nonnegativity of weights and $\sum_o w_o = 1$*. Geometrically, for a fixed* x *and fixed sign* $\pm$*, the values* $\{e_o^{\pm}(x)\}$ *are points on the real line, and* $F^{\pm}(x)$ *is their barycenter, which must lie between the extreme points.*

Lemma 121 (Monotonicity and commutation with p-bit negation)**.**

1. *(Monotonicity) If* $e'_o(x) \geq e_o(x)$ *componentwise for all* o, x*, then the induced* F' *satisfies* $F'(x) \geq F(x)$ *for all* x*.*
2. *(Negation commutes) Let* $\neg(a^+, a^-) := (a^-, a^+)$*. Define* $(\neg F)(x) := \neg(F(x))$ *and* $(\neg e_o)(x) := \neg(e_o(x))$*. Then*

$$\neg F \;=\; \sum_{o\in O} w_o\,(\neg e_o),$$

i.e. averaging and negation commute.

Remark 2010. *Part (1) is a monotonicity principle: if every observation is improved (never decreases support nor counter-support), then the aggregated predicate improves. This matters for comparative statics: it allows one to reason about how changes in sensors, data quality, or preprocessing flow through to* F*.*

Part (2) expresses a basic symmetry of this evidence representation: negation is just swapping the two components, and swapping commutes with averaging. This is one of the algebraic conveniences of the $[0,1]^2$ *"p-bit" style evidence used throughout the framework (cf. [23, 24]).*

Sketch. (1) Weighted sums preserve componentwise inequalities when weights are nonnegative. (2) Swap-of-components is linear: swapping after summing equals summing after swapping. □

Remark 2011. Proof sketch. *For (1), apply the inequality* $e'_o(x) \geq e_o(x)$ *componentwise, multiply by* $w_o \geq 0$*, and sum over* o*. For (2), write out both sides componentwise and use distributivity of summation over addition; swapping the* $+$ *and* $-$ *coordinates is an affine linear map.* □

The key step in (1) is the nonnegativity of weights: with negative weights the inequality could reverse. Visually, in (2), negation is a reflection across the diagonal in the (a^+, a^-) *plane, and averaging is taking a weighted centroid; reflecting a centroid equals taking the centroid of the reflected points.*

Lemma 122 (Conflict can increase under aggregation; determinacy can decrease)**.** *For $v = (v^+, v^-) \in [0,1]^2$, define*

$$\mathrm{conf}(v) := \min(v^+, v^-), \qquad \det(v) := \max(v^+, v^-).$$

Then for each $x \in X$,

$$\mathrm{conf}(F(x)) \;\geq\; \sum_{o \in O} w_o \,\mathrm{conf}(e_o(x)), \qquad \det(F(x)) \;\leq\; \sum_{o \in O} w_o \det(e_o(x)).$$

Thus disagreement across observations can raise "both-sides present" evidence, while the maximum certainty can shrink under averaging.

Remark 2012. *This lemma captures a subtle but practically important phenomenon in paraconsistent-style aggregation (Hyperseed-Concept 172). Suppose some observations strongly support x while others strongly oppose x. When we average these, we may end up with* both *moderately high support and moderately high counter-support, thereby increasing* $\mathrm{conf}(F(x)) = \min(F^+(x), F^-(x))$. *At the same time, the strongest one-sided certainty (*$\det = \max$*) can be diluted, because averaging pulls extremes toward the center. In an epistemic register: aggregation can make one more aware of conflict, and less confident in any single verdict. This is a mathematically clean articulation of why multi-source fusion does not necessarily "resolve" disagreement; it may instead faithfully represent it (cf. [24] for related motivations). In particular, because* conf *is* increasing *in each coordinate (holding the other fixed), any aggregation rule that can raise both F^+ and F^- simultaneously (as averaging can, when sources disagree) will typically increase* conf *even if each individual $e_o(x)$ is itself nearly one-sided. A concrete extreme illustrates the effect: if two equally weighted sources report $(e_1^+(x), e_1^-(x)) = (1, 0)$ and $(e_2^+(x), e_2^-(x)) = (0, 1)$, then $F(x) = (\frac{1}{2}, \frac{1}{2})$, so* $\mathrm{conf}(F(x)) = \frac{1}{2}$ *even though* $\mathrm{conf}(e_1(x)) = \mathrm{conf}(e_2(x)) = 0$, *while* $\det(F(x)) = \frac{1}{2}$ *is strictly smaller than* $\det(e_1(x)) = \det(e_2(x)) = 1$. *Thus, "more data" can mathematically force the representation to move from certainty-with-no-conflict at the source level to moderate-certainty-with-explicit-conflict at the fused level.*

Sketch. Let $A := \sum_o w_o e_o^+(x)$ and $B := \sum_o w_o e_o^-(x)$. For each o, $\min(e_o^+, e_o^-) \leq e_o^+$ and $\min(e_o^+, e_o^-) \leq e_o^-$, hence $\sum_o w_o \min(e_o^+, e_o^-) \leq A$ and also $\leq B$, so it is $\leq \min(A, B) = \mathrm{conf}(F(x))$. Similarly, $e_o^+ \leq \max(e_o^+, e_o^-)$ and $e_o^- \leq \max(e_o^+, e_o^-)$, so $A \leq \sum_o w_o \max(e_o^+, e_o^-)$ and $B \leq \sum_o w_o \max(e_o^+, e_o^-)$, hence $\max(A, B) = \det(F(x))$ is $\leq$ that same sum. The only structural properties used here are: (i) coordinatewise bounds of min / max, (ii) nonnegativity of the weights (so inequalities are preserved when multiplying by w_o), and (iii) additivity of sums (so inequalities are preserved under aggregation). In particular, no special smoothness or probabilistic assumptions are required beyond treating $F^{\pm}(x)$ as convex combinations of the corresponding $e_o^{\pm}(x)$. □

Remark 2013. Proof sketch. *Use the elementary inequalities* $\min(u, v) \leq u, v$ *and* $u, v \leq \max(u, v)$, *then multiply by w_o and sum. Finally, identify A and B with $F^+(x)$ and $F^-(x)$ and apply the definitions of* conf *and* det. □

The key device is bounding a min or max by each coordinate and then using linearity of weighted sums. Geometrically, $\mathrm{conf}(v)$ *is the distance of v from the boundary of the unit square in the direction of the diagonal: increasing both coordinates increases* conf. *Averaging conflicting points can move a point toward the diagonal, raising* $\min(v^+, v^-)$ *even if no individual observation had high conflict. Equivalently, level sets of* $\mathrm{conf}(v) = c$ *are the "L-shaped" regions $\{v^+ \geq c,\; v^- \geq c\}$; fusion that increases both coordinates (even slightly) can jump v into a higher-conflict region. By contrast,* $\det(v) = \max(v^+, v^-)$ *has level sets given by the union $\{v^+ \geq c\} \cup \{v^- \geq c\}$, and averaging* *tends to move points away from the corners* $(1, 0)$ *and* $(0, 1)$ *toward the interior, which decreases* $\max(v^+, v^-)$*. This makes precise the intuition that convex aggregation is "conflict-revealing" and "extremum-smoothing" at the same time.*

Corollary 16 (Complementary evidence collapses to ordinary fuzzy/probabilistic semantics)**.** *If each observation is complementary, i.e.* $e_o^-(x) = 1 - e_o^+(x)$ *for all* o, x*, then* $F^-(x) = 1 - F^+(x)$*. Hence* $F^+(x)$ *alone can be treated as a standard fuzzy membership/probability-of-satisfaction (Prop. 18). Moreover, under this complementarity constraint,* $\mathrm{conf}(F(x)) = \min(F^+(x), 1-F^+(x))$ *and* $\det(F(x)) = \max(F^+(x), 1 - F^+(x))$ *become functions of the single scalar* $F^+(x)$*, so the two-dimensional bookkeeping carries no additional degrees of freedom beyond the usual* $[0, 1]$ *semantics.*

Remark 2014. *This corollary identifies a special case in which the two-component evidence reduces to a single scalar: if every observation is internally consistent in the strong sense that negative evidence is exactly the complement of positive evidence, then the aggregation preserves this complementarity. In that regime, the machinery of soggy predicates is conservative: it does not conflict with, but rather extends, ordinary fuzzy or probabilistic semantics. The value of the general* $[0, 1]^2$ *framework is precisely that it can represent and propagate non-complementary (hence potentially conflicting) evidence when complementarity is not warranted [23]. One can also read the corollary as identifying the "classical submanifold" of the evidence square: complementary points form an affine line, so restricting to that line recovers the familiar one-parameter uncertainty calculus. Departures from the line quantify precisely how much "extra information" is present in the paraconsistent register, namely the possibility that both* F^+ *and* F^- *are simultaneously large (or simultaneously small) in a way that cannot be represented by a single probability-like number.*

Sketch. Componentwise computation (as in Prop. 18): linearity of sums and $\sum_o w_o = 1$. One may also emphasize that the complement map $t \mapsto 1 - t$ is affine, so it commutes with convex combination: applying the complement before averaging yields the same result as averaging first and then applying the complement. This is exactly the algebraic content of the identity $F^-(x) = 1 - F^+(x)$. □

Remark 2015. Proof sketch. *Compute* $F^-(x) = \sum_o w_o e_o^-(x) = \sum_o w_o(1 - e_o^+(x)) = 1 - \sum_o w_o e_o^+(x) = 1 - F^+(x)$. □

The only ingredient beyond algebra is the normalization $\sum_o w_o = 1$*, which ensures that averaging respects affine constraints like* $a^- = 1 - a^+$*. Geometrically, complementary evidence lives on the diagonal line* $v^- = 1-v^+$ *in the square; convex combinations of points on a line remain on the line. Said differently, the set* $\{(t, 1 - t) : t \in [0, 1]\}$ *is convex, and the aggregator* F *(being a weighted average) maps convex sets to themselves; thus complementarity is an invariant of the fusion rule. When the complementarity assumption fails for some sources, the fused point can leave this diagonal and thereby record non-classical evidence patterns, e.g. "both support and opposition are high" (a point near* $(1, 1)$*) or "both are low" (a point near* $(0, 0)$*), neither of which can be encoded by a single scalar* $t \in [0, 1]$.

35.6 Engine-friendly rule schemata (optional)

Useful forward rules. The following are direct corollaries of the definitions and lemmas above: In this subsection, "engine-friendly" means that the conclusions are stated in a way that supports monotone, forward-chaining saturation: once a premise is established, the corresponding consequences can be asserted and then used as premises for later rules without revisiting the original derivation. In particular, the rules below are formulated to expose (i) witness objects (such as a reference relation R) and (ii) simple numeric or preorder constraints (such as Sig_C-inequalities) that are cheap to store and combine.

- $\mathrm{Perceives}(M, A, t, C) \to \mathrm{Registers}(M, A, t, C)$ and $\exists R : \mathrm{Ref}_C(R, A|_C)$ nontrivial (Lemma 117). Here the first conjunct isolates the "recording" component of perception, while the existential conjunct isolates a concrete referential witness in context C. For implementations, the existential can be treated as a witness-producing rule (e.g., Skolemization to a fresh symbol $R_{M,A,t,C}$) so that later rules that require a specific $\mathrm{Ref}_C(\cdot, \cdot)$ instance can fire. The qualifier "nontrivial" is included to rule out degenerate witnesses (e.g., empty or information-free reference), ensuring that the obtained R actually supports downstream representational consequences rather than merely satisfying an existence claim vacuously.

- $\mathrm{Represents}_C(A, B) \to \mathrm{Sig}_C(A|_C) \leq \mathrm{Sig}_C(B|_C)$ (Def. 166). This inequality is useful operationally because it turns a qualitative claim (representation) into an order constraint that can be accumulated across many facts. In contexts where Sig_C is computable or estimable, this rule provides a quick consistency check: asserting $\mathrm{Represents}_C(A, B)$ commits one to the corresponding Sig_C-monotonicity, and contradictions can be detected by numeric reasoning even when the underlying cognitive predicates are opaque. Conceptually, it expresses that B is at least as "significant" (in C) as A when A is taken to represent B in that same context.

- $\mathrm{Icon}(A, B)$ or $\mathrm{Index}(A, B)$ or $\mathrm{Symbol}(A, B) \to \mathrm{Represents}_C(A, B)$ for some witnessing C of the appropriate kind (Cor. 15). This schema makes explicit the intended workflow: semiotic classifications (iconic, indexical, symbolic) are treated as higher-level tags, and the engine immediately lowers them to the uniform predicate $\mathrm{Represents}_C$ that participates in the order-theoretic machinery. The phrase "for some witnessing C" matters because the semiotic relation may determine constraints on C (e.g., what counts as an "appropriate kind" of context), and these constraints can be carried along as side-conditions or additional facts about C for later use. Practically, one may implement this as producing an existential context parameter when the context is not already fixed, or as selecting among already-known candidate contexts when the problem is posed with a finite menu of contexts.

- (Reference composes in C) and $\mathrm{Represents}_C(A, B)$ and $\mathrm{Represents}_C(B, D) \to \mathrm{Represents}_C(A, D)$ (Prop. 17 / Thm. 53). This is the main chaining rule: it allows an engine to close representational paths transitively, provided the ambient hypothesis that reference composes in the same context C is available. The explicit mention of the compositionality premise is important in mechanized settings, because it prevents accidental cross-context chaining: the rule is safe only when all representation claims live in the same C and the corresponding Ref_C structure supports composition. From an algorithmic perspective, this is the familiar "transitive closure" operation, but guarded by contextual coherence; from a conceptual perspective, it is the reason the earlier development yields a preorder-like structure on representational claims.

A typical forward-chaining use pattern is: first derive Registers-facts and reference witnesses from Perceives-facts; next normalize any semiotic predicates (Icon/Index/Symbol) into context-indexed representation claims; then saturate the resulting $\mathrm{Represents}_C$-graph using the compositional/transitive rule; and finally propagate any derived Sig_C constraints to support grading, comparison, or inconsistency detection. Stated differently, these schemata serve as a bridge between the "qualitative" front-end vocabulary and the "quantitative/order" back-end invariants.

Remark 2016. *These rule schemata are the "computational surface" of the preceding order-theoretic development. They are intentionally forward-chaining: each rule takes a high-level cognitive predicate (Perceives, Represents, Icon/Index/Symbol) and yields consequences that can be cached and reused by an inference engine. Philosophically, one may view them as a modest attempt to reconcile the fluidity of cognitive notions with the austerity of formal entailment: the semantics remain contextual and graded (via V, Sig_C, Ref_C), yet the inference skeleton is crisp. This style of extracting operational rules from abstract cognitive definitions aligns with the systems-engineering orientation of [19]. Moreover, because the conclusions are phrased as either (i) additional atomic predicates (e.g., Registers$(\cdot)$, Represents$_C(\cdot)$) or (ii) simple order constraints (e.g., $\leq$ on Sig_C-values), the resulting knowledge base admits efficient indexing and incremental maintenance under new observations. In particular, the separation between contextual parameters (the explicit C) and content parameters (the objects A, B, D) is deliberate: it encourages implementations to treat C as a key for partitioning the fact store, thereby avoiding unsound interactions between incompatible contexts while still allowing within-context closure. Finally, the explicit witness patterns (e.g., $\exists R : \mathrm{Ref}_C(R, A|_C)$) indicate where the formalism expects concrete supporting structure; in engines that track justifications, these witnesses can be stored as provenance artifacts that explain why a given Represents$_C$ fact was derived.*

36 Helper theorems for prediction, attraction, causality, and control

36.1 Standing notation

Fix an observer/context O, event-type set $\mathcal{E}$, proto-time carrier $(T, \prec)$, and lag windows Δ, Δ'. Let $\mathrm{Cond}_{O,\Delta} : \mathcal{E} \times \mathcal{E} \to [0,1]^2$ be the predictive conditional operator. Let $\pi : [0,1]^2 \to [0,1]$ be the plausibility projection

$$\pi(p, q) = \frac{1 + p - q}{2}.$$

Define predictive implication strength and predictive attraction by

$$PImp_{O,\Delta}(A, B) = \pi(\mathrm{Cond}_{O,\Delta}(B \mid A)), \qquad PAtt_{O,\Delta}(A, B) = PImp_{O,\Delta}(A, B) - PImp_{O,\Delta}(\neg A, B).$$

Define attraction by

$$\mathrm{Att}_{O,\Delta}(A, B) = \pi(\mathrm{Cond}_{O,\Delta}(B \mid A)) - \pi(\mathrm{Cond}_{O,\Delta}(B \mid \neg A)).$$

For a p-bit $v = (v^+, v^-) \in [0,1]^2$, write $\mathrm{bias}(v) = v^+ - v^-$. Write $\mathbf{1}[A \prec B] \in \{0,1\}$ for the temporal-precedence indicator used in causal implication.

The standing notation above is designed to keep track of three layers at once: (i) what is being talked about ($\mathcal{E}$ and proto-time), (ii) what information is available and how it is aggregated (the context O and the windows Δ, Δ'), and (iii) how that aggregated information is converted into scalar quantities that behave like the "strength" of a predictive relationship (the projection π and the derived operators $PImp$, $PAtt$, Att). The presence of two lag windows is intentional: typically one uses Δ for the primary predictive relation (e.g. A in a past window and B in a future window) and Δ' for auxiliary constructions (e.g. counterfactual or intervention-style comparisons, or multi-step chaining where the window of admissible temporal offsets changes).

Note that π is an affine map in the difference $(p - q)$, and it can be rewritten directly in terms of the bias of a p-bit. Indeed, for any $v = (v^+, v^-)$,

$$\pi(v) = \frac{1 + v^+ - v^-}{2} = \frac{1 + \mathrm{bias}(v)}{2}.$$

Thus $\pi(v) = \frac{1}{2}$ corresponds to zero net bias, $\pi(v) \to 1$ corresponds to maximally positive bias, and $\pi(v) \to 0$ corresponds to maximally negative bias. In particular, the projection does not measure the *amount* of evidence (since $(0.6, 0.0)$ and $(1.0, 0.4)$ have the same bias) but rather the net direction of support after positive and negative channels have been compared.

The two definitions $PAtt_{O,\Delta}$ and $\mathrm{Att}_{O,\Delta}$ are algebraically identical given the definition of $PImp_{O,\Delta}$, but it is useful to keep both names in view because they emphasize different conceptual decompositions. Writing attraction as $PAtt$ highlights the intermediate quantity "predictive implication strength" and makes later manipulations resemble standard calculi that first define implication-like quantities and then define contrasts. Writing attraction as Att emphasizes that it is fundamentally a *difference of plausibilities* between the A-conditioned and $\neg A$-conditioned regimes. In many arguments below, what matters is precisely this contrast structure: attraction is high when A shifts plausibility for B upward relative to its negation, and negative when A shifts plausibility downward.

Because $\pi(\cdot) \in [0, 1]$, the attraction quantities satisfy immediate range bounds:

$$PAtt_{O,\Delta}(A, B) \in [-1, 1], \qquad \mathrm{Att}_{O,\Delta}(A, B) \in [-1, 1].$$

Moreover, the sign of attraction encodes a qualitative direction: $\mathrm{Att}_{O,\Delta}(A, B) > 0$ indicates that, in the observer/context O and relative to the lag window Δ, conditioning on A makes B more plausible than conditioning on $\neg A$; $\mathrm{Att}_{O,\Delta}(A, B) < 0$ indicates a repulsive or inhibitory predictive relationship in the same comparative sense. This sign convention will later interact naturally with control notions, where one is interested not only in whether A predicts B, but whether steering toward A (rather than toward $\neg A$) tends to steer B up or down.

The precedence gate $\mathbf{1}[A \prec B]$ is introduced at the notation stage because it is the minimal device needed to separate mere predictive association from temporally directed causal implication. In proto-time settings where $\prec$ may be partial (e.g. concurrency, branching, or incomparable events), the indicator makes explicit that causal implication is only asserted (or only assigned a nonzero score) when a suitable precedence relation holds. In later formulas, such indicators serve as "hard" directionality constraints even when the underlying evidence and conditionals are "soft" and graded.

Remark 2017. *This section is, in effect, a small calculus for turning (possibly inconsistent) evidence about event-types into graded notions of prediction and control. The underlying philosophical move is to treat prediction not as a binary relation but as a quantity, and then to build more structured notions (attraction, causality, control) as derived quantities. In that sense it resembles the way one builds geometry from a metric: once a primitive numerical structure is fixed, many secondary concepts become algebraic corollaries.*

Remark 2018. *A few symbols deserve an explicit reading the first time through.*

- *O is an "observer/context": it determines what counts as an event-type, how evidence is gathered, and how conditionals are computed. This is aligned with the general contextual stance used throughout the document (compare the role of contexts in Section 13) and with a social-computational view of scientific inference [20].*

- *$\mathcal{E}$ is the set of event-types (not individual events). One may imagine $A, B \in \mathcal{E}$ as predicates or labels such as "rain" or "sensor_spike".*

- *$(T, \prec)$ is proto-time: $t \prec t'$ indicates temporal precedence without presuming a metric or linearity. This directly connects to the Hyperseed-Concept 142.*

- $\mathrm{Cond}_{O,\Delta}(B \mid A) \in [0,1]^2$ *is a p-bit (two-coordinate) conditional: its first coordinate v^+ and second coordinate v^- can be read as degrees of positive and negative support (or two-channel evidence), enabling paraconsistent bookkeeping [23, 24].*
- π *projects a p-bit to a single plausibility number in* $[0,1]$ *by comparing positive and negative support.*
- $\neg A$ *denotes the (event-type) negation used in the predictive attraction definitions.*
- $\mathbf{1}[A \prec B]$ *is the indicator that A precedes B in proto-time; it is used as a gate that enforces the temporal directionality characteristic of causation (Hyperseed-Concept 70).*

The definitions of predictive attraction and causal implication below are also closely aligned with the broader pattern-based account of prediction and emergence used elsewhere [5], and with the Hyperseed-Concept **??**.

Remark 2019. *It is helpful to keep in mind a few elementary properties of the plausibility projection π when reading later inequalities. Since $\pi(p,q) = \frac{1}{2} + \frac{1}{2}(p-q)$, it is monotone increasing in p and monotone decreasing in q, and it treats (p,q) and $(p+\epsilon, q+\epsilon)$ differently only insofar as the* difference *changes. In particular, $\pi(p,p) = \frac{1}{2}$ for any p, so perfectly balanced positive/negative support is always mapped to a neutral plausibility regardless of magnitude. This is one of the intended "paraconsistent" features: accumulating equally strong evidence on both sides yields uncertainty rather than collapse.*

Remark 2020. *A simple sanity-check example for the p-bit conventions: if $\mathrm{Cond}_{O,\Delta}(B \mid A) = (0.9, 0.1)$, then positive evidence outweighs negative, yielding high plausibility. If instead it is $(0.9, 0.9)$, this represents strong evidence for and against simultaneously (a paraconsistent situation), and the projection π will return a middling value, reflecting that the net bias is near zero even though the magnitude of evidence is large.*

Remark 2021. *A complementary sanity-check for attraction: suppose $\mathrm{Cond}_{O,\Delta}(B \mid A) = (0.8, 0.2)$ while $\mathrm{Cond}_{O,\Delta}(B \mid \neg A) = (0.4, 0.3)$. Then $\pi(\mathrm{Cond}(B \mid A)) = \frac{1+0.8-0.2}{2} = 0.8$ and $\pi(\mathrm{Cond}(B \mid \neg A)) = \frac{1+0.4-0.3}{2} = 0.55$, giving $\mathrm{Att}_{O,\Delta}(A,B) = 0.25$. The positive value indicates that A provides a net uplift in plausibility for B relative to $\neg A$, even though $\mathrm{Cond}(B \mid \neg A)$ may itself still have more positive than negative support. Conversely, if $\mathrm{Cond}_{O,\Delta}(B \mid \neg A)$ had the larger bias, attraction would be negative, capturing a net inhibitory relationship.*

36.2 Projection lemmas

Lemma HT14.1 (Plausibility as bias projection). For any $v \in [0,1]^2$, $\pi(v) = \frac{1}{2}(1 + \mathrm{bias}(v))$.

Remark 2022. *Intuitively, Lemma HT14.1 says: "plausibility" is just an affine rescaling of the* bias *between the positive and negative channels. Bias $\mathrm{bias}(v) = v^+ - v^-$ lives in $[-1,1]$, and π simply maps this symmetric interval to $[0,1]$ by shifting and halving. Philosophically, this is a decision to interpret plausibility as a* net leaning *rather than a total amount of evidence; the latter would require a different projection. This choice is well-suited to later algebraic manipulations because affine maps preserve order and convexity. In particular, the extreme states $(v^+, v^-) = (1,0)$ and $(0,1)$ have biases $+1$ and -1, and therefore map to plausibilities $\pi = 1$ and $\pi = 0$ respectively, while equal-channel states $v^+ = v^-$ have $\mathrm{bias}(v) = 0$ and hence $\pi(v) = \frac{1}{2}$, corresponding to a neutral midpoint. The formula can also be read backwards as $\mathrm{bias}(v) = 2\pi(v) - 1$, so π and* bias *carry exactly the same one-dimensional information, merely expressed in different coordinate conventions.*

Proof sketch. Expand $\pi(v^+, v^-) = \frac{1}{2}(1 + v^+ - v^-)$ and recognize $v^+ - v^- = \mathrm{bias}(v)$.

Remark 2023. Proof sketch. *The strategy is purely definitional: substitute the definitions and then factor out the quantity already named* $\mathrm{bias}(v)$*. The key step works because π was designed to be linear in each coordinate, so the difference $v^+ - v^-$ appears explicitly. A helpful visual is to think of v as a point in the unit square:* $\mathrm{bias}(v)$ *measures displacement along the $(1, -1)$ diagonal, and π renormalizes this diagonal coordinate into $[0, 1]$. Equivalently, the level sets of π (and of* bias*) are lines of the form $v^+ - v^- =$ constant, which run parallel to the $(1, 1)$ diagonal; moving in the $(1, 1)$ direction changes the* amount *in both channels but leaves the net leaning unchanged. This makes explicit why π should be viewed as a projection: it discards the orthogonal "total evidence" direction and retains only the signed contrast between channels, with the additive $\frac{1}{2}$ encoding the convention that zero bias corresponds to midpoint plausibility.* □

Lemma HT14.2 (Consistent-case recovery). If $v = (p, 1 - p)$ (consistent evidence), then $\pi(v) = p$.

Remark 2024. *This lemma verifies that π behaves like ordinary probability in the "consistent" submanifold where positive and negative coordinates sum to 1. In other words, when the two-channel evidence state is merely a redundant encoding of a single $p \in [0, 1]$, the projection returns p exactly. This matters because it ensures compatibility with standard probabilistic intuitions when inconsistency is absent, while still allowing the larger $[0, 1]^2$ space when inconsistency is present [23]. Geometrically, the consistent submanifold is the line segment $v^+ + v^- = 1$ joining $(0, 1)$ to $(1, 0)$, and Lemma HT14.2 states that π restricts on this segment to the identity coordinate $p = v^+$. Thus π can be seen as an extension of the usual identification of a Bernoulli parameter with its "positive mass," while permitting off-manifold points with $v^+ + v^- \neq 1$ to represent various forms of inconsistency, redundancy, or undercommitment.*

Proof sketch. $\pi(p, 1 - p) = \frac{1}{2}(1 + p - (1 - p)) = p$.

Remark 2025. Proof sketch. *The proof is a one-line substitution. The conceptual reason it works is that the constraint $v^- = 1 - v^+$ makes* $\mathrm{bias}(v) = 2p - 1$*, and the affine mapping in Lemma HT14.1 exactly inverts that relation. Geometrically, the line $v^+ + v^- = 1$ is mapped isomorphically onto $[0, 1]$ by π. It is also helpful to note how the endpoints behave under this restriction: $p = 0$ gives $v = (0, 1)$ and hence $\pi = 0$, while $p = 1$ gives $v = (1, 0)$ and hence $\pi = 1$, so the entire segment is carried to $[0, 1]$ with no distortion. In later applications, this recovery property is what permits one to interpret π as a conservative generalization of probability: whenever evidence is internally consistent, the projection agrees exactly with the familiar scalar representation.* □

36.3 Attraction and predictive implication

Lemma HT14.3 (Attraction via biases).

$$\mathrm{Att}_{O,\Delta}(A, B) = \tfrac{1}{2}\Big(\mathrm{bias}(\mathrm{Cond}_{O,\Delta}(B \mid A)) - \mathrm{bias}(\mathrm{Cond}_{O,\Delta}(B \mid \neg A))\Big).$$

Remark 2026. *Attraction* $\mathrm{Att}_{O,\Delta}(A, B)$ *is meant to quantify how much the presence of A "pulls" toward B relative to the absence of A. Lemma HT14.3 states that this pull is exactly the* difference of biases *in the p-bit conditionals, up to the constant factor $1/2$ coming from the projection π. The philosophical point is that attraction depends not on absolute belief in B, but on a counterfactual contrast: A versus $\neg A$. In particular, the use of* $\mathrm{bias}(\mathrm{Cond}_{O,\Delta}(B \mid A))$ *isolates the* directional tilt *in the conditional plausibility, while discarding any symmetric or baseline component that would be shared by both A and $\neg A$. This is why attraction functions as a signed quantity: positive values indicate a shift toward B under A, whereas negative values indicate that conditioning on A shifts away from B compared to conditioning on $\neg A$. Conceptually, one can read the right-hand side as: "how much more biased toward B is the conditional world where A holds than the conditional world where A fails," and the $\frac{1}{2}$ factor reflects the specific normalization induced by π (so that attraction remains on the same scale used elsewhere in the framework).*

Proof sketch. Use $\mathrm{Att} = \pi(\mathrm{Cond}(B \mid A)) - \pi(\mathrm{Cond}(B \mid \neg A))$ and Lemma HT14.1; the constant $\frac{1}{2}$ terms cancel.

Remark 2027. Proof sketch. *The strategy is: rewrite each $\pi(\cdot)$ using Lemma HT14.1, then subtract. The cancellation of constants reflects that attraction is a* difference *measure: any baseline plausibility independent of A disappears. One may picture two arrows in evidence-space: the conditional on A and the conditional on $\neg A$; attraction measures the signed displacement between their bias coordinates. Put differently, the role of Lemma HT14.1 is to express π as an affine function of the bias, and attraction retains only the linear part because affine offsets vanish under subtraction. This is also the formal reason that attraction is insensitive to any uniform recalibration that would add the same constant to both $\pi(\mathrm{Cond}(B \mid A))$ and $\pi(\mathrm{Cond}(B \mid \neg A))$.* □

Lemma HT14.4 (Predictive attraction equals attraction).

$$PAtt_{O,\Delta}(A, B) = \mathrm{Att}_{O,\Delta}(A, B).$$

Remark 2028. *This lemma says that the two notational routes to "attraction" are the same: whether one defines attraction directly via $\pi(\mathrm{Cond}(\cdot \mid \cdot))$ or indirectly via predictive implication $PImp$, the resulting scalar is identical. This is important because later sections may prefer one viewpoint or the other: $PImp$ emphasizes implication-like reasoning, while* Att *emphasizes differential influence. The equality ensures there is no hidden divergence between the two perspectives. It also ensures that sign conventions are compatible: e.g., any interpretation stated in terms of $PImp$ comparisons immediately transfers to* Att *without additional assumptions. From a modeling standpoint, Lemma HT14.4 is a coherence condition: the "predictive" layer ($PImp$ and $PAtt$) is not an extra structure but a repackaging of the same underlying conditional plausibility operations.*

Proof sketch. By definition $PImp_{O,\Delta}(A, B) = \pi(\mathrm{Cond}_{O,\Delta}(B \mid A))$; substitute into $PAtt$.

Remark 2029. Proof sketch. *The proof is definition chasing. The key step works because $PAtt$ is defined as a difference of two $PImp$ terms, and each $PImp$ is just a $\pi \circ \mathrm{Cond}$ term. So $PAtt$ collapses to the already-given formula for* Att*. Conceptually, this lemma is a consistency check: we have not accidentally introduced two different "attraction" notions. In practice, this means computations can be organized flexibly: one may compute $PImp$ values first (perhaps because they are reported or estimated), and then obtain* Att *automatically by subtraction, or one may work directly with* Att *when only contrasts matter.* □

Lemma HT14.5 (Negation antisymmetry).

$$\mathrm{Att}_{O,\Delta}(\neg A, B) = -\mathrm{Att}_{O,\Delta}(A, B), \qquad PAtt_{O,\Delta}(\neg A, B) = -PAtt_{O,\Delta}(A, B).$$

Remark 2030. *Negation antisymmetry formalizes the idea that if A attracts B, then "not A" repels B by the same amount, because attraction is defined as a contrast between A and $\neg A$. This is not a general truth about all causal or statistical measures; it is a special property of this particular contrastive definition. It is useful because it reduces many calculations: once* $\mathrm{Att}(A, B)$ *is known,* $\mathrm{Att}(\neg A, B)$ *is immediate. It also clarifies how to read negative values:* $\mathrm{Att}(A, B) < 0$ *is equivalent to* $\mathrm{Att}(\neg A, B) > 0$*, so one may always shift perspective to a positive-attraction statement by negating the antecedent. In settings where A is itself a decision or intervention variable, this antisymmetry also encodes a form of "two-sided" interpretability: the framework treats A and $\neg A$ as the two endpoints of the same binary contrast, rather than as unrelated predictors.*

Proof sketch. Swap A and $\neg A$ in the definition of Att and use involutivity $\neg\neg A = A$.

Remark 2031. Proof sketch. *The strategy is to rewrite* $\mathrm{Att}(\neg A, B)$ *as* $\pi(\mathrm{Cond}(B \mid \neg A)) - \pi(\mathrm{Cond}(B \mid \neg\neg A))$ *and observe this is the negative of* $\pi(\mathrm{Cond}(B \mid A)) - \pi(\mathrm{Cond}(B \mid \neg A))$*. The involutivity step is the formal reason the subtraction flips sign. Visually: attraction is an oriented difference between two points; swapping their order reverses the orientation. The same reasoning applies verbatim to $PAtt$ because Lemma HT14.4 identifies it with* Att*, but it is often convenient to state both identities explicitly since later manipulations may start from either notation.* ☐

Corollary HT14.6 (Sign rules).

$$\mathrm{Att}_{O,\Delta}(A, B) > 0 \iff PImp_{O,\Delta}(A, B) > PImp_{O,\Delta}(\neg A, B),$$

and $\mathrm{Att}_{O,\Delta}(A, B) = 0$ iff the two predictive implication strengths coincide.

Remark 2032. *This corollary translates the numerical sign of attraction into a plain-language comparison: A attracts B exactly when B is more plausible under A than under $\neg A$. The zero case is equally meaningful: it indicates predictive indifference with respect to the A versus $\neg A$ contrast. Such sign rules are the workhorses of interpretation, because they tell you what to look for in data or simulation output when assessing whether A is a "positive predictor" of B in this framework (Hyperseed-Concept* **??***). In particular, the statement emphasizes that attraction is* ordinally *interpretable: to determine the sign, one only needs to compare the two $PImp$ values, without needing their absolute magnitudes. The strictness of the inequality also matters: small positive attraction corresponds to a mild preference for B under A, while large positive attraction indicates a strong separation between the two conditional plausibilities. Conversely, if $PImp_{O,\Delta}(A, B) < PImp_{O,\Delta}(\neg A, B)$, then A is (predictively) adverse to B in precisely the same contrastive sense.*

Proof sketch. Immediate from $\mathrm{Att} = PImp(A, B) - PImp(\neg A, B)$.

Remark 2033. Proof sketch. *The proof is the elementary algebra of differences: $x - y > 0$ iff $x > y$, and $x - y = 0$ iff $x = y$. The deeper point is that attraction is defined as a* relative *measure, so it is inherently about comparisons rather than absolute magnitudes. This is also why attraction can be robust to uniform shifts in the underlying plausibility representation: anything that affects both $PImp(A, B)$ and $PImp(\neg A, B)$ equally will not change* $\mathrm{Att}(A, B)$*, whereas any asymmetric change will be detected immediately via the sign rule.* ☐

Lemma HT14.7 (Boundedness of attraction).

$$\mathrm{Att}_{O,\Delta}(A,B) \in [-1,1].$$

Remark 2034. *Attraction is bounded because plausibilities are bounded. The lemma is a small but essential piece of hygiene: it ensures that later constructions (such as mapping attraction to* $[0,1]$ *via* σ*) are always well-defined. It is also a reminder that the attraction score is not an unnormalized "force"; it is already scaled to a unit interval of influence, aligning with the broader theme of working inside compact, order-theoretic structures. The extremal values have a clear meaning in this scaling:* $\mathrm{Att}(A,B) = 1$ *corresponds to the largest possible separation where* $PImp(A,B) = 1$ *and* $PImp(\neg A,B) = 0$*, while* $\mathrm{Att}(A,B) = -1$ *corresponds to the opposite separation. Thus, boundedness is not merely technical; it encodes the intended interpretation that attraction represents a normalized contrast between two conditional plausibilities.*

Proof sketch. Each $PImp$ term lies in $[0,1]$, hence their difference lies in $[-1,1]$.

Remark 2035. Proof sketch. *The proof uses the elementary fact that the difference of two numbers in* $[0,1]$ *lies in* $[-1,1]$*. The reason this works is precisely the choice of* $\pi : [0,1]^2 \to [0,1]$*. If a different projection were used, the bound would change accordingly. A geometric picture:* $PImp(A,B)$ *and* $PImp(\neg A,B)$ *are points on the unit interval; attraction is the signed distance between them. One can also view this as a Lipschitz-type stability statement: since both terms are individually bounded and attraction is their difference, attraction cannot blow up under any admissible conditioning scenario within the framework.* □

36.4 Sequential temporal composition

Define the p-bit valued predictive relation

$$R_{O,\Delta}(A,B) := \mathrm{Cond}_{O,\Delta}(B \mid A) \in [0,1]^2.$$

Define two-step composition by

$$(R_{O,\Delta} \circ R_{O,\Delta'})(A,C) = \bigoplus_{B\in\mathcal{E}} R_{O,\Delta}(A,B) \otimes R_{O,\Delta'}(B,C),$$

where $\oplus$ is join and $\otimes$ is the quantale tensor.

Remark 2036. *The expression "p-bit valued" is meant in the sense that* $\mathrm{Cond}_{O,\Delta}(B \mid A)$ *returns a structured truth/evidence value rather than a single scalar. The display* $\in [0,1]^2$ *should be read as indicating a typical concrete instance (two coordinated degrees, e.g. a pair capturing positive/negative support, or support/anti-support); the abstract statements below use only the induced order, the join* $\oplus$ *(supremum), and the tensor* $\otimes$ *(monoidal combination), so the same arguments go through for other p-bit carriers as long as they form the relevant quantale. In particular, the results do not depend on interpreting the components probabilistically; they depend only on the order-theoretic and monoidal structure used to aggregate and propagate evidence.*

Remark 2037. *Here the parameters* (O,Δ) *should be viewed as fixing a predictive context:* O *determines which observational regime (or dataset, or filtration of information) is being used, while* Δ *selects a temporal offset. Thus* $R_{O,\Delta}(A,B)$ *is intended to encode "how much* A *at time* t *supports (in the p-bit sense) the occurrence of* B *at time* $t+\Delta$*" relative to the observation scheme* O*. The composition below then corresponds to chaining two temporal offsets, e.g.* Δ *followed by* Δ'*, with* *an intermediate event-type B serving as a latent bridge between the start and end. This aligns the algebra with the intuitive picture of multi-step prediction in time-indexed systems even when the underlying logic of evidence is paraconsistent or multi-valued.*

Remark 2038. *Here $R_{O,\Delta}$ is being treated as a V-relation (a weighted, p-bit valued kernel) from $\mathcal{E}$ to $\mathcal{E}$, exactly in the sense used elsewhere in the quantale-relational parts of the document. The notation $\circ$ is thus a generalized matrix product: multiply along an intermediate index with $\otimes$, then aggregate over intermediates with $\oplus$. In ordinary probabilistic Markov chains one would sum products; here one takes a join of tensor-products in the p-bit quantale, which is designed to be monotone and to respect evidence aggregation (compare the evidence-state algebra viewpoint in Section 38, and paraconsistent motivation [24]).*

Remark 2039. *It can be helpful to keep in mind two parallel readings of the same formula. Algebraically, $(R_{O,\Delta} \circ R_{O,\Delta'})(A, C)$ is the (A, C)-entry of the composite kernel. Operationally, it is the strongest two-step support for C obtainable from A by ranging over all possible intermediate event-types $B \in \mathcal{E}$. The role of $\otimes$ is to combine (compose) the support along a fixed two-step path $A \to B \to C$, while the role of $\oplus$ is to take the supremum over competing intermediates, i.e. to perform an evidence aggregation that is explicitly designed not to require additivity or exclusivity assumptions among the different B.*

Remark 2040. *A concrete example: if $\mathcal{E} = \{A, B, C\}$, then $(R_{O,\Delta} \circ R_{O,\Delta'})(A, C)$ is the join of the three terms obtained by choosing the intermediate event-type to be A, B, or C. Lemma HT14.8 below formalizes the obvious fact that this join must be at least as large as any one chosen two-step path.*

Remark 2041. *In such finite examples one can also see the analogy with ordinary matrix multiplication explicitly: if we arrange the values $R_{O,\Delta}(x, y)$ in a matrix with rows indexed by x and columns by y, then the composite has entries obtained by taking, for each fixed (A, C), the $\oplus$-aggregate over the intermediate index B of the "products" $R_{O,\Delta}(A, B) \otimes R_{O,\Delta'}(B, C)$. The only change from classical linear algebra is that the scalar multiplication and addition are replaced by the quantale operations $\otimes$ and $\oplus$, and the ambient order replaces numerical comparison.*

Lemma HT14.8 (Witnessed two-step lower bound). For any fixed intermediate event type $B_0 \in \mathcal{E}$,

$$R_{O,\Delta}(A, B_0) \otimes R_{O,\Delta'}(B_0, C) \ \leq\ (R_{O,\Delta} \circ R_{O,\Delta'})(A, C).$$

Remark 2042. *This lemma says: if you can predict B_0 from A and then predict C from B_0, you obtain a particular two-step predictive support for C from A. The full two-step composition takes the* best *(in the $\oplus$-join sense) among all possible intermediates, so it must dominate the support coming from any specific intermediate. This is the order-theoretic analogue of the familiar inequality "a sum over all paths is at least one particular path," adapted to join-based aggregation.*

Remark 2043. *In a paraconsistent or multi-evidence setting, the point is not merely that "more paths give more mass" (as in classical probability), but that allowing additional intermediates can only increase the composite in the underlying order, because $\oplus$ is a supremum operation. Thus HT14.8 is the minimal guarantee that any explicitly exhibited intermediate B_0 yields a sound lower bound on the two-step prediction; it legitimizes "witness-based" reasoning where one proposes a concrete mediator and obtains a provable bound without needing to analyze all alternatives in $\mathcal{E}$.*

Proof sketch. The right-hand side is the join over all intermediates, hence dominates any particular summand.

Remark 2044. Proof sketch. *The proof is a one-step lattice argument: $\bigoplus_B$ is a supremum, so each term is $\leq$ the supremum. The key step works because $\oplus$ is the join in the underlying quantale order. A useful picture is to imagine all intermediates B as offering candidate "bridges" from A to C; the composition selects their supremal evidence, so any chosen bridge yields a lower bound.* □

Lemma HT14.9 (Monotonicity of sequential composition). If $R \leq R'$ pointwise and $S \leq S'$ pointwise (in the p-bit order), then

$$S \circ R \leq S' \circ R'.$$

Remark 2045. *Monotonicity expresses a fundamental stability property: improving the evidence in R and S cannot make the composed evidence worse. This is the algebraic form of a commonsense epistemic principle: if every one-step predictive estimate is strengthened (never weakened), then every two-step estimate computed from them should also be strengthened. Such monotonicity is indispensable for comparative statics arguments and for reasoning about learning dynamics, where estimates are refined over time.*

Remark 2046. *Note that the pointwise order assumption is essential: it asserts that for every ordered pair of event-types, the replacement relation provides at least as much evidence (in the quantale order) as the original. Under that hypothesis, the composite inequality follows because the composition is built only from order-preserving primitives: first apply $\otimes$ to combine along a path, then apply $\oplus$ to aggregate over paths. No probabilistic normalization or additivity is needed, and no assumptions about independence of the intermediates are invoked; the statement is purely algebraic.*

Proof sketch. Each term $R(x, y) \otimes S(y, z)$ is monotone in each argument, and joins preserve monotonicity.

Remark 2047. Proof sketch. *The strategy is componentwise: for each triple (x, y, z), use monotonicity of $\otimes$ to show $R \otimes S \leq R' \otimes S'$, then use monotonicity of $\bigoplus$ to preserve the inequality after aggregating over y. The key step works because quantales are designed precisely so that both $\otimes$ and joins are order-preserving. One may visualize the composition as a funnel: inequalities enter at the leaves (the individual path terms) and are preserved through the funnel's aggregation.* □

Lemma HT14.10 (Associativity). Assuming $\otimes$ distributes over joins, relational composition is associative:

$$(R_1 \circ R_2) \circ R_3 = R_1 \circ (R_2 \circ R_3).$$

Remark 2048. *Associativity is what makes multi-step prediction well-defined without parentheses: it ensures that whether you first compose the first two steps or the last two steps, you get the same aggregate relation. In probabilistic terms, it is the analogue of associativity of matrix multiplication. Here it depends on the distributivity axiom: the ability to pull $\otimes$ through $\bigoplus$ is what permits the rearrangement of sums/joins needed to match the two expansions.*

Remark 2049. *Concretely, if one is interested in an n-step predictive relation built from successive lags $\Delta_1, \ldots, \Delta_n$, associativity ensures that the expression*

$$R_{O,\Delta_1} \circ R_{O,\Delta_2} \circ \cdots \circ R_{O,\Delta_n}$$

is unambiguous: any parenthesization yields the same V-relation. This matters for temporal semantics because it allows one to speak of "the" composite evidence transported through multiple intermediate event-types, without committing to a particular computational schedule for the aggregation of intermediates. In particular, it allows one to reorganize computations (e.g. dynamic programming versus direct expansion) while preserving the resulting p-bit valued predictions.

Proof sketch. Expand both sides, use distributivity to rearrange joins, and associativity of $\otimes$.

Remark 2050. Proof sketch. *The high-level strategy is "Fubini for joins": expand both sides into iterated joins over intermediate event-types, then use distributivity to swap the order of aggregation and regroup tensor-products using associativity. The key steps work because distributivity ensures that combining a fixed factor with a join of alternatives yields the join of the combined alternatives. A mental model is to think of three-step paths $A \to B \to C \to D$: both parenthesizations ultimately aggregate over the same set of paths, merely grouped differently.* $\square$

36.5 Extensional/intensional attraction and causality

Fix a finite property basis P, weights $w_{p,q} \geq 0$ with $\sum_{p,q} w_{p,q} = 1$, and a property map $\mathrm{Prop}_O : \mathcal{E} \to [0,1]^P$. Write $p \in_O A$ for the induced property-event types used in intensional attraction.

Let $PAtt^{\mathrm{ext}}_{O,\Delta}(A,B) := PAtt_{O,\Delta}(A,B)$ and

$$PAtt^{\mathrm{int}}_{O,\Delta}(A,B) := \sum_{p,q \in P} w_{p,q}\, PAtt_{O,\Delta}(p \in_O A,\ q \in_O B).$$

Remark 2051. *The extensional/intensional split is a formal way to separate two kinds of predictive regularity:*

- Extensional *attraction treats A and B as unanalyzed event-types.*
- Intensional *attraction looks inside the event-types via a property basis P and a mapping* Prop_O*, and then aggregates attraction between property-instantiations.*

This echoes the general theme that cognition (or an observer) can represent the same situation at multiple descriptive granularities: one may predict "storm" from "pressure drop" directly, or predict it by decomposing into component properties and recombining. In the broader Hyperseed vocabulary, this is a move across contexts and aspects (cf. Section 13), and it is closely related to the Hyperseed-Concept 157 and 134.

Remark 2052. *A simple example: take $P = \{red, round\}$ and consider event-types $A =$ "apple" and $B =$ "seen". Then $p \in_O A$ might denote "the current object has property p in observer O's representation." Intensional attraction can then express that "redness" is a stronger predictor of "seen" than "roundness," and the weights $w_{p,q}$ encode which property-pairs matter for the $A \to B$ relationship.*

Lemma HT14.11 (Intensional attraction is bounded). If each $PAtt_{O,\Delta}(p \in_O A, q \in_O B) \in [-1,1]$, then

$$PAtt^{\mathrm{int}}_{O,\Delta}(A,B) \in [-1,1].$$

Remark 2053. *This lemma is a convexity fact: intensional attraction is constructed as a weighted average (a convex combination) of bounded attraction values, so it must lie within the same bounds. This boundedness is crucial because causal implication below is built by feeding attraction values into $\sigma : [-1,1] \to [0,1]$; without the bound, σ would not necessarily be meaningful as a normalization.*

Proof sketch. $PAtt^{\text{int}}$ is a convex combination of numbers in $[-1,1]$.

Remark 2054. Proof sketch. *The strategy is: use that* $w_{p,q} \geq 0$ *and* $\sum_{p,q} w_{p,q} = 1$*, so the sum is an average. Any average of values in an interval remains in the interval. The key step works because the weights are normalized; without normalization, one would only get a scaled interval. Visually,* $PAtt^{\text{int}}$ *lies in the convex hull of the set of component attractions, which in one dimension is just the interval between the minimum and maximum (hence within* $[-1,1]$*).* □

Let $\sigma : [-1,1] \to [0,1]$ be $\sigma(x) = (1+x)/2$, and define causal implication

$$CImp_{O,\Delta}(A,B) = \Big(\lambda\sigma(PAtt^{\text{ext}}_{O,\Delta}(A,B)) + (1-\lambda)\sigma(PAtt^{\text{int}}_{O,\Delta}(A,B))\Big)\mathbf{1}[A \prec B].$$

Also define the strong product variant

$$CImp^{\times}_{O,\Delta}(A,B) = \sigma(PAtt^{\text{ext}}_{O,\Delta}(A,B)) \cdot \sigma(PAtt^{\text{int}}_{O,\Delta}(A,B)) \cdot \mathbf{1}[A \prec B].$$

Remark 2055. *The map* $\sigma(x) = (1+x)/2$ *is the same affine renormalization idea seen earlier: it converts an attraction score in* $[-1,1]$ *into a unit-interval score in* $[0,1]$*. The causal implication* $CImp_{O,\Delta}(A,B)$ *then blends extensional and intensional evidence using a convex weight* $\lambda \in [0,1]$*, and finally gates by temporal precedence* $\mathbf{1}[A \prec B]$*. This makes causality explicitly* asymmetric in proto-time*, aligning with the ordinary intuition that causes precede effects (Hyperseed-Concept 70), while also acknowledging that causal judgments are observer-dependent through O (a point compatible with process and relational perspectives [15, 14]).*

Remark 2056. *The product variant* $CImp^{\times}$ *is deliberately stricter: it is large only when* both *extensional and intensional attractions are large (after normalization). The mixture variant can be large if either component is large enough (depending on* λ*). This mirrors a familiar epistemic choice: do we accept "either kind of evidence suffices" (mixture) or "both must agree" (product)?*

Proposition HT14.12 (Causal implication range and gating). $CImp_{O,\Delta}(A,B) \in [0,1]$, and $CImp_{O,\Delta}(A,B) = 0$ whenever $\mathbf{1}[A \prec B] = 0$.

Remark 2057. *This proposition ensures that* $CImp$ *is a well-formed graded causal score: it always lies between* 0 *and* 1*, and it vanishes when the temporal-precedence constraint fails. The gating clause is particularly important: it prevents a purely predictive correlation (which might be symmetric in time) from being misread as a causal arrow. In scientific terms, it encodes a minimal causal discipline: time must point the right way [20].*

Proof sketch. σ maps $[-1,1]$ into $[0,1]$, the mixture is a convex combination, and multiplication by $\mathbf{1}[A \prec B] \in \{0,1\}$ gates the result.

Remark 2058. Proof sketch. *The strategy is bounding: map each attraction into* $[0,1]$*, average them, then multiply by a* $0/1$ *indicator. The key steps work because convex combinations preserve interval bounds. The visual intuition is that* $CImp$ *lives in the unit interval simplex: the mixture cannot escape* $[0,1]$*, and the indicator either leaves it unchanged (if precedence holds) or collapses it to* 0*.* □

Proposition HT14.13 (Affine simplification). Let $a = PAtt^{\text{ext}}_{O,\Delta}(A,B)$ and $b = PAtt^{\text{int}}_{O,\Delta}(A,B)$. Then

$$CImp_{O,\Delta}(A,B) = \sigma(\lambda a + (1-\lambda)b)\,\mathbf{1}[A \prec B] = \frac{1 + \lambda a + (1-\lambda)b}{2}\,\mathbf{1}[A \prec B].$$

Remark 2059. *This proposition reveals that the mixture-of-σ definition is equivalent to applying σ* after *mixing the attraction scores. That is, the causal implication is an affine function of the blended attraction $\lambda a + (1-\lambda)b$. This is algebraically convenient: it makes monotonicity and comparative statics transparent, and it clarifies that λ is literally interpolating between two attraction estimates before normalization. In particular, since $a, b \in [-1,1]$ (by construction of attraction), the blend $\lambda a + (1-\lambda)b$ also lies in $[-1,1]$, so the normalized score prior to gating lies in $[0,1]$ without any further clipping. The endpoint cases are also immediate: when $\lambda = 1$ the expression reduces to $\sigma(a)\mathbf{1}[A \prec B]$ (purely extensional), and when $\lambda = 0$ it reduces to $\sigma(b)\mathbf{1}[A \prec B]$ (purely intensional). Moreover, if $a = b$ then the causal implication is independent of λ, emphasizing that λ only matters insofar as the two attraction estimates disagree.*

Proof sketch. Use $\sigma(x) = (1+x)/2$ to compute $\lambda\sigma(a) + (1-\lambda)\sigma(b) = \sigma(\lambda a + (1-\lambda)b)$.

Remark 2060. Proof sketch. *The proof is an exercise in linearity: σ is affine, so it commutes with convex combinations. The key step works because σ has the form $x \mapsto \alpha x + \beta$. A geometric picture is that affine maps send line segments to line segments; blending a and b traces a segment in $[-1,1]$, and σ maps that segment to a segment in $[0,1]$. Concretely, expanding the left-hand side gives*

$$\lambda\frac{1+a}{2} + (1-\lambda)\frac{1+b}{2} = \frac{\lambda + \lambda a + 1 - \lambda + (1-\lambda)b}{2} = \frac{1 + \lambda a + (1-\lambda)b}{2} = \sigma(\lambda a + (1-\lambda)b),$$

and then multiplying by $\mathbf{1}[A \prec B]$ simply applies the same temporal (or precedence) gate to either equivalent form. This also makes clear that the only nonlinearity in $CImp_{O,\Delta}(A,B)$ comes from the indicator $\mathbf{1}[A \prec B]$, whereas the dependence on (a,b) is purely affine once $A \prec B$ is fixed. ☐

Proposition HT14.14 (Monotonicity in attraction components). Assume $\mathbf{1}[A \prec B]$ is fixed. Then $CImp_{O,\Delta}(A,B)$ is nondecreasing in $PAtt^{\text{ext}}_{O,\Delta}(A,B)$ and in $PAtt^{\text{int}}_{O,\Delta}(A,B)$ (holding the other fixed).

Remark 2061. *This proposition formalizes a basic rationality constraint on the scoring rule: if either the extensional or intensional evidence of attraction increases (with the temporal gate held fixed), then the causal implication score does not decrease. In practical terms, it guarantees that improving predictive evidence cannot perversely reduce the inferred causal strength. This kind of monotonicity is essential if one wishes to use these scores inside optimization or learning loops. Under the affine simplification of Proposition HT14.13 (when $A \prec B$ is fixed), the dependence is even transparent at the level of partial derivatives: for $\mathbf{1}[A \prec B] = 1$ one has $\partial CImp/\partial a = \lambda/2 \geq 0$ and $\partial CImp/\partial b = (1-\lambda)/2 \geq 0$, so the score is not merely monotone but linear in each component separately. The assumption that $\mathbf{1}[A \prec B]$ is fixed is crucial: if $A \prec B$ can flip from true to false, the score can drop discontinuously to 0 due to the gating, regardless of attraction values.*

Proof sketch. σ is increasing, and convex combinations preserve order.

Remark 2062. Proof sketch. *The strategy is composition of monotone maps: each attraction component is fed into the increasing function σ, and then combined linearly with nonnegative* *coefficients summing to* 1*. The key step is that both σ and convex combination operators are order-preserving. One can imagine sliders for a and b: pushing either slider upward cannot pull the output downward as long as the gate is constant. More explicitly, fix b and take $a' \geq a$; then $\lambda a' + (1-\lambda)b \geq \lambda a + (1-\lambda)b$, and since σ is increasing this implies $\sigma(\lambda a' + (1-\lambda)b) \geq \sigma(\lambda a + (1-\lambda)b)$, with the same argument holding when varying b while keeping a fixed. If one prefers an algebraic check, Proposition HT14.13 yields $CImp = \frac{1+\lambda a+(1-\lambda)b}{2}\mathbf{1}[A \prec B]$, which is visibly nondecreasing in a and b when $\mathbf{1}[A \prec B]$ is held constant.* □

Proposition HT14.15 (Product variant is a lower bound).

$$CImp^{\times}_{O,\Delta}(A, B) \ \leq\ CImp_{O,\Delta}(A, B).$$

Remark 2063. *This proposition makes precise the earlier intuition that the product variant is stricter. Algebraically, it says that xy is never larger than the weighted average $\lambda x + (1 - \lambda)y$ (indeed $xy \leq \min\{x, y\}$). Thus the product score is conservative: it penalizes disagreement between extensional and intensional evidence more strongly than the mixture does. This is a useful option when one wants causal claims to require corroboration across representational levels. The inequality also highlights when the two scores coincide: equality holds whenever one factor already enforces the bottleneck, e.g., if $x \leq y$ and $\lambda = 1$ then both reduce to x; more generally $xy = \min\{x, y\}$ occurs when $\min\{x, y\} \in \{0, 1\}$ (since if $x = 0$ or $y = 0$ then both product and mixture are* 0*, and if $x = y = 1$ then both are* 1 *before gating). For intermediate values, the gap between product and mixture quantifies how much the rule rewards having both components simultaneously high rather than merely one of them high.*

Proof sketch. Let $x = \sigma(PAtt^{\text{ext}})$ and $y = \sigma(PAtt^{\text{int}})$ so $x, y \in [0, 1]$. Then $xy \leq \min\{x, y\} \leq \lambda x + (1 - \lambda)y$. Multiply by $\mathbf{1}[A \prec B]$.

Remark 2064. Proof sketch. *The proof strategy is to compare product, minimum, and convex combination on* $[0, 1]$*. The key inequality $xy \leq \min\{x, y\}$ follows because both x and y are at most* 1*. The step $\min\{x, y\} \leq \lambda x + (1 - \lambda)y$ holds because a convex combination lies between its endpoints and hence above the minimum. Visually, the product curve lies below the line segment connecting* $(0, 0)$ *to* $(1, 1)$ *except at the endpoints; the convex combination is precisely such a line segment in the* (x, y) *plane. One can also verify the first inequality by cases: if $x \leq y$ then $xy \leq x \cdot 1 = x = \min\{x, y\}$, and symmetrically if $y \leq x$ then $xy \leq y$. For the second inequality, note that $\lambda x + (1 - \lambda)y \geq \lambda \min\{x, y\} + (1 - \lambda) \min\{x, y\} = \min\{x, y\}$, using $\lambda \in [0, 1]$. Finally, multiplying by $\mathbf{1}[A \prec B] \in \{0, 1\}$ preserves the inequality and makes explicit that the comparison is meaningful only on the temporally admissible pairs* (A, B)*.* □

36.6 Control and hierarchy

Let $\mathcal{A} \subseteq \mathcal{E}$ be the action event types. Define control strength for action $A \in \mathcal{A}$ and goal condition $G \in \mathcal{E}$ by

$$\mathrm{Ctrl}_{O,\Delta}(A \to G) := CImp_{O,\Delta}(A, G).$$

Remark 2065. *The restriction $A \in \mathcal{A}$ is doing conceptual work: it separates "things the agent does" from generic events in $\mathcal{E}$, while still letting goals G range over arbitrary event-types (including internal state conditions, external outcomes, or composite conditions). This is useful later when actions are treated as decision variables but goals are treated as desiderata or constraints.*

Remark 2066. *Because* $\mathrm{Ctrl}_{O,\Delta}(A \to G)$ *is observer-relative and* Δ*-gated, it should be read as an empirical, temporally localized notion of instrumental influence: an action-type may "control" a goal in one observational regime (or at one temporal scale) but not in another. In particular, the same physical actuator event-type can yield different* Ctrl *values for different observers* O *(e.g., due to sensor resolution, feature abstraction, or differing event-type definitions), and the same pair* (A, G) *can exhibit strong control at one time-lag but weak control at another (e.g., immediate effects versus delayed effects).*

Remark 2067. *Control is here defined as a special case of causal implication: an action* A *"controls" a goal-condition* G *to the extent that* A *causally implies* G *(in the observer-relative, temporally gated sense already defined). This matches the engineering stance common in agent architectures: control is causality-as-instrumentality [19]. In the Hyperseed core-concept set, this aligns directly with Hyperseed-Concept 87 (and the later use of hierarchical organization aligns with 88).*

Remark 2068. *It is also worth separating this usage from classical control theory:* $\mathrm{Ctrl}_{O,\Delta}(A \to G)$ *is not asserting the existence of a continuous feedback law or robustness margins, but rather providing a scalar, event-type level summary of how reliably* A *brings about* G *in the modeled distribution. In later applications, one can still layer feedback and policy concepts on top of this by interpreting policies as distributions over action event-types and evaluating their induced control of goal event-types.*

Corollary HT14.16 (Control inherits causal properties). For actions A and goals G:

$$\mathrm{Ctrl}_{O,\Delta}(A \to G) \in [0,1], \quad \mathrm{Ctrl}_{O,\Delta}(A \to G) = 0 \text{ if } \mathbf{1}[A \prec G] = 0,$$

and Ctrl is monotone in extensional and intensional predictive attraction (when precedence is fixed).

Remark 2069. *The indicator* $\mathbf{1}[A \prec G]$ *makes explicit the temporal-asymmetry constraint: without the stipulated precedence (within the gating window), "control" is defined to vanish, which prevents conflating mere correlation with actionable influence. This matters for later hierarchical constructions, since a controller at a higher level must be able to treat sub-actions as antecedents whose consequences occur after the sub-actions are executed, not before.*

Remark 2070. *The monotonicity clause is especially important operationally: increasing predictive attraction between* A *and* G *(whether by adding more supporting observations in the extensional sense, or by strengthening the intensional/structural component of attraction) cannot decrease the assessed control strength, so improvements in predictive coupling translate into non-worsening estimates of instrumentality. When precedence is fixed, this preserves the intuitive ordering "more predictively tied"* $\Rightarrow$ *"no less controllable," which is used implicitly when ranking candidate actions by anticipated goal achievement.*

Remark 2071. *This corollary is a packaging result: once control is defined as* $CImp$ *on action/goal pairs, all the previously established well-formedness and monotonicity properties come along automatically. It is useful because it lets later arguments about policy selection or hierarchical action selection reuse the causal math without re-proving basic bounds each time.*

Remark 2072. *In particular, the bounds* $\mathrm{Ctrl}_{O,\Delta}(A \to G) \in [0,1]$ *let* Ctrl *be used directly as a normalized objective component (e.g., in weighted sums with other* $[0,1]$*-valued criteria such as cost-normalized utility or constraint satisfaction). The precedence-zeroing property further ensures that any optimization procedure that searches over candidate actions will not be misled by "backwards" associations that are not implementable as control.*

Proof sketch. Immediate from $\mathrm{Ctrl} = CImp$ and Propositions HT14.12–HT14.14.

Remark 2073. Proof sketch. *The strategy is substitution: replace* Ctrl *with* $CImp$ *and cite the already proven propositions. The key step is recognizing that the definitions align exactly, so no additional structure is needed. The intuition is that "control" is not a new primitive here, but merely causality restricted to a designated subset of event-types called actions.* ☐

Remark 2074. *If one wants to see the substitution explicitly, each claimed property of* $\mathrm{Ctrl}_{O,\Delta}(A \to G)$ *is literally the corresponding property of* $CImp_{O,\Delta}(A, G)$ *with the argument-pair renamed from "(antecedent event-type, consequent event-type)" to "(action-type, goal-type)." The only extra premise is membership* $A \in \mathcal{A}$*, which does not affect the algebraic properties proved for* $CImp$*.*

Lemma HT14.17 (Score tradeoff rule for weak causal parent selection). For parent sets $S_1, S_2 \subseteq C$,

$$\mathsf{Score}(S_1 \to B) \geq \mathsf{Score}(S_2 \to B)$$

$$\mathrm{Fit}(S_1 \to B) - \mathrm{Fit}(S_2 \to B) \geq \eta\big(\mathrm{Weak}(S_2 \to B) - \mathrm{Weak}(S_1 \to B)\big).$$

In particular, if Fit ties, any maximizer of Weak is a maximizer of Score.

Remark 2075. *This lemma is a generic tradeoff principle: if a score* Score *is built by combining a "fit" term* Fit *and a "weakness" term* Weak *(penalized or rewarded with coefficient* η*), then comparing two candidates reduces to a linear inequality balancing fit improvement against weakness change. In many model-selection settings,* Fit *rewards explanatory adequacy while* Weak *rewards simplicity, regularity, or low commitment—a theme echoed in the quantale-weakness program [3] and in broader discussions of weak constraints in building cognitive systems [2]. The final sentence captures a common selection heuristic: if two models explain equally well, prefer the weaker (less committal / more transferable) one. This is closely related to the "tradeoff theorem" style of reasoning about dependency management [9].*

Remark 2076. *The direction of the inequality highlights the role of* η *as an exchange rate between the two desiderata: increasing* η *makes improvements in weakness more valuable (or, if* Weak *is interpreted as a penalty with a sign convention, makes reductions in weakness more valuable). Thus, the lemma can be read as specifying the minimum fit gain required to justify accepting a worse weakness value, namely* $\eta(\mathrm{Weak}(S_2 \to B) - \mathrm{Weak}(S_1 \to B))$*. This is the standard "regularization tradeoff" pattern, stated in a way that is independent of the internal definitions of* Fit *and* Weak*.*

Remark 2077. *Although the statement is phrased for parent sets* $S \subseteq C$ *predicting* B*, the same algebraic form applies whenever one compares two candidate explanations, controllers, or decompositions: one term measures performance on data or goals (fit), the other measures a structural preference (weakness as simplicity/portability/low-commitment), and* η *sets the operating point. This makes the lemma a convenient reusable tool in hierarchical selection arguments where one repeatedly chooses among alternative parent-sets at different levels of abstraction.*

Proof sketch. Subtract the two scores and rearrange.

Remark 2078. Proof sketch. *The strategy is algebraic normalization: write* $\mathsf{Score}(S_1 \to B) - \mathsf{Score}(S_2 \to B)$ *in terms of* Fit *and* Weak*, then isolate* $\mathrm{Fit}(S_1 \to B) - \mathrm{Fit}(S_2 \to B)$ *on one side. The key step works because the score is assumed to be affine in these components (as is typical for regularized objectives). One can picture a two-dimensional plane with axes* $(\mathrm{Fit}, \mathrm{Weak})$*: the inequality defines a half-space bounded by a line of slope determined by* η*; moving along that line preserves score, and maximizing score chooses points on the "northeast" side according to the sign convention.* ☐

Remark 2079. *Concretely, under the common form* $\mathsf{Score}(S \to B) = \mathrm{Fit}(S \to B) + \eta\,\mathrm{Weak}(S \to B)$*, the condition* $\mathsf{Score}(S_1 \to B) \ge \mathsf{Score}(S_2 \to B)$ *expands to*

$$\mathrm{Fit}(S_1 \to B) - \mathrm{Fit}(S_2 \to B) \;\ge\; \eta\big(\mathrm{Weak}(S_2 \to B) - \mathrm{Weak}(S_1 \to B)\big),$$

*which is exactly the displayed inequality. The "*Fit *ties" corollary then follows by setting the left-hand side to* 0*, leaving* $\mathrm{Weak}(S_1 \to B) \ge \mathrm{Weak}(S_2 \to B)$ *as the deciding criterion among equal-fit candidates.*

Lemma HT14.18 (No cycles in a control hierarchy). If $(\mathcal{M}, <)$ is a partial order used as a control hierarchy, there is no strict cycle $M_1 < M_2 < \cdots < M_k < M_1$.

Remark 2080. *This lemma is a structural axiom dressed as a theorem: a control hierarchy is modeled as a partial order, and partial orders cannot contain strict cycles. The usefulness is practical: if a hierarchy had cycles, then "higher" and "lower" control would become inconsistent, undermining modular decomposition and the intended semantics of top-down constraint and bottom-up reporting. In broader system-design terms, it is the mathematical reflection of the idea that a well-formed hierarchy should permit unambiguous escalation and delegation (Hyperseed-Concept 88; see also hierarchical organization themes in [5]).*

Remark 2081. *Interpreting* $<$ *as "is a higher-level controller than" (or "constrains/organizes"), acyclicity ensures that control influence cannot feed back into itself purely by the hierarchy relation. This does not forbid dynamical feedback in the underlying system; it only forbids representing the hierarchical* organizational *relation itself as cyclic. In practice, this means one can treat the hierarchy graph induced by* $<$ *as a DAG and safely perform top-down passes (for constraint propagation or goal decomposition) and bottom-up passes (for state aggregation or reporting) without encountering circular dependencies at the representation level.*

Proof sketch. A strict cycle contradicts transitivity and antisymmetry of a partial order.

Remark 2082. Proof sketch. *The proof strategy is contradiction: assume the cycle exists, then use transitivity to derive* $M_1 < M_1$*, which is impossible in a strict partial order (by irreflexivity, equivalently by antisymmetry when phrased via* $\le$*). The key step is that chaining* $<$*-relations via transitivity collapses the entire cycle into a self-relation. A visual intuition is to imagine "height levels": if* M_1 *is above* M_2 *and* M_k *is above* M_1*, then* M_1 *would have to be above itself, which is geometrically incoherent.* ◻

Remark 2083. *If the hierarchy is instead presented in a non-strict form* $(\mathcal{M}, \le)$*, the same conclusion can be phrased as: from* $M_1 \le M_2 \le \cdots \le M_k \le M_1$*, transitivity yields* $M_1 \le M_1$ *(always true) but antisymmetry then forces all* M_i *to be equal, so there is no* strict *cycle. This is often the convenient viewpoint when implementing hierarchies as reachability in a graph or as a refinement relation in a type lattice.*

37 Helper theorems for Attention and Cognitive Synergy

37.1 Standing notation from Section 15

Fix a system S with modules $M = \{1, \ldots, n\}$, patterns P (finite or countable), association map $\mathrm{Assoc} : M \to 2^P$, time index set T, genenergy expenditures $g : T \times M \to [0, \infty)$ with $g_t(i) := g(t, i)$, and total genenergy

$$G_t := \sum_{i \in M} g_t(i).$$

Assume each module i has weights $w_i : P \to [0,1]$ with $\sum_{p \in P} w_i(p) = 1$ and $w_i(p) = 0$ unless $p \in \mathrm{Assoc}(i)$. Define pattern-level attention mass

$$\mathrm{Attn}_t(p) := \sum_{i \in M} g_t(i)\, w_i(p), \qquad \mathrm{Attn}_t(X) := \sum_{p \in X} \mathrm{Attn}_t(p) \quad (X \subseteq P).$$

When $G_t > 0$, define the normalized attention distribution

$$\alpha_t(p) := \frac{\mathrm{Attn}_t(p)}{G_t}, \qquad \alpha_t(X) := \sum_{p \in X} \alpha_t(p).$$

Fix focus parameters (θ, k) with $\theta \in (0,1)$ and $k \in \mathbb{N}$. Let $P_{\mathrm{self}} \subseteq P$ be the current self-model pattern set.

Remark 2084. *It is helpful to read the foregoing as a precise bookkeeping scheme for "where the system is spending its effort" (Hyperseed-Concept 100) and, more specifically, for how that effort is apportioned across recognized patterns (Hyperseed-Concept 130; see also the pattern-oriented viewpoint in [5]). The set $M = \{1, \ldots, n\}$ is a finite index of modules; P is a (finite or countable) catalogue of patterns the system can attend to; and the association map $\mathrm{Assoc} : M \to 2^P$ assigns to each module the subset of patterns it is capable of touching. Here 2^P denotes the powerset of P, i.e. the set of all subsets of patterns.*

Remark 2085. *The quantity $g_t(i)$ is the (nonnegative) genenergy expenditure of module i at time t (Hyperseed-Concept **??**), and G_t is the total budget actually spent across modules at that time. Each w_i is a probability distribution over patterns (supported only on $\mathrm{Assoc}(i)$), which we may interpret as "if module i is active, what fraction of its activity is directed at each pattern." Then $\mathrm{Attn}_t(p)$ is the unnormalized attention mass landing on pattern p, while α_t is the corresponding normalized attention distribution, a genuine probability distribution when $G_t > 0$ (Hyperseed-Concept 60).*

Remark 2086. *Because the module index set M is finite, the total genenergy $G_t = \sum_{i \in M} g_t(i)$ is always well-defined as a finite sum, and similarly $\mathrm{Attn}_t(p) = \sum_{i \in M} g_t(i) w_i(p)$ is always a finite sum for each fixed pattern $p \in P$. Even when P is countably infinite, the expressions $\sum_{p \in P} w_i(p) = 1$ and $\mathrm{Attn}_t(X) = \sum_{p \in X} \mathrm{Attn}_t(p)$ may be understood as countable series with nonnegative terms, so there are no conditional-convergence issues: any reordering gives the same value, and $\mathrm{Attn}_t(X) \in [0, \infty]$ is determined unambiguously for every subset $X \subseteq P$. In typical applications one either restricts attention to subsets X for which the series converges, or simply notes that for $G_t < \infty$ (automatic here) and $G_t > 0$ one has $\mathrm{Attn}_t(P) = G_t$ and hence $\mathrm{Attn}_t(X) \le G_t$ for all $X \subseteq P$, so $\mathrm{Attn}_t(X)$ is automatically finite.*

Remark 2087. *A useful identity is that the attention distribution α_t can be viewed as a convex mixture of the module-specific pattern weights. Indeed, when $G_t > 0$,*

$$\alpha_t(p) = \frac{1}{G_t} \sum_{i \in M} g_t(i) w_i(p) = \sum_{i \in M} \Big(\frac{g_t(i)}{G_t} \Big) w_i(p),$$

so the mixing coefficients are precisely the normalized module expenditures $g_t(i)/G_t$. This makes explicit the separation between (i) how much *each module participates at time t (captured by $g_t(i)$) and (ii)* what *each module tends to point at when it participates (captured by w_i). The set-function $\alpha_t(X) = \sum_{p \in X} \alpha_t(p)$ may therefore be read as "the fraction of total effort allocated to the pattern subset X," and, in particular, $\alpha_t(P) = 1$ when $G_t > 0$.*

Remark 2088. *The $G_t > 0$ condition is included to avoid division by zero; intuitively, it corresponds to times at which the system is allocating a nontrivial amount of genenergy. When $G_t = 0$, all module expenditures are zero, hence $\mathrm{Attn}_t(p) = 0$ for every p and $\mathrm{Attn}_t(X) = 0$ for every $X \subseteq P$; in that case α_t is simply left undefined (or can be set arbitrarily by convention, depending on downstream use). Many later inequalities about attention are naturally stated either conditional on $G_t > 0$ or in a form that becomes vacuous when $G_t = 0$.*

Remark 2089. *The focus parameters (θ, k) are fixed global constants for the discussion that follows and are intended to parameterize a notion of "focused" attention: $\theta \in (0, 1)$ typically plays the role of a threshold for how much normalized attention mass must concentrate on a small set of patterns, and $k \in \mathbb{N}$ typically bounds the cardinality of such a set. Although the present subsection does not yet impose any focusing condition, it is convenient to fix (θ, k) in the standing notation so that subsequent helper theorems can state hypotheses such as "there exists a set $X \subseteq P$ with $|X| \leq k$ and $\alpha_t(X) \geq \theta$" without repeatedly restating the parameter choices.*

Remark 2090. *The distinguished subset $P_{\mathrm{self}} \subseteq P$ is meant to single out those patterns presently constituting the system's self-model. With the above notation, quantities like $\mathrm{Attn}_t(P_{\mathrm{self}})$ and $\alpha_t(P_{\mathrm{self}})$ become immediate ways of expressing how much absolute effort, or what fraction of total effort, is being allocated to self-model patterns at time t. This allows later arguments to treat "self-directed" attention as simply attention to a particular pattern subset, and to compare it directly against other relevant pattern classes using the same $\mathrm{Attn}_t(\cdot)$ and $\alpha_t(\cdot)$ bookkeeping.*

37.2 Genenergy and attention-mass identities

Lemma HT15.1 (Total pattern attention equals total genenergy). Assume $G_t < \infty$. Then

$$\mathrm{Attn}_t(P) = \sum_{p \in P} \mathrm{Attn}_t(p) = G_t.$$

Remark 2091. *Intuitively: if each module distributes all of its spent genenergy across patterns using weights that sum to 1, then no genenergy disappears and none is created when we "push it down" from modules to patterns. The lemma asserts that the total attention mass across all patterns is exactly the total genenergy expended across modules.*

Remark 2092. *Formally, this is a conservation law: the mapping $g_t(\cdot)$ on modules becomes $\mathrm{Attn}_t(\cdot)$ on patterns by redistribution through the kernels $w_i(\cdot)$. The condition $\sum_{p \in P} w_i(p) = 1$ for each module i is exactly the statement that w_i is* mass-preserving *on P, so summing over all patterns recovers the original mass $g_t(i)$ for each i, and then summing over i recovers G_t.*

Proof sketch. Expand and swap sums: $\sum_p \mathrm{Attn}_t(p) = \sum_p \sum_i g_t(i) w_i(p) = \sum_i g_t(i) \sum_p w_i(p) = \sum_i g_t(i) = G_t$.

Remark 2093. *The key move is the exchange of summation order, which is justified here by nonnegativity and the finiteness assumption $G_t < \infty$ (so no problematic $\infty - \infty$ expressions arise). Conceptually, we are summing the same collection of contributions $g_t(i) w_i(p)$ but first grouping by patterns and then by modules.*

Remark 2094. *If one wants a slightly more explicit justification: because $g_t(i) \geq 0$ and $w_i(p) \geq 0$, the double sum $\sum_{p \in P} \sum_{i \in M} g_t(i) w_i(p)$ is a sum of nonnegative terms, so Tonelli's theorem (or the monotone convergence theorem for series) permits reordering without changing the value (possibly $+\infty$). The additional assumption $G_t < \infty$ ensures the value is in fact finite, so the identity is an equality of real numbers rather than an extended-real statement.*

Corollary HT15.2 (Normalization). If $G_t > 0$ then α_t is a probability distribution on P:

$$\alpha_t(p) \geq 0, \qquad \sum_{p \in P} \alpha_t(p) = 1.$$

Remark 2095. *This makes precise the common informal statement that "attention is a distribution." The normalization step is not mere cosmetic algebra: it lets one compare attention profiles across times t even when the total spending G_t changes, and it prepares attention to be treated with probabilistic and information-theoretic tools (Hyperseed-Concept 139,* **??***).*

Remark 2096. *Note that the requirement $G_t > 0$ is essential: if $G_t = 0$ then every module spends zero genenergy, so $\mathrm{Attn}_t(p) = 0$ for all p, and there is no canonical way to normalize (one would be dividing 0 by 0). In that degenerate case, it is often convenient to treat Attn_t as the zero measure and leave α_t undefined (or define it by an external convention if a particular application needs a default).*

Proof sketch. Nonnegativity is immediate. The sum is $\mathrm{Attn}_t(P)/G_t = G_t/G_t = 1$ by Lemma HT15.1.

Remark 2097. *The corollary is an instance of a general pattern: once you prove a conserved-total identity (Lemma HT15.1), the normalized object inherits the axioms of probability almost for free.*

Remark 2098. *A useful way to read the conclusion is that α_t is the* relative *attention allocation: $\alpha_t(p)$ is the fraction of the total spent genenergy that ends up attributed to pattern p. In particular, statements such as "pattern p has twice the attention of pattern q" are meaningful at the level of α_t even when the absolute budget G_t varies with hardware, load, or scheduling.*

Lemma HT15.3 (Finite additivity and monotonicity). For disjoint $X, Y \subseteq P$,

$$\mathrm{Attn}_t(X \cup Y) = \mathrm{Attn}_t(X) + \mathrm{Attn}_t(Y), \qquad \alpha_t(X \cup Y) = \alpha_t(X) + \alpha_t(Y),$$

and if $X \subseteq Y$ then $\mathrm{Attn}_t(X) \leq \mathrm{Attn}_t(Y)$ and $\alpha_t(X) \leq \alpha_t(Y)$.

Remark 2099. *This lemma records the elementary but indispensable measure-like properties of attention mass: disjoint subsets do not "double count" attention, and larger sets can only receive more (or equal) mass. In practice, it lets one reason about attention to composite pattern-collections without repeatedly expanding sums.*

Remark 2100. *The statement is deliberately phrased in the language of sets to emphasize that $\mathrm{Attn}_t(\cdot)$ behaves like a (finitely additive) measure on the discrete space P. When P is finite or countable, one may equivalently view Attn_t as a measure determined by its point masses $\mathrm{Attn}_t(p)$, with $\mathrm{Attn}_t(X) = \sum_{p \in X} \mathrm{Attn}_t(p)$. The lemma then reduces to standard facts about sums over disjoint unions and inclusions.*

Proof sketch. All claims follow from the definitions as sums of nonnegative terms.

Remark 2101. *The proof relies on the fact that attention is constructed by summing nonnegative contributions. Once nonnegativity is present, monotonicity is almost tautological: adding patterns adds nonnegative summands.*

Remark 2102. *It is worth noting what is* not *asserted: countable additivity (a σ-additivity property) is not needed for most manipulations in this section, and it would require additional care if P were uncountable or if one allowed more general (non-discrete) pattern spaces. In the present discrete setting, whenever the relevant sums are well-defined, the same nonnegativity argument that gives finite additivity typically extends to countable disjoint unions by monotone convergence.*

Lemma HT15.4 (Scale invariance of normalized attention). If g_t is replaced by $g'_t(i) = c\, g_t(i)$ with $c > 0$, then α_t is unchanged.

Remark 2103. *Normalized attention is meant to capture* relative *allocation rather than absolute spending. This lemma formalizes that intention: multiplying the entire budget by a positive constant (e.g. due to a faster processor, or a higher clock rate) does not change the attention distribution.*

Remark 2104. *The restriction $c > 0$ matches the intended interpretation as a rescaling of available resources. If $c = 0$, then G_t collapses to 0 and the normalization in α_t is no longer defined; if $c < 0$ then g'_t would cease to be nonnegative and would no longer represent genenergy. Thus the lemma is a genuine invariance principle within the physically meaningful parameter range.*

Proof sketch. Both numerator $\mathrm{Attn}_t(p)$ and denominator G_t scale by c, so the ratio is invariant.

Remark 2105. *The invariance is a simple homogeneity property: α_t is of degree 0 in g_t, because it is defined as a quotient of two degree-1 quantities.*

Remark 2106. *In applications, this lemma is often invoked implicitly when comparing runs of the same system under different absolute budgets. It licenses treating α_t as a budget-free signature of what the system* cares about *at time t, while Attn_t and G_t quantify how strongly it can act on those cares in absolute terms.*

Lemma HT15.5 (Pushforward form of pattern attention). Define the module-level allocation $\hat{\alpha}_t(i) := g_t(i)/G_t$ (when $G_t > 0$). Then

$$\alpha_t(p) = \sum_{i \in M} \hat{\alpha}_t(i)\, w_i(p).$$

Remark 2107. *This expresses α_t as a two-stage mixture: first choose a module i according to the fraction of total genenergy it receives (the distribution $\hat{\alpha}_t$ on M), then choose a pattern p according to w_i. In probabilistic language, α_t is the pushforward/marginal distribution of this hierarchical sampling process (Hyperseed-Concept 139).*

Remark 2108. *Equivalently, if one thinks of $w_i(\cdot)$ as a stochastic "emission" profile from module i to patterns, then α_t is the mixture of these emissions with mixture weights given by the module budget shares $\hat{\alpha}_t(i)$. This makes clear that α_t aggregates two distinct design choices: (i) how genenergy is allocated across modules ($\hat{\alpha}_t$), and (ii) how each module maps its own activity to patterns (w_i).*

Proof sketch. Divide $\mathrm{Attn}_t(p) = \sum_i g_t(i) w_i(p)$ by G_t and rewrite $g_t(i)/G_t = \hat{\alpha}_t(i)$.

Remark 2109. *The identity is often used in the reverse direction: instead of computing α_t by first aggregating unnormalized attention, one may reason in terms of module shares $\hat{\alpha}_t(i)$, which can be treated as a higher-level control variable.*

Remark 2110. *This representation also highlights a useful conditional-expectation interpretation. If I is a random module with $\Pr(I = i) = \hat{\alpha}_t(i)$ and then P is drawn with $\Pr(P = p \mid I = i) = w_i(p)$, then $\alpha_t(p) = \Pr(P = p)$. In particular, for any test function f on patterns, $\sum_p f(p)\alpha_t(p) = \sum_i \hat{\alpha}_t(i) \sum_p f(p) w_i(p)$, so expectations under α_t can be computed by first taking expectations under each w_i and then averaging over modules.*

37.3 Attentional focus and self-reflective attention

Lemma HT15.6 (Top-k mass characterization). Let $\alpha_t^{\downarrow}(1) \geq \alpha_t^{\downarrow}(2) \geq \cdots$ be the nonincreasing rearrangement of $\{\alpha_t(p) : p \in P\}$. Then

$$\max\{\alpha_t(F) : F \subseteq P,\ |F| \leq k\} = \sum_{j=1}^{k} \alpha_t^{\downarrow}(j).$$

Remark 2111. *The statement says: if you may "choose at most k patterns to look at," the maximum total attention you can capture is obtained by choosing the k most-attended patterns. This is the formal underpinning of a very practical heuristic: focus is attention that is concentrated on a small set (Hyperseed-Concept 60).*

Remark 2112. *Two small edge-case clarifications are often useful in applications. First, if $k \geq |P|$ (when P is finite), then the maximizing set can be taken as $F = P$ and the right-hand side is simply the total attention mass $\sum_{p\in P} \alpha_t(p)$; the formula above remains consistent if one adopts the convention that $\alpha_t^{\downarrow}(j) = 0$ for $j > |P|$. Second, if there are ties among the k-th and $(k+1)$-st largest values, the maximizing F need not be unique: any choice that includes k elements achieving the top-k multiset of values attains the same maximum.*

Proof sketch. For any F of size $\leq k$, $\alpha_t(F)$ is a sum of $\leq k$ values from the multiset $\{\alpha_t(p)\}$. The maximum is obtained by choosing the k largest values.

Remark 2113. *The crucial observation is combinatorial: among all k-element subsets, the one that maximizes the sum is the one containing the top k entries. No deeper structure of α_t is required.*

Remark 2114. *One can also view the lemma as the discrete analogue of a rearrangement/majorization statement: the functional $F \mapsto \alpha_t(F)$ is maximized by taking a threshold cut on the sorted weights. This perspective can be helpful later when α_t is produced by a softmax (hence strictly positive and normalized) but the result itself does not depend on normalization, only on nonnegativity of the masses being summed.*

Corollary HT15.7 (Focus monotonicity in parameters). If focus holds at time t for (θ, k), then it holds for any (θ', k') with $\theta' \leq \theta$ and $k' \geq k$.

Remark 2115. *Once attention is sufficiently concentrated to clear a threshold θ using k patterns, it certainly clears any weaker requirement: allowing more patterns (larger k') or demanding less mass (smaller θ') cannot destroy the existence of a witnessing focus set.*

Proof sketch. Reuse the same witness set F_t (or enlarge it to size $\leq k'$); lowering θ or increasing k cannot break the inequality $\alpha_t(F_t) \geq \theta$.

Remark 2116. *This monotonicity is a small but important sanity check: the parameters (θ, k) behave in the direction one expects of a "focus" predicate.*

Remark 2117. *Equivalently, the set of parameter pairs for which focus holds is an "upper set" in k and a "lower set" in θ (with the natural product order). This matters when one later treats (θ, k) as hyperparameters: tuning k upward or θ downward cannot introduce false negatives for focus, so failure of focus at some (θ, k) automatically implies failure for any stricter pair (θ'', k'') with $\theta'' > \theta$ and $k'' < k$.*

Lemma HT15.8 (Self-reflective attention implies focus when the self-model is small). If $G_t > 0$ and $|P_{\text{self}}| \le k$, then

$$\alpha_t(P_{\text{self}}) \ge \theta_{\text{self}} \quad \Longrightarrow \quad \text{focus holds at level } (\theta_{\text{self}}, k) \text{ with witness } F_t = P_{\text{self}}.$$

Remark 2118. *Here P_{self} is the set of patterns constituting the system's current self-model. The lemma says that if the self-model is "small enough" (at most k patterns), then having a large fraction of attention on it automatically witnesses focus. This is one formal bridge from a self-model signal to an attentional-focus condition (Hyperseed-Concept 165, 60).*

Remark 2119. *The role of the size constraint $|P_{\text{self}}| \le k$ is purely structural: it ensures that the self-model is eligible to be a focus witness set under the cardinality cap. In particular, the implication does not claim that self-attention must be concentrated on a* subset *of the self-model; rather, it says that the entire self-model already fits within the allowed budget of k patterns, so its total attention mass can be used directly to certify focus.*

Proof sketch. Take $F_t = P_{\text{self}}$, which is allowed since $|P_{\text{self}}| \le k$.

Remark 2120. *The proof is a single substitution: the focus witness set is stipulated to be the self-model pattern set itself. Philosophically, it reflects that "self-attention" becomes a kind of "attention focus" whenever the self-representation is not too dispersed.*

Remark 2121. *The assumption $G_t > 0$ is not used in the set-theoretic part of the argument; it typically serves to rule out degenerate regimes in which self-related quantities are undefined or vacuous (e.g. when θ_{self} is only meaningful conditional on a nontrivial self-signal). Thus, in later dependencies, one can read this lemma as: whenever the self-signal is "active" and the self-model is small, any lower bound on total self-attention mass automatically instantiates a standard focus predicate.*

Lemma HT15.9 (Focused self-reflective attention implies nontrivial self salience). If focused self-reflective attention holds, i.e. there exists F_t with $|F_t| \le k$, $\alpha_t(F_t) \ge \theta$, and $F_t \cap P_{\text{self}} \ne \emptyset$, then

$$\max_{p \in P_{\text{self}}} \alpha_t(p) \;\ge\; \frac{\theta}{k}.$$

Remark 2122. *The conclusion is a pigeonhole principle: if a set of at most k patterns carries total mass at least θ, then some individual pattern in that set carries at least θ/k. The added hypothesis $F_t \cap P_{\text{self}} \ne \emptyset$ says that the focus set includes at least one self-model pattern; thus the self-model contains at least one pattern whose individual salience is bounded away from* 0.

Remark 2123. *Intuitively, the inequality θ/k quantifies the minimal "peakiness" forced by a (θ, k) focus certificate: if attention mass θ is compressed into at most k atoms, at least one atom must carry a non-negligible share. When α_t is a probability distribution (common in attention mechanisms), this also implies that a sufficiently large θ at modest k forces at least one pattern to have noticeable probability mass, which is often the operational notion of a salient token/pattern.*

Proof sketch. Since $\alpha_t(F_t) = \sum_{p \in F_t} \alpha_t(p) \ge \theta$ and $|F_t| \le k$, at least one $p \in F_t$ has $\alpha_t(p) \ge \theta/k$; if additionally $F_t \cap P_{\text{self}} \ne \emptyset$, then some self-model pattern can witness a lower bound of this form (or one can require a stronger variant where $F_t \subseteq P_{\text{self}}$).

Remark 2124. *The only subtlety is logical rather than algebraic: as written, the lower bound is guaranteed for some $p \in F_t$, and the intersection condition ensures there exists at least one self-pattern in F_t, but does not force the maximizer to lie in the intersection. The lemma therefore highlights a design choice: one may prefer a stronger condition (e.g. $F_t \subseteq P_{\text{self}}$) if one wants the lower bound to be witnessed explicitly by a self-pattern.*

Remark 2125. *A common strengthening (when appropriate) is to assume* self-focused *attention, meaning the witness set is required to be contained in P_{self}. Under that variant, the same pigeonhole argument yields the sharper and logically direct statement: there exists $p \in P_{\text{self}}$ with $\alpha_t(p) \geq \theta/k$, hence automatically $\max_{p \in P_{\text{self}}} \alpha_t(p) \geq \theta/k$. The present formulation is compatible with mixed focus sets (containing both self and non-self patterns) and is therefore weaker but sometimes more realistic in multi-object attention regimes.*

37.4 Attention as a bounded control policy

Let U be the set of internal update actions with costs $c : U \to [0, \infty)$, and let $B_t \geq 0$ be the attention budget at time t. An attention policy is $\pi_t : \Omega_t \to \Delta(U)$, and a draw $u \sim \pi_t(\omega)$ is feasible if $c(u) \leq B_t$.

Remark 2126. *The set Ω_t can be read as the space of internal "situations" the agent may be in at time t (e.g. working-memory contents, recent observations, latent beliefs, or any sufficient statistic used for decision-making). The time index emphasizes that the relevant state representation may itself evolve as the architecture accumulates memory or changes its internal model.*

Remark 2127. *The cost function c is intentionally abstract: it may represent compute time, number of primitive operations, energy consumption, memory bandwidth, or any bottlenecked resource. The scalar budget B_t can therefore be interpreted as a one-step resource envelope. In many settings one may also view B_t as a random variable or as an adapted process (e.g. fluctuating with fatigue or external constraints); the definitions below are pointwise in the realized B_t.*

Remark 2128. *Randomized policies $\pi_t(\omega) \in \Delta(U)$ allow the system to hedge among multiple update strategies when the state does not uniquely determine the best action, or when exploration is beneficial. Deterministic policies are recovered as the special case where $\pi_t(\omega)$ is a Dirac distribution concentrated on a single action $u(\omega)$.*

Remark 2129. *In this subsection, attention is treated as a control resource (Hyperseed-Concept 87): the system is in a state $\omega \in \Omega_t$ and stochastically selects an internal action $u \in U$ according to $\pi_t(\omega)$. The notation $\Delta(U)$ denotes the simplex of probability distributions over U (so $\pi_t(\omega)$ is a distribution). The budget constraint $c(u) \leq B_t$ plays the role of a feasibility condition familiar from constrained optimization and control.*

Remark 2130. *Formally, one may regard π_t as a Markov (stochastic) kernel from Ω_t to U. No additional measure-theoretic structure is required for the statements below beyond the interpretation that $\pi_t(\omega)(A)$ denotes the probability of drawing an action in the subset $A \subseteq U$ when in state ω. In finite or countable U, this reduces to ordinary sums of point probabilities.*

Lemma HT15.10 (Feasibility as support constraint). For a fixed state $\omega \in \Omega_t$, the policy is feasible at ω iff

$$\pi_t(\omega)\big(\{u \in U : c(u) \leq B_t\}\big) = 1,$$

equivalently $\pi_t(\omega)(u) = 0$ for all u with $c(u) > B_t$.

Remark 2131. *The lemma translates an operational statement ("every sampled action respects the budget") into a measure-theoretic one ("the policy assigns probability* 1 *to the feasible set"). This is useful because it turns feasibility into a simple algebraic property of distributions: the infeasible actions must have zero probability mass.*

Remark 2132. *Intuitively, the feasible set* $F_t := \{u \in U : c(u) \leq B_t\}$ *partitions actions into those that the architecture can "afford" at time* t *and those that it cannot. Lemma HT15.10 says that feasibility is equivalent to requiring* $\pi_t(\omega)$ *to live entirely on* F_t*—that is,* $\pi_t(\omega)$ *must not merely have high probability of affordable actions, but must exclude unaffordable ones altogether.*

Proof sketch. This is just the definition of feasibility for a random draw.

Remark 2133. *The equivalence between "probability* 1 *on a set" and "zero probability on its complement" is the essential step. Conceptually, feasibility is a statement about the support of* $\pi_t(\omega)$*.*

Remark 2134. *In the special case where* U *is finite, the condition* $\pi_t(\omega)(F_t) = 1$ *is simply* $\sum_{u \in F_t} \pi_t(\omega)(u) = 1$*, which forces* $\pi_t(\omega)(u) = 0$ *for each* $u \notin F_t$*. This highlights that feasibility is not an expectation constraint (such as* $\mathbb{E}[c(u)] \leq B_t$*) but a hard constraint ruling out any probability mass on infeasible actions.*

Lemma HT15.11 (Budget monotonicity). If $B_t \leq B'_t$ then any policy feasible under B_t is feasible under B'_t.

Remark 2135. *This is the expected monotonicity of constraints: relaxing a budget cannot invalidate an already-feasible policy. Such monotonicity is often used implicitly when comparing architectures or regimes with differing resource levels (Hyperseed-Concept 162; cf. [11] for broader discussion of resource-sensitive cognition).*

Remark 2136. *One operational interpretation is that an attention mechanism tuned to operate safely under scarcity will remain safe when additional compute or time becomes available; the extra budget can be used to enlarge the set of admissible actions, but it need not force any change in behavior. This is important when separating questions of* capability *(what could be done given more budget) from questions of* policy *(what is actually selected).*

Proof sketch. The feasible set $\{u : c(u) \leq B_t\}$ is contained in $\{u : c(u) \leq B'_t\}$.

Remark 2137. *The proof reduces to set inclusion. Once feasibility is expressed as a support constraint (Lemma HT15.10), the result becomes immediate.*

Remark 2138. *Equivalently, writing* $F_t = \{u : c(u) \leq B_t\}$ *and* $F'_t = \{u : c(u) \leq B'_t\}$*, the assumption* $B_t \leq B'_t$ *implies* $F_t \subseteq F'_t$*, so any distribution supported on* F_t *is automatically supported on* F'_t*. This makes clear that monotonicity is a purely set-theoretic consequence of the cost thresholding and does not depend on the particular form of* π_t*.*

Lemma HT15.12 (Feasible renormalization). Fix $\omega \in \Omega_t$ and let $F_t = \{u \in U : c(u) \leq B_t\}$. If $\pi_t(\omega)(F_t) > 0$, then the conditioned distribution

$$\pi_t^{\mathrm{feas}}(\omega)(A) \;:=\; \frac{\pi_t(\omega)(A \cap F_t)}{\pi_t(\omega)(F_t)}$$

defines a policy at ω that is feasible under budget B_t.

Remark 2139. *This lemma formalizes a simple "repair" operation: if a policy sometimes proposes infeasible actions, one can truncate to the feasible set and renormalize. In architectures, this corresponds to an attention controller that samples candidate updates and then vetoes those exceeding budget, resampling (or equivalently conditioning) until a feasible update is obtained.*

Proof sketch. By construction, $\pi_t^{\text{feas}}(\omega)$ assigns probability 1 to F_t, hence is feasible by Lemma HT15.10.

Remark 2140. *The condition $\pi_t(\omega)(F_t) > 0$ is necessary to avoid division by zero. When $\pi_t(\omega)(F_t) = 0$, feasibility cannot be restored by renormalization at that state without changing the policy more drastically (e.g. introducing new mass on F_t), which can be interpreted as the agent lacking any affordable option under its current proposal distribution.*

37.5 Priority scores and entropy-regularized allocation

Assume each pattern p has an event type A_p meaning "p is active/salient" and a goal set $\mathcal{G}$. Define the priority score

$$\mathrm{Prio}_t(p) := \max_{G \in \mathcal{G}} \left|\mathrm{Att}_{O,\Delta}(A_p, G)\right|.$$

Remark 2141. *The explicit time index t is a reminder that $\mathrm{Prio}_t(p)$ may vary as the underlying predictive/attraction quantities (and possibly the goal landscape) evolve. In particular, even if the set of patterns P is fixed, the allocation induced by Prio_t can shift rapidly as different A_p become more or less coupled to the currently-relevant goals. The use of a maximum over $G \in \mathcal{G}$ also implicitly defines a "most relevant goal witness" for each p (not necessarily unique), which can be useful conceptually when interpreting why a pattern is being attended to at time t.*

Fix $\beta > 0$ and (for this subsection) assume P is finite. Define

$$J(\alpha) := \sum_{p \in P} \alpha(p)\,\mathrm{Prio}_t(p) - \frac{1}{\beta} \sum_{p \in P} \alpha(p) \log \alpha(p), \qquad \alpha \in \Delta(P).$$

Remark 2142. *Because $\alpha \in \Delta(P)$, the first term is an expectation $\mathbb{E}_{p\sim\alpha}[\mathrm{Prio}_t(p)]$ while the second term is $(1/\beta)$ times the Shannon entropy $H(\alpha) = -\sum_p \alpha(p) \log \alpha(p)$. In this finite setting, $J(\alpha)$ is always well-defined under the usual convention $0 \log 0 := 0$, and the entropy term ensures that J does not reward degenerate allocations* unless *the priority landscape strongly supports them (i.e., unless β is sufficiently large relative to priority gaps).*

Remark 2143. *The priority score $\mathrm{Prio}_t(p)$ uses the attraction functional $\mathrm{Att}_{O,\Delta}$ from the prediction/attraction machinery (Section 14), turning predictive relationships into a scalar "how much should we care" signal. By taking an absolute value and then a maximum over goals $G \in \mathcal{G}$, we treat both positive and negative attraction as "attention-worthy," and we focus on whichever goal coupling is strongest. This is an explicit formalization of a common cognitive motif: attention is drawn to what is most consequential for current aims (Hyperseed-Concept **??**, **??**). Note that, operationally, the absolute value means that strong "repulsive" or cautionary predictive links can command attention just as much as strong "attractive" links, matching the intuition that both opportunities and threats can be salient drivers of cognitive resource allocation.*

Remark 2144. *The functional $J(\alpha)$ is the familiar tradeoff between expected priority and entropy regularization: the first term rewards placing mass on high-priority patterns, while the second penalizes overly-peaked distributions via $-\sum \alpha \log \alpha$ (Hyperseed-Concept **??**). The parameter β plays* *the role of an inverse temperature: small β enforces diffuse attention; large β permits sharp focus. The information-theoretic characterization via KL divergence below is standard and connects naturally to broader information-theoretic views of cognition (see e.g. [16] for general background). In particular, the entropic term can be read as a complexity/commitment penalty: allocating all mass to a single pattern is "cheap" in terms of expected priority only when the priority gaps justify the loss of entropy.*

Lemma HT15.11.5 (Softmax optimality and KL form). Assume P is finite and $\beta > 0$. Let

$$Z_\beta := \sum_{p \in P} \exp(\beta \operatorname{Prio}_t(p)), \qquad \alpha^*(p) := \frac{\exp(\beta \operatorname{Prio}_t(p))}{Z_\beta}.$$

Then α^* is the unique maximizer of $J(\alpha)$ over $\Delta(P)$, and moreover one has the decomposition

$$J(\alpha) = \frac{1}{\beta} \log Z_\beta \; - \; \frac{1}{\beta} D_{\mathrm{KL}}(\alpha \| \alpha^*), \qquad \alpha \in \Delta(P),$$

where $D_{\mathrm{KL}}(\alpha\|\alpha^*) = \sum_{p \in P} \alpha(p) \log \frac{\alpha(p)}{\alpha^*(p)}$.

Remark 2145. *This identity makes the optimization transparent: maximizing J is equivalent to minimizing the KL divergence to α^*, so J is maximized exactly at $\alpha = \alpha^*$. It also shows that the optimal value is $(1/\beta) \log Z_\beta$, i.e. a log-partition function familiar from statistical mechanics and exponential family theory. Conceptually, α^* is the "least committal" distribution (maximal entropy) among those that still give high expected priority, since any deviation from α^* incurs a precise KL penalty.*

Proof sketch. Rewrite

$$D_{\mathrm{KL}}(\alpha\|\alpha^*) = \sum_{p \in P} \alpha(p) \log \alpha(p) - \beta \sum_{p \in P} \alpha(p) \operatorname{Prio}_t(p) + \log Z_\beta,$$

and rearrange terms to obtain the displayed decomposition of $J(\alpha)$. Since $D_{\mathrm{KL}}(\alpha\|\alpha^*) \geq 0$ with equality iff $\alpha = \alpha^*$, the claims follow. Uniqueness holds because the negative entropy contribution makes J strictly concave on $\Delta(P)$ when $\beta > 0$.

Remark 2146. *The strict concavity point is often useful later: it guarantees stability of the attention allocation in the sense that small perturbations of the priority landscape lead to small perturbations of the optimizer, away from degenerate tie-breaking pathologies that can occur without entropy regularization.*

Lemma HT15.12 (Priority bounds). If $\operatorname{Att}_{O,\Delta}(\cdot,\cdot) \in [-1, 1]$, then $\operatorname{Prio}_t(p) \in [0, 1]$.

Remark 2147. *This lemma ensures priorities are numerically well-behaved: the optimization problem defining $J(\alpha)$ cannot be driven by unbounded scores. It also clarifies why the absolute value and the max do not introduce any pathological scaling: bounded inputs remain bounded outputs. In practical terms, this boundedness helps interpret β consistently across contexts, since β is not forced to compensate for arbitrarily large $\operatorname{Prio}_t(p)$ values.*

Proof sketch. Absolute values are in $[0, 1]$ and the max of numbers in $[0, 1]$ stays in $[0, 1]$.

Remark 2148. *Nothing deeper than interval arithmetic is needed; the lemma is a "range check" that makes later comparisons and limits cleaner. It also implies that any priority differences $\operatorname{Prio}_t(p) - \operatorname{Prio}_t(q)$ lie in $[-1, 1]$, which will bound softmax ratios and prevent extreme numerical blow-ups for moderate β.*

Lemma HT15.13 (Softmax ratio rule). Let $\alpha^*(p) \propto \exp(\beta \operatorname{Prio}_t(p))$ be the softmax distribution. Then for any $p, q \in P$,

$$\frac{\alpha^*(p)}{\alpha^*(q)} = \exp\big(\beta(\operatorname{Prio}_t(p) - \operatorname{Prio}_t(q))\big).$$

Remark 2149. *The ratio rule says that softmax converts* differences *in priority into* multiplicative *changes in probability. This is one reason softmax is such a natural mechanism for attention allocation: it is invariant to adding a constant to all priorities, and it turns relative advantage into relative mass in a controlled exponential way. A useful corollary of the displayed ratio is a Lipschitz-style bound: if* $|\operatorname{Prio}_t(p) - \operatorname{Prio}_t(q)| \leq \varepsilon$*, then the attention ratio lies in* $[e^{-\beta\varepsilon}, e^{\beta\varepsilon}]$*, so priorities close together yield comparable attention weights.*

Proof sketch. Divide the two softmax expressions; the normalizer cancels.

Remark 2150. *The cancellation of the normalizer is the key: it shows that softmax comparisons are local in (p, q) rather than global in P, which is often exploited in analysis and in implementations. It also highlights the scale role of β: the same priority difference produces a larger multiplicative change when β is larger.*

Corollary HT15.14 (Order preservation). If $\operatorname{Prio}_t(p) > \operatorname{Prio}_t(q)$ then $\alpha^*(p) > \alpha^*(q)$.

Remark 2151. *Softmax respects strict priority ordering: the most important patterns (by* Prio_t*) receive strictly more attention mass. This matters because it guarantees that the optimization does not perversely invert the intended ranking of priorities. In the non-strict case,* $\operatorname{Prio}_t(p) = \operatorname{Prio}_t(q)$ *implies $\alpha^*(p) = \alpha^*(q)$, so ties in priority translate exactly into ties in allocated attention.*

Proof sketch. Immediate from Lemma HT15.13 since exp is strictly increasing.

Remark 2152. *The proof reduces to a monotonicity fact about the exponential function, with* $\beta > 0$ *ensuring that "higher priority" really does mean "higher probability." If β were allowed to be negative, the order would reverse, which is another way to see why* $\beta > 0$ *is the natural parameter regime for attention allocation.*

Lemma HT15.15 (Temperature limits). Assume P is finite. As $\beta \to 0^+$, α^* approaches the uniform distribution on P. As $\beta \to \infty$, α^* concentrates on $\arg\max_{p\in P} \operatorname{Prio}_t(p)$.

Remark 2153. *These limits formalize the role of β as a knob between exploration and exploitation. When β is tiny, attention becomes nearly egalitarian (no strong preferences); when β is huge, attention becomes almost deterministic, selecting only the highest-priority patterns (a form of extreme focus). If the argmax set contains multiple maximizers, "concentrates" should be read as assigning vanishing mass to non-maximizers while distributing mass among maximizers; in the common case of equal maximal priorities, the limiting distribution over the maximizers is uniform because their exponentiated scores match.*

Proof sketch. Standard softmax asymptotics: small β flattens exponentials; large β makes the largest exponent dominate.

Remark 2154. *Geometrically, one may think of α^* as moving along a family of distributions from the center of the simplex (uniform) toward a vertex (an argmax selector) as β increases. Analytically, one can view the $\beta \to \infty$ limit as a "hardmax" approximation: for any fixed non-maximizer q and maximizer p, Lemma HT15.13 implies* $\alpha^*(q)/\alpha^*(p) = \exp(-\beta(\operatorname{Prio}_t(p) - \operatorname{Prio}_t(q))) \to 0$*, which forces mass away from q.*

Lemma HT15.16 (KL characterization of the optimum). Let $Z := \sum_{p \in P} \exp(\beta \operatorname{Prio}_t(p))$ and $\alpha^*(p) := \exp(\beta \operatorname{Prio}_t(p))/Z$. Then for any $\alpha \in \Delta(P)$,

$$J(\alpha) = \frac{1}{\beta} \log Z \; - \; \frac{1}{\beta} \operatorname{KL}(\alpha \| \alpha^*).$$

Hence α^* uniquely maximizes J.

Remark 2155. *This lemma reveals that maximizing $J(\alpha)$ is equivalent to minimizing $\operatorname{KL}(\alpha\|\alpha^*)$, i.e. making α as close as possible (in the KL sense) to the softmax distribution dictated by priorities. The constant $\frac{1}{\beta} \log Z$ is the optimum value of J and plays the role of a log-partition function, familiar from statistical mechanics and maximum-entropy reasoning (again aligning with information-theoretic viewpoints as in [16]). In particular, the identity is a finite-dimensional instance of the Gibbs variational principle: the "energy" term $\sum_p \alpha(p) \operatorname{Prio}_t(p)$ together with the entropy regularizer $-\frac{1}{\beta} \sum_p \alpha(p) \log \alpha(p)$ selects precisely the Gibbs/softmax law at inverse temperature β. Interpreting β as a precision (or inverse-noise) parameter makes explicit the exploration–exploitation tradeoff: smaller β yields a flatter α^*, while larger β concentrates mass on high-priority items.*

Proof sketch. Expand $\operatorname{KL}(\alpha\|\alpha^*) = \sum_p \alpha(p) \log \frac{\alpha(p)}{\alpha^*(p)}$ and substitute $\log \alpha^*(p) = \beta \operatorname{Prio}_t(p) - \log Z$. Equivalently,

$$\operatorname{KL}(\alpha\|\alpha^*) = \sum_p \alpha(p) \log \alpha(p) - \sum_p \alpha(p) \log \alpha^*(p) = \sum_p \alpha(p) \log \alpha(p) - \beta \sum_p \alpha(p) \operatorname{Prio}_t(p) + \log Z,$$

so rearranging gives

$$\beta \, J(\alpha) = \log Z - \operatorname{KL}(\alpha\|\alpha^*),$$

which is the claimed decomposition. Note that the argument uses only that $\alpha \in \Delta(P)$ and that Z is finite (automatic when P is finite and $\beta \operatorname{Prio}_t$ is real-valued), so no additional regularity assumptions are needed.

Remark 2156. *The key step is rewriting $-\sum \alpha \log \alpha$ together with $\sum \alpha \operatorname{Prio}$ into a KL divergence expression. Uniqueness follows because $\operatorname{KL}(\alpha\|\alpha^*) \geq 0$ with equality iff $\alpha = \alpha^*$, so $J(\alpha)$ is maximized exactly when the divergence term vanishes. It is also useful to keep in mind limiting cases: as $\beta \to 0^+$, one has $\alpha^* \to$ uniform on P (first-order expansion of $\exp(\beta \operatorname{Prio})$), reflecting maximal entropy; as $\beta \to +\infty$, α^* approaches the set of maximizers of Prio_t (mass concentrates on the argmax, with ties shared), reflecting greedy priority selection. These limits make concrete why the KL form is convenient: it separates a constant optimum value $\frac{1}{\beta} \log Z$ from a nonnegative "regret" term $\frac{1}{\beta}\operatorname{KL}(\alpha\|\alpha^*)$ measuring deviation from the optimal allocation.*

37.6 Varieties of knowledge: basic closure properties

A knowledge type T has carrier K_T, semantics map $\operatorname{Sem}_T : K_T \to S$, update operator(s) Upd_T, and an interface specifying exchange with other types. One can regard K_T as the internal state space of the representation (e.g. a set of formulas, probability tables, vector embeddings, or program traces), Sem_T as the interpretation map into a shared semantic domain S, and Upd_T as the admissible dynamical moves available to that representational substrate. The interface clause is included to ensure that the abstraction can express cross-type operations (translation, querying, consistency checking, feature extraction), which will be essential when discussing closure properties under interaction.

Remark 2157. *The phrase "knowledge type" is used here in an architectural sense: different representational formalisms (logical, probabilistic, neural, analogical, etc.) may have distinct carriers* K_T*, dynamics* Upd_T*, and semantics maps* Sem_T *(Hyperseed-Concept 200). The point of this abstraction is to reason about* collections *of types and about what becomes expressible when one allows them to interact—a central theme in cognitive synergy discussions (see [19]). In particular, the same semantic target* S *allows one to compare outputs of heterogeneous systems at the level of meaning (even if the internal carriers are incomparable), and the interfaces formalize the "glue" by which one type can supply partial structure that another type can refine, compress, or validate.*

Lemma HT15.17 (Monotone enrichment of the architecture cannot reduce expressivity). Let $T \subseteq T'$ be collections of knowledge types. If explanation sets satisfy $E_T(b) \subseteq E_{T'}(b)$, then for any fit threshold τ,

$$\mathrm{Con}^*(T'; \tau) \leq \mathrm{Con}^*(T; \tau).$$

Remark 2158. *This lemma says that adding representational resources cannot make your best achievable explanation* worse *in the contrivance sense: more expressive capacity means a larger feasible set of explanations, hence potentially lower minimal contrivance. It is a formal version of the methodological maxim that adding tools does not force you to use them, but can only enlarge your options (Hyperseed-Concept 191). The condition* $E_T(b) \subseteq E_{T'}(b)$ *captures "monotone enrichment" at the operational level: for each target* b *(e.g. a behavior trace, a dataset, or an observation bundle), every explanation constructible in the smaller architecture remains constructible in the enriched one. Thus the lemma is an immediate closure property: the class of admissible explanations is upward closed under adding types, and the optimal (minimal) contrivance is therefore downward monotone.*

Proof sketch. Con^* is an infimum over admissible explanations; enlarging the admissible set cannot increase the infimum. More explicitly, if $\mathcal{A} := \{e \in E_T(b) : \mathrm{Fit}(e, b) \geq \tau\}$ and $\mathcal{A}' := \{e \in E_{T'}(b) : \mathrm{Fit}(e, b) \geq \tau\}$ denote the feasible explanation sets at threshold τ, then $\mathcal{A} \subseteq \mathcal{A}'$ by hypothesis, and hence

$$\inf_{e \in \mathcal{A}'} \mathrm{Con}(e) \;\leq\; \inf_{e \in \mathcal{A}} \mathrm{Con}(e),$$

which is exactly the claim. The statement remains valid even when the infimum is not attained (i.e. there is no single best explanation), since the order property concerns infima rather than minima.

Remark 2159. *The proof is pure order theory: infima are monotone with respect to set inclusion of the domain over which one takes the infimum. A useful edge-case clarification is that if the feasible set is empty for some* (T, τ)*, then* $\mathrm{Con}^*(T; \tau)$ *is typically defined as* $+\infty$ *(or left undefined); under the* $+\infty$ *convention, the inequality still holds because enlarging the architecture can only turn infeasible instances into feasible ones, never the reverse, again consistent with monotonicity. Conceptually, the lemma isolates a minimal requirement for "synergistic expansion": to guarantee non-degradation of optimal contrivance, it suffices that the enlarged system preserves the smaller system's explanation repertoire as a subset (which is what one expects if* T' *literally contains* T *as a sub-architecture).*

37.7 Cognitive synergy and contrivance

Fix a task/dataset/behavior b. Candidate explanations $E \in E(b)$ have fit $\mathrm{Fit}(E) \in [0, 1]$ and weakness $\mathrm{Weak}(E) \in (0, 1]$. Define contrivance cost

$$\mathrm{Con}(E) := -\log(\mathrm{Weak}(E)) \in [0, \infty).$$

For a knowledge-type collection T and threshold $\tau \in (0,1]$, define

$$\mathrm{Con}^*(T;\tau) := \inf\{\mathrm{Con}(E) : E \in E_T(b),\ \mathrm{Fit}(E) \geq \tau\}.$$

For disjoint T_1, T_2, define synergy gain

$$\Sigma(T_1, T_2; \tau) := \Big(\mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau) - \mathrm{Con}^*(T_1 \cup T_2;\tau)\Big)_+.$$

Remark 2160. *The definition of* $\mathrm{Con}^*(T;\tau)$ *packages two distinct requirements into a single scalar objective: (i) a feasibility requirement* $\mathrm{Fit}(E) \geq \tau$*, and (ii) among feasible explanations, an optimality requirement minimizing contrivance. In particular,* T *only enters through the restricted explanation class* $E_T(b)$ *(i.e. the explanations accessible when one is allowed to draw on knowledge-types* T*), so* Con^* *is best read as a property of an* explanatory interface *rather than of the task* b *alone. When the feasible set* $\{E \in E_T(b) : \mathrm{Fit}(E) \geq \tau\}$ *is empty, the infimum is naturally understood as* $+\infty$*; under this convention, the subsequent lemmas and the definition of* Σ *remain meaningful (e.g. the absence of any adequate explanation is represented as an infinite contrivance requirement).*

Remark 2161. *Although* Fit *is left abstract, it is helpful to keep in mind common instantiations: predictive accuracy, likelihood, held-out log-score, or any task-appropriate performance surrogate normalized to* $[0,1]$*. Likewise,* $\mathrm{Weak}(E)$ *is intentionally* not *a measure of fit; rather it quantifies how "delicate" the explanation is with respect to perturbations, alternative datasets, re-encodings, or other context changes. This separation is important because one can easily achieve high fit with a brittle, over-tailored explanation (high* Fit *but low* Weak*), and the framework is designed to make such tradeoffs explicit.*

Remark 2162. *The pair* $(\mathrm{Fit}(E), \mathrm{Weak}(E))$ *separates* how well *an explanation matches the target behavior from* how contrived *or "fragile" that explanation is. The notion of weakness (Hyperseed-Concept 202) is treated as primary, with contrivance defined as its negative logarithm; this framing aligns with the idea that explanations come with degrees of robustness or "non-contrivance" and that these degrees compose multiplicatively in many settings [2, 3]. The use of* $-\log$ *turns multiplicative structure into additive accounting, much as in information theory and statistical mechanics.*

Remark 2163. *One can read* $\mathrm{Con}(E)$ *as the number of "nats of contrivance" required to support the explanation: a small change in weakness corresponds to an additive change in cost. For instance, if* $\mathrm{Weak}(E)$ *halves, contrivance increases by* $\log 2$*, so the scale is calibrated so that constant multiplicative penalties in weakness produce constant additive penalties in contrivance. This is often the correct qualitative behavior when weakness reflects independent "failure modes" whose probabilities multiply.*

Remark 2164. *Cognitive synergy (Hyperseed-Concept 73) is then the extent to which combining two disjoint knowledge collections* T_1, T_2 *yields an explanation that is* less contrived *than what one would obtain by solving separately and adding the contrivances. The* $(\cdot)_+$ *enforces that synergy is a nonnegative "gain" quantity: if combining yields no advantage (or is worse), the synergy score is* 0*. This style of formalization is consonant with the architectural notion of cognitive synergy in [19].*

Remark 2165. *The disjointness assumption for* T_1, T_2 *is a bookkeeping device ensuring that the "separate solving" baseline does not double-count shared resources; without disjointness, the expression* $\mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau)$ *can overestimate the true separate-solve cost simply because both terms may rely on the same underlying knowledge. In applications where overlap is unavoidable, one typically refines the formalism by splitting* T_1, T_2 *into common and private parts, or by introducing an explicit cost model for reusing shared components.*

Lemma HT15.18 (Basic contrivance calculus). For Weak $\in (0,1]$:

$$\mathrm{Con}(E) \geq 0, \qquad \mathrm{Weak}(E_1) \leq \mathrm{Weak}(E_2) \Longrightarrow \mathrm{Con}(E_1) \geq \mathrm{Con}(E_2), \qquad \mathrm{Con}(E) = 0 \iff \mathrm{Weak}(E) = 1.$$

Remark 2166. *This lemma states that contrivance behaves like a cost: it is nonnegative, it decreases when weakness increases (so "stronger" explanations are cheaper), and it vanishes exactly at maximal weakness* 1 *(i.e. no contrivance at all). The monotone reversal is the essential philosophical point: weakness is a kind of* capacity for non-accidentalness, *and contrivance is its additive shadow.*

Remark 2167. *Interpreting the boundary cases can also be helpful. As* $\mathrm{Weak}(E) \to 0^+$, $\mathrm{Con}(E) \to +\infty$*, representing that an explanation that is arbitrarily fragile carries an arbitrarily large "contrivance debt." Conversely,* $\mathrm{Weak}(E) = 1$ *acts as a normalization convention: it marks the ideal of complete non-contrivance in the chosen weakness scale, so that all other costs are measured relative to that baseline.*

Proof sketch. $-\log$ maps $(0,1]$ to $[0,\infty)$ and is monotone decreasing; $-\log 1 = 0$.

Remark 2168. *The proof uses only elementary calculus properties of* log*. The logarithm is the canonical bridge from multiplicative to additive reasoning, which is why it appears so often when one seeks a "cost" corresponding to a "strength" or "weight."*

Lemma HT15.19 (Threshold monotonicity). If $\tau_1 \leq \tau_2$ then $\mathrm{Con}^*(T;\tau_1) \leq \mathrm{Con}^*(T;\tau_2)$.

Remark 2169. *Demanding a higher fit threshold* τ_2 *restricts the set of admissible explanations, so the best achievable contrivance cannot improve; at best it stays the same, and typically it worsens. This lemma ensures that* τ *behaves as a genuine "difficulty dial" for explanation.*

Remark 2170. *It is worth noting that the inequality may be strict even when both thresholds are achievable: raising* τ *can force one to move from a simple, robust explanation to a more baroque one that chases residual errors. In that sense, the map* $\tau \mapsto \mathrm{Con}^*(T;\tau)$ *can be read as an "optimal fragility curve" (or cost–fit tradeoff frontier) for the knowledge collection* T *on behavior* b*.*

Proof sketch. The constraint $\mathrm{Fit}(E) \geq \tau_2$ is stricter, so the feasible set shrinks, making the infimum weakly larger.

Remark 2171. *Again the logic is order-theoretic: constraints create subset relations among feasible sets, and infima respect those inclusions.*

Lemma HT15.20 (Symmetry and nonnegativity of synergy gain).

$$\Sigma(T_1, T_2; \tau) = \Sigma(T_2, T_1; \tau) \geq 0.$$

Moreover, $\Sigma(T_1, T_2; \tau) > 0$ iff $\mathrm{Con}^*(T_1 \cup T_2; \tau) < \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau)$.

Remark 2172. *The statement says that synergy is not directional (swapping the two parties changes nothing), and that it is genuinely a* gain *rather than a signed difference. The "iff" clause clarifies that positive synergy is exactly strict subadditivity of optimal contrivance under union.*

Remark 2173. *The truncation* $(\cdot)_+$ *also makes the quantity robust to cases where combining knowledge introduces overheads (coordination costs, representational mismatch, or search complexity) that could in principle worsen the best achievable contrivance. In such cases, the framework does not force "negative synergy"; instead it records an absence of gain, aligning the interpretation of* Σ *as an advantage measure rather than as a signed interaction term.*

Proof sketch. Symmetry is by commutativity of $+$ and $\cup$. Nonnegativity is by definition of $(\cdot)_+$. Strictness equivalence follows from $(x)_+ > 0 \iff x > 0$.

Remark 2174. *The proof is essentially definitional, but this is not a defect: it shows that the chosen formalization of* Σ *faithfully encodes the intended informal meaning.*

Lemma HT15.21 (Multiplicative weakness ⇒ subadditive contrivance). Assume explanations can be composed $(E_1 \otimes E_2)$ and satisfy

$$\mathrm{Weak}(E_1 \otimes E_2) \geq \mathrm{Weak}(E_1)\mathrm{Weak}(E_2).$$

Then

$$\mathrm{Con}(E_1 \otimes E_2) \leq \mathrm{Con}(E_1) + \mathrm{Con}(E_2).$$

Remark 2175. *This lemma formalizes a key compositional intuition: if combining explanations does not make them disproportionately weaker (indeed, it is at least multiplicative in weakness), then the contrivance cost of the combination is at most the sum of the costs. This is precisely the reason to define contrivance using* $-\log$*: it linearizes multiplicative composition into additive bounds [3].*

Remark 2176. *The assumption* $\mathrm{Weak}(E_1 \otimes E_2) \geq \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$ *can be read as an "at-most-independent failure" postulate: the composed explanation fails no more often than would be predicted if the fragilities of the parts were independent. If in a given domain one expects positive interaction (composition is* more *robust than independence would suggest), then the inequality may hold with slack, strengthening the resulting subadditivity; conversely, if composition introduces new coupled failure modes, the inequality can fail, corresponding to superadditive contrivance.*

Proof sketch. Apply $-\log(\cdot)$ and use $-\log(xy) = -\log x - \log y$ plus monotonicity of $-\log$.

Remark 2177. *The inequality direction is worth lingering on: because* $-\log$ *is decreasing, a lower bound on weakness becomes an upper bound on contrivance. This reversal is the algebraic heart of the lemma.*

Remark 2178. *In many modeling contexts,* $\otimes$ *may correspond to conjunction of sub-claims, modular program composition, or assembling a pipeline whose stages provide complementary constraints on* b*. The lemma then reads as: if robustness (weakness) composes at least multiplicatively across stages, then the "extra contrivance" of the combined pipeline is controlled by the sum of stage-wise contrivances, making additive accounting of explanatory fragility consistent across levels of abstraction.*

Corollary HT15.22 (No-penalty combination bound). Under the same multiplicative weakness assumption, for all disjoint T_1, T_2 and all τ,

$$\mathrm{Con}^*(T_1 \cup T_2; \tau) \leq \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau).$$

Hence

$$\Sigma(T_1, T_2; \tau) = \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau) - \mathrm{Con}^*(T_1 \cup T_2; \tau).$$

Remark 2179. *This corollary says that if weakness composes multiplicatively, then combining knowledge sources cannot impose a contrivance penalty beyond the sum of separate solutions. In that case, the "*$(\cdot)_+$*" in* Σ *becomes unnecessary: the raw difference is already nonnegative.*

Remark 2180. *The disjointness condition $T_1 \cap T_2 = \varnothing$ is the formal way to say that the two sources are not literally reusing the same task instances (or the same labeled constraints) twice. Without disjointness, one can still derive related bounds, but they typically require additional bookkeeping for overlap (e.g., avoiding double-counting of shared constraints) or a modified notion of union that accounts for redundancy. The parameter τ plays the role of a fixed acceptability threshold for fit: the infimum defining $\mathrm{Con}^*(\cdot;\tau)$ is taken only over those explanations whose Fit clears the same bar in each term, so the comparison is genuinely "like-for-like" across T_1, T_2, and $T_1 \cup T_2$.*

Proof sketch. Use Lemma HT15.21 on ε-optimal explanations for T_1 and T_2, then take infima and let $\varepsilon \to 0$.

Remark 2181. *A slightly more explicit way to view the ε-optimal step is: fix $\varepsilon > 0$ and pick admissible E_1, E_2 with $\mathrm{Fit}(E_i) \geq \tau$ such that $\mathrm{Con}(E_i) \leq \mathrm{Con}^*(T_i;\tau) + \varepsilon$. Apply Lemma HT15.21 (the candidate-level multiplicative-weakness composition bound, translated into contrivance language) to obtain an admissible combined explanation E for $T_1 \cup T_2$ with $\mathrm{Fit}(E) \geq \tau$ and $\mathrm{Con}(E) \leq \mathrm{Con}(E_1) + \mathrm{Con}(E_2)$. Taking the infimum over all admissible E for $T_1 \cup T_2$ yields $\mathrm{Con}^*(T_1 \cup T_2;\tau) \leq \mathrm{Con}(E) \leq \mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau) + 2\varepsilon$, and then $\varepsilon \to 0$ gives the stated inequality.*

Remark 2182. *The ε-optimal trick is standard when passing from pointwise inequalities about candidates to inequalities about infima: you approximate the infimum by near-minimizers, apply the candidate-level inequality, and then let ε shrink away.*

Remark 2183. *One practical reason to emphasize the "infimum + ε" pattern is that Con^* need not be attained by any single explanation: there may be no true minimizer because one can construct a sequence of explanations that approach the optimal tradeoff but never quite reach it (e.g., by allocating ever-more parameters in a way that asymptotically improves fit while shaving contrivance). The corollary is therefore stated (and proved) in a way that remains valid even in non-compact hypothesis classes where minimizers fail to exist.*

Lemma HT15.23 (Synergy in "best-achievable weakness" form). Assume the supremum exists and define

$$\mathrm{Weak}^*(T;\tau) := \sup\{\mathrm{Weak}(E) : E \in E_T(b),\ \mathrm{Fit}(E) \geq \tau\}.$$

Then $\mathrm{Con}^*(T;\tau) = -\log(\mathrm{Weak}^*(T;\tau))$ and cognitive synergy (strict) is equivalent to

$$\mathrm{Weak}^*(T_1 \cup T_2;\tau) > \mathrm{Weak}^*(T_1;\tau)\,\mathrm{Weak}^*(T_2;\tau).$$

Remark 2184. *This lemma rewrites the entire story in terms of the* best achievable weakness *rather than the* least contrivance*. It is often psychologically and architecturally more natural to say "how robust can we get?" than "how contrived must we be?" Strict synergy then becomes strict supermultiplicativity of the best achievable weakness under union.*

Remark 2185. *Because $\mathrm{Con} = -\log(\mathrm{Weak})$, it is implicitly understood that $\mathrm{Weak}(E) \in (0,1]$ on the admissible set (or at least is strictly positive whenever $\mathrm{Con}(E)$ is finite). If $\mathrm{Weak}^*(T;\tau) = 0$, then the formula reads $\mathrm{Con}^*(T;\tau) = -\log 0 = +\infty$, matching the intuition that if no τ-fit explanation has any positive weakness score, then the "least contrivance" required at that threshold is effectively unbounded. When $\mathrm{Weak}^*(T;\tau) = 1$, the contrivance optimum becomes 0, i.e., maximal robustness corresponds to minimal contrivance in this scale.*

Proof sketch. Because $-\log$ is decreasing, $\inf_E(-\log \mathrm{Weak}(E)) = -\log(\sup_E \mathrm{Weak}(E))$. Translate strict subadditivity of $-\log$ into strict supermultiplicativity of Weak^*.

Remark 2186. *Unpacking the order reversal: for any set* $S \subseteq (0,1]$*, the map* $x \mapsto -\log x$ *satisfies* $\inf_{x\in S}(-\log x) = -\log(\sup_{x\in S} x)$ *because (i)* $\sup S$ *is the largest achievable (or approximable) weakness and (ii) decreasing monotonicity makes the largest weakness correspond to the smallest contrivance. The strict synergy inequality in contrivance form,*

$$\mathrm{Con}^*(T_1 \cup T_2; \tau) < \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau),$$

becomes, after exponentiating both sides with $\exp(-\cdot)$*, precisely the displayed strict supermultiplicativity of* Weak^*.

Remark 2187. *The proof is a clean example of a Galois-like duality between additive and multiplicative domains, mediated by the logarithm. It also clarifies why taking an infimum in one picture corresponds to taking a supremum in the other: the monotonicity reversal of* $-\log$ *swaps the order.*

Remark 2188. *This reformulation is also useful diagnostically: if one empirically estimates* Weak^* *(e.g., by best-of-N search over admissible explanations), then synergy can be tested as a multiplicative gap rather than an additive one. In particular, the inequality compares* ratios *of robustness rather than* differences *in contrivance, which can be numerically more stable when contrivance values are large but weakness values remain within a bounded scale.*

Lemma HT15.24 (Emergent compression implies multiplicative improvement). If an explanation E satisfies $\mathrm{Con}(E) < \mathrm{Con}(E_1)+\mathrm{Con}(E_2)$ for all admissible decompositions $E \approx E_1 \otimes E_2$, then equivalently

$$\mathrm{Weak}(E) > \mathrm{Weak}(E_1)\mathrm{Weak}(E_2) \quad \text{for all such decompositions.}$$

Remark 2189. *This lemma states the same phenomenon in two languages. The left-hand side is phrased as "emergent compression":* E *is cheaper (less contrived) than any decomposition into parts. The right-hand side says that, in weakness terms,* E *is stronger than any multiplicative combination of candidate parts. One may view this as a precise sense in which "the whole is more than the product of its parts," but expressed in a rigorously constrained way.*

Remark 2190. *The notation* $E \approx E_1 \otimes E_2$ *is intentionally permissive: it allows the decomposition relation to encode whatever equivalence or approximation notion is appropriate for the model class (e.g., functional equivalence on* T*, approximation in predictive distribution, or representational factorization up to a tolerated error). The lemma does not require that* every *explanation decomposes, only that whenever a decomposition is admissible under the chosen relation, the stated strict inequality compares* E *against that candidate factorization. Thus "emergent compression" is a universal quantification over the allowed factorization set, not a claim about existence of any particular* E_1, E_2.

Proof sketch. Exponentiate $-\mathrm{Con} = \log \mathrm{Weak}$ to turn $<$ on costs into $>$ on multiplicative weakness.

Remark 2191. *Formally:* $\mathrm{Con}(E) < \mathrm{Con}(E_1)+\mathrm{Con}(E_2)$ *is equivalent to* $-\mathrm{Con}(E) > -(\mathrm{Con}(E_1)+\mathrm{Con}(E_2))$*, and applying* $\exp$ *gives* $\exp(-\mathrm{Con}(E)) > \exp(-\mathrm{Con}(E_1))\exp(-\mathrm{Con}(E_2))$*. Using* $\mathrm{Weak} = \exp(-\mathrm{Con})$ *(equivalently,* $-\mathrm{Con} = \log \mathrm{Weak}$*) yields exactly* $\mathrm{Weak}(E) > \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$*. The converse direction reverses the steps with* $\log$*, noting again that strict inequalities are preserved because both* $\exp$ *and* $\log$ *are strictly increasing on the relevant domain.*

Remark 2192. *The strategy is the standard invertibility of the* $\log / \exp$ *transform: strict inequalities are preserved because* $\exp$ *is strictly increasing. Conceptually, the lemma explains why "compression" and "robustness gain" are the same structural claim viewed through different lenses (Hyperseed-Concept 169, 82).*

Remark 2193. *It is worth noting that the strictness matters: a non-strict inequality* $\mathrm{Con}(E) \leq \mathrm{Con}(E_1) + \mathrm{Con}(E_2)$ *would translate to* $\mathrm{Weak}(E) \geq \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$ *and would permit "no-emergence" cases where* E *is exactly as compressible/robust as the best factorization. The strict form isolates genuinely synergistic structure in* E *that cannot be matched even by the most favorable admissible decomposition into two parts.*

37.8 Synergy, budgets, and attention gating

Fix fit threshold τ and contrivance budget B. Define "solvable by T at budget B" as $\mathrm{Con}^*(T;\tau) \leq B$.

Remark 2194. *Here* τ *plays the role of a required level of adequacy (e.g. predictive accuracy, likelihood, or any task-specific fit functional fixed earlier), while* B *is the maximum admissible "engineering cost" as measured by contrivance. In particular,* $\mathrm{Con}^*(T;\tau)$ *should be read as the* minimum *contrivance over all explanations drawn from the model/explanation class associated with knowledge types* T *that achieve fit at least* τ*. Thus "solvable" is a feasibility statement about whether the architecture can meet the fit requirement without exceeding a resource cap, which mirrors how systems are typically evaluated under explicit compute/communication constraints.*

Remark 2195. *This subsection connects the abstract synergy notion to resource allocation: an architecture may in principle admit synergistic explanations, yet in practice those explanations may be unreachable without dedicating attention/bandwidth to cross-type communication (Hyperseed-Concept* **??***, 88). The language of budgets and gating mirrors engineering concerns in cognitive architectures [19].*

Remark 2196. *The central distinction is between (i)* representational possibility *(the union of types* $T_1 \cup T_2$ *contains enough ingredients to form a low-contrivance explanation) and (ii)* operational accessibility *(the system actually allocates sufficient routing/attention so that those ingredients can interact during construction or inference). The lemmas below formalize how changing the available types, the allowed budget, or the permitted cross-type bandwidth changes what can be achieved at or below the threshold* τ*.*

Lemma HT15.25 (Solvability monotonicity). If $T \subseteq T'$ and $B \leq B'$, then

$$\mathrm{Con}^*(T;\tau) \leq B \Longrightarrow \mathrm{Con}^*(T';\tau) \leq B'$$

(i.e. solvability is monotone in knowledge and budget).

Remark 2197. *If you can solve the task with fewer knowledge types and a smaller budget, then you can certainly solve it with more knowledge types and a larger budget. This is a basic monotonicity property, but it is essential for reasoning about developmental or iterative improvements: adding modules or resources cannot retroactively destroy solvability.*

Remark 2198. *Conceptually,* $T \subseteq T'$ *means the enlarged system can emulate the smaller one by simply ignoring the additional types, so the feasible set of explanations for* T' *contains (or at least does not exclude) the feasible set for* T*. Meanwhile* $B \leq B'$ *means the constraint is relaxed: any* *explanation that was admissible under the stricter cap remains admissible under the looser cap. The lemma packages these two ideas into a single implication that will be used repeatedly when comparing architectures across training stages or across different attention/communication configurations.*

Proof sketch. Use Lemma HT15.17 for $T \subseteq T'$ and the trivial monotonicity in B.

Remark 2199. *The proof composes two monotonicities: architectural enrichment improves (or preserves) the optimum, and increasing the allowed budget relaxes the feasibility condition.*

Remark 2200. *More explicitly, Lemma HT15.17 provides* $\mathrm{Con}^*(T';\tau) \le \mathrm{Con}^*(T;\tau)$ *when* $T \subseteq T'$*. If* $\mathrm{Con}^*(T;\tau) \le B$ *and* $B \le B'$*, then* $\mathrm{Con}^*(T';\tau) \le \mathrm{Con}^*(T;\tau) \le B \le B'$*, yielding solvability of* T' *at budget* B'*. The point is not the algebra but the interpretive consequence: any later-stage system that strictly adds capacities (types or resources) inherits the earlier feasibility guarantees.*

Lemma HT15.26 (Gating principle as a formal reduction; conditional). Let Γ be the inter-type communication graph on knowledge types with time-dependent bandwidth allocations $b_t(T \to T') \ge 0$ satisfying $\sum_{T,T'} b_t(T \to T') \le B_t$. Assume the model class $E_{T_1 \cup T_2}(b)$ is *restricted* by bandwidth as follows: if all cross edges between T_1 and T_2 have zero bandwidth, then only separable explanations $E \approx E_1 \otimes E_2$ are admissible (no emergent cross-type constructions). Then if all b_t between T_1 and T_2 are 0 for the relevant interval,

$$\mathrm{Con}^*(T_1 \cup T_2;\tau) = \mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau) \quad \text{and hence} \quad \Sigma(T_1, T_2;\tau) = 0.$$

Remark 2201. *This lemma articulates a precise "gating" thesis: synergy is not merely a property of having multiple knowledge types present, but of permitting them to communicate. When the bandwidth on cross edges is identically zero, the combined system is forced to behave like two independent subsystems, so the best combined explanation decomposes and no synergy can be expressed. The result provides a formal handle on the engineering intuition that attention and communication resources must be allocated to enable cognitive synergy [19].*

Remark 2202. *The bandwidth variables* $b_t(T \to T')$ *can be read as a quantitative proxy for attentional routing or message-passing capacity at time* t*: they indicate how much of the overall budget* B_t *is spent on transferring representations, constraints, or partial hypotheses from one type to another. The assumption "zero cross edges implies only separable explanations" is intentionally strong and therefore clearly identifiable as a conditional modeling choice: it encodes the claim that* constructing *cross-type structure is itself a communication-dependent act, not merely a post-hoc interpretation. Under this assumption, any emergent term that would mix variables, symbols, or latent factors across* T_1 *and* T_2 *is literally out of class when the gate is closed.*

Remark 2203. *The notation* $E \approx E_1 \otimes E_2$ *is meant to emphasize a factorization (or product) structure rather than exact equality of functions: the admissible joint explanations are those that can be represented as a composition of two explanations that do not exchange internal state. In concrete instantiations, this could correspond to additive decompositions of loss, block-diagonal parameterizations, independent latent variable models, or any architectural constraint that prevents cross-attention or cross-feature conjunctions between the two subsystems.*

Proof sketch. Under the restriction, $E_{T_1 \cup T_2}(b)$ coincides with the partition-limited class. Therefore the best achievable contrivance is exactly the sum of the two independent optima. This makes synergy latent but not expressible without allocating attention/bandwidth to communication.

Remark 2204. *At a high level, the proof strategy is to identify a collapse of hypothesis classes: with zero bandwidth, the admissible class for $T_1 \cup T_2$ becomes the same as the "separable" class. Once this identification is made, the equality of optima is not mysterious: the joint optimization problem literally factorizes into two independent ones. Visually, one may imagine Γ as a graph whose cut edges between T_1 and T_2 are removed; after the cut, there is no path by which information can flow, so emergent cross-type structure is prohibited by assumption.*

Remark 2205. *The equality* $\mathrm{Con}^*(T_1 \cup T_2; \tau) = \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau)$ *is best interpreted as an* additivity of minimal costs *when the admissible set of joint explanations is restricted to separable ones: any feasible E decomposes into E_1 and E_2, and the contrivance objective correspondingly decomposes (as in the earlier definition of synergy via additivity gaps). Consequently, the synergy quantity $\Sigma(T_1, T_2; \tau)$, defined as the amount by which the joint optimum beats the sum of separate optima, must vanish because the joint model class contains no mechanisms capable of achieving a strictly sub-additive construction. This clarifies the sense in which gating can turn a potentially synergistic pair of types into a merely parallel, non-interacting pair.*

38 Helper theorems for Section 17: consciousness, reflective will, and autonomy

This section collects small derived facts that are convenient for mechanized reasoning with the Section 17 interface (p-bit evidence states, attention-weighted aggregation, workspace closure, reflective introspection, coherence scores, will/autonomy, and the two-channel update).

Remark 2206. *A useful way to read this section is as a miniature "algebra of mindedness": we take evidence values in $V = [0,1]^2$ (a p-bit, i.e. a two-component degree of positive and negative support), assemble these into evidence states $E : L \to V$ over a proposition language L, and then show that the resulting constructions behave well under standard order-theoretic operations. This order-theoretic hygiene is not mere pedantry: it is what lets one use fixed-point theorems to speak precisely about stability of "workspace" closure and self-referential updates, in a way that can be automated and composed modularly.*

Remark 2207. *The underlying perspective is broadly compatible with a global-workspace style semantics for conscious content (in the weak, formal sense of* Con_θ *defined below), together with an explicitly paraconsistent representation of evidence via $[0,1]^2$ (cf. constructive/paraconsistent logics and evidence-pair semantics, e.g. [23, 24]). On the cognitive-architectural side, the interplay between attention-weighting, workspace closure, and reflective/introspective tokens echoes the "cognitive synergy" viewpoint [19, 5], while the use of quantale-like joins/tensors aligns with the "weakness" and quantale weakness viewpoint [3, 2] (see also Hyperseed-Concept 143).*

Concretely, the point of working with $V = [0,1]^2$ is that it supports an explicit separation between *support-for* and *support-against* the same proposition, without forcing these channels to collapse into a single probability-like quantity. In particular, one may have both high positive and high negative components simultaneously, representing the presence of strong but conflicting lines of evidence (a common situation in reflective or self-modeling settings), and the subsequent reasoning steps can be set up so that such conflict is carried rather than erased.

As a standing convention, evidence states $E : L \to V$ inherit most operations *pointwise* from V. Thus, whenever V is equipped with an order $\leq_V$ (typically the product order on $[0,1]^2$, unless explicitly stated otherwise), one obtains an induced order on evidence states by

$$E \leq E' \quad :\Longleftrightarrow \quad (\forall \varphi \in L)\ E(\varphi) \leq_V E'(\varphi),$$

and similarly for joins/meets (when they exist). This is the basic mechanism by which lattice- and quantale-flavored structure on V lifts to the "state space" V^L, letting one state the helper theorems in a uniform way: monotonicity, continuity (in the order-theoretic sense), existence of least/greatest fixed points, and closure properties can all be proved once and then reused across attention, workspace, and reflective operators.

The interface components mentioned above can also be read in this order-theoretic light. Attention-weighted aggregation is treated as an operator that combines multiple local pieces of evidence into a single evidence value (or state), with weights serving as a controlled bias in how joins or convex-like mixtures are taken. Workspace closure is treated as a closure operator (or, more generally, a monotone inflationary operator) on evidence states; its fixed points correspond to "stable workspaces" in which adding the closure consequences does not further change what is globally available. Reflective introspection introduces new tokens (e.g. propositions describing what is in the workspace, what is attended to, or what the agent takes itself to endorse), and the helper theorems ensure that introducing these tokens preserves the key monotonicity and fixed-point conditions needed for mechanized proofs.

Finally, the role of the two-channel update is to keep the positive and negative components synchronized with the dynamics of closure and reflection: the update rules can be analyzed as endomaps on V^L whose algebraic properties (e.g. monotonicity, compatibility with joins, or nonexpansiveness with respect to a chosen order/metric) determine whether iterative application converges, oscillates, or admits stable equilibria. The helper results in this section are therefore intended to function as "glue lemmas": they connect the cognitive reading (attention, workspace, will) to the mathematical backbone (orders, closure, fixed points), enabling later arguments to be stated compositionally rather than re-proved from scratch.

38.1 Evidence-state algebra and attention-weighted integration

Proposition 47 (Evidence-state lattices are complete). *Let $V = [0,1]^2$ with the componentwise order, and let L be any set of propositions. Then V^L (evidence states $E : L \to V$) is a complete lattice under the pointwise order. Joins and meets are computed pointwise, i.e.*

$$\Big(\bigvee_i E_i\Big)(\varphi) = \bigvee_i E_i(\varphi), \qquad \Big(\bigwedge_i E_i\Big)(\varphi) = \bigwedge_i E_i(\varphi).$$

Remark 2208. *Intuitively, an evidence state $E : L \to V$ assigns to each proposition $\varphi \in L$ a pair $E(\varphi) = (E^+(\varphi), E^-(\varphi)) \in [0,1]^2$, recording "support for" and "support against." The componentwise order on V means $(v^+, v^-) \le (w^+, w^-)$ iff $v^+ \le w^+$ and $v^- \le w^-$; the pointwise order on V^L then means $E \le E'$ iff this holds for every $\varphi \in L$. The claim that V^L is a complete lattice says, in effect, that we can coherently take suprema and infima of* arbitrary *families of evidence states, proposition-by-proposition.*

Remark 2209. *A simple example is when $L = \{\varphi, \psi\}$ is finite: then an evidence state is just a pair of p-bits, e.g. $E(\varphi) = (0.8, 0.1)$ and $E(\psi) = (0.3, 0.6)$. Given two evidence states E_1, E_2, their join $E_1 \vee E_2$ is computed by taking the join in V separately at φ and at ψ. Completeness matters because later we invoke Tarski-style fixed point results: these require complete lattices as the ambient domain on which closure operators (e.g. workspace closure) act.*

Remark 2210. What this is for. *In the later subsections, we treat workspace and reflective updates as monotone endomaps on evidence states. Completeness is the condition under which monotone dynamics have well-behaved fixed points and extremal fixed points (least/greatest), allowing one to speak rigorously about stable conscious content in the sense of Hyperseed-Concept 85.*

Remark 2211 (Intuition before the proof). *The proposition is a straightforward instance of a general idea: if your basic truth-values form a complete lattice, then functions assigning such values to propositions inherit completeness by taking suprema and infima pointwise. One may think of V^L as an L-indexed "field" of local degrees, and joins/meets as taking the best/worst local degrees everywhere at once.*

Sketch. V is a complete lattice because $[0,1]$ is complete and products of complete lattices are complete. Powers V^L inherit completeness under pointwise order, with pointwise sup/inf. □

Remark 2212. Proof sketch (high-level strategy). *The proof reduces the claim to two closure properties: (1) $[0,1]$ has all suprema/infima; (2) products and function spaces preserve completeness when ordered pointwise. This is a standard order-theoretic pipeline: build complex semantic domains from simpler ones, and transport lattice structure along the construction.*

Remark 2213. Why the key step works. *Completeness of $[0,1]$ is the analytic fact that every bounded set has a supremum and infimum. The rest is "coordinatewise reasoning": the supremum of a family of pairs is the pair of suprema, and the supremum of a family of functions is the function of suprema. Geometrically, $V = [0,1]^2$ is the unit square; a join takes the least upper bound in this square, and doing this pointwise over $\varphi \in L$ simply repeats that operation for each proposition.*

Proposition 48 (Explicit coordinate form of Γ_a in the canonical p-bit quantale). *Assume the canonical p-bit quantale operations on V:*

$$(v^+, v^-) \oplus (w^+, w^-) = (\max\{v^+, w^+\}, \max\{v^-, w^-\}), \qquad (v^+, v^-) \otimes (w^+, w^-) = (v^+w^+, v^-w^-),$$

and the scalar embedding $\alpha \mapsto (\alpha, \alpha)$ for $\alpha \in [0,1]$. Then for module evidence states $E_m : L \to V$ and attention $a : M \to [0,1]$, the integrated state $\Gamma_a(E_\bullet) : L \to V$ satisfies, for each $\varphi \in L$,

$$(\Gamma_a(E_\bullet))^+(\varphi) = \max_{m \in M} a(m)\, E_m^+(\varphi), \qquad (\Gamma_a(E_\bullet))^-(\varphi) = \max_{m \in M} a(m)\, E_m^-(\varphi).$$

Remark 2214. *Here the notation is doing a fair amount of work, so it is worth pausing over it.*

- *M is the set of modules (subsystems) contributing evidence.*
- *$a : M \to [0,1]$ is an attention weighting (Hyperseed-Concept 60): $a(m)$ is the degree to which module m is "listened to".*
- *$E_m : L \to V$ is module m's evidence state; we write $E_m(\varphi) = (E_m^+(\varphi), E_m^-(\varphi))$.*
- *$\Gamma_a(E_\bullet)$ denotes the integrated evidence state after attention-weighted aggregation.*
- *$\oplus$ is a join-like operator (here: componentwise* max*); $\otimes$ is a tensor-like operator (here: componentwise multiplication). This is a canonical quantale-style choice [3] (Hyperseed-Concept 143).*

Remark 2215. *Intuitively, the formula says: to compute overall positive evidence for φ, look at each module's positive evidence $E_m^+(\varphi)$, downweight it by attention $a(m)$, and then take the maximal such attended value across modules. The negative coordinate behaves analogously. Thus Γ_a implements a kind of attentional "winner-take-most" integration: the most attended, most confident module dominates, coordinatewise.*

Remark 2216. *A simple example: suppose $M = \{1,2\}$, $a(1) = 1$, $a(2) = 0.5$, and at proposition φ one has $E_1(\varphi) = (0.4, 0.2)$ and $E_2(\varphi) = (0.9, 0.1)$. Then*

$$(\Gamma_a(E_\bullet))^+(\varphi) = \max\{1 \cdot 0.4,\ 0.5 \cdot 0.9\} = \max\{0.4, 0.45\} = 0.45,$$

so module 2 wins on the positive coordinate only because its higher evidence compensates for lower attention. This kind of tradeoff is precisely what makes attention mathematically meaningful rather than merely a label. In the same numerical setup one can also read off the negative coordinate:

$$(\Gamma_a(E_\bullet))^-(\varphi) = \max\{1 \cdot 0.2,\ 0.5 \cdot 0.1\} = \max\{0.2, 0.05\} = 0.2,$$

so module 1 *still "wins" on the negative component. Thus even with only two modules the integrated state may be* coordinatewise *assembled from different sources, making explicit that Γ_a is not necessarily selecting a single module's whole pair (E_m^+, E_m^-), but rather forming the least upper bound in the product order by taking the best supported positive and best supported negative contributions separately.*

Remark 2217 (Intuition before the proof)**.** *The proof is essentially bookkeeping: start from the abstract definition of Γ_a as a join ($\oplus$) over modules of an attention-scaled tensor product ($\otimes$) with each module's evidence value, and then substitute the concrete forms of $\oplus$ and $\otimes$. Because $\oplus$ is* max *here, the whole expression collapses to a maximum over m of scaled coordinates. Concretely, the bookkeeping has two distinct "pointwise" aspects that are easy to conflate: the join is taken over* $m \in M$ for each fixed *proposition φ, and the resulting inequality/equality claims are then interpreted pointwise over propositions in L. This is why the explicit coordinate formulas are especially helpful: they avoid switching between these two pointwise readings mid-argument.*

Sketch. Unpack the definition $(\Gamma_a E_\bullet)(\varphi) = \bigoplus_{m\in M} a(m) \otimes E_m(\varphi)$. With the diagonal embedding, $a(m) \otimes (E_m^+, E_m^-) = (a(m)E_m^+, a(m)E_m^-)$. Then $\oplus$ takes componentwise maxima over m. ☐

Remark 2218. Proof sketch (high-level strategy). *The computation is a direct evaluation: (i) expand Γ_a at a proposition φ; (ii) apply the scalar embedding so attention becomes a p-bit; (iii) apply the tensor rule to obtain coordinatewise products; (iv) apply the join rule to obtain coordinatewise maxima. A useful way to remember the roles is that $\otimes$ models* within-module *attenuation (how a module's own evidence is weakened by limited access to the workspace), while $\oplus$ models* between-module *pooling (which module provides the strongest candidate contribution once attenuated).*

Remark 2219. Visual intuition. *For a fixed φ, each module contributes a point in the unit square $[0,1]^2$, which is then shrunk toward $(0,0)$ by the scalar factor $a(m)$. The operator $\oplus$ then takes the least upper bound in the product order, which for this canonical choice is exactly the coordinatewise upper envelope (componentwise maximum). One may picture $\Gamma_a(E_\bullet)(\varphi)$ as the "upper-right" corner of the set of attended points. Equivalently, if one draws axis-aligned rectangles from the origin to each attended point $(a(m)E_m^+(\varphi), a(m)E_m^-(\varphi))$, then the join is the unique point whose rectangle contains all the others and is minimal with that property. This makes clear why the join need not coincide with any single attended point when different modules dominate different coordinates.*

Proposition 49 (Monotonicity and uniform rescaling of attention)**.** *Assume the canonical operations above. If $a \le a'$ pointwise and $E_m \le E'_m$ pointwise for all m, then $\Gamma_a(E_\bullet) \le \Gamma_{a'}(E'_\bullet)$ pointwise on L. Moreover, for any $c \in [0,1]$,*

$$\Gamma_{ca}(E_\bullet) = c \otimes \Gamma_a(E_\bullet) \quad \textit{(pointwise on } L\textit{).}$$

Remark 2220. *Monotonicity here has a clear conceptual reading: giving* more *attention to each module (in the pointwise sense) and/or providing each module with* stronger *evidence (again pointwise) cannot make the integrated evidence weaker. This is a minimal sanity condition for any attention-weighted integrator purporting to model information flow into a workspace (Hyperseed-Concept 131). In terms of the product order on* $[0,1]^2$*, the claim is that increasing any input coordinate (either by raising attention or raising evidence) can only move the output "up and to the right" (or keep it fixed), never "down" on either coordinate. This aligns with the reading of* E^+ *and* E^- *as independent resource-like quantities that can be amplified or attenuated but not created by decreasing inputs.*

Remark 2221. *Uniform rescaling* $\Gamma_{ca} = c \otimes \Gamma_a$ *says that if one globally "turns down the volume" of attention by a factor* c*, the entire integrated evidence state is turned down by the same factor under the tensor operation. In the canonical multiplication tensor, this is literal scalar multiplication coordinatewise; in a more general quantale it is the appropriate notion of global weakening [3] (Hyperseed-Concept 202). Note that this identity is special to* uniform *rescaling: if different modules are rescaled by different constants, there is in general no factorization of the form* $\Gamma_{a'}(E_\bullet) = d \otimes \Gamma_a(E_\bullet)$ *with a single global* d*, precisely because the* max *aggregator can switch which module attains the maximum after rescaling. In other words, global rescaling commutes with integration, but heterogeneous rescaling can change the winner(s) coordinatewise.*

Remark 2222 (Intuition before the proof)**.** *Both claims are consequences of order-compatibility:* $\otimes$ *should respect increases in either argument, and* $\oplus$ *should respect increases in the joined terms. Once that is granted, the join over* m *of monotone expressions remains monotone. The rescaling claim then follows from factoring out the constant* c *through maxima/multiplication. A compact way to see the second claim is to view* $\Gamma_a(E_\bullet)(\varphi)$ *as built from the multiset* $\{a(m)E_m(\varphi)\}_{m\in M}$ *by taking a least upper bound; multiplying every element of that multiset by the same* c *multiplies its least upper bound by* c *when the order and tensor are compatible in the canonical way.*

Sketch. For monotonicity: each map $(a, E) \mapsto a \otimes E$ is monotone and $\oplus$ is monotone, so the pointwise join of monotone expressions is monotone. For rescaling: using the explicit coordinate form, $\max_m(ca(m)E^{\pm}_m) = c \max_m(a(m)E^{\pm}_m)$ since $c \geq 0$. □

Remark 2223. Proof sketch (high-level strategy). *The first part is a compositional argument: monotone pieces compose to give a monotone whole. The second part uses a concrete identity about maxima of scaled numbers when the scale is nonnegative, applied separately to the* $+$ *and* $-$ *coordinates. It may help to make the compositionality explicit: for fixed* φ*, each* m *induces a monotone map* $(a, E_\bullet) \mapsto a(m) \otimes E_m(\varphi)$*, and* $\Gamma_a(E_\bullet)(\varphi)$ *is obtained by applying a monotone* $\oplus$*-folding to these outputs. Monotonicity is then preserved under (i) evaluation at* m*, (ii) tensoring, and (iii) joining.*

Remark 2224. Why the key step works. *The only subtlety is that pulling* c *out of* $\max_m$ *requires* $c \geq 0$*, which is why* c *is restricted to* $[0,1]$*. If negative scalars were allowed, maxima would turn into minima after multiplication; but in this evidential setting negative attention has no intended interpretation. Also, restricting* $c \leq 1$ *matches the intended reading of attention as a* capacity limit *(attenuation), so that* $c \otimes x$ *never exceeds* x *in the canonical multiplication tensor. If one wished to model amplificatory gain (values* $c > 1$*), the same algebraic identity would remain valid for* $c > 0$*, but one would then leave the unit interval and would need a different codomain (or a different normalization convention) for evidence.*

Corollary 17 (Attentional gating)**.** *If* $a(m) = 0$*, module* m *contributes nothing to* $\Gamma_a(E_\bullet)$*. If there exists* m_0 *with* $a(m_0) = 1$*, then* $\Gamma_a(E_\bullet) \geq E_{m_0}$ *pointwise.*

Remark 2225. *This corollary formalizes two qualitative facts about attention (Hyperseed-Concept* **??***):* gating *(zero attention silences a module) and* guaranteed inclusion *(a fully attended module cannot be ignored by a max-based integrator). In computational terms, $a(m) = 0$ is like dropping a channel, while $a(m) = 1$ is like ensuring at least that module's evidence survives integration. Because the join is taken componentwise, "cannot be ignored" here should be read in the order-theoretic sense: every coordinate of $E_{m_0}(\varphi)$ is bounded above by the corresponding coordinate of the integrated state. This is weaker than saying the integrated state* equals E_{m_0} *(other modules may still raise some coordinates further), but strong enough to guarantee that fully attended information is never diminished by pooling.*

Remark 2226 (Intuition before the proof)**.** *With Γ_a computed by coordinatewise maxima of the values $a(m)E^{\pm}_m(\varphi)$, setting $a(m) = 0$ makes that module's candidate value identically* 0 *(hence never a new maximum unless everything else is also* 0*). Conversely, if some module has $a(m_0) = 1$, then its unscaled evidence appears among the candidates, so the maximum must be at least as large. One can also see the first claim directly from the tensor rule: $0 \otimes E_m(\varphi) = (0 \cdot E^+_m, 0 \cdot E^-_m)$, so the term for m is the bottom element $(0, 0)$ and therefore does not affect a join by coordinatewise maxima.*

Sketch. Fix $\varphi \in L$. If $a(m) = 0$, then $a(m) \otimes E_m(\varphi) = (0, 0)$ under the canonical tensor, so including this term in the join $\bigoplus_{m\in M} a(m) \otimes E_m(\varphi)$ does not change the componentwise maxima. For the second claim, if $a(m_0) = 1$ then $a(m_0) \otimes E_{m_0}(\varphi) = E_{m_0}(\varphi)$ appears among the joined terms, hence the componentwise maximum over all m is $\geq E_{m_0}(\varphi)$ in the product order. Since φ was arbitrary, the inequality holds pointwise on L. □

Sketch. If $a(m) = 0$, then $a(m)E^{\pm}_m(\varphi) = 0$ for all φ so it cannot increase the max. More explicitly, for any fixed $\varphi \in L$ we have $a(m)E^{\pm}_m(\varphi) = 0 \leq a(m')E^{\pm}_{m'}(\varphi)$ for some m' with $a(m') > 0$ (or else all terms are 0), hence the term with $a(m) = 0$ is never strictly maximal. If $a(m_0) = 1$, then $\max_m a(m)E^{\pm}_m(\varphi) \geq 1 \cdot E^{\pm}_{m_0}(\varphi)$. In other words, since the maximum is taken over a set of candidates that includes m_0, the supremum/maximum cannot be smaller than the particular candidate value contributed by m_0. □

Remark 2227. Visual intuition. *Think again of each module contributing an attended point in the square. Zero attention collapses the module's point to the origin, while full attention leaves it unchanged. The coordinatewise upper envelope cannot lie below an unchanged point that is included among the candidates. Equivalently: in each coordinate (positive or negative support), the upper envelope is obtained by taking the largest attended coordinate among the modules, so a module with weight* 0 *is geometrically irrelevant, while a module with weight* 1 *contributes its full coordinate as a lower bound on the envelope in that coordinate.*

Remark 2228. Order-theoretic restatement. *For each fixed φ, the map $m \mapsto a(m)E^{\pm}_m(\varphi)$ is a family in* $[0, 1]$*, and* $\max_m a(m)E^{\pm}_m(\varphi)$ *is its join in the usual order on* $[0, 1]$ *(when M is finite; in general one uses* $\sup$*). The two extreme-weight cases $a(m) = 0$ and $a(m) = 1$ are then immediate consequences of the facts that* 0 *is the least element and that joins are upper bounds for each summand.*

38.2 Workspace closure and conscious content thresholds

Proposition 50 (Monotonicity of thresholded conscious content)**.** *Fix $\theta \in (0, 1)$ and define $\mathrm{Con}_\theta(E) := \{\varphi \in L : \ E^+(\varphi) \geq \theta\}$. Then:*

1. *If $E \leq E'$, then $\mathrm{Con}_\theta(E) \subseteq \mathrm{Con}_\theta(E')$.*

2. If $\theta \leq \theta'$, then $\mathrm{Con}_{\theta'}(E) \subseteq \mathrm{Con}_{\theta}(E)$.

Remark 2229. *The set $\mathrm{Con}_\theta(E)$ is a deliberately austere formalization of "conscious content" (Hyperseed-Concept 85): it picks out those propositions whose* positive *support clears a fixed threshold θ. The parameter θ may be read as a minimal "broadcast strength" required for entry into the workspace, in the spirit of workspace-like theories in cognitive science [19]. One can also read θ as fixing a decision boundary for which contents become reportable, globally available, or able to drive downstream action selection, while leaving open the possibility that sub-threshold items still influence processing in a graded way.*

Remark 2230. *A simple example: if $E^+(\varphi) = 0.7$ and $E^+(\psi) = 0.2$, then with $\theta = 0.6$ we have $\mathrm{Con}_\theta(E) = \{\varphi\}$, while with $\theta = 0.1$ we have $\mathrm{Con}_\theta(E) = \{\varphi, \psi\}$. Thus lowering θ can only add content, never remove it, which is precisely the second monotonicity statement. In the same example, if we increase evidence to E' with $E'^+(\varphi) = 0.8$ and $E'^+(\psi) = 0.3$ (so $E \leq E'$), then $\mathrm{Con}_{0.6}(E') = \{\varphi\}$ still contains φ and does not "lose" it, illustrating the first monotonicity statement in a concrete way.*

Remark 2231 (Intuition before the proof)**.** *Both parts express a basic order-theoretic idea: increasing evidence should not make formerly conscious items unconscious, and raising the bar should not create new winners. The proof is a one-line unpacking of what it means for a coordinate to be larger and what it means to satisfy an inequality. Formally, (1) uses monotonicity of the predicate "$\geq \theta$" in its argument, while (2) uses antitonicity of the set $\{\varphi : x(\varphi) \geq \theta\}$ in the threshold parameter θ.*

Sketch. (1) $E \leq E'$ implies $E^+(\varphi) \leq E'^+(\varphi)$ for all φ. So if $\varphi \in \mathrm{Con}_\theta(E)$, meaning $E^+(\varphi) \geq \theta$, then by transitivity $E'^+(\varphi) \geq E^+(\varphi) \geq \theta$, hence $\varphi \in \mathrm{Con}_\theta(E')$. (2) If $E^+(\varphi) \geq \theta'$ and $\theta' \geq \theta$, then $E^+(\varphi) \geq \theta$. □

Remark 2232. Proof sketch (high-level strategy). *Translate set membership $\varphi \in \mathrm{Con}_\theta(E)$ into an inequality on real numbers, and then use transitivity of $\leq$ on $[0,1]$. Equivalently, the map $E \mapsto \mathrm{Con}_\theta(E)$ is monotone as a function from the evidence-state poset to the powerset lattice $(\mathcal{P}(L), \subseteq)$, while the map $\theta \mapsto \mathrm{Con}_\theta(E)$ is monotone in the reverse direction (larger thresholds yield smaller sets).*

Remark 2233. Boundary cases and conventions. *Taking $\theta \in (0,1)$ avoids trivial extremes: if $\theta \leq 0$ then $\mathrm{Con}_\theta(E) = L$ for all E, and if $\theta > 1$ then $\mathrm{Con}_\theta(E) = \varnothing$. Nothing in the monotonicity argument depends on excluding these cases; the restriction is mainly conceptual, to keep "thresholding" aligned with a non-degenerate notion of workspace admission.*

Theorem 54 (Existence of ordinary workspace-stable states)**.** *Assume $W : V^L \to V^L$ is monotone and V^L is a complete lattice (e.g. L finite). Then W admits at least one fixed point, and in fact has a least and a greatest fixed point.*

Remark 2234. *This is the first point at which the earlier completeness proposition pays off. A "workspace operator" W may be read as a closure-like transformation: it takes an evidence state and returns the evidence state after whatever internal propagation, consistency enforcement, and broadcast effects the workspace performs. Monotonicity is the formal version of a mild rationality condition: adding evidence should not reduce what the workspace ends up representing. Concretely, if $E \leq E'$ expresses that E' is at least as supportive (in every coordinate) as E, then $W(E) \leq W(E')$ says that the workspace update does not perversely invert this informativeness ordering.*

Remark 2235. *The theorem says that under these assumptions there exist stable evidence configurations E such that $W(E) = E$. These fixed points are the natural mathematical candidates for "workspace-stable" or "ordinary" conscious states. The existence of* least *and* greatest *fixed points is especially useful for mechanized reasoning: it provides canonical extremal solutions that can be approximated by iteration, and it connects naturally with stability analyses for self-modifying or goal-directed systems (cf. fixed-point tools in [10]). In particular, the least fixed point can be viewed as the minimal self-consistent broadcast state compatible with the update dynamics, while the greatest fixed point represents a maximally informative self-consistent state (relative to the underlying lattice order).*

Remark 2236 (Intuition before the proof)**.** *The result is a direct application of Tarski's fixed point theorem: in a complete lattice, a monotone map always has fixed points, and the fixed points themselves form a complete lattice. Philosophically, one might say: whenever you have a space of possible "belief/evidence states" rich enough to take all joins/meets, and an update rule that respects the ordering of informativeness, stable self-consistent states cannot fail to exist. This also clarifies why completeness is the key hypothesis: fixed points are obtained by taking meets/joins of certain invariant sets (e.g. pre-fixpoints or post-fixpoints), and those meets/joins must exist in the ambient structure for the construction to go through.*

Sketch. By Tarski's fixed point theorem, any monotone endomap on a complete lattice has a complete lattice of fixed points, in particular a least and greatest fixed point. More explicitly, the least fixed point is the meet of all pre-fixpoints $\{E : W(E) \leq E\}$ and the greatest fixed point is the join of all post-fixpoints $\{E : E \leq W(E)\}$, and completeness guarantees these extremal elements exist. $\square$

Remark 2237. Proof sketch (high-level strategy). *The proof is a citation of a deep structural theorem rather than a calculation: once the hypotheses match Tarski's theorem, the conclusion follows immediately. In applications, this means that substantial modeling work is pushed into verifying monotonicity and completeness; once those are in place, existence of stable workspace states is automatic.*

Remark 2238. Geometric intuition. *Even though V^L can be high-dimensional, one can imagine the update W as a deterministic "flow" that moves states upward in an order-respecting way. Tarski's theorem assures that such a flow must intersect the diagonal $E = W(E)$, and in fact does so in a way that has extremal intersection points (least and greatest). One should not over-interpret "upward" as literal numerical increase in every coordinate in time; rather, the order-respecting condition prevents W from creating cycles that strictly violate the partial order, and the lattice structure forces the family of approximants (formed by iterating from bottom or top) to have well-defined limits in the order sense.*

Lemma 123 (One-shot stability when W is idempotent)**.** *If the workspace operator additionally satisfies idempotence $W \circ W = W$, then for any integrated evidence state $E_0 := \Gamma_a(E_\bullet)$, the state $E := W(E_0)$ is workspace-stable: $W(E) = E$.*

Remark 2239. *Idempotence is the algebraic signature of a closure operator: applying the closure twice does not change what the first application already accomplished. When W is idempotent, the lemma says that a single workspace update already lands you in a stable configuration. This is useful when one wants a model in which "broadcast and settle" is a single step rather than an iterative convergence process. In cognitive terms, the extra assumption idealizes away oscillatory or multi-stage settling dynamics and treats the workspace as performing a full normalization/propagation in one pass.*

Remark 2240 (Intuition before the proof)**.** *The reasoning is essentially: if W is a closure, then $W(E_0)$ is already closed, so applying W again must return the same result. The proof is a single substitution into the idempotence equation. Equivalently, $W(E_0)$ is a fixed point because the image of an idempotent map consists entirely of fixed points.*

Sketch. $W(E) = W(W(E_0)) = W(E_0) = E$ by idempotence. Writing the equalities step-by-step: $E = W(E_0)$ by definition, hence $W(E) = W(W(E_0))$; idempotence gives $W(W(E_0)) = W(E_0)$, and the right-hand side is again E. □

Remark 2241. Proof sketch (high-level strategy). *Define E to be the image of E_0 under W; then stability is exactly the idempotence law instantiated at E_0.*

38.3 Reflective layer and introspection

Proposition 51 (Minimal introspection transfers base support to awareness tokens)**.** *Assume the "minimal" introspection choice for $I(\varphi)$:*

$$\mathrm{Int}(E)(I(\varphi)) = \big(E^+(\varphi),\ 1 - E^+(\varphi)\big).$$

Then $\mathrm{Int}(E)^+(I(\varphi)) = E^+(\varphi)$ *for all φ. Consequently, for any threshold θ,*

$$\varphi \in \mathrm{Con}_\theta(E) \quad \Longleftrightarrow \quad I(\varphi) \in \mathrm{Con}_\theta(\mathrm{Int}(E)).$$

Remark 2242. *This proposition concerns the reflective layer: we enrich the language with "awareness tokens" $I(\varphi)$, intended to represent something like "φ is present to awareness" (Hyperseed-Concept 203 and, more broadly, Hyperseed-Concept 85). The operator* Int *maps a base evidence state to a reflective evidence state. The displayed choice is called "minimal" because it transfers only the positive coordinate $E^+(\varphi)$ into the positive coordinate of the introspective token, and fills the negative coordinate so that the result is a* consistent *p-bit of the form $(p, 1-p)$ (cf. [23, 24]).*

Remark 2243. *Concretely, if $E^+(\varphi) = 0.8$, then* $\mathrm{Int}(E)(I(\varphi)) = (0.8, 0.2)$*, so the system's "degree of awareness of φ" tracks the base support for φ exactly, at least on the positive coordinate. The equivalence of thresholded conscious sets then says that the content thresholding commutes with this introspection: what is conscious at the base level is conscious at the awareness-token level and vice versa, under this minimal convention.*

Remark 2244 (Intuition before the proof)**.** *There is almost nothing to prove: the first coordinate of* $\mathrm{Int}(E)(I(\varphi))$ *is defined to be $E^+(\varphi)$. The equivalence for* Con_θ *then follows because* Con_θ *looks only at that first coordinate. The point is not difficulty, but transparency: we want the simplest reflective layer in which "being aware of φ" faithfully mirrors the degree to which φ is supported.*

Sketch. Immediate from the displayed definition: the positive coordinate of $\mathrm{Int}(E)(I(\varphi))$ is exactly $E^+(\varphi)$. □

Remark 2245. Proof sketch (high-level strategy). *Substitute the definition and read off the coordinate; then translate the thresholded set membership into inequalities. The argument is essentially definitional.*

Corollary 18 (Reflective-access collapses to self-presence under minimal introspection)**.** *In the reflective-self criterion, if "reflective access" is evaluated using the same threshold rule* Con_θ *on positive evidence, then under minimal introspection:*

$$E^+(\mathrm{Self}(p)) \geq \theta \implies \mathrm{Int}(E)^+(I(\mathrm{Self}(p))) \geq \theta.$$

So (for this particular Int *and thresholding convention) reflective-access adds no new constraint beyond self-presence; the remaining nontrivial requirement is stability under perturbation.*

Remark 2246. *The corollary is a small but conceptually clarifying point about the "reflective self" condition (Hyperseed-Concept 165): if introspection is defined minimally, then the system's positive support for a self-token* $\mathrm{Self}(p)$ *automatically lifts to positive support for the awareness-of-self token* $I(\mathrm{Self}(p))$ *at the same level. Thus, in this setting, the interesting work in reflective-selfhood is not done by a separate access test, but by robustness/stability conditions (continuity under perturbation), which capture something closer to the phenomenological idea of sustained self-presence rather than a momentary flicker.*

Remark 2247 (Intuition before the proof)**.** *This is exactly the previous proposition applied to the special case* $\varphi = \mathrm{Self}(p)$*: minimal introspection is designed so that the awareness-token inherits the same positive coordinate.*

Sketch. Apply the previous proposition with $\varphi := \mathrm{Self}(p)$. □

Remark 2248. Proof sketch (high-level strategy). *Reduce the corollary to a previously established identity by choosing the appropriate* φ*.*

Proposition 52 (Monotonicity of the reflective update map in inputs)**.** *Assume* Γ_a*,* Int*, and* W^{ref} *are monotone. Then the composite* $F := W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a$ *is monotone in module evidence and attention, i.e. if* $E_\bullet \leq E'_\bullet$ *and* $a \leq a'$*, then*

$$F(a, E_\bullet) \ \leq \ F(a', E'_\bullet) \quad \textit{(pointwise on } L^{\mathrm{ref}}\textit{).}$$

Remark 2249. *This proposition is the reflective analogue of earlier monotonicity facts: if you improve the informational inputs (stronger module evidence) and/or the attentional allocation (higher attention weights), then the final reflectively-closed state cannot become weaker. In other words, the reflective layer does not perversely punish additional evidence. Such monotonicity is a prerequisite for using lattice-theoretic fixed-point methods in the reflective space as well (cf. the stability tools emphasized in [10]).*

Remark 2250 (Intuition before the proof)**.** *The composite* F *consists of three stages: integrate (*Γ_a*), introspect (*Int*), and close reflectively (*W^{ref}*). If each stage is monotone, then their composition is monotone. This is the general categorical principle that order-preserving maps compose.*

Sketch. Γ_a is monotone in $(a, E_\bullet)$ (previous results), and Int and W^{ref} are monotone by assumption. Composition of monotone maps is monotone. In particular, if the relevant evidence/state space is ordered by the pointwise (product) order, then for any two inputs $X \leq Y$ we have $\Gamma_a(X) \leq \Gamma_a(Y)$, and applying Int and then W^{ref} preserves the inequality at each successive stage. Thus the full composite mapping inherits monotonicity without requiring any additional algebraic properties beyond order-preservation. □

Remark 2251. Proof sketch (high-level strategy). *Treat* F *as a pipeline and apply the lemma "monotone* $\circ$ *monotone is monotone" stage by stage. Concretely, one tracks how an increase in the input (in the chosen partial order) propagates forward: no stage can reverse an inequality, so the endpoint cannot reverse it either. This is the same proof pattern used for isotone maps on posets and for order-preserving program semantics (where sequential composition preserves monotonicity).*

38.4 Coherence score facts

Proposition 53 (Closed form, bounds, and negation-invariance of $|\sigma_{\mathbb{C}}|^2$)**.** *Using* $\sigma_{\mathbb{C}}(v^+, v^-) = (v^+ - v^-) + i\big((v^+ + v^-) - 1\big)$*, one has*

$$|\sigma_{\mathbb{C}}(v^+, v^-)|^2 = (v^+ - v^-)^2 + (v^+ + v^- - 1)^2 \in [0, 2].$$

Moreover $|\sigma_{\mathbb{C}}(v)| = |\sigma_{\mathbb{C}}(\neg v)|$ *where* $\neg(v^+, v^-) = (v^-, v^+)$*. Therefore, for* $\mathrm{Coh}(E) = \frac{1}{|K|}\sum_{\varphi \in K} |\sigma_{\mathbb{C}}(E(\varphi))|^2$*, we have* $\mathrm{Coh}(E) \in [0, 2]$ *and* $\mathrm{Coh}(E)$ *is invariant under p-bit negation applied pointwise to the evidence values of propositions in* K*. In addition, the lower bound* 0 *is achieved precisely when each* $E(\varphi)$ *satisfies* $v^+ = v^- = \frac{1}{2}$ *(so both the bias and the normalization deviation vanish), while the upper bound* 2 *is achieved when simultaneously* $|v^+ - v^-| = 1$ *and* $|v^+ + v^- - 1| = 1$*, which occurs at the corners* $(1, 0)$ *and* $(0, 1)$ *(maximal bias with classical normalization) and also at* $(1, 1)$ *and* $(0, 0)$ *(maximal over- or under-determination with zero bias).*

Remark 2252. *The map* $\sigma_{\mathbb{C}} : V \to \mathbb{C}$ *embeds a p-bit into the complex plane by treating two simple derived quantities as the real and imaginary parts:*

$$\mathrm{Re}(\sigma_{\mathbb{C}}(v)) = v^+ - v^- \quad \textit{(net bias)}, \qquad \mathrm{Im}(\sigma_{\mathbb{C}}(v)) = (v^+ + v^-) - 1 \quad \textit{(departure from classical normalization).}$$

The squared modulus $|\sigma_{\mathbb{C}}(v)|^2$ *is then a compact scalar "coherence" proxy that penalizes both large bias and large over/under-determination. Equivalently, it measures the squared Euclidean norm of the feature vector* $(v^+ - v^-,\ v^+ + v^- - 1)$*, so the two departures (directional bias versus total-evidence deviation) contribute additively and are not allowed to cancel each other. This complex-amplitude flavor is reminiscent of how interference-like quantities can arise in paraconsistent/quantum-inspired semantics [24], but here the role of the complex plane is strictly as a convenient two-dimensional representation with a rotationally symmetric magnitude.*

Remark 2253. *Two quick examples: if* $v = (0.5, 0.5)$*, then* $\sigma_{\mathbb{C}}(v) = 0 + i(0)$ *and* $|\sigma_{\mathbb{C}}(v)|^2 = 0$*, representing a perfectly balanced, classically-normalized evidence pair. If* $v = (1, 0)$*, then* $\sigma_{\mathbb{C}}(v) = 1 + i(0)$ *and* $|\sigma_{\mathbb{C}}(v)|^2 = 1$*. If* $v = (1, 1)$ *(maximal inconsistency), then* $\sigma_{\mathbb{C}}(v) = 0 + i(1)$ *and* $|\sigma_{\mathbb{C}}(v)|^2 = 1$ *again: the magnitude treats extreme bias and extreme inconsistency as comparable "departures" in orthogonal directions. As a further contrast, if* $v = (0, 0)$ *(maximal lack of support in either direction), then* $\sigma_{\mathbb{C}}(v) = 0 + i(-1)$ *and* $|\sigma_{\mathbb{C}}(v)|^2 = 1$*, showing that over-determination* $(1, 1)$ *and under-determination* $(0, 0)$ *are symmetric with respect to the baseline* $v^+ + v^- = 1$*.*

Remark 2254. *Negation-invariance expresses a philosophical symmetry: swapping "for" and "against" evidence changes the sign of the bias but not its magnitude, and does not change the normalization deviation. Hence the scalar* $|\sigma_{\mathbb{C}}|^2$ *(and therefore* Coh*) measures a kind of structural intensity rather than a directional preference. In particular,* Coh *cannot distinguish whether a system is coherently "pro" or coherently "con" with respect to propositions in* K*; it only records how far the evidence vectors sit from the neutral, classically-normalized midpoint.*

Remark 2255 (Intuition before the proof)**.** *The computation is the standard identity* $|x + iy|^2 = x^2 + y^2$*. The bound* $[0, 2]$ *comes from bounding each of* x *and* y *in* $[-1, 1]$*. Negation swaps* v^+ *and* v^-*, which flips* x *but leaves* y *unchanged, so the sum of squares is invariant. One can also read* x *as an antisymmetric component (it changes under swapping) and* y *as a symmetric component (it is preserved under swapping), so the modulus squares and adds precisely those two independent contributions.*

Sketch. Expand the modulus: for $x + iy$, $|x + iy|^2 = x^2 + y^2$. Here $x = v^+ - v^- \in [-1, 1]$ and $y = v^+ + v^- - 1 \in [-1, 1]$, so $x^2 + y^2 \le 2$. To justify the interval bounds explicitly: since $v^+, v^- \in [0, 1]$, the difference satisfies $-1 \le v^+ - v^- \le 1$, and the sum satisfies $0 \le v^+ + v^- \le 2$, hence $-1 \le (v^+ + v^-) - 1 \le 1$. Under negation, x flips sign and y is unchanged, so $x^2 + y^2$ is unchanged. Averaging over K preserves the bounds and invariance because a mean of numbers in $[0, 2]$ remains in $[0, 2]$, and because pointwise negation leaves each summand $|\sigma_{\mathbb{C}}(E(\varphi))|^2$ fixed. □

Remark 2256. Proof sketch (high-level strategy). *Reduce everything to bounds on two real scalars derived from v, then apply the elementary inequalities $x^2 \le 1$ and $y^2 \le 1$. If one wants a slightly sharper geometric phrasing, the argument is: the map $(v^+, v^-) \mapsto (x, y)$ sends the unit square into the square $[-1, 1]^2$, and the function $(x, y) \mapsto x^2 + y^2$ attains its maximum 2 on the corners, so no image point can exceed 2.*

Remark 2257. Geometric intuition. *The image of $V = [0, 1]^2$ under $(v^+, v^-) \mapsto (x, y)$ lies in the square $[-1, 1]^2$ in the (x, y)-plane; $|\sigma_{\mathbb{C}}(v)|^2$ is simply the squared Euclidean distance from the origin in this plane. Negation reflects points across the vertical axis $x \mapsto -x$, which preserves distance from the origin. Consequently, level sets of $|\sigma_{\mathbb{C}}(v)|^2$ correspond to circles centered at the origin intersected with the square: moving around such a circle changes the relative allocation between bias and normalization deviation while keeping the overall coherence proxy constant.*

38.5 Will, autonomy, and viability

Proposition 54 (Natural autonomy iff there exists a viable will policy on V)**.** *Let $D_u : S_{\mathrm{Ob}} \to S_{\mathrm{Ob}}$ be the update induced by intervention $u \in U$ and let $V \subseteq S_{\mathrm{Ob}}$ be a viable set. Then the following are equivalent:*

1. *(Natural autonomy)* $\forall s \in V\ \exists u \in U :\ D_u(s) \in V$.
2. *(Viable will policy)* $\exists\, \mathrm{Will} : V \to U\ \ \forall s \in V :\ D_{\mathrm{Will}(s)}(s) \in V$.

Remark 2258. *This proposition translates between two ways of stating autonomy (Hyperseed-Concept 119) and will (Hyperseed-Concept 203). Statement (1) is "pointwise possibility": from every viable state, there exists some intervention that keeps you viable for one step. Statement (2) is the existence of a* choice function*—a policy* Will *that actually selects such an intervention uniformly across states. Formally, the difference is the order of quantifiers: $(\forall s)(\exists u)$ versus $(\exists\, \mathrm{Will})(\forall s)$. The proposition says these are equivalent when one is allowed to define* Will *by choosing a witness $u(s)$ for each s.*

It is helpful to keep in mind that (2) is strictly stronger than merely asserting that the set

$$A(s) := \{u \in U :\ D_u(s) \in V\}$$

is nonempty for each $s \in V$: statement (2) asserts that one can simultaneously select an element of $A(s)$ for every *$s \in V$ in a way that packages these choices into a single object (a function). In other words, (2) turns a family of existence claims into a single policy object that can be reasoned about compositionally.*

Remark 2259. *In classical mathematics this is immediate, but it is still useful to record because it tells us what autonomy amounts to in this one-step setting: it is exactly the ability to define a viability-preserving will policy. In constructive or computable settings, the step (1) ⇒ (2) can conceal nontrivial choice principles; here, however, we remain in the usual classical mode, and* *the proposition functions as a notational bridge between an existential viability condition and a policy-existence statement.*

One practical way to see the "classical" nature is: if V is finite (or otherwise effectively enumerable) and for each $s \in V$ one can effectively find at least one u with $D_u(s) \in V$, then the construction of Will *is not only logically valid but algorithmic. In contrast, if V is infinite and the existence of u is nonconstructive, then* Will *may exist classically without being computable. This aligns with the conceptual distinction between* having *a will policy and* being able to compute *or* implement *one.*

Remark 2260 (Intuition before the proof)**.** *The forward direction is trivial: if you have a policy, you can always pick its action. The reverse direction is "define the policy by picking a witness action for each state." This is the archetype of a proof by choice-function construction.*

Another way to phrase the same intuition is: statement (1) says each viable state has at least one "safe" outgoing edge (labeled by an intervention) back into V; statement (2) says there is a way to choose one safe edge per node, producing a deterministic controller on V. The point is not that the chosen edge is unique, but that a single consistent selection suffices to define an explicit controller.

Sketch. (2) $\Rightarrow$ (1) is immediate by taking $u = \mathrm{Will}(s)$. (1) $\Rightarrow$ (2) choose, for each $s \in V$, some witnessing $u(s)$ with $D_{u(s)}(s) \in V$ and define $\mathrm{Will}(s) := u(s)$. ☐

Remark 2261. Proof sketch (high-level strategy). *Rearrange quantifiers by explicitly constructing the missing function: a will policy is nothing but a consistent selection of one admissible action per state.*

Equivalently, one can view (1) as stating that the relation

$$R \subseteq V \times U, \qquad (s,u) \in R \iff D_u(s) \in V$$

is left-total *(every $s \in V$ relates to at least one $u \in U$). Then (2) asserts the existence of a* selector *(or Skolem function) for this relation. In standard set-theoretic foundations, left-totality implies the existence of such a selector.*

Theorem 55 (Viability kernel as a greatest fixed point (useful for computing maximal autonomy))**.** *Define the one-step predecessor operator on subsets $X \subseteq S_{\mathrm{Ob}}$ by*

$$\mathrm{Pre}(X) := \{s \in S_{\mathrm{Ob}} : \ \exists u \in U,\ D_u(s) \in X\}.$$

Then Pre *is monotone on the powerset lattice $(\mathcal{P}(S_{\mathrm{Ob}}), \subseteq)$. Define $T(X) := V \cap \mathrm{Pre}(X)$. Then T is monotone and hence has a greatest fixed point K_V (by Tarski). This K_V is the* largest *subset of V such that*

$$\forall s \in K_V\ \exists u \in U : \ D_u(s) \in K_V.$$

In particular, K_V is the maximal "autonomy-sustaining" region inside V.

Remark 2262. *This theorem packages a standard viability-kernel construction into the fixed-point language used throughout these helper results. The operator* Pre *collects those states that can reach X in one step under* some *intervention. Intersecting with V enforces viability. A fixed point $X = T(X)$ is then a subset of V that is closed under one-step viability-preserving interventions: from every $s \in X$ one can stay within X.*

In control-theoretic terms, K_V is an invariant *(or* controlled-invariant*) subset of V for the controlled transition map D_u. The predecessor* Pre *is the usual controllable predecessor operator, and the intersection with V enforces state constraints. Thus the fixed-point formulation is a re-expression of the familiar "remove states that cannot be kept safe" computation.*

Remark 2263. *The greatest fixed point K_V is therefore the maximal region in which natural autonomy holds (Hyperseed-Concept 119): it is the largest subset of V from which one can keep choosing actions so as to remain in that subset indefinitely (in the one-step unfolding sense encoded by T). This is not only conceptually neat but computationally useful: greatest fixed points can often be approximated by descending iteration from the top element, which aligns with mechanized analyses of stable goal/viability systems [10].*

Concretely, once K_V has been identified, Proposition 54 applied to the set K_V shows that there exists a will policy $\mathrm{Will} : K_V \to U$ *that keeps the system inside K_V. So the fixed-point theorem is not merely about identifying a region; it also supports the existence of an explicit* closed-loop *viability-preserving will policy when one restricts attention to K_V.*

Remark 2264 (Intuition before the proof)**.** *The key ideas are (i) monotonicity of* Pre*: enlarging the target set can only enlarge the set of states that can reach it; and (ii) Tarski: monotone maps on complete lattices have greatest fixed points. The characterization of K_V as "largest autonomy-sustaining region" then follows from the universal property of the greatest fixed point as the largest post-fixed point.*

Here "post-fixed point" means a set X with $X \subseteq T(X)$. Intuitively, $X \subseteq T(X)$ says: if you start in X, you can choose an action to remain in X (and of course remain in V), so X is already self-sustaining. The greatest fixed point is then the union (supremum) of all such self-sustaining sets, which explains why it captures the maximal autonomy-sustaining region.

Sketch. If $X \subseteq Y$, then $\mathrm{Pre}(X) \subseteq \mathrm{Pre}(Y)$ (witnesses into X are also witnesses into Y), so Pre is monotone; intersecting with fixed V preserves monotonicity, hence T is monotone. Tarski gives a greatest fixed point K_V. If X is any subset of V with the autonomy property, then $X \subseteq \mathrm{Pre}(X)$, hence $X \subseteq V \cap \mathrm{Pre}(X) = T(X)$; thus X is a post-fixed point of T and must lie below the greatest fixed point K_V. □

Remark 2265. Proof sketch (high-level strategy). *Establish monotonicity, invoke Tarski to obtain K_V, and then prove maximality by showing that any autonomy-sustaining set is a post-fixed point and therefore contained in the greatest fixed point.*

A complementary viewpoint (often used in algorithms) is to start from the "largest possible candidate" $X_0 := V$ and iteratively eliminate states that fail the one-step autonomy condition relative to the current candidate:

$$X_{n+1} := T(X_n) = V \cap \mathrm{Pre}(X_n).$$

This produces a descending chain $X_0 \supseteq X_1 \supseteq X_2 \supseteq \cdots$ whose limit (in finite-state systems, after finitely many steps) is precisely K_V. This operationalizes the greatest-fixed-point interpretation: keep pruning until stability.

Remark 2266. Visual intuition. *Imagine V as the "safe region" in state space. Starting from a candidate set X, the operator T throws away points in V that cannot be kept within X in one step, retaining only those that have at least one safe continuation. Iterating T refines X by removing states that lead inevitably outside; the limit of this refinement is K_V. In viability terms, this iterative "pruning" is the dynamic counterpart of the static requirement that a state admit at least one indefinitely safe policy: the limiting set collects precisely those states for which safety can be maintained by appropriate choices over time.*

38.6 Two-channel (intuition/reason) update facts

Proposition 55 (Monotonicity and channel specializations of the two-channel update)**.** *Assume* $U_{\mathrm{int}}, U_{\mathrm{rea}} : V^{L^{\mathrm{ref}}} \to V^{L^{\mathrm{ref}}}$ *and* $W^{\mathrm{ref}} : V^{L^{\mathrm{ref}}} \to V^{L^{\mathrm{ref}}}$ *are monotone. Define*

$$U(E) := W^{\mathrm{ref}}(E \;\oplus\; \beta \otimes U_{\mathrm{int}}(E) \;\oplus\; \gamma \otimes U_{\mathrm{rea}}(E))\,, \qquad \beta, \gamma \in [0,1].$$

Then U *is monotone. Moreover:*

$$\beta = 0 \Rightarrow U(E) = W^{\mathrm{ref}}(E \oplus \gamma \otimes U_{\mathrm{rea}}(E))\,, \quad \gamma = 0 \Rightarrow U(E) = W^{\mathrm{ref}}(E \oplus \beta \otimes U_{\mathrm{int}}(E))\,, \quad \beta = \gamma = 0 \Rightarrow U(E) = W^{\mathrm{r}}$$

Remark 2267. *This proposition abstracts a common cognitive-architectural trope: two different update "voices" contribute to the reflective state—an intuition-like channel and a reason-like channel (Hyperseed-Concept* **??** *and Hyperseed-Concept 150). The weights* $\beta, \gamma \in [0,1]$ *control how strongly each channel's suggested update is injected, and* W^{ref} *then enforces the reflective workspace closure. The overall form is compatible with the cognitive-synergy view that multiple inference processes, each partial, can be combined via a common representational substrate [19]. In particular, the formula separates (i) proposal generation (*$U_{\mathrm{int}}(E)$ *and* $U_{\mathrm{rea}}(E)$*) from (ii) proposal integration (the* $\oplus$*-aggregation, modulated by* $\otimes$*-weights) and (iii) admissibility/consistency restoration (the* W^{ref} *step), mirroring a "suggest–merge–stabilize" pattern.*

Remark 2268. *The specialization identities are not deep, but they are operationally important: they tell us that setting* β *or* γ *to* 0 *literally disables the corresponding channel, and setting both to* 0 *reduces the update to pure reflective closure. This makes it easy to compare the full two-channel dynamics against single-channel baselines, a useful move in both theoretical analysis and mechanized verification. In many applications one additionally compares intermediate regimes (e.g.* $\beta \ll \gamma$*) to express dominance of one channel over the other, but the boundary cases already provide clean "knobs" for ablation-style arguments about causal contribution.*

Remark 2269 (Intuition before the proof)**.** *Monotonicity again comes from compositionality:* $E \mapsto U_{\mathrm{int}}(E)$ *and* $E \mapsto U_{\mathrm{rea}}(E)$ *are monotone by assumption; scaling by* β, γ *via* $\otimes$ *is monotone; joining via* $\oplus$ *is monotone; and finally applying the monotone closure* W^{ref} *preserves monotonicity. The specializations are simply what remains after substituting* $\beta = 0$ *and/or* $\gamma = 0$*. Conceptually, monotonicity expresses a stability property: strengthening the current reflective state (in the underlying order on* $V^{L^{\mathrm{ref}}}$*) cannot lead to a* weaker *updated state when all components of the update rule are themselves order-preserving.*

Sketch. $\oplus$ and $\otimes$ are monotone in each argument, and scaling by fixed $\beta, \gamma \in [0,1]$ is monotone. Thus the inner expression is monotone in E, and composing with monotone W^{ref} preserves monotonicity. The specialization identities follow by substitution. □

Remark 2270. Proof sketch (high-level strategy). *Prove monotonicity by checking each syntactic constructor in the definition of* U *is order-preserving, and then use the fact that order-preserving maps are closed under composition. Prove the special cases by direct substitution. Equivalently, one can view the inner map*

$$E \longmapsto E \;\oplus\; \beta \otimes U_{\mathrm{int}}(E) \;\oplus\; \gamma \otimes U_{\mathrm{rea}}(E)$$

as a monotone "pre-closure" aggregator, and then appeal to monotonicity of W^{ref} *as a final post-processing step.*

Remark 2271. On the underlying order and operators. *The statement is intentionally agnostic about the concrete instantiation of $V^{L^{\mathrm{ref}}}$ and the order $\leq$ it carries. In typical uses, $V^{L^{\mathrm{ref}}}$ is a product of value-domains (one per reflective label), ordered pointwise, $\oplus$ acts like a join (adding information or taking a supremum), and $\otimes$ acts like a weight-modulated injection that is order-preserving for fixed weights. Under such readings, monotonicity of U formalizes the intuition that adding evidence or constraints to E cannot make the next reflective state "forget" admissible content, except insofar as W^{ref} enforces whatever closure conditions the model requires.*

Remark 2272. Why monotonicity matters for iteration. *In later fixed-point or convergence arguments, one often studies the iterates $E_{n+1} = U(E_n)$. When U is monotone (and when the ambient order supports the needed completeness assumptions), standard lattice-theoretic tools become available for reasoning about least/greatest fixed points and about the effect of channel ablations. The present proposition isolates the monotonicity check so that such arguments can treat the two-channel update as a single well-behaved operator.*

39 Helper theorems for Section 18: evaluative fields, resonance, and ethics

Remark 2273. *This section collects small but structurally telling facts about the evaluative interface used in Section 18. They are "helper" results in the sense that none of them introduces new primitives; rather, they clarify how the primitives behave when combined. In particular, the p-bit representation of evaluation (pairs $(v^+, v^-) \in [0,1]^2$) reflects a paraconsistent attitude: positive and negative support can coexist without forcing collapse to a single scalar [23, 24]. This makes it natural to separate "joy" and "woe" (Hyperseed-Concepts* **??** *and 204) and then reason about orderings, aggregation, and resonance (Hyperseed-Concept 159) in a way that keeps both aspects explicit. For the broader conceptual framing of evaluative fields and resonance within the Hyperseed ontology, see [1].*

39.1 Helper results

1. **(H1) The p-bit preference order is a partial order.**

   Statement. Let $V := [0,1]^2$. Define, for $v = (v^+, v^-)$ and $u = (u^+, u^-)$,

   $$v \preceq_{\mathrm{pref}} u \quad :\iff \quad v^+ \leq u^+ \ \text{ and } \ v^- \geq u^-.$$

   Then $\preceq_{\mathrm{pref}}$ is reflexive, antisymmetric, and transitive (hence a partial order).

   Remark 2274. ***Notation and intuition.*** *The ambient space $V = [0,1]^2$ is the set of p-bit values: a pair (v^+, v^-) where v^+ is "positive evidence / positive valence" and v^- is "negative evidence / negative valence." The order $\preceq_{\mathrm{pref}}$ points "toward the better" in the following sense: u is at least as preferred as v precisely when u has* no less *positive component and* no more *negative component. Equivalently, $\preceq_{\mathrm{pref}}$ is the product order on $[0,1] \times [0,1]$ with the second coordinate reversed. This is the simplest order that takes the two-channel nature of evaluation seriously (Hyperseed-Concept 198).*

   Examples. *If $v = (0.6, 0.2)$ and $u = (0.7, 0.1)$, then $v \preceq_{\mathrm{pref}} u$ because $0.6 \leq 0.7$ and $0.2 \geq 0.1$; thus u is unambiguously preferable. On the other hand, $v' = (0.9, 0.9)$ and $u' = (0.1, 0.1)$ are* incomparable*: v'^+ is larger but v'^- is also larger, so neither dominates the other. This incomparability is not a bug but a formal way of registering ethical and affective conflict (Hyperseed-Concept 97).*

Proof sketch. Reflexivity is immediate since $v^+ \le v^+$ and $v^- \ge v^-$. Antisymmetry: if $v \preceq_{\text{pref}} u$ and $u \preceq_{\text{pref}} v$ then $v^+ = u^+$ and $v^- = u^-$, so $v = u$. Transitivity: $v \preceq_{\text{pref}} u$ and $u \preceq_{\text{pref}} w$ imply $v^+ \le u^+ \le w^+$ and $v^- \ge u^- \ge w^-$, so $v \preceq_{\text{pref}} w$.

Remark 2275. ***Why this matters.*** *Many later constructions (Pareto dominance, feasible choice sets, and monotone aggregators) quietly rely on the fact that we are reasoning inside a partial order rather than an arbitrary binary relation. Reflexivity and transitivity are what let us "chain" improvements without logical friction; antisymmetry is what prevents the order from identifying distinct evaluative states as mutually preferable.*

Proof strategy (high level). *The proof is an instance of a general theme: whenever an order is defined by coordinatewise inequalities, the order axioms reduce to the corresponding axioms of* $\le$ *on* $[0,1]$, *with the only subtlety being that the second coordinate uses* $\ge$ *rather than* $\le$. *Geometrically, the upper contour set of a point* v *is a rectangle extending to the northeast in the* v^+*-direction and to the* southeast *in the* v^-*-direction because "less woe" means moving downward in the usual plane.*

2. **(H2) Joy/woe projections are monotone (and bias is monotone).**

   Statement. Define $\mathrm{J}(v) := v^+$ and $\mathrm{W}(v) := v^-$. If $v \preceq_{\text{pref}} u$ then

   $$\mathrm{J}(v) \le \mathrm{J}(u), \qquad \mathrm{W}(v) \ge \mathrm{W}(u), \qquad \mathrm{J}(v) - \mathrm{W}(v) \le \mathrm{J}(u) - \mathrm{W}(u).$$

   Remark 2276. ***Conceptual role.*** *The maps* $\mathrm{J}, \mathrm{W} : V \to [0,1]$ *are simply coordinate projections, but they encode a philosophical stance: evaluation is not assumed to be reducible to a single scalar; instead, a mind may carry simultaneously a measure of attraction and a measure of aversion (Hyperseed-Concepts* **??**, *204). The final inequality concerns the "bias" or net valence* $\mathrm{J} - \mathrm{W}$*: it says that if* u *is preferable to* v *in the p-bit sense, then* u *is also at least as good by this naive scalarization. This is useful because it lets us connect the two-channel order to scalar decision rules when needed (cf. (H4)–(H5)).*

   Proof sketch. Unpack $v \preceq_{\text{pref}} u$: the first two inequalities are exactly the definition. Subtracting gives $(v^+ - v^-) \le (u^+ - u^-)$.

   Remark 2277. ***Proof sketch commentary.*** *Nothing deep happens here; the point is that the order* $\preceq_{\text{pref}}$ *was* chosen *so that* J *increases and* W *decreases along it. One may view this as a compatibility condition between the order-theoretic structure and the semantic interpretation of the coordinates. Visually, the function* $\mathrm{J} - \mathrm{W}$ *is constant on lines of slope* 1, *and moving in the preferred direction moves you toward larger* $\mathrm{J} - \mathrm{W}$.

3. **(H3) Pareto dominance on value-vectors is a partial order.**

   Statement. For a dimension set D, consider value-vectors in V^D. Define $x \preceq_{\text{Pareto}} y$ iff $x_d \preceq_{\text{pref}} y_d$ for all $d \in D$. Then $\preceq_{\text{Pareto}}$ is a partial order on V^D.

   Remark 2278. ***Notation.*** *An element* $x \in V^D$ *assigns to each ethical/evaluative dimension* $d \in D$ *a p-bit value* $x_d = (x_d^+, x_d^-)$. *The relation* $x \preceq_{\text{Pareto}} y$ *means that* y *is at least as good as* x in every dimension simultaneously. *This is the standard Pareto idea from multiobjective optimization, but instantiated in a two-channel evaluative codomain rather than in* $\mathbb{R}^D$.

   Why useful. *Pareto dominance is the canonical way to express "no tradeoff is being made": if* y *dominates* x, *then* y *improves (or matches)* x *on every criterion, so preferring* x *would be perverse unless some additional constraint intervenes. This underlies the "sanity" of later selection rules that first discard dominated options before applying more delicate criteria such as resonance or resource costs (see (H6)–(H7)).*

Proof sketch. Apply (H1) coordinatewise: reflexivity, antisymmetry, and transitivity lift to the product order.

Remark 2279. ***Proof strategy.*** *This is a general product-construction principle: orders on coordinates induce an order on tuples by pointwise comparison. The reason antisymmetry lifts is that equality in every coordinate forces equality of the whole function $D \to V$. Geometrically, $\preceq_{\text{Pareto}}$ defines a partial order on a high-dimensional orthant-like region, where comparability becomes rarer as $|D|$ grows—a formal reflection of the lived fact that ethical life in many dimensions is often not totally orderable.*

4. **(H4) Pareto dominance implies monotone scalar reward ordering.**

   Statement. Fix weights $\lambda_d \geq 0$ and $\beta \geq 0$. Define a scalar score

$$\text{Score}(a) := \sum_{d \in D} \lambda_d \big(\text{J}(\text{EO}(s, a; d)) - \text{W}(\text{EO}(s, a; d))\big) \ - \ \beta R_O(s, a).$$

   If $\text{EO}(s, a) \preceq_{\text{Pareto}} \text{EO}(s, a')$ *and* $R_O(s, a') \leq R_O(s, a)$, then $\text{Score}(a') \geq \text{Score}(a)$. Moreover, if (i) $\lambda_d > 0$ for all d, and (ii) at least one dimension improves strictly ($\text{EO}(s, a; d) \prec_{\text{pref}} \text{EO}(s, a'; d)$) or $R_O(s, a') < R_O(s, a)$ with $\beta > 0$, then $\text{Score}(a') > \text{Score}(a)$.

   Remark 2280. ***Intuitive reading.*** *The score is a weighted sum of per-dimension net valence* $\text{J} - \text{W}$*, penalized by a resistance/cost term* R_O *(Hyperseed-Concept 158). The claim says: if action a' is at least as good as a in every evaluative dimension (Pareto), and it costs no more resistance, then any nonnegative linear scalarization will not rank a' below a. This is the minimal compatibility one expects between multiobjective improvement and scalar scoring.*

   Connection to later items. *(H4) is the monotonicity lemma that makes (H5) believable: if scalar maximizers could be Pareto-dominated, the scalar would be violating the basic direction of ethical improvement. In decision-theoretic terms, this is a sanity check on scalarization as a tool for rational decision (Hyperseed-Concept 147) as used broadly in AGI-style action selection [19].*

   Proof sketch. From $\text{EO}(s, a) \preceq_{\text{Pareto}} \text{EO}(s, a')$ and (H2), each term $\text{J} - \text{W}$ is $\leq$ the corresponding term for a'. Multiply by $\lambda_d \geq 0$ and sum. Also $-\beta R_O(s, a') \geq -\beta R_O(s, a)$ if $R_O(s, a') \leq R_O(s, a)$. Strictness follows if any positively weighted term increases strictly and/or the resistance term improves with $\beta > 0$.

   Remark 2281. ***Proof sketch commentary.*** *The essential move is order preservation under two operations: (1) multiplying an inequality by a nonnegative weight preserves direction; and (2) summing coordinatewise inequalities preserves direction. The resistance part is the same idea after multiplying by $-\beta$ (which flips nothing because we pre-assume $R_O(s, a') \leq R_O(s, a)$). One may view this as an instance of the general fact that monotone maps between ordered sets compose well:* $\text{J} - \text{W}$ *is monotone w.r.t. $\preceq_{\text{pref}}$ by (H2), and the weighted sum is monotone w.r.t. the usual order on $\mathbb{R}$.*

5. **(H5) Score maximizers are Pareto-undominated (sanity of scalarization).**

   Statement. Assume A is finite and $\lambda_d > 0$ for all d, $\beta \geq 0$. If $a^\star \in \arg\max_{a \in A} \text{Score}(a)$, then $a^\star$ is Pareto-undominated w.r.t. the coordinates $(\text{J}(\text{EO}(s, a; d)), -\text{W}(\text{EO}(s, a; d)))_{d \in D}$ and $-R_O(s, a)$.

Remark 2282. ***What it says in plain language.*** *If you pick an action by maximizing a strictly positively weighted score (and optionally penalizing resistance), you cannot end up with an action that is* strictly worse in every respect *than some other available action. In other words, scalarization does not force you to choose an obviously dominated option, provided you truly care (positively) about each dimension included in the score.*

Why the strict positivity matters. *If some $\lambda_d = 0$, then the scalar score is blind to dimension d, and it may select an action that is dominated only via improvements in that ignored dimension. The hypothesis $\lambda_d > 0$ is precisely what closes this loophole.*

Proof sketch. This is the usual scalarization argument: if some a dominates $a^\star$, then every positively weighted term in Score is $\geq$ and at least one is $>$, contradicting maximality. (This is exactly the logic behind the document's Proposition 24.)

Remark 2283. ***Proof strategy and geometric intuition.*** *The argument is a contradiction proof: assume there is an action a that dominates the maximizer $a^\star$, and then use strict positivity of weights to show that the scalar functional must strictly increase from $a^\star$ to a. Geometrically, the level sets of a positive linear functional are supporting hyperplanes: a maximizer lies on an exposed face of the feasible set, and a point strictly dominated cannot be exposed in that direction.*

6. **(H6) Finite action sets always have Pareto-undominated actions.**

   Statement. If A is finite, then the Pareto-undominated set

   $$A_{\text{Pareto}}(s) := \{a \in A : \nexists a' \in A \text{ with } \mathrm{EO}(s,a) \prec_{\text{Pareto}} \mathrm{EO}(s,a')\}$$

   is nonempty.

   Remark 2284. ***Why this is nontrivial.*** *In infinite sets, one can have infinite ascending chains with no maximal elements. In a finite set, however, strict improvement cannot proceed forever. Thus some actions must sit at the "top" of the strict dominance relation even if many actions are incomparable.* ***Use in the architecture.*** *This guarantees that a choice rule that first filters to Pareto frontiers (as in (H7)) will not face an empty feasible set merely because of the ordering formalism. It is a small theorem with a large practical role: it prevents non-existence pathologies in finite decision modules.*

   Proof sketch. Strict Pareto dominance $\prec_{\text{Pareto}}$ is a strict partial order on a finite set, so it has at least one maximal element; any maximal element is undominated.

   Remark 2285. ***Proof sketch commentary.*** *The proof is a finiteness principle: starting from any action, if it is dominated we move to a dominator; since there are only finitely many actions, this process must terminate, and termination means "undominated." One can also view this as the existence of maximal elements in finite partially ordered sets, a discrete cousin of Zorn-style reasoning but requiring no choice principles.*

7. **(H7) The resonant-feasible rule satisfies key rationality axioms.**

   Statement. Define the resonant-feasible choice set

   $$A_{\text{res}}(s) := \underset{a \in A_{\text{Pareto}}(s)}{\arg\max} \Big(\kappa_O(s,a) - \beta\, R_O(s,a)\Big).$$

   Then:

(a) *Non-explosion:* $A_{\text{res}}(s) \neq \emptyset$ for finite A.

(b) *Dominance respect:* if a is strictly Pareto-dominated then $a \notin A_{\text{res}}(s)$.

(c) *Resource sensitivity:* if $\kappa_O(s,a) = \kappa_O(s,a')$ and $R_O(s,a) < R_O(s,a')$ with $\beta > 0$, then no maximizer can prefer a' over a within the feasible set.

Remark 2286. ***Interpretation.*** *This choice rule has two stages: first, exclude actions that are ethically worse in every relevant dimension (Pareto filtering); second, among the survivors, prefer higher resonance κ_O (Hyperseed-Concept 159) while optionally penalizing resistance R_O (Hyperseed-Concept 158). The rule thereby attempts to encode a "non-fanatical" rationality: it refuses dominated choices, but it does not insist on a single global scalar utility before it has removed the plainly inferior options.*

Why it connects to Section 18. *Section 18 treats resonance as a structurally meaningful summary of multi-dimensional evaluative alignment. The present item shows that using resonance in a decision rule does not violate basic rationality desiderata such as non-emptiness and dominance respect. In the broader cognitive-systems context, this aligns with the methodological stance that decision rules must be order-theoretically well-posed before one asks whether they are psychologically or ethically apt (Hyperseed-Concept 147).*

Proof sketch. (1) By (H6), $A_{\text{Pareto}}(s) \neq \emptyset$, and an argmax over a nonempty finite set is nonempty. (2) The maximization domain explicitly excludes dominated actions. (3) With equal κ_O, the objective differs only by $-\beta R_O$, so smaller resistance yields a larger objective if $\beta > 0$.

Remark 2287. ***Proof sketch commentary.*** *The structure mirrors a common pattern in formal ethics and multiobjective control: establish feasibility/nonemptiness first (a logical precondition for any normative rule), and then verify that the selection criterion respects the ordering constraints it claims to respect. Geometrically, (2) says we restrict attention to the Pareto frontier; (3) says that on a contour of equal resonance, resistance tilts the objective toward lower-cost points.*

8. **(H8) Resonance magnitude bounds (triangle inequality + bounded σ_C).**

Statement. Let $z_d := \sigma_C(\mathrm{EO}(s,a;d)) \in \mathbb{C}$ and $Z := \sum_{d \in D} w_d z_d$ with $w_d \geq 0$. Then

$$\kappa_O(s,a) = |Z| \leq \sum_{d \in D} w_d |z_d|.$$

Moreover, since $\sigma_C(v^+, v^-) = (v^+ - v^-) + i((v^+ + v^-) - 1)$ has real and imaginary parts in $[-1,1]$, one has $|z_d| \leq \sqrt{2}$ for all d, hence

$$\kappa_O(s,a) \leq \sqrt{2} \sum_{d \in D} w_d.$$

(In particular, if $\sum_d w_d = 1$, then $\kappa_O(s,a) \leq \sqrt{2}$.)

Remark 2288. ***Notation and first-time symbols.*** *Here $\sigma_C : V \to \mathbb{C}$ embeds a p-bit value into the complex plane. The complex numbers z_d are the dimension-wise "phasors," and Z is their weighted superposition. The resonance magnitude $\kappa_O(s,a)$ is defined as $|Z|$, the modulus of this complex sum. Intuitively, each evaluative dimension contributes a vector in the plane; resonance is large when these vectors align and small when they cancel (Hyperseed-Concepts*

159, 97). This complex-geometry viewpoint is also central in paraconsistent resonance approaches [24].

Why the bound is useful. *Bounds like $\kappa_O(s,a) \le \sqrt{2}\sum_d w_d$ are not merely aesthetic: they make κ_O scale-controlled, which is essential if κ_O is to be compared across states, combined with other terms (like R_O), or placed into probabilistic/optimization pipelines.*

Proof sketch. The first inequality is the triangle inequality. For the bound on $|z_d|$, note $v^+ - v^- \in [-1,1]$ and $(v^+ + v^-) - 1 \in [-1,1]$, so $|z_d|^2 \le 1^2 + 1^2 = 2$.

Remark 2289. ***Geometric intuition.*** *The triangle inequality is the formal statement that the length of a resultant vector cannot exceed the sum of lengths of the vectors being added. The second part says: the embedding σ_C maps the square $[0,1]^2$ into a bounded region of the complex plane, specifically into the box $[-1,1]\times[-1,1]$. Thus every contribution has bounded magnitude, so the total resonance grows at most linearly with the total weight.*

9. **(H9) Interference expands into pairwise alignment terms.** *Statement.* With $Z = \sum_d w_d z_d$ and

$$I_O(s,a) := |Z|^2 - \sum_{d\in D} |w_d z_d|^2,$$

one has the exact expansion

$$I_O(s,a) = 2\sum_{d<d'} w_d w_{d'}\ \mathrm{Re}\big(z_d\,\overline{z_{d'}}\big).$$

Remark 2290. ***What I_O measures.*** *The quantity $I_O(s,a)$ is the difference between the squared magnitude of the sum and the sum of squared magnitudes of the terms. If there were no cross-terms, these would coincide. Thus I_O isolates the purely relational component: how much the dimensions help or hinder each other when superposed. This is precisely why it is natural to call it "interference": it is made entirely of pairwise overlaps.*

Why this connects. *In Section 18, resonance is meant to capture alignment across ethical dimensions, not merely goodness in each dimension separately. (H9) gives the algebraic lens for that claim: alignment is encoded in the real parts $\mathrm{Re}(z_d\overline{z_{d'}})$, i.e. in the cosines of pairwise phase differences.*

Proof sketch. Expand: $|\sum_d w_d z_d|^2 = \sum_d |w_d z_d|^2 + \sum_{d\ne d'} w_d w_{d'} z_d \overline{z_{d'}}$, then take the real part; the $d \ne d'$ sum collapses to $2\sum_{d<d'}\mathrm{Re}(\cdot)$.

Remark 2291. ***Proof sketch commentary.*** *The key identity is $|Z|^2 = Z\overline{Z}$. Writing $Z = \sum_d w_d z_d$ and distributing the product produces diagonal terms (the $|w_d z_d|^2$ terms) and off-diagonal terms (the interference terms). Pairing d, d' with d', d is what produces the factor of 2 and forces the final expression to be real, since each pair contributes complex conjugates whose sum is $2\,\mathrm{Re}(\cdot)$.*

Remark 2292. ***Useful normal form.*** *Writing each complex contribution in polar form $z_d = r_d e^{i\phi_d}$ (with $r_d \ge 0$ and $\phi_d \in \mathbb{R}$) makes the structure of (H9) completely explicit:*

$$\mathrm{Re}\big(z_d\overline{z_{d'}}\big) = \mathrm{Re}\big(r_d e^{i\phi_d}\cdot r_{d'} e^{-i\phi_{d'}}\big) = r_d r_{d'}\cos(\phi_d - \phi_{d'}).$$

Hence I_O is a weighted sum of pairwise cosine similarities:

$$I_O(s,a) = 2\sum_{d<d'} w_d w_{d'}\, r_d r_{d'}\cos(\phi_d - \phi_{d'}).$$

This makes clear (i) why I_O depends only on relative *phases $\phi_d - \phi_{d'}$ (global phase cancels), and (ii) why I_O can be positive or negative even when all magnitudes r_d are large: the sign is controlled by the geometry of the phase differences.*

On weights and symmetry. *The algebraic expansion itself does not require $w_d \geq 0$, only that the weights be real so that $|w_d z_d|^2 = w_d^2 |z_d|^2$ behaves as expected; however, in evaluative interpretations the natural regime is typically $w_d \geq 0$ (importance weights). In that regime, the sign of each pairwise contribution to I_O is driven directly by $\mathrm{Re}(z_d \overline{z_{d'}})$, so "alignment" and "misalignment" have a clean monotone relationship with constructive versus destructive interference.*

10. **(H10) Coherence/cancellation criteria via phase geometry.**

Statement. If for all pairs $d \neq d'$ the phase alignment is nonnegative, i.e. $\mathrm{Re}(z_d \overline{z_{d'}}) \geq 0$, then $I_O(s, a) \geq 0$ (constructive interference). If many pairs have $\mathrm{Re}(z_d \overline{z_{d'}}) < 0$, then $I_O(s, a)$ can be negative (cancellation).

A sufficient condition for maximal coherence is: there exists $\theta \in \mathbb{R}$ with $z_d = e^{i\theta} r_d$ for all d and all $r_d \geq 0$. Then

$$\kappa_O(s, a) = \sum_d w_d r_d, \qquad I_O(s, a) = \Big(\sum_d w_d r_d\Big)^2 - \sum_d w_d^2 r_d^2 \geq 0.$$

Remark 2293. ***Plain-English meaning.*** *When all the complex contributions z_d point roughly in the same direction (pairwise angles at most 90°), then their sum is amplified: the cross-terms in (H9) are nonnegative, so interference is constructive. When many pairs point in opposing directions (angles greater than 90°), cross-terms can be negative and the sum can shrink: cancellation.*

A simple special case. *If all z_d share a common phase θ, then every contribution is a nonnegative scalar multiple of the same unit complex number $e^{i\theta}$. In that case, resonance reduces to an ordinary weighted sum $\sum_d w_d r_d$, and interference is automatically nonnegative. This shows that the complex formalism strictly generalizes classical additive aggregation: additivity is recovered when the system is perfectly coherent.*

Proof sketch. First part: combine (H9) with the sign assumptions. Second part: under common phase, $Z = e^{i\theta} \sum_d w_d r_d$, so $|Z| = \sum_d w_d r_d$, and I_O is just the cross-term sum (hence nonnegative).

Remark 2294. ***Geometric intuition.*** *The condition $\mathrm{Re}(z_d \overline{z_{d'}}) \geq 0$ is equivalent to saying that the angle between z_d and $z_{d'}$ is in $[-\pi/2, \pi/2]$ modulo 2π. Thus (H10) is really a theorem about polygonal addition of vectors: if every pair of edges has acute or right angle, the polygon tends to "open" rather than fold back on itself. In ethical terms, this is a formal analogue of the idea that virtues can mutually reinforce when they are not pulling in contradictory directions.*

Remark 2295. ***Sufficiency versus necessity.*** *The pairwise nonnegativity condition $\mathrm{Re}(z_d \overline{z_{d'}}) \geq 0$ for all $d \neq d'$ is a strong* sufficient *criterion for $I_O \geq 0$, but it is not necessary: I_O is a* sum *of pairwise contributions, so some negative pairs can be outweighed by sufficiently many positive pairs (or by larger magnitudes/weights on aligned pairs). Put differently, constructive global interference is compatible with local pockets of cancellation, as long as the total weighted cosine-similarity mass is nonnegative.*

Interference as a deviation-from-additivity diagnostic. *Under common phase,* $I_O(s,a) = \big(\sum_d w_d r_d\big)^2 - \sum_d w_d^2 r_d^2$ *is exactly the difference between the squared* ℓ^1*-type aggregate and the squared* ℓ^2*-type aggregate (with weights). This emphasizes that* I_O *can be read as a quantitative measure of "how non-orthogonal" the set of contributions is: perfect coherence pushes the system toward* ℓ^1*-like accumulation, while strong phase dispersion moves it toward cancellation and smaller resultant magnitude.*

11. **(H11) Compassion aggregation is monotone in its inputs.**

Statement. Let compassion be defined by an aggregation of the form

$$\mathrm{EO}(s,a;\mathrm{comp}) := \bigoplus_{j\neq\mathrm{self}} \Big(\mathrm{EO}_{\to j}(s,a;\mathrm{joy}) \;\otimes\; \neg\mathrm{EO}_{\to j}(s,a;\mathrm{woe}) \Big).$$

Assume $\oplus$ and $\otimes$ are monotone in each argument (as in the quantale core). If for every $j \neq \mathrm{self}$,

$$\mathrm{EO}_{\to j}(s,a;\mathrm{joy}) \le \mathrm{EO}'_{\to j}(s,a;\mathrm{joy}) \quad \text{and} \quad \neg\mathrm{EO}_{\to j}(s,a;\mathrm{woe}) \le \neg\mathrm{EO}'_{\to j}(s,a;\mathrm{woe}),$$

then $\mathrm{EO}(s,a;\mathrm{comp}) \le \mathrm{EO}'(s,a;\mathrm{comp})$.

Remark 2296. ***Conceptual role (Compassion).*** *Compassion (Hyperseed-Concept 81) is being modeled here as an aggregation across other agents* $j \neq \mathrm{self}$ *of a conjunction-like term: "joy for* j*" combined with "not woe for* j*." The logical moral is: if you increase each input in a monotone way, the aggregate cannot decrease. This is a basic coherence requirement for any ethical aggregator: improvements in the welfare-related ingredients should not perversely reduce the compassion score.*

Connection to prosocial efficiency. *This monotonicity property is one of the algebraic preconditions for turning compassion-like terms into decision criteria without logical traps, a theme that aligns with arguments that prosocial structure can increase group efficiency in multi-agent settings [6].*

Proof sketch. Monotonicity of $\otimes$ gives monotonicity for each j-term; monotonicity of $\oplus$ (and of joins over index sets) lifts this to the aggregate.

Remark 2297. ***Where the assumption on*** $\neg$ ***enters.*** *The statement is phrased in terms of the inputs* $\mathrm{EO}_{\to j}(s,a;\mathrm{joy})$ *and* $\neg\mathrm{EO}_{\to j}(s,a;\mathrm{woe})$*, so it only needs monotonicity of* $\otimes$ *and* $\oplus$ *in their arguments. If one prefers to phrase conditions directly in terms of* $\mathrm{EO}_{\to j}(s,a;\mathrm{woe})$*, then a separate property of the negation-like operator becomes relevant: in many ordered settings,* $\neg$ *is antitone (order-reversing), so a decrease in woe (*$\mathrm{EO}_{\to j}(\mathrm{woe}) \ge \mathrm{EO}'_{\to j}(\mathrm{woe})$*) corresponds to an increase in* $\neg\mathrm{EO}_{\to j}(\mathrm{woe}) \le \neg\mathrm{EO}'_{\to j}(\mathrm{woe})$*. The present formulation avoids baking in any particular order-theoretic behavior of* $\neg$*, while still capturing the intended monotone comparative statics.*

Index-set aggregation. *The step "lift to the aggregate" uses the standard fact that if each term in a family increases (pointwise in* j*), then any join/aggregate operator* $\bigoplus$ *that is monotone with respect to that pointwise order must also increase. Concretely, defining* $t_j := \mathrm{EO}_{\to j}(\mathrm{joy}) \otimes \neg\mathrm{EO}_{\to j}(\mathrm{woe})$ *and similarly* t'_j*, one has* $t_j \le t'_j$ *for all* j*, hence* $\bigoplus_j t_j \le \bigoplus_j t'_j$*.*

Optional concrete p-bit case. If $\oplus$ is componentwise max, $\otimes$ is componentwise multiplication, and $\neg(p,q) = (q,p)$, then each j-term equals

$$(\, \mathrm{J}_j^+ \cdot \mathrm{W}_j^- \;,\; \mathrm{J}_j^- \cdot \mathrm{W}_j^+ \,),$$

and the outer $\oplus$ is max over j. In particular, the aggregation can be read as taking, separately in each coordinate, the worst-case (largest) composite contribution across j, which makes the dependence on each input coordinate explicit and reduces the abstract order-theoretic claim to familiar scalar monotonicity facts.

Remark 2298. ***Proof strategy and intuition.*** *The proof is a textbook application of monotonicity preservation under composition: each inner term is a monotone function of its inputs (because $\otimes$ is monotone), and then the outer aggregation is monotone because $\oplus$ (and hence finite or indexed joins built from it) is monotone. In the concrete p-bit case, the max/multiply structure makes this especially transparent: increasing inputs can only increase products and can only increase maxima. One can also view the j-term as a coordinatewise bilinear map on nonnegative reals (or on $[0,1]$ if probabilities are intended), so the only potential failure modes for monotonicity would come from allowing negative values; the intended p-bit semantics rules these out, leaving plain isotonicity.*

Why the quantale hypothesis appears. *The reliance on monotone $\oplus, \otimes$ is exactly the kind of "weakness is all you need" condition one encounters in quantale-based semantics: monotonicity is the minimal structural demand for stable inference and aggregation [3] (Hyperseed-Concept 143). In other words, the result is deliberately insensitive to whether $\oplus$ is idempotent, whether $\otimes$ distributes over arbitrary joins, or whether one has a residuation operator: those stronger axioms are useful for richer algebraic manipulations, but the present claim only needs that the order is respected by the constructors used to build the evaluation functional.*

Interpretive note (why the swap $\neg(p,q) = (q,p)$ matters). *In this p-bit instance, $\neg$ merely exchanges the roles of the two coordinates, so the two products $\mathrm{J}_j^+\mathrm{W}_j^-$ and $\mathrm{J}_j^-\mathrm{W}_j^+$ can be read as the two cross-couplings induced by pairing "positive" evidence on one side with "negative" evidence on the other. The monotonicity claim then says that strengthening any of these evidence streams (in the intended direction) cannot reduce the corresponding aggregated cross-coupled score.*

12. **(H12) Explicit goals can be treated as implicit goals when enforced by policy.**

    Statement. Let an explicit goal be a constraint $G \subseteq \Omega$ on outcomes/trajectories. Suppose a policy π is implemented with a hard constraint that assigns zero probability to $\Omega \setminus G$. Then the induced action-selection regularity of π is consistent with treating G as an implicit goal. Equivalently, the support of the induced trajectory distribution is contained in G, so every realized history witnesses the same feasibility regularity regardless of whether G is internally represented as a goal object.

    Conversely, if an implicit goal regularity is representable by some set $G \subseteq \Omega$ in O's language (e.g. as a definable predicate on histories), then G can be promoted to an explicit goal without changing behavior. Here "representable" can be read in the weakest sense needed for the surrounding framework (e.g. first-order definability, programmatic recognizability, or measurability relative to the trajectory σ-algebra), since the behavioral content is only that the same subset of trajectories is selected/ruled out.

    Remark 2299. ***Intuition (explicit vs. implicit).*** *An explicit goal is a named constraint, something the system can in principle cite or reason about (Hyperseed-Concept* **??***). An* *implicit goal is visible only through regularities in action selection (Hyperseed-Concept* **??***). This item states a kind of definitional equivalence: if behavior is* hard-constrained *to respect G, then regardless of whether the system says "I have goal G," it behaves as though G were a goal; and if an implicit regularity is definable, it can be reified as an explicit constraint. In the language of constraint satisfaction, the first direction says that a feasibility region enforced at execution time functions indistinguishably from a goal-set, while the second direction says that any stable, nameable regularity can be moved from the level of emergent behavior to the level of specification.*

Why this is useful. *For formal analysis, it allows one to swap between a behavioral and a declarative representation of goals depending on which is technically convenient. When proving theorems about behavior, implicit goals may be simpler; when specifying systems or verifying constraints, explicit goals may be clearer. This is particularly helpful when discussing alignment or ethics: many safety properties are naturally stated as explicit constraints on trajectories, but may in practice be implemented as architectural or training-induced regularities; (H12) licenses treating these two presentations as equivalent provided the regularity is hard (support-level) rather than merely soft (preference-level).*

Boundary case (hard vs. soft enforcement). *The hypothesis uses "zero probability" precisely to rule out the ambiguous case where violations are merely unlikely. If $\pi(\Omega \setminus G) > 0$, then G is at best a soft preference, and the corresponding implicit goal is only approximate; the present helper result intentionally addresses the crisp case needed for clean logical substitution in later arguments.*

Proof sketch. The first direction is by definition: hard feasibility constraints generate stable behavioral regularities. The second direction is reification: if a behavioral regularity is definable as a set of allowed trajectories, encode it as G. Concretely, for the first direction one takes the induced distribution $\mathbb{P}_\pi$ on Ω and observes $\mathbb{P}_\pi(G) = 1$, so every conditional action distribution along realized histories is computed under the invariant that the continuation lies in G. For the second direction, one defines $G := \{\omega \in \Omega : \omega \text{ satisfies the implicit regularity}\}$ and notes that restricting π to G does not alter any action choices on histories that already occur under π.

Remark 2300. ***Proof sketch commentary.*** *The first direction is essentially semantic: "probability zero" is a strong form of impossibility inside the model, hence a robust invariance of behavior. The second direction is syntactic/representational: definability in O's language allows the system to name the regularity. Philosophically, this echoes a Peircean movement from habit (regularity) to symbol (explicit representation) [14], while keeping the formal content purely set-theoretic. One can also read the two directions as a correspondence between* support constraints *(properties of the realized set of trajectories) and* specification constraints *(properties stated in the observer's or agent's representational language): the helper lemma is a bridge allowing later arguments to move freely between these levels without changing the predicted behavior.*

13. **(H13) Categorical-imperative test is monotone in thresholds; evil tends to fail it (when compassion is included).**

   Statement (monotonicity). Fix ethical dimensions D_{eth}. A maxim m passes the CI-test with thresholds $(\tau_d, \eta_d)_{d \in D_{\text{eth}}}$ if

$$\forall d \in D_{\text{eth}} : \quad \mathrm{J}(E_{\text{univ}}(m; d)) \geq \tau_d \;\; \text{and} \;\; \mathrm{W}(E_{\text{univ}}(m; d)) \leq \eta_d.$$

If m passes for (τ_d, η_d), then it also passes for any (τ'_d, η'_d) with $\tau'_d \le \tau_d$ and $\eta'_d \ge \eta_d$ for all d. This is just upward-closure of the set of passing maxims under relaxation of requirements: lowering the required joy or raising the allowed woe cannot invalidate inequalities that already hold.

Remark 2301. ***CI-test as a thresholded universalization check.*** *The CI-test (Hyperseed-Concept 71) is formulated here as a pair of threshold requirements applied to the universalized evaluation* $E_{\text{univ}}(m; d)$*: sufficient joy and limited woe in each ethical dimension. Monotonicity in thresholds is a basic sanity property: if a maxim passes a stringent test, it should pass any more permissive test. Formally, for each* d *one is intersecting two half-spaces in the* (J, W)*-plane; relaxing* (τ_d, η_d) *enlarges the feasible region, so membership is preserved.*

Connection to order structure. *This is another instance where the direction of inequalities is doing conceptual work: increasing the required joy* (τ_d) *makes the test harder, whereas increasing the tolerated woe* (η_d) *makes it easier. This matches the intended polarity of* J *and* W *as order-preserving summaries:* J *is to be pushed up,* W *is to be pushed down, so the CI-test is monotone in the natural product order on the space of threshold pairs.*

Proof sketch. Immediate from the direction of the inequalities. Expanding the quantifiers: for each fixed d, from $\mathrm{J}(E_{\text{univ}}(m; d)) \ge \tau_d$ and $\tau'_d \le \tau_d$ we get $\mathrm{J}(E_{\text{univ}}(m; d)) \ge \tau'_d$; similarly, from $\mathrm{W}(E_{\text{univ}}(m; d)) \le \eta_d$ and $\eta'_d \ge \eta_d$ we get $\mathrm{W}(E_{\text{univ}}(m; d)) \le \eta'_d$. Since this holds for all $d \in D_{\text{eth}}$, the conjunction defining CI-passage is preserved.

Statement (evil ⇒ likely CI-failure under compassion). Assume comp $\in D_{\text{eth}}$ and its CI-threshold η_{comp} is meaningfully restrictive. If a policy π is "evil" in the sense that under universalized/repeated application it yields persistently high W in compassion-related dimensions, then the maxim corresponding to π fails the CI-test for comp. More explicitly, if universalization drives $\mathrm{W}(E_{\text{univ}}(m; \text{comp}))$ above η_{comp} (either deterministically or in a robust expectation/liminf sense, depending on how E_{univ} is instantiated), then the CI-inequality for the compassion coordinate is violated, hence the conjunction over $d \in D_{\text{eth}}$ cannot hold.

Remark 2302. ***Interpretation (Evil and Compassion).*** *"Evil" (Hyperseed-Concept* **??***) is not being treated as a metaphysical primitive here, but as a pattern of outcomes under universalization: persistently high woe in compassion-relevant dimensions (Hyperseed-Concept 81). Given this, failure of a compassion woe-threshold is almost tautological. The value of stating it explicitly is architectural: it clarifies that, once compassion is included among ethical dimensions, a policy that systematically harms others is filtered out by the CI-test rather than being left to downstream heuristics.*

A further interpretive point is that this usage of "evil" is operational and comparative: the label is attached to policies or action-rules insofar as *they induce high compassion-woe when scaled up (via* E_{univ}*) or iterated, not insofar as they have any special ontological status. This keeps the concept compatible with pluralistic ethical modeling: one can disagree about metaphysics while still agreeing that certain universalized patterns reliably generate severe compassion-relevant harm.*

Relation to broader arguments. *This sits naturally beside arguments that prosociality tends to yield more stable and efficient multi-agent outcomes [6]: if universalized harm is self-defeating at the group level, then the CI-test will tend to reject it when compassion is operationalized. One can also read this as a minimal interface between qualitative ethical vocabulary and quantitative screening: the CI-test does not need a rich theory of vice; it* *only needs a way to detect that a candidate policy, when generalized, drives* W *beyond the compassion tolerance encoded by* η_{comp}*.*

Proof sketch. "Evil" provides (by assumption/definition) that universalized/repeated application produces large $\mathrm{W}(E_{\text{univ}}(m_\pi; \text{comp}))$, hence it exceeds η_{comp}, violating CI. Equivalently, if the defining property of the "evil" pattern is that the compassion-relevant woe remains high *under the same transformation* E_{univ} used by the CI-test, then the CI constraint is breached by construction; the proof is essentially a type-check that the quantities compared are identical.

Remark 2303. ***Proof sketch commentary.*** *This is a direct implication: the premise is already phrased in terms of the same quantitative observable (*W*) that the CI-test constrains. The only subtlety is extra-mathematical: what counts as "meaningfully restrictive" depends on how the thresholds are set, which is a modeling choice rather than a theorem. Another subtlety (still extra-mathematical) is the scope of the universalization operator* E_{univ}*: different formalizations of "universal adoption" can vary in how they model feedback, retaliation, and long-run equilibrium. The remark here is therefore best read as a robustness claim about the* architecture*: whatever specific* E_{univ} *is chosen in Section 18, if "evil" is defined as producing persistently high compassion-woe under that same operator, then CI-rejection follows immediately.*

14. **(H14) Humility-calibrated confidence simplifies and is monotone.**

Statement. Let $C_O(\Pi, c) = (C_O^+(\Pi, c), C_O^-(\Pi, c)) \in [0,1]^2$ and $h \in [0,1]$. Define

$$\mathrm{Conf}_h(\Pi, c) := C_O^+(\Pi, c) \cdot (1-h) \; + \; C_O^+(\Pi, c) \cdot h \cdot (1 - C_O^-(\Pi, c)).$$

Then the expression simplifies to

$$\mathrm{Conf}_h(\Pi, c) = C_O^+(\Pi, c) \cdot \big(1 - h\, C_O^-(\Pi, c)\big).$$

In particular, $0 \le \mathrm{Conf}_h(\Pi, c) \le C_O^+(\Pi, c) \le 1$, and for fixed C_O^+, Conf_h is weakly decreasing in h and in C_O^-. It is also useful to note the boundary cases encoded by the formula: when $h = 0$, one recovers the undamped confidence $\mathrm{Conf}_0 = C_O^+$; when $h = 1$, one obtains the maximally humility-weighted confidence $\mathrm{Conf}_1 = C_O^+(1 - C_O^-)$, i.e. the positive support is discounted directly by the negative support. Likewise, if $C_O^- = 0$ (no negative evidence) then $\mathrm{Conf}_h = C_O^+$ for all h, whereas if $C_O^- = 1$ (maximal negative evidence) then $\mathrm{Conf}_h = C_O^+(1-h)$, so humility alone sets how strongly confidence is reduced.

Remark 2304. ***Intuition (humility as a regulator).*** *The parameter* $h \in [0,1]$ *plays the role of humility: how strongly negative evidence* C_O^- *should damp positive confidence* C_O^+*. This is an algebraic counterpart of "weakness" principles: confidence ought not be a blunt affirmation but a moderated stance responsive to counterevidence (Hyperseed-Concepts 202,* **??***) [2, 3]. The simplified form* $C_O^+(1 - hC_O^-)$ *makes the damping effect explicit: the discount factor lies in* $[0,1]$ *and decreases as either humility* h *or negative support* C_O^- *increases.*

One can also interpret the expression as a two-stage gating: C_O^+ *sets the* ceiling *on how confident the system is permitted to be (given its positive support), while* $(1 - hC_O^-)$ *is a multiplicative* veto strength *determined jointly by humility and counterevidence. Because the damping is multiplicative, the rule preserves scale: if positive support doubles (within* $[0,1]$*), the humility-adjusted confidence doubles as well, but always subject to the same fractional discount induced by* h *and* C_O^-*.*

Why it is useful. *Monotonicity properties like these are crucial when confidence is used inside control loops: they guarantee that increasing humility cannot paradoxically increase confidence, and that increasing negative evidence cannot increase confidence. Such guarantees are small but stabilizing. In particular, if downstream decision rules threshold on* Conf_h *(e.g. to authorize risky actions), then these monotonicities ensure that adding counterevidence or increasing humility cannot create a new authorization that was previously absent, which is the directionality one typically wants for safety-leaning calibration.*

Proof sketch. Algebra:

$$C^+(1-h) + C^+h(1-C^-) = C^+\big((1-h) + h - hC^-\big) = C^+(1-hC^-).$$

Bounds follow since $hC^- \in [0,1]$. Monotonicity follows because $1 - hC^-$ decreases as either h or C^- increases. More explicitly, holding C^+ fixed, $\partial\mathrm{Conf}_h/\partial h = -C^+C^- \le 0$ and $\partial\mathrm{Conf}_h/\partial C^- = -C^+h \le 0$, so the weak monotonicity statements follow immediately (where the "weak" qualifier covers the cases $C^+ = 0$, $C^- = 0$, or $h = 0$ in which the derivative vanishes).

Remark 2305. ***Proof sketch commentary.*** *The simplification is a distributive-law exercise, but it reveals the core structure: a product of a "pro" term and a "discount" term. The inequalities then become immediate because both factors live in* $[0,1]$*. Visually, for fixed* C^+*,* Conf_h *is an affine function of* h *with nonpositive slope* $-C^+C^-$*, so it decreases linearly as humility increases. Additionally, the factorization makes it clear that* Conf_h *is jointly continuous (indeed, bilinear) in* (h, C^-) *for fixed* C^+*, which is often convenient when* Conf_h *is embedded into optimization or learning procedures: small changes in humility or assessed negative support yield proportionally small changes in the resulting confidence.*

40 Helper theorems for life, energy, and genenergy

Remark 2306. *This section collects "small" facts that are logically routine but conceptually clarifying: one often wants to rewrite a homeostasis condition, to weaken parameters without redoing an argument, or to turn a capacity definition into a usable inequality. The role of these helper results is analogous to what Russell would call the housekeeping of a theory: we separate the essential conceptual commitments (in the main definitions and propositions) from the algebraic consequences that are repeatedly invoked. In practice, these lemmas function as a shared toolbox: once proved, they can be cited whenever a later argument needs (i) a change of quantifiers (e.g. from "for all times" to "for a finite horizon" and then to a limit), (ii) a standard probabilistic relaxation (e.g. using a union bound or Markov-type inequality), or (iii) a monotonicity claim (e.g. showing that a property becomes easier to satisfy when viability thresholds are relaxed or when available budget is increased). The intent is not to introduce new conceptual primitives here, but to make repeated manipulations explicit, so that downstream sections can focus on the substantive modeling choices rather than on routine rearrangements.*

Remark 2307. *A small piece of notation that recurs throughout:*

- $\mathbb{P}_\pi(\cdot)$ *is probability under the trajectory distribution induced by policy* π *(Hyperseed-Concept 139). Concretely,* π *together with the environment dynamics determines a measure over full histories* $(s_0, g_0, s_1, g_1, \dots)$ *(and, where relevant, observations and actions), and* $\mathbb{P}_\pi$ *denotes probability computed with respect to that induced measure.*

- $\mathbb{E}_\pi[\cdot]$ *is expectation under the same distribution. Unless otherwise stated, expectations are taken over all sources of randomness in the closed-loop system (environment stochasticity and any stochasticity internal to* π*), and the subscript* π *is retained to make explicit which controller induces the distribution.*

- $\mathbb{I}[\cdot]$ *is an indicator variable:* $\mathbb{I}[\textit{statement}] = 1$ *if the statement is true and* 0 *otherwise. This device is useful for converting event probabilities into expectations (e.g.* $\mathbb{P}_\pi(A) = \mathbb{E}_\pi[\mathbb{I}[A]]$*), and for expressing quantities like time-in-viability as sums of indicators.*

- (s_t, g_t) *denotes the state and budget (or resource) variables at time* t*. Here* t *is typically discrete and ranges over* $\{0, 1, 2, \dots\}$*, but the helper statements are generally organized so that finite-horizon and infinite-horizon variants can be related by standard limiting arguments. The pairing emphasizes that "life" conditions are usually constraints on joint configurations of external state and internal resource.*

- V_O *is the viability set: the set of acceptable pairs* (s, g) *(Hyperseed-Concept 101,* **??***). Depending on the modeling choice,* V_O *may encode hard safety constraints, soft operational envelopes, or time-dependent requirements (e.g. seasonal or mission-phase constraints); when such dependence matters, it will be made explicit, but the default notation suppresses it for readability.*

The intuition is that homeostasis means "staying in the acceptable region" with high probability, while genenergy formalizes how much controllable influence an agent has on future observations under resource constraints [1]. Put differently, the homeostasis conditions are typically statements about events of the form $\{(s_t, g_t) \in V_O \text{ for all } t \leq T\}$ *or about the frequency with which* (s_t, g_t) *remains viable, whereas genenergy-type quantities are typically statements about how the policy can shape distributions of future trajectories (often measured through information-theoretic or capacity-like expressions) subject to the feasible evolution of* g_t*. Many of the "small" results below are therefore bridges between these viewpoints: they translate viability events into expectation bounds, relate worst-case and average-case variants, and make explicit how resource constraints enter the probability calculus.*

Remark 2308. *It is worth emphasizing the scope of the helper results in this section. They are intended to be policy-agnostic in the sense that they apply uniformly to any fixed* π *for which the induced trajectory distribution is well-defined, and they are typically insensitive to the particular parametrization of the environment beyond basic measurability and boundedness assumptions stated locally when needed. When a lemma introduces a technical condition (for example, bounded indicators, nonnegativity of a resource variable, or a nesting relation between viability sets), the point is usually to enable a standard inequality step that would otherwise be repeated verbatim across proofs. In particular, several statements will exploit elementary facts such as: if* $V_{O1} \subseteq V_{O2}$ *then any homeostasis guarantee relative to* V_{O1} *implies the corresponding guarantee relative to* V_{O2}*; or if a budget lower bound is relaxed then feasibility (and thus achievable genenergy) is monotone nondecreasing. These are conceptually straightforward, but writing them once avoids ambiguity about which direction the implication runs.*

40.1 Homeostasis (Defs. 270–271): event rewrites and monotonicity

Remark 2309. *Homeostasis is, in this framework, a probabilistic notion of viability: rather than demanding that* every *trajectory remain viable, one demands that* most *trajectories do, up to a tolerated failure probability* δ*. Conceptually this mirrors the way real organisms (or robust engineered systems) tolerate rare shocks while maintaining an overall regime of functioning (Hyperseed-Concept 101, 109).*

Helper Lemma 40.1 (Complement form of homeostasis)**.** *If a policy π is homeostatic for (V_O, T, δ), i.e.*

$$\mathbb{P}_\pi\Big((s_t, g_t) \in V_O \text{ for all } t = 0, \dots, T\Big) \ \geq\ 1 - \delta,$$

then equivalently

$$\mathbb{P}_\pi\Big(\exists t \in \{0, \dots, T\} \text{ with } (s_t, g_t) \notin V_O\Big) \ \leq\ \delta.$$

Remark 2310. *Intuitively, the lemma says: "staying viable with probability at least $1-\delta$" is the same as "failing viability at some time with probability at most δ." This is not deep mathematics; it is the basic logical duality between a safety property and its violation event. Its utility is that many bounds (union bounds, Markov bounds, and other risk controls) are more naturally phrased in terms of the probability of* violation*, rather than the probability of* success*.*

Sketch. Take complements: $\{\forall t \leq T : (s_t, g_t) \in V_O\}^c = \{\exists t \leq T : (s_t, g_t) \notin V_O\}$, and use $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. □

Remark 2311. Proof sketch (high level). *The strategy is to recognize that "for all t" and "there exists t" are De Morgan duals. The probability identity $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ then transfers the $1 - \delta$ bound to a δ bound.*

Why the key step works. *The complement of "all times are viable" is precisely "some time is not viable." Nothing about Markov decision processes is used here; only elementary logic and probability.*

Visual intuition. *Imagine a bundle of trajectories drawn as curves in a time–state diagram. The "good" bundle consists of curves staying in the shaded region V_O for the whole time interval; the complement bundle consists of curves that cross the boundary at least once.*

Remark 2312. Equivalent union form. *The violation event can be written explicitly as a finite union:*

$$\Big\{\exists t \in \{0, \dots, T\} : (s_t, g_t) \notin V_O\Big\} = \bigcup_{t=0}^{T} \Big\{(s_t, g_t) \notin V_O\Big\}.$$

This rewriting is often the point at which one can apply standard probability tools. For example, without assuming any independence across time, the union bound yields

$$\mathbb{P}_\pi\Big(\exists t \leq T : (s_t, g_t) \notin V_O\Big) \leq \sum_{t=0}^{T} \mathbb{P}_\pi\big((s_t, g_t) \notin V_O\big),$$

so any way of controlling the per-time violation probabilities automatically controls the global homeostasis risk, albeit sometimes conservatively.

Helper Corollary 40.2 (Per-time viability and expected violations)**.** *Under the same hypothesis, for every fixed $t \in \{0, \dots, T\}$,*

$$\mathbb{P}_\pi\big((s_t, g_t) \in V_O\big) \ \geq\ 1 - \delta, \quad \text{so} \quad \mathbb{P}_\pi\big((s_t, g_t) \notin V_O\big) \ \leq\ \delta.$$

Consequently the expected number of violations satisfies

$$\mathbb{E}_\pi\Big[\sum_{t=0}^{T} \mathbb{I}[(s_t, g_t) \notin V_O]\Big] \ \leq\ (T+1)\delta,$$

and by Markov's inequality, $\mathbb{P}_\pi(\#\text{violations} \geq k) \leq (T+1)\delta/k$ for any $k \geq 1$.

Remark 2313. *This corollary extracts two kinds of "local" consequences from a "global" homeostasis guarantee. First, if you are safe for* all *times with high probability, then you are certainly safe at any* particular *time with at least the same probability. Second, once we rewrite violations using indicators, linearity of expectation turns a logical statement into an additive quantitative bound. The Markov inequality bound then says that frequent violations are unlikely when the expected number is small.*

Remark 2314. *In applications, this is often how one turns a desired high-level safety requirement into a tractable performance objective: one controls (or estimates) the expected count of boundary crossings. This is reminiscent of how one manages risk in stochastic control: one may not be able to guarantee perfect safety, but one can constrain the probability (or expected frequency) of failure events (Hyperseed-Concept 87, 195).*

Sketch. The event "viable for all times" is a subset of the event "viable at time t", so probabilities inherit the lower bound. Then linearity of expectation gives the bound on the sum of indicators; Markov gives the tail bound. □

Remark 2315. Proof sketch (high level). *The strategy is: (i) use set inclusion to move from a conjunction over times to a single time; (ii) sum indicator variables to count violations and take expectations; (iii) use Markov to convert an expectation bound into a tail-probability bound.*

Key steps. *The inclusion step is the fundamental monotonicity of probability: if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$. Linearity of expectation is what turns the random count $\sum_t \mathbb{I}[\cdot]$ into a sum of per-time probabilities. Markov is the final bridge from average-case to worst-case frequency.*

Geometric intuition. *Think of each time t as a "slice" through trajectory space. Homeostasis controls the measure of trajectories that ever leave the viable tube; the corollary bounds the measure of trajectories outside the tube at any slice, and the expected number of slices where a trajectory lies outside the tube.*

Remark 2316. On tightness and interpretation. *The bound $\mathbb{P}_\pi\big((s_t, g_t) \notin V_O\big) \leq \delta$ is generally loose as a statement about a fixed time t: it follows only from the fact that violating at time t implies violating at* some *time. In many models, the per-time violation probability can be much smaller than δ (for instance, when failures are concentrated near the end of the horizon, or when leaving V_O is a rare transient effect). Similarly, $\mathbb{E}[\#\text{violations}] \leq (T+1)\delta$ follows from applying the same uniform bound to each term; it is a conservative but assumption-free way to convert the global requirement into an additive quantity.*

Helper Proposition 40.3 (Parameter weakening rules)**.** *If π is homeostatic for (V_O, T, δ) then:*

1. *for any $T' \leq T$, π is homeostatic for (V_O, T', δ);*
2. *for any $V_O \subseteq V_O'$ (a weaker viability set), π is homeostatic for (V_O', T, δ);*
3. *for any $\delta' \geq \delta$, π is homeostatic for (V_O, T, δ').*

Remark 2317. *This proposition records a basic monotonicity principle: homeostasis becomes easier to satisfy when we (i) ask for it over a shorter time horizon, (ii) enlarge the set considered "viable," or (iii) tolerate a larger failure probability. These weakening rules are useful when comparing designs or proofs across different parameterizations: one often establishes a strong guarantee (small δ, long T, strict V_O) and then inherits a family of weaker guarantees for free.*

Sketch. Let

$$A_T \;:=\; \Big\{(s_t, g_t) \in V_O \text{ for all } t = 0, \ldots, T\Big\}.$$

(i) If $T' \leq T$, then $A_T \subseteq A_{T'}$ because satisfying viability for all times up to T in particular implies satisfying it for all times up to T'. Hence $\mathbb{P}_\pi(A_{T'}) \geq \mathbb{P}_\pi(A_T) \geq 1 - \delta$.

(ii) If $V_O \subseteq V'_O$, then

$$\Big\{(s_t, g_t) \in V_O \text{ for all } t \leq T\Big\} \subseteq \Big\{(s_t, g_t) \in V'_O \text{ for all } t \leq T\Big\},$$

so the probability lower bound transfers by monotonicity.

(iii) If $\delta' \geq \delta$, then $1 - \delta' \leq 1 - \delta$, so any inequality of the form $\mathbb{P}_\pi(A_T) \geq 1 - \delta$ immediately implies $\mathbb{P}_\pi(A_T) \geq 1 - \delta'$. $\square$

Remark 2318. *The proposition states three monotonicities that match the moral shape of the definitions: it is easier to be safe for fewer steps, easier to be safe in a larger allowed region, and easier to satisfy a weaker probability requirement. These rules are frequently used implicitly; it is therefore helpful to isolate them as explicit "weakening" principles (Hyperseed-Concept 202, 101).*

Remark 2319. *Connection to later results: these weakening rules will pair naturally with budget dynamics (below). For example, once one proves homeostasis for a strict minimal-budget viability set and a long horizon, one automatically inherits homeostasis for any shorter horizon or any relaxed notion of viability.*

Sketch. Each item is immediate by event inclusion (smaller horizon / larger viable set / weaker probability requirement).

To make the inclusions explicit, one may rewrite the underlying "good event" in the definition of homeostasis as an intersection over time. For instance, if the viability requirement is that the state remains in a set V for times $t = 0, 1, \ldots, T$, the good event can be written as

$$E_{T,V} \;:=\; \bigcap_{t=0}^{T} \{\, x_t \in V \,\}.$$

Then the shortening-of-horizon monotonicity is simply the containment

$$E_{T,V} \subseteq E_{T',V} \qquad \text{whenever } T' \leq T,$$

because intersecting fewer constraints produces a larger event. Likewise, if $V \subseteq V'$ then for each t one has $\{x_t \in V\} \subseteq \{x_t \in V'\}$, and hence

$$E_{T,V} \subseteq E_{T,V'}.$$

Finally, for the probability-threshold weakening, if the homeostasis condition is of the form $\mathbb{P}(E_{T,V}) \geq 1 - \delta$ and one replaces δ by a larger tolerance $\delta' \geq \delta$, then $1 - \delta' \leq 1 - \delta$, so the same inequality automatically implies $\mathbb{P}(E_{T,V}) \geq 1 - \delta'$.

In all cases, the probability comparison uses only monotonicity of $\mathbb{P}$ with respect to event inclusion: if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$, so a lower bound for $\mathbb{P}(A)$ is also a lower bound for $\mathbb{P}(B)$. $\square$

Remark 2320. Proof sketch (high level). *Each clause is a one-line containment argument: if you demand less, you get a superset of acceptable trajectories.*

Why it works. *Homeostasis is defined by a single inequality* $\mathbb{P}(\textit{good event}) \geq 1 - \delta$. *Each weakening moves from a smaller good event to a larger good event, or lowers the required threshold* $1 - \delta$.

Visual intuition. *In a picture of a viability tube over time, item (1) shortens the tube, item (2) widens it, and item (3) accepts a larger fraction of trajectories that may leave it.*

Optional formalism. *If one prefers to keep the argument entirely at the level of set-theoretic operations, the three weakenings correspond to: (i) dropping factors from an intersection over time, (ii) enlarging the sets inside that intersection, and (iii) lowering the right-hand side in the defining inequality. No particular structure of the dynamics is needed beyond the fact that the events in question are measurable (so that* $\mathbb{P}(\cdot)$ *is defined) and that* $\mathbb{P}$ *is monotone under inclusion.*

Operational reading. *These monotonicities can also be read as closure properties of the set of policies (or initial conditions) that satisfy homeostasis: if a policy achieves homeostasis for a stringent specification, then the same policy remains valid after any of the three relaxations. This is the precise sense in which they function as "weakening principles" and explains why they can be invoked without re-checking the dynamics in later budget-coupled arguments.*

40.2 Genenergy as constrained capacity (Defs. 272–276, Props. 25–26)

Remark 2321. *Genenergy is defined via an information-theoretic constrained capacity: one asks how many bits of influence an agent's action* A *can transmit to the next observation* Y_{t+1}*, given a cost budget* g*. In classical information theory this is a channel capacity with an input cost constraint; here it is reinterpreted as a measure of "available effective agency" under resource limits (Hyperseed-Concept* **??***, 117,* **??***). For background on mutual information and capacity as measures of influence, see [16]; for alternative but related measures of distinction/information, compare logical entropy [17]. In particular, one may read* $I(A; Y_{t+1} \mid h_{0:t})$ *as quantifying how strongly the distribution of the next observation can be* steered *by choosing different action distributions, holding fixed the historical context* $h_{0:t}$ *(so that the "channel" is the conditional map* $p(y_{t+1} \mid a, h_{0:t})$ *induced by the world model at time* t*). The budget constraint* $\mathbb{E}[c_O(A)] \leq g$ *then expresses that steering is only counted when it is achieved with affordable mixtures of actions, which matches the intended "agency under resource limits" reading.*

Remark 2322. *It is useful to keep in mind the standard constrained-capacity template: for each fixed* $(O, t, h_{0:t})$*, the function* $g \mapsto \mathrm{Cap}_{O,t,h}(g)$ *is the value of an optimization problem where the feasible set expands monotonically with* g*. Consequently, even before proving any regularity properties, one should expect (and later can show under mild conditions) that* $\mathrm{Cap}_{O,t,h}(g)$ *is nondecreasing in* g*, with saturation once the constraint stops binding. The next lemma isolates exactly that saturation regime in the simplest possible way.*

Helper Lemma 40.4 (Unconstrained-capacity regime once budget exceeds max effort)**.** *Fix* $(O, t, h_{0:t})$ *and cost* $c_O : A \to \mathbb{R}_{\geq 0}$*. If* $g \geq \max_{a \in A} c_O(a)$*, then the constraint* $\mathbb{E}[c_O(A)] \leq g$ *is vacuous for all* $p \in \Delta(A)$*, hence*

$$\mathrm{Cap}_{O,t,h}(g) \;=\; \max_{p \in \Delta(A)} I(A; Y_{t+1} \mid h_{0:t}).$$

Remark 2323. *This lemma says that once the budget is large enough to afford even the most expensive action, the optimization ceases to be a constrained one: every action distribution is feasible, so capacity becomes the ordinary (unconstrained) mutual-information maximum. Philosophically, it draws a sharp line between regimes where "energy matters" and regimes where it does not: beyond a threshold, the resource variable* g *stops being a binding constraint. A further conceptual point* *is that this threshold depends only on the* cost model c_O *and the action set* A*, and not on any dynamics or observation model; it is therefore a purely "accounting" boundary between constrained and unconstrained agency.*

Remark 2324. *A simple example: if* $A = \{a_1, a_2\}$ *with costs* $c_O(a_1) = 0.2$ *and* $c_O(a_2) = 0.9$*, then any budget* $g \geq 0.9$ *makes all mixtures* $p(a_1), p(a_2)$ *feasible. For* $g < 0.9$*, distributions placing too much mass on* a_2 *are excluded, and the achievable influence may drop. One can also see explicitly how the constraint cuts out a half-space in the simplex: the feasible set is* $\{p \in \Delta(A) : 0.2\, p(a_1) + 0.9\, p(a_2) \leq g\}$*, i.e.* $p(a_2) \leq (g - 0.2)/(0.7)$ *when* $g \geq 0.2$*, and no nontrivial distribution is feasible if* $g < 0.2$ *(unless one allows zero-cost slack actions). This geometric view will be helpful when thinking about how genenergy changes continuously as the budget parameter is varied.*

Sketch. If $g \geq \max_a c_O(a)$, then for any distribution p, $\mathbb{E}_{A\sim p}[c_O(A)] \leq \max_a c_O(a) \leq g$. So the feasible set is all of $\Delta(A)$. □

Remark 2325. Proof sketch (high level). *The strategy is to upper bound an average by a maximum.*

Key step. *The inequality* $\mathbb{E}[c_O(A)] \leq \max_a c_O(a)$ *holds for any distribution because an expectation is a convex combination of pointwise values.*

Intuition. *If even the priciest action is affordable, then affordability does not restrict choice. One may also interpret the same step as a one-line application of the fact that* $\max_a c_O(a)$ *is an (almost sure) upper bound on the random variable* $c_O(A)$*, and taking expectations preserves upper bounds.*

Remark 2326. *In later applications it is sometimes convenient to rewrite the constrained maximization defining* $\mathrm{Cap}_{O,t,h}(g)$ *using a Lagrange multiplier* $\lambda \geq 0$*, maximizing* $I(A; Y_{t+1} \mid h_{0:t}) - \lambda(\mathbb{E}[c_O(A)] - g)$ *over* $p \in \Delta(A)$*. The present lemma then corresponds to the situation where an optimal multiplier can be taken as* $\lambda = 0$*, because the constraint is inactive; this is another standard way of recognizing the "unconstrained" regime.*

Helper Lemma 40.5 (Information-theoretic upper bounds from finiteness of Y_O)**.** *Let* Y_O *be the finite observation alphabet for* O *(Def. 273). Then for any* (O, t, h, g)*,*

$$\mathrm{Cap}_{O,t,h}(g) \;\leq\; \log |Y_O| \qquad \text{and} \qquad \mathrm{Cap}_{O,t,h}(g) \;\leq\; \log |A|.$$

(Logs are base 2 if mutual information is measured in bits.)

Remark 2327. *This lemma states two universal ceilings on genenergy that come purely from finiteness: you cannot extract more than* $\log |Y_O|$ *bits of information from a variable that only takes* $|Y_O|$ *possible values, and you cannot transmit more than* $\log |A|$ *bits if your input itself only has* $|A|$ *possibilities. The importance is practical: it tells us that even without knowing any dynamics, genenergy is bounded above by a simple combinatorial quantity. In particular, the bound does not depend on the budget* g *at all; increasing the budget can only help until these alphabet-size ceilings are met, at which point no additional resources can increase the measured action-to-observation information flow.*

Remark 2328. *The bounds also reinforce an epistemological point: an observer with a coarse alphabet* Y_O *cannot, by definition, register arbitrarily fine distinctions, so the measured influence of actions on observations is limited (Hyperseed-Concept 134,* **??***). This prepares the ground for the data-processing theorem below. Equivalently, if the "true" underlying world state is high-dimensional but* O *records it through a many-to-one perception map into* Y_O*, then any influence actions exert on microscopic details that are subsequently discarded by perception is invisible to* $I(A; Y_{t+1} \mid h_{0:t})$ *and therefore invisible to genenergy as defined here.*

Sketch. For any feasible p, $I(A; Y \mid h) \leq H(Y \mid h) \leq \log |Y_O|$, and also $I(A; Y \mid h) \leq H(A) \leq \log |A|$. Taking the maximum preserves the bound. □

Remark 2329. Proof sketch (high level). *The strategy is to sandwich mutual information between entropies and then use the trivial entropy upper bound for finite alphabets.*

Why it works. *Mutual information satisfies $I(A; Y \mid h) \leq H(Y \mid h)$ and also $I(A; Y \mid h) \leq H(A)$, because conditioning cannot increase uncertainty and I is a reduction in uncertainty. Finally, $H(Z) \leq \log |Z|$ for a finite random variable Z.*

Visual intuition. *If Y has only $|Y_O|$ bins, it is like a measuring device with only $|Y_O|$ display states; it cannot convey more than $\log |Y_O|$ bits, regardless of how intricate the underlying world is. It can also be helpful to remember the extreme case: if $|Y_O| = 1$ (the observer always sees the same symbol), then $\log |Y_O| = 0$ and necessarily $I(A; Y_{t+1} \mid h_{0:t}) = 0$ for all choices of p, so genenergy is identically zero because nothing observable can ever change.*

Remark 2330. *A concrete illustration of the $\log |A|$ ceiling: if A has only two actions, then even in the best possible world model the action can carry at most $\log 2 = 1$ bit of information about the next observation (conditional on history). Thus any reported genenergy value above 1 bit per step would signal a mismatch between the formal action alphabet and the effective degrees of freedom actually being optimized over.*

Helper Corollary 40.6 (Uniform bound on realizable genenergy over horizon T)**.** *For any policy π (Def. 276),*

$$G_O(\pi; T) \;=\; \mathbb{E}\Big[\frac{1}{T} \sum_{t=0}^{T-1} \mathrm{Cap}_{O,t,h}(g_t)\Big] \;\leq\; \log |Y_O|.$$

Therefore $G_O^{\mathrm{real}}(T) = \sup_\pi G_O(\pi; T) \leq \log |Y_O|$.

Remark 2331. *The corollary lifts a pointwise capacity bound to an averaged-over-time and averaged-over-trajectories bound. Its conceptual message is sobering and useful: regardless of policy ingenuity, the observer's alphabet limits the long-run average genenergy. In other words, the "ceiling of agency as measured through observation" is partly a ceiling of* representation *(Hyperseed-Concept 157, 134). Note that the bound is uniform in T: increasing the time horizon does not allow one to exceed* $\log |Y_O|$ per time step *on average, because each step produces at most one symbol in Y_O. This keeps the scale of genenergy comparable across horizons and prevents spurious growth that would come from simply aggregating influence over longer runs without normalization.*

Sketch. Apply the previous lemma inside the average and expectation. □

Remark 2332. Proof sketch (high level). *Bound each summand by $\log |Y_O|$, then average.*

Key step. *Expectations and averages preserve inequalities when applied to pointwise-bounded random variables.*

Intuition. *If each day you can gain at most B bits of influence, then your average influence per day is also at most B. Formally, if $X_t \leq B$ almost surely for each t, then $\frac{1}{T} \sum_{t=0}^{T-1} X_t \leq B$ almost surely, and taking $\mathbb{E}[\cdot]$ preserves the inequality; here one takes $X_t = \mathrm{Cap}_{O,t,h}(g_t)$ and $B = \log |Y_O|$.*

Helper Theorem 40.7 (Observer coarsening reduces genenergy (data processing))**.** *Let two observers have observation maps $\phi_1 : E \to Y_1$ and $\phi_2 : E \to Y_2$ (Def. 273). If there exists a deterministic ψ with $\phi_1 = \psi \circ \phi_2$, then for all (t, h, g),*

$$\mathrm{Cap}_{1,t,h}(g) \;\leq\; \mathrm{Cap}_{2,t,h}(g).$$

Remark 2333. *In plain terms: if observer O_1 sees only a deterministic coarsening of what O_2 sees, then O_1 can never register* more *action-to-observation influence than O_2. This is a direct application of the data-processing inequality, one of the central theorems of information theory [16]. It is important because it makes "observer dependence" mathematically explicit: genenergy is not an intrinsic property of the world alone, but of the world* as perceived through *an observation map (Hyperseed-Concept 134, 112).*

Remark 2334. *Connection to the previous bounds: finiteness gave absolute ceilings; data processing gives* comparative *ceilings across observers. Together they support a disciplined notion of refinement: adding observational resolution cannot reduce maximal attainable information flow, while coarse-graining cannot increase it.*

Sketch. For any feasible input distribution $p(a)$, $Y^{(1)}_{t+1} = \psi(Y^{(2)}_{t+1})$ so by data processing $I(A; Y^{(1)}_{t+1} \mid h) \le I(A; Y^{(2)}_{t+1} \mid h)$. Maximizing over the same feasible set yields the inequality. □

Remark 2335. Proof sketch (high level). *Show the inequality for each fixed feasible input distribution $p(a)$, then take maxima.*

Key step. *The data-processing inequality says that applying a deterministic function ψ to an observation cannot increase its mutual information with the input. Deterministic post-processing can only forget distinctions, not create them.*

Visual intuition. *Think of Y_2 as a high-resolution image and Y_1 as the same image after blurring. No choice of action can make the blurred image reveal more about the action than the original image could.*

Helper Corollary 40.8 (Positive genenergy iff some feasible distribution has influence)**.** *Fix (O, t, h) and budget g. Then $\mathrm{Cap}_{O,t,h}(g) > 0$ iff there exists some feasible $p \in \Delta(A)$ with $\mathbb{E}[c_O(A)] \le g$ and $I(A; Y_{t+1} \mid h_{0:t}) > 0$. Equivalently (Prop. 25), $\mathrm{Cap}_{O,t,h}(g) = 0$ iff $A \perp Y_{t+1} \mid h_{0:t}$ for all feasible p.*

Remark 2336. *This corollary translates the definition of constrained capacity into an existential criterion: positive genenergy means that* some *affordable randomized choice of actions genuinely influences what the observer can see next (in the precise sense of mutual information). If capacity is zero, then within the budget constraint the action is informationally irrelevant to the next observation, given history: the agent may still act, but it cannot make a detectable difference to Y_{t+1} as measured by O (Hyperseed-Concept* **??**, *87).*

Sketch. Immediate from the definition of Cap as a maximum and the characterization of the zero case. □

Remark 2337. Proof sketch (high level). *A maximum is positive exactly when at least one feasible point attains a positive value.*

Key step. *The "only if" direction uses attainment in the finite-dimensional optimization (or, more generally, the definition of* max*); the "if" direction uses monotonicity of* max *with respect to the feasible set.*

Intuition. *Capacity is the best achievable influence; it is nonzero exactly when influence is achievable at all.*

Helper Proposition 40.9 (Concavity calculus: diminishing returns and baseline-subadditivity)**.** *Assume the concavity from Prop. 26: $g \mapsto \mathrm{Cap}_{O,t,h}(g)$ is concave and nondecreasing. Then chord slopes are decreasing: for $0 \le g_1 < g_2 < g_3$,*

$$\frac{\mathrm{Cap}(g_2) - \mathrm{Cap}(g_1)}{g_2 - g_1} \ge \frac{\mathrm{Cap}(g_3) - \mathrm{Cap}(g_2)}{g_3 - g_2}.$$

Moreover the baseline-adjusted map $\widehat{\mathrm{Cap}}(g) := \mathrm{Cap}(g) - \mathrm{Cap}(0)$ *is subadditive:*

$$\widehat{\mathrm{Cap}}(g_1 + g_2) \ \leq \ \widehat{\mathrm{Cap}}(g_1) + \widehat{\mathrm{Cap}}(g_2) \qquad (g_1, g_2 \geq 0).$$

Remark 2338. *The first statement is the familiar "diminishing returns" principle: as you invest more budget* g*, the additional genenergy per additional unit of budget cannot increase when the capacity curve is concave. In other words, the marginal gain (interpretable as a slope, derivative, or subderivative depending on regularity) is nonincreasing in* g*, so each extra unit of budget buys weakly less additional baseline-adjusted capacity than the previous unit. Geometrically, for a concave graph the slopes of secant lines between points on the curve decrease as the points move to the right, and this is exactly the comparison of incremental gains over successive budget intervals.*

The second statement is slightly subtler: if we measure capacity above baseline $\mathrm{Cap}(0)$*, then splitting a total budget* g_1+g_2 *into two separate "allocations" cannot yield less total baseline-adjusted genenergy than spending it all at once. Here one is not claiming that splitting produces more raw capacity* $\mathrm{Cap}(\cdot)$ *(which would be false in general), but rather that after subtracting the common baseline* $\mathrm{Cap}(0)$ *the resulting function behaves subadditively under a concavity assumption. This kind of inequality is often what one needs to justify modular budgeting arguments. Concretely, it supports the idea that if two subsystems each receive their own budget and each is evaluated relative to its own baseline, then combining the resulting baseline-adjusted contributions does not underperform a single monolithic expenditure evaluated relative to the same baseline.*

Remark 2339. *Connection to the metabolic model below: concavity is precisely what makes resource allocation nontrivial. If returns were increasing, the rational strategy would be to concentrate all budget; with diminishing returns, diversification and distributed spending can become sensible. In this way, a simple analytic property (concavity) expresses a strategic fact about agency under constraints (Hyperseed-Concept 100,* **??***). Put differently, concavity is the mathematical encoding of "saturation": once some pathway is well-funded, pouring additional effort into the same pathway yields progressively smaller improvements, which creates room for parallel investments to be competitive.*

Sketch. The slope statement is a standard equivalent form of concavity. More explicitly, for a concave function f on an interval, the secant slope

$$\frac{f(g+h) - f(g)}{h}$$

is nonincreasing as g increases (for fixed $h > 0$), and equivalently the left and right derivatives (when they exist) satisfy $f'(g_1) \geq f'(g_2)$ whenever $g_1 \leq g_2$. This expresses precisely that the incremental benefit of additional budget weakly decreases.

For subadditivity: $\widehat{\mathrm{Cap}}$ is concave, nondecreasing, and $\widehat{\mathrm{Cap}}(0) = 0$. Recall the concavity inequality in the usual form

$$f(\lambda a + (1-\lambda)b) \ \geq \ \lambda f(a) + (1-\lambda) f(b) \qquad (0 \leq \lambda \leq 1).$$

Using concavity with $x = \frac{g_1}{g_1+g_2}(g_1+g_2) + \frac{g_2}{g_1+g_2} \cdot 0$ (and similarly for g_2), one gets $\widehat{\mathrm{Cap}}(g_1) + \widehat{\mathrm{Cap}}(g_2) \geq \widehat{\mathrm{Cap}}(g_1 + g_2)$. To see the algebra: set $f = \widehat{\mathrm{Cap}}$, $a = g_1 + g_2$, $b = 0$, and $\lambda = \frac{g_1}{g_1+g_2}$, so that $\lambda a + (1-\lambda) b = g_1$. Concavity then yields

$$f(g_1) \ \geq \ \frac{g_1}{g_1+g_2} f(g_1+g_2) + \frac{g_2}{g_1+g_2} f(0) \ = \ \frac{g_1}{g_1+g_2} f(g_1+g_2),$$

where the last equality uses $f(0) = 0$. Similarly, with $\lambda = \frac{g_2}{g_1+g_2}$ one obtains

$$f(g_2) \;\geq\; \frac{g_2}{g_1+g_2} f(g_1+g_2).$$

Adding these two inequalities gives

$$f(g_1) + f(g_2) \;\geq\; \Big(\frac{g_1}{g_1+g_2} + \frac{g_2}{g_1+g_2}\Big) f(g_1+g_2) \;=\; f(g_1+g_2),$$

which is the desired subadditivity. The nondecreasing hypothesis ensures that interpreting f as "baseline-adjusted capacity" is consistent with budget monotonicity, but the normalization $f(0) = 0$ is the crucial point that prevents additive constants from interfering with the inequality. □

Remark 2340. Proof sketch (high level). *The first part uses the geometric characterization of concavity: chords lie below tangents, so chord slopes decrease as you move right. Equivalently, moving a fixed horizontal distance to the right yields a smaller vertical increase when you start further to the right. The second part uses concavity in the special case of convex combinations with* 0 *to relate values at* g_1*,* g_2*, and* $g_1 + g_2$*. The role of* 0 *is not incidental: it anchors the comparison at a common reference point, which is exactly what "baseline-adjusted" means.*

Key step. *Writing* g_1 *as a convex combination of* $(g_1 + g_2)$ *and* 0 *is the algebraic trick that makes concavity apply. Concretely,* $g_1 = \lambda(g_1 + g_2) + (1 - \lambda)0$ *with* $\lambda = \frac{g_1}{g_1+g_2} \in [0, 1]$*, and the same pattern holds for* g_2*. The baseline adjustment* $\widehat{\mathrm{Cap}}(0) = 0$ *is what prevents constant offsets from breaking subadditivity. Without this normalization, concavity would only yield an "affine-subadditivity" statement up to an additive constant, and that constant would precisely be the baseline being subtracted off.*

Visual intuition. *For a concave curve starting at* $(0, \mathrm{Cap}(0))$*, the "height above baseline" bends downward; two shorter horizontal moves can reach at least as much total height as one longer move, once you compare everything to the same baseline. Another way to picture it is to imagine that the first units of budget climb the steep part of the curve, while later units climb a flatter part; splitting lets both portions benefit from steeper initial segments when evaluated relative to the shared baseline.*

40.3 Entity-sets and potential genenergy (Defs. 277–278, Prop. 27)

Remark 2341. *Potential genenergy* $g_O(E; T)$ *(as defined earlier) aggregates what may be achievable when one considers a set* E *of entities (Hyperseed-Concept 175). The monotonicity property in Prop. 27 formalizes an obvious idea: enlarging the set of entities under consideration can only increase (or preserve) what is possible, because it relaxes constraints on what counts as an admissible entity-system.*

Remark 2342. *One convenient way to read Prop. 27 is as an order-preservation statement: the mapping*

$$E \longmapsto g_O(E; T)$$

is monotone with respect to set inclusion. In particular, whenever one argues that a certain capability is achievable using some collection of entities, that argument automatically remains valid when additional entities are allowed, since the admissible class of entity-systems has only expanded.

Helper Lemma 40.10 (Union lower bound from monotone potential genenergy)**.** *Let* $E_1, E_2 \subseteq \mathrm{Ent}$ *be entity-sets. Then for any* T*,*

$$g_O(E_1 \cup E_2; T) \;\geq\; \max\{g_O(E_1; T),\; g_O(E_2; T)\}.$$

Remark 2343. *The lemma says that the union is at least as powerful as either component set. This is a minimal but frequently used inequality: it lets one bound a complicated set from below by any simpler subset one can analyze. In proofs, it often functions as a controlled way of saying "we may restrict attention to a convenient subcollection without increasing difficulty."*

Remark 2344. *Equivalently, the displayed inequality can be split into the two statements*

$$g_O(E_1 \cup E_2; T) \;\geq\; g_O(E_1; T) \qquad \text{and} \qquad g_O(E_1 \cup E_2; T) \;\geq\; g_O(E_2; T),$$

and the "max" packaging simply records both at once. This way of writing it is often helpful when one later takes successive unions: each union step can be seen as adding more admissible entities without reducing the best-achievable potential genenergy.

Sketch. $E_i \subseteq E_1 \cup E_2$ so monotonicity (Prop. 27) gives $g_O(E_i; T) \leq g_O(E_1 \cup E_2; T)$. □

Remark 2345. Proof sketch (high level). *Reduce to monotonicity: subset inclusion implies an inequality.*

Key step. *The only substantive input is Prop. 27 (monotonicity). Once that is granted, the union bound is immediate.*

Intuition. *Having more entities available cannot make your best achievable genenergy worse.*

Remark 2346. *Two small boundary cases are worth keeping in mind for later use. First, if $E_1 = \varnothing$ then the lemma reduces to $g_O(E_2; T) \leq g_O(E_2; T)$ (together with the implicit inequality $g_O(\varnothing; T) \leq g_O(E_2; T)$ coming from monotonicity). Second, if $E_1 \subseteq E_2$, then $E_1 \cup E_2 = E_2$ and the lemma collapses to the monotonicity inequality already supplied by Prop. 27. In this sense, the union lower bound is best viewed as a direct reformulation of monotonicity that is tailored to the common situation where one builds larger entity-sets by unions.*

Corollary 19 (Finite union lower bound)**.** *Let $E_1, \ldots, E_n \subseteq \mathrm{Ent}$ be entity-sets. Then for any T,*

$$g_O\Big(\bigcup_{k=1}^{n} E_k; T\Big) \;\geq\; \max_{1 \leq k \leq n} g_O(E_k; T).$$

Sketch. Apply the lemma iteratively (or argue by induction on n), using that $E_j \subseteq \bigcup_{k=1}^{n} E_k$ for each j and then invoking Prop. 27. □

Remark 2347. *The inequality is one-sided because combining entity-sets can introduce new admissible entity-systems that were not available when restricted to either component alone. Hence $g_O(E_1 \cup E_2; T)$ can be strictly larger than both $g_O(E_1; T)$ and $g_O(E_2; T)$. Equality holds in particular when one of the two sets already contains the other, but strict improvement is the generic case when the definition of g_O permits "hybrid" systems drawing on entities from both sets.*

40.4 Metabolism and budget dynamics (Defs. 279–280, Prop. 28)

Remark 2348. *The metabolic accounting model treats budget g_t as a leaky integrator: it decays by a factor $(1 - \lambda)$, receives income $\Gamma_O(r_t)$ from resources r_t, and pays costs $c_O(a_t)$ for actions. This is a stylized but mathematically transparent way to formalize "maintenance" and "effort" (Hyperseed-Concept 109, 100,* **??***).*

Remark 2349. *Two parameters/choices deserve emphasis.*

- *The leakage rate $\lambda \in [0,1]$ encodes the timescale of decay: small λ corresponds to slow forgetting/slow dissipation, while λ near 1 corresponds to almost complete reset each step. In particular, when $\lambda = 1$ we have $g_{t+1} = \Gamma_O(r_t) - c_O(a_t)$, so the state carries no budget memory beyond one step.*

- *The splitting of the update into an "income" term $\Gamma_O(r_t)$ and a "cost" term $c_O(a_t)$ is conceptually useful because it separates what the environment provides (resources) from what the agent chooses (actions), even though both can be random and coupled through the policy/environment dynamics.*

Helper Lemma 40.11 (Closed form of the budget recursion)**.** *Under the metabolic accounting model (Def. 279),*

$$g_{t+1} = (1-\lambda)g_t + \Gamma_O(r_t) - c_O(a_t),$$

we have for all $t \geq 1$:

$$g_t = (1-\lambda)^t g_0 + \sum_{k=0}^{t-1} (1-\lambda)^{t-1-k} \big(\Gamma_O(r_k) - c_O(a_k)\big).$$

Remark 2350. *This lemma simply unrolls the recursion. The point is not that it is hard, but that it reveals structure: current budget is the discounted sum of past net incomes plus a decayed initial budget. In other words, the past persists, but with exponentially fading influence. Such explicit formulas are often indispensable when taking limits (steady states) or bounding expectations.*

Remark 2351. *It is also helpful to read the formula as a discrete-time convolution: the "impulse response" of the budget dynamics is the geometric kernel $(1-\lambda)^{t-1-k}$. This makes it immediate, for example, that if the net income sequence $\Gamma_O(r_k) - c_O(a_k)$ is uniformly bounded, then g_t is bounded up to a term that depends on λ (and similarly for moment bounds when integrability is available). When $\lambda > 0$, the kernel is summable, reflecting that very old contributions become negligible.*

Sketch. Induction on t. □

Remark 2352. Proof sketch (high level). *Prove the formula holds at $t = 1$, then assume it holds at t and substitute into the recursion to obtain the $t+1$ case.*

Key step. *The geometric factor $(1-\lambda)^{t-1-k}$ arises because each net income term must survive decay for the number of steps between k and t.*

Visual intuition. *Imagine pouring water into a leaky bucket each step: earlier pours contribute less to the current water level because of leakage.*

Remark 2353. *A useful special case check is $\lambda = 0$, in which case there is no leakage and the formula reduces to*

$$g_t = g_0 + \sum_{k=0}^{t-1} \big(\Gamma_O(r_k) - c_O(a_k)\big),$$

i.e. a pure cumulative sum. This makes clear why $\lambda > 0$ changes qualitative behavior: it turns an accumulating process into a mean-reverting one (in expectation under stationarity assumptions).

Helper Corollary 40.12 (Expected steady state under stationarity)**.** *Assume $\lambda > 0$ and there exists μ with $\mathbb{E}[\Gamma_O(r_t) - c_O(a_t)] = \mu$ for all t (e.g. stationary regime). Then*

$$\mathbb{E}[g_t] = (1-\lambda)^t \mathbb{E}[g_0] + \frac{\mu}{\lambda}\big(1 - (1-\lambda)^t\big), \quad \textit{so} \quad \mathbb{E}[g_t] \to \mu/\lambda.$$

Remark 2354. *This corollary is the expected-value analogue of a steady-state calculation: if on average you earn μ net budget per time-step, and you leak a fraction λ per step, then the expected equilibrium budget is μ/λ. The condition $\lambda > 0$ is essential: without leakage, budget would drift linearly rather than converge. This is a precise version of the intuitive idea that maintenance costs (modeled here by leakage) stabilize resource levels (Hyperseed-Concept 109).*

Remark 2355. *Note that the claim concerns $\mathbb{E}[g_t]$ and not almost-sure convergence of g_t. Even in a stationary regime, the random fluctuations of $\Gamma_O(r_t) - c_O(a_t)$ can keep g_t wandering, while the expectation settles to μ/λ. To upgrade from an expectation statement to concentration or almost-sure statements typically requires additional assumptions (e.g. bounded increments, martingale arguments, or ergodicity/mixing). The explicit formula from the lemma is often the starting point for such refinements because it expresses g_t as a weighted sum of past shocks.*

Sketch. Take expectations in the recursion and solve the resulting linear difference equation. □

Remark 2356. Proof sketch (high level). *The expectation of the recursion is a first-order linear difference equation with constant forcing term μ. Solve it by standard methods (homogeneous solution plus particular solution).*

Key step. *Stationarity is used only to make the forcing term constant across t.*

Intuition. *The long-run expected budget is where average inflow balances average leakage: $\lambda\,\mathbb{E}[g] \approx \mu$.*

Remark 2357. *It can be instructive to see the "balance" directly: if $\mathbb{E}[g_t] \to \bar{g}$ exists, then taking limits in*

$$\mathbb{E}[g_{t+1}] = (1 - \lambda)\mathbb{E}[g_t] + \mu$$

forces $\bar{g} = (1-\lambda)\bar{g} + \mu$, hence $\lambda\bar{g} = \mu$ and $\bar{g} = \mu/\lambda$. This highlights why the limit value is insensitive to $\mathbb{E}[g_0]$: initial conditions are forgotten at the exponential rate $(1 - \lambda)^t$.

Helper Proposition 40.13 (Iterated invariant lower bound (Prop. 28 as an induction rule))**.** *Let $\underline{g} > 0$ be such that $(s, g) \in V_O \Rightarrow g \geq \underline{g}$ and suppose there is a policy regime satisfying the Prop. 28 inequality $\mathbb{E}[\Gamma_O(r_t)] \geq \mathbb{E}[c_O(a_t)] + \lambda\underline{g}$ whenever $g_t \geq \underline{g}$. Then the condition $\mathbb{E}[g_t] \geq \underline{g}$ is inductive: if it holds at time t it holds at $t + 1$. Hence if $\mathbb{E}[g_0] \geq \underline{g}$ then $\mathbb{E}[g_t] \geq \underline{g}$ for all t.*

Remark 2358. *The proposition isolates the logical heart of many viability arguments: identify a lower bound $\underline{g}$ required by the viability set, then show that the expected budget update cannot push you below $\underline{g}$ provided you are already at or above it. This transforms Prop. 28 into a reusable induction principle: once above the floor (in expectation), always above the floor (in expectation). It is a formal version of "metabolic sustainability" (Hyperseed-Concept 109, 101).*

Remark 2359. *One should also notice what is* not *assumed: the proposition does not require the net income to be nonnegative almost surely, nor does it require $\Gamma_O(r_t)$ and $c_O(a_t)$ to be independent of g_t. The condition is framed to be robust under coupling, because it is stated as a one-step expected inequality under the regime of interest. This is often exactly what is available from a control design or a Lyapunov-style estimate, where one can guarantee that some expected drift is favorable above a threshold.*

Remark 2360. *Connection back to homeostasis: homeostasis concerns the probability of staying in V_O; this proposition concerns maintaining a budget floor that is* necessary *for membership in V_O. In larger arguments, one often uses budget invariance to help establish viability, and then applies the event-rewrite lemmas to reason about failure probability.*

Remark 2361. *There is also a useful conceptual separation between* viability constraints *and* budget dynamics*. The assumption $(s, g) \in V_O \Rightarrow g \geq \underline{g}$ says the viability set imposes a hard feasibility requirement on budget; the Prop. 28 inequality then provides a* sufficient *drift condition ensuring the dynamics respect that requirement (in expectation). This is the same pattern as "safe set implies barrier function lower bound" plus "controller keeps barrier from decreasing" in other safety/control formalisms, except specialized here to a linear leaky-integrator state variable.*

Sketch. This is exactly the one-step bound shown in the proof of Prop. 28, iterated by induction. □

Remark 2362. Proof sketch (high level). *Use the one-step inequality to show $\mathbb{E}[g_{t+1}] \geq \underline{g}$ whenever $\mathbb{E}[g_t] \geq \underline{g}$, then iterate.*

Key step. *The hypothesis $\mathbb{E}[\Gamma_O(r_t)] \geq \mathbb{E}[c_O(a_t)] + \lambda \underline{g}$ is precisely chosen to cancel the expected leakage term λg_t at the floor value $\underline{g}$.*

Intuition. *If your expected net intake covers both your expected expenses and the leakage needed to maintain the floor, then you do not (in expectation) fall through the floor.*

Remark 2363. *Algebraically, the one-step calculation behind the induction is*

$$\mathbb{E}[g_{t+1}] = \mathbb{E}[(1-\lambda)g_t] + \mathbb{E}[\Gamma_O(r_t)] - \mathbb{E}[c_O(a_t)] \geq (1-\lambda)\underline{g} + \lambda \underline{g} = \underline{g},$$

where the middle inequality uses the regime assumption together with $\mathbb{E}[g_t] \geq \underline{g}$. Written this way, it is clear that the proof is a drift condition at a boundary: the expected update at the floor does not point downward. In applications, strengthening the inequality to a strict margin (e.g. $\geq \mathbb{E}[c_O(a_t)] + \lambda \underline{g} + \epsilon$) yields a cushion that can be useful when translating expectation bounds into probability bounds.

40.5 Reproduction and species (Defs. 281–286): closure and cycle facts

Remark 2364. *The reproduction and species layer formalizes persistence through time by allowing an entity-state S to "generate" future states S' under a reproduction relation. Similarity relations $\approx_{O,\varepsilon}$ define what counts as "the same kind" up to tolerance, while species axioms impose closure conditions under reproduction reachability (Hyperseed-Concept 156, 174). The use of pattern webs as the substrate for similarity is discussed in the broader pattern framework [5] (Hyperseed-Concept 132, 130).*

Remark 2365. *In this subsection, it is useful to keep three distinct ingredients conceptually separate: (i) the* dynamics $\Rightarrow_O$ *(what reproduces from what, relative to observer/context O), (ii) the* typing/identity up to tolerance $\approx_{O,\varepsilon}$ *(what counts as "the same kind" for purposes of continuity), and (iii) the* closure principle *defining a species C (what sets of states are stable under the dynamics). The results below are elementary but serve as reusable "glue" facts: they permit moving between local statements about one-step reproduction and global statements about multi-step reachability, and they quantify how similarity degrades under chaining.*

Helper Lemma 40.14 (Similarity relation: reflexive and symmetric; approximate transitivity under a metric)**.** *Let $\approx_{O,\varepsilon}$ be the within-species similarity relation (Def. 282) induced by a symmetric dissimilarity d_O on pattern webs. Then $\approx_{O,\varepsilon}$ is reflexive and symmetric. If additionally d_O satisfies the triangle inequality, then*

$$S_0 \approx_{O,\varepsilon} S_1 \ \wedge \ S_1 \approx_{O,\varepsilon} S_2 \ \Rightarrow \ S_0 \approx_{O,2\varepsilon} S_2,$$

and along an n-step chain, the tolerance blows up to $n\varepsilon$.

Remark 2366. *This lemma makes explicit what one should expect of any reasonable similarity notion: it is always self-similar (reflexive) and similarity is symmetric. True transitivity is not guaranteed unless d_O is a metric and ε is treated carefully; instead we get a controlled weakening: similarity composes at the cost of increasing tolerance. This is philosophically apt: as Peirce would emphasize, categories of resemblance are habits of generalization, and in chaining them we often accumulate error (Hyperseed-Concept 59, 96).*

Remark 2367. *Example: if d_O is Euclidean distance on an embedding of pattern webs, and $\varepsilon = 0.1$, then two successive "0.1-close" steps imply "0.2-close" overall. The lemma tells you precisely how tolerance degrades under chaining, which matters when defining species via repeated reproduction steps.*

Remark 2368. *A practical reading is that $\approx_{O,\varepsilon}$ behaves like a family of relations indexed by tolerance, and the triangle inequality provides a monotone "composition rule":*

$$(\approx_{O,\varepsilon_1}) \circ (\approx_{O,\varepsilon_2}) \subseteq \approx_{O,\varepsilon_1+\varepsilon_2} .$$

In particular, when ε is chosen to reflect measurement or modeling noise, the $n\varepsilon$ blow-up along an n-step chain is a quantitative reminder that long chains of approximate identifications can drift. This often motivates either (a) keeping chain lengths bounded in later arguments, or (b) re-normalizing by periodically re-anchoring similarity to a canonical representative within a species.

Sketch. Reflexive: $d_O(W, W) = 0 \leq \varepsilon$. Symmetric: symmetry of d_O. Approximate transitivity: apply triangle inequality to the witnessing web-states. □

Remark 2369. Proof sketch (high level). *The first two properties use the axioms of d_O directly; the third uses triangle inequality and adds the error terms.*

Key step. *Triangle inequality yields $d_O(S_0, S_2) \leq d_O(S_0, S_1) + d_O(S_1, S_2) \leq \varepsilon + \varepsilon = 2\varepsilon$.*

Intuition. *Similarity is like being within a radius ε ball; chaining two balls gives a ball of doubled radius.*

Remark 2370. *One can also view the n-step statement as an immediate induction: if $d_O(S_0, S_k) \leq k\varepsilon$ and $d_O(S_k, S_{k+1}) \leq \varepsilon$, then the triangle inequality gives $d_O(S_0, S_{k+1}) \leq (k + 1)\varepsilon$. This form is sometimes more directly usable when a later proof naturally proceeds by stepping through time indices.*

Helper Lemma 40.15 (Forward closure along reproduction reachability)**.** *Let C be a species (Def. 284), and write $\Rightarrow_O^*$ for the reflexive transitive closure of the reproduction relation $\Rightarrow_O$. If $S \in C$ and $S \Rightarrow_O^* S'$, then $S' \in C$.*

Remark 2371. *This lemma states that species membership is stable under any finite number of reproduction steps. It is essentially what it means to treat C as a "closed" class under the reproduction dynamics: once you are in the species, your descendants (in the reachability sense) remain in it. This is the minimal formal backbone of any notion of species as a persistent kind (Hyperseed-Concept 174,* **??***).*

Remark 2372. *Recall that $\Rightarrow_O^*$ is the least relation containing $\Rightarrow_O$ that is both reflexive and transitive: informally, $S \Rightarrow_O^* S'$ means that there exists a finite sequence*

$$S = S_0 \Rightarrow_O S_1 \Rightarrow_O \cdots \Rightarrow_O S_n = S'$$

for some $n \geq 0$. The case $n = 0$ is exactly reflexivity ($S = S'$), and the closure property in the species definition is what lets one extend membership from S_k to S_{k+1} repeatedly.

Sketch. Induct on the length of the $\Rightarrow_O$-chain using the closure axiom of species. $\square$

Remark 2373. Proof sketch (high level). *Induction on chain length: length* 0 *is trivial, length* $n+1$ *reduces to length* n *plus one application of the closure axiom.*

Key step. *The definition of reflexive transitive closure* $\Rightarrow_O^*$ *is precisely designed so that induction applies.*

Visual intuition. *View reproduction as edges in a directed graph. The lemma says: if you start inside* C *and walk along directed edges, you never leave* C*.*

Remark 2374. *In later uses, it is often helpful to keep track of* which *axiom of species is being invoked: the lemma does not require any similarity assumptions, nor any finiteness assumptions, only the forward-closure clause of Def. 284. Thus, it applies equally well to infinite species, branching reproduction, and nondeterministic reproduction (multiple possible* S' *for a given* S*), as long as the closure axiom is stated universally over* $\Rightarrow_O$*-successors.*

Helper Proposition 40.16 (Finite-species regeneration implies a reproduction cycle)**.** *Assume* C *is a finite species and ignore any designated "initial" exception to regeneration. If for every* $S' \in C$ *there exists* $S \in C$ *with* $S \Rightarrow_O S'$*, then the directed graph* $(C, \Rightarrow_O)$ *contains a directed cycle.*

Remark 2375. *This proposition is a graph-theoretic inevitability: in a finite directed graph where every node has at least one incoming edge from within the graph (a "regeneration" condition), you cannot keep tracing predecessors forever without repeating a node. Repetition creates a directed cycle. The result matters because cycles correspond to recurring lineages or steady regenerative loops, which can be interpreted as a structural signature of ongoing reproduction within a bounded repertoire of states (Hyperseed-Concept 155, 156).*

Remark 2376. *Connection to the closure lemma: forward closure says you remain in* C *when moving forward along* $\Rightarrow_O$*; here we reason backward via predecessors. Together they show how finiteness forces recurrence once regeneration is ubiquitous.*

Remark 2377. *The hypothesis "ignore any designated 'initial' exception to regeneration" can be read as: treat regeneration as holding* uniformly *on all of* C *(no privileged seed-state allowed to lack an in-species predecessor). If such an exception were permitted, the conclusion can fail in the obvious way: a finite directed graph may be a rooted in-arborescence with a unique root of in-degree* 0*, producing no directed cycle. Thus, the proposition isolates exactly the condition under which one cannot have a strictly acyclic "ancestry" structure within a finite state-set.*

Sketch. Pick any $S_0 \in C$ and choose predecessors $S_{k+1} \Rightarrow_O S_k$ indefinitely. Since C is finite, some $S_i = S_j$ with $i < j$; the segment $S_{j-1} \Rightarrow \cdots \Rightarrow S_i$ forms a directed cycle. $\square$

Remark 2378. Proof sketch (high level). *Construct an infinite backward path by repeatedly choosing an in-species predecessor; finiteness forces repetition; repetition yields a cycle.*

Remark 2379. *A slightly more explicit way to see the "repetition yields a cycle" step is: from* $S_i = S_j$ *(with* $i < j$*) we have a nonempty chain*

$$S_j = S_i \Leftarrow_O S_{i+1} \Leftarrow_O \cdots \Leftarrow_O S_j$$

when read in the predecessor direction, and hence, when read in the forward $\Rightarrow_O$ *direction, a directed cycle*

$$S_i \Rightarrow_O S_{j-1} \Rightarrow_O \cdots \Rightarrow_O S_i.$$

This is just the pigeonhole principle applied to states visited along the infinite predecessor sequence. The argument does not require determinism: even if there are many possible predecessors at each step, choosing some *predecessor repeatedly is enough to force eventual repetition in a finite set.*

Remark 2380. *Interpreted in reproduction terms, the cycle can be read as the existence of a finite lineage of states in C that "comes back" (possibly through a sequence of distinct intermediate states). In many applications this is the minimal formal shadow of an invariant or steady regime: although individual reproduction events may change the state, the system cannot wander indefinitely through new states if C is finite and internally regenerating.*

Remark 2381. Key step. *The pigeonhole principle: an infinite sequence in a finite set must repeat. Imagine for instance: You are walking backward through ancestors in a village with only finitely many people; if everyone has an ancestor in the village, eventually you meet someone you have already met, closing a loop.*

40.6 Alive/dead p-bits (Defs. 287–289): rewrite rules and threshold entailments

Remark 2382. *The alive/dead layer assigns to an entity-state S a p-bit: a two-coordinate value $(v^+, v^-) \in [0,1]^2$ encoding positive and negative evidence. This is an explicitly paraconsistent style of representation: one can simultaneously have evidence-for and evidence-against, without explosion. Such multi-valued logics are studied systematically in paraconsistent frameworks such as Constructible Duality Logic [23] and related resonance-motivated uses [24]; in the Hyperseed vocabulary it is naturally linked to Alive/Dead (Hyperseed-Concept 54) and to Weakness as graded entailment (Hyperseed-Concept 202) [3, 2].*

Helper Lemma 40.17 (Clamp in Alive is redundant)**.** *Alive is defined (Def. 288) using* $\min\{1, \cdot\}$ *applied to a weighted average of numbers in* $[0,1]$*. Because any weighted average of* $[0,1]$*-values lies in* $[0,1]$*, the* $\min\{1, \cdot\}$ *does not change the value.*

Remark 2383. *The lemma is a small algebraic reassurance: the definition includes a clamp to stay within* $[0,1]$*, but because the computation is already a convex combination of* $[0,1]$ *values, it cannot leave the interval. Such redundancies are common in definitions meant to be robust to generalization; the lemma tells us that in the present setting no special case occurs.*

Sketch. If $x, y \in [0,1]$ and $\alpha, \beta \geq 0$ with $\alpha + \beta > 0$, then $\frac{\alpha x + \beta y}{\alpha + \beta} \leq \frac{\alpha + \beta}{\alpha + \beta} = 1$ and ≥ 0. □

Remark 2384. Proof sketch (high level). *Show a weighted average is bounded by the max and min of its inputs.*
Key step. *Bounding the numerator* $\alpha x + \beta y \leq \alpha + \beta$ *uses* $x, y \leq 1$*; the lower bound uses* $x, y \geq 0$.
Intuition. *Averaging cannot create values outside the range of what is being averaged.*

Helper Proposition 40.18 (Alive monotonicity and bounds)**.** *Let* $\mathrm{Alive}_O(S,t)$ *be the weighted aggregation of* $\mathrm{Met}_O(S,t)$ *and* $\mathrm{Rep}_O(S,t)$ *(Def. 288). Then* $\mathrm{Alive}_O(S,t)$ *is componentwise monotone in each input p-bit. Also, for the positive coordinate,*

$$\min\{\mathrm{Met}_O^+(S,t), \mathrm{Rep}_O^+(S,t)\} \;\leq\; \mathrm{Alive}_O^+(S,t) \;\leq\; \max\{\mathrm{Met}_O^+(S,t), \mathrm{Rep}_O^+(S,t)\},$$

and similarly for the negative coordinate.

Remark 2385. *Monotonicity says: if you increase evidence-for (or evidence-against) in either metabolism or reproduction, you do not decrease the corresponding alive evidence. The min/max bounds say: the aggregation is genuinely an average-like synthesis, not an exotic operation that could produce an "alive" score outside the range suggested by its two inputs. This makes the alive p-bit behave like a reasonable mediator between the metabolic and reproductive aspects of life (Hyperseed-Concept 109, 156, 54).*

Sketch. Weighted averages are monotone and lie between the min and max of their inputs. □

Remark 2386. Proof sketch (high level). *Use elementary properties of convex combinations: coordinatewise,* Alive *is a convex combination of the two coordinates of* Met *and* Rep*.*

Key step. *Componentwise monotonicity follows because increasing one input term increases the convex combination when weights are nonnegative.*

Intuition. Alive *is literally an interpolation between* Met *and* Rep *in each coordinate.*

Helper Corollary 40.19 (Threshold lifting for Alive)**.** *Fix $\theta \in (0,1]$. If $\mathrm{Met}^+_O(S,t) \geq \theta$ and $\mathrm{Rep}^+_O(S,t) \geq \theta$ then $\mathrm{Alive}^+_O(S,t) \geq \theta$. Likewise, if $\mathrm{Met}^-_O(S,t) \geq \theta$ and $\mathrm{Rep}^-_O(S,t) \geq \theta$ then $\mathrm{Alive}^-_O(S,t) \geq \theta$.*

Remark 2387. *This is the familiar stability of lower bounds under averaging: if both contributing signals clear a threshold, so does their weighted blend. It is useful when one reasons in thresholded terms (e.g. "alive at level θ"), because it lets one infer alive-threshold facts from stronger componentwise evidence.*

Sketch. A weighted average of numbers each at least θ is at least θ. □

Remark 2388. Proof sketch (high level). *The average is a convex combination of numbers $\geq \theta$, hence $\geq \theta$.*

Key step. *Multiply the inequality $x \geq \theta$ by a nonnegative weight and sum.*

Intuition. *If every ingredient is at least θ, the mixture cannot fall below θ.*

Helper Lemma 40.20 (WasAlive monotonicity in window length)**.** *Let $\mathrm{WasAlive}_O(S,t)$ be the p-bit join over the lookback window (Def. 289) of length τ. If $\tau' \geq \tau$, then $\mathrm{WasAlive}^{(\tau')}_O(S,t) \geq \mathrm{WasAlive}^{(\tau)}_O(S,t)$ componentwise.*

Remark 2389. *Here "was alive" is an aggregation over past evidence, and enlarging the window can only increase what you have "ever seen" in that window. This is a formal version of the obvious epistemic principle: remembering more cannot reduce the maximum strength of remembered evidence (Hyperseed-Concept 63, 52, 155).*

Sketch. A join over a larger index set is $\geq$ the join over a subset. □

Remark 2390. Proof sketch (high level). *Larger window means more terms in the join; joins are monotone in their index sets.*

Key step. *Any set is a subset of a larger set; the join over the larger set must dominate the join over the subset by definition of $\oplus$ as a join.*

Intuition. *Taking a maximum over more numbers cannot decrease the maximum.*

Helper Corollary 40.21 (Deadness monotone in lookback window)**.** *Since $\mathrm{Dead}_O(S,t) = \mathrm{WasAlive}_O(S,t) \otimes \neg\,\mathrm{Alive}_O(S,t)$ (Def. 289) and $\otimes$ is monotone, $\tau' \geq \tau$ implies $\mathrm{Dead}^{(\tau')}_O(S,t) \geq \mathrm{Dead}^{(\tau)}_O(S,t)$ componentwise.*

Remark 2391. *Deadness is defined as a conjunction-like combination: evidence of having been alive (recently) together with evidence of not being alive now. If the "having been alive" part can only grow when you look back further, then deadness can only grow as well (since the other factor is unchanged). This matches the intended semantics: extending memory makes it easier to accumulate evidence that something* used to be *alive.*

Sketch. Combine the previous lemma with monotonicity of $\otimes$. □

Remark 2392. Proof sketch (high level). *Monotonicity composed with monotonicity: increase* WasAlive*, keep* $\neg$ Alive *fixed, then the tensor product increases.*

Key step. *The property that* $\otimes$ *is monotone in each argument is what makes the inequality propagate.*

Intuition. *If deadness requires two ingredients and one ingredient increases while the other stays the same, the combined score should not decrease.*

Helper Proposition 40.22 (Canonical p-bit expansion of Dead (optional, when using the canonical core))**.** *Assume the canonical p-bit operations used elsewhere in the document:*

$$(p^+,p^-)\oplus(q^+,q^-) = (\max\{p^+,q^+\},\max\{p^-,q^-\}), \quad (p^+,p^-)\otimes(q^+,q^-) = (p^+q^+,p^-q^-), \quad \neg(p^+,p^-) = (p^-,p^+).$$

Then Def. 289 yields:

$$\mathrm{WasAlive}^+_O(S,t) = \max_{u=t-\tau,\dots,t-1} \mathrm{Alive}^+_O(S,u), \qquad \mathrm{WasAlive}^-_O(S,t) = \max_{u=t-\tau,\dots,t-1} \mathrm{Alive}^-_O(S,u),$$

and

$$\mathrm{Dead}^+_O(S,t) = \mathrm{WasAlive}^+_O(S,t)\cdot \mathrm{Alive}^-_O(S,t), \qquad \mathrm{Dead}^-_O(S,t) = \mathrm{WasAlive}^-_O(S,t)\cdot \mathrm{Alive}^+_O(S,t).$$

Remark 2393. *This proposition is a "coordinate unpacking" result: it turns the abstract quantale formula for deadness into explicit max-and-multiply expressions. The value is not merely computational; it makes the semantics transparent. In the canonical core, the positive deadness coordinate is literally "(max past aliveness) times (current not-aliveness)," i.e. a conjunction of two evidential conditions expressed multiplicatively.*

Remark 2394. *This also highlights how the framework combines temporal aggregation (via* max *over the window) with present-time contradiction-tolerant evaluation (via the two coordinates). One may read this as a small instance of the broader theme that time (memory) and logic (evidence combination) are co-constitutive of an observer's ontology [1, 15, 14].*

Sketch. Unfold the join $\oplus$ and then apply coordinatewise multiplication and negation-swap. ☐

Remark 2395. Proof sketch (high level). *Replace* $\oplus$ *by coordinatewise maxima, replace* $\otimes$ *by coordinatewise multiplication, and replace* $\neg$ *by swapping coordinates.*

Key step. *The join over a finite window becomes a maximum because* $\oplus$ *is defined using* max *in each coordinate.*

Intuition. *The abstract expression* WasAlive $\otimes\neg$ Alive *becomes a literal arithmetic formula once the canonical operations are chosen.*

Helper Corollary 40.23 (Deadness entails "was alive" and "not alive now" at thresholds (canonical case))**.** *Assume the canonical p-bit expansion above and fix* $\theta \in (0,1]$*. If* $\mathrm{Dead}^+_O(S,t) \geq \theta$*, then:*

1. $\mathrm{Alive}^-_O(S,t) \geq \theta$ *(strong evidence of not-alive now), and*
2. *there exists some* $u \in \{t-\tau,\dots,t-1\}$ *with* $\mathrm{Alive}^+_O(S,u) \geq \theta$ *(strong evidence of being alive recently).*

Remark 2396. *This corollary spells out a useful threshold entailment: to have strong positive evidence of deadness, one must simultaneously have strong present evidence of not being alive and strong past evidence of being alive. This is exactly how one would want a deadness predicate to behave, and it is reassuring that the canonical p-bit arithmetic makes this precise.*

Remark 2397. *Note the asymmetry: we reason from* Dead^+ *(positive evidence) to conclusions about* Alive^- *and past* Alive^+*. This is a small example of how p-bits let one keep track of which direction of evidence is doing the work, rather than collapsing everything into a single scalar.*

Sketch. If $xy \geq \theta$ with $x, y \in [0,1]$, then $x \geq \theta$ and $y \geq \theta$. Also $\mathrm{WasAlive}^+$ is a maximum over the window. □

Remark 2398. Proof sketch (high level). *Use the fact that in* $[0,1]$ *a product cannot exceed* θ *unless each factor is at least* θ*, and interpret* $\mathrm{WasAlive}^+$ *as a maximum.*

Key step. *From* $\mathrm{Dead}^+_O(S,t) = \mathrm{WasAlive}^+_O(S,t) \cdot \mathrm{Alive}^-_O(S,t) \geq \theta$ *with both factors in* $[0,1]$*, infer both* $\mathrm{WasAlive}^+_O(S,t) \geq \theta$ *and* $\mathrm{Alive}^-_O(S,t) \geq \theta$*. Then* $\mathrm{WasAlive}^+_O(S,t) \geq \theta$ *implies some window time achieves the maximum at least* θ*.*

Intuition. *Strong deadness requires both a strong "used to be alive" witness and a strong "not alive now" witness; the product encodes conjunction, and the maximum encodes existential witness over time.*

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41 Helper theorems for the epistemic layer (Section 20)

41.1 Notation and lightweight macros

Throughout, L is the claim language and $V = [0,1]^2$ is the p-bit quantale. Belief states are valuations $\beta : L \to V$ ordered pointwise by the quantale order. We write $\oplus$ for the pointwise join and $\otimes$ for the quantale tensor. We use the inference operator F_R and closure Cl_R from Definition 298 / Theorem 17, the hypothesis-injection $\beta \uparrow_s h$ from Definition 296, and the leave-one-out state $\beta \setminus \varphi$ from Definition 302.

In concrete terms, since $V = [0,1]^2$ is a product quantale, the order on V is the product order coming from $[0,1]$ (i.e. $(x_1, x_2) \leq (y_1, y_2)$ iff $x_1 \leq y_1$ and $x_2 \leq y_2$), and the induced order on belief states is the corresponding pointwise lifting: $\beta \leq \beta'$ iff $\beta(\varphi) \leq \beta'(\varphi)$ for every $\varphi \in L$. This pointwise viewpoint is used implicitly in all arguments below: inequalities and equations of belief states are always to be read coordinatewise on L and componentwise in $[0,1]^2$.

The pointwise join $\beta \oplus \beta'$ is the belief state defined by $(\beta \oplus \beta')(\varphi) = \beta(\varphi) \oplus \beta'(\varphi)$ for each $\varphi \in L$, where $\oplus$ on the right-hand side is the quantale join in V. Similarly, if γ is another belief state, then $(\beta \otimes \gamma)(\varphi)$ denotes the pointwise tensor $\beta(\varphi) \otimes \gamma(\varphi)$. We will only use the standard order-theoretic facts about these operations: joins are monotone in each argument and satisfy $x \leq x \oplus y$, and the tensor is monotone in each argument (as assumed for a quantale).

When working with $\beta \uparrow_s h$, it is convenient to name the "singleton" valuation $\delta_{h,s}$ that appears in Definition 296: it is the belief state satisfying $\delta_{h,s}(h) = s$ and $\delta_{h,s}(\varphi) = \bot$ for $\varphi \neq h$ (where $\bot$ is the bottom element of V). Thus $\beta \uparrow_s h = \beta \oplus \delta_{h,s}$ is literally "β with h joined up to at least s". Likewise, $\beta \setminus \varphi$ should be read as a valuation identical to β except that the coordinate at φ is reset/removed as specified in Definition 302; this chunk will not use any special property of $\setminus$ beyond the idea that it is a coordinatewise modification.

For the axiomaticity lemma below, recall that β^+ denotes the (epistemic) reading or projection of β used in Definition 295 (e.g. the "acceptance" or "positive" component, depending on the chosen semantics of the p-bit valuation). The only property used is that $\beta \leq \beta'$ implies $\beta^+ \leq \beta'^+$ pointwise, so that lower bounds on $\beta^+(\varphi)$ persist under pointwise increases of β.

41.2 Belief-state update algebra

Lemma 124 (Hypothesis injection is monotone and redundant when already present)**.** *Fix $h \in L$. For belief states $\beta \le \beta'$ and hypothesis strengths $s \le s'$ (in the quantale order),*

$$\beta \uparrow_s h \ \le\ \beta' \uparrow_{s'} h.$$

Moreover, if $\beta(h) \ge s$ then $\beta \uparrow_s h = \beta$.

Sketch. By Definition 296, $\beta \uparrow_s h = \beta \oplus \delta_{h,s}$. Join is monotone in each argument, so increasing either β or s can only increase the result. If $\beta(h) \ge s$ then at the coordinate h we have $\beta(h) \oplus s = \beta(h)$ and all other coordinates are unchanged, hence $\beta \oplus \delta_{h,s} = \beta$. □

The preceding lemma is often used in "idempotence" arguments: injecting the same hypothesis twice with the same strength has no further effect after the first injection, and injecting a weaker strength after a stronger one is redundant by monotonicity. Formally, if $s \le s'$ then

$$(\beta \uparrow_{s'} h) \uparrow_s h \ =\ \beta \uparrow_{s'} h,$$

since $\beta \uparrow_{s'} h$ already has $(\beta \uparrow_{s'} h)(h) \ge s' \ge s$ and thus the second injection collapses to the identity case of the lemma.

Lemma 125 (Join-updates are monotone and inflationary)**.** *For any belief states $\beta, \varepsilon, \varepsilon'$ with $\varepsilon \le \varepsilon'$, we have*

$$\beta \oplus \varepsilon \ \le\ \beta \oplus \varepsilon'.$$

Moreover, $\beta \le \beta \oplus \varepsilon$ for all β, ε.

Sketch. Both statements are coordinatewise consequences of the corresponding facts in the quantale V: join is monotone in each argument, and $x \le x \oplus y$ holds because $x \oplus y$ is the least upper bound of $\{x, y\}$. □

Lemma 126 (Closure is monotone and inflationary)**.** *For any rule set R and belief states $\beta \le \beta'$, one has*

$$\mathrm{Cl}_R(\beta) \ \le\ \mathrm{Cl}_R\big(\beta'\big).$$

In addition, $\beta \le \mathrm{Cl}_R(\beta)$ for all β.

Sketch. By Definition 298 / Theorem 17, Cl_R is the closure induced by the inference operator F_R (typically as an iterated or least fixed-point construction). The standard closure properties established there imply monotonicity of F_R and hence of Cl_R, as well as extensivity (inflationarity) $\beta \le \mathrm{Cl}_R(\beta)$. Concretely, both inequalities can be read pointwise: the closure adds consequences but does not decrease any coordinate. □

Lemma 127 (Axiomaticity is upward closed)**.** *Let (A, θ) be an axiom system (Definition 295). If β treats (A, θ) as axiomatic and $\beta \le \beta'$, then β' also treats (A, θ) as axiomatic. In particular, if updates are of the form $\beta \leftarrow \beta \oplus \varepsilon$ and/or $\beta \leftarrow \mathrm{Cl}_R(\beta)$, then once axiomatic, always axiomatic.*

Sketch. Treating (A, θ) as axiomatic means $\beta^+(\varphi) \ge \theta$ for all $\varphi \in A$. Since $\beta \le \beta'$ pointwise, $\beta'^+(\varphi) \ge \beta^+(\varphi) \ge \theta$. Both $\oplus$-updates and Cl_R are inflationary (see below), so they preserve the pointwise lower bound. □

The "once axiomatic, always axiomatic" clause is intended to be read operationally: if an algorithm maintains a belief state and interleaves (i) additive evidence updates $\beta \leftarrow \beta \oplus \varepsilon$ with (ii) logical saturation $\beta \leftarrow \mathrm{Cl}_R(\beta)$, then the set of axioms whose acceptance has crossed the threshold θ is monotone in time. The earlier lemmas isolate the only ingredients needed for that monotonicity: pointwise order on belief states and inflationary update operators.

41.3 Inference and closure

Lemma 128 (F_R is monotone and inflationary)**.** *For any rule set R, the operator $F_R((\cdot)) : V^L \to V^L$ is monotone: $\beta \le \beta' \Rightarrow F_R(\beta) \le F_R(\beta')$. Also it is inflationary: $\beta \le F_R(\beta)$.*

Sketch. Recall that the order on V^L is the pointwise order: $\beta \le \beta'$ means $\beta(\psi) \le \beta'(\psi)$ for all formulas $\psi \in L$. Thus to show $F_R(\beta) \le F_R(\beta')$ it suffices to fix an arbitrary ψ and show $F_R(\beta)\,(\psi) \le F_R(\beta')\,(\psi)$ in V.

Definition 298 builds $F_R(\beta)\,(\psi)$ from $\beta(\psi)$ by joining it with rule-contributions, each of which is a monotone expression in β using $\otimes$ and $\oplus$. More explicitly, each contribution has the form

$$\lambda_r \otimes \Big(\bigotimes_{\varphi \in \Gamma} \beta(\varphi)\Big)$$

for a rule $r = (\Gamma, \psi, \lambda_r) \in R$, and $F_R(\beta)\,(\psi)$ is obtained by taking $\beta(\psi)$ and then joining (via $\oplus$ / the relevant join operation) all such contributions that target ψ. If $\beta \le \beta'$, then for every premise $\varphi \in \Gamma$ we have $\beta(\varphi) \le \beta'(\varphi)$, so by monotonicity of $\otimes$ in each argument we get

$$\bigotimes_{\varphi \in \Gamma} \beta(\varphi) \;\le\; \bigotimes_{\varphi \in \Gamma} \beta'(\varphi), \qquad \text{hence} \qquad \lambda_r \otimes \Big(\bigotimes_{\varphi \in \Gamma} \beta(\varphi)\Big) \;\le\; \lambda_r \otimes \Big(\bigotimes_{\varphi \in \Gamma} \beta'(\varphi)\Big).$$

Joining these inequalities with the (also monotone) $\oplus$ operation preserves the order, and additionally $\beta(\psi) \le \beta'(\psi)$ contributes the base-term monotonicity. Therefore the entire join defining $F_R(\beta)\,(\psi)$ is $\le$ the corresponding join defining $F_R(\beta')\,(\psi)$, establishing monotonicity pointwise and hence globally.

Inflationary holds because $\beta(\psi)$ is one of the joined terms, so $\beta(\psi) \le F_R(\beta)\,(\psi)$ for all ψ. Equivalently, the defining join for $F_R(\beta)\,(\psi)$ includes $\beta(\psi)$ as a summand, and any join is above each of its components. □

Theorem 56 (Cl_R is a closure operator)**.** *Assume $\mathrm{Cl}_R(\beta)$ is defined as in Theorem 17 (least fixed point of F_R above β). Then Cl_R satisfies the standard closure-operator laws:*

$$\textit{(Extensive)}\ \beta \le \mathrm{Cl}_R(\beta)\,, \qquad \textit{(Monotone)}\ \beta \le \beta' \Rightarrow \mathrm{Cl}_R(\beta) \le \mathrm{Cl}_R\big(\beta'\big)\,, \qquad \textit{(Idempotent)}\ \mathrm{Cl}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)\,.$$

Moreover, $\mathrm{Cl}_R(\beta)$ *is the* least R-closed *belief state γ with $\beta \le \gamma$.*

Sketch. Extensiveness follows since $\beta \le F_R(\beta) \le F_R^2(\beta) \le \cdots$ and $\mathrm{Cl}_R(\beta)$ is the join of this ascending chain (Theorem 17). Here the inequalities $\beta \le F_R(\beta) \le F_R^2(\beta) \le \cdots$ come from inflationarity of F_R together with its monotonicity: from $\beta \le F_R(\beta)$ and monotonicity we obtain $F_R(\beta) \le F_R(F_R(\beta)) = F_R^2(\beta)$, and so on by induction. Since a join (supremum) of an ascending chain is above every element of the chain, it is in particular above the first element β.

Monotonicity follows because F_R is monotone, hence so are all iterates, and joins preserve order. Concretely, if $\beta \le \beta'$, then by monotonicity of F_R we have $F_R^n(\beta) \le F_R^n(\beta')$ for each $n \in \mathbb{N}$ (induction on n). Taking the join over n on both sides preserves $\le$ because joins are computed pointwise in V^L and each pointwise join in V is monotone in its indexed family.

Idempotence holds because $\mathrm{Cl}_R(\beta)$ is a fixed point of F_R; applying Cl_R again yields the least fixed point above an already fixed point, hence itself. More explicitly, Theorem 17 ensures that $\mathrm{Cl}_R(\beta)$ is R-closed, i.e. $F_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)$, and also that $\mathrm{Cl}_R(\beta) \ge \beta$. Thus, when forming $\mathrm{Cl}_R(\mathrm{Cl}_R(\beta))$, we are taking the least fixed point of F_R above $\mathrm{Cl}_R(\beta)$; but $\mathrm{Cl}_R(\beta)$ is itself a fixed point above $\mathrm{Cl}_R(\beta)$, so by minimality the least such fixed point cannot be strictly smaller or larger, forcing equality.

Leastness is the standard Knaster–Tarski/Kleene fixed-point argument used in Theorem 17. In particular, if γ is any fixed point of F_R with $\beta \le \gamma$, then (by monotonicity) the entire Kleene chain above β is bounded by γ, i.e. $F_R^n(\beta) \le \gamma$ for all n, and therefore its join $\mathrm{Cl}_R(\beta)$ is also $\le \gamma$. This shows that $\mathrm{Cl}_R(\beta)$ is the least among all R-closed γ extending β. □

Corollary 20 (Characterizations of R-closed states)**.** *For a belief state β, the following are equivalent: (i) β is R-closed, i.e. $F_R(\beta) = \beta$; (ii) $\mathrm{Cl}_R(\beta) = \beta$; (iii) for every rule $r = (\Gamma, \psi, \lambda_r) \in R$,*

$$\lambda_r \otimes \Big(\bigotimes_{\varphi\in\Gamma} \beta(\varphi)\Big) \;\le\; \beta(\psi).$$

Sketch. (i)⇔(ii) is immediate from the definition of Cl_R as the least fixed point above β. Indeed, if β is already a fixed point of F_R, then the least fixed point above β is β itself; conversely, if $\mathrm{Cl}_R(\beta) = \beta$, then β must be a fixed point because $\mathrm{Cl}_R(\beta)$ is R-closed by construction.

(i)⇔(iii) follows by unpacking Definition 298: $F_R(\beta)\,(\psi)$ is $\beta(\psi)$ joined with all contributions to ψ, so equality holds iff each contribution is already $\le \beta(\psi)$. More detail: for the forward direction, assume $F_R(\beta) = \beta$. Fix a rule $r = (\Gamma, \psi, \lambda_r) \in R$ and consider the corresponding contribution term $\lambda_r \otimes \big(\bigotimes_{\varphi\in\Gamma} \beta(\varphi)\big)$, which appears among the joined terms defining $F_R(\beta)\,(\psi)$. Since a join is an upper bound of its components, we have

$$\lambda_r \otimes \Big(\bigotimes_{\varphi\in\Gamma} \beta(\varphi)\Big) \;\le\; F_R(\beta)\,(\psi) = \beta(\psi),$$

which is exactly (iii). For the reverse direction, assume (iii) holds for every rule. Then for each ψ, every rule-contribution targeting ψ is $\le \beta(\psi)$, and of course the base term $\beta(\psi) \le \beta(\psi)$. Hence the join of $\beta(\psi)$ with all these contributions is still $\beta(\psi)$, so $F_R(\beta)\,(\psi) = \beta(\psi)$ for all ψ, i.e. $F_R(\beta) = \beta$. □

Lemma 129 (Adding rules increases closure)**.** *If $R \subseteq R'$, then for all belief states β,*

$$\mathrm{Cl}_R(\beta) \;\le\; \mathrm{Cl}_{R'}(\beta)\,.$$

Sketch. For each β, Definition 298 gives $F_R(\beta) \le F_{R'}(\beta)$ pointwise because R' has (at least) all the join-terms that R has. To see this pointwise, fix ψ and observe that the set of contributions to ψ generated from rules in R is a subset of the contributions generated from rules in R'; joining over a larger family can only increase (or preserve) the value in the lattice order. Since the base term $\beta(\psi)$ is the same in both $F_R(\beta)\,(\psi)$ and $F_{R'}(\beta)\,(\psi)$, the only difference is the presence of additional contribution terms coming from $R' \setminus R$, which can only enlarge the join.

Iterating preserves the inequality, and joining over all iterates preserves it, yielding the claim. More explicitly, from $F_R(\beta) \le F_{R'}(\beta)$ and monotonicity of $F_R((\cdot))$ and $F_{R'}((\cdot))$ (as operators on V^L) one obtains by induction that $F_R^n(\beta) \le F_{R'}^n(\beta)$ for each n. Taking joins of these ascending chains (as in Theorem 17) then gives $\mathrm{Cl}_R(\beta) \le \mathrm{Cl}_{R'}(\beta)$. Equivalently, since $\mathrm{Cl}_{R'}(\beta)$ is R'-closed it is in particular closed under all rules in R, so it is an R-closed extension of β; by the leastness clause of the previous theorem, the least such extension is $\mathrm{Cl}_R(\beta)$, hence $\mathrm{Cl}_R(\beta) \le \mathrm{Cl}_{R'}(\beta)$. □

41.4 Batching and order-independence of evidence

The following facts formalize that (a) repeatedly "closing" a belief state does not interfere with subsequently adding evidence, and (b) a sequence of evidence joins interleaved with closures can be equivalently performed as one big join followed by one final closure. Intuitively, Cl_R behaves like a closure operator on belief states (extensive, monotone, idempotent) that enforces the R-constraints, while $\oplus$ aggregates information by pointwise joining in V.

Lemma 130 (Closure commutes with further evidence joins)**.** *For any belief state β and any evidence increment $\varepsilon : L \to V$,*

$$\mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \ \oplus \ \varepsilon) \ = \ \mathrm{Cl}_R(\beta \ \oplus \ \varepsilon) \,.$$

Sketch. Recall that $\mathrm{Cl}_R(\beta)$ is the *least* (with respect to $\leq$) R-closed belief state that lies above β; in particular, Cl_R is extensive ($\beta \leq \mathrm{Cl}_R(\beta)$), monotone, and idempotent ($\mathrm{Cl}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)$). Also, $\beta \oplus \varepsilon$ denotes the join of two belief states (or of a belief state with an increment) in the underlying pointwise order, so it is monotone in each argument.

Since $\beta \leq \mathrm{Cl}_R(\beta)$, monotonicity gives $\beta \oplus \varepsilon \leq \mathrm{Cl}_R(\beta) \oplus \varepsilon$, and then applying monotonicity of Cl_R yields

$$\mathrm{Cl}_R(\beta \oplus \varepsilon) \leq \mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \oplus \varepsilon).$$

This is the first inequality.

Conversely, $\mathrm{Cl}_R(\beta \oplus \varepsilon)$ is R-closed and dominates both β and ε, hence it also dominates $\mathrm{Cl}_R(\beta)$ (because $\mathrm{Cl}_R(\beta)$ is the least R-closed state above β). Concretely: from $\beta \leq \beta \oplus \varepsilon \leq \mathrm{Cl}_R(\beta \oplus \varepsilon)$ and R-closedness of $\mathrm{Cl}_R(\beta \oplus \varepsilon)$, minimality of $\mathrm{Cl}_R(\beta)$ implies $\mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_R(\beta \oplus \varepsilon)$. Together with $\varepsilon \leq \beta \oplus \varepsilon \leq \mathrm{Cl}_R(\beta \oplus \varepsilon)$, we obtain

$$\mathrm{Cl}_R(\beta) \oplus \varepsilon \ \leq \ \mathrm{Cl}_R(\beta \oplus \varepsilon).$$

Applying Cl_R to both sides and using idempotence on the right-hand side (since $\mathrm{Cl}_R(\beta \oplus \varepsilon)$ is already R-closed) gives

$$\mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \oplus \varepsilon) \ \leq \ \mathrm{Cl}_R(\beta \oplus \varepsilon),$$

which is the reverse inequality. Combining both directions yields equality. □

Corollary 21 (Order-independence and batching of multiple updates)**.** *Let $\varepsilon_1, \ldots, \varepsilon_n : L \to V$ be evidence increments. Define the iterative update $\beta_{k+1} := \mathrm{Cl}_R(\beta_k \oplus \varepsilon_{k+1})$. Then*

$$\beta_n \ = \ \mathrm{Cl}_R(\beta_0 \ \oplus \ \varepsilon_1 \ \oplus \ \cdots \ \oplus \ \varepsilon_n) \,.$$

In particular, the final result is independent of the order of the ε_k (since $\oplus$ is commutative/associative).

Sketch. The key point is that the previous lemma allows us to "slide" any intermediate closure past subsequent joins, so that all closures can be postponed until after all evidence increments have been aggregated.

Induct on n. For $n = 1$ the statement is exactly the definition of β_1. For the induction step, assume

$$\beta_n = \mathrm{Cl}_R(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n).$$

Then

$$\beta_{n+1} = \mathrm{Cl}_R(\beta_n \oplus \varepsilon_{n+1}) = \mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n) \oplus \varepsilon_{n+1}),$$

and by the lemma this equals

$$\mathrm{Cl}_R(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n \oplus \varepsilon_{n+1}).$$

Associativity of $\oplus$ justifies omitting parentheses in the accumulated join. Finally, commutativity/associativity of $\oplus$ implies that the joined term $\varepsilon_1 \oplus \cdots \oplus \varepsilon_n$ depends only on the multiset of increments, so the same closed result is obtained regardless of the order in which the ε_k are processed. □

Corollary 22 (Thinking episodes collapse to "closure of accumulated evidence"). *In a thinking episode (Definition 300), suppose we perform a finite sequence of steps that (i) joins in evidence increments $\varepsilon_1, \dots, \varepsilon_n$, and (ii) optionally injects hypotheses δ_{h_j,s_j} (Definition 296), with closure Cl_R performed any number of times in-between. Then the final belief state equals*

$$\mathrm{Cl}_R(\beta_0 \ \oplus \ \varepsilon_1 \ \oplus \cdots \oplus \ \varepsilon_n \ \oplus \ \delta_{h_1,s_1} \ \oplus \cdots \oplus \ \delta_{h_m,s_m}) \, .$$

Sketch. View each operation in the episode as producing a new belief state from the old one by one of two kinds of moves: either (a) joining in some increment (one of the ε_i or one of the hypothesis-injections δ_{h_j,s_j}), or (b) applying Cl_R. The lemma "Closure commutes with further evidence joins" shows that whenever a closure is followed later by additional joins, we may equivalently postpone that closure until after those joins have occurred, without changing the end result.

More explicitly, repeatedly use the lemma to push every intermediate $\mathrm{Cl}_R(\,\cdot\,)$ to the right, step-by-step, past subsequent $\oplus$-updates, until only a single final closure remains. At that point the belief state is Cl_R of β_0 joined with all increments that were ever added during the episode. Since $\oplus$ is associative (and, when relevant, commutative), all joins can be batched into the single accumulated expression displayed in the statement, regardless of how the joins and closures were interleaved during the episode. □

41.5 Community aggregation and belief-system scores

Lemma 131 (Dominance and monotonicity of community aggregation). *Let M be a finite set of agents with weights $w_i \in V$ and belief states $\beta_i : L \to V$, and let β_M be as in Definition 301. Then:*

$$w_i \otimes \beta_i(\varphi) \ \le \ \beta_M(\varphi) \quad \text{for all } i \in M, \ \varphi \in L.$$

If $M \subseteq M'$ and the weights/beliefs on M are unchanged, then $\beta_M \le \beta_{M'}$. If $\beta_i \le \beta'_i$ for all i, then the aggregate increases: $\beta_M \le \beta'_M$.

Sketch. All statements follow from the fact that $\beta_M(\varphi)$ is a join over $i \in M$ of the terms $w_i \otimes \beta_i(\varphi)$, and join is monotone and dominates each joined term. More explicitly, Definition 301 has the form

$$\beta_M(\varphi) = \bigvee_{i \in M} \big(w_i \otimes \beta_i(\varphi)\big),$$

so for any fixed $i \in M$ the element $w_i \otimes \beta_i(\varphi)$ is one of the joinands and hence $w_i \otimes \beta_i(\varphi) \le \bigvee_{j \in M}(w_j \otimes \beta_j(\varphi)) = \beta_M(\varphi)$ by the universal property of joins. For set monotonicity, if $M \subseteq M'$ and the terms for indices in M are unchanged, then the join for M' ranges over a superset of joinands, so $\bigvee_{i \in M}(\cdots) \le \bigvee_{i \in M'}(\cdots)$. For belief monotonicity, if $\beta_i \le \beta'_i$ pointwise, then for each φ we have $w_i \otimes \beta_i(\varphi) \le w_i \otimes \beta'_i(\varphi)$ (by monotonicity of $\otimes$ in its second argument, as assumed in the underlying V-structure), and taking joins over $i \in M$ preserves $\le$. Consequently $\beta_M(\varphi) \le \beta'_M(\varphi)$ for all φ, i.e. $\beta_M \le \beta'_M$. □

Lemma 132 (Coherence score bounds and a simple "support witness" property). *Let $\mathrm{Coh}_R(B; \beta)$ be as in Definition 302. Then $0 \le \mathrm{Coh}_R(B; \beta) \le 1$. Moreover, there exists some $\varphi^\star \in B$ such that*

$$\pi(\mathrm{Cl}_R(\beta \backslash \varphi^\star)\,(\varphi^\star)) \ \ge \ \mathrm{Coh}_R(B; \beta) \, .$$

Sketch. Each summand is a π-projection of a p-bit, hence lies in $[0, 1]$, so the average lies in $[0, 1]$. For the witness: the maximum of finitely many numbers is at least their average. To make the bounds explicit, write Definition 302 in the schematic form

$$\mathrm{Coh}_R(B; \beta) = \frac{1}{|B|} \sum_{\varphi \in B} \pi(\mathrm{Cl}_R(\beta \backslash \varphi)\,(\varphi)) \, ,$$

where $|B| < \infty$ (as B is a finite belief set in the intended use of the score). Since $\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi) \in V$ is a p-bit (or, more generally, lies in the domain on which π takes values in $[0,1]$), each term $\pi(\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi)) \in [0,1]$. A finite average of numbers in $[0,1]$ also lies in $[0,1]$, establishing $0 \leq \mathrm{Coh}_R(B;\beta) \leq 1$. For the witness property, consider the finite set of real numbers

$$a_\varphi := \pi(\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi)) \qquad (\varphi \in B).$$

Let $\varphi^\star$ be an element of B attaining $\max_{\varphi\in B} a_\varphi$ (existence follows from finiteness of B). Then $a_{\varphi^\star} \geq \frac{1}{|B|}\sum_{\varphi\in B} a_\varphi = \mathrm{Coh}_R(B;\beta)$, which is exactly the displayed inequality. Intuitively, the lemma states that the coherence score cannot exceed the best single leave-one-out self-support value among the sentences in B. $\square$

Lemma 133 (Surprise-adjusted implication is bounded and prefers simpler claims at equal support)**.** *Let* $\mathrm{Prior}(\varphi) = e^{-\lambda \mathrm{Comp}(\varphi)}$ *and* $\mathrm{SAI}(\varphi;\beta,R)$ *be as in Definition 303. Then* $0 \leq \mathrm{SAI}(\varphi;\beta,R) \leq \mathrm{Prior}(\varphi) \leq 1$*. If two claims* φ,ψ *have equal implied plausibility under leave-one-out closure, i.e.*

$$\pi(\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi)) = \pi(\mathrm{Cl}_R(\beta\backslash\psi)\,(\psi))\,,$$

and $\mathrm{Comp}(\varphi) < \mathrm{Comp}(\psi)$*, then* $\mathrm{SAI}(\varphi;\beta,R) > \mathrm{SAI}(\psi;\beta,R)$*.*

Sketch. Boundedness is immediate since $\pi(\cdot) \in [0,1]$ and $\mathrm{Prior}(\varphi) \in (0,1]$. For the preference: equal plausibility makes the $\pi(\cdot)$ factors equal, while Prior is strictly decreasing in Comp when $\lambda > 0$. In more detail, Definition 303 has the multiplicative structure

$$\mathrm{SAI}(\varphi;\beta,R) = \pi(\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi)) \cdot \mathrm{Prior}(\varphi),$$

where $\pi(\cdot) \in [0,1]$ by construction and $\mathrm{Prior}(\varphi) = e^{-\lambda\mathrm{Comp}(\varphi)}$. Assuming $\lambda \geq 0$ and $\mathrm{Comp}(\varphi) \geq 0$, we have $\mathrm{Prior}(\varphi) \in (0,1]$, and hence $0 \leq \mathrm{SAI}(\varphi;\beta,R) \leq 1$. Moreover, since $\pi(\cdot) \leq 1$, the product is bounded above by $\mathrm{Prior}(\varphi)$, yielding $0 \leq \mathrm{SAI}(\varphi;\beta,R) \leq \mathrm{Prior}(\varphi) \leq 1$ as stated. For the comparative claim, set

$$s := \pi(\mathrm{Cl}_R(\beta\backslash\varphi)\,(\varphi)) = \pi(\mathrm{Cl}_R(\beta\backslash\psi)\,(\psi))\,.$$

Then $\mathrm{SAI}(\varphi;\beta,R) = s \cdot e^{-\lambda\mathrm{Comp}(\varphi)}$ and $\mathrm{SAI}(\psi;\beta,R) = s \cdot e^{-\lambda\mathrm{Comp}(\psi)}$. If $\lambda > 0$ and $\mathrm{Comp}(\varphi) < \mathrm{Comp}(\psi)$, then $e^{-\lambda\mathrm{Comp}(\varphi)} > e^{-\lambda\mathrm{Comp}(\psi)}$, and multiplying by the common nonnegative factor s preserves the strict inequality, giving $\mathrm{SAI}(\varphi;\beta,R) > \mathrm{SAI}(\psi;\beta,R)$. This captures the intended "simplicity bonus": at equal implied plausibility (support), the less complex formula receives the higher surprise-adjusted score. $\square$

41.6 Paraconsistent "non-explosion" helper facts

Lemma 134 (Irrelevance: claims with no incoming rules do not change under closure)**.** *If no rule in* R *has conclusion* ψ*, then* $F_R(\beta)\,(\psi) = \beta(\psi)$ *and hence* $\mathrm{Cl}_R(\beta)\,(\psi) = \beta(\psi)$*. In particular, if* $\beta(\psi) = (0,0)$ *initially and* ψ *is never a conclusion, then* ψ *stays at* $(0,0)$ *forever.*

Sketch. In Definition 298, the join over rules concluding ψ is empty, so the update leaves $\beta(\psi)$ unchanged. All iterates leave it unchanged, hence so does their join (Theorem 17). Moreover, the pointwise nature of $F_R(\beta)$ ensures that the value at ψ depends only on rules whose conclusion is exactly ψ: when there are none, neither supporting nor opposing components for ψ can be generated "indirectly" by the update operator itself. Equivalently, for this particular coordinate ψ, $F_R(\beta)$ acts as the identity map, so every iterate $F_R(\beta)^k\,(\psi)$ equals $\beta(\psi)$, and the closure (defined as the join/supremum of these iterates in Theorem 17) must therefore coincide with $\beta(\psi)$ as well. $\square$

Remark 2399 (Why this expresses paraconsistent "non-explosion" at a coordinate). *The lemma isolates a minimal "non-explosion" guarantee: absent an explicit rule concluding ψ, the closure process cannot manufacture any nonzero support against ψ out of unrelated contradictions elsewhere in the theory. In particular, even if the ambient belief state contains inconsistent information about other claims (e.g., some φ has both positive and negative evidence), such inconsistency does not propagate to ψ unless R contains a rule whose conclusion is ψ. This is a localized form of irrelevance: contradiction does not automatically entail arbitrary conclusions, because entailment is mediated by the presence of a rule with the relevant conclusion.*

Remark 2400 (A useful contrapositive reading). *If one observes that $\mathrm{Cl}_R(\beta)(\psi) \neq \beta(\psi)$ for some β, then necessarily there exists at least one rule in R whose conclusion is ψ (or, more precisely, there exists a nonempty set of rules concluding ψ that contributes nontrivially to the join in Definition 298). Thus any change at ψ can be traced to an explicit "incoming edge" in the rule graph, which is often convenient when diagnosing unexpected belief propagation.*

41.7 Scientific update as a special case of the evidence-batching laws

Corollary 23 (Scientific updates batch and commute). *Let $\varepsilon_{e,o}$ be the empirical evidence increment from Definition 307. After a finite sequence of experiments (e_k, o_k), repeated scientific update steps yield*

$$\beta_n \;=\; \mathrm{Cl}_R(\beta_0 \;\oplus\; \varepsilon_{e_1,o_1} \;\oplus\; \cdots \;\oplus\; \varepsilon_{e_n,o_n})\,.$$

Thus, within a fixed methodological rule set R, the final closed belief state depends only on the (join-)aggregate of the empirical increments, not on their order.

Sketch. This is exactly the batching corollary above, instantiated with $\varepsilon_k = \varepsilon_{e_k,o_k}$. Concretely, each scientific update step adds an increment via $\oplus$ and then re-closes under the fixed rule set R; batching states that repeated alternation of "add evidence" and "close" is equivalent to first aggregating all increments by repeated $\oplus$ and then applying a single closure. Commutativity with respect to the order of experiments follows because the right-hand side depends only on the $\oplus$-combination of increments, so any permutation of (e_k, o_k) yields the same aggregate and hence the same $\mathrm{Cl}_R(\cdot)$. $\square$

Remark 2401 (Interpretation and scope of the commutation claim). *The corollary should be read as an "order-independence" result under two fixed ingredients: the methodological rule set R (kept constant across updates) and the empirical increments ε_{e_k,o_k} (treated as exogenous inputs). Under these conditions, the closure operator $\mathrm{Cl}_R(\cdot)$ is applied to the same aggregated pre-closure state regardless of the chronological order in which the increments arrived. This captures a standard methodological idealization: when evidence items are merely accumulated (by $\oplus$) and then the same closure rules are applied, the final closed state is a function of the multiset of evidence items, not of their temporal sequencing.*

41.8 Helper theorems for practical reasoning with tasks, intelligence, and agency

Lemma 135 (Bounds and monotonicity of plausibility scalarization). *Let $v = (v^+, v^-) \in [0,1]^2$ and $\mathrm{pl}(v) := \frac{1+v^+-v^-}{2}$ (Definition 314). Then:*

1. $0 \le \mathrm{pl}(v) \le 1$.
2. *If $u^+ \le v^+$ and $u^- \ge v^-$ then* $\mathrm{pl}(u) \le \mathrm{pl}(v)$.
3. pl *is affine in each coordinate:* $\mathrm{pl}(\alpha v + (1-\alpha)w) = \alpha \mathrm{pl}(v) + (1-\alpha)\mathrm{pl}(w)$.

Proof sketch. (1) Since $v^+ - v^- \in [-1, 1]$, we have $\mathrm{pl}(v) \in [0, 1]$. (2) Immediate from the formula. (3) Immediate by linearity of addition and scalar multiplication.
For (1), note explicitly that $\mathrm{pl}(v) = \frac{1}{2} + \frac{1}{2}(v^+ - v^-)$, i.e. pl is the midpoint-shifted, halved version of the signed difference $v^+ - v^-$. Since $v^+, v^- \in [0, 1]$, the extreme cases are $(v^+, v^-) = (0, 1)$ giving $\mathrm{pl} = 0$ and $(1, 0)$ giving $\mathrm{pl} = 1$, with all other cases interpolating linearly.
For (2), the monotonicity statement is with respect to the partial order where increasing the "positive" coordinate and decreasing the "negative" coordinate can only increase plausibility; the condition $u^+ \leq v^+$ and $u^- \geq v^-$ yields $(u^+ - u^-) \leq (v^+ - v^-)$ and hence $\mathrm{pl}(u) \leq \mathrm{pl}(v)$.
For (3), the displayed identity can also be checked componentwise: if $x = \alpha v + (1 - \alpha)w$, then $x^+ = \alpha v^+ + (1 - \alpha)w^+$ and $x^- = \alpha v^- + (1 - \alpha)w^-$, so substitution into $\mathrm{pl}(x) = \frac{1+x^+-x^-}{2}$ yields the claimed affine combination. ☐

Corollary 24 (Competence is always a valid $(0, 1)$ score)**.** *For any task T (Definition 312) and any policy π (Definition 311),* $\mathrm{pl}(J_T(\pi)) \in [0, 1]$*. Hence for any agent Ag (Definition 317),* $\mathrm{Comp}(Ag; T) := \mathrm{pl}(J_T(\pi_{Ag,T})) \in [0, 1]$ *(Definition 318).*

Proof sketch. $J_T(\pi)$ is an expectation of a $[0, 1]^2$-valued random variable, hence remains in $[0, 1]^2$. Apply the previous lemma.
Concretely, if the per-episode score is a random vector $X = (X^+, X^-)$ with $X^+, X^- \in [0, 1]$ almost surely, then $J_T(\pi) = \mathbb{E}[X] = (\mathbb{E}[X^+], \mathbb{E}[X^-])$ and each expectation lies in $[0, 1]$ by basic properties of expectation (e.g. $\mathbb{E}[X^+] \leq \mathbb{E}[1] = 1$ and $\mathbb{E}[X^+] \geq \mathbb{E}[0] = 0$). The lemma then converts the vector-valued score to a scalar in a way that preserves the unit interval bounds. ☐

Proposition 56 (Value preservation under task reduction implies competence preservation)**.** *Let $F : T \to T'$ be a reduction (Definition 316). For any policy π' on T', let π be the induced policy on T from Proposition 31 ("Transfer by reduction"). Then:*

$$J_T(\pi) = J_{T'}(\pi') \quad \textit{and hence} \quad \mathrm{pl}(J_T(\pi)) = \mathrm{pl}(J_{T'}(\pi')).$$

Proof sketch. This is Proposition 31 for the p-bit value equality, followed by applying pl.
The second equality is simply the functoriality of composing functions: once the reduction guarantees that the underlying two-coordinate value vector is exactly preserved (not merely bounded), applying any deterministic scalarization map—here pl—to both sides yields equality of scalar scores. In particular, no additional assumptions (such as continuity or monotonicity) are needed beyond pl being a well-defined function on the codomain of $J_T(\pi)$. ☐

Corollary 25 (Agent transfer along a reduction (compile-time porting rule))**.** *Let $F : T \to T'$ be a reduction with maps (f_O, f_A) as in Definition 316. Given any agent Ag' intended to operate on T', define a ported agent F^*Ag' for T by:*

1. *encode each perceived observation $o \in O$ as $f_O(o) \in O'$ before feeding it to Ag',*
2. *sample an abstract action $a' \sim \pi_{Ag',T'}(\cdot)$,*
3. *execute the decoded action $a := f_A(o, a')$ in T.*

Then $\mathrm{Comp}(F^*Ag'; T) = \mathrm{Comp}(Ag'; T')$.

Proof sketch. The ported closed-loop interaction implements exactly the induced policy construction of Proposition 31; the value equality then gives competence equality.
Intuitively, the ported agent never needs to "understand" T directly: it only needs the ability to (i) reinterpret concrete observations as abstract observations, and (ii) reinterpret abstract actions

as concrete actions in a way that is allowed to depend on the current concrete observation (as encoded in $f_A(o, a')$). Because the reduction ensures that this interface preserves the trajectory distribution relevant to the value definition, the resulting competence, after scalarization by pl, is identical across T and T'. $\square$

Proposition 57 (Functoriality of policy lifting). *Let $F : T \to T'$ and $G : T' \to T''$ be reductions. For any policy π'' on T'', let $(G^*\pi'')$ be the policy on T' induced by Proposition 31, and similarly define $F^*(\cdot)$. Then:*

$$(G \circ F)^*\pi'' \ = \ F^*(G^*\pi'').$$

Proof sketch. Both sides encode histories by composing observation maps and decode actions by the composite decoder from Proposition 30. One checks inductively on history length that the sampled abstract actions and executed concrete actions have the same conditional distributions, hence define the same policy on T.

The key point is that the "lift" operation $(\cdot)^*$ does not introduce any additional randomness beyond that already present in π'': at each time step, it applies a deterministic observation transform (possibly on the whole history, depending on the formalism) and then pushes forward the sampled action through the corresponding decoder. Because reduction composition (Proposition 30) composes these deterministic transforms in the expected way, lifting along $G \circ F$ agrees with lifting along G and then along F. $\square$

Lemma 136 (Reductions form a preorder; equivalence is an equivalence relation). *Define $T \le_{\mathrm{red}} T'$ iff there exists a reduction $F : T \to T'$ (Definition 316). Then $\le_{\mathrm{red}}$ is reflexive and transitive. Define $T \simeq_{\mathrm{red}} T'$ iff $T \le_{\mathrm{red}} T'$ and $T' \le_{\mathrm{red}} T$. Then $\simeq_{\mathrm{red}}$ is an equivalence relation.*

Proof sketch. Reflexivity uses the identity reduction (Corollary 2). Transitivity uses composition (Proposition 30). Equivalence relation properties are immediate from preorder properties.

More explicitly, reflexivity means every task reduces to itself, witnessed by choosing $f_O = \mathrm{id}$ and $f_A(o, a) = a$ (in the notation of Definition 316), which is exactly the identity reduction. Transitivity means that if one can simulate T' inside T and simulate T'' inside T', then one can simulate T'' inside T by composing the simulators. Finally, $\simeq_{\mathrm{red}}$ is symmetric by definition (it requires reductions in both directions), reflexive because $\le_{\mathrm{red}}$ is reflexive, and transitive because the two required reductions compose in each direction. $\square$

Proposition 58 (Breadth is monotone in tolerance). *Fix a task environment $(\mathcal{T}, D)$ (Definition 315) and agent Ag. If $0 \le \varepsilon_1 \le \varepsilon_2 \le 1$, then*

$$\mathrm{Br}_{\varepsilon_1}(Ag; \mathcal{T}, D) \ \le \ \mathrm{Br}_{\varepsilon_2}(Ag; \mathcal{T}, D)$$

(Definition 320).

Proof sketch. The set $\{T : \kappa_{Ag}(T) \ge 1 - \varepsilon\}$ grows as ε increases. Taking probability under D preserves inclusion.

Equivalently, the threshold $1 - \varepsilon$ decreases as ε increases, so it becomes (weakly) easier for a task to qualify as "good" under the predicate $\kappa_{Ag}(T) \ge 1 - \varepsilon$. If $A_\varepsilon := \{T \in \mathcal{T} : \kappa_{Ag}(T) \ge 1 - \varepsilon\}$, then $\varepsilon_1 \le \varepsilon_2$ implies $A_{\varepsilon_1} \subseteq A_{\varepsilon_2}$, and hence $D(A_{\varepsilon_1}) \le D(A_{\varepsilon_2})$ by monotonicity of measures. $\square$

Proposition 59 (Capability dominance implies breadth dominance). *Let Ag_1, Ag_2 be agents on the same task set $\mathcal{T}$ and suppose $\kappa_{Ag_1}(T) \ge \kappa_{Ag_2}(T)$ for all $T \in \mathcal{T}$ (Definition 319). Then for every tolerance $\varepsilon \in [0, 1]$ and every task distribution D,*

$$\mathrm{Br}_\varepsilon(Ag_1; \mathcal{T}, D) \ \ge \ \mathrm{Br}_\varepsilon(Ag_2; \mathcal{T}, D).$$

Proof sketch. If $\kappa_{Ag_2}(T) \geq 1-\varepsilon$ then also $\kappa_{Ag_1}(T) \geq 1-\varepsilon$. So the "good task" set for Ag_2 is contained in that for Ag_1; apply $D(\cdot)$.
This proposition formalizes the intuition that "being at least as capable on every individual task" implies "covering at least as much of the task distribution" for any fixed success tolerance. The proof is a direct application of set inclusion: dominance yields pointwise implication of the threshold event, which lifts to probability under any distribution D without requiring additional structure on $\mathcal{T}$. □

Proposition 60 (Monotonicity of intellectuality under monotone learning). *Assume for each $T \in \mathcal{T}$ the learning-curve competence $\mathrm{Comp}^{(n)}(Ag;T)$ (Definition 321) is nondecreasing in budget n. Then the intellectuality curve $\mathrm{Int}^{(n)}(Ag;\mathcal{T},D) := \mathbb{E}_{T\sim D}[\mathrm{Comp}^{(n)}(Ag;T)]$ is nondecreasing in n.*

Proof sketch. Expectation preserves pointwise inequalities: if $X_n(T) \leq X_{n+1}(T)$ for all T, then $\mathbb{E}[X_n] \leq \mathbb{E}[X_{n+1}]$.
In detail, define $X_n := \mathrm{Comp}^{(n)}(Ag;T)$ as a random variable over $T \sim D$. The hypothesis gives $X_n(T) \leq X_{n+1}(T)$ for every T, so $X_{n+1} - X_n$ is pointwise nonnegative. Taking expectations yields $\mathbb{E}[X_{n+1} - X_n] \geq 0$, i.e. $\mathrm{Int}^{(n+1)} - \mathrm{Int}^{(n)} \geq 0$. Thus any per-task learning monotonicity property aggregates to monotonicity in the average case. □

Lemma 137 (Transducer agents induce history-based policies). *Let $Ag = (S, s_0, U, G)$ be an agent as a controlled transducer (Definition 322) and T be any task on the same interface (O, A). Then Ag induces a policy $\pi_{Ag,T} : H \to \Delta(A)$ (Definition 317) via*

$$\pi_{Ag,T}(h_t) := G(s_t), \quad \text{where } s_t \text{ is obtained by iterating } s_{k+1} = U(s_k, o_k) \text{ along } h_t.$$

Moreover, the coupled process on $S \times O$ is Markov with kernel given in Definition 323.

Proof sketch. Given a history, the internal state is determined (or distributed, if U is stochastic) by iterating the update rule; the action distribution is then $G(s_t)$. Markovicity holds because the next (s_{t+1}, o_{t+1}) depends on the past only through the current (s_t, o_t). □

Lemma 138 (Expressive completeness: any history policy is a transducer). *Given any history-based policy $\pi : H \to \Delta(A)$ (Definition 311), there exists a controlled transducer $Ag_\pi = (S, s_0, U, G)$ whose induced policy equals π. One canonical construction is $S := H$, $s_0 := (o_0)$, $U(h, o) := h \cdot o$ (append observation), and $G(h) := \pi(h)$.*

Proof sketch. By construction, the internal state at time t is exactly the interaction history h_t, so the action map reproduces $\pi(h_t)$ at every step. □

Proposition 61 (Intent dynamics yields a Markov state-space model). *For an intent-augmented agent (Definition 324) with intent update rule Φ as in Definition 325, the closed-loop coupling to any task can be written as a Markov process on $S_e \times \mathcal{G} \times O$ (environmental state, intent, observation).*

Proof sketch. At time t, the next intent is sampled from $\Phi(\tilde{s}_t, o_t)$, the next internal state is updated from $(\tilde{s}_t, g_t, o_t)$, and the next observation is sampled from $P(o_t, a_t)$. All depend on the past only through the current triple, hence Markov. □

Lemma 139 (Operational witness for "nontrivial" autonomy). *Let $\Phi : S_e \times O \to \Delta(\mathcal{G})$ be an intent update rule (Definition 325). A sufficient (and operationally checkable) condition that Φ depends nontrivially on internal state is:*

$$\exists\, \tilde{s}_1, \tilde{s}_2 \in S_e,\ \exists\, o \in O \text{ such that } \Phi(\tilde{s}_1, o) \neq \Phi(\tilde{s}_2, o).$$

Conversely, if $\Phi(\tilde{s}, o) = \Psi(o)$ for some $\Psi : O \to \Delta(\mathcal{G})$, then intent dynamics are fully externally scripted by perceptual input (no internal dependence).

Proof sketch. The forward direction just restates "depends on internal state" as existence of a separating pair. The converse direction exhibits an explicit factorization through O alone. $\square$

Proposition 62 (Stimulate/inhibit trichotomy). *Fix a claim $\varphi \in L$ and context o (Definition 326). For any action a exactly one holds:*

$$\epsilon^{+}_{a,o}(\varphi) > \epsilon^{-}_{a,o}(\varphi) \;\; (a \textit{ stimulates } \varphi), \quad \epsilon^{+}_{a,o}(\varphi) < \epsilon^{-}_{a,o}(\varphi) \;\; (a \textit{ inhibits } \varphi), \quad \epsilon^{+}_{a,o}(\varphi) = \epsilon^{-}_{a,o}(\varphi) \;\; (a \textit{ is neutral}).$$

Proof sketch. Trichotomy of the strict order on real numbers applied to the two scalar components. $\square$

Proposition 63 (Join-closure of stimulate and inhibit). *Let $\epsilon^{(1)}, \epsilon^{(2)}$ be two effect increments (Definition 326) and define their join $\epsilon := \epsilon^{(1)} \oplus \epsilon^{(2)}$ pointwise in V. Fix (a, o, φ). If both $\epsilon^{(1)}_{a,o}$ and $\epsilon^{(2)}_{a,o}$ stimulate φ, then $\epsilon_{a,o}$ stimulates φ. If both inhibit, then $\epsilon_{a,o}$ inhibits.*

Proof sketch. Write $p_i := \epsilon^{(i)+}_{a,o}(\varphi)$ and $m_i := \epsilon^{(i)-}_{a,o}(\varphi)$. Under join, $p = \max\{p_1, p_2\}$ and $m = \max\{m_1, m_2\}$. If $p_1 > m_1$ and $p_2 > m_2$ then $p > m$: otherwise $p \leq m$ would force (for the index attaining m) a contradiction with $p_i > m_i$. The inhibit case is symmetric. $\square$

Lemma 140 (Belief updates by join are monotone). *Assume belief update has the form $\beta' := \beta \oplus \epsilon_{a,o}$ (Definition 326), possibly followed by any monotone closure operator* Cl *(i.e., $\beta_1 \leq \beta_2 \Rightarrow \mathrm{Cl}(\beta_1) \leq \mathrm{Cl}(\beta_2)$). Then $\beta' \geq \beta$ pointwise in the belief order.*

Proof sketch. Join satisfies $\beta \leq \beta \oplus \epsilon$ componentwise. If a monotone closure is applied, monotonicity preserves the inequality. $\square$

Proposition 64 (Synergy for finitely many subsystems under monotone control). *Let $\{\beta^{(i)}\}_{i=1}^{k}$ be belief states over the same claim language and define $\beta := \bigoplus_{i=1}^{k} \beta^{(i)}$ (pointwise join). Suppose the downstream action-selection mechanism is monotone in the sense of Definition 329, and interpret "no more restrictive" concretely as:*

$$\beta_1 \leq \beta_2 \;\Rightarrow\; \mathrm{supp}(A(\beta_1)) \subseteq \mathrm{supp}(A(\beta_2)).$$

Then the combined system can achieve competence at least as high as the best subsystem:

$$\mathrm{Comp}(Ag_{\oplus}; T) \;\geq\; \max_{1 \leq i \leq k} \mathrm{Comp}(Ag_i; T),$$

generalizing Proposition 32 from $k = 2$ to arbitrary finite k.

Proof sketch. Induct on k. The case $k = 2$ is Proposition 32. For the induction step, join the first $k - 1$ beliefs into $\beta^{(\leq k-1)}$ and apply the $k = 2$ case to $(\beta^{(\leq k-1)}, \beta^{(k)})$. $\square$

Remark 2402 (About abstractions and well-defined quotient rewards). *Theorem 18 defines $r_Q(q(o), a, q(o^+)) := r(o, a, o^+)$ and asserts well-definedness from the "evaluation invariance" clause of Definition 328. If, in an implementation, your reward depends on finer distinctions of o^+ inside a fiber of q, you may want to strengthen the abstraction hypothesis by additionally requiring invariance in the o^+ argument (or restrict to rewards that factor through $q(o^+)$). This avoids ambiguity when reasoning mechanically about quotients.*

Concretely, the potential ambiguity is the following: if $q(o_1) = q(o_2)$ and $q(o_1^+) = q(o_2^+)$ but $r(o_1, a, o_1^+) \neq r(o_2, a, o_2^+)$, then the right-hand side depends on the choice of representative of the equivalence classes induced by q, so r_Q would not be a function of $(q(o), a, q(o^+))$ alone. In that case, one either (i) strengthens Definition 328 so that evaluation invariance explicitly quantifies over pairs of next-observations o^+ in the same fiber (i.e. invariance in the o^+ argument), or (ii) redefines the quotient reward as an aggregate over fibers (e.g. conditional expectation or worst-case), which produces a well-defined object but changes the semantics of the quotient task. For purposes of the later epistemic layer, option (i) is typically preferable when the quotient is intended as a semantics-preserving abstraction rather than a change of objective.

Proposition 65 (Compositionality of task abstractions (hierarchical coarse-graining))**.** *Let $q : O \to Q$ be a task abstraction for T (Definition 328), and let $q' : Q \to R$ be a task abstraction for the quotient task T_Q constructed in Theorem 18. Then (under the same well-definedness assumptions used to form T_Q) the composite $q' \circ q : O \to R$ is a task abstraction for T.*

Proof sketch. Lumpable dynamics compose because pushforwards compose: $(q' \circ q)_* = q'_* \circ q_*$. Evaluation invariance for $q' \circ q$ follows by applying invariance for q (to pass from o_1 to o_2) and then invariance for q' on the quotient side. The required well-definedness matches the same conditions needed in Theorem 18 to define quotient dynamics/evaluation.

To make the compositionality fully explicit, fix $o \in O$ and $a \in A$, and write $P(\cdot \mid o, a)$ for the (original) next-observation kernel on O. Lumpability of q asserts that there is a kernel $P_Q(\cdot \mid q(o), a)$ on Q such that $q_* P(\cdot \mid o, a) = P_Q(\cdot \mid q(o), a)$, i.e. for every measurable $B \subseteq Q$,

$$P_Q(B \mid q(o), a) = P(q^{-1}(B) \mid o, a),$$

and this value depends on o only through $q(o)$. Likewise, lumpability of q' for T_Q provides a kernel $P_R(\cdot \mid q'(q(o)), a)$ on R such that $q'_* P_Q(\cdot \mid q(o), a) = P_R(\cdot \mid q'(q(o)), a)$. Combining the two equalities yields $(q' \circ q)_* P(\cdot \mid o, a) = P_R(\cdot \mid (q' \circ q)(o), a)$, which is exactly the lumpability requirement for $q' \circ q$.

For evaluation invariance, one checks the representative-independence of any quotient-defined quantity along the same two-step chain. If evaluation invariance for q says that whenever $q(o_1) = q(o_2)$ (and a is fixed), the evaluation-related expressions (e.g. reward, value contributions, or whatever Definition 328 packages) computed from o_1 and o_2 agree after passing through q, then evaluation invariance for q' guarantees that the corresponding expressions computed from $q(o_1)$ and $q(o_2)$ agree after passing through q'. Chaining these two invariances yields invariance under $q' \circ q$. In particular, any construction in Theorem 18 that uses a choice of representative in $q^{-1}(q(o))$ remains invariant under replacing that representative by another element of the same fiber, and then the subsequent quotienting from Q to R is likewise invariant under changes within fibers of q'.

Finally, note that the "same well-definedness assumptions" clause is not merely a formality: if T_Q is defined by quotienting both dynamics and evaluation, then q' being a task abstraction for T_Q presupposes that those quotient components are already functions on Q (rather than on representatives in O). The point of the proposition is that once T_Q is well-defined, any further abstraction performed at the quotient level lifts back to a valid abstraction of the original task. □

Proposition 66 (Engineering entails an intent-aligned causal pathway)**.** *If an artifact/pattern Y is engineered by agent Ag over $[t_0, t_1]$ (Definition 327), then:*

1. *there exists an intent token g active for a nontrivial fraction of times in $[t_0, t_1]$ that entails a positive commitment to the existence-claim φ_Y, and*

2. *the agent's actions in* $[t_0, t_1]$ *are causally implicated (in the control/causal sense of Section 14) in the increase of* $\beta^+(\varphi_Y)$*.*

Proof sketch. Unpack clauses (b) and (c) of Definition 327; this proposition is a "reasoner-friendly" restatement.

More explicitly, Definition 327(b) provides (by hypothesis of "engineered") an intent-bearing component of the agent's internal process: there is a goal/intent token g whose activation is sustained on a set of times of nonzero (and typically bounded away from zero) measure within $[t_0, t_1]$, and whose content entails the agent's endorsement of φ_Y as a target condition (hence "positive commitment" to the existence-claim). This is the formal bridge from "engineering" as an external description to an internal, time-extended intentional state.

Definition 327(c) provides the causal-control component: the agent's policy-driven interventions during $[t_0, t_1]$ make a difference, in the Section 14 sense, to the downstream probability/credence update supporting φ_Y. In the notation of the epistemic layer, this is expressed as causal implication in an increase of $\beta^+(\varphi_Y)$, i.e. the agent's actions are not merely correlated with the outcome but participate in a directed pathway (via controllable degrees of freedom) that raises the post-update belief/attainment measure of the claim that Y exists (or is instantiated) relative to the relevant baseline. Thus (1) extracts the intent witness g, and (2) extracts the causal witness that the agent's action-selection is among the explanatory variables for the increase in $\beta^+(\varphi_Y)$, which is exactly what makes the notion operational for later mechanical reasoning. $\square$

Remark 2403 (On the role of "nontrivial fraction of times")**.** *The "nontrivial fraction" condition in item (1) should be read as excluding degenerate witnesses where the intent token* g *is present only instantaneously (e.g. a single timepoint) or only as a fleeting artifact unrelated to action-selection. In typical dynamical or discrete-time formalisms, this corresponds respectively to requiring a positive-measure subset of* $[t_0, t_1]$ *or a non-negligible subset of timesteps on which* g *is active, so that the token can meaningfully constrain/plausibly explain the agent's sequence of actions over the interval.*

41.9 Helper theorems for the general intelligence measures

Standing notation. Let $\mathcal{E}$ be a set of (computable) environments and $\mathcal{G}$ a set of goals. When relevant, let $\mathcal{T}$ denote the admissible time-scales/intervals used in the context construction in Def. 340. Let V^π_μ or $V^\pi_{\mu,g,T}$ denote the corresponding expected performance quantities appearing in Defs. 336 and 340. All sums below are interpreted only when convergent (as already stipulated in the defining equations).

Lemma 141 (Bounds for universal intelligence)**.** *Assume Def. 336. Suppose:*

1. *(reward-summability bound)* $0 \le V^\pi_\mu \le 1$ *for all* $\mu \in \mathcal{E}$*, and*

2. *(prefix-code/Kraft bound)* $\sum_{\mu\in\mathcal{E}} 2^{-K(\mu)} \le 1$*.*

Then $0 \le \Upsilon(\pi) \le 1$*.*

Sketch. Nonnegativity is immediate from nonnegative weights and rewards. For the upper bound:

$$\Upsilon(\pi) = \sum_{\mu\in\mathcal{E}} 2^{-K(\mu)} V^\pi_\mu \le \sum_{\mu\in\mathcal{E}} 2^{-K(\mu)} \cdot 1 \le 1.$$

$\square$

Lemma 142 (Bounds for pragmatic general intelligence). *Assume Def. 340. Suppose the context weights form a probability measure; concretely, assume:*

1. *ρ is a probability distribution on $\mathcal{E}$, and for each μ, $\gamma(\cdot \mid \mu)$ is a probability distribution on $\mathcal{G}$;*
2. *any time-scale indicator $\tau_{g,\mu}(|T|)$ (if present in your definition) lies in $[0,1]$; and*
3. *$0 \leq V^{\pi}_{\mu,g,T} \leq 1$ for all (μ, g, T).*

Then $0 \leq \Pi(\pi) \leq 1$.

Sketch. $\Pi(\pi)$ is a convex combination (possibly with additional $[0,1]$ gates such as $\tau_{g,\mu}$) of numbers in $[0,1]$. □

Lemma 143 (Pointwise dominance implies GI dominance). *Let π_1, π_2 be two agents.*

1. *If $V^{\pi_1}_{\mu} \geq V^{\pi_2}_{\mu}$ for all $\mu \in \mathcal{E}$, then $\Upsilon(\pi_1) \geq \Upsilon(\pi_2)$.*
2. *If $V^{\pi_1}_{\mu,g,T} \geq V^{\pi_2}_{\mu,g,T}$ for all contexts (μ, g, T), then $\Pi(\pi_1) \geq \Pi(\pi_2)$ (and likewise for any variant obtained by reweighting the same $V_{\mu,g,T}$ values with nonnegative weights, e.g. Def. 342 in the common "nonnegative weight" form).*

Proof. Each GI measure in question is a sum of nonnegative weights times the corresponding V values. Replacing each V^{π_2} by a (weakly) larger V^{π_1} cannot decrease the sum. □

Lemma 144 (Linearity in mixture policies (episode-level mixtures)). *Fix any of the scalar GI measures defined in Defs. 336–342, and suppose the relevant V-functionals are linear in randomized choice of policy at the start of an episode/context (i.e., law of total expectation applies in the obvious way).*

Let π_λ be the agent that, at the beginning of each episode/context, samples $i \in \{1, 2\}$ with $\mathbb{P}[i = 1] = \lambda$ and then behaves as π_i for that whole episode. Then for universal intelligence and pragmatic general intelligence:

$$\Upsilon(\pi_\lambda) = \lambda\Upsilon(\pi_1) + (1-\lambda)\Upsilon(\pi_2), \qquad \Pi(\pi_\lambda) = \lambda\Pi(\pi_1) + (1-\lambda)\Pi(\pi_2),$$

and analogously for Π^{eff} whenever it is also defined as a nonnegative-weighted sum of the same underlying V values.

Sketch. For each environment/context, V^{π_λ} is a convex combination $\lambda V^{\pi_1} + (1-\lambda)V^{\pi_2}$ by conditioning on the policy-choice coin flip. Then substitute into the defining sums. □

Lemma 145 (Linearity in priors / context weightings). *Fix a policy π. View pragmatic GI as a functional of the context weighting (e.g., via ρ and γ as in Def. 340). If a new environment prior is formed by convex combination $\rho_\lambda := \lambda\rho_1 + (1-\lambda)\rho_2$, then (for fixed γ)*

$$\Pi_{\rho_\lambda,\gamma}(\pi) = \lambda\Pi_{\rho_1,\gamma}(\pi) + (1-\lambda)\Pi_{\rho_2,\gamma}(\pi).$$

Likewise, if goal-weightings are convex-combined pointwise (for each μ), then Π is affine in those mixtures as well.

Proof. Expand the defining sum and distribute over addition. □

Proposition 67 (Universal intelligence as a special case of pragmatic GI). *Assume Defs. 336 and 340. Let $\xi(\mu) := 2^{-K(\mu)}$ be the (possibly subnormalized) universal weight. Let $Z := \sum_{\mu\in\mathcal{E}} \xi(\mu)$ (so $0 < Z \leq 1$ under the Kraft bound).*

Assume the pragmatic setup is chosen so that:

1. $\rho(\mu) = \xi(\mu)/Z$;

2. *the goal mechanism is "degenerate" in the sense that for each μ there is a distinguished g_μ that reproduces the reward signal used in Def. 336, and $\gamma(g \mid \mu) = \mathbf{1}[g = g_\mu]$; and*

3. *any time-scale indicator used in Def. 340 is identically 1 (or the time-scale set $\mathcal{T}$ is chosen so that it reduces to the universal-reward evaluation).*

Then

$$\Pi(\pi) = \frac{1}{Z}\,\Upsilon(\pi), \qquad \textit{equivalently} \qquad \Upsilon(\pi) = Z\,\Pi(\pi).$$

Thus Υ and Π differ here only by a constant normalization factor.

Sketch. Under the stated specialization, the pragmatic per-context performance values coincide with the universal V_μ^π, and the context weights differ from $2^{-K(\mu)}$ only by the constant factor $1/Z$. □

Lemma 146 (Efficient pragmatic GI as pragmatic GI on an augmented context space)**.** *Assume Defs. 340 and 342 in the common GTGI form where efficiency is implemented by reweighting each context by an expected (or sampled) inverse resource-usage factor.*

Let $\mathcal{R} \subset \mathbb{R}_{>0}$ be the resource domain, and let $D_{\mu,g,T}$ be the resource-usage distribution from Def. 342. Define the augmented context space

$$\mathcal{C}^{\mathrm{aug}} := \mathcal{E} \times \mathcal{G} \times \mathcal{T} \times \mathcal{R}.$$

Define augmented weights

$$w^{\mathrm{aug}}(\mu, g, T, r) := w(\mu, g, T)\, D_{\mu,g,T}(r)\, \frac{1}{r},$$

where $w(\mu, g, T)$ denotes the (nonnegative) context weighting used in Def. 340 (e.g., $w(\mu, g, T) = \rho(\mu)\gamma(g \mid \mu)\tau_{g,\mu}(|T|)$ if you use an explicit time-scale gate).

Then Def. 342 can be rewritten as a pragmatic-GI-style weighted sum over the augmented contexts:

$$\Pi^{\mathrm{eff}}(\pi) = \sum_{(\mu,g,T,r)\in\mathcal{C}^{\mathrm{aug}}} w^{\mathrm{aug}}(\mu, g, T, r)\, V^\pi_{\mu,g,T}.$$

Sketch. This is just a change of indexing: expand the efficiency-normalized definition as a sum/integral over resource outcomes r, and absorb the extra factor r^{-1} into the context weight. □

Corollary 26 (Constant or bounded resources yield simple relations between Π and Π^{eff})**.** *Assume the hypotheses of Lem. 146 and that the (per-context) inverse-resource factor is well-defined.*

1. *If $D_{\mu,g,T}$ is degenerate at a constant $r_0 > 0$ for all contexts, then $\Pi^{\mathrm{eff}}(\pi) = \Pi(\pi)/r_0$.*

2. *If there exist constants $0 < r_{\min} \le r_{\max} < \infty$ such that $\operatorname{supp}(D_{\mu,g,T}) \subseteq [r_{\min}, r_{\max}]$ for all contexts, then*
$$\frac{1}{r_{\max}}\Pi(\pi) \;\le\; \Pi^{\mathrm{eff}}(\pi) \;\le\; \frac{1}{r_{\min}}\Pi(\pi).$$

Sketch. Use $r^{-1} = r_0^{-1}$ in the degenerate case. In the bounded case, use $1/r_{\max} \le 1/r \le 1/r_{\min}$ pointwise inside the efficiency-weighted sum. □

Lemma 147 (Breadth is invariant under positive rescaling of fuzziness)**.** *Assume Def. 343. Let $\Phi_\pi(\mu, g, T)$ denote the fuzzy membership (or "intelligence degree") assigned to contexts before normalization, and let p_π be the probability distribution obtained by normalizing Φ_π (as in the definition).*

If Φ_π is replaced by $\Phi'_\pi := c\,\Phi_\pi$ for any constant $c > 0$, then the normalized distribution is unchanged ($p'_\pi = p_\pi$) and hence the intellectual breadth is unchanged.

Proof. Normalization divides by $\sum \Phi_\pi$, and $(c\,\Phi_\pi)/\sum(c\,\Phi_\pi) = \Phi_\pi/\sum \Phi_\pi$. □

Lemma 148 (Breadth bounds via Shannon entropy)**.** *Assume Def. 343 and that the induced context distribution p_π has finite support of size N. Let H denote Shannon entropy as in Def. 89 (with the same log base convention). Then the intellectual breadth $\mathrm{Br}(\pi) = H(p_\pi)$ satisfies*

$$0 \leq \mathrm{Br}(\pi) \leq \log N.$$

Moreover, $\mathrm{Br}(\pi) = 0$ iff p_π is a point mass (all weight on one context), and $\mathrm{Br}(\pi) = \log N$ iff p_π is uniform on its support.

Sketch. Standard Shannon-entropy bounds on a finite alphabet. □

Lemma 149 (Concavity of breadth under mixtures (when the induced context law mixes linearly))**.** *Assume Def. 343. Suppose that for episode-level mixtures π_λ (as in Lem. 144) the induced normalized context distribution satisfies*

$$p_{\pi_\lambda} = \lambda p_{\pi_1} + (1-\lambda) p_{\pi_2}.$$

Then breadth is concave:

$$\mathrm{Br}(\pi_\lambda) \;\geq\; \lambda \mathrm{Br}(\pi_1) + (1-\lambda)\mathrm{Br}(\pi_2).$$

Sketch. Shannon entropy is concave on the simplex; apply concavity to the assumed affine mixing of p_π. □

Further explanation. If $\mathrm{Br}(\pi)$ is instantiated as the Shannon entropy of the induced normalized context law (as in Def. 343), i.e.,

$$\mathrm{Br}(\pi) = H(p_\pi) \quad \text{with} \quad H(p) := -\sum_c p(c)\log p(c),$$

then the claim is exactly the standard concavity inequality

$$H\big(\lambda p + (1-\lambda) q\big) \;\geq\; \lambda H(p) + (1-\lambda) H(q) \qquad (\lambda \in [0,1]).$$

The assumption $p_{\pi_\lambda} = \lambda p_{\pi_1} + (1-\lambda) p_{\pi_2}$ is the crucial step that transfers concavity of $H(\cdot)$ (a property of distributions) into concavity of $\mathrm{Br}(\cdot)$ (a property of policies via their induced context laws). In particular, this highlights that concavity is guaranteed under linear mixing at the level of the induced distribution; if the mapping $\pi \mapsto p_\pi$ were nonlinear under mixtures, the inequality need not hold without additional regularity assumptions.

Remark 2404 (When the mixing assumption is satisfied)**.** *The condition $p_{\pi_\lambda} = \lambda p_{\pi_1} + (1-\lambda) p_{\pi_2}$ holds in common cases where the mixture policy π_λ is implemented by first sampling an episode-level coin flip $Z \sim \mathrm{Bernoulli}(\lambda)$ and then executing π_Z for the entire episode, and the context random variable is measurable with respect to episode-level exogenous randomness. In such cases, the resulting context law is literally a mixture distribution, hence affine in λ.*

Proposition 68 (Pareto preorder induced by multiple-criterion GI)**.** *Assume Def.* **??** *yields a vector-valued score* $\mathbf{GI}(\pi) \in \mathbb{R}^k$ *where larger is better in each component (if some criteria are "lower is better", flip their sign first).*

Define $\pi_1 \preceq_{\mathrm{P}} \pi_2$ *iff* $\mathbf{GI}(\pi_1)_j \le \mathbf{GI}(\pi_2)_j$ *for all components* $j = 1, \ldots, k$*.*

Then $\preceq_{\mathrm{P}}$ *is a preorder (reflexive and transitive). If one quotients by the equivalence* $\pi_1 \sim \pi_2$ *iff* $\pi_1 \preceq_{\mathrm{P}} \pi_2$ *and* $\pi_2 \preceq_{\mathrm{P}} \pi_1$*, the induced relation on equivalence classes is a partial order.*

Proof. Reflexivity and transitivity are inherited from $\le$ on $\mathbb{R}$ componentwise. □

Further explanation. Reflexivity is immediate since for any π and each j, one has $\mathbf{GI}(\pi)_j \le \mathbf{GI}(\pi)_j$, hence $\pi \preceq_{\mathrm{P}} \pi$. Transitivity follows because if $\pi_1 \preceq_{\mathrm{P}} \pi_2$ and $\pi_2 \preceq_{\mathrm{P}} \pi_3$, then for every component j,

$$\mathbf{GI}(\pi_1)_j \le \mathbf{GI}(\pi_2)_j \le \mathbf{GI}(\pi_3)_j,$$

so $\pi_1 \preceq_{\mathrm{P}} \pi_3$. The relation need not be antisymmetric on policies (distinct policies can tie on all criteria), which is why it is only a preorder. After quotienting by $\sim$ (i.e., identifying all policies with identical criterion-vectors), the induced relation becomes antisymmetric by construction: if $[\pi_1] \preceq_{\mathrm{P}} [\pi_2]$ and $[\pi_2] \preceq_{\mathrm{P}} [\pi_1]$, then $\pi_1 \sim \pi_2$, hence $[\pi_1] = [\pi_2]$.

Remark 2405 (Interpretation: nondominated front)**.** *Under* $\preceq_{\mathrm{P}}$*, the set of Pareto-efficient (nondominated) policies is the subset of policies* π *for which there is no* π' *such that* $\pi \preceq_{\mathrm{P}} \pi'$ *and* $\pi' \not\preceq \pi$*. This is the natural optimality notion when Def.* **??** *is treated as genuinely multiobjective rather than scalarized a priori.*

Proposition 69 (Positive scalarizations respect Pareto dominance)**.** *In the setting of Prop. 68, fix weights* $\alpha \in \mathbb{R}^k$ *with* $\alpha_j \ge 0$ *for all* j *and define a scalarization*

$$S_\alpha(\pi) := \sum_{j=1}^{k} \alpha_j \, \mathbf{GI}(\pi)_j.$$

If $\pi_1 \preceq_{\mathrm{P}} \pi_2$*, then* $S_\alpha(\pi_1) \le S_\alpha(\pi_2)$*.*

Proof. Componentwise $\mathbf{GI}(\pi_1)_j \le \mathbf{GI}(\pi_2)_j$ and $\alpha_j \ge 0$ imply $\alpha_j \mathbf{GI}(\pi_1)_j \le \alpha_j \mathbf{GI}(\pi_2)_j$ for each j; sum over j. □

Further explanation. This is simply monotonicity of a nonnegative linear functional with respect to the product order on $\mathbb{R}^k$. Note that if at least one weight is strictly positive on a component where $\mathbf{GI}(\pi_1)_j < \mathbf{GI}(\pi_2)_j$, then the inequality becomes strict: $S_\alpha(\pi_1) < S_\alpha(\pi_2)$. Conversely, if some $\alpha_j = 0$, then the scalarization deliberately ignores the jth criterion, so Pareto improvements exclusively in that component need not be reflected in S_α.

Remark 2406 (Scope and limitation of linear scalarizations)**.** *Positive linear scalarizations are convenient, but they do not in general recover every Pareto-efficient policy unless additional convexity assumptions hold for the achievable criterion set* $\{\mathbf{GI}(\pi) : \pi \in \Pi\}$*. The proposition here is one-way: Pareto dominance implies monotonic improvement under any positive weights, but not every nondominated solution must maximize a given linear scalarization.*

Corollary 27 (Efficient pragmatic GI as a particular scalarization of goal-achievement and resource frugality)**.** *Assume Def. 342 is implemented via an inverse-resource reweighting (as in Lem. 146). Then* Π^{eff} *is monotone in the Pareto sense with respect to the pair of per-context objectives*

$$\Big(V^{\pi}_{\mu,g,T},\ \frac{1}{r}\Big),$$

and thus can be treated as a (context-weighted) positive scalarization of these two criteria.

Sketch. In the augmented-context form, Π^{eff} is a nonnegative-weighted sum of products of (goal-achievement) and (inverse resource) factors. Increasing either factor pointwise cannot decrease the total. □

Further explanation. Writing Π^{eff} in the augmented-context representation of Lem. 146 makes the monotonicity transparent: each contextual contribution is weighted by a nonnegative context probability (or density) and then combined using nonnegative coefficients, so the overall functional is increasing in each argument when the other is held fixed. Concretely, for fixed (μ, g, T), if policy π_2 achieves weakly higher value $V^{\pi}_{\mu,g,T}$ than π_1 and uses weakly less resource (so that $1/r$ is weakly higher), then every term entering the efficient pragmatic score is weakly larger under π_2 than under π_1, hence the aggregate Π^{eff} cannot decrease. This is exactly the two-criterion instance of the Pareto monotonicity statement in Prop. 69, with the understanding that the weights arise from the context law and the particular reweighting scheme.

Remark 2407 (Two-objective interpretation)**.** *The pair $\left(V^{\pi}_{\mu,g,T}, 1/r\right)$ can be read as (i) effectiveness at achieving the goal in the given environment over horizon T, and (ii) frugality, expressed as an increasing transform of a resource usage variable. The corollary states that, under the stated implementation, Π^{eff} behaves like a standard multiobjective aggregation that never penalizes simultaneous improvements in effectiveness and frugality, aligning the efficient pragmatic construction with the Pareto order of these two desiderata.*

42 Helper theorems for society, culture, and collective mind systems

Remark 2408. *This section collects small but frequently used monotonicity and fixed-point facts for the formal objects introduced in Section 22: interaction graphs, message passing, cultural aggregation, collective belief, and the nested notions of collective mind systems. The common pattern is that each definition has been engineered so that "adding resources" (more edges, lower thresholds, larger agent sets, stronger couplings, richer approximator classes) cannot make a relevant quantity worse. This makes the mathematics hospitable to automated reasoning: many arguments reduce to event inclusion, order-theoretic monotonicity, or least-fixed-point folklore.*

Conceptually, these helper results touch several core notions in the Hyperseed core-concept set: Society *(Hyperseed-Concept 170),* Tribe *(Hyperseed-Concept 194),* Culture *(Hyperseed-Concept 91),* Mindplex *(Hyperseed-Concept 113), and* Global Brain *(Hyperseed-Concept* **??***). The underlying algebraic substrate is quantale-like: belief/messaging values live in an ordered space V with a join $\oplus$ and a tensor $\otimes$; this resonates with the "weakness as structure" perspective in [3] and with paraconsistent logics such as [23, 24] when $V = [0,1]^2$.*

To make the "adding resources" slogan precise, one typically fixes a preorder on each design dimension (graphs ordered by edge inclusion, threshold parameters ordered by $\leq$ but used contravariantly so that smaller thresholds yield larger reachable sets, agent populations ordered by inclusion, coupling strengths ordered pointwise, hypothesis/approximator classes ordered by inclusion). Each construction in Section 22 then induces an operator that is monotone with respect to these preorders and with respect to the intrinsic order on V. For example, a message-passing update can be viewed as a monotone map on a product poset such as V^E or $V^{A\times A}$, while a cultural aggregation rule is a monotone map from a multiset (or measure) of individual belief-states into a collective belief-state in $V^{\mathcal{L}}$ (for an appropriate language $\mathcal{L}$). The technical content of many helper lemmas is therefore that the semantics is functorial in the resource parameters: inclusion of structure yields inclusion (or order increase) of outcomes.

The fixed-point facts are used in precisely this order-theoretic style. Once an update/closure operator F is known to be monotone on a complete lattice (or at least on a directed-complete partial order, depending on the variant), standard results guarantee the existence of least and greatest fixed points, and provide constructive approximations via iteration from bottom or top. In the intended applications, the least fixed point corresponds to the "minimal" collective belief state (or minimal stable message configuration) compatible with the available interaction structure and update rules, while the greatest fixed point corresponds to a "maximal" or most permissive equilibrium. When thresholds are lowered, edges are added, or couplings are strengthened, monotonicity ensures that these fixed points move in the expected direction (least fixed points can only rise, reachable event sets can only expand, and admissible equilibria can only widen, depending on the specific order used).

The quantale-like assumptions on $(V, \oplus, \otimes)$ supply the algebraic glue needed for compositional arguments. In particular, distributivity properties (informally: $\otimes$ is compatible with joins) allow one to push order bounds through multi-step influence paths: combining messages along a path (via $\otimes$) and aggregating competing paths (via $\oplus$) yields an overall influence estimate that is monotone in each input. In the paraconsistent example $V = [0,1]^2$ with its natural product order, one can interpret the two coordinates as (degrees of) support and refutation; then $\oplus$ can be taken componentwise to reflect pooling of evidence, while $\otimes$ models composition of influence or entailment strength. The helper lemmas in this section should therefore be read as small "calculus rules" that let one rearrange such expressions without changing meaning, while preserving or strengthening bounds whenever the underlying resources are increased.

Finally, these monotonicity and fixed-point patterns are not merely aesthetic: they are what let higher-level notions such as "collective mind system" be stratified by inclusion and closure conditions. Nested mindplex constructions, for instance, can often be expressed as repeated application of monotone operators (add agents, close under communication, close under aggregation, close under meta-belief formation), so that the resulting hierarchy is well-behaved under refinement. In this sense, the helper theorems form the connective tissue between the concrete combinatorics of interaction graphs and the abstract semantics of collective belief and culture, ensuring that the overall framework admits principled comparison, extension, and reuse across different societal and cultural modeling choices.

42.1 Strong ties and tribes (Defs. 333–334): monotonicity rules

Helper Lemma 42.1 (Strong-tie edge monotonicity in the threshold)**.** *Let $G = (\mathcal{A}, E)$ be an interaction graph and let $K : E \to [0,1]$ be scalar couplings. For $\tau \in (0,1]$ define the strong-tie edge set*

$$E_\tau := \{(i,j) \in E : K(i,j) \geq \tau\}, \quad \text{and} \quad G_\tau := (\mathcal{A}, E_\tau).$$

If $\tau_1 \geq \tau_2$ then $E_{\tau_1} \subseteq E_{\tau_2}$, hence G_{τ_1} is a subgraph of G_{τ_2}.

Remark 2409. *Intuitively, $G = (\mathcal{A}, E)$ is a directed graph of agents $\mathcal{A}$ with edges E representing potential influence or interaction, while $K(i,j) \in [0,1]$ measures how strongly i can affect j along an edge. The parameter τ is a* threshold*: we keep only edges whose coupling is at least τ, thereby extracting the "strong-tie" subgraph G_τ. This formalizes the sociological idea that if we demand stronger ties, fewer edges qualify; if we relax the standard, more edges qualify (Hyperseed-Concept 194).*

A simple example: suppose $E = \{(1,2), (2,3)\}$ with $K(1,2) = 0.9$ and $K(2,3) = 0.4$. Then at $\tau = 0.8$, we keep only $(1,2)$; at $\tau = 0.3$, we keep both edges. The lemma states precisely this nesting behavior, a basic prerequisite for monotonicity arguments about tribes and connectivity in thresholded graphs, and a staple move in network-based emergence arguments [5].

One can also view the family $\{G_\tau\}_{\tau\in(0,1]}$ as a decreasing filtration in τ: as τ moves downward, the edge set can only grow. In weighted-network terms, this is the standard "thresholding" operation turning a weighted digraph into an unweighted digraph at level τ, and the lemma says that thresholding is order-preserving with respect to the parameter.

Remark 2410. *What the lemma says in plain language is:* raising the bar for what counts as a strong tie can only delete edges, never add them. *This is important because virtually every downstream "tribe" or "community" definition depends on connectivity in G_τ; once we know how G_τ changes with τ, we can reason about how group structure changes as we shift thresholds.*

It is sometimes helpful to keep the extreme cases in mind. At $\tau = 1$, only edges with maximal coupling survive, so G_1 is typically very sparse; as τ decreases toward 0, more and more of the original edge set E is recovered (indeed, if one allowed $\tau = 0$ formally, then $E_0 = E$). The monotonicity statement is exactly what makes it meaningful to speak of edges and communities "appearing" as the threshold is weakened, since once an edge appears at some τ it persists for all smaller τ.

Sketch. If $K(i,j) \geq \tau_1$ and $\tau_1 \geq \tau_2$ then $K(i,j) \geq \tau_2$. Thus any edge in E_{τ_1} is also in E_{τ_2}.

Equivalently, the inclusion $E_{\tau_1} \subseteq E_{\tau_2}$ is simply the statement that the indicator function of the event "$K(i,j) \geq \tau$" is monotone nonincreasing in τ for each fixed edge $(i,j) \in E$. □

Remark 2411. Proof sketch. *The strategy is a one-line order argument: the predicate "$K(i,j) \geq \tau$" becomes easier to satisfy when τ is smaller. Geometrically, one may picture τ as a horizontal cut through edge weights; lowering the cut includes more edges.*

In this picture, each edge (i,j) has a "birth time" at $\tau = K(i,j)$: for thresholds $\tau \leq K(i,j)$ the edge is present, and for $\tau > K(i,j)$ it is absent. The lemma is the global statement obtained by taking the union of these per-edge birth rules across all edges in E. □

Helper Corollary 42.2 (Tribe monotonicity under weakening the strong-tie threshold)**.** *Assume a tribe at threshold τ is a subset $T \subseteq \mathcal{A}$ such that: (i) the induced subgraph $G_\tau[T]$ is (strongly) connected, and (ii) a boundedness/density condition holds that does not depend on τ. If T is a tribe at threshold τ_1 and $\tau_2 \leq \tau_1$, then T is also a tribe at threshold τ_2.*

Remark 2412. *In ordinary speech: if a set of agents T is already knitted together by sufficiently strong ties, then it remains knitted together when we decide to count* weaker *ties as admissible strong ties. Condition (i) says that within T, there is a directed path between any two members using only edges of strength at least τ. Condition (ii) represents whatever additional finiteness or density constraint the tribe definition imposes; the point here is that it does not change as τ changes, so only connectivity is at stake.*

This matches a natural conceptual expectation about Tribe *(Hyperseed-Concept 194): tribes are robust under making the social microscope less demanding. It is also a useful computational fact: one can search tribes at a high threshold, and know that any tribe found persists at lower thresholds (though new tribes may appear).*

For directed graphs, the parenthetical "(strongly)" is important: if one adopts strong connectivity, then the statement reads "directed reachability in both directions cannot be destroyed by adding directed edges"; if one adopts weak connectivity (connectivity of the underlying undirected graph), the statement is even more immediate. In either interpretation, the relevant principle is the same: adding admissible edges cannot eliminate existing paths inside the induced subgraph on T.

Remark 2413. *The corollary is an immediate companion to the previous lemma: the lemma gives edge-set nesting, and this corollary translates that nesting into* connectivity *preservation. This is* *the kind of monotonicity one leans on when proving "phase transition" statements for communities as thresholds vary, a motif common in network-centric treatments of collective cognition [19, 5].*

More concretely, as τ decreases one may see previously separate strongly connected components of G_τ merge into larger components, but one cannot see a strongly connected component split purely as a result of lowering τ (splitting would require deleting edges, not adding them). This "merging but not splitting" intuition is often what underlies persistence-style arguments about the stability of group-level structures across ranges of thresholds.

Sketch. By Lemma above, $G_{\tau_1}[T]$ is a subgraph of $G_{\tau_2}[T]$. Connectivity is preserved when adding edges. The boundedness condition is unchanged by assumption.

Spelled out slightly more: any path in $G_{\tau_1}[T]$ is still a path in $G_{\tau_2}[T]$, because every edge used by the path at threshold τ_1 remains present at threshold $\tau_2 \leq \tau_1$. Hence whatever reachability witnesses connectivity at τ_1 also witnesses it at τ_2. ☐

Remark 2414. Proof sketch. *The proof separates the two defining clauses of "tribe": only the connectivity clause depends on τ. Since lowering τ adds edges, any path that existed before still exists, and possibly many more; hence connectivity cannot be lost.*

Condition (ii) is treated as an invariant constraint across τ; typical examples in applications include cardinality bounds (e.g. $|T| \leq M$), minimum in/out-degree requirements computed in the original *graph rather than the thresholded one, or density constraints formulated in terms that do not tighten when τ changes. Under such hypotheses, the corollary reduces to a single monotonicity fact about paths.* ☐

42.2 Messages and assimilation (Defs. 335–336): algebra and batching

Helper Lemma 42.3 (Scaling of messages is monotone)**.** *Let $m : L \to V$ be a message and let k be a coupling weight (either scalar $k \in [0, 1]$ embedded as (k, k), or $k \in V$). Define scaling pointwise by*

$$(k \odot m)(\varphi) := k \otimes m(\varphi) \quad \textit{(interpreting scalar } k \textit{ via diagonal embedding).}$$

Then:

1. *If $m \leq m'$ pointwise, then $k \odot m \leq k \odot m'$.*
2. *If $k \leq k'$ (in the scalar order or the V-order), then $k \odot m \leq k' \odot m$.*

Remark 2415. *Here L is the claim language, V is the value space (often the p-bit quantale $[0, 1]^2$), and a message $m : L \to V$ assigns to each formula $\varphi \in L$ some graded evidence value. The symbol $\otimes$ is the quantale tensor (a kind of "and/combination" operation), while $\odot$ denotes scalar (or value) scaling of messages by a coupling k, i.e. attenuating or amplifying evidence before it is assimilated. The phrase "scalar $k \in [0, 1]$ embedded as (k, k)" means that we treat a real coupling strength as acting equally on the positive and negative coordinates of a p-bit value.*

The lemma says that scaling respects order: if you make the message larger, the scaled message becomes larger; and if you make the coupling larger, the scaled message becomes larger. This is one of the minimal algebraic sanity checks needed for any compositional account of social information flow (Hyperseed-Concept **??***), and it is exactly the kind of monotonicity guaranteed by quantale axioms [3].*

Remark 2416. *It is worth making explicit the underlying order conventions used implicitly throughout this subsection. The order on messages $m, m' : L \to V$ is the* pointwise *order:*

$$m \leq m' \quad :\Longleftrightarrow \quad \forall \varphi \in L,\ m(\varphi) \leq_V m'(\varphi).$$

Likewise, when k is a scalar $k \in [0, 1]$ viewed inside V by the diagonal map $k \mapsto (k, k)$, the inequality $k \le k'$ is just the usual real order transported along that embedding, so the comparison in (2) reduces to a comparison in V. Under these conventions, both items of the lemma are simply the statement that the map

$$V \times V \to V, \qquad (a, b) \mapsto a \otimes b$$

is monotone in each argument, which is one of the standard quantale requirements (and is also the key property that makes $\otimes$ behave like a generalized product rather than an arbitrary binary operation).

Remark 2417. *In plain language:* stronger evidence in, stronger evidence out; stronger channel, stronger delivered message. *This matters because all later batching and path-composition facts tacitly assume that local improvements cannot perversely reduce propagated evidence. It also connects this section to the epistemic-layer helper theorems (Section 42) where the same monotonicity of $\otimes$ and joins underwrites closure constructions.*

Sketch. Both are immediate from monotonicity of the tensor $\otimes$ in each argument (quantale axiom). □

Remark 2418. Proof sketch. *The proof is pointwise in φ: compare $k \otimes m(\varphi)$ and $k \otimes m'(\varphi)$ using monotonicity in the second input, and compare $k \otimes m(\varphi)$ and $k' \otimes m(\varphi)$ using monotonicity in the first input. Visually, one may imagine $\otimes$ as a generalized multiplication: increasing either factor cannot decrease the product.* □

Remark 2419. *In applications, the coupling weight k often summarizes multiple "channel" factors at once (attention, trust, bandwidth, proximity, institutional filtering). The lemma then says that any increase in such channel factors (modeled as an increase of k in the underlying order) can only increase the transmitted/usable evidence values in the resulting scaled message. This monotonicity is precisely what allows one to compare alternative social/organizational designs by order-theoretic dominance: if one design has uniformly larger couplings (or produces uniformly larger raw messages), then it cannot yield weaker delivered evidence after scaling.*

Helper Lemma 42.4 (Assimilation update is inflationary and idempotent under repeat)**.** *Let $\beta : L \to V$ be a belief state and update by $\beta \leftarrow \beta \oplus (k \odot m)$. Then $\beta \le \beta \oplus (k \odot m)$ pointwise. Moreover, applying the* same *assimilated message twice has no further effect:*

$$(\beta \oplus (k \odot m)) \oplus (k \odot m) = \beta \oplus (k \odot m),$$

since $\oplus$ is idempotent.

Remark 2420. *This lemma is the algebraic analogue of two everyday intuitions about communication. First: assimilating a message by joining it into your current belief state cannot make you* less *informed in the order-theoretic sense (inflationary update). Second: if you hear precisely the same message twice through exactly the same coupling, and assimilation is modeled by a join, then the second hearing adds no new information (idempotence). In the canonical p-bit setting, where $\oplus$ is componentwise* max*, this corresponds to the simple fact that* $\max(x, y)$ *does not change if you take* $\max(\max(x, y), y)$.

This is useful because it turns message passing into a kind of set-like *accumulation of evidence: duplicates are harmless, and updates can be reasoned about without tracking the detailed history of repeated identical transmissions. Such idempotent accumulation is characteristic of many closure-style and fixed-point constructions, including those used to analyze stability of self-updating systems [10].*

Remark 2421. *The result is not surprising once one sees that $\oplus$ is behaving like logical "or" or like taking a supremum. But it is important because it is exactly what makes batching (next proposition) and optional closure (next corollary) tractable: if joins were not inflationary and idempotent, then order and repetition effects would contaminate every multi-message calculation.*

Remark 2422. *One may also read the idempotence clause as a modeling decision about what counts as "new" information. Joining treats repeated identical inputs as redundant, which matches many evidential and logical intuitions (a proposition does not become truer by restating it verbatim). If one wanted instead to model* reinforcement by repetition *(e.g. repeated exposure increasing salience), then the update operator would typically not be idempotent; the present lemma clarifies that the batching results below rely specifically on the join-semilattice style of assimilation.*

Sketch. Inflationary: $\beta(\varphi) \leq \beta(\varphi) \oplus (k \odot m)(\varphi)$ for each φ. Idempotence: $x \oplus x = x$ in any join-semilattice. $\square$

Remark 2423. Proof sketch. *The proof is entirely local to each formula φ: at that coordinate, the update replaces $\beta(\varphi)$ by a join with an extra term, which cannot decrease it. Re-applying the same join is redundant because joining an element with itself does nothing. A helpful picture is to view β as a "landscape" of values; joining in a message raises some points of the landscape, never lowers them, and repeating the same raising operation cannot raise further.* $\square$

Helper Proposition 42.5 (Batching and order-independence of multiple assimilations)**.** *Let $(k_1, m_1), \ldots, (k_n, m_n)$ be messages received by an agent, and define the iterated update*

$$\beta_0 := \beta, \qquad \beta_{t+1} := \beta_t \oplus (k_{t+1} \odot m_{t+1}).$$

Then

$$\beta_n = \beta \oplus (k_1 \odot m_1) \oplus \cdots \oplus (k_n \odot m_n),$$

so the final state does not depend on the order in which messages are processed (associativity/commutativity of $\oplus$).

Remark 2424. *Equivalently, for each $\varphi \in L$ the final value $\beta_n(\varphi)$ is the join (supremum) of the initial belief value $\beta(\varphi)$ together with all delivered message-values $(k_i \odot m_i)(\varphi)$. In the common p-bit/*max *instantiation, this is simply the statement that the coordinatewise maximum over a finite set of inputs is independent of the order in which those inputs are processed. Conceptually, batching says that assimilation is "history-free" with respect to permutations of the same multiset of incoming (coupling,message) pairs: only the set of delivered evidence values matters, not the temporal order in which they arrived.*

Sketch. By induction on n, using associativity of $\oplus$ to reassociate iterated joins, and commutativity of $\oplus$ to permute the join-factors. $\square$

Remark 2425. Proof sketch. *Unfold the recursion once:*

$$\beta_1 = \beta \oplus (k_1 \odot m_1), \qquad \beta_2 = (\beta \oplus (k_1 \odot m_1)) \oplus (k_2 \odot m_2),$$

and so on. Associativity ensures that parentheses can be dropped without changing the value, so the iterated update is equal to the single expression obtained by joining all scaled messages into β. Commutativity then implies that any permutation of $(k_1 \odot m_1), \ldots, (k_n \odot m_n)$ yields the same final join, hence the same β_n. If some of the delivered terms coincide, idempotence (Lemma above) shows that duplicates do not affect the result, which is often convenient when messages are received through multiple redundant channels. $\square$

Remark 2426. *This proposition says that when assimilation is modeled as repeated joins, the agent may treat the incoming stream of messages as a* multiset *whose final effect is just the join of all scaled messages—and since $\oplus$ is commutative and associative, the order does not matter. Practically, this means one can batch messages, parallelize assimilation, or reorder them for computational convenience without changing the resulting belief state. A useful way to read this is that the* only *information the update dynamics keeps from a finite message history is the "envelope" given by $\oplus$ of the scaled inputs. Equivalently, the update map factors through the free commutative idempotent monoid on messages (i.e. multisets under union), because idempotence of $\oplus$ also implies that receiving the* same *scaled message twice does not further change the result beyond the first receipt. In this sense, the repeated-join assimilation law behaves like a deterministic, confluent rewrite system: different schedules of incorporating evidence all converge to the same normal form. One can also think of this as a kind of weak path-independence: the belief state β evolves monotonically (with respect to the lattice order induced by $\oplus$), and the final state depends only on the set of constraints/messages eventually imposed, not on the temporal order in which they arrive.*

In the canonical p-bit case, imagine two messages about the same claim φ: one provides 0.6 positive evidence and another provides 0.7 negative evidence (appropriately scaled). Joining them simply takes the coordinatewise maxima, so doing "positive then negative" or "negative then positive" yields the same final pair. This is one of the algebraic reasons paraconsistent evidence aggregation is computationally neat [23, 24]. In more detail, if a belief state stores a pair $(b^+(\varphi), b^-(\varphi)) \in [0,1]^2$ and $\oplus$ is coordinatewise max*, then the update with a message contributing $(0.6, 0)$ and then one contributing $(0, 0.7)$ yields*

$$(0,0) \oplus (0.6,0) \oplus (0,0.7) = (\max\{0,0.6,0\}, \max\{0,0,0.7\}) = (0.6,0.7),$$

and the same expression appears in the opposite order. This makes explicit that, even when evidence is "in tension" (both positive and negative components are large), the aggregation rule still stays purely algebraic: it is not forced to resolve the tension immediately, and the computation does not branch on arrival order.

Remark 2427. *The importance of this result is structural: it lets us treat a communicative system as defining a* commutative semilattice action *on belief states. That, in turn, is what makes fixed-point analyses of social belief propagation feasible later (compare the reinforcement operator F below), and it aligns with the broader quantale-based viewpoint [3]. Concretely, the space of "aggregated message-bundles" (finite joins of scaled messages) itself forms a commutative idempotent semigroup under $\oplus$, and assimilation becomes an action $(x, \beta) \mapsto \beta \oplus x$ of this semilattice on states. The action laws are exactly the algebraic equalities one wants for modular reasoning: joining two bundles and then acting equals acting by each bundle in either order, i.e.*

$$\beta \oplus (x \oplus y) = (\beta \oplus x) \oplus y = (\beta \oplus y) \oplus x,$$

so "communication composition" and "belief update composition" agree. This turns a dynamic process into an object that can be studied with standard lattice/semigroup tools: invariants, monotone iteration, and eventual stabilization can be phrased as statements about joins and fixed points rather than about time indices.

Sketch. Induction on n, using associativity/commutativity of $\oplus$. More explicitly, for the induction step one expands the definition of the $n+1$-step assimilation, uses the induction hypothesis to replace the first n steps by the join of the first n scaled messages, and then uses associativity to flatten parentheses. Commutativity is what allows any permutation of the messages to be rewritten to the same join expression, showing that the normal form depends only on the multiset of inputs rather than on their sequence. □

Remark 2428. Proof sketch. *The high-level strategy is to expand the recurrence* $\beta_{t+1} = \beta_t \oplus (\cdots)$ *repeatedly until all the joins are explicit, then use associativity to remove parentheses. Commutativity justifies reordering. One can think of each update as "adding a stone" to a pile; in a commutative monoid, the pile does not remember the order in which stones were added. If one additionally recalls that* $\oplus$ *is typically idempotent in these settings (as a join operation in a semilattice), the "pile" metaphor can be sharpened: adding the* same *stone twice does not change the pile after the first addition. This is exactly the multiset-to-set collapse at the level of effect: the update is sensitive to the maximum delivered strength along each coordinate, not to the number of times the same strength is repeated.* □

Helper Corollary 42.6 (Optional inference closure can be pushed to the end)**.** *Assume a closure operator* Cl *on belief states is monotone, extensive, and idempotent (as in the inference-closure construction of Section 20). If after each message one optionally applies* $\beta \leftarrow \mathrm{Cl}(\beta)$*, then the final closed belief state equals*

$$\mathrm{Cl}\Big(\beta \oplus (k_1 \odot m_1) \oplus \cdots \oplus (k_n \odot m_n)\Big).$$

Remark 2429. *A closure operator* Cl *(as in the epistemic layer) represents "closing under inference": after assimilating raw evidence, one may apply rules to derive implied beliefs. This corollary states that if* Cl *is a genuine closure operator (extensive, monotone, idempotent), then repeatedly closing after every single message is equivalent to first joining all the messages in any order, and then closing once at the end. Intuitively, closure is a "rounding up" operation to the nearest inference-complete state. If one performs many small rounds of inference after each communication event, the axioms guarantee that nothing essentially new is gained compared to postponing the rounding until all evidence has been incorporated: extensiveness prevents closure from discarding information, monotonicity prevents later evidence from being "blocked" by earlier closure, and idempotence prevents overcounting the effect of repeated closure. In particular, the corollary can be read as an* interleaving invariance *principle: communication steps (joins with message contributions) and reasoning steps (applications of* Cl*) can be freely interleaved without changing the final closed outcome, so long as one ends in a closed state.*

This is valuable computationally and conceptually. Computationally, it means we can defer expensive inference until after batching evidence. Conceptually, it means that closure is a kind of consolidation *rather than an event that must occur at every moment; the algebra guarantees no loss of rigor by postponement. This resonates with the general theme that when inference is modeled as least fixed points of monotone operators, the order of "small steps" often collapses to a single fixed-point computation (compare Knaster–Tarski style arguments discussed around Section 42). From a systems viewpoint, this also clarifies a design separation: message handling can remain a lightweight, streaming operation (a join-semilattice update), while inference can be scheduled as a background task or a periodic batch job. The corollary states that this engineering choice is not merely heuristic; under the closure axioms, it is provably semantics-preserving.*

Remark 2430. *The statement is important because it connects the social layer back to the epistemic-layer closure theorems: the same abstract conditions (monotone, extensive, idempotent) ensure that closure behaves like a projection onto a subspace of "fully reasoned" states. The surprise, for students encountering closure operators for the first time, is how much control these three axioms give over the interleaving of communication and inference. In categorical or order-theoretic language, a closure operator is a (monotone) idempotent inflationary endomorphism, and its image consists of the "closed" elements. The corollary is then a statement that the endomorphism can be applied after forming the join of a finite family of updates, without changing the resulting closed element. This makes it possible to reason about the long-run behavior of social-epistemic dynamics by first* *analyzing the easier semilattice accumulation, and only afterward restricting attention to the closed subspace.*

Sketch. Extensiveness gives $\beta \le \mathrm{Cl}(\beta)$; monotonicity lets closure commute with joins up to equality, and idempotence lets repeated closures collapse to one final closure. A common way to formalize the "collapse" is to show, by induction on the number of steps, that after any interleaving of joins-with-messages and closures, the current state is always bounded above by the state obtained by joining all message contributions so far and then applying Cl, and also bounded below by the unclosed join-state. These two bounds tighten to equality once one applies Cl at the end, because applying Cl to the lower bound lifts it into the closed region while idempotence ensures no further change. ☐

Remark 2431. Proof sketch. *The strategy is to compare the step-by-step process with the "batch then close" process by sandwiching each intermediate state between the unclosed join-state and its closure, using monotonicity. Idempotence then removes redundant closures. A useful mental image is that* Cl *pulls any state upward to the nearest closed state above it; doing this repeatedly is no different from doing it once at the end. This "nearest closed state" picture is especially intuitive in finite examples: one can imagine the lattice of states and the subset of closed points as a region, with* Cl *acting like a projection that moves each point upward. Joins move states upward as well, and monotonicity ensures these upward motions do not interfere destructively. Thus, alternating "move upward by evidence" and "project to closed" is equivalent to doing all evidence-moves first and then projecting once.* ☐

Helper Lemma 42.7 (Redundant channels combine by join of couplings (quantale distributivity))**.** *Suppose the* same *message m reaches a recipient through two links with couplings k_1, k_2. Assuming $\otimes$ distributes over $\oplus$ (quantale axiom), we have*

$$(k_1 \odot m) \oplus (k_2 \odot m) \;=\; (k_1 \oplus k_2) \odot m.$$

Remark 2432. *This lemma formalizes the idea that two redundant transmissions of the same content combine into a single transmission whose effective coupling is the join (supremum) of the two couplings. In the canonical p-bit setting with $\oplus = \max$ and $\otimes$ behaving like multiplication by a scalar coupling, the identity reduces to*

$$\max(k_1 m,\, k_2 m) = \max(k_1, k_2)\, m,$$

which is exactly what one expects: the stronger of the two channels dominates the delivered strength when one aggregates by maxima. Put differently, the recipient does not need to remember how many *copies of the same message arrived, nor from which path, so long as the aggregation rule is supremum-like: what matters is the best-supported (highest-coupling) delivery of that content. This is consistent with many "max-confidence" and "best-signal" heuristics, but here it is not a heuristic; it is an algebraic consequence of the distributive law in the underlying quantale.*

The assumption that $\otimes$ distributes over $\oplus$ is part of the quantale structure, and is precisely what makes such "algebra of evidence flow" clean and compositional [3]. It matters for simplifying network calculations: rather than tracking multiple redundant edges carrying the same message, one can collapse them into an effective coupling. At the level of symbolic manipulation, the lemma is the direct specialization of distributivity:

$$(k_1 \oplus k_2) \otimes m \;=\; (k_1 \otimes m) \oplus (k_2 \otimes m),$$

together with the convention that $k \odot m$ abbreviates $k \otimes m$. In graph terms, if two parallel edges carry the same label m, then the parallel composition of edges corresponds to $\oplus$ on couplings, so one may replace the parallel edges by a single edge whose coupling is $k_1 \oplus k_2$ without changing any downstream assimilation results. This kind of reduction is particularly useful when analyzing dense social graphs where redundancy is common (multiple acquaintances relaying the same claim), because it allows one to normalize the network before running belief-propagation computations.

Remark 2433. *The result is a small but powerful rewrite rule: it lets you move the join from the message-side to the coupling-side. This is particularly helpful when analyzing effective communication strength along multiple parallel paths (next subsection), where the combinatorics of paths would otherwise proliferate. In practice, it means that when a fixed message m is being broadcast through several competing couplings $k_1, k_2, \ldots$, one may first aggregate the couplings (by $\oplus$) and then apply the message once, rather than repeatedly scaling and joining on the message side. Algebraically, this is the familiar distributivity pattern $(a \otimes x) \vee (b \otimes x) = (a \vee b) \otimes x$ (for $\vee = \oplus$), applied pointwise to the message coordinates; conceptually, it is a way of factoring out a common "content" from multiple delivery channels.*

Sketch. Pointwise in φ: $(k_1 \otimes m(\varphi)) \oplus (k_2 \otimes m(\varphi)) = ((k_1 \oplus k_2) \otimes m(\varphi))$. Here the pointwise qualification is essential: the join and tensor are applied in the underlying value structure at each feature/coordinate φ, so the identity follows immediately from distributivity in that base algebra (and then lifts to messages by extensional equality). If one prefers, the same argument can be read as a two-term instance of the more general finite-join law $\left(\bigvee_i k_i\right) \otimes x = \bigvee_i (k_i \otimes x)$, specialized to a common $x = m(\varphi)$. □

Remark 2434. Proof sketch. *The proof is a direct application of distributivity at each coordinate φ: factor out $m(\varphi)$ from the join of two scaled values. Geometrically, think of $m(\varphi)$ as a "shape" and k as a "brightness" multiplier; taking the brighter of two renderings is equivalent to rendering once at the maximum brightness. This picture also highlights why $\oplus$ being idempotent (as a supremum-like operation) is natural for evidence aggregation: repeating the same rendering does not compound brightness, whereas improving the coupling increases it monotonically. In particular, if $k_1 \le k_2$ then $k_1 \otimes m(\varphi) \oplus k_2 \otimes m(\varphi) = k_2 \otimes m(\varphi)$, so weaker redundant channels are absorbed by stronger ones at that coordinate.* □

42.3 Communicative systems (Def. 338): path composition heuristics

Helper Proposition 42.8 (Effective coupling along a communication path (scalar case))**.** *Assume scalar couplings $k \in [0, 1]$ embedded diagonally and the canonical tensor corresponds to multiplying by k. Consider a directed path*

$$i_0 \to i_1 \to \cdots \to i_\ell$$

and suppose the same message m is forwarded along the path, with assimilation at each hop. Then the net scaling at the end of the path is bounded below by multiplication of couplings:

$$\textit{effective scale} \;\ge\; \prod_{t=0}^{\ell-1} k(i_t, i_{t+1}),$$

i.e. the forwarded evidence is at least $(\prod_t k_t) \odot m$ before joining into beliefs. If there are multiple paths, the final assimilated contribution is the join over path contributions. In particular, when $\oplus$ is realized as pointwise maximum and the canonical tensor is ordinary multiplication, each path yields a pointwise attenuated copy of m, and the receiver takes the pointwise best-supported copy *across paths. Edge cases behave as expected: any hop with $k(i_t, i_{t+1}) = 0$ blocks that path entirely, while $k(i_t, i_{t+1}) = 1$ leaves the message unchanged at that hop, so the product expresses cumulative attenuation along the route.*

Remark 2435. *The statement translates a familiar engineering principle into the present algebra: if a signal passes through multiple imperfect links, its strength attenuates multiplicatively. Here the "signal" is the message m, and each hop (i_t, i_{t+1}) has coupling $k(i_t, i_{t+1}) \in [0, 1]$; under canonical scaling, the message is multiplied by k at each hop. Thus a two-hop path with couplings 0.8 and 0.5 yields an overall factor 0.4.*

The lower-bound phrasing "$\geq$" reflects that, depending on the implementation, an agent might preserve some of the message strength locally or combine it with other evidence; but in the basic model where forwarding consists of successive scalings and joins, the product is the natural minimal effective factor. The join over multiple paths expresses a second natural idea: independent deliveries do not cancel; they accumulate by supremum (compare redundant channels in the previous lemma). This is one of the simplest mathematical embodiments of communication as compositional propagation *in a society (Hyperseed-Concept 170). One can also view the product $\prod_t k(i_t, i_{t+1})$ as a path-score: longer paths are not automatically worse, but they are penalized unless intermediate couplings are near 1. This makes explicit a trade-off between reachability (existence of paths) and reliability (strength of the best path), which will be reused when turning network structure into effective influence bounds.*

Remark 2436. *This proposition is important as a bridge between local update rules (scaling and joining) and global network effects (path reachability and attenuation). It anticipates the later reinforcement-operator expansion, where max/min composition replaces multiplication/min in a fuzzy-logic regime. Both are instances of a broader pattern-web style analysis of distributed cognition [5, 19]. In that later regime, "composition along a path" remains the central heuristic, but the specific aggregator (product versus a t-norm, max/min, etc.) changes with the semantic interpretation of evidence. The present scalar-product case can be read as the baseline against which alternative path-composition laws are compared.*

Sketch. Each hop multiplies the message by the next coupling factor, so scales multiply. Independently delivered contributions are joined by $\oplus$ (as in the message update rule). Formally, one can argue by induction on ℓ: for $\ell = 1$ the claim is immediate; assuming the scaling after $\ell - 1$ hops is at least $\prod_{t=0}^{\ell-2} k(i_t, i_{t+1})$, applying the next hop multiplies by $k(i_{\ell-1}, i_\ell)$ and yields the product for ℓ. The multi-path statement then follows by applying the same single-path computation to each path and using the definition of assimilation as a join of incoming contributions. □

Remark 2437. Proof sketch. *The proof strategy is to unroll the sequential forwarding: after one hop the message has factor k_0, after two hops $k_0 k_1$, and so on. For multiple paths, one applies the batching/order-independence proposition: each path contributes a scaled version of m, and the receiver's belief state takes the join of all contributions. A visual metaphor is light passing through successive tinted panes (multiplication), with multiple windows shining on the same wall (join). Equivalently, for each path p one computes a path weight $w(p) = \prod_{e \in p} k(e)$ and the receiver aggregates $\bigvee_p \big(w(p) \odot m\big)$; the earlier rewrite rule justifies pulling joins across shared messages when comparing alternative couplings or routes. This makes it clear why redundant or partially overlapping routes can be analyzed without enumerating all interleavings: path contributions remain separable up to a final join.* □

43 Helper theorems for Hyperseed-grounded Universal Grammar

This section collects small "helper" results intended to support automated reasoning over the Hyperseed-grounded TyLAA/TUG formalization of Universal Grammar. The key objects referenced are: (i) the label-independence relation (Definition **??**; Hyperseed-Concept **??**), (ii) the swap congruence / trace skeleton (Definitions **??**–**??**; Hyperseed-Concepts **??**, **??**), (iii) the symmetry group of labels (Definition **??**; Hyperseed-Concept **??**), (iv) the induced TUG congruence (Definition **??**; Hyperseed-Concept **??**), and (v) the dependency topology extracted from failures of independence (Definition **??**; Hyperseed-Concept **??**).

Helper Lemma 43.1 (Basic properties of label-independence)**.** *Let* $\perp \subseteq \mathsf{Lab}_V \times \mathsf{Lab}_V$ *be the label-independence relation from Definition* **??***, i.e.*

$$\ell_1 \perp \ell_2 \quad \Longleftrightarrow \quad \mathsf{disjointPos}(\mathsf{pos}(\ell_1), \mathsf{pos}(\ell_2)) \;\wedge\; (\mathsf{rely}(\ell_1) \cap \mathsf{rely}(\ell_2) = \emptyset).$$

Then:

1. $\perp$ *is symmetric: if* $\ell_1 \perp \ell_2$ *then* $\ell_2 \perp \ell_1$*.*
2. $\perp$ *is irreflexive: for all* $\ell \in \mathsf{Lab}_V$*, not(*$\ell \perp \ell$*).*

Proof. (1) Symmetry follows because $\mathsf{disjointPos}(\cdot,\cdot)$ is symmetric and because set intersection is commutative: $\mathsf{rely}(\ell_1) \cap \mathsf{rely}(\ell_2) = \emptyset$ iff $\mathsf{rely}(\ell_2) \cap \mathsf{rely}(\ell_1) = \emptyset$.

(2) For any position p, $\mathsf{disjointPos}(p, p)$ fails because p is a prefix of itself. Hence the conjunction defining $\ell \perp \ell$ fails for every ℓ. □

Helper Lemma 43.2 (Independence preserved by structural symmetries)**.** *Assume* Γ *is a symmetry group acting on visible labels as in Definition* **??***, and assume each* $\gamma \in \Gamma$ *preserves the structural annotations used by Definition* **??***, i.e.*

$$\mathsf{pos}(\gamma\ell) = \mathsf{pos}(\ell) \quad \textit{and} \quad \mathsf{rely}(\gamma\ell) = \mathsf{rely}(\ell) \qquad \forall \ell \in \mathsf{Lab}_V.$$

Then for all $\ell_1, \ell_2 \in \mathsf{Lab}_V$*,*

$$\ell_1 \perp \ell_2 \quad \Longleftrightarrow \quad (\gamma\ell_1) \perp (\gamma\ell_2).$$

Proof. Immediate from the defining conjunction for $\perp$ and the preservation equalities for pos and rely. □

Helper Proposition 43.3 (Swap-equivalence is a monoid congruence)**.** *Let* $\equiv_\perp$ *be the swap congruence on* Lab_V^* *from Definition* **??***. Then:*

1. $\equiv_\perp$ *is an equivalence relation on* Lab_V^**.*
2. $\equiv_\perp$ *is a congruence for concatenation: if* $\pi_1 \equiv_\perp \pi_2$ *then for all* $\alpha, \beta \in \mathsf{Lab}_V^*$*,*

$$\alpha\, \pi_1\, \beta \;\equiv_\perp\; \alpha\, \pi_2\, \beta.$$

Proof. (1) By construction, $\equiv_\perp$ is the equivalence closure of the elementary adjacent-swap rewrites generated by pairs $\ell_1 \perp \ell_2$.

(2) Each elementary rewrite step has the form $\alpha\,(\ell_1\ell_2)\,\beta \mapsto \alpha\,(\ell_2\ell_1)\,\beta$ for some context words α, β and some independent pair $\ell_1 \perp \ell_2$. Thus closure under adding the same prefix/suffix is built into the generator. Stability under finite sequences of generator steps follows by induction on the length of the derivation witnessing $\pi_1 \equiv_\perp \pi_2$. □

Helper Proposition 43.4 (The trace skeleton forms a monoid). *Let* $\mathsf{Tr}_\perp := \mathsf{Lab}_V^*/{\equiv_\perp}$ *be the trace skeleton (Definition* **??***; Hyperseed-Concept* **??***). Define multiplication on equivalence classes by*

$$[\pi] \cdot [\rho] := [\pi\rho],$$

where $\pi\rho$ *is concatenation and* $[\cdot]$ *denotes the* $\equiv_\perp$*-class. Then* $(\mathsf{Tr}_\perp, \cdot, [\epsilon])$ *is a well-defined monoid.*

Proof. Well-definedness: if $\pi \equiv_\perp \pi'$ and $\rho \equiv_\perp \rho'$ then $\pi\rho \equiv_\perp \pi'\rho'$ by Proposition 43.3(2), so the product does not depend on representatives.

Associativity and the identity law follow from associativity of word concatenation and the fact that ϵ is the identity for concatenation. □

Helper Proposition 43.5 (Monotonicity under enlarging independence or symmetry). *Let* $(\perp_1, \Gamma_1)$ *and* $(\perp_2, \Gamma_2)$ *be two independence/symmetry choices with* $\perp_1 \subseteq \perp_2$ *and* $\Gamma_1 \subseteq \Gamma_2$*. Let* $\equiv_{\perp_i,\Gamma_i}$ *be the corresponding TUG congruences (Definition* **??***). Then, as relations on* Lab_V^**,*

$$\equiv_{\perp_1,\Gamma_1} \subseteq \equiv_{\perp_2,\Gamma_2} .$$

Proof. $\equiv_{\perp,\Gamma}$ is generated by (i) swaps of adjacent independent labels and (ii) Γ-identifications. If $\perp_1 \subseteq \perp_2$ and $\Gamma_1 \subseteq \Gamma_2$, then every generator step allowed for $(\perp_1, \Gamma_1)$ is also allowed for $(\perp_2, \Gamma_2)$. Hence any finite derivation witnessing $\pi \equiv_{\perp_1,\Gamma_1} \pi'$ is also a derivation witnessing $\pi \equiv_{\perp_2,\Gamma_2} \pi'$. Equivalently, if one views $\equiv_{\perp,\Gamma}$ as the smallest equivalence relation closed under these generator steps, then enlarging $\perp$ and Γ can only enlarge the set of admissible steps, and therefore can only enlarge the generated relation. Formally, one may argue by induction on the length of the derivation chain $\pi = \pi_0 \to \pi_1 \to \cdots \to \pi_n = \pi'$, noting that each rewrite $\pi_i \to \pi_{i+1}$ is a single instance of either (i) or (ii), and the same instance is still valid when $\perp$ and Γ are replaced by supersets. □

Helper Corollary 43.6 (UG coarsening increases quantale weakness). *Fix an evidence quantale* $(V, \leq, \oplus, \otimes)$ *and a weight map* $\mu : \mathsf{Lab}_V^* \to V$*. For any congruence* $\equiv$ *on* Lab_V^**, view it as an "undistinguished pairs" relation* $H_\equiv \subseteq \mathsf{Lab}_V^* \times \mathsf{Lab}_V^*$ *via* $(\pi, \pi') \in H_\equiv \iff \pi \equiv \pi'$*. If* $\equiv_1 \subseteq \equiv_2$ *then*

$$w(H_{\equiv_1}) \leq w(H_{\equiv_2}),$$

where $w(\cdot)$ *is the quantale weakness functional (Hyperseed-Concept 143). In particular, the inclusion from Proposition 43.5 implies weakness monotonicity for UG coarsenings. Moreover, this corollary packages the general principle that declaring* more *pairs undistinguished (i.e. passing to a coarser congruence) can only make the resulting comparison criterion weaker in the sense measured by* w*, since fewer distinctions remain available to a test.*

Proof. If $\equiv_1 \subseteq \equiv_2$ then $H_{\equiv_1} \subseteq H_{\equiv_2}$ by definition. Apply Theorem 1 (weakness monotonicity under enlarging the undistinguished-pairs relation). Concretely, the hypothesis of Theorem 1 is exactly set inclusion of undistinguished-pairs relations, and the conclusion is the desired inequality in V; the role of the weight map μ is already built into the definition of $w(\cdot)$, so no additional argument beyond the inclusion check is required here. □

Helper Proposition 43.7 (The trace quotient is a Hyperseed abstraction). *Let* $q_{\perp,\Gamma} : \mathsf{Lab}_V^* \twoheadrightarrow \mathsf{Lab}_V^*/{\equiv_{\perp,\Gamma}}$ *be the quotient map* $\pi \mapsto [\pi]_{\perp,\Gamma}$*. Then* $q_{\perp,\Gamma}$ *is a surjection, and the induced equivalence relation*

$$\pi \sim_{q_{\perp,\Gamma}} \pi' \iff q_{\perp,\Gamma}(\pi) = q_{\perp,\Gamma}(\pi')$$

is exactly $\equiv_{\perp,\Gamma}$*. Hence* $q_{\perp,\Gamma}$ *is an abstraction in the sense of Hyperseed-Concept 51. In other words, the quotient construction realizes the standard "forgetful" abstraction that collapses precisely those traces that UG declares observationally interchangeable (up to swaps and* Γ*-actions), and nothing more.*

Proof. Surjectivity is immediate: every equivalence class has a representative. The induced relation $\sim_{q_{\perp,\Gamma}}$ is equality of equivalence classes, which is precisely $\equiv_{\perp,\Gamma}$ by definition of the quotient. This matches the Hyperseed abstraction schema: a surjection whose fibers define an equivalence relation (Hyperseed-Concept 51). Equivalently, $\sim_{q_{\perp,\Gamma}}$ is the kernel relation of the map $q_{\perp,\Gamma}$, and for a quotient map the kernel is exactly the congruence used to form the quotient. Thus the "information discarded" by $q_{\perp,\Gamma}$ is exactly the information modded out by $\equiv_{\perp,\Gamma}$. ☐

Helper Lemma 43.8 (Dependency topology invariance under swap and symmetry). *Let* $\pi, \pi' \in \mathsf{Lab}_V^*$. *Assume:*

1. $\pi \equiv_\perp \pi'$ *(swap-equivalent), and*
2. *the symmetry group* Γ *preserves independence as in Lemma 43.2.*

Then the dependency topology extracted from π *(Definition* **??***) is isomorphic to the dependency topology extracted from* π'*. Consequently, any dependency-topology invariant (e.g. undirected dependency graph, dependence partial order up to renaming of event indices) is constant on* $\equiv_{\perp,\Gamma}$*-classes. Here "isomorphic" is meant in the concrete sense appropriate to the construction of Definition* **??***: there is a bijection on event occurrences/indices that preserves the dependence relation and therefore induces a homeomorphism (or graph isomorphism, as appropriate) of the resulting topological/relational structure.*

Proof. It suffices to check invariance under one generator step and then use induction over the finite sequence of generator steps witnessing $\pi \equiv_\perp \pi'$.

A generator step replaces a factor $\ell_1\ell_2$ with $\ell_2\ell_1$ where $\ell_1 \perp \ell_2$. Because ℓ_1 and ℓ_2 are independent, there is no dependence edge between these two event occurrences in the extracted dependency topology, and swapping their order does not create one. For any other label occurrence ℓ_3, the dependence/independence relation between ℓ_3 and ℓ_1 (resp. ℓ_2) is unchanged by the swap; only the adjacent indices are exchanged. Thus the dependency structure is preserved up to the obvious renaming of the two swapped indices. More explicitly, if the swap exchanges occurrences at positions i and $i+1$, then the bijection on event indices that transposes i and $i+1$ (and fixes all other indices) preserves the dependence relation, since the only potential new comparisons would involve ℓ_1 and ℓ_2, which are independent by hypothesis.

For symmetry steps, Lemma 43.2 guarantees that dependence/independence is preserved under the label map $\ell \mapsto \gamma\ell$. In that case the underlying event-index set is unchanged and one may take the identity bijection on indices; the isomorphism condition reduces to the fact that every dependence edge is defined in terms of (in)dependence of labels, which is invariant under the Γ-action by the cited lemma. ☐

Helper Theorem 43.9 (Observational soundness of TUG congruence). *Assume:*

1. *(Local commutation) whenever two visible labels* ℓ_1, ℓ_2 *are independent (*$\ell_1 \perp \ell_2$*), the corresponding visible steps commute in the LTS in the sense of Assumption* **??***.*
2. *(Symmetry soundness) for every* $\gamma \in \Gamma$*, replacing each visible label* ℓ *in a derivation by* $\gamma\ell$ *does not change the observation at the interface, in the sense of Assumption* **??***.*
3. *(State-determined observation) the observation functor/interface reads off an invariant from the derivation's state trajectory as specified in Definition* **??***.*

Then for all visible derivations $\pi, \pi' \in \mathsf{Lab}_V^*$,

$$\pi \equiv_{\perp,\Gamma} \pi' \quad \Longrightarrow \quad \pi \equiv_{\text{obs}} \pi'.$$

In particular, $\equiv_{\perp,\Gamma}$ is a sound proof principle *for observational equivalence: any equation derivable from the commutation and symmetry generators denotes traces that cannot be told apart by the chosen observation interface.*

Proof. By Definition **??**, $\equiv_{\perp,\Gamma}$ is generated by two kinds of elementary rewrites: (i) swapping adjacent independent labels, and (ii) applying a symmetry $\gamma \in \Gamma$ to a label (or to a factor) as allowed by the symmetry action.

For generator type (i), local commutation ensures that the swapped derivations lead to the same relevant state/trace-as-state information, hence have the same observation by state-determined observation. More concretely, Assumption **??** provides a commuting square (or diamond) in the LTS for the two-step fragment labeled by $\ell_1\ell_2$ versus $\ell_2\ell_1$, showing that the two orderings reach matching states; since the observation is a function of the state trajectory in the sense of Definition **??**, the observation computed from either ordering is identical.

For generator type (ii), symmetry soundness gives observation preservation directly. That is, applying γ pointwise to the labels of a derivation changes only the names of visible actions within a Γ-orbit and, by Assumption **??**, leaves the interface-level observation invariant.

Since $\equiv_{\perp,\Gamma}$ is the equivalence closure of these generators, any derivation witnessing $\pi \equiv_{\perp,\Gamma} \pi'$ is a finite chain of observation-preserving steps. Thus π and π' have equal observations, i.e. $\pi \equiv_{\text{obs}} \pi'$. Equivalently, $\equiv_{\text{obs}}$ contains each generator instance, and therefore contains the smallest equivalence relation generated by them, namely $\equiv_{\perp,\Gamma}$. ☐

Helper Proposition 43.10 (Independence implies commutation of compiled tensor operations)**.** *Assume there is a compilation map* Comp *from visible labels (or visible derivations) into tensor programs as in Definition* **??***, and assume the compilation has the property that:*

1. *disjoint redex positions imply disjoint tensor index sets touched by the compiled operation, and*
2. *disjoint rely footprints imply disjoint mutable tensor operands touched by the compiled operation.*

In other words, the compilation is required to preserve the two standard "non-interference" facets of independence: (i) independence at the level of where *a step acts in the visible syntax is reflected as separation of the* index-level *access pattern in the compiled program, and (ii) independence at the level of* what a step assumes about shared state *(its rely information) is reflected as separation of the* operand-level *write targets in the compiled program. This is the minimal structure needed to turn syntactic independence into operational commutativity. Then for any independent visible labels $\ell_1 \perp \ell_2$,*

$$\mathsf{Comp}(\ell_1) \circ \mathsf{Comp}(\ell_2) \;=\; \mathsf{Comp}(\ell_2) \circ \mathsf{Comp}(\ell_1).$$

Equivalently, the compiled tensor programs are permutation-invariant with respect to swapping adjacent independent visible steps, which is the tensor-program counterpart of the usual swap/diamond principle for independent reductions.

Proof. Under the stated hypotheses, $\ell_1 \perp \ell_2$ implies the compiled tensor computations share neither (a) contraction indices nor (b) mutable tensor operands. More explicitly, by (1) the sets of indices whose entries are read, contracted over, or otherwise addressed by $\mathsf{Comp}(\ell_1)$ and $\mathsf{Comp}(\ell_2)$ are disjoint; thus any index-scoped effects (such as reductions over contracted dimensions, scatter/gather patterns, or pointwise updates parameterized by indices) cannot overlap. By (2) the sets of mutable tensor operands (the concrete storage locations designated as being updated in-place) are disjoint; thus neither compiled operation can overwrite values that the other operation might subsequently read or write as part of its own in-place behavior.

Tensor computations with disjoint index sets and disjoint mutable operands commute under composition, yielding the stated equality. One way to view this is to model each compiled operation P as carrying a read-set $\mathsf{Rd}(P)$ and write-set $\mathsf{Wr}(P)$ of tensor-addressable locations (with index sets determining which locations are actually touched), so that disjointness of the relevant index sets plus disjointness of mutable operands entails $\mathsf{Wr}(\mathsf{Comp}(\ell_1)) \cap \mathsf{Wr}(\mathsf{Comp}(\ell_2)) = \varnothing$ and also rules out write–read interference in either direction. Under such a non-interference condition, both compositions compute the same resulting tensor store, since each operation acts as the identity outside its own footprint and their footprints do not overlap. This is exactly the situation in which sequential order is observationally irrelevant, so the two composites coincide as tensor programs. $\square$

Remark 2438 (Automated reasoning usage)**.** *The helper results above support three routine proof moves that are valuable for automated reasoning over the UG/TUG formalization:*

(1) Normalization by commuting independent steps: *use Proposition 43.3 to commute independent steps inside arbitrary contexts, and Proposition 43.4 to reason at the level of trace classes rather than raw derivation words. Concretely, this permits a proof automation tactic to reorder a derivation into a canonical form (e.g. grouping all steps acting on one syntactic region together) without changing its trace class, thereby reducing the number of distinct cases the prover must consider. It is also the mechanism by which local commutations justify global rearrangements of long derivations when the dependence relation is sparse.*

(2) Invariant extraction: *use Lemma 43.8 to compute dependency topology from any convenient representative of a trace class. In practice, this means one can pick a representative derivation word that is easy to analyze (e.g. already normalized, or produced by a deterministic strategy) and be confident that any derived dependence graph, partial order, or causal structure is an invariant of the entire trace class. This is particularly useful when a decision procedure needs only the dependence information, not the exact syntactic interleaving of independent steps.*

(3) Quotient-based abstraction and weakness accounting: *use Proposition 43.7 to treat the trace map as a Hyperseed abstraction (Hyperseed-Concept 51), and Corollary 43.6 to relate UG coarsenings to quantale weakness (Hyperseed-Concept 143). Operationally, this supports a common workflow in which one first reasons in a coarser, more quotiented semantics to simplify a proof obligation, and then transfers results back to a finer semantics while explicitly tracking the induced loss of discrimination as an increase in weakness. In automated settings, the same pattern enables sound over-approximations: the prover can work in the quotient where fewer distinct executions exist, while the weakness relation records precisely what has been forgotten.*

43.1 Culture (Defs. 343–344, Thm. 19): monotonicity and propagation

Helper Lemma 43.1 (Cultural aggregation is monotone; threshold cultures are nested)**.** *Let $C : P \to [0,1]$ or $C : P \to V$ be a cultural state formed by aggregating agent intensities and artifact-embedded intensities using $\oplus$ and scaling $\odot$. Then for each pattern $p \in P$:*

1. *Increasing any weight or any contributing intensity cannot decrease $C(p)$.*
2. *For a threshold culture $\mathrm{Cult}_\theta := \{p \in P : \pi(C(p)) \geq \theta\}$, if $\theta' \leq \theta$ then $\mathrm{Cult}_\theta \subseteq \mathrm{Cult}_{\theta'}$.*

Remark 2439. *A cultural state C assigns to each pattern $p \in P$ an aggregate intensity—roughly, "how present is p in the culture?" (Hyperseed-Concept 91, and compare* Pattern / Pattern Web *in Hyperseed-Concepts 130, 132). The construction is intentionally modular: many contributors (agents, artifacts) each supply some intensity for p, these are scaled by weights (importance, exposure, coupling), and then aggregated by a join $\oplus$. Because join and tensor are monotone, aggregation inherits monotonicity: adding stronger contributors cannot weaken the result.*

Part (2) defines a thresholded culture Cult_θ*, i.e. the set of patterns whose cultural score exceeds a threshold after applying a projection π (for p-bits, π might map (v^+, v^-) to a scalar). Nestedness is the expected behavior: lowering the admission threshold yields a superset. This is the same structural monotonicity as in the tribe threshold corollary, but now applied to patterns rather than edges.*

It is worth keeping the order-theoretic content explicit: the lemma does not rely on a particular analytic form of C, only on the fact that $\odot$ and $\oplus$ are monotone maps in the underlying order of $[0, 1]$ or of the valuation lattice V. In particular, if V is taken as a product order (as in the p-bit case) then "cannot decrease" is understood coordinatewise. Conceptually, the lemma says that cultural formation behaves like an isotone (order-preserving) data-fusion operator: more evidence or greater coupling pushes patterns upward in the cultural order, never downward.

Finally, the presence of π in (2) is a reminder that one often thresholds a derived *scalar score even when the internal representation is richer. For example, one may keep a two-coordinate record (v^+, v^-) for reasoning, but decide membership in* Cult_θ *by a scalar "net acceptance" or "salience" function π. The monotonicity conclusion only depends on the fact that the threshold predicate is monotone in the threshold parameter; no special properties of π beyond being well-defined on $C(p)$ are required for the set-inclusion direction stated.*

Remark 2440. *The lemma is a quiet workhorse: it guarantees that cultural formation, as modeled here, is an order-preserving aggregation process. This is a common theme in formal treatments of emergence: once one fixes a notion of "presence" as a supremum-like accumulation, then all the interesting dynamical questions can be asked without fear that merely adding evidence could reduce a cultural feature [5].*

One practical consequence is stability under enrichment of the contributor set: if one extends the model by adding additional agents, additional artifacts, or additional sensing channels (e.g. new modalities that supply further intensity estimates), then the resulting culture can only become at least as strong on every pattern, relative to the same weights and intensity assignments. This supports modular model-building: one can refine the representation of cultural sources without introducing pathological reversals in what the model regards as culturally present.

Sketch. (1) is by monotonicity of $\otimes$ and $\oplus$ in each argument. (2) is immediate from the definition: the predicate $\pi(C(p)) \geq \theta$ becomes easier as θ decreases.

More explicitly, a typical aggregated term contributing to $C(p)$ has the form $w \odot I(p)$ (for some contributor intensity I and weight w), and $C(p)$ is obtained by applying $\oplus$ over all such terms. Since $\odot$ is monotone in both w and $I(p)$ (in the relevant order) and $\oplus$ is monotone in each input, replacing any one term by a larger one forces the join to be larger as well. For (2), if $\theta' \leq \theta$ and $\pi(C(p)) \geq \theta$, then automatically $\pi(C(p)) \geq \theta'$, hence membership propagates from the stricter threshold set to the looser one. □

Remark 2441. Proof sketch. *Part (1) reduces to the observation that each contributor enters $C(p)$ through monotone operations, so increasing any input can only increase the output. Part (2) is basic order logic: the set of elements above a lower threshold contains the set above a higher* *threshold. One can picture $\pi(C(p))$ as a height function on patterns; lowering the waterline exposes more peaks.*

In contexts where V is not totally ordered, it can be helpful to remember that the "height function" picture is being applied after the projection π: $C(p)$ may live in a multi-dimensional or partially ordered space, while $\pi(C(p))$ yields a one-dimensional scalar that can be directly thresholded. Thus the nestedness statement is really about the scalar scores induced by π, and it is compatible with the idea that different projections could induce different, but still nested-by-threshold, cultural snapshots. □

Helper Corollary 43.2 (Paraconsistent culture witness (canonical p-bit case))**.** *Assume $V = [0,1]^2$ with $\oplus$ as componentwise* max*, scalar scaling $w \odot v = (w,w) \otimes v$ as coordinatewise multiplication, and let $\pi(v^+, v^-) = \frac{1+v^+-v^-}{2}$. If for some pattern p there exist contributors x, y (agents and/or artifacts) such that*

$$w_x\, I_x^+(p) \geq \theta \quad \text{and} \quad w_y\, I_y^-(p) \geq \theta,$$

then the aggregate satisfies $C^+(p) \geq \theta$ and $C^-(p) \geq \theta$. Thus p can be simultaneously "in-culture" and "contested" at comparable strength.

Remark 2442. *This corollary makes vivid why p-bit (two-coordinate) valuations are useful for culture: the same pattern p may receive strong positive endorsement and strong negative rejection from different parts of a society, and the aggregate culture can preserve both facts rather than forcing an artificial collapse into a single number. In the canonical case, $C^+(p)$ is simply the maximum (over contributors) of weighted positive intensities, and $C^-(p)$ is the maximum of weighted negative intensities. So if one contributor pushes p strongly positive and another pushes it strongly negative, the aggregate records both.*

This is a precise mathematical witness of paraconsistent cultural states, aligning with paraconsistent logics and evidence semantics [23, 24]. Philosophically, it also reflects a realistic view of culture: contested symbols and practices are often not "partly present" but strongly present in incompatible ways *(Hyperseed-Concept 198).*

The specific projection $\pi(v^+, v^-) = \frac{1+v^+-v^-}{2}$ can be read as a normalized "net valence" score: holding v^- fixed, increasing v^+ raises the score, while holding v^+ fixed, increasing v^- lowers it. Importantly, the corollary does not assert that $\pi(C(p))$ is large; rather, it asserts that both coordinates of $C(p)$ can be large simultaneously. This separation is exactly what allows one to model a pattern as both saliently present (high v^+) and saliently resisted (high v^-), without erasing either component during aggregation.

Remark 2443. *The result connects to later notions of collective belief and mindplexes: if disagreement can be preserved at the cultural layer, then a collective mind can have internal tension without logical explosion. This is one of the conceptual motivations for using $V = [0,1]^2$ rather than $[0,1]$ in social cognition models.*

A related modeling point is that the use of max *for $\oplus$ implements a "there exists a strong witness" semantics: a single highly influential contributor can set the cultural coordinate for p (positive or negative). Other join operators (e.g. probabilistic sums, t-conorms, or softmax-like joins) would instead implement "many weak witnesses" semantics. The corollary isolates a minimal and interpretable case where the paraconsistent effect is unavoidable: even with the simplest possible aggregation rule, independent positive and negative maxima can coexist.*

Sketch. In the canonical case, $C^+(p) = \max_x w_x I_x^+(p)$ and $C^-(p) = \max_x w_x I_x^-(p)$, so each coordinate lower bound is witnessed by the corresponding contributor.

Formally, from $w_x I_x^+(p) \geq \theta$ we have $\max_{x'} w_{x'} I_{x'}^+(p) \geq w_x I_x^+(p) \geq \theta$, and similarly for the negative coordinate using y. No interaction term couples the two coordinates under componentwise max, so the positive witness and negative witness can be different contributors without weakening either inequality. $\square$

Remark 2444. Proof sketch. *The proof is a witness argument: because the aggregate is a coordinatewise maximum, it must be at least as large as any particular contributor's scaled coordinate. The geometry is literally that of two heights (v^+, v^-); taking maxima preserves the highest positive pillar and the highest negative pillar, even if they come from different sources.*

This geometric picture also clarifies why the phenomenon is robust: as long as the aggregation operator is monotone and does not forcibly trade off v^+ against v^-, strong evidence in each coordinate can survive aggregation. The componentwise maximum is the clearest case, but the key intuition is that "positive presence" and "negative presence" are tracked as separate conserved signals rather than being collapsed into a single axis at the point of cultural formation. $\square$

Helper Lemma 43.3 (Social reinforcement operator is inflationary; Kleene chain is ascending)**.** *Let F be the social reinforcement operator on intensities $\mathbb{I} \in [0,1]^{\mathcal{A}\times P}$ given by*

$$(F(\mathbb{I}))_{i,p} := \mathbb{I}_{i,p} \ \vee \bigvee_{(j,i)\in E} \big(K(j,i) \wedge \mathbb{I}_{j,p}\big) \quad (\vee = \max,\ \wedge = \min).$$

Then $\mathbb{I} \leq F(\mathbb{I})$ pointwise, so the Kleene iterates $\mathbb{I}^{(0)} := \mathbb{I}(0)$, $\mathbb{I}^{(n+1)} := F(\mathbb{I}^{(n)})$ form an ascending chain.

Remark 2445. *The operator F captures a stylized reinforcement dynamic: an agent i retains its current intensity $\mathbb{I}_{i,p}$ for pattern p, but may increase it if influential neighbors j have intensity for p and the coupling $K(j,i)$ is strong enough. The use of $\wedge = \min$ models a bottleneck: what i can absorb from j is limited by both the strength of j's intensity and the coupling strength. The use of $\vee = \max$ models accumulation by taking the strongest available support.*

Inflationarity $\mathbb{I} \leq F(\mathbb{I})$ says reinforcement does not weaken intensities. Consequently, the Kleene iteration $\mathbb{I}^{(n+1)} = F(\mathbb{I}^{(n)})$ climbs monotonically toward a least fixed point (as referenced from Thm. 19). This is the standard order-theoretic posture that makes social propagation analyzable, and it fits neatly with the general emergence/pattern-web picture [5, 19].

Remark 2446. *The lemma is the key enabling fact for using fixed-point machinery: once iterates are guaranteed ascending, one can take their pointwise supremum and obtain the least stabilized state. This is the same skeleton as in inference closure (Section 42), but applied to social intensities rather than logical beliefs.*

Sketch. For each (i,p), $(F(\mathbb{I}))_{i,p}$ is a max that includes $\mathbb{I}_{i,p}$ as one of its terms, so it is at least $\mathbb{I}_{i,p}$. $\square$

Remark 2447. Proof sketch. *The proof is a direct "included term" argument: $\mathbb{I}_{i,p}$ appears explicitly among the candidates in the max defining $(F(\mathbb{I}))_{i,p}$. Therefore the max cannot be smaller. Visually, F overlays an extra set of candidate reinforcements on top of the existing value and then takes the maximum.* $\square$

Helper Proposition 43.4 (Path-expansion of iterated reinforcement (fuzzy reachability form))**.** *Under the same max/min definition of F, for each $n \geq 0$ and each (i,p),*

$$\mathbb{I}_{i,p}^{(n)} = \max_{j\in\mathcal{A}} \ \max_{\textit{paths } j\to i \textit{ of length } \leq n} \ \min\Big(\mathbb{I}_{j,p}^{(0)},\ \min_{e\in \textit{path}} K(e)\Big),$$

with the convention that the empty path $i \to i$ has edge-minimum 1.

Remark 2448. *This proposition is a "semantic unraveling" of the iteration: it tells us what* $\mathbb{I}_{i,p}^{(n)}$ means. *After* n *reinforcement steps,* i *can be influenced by any agent* j *that can reach* i *by a path of length at most* n*. The strength of influence along a particular path is the minimum of the initial intensity at the source,* $\mathbb{I}_{j,p}^{(0)}$*, and the weakest coupling along the path (the bottleneck* $\min_{e\in path} K(e)$*). Then we take the maximum across all paths and all sources, representing selection of the strongest available transmission channel.*

The empty-path convention says that i *always has access to its own initial intensity without attenuation. This is a standard fuzzy-graph reachability formula:* min *composes along edges,* max *aggregates alternative routes. It provides the essential intuition for why stabilization corresponds to a form of "best bottleneck path" from the strongest initial sources, a recurring motif in network propagation models [5].*

Remark 2449. *The result is important because it connects the algebraic definition of* F *to graph structure. It is, in effect, a closed form for the* n*th iterate: it explains exactly which paths matter and how their couplings interact. This connects back to the earlier "effective coupling along a path" proposition, but with max/min rather than product; the two are different regimes of composition, each natural in its setting.*

Sketch. Induct on n. The update step takes the current value and also the best one-step incoming propagation. Max/min propagation composes by taking min along edges (bottleneck strength) and max across alternative paths. □

Remark 2450. Proof sketch. *The proof strategy is induction: assume the path formula holds for length* $\leq n$*, then show applying* F *once more extends paths by one edge. The key step is that* min *implements concatenation of bottlenecks: the bottleneck of an extended path is the minimum of the previous bottleneck and the new edge coupling. The outer* max *simply records the best among all such extended paths. A helpful picture is water flowing through pipes: the flow is limited by the narrowest pipe (min), and among many routes you care about the route allowing maximal flow (max).* □

Helper Corollary 43.5 (Two useful special cases of the stabilized state)**.** *Let* $\mathbb{I}^{(\infty)}$ *be the least fixed point above* $\mathbb{I}^{(0)}$ *(Thm. 19).*

1. *(No incoming edges) If agent* i *has no incoming edges, then* $\mathbb{I}_{i,p}^{(n)} = \mathbb{I}_{i,p}^{(0)}$ *for all* n*, hence* $\mathbb{I}_{i,p}^{(\infty)} = \mathbb{I}_{i,p}^{(0)}$.
2. *(Unit couplings on a strongly connected graph) If* $K(j,i) = 1$ *for all edges and the graph is strongly connected, then for each pattern* p *and each agent* i*,*

$$\mathbb{I}_{i,p}^{(\infty)} = \max_{j\in\mathcal{A}} \mathbb{I}_{j,p}^{(0)},$$

i.e. everyone converges to the global maximum initial intensity for each pattern.

Remark 2451. *These two cases serve as "boundary conditions" for intuition about social reinforcement. In case (1), the agent is informationally isolated on the incoming side: nothing flows into it, so reinforcement cannot change it. In case (2), information flow is perfect (no attenuation, since* $\min(1,x) = x$*) and the network is richly connected (strongly connected), so the strongest initial advocate for each pattern eventually dominates everyone else's intensity for that pattern. The stabilized state is then simply the global maximum, replicated across all agents.*

Both extremes are conceptually clarifying for Culture *formation (Hyperseed-Concept 91): isolation yields cultural idiosyncrasy, while perfect connectivity yields homogenization to the strongest signals. Real systems live between these extremes, with attenuation and partial connectivity producing plural, locally clustered cultures.*

One can also read these extremes as "degenerate" instances of the usual tension between persistence *and* diffusion*: with no in-neighbors, persistence is absolute because the update at i has no external inputs; with unit couplings on a strongly connected graph, diffusion is maximally effective because every agent is ultimately downstream of every other agent. In particular, case (2) illustrates that when attenuation is removed, the only remaining obstruction to global homogenization is purely graph-theoretic (reachability), so the strongly connected hypothesis is exactly what eliminates the possibility of permanently separated cultural basins.*

Remark 2452. *The corollary connects directly back to the path-expansion proposition: (1) says there are no nontrivial paths into i, so only the empty path matters; (2) says all paths have bottleneck* 1*, so only the source intensity matters. These specializations are often used as sanity checks when implementing or reasoning about propagation algorithms.*

In the language of monotonic propagation operators, (1) is the case where the update at i is pointwise constant in all coordinates except those already present at i, while (2) is the case where the update is governed entirely by taking joins of already-existing values transported along paths. Thus both situations are useful not only for debugging but also for building intuition about why repeated application of F is nondecreasing and why fixed points can often be characterized by simple "no-improvement along any path" conditions.

Sketch. (1) With no incoming edges the big join over $(j, i) \in E$ is empty, so $F(\mathbb{I})_{i,p} = \mathbb{I}_{i,p}$. Concretely, the reinforcement term is a join over an empty index set, hence contributes the neutral element for joins (so nothing is added), and the update reduces to the identity on the (i, p)-entry.

(2) With all couplings 1, each edge transmits the source intensity without attenuation (min with 1 does nothing), and strong connectivity ensures every agent can be reached from every other agent by some finite path. In particular, for any fixed pattern p and any pair of agents $j \to i$, the path-expansion formula shows that after iterating F enough times to cover the length of a j–to–i path, the value at (i, p) is at least $\mathbb{I}_{j,p}$; taking the maximum over all possible sources j yields the global maximum. The value cannot exceed the global maximum because each update is formed by taking joins of previously existing intensities (and in this case transports them without amplification), so the global maximum is an upper bound throughout the dynamics. □

Remark 2453. Proof sketch. *For (1), the operator F has no reinforcement term to add, so it is the identity on i's coordinates. For (2), because every edge transmits perfectly, the only question is reachability; strong connectivity guarantees every source can reach every target, so each agent eventually attains at least the maximum source value, and cannot exceed it because updates are built from maxima of existing values.*

Equivalently, letting $\mathbb{I}^{(t)} = F^t(\mathbb{I})$, one can argue by induction on t (using the path-expansion proposition) that $\mathbb{I}^{(t)}_{i,p}$ equals the maximum of $\mathbb{I}_{j,p}$ over all j that can reach i by a path of length at most t; when the graph is strongly connected, there exists some finite t (depending on i and j) for which every j reaches i, so the expression stabilizes at the global maximum. This also makes clear why the fixed point in case (2) is not merely an equilibrium but the least fixed point above the initial condition under the monotone iteration of F. □

43.2 Collective belief and pattern webs (Defs. 345–347): lattice facts

Helper Lemma 43.6 (Collective belief is the pointwise supremum of individual beliefs)**.** *Let $\{\beta_i : L \to V\}_{i\in\mathcal{A}}$ be individual belief states and define*

$$\beta_{\mathrm{col}}(\varphi) := \Join_{i\in\mathcal{A}} \beta_i(\varphi) \quad \textit{(pointwise join).}$$

Then:

1. *For each i, $\beta_i \le \beta_{\mathrm{col}}$ pointwise.*
2. *If γ is any belief state with $\beta_i \le \gamma$ for all i, then $\beta_{\mathrm{col}} \le \gamma$.*

Thus β_{col} is the least belief state that preserves all individuals' evidence.

Remark 2454. *The definition of β_{col} is the simplest possible model of a "collective belief state": at each claim φ, we aggregate the community's evidence by taking the join (supremum) across agents. If V is the p-bit space $[0,1]^2$ with componentwise maxima, this means the collective retains the strongest observed positive evidence and the strongest observed negative evidence for each claim. This is a principled paraconsistent move: disagreement is not averaged away but preserved, making the collective belief state a record of the group's epistemic potential rather than a forced consensus [23, 24].*

The lemma states that this construction has the universal property of a supremum: it is above each individual, and it is the least such upper bound. In order-theoretic language, β_{col} is the pointwise least upper bound in the poset V^L. This is the right abstraction if one wants the collective to "contain" (in the order sense) the evidence of all members without privileging any particular aggregation beyond join.

It is also worth making explicit the mild lattice-theoretic precondition implicit in the notation $\Join_{i\in\mathcal{A}}$: for each fixed $\varphi \in L$, the join of the family $\{\beta_i(\varphi)\}_{i\in\mathcal{A}}$ must exist in V. In most intended applications, V is at least a (complete) join-semilattice, and often a complete lattice or a quantale, so arbitrary (possibly infinite) joins are defined. Under these standard assumptions, the function space V^L inherits joins pointwise, and the expression "pointwise join" is not merely suggestive but literally the join in the product/order-enriched structure.

Remark 2455. *What the lemma says in plain English is:* the collective belief state is exactly the minimal state that does not forget anyone's evidence. *This is important because later notions like pattern webs and mindplex structure often require a collective substrate; by choosing the supremum construction, one ensures that reasoning about the collective is monotone in the agent set and compatible with the rest of the quantale algebra [3]. It also aligns with the "mindplex" ethos of preserving distributed cognitive content [12].*

Concretely, if one treats each β_i as a constraint or information-bearing profile over L, then β_{col} is the least profile that simultaneously satisfies (dominates) all those constraints in the information order. This is precisely the kind of construction that later allows pattern webs (Defs. 345–347) to be formed without committing to a single "resolved" stance at points of disagreement: the web-building operations can be applied to β_{col} and will see, at each φ, the totality of available supports and counter-supports rather than a prematurely collapsed summary.

Sketch. (1) Each $\beta_i(\varphi)$ is one of the joined terms in $\beta_{\mathrm{col}}(\varphi)$. (2) A join is the least upper bound: if every term is below $\gamma(\varphi)$ then their join is too, for each φ. $\square$

Remark 2456. Proof sketch. *The proof is the defining property of joins, applied coordinatewise in φ. The key step is recognizing that V^L is ordered pointwise, so "least upper bound" decomposes into independent least upper bounds in V for each claim. One may visualize each claim φ as its own little lattice; the collective belief simply takes the supremum in each lattice separately.*

Equivalently (and this is often useful later when manipulating pattern-web constructions), one can regard β_{col} as the join $\bowtie_{i\in\mathcal{A}} \beta_i$ inside the poset V^L itself: for each φ, evaluation at φ is an order-preserving map $V^L \to V$, so it preserves the order inequalities needed to verify the universal property. This makes the lemma a direct instance of the general fact that products (and function spaces with pointwise order) compute joins componentwise whenever the codomain admits the relevant joins. □

Helper Corollary 43.7 (Adding agents cannot reduce collective belief)**.** *If $\mathcal{A} \subseteq \mathcal{A}'$ and β_{col} is computed by joining over the agent set, then $\beta_{\text{col}}^{\mathcal{A}} \leq \beta_{\text{col}}^{\mathcal{A}'}$ pointwise.*

Remark 2457. *This corollary is the monotonicity of the collective with respect to membership: adding more agents can only add more evidence, never remove it, when aggregation is by join. It expresses a basic intuition about* Society *(Hyperseed-Concept 170): an enlarged society has (weakly) greater epistemic capacity, at least in the sense of possessing more available evidence to draw upon, even if it also contains more conflict.*

This is also an algebraic analogue of arguments about group efficiency and prosocial coordination: in settings where contributions can be integrated without destructive interference, larger groups can in principle do more [6].

From the perspective of later pattern-web constructions, this monotonicity has a useful stability consequence: any derived structure that is monotone in its belief-state input (e.g. closure operators, reachability or entailment-like transforms defined via isotone maps on V^L) will also be monotone in the agent set. Thus, enlarging the agent set cannot delete already-available patterns; at most it can add new ones or strengthen existing ones in the lattice order.

Remark 2458. *The result is a direct corollary of the supremum characterization in the previous lemma, so it tightly connects the collective-belief definition to its order-theoretic consequences.*

In particular, one can read the inequality $\beta_{\text{col}}^{\mathcal{A}} \leq \beta_{\text{col}}^{\mathcal{A}'}$ as an instance of functoriality of the join operator with respect to inclusion of index sets: the map $\mathcal{A} \mapsto \bowtie_{i\in\mathcal{A}} \beta_i(\varphi)$ is an order-preserving set-function when the codomain is ordered by $\leq$ and joins exist. This observation is frequently used implicitly when reasoning about "adding voices" to an epistemic aggregate in a lattice-valued setting.

Sketch. A join over a larger index set is greater than or equal to the join over a subset. □

Remark 2459. Proof sketch. *The strategy is inclusion of index sets: the join over $\mathcal{A}$ is a join over fewer terms than the join over $\mathcal{A}'$, hence cannot exceed it. One can picture taking a maximum: the maximum of a set cannot decrease when you add more elements.*

Spelling out the pointwise step: fix $\varphi \in L$. Since $\mathcal{A} \subseteq \mathcal{A}'$, the family $\{\beta_i(\varphi)\}_{i\in\mathcal{A}}$ is a subfamily of $\{\beta_i(\varphi)\}_{i\in\mathcal{A}'}$. Therefore the join of the smaller family is $\leq$ the join of the larger family in V, and this holds for every φ, yielding the claimed pointwise inequality in V^L. □

Helper Corollary 43.8 (Paraconsistent disagreement witness (canonical p-bit case))**.** *In the canonical p-bit join, if there exist agents i, j such that $\beta_i^+(\varphi) \geq \theta$ and $\beta_j^-(\varphi) \geq \theta$, then $\beta_{\text{col}}(\varphi)$ has both coordinates at least θ.*

Remark 2460. *This is the collective-belief analogue of the paraconsistent culture witness above: it shows that if one agent strongly supports φ while another strongly opposes it, the collective belief* *state records both facts simultaneously. In other words, the collective can* know *that it disagrees. In paraconsistent epistemology, this is not a defect but a feature: it preserves the informational situation from which further inquiry can proceed without collapsing into triviality [23, 24].*

This connects naturally to Belief System *and* Intersubjective Reality *(Hyperseed-Concepts 64,* **??***): a society's shared epistemic object may include tensions that are themselves stable social facts.*

In the canonical p-bit setting (with $V = [0,1]^2$ and order/join taken componentwise), the statement is almost tautological but conceptually important: the join operation does not impose any normalization constraint like "the positive and negative weights must sum to at most 1". Hence it can represent "high positive" and "high negative" evidence simultaneously. This is precisely the algebraic room needed for later pattern-web and mindplex formalisms to distinguish (i) absence of information from (ii) presence of balanced but conflicting information, since both would be collapsed by scalar averaging but remain distinct in the product lattice.

Remark 2461. *The corollary is important because it demonstrates, in the simplest possible way, how the chosen lattice operations make disagreement first-class. It thereby provides an algebraic foundation for later collective-mind constructions that must tolerate internal inconsistency (e.g. mindplexes spanning diverse subcultures) [12].*

Sketch. Coordinatewise maximum preserves each witnessed coordinate lower bound. □

Remark 2462. Proof sketch. *The proof is again a witness argument: since $\beta^+_{\mathrm{col}}(\varphi)$ is the maximum of all agents' positive coordinates, it is at least $\beta^+_i(\varphi)$; similarly for the negative coordinate. The geometry is two independent axes; maxima act independently on each axis.* □

43.3 Coherence and mindplex structure (Defs. 348–351): useful rewrites

Helper Lemma 43.9 (Coherence bounds and an equivalent per-component form)**.** *Assume* $\mathrm{Err}_O(x;S) \in [0,1]$ *after normalization and define*

$$\mathrm{Coh}_O(S) := 1 - \max_{x\in S} \mathrm{Err}_O(x;S).$$

Then $0 \le \mathrm{Coh}_O(S) \le 1$*. Moreover, for any* $\varepsilon \in [0,1]$*,*

$$\mathrm{Coh}_O(S) \ge 1-\varepsilon \quad \Longleftrightarrow \quad \forall x \in S:\ \mathrm{Err}_O(x;S) \le \varepsilon.$$

Remark 2463. *Coherence here is defined via a* worst-case error*: $\mathrm{Err}_O(x;S)$ measures (for observer O) how well the behavior of x can be approximated from the rest of the set $S \backslash x$; then $\mathrm{Coh}_O(S)$ is one minus the maximum of these errors. Thus high coherence means no member of S is "too surprising" relative to the others, under the observer's admissible approximations. This is one formal route toward capturing the idea that a mindplex is not merely a heap of agents, but a set whose components are mutually predictable and hence capable of higher-order integration (Hyperseed-Concept 113; see also [12]).*

The equivalence with the per-component bound is especially useful: a single inequality about $\mathrm{Coh}_O(S)$ is exactly the same as uniform control of each $\mathrm{Err}_O(x;S)$. In practice, this allows proofs about coherence to proceed by checking all components individually, rather than manipulating maxima abstractly.

Remark 2464. *The lemma says, in plain terms:* coherence is always between 0 and 1, and being at least $1-\varepsilon$ means every component has error at most ε*. This is important because later threshold-style definitions of mindplex structure often quantify coherence by demanding it exceed some bound; the lemma converts that global demand into local constraints that are easier to verify and reason about.*

Sketch. Bounds: $\max_x \mathrm{Err}_O(x;S) \in [0,1]$ so $1 - \max_x \mathrm{Err} \in [0,1]$. Equivalence: $\mathrm{Coh}_O(S) \geq 1 - \varepsilon$ iff $\max_x \mathrm{Err}_O(x;S) \leq \varepsilon$, which holds iff every $\mathrm{Err}_O(x;S) \leq \varepsilon$. □

Remark 2465. Proof sketch. *The proof uses elementary properties of maxima: bounding the maximum is equivalent to bounding each element. The "geometry" is that* $\mathrm{Coh}_O(S)$ *is determined by the single worst offender in* S*; raising coherence above a threshold means eliminating any offender worse than* ε*.* □

Helper Lemma 43.10 (Coherence monotone in approximator power and distance)**.** *Let* $\mathcal{F}$ *be the admissible approximator class used in* $\mathrm{Err}_O(x;S) := \inf_{f\in\mathcal{F}} d_O(\mathrm{Beh}_O(x), f(\mathrm{Beh}_O(S \setminus x)))$*. If* $\mathcal{F} \subseteq \mathcal{F}'$*, then* $\mathrm{Err}_{O,\mathcal{F}'}(x;S) \leq \mathrm{Err}_{O,\mathcal{F}}(x;S)$ *for all* x, S*, hence* $\mathrm{Coh}_{O,\mathcal{F}'}(S) \geq \mathrm{Coh}_{O,\mathcal{F}}(S)$*. Also, if* $\mathrm{d}'_O \leq \mathrm{d}_O$ *pointwise, then coherence computed with* d'_O *is at least that computed with* d_O*.*

Remark 2466. *This lemma expresses a methodological principle: coherence depends on the observer's modeling power and metric choice. If the observer allows a richer approximator class* $\mathcal{F}'$ *(more expressive models), then each error* Err *can only go down, because the infimum is taken over a larger set. Likewise, if the observer uses a more permissive distance* d'_O *that judges discrepancies smaller (*$\mathrm{d}'_O \leq \mathrm{d}_O$ *pointwise), then errors shrink. In both cases coherence increases.*

This matters philosophically: coherence is not purely "in the system" but is defined relative to an observer O *and its representational/measurement scheme (Hyperseed-Concept 112). Mathematically, it supplies monotonicity levers: if you want a lower bound on coherence, you may safely enlarge* $\mathcal{F}$ *or relax the metric without breaking the inequality.*

Remark 2467. *The lemma is also important for connecting mindplex structure to computational resources: a more powerful modeling class can recognize stronger interdependence among components, raising measured coherence. This resonates with views in which richer-resource minds can exploit more sophisticated internal simulation and modeling strategies [11].*

Sketch. Enlarging the infimum domain can only decrease an infimum, hence errors decrease and coherence increases. If $d'_O \leq d_O$, every candidate distance shrinks, so the infimum shrinks as well. □

Remark 2468. Proof sketch. *The core move is order-theoretic:* inf *is monotone with respect to set inclusion of its domain, and also monotone with respect to pointwise reduction of the objective function. One can picture* $\inf_{f\in\mathcal{F}}$ *as "best achievable error"; giving oneself more candidates or a more forgiving error measure cannot worsen the best achievable error.* □

Helper Proposition 43.11 (Type implications among collective mind notions)**.** *From the definitions:*

$$\textit{Global brain} \Rightarrow \textit{Mindplex} \Rightarrow \textit{Society of mind} \Rightarrow \textit{Collective mind system} \Rightarrow \textit{Society.}$$

Remark 2469. *This proposition states a chain of definitional containment among increasingly weak notions. At the top is a* global brain *(Hyperseed-Concept* **??***), intended as a planet-scale cognitive integration; this is, by definition, a special kind of* mindplex *(Hyperseed-Concept 113), which is itself a constrained kind of* society of mind*, and so on down to the bare notion of* society *(Hyperseed-Concept 170). The arrows are not deep theorems; they are reminders that the terminology is layered: each higher term simply includes all the structure of the lower term plus additional axioms. In particular, each node in the chain can be read as a* structure *(in the sense of "a carrier with operations/relations satisfying axioms"), and each arrow corresponds to a canonical "forgetful" passage that drops some extra fields/axioms while retaining the underlying society. Thus the* *content is not that some emergent phenomenon always exists, but that whenever you have chosen to work with the stronger notion, you automatically have an instance of every weaker notion by forgetting additional structure.*

This is useful in proofs because it licenses "weakening": if you have established a property for all societies, you automatically get it for all collective mind systems, societies of mind, mindplexes, and global brains. Conversely, if you assume you have a global brain, you may freely use every lemma that only requires society-hood. Operationally, this is the common pattern "prove at the bottom, reuse at the top": one proves invariants about interaction graphs, information-flow constraints, or agent counts at the level of societies, and then reuses them unchanged when later specializing to mindplexes or global brains. The direction does matter: results proved only *for mindplexes will not automatically apply to arbitrary societies unless their hypotheses are explicitly weakened.*

Remark 2470. *The importance is practical and conceptual. Practical, because it organizes lemmas by the weakest hypotheses needed. Conceptual, because it mirrors an ontological ladder: from mere aggregation of agents to increasingly integrated wholes, as explored in discussions of mindplexes and collective locations [12] and in broader AGI-oriented analyses of multi-agent cognition [19]. One can also view the ladder as clarifying which aspects of "mind" are being idealized at each stage: at the society level one keeps only plurality and interaction; at a society-of-mind level one adds an explicit decomposition into subagents with roles; at a mindplex level one adds coherence constraints (e.g. compatibility of decompositions, stability of interfaces, or admissible refinements); and at a global-brain level one adds scale and boundary conditions that tie the system to an entire planetary society. In that sense, the chain functions as a conceptual bookkeeping device: when a later argument invokes "coherence," the reader can immediately locate it at the mindplex/global-brain end rather than mistaking it for a generic property of arbitrary societies.*

Sketch. Each arrow is a direct definitional implication: a global brain is defined to be a mindplex on a planet-scale society; a mindplex is defined to be a society of mind with additional coherence/decomposition properties; and a society of mind is defined to be a collective mind system whose components are autonomous agents; etc. Formally, if the concepts are presented as predicates on structured tuples (or as types with fields), then each implication is discharged by taking an instance of the stronger predicate/type and projecting to the fields demanded by the weaker one, observing that the weaker axioms are literally included among the stronger axioms (cf. Defs. 348–351). No additional construction is required beyond this projection/forgetting step. ☐

Remark 2471. Proof sketch. *The proof is a sequence of "unpack the definition" steps. The key idea is that the vocabulary has been designed hierarchically: every time you move left in the chain you are adding constraints, never removing them. A helpful picture is nested sets: global brains are a subset of mindplexes, which are a subset of societies of mind, and so on. Equivalently, one can picture a chain of inclusions of classes of models, or a chain of coercions in a proof assistant: whenever a term inhabits the stronger class, it can be coerced to inhabit the weaker class by forgetting structure. This is precisely why the arrows should be read as "is in particular a" rather than as an existential claim that every society can be completed into a mindplex/global brain; the containment is one-way and definitional. In applications, this viewpoint prevents category mistakes: any lemma whose statement mentions only society-level primitives is automatically stable under these forgetful maps, whereas lemmas mentioning coherence or decomposition data are not.* ☐

43.4 Role templates and fit scores (Defs. 339–342): threshold monotonicities

Helper Lemma 43.12 (Fit-threshold monotonicity)**.** *Let* $\mathrm{Fit}(R, \rho; \mathrm{Ep}) \in [0, 1]$ *be a role-fit score and say the role is present if* $\mathrm{Fit} \geq \theta$*. If* $\theta' \leq \theta$ *and the role is present at threshold* θ*, then it is* *present at threshold θ'.*

Remark 2472. *This is the same threshold monotonicity pattern seen earlier for tribes and cultures, now applied to social roles (Hyperseed-Concept 171). A role-fit score* $\mathrm{Fit}(R, \rho; \mathrm{Ep}) \in [0, 1]$ *measures how well an agent or sub-system matches a role template R with parameters ρ in an episode* Ep*. Declaring the role "present" when* $\mathrm{Fit} \geq \theta$ *turns a graded measure into a crisp predicate; lowering θ can only make it easier to satisfy.*

As a concrete example: if a "mediator" role requires fit at least 0.8*, and an agent scores* 0.85*, then it is a mediator at threshold* 0.8*; it is automatically also a mediator at threshold* 0.7*. This lemma is useful whenever one wants to reason about role detection across multiple sensitivity settings without re-running the detection logic.*

Remark 2473. *One convenient way to package the lemma is to explicitly name the induced predicate. For a fixed* (R, ρ, Ep)*, define*

$$\mathrm{Present}_\theta(R, \rho; \mathrm{Ep}) \ :\Longleftrightarrow\ \big(\mathrm{Fit}(R, \rho; \mathrm{Ep}) \geq \theta\big).$$

Then the lemma states that $\theta' \leq \theta$ *implies* $\mathrm{Present}_\theta(R, \rho; \mathrm{Ep}) \Rightarrow \mathrm{Present}_{\theta'}(R, \rho; \mathrm{Ep})$*; i.e., the map* $\theta \mapsto \mathrm{Present}_\theta$ *is antitone in the threshold parameter (smaller threshold yields a larger truth set). Equivalently, for any fixed θ, the set of triples* (R, ρ, Ep) *(or agents, if* Fit *is agent-indexed) satisfying the predicate is* upward closed *in the fit score and* downward closed *in the threshold.*

Remark 2474. *The lemma is not surprising, but it is structurally important: it means the collection of agents satisfying a role predicate is nested as thresholds vary. This supports monotone searches over thresholds and makes role-based decompositions stable under weakening criteria.*

Remark 2475. *A related contrapositive form is often what gets used in practice: if the role is* absent *at a lower threshold θ' (i.e.,* $\mathrm{Fit} < \theta'$*), then it is absent at every higher threshold $\theta \geq \theta'$. In algorithmic terms, once an agent fails to meet an easy criterion, it cannot meet a stricter one; this is what permits early stopping in threshold sweeps, and it underlies the typical "tune θ upward until false positives disappear" workflow.*

Sketch. If $\mathrm{Fit} \geq \theta$ and $\theta' \leq \theta$, then $\mathrm{Fit} \geq \theta'$. ☐

Remark 2476. Proof sketch. *The argument is pure order: a real number above a larger threshold is also above any smaller threshold. The "geometry" is a number line with two cut points; crossing the higher cut implies crossing the lower.* ☐

Remark 2477. *Although the statement is formulated for* $\mathrm{Fit}(R, \rho; \mathrm{Ep}) \in [0, 1]$*, only the ambient order is used; the same monotonicity holds for any totally ordered codomain (or, more generally, any poset) in which the predicate "$\geq \theta$" makes sense. The* $[0, 1]$ *normalization is thus best read as a modeling convenience (comparability across roles and episodes) rather than a proof-critical assumption.*

Helper Lemma 43.13 (Colleague relation is symmetric under the operational definition)**.** *If "colleagues" are defined as co-participants in the same collaborative project episode (Def. 342), then for any episode,* $\mathrm{Colleague}(i, j) \Rightarrow \mathrm{Colleague}(j, i)$*.*

Remark 2478. *This lemma records a basic symmetry built into the operational notion of colleague (Hyperseed-Concept 74). If the definition is simply "i and j participated in the same collaborative project episode," then the relation is symmetric because the underlying fact is symmetric: the set of participants does not distinguish ordered pairs. This is a small logical convenience: it lets one treat* *colleague graphs as undirected (at least at the level of adjacency), even if other interaction graphs remain directed.*

In broader social-cognitive modeling, such symmetry facts matter because they allow simplifications: for instance, when reasoning about mutual trust or shared context between colleagues, one may freely swap the order of agents without changing truth values.

Remark 2479. *Unpacking the operational definition slightly, one typically has something like*

$$\text{Colleague}(i,j) \;:\Longleftrightarrow\; \exists\,\text{Ep}\;\big(\text{CollaborativeProject}(\text{Ep})\wedge i\in\text{Participants}(\text{Ep})\wedge j\in\text{Participants}(\text{Ep})\big).$$

Under this reading, symmetry is immediate because the conjunction is invariant under exchanging i and j. This also clarifies what is not *being claimed: symmetry does not by itself imply transitivity (two people can each work with a common third person on different projects without ever working together), and it does not force reflexivity unless one separately stipulates that every agent is trivially a co-participant with itself in an episode.*

Remark 2480. *The result is important mainly as a rewrite rule: it prevents accidental orientation from creeping into arguments that should be purely about co-membership. It also complements the earlier directed-edge results by distinguishing which relations are inherently directional (influence, communication) and which are inherently mutual (co-participation).*

Remark 2481. *One additional payoff is representational: in any adjacency-matrix encoding of* Colleague *for a fixed time window, the lemma justifies symmetry of the matrix (or, equivalently, use of a simple undirected graph). When colleague edges are later combined with directed edges (e.g., "mentors" or "reports-to" relations), this symmetry fact helps keep the mixed graph semantics clean by ensuring that undirected components arise only from definitions that genuinely ignore order.*

Sketch. Co-participation in the same project is symmetric in the participants. □

Remark 2482. Proof sketch. *The proof strategy is to unpack "co-participation": it asserts existence of an episode containing both i and j. But set membership is symmetric: if both are in the same set, the same fact certifies both directions.* □

44 Helper theorems for tools, machines, computation, and embodiment

Remark 2483. *This section collects "small" algebraic facts that become powerful once one begins to compose them: joins and tensors in the p-bit quantale, relational composition of affordances, and categorical composition of semantics/realizations. Conceptually, it is a bridge between (i) the formal notion of a* tool *and its affordance profile (Hyperseed-Concept 191) and (ii) the formal notion of* machines *and* computers/programs *(Hyperseed-Concepts 106 and 80), and (iii) the embodied constraint that mind–world influence is mediated by a* body *(Hyperseed-Concept 135 and 112). The intent is not to introduce new theory, but to make explicit the algebra that an automated reasoner (or a patient human) will repeatedly invoke.*

Remark 2484. *A brief reminder of the ambient algebra may help orient the eye. The symbol V denotes the truth-value domain for graded evidence; in the surrounding text this is typically the p-bit quantale $V=[0,1]^2$ (positive/negative evidence pairs) ordered componentwise. The join $\oplus$ is the quantale join (often componentwise max), while the tensor $\otimes$ (written here as $\otimes$) is the "and/compose"-like operation that distributes over joins. Thus, expressions like $\bigoplus_{a\in A}\mathsf{Aff}(a,e)$*

mean "aggregate across alternatives" (existential aggregation over actions), and expressions like $\mathsf{Aff}_1(a,m) \otimes \mathsf{Aff}_2(m,e)$ *mean "compose the support for a producing m with the support for m producing e." This style of reasoning is standard in quantale-valued semantics and is used elsewhere in the theory to speak about Quantale Weakness (Hyperseed-Concept 143; see also [3]).*

Definition 445 (Achievability induced by an affordance profile)**.** *Let* $\mathsf{Aff} : A \times E \to V$ *be an affordance profile. Define the (existential) achievability of an effect token* $e \in E$ *by*

$$\mathsf{Ach}_{\mathsf{Aff}}(e) \ := \ \bigoplus_{a \in A} \mathsf{Aff}(a,e).$$

Intuition: $\mathsf{Ach}_{\mathsf{Aff}}(e)$ *aggregates evidence that* some *action can produce* e*.*

Remark 2485. *The definition extracts from the* action-to-effect *table* Aff *a simpler object: a single value per effect e, summarizing "how achievable e is, in principle, by some choice of action." Philosophically, it is the shift from a conditional statement (if we do a we may get e) to an existential one (there exists an a such that e is attainable). This is precisely the kind of compression that turns a detailed model of agency into a crude but useful proxy for "capability." When tools are treated as capability amplifiers (Hyperseed-Concept 191), such a proxy often suffices to establish monotone-improvement claims.*

A simple example is the Boolean special case: if $V = \{0,1\}$ *with* $\oplus = \vee$*, then* $\mathsf{Ach}_{\mathsf{Aff}}(e) = 1$ *iff at least one action achieves e. In the p-bit case* $[0,1]^2$*, if* $\mathsf{Aff}(a,e) = (p^+_{a,e}, p^-_{a,e})$*, then* $\mathsf{Ach}_{\mathsf{Aff}}(e)$ *takes the componentwise maximum across actions, capturing the idea that the best-supported action dominates the existential claim. In engineering terms (cf. [19]),* $\mathsf{Ach}_{\mathsf{Aff}}$ *is a coarse "capability envelope" derived from the more detailed affordance model.*

Lemma 150 (Monotonicity of achievability)**.** *If* $\mathsf{Aff}_1, \mathsf{Aff}_2 : A \times E \to V$ *satisfy* $\mathsf{Aff}_1(a,e) \le \mathsf{Aff}_2(a,e)$ *for all* (a,e)*, then* $\mathsf{Ach}_{\mathsf{Aff}_1}(e) \le \mathsf{Ach}_{\mathsf{Aff}_2}(e)$ *for all* $e \in E$*.*

Remark 2486. *In plain terms: if every individual action looks at least as effective under* Aff_2 *as it does under* Aff_1*, then the best available action (as judged by the join* $\oplus$*) cannot become worse. This lemma is important because many later comparisons ("tool T helps"; "model* M' *is more capable") reduce to pointwise inequalities, and we then want those inequalities to survive the existential aggregation that defines achievability.*

Proof sketch. For fixed e, $\mathsf{Ach}_{\mathsf{Aff}}(e)$ is a join over actions. Joins are monotone in each argument, so pointwise domination of Aff_2 over Aff_1 implies domination of the joins. □

Remark 2487. Proof sketch. *The strategy is purely order-theoretic: we use that* $\bigoplus$ *is computed by repeatedly applying* $\oplus$*, and* $\oplus$ *respects the order in each coordinate. No special structure of actions or effects is needed.* □

The key step is recognizing that $\mathsf{Ach}_{\mathsf{Aff}}(e)$ *is a monotone functional of the entire family* $\{\mathsf{Aff}(a,e)\}_{a \in A}$*: increasing any term cannot decrease the join. Visually, one may imagine each* $\mathsf{Aff}(a,e)$ *as a point in the ordered set* V*; the join is then the "upper envelope" of these points, and pushing all points upward can only push the envelope upward as well.*

Definition 446 (Quantale-relational composition of affordances)**.** *Let* $\mathsf{Aff}_1 : A \times M \to V$ *and* $\mathsf{Aff}_2 : M \times E \to V$ *be two affordance profiles where* M *is an intermediate token set (e.g. sub-effects, internal states, or interface signals). Define the composed affordance profile* $\mathsf{Aff}_2 \circledast \mathsf{Aff}_1 : A \times E \to V$ *by*

$$(\mathsf{Aff}_2 \circledast \mathsf{Aff}_1)(a,e) \ := \ \bigoplus_{m \in M} \mathsf{Aff}_1(a,m) \otimes \mathsf{Aff}_2(m,e).$$

Remark 2488. *This is the quantale-valued analogue of ordinary relational composition: in classical relations, (a, e) is related if there exists an intermediate m such that aRm and mSe. Here, existence is replaced by the join $\bigoplus$, and conjunction/sequence is replaced by the tensor $\otimes$. One may also read it as a kind of "matrix multiplication" over a semiring/quantale: $A \times M$ values are multiplied (via $\otimes$) and summed (via $\oplus$) over m.*

A concrete example: let A be a set of button-presses, M a set of device internal states, and E a set of externally visible outcomes. $\mathsf{Aff}_1(a, m)$ measures how strongly action a puts the device into state m, while $\mathsf{Aff}_2(m, e)$ measures how strongly state m yields outcome e. The composed profile measures how strongly a yields e through *some internal state. This is useful because tool-use often has hidden intermediate structure (interfaces, protocols, bodily channels), yet the agent wants an effective external affordance profile to reason with (Hyperseed-Concept 191; cf. [1] for the broader ontological framing).*

Lemma 151 (Lower bounds for composed affordances)**.** *For all $a \in A$, $e \in E$, and $m \in M$,*

$$(\mathsf{Aff}_2 \circledast \mathsf{Aff}_1)(a, e) \;\geq\; \mathsf{Aff}_1(a, m) \otimes \mathsf{Aff}_2(m, e).$$

Remark 2489. *The lemma says: the composite affordance is at least as strong as any particular "path" through a chosen intermediate token m. In other words, the join over m does not lose information; it retains every specific witness-path as a lower bound. This is the basic inequality that lets one prove things about the composite by exhibiting a single good intermediate realization.*

Proof sketch. The join over m includes the particular term indexed by m, and joins are upper bounds on their joined elements. ☐

Remark 2490. Proof sketch. *The proof is the tautology of joins: $\bigoplus_{m \in M} x_m$ is, by definition, an upper bound of each x_m.* ☐

The key mechanism is that quantale joins behave like suprema. If one pictures each intermediate m as a possible conduit from a to e, then the composite takes the best-supported conduit; any specific conduit therefore provides a guaranteed floor.

Lemma 152 (Monotonicity of affordance composition)**.** *If $\mathsf{Aff}_1 \leq \mathsf{Aff}_1'$ pointwise, then $\mathsf{Aff}_2 \circledast \mathsf{Aff}_1 \leq \mathsf{Aff}_2 \circledast \mathsf{Aff}_1'$. If $\mathsf{Aff}_2 \leq \mathsf{Aff}_2'$ pointwise, then $\mathsf{Aff}_2 \circledast \mathsf{Aff}_1 \leq \mathsf{Aff}_2' \circledast \mathsf{Aff}_1$.*

Remark 2491. *This records the expected monotonicity of "piping" affordances through an interface: strengthening either stage cannot weaken the end-to-end profile. It matters because it allows modular engineering arguments: improve a component (better sensor, better actuator, better protocol) and infer improvement of the overall composed affordance without reanalyzing from scratch.*

Proof sketch. Use monotonicity of $\otimes$ in each argument, and monotonicity of $\oplus$ over m. ☐

Remark 2492. Proof sketch. *Pointwise increase of Aff_1 or Aff_2 increases each term $\mathsf{Aff}_1(a, m) \otimes \mathsf{Aff}_2(m, e)$, and then the join over m preserves that increase.* ☐

What does the work is the two-layered monotonicity: first at the level of each path (tensor), then at the level of aggregating paths (join). Visually: if every edge label in a two-hop graph is raised, then every two-hop path weight is raised, hence the supremum over paths is raised.

Proposition 70 (Associativity of affordance composition (when defined))**.** *Given $\mathsf{Aff}_1 : A \times M \to V$, $\mathsf{Aff}_2 : M \times N \to V$, $\mathsf{Aff}_3 : N \times E \to V$, one has*

$$\mathsf{Aff}_3 \circledast (\mathsf{Aff}_2 \circledast \mathsf{Aff}_1) \;=\; (\mathsf{Aff}_3 \circledast \mathsf{Aff}_2) \circledast \mathsf{Aff}_1,$$

assuming the quantale laws (associativity of $\otimes$ and distributivity of $\otimes$ over joins).

Remark 2493. *Associativity says that multi-stage affordance pipelines have an unambiguous end-to-end meaning: whether one first composes stages 1 and 2, or stages 2 and 3, the resulting $A \times E$ profile is the same. This is the affordance-level analogue of the familiar fact that function composition is associative; it is also the algebraic reason one can speak of a complex toolchain as a single tool interface.*

This result connects forward to the later "pipeline rule" for program realizations: in both cases, the content is that composition is well-behaved*, so that certification and reasoning can be performed modularly and then composed.*

Proof sketch. Expand both sides:

$$(\mathsf{Aff}_3 \circledast (\mathsf{Aff}_2 \circledast \mathsf{Aff}_1))(a,e) = \bigoplus_{n \in N} \Big(\bigoplus_{m \in M} \mathsf{Aff}_1(a,m) \otimes \mathsf{Aff}_2(m,n) \Big) \otimes \mathsf{Aff}_3(n,e).$$

Distribute $\otimes$ over $\oplus$ and reassociate $\otimes$ to obtain a double join

$$\bigoplus_{m \in M} \bigoplus_{n \in N} \mathsf{Aff}_1(a,m) \otimes \mathsf{Aff}_2(m,n) \otimes \mathsf{Aff}_3(n,e),$$

which is symmetric in the bracketing and equals the expansion of $((\mathsf{Aff}_3 \circledast \mathsf{Aff}_2) \circledast \mathsf{Aff}_1)(a,e)$. □

Remark 2494. Proof sketch. *The proof normalizes both bracketings into the same "sum over all length-3 paths" expression by (i) expanding definitions, (ii) distributing $\otimes$ over joins to flatten nested joins, and (iii) using associativity of $\otimes$ to remove parenthesization differences.* □

The crucial step is distributivity: it licenses moving from "join over n of (join over m of something) tensored with something" to a joint join over (m,n). A geometric intuition is to picture a three-layer directed graph $A \to M \to N \to E$; either bracketing corresponds to choosing how to group the internal layers, but the end-to-end weight is determined by the same set of paths.

Remark 2495. *One can view the normalization step as a quantale-valued analogue of a Fubini/Tonelli exchange of "sums": nested joins are flattened into a single join over the product index set, and the associativity of $\otimes$ ensures that the weight assigned to a composite path $a \to m \to n \to e$ is independent of where one places parentheses. In particular, if the two bracketings correspond to two different composite affordance constructions, distributivity asserts that both compute the same global path-aggregate because the internal layer choices (m,n) are ultimately being ranged over in both cases.*

*In the graph picture, each edge carries a V-weight (evidence/feasibility/cost, depending on the chosen quantale semantics). Distributivity guarantees that aggregating over intermediate nodes commutes with composing along edges: whether one first aggregates over $m \in M$ and then over $n \in N$, or aggregates over pairs (m,n) directly, the resulting end-to-end evaluation is the join over the same set of length-*3 *paths.*

Proposition 71 (A sufficient condition for purpose-supporting tools via monotone objectives)**.** *Let an objective functional be a map*

$$U : V^E \to \mathbb{R}$$

that is monotone w.r.t. the pointwise order on V^E. Assume that, without tool T, the agent has affordance profile $\mathsf{Aff}_0 : A \times E \to V$, and with tool T it has $\mathsf{Aff}_T : A \times E \to V$. If $\mathsf{Ach}_{\mathsf{Aff}_0}(e) \leq \mathsf{Ach}_{\mathsf{Aff}_T}(e)$ for all $e \in E$ and the inequality is strict for at least one e that has nonzero marginal utility under U, then

$$U(\mathsf{Ach}_{\mathsf{Aff}_T}) \; > \; U(\mathsf{Ach}_{\mathsf{Aff}_0}).$$

Thus T is purpose-supporting under this monotone-achievability proxy.

Remark 2496. *Read informally: if the tool weakly increases what you can achieve (effect-by-effect), and it strictly increases at least one effect that the objective actually cares about, then the objective score strictly improves. There is nothing mysterious here; it is the familiar monotone-utility argument, transplanted into the quantale-valued achievability setting.*

This proposition is useful because "purpose-supporting" in the sense of tools (Hyperseed-Concept 191) is often difficult to certify directly, whereas monotone proxies are tractable. In cognitive-engineering practice (cf. [19]), one often chooses objectives that are monotone with respect to some capability representation precisely so that such conclusions can be drawn compositionally.

Remark 2497. *The phrase "nonzero marginal utility" can be read in an order-theoretic way that avoids any differential structure: there exists a coordinate $e \in E$ and two profiles $x, y \in V^E$ with $x \leq y$, $x(e) < y(e)$, and $x(e') = y(e')$ for all $e' \neq e$, such that $U(x) < U(y)$. This expresses that U is not merely monotone but* strictly sensitive *to that coordinate in some region of interest. Under this reading, the strictness condition in the proposition precisely rules out the case where U collapses all improvements in* Ach *to the same real score.*

In applications, a common sufficient condition is that U decomposes as an increasing aggregator over coordinates (e.g. a weighted sum, a max, or any monotone norm-like functional on V^E once V is embedded into $\mathbb{R}$), with at least one coordinate receiving genuinely positive weight. The proposition then says that any tool that improves achievability in a positively-weighted dimension is certified as purpose-supporting under the proxy.

Proof sketch. Pointwise domination of $\mathsf{Ach}_{\mathsf{Aff}_T}$ over $\mathsf{Ach}_{\mathsf{Aff}_0}$ and monotonicity of U gives $U(\mathsf{Ach}_{\mathsf{Aff}_T}) \geq U(\mathsf{Ach}_{\mathsf{Aff}_0})$. The strictness assumption yields strict improvement. □

Remark 2498. Proof sketch. *The proof splits into (i) a weak inequality from monotonicity and (ii) a strictness upgrade from a single effect whose improvement is not utility-neutral.* □

The key point is that monotonicity alone only yields $\geq$; to obtain $>$ one must rule out the degenerate case where U ignores all improved coordinates. In more visual terms: U is a monotone "preference gradient" on the partially ordered space V^E; moving upward in at least one direction of nonzero sensitivity forces the value to increase.

Remark 2499. *It is worth emphasizing where the quantale structure enters and where it does not. The proposition itself only needs the pointwise order on V^E together with the definition of* $\mathsf{Ach}_{\mathsf{Aff}}$ *(as a derived profile in V^E) and does not require any additional algebraic properties of V. The quantale assumptions become relevant earlier, in ensuring that* $\mathsf{Ach}_{\mathsf{Aff}}$ *is well-defined and behaves compositionally under changes of affordances (e.g. when tools are composed, or when affordances are induced from multi-step interaction graphs).*

Proposition 72 (Implication chain: computer =¿ machine =¿ engineered)**.** *If C is a computer in the sense of Def. 358, then C is a machine (Def. 354), hence engineered (Def. 355).*

Remark 2500. *This proposition is logically elementary but methodologically important: it makes explicit that the notion of a* computer *used here is not an abstract mathematical object floating free of the world, but a particular kind of* engineered physical system *(Hyperseed-Concepts 80, 106,* **??***). In other words, the chain of definitions forces computation, in this context, to be an embodied and built phenomenon rather than a mere formalism.*

Remark 2501. *The usefulness of making the definitional implication explicit is that later arguments can freely appeal to machine-level or engineering-level properties (robustness, interfaces, failure modes, resource bounds, manufacturability, calibration, etc.) whenever "computer" is assumed, without re-proving that those considerations are in scope. This is especially relevant when* *connecting semantic notions of computation (programs, realizations, interpretations) to causal and physical notions (actuation, sensing, and environmental coupling): the definitional chain prevents an equivocation between abstract computation and realized computation.*

Proof sketch. Def. 358 states a computer is a machine with additional program/realization structure. Def. 354 states a machine is an engineered physical system. Therefore computer implies engineered. □

Remark 2502. Proof sketch. *Unpack the definitional nesting: "computer" includes "machine" as a conjunct, and "machine" includes "engineered" as a conjunct.* □

No algebra is needed; the proof is the transitivity of "is-a" along a definitional hierarchy. The intuitive picture is a set-inclusion chain of concepts: computers sit inside machines, which sit inside engineered systems.

Remark 2503. *Although presented as a simple implication chain, the proposition also functions as a scope marker: it indicates that whenever later sections speak about programs, compilation, or execution, those notions are to be interpreted against a background of engineered constraints and interfaces. In particular, "program/realization structure" is not being treated as a purely Platonic relation, but as something that is established by design choices, fabrication, and the maintenance of stable input–output behavior under relevant operating conditions.*

Proposition 73 (Machine induces an external tool interface (sufficient condition))**.** *Let M be a machine decomposed into components (M_i) with stable input-output behavior. Assume an observer O fixes an external action vocabulary A_{ext} for manipulating M and an external effect vocabulary E_{ext} for outcomes attributable to M. If one can empirically or model-theoretically extract an affordance profile*

$$\mathsf{Aff}_M : A_{\text{ext}} \times E_{\text{ext}} \to V$$

with (i) usability: $\exists(a, e)$ with substantial positive evidence, and (ii) a monotone objective proxy as above is strictly improved by using M, then (M, Aff_M) satisfies the tool criteria.

Proof sketch. Stability of the input–output behavior across the decomposition ensures that the mapping $\mathsf{Aff}_M(a, e)$ is well-defined as a reproducible (or at least model-identifiable) quantale-valued assessment of how action $a \in A_{\text{ext}}$ supports effect $e \in E_{\text{ext}}$. The usability condition prevents the extracted interface from being vacuous (e.g. assigning bottom evidence to every pair), so that M is not a "tool" only in name. Finally, if incorporating the affordances of M yields an achievability profile whose value under a monotone objective proxy is strictly higher, then by Proposition 71 (applied with Aff_0 the baseline and Aff_T the baseline augmented by access to M through A_{ext} and E_{ext}), M is purpose-supporting under that proxy, hence meets the stated tool criterion. □

Remark 2504. *This proposition formalizes a common engineering move: treat a complex artifact as a black-box interface at the boundary of observation, choose a control vocabulary (the external actions) and a measurement vocabulary (the external effects), and then fit or elicit an affordance model relative to those vocabularies. The decomposition into components (M_i) is not used to expose internal details to the agent, but to justify that the boundary behavior is stable enough that an affordance-style summary is meaningful.*

The two conditions (usability and proxy improvement) also separate two failure modes. A system may be stable yet unusable for the agent in the specified action vocabulary, in which case it does not function as a tool. Conversely, a system may be usable but irrelevant to the objective (or improve only utility-neutral effects), in which case it is an "instrument" without being purpose-supporting for the chosen task. The monotone-objective condition provides a lightweight certificate that avoids *modeling the full downstream consequences of using M while still connecting the interface to an explicit notion of benefit.*

Remark 2505. *This statement is a formal "interface theorem": under stable input–output behavior, and relative to a chosen observer vocabulary, a machine can be treated as a tool by extracting an affordance profile and checking the two tool clauses (usable; purpose-supporting). The role of the observer O is philosophically significant: what counts as an "action" and an "effect" is not written into nature in Platonic ink, but carved out by a modeling stance. One can read this in a Peircean spirit as emphasizing the triadic relation among sign (the vocabulary), object (the machine), and interpretant (the observer's inferential practices) [14].*

In particular, the "stable input–output behavior" assumption is doing the technical work of ensuring that the extracted profile is not an artifact of transient internal conditions: repeated applications of the same observed action (in the vocabulary of O) lead to sufficiently consistent observed effects for the profile to be well-defined for downstream reasoning. The theorem therefore functions as a disciplined bridge from an embodied *system with potentially complicated internal dynamics to an* external *interface object suitable for compositional analysis.*

This proposition connects directly back to the earlier achievability monotonicity results: once Aff_M is extracted, one can reason about usefulness via $\mathsf{Ach}_{\mathsf{Aff}_M}$ and monotone objectives, without needing to revisit the internal component structure (M_i). In other words, the same monotonicity machinery that supports "more options cannot hurt" (in the appropriate preorder of objectives) can be applied at the affordance level, where the object of study is the observable capability set rather than the mechanism that generates it.

Proof sketch. This is a packaging theorem: once an external affordance profile exists and improves a suitable objective proxy, Def. 353's (a) usable and (b) purpose-supporting clauses are met. □The only substantive content is the alignment of hypotheses with the two clauses: the stability assumptions guarantee that "usable" is meaningful as a property of the interface, while the objective-improvement assumption instantiates "purpose-supporting" for the chosen purpose proxy.

Remark 2506. Proof sketch. *The proof is definitional: verify the tool axioms by supplying Aff_M as the witness affordance profile and invoking (i) usability and (ii) objective improvement as the two required premises.* □

The key step is recognizing that the internal decomposition (M_i) is only used to justify that an external affordance profile can *be extracted; once Aff_M is in hand, the tool criterion speaks purely in terms of the external action/effect interface. This makes explicit a methodological point: proofs about tool-hood are, after extraction, proofs in the "interface language" of O, and therefore remain valid under any internal redesign that preserves the same Aff_M (or improves it in the relevant preorder). Put differently, the tool predicate is intentionally invariant under internal refactoring, so long as the observed interface contract is preserved.*

Lemma 153 (Functoriality of program semantics)**.** *Let* $\mathrm{J}\cdot\mathrm{K} : \mathsf{Syn} \to \mathsf{Abs}$ *be the semantics functor. Then for any composable programs p, q in* Syn*:*

$$\mathrm{JidK} = \mathrm{id}, \qquad \mathrm{J}q \circ p\mathrm{K} = \mathrm{J}q\mathrm{K} \circ \mathrm{J}p\mathrm{K}.$$

Remark 2507. *This lemma states that the semantics mapping respects the two most basic structural facts about programs: (i) doing nothing means doing nothing, and (ii) the meaning of running p and then q is the composition of the meanings. Here* Syn *is the category of syntactic programs,* Abs *the category of abstract behaviors (or computations), and* $\llbracket \cdot \rrbracket$ *is the interpretation function that forgets syntax and keeps only meaning (Hyperseed-Concept 80).*

It is useful to read the equalities as coherence conditions *on any semantics one is willing to use for certification: if the syntactic category has a designated identity program and a notion of sequencing, then the semantic world must mirror these operations in a way that is compatible with equational reasoning. Concretely, if one proves in* Syn *that two programs are equal by rewriting using associativity/identity laws, then functoriality guarantees that their denotations in* Abs *match, so that one can transport correctness arguments from code structure to abstract behavior.*

While mathematically routine, functoriality is the hinge on which compositional certification turns: it lets one transfer equational reasoning from syntax to semantics without re-proving it case by case. The later pipeline realization theorem depends on this identity explicitly. In particular, the pipeline rule relies on the fact that "compose programs then interpret" agrees with "interpret then compose behaviors," so that the physical composition theorem can be applied to the correct abstract target.

Proof sketch. This is the defining property of a functor. □Equivalently, once $\llbracket \cdot \rrbracket$ is stipulated to be a functor (rather than an arbitrary map), the two displayed equalities are immediate from the functor axioms.

Remark 2508. Proof sketch. *Invoke the axioms of a functor: preservation of identities and composition.* □

The proof works because functoriality is not an empirical claim but a contract: to call $\llbracket \cdot \rrbracket$ *a semantics functor is precisely to assert these two equalities. Intuitively, it is the assertion that meaning is compositional rather than holistic in an uncontrolled way. This is also the point where the categorical setup earns its keep: the axioms are exactly the minimum needed to justify "local reasoning" about program fragments and their assembly into larger systems, without needing to inspect the full global program each time.*

Theorem 57 (Compositional realizations for composed programs (end-to-end pipeline rule))**.** *Let* $p : A \to B$ *and* $q : B \to C$ *be programs in* Syn*, with abstract semantics* $f := \llbracket p \rrbracket$ *and* $g := \llbracket q \rrbracket$ *in* Abs*. Assume there exist physical realizations* R_f *of* f *and* R_g *of* g *(Def. 360), and assume interface compatibility as in Prop. 33. Then there exists a composite realization* $R_{g\circ f}$ *of* $g \circ f$*, hence also of* $\llbracket q \circ p \rrbracket$*, such that*

$$\mathsf{Score}(R_{g\circ f}, \llbracket q \circ p \rrbracket) \;=\; \mathsf{Score}(R_{g\circ f}, g \circ f) \;\geq\; \mathsf{Score}(R_g, g) \otimes \mathsf{Score}(R_f, f).$$

Remark 2509. *In plain English: if you have a physical device that realizes* f *well, and another that realizes* g *well, and you can connect them correctly, then the composite device realizes* $g \circ f$ *with a score at least as good as the tensor-combination of the component scores. This is the formal counterpart of the engineering intuition that reliable modules can be chained to obtain a reliable pipeline, with an explicit lower bound on end-to-end quality.*

The inequality should be read as a guarantee *rather than an exact prediction: depending on the meaning of* Score *and the interaction of components, the composite might perform better than the bound, but it cannot be worse than the bound provided the stated interface assumptions hold. Moreover, the appearance of* $\otimes$ *emphasizes that the way scores combine is part of the theory: one does not merely multiply numbers by convention, but uses the monoidal structure appropriate to the chosen scoring semantics (for example, conjunction of safety guarantees, product of independent probabilities, or addition of log-scores, depending on earlier definitions).*

This theorem ties together several strands: the categorical view of programs/semantics, the notion of physical realization (Hyperseed-Concept 136), and the affordance composition intuition *earlier in the section. It is also one of the basic compositionality principles that makes large engineered systems tractable, rather than a mere heap of parts. One can also view it as an "abstraction barrier" result: once each module is certified against an abstract spec (f and g) with a score, the end-to-end certification can be obtained without re-opening the modules, provided the connection respects the specified interface.*

Proof sketch. By functoriality, $\mathrm{J}q \circ p\mathrm{K} = \mathrm{J}q\mathrm{K} \circ \mathrm{J}p\mathrm{K} = g \circ f$. Then Prop. 33 supplies a composite physical realization with the stated score lower bound. □The proof is therefore a two-line composition of previously established principles: first transport syntactic composition into Abs, then transport physical composition into a score guarantee.

Remark 2510. Proof sketch. *First identify the abstract behavior of the composed program using functoriality. Then apply the existing composition theorem for realizations (Prop. 33) to construct the physical composite and obtain the score inequality.* □

The key step is separating semantic composition *(a statement in* Abs*) from* physical composition *(a statement about realizers and interfaces). The interface-compatibility assumption is what prevents category theory from becoming empty poetry here: it is the concrete requirement that outputs of R_f can serve as inputs of R_g. Stated differently, without compatibility one might still form the formal composite $g \circ f$ in* Abs*, but there would be no corresponding composite artifact, or the artifact would incur unmodeled adapters that could degrade* Score *in ways not captured by the bound. The theorem thus pinpoints exactly where "plumbing" constraints enter the correctness story: not in the semantic equation, but in the existence of the physical composite and the preservation (or controlled degradation) of performance guarantees.*

Corollary 28 (Thresholded pipeline certification)**.** *Fix a threshold $\tau \in V$. If $\mathsf{Score}(R_f, f) \geq \tau_f$ and $\mathsf{Score}(R_g, g) \geq \tau_g$ and $\tau_g \otimes \tau_f \geq \tau$, then the composed realization satisfies*

$$\mathsf{Score}(R_{g\circ f}, \mathrm{J}q \circ p\mathrm{K}) \geq \tau.$$

Remark 2511. *This corollary packages the previous theorem into a certification rule: if each stage meets its own threshold, and the tensor of those thresholds clears the desired end-to-end bar, then the composed system is certified at level τ. In practice, this is how one turns a compositional lower bound into an actionable engineering test: certify components, then certify the pipeline by arithmetic in V.*

One should read τ_f and τ_g as local acceptance criteria *that may be chosen for different reasons (safety margin, regulatory constraint, validation coverage, etc.), while τ is the* system-level *criterion. The condition $\tau_g \otimes \tau_f \geq \tau$ is then a compatibility constraint: the local bars must be set so that their compositional guarantee meets (or exceeds) the global requirement. In particular, this makes explicit that raising the global bar τ forces a corresponding strengthening of (at least one of) the component bars, unless the tensor operation $\otimes$ has slack (e.g. saturating behavior) in the relevant region of V.*

Proof sketch. Combine the score lower bound from the previous theorem with monotonicity of $\otimes$ and transitivity of $\geq$. □

Remark 2512. Proof sketch. *Use the theorem's inequality and then substitute the component lower bounds τ_f, τ_g; finally apply $\tau_g \otimes \tau_f \geq \tau$ to reach the desired threshold.* □

The proof works because $\otimes$ is monotone: improving either component score cannot reduce the product-like combined guarantee. One may visualize this as a "budget" of reliability passing down a pipeline: each stage consumes some reliability, but if the remaining budget stays above τ the whole chain is acceptable.

It is also worth noting what the corollary does not *assert: it does not claim that the bound $\tau_g \otimes \tau_f$ is tight, nor that the combined score factors exactly as a tensor product in general. Rather, it provides a conservative certification rule derived from a compositional lower bound. Consequently, failure of the arithmetic test $\tau_g \otimes \tau_f \geq \tau$ does not by itself imply the pipeline fails at level τ; it only implies that this particular sufficient condition does not certify it, motivating either stronger component guarantees, a refined analysis, or a different compositional inequality more adapted to the situation.*

Proposition 74 (Body-channel mediation as a Hom-factorization constraint)**.** *Let (M, B) be an embodied mind and body, and let R_{phys} be a physical reality-system. Assume body-channel mediation holds in the factorization form: for most attributable effects, the corresponding influence morphism factors*

$$f : M \to R_{\text{phys}} \quad \textit{as} \quad f \approx h \circ g \ \textit{ with } \ g : M \to B,\ h : B \to R_{\text{phys}}.$$

Then (for those effects) the set of mind-to-physics influence morphisms is (approximately) contained in the image of postcomposition by morphisms out of B.

Remark 2513. *The claim is simple: if the mind's influence on the physical reality-system is mediated by the body, then every admissible influence map f essentially factors through B. In categorical language, B behaves like a universal "bottleneck" (up to approximation) for mind-to-world causation. This makes explicit, in the formalism, the ordinary scientific stance that action on the world is carried by bodily channels (Hyperseed-Concept 135), and it aligns with process-oriented readings in which concrete causation is routed through actual physical occasions [15].*

This proposition also provides a clean logical handle: it lets one rule in *or* rule out *hypothesized channels by checking whether they admit the required factorization, as formalized in the next corollary.*

Formally, one may read "image of postcomposition by morphisms out of B" as the set of composites

$$\{\, h \circ g \mid g \in \mathrm{Hom}(M, B),\ h \in \mathrm{Hom}(B, R_{\text{phys}}) \,\} \subseteq \mathrm{Hom}(M, R_{\text{phys}}),$$

with the inclusion weakened to an $\approx$-containment when empirical or modeling approximations are allowed. In this sense, the proposition is a statement about which region of the Hom-set $\mathrm{Hom}(M, R_{\text{phys}})$ *is even eligible to count as an M-attributed causal pathway under the mediation assumption.*

Proof sketch. This restates Def. 362 in categorical language: B is a (near-)universal mediator for M's effects on R_{phys}. □

Remark 2514. Proof sketch. *The proof is definitional: interpret "body-channel mediation" as the existence of a factorization $f \approx h \circ g$ through B, and then observe that this says precisely that f lies in the image of postcomposition by maps out of B (up to $\approx$).* □

The essential step is translating an intuitive constraint ("influence goes through the body") into a compositional constraint on morphisms. The approximation symbol $\approx$ carries the empirical messiness: mediation need not be exact to be structurally dominant.

Depending on the ambient framework, $\approx$ might be instantiated as observational equivalence, bounded error in a metric on processes, statistical indistinguishability relative to O, or a coarser relation induced by measurement and intervention limits. The proposition itself does not require a specific choice; it only uses the idea that "mediated" effects are those that can be matched (to the relevant tolerance) by some factorization through B.

Corollary 29 (Ruling out direct mind-to-physics channels in a body-mediated model)**.** *Assume body-channel mediation holds for a class of effects under an observer O. If a hypothesized influence morphism $f^\star : M \to R_{\mathrm{phys}}$ cannot be approximated by any composition $h \circ g$ through B in the chosen approximation notion, then $f^\star$ does not belong to the model's "attributable-to-M" effect class (relative to O).*

Remark 2515. *This is the negation test: under a body-mediated model, a genuinely "direct" mind-to-physics map that evades all body factorization is, by that very fact, outside the model's attribution boundary. In a Russellian spirit one may say: the boundary is not a metaphysical decree, but a logical consequence of the adopted explanatory form; change the form and the boundary moves.*

Operationally, this can be read as follows: if the only experimentally supported influence patterns from M to R_{phys} are those reproducible (up to $\approx$) by first mapping into B and then mapping out of B, then positing an additional $f^\star$ that cannot be so reproduced is not a mere extension of the same model-class; it is a departure from the mediation hypothesis itself, and should be evaluated as such (e.g. by proposing new observables, interventions, or auxiliary structure that would make $f^\star$ testable and comparable).

Proof sketch. Contrapositive of the mediation criterion: attributable effects are (by assumption) those admitting a factorization through B up to approximation. ☐

Remark 2516. Proof sketch. *Assume $f^\star$ were attributable; then by mediation it would admit an approximate factorization through B. Since it does not, it cannot be attributable.* ☐

What matters is that the corollary does not deny that $f^\star$ exists as a formal morphism; it only denies its membership in the privileged class of attributed *effects, which is observer- and model-relative (Hyperseed-Concept* **??** *is a helpful nearby lens).*

In particular, the observer index O can matter both through what counts as "most attributable effects" and through the granularity of $\approx$: an $f^\star$ might fail to factor through B under a fine-grained approximation (high-resolution instrumentation) while being indistinguishable from some mediated composite under a coarser approximation (limited instrumentation). The logical form of the corollary remains the same; only the induced attribution boundary shifts with the observational regime.

Proposition 75 (Non-uniqueness of physical lift under a forgetful map (sufficient condition))**.** *Let $U : \mathsf{Alg}(\mathsf{Alg}_{\mathrm{phys}}) \to \mathsf{Alg}(\mathsf{Alg}_{\mathrm{mind}})$ be the forgetful functor. If U is not injective on isomorphism classes of objects, i.e. there exist non-isomorphic P_1, P_2 with $U(P_1) \cong U(P_2)$, then the corresponding mental object $m := U(P_1) \cong U(P_2)$ does not determine a unique physical structure.*

Remark 2517. *The statement formalizes a familiar theme: the same "mental" or "functional" organization may be realized by multiple distinct physical organizations. Here the functor U forgets some physical structure, retaining only what is deemed mentally salient; non-injectivity then means that the forgetting collapses genuinely different physical possibilities to the same mental equivalence class. This is one crisp way to express Algebraically Asymmetric Physical Reality (Hyperseed-Concept 55): the map from physical to mental can be many-to-one, so the reverse direction is underdetermined.*

This matters because any attempt to "lift" mental descriptions back to physical ones must then introduce extra choices, constraints, or evidence. In other words, the lifting problem is not merely difficult; it is ill-posed without additional structure.

One can restate the hypothesis as saying that some isomorphism class $[m] \in \mathsf{Alg}(\mathsf{Alg}_{\mathrm{mind}})$ has a fiber under U containing at least two distinct isomorphism classes in $\mathsf{Alg}(\mathsf{Alg}_{\mathrm{phys}})$. Concretely, *the datum of m specifies at most the* equivalence class of physical realizers modulo the forgotten structure*, not a canonical representative of that class.*

The qualifier "sufficient condition" is important: non-injectivity on isomorphism classes is an easy to check criterion guaranteeing non-uniqueness, but uniqueness can also fail in subtler ways (e.g. failure to choose a canonical section, or the presence of multiple non-isomorphic lifts even when a given representative is fixed). The present proposition isolates the simplest categorical mechanism: identification in the image forces underdetermination in any inverse reconstruction.

A toy example to keep in mind is the usual many-realizations picture for computation: two distinct hardware implementations (say, different circuit layouts or different physical substrates) can present isomorphic state-transition structure at the level of an abstract finite automaton. If U forgets substrate details and remembers only the automaton, then $U(P_1) \cong U(P_2)$ while $P_1 \ncong P_2$, and the automaton-level description does not pick out a unique hardware realizer.

Proof sketch. Immediate: two distinct physical realizers share the same forgotten mental structure. Thus any "lifting" from mental to physical requires extra choices beyond the mental description. □

Remark 2518. Proof sketch. *Exhibit the counterexample to uniqueness directly: $P_1 \ncong P_2$ but $U(P_1) \cong U(P_2)$.* □

The decisive step is the appeal to isomorphism classes rather than raw objects: it ensures the claim is invariant under mere renaming of structure, and really concerns "physical form" versus "mental form" in the categorical sense.

Equivalently, the proposition is asserting the nonexistence of a unique lift up to unique isomorphism *of m along U in general: if the fiber over m contains two non-isomorphic objects, then even granting that "a lift exists," it cannot be unique in the only sense that matters in a category, namely uniqueness up to isomorphism. Said differently, the usual categorical notion of determinacy here is not "there exists some P with $U(P) \cong m$," but rather "all such P are isomorphic," and the hypothesis explicitly negates that condition.*

Proposition 76 (Coarse-graining increases weakness (weakness monotonicity rule))**.** *If mental/intersubjective structure is modeled as a coarse-graining of physical structure by identifying previously distinct physical states, then the passage from physical to mental structure monotonically increases quantale weakness (relative to the finest-grained physical observer).*

Remark 2519. *This proposition says that* forgetting distinctions *makes the resulting description weaker: the more you identify, the less you can discriminate, and thus the more "undistinguished pairs" you carry. Quantale weakness (Hyperseed-Concepts 202 and 143) is designed precisely to measure this loss of discriminatory power, in a way compatible with the quantale structure used throughout the epistemic layer.*

The result is important because coarse-graining is ubiquitous: sensors blur, abstractions compress, languages omit, communities negotiate shared categories. The proposition ensures that weakness behaves in the direction one expects under these operations; it is a sanity theorem in the sense emphasized in [3], and it resonates with the broader idea (cf. [2]) that constraint and partiality, rather than perfect specification, are the norm for real cognitive systems.

In the concrete "partition" picture, an observer at the physical level distinguishes states up to some fine equivalence relation $\sim_{\text{phys}}$, whereas the mental/intersubjective level uses a coarser relation $\sim_{\text{mind}}$ obtained by merging equivalence classes (so $\sim_{\text{phys}} \subseteq \sim_{\text{mind}}$ as relations). The monotonicity claim is then the order-theoretic statement that enlarging the indistinguishability relation moves one upward in the weakness preorder induced by the quantale-valued weakness functional.

The phrase "relative to the finest-grained physical observer" also matters: weakness is evaluated with respect to a baseline notion of distinguishability, and coarse-graining is defined as a loss relative to that baseline. *Thus the monotonicity is not a metaphysical claim that "mental is always weaker," but a structural claim that* given a fixed physical discrimination structure, *any additional identifications (whether due to abstraction, measurement limits, or negotiated intersubjective categories) cannot improve discriminatory power in the formalism.*

When the quantale weakness assigns weights rather than booleans to undistinguished pairs, the same intuition applies: coarse-graining can be viewed as either adding new undistinguished pairs or increasing the weight/degree to which certain pairs are treated as indistinguishable. Under either interpretation, the order-preserving property ensures the weakness value can only stay the same or increase.

Proof sketch. Coarse-graining adds (or increases weight on) undistinguished pairs. Weakness is monotone under adding undistinguished pairs (Theorem 1 in the paper's core sanity theorems), hence weakness increases. ☐

Remark 2520. Proof sketch. *Translate coarse-graining into the operation "add more pairs that the observer treats as not distinguishable." Then apply monotonicity of the weakness functional with respect to adding such pairs.* ☐

The proof works because weakness is, by construction, order-preserving in the lattice of distinguishability relations: adding identifications moves upward in "weakness order." A visual intuition is to imagine partitioning a set of physical states: coarse-graining merges partition cells, producing a strictly coarser partition, hence fewer distinctions and therefore more weakness.

It is sometimes helpful to note the two boundary cases: at one extreme, the identity (no coarse-graining) leaves weakness unchanged; at the other extreme, collapsing all states to one class maximizes weakness by making every pair undistinguished. The proposition asserts that all intermediate coarse-grainings interpolate monotonically between these extremes.

The monotonicity is also stable under "mixed" coarse-grainings that only identify some states but leave others intact: the weakness functional need not be linear in the number of identifications, only order-preserving with respect to inclusion (or weight-increase) of the undistinguished-pairs data.

Corollary 30 (Iterated forgetting yields iterated weakness increase)**.** *If U_1 and U_2 are successive coarse-grainings (forgetful maps), then the induced weakness is monotone along the chain:*

$$w \ \leq \ w \circ U_1 \ \leq \ w \circ (U_2 \circ U_1),$$

interpreting $w \circ U$ as "compute weakness after coarse-graining by U".

Remark 2521. *This is the compositional form of the previous proposition: repeated forgetting yields repeated (or at least nondecreasing) weakness. It is the formal analogue of the commonsense idea that if you first compress a description and then compress it again, you do not regain lost detail. It also aligns with the earlier associativity/compositionality themes in this section: here the composed map $U_2 \circ U_1$ behaves as the "combined forgetting," and weakness tracks it monotonically.*

In categorical terms, the point is that the class of coarse-grainings under consideration is closed under composition: $U_2 \circ U_1$ is itself a coarse-graining capturing the net identifications made by the two stages. The inequality chain then expresses that w is monotone with respect to this composition, so that the ordering of "informational strength" is functorial along the forgetful/coarse-graining direction.

One can also read the inequalities as a "data processing" principle for the weakness measure: post-processing by a forgetful map cannot reduce the weakness induced by prior processing, because no new distinctions are introduced by additional forgetting.

Proof sketch. Each coarse-graining adds undistinguished pairs; apply weakness monotonicity twice. $\square$

Remark 2522. Proof sketch. *Apply the monotonicity rule once to U_1 to get $w \leq w \circ U_1$, and again to U_2 (viewed as a coarse-graining on the already coarse-grained structure) to get $w \circ U_1 \leq w \circ (U_2 \circ U_1)$.* $\square$

The key step is recognizing that "successive coarse-grainings" are again a coarse-graining when composed, so the same monotonicity argument iterates without alteration.

This corollary is also a reminder that any multi-stage modeling pipeline (physical $\to$ perceptual $\to$ linguistic $\to$ intersubjective, etc.) will accumulate weakness unless explicit "side channels" of additional information are introduced. In the present formalism, such side channels would correspond not to further forgetful functors but to added structure (constraints, measurements, or priors) that refine, rather than coarsen, the distinguishability relation.

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45 Helper theorems for Section 24 (Communion, Explicit Communion, Beauty)

This section states helper theorems that follow immediately from the formal definitions in Section 24 (Definitions 364–385, Propositions 35–37, Theorems 20–21). They are intended as "ready-to-fire" lemmas for an automated reasoner.

Remark 2523. *Philosophically, the purpose of this section is to distill a large family of qualitative notions (*communion*,* explicit communion*,* beauty*,* spiritual communion*, etc.) into algebraic invariants and monotonicity rules that an automated system can reliably use. The underlying approach is broadly "pattern-centric" in the sense of Section 24 and the broader Hyperseed framework: intensities are assigned to patterns over individual and collective supports, and then various higher-level predicates are defined by comparing aggregated intensities. This pattern-emergence orientation aligns with the general account of patterns and their networks in [5] and with the initial Hyperseed ontology presentation [1]. When the results touch prosocial coordination (e.g. the way cross-supported structure can dominate private structure), they also resonate with arguments that prosociality can be efficiency-enhancing in multi-agent settings [6].*

45.1 Algebra of within-/cross-mind intensity and communion

Remark 2524. *We will repeatedly use the following intuition (without re-stating it each time):*

- $\mathrm{Pat}_U(I)$ *denotes a set of patterns whose* support *is exactly the subcommunity $U \subseteq C$ over the interval I (as defined in Section 24). Thus $|U| = 1$ corresponds to "private" or within-mind patterns, while $|U| \geq 2$ corresponds to genuinely cross-mind patterns.*
- $w_U(p) \in [0, \infty]$ *is an intensity assigned to the pattern $p \in \mathrm{Pat}_U(I)$.*
- $\mathrm{Win}(C, I; w)$ *aggregates the singleton-support intensities (within-mind total);* $\mathrm{Wcross}(C, I; w)$ *aggregates the $|U| \geq 2$ intensities (cross-mind total); and* $\mathrm{DComm}(C, I; w)$ *is the difference* $\mathrm{Wcross} - \mathrm{Win}$*, the "margin" by which the shared exceeds the private.*

This is, in effect, a bookkeeping calculus for the distinction between Self *and* Other *at the level of pattern support; compare the Hyperseed core-concept set entries 165, 130, and* **??**.

HT24.1 (Communion margin decomposition). For any community intensity datum w on (C, I),

$$\mathrm{DComm}(C, I; w) \;=\; \mathrm{Wcross}(C, I; w) \;-\; \mathrm{Win}(C, I; w),$$

and expanding the definitions,

$$\mathrm{DComm}(C, I; w) = \sum_{\substack{U \subseteq C \\ |U| \geq 2}} \sum_{p \in \mathrm{Pat}_U(I)} w_U(p) \;-\; \sum_{m \in C} \sum_{p \in \mathrm{Pat}_{\{m\}}(I)} w_{\{m\}}(p).$$

Moreover, $\mathrm{Comm}(C, I; w)$ holds iff $\mathrm{DComm}(C, I; w) > 0$.

Remark 2525. *Intuitively, this says that communion is not a mysterious extra primitive: it is a* comparison *between two kinds of total intensity. The cross-supported total* Wcross *counts intensities of patterns that "span minds," while* Win *counts intensities that remain confined to individual members. Communion holds precisely when the shared structure dominates the private structure in the aggregated sense. One can view* DComm *as an "intersubjective surplus" (Hyperseed-Concept* **??***).*

Proof sketch. Immediate from Definitions 368–369: Wcross sums intensities for $|U| \geq 2$, Win sums singleton-support intensities, and communion is defined as Wcross > Win.

Remark 2526. *A useful way to picture the proof is as a partition of a sum: patterns are sorted by the cardinality of their support. Nothing deeper is required because Section 24 defines all the operators explicitly in terms of these sums. Geometrically, one may imagine an axis with "within" intensity on one side and "cross" intensity on the other;* DComm *is then simply the signed displacement from within-dominance to cross-dominance.*

HT24.2 (Nonnegativity and trivial bounds). Always $\mathrm{Win}(C, I; w) \geq 0$ and $\mathrm{Wcross}(C, I; w) \geq 0$, hence

$$-(\mathrm{Win} + \mathrm{Wcross}) \;\leq\; \mathrm{DComm} \;\leq\; \mathrm{Win} + \mathrm{Wcross}.$$

In particular, if $\mathrm{Wcross}(C, I; w) = 0$ then $\neg\mathrm{Comm}(C, I; w)$.

Remark 2527. *This is the "sanity-check" lemma: since intensities live in* $[0, \infty]$*, the aggregated totals cannot be negative. The bounds on* DComm *are just the obvious bounds on a difference of two nonnegative quantities. The final sentence captures the intended meaning of communion: if there is literally no cross-supported intensity, then there is nothing upon which communion could be based (Hyperseed-Concept* **??***).*

Proof sketch. Each $w_U(p) \in [0, \infty]$ and totals are (finite or extended) sums. If $\mathrm{Wcross} = 0$ then $\mathrm{Wcross} > \mathrm{Win}$ is impossible since $\mathrm{Win} \geq 0$.

Remark 2528. *The high-level strategy is to reduce everything to order properties of* $[0, \infty]$*. The only "key step" is noticing that the definition of* Comm *is a strict inequality; thus even if* $\mathrm{Win} = 0$, $\mathrm{Wcross} = 0$ *fails communion. Visually: if you draw the nonnegative quadrant with coordinates* $(\mathrm{Wcross}, \mathrm{Win})$*, communion lives strictly above the diagonal* $\mathrm{Wcross} = \mathrm{Win}$*, and the line* $\mathrm{Wcross} = 0$ *is entirely outside this region.*

HT24.3 (Scaling invariance of communion). Fix $c > 0$ and define a rescaled datum $(c \cdot w)_U(p) := c\, w_U(p)$. Then

$$\mathrm{Win}(C, I; c{\cdot}w) = c\,\mathrm{Win}(C, I; w), \quad \mathrm{Wcross}(C, I; c{\cdot}w) = c\,\mathrm{Wcross}(C, I; w), \quad \mathrm{DComm}(C, I; c{\cdot}w) = c\,\mathrm{DComm}(C, I; w)$$

and therefore $\mathrm{Comm}(C, I; c \cdot w)$ iff $\mathrm{Comm}(C, I; w)$.

Remark 2529. *The content is that communion is* scale-free*: it depends on a comparison between cross and within totals, not on the absolute units in which intensities are measured. This matches the idea that intensity values are often proxies for heterogeneous empirical or phenomenological measurements (Hyperseed-Concept* **??** *is not listed, but compare* **??** *and* **??***). A simple example: if all intensities are multiplied by* 10 *because one changes the measuring instrument, the truth of* Comm *is unchanged.*

Proof sketch. Linearity of finite sums over $[0, \infty]$; inequality is preserved by multiplication by $c > 0$.

Remark 2530. *The proof is a one-line homogeneity argument: sums commute with multiplication by a scalar, and strict inequalities are preserved by multiplying by a positive constant. The geometric intuition is that scaling moves a point* $(\mathrm{Wcross}, \mathrm{Win})$ *along a ray from the origin, and the communion region is a cone bounded by the diagonal; rays cannot cross the diagonal unless they start on it.*

HT24.4 (Cross-monotonicity of communion margin). Let w, w' be two intensity data on (C, I) such that: (i) for all $m \in C$ and all $p \in \mathrm{Pat}_{\{m\}}(I)$, $w'_{\{m\}}(p) = w_{\{m\}}(p)$; (ii) for all $U \subseteq C$ with $|U| \geq 2$ and all $p \in \mathrm{Pat}_U(I)$, $w'_U(p) \geq w_U(p)$. Then $\mathrm{DComm}(C, I; w') \geq \mathrm{DComm}(C, I; w)$. In particular, if $\mathrm{Comm}(C, I; w)$ then $\mathrm{Comm}(C, I; w')$.

Remark 2531. *This formalizes an intuitive principle: if you do not change anyone's private intensity landscape, but you strengthen (or add) the shared patterns, then communion cannot get worse. As a toy example, suppose a dyad* $\{m, n\}$ *gains a new cross-supported pattern of intensity* 1 *while their private patterns remain unchanged; then* Wcross *increases and* Win *does not, so the communion margin increases. This lemma is useful because many interventions (education, ritual, coordinated practice) are modeled precisely as boosting cross-supported patterns without directly inflating private noise.*

Proof sketch. Win is identical under (i), while Wcross can only increase under (ii); subtract.

Remark 2532. *The strategy is monotonicity of sums: if every summand in the cross part increases (or stays the same), then the total increases. The key step is the separation of within- and cross-summands, which is already built into the definition of* DComm*. One may visualize this as keeping the "within" bar fixed while raising the "cross" bar; the difference must grow.*

HT24.5 (Within-monotonicity: increasing private intensity hurts communion). Let w, w' be such that cross intensities are unchanged, but singleton intensities increase: (i) $w'_U = w_U$ for all U with $|U| \geq 2$; (ii) $w'_{\{m\}}(p) \geq w_{\{m\}}(p)$ for all singletons. Then $\mathrm{DComm}(C, I; w') \leq \mathrm{DComm}(C, I; w)$. Hence if $\mathrm{Comm}(C, I; w')$ then $\mathrm{Comm}(C, I; w)$.

Remark 2533. *This is the mirror image of HT24.4: if the shared world stays as it is but private intensity grows, the communion margin shrinks. In everyday terms, if each participant's inner* *monologue gets louder while the shared conversation does not intensify, then the collective unity becomes harder to sustain. This is important for reasoning about failure modes: communion can be lost not only by weakening shared patterns but also by amplifying private patterns (Hyperseed-Concept* **??** *via 60 is often implicated).*

Proof sketch. Now Wcross is identical, while Win can only increase.

Remark 2534. *Again, the proof is a direct application of monotonicity of sums and the definition of* DComm *as a difference. The key step is noticing that the strictness in the communion condition flows "backwards" here: if w' still achieves communion despite a larger within component, then certainly w achieves communion. Geometrically: you move vertically upward in the* $(\mathrm{Wcross}, \mathrm{Win})$ *plane, away from the communion region; if you are still in communion after moving up, you must have started even deeper inside the region.*

HT24.6 (Add-a-member update formula). Let $C' = C \cup \{n\}$ with $n \notin C$. Then

$$\mathrm{DComm}(C', I; w) = \mathrm{DComm}(C, I; w) \;+\; \Big(\sum_{\substack{U\subseteq C\\ U\neq\emptyset}} \sum_{p\in\mathrm{Pat}_{U\cup\{n\}}(I)} w_{U\cup\{n\}}(p)\Big) \;-\; \sum_{p\in\mathrm{Pat}_{\{n\}}(I)} w_{\{n\}}(p).$$

(Interpret the first big sum as the total cross-supported intensity of patterns whose support includes n and at least one member of C.)

Remark 2535. *This provides a modular way to update communion calculations when a community grows by one member. The additional term is precisely "what the newcomer brings to shared structure" minus "how much private structure the newcomer has", all measured in the same intensity currency. As a simple example, if n has no private patterns in the relevant vocabulary (or they are assigned negligible intensity) but creates strong shared patterns with members of C, then adding n can increase communion margin. This lemma connects naturally to questions of recruitment and assimilation in collective systems (Hyperseed-Concept 170).*

Proof sketch. Expand $\mathrm{Wcross}(C', \cdot)$ into (a) terms entirely inside C plus (b) terms whose support includes n. Expand $\mathrm{Win}(C', \cdot)$ into $\mathrm{Win}(C, \cdot)+$ the singleton sum for n. Subtract.

Remark 2536. *The proof is a careful but elementary partition of the defining sums according to whether their support contains n. The key step is that supports are sets, so "contains n" cleanly distinguishes the new terms from the old ones. A visual intuition is bookkeeping in a ledger: the old balance is* $\mathrm{DComm}(C, I; w)$*, and the new balance is old plus new cross credits minus new within debits.*

HT24.7 (Independent-subcommunity additivity). Let $C = C_1 \cup C_2$ with $C_1 \cap C_2 = \emptyset$. Assume *no* pattern has support spanning both parts, i.e. for every nonempty $U \subseteq C$ with $U \cap C_1 \neq \emptyset$ and $U \cap C_2 \neq \emptyset$, we have $\mathrm{Pat}_U(I) = \emptyset$ (or equivalently, their total intensity is 0). Then

$$\mathrm{DComm}(C, I; w) = \mathrm{DComm}(C_1, I; w) + \mathrm{DComm}(C_2, I; w),$$

and similarly for Win and Wcross.

Remark 2537. *This is an additivity principle under a strong independence hypothesis: if the community decomposes into two parts with no cross-supported patterns linking them, then communion margin is just the sum of the margins inside each part. One may read this as an algebraic reflection* *of the idea that an intersubjective field cannot exist across a boundary where there are no shared patterns (Hyperseed-Concept* **??** *is not listed, but compare* **??***). This lemma is useful when analyzing large communities that factor into disconnected "tribes" in the pattern-support sense. In particular, it formalizes a "no interaction, no synergy" slogan: the net surplus of cross-mind intensity over within-mind intensity cannot be created by simply juxtaposing two groups if absolutely no pattern is jointly supported across the split. Equivalently, the hypothesis is not merely that interactions are weak, but that they are* absent *at the level of the pattern-support relation, so that every support-set U contributing to cross terms is forced to lie entirely in one side.*

Proof sketch. Under the hypothesis, all sums split into disjoint contributions supported entirely in C_1 or entirely in C_2. More explicitly, for each indexing set $U \subseteq C$ appearing in the definitions of Wcross and Win, either $U \subseteq C_1$ or $U \subseteq C_2$, since the case $U \cap C_1 \neq \emptyset$ and $U \cap C_2 \neq \emptyset$ forces $\mathrm{Pat}_U(I) = \emptyset$ and hence contributes zero. Thus, both the cross-supported total and the within-supported total decompose as a sum of two independent blocks, and so does their difference.

Remark 2538. *The strategy is to split the index set of each sum. The crucial step is the hypothesis $\mathrm{Pat}_U(I) = \emptyset$ whenever U intersects both sides: it eliminates all mixed terms, leaving only terms wholly in C_1 or wholly in C_2. Visually, the community is two islands with no bridges; totals computed on the archipelago are totals computed on each island and added. One can also view this as a bookkeeping statement about supports: the decomposition $C = C_1 \sqcup C_2$ induces a partition of all relevant supports into those contained in C_1 and those contained in C_2, and the hypothesis says that the "straddling" supports contribute identically zero. Because the weights $w_U(p)$ are nonnegative, there is no cancellation to worry about; the additivity is purely set-theoretic and does not depend on delicate sign arguments.*

HT24.8 (Pairwise lower bound suffices for communion). Define the *pairwise* cross total

$$\mathrm{Wcross}^{(2)}(C, I; w) := \sum_{\{m,n\}\subseteq C} \; \sum_{p\in \mathrm{Pat}_{\{m,n\}}(I)} w_{\{m,n\}}(p).$$

Then $\mathrm{Wcross}^{(2)}(C, I; w) \leq \mathrm{Wcross}(C, I; w)$. Consequently, if $\mathrm{Wcross}^{(2)}(C, I; w) > \mathrm{Win}(C, I; w)$ then $\mathrm{Comm}(C, I; w)$.

Remark 2539. *This says that, to certify communion, it can suffice to look only at* pairwise *cross-patterns. Since* Wcross *includes all higher-arity supports ($|U| \geq 3$) in addition to pairs, the pairwise sum is a lower bound. Thus, if even the pairwise shared intensity already dominates all singleton intensity, then communion is guaranteed. This is practically useful because pairwise statistics are often far easier to estimate than higher-order ones. In terms of hypothesis-testing language:* $\mathrm{Wcross}^{(2)}$ *is a conservative proxy for the full cross total, so a strict inequality* $\mathrm{Wcross}^{(2)} > \mathrm{Win}$ *is a strong certificate that cannot be invalidated by discovering additional higher-order structure. Conversely, failure of the pairwise test does* not *refute communion, because genuinely communal structure may reside predominantly in $|U| \geq 3$ supports; the lemma only provides a sufficient condition, not a necessary one.*

Proof sketch. Wcross sums over all $|U| \geq 2$, including all pairs; all terms are nonnegative. More formally, writing $\mathrm{Wcross}(C, I; w) = \sum_{U\subseteq C,\, |U|\geq 2} \sum_{p\in \mathrm{Pat}_U(I)} w_U(p)$, the subcollection of indices with $|U| = 2$ is precisely what appears in $\mathrm{Wcross}^{(2)}$. Removing all other $|U| \geq 3$ terms yields $\mathrm{Wcross}^{(2)}$, and since each removed term is ≥ 0, the inequality follows.

Remark 2540. *The proof is containment of index sets: the set of pairs is a subset of the set of all supports of size at least* 2*. Nonnegativity ensures that throwing away terms can only decrease the sum, never increase it. Geometrically: if you approximate a high-dimensional shared structure by its edges, you get a conservative estimate. A useful way to picture the indexing is as a hypergraph on C:* Wcross *sums contributions over all hyperedges of size ≥ 2, whereas* $\mathrm{Wcross}^{(2)}$ *keeps only the ordinary edges. The lemma then says that the total weight of edges is bounded above by the total weight of all hyperedges, which is immediate when weights are nonnegative and no term is double-counted outside its own support size.*

HT24.9 (Normalized communion score is bounded). Define (for any fixed $\varepsilon > 0$)

$$\mathrm{NComm}_\varepsilon(C, I; w) := \frac{\mathrm{DComm}(C, I; w)}{\mathrm{Wcross}(C, I; w) + \mathrm{Win}(C, I; w) + \varepsilon}.$$

Then $|\mathrm{NComm}_\varepsilon(C, I; w)| < 1$ and $\mathrm{sign}(\mathrm{NComm}_\varepsilon) = \mathrm{sign}(\mathrm{DComm})$.

Remark 2541. *This introduces a dimensionless score in $(-1, 1)$ that is robust to overall scaling of intensities (compare HT24.3), and also avoids division by zero through $\varepsilon > 0$. Conceptually, it plays a role similar to a normalized "effect size" or signed correlation: the numerator measures net surplus, and the denominator measures total mass. Such normalizations are often helpful when comparing communities of very different sizes or intensity scales. Note that ε may be interpreted as a regularization parameter: it prevents instability when both* Wcross *and* Win *are tiny (or zero), and it ensures strict inequality $|\mathrm{NComm}_\varepsilon| < 1$ even in extreme cases where* DComm *nearly matches* $\mathrm{Wcross} + \mathrm{Win}$*. In applications, one can choose ε to reflect a baseline noise floor or a desired numerical tolerance, without changing the sign of the score.*

Proof sketch. From HT24.2, $|\mathrm{DComm}| \leq \mathrm{Wcross} + \mathrm{Win}$. Adding $\varepsilon > 0$ makes the denominator strictly larger than $|\mathrm{DComm}|$ unless both totals are infinite; in the extended-real case, interpret as the usual limiting inequality. For the sign claim, note that the denominator is strictly positive in the ordinary real-valued setting (and nonnegative in the extended-real setting with $\varepsilon > 0$), so dividing by it cannot change the sign of the numerator.

Remark 2542. *The proof is a direct inequality sandwich: the denominator dominates the numerator in absolute value. The subtle point is extended-real arithmetic; the intended reading is the standard one that when both totals diverge, the ratio is interpreted via limits (or as an indeterminate form resolved by a convention). Visually: the score is a signed ratio whose magnitude cannot reach* 1 *unless $\varepsilon = 0$ and one side overwhelms the other completely. Equivalently,* $\mathrm{NComm}_\varepsilon$ *behaves like a "normalized difference":*

$$-1 < \frac{\mathrm{Wcross} - \mathrm{Win}}{\mathrm{Wcross} + \mathrm{Win} + \varepsilon} < 1,$$

since the numerator is bounded in magnitude by the sum of the two nonnegative totals, and ε enforces strictness. This places $\mathrm{NComm}_\varepsilon$ *in the same family as other bounded contrast scores used to compare two nonnegative quantities, with the additional interpretation that the two quantities here are "cross-mind" and "within-mind" intensity masses.*

45.2 Paraconsistent designation calculus (for explicit communion and beauty)

Fix the truth-value set $B = [0, 1] \times [0, 1]$ with $\neg(t, f) = (f, t)$, $(t_1, f_1) \wedge (t_2, f_2) = (\min(t_1, t_2), \max(f_1, f_2))$, $(t_1, f_1) \vee (t_2, f_2) = (\max(t_1, t_2), \min(f_1, f_2))$, and designation $\mathrm{Desig}_\tau(t, f) \iff t \geq \tau$.

Remark 2543. *Here $B = [0,1] \times [0,1]$ should be read as a* two-component *truth value: the first coordinate t is "degree of support for truth" and the second coordinate f is "degree of support for falsity." The negation $\neg(t,f) = (f,t)$ swaps these roles. The connectives $\wedge, \vee$ are the standard min–max / max–min choices that preserve information from both components: conjunction keeps the weaker truth evidence (min) and the stronger falsity evidence (max), while disjunction keeps the stronger truth evidence (max) and the weaker falsity evidence (min). Designation $\mathrm{Desig}_\tau(t,f) \iff t \geq \tau$ is a* threshold rule *declaring statements "acceptable" when their truth-evidence is high enough. This style of four-valued / paraconsistent semantics is closely aligned with Constructible Duality Logic [23] and with paraconsistent treatments of resonance-like phenomena [24]; it also matches the Hyperseed emphasis on Value Paraconsistency (Hyperseed-Concept 198).*

Remark 2544. *A convenient auxiliary structure on B (not required for the proofs below, but helpful for intuition) is the information/order-theoretic reading: one often compares values by saying that (t_1, f_1) carries "at least as much truth support" as (t_2, f_2) when $t_1 \geq t_2$, and "at least as much falsity support" when $f_1 \geq f_2$. On this reading, $\wedge$ and $\vee$ behave as lattice operations that are conservative with respect to evidence: $\wedge$ does not invent truth support (it uses* min*), and $\vee$ does not invent falsity support (it uses* min *on f). This is one precise sense in which the calculus is paraconsistent: high truth support and high falsity support can co-exist (e.g. $(0.9, 0.9)$), and the algebra keeps both components available rather than forcing an immediate collapse to a single classical value.*

Remark 2545. *It is also useful to separate two distinct "crispifications" of a two-component value: (i) the designatedness predicate $\mathrm{Desig}_\tau(t,f)$, which depends only on t, and (ii) a potential (not used here) "anti-designation" predicate based on f. The present section deliberately commits only to (i), because explicit communion and beauty (as used in Section 24) are controlled by whether the episode clears a truth-support threshold, while falsity-support remains available to model tension, ambiguity, or countervailing evidence without immediately destroying designatedness.*

HT24.10 (Designation of conjunction/disjunction). For any $b_1, b_2 \in B$,

$$\mathrm{Desig}_\tau(b_1 \wedge b_2) \iff \mathrm{Desig}_\tau(b_1) \;\wedge\; \mathrm{Desig}_\tau(b_2),$$

$$\mathrm{Desig}_\tau(b_1 \vee b_2) \iff \mathrm{Desig}_\tau(b_1) \;\vee\; \mathrm{Desig}_\tau(b_2).$$

Remark 2546. *In plain terms: at the level of* designatedness *(what we will treat as "passing the threshold"), conjunction behaves like logical "and" and disjunction behaves like logical "or," even though the underlying truth values are two-component objects. This is precisely why threshold-designation is useful: it turns graded, paraconsistent truth values into crisp predicates for subsequent reasoning.*

Remark 2547. *A concrete numerical example can be helpful. Let $\tau = 0.7$, $b_1 = (0.8, 0.1)$ and $b_2 = (0.6, 0.9)$. Then $b_1 \wedge b_2 = (\min(0.8, 0.6), \max(0.1, 0.9)) = (0.6, 0.9)$, which is* not *designated at $\tau = 0.7$; indeed b_2 was not designated. Meanwhile $b_1 \vee b_2 = (\max(0.8, 0.6), \min(0.1, 0.9)) = (0.8, 0.1)$, which* is *designated; indeed b_1 was designated. This mirrors the equivalences in HT24.10 while illustrating that the falsity component may move in the opposite direction under $\wedge$ versus $\vee$, without affecting designatedness.*

Proof sketch. The truth component of $b_1 \wedge b_2$ is $\min(t_1, t_2)$, so it exceeds τ iff both t_1, t_2 do. The truth component of $b_1 \vee b_2$ is $\max(t_1, t_2)$, so it exceeds τ iff at least one does.

Remark 2548. *The strategy is to ignore the falsity component because* Desig_τ *depends only on the truth component. Then the proof becomes the elementary equivalences:* $\min(t_1, t_2) \geq \tau \Leftrightarrow (t_1 \geq \tau \wedge t_2 \geq \tau)$ *and* $\max(t_1, t_2) \geq \tau \Leftrightarrow (t_1 \geq \tau \vee t_2 \geq \tau)$. *Visually, in the unit square for* (t, f), *designatedness is the half-strip* $t \geq \tau$; *min and max correspond to intersections and unions of such strips.*

Remark 2549. *Two immediate corollaries (used implicitly when chaining "episode conditions" in Section 24) are:*

$$\mathrm{Desig}_\tau(b \wedge b) \iff \mathrm{Desig}_\tau(b), \qquad \mathrm{Desig}_\tau(b \vee b) \iff \mathrm{Desig}_\tau(b),$$

and the monotonicity principle

$$\mathrm{Desig}_\tau(b_1) \;\Rightarrow\; \mathrm{Desig}_\tau(b_1 \vee b_2), \qquad \mathrm{Desig}_\tau(b_1 \wedge b_2) \;\Rightarrow\; \mathrm{Desig}_\tau(b_1),$$

which are just the idempotence/weakening patterns familiar from classical reasoning, recovered at the level of designatedness even though the underlying values remain two-component. These are frequently the only properties needed when the formal role of $\wedge, \vee$ *is to combine multiple "support conditions" into a single acceptability judgment.*

HT24.11 (De Morgan laws hold at the level of truth values). For any $b_1, b_2 \in B$,

$$\neg(b_1 \wedge b_2) = (\neg b_1) \vee (\neg b_2), \qquad \neg(b_1 \vee b_2) = (\neg b_1) \wedge (\neg b_2).$$

Remark 2550. *This asserts that the algebra of truth values is well-behaved: even though we are not in classical two-valued logic, the familiar De Morgan transformations remain exactly correct* as equalities in B. *This matters because explicit communion and beauty are built using these connectives, and we want standard rewrites to remain valid when manipulating formulas symbolically.*

Remark 2551. *Note that HT24.11 is stronger than a mere claim about designatedness. For example, it is* not *generally true that* $\mathrm{Desig}_\tau(\neg b)$ *is determined by* $\mathrm{Desig}_\tau(b)$ *alone, because* $\mathrm{Desig}_\tau(\neg(t, f))$ *depends on* f, *not on* t. *Thus, having De Morgan at the level of full values (rather than merely at the level of designatedness) is what allows one to push negations inward when one later wants to track both supportive and counter-supportive evidence through a construction.*

Proof sketch. Expand both sides componentwise using swap/min/max identities: $\neg(\min t, \max f) = (\max f, \min t)$ etc.

Remark 2552. *The proof is pure algebra on min/max operations plus the swap defining negation. The key observation is that swap interchanges the roles of min and max in exactly the way De Morgan requires. A helpful mental picture is to regard* (t, f) *as a point and negation as reflection across the diagonal* $t = f$; *then the lattice operations interact with this reflection in the familiar De Morgan fashion.*

Remark 2553. *One can also view HT24.11 as the statement that* $\neg$ *is an involutive order-reversing symmetry of the* $(\wedge, \vee)$*-structure on* B: *indeed* $\neg\neg(t, f) = (t, f)$, *and the equalities in HT24.11 say precisely that* $\neg$ *converts meets to joins and joins to meets. This is the algebraic reason that "dual" formulations of a condition (e.g. phrasing an episode criterion negatively versus positively) can be shown equivalent without changing the intended two-component semantics.*

HT24.12 (Explicit communion implies communion; threshold monotonicity). By definition,

$$\mathrm{ExplComm}(C, I; w) \;\Rightarrow\; \mathrm{Comm}(C, I; w).$$

Moreover, if $\tau' \leq \tau$ then ExplComm at threshold τ implies ExplComm at threshold τ'.

Remark 2554. *Explicit communion is designed as a strengthening of communion: it requires communion* and *additional designated statements about the episode (in Section 24). So the first implication is a formal reminder that explicit communion cannot hold in the absence of communion. Threshold monotonicity is equally intuitive: if something clears a higher bar τ, it clears any lower bar τ'. This is a typical pattern when using designatedness as a control knob for strictness (Hyperseed-Concept 87).*

Remark 2555. *The second clause (monotonicity in τ) is frequently used operationally as follows: one proves an episode is explicitly communal at some "strict" setting of the model, and then immediately inherits explicit communion at all weaker settings. This allows later arguments to fix a convenient threshold for a technical lemma without losing the ability to apply the result in a more permissive regime. In applications to beauty in Section 24, this is the formal counterpart of the intuition that raising standards can rule out borderline cases, but cannot create new ones.*

Proof sketch. ExplComm includes Comm as a conjunct. Also, $\mathrm{Desig}_\tau(b)$ implies $\mathrm{Desig}_{\tau'}(b)$ whenever $\tau' \leq \tau$.

Remark 2556. *The proof is a direct unwrapping of definitions: one part is propositional logic on the conjunctive definition, the other part is the monotonicity of an inequality $t \geq \tau$ with respect to τ. A visual intuition is that as τ decreases, the designated region $t \geq \tau$ in the unit square expands.*

Remark 2557. *For later use, it is often convenient to isolate the underlying monotonicity fact as a stand-alone inference: for any $b \in B$ and any $\tau' \leq \tau$,*

$$\mathrm{Desig}_\tau(b) \Rightarrow \mathrm{Desig}_{\tau'}(b).$$

When explicit communion is defined by a conjunction of multiple designated constraints, HT24.10 then propagates this monotonicity through the whole definition: lowering the threshold can only add designated conjuncts (never remove them), so the explicit-communion condition becomes easier to satisfy. This is the formal mechanism by which the threshold parameter acts as a single global knob controlling strictness across all component criteria.

45.3 Scope and spiritual communion

HT24.13 (Scope is monotone in its components). For fixed $\alpha \in [0, 1]$, if $H_n(I) \geq H_m(I)$ and $D_n(I) \geq D_m(I)$ then

$$\mathrm{Scope}(n, I) = \alpha H_n(I) + (1 - \alpha) D_n(I) \;\geq\; \alpha H_m(I) + (1 - \alpha) D_m(I) = \mathrm{Scope}(m, I).$$

Remark 2558. Scope *is an affine blend of two quantities, H and D, with weights α and $1-\alpha$. Thus it preserves the partial order: increasing either component cannot decrease scope. Conceptually, this lets one reason about spiritual communion by independently bounding the "height"-like and "depth"-like aspects that contribute to scope (Hyperseed-Concept 86 and 178 are relevant framings).*

Proof sketch. Affine monotonicity in each argument when $\alpha \in [0, 1]$.

Remark 2559. *Unpacking the "affine monotonicity" statement explicitly: since $\alpha \geq 0$, the inequality $H_n(I) \geq H_m(I)$ implies*

$$\alpha H_n(I) \;\geq\; \alpha H_m(I),$$

and since $1-\alpha \geq 0$, the inequality $D_n(I) \geq D_m(I)$ implies

$$(1-\alpha)D_n(I) \;\geq\; (1-\alpha)D_m(I).$$

Adding these two inequalities yields the displayed conclusion. This is the only reasoning used, and it highlights that no special structural properties of H or D are required beyond being real-valued.

Remark 2560. *The strategy is to use that multiplying an inequality by a nonnegative scalar preserves it, and adding inequalities preserves it. The only role of $\alpha \in [0,1]$ is to guarantee nonnegativity of both coefficients α and $1-\alpha$. Geometrically,* Scope *is a projection onto a line segment in the (H,D)-plane with nonnegative slope in each coordinate.*

Remark 2561. *Two endpoint cases are worth keeping in mind when interpreting the lemma. If $\alpha = 0$ then $\mathrm{Scope}(n,I) = D_n(I)$ and the statement reduces to monotonicity of D alone; if $\alpha = 1$ then $\mathrm{Scope}(n,I) = H_n(I)$ and the statement reduces to monotonicity of H alone. For intermediate $\alpha \in (0,1)$, both components contribute, but the argument remains componentwise because the combination is coordinatewise nondecreasing. In particular, the lemma is robust under ties: if $H_n(I) = H_m(I)$ (or $D_n(I) = D_m(I)$), the conclusion still holds with equality contributed by that component.*

Remark 2562. *One can also phrase HT24.13 as: the map $(H,D) \mapsto \alpha H + (1-\alpha)D$ is order-preserving with respect to the product order on $\mathbb{R}^2$ when $\alpha \in [0,1]$. Equivalently, its gradient vector $(\alpha, 1-\alpha)$ lies in the nonnegative orthant, so moving "northeast" in the (H,D)-plane cannot decrease* Scope. *This viewpoint clarifies why the lemma would fail if α were allowed outside $[0,1]$: a negative coefficient would reverse the inequality for its component.*

HT24.14 (Sufficient conditions for spiritual communion). Assume $\alpha \in [0,1)$ and $\delta > 0$. If $\mathrm{Comm}(\{m,n\}, I; w)$ and

$$D_n(I) - D_m(I) \;\geq\; \frac{\delta}{1-\alpha} \quad \text{and} \quad H_n(I) \geq H_m(I),$$

then $\mathrm{Scope}(n,I) \geq \mathrm{Scope}(m,I) + \delta$, hence (m,n) satisfies the scope-inequality condition in the definition of spiritual communion.

Remark 2563. *This lemma isolates a simple sufficient condition for the "scope gap" part of spiritual communion: if n exceeds m in both components, and the D-advantage is large enough to compensate for the weighting $(1-\alpha)$, then n's scope exceeds m's by at least δ. The $\alpha \in [0,1)$ assumption matters because if $\alpha = 1$ then D would carry no weight and no D-gap could help. This supports algorithmic reasoning: one can check the displayed inequalities to certify the scope condition without re-deriving it.*

Remark 2564. *The displayed inequality*

$$D_n(I) - D_m(I) \;\geq\; \frac{\delta}{1-\alpha}$$

should be read as a "required raw D-margin" given a target scope margin δ and a mixing weight $(1-\alpha)$. As α increases toward 1, the coefficient on D decreases and the required D-margin blows up, reflecting that D becomes less influential in scope. Conversely, when α is small, the condition becomes close to $D_n(I) - D_m(I) \geq \delta$, since then $(1-\alpha) \approx 1$.

Proof sketch. Use $\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I) = \alpha(H_n-H_m)+(1-\alpha)(D_n-D_m)$ and lower-bound the RHS by $(1-\alpha)\cdot\delta/(1-\alpha)=\delta$.

Remark 2565. *Spelling out the algebra in the proof sketch: start from*

$$\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I) = \big(\alpha H_n(I)+(1-\alpha)D_n(I)\big) - \big(\alpha H_m(I)+(1-\alpha)D_m(I)\big),$$

then regroup terms to obtain

$$\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I) = \alpha\big(H_n(I)-H_m(I)\big)+(1-\alpha)\big(D_n(I)-D_m(I)\big).$$

Now $\alpha \geq 0$ and $H_n(I)-H_m(I)\geq 0$ imply $\alpha(H_n(I)-H_m(I))\geq 0$, hence

$$\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I) \;\geq\; (1-\alpha)\big(D_n(I)-D_m(I)\big).$$

Finally, substitute the assumption $D_n(I)-D_m(I)\geq \delta/(1-\alpha)$ and use $1-\alpha>0$ (which is why $\alpha\in[0,1)$ is required) to conclude

$$\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I) \;\geq\; (1-\alpha)\cdot\frac{\delta}{1-\alpha}=\delta,$$

which is equivalent to $\mathrm{Scope}(n,I)\geq \mathrm{Scope}(m,I)+\delta$.

Remark 2566. *The proof is a standard "difference-of-affine-expressions" manipulation. The key step is bounding $\alpha(H_n-H_m)$ below by 0 using $H_n\geq H_m$ and $\alpha\geq 0$, so that the entire difference is controlled by the D-term. A helpful picture is that H can only help, so we ignore it and demand that D alone guarantees the desired margin.*

Remark 2567. *A concrete numerical sanity check can help fix intuition. Take $\alpha=\frac{1}{2}$ and $\delta=1$. Then the condition requires $D_n(I)-D_m(I)\geq\delta/(1-\alpha)=2$, together with $H_n(I)\geq H_m(I)$. Indeed, if $H_n(I)=H_m(I)$ and $D_n(I)=D_m(I)+2$, then*

$$\mathrm{Scope}(n,I)-\mathrm{Scope}(m,I)=\tfrac{1}{2}\cdot 0+\tfrac{1}{2}\cdot 2=1=\delta.$$

If, in addition, $H_n(I)>H_m(I)$, the scope gap becomes strictly larger than δ. This illustrates the lemma's role as a sufficient condition: it is intentionally conservative by allowing the H-term to be dropped.

Remark 2568. *The assumption* $\mathrm{Comm}(\{m,n\},I;w)$ *is carried as a hypothesis because the definition of spiritual communion typically combines a scope-inequality with other non-scope conditions (e.g. compatibility, mutual accessibility, or w-dependent constraints). HT24.14 only addresses the scope-inequality component: it says that once the non-scope parts of* Comm *are already satisfied, the displayed inequalities suffice to force the required scope gap. In that sense, the lemma modularizes the verification of spiritual communion into (i) checking non-scope constraints and (ii) checking a simple numeric margin condition on (H,D).*

45.4 Entheogens, emotion, compassion

HT24.15 (Margin-crossing intervention implies communion). Let w be baseline and w_E the intensity datum after an intervention E. If $\mathrm{DComm}(C,I;w)\leq 0$ but $\mathrm{DComm}(C,I;w_E)>0$, then $\mathrm{Comm}(C,I;w_E)$ holds.

Remark 2569. *This formalizes a very direct causality pattern: an intervention that pushes the system across the "communion boundary" (turning nonpositive margin into positive margin) yields communion by definition. In Section 24, interventions may include entheogens, emotional catalysts, or structured practices (Hyperseed-Concept* **??**, **??**, *and 176). The lemma is useful because it allows a reasoner to treat the sign of* DComm *as the single decisive quantity.*

Remark 2570. *Note that the hypothesis uses a weak inequality at baseline and a strict inequality after the intervention. This matches the intended boundary semantics:* $\mathrm{DComm}(C, I; w) = 0$ *represents a knife-edge case (exactly on the boundary), and the theorem asserts that any strict improvement into* $\mathrm{DComm} > 0$ *suffices for communion. In particular, no assumptions are needed about how E produces the change (e.g. whether by increasing* Wcross*, decreasing* Win*, or both); only the sign-change matters.*

Remark 2571. *When modeling interventions such as entheogens or emotional catalysts, it is common that w_E changes multiple ingredients at once: attention allocation, salience weighting of shared cues, suppression of distractors, or re-interpretation of the same stimulus I. HT24.15 is intentionally agnostic about mechanism, so that one may first establish* $\mathrm{DComm}(C, I; w_E) > 0$ *by any available argument (empirical, phenomenological, or derived from lower-level lemmas) and then immediately conclude communion at the level of* Comm*.*

Proof sketch. By definition, communion is equivalent to $\mathrm{DComm} > 0$.

Remark 2572. *The proof is definition-chasing: once* $\mathrm{DComm}(C, I; w_E) > 0$ *is known,* $\mathrm{Comm}(C, I; w_E)$ *follows immediately. Geometrically, the intervention moves the point* $(\mathrm{Wcross}, \mathrm{Win})$ *from on/below the diagonal to strictly above it.*

Remark 2573. *A helpful way to read the geometry is to interpret the diagonal* $\mathrm{Wcross} = \mathrm{Win}$ *as the set where the cross-supported and within-supported contributions are exactly balanced. The conclusion does not require knowing the location of the baseline point beyond "not above the diagonal," and the intervention conclusion uses only that the post-intervention point lies in the open half-plane* $\mathrm{Wcross} > \mathrm{Win}$*. Thus the lemma isolates a binary predicate (above/below a boundary) from any finer-grained quantitative story.*

HT24.16 (Sufficient condition for an intervention to increase communion margin). If an intervention E satisfies

$$\mathrm{Wcross}(C, I; w_E) \geq \mathrm{Wcross}(C, I; w) \quad \text{and} \quad \mathrm{Win}(C, I; w_E) \leq \mathrm{Win}(C, I; w),$$

then $\mathrm{DComm}(C, I; w_E) \geq \mathrm{DComm}(C, I; w)$.

Remark 2574. *This is a clean sufficient condition for "communion-improving" interventions: make cross-supported intensity larger (or equal) while making within-supported intensity smaller (or equal). It matches the intuitive goal of many practices: cultivate shared attention and shared meaning while quieting isolating internal noise. As a simple example, synchrony rituals could increase pairwise and group patterns while decreasing idiosyncratic rumination.*

Remark 2575. *The condition is intentionally one-sided (sufficient, not necessary). Many real interventions plausibly raise both* Wcross *and* Win*; for instance, heightened affect may increase both shared resonance and private imagery. In such cases, HT24.16 may not apply directly, but it still provides a baseline comparison tool: if one can bound the change in* Wcross *from below and the change in* Win *from above, then a net margin increase follows.*

Remark 2576. *The weak inequalities also matter: an intervention that leaves* Wcross *unchanged while reducing* Win *is still margin-improving, and conversely leaving* Win *unchanged while increasing* Wcross *is also margin-improving. Thus the lemma captures two conceptually distinct routes to increasing communion margin: "build bridges" (increase cross support) and "reduce self-noise" (decrease within support).*

Proof sketch. Subtract the two inequalities: $(\mathrm{Wcross}_E - \mathrm{Win}_E) \geq (\mathrm{Wcross} - \mathrm{Win})$.

Remark 2577. *The proof strategy is to recognize* DComm *as a difference and apply monotonicity of subtraction in the ordered reals (or extended reals). The key step is that the hypotheses are already exactly the two components needed to compare the differences. Visually: move right (increase* Wcross*) and move down (decrease* Win*) in the* $(\mathrm{Wcross}, \mathrm{Win})$ *plane; both motions increase the vertical distance above the diagonal.*

Remark 2578. *If* Wcross *and* Win *are allowed to take values in an extended ordered domain (e.g.* ∞ *as a formal supremum of intensities), then the monotonicity reasoning still goes through provided the difference* $\mathrm{Wcross} - \mathrm{Win}$ *is interpreted in the same domain where it is defined. In applications, one typically ensures finiteness by construction (e.g. by summability of pattern intensities), but the lemmas algebraic form makes clear that only the order properties are used, not any measure-theoretic subtleties.*

Remark 2579. *Combining HT24.16 with HT24.15 yields a practical two-step template: (i) prove the intervention weakly increases margin, and (ii) check that the post-intervention margin is strictly positive (or that the weak increase carries a nonpositive baseline into positivity). This mirrors how one might argue about structured practices: show they systematically push in the "right direction," then argue that, for some range of parameters or some subjects, the push is large enough to cross the communion boundary.*

HT24.17 (Compassion forces nonzero cross-supported intensity). If m has a compassion episode toward m_0 on I (Definition 377), then for the dyad $\{m, m_0\}$,

$$\mathrm{Wcross}(\{m, m_0\}, I; w) > 0.$$

Remark 2580. *Compassion is treated here not merely as an inner sentiment but as something that* couples *minds through cross-supported patterns. Thus, in the formalism, a compassion episode entails the existence of at least one positive-intensity pattern supported on the dyad. This links the ethical-phenomenological notion of compassion (Hyperseed-Concept 81) to a crisp algebraic consequence.*

Remark 2581. *The dyadic restriction* $\{m, m_0\}$ *is important: the lemma does not claim that compassion instantly creates cross-supported intensity at the level of a larger community* C*, only that it forces at least one nontrivial cross-supported contribution on the specific pair involved. In later uses, this dyadic guarantee can serve as a "seed" for larger communion arguments, e.g. by aggregation lemmas that lift pairwise cross-supported patterns to group-level* $\mathrm{Wcross}(C, I; w)$ *under additional connectivity or propagation assumptions.*

Remark 2582. *Conceptually, the conclusion* $\mathrm{Wcross} > 0$ *distinguishes compassion from purely solipsistic concern. One might imagine a private affective episode that involves thoughts* about m_0 *while remaining entirely within-supported; Definition 377 rules this out by building in an explicit cross-mind coupling requirement. HT24.17 is the algebraic footprint of that design choice.*

Proof sketch. Definition 377 requires coupling to cross-mind patterns supported on $\{m, m_0\}$ with positive intensity; summing yields a positive cross total.

Remark 2583. *The proof is existential-to-sum: if at least one summand in a nonnegative sum is strictly positive, then the sum is strictly positive. The key step is that Definition 377 guarantees such a positive-intensity dyadic pattern exists. A visual intuition is that compassion creates at least one "bridge" edge between two nodes; the total cross weight on that dyad cannot be zero.*

Remark 2584. *If* $\mathrm{Wcross}(\{m, m_0\}, I; w)$ *is defined as a sum (or integral) over cross-supported patterns with nonnegative intensities, then the argument can be phrased as: Definition 377 provides a witness pattern* p *with intensity* $\mathrm{W}(p) > 0$ *and support contained in* $\{m, m_0\}$*, hence* p *contributes strictly positively to* $\mathrm{Wcross}(\{m, m_0\}, I; w)$*, and nonnegativity of all other contributions prevents cancellation. This makes explicit that the lemma depends on the "no negative intensity" convention for pattern weights.*

Remark 2585. *HT24.17 can be paired with HT24.16 as follows: a compassion episode can be seen as an intervention that (at least locally, on the dyad) increases* Wcross *from* 0 *to a positive value, thereby tending to increase* DComm *on that dyad. Additional hypotheses would be needed to conclude full communion* $\mathrm{Comm}(\{m, m_0\}, I; w)$*, since* Win *might also be large; nevertheless, the lemma supplies a guaranteed positive cross term that later bounds can leverage.*

45.5 Beauty as paraconsistent "surprising fulfillment"

Fix a designation threshold $\tau \in (0, 1]$. Recall the valuation assignments:

$$v_m(\mathrm{Ful}(B)) = (\min\{1, \mathrm{Ful}_m(B; I)\},\ 1 - \min\{1, \mathrm{Ful}_m(B; I)\}),$$

$$v_m(\mathrm{Sur}(B)) = \Big(\min\{1, \frac{\mathrm{Sur}_m(B; I)}{\mathrm{Sur}_m(B; I) + 1}\},\ 1 - \min\{1, \frac{\mathrm{Sur}_m(B; I)}{\mathrm{Sur}_m(B; I) + 1}\}\Big),$$

and $\mathrm{Beaut}(B) := \mathrm{Ful}(B) \wedge \mathrm{Sur}(B)$.

Remark 2586. *The notation* $v_m(\cdot)$ *denotes the paraconsistent truth value in* $B = [0, 1]^2$ *assigned by mind* m *to a predicate. Here,* $\mathrm{Ful}_m(B; I) \in (0, 1]$ *is a numeric fulfillment score and* $\mathrm{Sur}_m(B; I) \in [0, \infty]$ *is a numeric surprise score (see Section 24). The maps* $x \mapsto (\min\{1, x\}, 1 - \min\{1, x\})$ *and* $s \mapsto (\min\{1, s/(s + 1)\}, 1 - \min\{1, s/(s + 1)\})$ *embed these numeric scores into the* B*-truth-value space. The operator* $\mathrm{Beaut}(B) := \mathrm{Ful}(B) \wedge \mathrm{Sur}(B)$ *then expresses the intended conjunction: beauty is "fulfilled and surprising" (Hyperseed-Concept 53,* **??** *is not listed but compare* **??***, and 130). The paraconsistent framing echoes Peirce's triadic sensibility about qualities, reactions, and mediations [14], and Whitehead's emphasis on the processual synthesis of feeling and form [15], while remaining technically grounded in the designation calculus above [23].*

HT24.18 (Designated fulfillment is a direct numeric threshold). Since $\mathrm{Ful}_m(B; I) \in (0, 1]$, we have

$$\mathrm{Desig}_\tau\big(v_m(\mathrm{Ful}(B))\big) \iff \mathrm{Ful}_m(B; I) \geq \tau.$$

Remark 2587. *This lemma says that the somewhat elaborate-looking valuation* $v_m(\mathrm{Ful}(B))$ *does not distort thresholds: designatedness at level* τ *is exactly the underlying numeric condition* $\mathrm{Ful}_m \geq \tau$*. So the paraconsistent embedding is, in this case, simply a convenient representation that preserves the operative inequality.*

Proof sketch. Here $\min\{1, \mathrm{Ful}_m\} = \mathrm{Ful}_m$, so the designatedness condition $t \geq \tau$ is exactly $\mathrm{Ful}_m \geq \tau$.

Remark 2588. *The proof strategy is simplification using the range restriction* $\mathrm{Ful}_m(B; I) \in (0, 1]$. *The key step is observing that* $\min\{1, x\} = x$ *on this interval. Visually: the embedding is the identity on the first coordinate for fulfillment, so thresholding is unchanged.*

HT24.19 (Designated surprise is a rational inequality in Sur_m). Assume $\tau \in (0, 1)$. Then

$$\mathrm{Desig}_\tau\big(v_m(\mathrm{Sur}(B))\big) \iff \frac{\mathrm{Sur}_m(B; I)}{\mathrm{Sur}_m(B; I) + 1} \geq \tau \iff \mathrm{Sur}_m(B; I) \geq \frac{\tau}{1 - \tau}.$$

(If $\tau = 1$, then designatedness requires $\mathrm{Sur}_m(B; I) = +\infty$.)

Remark 2589. *Unlike fulfillment, surprise is mapped through a saturating transform* $s \mapsto s/(s+1)$ *that converts* $[0, \infty]$ *to* $[0, 1)$*, so a threshold on the transformed value corresponds to a* nonlinear *threshold on* Sur_m*. The inequality* $\mathrm{Sur}_m \geq \tau/(1-\tau)$ *shows how demanding large* τ *becomes: as* $\tau \uparrow 1$*, the required surprise diverges. This matches the intuition that "maximally designated surprise" is only achievable in an infinite-surprise limit.*

Proof sketch. Solve $\mathrm{Sur}/(\mathrm{Sur} + 1) \geq \tau$: $\mathrm{Sur} \geq \tau(\mathrm{Sur} + 1) = \tau\mathrm{Sur} + \tau$, so $(1 - \tau)\mathrm{Sur} \geq \tau$ hence $\mathrm{Sur} \geq \tau/(1 - \tau)$.

Remark 2590. *The proof is algebraic rearrangement of a rational inequality. The key step is that* $1 - \tau > 0$ *when* $\tau \in (0, 1)$*, allowing division without flipping the inequality. Geometrically,* $s/(s+1)$ *is an increasing concave curve approaching* 1 *asymptotically, so hitting a high threshold requires disproportionately large* s*.*

HT24.20 (Beauty designatedness factorizes into two numeric thresholds). For $\tau \in (0, 1)$,

$$\mathrm{Desig}_\tau\big(v_m(\mathrm{Beaut}(B))\big) \iff \big(\mathrm{Ful}_m(B; I) \geq \tau\big) \ \wedge\ \big(\mathrm{Sur}_m(B; I) \geq \tau/(1 - \tau)\big).$$

Remark 2591. *This is the operational heart of the beauty formalism: once one fixes* τ*, beauty becomes a pair of ordinary numeric inequalities. Thus the paraconsistent connective* $\wedge$ *plus designatedness translates to a crisp conjunction of bounds on fulfillment and surprise. In other words, the "mystery" of beauty is not placed in the logic; it is placed in how* Ful_m *and* Sur_m *are computed in Section 24.*

Proof sketch. By HT24.10, designatedness of $\wedge$ is equivalent to designatedness of both conjuncts; then apply HT24.18–HT24.19.

Remark 2592. *The proof strategy is compositional: (i) reduce designatedness of a conjunction to conjunction of designatedness, then (ii) rewrite each designatedness statement into a numeric inequality using the previous lemmas. This is a typical pipeline in paraconsistent threshold logics: algebraic laws + threshold rules yield executable conditions.*

HT24.21 (Beauty-to-OEI rule schema). Suppose a system provides monotone functionals $\mathrm{GI}, \mathrm{GT}, \mathrm{SelfC}$ satisfying: (i) GI is nondecreasing in Ful_m; (ii) GT is nondecreasing in Sur_m and in SelfC; and thresholds $\theta_F, \theta_S, \theta_C$ such that $\mathrm{Ful}_m \geq \theta_F \Rightarrow \mathrm{GI} > 0$ and $\mathrm{Sur}_m \geq \theta_S \ \wedge\ \mathrm{SelfC} \geq \theta_C \Rightarrow \mathrm{GT} > 0$. Then a beauty episode together with $\mathrm{SelfC} \geq \theta_C$ implies OEI-activity (i.e. $\mathrm{GI} > 0$ and $\mathrm{GT} > 0$).

Remark 2593. *This is an abstract "rule schema" connecting the phenomenology of beauty to downstream system-level activity (OEI), assuming only monotonicity and threshold-trigger behavior of certain functionals. Its conceptual role is to make Theorem 21 usable in automated reasoning: rather than re-proving the entire theorem, one checks monotonicity and threshold premises, then applies the schema. In the broader cognitive-systems literature, such monotonicity-based triggers are common when modeling attention allocation, intrinsic motivation, and self-model consistency; compare related perspectives in [19]. In particular, the point is that the premises are deliberately* structural*: they do not depend on the internal construction of the OEI variables, only on their order-respecting response to increases in the relevant "driving" quantities (here: fulfillment and surprise). This matches how such subsystems are often implemented in practice (or idealized in theory): a downstream module does not need to know* why *an upstream signal is high, only that once it is high enough it reliably crosses an activation boundary. Formally, the schema can be read as a reusable inference template: from designatedness of the relevant upstream predicates one extracts real-valued lower bounds, and from those lower bounds one invokes threshold axioms to conclude positivity (or activation) of the corresponding OEI channels. Finally, the emphasis on "designated" (rather than, say, classical truth) is intentional: the schema is compatible with paraconsistent settings in which a state may carry conflicting information while still permitting forward progress on designated consequences.*

Proof sketch. This is exactly the reasoning pattern of Theorem 21: beauty designated implies both $\mathrm{Ful}(B)$ and $\mathrm{Sur}(B)$ designated (HT24.10), giving numeric lower bounds (HT24.18–HT24.19), which trigger $\mathrm{GI} > 0$ and $\mathrm{GT} > 0$ by the threshold assumptions. Equivalently, the proof has the canonical "decode then activate" shape: one first *decodes* logical/phenomenological predicates into quantitative constraints, and then *activates* the downstream consequences by applying monotone threshold rules. A useful way to view the threshold assumptions is as implications of the form

$$\mathrm{Ful}(B) \text{ designated} \wedge \mathrm{Ful}(B) \geq \theta_I \ \Rightarrow\ \mathrm{GI} > 0, \qquad \mathrm{Sur}(B) \text{ designated} \wedge \mathrm{Sur}(B) \geq \theta_T \ \Rightarrow\ \mathrm{GT} > 0,$$

for fixed thresholds θ_I, θ_T (the exact constants are immaterial to the schema as long as they exist and are used consistently). Monotonicity enters by ensuring that once the decoded value meets or exceeds the threshold, any strengthening of the antecedent (e.g. higher fulfillment/surprise) cannot "turn off" the activation conclusion; this is the minimal stability property needed for reliable downstream inference.

Remark 2594. *The proof strategy is a chain of implications:*

$$\textit{beauty} \Rightarrow (\textit{fulfillment} \wedge \textit{surprise}) \Rightarrow (\textit{numeric bounds}) \Rightarrow (\mathrm{GI} > 0 \wedge \mathrm{GT} > 0).$$

The key steps are exactly where earlier helper theorems are invoked, showing the internal connectivity of this section: HT24.10 provides logical factorization, and HT24.18–HT24.19 provide numeric decoding. As a visual intuition, imagine two gauges (fulfillment and surprise) with needles; beauty means both needles exceed their marks, and the OEI subsystems are wired to activate when their respective gauges exceed pre-set thresholds. One can also read the chain as separating semantic *from* operational *content: the first arrow is a semantic axiom about how "beauty" decomposes into the two constituent phenomenological modes, while the latter arrows are operational axioms describing how numeric estimates drive internal activity. This separation is what makes the rule schema portable: any alternative model that preserves the same factorization (beauty entails both components) and the same decoding/threshold interface will admit the same conclusion, even if the internal realization of* GI *and* GT *differs. In automated reasoning terms, HT24.10 plays the role of introducing two subgoals from a single goal (a conjunction introduction in the forward direction),* *whereas HT24.18–HT24.19 discharge those subgoals by replacing them with inequalities that are compatible with arithmetic solvers. The "two gauges" picture can be sharpened slightly: the needles need not represent conscious reports; they may represent any internal scores whose increase is monotone with respect to accumulating evidence for fulfillment and surprise, so that the activation can be robust even under partial, noisy, or inconsistent inputs.*

45.6 Artifacts, archetypes, and communion capacity

HT24.22 (From "art object" to positive communion capacity under joint-attention dominance). Let A be an art object for a population X (Definition 383), and assume joint attention to A creates cross-supported patterns whose intensities increase with the number of participants who undergo beauty episodes, while within-mind intensities do not increase faster than cross-mind intensities. Then the communion capacity $\Gamma_{\mathrm{Comm}}(A; C, I)$ is positive for a substantial measure of (C, I).

Remark 2595. *This statement formalizes a familiar observation from aesthetics and social psychology: shared attention to an artifact can synchronize minds, generating shared patterns that outweigh purely private responses. In the Hyperseed vocabulary, this links Art and Aesthetics (Hyperseed-Concept 53) to Attention/Attentional Focus (60) and to Archetypal resonance (58) insofar as art objects often function by evoking widely shared templates. The lemma is also a bridge between the formal notion of beauty episodes (previous subsection) and the aggregate notion of communion capacity, i.e. the ability of an artifact to reliably induce communion across many contexts.*

Beyond the informal reading, it is helpful to keep the quantifier structure in view: the hypothesis is not merely that some individuals have beauty episodes, but that, conditioned on *joint attention* to A, the resulting cross-supported patterns admit an intensity scaling with the number of beauty-episode participants. In other words, there is an assumed population-level reinforcement effect: as more agents undergo beauty episodes *together with* attention to A, the cross-mind component (captured by Wcross in the later proof sketch) aggregates in a way that is not matched (or is matched only subdominantly) by purely within-mind amplification. This is precisely the kind of "dominance" condition needed for a robust, measure-theoretic capacity statement, because it converts individual-level events (beauty episodes) into a consistent bias toward DComm > 0 on a non-negligible set of contextual parameters.

The conclusion refers to $\Gamma_{\mathrm{Comm}}(A; C, I)$ being positive for a *substantial measure* of (C, I). Here "substantial" should be read as ruling out a thin exceptional set: the intended content is that the set of contexts (C, I) in which A induces communion is not merely nonempty but occupies positive measure in the ambient context space used to define capacity. This is stronger than an existence statement and weaker than a uniform-for-all statement; it matches the empirical intuition that artworks reliably generate shared experience across many (but not necessarily all) conditions of audience, setting, and interpretive frame.

Proof sketch. Under the hypotheses, joint participation shifts w to w_A in a way that increases Wcross more than Win in expectation, hence increases DComm in expectation. This is the direct proof template of Proposition 37 combined with the definition of Γ_{Comm}.

One way to read the sketch is as an "expected-sign" argument: the joint-attention update $w \mapsto w_A$ is assumed to have an asymmetric effect on the cross-terms versus the within-terms. If one writes (conceptually) the communion-relevant difference as something like

$$\mathrm{DComm}(w_A) = \Phi(\mathrm{Wcross}(w_A), \mathrm{Win}(w_A))$$

for an increasing dependence on cross-supported structure and a non-increasing (or less rapidly increasing) dependence on within-mind structure, then the scaling hypothesis implies that, as the number of beauty-episode participants increases, the expectation of $\mathrm{Wcross}(w_A)$ is driven upward at a rate that outpaces any corresponding increase in $\mathrm{Win}(w_A)$. The proof template of Proposition 37 is then invoked to convert this directional inequality (cross dominates within) into a positive expectation for DComm, and the definition of Γ_{Comm} converts that positive-bias condition into a positive-measure set of contexts where communion is actually achieved. In this sense, Proposition 37 supplies the generic "dominance ⇒ communion" inference pattern, while the present lemma supplies a concrete route to the dominance condition via joint attention to an artifact.

Remark 2596. *The proof strategy is to combine (i) a monotonicity-type argument (cross grows faster than within) with (ii) the definition of capacity as a measure of contexts where communion is achieved. The key step is recognizing that expected growth inequalities translate into a positive-measure set of (C, I) where* $\mathrm{DComm} > 0$*, hence communion holds. A useful image is that the artifact creates a "shared field" whose strength scales superlinearly with the number of jointly attending participants, a dynamic reminiscent of resonance metaphors (Hyperseed-Concept 159) and discussed in broader speculative contexts such as morphic resonance [13], though the present lemma remains purely within the formal definitions of Section 24.*

A further clarification of the monotonicity component is that it is not claiming cross-supported patterns always dominate within-mind patterns, but rather that *the marginal effect of additional jointly attending, beauty-episode participants* preferentially increases cross-mind intensity. This "marginal dominance" is what allows the argument to survive heterogeneity: different viewers may have different baseline within-mind responses, but the incremental contribution from synchronizing attention is assumed to attach mainly to cross-supported structure (e.g. shared interpretive anchors, temporally aligned salience peaks, or convergent affective appraisal), thereby pushing the communion-relevant difference DComm in the positive direction often enough to register in the capacity measure.

It is also useful to note why the lemma is formulated in terms of beauty episodes rather than generic engagement. Within the internal logic of Section 24, beauty episodes are singled out as events that already satisfy certain coherence or integrative criteria (from the previous subsection), making them natural carriers of robust pattern formation. The lemma then asserts that when such episodes occur in a jointly attending group, the resulting cross-supported patterns are not merely additive noise but have a structured intensification property. This is the formal bridge from "individual integrative experience" to "group-level communion" that one expects from artifacts functioning as coordination devices.

Finally, the role of A being an "art object" (Definition 383) is that it supplies a principled restriction on the class of artifacts under discussion: A is not an arbitrary stimulus but one that, by definition, is situated to elicit the relevant episodes and interpretive stabilization across a population X. Under that restriction, the scaling hypothesis can be seen as a formal proxy for familiar mechanisms such as shared conventions, culturally transmitted archetypes, and mutual predictability of attention trajectories while jointly engaging the work. This makes the conclusion—positive $\Gamma_{\mathrm{Comm}}(A; C, I)$ on a substantial set of contexts—a mathematically crisp rendering of the idea that artworks often operate as repeatable generators of communion rather than as purely idiosyncratic, private experiences.

45.7 State-dependent science (useful conditionalization schema)

HT24.23 (Conditioning viewpoint on state-dependent science). Let Σ be a state predicate selecting a class of intervals I (Definition 385). Any predictive or causal claim produced by a "Σ-science" can be treated formally as a *state-conditional* claim (e.g. "$P(\cdot \mid \Sigma)$" or "$P(\cdot \mid \mathrm{do}(\cdot), \Sigma)$"). In particular, if Σ_1 implies Σ_2 (every Σ_1-interval is a Σ_2-interval), then Σ_1-science is a refinement of Σ_2-science (it conditions on a smaller class). Equivalently, one may regard a Σ-science as fixing a context variable (explicitly or implicitly) and then performing ordinary statistical or causal inference within that context; on this reading, the "state" does not add a new inferential primitive, but supplies a conditioning event that sharpens the reference class. Concretely, if a Σ-science asserts a regularity of the form "Y is predictable from X" it is naturally rendered as $P(Y \mid X, \Sigma)$; if it asserts an invariance under intervention, it is naturally rendered as $P(Y \mid \mathrm{do}(X{=}x), \Sigma)$, with Σ functioning as the background regime in which the intervention is contemplated.

Remark 2597. *The point is methodological: "state-dependent science" is not a fundamentally different kind of science, but rather ordinary inference carried out under a restricted regime of system-states. Thus, from the standpoint of probability and causality, it is natural to treat it as conditioning on an event Σ. The refinement claim matches the common Bayesian intuition: stronger conditions yield more specific (and potentially less statistically supported) claims. This connects to the social-computational-probabilistic view of scientific practice in [20] and to the Hyperseed framing of contexts and aspects (Hyperseed-Concept 86). One can also read Σ as bundling together what, in practice, would otherwise appear as scattered "ceteris paribus" clauses, selection criteria, and instrumentation constraints: the formalization makes explicit that the resulting claims are indexed to a regime. In this sense, a Σ-science is not merely a claim about the world but a claim about the world* as encountered under *a controlled (or at least characterized) subset of states; this is the same move by which laboratory science distinguishes "in vitro" from "in vivo" regularities, or by which psychology distinguishes performance "under sleep deprivation" from performance "while rested." The refinement statement then expresses a basic reference-class inclusion: if Σ_1 is a stricter operationalization (say, "rested and caffeinated") than Σ_2 (say, "rested"), then results valid in the stricter class are automatically results in the broader class* as conditionalized claims*, even though their empirical support may be weaker due to fewer admissible intervals.*

Proof sketch. The definition of state-dependent science is a restriction of the epistemic practice to intervals satisfying Σ. Formally, restriction corresponds to conditioning the data-generating process and the accepted-observation regime on Σ. Monotonicity under implication of predicates is then immediate: conditioning on Σ_1 is conditioning on a stronger event than Σ_2. More explicitly, if D denotes the stream of admissible observations/records extracted from intervals, then "doing science under Σ" amounts to working with the conditional model $P(D \mid \Sigma)$ (and, when causal language is used, with $P(D \mid \mathrm{do}(\cdot), \Sigma)$). Any learned parameter θ or derived predictive distribution is thereby indexed to Σ (e.g. $P_\theta(Y \mid X, \Sigma)$). If $\Sigma_1 \Rightarrow \Sigma_2$, then $\{\Sigma_1\text{-intervals}\} \subseteq \{\Sigma_2\text{-intervals}\}$, so the Σ_1-conditioned regime is a sub-regime of the Σ_2-conditioned regime; in the standard measure-theoretic sense, Σ_1 is a stronger conditioning event. (As usual, one tacitly assumes $\mathbb{P}(\Sigma) > 0$ or otherwise works with an appropriate regular conditional probability so that these expressions are well-defined.) The "refinement" language is then simply the statement that the inferential target has been restricted to a smaller reference class, which yields narrower generalizations but typically increases variance/uncertainty due to reduced effective sample size.

Remark 2598. *The strategy is to translate an informal "restrict attention to these intervals" instruction into the formal operation of conditioning. The key step is recognizing that both data* *selection and acceptance criteria can be modeled as parts of the same conditional distribution. Visually: $\Sigma_1 \Rightarrow \Sigma_2$ means Σ_1 carves out a smaller region inside the region carved out by Σ_2; a science conditioned on the smaller region is, by definition, more specialized. In causal terms, the same picture applies: a claim like $P(Y \mid \mathrm{do}(X{=}x), \Sigma)$ should be read as "the effect of setting X to x* within *the Σ-regime," which is often exactly what experimental protocols intend (holding fixed the background state while varying X). This helps prevent a common confusion in informal discussion, where a state-restricted causal claim is mistakenly interpreted as unconditional. The refinement implication also clarifies a practical hierarchy: broad predicates Σ_2 support more data and hence more stable estimates, while narrow predicates Σ_1 support finer-grained, context-sensitive laws; the trade-off is not conceptual but statistical and operational.*

46 Helper theorems for Section 25 (Reality-systems, universe, multiverse, eurycosm)

These helper results are small "algebraic moves" that recur when one reasons about reality-systems, universes, multiverses, and eurycosmic constructions in the Hyperseed-style stack (e.g. [1, 18]). Most of the statements are deliberately elementary: their value is not that they surprise an expert, but that they stabilize a vocabulary of monotonicity, fixed points, and prediction-theoretic certificates that an automated prover (or a tired human) can reuse without re-deriving each time.

Remark 2599. *A recurring philosophical theme in this section is that "realness" is treated operationally: an entity (or hypothesis) counts as more empirically real insofar as conditioning on it improves prediction—a pragmatic criterion reminiscent of a social-computational view of science ([20]) and closely aligned with the Hyperseed core-concepts of Empirical Reality and Reality-System (Hyperseed-Concepts* **??** *and 149).*

46.1 Empirical realness as prediction-loss reduction

Lemma 46.1 (Basic bounds and sign conditions)**.** *Assume the loss is nonnegative and bounded: $0 \le L \le L_{\max}$, so $0 \le \mathrm{Err}_I(\Pi_M) \le L_{\max}$ and likewise for $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x))$. Then for $\delta > 0$,*

$$-\frac{L_{\max}}{\delta} \;\le\; \rho_{M,I}(x) \;\le\; \frac{L_{\max}}{\delta}.$$

Moreover:

1. *If conditioning on x does not worsen prediction error, i.e. $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x)) \le \mathrm{Err}_I(\Pi_M)$, then $0 \le \rho_{M,I}(x) < 1$.*
2. *$\rho_{M,I}(x) = 0$ iff $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x)) = \mathrm{Err}_I(\Pi_M)$.*
3. *If conditioning strictly improves prediction error, then $\rho_{M,I}(x) > 0$; if it strictly worsens it, then $\rho_{M,I}(x) < 0$.*

Remark 2600. Intuitive reading. *The quantity $\rho_{M,I}(x)$ is a normalized "error-reduction score" for treating x as an evidential condition: it compares the baseline predictive error $\mathrm{Err}_I(\Pi_M)$ against the conditional error $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x))$, and rescales by $\mathrm{Err}_I(\Pi_M) + \delta$ to keep the fraction well-behaved even when the baseline error is near 0. The sign is the key: positive means "x helps prediction", negative means "x misleads prediction." This is a formal operationalization of the idea that empirical realness is tied to predictive traction (Hyperseed-Concept* **??** *and also* **??***).*

Notation unpacking. *Here M indexes a model-class or mind-model, I indexes an information context or dataset/interaction regime, Π_M is the predictor induced by M (or chosen within M), and $\mathrm{obs}_I(x)$ denotes the observation/event "x occurs" as registered in context I. The notation $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x))$ is the prediction loss of Π_M when conditioned on that event. This way of coding "realness as error-reduction" is consonant with the pragmatic stance taken in [20].*

Remark 2601. Why the lemma matters. *Boundedness and sign-conditions are the minimal safety checks that let one use $\rho_{M,I}$ inside larger constructions (e.g. thresholds for predictive adjacency) without having to carry pathological cases everywhere. In proofs downstream, one typically needs only: (i) ρ cannot explode when loss is bounded, and (ii) improvement vs. worsening corresponds exactly to the sign.*

Sketch. Boundedness gives $\mathrm{Err}, \mathrm{Err}(\cdot \mid \cdot) \in [0, L_{\max}]$, so the numerator $\mathrm{Err}_I(\Pi_M) - \mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x))$ lies in $[-L_{\max}, L_{\max}]$ and the denominator is at least δ. For (1), the numerator is in $[0, \mathrm{Err}_I(\Pi_M)]$ and the denominator is $\mathrm{Err}_I(\Pi_M) + \delta > \mathrm{Err}_I(\Pi_M)$. (2) and (3) are immediate from the defining fraction. □

Remark 2602. Proof sketch (high-level strategy). *The argument is a bounding exercise: constrain the numerator by the range of the loss, constrain the denominator away from 0 using $\delta > 0$, and then interpret the sign of a fraction via the sign of its numerator.*

Key step commentary. *The strict inequality < 1 in (1) comes from the fact that the denominator $\mathrm{Err}_I(\Pi_M) + \delta$ is strictly larger than $\mathrm{Err}_I(\Pi_M)$, so even a maximal non-worsening improvement cannot yield ratio 1. This is conceptually important: δ is not merely technical; it builds in a kind of epistemic humility that prevents declaring "perfect" empirical realness from a single conditioning event.*

Geometric intuition. *Think of $(\mathrm{Err}_I(\Pi_M), \mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x)))$ as a point in a square $[0, L_{\max}]^2$. The numerator is the signed vertical difference from the diagonal $\{e = e'\}$, and δ rescales the slope of the score so the diagonal remains the zero-set while keeping the score finite.*

Proposition 46.2 (Monotonicity in the conditional error)**.** *Fix M, I, Π_M, δ. If two entities x, x' satisfy $\mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x)) \leq \mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(x'))$, then $\rho_{M,I}(x) \geq \rho_{M,I}(x')$.*

Remark 2603. Intuitive reading. *Holding the baseline model and baseline error fixed, the only way ρ changes from one entity to another is through the conditional error. If conditioning on x yields a smaller conditional error than conditioning on x', then x is (weakly) more empirically real in this operational sense.*

Connection. *This proposition is the monotonicity principle that later justifies thresholding: if we pick an "empirical-realness threshold" and then refine our conditioning event to get* better *predictive performance, we cannot drop below threshold by accident.*

Sketch. The map $e \mapsto (\mathrm{Err}_I(\Pi_M) - e)/(\mathrm{Err}_I(\Pi_M) + \delta)$ is affine and strictly decreasing in e. □

Remark 2604. Proof sketch. *Reduce the statement to a one-dimensional inequality: an affine decreasing function reverses order. This is the simplest possible monotonicity argument.*

Visual intuition. *Plot ρ as a function of $e = \mathrm{Err}_I(\Pi_M \mid \mathrm{obs}_I(\cdot))$; it is a straight line with negative slope. Moving left (smaller conditional error) moves up (larger ρ).*

Lemma 46.3 (Optional p-bit lift for empirical realness)**.** *Define a helper p-bit realness value*

$$\rho^{\pm}_{M,I}(x) := \big(\max(\rho_{M,I}(x), 0),\ \max(-\rho_{M,I}(x), 0)\big) \in [0,1]^2$$

(after clipping to $[0,1]$ if desired). Then:

1. *$\rho^{\pm}(x)$ records evidence for improvement vs evidence for worsening (as two coordinates).*
2. *$\rho_{M,I}(x) = \rho^{\pm,+}(x) - \rho^{\pm,-}(x)$ recovers the signed scalar.*

Remark 2605. Intuitive reading. *This is a bookkeeping device: rather than carrying a signed scalar, we store two nonnegative "evidence masses"—one supporting that x helps prediction, and one supporting that x harms it. This mirrors the practice, common in paraconsistent and multi-valued logics, of representing positive and negative support separately before any final aggregation (compare the general style of evidence-pairs in [23, 24]).*

Why it is useful. *In downstream ontology-building, one often wants to combine heterogeneous evidences without prematurely canceling them. The p-bit form delays cancellation until one explicitly subtracts coordinates, and it also interfaces naturally with vector-valued truth/evidence calculi.*

Sketch. Immediate from the definition of $\max(\cdot, 0)$. □

Remark 2606. Proof sketch. *The decomposition is literally the standard identity $z = \max(z, 0) - \max(-z, 0)$ applied to $z = \rho_{M,I}(x)$.*

Geometric intuition. *The map $z \mapsto (\max(z, 0), \max(-z, 0))$ folds the real line onto the two coordinate axes in the first quadrant: positive values lie on the x-axis, negative values lie on the y-axis.*

46.2 Reality-systems as fixed points of mutual predictive closure

Proposition 46.4 (Threshold monotonicity of predictive adjacency). *Let $\theta_1 \geq \theta_2$. If $x \to_{\theta_1} y$ then $x \to_{\theta_2} y$. Consequently, for all $R \subseteq U$,*

$$\mathrm{Pred}_{\to}\theta_1(R) \subseteq \mathrm{Pred}_{\to}\theta_2(R), \quad \mathrm{Pred}_{\leftarrow}\theta_1(R) \subseteq \mathrm{Pred}_{\leftarrow}\theta_2(R), \quad F\theta_1(R) \subseteq F\theta_2(R).$$

Remark 2607. Intuitive reading. *A higher threshold θ means we are more demanding about what counts as "x predicts y." If a relation passes a strict threshold, it automatically passes any weaker one. Thus, as one raises θ, the predictive neighborhoods shrink; as one lowers θ, they expand.*

Notation note. *The symbols* $\mathrm{Pred}_{\to}, \mathrm{Pred}_{\leftarrow}$ *denote forward- and backward-predictive expansions of a set, and F denotes their intersection (mutual predictive closure), which is the key operator whose fixed points define reality-systems (Hyperseed-Concept 149). The text writes* $\mathrm{Pred}_{\to}\theta(R)$ *etc.; this can be read as "the θ-parameterized version" (i.e. what one might elsewhere denote by* $\mathrm{Pred}_{\to,\theta}(R)$*), with the parameter carried as a visible tag.*

Sketch. If $x \to_{\theta_1} y$ means a score is at least θ_1, it is also at least $\theta_2 \leq \theta_1$. The set inclusions follow by unwinding the existential witness definition of $\mathrm{Pred}_{\to}, \mathrm{Pred}_{\leftarrow}$ and taking intersections. □

Remark 2608. Proof sketch. *The proof is a single monotonicity step at the level of the underlying predicate "score$\geq \theta$", lifted through the existential quantifiers defining neighborhood expansions, and finally through intersection.*

Why it works. *Existential definitions preserve implication: if every witness for θ_1 is also a witness for θ_2, then every element reachable at θ_1 remains reachable at θ_2.*

Lemma 46.5 (Distributivity over unions). *For all $R, S \subseteq U$,*

$$\mathrm{Pred}_{\to}(R \cup S) = \mathrm{Pred}_{\to}(R) \cup \mathrm{Pred}_{\to}(S), \qquad \mathrm{Pred}_{\leftarrow}(R \cup S) = \mathrm{Pred}_{\leftarrow}(R) \cup \mathrm{Pred}_{\leftarrow}(S).$$

Hence

$$F(R \cup S) \supseteq F(R) \cup F(S).$$

Remark 2609. Intuitive reading. *Forward (or backward) predictive expansion is "existential in the source": an element is forward-reachable from* $R \cup S$ *iff it is forward-reachable from* R *or from* S*. The mutual closure* F *is an* intersection *of a forward and a backward expansion, and intersections do not distribute over unions in the same direction; hence we get only the inclusion* $\supseteq$*.*

Why it matters. *This lemma is a small algebraic fact that frequently appears when constructing candidate reality-systems by seeding with multiple sources and then closing under mutual predictability.*

Sketch. A witness $x \in R \cup S$ is equivalent to $(x \in R) \vee (x \in S)$, giving the union formulas. Then $F(R \cup S)$ is an intersection of two unions, which always contains each of the two intersections $F(R)$ and $F(S)$. □

Remark 2610. Proof sketch. *Reduce to propositional logic:* $(\exists x \in R \cup S)\, \Phi(x)$ *is equivalent to* $((\exists x \in R)\, \Phi(x)) \vee ((\exists x \in S)\, \Phi(x))$*. Then apply the elementary inclusion* $(A \cup B) \cap (C \cup D) \supseteq (A \cap C) \cup (B \cap D)$*.*

Visual intuition. *Imagine* $\mathrm{Pred}_{\rightarrow}$ *and* $\mathrm{Pred}_{\leftarrow}$ *as two "halo" operators around a set. The halo of a union is the union of halos; but the overlap of two halos (mutuality) can be larger for the union than the union of overlaps, because points may be forward-connected to one seed and backward-connected to the other.*

Helper Theorem 46.6 (Membership criterion inside a reality-system)**.** *Let* R *be a reality-system at threshold* θ*, i.e.* $R \neq \emptyset$ *and* $F(R) = R$*. Then for any* $y \in U$*,*

$$y \in R \iff \big(\exists x_{\text{in}} \in R:\ x_{\text{in}} \rightarrow_\theta y\big) \ \wedge\ \big(\exists x_{\text{out}} \in R:\ y \rightarrow_\theta x_{\text{out}}\big).$$

In particular, if there exist $x_1, x_2 \in R$ *with* $x_1 \rightarrow_\theta y$ *and* $y \rightarrow_\theta x_2$*, then* $y \in R$*.*

Remark 2611. What the theorem says (plain English). *Inside a reality-system* R*, membership of an entity* y *is exactly the two-sided condition:* y *must be* predictable from *something already in* R *and must* predict *something already in* R*. So* R *behaves like a mutually-supporting predictive enclave.*

Why it is important. *This gives a usable certificate for membership without recomputing an entire fixed point: to show* $y \in R$*, it suffices to exhibit two witnesses in* R *(one pointing into* y*, one pointed to by* y*). This is precisely the kind of local criterion automated reasoning benefits from.*

Conceptual connection. *The statement explicates the Reality-System idea (Hyperseed-Concept 149) as a fixed point of a closure operator defined by mutual predictability, aligning with the broader fixed-point methodology used elsewhere (cf. the fixed-point schema in Section 48 and related stability arguments in [10]).*

Sketch. Unwind $F(R) = \mathrm{Pred}_{\rightarrow}(R) \cap \mathrm{Pred}_{\leftarrow}(R)$ and the definitions of $\mathrm{Pred}_{\rightarrow}, \mathrm{Pred}_{\leftarrow}$. □

Remark 2612. Proof sketch. *Replace the fixed point equation* $R = \mathrm{Pred}_{\rightarrow}(R) \cap \mathrm{Pred}_{\leftarrow}(R)$ *by the equivalent statement "*$y \in R$ *iff* y *is in both expansions," and then expand each membership condition into the existence of a witness.* Why the key step works. *The operator* $\mathrm{Pred}_{\rightarrow}(R)$ *is defined by an existential condition "*$\exists x \in R$ *with* $x \rightarrow_\theta y$*," and similarly* $\mathrm{Pred}_{\leftarrow}(R)$ *by "*$\exists x \in R$ *with* $y \rightarrow_\theta x$*." Intersections correspond to conjunction, so we get exactly the two-witness criterion. In particular, when* $y \in \mathrm{Pred}_{\rightarrow}(R) \cap \mathrm{Pred}_{\leftarrow}(R)$ *we may (and generally must) choose* potentially different *witnesses* $x_F, x_B \in R$ *to certify the two existential statements; the intersection does not force a single common witness unless one separately assumes a stronger property of* $\rightarrow_\theta$*. Conversely, if we are given witnesses* $x_F, x_B \in R$ *with* $x_F \rightarrow_\theta y$ *and* $y \rightarrow_\theta x_B$*, then the two existential conditions* *hold independently, which is exactly what is needed for membership in the intersection. This is the precise logical reason that the construction behaves like a "mutual" closure: it is a conjunction of two reachability requirements, each expressed as an existential predicate over the same seed set R.*

Network intuition. *Picture a directed graph with edges $x \to_\theta y$. Then* $\mathrm{Pred}_\to(R)$ *is the set of out-neighbors of R, and* $\mathrm{Pred}_\leftarrow(R)$ *is the set of in-neighbors of R. A reality-system is a set that equals the intersection of its in- and out-neighborhoods; membership of y means y has at least one incoming edge from R and at least one outgoing edge to R. Equivalently, y is simultaneously "supported by" R (some $x \in R$ points to y) and "supporting" R (y points back to some element of R), so the set is closed under a bidirectional consistency check. This bidirectionality is what prevents purely feed-forward expansion (which could accumulate nodes that are only predicted* by *R) and also prevents purely feed-backward contraction (which could keep only nodes that predict* into *R): the intersection enforces both constraints at once.*

Proposition 46.7 (Reflexive closure makes F inflationary (implementation convenience))**.** *Define the reflexive closure of adjacency: $x \to_\theta^{\mathrm{refl}} y$ iff $(x \to_\theta y) \vee (x = y)$, and let F^{refl} be the corresponding mutual predictive operator. Then F^{refl} is inflationary: $R \subseteq F^{\mathrm{refl}}(R)$ for all R. A useful corollary (often relied on implicitly in implementations) is that the iterates of F^{refl} started from a seed set R never "oscillate" by losing elements: once an element is present, it persists in all subsequent iterates.*

Remark 2613. What this buys you. *Many fixed-point iteration schemes are simplest when the operator is inflationary (never deletes existing elements). The reflexive closure forces every node to predict itself, so any set R is automatically included in both its forward and backward expansions, hence included in the mutual closure. Concretely, inflationarity lets one treat iteration as a monotone growth process: one can maintain a current set and only add newly implied tokens, rather than having to recompute from scratch while handling possible removals. This is especially convenient when F is evaluated via local graph operations (neighbor queries), because each step can be implemented as a union with newly discovered vertices.*

Interpretive aside. *Declaring x to "predict itself" is not a claim about the world; it is a bookkeeping choice about the closure operator. One may read it as encoding the minimal self-consistency of reference (Hyperseed-Concept 152) required to make iterative construction behave monotonically. One can also view reflexive closure as the smallest modification of $\to_\theta$ that guarantees inflationarity: it changes nothing about predictions between* distinct *tokens, but it prevents the closure process from discarding the very tokens that supplied the witnesses for the existential conditions. In applications, this is often the difference between a closure that behaves like "propagation" and one that behaves like "filtering": adding self-loops ensures that the seed evidence is never filtered out merely for lacking an explicit self-edge in the raw adjacency relation.*

Sketch. For any $y \in R$, take the witness $x = y$ and use reflexivity $y \to_\theta^{\mathrm{refl}} y$ to show $y \in \mathrm{Pred}_\to^{\mathrm{refl}}(R)$ and $y \in \mathrm{Pred}_\leftarrow^{\mathrm{refl}}(R)$, hence $y \in F^{\mathrm{refl}}(R)$. Spelling out the quantifiers: since $y \in R$ and $y \to_\theta^{\mathrm{refl}} y$, the defining condition $\exists x \in R\ (x \to_\theta^{\mathrm{refl}} y)$ holds (take $x = y$), proving $y \in \mathrm{Pred}_\to^{\mathrm{refl}}(R)$; similarly, $\exists x \in R\ (y \to_\theta^{\mathrm{refl}} x)$ holds (again take $x = y$), proving $y \in \mathrm{Pred}_\leftarrow^{\mathrm{refl}}(R)$. Membership in the mutual operator then follows from the definition $F^{\mathrm{refl}}(R) = \mathrm{Pred}_\to^{\mathrm{refl}}(R) \cap \mathrm{Pred}_\leftarrow^{\mathrm{refl}}(R)$. □

Remark 2614. Proof sketch. *Prove inflationarity pointwise: fix $y \in R$ and show y survives one application of the operator by using y itself as the witness on both the forward and backward sides. The "pointwise" style is often clearer than set-inclusion algebra here because the defining clauses for* $\mathrm{Pred}_\to$ *and* $\mathrm{Pred}_\leftarrow$ *are existential: you literally exhibit the element that witnesses the existential. Notice that no property of $\to_\theta$ beyond reflexivity is needed; in particular, transitivity, symmetry, or any probabilistic interpretation of θ play no role in this argument.*

Graph intuition. *Adding self-loops to a directed graph ensures every vertex is in its own in- and out-neighborhood, hence any vertex already in R cannot be lost when taking neighborhoods and intersecting. Another way to say this: without self-loops, the neighborhood-of-R operation can behave like a "one-step boundary" operator that forgets interior points; adding self-loops converts it into a genuine closure-like operator that retains the interior while still expanding to include adjacent boundary points.*

Helper Theorem 46.8 (Finite-step computation of the least fixed point (under inflationarity))**.** *Assume F is inflationary (e.g. by using the reflexive closure above). If U is finite and $R_0 \subseteq U$, define $R_{n+1} := F(R_n)$. Then*

$$R_0 \subseteq R_1 \subseteq R_2 \subseteq \cdots$$

stabilizes in at most $|U|$ steps at a fixed point R_∞, and R_∞ is the least fixed point containing R_0. In other words, the iterative procedure computes the closure of R_0 under F by repeated application, and on a finite universe this closure is reached after finitely many iterations with an explicit worst-case bound.

Remark 2615. What the theorem says. *On a finite universe of tokens U, repeatedly closing a seed set R_0 under an inflationary operator cannot strictly grow forever; it must stabilize after at most $|U|$ strict inclusions. The stabilized set is a fixed point, and it is the smallest fixed point that still contains the initial seed. Equivalently, there is some $N \leq |U|$ such that $R_N = R_{N+1}$, and then $R_n = R_N$ for all $n \geq N$ by repeated application of F. It is worth emphasizing that the bound is a* worst-case *guarantee: many concrete graphs stabilize much earlier, for example when R_0 is already a fixed point, or when the closure process quickly saturates a small connected component of U.*

Why it is important. *This is the computational heart of reality-system construction in finite settings: one can build the "minimal mutually predictive world containing these seed observations" by iterating closure, rather than invoking a nonconstructive fixed-point theorem. It also connects directly to the general fixed-point schema used in Section 48 and to stability-of-goals applications of fixed points ([10]). Operationally, the theorem justifies the common practice of running the update rule until convergence and then treating the resulting set as the constructed reality-system; the least-fixed-point property ensures that no smaller fixed point could have contained the same initial evidential seed.*

Sketch. Inflationarity gives an ascending chain in a finite powerset lattice, hence eventual stabilization. At stabilization, $R_{n+1} = R_n$ is a fixed point. Minimality follows from standard Kleene/Knaster-Tarski reasoning: any fixed point containing R_0 must contain all iterates, hence contains the limit. More explicitly, inflationarity gives $R_n \subseteq R_{n+1}$ for all n by induction on n. Since there are only finitely many subsets of U, and since strict inclusion can occur at most $|U|$ times along a chain that starts at R_0 and only adds elements of U, there must be some stage N where $R_N = R_{N+1}$. For minimality, let S be any fixed point with $R_0 \subseteq S$; then $R_1 = F(R_0) \subseteq F(S) = S$ (using that applying F to a superset cannot invalidate membership in S once S is a fixed point), and iterating this argument yields $R_n \subseteq S$ for all n, so the stabilized value R_∞ is also a subset of S. ☐

Remark 2616. Proof sketch. *The strategy is: (i) show the iterates form an ascending chain; (ii) invoke finiteness to conclude stabilization; (iii) argue minimality by induction: any fixed point above R_0 contains each iterate. In step (iii), the induction uses the simple observation that if $R_n \subseteq S$ and S is a fixed point, then $R_{n+1} = F(R_n) \subseteq F(S) = S$; this is the standard "post-fixed point contains the Kleene chain" argument specialized to the inflationary finite case.*

Key step commentary. *The "at most $|U|$ steps" bound comes from the fact that each strict inclusion adds at least one new element of U, and there are only $|U|$ available. A slightly sharper statement is that stabilization occurs in at most $|U| - |R_0|$ strict growth steps, since no step can add elements outside of U and R_0 already accounts for $|R_0|$ elements at the start; the displayed $|U|$ bound is a clean uniform upper bound that does not depend on the particular seed.*

Intuition. *Think of pouring ink into a finite sponge: inflationarity says the ink never retracts, and finiteness says eventually every pore that can be reached under the closure rule has been filled. In the graph picture, each iteration can be seen as expanding the set by adding vertices that are mutually supported by the current set; once no new vertices satisfy the support condition relative to the current set, the process has reached self-consistency and further updates do nothing.*

46.3 Physical reality and body-channel mediation

Lemma 46.9 (Bounds and certificates for graded physicalness)**.** *Assume the distance $d(\cdot,\cdot)$ used in the physicalness score satisfies $0 \le d \le 1$. Then the graded physicalness*

$$\mathrm{phys}(M, R; B) := 1 - \sup_{f:M\to R} \inf_{g:M\to B,\, h:B\to R} d(f, h \circ g)$$

satisfies $0 \le \mathrm{phys}(M, R; B) \le 1$. Moreover, if for some $\varepsilon \in [0, 1]$ every $f : M \to R$ admits g, h with $d(f, h \circ g) \le \varepsilon$, then $\mathrm{phys}(M, R; B) \ge 1 - \varepsilon$.

Remark 2617. Equivalent re-parameterization. *It is often convenient to name the "worst-case best mediation error"*

$$E(M, R; B) \;:=\; \sup_{f:M\to R} \; \inf_{g:M\to B,\, h:B\to R} d(f, h \circ g),$$

so that $\mathrm{phys}(M, R; B) = 1 - E(M, R; B)$. In this notation, the lemma simply states that $E(M, R; B) \in [0, 1]$ and that a uniform approximation guarantee $E(M, R; B) \le \varepsilon$ immediately yields $\mathrm{phys}(M, R; B) \ge 1 - \varepsilon$. This separation can be useful when one wants to reason about error bounds first and then convert them to a graded score.

Remark 2618. Intuitive reading. *The quantity $\mathrm{phys}(M, R; B)$ measures how well the "mind-to-reality" mappings $f : M \to R$ can be* mediated *by a body-channel B. If every mapping from M to R can be well-approximated (by the distance d) by a composition $h \circ g$ that passes through B, then the body-channel explains the mind–reality coupling, and physicalness is high. If some mappings resist such factorization, physicalness drops. This relates directly to Physical Reality / Body (Hyperseed-Concept 135) and Physical Realizations (Hyperseed-Concept 136).*

What varies in applications. *The role of d is to formalize what counts as "close" between two maps $M \to R$. For instance, if R has a metric ρ, one may take a uniform distance $d(f_1, f_2) := \sup_{m\in M} \min(1, \rho(f_1(m), f_2(m)))$, or (when M is equipped with a probability measure) an average distortion $d(f_1, f_2) := \mathbb{E}_m[\min(1, \rho(f_1(m), f_2(m)))]$. The definition then reads as a worst-case (over f) minimal achievable distortion (over mediations $h \circ g$), turned into a "higher-is-better" score by subtracting from* 1.

Simple example. *If d is the discrete distance (0 if equal, 1 otherwise), then $\inf_{g,h} d(f, h \circ g) = 0$ means f* exactly *factors through B. In that case, $\mathrm{phys}(M, R; B) = 1$ iff every f factors through B. If some f fails to factor, the supremum becomes 1 and* phys *becomes* 0.

Exact factorization in the discrete case. *When d is discrete, the condition "$f = h \circ g$ for some g, h" can be unpacked: g partitions M into fibers $g^{-1}(b)$, and f must be constant on each fiber (so that $h(b)$ can be defined consistently as the common f-value on that fiber). Thus, exact mediation* *through B means that the body-channel B carries enough labels to represent a partition of M that respects the distinctions made by f.*

Remark 2619. Why it matters. *The bounds are sanity checks: the definition is a nested* $\sup\inf$ *of a bounded distance, so without such a lemma one would repeatedly re-verify that the resulting score is a legitimate* $[0, 1]$ *grade. The second clause provides a usable certificate: an explicit approximation guarantee immediately yields a lower bound on physicalness.*

Upper-bound certificates (dual viewpoint). *While the lemma phrases a convenient* lower *bound via a uniform ε-approximation, the same structure suggests how to get* upper *bounds in practice: if one can exhibit a specific $f : M \to R$ such that* every *factorization through B satisfies $d(f, h \circ g) \geq \delta$, then*

$$\sup_f \inf_{g,h} d(f, h \circ g) \;\geq\; \delta \quad \textit{and hence} \quad \mathrm{phys}(M, R; B) \;\leq\; 1 - \delta.$$

In other words, a single "hard-to-mediate" f witnesses low physicalness, just as a uniform construction of (g, h) witnesses high physicalness.

Sketch. Since $0 \leq d \leq 1$, we have $0 \leq \inf d \leq 1$ and thus $0 \leq \sup\inf d \leq 1$. The second claim is immediate from $\sup_f \inf_{g,h} d(f, h \circ g) \leq \varepsilon$.

More explicitly, for each fixed $f : M \to R$ define

$$A(f) \;:=\; \inf_{g:M\to B,\, h:B\to R} d\big(f, h \circ g\big).$$

The assumption $0 \leq d \leq 1$ implies $0 \leq A(f) \leq 1$ for every f (an infimum of numbers in $[0, 1]$ remains in $[0, 1]$). Taking a supremum over all f preserves the bounds, so $0 \leq \sup_f A(f) \leq 1$. Subtracting from 1 yields $0 \leq 1 - \sup_f A(f) \leq 1$, i.e. $0 \leq \mathrm{phys}(M, R; B) \leq 1$. For the certificate statement, if for all f there exist g, h with $d(f, h \circ g) \leq \varepsilon$, then $A(f) \leq \varepsilon$ for all f, hence $\sup_f A(f) \leq \varepsilon$, and thus $\mathrm{phys}(M, R; B) \geq 1 - \varepsilon$. □

Remark 2620. Proof sketch. *The argument is order-theoretic: infima and suprema preserve bounds. Then the definition* $\mathrm{phys} = 1 - (\sup\inf d)$ *converts a small worst-case mediation error into a large physicalness score.*

Geometric intuition. *Think of the set of all maps $M \to R$ as a space, and the maps that factor through B as a subset. The inner* inf *measures distance to that subset; the outer* sup *takes the worst-case distance. Physicalness is "one minus" that worst-case deviation.*

Communication-channel intuition. *Interpreting $g : M \to B$ as an* encoding *of mental states into body states and $h : B \to R$ as a* decoding *into reality states, the inner optimization asks for the best encoder–decoder pair for a given target mapping f. The outer supremum then asks how bad the hardest target mapping can be under the best available mediation through B under the chosen distortion measure d.*

46.4 Algebraic asymmetry and free lifts (adjunction rules)

Helper Theorem 46.10 (Adjunction factoring rule (a useful reasoning principle))**.** *Let $U : \mathbf{R}_{phys} \to \mathbf{R}_{ment}$ be a forgetful functor and suppose it has a left adjoint $F : \mathbf{R}_{ment} \to \mathbf{R}_{phys}$ with unit $\eta_m : m \to U(F(m))$. Then for any $m \in \mathrm{Ob}(\mathbf{R}_{ment})$ and any $p \in \mathrm{Ob}(\mathbf{R}_{phys})$, every mental-to-physical comparison map $\alpha : m \to U(p)$ factors uniquely through the unit: there exists a unique $\bar{\alpha} : F(m) \to p$ with*

$$U(\bar{\alpha}) \circ \eta_m = \alpha.$$

Operationally: any physical realization consistent with the mental description must at least pass through the free lift $F(m)$ (plus extra constraints in p).

Remark 2621. *A useful way to picture the statement is as the existence of a unique "completion" of the triangle*

$$\begin{array}{ccc} & m & \\ & \eta_m \searrow \quad \swarrow \alpha & \\ U(F(m)) & \xrightarrow{U(\bar{\alpha})} & U(p) \end{array}$$

where the diagonal arrow α is given, and the base arrow $U(\bar{\alpha})$ is forced by universality. In particular, the content is not merely that α can *be written as $U(\bar{\alpha}) \circ \eta_m$, but that the $\bar{\alpha}$ witnessing this is determined uniquely (hence can be treated as canonical data).*

Remark 2622. What the theorem says. *An adjunction $F \dashv U$ asserts that "freely realizing" a mental object m as a physical object $F(m)$ is universal: any way of comparing m to the underlying-mental aspect of a physical object p must pass uniquely through this free realization. In short, $F(m)$ is the most general physical realization compatible with m.*

Equivalently (and sometimes more technically useful), $F(m)$ is an initial object in the comma category $(m \downarrow U)$: objects are comparison maps $m \to U(p)$, and a morphism from $\alpha : m \to U(p)$ to $\alpha' : m \to U(p')$ is a physical arrow $h : p \to p'$ with $U(h) \circ \alpha = \alpha'$. The initiality of $\eta_m : m \to U(F(m))$ is exactly the statement that every α receives a unique such morphism from η_m, i.e. a unique $\bar{\alpha} : F(m) \to p$ with $U(\bar{\alpha}) \circ \eta_m = \alpha$.

Why it is important (algebraic asymmetry). *The existence of a left adjoint expresses a one-way generosity: from mental to physical there is a canonical* free lift*, but from physical to mental we often only have forgetting/projection. This articulates the idea of Algebraically Asymmetric Physical Reality (Hyperseed-Concept 55) in categorical terms, and it matches the Hyperseed style of treating ontology-building steps as universal constructions ([1]).*

A further way to read the "one-way generosity" is: U preserves constraints by forgetting them, while F creates a maximally unconstrained physical carrier compatible with the given mental specification. Any additional structure or constraints characteristic of a specific p are exactly what the morphism $\bar{\alpha} : F(m) \to p$ supplies; the universal property isolates what is forced by m alone (encoded in η_m) from what is extra (choice of p and of $\bar{\alpha}$).

Concrete example. *If $\mathbf{R}_{phys}$ were a category of "structured" physical systems and $\mathbf{R}_{ment}$ a category of "unstructured" specifications, U forgets structure and F freely adds it. Then the theorem says: any implementation of a specification factors uniquely through the free implementation.*

A standard mathematical instance is $U : \mathbf{Grp} \to \mathbf{Set}$ with F the free-group functor: a set-map $\alpha : X \to U(G)$ into the underlying set of a group G uniquely extends to a group homomorphism $\bar{\alpha} : F(X) \to G$. The unit $\eta_X : X \to U(F(X))$ is the inclusion of generators, and the factorization condition $U(\bar{\alpha}) \circ \eta_X = \alpha$ says precisely that the extension agrees with α on generators.

Proof. **Sketch** This is the defining universal property of an adjunction $F \dashv U$: $\mathrm{Hom}_{phys}(F(m), p) \cong \mathrm{Hom}_{ment}(m, U(p))$, natural in m, p. □

Remark 2623. Proof sketch. *Invoke the adjunction bijection of hom-sets and interpret η_m as the image of $\mathrm{id}_{F(m)}$ under that bijection. The required factorization is precisely what the bijection encodes.*

More explicitly: let $\Phi_{m,p} : \mathrm{Hom}_{phys}(F(m), p) \to \mathrm{Hom}_{ment}(m, U(p))$ denote the adjunction bijection. By definition of the unit, $\eta_m = \Phi_{m,F(m)}(\mathrm{id}_{F(m)})$. Given $\alpha : m \to U(p)$, let $\bar{\alpha} := \Phi^{-1}_{m,p}(\alpha)$. Then $\Phi_{m,p}(\bar{\alpha}) = \alpha$, which is exactly the displayed equation $U(\bar{\alpha}) \circ \eta_m = \alpha$ when Φ is written in the usual "unit-based" form.

Why the factorization is unique. *Uniqueness is not an extra condition; it is the content of "$\cong$": each α corresponds to exactly one $\bar{\alpha}$. In a reasoning engine, this uniqueness is what licenses canonical rewrites: one can replace an arbitrary α by $U(\bar{\alpha}) \circ \eta_m$ without ambiguity.*

It can also be related to the triangle identities for $(F \dashv U)$: the unit η and counit $\varepsilon : F(U(p)) \to p$ satisfy $(\varepsilon_{F(m)} \circ F(\eta_m)) = \mathrm{id}_{F(m)}$ and $U(\varepsilon_p) \circ \eta_{U(p)} = \mathrm{id}_{U(p)}$. These identities ensure that the "free lift then realize" pipeline behaves coherently, i.e. free lifting does not introduce spurious degrees of freedom once a realization map has been chosen.

46.5 Signals, universes, multiverse, and Yverse

Proposition 46.11 (Universe monotonicity in signal-class and in threshold)**.** *Let* $\mathrm{Univ}_{C,\eta}(S)$ *be the observer-indexed universe defined by* C*-signal reachability at threshold* η*. Then:*

1. *If* $C \subseteq C'$*, then* $\mathrm{Univ}_{C,\eta}(S) \subseteq \mathrm{Univ}_{C',\eta}(S)$.
2. *If* $\eta' \geq \eta$*, then* $\mathrm{Univ}_{C,\eta'}(S) \subseteq \mathrm{Univ}_{C,\eta}(S)$.

Remark 2624. Intuitive reading. *A universe* $\mathrm{Univ}_{C,\eta}(S)$ *is what an observer* S *can reach by passing signals of allowed types* C *that meet an evidential threshold* η*. If we allow* more *signal types, reachability can only expand; if we demand* higher *evidential strength, reachability can only shrink. This matches the Universe/Eurycosm framing (Hyperseed-Concept 89) where a "world" is indexed both by channels of interaction and by standards of evidential admissibility.*

Connection. *This monotonicity is a prerequisite for any lattice- or fixed-point reasoning about universes, since it guarantees that tuning parameters behaves predictably (pun intended).*

Sketch. (1) Any C-signal path is also a C'-signal path. (2) Any path that meets the stricter evidence threshold η' also meets the weaker threshold η. □

Remark 2625. Proof sketch. *Both parts are inclusion-by-relaxation arguments: relax constraints on admissible paths (by enlarging* C *or lowering* η*) and you can only gain reachability, never lose it.*

Path intuition. *Visualize a graph whose edges are labeled by signal types and weights/evidences. Then (1) allows more edge-labels, (2) allows more edges by lowering the weight cutoff.*

Lemma 46.12 (Concatenation of signal paths)**.** *If there is a finite* C*-signal path* $S \rightsquigarrow X$ *and a finite* C*-signal path* $X \rightsquigarrow Y$*, then there is a finite* C*-signal path* $S \rightsquigarrow Y$ *(by concatenation). Consequently, reachability is transitive, and universe membership is closed under reachable extensions.*

Remark 2626. What the lemma says. *Reachability behaves like any sane notion of "can get from here to there": if you can reach* X *from* S *and then reach* Y *from* X*, you can reach* Y *from* S *by performing the two journeys consecutively.*

Why it matters. *Transitivity is what turns reachability into a closure concept, allowing one to talk about universes as closed sets under signal propagation. Without this, universe membership would not behave stably under intermediate constructions.*

Sketch. Concatenate the two finite sequences of C-signals. In a categorical model, use closure of morphisms under composition. □

Remark 2627. Proof sketch. *The strategy is literal concatenation: append the second finite list of signal-steps to the first. In categorical language, the same fact is expressed as: the composite of two morphisms is again a morphism.*

Intuition. *A universe is a region of the graph closed under following edges; this lemma is the statement that "following edges" is compositional.*

Proposition 46.13. ***Guidable multiverse as an MDP , and reduction to ordinary multiverse*** *A guidable multiverse with state space $\mathcal{U}$, action set A, and controlled kernel $P(\cdot \mid U, a)$ is exactly a Markov decision process on $\mathcal{U}$. If A is a singleton, the guidable multiverse reduces to an ordinary multiverse kernel. Conversely, any multiverse kernel $P(\cdot \mid U, e)$ can be regarded as guidable by taking A to encode a designated set of event labels (possibly with a policy restricting actions to match feasible events).*

Remark 2628. Plain-English statement. *"Guidability" does not add new mathematics beyond standard control theory: it is precisely the addition of actions to a state-transition kernel, i.e. an MDP. Removing actions collapses back to an uncontrolled stochastic process.*

Why it is useful. *This identification allows one to import the full toolbox of planning and optimal control when reasoning about multiverse navigation and policy-conditioned branching (cf. the general AGI/control framing in [19]). It also clarifies that "multiverse" here is not (only) metaphysical rhetoric; it is a formal object (Hyperseed-Concept 116) with standard computational semantics.*

Sketch. This is a repackaging of the data: (states, actions, transition kernel) is the definition of an MDP. The singleton-action case collapses to an uncontrolled Markov chain. □

Remark 2629. Proof sketch. *No derivation is required: the proof is definitional equivalence. The only "step" is recognizing that the symbols match the standard tuple defining an MDP.*

Intuition. *A multiverse kernel says "what happens"; adding actions says "what you can choose to do before what happens." Everything else is bookkeeping.*

Helper Theorem 46.14 (Bounding least/greatest Yverse fixed points via pre/post-fixed points)**.** *Let* $\mathrm{Multi} : D \to D$ *be monotone on a complete lattice. If X is pre-fixed (*$\mathrm{Multi}(X) \leq X$*), then* $\mathrm{lfp}(\mathrm{Multi}) \leq X$*. If Y is post-fixed (*$Y \leq \mathrm{Multi}(Y)$*), then* $Y \leq \mathrm{gfp}(\mathrm{Multi})$.

Remark 2630. What the theorem says. *Pre-fixed points are* upper bounds *on the least fixed point, and post-fixed points are* lower bounds *on the greatest fixed point. So to bound the extremal stable regimes of* Multi*, it suffices to exhibit any pre- or post-fixed candidate.*

Where this sits in fixed-point theory. *This is the standard "sandwiching" move behind Knaster–Tarski: on a complete lattice, monotone maps have least and greatest fixed points, and inequalities of the form* $\mathrm{Multi}(X) \leq X$ *or* $Y \leq \mathrm{Multi}(Y)$ *let one compare arbitrary candidates to those extremal points without solving the fixed-point equations explicitly.*

Operational interpretation for Yverse. *When the Yverse is defined as a fixed point of a multiverse operator* Multi *(e.g. an operator that closes a set of states under admissible branching, or under signal-validated extensions), a pre-fixed X is a "conservative envelope" that is already stable under* Multi *in the sense that applying* Multi *cannot push you outside X. Dually, a post-fixed Y is a "seed" that is guaranteed to expand (or at least not contract) under* Multi*, hence it must lie beneath the maximal stable regime.*

Typical instantiation of D. *In many uses, D will be a powerset lattice $(\mathcal{P}(\mathcal{U}), \subseteq)$ of multiverse states, or a lattice of predicates/properties on $\mathcal{U}$. Then* $\mathrm{lfp}(\mathrm{Multi})$ *and* $\mathrm{gfp}(\mathrm{Multi})$ *can be read as the smallest and largest sets (or properties) that are closed under the multiverse evolution/closure rule encoded by* Multi.

Sketch. Let $\mathrm{Fix}(\mathrm{Multi}) = \{Z \in D : \mathrm{Multi}(Z) = Z\}$ be the set of fixed points. By the Knaster–Tarski theorem (or by the standard lattice argument using monotonicity), $\mathrm{lfp}(\mathrm{Multi})$ exists and is the infimum of all pre-fixed points, while $\mathrm{gfp}(\mathrm{Multi})$ exists and is the supremum of all post-fixed points. Hence if X is pre-fixed, then $\mathrm{lfp}(\mathrm{Multi}) \leq X$. If Y is post-fixed, then $Y \leq \mathrm{gfp}(\mathrm{Multi})$. □

Remark 2631. Proof sketch, unpacked. *Monotonicity ensures that the collection of pre-fixed points is downward-directed in the appropriate sense (closed under meets) and the collection of post-fixed points is upward-directed (closed under joins), so taking the infimum/supremum yields extremal fixed points. The inequalities in the theorem are then immediate because infima are below every element of the set they bound, and suprema are above every element of the set they bound.*

Practical use. *In applications, one often cannot compute* lfp(Multi) *or* gfp(Multi) *in closed form. This theorem shows that it is nevertheless meaningful to propose candidate "outer" and "inner" approximations: any invariant upper approximation X (pre-fixed) must contain the least stable core, and any expansive lower approximation Y (post-fixed) must be contained in the greatest stable closure.*

Link to iterative constructions. *If one defines the least fixed point via iteration from a bottom element $\perp$ (e.g.* $\mathrm{Multi}^0(\perp) = \perp$, $\mathrm{Multi}^{n+1}(\perp) = \mathrm{Multi}(\mathrm{Multi}^n(\perp))$*), then any pre-fixed X bounds all iterates: by monotonicity,* $\mathrm{Multi}^n(\perp) \leq X$ *for all n, so the limit (in the appropriate lattice sense) also satisfies* $\mathrm{lfp}(\mathrm{Multi}) \leq X$*. Dually, iterating from a top element $\top$ yields descending approximations to* gfp(Multi) *that are bounded below by any post-fixed Y.*

Remark 2632. Why it matters. *In practice one rarely computes* lfp *or* gfp *exactly; one certifies them by inequalities. This is the basic "bounding move" behind many stability arguments, including meta-goal stability analyses ([10]) and the general fixed-point package recalled in Section 48.*

Sketch. Standard Knaster-Tarski: lfp is the meet of pre-fixed points; gfp is the join of post-fixed points. □

Remark 2633. Proof sketch. *The proof is a direct one-line application of the Knaster–Tarski characterization of extremal fixed points (cf. Theorem 58 for the general package). "Meet of pre-fixed points" immediately implies it is $\leq$ each pre-fixed point, and similarly for* gfp.

Order intuition. *In a lattice,* lfp *is the greatest lower bound of all safe over-approximations (pre-fixed points), while* gfp *is the least upper bound of all safe under-approximations (post-fixed points).*

Corollary 46.15 (Iterative construction under omega-continuity (optional))**.** *If* Multi *is Scott-continuous (or at least omega-continuous), then*

$$\mathrm{lfp}(\mathrm{Multi}) = \bigvee_{n \geq 0} \mathrm{Multi}^n(\perp), \qquad \mathrm{gfp}(\mathrm{Multi}) = \bigwedge_{n \geq 0} \mathrm{Multi}^n(\top).$$

Remark 2634. Plain-English statement. *Under a continuity hypothesis, one can compute the least fixed point by starting from the bottom element $\perp$ and iterating* Multi *upward, taking the limit (join) of the chain; dually, compute the greatest fixed point by iterating downward from $\top$ and taking the limit (meet).*

Why it is useful. *This corollary turns an existence theorem into an algorithmic recipe. In Yverse-style constructions (where* Multi *encodes a multiverse-closure operator), it provides a canonical iterative construction whenever the continuity assumption is justified.*

Sketch. Kleene fixed-point theorem for omega-continuous maps on complete lattices. □

Remark 2635. Proof sketch. *Apply the Kleene theorem: omega-continuity ensures that* Multi *commutes with countable joins of increasing chains (and dually with countable meets of decreasing chains), so the limit of iteration is a fixed point and is extremal.*

Intuition. *Continuity is the promise that no "new behavior" appears only at the limit; everything that is true at the fixed point is already approximable by finite iteration depth.*

Remark 2636. Bounding move, made explicit. *Write* $f := \mathrm{Multi}$*. A point* x *with* $f(x) \le x$ *is a pre-fixed point, so by the Knaster–Tarski characterization one always has* $\mathrm{lfp}(f) \le x$*. Dually, a point* y *with* $y \le f(y)$ *is a post-fixed point, hence* $y \le \mathrm{gfp}(f)$*. These two one-sided certificates are typically what one can prove in applications: one exhibits a candidate region and checks that applying* f *stays inside it (or, dually, expands it), thereby obtaining a provable enclosure of the true extremal fixed points without computing them.*

Remark 2637. Notation and chain picture. *The iterate* f^n *denotes* n*-fold composition with* $f^0 = \mathrm{id}$*, so* $f^0(\perp) = \perp$*,* $f^1(\perp) = f(\perp)$*,* $f^2(\perp) = f(f(\perp))$*, etc. Under monotonicity, the sequence*

$$\perp \;\le\; f(\perp) \;\le\; f^2(\perp) \;\le\; \cdots$$

is an increasing chain, and the join $\bigvee_{n\ge 0} f^n(\perp)$ *is simply its least upper bound. Likewise,*

$$\top \;\ge\; f(\top) \;\ge\; f^2(\top) \;\ge\; \cdots$$

is a decreasing chain whose meet $\bigwedge_{n\ge 0} f^n(\top)$ *is its greatest lower bound. The continuity hypothesis is what lets these chain-limits be* fixed *points rather than merely pre-/post-fixed points.*

Remark 2638. On Scott- vs. omega-continuity. *Scott-continuity means that for every directed set* D *one has* $f(\bigvee D) = \bigvee f[D]$*, i.e.,* f *preserves directed joins. Omega-continuity is the countable-chain special case, requiring preservation of joins of increasing* ω*-chains (and, for the dual statement about* gfp*, the corresponding preservation of meets of decreasing* ω*-chains). In many "constructive" settings one proves only the* ω*-chain version, which is already enough to justify the iterative formulas displayed in the corollary.*

Remark 2639. Concrete model to keep in mind. *A common complete-lattice instance is the powerset lattice* $(\mathcal{P}(S), \subseteq)$ *with* $\perp = \varnothing$ *and* $\top = S$*. If* f *adds all elements reachable by one step of a rule (or all "consistent extensions" of a signal set), then* $f^n(\varnothing)$ *is the set of elements reachable in* $\le n$ *steps, and* $\bigvee_{n\ge 0} f^n(\varnothing) = \bigcup_{n\ge 0} f^n(\varnothing)$ *is the set reachable in finitely many steps. Under the relevant continuity assumption, this finite-step closure coincides with the least fixed point, matching the idea that the "universe" generated by repeatedly applying multiverse-extension rules is the smallest closed object containing the seed.*

46.6 Eurycosm, near eurycosm, anthropic conditioning

Proposition 46.16 (Monotonicity of eurycosm construction)**.** *Let* $\mathrm{Eury}(\mathrm{Mind}, \mathrm{SigClass}) := \bigcup_{M\in\mathrm{Mind}} \bigcup_{C\in\mathrm{SigClass}} \mathrm{U}$
If $\mathrm{Mind} \subseteq \mathrm{Mind}'$ *and* $\mathrm{SigClass} \subseteq \mathrm{SigClass}'$*, then*

$$\mathrm{Eury}(\mathrm{Mind}, \mathrm{SigClass}) \subseteq \mathrm{Eury}(\mathrm{Mind}', \mathrm{SigClass}').$$

Remark 2640. Intuitive reading. *The eurycosm is a "big world" built by taking the union of universes generated by many minds and many signal-classes (Hyperseed-Concepts* **??** *and 89; see also [18]). If we enlarge the set of minds considered, or enlarge the palette of admissible signal-classes, we are taking unions over larger index sets; the result can only grow.*

Why it matters. *Monotonicity is the minimal structural fact needed to treat eurycosm formation as a functorial or lattice-like construction, and to reason about nested eurycosms associated with expanding civilizations, models, or observational modalities.*

Sketch. Both sides are unions; enlarging the index sets can only add elements. $\square$

Remark 2641. Proof sketch. *Use the basic set-theoretic fact: if* $I \subseteq I'$ *then* $\bigcup_{i\in I} A_i \subseteq \bigcup_{i\in I'} A_i$.

Intuition. *The eurycosm is an envelope: adding more contributors (minds) or more channels (signals) cannot shrink an envelope formed by union.*

Remark 2642. Componentwise monotonicity. *The statement decomposes into two independent monotonicities: (i) holding* SigClass *fixed, the map* Mind $\mapsto$ Eury(Mind, SigClass) *is monotone under inclusion of mind-sets; (ii) holding* Mind *fixed, the map* SigClass $\mapsto$ Eury(Mind, SigClass) *is monotone under inclusion of signal-class families. The proposition is simply the simultaneous application of (i) and (ii).*

When equality holds. *The inclusion may be strict or may be an equality depending on redundancy of generators: if every additional* $M \in \mathrm{Mind}' \setminus \mathrm{Mind}$ *and every additional* $C \in \mathrm{SigClass}' \setminus \mathrm{SigClass}$ *contributes only universes already present in the smaller union, then the two eurycosms coincide. In applications this corresponds to enlarging the index sets without adding genuinely new observational/constructive capacity.*

Canonical inclusions. *The monotonicity induces a natural embedding* $\mathrm{Eury}(\mathrm{Mind}, \mathrm{SigClass}) \hookrightarrow \mathrm{Eury}(\mathrm{Mind}', \mathrm{SigClass}')$ *whenever* $(\mathrm{Mind}, \mathrm{SigClass}) \subseteq (\mathrm{Mind}', \mathrm{SigClass}')$. *This is the set-theoretic backbone behind "nested eurycosms": one may regard the smaller eurycosm as a substructure of the larger without additional identification work.*

Lemma 46.17 (Near eurycosm monotonicity in the budget)**.** *Let* $\mathrm{NearEury}_B(M) := \{X \in \mathrm{Eury} : \mathrm{Comp}_M(X) \le B\}$. *If* $B_1 \le B_2$ *then* $\mathrm{NearEury}_{B_1}(M) \subseteq \mathrm{NearEury}_{B_2}(M)$. *Moreover,* $\bigcup_{B\ge 0} \mathrm{NearEury}_B(M) = \{X \in \mathrm{Eury} : \mathrm{Comp}_M(X) < \infty\}$.

Remark 2643. Intuitive reading. *A near-eurycosm is the part of the eurycosm that a mind* M *can access/represent within a resource budget* B. *Increasing* B *relaxes the resource constraint and therefore can only include more elements. The union over all finite budgets recovers exactly the objects of finite* M*-relative complexity.*

Connection to complexity. *The predicate* $\mathrm{Comp}_M(X) \le B$ *is a generic resource bound; in many instantiations it is informed by algorithmic information/description complexity ideas (Hyperseed-Concept* **??***; see [16]). The lemma is thus a monotonicity statement about resource-bounded access to worlds, which is central when contrasting rich-resource strategies with bounded ones ([11]).*

Sketch. Monotonicity is immediate from the inequality $\mathrm{Comp}_M(X) \le B$. The union identity is the definition of finiteness of $\mathrm{Comp}_M(X)$. □

Remark 2644. Proof sketch. *The first part is a direct subset relation between sublevel sets of a function. The second part simply restates that "finite" means "bounded by some finite* B*," i.e. belonging to at least one sublevel set.*

Geometric intuition. *Think of* Comp_M *as a "height function" over* Eury*; then* $\mathrm{NearEury}_B(M)$ *is the region below height* B. *Raising the waterline includes more terrain.*

Remark 2645. Filtration viewpoint. *The family* $\big(\mathrm{NearEury}_B(M)\big)_{B\ge 0}$ *is an increasing chain (a filtration) of subsets of* Eury *indexed by the budget parameter. This makes precise the idea that "better resourced" versions of the same mind (or the same mind at later times) form a nested sequence of accessible world-fragments, and it allows one to speak unambiguously about limits such as* $\lim_{B\to\infty} \mathrm{NearEury}_B(M)$ *as a union rather than as an analytic limit.*

Budget granularity. *No special structure on the index set of budgets is required beyond an order: the lemma holds equally for integer budgets, real-valued budgets, or multi-criteria budgets partially ordered componentwise (e.g. time, memory, and sample complexity), in which case "monotone in the budget" should be read as isotone with respect to that partial order.*

Finite complexity as eventual entry. *The identity* $\bigcup_{B\geq 0} \mathrm{NearEury}_B(M) = \{X : \mathrm{Comp}_M(X) < \infty\}$ *can be read operationally: an object has finite M-complexity iff there exists some finite resource cap at which it becomes accessible. This "eventual entry" interpretation is often the most useful when comparing two minds M, M' with different complexity functionals.*

Helper Theorem 46.18 (Anthropic posterior odds (Bayes factor form))**.** *Let $\pi(\theta)$ be a prior on Θ and A an anthropic event. Then for any θ_1, θ_2 with $\pi(\theta_i) > 0$,*

$$\frac{\pi(\theta_1 \mid A)}{\pi(\theta_2 \mid A)} = \frac{\pi(\theta_1)}{\pi(\theta_2)} \cdot \frac{P(A \mid \theta_1)}{P(A \mid \theta_2)}.$$

Thus anthropic conditioning updates model odds by the likelihood ratio of observer-existence.

Remark 2646. What the theorem says. *Conditioning on an anthropic event A (e.g. "observers like us exist") changes the odds between two hypotheses exactly by multiplying by the Bayes factor $P(A \mid \theta_1)/P(A \mid \theta_2)$.*

Why it matters. *This isolates anthropic reasoning to a single likelihood ratio, making explicit what is often left implicit in philosophical debate: the whole force of anthropic updating lies in the comparative probabilities of observer-existence under competing models (Hyperseed-Concept 57; see also [18] for the broader cosmological/eurycosmic framing).*

Connection. *In eurycosm discussions, A often encodes selection effects about which universes/multiverses are actually observed. This theorem gives the exact algebra for how those selection effects reweight priors.*

Sketch. From $\pi(\theta \mid A) \propto \pi(\theta)P(A \mid \theta)$, divide the two posteriors and cancel the shared normalizer. □

Remark 2647. Proof sketch. *Write Bayes' rule for each θ_i; both posteriors share the same evidence term $P(A)$, so it cancels in the ratio.*

Intuition. *Odds are what remain after common normalizers are stripped away. Anthropic conditioning is therefore best expressed in odds form: it reveals precisely what changes (the likelihood ratio) and what does not (the prior odds structure except for that multiplicative update).*

Remark 2648. Support and degeneracy. *The hypothesis-level assumption $\pi(\theta_i) > 0$ ensures that the prior odds are well-defined. If, in addition, $P(A \mid \theta_2) = 0$ while $P(A \mid \theta_1) > 0$, then the Bayes factor is infinite and $\pi(\theta_2 \mid A) = 0$ (anthropic conditioning rules out θ_2 outright). Conversely, if $P(A \mid \theta_1) = P(A \mid \theta_2)$, then A is evidentially irrelevant for distinguishing θ_1 from θ_2 and posterior odds equal prior odds.*

Log-odds form. *Taking logarithms yields*

$$\log \frac{\pi(\theta_1 \mid A)}{\pi(\theta_2 \mid A)} = \log \frac{\pi(\theta_1)}{\pi(\theta_2)} + \log \frac{P(A \mid \theta_1)}{P(A \mid \theta_2)},$$

so the anthropic update contributes additively in log-odds space. This form is convenient when anthropic information is combined with non-anthropic data, because independent likelihood contributions add on the log scale.

Conditioning on data and anthropics. *If one has ordinary data D in addition to anthropic information A, then (without any special anthropic assumptions) $\pi(\theta \mid D, A) \propto \pi(\theta)\, P(D, A \mid \theta)$. Under conditional independence $D \perp A \mid \theta$ this factorizes into $\pi(\theta)\, P(D \mid \theta)P(A \mid \theta)$, exhibiting anthropic conditioning as just another likelihood factor; the theorem is the two-hypothesis odds specialization of that general factorization.*

46.7 Big-bounce bounce operator (speculative helper bounds)

Lemma 46.19 (Lambda=0 reduction and a crude KL deviation bound)**.** *Consider the bounce operator objective*

$$B(\mu_T) := \arg\min_{\nu}\Big(\mathrm{KL}(\nu\|\mu^*) + \lambda\,\mathrm{Res}(\nu,\mu_T)\Big),$$

where Res *is bounded:* $\mathrm{Res}_{\min} \le \mathrm{Res}(\nu,\mu_T) \le \mathrm{Res}_{\max}$ *for all* ν*. Assume* μ^* *is feasible and* $\mathrm{KL}(\nu\|\mu^*) \ge 0$ *with equality at* $\nu = \mu^*$*. In particular, this implicitly restricts attention to* ν *for which* $\mathrm{KL}(\nu\|\mu^*)$ *is well-defined (e.g.* $\nu \ll \mu^*$ *in standard measure-theoretic settings), since otherwise the KL term is* $+\infty$ *and cannot be minimizing. Also,* arg min *may be set-valued; throughout, "any minimizer" means any element of this argmin set, and existence is taken as an assumption (or, in applications, one may read the inequalities below as applying to any sufficiently good approximate minimizer). Then:*

1. *If* $\lambda = 0$*, any minimizer satisfies* $B(\mu_T) = \mu^*$ *(unique if the KL minimizer is unique). In other words, at* $\lambda = 0$ *the operator becomes* state-independent*: the input* μ_T *plays no role because* Res *is switched off entirely.*

2. *For* $\lambda > 0$*, any minimizer* $\nu^\star = B(\mu_T)$ *satisfies the crude bound*

$$\mathrm{KL}(\nu^\star\|\mu^*) \le \lambda(\mathrm{Res}_{\max} - \mathrm{Res}_{\min}).$$

Equivalently, the deviation penalty contributed by the KL term cannot exceed the maximal possible gain (up to the factor λ*) that could be achieved by moving from the worst-case resistance value to the best-case resistance value.*

Remark 2649. Intuitive reading. *The "bounce operator" trades off closeness to a reference distribution* μ^* *(measured by KL divergence) against a resistance/penalty term* $\mathrm{Res}(\nu,\mu_T)$ *that depends on the current state* μ_T*. When* $\lambda = 0$*, only KL matters, so the minimizer is the reference* μ^**. When* $\lambda > 0$*, the minimizer may deviate from* μ^**, but the lemma provides a coarse upper bound on how far it can deviate in KL units, assuming* Res *is bounded. One can also read the bound as a simple scaling law: increasing* λ *linearly increases the largest KL deviation that could be "worth it," while shrinking the range* $\mathrm{Res}_{\max} - \mathrm{Res}_{\min}$ *linearly tightens the permitted deviation.*

Context and caution. *The "Big Bounce" language is speculative (Hyperseed-Concept 66), but the inequality is purely variational and uses only the order properties of KL and boundedness of* Res*. This sort of bound is typical when one introduces a regularizer: it quantifies how much the regularizer can pull an optimizer away from a baseline. Related KL-regularized variational identities appear throughout control and inference (compare the Gibbs-style KL regularization in Section 47, and the general euryphysical framing in [18]). Because the inequality does not depend on* μ_T *at all (except through the assumed uniform bounds on* Res*), it is a worst-case certificate: it rules out arbitrarily large KL excursions even if the state-dependent term is chosen adversarially, provided only that* Res *itself stays within a fixed numerical range.*

From KL to other distances (optional intuition). *If one wants a more "metric" notion of deviation, standard inequalities can convert the KL bound into bounds on other discrepancies under additional assumptions; for instance, Pinsker's inequality yields* $\|\nu^\star - \mu^*\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\,\mathrm{KL}(\nu^\star\|\mu^*)}$*, so the lemma immediately implies a (still crude) total-variation bound scaling like* $\sqrt{\lambda(\mathrm{Res}_{\max} - \mathrm{Res}_{\min})}$*.*

Sketch. (1) With $\lambda = 0$, the objective is $\mathrm{KL}(\nu\|\mu^*)$ minimized at μ^*. More explicitly, since $\mathrm{KL}(\nu\|\mu^*) \ge 0$ for all feasible ν and $\mathrm{KL}(\mu^*\|\mu^*) = 0$, every minimizer must achieve the value

0, which forces $\nu = \mu^*$ whenever the KL minimizer is unique (and otherwise restricts ν to the KL-minimizing set).

(2) Compare the objective at $\nu^\star$ to the objective at μ^*:

$$\mathrm{KL}(\nu^\star \| \mu^*) + \lambda \mathrm{Res}(\nu^\star, \mu_T) \le \mathrm{KL}(\mu^* \| \mu^*) + \lambda \mathrm{Res}(\mu^*, \mu_T) \le \lambda \mathrm{Res}_{\max}.$$

The first inequality is simply optimality of $\nu^\star$ against the feasible competitor μ^*, while the second uses $\mathrm{KL}(\mu^* \| \mu^*) = 0$ and the uniform upper bound $\mathrm{Res}(\mu^*, \mu_T) \le \mathrm{Res}_{\max}$. Rearrange and use $\mathrm{Res}(\nu^\star, \mu_T) \ge \mathrm{Res}_{\min}$. □

Remark 2650. Proof sketch. *The key trick is "compare the minimizer to a known feasible competitor" (here, μ^*). Optimality says the objective at $\nu^\star$ is no larger than at μ^*. Then boundedness of* Res *lets one isolate* $\mathrm{KL}(\nu^\star \| \mu^*)$ *by squeezing* Res *between* $\mathrm{Res}_{\min}$ *and* $\mathrm{Res}_{\max}$*. This is the same comparison principle that underlies many stability and sensitivity estimates: one often gets a universal bound by pitting the unknown optimizer against a fixed reference point whose objective value can be controlled.*

Why the inequality is crude. *No structure of* Res *is used beyond boundedness, so the result cannot reflect fine geometry of the optimization landscape. Its role is primarily as a robustness certificate: no matter how adversarial μ_T is, the optimizer cannot wander arbitrarily far from μ^* in KL distance when λ and the* Res*-range are bounded. In particular, the lemma does not claim that the bound is tight, nor that deviations of that size actually occur; it only states that such deviations are never* necessary *for optimality given the capped benefit obtainable from* Res.

Variational intuition. *One may picture KL as a "spring" anchoring ν to μ^* and* $\lambda\,\mathrm{Res}$ *as an external force. Bounded force over bounded range yields bounded displacement (in the KL metric). A complementary way to say this is that the optimizer can only "buy" KL deviation by "selling" resistance reduction, and the most it can possibly sell is the finite amount* $\mathrm{Res}_{\max} - \mathrm{Res}_{\min}$.

Limit behavior in λ. *Although not proved here, the lemma is consistent with the heuristic that as $\lambda \to 0^+$ the optimizer should return toward μ^* (at least in the sense that the KL bound forces* $\mathrm{KL}(\nu^\star \| \mu^*) \to 0$ *whenever* $\mathrm{Res}_{\max} - \mathrm{Res}_{\min}$ *is fixed), while larger λ permits larger deviations; the displayed inequality makes this scaling explicit without requiring differentiability or convexity.*

47 Helper theorems for Wu Wei dynamics and optimal control

Remark 2651. *This section collects "ready-to-fire" algebraic facts for the Wu Wei control formalism, in both discrete-time KL-regularized control and its continuous-time Schrödinger bridge analogue. Philosophically, the guiding idea is that effortful control is measured not only by external cost but also by how far one* forces *the world away from its passive dynamics. This connects the intuitive notion of* Wu Wei *(non-forcing, or action that harmonizes with the underlying flow; Hyperseed-Concept 206) to a precise calculus: deviate from a baseline kernel P_0 or baseline policy π_0 only when the gained utility warrants paying an information-theoretic "resistance" penalty (Hyperseed-Concept 158, 100, 87). In particular, one may think of P_0 (or π_0) as encoding "what would happen anyway" in the absence of intervention—for example a nominal model, a default behavior, or a passive diffusion—and the KL term as enforcing a soft constraint that discourages gratuitous changes to that baseline. This perspective makes the tradeoff parameter (often written $\lambda > 0$ below) interpretable as an inverse* control temperature*: small λ permits more aggressive deviations from the baseline in pursuit of reward, while large λ enforces near-passive behavior where only sufficiently high-value changes are worth the informational effort.*

Remark 2652. *Technically, the core recurring move is the Gibbs/exponential tilt: objectives of the form*

$$\text{(expected cost or negative utility)} \;+\; \lambda\,\mathrm{KL}(\text{chosen law}\,\|\,\text{baseline law})$$

(or the sign-flipped maximization analogue) are solved exactly by a normalized exponential reweighting of the baseline. This is the discrete-time counterpart of the Schrödinger bridge viewpoint developed in the Wu Wei geodesics framework [21] (Hyperseed-Concept 207, **??***). Equivalently, the optimizer takes the baseline law and assigns higher probability to trajectories (or actions, or transitions) with lower cost / higher utility, with the KL term ensuring that this reweighting remains "close" in the information-geometric sense. In the continuous-time Schrödinger bridge setting, the same mechanism appears as an entropic regularization of optimal transport: among all controlled dynamics that match endpoint constraints (or marginals), the bridge selects the path measure closest in KL to the uncontrolled diffusion, which yields a controlled drift that can be expressed via logarithmic gradients of certain space–time harmonic (or Schrödinger) potentials. Thus, throughout the helper results, it is useful to keep in mind the unifying template:* linear *expectations plus a* convex *KL penalty lead to closed-form exponential-family solutions and, after taking logarithms, to "soft" Bellman or Hamilton–Jacobi–Bellman-type equations.*

Remark 2653. *To avoid ambiguity in later statements, we adopt the following informal dictionary (made formal in the lemmas and propositions below). In discrete time, a "law" may refer either to (i) an action distribution $\pi(\cdot \mid s)$ relative to a baseline $\pi_0(\cdot \mid s)$ at a fixed state s, or to (ii) a controlled transition kernel $P(\cdot \mid s)$ relative to a passive kernel $P_0(\cdot \mid s)$, or to (iii) an entire trajectory distribution over paths $(x_0, x_1, \ldots, x_T)$. In each case, the KL term is taken with respect to the same reference object (policy-to-policy, kernel-to-kernel, or path-measure-to-path-measure), and the exponential tilt can be applied either pointwise (statewise / timewise) or globally (on path space), yielding equivalent characterizations under standard factorization assumptions. When the baseline is Markovian, the path-space KL decomposes as a sum of one-step KL terms, which is the algebraic reason that KL-regularized dynamic programming admits tractable recursions.*

Remark 2654. *The theorem-like environments below are intentionally lightweight and computational: they are meant to be invoked as algebraic subroutines (e.g., "apply the exponential-tilt lemma" or "use the log-partition identity") rather than as standalone conceptual milestones. Where relevant, we will also highlight the role of normalization constants (log-partition functions), since they encode both the optimal value and the induced "soft" potentials that appear in Bellman and Schrödinger systems.*

47.1 One-step wu wei (Gibbs tilting) as a reusable inference rule

Helper Theorem 47.1 (Gibbs variational identity with exact optimality gap)**.** *Fix a state x, a passive distribution $P_0(\cdot \mid x)$ with full support, a cost $c : X \to \mathbb{R}$, and $\lambda > 0$. Define the one-step regularized objective*

$$J_x(P) \;:=\; \mathbb{E}_{x' \sim P(\cdot|x)}[c(x')] + \lambda\,\mathrm{KL}\big(P(\cdot \mid x) \,\|\, P_0(\cdot \mid x)\big).$$

Let

$$Z(x) \;:=\; \sum_{y \in X} P_0(y \mid x)\exp(-c(y)/\lambda), \qquad P^\star(y \mid x) \;:=\; \frac{P_0(y \mid x)\exp(-c(y)/\lambda)}{Z(x)}.$$

Then:

1. *(Minimum value)* $\min_P J_x(P) = -\lambda \log Z(x)$.

2. *(Exact gap) For every P, $J_x(P) = -\lambda \log Z(x) + \lambda \,\mathrm{KL}(P \,\|\, P^\star)$.*

In particular, $J_x(P) \geq -\lambda \log Z(x)$ with equality iff $P = P^\star$.

Remark 2655. ***Notation and conventions.*** *Here $P(\cdot \mid x)$ ranges over probability distributions on X (the next-state law given current state x). The symbol $\mathbb{E}_{x' \sim P(\cdot|x)}[\cdot]$ denotes expectation with respect to that distribution (Hyperseed-Concept 139). The divergence $\mathrm{KL}(P\|Q)$ is the Kullback–Leibler divergence, measuring how many "nats" of information are spent to deform Q into P; it is always ≥ 0 and vanishes exactly when $P = Q$. The parameter $\lambda > 0$ plays the role of a temperature or softness parameter: small λ makes the optimizer more sharply concentrated on cost minimizers; large λ makes it stay close to P_0.*

Remark 2656. ***Intuitive content.*** *The theorem says: if you minimize "expected cost + (effort to deviate from baseline)," then the optimal choice is not an arbitrary distribution but a* Gibbs tilt *of the passive distribution P_0 by $\exp(-c/\lambda)$. The normalizer $Z(x)$ is a partition function: it measures, in aggregate, how much low-cost mass is available under P_0. The "exact optimality gap" identity is especially useful for engines: it gives a quantitative certificate of suboptimality, namely the KL distance to the optimum.*

Remark 2657. ***Simple example.*** *Let $X = \{0,1\}$, $P_0(0 \mid x) = P_0(1 \mid x) = 1/2$, and costs $c(0) = 0$, $c(1) = 1$. Then*

$$P^\star(1 \mid x) = \frac{e^{-1/\lambda}}{1 + e^{-1/\lambda}}, \qquad P^\star(0 \mid x) = \frac{1}{1 + e^{-1/\lambda}}.$$

As $\lambda \to 0^+$ the policy becomes nearly deterministic on the cheaper outcome 0; as λ grows, it returns toward the unbiased baseline $1/2$.

Proof sketch. Compute $\mathrm{KL}(P\|P^\star)$ using $P^\star \propto P_0 \exp(-c/\lambda)$:

$$\lambda \mathrm{KL}(P\|P^\star) = \lambda \mathrm{KL}(P\|P_0) + \mathbb{E}_P[c] + \lambda \log Z(x).$$

Rearrange to obtain $J_x(P) = -\lambda \log Z(x) + \lambda \mathrm{KL}(P\|P^\star)$. Nonnegativity of KL gives the inequality and uniqueness. $\square$

Remark 2658. ***Commentary on the key steps.*** *The entire argument is an "algebraic completion of the square," except the square is replaced by KL divergence. One rewrites the objective in terms of KL to the candidate optimizer $P^\star$. The partition function $Z(x)$ appears because probabilities must normalize: $P^\star$ is defined only up to proportionality until $Z(x)$ is introduced.*

Remark 2659. ***Geometric intuition.*** *The set of distributions is a curved space; KL divergence behaves like a directed distance on it. This theorem shows the objective J_x is (up to an additive constant) exactly λ times the KL distance to a distinguished point $P^\star$. Thus minimizing J_x is literally moving as little as possible (in KL geometry) while still biasing toward low-cost outcomes. This is a formal rendering of "least forcing" (Hyperseed-Concept 206, 100).*

Corollary 47.2 (Easy bounds on the optimal cost)**.** *With the same setup,*

$$\min_{y \in X} c(y) \;\leq\; -\lambda \log Z(x) \;\leq\; \mathbb{E}_{y \sim P_0(\cdot|x)}[c(y)].$$

Remark 2660. ***Intuitive content and importance.*** *The optimal regularized value $-\lambda \log Z(x)$ is a "soft minimum" of the cost landscape seen through P_0. The corollary brackets this soft minimum between two familiar quantities: the hard minimum cost and the baseline expected cost. It reassures us that regularization does not invent values outside the meaningful range: one cannot do better than the best achievable cost, and one can always fall back to acting passively.*

Proof sketch. Lower bound: $Z(x) \le \exp(-\min c/\lambda)$ so $-\lambda \log Z(x) \ge \min c$. Upper bound: choose $P = P_0$ so $J_x(P_0) = \mathbb{E}_{P_0}[c]$ and $\min_P J_x(P) \le J_x(P_0)$. $\square$

Remark 2661. ***Why the inequalities work.*** *For the lower bound, the exponential weights never exceed the weight of the minimum-cost element, hence the sum $Z(x)$ cannot exceed what you would get if every term had that maximal weight. For the upper bound, we use the minimal principle itself: any candidate policy (in particular, the baseline) upper-bounds the minimum.*

Lemma 47.3 (Soft-min limits (temperature limits))**.** *Assume X is finite. Then:*

$$\lambda \to 0^+ \implies -\lambda \log Z(x) \to \min_{y \in X} c(y), \qquad \lambda \to \infty \implies P^\star(\cdot \mid x) \to P_0(\cdot \mid x).$$

Remark 2662. ***Intuition.*** *This lemma formalizes the two extreme temperaments of control. At low temperature ($\lambda \to 0^+$), the system becomes ruthlessly cost-minimizing: only the minimizers of c matter, and the value becomes the hard minimum. At high temperature ($\lambda \to \infty$), the exponential tilt flattens, and the optimal behavior becomes indistinguishable from passivity. In Wu Wei language, large λ corresponds to maximal non-forcing, while small λ corresponds to decisive forcing. A useful quantitative mnemonic is the "soft minimum" identity*

$$-\lambda \log \sum_x P_0(x) \exp\Big(-\frac{c(x)}{\lambda}\Big),$$

which interpolates between $\min_x c(x)$ as $\lambda \to 0^+$ and the passive baseline (up to an additive normalization determined by P_0) as $\lambda \to \infty$. In KL-regularized control, λ thus plays the role of a tradeoff parameter: it sets how expensive it is (in relative-entropy units) to deviate from P_0 in order to reduce expected cost.

Proof sketch. As $\lambda \to 0^+$, $Z(x)$ is dominated by minimizers of c (Laplace principle / softmin). As $\lambda \to \infty$, $\exp(-c/\lambda) \to 1$ pointwise so $P^\star \propto P_0$ and normalizes to P_0. More explicitly, writing the one-step optimizer in the Gibbs form (as in the preceding lemma)

$$P^\star(x) \;=\; \frac{P_0(x) \exp(-c(x)/\lambda)}{\sum_y P_0(y) \exp(-c(y)/\lambda)},$$

one sees that as $\lambda \to 0^+$ the denominator is asymptotically controlled by the set $\arg\min c$, so $P^\star$ concentrates its mass there (with relative weights proportional to P_0 restricted to the minimizers). Conversely, as $\lambda \to \infty$, the expansion $\exp(-c/\lambda) = 1 - \frac{c}{\lambda} + o(\lambda^{-1})$ shows the tilt becomes negligible, so the normalized distribution approaches P_0. $\square$

Remark 2663. ***Proof strategy (high-level).*** *The key is to observe which terms dominate a sum of exponentials. As λ shrinks, differences in cost are amplified exponentially, so the lowest-cost terms dominate. As λ grows, cost differences are suppressed, so the weighting becomes nearly uniform and thus returns to the baseline. Equivalently, one can view the normalizer $\sum_x P_0(x) e^{-c(x)/\lambda}$ as a partition function: at low temperature it is governed by ground states (minimum-cost states), while at high temperature it becomes insensitive to energetic differences and primarily reflects the baseline measure.*

47.2 Multi-step wu wei: desirability recursion and path-integral form

Helper Theorem 47.4 (Feynman–Kac / path-integral unrolling of desirability)**.** *Assume the finite-horizon KL-regularized control setup of Defs. 407–410 and Thm. 23, with desirability $z_t(x) =$*

$\exp(-V_t(x)/\lambda)$ *and terminal* $z_T(x) = \exp(-q_T(x)/\lambda)$*. Then* z_t *admits the passive-trajectory expectation form:*

$$z_t(x) = \mathbb{E}\Big[\exp\Big(-\frac{1}{\lambda}\sum_{s=t}^{T-1} q(x_s)\Big)\, z_T(x_T) \;\Big|\; x_t = x,\ x_{s+1} \sim P_0(\cdot \mid x_s)\Big].$$

In particular, in the finite-state case this conditional expectation is an explicit finite sum over all length-$(T-t)$ *paths starting at* x*, with the product of transition probabilities* $\prod_{s=t}^{T-1} P_0(x_{s+1} \mid x_s)$ *and the multiplicative weight* $\exp(-\sum q/\lambda)$ *appearing as factors in each term. This is exactly the discrete-time analogue of a Feynman–Kac representation: a linear evolution (for* z_t*) can be expressed as an average over random trajectories under a reference (passive) process with an exponential running weight.*

Remark 2664. ***Notation and meaning.*** *The sequence* $(x_t, x_{t+1}, \ldots, x_T)$ *is a random trajectory generated by repeatedly sampling from the passive kernel* $P_0(\cdot \mid x_s)$*. The term*

$$\exp\Big(-\frac{1}{\lambda}\sum_{s=t}^{T-1} q(x_s)\Big)$$

is a multiplicative weight (a Boltzmann factor) penalizing trajectories that incur large cumulative running cost q*. The theorem asserts that the desirability* $z_t(x)$ *is exactly the expectation of this weight times the terminal desirability* $z_T(x_T)$ *under passive dynamics. It is often helpful to read this as a "soft survival" quantity: trajectories that accumulate large cost are exponentially downweighted, and the remaining weighted mass is summarized by* $z_t(x)$*. Because* $z_T(x_T) = \exp(-q_T(x_T)/\lambda)$*, the terminal cost enters the same multiplicative structure, so both running and terminal costs contribute additively in the exponent along the entire path.*

Remark 2665. ***Intuitive content and importance.*** *This is the bridge from dynamic programming to "path integrals." Instead of solving a nonlinear Bellman recursion for* V_t*, one solves a* linear *recursion for* z_t *and can interpret* $z_t(x)$ *as a passive average over exponentially reweighted futures. For an automated reasoner, this is a powerful inference rule: it allows one to substitute reasoning about controlled trajectories with reasoning about passive trajectories plus explicit weights, aligning with the Wu Wei perspective [21]. A further conceptual benefit is that linearity unlocks standard tools (e.g. matrix products in finite state spaces, spectral viewpoints in infinite-horizon limits, and unbiased Monte Carlo estimation under* P_0 *with importance weights). In practice, the expectation form suggests simulation-based approximations: sample passive rollouts from* x*, compute the weight* $\exp(-\sum q/\lambda)z_T(x_T)$*, and average, which directly estimates* $z_t(x)$ *without explicitly enumerating actions.*

Proof sketch. Induct backward on t using the linear recursion $z_t(x) = \exp(-q(x)/\lambda)\sum_{x'} P_0(x' \mid x)z_{t+1}(x')$ and tower property of conditional expectation. Concretely, conditioning on the first passive step x_{t+1} gives

$$\mathbb{E}\Big[\exp\Big(-\frac{1}{\lambda}\sum_{s=t}^{T-1} q(x_s)\Big)z_T(x_T) \;\Big|\; x_t = x\Big] = \exp\Big(-\frac{q(x)}{\lambda}\Big)\sum_{x'} P_0(x' \mid x)\,\mathbb{E}\Big[\exp\Big(-\frac{1}{\lambda}\sum_{s=t+1}^{T-1} q(x_s)\Big)z_T(x_T) \;\Big|\; x_{t+1} = x'\Big],$$

and the inductive hypothesis identifies the inner conditional expectation with $z_{t+1}(x')$. ☐

Remark 2666. ***Proof sketch (strategy).*** *One first recognizes that the desirability recursion is linear in* z_{t+1}*. Then, by repeatedly substituting the recursion forward (or inducting backward), one* *obtains an iterated sum over paths. The tower property packages that iterated sum into a single conditional expectation over passive trajectories. Another way to view the same step is as an application of iterated conditioning: "peel off" one time step at a time, factor out the current-stage weight* $\exp(-q(x_t)/\lambda)$*, and propagate the remaining expected future weight through* P_0*. This is exactly the mechanism by which products of random weights along a Markov chain become expectations of exponentials of additive functionals.*

Remark 2667. ***Visual intuition.*** *Imagine all passive trajectories fan out from* x *like a tree. Each path gets a weight that shrinks exponentially with accumulated cost. The desirability is the total weighted mass reaching time* T*. High desirability means "many reasonably low-cost futures exist under the passive flow," which is precisely the sense in which forcing is unnecessary. Equivalently,* $-\lambda \log z_t(x) = V_t(x)$ *can be read as a "soft" cost-to-go: if* $z_t(x)$ *is large, then even without steering the dynamics aggressively, the passive process already tends to drift toward low-cost regions; if* $z_t(x)$ *is small, then most passive futures are costly, so meaningful improvement typically requires a sharper tilt (stronger forcing).*

Proposition 47.5 (Doob h-transform form of the optimal controlled kernel)**.** *Let* z_{t+1} *be as above and define*

$$P_t^\star(x' \mid x) \;:=\; \frac{P_0(x' \mid x)\, z_{t+1}(x')}{\sum_{y\in X} P_0(y \mid x)\, z_{t+1}(y)}.$$

Then $P_t^\star(\cdot \mid x)$ *is a Markov kernel with:*

1. *Normalization:* $\sum_{x'} P_t^\star(x' \mid x) = 1$.
2. *Absolute continuity:* $P_0(x' \mid x) = 0 \Rightarrow P_t^\star(x' \mid x) = 0$.

The denominator is exactly the one-step prediction of desirability under the passive dynamics,

$$\sum_{y\in X} P_0(y \mid x)\, z_{t+1}(y) = \mathbb{E}\big[z_{t+1}(x_{t+1}) \mid x_t = x,\ x_{t+1} \sim P_0(\cdot \mid x)\big],$$

so the formula can be read as a normalized reweighting of passive next-state proposals by their expected future "goodness".

Remark 2668. ***Intuitive content.*** *The optimal controlled kernel* $P_t^\star$ *is obtained by reweighting the passive kernel* P_0 *by the "promise of the future" encoded in* z_{t+1}*. Transitions leading to states with high* z_{t+1} *are amplified. This is the multi-step analogue of one-step Gibbs tilting: the immediate action is chosen so as to align with a future-looking desirability potential. The name "Doob* h*-transform" emphasizes that this reweighting preserves the Markov structure while conditioning the process toward favorable regions. One can also regard this as a Bayesian-looking update:* $P_0(\cdot \mid x)$ *plays the role of a proposal distribution over next states, while* $z_{t+1}(x')$ *is an (unnormalized) "likelihood" score for how desirable it is to land in* x'*. The controlled kernel* $P_t^\star$ *is then the corresponding normalized posterior over* x_{t+1}*.*

Proof sketch. Normalization is by construction. Absolute continuity follows because $P_t^\star$ is P_0 times a nonnegative weight, divided by a positive normalizer (full support ensures positivity). In addition, to see positivity of the denominator, note that if $P_0(\cdot \mid x)$ has full support and z_{t+1} is not identically zero, then there exists at least one y with $z_{t+1}(y) > 0$ and thus $P_0(y \mid x)z_{t+1}(y) > 0$, implying $\sum_y P_0(y \mid x)z_{t+1}(y) > 0$. □

Remark 2669. ***Commentary.*** *The nonnegativity of z_{t+1} is essential: it ensures we do not create negative transition weights. Full support of $P_0(\cdot \mid x)$ (as assumed in the one-step identity) ensures the denominator is strictly positive whenever z_{t+1} is not identically zero, so the formula defines a legitimate probability distribution. From the control perspective, absolute continuity encodes a "non-teleportation" principle: the controller cannot assign positive probability to transitions that are impossible under the passive physics. The only freedom is to* reallocate *probability mass among feasible next states, paying KL cost for deviating from P_0; the h-transform makes this reallocation explicit and computable from z_{t+1}.*

47.3 Resistance, acceptance, and opening: algebra useful for engines

Lemma 47.6 (Acceptance is an exponential similarity (and is lambda-free))**.** *Let* $\mathrm{Resist}_\lambda(\pi; x) = \lambda \mathrm{KL}(P_\pi(\cdot \mid x) \| P_0(\cdot \mid x))$ *and* $\mathrm{Accept}_\lambda(\pi; x) = \exp(-\mathrm{Resist}_\lambda(\pi; x)/\lambda)$ *as in Def. 414. Then*

$$\mathrm{Accept}_\lambda(\pi; x) = \exp\Big(-\mathrm{KL}(P_\pi(\cdot \mid x) \| P_0(\cdot \mid x))\Big) \in (0, 1],$$

so $\mathrm{Accept}_\lambda(\pi; x) = 1$ *iff* $P_\pi(\cdot \mid x) = P_0(\cdot \mid x)$*. In particular,* Accept_λ *does not depend on* λ*.*

Remark 2670. ***Intuition and usefulness.*** *"Resistance" is the λ-scaled KL cost of deviating from the passive kernel; "acceptance" is a similarity score obtained by exponentiating the negative KL. This lemma notes a pleasing cancellation: acceptance depends only on the* shape *of the deviation (the KL), not on the temperature/regularization scale λ. For an engine, this means acceptance can serve as a stable, scale-free diagnostic of non-forcing (Hyperseed-Concept 158, 206).*

Remark 2671. ***Operational reading.*** *The quantity* $\mathrm{Accept}_\lambda(\pi; x)$ *can be interpreted as a multiplicative penalty attached to the entire next-state distribution under π relative to the passive dynamics: a small change in KL produces a proportional change in* $\log \mathrm{Accept}$ *since*

$$\log \mathrm{Accept}_\lambda(\pi; x) \;=\; -\mathrm{KL}(P_\pi(\cdot \mid x) \| P_0(\cdot \mid x)).$$

Thus acceptance is naturally a "log-linear" score: differences in acceptance are easiest to compare on the log scale, where they coincide with differences in KL divergence.

Remark 2672. ***Range and edge cases.*** *The inclusion* $\mathrm{Accept}_\lambda(\pi; x) \in (0, 1]$ *uses that* $\mathrm{KL} \geq 0$*, hence* $\exp(-\mathrm{KL}) \leq 1$*. When* KL *is finite but large, acceptance can be extremely small, which is often desirable in diagnostics: a small acceptance cleanly flags that $P_\pi(\cdot \mid x)$ has become highly concentrated on events that are improbable under $P_0(\cdot \mid x)$. If* $\mathrm{KL}(P_\pi \| P_0) = +\infty$ *(e.g. P_π assigns positive probability to an event of P_0-probability 0), then* $\mathrm{Accept}_\lambda(\pi; x) = 0$ *in the extended-real sense; in many engine designs this corresponds to "hard incompatibility" with the passive flow and is treated as disallowed forcing.*

Remark 2673. ***Simple example.*** *If $P_\pi(\cdot \mid x)$ and $P_0(\cdot \mid x)$ are identical, then* $\mathrm{KL} = 0$ *and* $\mathrm{Accept} = 1$*. If P_π concentrates on outcomes that are unlikely under P_0, KL becomes large and* Accept *decays rapidly toward 0, reflecting that the policy is "fighting the flow."*

Remark 2674. ***Two-outcome sanity check.*** *Consider a binary next-state outcome with $P_0(\cdot \mid x) = (q, 1-q)$ and $P_\pi(\cdot \mid x) = (p, 1-p)$. Then*

$$\mathrm{KL}(P_\pi \| P_0) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q},$$

so acceptance is $\exp(-\mathrm{KL})$*, a smooth function peaked at $p = q$ and decreasing as p deviates from q. This makes explicit that acceptance is not a linear penalty in $p - q$ but a curvature-sensitive penalty that grows sharply as p approaches 0 or 1 while q stays away from that boundary.*

Proof sketch. KL is nonnegative and equals 0 iff the two distributions coincide. The displayed expression cancels λ. $\square$

Remark 2675. ***Key step.*** *The proof is essentially dimensional analysis:* Resist_λ *is defined as* λ *times KL, while acceptance exponentiates* $-\mathrm{Resist}_\lambda/\lambda$*, so the scale* λ *drops out, leaving a pure KL-based similarity.*

Remark 2676. ***Consequences for optimization.*** *Because* $\log \mathrm{Accept}_\lambda(\pi; x)$ *is exactly* $-\mathrm{KL}(P_\pi(\cdot \mid x)\|P_0(\cdot \mid x))$*, maximizing acceptance is equivalent to minimizing the KL deviation from the passive kernel at each state. In constrained or regularized control problems, acceptance can therefore be used as a proxy constraint (e.g. enforce* $\mathrm{Accept}_\lambda(\pi; x) \geq \alpha$*) that is independent of the particular numerical choice of* λ*.*

Lemma 47.7 (Factorization rule: resistance adds, acceptance multiplies)**.** *Suppose the next-state distribution factors over independent components:* $P_0(x_1', x_2' \mid x_1, x_2) = P_0^1(x_1' \mid x_1)P_0^2(x_2' \mid x_2)$ *and similarly* $P_\pi = P_\pi^1 \otimes P_\pi^2$*. Then*

$$\mathrm{KL}(P_\pi\|P_0) = \mathrm{KL}(P_\pi^1\|P_0^1) + \mathrm{KL}(P_\pi^2\|P_0^2),$$

hence Resist_λ *is additive and* Accept *is multiplicative across components.*

Remark 2677. ***Intuition and importance.*** *Independence means "separable worlds" incur separable forcing costs: deviating from the baseline in subsystem* 1 *does not interact (in KL) with deviation in subsystem* 2*. Consequently, acceptance behaves like a product of similarities, mirroring how independent evidences combine multiplicatively. This add/multiply duality is often the algebraic backbone behind modular control architectures.*

Remark 2678. ***Explicit acceptance factorization.*** *Under the stated product assumptions,*

$$\mathrm{Accept}(\pi; x_1, x_2) = \exp\big(-\mathrm{KL}(P_\pi^1\|P_0^1) - \mathrm{KL}(P_\pi^2\|P_0^2)\big) = \mathrm{Accept}(\pi^1; x_1)\,\mathrm{Accept}(\pi^2; x_2),$$

so one can read acceptance as a "probabilistic compatibility" score that composes exactly like independent likelihood factors. This is often more numerically stable than working directly with KL sums when many components are present, since products can be accumulated via logs.

Remark 2679. ***Modularity caveat.*** *The lemma relies on true factorization of both* P_0 *and* P_π*. If the passive dynamics couple components (or the policy induces coupling), KL generally contains cross-terms and the forcing cost is no longer separable. In engine terms, coupling corresponds to "shared channels" through which forcing in one module can increase resistance in another, so the product rule is a diagnostic for when clean modularization is justified.*

Proof sketch. KL divergence is additive for product measures; substitute into definitions of Resist_λ and Accept. $\square$

Remark 2680. ***Geometric intuition.*** *Product structure corresponds to an orthogonality of coordinates in information geometry: one can move (in distribution space) along factor* 1 *and factor* 2 *independently, and the total "distance" (KL) is the sum of the factor distances. Exponentiating turns sums into products, hence acceptance multiplies.*

Remark 2681. ***Many-component extension.*** *The same calculation extends immediately to* n *independent components. If* $P_0 = \bigotimes_{i=1}^n P_0^i$ *and* $P_\pi = \bigotimes_{i=1}^n P_\pi^i$*, then*

$$\mathrm{KL}(P_\pi\|P_0) = \sum_{i=1}^n \mathrm{KL}(P_\pi^i\|P_0^i), \qquad \mathrm{Accept}(\pi; x) = \prod_{i=1}^n \mathrm{Accept}(\pi^i; x_i).$$

This "sum/product" structure is what enables distributed control laws to track a global forcing budget by tracking local forcing budgets and aggregating them additively.

Lemma 47.8 (Entropy bounds and KL-to-uniform identity for opening)**.** *Let* $\mathrm{Open}(\pi; x)$ *be Shannon entropy on a finite action set* A *(Def. 415). Then*

$$0 \leq \mathrm{Open}(\pi; x) \leq \log |A|,$$

and if Unif *is the uniform distribution on* A *then*

$$\mathrm{Open}(\pi; x) = \log |A| - \mathrm{KL}(\pi(\cdot \mid x) \,\|\, \mathrm{Unif}).$$

Remark 2682. ***Notation and meaning.*** $\mathrm{Open}(\pi; x)$ *is an "opening" measure: the entropy of the action distribution at state* x *(Hyperseed-Concept* **??***, 125). The bounds express that entropy is minimized by determinism (value* 0*) and maximized by the uniform distribution (value* $\log |A|$*). The KL-to-uniform identity says that being open is exactly being close to uniform, quantitatively.*

Remark 2683. ***Concrete extremizers.*** *The lower bound* 0 *is attained when* $\pi(\cdot \mid x)$ *is a point mass (choose one action with probability* 1*), while the upper bound* $\log |A|$ *is attained exactly at* $\pi(\cdot \mid x) = \mathrm{Unif}$*. The identity with* $\mathrm{KL}(\pi \| \mathrm{Unif})$ *makes these extremizers immediate: KL is minimized at* 0 *only when the two distributions coincide, so maximal opening occurs only under uniform randomized choice.*

Remark 2684. ***Why this is useful for engines.*** *In a Wu Wei-oriented controller, opening can be used as a knob to avoid brittle overcommitment: it measures how much the policy keeps possibilities alive. The identity expresses opening as a constant minus a KL term, so many monotonicity facts about KL immediately translate into monotonicity facts about opening.*

Remark 2685. ***Interpretation as a deficit-from-max formula.*** *The equality*

$$\log |A| - \mathrm{Open}(\pi; x) = \mathrm{KL}(\pi(\cdot \mid x) \,\|\, \mathrm{Unif})$$

can be read as: the gap *between maximal opening and current opening is exactly the information divergence from uniformity. Thus, opening is not merely a heuristic exploration score; it is a calibrated measure whose shortfall from* $\log |A|$ *has a precise information-theoretic meaning.*

Proof sketch. Standard entropy bounds. For the identity: $\mathrm{KL}(\pi \| \mathrm{Unif}) = \sum_a \pi(a) \log(\pi(a)/(1/|A|)) = \sum_a \pi(a) \log \pi(a) + \log |A|$. Rearrange. □

Remark 2686. ***Key step.*** *The proof uses the definition of KL divergence and the fact that* $\mathrm{Unif}(a) = 1/|A|$*. The term* $\log |A|$ *appears as the logarithm of the reciprocal of uniform probability, and the remaining term is (minus) the Shannon entropy.*

Remark 2687. ***Common corollaries.*** *Since* $\mathrm{KL}(\pi \| \mathrm{Unif}) \geq 0$*, the identity immediately yields* $\mathrm{Open}(\pi; x) \leq \log |A|$ *without separate bounding arguments. Moreover, if a sequence of policies* $\pi_k(\cdot \mid x)$ *converges to uniform in KL (i.e.* $\mathrm{KL}(\pi_k \| \mathrm{Unif}) \to 0$*), then* $\mathrm{Open}(\pi_k; x) \to \log |A|$*. Conversely, any systematic reduction in opening corresponds to an increase in KL-to-uniform, i.e. a quantitative move toward concentration.*

47.4 Resonance reward: algebraic expansions and bounds

Lemma 47.9 (Amplitude magnitude bounds)**.** *With $z_k(a \mid x) = (2E_k^+(a \mid x)-1)+i(2E_k^-(a \mid x)-1)$ as in Def. 416, we have $|z_k(a \mid x)| \le \sqrt{2}$ for all k, a, x. Consequently, with weights $w_k \ge 0$ and $z_{\text{tot}} = \sum_k w_k z_k$,*

$$0 \le r_{\text{res}}(a \mid x) = |z_{\text{tot}}(a \mid x)|^2 \le 2\Big(\sum_k w_k\Big)^2.$$

Remark 2688. ***Intuition and context.*** *The complex quantity z_k encodes, in its real and imaginary parts, two signed "evidence" coordinates derived from $(E_k^+, E_k^-) \in [0,1]^2$. Each coordinate is rescaled to lie in $[-1,1]$, so z_k lies in the square $[-1,1] \times [-1,1]$ in the complex plane. The lemma simply states the Euclidean radius of that square is $\sqrt{2}$, hence the magnitude bound. The resulting reward $r_{\text{res}} = |z_{\text{tot}}|^2$ is a nonnegative scalar measuring the strength of coherent summed resonance (Hyperseed-Concept 159, 160).*

Remark 2689. ***Tightness and normalization.*** *The bound $|z_k| \le \sqrt{2}$ is tight: equality holds precisely when both coordinates hit an extreme, i.e. when $2E_k^+ - 1 \in \{\pm 1\}$ and $2E_k^- - 1 \in \{\pm 1\}$ (equivalently $E_k^\pm \in \{0,1\}$). In that case z_k sits on a corner of the square and has maximal magnitude. In contrast, if $(E_k^+, E_k^-) = (\frac{1}{2}, \frac{1}{2})$ then $z_k = 0$ and the channel contributes no resonance. If one desires a per-channel amplitude constrained by 1 rather than $\sqrt{2}$, a conventional alternative is to consider the normalized amplitude $\tilde{z}_k := z_k/\sqrt{2}$, which would rescale r_{res} by a factor of $1/2$; the present convention keeps the mapping $(E_k^+, E_k^-) \mapsto z_k$ affine with unit coordinate ranges.*

Remark 2690. ***Why this matters.*** *Upper bounds are essential for stability arguments and for safely combining resonance reward with other bounded utilities. The inequality*

$$r_{\text{res}}(a \mid x) \le 2\Big(\sum_k w_k\Big)^2$$

gives an explicit global cap determined only by the weights, independent of the particular evidences. This kind of "worst-case amplitude control" is also consonant with resonance-based constructions in paraconsistent settings [24].

Remark 2691. ***Immediate corollaries for weight choices.*** *If the weights are normalized so that $\sum_k w_k = 1$ (a common choice when w_k represent a convex mixture of channels), then the lemma yields the simple global bound $r_{\text{res}}(a \mid x) \le 2$. More generally, if only an ℓ_1-budget is imposed, $\sum_k w_k \le W$, then $r_{\text{res}} \le 2W^2$. These restatements are often the form used in later control arguments because they separate (i) the design-time weight budget from (ii) the run-time evidence values.*

Proof sketch. Each real/imag part of z_k lies in $[-1,1]$, so $|z_k|^2 \le 1^2 + 1^2 = 2$. Then $|z_{\text{tot}}| \le \sum_k w_k |z_k| \le \sqrt{2} \sum_k w_k$ by triangle inequality. Square to get the bound. $\square$

Remark 2692. ***Proof strategy.*** *First bound each component amplitude, then bound the weighted sum by the triangle inequality, and finally square. The step from a bound on $|z_{\text{tot}}|$ to a bound on r_{res} is immediate because r_{res} is defined as the squared magnitude.*

Remark 2693. ***Algebraic detail of the per-channel bound.*** *Writing $z_k = (2E_k^+ - 1) + i(2E_k^- - 1)$ gives*

$$|z_k|^2 = (2E_k^+ - 1)^2 + (2E_k^- - 1)^2.$$

Since each factor satisfies $(2E-1)^2 \le 1$ for $E \in [0,1]$, one has $|z_k|^2 \le 2$ and hence $|z_k| \le \sqrt{2}$. This makes explicit that no probabilistic or independence assumptions are required: the estimate is purely geometric and follows only from the range constraint $E_k^\pm \in [0,1]$.

Proposition 47.10 (Coherence decomposition of resonance reward)**.** *Let $\bar{z}$ denote complex conjugate. Then*

$$r_{\text{res}}(a \mid x) = \Big|\sum_k w_k z_k(a \mid x)\Big|^2 = \sum_k w_k^2 |z_k(a \mid x)|^2 + 2\sum_{i<j} w_i w_j \Re\big(z_i(a \mid x)\, \bar{z}_j(a \mid x)\big).$$

Thus the cross-terms $\Re(z_i \bar{z}_j)$ encode agreement (positive) vs cancellation/conflict (negative).

Remark 2694. ***Intuitive content.*** *This is the familiar "sum of squares plus interference" decomposition. The diagonal terms $\sum_k w_k^2 |z_k|^2$ measure how strong each resonance channel is on its own. The off-diagonal terms measure how channels combine: if two channels point in the same direction in the complex plane, the real part $\Re(z_i \bar{z}_j)$ is positive and the total reward grows superadditively; if they point oppositely, the term is negative and they cancel.*

Remark 2695. ***Signed coherence as an angle.*** *For nonzero z_i, z_j, one may write*

$$\Re(z_i \bar{z}_j) = |z_i|\,|z_j|\, \cos(\theta_{ij}),$$

where θ_{ij} is the oriented angle between the two complex vectors. In this form, $\cos(\theta_{ij})$ acts as a signed coherence coefficient: it equals 1 for perfect alignment, -1 for perfect opposition, and 0 for orthogonality. This makes it clear that the reward can increase either by increasing per-channel magnitudes $|z_k|$ or by increasing alignment among channels.

Remark 2696. ***Why it connects.*** *In the subsequent Gibbs policy, resonance enters inside an exponential. There, constructive interference increases utility and hence probability mass; destructive interference suppresses it. This proposition is therefore the algebraic hinge between multi-channel evidence and action selection (Hyperseed-Concept 159, 59).*

Remark 2697. ***Bounding the interference contribution.*** *The cross-term always satisfies the sharp inequality*

$$-\,|z_i|\,|z_j| \;\le\; \Re(z_i \bar{z}_j) \;\le\; |z_i|\,|z_j|,$$

since $\Re(u) \le |u|$ and $|z_i \bar{z}_j| = |z_i|\,|z_j|$. Combined with $|z_k| \le \sqrt{2}$, this yields the uniform bound

$$\Re(z_i \bar{z}_j) \in [-2, 2],$$

and therefore each weighted interference term obeys

$$2w_i w_j\, \Re(z_i \bar{z}_j) \in [-4w_i w_j,\ 4w_i w_j].$$

These estimates clarify how negativity can enter r_{res} only through cancellation at the level of the inner sum z_{tot}; the final quantity $r_{\text{res}} = |z_{\text{tot}}|^2$ remains nonnegative, but destructive interference can substantially reduce it relative to the sum of diagonal contributions.

Proof sketch. Expand $|z_{\text{tot}}|^2 = z_{\text{tot}} \bar{z}_{\text{tot}}$ and collect diagonal and off-diagonal terms. □

Remark 2698. ***Expansion spelled out.*** *Writing $z_{\text{tot}} = \sum_k w_k z_k$ gives*

$$|z_{\text{tot}}|^2 = \Big(\sum_i w_i z_i\Big)\Big(\sum_j w_j \bar{z}_j\Big) = \sum_i w_i^2 z_i \bar{z}_i + \sum_{i \ne j} w_i w_j z_i \bar{z}_j.$$

Since $z_i \bar{z}_i = |z_i|^2$ and the $i \ne j$ terms can be paired as $z_i \bar{z}_j + z_j \bar{z}_i = 2\Re(z_i \bar{z}_j)$, the stated decomposition follows by grouping $i < j$.

Remark 2699. ***Geometric intuition.*** *Think of each z_k as a vector in the plane; z_{tot} is the weighted vector sum. The squared norm of a vector sum equals the sum of squared norms plus twice the pairwise dot products. Here $\Re(z_i \bar{z}_j)$ is exactly that dot product in complex notation.*

47.5 Resonance-modulated wu wei: Gibbs form, gap, and sensitivity

Helper Theorem 47.11 (Resonant Gibbs policy and exact optimality gap)**.** *Fix state x, baseline action distribution $\pi_0(\cdot \mid x)$ with full support, bounded cost $c(a \mid x)$, resonance weight $\beta \geq 0$, and $\lambda > 0$. Define utility*

$$u(a \mid x) := \beta\, r_{\mathrm{res}}(a \mid x) - c(a \mid x),$$

and the regularized objective

$$F_x(\pi) := \mathbb{E}_{a\sim\pi(\cdot|x)}[u(a \mid x)] - \lambda\, \mathrm{KL}(\pi(\cdot \mid x) \,\|\, \pi_0(\cdot \mid x)).$$

Let

$$Z_x := \sum_{b\in A} \pi_0(b \mid x) \exp(u(b \mid x)/\lambda), \qquad \pi^\star(a \mid x) := \frac{\pi_0(a \mid x)\exp(u(a \mid x)/\lambda)}{Z_x}.$$

Then:

1. *(Maximum value)* $\max_\pi F_x(\pi) = \lambda \log Z_x$*.*
2. *(Exact gap) For every π, $F_x(\pi) = \lambda \log Z_x - \lambda\, \mathrm{KL}(\pi \,\|\, \pi^\star)$.*

Hence $\pi^\star$ is the unique maximizer.

Remark 2700. ***Notation and sign conventions.*** *Compared to the earlier one-step theorem, we now* maximize *an objective $F_x(\pi)$, and the KL term appears with a negative sign. Thus KL penalizes deviation from the baseline π_0, while the utility u rewards resonance and penalizes cost. The normalizer Z_x is again a partition function, now over actions A rather than next-states X.*

Remark 2701. ***Intuitive content and importance.*** *The theorem says the optimal policy is again a Gibbs tilt, now favoring actions with high $(\beta\, r_{\mathrm{res}} - c)$. The exact gap identity provides a rigorous notion of "how far from wu wei" a candidate policy is: the performance loss is exactly λ times a KL divergence to the optimal policy. This makes the result especially suitable for meta-control and learning loops: it yields a scalar certificate that is both computable and interpretable.*

Remark 2702. ***Connection to earlier results.*** *This is the action-space analogue of the one-step kernel tilt: the algebra is identical once one recognizes the structural pattern "linear term in expectation plus KL regularizer." The resonance bounds and coherence decomposition above feed into this theorem by ensuring u is well-controlled and interpretable in terms of constructive/destructive interference.*

Proof sketch. Same algebra as the one-step control case, but for maximization: write $\mathrm{KL}(\pi\|\pi^\star)$ using $\pi^\star \propto \pi_0 \exp(u/\lambda)$ and rearrange to isolate $F_x(\pi)$. □

Remark 2703. ***Proof strategy.*** *One guesses the Gibbs form by analogy, then verifies it by rewriting* $\mathrm{KL}(\pi\|\pi^\star)$ *and solving for $F_x(\pi)$. The sign flip (minimization vs maximization) only changes whether $-\lambda \log Z$ or $+\lambda \log Z$ appears.*

Lemma 47.12 (Monotonicity of forcing-deviation with temperature)**.** *Let $\pi^\star_\lambda(a) \propto \pi_0(a)\exp(u(a)/\lambda)$ on a finite A with full-support π_0. Then* $\mathrm{KL}(\pi^\star_\lambda\|\pi_0)$ *is nonincreasing in λ. Equivalently, with $s = 1/\lambda$,* $\mathrm{KL}(\pi^\star_{1/s}\|\pi_0)$ *is nondecreasing in s.*

Remark 2704. ***Intuition.*** *As temperature λ increases, the Gibbs policy becomes less sharp and stays closer to the baseline π_0. Therefore the KL divergence to π_0 should drop. This lemma makes that intuition exact, and thereby justifies reading* $\mathrm{KL}(\pi^\star_\lambda\|\pi_0)$ *as an "amount of forcing" that is controlled monotonically by the temperature knob (Hyperseed-Concept 100, 88).*

Remark 2705. ***Why it connects.*** *Together with the KL-to-uniform identity for opening, this lemma yields monotonicity of openness under temperature changes (next corollary). Thus a single analytic inequality about* KL *propagates into a behavioral principle about exploration vs commitment.*

Proof sketch. Let $\pi_s(a) \propto \pi_0(a)\exp(su(a))$. Then $\mathrm{KL}(\pi_s\|\pi_0) = s\,\mathbb{E}_{\pi_s}[u] - \log Z(s)$ with $Z(s) = \sum_a \pi_0(a)\exp(su(a))$. Differentiate w.r.t. s and use $\frac{d}{ds}\mathbb{E}_{\pi_s}[u] = \mathrm{Var}_{\pi_s}(u) \geq 0$ to get $\frac{d}{ds}\mathrm{KL}(\pi_s\|\pi_0) = s\,\mathrm{Var}_{\pi_s}(u) \geq 0$. □

Remark 2706. ***Key step commentary.*** *The identity $\frac{d}{ds}\mathbb{E}_{\pi_s}[u] = \mathrm{Var}_{\pi_s}(u)$ is the classical exponential-family calculus: differentiating the log-partition function yields moments, and differentiating again yields variances. The inequality then becomes almost tautological: variance is non-negative, hence KL increases with sharpness $s = 1/\lambda$.*

Corollary 47.13 (If baseline is uniform, opening increases with lambda for Gibbs policies)**.** *If* $\pi_0 = \mathrm{Unif}$ *(uniform on A), then for $\pi^\star_\lambda$ as above,*

$$\mathrm{Open}(\pi^\star_\lambda; x) = \log|A| - \mathrm{KL}(\pi^\star_\lambda\|\mathrm{Unif})$$

is nondecreasing in λ.

Remark 2707. ***Intuition.*** *With a uniform baseline, "openness" is literally the complement of KL-to-uniform. Since higher temperature reduces KL-to-baseline, it increases openness. This expresses, in a clean inequality, the common-sense idea that softer control keeps more options available (Hyperseed-Concept 125, 195).*

Proof sketch. Combine the KL-to-uniform identity for Open with the monotonicity of $\mathrm{KL}(\pi^\star_\lambda\|\pi_0)$. □

Remark 2708. ***Proof strategy.*** *This is a direct substitution: rewrite opening in terms of KL, then apply the previous lemma and note that subtracting a nonincreasing function yields a nondecreasing one.*

Lemma 47.14 (Sensitivity of the optimal value to resonance weight)**.** *Let* $\Phi(\beta) := \max_\pi F_x(\pi) = \lambda \log Z_x(\beta)$ *with* $Z_x(\beta) = \sum_a \pi_0(a \mid x)\exp((\beta\, r_{\mathrm{res}}(a \mid x) - c(a \mid x))/\lambda)$*. Then*

$$\frac{d}{d\beta}\Phi(\beta) = \mathbb{E}_{a\sim\pi^\star(\cdot|x)}[r_{\mathrm{res}}(a \mid x)].$$

Remark 2709. ***Intuition and usefulness.*** *This is an envelope-theorem style statement: the derivative of the optimal value with respect to the resonance weight β equals the expected resonance reward under the* optimal *policy. It provides an interpretable gradient for meta-optimization: increasing β increases the optimum value at a rate equal to "how much resonance the current optimum is already harvesting." This is precisely the kind of sensitivity rule an automated reasoner can reuse to tune hyperparameters.*

Proof sketch. Differentiate $\lambda\log Z_x(\beta)$: $\frac{d}{d\beta}\log Z_x = (1/\lambda)\mathbb{E}_{\pi^\star}[r_{\mathrm{res}}]$, then multiply by λ. □

Remark 2710. ***Key step.*** *Differentiating $\log Z_x$ brings down the inside term r_{res}/λ and simultaneously normalizes by Z_x, which is exactly what constructs the Gibbs distribution $\pi^\star$. Thus the gradient naturally appears as an expectation under $\pi^\star$.*

47.6 Continuous-time wu wei (Schrodinger bridge): monotone knobs

Proposition 47.15 (WSB value is nondecreasing in epsilon)**.** *Let* $\mathrm{WSB}_\epsilon[\rho, v] = \iint \rho\big(\frac{1}{2}\|v\|^2 + \epsilon \log(\rho/p)\big)\, dx\, dt$ *as in Def. 411, with a fixed feasible set (continuity equation + boundary densities). Let* $J^\star(\epsilon)$ *be the minimal value over feasible* (ρ, v)*. Then for* $0 \le \epsilon_1 \le \epsilon_2$*,*

$$J^\star(\epsilon_1) \le J^\star(\epsilon_2).$$

Remark 2711. ***Intuition and conceptual role.*** *In the continuous-time Wu Wei Schrödinger bridge (WSB) formulation [21],* ϵ *scales the entropic regularization term* $\iint \rho \log(\rho/p)$*, i.e. the penalty for deviating from a reference path measure encoded by* p*. Increasing* ϵ *means "charging more" for informational deviation. It is therefore unsurprising, yet important to state explicitly, that the best achievable objective value cannot decrease as* ϵ *increases.*

Remark 2712. ***Connection to discrete time.*** *This monotonicity is the continuous analogue of how, in discrete-time Gibbs tilting, increasing* λ *makes control softer and reduces forcing; here, increasing* ϵ *increases the cost of deviation and thus weakly worsens the optimal value. Both are instances of the same structural principle: enlarging a penalty coefficient cannot improve the penalized optimum when the feasible set is fixed.*

Proof sketch. For any fixed feasible (ρ, v), $\mathrm{WSB}_{\epsilon_2}[\rho, v] \ge \mathrm{WSB}_{\epsilon_1}[\rho, v]$ because the only difference is $(\epsilon_2 - \epsilon_1) \iint \rho \log(\rho/p)$. Taking minima over the same feasible set preserves the inequality. □

Remark 2713. ***Proof strategy.*** *This is a pointwise comparison: for each feasible candidate, the objective increases with* ϵ*. The minimum over candidates therefore also increases (or stays the same). No deeper structure is required beyond feasibility not depending on* ϵ*.*

Lemma 47.16 (Cognitive Reynolds knob correspondence)**.** *In Def. 413,* $\mathrm{Re}_c = \|v\|_{\mathrm{typ}} L/\epsilon$ *decreases as* ϵ *increases. In discrete-time KL control, the analogous "low-regularization" knob is* $1/\lambda$*: larger* $1/\lambda$ *corresponds qualitatively to larger* Re_c *(sharper commitment / more brittle forcing).*

Remark 2714. ***Intuition.*** *The cognitive Reynolds number* Re_c *is a metaphorical measure of how inertial/ballistic the cognitive flow is compared to the smoothing effect of entropic regularization. When* ϵ *is small, the regularization is weak,* Re_c *is large, and the optimizer can form sharp, high-velocity "jets" of behavior (strong forcing). When* ϵ *is large, regularization dominates and the flow becomes smooth and conservative. Discrete-time temperature* λ *plays the same qualitative role: small* λ *(large* $1/\lambda$*) yields sharp Gibbs tilts, hence strong forcing.*

Remark 2715. ***Why this correspondence is helpful.*** *It allows one to transfer intuitions and even heuristics between the discrete-time inference rule (Gibbs tilting with* λ*) and the continuous-time bridge (WSB with* ϵ*), aligning two computational regimes under a single interpretive knob (Hyperseed-Concept* **??***, 207).*

Proof sketch. Immediate from the definitions: Re_c is inversely proportional to ϵ, and one-step/multi-step Gibbs tilting becomes sharper as λ decreases (equivalently $1/\lambda$ increases). □

Remark 2716. ***Commentary.*** *While the lemma is algebraically immediate, its conceptual content is not: it asserts that two superficially different regularization parameters (*ϵ *in a continuum variational problem,* λ *in a discrete KL control problem) should be read through the same lens of "how much the system is permitted to force." This is one of the reasons the Wu Wei formalism is a coherent unification rather than a mere collection of tricks [21].*

Part III
Theoretical Developments Regarding Hyperseed Concepts

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48 A Fixed Point + Selection Meta-Theorem Schema

Many constructions in compositional ontologies (including Hyperseed-style stacks) define “stable regimes” as fixed points of monotone operators on complete lattices: closures, viability kernels, mutually predictive reality-systems, workspace stabilization, etc. A reasoning engine typically needs more than existence: it must also *select* a specific stable regime using coherence/utility scores, projectors/regularizers, or meta-control policies. This note packages a reusable proof schema: (i) a fixed-point existence theorem and (ii) a selection principle that itself can be encoded as a monotone operator.

Concretely, the ambient order structure is meant to capture “informativeness” or “constraint”: elements higher in the order represent stronger commitments (e.g. larger closures, tighter invariant sets, more determinate world-models), and the join/meet operations of a complete lattice represent aggregation of compatible commitments and extraction of common content. The monotone operator then formalizes one step of a stabilization procedure: given a provisional regime, apply an update map that (by design) does not reverse the underlying notion of commitment. In many applications, the operator is built from compositional pieces (e.g. a perception update, a predictive closure, and a consistency filter), and monotonicity is precisely the algebraic condition that makes those pieces compatible with order-theoretic convergence arguments.

The fixed-point layer can be read as a pure existence-and-structure claim: under completeness and monotonicity, the set of fixed points is itself well-behaved (in particular, closed under meets and joins in standard settings), and there are canonical extremal stable regimes (least and greatest fixed points) corresponding to “minimal” and “maximal” consistent solutions. This is the level at which Knaster–Tarski style arguments typically enter: instead of analyzing a dynamical trajectory, one reasons about the order structure of post-fixed points and pre-fixed points, and obtains stable regimes as lattice-theoretic limits. In engineering terms, this tells us that if the update rule is order-preserving and the state space is closed under the relevant combinations of information, then some stable operating points *must* exist, even when they are not unique.

The selection layer is a distinct requirement: when multiple fixed points exist (as is typical in highly compositional systems), the mere existence of equilibria is not yet a usable specification. Selection can be implemented by introducing an auxiliary ordering on “candidates” (for example, a preference preorder derived from coherence scores, cost functionals, resource bounds, or safety constraints) and then building an operator that maps a set (or ideal/filter) of candidates to the subset that survive a selection rule. The key technical move in this note is that the selection rule itself can often be expressed as a monotone operator on an appropriate complete lattice of *collections* of regimes (e.g. subsets ordered by inclusion, down-sets ordered by inclusion, or pairs consisting of a regime and a certificate ordered componentwise). In that way, selecting a particular stable regime becomes another fixed-point problem: one seeks a fixed point of the selection operator that represents a self-consistent “chosen” candidate (or a minimal/maximal choice set) under the policy encoded by the selection map.

This separation—first prove that stable regimes exist as fixed points of an update operator, then prove that a choice mechanism exists as a fixed point of a selection operator—is intended as a reusable meta-theorem schema. It allows one to swap in different domain theories (different lattices and update operators) while reusing the same abstract proof skeleton, and it also makes explicit which assumptions belong to "physics" (the update dynamics) versus "governance" (the selection policy). In particular, by requiring monotonicity for both layers, the schema keeps the reasoning purely order-theoretic: existence, extremality, and (when present) canonicality of the selected regime are derived from lattice properties rather than from metric convergence or probabilistic mixing.

Remark 2717. *One can read the present section as a small piece of "mathematical political philosophy" for formal systems: we do not merely ask whether equilibria exist, but how a system may come to* legitimately settle *on one equilibrium among many. In Hyperseed-style compositional stacks, this corresponds to the engineering question of how an ontology, a workspace, or a goal-system stabilizes* and *how it breaks symmetry among multiple stable regimes [1]. The abstract results here are also the canonical tools used when fixed points are invoked to model meta-goal and self-modification stability [10].*

One may also view "legitimacy" here in a narrowly formal sense: the selection outcome should be *internally certified* by the same order-theoretic constraints that define stability, rather than imposed as an external ad hoc choice. For example, a meta-control policy may be required to be invariant under refinements of information (monotonicity), or to respect a dominance order induced by safety constraints; in such cases, the selected equilibrium is not merely a picked element, but a fixed point of a policy operator that can be audited against explicit axioms. This makes it possible to separate questions of value (which policy operator is endorsed) from questions of correctness (what fixed points that operator entails), while still working in a uniform mathematical idiom.

Remark 2718. *Conceptually, "stability" is being represented in the most austere Russellian way: as a fixed point of an order-preserving transformation. Yet the "selection" layer gestures toward the more process-oriented sensibility of Whitehead, in which actuality is not merely what is consistent, but what is* selected *by a process of concrescence; and toward Peirce's emphasis on habits (regularities) as the outcome of lawful formation [15, 14]. The mathematics itself, however, remains purely order-theoretic.*

In practical terms, the austerity is a feature: by working only with order structure, the schema applies equally to symbolic closures, semantic entailment operators, constraint propagation on workspaces, and abstract viability constructions, without committing to a particular notion of distance, probability, or gradient. When additional structure is available (e.g. topology, convexity, or metrics), it can often be layered on top to obtain stronger uniqueness or continuity properties; but the baseline fixed-point-plus-selection pattern already delivers a robust, compositional way to argue that "a stable regime exists" and that "a principled choice among stable regimes exists" under assumptions that are easy to state and easy to preserve under composition.

48.1 Standing assumptions

Definition 447 (Complete lattice, monotone map)**.** *A* complete lattice *is a poset $(L, \leq)$ in which every subset $A \subseteq L$ has a meet $\bigwedge A$ and a join $\bigvee A$. A map $F : L \to L$ is* monotone *if $x \leq y$ implies $F(x) \leq F(y)$.*

Remark 2719. *A useful immediate consequence of completeness is that L has distinguished extremal elements: the meet of the empty set $\bigwedge \varnothing$ is the* top *element (often written $\top$), and the join* *of the empty set $\bigvee \varnothing$ is the* bottom *element (often written $\bot$). Thus completeness is stronger than merely having binary meets/joins; it asserts that arbitrary intersections and unions (in the powerset case), or arbitrary infima and suprema (in the real-interval case), exist inside L without leaving the universe of admissible states.*

Remark 2720. *Intuitively, a complete lattice is a space of "states" where one can always form the* most conservative common refinement *(the meet, $\bigwedge A$) and the* least conservative common generalization *(the join, $\bigvee A$) of any collection of states, even an infinite one. The order $\le$ is the comparison of information, constraint, or commitment: $x \le y$ often reads as "y contains (at least) as much commitment/information/structure as x" depending on the application. In this reading, meets correspond to combining constraints (keeping only what all states in A jointly enforce), while joins correspond to relaxing constraints just enough to cover all states in A.*

Remark 2721. *Monotonicity is the minimal structural hypothesis one typically needs to make order-theoretic iteration meaningful: if $x \le y$ represents "y is at least as informative as x," then monotonicity says that applying the update rule does not invert information ordering. In particular, it allows one to compare runs started from different initial states, and it is exactly the hypothesis under which classical fixed-point theorems (e.g. Knaster–Tarski) apply in complete lattices.*

Remark 2722. *Two standard examples help fix ideas. (1) The powerset lattice $(\mathcal{P}(U), \subseteq)$ is complete, with $\bigwedge A = \bigcap A$ and $\bigvee A = \bigcup A$. Monotone maps here include closure operators, forward/backward reachability maps, and the mutual-prediction closure operators used to define reality-systems (cf. Hyperseed-Concept 149). A common pattern is that $F(S)$ is obtained from S by adding all elements forced by some rule; monotonicity then says that starting from a larger set cannot lead to fewer forced consequences. (2) Any interval like $([0,1], \le)$ is a complete lattice where $\bigwedge A = \inf A$ and $\bigvee A = \sup A$. In such a setting, monotone maps include many "update rules" for bounded confidence or graded evidence. The completeness assumption is precisely what prevents the pathological situation in which a limiting process has nowhere to go. One should keep in mind that completeness here refers to order-theoretic completeness (existence of all $\inf / \sup$), not necessarily metric completeness; the point is that order-limits of increasing/decreasing families remain representable as elements of the same state space.*

Remark 2723. *On notation: $\bigwedge A$ and $\bigvee A$ denote meet/join of an* arbitrary subset *$A \subseteq L$. When we write expressions like $x \wedge \alpha$ or $x \vee \beta$ later, these are the* binary *meet/join, i.e. $x \wedge \alpha := \bigwedge\{x, \alpha\}$ and $x \vee \beta := \bigvee\{x, \beta\}$. In particular, binary meets/joins are just a special case of arbitrary meets/joins, and the latter are the ones that enable taking "limits" of potentially infinite collections of approximants in a purely order-theoretic sense.*

Definition 448 (Prefix/postfix points, fixed points)**.** *Let $F : L \to L$.*

$$\begin{aligned}
\mathrm{Pre}(F) &:= \{x \in L : F(x) \le x\} \quad \textit{(prefix points)},\\
\mathrm{Post}(F) &:= \{x \in L : x \le F(x)\} \quad \textit{(postfix points)},\\
\mathrm{Fix}(F) &:= \{x \in L : F(x) = x\} \quad \textit{(fixed points)}.
\end{aligned}$$

Remark 2724. *A prefix point is a state x that is "already at least as strong as what F would produce from it": applying F does not add anything beyond x itself. A postfix point is the dual notion: x is "not stronger than its own update," i.e. F moves x upward (or at least not downward). A fixed point is where these two meet: the update rule leaves the state unchanged. Equivalently, $\mathrm{Fix}(F) = \mathrm{Pre}(F) \cap \mathrm{Post}(F)$, so fixed points are precisely the states that are simultaneously stable against strengthening (prefix) and stable against weakening (postfix) under the action of F.*

Remark 2725. *In the "information order" reading, the inequalities in* $\mathrm{Pre}(F)$ *and* $\mathrm{Post}(F)$ *can be interpreted as one-step invariants. For* $x \in \mathrm{Pre}(F)$*, the state already entails all consequences that the update rule would infer (so* F *is conservative relative to* x*); for* $x \in \mathrm{Post}(F)$*, the update rule can only add information to* x *(so* x *is an under-approximation of its own update). These two notions are especially useful when* F *is used to define a notion of consistency or closure: prefix points act like candidates that are closed "from above," while postfix points act like candidates that are closed "from below."*

Remark 2726. *Simple examples clarify the three sets. If* $L = \mathcal{P}(U)$ *and* F *is a closure operator (e.g. reachability closure, or mutual predictive closure), then* $\mathrm{Fix}(F)$ *is the family of closed sets;* $\mathrm{Pre}(F)$ *are sets that already contain their closure (which is the same as being closed if* F *is extensive), and* $\mathrm{Post}(F)$ *are sets whose closure contains them (which is always true for extensive* F*). In many applications,* F *is not necessarily extensive or contractive, so separating* $\mathrm{Pre}(F)$ *and* $\mathrm{Post}(F)$ *is not pedantry: it is exactly what allows the order-theoretic machinery to work without extra assumptions. In particular, one can reason about* upper bounds *on fixed points via prefix points and about* lower bounds *via postfix points, even when* F *does not behave like a classical closure (extensive and idempotent) or like a contraction.*

Remark 2727. *A key motivation for recording* $\mathrm{Pre}(F)$ *and* $\mathrm{Post}(F)$ *explicitly is that, under the standing assumptions (complete lattice and monotone* F*), the collection of fixed points is itself well-behaved: Knaster–Tarski implies that* $\mathrm{Fix}(F)$ *forms a complete lattice, with the least fixed point given by* $\bigwedge \mathrm{Pre}(F)$ *and the greatest fixed point given by* $\bigvee \mathrm{Post}(F)$*. Even when one does not immediately invoke the theorem, these formulas serve as a guiding heuristic for how "maximally conservative" and "maximally liberal" solutions are extracted from families of one-step invariants.*

48.2 Fixed points as a complete "solution space"

Theorem 58 (Knaster–Tarski package)**.** *Let* $(L, \leq)$ *be a complete lattice and let* $F : L \to L$ *be monotone. Define*

$$\mu := \bigwedge \mathrm{Pre}(F), \qquad \nu := \bigvee \mathrm{Post}(F).$$

Then:

1. μ *and* ν *are fixed points:* $F(\mu) = \mu$ *and* $F(\nu) = \nu$*.*
2. μ *is the least fixed point and* ν *is the greatest fixed point: for all* $x \in \mathrm{Fix}(F)$*,* $\mu \leq x \leq \nu$*.*
3. $\mathrm{Fix}(F)$ *is a complete lattice under the inherited order. Moreover, for any nonempty* $S \subseteq \mathrm{Fix}(F)$*, if we set*

   $$\alpha := \bigwedge S, \qquad \beta := \bigvee S,$$

   and define two auxiliary monotone maps

   $$H_\alpha(x) := F(x) \wedge \alpha, \qquad K_\beta(x) := F(x) \vee \beta,$$

   then:

   $$\bigwedge_{\mathrm{Fix}(F)} S \;=\; \mathrm{gfp}(H_\alpha), \qquad \bigvee_{\mathrm{Fix}(F)} S \;=\; \mathrm{lfp}(K_\beta).$$

   (Here $\bigwedge_{\mathrm{Fix}(F)}$ *and* $\bigvee_{\mathrm{Fix}(F)}$ *denote meet/join computed* inside $\mathrm{Fix}(F)$*.)*

Remark 2728. *For clarity of notation:* $\mathrm{Pre}(F) := \{x \in L : F(x) \le x\}$ *is the set of* pre-fixed points *(also called* prefixpoints*),* $\mathrm{Post}(F) := \{x \in L : x \le F(x)\}$ *is the set of* post-fixed points *(or* postfixpoints*), and* $\mathrm{Fix}(F) := \{x \in L : F(x) = x\}$ *is the set of fixed points. In a complete lattice* $\mathrm{Pre}(F)$ *and* $\mathrm{Post}(F)$ *are automatically nonempty:* $\top \in \mathrm{Pre}(F)$ *because* $F(\top) \le \top$*, and* $\perp \in \mathrm{Post}(F)$ *because* $\perp \le F(\perp)$*. Thus the meets/joins defining* μ *and* ν *are always defined.*

Remark 2729. *One useful way to read the theorem is as a "three-layer package": (i) fixed points exist (indeed, extremal ones exist), (ii) all fixed points are trapped between the extremal ones, and (iii) the collection of fixed points inherits enough structure to support arbitrary meets/joins* internally*. The third item is stronger than mere existence: it says the space of solutions is closed under the natural lattice operations once those operations are interpreted inside* $\mathrm{Fix}(F)$*.*

Remark 2730. *In plain language, the theorem says: if your update rule* F *never reverses the order of states, then there is always a* smallest *stable regime* μ *and a* largest *stable regime* ν*, and all other stable regimes sit between them. Moreover, the* set of all stable regimes *is itself structured as a complete lattice, meaning that stable regimes can be combined (by meet/join) and the combination is still stable when computed in the right way.*

Remark 2731. *This result is important because it turns existence of equilibria from a fragile analytic question into a structural certainty: no metric, no convexity, no continuity are required—only completeness of the state-space and monotonicity of the update rule. In systems terms, this is why fixed-point reasoning appears so pervasively when defining closures ("what is implied by what we already have"), viability kernels ("what can remain viable under constraints"), and mutually predictive closures (cf. reality-systems as fixed points, Hyperseed-Concept 149). In the Hyperseed context, this style of reasoning is explicitly leveraged in the analysis of stable goal configurations and self-modification [10].*

Remark 2732. *A small notational comment for later use:* $\mathrm{lfp}(\cdot)$ *and* $\mathrm{gfp}(\cdot)$ *denote the least and greatest fixed points of a monotone operator, respectively, computed in the ambient complete lattice. The theorem shows how to compute* meets/joins internal to $\mathrm{Fix}(F)$ *using* lfp/gfp *of auxiliary operators that force the result to stay below (or above) a target bound.*

Remark 2733. *The auxiliary operators in (3) can be viewed as "clamped" versions of* F*. The map* $H_\alpha(x) = F(x) \wedge \alpha$ *runs one step of* F *and then caps the result at* α*; fixed points of* H_α *are precisely fixed points of* F *that do not exceed* α*. Dually,* $K_\beta(x) = F(x) \vee \beta$ *runs one step of* F *and then enforces a floor at* β*; its fixed points are exactly fixed points of* F *that lie above* β*. This is why taking* $\mathrm{gfp}(H_\alpha)$ *and* $\mathrm{lfp}(K_\beta)$ *produces, respectively, the greatest fixed point below all of* S *and the least fixed point above all of* S*.*

Proof. (1) Let $\mu := \bigwedge \mathrm{Pre}(F)$. For any $p \in \mathrm{Pre}(F)$ we have $\mu \le p$, so by monotonicity $F(\mu) \le F(p) \le p$. Thus $F(\mu)$ is a lower bound of $\mathrm{Pre}(F)$, hence $F(\mu) \le \mu$. Now apply F again: since $F(\mu) \le \mu$, monotonicity implies $F(F(\mu)) \le F(\mu)$, i.e. $F(\mu) \in \mathrm{Pre}(F)$. Because μ is the greatest lower bound of $\mathrm{Pre}(F)$, we get $\mu \le F(\mu)$. So $F(\mu) = \mu$.

Similarly, let $\nu := \bigvee \mathrm{Post}(F)$. For any $q \in \mathrm{Post}(F)$ we have $q \le \nu$, so $F(q) \le F(\nu)$ by monotonicity, i.e. $q \le F(\nu)$ because $q = F(q)$ need not hold but $q \le F(q)$ does. Hence $F(\nu)$ is an upper bound of $\mathrm{Post}(F)$, so $\nu \le F(\nu)$. Also, since ν is an upper bound of $\mathrm{Post}(F)$, it is itself in $\mathrm{Post}(F)$: indeed, for each $q \in \mathrm{Post}(F)$, $q \le \nu$, and therefore $q \le F(\nu)$, so $\nu \le F(\nu)$. Then applying F to $\nu \le F(\nu)$ gives $F(\nu) \le F(F(\nu))$, i.e. $F(\nu) \in \mathrm{Post}(F)$. Since ν is the least upper bound of $\mathrm{Post}(F)$, we obtain $F(\nu) \le \nu$. Thus $F(\nu) = \nu$.

To make the "$q \le F(\nu)$" step fully explicit: $q \in \mathrm{Post}(F)$ means $q \le F(q)$, and $q \le \nu$ implies $F(q) \le F(\nu)$ by monotonicity, hence $q \le F(q) \le F(\nu)$. This is the only place where we use post-fixedness rather than fixedness.

(2) If $x \in \mathrm{Fix}(F)$, then $F(x) = x$ implies both $F(x) \le x$ and $x \le F(x)$, so $x \in \mathrm{Pre}(F) \cap \mathrm{Post}(F)$. Therefore $\mu = \bigwedge \mathrm{Pre}(F) \le x$ and $x \le \bigvee \mathrm{Post}(F) = \nu$. In particular, μ and ν are respectively the bottom and top elements of the poset $(\mathrm{Fix}(F), \le)$, so every fixed point is "sandwiched" between them. (This is often how one proves bounds on equilibria: show a candidate is pre-fixed or post-fixed to trap it between μ and ν.)

(3) We sketch the completeness construction with the stated formulas. Fix nonempty $S \subseteq \mathrm{Fix}(F)$ and put $\alpha := \bigwedge S$ (meet in L). First observe $F(\alpha) \le \alpha$: for each $s \in S$ we have $\alpha \le s$, hence $F(\alpha) \le F(s) = s$ by monotonicity. So $F(\alpha)$ is a lower bound of S, hence $F(\alpha) \le \alpha$.

Now define $H_\alpha(x) = F(x) \wedge \alpha$. This map is monotone (meet with a constant preserves monotonicity). Any fixed point y of H_α satisfies $y \le \alpha$, hence $F(y) \le F(\alpha) \le \alpha$, so

$$y = H_\alpha(y) = F(y) \wedge \alpha = F(y).$$

Thus every fixed point of H_α is a fixed point of F lying below α, hence below every element of S. Conversely, if $y \in \mathrm{Fix}(F)$ and $y \le \alpha$, then $H_\alpha(y) = F(y) \wedge \alpha = y \wedge \alpha = y$, so $y \in \mathrm{Fix}(H_\alpha)$. Therefore $\mathrm{Fix}(H_\alpha) = \{y \in \mathrm{Fix}(F) : y \le \alpha\}$, and its greatest element $\mathrm{gfp}(H_\alpha)$ is exactly the meet of S computed inside $\mathrm{Fix}(F)$.

Equivalently (spelling out the "internal meet" viewpoint), $\bigwedge_{\mathrm{Fix}(F)} S$ is characterized as the *greatest* fixed point that is a lower bound of S within $\mathrm{Fix}(F)$. Since $\alpha = \bigwedge S$ is the greatest lower bound in the ambient lattice L, the condition "y is a lower bound of S" is exactly $y \le \alpha$, so taking the greatest fixed point among those y is precisely the gfp of the clamped operator H_α. This is why the formula uses $\mathrm{gfp}(H_\alpha)$ rather than simply α (which may fail to be a fixed point of F).

The join formula is dual. Let $\beta := \bigvee S$ (join in L). For each $s \in S$, $s \le \beta$ implies $F(s) = s \le F(\beta)$, so $F(\beta)$ is an upper bound of S. By minimality of β as the least upper bound, $\beta \le F(\beta)$, i.e. $\beta \in \mathrm{Post}(F)$. Define $K_\beta(x) = F(x) \vee \beta$, which is monotone. Any fixed point y of K_β satisfies $y \ge \beta$, hence $F(y) \ge F(\beta) \ge \beta$, so

$$y = K_\beta(y) = F(y) \vee \beta = F(y).$$

Thus fixed points of K_β are exactly fixed points of F lying above β, and the least such point $\mathrm{lfp}(K_\beta)$ is the join of S inside $\mathrm{Fix}(F)$.

As with meets, the point is that $\beta = \bigvee S$ computed in L need not itself be fixed; instead, $\mathrm{lfp}(K_\beta)$ selects the *least* fixed point that still dominates all of S. The restriction to nonempty S in the statement is only to avoid notational distraction: completeness of $\mathrm{Fix}(F)$ also covers the empty meet/join, yielding the top and bottom elements of $\mathrm{Fix}(F)$, namely ν and μ from (1)–(2), respectively. □

Remark 2734. Proof sketch (high level). *The least fixed point μ is obtained by taking the "most informative" state that is still below* every *prefix point; monotonicity then forces $F(\mu)$ to be both below and above μ, hence equal. The greatest fixed point ν is dual. For completeness of* Fix(F), *the idea is that the naive meet $\alpha = \bigwedge S$ might fail to be a fixed point; one therefore applies a constrained fixed-point computation: iterate F while never exceeding the bound α (via H_α), and take the greatest fixed point compatible with the bound. The join construction is the same story with an upward bound.*

Remark 2735. *A useful geometric intuition is to picture L as a landscape of partially ordered regimes, and F as a "flow" that can only move points upward when inputs move upward. Prefix points are those above their image, postfix points below their image. The meet of prefix points is the lowest "ceiling" compatible with all ceilings; applying F cannot jump above any ceiling, so it must land under that ceiling, and then the self-referential step closes the trap.*

48.3 Selection as an operator: "stable after dynamics + projection"

Definition 449 (Selection projector)**.** *A* selection projector *is a map $S : L \to L$ such that:*

1. *S is monotone;*
2. *S is idempotent: $S \circ S = S$.*

Intuition: S maps arbitrary states to a "selected" subset of states, and idempotence means selected states are already stable under selection.

Remark 2736. *Philosophically, S formalizes a familiar move: from the profusion of possible states, we admit only those states that satisfy some criterion of admissibility, coherence, feasibility, or economy. The monotonicity requirement says that if a state becomes "stronger" in the order-theoretic sense, then selection does not behave perversely with respect to this strengthening. Idempotence says selection is not an endless ritual: once a state has been brought into the selected form, re-selecting changes nothing.*

Remark 2737. *Examples. (1) Feasibility projection: if $A \subseteq L$ is a set of admissible states and $S(x)$ is the least element of A above x (when it exists), then S is idempotent and monotone. (2) Normalization/projection for resource-bounded reasoning: S may delete components exceeding a budget, or collapse distinctions deemed too costly. This connects directly to the use of weakness/simplicity normalization (Hyperseed-Concept 202 and Hyperseed-Concept 143) as an operator that enforces cognitive economy [3, 2].*

Remark 2738. *In applications, S can model: coherence regularization, simplicity/weakness normalization, budget-feasible projection, policy-class restriction, "make explicit" compilation, etc. Nothing here requires S to be extensive or contractive.*

Theorem 59 (Existence of selected equilibria)**.** *Let $(L, \leq)$ be a complete lattice, $F : L \to L$ monotone, and $S : L \to L$ a selection projector. Define the combined update*

$$T := S \circ F.$$

Then:

1. *T is monotone and therefore has a least and a greatest fixed point in L.*
2. *Every fixed point of T is automatically a fixed point of S:*

$$x \in \mathrm{Fix}(T) \;\Rightarrow\; x \in \mathrm{Fix}(S).$$

3. *Any state that is simultaneously F-stable and selected is T-stable:*

$$\mathrm{Fix}(F) \cap \mathrm{Fix}(S) \subseteq \mathrm{Fix}(T).$$

Remark 2739. *This theorem says that if you first apply dynamics (F) and then apply selection (S), the composite update rule T is still monotone, hence still possesses extremal equilibria by the Knaster–Tarski package (Theorem 58). Moreover, any equilibrium of the composite is necessarily already "in selected form," and any state that is both dynamically stable and already selected is an equilibrium of the composite. In other words: "stability after projection" is itself a stable regime.*

Remark 2740. *This is the basic bridge between existence and design: it legitimizes the common engineering move of taking a system with messy dynamics and then repeatedly projecting back to a manageable subspace (feasible, coherent, simple, budgeted). In AGI architecture language, it is a formal template for repeatedly applying a cognitive update and then applying a meta-control or resource-allocation constraint (cf. Hyperseed-Concept 87 and Hyperseed-Concept 60), a theme recurring in cognitive synergy approaches [19].*

Proof. (1) T is a composition of monotone maps, hence monotone. Apply Theorem 58(1).

(2) If $x \in \mathrm{Fix}(T)$, then $x = T(x) = S(F(x))$ lies in the image of S. Apply idempotence:

$$S(x) = S(S(F(x))) = S(F(x)) = x.$$

So $x \in \mathrm{Fix}(S)$.

(3) If $x \in \mathrm{Fix}(F) \cap \mathrm{Fix}(S)$, then

$$T(x) = S(F(x)) = S(x) = x,$$

so $x \in \mathrm{Fix}(T)$. □

Remark 2741. Proof sketch (high level). *The only substantive ingredient is that monotonicity is preserved under composition, so Theorem 58 applies to T. The rest is algebraic: if x is fixed by T, then x is literally of the form $S(\cdot)$, and idempotence of S forces $S(x) = x$; conversely, if x is fixed by both F and S, then substituting these equalities into $T(x) = S(F(x))$ yields $T(x) = x$.*

Remark 2742. *A helpful picture is to imagine F as moving points around and S as a "snap-to-grid" operator. A T-fixed point is a point that, after being moved and snapped, lands back on itself. The theorem says such a point must already lie on the grid (be S-fixed), and that any point that is simultaneously unmoved by F and already on the grid is also a T-fixed point.*

48.4 Meta-selection: selecting the update rule (policy/parameter) and the state together

Theorem 60 (Self-consistent selection fixed point on a product lattice)**.** *Let $(P, \preceq)$ and $(L, \leq)$ be complete lattices. Assume:*

- *$F : P \times L \to L$ is monotone in both arguments;*
- *$S : L \to L$ is a selection projector (monotone, idempotent);*
- *$G : L \to P$ is monotone (a "policy/parameter selection" map).*

Define $H : P \times L \to P \times L$ by

$$H(p, x) := \big(G(x),\, S(F(p, x))\big).$$

Then H has a least and a greatest fixed point (p^*, x^*), and any fixed point satisfies:

$$p^* = G(x^*), \qquad x^* = S(F(p^*, x^*)), \qquad x^* \in \mathrm{Fix}(S).$$

Moreover, the fixed point condition $H(p^*, x^*) = (p^*, x^*)$ is exactly the coupled "simultaneous equation" system consisting of a meta-selection equation for the update rule ($p^* = G(x^*)$) together with a selected equilibrium condition for the state ($x^* = S(F(p^*, x^*))$). In particular, the result is not merely an existence statement for equilibria in L for a given *p, but an existence statement for equilibria of the* joint *closed-loop map that chooses p from x and then updates x using that p.*

Remark 2743. *The theorem expresses a clean notion of* self-consistency*: the "policy/parameter" p^* is exactly what the policy-selection map G would choose when observing the resulting stable state x^*, and the state x^* is stable under the dynamics induced by that very policy (with selection applied). This is a formal schema for feedback loops in which the system chooses how it will update itself based on its current condition, and then settles into a regime compatible with that choice (Hyperseed-Concept 110 and Hyperseed-Concept 203). One useful way to read the equations is as a closure condition: starting from x^*, the system "reads" a policy p^* via G, then applies the corresponding update $F(p^*, \cdot)$, and finally enforces admissibility via S, landing back on the same x^*. The extra statement $x^* \in \mathrm{Fix}(S)$ emphasizes that the stabilized state lies in the selected (admissible) region, so that selection is not merely applied once but is consistent with the fixed point.*

Remark 2744. *Why this matters: without such a product fixed point, one could have an endless regress in which the system's state determines its policy, which changes its state, which changes its policy, and so on, without convergence. The theorem guarantees that, at least at the level of order-theoretic abstraction, there are extremal self-consistent solutions. This is one of the clean mathematical backbones behind fixed-point analyses of meta-goal stability in self-modifying agents [10], and it resonates with the broader theme that "least-effort" or constraint-regularized selection rules can be treated as operators in their own right [21]. Concretely, the existence of a* least *and* greatest *fixed point provides two canonical boundaries for the space of self-consistent regimes: one can interpret the least fixed point as the most conservative (least committing) self-consistent configuration compatible with the order, and the greatest fixed point as the most expansive one. Even when intermediate fixed points exist, these extremal fixed points supply well-defined endpoints for comparative statics (e.g., how changing F, G, or S shifts the minimal or maximal stable closed-loop behavior).*

Proof. The product poset $(P \times L, \preceq \times \leq)$ is a complete lattice with coordinatewise meets and joins. Explicitly, if $\{(p_i, x_i)\}_{i\in I} \subseteq P \times L$, then

$$\bigvee_{i\in I}(p_i, x_i) = \Big(\bigvee_{i\in I} p_i,\ \bigvee_{i\in I} x_i\Big), \qquad \bigwedge_{i\in I}(p_i, x_i) = \Big(\bigwedge_{i\in I} p_i,\ \bigwedge_{i\in I} x_i\Big),$$

which exist by completeness of P and L. The map H is monotone because G, S, and F are monotone and products preserve monotonicity. More explicitly, if $(p, x) \preceq \times \leq (p', x')$ (i.e. $p \preceq p'$ and $x \leq x'$), then monotonicity of G gives $G(x) \preceq G(x')$, and monotonicity of F in both arguments gives $F(p, x) \leq F(p', x')$, hence monotonicity of S yields $S(F(p, x)) \leq S(F(p', x'))$. Thus $H(p, x) \preceq \times \leq H(p', x')$. By Theorem 58(1), H has least and greatest fixed points, and any fixed point satisfies the displayed equalities by unwinding $H(p^*, x^*) = (p^*, x^*)$. Finally, $x^* = S(F(p^*, x^*))$ implies $x^* \in \mathrm{Fix}(S)$ by the same idempotence argument as in Theorem 59(2): applying S to both sides gives $S(x^*) = S(S(F(p^*, x^*))) = S(F(p^*, x^*)) = x^*$, so indeed $S(x^*) = x^*$. □

Remark 2745. Proof sketch (high level). *The core move is to shift perspective: instead of treating "policy choice" and "state update" as two separate processes, we bundle them into a single update on the product space $P \times L$. Completeness and monotonicity then re-enter exactly as in Theorem 58.*

The self-consistency equalities are not extra work: they are simply what it means to be a fixed point of H. Equivalently, one can think of the coupled iteration

$$(p_{t+1}, x_{t+1}) = \big(G(x_t),\ S(F(p_t, x_t))\big)$$

as a single monotone dynamical system on a complete lattice; Tarski's theorem guarantees extremal stationary points for this iteration even when no metric contraction or continuity assumptions are available.

Remark 2746. *Visually, one can imagine x living in a landscape of states and p as choosing which "vector field" on that landscape applies. The map G reads off from the current position which vector field to use next; F moves under the chosen field; S projects the result into a selected subspace. The theorem says there exist equilibria of this coupled system: a point on the landscape together with a choice of vector field that mutually sustain one another. In settings where S represents feasibility (hard constraints) or attention (projection to a subspace of represented features), the condition $x^* \in \mathrm{Fix}(S)$ can be read as "the equilibrium is internally representable/admissible": re-applying the feasibility/attention filter does not further change the equilibrium.*

Remark 2747 (Hyperseed reading)**.** *This theorem captures the "fixed point + selection" picture where the system selects an intervention, attention regime, or update-operator parameter* as a function of its own current state*, and then stabilizes under the resulting dynamics + selection. It is a direct abstract analog of "reflective will as meta-selection". The product structure is doing the conceptual work: it lets one treat "choosing how to update" as part of the state to be stabilized, rather than as an external controller. In particular, the meta-level variable p is not assumed fixed during convergence; it is itself subject to a monotone self-consistency constraint mediated by G.*

Remark 2748. *In Hyperseed terms, one may view G as a meta-level attentional or policy-selection habit, hence a Peircean "Third" organizing a family of possible updates; while S plays the role of the admissibility filter that keeps the resulting state within a coherent sub-ontology or feasible workspace [14, 15]. The point is not to import metaphysics into the proof, but to clarify why this particular abstract pattern recurs so widely in cognitive and ontological engineering [1, 19]. From a purely mathematical standpoint, the same pattern also appears whenever one has (i) a family of monotone operators $x \mapsto F(p, x)$ indexed by a parameter p, (ii) a monotone "index selection" rule $x \mapsto G(x)$, and (iii) a monotone projection enforcing a constraint set $\mathrm{Fix}(S)$. The theorem isolates exactly the minimal order-theoretic hypotheses under which the closed-loop operator admits extremal equilibria.*

48.5 Comparative statics: how selected equilibria move when the dynamics are strengthened

Definition 450 (Pointwise order on operators)**.** *For $F, G : L \to L$, write $F \leq_{\mathrm{pt}} G$ if $F(x) \leq G(x)$ for all $x \in L$.*

Remark 2749. *The pointwise order $\leq_{\mathrm{pt}}$ is the natural way to say "G is everywhere at least as strong as F." In many applications F and G are closure-like updates, reachability operators, or inference operators; then $F \leq_{\mathrm{pt}} G$ expresses that G adds (weakly) more consequences than F from any starting point. This definition is useful because it allows comparative statics arguments without committing to any metric notion of distance between operators.*

Remark 2750. *Two small but useful clarifications about $\leq_{\mathrm{pt}}$: (i) it is itself a partial order on the set of all endomaps $L \to L$ (reflexive, antisymmetric, transitive), inherited pointwise from $\leq$ on L;*

(ii) it is tailored to order-theoretic iteration arguments because it interacts well with composition by monotone maps: if $F \le_{\mathrm{pt}} G$ and S is monotone, then $S \circ F \le_{\mathrm{pt}} S \circ G$, exactly the comparison needed below. In particular, no continuity, contractiveness, or topology on L is required for these comparisons.

Theorem 61 (Monotone comparative statics for extremal selected equilibria)**.** *Let $(L, \le)$ be a complete lattice, let $S : L \to L$ be monotone, and let $F, G : L \to L$ be monotone. Assume $F \le_{\mathrm{pt}} G$. Define $T_F := S \circ F$ and $T_G := S \circ G$. Then:*

$$\mathrm{lfp}(T_F) \le \mathrm{lfp}(T_G), \qquad \mathrm{gfp}(T_F) \le \mathrm{gfp}(T_G).$$

Remark 2751. *This theorem says: if you strengthen the underlying dynamics from F to G (in the sense that G produces larger updates at every point), and you keep the same monotone selection layer S, then both the least and the greatest selected equilibria can only move upward. It is a formal version of a common-sense principle: more supportive/stronger dynamics cannot yield a smaller extremal stable regime when selection preserves order.*

Remark 2752. *It is worth noting what* moves upward *means here. Since L is an abstract complete lattice, "upward" is with respect to $\le$ and need not refer to any numerical increase. For example, on a powerset lattice $(\mathcal{P}(U), \subseteq)$, the conclusion reads* $\mathrm{lfp}(T_F) \subseteq \mathrm{lfp}(T_G)$ *and* $\mathrm{gfp}(T_F) \subseteq \mathrm{gfp}(T_G)$*, i.e., strengthening the dynamics can only enlarge the least and greatest selected invariant sets. On a product lattice of features ordered componentwise, it means every component of the extremal equilibria weakly increases.*

Remark 2753. *This result connects directly back to Theorem 58 via the characterization of* lfp *and* gfp *as meets of prefix points and joins of postfix points. It also underwrites the practical methodology of parameter sweeps: when a reasoning engine strengthens couplings, adds evidence, increases communication, or broadens admissible inferences, one obtains monotone bounds on how equilibria can shift. In the language of cognitive architectures, it gives a rigorous backbone to "turn the knob and see how the stable workspace grows" arguments [19].*

Remark 2754. *A standard comparative-statics perspective is to view F and G as two parameter values of a family $(F_\theta)_{\theta \in \Theta}$ ordered by strength. If $\theta \le \theta'$ implies $F_\theta \le_{\mathrm{pt}} F_{\theta'}$, then Theorem 61 implies that both $\theta \mapsto \mathrm{lfp}(S \circ F_\theta)$ and $\theta \mapsto \mathrm{gfp}(S \circ F_\theta)$ are monotone maps from $(\Theta, \le)$ into $(L, \le)$. This is exactly the order-theoretic analogue of "monotone policy functions" in parametric fixed-point problems, except that the present statement requires only lattice completeness and monotonicity.*

Proof. Because $F \le_{\mathrm{pt}} G$ and S is monotone, we have $T_F \le_{\mathrm{pt}} T_G$. Let $\mathrm{Pre}(T) := \{x : T(x) \le x\}$ as before. If $x \in \mathrm{Pre}(T_G)$ then $T_G(x) \le x$, and since $T_F(x) \le T_G(x)$ we also get $T_F(x) \le x$. Hence $\mathrm{Pre}(T_G) \subseteq \mathrm{Pre}(T_F)$, so

$$\mathrm{lfp}(T_F) = \bigwedge \mathrm{Pre}(T_F) \le \bigwedge \mathrm{Pre}(T_G) = \mathrm{lfp}(T_G).$$

Similarly, if $x \in \mathrm{Post}(T_F)$ then $x \le T_F(x) \le T_G(x)$, so $x \in \mathrm{Post}(T_G)$. Thus $\mathrm{Post}(T_F) \subseteq \mathrm{Post}(T_G)$ and therefore

$$\mathrm{gfp}(T_F) = \bigvee \mathrm{Post}(T_F) \le \bigvee \mathrm{Post}(T_G) = \mathrm{gfp}(T_G).$$

□

Remark 2755. *The proof uses only order-theoretic comparisons of sets of prefix/postfix points, so it is robust to substantial changes in interpretation. In particular, the argument does not require* *that F or G be inflationary ($x \leq F(x)$) or deflationary ($F(x) \leq x$) in general; it only uses monotonicity plus the pointwise comparison $F \leq_{\mathrm{pt}} G$. If, in an application, one does have inflationarity (typical for closure-like "add information" updates), then* $\mathrm{Post}(T)$ *is automatically nonempty (e.g. it contains $\perp$), which can make the existence of* $\mathrm{gfp}(T)$ *feel especially concrete, but the theorem does not rely on such additional structure.*

Remark 2756. Proof sketch (high level). *The strategy is to compare the sets of prefix points and postfix points. If $T_F \leq_{\mathrm{pt}} T_G$, then it is* easier *to satisfy $T_G(x) \leq x$ than to satisfy $T_F(x) \leq x$, so* $\mathrm{Pre}(T_G) \subseteq \mathrm{Pre}(T_F)$*; taking meets reverses inclusions into inequalities. Dually, it is* easier *to satisfy $x \leq T_F(x)$ than $x \leq T_G(x)$ when $T_F(x) \leq T_G(x)$, so postfix points include the other way; taking joins again yields the desired inequality.*

Remark 2757. *One can picture* $\mathrm{Pre}(T)$ *as a family of "upper barriers" and* $\mathrm{Post}(T)$ *as "lower supports" for the dynamics. Strengthening the dynamics shrinks the set of barriers and expands the set of supports; the extremal equilibria, defined as the tightest barrier and the loosest support, therefore move monotonically.*

Remark 2758. *A concrete mental model is: if x is an "upper barrier" for T_G (meaning $T_G(x) \leq x$), then it is automatically an upper barrier for any weaker update T_F because $T_F(x) \leq T_G(x)$. Thus, as the dynamics become stronger, some previously valid barriers cease to be barriers, so the meet of all barriers can only increase. Conversely, if x is a "lower support" for T_F (meaning $x \leq T_F(x)$), then it remains a support for T_G, so supports become more abundant and their join can only increase. This restates the same inclusions but emphasizes why the direction of the inequalities in the theorem is the "same" for* lfp *and* gfp *even though meets and joins react oppositely to set inclusion.*

Remark 2759. *For reasoning engines, this is the basic "if you strengthen supports/couplings/inputs, the least stable selected regime can only increase" principle. It is often the engine-level justification for doing parameter sweeps and sensitivity bounds.*

Remark 2760. *An illustrative example on a powerset lattice: let $L = \mathcal{P}(U)$, let S be a monotone "selection" operator (e.g. enforcing admissibility constraints), and let $F(X)$ be the set of states reachable from X under some transition relation. If G corresponds to adding transitions (or adding inference rules), then $F \leq_{\mathrm{pt}} G$ holds as $\subseteq$-inclusion for every starting set X. Theorem 61 then states that both the least and greatest S-selected invariant regions under reachability expand when transitions are added: one cannot obtain a smaller extremal stable region by allowing more reachability, provided the selection stage respects inclusion.*

48.6 Score-based selection among selected equilibria (existence of an optimizer)

Theorem 62 (Existence of score-maximizing selected equilibria under compactness)**.** *Let L be a compact Hausdorff topological space and let $T : L \to L$ be continuous. Assume* $\mathrm{Fix}(T) \neq \emptyset$. *Then* $\mathrm{Fix}(T)$ *is compact. In particular, for any continuous score function $J : L \to \mathbb{R}$, there exists $x^\star \in \mathrm{Fix}(T)$ such that*

$$J(x^\star) = \max\{J(x) : x \in \mathrm{Fix}(T)\}.$$

Moreover, the set of maximizers

$$\arg\max_{\mathrm{Fix}(T)} J \ := \ \{x \in \mathrm{Fix}(T) : J(x) = \max_{y \in \mathrm{Fix}(T)} J(y)\}$$

is a nonempty compact subset of $\mathrm{Fix}(T)$ *(and hence of L).*

Remark 2761. *Up to this point, selection has been treated as an* operator *(S) whose fixed points define what is admissible. But in many applications one also wants an* optimizer *among admissible equilibria: maximize coherence, utility, elegance, stability margin, or some other score. This theorem says that when the state space is compact and the update map is continuous, the set of equilibria is itself compact, hence any continuous score function achieves a maximum on it. A further modeling takeaway is that the maximizer need not be unique: the theorem guarantees existence, not single-valuedness. When multiple equilibria share the top score, the "selected-by-score" outcome is naturally a set-valued choice (the argmax set above), which is still well-behaved topologically due to compactness.*

Remark 2762. *This result is not order-theoretic but topological: it complements Knaster–Tarski rather than competing with it. Knaster–Tarski guarantees extremal fixed points under monotonicity and completeness; the present theorem guarantees existence of a score-maximizing fixed point under compactness and continuity. In practice, a system designer may use the order-theoretic results to prove that equilibria exist robustly, and then use compactness to justify a subsequent maximization step as well-posed. One can view these as two different "existence engines": order-theoretic hypotheses often align with comparative statics and monotone updates, while compactness/continuity hypotheses align with geometric or analytic models where scores and equilibria vary continuously with state.*

Remark 2763. *The assumption* $\mathrm{Fix}(T) \neq \emptyset$ *is logically independent of compactness and continuity: compactness alone does not force a fixed point to exist without additional structure (e.g. a convexity hypothesis as in Schauder-type theorems). Here the role of compactness is not to* create *fixed points but to make the set of already-existing fixed points closed and hence compact, so that maximization over them is meaningful and attained.*

Proof. Consider the continuous map $h : L \to L \times L$ defined by $h(x) = (x, T(x))$. In a Hausdorff space, the diagonal $\Delta := \{(x, x) : x \in L\}$ is closed in $L \times L$. We have

$$\mathrm{Fix}(T) = \{x : T(x) = x\} = h^{-1}(\Delta),$$

so $\mathrm{Fix}(T)$ is closed in L. Since L is compact, $\mathrm{Fix}(T)$ is compact. A continuous real-valued function on a compact set attains its maximum, so J has a maximizer on $\mathrm{Fix}(T)$. For the additional claim about $\arg\max_{\mathrm{Fix}(T)} J$, note that $\mathrm{Fix}(T)$ is compact and $J|_{\mathrm{Fix}(T)}$ is continuous, so the maximum value $m := \max_{x\in\mathrm{Fix}(T)} J(x)$ is finite and the level set

$$\arg\max_{\mathrm{Fix}(T)} J = \{x \in \mathrm{Fix}(T) : J(x) = m\} = (J|_{\mathrm{Fix}(T)})^{-1}(\{m\})$$

is closed in $\mathrm{Fix}(T)$; hence it is compact, and it is nonempty because m is attained. □

Remark 2764. Proof sketch (high level). *The fixed point set is the preimage of the diagonal under the graph map $x \mapsto (x, T(x))$. In Hausdorff spaces the diagonal is closed, so continuity makes the fixed point set closed; compactness of L then forces compactness of* $\mathrm{Fix}(T)$*. Finally, the Weierstrass maximum principle gives existence of a maximizer for J. Equivalently, one can say that the graph of a continuous map T is closed in $L \times L$ and fixed points are exactly the intersection of this graph with the diagonal; the Hausdorff hypothesis is what ensures the diagonal is closed, making the intersection behave cleanly.*

Remark 2765. *A visual intuition: draw the graph of T inside $L \times L$; fixed points are where that graph intersects the diagonal. Compactness prevents "escaping to infinity" and ensures the* *intersection structure behaves well enough that closedness and maximum-attainment follow. One can also interpret the score J as a "height function" on the equilibrium locus: compactness prevents taking an improving sequence of equilibria whose scores increase without limit while the equilibria themselves drift toward a non-equilibrium boundary point.*

Remark 2766 (Hyperseed reading)**.** *When L is a finite product of $[0,1]^k$ (e.g. a p-bit evidence state space for a finite language), L is compact. If the combined "dynamics + selection" operator T is continuous (often true when built from joins/meets and continuous t-norm-like operations), then there exists a coherence-maximizing selected stable state, formalizing "resonance/coherence selects among fixed points". In such cases, compactness is typically inherited from the product topology, and continuity can often be checked componentwise (e.g. if each coordinate update is a continuous formula in the coordinates). This makes the theorem a convenient "plug-in" lemma: once fixed points have been established by other means, optimizing a continuous coherence functional over them requires no additional existence work.*

Remark 2767. *In contexts where "selection" is interpreted as a tendency toward least-effort trajectories or geodesic-like regularization (Hyperseed-Concept 206 and Hyperseed-Concept 207), continuity is not merely a technical assumption but a modeling claim: small changes in state produce small changes in the selected update, allowing a score landscape to be meaningfully optimized [21]. From this perspective, discontinuities in T can be read as "phase transitions" in the update rule; the present theorem clarifies that if such transitions occur, maximization over equilibria may fail to be attained unless one adds further structure (for instance, by enlarging the state space to a compactification, or by relaxing the score from continuous to upper semicontinuous while retaining compactness).*

49 Coarse-Graining and Refinement as Monotone Inference Principles

This section collects several related theorems: invariance/monotonicity principles for passing between fine and coarse descriptions (contexts, resolutions, coarse-grainings). The key construction is a sup-pushforward (a left Kan extension in a thin/posetal setting), which yields sound "fine-to-coarse" inference rules. We then show monotone consequences for (i) pattern intensity, (ii) weakness/simplicity as failed distinctions, and (iii) effability/ineffability as resource-bounded fidelity under indistinction.

Concretely, the guiding picture is that a *fine* description distinguishes more cases (more states, more features, more measurement outcomes), whereas a *coarse* description identifies some of those cases. An assertion verified at the fine level can be transported to the coarse level by asking for the *least* coarse assertion that is implied by it (or, dually, the *greatest* coarse assertion consistent with it, depending on the direction of inference under study). In an order-theoretic setting this transport is most naturally expressed by adjoint-style constructions; the "sup-pushforward" mentioned above is the order-enriched analogue of taking an image along a map by collecting evidence via a supremum over fibers.

The phrase "left Kan extension in a thin/posetal setting" is meant literally: if one regards a preorder as a category with at most one morphism between any two objects, then many categorical universal properties reduce to order-theoretic ones. In particular, left Kan extension becomes a pointwise supremum formula, and the soundness of "forgetting details" becomes an instance of monotonicity together with the fact that suprema compute least upper bounds. This is the technical mechanism behind the invariance/monotonicity principles collected here.

Remark 2768. *The philosophical motive is straightforward: whenever our epistemic access is limited, we work with coarse descriptions. The central question is then: which inferences remain valid when details are forgotten? The results below answer this by giving* order-theoretic *conditions under which truths established at a fine resolution remain true (or become easier to satisfy) at a coarser one. In Hyperseed terms, this is precisely the kind of "safe abstraction" one wants when moving between* Contexts and Aspects *(Hyperseed-Concept 86) and when reasoning under limited* Effort *(Hyperseed-Concept 100) about* Patterns *(Hyperseed-Concept 130); see also [1, 5, 19].*

A useful way to keep the direction of "safety" in mind is the following. If x is a fine-grained claim and $\gamma(x)$ is its coarse-grained counterpart, then "safe abstraction" typically means that whenever x holds, $\gamma(x)$ also holds, i.e. $x \le \gamma(x)$ in an information order where larger elements represent weaker (or less informative) conditions. Many of the results below can be read as systematic ways of producing such γ (often as a left adjoint / left Kan extension), together with monotonicity statements ensuring that increasing coarseness cannot invalidate what has already been established at finer resolution.

49.1 Standing assumptions

Definition 451 (Quantale-lite assumptions)**.** *Fix a complete lattice $(V, \le)$ with joins of all subsets. We write $\oplus$ for joins. Assume also a monotone binary operation $\otimes : V \times V \to V$ (monotone in each argument).*

In particular, completeness means that for every subset $S \subseteq V$ the element $\oplus S \in V$ exists and is characterized by the usual universal property: (a) $s \le \oplus S$ for all $s \in S$, and (b) if $u \in V$ satisfies $s \le u$ for all $s \in S$, then $\oplus S \le u$. Dually, because V is a complete lattice (not merely a complete join-semilattice), arbitrary meets also exist; while meets will typically remain in the background in what follows, they are available whenever a dual formulation is convenient (e.g. when expressing "must hold under all refinements" rather than "holds under some refinement").

Remark 2769. *This definition deliberately asks for less than a full quantale: we do* not *assume associativity, commutativity, a unit, or distributivity of $\otimes$ over joins. The proofs below only need: (i) we can form a join $\oplus S$ for* any *subset $S \subseteq V$ (so V is complete as a join-semilattice, indeed a complete lattice), and (ii) if $a \le a'$ and $b \le b'$ then $a \otimes b \le a' \otimes b'$. In particular, $\oplus\{\cdots\}$ denotes the least upper bound (supremum) of the displayed set, and the symbol $\perp$ (used later) denotes the least element of V (the join of the empty set).*

Intuitively, one may read V as a space of "degrees" (truth degrees, intensities, plausibilities, costs-as-ordered, etc.), and $\otimes$ as some way of composing two degrees. The link to the Hyperseed notion of Quantale Weakness *(Hyperseed-Concept 143) and* Weakness *(Hyperseed-Concept 202) is conceptual rather than syntactic here: we are using the minimal order-theoretic skeleton that makes the later monotonicity arguments go through; compare [3, 2].*

It is worth emphasizing what monotonicity of $\otimes$ gives (and what it does not). Monotonicity guarantees that if one replaces an input degree by a weaker/greater one (depending on the chosen interpretation of $\le$), then the composed degree cannot become strictly stronger/smaller; this is precisely the kind of "no miracles under weakening" property needed when transporting statements along coarse-grainings. By contrast, associativity or distributivity would enable algebraic reshuffling of multi-step compositions or commuting joins past $\otimes$; these are powerful but unnecessary for the basic fine-to-coarse soundness arguments, which rely only on order and supremum constructions.

Remark 2770. *Nothing below uses special properties of the canonical p-bit quantale beyond: (1) joins exist; (2) monotonicity of $\otimes$.*

One can read this as a robustness claim: the monotone inference principles proved later are not artifacts of a particular numerical scale (such as probabilities, p-bits, or log-odds), but follow from much weaker structural assumptions. In practice, this flexibility lets one instantiate V by a variety of ordered domains (e.g. bounded truth degrees, resource bounds, or penalty scores) as long as "combining evidence/resources" is monotone. This is also why the section speaks of "quantale-lite" rather than committing to a fully quantalic semantics at the outset: the later constructions are meant to apply uniformly across many choices of degree domain, provided only that the relevant suprema and monotone compositions exist.

49.2 Coarse-graining as sup-pushforward

Definition 452 (Coarse-graining data)**.** *Let X be a set of fine states and X_0 a set of coarse states. A coarse-graining map is a surjection $\alpha : X \to X_0$.*

Let P be a set of fine patterns and P_0 a set of coarse patterns, with an arbitrary map $\beta : P \to P_0$. An intensity is any function $I : P \times X \to V$.

Remark 2771. *Although the definition above treats V as an arbitrary codomain, the constructions below use $\oplus$ (and a bottom element $\perp$ for empty joins). Thus, minimally, one may take V to be a poset equipped with the relevant joins of subsets that arise as fibers*

$$\{ I(p, x) : \beta(p) = p_0,\ \alpha(x) = x_0 \}.$$

In many applications it is convenient to assume V is a complete join-semilattice (or a complete lattice), so that every *subset has a join. This ensures that the sup-pushforward is always defined without side conditions, and it clarifies why the empty-fiber value is naturally $\perp$ (the join of the empty set).*

Remark 2772. *The picture is: X is the world described in high resolution, while X_0 is what remains after we "forget" some distinctions (Hyperseed-Concept 98). The surjection $\alpha : X \to X_0$ identifies many fine states with a single coarse state; surjectivity ensures every coarse state is genuinely represented by at least one fine state. Likewise, $\beta : P \to P_0$ allows us to map fine-grained pattern descriptions to coarser pattern descriptions (or to merge multiple fine patterns into one coarse label). The intensity function $I(p, x) \in V$ may be read as "how strongly pattern p applies to state x," generalizing the scalar intensities used earlier for patterns and emergence (Hyperseed-Concept 130,* **??***; see [5]).*

A simple example: let X be pixel images and X_0 be their downsampled versions; α is the downsampling. Let P be a set of fine edge-detectors, and P_0 a set of coarse edge-detectors; β maps each fine detector to its coarse "family." Then $I(p, x)$ could be a graded edge strength. The point of the formalism is that it does not depend on the particulars of images or edges: it is a general abstraction mechanism.

Remark 2773. *It is useful to keep two kinds of "many-to-one" effects distinct. The surjection α guarantees that every coarse state x_0 has at least one fine representative x (so every x_0 corresponds to a nonempty fiber $\alpha^{-1}(x_0)$). In contrast, β is not required to be surjective: some coarse pattern labels $p_0 \in P_0$ may have no fine patterns mapping to them, in which case the coarse intensity at (p_0, x_0) will typically collapse to $\perp$ under sup-pushforward. This is often desirable: it records that the coarse label is* unrealized *(or unsupported) by the current fine vocabulary.*

Definition 453 (Sup-pushforward intensity)**.** *Given (α, β) and $I : P \times X \to V$, define $I_0 : P_0 \times X_0 \to V$ by*

$$I_0(p_0, x_0) \ :=\ \oplus\{ I(p, x) \ :\ \beta(p) = p_0,\ \alpha(x) = x_0 \}.$$

(If the constraint set is empty, interpret the join as $\perp$.)

Remark 2774. *This is the central construction. The set*

$$\{\, I(p,x) \;:\; \beta(p) = p_0,\ \alpha(x) = x_0 \,\}$$

collects all *fine-level intensities consistent with the coarse labels (p_0, x_0). Taking their join (supremum) makes $I_0(p_0, x_0)$ the* best-case *intensity available among the fine witnesses.*

Two small examples clarify the semantics.

- *If $V = [0,1]$ ordered by $\leq$ and $\oplus$ is* sup*, then $I_0(p_0, x_0)$ is the maximum fine intensity compatible with (p_0, x_0). Thus coarse intensity records that "there exists a refinement where the pattern is this strong."*

- *If $V = \{0,1\}$ with $\oplus = \vee$, then $I_0(p_0, x_0) = 1$ iff there exists some refinement (p, x) mapping to (p_0, x_0) with $I(p,x) = 1$. This is the ordinary existential pushforward of a predicate along a map.*

This "$\oplus$-over-fibers" operation is the posetal analogue of a left Kan extension: we extend the fine intensity along the coarse identification by taking the least upper bound compatible with the coarse data. In a reasoning engine, this is the canonical way to compute a coarse observable that is sound for existential-style inferences (and is aligned with the general Hyperseed ontology stance that many statements are resolution-relative; compare [1, 19]).

Remark 2775. *One can also read sup-pushforward as an "existential quantification" over refinements, generalized from booleans to an arbitrary join-structure: fixing (p_0, x_0), we range over all (p, x) that refine it, and aggregate their contributions by $\oplus$. In particular, when V supports graded notions of evidence, $I_0(p_0, x_0)$ expresses the strongest evidence achievable by choosing an appropriate fine witness consistent with the coarse observation.*

Concretely, if one forms the relation (or correspondence)

$$R \subseteq (P \times X) \times (P_0 \times X_0), \qquad ((p,x),(p_0,x_0)) \in R \iff \beta(p) = p_0 \ \wedge\ \alpha(x) = x_0,$$

then the definition is exactly "push I forward along R by taking joins along the R-fibers." This viewpoint often makes later monotonicity statements feel like familiar facts about existential images.

Remark 2776. *The choice of $\oplus$ encodes an* optimistic *(best-case, witness-based) abstraction principle. Other choices are possible in other inference regimes: for instance, an $\bigwedge$-over-fibers construction would encode a* pessimistic *(worst-case, all-refinements) principle, corresponding to universal rather than existential reasoning. The present section focuses on $\oplus$ because it is the one that preserves witnesses under forgetting, which is the monotone inference behavior used repeatedly later.*

Theorem 63 (Dominance of sup-pushforward)**.** *For all $p \in P$ and $x \in X$,*

$$I_0(\beta(p), \alpha(x)) \ \geq\ I(p,x).$$

Remark 2777. *Intuitively, the theorem says:* coarsening by sup-pushforward can only increase (or preserve) intensity*. Whatever strength a pattern has at a fine state remains attainable when we forget detail, because the coarse value is defined by taking a join over all consistent fine values and therefore cannot fall below any one of them.*

This is the basic "soundness" guarantee that underwrites later monotonicity results: once we accept that coarse intensity is defined by a supremum over fine witnesses, then any fine witness remains a witness after coarsening. This connects directly to the later "threshold persistence" corollary and, in a different guise, to the effability and registration results, which are also essentially "witness persistence under forgetting."

Remark 2778. *It can also be helpful to view the inequality as a form of* extensiveness *of an abstraction operator: starting from a particular fine datum (p, x), mapping it to the coarse pair $(\beta(p), \alpha(x))$ and then evaluating I_0 yields a value that dominates the original intensity. Equality holds exactly when the chosen witness (p, x) already attains the fiber-maximum, i.e. when $I(p, x)$ is a join-maximal element among all $I(p', x')$ with the same coarse image. Thus the strictness of the inequality measures how much "hidden headroom" is present inside the coarse equivalence class.*

Proof. Fix $p \in P$ and $x \in X$. In the defining join for $I_0(\beta(p), \alpha(x))$, the pair (p, x) is a valid witness because $\beta(p) = \beta(p)$ and $\alpha(x) = \alpha(x)$. Therefore the join is at least $I(p, x)$. □

Remark 2779. Proof sketch. *The proof uses only one idea: a join is an upper bound for each element being joined. Since (p, x) itself appears among the candidates for $I_0(\beta(p), \alpha(x))$, the supremum cannot be smaller than $I(p, x)$.*

The key step is recognizing that the "fiber constraints" $\beta(p) = p_0$ and $\alpha(x) = x_0$ are satisfied tautologically when $p_0 = \beta(p)$ and $x_0 = \alpha(x)$. Visually, if one imagines α and β as partitioning X and P into coarse equivalence classes, the point is that each fine element sits inside its own class, so it is always available as a representative when we compute the coarse supremum.

Remark 2780. *Categorically inclined readers may recognize the same mechanism in the adjunction between pullback and $\oplus$-pushforward for V-valued relations: precomposition along (β, α) (a pullback of intensities from coarse to fine) has a left adjoint given by $\oplus$ over the fibers. The dominance inequality is then the unit inequality of that adjunction specialized to the posetal setting. This perspective is optional here, but it helps explain why the construction is "canonical": it is forced by the requirement that the coarse observable be the least one that still soundly supports existential witness-lifting.*

Corollary 31 (Threshold persistence (sound fine-to-coarse rule))**.** *Fix any threshold $\theta \in V$. If $I(p, x) \geq \theta$, then $I_0(\beta(p), \alpha(x)) \geq \theta$.*

Remark 2781. *This corollary is the operational form of the previous theorem: it tells a reasoning engine that "if I can certify intensity $\geq \theta$ at the fine level, then I may assert intensity $\geq \theta$ at the coarse level." In many applications, θ encodes an activation threshold or a decision boundary, so this is precisely a* sound abstraction rule*. One may read it as a formal instance of "truth persists under forgetting" when truth is expressed as crossing an order-threshold.*

Remark 2782. *From the standpoint of automated reasoning, threshold persistence is a "monotone compilation" principle: any rule or classifier validated at fine resolution remains valid after abstraction, provided it is phrased in terms of inequalities that are preserved by $\geq$-monotonicity. In particular, when V is numeric and θ is tuned to control false positives/false negatives, this corollary explains why coarse-grained systems often become* more permissive*: they can inherit any strong fine witness hidden inside the coarse bin, but they cannot distinguish whether that witness is typical or exceptional without additional structure.*

Proof. By the previous theorem, $I_0(\beta(p), \alpha(x)) \geq I(p, x) \geq \theta$. □

Remark 2783. Proof sketch. *This is a one-line transitivity argument: dominance gives $I_0 \geq I$, and the assumption gives $I \geq \theta$, hence $I_0 \geq \theta$. In other words, the order $\geq$ on V is being used only through its transitivity: if an abstracted quantity sits above the concrete one, and the concrete one sits above a fixed threshold, then the abstracted quantity must also sit above that same threshold. No additional algebraic structure (beyond the order) is needed for this step.*

The conceptual point is that thresholds behave well with respect to monotone maps and joins: once we have $a \leq b$, any lower bound $\theta \leq a$ is also a lower bound for b. Equivalently, the set of elements above θ is upward closed: enlarging information (moving upward in the order) cannot invalidate a previously satisfied lower-bound constraint. This upward-closedness viewpoint is what later makes it natural to speak of "safe abstraction" under coarse-graining: we may lose distinctions, but we do not lose the validity of monotone consequences expressible as lower bounds.

49.2.1 Compositionality

Definition 454 (Successive coarse-grainings). *Let $X \xrightarrow{\alpha_1} X_0 \xrightarrow{\alpha_2} X_{00}$ be surjective maps. Let $P \xrightarrow{\beta_1} P_0 \xrightarrow{\beta_2} P_{00}$ be any maps. The surjectivity assumption on the α_i emphasizes that every coarse state has at least one refinement; this is the typical situation when α_i is a "forgetful" map induced by a partition of X. (If one drops surjectivity, one must adopt a convention for joins over empty fibers, e.g. taking them to be a bottom element when it exists.)*

Given $I : P \times X \to V$, define:

- *I_1 = sup-pushforward of I along (α_1, β_1),*
- *I_2 = sup-pushforward of I_1 along (α_2, β_2),*
- *I_{12} = sup-pushforward of I along $(\alpha_2 \circ \alpha_1, \beta_2 \circ \beta_1)$.*

Here the phrase "sup-pushforward" implicitly assumes that the relevant joins exist in V for the fibers determined by the constraints $\alpha_i(x) = x_i$ and $\beta_i(p) = p_i$; in many applications V is a complete lattice or at least admits the required (possibly finite) joins. Note also that we do not require the β_i to be surjective: parameter reindexing may collapse or discard parameters, and the compositionality statement below only compares the values at (p_{00}, x_{00}) where the defining joins are taken over whatever preimages exist.

Remark 2784. *This definition formalizes the idea that coarsening can happen in stages: perhaps we first forget fine pixel-level detail, and later forget color channels; or first aggregate individuals into households, then households into regions. The two procedures for producing a twice-coarsened intensity are:*

(a) push forward once, then push forward again; or

(b) push forward directly along the composite forgetting map.

The theorem below says these agree exactly, which is a kind of associativity law for abstraction. This is the reason one may regard sup-pushforward as a genuine "inference principle": it composes cleanly, like a functorial construction in a posetal/categorical setting (compare the thin Kan-extension remark above, and the general compositional stance of [1]). From the poset viewpoint, the content is that taking a left Kan extension along a composite map agrees with composing the corresponding left Kan extensions; in the thin (order-enriched) setting the proof reduces to manipulating joins and index sets. Concretely, this compositionality is what allows one to build multi-resolution pipelines without having to re-justify each new "level" as an ad hoc construction: the same abstraction operator can be iterated without changing its meaning.

Theorem 64 (Sup-pushforward is compositional). *For all $p_{00} \in P_{00}$ and $x_{00} \in X_{00}$,*

$$I_2(p_{00}, x_{00}) \;=\; I_{12}(p_{00}, x_{00}).$$

Remark 2785. *In plain terms:* it does not matter whether we coarse-grain in one step or many; the resulting coarse intensity is the same. *This is important because practical systems often maintain multiple levels of representation and may switch between them dynamically. Without compositionality, the meaning of a coarse statement would depend on the path taken to get there, which would be an undesirable kind of epistemic relativity. One can also read the equation $I_2 = I_{12}$ as a path-independence property on the "abstraction graph" of representations: any two routes that forget the same information yield the same abstracted value when the forgetting maps agree on the nose.*

This theorem connects back to "dominance" because dominance is what makes each pushforward a safe abstraction; compositionality is what makes sequences of safe abstractions themselves coherent. In particular, dominance is a pointwise inequality that ensures no coarse value undershoots any of its refinements, while compositionality ensures that repeatedly taking such safe upper envelopes does not accidentally introduce extra slack beyond what the final forgetting map already enforces.

Proof. Fix (p_{00}, x_{00}). By definition,

$$I_2(p_{00}, x_{00}) = \oplus_{\substack{\beta_2(p_0)=p_{00} \\ \alpha_2(x_0)=x_{00}}} I_1(p_0, x_0).$$

Expand I_1:

$$I_1(p_0, x_0) = \oplus_{\substack{\beta_1(p)=p_0 \\ \alpha_1(x)=x_0}} I(p, x).$$

Substitute into the expression for $I_2(p_{00}, x_{00})$ and use associativity/idempotence of joins (joining over a join) to get a single join over all (p, x) satisfying $\beta_2(\beta_1(p)) = p_{00}$ and $\alpha_2(\alpha_1(x)) = x_{00}$, i.e. $(\beta_2 \circ \beta_1)(p) = p_{00}$ and $(\alpha_2 \circ \alpha_1)(x) = x_{00}$. But this is exactly the definition of $I_{12}(p_{00}, x_{00})$. More explicitly, one is using the elementary fact that for a family $\{a_{i,j}\}$ indexed by pairs (i, j) in some relation $R \subseteq I \times J$, one has

$$\oplus_{i\in I} \oplus_{\substack{j\in J \\ (i,j)\in R}} a_{i,j} = \oplus_{(i,j)\in R} a_{i,j},$$

provided the relevant joins exist; this is precisely the "flattening" of a nested join into a join over a combined indexing set. No distributivity assumptions are invoked: only the universal property of $\oplus$ as the least upper bound (and hence its associativity/commutativity/idempotence as an n-ary operation on finite index sets, or the corresponding coherence for the existing joins in the general case). $\square$

Remark 2786. Proof sketch. *The strategy is "flattening": write I_2 as a join of I_1's, then write each I_1 as a join of I's. A join of joins can be flattened into a single join over the combined index set; this uses only the universal property of joins (equivalently, that taking joins is associative and idempotent at the level of indexing). One can also regard this as a Fubini-type principle for suprema: iterated suprema over successive fibers coincide with a single supremum over the total fiber of the composite map.*

The key step is matching the indexing constraints: the nested constraints

$$\beta_2(p_0) = p_{00},\ \beta_1(p) = p_0 \quad \textit{and} \quad \alpha_2(x_0) = x_{00},\ \alpha_1(x) = x_0$$

are equivalent to the single composite constraints

$$(\beta_2 \circ \beta_1)(p) = p_{00}, \qquad (\alpha_2 \circ \alpha_1)(x) = x_{00}.$$

This equivalence is simply the elimination of the intermediate variables p_0 and x_0: choosing an intermediate representative and then requiring consistency is the same as directly requiring that the *coarse labels match after both forgetting steps. In relational terms, the set of admissible pairs* (p, x) *is the pullback of the singleton* $\{(p_{00}, x_{00})\}$ *along* $(\beta_2 \circ \beta_1, \alpha_2 \circ \alpha_1)$*, and the nested join computes the supremum over that pullback by first grouping elements according to their intermediate images.*

Geometrically, one can picture the refinement maps as partitions: refining in two stages means first grouping fine elements into intermediate blocks, then grouping those blocks into larger blocks. The theorem says that taking a supremum over fine elements inside the final large block can be done either directly or by first taking suprema inside intermediate blocks and then taking the supremum over those intermediate suprema. In particular, the intermediate blocks function only as a book-keeping device: they may be convenient computationally or conceptually, but they do not alter the final abstract value determined by the coarsest partition.

49.3 Weakness under coarse-graining

Definition 455 (Crisp indistinction relation and weakness)**.** *Let* $H \subseteq X \times X$ *be a (crisp) indistinction relation (failed distinctions). Let* $\mu : X \to V$ *be an importance weight.*

Define the weakness of H *(relative to* μ*) by*

$$w_X(H) \ := \ \oplus_{(u,v)\in H}\ \mu(u) \otimes \mu(v).$$

Remark 2787. *Here* H *encodes which pairs of states the system fails to distinguish; this is an explicit formal avatar of the Hyperseed notion of* Distinction *and its failure (Hyperseed-Concept 98). Calling* H *"crisp" means membership is Boolean (either a pair is indistinct or not), in contrast to a graded indistinction weight used later.*

It is intentional that no further axioms are imposed on H *here:* H *need not be an equivalence relation, need not be symmetric, and need not include the diagonal. In applications one often considers symmetric* H *(mutual confusion) or reflexive* H *(each state is trivially indistinct from itself), but the definition of weakness does not require these. Notably, whether* $(u, u) \in H$ *is included affects whether "self-confusions" contribute to weakness; excluding the diagonal focuses the measure on* failures to separate distinct states*, while including it can be used to model degenerate sensors that sometimes fail even to stably identify a state with itself.*

The weakness $w_X(H)$ *aggregates the "importance" of indistinguishable pairs. If* $\mu(u)$ *measures how consequential state* u *is (in attention, reward, probability mass, salience, etc.), then* $\mu(u)\otimes\mu(v)$ *measures the combined seriousness of failing to distinguish* u *from* v*, and the join over* H *reports the overall weakness level. In many quantale-weakness instantiations (Hyperseed-Concept 143),* $\oplus$ *is a supremum and* $\otimes$ *behaves like a conjunction or product; but theorems below only require monotonicity. This aligns with the stance in [3, 2] that "weakness" is a structurally robust notion across many concrete choices of semantics.*

The definition also makes sense for edge cases: if $H = \varnothing$*, then* $w_X(H)$ *is the join over an empty set, i.e. the bottom element of the* V*-order (when such a bottom exists, as in standard complete lattices). Thus "no failed distinctions recorded" corresponds to minimal weakness, which matches the intended reading.*

A toy example: let $V = [0, 1]$ *with* $\oplus = \sup$ *and* $\otimes = \min$*. Then* $w_X(H)$ *is simply the maximum, over indistinct pairs, of* $\min(\mu(u), \mu(v))$*; so it reports whether there exists a highly important pair that is being conflated. If instead* $\otimes$ *were taken as ordinary multiplication, then* $\mu(u) \otimes \mu(v) = \mu(u)\mu(v)$ *would penalize confusions between two simultaneously high-importance states more sharply, while still preserving the monotonicity-based arguments used below.*

Definition 456 (Pushforward of weights and relations)**.** *Let* $\alpha : X \to X_0$ *be surjective. Define pushed-forward importance* $\mu_0 : X_0 \to V$ *by*

$$\mu_0(x_0) \ := \ \oplus\{\mu(x) : \alpha(x) = x_0\}.$$

Define pushed-forward indistinction $H_0 \subseteq X_0 \times X_0$ by:

$$(x_0, y_0) \in H_0 \iff \exists (x, y) \in H \text{ with } \alpha(x) = x_0,\ \alpha(y) = y_0.$$

Equivalently, $H_0 = (\alpha \times \alpha)[H]$.

Remark 2788. *The pushforward μ_0 is the same "sup over fibers" idea used for intensities: the importance of a coarse state x_0 is taken as the join of the importances of its fine refinements. This is a natural choice when one wants the coarse system to remain conservative: if any fine refinement is important, then the coarse representative should be regarded as important too.*

Surjectivity of α makes X_0 genuinely a quotient/coarsening of X: every coarse state has at least one fine representative, so the fiber join is always taken over a nonempty set. If α were not surjective, one would have to decide what importance to assign to coarse states with empty fibers (typically the bottom of V), but the intended "coarse-graining by forgetting distinctions" picture is clearest in the surjective case.

The use of $\oplus$ rather than, say, addition or averaging is deliberate: it produces an upper envelope *abstraction. In probabilistic interpretations this is not the usual marginalization (which would sum probabilities), but it is the right operation for worst-case/conservative propagation of importance and for monotone inference principles: the coarse system is required not to understate the importance of any fine possibility that it collapses.*

The pushed-forward relation H_0 says that two coarse states are indistinct if they have some *indistinct fine representatives. This is again an existential definition: coarse indistinction holds whenever the fine system contains a witness of indistinction that survives forgetting. In terms of* Self vs. Other *boundaries and perceptual confusions (Hyperseed-Concept 165), this is the minimal condition under which a confusion at high resolution still counts as a confusion after merging states.*

It is also worth noting what information is intentionally discarded by this pushforward: multiple distinct fine witnesses of indistinction between the same coarse pair (x_0, y_0) do not accumulate as multiple counts; H_0 only records that indistinction is possible*. This choice aligns with the subsequent weakness functional, which already aggregates contributions via $\oplus$ rather than by counting or summing occurrences.*

Theorem 65 (Coarse-graining does not decrease weakness)**.** *With $w_X(H)$ and $w_{X_0}(H_0)$ computed using μ and μ_0 as above,*

$$w_X(H) \ \leq \ w_{X_0}(H_0).$$

Remark 2789. *The theorem states:* when we coarse-grain by taking joins over refinements, the measured weakness cannot go down*. This matches intuition: forgetting distinctions cannot make you* less *weak with respect to the distinctions you were already failing to draw. If anything, coarsening tends to make weakness more visible, because more fine importance mass is aggregated into fewer coarse bins, and indistinction witnesses are preserved.*

One can also read this as a "soundness under abstraction" statement: $w_{X_0}(H_0)$ is a safe upper bound on the fine weakness $w_X(H)$ computed before forgetting detail. Thus any downstream principle that enforces a bound on weakness at the coarse level (e.g. "keep weakness $\leq \theta$") is automatically conservative with respect to the fine level.

A concrete micro-example: suppose $X = \{a, b, c\}$, $X_0 = \{A, C\}$ with $\alpha(a) = \alpha(b) = A$ and $\alpha(c) = C$. Let $H = \{(a, c)\}$ record a single failed distinction between a and c. Then $H_0 = \{(A, C)\}$. If $\mu(a)$ is small but $\mu(b)$ is large, the pushforward makes $\mu_0(A) = \mu(a) \oplus \mu(b)$, so the coarse weakness for the pair (A, C) can become large even though the original witness involved only a rather than b. This captures the intended conservatism: after merging a and b, any confusion between some *representative of A and C becomes a potential confusion involving the whole coarse bin A, and the bin inherits the maximum importance among its refinements.*

This is a monotonicity fact of the same logical type as the dominance theorem for intensities. It also connects to the later effability monotonicity theorem: there, coarsening the indistinction weight makes fidelity easier *to achieve, hence effability sets grow; here, coarsening states makes weakness* larger*, hence weakness bounds become more conservative. Both are instances of the general principle: "sup-pushforward turns fine information into coarse information by taking an upper envelope." See [3] for broader discussion of weakness under abstraction.*

Finally, observe that the inequality direction depends on two design choices acting together: (i) μ_0 uses a join over fibers (so importance can only increase when passing to the quotient), and (ii) H_0 uses existential pushforward (so indistinction witnesses can only be added, not removed). Changing either choice changes which quantities become monotone under coarse-graining.

Proof. Fix any $(u, v) \in H$. Then by definition $(\alpha(u), \alpha(v)) \in H_0$. Also by construction of μ_0, $\mu_0(\alpha(u)) \geq \mu(u)$ and $\mu_0(\alpha(v)) \geq \mu(v)$. By monotonicity of $\otimes$,

$$\mu(u) \otimes \mu(v) \ \leq\ \mu_0(\alpha(u)) \otimes \mu_0(\alpha(v)).$$

Now take joins. The left join ranges over all $(u, v) \in H$. The right join ranges over all pairs in H_0, which in particular includes all $(\alpha(u), \alpha(v))$ with $(u, v) \in H$. Therefore

$$\oplus_{(u,v)\in H}\mu(u) \otimes \mu(v) \ \leq\ \oplus_{(x_0,y_0)\in H_0}\mu_0(x_0) \otimes \mu_0(y_0).$$

□

Remark 2790. Proof sketch. *The proof transports each fine indistinction witness $(u, v) \in H$ to a coarse witness $(\alpha(u), \alpha(v)) \in H_0$, and shows its contribution can only increase because $\mu_0(\alpha(u))$ and $\mu_0(\alpha(v))$ are defined as joins that dominate $\mu(u), \mu(v)$. Monotonicity of $\otimes$ then propagates this domination to the pairwise combined importance, and taking joins preserves the inequality.*

The key step is that the definition of H_0 is existential: every fine indistinction produces a coarse indistinction. If H_0 were defined universally (e.g. requiring all *representatives to be indistinct), the inequality could fail. So the theorem is also a lesson in how the choice of pushforward definition encodes a philosophy of abstraction: existential pushforward yields safe upper bounds on weakness.*

Technically, the argument uses only three monotonicity facts: (1) each fiber join satisfies $\mu(x) \leq \mu_0(\alpha(x))$, (2) $\otimes$ is monotone in each argument, and (3) $\oplus$ is monotone with respect to enlarging the index set (here, H mapping into H_0). No commutativity or associativity properties of $\otimes$ are needed for this particular inequality, and no algebraic properties of H beyond being a subset are required. This minimality is important for later reuse when H is replaced by more structured, graded, or context-dependent notions of indistinction.

Corollary 32 (Iterated coarse-graining yields a nondecreasing weakness chain)**.** *For a chain of surjections $X \to X_0 \to X_{00} \to \cdots$ with corresponding pushed-forward relations and pushed-forward weights, the sequence of weaknesses is nondecreasing.*

Remark 2791. *This corollary says that as we ascend a hierarchy of ever coarser descriptions (Hyperseed-Concept* **??***), the weakness measure forms an ascending chain. In decision terms, once a weakness bound is exceeded at some coarse level, it will remain exceeded at all coarser levels. This is useful for pruning: one can detect unacceptable weakness early without needing to compute fine-grained details.*

Proof. Apply the previous theorem at each step and compose the inequalities. □

Remark 2792. Proof sketch. *Each coarse-graining step yields an inequality $w_{fine} \le w_{coarse}$. Chaining these inequalities yields monotonicity along the entire tower.*

Conceptually, this is the order-theoretic analogue of a renormalization flow that is monotone in a chosen statistic: weakness is a quantity that cannot be "improved" merely by forgetting structure.

49.4 Effability monotonicity under coarsening

Definition 457 (Fidelity under indistinction weights)**.** *Let X be finite. An indistinction weight function is any $\iota : X \times X \to [0,1]$. Given a target $x \in X$ and a distribution $q \in \Delta(X)$, define fidelity:*

$$\mathrm{Fid}_\iota(x, q) \;:=\; \sum_{y \in X} q(y)\, \iota(x, y).$$

Remark 2793. *This definition introduces a* graded *notion of indistinction, in contrast to the crisp relation H above. The function $\iota(x, y)$ quantifies how acceptable it is to confuse y with x at a given resolution: $\iota(x, y) = 1$ means "y is indistinguishable from x" (perfectly acceptable), and $\iota(x, y) = 0$ means "completely wrong." This corresponds directly to the Hyperseed idea that* Ineffability *(Hyperseed-Concept* **??***) is not absolute but resolution- and resource-relative.*

The distribution $q \in \Delta(X)$ represents what an observer's description decodes to: it is a probabilistic hypothesis over fine states. The fidelity $\mathrm{Fid}_\iota(x, q)$ *is then the expected indistinction weight under q. Two simple examples:*

- *If $\iota(x, y) = \mathbf{1}[x = y]$, then* $\mathrm{Fid}_\iota(x, q) = q(x)$*, the ordinary probability of being exactly correct.*
- *If $\iota(x, y)$ is* 1 *whenever x and y share a coarse label (say $\alpha(x) = \alpha(y)$) and* 0 *otherwise, then fidelity measures the probability that q lands in the correct coarse class.*

Thus, ι formalizes a "metric of adequacy" for descriptions, connecting naturally with Perception *and* Presentational Immediacy *(Hyperseed-Concept 134, 138) insofar as these are always mediated by permissible confusions.*

Definition 458 ((epsilon,K)-effability)**.** *Fix an observer O equipped with: a description space D_O, a cost $\sigma_O : D_O \to [0, \infty)$, and a decoder* $\mathrm{dec}_O : D_O \to \Delta(X)$*.*

Fix $\epsilon \in (0, 1)$ and $K > 0$. An element $x \in X$ is (ϵ, K)-effable (relative to ι) iff there exists $d \in D_O$ such that $\sigma_O(d) \le K$ and $\mathrm{Fid}_\iota(x, \mathrm{dec}_O(d)) \ge 1 - \epsilon$*. Let* $\mathrm{Eff}_O(\epsilon, K; \iota) \subseteq X$ *denote the effable set.*

Remark 2794. *This definition makes* effability *explicitly a function of two constraints:*

- *an error tolerance ϵ (how close we must get), and*
- *a resource budget K (how costly a description we are allowed).*

In Hyperseed language, this is a precise way to talk about Ineffability *and its negation under bounded* Effort *(Hyperseed-Concept* **??***, 100) and bounded representational capacity (Hyperseed-Concept 157). The observer model* $(D_O, \sigma_O, \mathrm{dec}_O)$ *is intentionally abstract; it can represent symbolic language, neural codes, scientific theories, or internal cognitive encodings (compare [19, 20]).*

A concrete example: let D_O be bitstrings, $\sigma_O(d)$ be length, and $\mathrm{dec}_O(d)$ *be a probabilistic decoder (e.g. a Bayesian posterior over X). Then (ϵ, K)-effable means: "there exists a K-bit message whose decoded hypothesis assigns at least $1 - \epsilon$ fidelity to x under the indistinction weighting ι." This resembles algorithmic information viewpoints ([16]) but with an explicitly resolution-relative success criterion.*

Theorem 66 (Monotonicity in resources and resolution)**.** *Let $K' \geq K$. Then*

$$\mathrm{Eff}_O(\epsilon, K; \iota) \subseteq \mathrm{Eff}_O(\epsilon, K'; \iota).$$

If ι and ι' satisfy $\iota(x, y) \leq \iota'(x, y)$ for all x, y (i.e. ι' is coarser), then

$$\mathrm{Eff}_O(\epsilon, K; \iota) \subseteq \mathrm{Eff}_O(\epsilon, K; \iota').$$

Remark 2795. *The theorem says two things, both unsurprising yet foundational for disciplined reasoning:*

1. *If you are allowed to spend more description cost, you can effably describe at least as many things.*
2. *If you relax the fidelity criterion by making more confusions acceptable (i.e. you coarsen the indistinction weights), then effability becomes easier, so the effable set can only grow.*

In short: more resources and lower resolution both increase effability*.*

This result complements the earlier weakness monotonicity result. There, coarsening increased a defect measure *(weakness). Here, coarsening increases a* capability set *(effability). Both are monotone phenomena, and both are indispensable if one wants stable meta-reasoning about what can or cannot be expressed under bounded resources (cf. [19]).*

Proof. (Resource monotonicity.) If x has a witnessing description d with $\sigma_O(d) \leq K$, then also $\sigma_O(d) \leq K'$.

(Resolution monotonicity.) Fix x and any distribution q. If $\iota \leq \iota'$ pointwise, then

$$\mathrm{Fid}_{\iota'}(x, q) = \sum_y q(y)\iota'(x, y) \;\geq\; \sum_y q(y)\iota(x, y) = \mathrm{Fid}_\iota(x, q),$$

since $q(y) \geq 0$ and sums preserve inequality. Therefore any d that meets the $1 - \epsilon$ threshold under ι also meets it under ι'. □

Remark 2796. Proof sketch. *For resource monotonicity, keep the same witness description: a larger budget cannot invalidate a previously affordable description. For resolution monotonicity, observe that fidelity is a nonnegative weighted sum of $\iota(x, y)$ values, so increasing ι pointwise can only increase fidelity.*

The key step is the use of pointwise order $\iota \leq \iota'$ and nonnegativity of q: this is exactly the kind of elementary inequality that underlies many "data processing" principles in information theory, but here it is presented in its bare order-theoretic form.

Corollary 33 (Safe contrapositive rule for reasoning engines)**.** *If $x \notin \mathrm{Eff}_O(\epsilon, K; \iota')$ for some coarser $\iota' \geq \iota$, then $x \notin \mathrm{Eff}_O(\epsilon, K; \iota)$. Equivalently: ineffability at an easier (coarser) resolution implies ineffability at a harder (finer) one.*

Remark 2797. *This corollary is often the form one actually uses: it gives a* safe negative inference*. If an entity cannot be described even when we permit generous confusions (coarse ι'), then it certainly cannot be described when we demand stricter fidelity (fine ι). For automated systems, this supports early stopping and conservative claims of ineffability (Hyperseed-Concept* **??***) without having to search at the finest resolution first.*

Proof. Contrapositive of $\mathrm{Eff}_O(\epsilon, K; \iota) \subseteq \mathrm{Eff}_O(\epsilon, K; \iota')$. □

Remark 2798. Proof sketch. *If $A \subseteq B$, then "not in B" implies "not in A"; apply this with $A = \mathrm{Eff}_O(\epsilon, K; \iota)$ and $B = \mathrm{Eff}_O(\epsilon, K; \iota')$.*

The deeper intuition is that effability sets are upward-closed with respect to both budgets and coarsening of indistinction: a failure at the top propagates downward.

49.5 Downward closure of registration (optional, but useful)

Definition 459 (Context refinement and forgetting)**.** *Let $\mathcal{C}$ be a poset of contexts, where $C \preceq D$ means "D refines C". Assume for each C a set U_C and a map $\mathrm{asp}_C : U \to U_C$. Write $x|_C := \mathrm{asp}_C(x)$.*

For $C \preceq D$, assume a forgetful map $\pi_{D\to C} : U_D \to U_C$ with $x|_C = \pi_{D\to C}(x|_D)$ for all $x \in U$.

Remark 2799. *This definition introduces a general apparatus for moving between contexts (Hyperseed-Concept 86). The poset $\mathcal{C}$ encodes a refinement order: D contains more distinctions than C. The map $\mathrm{asp}_C : U \to U_C$ extracts the "aspect" of a global entity $x \in U$ as seen in context C; notation $x|_C$ emphasizes that we are restricting attention.*

The coherence condition $x|_C = \pi_{D\to C}(x|_D)$ states that forgetting commutes with taking aspects: if you view x in the fine context D and then forget down to C, you get the same result as viewing x in C directly. This is the minimal algebra one needs for consistent multi-resolution reasoning, and it is one of the standard background assumptions in compositional ontologies (cf. [1]).

A simple example: U could be full physical world-histories; U_C could be coarse summaries such as "macrostates" in thermodynamics; and $\pi_{D\to C}$ could be marginalization/aggregation.

Definition 460 (Registration schema)**.** *Fix a system X and time t. In context C, assume: (i) an equivalence relation $\sim_C$ on U_C, (ii) a pattern set P_C, (iii) an intensity map $\mathrm{Int}_{C,t} : P_C \times U_C \to V$, (iv) a reference predicate $\mathrm{Ref}_C(\cdot,\cdot)$.*

Fix a threshold $\theta \in V$.

We say "X registers A at time t in context C" if there exist: an entity $A_0 \in U$ and a pattern $p_C \in P_C$ such that

$$A_0|_C \sim_C A|_C, \quad \mathrm{Int}_{C,t}(p_C, A_0|_C) \not\leq \theta, \quad \mathrm{Ref}_C(p_C, A|_C) \textit{ is nontrivial.}$$

Remark 2800. *This schema formalizes a familiar epistemic pattern: a system "registers" an entity A when it can bring to bear some pattern p_C that applies with intensity above threshold, and when that pattern actually* refers *to A in a nontrivial way (Hyperseed-Concept 153, 152). The auxiliary witness A_0 allows registration up to contextual equivalence: it may suffice to register something equivalent to A under $\sim_C$ (Hyperseed-Concept* **??***), rather than A itself with absolute precision.*

A basic example: in a visual context C, A is a person, $\sim_C$ identifies people up to clothing color, p_C is a face template, Int *measures match strength, and Ref_C encodes that the template genuinely tracks the person rather than some spurious correlation. The threshold θ is then an attention/recognition cutoff (Hyperseed-Concept 60, 151).*

The usefulness of the schema is that it separates three issues: (1) identity up to a chosen equivalence, (2) strength/intensity above a threshold, and (3) referential nontriviality. This separation is essential when one wants to transport registration statements across contexts, as done in the theorem below.

Theorem 67 (Registration is downward closed under forgetting, under coherence hypotheses)**.** *Let $C \preceq D$. Assume:*

1. *(Equivalence coherence) For all $u, v \in U_D$, $u \sim_D v \Rightarrow \pi_{D\to C}(u) \sim_C \pi_{D\to C}(v)$.*
2. *(Pattern map) A function $\beta : P_D \to P_C$.*
3. *(Intensity is sup-pushforward) For all $p_C \in P_C$ and $u_C \in U_C$,*

$$\mathrm{Int}_{C,t}(p_C, u_C) = \oplus\{\mathrm{Int}_{D,t}(p_D, u_D) : \beta(p_D) = p_C,\ \pi_{D\to C}(u_D) = u_C\}.$$

4. *(Reference coherence) For all $p_D \in P_D$,*

$$\mathrm{Ref}_D(p_D, A|_D) \textit{ nontrivial} \Rightarrow \mathrm{Ref}_C(\beta(p_D), A|_C) \textit{ nontrivial.}$$

Then:

If X registers A at time t in context D, then X registers A at time t in context C.

Remark 2801. *What the theorem asserts is a conservative stability principle:* if registration holds at a fine resolution, then it still holds after forgetting detail*, provided the ingredients of registration (equivalences, patterns, intensities, reference) cohere with forgetting in the stated ways.*

Why this matters is pragmatic and philosophical at once. Pragmatically, a system often proves a claim using detailed internal structures but must communicate or act using coarser summaries; the theorem guarantees it does not lose what it has already secured. Philosophically, it articulates a modest form of realism about registration: the act of "seeing that A is there" should not evaporate merely because we choose to speak in broader strokes, as long as our coarsening is coherent.

This connects directly to the earlier sup-pushforward dominance theorem: hypothesis (3) ensures that coarse intensities are defined by the same supremum-over-witnesses construction, so that a fine witness remains a coarse witness. The extra hypotheses (1) and (4) ensure that identity-up-to-equivalence and reference also survive forgetting. In the broader Hyperseed setting this is an instance of "downward closure" of cognitive commitments across contexts; compare [1, 19].

Proof. Assume X registers A at time t in context D. By the registration schema (in D), there exist $A_0 \in U$ and $p_D \in P_D$ such that

$$A_0|_D \sim_D A|_D, \quad \mathrm{Int}_{D,t}(p_D, A_0|_D) \not\leq \theta, \quad \mathrm{Ref}_D(p_D, A|_D) \text{ nontrivial.}$$

Define $p_C := \beta(p_D) \in P_C$ and $u_C := A_0|_C \in U_C$. By forgetting, $u_C = \pi_{D\to C}(A_0|_D)$ and also $A|_C = \pi_{D\to C}(A|_D)$. By equivalence coherence, $A_0|_D \sim_D A|_D$ implies $A_0|_C \sim_C A|_C$. In particular, the coarse identity/equivalence requirement is satisfied by the *same* underlying object A_0 after applying the restriction/forgetting map; no new matching argument is needed at level C beyond the coherence axiom.

By reference coherence, $\mathrm{Ref}_C(p_C, A|_C)$ is nontrivial. Here "nontrivial" is inherited rather than re-proved: the role of reference coherence is precisely to ensure that pushing a pattern forward along β does not collapse a previously meaningful reference relation into a degenerate one when the associated ambient content is forgotten to C.

Finally, apply sup-pushforward dominance: in the defining join for $\mathrm{Int}_{C,t}(p_C, u_C)$, the witness $(p_D, A_0|_D)$ is feasible since $\beta(p_D) = p_C$ and $\pi_{D\to C}(A_0|_D) = u_C$. Hence $\mathrm{Int}_{C,t}(p_C, u_C) \geq \mathrm{Int}_{D,t}(p_D, A_0|_D)$. The point of the feasibility check is that the supremum/join at the coarse level ranges over *fine-level* pairs that project to the prescribed coarse arguments; thus it is enough to exhibit a single fine pair lying above (p_C, u_C) under $(\beta, \pi_{D\to C})$. Since $\mathrm{Int}_{D,t}(p_D, A_0|_D) \not\leq \theta$, also $\mathrm{Int}_{C,t}(p_C, u_C) \not\leq \theta$ (monotonicity of $\leq$). Equivalently, if the coarse intensity were $\leq \theta$, then by transitivity with the displayed inequality one would obtain $\mathrm{Int}_{D,t}(p_D, A_0|_D) \leq \theta$, contradicting the fine-level registration hypothesis; this is the contrapositive form of the same order-theoretic monotonicity step. Thus A_0 and p_C witness registration in C. $\square$

Remark 2802. Proof sketch. *The proof transports the fine-level witnesses (A_0, p_D) down to the coarse context. For identity, it uses equivalence coherence: if A_0 matches A in D, then their forgotten images match in C. For reference, it uses reference coherence: nontrivial reference persists when patterns are mapped by β. For intensity, it uses the same dominance principle as earlier:*

since coarse intensity is a supremum over all fine witnesses, the specific fine witness we already have provides a lower bound on the coarse intensity. Put differently, the argument is modular: (i) identity compatibility is preserved by $\pi_{D\to C}$, (ii) reference compatibility is preserved by β (together with forgetting on the object side), and (iii) intensity compatibility is preserved because the coarse score is constructed to be an upper envelope of all consistent fine scores.

The key step is the last one: turning $\mathrm{Int}_{D,t}(p_D, A_0|_D) \not\leq \theta$ into $\mathrm{Int}_{C,t}(p_C, u_C) \not\leq \theta$ via the inequality $\mathrm{Int}_{C,t}(p_C, u_C) \geq \mathrm{Int}_{D,t}(p_D, A_0|_D)$. This is precisely where the choice of sup-pushforward matters: it is engineered to preserve witnesses. Intuitively, the coarse system is not allowed to "forget away" intensity evidence, because any piece of fine evidence compatible with the coarse description is still admissible in the coarse supremum. In that sense, the construction implements a monotone inference rule: projecting information downward cannot invalidate an above-threshold intensity that was already achieved by some compatible fine description.

One can visualize the situation as a commutative diagram: U_D maps down to U_C by forgetting, and P_D maps down to P_C by β. Registration at level D is a point in the product $P_D \times U_D$ above threshold; the theorem says its image in $P_C \times U_C$ remains above threshold. Concretely, the pair $(p_D, A_0|_D)$ sits over (p_C, u_C) in the fiber of the map $(\beta, \pi_{D\to C})\colon P_D \times U_D \to P_C \times U_C$, and the coarse intensity is evaluated by taking the join over that fiber. The proof therefore amounts to observing that the fiber is nonempty (witnessed by $(p_D, A_0|_D)$) and that taking a join cannot yield a value below one of its contributing elements.

Remark 2803 (Why this is useful)**.** *This theorem gives a safe inference rule: registration/perceptual claims proved at a detailed resolution remain valid at coarser resolutions, provided intensities and reference are defined by coherent pushforward/forgetting laws. This is a common need in resource-bounded reasoning. It is also useful for multi-scale modeling: one may establish registration in a fine model where the relevant discriminations are explicit, and then soundly reuse the conclusion in a reduced model (lower-dimensional, aggregated, or otherwise compressed) without re-running the full fine analysis. The proof shows exactly which axioms are doing the work: coherence for identity and reference, and supremum-based pushforward for intensity.*

50 Weakness/Effort/Pattern-Intensity Tradeoff Theorems as Decision Tools

Here we collect a small family of "tradeoff" theorems linking (i) effort/weakness monotonicity for distinction policies, (ii) effort-based simplicity, and (iii) pattern intensity as normalized compression gain. The results are stated so they can be used as decision tools: convert intensity thresholds into explicit effort budgets, bound sensitivity to modeling overhead, and justify Pareto-style pruning and "maximally open adequate" selection rules.

Concretely, the intended workflow is that a designer begins with operational constraints (available compute, limited supervision, bounded attention, restricted witness language) and expresses these as an *effort budget*; the theorems then give sufficient conditions ensuring that certain distinctions are infeasible, unstable under overhead, or dominated by other policies. Conversely, if a minimum pattern-intensity threshold is required for downstream use (e.g. to trigger memory consolidation, to justify adding a concept, or to promote a hypothesis into a reusable module), the tradeoffs provide a principled way to translate this threshold into a constraint on allowable weakness and on the complexity of the corresponding representations.

The decision-theoretic reading also clarifies why the statements are phrased as inequalities rather than as exact characterizations. In practice, the quantities being compared—effort, weakness, description length, and compression gain—are rarely measured in perfectly commensurate units, and they are often estimated with modeling-dependent slack. The goal, therefore, is to supply bounds that remain informative when the cost model is coarsened, when the witness language is refactored, or when a policy class is enlarged, while still supporting reliable choices such as "this family of distinctions is not worth paying for" or "any policy achieving this intensity must incur at least this much effort."

Remark 2804. *Philosophically, the guiding idea is that an ontology or inference engine is always bargaining with its own finitude: to draw more distinctions requires more effort, and to describe things more simply requires representational resources. The notion of "weakness" used here (Hyperseed-Concept 202) is a formal way to talk about deliberately* failing *to distinguish (Hyperseed-Concept 98) certain differences, a stance that appears in multiple places in the Hyperseed stack [1] and in quantale-based treatments of weakness [3, 2].*

This "bargaining" viewpoint is also meant to be operational rather than merely metaphorical. A policy that is weaker is not simply "less accurate"; it is a choice to refrain from spending effort on certain refinements, often because their expected return in compression, prediction, or control is low relative to the resources they consume. In this sense, weakness functions as a knob controlling granularity: turning the knob toward greater weakness collapses distinctions, reduces the burden on representation, and can improve robustness by ignoring brittle micro-differences; turning it toward greater strength resolves finer differences, but may demand more effort and may increase exposure to modeling overhead.

Remark 2805. *The emphasis on tradeoffs is aligned with the "route A / route B" perspective on managing dependency and modeling overhead in compositional reasoning [9]. The specific choice here is to keep the mathematics austere: we work with inequalities that are robust under changes of modeling assumptions, so that a system designer can rely on them even when the internal witness language, cost model, or policy space is being revised.*

In particular, the phrase "pattern intensity as normalized compression gain" is meant to signal a quantity that is comparable across domains and scales: it measures how much shorter (or simpler) a description becomes when a pattern is used, relative to a baseline that encodes the same data or task without that pattern. Normalization matters because raw compression gain can be inflated by trivial changes of units or by scaling the dataset, whereas an intensity-like ratio (or suitably offset-normalized difference) can be interpreted as a stable notion of "signal-to-overhead" for pattern utility. The tradeoff theorems in this section therefore aim to separate three effects that otherwise get conflated in informal reasoning: (a) genuine structure captured by a pattern, (b) the effort required to discover, maintain, or apply the pattern under a given distinction policy, and (c) modeling overhead due to the representational apparatus itself.

The references to "Pareto-style pruning" and "maximally open adequate" selection rules should be read as practical corollaries: once effort and intensity are placed in the same inequality framework, one can rule out policies that are jointly worse (higher effort and lower intensity) and prefer policies that remain as weak as possible while still meeting adequacy constraints (e.g. a minimum discrimination requirement for a task). This supports an engineering discipline in which policy classes are not only compared by objective value, but also organized by dominance relations, with explicit attention to the stability of these relations under reasonable changes in overhead assumptions.

50.1 Setup

50.1.1 Effort-based simplicity and pattern intensity

Definition 461 (Baseline cost, witnesses, and simplicity)**.** *Fix a context C and an entity x.*

A witness *of x is any description/representation w that decodes to x. Write $u(w;x) \in [0,\infty]$ for the cost of witness w (e.g. description cost or a compositional cost). Fix a* baseline *witness* $\mathrm{base}_C(x)$ *with baseline cost*

$$s_C^{\mathrm{base}}(x) := u(\mathrm{base}_C(x); x).$$

Define the (contextual) simplicity of x as the infimal witness cost

$$s_C(x) := \inf\{u(w;x) : w \text{ is a witness of } x\}.$$

Remark 2806. *The symbols in this definition play distinct conceptual roles. The object x is whatever the context C takes to be an "entity"; the witness w is a* representation *(Hyperseed-Concept 157) of x in some description language; and $u(w;x)$ is a cost (length, energy, time, number of primitives, etc.) assigned to using w to obtain x. Allowing the value ∞ makes it possible to treat "inapplicable" or "ill-formed" witnesses uniformly: they simply never matter to infima. The baseline witness* $\mathrm{base}_C(x)$ *is a conventional point of comparison: it is not assumed optimal, only available.*

Remark 2807. *Intuitively, $s_C(x)$ measures how simply x* can *be described within the representational resources of the context C (Hyperseed-Concept 169). In algorithmic information theory, a canonical example would be: x is a finite string, w is a program that prints x, and $u(w;x)$ is program length; then $s_C(x)$ is essentially Kolmogorov complexity [16]. In a more structural example aligned with pattern ontologies [5], x might be a composite object (a graph, a scene, an interaction trace), w a decomposition plus glue, and u the sum of part costs plus coordination overhead.*

Remark 2808. *This definition is useful because it lets later tradeoff theorems speak in the simplest possible currency:* cost. *Rather than trying to reason directly about the metaphysical status of patterns, we reason about whether a candidate pattern witness reduces description cost relative to a baseline. This is deliberately operational: an inference engine can compute or approximate costs and use them to make decisions.*

Remark 2809. *Note that $s_C(x)$ is defined via an infimum and therefore supports approximation: even when the true optimum witness is unknown, any candidate witness w with finite $u(w;x)$ provides an upper bound $s_C(x) \le u(w;x)$. Conversely, lower bounds on $s_C(x)$ (e.g. information-theoretic bounds, resource-bounded adversary arguments, or minimality certificates in a restricted witness language) translate directly into limits on attainable compression, which is precisely what later decision tools will need.*

Definition 462 (Scalar pattern intensity)**.** *Assume $s_C^{\mathrm{base}}(x) > 0$. Define the scalar intensity of witness w for x by*

$$I_C(w;x) := \max\left\{0,\ \frac{s_C^{\mathrm{base}}(x) - u(w;x)}{s_C^{\mathrm{base}}(x)}\right\} = \max\left\{0,\ 1 - \frac{u(w;x)}{s_C^{\mathrm{base}}(x)}\right\}.$$

If $s_C^{\mathrm{base}}(x) = 0$, set $I_C(w;x) = 0$ for all w by convention.

Remark 2810. *The expression $1 - u(w;x)/s_C^{\mathrm{base}}(x)$ is the* fractional savings *achieved by using w instead of the baseline witness. The outer* $\max\{0,\cdot\}$ *clamps negative savings to 0, which enforces a* *conceptual discipline: intensity is meant to measure the presence of a pattern (Hyperseed-Concept 130) as a* compression gain *relative to a chosen reference, not as a penalty for doing worse than baseline. Thus, intensity lives in* $[0,1]$ *and can be read as "how much of the baseline description has been made unnecessary by this witness."*

Remark 2811. *Two quick examples:*

- *(a) If* $u(w;x) = \frac{1}{2}s_C^{\text{base}}(x)$*, then* $I_C(w;x) = \frac{1}{2}$*: the witness compresses the baseline by 50%.*
- *(b) If* $u(w;x) \geq s_C^{\text{base}}(x)$*, then* $I_C(w;x) = 0$*: the witness has no positive intensity in this context (it may still be useful for other reasons, but not as a compression-based pattern witness).*

The definition is useful precisely because it translates the somewhat qualitative idea of "pattern strength" into a normalized quantity that supports thresholding and budgeting.

Remark 2812. *The normalization by* $s_C^{\text{base}}(x)$ *makes intensities comparable* within *a fixed baseline convention, but it also makes explicit that intensity is baseline-relative: changing* $\text{base}_C(x)$ *can change* $I_C(w;x)$ *even when* $u(w;x)$ *is unchanged. This is intentional, because later decision procedures will often be framed as "is this witness worth adopting relative to what we already do by default?" rather than "is this witness absolutely optimal?"*

Remark 2813. *For fixed* x *and* C *with* $s_C^{\text{base}}(x) > 0$*, the map* $u \mapsto \max\{0, 1 - u/s_C^{\text{base}}(x)\}$ *is monotone non-increasing in* u*. Thus, any improvement in witness cost (lower* $u(w;x)$*) can only increase intensity, and witnesses with the same cost have the same intensity regardless of their internal structure. This ensures that later theorems can treat intensity as a pure function of resource use, leaving representational details to the cost model* u*.*

Definition 463 (Optimal attainable intensity (supremum form))**.** *Assume* $s_C^{\text{base}}(x) > 0$*. Define the* attainable intensity bound *of* x *by*

$$I_C^\star(x) := \sup\{I_C(w;x) : w \text{ is a witness of } x\}.$$

Remark 2814. *The supremum* sup *is chosen instead of* max *because there need not exist a single best witness: a sequence of witnesses may approach the infimal cost without ever attaining it. In such cases* $I_C^\star(x)$ *expresses an idealized "capacity" for compression-based pattern intensity, given the representational regime of* C*. This quantity becomes a convenient upper bound for decision procedures: any concrete witness must lie below it.*

Remark 2815. *Conceptually,* $I_C^\star(x)$ *is to intensity what* $s_C(x)$ *is to simplicity: it is a best-possible value under the current rules of representation, and it changes when we expand the witness language. This fits neatly with the Hyperseed view that pattern and representation are co-constituted by a context [5]: change the context, and what counts as an intense (i.e. compressive) pattern can change as well.*

Proposition 77 (Attainable intensity in terms of simplicity)**.** *Assume* $s_C^{\text{base}}(x) > 0$*. Then*

$$I_C^\star(x) \;=\; \max\left\{0,\; 1 - \frac{s_C(x)}{s_C^{\text{base}}(x)}\right\}.$$

Remark 2816. *This identity unpacks the definitions: since* $I_C(w;x) = \max\{0, 1-u(w;x)/s_C^{\text{base}}(x)\}$*, taking a supremum over witnesses is equivalent to minimizing* $u(w;x)$ *over witnesses, i.e. passing to* $s_C(x)$*. In particular, whenever* $s_C(x) \geq s_C^{\text{base}}(x)$*, no witness can achieve positive compression relative to baseline and* $I_C^\star(x) = 0$*; and whenever* $s_C(x) \ll s_C^{\text{base}}(x)$*, the context admits high potential intensity. This is one reason the later tradeoff theorems can be stated either in the language of "simplicity gaps" (*$s_C^{\text{base}}(x) - s_C(x)$*) or in the normalized language of intensity without changing substantive content.*

50.1.2 Distinction policies, weakness, and effort

Definition 464 (Policies and opening/closing)**.** *Let H range over admissible indistinction policies (each H can be thought of as a set of undistinguished pairs). Write $H \subseteq H'$ to mean that H' is an* opening *of H (it collapses weakly more distinctions), and $H' \subseteq H$ to mean that H' is a* closing.

Remark 2817. *The inclusion order here is a convenient way to treat "how much the agent chooses to ignore" as a lattice-like control knob: moving upward (an opening) allows more pairs to be treated as the same, while moving downward (a closing) forces the agent to preserve more distinctions. In later sections, this order will serve as the formal backbone for statements of the form "if you open the policy, effort can decrease but weakness can increase," i.e. monotonic tradeoffs along the inclusion axis.*

Remark 2818. *It is often helpful to keep two extremes in mind. The minimal policy $H = \varnothing$ corresponds to collapsing no pairs (maximal discrimination), while a maximally open policy (when admissible) would collapse every pair the context allows, yielding a very coarse view of entities. Many practical settings restrict admissibility so that H must satisfy consistency constraints (for example, being symmetric, or being closed under some form of transitivity), but the definition intentionally does not bake in a particular constraint set: different applications (perception, clustering, theorem proving, abstraction in planning) impose different closure requirements on what counts as a permissible "indistinction."*

Remark 2819. *Thinking of H as a set of undistinguished pairs also anticipates quotient-style constructions: an opening $H \subseteq H'$ can be read as moving to a coarser quotient in which more items are identified. Even when H is not literally an equivalence relation, it can be completed to one (for instance by taking an admissible closure) and the resulting quotient intuition explains why openings typically reduce representational burden: fewer distinctions mean fewer bits, fewer cases, and fewer branches to handle, at the possible cost of conflating situations that matter downstream.*

Remark 2820. *Here H is an abstract stand-in for a "policy of indistinction": which pairs of fine-grained states or entities are treated as the same for current purposes. The partial order by inclusion captures a natural qualitative notion: an* opening *(Hyperseed-Concept 125) throws away more distinctions (Hyperseed-Concept 98) and thus yields a coarser view; a* closing *(Hyperseed-Concept 72) reinstates distinctions and yields a finer view. This is the same order-theoretic stance used in coarse-graining arguments: one can often reason monotonically as one moves from fine to coarse. In particular, thinking of H as a binary relation of "treated-as-equal" pairs, the condition $H \subseteq H'$ means that every identification licensed by H remains licensed by H', while H' may additionally identify pairs that H still keeps distinct. When H happens to be an equivalence relation (or a congruence in an algebraic setting), opening corresponds to passing to a coarser partition (or quotient) and closing corresponds to refining that partition. This makes the inclusion order a convenient proxy for the intuitive idea of "more or fewer distinctions enforced" without committing to a particular ontology of states, features, or representational primitives.*

Remark 2821. *A simple example is perceptual resolution: H might equate colors within 1 unit in RGB distance, whereas an opening H' equates colors within 5 units; the latter policy collapses more pairs, and hence $H \subseteq H'$. In a symbolic setting, H might declare two terms indistinguishable when they are alpha-equivalent, while H' also identifies terms under a wider congruence. One can also view these examples as moving from stricter to looser admissibility constraints on downstream judgments: under H', more stimuli count as "the same input," and more syntactic forms count as "the same expression," so fewer case distinctions must be carried through later steps. The intended* *reading is not that H' is "more correct," but that it is a deliberate relaxation whose costs and benefits can be reasoned about monotonically.*

Assumption 1 (Weakness and effort monotonicity)**.** *There is a weakness functional $w(H)$ and an effort functional $\mathrm{Eff}_C(H)$ such that:*

$$H \subseteq H' \implies w(H) \leq w(H') \quad \textit{and} \quad \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H').$$

That is: opening increases weakness and decreases policy effort.

Remark 2822. *This assumption formalizes a widely encountered engineering and epistemic motif: it is easier to act, compute, and decide when one is willing to lump more things together, and the corresponding "weakness" is precisely the degree to which distinctions are not enforced. The effort functional (Hyperseed-Concept 100) may represent time, energy, number of measurements, or attention allocation; the weakness functional (Hyperseed-Concept 202) may measure how much indistinction is permitted. The mathematical content is simply antitonicity of effort and monotonicity of weakness with respect to inclusion, which is the minimal structure needed for the later "open if you can" theorems. See [3] for quantale-centric variants and interpretations. A useful intuition is that $w(H)$ can be read as a "slack" or "tolerance" parameter attached to the policy: as H opens, the system tolerates more equivalences and therefore becomes less brittle to micro-variation, but also potentially less discriminative. Conversely, $\mathrm{Eff}_C(H)$ can be read as the cost of enforcing the policy—e.g. the cost of gathering the information needed to separate cases that H insists on separating, or the computational cost of maintaining invariants that are only meaningful at a finer granularity. The assumption deliberately avoids specifying any functional form (linearity, convexity, etc.); only the direction of change under opening is fixed, so later results can be stated without overfitting to a particular application domain.*

Assumption 2 (Representational monotonicity under opening)**.** *Each policy H induces an admissible witness family $\mathcal{W}_H(x)$ for each x, and opening weakly enlarges it:*

$$H \subseteq H' \implies \mathcal{W}_H(x) \subseteq \mathcal{W}_{H'}(x) \;\; \textit{for all } x.$$

Define the policy-indexed simplicity

$$s_{C,H}(x) := \inf\{u(w; x) : w \in \mathcal{W}_H(x)\},$$

and the policy-indexed attainable intensity bound

$$I^{\star}_{C,H}(x) := \sup\{I_C(w; x) : w \in \mathcal{W}_H(x)\}.$$

Remark 2823. *This is the representational counterpart of weakness/effort monotonicity: a coarser policy allows more descriptions to count as legitimate witnesses, because fewer distinctions need to be honored. Formally, $\mathcal{W}_H(x)$ is a* constrained *witness set; opening removes constraints and thus can only enlarge the set. The derived quantities $s_{C,H}(x)$ and $I^{\star}_{C,H}(x)$ let us talk about simplicity and intensity* under the policy*, rather than in the unconstrained ideal. Concretely, $\mathcal{W}_H(x)$ can be thought of as the set of admissible encodings, explanations, models, certificates, or constructions of x that are compatible with the equivalences imposed by H. When H is stricter, some candidate witnesses are disallowed because they implicitly rely on identifications that the policy forbids, or because verifying them would require resolving distinctions that the policy insists must remain explicit. When H opens, these requirements relax, so witnesses that were previously "invalid" (or too costly to validate under the stricter distinction regime) can become admissible. In this way, the enlargement $\mathcal{W}_H(x) \subseteq \mathcal{W}_{H'}(x)$ expresses a permissiveness ordering that mirrors the earlier ordering on effort and weakness.*

Remark 2824. *A concrete picture is: a stricter policy H might require a witness to preserve exact structure (e.g. exact graph isomorphism), while an opening H' permits approximate structure (e.g. graph edit distance within a threshold). Then $\mathcal{W}_{H'}(x)$ contains witnesses that would be rejected under H, so the infimal achievable cost can drop, and the achievable intensity can rise. This is one formal route to the idea that "letting go of distinctions" can unlock simpler representations, a recurring theme in pattern/emergence accounts [5]. It is also helpful to separate two effects that this example conflates: (i)* search *becomes easier because more candidates qualify as witnesses, and (ii)* validation *becomes easier because approximate constraints are cheaper to check than exact ones. Both effects are compatible with the single inclusion statement for witness families, because admissibility can be defined to include whatever verification obligations are part of the policy. In applications, the same opening may simultaneously lower description cost $u(w; x)$ (by permitting shorter or more compressive encodings) and raise attainable intensity $I_C(w; x)$ (by permitting witnesses that capture stronger regularities once irrelevant micro-structure is ignored).*

Remark 2825. *The inclusion $\mathcal{W}_H(x) \subseteq \mathcal{W}_{H'}(x)$ immediately yields monotonicity of the derived extrema: if $H \subseteq H'$, then*

$$s_{C,H'}(x) \leq s_{C,H}(x) \quad \textit{and} \quad I^{\star}_{C,H}(x) \leq I^{\star}_{C,H'}(x),$$

since the infimum is taken over a larger set and the supremum is taken over a larger set. Thus, opening weakly improves the best achievable simplicity (in the sense of lowering the minimal cost) and weakly improves the attainable intensity bound (in the sense of allowing higher intensity), even before invoking any additional properties of $u(\cdot; x)$ or $I_C(\cdot; x)$. These monotone relationships are the representational analogue of "opening reduces effort" and will be used later to compare decision quality across policies without requiring any explicit optimization algorithm.

50.2 Core intensity algebra: converting intensity into budgets

Lemma 154 (Intensity is truncated relative savings)**.** *Assume $s_C^{\text{base}}(x) > 0$ and let $s := s_C^{\text{base}}(x)$. For any witness w with cost $u := u(w; x)$,*

$$I_C(w; x) = \max\left\{0,\ 1 - \frac{u}{s}\right\}.$$

In particular, $I_C(w; x) > 0$ iff $u < s$ and $I_C(w; x) = 0$ iff $u \geq s$.

Remark 2826. *This lemma states that the intensity formula is nothing more (and nothing less) than a normalized notion of savings relative to baseline, truncated to avoid negative values. The key operational consequence is the "phase transition" at $u = s$: once a witness costs at least as much as the baseline, it is intensity-zero and can be ignored by any routine that is specifically searching for compressive patterns.*

Remark 2827. *A useful rearrangement (valid on the positive-intensity region $u < s$) is the inversion*

$$u = (1 - I_C(w; x))\, s_C^{\text{base}}(x).$$

Thus, once intensity is interpreted as "fraction of baseline saved," the witness cost is exactly "fraction of baseline not saved.*" This also makes clear why the truncation at 0 is natural: if $u > s$, then the formula $1 - u/s$ would correspond to a* negative *savings fraction, which is typically not treated as a separate regime but simply regarded as "no compression."*

Proof. Immediate algebra from the definition of $I_C(w; x)$ and the clamp at 0. □

Remark 2828. Proof sketch and intuition. *One expands the definition and notes that* $s_C^{\mathrm{base}}(x)$ *has been abbreviated as* s*. The "iff" statements are simply the observation that* $\max\{0, 1 - u/s\}$ *is positive exactly when its unclamped argument is positive, i.e. when* $u/s < 1$*. Geometrically,* $I_C(w; x)$ *is the line* $1 - u/s$ *as a function of* u*, chopped off at* 0*.*

Remark 2829. *Concrete calibration can be helpful. If* $s_C^{\mathrm{base}}(x) = 100$ *and a candidate witness has cost* $u = 30$*, then* $I_C(w; x) = 1 - 30/100 = 0.7$*, meaning it realizes* 70% *of the baseline cost as relative savings. If instead* $u = 120$*, then* $I_C(w; x) = \max\{0, 1 - 1.2\} = 0$*, so this candidate is treated identically (for compression purposes) to any even more expensive candidate.*

Theorem 68 (Intensity threshold $\Leftrightarrow$ explicit cost budget)**.** *Assume* $s_C^{\mathrm{base}}(x) > 0$ *and fix* $\tau \in (0, 1]$*. For a witness* w *of* x *with cost* $u := u(w; x)$*, the following are equivalent:*

$$I_C(w; x) \geq \tau \quad \Longleftrightarrow \quad u \leq (1 - \tau)\, s_C^{\mathrm{base}}(x).$$

Remark 2830. *In plain language, this theorem says:* having intensity at least τ is exactly the same as being below a linear cost budget*. The budget* $(1-\tau)s_C^{\mathrm{base}}(x)$ *is the baseline cost times the fraction of baseline you are still allowed to spend. Thus the theorem converts an epistemic or heuristic condition ("pattern intensity is high enough") into an explicit resource constraint (Hyperseed-Concept 100) on witness search.*

Remark 2831. *The parameter endpoints have clear interpretations. If* $\tau = 1$*, then the budget is* $(1-\tau)s_C^{\mathrm{base}}(x) = 0$*, so only witnesses of zero cost can meet the threshold, matching* $I_C(w; x) = 1 - u/s$ *on* $u < s$*. If* τ *is very small (approaching* 0 *from above), then the budget approaches* $s_C^{\mathrm{base}}(x)$*, meaning that even very mild compression counts as meeting the threshold, but still excluding witnesses that do not beat baseline. The exclusion of* $\tau = 0$ *in the statement is aligned with this: at* $\tau = 0$ *one would allow all* $u \leq s$*, but the "clamp is inactive" step used in the proof would no longer be forced by the inequality.*

Remark 2832. *This equivalence is important because it makes intensity thresholds actionable. Many systems can prune candidates quickly based on cost (e.g. partial evaluation, admissible heuristics, early termination), whereas intensity is often computed only after substantial semantic checking. Therefore the theorem gives a principled "do not even consider" cutoff that is guaranteed sound.*

Remark 2833. *Operationally, the theorem can be read as a compilation rule for search: replace the semantic predicate "*$I_C(w; x) \geq \tau$*" by the syntactic (or at least cheaper-to-check) predicate "*$u(w; x) \leq (1 - \tau)s_C^{\mathrm{base}}(x)$*," and postpone any expensive validation steps until after a candidate has passed this budget gate. This is precisely the kind of conversion needed to turn intensity from a descriptive statistic into a decision tool.*

Proof. Write $s := s_C^{\mathrm{base}}(x)$. Since $\tau > 0$, the inequality $I_C(w; x) \geq \tau$ forces $I_C(w; x) > 0$, hence the clamp is inactive and

$$I_C(w; x) = 1 - \frac{u}{s}.$$

Thus $I_C(w; x) \geq \tau$ iff $1 - u/s \geq \tau$ iff $u \leq (1 - \tau)s$. The converse is the same algebra. $\square$

Remark 2834. Proof sketch and intuition. *The proof consists of two moves. First, because* $\tau > 0$*, any witness meeting the threshold must lie in the region where the clamp* $\max\{0, \cdot\}$ *does nothing. Second, on that region the intensity is an affine function of the cost, so solving the inequality* $I_C(w; x) \geq \tau$ *is a single line of algebra. One may picture the graph of* $u \mapsto I_C(w; x)$ *as a line descending from* 1 *at* $u = 0$ *to* 0 *at* $u = s$*, and then staying flat at* 0 *thereafter; the threshold* τ *corresponds to a horizontal line cutting the descending segment at exactly* $u = (1 - \tau)s$*.*

Remark 2835. *It is sometimes convenient to express the same cutoff in relative terms by dividing through by baseline:*

$$I_C(w;x) \geq \tau \quad \Longleftrightarrow \quad \frac{u(w;x)}{s_C^{\text{base}}(x)} \leq 1-\tau.$$

That is, meeting intensity τ means the witness must cost at most a $(1-\tau)$ fraction of the baseline cost. This ratio form emphasizes that the criterion is scale-free in the baseline: if all costs are multiplied by a constant, the threshold decision is unchanged.

Corollary 34 (Decision tool: "intensity at least τ" becomes a hard budget)**.** *To guarantee $I_C(w;x) \geq \tau$, it suffices to search only among witnesses with $u(w;x) \leq (1-\tau)s_C^{\text{base}}(x)$. Equivalently: any candidate witness exceeding that budget can be discarded without further analysis.*

Remark 2836. *This corollary is the computational face of the previous theorem: it licenses a strict pruning rule in any witness-generation or pattern-mining loop. In particular, it supports beam search and A* style procedures in which cost bounds are maintained incrementally: once a partial witness already exceeds $(1-\tau)s_C^{\text{base}}(x)$, it cannot be completed into a witness of intensity at least τ.*

Remark 2837. *One can also view the pruning rule as enforcing a search invariant: throughout the computation, the current frontier of candidates is restricted to those whose* best possible completion cost *remains below the budget. Any admissible lower bound $\underline{u}$ on the final witness cost therefore yields a sound early rejection criterion $\underline{u} > (1-\tau)s_C^{\text{base}}(x)$, even when the exact $u(w;x)$ is not yet known. This is where the theorem interfaces naturally with standard branch-and-bound logic.*

Lemma 155 (Lipschitz sensitivity of intensity in witness cost)**.** *Assume $s_C^{\text{base}}(x) = s > 0$. For any two witnesses with costs $u, v \in [0, \infty]$,*

$$\big|I(u;x) - I(v;x)\big| \leq \frac{|u-v|}{s},$$

where $I(\cdot;x) := \max\{0, 1 - \cdot/s\}$.

Remark 2838. *This lemma quantifies robustness: small additive errors in cost estimation produce proportionally small errors in intensity, with proportionality constant $1/s$. It is particularly relevant when $u(w;x)$ is not computed exactly but approximated (e.g. by an estimator, a surrogate model, or a partial evaluation). The notation $I(u;x)$ is a deliberate abbreviation: once x and s are fixed, intensity depends on a witness only through its cost.*

Remark 2839. *The constant $1/s$ has an immediate interpretation: larger baselines make intensity* less *sensitive to absolute cost perturbations. For instance, an absolute error of ± 1 in the cost changes intensity by at most 0.01 when $s = 100$, but by at most 0.1 when $s = 10$. This helps separate two regimes: when the baseline is large, coarse cost approximations may be sufficient for stable thresholding; when the baseline is small, intensity-based decisions may require more precise cost accounting.*

Proof. The map $g(t) := 1 - t/s$ is $(1/s)$-Lipschitz. The clamp map $h(t) := \max\{0, t\}$ is 1-Lipschitz. Thus $I = h \circ g$ is $(1/s)$-Lipschitz. □

Remark 2840. Proof sketch and intuition. *The unclamped intensity $g(t) = 1 - t/s$ is a line of slope $-1/s$, so changing t by Δ changes $g(t)$ by exactly Δ/s in magnitude. The clamp $h(t)$ can only* reduce *variation (it never amplifies differences), which is captured by 1-Lipschitzness. Composing two Lipschitz maps multiplies the constants, giving $1/s$. Visually: away from the clamp point the graph is a straight line, and at the clamp point it "bends" into a flat segment, which cannot increase sensitivity.*

Remark 2841. *A complementary way to see the same bound is to consider cases. If both $u, v < s$, then $I(u;x) = 1 - u/s$ and $I(v;x) = 1 - v/s$, giving equality $\big|I(u;x) - I(v;x)\big| = |u-v|/s$. If both $u, v \geq s$, then both intensities are 0 and the left-hand side is 0. If one cost is below s and the other above, then the intensity difference is at most the below-s intensity, which is itself at most the normalized distance to the clamp point s; this is again bounded by $|u-v|/s$ because crossing the clamp requires $|u-v| \geq |s - \min\{u,v\}|$. This case analysis makes explicit how the truncation creates a "saturation" region in which changes in cost do not change intensity at all.*

50.3 Overhead and compositional witnesses: robustness bounds

Definition 465 (Compositional witness overhead)**.** *A compositional witness has the form $P = (y, z, t)$ with $t \geq 0$ and cost*

$$u(P;x) := h_C(P) := s_C(y) + s_C(z) + t,$$

so that increasing glue/interaction overhead t increases the witness cost additively.

Remark 2842. *This definition isolates a common structure in compositional description: one pays for the parts ($s_C(y)$ and $s_C(z)$) and for the "glue" t that binds them into x. The glue may represent alignment, interface constraints, communication bandwidth, synchronization, or any interaction term that is not reducible to the isolated part costs. In the language of pattern theory, P is a candidate* explanation *of x by subpatterns plus relations [5]. In particular, t is meant to capture any* nonseparable *burden that arises only when the parts are forced to cohere as a single description of x, even if each part is individually cheap.*

Remark 2843. *A simple example is a string x witnessed by concatenation of substrings y, z: the glue cost t might encode the boundary information and the instruction "concatenate." In a causal or combinatorial witness, t might encode the coupling rule that turns components into a whole. The definition is useful because it lets us reason cleanly about how modeling overhead degrades intensity, without needing to assume anything deep about $s_C(\cdot)$. One can also view t as a stand-in for "engineering overhead" in representation learning: even if two latent factors are cheap to encode, making them interact (e.g., via attention, cross-terms, or constraints) typically requires additional parameters, data, or coordination that should be accounted for explicitly.*

Remark 2844. *Although we write the witness as a pair (y, z) plus glue t, the same bookkeeping extends to a k-part decomposition $P = (y_1, \ldots, y_k, t)$ with cost $u(P;x) = \sum_i s_C(y_i) + t$; all bounds below continue to hold verbatim because they depend only on the fact that t enters additively. In this sense, the present two-part notation is a minimal template for compositionality rather than a restriction to binary factoring.*

Theorem 69 (Overhead penalty bound)**.** *Assume $s_C^{\mathrm{base}}(x) = s > 0$ and let $P = (y, z, t)$ be a compositional witness of x. For any $\Delta \geq 0$ let $P' = (y, z, t + \Delta)$ (same parts, larger overhead). Then*

$$I_C(P';x) \geq I_C(P;x) - \frac{\Delta}{s}.$$

In particular, increasing overhead by Δ cannot reduce intensity by more than Δ/s.

Remark 2845. *This theorem says that overhead hurts in a controlled way: intensity cannot collapse faster than linearly in the added overhead, and the scale factor is set by the baseline cost s. In practical terms, if the baseline cost is large, then modest overhead changes have little relative effect; if the baseline cost is small, overhead is proportionally more dangerous. This is exactly* *the kind of engineering heuristic one would like to promote from folklore to theorem. Note also that the assumption $s > 0$ is essential only to avoid division by zero; conceptually, s serves as the normalization that turns an absolute cost increase Δ into a dimensionless intensity drop Δ/s.*

Remark 2846. *The result connects directly to the Lipschitz lemma: overhead enters additively into witness cost, so the same Lipschitz constant yields the bound. It also anticipates later "tradeoff" uses: when an engine considers making a model more compositional (hence paying more glue), this inequality bounds how much intensity can be sacrificed for the compositional restructuring. Equivalently, it gives a simple sensitivity guarantee: if the glue term is uncertain up to $\pm\Delta$ due to implementation details, then the resulting intensity uncertainty is at most Δ/s.*

Remark 2847. *For the common linear intensity profile (as in earlier sections), $I(u;x)$ behaves like a clipped affine function of u (e.g., $I(u;x) = \max\{0, 1 - u/s\}$), in which case the bound is tight whenever both $u(P;x)$ and $u(P;x) + \Delta$ lie in the linear (unclipped) regime. When $u(P;x) + \Delta$ pushes the cost into the saturated regime where intensity is already near 0, the inequality remains valid but becomes conservative, reflecting the fact that intensity cannot fall below 0.*

Proof. We have $u(P';x) = u(P;x) + \Delta$. Apply the Lipschitz bound with $u = u(P;x)$ and $v = u(P;x) + \Delta$:

$$|I(u;x) - I(u + \Delta;x)| \le \Delta/s.$$

Since I is nonincreasing in cost, $I(u + \Delta;x) \le I(u;x)$, hence $I(u + \Delta;x) \ge I(u;x) - \Delta/s$, which is the stated inequality. Interpreting $I_C(P;x)$ as $I(u(P;x);x)$, the same one-dimensional cost perturbation argument applies because changing t by Δ leaves the part costs fixed and alters the total cost only through an additive shift. □

Remark 2848. Proof sketch and intuition. *The proof has two ingredients: (i) overhead Δ is exactly an additive increase in cost, and (ii) intensity changes at most Δ/s under a Δ change in cost (Lipschitz). The final step uses monotonicity of I in cost to remove absolute values and get a one-sided bound. One may imagine sliding along the descending segment of the intensity-vs-cost line: increasing cost by Δ slides rightward by Δ and therefore moves downward by at most Δ/s. In decision terms, this provides a "worst-case penalty" certificate: even if the glue turns out to be less efficient than expected by Δ, the intensity cannot be worse by more than Δ/s.*

Corollary 35 (Design inequality for keeping intensity above a threshold)**.** *Assume $s_C^{\mathrm{base}}(x) = s > 0$ and fix $\tau \in (0, 1]$. If a compositional witness $P = (y, z, t)$ satisfies*

$$t \le (1 - \tau)s - s_C(y) - s_C(z),$$

then $I_C(P;x) \ge \tau$.

Remark 2849. *This corollary is a direct "specification inequality": if you want intensity at least τ, then the admissible overhead is what remains of the budget after paying for the parts. It is useful in synthesis settings: one can select candidate components y, z and immediately see whether any glue budget remains, before attempting to construct the detailed coupling. When the right-hand side is negative, the inequality simply reports infeasibility under this two-part factoring: even with $t = 0$, the parts alone already exceed the $(1 - \tau)s$ budget, so no amount of clever interfacing can meet the target intensity without changing the parts (or the baseline scale s).*

Proof. The condition is exactly $u(P;x) = s_C(y) + s_C(z) + t \le (1 - \tau)s$. Apply the threshold–budget theorem. □

Remark 2850. Proof sketch and intuition. *The inequality on t is simply the rearranged statement "total cost is within the $(1-\tau)s$ budget." The earlier threshold theorem then converts that budget condition into the desired intensity condition. In other words: the corollary is not new algebra, but a useful way of presenting the same algebra in the language of "overhead capacity." One can also read it as a margin statement: the quantity $(1-\tau)s - s_C(y) - s_C(z)$ is the remaining slack available for interactions, and the corollary says that keeping glue within this slack is sufficient to preserve the desired intensity.*

Corollary 36 (Overhead robustness budget for a tolerated intensity drop)**.** *Assume $s_C^{\mathrm{base}}(x) = s > 0$ and let $P = (y, z, t)$ be a compositional witness of x. Fix $\varepsilon \geq 0$. If the glue overhead is increased by at most $\Delta \leq \varepsilon s$, then the resulting witness $P' = (y, z, t + \Delta)$ satisfies*

$$I_C(P'; x) \geq I_C(P; x) - \varepsilon.$$

Remark 2851. *This restates the overhead penalty bound in the most operational form: to guarantee that intensity degrades by at most ε, it suffices to bound unforeseen glue overhead by εs. In optimization or design loops, this gives a simple acceptance test for refactorings that incur additional interface machinery: if the added machinery is known (or bounded) to cost at most εs, then the refactoring is intensity-safe up to tolerance ε.*

50.4 Attainable intensity capacity and observer upgrades

Theorem 70 (Closed form for attainable intensity bound)**.** *Assume $s_C^{\mathrm{base}}(x) > 0$. Then*

$$I_C^{\star}(x) = \max\left\{0,\ 1 - \frac{s_C(x)}{s_C^{\mathrm{base}}(x)}\right\} = \max\left\{0,\ \frac{s_C^{\mathrm{base}}(x) - s_C(x)}{s_C^{\mathrm{base}}(x)}\right\}.$$

Moreover, for every witness w of x, one has $I_C(w; x) \leq I_C^{\star}(x)$.

Remark 2852. *The closed form makes explicit that $I_C^{\star}(x) \in [0, 1]$ whenever $s_C(x) \geq 0$ and $s_C^{\mathrm{base}}(x) > 0$: the outer $\max\{0, \cdot\}$ truncates negative values, and the ratio cannot exceed 1 when the infimal cost is nonnegative. In particular, the two boundary cases are conceptually important: (i) if $s_C(x) \geq s_C^{\mathrm{base}}(x)$, then $I_C^{\star}(x) = 0$, meaning that even the best witness does not beat the baseline in cost; and (ii) if $s_C(x) = 0$, then $I_C^{\star}(x) = 1$, meaning (on this scale) that one can in principle achieve full intensity relative to the baseline. This provides a direct way to interpret intensity as a* normalized cost savings *relative to the baseline, namely*

$$I_C^{\star}(x) = \frac{\text{(baseline cost)} - \text{(best achievable cost)}}{\text{(baseline cost)}} \quad \text{truncated below at } 0.$$

Remark 2853. *This theorem identifies the best possible intensity in terms of the best possible witness cost. Thus, intensity capacity is nothing but (truncated) normalized "compressibility gap" between the baseline and the simplest description. The result is important because it reduces a supremum over intensities to a single ratio involving $s_C(x)$, which is often the quantity one estimates or bounds in complexity arguments [16]. It also clarifies the role of the baseline: a poor baseline inflates the apparent available intensity, while a strong baseline makes high intensity harder to achieve.*

Remark 2854. *A practical corollary is that comparing attainable intensities across systems is only meaningful once the baseline choice is pinned down. For example, changing $s_C^{\mathrm{base}}(x)$ rescales the same underlying best cost $s_C(x)$ into a different normalized gap. This is exactly the same phenomenon one sees in approximation ratios in algorithms: the denominator sets the reference difficulty, and the numerator measures improvement relative to that reference.*

Remark 2855. *The connection to the broader Hyperseed pattern story [5] is that emergent or meaningful patterns are those that* explain *by compressing. Here the explanation is measured against an explicit baseline. The formula shows exactly what must be true for nonzero attainable intensity: we must have $s_C(x) < s_C^{\mathrm{base}}(x)$, i.e. some witness is strictly simpler than the baseline.*

Remark 2856. *It is also worth separating two notions that can otherwise be conflated: (a) the* existence *of a witness with high intensity, and (b) the* attainment *of the supremal intensity. The theorem gives the value of the supremum $I_C^\star(x)$ in closed form, but does not require that there is an actual witness $w^\star$ with $u(w^\star; x) = s_C(x)$ achieving it. In many complexity settings, the infimum is not necessarily a minimum; nonetheless, one can approximate the optimum cost arbitrarily well, which is sufficient to approximate the attainable intensity arbitrarily well.*

Proof. Write $s := s_C^{\mathrm{base}}(x) > 0$ and $m := s_C(x) = \inf_w u(w; x)$.

For any witness w, $u(w; x) \geq m$, and since $u \mapsto \max\{0, 1 - u/s\}$ is nonincreasing,

$$I_C(w; x) \leq \max\{0, 1 - m/s\}.$$

Hence $I_C^\star(x) \leq \max\{0, 1 - m/s\}$.

For the reverse inequality: by the definition of infimum, for each $n \in \mathbb{N}$ there exists a witness w_n with $u(w_n; x) \leq m + \frac{1}{n}$. Then

$$I_C(w_n; x) \geq \max\left\{0, 1 - \frac{m + 1/n}{s}\right\}.$$

Taking $n \to \infty$ shows $\sup_n I_C(w_n; x) \geq \max\{0, 1 - m/s\}$, hence $I_C^\star(x) \geq \max\{0, 1 - m/s\}$. ☐

Remark 2857. *The proof uses only two structural facts: (i) $I_C(w; x)$ depends on w only through the scalar cost $u(w; x)$, and (ii) the mapping from cost to intensity is monotone (and continuous from the right on $(0, \infty)$, with truncation at 0). This is why essentially the same reasoning applies to many alternative "intensity" parametrizations, as long as they are monotone transforms of cost relative to a fixed baseline.*

Remark 2858. Proof sketch and intuition. *The upper bound is monotonicity: since intensity decreases with cost, the best intensity cannot exceed what you would get if you could plug in the minimal possible cost m. The lower bound uses the defining property of an infimum: one can approximate m with witness costs $m + 1/n$. Applying the intensity function and taking $n \to \infty$ forces the supremum of intensities up to the same value. Visually, one is pushing the witness costs leftward toward the point $u = m$ on the intensity graph.*

Remark 2859. *A concrete sanity check is the "do-nothing" witness: if the witness class contains a baseline-like encoding whose cost is exactly $s_C^{\mathrm{base}}(x)$, then one always has at least intensity 0, and one only gains positive intensity by finding witnesses that reduce cost below the baseline. This aligns with the interpretation that intensity measures* improvement *rather than absolute simplicity.*

Theorem 71 (Representation expansion improves simplicity and intensity capacity)**.** *Fix x and assume $s_C^{\mathrm{base}}(x) > 0$. If $\mathcal{W}(x) \subseteq \mathcal{W}'(x)$ (same cost function u), then*

$$\inf_{w \in \mathcal{W}'(x)} u(w; x) \leq \inf_{w \in \mathcal{W}(x)} u(w; x),$$

and consequently the attainable intensity bound satisfies

$$\sup_{w \in \mathcal{W}'(x)} I_C(w; x) \geq \sup_{w \in \mathcal{W}(x)} I_C(w; x).$$

Remark 2860. *The condition "same cost function u" matters: the statement is about enlarging the feasible set while holding fixed the evaluation criterion. If an observer upgrade also changes how costs are measured (e.g. by reweighting primitive operations, changing encoding length conventions, or altering what counts as a valid derivation), then the monotonicity comparison becomes a comparison of two different objective functions and need not hold without additional assumptions.*

Remark 2861. *This theorem states an elementary but powerful monotonicity principle: enlarging the space of allowable representations cannot make the best achievable description worse. In the engineering idiom, adding new primitives, new rewrite rules, new compositional schemas, or new sensors cannot force you to lose previously available solutions; you can always ignore the upgrade. Consequently, the system's "pattern capacity" (in the compressive sense) can only increase.*

Remark 2862. *One can also read the conclusion in a strictness-sensitive way: if the expansion $\mathcal{W}(x) \subseteq \mathcal{W}'(x)$ introduces at least one genuinely new witness that attains a strictly smaller cost than all witnesses in $\mathcal{W}(x)$, then the infimum strictly decreases and the attainable intensity strictly increases (provided the truncation at 0 is not already active). Thus, observer upgrades are only* detectable *in intensity when they actually enable new, more economical descriptions for the particular x under consideration.*

Remark 2863. *This result connects tightly to the preceding closed-form theorem: intensity capacity depends only on the infimal achievable cost. So any mechanism that decreases infimal cost (by expanding the witness set) increases attainable intensity. This is one formal justification for the common practice of "representation learning" as a means to reveal stronger patterns: by changing what counts as a witness, one changes what can be compressed.*

Remark 2864. *Equivalently, if the baseline is fixed, then the upgrade comparison can be expressed purely at the level of ratios:*

$$I_C^{\star\prime}(x) - I_C^{\star}(x) = \max\left\{0, 1 - \frac{s'_C(x)}{s_C^{\mathrm{base}}(x)}\right\} - \max\left\{0, 1 - \frac{s_C(x)}{s_C^{\mathrm{base}}(x)}\right\},$$

where $s'_C(x) := \inf_{w \in \mathcal{W}'(x)} u(w; x)$. This makes explicit that the only lever the upgrade has (in this fixed-baseline setting) is to reduce the achievable infimal cost; everything else is a monotone postprocessing.

Proof. The first inequality is the basic fact that an infimum over a larger set cannot increase. For the second, use the closed form for $I^\star$ in terms of the infimal cost: decreasing the infimal cost weakly increases $\max\{0, 1 - \cdot / s_C^{\mathrm{base}}(x)\}$. □

Remark 2865. *In optimization language, $\mathcal{W}(x) \subseteq \mathcal{W}'(x)$ is a feasible-set expansion, so the optimal value (here, the infimum of costs) is monotone nonincreasing. The intensity statement is then immediate because intensity is an order-reversing transform of cost (up to truncation). This perspective is useful when chaining multiple upgrades: any sequence of feasible-set expansions yields a monotone nondecreasing sequence of attainable intensity capacities for each fixed x and fixed baseline.*

Remark 2866. Proof sketch and intuition. *The first claim is order theory: $\mathcal{W}(x) \subseteq \mathcal{W}'(x)$ means you are minimizing over more options, so the minimum (or infimum) cannot go up. The second claim is calculus-free: the function $m \mapsto \max\{0, 1 - m/s\}$ is nonincreasing in m, so if the best cost drops, the best intensity rises. The point is not subtlety but reusability: the same argument appears repeatedly when comparing reasoning regimes.*

50.5 Coupling to weakness/effort policies: "open if you can"

Theorem 72 (Opening is a triple win under representational monotonicity)**.** *Assume effort/weakness monotonicity and representational monotonicity under opening. Then for any opening $H \subseteq H'$ and any x:*

$$w(H) \leq w(H'), \qquad \mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(H'), \qquad I^{\star}_{C,H}(x) \leq I^{\star}_{C,H'}(x).$$

Remark 2867. *The ordering $H \subseteq H'$ should be read as "H' is at least as open as H," i.e., H' permits at least the same identifications/indistinctions as H (and possibly more). In that sense, weakness is naturally monotone in this order (more openness means more allowed indistinction), while effort is naturally antitone (more openness typically means fewer constraints to enforce, hence less work). Representational monotonicity then asserts that opening does not remove any representational options available under H: it can only add witnesses or constructions.*

Remark 2868. *The theorem says: if opening a policy both (i) reduces effort and (ii) enlarges the allowed witness family, then opening is simultaneously good for three quantities: it increases weakness (you accept more indistinction), decreases effort (you work less), and increases pattern capacity (you can achieve at least as much intensity). It is not that opening is always normatively preferred, but that* under these assumptions *there is no internal tradeoff between these three dimensions.*

Remark 2869. *A useful way to read the last inequality is as a* pointwise dominance *statement in x: for each fixed instance x, the best attainable intensity when using the more open policy H' is at least the best attainable intensity under H. This is stronger than a mere statement about averages or typical cases: it rules out the possibility that opening helps some inputs but hurts others, provided representational monotonicity holds as stated for every x.*

Remark 2870. *This is a formal articulation of the stance "open if you can": if feasibility constraints do not forbid openness, and if openness expands representational options, then openness is a rational default. The assumptions are strong enough to be useful and weak enough to hold in many coarse-graining or abstraction regimes.*

Remark 2871. *The assumptions can be viewed as separating* engineering *effects from* mathematical *effects. Effort/weakness monotonicity are policy-level regularities about the cost and permissiveness of enforcement; representational monotonicity is a structural regularity about the modeling language or witness mechanism. The theorem then says that once these are granted, the intensity comparison becomes purely set-theoretic: enlarging the feasible witness family cannot make the supremum of an objective smaller.*

Proof. The weakness and effort inequalities are assumptions. For intensity capacity: $H \subseteq H'$ implies $\mathcal{W}_H(x) \subseteq \mathcal{W}_{H'}(x)$ by representational monotonicity, so the attainable intensity bound can only increase by the representation-expansion theorem. □

Remark 2872. Proof sketch and intuition. *Two of the three inequalities are simply unpacking the assumptions. The third inequality is a one-line reduction: opening enlarges $\mathcal{W}_H(x)$, and the best achievable intensity over a larger set cannot be smaller. The picture is: opening "widens the funnel" through which witnesses may pass, so it cannot reduce the best witness you could find.*

Remark 2873. *One can also interpret the third inequality as an "option value" principle: even if the agent ultimately chooses to use only witnesses that were already available under H, the mere availability of additional witnesses under H' cannot decrease the optimum. Any intensity-achieving* *(or approximately intensity-achieving) witness feasible under H remains feasible under H' and can be reused verbatim, while H' may additionally permit more structured or more compressive witnesses that achieve higher intensity.*

Theorem 73 (Maximally open adequate policy maximizes pattern capacity pointwise)**.** *Let $\mathcal{F}_C$ be a feasible (adequate) set of policies, and assume it is upward closed:*

$$H \in \mathcal{F}_C,\ H \subseteq H' \implies H' \in \mathcal{F}_C.$$

Define the union (maximally open) feasible policy

$$\overline{H} := \bigcup_{H \in \mathcal{F}_C} H.$$

Assume effort/weakness monotonicity and representational monotonicity under opening. Then:

1. *$\overline{H} \in \mathcal{F}_C$ and $H \subseteq \overline{H}$ for all $H \in \mathcal{F}_C$;*
2. *$\overline{H}$ minimizes effort on $\mathcal{F}_C$ and maximizes weakness on $\mathcal{F}_C$;*
3. *for every x, $\overline{H}$ maximizes attainable intensity capacity pointwise:*

$$I^\star_{C,\overline{H}}(x) \geq I^\star_{C,H}(x) \quad \textit{for all } H \in \mathcal{F}_C.$$

Remark 2874. *The upward-closed condition is the formal expression of "if you can open, you remain adequate." In many applications, adequacy/feasibility expresses resource, safety, or consistency constraints that become* easier *(not harder) to satisfy as one relaxes distinctions: e.g., if a strict discrimination requirement is satisfiable within a budget, then a weaker discrimination requirement is also satisfiable within the same budget. When this monotonicity of feasibility fails, the union construction may no longer produce an admissible policy, and the selection rule would require modification (for example, selecting maximal elements within a non-upward-closed feasible region).*

Remark 2875. *This theorem constructs a canonical "most open" policy consistent with feasibility: take the union of all feasible policies. The upward-closed condition says feasibility is preserved under opening, so if some level of indistinction is allowed, then any weaker demand for distinctions is also allowed. In that case, the union is feasible and dominates every other feasible policy in openness. This is an instance of a familiar lattice-theoretic idea: maximal elements arise by joining (here, unioning) all admissible approximations. It resonates with fixed-point-and-selection themes (cf. Section 48) in that one often selects a "largest" stable regime subject to constraints.*

Remark 2876. *It may be helpful to emphasize why the union is the right "join" operation here. Inclusion equips the policy space with a natural partial order, and union is the least upper bound with respect to this order: it is the smallest policy that is at least as open as every feasible $H \in \mathcal{F}_C$, while still containing all distinctions that any feasible policy is willing to drop. Thus $\overline{H}$ is not an arbitrary choice; it is forced once one insists on a single policy that upper-bounds the entire feasible family.*

Remark 2877. *The conclusion is a precise, pointwise version of "maximally open adequate." It says that within the feasible region, $\overline{H}$ is extremal in effort and weakness, and it is also extremal in the capacity to support intense witnesses. This is a principled selection rule for a reasoning engine: if the only objective is to maximize compressive pattern capacity while remaining feasible, then select $\overline{H}$. The conceptual background aligns with tradeoff-based policy selection discussed in [9].*

Remark 2878. *Note the separation between extremality statements: item (2) is a global extremality over $\mathcal{F}_C$ (a single scalar effort and scalar weakness per policy), while item (3) is an extremality for each x separately. In particular, even if one later aggregates $I^{\star}_{C,H}(x)$ over x (e.g., by expectation under a data distribution), the pointwise dominance immediately implies dominance under any monotone aggregation functional.*

Proof. (1) Upward closure implies $\overline{H}$ is feasible, and the union contains each member. (2) If $H \subseteq \overline{H}$ and effort is antitone, then $\mathrm{Eff}_C(H) \geq \mathrm{Eff}_C(\overline{H})$; if weakness is monotone, then $w(H) \leq w(\overline{H})$. (3) For any feasible H, we have $H \subseteq \overline{H}$, hence $\mathcal{W}_H(x) \subseteq \mathcal{W}_{\overline{H}}(x)$ for all x. Therefore $I^{\star}_{C,\overline{H}}(x) \geq I^{\star}_{C,H}(x)$ by representation expansion. □

Remark 2879. Proof sketch and intuition. *The proof is a chain of monotonicities. Because feasibility is upward closed, taking the union of feasible policies stays feasible. Union is the largest element with respect to inclusion, so every feasible H is contained in $\overline{H}$. Antitonicity of effort and monotonicity of weakness then make $\overline{H}$ extremal for those quantities. Finally, representational monotonicity converts $H \subseteq \overline{H}$ into inclusion of witness families, and the representation-expansion theorem converts that into the pointwise maximality of intensity capacity.*

Remark 2880. *A concrete mental model is: $\mathcal{F}_C$ encodes all "acceptable" abstraction levels under a fixed budget or constraint set C, and $\overline{H}$ is the coarsest (most open) abstraction among them. The theorem then asserts that, under the monotonicities, the coarsest acceptable abstraction simultaneously (i) costs the least to operate, (ii) tolerates the most indistinction, and (iii) enables the richest attainable patterns (as measured by intensity capacity) at every input x. This provides a decision tool: rather than tuning openness by hand, one can search for feasibility and then take the join of what is feasible.*

50.6 Pareto pruning in tradeoff spaces (finite candidate sets)

Definition 466 (Dominance for witnesses at fixed x)**.** *Fix x and define a quality vector for a witness w by*

$$q(w) := \big(u(w;x),\ -I_C(w;x)\big) \in [0,\infty] \times [-1,0].$$

Say w' dominates *w if $q(w') \leq q(w)$ coordinatewise and $q(w') \neq q(w)$. Equivalently: $u(w';x) \leq u(w;x)$ and $I_C(w';x) \geq I_C(w;x)$ with at least one strict.*

Remark 2881. *The codomain $[0,\infty] \times [-1,0]$ allows for "infinite" cost (e.g. an infeasible or undefined witness under the current resource model) while keeping the dominance relation well-defined: any finite-cost alternative with comparable intensity will dominate an infinite-cost candidate. Similarly, if two witnesses coincide in both coordinates (a genuine tie), neither dominates the other; they are interchangeable as far as the two criteria are concerned, and downstream logic may keep either or both depending on representational needs (e.g. different internal structures producing the same tradeoff point).*

Remark 2882. *This is the standard Pareto order (Hyperseed-Concept 147) specialized to two criteria: lower cost is better, and higher intensity is better. The sign flip in the second coordinate is merely a notational convenience so that "better" corresponds to "smaller" in both coordinates. Dominance captures the idea of an option that is unambiguously inferior: it is more expensive and no more intense, or less intense and no cheaper.*

Remark 2883. *It can be helpful to view dominance as the strict part of a preorder induced by coordinatewise comparison. Concretely, if we define the weak relation $w' \preceq w$ by $q(w') \leq q(w)$*

coordinatewise, then w' dominates w exactly when $w' \preceq w$ and not $w \preceq w'$. In the plane $(u, -I)$, the set of points weakly improving on $q(w)$ is the south-west closed quadrant with corner at $q(w)$; dominance means that $q(w')$ lies in that quadrant and is not identical to $q(w)$.

Remark 2884. *In practice, this definition is useful because it decouples local pruning from global objectives. An engine might later apply any scalarization of the two objectives (weighted sums, lexicographic orders, constraints like $I \geq \tau$ or $u \leq B$), but dominance pruning is always safe for any monotone scalarization: a dominated witness cannot become optimal under any reasonable way of combining the criteria.*

Remark 2885. *One way to formalize "reasonable" here is: a scalar objective $F(u, I)$ is* monotone *if it is nondecreasing in u and nonincreasing in $-I$ (equivalently, nonincreasing in I when interpreted as a maximization). Then $q(w') \leq q(w)$ implies $F\big(u(w'; x), I_C(w'; x)\big) \leq F\big(u(w; x), I_C(w; x)\big)$ for minimization-style scalarizations, so a dominated witness can never strictly beat its dominator. This covers common choices such as weighted sums $\lambda u - (1 - \lambda)I$, Lagrangian relaxations, and feasibility-first rules like "minimize u subject to $I \geq \tau$" (which is monotone in the sense that improving intensity or reducing cost cannot turn a feasible better point into a worse one).*

Theorem 74 (Finite Pareto frontier is nonempty)**.** *Fix x and let $\mathcal{A}$ be a finite set of candidate witnesses for x. Then there exists at least one undominated witness in $\mathcal{A}$ (a Pareto frontier element).*

Remark 2886. *The theorem asserts that in a finite candidate set, there is always at least one "maximally good" element under the dominance order. This is not deep, but it is foundational for designing bounded search procedures: if you prune dominated candidates, you will not prune everything. The result is a small but essential guarantee that supports iterative refinement, beam maintenance, and anytime algorithms.*

Remark 2887. *The statement is the two-criterion instance of a general fact: in any finite set equipped with a strict partial order, there exists at least one* maximal *element (an element with no strictly larger element). Here "larger" corresponds to "better" under the Pareto order. This perspective is useful because it separates the combinatorial content (finiteness implies existence of maximals) from the semantic content (what we chose as the dominance relation).*

Proof. Assume for contradiction that every $w \in \mathcal{A}$ is dominated by some $w' \in \mathcal{A}$. Pick any $w_0 \in \mathcal{A}$ and iteratively choose a strict dominator w_{k+1} of w_k. Each step strictly improves at least one coordinate while not worsening the other, so w_k are all distinct. This yields an infinite sequence of distinct elements of $\mathcal{A}$, contradicting finiteness. $\square$

Remark 2888. *The "all distinct" claim can be unpacked as follows. If w_{k+1} dominates w_k, then $q(w_{k+1}) \leq q(w_k)$ coordinatewise and $q(w_{k+1}) \neq q(w_k)$. In particular, at least one coordinate strictly decreases, so returning to any previously visited w_j would require a later step to increase that coordinate to match $q(w_j)$, which is impossible because every step is coordinatewise nonincreasing. Thus the construction cannot cycle, and strict improvement forces an ever-descending chain in a set that was assumed finite.*

Remark 2889. Proof sketch and intuition. *The argument is a classic "infinite descent" in a finite set. If everything is dominated, then starting anywhere you can always step to something strictly better. Strict improvement prevents cycles, so you obtain an infinite strictly improving chain, which cannot fit inside a finite set. Geometrically, points $(u, -I)$ in the plane cannot have infinite strictly decreasing sequences if there are only finitely many points available.*

Remark 2890. *A concrete geometric way to visualize the frontier: plot each candidate as a point* $(u(w;x), -I_C(w;x))$. *The undominated points are exactly those for which the south-west quadrant contains no other point. For example, if candidates yield* $(u, -I)$ *values* $\{(1, -0.2), (2, -0.9), (1.5, -0.2)\}$, *then* $(1, -0.2)$ *dominates* $(1.5, -0.2)$ *(same intensity, lower cost), while* $(2, -0.9)$ *may remain undominated because it trades higher cost for much higher intensity. The frontier is then* $\{(1, -0.2), (2, -0.9)\}$, *reflecting two qualitatively different tradeoffs.*

Remark 2891 (How an engine uses this)**.** *If a reasoning system maintains a finite beam of candidate patterns, it can safely discard any candidate dominated in* (*cost*, *intensity*) *space. The remaining beam is guaranteed nonempty and contains all candidates that could be optimal under any monotone scalarization (e.g. minimize cost subject to intensity* $\geq \tau$*, or maximize intensity subject to cost* $\leq B$*).*

Remark 2892. *From an implementation standpoint, dominance pruning over a finite beam of size* n *can be done by a simple* $O(n^2)$ *pairwise check, which is often acceptable when the beam is small and the per-candidate evaluation is the expensive part. When* n *is large, one can exploit the two-dimensional structure: sorting by cost and scanning while maintaining the best intensity seen so far yields a skyline/frontier computation that is typically faster (and in many cases* $O(n \log n)$ *due to sorting). These algorithmic considerations matter because they make Pareto maintenance a low-overhead invariant that can be enforced at every iteration of a search loop.*

Remark 2893. *A further pragmatic reading is that Pareto pruning is a way of maintaining epistemic humility: one does not commit prematurely to a particular tradeoff between cost and intensity. Instead one keeps the frontier, which represents the live options consistent with multiple possible downstream preferences, goals, or meta-policies (Hyperseed-Concept 110). This resonates with the broader project of building reasoning systems that can revise their objective functions while preserving sound intermediate commitments.*

Remark 2894. *Finally, note that the same pruning logic generalizes immediately to more than two criteria (e.g. cost, intensity, and a third axis such as robustness or interpretability). The finite-frontier-nonempty theorem remains true for any fixed finite number of coordinates: in a finite set, there is always at least one undominated element under coordinatewise dominance. The two-dimensional case is emphasized here because it admits particularly clear geometric intuition and efficient frontier-maintenance routines, making it a convenient "decision tool" module inside larger Weakness/Effort/Pattern-Intensity tradeoff workflows.*

51 Emergence and Synergy Theorems for Explanation and Discovery

We prove a small family of emergence/synergy theorems designed to be useful for inference engines leveragint Hyperseed ontology [1]: (1) selection/witness results that guarantee the existence of a specific emergent pattern to point to, (2) a cost-level characterization of emergent surplus as strict subadditivity (emergent compression), (3) quantitative lower bounds connecting a single emergent-compression explanation to synergy gain, (4) an operational budget theorem showing synergy expands solvable tasks, (5) witness theorems for collective (interaction-only) properties.

Remark 2895. *Philosophically, this section treats "emergence" and "synergy" as two sides of one logical coin. At the entity level, emergence is a statement about how the joint object* $A *_C B$ *admits a pattern witness* Q *that is not merely a weighted average of how* Q *behaves on* A *and on*

*B separately (see Hyperseed-Concept **??** and 130). At the explanation level, synergy is a statement about how combining distinct inference resources can reduce the contrivance needed to achieve a given fit (see Hyperseed-Concept 73 and 202), echoing the "cognitive synergy" viewpoint in [19]. The core technical move is to relate these semantic notions to inequalities on costs, so that an engine can turn a qualitative slogan ("the whole is more than the parts") into a quantitative decision rule.*

Concretely, the role of the "witness" language in items (1) and (5) is operational: rather than asserting that some global regularity exists in an abstract sense, we aim to certify the existence of a particular pattern variable Q (or a bounded family of candidates) that an inference engine can explicitly track, score, and later reuse as an explanatory handle. In this sense, a witness theorem is also a pointer theorem: it justifies searching for, caching, and communicating a finite artifact that stands for an otherwise diffuse emergent effect.

Items (2) and (3) translate that witness perspective into inequalities. Strict subadditivity corresponds to a regime in which a single joint description of the composite $A *_C B$ (or of the data generated by it) is cheaper than any decomposition into separate descriptions of A and B plus whatever bookkeeping is needed to coordinate them through C. This is the formal sense in which "emergent surplus" becomes "emergent compression": the surplus is measured not by an extra ontological ingredient but by a gap between best joint cost and best separable cost. The quantitative bounds in (3) then show how a certified compression gap can be cashed out as a lower bound on synergy gain, i.e. on the reduction in contrivance achievable by mixing inference resources instead of using them in isolation.

Item (4) makes the preceding correspondence actionable in a resource-bounded setting. Rather than treating synergy as an abstract property of agents, we treat it as an effect on feasible sets of tasks under explicit budgets: a fixed contrivance or cost budget can certify a strictly larger set of achievable fits when resources are allowed to interact. In this way, the theorems support a pipeline where an engine (i) detects an emergent witness Q for $A *_C B$, (ii) verifies a strict subadditivity gap at the cost level, and (iii) upgrades that gap into a guarantee that a broader class of explanatory targets becomes solvable under the same operational constraints.

Finally, the "collective (interaction-only)" emphasis in (5) isolates properties that provably do not decompose into attributes of A alone or B alone. Such theorems are intended to prevent a common failure mode in automated discovery: misattributing a relational regularity to one component due to an overly separable model class. By requiring that the witness depend essentially on the joint structure $A *_C B$ (and not merely on marginal behavior), we obtain a principled certificate that the identified pattern is genuinely collective in the technical sense required for downstream reasoning and intervention.

51.1 Standing setup

51.1.1 Pattern intensity and emergence (entity-level)

Fix a context C with:

- a set of entities $\mathcal{E}_C$,
- a (partial) binary combination operation $*_C : \mathcal{E}_C \times \mathcal{E}_C \rightharpoonup \mathcal{E}_C$,
- a baseline description cost $s_C^{\text{base}} : \mathcal{E}_C \to \mathbb{R}_{\geq 0}$,
- a pattern vocabulary $\mathcal{P}_C$ (finite or countable).

We treat each $Q \in \mathcal{P}_C$ as a *candidate pattern witness* that, when applicable to $x \in \mathcal{E}_C$, comes with a nonnegative cost $u_C(Q;x) \in [0,\infty]$ (if Q is not applicable to x, take $u_C(Q;x) = \infty$).

Remark 2896. *A few pieces of notation are doing most of the work:*

- $\mathcal{E}_C$ *is the domain of things we may wish to explain or compress (data items, objects, states), relative to a context* C.
- *The partial operation* $*_C$ *formalizes "composition" or "interaction" inside the context; the arrow* $\rightharpoonup$ *indicates that not all pairs* (A, B) *can be composed.*
- $s_C^{\mathrm{base}}(x)$ *is a reference cost for* x*—the cost of an unstructured, default, or otherwise agreed-upon representation (cf. Hyperseed-Concept 169 and 82).*
- $\mathcal{P}_C$ *is a vocabulary of candidate pattern-witnesses, and* $u_C(Q;x)$ *is the cost of using witness* Q *to represent or account for* x.

This overall framing is close in spirit to description-length and compression viewpoints from algorithmic information theory [16], and to the general Hyperseed perspective that patterns are, operationally, economical representations [5] (see Hyperseed-Concept 130).

Definition 467 (Scalar intensity)**.** *Assume* $s_C^{\mathrm{base}}(x) > 0$. *Define the (scalar) intensity of* Q *for* x *by*

$$I_C(Q;x) := \max\left\{0,\ 1 - \frac{u_C(Q;x)}{s_C^{\mathrm{base}}(x)}\right\} \in [0,1].$$

If $s_C^{\mathrm{base}}(x) = 0$, *set* $I_C(Q;x) = 0$ *by convention.*

Remark 2897. *Intuitively,* $I_C(Q;x)$ *measures the* fractional savings *achieved by explaining/encoding* x *using* Q *instead of the baseline. If* $u_C(Q;x)$ *is smaller than* $s_C^{\mathrm{base}}(x)$, *then* Q *is genuinely compressive and the intensity is positive. If* $u_C(Q;x) \geq s_C^{\mathrm{base}}(x)$, *then the witness is not better than baseline and intensity is clamped to* 0. *This makes intensity a normalized indicator of "pattern usefulness" (Hyperseed-Concept 130) rather than merely a raw cost difference.*

For a simple example, if $s_C^{\mathrm{base}}(x) = 10$ *and* $u_C(Q;x) = 7$, *then* $I_C(Q;x) = 1 - 7/10 = 0.3$. *If* $u_C(Q;x) = 12$, *then* $I_C(Q;x) = \max\{0, 1 - 12/10\} = 0$. *The clamp is important for downstream reasoning: it prevents negative values from being misread as "anti-patterns" and keeps intensities comparable across witnesses.*

Definition 468 (Emergence weights and emergent surplus/intensity)**.** *Fix* $A, B \in \mathcal{E}_C$ *with* $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) > 0$ *and assume* $A *_C B$ *is defined. Define weights*

$$w_C(A) := \frac{s_C^{\mathrm{base}}(A)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}, \qquad w_C(B) := \frac{s_C^{\mathrm{base}}(B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

For $Q \in \mathcal{P}_C$, *define the emergent surplus*

$$\mathrm{Sur}_C(Q;A,B) := I_C\big(Q; A *_C B\big) - \Big(w_C(A)\, I_C(Q;A) + w_C(B)\, I_C(Q;B)\Big),$$

and the emergent intensity

$$\mathrm{Emer}_C(Q;A,B) := \max\{0, \mathrm{Sur}_C(Q;A,B)\}.$$

Remark 2898. *The weights $w_C(A), w_C(B)$ sum to 1 and bias the comparison by baseline description size: if A is "larger" than B in baseline cost, then A's intensity contributes more heavily to the "non-emergent" expectation. Thus $\mathrm{Sur}_C(Q; A, B)$ compares the pattern intensity on the composite $A *_C B$ against what one would predict by treating A and B as merely co-present but not truly interacting. When $\mathrm{Sur}_C(Q; A, B) > 0$, Q is more intense on the composite than this weighted baseline, and we call that an emergent contribution (Hyperseed-Concept* **??***).*

Geometrically (in the very mild sense of convexity), the term $w_C(A)\, I_C(Q; A) + w_C(B)\, I_C(Q; B)$ is a point on the line segment between the "part intensities." Emergence asserts that $I_C(Q; A *_C B)$ lies strictly above that segment. A toy example: suppose $w_C(A) = w_C(B) = 1/2$, $I_C(Q; A) = 0.1$, $I_C(Q; B) = 0.1$, but $I_C(Q; A *_C B) = 0.6$. Then $\mathrm{Sur}_C(Q; A, B) = 0.6 - 0.1 = 0.5$ and $\mathrm{Emer}_C(Q; A, B) = 0.5$: Q becomes salient only at the interaction level.

Definition 469 (Collective-property-set)**.** *Fix a reference class $\mathcal{B} \subseteq \mathcal{E}_C$. Define*

$$\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q) := \sup_{B \in \mathcal{B}} \mathrm{Emer}_C(Q; A, B).$$

Remark 2899. *This definition elevates an interaction effect into an "attribute" of A relative to a reference class $\mathcal{B}$. It is deliberately supremal: $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$ asks for the best interaction partner $B \in \mathcal{B}$ that makes Q emerge with A. In this sense it is a controlled way to "lift from instances to groups" (Hyperseed-Concept 103) and to formalize what one might otherwise describe informally as a* collective *or* relational *property.*

As a simple example, let A be a molecule and $\mathcal{B}$ a set of solvents. A pattern witness Q might represent "forms a stable complex." Then $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$ measures the maximal emergent stability obtainable by pairing A with some solvent in the class, even if A alone does not display that property.

51.1.2 Synergy (explanation-level)

Fix a task/dataset/behavior b. Let $\mathcal{E}(b)$ be a space of candidate explanations for b. Each explanation $E \in \mathcal{E}(b)$ has:

- a fit score $\mathrm{Fit}(E) \in [0, 1]$,
- a weakness $\mathrm{Weak}(E) \in (0, 1]$,
- a contrivance cost $\mathrm{Con}(E) := -\log(\mathrm{Weak}(E)) \in [0, \infty)$.

In particular, $\mathrm{Weak}(E) = 1$ corresponds to $\mathrm{Con}(E) = 0$ (no contrivance penalty), while $\mathrm{Weak}(E) \downarrow 0$ corresponds to $\mathrm{Con}(E) \uparrow \infty$, encoding the idea that extremely fragile or ad-hoc explanations incur arbitrarily large cost.

Fix a collection of knowledge types (or inference resources) T. Let $\mathcal{E}_T(b) \subseteq \mathcal{E}(b)$ be the set of explanations constructible using only T. One can regard T either as a set of primitive operations available to an agent (proof rules, learned heuristics, simulators, retrieval tools, etc.) or as a restriction on allowable internal representations; in either case, $\mathcal{E}_T(b)$ induces a feasibility constraint on what counts as a candidate explanation.

Remark 2900. *Here* $\mathrm{Fit}(E)$ *is an adequacy measure (how well E accounts for b), while* $\mathrm{Weak}(E)$ *is a penalty-like score: higher weakness means less contrived, more natural, or more "easily supported" explanation. Defining* $\mathrm{Con}(E) = -\log(\mathrm{Weak}(E))$ *merely turns multiplicative weakness into additive cost, as is standard when moving between likelihood-like quantities and description-length-like quantities [16]. This weakness-based formalism aligns with the central role of weakness* *in Hyperseed-style quantale reasoning [3, 2] (see Hyperseed-Concept 202 and 143). Intuitively, if an explanation can be factored into several independent-ish supporting components whose weaknesses multiply, then their contrivances add; this is the operational reason* log *is convenient when comparing composite explanations and when attributing "where the cost went" across sub-claims.*

Also, the restriction $\mathcal{E}_T(b)$ formalizes architectural constraints: one may only build explanations using a chosen set of inference resources T (Hyperseed-Concept 191 and 186). Synergy, then, is the phenomenon that the union $T_1 \cup T_2$ can support explanations of lower contrivance than what is reachable by using T_1 and T_2 separately, reflecting the cognitive-synergy view of AGI system design [19]. Put differently, T controls the search space *of explanations, while* Con *ranks explanations inside that space by how non-accidental they are; synergy is about how enlarging the search space via tool-combination changes the best attainable ranking at a fixed adequacy requirement.*

Definition 470 (Best achievable contrivance, synergy gain)**.** *Fix a fit threshold $\tau \in (0, 1]$. Define*

$$\mathrm{Con}^*(T;\tau) := \inf\{\mathrm{Con}(E) : E \in \mathcal{E}_T(b),\ \mathrm{Fit}(E) \geq \tau\}.$$

For disjoint T_1, T_2, define the synergy gain

$$\Sigma(T_1, T_2; \tau) := \Big(\mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau) - \mathrm{Con}^*(T_1 \cup T_2;\tau)\Big)_+.$$

In words, $\mathrm{Con}^*(T;\tau)$ is the minimal contrivance cost needed to reach fit at least τ using only resources T, and Σ records the extent to which joint access $T_1 \cup T_2$ improves upon the "separate bests" baseline.

Remark 2901. *The notation $(\cdot)_+$ denotes the* positive part*: $(x)_+ := \max\{0, x\}$. Thus $\Sigma(T_1, T_2; \tau)$ is, by definition, never negative: if combining knowledge types does not help, synergy is recorded as 0 rather than as "negative synergy." Operationally this makes Σ a conservative certificate: $\Sigma > 0$ is strong evidence that integrated reasoning provides a genuine efficiency gain, whereas $\Sigma = 0$ does not claim that integration is harmful.*

A concrete numerical example: if $\mathrm{Con}^*(T_1;\tau) = 10$, $\mathrm{Con}^*(T_2;\tau) = 9$, and $\mathrm{Con}^*(T_1 \cup T_2;\tau) = 16$, then $\Sigma = (10+9-16)_+ = 3$. Interpreting contrivance as an effort-like resource (Hyperseed-Concept 100), this says that integration saves "3 units" of contrivance at the target fit threshold. One may also read the same example in weakness terms: since $\mathrm{Weak} = \exp(-\mathrm{Con})$, the separate best weaknesses multiply to $\exp(-(10+9))$, while the integrated best achieves weakness $\exp(-16)$, i.e., a factor $\exp(3)$ less contrived than the multiplicative baseline.

Two further technical clarifications are often useful in applications. First, the use of inf *(rather than* min*) allows the possibility that there is no single optimal explanation, but there exists a sequence of explanations in $\mathcal{E}_T(b)$ whose contrivance approaches the optimum arbitrarily closely; this matches common situations in continuous model classes or in asymptotic regimes. Second, feasibility depends on τ: if τ is increased, the constraint $\mathrm{Fit}(E) \geq \tau$ becomes stricter, so $\mathrm{Con}^*(T;\tau)$ is (weakly) nondecreasing in τ whenever the feasible set is nonempty. In the extreme case where no explanation in $\mathcal{E}_T(b)$ reaches fit τ, the infimum is taken over the empty set; adopting the standard convention $\inf \emptyset = +\infty$ makes $\mathrm{Con}^*(T;\tau) = +\infty$ in that case, which in turn ensures $\Sigma(T_1, T_2; \tau) = 0$ whenever either T_1 or T_2 cannot meet the threshold at all (i.e., synergy is not credited merely for turning "impossible" into "possible" unless one separately tracks feasibility).*

Finally, the assumption that T_1 and T_2 are disjoint is a bookkeeping convenience: it prevents double-counting the same knowledge type in the baseline $\mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau)$. If T_1 and T_2 overlap, one may either replace them by disjoint copies (tagging shared resources) or interpret the definition as measuring synergy relative to a chosen partition of capabilities into "modules."

51.2 Emergence as a selection signal (discovery by pointing)

Definition 471 (Total emergent mass)**.** *Assume $\mathcal{P}_C$ is finite. Define*

$$M_C^{\mathrm{em}}(A,B) := \sum_{Q\in\mathcal{P}_C} \mathrm{Emer}_C(Q;A,B).$$

Remark 2902. *$M_C^{\mathrm{em}}(A,B)$ aggregates, over a finite pattern vocabulary, how much emergent intensity the pair (A,B) exhibits in total. The finiteness assumption is crucial: it makes $M_C^{\mathrm{em}}(A,B)$ a genuine finite sum rather than a supremum or integral, and allows a simple averaging argument to guarantee a concrete witness. From the standpoint of an inference engine, $M_C^{\mathrm{em}}(A,B)$ is a coarse global signal, while the individual $\mathrm{Emer}_C(Q;A,B)$ values are fine-grained, interpretable hypotheses to point at (Hyperseed-Concept 56 and 144). In particular, $M_C^{\mathrm{em}}(A,B)$ can be read as the "total budget" of emergent evidence distributed across candidate patterns, so that the subsequent witness theorem formalizes a basic conversion principle: a nontrivial total budget implies at least one nontrivial, human-inspectable contributor. A useful normalization that will implicitly appear in applications is the* average emergent mass per pattern*, $M_C^{\mathrm{em}}(A,B)/|\mathcal{P}_C|$, which can be interpreted as the expectation of $\mathrm{Emer}_C(Q;A,B)$ under the uniform distribution on $\mathcal{P}_C$; the witness theorem can then be seen as the elementary fact that a maximum dominates an average.*

Theorem 75 (Finite-vocabulary witness of emergence)**.** *Assume $\mathcal{P}_C$ is finite with $|\mathcal{P}_C| = N$. If $M_C^{\mathrm{em}}(A,B) \geq \varepsilon > 0$, then there exists $Q \in \mathcal{P}_C$ such that*

$$\mathrm{Emer}_C(Q;A,B) \;\geq\; \frac{\varepsilon}{N}.$$

Remark 2903. *Intuitively, this theorem says: if the* total *amount of emergence is nontrivial, then at least one candidate pattern witness must carry a nontrivial share of it. There is nothing mysterious here; it is an instance of the pigeonhole principle applied to emergent mass. But its importance is methodological: it justifies "discovery by pointing," i.e. replacing an aggregate signal with a specific explainable hypothesis Q.*

This result connects forward to the later cost-subadditivity theorem: once a particular Q is singled out, one may ask whether its emergence is explained by strict joint compression (Section 76 below), and then propagate that fact into explanation-level synergy bounds. A further interpretive point is that the bound ε/N is the best possible uniform guarantee given only the sum constraint: if the emergent mass were spread perfectly evenly across the N patterns, one would have $\mathrm{Emer}_C(Q;A,B) = \varepsilon/N$ for all Q, so no larger lower bound could be certified without additional structure (e.g. sparsity, heavy-tailedness, or a prior over $\mathcal{P}_C$). For concrete intuition, if $N = 100$ and $M_C^{\mathrm{em}}(A,B) \geq 5$, the theorem guarantees at least one pattern with $\mathrm{Emer}_C(Q;A,B) \geq 0.05$; the "pointing" step then amounts to surfacing such a pattern rather than treating the value 5 as an opaque scalar.

Proof. Suppose for contradiction that $\mathrm{Emer}_C(Q;A,B) < \varepsilon/N$ for every $Q \in \mathcal{P}_C$. Summing over N elements yields

$$M_C^{\mathrm{em}}(A,B) = \sum_{Q\in\mathcal{P}_C} \mathrm{Emer}_C(Q;A,B) < \sum_{Q\in\mathcal{P}_C} \frac{\varepsilon}{N} = \varepsilon,$$

contradicting $M_C^{\mathrm{em}}(A,B) \geq \varepsilon$. Equivalently (and without contradiction), dividing by N shows that the average value of $\mathrm{Emer}_C(Q;A,B)$ over $\mathcal{P}_C$ is at least ε/N, and hence the maximum over Q must be at least ε/N. $\square$

Proof sketch. The argument is pure averaging: if all N terms were $< \varepsilon/N$, their sum would be $< \varepsilon$, contradicting the assumed lower bound on the sum. A one-line reformulation is $\max_{Q\in\mathcal{P}_C} \mathrm{Emer}_C(Q; A, B) \geq \frac{1}{N}\sum_{Q\in\mathcal{P}_C} \mathrm{Emer}_C(Q; A, B) = \frac{M_C^{\mathrm{em}}(A,B)}{N}$. $\square$

Remark 2904. *The key step is the contrapositive use of linearity of finite summation: a lower bound on the sum forces at least one term to be large. Visually, one may imagine distributing ε units of emergent "mass" across N bins; if every bin held less than ε/N, the total could not reach ε.*

In an implementation-oriented reading, the theorem gives a certificate that $\arg\max_Q \mathrm{Emer}_C(Q; A, B)$ *is meaningful in the strong sense that it must return a witness whose emergent intensity is bounded below whenever total emergence is. Operationally, this means that any procedure that (even approximately) ranks candidates by* $\mathrm{Emer}_C(Q; A, B)$ *is justified as a* selection mechanism*: it is guaranteed to surface at least one pattern above a quantitative floor whenever the global statistic crosses the corresponding threshold. This is also the place where finiteness is doing real work: for infinite $\mathcal{P}_C$, one can have positive "total mass" only after replacing the sum by a different aggregator (e.g. an integral or supremum), and then the existence of a pointwise witness generally requires additional compactness, measurability, or regularity assumptions that are deliberately avoided here.*

Corollary 37 (Pointing corollary)**.** *If $M_C^{\mathrm{em}}(A, B) > 0$ and $\mathcal{P}_C$ is finite, then there exists at least one pattern Q with* $\mathrm{Emer}_C(Q; A, B) > 0$.

Remark 2905. *This is the qualitative version of Theorem 75: "somewhere" in the pattern vocabulary there is a genuinely emergent witness. When used as a sanity-check in a reasoning system, it says that a strictly positive global emergence statistic cannot be a numerical artifact spread across infinitely many infinitesimal contributions; in a finite vocabulary, it must show up as a positive value at some concrete Q. Equivalently, the corollary rules out the possibility that* every *candidate pattern is non-emergent (has* $\mathrm{Emer}_C(Q; A, B) \leq 0$*) while the aggregated score is still strictly positive.*

Remark 2906 (Engine reading)**.** *If a reasoning system can estimate* $\mathrm{Emer}_C(Q; A, B)$ *over a finite candidate set, then Theorem 75 guarantees there is a concrete Q whose emergent intensity is nontrivial whenever the* aggregate *emergence is nontrivial. This legitimizes using* $\arg\max_Q \mathrm{Emer}_C(Q; A, B)$ *as a hypothesis-generation operator. In practice, this supports a two-stage pipeline: (i) monitor $M_C^{\mathrm{em}}(A, B)$ (or a computable proxy) as a coarse trigger that "something interesting is happening," and (ii) upon triggering, output a short list of top-k patterns by* $\mathrm{Emer}_C(Q; A, B)$ *as candidate explanations. The witness bound provides a minimal quantitative target for such a shortlist: if $M_C^{\mathrm{em}}(A, B) \geq \varepsilon$, then at least one returned pattern must clear ε/N provided the maximizer is found (or approximated tightly enough), turning an otherwise diffuse aggregate into a checkable, pattern-level claim.*

51.3 Emergent surplus as strict cost subadditivity (compression bridge)

We now prove a useful algebraic bridge: under an additive-baseline assumption, emergent surplus becomes exactly a *normalized strict subadditivity* inequality on witness costs.

Definition 472 (Additive baseline assumption)**.** *We say the baseline is additive if, whenever $A *_C B$ is defined,*

$$s_C^{\mathrm{base}}(A *_C B) = s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B).$$

Remark 2907. *The additive baseline assumption says that, in the absence of pattern structure, the cost of describing the composite is just the sum of the costs of describing the parts. This is a natural "null hypothesis" in many description-length settings: if you do not exploit any interaction, you pay separately for each component [16]. In cognitive terms, it models a representational regime where concatenating two default descriptions incurs no additional overhead and provides no discount. Against this background, any strict improvement in the joint witness cost can be interpreted as emergent compression, hence emergence (Hyperseed-Concept* **??** *and 82).*

Remark 2908. *Two small clarifications about what is (and is not) being assumed are useful. First, additivity is imposed only on the* baseline s_C^{base}*, not on the witness cost* $u_C(Q;\cdot)$*: the whole point of emergence is that a good* Q *can exploit cross-part regularities so that* $u_C(Q; A *_C B)$ *can drop below* $u_C(Q;A) + u_C(Q;B)$*. Second, the additivity requirement is conditional on* $A *_C B$ *being defined: the algebra below never needs a global monoid structure, only that the particular composite under study is meaningful in context* C*.*

Remark 2909. *The strict-positivity condition* $s_C^{\text{base}}(A), s_C^{\text{base}}(B) > 0$ *plays two roles. Algebraically it ensures the intensities* $I_C(Q;A)$ *and weights* $w_C(A), w_C(B)$ *are well-defined and the normalization by* $s_C^{\text{base}}(A) + s_C^{\text{base}}(B)$ *is nondegenerate. Conceptually it excludes pathological "free" objects for which baseline description length is zero, in which case any ratio-based comparison would collapse and the notion of a fractional improvement would be ill-posed.*

Theorem 76 (Emergent surplus equals normalized cost subadditivity (when unclamped))**.** *Assume:*

1. $s_C^{\text{base}}(A), s_C^{\text{base}}(B) > 0$ *and* $A *_C B$ *is defined,*
2. *the baseline is additive,*
3. *for some* $Q \in \mathcal{P}_C$ *the clamp is inactive on* $A, B, A *_C B$*, i.e.*

$$u_C(Q;A) < s_C^{\text{base}}(A), \quad u_C(Q;B) < s_C^{\text{base}}(B), \quad u_C(Q;A *_C B) < s_C^{\text{base}}(A *_C B).$$

Then

$$\mathrm{Sur}_C(Q;A,B) = \frac{u_C(Q;A) + u_C(Q;B) - u_C(Q;A *_C B)}{s_C^{\text{base}}(A) + s_C^{\text{base}}(B)}.$$

Remark 2910. *This theorem states that, once the max-clamp is not interfering, emergent surplus becomes exactly "how much cheaper the joint witness is than the sum of separate witnesses," normalized by the total baseline size. Thus emergence is not an ineffable metaphysical extra; it is literally strict subadditivity of costs. The clamp-inactive condition is a technical guardrail: it ensures intensities are genuinely linear functions of costs, rather than truncated at* 0*.*

The result is important because it connects the semantic definition of emergence (a comparison of intensities) to a directly checkable algebraic inequality on costs. It also anticipates the later explanation-level synergy results: strict subadditivity of costs is the same structural shape as a synergy gain inequality (compare with Theorem 77).

Remark 2911. *The normalization by* $s_C^{\text{base}}(A)+s_C^{\text{base}}(B)$ *makes the surplus dimensionless and comparable across pairs* (A,B) *of very different absolute sizes. In particular, the numerator* $u_C(Q;A)+u_C(Q;B) - u_C(Q;A *_C B)$ *is an* absolute *compression gap (measured in the same units as the cost function), while* $\mathrm{Sur}_C(Q;A,B)$ *is a* relative *gap scaled to the baseline. This scaling is what allows one to interpret the quantity as "fractional emergent savings per baseline unit," and it is also what makes the weighted average in the definition of surplus the correct comparator: it matches the same baseline scale as the composite.*

Remark 2912. *The clamp-inactive hypothesis should be read as a statement about the regime in which the witness Q is genuinely informative relative to baseline. If, say, $u_C(Q;A) \geq s_C^{\mathrm{base}}(A)$, then $I_C(Q;A)$ would be clamped to 0 (or otherwise truncated by the max), and the intensity would no longer carry linear information about* how much worse *than baseline the witness is. In that clamped regime,* $\mathrm{Sur}_C(Q;A,B)$ *can still be defined, but it becomes a mixture of linear and saturated terms; the clean equivalence with strict subadditivity is therefore a property of the "unclamped" interior where intensities reflect true ratios.*

Proof. Under the clamp-inactive assumption, the max operation in I_C does nothing, so

$$I_C(Q;A) = 1 - \frac{u_C(Q;A)}{s_C^{\mathrm{base}}(A)}, \quad I_C(Q;B) = 1 - \frac{u_C(Q;B)}{s_C^{\mathrm{base}}(B)}.$$

Since $w_C(A) = \frac{s_C^{\mathrm{base}}(A)}{s_C^{\mathrm{base}}(A)+s_C^{\mathrm{base}}(B)}$, we get

$$w_C(A)\, I_C(Q;A) = \frac{s_C^{\mathrm{base}}(A)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)} \left(1 - \frac{u_C(Q;A)}{s_C^{\mathrm{base}}(A)}\right) = \frac{s_C^{\mathrm{base}}(A) - u_C(Q;A)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

Similarly,

$$w_C(B)\, I_C(Q;B) = \frac{s_C^{\mathrm{base}}(B) - u_C(Q;B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

Therefore the weighted baseline term is

$$w_C(A) I_C(Q;A) + w_C(B) I_C(Q;B) = \frac{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) - u_C(Q;A) - u_C(Q;B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

By additivity, $s_C^{\mathrm{base}}(A *_C B) = s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)$, and clamp-inactive gives

$$I_C(Q;A *_C B) = 1 - \frac{u_C(Q;A *_C B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)} = \frac{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B) - u_C(Q;A *_C B)}{s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)}.$$

Subtracting the baseline term from $I_C(Q;A *_C B)$ yields the claimed identity. □

Remark 2913. *A slightly different way to read the same algebra is as an invariance statement: the particular weights $w_C(A), w_C(B)$ ensure that the baseline comparator*

$$w_C(A)\, I_C(Q;A) + w_C(B)\, I_C(Q;B)$$

is exactly the intensity one would obtain by evaluating the concatenated baseline description *of A and B on the shared denominator $s_C^{\mathrm{base}}(A) + s_C^{\mathrm{base}}(B)$. In other words, the weights are not an aesthetic choice but the unique choice (given this linear form) that puts all three intensities on the same baseline scale so that subtraction isolates a pure cost gap.*

Proof sketch. Expand all intensities into linear functions of the costs (possible because the clamp is inactive), rewrite the weighted average using the explicit weights $w_C(A), w_C(B)$, and then simplify using the additive baseline identity. The remaining numerator is exactly the strict subadditivity gap $u(Q;A) + u(Q;B) - u(Q;A *_C B)$. □

Remark 2914. *The proof is essentially a bookkeeping exercise: the weights were chosen precisely so that*

$$w_C(A)\left(1-\frac{u(Q;A)}{s^{\mathrm{base}}(A)}\right)+w_C(B)\left(1-\frac{u(Q;B)}{s^{\mathrm{base}}(B)}\right)$$

collapses into one fraction with denominator $s^{\mathrm{base}}(A)+s^{\mathrm{base}}(B)$*. The additive baseline assumption then aligns the denominator for the composite intensity with the same quantity, allowing the difference to telescope.*

A helpful visualization is to imagine costs as lengths: s_C^{base} *gives a "ruler" for scale, while* u_C *gives the length of a particular witness description. Emergence corresponds to a joint description that is shorter than putting two separate descriptions end-to-end.*

Remark 2915. *Interpreted in minimum-description-length terms, the numerator*

$$u_C(Q;A)+u_C(Q;B)-u_C(Q;A*_C B)$$

is the amount of description length saved by using a single *model* Q *to account for the composite rather than fitting* Q *separately to each part. The theorem therefore makes precise the idea that "emergent structure" is exactly the presence of shared code, shared parameters, or shared regularities that amortize across components. In this sense, the bridge is not merely algebraic: it identifies emergence with the same kind of reuse that underlies generalization and transfer in learning systems.*

Corollary 38 (Emergent surplus detects strict joint compression)**.** *Under the assumptions of Theorem 76:*

$$\mathrm{Sur}_C(Q;A,B)>0 \quad\Longleftrightarrow\quad u_C(Q;A*_C B)<u_C(Q;A)+u_C(Q;B).$$

Hence $\mathrm{Emer}_C(Q;A,B)>0$ *iff the witness cost is strictly subadditive.*

Remark 2916. *The strictness matters:* $\mathrm{Sur}_C(Q;A,B)=0$ *corresponds exactly (in the unclamped regime) to* additivity *of witness costs, i.e. the joint witness is no better than treating* A *and* B *independently with the same* Q*. Negative surplus corresponds to superadditivity, which can be read as "anti-emergence" in the purely compressive sense: representing the interaction jointly is actually more expensive than representing the parts separately. This can happen when* Q *is mismatched to the composite (so that coupling introduces additional descriptive burden) even if* Q *is individually decent on* A *and* B*.*

Remark 2917. *One can also see why the statement is framed in terms of* $\mathrm{Sur}_C(Q;A,B)$ *rather than raw cost differences. Because the denominator is the total baseline size, a fixed absolute gap (say, saving 10 bits) yields a larger surplus for smaller baselines and a smaller surplus for larger baselines. Thus* Sur_C *is tailored to comparisons "at equal baseline scale": it encodes a notion of* proportional *emergence rather than absolute emergence, which is often the relevant quantity when comparing across domains with different representational granularities.*

Remark 2918. *This corollary is the operational heart of the section: it says that, at least in the unclamped additive-baseline regime, emergence is* equivalent *to strict joint compression. Thus one may detect emergence by comparing three costs, without ever computing intensities explicitly. Conversely, one may interpret strict subadditivity not merely as an accident of encoding, but as a formal witness of emergence (Hyperseed-Concept* **??***). In particular, the equivalence highlights that "emergent surplus" is not being postulated as a new primitive: it is exactly the residual gap that remains once one accounts for the best separate compressions, and strictness is what rules out the degenerate case where the joint description is merely a concatenation in disguise. Operationally,* *the "three costs" are precisely the two marginal witness-costs and the joint witness-cost, so the test is implementable whenever one can approximate those costs (even coarsely) in a shared coding scheme.*

Remark 2919 (Engine reading)**.** *This is the core "discovery" bridge: if you can estimate witness-costs for candidate patterns, then* emergence *is exactly the signal that a joint hypothesis Q compresses the composite better than any explanation that keeps the two sides separate at the witness level. In other words, the engine does not need to enumerate or optimize over all possible intensities or decompositions: it only needs a reliable comparison oracle on candidate witnesses, and strict improvement is the discriminant. This also clarifies what can go wrong in practice: if witness-cost estimates are biased (e.g., by mismatched priors or incomparable encoding conventions between the separate and joint cases), then apparent "emergence" may reflect measurement artifacts rather than genuine joint structure.*

51.4 From a single emergent-compression explanation to synergy gain (quantitative)

Theorem 77 (Quantitative lower bound on synergy gain from one nonseparable explanation)**.** *Fix disjoint knowledge sets T_1, T_2 and fit threshold τ. Assume there exist explanations $E_1 \in \mathcal{E}_{T_1}(b)$, $E_2 \in \mathcal{E}_{T_2}(b)$, and $E \in \mathcal{E}_{T_1 \cup T_2}(b)$ such that*

$$\mathrm{Fit}(E_1) \geq \tau, \quad \mathrm{Fit}(E_2) \geq \tau, \quad \mathrm{Fit}(E) \geq \tau,$$

and for some $\delta > 0$,

$$\mathrm{Con}(E) \;\leq\; \mathrm{Con}(E_1) + \mathrm{Con}(E_2) - \delta.$$

Then

$$\Sigma(T_1, T_2; \tau) \;\geq\; \delta,$$

and in particular $\Sigma(T_1, T_2; \tau) > 0$.

Remark 2920. *In plain language: if you can exhibit* one *integrated explanation E using $T_1 \cup T_2$ that beats the best-known separate explanations by an additive margin δ in contrivance, then synergy is at least δ. This is important because Σ is defined using infima over large search spaces, which is often hard to compute directly. The theorem says we can lower-bound Σ by producing a single certificate explanation. Note that δ is a* verifiable *margin: it is computed from explicit witness values $\mathrm{Con}(E), \mathrm{Con}(E_1), \mathrm{Con}(E_2)$ rather than from unknown optima, so any valid lower bound is actionable even if the global search remains incomplete.*

Conceptually, this is the explanation-level analogue of Theorem 75: there, positive aggregate emergence guarantees some witness Q; here, one good witness E guarantees positive synergy. This connects naturally to Hyperseed's emphasis on explainable "pointing" rather than only scalar global scores [1]. One can also read the theorem as a robustness statement: because it only uses the monotonicity property of an infimum, it continues to hold even when the feasible explanation classes are extremely complex or only implicitly specified, provided E_1, E_2, E genuinely belong to the stated classes and meet the same fit threshold.

Proof. By definition of Con^* as an infimum over feasible explanations,

$$\mathrm{Con}^*(T_1; \tau) \leq \mathrm{Con}(E_1), \quad \mathrm{Con}^*(T_2; \tau) \leq \mathrm{Con}(E_2), \quad \mathrm{Con}^*(T_1 \cup T_2; \tau) \leq \mathrm{Con}(E).$$

Therefore,

$$\begin{aligned}\mathrm{Con}^*(T_1;\tau) + \mathrm{Con}^*(T_2;\tau) - \mathrm{Con}^*(T_1 \cup T_2;\tau) &\geq \mathrm{Con}(E_1) + \mathrm{Con}(E_2) - \mathrm{Con}(E)\\ &\geq \delta.\end{aligned}$$

Taking positive part does not change the inequality (since $\delta > 0$), hence $\Sigma(T_1, T_2; \tau) \geq \delta$. A small but sometimes practically relevant point is that the argument does not require the infimum to be attained (i.e., there need not exist an optimizer achieving Con^*); the comparison holds for infima exactly because they are lower envelopes over all feasible witnesses. □

Proof sketch. Use the defining property of an infimum: $\mathrm{Con}^*(\cdot;\tau)$ is bounded above by the contrivance of any specific feasible explanation at fit $\geq \tau$. Plug in E_1, E_2, E, rearrange, and observe that $(\cdot)_+$ is irrelevant once a strictly positive lower bound is obtained. Equivalently: the certificate E forces a definite gap between the joint optimum and the separate optima because the joint optimum can only be *no worse* than any particular feasible joint construction. □

Remark 2921. *The inequality direction is the entire story: infima make* Con^* *smaller than any particular witness value, so subtracting* $\mathrm{Con}^*(T_1 \cup T_2; \tau)$ *is at least as favorable as subtracting* $\mathrm{Con}(E)$*. The positive-part operation is a guard against negative results, but here* $\delta > 0$ *forces positivity. Said differently,* Σ *is defined to ignore "anti-synergy" (cases where joint explanations are more contrived than separate ones), whereas the theorem shows that any strictly nonseparable certificate immediately escapes that truncation.*

If one prefers a geometric intuition, imagine a "cost landscape" where each knowledge set T *defines a reachable region of explanations; the theorem says that finding a single point in the union-region that lies* δ *below the sum of the two separate best heights forces a corresponding gap between the optimum of the union and the sum of separate optima. The disjointness of* T_1, T_2 *can be viewed as ensuring that the two separate regions represent genuinely distinct "axes of knowledge" rather than double-counting the same explanatory resource; if there is overlap, the same reasoning can be adapted but the baseline of "separate" costs may need an explicit correction for shared components.*

Corollary 39 (Emergent weakness factor implies synergy)**.** *Suppose explanations compose as* $E_1 \otimes E_2$*, fit is preserved at threshold* τ*, and*

$$\mathrm{Weak}(E_1 \otimes E_2) = g\ \mathrm{Weak}(E_1)\mathrm{Weak}(E_2)$$

for some factor $g > 1$ *(necessarily with* $g \leq 1/(\mathrm{Weak}(E_1)\mathrm{Weak}(E_2))$ *so the RHS stays* ≤ 1*). Then*

$$\mathrm{Con}(E_1 \otimes E_2) = \mathrm{Con}(E_1) + \mathrm{Con}(E_2) - \log g,$$

so Theorem 77 applies with $\delta = \log g$ *and yields* $\Sigma(T_1, T_2; \tau) \geq \log g > 0$*.*

Remark 2922. *The corollary translates multiplicative "extra strength" in weakness into additive synergy in contrivance. Because contrivance is* $-\log(\mathrm{Weak})$*, any multiplicative gain* g *in weakness becomes an additive gain* $\log g$ *in contrivance. This echoes the general theme in [3] that weakness-like quantities often behave most naturally under multiplication, while decision rules tend to be additive after taking logs. It is also worth noting that the constraint* $g > 1$ *is exactly the strictness needed to get a nontrivial lower bound:* $g = 1$ *recovers the separable baseline where the composed explanation is neither better nor worse (in contrivance) than the sum of its parts, and* $\log g = 0$ *makes the certificate vacuous.*

Remark 2923 (Engine reading)**.** *Theorem 77 is a "single-witness certificate" for synergy: to prove nontrivial synergy it suffices to exhibit one integrated explanation E whose contrivance beats the sum of two separate explanations by a margin δ. The corollary makes this margin interpretable as a multiplicative emergent-weakness factor g. In practice, this suggests a workflow: propose a candidate joint mechanism (yielding E), compute its fit and contrivance, and compare against the best available modular baselines (E_1, E_2); any strict improvement gives a quantitative floor on Σ without requiring full optimization over $\mathcal{E}_{T_1 \cup T_2}(b)$.*

51.5 Cross-type emergent compression and budgeted solvability

We restate and prove two operational results (matching the document's intent) in a form that fits the present note.

Definition 473 (Cross-type emergent compression pattern)**.** *Fix disjoint knowledge sets T_1, T_2 and threshold τ. We say there is a cross-type emergent compression pattern for b if there exists $E \in \mathcal{E}_{T_1 \cup T_2}(b)$ with* $\mathrm{Fit}(E) \geq \tau$ *such that for every decomposition $E \approx E_1 \otimes E_2$ with $E_i \in \mathcal{E}_{T_i}(b)$ and* $\mathrm{Fit}(E_i) \geq \tau$*,*

$$\mathrm{Con}(E) < \mathrm{Con}(E_1) + \mathrm{Con}(E_2).$$

Remark 2924. *This definition formalizes an especially strong kind of nonseparability: not merely that there exists* some *pair (E_1, E_2) that is worse than E, but that* every *admissible partitioned explanation is worse. The approximation symbol $E \approx E_1 \otimes E_2$ is intentionally left abstract: it stands for whatever notion of compositional decomposition is appropriate in the application domain. The intent is to capture the situation where the integrated structure of E is essential, rather than merely a convenient alternative.*

This notion anticipates later independence-vs-dependence tradeoffs (cf. the document's dependence dichotomy discussion elsewhere): if a phenomenon is truly cross-type emergent in this sense, then any architecture that insists on separability is structurally mis-specified.

Theorem 78 (Cross-type emergent compression implies positive synergy)**.** *If there exists a cross-type emergent compression pattern for b at level τ, then $\Sigma(T_1, T_2; \tau) > 0$.*

Remark 2925. *The theorem says that the strong nonseparability condition in the previous definition forces strictly positive synergy. This is important because it turns a qualitative modeling diagnosis ("no separable explanation can match the integrated one") into a quantitative, scalar conclusion ($\Sigma > 0$), which can then be used in budgeting and architecture selection. It also connects directly to Theorem 79: once $\Sigma > 0$ is established, one can exhibit budgets where integrated reasoning succeeds and partition-limited reasoning necessarily fails.*

Proof. Let E witness cross-type emergent compression. By definition, for every admissible decomposition $E \approx E_1 \otimes E_2$ with $\mathrm{Fit}(E_i) \geq \tau$ we have

$$\mathrm{Con}(E) < \mathrm{Con}(E_1) + \mathrm{Con}(E_2).$$

In particular, taking the infimum over all admissible $E_1 \in \mathcal{E}_{T_1}(b)$ with fit $\geq \tau$ and similarly for E_2, we obtain

$$\mathrm{Con}(E) < \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau).$$

Also $\mathrm{Con}^*(T_1 \cup T_2; \tau) \leq \mathrm{Con}(E)$ since E is feasible for $T_1 \cup T_2$. Therefore

$$\mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau) - \mathrm{Con}^*(T_1 \cup T_2; \tau) > 0,$$

so $\Sigma(T_1, T_2; \tau) > 0$. $\square$

Proof sketch. Start from the strict inequality $\mathrm{Con}(E) < \mathrm{Con}(E_1) + \mathrm{Con}(E_2)$, then make the right-hand side as small as possible by taking infima over all feasible E_1, E_2. Finally, compare $\mathrm{Con}(E)$ to the infimum over all union-allowed explanations to conclude the strict gap persists at the optimum level. $\square$

Remark 2926. *The essential step is that strict inequality is preserved when we replace* $\mathrm{Con}(E_1)$ *and* $\mathrm{Con}(E_2)$ *by their best possible values, because E is assumed to beat* every *admissible decomposition. If the inequality held only for some particular decomposition, we would generally only obtain a nontrivial* lower bound *(as in Theorem 77), not necessarily strict positivity of synergy. Thus the present theorem highlights why the cross-type condition is stronger than mere existence of one good integrated witness.*

One may picture a two-column architecture where T_1 and T_2 are processed independently; the theorem says: if no pair of column-wise explanations can match E, then independent processing has a provable penalty in contrivance.

Definition 474 (Budgeted solvability)**.** *Fix τ and a contrivance budget $B \geq 0$. We say b is solvable by T at budget B if* $\mathrm{Con}^*(T; \tau) \leq B$*.*

Remark 2927. *This definition turns contrivance into an explicit resource constraint: one fixes a required adequacy level τ and asks whether the best achievable contrivance is within a budget B (Hyperseed-Concept 100). In practice, B may correspond to allowed model size, allowed search depth, allowed proof length, or any other proxy for representational and computational expense. The point is to express synergy not merely as an abstract efficiency gain but as a difference between what is solvable and unsolvable under a given resource cap.*

Theorem 79 (Synergy expands solvable set relative to partition-limited architectures)**.** *Let T_1, T_2 be disjoint. Suppose $\Sigma(T_1, T_2; \tau) > 0$ for the given b. Define the partition-limited contrivance level*

$$\mathrm{Con}^{\mathrm{part}}(T_1, T_2; \tau) := \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau).$$

Then there exists a budget B such that:

1. *b is solvable by $T_1 \cup T_2$ at budget B, and*
2. *b is not solvable by any architecture that restricts itself to partition-limited (separable) explanations at the same budget B.*

Remark 2928. *This theorem is the operational payoff: synergy creates a* budget gap *where integrated reasoning can solve b but any separable architecture cannot. Thus synergy is not merely a matter of elegance; it changes the boundary of the feasible (Hyperseed-Concept 126 and 186). More concretely, the claim is not that the separable architecture is* always *worse, but that once $\Sigma > 0$ holds there exists a regime of resource budgets in which the separability constraint becomes the decisive bottleneck. In that regime, the difference between architectures is measurable as a binary feasibility distinction at fixed fit threshold τ.*

The result is also a conceptual bridge to broader architectural tradeoff discussions: if a system is built to enforce separability between two subsystems (here modeled by disjoint T_1 and T_2), then positive synergy implies there are tasks for which that design is provably suboptimal under fixed resource budgets. Equivalently, separability is not a neutral modeling choice: it can convert what would be a solvable task (at a given budget and fit criterion) into an unsolvable one by forbidding cross-term structure that reduces contrivance.

Proof. Let $C_{12} := \mathrm{Con}^*(T_1 \cup T_2; \tau)$ and $C_{\mathrm{part}} := \mathrm{Con}^{\mathrm{part}}(T_1, T_2; \tau)$. Since $\Sigma > 0$, we have $C_{12} < C_{\mathrm{part}}$. This strict inequality is the only ingredient needed to manufacture a nonempty interval of budgets that separates the two architectures: any B in that interval will certify feasibility for the integrated model and infeasibility for the partition-limited one. Choose any budget B with

$$C_{12} \leq B < C_{\mathrm{part}}.$$

(Such a B exists because the interval $[C_{12}, C_{\mathrm{part}})$ is nonempty when $C_{12} < C_{\mathrm{part}}$; for instance, one may take $B = (C_{12} + C_{\mathrm{part}})/2$ if budgets are treated as real-valued, or the appropriate nearby admissible value if budgets are discretized.) Then b is solvable by $T_1 \cup T_2$ at budget B by definition of solvability. Here the relevant point is that C_{12} is the *minimum* contrivance needed by any explanation built from $T_1 \cup T_2$ to reach fit at least τ, so any budget $B \geq C_{12}$ suffices for some integrated explanation to meet the threshold. On the other hand, any separable/partition-limited architecture that insists on independent T_1 and T_2 explanations needs contrivance at least C_{part} to reach fit $\geq \tau$, hence cannot meet budget $B < C_{\mathrm{part}}$. In other words, the separability constraint imposes an unavoidable lower bound on required contrivance, and the chosen B is intentionally placed below that bound so that no amount of optimization *within* the separable class can recover feasibility. □

Proof sketch. Synergy $\Sigma > 0$ means the union-optimum contrivance C_{12} is strictly smaller than the sum of separate optima C_{part}. Pick any budget B strictly between these two numbers; then $T_1 \cup T_2$ can meet it, while any partition-limited approach cannot. Intuitively, the integrated architecture is allowed to reuse shared structure and encode cross-dependencies once, whereas the partitioned architecture must pay for them (or simulate them) separately inside each subsystem, which raises the minimum description length needed to reach the same fit level. □

Remark 2929. *The proof is a one-dimensional separation argument: the strict inequality $C_{12} < C_{\mathrm{part}}$ guarantees there is an interval of budgets between them. Choosing B in this interval turns the inequality into an observable behavioral difference between architectures. This is the same logical shape as a margin argument: once two feasibility thresholds are separated, any operating point in the margin witnesses a qualitative difference in capability under identical evaluation criteria (same b, same τ, same budget accounting).*

A visual intuition is to imagine two thresholds on a line: the integrated architecture has a lower threshold C_{12} for feasibility, while the partitioned architecture has a higher threshold C_{part}. Any budget lying between them separates the possible from the impossible, relative to the architectural constraint. One can also read the picture dynamically: as the budget B increases from 0, the integrated system becomes feasible first (at C_{12}), and only later does the partitioned system become feasible (at C_{part}); the interval between these activation points is precisely the "synergy-only" region.

Remark 2930 (Engine reading)**.** *Theorem 79 is an operational "why synergy matters" statement: there are tasks for which integrated reasoning is not just nicer, but strictly necessary under a given resource (contrivance/description-length) budget. In deployment terms, if one fixes a maximum allowable complexity (e.g., memory footprint, parameter budget, message length, or any proxy captured by* Con*), then positive synergy implies there exist achievable performance targets that require allowing the subsystems to co-explain rather than forcing them to explain in isolation.*

51.6 Collective properties (interaction-only features) have witnesses

Theorem 80 (Approximate witness for collective properties)**.** *Fix $\mathcal{B} \subseteq \mathcal{E}_C$ and $A \in \mathcal{E}_C$. For any $Q \in \mathcal{P}_C$ and any $\varepsilon > 0$, there exists $B_\varepsilon \in \mathcal{B}$ such that*

$$\mathrm{Emer}_C(Q; A, B_\varepsilon) \;\geq\; \mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q) \;-\; \varepsilon.$$

Remark 2931. *It is helpful to read the statement as an "ε-maximizer exists" claim: once A and the query/property Q are fixed, the collective score* $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$ *(which is defined via a supremum over partners in $\mathcal{B}$) can always be nearly realized by at least one explicit interaction partner B_ε in the same reference class. Equivalently, the supremal definition does not merely certify that* some *partners do well; it certifies that, for every tolerance level, there is a concrete partner whose emergent interaction score is within that tolerance of optimal.*

This theorem says that collective properties, defined via a supremum over a reference class, are not merely abstract upper bounds: they can be approximated arbitrarily well by some concrete partner B_ε. In other words, even if the "best possible partner" is not uniquely defined or not attained, it can be approached. This matters for discovery and explanation because it means that the supremal definition of a collective property remains operational: a search procedure can always find an explicit near-witness.

The result is conceptually aligned with the idea that relational attributes should be discoverable by interaction experiments (Hyperseed-Concept 59 and **??***), and, in broader Hyperseed discussions of collective mind-like phenomena, with the operational emphasis on identifying concrete interaction partners rather than positing purely global collectives [12].*

Remark 2932. *A minor but practically relevant edge case concerns whether the reference class $\mathcal{B}$ is nonempty. When $\mathcal{B} \neq \varnothing$, the set of attainable scores* $\{\mathrm{Emer}_C(Q;A,B) : B \in \mathcal{B}\}$ *is nonempty and the supremum has its usual approximation-from-below meaning. If one allows $\mathcal{B} = \varnothing$, then one must adopt a convention for* $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$ *(often* $\sup\varnothing := -\infty$ *in general order theory, or* 0 *if intensities are constrained to* $[0,1]$ *and the intention is "no partner, no collective effect"); under such a convention, the theorem is either vacuous or trivially satisfied. The intended discovery-theoretic reading typically assumes a nonempty candidate pool, so that witnesses correspond to actual interaction experiments.*

Proof. Let $S := \{\mathrm{Emer}_C(Q;A,B) : B \in \mathcal{B}\} \subseteq [0,1]$ and let $\alpha := \sup S$. By the defining property of supremum, for any $\varepsilon > 0$ there exists an element of S strictly greater than $\alpha - \varepsilon$. Thus there exists $B_\varepsilon \in \mathcal{B}$ with $\mathrm{Emer}_C(Q;A,B_\varepsilon) > \alpha - \varepsilon$, which implies the desired inequality. In particular, since $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$ is precisely the supremum of the same set of scores (by definition of the collective property functional), one may read α as $\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q)$, so the proof is exactly the standard "approximate maximizer from the supremum" argument. ☐

Proof sketch. Unpack the definition of sup: for any target slack $\varepsilon > 0$, some element of the set must lie within ε of the supremum from below. Choose the corresponding partner B_ε. In algorithmic terms, the sketch corresponds to the guarantee that an iterative search that improves $\mathrm{Emer}_C(Q;A,B)$ cannot be blocked by a purely definitional artifact: if the current candidate is more than ε below the supremum, there exists some other candidate in $\mathcal{B}$ that beats it by at least some margin. ☐

Remark 2933. *The only "engine" inside the proof is the completeness property of real numbers (or, more generally, of the order structure used for intensities): suprema are approximable from below by elements of the set. The strict inequality $>$ produced by the supremum property is then relaxed to $\geq$ in the statement.*

Visually, if you plot the values $\mathrm{Emer}_C(Q;A,B)$ *as points on the* $[0,1]$ *axis, the supremum is the least upper bound. The theorem guarantees a point within ε of that bound, so the supremum is not a distant ideal but a limit point approachable by explicit choices of B. A further interpretive point is that the witness B_ε may depend on (A,Q,ε): the "best partner" for explaining one property Q need not be the best partner for a different Q', and the partner that is sufficient for a coarse-grained explanation (large ε) need not be the one needed for a fine-grained explanation (small ε).*

Corollary 40 (Exact witness when $\mathcal{B}$ is finite)**.** *If $\mathcal{B}$ is finite, then the supremum is a maximum: for every Q there exists $B^* \in \mathcal{B}$ such that*

$$\mathrm{Prop}^{\mathrm{coll}}_{C,\mathcal{B}}(A)(Q) = \mathrm{Emer}_C(Q; A, B^*).$$

Remark 2934. *Finiteness eliminates limit behavior: among finitely many real numbers, the supremum is attained. From an implementation viewpoint, this says that if $\mathcal{B}$ is a finite candidate set (e.g. a finite cohort of interaction partners), then one can always point to an exact maximizing witness B^* rather than merely an approximation. In addition, the finite case supports a simple explanation protocol: compute* $\mathrm{Emer}_C(Q; A, B)$ *for each $B \in \mathcal{B}$ and select an argmax. The distinction between the theorem and this corollary mirrors the practical distinction between (i) searching in a large, possibly unstructured hypothesis space and (ii) selecting among a finite menu of experimental conditions or partner prototypes.*

Remark 2935. *More generally, exact witnesses can also arise in infinite families $\mathcal{B}$ when additional structure ensures attainment of the supremum (for example, compactness of the partner space together with appropriate continuity or upper semicontinuity of $B \mapsto \mathrm{Emer}_C(Q; A, B)$). This highlights that the approximate-witness theorem is the minimal guarantee (needing only the order-completeness of $[0, 1]$), while exact attainment is an extra regularity phenomenon rather than a definitional necessity.*

Remark 2936 (Engine reading)**.** *This legitimizes an "interaction partner search" routine: to explain why A has a collective property Q, search for B that maximizes* $\mathrm{Emer}_C(Q; A, B)$ *(or approximately maximizes it). One can also read the theorem as justifying stopping criteria: if an empirical procedure produces a B with* $\mathrm{Emer}_C(Q; A, B)$ *within a pre-specified tolerance of the best attainable value over $\mathcal{B}$, then the procedure has already produced a valid explanatory witness at that tolerance level.*

52 Exploring Emergence, Weakness, and McBride-Pattern Sensitivities

This section states and proves several additional theorems suggested by the quantale pattern theory and McBride-derivative material in *Weakness is All You Need*, as related to Hyperseed. The goal is to produce results that are (i) structurally "meaty" and (ii) directly usable as decision tools for reasoning systems: tests for emergence, invariances, sensitivity and interaction diagnostics, greedy-selection guarantees in important quantales, and a clean link between delay-logistic dynamics and bursty/chaotic pattern intensities.

Remark 2937. *The underlying point of view is that many quantities used in reasoning systems—"how much evidence supports a predicate," "how strongly a pattern is present," "how emergent a joint description is"—can be represented in a single algebraic language: a commutative quantale with residuation. This is the quantale-weakness stance developed in [3], and it is one of the mathematical backbones of the Hyperseed ontology program [1]. The present section focuses on the operational payoffs: when the semantics is quantalic, one can derive robust invariances, testable emergence criteria, and sensitivity/interaction diagnostics that are lightweight enough to be embedded in automated decision-making.*

52.1 Preliminaries (definitions and standing notation)

Definition 475 (Commutative quantale with residuation)**.** *A* commutative quantale *is a complete lattice* $(V, \leq)$ *equipped with a commutative, associative binary operation* $\otimes$ *with unit* e*, such that* $\otimes$ *distributes over arbitrary joins. The* residuated implication $\multimap$ *is defined by the adjunction:*

$$a \otimes b \leq c \quad \Longleftrightarrow \quad b \leq a \multimap c.$$

Remark 2938 (Intuition, notation, and examples)**.** *A commutative quantale is a way to do "graded logic" where degrees of truth/evidence live in a lattice* V*, and where* $\otimes$ *plays the role of combining evidence (or composing resources), while* $\oplus$ *(i.e. join) plays the role of taking the least upper bound of alternatives. Completeness of the lattice means that arbitrary joins* $\bigvee S$ *exist (here* $\bigvee$ *is the lattice join* $\bigvee$ *written with the macro* `\bigjoin`*), so one can aggregate evidence across arbitrarily large families. Distributivity of* $\otimes$ *over joins is the algebraic expression of a familiar computational idea: combining with a fixed resource should commute with taking suprema over choices.*

The residuation law $a \otimes b \leq c \iff b \leq a \multimap c$ *defines* $\multimap$ *as the "largest* b*" that, when combined with* a*, stays below* c*. If* $\otimes$ *is thought of as multiplying plausibilities, then* $\multimap$ *behaves like a division/normalization; if* $\otimes$ *is like conjunction, then* $\multimap$ *behaves like an implication. Concrete examples include: (i)* $V = \{0, 1\}$ *with* $\otimes = \wedge$ *and* $\multimap$ *the usual material implication; (ii)* $V = [0, 1]$ *with* $\otimes = \times$*, join* $= \max$*, and* $\multimap$ *the truncated ratio* $\min(1, b/a)$*; and (iii)* $V = \mathbb{R}_{\geq 0}$ *with* $\otimes = \times$ *and* $\multimap$ *literal division* b/a *when* $a > 0$*. Quantales of this sort are the ambient setting for quantale weakness (Hyperseed-Concept 143) and for formalized notions of* Weakness *(Hyperseed-Concept 202) and* Pattern *(Hyperseed-Concept 130).*

Definition 476 (Weakness, intensity, synergy)**.** *Fix a universe* X*, a valuation* $\phi : X \to V$*, and predicates* $m, y, z : X \to V$*. Define the weakness*

$$\mathrm{Weak}(p) \;:=\; \bigvee_{x \in X} \phi(x) \otimes p(x).$$

Define the composite weakness $h(y, z) := \mathrm{Weak}(y) \otimes \mathrm{Weak}(z)$ *and the (joint) pattern intensity*

$$I_{y,z}(m) \;:=\; (\mathrm{Weak}(y) \otimes \mathrm{Weak}(z)) \multimap \mathrm{Weak}(m).$$

Define individual intensities $I_y(m) := \mathrm{Weak}(y) \multimap \mathrm{Weak}(m)$ *and* $I_z(m) := \mathrm{Weak}(z) \multimap \mathrm{Weak}(m)$*. Define emergent synergy*

$$\sigma_{y,z}(m) \;:=\; (I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m).$$

We say (y, z) *is a* pattern *in* m *when* $I_{y,z}(m) \geq e$*; and* emergent *in* m *when* $\sigma_{y,z}(m) \geq e$*.*

Remark 2939 (What the symbols mean, and why these definitions are natural)**.** *Here* X *is the universe of "items" being aggregated (examples: events, links, data-points, micro-features),* $\phi(x) \in V$ *is a weight/valuation assigning importance or availability to each* x*, and a predicate* $p : X \to V$ *is a* V*-valued membership function (so* $p(x)$ *is the degree to which* x *supports or belongs to* p*). The weakness* $\mathrm{Weak}(p)$ *is then the join-aggregation over* x *of the combined contribution* $\phi(x) \otimes p(x)$*. When* $V = [0, 1]$ *with join* $= \max$ *and* $\otimes = \times$*, this is literally "the strongest weighted supporting element":* $\mathrm{Weak}(p) = \max_x \phi(x)p(x)$*. When* $V = \mathbb{R}_{\geq 0}$ *with join as* $\sup$ *and* $\otimes = \times$*, it becomes a supremum-weighted score.*

The intensities are residuated normalizations of weakness: $I_y(m) = \mathrm{Weak}(y) \multimap \mathrm{Weak}(m)$ *answers the question "relative to how strong* y *is, how strong is* m*?" The joint intensity* $I_{y,z}(m)$ *asks* *the analogous question but with the combined baseline* Weak$(y) \otimes$ Weak(z)*. Finally, the synergy* $\sigma_{y,z}(m)$ *compares the joint intensity to what would be expected from the individual intensities combined via* $\otimes$*; thus* $\sigma_{y,z}(m) \geq e$ *formalizes "the whole exceeds the multiplicative baseline." This is an algebraic capture of* Emergence *(Hyperseed-Concept* **??***) in the sense emphasized in pattern-based accounts [5].*

Remark 2940 (Simple examples (Boolean and max-product))**.** *If* $V = \{0, 1\}$ *with* $\otimes = \wedge$ *and join* $= \vee$*, then* Weak$(p) = 1$ *iff there exists some* x *with* $\phi(x) = 1$ *and* $p(x) = 1$*. In this case* $I_y(m) = 1$ *expresses that whenever* y *is witnessed (in the weighted sense), so is* m*; the joint version similarly matches a conjunctive entailment. At the other extreme, if* $V = [0, 1]$ *with* $\otimes = \times$ *and join* $= \max$*, then* Weak(p) *picks the best single* x *supporting* p *under* ϕ*, and intensities become truncated ratios. These are precisely the kinds of semantics that make the subsequent greedy/submodular results computationally useful.*

Remark 2941 (McBride derivative conventions used here)**.** *We will use a McBride-style "partial derivative w.r.t. the inclusion of* $x \in X$*" denoted* $\partial_x(\cdot)$*, consistent with the derivative identities for* Weak*,* I_y*,* $I_{y,z}$*, and* $\sigma_{y,z}$ *given in Weakness Theory 10. Separately, we will also use ordinary multivariate calculus derivatives* $\frac{\partial}{\partial \theta_i}$ *for the* parameterized soft model *setting (the curvature/Hessian results).*

Remark 2942. *The notational coexistence is deliberate. The McBride-style* ∂_x *is a derivative with respect to a* discrete structural perturbation *(whether a particular* x *is present/active), whereas* $\frac{\partial}{\partial \theta_i}$ *is the ordinary analytic derivative with respect to a real parameter in a soft relaxation. The former is natural when one is doing symbolic or combinatorial sensitivity analysis (as in [22]), and the latter is natural when one is optimizing a differentiable surrogate. Theorems later in this section explain when first-order (one-hole) information is reliable, and when higher-order interaction terms must be accounted for.*

52.2 Shape-2-style: Conditional-intensity factorization and an emergence test

Lemma 156 (Currying for residuation)**.** *In any commutative quantale,*

$$(a \otimes b) \multimap c \;=\; b \multimap (a \multimap c).$$

Remark 2943 (Intuition and why this matters)**.** *This lemma says that residuation "curries" exactly as implication curries in ordinary logic: an implication out of a tensor-product* $a \otimes b$ *can be re-read as an implication out of* b *into an implication out of* a*. In computational terms, it is the algebraic justification for rewriting a joint-normalization into a nested normalization, which is the key step behind conditional intensities. Once this is available, many emergence conditions become one-line comparisons rather than opaque ratios.*

Proof. Let $t \in V$. Using the residuation adjunction twice and associativity/commutativity:

$$\begin{aligned} t \leq (a \otimes b) \multimap c &\iff (a \otimes b) \otimes t \leq c \\ &\iff b \otimes (a \otimes t) \leq c \\ &\iff a \otimes t \leq b \multimap c \\ &\iff t \leq a \multimap (b \multimap c). \end{aligned}$$

By commutativity of $\otimes$, $a \multimap (b \multimap c) = b \multimap (a \multimap c)$, hence the claim. $\square$

Proof sketch. The strategy is purely order-theoretic: characterize each side as the set of all t such that a certain inequality holds, then use the defining adjunction of $\multimap$ to rewrite the inequality step-by-step until it matches the other side. The only "content" is associativity/commutativity of $\otimes$, which lets us re-bracket and swap a and b. $\square$

Remark 2944 (Geometric/order intuition). *In a residuated structure, each map $b \mapsto a \otimes b$ has a right adjoint $c \mapsto a \multimap c$. The lemma is then the statement that the right adjoint of the composite $b \mapsto (a \otimes b) \otimes (\cdot)$ can be computed in two stages, corresponding to two successive adjunctions. This is the lattice-theoretic analogue of "integrating out" one variable at a time.*

Definition 477 (Conditional intensity and conditional amplification). *Define the conditional intensity of z "given y" in context m as*

$$I(z \mid y; m) \;:=\; \mathrm{Weak}(z) \multimap I_y(m).$$

Define the conditional amplification (a "lift" measure) of z induced by y as

$$A(y \to z; m) \;:=\; I_y(m) \multimap I_{y,z}(m).$$

Remark 2945 (Intuition and simple reading). *The quantity $I(z \mid y; m)$ is a nested normalization: first measure how intense m is relative to y (via $I_y(m)$), then measure how that quantity compares to the baseline strength of z (via $\mathrm{Weak}(z) \multimap$). In probabilistic language one might think of it as a conditional score; in resource language it is the "residual slack" for z after accounting for y.*

The amplification $A(y \to z; m)$ measures how much stronger the joint *story (y, z) makes m look, compared to the y-only baseline. When $V = \mathbb{R}_{>0}$ with $\otimes = \times$ and $\multimap$ division, $A(y \to z; m) = I_{y,z}(m)/I_y(m)$ is literally a multiplicative lift factor.*

Remark 2946 (Why these are useful (decision-tool perspective)). *In a reasoning system, one often wants to ask: "Does adding z to an existing explanatory component y actually buy me anything?" The quantity $A(y \to z; m)$ is designed to answer exactly this, and the next theorem shows that synergy can be decided by comparing $A(y \to z; m)$ to the baseline intensity $I_z(m)$. This turns emergence detection (Hyperseed-Concept* **??***) into a single inequality check.*

Theorem 81 (Conditional-intensity factorization and an emergence test). *For any $m, y, z : X \to V$:*

1. *(Conditional factorization)*

$$I_{y,z}(m) \;=\; \mathrm{Weak}(z) \multimap I_y(m) \;=\; \mathrm{Weak}(y) \multimap I_z(m).$$

   Equivalently, $I_{y,z}$ is a conditional intensity in either direction.

2. *(Synergy as "amplification beyond baseline")*

$$\sigma_{y,z}(m) \;=\; I_z(m) \multimap A(y \to z; m) \;=\; I_y(m) \multimap A(z \to y; m).$$

3. *(Emergence test) The following are equivalent:*

$$\sigma_{y,z}(m) \geq e \quad \iff \quad A(y \to z; m) \geq I_z(m) \quad \iff \quad A(z \to y; m) \geq I_y(m).$$

Remark 2947 (Plain-language meaning and connections)**.** *Part (1) says: the intensity of the joint pattern (y, z) in m can be read as a* conditional *intensity, either "z given y" or "y given z." Part (2) then expresses synergy as the gap between the conditional amplification and the baseline. Part (3) is the operational punchline: emergence holds exactly when the amplification provided by y is at least as large as the baseline intensity of z (and symmetrically).*

This theorem connects directly to the definition of synergy in [3], but it packages the algebra in a form suited to fast diagnostics. It is also conceptually consonant with pattern/emergence discussions in [5]: "emergence" is not merely that a joint description exists, but that the joint description improves the explanatory leverage beyond what the parts already warrant.

Proof. (1) By Lemma 156 with $a = \mathrm{Weak}(y)$, $b = \mathrm{Weak}(z)$, $c = \mathrm{Weak}(m)$:

$$(\mathrm{Weak}(y) \otimes \mathrm{Weak}(z)) \multimap \mathrm{Weak}(m) \;=\; \mathrm{Weak}(z) \multimap (\mathrm{Weak}(y) \multimap \mathrm{Weak}(m)) \;=\; \mathrm{Weak}(z) \multimap I_y(m).$$

The symmetric identity follows by commutativity.

(2) Apply Lemma 156 to $a = I_y(m)$, $b = I_z(m)$, $c = I_{y,z}(m)$:

$$(I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m) \;=\; I_z(m) \multimap (I_y(m) \multimap I_{y,z}(m)).$$

But $I_y(m) \multimap I_{y,z}(m) = A(y \to z; m)$ by definition. (3) Using the adjunction with $a = I_z(m)$, $b = e$, $c = A(y \to z; m)$:

$$\sigma_{y,z}(m) \geq e \iff e \leq I_z(m) \multimap A(y \to z; m) \iff I_z(m) \otimes e \leq A(y \to z; m) \iff I_z(m) \leq A(y \to z; m).$$

Here the first equivalence is simply an order-rewriting into the internal $\leq$ relation (so that the adjunction can be applied uniformly), the middle equivalence is the residuation law $b \leq a \multimap c \iff a \otimes b \leq c$, and the last equivalence uses that e is the tensor unit (hence $I_z(m) \otimes e = I_z(m)$). In particular, this chain exhibits the emergence test as a pure comparison between two "local" quantities associated to m: the baseline $I_z(m)$ and the conditional gain $A(y \to z; m)$. The symmetric direction is identical with roles swapped. Concretely, interchanging y and z replaces $I_z(m)$ by $I_y(m)$ and $A(y \to z; m)$ by $A(z \to y; m)$, and the same adjunction steps go through verbatim because they only depend on the monoidal closed structure, not on any asymmetry of the symbols. ☐

Proof sketch. Everything is driven by currying (Lemma 156). First, rewrite the joint residuation $(\mathrm{Weak}(y) \otimes \mathrm{Weak}(z)) \multimap \mathrm{Weak}(m)$ as a nested residuation to obtain the conditional-factorization. Next, apply the same currying trick to the definition of $\sigma_{y,z}$ to expose $I_y(m) \multimap I_{y,z}(m)$, which is precisely the amplification $A(y \to z; m)$. Finally, translate $\sigma_{y,z}(m) \geq e$ into an inequality by unrolling the adjunction with e as the "do-nothing" tensor unit. At a slightly more granular level, the first step uses that in a monoidal closed setting one has

$$(\alpha \otimes \beta) \multimap \gamma \;=\; \beta \multimap (\alpha \multimap \gamma),$$

so one can read a joint-to-target comparison as an "incremental" comparison: first account for α, then ask how much β is needed to reach γ thereafter. In the present notation, this incremental reading is exactly what makes the subsequent appearance of a conditional term like $I_y(m) \multimap I_{y,z}(m)$ unsurprising: it is the residuated counterpart of "holding y fixed and asking what changes when z is added." Likewise, the last step is not merely a change of symbols: expressing $\sigma_{y,z}(m) \geq e$ as $e \leq I_z(m) \multimap A(y \to z; m)$ forces the quantity $A(y \to z; m)$ into the right-hand position of an implication, where residuation converts it into a tensor-inequality and then into a plain comparison against $I_z(m)$ after using $\otimes e$-neutrality. ☐

Remark 2948 (Key step intuition). *The repeated use of the adjunction is not mere algebraic cleverness; it is the mechanism by which a global comparison is reduced to a local budget check. If one thinks of* $\multimap$ *as "how much more do I need," then* $A(y \to z; m) = I_y(m) \multimap I_{y,z}(m)$ *asks "how much extra does* z *contribute once* y *is already in place." The emergence test then says: emergence occurs precisely when this "extra" meets or exceeds the baseline* $I_z(m)$ *that* z *would have had without the help of* y*. Equivalently,* $I_z(m) \leq A(y \to z; m)$ *can be read as a dominance condition: the conditional gain attributable to adding* z *after* y *is at least as large as the standalone "credit" of* z*. The adjunction makes this dominance test compositional: it compares* $I_z(m)$ *and* $A(y \to z; m)$ *without requiring one to explicitly expand* $\sigma_{y,z}$ *into a multi-term expression and then simplify. This also clarifies why* e *is the right threshold: since* e *is the neutral element for* $\otimes$*, the inequality* $\sigma_{y,z}(m) \geq e$ *is exactly the point at which the synergy score becomes non-deficient relative to the monoidal baseline, so that converting it via residuation yields a statement with no remaining reference to any extraneous scale factor.*

Remark 2949 (Why this is a useful reasoning-system tool). *Theorem 81(3) says you can test emergence by comparing a conditional-amplification quantity ("how much does* y *boost* z *jointly?") against the baseline intensity of* z*, without directly computing the full ratio-style synergy expression. This is often numerically and conceptually simpler, and it suggests causal/diagnostic interpretations: emergence means* y *makes* z *"more pattern-like than it otherwise is." In practice, this separation is helpful because* $I_z(m)$ *and* $A(y \to z; m)$ *typically come from different kinds of estimates:* $I_z(m)$ *is a marginal intensity (a single-input assessment), whereas* $A(y \to z; m)$ *is a conditional change (a two-stage assessment). The theorem therefore supports modular diagnostics: one can keep track of baseline intensities for many* m *and then, only when needed, evaluate which pairs* (y, z) *yield conditional gains large enough to surpass those baselines. Moreover, because the criterion is an order comparison rather than an equality, it is robust under approximation: if one can bound* $A(y \to z; m)$ *from below (or* $I_z(m)$ *from above), one can certify emergence without needing exact values. This is especially useful when the underlying computations occur in a qualitative or partially ordered setting where the emphasis is on entailment/feasibility rather than on precise numeric magnitudes.*

52.3 Shape-3-style: Gauge invariance and robustness under rescaling

Definition 478 (Tensor-cancellative scale elements). *An element* $k \in V$ *is* tensor-cancellative *if for all* $u, v \in V$,

$$k \otimes u \leq k \otimes v \iff u \leq v.$$

Remark 2950 (Intuition and examples). *Tensor-cancellativity is the quantalic analogue of multiplying both sides of an inequality by a positive scalar: in* $\mathbb{R}_{\geq 0}$ *with* $\otimes = \times$*, it holds exactly when* $k > 0$*. Conceptually,* k *is then a "pure change of units" rather than a genuine distortion. This condition is what allows us to treat intensities and synergies as* dimensionless *quantities: they depend on ratios/implications, not on the absolute scale of weights.*

Lemma 157 (Uniform rescaling of valuation rescales weakness). *Let* $\phi_k(x) := k \otimes \phi(x)$*. Then for any predicate* p,

$$w_{\phi_k}(p) \;=\; k \otimes w_\phi(p).$$

Remark 2951 (Intuition and why it is the right lemma). *Weakness is defined by aggregating* $\phi(x) \otimes p(x)$ *across* x*. If every* $\phi(x)$ *is uniformly scaled by the same factor* k*, then the entire aggregation should scale by* k *as well; this is a minimal sanity requirement for any notion of "weighted support."*

The lemma formalizes that sanity check using exactly the quantale axiom that $\otimes$ distributes over joins.

Proof. Using distributivity of $\otimes$ over joins,

$$w_{\phi_k}(p) = \bigvee_x (k \otimes \phi(x)) \otimes p(x) = \bigvee_x k \otimes (\phi(x) \otimes p(x)) = k \otimes \bigvee_x \phi(x) \otimes p(x) = k \otimes w_\phi(p).$$

□

Proof sketch. Pull the constant k outside the join: distributivity lets one rewrite the join of $k \otimes (\cdot)$ as $k\otimes$ (join of $\cdot$). The rest is associativity of $\otimes$. □

Remark 2952 (Operational intuition)**.** *If ϕ encodes "attention weights" or "resource availability" and we re-normalize those weights, the weakness score changes predictably. The deeper question, addressed next, is whether* intensities *(which are comparative) stay fixed.*

Lemma 158 (Cancellation passes through residuation)**.** *If k is tensor-cancellative, then for all $a, b \in V$,*

$$(k \otimes a) \multimap (k \otimes b) \ = \ a \multimap b.$$

Remark 2953 (Why this is the heart of gauge invariance)**.** *Residuation $\multimap$ is the operation that turns "absolute" quantities into "relative" ones. This lemma states that if k is a pure rescaling (cancellative), then relative comparisons are unaffected: scaling both the antecedent and consequent by the same k does not change the implication value. This is precisely the condition one expects if intensities are to behave like ratios rather than raw magnitudes.*

Proof. For any $t \in V$,

$$\begin{aligned} t \le (k \otimes a) \multimap (k \otimes b) &\iff (k \otimes a) \otimes t \le k \otimes b \\ &\iff k \otimes (a \otimes t) \le k \otimes b \\ &\iff a \otimes t \le b \qquad \text{(by tensor-cancellativity)} \\ &\iff t \le a \multimap b. \end{aligned}$$

Thus the two elements have the same lower set, hence are equal. □

Proof sketch. Use the adjunction definition of $\multimap$ on both sides: both are characterized by which t satisfy a certain inequality. Tensor-cancellativity lets one cancel the leading k from that inequality, reducing it to the unscaled condition. □

Remark 2954 (Order-theoretic picture)**.** *A right adjoint is determined by the preimage of principal down-sets. The proof shows that scaling by a cancellative k preserves these down-sets exactly, so the right adjoint (residuation) is unchanged. This is the lattice version of the familiar fact that dividing kb by ka yields b/a.*

Theorem 82 (Gauge invariance of intensities and synergy under uniform valuation scaling)**.** *Assume $k \in V$ is tensor-cancellative and define $\phi_k(x) := k \otimes \phi(x)$. Then for any m, y, z,*

$$I_y^{(\phi_k)}(m) = I_y^{(\phi)}(m), \qquad I_{y,z}^{(\phi_k)}(m) = I_{y,z}^{(\phi)}(m), \qquad \sigma_{y,z}^{(\phi_k)}(m) = \sigma_{y,z}^{(\phi)}(m).$$

Remark 2955 (Intuitive statement and significance)**.** *The theorem says that if we uniformly rescale the valuation ϕ (e.g. by changing units, renormalizing weights, or globally adjusting confidence), then intensities and synergies do not change. This is important because it means that these quantities are not artifacts of calibration: they are invariant features of the relational structure between y, z, and m. In the language of Hyperseed, it supports treating these values as stable indicators of* Pattern *and* Emergence *(Hyperseed-Concepts 130,* **??***) rather than as numerology dependent on a particular scaling convention.*

Proof. By Lemma 157, $\mathrm{Weak}_{\phi_k}(p) = k \otimes \mathrm{Weak}_\phi(p)$ for any p. Thus

$$I_y^{(\phi_k)}(m) = \mathrm{Weak}_{\phi_k}(y) \multimap \mathrm{Weak}_{\phi_k}(m) = (k \otimes \mathrm{Weak}_\phi(y)) \multimap (k \otimes \mathrm{Weak}_\phi(m)) = \mathrm{Weak}_\phi(y) \multimap \mathrm{Weak}_\phi(m)$$

by Lemma 158. The joint-intensity case is identical. Synergy depends only on these intensities via the same residuation operation, so it is also invariant. □

Proof sketch. First show weakness scales linearly with k (Lemma 157). Then observe that each intensity is an implication between two weaknesses, so scaling both inputs by k cancels out inside $\multimap$ (Lemma 158). Since synergy is built from intensities using the same operations, it inherits the invariance. □

Remark 2956 (Geometric/measurement intuition)**.** *One may regard ϕ as choosing a "measuring instrument" for the universe X. Uniformly changing the instrument's gain changes all raw readings, but it should not change dimensionless ratios. The theorem formalizes exactly that expectation: intensity and synergy are ratio-like, hence gauge invariant.*

Remark 2957 (Interpretation)**.** *Under mild cancellativity, intensities and synergies are* dimensionless*: they do not change if you uniformly rescale the valuation weights. This is a robustness theorem that supports cross-domain reuse: a reasoning system can compare intensities across settings whose raw scales differ (units, normalization conventions, etc.).*

52.4 Emergence via interactions: curvature, multi-hole effects, and reliability of first-order McBride sensitivities

Definition 479 (Soft-link weakness (parameterized model))**.** *Let $\theta \in \mathbb{R}^d$ parameterize a family of soft inclusion probabilities $p_\theta(e) \in [0,1]$ over a finite link set E. Given link weights $\phi(e) \geq 0$, define*

$$w(\theta) := \sum_{e \in E} \phi(e)\, p_\theta(e).$$

Assume w is twice continuously differentiable.

Remark 2958 (Intuition and relationship to "holes")**.** *This is a smooth relaxation of the discrete setting in which one either includes or excludes links $e \in E$. The parameter vector θ encodes how willing the model is to include each link, and $p_\theta(e)$ is the induced soft inclusion probability. The resulting weakness $w(\theta)$ is the expected weighted mass of included links. In McBride-derivative language [22], removing a link corresponds to creating a "hole"; here, the edit $\theta \mapsto \theta - \varepsilon e_i$ is a small differentiable analogue of carving such a hole.*

Remark 2959 (Notation clarification)**.** *e_i denotes the ith standard basis vector of $\mathbb{R}^d$. Later, for a set of indices $S \subseteq \{1, \ldots, d\}$, $\mathbf{1}_S \in \mathbb{R}^d$ denotes the indicator vector with $(\mathbf{1}_S)_i = 1$ if $i \in S$ and 0 otherwise.*

Definition 480 (Interaction gap between two "holes")**.** *For coordinate directions i, j and step size $\varepsilon > 0$, define*

$$\Delta_{ij}(\varepsilon) := w(\theta - \varepsilon e_i - \varepsilon e_j) - w(\theta - \varepsilon e_i) - w(\theta - \varepsilon e_j) + w(\theta).$$

Remark 2960 (Intuition and a sanity check)**.** *$\Delta_{ij}(\varepsilon)$ measures whether the effect of two simultaneous edits is additive. If the two edits were independent in their impact on w, then the inclusion–exclusion combination above would be 0. Thus Δ_{ij} is a quantitative probe for interaction effects, i.e. a second-order signature of emergence among edits. This is the smooth analogue of "multi-hole" effects discussed in the quantale/McBride setting [3, 22].*

Theorem 83 (Mixed curvature is the leading-order interaction term)**.** *Let $H(\theta)$ be the Hessian of w at θ, with entries $H_{ij}(\theta)$. Then*

$$\Delta_{ij}(\varepsilon) \;=\; \varepsilon^2 \, H_{ij}(\theta) \;+\; o(\varepsilon^2) \qquad \text{as } \varepsilon \to 0.$$

In particular, the mixed second derivative H_{ij} is a precise second-order measure of non-additive interaction between the ith and jth one-hole edits.

Remark 2961 (What the theorem is saying, in plain terms)**.** *As ε becomes small, the interaction gap is dominated by the mixed Hessian entry $H_{ij}(\theta)$ times ε^2. So the Hessian is not just a generic curvature object; it is exactly the infinitesimal measurement of "how non-additive" two edits are. If one wants to detect emergent effects that only appear when two components are altered together, H_{ij} is the first place to look.*

Proof. Apply the second-order Taylor expansion of w at θ in the three directions $-\varepsilon e_i - \varepsilon e_j$, $-\varepsilon e_i$, and $-\varepsilon e_j$. Write $g = \nabla w(\theta)$. Then

$$\begin{aligned} w(\theta - \varepsilon e_i - \varepsilon e_j) &= w(\theta) - \varepsilon(g_i + g_j) + \frac{\varepsilon^2}{2}(H_{ii} + 2H_{ij} + H_{jj}) + o(\varepsilon^2), \\ w(\theta - \varepsilon e_i) &= w(\theta) - \varepsilon g_i + \frac{\varepsilon^2}{2} H_{ii} + o(\varepsilon^2), \\ w(\theta - \varepsilon e_j) &= w(\theta) - \varepsilon g_j + \frac{\varepsilon^2}{2} H_{jj} + o(\varepsilon^2). \end{aligned}$$

Subtracting according to the definition of $\Delta_{ij}(\varepsilon)$ cancels the linear and diagonal quadratic terms, leaving

$$\Delta_{ij}(\varepsilon) = \varepsilon^2 H_{ij} + o(\varepsilon^2).$$

□

Proof sketch. Compute Taylor expansions for the three perturbed points and combine them via the inclusion–exclusion pattern defining $\Delta_{ij}(\varepsilon)$. The linear terms cancel automatically, and the pure second-order terms H_{ii}, H_{jj} also cancel, leaving only the cross term $2H_{ij}$, hence the leading $\varepsilon^2 H_{ij}$ contribution. □

Remark 2962 (Visual intuition)**.** *One can picture w as a landscape over parameter space. The term H_{ij} measures how the slope in direction i changes as one moves in direction j. If the landscape is locally "separable" in those coordinates, the mixed curvature is near zero; if it is twisted, the mixed curvature is large, indicating genuine interaction.*

Remark 2963 (Decision-tool reading: emergence of edits)**.** *If $H_{ij}(\theta)$ is large in magnitude, then the combined effect of simultaneously changing parameters i and j differs substantially from the sum of their individual effects. This is exactly the "non-additive interaction between holes" intuition emphasized in the one-hole-context curvature discussion.*

Theorem 84 (Curvature-controlled near-optimality of top-k McBride pruning)**.** *Assume that on a convex region $\mathcal{R}$ containing θ and all points $\theta - \varepsilon \mathbf{1}_S$ with $|S| = k$, the Hessian operator norm is bounded: $\|H(\xi)\|_2 \le M$ for all $\xi \in \mathcal{R}$. Define the first-order sensitivities $s_i := \frac{\partial w}{\partial \theta_i}(\theta)$. For any size-$k$ set S, define the (soft) decrease*

$$D(S) := w(\theta) - w(\theta - \varepsilon \mathbf{1}_S).$$

Let S^{FO} be the set of k indices with largest s_i (the "top-k McBride rule"), and let $S^\star$ maximize $D(S)$ among all $|S| = k$. Then

$$D(S^{\mathrm{FO}}) \;\ge\; D(S^\star) \;-\; M\,\varepsilon^2\,k.$$

Remark 2964 (Intuitive statement and why it is important)**.** *The theorem formalizes a common engineering heuristic: "pick the k parameters with largest first-order effect." It says this greedy-by-gradient choice is near-optimal, up to an explicit error term controlled by curvature M, step size ε, and budget k. Thus, first-order McBride sensitivities are reliable precisely when the model is locally close to linear along the relevant multi-edit directions—and unreliable exactly when strong second-order interactions are present (cf. Theorem 83). This provides a principled trigger for when to upgrade from one-hole diagnostics to multi-hole interaction mining [22]. In particular, the bound distinguishes two separate failure modes of the heuristic: (i) large edit magnitudes ε (which amplify quadratic terms even when curvature is moderate) and (ii) genuinely large local curvature M (where even small edits can interact nontrivially). From an experimental-design viewpoint, it also clarifies what the greedy rule is implicitly optimizing: it is exactly optimal for the linearized surrogate objective, and the theorem quantifies how far the true nonlinear objective can deviate from that surrogate along the k-sparse edit directions that matter operationally.*

Proof. Fix any S with $|S| = k$ and consider the Taylor expansion along $-\varepsilon \mathbf{1}_S$:

$$w(\theta - \varepsilon \mathbf{1}_S) = w(\theta) - \varepsilon \sum_{i \in S} s_i + \frac{\varepsilon^2}{2}\, \mathbf{1}_S^\top H(\xi_S) \mathbf{1}_S$$

for some ξ_S on the segment between θ and $\theta - \varepsilon \mathbf{1}_S$. (Here the remainder is written in the standard second-order mean-value form, i.e., the Hessian is evaluated at an intermediate point ξ_S on the line segment; this is the multivariate analogue of the one-dimensional Taylor theorem with Lagrange remainder.) Thus

$$D(S) = \varepsilon \sum_{i \in S} s_i - \frac{\varepsilon^2}{2}\, \mathbf{1}_S^\top H(\xi_S) \mathbf{1}_S.$$

The first term $\varepsilon \sum_{i \in S} s_i$ is the linear (additive) prediction obtained by summing the individual first-order effects over the chosen indices, while the second term captures all within-set interactions contributed by curvature along the joint direction $\mathbf{1}_S$. By the operator norm bound,

$$\left| \mathbf{1}_S^\top H(\xi_S) \mathbf{1}_S \right| \le \|H(\xi_S)\|_2 \, \|\mathbf{1}_S\|_2^2 \le M \cdot k.$$

The inequality $\left| x^\top A x \right| \le \|A\|_2 \|x\|_2^2$ is a direct consequence of the spectral characterization of the operator norm, and it is precisely the step that collapses potentially complicated cross-terms in the quadratic form into a single worst-case curvature constant M. Hence for all S,

$$\varepsilon \sum_{i \in S} s_i - \frac{\varepsilon^2}{2} M k \le D(S) \le \varepsilon \sum_{i \in S} s_i + \frac{\varepsilon^2}{2} M k.$$

This pair of inequalities can be read as a uniform sandwiching of the true multi-edit gain $D(S)$ between its linear surrogate and an additive tolerance of size $\frac{\varepsilon^2}{2}Mk$, independent of the particular combinatorial choice of S. Since S^{FO} maximizes $\sum_{i\in S} s_i$ among $|S| = k$,

$$\sum_{i\in S^{\mathrm{FO}}} s_i \geq \sum_{i\in S^\star} s_i.$$

Combine the lower bound on $D(S^{\mathrm{FO}})$ with the upper bound on $D(S^\star)$:

$$D(S^{\mathrm{FO}}) \geq \varepsilon \sum_{i\in S^{\mathrm{FO}}} s_i - \frac{\varepsilon^2}{2}Mk \geq \varepsilon \sum_{i\in S^\star} s_i - \frac{\varepsilon^2}{2}Mk \geq D(S^\star) - M\varepsilon^2 k.$$

The final constant $M\varepsilon^2 k$ arises because one pays a quadratic remainder once for the chosen set and once for the comparator set, yielding two halves that add to a full $M\varepsilon^2 k$ gap. □

Proof sketch. Approximate $w(\theta - \varepsilon \mathbf{1}_S)$ by its linear term $w(\theta) - \varepsilon \sum_{i\in S} s_i$ and bound the Taylor remainder using the Hessian norm. The top-k rule is optimal for the linear surrogate because it maximizes the sum of chosen sensitivities. The only loss comes from the bounded quadratic remainder, giving the explicit $M\varepsilon^2 k$ gap. Equivalently, if the gradient is M-Lipschitz in the relevant neighborhood, then the discrepancy between the true objective and its linearization along any k-sparse step of size ε is uniformly controlled, so the combinatorial selection is stable under this controlled nonlinearity. □

Remark 2965 (Why the bound scales as k)**.** *The factor $\|\mathbf{1}_S\|_2^2 = k$ is the geometric reason the penalty grows with the number of simultaneous edits: moving along a k-sparse direction has Euclidean length $\sqrt{k}$, and second-order errors scale like (length)^2. Thus even small curvature can accumulate if one performs many edits at once; this is the smooth analogue of many-hole interference. Another way to view this is that the quadratic form $\mathbf{1}_S^\top H(\xi_S)\mathbf{1}_S$ contains k diagonal contributions and $k(k-1)$ off-diagonal contributions; the operator-norm bound suppresses the detailed combinatorics, but the net effect still grows with the size of the joint move through the $\|\mathbf{1}_S\|_2^2$ factor.*

Remark 2966 (When to trust first-order vs. mine higher-order interactions)**.** *Theorem 84 makes the qualitative curvature guidance quantitative: when curvature is small (small M) and/or step sizes are small, first-order McBride sensitivities give near-optimal multi-edit choices. When M is large, Theorem 83 suggests mining higher-order (two-hole, three-hole) interactions to capture emergence among edits. Operationally, the bound recommends a simple diagnostic: if the estimated interaction scale $M\varepsilon^2 k$ is negligible relative to the achieved improvement $D(S^{\mathrm{FO}})$ (or relative to typical differences among candidate sets), then the top-k selection is robust; if not, then one should expect ranking instabilities where sets with smaller first-order scores can win due to synergistic (or antagonistic) curvature effects, motivating explicit multi-hole searches.*

52.5 A concrete greedy guarantee in an important quantale (max-product)

Definition 481 (Max-product quantale and crisp contexts)**.** *Let $V = [0,1]$ with $\oplus = \max$, $\otimes = \times$, and unit $e = 1$. For a crisp subset $M \subseteq X$, represent its predicate by $m_M(x) = 1$ if $x \in M$ and 0 otherwise. Assume $\phi(x) \in [0,1]$, so*

$$\mathrm{Weak}(m_M) = \max_{x\in M} \phi(x).$$

Fix y, z with constants $A := \mathrm{Weak}(y)$ and $B := \mathrm{Weak}(z)$, and define

$$f(M) := I_{y,z}(m_M) = (AB) \multimap \mathrm{Weak}(m_M).$$

In this quantale, $(a \multimap b) = \min(1, b/a)$ for $a > 0$ (and $0 \multimap b := 1$).

Remark 2967 (Interpreting the residuum in the max-product setting)**.** *Because $\otimes = \times$ on $[0,1]$, the residuum $a \multimap b$ is the largest $c \in [0,1]$ such that $a \times c \leq b$. When $a > 0$, solving $ac \leq b$ gives $c \leq b/a$, hence $a \multimap b = \min(1, b/a)$. Thus $f(M) = (AB) \multimap \mathrm{Weak}(m_M)$ can be read as a* normalized sufficiency score*: it measures how large a multiplicative factor one can apply to the baseline AB while still not exceeding the evidence level $\mathrm{Weak}(m_M)$, capped at 1. In particular, $f(M) = 1$ exactly when $\mathrm{Weak}(m_M) \geq AB$; otherwise $f(M) = \mathrm{Weak}(m_M)/(AB)$.*

Remark 2968 (Degenerate baseline and saturation edge cases)**.** *If $AB = 0$, then by the convention $0 \multimap b = 1$ we have $f(M) = 1$ for all M, i.e. the objective is trivial and every set is optimal. In the nondegenerate regime $AB > 0$ used below, $f(M)$ is a scaled version of $\mathrm{Weak}(m_M)$ that saturates at 1 once the threshold AB is met. This saturation is important conceptually: once some element of M is "good enough," adding more elements cannot further increase the intensity beyond 1.*

Remark 2969 (Why this quantale is "important" and what is being optimized)**.** *The max-product quantale is a natural model for reasoning in which one strong exemplar dominates the evidence: the weakness $\mathrm{Weak}(m_M)$ is the best (largest) weighted support among elements of M. The induced intensity $f(M)$ then asks whether the set M contains an element strong enough to meet the joint baseline AB, up to truncation at 1. This is precisely the sort of objective one encounters when a system can attend to only k items and wants to maximize a pattern intensity score quickly.*

In particular, if AB is viewed as a required joint-strength threshold, then maximizing $f(M)$ is essentially maximizing the best available $\phi(x)$ inside M (but expressed in the intensity language consistent with quantale weakness [3]).

Remark 2970 (A useful explicit form for later inequalities)**.** *When $AB > 0$, combining the definition of $\mathrm{Weak}(m_M)$ with the residuum formula yields the concrete closed form*

$$f(M) = \min\left(1, \frac{\max_{x \in M} \phi(x)}{AB}\right).$$

This makes it clear that the objective depends on M only through the single statistic $\max_{x \in M} \phi(x)$. Equivalently, any two sets with the same maximum ϕ-value have the same intensity score.

Lemma 159 (Submodularity of max-weight)**.** *Let $g(M) = \max_{x \in M} \phi(x)$ (with $g(\emptyset) = 0$). Then g is monotone and submodular: for $S \subseteq T$ and $x \notin T$,*

$$g(S \cup \{x\}) - g(S) \ \geq\ g(T \cup \{x\}) - g(T).$$

Remark 2971 (Equivalent "lattice" form of submodularity in this special case)**.** *For completeness, note that submodularity is also equivalent to $g(S) + g(T) \geq g(S \cup T) + g(S \cap T)$ for all S, T. For the max function this inequality is consistent with the idea that $g(S \cup T) = \max(g(S), g(T))$ while $g(S \cap T) \leq \min(g(S), g(T))$, so the left-hand side dominates the right-hand side. The marginal-gain formulation in Lemma 159 is the one needed for the greedy analysis in Theorem 85.*

Remark 2972 (Intuition and why submodularity appears here)**.** *Submodularity is a formal version of diminishing returns. For a max function it is almost tautological: if you already have a large set T whose maximum $g(T)$ is high, then adding a new element x is less likely to increase the max than if you start from a smaller set S whose max is lower. The lemma packages this intuition into the standard submodularity inequality, which will unlock a canonical greedy approximation theorem.*

Proof. Let $a = g(S)$ and $b = g(T)$ so $a \le b$. Let $c = \phi(x)$. If $c \le b$, then $g(T \cup \{x\}) = b$ so the RHS is 0, while the LHS is nonnegative. If $c > b$, then $g(S \cup \{x\}) = g(T \cup \{x\}) = c$, so the inequality becomes $c - a \ge c - b$, which holds since $a \le b$. □

Proof sketch. Reduce to two cases depending on whether the new element x beats the current maximum of T. If it does not, the marginal gain for T is 0 and the inequality is automatic. If it does, both sets' maxima jump to the same value c, and the inequality reduces to $a \le b$. □

Remark 2973 (Visual intuition)**.** *Imagine $g(M)$ as the height of the tallest pillar among those indexed by M. Adding a new pillar can only increase the height if the pillar is taller than the current tallest. The more pillars you already have, the taller your current tallest tends to be, hence the less marginal benefit you typically get from a new pillar.*

Lemma 160 (Submodularity of the induced joint intensity)**.** *The set function $f(M) = \min(1, g(M)/(AB))$ is monotone and submodular.*

Remark 2974 (Why truncation does not break submodularity here)**.** *The function $t \mapsto \min(1, t/(AB))$ is monotone and saturates at 1. Because g itself only changes by either 0 or a jump to the new value c when an element is added, the truncation cannot invert the diminishing-returns comparison: it can only compress gains further. Thus the same case split from Lemma 159 still works.*

Remark 2975 (A slightly more structural view: "piecewise linear" composition)**.** *On the range $t \in [0, AB]$, the map $t \mapsto t/(AB)$ is linear, hence it preserves the ordering of marginal gains exactly. On the range $t \ge AB$, the map is constant ($= 1$), hence all further marginal gains become 0. Since g is itself a max operator (hence its marginals are already of the form "0 or a single jump"), composing with a function that is linear-then-constant can only maintain or reduce marginal gains, never create a new violation of diminishing returns.*

Proof. Monotonicity is immediate from monotonicity of g and of $t \mapsto \min(1, t/(AB))$. For submodularity, use the same case split as in Lemma 159, observing that $g(S \cup \{x\})$ and $g(T \cup \{x\})$ are either unchanged (marginal 0) or both become c, and the outer truncation $\min(1, \cdot)$ is monotone, so it preserves the inequality of marginal gains. □

Proof sketch. Compose a submodular monotone function g with a monotone saturating transform that does not introduce new kinds of marginal-gain behavior (only 0 vs. jump-to-c). Check the submodularity inequality by the same two cases as for g. □

Remark 2976 (What greedy looks like in this particular objective)**.** *Because $f(M)$ depends only on $\max_{x \in M} \phi(x)$, any set containing an element that attains the global maximum of ϕ achieves the best possible f-value under* any *size constraint $k \ge 1$. Consequently, in this max-product instantiation, the greedy algorithm will select (one of) the highest-ϕ elements at its first step (since that maximizes the marginal gain from $\emptyset$), and after that either (i) the score is already saturated at 1 so all remaining marginals are 0, or (ii) the score is unsaturated but no additional element can beat the current maximum, so again marginals are 0. Thus, while Theorem 85 states the standard $(1 - 1/e)$ guarantee, the realized approximation factor here can in fact be 1 whenever ties and the $k \ge 1$ constraint allow inclusion of an argmax of ϕ.*

Theorem 85 (Greedy maximization of intensity has a $(1 - 1/e)$ guarantee)**.** *Let f be as above. Among all subsets $M \subseteq X$ of size at most k, let $M^\star$ maximize $f(M)$. Let M^{gr} be the result of the greedy algorithm that starts from $\emptyset$ and repeatedly adds the element with the largest marginal increase in f until k elements are chosen. Then*

$$f(M^{\mathrm{gr}}) \;\ge\; \left(1 - \frac{1}{e}\right) f(M^\star).$$

Remark 2977 (How the standard theorem applies (and what hypotheses were proved above))**.** *The classical greedy approximation theorem requires exactly the properties established in Lemma 160: f must be normalized ($f(\emptyset) = 0$), monotone, and submodular, and the feasible sets are those of cardinality at most k. Here normalization holds because $g(\emptyset) = 0$, hence $f(\emptyset) = \min(1, 0/(AB)) = 0$ when $AB > 0$ (and in the $AB = 0$ degenerate case the objective is constant, so the inequality is trivially satisfied). Therefore the hypotheses line up directly with the standard analysis (e.g. the Nemhauser–Wolsey–Fisher guarantee for greedy under a cardinality constraint).*

Remark 2978 (Plain-English meaning and where it fits)**.** *This is the classic guarantee for greedy maximization of a monotone submodular set function under a cardinality constraint. In our setting, it says: if you have a budget of k elements to select in order to maximize a quantale-derived pattern intensity, then greedy selection achieves at least a $(1 - 1/e)$ fraction of the optimum. Within this document's broader arc, this is an explicit bridge from quantale pattern-intensity semantics to an algorithmic decision rule with a worst-case performance certificate.*

Proof. By Lemma 160, f is monotone submodular and $f(\emptyset) = 0$. In particular, for $A \subseteq B$ and $x \notin B$ we have the diminishing-returns property

$$f(A \cup \{x\}) - f(A) \;\geq\; f(B \cup \{x\}) - f(B),$$

and monotonicity gives $f(A) \leq f(B)$ whenever $A \subseteq B$. These two facts are the only structural inputs used below.

Let S_t be the greedy set after t steps ($|S_t| = t$). Equivalently, $S_{t+1} = S_t \cup \{g_t\}$ where g_t is chosen to maximize the marginal gain $f(S_t \cup \{x\}) - f(S_t)$ among all available elements $x \notin S_t$.

Let OPT be an optimal size-k set. By submodularity and monotonicity,

$$f(OPT) - f(S_t) \leq \sum_{x \in OPT \setminus S_t} \big(f(S_t \cup \{x\}) - f(S_t)\big).$$

To see why this inequality holds, list the elements of $OPT \setminus S_t$ as $x_1, \ldots, x_m$ (with $m \leq k$) and write a telescoping expansion

$$f(S_t \cup (OPT \setminus S_t)) - f(S_t) = \sum_{i=1}^{m} \Big(f\big(S_t \cup \{x_1, \ldots, x_i\}\big) - f\big(S_t \cup \{x_1, \ldots, x_{i-1}\}\big)\Big).$$

Submodularity (diminishing returns) upper-bounds each term in this telescoping sum by the corresponding singleton marginal $f(S_t \cup \{x_i\}) - f(S_t)$, yielding

$$f(S_t \cup (OPT \setminus S_t)) - f(S_t) \;\leq\; \sum_{x \in OPT \setminus S_t} \big(f(S_t \cup \{x\}) - f(S_t)\big).$$

Finally, monotonicity implies $f(OPT) \leq f(S_t \cup OPT) = f(S_t \cup (OPT \setminus S_t))$, which gives the displayed inequality above.

There are at most k terms on the RHS, so at least one x has marginal gain at least $\frac{1}{k}(f(OPT) - f(S_t))$. (If there are fewer than k terms, the same conclusion still holds because dividing by k only weakens the lower bound.) Greedy picks an element with marginal at least this large, hence

$$f(S_{t+1}) - f(S_t) \;\geq\; \frac{1}{k}\big(f(OPT) - f(S_t)\big).$$

It is sometimes helpful to rearrange this inequality into a "gap-shrinking" form:

$$f(OPT) - f(S_{t+1}) \;\leq\; \Big(1 - \frac{1}{k}\Big)\big(f(OPT) - f(S_t)\big).$$

Let $R_t := f(OPT) - f(S_t)$. Then $R_{t+1} \le (1 - \frac{1}{k})R_t$, so iterating gives

$$R_t \le \left(1 - \frac{1}{k}\right)^t R_0.$$

Since $S_0 = \emptyset$, we have $R_0 = f(OPT) - f(\emptyset) = f(OPT)$, and therefore $R_k \le (1 - \frac{1}{k})^k f(OPT) \le e^{-1} f(OPT)$. The final inequality uses the standard estimate $(1 - \frac{1}{k})^k \le e^{-1}$, which follows for example from $\ln(1-u) \le -u$ (with $u = 1/k$), or from $(1 + 1/(k-1))^{k-1} \ge e$.

Thus $f(S_k) = f(OPT) - R_k \ge (1 - e^{-1}) f(OPT)$. In other words, after k greedy steps the achieved value is within a $(1 - 1/e)$ fraction of optimal for this monotone submodular objective. □

Proof sketch. Submodularity implies that the total remaining gap to optimum can be distributed across at most k elements of an optimal set, so one of them must close at least a $1/k$ fraction of the gap. Greedy picks an element that does at least as well, leading to a multiplicative decay of the gap and hence the $(1 - 1/e)$ bound after k steps. Equivalently, the "regret" $R_t = f(OPT) - f(S_t)$ contracts by a factor of at most $(1 - 1/k)$ at each step; after k steps one has $R_k \le (1 - 1/k)^k R_0 \le e^{-1} f(OPT)$, so the achieved value is at least $(1 - e^{-1}) f(OPT)$. □

Remark 2979 (Why this guarantee is philosophically comforting)**.** *Greedy selection is a local method: it trusts immediate marginal gains. The theorem shows that for the max-product quantale intensity objective, such local trust is not blind faith but is justified by the global convexity-like property encoded by submodularity. In that sense, the lattice order provides a disciplined middle ground between brute-force search and mere heuristic. One can also read the proof as saying that "local evidence" (marginals at S_t) cannot systematically mislead, because diminishing returns prevents the optimum from hiding all of its value in complicated multiway synergies that are invisible to single-step gains. The guarantee is therefore a structural statement about the objective: it rules out pathological landscapes in which every locally best move is globally disastrous.*

Remark 2980 (Why this matters)**.** *This gives a formal justification for a greedy pattern-mining heuristic (under a natural quantale): a reasoning system that must allocate a limited budget of "attended elements" to maximize a target pattern intensity can do so with a principled approximation guarantee. In particular, even when exact optimization is computationally expensive, the greedy strategy comes with a worst-case bound that is independent of the ambient ground-set size, depending only on the budget k through the universal constant $1 - 1/e$. This separates the method from purely ad hoc heuristics: the approximation factor is a theorem-level consequence of monotonicity and submodularity of the intensity objective in the max-product setting.*

52.6 Dynamics: delay-logistic and "bursty" emergence

Theorem 86 (Delay-logistic inheritance for pattern intensity)**.** *Work in $V = \mathbb{R}_{\ge 0}$ with $\otimes = \times$ and assume* $\mathrm{Weak}(m(t)) = W(t)$ *satisfies the classical delay-logistic equation*

$$\dot{W}(t) = r\, W(t) \left(1 - \frac{W(t-\tau)}{K}\right), \qquad r > 0,\ K > 0.$$

Fix any predicate y with constant $c := \mathrm{Weak}(y) > 0$, and define $I_y(t) := I_y(m(t)) = c \multimap W(t) = W(t)/c$. Then $I_y(t)$ satisfies a delay-logistic equation of the same form:

$$\dot{I}_y(t) = r\, I_y(t) \left(1 - \frac{I_y(t-\tau)}{K/c}\right).$$

Remark 2981 (Further technical context: initial histories and well-posedness)**.** *A delay differential equation is specified not only by parameters (r, τ, K) but also by an initial* history *function (e.g. $W_0 \colon [-\tau, 0] \to \mathbb{R}_{\geq 0}$) so that $W(t) = W_0(t)$ for $t \in [-\tau, 0]$. Under the change of variables $W = c\, I_y$, the corresponding initial history is simply $I_{y0} = W_0/c$. Thus the "inheritance" statement should be read as: for any admissible history W_0, the induced history I_{y0} generates a unique trajectory $I_y(t)$ whose dynamics is the rescaled delay-logistic system. This clarifies that the result is not merely pointwise in t, but holds at the level of solution operators determined by the underlying delay system.*

Remark 2982 (Intuitive meaning and why it belongs in this section)**.** *If weakness* $\mathrm{Weak}(m(t))$ *evolves according to a known dynamical law, then intensities derived from weakness inherit a related law. Here the transformation is a simple scaling $W(t) = c\, I_y(t)$ (because $\multimap$ is division in this quantale), so the entire delay-logistic structure is preserved, with only the carrying capacity rescaled. This is a dynamical analogue of the earlier gauge-invariance theme: relative quantities evolve by the same shape of equation.*

Remark 2983 (On the role of $c > 0$ and the quantale residual)**.** *The assumption $c = \mathrm{Weak}(y) > 0$ ensures that $c \multimap W = W/c$ is well-defined as a finite real quantity and that the map $W \mapsto I_y$ is invertible on $\mathbb{R}_{\geq 0}$ (with inverse $I_y \mapsto c\, I_y$). In $V = \mathbb{R}_{\geq 0}$ with $\otimes = \times$, the residual $a \multimap b$ is the largest x such that $ax \leq b$; for $a > 0$ this indeed coincides with b/a. This makes explicit that the theorem is really about* residual-based normalization *of weakness by a fixed reference predicate, which is precisely the kind of "pattern-sensitive" renormalization that arises in McBride-style comparisons.*

Proof. Since $W(t) = c\, I_y(t)$, we have $\dot{W}(t) = c\, \dot{I}_y(t)$ and $W(t - \tau) = c\, I_y(t - \tau)$. Substitute into the equation for $\dot{W}(t)$:

$$c\, \dot{I}_y(t) = r\, (c\, I_y(t)) \left(1 - \frac{c\, I_y(t - \tau)}{K}\right).$$

Divide by $c > 0$ to obtain

$$\dot{I}_y(t) = r\, I_y(t) \left(1 - \frac{I_y(t - \tau)}{K/c}\right).$$

In particular, the delay enters only through evaluation at $t - \tau$, and the scaling commutes with this evaluation, so no additional terms appear. $\square$

Proof sketch. The proof is a change of variables: define $I_y(t) = W(t)/c$, substitute $W = cI_y$ into the differential equation, and simplify. Because the delay term is linear in $W(t - \tau)$, the same substitution works there as well. $\square$

Remark 2984 (Dynamical-systems intuition)**.** *A delay-logistic equation can be viewed as a feedback loop with memory τ. Multiplying the state variable by a constant does not alter the qualitative feedback structure; it only rescales the level at which saturation occurs. Thus the same mechanism that yields oscillations or chaos in W can do so in I_y as well.*

Remark 2985 (Phase-space viewpoint for delay equations)**.** *Although written as a scalar equation, a delay system is naturally a dynamical system on a function space, where the "state" at time t is the history segment $W_t(\theta) := W(t + \theta)$ for $\theta \in [-\tau, 0]$. The scaling map $W_t \mapsto I_{yt} := W_t/c$ is then a homeomorphism of the history space (on the appropriate subset where the solutions live), so the inheritance claim can be interpreted as an equivalence of semiflows on these history spaces. This is the precise sense in which the qualitative behaviors (e.g. recurrent dynamics) are transported between W and I_y.*

Corollary 41 (Chaotic regimes transfer through intensity transforms)**.** *If $W(t)$ exhibits chaotic dynamics for some (r, τ, K), then $I_y(t) = W(t)/c$ exhibits chaotic dynamics for the same (r, τ) and carrying capacity K/c. Moreover, any invertible monotone transform of $W(t)$ (e.g. $\sigma(t) = 1/W(t)$ on $(0, \infty)$) inherits chaos via topological conjugacy.*

Remark 2986 (Domain restrictions for monotone transforms)**.** *The example $\sigma(t) = 1/W(t)$ implicitly requires $W(t) > 0$ along the trajectory (or at least on the time interval of interest), since inversion is not continuous at 0 on $\mathbb{R}_{\geq 0}$. More generally, for a monotone transform to yield a conjugacy, it must be a homeomorphism on an invariant subset of the state space (or of the history space in the delay setting). Operationally, this means that "score transforms" are safest when they preserve the relevant invariant region and do not introduce singularities that could masquerade as additional burstiness.*

Remark 2987 (Why the corollary is not merely formal)**.** *The corollary warns against a naive expectation that "scores" derived from weakness should be smoother than weakness itself. If the underlying weakness dynamics is bursty or chaotic, then intensities (and other monotone transforms) can be equally bursty. Thus, a reasoning system that monitors pattern intensity (Hyperseed-Concept 130) should treat volatility as a possible* structural *phenomenon rather than as mere measurement noise.*

Remark 2988 (Concrete dynamical implication for "bursts")**.** *In practice, delay-logistic systems are known to admit sustained oscillations and more complicated behavior for certain ranges of $r\tau$ (with the onset of oscillatory instability occurring when the feedback delay is large relative to the intrinsic growth rate). The corollary therefore supports a mechanistic picture in which long-lag feedback in global weakness can generate intermittent surges in I_y even when the reference predicate y (hence c) is time-independent. This connects the qualitative language of "bursty emergence" to a specific and testable dynamical mechanism: delayed negative feedback combined with amplification.*

Proof. Scaling $W \mapsto W/c$ is an invertible continuous map with continuous inverse. Thus the flow on W is conjugate to the flow on I_y. The same holds for any invertible monotone transform on the relevant state domain (e.g. inversion on $(0, \infty)$), which preserves qualitative dynamical complexity such as chaos. In the delay setting, the same argument applies to the induced map on history segments (sending W_t to I_{y_t}), giving conjugacy of the associated semiflows on the history space, which is the standard framework in which "chaos" for delay equations is stated. □

Proof sketch. Use the dynamical-systems principle that invertible continuous reparameterizations of state space yield conjugate flows. Conjugacy preserves qualitative behaviors like chaos. Scaling and monotone inversion (on the appropriate domain) are examples of such reparameterizations. □

Remark 2989 (Operational reading)**.** *This supports a concrete "bursty emergence" picture: if the global weakness evolves by a known chaotic delay-feedback mechanism, then intensities (and any monotone score derived from them) can inherit chaotic fluctuations, providing a mathematical basis for unpredictable surges in detected patterns.*

Remark 2990 (Connection to McBride-pattern sensitivities)**.** *Because $I_y = \mathrm{Weak}(y) \multimap \mathrm{Weak}(m)$ is a relative quantity, it can be viewed as a "pattern-sensitive" normalization of weakness: different choices of predicate y change only the scale, not the qualitative dynamics. Thus, when comparing intensities across predicates (a typical McBride-style sensitivity analysis), differences in observed burstiness should not automatically be attributed to different underlying mechanisms; they may simply reflect different normalizing constants or transforms applied to the same underlying weak* *dynamics. This is especially relevant when one predicate yields values near a transform singularity (e.g. inversion near* 0*), in which case apparent bursts can be amplified artifacts of the chosen representation rather than changes in the underlying system.*

52.7 Log-linearization: emergence as an information-like inequality

Theorem 87 (Log-synergy and an additive "mutual-emergence" quantity)**.** *Assume $V = \mathbb{R}_{>0}$ with $\otimes = \times$ and $\multimap$ given by division: $a \multimap b = b/a$. Assume $I_y(m), I_z(m), I_{y,z}(m) > 0$. Then*

$$\sigma_{y,z}(m) = \frac{I_{y,z}(m)}{I_y(m)\, I_z(m)}.$$

Define the log-synergy (mutual-emergence) quantity

$$\mathrm{ME}(y, z; m) := \log \sigma_{y,z}(m).$$

Then

$$\mathrm{ME}(y, z; m) = \log I_{y,z}(m) - \log I_y(m) - \log I_z(m),$$

and the emergence criterion $\sigma_{y,z}(m) \geq 1$ is equivalent to $\mathrm{ME}(y, z; m) \geq 0$, i.e.

$$\log I_{y,z}(m) \geq \log I_y(m) + \log I_z(m).$$

Moreover, since $\sigma_{y,z}(m) \geq 1$ is equivalent to $I_{y,z}(m) \geq I_y(m)I_z(m)$, one can view $\mathrm{ME}(y, z; m)$ as a signed "gap" (in log-units) measuring the extent to which the joint intensity exceeds (or falls below) what would be predicted by a simple multiplicative baseline. In particular, $\mathrm{ME}(y, z; m) = 0$ corresponds exactly to the multiplicative factorization $I_{y,z}(m) = I_y(m)I_z(m)$, while $\mathrm{ME}(y, z; m) > 0$ and $\mathrm{ME}(y, z; m) < 0$ represent (respectively) super-multiplicative and sub-multiplicative joint behavior relative to that baseline.

Remark 2991 (Intuition and conceptual connection)**.** *When $\otimes$ is multiplication and $\multimap$ is division, synergy becomes a familiar multiplicative interaction ratio: joint intensity divided by the product of individual intensities. Taking logs converts products into sums, yielding an additive inequality. This resembles the way information theory turns multiplicative likelihood ratios into additive log-likelihoods, and it resonates with the general use of log-costs (description length, contrivance) elsewhere in the framework [3]. In particular, the expression*

$$\log I_{y,z}(m) - \log I_y(m) - \log I_z(m)$$

is formally analogous to pointwise mutual information (PMI) when I-terms are interpreted as probability-like or likelihood-like quantities; the sign of $\mathrm{ME}(y, z; m)$ then indicates whether the (y, z) joint is more or less "surprising" than would be expected under multiplicative independence. This analogy is purely structural here (no probabilistic assumptions are required beyond positivity), but it motivates treating ME as an "information-like" score in downstream selection and comparison tasks.

Proof. With $a \multimap b = b/a$, the definition of $\sigma_{y,z}$ gives the displayed ratio directly. Taking logs yields the additive identity. Since log is monotone increasing, $\sigma_{y,z}(m) \geq 1 \iff \log \sigma_{y,z}(m) \geq 0$. The assumption $I_y(m), I_z(m), I_{y,z}(m) > 0$ is exactly what ensures $\sigma_{y,z}(m) > 0$ and makes $\log \sigma_{y,z}(m)$ well-defined without additional case distinctions. ☐

Proof sketch. Unroll the definitions using the concrete form of $\multimap$ as division, then apply log to linearize multiplication into addition. Monotonicity of log preserves the direction of the emergence inequality. The positivity assumptions guarantee that both the division and the logarithm are valid operations in $V = \mathbb{R}_{>0}$, so the derivation requires no further algebraic side conditions. $\square$

Remark 2992 (Log base, units, and scale invariances)**.** *The base of the logarithm in* $\mathrm{ME}(y, z; m)$ *can be chosen arbitrarily (e.g. natural log or base-2); changing the base only rescales* ME *by a positive constant factor and therefore does not affect the sign test* $\mathrm{ME}(y, z; m) \geq 0$*. Because* $\sigma_{y,z}(m)$ *is a ratio, it is dimensionless even when the intensities* $I_y, I_z, I_{y,z}$ *carry units; correspondingly,* $\mathrm{ME}(y, z; m)$ *is naturally interpreted as a "log-unit" gap. In addition, if each intensity is rescaled multiplicatively by fixed positive constants (e.g. due to a change of overall normalization), the ratio* $\sigma_{y,z}(m)$*, and hence* $\mathrm{ME}(y, z; m)$*, remains unchanged whenever the same scaling convention is applied consistently to* $I_y, I_z, I_{y,z}$ *so that the joint-to-marginal comparison is preserved.*

Remark 2993 (Why logs help in practice)**.** *Many reasoning systems already operate in log-space to avoid numerical underflow and to make additive budgets natural (e.g. one can add costs rather than multiply probabilities). The quantity* $\mathrm{ME}(y, z; m)$ *is then directly comparable to other additive scores: one can threshold emergence by a single scalar and combine it with other terms in a multi-objective selection rule without leaving log-space. An additional practical benefit is interpretability: differences in* ME *correspond to multiplicative factors in* $\sigma_{y,z}$*. For example,* $\mathrm{ME}(y, z; m)$ *increasing by* $\log 2$ *means the synergy ratio doubled, regardless of the absolute scale of the underlying intensities.*

Remark 2994 (A small numerical illustration)**.** *If (for a fixed* m*) one has* $I_y(m) = 2$*,* $I_z(m) = 3$*, and* $I_{y,z}(m) = 12$*, then* $\sigma_{y,z}(m) = 12/(2 \cdot 3) = 2$*, hence* $\mathrm{ME}(y, z; m) = \log 2 > 0$*, indicating emergence under the* $\sigma_{y,z}(m) \geq 1$ *criterion. Conversely, if* $I_{y,z}(m) = 5$ *with the same* $I_y(m), I_z(m)$*, then* $\sigma_{y,z}(m) = 5/6 < 1$ *and* $\mathrm{ME}(y, z; m) = \log(5/6) < 0$*, indicating a shortfall of the joint intensity relative to the multiplicative baseline. Such examples emphasize that the sign of* ME *captures the direction of deviation, while its magnitude quantifies the deviation on a logarithmic (multiplicative) scale.*

Remark 2995 (Why this is practically helpful)**.** *In domains where a reasoning system already uses log-scores (description lengths, contrivance, clarity), Theorem 87 turns a multiplicative emergence condition into an additive inequality. This often simplifies thresholding, budgeting, and multi-objective tradeoffs. It also makes sensitivity analyses more straightforward: a small additive perturbation to* $\log I_{y,z}(m)$*,* $\log I_y(m)$*, or* $\log I_z(m)$ *produces an equally small additive perturbation to* $\mathrm{ME}(y, z; m)$*, which is often easier to track than the corresponding multiplicative perturbations in the original ratio* $\sigma_{y,z}(m)$*.*

53 Independence as the Pivot: Connecting Hyperseed Pattern-Intensity Tradeoffs to the Dependence Dichotomy and the Tradeoff Theorem

We formalize several cross-links between (i) Hyperseed-style tradeoffs among effort, weakness, and pattern intensity, and (ii) the Dependence Dichotomy / tradeoff theorem framework in which any attempt to increase evidential quality must choose between enforcing independence (Route A) and explicitly modeling dependence (Route B) [9, 8]. More concretely, the Hyperseed viewpoint treats "pattern intensity" as a measure of how strongly a regularity asserts itself across observations and internal representations, while "effort" captures the resource expenditure required to find, test, and maintain that regularity, and "weakness" captures the degree to which the pattern is fragile, approximate, or contingent on hidden context. The dependence framework, in turn, treats evidential quality as a function not only of the amount of data but of whether distinct pieces of evidence behave as genuinely independent samples (so that aggregation yields near-additive information) or whether they are entangled by common causes, feedback, or shared construction (so that naive aggregation overstates confidence). The bridge between the two is that high pattern intensity can arise either from many weakly-related confirmations (an independence-friendly regime) or from a smaller number of tightly coupled mechanisms that reinforce each other (a dependence-dominated regime), and these two sources have very different costs and failure modes.

The results emphasize independence as a unifying latent variable, yielding practical decision rules: when synergy is observed, Route B is forced; when residual dependence is small, conditional independence approximations are sound; when double-counting exists, redundant couplings can be pruned; and on the Pareto frontier, marginal gains per unit effective cost must equalize across all active "independence vs coupling" knobs. Here "synergy" is meant in the operational sense that the combined evidential impact of two sources exceeds what would be predicted by treating them as conditionally independent given the current model; this indicates that there is unmodeled shared structure (a coupling) that is carrying information and must be represented if one wishes to claim calibrated confidence. "Residual dependence is small" refers to the regime in which, after conditioning on the model's latent variables and known confounders, the remaining correlations among evidential channels are weak enough that the error introduced by a factorized approximation falls below the decision-relevant threshold. "Double-counting" arises when correlated evidence is fed into an inference pipeline as if independent, producing artificially high effective sample size; pruning redundant couplings means either removing duplicate channels, collapsing them into a single sufficient statistic, or introducing an explicit dependence term so that the combined update reflects the true degrees of freedom. Finally, the Pareto-frontier statement can be read as a Lagrange-multiplier condition: once one trades off evidential quality against computational and modeling cost, optimal allocation requires equalizing the marginal return of spending effort on independence-enforcing transformations (e.g., whitening, deconfounding, decorrelation, or experimental design) versus spending effort on richer coupling structure (e.g., additional latent variables, interaction terms, or explicit graph edges). This makes the "independence vs coupling" dial analogous to the Hyperseed allocation between pursuing stronger patterns and paying the price in complexity and maintenance.

Remark 2996. *Philosophically, the theme is that "independence" is not merely a modeling convenience; it is a kind of hidden coordinate for the whole enterprise. If the world (or the agent's representation of it) is close to independent, then explanation can be modular and cheap; if it is not, then any claim to high-quality evidence must purchase its legitimacy by paying for coupling. This is the same tension that appears, in different vocabulary, in quantale weakness theory [3, 2] and in the Hyperseed treatment of patterns and emergent structure [5, 1]. In this sense, independence functions like a "gauge choice" for reasoning: one may either (a) transform the representation so that interactions are pushed into the background and the remaining components behave nearly separably, or (b) keep the representation closer to the raw generative process and account explicitly for the interaction terms that remain. The former yields reusable modules and clearer accounting of evidence, but risks systematic bias if the transformations only partially remove coupling; the latter yields higher fidelity but can suffer from combinatorial explosion and overfitting if the dependence structure is expanded without adequate constraints. The unifying claim is not that independence is always attainable, but that reasoning systems must track how far they are from it, because this distance controls both the effective strength of patterns (how much signal truly accumulates) and the* *effective cost of using them (how much structure must be represented to avoid illusory confidence).*

53.1 Preliminaries (minimal)

53.1.1 Hyperseed-style compositional patterns and intensity

Fix a context C. A compositional pattern witness for x is a triple $P = (y, z, t)$ with $x = y *_C z$ and glue/interaction overhead $t \geq 0$. Define composite cost

$$h_C(P) := s_C(y) + s_C(z) + t.$$

With baseline cost $s_C^{base}(x) > 0$, define scalar pattern intensity

$$I_C(P; x) := \max\left\{0, \frac{s_C^{base}(x) - h_C(P)}{s_C^{base}(x)}\right\} \in [0, 1].$$

Remark 2997 (Notation and intuition). *Here $*_C$ is a (possibly partial) composition operator internal to the context C (so $y *_C z$ may or may not be defined). The quantity $s_C(\cdot)$ is a description or construction cost (one may read it as "how many resources it takes to specify/build the entity"), and $s_C^{base}(x)$ is the baseline cost used to normalize intensity. The overhead t is the "price of interaction": everything that cannot be attributed cleanly to describing y and z separately. This overhead is the informal locus of dependence: if y and z were perfectly independent, one expects little or no glue.*

The intensity $I_C(P; x)$ simply measures relative compression: if the composite description $h_C(P)$ is close to the baseline $s_C^{base}(x)$, then the witness does not help much and intensity is near 0; if $h_C(P)$ is much smaller, intensity approaches 1. This matches the Hyperseed view of Pattern *and* Emergence *as forms of compressive explanatory power (Hyperseed-Concept 130, Hyperseed-Concept* **??***) [5].*

Remark 2998 (Examples). *A simple example is string composition: let x be a long string, let y and z be substrings, and let $*_C$ be concatenation with a small delimiter. Then t corresponds to delimiter/format overhead. If x is literally y followed by z, then t can be tiny and intensity can be high. If instead x is an intricate interleaving of bits of y and z, then any decomposition forces large t, and intensity collapses.*

Another example is a model composed of two submodels: y predicts one subset of variables, z predicts another, and t represents the extra parameters needed to coordinate them (shared latent variables, attention links, synchronization constraints). In this reading, t is a quantitative avatar of Dependency *(Hyperseed-Concept 93) and of the* Effort *needed to manage it (Hyperseed-Concept 100).*

53.1.2 Quantale pattern intensity and synergy (emergence witness)

Let $(V, \otimes, \oplus, e)$ be a commutative quantale with residuation $\multimap$ satisfying $a \otimes b \leq c \iff b \leq a \multimap c$. For a valuation $\varphi : X \to V$ and predicates $y, z, m : X \to V$, define

$$w_\varphi(p) := \bigoplus_{x \in X} \varphi(x) \otimes p(x), \qquad h(y, z) := w_\varphi(y) \otimes w_\varphi(z).$$

Define intensities

$$I_y(m) := w_\varphi(y) \multimap w_\varphi(m), \quad I_z(m) := w_\varphi(z) \multimap w_\varphi(m), \quad I_{y,z}(m) := h(y, z) \multimap w_\varphi(m).$$

Define synergy

$$\sigma_{y,z}(m) := (I_y(m) \otimes I_z(m)) \multimap I_{y,z}(m).$$

Call (y, z) emergent in m if $\sigma_{y,z}(m) \geq e$.

Remark 2999 (Notation and conventions)**.** *The symbol* $\oplus$ *denotes join (supremum) in the quantale, while* $\otimes$ *is the commutative monoidal product. The unit* e *is the multiplicative identity for* $\otimes$*. The residuation* $\multimap$ *is the (right) adjoint to left-multiplication:* $a \otimes b \leq c$ *iff* $b \leq a \multimap c$*. Intuitively,* $a \multimap c$ *measures "how much* b *you can afford" given budget* a *to stay under* c*.*

The map $w_\varphi(p)$ *is the quantale-theoretic analogue of "aggregated support" or "weakness" for predicate* p *under weighting* φ*, as in quantale weakness theory (Hyperseed-Concept 143, Hyperseed-Concept 202) [3, 2]. In the Boolean quantale,* w_φ *reduces to existence-of-a-witness; in probabilistic quantales, it resembles a weighted expectation; in cost/log semirings, it resembles soft-min or log-sum-exp aggregation.*

Remark 3000 (Intuition: why residuated intensities encode "pattern" and "emergence")**.** *The intensity* $I_y(m) = w_\varphi(y) \multimap w_\varphi(m)$ *reads: "relative to the baseline support of* y*, how strongly does* m *hold?". The joint intensity* $I_{y,z}(m)$ *compares* m *to the joint baseline* $h(y,z) = w_\varphi(y) \otimes w_\varphi(z)$*, i.e. the independent product baseline. Synergy* $\sigma_{y,z}(m)$ *then measures whether the joint intensity exceeds what one would expect from simply multiplying the individual intensities: if* $\sigma_{y,z}(m) > e$*, then* m *looks "more true" given the pair than one can justify from the individuals alone. This is the quantale version of an emergence witness: the whole is stronger than the parts under the chosen aggregation semantics (Hyperseed-Concept* **??***) [3, 5].*

53.1.3 Tradeoff-theorem viewpoint: Route A vs Route B

Abstractly, we have:

- a cost lattice/quantale Q_{comp} (effort/resources),
- an evidence lattice/quantale Q_{evid} (accuracy / evidential adequacy),
- a product order on $Q := Q_{comp}^{op} \times Q_{evid}$,
- circuits/architectures f with cost $\kappa(f) \in Q_{comp}$ and evidence $\iota(f) \in Q_{evid}$,
- Route A = independence-enforcing transforms (decorrelation/comonad action),
- Route B = explicit multi-ary couplings (arity ≥ 2 generators).

A penalty-bridge (quantale morphism) $\phi : Q_{comp}^{op} \to Q_{evid}$ yields

$$W(f) := \iota(f) \otimes \phi(\kappa(f)) \in Q_{evid},$$

interpreted as a weakness-regularized evidence score.

Remark 3001 (What the objects mean, operationally)**.** *The product order* $Q := Q_{comp}^{op} \times Q_{evid}$ *is the formal way to encode the usual engineering preference:* lower *computational cost and* higher *evidence. Taking* Q_{comp}^{op} *flips the order on costs so that "better" means "smaller cost". The map* $\kappa(f)$ *is whatever cost model one uses (time, memory, samples, energy), relating to Hyperseed-Concept 100. The map* $\iota(f)$ *is an evidential adequacy measure (accuracy, likelihood, calibration, constraint satisfaction), relating to Hyperseed-Concept* **??** *and Hyperseed-Concept 139. In particular, writing the target as a* product *order makes explicit that we are not assuming a single scalar utility a priori: we are tracking a two-axis (or more generally multi-axis) preference and thereby working with a Pareto-style comparison where improvements in one coordinate do not automatically compensate for deterioration in the other. The choice of* Q_{comp} *and* Q_{evid} *as posets (rather than* $\mathbb{R}$*) also allows one to encode coarse-grained or partially comparable resource budgets (e.g. different kinds* *of memory, heterogeneous hardware, or hard-vs-soft constraints) and similarly structured notions of evidence (e.g. accuracy vs. calibration, likelihood vs. coverage), while still retaining a disciplined notion of "no worse than" via the order. Operationally, an arrow f (model, procedure, component, or sub-system) is thus judged by the pair $(\kappa(f), \iota(f))$, and the product order asserts that a design is preferred when it is cheaper* and *at least as evidentially adequate, with the op ensuring that "cheaper" is aligned with "greater" in the combined preference. One typically also expects κ and ι to be monotone with respect to the relevant refinement/implementation order on artifacts f (when such an order is present), so that a refinement that genuinely increases evidence or decreases cost is reflected as an improvement in Q.*

Route A is "make things independent enough that simple composition is legitimate"; Route B is "admit dependence and represent it explicitly". Both are manifestations of Hyperseed-Concept 93 and Hyperseed-Concept 88 at the architectural level. The general Route A/Route B dichotomy is treated as a central organizing device in [9, 8]. The operational distinction is that Route A spends design effort on interventions (modularization, isolation, randomization, buffering, architectural separation, careful interface contracts) that reduce unwanted couplings so that composed performance is close to the performance predicted by simpler factorizations. Route B, by contrast, treats the couplings as first-class: the system carries extra representational and computational machinery (joint models, graphical structure, dependence-aware estimators, coordination protocols, explicit shared-state) so that composition remains sound even when independence is false. In terms of the order Q, Route A often lowers κ for downstream inference/composition by enforcing simpler semantics at interfaces, whereas Route B often increases κ (and sometimes ι) by paying for the additional structure needed to track dependencies explicitly; the point of the framework is to make that cost–evidence exchange legible rather than implicit.

Remark 3002 (Why the penalty bridge ϕ is the right abstraction)**.** *The weakness-regularized score $W(f) = \iota(f) \otimes \phi(\kappa(f))$ expresses the idea that evidential adequacy is meaningful only when paid for honestly. One may think of ϕ as converting cost into an evidential penalty (or "contrivance tax"), matching the spirit of weakness-as-regularization in [3, 2]. In the simplest numeric setting, ϕ could be $k \mapsto e^{-\lambda k}$ so that multiplying by $\phi(\kappa)$ discounts evidence by an exponential in cost. The advantage of the quantale formulation is that it remains valid when evidence and cost are not scalars but structured quantities (multi-objective, partially ordered, or compositional). More concretely, the bridge $\phi : Q_{comp} \to Q_{evid}$ (or into the evidential side of the ambient quantale, depending on the chosen encoding) is the abstraction that enforces a disciplined* exchange rate *between resources spent and evidential credit received: higher cost should not be able to masquerade as "free" evidence, and conversely a cost reduction should not be treated as a loss of evidence unless the application explicitly chooses such a trade. The tensor $\otimes$ is deliberately left abstract because the appropriate combination law depends on what "evidence" means: in a multiplicative setting it behaves like weighting/discounting, while in a log-domain or additive quantale it can correspond to adding a penalty term (e.g. $-\lambda k$) to a score. The quantale perspective guarantees that W composes coherently with whatever notion of combination is already built into the evidential semantics (for instance, conjunction of constraints, fusion of independent likelihood factors, or aggregation of multiple validation criteria), while still respecting monotonicity of the order. Operationally, ϕ is also where one can encode hard budgets and infeasibility: a design whose $\kappa(f)$ exceeds an allowable budget can be sent by ϕ to a bottom/absorbing penalty (so that $W(f)$ becomes decisively poor), whereas modest costs can be penalized gently to reflect a "pay as you go" engineering regime. In this way, ϕ isolates the normative choice (how strongly to punish complexity or resource use) from the descriptive choice of ι (how to measure evidential adequacy), which is precisely why it is the right abstraction for comparing routes and architectures that may differ widely in where they spend* *computation to buy robustness, calibration, or other forms of evidence.*

53.2 New theorems connecting tradeoffs to the tradeoff theorem

53.2.1 Glue overhead as an independence bottleneck

Lemma 161 (Intensity ceiling induced by glue overhead)**.** *Fix C and x with $s_C^{base}(x) > 0$. For any compositional witness $P = (y, z, t)$ of x,*

$$I_C(P; x) \le \max\left\{0,\, 1 - \frac{t}{s_C^{base}(x)}\right\}.$$

Equivalently, if $I_C(P; x) \ge \alpha$ for some $\alpha \in [0, 1]$, then

$$t \le (1 - \alpha)\, s_C^{base}(x).$$

Remark 3003 (What the lemma is saying, in plain language)**.** *This lemma isolates the simplest possible obstruction to strong compositional explanation: even if y and z were "free" to describe, the glue t alone already caps the intensity. So intensity cannot be high unless glue is small. One may read t as a numerical proxy for dependence: high dependence forces high glue, and high glue suppresses intensity.*

This is important because it turns the somewhat qualitative slogan "independence helps compositionality" into a quantitative inequality. In the Route A vs Route B framing [9], it says that Route A (decorrelation, factorization, abstraction) is not optional ornamentation: it is precisely the attempt to reduce the effective t so that high intensity becomes possible.

Proof. By definition, $I_C(P; x) = \max\{0, (s_C^{base}(x) - [s_C(y) + s_C(z) + t])/s_C^{base}(x)\}$. Dropping the nonnegative terms $s_C(y) + s_C(z) \ge 0$ can only increase the bracketed expression, so

$$\frac{s_C^{base}(x) - [s_C(y) + s_C(z) + t]}{s_C^{base}(x)} \le \frac{s_C^{base}(x) - t}{s_C^{base}(x)} = 1 - \frac{t}{s_C^{base}(x)}.$$

Clamping with $\max\{0, \cdot\}$ preserves the inequality, giving the first claim. The second claim is immediate by rearranging $I_C(P; x) \ge \alpha$ in the unclamped regime.

It is also worth noting that the positivity assumption $s_C^{base}(x) > 0$ is exactly what makes the normalization meaningful: the statement is a bound on a *fractional* (relative) gain compared to a nontrivial baseline. When the baseline were 0, the normalization would be undefined and one would instead work with the unnormalized slack $s_C^{base}(x) - [s_C(y) + s_C(z) + t]$. □

Proof sketch. The proof uses a single monotonicity idea: removing nonnegative costs (here $s_C(y)$ and $s_C(z)$) can only make the witness look better, hence can only increase the un-clamped intensity expression. Therefore the largest intensity compatible with glue t occurs in the fictitious best case $s_C(y) = s_C(z) = 0$, yielding $1 - t/s_C^{base}(x)$. Geometrically, one may picture the baseline cost as a fixed "budget" and t as a mandatory tax; the remaining budget fraction is an absolute upper bound on relative compression. □

Remark 3004. *Lemma 161 formalizes a key intuition: high pattern intensity forces small "interaction overhead" t, i.e., the composite must be close to independent/low-glue. This is where Route A typically enters: it is a mechanism whose purpose is to* reduce *effective glue by enforcing independence.*

Remark 3005 (A few immediate boundary cases)**.** *The inequality has two useful edge-case readings that are easy to miss:*

1. *If $t \geq s_C^{base}(x)$, then the right-hand side is ≤ 0, hence $I_C(P;x) = 0$. In words: if the glue alone costs at least as much as the entire baseline description of x, then* no *compositional explanation of this form can have positive intensity.*

2. *If $t = 0$, then the lemma only yields the trivial ceiling $I_C(P;x) \leq 1$. This is as it should be: eliminating glue removes the* dependence-induced *obstruction, but the remaining terms $s_C(y) + s_C(z)$ may still prevent high intensity.*

3. *Solving $I_C(P;x) \geq \alpha$ for t emphasizes a linear "price of intensity": each additional $\Delta\alpha$ of guaranteed intensity requires shrinking allowable glue by $\Delta\alpha \cdot s_C^{base}(x)$. Thus, when $s_C^{base}(x)$ is large, even modest target intensities can impose large absolute reductions in admissible glue.*

These are merely restatements of Lemma 161, but they often serve as quick sanity checks when applying the bound.

53.2.2 Synergy forces explicit coupling (independence is not free)

Definition 482 (Strong monoidality on the independence fragment)**.** *Let $\mathcal{A}$ denote the "independence fragment" (Route A-only) of circuits: those built without any multi-ary generator of arity ≥ 2. Assume the evidence evaluator is* strong monoidal *on $\mathcal{A}$: for all $f, g \in \mathcal{A}$,*

$$\iota(f \otimes g) = \iota(f) \otimes \iota(g).$$

Remark 3006 (Intuition and why this assumption is natural)**.** *Strong monoidality says: if you build a system by placing two independent subcircuits side-by-side (via $\otimes$), then the evidence of the whole is exactly the monoidal product of the evidences of the parts. In more everyday terms, there is no hidden bonus (and no hidden penalty) from mere juxtaposition. Any departure from this multiplicativity must come from explicit coupling.*

This is precisely the formalization needed to make "synergy" a witness of dependence. Without it, one could not rule out the possibility that the evaluator itself secretly bakes in cross-channel terms. Conceptually, this assumption is the mathematical counterpart of the idea that independence is a modeling constraint with bite, not a vague aspiration (Hyperseed-Concept 93, Hyperseed-Concept 202) [9, 3].

Remark 3007 (On the role of residuation in the definition of synergy)**.** *The expression $I_{ind} \multimap \iota(f)$ uses the residuated implication $\multimap$ of the underlying quantale (or, more generally, a residuated monoidal poset). Operationally, $\Sigma(f) = I_{ind} \multimap \iota(f)$ is intended to be the* largest *element σ such that*

$$I_{ind} \otimes \sigma \;\leq\; \iota(f),$$

when such a largest σ exists. This makes $\Sigma(f)$ a canonical "multiplicative gap" between the independence baseline and the actual evidence: it is exactly the missing factor one must supply to the baseline in order to reach (or upper-bound) the observed evidence. In particular, $\Sigma(f) = e$ means "no multiplicative surplus beyond baseline," whereas $\Sigma(f) > e$ means there is a strictly positive surplus that cannot be attributed to mere juxtaposition under strong monoidality.

Theorem 88 (Synergy is a dependence witness: emergent synergy forces Route B)**.** *Assume strong monoidality on the independence fragment $\mathcal{A}$. Let a system decompose into two subchannels, and* *let f be the overall circuit. Define the "independence baseline" evidence as $I_{ind} := \iota(f_A) \otimes \iota(f_B)$, where f_A, f_B are the induced subcircuits if f factorizes (i.e., $f = f_A \otimes f_B$). Define a synergy factor $\Sigma(f)$ (when the residual exists) by*

$$\Sigma(f) := I_{ind} \multimap \iota(f).$$

Then:

1. *If $f \in \mathcal{A}$ (no multi-ary couplings), then $\Sigma(f) = e$.*

2. *If $\Sigma(f) > e$, then $f \notin \mathcal{A}$, hence any generator decomposition of f must contain at least one multi-ary (arity ≥ 2) coupling (Route B activation).*

Proof. For (1), if $f \in \mathcal{A}$ and the system is presented as two independent subchannels, then f must be representable (within the fragment) as a monoidal juxtaposition $f = f_A \otimes f_B$ with $f_A, f_B \in \mathcal{A}$. By strong monoidality on $\mathcal{A}$,

$$\iota(f) = \iota(f_A \otimes f_B) = \iota(f_A) \otimes \iota(f_B) = I_{ind}.$$

Therefore $\Sigma(f) = I_{ind} \multimap \iota(f) = I_{ind} \multimap I_{ind} = e$, using the standard residuation identity $a \multimap a = e$.

For (2), we argue by contrapositive. Suppose $f \in \mathcal{A}$. Then by (1) we have $\Sigma(f) = e$, which implies it is *not* the case that $\Sigma(f) > e$. Hence if $\Sigma(f) > e$, it must be that $f \notin \mathcal{A}$. By definition of $\mathcal{A}$, leaving $\mathcal{A}$ means that any construction of f from generators must invoke at least one multi-ary (arity ≥ 2) generator, i.e., some explicit coupling, corresponding to Route B. □

Remark 3008 (Plain-English reading and why it matters)**.** *The theorem says: under a clean independence semantics, any "extra" evidence beyond the factorized baseline is a certificate of dependence. If you compute a synergy residual $\Sigma(f)$ and it is strictly above the neutral element e, then the circuit cannot live in the independence-only fragment; some explicit coupling is logically necessary.*

This result is important because it is a one-bit decision procedure*: synergy observed $\Rightarrow$ Route B is forced. It connects the quantale-emergence notion (synergy as residuated surplus) to the architecture-level Route A/Route B dichotomy [9]. In the language of Hyperseed-Concept* **??***, it tells us that emergence is not a decorative label: it has a sharp computational implication about which class of explanatory mechanisms must be invoked.*

Remark 3009 (Caveats and interpretive notes)**.** *Two small interpretive points help avoid over-reading Theorem 88:*

1. *The conclusion is logically one-way: $\Sigma(f) > e$ forces coupling, but $\Sigma(f) = e$ does not, by itself, certify that the mechanism is independent in any stronger empirical sense. It only says that, under the chosen evaluator ι and within the residuation structure being used, no* multiplicative surplus *was detected relative to the factorized baseline.*

2. *The phrase "when the residual exists" matters in practice. In fully residuated settings (e.g., standard quantales) it always exists, but in weaker settings one may only have partial residuals. In such cases, the theorem should be read as applying on the domain where $\multimap$ is defined, i.e., where a canonical "gap element" can be formed.*

These points do not weaken the theorem; they clarify that the force of the result lies in turning a surplus in evidence into a structural claim about the necessity of explicit coupling mechanisms.

Proof. (1) If $f \in \mathcal{A}$, it has no cross-channel coupling and therefore factorizes as $f = f_A \otimes f_B$. By strong monoidality on $\mathcal{A}$, $\iota(f) = \iota(f_A \otimes f_B) = \iota(f_A) \otimes \iota(f_B) = I_{ind}$. Thus $\Sigma(f) = I_{ind} \multimap I_{ind} = e$ by the residuation law.

(2) Contrapositive: if $f \in \mathcal{A}$ then $\Sigma(f) = e$ by (1). Therefore if $\Sigma(f) > e$, $f \notin \mathcal{A}$, i.e., f uses at least one multi-ary generator (Route B). □

Remark 3010 (What is doing the work in the proof)**.** *The key step in (1) is that "no cross-channel coupling" is not merely an intuitive claim about separability, but is captured syntactically by membership in the independence fragment $\mathcal{A}$, where every term is built from unary operations and tensoring, never from genuinely multi-ary generators. In that fragment the tensor product $f_A \otimes f_B$ is not an approximation but an exact normal form.*

The step $\iota(f_A \otimes f_B) = \iota(f_A) \otimes \iota(f_B)$ is exactly where strong monoidality enters: it asserts that the evidence translation ι respects the independence structure on the nose *(rather than only up to inequality or up to a slack variable). Once this holds, the residual $\Sigma(f)$ becomes a purely order-theoretic "difference" between identical points, and residuation forces it to be the monoidal unit e.*

Finally, the contrapositive direction (2) is best read as a certificate *statement: a strictly non-trivial residual $\Sigma(f) > e$ witnesses that some coupling operation must have been used somewhere in the construction of f, because the fragment $\mathcal{A}$ has no mechanism for producing such residuals. In other words, $\Sigma(f)$ is an algebraic obstruction to explaining the behavior with independence-only structure.*

Proof sketch. The strategy is to show that, on the independence fragment, evidence is exactly multiplicative, hence the residual between the factorized baseline and the actual evidence must be the identity element e. The second part is then immediate by contrapositive reasoning: anything that produces a nontrivial residual cannot lie in the fragment. One can visualize $\Sigma(f)$ as the "gap" between two points in an ordered evidence space: the factorized point and the true point; strong monoidality collapses this gap to zero unless a coupling is present. □

Remark 3011 (Operational reading of $\Sigma(f)$)**.** *One can think of $\Sigma(f)$ as measuring the extent to which a purportedly modular explanation fails to account for observed joint structure. If $\Sigma(f) = e$, then the "independence baseline" I_{ind} is already sufficient to explain the evidence produced by f under ι. If $\Sigma(f) > e$, then there is irreducible joint structure not captured by the baseline, and any attempt to fit the phenomenon using only tensor-separable components necessarily leaves a positive residual. This is the formal sense in which synergy/emergence can be treated as* detectable dependence*: it is not just that a coupling might exist, but that the algebra forces its presence in any generating explanation.*

Remark 3012. *Theorem 88 is the clean logical bridge:* observed synergy/emergence *is evidence that an independence-only explanation is impossible. This is exactly the "independence pivot" shared by our Hyperseed based tradeoffs and the tradeoff theorem: to explain synergy you must pay for dependence (Route B), while to preserve broad weakness you prefer independence (Route A).*

53.2.3 Residual dependence as a global certificate for approximate independence

Definition 483 (Residual dependence as total correlation (numeric instantiation))**.** *Let random variables $X_1, \ldots, X_n$ have joint law P. Define the total correlation (multi-information)*

$$TC(X_1, \ldots, X_n) := D\left(P \,\|\, \prod_{i=1}^{n} P_{X_i}\right),$$

where $D(\cdot\|\cdot)$ is KL divergence. Interpret TC as a numeric instance of residual dependence ρ / ResDep (up to scaling).

Remark 3013 (Equivalent forms and basic properties)**.** *Besides its KL form, TC admits the familiar entropy identity*

$$TC(X_1, \dots, X_n) = \sum_{i=1}^{n} H(X_i) - H(X_1, \dots, X_n),$$

whenever the entropies are well-defined. This makes explicit why $TC \geq 0$: joint entropy can never exceed the sum of marginal entropies, and equality holds exactly in the product case. For $n = 2$, $TC(X_1, X_2) = I(X_1; X_2)$, so total correlation really is the natural multi-variate extension of mutual information.

From the present viewpoint, these identities clarify the role of TC as a single global slack variable *for independence: it quantifies how much "extra compression" becomes possible when encoding the tuple jointly rather than independently. This aligns with interpreting ResDep as the residual joint structure left over after applying independence-enforcing design choices (Route A), because any such choices are precisely attempts to make the joint behave as though it were near-product.*

Remark 3014 (Intuition and relationship to independence)**.** *The quantity TC measures how far the joint law P is from being a pure product of its marginals. If $TC = 0$, the variables are mutually independent; as TC grows, the joint law carries more irreducible dependence that cannot be explained by looking at each marginal alone. Thus TC is a scalar certificate of "how much dependence remains" after any independence-enforcing transforms (Route A) have been applied.*

This definition is useful here because it turns the vague phrase "residual dependence is small" into a concrete numeric statement that can be estimated, bounded, or optimized. It anchors the conceptual notion of Dependency *(Hyperseed-Concept 93) in standard information-geometric distance, and it connects naturally to mutual information bounds (Hyperseed-Concept 117).*

Theorem 89 (Residual dependence upper-bounds every cross-partition dependence)**.** *For any partition of indices into disjoint sets A, B,*

$$I(X_A; X_B) \leq TC(X_1, \dots, X_n).$$

Remark 3015 (What the theorem says and why it is powerful)**.** *The theorem says that TC is a* uniform *dependence bound: once you know the total correlation is small, you automatically know that* every *bipartition has small mutual information. This is stronger than bounding $I(X_A; X_B)$ for one particular split: it gives a global "permission slip" to treat any chosen modularization as approximately independent.*

In the Route A vs Route B dialectic [9], this means that one can spend effort to reduce a single global quantity TC, and thereby justify a whole family of independence approximations used by tractable inference and modular architectures. It is a particularly clean example of how independence acts as a pivot variable between explanation quality and computational feasibility.

Remark 3016 (A convenient corollary for "approximate modularity")**.** *A direct corollary is: if $TC(X_1, \dots, X_n) \leq \varepsilon$, then for every split (A, B) we have $I(X_A; X_B) \leq \varepsilon$. Thus a single certificate controls* all *modular decompositions simultaneously. When combined with standard inequalities relating KL divergence to other discrepancy measures (e.g. via Pinsker-type bounds), one can further convert a small-TC assumption into quantitative statements that the joint P_{X_A, X_B} is close to the product $P_{X_A} P_{X_B}$ in stronger norms, giving explicit error bars for independence-based approximations. This is exactly the kind of bound needed to make "Route A works well" into a falsifiable, checkable condition rather than a qualitative slogan.*

Proof. Let $Q := \prod_{i=1}^{n} P_{X_i}$ be the fully factorized product. Let π be the marginalization map onto the block variables (X_A, X_B). Then $\pi(P) = P_{X_A,X_B}$ and $\pi(Q) = P_{X_A} P_{X_B}$. By the data-processing inequality for KL divergence,

$$D(\pi(P) \,\|\, \pi(Q)) \le D(P \,\|\, Q).$$

The left-hand side is $D(P_{X_A,X_B} \,\|\, P_{X_A} P_{X_B}) = I(X_A; X_B)$ and the right-hand side is $TC(X_1, \dots, X_n)$. Hence $I(X_A; X_B) \le TC(X_1, \dots, X_n)$. □

Remark 3017 (Why marginalization is the right map)**.** *The map π forgets internal structure within A and within B and retains only the two-block joint. If dependence is already small in the stronger sense of being close to the* fully *factorized product Q, then it must remain small after any such forgetting operation. The data-processing inequality formalizes this monotonicity: coarse-graining cannot create information-geometric separation that was not already present. This is the same monotonic intuition underlying many "no free dependence" principles: one cannot manufacture cross-module coupling merely by relabeling or discarding variables; coupling has to come from the underlying joint law.*

Proof sketch. The proof is a one-line application of functoriality of information: pushing a distribution through a map (here, marginalization π) cannot increase KL divergence. Since mutual information is exactly the KL divergence between a joint and the corresponding product of marginals, and since TC is the KL divergence to the fully factorized product, the inequality follows immediately. Visually: TC is a distance from P to the product manifold; projecting onto any bipartition cannot increase that distance, so each bipartition's distance (mutual information) is bounded. □

Remark 3018 (Decision tool)**.** *If a reasoning system can maintain (or estimate) a small residual-dependence budget, Theorem 89 justifies treating* any *chosen partition of the world-model as approximately independent (bounding the error by the same certificate), which is exactly the Route-A logic behind many tractable approximations.*

Remark 3019 (Connection back to synergy and the upcoming incompatibility bound)**.** *In the present section, TC plays the numeric role analogous to $\Sigma(f)$: both are "residual" quantities that vanish under perfect independence and become strictly positive exactly when some coupling is present. This makes TC a natural quantitative knob for the next step, where we want an explicit statement of incompatibility between (i) enforcing broad weakness via near-independence (small residual dependence) and (ii) realizing genuine synergy/emergence, which requires a nontrivial residual. The next subsubsection makes this tension explicit by turning the independence pivot into a concrete inequality of the form "high synergy implies non-negligible dependence," thereby linking the abstract tradeoff-theorem logic to an information-theoretic certificate.*

53.2.4 Weakness vs synergy: an explicit incompatibility bound

Theorem 90 (Weakness–synergy incompatibility (quantale form))**.** *Let f be a circuit, with weakness $W(f) = \iota(f) \otimes \phi(\kappa(f))$. Fix an evidence target $\iota(f) \ge I_\infty \otimes \varepsilon^{-1}$ (tolerance ε). Assume (as in the Dichotomy) that meeting this target forces either:*

- *Route A: $\kappa(f) \ge K_{transform}(\varepsilon)$, or*
- *Route B: f uses multi-ary couplings, with per-use costs c_j and counts $N_j(f)$.*

Then any circuit meeting the evidence target must satisfy the upper bound

$$W(f) \leq \left(I_\infty \otimes \varepsilon^{-1}\right) \otimes \max \left\{ \phi(K_{transform}(\varepsilon)),\ \bigotimes_{j \geq 2} \phi(c_j \otimes N_j(f)) \right\}.$$

In particular, for any desired weakness level W_0 strictly above the right-hand side, no circuit can simultaneously achieve evidence $\geq I_\infty \otimes \varepsilon^{-1}$ and weakness $\geq W_0$ without violating the Dichotomy constraints.

Remark 3020 (Quantale-level reading of the bound)**.** *The displayed inequality should be read as an "envelope" constraint: the achievable weakness at evidence level ε is bounded above by the larger of two ceilings, one induced by transform cost and one induced by explicit coupling cost. The $\max\{\cdot,\cdot\}$ here is the join in the ambient order used to compare penalties, so the statement is invariant under the choice of concrete representation (e.g., max-plus, min-plus, log-domain, or multiplicative semiring) as long as $\otimes$ and the order are the ones used to define W. In particular, if one works in a log-parameterization where $\otimes$ becomes addition and the order is the usual $\leq$, then the bound takes the familiar "evidence term plus cost-penalty term" shape, with the join becoming an ordinary maximum of additive penalties.*

Remark 3021 (Intuitive content and connection to weakness theory)**.** *This theorem says that once you fix a stringent evidence target, the weakness $W(f)$ becomes upper-bounded by whichever route you take to achieve that target. If you enforce independence (Route A), you pay in transform cost; if you model dependence (Route B), you pay in coupling cost. Either way, the penalty bridge ϕ converts that cost into a weakness discount. Thus there is a genuine incompatibility between "very demanding evidence" and "very broad weakness", unless one relaxes the dichotomy assumptions.*

This formalizes, in the route framework, the general moral of weakness-based regularization [3, 2]: pushing for sharper fit (here encoded by ε^{-1}) tends to require more contrivance, and contrivance lowers weakness. The statement is not about psychological weakness but about an order-theoretic tradeoff between evidential strength and the cost of the mechanism producing it (Hyperseed-Concept 202, Hyperseed-Concept 100).

One useful way to connect this to the dependence dichotomy is to interpret "synergy" as the part of ι that cannot be produced by a purely factorized mechanism without either (i) increasing the complexity of the factorization itself (Route A), or (ii) admitting explicit cross-factor couplings (Route B). On this reading, the theorem is precisely a statement that synergy-driven evidence requirements cannot coexist with high weakness unless one is willing to "spend" the needed resources somewhere, and ϕ is the formal device that ensures such spending reduces achievable weakness.

Proof. If Route A holds, monotonicity of ϕ on Q_{comp}^{op} gives $\phi(\kappa(f)) \leq \phi(K_{transform}(\varepsilon))$ and hence $W(f) = \iota(f) \otimes \phi(\kappa(f)) \leq \iota(f) \otimes \phi(K_{transform}(\varepsilon)) \leq (I_\infty \otimes \varepsilon^{-1}) \otimes \phi(K_{transform}(\varepsilon))$. If Route B holds, similarly $\phi(\kappa(f))$ is bounded by the (multiplicative) penalty of the used couplings, yielding $W(f) \leq (I_\infty \otimes \varepsilon^{-1}) \otimes \bigotimes_{j \geq 2} \phi(c_j \otimes N_j(f))$. Taking the maximum of the two upper bounds yields the displayed inequality. The final sentence is immediate: if W_0 exceeds both route-wise upper bounds, neither route can realize $(\iota \geq I_\infty \otimes \varepsilon^{-1}) \wedge (W \geq W_0)$.

For avoidance of doubt about the order direction: since ϕ is taken monotone on Q_{comp}^{op}, larger underlying cost (in Q_{comp}) produces smaller (or equal) penalty factors in the target order used for weakness. Thus Route A's lower bound $\kappa(f) \geq K_{transform}(\varepsilon)$ becomes an upper bound after applying ϕ. Likewise, in Route B the aggregate coupling contribution (here written as $\bigotimes_{j \geq 2} \phi(c_j \otimes N_j(f))$) is the penalty factor corresponding to the total coupling resources consumed, and this penalty factor plays the same formal role as $\phi(\kappa(f))$ in the definition of $W(f)$. $\square$

Proof sketch. The proof is a case split on the dichotomy: either the system paid transform cost (Route A) or it paid coupling cost (Route B). In each case, monotonicity of the penalty bridge ϕ yields an upper bound on $\phi(\kappa(f))$, and multiplying by the evidence target gives an upper bound on $W(f)$. Conceptually, one draws two ceiling curves for achievable weakness as a function of ε, one per route, and then takes their maximum: above that envelope, the constraints are jointly infeasible. □

Remark 3022. *Theorem 90 is a direct formalization of the above "decision tool" idea: you cannot demand simultaneously (i) very broad independence commitments (high weakness) and (ii) very strong evidential constraints (often driven by synergy/dependence) without paying either transform cost (Route A) or coupling cost (Route B).*

Remark 3023 (A convenient corollary for "hard" evidence targets)**.** *If the evidence target is made more stringent (smaller ε) in a regime where either $K_{transform}(\varepsilon)$ is increasing or the required coupling multiplicities $N_j(f)$ must increase, then the right-hand side decreases (in the weakness order) provided ϕ is strictly order-reversing on the relevant range. In that common regime, demanding asymptotically sharp evidence forces an asymptotic collapse of achievable weakness, making the incompatibility quantitative rather than merely qualitative. This is the sense in which the theorem packages the tradeoff theorem intuition into an architectural constraint: the frontier cannot be pushed outward in both coordinates without violating at least one route constraint.*

53.2.5 Equalized-marginals beyond two routes

Theorem 91 (Multi-route equalized marginals (a k-way control law))**.** *Suppose a system has $k \geq 2$ disjoint "knobs" (fragments) that can receive resource budget, where knob i produces an attainable tradeoff curve $(K_i(u_i), I_i(u_i))$ as a function of its allocated budget u_i, and overall (K, I) is obtained by composing the knobs in the product quantale order (so that increasing any u_i weakly increases cost and weakly increases evidence). Assume a single total-budget constraint $\sum_{i=1}^{k} u_i = U$.*

At any Pareto-efficient operating point with all k knobs active, there exist positive multipliers μ, λ such that for every i and j,

$$\mu \cdot \frac{\partial K}{\partial u_i} = \lambda \cdot \frac{\partial I}{\partial u_i}, \qquad \text{and hence} \qquad \frac{\partial I/\partial u_i}{\partial K/\partial u_i} = \frac{\partial I/\partial u_j}{\partial K/\partial u_j}.$$

Equivalently: marginal evidence per unit marginal effective cost is equalized across all active knobs.

Remark 3024 (Regularity conditions and relation to KKT)**.** *The derivative notation implicitly assumes a local smoothness/regularity regime: $u \mapsto K(u)$ and $u \mapsto I(u)$ should be differentiable (or at least admit appropriate subgradients), and "all k knobs active" means the solution is interior with respect to any nonnegativity constraints $u_i \geq 0$ (if such constraints are present). Under these conditions, the statement is a direct KKT/Lagrange-multiplier condition for Pareto efficiency: one may view Pareto points as solutions of maximizing $I(u)$ subject to fixed $K(u)$ and $\sum_i u_i = U$ (or dually minimizing $K(u)$ at fixed $I(u)$ and budget). The positivity of multipliers corresponds to the economically meaningful regime in which increased budget can trade cost for evidence rather than reversing direction. If some knob is inactive (e.g., $u_i = 0$ at optimum), the equalization becomes an inequality: its marginal evidence-per-cost cannot exceed the common active value, otherwise it would become profitable to turn that knob on.*

Remark 3025 (Interpretation as a design principle)**.** *This theorem is the familiar equal-marginal-returns principle, stated in the language of multi-route architecture choice. It says: if you are truly* *on the Pareto frontier, you cannot improve evidence more cheaply by reallocating an infinitesimal budget from one knob to another. So the "exchange rate" between cost and evidence must be the same everywhere you are actively investing.*

Its relevance to independence is direct: in practice, the knobs include multiple independence-promoting moves (Route A transforms) and multiple dependence-admitting moves (Route B couplings). Thus equalized marginals becomes a principled control law for how much independence to enforce, and where to place explicit couplings, when resources are limited (Hyperseed-Concept 87, Hyperseed-Concept 100).

Remark 3026 (How this extends the two-route picture)**.** *In the two-route setting, one informally compares which route yields the better weakness ceiling at a given ε and chooses accordingly. The multi-knob formulation makes that comparison local and compositional: instead of a discrete choice between two routes, one can continuously allocate resources among many route-fragments (some implementing independence constraints, others implementing controlled couplings). Equalized marginals then states that, at an efficient configuration, each active fragment has been tuned until its next unit of resource buys exactly as much marginal evidence-per-cost as every other active fragment. This provides an explicit mechanism-level notion of "balancing independence against dependence": not as a binary decision, but as a distribution of budget across constraint-enforcing and coupling-enabling components.*

Proof. This is the classical KKT condition for maximizing evidence $I(u_1, \ldots, u_k)$ subject to a cost/budget constraint, but it can be derived in the same "pairwise reallocation" spirit as the two-route case: consider feasible perturbations that move an infinitesimal δ budget from knob j to knob i while preserving $\sum u_i$. Pareto efficiency implies no such perturbation can increase evidence without increasing cost in the ordered sense. In smooth numeric regions, this yields a supporting hyperplane with multipliers (μ, λ), and stationarity forces $\mu\, \partial K/\partial u_i = \lambda\, \partial I/\partial u_i$ for each active coordinate i. Dividing gives the equalized ratio across i. □

Proof sketch. The argument is the standard variational one: at an optimal (Pareto-efficient) allocation under a single linear budget constraint, the gradient of the objective must be proportional to the gradient of the constraint. Here, that means ∇I is proportional to ∇K on active coordinates, yielding identical ratios $(\partial I/\partial u_i)/(\partial K/\partial u_i)$ across i. A helpful picture is the tangency of a level set of evidence to a level set of cost: if the tangency failed, one could slide along the cost surface to improve evidence. □

Remark 3027 (Why this matters for independence)**.** *In practice, "knobs" include multiple Route-A transforms (whitening, abstraction, pruning, masking, coarse-graining) and multiple Route-B couplings (attention heads, constraint joins, factor-graph edges, cross-module messaging). Theorem 91 says a system that is near the frontier should allocate effort so that no knob has strictly better marginal return than another.*

53.2.6 From residual dependence to "ignore only weak dependencies"

Definition 484 (Residual dependence with coverage weights)**.** *Let interaction sets S have strengths $\|\kappa_S\|_1 \geq 0$ and coverage factors $\gamma_S \in [0, 1]$. Define*

$$\mathit{ResDep} := \sum_{|S| \geq 2} (1 - \gamma_S)\, \|\kappa_S\|_1.$$

Remark 3028 (Intuition and role in practical reasoning)**.** *This definition abstracts the common situation where a system covers some interactions explicitly and others only partially (or not at* *all). Here γ_S measures how much of interaction S is "accounted for" by the model (so $1-\gamma_S$ is the miss fraction), and $\|\kappa_S\|_1$ measures how strong that interaction is in the underlying domain. The sum is then the total mass of uncovered dependence.*

It is useful because it connects a high-dimensional combinatorics of dependencies into a single scalar certificate. In the Route A vs Route B story, ResDep can be interpreted as the remaining dependence after some combination of Route A transforms and Route B couplings has been applied. It is thus a numerically manageable proxy for Hyperseed-Concept 93 and for the costs of neglecting it.

Theorem 92 (Thresholding lemma: small residual dependence implies only a small mass of strong misses)**.** *Fix $\alpha \in (0, 1]$. Let*

$$\mathcal{M}_\alpha := \{S : |S| \geq 2 \text{ and } (1 - \gamma_S) \geq \alpha\}.$$

Then

$$\sum_{S \in \mathcal{M}_\alpha} \|\kappa_S\|_1 \leq \frac{ResDep}{\alpha}.$$

More generally, for any $\tau > 0$,

$$\#\{S : |S| \geq 2,\ (1 - \gamma_S) \geq \alpha,\ \|\kappa_S\|_1 \geq \tau\} \leq \frac{ResDep}{\alpha\,\tau}.$$

Remark 3029 (What the theorem guarantees)**.** *If ResDep is small, then the set of interactions that are both (i) substantially missed and (ii) collectively strong must itself be small. The first inequality bounds the* total strength *of substantially missed interactions; the second bounds the* count *of missed interactions above a strength threshold. So one can justify a working maxim: "assume independence except for a small exceptional set".*

This is precisely the sort of statement an inference engine wants: it translates a global budget into a sparse set of "must-handle" dependencies, which is a practical guide for when to activate Route B couplings and when to remain in an independence approximation (Route A) [9].

Proof. If $(1 - \gamma_S) \geq \alpha$ then each term contributes at least $(1 - \gamma_S)\|\kappa_S\|_1 \geq \alpha\|\kappa_S\|_1$ to $ResDep$. Summing over $S \in \mathcal{M}_\alpha$ gives $ResDep \geq \sum_{S \in \mathcal{M}_\alpha} \alpha\|\kappa_S\|_1$, hence $\sum_{S \in \mathcal{M}_\alpha} \|\kappa_S\|_1 \leq ResDep/\alpha$.

For the counting bound, each S in the set has contribution at least $\alpha\tau$, so $ResDep \geq (\alpha\tau) \cdot \#(\cdots)$, proving the inequality. ☐

Proof sketch. The proof is a bookkeeping argument: each interaction in $\mathcal{M}_\alpha$ must contribute at least an α fraction of its strength to $ResDep$, so the total strength of such interactions cannot exceed $ResDep/\alpha$. For the counting bound, if each such interaction has strength at least τ, then each contributes at least $\alpha\tau$, so there cannot be more than $ResDep/(\alpha\tau)$ of them. Visually, $ResDep$ is a fixed "mass budget", and the theorem says you cannot pack too many heavy, badly-covered interactions into a small mass. ☐

Remark 3030 (Decision tool)**.** *Theorem 92 turns a scalar residual-dependence certificate into a sparse exception-set guarantee: if ResDep is small, then* either *you covered almost all large interactions,* or *the ones you missed have small total strength and can be ignored. This is exactly the operational content of "assume independence except for a small set of strongly dependent interactions".*

53.2.7 Double-counting exposes pruneable couplings

Definition 485 (Double-counting penalty)**.** *Let $m_S \in \mathbb{N}$ be the number of factors containing interaction set S and let $\gamma_S \in [0,1]$ be effective coverage. Define*

$$DC := \sum_{|S|\geq 2} (m_S - \gamma_S)_+ \, \|\kappa_S\|_1.$$

In particular, $(x)_+ := \max\{x, 0\}$ ensures that only excess *modeling beyond effective coverage contributes to the penalty, and $\|\kappa_S\|_1$ serves as an interaction-specific weight capturing the magnitude/importance of the true (or target) dependence at order S.*

Remark 3031 (Intuition and why this definition is necessary)**.** *The quantity DC penalizes paying for the same dependence more than once. If m_S counts how many architectural components attempt to model interaction S, and γ_S measures the effective fraction of S actually covered, then $(m_S - \gamma_S)_+$ is the "redundant excess" modeling devoted to S. Multiplying by $\|\kappa_S\|_1$ weights redundancy by importance.*

This is a formal counterpart of a familiar engineering phenomenon: adding extra couplings can feel like it should help, but in a well-posed tradeoff model, redundant couplings often only add cost and overfitting risk without increasing true coverage. Thus DC supports a principled pruning criterion, mirroring Occam-style pressure encoded via weakness/regularization (Hyperseed-Concept 202) [3].

It is also important that $\gamma_S \in [0,1]$ be interpreted as effective *(not nominal) coverage: two different factors might each claim to address S, but if they are highly correlated in what they capture, the combined effective gain can be much less than the raw count m_S. The positive-part form then implements the modeling principle "you only get paid once" for genuinely covering an interaction set S; additional attempts only avoid penalty to the extent that they increase γ_S.*

Finally, weighting by $\|\kappa_S\|_1$ expresses that redundancy is not uniformly bad: duplicating effort on a near-zero interaction set is nearly harmless, while duplicating effort on a strong interaction is more costly because it invites larger downstream distortion (e.g., conflicting parameterizations, wasted capacity, and brittle coordination among factors).

Remark 3032 (Basic properties of DC (for later use))**.** *The penalty DC is always nonnegative, and it is monotone nondecreasing in each m_S holding γ_S fixed. Conversely, for fixed m_S, increasing γ_S can only decrease (or leave unchanged) the associated term $(m_S - \gamma_S)_+ \, \|\kappa_S\|_1$. In particular, if $\gamma_S = 1$ (full effective coverage) then any additional modeling multiplicity $m_S > 1$ is* pure *double-counting in the sense that it contributes $(m_S - 1)\|\kappa_S\|_1$ to DC without any available "coverage credit" to offset it.*

Theorem 93 (Redundant coupling elimination)**.** *Suppose an architecture A contains a factor F such that removing F strictly decreases m_S for at least one interaction set S but does not decrease any effective coverage γ_T. Let A' be the architecture with F removed. Then:*

1. *$ResDep(A') = ResDep(A)$,*
2. *$DC(A') < DC(A)$,*
3. *for any information-efficiency objective of the form $I(A) = \sum_S \gamma_S G_S - \lambda_{dc} DC(A) - \lambda_{res} ResDep(A)$ with $\lambda_{dc} > 0$, we have $I(A') > I(A)$.*

Hence A cannot be Pareto-efficient in (cost, evidence) order if F has positive cost.

Remark 3033 (Plain-English reading)**.** *If a coupling does not increase what you cover (the γ_T do not improve), then it is pure redundancy. Removing it keeps residual dependence unchanged (you are not missing anything new), strictly reduces double-counting, and therefore strictly improves any objective that penalizes redundancy. If the coupling also costs resources, you improve cost as well, so the original architecture cannot be on the Pareto frontier.*

This theorem supplies a clean decision rule that sits naturally alongside the Route A vs Route B dichotomy [9]: Route B couplings are not sacred; they must earn their keep by increasing coverage, otherwise they are dominated and should be pruned.

One useful way to read the conclusion is that the pair $(\{\gamma_S\}, \{m_S\})$ contains all *the information relevant to whether F can be justified under objectives of the displayed form: if removing F leaves γ unchanged, then F is only moving m in a direction that increases redundancy. In this sense the theorem isolates a minimal "certificate of non-utility" for a coupling: it is enough to show that it fails to increase effective coverage for* any *interaction set, not necessarily that it harms performance.*

In the language of dependence modeling, the hypothesis "does not decrease any effective coverage γ_T" encodes that F is not providing a unique pathway to any interaction set; equivalently, every dependence contribution that F was intended to capture is already captured (up to the model's tolerance) by other factors. The conclusion then formalizes the modeling heuristic that non-identifiable or linearly dependent couplings should be removed even if they are individually expressive, because expressiveness that does not translate into new effective coverage is exactly what produces double-counting.

Proof. By assumption, all γ_T are unchanged, so $ResDep$ (a function of the γ_T) is unchanged. Also, $(m_S - \gamma_S)_+$ strictly decreases for at least one S (because m_S decreases and γ_S is fixed) and does not increase for any S, hence DC strictly decreases. With $\lambda_{dc} > 0$ and $ResDep$ unchanged, the objective $I(\cdot)$ strictly increases. If F has positive cost, then A' also weakly improves cost. Thus A is dominated.

To see the strictness of the DC decrease explicitly: there exists at least one set S_0 such that $m_{S_0}(A') < m_{S_0}(A)$ while $\gamma_{S_0}(A') = \gamma_{S_0}(A)$. For this S_0, the map $m \mapsto (m - \gamma_{S_0})_+$ is strictly increasing on $\mathbb{N}$ whenever $m > \gamma_{S_0}$, and it is nonincreasing only when both values lie at or below γ_{S_0}. But the hypothesis that m_{S_0} strictly decreases means the associated summand cannot increase, and because it decreases for at least one S_0 with $\|\kappa_{S_0}\|_1 > 0$ (the only case in which redundancy matters), we obtain $DC(A') < DC(A)$. ☐

Proof sketch. The proof follows the dependencies of the objective on the summaries (γ_S, m_S): if γ_S is unchanged, then $ResDep$ is unchanged. If some m_S decreases while no γ_S decreases, then the positive-part redundancy $(m_S - \gamma_S)_+$ cannot increase and must decrease somewhere, so DC strictly drops. Therefore any objective that subtracts $\lambda_{dc} DC$ strictly improves. The geometric intuition is that redundant couplings move you "inward" from the Pareto frontier: they increase cost without moving the evidence coordinate outward. ☐

Remark 3034. *Theorem 93 is a formal "anti-overfitting" rule: if a coupling does not buy you additional coverage (evidence) it must be removed at any frontier point. This is the Route-B analogue of Route-A simplification/pruning.*

Equivalently, for Pareto-efficient architectures under objectives that penalize DC, every retained factor must be witnessed *by at least one interaction set whose effective coverage would strictly drop if the factor were removed. This witness condition is a convenient design-time diagnostic: factors that have no witness are precisely those that can be eliminated without sacrificing evidence, and thus correspond to "spurious dependence structure" rather than authentic interaction modeling.*

Remark 3035 (Connection to the tradeoff theorem viewpoint). *Within the dependence dichotomy, Route A aims to make γ_S large by pushing dependence into a representation change, whereas Route B aims to make γ_S large by adding explicit coupling structure. Theorem 93 clarifies that Route B must still obey a tradeoff-theorem-style accounting: if adding structure increases m_S without increasing γ_S, then it worsens the balance of cost versus evidence in a way that cannot be recovered elsewhere in the objective. In this sense, the* independence pivot *shows up as a bookkeeping invariance: once effective coverage is fixed, additional dependence machinery has no principled justification and is guaranteed to be dominated.*

53.2.8 A quantitative route-choice threshold (when should a system enforce independence vs model dependence?)

Theorem 94 (Critical-order threshold from arity-growth vs transform costs). *Consider a size-n system with interaction order r and per-arity coupling costs producing a lower bound*

$$K_B(r) := \Omega\left(\sum_{j=2}^{r} \binom{n}{j}\right)$$

for Route B (explicitly realizing interactions up to order r). Assume Route A (decorrelation/independence transform) yields a lower bound

$$K_A := \Omega(nd) + K_{transform}(\varepsilon)$$

for some representation dimension d and tolerance ε.

Then any algorithm achieving the evidence target must incur cost at least

$$K_{\min} \geq \min\{K_A,\, K_B(r)\}.$$

Moreover, if $K_{transform}(\varepsilon)$ and d are fixed while r grows, there exists a (problem-dependent) critical order $r^\star$ such that:

$$K_B(r) < K_A \text{ for } r < r^\star, \qquad K_B(r) > K_A \text{ for } r > r^\star.$$

Remark 3036 (Meaning: when dependence becomes too expensive). *This theorem distills an architectural threshold phenomenon: representing dependence explicitly becomes combinatorially expensive as interaction order grows. At low order r, explicit couplings (Route B) can be economical; at high order, the number of possible interactions explodes (the binomial sum), and it becomes cheaper to pursue independence-enforcing representations (Route A) that reduce the effective dependence order.*

The result is not a fine-grained performance bound but a structural compass: it tells you which side of the dichotomy is likely to dominate as the domain's required interaction order increases. This connects naturally to Dependency-Tower reasoning (Hyperseed-Concept 94) and to the general tradeoff theorem heuristics developed in [8, 9].

A useful quantitative reading of the Route B lower bound is that, for moderate $r \ll n$, the leading term of $\sum_{j=2}^{r} \binom{n}{j}$ scales on the order of $n^r/r!$ (up to polynomial factors in r), so even when each individual coupling is "cheap," the sheer count of distinct interaction subsets makes high-order dependence modeling expensive. By contrast, Route A aggregates the burden into (i) a linear-in-n representational cost $\Omega(nd)$ and (ii) a transform-dependent overhead $K_{transform}(\varepsilon)$ capturing how hard it is to achieve approximate independence to tolerance ε.

The phrase "problem-dependent" for $r^\star$ is essential: $K_{transform}(\varepsilon)$ may itself be large in domains where decorrelation requires iterative optimization, global coordination, or expensive normalizations, and d may need to scale with n to preserve sufficient information. In such regimes, K_A can grow, moving the threshold $r^\star$ upward and making Route B favorable over a wider range. Conversely, when a domain admits a fast independence transform (small $K_{transform}$) and a compact representation (small d), the threshold $r^\star$ can be low, making Route A preferable already at modest interaction orders.

Interpreted through the independence pivot lens, the theorem explains why *enforcing independence can be the correct move even in a highly dependent world: once the required interaction order crosses the critical point, the cheapest way to meet an evidence target is often to restructure the system so that the remaining modeled interactions are effectively lower-order, thereby avoiding the combinatorial explosion inherent in explicit high-arity coupling families.*

Proof. The first inequality is immediate from the dichotomy: any feasible method must satisfy at least one route-wise lower bound, hence must pay at least their minimum. More explicitly, if Route A yields a necessary cost K_A for any method that proceeds by reducing effective dependence (e.g. via factorization, representation changes, or independence-promoting transforms), and Route B yields a necessary cost $K_B(r)$ for any method that directly models dependence of order r, then every admissible method falls into (at least) one of these regimes; consequently its cost is bounded below by K_A or by $K_B(r)$, and therefore by $\min\{K_A, K_B(r)\}$.

For the threshold statement: $K_B(r)$ is monotone increasing in r and grows combinatorially in r for fixed n, while K_A is independent of r in this simplified comparison. Therefore the inequality $K_B(r) \le K_A$ holds for r in an initial interval and fails beyond some largest $r^\star$ (or never holds / always holds in degenerate cases). To make the monotonicity implication explicit: since $K_B(r)$ is nondecreasing, the set

$$S := \{r \in \{0, 1, 2, \dots\} : K_B(r) \le K_A\}$$

is a (possibly empty) initial segment of the nonnegative integers; hence either $S = \varnothing$, or S is all admissible r, or there exists a maximal element $r^\star := \max S$ which is the claimed threshold. The "combinatorial growth" condition ensures that, in typical nondegenerate settings, $K_B(r)$ eventually exceeds any fixed K_A for sufficiently large r (e.g. if $K_B(r)$ scales like $\binom{n}{r}$, n^r, or related families), which guarantees finiteness of $r^\star$ unless K_A is itself large enough to dominate throughout. $\square$

Proof sketch. The proof is an order-of-growth comparison. The dichotomy implies you must pay at least one of the route-wise costs, hence at least their minimum. A useful way to view this is to regard K_A and $K_B(r)$ as two lower envelopes induced by two incompatible design commitments: Route A commits resources to suppressing or "explaining away" dependence so that downstream reasoning can be done with (approximate) independence, whereas Route B commits resources to representing and exploiting dependence explicitly up to order r.

Then, as r increases, the Route B lower bound increases monotonically and rapidly (combinatorial growth), while the Route A bound is held fixed in r; so a crossing point (critical order) must occur unless one curve dominates everywhere. One can picture two curves in r: a flat (or gently rising) Route A curve and a steep Route B curve; the intersection defines $r^\star$.In more concrete terms, if $K_B(r)$ behaves like the number of r-wise interaction terms that must be tracked (e.g. proportional to $\binom{n}{r}$ times a per-interaction cost), then even modest increases in r can cause a sharp escalation in $K_B(r)$, whereas the Route A cost is governed primarily by the difficulty of finding an independence-inducing representation and does not scale directly with the requested interaction order in this stylized comparison. $\square$

Remark 3037 (Decision tool)**.** *Theorem 94 is a direct "architectural switch" criterion: if the dependence order required by the task is low, it is cheaper to model it (Route B); if it is high, it is cheaper to reduce effective dependence via representation/independence transforms (Route A). Equivalently, $r^\star$ marks the point at which it becomes rational (in a cost-minimizing sense) to spend effort* upstream *on making the problem closer to independent rather than spending it* downstream *on enumerating ever higher-order interactions. In practice, the criterion can be applied qualitatively even when the exact constants are unknown: one estimates whether the task's necessary interaction order lies in a regime where $K_B(r)$ is still modest (so explicit dependence modeling is viable) or in a regime where it already exhibits the rapid growth that typically makes dependence-first designs brittle.*

53.3 Summary

Independence is the common latent variable connecting Hyperseed-ish tradeoffs and the tradeoff theorem:

- Glue overhead bounds attainable intensity (Lemma 161).
- Synergy/emergence is a witness that independence-only explanations are impossible (Thm. 88).
- Small residual dependence globally certifies approximate independence across any partition (Thm. 89).
- Weakness and strong evidence targets conflict unless one pays a route-specific cost (Thm. 90).
- Frontier behavior demands equalized marginal returns across all independence/coupling knobs (Thm. 91).
- Double-counting reveals pruneable couplings (Thm. 93).
- A critical interaction-order threshold determines when Route A vs Route B dominates (Thm. 94).

Read as a single thread, these items say that "independence" is not merely a modeling convenience but a quantity that can be budgeted, certified, and traded against task demands: it can be enforced (at a cost), violated (and detected by synergy), and approximately guaranteed (via global dependence control). In particular, the route-wise lower bounds provide a way to translate qualitative design choices (factorize vs. couple) into quantitative commitments, and the threshold theorem turns that translation into a concrete switching rule once a task-implied interaction order is estimated.

Remark 3038. *Taken together, these results aim to turn a conceptual duality into operational guidance. Route A and Route B are not rival ideologies but complementary instruments; what changes from problem to problem is the price of independence and the price of dependence. The theorems above give certificates and thresholds that let a reasoning system decide when to factorize, when to couple, when to prune, and how to allocate marginal effort. In this sense, independence functions as a regulative ideal in the Russellian sense—a constraint that disciplines inference—yet it is also, in a more Peircean mode, a fallible hypothesis whose failure is detected by emergent synergy (Hyperseed-Concept 190, Hyperseed-Concept* **??***) [14]. One can also view the collection as a calibration of where "independence assumptions" should live in the stack: Route A makes independence an* achievement *of representation (thereby paying for it explicitly), while Route B makes dependence an* object *of representation (thereby paying for it combinatorially as order increases).*

The pivot is that both perspectives meet at the same diagnostic quantities—residual dependence, synergy, and marginal tradeoff conditions—so that the choice between them can be made on the basis of observable certificates rather than on stylistic preference.

54 Habit Webs + Morphic Resonance: Fixed Points, Polarization, and Diffusion Thresholds (with Random-Graph Corollaries)

We treat a habit web as a quantale-valued reinforcement relation (kernel) on a pattern class, with state evolution given by an order-theoretic update operator. Morphic resonance couples multiple contexts by adding cross-context kernels. We prove: (i) existence and computation of least fixed points for time-homogeneous habit dynamics (single- and multi-context), (ii) a block-kernel reduction that turns morphic resonance into ordinary closure on a single enlarged web (hence diffusion becomes reachability/closure), (iii) a nucleation theorem: sufficient conditions under which resonance seeds a self-weaving (autocatalytic) core that persists even after coupling ceases, (iv) a synchronization/no-polarization criterion under global contractivity, and (v) random-graph thresholds for (minimal) autocatalytic cores and for macroscopic diffusion of habit support.

Concretely, the ambient value structure is a quantale $(V, \oplus, \otimes, \leq)$: $\oplus$ is an (often complete) join encoding accumulation of reinforcing evidence, while $\otimes$ propagates (and typically attenuates or transforms) support along a directed interaction. A "kernel" is thus a V-weighted relation on patterns, and a state $s : P \to V$ is a valuation assigning to each pattern its current degree of activation, credibility, or habit-strength. The update operator is monotone in the order on V^P and is designed so that one update-step corresponds to one round of reinforcement propagation through the web. Time-homogeneity means that the kernel (and any parameters of the update rule) remain fixed across steps, allowing the asymptotic behavior to be captured by fixed points and their least solutions.

Items (i)–(v) can be read as a progression from purely order-theoretic guarantees to structural reductions and then to probabilistic corollaries. The fixed-point results formalize "settled habits" as post-update states s satisfying $s = F(s)$, with least fixed points representing minimal stable support compatible with the reinforcement structure. The block-kernel reduction provides a way to treat a family of contexts (each with its own pattern set and internal kernel) as a single enlarged web, so that questions about cross-context diffusion reduce to closure/reachability computations in an ordinary single kernel. The nucleation theorem isolates conditions under which a transient coupling can create a persistent core: after the coupling is removed, internal reinforcement within the core is sufficient to maintain nontrivial support. The contractivity criterion addresses when repeated updates erase initial differences (synchronization) rather than amplifying them into incompatible attractors (polarization). Finally, the random-graph thresholds connect these qualitative phenomena to typical-case behavior in sparse regimes, identifying critical densities at which autocatalytic cores and large-scale diffusion emerge with high probability.

Remark 3039. *Philosophically, the guiding idea is that "habit" is not here a psychological metaphor but a mathematical dynamical object: a state $s : P \to V$ assigning graded support to patterns, and a kernel A that propagates and reinforces that support through a web of interactions. The quantale structure supplies the two basic moves: (i)* aggregation *of evidence/support (via $\oplus$ as a join), and (ii)* combination *of supports along a causal/reinforcing link (via $\otimes$). This is aligned with the general quantale weakness viewpoint developed in [3], and with Hyperseed's pattern-centric ontology [1, 5] (Hyperseed-Concept 130, 132, 131). In particular, treating s as a map into an ordered* *support domain makes comparison and convergence statements precise: one can ask whether a later state is* at least as supportive *as an earlier one, whether an update operator is inflationary or deflationary, and whether iterative closure stabilizes in finitely or transfinitely many steps. This formalization also clarifies how "weak" signals can accumulate through repeated joins while still requiring compatible pathways (via $\otimes$) to travel across the web.*

Remark 3040. *The phrase "morphic resonance" is used in a deliberately formalized sense: cross-context kernels $K_{\ell' \to \ell}$ are treated as additional reinforcement channels, mathematically no different from internal reinforcement except for their block structure. This abstraction echoes Sheldrake's original proposal [13] but is deployed here as a rigorous order-theoretic dynamical system (Hyperseed-Concept 115, 159, 86). Operationally, the block structure is what preserves interpretability: each context ℓ retains its own internal dynamics, while cross-context influence is mediated by explicitly typed maps $\ell' \to \ell$ rather than by an undifferentiated global kernel. The reduction in (ii) shows that, despite this bookkeeping, the coupled system can be analyzed using the same closure and fixed-point tools as a single-context web by passing to a suitable direct sum (or coproduct-like) pattern space. In this sense, "resonance" is not an extra dynamical principle but a compositional way to assemble kernels so that inter-context diffusion and persistence questions become statements about the existence of closed, self-supporting substructures in the enlarged web.*

54.1 Background

54.1.1 Quantale-valued habit webs

Fix a commutative quantale $(V, \leq, \oplus, \otimes, e)$. Let P be a (finite) pattern class. A pattern-support state is a function $s : P \to V$. A reinforcement relation (kernel) is a map $A : P \times P \to V$.

Remark 3041. *A few notational conventions are doing substantial work here. The set P is the universe of "patterns" under discussion; one can picture P as the vertex set of a directed weighted graph. The quantale value V is the space of support/evidence/activation values, ordered by $\leq$; the operation $\oplus$ is the join (think: "take the strongest support available"), and $\otimes$ combines supports along a link (think: "support transmits through an edge, attenuated or transformed by the edge weight"). The unit $e \in V$ is the neutral element for $\otimes$ (so $e \otimes v = v$).*

Remark 3042. *Concrete examples to keep in mind:*

- *Boolean case: $V = \{0, 1\}$, $\oplus = \vee$, $\otimes = \wedge$, $e = 1$. Then $s(p) = 1$ means "pattern p is present/active", and $(A?s)(q) = 1$ iff there exists some p with $A(p, q) = 1$ and $s(p) = 1$.*
- *Max-times case: $V = [0, 1]$, $\oplus = \max$, $\otimes = \cdot$, $e = 1$. Then $A(p, q)$ is a strength of reinforcement from p to q, and $(A?s)(q)$ is the strongest incoming reinforced support for q.*

These examples recur below when we discuss thresholds, diffusion, and random graphs.

Definition 486 (Relational image)**.** *Given $A : P \times P \to V$ and $s : P \to V$, define the image (one-step propagation)*

$$(A?s)(q) := \bigoplus_{p \in P} A(p, q) \otimes s(p).$$

Remark 3043. *Intuitively, $(A?s)(q)$ is the amount of support that flows into pattern q in one step, assuming each pattern p currently holds support $s(p)$ and transmits it along the link $p \to q$ with strength $A(p, q)$. The symbol "?" is meant to evoke a relational action: A acts on s much like a matrix acts on a vector, except with $\oplus$ and $\otimes$ playing the roles of addition and multiplication.*

Remark 3044. *A simple example: in the Boolean case, $(A?s)(q) = 1$ iff q has at least one active predecessor p with an edge into q. In the max-times case, $(A?s)(q) = \max_p A(p,q)\,s(p)$, i.e. the best incoming reinforcement signal. This operator is useful because it isolates the purely structural part of the dynamics (the kernel A) from the contingent part (the present state s), letting us study fixed points, closure, and diffusion in a uniform algebraic language.*

Definition 487 (Habit update operator)**.** *Fix decay/persistence $d \in V$ and an external input $u : P \to V$. Define*

$$\mathrm{Upd}_{A,d,u}(s) := u \oplus (d \odot s) \oplus (A?s),$$

where $(d \odot s)(p) := d \otimes s(p)$ and $\oplus$ on functions is pointwise join.

Remark 3045. *The update $\mathrm{Upd}_{A,d,u}$ separates three influences on the next state. The term u is exogenous stimulation (new evidence, new rewards, new experiences). The term $(d \odot s)$ is persistence/decay: it scales the old support s by a factor d using $\otimes$ (so in $[0,1]$ with $\otimes = \cdot$, this is literal multiplication by d). Finally $(A?s)$ is reinforcement by the web itself. The pointwise join $\oplus$ expresses the principle "keep whichever source provides stronger support" (or more generally, take the least upper bound in the lattice ordering).*

Remark 3046. *Two small examples illustrate why this definition is the right granularity for later theorems.*

- *If $u = \bot$ and $d = e$, then $\mathrm{Upd}(s) = s \oplus (A?s)$, i.e. pure inflationary closure under the web: once support appears, it can only accumulate via propagation.*
- *If $A = \bot$ and u is constant, then $\mathrm{Upd}(s) = u \oplus (d \odot s)$ describes a decaying memory forced by a baseline signal u.*

Formally, these cases are precisely those in which least fixed points can be read as closures or steady states, which connects directly to the Kleene iteration theorem below and to fixed-point stability ideas familiar from self-modifying goal dynamics [10].

54.1.2 Self-weaving (autocatalytic) cores

Definition 488 (θ-autocatalytic subset)**.** *Fix a threshold $\theta \in V$. A nonempty $S \subseteq P$ is called θ-autocatalytic if there exists a state $s : P \to V$ such that: (i) for all $p \in S$, $s(p) \geq \theta$; and (ii) for all $p \in S$, $(A?s)(p) \geq \theta$. A web (P, A) is self-weaving at threshold θ if it contains a θ-autocatalytic subset.*

Remark 3047. *This definition formalizes the notion of a "self-sustaining loop" (Hyperseed-Concept 168, 61). The threshold θ expresses a minimum viable level of activation/support: patterns below θ are treated as inert for the purposes of the autocatalytic criterion. Condition (i) says the subset S is jointly active at level θ under some state s; condition (ii) says that the web reinforcement, applied to that same state, regenerates support for every element of S at level θ. In other words, S is not merely co-present; it is collectively maintained by the kernel A.*

Remark 3048. *In the Boolean case $V = \{0,1\}$ and $\theta = 1$, a θ-autocatalytic subset is one where every node in S has at least one incoming edge from within S (since $(A?s)(p) = 1$ demands some predecessor in S). Thus a directed cycle gives a minimal example. In max-times $V = [0,1]$ with $\theta \in (0,1]$, one can think of each node in S receiving an incoming edge of strength at least θ from a node in S that itself is at least θ. The usefulness of the definition is that it pinpoints precisely what must be "seeded" from outside in order for the web to take over and sustain itself: once S is above θ, the internal dynamics can keep it there.*

In slightly more detail, the Boolean condition can be rewritten as $S \subseteq \mathrm{Pre}(S)$, where $\mathrm{Pre}(S)$ is the set of nodes with at least one incoming edge from S. This makes clear that θ-autocatalysis is an *internal support* condition: S does not require external parents to keep its members active once the threshold is met. In graded settings (such as max-times), the same idea appears as a *self-reinforcing lower bound*: if $s(p) \geq \theta$ for all $p \in S$, then the internal image $(A?s)(p)$ provides (via at least one sufficiently strong incoming edge from within S) enough reinforcement to prevent p from falling below θ in the absence of adverse decay or competing rules. Put differently, θ isolates the part of the state space that is forward-invariant under the internal propagation map once it has been reached, so that the remaining problem is to understand what exogenous inputs u (or cross-context inputs) are needed to drive the system into that region.

54.1.3 Morphic resonance as cross-context coupling

Let L index contexts. For each $\ell \in L$ fix: a pattern class P_ℓ, an internal kernel $A_\ell : P_\ell \times P_\ell \to V$, a decay scalar $d_\ell \in V$, and a (possibly time-dependent) input $u_{\ell,t} : P_\ell \to V$.

For $\ell' \neq \ell$, a morphic coupling kernel is a V-relation $K_{\ell' \to \ell} : P_{\ell'} \times P_\ell \to V$ with induced input

$$(K_{\ell' \to \ell}?s_{\ell',t})(q) := \bigoplus_{p \in P_{\ell'}} K_{\ell' \to \ell}(p,q) \otimes s_{\ell',t}(p).$$

The coupled update (morphic resonance dynamics) is:

$$s_{\ell,t+1} = u_{\ell,t} \oplus (d_\ell \odot s_{\ell,t}) \oplus (A_\ell ? s_{\ell,t}) \oplus \bigoplus_{\ell' \neq \ell} (K_{\ell' \to \ell}?s_{\ell',t}).$$

Equivalently, the quantity $\bigoplus_{\ell' \neq \ell}(K_{\ell' \to \ell}?s_{\ell',t})$ can be read as a context-dependent "import" term: it aggregates the images of all other context-states through their respective coupling kernels, using the same $\oplus$-aggregation rule as internal influence. This ensures that cross-context effects are not an extra mechanism but the same propagation calculus applied on a larger typed graph whose nodes are the disjoint union $\bigsqcup_{\ell \in L} P_\ell$, with edge-weights given by the block-kernels A_ℓ (within blocks) and $K_{\ell' \to \ell}$ (off-diagonal blocks).

Remark 3049. *The new ingredient relative to a single web is the family of cross-context kernels $K_{\ell' \to \ell}$. These kernels are typed: they map patterns from $P_{\ell'}$ into patterns of P_ℓ, and their action $(K_{\ell' \to \ell}?s_{\ell',t})$ is literally the same relational-image construction as for internal propagation, only across contexts. One can read this as a formal account of "resonance" in the minimal sense: activations in one context contribute to activations in another, with strengths specified by $K_{\ell' \to \ell}$ (Hyperseed-Concept 115, 159, 192).*

A useful intuition is that the kernels $K_{\ell' \to \ell}$ play the role of (possibly asymmetric) "translation" or "analogy" maps between pattern vocabularies: a pattern $p \in P_{\ell'}$ can support a pattern $q \in P_\ell$ to the extent that $K_{\ell' \to \ell}(p,q)$ is large in the V-order. Because the induced input uses $\bigoplus_p$, multiple source patterns in $P_{\ell'}$ can jointly provide support, and because it uses $\otimes$, a source pattern that is only weakly active cannot produce strong downstream activation unless the kernel-weight compensates for it (as appropriate to the chosen quantale). In this way, the same algebraic primitives simultaneously encode (i) how evidence combines within a context (via A_ℓ) and (ii) how evidence transfers between contexts (via $K_{\ell' \to \ell}$), so that "cross-context learning" is expressed as a property of kernels rather than as a separate learning rule.

Remark 3050. *It is worth emphasizing what is* not *assumed: no probabilistic semantics are required, and no symmetry is required (one may have $K_{\ell' \to \ell} \neq K_{\ell \to \ell'}$). The only essential structure* *is order-theoretic and algebraic, which is why fixed-point methods apply cleanly. This is one reason quantales are convenient: they provide a uniform calculus for propagation and aggregation while keeping enough generality to represent max-times, Boolean reachability, p-bit evidence pairs, and other "graded logic" settings [3].*

One consequence of this order-theoretic framing is that, under the standard monotonicity properties of $\oplus$, $\otimes$, and the relational image $(\cdot\ ?\ \cdot)$, each per-context update map is monotone in its arguments, and the full coupled system is monotone in the joint state $s_t = (s_{\ell,t})_{\ell \in L}$. This is the basic reason least and greatest fixed points become natural objects: for time-independent inputs $u_{\ell,t} \equiv u_\ell$ (and time-independent kernels), equilibria are precisely fixed points of the combined operator, and iterative dynamics provide constructive approximations under mild continuity assumptions. Moreover, the block-graph view makes it clear how "resonant loops" can occur: even if each A_ℓ alone is subcritical, the off-diagonal kernels can create directed cycles across contexts, yielding cross-context autocatalytic structures in the enlarged graph and thereby lowering the seeding needed for global activation.

54.2 Deterministic fixed-point theorems

54.2.1 Least fixed point for time-homogeneous habit dynamics

Assume P is finite and V is complete, so V^P is a complete lattice. Fix A, d, u and define $F : V^P \to V^P$ by $F(s) = \mathrm{Upd}_{A,d,u}(s)$.

Theorem 95 (Scott-continuity and Kleene computation of the least fixed point)**.** *Assume $(V, \leq, \oplus, \otimes, e)$ is a commutative quantale and P is finite. Then:*

1. *F is monotone and Scott-continuous (preserves directed joins).*
2. *The least fixed point $\mathrm{lfp}(F)$ exists and is given by the Kleene chain*

$$\mathrm{lfp}(F) = \bigvee_{n \geq 0} F^n(\bot),$$

   where $\bot$ is the bottom state (pointwise bottom element of V).

3. *For any fixed point s^* of F, one has $F^n(\bot) \leq s^*$ for all n, hence the above supremum is indeed the least fixed point.*

Remark 3051. *In plain terms: if one repeatedly applies the update rule starting from "no support anywhere" (the bottom state $\bot$), then the state monotonically accumulates support, and the limiting state is the smallest stable configuration compatible with the kernel A, the decay d, and the input u. The result is important because it converts an abstract existence claim (there is some fixed point) into a concrete computation rule (iterate F). This is the basic reason closure-based reasoning systems can treat fixed points as operational objects rather than metaphysical ones. It also sets up the later block-kernel reduction: once multi-context dynamics are compiled into a single operator, the same Kleene computation applies.*

Remark 3052. *The condition "Scott-continuous" is an order-theoretic analogue of continuity tailored to complete lattices: it means that F respects directed suprema, i.e. suprema of consistent increasing approximations. Conceptually, this is what legitimizes taking a limit of iterated approximations, as one does in Kleene's theorem and in many semantics of programs and logics.*

Proof. (1) Monotonicity follows because F is built from $\oplus$, $\otimes$, and joins, each monotone in a quantale. For Scott-continuity, let $D \subseteq V^P$ be directed. Pointwise:

$$F\Big(\bigvee D\Big) = u \oplus (d \odot \bigvee D) \oplus (A?(\bigvee D)).$$

Since $\otimes$ distributes over arbitrary joins, $d \odot (\bigvee D) = \bigvee_{s \in D}(d \odot s)$. Also, for each $q \in P$,

$$(A?(\bigvee D))(q) = \bigoplus_{p \in P} A(p,q) \otimes \Big(\bigvee_{s \in D} s(p)\Big) = \bigoplus_{p \in P} \bigvee_{s \in D} \big(A(p,q) \otimes s(p)\big).$$

Because P is finite and joins distribute, this equals $\bigvee_{s \in D}(A?s)(q)$. Thus $F(\bigvee D) = \bigvee_{s \in D} F(s)$.

(2) By completeness, the supremum $\bigvee_{n \geq 0} F^n(\perp)$ exists. Since $\perp \leq F(\perp)$ (always true because $\perp$ is least), the Kleene iterates form an ascending chain. Scott-continuity implies $F(\bigvee_n F^n(\perp)) = \bigvee_n F^{n+1}(\perp)$, so the supremum is a fixed point.

(3) If s^* is any fixed point, then $\perp \leq s^*$ and monotonicity implies $F^n(\perp) \leq F^n(s^*) = s^*$ for all n. Taking the join over n yields the claim. □

Remark 3053. Proof sketch (high level). *The strategy is: (i) show F respects the order in the obvious way (monotonicity), (ii) show F respects increasing "limits" (Scott-continuity), and then (iii) use the standard Kleene fixed-point construction: iterate from the least element and take the join of all iterates. Scott-continuity ensures that applying F after taking this join is the same as taking the join after applying F to each iterate, which is exactly what makes the supremum a fixed point.*

Remark 3054. *The key technical step is the interchange of $\bigoplus$ (join over $p \in P$) and $\bigvee$ (join over $s \in D$). Finiteness of P makes this interchange straightforward: one can push the directed join inside a finite join without losing equality. Visually, one can picture the kernel as a finite matrix; pushing suprema inside corresponds to swapping the order of "take the best predecessor" and "take the limit of better and better approximations to s".*

54.2.2 Block-kernel reduction: morphic resonance is closure on an enlarged web

Assume time-homogeneous inputs and no decay for clarity: $d_\ell = e$ and $u_{\ell,t} \equiv u_\ell$.

Let the global pattern set be the disjoint union

$$P_{\text{tot}} := \bigsqcup_{\ell \in L} P_\ell.$$

Define the global kernel $G : P_{\text{tot}} \times P_{\text{tot}} \to V$ by:

$$G(p,q) := \begin{cases} A_\ell(p,q) & \text{if } p, q \in P_\ell, \\ K_{\ell' \to \ell}(p,q) & \text{if } p \in P_{\ell'},\ q \in P_\ell,\ \ell' \neq \ell. \end{cases}$$

Let $u_{\text{tot}} : P_{\text{tot}} \to V$ be the disjoint union of the u_ℓ, and similarly s_t the disjoint union of $(s_{\ell,t})_{\ell \in L}$.

Remark 3055. *The disjoint union symbol $\bigsqcup$ is used to keep contexts from accidentally identifying patterns with the same "name": even if P_{ℓ_1} and P_{ℓ_2} share labels, in P_{tot} they become distinct tagged elements. This is the correct categorical move for "many worlds in parallel" before we add couplings between them (Hyperseed-Concept 112,* **??***).*

Remark 3056. *The block kernel G is literally a block matrix: diagonal blocks are the internal kernels A_ℓ, and off-diagonal blocks are the coupling kernels $K_{\ell'\to\ell}$. The virtue of writing G is that it turns the multi-context system into an ordinary single-kernel system, so all theorems about closure, fixed points, and reachability become immediately reusable.*

Theorem 96 (Morphic resonance as ordinary habit closure on (P_{tot}, G))**.** *Under the assumptions above (no decay, constant inputs), the coupled morphic-resonance update is equivalent to the single-web update*

$$s_{t+1} = u_{\mathrm{tot}} \oplus s_t \oplus (G?s_t).$$

Consequently, the least fixed point of the global dynamics is

$$s_\infty = G^*?u_{\mathrm{tot}},$$

where $G^* = \bigvee_{n\geq 0} G^n$ is the Kleene star (path-closure) of G.

Remark 3057. *What this theorem says, informally, is that "morphic resonance adds no new mathematics" once we allow ourselves to enlarge the state space: cross-context propagation is just propagation in a bigger graph. This matters because it turns questions about diffusion across contexts into ordinary closure computations. In a reasoning system, this is the difference between needing bespoke multi-agent/multi-context machinery versus compiling the problem down to one closure operator.*

Remark 3058. *The Kleene star $G^* = \bigvee_{n\geq 0} G^n$ should be read as "all paths of all lengths": G^n captures the best n-step reinforcement along n edges, and the join over all n aggregates paths of arbitrary length. Thus $G^*?u_{\mathrm{tot}}$ is the closure of the base input under all possible resonance-and-reinforcement cascades. This is a precise algebraic analogue of reachability in directed graphs, generalized to quantale weights.*

Proof. Fix $\ell \in L$ and $q \in P_\ell$. By definition of disjoint union and of G,

$$(G?s_t)(q) = \bigoplus_{p\in P_{\mathrm{tot}}} G(p,q) \otimes s_t(p) = \Big(\bigoplus_{p\in P_\ell} A_\ell(p,q) \otimes s_{\ell,t}(p)\Big) \oplus \Big(\bigoplus_{\ell'\neq\ell}\bigoplus_{p\in P_{\ell'}} K_{\ell'\to\ell}(p,q) \otimes s_{\ell',t}(p)\Big).$$

The first parenthesis is $(A_\ell?s_{\ell,t})(q)$ and the second is $\bigoplus_{\ell'\neq\ell}(K_{\ell'\to\ell}?s_{\ell',t})(q)$. Thus restricting $u_{\mathrm{tot}} \oplus s_t \oplus (G?s_t)$ to P_ℓ yields exactly $u_\ell \oplus s_{\ell,t} \oplus (A_\ell?s_{\ell,t}) \oplus \bigoplus_{\ell'\neq\ell}(K_{\ell'\to\ell}?s_{\ell',t})$.

For the fixed point formula, apply the closure/Kleene-star identity to the inflationary update $s \mapsto u_{\mathrm{tot}} \oplus s \oplus (G?s)$: iterating from bottom yields the least fixed point, and the standard unfolding gives $s_\infty = G^*?u_{\mathrm{tot}}$. □

Remark 3059. Proof sketch (high level). *The argument is bookkeeping: show that the global image operator $G?$ decomposes, on each component P_ℓ, into the internal image $A_\ell?$ plus the sum of coupling images $K_{\ell'\to\ell}?$. Once this is established, the multi-context update is exactly the single-context update on the disjoint union. The fixed point formula then follows from the same Kleene iteration principle as before, since the update is inflationary when $d_\ell = e$ and the s_t term is included.*

Remark 3060. *A useful geometric picture is to imagine stacking the context graphs into layers. The block kernel G simply adds inter-layer edges. Then morphic resonance is nothing but ordinary diffusion on the resulting layered directed graph. The lattice join $\oplus$ plays the role of combining multiple incoming paths, and $\otimes$ composes edge strengths along a path.*

54.2.3 Resonant nucleation: seeding a self-weaving core via coupling

Intuition: a context can have a latent autocatalytic core S that does not activate by itself, but once resonance pushes all patterns in S above threshold θ, the internal kernel keeps them alive (and possibly expands them via closure), so the habit persists even if coupling later vanishes.

Remark 3061. *This "one-shot seeding" phenomenon is one of the most operationally relevant consequences of self-weaving structure: it tells us that a transient cross-context influence can have a permanent effect, not by continuing to act, but by pushing the system across a threshold into a region where internal feedback loops do the maintenance. In the language of dynamical systems, S defines (at least) a basin condition; in the language of Hyperseed concepts, it is a seed for a self-weaving web (Hyperseed-Concept 168, **??**, **??**).*

Theorem 97 (One-shot seeding of an autocatalytic core)**.** *Fix a context ℓ with internal kernel A_ℓ and threshold $\theta \in V$. Assume $S \subseteq P_\ell$ is θ-autocatalytic. Consider the no-decay, no-input internal reinforcement dynamics on ℓ:*

$$x_{t+1} = x_t \oplus (A_\ell ? x_t).$$

If there exists a time t_0 such that $x_{t_0}(p) \geq \theta$ for all $p \in S$, then for all $t \geq t_0$ one has $x_t(p) \geq \theta$ for all $p \in S$. Moreover the trajectory converges (as an ascending join) to a fixed point x_∞ with $x_\infty(p) \geq \theta$ for all $p \in S$.

In particular, in the full morphic resonance dynamics, if for some source context ℓ' the coupling input satisfies

$$(K_{\ell' \to \ell} ? s_{\ell', t_0})(p) \geq \theta \quad \text{for all } p \in S,$$

then after one update step, context ℓ reaches a state that activates S at level θ, and internal reinforcement can sustain it thereafter.

Remark 3062. *The theorem asserts a form of hysteresis: once the core is "lit" above threshold, the subsequent internal dynamics cannot extinguish it, because the update is inflationary (support only grows via $\oplus$) and the autocatalytic witness guarantees reinforcement at the same threshold. This links the structural notion of autocatalysis directly to a temporal persistence guarantee, which is why autocatalytic sets are so central in many theories of self-organization (Hyperseed-Concept 61, **??**).*

Proof. Let $x_{t+1} = x_t \oplus (A_\ell ? x_t)$. This map is inflationary, so $x_{t+1} \geq x_t$ pointwise. (Here and throughout, inequalities are taken in the pointwise order on V^{P_ℓ}, i.e. $u \geq v$ means $u(p) \geq v(p)$ for every $p \in P_\ell$; inflationarity follows because a join satisfies $u \oplus v \geq u$ in any join-semilattice.)

Because S is θ-autocatalytic, there exists a witness state w such that $w(p) \geq \theta$ and $(A_\ell ? w)(p) \geq \theta$ for all $p \in S$. Assume $x_{t_0}(p) \geq \theta$ for $p \in S$. Define $y := x_{t_0} \oplus w$. Then $y \geq w$ and $y \geq x_{t_0}$, hence $y(p) \geq \theta$ on S. (Equivalently, for each $p \in S$ we have $y(p) = x_{t_0}(p) \oplus w(p) \geq w(p) \geq \theta$, so y simultaneously dominates the current state and the witness on the entire subset S.) By monotonicity of the image operator, $(A_\ell ? y) \geq (A_\ell ? w)$, so $(A_\ell ? y)(p) \geq \theta$ on S. (This is the step where θ-autocatalysis becomes "self-sustaining": once the state dominates a witness, the activation contributed by $A_\ell ?$ cannot be weaker than the witnesss contributed activation.)

Now consider the trajectory starting from y: $y_0 := y$, $y_{n+1} := y_n \oplus (A_\ell ? y_n)$. Inductively, $y_n \geq y$ so $y_n(p) \geq \theta$ on S for all n. Indeed, the induction is immediate from inflationarity: $y_{n+1} \geq y_n$ for all n, hence $y_n \geq y_0 = y$, and thus $y_n(p) \geq y(p) \geq \theta$ on S. Moreover, using the previous paragraph one also has $(A_\ell ? y_n) \geq (A_\ell ? y)$ for all n by monotonicity, so the reinforcement term itself remains above θ on S once it is above θ at $n = 0$.

Since $x_{t_0} \le y$ and the update is monotone, the original trajectory satisfies $x_{t_0+n} \le y_n$ but also (by inflationarity) $x_{t_0+n} \ge x_{t_0}$, hence $x_{t_0+n}(p) \ge \theta$ on S for all n. (For the desired lower bound on S, the key inequality is $x_{t_0+n} \ge x_{t_0}$: once a coordinate is at least θ at time t_0, it can never decrease below θ under an inflationary update. The comparison $x_{t_0+n} \le y_n$ is compatible with this and can be useful to show that x stays within the forward-closed region generated by y, but the persistence above θ on S already follows from inflationarity together with the hypothesis on x_{t_0}.)

Because P_ℓ is finite and V complete, the join $x_\infty := \bigvee_{t \ge t_0} x_t$ exists. (Concretely, since P_ℓ is finite, this join can be formed coordinatewise: for each $p \in P_\ell$, $(x_\infty)(p) = \bigvee_{t \ge t_0} x_t(p)$ exists in V by completeness, and finiteness ensures these coordinatewise joins assemble into an element of V^{P_ℓ}.) Monotonicity and distributivity yield that x_∞ is a fixed point of $x \mapsto x \oplus (A_\ell ? x)$, and we already proved all x_t are $\ge \theta$ on S, hence so is x_∞. (To see the fixed-point claim explicitly, note first that $x_t \le x_\infty$ for all t, so by monotonicity $A_\ell ? x_t \le A_\ell ? x_\infty$, hence $x_{t+1} = x_t \oplus (A_\ell ? x_t) \le x_\infty \oplus (A_\ell ? x_\infty)$. Taking the join over $t \ge t_0$ gives $x_\infty \le x_\infty \oplus (A_\ell ? x_\infty)$. Conversely, distributivity/continuity of $A_\ell ?$ over the directed join (together with $\oplus$ distributing over joins in V^{P_ℓ}) yields

$$A_\ell ? x_\infty \;=\; A_\ell ? \Big(\bigvee_{t \ge t_0} x_t \Big) \;=\; \bigvee_{t \ge t_0} (A_\ell ? x_t),$$

so

$$x_\infty \oplus (A_\ell ? x_\infty) = \Big(\bigvee_{t \ge t_0} x_t \Big) \oplus \Big(\bigvee_{t \ge t_0} (A_\ell ? x_t) \Big) = \bigvee_{t \ge t_0} \big(x_t \oplus (A_\ell ? x_t) \big) = \bigvee_{t \ge t_0} x_{t+1} = x_\infty,$$

where the last equality uses that $\{x_t\}$ is increasing, so shifting the index does not change the supremum.)

For the morphic resonance clause: the coupled update includes the join term $(K_{\ell' \to \ell} ? s_{\ell', t_0})$, so $s_{\ell, t_0+1} \ge (K_{\ell' \to \ell} ? s_{\ell', t_0})$. Thus if that term is $\ge \theta$ on S, the previous argument applies with $x_{t_0+1} = s_{\ell, t_0+1}$. (In other words, the resonance term provides an explicit lower bound at the next time step; once s_{ℓ, t_0+1} clears the threshold on S, the subsequent within-layer inflationary dynamics preserve that clearance forever. No further assumptions about continued resonance are needed beyond ensuring the initial lift above θ on S.) $\square$

Remark 3063. Proof sketch (high level). *The core idea is to exploit monotonicity and inflationarity. Once x_{t_0} is above θ on S, joining it with a witness state w produces a state y that is definitely above θ and is guaranteed to reproduce θ via $A_\ell ? y$. Iterating from y can never fall below θ on S, and the original trajectory, being monotone and inflationary, inherits this lower bound. A second, lattice-theoretic idea is that increasing trajectories in a complete lattice admit a supremum x_∞ that behaves like a limit; under mild distributivity/continuity hypotheses, applying the update map commutes with taking this supremum, which forces x_∞ to satisfy the fixed-point equation.*

Remark 3064. *One can visualize S as a small knot of mutually reinforcing edges. The theorem says: if resonance (or any external push) lifts every node in the knot above the activation line θ, then the knot becomes a permanent source of reinforcement for itself. In this sense, resonance acts less as a continuous force and more as a catalyst that enables the web's own feedback to take over. Equivalently, S behaves like a "memory cell" cut out of the larger network: once it is written (all coordinates in S exceed θ), the inflationary update prevents erasure, and the autocatalytic witness condition ensures that the networks internal propagation mechanism is compatible with maintaining that written state.*

54.3 Polarization vs synchronization

We interpret *polarization* as the presence of multiple stable fixed points (or long-lived metastable basins) in the coupled dynamics, so that different initial conditions lead to qualitatively different cross-context habit configurations. In particular, polarization is a statement about *macroscopic* dependence on initial conditions: two trajectories can be driven by the same inputs yet settle into distinct self-consistent global patterns. In contrast, *synchronization* means that, after transients, the global configuration becomes effectively independent of initialization (up to the obvious symmetries of the model, if any), so that all trajectories collapse to a single asymptotic regime. While we focus here on fixed points (and their basins), the same intuition extends to long-lived periodic or wandering attractors: the key distinction is whether there are *multiple* stable asymptotic behaviors that persist under small perturbations.

A clean sufficient condition for *no polarization* is global contractivity. It is worth emphasizing that this condition is deliberately strong: it gives an easily checkable certificate that rules out *all* forms of multi-attractor behavior, not only multiple fixed points. Conversely, failing contractivity does not by itself imply polarization; it merely leaves open the possibility of multiple attractors, slow mixing, or other non-synchronizing phenomena.

Remark 3065. *The contrast between polarization and synchronization can be read as the contrast between a world with many self-consistent habit-realities versus a world with a single inevitable attractor. In order-theoretic language, multiple fixed points correspond to multiple consistent closures; in metric language, contractivity collapses all trajectories to one point. The next theorem gives a simple certificate for the latter regime. A further way to read the certificate is as a* gain bound*: if the effective gain of every feedback path (within and across contexts) is uniformly below* 1*, then the system cannot amplify differences, so it cannot sustain incompatible global closures as stable alternatives.*

54.3.1 A contractive regime (p-bit toy quantale): unique global attractor

In the p-bit toy setting $V = [0,1]^2$ with pointwise order, $\oplus$ = max componentwise, and $\otimes$ componentwise multiplication, define for $v \in V$: $|v|_\infty := \max(v^+, v^-)$ and for a state $s : P \to V$: $|s|_\infty := \max_{p\in P} |s(p)|_\infty$. The choice of the ∞-norm is not essential, but it is convenient: it matches the use of max as aggregation, and it yields clean nonexpansiveness statements for joins and for coordinatewise updates. Conceptually, $|s|_\infty$ measures the strongest evidence magnitude present anywhere in the habit field.

For two states s, s' define the metric $\|s-s'\|_\infty := \max_{p\in P} \max(|s^+(p)-s'^+(p)|, |s^-(p)-s'^-(p)|)$. Note that $x \mapsto \max(x, y)$ is 1-Lipschitz and $x \mapsto ax$ is a-Lipschitz on $[0, 1]$. More explicitly, for scalar x, x', y one has

$$|\max(x, y) - \max(x', y)| \le |x - x'|,$$

and applying this coordinatewise implies that componentwise max is nonexpansive under the supremum metric. Similarly, for $a \in [0, 1]$, $|ax - ax'| = a|x - x'|$, giving the stated Lipschitz bound.

Remark 3066. *Here a value $v \in [0,1]^2$ can be read as a two-sided evidence pair (v^+, v^-), a common trick in paraconsistent and bipolar evidence formalisms (compare [23, 24]). The pointwise order and componentwise max/multiplication make the analysis transparent: each update is assembled from simple nonexpansive operations. The point is not that this is the only quantale of interest, but that it demonstrates a general phenomenon: if all propagation and persistence weights are uniformly small, then the system cannot sustain multiple incompatible basins. In this toy quantale, "uniformly* *small" is expressed as an ℓ_∞ bound on the relevant kernels; in other settings one typically replaces this with an operator norm or a suitable enriched-metric bound that plays the same role.*

Theorem 98 (Global contractivity implies synchronization and uniqueness)**.** *Consider a multi-context morphic resonance dynamics in the p-bit toy setting, with constant inputs u_ℓ and time-homogeneous kernels. Let F be the induced global update map on the product state space. Define*

$$\rho := \max\Big(\max_\ell |d_\ell|_\infty,\ \max_\ell |A_\ell|_\infty,\ \max_{\ell'\neq\ell} |K_{\ell'\to\ell}|_\infty\Big),$$

where $|A|_\infty := \max_{p,q} |A(p,q)|_\infty$ and similarly for K. If $\rho < 1$, then:

1. *F is a contraction: for all global states S, S', $\|F(S) - F(S')\|_\infty \le \rho\,\|S - S'\|_\infty$.*
2. *F has a unique fixed point S^*.*
3. *For every initial state S_0, the trajectory converges to S^* at rate ρ^t. In particular, there is no polarization into multiple stable fixed points.*

Remark 3067. *The theorem says: if every "memory factor" d_ℓ, every internal reinforcement edge, and every cross-context coupling edge has magnitude strictly less than 1 (in the supremum norm sense), then the entire system is globally contractive. Contractive systems cannot remember their initial conditions in the large; they synchronize to a unique attractor. This provides a crisp sufficient condition for "no polarization" that can be checked by simple max-norm bounds on kernels. Operationally, one can treat ρ as the worst-case one-step amplification factor for discrepancies: if two global states differ by at most ε everywhere, then after one update they differ by at most $\rho\varepsilon$. The strict inequality $\rho < 1$ is exactly the requirement that every update step strictly reduces the maximal discrepancy.*

Proof. Fix two global states S, S'. Consider one context component ℓ and one pattern $q \in P_\ell$. Write the update as a componentwise max of four terms: input, decay term, internal propagation, and coupling propagation.

The input term cancels when subtracting because it is the same constant u_ℓ in both updates. For the decay term, componentwise multiplication by d_ℓ is $|d_\ell|_\infty$-Lipschitz, so

$$\|(d_\ell \odot s_\ell) - (d_\ell \odot s'_\ell)\|_\infty \le |d_\ell|_\infty \|s_\ell - s'_\ell\|_\infty.$$

For the internal propagation term $(A_\ell ? s_\ell)(q) = \max_p A_\ell(p,q)\cdot s_\ell(p)$ (componentwise). Each map $s \mapsto A_\ell(p,q)\cdot s(p)$ is $|A_\ell(p,q)|_\infty$-Lipschitz, and taking max over p preserves the largest Lipschitz constant, hence

$$\|(A_\ell ? s_\ell) - (A_\ell ? s'_\ell)\|_\infty \le |A_\ell|_\infty \|s_\ell - s'_\ell\|_\infty.$$

The key point in the max step is that, for any family of scalar functions $\{f_p\}_p$ with Lipschitz constants bounded by L, the aggregate $x \mapsto \max_p f_p(x)$ is again L-Lipschitz under the same metric; here we apply this coordinatewise to the $(+)$ and $(-)$ components. Similarly for each coupling term $(K_{\ell'\to\ell} ? s_{\ell'})(q)$:

$$\|(K_{\ell'\to\ell} ? s_{\ell'}) - (K_{\ell'\to\ell} ? s'_{\ell'})\|_\infty \le |K_{\ell'\to\ell}|_\infty \|s_{\ell'} - s'_{\ell'}\|_\infty.$$

Finally, the full update at (ℓ, q) is the componentwise max of these terms. Max is 1-Lipschitz, so the difference of the maxima is bounded by the maximum of the differences. Taking maxima over all components yields

$$\|F(S) - F(S')\|_\infty \le \rho\,\|S - S'\|_\infty.$$

If $\rho < 1$, Banach's fixed point theorem gives existence and uniqueness of S^* and global convergence with rate ρ^t. Note that the use of a supremum metric is what makes the final step particularly direct: bounding each componentwise discrepancy and then taking the maximum immediately yields a global bound on the full product state. $\square$

Remark 3068. Proof sketch (high level). *The update is built by composing and joining simple maps, each of which has an explicit Lipschitz constant: multiplication by a bounded coefficient is contractive, and max is nonexpansive. The overall Lipschitz constant is therefore the maximum of the constants appearing in the pieces. If that maximum ρ is < 1, Banach's theorem forces a unique fixed point and geometric convergence. Equivalently, the update map F is a strict contraction on a complete metric space (a finite product of compact cubes), so the fixed point is not only unique but also* globally stable*: every orbit is pulled into it at a uniform rate.*

Remark 3069. *Visually: imagine each update step as taking the strongest of several "signals" reaching a node. If every signal is uniformly attenuated by at most $\rho < 1$, then discrepancies between two initial signals decay exponentially as they propagate, so the system cannot support distinct long-lived basins. In this regime, resonance cannot amplify differences into polarization; it can only dampen them into synchronization. This also clarifies why the condition is robust to network size: even if there are many nodes and many paths, the max-based aggregation means the update at each node only tracks the strongest incoming influence, and the uniform attenuation bound prevents any path (internal or cross-context) from acting as a net amplifier of disagreement.*

54.3.2 A simple polarization mechanism: below-threshold cross-coupling

Even if internal webs are self-weaving, morphic coupling may be too weak to seed habits across community boundaries. A crisp statement is easiest in scalar $V = [0, 1]$ with $\oplus = \max$, $\otimes = \cdot$. In this specialization, $K : P_{\ell'} \times P_\ell \to [0, 1]$ can be read as a weighted bipartite link (or transfer kernel) from a source pattern-set $P_{\ell'}$ to a target pattern-set P_ℓ, and $(K?s)(q)$ is the max-times analogue of "how much support reaches q" when all incoming supports compete by max and attenuate by multiplication.

Theorem 99 (Below-threshold coupling cannot seed above-threshold habits)**.** *Assume $V = [0, 1]$ with $\oplus = \max$ and $\otimes = \cdot$. Fix a threshold $\theta \in (0, 1]$. Let $K : P_{\ell'} \times P_\ell \to [0, 1]$ satisfy $K(p, q) < \theta$ for all (p, q). Then for every source state $s : P_{\ell'} \to [0, 1]$ with $0 \leq s(p) \leq 1$, one has*

$$(K?s)(q) < \theta \quad \text{for all } q \in P_\ell.$$

Consequently, if a target autocatalytic core requires all its patterns to reach level $\geq \theta$ to nucleate, then this coupling K can never nucleate that core, no matter how strong the source state is (as long as $s \leq 1$).

Remark 3070. *The claim is conceptually simple: max-times propagation cannot exceed the largest edge weight when the source is bounded by 1. Thus a uniformly sub-threshold bridge is not merely "weak"; it is structurally incapable of crossing the activation boundary. This provides a clean sufficient condition for polarization between communities: if all inter-community links fall below the activation threshold, then no community can ignite another's autocatalytic core.*

Remark 3071. *In particular, the hypothesis $s \leq 1$ is doing real work: it formalizes the idea that a source pattern cannot contribute more than its full activation. Under max-times, this means each incoming contribution to q is bounded by the corresponding coupling weight, so the bridge itself* *becomes the bottleneck. If one were to allow an unbounded source scale (e.g., $s(p) > 1$), then a sub-threshold edge weight could in principle be compensated by super-unit source magnitude; the theorem rules out that regime by design, aligning the semantics with probabilistic or normalized activation levels.*

Proof. For any $q \in P_\ell$,

$$(K?s)(q) = \max_{p \in P_{\ell'}} K(p, q) \cdot s(p).$$

Since $0 \leq s(p) \leq 1$, we have $K(p, q) \cdot s(p) \leq K(p, q) < \theta$ for every p. Taking max over p preserves $< \theta$. The final clause is immediate. □

Remark 3072. Proof sketch (high level). *Each term in the max is bounded by $K(p, q)$ because $s(p) \leq 1$. So the entire max is bounded by the largest $K(p, q)$, which is assumed $< \theta$.*

Remark 3073. *A useful corollary is that the obstruction is not merely one-step. If one iterates the same style of propagation across multiple stages, sub-threshold attenuation compounds rather than accumulates. Concretely, for any composable kernels K_1, K_2 with all entries $< \theta$ and any $s \leq 1$, the max-times composite satisfies*

$$(K_2?(K_1?s))(q) = \max_{p,r} K_2(r, q)\, K_1(p, r)\, s(p) \ < \ \theta,$$

since every product term is $\leq K_2(r, q) < \theta$ (and also $\leq K_1(p, r) < \theta$). Thus, even allowing arbitrarily many intermediate "relay" patterns does not create a super-threshold signal in this semiring when all couplings remain uniformly sub-threshold and sources remain normalized. In diffusion language: there is no multi-hop amplification mechanism available; every path weight is a product of numbers in $[0, 1]$ and hence cannot exceed any single factor on that path.

Remark 3074. *This theorem makes polarization intuitive: if the context network has weak (sub-threshold) bridges, then different context communities can remain in different habit basins, because they cannot seed each other above the activation threshold. In Hyperseed language, "resonance" exists as a formal channel, but it is filtered by an activation threshold, and thus can fail to induce intersubjective convergence (Hyperseed-Concept* **??**, *158,* **??***).*

Remark 3075. *Interpreted as "polarization vs. synchronization," the result identifies a sufficient condition for* failure *of cross-community synchronization: even perfect internal synchronization within each community does not help if every bridge edge lies below the activation floor needed for nucleation. Conversely, to even make synchronization* possible *in this max-times model (with normalized sources), at least one bridge into each required target pattern must meet or exceed θ; otherwise the target cannot be pushed into the basin where its own internal autocatalysis sustains it. This highlights an asymmetry between strong internal ties (which maintain a habit once present) and strong inter-community ties (which are required to* create *the habit elsewhere).*

54.4 Random-graph theorems for habit webs

We now connect habit-web properties to classical random graph thresholds. The key move is to analyze a *thresholded skeleton* of a weighted web.

54.4.1 Thresholded skeleton

Let $A : P \times P \to [0, 1]$ be a weighted kernel. Fix $\theta \in (0, 1]$ and define the Boolean θ-skeleton adjacency relation

$$E_\theta(p, q) := \mathbf{1}[A(p, q) \geq \theta].$$

In many max-times models, reaching support $\geq \theta$ along an edge requires $A(p, q) \geq \theta$ (or at least some uniform lower bound of that form), so E_θ is a conservative predictor of above-threshold diffusion and self-weaving.

Remark 3076. *The skeleton E_θ is a deliberate loss of information: it forgets* how far above θ *an edge is, and remembers only whether it clears the bar. The payoff is that questions about activation at level θ reduce to questions about reachability and cycles in an ordinary directed graph. This is a standard analytic move: discretize a graded process at the threshold where the qualitative behavior changes.*

Remark 3077. *In the max-times model, the condition $A(p, q) \geq \theta$ is sufficient (though not always necessary) for the possibility that a node at level $\geq \theta$ pushes its successor to level $\geq \theta$ in one step. Thus E_θ is conservative: if the skeleton has no relevant paths/cycles, then the weighted system cannot exhibit the corresponding above-threshold phenomena either.*

54.4.2 Minimal autocatalytic cores: 2-cycles in a random directed web

Consider a random directed graph $D(n, p)$ on n patterns: each ordered edge $(i \to j)$, $i \neq j$, is present independently with probability p. In the Boolean setting $V = \{0, 1\}$ with $\oplus = \vee$, $\otimes = \wedge$, any directed 2-cycle is a (minimal) autocatalytic subset of size 2.

Theorem 100 (Sharp threshold for a minimal self-weaving core via 2-cycles)**.** *Let $D(n, p)$ be a random directed graph with independent directed edges. Let X be the number of directed 2-cycles (unordered pairs $\{i, j\}$ such that $i \to j$ and $j \to i$ are both present). Then $X \sim \mathrm{Bin}(N, p^2)$ with $N = \binom{n}{2}$ and*

$$\mathbb{P}[X = 0] = (1 - p^2)^N.$$

Consequently:

1. *If $p = o(1/n)$ then $\mathbb{P}[X > 0] \to 0$.*
2. *If $p = \omega(1/n)$ then $\mathbb{P}[X > 0] \to 1$.*

Therefore the emergence of a minimal autocatalytic core (a 2-cycle) has threshold $p \asymp 1/n$.

Remark 3078. *The theorem quantifies a basic intuition: a minimal self-sustaining habit requires at least a loop. In a random directed web, the simplest loop is a 2-cycle, and it appears right when the expected number of such loops passes from negligible to large. The result is important for priors: below $p \asymp 1/n$, one should not expect even the smallest self-weaving structure to exist by chance, whereas above that density, such loops become typical.*

Proof. For each unordered pair $\{i, j\}$, let I_{ij} be the indicator that both edges $i \to j$ and $j \to i$ are present. Since edges are independent, $\mathbb{P}[I_{ij} = 1] = p^2$. Moreover the sets of edges used by distinct unordered pairs are disjoint, hence the indicators are independent. Thus $X = \sum_{\{i,j\}} I_{ij} \sim \mathrm{Bin}(N, p^2)$ and $\mathbb{P}[X = 0] = (1 - p^2)^N$.

Now $Np^2 \sim (n^2/2)p^2$. If $p = o(1/n)$ then $Np^2 \to 0$ so $(1 - p^2)^N \to 1$ and $\mathbb{P}[X > 0] \to 0$. If $p = \omega(1/n)$ then $Np^2 \to \infty$ and $(1 - p^2)^N \to 0$, hence $\mathbb{P}[X > 0] \to 1$. $\square$

Remark 3079. Proof sketch (high level). *Count candidate 2-cycles (there are* $\binom{n}{2}$ *unordered pairs), note each becomes a 2-cycle with probability* p^2*, and observe independence across pairs because disjoint edge sets are involved. Then the probability of no 2-cycles is a simple binomial tail at* 0*, which is well-approximated by* $\exp(-Np^2)$*, giving the threshold* $p \asymp 1/n$*.*

Remark 3080. *A visual way to remember the threshold is: there are* $\Theta(n^2)$ *possible mutual pairs, so one needs edge probability* $\Theta(1/n)$ *for a constant expected number of mutual pairs. Below that, the web is too sparse to close even the smallest feedback loop.*

54.4.3 Macroscopic diffusion: out-component threshold via branching processes

In the Boolean no-decay/no-input closure dynamics, starting from a seed set S_0 and repeatedly adding out-neighbors yields the reachability closure, i.e. the directed out-component.

Theorem 101 (Diffusion threshold for downstream closure in $D(n, c/n)$ (branching heuristic))**.** *Let* $D(n,p)$ *with* $p = c/n$ *for constant* $c > 0$*. Start from one seed node* v *and let* $C_{\text{out}}(v)$ *be its directed out-component. Then:*

1. *If* $c < 1$*, the exploration process is stochastically dominated by a subcritical Galton–Watson process with mean c, hence* $|C_{\text{out}}(v)| = O(\log n)$ *with high probability.*
2. *If* $c > 1$*, the corresponding Galton–Watson process is supercritical, with survival probability* $1-q$ *where* $q \in (0,1)$ *solves* $q = e^{c(q-1)}$*. With probability bounded away from 0, the exploration does not die out quickly and reaches* $\Theta(n)$ *nodes; in particular a macroscopic diffusion event becomes possible.*

Remark 3081. *This is the familiar mean-degree-*1 *phase transition, here interpreted as a diffusion threshold for closure dynamics. In the subcritical regime, reinforcement paths die out quickly; in the supercritical regime, a seed can ignite a giant downstream set. The connection back to habit webs is direct in the Boolean/skeleton setting: diffusion is reachability, and reachability is governed by out-components.*

Proof sketch (standard exploration coupling). Reveal out-neighbors of newly discovered nodes. For early times when the explored set is small relative to n, each discovered node has approximately Poisson(c) fresh out-neighbors, nearly independent, because each of the n possible targets is hit with probability c/n.

Thus the number of newly discovered nodes per step is well-approximated by a Galton–Watson branching process with offspring distribution Poisson(c). If $c < 1$ the branching process becomes extinct quickly, and standard tail bounds yield logarithmic component sizes. If $c > 1$ the branching process survives with probability $1 - q > 0$ and grows geometrically until finite-size effects matter, at which point it typically reaches linear size. □

Remark 3082. Commentary on key steps. *The crucial approximation is that, while the explored set is small, edges from newly explored nodes point to "fresh" nodes with near-independence. This is precisely the regime where a branching process is an accurate proxy. The fixed-point equation* $q = e^{c(q-1)}$ *is the extinction probability equation for a Poisson(c) Galton–Watson process.*

Remark 3083. *One can make the "freshness" heuristic quantitative as follows: if the explored set has size* $s \ll n$*, then a newly revealed out-edge hits a previously explored vertex with probability approximately* s/n*, so the expected number of collisions among* $O(1)$ *out-edges is* $O(s/n)$*, hence negligible until the exploration reaches size* n^γ *for any fixed* $\gamma < 1$*. In that regime, the exploration* *is locally tree-like and the breadth-first queue length evolves like a random walk with increment distribution* Poisson$(c) - 1$*, which is exactly the branching-process coupling. In particular, the same fixed point q that gives extinction probability also controls the asymptotic fraction of vertices with "small" out-components; when $c > 1$, the probability of a linear-size out-component is $1 - q$ under the branching heuristic, aligning with the emergence of a giant out-component in the directed setting.*

Remark 3084. *Geometrically, one can picture the exploration as a wavefront moving along directed edges. When $c < 1$, the wavefront shrinks on average; when $c > 1$, it expands on average until constrained by finite size. This is the graph-theoretic analogue of a reproduction number crossing* 1 *in epidemic models, which anticipates the two-level cascade theorem below.*

Remark 3085. *From the perspective of fixed points and polarization, the dichotomy at $c = 1$ can be read as a change in the qualitative stability of the "inactive" state in the exploration dynamics: in the subcritical regime the only consistent macroscopic outcome is extinction (return to zero queue), whereas in the supercritical regime there is a second, self-consistent macroscopic outcome in which the exploration sustains itself long enough to occupy a positive fraction of the graph. The branching-process fixed point q therefore plays the role of a coarse order parameter: it is both the extinction probability and (heuristically) the complement of the giant reachability fraction.*

54.4.4 Two-level diffusion: random graph of contexts + random internal habit webs

Now consider N contexts. Let the *context coupling graph* be an undirected (or directed) random graph on contexts, say $G(N, p_L)$ with $p_L = c_L/N$.

Within each context ℓ, suppose the internal θ-skeleton of its habit kernel behaves like a random directed graph $D(n_\ell, p_{\text{int}})$ for some effective p_{int}. Let α_ℓ be the probability that context ℓ contains a minimal autocatalytic core (a 2-cycle) in its θ-skeleton. By the 2-cycle theorem,

$$\alpha_\ell = 1 - (1 - p_{\text{int}}^2)^{\binom{n_\ell}{2}} \approx 1 - \exp\Big(- \tfrac{1}{2} n_\ell^2 p_{\text{int}}^2 \Big).$$

Assume that when a neighboring context is already in a high-support habit state, the coupling successfully *seeds* the target above threshold θ with probability β (a coarse summary of kernel strength, alignment, and thresholding).

Remark 3086. *This construction makes explicit a two-level architecture: within each context there is a habit web that may or may not be able to sustain itself once activated (captured by α_ℓ), and between contexts there is a coupling network that may or may not successfully transmit activation (captured by β and the random context graph). Such coarse-graining is common in reasoning systems: one replaces detailed internal dynamics by a small number of effective parameters that control whether an influence is transient or persistent.*

Remark 3087. *Although α_ℓ is defined here via the presence of a 2-cycle (as a minimal sufficient structure for self-support in the θ-skeleton), it can also be interpreted more broadly as an* effective persistence probability *of internal dynamics conditional on seeding. In contexts where larger autocatalytic sets are the typical sustaining mechanism, α_ℓ may be replaced by the probability of having* some *self-sustaining strongly connected subgraph above threshold; the two-level analysis below only uses α as the coarse probability that seeding produces a persistent source. If contexts are exchangeable (e.g. $n_\ell \equiv n$ and common p_{int}), one may set $\alpha_\ell \equiv \alpha$, while in heterogeneous settings one typically works with an average (or size-biased average) persistence parameter that reflects which contexts are more often reached by edges in the coupling graph.*

Theorem 102 (Cascade threshold for morphic resonance on a random context graph)**.** *Consider the following coarse-grained process on contexts: an "infected" context attempts to seed each neighboring context independently with success probability β, and a successfully seeded context becomes persistently infected with probability α (internal self-weaving once seeded). On $G(N, c_L/N)$ this is approximated by a Galton–Watson process with mean reproduction number*

$$R := \alpha\,\beta\,c_L.$$

If $R < 1$ then the infected set remains $O(\log N)$ with high probability. If $R > 1$ then with positive probability the infected set reaches $\Theta(N)$ contexts (i.e. a macroscopic morphic-resonance diffusion).

Remark 3088. *In the supercritical case $R > 1$, the branching approximation also predicts a nontrivial survival probability $1 - \rho > 0$, where ρ is the smallest solution in $[0, 1]$ of the fixed-point equation $\rho = \exp(R(\rho - 1))$ (the extinction equation for a Poisson(R) offspring law). Under the same heuristics, conditional on survival the cascade grows roughly geometrically for many generations before saturation, and the eventual macroscopic infected fraction is controlled by the same fixed-point structure that appears in one-level random-graph giant-component theory. This makes the analogy to the $c = 1$ transition not merely qualitative: it is encoded by an explicit scalar fixed-point equation at the coarse-grained level.*

Remark 3089. *This theorem is the contextual analogue of the $c = 1$ out-component threshold: the mean reproduction number R plays the same role, but now it factors into (i) network connectivity c_L, (ii) the probability β that resonance successfully seeds across an edge, and (iii) the probability α that what is seeded becomes self-sustaining. The factorization is conceptually useful because it separates three failure modes: sparse coupling, weak bridges, or lack of internal autocatalysis.*

Proof sketch. In $G(N, c_L/N)$ the number of neighbors of a typical context is approximately $\mathrm{Poisson}(c_L)$. Each neighbor is successfully seeded with probability β, and conditional on seeding, becomes a persistent source with probability α. Thus the number of newly persistent contexts caused by one persistent context is approximately $\mathrm{Poisson}(\alpha\beta c_L)$. Branching-process theory yields the stated subcritical/supercritical dichotomy. □

Remark 3090. *A slightly more explicit way to see the Poisson offspring law is by successive thinning: if $D \sim \mathrm{Poisson}(c_L)$ is the number of neighbors, and each edge transmits with probability β, then the number of successfully seeded neighbors is $\mathrm{Poisson}(\beta c_L)$; then, keeping each seeded neighbor with probability α (persistence conditional on seeding) yields a final thinned count $\mathrm{Poisson}(\alpha\beta c_L)$. This repeated thinning viewpoint makes clear why the product $\alpha\beta c_L$ is the only parameter that appears in the mean-field cascade threshold.*

Remark 3091. Commentary on key steps. *The Poisson approximation enters twice: first for the degree distribution of $G(N, c_L/N)$, and then for the thinning of neighbors by independent success probabilities β and α. The product $\alpha\beta c_L$ is exactly the mean of the thinned offspring distribution, hence the effective R.*

Remark 3092. *As in one-level exploration, the independence assumptions are most accurate during the early phase when the infected set is small compared to N, so that attempted infections largely target previously uninfected contexts and different infection paths rarely collide. Once a linear number of contexts is reached (when $R > 1$ and the cascade survives), collisions and depletion become non-negligible, but at that point the key question (survival versus extinction) has already been resolved by the initial branching phase. This is why the branching-process approximation is well-suited for identifying the diffusion threshold even when later-time dynamics require more detailed finite-size analysis.*

Remark 3093. *A helpful picture is to view each context as a "reactor" that, once lit, can light others. The parameter β is the transmissibility of flame across pipes; α is whether the receiver contains enough fuel arranged in a self-sustaining configuration. When $R > 1$, the fire can spread through a macroscopic fraction of the network; when $R < 1$, it sputters locally. This metaphor is mathematically exact in the branching approximation and ties back to the one-shot nucleation theorem: the difference between transient seeding and persistent ignition is precisely internal self-weaving.*

54.5 Takeaways for reasoning systems (informal)

- Fixed points are not an extra assumption: in the finite-pattern regime, least fixed points exist and can be computed by iterating the update from bottom. Here "from bottom" means starting from the least informative state (the empty/false assignment, or the least element of the underlying lattice/quantale module), and repeatedly applying the monotone update operator until no new facts are added. In practice this provides an implementable closure algorithm whose intermediate iterates are themselves meaningful approximations (each step is a sound under-approximation of the eventual closure), and termination is guaranteed by finiteness of the pattern space.

- Morphic resonance can be compiled away into an enlarged single habit web (a block kernel), making diffusion queries reducible to a single closure computation $G^*?u$. Operationally, the block construction replaces "multiple interacting contexts" by a single augmented state space in which cross-context couplings become ordinary edges (blocks) of one composite kernel. This matters for engineering because it eliminates the need for bespoke multi-stage simulation: once the block kernel is assembled, all reachability / activation questions reduce to the same standard closure primitive, and comparisons across scenarios become comparisons of kernels and initial seeds.

- The nucleation theorem yields an actionable test: can coupling drive push a latent autocatalytic core above its activation threshold? Concretely, one searches for a small subweb (candidate core) whose internal reinforcement is near-critical, then evaluates whether external inputs or cross-context couplings supply enough additional weight to trigger self-sustaining growth. This frames "emergence" as a finite check on a localized structure, rather than as an unstructured global phenomenon, and motivates scanning for minimal cores as a diagnostic routine.

- Contractivity gives a clean no-polarization certificate; below-threshold bridges give a clean polarization certificate (communities cannot seed each other). The first condition says that the global update map shrinks differences so that distinct initial states converge toward a unique attractor, ruling out long-lived divergence of beliefs/activations. The second condition isolates a dual extreme: even if each community can internally amplify signals, if the inter-community coupling remains sub-critical then activation cannot cross the bridge, so distinct fixed points (or distinct basins of attraction) persist and polarization is structurally stable against small perturbations.

- Random-graph thresholds provide priors: sparse webs below $p \asymp 1/n$ are unlikely to contain even minimal self-weaving cores; above that, autocatalytic loops appear rapidly, and above mean degree 1, macroscopic diffusion becomes plausible. Interpreting this in an Erdős–Rényi heuristic, $p \asymp 1/n$ is the scale at which cycles begin to occur with non-negligible probability, so the absence/presence of feedback motifs can often be predicted from density alone. The "mean degree 1" landmark aligns with the emergence of giant components, suggesting that once a large connected substrate exists, supercritical local amplification has a pathway to spread widely, whereas below it diffusion is typically confined to small components even when local reinforcement is present.

Remark 3094. *From an engineering perspective, the results in this section can be read as a toolkit for deciding which computational questions to ask and in which order. The fixed-point theorems say that closure computations are principled and converge; the block-kernel theorem says that multi-context resonance can be reduced to one closure problem; the nucleation theorem says to search for small candidate autocatalytic cores and test whether couplings can push them above threshold; the polarization criteria provide two extremal certificates (global contraction versus sub-threshold bridges) that bound what the system can do. This style of turning qualitative narrative into testable algebra is characteristic of the quantale approach to weakness and emergence [3, 5]. A useful procedural reading is: (i) compute closures to identify stable consequences of a seed; (ii) if multiple contexts are present, first compile them into the block kernel so that subsequent queries are uniform; (iii) if surprising global spread is observed or desired, localize the explanation by searching for near-critical cores and checking whether cross-couplings supply the missing gain; (iv) when the system exhibits persistent disagreement, attempt to certify either global contractivity (which forces eventual agreement) or the existence of sub-threshold cuts/bridges (which explain why agreement is impossible without strengthening specific links). In this sense the theorems do not merely predict behavior; they also suggest which measurements to take (e.g., local gain, bridge strength, component sizes) and which interventions are most leverageful (strengthen a specific coupling, weaken a feedback loop, or alter the seed) when attempting to control diffusion.*

54.6 Conclusion

Remark 3095 (What these theorems add beyond the base document)**.** *The base Weakness Theory material provides the core definitions and several algebraic laws, plus first-order McBride sensitivities. The additional results here emphasize: (i) a practical emergence test via conditional amplification (Theorem 81), (ii) robustness/invariance properties important for transfer (Theorem 82), (iii) a principled second-order interaction diagnostic tied to curvature (Theorem 83), (iv) a curvature-controlled guarantee for when first-order McBride pruning is near-optimal (Theorem 84), (v) a concrete submodularity/greedy guarantee in the max-product quantale (Theorem 85), (vi) a clean route from delay-feedback dynamics to bursty/chaotic pattern intensity (Theorem 86), and (vii) log-linearization for information-like reasoning (Theorem 87). In more operational terms, the new material turns the base algebra into a usable toolkit: one can (i)* detect *when a small conditional perturbation triggers a qualitatively different closure outcome, (ii)* compare *models under reparameterizations without changing the induced ordering of supports, (iii)* separate *genuine synergistic effects from additive explanations by looking at curvature-like deviations, and (iv)* justify *greedy or top-k approximations in regimes where the second-order terms are provably small. Items (v)–(vii) further supply algorithmic and dynamical interpretations: submodularity connects the max-product semantics to standard approximation guarantees, delay-feedback provides an explicit dynamical mechanism for intermittent intensity that complements fixed-point language, and log-linearization places the same phenomena into an additive scale where ratios and products become differences and sums, making "information-like" arguments (e.g., about gains, gaps, and effective contributions) transparent.*

Remark 3096. *Read in the broader Hyperseed frame, this section supplies a bridge between three* *levels of description: (1) local algebra of support propagation in a quantale, (2) mesoscopic structure in the form of autocatalytic/self-weaving subsets, and (3) macroscopic diffusion and phase transitions in random coupling regimes. The connecting thread is the fixed point: whether one thinks in terms of closure in a web, persistent activation in a core, or global attractors versus polarization, the mathematics is the mathematics of monotone/contractive operators on structured state spaces (Hyperseed-Concept 168, 115, 61,* **??***). A useful way to parse this bridge is to treat each level as a different "resolution" of the same underlying operator: the local view studies how individual edges/constraints compose under the quantale product, the mesoscopic view asks when certain subsets become approximately closed under that composition (thereby behaving like internally reinforcing modules), and the macroscopic view averages these mechanisms over random couplings to locate thresholds where reinforcement becomes typical rather than exceptional. In that sense, fixed points and their stability act as the shared currency: they encode both* existence *(a closure/attractor is present) and* robustness *(it persists under perturbations, rescalings, or pruning), which is exactly what one needs to connect "habit webs" to diffusion/polarization narratives without changing mathematical language.*

Remark 3097. *Finally, the random-graph corollaries can be read as giving an explicit "thermodynamic" intuition for when the web-level closure picture becomes predictive: below the relevant coupling/intensity threshold, amplification chains die out quickly and fixed points tend to be trivial or fragmented; above it, autocatalytic cores and large closures appear with high probability, and the main qualitative uncertainty shifts from* whether *emergence occurs to* which *stable configuration (or polarized basin) is selected by initial conditions and perturbations. This makes the section's themes compatible with both engineering usage (designing for or against runaway reinforcement) and interpretive usage (explaining why similar local rules can yield either weak diffusion or strong resonance depending on global coupling).*

55 Belief Closure + Question Networks + Science: Formal Theorems that Guide Practical Reasoning Loops

Here we formalize and prove a collection of theorems about (i) belief closure under weighted inference rules, (ii) question networks and thinking episodes as controlled closure, and (iii) empirically coupled scientific update loops. The results are chosen to be directly useful for a reasoning system: incremental recomputation laws, confluence/order-independence, explanation/provenance semantics, relevance pruning, modular computation via SCC decomposition, hypothesis sensitivity monotonicity, and reproducibility invariances.

Operationally, the intent is that a reasoning system maintains a graded belief state β, receives an external increment ε (from observation, user input, or measurement), and then runs a bounded or convergent closure procedure to compute consequences under a rule set R. Later subsections will formalize "thinking episodes" as controlled (possibly partial) closure steps and will use the fixed-point structure below to state correctness and invariance properties of those episodes.

55.1 Setup

Definition 489 (Quantale assumptions)**.** *A (commutative) quantale is a tuple $(V, \leq, \oplus, \otimes, e)$ where: (i) $(V, \leq)$ is a complete lattice with join $\oplus$ (and arbitrary joins); (ii) $(V, \otimes, e)$ is a commutative monoid; (iii) $\otimes$ distributes over arbitrary joins in each argument: $a \otimes \oplus_i b_i = \oplus_i (a \otimes b_i)$. We write $\perp$ for the bottom element of V.*

Intuitively, V is a space of "support values" (or degrees of belief, evidence, plausibility, or score). The join $\oplus$ acts as an aggregation of alternative sources of support (taking the least upper bound), while $\otimes$ acts as a combination of jointly required supports (e.g. conjunction-like composition of premise strengths). Commutativity of $\otimes$ encodes that premises are treated as an unordered multiset at the semantic level, which is convenient for confluence and order-independence results later. The distributivity condition is the key algebraic property that permits moving between "compute all alternatives, then combine" and "combine, then take alternatives," which will underwrite several incremental recomputation laws.

Concrete instances to keep in mind include: (a) the Boolean quantale $(\{0, 1\}, \leq, \vee, \wedge, 1)$ recovering ordinary Horn closure; (b) fuzzy-style quantales where $\oplus = \max$ and $\otimes$ is a t-norm; (c) cost or log-score quantales where $\oplus$ is min (interpreting "better" as smaller) and $\otimes$ is addition, provided the lattice/completeness requirements are met by the chosen completion. The bottom element $\perp$ represents "no support" (or the least informative value).

Definition 490 (Belief states, evidence increments)**.** *Fix a claim language L (a set). A belief state is a function $\beta : L \to V$. We order belief states pointwise: $\beta \leq \gamma$ iff $\forall \varphi \in L,\ \beta(\varphi) \leq \gamma(\varphi)$. For belief states β, γ define their join pointwise: $(\beta \oplus \gamma)(\varphi) := \beta(\varphi) \oplus \gamma(\varphi)$. An evidence increment is any $\varepsilon : L \to V$ (same type as a belief state).*

Thus V^L is itself a complete lattice under the pointwise order, with pointwise joins and bottom element $\perp_L$ given by the constant function $\varphi \mapsto \perp$. This makes it natural to treat belief revision steps as monotone operators on a lattice of states. Evidence increments are typed the same way as belief states so that "incorporating evidence" can be expressed uniformly as a join $\beta \oplus \varepsilon$ (or, in variants, as an operator built from $\oplus$ and $\otimes$). In the later "empirically coupled" loop, ε will represent the mapping from an experimental readout to updated support on measurement claims, and the theorems will describe when closure can be recomputed locally.

Definition 491 (Weighted inference rules and one-step inference operator)**.** *A weighted inference rule is a triple $r = (\Gamma, \psi, \lambda_r)$ where $\Gamma \subseteq L$ is finite (premises), $\psi \in L$ (conclusion), and $\lambda_r \in V$ (strength). Define the rule contribution on belief state β by*

$$\mathrm{contrib}_r(\beta) \ := \ \lambda_r \otimes \otimes_{\varphi \in \Gamma} \beta(\varphi).$$

For a rule set R, define the one-step inference operator $\mathrm{F}_R : V^L \to V^L$ by

$$\mathrm{F}_R(\beta)(\psi) \ := \ \beta(\psi) \ \oplus \ \oplus_{r=(\Gamma,\chi,\lambda)\in R:\ \chi=\psi} \Big(\lambda \otimes \otimes_{\varphi\in\Gamma} \beta(\varphi)\Big).$$

The expression $\otimes_{\varphi\in\Gamma}\beta(\varphi)$ is the monoid product over the finite premise set Γ; by convention, when $\Gamma = \emptyset$ this product is e, so a rule with no premises contributes $\lambda_r \otimes e = \lambda_r$ directly as an "axiom" or base-rate support for its conclusion. The finiteness of Γ ensures that the product is well-defined without requiring any additional completeness of $\otimes$.

The one-step operator F_R has two conceptual parts: it preserves existing support $\beta(\psi)$ and then adds (via $\oplus$) whatever new support is derivable in one application of some rule concluding ψ. In particular, F_R is inflationary in the sense that $\beta \leq \mathrm{F}_R(\beta)$ pointwise, because each coordinate is joined with its prior value. Later confluence and incremental-update theorems will leverage the fact that F_R is also monotone (and, under suitable hypotheses, continuous) as a map on the complete lattice V^L.

Definition 492 (Closure as least fixed point above an initial state)**.** *Assume either: (A) the domain effectively used is finite (e.g. finite L and values in a finite subquantale), so ascending chains stabilize; or (B) F_R is Scott-continuous so ω-iteration reaches a fixed point.*

Define the closure of β under R by

$$\mathrm{Cl}_R(\beta) \ := \ \oplus_{n\geq 0}\mathrm{F}_R^n(\beta),$$

where F_R^0 is the identity and $\mathrm{F}_R^{n+1} = \mathrm{F}_R \circ \mathrm{F}_R^n$.

This definition implements the standard saturation construction: repeatedly apply one-step inference, accumulating results by joining all iterates. Under (A), the ascending chain $\beta \leq \mathrm{F}_R(\beta) \leq \mathrm{F}_R^2(\beta) \leq \cdots$ stabilizes after finitely many steps in the effectively used portion of the state space; under (B), Scott-continuity guarantees that the join of the ω-chain is a fixed point. In either case, $\mathrm{Cl}_R(\beta)$ is the least post-fixed point above β and (under the usual monotonicity conditions) coincides with the least fixed point above β in the sense of Knaster–Tarski; later proofs will make this "least solution" property explicit and use it to justify modular recomputation and pruning.

From an implementation perspective, $\mathrm{Cl}_R(\beta)$ represents the fully closed belief state for the rule set R starting from initial support β. In later sections, we will also consider truncated or focused closures (e.g. only within a strongly connected component of the rule dependency graph, or only along questions that are currently active), and the laws proved for Cl_R will serve as the reference semantics those controlled procedures approximate.

Remark 3098. *In the Hyperseed reconstruction, $V = [0,1]^2$ (p-bits) with $\oplus$ as coordinatewise max and $\otimes$ as coordinatewise multiplication; no consistency constraint is imposed. The theorems below are stated at quantale level so they can be re-used with other graded/paraconsistent domains.*

In that p-bit instance, a claim φ may carry separate degrees for distinct evidential channels (e.g. "support" and "counter-support"), and coordinatewise max corresponds to retaining whichever channel has stronger accumulated evidence from any source. Coordinatewise multiplication makes the premises behave like independent attenuations along each channel, so that weak premise support proportionally limits the derived conclusion support in that coordinate. The absence of an imposed consistency constraint is precisely what permits paraconsistent behavior: a claim and its negation may both receive high support without forcing collapse, which is useful when modeling incomplete or conflicting data streams. The quantale-level abstraction here ensures that subsequent theorems can be instantiated equally well for Boolean, fuzzy, probabilistic, or paraconsistent graded logics, provided their combination and aggregation operations satisfy the algebraic conditions above.

55.2 Closure operator laws and incremental update laws

The results in this subsection record the basic algebraic properties that make Cl_R behave like an ordinary closure operator (in the sense of order/lattice theory), while also supporting "incremental" reasoning patterns in which one alternates between incorporating new evidence and re-closing under the rules. Throughout, the order $\leq$ is the pointwise order on belief states (so $\beta \leq \gamma$ means $\beta(\varphi) \leq \gamma(\varphi)$ for every $\varphi \in L$), and joins/meets are understood pointwise as well.

Lemma 162 (Monotonicity and inflationarity of F_R)**.** *For any rule set R, F_R is monotone and inflationary:*

$$\beta \leq \gamma \ \Rightarrow \ \mathrm{F}_R(\beta) \leq \mathrm{F}_R(\gamma), \qquad \beta \leq \mathrm{F}_R(\beta).$$

Proof. Fix $\psi \in L$. Monotonicity: if $\beta \leq \gamma$ then for each premise $\varphi \in \Gamma$, $\beta(\varphi) \leq \gamma(\varphi)$. By monotonicity of $\otimes$ in each argument (a consequence of join-distributivity), $\otimes_{\varphi\in\Gamma}\beta(\varphi) \leq \otimes_{\varphi\in\Gamma}\gamma(\varphi)$. Multiplying by λ preserves order, and joining over rules preserves order, hence $\mathrm{F}_R(\beta)(\psi) \leq \mathrm{F}_R(\gamma)(\psi)$. Inflationary: $\beta(\psi)$ is one of the joined terms in the definition of $\mathrm{F}_R(\beta)(\psi)$, so $\beta(\psi) \leq \mathrm{F}_R(\beta)(\psi)$ for all ψ.

It may be helpful to keep the following reading in mind: F_R performs one "forward-chaining" sweep of rule application. Monotonicity expresses that strengthening any premise degrees in β cannot weaken the derived degrees after one sweep, because every operation used to compute $\mathrm{F}_R(\beta)(\psi)$ (tensoring premise degrees, scaling by λ, and finally taking a join across applicable rules and the original value) is order-preserving. Inflationarity expresses that F_R never retracts what is already believed: the definition explicitly includes the previous value $\beta(\psi)$ as a candidate in the join. $\square$

Theorem 103 (Existence of closure as least fixed point above β)**.** *Under the standing assumption in Definition 4,* $\mathrm{Cl}_R(\beta)$ *is a fixed point of* F_R*, dominates* β*, and is the least fixed point above* β*:*

$$\beta \leq \mathrm{Cl}_R(\beta), \quad \mathrm{F}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta), \quad (\mathrm{F}_R(\gamma) = \gamma \ \wedge \ \beta \leq \gamma) \Rightarrow \mathrm{Cl}_R(\beta) \leq \gamma.$$

Proof. By Lemma 1, the sequence $\beta \leq \mathrm{F}_R(\beta) \leq \mathrm{F}_R^2(\beta) \leq \cdots$ is ascending. Thus $\mathrm{Cl}_R(\beta) = \oplus_{n\geq 0}\mathrm{F}_R^n(\beta)$ exists and satisfies $\beta \leq \mathrm{Cl}_R(\beta)$. In other words, $\mathrm{Cl}_R(\beta)$ collects (by pointwise join) all information obtainable by repeatedly applying the one-step inference operator F_R.

If (A) holds, the ascending chain stabilizes at some N, hence $\mathrm{Cl}_R(\beta) = \mathrm{F}_R^N(\beta)$ and applying F_R yields $\mathrm{F}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)$. This is the familiar "finite convergence" situation: after enough iterations no further increases occur, so the stabilized state is already closed under one more application of F_R.

If (B) holds, Scott-continuity gives $\mathrm{F}_R(\mathrm{Cl}_R(\beta)) = \mathrm{F}_R(\oplus_n \mathrm{F}_R^n(\beta)) = \oplus_n \mathrm{F}_R^{n+1}(\beta) = \mathrm{Cl}_R(\beta)$. Here Scott-continuity is exactly the property that allows F_R to commute with joins of directed (in particular, ascending) chains, which is why the join of the iterates is again a fixed point even when the chain does not literally stabilize at a finite stage.

For leastness: if γ is a fixed point with $\beta \leq \gamma$, then by monotonicity $\mathrm{F}_R^n(\beta) \leq \mathrm{F}_R^n(\gamma) = \gamma$ for all n. Taking the join over n yields $\mathrm{Cl}_R(\beta) \leq \gamma$. This "least fixed point above β" characterization is the precise sense in which $\mathrm{Cl}_R(\beta)$ is the minimal conservative extension of β that is closed under the rules: any other R-closed state that already contains β must also contain all iterated consequences, hence must dominate their join. $\square$

Theorem 104 (Closure operator laws)**.** Cl_R *is extensive, monotone, and idempotent:*

$$\beta \leq \mathrm{Cl}_R(\beta), \qquad \beta \leq \gamma \Rightarrow \mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_R(\gamma), \qquad \mathrm{Cl}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta).$$

Proof. Extensive holds by Theorem 2. Monotone: if $\beta \leq \gamma$, then $\mathrm{F}_R^n(\beta) \leq \mathrm{F}_R^n(\gamma)$ for all n by Lemma 1, so joining over n preserves the inequality. Equivalently, starting from a stronger base state can only enlarge (never shrink) the set of derivable consequences, because the entire iteration is built from monotone steps and the final closure is obtained by a join.

Idempotent: $\mathrm{Cl}_R(\beta)$ is a fixed point by Theorem 2, so it is the least fixed point above itself, hence applying Cl_R again yields the same state. Operationally, idempotence says that once a belief state has been fully closed under R, running the closure procedure again produces no additional increase: closure is a "saturation" operation. $\square$

Lemma 163 (Warm-start / incremental closure)**.** *For any belief state* β *and evidence increment* ε*,*

$$\mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \oplus \varepsilon) \ = \ \mathrm{Cl}_R(\beta \oplus \varepsilon).$$

Proof. Because $\beta \leq \mathrm{Cl}_R(\beta)$ and $\oplus$ is monotone, $\beta \oplus \varepsilon \leq \mathrm{Cl}_R(\beta) \oplus \varepsilon$. Monotonicity of Cl_R gives $\mathrm{Cl}_R(\beta \oplus \varepsilon) \leq \mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \oplus \varepsilon)$. This direction formalizes the idea that if one first closes β and then adds ε, one cannot end up with less after reclosing than if one had added ε earlier.

Conversely, $\mathrm{Cl}_R(\beta \oplus \varepsilon)$ is R-closed and dominates both β and ε. Since $\mathrm{Cl}_R(\beta)$ is the least R-closed state dominating β, we have $\mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_R(\beta \oplus \varepsilon)$. Hence $\mathrm{Cl}_R(\beta) \oplus \varepsilon \leq \mathrm{Cl}_R(\beta \oplus \varepsilon)$, and applying Cl_R and idempotence yields $\mathrm{Cl}_R(\mathrm{Cl}_R(\beta) \oplus \varepsilon) \leq \mathrm{Cl}_R(\mathrm{Cl}_R(\beta \oplus \varepsilon)) = \mathrm{Cl}_R(\beta \oplus \varepsilon)$. Intuitively, the key step is the minimality property of $\mathrm{Cl}_R(\beta)$: any closed state that already contains β (in particular $\mathrm{Cl}_R(\beta \oplus \varepsilon)$) must be at least as large as $\mathrm{Cl}_R(\beta)$, so "pre-closing" cannot provide any extra advantage beyond what reclosing after adding ε would already compute. □

Corollary 42 (Batching and order-independence of multiple evidence updates)**.** *Let $\varepsilon_1, \ldots, \varepsilon_n : L \to V$ and define $\beta_{k+1} := \mathrm{Cl}_R(\beta_k \oplus \varepsilon_{k+1})$. Then*

$$\beta_n \;=\; \mathrm{Cl}_R(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n),$$

so the final closed belief state depends only on the join-aggregate of increments, not the order or interleaving of closure steps.

Proof. Induct on n. For the step $n \to n+1$:

$$\beta_{n+1} = \mathrm{Cl}_R(\beta_n \oplus \varepsilon_{n+1}) = \mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus \cdots \oplus \varepsilon_n) \oplus \varepsilon_{n+1}) = \mathrm{Cl}_R(\beta_0 \oplus \cdots \oplus \varepsilon_{n+1}),$$

using Lemma 4 in the middle equality. Commutativity/associativity of $\oplus$ yields order-independence. Concretely, this shows that alternating "close, then incorporate the next increment" is equivalent to a single batch update followed by one closure, which is exactly the property exploited by warm-start implementations: one may cache an already closed state and safely reuse it when new evidence arrives, without worrying about the precise sequencing of previous closure steps. □

55.3 Thinking episodes as confluent reasoning loops

Definition 493 (Hypothesis injection)**.** *Fix $h \in L$ and a strength $s \in V$. Define $\delta_{h,s} : L \to V$ by $\delta_{h,s}(h) = s$ and $\delta_{h,s}(\varphi) = \bot$ for $\varphi \neq h$. Define hypothesis injection by $\beta \uparrow^s h := \beta \oplus \delta_{h,s}$.*

In words, $\delta_{h,s}$ is the "minimal" increment that touches only a single sentence h, leaving all other sentences at the null contribution $\bot$. Thus $\beta \uparrow^s h$ is not a special-purpose operator so much as a convenient name for a particular kind of evidence join: it adds support of magnitude s to h and does nothing elsewhere. In particular, if $\oplus$ is interpreted pointwise (as in the standard product order on V^L), then for any $\varphi \neq h$ we have $(\beta \uparrow^s h)(\varphi) = \beta(\varphi) \oplus \bot$, while at h we obtain $(\beta \uparrow^s h)(h) = \beta(h) \oplus s$. This makes explicit that "injecting a hypothesis" is operationally the same kind of update as joining ordinary evidence, except that the increment is concentrated on a single claim.

Definition 494 (Abstract reasoning loop)**.** *A (finite) reasoning loop is any finite sequence of operations applied to an initial β_0 where each step is one of: (i) evidence join $\beta \leftarrow \beta \oplus \varepsilon$, (ii) hypothesis injection $\beta \leftarrow \beta \uparrow^s h$, (iii) closure $\beta \leftarrow \mathrm{Cl}_R(\beta)$.*

The finiteness condition is important here because the conclusion below is stated for a completed episode with a definite final state. Conceptually, however, the definition is designed so that intermediate closures are allowed at arbitrary times (e.g. after each observation, after each derived consequence, or on demand for query answering), and the result will state that these scheduling choices do not affect the final outcome once all increments have been incorporated. Note also that (ii) is syntactic sugar for a particular instance of (i); it is separated only because it is a common kind of step in practical loops, where one temporarily posits a conjecture h with a chosen "working strength" s and then explores its consequences under closure.

Theorem 105 (Confluence of reasoning loops: "close once at the end")**.** *Let a reasoning loop start at β_0 and incorporate evidence increments $\varepsilon_1, \ldots, \varepsilon_n$ and hypothesis injections $\delta_{h_1,s_1}, \ldots, \delta_{h_m,s_m}$, in any order, with Cl_R applied any number of times in between. Then the final belief state equals*

$$\mathrm{Cl}_R\Big(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n \oplus \delta_{h_1,s_1} \oplus \cdots \oplus \delta_{h_m,s_m}\Big).$$

This is a confluence/normal-form statement: among all "interleavings" of join-updates and closures, every complete finite episode reaches the same result as the canonical schedule "accumulate all increments by $\oplus$, then apply closure once." Equivalently, the closure operator Cl_R behaves like a saturating compilation step that can be cached and postponed without loss, provided no other kind of revision operator is present. The theorem therefore justifies a common implementation pattern: perform cheap incremental joins as data arrive, and only occasionally (or at query time) run the more expensive closure/saturation routine, knowing that doing so earlier would not have changed the eventual closed state.

Proof. Every step is either a join with some increment or an application of Cl_R. By Lemma 4, any application of Cl_R can be commuted past later joins (up to equality) and pushed to the end. By associativity/commutativity of $\oplus$, all joins can be batched. Finally, idempotence collapses multiple closures to one. □

Unpacking the commutation argument: if at some stage we have β and later we will join an increment γ, Lemma 4 provides $\mathrm{Cl}_R(\beta) \oplus \gamma = \mathrm{Cl}_R(\beta \oplus \gamma)$ (or, depending on the precise statement of Lemma 4 used earlier, an equivalent equality that allows rewriting a "close-then-join" segment into a "join-then-close" segment). Repeatedly applying this rewrite to each closure occurrence moves that closure step rightward until no joins remain to its right. Performing this for each closure step yields a sequence where all joins occur first and all closures occur last. Since $\oplus$ is associative and commutative, the order of the join increments is irrelevant to the resulting batch-join. Since Cl_R is idempotent, any suffix $\mathrm{Cl}_R(\mathrm{Cl}_R(\cdot))$ reduces to a single $\mathrm{Cl}_R(\cdot)$, yielding the claimed normal form.

Theorem 106 (Monotone-update "no retraction" principle)**.** *Suppose a belief dynamics uses only operations of the form $\beta \leftarrow \beta \oplus \varepsilon$ and $\beta \leftarrow \mathrm{Cl}_R(\beta)$ (no other revision operator). Then belief states are monotone in time: once a value is achieved at a claim, it never decreases. Formally, along the trajectory $\beta_0, \beta_1, \ldots$, we have $\beta_t \leq \beta_{t+1}$ for all t.*

The pointwise inequality $\beta_t \leq \beta_{t+1}$ should be read as: for every sentence $\varphi \in L$, the strength assigned to φ at time $t+1$ is at least its strength at time t under the order on V. In particular, even if closure introduces additional derived support for some sentences, closure cannot reduce any previously held support. Thus, without an explicit operator that can lower values, the dynamics is inherently cumulative. This captures formally the intuition that join-based updates behave like adding information to a ledger: later steps may add new entries or strengthen existing ones, but they never erase.

Proof. A join update is inflationary since $\beta \leq \beta \oplus \varepsilon$ pointwise. Closure is extensive: $\beta \leq \mathrm{Cl}_R(\beta)$ by Theorem 3. Thus each step is inflationary, hence the whole trajectory is monotone. □

Concretely, for the join step, inflationarity means $\beta(\varphi) \leq (\beta \oplus \varepsilon)(\varphi)$ for each φ, which follows from the defining property of $\oplus$ as a least upper bound / join-like operator in the underlying order. For the closure step, extensivity means closure only adds what is required by R-consequence and never subtracts: $\mathrm{Cl}_R(\beta)$ is a completion of β that contains β as a substate. Chaining these inequalities along the update sequence yields the trajectory monotonicity.

Remark 3099 (Practical meaning). *This explains a standard design choice: if you want genuine retraction, you must represent it explicitly (e.g. add counter-evidence in a paraconsistent negative channel, or move to a time-indexed belief state and reason about change). In the pure join+closure dynamics, you get strong "anytime" and caching properties, but not deletion.*

A related operational consequence is that intermediate results are safe to memoize: because later updates only increase the belief state, any previously computed closed state can be reused as a lower bound and incrementally strengthened rather than recomputed from scratch, provided the implementation supports incremental maintenance of joins and closure. Conversely, domains that require non-monotone belief change (e.g. belief revision with inconsistency repair, or model selection where hypotheses are discarded) must extend the loop with an explicit weakening step or a separate bookkeeping structure (such as provenance, timestamps, or competing "possible worlds"), since the present loop semantics alone cannot express "undo" or "forget."

55.4 Explanation/provenance semantics and relevance pruning

Definition 495 (Derivation trees and their weights). *Fix R and an initial belief state β_0. A derivation tree for a claim ψ is a finite rooted tree whose nodes are labeled by claims, such that: (i) each internal node labeled χ is annotated by a rule $r = (\Gamma, \chi, \lambda_r) \in R$, and has one child for each premise in Γ, labeled by that premise; (ii) leaves are labeled by claims but carry no rule annotation.*

Define the weight of a derivation tree T recursively: if T is a leaf labeled φ, then $w_{\beta_0}(T) := \beta_0(\varphi)$; if T has root rule $r = (\Gamma, \psi, \lambda)$ and subtrees $(T_\varphi)_{\varphi\in\Gamma}$, then

$$w_{\beta_0}(T) \ := \ \lambda \otimes \otimes_{\varphi\in\Gamma} w_{\beta_0}(T_\varphi).$$

Let $\mathrm{Deriv}_n(\psi)$ *denote derivation trees for ψ of height at most n.*

The intended reading is that each internal node contributes the "strength" λ of the rule instance, while each leaf contributes whatever initial support β_0 assigns to the corresponding claim. Thus, T serves simultaneously as (i) an *explanation object* (it records which rules and premises were used) and (ii) a *provenance object* (it supports quantitative ranking via the induced weight). The height bound in $\mathrm{Deriv}_n(\psi)$ corresponds to limiting the number of rule-application layers along any branch, i.e., a finite-depth or bounded-resource reasoning budget.

When a rule has an empty premise set $\Gamma = \varnothing$ (a "fact rule"), the expression $\otimes_{\varphi\in\Gamma} w_{\beta_0}(T_\varphi)$ should be understood as the empty tensor product, i.e., the multiplicative unit of the underlying $\otimes$-structure; then the tree weight reduces to λ. This convention matches the operational reading that a nullary rule provides direct support to its conclusion.

Theorem 107 (Derivation-tree semantics for finite-depth inference). *For every $n \geq 0$ and every claim $\psi \in L$,*

$$\mathrm{F}_R^n(\beta_0)(\psi) \ = \ \oplus_{T\in\mathrm{Deriv}_n(\psi)} w_{\beta_0}(T).$$

Consequently,

$$\mathrm{Cl}_R(\beta_0)(\psi) \ = \ \oplus_{T\in\mathrm{Deriv}(\psi)} w_{\beta_0}(T),$$

where $\mathrm{Deriv}(\psi) := \bigcup_{n\geq 0} \mathrm{Deriv}_n(\psi)$ *is the set of all finite derivation trees.*

Beyond giving a proof-theoretic description of F_R^n, the statement can be read as an "all-derivations" semantics: the value assigned to ψ after n rounds is exactly the aggregation (via $\oplus$) of all ways of deriving ψ using at most n layers of rule applications, where each way is scored by the tree weight. In particular, this isolates two orthogonal design choices: (a) the algebra $(\oplus, \otimes)$ determining how supports combine and (b) the rule set R determining which derivations exist.

Proof. Induct on n. Base $n = 0$: $\mathrm{F}_R^0(\beta_0) = \beta_0$, and $\mathrm{Deriv}_0(\psi)$ contains only the leaf tree labeled ψ, so both sides equal $\beta_0(\psi)$. Inductive step: assume the claim holds for n. Then

$$\mathrm{F}_R^{n+1}(\beta_0)(\psi) = \mathrm{F}_R(\mathrm{F}_R^n(\beta_0))(\psi) = \mathrm{F}_R^n(\beta_0)(\psi) \oplus \oplus_{r=(\Gamma,\psi,\lambda)\in R}\Big(\lambda \otimes \otimes_{\varphi\in\Gamma}\mathrm{F}_R^n(\beta_0)(\varphi)\Big).$$

By the induction hypothesis, each premise value $\mathrm{F}_R^n(\beta_0)(\varphi)$ is a join over weights of height-$\leq n$ derivations of φ. Since $\otimes$ distributes over joins, the tensor of premise-joins expands as a join over all choices of premise-derivations, which is exactly the join over weights of height-$\leq n+1$ derivations of ψ whose root uses rule r. Including the leftmost $\mathrm{F}_R^n(\beta_0)(\psi)$ term accounts for derivations where the root is a leaf. Thus the join equals $\oplus_{T\in\mathrm{Deriv}_{n+1}(\psi)} w_{\beta_0}(T)$.

Taking the join over n yields the closure formula. □

The distributivity step used above is precisely where the algebraic assumptions connect to explainability: it licenses moving from "a rule applied to aggregated premise-support" to "an aggregation over all concrete premise-explanations." Concretely, if $\mathrm{F}_R^n(\beta_0)(\varphi) = \oplus_i a_i$ and $\otimes$ distributes, then

$$\lambda \otimes \Big(\otimes_{\varphi\in\Gamma} \oplus_i a_{\varphi,i}\Big) \;=\; \oplus_{\text{choices } i(\cdot)} \; \lambda \otimes \otimes_{\varphi\in\Gamma} a_{\varphi,i(\varphi)},$$

so each selection of indices $i(\cdot)$ corresponds to choosing one derivation for each premise, i.e., to choosing one concrete derivation tree for the conclusion. This is the formal bridge from numeric/graded belief values to explicit provenance objects.

Remark 3100 (Why this is a decision tool)**.** *The theorem gives a canonical explanation/provenance object: every inferred belief value is an aggregate of derivation weights. A reasoning system can (a) store top-k derivations, (b) compute minimal-commitment explanations, or (c) locate which premises/rules are responsible for conflict by inspecting high-weight trees.*

In more operational terms, one can treat derivation trees as a search space and the weight function as an objective. For instance, when $\oplus$ is max and $\otimes$ is $+$ (or $\times$), the closure value is the score of the best-scoring derivation; when $\oplus$ is a sum-like aggregation, it behaves like total support accumulated from all derivations. In either case, the tree representation makes it possible to (i) return a human-readable justification (the subtree rooted at the conclusion), (ii) perform sensitivity analysis (how much a premise's initial value affects the conclusion), and (iii) implement debugging tools (identify a small set of high-impact rules contributing to an unexpected conclusion).

Definition 496 (Rule dependency graph and ancestor set)**.** *Define the (directed) rule dependency graph G_R on vertex set L with an edge $\varphi \to \psi$ iff there exists a rule $r = (\Gamma, \psi, \lambda) \in R$ with $\varphi \in \Gamma$. For $S \subseteq L$, define $\mathrm{Anc}_R(S)$ as the set of vertices that can reach some element of S via a directed path in G_R, together with S itself.*

The graph G_R is a purely syntactic abstraction of potential information flow: an edge $\varphi \to \psi$ asserts that changing support for φ *may* affect support for ψ via at least one rule, independent of weights and independent of whether φ is actually derivable in a given run. Accordingly, $\mathrm{Anc}_R(S)$ is a conservative relevance superset: it contains every claim that could appear as a leaf of a derivation tree for some $\psi \in S$. Computing $\mathrm{Anc}_R(S)$ amounts to a backward reachability query (reverse graph search) from S in G_R; this supports a standard relevance-pruning optimization in which one restricts attention to the sublanguage induced by $\mathrm{Anc}_R(S)$ when answering queries about S.

Theorem 108 (Ancestor locality / relevance pruning)**.** *Let $S \subseteq L$ and let β_0, β_0' be initial belief states that agree on $\mathrm{Anc}_R(S)$. Then the closures agree on S:*

$$\forall\psi \in S, \quad \mathrm{Cl}_R(\beta_0)(\psi) = \mathrm{Cl}_R(\beta_0')(\psi).$$

Proof. Fix $\psi \in S$. By Theorem 8, $\mathrm{Cl}_R(\beta_0)(\psi)$ is the join over weights of derivation trees for ψ. Every leaf label appearing in any derivation tree for ψ must lie in $\mathrm{Anc}_R(\{\psi\}) \subseteq \mathrm{Anc}_R(S)$ because each application of a rule introduces premises connected by edges to the conclusion. Hence, for every derivation tree T for ψ, its weight $w_{\beta_0}(T)$ depends only on β_0 restricted to $\mathrm{Anc}_R(S)$, and therefore equals $w_{\beta'_0}(T)$. Thus the joins over all trees are equal. $\square$

A useful way to internalize the proof is: relevance is determined by *which leaves can ever appear* in a derivation of the target. Since internal nodes are created only by applying rules, and each rule application moves along edges of G_R from premises to conclusion, any leaf that contributes to a derivation of ψ must have a path to ψ and thus lies in the ancestor set. The theorem therefore justifies a correctness-preserving implementation trick: if the goal is to answer queries in S, then one may replace β_0 by its restriction to $\mathrm{Anc}_R(S)$ (setting all other claims to arbitrary values, including $\perp$) without changing the answers on S. This is the formal sense in which "unrelated" parts of the belief base are operationally irrelevant.

Corollary 43 (Non-explosion as operational irrelevance)**.** *If $\psi \in L$ is not reachable from any claim with non-bottom initial value (and not reachable from any injected hypothesis), then $\mathrm{Cl}_R(\beta_0)(\psi) = \beta_0(\psi)$. In particular, if $\beta_0(\psi) = \perp$ and ψ has no derivation tree, then $\mathrm{Cl}_R(\beta_0)(\psi) = \perp$.*

Proof. If there is no derivation tree for ψ other than the trivial leaf, then Theorem 8 yields $\mathrm{Cl}_R(\beta_0)(\psi) = \beta_0(\psi)$. If additionally $\beta_0(\psi) = \perp$ then the value remains bottom. $\square$

This corollary is best read as a guardrail for scaling: even if the global rule base R is large, only the portion that is (i) upstream of the query and (ii) seeded by non-bottom information can affect the outcome. In particular, one avoids a form of "explosion" in which irrelevant axioms or rules inadvertently influence unrelated queries: if there is no derivational route from supported claims to ψ, then ψ cannot gain additional support. Equivalently, if ψ is outside the forward reachability region of the initially supported set, then the closure operator leaves ψ unchanged.

55.5 Cycles, modularity, and question networks

Definition 497 (Strongly connected components (SCCs) and condensation DAG)**.** *Let G_R be the dependency graph. Let $\mathcal{C} = \{C_1, \dots, C_k\}$ be its SCC partition. The condensation graph $\mathrm{Cond}(G_R)$ is the DAG whose nodes are SCCs and with an edge $C_i \to C_j$ if there exists $\varphi \in C_i$ and $\psi \in C_j$ with an edge $\varphi \to \psi$ in G_R. Fix a topological order of SCCs so edges go from lower to higher indices. Let $P_i := \bigcup_{j \le i} C_j$ be the prefix set. Intuitively, each SCC C_i is a "cycle-cluster" of mutually interdependent formulas, while the condensation DAG records one-way informational flow between these clusters. Because $\mathrm{Cond}(G_R)$ is acyclic, the topological order ensures that any dependency path that enters an SCC C_i can only come from SCCs among $\{C_1, \dots, C_{i-1}\}$, never from future SCCs.*

Theorem 109 (SCC modularization of closure)**.** *Assume L is finite (or effectively finite for the active reasoning slice). Let SCCs be topologically ordered as above. In particular, the finiteness/effective-finiteness assumption ensures that least fixed points used below exist and are attainable by iteration within each SCC (e.g. via a standard monotone-operator convergence argument on a finite product order).*

Define an iterative construction:

- *Let $\beta^{(0)} := \beta_0$.*

- *For $i = 1, \dots, k$, compute the least R-closed extension of $\beta^{(i-1)}$ on the SCC C_i while keeping values on $L \setminus C_i$ fixed, i.e. compute*

$$\beta^{(i)} := \min\{\gamma : L \to V \ : \ \beta^{(i-1)} \le \gamma,\ \gamma|_{L\setminus C_i} = \beta^{(i-1)}|_{L\setminus C_i},\ \gamma \textit{ is } R\textit{-closed on conclusions in } C_i\}.$$

Then $\beta^{(k)} = \mathrm{Cl}_R(\beta_0)$.

Equivalently: closure can be computed SCC-by-SCC in condensation topological order; cycles are isolated to SCC-local fixed point computations. Equivalently again, if one views R-closure as the least fixed point of a global (monotone) consequence operator on V^L, then the condensation ordering provides a factorization of that global fixed point into a sequence of local fixed points, each taken under boundary conditions established by earlier SCCs.

Proof. Key fact: If $\psi \in P_i$, then every ancestor of ψ lies in P_i by topological ordering of SCCs, so $\mathrm{Anc}_R(\{\psi\}) \subseteq P_i$. To see this, note that any directed path in G_R from some φ to ψ induces a path in $\mathrm{Cond}(G_R)$ from the SCC of φ to the SCC of ψ; by topological ordering, that SCC index cannot exceed the index of the SCC containing ψ, so φ must lie in P_i.

Let $\beta^* := \mathrm{Cl}_R(\beta_0)$. We prove by induction on i that $\beta^{(i)}$ agrees with β^* on P_i.

Base $i = 0$ is trivial for the empty prefix. (Equivalently, $\beta^{(0)} = \beta_0$ provides the initial boundary condition before any SCC-local closure is performed.)

Inductive step: assume agreement on P_{i-1}. Consider any $\psi \in C_i$. All rules concluding ψ have premises in $\mathrm{Anc}_R(\{\psi\}) \subseteq P_i$, so by Theorem 9 the value $\beta^*(\psi)$ depends only on β_0 restricted to P_i and on closure within P_i. When computing $\beta^{(i)}$, we keep $L \setminus C_i$ fixed at $\beta^{(i-1)}$, which by induction matches β^* on P_{i-1}. Thus the SCC-local closure computation within C_i is exactly the least fixed point consistent with the already-correct boundary conditions from P_{i-1}, hence matches β^* on C_i by minimality/least-fixed-point uniqueness on the finite slice. More explicitly, restricting attention to the coordinates in C_i, the closure conditions induced by rules with conclusions in C_i define a monotone update operator on V^{C_i} whose parameters are the already-fixed values on P_{i-1}; the construction of $\beta^{(i)}$ chooses the least post-fixed point extending those parameters, and β^* is itself such a post-fixed point, so the least one must coincide with β^* on C_i.

Therefore $\beta^{(i)}$ matches β^* on P_i. At $i = k$, $P_k = L$, giving $\beta^{(k)} = \beta^*$. □

Remark 3101 (Practical meaning)**.** *This theorem is a blueprint for scalable reasoning: build the rule dependency graph, condense SCCs, and run fixed point solvers only inside SCCs; outside SCCs, do a single topological pass. Cycles become localized engineering objects. In implementations, this often translates into (i) a one-time graph analysis step (SCC decomposition is linear-time in $|V|+|E|$), followed by (ii) per-SCC propagation that can use specialized solvers depending on the rule fragment (e.g. Horn-like propagation, constraint propagation, or abstract interpretation) without re-solving already-settled acyclic regions. A further consequence is that independent SCCs with no edges between them in $\mathrm{Cond}(G_R)$ can be processed in parallel once their respective predecessor SCCs have stabilized.*

Definition 498 (Question networks and cycle breaking)**.** *A question network is a directed graph $Q = (V, E)$ whose nodes are questions and edges $u \to v$ mean "answering u requires (or strongly benefits from) answering v". A feedback vertex set (FVS) is $H \subseteq V$ such that removing H makes the graph acyclic. This notion is a direct analogue of SCC condensation for reasoning tasks: SCCs identify mutually entangled questions, while an FVS provides an explicit set of "cycle breakers" that, once provisionally resolved, turns the remaining agenda into a schedulable acyclic plan.*

Theorem 110 (Cycle-breaking by hypotheses yields schedulable question answering)**.** *Let $Q = (V, E)$ be a finite question network and let H be any feedback vertex set. Then $Q \setminus H$ is acyclic* *and admits a topological order, so questions in $V \setminus H$ can be answered in a dependency-respecting sequence. If hypotheses are injected to provide provisional answers for H, then any interleaving of the remaining answers with closure yields the same final state as "batch then close" (Theorem 6). In other words, choosing H turns an otherwise cyclic information-gathering process into an iterative loop with well-defined stages: (a) posit tentative values on H, (b) propagate consequences through the acyclic remainder, (c) optionally revise hypotheses, and (d) repeat, with Theorem 6 guaranteeing order-independence of the propagation steps within each stage.*

Proof. By definition of feedback vertex set, removing H yields an acyclic digraph. Every finite acyclic digraph admits a topological ordering (standard). The second claim follows by treating each provisional hypothesis and each answered-question datum as an increment and applying Theorem 6. Concretely, once the non-hypothesis questions are arranged in a topological order, each answered question can be added as an update that is immediately closed under R; Theorem 6 ensures that whether one closes after each individual answer or after collecting a batch of answers, the eventual R-closed result is the same (given the same injected hypotheses on H). □

55.6 Hypothesis sensitivity as a controlled exploration knob

Theorem 111 (Monotone hypothesis sensitivity)**.** *Fix R, an initial β, a claim h, and strengths $s \le s'$ in V. Then for all $\psi \in L$,*

$$\mathrm{Cl}_R(\beta \uparrow^s h)(\psi) \;\le\; \mathrm{Cl}_R(\beta \uparrow^{s'} h)(\psi).$$

Proof. Since $s \le s'$, we have $\delta_{h,s} \le \delta_{h,s'}$ pointwise, hence $\beta \oplus \delta_{h,s} \le \beta \oplus \delta_{h,s'}$. Apply monotonicity of Cl_R (Theorem 3). In other words, raising the strength parameter cannot lower the pre-closure belief value of any formula, and closure preserves this ordering because it is an order-preserving map on belief states. Equivalently, the operator $\beta \mapsto \mathrm{Cl}_R(\beta \uparrow^s h)$ is monotone in the knob s when all other inputs are held fixed. □

Remark 3102 (Practical meaning)**.** *This supports "turn the hypothesis dial" search strategies: when strengths live on a chain (e.g. $s = (t, \bot)$ with scalar t), the induced consequences are monotone, enabling threshold search. Concretely, if one is looking for the least strength s that makes a target consequence ψ reach a desired acceptance level (e.g. $\mathrm{Cl}_R(\beta \uparrow^s h)(\psi) \ge \tau$ for some $\tau \in V$), then monotonicity guarantees that the set of strengths satisfying the constraint is upward-closed. This is precisely the condition needed for efficient one-dimensional search procedures such as bisection (when the chain is discrete or admits suitable midpoints) or simple incremental stepping (when the chain is finite). It also justifies sensitivity analysis: one can rank downstream consequences by how early (at what knob setting) they appear, and interpret earlier appearance as greater logical dependence on h under R and β.*

55.7 Science as empirically coupled evidence-batching

Definition 499 (Empirical increment and scientific update operator)**.** *Let E be experiments and O outcomes. Suppose each observed result (e, o) produces an increment $\varepsilon_{e,o} : L \to V$. Define the scientific update step (given rule set R) by*

$$U_{e,o}(\beta) \;:=\; \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o}).$$

The intended reading is that $\varepsilon_{e,o}$ records the direct empirical push supplied by the observation (e, o), while $\mathrm{Cl}_R(\cdot)$ propagates that push through the inferential constraints in R. Thus $U_{e,o}$ separates "data entry" (via $\oplus$ with $\varepsilon_{e,o}$) from "theory-driven closure" (via Cl_R), mirroring the practical loop of accumulating measurements and then recomputing derived commitments.

Theorem 112 (Reproducibility invariance: experiments batch and commute)**.** *For any finite sequence* $(e_1, o_1), \ldots, (e_n, o_n)$*,*

$$U_{e_n,o_n} \circ \cdots \circ U_{e_1,o_1}(\beta_0) \;=\; \mathrm{Cl}_R\Big(\beta_0 \oplus \varepsilon_{e_1,o_1} \oplus \cdots \oplus \varepsilon_{e_n,o_n}\Big).$$

Hence the final closed belief state depends only on the join-aggregate of empirical increments, not on the order of experiments or how frequently closure is re-run.

Proof. This is exactly Corollary 1 (batching/order-independence) applied with increments $\varepsilon_k := \varepsilon_{e_k,o_k}$. Intuitively, $\oplus$ acts as an order-insensitive accumulator of the raw experimental contributions, while the closure operator Cl_R can be postponed until after accumulation without changing the result. Operationally, this means one may either (i) close after each new datum arrives, or (ii) store increments and close only periodically, and both pipelines coincide at the level of the final closed state. □

A useful special case is when multiple outcomes for the same experimental condition are merged into a single increment (e.g. by taking their join as a coarse-grained summary); the theorem shows that any such batching scheme is equivalent to processing each outcome individually and closing at the end, provided the same overall join-aggregate is obtained.

Corollary 44 (Idempotence of repeating the same experimental result)**.** *If the same empirical increment* ε *is joined twice, it has no further effect:*

$$\mathrm{Cl}_R\big(\beta \oplus \varepsilon \oplus \varepsilon\big) = \mathrm{Cl}_R(\beta \oplus \varepsilon).$$

Proof. $\oplus$ is idempotent: $\varepsilon \oplus \varepsilon = \varepsilon$ pointwise. Apply Theorem 14. Equivalently, the second occurrence of ε does not contribute any additional information beyond what is already present after the first join, so closure sees identical inputs on both sides. □

Remark 3103 (Interpretation and scope)**.** *The corollary formalizes a minimal "reproducibility sanity check": exact repetition of the same encoded outcome cannot, by itself, further change the closed beliefs. This is not a claim that repeating a physical experiment is uninformative; rather, it says that if repetition yields the same increment object* ε *(i.e. the same update signal in* V^L*), then the algebra of evidence accumulation treats it as a duplicate. When one wishes repeated trials to strengthen confidence, that strengthening must be represented at the level of the increment map itself—for example, by having multiple trials produce distinct increments whose join is strictly larger, or by defining* $\varepsilon_{e,o}$ *to already encode sample size or reliability.*

55.8 Methodology refinement and collective synergy

In this subsection we make precise two closely related ideas that often occur in practical reasoning loops: (a) refining a methodology by strengthening existing inferential links and/or adding new ones should not reduce what can be concluded, and (b) pooling partially informative belief states can unlock conclusions that neither agent can reach alone. Both effects are captured abstractly by the order-theoretic behavior of the one-step operator F_R and its induced closure Cl_R, with all inequalities understood pointwise in the underlying truth-value / confidence order.

Definition 500 (Rule-set refinement preorder)**.** *Given rule sets* R, R'*, write* $R \preceq R'$ *if: (i) for every rule* $(\Gamma, \psi, \lambda) \in R$ *there is a rule* $(\Gamma, \psi, \lambda') \in R'$ *with* $\lambda \leq \lambda'$*, and (ii)* R' *may contain additional rules not in* R*.*

Intuitively, $R \preceq R'$ means that R' is at least as "methodologically strong" as R: no previously available inference is weakened (the same premises Γ still support the same conclusion ψ with at least the same strength), and one is permitted to introduce additional inferential routes. The relation is a preorder rather than a partial order because different rule sets can be mutually refining if they contain duplicates or different collections of redundant rules that nonetheless yield the same operational behavior under F and Cl.

Theorem 113 (Methodology refinement monotonicity). *If $R \preceq R'$, then for all belief states β,*

$$\mathrm{F}_R(\beta) \leq \mathrm{F}_{R'}(\beta) \qquad \text{and} \qquad \mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_{R'}(\beta).$$

This theorem formalizes the idea that improving a protocol, measurement pipeline, or background theory—modeled here by strengthening rule weights λ and adding rules—cannot decrease the immediate inferential update (F) nor the eventual stable conclusions reached by iterating that update (Cl). In particular, it separates two mechanisms of refinement: increasing λ for existing rules increases the contribution of each such rule, while adding rules enlarges the join over available derivations.

Proof. Fix ψ. In $\mathrm{F}_{R'}(\beta)(\psi)$, the join ranges over (at least) all rules concluding ψ that appear in R, and for each such rule the contribution is multiplied by a strength $\lambda' \geq \lambda$, so each corresponding contribution in R' is $\geq$ the one in R. Thus the entire join is larger, hence $\mathrm{F}_R(\beta) \leq \mathrm{F}_{R'}(\beta)$ pointwise. By monotonicity of iteration and join over iterates, $\mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_{R'}(\beta)$. □

To unpack the last sentence: if we write $\mathrm{F}_R^{(n)}$ for the n-fold iterate of F_R, then the already-established pointwise inequality $\mathrm{F}_R \leq \mathrm{F}_{R'}$ (as operators at the fixed input β) propagates to $\mathrm{F}_R^{(n)}(\beta) \leq \mathrm{F}_{R'}^{(n)}(\beta)$ for all n by induction using monotonicity of F in its argument. Taking the join over all iterates (the definition of closure in this framework) preserves the inequality because joins are monotone in each component. Conceptually, refinement cannot remove reachable states from the iteration chain; it can only push some coordinates upward and possibly accelerate convergence.

Theorem 114 (Pooling beliefs is super-additive; distributed-premise synergy criterion). *For any belief states β, γ,*

$$\mathrm{Cl}_R(\beta \oplus \gamma) \;\geq\; \mathrm{Cl}_R(\beta) \oplus \mathrm{Cl}_R(\gamma).$$

Moreover, suppose there exists a rule $r = (\Gamma, \psi, \lambda) \in R$ and a partition $\Gamma = \Gamma_1 \cup \Gamma_2$ such that:

$$\forall \varphi \in \Gamma_1 : \; \beta(\varphi) > \perp \;\wedge\; \gamma(\varphi) = \perp, \qquad \forall \varphi \in \Gamma_2 : \; \gamma(\varphi) > \perp \;\wedge\; \beta(\varphi) = \perp,$$

and $\beta(\psi) = \gamma(\psi) = \perp$ and no other rule derives ψ. Then

$$\mathrm{Cl}_R(\beta)(\psi) = \mathrm{Cl}_R(\gamma)(\psi) = \perp \quad \text{but} \quad \mathrm{Cl}_R(\beta \oplus \gamma)(\psi) \;\geq\; \lambda \otimes \otimes_{\varphi \in \Gamma}(\beta \oplus \gamma)(\varphi) \;>\; \perp,$$

so the inequality is strict at ψ.

The first inequality is a general "super-additivity" (more precisely, super-join-preservation) statement: closing after pooling is at least as informative as pooling after closing each agent separately. The second part gives an explicit and checkable sufficient condition for *strict* improvement, capturing a familiar pattern in collaborative science: one group has evidence for some premises, another group has evidence for the complementary premises, and only the pooled record supports the full premise set Γ required by the rule r. The assumptions $\beta(\psi) = \gamma(\psi) = \perp$ and "no other rule derives ψ" isolate the effect, ensuring that ψ cannot become supported except through this particular distributed-premise pathway.

Proof. Since $\beta \le \beta \oplus \gamma$ and $\gamma \le \beta \oplus \gamma$, monotonicity of Cl_R gives $\mathrm{Cl}_R(\beta) \le \mathrm{Cl}_R(\beta \oplus \gamma)$ and $\mathrm{Cl}_R(\gamma) \le \mathrm{Cl}_R(\beta \oplus \gamma)$. Joining these inequalities yields $\mathrm{Cl}_R(\beta) \oplus \mathrm{Cl}_R(\gamma) \le \mathrm{Cl}_R(\beta \oplus \gamma)$.

For strictness: under the stated premises, in $\mathrm{F}_R(\beta)(\psi)$ the contribution of rule r includes a tensor factor $\beta(\varphi) = \bot$ for some $\varphi \in \Gamma_2$, so the rule contribution is $\bot$; with no other rules and base $\beta(\psi) = \bot$, ψ remains $\bot$ under all iterates, hence under closure. Similarly for γ. But in $\beta \oplus \gamma$, every premise in Γ is $> \bot$ by construction, so the one-step contribution of r to ψ is $> \bot$, hence closure yields $> \bot$ at ψ. □

One can view the strictness argument as identifying a minimal "cut" in the support graph of ψ: each individual belief state fails to activate r because at least one prerequisite premise sits at $\bot$, and because there is no alternative derivation, the iterate sequence cannot raise ψ above $\bot$. Pooling via $\oplus$ removes these bottlenecks by taking the least upper bound of the two information sources premise-wise, after which the tensor aggregation across Γ yields a strictly positive combined premise support, which is then scaled by λ. In methodological terms, this provides a formal criterion for when collaboration yields genuinely new conclusions rather than merely averaging or redundantly confirming what was already derivable.

Corollary 45 (Protocol quality monotonicity)**.** *If two experimental protocols yield increments $\varepsilon \le \varepsilon'$ (pointwise), then for any β,*

$$\mathrm{Cl}_R(\beta \oplus \varepsilon) \;\le\; \mathrm{Cl}_R(\beta \oplus \varepsilon').$$

Here ε and ε' can be read as external evidence injections (e.g., measurement results, curated dataset additions, or validated intermediate findings) that are pooled into the current belief state β via $\oplus$. The pointwise comparison $\varepsilon \le \varepsilon'$ models the idea that the improved protocol delivers evidence that is at least as informative at every proposition, so subsequent closure under the same inferential rules should not be weakened.

Proof. $\beta \oplus \varepsilon \le \beta \oplus \varepsilon'$ by monotonicity of $\oplus$. Apply monotonicity of Cl_R. □

56 Mind–World Correspondence and Reality-Systems: Transfer, Shared Reality, and Robustness

We state and prove reusable theorems for reasoning engines that represent internal and external structure as quantale-weighted directed graphs and relate them via approximate morphisms. The results provide: (i) quantitative transfer bounds for multi-step inferences and pattern constraints under mind–world correspondence; (ii) fixed-point constructions for shared reality across multiple minds; and (iii) robustness criteria ensuring that correspondences and reality-systems persist under perturbations of weights or thresholds.

Remark 3104. *This section is best read as a small "transport calculus": we have a graph-like structure inside a mind (or model) and a graph-like structure outside (or in another model), and we track what can be reliably carried across the correspondence. This is a formalization of Mind–World Correspondence in the sense of Hyperseed-Concept 112, with a deliberately quantitative flavor (the correspondence has "slack" or "quality"). Related motivations appear in the Hyperseed presentation [1] and in transfer-learning formalisms such as [7].*

56.1 Standing setting

Definition 501 (Quantale with residuation)**.** *Let $(V, \leq, \otimes, e)$ be a commutative quantale that is also a complete lattice. Assume it is* residuated*: for each $u \in V$, the map $u \otimes (-)$ has a right adjoint, written $(u \to -)$, characterized by*

$$u \otimes w \leq v \quad \textit{iff} \quad w \leq (u \to v).$$

Remark 3105. *The residuation operation $(u \to v)$ is an "implication-like" quantity internal to the quantale: it measures how much additional weight w one must supply, when combined with u via $\otimes$, to stay below (or within) v. In logical terms, the adjunction says that multiplying evidence/resources by u and then comparing to v is equivalent to directly comparing the remaining allowance w to the residuum $u \to v$. This is one of the algebraic devices by which graded inference and graded constraint satisfaction can be carried out without leaving the lattice-theoretic world; see also the quantale-based treatment of graded "weakness" and slack in [3] (Hyperseed-Concept 202, Hyperseed-Concept 143).*

A simple anchor example is $V = \{0, 1\}$ with $\otimes = \wedge$ and $\leq$ the usual order: then $u \to v$ is the ordinary Boolean implication. Another common example is $V = [0, 1]$ ordered by $\leq$ with $\otimes$ a t-norm (e.g. multiplication); the residuum then encodes a familiar notion of "how much more confidence is permitted." The utility of residuation in this section is that it allows us to speak about degree of structure-preservation via a single meet of implications, and then to manipulate that degree monotonically.

Definition 502 (V-graphs and approximate morphisms)**.** *A V-graph is a pair $G = (X, G)$ where X is a set of nodes and $G : X \times X \to V$. Given V-graphs $G = (X, G)$ and $H = (Y, H)$ and a function $f : X \to Y$, define the* preservation degree

$$\deg(f; G, H) \;:=\; \bigwedge_{x, x' \in X} \Big(G(x, x') \to H(f(x), f(x')) \Big).$$

*For $q \in V$, we say f is a q-*approximate morphism $G \to H$ *if $q \leq \deg(f; G, H)$, equivalently*

$$G(x, x') \otimes q \leq H(f(x), f(x')) \qquad \forall x, x' \in X.$$

Remark 3106. *A V-graph is simply a directed graph whose edges are labeled by elements of V rather than by booleans. Intuitively, $G(x, x')$ is the strength with which the structure asserts (or supports) the directed relation $x \to x'$. When $V = \{0, 1\}$ we recover ordinary directed graphs, and when $V = [0, 1]$ we recover weighted graphs where larger numbers indicate stronger edges (Hyperseed-Concept 132, Hyperseed-Concept 131; compare [5] for pattern-web motivations).*

The preservation degree $\deg(f; G, H)$ is defined as a meet $\bigwedge$ (infimum) over all pairs (x, x') of the implication-like term $G(x, x') \to H(f(x), f(x'))$. Reading this in plain language: for each edge (or potential edge) in G, we ask "how well is it respected after mapping into H?"; then we take the worst case over all edges. The condition that f is a q-approximate morphism says that every *internal edge, after paying a uniform slack factor q, is still supported externally. In the frequent probabilistic intuition where $\otimes$ is multiplication, the inequality $G(x, x') \otimes q \leq H(f(x), f(x'))$ says: "external strength is at least internal strength times a global fidelity factor q."*

This notion is the algebraic core of mind–world correspondence: the mind's internal graph G can be transported to the world's graph H only up to a quantifiable correspondence quality q (Hyperseed-Concept 112). The use of a single scalar q is a deliberate discipline: it gives conservative bounds that can be composed over paths, yielding depth-dependent transfer penalties in the next subsection (a transfer-learning flavor; cf. [7]).

Definition 503 (Relational composition and path powers)**.** *For a V-graph $G = (X, G)$ define the (quantale) matrix composition*

$$(G \circ G)(x, z) := \bigvee_{y \in X} G(x, y) \otimes G(y, z).$$

Inductively define $G^{(1)} := G$ and $G^{(n+1)} := G^{(n)} \circ G$. (Thus $G^{(n)}(x, z)$ is the best weight of an n-step path from x to z.)

Remark 3107. *The notation $\bigvee$ denotes join (supremum) in the complete lattice V, so the formula*

$$(G \circ G)(x, z) = \bigvee_{y} G(x, y) \otimes G(y, z)$$

is the usual "sum over intermediates" familiar from matrix multiplication, except that addition is replaced by join $\bigvee$ and multiplication is replaced by $\otimes$. When $V = \{0, 1\}$ and $\otimes = \wedge$, this is exactly the standard composition of relations: there is an edge from x to z in $G \circ G$ iff there exists an intermediate y with edges $x \to y$ and $y \to z$ in G. When $V = [0, 1]$ with $\otimes$ multiplication and $\bigvee$ as max, it says that the best two-step path weight is the maximum over intermediates of the product of the two edge weights.

The path powers $G^{(n)}$ let us speak about multi-step inference or multi-step constraint propagation: $G^{(n)}(x, z)$ is the strongest support obtainable for a directed connection from x to z using exactly n hops. This "best path" semantics is the bridge between graph structure and reasoning depth that will appear explicitly in the transfer theorems.

Lemma 164 (Monotonicity of residuum (rewrite rule))**.** *For all $u, u', v, v' \in V$:*

$$u \le u' \Rightarrow (u' \to v) \le (u \to v), \qquad v \le v' \Rightarrow (u \to v) \le (u \to v').$$

Remark 3108. *This lemma records the two basic monotonicity directions of the residuum. It says that the implication-like quantity $u \to v$ is* antitone *in its left argument and* monotone *in its right argument: making the antecedent u larger (harder to satisfy) can only shrink the implication, while making the consequent v larger (easier to satisfy) can only enlarge it.*

The reason this matters is conceptual as well as technical: $\deg(f; G, H)$ *is built from many terms of the form $G(x, x') \to H(f(x), f(x'))$. Thus, if we later prune internal edges (make G smaller) or strengthen the external graph (make H larger), we can predict how the overall degree changes. This monotonicity will be used verbatim in the robustness theorems of Section 4.*

It is also helpful to register the order-theoretic content in a slightly more structural way: the residuum is the right adjoint of the map $w \mapsto u \otimes w$. Right adjoints are automatically monotone in the argument they act on (here, the consequent v), while the parameter u appears in the left-multiplication operator $(-) \mapsto u \otimes (-)$, whose strengthening makes the corresponding right adjoint smaller. In that sense, the lemma is not a special trick but a basic "variance rule" for implication in any residuated setting.

Finally, note that these directions match the familiar truth-functional intuition: to satisfy an implication, making the premise stronger can only make the implication harder to satisfy, while making the conclusion stronger (in the order of truth values) can only make the implication easier to satisfy. The lemma formalizes exactly this intuition in the general lattice-valued framework used throughout the paper.

Proof. By definition, $(u \to v) = \bigvee\{w : u \otimes w \le v\}$. If $u \le u'$, then for any w, $u \otimes w \le u' \otimes w$ (monotonicity of $\otimes$), so the constraint $u' \otimes w \le v$ is stronger than $u \otimes w \le v$; hence the feasible set for $u' \to v$ is contained in that for $u \to v$, giving $(u' \to v) \leq (u \to v)$. If $v \leq v'$, then $\{w : u \otimes w \leq v\} \subseteq \{w : u \otimes w \leq v'\}$, hence $(u \to v) \leq (u \to v')$.

To make the order reasoning completely explicit: in the first case, let $S(u, v) := \{w : u \otimes w \leq v\}$. From $u \leq u'$ and monotonicity of $\otimes$ in the first coordinate, we obtain $u \otimes w \leq u' \otimes w$ for each w, hence $u' \otimes w \leq v$ implies $u \otimes w \leq v$, i.e. $S(u', v) \subseteq S(u, v)$. Since $\bigvee S$ is the least upper bound of S (and $\bigvee \emptyset$ is well-defined in our ambient complete lattice), inclusion of feasible sets yields $\bigvee S(u', v) \leq \bigvee S(u, v)$, which is precisely $(u' \to v) \leq (u \to v)$. The second case is analogous: $v \leq v'$ makes the constraint $u \otimes w \leq v'$ weaker, so $S(u, v) \subseteq S(u, v')$, and taking suprema preserves the inequality in the expected direction.

Equivalently (and compatibly with the definition used here), one may restate residuation as the adjunction $u \otimes w \leq v \iff w \leq (u \to v)$. With that characterization, the same monotonicity directions follow by elementary order manipulation: increasing u makes $u \otimes w$ larger and thus the inequality $u \otimes w \leq v$ harder to satisfy, while increasing v makes it easier to satisfy. □

Remark 3109 (Proof sketch and intuition)**.** *The strategy is to unpack residuation as a supremum of all "admissible slacks" w such that $u \otimes w \leq v$. If you increase u to u', fewer w will remain admissible, so the supremum (the best admissible slack) cannot increase; this is the antitonicity in the left argument. If you increase v to v', more w become admissible, so the supremum cannot decrease; this is monotonicity in the right argument.*

One can picture $u \otimes w \leq v$ as a constraint region in the (u, w) plane: pushing u up tightens the region, pushing v up loosens it. The lemma is exactly the order-theoretic shadow of that geometric picture.

A complementary way to phrase the same intuition is to think of $u \to v$ as the largest *truth value w that can be conjoined (via $\otimes$) with u without exceeding v. Under this reading, a stronger antecedent u leaves less "room" for such a w, while a larger consequent v leaves more room. This is why the directions are sometimes summarized as contravariance in the antecedent and covariance in the consequent.*

In concrete t-norm semantics this behavior is familiar. For example, in the Gödel case ($\otimes = \min$) one has $u \to v = 1$ when $u \leq v$ and $u \to v = v$ otherwise, so increasing u can only move one from the 1-regime to the v-regime (never upward), while increasing v can only increase the output. In the Lukasiewicz case ($u \to v = \min(1, 1 - u + v)$), the same monotonicity is visible by inspection: the expression is decreasing in u and increasing in v. The abstract lemma ensures we do not need to commit to any particular t-norm; the robustness arguments later rely only on these order-theoretic directions.

56.2 Transfer theorems for inference and pattern constraints

A central use-case is: a reasoning engine performs multi-step inference internally, but wants to transport conclusions to (or from) the world graph using a correspondence map.

56.2.1 Pushforward/pullback as an adjoint pair

Definition 504 (Pullback and pushforward of V-relations along a map)**.** *Let $f : X \to Y$. For any V-relation $R : X \times X \to V$, define the* pushforward $f_! R : Y \times Y \to V$ *by*

$$(f_! R)(y, y') := \bigvee_{\substack{x, x' \in X: \\ f(x) = y,\ f(x') = y'}} R(x, x').$$

For any V-relation $S : Y \times Y \to V$, define the pullback $f^*S : X \times X \to V$ *by*

$$(f^*S)(x, x') := S(f(x), f(x')).$$

Remark 3110. *The definition implicitly uses that the join $\bigvee$ over (possibly many) witnesses exists in V. In most applications here, V is taken to be a complete lattice (or a quantale, so that joins interact well with $\otimes$), ensuring that $f_!R$ is always well-defined even when fibers of f are infinite. When $V = \mathbf{2} = \{\bot, \top\}$ with $\bigvee$ as logical "or," $(f_!R)(y, y')$ is true exactly when there exists at least one internal witness (x, x') with $f(x) = y, f(x') = y'$ and $R(x, x') = \top$; when $V = [0, 1]$ with $\bigvee = \sup$, $(f_!R)(y, y')$ retains the strongest available degree among all internal witnesses consistent with the external pair.*

Remark 3111. *The pullback f^*S is the easy direction: it simply re-labels S along f by substitution. If S is a relation on Y, then f^*S is the induced relation on X obtained by first mapping x and x' into Y and then reading off S there. This is the usual operation of "viewing external relations in internal coordinates."*

The pushforward $f_!R$ is the more informative direction: it aggregates internal relations to the external level by taking a join over all *internal pairs that map to a given external pair. If f identifies several internal nodes with one external node (many-to-one), then $f_!R$ keeps the strongest internal evidence compatible with that external pair. This is precisely the kind of operation one wants when exporting conclusions: it is safe (because it never invents support not witnessed internally), and it is maximally permissive given that safety (because it uses join to keep the best witness).*

Remark 3112. *Two basic monotonicity facts will be used implicitly in later transfer arguments. First, both $f_!$ and f^* are order-preserving: if $R \le R'$ pointwise on $X \times X$, then $f_!R \le f_!R'$ because joins preserve order; and if $S \le S'$ pointwise on $Y \times Y$, then $f^*S \le f^*S'$ by substitution. Second, pullback is strictly functorial in the sense that $(g \circ f)^* = f^* \circ g^*$ holds on the nose, while pushforward is functorial when defined by joins as above, i.e., $(g \circ f)_! = g_! \circ f_!$; this reflects that both constructions are computed by explicit formulas that compose by regrouping quantifiers over witnesses. These elementary properties are often the hidden "bookkeeping" behind multi-stage transport across several correspondence maps.*

Theorem 115 (Galois connection: pushforward/pullback)**.** *For all $R : X \times X \to V$ and $S : Y \times Y \to V$,*

$$f_!R \le S \quad \textit{iff} \quad R \le f^*S.$$

Remark 3113. *This theorem says that pushforward and pullback form an adjoint pair (a Galois connection) between the posets of V-relations on X and on Y. In plain terms: "exporting R and asking whether it fits inside S" is equivalent to "importing S and asking whether it dominates R." This is important because it makes many transfer arguments one-line: once you have an inequality on one side, you can mechanically move it to the other side via the adjunction. In reasoning-engine terms, it supplies a sound algebra for translating between internal and external vocabularies.*

This adjointness is the categorical skeleton behind the later transfer bounds: it explains why we can prove external soundness from internal structure plus correspondence slack with no ad hoc case analysis. Such adjunction-based transport is a recurring theme in structured transfer learning; compare [7].

Remark 3114. *A useful way to unpack the adjunction is via the associated unit/counit inequalities, which hold for every adjoint pair. Taking $S = f_!R$ in the theorem yields*

$$R \le f^*f_!R,$$

meaning that if one exports R and then re-imports it, one obtains a relation on X that is at least as permissive as the original R (information can be lost by collapsing fibers, but not fabricated in a way that would violate this inequality). Dually, taking $R = f^*S$ yields

$$f_! f^* S \leq S,$$

meaning that exporting a pulled-back external relation cannot exceed the original external relation. In many concrete settings, these inequalities formalize "no overstatement upon export" and "no strengthening by round-tripping," which are precisely the safety properties desired when using f as a correspondence map between an internal reasoning state and an external world model.

Proof. ($\Rightarrow$) Assume $f_! R \leq S$. Fix $x, x' \in X$ and set $y = f(x)$, $y' = f(x')$. Then

$$R(x, x') \leq \bigvee_{u,u' :\, f(u)=y,\ f(u')=y'} R(u, u') = (f_! R)(y, y') \leq S(y, y') = (f^* S)(x, x').$$

Thus $R \leq f^* S$.

($\Leftarrow$) Assume $R \leq f^* S$. Fix $y, y' \in Y$. For all x, x' with $f(x) = y, f(x') = y'$,

$$R(x, x') \leq (f^* S)(x, x') = S(y, y').$$

Taking the join over all such (x, x') yields $(f_! R)(y, y') \leq S(y, y')$. $\square$

Remark 3115 (Proof sketch and intuition)**.** *The proof proceeds by chasing definitions. For the forward direction, one observes that a particular internal pair (x, x') is one of the witnesses included in the join defining $(f_! R)(f(x), f(x'))$, so its value cannot exceed that join; if the join is bounded by S, then R is bounded by the pulled-back S. For the reverse direction, one uses that if* every *witness of the join is bounded by $S(y, y')$, then the join itself is bounded by $S(y, y')$.*

A visual metaphor is helpful: pullback draws a rectangle on $X \times X$ by projecting it through $f \times f$ into $Y \times Y$; pushforward collapses the fibers $(f \times f)^{-1}(y, y')$ by taking the supremum. The Galois connection says these operations are precisely dual in the order-theoretic sense.

Remark 3116. *In the intended "mind–world" reading, X can be the internal state space of a reasoner and Y a space of externally named entities or observations, with f the current grounding map. Then f^* acts like a* query *operator: it answers questions about internal states by looking at how they are named externally. Conversely, $f_!$ acts like a* projection *or* publishing *operator: it takes what is known internally and expresses it in the external naming scheme, merging indistinguishable internal states via joins. The adjunction guarantees that these two directions of translation fit together coherently, so that verification can be performed either before export (internally) or after export (externally) with equivalent results.*

Corollary 46 (External soundness from internal structure plus correspondence slack)**.** *If $f : X \to Y$ is a q-approximate morphism from $G = (X, G)$ to $H = (Y, H)$, then*

$$f_!(G \otimes q) \leq H,$$

where $(G \otimes q)(x, x') := G(x, x') \otimes q$.

Remark 3117. *Intuitively, if f preserves edges up to slack q, then after we explicitly "pay" that slack by scaling the internal graph to $G \otimes q$, the resulting relation can be exported to Y without exceeding what the external graph H already supports. This is a precise form of soundness: no transported claim is stronger than what the world-graph warrants.*

This statement is also the point where the residuated definition of q-approximate morphism and the Galois connection click together: the former gives $G \otimes q \leq f^*H$, and the latter turns this into a pushforward inequality. The resulting rule is what one would want in a conservative reasoning loop that must avoid overconfident export of internal inferences.

Remark 3118. *Two limiting cases help calibrate the meaning of the corollary. If q is the top element (e.g. $q = 1$ in $[0,1]$-valued semantics, or $q = \top$ in Boolean semantics), the statement reduces to $f_!G \leq H$, i.e. exact external soundness: exporting the internal edge relation does not overshoot the external one. If q is smaller (more stringent), the hypothesis that f is a q-approximate morphism is weaker, but the conclusion compensates by exporting only the down-weighted relation $G \otimes q$. In other words, poorer correspondence quality forces more conservative export, and the formula makes that tradeoff explicit and algebraic rather than heuristic.*

Proof. q-approximate morphism means $G \otimes q \leq f^*H$. Apply the Galois connection with $R = G \otimes q$ and $S = H$ to obtain $f_!(G \otimes q) \leq H$. ☐

Remark 3119 (Proof sketch and intuition)**.** *The proof is a single adjunction step. The high-level strategy is: (i) rewrite "q-approximate morphism" as an inequality on $X \times X$, and then (ii) transport that inequality to $Y \times Y$ using the pushforward/pullback Galois connection.*

Geometrically, you can imagine the internal edges as lying above external edges along fibers of $f \times f$. Paying the uniform slack q ensures each internal edge fits below its corresponding external edge; taking joins along fibers (pushforward) cannot break that fit.

Remark 3120 (Reasoning-engine use)**.** *This gives a mechanically checkable rule: to export an internally inferred relation G into the world vocabulary, first scale it by the correspondence quality q, then push it forward along f.*

56.2.2 Depth-n transfer bounds

Theorem 116 (Depth-n inference transfer under a q-approximate morphism)**.** *Let $f : X \to Y$ be a q-approximate morphism $G \to H$. Then for all $n \geq 1$ and all $x, z \in X$,*

$$G^{(n)}(x,z) \otimes q^{\otimes n} \leq H^{(n)}(f(x), f(z)),$$

where $q^{\otimes n} := \underbrace{q \otimes q \otimes \cdots \otimes q}_{n \text{ times}}$.

Remark 3121. *This theorem makes explicit a philosophical constraint that is easy to ignore in informal reasoning: if the mind–world correspondence has limited fidelity q, then every inferential step spends some of that fidelity. The n-step conclusion is therefore discounted by $q^{\otimes n}$. In the common multiplicative case, this is literally an exponential decay q^n, a formal warning against exporting long internal chains through a weak correspondence.*

The result is central for transfer learning and for any system that must manage the tradeoff between depth and trust: it tells you how to compute conservative external lower bounds from internal multi-step inferences. In Hyperseed language, this is a mathematically crisp version of depth-sensitive Transfer Learning (Hyperseed-Concept 192, Hyperseed-Concept 193), aligning with the general theme of [7].

Proof. First prove by induction on n that for every length-n path $p = (x = x_0, x_1, \ldots, x_n = z)$,

$$\Big(\bigotimes_{i=1}^{n} G(x_{i-1}, x_i)\Big) \otimes q^{\otimes n} \leq \bigotimes_{i=1}^{n} H(f(x_{i-1}), f(x_i)).$$

For $n = 1$ this is exactly $G(x_0, x_1) \otimes q \le H(f(x_0), f(x_1))$. Assuming the claim for n, multiply the last inequality by $G(x_n, x_{n+1}) \otimes q$ and use commutativity/associativity of $\otimes$ plus the edge-wise constraint again.

Now take joins over all length-n paths. Because $\otimes$ distributes over arbitrary joins,

$$G^{(n)}(x,z)\otimes q^{\otimes n} = \Big(\bigvee_{p\colon |p|=n} w_G(p)\Big)\otimes q^{\otimes n} = \bigvee_{p\colon |p|=n} (w_G(p)\otimes q^{\otimes n}) \le \bigvee_{p\colon |p|=n} w_H(f(p)) = H^{(n)}(f(x), f(z)).$$

□

Remark 3122 (Proof sketch and intuition)**.** *The proof has two layers. First, it shows that each individual path weight transports: if every edge transports with penalty q, then an n-edge product transports with penalty $q^{\otimes n}$ simply by multiplying inequalities and regrouping factors. Second, it upgrades from a single path to the optimal n-step connection by taking joins over all paths; the quantale distributivity guarantees that scaling by $q^{\otimes n}$ commutes with the join.*

A helpful mental image is to view q as a per-hop "friction" or "dissipation" in the act of correspondence. The best n-hop internal route remains a route externally, but each hop pays the friction; taking the best among routes (the join) cannot undo that accumulated cost.

Corollary 47 (A safe bounded-depth transfer rule (decision tool))**.** *Fix a threshold $\tau \in V$ and depth n. If an internal n-step inference yields $G^{(n)}(x,z) \ge \tau$, then the corresponding external n-step inference satisfies*

$$H^{(n)}(f(x), f(z)) \ge \tau \otimes q^{\otimes n}.$$

In particular, to guarantee $H^{(n)}(f(x), f(z)) \ge \tau_{\text{out}}$, it suffices to require $\tau \otimes q^{\otimes n} \ge \tau_{\text{out}}$.

Remark 3123. *This corollary is the operational form of the depth-n theorem: it tells you how to set internal thresholds so that, after exporting through correspondence quality q, you still exceed a desired external threshold. Conceptually, it is the familiar engineering move of budgeting for losses: decide what you need in the world, then work backwards to what you must establish internally.*

The connection to the earlier statement is direct: the theorem gives an inequality of the form "discounted internal $\le$ external". The corollary simply applies monotonicity to replace $G^{(n)}(x,z)$ by a known lower bound τ, producing a computable external lower bound. This is exactly the sort of conservative rule one wants in a reasoning loop that must avoid hallucinated certainty (Hyperseed-Concept **??***).*

Proof. From the theorem, $G^{(n)}(x,z) \otimes q^{\otimes n} \le H^{(n)}(f(x), f(z))$. If $G^{(n)}(x,z) \ge \tau$, monotonicity of $\otimes$ gives $\tau \otimes q^{\otimes n} \le G^{(n)}(x,z) \otimes q^{\otimes n} \le H^{(n)}(f(x), f(z))$. □

Remark 3124 (Proof sketch and intuition)**.** *The entire argument is a single monotonicity step: once we know that external support dominates discounted internal support, any internal lower bound can be substituted into the discounted term. One may read this as a safe "certificate" mechanism: internal evidence above τ certifies external evidence above the discounted $\tau \otimes q^{\otimes n}$.*

In terms of intuition, the corollary says: if you want a fact to remain true after translation through an imperfect dictionary, you must state it more strongly (or more conservatively) in the source language, by exactly the factor that accounts for translation loss.

Remark 3125 (Interpretation)**.** *Transfer quality decays with inferential depth (via $q^{\otimes n}$). This suggests a general reasoning heuristic: when exporting conclusions across a weak correspondence, prefer shallow chains, or raise internal thresholds to compensate.*

56.2.3 Constraint-set (pattern) transfer

Definition 505 (Constraint-set pattern score)**.** *Let $G = (X, G)$ and let $P \subseteq X \times X$ be a finite constraint set (a "pattern"). Define its conjunctive score by*

$$\mathrm{Score}_G(P) := \bigotimes_{(u,v)\in P} G(u, v).$$

Remark 3126. *Here a "pattern" is treated as a finite set of directed constraints (u, v) that are supposed to cohere together. The score $\mathrm{Score}_G(P)$ is the joint support for the whole pattern, formed by multiplying all the edge weights. This is the natural conjunction operation when $\otimes$ plays the role of "and / joint satisfaction / joint evidence." It connects directly with the Hyperseed notion of Pattern (Hyperseed-Concept 130) as a set of mutually reinforcing constraints (cf. [5]).*

In the boolean case, $\mathrm{Score}_G(P) = 1$ iff all *required edges are present; otherwise it is 0. In a probabilistic-style case with $V = [0, 1]$ and $\otimes$ multiplication, the score is the product of edge strengths, penalizing patterns that rely on many moderately uncertain constraints. This is precisely why the subsequent transfer theorem has a cardinality-dependent penalty: larger patterns have more places where correspondence can leak.*

It is also useful to note that this "multiplicative" aggregation is sensitive to the weakest links: if any single constraint has very low weight, the entire score is driven down. This matches the intended reading of a pattern as a joint *requirement rather than a disjunction. In settings where V is a general commutative semiring or a t-norm algebra, the same conceptual role is played by whatever $\otimes$ is chosen: it is the operation that encodes how independent (or jointly required) constraints compose.*

Theorem 117 (Pattern constraint transfer with cardinality penalty)**.** *Let $f : X \to Y$ be a q-approximate morphism $G \to H$ and let $P \subseteq X \times X$ be finite. Then*

$$\mathrm{Score}_G(P) \otimes q^{\otimes|P|} \leq \mathrm{Score}_H\big(f(P)\big),$$

where $f(P) := \{(f(u), f(v)) : (u, v) \in P\}$ (as a multiset, if duplicates are tracked).

Remark 3127. *A helpful boundary case is $q = 1$ (the "no slack" or exact morphism case): then the penalty term is neutral, and one recovers the expectation that a structure-preserving map transfers the full pattern score without degradation. At the other extreme, when $q < 1$ and $|P|$ is large, the factor $q^{\otimes|P|}$ can become small rapidly (e.g. exponentially for ordinary multiplication), making explicit that high-cardinality patterns are brittle under imperfect correspondence. This is not merely a technical artifact: it formalizes the intuition that each additional conjunct introduces another opportunity for mismatch between the internal constraint graph and the external one.*

If duplicates in $f(P)$ are tracked (multiset semantics), then multiple distinct constraints in P may map to the same target edge in H. In that case $\mathrm{Score}_H(f(P))$ multiplies the weight of that target edge multiple times, which is appropriate when the original pattern truly contained multiple independent "reasons" or "requirements" that happen to coincide after mapping. If instead one wishes to quotient by duplicates (set semantics), then one is implicitly imposing an idempotence convention for repeated constraints; the multiset phrasing makes the bookkeeping choice explicit.

Remark 3128. *This theorem is the constraint-set analogue of the depth-n path-transfer result. Instead of an n-step chain, we have a finite conjunction of constraints; each constraint pays a q penalty, hence the joint penalty $q^{\otimes|P|}$. The message is again one of discipline: exporting large, intricate internally-recognized patterns requires correspondingly strong mind–world alignment.*

The result matters computationally because pattern recognition is often how a reasoning system packages hypotheses: a hypothesis is a bundle of constraints that jointly describe a situation. This theorem provides a safe rule for translating such bundles into an external hypothesis space, which is a recurring need in transfer learning and in model-to-world mapping (Hyperseed-Concept 192; cf. [7]).

In probabilistic practice, one often works in a log domain, where products become sums. Under that reading (when $\otimes$ is multiplication), the theorem corresponds to an additive penalty $|P| \cdot \log q$ on the log-score. This makes the cardinality dependence especially transparent: each additional constraint contributes a fixed "alignment cost" determined by q.

Proof. For each $(u, v) \in P$, $G(u, v) \otimes q \leq H(f(u), f(v))$. Multiply these inequalities over all constraints in P and use commutativity/associativity to collect the q factors into $q^{\otimes |P|}$.

More explicitly, writing the product over P as $\bigotimes_{(u,v)\in P}(\cdot)$, one has

$$\bigotimes_{(u,v)\in P} \big(G(u, v) \otimes q\big) \;\leq\; \bigotimes_{(u,v)\in P} H(f(u), f(v)),$$

and then commutativity/associativity yields

$$\left(\bigotimes_{(u,v)\in P} G(u, v)\right) \otimes \left(\bigotimes_{(u,v)\in P} q\right) = \mathrm{Score}_G(P) \otimes q^{\otimes |P|}$$

on the left-hand side. The inequality direction is preserved because $\otimes$ is taken to be order-preserving in each argument (as is standard for the conjunction-like operations used here). □

Remark 3129 (Proof sketch and intuition)**.** *The strategy is pointwise: transport each constraint individually using the approximate morphism inequality, then combine them multiplicatively. Because $\otimes$ is commutative and associative, we may regroup all the q factors into a single $q^{\otimes |P|}$ multiplier.*

Visually, one may imagine a pattern as a small subgraph with several edges. The map f sends each edge to a corresponding edge in H (perhaps with collisions); the theorem asserts that if each edge survives translation up to slack q, then the whole subgraph survives translation up to a slack that grows with the number of edges.

A concrete mental model is to take P with three constraints. If $q = 0.9$ and $\otimes$ is ordinary multiplication, then even if the internal pattern score is near 1, the guaranteed transferred score is discounted by at least $0.9^3 = 0.729$. For larger P, the bound becomes correspondingly more demanding, capturing the sense in which "many-small-errors" can accumulate into a large overall mismatch.

Remark 3130 (Use)**.** *This turns mind–world correspondence into a quantitative rule for exporting internally recognized patterns into external hypotheses: larger constraint-sets require stronger correspondence or higher internal confidence.*

In particular, for systems that search over candidate patterns/hypotheses, the theorem suggests a regularization heuristic: all else equal, prefer smaller patterns unless one has either (i) high internal scores on all constituent constraints, or (ii) a demonstratedly high q for the relevant transfer map. In this way, the formal penalty term aligns with common computational tradeoffs between expressivity (large patterns) and robustness (transfer under mismatch).

56.3 Reality-systems and shared reality as fixed points

We now move from "structure transport" to "what counts as real," modeled as a fixed point of mutual predictive closure. Concretely, the guiding intuition is that an entity counts as "in" a candidate reality-system not merely when it is predicted by something already in the system, but when it also participates in predictive constraints that flow back into the system, so that membership is supported by a two-way web of model-mediated expectations.

Definition 506 (Predictive adjacency and mutual closure operator)**.** *Let U be a set of entities and let $\to$ be a directed predictive adjacency relation on U (depending on a mind M, an interval I, and a threshold). For $R \subseteq U$ define*

$$\mathrm{Pred}_{\to}(R) := \{y \in U : \exists x \in R \text{ s.t. } x \to y\}, \qquad \mathrm{Pred}_{\leftarrow}(R) := \{y \in U : \exists x \in R \text{ s.t. } y \to x\}.$$

Define

$$F(R) := \mathrm{Pred}_{\to}(R) \cap \mathrm{Pred}_{\leftarrow}(R).$$

A (nonempty) reality-system *is a set $R \subseteq U$ such that $F(R) = R$.*

Remark 3131. *The notation here is set-theoretic rather than quantale-valued: 2^U denotes the power set of U ordered by inclusion $\subseteq$. The operation $\mathrm{Pred}_{\to}(R)$ collects everything that is predicted* from *something in R (one step forward), while $\mathrm{Pred}_{\leftarrow}(R)$ collects everything that predicts* into *R (one step backward). The mutual-closure operator $F(R)$ keeps precisely those entities that participate in both directions of predictive coupling with R.*

Calling a fixed point $R = F(R)$ a "reality-system" is a formal way to express a pragmatic criterion for reality: the real is what is stably entangled in mutual prediction. This aligns with the Hyperseed concept of Reality-System (Hyperseed-Concept 149) and with the broader intersubjective framing (Hyperseed-Concept **??***). Philosophically, one may hear in $F(R)$ a Peircean emphasis on lawlike mediation and mutual constraint (Thirdness) [14], and a Whiteheadian insistence that "actuality" is processual and relational rather than atomistic [15].*

As a simple example: if $\to$ is the edge relation of a directed graph, then $F(R)$ is the set of nodes that are either reached from R in one step and also point into R in one step. A fixed point R is then a set that already contains all such mutually linked neighbors, so it is closed under this two-sided predictive neighborhood operation.

Remark 3132 (On the role of $(M, I, \text{threshold})$ in $\to$)**.** *Although $\to$ is written as a single relation, it is helpful to read it as a family $\to_{M,I,\tau}$ parameterized by a mind/model M, a time window I, and a decision threshold τ (for example, a minimum predictive accuracy, a minimum mutual information, or a statistical significance level). Under this reading, a reality-system is not an absolute subset of U but a stability notion relative to a chosen epistemic regime: changing sensors, priors, or thresholds changes the adjacency relation and thus changes which fixed points exist and which are selected. This makes explicit the intended "operational" character of the construction: the same world U can support multiple reality-systems depending on what predictive couplings are actually available.*

Remark 3133 (Nonemptiness and trivial fixed points)**.** *The definition excludes the empty set to avoid counting the vacuous fixed point that often arises in monotone fixed-point settings. Indeed, when $\mathrm{Pred}_{\to}(\varnothing) = \varnothing$ and $\mathrm{Pred}_{\leftarrow}(\varnothing) = \varnothing$ (as is typical for witness-based definitions), one has $F(\varnothing) = \varnothing$, so $\varnothing$ is a fixed point but is not informative as a candidate "reality." Requiring $R \neq \varnothing$ encodes the minimal phenomenological demand that a reality-system must contain at least one entity participating in predictive coupling.*

Lemma 165 (Monotonicity)**.** *$F : 2^U \to 2^U$ is monotone: $R \subseteq R' \Rightarrow F(R) \subseteq F(R')$.*

Remark 3134. *Monotonicity is the minimal structural property needed to apply the standard fixed-point apparatus (Knaster–Tarski) later when we discuss shared reality. Intuitively, enlarging the set R gives you more potential predictors and more predicted entities, so the forward and backward prediction neighborhoods can only grow, and their intersection can only grow as well.*

This lemma is also a small statement about epistemic humility: if you admit more entities into your candidate reality-system, you can only increase (never decrease) what is mutually predicted in one step. The later lattice theorem formalizes the idea that there are canonical least and greatest stable "realities" compatible with a family of predictive closures.

Proof. If $R \subseteq R'$, then any witness $x \in R$ to $x \to y$ is also in R', so $\mathrm{Pred}_{\to}(R) \subseteq \mathrm{Pred}_{\to}(R')$ and similarly for $\mathrm{Pred}_{\leftarrow}$. Intersecting preserves inclusion. □

Remark 3135 (Proof sketch and intuition). *The proof simply observes that both* $\mathrm{Pred}_{\to}$ *and* $\mathrm{Pred}_{\leftarrow}$ *are monotone set transformers: witnesses for membership persist when the domain set is enlarged. Since intersection is itself monotone, the composite F is monotone.*

In graph terms: starting from a larger set of source nodes cannot reduce the set of nodes with an incoming edge from the source set, nor can it reduce the set of nodes with an outgoing edge to the source set.

Remark 3136 (Fixed points as two-sided predictive invariants). *The equation $F(R) = R$ can be read as an invariance condition: R is exactly the set of entities that are simultaneously (i) generated forward by prediction from R and (ii) supported backward by predictive influence into R. In this sense, reality-systems are not merely "closed" sets but self-sustaining predictive substructures. When $\to$ arises from a learned dynamics model, one may view R as the subset of entities whose predicted interactions are internally coherent under the model's own one-step predictive coupling in both causal directions (outgoing effects and incoming constraints).*

Remark 3137 (Graph-theoretic intuition beyond one step). *While F is defined using one-step neighborhoods, fixed points often correspond to the persistence of cycles and reciprocal connectivity patterns. In a directed graph, mutual one-step coupling is a local proxy for membership in a bidirectionally constrained subgraph; iterating F can progressively remove nodes that are only "one-way" attached. This perspective is useful when thinking about robustness: entities that remain after repeated application of F are those that cannot be peeled away without breaking mutual predictive support.*

56.3.1 Shared reality across multiple minds

Definition 507 (Shared reality operator). *Let $\{F_i\}_{i \in I}$ be mutual predictive closure operators (one per mind, or per viewpoint). Define the* shared reality operator

$$J(R) := \bigcap_{i \in I} F_i(R).$$

Call any nonempty fixed point $R = J(R)$ a shared reality-system.

Remark 3138. *The operator J takes a candidate set R and retains only what survives every viewpoint's mutual-prediction filter. In this sense J is an algebraic representation of "common ground": the intersection enforces unanimity across the family of minds/viewpoints (Hyperseed-Concept **??**, Hyperseed-Concept **??**). When I indexes multiple agents, this resonates with the idea that stable shared structure is what remains invariant under multiple partial perspectives.*

In practical terms, F_i can vary because different minds have different models, sensors, or thresholds. The shared fixed point notion then becomes a tool for multi-agent or multi-model systems: compute what is robustly real for the group, not merely for an individual. This also connects naturally to work that distinguishes individual vs. collective loci of cognition (e.g. [4] and the collective-location framing in [12]).

Remark 3139 (Interpretation as robustness under viewpoint variation)**.** *Because J is defined by intersection, it is inherently conservative: it retains only those entities that are mutually predictively closed* for all $i \in I$. *Thus, a shared reality-system represents an invariance class under the family of predictive adjacencies induced by the group. If one views each F_i as encoding a different inductive bias or access channel, then J formalizes the idea that shared reality is what persists under such variations—a form of robustness that is social/epistemic rather than purely statistical.*

Remark 3140 (Edge cases: disagreement and the empty intersection)**.** *The insistence on nonempty fixed points is especially relevant in the multi-mind case: if the family $\{F_i\}$ is sufficiently incompatible, J may have only the empty fixed point, reflecting a failure to achieve any nontrivial common ground at the chosen thresholds. This is not a defect of the formalism but a way to represent, within the same algebra, the phenomenon that sufficiently misaligned models (or insufficient communication) can collapse shared reality to a null consensus. In later applications, one can soften unanimity by replacing $\bigcap$ with majority/weighted aggregators, but the present definition isolates the strictest notion of intersubjective agreement.*

Theorem 118 (Existence of minimal/maximal shared reality-systems (lattice theorem))**.** *Assume each F_i is monotone. Then J is monotone, and the set of fixed points of J is a complete lattice under inclusion. In particular, J has a least fixed point* $\mathrm{lfp}(J)$ *and a greatest fixed point* $\mathrm{gfp}(J)$.

Remark 3141 (Why completeness matters)**.** *The complete lattice structure is not merely an abstract convenience: it supports principled selection of canonical shared realities. The least fixed point* $\mathrm{lfp}(J)$ *can be interpreted as the smallest set of entities that is self-sustaining under all viewpoints' mutual predictive closures (a "minimal consensus core"), while the greatest fixed point* $\mathrm{gfp}(J)$ *is the largest such set (a "maximal common world" consistent with the group). These bounding objects allow one to reason about refinement (moving upward from* $\mathrm{lfp}(J)$*) and skepticism (moving downward from* $\mathrm{gfp}(J)$*) while remaining within fixed points.*

Remark 3142 (Standard proof route via Knaster–Tarski)**.** *Given monotonicity of each F_i, monotonicity of J follows because intersections of monotone images preserve inclusion: if $R \subseteq R'$ then $F_i(R) \subseteq F_i(R')$ for all i, hence $\bigcap_i F_i(R) \subseteq \bigcap_i F_i(R')$. The existence of* $\mathrm{lfp}(J)$ *and* $\mathrm{gfp}(J)$ *then follows from the Knaster–Tarski theorem on monotone endomaps of complete lattices (here, $(2^U, \subseteq)$). In particular,*

$$\mathrm{lfp}(J) = \bigcap\{R \subseteq U : J(R) \subseteq R\}, \qquad \mathrm{gfp}(J) = \bigcup\{R \subseteq U : R \subseteq J(R)\},$$

so the extremal shared reality-systems can be characterized respectively as the intersection of all pre-fixed points and the union of all post-fixed points.

Remark 3143 (Computational reading: iteration toward consensus)**.** *When U is finite (or otherwise effectively representable), one can often approximate fixed points by iteration. For example, starting from a large candidate set (such as U itself) and repeatedly applying J yields a descending chain $U \supseteq J(U) \supseteq J^2(U) \supseteq \cdots$ that stabilizes at a post-fixed point and, under common finiteness conditions, reaches* $\mathrm{gfp}(J)$. *Dually, starting from a small seed (when a suitable bottom element is* *chosen, often* $\varnothing$*) and iterating upward can approach* $\mathrm{lfp}(J)$*. This connects the formal lattice result to a concrete procedure: shared reality can be viewed as what remains after repeatedly removing entities that fail some viewpoint's mutual predictive closure.*

Remark 3144. *This theorem says there are canonical extremes of shared reality: a smallest shared reality (the least fixed point) and a largest shared reality (the greatest fixed point). The surprise, if one expects "reality" to be unique, is that the mathematics does not force uniqueness; instead it offers a lattice of stable candidates, ordered by inclusion, within which one can choose according to additional criteria (e.g. minimality, maximality, or pragmatic constraints). In particular, even if every individual viewpoint has a well-defined "closure" behavior, the act of taking a joint closure across multiple viewpoints can admit multiple consistent equilibria. The lattice picture makes this multiplicity orderly: different equilibria are not arbitrary, but sit between canonical bounds and can be compared by refinement (*$R \subseteq R'$*) as a notion of "adding commitments" or "adding jointly endorsed structure."*

The importance is twofold. Conceptually, it makes "shared reality" into an object that can be computed and compared. Technically, it gives us access to the whole fixed-point toolkit on complete lattices. Fixed-point arguments of this kind recur throughout Hyperseed-style analyses of stability (e.g. in goal-system dynamics [10]), and here they support a precise account of intersubjective closure. One can also read the least and greatest fixed points as "minimal common ground consistent with mutual prediction" versus "maximal common ground that remains self-stabilizing under mutual prediction." In applications, these extremes can serve as conservative and liberal baselines, with intermediate fixed points representing different compatible ways of settling ambiguity.

Proof. Each F_i monotone implies $R \subseteq R' \Rightarrow F_i(R) \subseteq F_i(R')$. Intersecting over i preserves inclusion, so J is monotone. Since $(2^U, \subseteq)$ is a complete lattice and J is monotone, the Knaster–Tarski theorem implies that the fixed points of J form a complete lattice and least/greatest fixed points exist. For clarity: completeness of 2^U means that for any family $\{R_\alpha\}_{\alpha \in A} \subseteq 2^U$, both $\bigcap_{\alpha \in A} R_\alpha$ and $\bigcup_{\alpha \in A} R_\alpha$ exist in 2^U and serve as the infimum and supremum under $\subseteq$. Knaster–Tarski then applies directly to the endomap $J : 2^U \to 2^U$, yielding not only existence but also the order structure of the set of fixed points. $\square$

Remark 3145 (Proof sketch and intuition)**.** *The strategy is purely order-theoretic: (i) show* J *is monotone because it is an intersection of monotone maps, then (ii) invoke Knaster–Tarski, which guarantees that any monotone endomap on a complete lattice has a complete lattice of fixed points. Equivalently, there are always stable sets* R *satisfying* $R = J(R)$*, and among them there is a distinguished smallest and largest one with respect to inclusion. This is precisely what makes the fixed-point set itself behave like a "space of candidates" rather than a single solution.*

A useful image is that each mind/viewpoint i *supplies a "filter"* F_i*. Applying* J *applies all filters simultaneously by intersection. Monotonicity ensures filters behave coherently under enlarging hypotheses, and Knaster–Tarski then ensures stable hypotheses exist and can be ordered. In that image, a fixed point* $R = J(R)$ *is a hypothesis set that survives all filters unchanged. The least fixed point can be thought of as what remains after iterating filtering from the bottom (starting at* $\varnothing$*), while the greatest fixed point corresponds to what survives when starting from the top (starting at* U*) and repeatedly enforcing mutual predictability, whenever those iterative characterizations are valid for the operators at hand.*

Remark 3146 (Interpretation)**.** *A shared reality-system is "what all considered minds mutually predict with one another," formalized as a joint fixed point. This is a direct mathematical handle on intersubjective "common ground". More explicitly,* $R = J(R)$ *means that if one assumes* R

as the current candidate for common ground, then each viewpoint's predictive closure does not introduce anything outside R that would still be jointly endorsed after intersecting across minds. Thus R is not merely mutually compatible, but stable under the operation of "take what everyone would infer/predict from the current common ground and keep only what is jointly retained." The existence of extremal fixed points then says there are canonical choices of common ground even when agreement is underdetermined.

Theorem 119 (Monotonicity of shared reality in predictive power)**.** *Suppose two families of operators satisfy $F_i(R) \subseteq F_i'(R)$ for all i and all $R \subseteq U$ (i.e., every mind's predictive closure becomes at least as expansive). Let $J = \cap_i F_i$ and $J' = \cap_i F_i'$. Then*

$$\mathrm{gfp}(J) \subseteq \mathrm{gfp}(J') \qquad \textit{and} \qquad \mathrm{lfp}(J) \subseteq \mathrm{lfp}(J').$$

Remark 3147. *This theorem formalizes a sober monotonicity principle: if every mind becomes "more predictively capable" (or less restrictive) in the sense that its closure operator expands, then the extremal shared realities expand as well. In other words, increasing predictive power cannot shrink what is stably shareable. A useful way to read this is comparative statics: when all individual closures are relaxed (pointwise), the joint equilibrium set shifts upward in the lattice order. The conclusion is stated only for the least and greatest fixed points because those are canonical selections, but the same pointwise order $J \leq J'$ also implies inclusion relationships between the full sets of pre/post-fixpoints that underlie intermediate fixed points.*

This connects to the lattice theorem by describing how the lattice of shared realities changes when we perturb the underlying operators. Later, in the robustness subsection, we will perform a similar analysis but at the level of graph weights and thresholded edges. The present result is also a sanity check: if one interprets each F_i as encoding what is predictively reachable or inferable for agent i, then granting agents additional inferential resources should not paradoxically remove stable joint conclusions; it can only keep them or add more, at least at the level of these extremal equilibria.

Proof. Pointwise inclusion $F_i \subseteq F_i'$ implies $J(R) = \cap_i F_i(R) \subseteq \cap_i F_i'(R) = J'(R)$ for all R, i.e., $J \leq J'$ pointwise. For monotone maps on a complete lattice, pointwise order implies monotonicity of least and greatest fixed points: the set of post-fixpoints of J is contained in that of J', yielding $\mathrm{gfp}(J) \subseteq \mathrm{gfp}(J')$, and similarly for least fixed points. Concretely, recall the standard characterizations (Knaster–Tarski):

$$\mathrm{gfp}(J) = \bigcup\{R \subseteq U : R \subseteq J(R)\}, \qquad \mathrm{lfp}(J) = \bigcap\{R \subseteq U : J(R) \subseteq R\}.$$

If $J \leq J'$, then any R satisfying $R \subseteq J(R)$ also satisfies $R \subseteq J'(R)$, so the union defining $\mathrm{gfp}(J)$ is taken over a subset of the family used for $\mathrm{gfp}(J')$, giving the desired inclusion. Dually, any R satisfying $J'(R) \subseteq R$ also satisfies $J(R) \subseteq R$, so the intersection defining $\mathrm{lfp}(J)$ is over a (weakly) larger family, yielding $\mathrm{lfp}(J) \subseteq \mathrm{lfp}(J')$. □

Remark 3148 (Proof sketch and intuition)**.** *The proof compares J and J' pointwise: $J(R) \subseteq J'(R)$ for every R. Fixed points can be characterized as intersections/unions of pre-fixpoints or post-fixpoints; since J is everywhere smaller than J', the collection of post-fixpoints for J is more constrained, hence the greatest fixed point cannot exceed that of J'. The least fixed point comparison is analogous. This is an instance of the general monotonicity of the operators* $\mathrm{lfp}(\cdot)$ *and* $\mathrm{gfp}(\cdot)$ *with respect to the pointwise order on monotone endomaps of a complete lattice.*

An intuition in terms of "constraints": each F_i imposes constraints on what counts as mutually predictive. Replacing F_i by a larger F_i' relaxes constraints, so the stable solutions can only get larger. Said differently, J is the "strict" joint filter and J' is the "lenient" joint filter; a set that *is stable under the strict filter is automatically stable (or at least extendable to stability) under the lenient one, so the maximal stable set under lenience must contain the maximal stable set under strictness. The same reasoning holds at the minimal end: once pre-fixpoint conditions are relaxed, the intersection of all admissible supersets can only move upward.*

56.3.2 A computable graph-theoretic picture (optional but useful)

Definition 508 (Two-sided reachability closure)**.** *Given a directed graph $(U, \to)$, define reachability sets*

$$\mathrm{Reach}^+(R) := \{y : \exists x \in R \text{ with a directed path } x \rightsquigarrow y\}, \quad \mathrm{Reach}^-(R) := \{y : \exists x \in R \text{ with a directed path } y \rightsquigarrow x\}.$$

Define $F^*(R) := \mathrm{Reach}^+(R) \cap \mathrm{Reach}^-(R)$.

Remark 3149. *This definition replaces one-step prediction neighborhoods with arbitrary-length reachability. The notation $x \rightsquigarrow y$ denotes the existence of a directed path from x to y (possibly of length 0 or positive length, depending on convention; here it suffices that it is a preorder). Then* $\mathrm{Reach}^+(R)$ *is everything downstream of R, while* $\mathrm{Reach}^-(R)$ *is everything upstream of R, and $F^*(R)$ keeps what lies in both the upstream and downstream cones. Note that* Reach^+ *and* Reach^- *are themselves monotone in R (adding starting vertices can only add reachable vertices), so F^* is monotone as an intersection of monotone maps; therefore the earlier fixed-point existence results apply directly to F^* as well.*

Computationally, this is valuable because reachability is a standard graph primitive, and fixed points of F^* can be related to strongly connected structure. Conceptually, moving from F to F^* corresponds to allowing multi-step mutual predictability, which is often closer to the way reasoning systems actually propagate implications. In particular, membership $y \in F^*(R)$ means that y is both reachable from some element of R and can reach back to some element of R, so y lies in the same "mutual influence zone" as R up to directed paths. When R is a singleton $\{x\}$, $F^*(\{x\})$ collects exactly those vertices that are mutually reachable with x, i.e. the strongly connected component of x (under the convention that reachability includes the length-0 path). More generally, $F^*(R)$ aggregates the union of strongly connected components that intersect R, together with any vertices that sit in directed cycles connecting into and out of R; fixed points $R = F^*(R)$ therefore correspond to sets that are closed under this two-sided reachability notion.

Theorem 120 (Fixed points of F^* are "path-convex" (interval-closed))**.** *Let $\preceq$ be the reachability preorder on U (so $a \preceq b$ iff $a \rightsquigarrow b$). Then for any $R \subseteq U$,*

$$F^*(R) = \{z \in U : \exists a, b \in R \text{ with } a \preceq z \preceq b\}.$$

Hence R is a fixed point of F^* iff R is order-convex with respect to $\preceq$: whenever $a, b \in R$ and $a \preceq z \preceq b$, then $z \in R$.

Equivalently: $F^*(R)$ consists of all nodes that lie on a directed "two-sided" reachability chain starting in R and ending in R. When $\preceq$ happens to be antisymmetric (so it is a partial order), this is the familiar notion of order-convexity; when $\preceq$ is merely a preorder, the same condition should be understood up to mutual reachability (i.e. within and between $\preceq$-equivalence classes).

Remark 3150. *The theorem says that the closure $F^*(R)$ is exactly the set of points that lie on some directed "interval" between two points of R (with respect to the reachability preorder). Thus, F^* is an interval hull operator, and its fixed points are precisely the interval-closed (order-convex) subsets.*

This is important because it translates a fixed-point definition of reality into a geometric one: a reality-system is not an arbitrary set of nodes, but an "unbroken region" in the reachability order. In explanatory terms, it gives a crisp style of justification: to explain why z belongs to the closure of R, exhibit $a, b \in R$ such that a can reach z and z can reach b.

It is also useful to keep in mind the directed-graph picture. If U is the vertex set of a directed graph and $\rightsquigarrow$ denotes existence of a directed path, then $a \preceq b$ means "there is a path from a to b." In that setting, $z \in F^*(R)$ precisely when z lies on a directed path that starts somewhere in R and can be continued to end somewhere in R (not necessarily on the same path segment, but with the same intermediate vertex z witnessing both reachability conditions).

Finally, the preorder perspective clarifies what "interval" means when cycles are present. If $a \preceq b$ and $b \preceq a$, then a and b are mutually reachable and are indistinguishable by $\preceq$; the corresponding equivalence classes (strongly connected components in the graph interpretation) behave like the "points" of an induced partial order. The order-convex sets in the theorem can then be read as unions of equivalence classes that are convex in this induced partial order.

Proof. $z \in \mathrm{Reach}^+(R)$ iff $\exists a \in R$ with $a \preceq z$. Also $z \in \mathrm{Reach}^-(R)$ iff $\exists b \in R$ with $z \preceq b$. Thus $z \in F^*(R)$ iff both conditions hold, i.e. iff $\exists a, b \in R$ with $a \preceq z \preceq b$. The characterization of fixed points is immediate.

In particular, the forward direction uses only that $F^*(R)$ is defined as an intersection: membership in $F^*(R)$ means simultaneously satisfying a forward-reachability witness $a \in R$ and a backward-reachability witness $b \in R$. Conversely, if witnesses $a, b \in R$ exist with $a \preceq z \preceq b$, then z lies in both reachability sets, hence in their intersection.

As a small boundary check: if $R = \varnothing$ then $F^*(R) = \varnothing$ by the displayed characterization, and if $R = \{a\}$ then $F^*(R) = \{z \in U : a \preceq z \preceq a\}$, i.e. the $\preceq$-equivalence class of a (all nodes mutually reachable with a). This illustrates how F^* treats cyclic structure as "collapsed" into equivalence classes even before any further convexity across classes is considered. □

Remark 3151 (Proof sketch and intuition)**.** *The proof is a direct logical unpacking: membership in the intersection* $\mathrm{Reach}^+(R) \cap \mathrm{Reach}^-(R)$ *is exactly the conjunction of "reachable from R" and "can reach R", which is equivalent to lying between some $a \in R$ and some $b \in R$ in the preorder $\preceq$.*

One can picture $\preceq$ as defining a directed landscape. The set $F^*(R)$ is the region that is simultaneously downstream of R and upstream of R. Fixed points are regions that contain every node that lies between any two of their nodes, hence they are "path-convex."

A complementary way to say this is that $F^*(R)$ supplies all "bridge" nodes needed to make R closed under interpolation along reachability: whenever R contains endpoints of a reachability interval, the entire interval must also be present. This is exactly the directed analogue of taking the convex hull in Euclidean geometry, except that "straight lines" are replaced by preorder intervals.

This intuition also makes clear why the operator is well-suited to multi-step mutual predictability: a node counts as stabilized by R only if it is both supported by (reachable from) some element of R and can, in turn, support (reach) some element of R. The two-sidedness is what forces an "interval" rather than merely a forward closure.

Remark 3152 (Use)**.** *This provides a clean structural reading: if reality is defined via multi-step mutual predictability, then reality-systems are exactly the "interval-closed" subsets of the predictive reachability order. This supports explanation: to justify why z is in reality-system R, it suffices to exhibit $a, b \in R$ with $a \rightsquigarrow z \rightsquigarrow b$.*

Moreover, the theorem identifies $F^*(R)$ as the smallest reality-system (fixed point) that contains R, in the following sense: if S is any fixed point with $R \subseteq S$, then the interval condition forces $F^*(R) \subseteq S$. This "least closed superset" reading is exactly what makes F^* behave like a closure *operator relative to the interval-closed sets, and it is the property that underwrites robust inference: once R is accepted, any intermediate node on a two-sided reachability chain between accepted nodes is also forced to be accepted.*

When applying the criterion in practice, it is often helpful to choose a as a node in R that most directly generates z (a minimal witness for $a \preceq z$) and choose b as a node in R that z most directly generates (a minimal witness for $z \preceq b$). This makes the justification not just existence-based but also explanatorily economical: it displays z as an interpolant that transmits influence from a to b.

56.4 Robustness theorems

56.4.1 Robustness of approximate morphism degrees

Theorem 121 (Degree monotonicity under graph perturbations)**.** *Let $f : X \to Y$. Suppose G, G' are V-graphs on X and H, H' are V-graphs on Y with $G' \leq G$ and $H \leq H'$ pointwise. Then*

$$\deg(f; G', H') \geq \deg(f; G, H).$$

Remark 3153. *This is a robustness statement about correspondence quality. It says: if we make the internal graph* less demanding *(by reducing edge weights: $G' \leq G$) and/or make the external graph* more supportive *(by increasing edge weights: $H \leq H'$), then the measured degree of preservation cannot get worse; indeed, it can only improve.*

In a reasoning system, $G' \leq G$ often arises when we prune weak internal edges or lower our internal claims; $H \leq H'$ arises when additional external evidence strengthens the world model. The theorem therefore formalizes a safe engineering expectation: conservative internalization and improved external evidence should not break previously established correspondences. This is a quantitative stability property of Mind–World Correspondence (Hyperseed-Concept 112) under refinement and noise.

Concretely, the pointwise inequalities mean that for every ordered pair (x, x') one has $G'(x, x') \leq G(x, x')$, and for every ordered pair (y, y') one has $H(y, y') \leq H'(y, y')$. Thus the comparison is uniform across the entire relational structure, not merely on average. If $V = [0, 1]$ with the usual order, the situation is especially intuitive: lowering G relaxes the demanded internal similarity/entailment strengths, while raising H increases the degree to which the world-side relation is deemed satisfied.

Proof. Fix $x, x' \in X$. By $G'(x, x') \leq G(x, x')$ and Lemma 1 (antitonicity in the left argument),

$$(G'(x, x') \to H(f(x), f(x'))) \geq (G(x, x') \to H(f(x), f(x'))).$$

Also $H(f(x), f(x')) \leq H'(f(x), f(x'))$ and Lemma 1 (monotonicity in the right argument) give

$$(G'(x, x') \to H'(f(x), f(x'))) \geq (G'(x, x') \to H(f(x), f(x'))).$$

Chaining yields

$$(G'(x, x') \to H'(f(x), f(x'))) \geq (G(x, x') \to H(f(x), f(x'))).$$

Taking meets over all pairs (x, x') preserves the inequality, giving the claim.

Equivalently: the degree $\deg(f; G, H)$ is defined as the meet (infimum, or "worst-case" aggregate) of the individual implication values $(G(x, x') \to H(f(x), f(x')))$ as (x, x') ranges over $X \times X$. Since the residuum $\to$ is antitone in its left input and monotone in its right input, each term in this family is pointwise increased when passing from (G, H) to (G', H') under the stated inequalities. The meet operator is monotone with respect to pointwise order on families, so an everywhere-increase of terms yields an increase of the overall meet. □

Remark 3154 (Proof sketch and intuition)**.** *The proof compares each individual implication term in the meet defining* $\deg(f; G, H)$*. Reducing* G *to* G' *increases the implication because the antecedent becomes easier (residuum is antitone on the left). Increasing* H *to* H' *also increases the implication because the consequent becomes easier (residuum is monotone on the right). Thus every term in the meet increases, so the meet increases.*

A visual intuition: deg *is governed by the worst edge. If you weaken the internal edges or strengthen the external ones, then no edge becomes harder to match, so the "worst mismatch" cannot worsen.*

Another way to read this is as a form of order-theoretic Lipschitz behavior with respect to the pointwise order on V*-graphs: the degree functional is monotone in* H *and antitone in* G*, so "pessimistic" changes to internal commitments (lowering* G*) and "optimistic" changes to external support (raising* H*) can only move* deg *in the favorable direction.*

Corollary 48 (Robust correspondence lower bound under pruning/noise)**.** *If a reasoning system prunes weak edges from its internal graph (making* G' *smaller) and/or its estimate of the external graph becomes more supportive (making* H' *larger), then any previously established correspondence degree cannot decrease.*

Remark 3155. *This is the direct operational reading of the preceding theorem. It is especially useful in incremental systems: you may cache a previously computed correspondence degree, and later updates that only prune internal commitments or add external support do not require recomputing that degree just to ensure it has not degraded.*

In practice, this supports a common workflow: maintain a conservative lower bound on alignment between an internal representation and a world model while iteratively editing the internal graph. As long as edits are of the form "remove or downweight dubious internal edges" and/or "add or upweight supported external edges," the cached degree remains valid as a lower bound, even if the exact numerical value might become higher.

56.4.2 Robustness of reality-systems under threshold perturbations

Definition 509 (Edge stability margin for thresholded adjacency)**.** *Let* $s : U \times U \to \mathbb{R}$ *be a predictive score and fix a threshold* θ*. Define the adjacency* $\to_\theta$ *by* $x \to_\theta y$ *iff* $s(x, y) \geq \theta$*. Given a perturbation* $\tilde{s}$*, define the uniform deviation* $\|s - \tilde{s}\|_\infty := \sup_{x,y} |s(x, y) - \tilde{s}(x, y)|$*. An edge* (x, y) *has margin* $m(x, y) := |s(x, y) - \theta|$*.*

Remark 3156. *This definition separates two layers: a real-valued score function* s *(often learned or estimated) and a hard adjacency relation* $\to_\theta$ *obtained by thresholding. The sup norm* $\|s - \tilde{s}\|_\infty$ *measures worst-case perturbation over all ordered pairs, and the margin* $m(x, y)$ *measures how close a particular score is to the decision boundary* θ*.*

The philosophical point is that many realities are built by thresholding uncertain evidence, and therefore their stability depends not on the scores alone but on how decisively those scores clear (or fail to clear) the threshold. Margins quantify decisiveness. This is a standard robustness idea transposed into the present predictive-closure setting (Hyperseed-Concept 195).

Note that the adjacency is defined with a weak inequality ($\geq \theta$*), so the exact tie case* $s(x, y) = \theta$ *has margin* 0 *and is maximally fragile: an arbitrarily small negative perturbation will remove the edge, and an arbitrarily small positive perturbation will keep it (or create it if it was previously just below threshold). This makes the margin a precise local certificate of discrete stability for each ordered pair.*

Lemma 166 (Margins prevent edge flips)**.** *If* $\|s - \tilde{s}\|_\infty \leq \varepsilon$ *and* $m(x, y) > \varepsilon$*, then* $x \to_\theta y$ *iff* $x \tilde{\to}_\theta y$*.*

Remark 3157. *This lemma states a simple but powerful principle: if a score is farther than ε from the threshold, then any perturbation bounded by ε cannot change which side of the threshold it lies on. In practice, this says that only near-threshold edges are fragile; far-from-threshold edges are stable.*

The lemma is the hinge between numerical noise and discrete graph structure. Once one has it, one can reason about stability of fixed points of F by ensuring stability of the edges that actually matter for $F(R)$.

In particular, when a reality-system is defined by iterated application of an update/closure operator that depends only on the thresholded adjacency, the lemma isolates the only potential source of qualitative change under bounded score perturbations: the subset of pairs whose scores lie in the ε-band around θ. This can be used to focus verification or data collection on precisely those "knife-edge" relations.

Proof. If $s(x,y) \geq \theta$ and $s(x,y) - \theta > \varepsilon$, then $\tilde{s}(x,y) \geq s(x,y) - \varepsilon > \theta$. If $s(x,y) < \theta$ and $\theta - s(x,y) > \varepsilon$, then $\tilde{s}(x,y) \leq s(x,y) + \varepsilon < \theta$. □

Remark 3158 (Proof sketch and intuition)**.** *The strategy is to use the triangle inequality in its simplest form: a bounded perturbation can shift a score by at most ε. If the original score is at least ε away from the threshold, it cannot cross the threshold after perturbation.*

Geometrically, the interval $(\theta - \varepsilon, \theta + \varepsilon)$ is the "danger zone": only scores in that band can flip their thresholded decision under ε noise.

A useful global corollary (often applied implicitly) is: if all margins satisfy $\inf_{x,y} m(x,y) > \varepsilon$, then no *edges flip anywhere in the graph, so the entire thresholded adjacency relation $\to_\theta$ is invariant under all perturbations $\tilde{s}$ with $\|s - \tilde{s}\|_\infty \leq \varepsilon$. This is precisely the condition under which a reality-system built from $\to_\theta$ is structurally stable against uniform score noise.*

Theorem 122 (Fixed-point robustness from boundary-edge stability)**.** *Let F be defined from $\to_\theta$ as in Section 3, and let $\tilde{F}$ be defined from $\tilde{\to}_\theta$. Let $R \subseteq U$. If $\to_\theta$ and $\tilde{\to}_\theta$ agree on all edges in*

$$(R \times U) \ \cup \ (U \times R),$$

then $F(R) = \tilde{F}(R)$. In particular, if R is a reality-system for F, then it is also a reality-system for $\tilde{F}$.

In other words, the value of F at the argument R is invariant under any perturbation of the underlying directed graph that leaves unchanged (i) every arrow with tail in R (all potential "predictions made from within R") and (ii) every arrow with head in R (all potential "predictions landing inside R"). Note that this hypothesis permits arbitrary changes on $R \times R$ (edges internal to R) and on $(U \setminus R) \times (U \setminus R)$ (edges wholly outside R), as well as any changes on edges that do not touch R at all.

Equivalently, if one views $\to_\theta$ and $\tilde{\to}_\theta$ as adjacency matrices on U, then the theorem requires agreement only on the rows indexed by R and the columns indexed by R; the remaining submatrices may differ freely without affecting $F(R)$.

Remark 3159. *This theorem isolates exactly where $F(R)$ "looks" in the adjacency matrix: it only cares about edges leaving R and edges entering R. Thus, to ensure the evaluation of F on a candidate set R is robust, it is not necessary to stabilize the entire graph—only the boundary edges touching R.*

Here "boundary" should be read literally in the directed sense: the set of ordered pairs (r, u) with $r \in R$ and $u \in U$ (outgoing boundary), together with the ordered pairs (u, r) with $u \in U$ and $r \in R$ (incoming boundary). Any changes that remain entirely within the interior of R do not affect *whether R is closed under the predecessor conditions encoded by F, because those conditions are evaluated by looking outward from R and inward to R, not by inspecting internal connectivity.*

The result is important for computation and for epistemology. Computationally, it yields local robustness checks. Epistemologically, it says that the stability of a candidate reality-system depends on the stability of its boundary relations: what can predict from it and what can predict into it. One might say, with Russellian sobriety, that "reality" here is determined not by intrinsic essence but by relational closure conditions; with Whitehead, that it is maintained by patterns of ingress and egress of process [15].

A concrete way to read the conclusion is: if two agents (or two measurement runs) disagree about the world-graph only in regions that do not touch R (or only about internal relations among elements of R), then their induced closure computations agree on the status of R as a reality-system. Thus the theorem supports a notion of "shared reality" that is mediated by stable interface relations rather than by full global agreement.

Proof. $\mathrm{Pred}_{\to}(R)$ depends only on edges leaving R, i.e. pairs in $R \times U$. $\mathrm{Pred}_{\leftarrow}(R)$ depends only on edges entering R, i.e. pairs in $U \times R$. If $\to$ and $\tilde{\to}$ agree on these pairs, then both predecessor sets agree, hence their intersection agrees: $F(R) = \tilde{F}(R)$. If $F(R) = R$, then $\tilde{F}(R) = R$ as well.

More explicitly (unfolding the dependency that is implicit in the notation of Section 3), the membership of a point $u \in U$ in $\mathrm{Pred}_{\to}(R)$ is determined by whether u is reachable from (or predicted by) some $r \in R$ via the relation $\to_\theta$; this is a statement about which ordered pairs (r, u) are present. Dually, membership of u in $\mathrm{Pred}_{\leftarrow}(R)$ is determined by whether some $r \in R$ is reachable from u via $\to_\theta$, i.e. about which ordered pairs (u, r) are present. Since $F(R)$ is computed by taking the intersection of these two predecessor sets, no other edge-information can enter the calculation of $F(R)$ at all. $\square$

Remark 3160 (Proof sketch and intuition)**.** *The proof is an exercise in dependency-tracing: identify the exact input data needed to compute $F(R)$. Forward prediction depends on edges out of R; backward prediction depends on edges into R. If those edges are unchanged, then the computed predecessor sets are unchanged, hence their intersection is unchanged.*

A picture to hold: draw the subset R as a region in a directed graph. The operator F only consults arrows crossing the boundary of this region (incoming and outgoing). If the boundary arrows are stable, then the closure of the region is stable.

In particular, perturbations that "rewire" the world outside R may drastically alter global reachability, clustering, or other graph-theoretic properties, and yet still leave $F(R)$ completely unchanged so long as those perturbations do not modify any arrow that starts in R or ends in R. This clarifies why the robustness is genuinely local: it is pinned to the interface between R and its complement, not to any global feature of the ambient network.

Remark 3161 (Use)**.** *This yields an implementable robustness test: a candidate reality-system R is stable under score noise ε if all edges that touch R and matter for closure have margins $> \varepsilon$.*

Concretely, when $\to_\theta$ is obtained by thresholding an underlying score function (as in the construction leading to $\to_\theta$), an "edge margin" may be taken to mean the distance of an edge's score from the threshold θ. If every boundary pair in $(R \times U) \cup (U \times R)$ has score at least ε away from θ, then any perturbation of size at most ε cannot flip the truth value of those boundary edges, and therefore cannot change $F(R)$. In that sense, the theorem upgrades margin information on the boundary to a certificate of invariance for the fixed-point test $F(R) = R$.

56.5 Optional: tying back to "mind–world correspondence" and "reality"

Remark 3162 (How these results operationalize mind–world correspondence and reality-systems)**.** *Mind–world correspondence provides a quantitative slack q for transporting internal structure to external structure (Sections 2.2–2.3). Reality-systems provide a fixed-point criterion for what entities cohere as "real" relative to predictive coupling (Section 3). Robustness theorems show when (i) correspondence degrees are monotone under refinement/pruning, and (ii) reality-systems persist under bounded perturbations of predictive scores. Together they support a reasoning loop: maintain internal inferences, transport them outward with depth-aware penalties, update predictive edges/scores, and track stable fixed points as candidate "shared realities".*

To make the operational role of q more explicit, one may regard q as an "allowable distortion budget" that quantifies how much inferential content can survive translation across representational interfaces. In particular, when a transported statement depends on deeper chains of inference or larger bundles of constraints, the theorems justify treating it as more fragile: the same raw internal confidence should yield a smaller externally warranted confidence unless q is large enough to cover the additional structural load.

The monotonicity clause (i) can be read as a minimal sanity condition for disciplined updating: if one refines a model (e.g., adds resolution, splits latent states, or introduces additional distinctions) or prunes it (e.g., discards unsupported branches), then the induced correspondence degree should move in the expected direction rather than oscillating pathologically. This prevents "correspondence inflation" by gratuitous complexity and, conversely, prevents "correspondence collapse" from harmless simplifications, so long as the refinement/pruning respects the stated hypotheses.

Likewise, persistence under bounded perturbations in (ii) upgrades the fixed-point notion from a purely formal equilibrium to a stability claim: if predictive scores change slightly (through noise, finite-sample effects, or small environmental drift), then the reality-system should not change discontinuously. Conceptually, this is what licenses treating a fixed point as tracking a world-facing regularity rather than an artifact of a particular threshold, parametrization, or numerical idiosyncrasy.

Finally, the phrase "depth-aware penalties" can be understood as the mathematical mechanism that prevents the loop from over-exporting internal coherence: deeper derivations may remain perfectly valid internally, but their external import is tempered unless the correspondence slack q supports the additional inferential distance. In that sense, the loop enforces a separation between "coherence within a map" and "robust contact with the territory," while still allowing the two to communicate under quantitatively stated conditions.

Remark 3163. *In the Hyperseed idiom, the transfer results formalize a disciplined version of "do not confuse your map with the territory": the map-to-territory correspondence is parameterized by q, and deep inferences or large constraint-bundles must be discounted accordingly (Hyperseed-Concept 112). The shared-reality fixed points formalize how multiple maps may still overlap in a stable intersubjective core (Hyperseed-Concept* **??***), and the robustness criteria clarify when such cores are not mere artifacts of thresholding noise (Hyperseed-Concept 149).*

This discounting is not an invitation to skepticism; rather, it supplies a calibratable bridge between internal entailment and external warrant. A map may support far-reaching consequences, but the parameterization by q forces one to state (and, where possible, estimate) how reliably those consequences survive translation into externally checkable constraints. When q is small, the framework predicts that only shallow, low-complexity consequences are stably exportable; when q is larger, one can legitimately transport more structure without conflating internal elegance with external truth.

The intersubjective core, in turn, is characterized not by unanimous agreement on full models, but by convergence on the fixed points that remain stable under coupling: different agents may *disagree about hidden variables, ontological commitments, or explanatory narratives, yet still share a reality-system insofar as their predictive edges mutually stabilize the same entities as reliable anchors. The robustness requirements then express a further constraint: such overlap must survive reasonable perturbations (measurement error, shifting priors, finite data), otherwise the apparent core is better treated as a coordination artifact than as a candidate "shared reality."*

One may also read the overall structure as a pragmatic reconciliation of two impulses: the Peircean urge to define meaning by stable inferential consequences [14], and the Sagan/Dyson-style insistence that any such stability must be tested against perturbation and noise. In this sense, the mathematics here is not merely bookkeeping; it is a compact ethics of epistemic caution. Concretely, it asks that claims earn their status not only by fitting a self-consistent inferential role, but also by remaining invariant (up to controlled slack) when the coupling signals are stressed, reweighted, or partially corrupted.

Remark 3164. *A useful way to connect these ideas back to practice is to view q and the perturbation bounds as explicit "audit parameters" for epistemic workflows. For example, if two agents (or two models) agree on a predicted regularity only when q is taken unrealistically large, then the agreement is diagnostic of a shared internal encoding rather than of a shared external constraint. Conversely, if a putative reality-system persists across a range of plausible perturbation levels, then it merits being treated as a stable target for further inquiry (e.g., as a hypothesis class to be tested, or as an ontology to be used for coordination), even if the agents retain substantial disagreement about the surrounding explanatory superstructure.*

In this reading, "shared reality" is not a binary declaration but a graded outcome: one can track how the fixed points move (or fail to move) as q is tightened and as perturbation budgets are varied. The framework thereby encourages a disciplined form of pluralism: multiple maps may be kept in play, while the stable overlaps that survive transport and robustness checks are promoted as the most defensible candidates for world-facing commitments.

57 Value Conflict, Resonance, and Social Coordination

We give formal theorems that connect (i) paraconsistent value evidence, (ii) resonance/interference as a coherence proxy, and (iii) social coordination phenomena (agreement, polarization, diffusion, and stabilization of norms). The results are biased toward computational use in reasoning loops: diagnosing value conflict, measuring polarization, predicting diffusion thresholds, and selecting coordination actions without collapsing contradictions.

The organizing idea is that "value evidence" is treated as potentially inconsistent data rather than as a single real-valued utility, and that aggregation across value dimensions is performed in a way that makes both alignment and conflict measurable. The complex embedding below is a convenience: it packages two kinds of information (directional support and degree of determination) into a single signal so that geometric quantities (e.g., magnitude and cross-terms) can serve as proxies for coherence, tension, or cancellation. The social layer then treats communication as transmission of such evidence bundles through a coupling graph, so that standard coordination phenomena can be stated as properties of repeated evidence pooling under heterogeneous couplings.

57.1 Preliminaries: p-bits, resonance, and social coupling

57.1.1 p-bits and the complex embedding

Definition 510 (p-bit domain and join)**.** *Let* $V := [0,1]^2$*. Write* $v = (v^+, v^-) \in V$*. We use the componentwise (information-accumulating) partial order*

$$(v^+, v^-) \le (u^+, u^-) \iff v^+ \le u^+ \text{ and } v^- \le u^-.$$

Define join (evidence pooling)

$$v \oplus u := (\max(v^+, u^+), \max(v^-, u^-)).$$

Intuitively, a p-bit $v = (v^+, v^-)$ stores two independent pieces of evidence: support-for (v^+) and support-against (v^-) a proposition, action, or evaluative predicate. The partial order is "information-accumulating" in the sense that moving upward can only increase (never decrease) either kind of evidence; this makes $\oplus$ a natural pooling operator when evidence arrives from multiple sources and we wish to retain the strongest pro and strongest con signals simultaneously. Note that this join does not force normalization (so $v^+ + v^-$ may exceed 1), which is precisely what allows explicit representation of contradiction (v^+ and v^- both large) and ignorance (both small) without collapse to a single scalar.

A useful boundary intuition is: $(1, 0)$ corresponds to maximally supported with no counterevidence, $(0, 1)$ to maximally refuted, $(0, 0)$ to no evidence either way, and $(1, 1)$ to maximal conflict (strong evidence for and against at once). The operator $\oplus$ keeps these extremes stable under pooling: for instance $(1, 0) \oplus (0, 1) = (1, 1)$ records the emergence of explicit conflict rather than resolving it by averaging.

Definition 511 (logic-to-complex map)**.** *Define* $\sigma_C : V \to \mathbb{C}$ *by*

$$\sigma_C(v^+, v^-) \; := \; (v^+ - v^-) \; + \; i\big((v^+ + v^-) - 1\big).$$

Interpretation: $\mathrm{Re}(\sigma_C)$ *is net pro-vs-con bias;* $\mathrm{Im}(\sigma_C)$ *is over/under-determination (contradiction/ignorance axis).*

The map σ_C separates two conceptually distinct degrees of freedom. The real part $v^+ - v^-$ captures directional tilt (which side has more support), while the imaginary part $(v^+ + v^-) - 1$ captures how "determined" the proposition is relative to the reference level $v^+ + v^- = 1$. In particular, $\mathrm{Im}(\sigma_C) > 0$ indicates over-determination (simultaneous strong pro and con evidence, i.e., contradiction), whereas $\mathrm{Im}(\sigma_C) < 0$ indicates under-determination (low total evidence, i.e., ignorance or indifference). This choice centers the determination axis so that "balanced but fully determined" cases (e.g., $v^+ = v^- = 1/2$) map to the real line.

Geometrically, σ_C embeds the unit square into the complex plane as a diamond-shaped region: the corners map to $\sigma_C(1, 0) = 1 + 0i$, $\sigma_C(0, 1) = -1 + 0i$, $\sigma_C(0, 0) = 0 - i$, and $\sigma_C(1, 1) = 0 + i$. Thus, purely directional certainty lies on the real axis, while pure "status" about (over/under)-determination lies on the imaginary axis. This will let us interpret constructive vs. destructive interactions across multiple value dimensions as vector addition in $\mathbb{C}$.

57.1.2 Evaluative fields and resonance

Definition 512 (paraconsistent evaluative field)**.** *Fix a state space* S*, action set* $\mathcal{A}$*, and value-dimension set* D*. An (observer-relative) paraconsistent evaluative field is a map*

$$E : S \times \mathcal{A} \times D \to V, \quad E(s, a; d) = (E^+(s, a; d), E^-(s, a; d)).$$

One can read $E(s, a; d)$ as the evidence profile (pro/con) that an observer associates with taking action a in state s under value dimension d (e.g., fairness, harm, autonomy, loyalty). The "observer-relative" qualifier matters because different agents (or different models used by the same agent at different times) can encode different evidential bases for the same $(s, a; d)$. Importantly, paraconsistency is represented locally at the level of each dimension: a single d can carry both high E^+ and high E^-, enabling explicit modeling of internal value conflict that is not immediately resolved by choosing a single score.

Definition 513 (complex value signals, resonance magnitude, interference)**.** *Fix weights $w_d \geq 0$. For each (s, a) and $d \in D$, define*

$$z_d(s, a) := \sigma_C(E(s, a; d)) \in \mathbb{C}, \qquad Z(s, a) := \sum_{d \in D} w_d \, z_d(s, a).$$

Define the resonance magnitude $\kappa(s, a) := |Z(s, a)|$ and interference

$$I(s, a) \;:=\; |Z(s, a)|^2 \;-\; \sum_{d \in D} |w_d z_d(s, a)|^2 .$$

The complex signal $z_d(s, a)$ turns each value dimension into a 2D vector whose direction encodes whether the dimension is net approving or disapproving (real axis), and whether it is over- or under-determined (imaginary axis). The weighted sum $Z(s, a)$ then behaves like a resultant: if dimensions "point" in similar directions, they add constructively and $\kappa(s, a)$ becomes large; if they oppose, cancellation reduces $\kappa(s, a)$. In this sense, κ serves as a proxy for cross-dimensional coherence of the evaluation of (s, a), not merely the strength of any single dimension.

The interference term $I(s, a)$ isolates the cross-terms that appear when squaring a sum. Writing $Z = \sum_d \alpha_d$ with $\alpha_d := w_d z_d$, we have

$$|Z|^2 \;=\; \sum_d |\alpha_d|^2 \;+\; \sum_{d \neq d'} \mathrm{Re}\big(\alpha_d \, \overline{\alpha_{d'}}\big),$$

so $I(s, a)$ is exactly the sum of pairwise interactions $\sum_{d \neq d'} \mathrm{Re}(\alpha_d \overline{\alpha_{d'}})$. Positive interference indicates alignment (net constructive combination across dimensions), while negative interference indicates antagonism (net cancellation), which operationalizes "value conflict" at the aggregate level even when each dimension is itself internally paraconsistent. Because z_d includes the determination axis, two dimensions can also interfere through shared contradiction/ignorance structure, not only through pro/con direction.

A small sanity check: if D has a single dimension, then $I(s, a) = 0$ by construction; interference is purely relational across dimensions. Likewise, if all z_d are mutually orthogonal in the complex plane (in the sense that $\mathrm{Re}(z_d \overline{z_{d'}}) = 0$ after weighting), then $I(s, a) = 0$ and the magnitude squared is additive.

57.1.3 Social coupling and messages (minimal)

Definition 514 (interaction graph and couplings)**.** *A society is modeled (minimally) by a directed graph $G = (A, E)$ of agents and couplings $K : E \to [0, 1]$ (scalar case) or $K : E \to V$ (ambivalent case).*

The edge set E represents who influences whom; direction matters because attention, trust, authority, or exposure can be asymmetric. A scalar coupling $K(i \to j) = k$ can be read as an overall assimilation rate or channel strength from agent i to agent j. The ambivalent case $K : E \to V$ is included to allow couplings that themselves carry pro/con structure (e.g., simultaneous attraction and distrust), although the concrete update rule below specializes to the scalar case for simplicity.

Definition 515 (messages as paraconsistent evidence bundles and assimilation (scalar coupling))**.** *Fix a claim language L. A message is a finitely supported $m : L \to V$. Given scalar coupling $k \in [0,1]$, define scaled message $(k \odot m)(\varphi) := (k,k) \otimes m(\varphi)$, where (for concreteness) we take $\otimes$ to be coordinatewise multiplication: $(x^+, x^-) \otimes (y^+, y^-) = (x^+y^+, x^-y^-)$. Belief update is*

$$\beta \leftarrow \beta \oplus (k \odot m),$$

pointwise on L.

Finite support means that each message asserts only finitely many claims in L, which is convenient for algorithmic updates and matches the idea that communication episodes are sparse relative to the full space of possible propositions. The scaling rule $(k,k)\otimes m(\varphi)$ implements a simple attenuation model: lower coupling uniformly dampens both supportive and opposing evidence transmitted by the message, while $k = 1$ passes the evidence through unchanged. Because scaling is applied before the join, strong couplings can quickly propagate high evidence (either pro or con), whereas weak couplings slow diffusion.

The update $\beta \leftarrow \beta \oplus (k \odot m)$ is monotone in the information order: neither E^+ nor E^- for any claim ever decreases under repeated pooling. This captures an "accumulation" regime appropriate for settings where agents store received evidence rather than averaging it away. In later results, this monotonicity will allow diffusion and stabilization statements to be phrased as reachability and fixed-point properties on V^L under iterated message passing, while paraconsistency ensures that contradictory streams can coexist as explicit structure rather than forcing immediate resolution.

57.2 Value-conflict geometry: interference, contradiction growth, and decision diagnostics

57.2.1 Conflict accumulation under social evidence pooling

Definition 516 (two simple contradiction/conflict readouts)**.** *For $v = (v^+, v^-) \in V$, define*

$$\mathrm{contr}(v) := (v^+ + v^-) - 1 = \mathrm{Im}(\sigma_C(v)), \qquad \mathrm{conf}(v) := \min(v^+, v^-).$$

Theorem 123 (Pooling never reduces contradiction or conflict)**.** *For all $u, v \in V$,*

$$\mathrm{contr}(u \oplus v) \;\geq\; \max(\mathrm{contr}(u), \mathrm{contr}(v)), \qquad \mathrm{conf}(u \oplus v) \;\geq\; \max(\mathrm{conf}(u), \mathrm{conf}(v)).$$

Proof. Write $u = (a,b)$, $v = (c,d)$ with $a,b,c,d \in [0,1]$. Then $u \oplus v = (a \vee c, b \vee d)$ where $\vee = \max$. Since $a \vee c \geq a$ and $b \vee d \geq b$, we have $(a \vee c) + (b \vee d) - 1 \geq a + b - 1$, and similarly $\geq c + d - 1$, proving the first inequality.

For conf, note

$$\mathrm{conf}(u \oplus v) = (a \vee c) \wedge (b \vee d)$$

with $\wedge = \min$. In the distributive lattice $([0,1], \vee, \wedge)$,

$$(a \vee c) \wedge (b \vee d) \;\geq\; (a \wedge b) \vee (c \wedge d),$$

hence $\mathrm{conf}(u \oplus v) \geq \max(\mathrm{conf}(u), \mathrm{conf}(v))$. ∎

Remark 3165 (Reasoning use)**.** *In a join-based social epistemic/evaluative layer, contradictions naturally* accumulate *when aggregating many sources. Thus "resolve by pooling more" is structurally incapable of reducing contradiction; one needs explicit revision/forgetting/discounting operators to do that.*

57.2.2 Interference decompositions that explain value conflict

Definition 517 (pairwise phase-alignment)**.** *For nonzero $x, y \in \mathbb{C}$, define* $\operatorname{align}(x,y) := \operatorname{Re}(x\overline{y})/(|x|\,|y|) \in [-1,1]$*. If $x=0$ or $y=0$, set* $\operatorname{align}(x,y)=0$*.*

Theorem 124 (Interference is a weighted sum of pairwise alignments)**.** *Let $(z_d)_{d\in D} \subset \mathbb{C}$ and weights $w_d \ge 0$. Set $Z = \sum_d w_d z_d$ and $r_d := |w_d z_d|$. Then*

$$|Z|^2 = \sum_{d\in D} r_d^2 \; + \sum_{\substack{d,e\in D\\ d\neq e}} w_d w_e \operatorname{Re}\bigl(z_d\overline{z_e}\bigr),$$

and equivalently

$$I = \sum_{d<e} 2\, r_d r_e \operatorname{align}(z_d, z_e).$$

Proof. Expand:

$$|Z|^2 = Z\overline{Z} = \left(\sum_d w_d z_d\right)\left(\sum_e w_e \overline{z_e}\right) = \sum_{d,e} w_d w_e z_d \overline{z_e}.$$

Split $d=e$ from $d\neq e$: $\sum_d w_d^2 |z_d|^2 = \sum_d |w_d z_d|^2 = \sum_d r_d^2$, and for $d \neq e$ take real parts since $|Z|^2 \in \mathrm{R}$ and cross terms pair up: $z_d\overline{z_e}$ and $z_e\overline{z_d}$ have the same real part. Thus

$$|Z|^2 = \sum_d r_d^2 + \sum_{d\neq e} w_d w_e \operatorname{Re}(z_d\overline{z_e}).$$

Subtracting $\sum_d r_d^2$ yields the first form for I; dividing each term by $r_d r_e$ (where nonzero) yields the alignment form. $\square$

Remark 3166 (Reasoning use)**.** *This gives an "explanation object" for value conflict: negative interference $I<0$ is literally net negative pairwise alignment among value-dimension complex signals, i.e. cancellation in the complex embedding.*

Theorem 125 (Resonance gap quantifies phase dispersion)**.** *Let $Z = \sum_d w_d z_d$ and $r_d = |w_d z_d|$. Define $S := \sum_d r_d$. Then*

$$0 \le S^2 - |Z|^2 = \sum_{d<e} 2\, r_d r_e \bigl(1-\cos(\theta_d-\theta_e)\bigr) = \sum_{d<e} 4\, r_d r_e \sin^2\!\left(\frac{\theta_d-\theta_e}{2}\right),$$

where θ_d is any argument of $w_d z_d$ (when $r_d>0$). Moreover, $S=|Z|$ iff all nonzero $w_d z_d$ share the same phase.

Proof. Write $w_d z_d = r_d e^{i\theta_d}$ with $r_d \ge 0$. Then

$$|Z|^2 = \left|\sum_d r_d e^{i\theta_d}\right|^2 = \sum_d r_d^2 + \sum_{d<e} 2 r_d r_e \cos(\theta_d-\theta_e).$$

Also $S^2 = (\sum_d r_d)^2 = \sum_d r_d^2 + \sum_{d<e} 2 r_d r_e$. Subtract to get

$$S^2 - |Z|^2 = \sum_{d<e} 2 r_d r_e \bigl(1-\cos(\theta_d-\theta_e)\bigr) = \sum_{d<e} 4 r_d r_e \sin^2\!\left(\frac{\theta_d-\theta_e}{2}\right) \ge 0.$$

Equality $S^2 = |Z|^2$ holds iff every $\sin^2((\theta_d-\theta_e)/2) = 0$ for pairs with $r_d r_e > 0$, i.e. all relevant phases coincide. Finally $S = |Z|$ follows since both are nonnegative. $\square$

Remark 3167 (Decision tool)**.** *A reasoning system can treat* $S^2 - |Z|^2$ *as a* quantitative *dissonance measure: it decomposes into pairwise phase-dispersion penalties weighted by salience magnitudes* $r_d r_e$*.*

57.3 Social coordination: group resonance, polarization identities, and decompositions

57.3.1 Agents as complex value-signal generators

Definition 518 (agent and group value signals)**.** *Fix an option* (s, a) *and let each agent* $i \in A$ *have complex value signal*

$$Z_i := \sum_{d \in D} w_{i,d}\, \sigma_C(E_i(s, a; d)) \in \mathbb{C}.$$

Let $(\alpha_i)_{i \in A}$ *be nonnegative weights with* $\sum_i \alpha_i = 1$*. Define group-mean signal*

$$\overline{Z} := \sum_{i \in A} \alpha_i Z_i, \quad \textit{and group resonance} \quad \kappa_G := \left|\overline{Z}\right|.$$

The object $Z_i \in \mathbb{C}$ can be read geometrically as a point (or vector) in a two-dimensional value-signal plane, where both *direction* (phase) and *strength* (magnitude) matter for coordination. The map $\sigma_C(\cdot)$ is left abstract here, but its role is to turn each domain-specific evaluation $E_i(s, a; d)$ into a bounded complex contribution that can represent, for example, (i) the sign and intensity of valence, (ii) a tradeoff between two partially incommensurable desiderata via the real and imaginary axes, or (iii) a circular/periodic attitude (phase) together with confidence (magnitude). The weights $w_{i,d}$ encode the agents internal salience over domains, while the social weights α_i encode the aggregation rule (e.g., equal agents, population shares, institutional voting weights, or attention weights). Under this reading, $\overline{Z}$ is a weighted centroid of signals, and $\kappa_G = |\overline{Z}|$ measures how far that centroid lies from the origin: high resonance means the groups signals align sufficiently to produce a strong resultant, while low resonance can arise either from weak individual signals or from strong but mutually canceling signals.

Theorem 126 (Polarization identity: resonance vs disagreement variance)**.** *Define polarization (complex variance)*

$$\mathrm{Pol} := \sum_{i \in A} \alpha_i \left|Z_i - \overline{Z}\right|^2.$$

Then

$$\mathrm{Pol} = \sum_{i \in A} \alpha_i |Z_i|^2 - \left|\overline{Z}\right|^2.$$

Proof. Expand $\left|Z_i - \overline{Z}\right|^2 = |Z_i|^2 - 2\,\mathrm{Re}(Z_i\overline{\overline{Z}}) + \left|\overline{Z}\right|^2$. Multiply by α_i and sum:

$$\mathrm{Pol} = \sum_i \alpha_i |Z_i|^2 - 2\,\mathrm{Re}\left(\sum_i \alpha_i Z_i \overline{\overline{Z}}\right) + \left(\sum_i \alpha_i\right) \left|\overline{Z}\right|^2.$$

But $\sum_i \alpha_i Z_i = \overline{Z}$ and $\sum_i \alpha_i = 1$, so

$$\mathrm{Pol} = \sum_i \alpha_i |Z_i|^2 - 2\,\mathrm{Re}(\overline{Z}\overline{\overline{Z}}) + \left|\overline{Z}\right|^2 = \sum_i \alpha_i |Z_i|^2 - \left|\overline{Z}\right|^2.$$

$\square$

This identity is exactly the familiar variance decomposition $\mathrm{Var}(Z) = \mathbb{E}|Z|^2 - |\mathbb{E}Z|^2$, with $\mathbb{E}$ realized by the weighted average $\sum_i \alpha_i(\cdot)$ and with $\mathbb{C}$ viewed as R^2. In particular, $\mathrm{Pol} \geq 0$ automatically, and $\mathrm{Pol} = 0$ holds if and only if all agents have the same signal $Z_i = \overline{Z}$ for all i with $\alpha_i > 0$ (perfect alignment). The term $\sum_i \alpha_i |Z_i|^2$ can be interpreted as the groups total "signal energy" (a second moment), while $|\overline{Z}|^2$ is the squared magnitude of the coherent component. Thus the theorem formalizes a basic coordination intuition: holding individual signal energy fixed, any increase in coherent alignment (resonance) must come from a decrease in incoherent dispersion (polarization).

Corollary 49 (Resonance-maximizing is polarization-minimizing under fixed energy)**.** *If $\sum_i \alpha_i |Z_i|^2$ is the same constant across a candidate set of actions, then maximizing group resonance $\left|\overline{Z}\right|$ is equivalent to minimizing* Pol*.*

Proof. Immediate from Theorem 126, since $\mathrm{Pol} = \mathrm{const} - \left|\overline{Z}\right|^2$. □

The "fixed energy" condition is a useful modeling idealization: it holds, for instance, when σ_C normalizes contributions to have fixed magnitude (so only phases/directions change across actions), or when actions are compared after rescaling to equalize overall intensity so that only alignment structure differs. When energy is not fixed, resonance and polarization become partially decoupled: one action can raise $|\overline{Z}|$ simply by increasing many $|Z_i|$ (stronger opinions) even if disagreement also grows. The corollary isolates the clean case in which resonance is purely an alignment objective rather than an intensity objective.

Theorem 127 (Pairwise-distance form of polarization)**.** *With $\sum_i \alpha_i = 1$,*

$$\mathrm{Pol} = \frac{1}{2} \sum_{i,j \in A} \alpha_i \alpha_j \left|Z_i - Z_j\right|^2.$$

Proof. Expand $|Z_i - Z_j|^2 = |Z_i|^2 + |Z_j|^2 - 2\,\mathrm{Re}(Z_i \overline{Z_j})$. Then

$$\frac{1}{2} \sum_{i,j} \alpha_i \alpha_j \left|Z_i - Z_j\right|^2 = \sum_i \alpha_i \left|Z_i\right|^2 - \mathrm{Re}\left(\sum_{i,j} \alpha_i \alpha_j Z_i \overline{Z_j} \right).$$

But $\sum_{i,j} \alpha_i \alpha_j Z_i \overline{Z_j} = (\sum_i \alpha_i Z_i) \overline{\left(\sum_j \alpha_j Z_j\right)} = \overline{Z}\,\overline{\overline{Z}} = \left|\overline{Z}\right|^2$ (real). So the expression equals $\sum_i \alpha_i \left|Z_i\right|^2 - \left|\overline{Z}\right|^2 = \mathrm{Pol}$ by Theorem 126. □

Remark 3168 (Reasoning use)**.** *Theorem 127 makes polarization an explicit "explanation object": it is exactly the mean pairwise disagreement (squared distance) in the complex value-signal plane.*

The pairwise form emphasizes that polarization is not merely deviation from a central tendency, but an aggregate of all bilateral frictions. This makes it naturally compatible with network and bargaining interpretations: if agents interact pairwise with "cost" proportional to $|Z_i - Z_j|^2$, then Pol is (up to the $\frac{1}{2}$ convention) the expected cost under independently sampling two agents according to (α_i). It also clarifies how a small minority with small total weight can have limited contribution to Pol even if its signal is extreme, while two large blocs with opposed signals can dominate Pol through many cross-bloc pairs.

Theorem 128 (Two-bloc decomposition: within vs between polarization)**.** *Partition agents into two blocs A_1, A_2 with total weights $a := \sum_{i\in A_1} \alpha_i$ and $b := \sum_{j\in A_2} \alpha_j$ ($a + b = 1$). Define bloc means $Z_{A_1} := \frac{1}{a}\sum_{i\in A_1} \alpha_i Z_i$ (if $a > 0$) and similarly Z_{A_2}. Define within-bloc polarizations*

$$\mathrm{Pol}_{A_1} := \sum_{i\in A_1} \frac{\alpha_i}{a} |Z_i - Z_{A_1}|^2, \qquad \mathrm{Pol}_{A_2} := \sum_{j\in A_2} \frac{\alpha_j}{b} |Z_j - Z_{A_2}|^2.$$

Then the global polarization decomposes as

$$\mathrm{Pol} = a\,\mathrm{Pol}_{A_1} + b\,\mathrm{Pol}_{A_2} + ab\,|Z_{A_1} - Z_{A_2}|^2.$$

Proof. This is the law of total variance in $\mathbb{C} \cong \mathrm{R}^2$. Write $\overline{Z} = aZ_{A_1} + bZ_{A_2}$. For $i \in A_1$,

$$Z_i - \overline{Z} = (Z_i - Z_{A_1}) + (Z_{A_1} - \overline{Z}),$$

and summing $\alpha_i(Z_i - Z_{A_1})$ over A_1 gives 0, so cross terms vanish:

$$\sum_{i\in A_1} \alpha_i \left|Z_i - \overline{Z}\right|^2 = \sum_{i\in A_1} \alpha_i |Z_i - Z_{A_1}|^2 + a\left|Z_{A_1} - \overline{Z}\right|^2.$$

Similarly for A_2. Add the two equalities and note $a\left|Z_{A_1} - \overline{Z}\right|^2 + b\left|Z_{A_2} - \overline{Z}\right|^2 = ab\,|Z_{A_1} - Z_{A_2}|^2$. Finally rewrite $\sum_{i\in A_1} \alpha_i |Z_i - Z_{A_1}|^2 = a\,\mathrm{Pol}_{A_1}$ and similarly for A_2. □

The decomposition separates two qualitatively different sources of disagreement. The terms $a\,\mathrm{Pol}_{A_1}$ and $b\,\mathrm{Pol}_{A_2}$ measure dispersion *inside* each bloc after reweighting to a within-bloc probability distribution (so Pol_{A_k} is a variance relative to the bloc mean). The term $ab\,|Z_{A_1} - Z_{A_2}|^2$ measures *between*-bloc separation: it scales with both the distance between bloc centroids and with the product ab, meaning it is largest when both blocs are large and sharply separated. Edge cases behave as expected: if $a = 0$ or $b = 0$ (one bloc empty), the between term vanishes and Pol reduces to the within term of the nonempty bloc; if each bloc is internally coherent (near-zero $\mathrm{Pol}_{A_1}, \mathrm{Pol}_{A_2}$), then global polarization is essentially the squared distance between two representative signals times ab. This is the complex-plane analogue of a two-group ANOVA identity, and it can be iterated (or generalized) to multi-bloc partitions when one wants to attribute polarization to nested levels of social structure.

Remark 3169 (Reasoning use)**.** *This theorem supports causal explanation of polarization: "is disagreement mostly within-group noise, or primarily between two coherent camps?"*

57.4 Communication and diffusion: from couplings to reachability of value evidence

The goal of this subsection is to make explicit how local pairwise couplings $K(u, v)$ induce global, multi-hop reachability of value-relevant evidence. In particular, the following statements isolate a *path-closure* perspective: forwarding along edges generates contributions indexed by directed paths, and the recipient's state is bounded below by joining all such path contributions (up to the number of rounds elapsed). This is useful when couplings are heterogeneous and when one wishes to reason about attenuation as a function of distance, bottlenecks, and alternative routes.

57.4.1 Path-closure form of forwarded evidence

Definition 519 (path attenuation)**.** *Let $G = (A, E)$ be a directed graph with scalar couplings $K : E \to [0, 1]$. For a directed path $p : i_0 \to i_1 \to \cdots \to i_\ell$, define its attenuation*

$$\mathrm{att}(p) := \prod_{t=0}^{\ell-1} K(i_t, i_{t+1}) \in [0, 1].$$

The quantity $\mathrm{att}(p)$ can be read as the multiplicative survival or transmission factor of a signal traversing the particular route p. The choice of product corresponds to assuming independent per-hop attenuation and is compatible with the intended interpretation of each $K(u, v) \in [0, 1]$ as a (possibly asymmetric) strength of influence or fidelity of transmission. Note that the definition applies to any finite directed path; in particular, if one wishes to include the trivial path of length 0 (from a node to itself), then the empty product convention yields $\mathrm{att}(p) = 1$, which matches the idea of no attenuation when no forwarding occurs.

Theorem 129 (Forwarding + assimilation yields join over path contributions)**.** *Fix a message $m : L \to V$ sent by a source $s \in A$ and suppose it is forwarded along directed edges, with assimilation at each hop: at each hop $(u \to v)$ the forwarded bundle is scaled by $K(u, v)$ and joined into the recipient belief. Let $\beta^{(0)}$ be the initial belief states and $\beta^{(t)}$ the states after t synchronous forwarding rounds. Then for each agent j and proposition $\varphi \in L$,*

$$\beta_j^{(t)}(\varphi) \;\geq\; \beta_j^{(0)}(\varphi) \;\oplus\; \bigoplus_{\substack{p:s\to j \\ \mathrm{len}(p)\leq t}} \left(\mathrm{att}(p) \odot m\right)(\varphi),$$

where $\odot$ is scalar-to-V scaling and the join is taken in V (pointwise on φ).

The inequality form emphasizes that $\beta^{(t)}$ may contain additional information beyond forwarded m (e.g. prior evidence, other exogenous messages, or internal inference/closure). The statement is therefore a robust lower bound: it isolates what must be present purely due to forwarding and assimilation, independent of other update mechanisms. The join $\oplus$ and big-join $\bigoplus$ should be read as the aggregator for evidential bundles in V; in many instantiations, $\oplus$ is the coordinatewise supremum or an idempotent/commutative merge, so that receiving the same contribution via multiple redundant paths does not artificially inflate evidence beyond what the merge operation permits. When $\oplus$ is idempotent, repeated reception of identical (or dominated) contributions has no further effect, which is a natural way to represent saturation or non-additive assimilation.

The restriction to paths of length at most t corresponds to the synchronous-rounds assumption: in each round, an agent forwards what it currently has (or the designated forwarded bundle), and thus information can traverse at most one edge per round. In asynchronous or continuous-time variants, the same path decomposition typically reappears, but length is replaced by a time budget and edge traversal delays; the essential combinatorics of accumulating over directed routes remains the same.

Proof. We argue by induction on t. For $t = 0$ there are no nontrivial paths, so the statement reduces to $\beta_j^{(0)} \geq \beta_j^{(0)}$.

Assume true for t. Consider $t + 1$. In one more round, any path $p : s \to j$ of length $\leq t + 1$ is either: (i) length $\leq t$ already covered by the inductive hypothesis, or (ii) of the form $p = p' \cdot (u \to j)$ where $p' : s \to u$ has length $\leq t$ and $(u \to j) \in E$. By the forwarding rule, the contribution along p is

$$\mathrm{att}(p) \odot m = \left(\mathrm{att}(p')K(u, j)\right) \odot m = K(u, j) \odot \left(\mathrm{att}(p') \odot m\right),$$

and it is joined into $\beta_j^{(t+1)}$. Since join is monotone and associative/commutative, joining all such new contributions yields the displayed lower bound for $t + 1$. $\square$

The inductive step implicitly uses two structural properties that are typically assumed in evidential algebras used for such diffusion models: (a) scalar scaling $\odot$ is monotone in the scalar (so larger couplings do not decrease the scaled message), and (b) join $\oplus$ is monotone in each argument (so adding more incoming contributions cannot reduce belief). Together these ensure that the enumerate all paths and join their attenuated messages construction behaves as a genuine lower envelope of what the dynamics can accumulate. Note also that cycles in G do not cause a problem at finite t: only finitely many paths have length $\leq t$, so the big-join is over a finite set; moreover, attenuation products along longer cyclic paths tend to shrink when $K < 1$ on at least one edge in the cycle, making explicit the intuition that repeated forwarding through imperfect channels does not create unbounded amplification.

Corollary 50 (Strong-tie diffusion at threshold)**.** *Fix a threshold $\tau \in (0, 1]$ and suppose the τ-strong-tie graph $G_\tau = (A, E_\tau)$ with $E_\tau = \{(u, v) \in E : K(u, v) \geq \tau\}$ has a directed path from s to j. If a message m has coordinatewise evidence at least θ on some claim φ, i.e. $m(\varphi) \geq (\theta, \theta)$, then after ℓ rounds (where ℓ is the path length), agent j has at least $(\tau^\ell\theta, \tau^\ell\theta)$ additional evidence on φ (before any extra closure).*

This corollary packages the path-based statement into an easily checkable condition: connectivity in the thresholded graph G_τ guarantees a minimum-strength route, and the minimum coupling τ along that route yields a clean exponential lower bound τ^ℓ. The coordinatewise expression (θ, θ) is intended to match the common case where V is a product order (e.g. a two-sided evidence representation, or a pair of lower/upper supports), and where scaling by a scalar acts componentwise. In such settings, the inequality $m(\varphi) \geq (\theta, \theta)$ is simply a concise way of saying that the message provides at least θ units of evidence in each relevant coordinate, and attenuation preserves this ordering by multiplying both coordinates by the same factor.

Proof. Take the path p of length ℓ with all edges $\geq \tau$. Then $\mathrm{att}(p) \geq \tau^\ell$ and scaling is coordinatewise multiplication, so $(\mathrm{att}(p) \odot m)(\varphi) \geq (\tau^\ell\theta, \tau^\ell\theta)$. Now apply Theorem 129. $\square$

The phrase before any extra closure is included to separate two effects: (i) the guaranteed imported evidence due to forwarding/assimilation, and (ii) any additional strengthening that might occur because the recipient applies local inference rules, consistency closure, aggregation with other messages, or value-laden interpretive steps after the round. In other words, the bound refers to the direct contribution attributable to the communication channel itself, without assuming anything about the agents subsequent reasoning pipeline.

Remark 3170 (Reasoning use)**.** *These results let a reasoning system answer: "who can influence whom, at what minimum strength, given couplings?" and "how long until a value signal reaches a target subgroup?"*

More concretely, one can treat Theorem 129 as a basis for *reachability queries* that are sensitive not only to graph distance but also to coupling geometry: two agents can be equally many hops away from s yet differ dramatically in guaranteed received evidence due to bottleneck edges (small K) or, conversely, due to the availability of multiple medium-strength routes whose joined contributions exceed what any single path would provide. Similarly, Corollary 50 provides a conservative planning heuristic: choosing τ selects a trusted backbone of communication, and the resulting ℓ-step bound quantifies how quickly an initially strong value signal may decay as it propagates, which is central when assessing whether coordination requires short pathways, high-trust intermediaries, or repeated re-injection of evidence from the source.

57.5 Norm stabilization: explicit fixed points and contested culture propagation

We now connect value conflict to stabilized social patterns (norms/rituals/institutions) via max/min reinforcement (a toy but structurally revealing model). In words, we treat the intensity of a single social pattern P as a *graded* state at each agent, and we model reinforcement as a one-step update that (i) preserves what the agent already has and (ii) admits influence from in-neighbors, but only to the extent permitted by both the neighbor's current intensity and the coupling strength on the edge. This yields a convenient "fuzzy reachability" semantics: the pattern can travel along directed paths, yet along each path it is limited by the weakest link (a bottleneck) and by the initial seed intensity at the source.

Definition 520 (scalar social reinforcement operator for a single pattern)**.** *Fix a directed graph $G = (A, E)$, couplings $K : E \to [0, 1]$, and a single pattern P. Let $x \in [0, 1]^A$ be intensities x_i of pattern P at each agent. Define $F : [0, 1]^A \to [0, 1]^A$ by*

$$(F(x))_i := x_i \;\vee \bigvee_{(j,i)\in E} \big(K(j, i) \wedge x_j\big).$$

The lattice operations $\vee$ and $\wedge$ should be read as max and min (respectively), so that each update is "conservative upward": intensities do not decrease. The term $K(j, i) \wedge x_j$ is the amount of P that can be transmitted from j to i in one step, limited simultaneously by the sender's current level x_j and by the edge's coupling $K(j, i)$. The outer $\vee$ aggregates multiple possible influences (multiple in-neighbors) by taking the strongest available reinforcement. Thus F can be interpreted as a one-step closure operator for a weighted directed graph, akin to a max–min (a.k.a. bottleneck) composition rule familiar from fuzzy relations.

Theorem 130 (Closed form for the least stabilized intensity (max-min path formula))**.** *Let $x^{(0)} \in [0, 1]^A$ and define $x^{(t+1)} := F(x^{(t)})$. For each agent i define*

$$x_i^{(\infty)} := \sup_{t\geq 0} x_i^{(t)}.$$

Then $x^{(\infty)}$ is the least fixed point of F above $x^{(0)}$, and moreover

$$x_i^{(\infty)} = \sup_{p:s\to i} \min\Big(x_s^{(0)},\ \min_{(u\to v)\in p} K(u, v)\Big),$$

where the supremum ranges over all directed paths p ending at i (including length 0 paths).

Proof. Monotonicity of F implies the Kleene chain $x^{(0)} \leq x^{(1)} \leq \cdots$ is ascending, hence $x^{(\infty)} = \sup_t x^{(t)}$ exists coordinatewise and is a post-fixed point; standard complete-lattice arguments give it is the least fixed point above $x^{(0)}$.

For the explicit formula, prove by induction that $x_i^{(t)}$ equals the supremum over all paths of length $\leq t$ of the stated bottleneck value. For $t = 0$ only length-0 paths contribute, giving $x_i^{(0)}$. Assume true for t. Then $x_i^{(t+1)}$ takes the max of (i) the previous value (paths of length $\leq t$) and (ii) terms $K(j, i) \wedge x_j^{(t)}$, which correspond exactly to extending any path ending at j by one edge into i, producing paths of length $\leq t+1$ and updating the bottleneck by an extra min with $K(j, i)$. Taking $\sup_t$ yields the all-path supremum. $\square$

The closed form makes clear that $x_i^{(\infty)}$ is a *widest-path* value (maximizing the minimal edge weight along the route), except that the source node contributes an additional cap via $x_s^{(0)}$. Equivalently, each source s offers a candidate influence level to i equal to the best bottleneck coupling from s to i, then clipped by the seed intensity at s, and finally all sources compete via a supremum. This is precisely the sense in which the model isolates a structural mechanism of norm stabilization: stable intensities are determined by weighted reachability and bottlenecks, not by additive accumulation.

Remark 3171 (Finite-time stabilization on finite graphs)**.** *If A is finite, then the path-based supremum in Theorem 130 may be taken over simple paths (any cycle can only weakly decrease the bottleneck via an additional* min*), so the relevant path lengths are bounded by $|A|-1$. Consequently, for finite A the ascending chain $(x^{(t)})_{t\geq 0}$ stabilizes after finitely many steps (at most $|A|-1$ iterations suffice to realize all simple-path contributions), and $x^{(\infty)} = x^{(T)}$ for some finite T. This observation connects the abstract least-fixed-point statement to an operational viewpoint: repeated local reinforcement computes a global closure in bounded time on finite networks.*

Corollary 51 (Global spread criterion at threshold θ)**.** *Fix $\theta \in (0,1]$. Let $S_\theta := \{s : x_s^{(0)} \geq \theta\}$. Then an agent i satisfies $x_i^{(\infty)} \geq \theta$ iff there exists a path from some $s \in S_\theta$ to i such that every edge on the path has coupling at least θ.*

Proof. By Theorem 130, $x_i^{(\infty)} \geq \theta$ iff there exists a path $p : s \to i$ with $\min(x_s^{(0)}, \min_{e\in p} K(e)) \geq \theta$, i.e. $x_s^{(0)} \geq \theta$ and $K(e) \geq \theta$ for all $e \in p$. □

This threshold view can be read as a crisp percolation statement on a θ-filtered subgraph: delete all nodes with sub-threshold initial intensity (i.e. restrict to seeds in S_θ) and delete all edges with coupling $< \theta$; then $x_i^{(\infty)} \geq \theta$ exactly when i is reachable from the surviving seeds. Thus the graded dynamics reduce, at any fixed θ, to reachability in a deterministic graph, which is often the right lens for discussing "whether the norm takes hold" at a socially meaningful strength level.

57.5.1 Contested norms (paraconsistent cultural states)

Definition 521 (p-bit norm intensities and coordinatewise reinforcement)**.** *Let each agent have p-bit intensity $X_i = (X_i^+, X_i^-) \in V$ for a pattern P (support/opposition). Define reinforcement coordinatewise:*

$$X^{(t+1),+} := F(X^{(t),+}), \qquad X^{(t+1),-} := F(X^{(t),-}),$$

with F as in the scalar case.

The intended semantics is that X_i^+ and X_i^- are not constrained to sum to 1 or to be complementary: an agent can simultaneously exhibit strong pro- and strong anti- orientations toward the same pattern P. This "two-channel" representation is a minimal paraconsistent encoding of value conflict: it allows contradiction to be a *state* rather than an error, and it permits social reinforcement to act on each pole without forcing premature resolution. Coordinatewise reinforcement further isolates a structural claim: even if individuals do not reconcile support and opposition internally, each pole can still propagate through the network by the same bottleneck mechanism as in the scalar case.

Theorem 131 (Contestation percolation criterion)**.** *Fix thresholds $\theta_+, \theta_- \in (0,1]$. Assume there exist seeds s_+, s_- with $X_{s_+}^{(0),+} \geq \theta_+$ and $X_{s_-}^{(0),-} \geq \theta_-$. If for every agent i there is a path from s_+*

to i whose edges all have coupling at least θ_+, and also a path from s_- to i whose edges all have coupling at least θ_-, then in the stabilized state:

$$X_i^{(\infty)} \geq (\theta_+, \theta_-) \quad \textit{for all } i.$$

In particular, the pattern becomes society-wide contested *(strong support and strong opposition everywhere).*

Proof. Apply Corollary 51 to the $+$ channel with threshold θ_+ to get $X_i^{(\infty),+} \geq \theta_+$ for all i. Apply it to the $-$ channel with threshold θ_- to get $X_i^{(\infty),-} \geq \theta_-$ for all i. Combine coordinates. □

The hypothesis allows $\theta_+ \neq \theta_-$, so the two channels may have different "propagation demands": for example, support might require high-coupling ties (a high θ_+) while opposition spreads through weaker ties (a lower θ_-), or vice versa. The conclusion shows that if both channels have sufficiently robust reachability at their own thresholds, then contradiction is not localized but becomes an invariant feature of the stabilized culture. Notably, nothing in the argument requires the pro- and anti- seeds to be the same individual, nor does it require the paths to coincide; contestation can emerge from distinct subpopulations whose influence simultaneously reaches everyone.

Remark 3172 (Reasoning use)**.** *This theorem makes "how contradictions become cultural" fully explicit: it is a reachability condition in two channels with possibly different thresholds.*

Remark 3173 (Interpretive consequence)**.** *Because each coordinate is a least fixed point above its respective initialization, the stabilized contested state is in this sense* minimal*: it is the weakest society-wide configuration that is still closed under reinforcement given the initial pro/anti seeds and couplings. So, when the criterion holds, contestation is not an artifact of assuming maximal polarization; it is an inevitable closure outcome of the network's two-threshold connectivity structure.*

57.6 Non-explosive social choice: existence and a polarization-aware resonance rule

57.6.1 Pareto safety and existence (finite action sets)

Definition 522 (p-bit preference order and Pareto order)**.** *On V define $v \preceq_{\mathrm{pref}} u$ iff $v^+ \leq u^+$ and $v^- \geq u^-$ ("more pro and less con"). For value-vectors $(E(s,a;d))_{d\in D}$ define Pareto dominance coordinatewise: $a \preceq_{\mathrm{Pareto}} a'$ iff $E(s,a;d) \preceq_{\mathrm{pref}} E(s,a';d)$ for all $d \in D$.*

The order $\preceq_{\mathrm{pref}}$ is the natural product order on the "pro" coordinate together with the reverse order on the "con" coordinate: moving upward in $\preceq_{\mathrm{pref}}$ means weakly increasing the pro component while weakly decreasing the con component. In particular, $v \preceq_{\mathrm{pref}} u$ allows equality in either coordinate, so it represents a *weak* improvement (not necessarily strict), and it treats the pro and con channels as simultaneously relevant rather than collapsing them into a single scalar. The induced $\preceq_{\mathrm{Pareto}}$ relation then asserts that an action a' is socially at least as good as a when it is weakly better for *every* dimension $d \in D$ under this two-channel comparison.

Theorem 132 (Non-explosion: Pareto-undominated actions exist in finite sets)**.** *If the action set $\mathcal{A}$ is finite, then the set of Pareto-undominated actions (with respect to $\preceq_{\mathrm{Pareto}}$) is nonempty.*

Proof. $\preceq_{\mathrm{Pareto}}$ is a partial order on the finite set $\mathcal{A}$. Every finite partially ordered set has at least one maximal element (take any element and iteratively move to a strictly larger one until termination). Any maximal element is undominated. □

It is useful to emphasize the content of "undominated" in this setting: an action is Pareto-undominated when there is no alternative action that is (weakly) more pro and (weakly) less con in *every* relevant dimension. Thus the theorem rules out an "explosion" in which the requirement of Pareto safety would empty the feasible set; finiteness of $\mathcal{A}$ suffices to guarantee at least one maximal (hence undominated) element even when many actions are incomparable. Note also that maximal elements need not be unique: the set of Pareto-undominated actions can contain multiple actions corresponding to distinct trade patterns across $d \in D$.

Corollary 52 (Resonant feasible choice exists (finite $\mathcal{A}$)). *Let $\kappa(s,a)$ be the resonance magnitude and $R(s,a) \geq 0$ any resistance cost. If $\mathcal{A}$ is finite, then the rule "choose any Pareto-undominated maximizer of $\kappa(s,a) - \beta R(s,a)$" yields a nonempty choice set.*

Proof. By Theorem 132, the Pareto-undominated set is nonempty and finite. A real-valued function on a finite nonempty set attains a maximum. □

Here the Pareto filter enforces a minimal safety constraint (no avoidable coordinatewise harm), while the objective $\kappa(s,a) - \beta R(s,a)$ provides a secondary ranking among those safe actions. The parameter $\beta \geq 0$ controls how strongly the selector accounts for resistance or implementation cost: when $\beta = 0$ the rule reduces to choosing a Pareto-undominated maximizer of resonance magnitude alone, while larger β shifts weight toward feasibility and away from raw resonance. Because the maximization is performed *after* restricting to $\mathcal{A}_{\text{Pareto}}$, the existence conclusion is purely combinatorial: it does not rely on continuity, convexity, or any cardinal comparability across dimensions beyond the partial-order structure.

57.6.2 A coordination selector that explicitly targets polarization

Definition 523 (polarization-aware resonant coordination rule). *Fix a finite action set $\mathcal{A}$. Let $\overline{Z}(a)$ be the group-mean complex signal and* $\mathrm{Pol}(a)$ *the polarization for action a. Define a selector*

$$a^{\star} \in \arg \max_{a \in \mathcal{A}_{\text{Pareto}}} \left(\left|\overline{Z}(a)\right|^2 - \gamma \,\mathrm{Pol}(a) - \beta R(a) \right),$$

where $\mathcal{A}_{\text{Pareto}}$ is the (socially) Pareto-undominated set and $\gamma, \beta \geq 0$.

The objective separates three considerations: (i) $\left|\overline{Z}(a)\right|^2$ rewards coherent alignment in the aggregated (mean) complex signal, (ii) $\mathrm{Pol}(a)$ penalizes dispersion or disagreement across individuals (as formalized earlier), and (iii) $R(a)$ captures any additional resistance term (institutional friction, switching cost, or similar). The squared magnitude $|\cdot|^2$ is convenient because it is non-negative, scales quadratically with signal strength, and interacts algebraically with polarization via the identity in Theorem 133; operationally, it treats phase disagreement as directly reducing group-level coherence even when individuals have large magnitudes.

Theorem 133 (The selector is an explicit trade between coherence and polarization). *Assume $\sum_i \alpha_i = 1$. For each action a,*

$$\left|\overline{Z}(a)\right|^2 - \gamma \,\mathrm{Pol}(a) = (1+\gamma) \left|\overline{Z}(a)\right|^2 - \gamma \sum_i \alpha_i \left|Z_i(a)\right|^2.$$

Hence increasing γ penalizes actions whose apparent group resonance is "faked" by high individual magnitudes with large disagreement.

Proof. By Theorem 126, $\mathrm{Pol}(a) = \sum_i \alpha_i |Z_i(a)|^2 - \left|\overline{Z}(a)\right|^2$. Substitute:

$$\left|\overline{Z}\right|^2 - \gamma\,\mathrm{Pol} = \left|\overline{Z}\right|^2 - \gamma\left(\sum_i \alpha_i |Z_i|^2 - \left|\overline{Z}\right|^2\right) = (1+\gamma)\left|\overline{Z}\right|^2 - \gamma\sum_i \alpha_i |Z_i|^2 .$$

□

The rearrangement makes clear that γ changes the *relative* valuation of two quantities: the aggregate coherence $\left|\overline{Z}(a)\right|^2$ and the average individual energy $\sum_i \alpha_i |Z_i(a)|^2$. When individuals are strongly activated but point in conflicting directions, $\sum_i \alpha_i |Z_i(a)|^2$ can remain large while $\left|\overline{Z}(a)\right|^2$ collapses due to cancellation; the $-\gamma\sum_i \alpha_i |Z_i(a)|^2$ term then ensures such actions are down-ranked as γ increases. Conversely, when individuals are both strong *and* aligned, the penalty term is offset by the boost on $\left|\overline{Z}(a)\right|^2$, so high-coherence actions remain attractive even for moderate γ.

Remark 3174 (Reasoning use)**.** *This gives a concrete knob: γ interpolates between "maximize shared alignment" (small γ) and "avoid high-magnitude but divisive actions" (large γ), all while preserving a Pareto safety filter.*

A useful way to read the interpolation is through limiting cases: at $\gamma = 0$ the selector reduces (up to the resistance term) to maximizing $\left|\overline{Z}(a)\right|^2$, i.e. pure coherence among Pareto-safe actions, whereas for large γ the selector increasingly prefers actions that keep individual magnitudes modest unless they translate into genuine mean alignment. Because the maximization is restricted to $\mathcal{A}_{\mathrm{Pareto}}$, the polarization-aware trade is applied only after eliminating actions that are avoidably worse for everyone in the p-bit sense; thus polarization control is not allowed to justify selecting an action that is coordinatewise dominated by another available option.

57.7 Summary of what these theorems buy a reasoning system

- (Thm. 123) **A structural diagnosis of contradiction growth** under join-based social evidence aggregation: contradiction will not shrink without explicit revision machinery. This means that any architecture that only "adds" information from multiple sources (e.g., by monotone pooling, union, or join-style merges) will accumulate inconsistencies as a predictable byproduct of scale, rather than as a rare bug. Operationally, the theorem motivates separating (i) an aggregation layer that is allowed to be inconsistency-tolerant from (ii) an explicit revision layer that can retract, downweight, or quarantine claims, and it clarifies that merely collecting more evidence is not, by itself, a remedy for contradiction.

- (Thm. 124, Thm. 125) **Explainable value conflict**: negative interference decomposes into pairwise anti-alignment; dissonance is phase dispersion energy. The key gain here is interpretability: when global conflict is observed, the decomposition identifies which pairs of value components (or agents, depending on the representation) are responsible for the negative terms, rather than attributing the phenomenon to an opaque aggregate score. Treating dissonance as a dispersion energy also yields a diagnostic handle: one can distinguish "low resonance because everyone is weak" from "low resonance because strong components are out of phase," which in turn suggests different interventions (strengthening shared commitments versus reducing misalignment among salient commitments).

- (Thm. 126–128) **A principled polarization metric** with (1) resonance–variance identity, (2) pairwise distance meaning, and (3) two-bloc causal decomposition. These properties jointly make polarization measurable in a way that is both mathematically constrained and socially legible: the resonance–variance identity ties polarization to a familiar notion of spread, the pairwise interpretation supports local explanations ("which pairs are far apart"), and the two-bloc decomposition separates within-bloc coherence from between-bloc separation. For a reasoning system, this matters because it enables monitoring and control: one can penalize polarization while still allowing diversity, and one can attribute changes in a polarization score to identifiable shifts in group structure rather than to arbitrary metric artifacts.

- (Thm. 129–Cor. 50) **Diffusion predictions** from couplings: who influences whom, with what attenuation, and under what strong-tie thresholds. In concrete terms, the coupling/path characterization turns a qualitative influence story into a quantitative forecast: influence is traced along paths, aggregated by the relevant join/closure, and discounted by attenuation so that long or weak chains contribute less. The strong-tie thresholds provide a crisp criterion for when influence is expected to persist versus die out, which supports design choices such as pruning unreliable edges, identifying bottlenecks, or targeting interventions to edges that dominate the effective diffusion.

- (Thm. 130–131) **Stabilization and contested culture** as explicit max-min path closures in two channels. The explicit max–min structure provides a computationally implementable closure operator: stabilization emerges from maximizing over paths while respecting the weakest link along each path, and contestation arises when the two channels (e.g., supportive vs. opposing, or affirmation vs. negation) percolate differently. For a reasoning system, this yields a way to represent "cultural" or "normative" persistence without assuming consensus: stable patterns can be traced to robust pathways, while contested regions are precisely those where comparable-strength pathways exist for incompatible signals.

- (Thm. 132–133) **Non-explosive coordination choices** that keep conflicts visible yet select robustly: Pareto safety + resonance + polarization penalty + effort. The point is not merely that a selector exists, but that it can be tuned to avoid two common failure modes: (i) suppressing disagreement by collapsing to a brittle single objective, or (ii) amplifying disagreement by optimizing a criterion that rewards extremity. By combining Pareto safety (no dominated moves), resonance (coordination quality), a polarization penalty (social risk), and an effort term (implementation cost), the system can make choices that remain justifiable under multiple evaluative lenses, while still surfacing the underlying tradeoffs rather than hiding them.

These results function together as an end-to-end toolkit: aggregation theorems clarify what information fusion will inevitably do without revision; interference and polarization identities make conflicts diagnosable rather than mysterious; diffusion and closure results predict how signals and norms propagate and stabilize; and the selection theorems provide a disciplined way to act under plural objectives without either erasing disagreement or letting it spiral. As a consequence, the reasoning system can (i) represent incompatible evidence without immediate collapse, (ii) explain *why* conflict appears in terms of specific anti-alignments and dispersion, (iii) anticipate network-mediated amplification or attenuation, and (iv) commit to coordination decisions that are robust, auditable, and sensitive to polarization externalities.

58 Embedding the Cultural-Probabilistic-Pragmatic Model of Science in Hyperseed

We formalize the Cultural/Pragmatic Probabilism (CPP) philosophy and model of science — "science as weakest selection across evidential, cultural-simplicity, and pragmatic-usefulness channels" — inside the Hyperseed formalization of science as empirically coupled belief-system dynamics [1, 20]. We then prove a family of structural theorems useful to reasoning systems: (1) CPP is representable as Hyperseed science plus two additional weakness channels; (2) scientific updates are path-independent (batching/commutativity), enabling caching and method flexibility; (3) Pareto-dominance and robustness regions yield decision tools under uncertain channel weights; (4) modularity/independence yields domain decomposition theorems; (5) state-dependent science is reducible to ordinary science on an expanded experiment space; and (6) "paradigm innovation" reduces to ordinary model selection on a lifted hypothesis space.

In this embedding, the key interpretive move is to treat "acceptance" or "preferred status" of a hypothesis not as a primitive binary judgment but as an emergent consequence of iterative update dynamics operating on a structured belief state. Hyperseed provides the ambient machinery for this: a scientific community (or an individual reasoner) maintains a population of candidate hypotheses/models together with degrees of endorsement, and these endorsements are repeatedly revised in response to interaction with an experiment space that is coupled to the world via observation. The CPP proposal then enters by specifying *multiple* concurrent evaluation channels, each channel inducing its own notion of weakness/selection pressure, and the operative scientific update is the resulting weakest-selection aggregation across these channels.

Concretely, the evidential channel captures ordinary empirical adequacy (e.g., fit to data, predictive accuracy, calibration, likelihood-based scoring, or other statistical loss proxies); the cultural-simplicity channel captures pressures toward communicability and compressibility (e.g., shorter descriptions, fewer free parameters, alignment with shared conceptual schemas, and lower cognitive/organizational cost of transmission); and the pragmatic-usefulness channel captures goal-directed performance (e.g., actionable interventions, engineering effectiveness, decision quality, and downstream utility in situated tasks). The philosophy of CPP is that these three channels are jointly constitutive of scientific practice as actually executed: in real scientific ecosystems, hypotheses survive not only by fitting existing data, but by being adoptable, teachable, and useful under constraints of limited time, attention, and resources.

Hyperseed's language of *weakness channels* is particularly apt here because it allows each criterion to be represented as a morphism from hypotheses into an ordered space of "weakness" values, with selection realized by comparing hypotheses according to the induced order. The pluralism of CPP thus becomes a statement about a *vector* (or family) of weakness functionals rather than a single scalar score. This matters for reasoning systems because different deployments may instantiate the channels differently: e.g., evidential weakness might be computed from Bayesian posterior risk, cultural weakness from description length or program-size priors, and pragmatic weakness from expected utility or regret on a task distribution. The embedding theorem in (1) is therefore not merely a representational convenience: it guarantees that a reasoning system can treat CPP-style multi-criterion assessment as a standard Hyperseed object by adding channels, rather than rewriting the underlying notion of science.

The remaining theorems articulate computational and conceptual consequences of this representation. Path-independence in (2) is read as an algebraic statement that the order in which evidence and other channel-relevant updates are processed does not affect the resulting closure (under the specified update operator), which legitimizes batching, streaming, parallelization, and caching strategies in implementations. The Pareto and robustness results in (3) formalize the commonplace observation that channel trade-offs are often underdetermined: if the relative weights among evidential, cultural, and pragmatic criteria are uncertain or context-sensitive, then one should reason in terms of Pareto-frontiers and regions of weight-space in which a hypothesis remains admissible or optimal. The modularity results in (4) capture when and how domains can be decomposed: if the experiment space and hypothesis structure factorize, then multi-channel weakness can factorize as well, allowing separate sub-sciences to be updated independently and then recombined without loss. The state-dependent reduction in (5) addresses the fact that in many realistic settings, what counts as an "experiment" depends on prior commitments (e.g., instrument choice, measurement protocol, or adaptive experimental design); the reduction shows that such endogenous experiment selection can be simulated by ordinary Hyperseed science on a suitably expanded space that treats state as part of the experimental context. Finally, the paradigm-innovation result in (6) makes precise a key aspiration of CPP-friendly meta-science: even changes that look like shifts in the space of admissible models can be represented as ordinary selection once one moves to a lifted hypothesis space that contains both models and model-generators (or representational frameworks) as first-class candidates.

Remark 3175. *The philosophical stance here is intentionally pluralistic: the "scientific quality" of a model is not reduced to a single scalar such as likelihood, but is treated as a synthesis of (at least) three partially incommensurable criteria. In the idiom of Hyperseed, these criteria are implemented as* weakness channels *(cf. Hyperseed-Concept 202) [3, 2]. The formal results below show that this pluralism can nevertheless be made algebraically tame: we can compute closures, batch evidence, factorize domains, and reason about robustness without sacrificing rigor.*

A further motivation for making the channels explicit (rather than folding them into a single aggregate score from the outset) is that scientific disagreement often arises from channel emphasis *rather than from factual dispute: two communities may share the same evidence yet differ in how much cultural simplicity or pragmatic usefulness should constrain acceptance. The multi-channel formalization keeps these sources of disagreement visible and manipulable, so that one can ask principled questions such as: which hypotheses are stable across a range of cultural/pragmatic pressures; which are fragile to small changes in the balance of criteria; and which are non-dominated in the sense that improving one channel necessarily worsens another. In this sense, the CPP-in-Hyperseed embedding is meant not only to generalize standard evidentialist accounts, but also to support transparent, auditable reasoning about trade-offs that are ubiquitous in real scientific and engineering workflows.*

58.1 Minimal Hyperseed science machinery (from above)

Definition 524 (Belief values and belief states)**.** *Fix a claim language L and a commutative unital quantale $(V, \leq, \oplus, \otimes, 0, 1)$. A belief state is a valuation $\beta : L \to V$, ordered pointwise.*

Remark 3176. *Intuitively, L is a set of statements an agent (or community) can talk about, and V is the space of* graded *truth-values, supports, or plausibilities (Hyperseed-Concepts 65, 64). A map $\beta : L \to V$ assigns to each claim $\varphi \in L$ a value $\beta(\varphi) \in V$ representing how strongly the agent currently endorses it. The pointwise order means: $\beta \leq \beta'$ iff for every claim φ, $\beta(\varphi) \leq \beta'(\varphi)$, i.e. β' contains at least as much support as β everywhere.*

A simple example is the "Boolean" case $V = \{0, 1\}$ with $\oplus = \vee$ and $\otimes = \wedge$, where a belief state is just a set of believed sentences. A more informative example is a numeric quantale such as $V = [0, 1]$ with $\oplus = \max$ and $\otimes = \times$, where $\beta(\varphi)$ can be read as a confidence and $\oplus$ as evidence pooling. This quantale structure is useful because it lets us define inference and closure abstractly, *while still admitting many concrete semantics (probabilistic, fuzzy, possibilistic, etc.; Hyperseed-Concept 139).*

Remark 3177. *The quantale assumptions encode two distinct design goals that matter for "science machinery." First, $(V, \leq)$ supplies an order in which "more supported" (or "less refuted") beliefs lie above weaker ones; this is what makes iteration and fixed points meaningful. Second, $\oplus$ and $\otimes$ separate two ways of aggregating information: $\oplus$ is typically used to combine* independent contributions to the same claim *(e.g. multiple lines of evidence), whereas $\otimes$ is typically used to propagate support* through *an inference step (e.g. a rule weight times the support of its premise). Commutativity and unitality make these aggregations robust under re-ordering and the presence of neutral elements, which is important when the "community" is distributed and updates arrive asynchronously.*

Definition 525 (Weighted inference and closure)**.** *Fix a set R of weighted inference rules. Let $F_R : V^L \to V^L$ be the one-step inference operator and let $\mathrm{CI}_R(\beta)$ be the least fixed point of F_R above β.*

Remark 3178. *The notation V^L denotes the set of all functions $L \to V$, i.e. all belief states; thinking set-theoretically, this is an exponential object (a "function space"). The operator F_R is the engine that applies the rules in R once: from current beliefs, it produces an expanded belief state containing whatever the rules can infer in one step, with weights combined using the quantale operations. The closure $\mathrm{CI}_R(\beta)$ is then the result of iterating this one-step procedure to saturation, taking the least fixed point that extends the starting information β.*

Concretely: if L contains claims φ, ψ and R contains a rule "from φ infer ψ with weight w", then $F_R(\beta)(\psi)$ will be at least $\beta(\varphi) \otimes w$, and closure iterates until no new support can be added. This construction is necessary because scientific reasoning is not merely a sequence of one-off updates; it is a process of repeatedly integrating evidence and its inferential consequences until reaching a stable state (Hyperseed-Concepts 150, 140).

Remark 3179. *The phrase "least fixed point above β" is doing two jobs at once. "Fixed point" captures the idea of* stability under the currently endorsed inference rules*: once closed, applying F_R again produces no change. "Least above β" captures* conservativeness*: closure should add exactly what is forced (by R and the algebra of V) and no more, so that $\mathrm{CI}_R(\beta)$ is the minimally-committal stable extension of β. This is the order-theoretic analogue of "close under consequences, but do not invent new premises."*

Theorem 134 (Closure exists; Cl_R is a closure operator)**.** *F_R is monotone on the complete lattice V^L, hence $\mathrm{Cl}_R(\beta)$ exists for all β (as the join of the Kleene chain). Moreover Cl_R is extensive, monotone, and idempotent.*

Remark 3180. *The notation in this subsection is meant to be read as follows: $\mathrm{CI}_R(\beta)$ names the least fixed point of F_R above β (as in the definition), and $\mathrm{Cl}_R(\beta)$ names the corresponding* closure operation *applied to β (as in the theorem). In standard lattice-theoretic treatments these coincide as maps $V^L \to V^L$; i.e. one may set $\mathrm{Cl}_R(\beta) \coloneqq \mathrm{CI}_R(\beta)$, with Cl_R emphasizing the algebraic "closure-operator" viewpoint and CI_R emphasizing the "closure by iterated inference" viewpoint.*

Remark 3181. *What this theorem says, in plain language, is that "closing under inference" is always well-defined: if you repeatedly apply your inference rules, you do not wander endlessly without a limit; instead you converge (in the lattice-theoretic sense) to a smallest stable belief state that contains your starting beliefs. The surprise here is not logical but architectural: once the* *algebraic conditions (complete lattice + monotone inference operator) are in place, closure becomes a reusable component in a reasoning system, independent of the particular content of L.*

This result is the backbone for later batching/commutativity claims: once closure is idempotent, intermediate closures can be postponed without changing the end result. It also connects to the broader fixed-point style used elsewhere in Hyperseed (e.g. stability analyses) [10].

Remark 3182. *The "complete lattice" hypothesis is automatically satisfied for many quantales used in practice: if V is complete (has all joins/meets), then the function space V^L inherits completeness pointwise (joins and meets are computed claim-by-claim). Under that identification, the Kleene chain mentioned in the theorem is the ascending sequence*

$$\beta \ \leq \ F_R(\beta) \ \leq \ F_R^2(\beta) \ \leq \ \cdots,$$

and its join is computed pointwise in V. This is one reason Hyperseed formulates the machinery in V^L: it reduces global questions about "community belief" to local aggregation at each claim $\varphi \in L$ while keeping the global fixed-point semantics intact.

Proof. To make the existence claim explicit, define the Kleene iterates $\beta_0 := \beta$ and $\beta_{n+1} := F_R(\beta_n)$ for $n \geq 0$. By monotonicity of F_R one has $\beta_n \leq \beta_{n+1}$ for all n, hence $(\beta_n)_{n\in\mathbb{N}}$ is an increasing chain in the complete lattice V^L. Let $\beta_\infty := \bigvee_{n\in\mathbb{N}} \beta_n$ (join taken in V^L, i.e. pointwise in V). Under the standard continuity conditions used in Kleene-style constructions, $F_R(\beta_\infty) = \beta_\infty$, so β_∞ is a fixed point above β, and by construction it is the least such fixed point; one may thus take $\mathrm{Cl}_R(\beta) := \beta_\infty$.

Extensiveness ($\beta \leq \mathrm{Cl}_R(\beta)$) follows from $\beta = \beta_0 \leq \bigvee_n \beta_n$. Monotonicity of Cl_R follows because $\beta \leq \beta'$ implies $\beta_n \leq \beta'_n$ for all n (by induction using monotonicity of F_R), hence $\bigvee_n \beta_n \leq \bigvee_n \beta'_n$. Idempotence ($\mathrm{Cl}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)$) holds because $\mathrm{Cl}_R(\beta)$ is already a fixed point of F_R, so closing it again does not add anything further. □

Proof sketch. The strategy is the standard Knaster–Tarski/Kleene fixed-point picture: monotonicity guarantees that iterating F_R from β produces an increasing chain, and completeness guarantees that taking its join yields a fixed point. Extensiveness means closure contains the starting beliefs; monotonicity means more input evidence yields more closed beliefs; idempotence means "closing twice" is the same as "closing once," which is what makes batching and modularity work smoothly. □

Remark 3183. *A useful visual metaphor is to picture V^L as a landscape ordered by "information content," and F_R as a gravity-like operator that pulls a belief state upward to include consequences. Iteration follows a monotone ascent; the join of the Kleene chain is the least plateau where applying F_R no longer moves you.*

Remark 3184. *This "landscape" picture also clarifies why idempotence matters operationally. In a distributed setting, agents may alternate between (i) incorporating new evidence and (ii) propagating consequences. If closure were not idempotent, then the order and frequency of these phases would matter, and the system could exhibit artificial dependence on scheduling. Idempotence says that once a community has done the inferential bookkeeping to reach the plateau, repeating the same bookkeeping step is harmless; this is precisely the property that enables caching, memoization of derived claims, and stable modular composition of subtheories.*

Definition 526 (Hyperseed science: observation sets, models, empirical update)**.** *A (community-accepted) observation set consists of an experiment class E, an observation set O, and an observation map* $\mathrm{Obs} : E \to O$*, with a community acceptance criterion. A scientific model class is a set*

$\mathcal{M}$ *of candidate models whose predictive/causal claims are encoded in* L*. An empirical evidence increment is a map* $\varepsilon_{e,o} : L \to V$ *induced by experiment* e *and observation* $o = \mathrm{Obs}(e)$*. A scientific update step is:*

$$\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o}).$$

Remark 3185. *This definition separates three layers that are often conflated in informal discussions of "the scientific method." The observation set* (E, O, Obs) *fixes what counts as a* repeatable *experimental intervention (*$e \in E$*) and what counts as an admissible readout (*$o \in O$*). The "community acceptance criterion" is included to acknowledge that real scientific communities do not treat all reported* (e, o) *pairs equally: issues of calibration, provenance, replication, and protocol compliance can gate whether a datum is allowed to enter the shared record at all.*

The model class $\mathcal{M}$ *specifies the hypothesis space whose members are intended to explain and predict observations; encoding predictive/causal claims in* L *means that model evaluation can be mediated through the same belief language used for other claims (background theory, methodological constraints, auxiliary hypotheses, etc.). Finally,* $\varepsilon_{e,o} : L \to V$ *is the bridge from raw outcomes to graded support over claims: it can be sparse (affecting only a small fragment of* L*) or rich (updating many linked claims at once), and it can represent positive evidence, negative evidence, or partial/ambiguous evidence depending on the structure of* V*.*

The update equation $\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o})$ *makes explicit that "learning from data" is not just adding the evidence increment; it is adding it* and then propagating its consequences *via closure under* R*. In particular, the use of* $\oplus$ *emphasizes that evidence is aggregated with prior support in a way appropriate to the chosen quantale semantics, while* Cl_R *enforces the community's inferential commitments (methodological and theoretical) after each accepted observation.*

Remark 3186. *This definition formalizes "science" as a loop with three kinds of structure: (i)* what can be done *(experiments* E*), (ii)* what is registered *(observations* O *and the mapping* Obs*), and (iii)* how observations affect belief *(the increment* $\varepsilon_{e,o}$*), all mediated through communal norms about what counts as acceptable evidence (Hyperseed-Concepts* **??**, **??**, *91) [20]. The update equation says: merge the new empirical increment into the current belief state using* $\oplus$ *(an "evidence pooling" operator), then close under inference rules* R*.*

A simple example is: E *is a class of measurement procedures,* O *is a set of recorded outcomes, and* $\varepsilon_{e,o}$ *assigns high support in* V *to claims like "the outcome was* o*" and to model adequacy claims consistent with* o*. The closure step is what propagates the observation through the inferential web: from local measurement reports to broader theoretical commitments. This division of labor is useful because it separates* data acquisition *from* inferential assimilation.

One can also read the loop as a minimal abstraction of the social-epistemic workflow: E *encodes not only physical interventions but also protocolized practices (what a lab is allowed to do),* Obs *encodes both instrumentation and reporting conventions (what counts as a record), and* $\varepsilon_{e,o}$ *encodes the community's tacit "likelihood model" of which claims should be strengthened by which outcomes. In particular,* $\oplus$ *need not be additive or Bayesian: it can represent negotiated aggregation across agents or institutions (e.g., replication-weighted pooling), while the closure under* R *enforces whatever inferential commitments the community treats as mandatory (e.g., deductive closure, default rules, or domain-specific calculi). In this way the same formal skeleton can represent both individual updating and intersubjective stabilization, depending on how* $V, \oplus$*, and* R *are instantiated.*

58.2 CPP view of science (informal-to-formal bridge)

CPP characterizes scientific quality via three "weakness channels": (i) evidential weakness (no unwarranted predictive distinctions), (ii) cultural weakness (simplicity relative to paradigm primitives), (iii) pragmatic weakness (no practically irrelevant distinctions), and views a *paradigm* as a package specifying the semantics of evidence, simplicity, and pragmatic relevance.

The informal slogan "no unwarranted distinctions" can be read as a form of invariance requirement: scientific representations should identify (treat as equivalent) any differences that are not licensed by the relevant channel. In the evidential channel, the identification is driven by what can be discriminated by the experiment-observation interface; in the cultural channel, it is driven by what can be expressed economically in the community's representational repertoire; and in the pragmatic channel, it is driven by what makes a difference to intervention, decision, or control. On this reading, a paradigm does not merely choose *which theories are entertained*, but also chooses the *grain* at which the world is described and evaluated.

Remark 3187. *The word "weakness" is meant in a technical sense: a good scientific claim is often one that refrains from making unnecessary distinctions, thereby remaining robust under variation and resistant to overfitting [3, 2]. Evidential weakness corresponds to not asserting predictive structure beyond what the evidence warrants (Hyperseed-Concept 202). Cultural weakness corresponds to parsimony relative to the representational primitives a community treats as natural (Hyperseed-Concepts 91, 82). Pragmatic weakness corresponds to ignoring distinctions that do not alter action, control, or usefulness (Hyperseed-Concepts 191, 186).*

The notion of a paradigm as a "package" is not merely sociological decoration: formally, it is the claim that simplicity and usefulness are not absolute, but are functions of representational and practical context. This resonates with process-oriented and semiotic perspectives in which meaning is enacted by a community of interpreters rather than residing in symbols alone [15, 14].

In particular, a paradigm package fixes (at least implicitly) three kinds of typing information: (a) which observational reports are admissible and how they constrain models (the evidential semantics), (b) which descriptions count as short, natural, or "not ad hoc" (the cultural semantics), and (c) which distinctions are decision-relevant and at what scales costs and benefits are measured (the pragmatic semantics). When those background typings shift, two agents can agree on raw outcomes $o \in O$ yet disagree on which model comparisons are meaningful, because they are effectively computing weakness with respect to different equivalence relations over models. This is one formal route to the familiar phenomenon of incommensurability: disagreement can persist even when there is no disagreement about the syntactic data stream, because the paradigm changes the evaluative and inferential interpretation of that stream.

We now make this precise inside Hyperseed.

Definition 527 (CPP-paradigm (as data on top of Hyperseed science))**.** *A CPP-paradigm is a tuple*

$$P = (E, O, \mathrm{Obs}, \mathcal{M}, R, \mathrm{Comp}_P, \mathrm{Prag}_P, \lambda_c, \lambda_p, \mathrm{Adeq})$$

where:

- $(E, O, \mathrm{Obs}, \mathcal{M}, R)$ *is a Hyperseed science instance;*
- $\mathrm{Comp}_P : \mathcal{M} \to [0, \infty)$ *is a* cultural complexity cost *(paradigm-relative);*
- $\mathrm{Prag}_P : \mathcal{M} \to [0, \infty)$ *is a* pragmatic cost *(lower is more useful/actionable);*
- $\lambda_c, \lambda_p > 0$ *are channel weights;*
- $\mathrm{Adeq} : \mathcal{M} \to L$ *maps a model to a claim in* L *encoding "predictive adequacy."*

Given a belief state β, define evidential support

$$\mathrm{Supp}_\beta(m) \in [0,1]$$

as some monotone scalarization of $\beta(\mathrm{Adeq}(m))$ *(e.g. the plausibility projection). Define the CPP score:*

$$\mathsf{Score}_{P,\beta}(m) := \mathrm{Supp}_\beta(m)\; e^{-\lambda_c \mathrm{Comp}_P(m)}\; e^{-\lambda_p \mathrm{Prag}_P(m)}.$$

Remark 3188. *The tuple P is best read as "science plus two extra semantics." The Hyperseed core gives us evidence and inference $(E, O, \mathrm{Obs}, \mathcal{M}, R)$. CPP then adds two additional evaluative maps:* Comp_P *quantifies how complex a model is* in the representational language the paradigm finds natural *(Hyperseed-Concept 82), and* Prag_P *quantifies how costly or irrelevant the model is from an action-oriented perspective (Hyperseed-Concepts 191, 147). The weights λ_c, λ_p set the current "severity" of simplicity and usefulness constraints.*

The support functional $\mathrm{Supp}_\beta(m)$ *is deliberately abstract: it is any monotone way to turn the belief value $\beta(\mathrm{Adeq}(m)) \in V$ into a scalar in $[0,1]$. For example, if $V = [0,1]$ already, one can take* $\mathrm{Supp}_\beta(m) = \beta(\mathrm{Adeq}(m))$*. If V is multi-dimensional,* Supp_β *can pick out a plausibility coordinate. The product form of* $\mathsf{Score}_{P,\beta}$ *is a familiar log-linear "energy" model: the exponentials turn additive costs into multiplicative penalties, so that small cost differences are weighted smoothly but decisively.*

As a toy example, suppose two models m_1, m_2 explain the same data equally well so $\mathrm{Supp}_\beta(m_1) = \mathrm{Supp}_\beta(m_2)$*, but m_1 is far simpler in the paradigm's representational vocabulary: then m_1 wins by an Occam-like preference. Conversely, a slightly less-supported but dramatically simpler model can win if λ_c is large. This is precisely the intended formalization of CPP as a tri-channel "weakest selection" principle [20, 3].*

It is also useful to note what, concretely, can vary between paradigms in this formalization. Changing Comp_P *corresponds to changing which descriptions are treated as primitive or cheaply composable (for instance, preferring geometric over algebraic primitives, or preferring mechanistic modules over black-box function classes). Changing* Prag_P *corresponds to changing the task environment and the decision interface (for instance, whether interpretability, real-time computability, or controllability is prioritized). Changing the weights λ_c, λ_p corresponds to shifting how strongly simplicity and usefulness trade off against raw predictive support, which can encode differences between exploratory and production contexts. Finally, changing* Adeq *(or, upstream, the inference rules R that relate adequacy claims to other claims) corresponds to changing what counts as "the same predictions," e.g. whether calibration, counterfactual robustness, or mechanistic faithfulness is being demanded. In this sense, a paradigm shift can be modeled not only as a change of favored models in $\mathcal{M}$, but as a change in the scoring geometry over $\mathcal{M}$ induced by* $\mathrm{Comp}_P, \mathrm{Prag}_P, \lambda_c, \lambda_p$*, and the adequacy semantics.*

The log form makes the channel decomposition explicit: $\log \mathsf{Score}_{P,\beta}(m) = \log \mathrm{Supp}_\beta(m) - \lambda_c \mathrm{Comp}_P(m) - \lambda_p \mathrm{Prag}_P(m)$*. Thus evidential weakness is represented by the extent to which* Supp_β *refuses to separate models not distinguished by the evidence encoded in β and* Adeq*, while cultural and pragmatic weakness are represented by linear penalties that suppress distinctions that are expensive in the paradigm's description language or irrelevant to its task landscape. This makes contact with familiar Bayesian/MDL intuitions (support times an Occam factor) without committing the framework to a specific probabilistic semantics for β.*

58.3 Crosswalk theorem: CPP is Hyperseed science + two weakness channels

Theorem 135 (CPP-to-Hyperseed crosswalk)**.** *Every CPP-paradigm P determines:*

1. *a Hyperseed science instance $(E, O, \mathrm{Obs}, \mathcal{M}, R)$ with empirical updates $\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o})$;*

2. *two additional (commutative) weakness channels* $m \mapsto e^{-\lambda_c \mathrm{Comp}_P(m)}$ *and* $m \mapsto e^{-\lambda_p \mathrm{Prag}_P(m)}$*;*

3. *a model selection operator* $\arg\max_{m\in\mathcal{M}} \mathsf{Score}_{P,\beta}(m)$ *implementing CPP "weakest selection" (maximally weak across the three channels). Conversely, any Hyperseed science instance equipped with two such channel maps and a monotone support functional* Supp_β *induces a CPP-paradigm.*

In particular, the crosswalk is functorial at the level of "data packaging": the CPP object P contains (or canonically generates) the Hyperseed tuple, and the only extra degrees of freedom needed to recover CPP from Hyperseed are (i) two scalar penalty/weight maps on $\mathcal{M}$ and (ii) a rule that converts the current belief state β (and the update history it encodes) into a scalar notion of evidential adequacy for each model. The adjective "commutative" here means that, when the channels are combined multiplicatively into a single score, their order of application is irrelevant: multiplying by $e^{-\lambda_c \mathrm{Comp}_P(m)}$ *and then by* $e^{-\lambda_p \mathrm{Prag}_P(m)}$ *yields the same weight as multiplying in the opposite order, so the two additional channels can be treated as symmetric modifiers of evidential support rather than as sequential filters.*

Remark 3189. *This theorem states that CPP is not an alien add-on to Hyperseed science: it is obtained by taking the Hyperseed empirical/inferential loop and multiplying in two further factors that encode cultural simplicity and pragmatic usefulness. The importance is practical: once CPP is expressed as "Hyperseed + extra channels," all the fixed-point/closure theorems and batching/modularity results that hold for Hyperseed can be reused for CPP-aligned scientific reasoning systems [19]. A useful way to read the statement is that the empirical channel provides a* support-like *term (coming from β and the observation process), while* Comp_P *and* Prag_P *provide* regularizing *terms; the CPP score* $\mathsf{Score}_{P,\beta}(m)$ *is then an aggregation that ranks models by simultaneously (a) fitting/being supported by evidence and (b) remaining culturally compressible and pragmatically serviceable. The exponential form* $e^{-\lambda\cdot}$ *is also conceptually clarifying: it turns an additive cost scale (complexity or pragmatics) into a multiplicative attenuation, so "more costly" models are* down-weighted *rather than categorically excluded, and the sensitivity of that downweighting is tuned by* λ_c, λ_p*.*

Conceptually, the crosswalk is also a modest unification claim: it says that disputes about "what counts as good science" can often be formalized as disputes about (i) the channel maps $(\mathrm{Comp}_P, \mathrm{Prag}_P)$ *and (ii) their weights, rather than disputes about the existence of a scientific update loop at all. Equivalently, within a fixed Hyperseed instance* $(E, O, \mathrm{Obs}, \mathcal{M}, R)$*, different CPP paradigms can be seen as different* prioritizations *of non-empirical constraints, and the theorem asserts that these prioritizations can be represented without changing the underlying update mechanism, only the selection criterion built on top of it.*

Proof. Immediate by unpacking the data. The Hyperseed part is $(E, O, \mathrm{Obs}, \mathcal{M}, R)$ and the empirical-update loop. The two exponentiated costs are $[0,1]$-valued weakness weights. The product score is a concrete monotone aggregation of the three channels. Conversely, given the Hyperseed instance plus channel maps plus Supp_β, one reconstructs $\mathsf{Score}_{P,\beta}$ and hence a CPP-paradigm structure. More explicitly, the "unpacking" direction amounts to observing that a CPP-paradigm already supplies (at minimum) (i) a space of possible evidential events (e, o), (ii) an observational interface Obs that yields update increments $\varepsilon_{e,o}$, (iii) a belief/update state β closed under Cl_R relative to the relevant constraint set R, and (iv) a model class $\mathcal{M}$ to be scored. The two additional channel maps are simply functions $\mathcal{M} \to (0,1]$ once exponentiated; their codomain and monotonicity properties ensure they can be multiplied into a total score without leaving the scalar range used for comparison. In the reconstruction direction, the monotone support functional Supp_β

plays the role of extracting the "empirical channel" contribution to $\mathsf{Score}_{P,\beta}(m)$; together with the two penalty factors, one obtains a well-defined model ranking on $\mathcal{M}$, and packaging this ranking together with the already-given Hyperseed update loop yields a CPP-paradigm in the sense used throughout the section. □

Proof sketch. The proof is a bookkeeping argument: a CPP-paradigm already contains the underlying Hyperseed science data, and the score is literally a product of three terms, two of which are exponentiated costs. Conversely, if you start with Hyperseed science and provide two extra channel penalties and a way to read evidential adequacy as a scalar support, you can package these ingredients back into the tuple P. One can also view the crosswalk as a change of emphasis: Hyperseed specifies how β changes when (e, o) arrives (via $\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o})$), whereas CPP specifies how, at any given β, models $m \in \mathcal{M}$ are comparatively preferred once *all* three channels are accounted for. □

Remark 3190. *The key idea is that "weakness channels" are operationally just additional multiplicative weights in model selection. This is why they can be treated symmetrically with evidential support, even if their interpretation is sociocultural or pragmatic. In particular, multiplying by weakness weights implements a soft form of constraint: models are not rejected outright for being culturally complex or pragmatically unhelpful, but they are penalized in proportion to* $\mathrm{Comp}_P(m)$ *and* $\mathrm{Prag}_P(m)$*, and the hyperparameters* λ_c, λ_p *control how quickly those penalties dominate the overall score. This symmetry is also the technical reason that many Hyperseed arguments transfer: if a theorem only uses monotonicity and closure properties of the update-and-score pipeline (rather than any domain-specific interpretation of the score), then adding extra multiplicative channels preserves the relevant order structure needed by those arguments.*

58.4 Quantale unification theorem (optional but structurally useful)

Theorem 136 (Product quantale for multi-channel semantics)**.** *Let* $(V, \leq, \oplus, \otimes, 0, 1)$ *be a commutative unital quantale (evidence values). Let* $(W_c, \leq_c, \oplus_c, \otimes_c, 0_c, 1_c)$ *and* $(W_p, \leq_p, \oplus_p, \otimes_p, 0_p, 1_p)$ *be commutative unital quantales (cultural/pragmatic weakness values). Define* $Q := V \times W_c \times W_p$ *with pointwise order and operations:*

$$(a_0, a_1, a_2) \oplus (b_0, b_1, b_2) := (a_0 \oplus b_0,\ a_1 \oplus_c b_1,\ a_2 \oplus_p b_2),$$

$$(a_0, a_1, a_2) \otimes (b_0, b_1, b_2) := (a_0 \otimes b_0,\ a_1 \otimes_c b_1,\ a_2 \otimes_p b_2).$$

Then Q *is a commutative unital quantale. Hence all Hyperseed closure/fixed-point machinery lifts verbatim to tri-channel belief states* $\beta : L \to Q$*.*

More explicitly, the pointwise order is

$$(a_0, a_1, a_2) \leq (b_0, b_1, b_2) \quad \textit{iff} \quad a_0 \leq b_0,\ a_1 \leq_c b_1,\ a_2 \leq_p b_2,$$

and the bottom/top and unit elements in Q *are given by*

$$0_Q := (0, 0_c, 0_p), \qquad 1_Q := (1, 1_c, 1_p),$$

so that "doing nothing" in each channel corresponds to doing nothing in the product. In particular, arbitrary joins in Q *are computed coordinatewise: for any family* $\{(a_0^i, a_1^i, a_2^i)\}_{i \in I} \subseteq Q$ *one has*

$$\bigoplus_{i \in I} (a_0^i, a_1^i, a_2^i) = \left(\bigoplus_{i \in I} a_0^i,\ \bigoplus_{i \in I} a_1^i,\ \bigoplus_{i \in I} a_2^i \right),$$

where each $\bigoplus$ is taken in its respective coordinate quantale. This is the structural reason that the fixed-point/closure constructions (which only ever require monotonicity, joins, and $\otimes$ distributing over joins) can be reused without alteration.

Remark 3191. *Informally: instead of keeping evidence, cultural simplicity, and pragmatic usefulness as separate scoring systems glued together ad hoc, we can bundle them into a single algebraic value space Q and run one unified inference/closure pipeline. The pointwise definition is exactly what one would guess: to combine two tri-channel values, combine each channel in its own way.*

This matters because it allows the entire "science as closure under rules" apparatus to operate on richer value types without modification. In other words, once we know how to do closure for beliefs valued in a quantale, we automatically know how to do closure for beliefs valued in a product of quantales. This is a common categorical theme: products preserve structure, and here the preserved structure is precisely what fixed-point iteration needs.

A useful interpretive detail is that the pointwise order on Q is a Pareto-style order: $(a_0, a_1, a_2) \leq (b_0, b_1, b_2)$ precisely when every *channel is at least as good (in the appropriate sense) as before. This avoids prematurely collapsing multi-channel comparisons into a single scalar; if a later stage of the framework needs an overall preference, it can be obtained by adding an external aggregator (e.g. a weighted map $Q \to V$), but the closure computation itself remains purely structural and monotone in each coordinate.*

Concretely, one can instantiate V as an "evidence" quantale such as $([0,1], \leq, \max, \cdot, 0, 1)$ or $([0,\infty], \geq, \min, +, \infty, 0)$ (depending on whether one reads values as strength or cost), while W_c and W_p may encode cultural/pragmatic penalties or weaknesses with their own combination laws. The theorem guarantees that, regardless of these modeling choices, the combined tri-channel semantics stays within the same algebraic class needed by Hyperseed.

Proof. Products of complete lattices are complete lattices with pointwise joins. Commutativity/associativity/unit for $\otimes$ hold coordinatewise. Distributivity of $\otimes$ over arbitrary joins holds coordinatewise because each factor is a quantale.

For clarity on the distributivity step: if $x = (x_0, x_1, x_2) \in Q$ and $\{y^i\}_{i\in I} \subseteq Q$ with $y^i = (y_0^i, y_1^i, y_2^i)$, then using coordinatewise definitions,

$$x \otimes \left(\bigoplus_{i\in I} y^i\right) = \left(x_0 \otimes \bigoplus_{i\in I} y_0^i,\ x_1 \otimes_c \bigoplus_{i\in I} y_1^i,\ x_2 \otimes_p \bigoplus_{i\in I} y_2^i\right),$$

and each coordinate equals the join of the corresponding products because $\otimes$, $\otimes_c$, and $\otimes_p$ distribute over arbitrary joins in their respective quantales. Reassembling the coordinates yields

$$x \otimes \left(\bigoplus_{i\in I} y^i\right) = \bigoplus_{i\in I}(x \otimes y^i),$$

and similarly on the other side (commutativity makes left/right distribution equivalent here, but the argument is coordinatewise in any case). ☐

Proof sketch. One checks the quantale axioms component-by-component. Completeness holds because joins/meets in a product are computed coordinatewise. The distributivity axiom holds because each coordinate distributes over joins in its own quantale, so the triple does as well. The lifting claim then follows because the closure construction only uses these axioms.

In Hyperseed terms: if the closure operator (or rule-application operator) is built from monotone combinations of $\oplus$ (joins) and $\otimes$ (rule composition, aggregation, or propagation), then replacing V

by Q simply means doing the same computation three times in parallel, one per channel. Thus any Knaster–Tarski-style fixed-point existence and any iterative approximation scheme (e.g. iterating from 0_Q and taking joins at limits) carries over immediately. $\square$

Remark 3192. *Geometrically, you can imagine Q as a three-axis space: evidence on one axis, cultural simplicity on another, pragmatic utility on a third. Closure and inference then become motions that are monotone in each coordinate, so fixed-point convergence is preserved.*

It can also be helpful to view Q as encoding a frontier *of admissible beliefs rather than a single score: rule closure will propagate constraints and supports across the three coordinates simultaneously, and the pointwise order ensures that adding information in one channel cannot "hide" a loss in another channel. When later projecting to decisions (e.g. selecting hypotheses), one may impose additional selection criteria, but the internal logical/algebraic step of achieving closure stays faithful to the multi-dimensional semantics.*

58.5 Decision-theoretic and paradigm-shift theorems

Theorem 137 (Log-linear tradeoff form)**.** *Assume* $\mathrm{Supp}_\beta(m) \in (0,1]$ *for all* m*. Then maximizing* $\mathsf{Score}_{P,\beta}(m)$ *is equivalent to minimizing*

$$\mathcal{L}_{P,\beta}(m) := -\log \mathrm{Supp}_\beta(m) \;+\; \lambda_c \mathrm{Comp}_P(m) \;+\; \lambda_p \mathrm{Prag}_P(m).$$

Remark 3193. *This theorem says that CPP selection is an "energy minimization" problem: evidential support contributes a term* $-\log \mathrm{Supp}$*, and the two weakness channels contribute additive penalties. This is important because additive objectives are often easier to analyze, optimize, and differentiate than multiplicative ones; it also aligns CPP with familiar statistical mechanics and Bayesian model selection heuristics (where* $-\log$ *turns products into sums).*

It connects directly to the definition of $\mathsf{Score}_{P,\beta}$*: the product structure is not a mere convenience; it implies that the tradeoff surface is linear in* (λ_c, λ_p) *once we take logs. This observation will also underlie the robustness region theorem, where continuity in* (λ_c, λ_p) *is central.*

Proof. For $x > 0$, log is strictly increasing, so

$$\arg\max_m \mathsf{Score}(m) = \arg\max_m \log \mathsf{Score}(m) = \arg\max_m \big(\log \mathrm{Supp}(m) - \lambda_c \mathrm{Comp}(m) - \lambda_p \mathrm{Prag}(m)\big),$$

equivalently $\arg\min_m \mathcal{L}(m)$. $\square$

Proof sketch. Apply a strictly increasing transform (log) to preserve argmax, then expand log of the product into a sum. Move the minus signs to convert a maximization to a minimization. $\square$

Remark 3194. *The key step is the monotonicity of* log*, which preserves preference order. The rest is algebra: exponentials and logarithms are inverse, so costs become linear penalties. Visually, one goes from multiplying three "dials" to adding three "weights".*

Theorem 138 (Multi-channel Occam principle)**.** *Fix* P, β*. If two models* m_1, m_2 *satisfy:*

$$\mathrm{Supp}_\beta(m_1) = \mathrm{Supp}_\beta(m_2), \qquad \mathrm{Prag}_P(m_1) = \mathrm{Prag}_P(m_2), \qquad \mathrm{Comp}_P(m_1) < \mathrm{Comp}_P(m_2),$$

then $\mathsf{Score}_{P,\beta}(m_1) > \mathsf{Score}_{P,\beta}(m_2)$*.*

Remark 3195. *This is the expected Occam-style conclusion, but now explicitly "multi-channel": holding evidence and pragmatic usefulness fixed, cultural simplicity breaks ties. The result is not surprising, yet it is important because it clarifies exactly what kind of simplicity principle is being assumed: simplicity is not absolute but indexed by P (the paradigm), i.e. by what the culture takes as primitive and cheap to express (Hyperseed-Concepts 91, 82) [20].*

In the broader structure of this section, this theorem is the local (pairwise) version of the later Pareto-pruning theorem, which generalizes "break ties by simplicity" to "discard anything dominated in all channels."

Proof. With equal support and equal pragmatic factor,

$$\frac{\mathsf{Score}(m_1)}{\mathsf{Score}(m_2)} = \frac{e^{-\lambda_c \mathrm{Comp}(m_1)}}{e^{-\lambda_c \mathrm{Comp}(m_2)}} = e^{-\lambda_c(\mathrm{Comp}(m_1)-\mathrm{Comp}(m_2))} > 1$$

since $\lambda_c > 0$ and $\mathrm{Comp}(m_1) - \mathrm{Comp}(m_2) < 0$. □

Proof sketch. Cancel the equal multiplicative factors, compare the remaining exponential penalties, and use that $e^{-\lambda x}$ decreases strictly in x when $\lambda > 0$. □

Remark 3196. *The ratio trick isolates the only relevant difference between the two models. Once reduced to comparing two exponentials, the inequality follows immediately from the monotonicity of the exponential function.*

Theorem 139 (Pareto-dominance pruning (weight-robust selection))**.** *Fix P, β and assume $\lambda_c, \lambda_p > 0$. If m_1, m_2 satisfy*

$$\mathrm{Supp}_\beta(m_1) \geq \mathrm{Supp}_\beta(m_2), \quad \mathrm{Comp}_P(m_1) \leq \mathrm{Comp}_P(m_2), \quad \mathrm{Prag}_P(m_1) \leq \mathrm{Prag}_P(m_2),$$

with at least one inequality strict, then $\mathsf{Score}_{P,\beta}(m_1) > \mathsf{Score}_{P,\beta}(m_2)$. In particular, m_2 can be discarded without knowing the exact relative tradeoff magnitudes, so long as channel weights are positive.

Remark 3197. *This theorem says: if one model is at least as good in evidential support and no worse in either cultural or pragmatic cost (and strictly better in at least one channel), then it wins for* every *positive choice of weights. This is a crucial computational principle: it provides a safe pruning rule that reduces the search space before any delicate calibration of λ_c, λ_p.*

The connection to "weakest selection" is direct: Pareto-dominance is the partial order induced by simultaneously pursuing weakness in multiple senses. In practice, it gives a robust notion of "dominated hypotheses" that can be discarded even under uncertainty about community preferences.

Proof. The score is monotone increasing in Supp and monotone decreasing in both costs (because $m \mapsto e^{-\lambda m}$ is strictly decreasing for $\lambda > 0$). Thus each weakly improved coordinate weakly improves Score, and any strict improvement yields a strict improvement in the product. □

Proof sketch. Use coordinatewise monotonicity: increasing Supp helps, decreasing costs helps. Because the score is a product of positive terms, any strict improvement in any factor yields a strict improvement in the product. □

Remark 3198. *One may picture the triple* $(\mathrm{Supp}, -\mathrm{Comp}, -\mathrm{Prag})$ *as a point in $\mathbb{R}^3$ ordered coordinatewise. Pareto-dominance means lying "northeast" of another point; the theorem says the scoring function respects this order whenever the weights are positive.*

Theorem 140 (Paradigm-shift as preference reversal)**.** *Let P and P' be two paradigms with the same $(E, O, \mathrm{Obs}, \mathcal{M}, R, \mathrm{Adeq})$ and the same belief state β, but different cultural semantics $\mathrm{Comp}_P, \mathrm{Comp}_{P'}$ (or weights). Assume m_1, m_2 satisfy $\mathrm{Supp}_\beta(m_1) = \mathrm{Supp}_\beta(m_2)$ and $\mathrm{Prag}_P = \mathrm{Prag}_{P'}$ on $\{m_1, m_2\}$. If*

$$\mathrm{Comp}_P(m_1) < \mathrm{Comp}_P(m_2) \quad \textit{and} \quad \mathrm{Comp}_{P'}(m_1) > \mathrm{Comp}_{P'}(m_2),$$

then P prefers m_1 while P' prefers m_2 (i.e. the argmax flips).

Remark 3199. *Here "paradigm shift" is rendered as a precise mathematical possibility: two communities can agree on the evidence, the experiment class, the model class, and the inferential rules, and yet disagree on which model is preferable because they disagree on what counts as "simple." This is important because it shows how incommensurability can arise* without *abandoning formal rationality: it is implemented as a change of cost semantics (Hyperseed-Concept 91) [20]. More explicitly, the hypotheses $\mathrm{Supp}_\beta(m_1) = \mathrm{Supp}_\beta(m_2)$ and $\mathrm{Prag}_P = \mathrm{Prag}_{P'}$ on $\{m_1, m_2\}$ ensure that (for this pair) there is no disagreement about evidential fit or action-guiding utility; the only remaining discriminator in the score is the cultural-complexity channel. In this sense, the theorem isolates a "pure" paradigm effect: the preference reversal is not driven by new data, new goals, or changes in the inferential machinery, but by a re-encoding of which descriptions are treated as costly.*

The result links back to the CPP-paradigm definition: Comp_P is explicitly paradigm-relative, so preference reversals are not anomalies but expected phenomena when Comp *changes. Equivalently, even if two agents share a common β (so they update identically under R on* Obs*) and share the same pragmatic evaluation functional on the candidate set, a shift in* Comp *can still change the ordering induced by* Score *and hence change the selected model. This recovers, in a deliberately minimal formal setting, the familiar methodological phenomenon that "simplicity" (or "naturalness," "elegance," "explanatory unity") is not a purely syntactic property but depends on background representational commitments.*

Proof. By the multi-channel Occam theorem applied in each paradigm separately: under P the smaller Comp_P wins; under P' the smaller $\mathrm{Comp}_{P'}$ wins. Since $\mathrm{Supp}_\beta(m_1) = \mathrm{Supp}_\beta(m_2)$ and $\mathrm{Prag}_P = \mathrm{Prag}_{P'}$ on $\{m_1, m_2\}$, the score comparison between m_1 and m_2 reduces to comparing the terms involving Comp alone; the stated strict inequalities therefore force opposite strict score orderings in P and P'. □

Proof sketch. Reduce each paradigm's choice to the Occam tie-breaker because support and pragmatics are equal on the pair; then apply the opposite inequalities for Comp_P and $\mathrm{Comp}_{P'}$. The argmax flips because each paradigm induces a different strict total order on $\{m_1, m_2\}$ once all other channels are held fixed. □

Remark 3200. *The proof is short because the conceptual work was already done in the definitions: by explicitly factoring out the paradigm dependence into Comp_P, we make preference reversal an immediate corollary rather than a mysterious sociological exception. In particular, the theorem should be read as a structural claim about the scoring functional: whenever the decision rule is multiplicatively (or additively in log-space) separable into evidential support and cost-like channels, any channel whose semantics is allowed to vary with culture can induce reversals even under full agreement elsewhere. This clarifies why debates over "simplicity" can be both persistent and rational: the disagreement may concern the mapping from descriptions to costs, not the correctness of the underlying updating rule.*

Theorem 141 (Robustness region under small channel-weight changes)**.** *Assume $\mathcal{M}$ is finite and $\mathrm{Supp}_\beta(m) \in (0, 1]$ for all m. Fix a paradigm with fixed costs* Comp, Prag *but variable weights*

$(\lambda_c, \lambda_p) \in (0,\infty)^2$. *If* $m^\star$ *is a* unique *maximizer of* $\mathsf{Score}(m; \lambda_c, \lambda_p)$ *at* $(\lambda_c^0, \lambda_p^0)$*, then there exists* $\delta > 0$ *such that* $m^\star$ *remains the unique maximizer for all* (λ_c, λ_p) *with* $|\lambda_c - \lambda_c^0| < \delta$ *and* $|\lambda_p - \lambda_p^0| < \delta$.

Remark 3201. *This is a stability theorem: if the currently chosen model is not merely best but* separated *from its competitors, then small changes in how much we care about simplicity vs. pragmatics will not change the choice. The importance is operational: it tells a reasoning system when it can safely act without finely tuning the channel weights, because the decision lies in a robust region of parameter space. One can think of* (λ_c, λ_p) *as "attention" or "regularization" parameters for the cultural and pragmatic channels: increasing* λ_c *penalizes complexity more heavily, while increasing* λ_p *penalizes pragmatic cost more heavily. The theorem then says that if* $m^\star$ *wins with a strict margin at* $(\lambda_c^0, \lambda_p^0)$*, there is an open neighborhood of weights around that point in which no competitor overtakes it. In practice, this is the mathematical content of the intuition that only near-ties are sensitive to minor reweightings.*

It connects naturally to the Pareto-pruning theorem: pruning removes dominated alternatives, which often increases the "margin" by which the best model wins, thereby enlarging robustness neighborhoods. Stated differently, removing models that can never win for any admissible weight choice reduces the number of potential "boundary crossings" in weight space, so the region in which $m^\star$ *is stable typically becomes easier to characterize and, in many cases, larger.*

Proof. For each fixed m, the function

$$(\lambda_c, \lambda_p) \mapsto \mathsf{Score}(m; \lambda_c, \lambda_p) = \mathrm{Supp}(m) \exp(-\lambda_c \mathrm{Comp}(m)) \exp(-\lambda_p \mathrm{Prag}(m))$$

is continuous. Since $\mathcal{M}$ is finite, the function $\Delta(\lambda_c, \lambda_p) := \mathsf{Score}(m^\star) - \max_{m \neq m^\star} \mathsf{Score}(m)$ is continuous and satisfies $\Delta(\lambda_c^0, \lambda_p^0) > 0$. By continuity, Δ remains > 0 in some neighborhood of $(\lambda_c^0, \lambda_p^0)$. Concretely, letting $\varepsilon := \Delta(\lambda_c^0, \lambda_p^0)/2 > 0$, continuity implies the existence of $\delta > 0$ such that whenever (λ_c, λ_p) lies within δ (in the stated coordinate-wise sense), one has $|\Delta(\lambda_c, \lambda_p) - \Delta(\lambda_c^0, \lambda_p^0)| < \varepsilon$, hence $\Delta(\lambda_c, \lambda_p) > \varepsilon > 0$, which preserves strict optimality. ☐

Proof sketch. Define a positive "gap" between the best model and the best runner-up. The gap is continuous in the weights (finite max of continuous functions). A positive value at the reference point implies positivity in a neighborhood, preserving uniqueness of the maximizer. Equivalently, the argmax map is locally constant at any point where the best model strictly beats all others. ☐

Remark 3202. *The finiteness of* $\mathcal{M}$ *is what prevents pathological "near-ties" accumulating from infinitely many alternatives. Geometrically, in the* (λ_c, λ_p) *plane, the boundaries where the argmax changes are curves; a strict argmax point lies away from these curves, hence has an open neighborhood with constant argmax. A useful equivalent viewpoint is obtained by taking logs:* $\log \mathsf{Score}(m) = \log \mathrm{Supp}(m) - \lambda_c \mathrm{Comp}(m) - \lambda_p \mathrm{Prag}(m)$*, so comparisons between any two models become linear inequalities in* (λ_c, λ_p)*. For a fixed pair* $(m^\star, m)$*, the locus where they tie is (typically) a line in weight space (or empty, depending on parameters), and the region where* $m^\star$ *beats* m *is a half-plane. When* $\mathcal{M}$ *is finite, the robustness region for* $m^\star$ *can be seen as the intersection of finitely many open half-planes, hence itself open, matching the neighborhood conclusion of the theorem. The condition* $\mathrm{Supp}_\beta(m) \in (0,1]$ *avoids degenerate cases where a model has zero support and is therefore irrelevant to the argmax regardless of weights; it ensures the score remains strictly positive so that the continuity argument applies cleanly without having to treat* 0 *as a boundary value.*

58.6 Method/path-independence and non-explosion theorems

Theorem 142 (Batching and order-independence of scientific updates)**.** *Let* $\varepsilon_1, \ldots, \varepsilon_n : L \to V$ *be empirical evidence increments produced by a finite sequence of experiments in a fixed methodological* *rule set R. Define iterative updates $\beta_{k+1} := \mathrm{Cl}_R(\beta_k \oplus \varepsilon_{k+1})$. Then*

$$\beta_n = \mathrm{Cl}_R(\beta_0 \oplus \varepsilon_1 \oplus \cdots \oplus \varepsilon_n),$$

hence the final closed belief state depends only on the join-aggregate of increments, not on their order.

Remark 3203. *The theorem asserts a kind of "path-independence" for scientific updating: you may run experiments in any order, or even batch their evidence together and update once, and the eventual closed belief state is the same. This is important for both epistemology and engineering. Epistemologically, it expresses a minimal ideal of method-independence: what matters is the total evidential import, not the accident of sequence. Engineer-wise, it means you can cache increments, merge them asynchronously, and close only periodically, which is essential in distributed or resource-bounded reasoning systems [19].*

Formally, the commutativity/associativity of $\oplus$ captures the idea that evidence pooling is insensitive to order, while idempotence of closure captures the idea that "once you have fully propagated consequences, doing it again adds nothing."

Remark 3204. *It is helpful to make explicit the algebraic assumptions silently used by the statement. The operator $\oplus$ is the pointwise join on belief states (so $(\beta \oplus \varepsilon)(\varphi) = \beta(\varphi) \vee \varepsilon(\varphi)$ in the valuation lattice V), and Cl_R is a closure operator associated to the rule set R (in particular, extensive $\beta \leq \mathrm{Cl}_R(\beta)$, monotone $\beta \leq \beta' \Rightarrow \mathrm{Cl}_R(\beta) \leq \mathrm{Cl}_R(\beta')$, and idempotent $\mathrm{Cl}_R(\mathrm{Cl}_R(\beta)) = \mathrm{Cl}_R(\beta)$). Under these standard "methodological" constraints, the update map $\beta \mapsto \mathrm{Cl}_R(\beta \oplus \varepsilon)$ behaves like a confluent saturation procedure: evidence can be accumulated in any order, and closure merely computes the least R-stable belief state above the accumulated evidence.*

One can also view the theorem as a mild "confluence" or "Church–Rosser"-style property for the operational process of updating. Although no rewriting system is assumed here, the practical reading is analogous: intermediate normalization steps (closures) can be postponed, and different operational interleavings of evidence collection lead to the same normal form (the final closed belief state).

Proof. This is an induction on n using the closure-operator laws and associativity/commutativity of $\oplus$. For the inductive step, use idempotence $\mathrm{Cl}_R(\mathrm{Cl}_R(\cdot)) = \mathrm{Cl}_R(\cdot)$ to push intermediate closures to the end, and then use associativity of $\oplus$ to batch evidence. $\square$

Proof. For clarity, write $E_k := \varepsilon_1 \oplus \cdots \oplus \varepsilon_k$ (with E_0 the neutral element for $\oplus$, or simply omit it if $\oplus$ is defined as join so that $\beta_0 \oplus E_0 = \beta_0$). The claim is that for each $k \leq n$,

$$\beta_k = \mathrm{Cl}_R(\beta_0 \oplus E_k).$$

Base case $k = 0$ is immediate if we take β_0 already as the initial state; otherwise it holds after applying Cl_R once since Cl_R is extensive and idempotent.

Inductive step: assume $\beta_k = \mathrm{Cl}_R(\beta_0 \oplus E_k)$. Then

$$\beta_{k+1} = \mathrm{Cl}_R(\beta_k \oplus \varepsilon_{k+1}) = \mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus E_k) \oplus \varepsilon_{k+1}).$$

By monotonicity and extensiveness, $\mathrm{Cl}_R(\beta_0 \oplus E_k) \leq \mathrm{Cl}_R(\beta_0 \oplus E_k \oplus \varepsilon_{k+1})$, and hence

$$\mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus E_k) \oplus \varepsilon_{k+1}) \leq \mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus E_k \oplus \varepsilon_{k+1})).$$

Conversely, since $\beta_0 \oplus E_k \leq \mathrm{Cl}_R(\beta_0 \oplus E_k)$, we have

$$\beta_0 \oplus E_k \oplus \varepsilon_{k+1} \leq \mathrm{Cl}_R(\beta_0 \oplus E_k) \oplus \varepsilon_{k+1},$$

and by monotonicity of Cl_R,

$$\mathrm{Cl}_R(\beta_0 \oplus E_k \oplus \varepsilon_{k+1}) \le \mathrm{Cl}_R(\mathrm{Cl}_R(\beta_0 \oplus E_k) \oplus \varepsilon_{k+1}).$$

Combining these inequalities and using idempotence $\mathrm{Cl}_R(\mathrm{Cl}_R(\cdot)) = \mathrm{Cl}_R(\cdot)$ yields

$$\beta_{k+1} = \mathrm{Cl}_R(\beta_0 \oplus E_k \oplus \varepsilon_{k+1}) = \mathrm{Cl}_R(\beta_0 \oplus E_{k+1}),$$

where the last equality uses associativity (and commutativity if the increments are permuted). Setting $k = n$ proves the stated batching identity. □

Proof sketch. Induct on the number of increments. At each step, rewrite $\mathrm{Cl}_R(\mathrm{Cl}_R(\cdot)\oplus\varepsilon)$ as $\mathrm{Cl}_R(\cdot\oplus\varepsilon)$ using idempotence and monotonicity, then reassociate/commute joins to collect all increments together before a final closure. □

Remark 3205. *The key maneuver is "postponing closure": because Cl_R is idempotent, closing after each increment or only at the end yields the same result. Visually, you can imagine evidence increments as adding paint to a canvas and closure as letting the paint diffuse along rule-edges; diffusion can be done after each addition or after all additions, producing the same final saturation pattern.*

Remark 3206. *In the cultural-probabilistic-pragmatic framing, this captures a separation between (i) the contingent temporal history of laboratory practice (the order in which experiments are performed) and (ii) the stable methodological constraints encoded by R. The theorem does not claim that scientists in fact update in an order-free way, but that the* methodological idealization *represented by Cl_R and $\oplus$ yields an order-invariant fixed point when evidence is pooled. This is exactly the behavior one wants if R is meant to represent publicly shareable inferential norms: two communities that differ only in scheduling of experiments, but agree on R and on the multiset of increments, converge to the same closed state.*

Operationally, the result is also a robustness guarantee: intermediate closures can be viewed as "checkpointing" or "garbage collection" steps in a belief-maintenance engine. Since the final state is invariant under the placement of these checkpoints, implementations are free to trade off immediacy of closure against cost, without changing eventual conclusions (only the transient states).

Theorem 143 (Irrelevance / localized non-explosion)**.** *Fix rule set R. If no rule in R has conclusion $\psi \in L$, then for every belief state β,*

$$\mathrm{Cl}_R(\beta)(\psi) = \beta(\psi).$$

In particular, evidence and anomalies elsewhere cannot induce belief in ψ unless there is an explicit inferential path to ψ via R.

Remark 3207. *This theorem formalizes a basic sanity property: your inference rules cannot magically produce support for claims they never conclude. In paraconsistent and nonclassical settings this is sometimes called a non-explosion or locality property, but here it is a straightforward structural statement about the shape of the rule set (Hyperseed-Concept 98). It matters computationally: it guarantees that belief propagation is localized to the part of the language actually targeted by rules, enabling sparse implementations.*

It also provides a clean criterion for when "anomalies elsewhere" can matter: only if the rule graph contains a directed path of conclusions leading to ψ. Thus relevance is a property of the inferential wiring, not merely of the data.

Remark 3208. *The condition "no rule in R has conclusion ψ" can be read as a graph-theoretic boundary condition on the dependency network induced by R: ψ has in-degree 0 with respect to the directed edges "premises $\to$ conclusion." The theorem then says that ψ is a* sinkless *coordinate for the closure dynamics, so whatever happens elsewhere in the network (including inconsistent or highly activated regions) cannot leak into ψ without an explicit channel. This is the precise sense in which the framework prevents a global "blast radius" from contradictions: abnormality does not automatically propagate to unrelated claims unless the methodology actually links them.*

From the perspective of modular scientific models, this yields a formal justification for treating sublanguages as quasi-independent modules: if a module contains symbols never concluded by rules outside the module, then closure computations for those symbols can be performed locally and are invariant under external updates.

Proof. If ψ is never a rule conclusion, the one-step inference operator F_R leaves ψ unchanged (because the join over contributions to ψ is empty). Hence every iterate $F_R^k(\beta)(\psi) = \beta(\psi)$. Taking the join over k yields $\mathrm{Cl}_R(\beta)(\psi) = \beta(\psi)$. □

Proof. Unpacking the "empty join" point: by definition, $F_R(\beta)(\psi)$ is obtained by joining $\beta(\psi)$ with all supports contributed by rules in R whose conclusion is exactly ψ. If there are no such rules, the additional join is taken over an empty index set, which evaluates to the neutral (least) element for join in V, and thus does not change the value. Therefore $F_R(\beta)(\psi) = \beta(\psi)$, and by straightforward induction on k we obtain $F_R^k(\beta)(\psi) = \beta(\psi)$ for all $k \in \mathbb{N}$.

Finally, since $\mathrm{Cl}_R(\beta)$ is the (directed) join of these iterates (or, equivalently, the least fixed point above β), and since the ψ-coordinate is constant across all iterates, the join is the same constant. This shows that closure does not introduce any new support at ψ when R provides no inferential mechanism targeting ψ. □

Proof sketch. Observe that F_R updates each claim by joining the supports generated by rules concluding that claim. If there are no such rules for ψ, the update is vacuous, so iteration never changes ψ and neither does the limit. □

Remark 3209. *The crucial step is recognizing that the closure is built from iterating a local update operator. If the local update has no input channel for ψ, then ψ is an invariant coordinate of the dynamics.*

Remark 3210. *This invariance viewpoint also clarifies how the result interacts with methodological change. The theorem is parameterized by a fixed R: if one enlarges R by adding even a single rule with conclusion ψ, then ψ ceases to be an invariant coordinate and may begin to respond to evidence elsewhere (via the premises of the new rule). Thus, what counts as "irrelevant" is not an absolute property of the language L but a relative property of the chosen inferential practice. In the broader Hyperseed program, this matches the intended reading that relevance and possible propagation pathways are encoded in, and constrained by, explicit methodological commitments.*

58.7 Modularity/independence theorems (useful for decomposing science)

Theorem 144 (Rule-separable modularity (independence of sublanguages))**.** *Suppose the claim language decomposes as a disjoint union $L = L_1 \sqcup L_2$. Suppose the rule set decomposes as $R = R_1 \sqcup R_2$ where every rule in R_i has all premises and its conclusion in L_i (no cross-rules). Then for any belief state β,*

$$\mathrm{Cl}_R(\beta)\big|_{L_1} = \mathrm{Cl}_{R_1}(\beta|_{L_1}), \qquad \mathrm{Cl}_R(\beta)\big|_{L_2} = \mathrm{Cl}_{R_2}(\beta|_{L_2}).$$

Consequently, evidence increments supported only on L_1 do not affect closed beliefs on L_2, and vice versa. Moreover, the statement is robust under iterated updates: if $(\beta_t)_{t\geq 0}$ evolves by any sequence of updates that at each step modifies only coordinates in L_1, then the entire trajectory of closed beliefs on L_2 remains fixed, i.e. $\mathrm{Cl}_R(\beta_t)|_{L_2}$ is constant in t.

Remark 3211. *This is a domain decomposition theorem: if the language and rules split into two noninteracting parts, then scientific inference splits exactly as well. The importance is obvious for large systems: it justifies modular architectures where different subdisciplines (or different subsystems of a scientist-agent) can update internally without risking unintended contamination across unrelated domains (Hyperseed-Concept 93).*

One useful way to read the hypotheses is operational: a module boundary is not merely a partition of vocabulary, but a partition of inferential pathways. *Even if L_1 and L_2 are conceptually distinct, independence fails as soon as some rule can transport support across the boundary (e.g. a bridge principle, a calibration equation, or a measurement rule that links theoretical quantities in L_2 to observational claims in L_1). The theorem therefore singles out a formally checkable condition: scan the rule base for any premise–conclusion pattern whose symbols touch both sides.*

The result also provides a crisp criterion for when modularization is not *safe: the existence of even a single cross-rule creates a coupling that defeats exact independence, and then one needs approximate or bounded-dependency analyses (as developed elsewhere in the broader document). In practice, such "approximate" modularity can arise when cross-rules are present but weak, rarely activated, or quarantined behind explicit interface claims; the present theorem describes the idealized limit in which the interface is empty.*

Proof. Identify $V^L \cong V^{L_1} \times V^{L_2}$ via restriction. Under the "no cross-rules" assumption, F_R decomposes as $F_{R_1} \times F_{R_2}$ on this product. Least fixed points in a product of complete lattices are computed componentwise, so the least fixed point above β is the pair of least fixed points above each restriction. This yields the equalities. To make the componentwise computation explicit: if $F := F_R$ and $F_i := F_{R_i}$, then for $(x_1, x_2) \in V^{L_1} \times V^{L_2}$ we have

$$F(x_1, x_2) = (F_1(x_1), F_2(x_2)),$$

hence any post-fixed point condition $(x_1, x_2) \leq F(x_1, x_2)$ splits into $x_i \leq F_i(x_i)$, and the infimum of all such post-fixed points above $(\beta|_{L_1}, \beta|_{L_2})$ splits similarly. Equivalently, the Kleene iteration starting from β evolves independently on each coordinate. □

Proof sketch. Rewrite the global belief state as a pair of belief states on L_1 and L_2. With no cross-rules, one-step inference acts independently on each coordinate, so the closure (least fixed point above β) is just the pair of closures in each module. In particular, there is no stage of the iteration at which an L_1-supported change can alter the L_2-component, because rule firing in R_2 only ever inspects and writes L_2-claims. □

Remark 3212. *The core idea is that "no cross-rules" makes the inference operator block-diagonal. In linear algebra one would say the dynamics decouple; here, in order theory, the fixed points decouple. A further intuition is categorical: the closure construction behaves like a product construction when the rule base factors, so global inference is the product of local inferences. This matters for implementation: one may compute Cl_{R_1} and Cl_{R_2} in parallel and splice the results without any reconciliation step, because reconciliation would only be needed to enforce cross-constraints that do not exist in this setting.*

Theorem 145 (Collective closure dominance (social science-as-belief-system fact))**.** *Let $\{\beta_i\}_{i\in I}$ be belief states and define the collective join-aggregate $\beta_{\mathrm{col}} := \bigoplus_{i\in I} \beta_i$ (pointwise join). Then for every $i \in I$,*

$$\mathrm{Cl}_R(\beta_i) \le \mathrm{Cl}_R(\beta_{\mathrm{col}}).$$

Thus any inferential consequence available to an individual (under R) is also available to the collective when the collective retains all individuals' evidence. In particular, for any claim $\varphi \in L$ and any index i, if $\mathrm{Cl}_R(\beta_i)(\varphi)$ attains some support level, then the collective closure attains at least that level on the same claim.

Remark 3213. *This theorem articulates a minimal sense in which science is "social": if a community aggregates the evidence held by its members without loss, then after running the same inference rules it will contain at least what any one member could infer. It is not a claim about wisdom of crowds in any strong sense; it is simply the monotonicity of closure applied to an aggregated evidence base (Hyperseed-Concepts 170, **??**).*

The pointwise join $\bigoplus_{i\in I} \beta_i$ should be read as an "archival" idealization: every proposition receives whatever maximal degree of support any member assigns it. Under that reading, the theorem says that the community, viewed as a single epistemic agent that keeps a lossless record, cannot be inferentially *weaker than its strongest member with respect to the fixed rule base R. This is the precise sense in which distribution of labor helps: different members may contribute evidence on different parts of L, and the join combines these contributions before closure propagates them along rules.*

This structural fact is a prerequisite for more delicate analyses of cooperation vs. conflict, incentives, and norm formation; here we only need the order-theoretic monotonicity. In particular, the theorem does not address whether the collective will be better calibrated *or* less biased, *because those are properties not captured by the mere partial order on belief states unless additional scoring or semantics are imposed.*

Proof. Since $\beta_i \le \beta_{\mathrm{col}}$ pointwise and Cl_R is monotone, we get $\mathrm{Cl}_R(\beta_i) \le \mathrm{Cl}_R(\beta_{\mathrm{col}})$. Concretely, the inequality $\beta_i \le \bigoplus_{j\in I} \beta_j$ holds at each coordinate $\varphi \in L$ by the defining property of join, and monotonicity transports this coordinatewise domination through the closure operator. □

Proof sketch. The collective state is an upper bound of each individual state. Monotonicity of closure preserves the ordering, so the collective closure dominates each individual closure. If one imagines the closure as "running" the rule engine until saturation, then starting from a state with (weakly) more support everywhere cannot lead to a saturated state with less support anywhere. □

Remark 3214. *The proof is one line because the theorem is really a definition: "collective" here means "keeping everything anyone has" via join. The interesting questions begin when aggregation is lossy, weighted, strategic, or inconsistent; those require extra structure beyond this minimal lemma. One immediate extension, for example, is to replace the pure join by a constrained aggregator (e.g. bounded confidence, trust networks, or resource-limited communication) and then ask for sufficient conditions under which a weaker dominance relation still holds approximately.*

58.8 State-dependent science and observer-state paradigms

Theorem 146 (State-dependent science as ordinary science on an expanded experiment space)**.** *Let S be a set of community-accessible states and for each $s \in S$ let $\mathrm{Obs}_s : E_s \to O$ be a state-indexed observation map. Define the expanded experiment space*

$$\widetilde{E} := \{(s, e) : s \in S,\ e \in E_s\}$$

and observation map $\widetilde{\mathrm{Obs}} : \widetilde{E} \to O$ *by* $\widetilde{\mathrm{Obs}}(s,e) := \mathrm{Obs}_s(e)$. *Then any state-dependent science family* $\{(B_s, R_s, \mathrm{Obs}_s)\}_{s\in S}$ *can be represented as an ordinary Hyperseed science instance on* $\widetilde{E}$ *by treating "state" as part of experimental context. Conversely, any ordinary science instance on* $\widetilde{E}$ *decomposes into a state-dependent family by restriction.*

In particular, one may view $\widetilde{E}$ *as a "tagged union" (a disjoint union) of the state-specific experiment spaces:*

$$\widetilde{E} \;\cong\; \bigsqcup_{s\in S} \big(\{s\} \times E_s\big),$$

so that the state label plays the same role as any other experimental descriptor (instrument setting, protocol version, lab condition), and the observation rule is uniformly a single map defined on the union.

Remark 3215. *The theorem says that "state-dependent" science is not a fundamentally different beast: it is ordinary science performed on a richer experiment space where the state label is included as part of the experimental description. This is a common mathematical move: contextual variation is represented by enlarging the domain so that context becomes explicit data. In epistemic terms, it formalizes the idea that observer-state (or environmental state) is part of what makes an experiment what it is (Hyperseed-Concept 86) [20].*

Practically, this means we can reuse the same update/closure algorithms even when observations depend on state (e.g. different calibration regimes, sensor modes, or community conditions), simply by lifting experiments to pairs (s, e).

One can also read the construction as an explicit bookkeeping device for what is sometimes informally described as "conditioning on the observer" or "conditioning on the measurement context." Once s *is included in the experimental description, statements that would otherwise appear to conflict across states become compatible as statements about different regions (fibers) of a single space* $\widetilde{E}$.

Proof. This is a set-theoretic repackaging: the data of maps $\mathrm{Obs}_s : E_s \to O$ is equivalent to a single map on the disjoint union (or subset of the product) of all (s, e) pairs. All update/closure constructions then apply verbatim. Restriction along the inclusion $E_s \hookrightarrow \widetilde{E}$ recovers each state-specific instance.

Concretely, let $\iota_s : E_s \to \widetilde{E}$ be the injection $\iota_s(e) = (s, e)$. Then $\widetilde{\mathrm{Obs}} \circ \iota_s = \mathrm{Obs}_s$ for each s, so every state-indexed observation family factors through the single map $\widetilde{\mathrm{Obs}}$. Conversely, given $\widetilde{\mathrm{Obs}} : \widetilde{E} \to O$, define $E_s := \{e : (s, e) \in \widetilde{E}\}$ and $\mathrm{Obs}_s(e) := \widetilde{\mathrm{Obs}}(s, e)$; this recovers the original family by restriction to each fiber $\{s\} \times E_s$. The same fiberwise restriction applies to any additional structure carried by the instance (e.g. belief/state objects B_s and rule components R_s) provided they are likewise indexed by s or induced from data on $\widetilde{E}$. □

Proof sketch. Bundle all state-indexed experiment sets into one disjoint union indexed by s. Define a single observation map on the union by applying the appropriate Obs_s. Then "state-dependent updates" become ordinary updates on the enlarged set, and conversely restricting to each fiber recovers the original family.

Equivalently, think of s as an extra input to the experimental protocol: the update mechanism does not need to know anything metaphysically special about states; it simply operates on experiments with an additional label. □

Remark 3216. *The key step is recognizing that a family of functions* $\{\mathrm{Obs}_s\}$ *is the same thing as a single function on a tagged union. Visually, one stacks the different state-specific experiment spaces into layers and regards the stack as one big space.*

From the point of view of observer-state paradigms, this stacking clarifies what varies and what does not: what varies across s is the observational interface (what is registered as O for a given underlying trial e), while the global representation keeps a fixed codomain O and a uniform update story. In CPP terms, the "observer state" becomes part of the evidential context rather than an external qualifier attached after the fact, so comparisons across communities or modalities are comparisons across fibers inside $\widetilde{E}$ rather than comparisons of incommensurable experiment spaces.

58.9 Three-tier artificial scientist: a lifting theorem

Definition 528 (Tiered search spaces)**.** *Let $\mathcal{P}$ be a set of paradigms. For each $P \in \mathcal{P}$ let $\mathcal{M}_P$ be its model class. Define the lifted hypothesis space*

$$\widehat{\mathcal{M}} := \{(P, m) : P \in \mathcal{P},\ m \in \mathcal{M}_P\}.$$

Define a meta-complexity $\mathrm{Comp}_{\mathrm{meta}} : \mathcal{P} \to [0, \infty)$ *and a lifted score*

$$\widehat{\mathsf{Score}}_\beta(P, m) := \mathsf{Score}_{P,\beta}(m)\ e^{-\lambda_{\mathrm{meta}}\mathrm{Comp}_{\mathrm{meta}}(P)}.$$

In this notation, $\mathsf{Score}_{P,\beta}(m)$ is the within-paradigm score (or utility) assigned to a model m when evidence and inductive preferences are interpreted under P; the parameter β may be read as an inverse-temperature or rationality/selection-strength parameter that controls how sharply the scoring function concentrates on high-performing models. The coefficient $\lambda_{\mathrm{meta}} \geq 0$ controls the strength of the meta-level penalty and makes explicit that "paradigm cost" is conceptually distinct from ordinary within-paradigm complexity penalties: it acts uniformly on all models inside the same P.

Remark 3217. *This definition makes explicit a hierarchy of choices: not only do we choose a model m, we may also choose the paradigm P that supplies the semantics of evidence, simplicity, and usefulness (Hyperseed-Concepts 91, 64). The lifted space $\widehat{\mathcal{M}}$ is simply the set of all such pairs; it is the natural search space for an "artificial scientist" that can revise its methodological commitments.*

The meta-complexity $\mathrm{Comp}_{\mathrm{meta}}$ *expresses that paradigms themselves can be more or less costly: adopting a paradigm may require learning new primitives, instruments, or practices. The exponential penalty mirrors the within-paradigm penalties: we treat paradigm complexity as another weakness channel, now at a higher level of organization [3, 19].*

One way to read the construction is as an explicit three-tier separation of concerns: (Tier 1) evaluate candidate models m given a fixed P; (Tier 2) compare models across candidates within a single sheet $\mathcal{M}_P$ using $\mathsf{Score}_{P,\beta}$; (Tier 3) compare entire sheets (paradigms) using $\mathrm{Comp}_{\mathrm{meta}}$ *as a higher-order regularizer that captures the cost of adopting the sheet itself. The "lifting" step turns this nested decision into a single optimization over a product-like space, while still preserving the interpretive fact that $\mathsf{Score}_{P,\beta}$ depends on P through the meaning of data, observables, and acceptable forms of explanation.*

Theorem 147 (Paradigm innovation reduces to model selection on a lifted space)**.** *Assume $\widehat{\mathcal{M}}$ is finite. Then there exists at least one pair $(P^\star, m^\star)$ maximizing $\widehat{\mathsf{Score}}_\beta(P, m)$ over $\widehat{\mathcal{M}}$. Moreover, searching over paradigms and models (Tier 3) is formally identical to searching over the lifted hypothesis space $\widehat{\mathcal{M}}$ (Tier 2) with an additional penalty on paradigm complexity.*

Remark 3218. *The theorem says that "paradigm innovation" can be treated as ordinary model selection, provided we lift the hypothesis space to include paradigms as part of the hypothesis. This is* *important because it demystifies a notion often treated as extra-rational: here it becomes an explicit optimization problem with a well-defined objective and (in the finite case) guaranteed solutions.*

In terms of the earlier crosswalk, this is the same structural move again: add a new weakness channel (meta-complexity) and then apply the same selection logic. This is precisely the kind of reuse that motivates encoding science in algebraic terms.

Operationally, this also clarifies what changes when an agent "switches paradigms": the agent is not merely changing parameter values within a fixed model class, but changing the very interpretation function that maps raw experience into evidence, as well as the catalogue of admissible hypotheses $\mathcal{M}_P$. In the lifted formulation, such a switch is represented by moving to a different value of P in the pair (P, m), with the cost of that move summarized by $\mathrm{Comp}_{\mathrm{meta}}(P)$. When λ_{meta} is large, the system is conservative and tends to stay within a paradigm unless the within-paradigm score gain is decisive; when λ_{meta} is small, the system is more willing to explore methodologically distant hypotheses.

Proof. A real-valued function on a finite set attains a maximum. The second claim holds because "choose a paradigm then choose a model" is equivalent to "choose a pair" and the lifted score makes the meta-objective explicit. In particular, the map $(P, m) \mapsto \widehat{\mathsf{Score}}_\beta(P, m)$ is a single objective on $\widehat{\mathcal{M}}$ whose factors separate into a within-paradigm component (depending on both P and m) and a paradigm-only component (depending only on P). This separation ensures that the two-stage reasoning about "first decide whether a paradigm is worth its adoption cost, then decide which model is best inside it" is representable without loss as a single argmax computation on pairs. □

Proof sketch. Finiteness implies the maximum exists. The equivalence of two-stage choice and pair choice is a basic set-theoretic identification: $\bigcup_{P \in \mathcal{P}} \{P\} \times \mathcal{M}_P$ is exactly the space of "paradigm then model" decisions. Under this identification, any algorithm that enumerates paradigms and then, for each paradigm, enumerates models, is simply enumerating $\widehat{\mathcal{M}}$ in a particular order; conversely, any procedure that searches $\widehat{\mathcal{M}}$ implicitly induces a (possibly interleaved) search schedule over paradigms and their internal model classes. □

Remark 3219. *The proof works because the lifted space has no hidden structure: it is literally a disjoint union of the model classes, tagged by paradigm. Visually, each paradigm contributes a sheet of models; we search across the stack of sheets with an added penalty depending only on which sheet we choose.*

This visualization also highlights an important modeling choice: because the penalty is paradigm-only, the relative ranking of models within *a paradigm is unaffected by* $\mathrm{Comp}_{\mathrm{meta}}(P)$*, while the relative ranking* across *paradigms is shifted by the paradigm penalty. That is,* $\mathrm{Comp}_{\mathrm{meta}}$ *behaves like an offset applied to all models in the sheet. If one wished to represent heterogeneity of transition costs inside a paradigm (e.g., some models requiring new instruments while others do not), that would correspond to pushing part of the meta-complexity down into* $\mathsf{Score}_{P,\beta}(m)$ *or refining P into sub-paradigms; the present formulation deliberately keeps the meta-level penalty coarse to isolate the lifting move itself.*

58.10 Summary (operational takeaways, stated as corollaries)

Corollary 53 (Caching and method flexibility)**.** *Because scientific updates batch and commute, a reasoning system can store and merge evidence increments in any order and close only at chosen times, without changing the final closed state (for fixed R).*

Remark 3220. *Operationally: evidence increments $\varepsilon_{e,o}$ can be treated as cacheable objects. One may accumulate them in a database, merge them by $\oplus$ (commutatively), and apply closure when* *computationally convenient. This is the algebraic justification for asynchronous and distributed "scientist agents" that do not require a single canonical experimental sequence [19]. In more explicit algebraic terms, if $\oplus$ is associative and commutative, then for any finite multiset of increments $\{\varepsilon_i\}_{i=1}^n$ one has*

$$\varepsilon_{\text{tot}} \;=\; \bigoplus_{i=1}^{n} \varepsilon_i$$

independent of evaluation order, so the system can implement $\oplus$ via streaming updates, batch jobs, or federated merges. The clause "for fixed R" matters operationally: the invariance claim assumes the same rule set R (and hence the same closure operator) is applied at the end, so that only the scheduling *of updates changes, not the inferential policy. Practically, this supports architectures in which agents publish signed increments (provenance-preserving), a coordinator (or peer network) reconciles them via $\oplus$, and closure is applied only when a query arrives or when compute budgets allow. When closure is expensive, one can also maintain intermediate partially-closed states, provided the eventual closure result is taken with respect to the full merged increment and the same R.*

Corollary 54 (Model pruning under uncertain cultural/pragmatic weights)**.** *If a candidate model is Pareto-dominated in the three channels (support, cultural cost, pragmatic cost), it can be discarded regardless of the precise positive channel weights.*

Remark 3221. *This is a concrete decision tool: before debating the exact values of λ_c and λ_p, one can eliminate any model that is strictly worse in at least one channel and no better in any channel. The corollary thus separates* structural *inferiority (dominance) from* parametric *disagreement (tradeoff tuning). Concretely, write each model M as a triple*

$$\mathbf{v}(M) \;=\; \big(s(M),\, c(M),\, p(M)\big),$$

where s is to be maximized (support) while c, p are to be minimized (costs). Then M_1 Pareto-dominates M_2 if $s(M_1) \geq s(M_2)$, $c(M_1) \leq c(M_2)$, $p(M_1) \leq p(M_2)$, and at least one inequality is strict. Under any aggregator that is strictly increasing in s and strictly decreasing in c, p (e.g., a weighted objective with positive weights), dominance implies M_1 scores strictly better than M_2 without needing to know the particular weight values. Operationally, this means one can compute a Pareto frontier early (as a preprocessing step), shrinking the candidate set before engaging in culturally or pragmatically sensitive deliberation about acceptable tradeoffs. It also clarifies what kinds of disagreement remain after pruning: only models on the frontier can be selected differently by different communities or deployment contexts via different (λ_c, λ_p).

Corollary 55 (Safe modularization)**.** *If the rule graph splits into independent sublanguages, scientific inference decomposes exactly; updates in one module cannot contaminate beliefs in the other.*

Remark 3222. *This provides an exact criterion for when disciplinary or subsystem boundaries are formally justified: if there are no cross-rules, then there is no cross-inference. It is, in effect, a theorem about the integrity of partitions in a belief system (Hyperseed-Concepts 64, 93). More precisely, if the language (or symbol set) factors as $L = L_1 \cup L_2$ with $L_1 \cap L_2 = \varnothing$, and the rule set factors as $R = R_1 \cup R_2$ where every rule in R_i mentions only symbols from L_i, then the dependency (rule) graph has no edges between the components, and closure on the union decomposes into separate closures on each component. Under these conditions, an increment whose content lies wholly in L_1 can change only the L_1-restricted beliefs, because there exists no derivational path (via R) into L_2. This gives a practical test for modular system design: audit whether any* *rule, bridge principle, shared latent variable, or shared measurement predicate straddles the two vocabularies; if so, strict non-contamination no longer holds and one must treat the modules as coupled. Conversely, when the criterion is satisfied, one can parallelize inference, store module-local caches, and even enforce access control or governance boundaries per module without risking hidden inferential leakage across the partition.*

59 Surprisingness, Weakness, and Science in the Hyperseed Ontology: A Quantale-Compositional Bridge

We connect (A) Hyperseed's "surprise-adjusted implication" score for beliefs (surprisingness given complexity) and its formalization of science as empirically coupled belief-system dynamics, with (B) an inference-centric quantale calculus of surprisingness in which surprise is the marginal value of additional inference budget. This yields concrete theorems showing that Hyperseed's belief surprise is an instance of budget-residual surprise (ratio and tropical forms), inherits telescoping and chain rules, and can be robustified across contexts/experiments via a weakness (infimum) operator. We also formalize the sense in which both scientists and ordinary curious minds are engaged in a surprise-seeking loop: both are special cases of question/answer/belief dynamics that aim (among other goals) to generate high-surprise beliefs and hypotheses without collapsing coherence.

Concretely, the intended reading is that a belief (or hypothesis) is evaluated under two coupled pressures: it should be supported by the available empirical and inferential process, yet it should not be "too easy" in the sense of being fully predicted by its descriptive simplicity alone. In Hyperseed language, the "surprise-adjusted implication" score is designed to reward beliefs whose support rises sharply when additional inference is performed, relative to what one would expect from their complexity penalty. In the quantale reading, this same intuition is captured by comparing two evaluations (e.g. "support after spending budget b" vs. "baseline support from prior simplicity"), and taking a residuated difference that measures the incremental budget required to bridge the gap. The result is not merely an analogy: it is a way to translate between a cognitive/ scientific narrative and a compositional algebra of updates.

The phrase "marginal value of additional inference budget" is meant operationally: if one views inference as a resource-bounded process (time, compute, attention, experimental cost, or social coordination), then surprise is what is gained when that resource bound is relaxed. This makes the budget parameter explicit in the semantics of belief revision: surprise is not only a property of a statement, but of a statement *relative to* a process and a constraint. The quantale framework packages this dependence in a way that supports composition: sequential inference steps combine, and so do their associated surprise contributions, which is precisely what later yields telescoping and chain rules.

The two advertised forms—ratio and tropical—correspond to two common ways of representing this budget-residual comparison. In the ratio form, surprise is expressed as a multiplicative residual: one asks for the least factor by which the budget (or the evaluative strength) must be scaled so that the post-inference evaluation dominates the baseline. In the tropical form, multiplicative structure is log-transformed into addition, so surprise becomes an additive residual (a minimal increment), which is often more interpretable as "extra effort" or "extra evidence" required. The point of including both is that Hyperseed-like scores are frequently reported in multiplicative language (implication strength adjusted by complexity), while compositional reasoning about sequential updates often becomes cleaner in additive (tropical) coordinates.

The robustness claim via weakness (infimum) is likewise intended in an operational sense. When a belief is tested across contexts—different experimental conditions, different background assumptions, different communities, or different data sources—one wants a conservative aggregate that does not overfit to a single favorable setting. The weakness operator acts as a least-commitment aggregator: by taking an infimum over contexts, it selects what is uniformly supported (and uniformly surprising) rather than what is maximized by cherry-picked circumstances. In later results, this infimum-based robustness interacts well with compositional laws, so that robust surprise bounds can be propagated through chains of reasoning without having to re-derive everything from scratch.

Remark 3223. *Philosophically, the bridge in this section is a bridge between two notions of "novelty." One notion is epistemic and social: a community (or a mind) is surprised when, after it performs more inference and integrates more evidence, certain claims become notably more supported than one would expect from their simplicity alone. The other notion is algebraic: "surprise" is a residuated difference between two evaluations living in a quantale, and therefore inherits compositional laws (telescoping, factorization, robustness via infima) almost for free.*

Remark 3224. *The epistemic/social notion of novelty implicitly involves a dynamics: questions are posed, candidate answers are proposed, and beliefs are updated under constraints of coherence and limited resources. What is "surprising" in this sense is not merely what is improbable under a static prior, but what resists being explained away by cheap descriptions, routine heuristics, or low-effort inference. The algebraic notion makes this dynamic explicit by treating evaluation as something that changes with budget, and by measuring surprise as a residual that tracks the* least additional *budget needed to obtain a given improvement in evaluation. This makes it natural to speak about the distribution of surprise across a multi-step inquiry: telescoping identities state that total surprise over a path can be decomposed into stepwise contributions, while chain rules state that conditioning on intermediate results factors the surprise in a way compatible with sequential question answering.*

Remark 3225. *In Hyperseed terms [1], this section touches several core ideas: belief systems (Hyperseed-Concept 64), questions and answers (Hyperseed-Concept 144, 56), patterns (Hyperseed-Concept 130), and weakness as a robustness/least-commitment principle (Hyperseed-Concept 202, 143). On the philosophy-of-science side, the operational reading of science as an empirically coupled communal belief dynamics aligns with the social-computational-probabilistic view [20].*

59.1 Background

Inference-centric surprisingness. The "Calculus of Surprisingness" defines a budget-indexed evaluation map $B \mapsto TV_B(\varphi)$ and defines surprise as a residual/distance between $TV_{B_0}(\varphi)$ and $TV_{B_1}(\varphi)$. In particular, in the residual-form one defines

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) := TV_{B_0}(\varphi) \Rightarrow TV_{B_1}(\varphi),$$

in a residuated quantale. Surprise then composes across budgets (via-point / telescoping laws) and families can be robustified via a weakness operator (infimum over families of patterns).

Remark 3226. *The motivating picture is mundane but deep: one runs a weak reasoning procedure (small budget B_0), obtains some evaluation of φ, then permits a stronger reasoning procedure (larger budget B_1), and asks: how much did the evaluation move? If it moved a lot, φ was "surprising" relative to what was available under the weaker regime. This is a direct formalization of the common experience that many ideas look implausible until one performs the extra reasoning steps that reveal a hidden constraint or a previously unseen connection (compare the broad pattern-emergence viewpoint in [5]).*

A key technical point is that the use of the residual $\Rightarrow$ is not merely a convenient choice of "distance-like" primitive: in a residuated quantale it is exactly the structure that makes budget-composition algebraic. Concretely, if $\otimes$ is the quantale multiplication, then residuation is characterized by the adjunction

$$a \otimes c \leq b \quad \text{iff} \quad c \leq (a \Rightarrow b),$$

so that $a \Rightarrow b$ can be read as the least multiplicative "effort" c that, when combined with a, reaches (or exceeds) b. Under the intended reading $a = TV_{B_0}(\varphi)$ and $b = TV_{B_1}(\varphi)$, the surprise term $\mathrm{Surp}_{B_0 \to B_1}(\varphi)$ is thus the minimal factor (in the $\otimes$-sense) required to transform the weaker-budget evaluation into the stronger-budget evaluation.

The telescoping/via-point law alluded to above can be made explicit: for a chain of budgets $B_0 \leq B_1 \leq B_2$, one typically seeks inequalities (often equalities in well-behaved cases) of the general form

$$\mathrm{Surp}_{B_0 \to B_2}(\varphi) \;\leq\; \mathrm{Surp}_{B_0 \to B_1}(\varphi) \otimes \mathrm{Surp}_{B_1 \to B_2}(\varphi),$$

capturing the intuition that the total movement from B_0 to B_2 factors through intermediate budgets. This is the algebraic expression of the operational story: upgrades of inference power can be staged, and the surprisingness of the final outcome is compositional with respect to those stages.

Quantale examples. The same document highlights the ratio quantale and the tropical (max-plus) quantale:

$$Q_\times = ([0, \infty], \leq, \otimes = \times, \oplus = \max, 1 = 1, 0 = 0,\ a \Rightarrow b = b/a),$$

and

$$Q_+ = (\mathbb{R} \cup \{-\infty\}, \leq, \otimes = +, \oplus = \max, 1 = 0, 0 = -\infty,\ a \Rightarrow b = b - a).$$

These instantiate multiplicative lift and additive log-deltas respectively.

Remark 3227. *Here $Q_\times$ and Q_+ are the same information viewed in two coordinate systems. If one thinks of $TV_B(\varphi)$ as a nonnegative "score" (plausibility, utility, adequacy), then ratios naturally encode multiplicative improvement: "how many times larger did the score become?" If one takes logs, multiplication turns into addition and the residual becomes a difference. The tropical quantale Q_+ is precisely this log-domain picture, and it is often computationally natural because additive decompositions are easier to accumulate and to optimize.*

It is also helpful to record what "surprise" looks like in these two running examples. In the ratio quantale, assuming $TV_{B_0}(\varphi) > 0$, one has

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) = \frac{TV_{B_1}(\varphi)}{TV_{B_0}(\varphi)},$$

so that surprise is literally a gain factor. In the tropical quantale, one has

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) = TV_{B_1}(\varphi) - TV_{B_0}(\varphi),$$

so that surprise is a delta. Both are compatible with the same underlying idea: the residual measures the minimal increment (multiplicative or additive) that turns the earlier valuation into the later one, and the choice between $Q_\times$ and Q_+ amounts to a choice of scale on which "movement" should be linearized.

Hyperseed: belief surprise and productive belief systems. Hyperseed's formalization introduces a complexity cost $\mathrm{Comp}(\varphi)$ and a simplicity-biased prior weight $\mathrm{Prior}(\varphi) = e^{-\lambda \mathrm{Comp}(\varphi)}$, then defines a "surprise-adjusted implication" score

$$\mathrm{SAI}(\varphi; \beta, R) := \frac{\pi(\mathrm{Cl}_R(\beta \setminus \varphi)(\varphi))}{\mathrm{Prior}(\varphi)}.$$

A productive belief system is required (among other things) to regularly generate new high-SAI beliefs or new questions whose answers yield novel patterns, without collapsing coherence.

Remark 3228. *This is recognizably in the lineage of algorithmic and description-length ideas: complexity costs behave like code-lengths or description lengths, and priors like* $e^{-\lambda \mathrm{Comp}}$ *echo standard simplicity biases (cf. algorithmic information theory [16]). The novelty here is not the presence of a simplicity penalty but the way it is made to interact with inference closure* Cl_R *and with explicitly community-relative support.*

To connect this to the inference-centric picture, note that SAI is structurally a ratio of a support-like quantity to a simplicity prior. The numerator $\pi(\mathrm{Cl}_R(\beta \setminus \varphi)(\varphi))$ can be read as the degree to which φ becomes supported after closing a context in which φ has been hypothetically removed (or held out) from β, using a rule set R; in other words, it aims to isolate whether φ is redundant (derivable anyway) or whether it is a genuinely "productive" addition that changes what the system can infer. The denominator $\mathrm{Prior}(\varphi)$ then normalizes by simplicity, so that a complex φ must earn correspondingly higher support to achieve the same SAI. Under a ratio-quantale reading, this normalization already resembles a residual: it is a multiplicative comparison between an achieved evaluation and a baseline.

One can also interpret SAI as implementing a specific orientation convention for surprise: holding complexity fixed, higher $\pi(\cdot)$ increases SAI; holding support fixed, higher complexity decreases SAI. This makes explicit that Hyperseed's "surprisingness" is not only about inferential movement across budgets, but also about a tradeoff between inferential payoff and representational cost.

Hyperseed: science and thinking. Hyperseed models science as a community belief system with explicit empirical coupling: questions are experiments, answers are shared observations, and the update step is $\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o})$ for an evidence increment $\varepsilon_{e,o}$. It also states an operational view of cognition: a system "knows" to the extent it can answer questions and "thinks" to the extent it can generate/select/refine questions, answers, and beliefs.

Remark 3229. *The conceptual move is to treat science not primarily as a list of doctrines, but as a disciplined variant of the question–answer loop (Hyperseed-Concept 144, 56) implemented by a community (Hyperseed-Concept 170) in contact with empirical reality (Hyperseed-Concept* **??***). This is also consonant with the engineering orientation in [19], where cognitive dynamics are characterized by iterative cycles of hypothesis formation, evaluation, and revision rather than by static representations.*

The update $\beta \leftarrow \mathrm{Cl}_R(\beta \oplus \varepsilon_{e,o})$ can be viewed as a two-stage process: first, $\oplus$ injects a new evidence token (indexed by experimental setup e and outcome o) into the current belief state; then Cl_R propagates the consequences of that addition through the inferential fabric determined by R. This makes "empirical coupling" precise in an operational way: the community does not merely store observations, but continuously re-derives what follows from them under its standing inferential commitments.

From the standpoint of budget-indexed surprisingness, one can regard empirical work as supplying new constraints that effectively increase the reachable evaluation of some hypotheses while decreasing others. In that sense, the "budget" is not only internal computation time but also the availability of external informational resources (instruments, protocols, replication capacity), so that moving from B_0 to B_1 may correspond to permitting additional experimental channels as well as additional inference steps.

Weakness-style robustness. In the surprisingness calculus, weakness is explicitly treated as an infimum over a family:

$$w(H) := \bigwedge_{h \in H} \mu(h),$$

with the intended reading "robust worst-case guarantee" (after choosing an orientation convention).

Remark 3230. *This formalizes a familiar scientific scruple: do not celebrate a result that holds only in one laboratory, one dataset, one social subgroup, or one lucky regime. Instead, one seeks claims whose quality persists across a family of contexts. In the quantale setting, the infimum is the canonical notion of "worst case" compatible with compositional algebra; see [3] for a systematic development, and [2] for related motivational discussion of weakness as a design principle for robust cognition.*

The use of $\bigwedge$ is particularly natural when the family H is interpreted as a set of "stress tests" (datasets, environments, submodels, perturbations, or alternative background assumptions), and $\mu(h)$ as a valuation of performance, truthlikeness, or adequacy within that test. Then $w(H)$ forces attention to the minimum-supported member of the family, preventing a single exceptionally favorable context from dominating the assessment. This connects directly to scientific practice: robustness is not merely an aesthetic preference but an explicit aggregation rule for evidence across heterogeneity.

In budget-indexed terms, one may also treat H as indexing multiple inference regimes (or multiple agents) that each yield a $TV_B(\varphi)$-like assessment. Applying weakness before measuring surprise amounts to asking for "surprisingness that survives the whole family," i.e. surprise that is not an artifact of one special evaluator.

A second Hyperseed notion of surprise (compression gain). Hyperseed also defines a "compression surprise" for an encounter B over interval I by

$$\mathrm{Sur}_m(B; I) := \max\{0, L_m^{old}(B) - L_m^{new}(B)\},$$

where L_m is a code-length functional (before/after updating internal codes).

This definition makes explicit that at least one kind of surprise is identified with a learning-driven drop in description length: the encounter is surprising exactly to the extent that it enables a shorter encoding under the updated model class or coding scheme m. The truncation at 0 enforces an orientation convention in which surprise is nonnegative and only counts "compression improvements" rather than degradations, aligning the quantity with a gain interpretation.

A useful bridge to the quantale examples above is to note that code lengths are typically additive across independent concatenations, so differences of code lengths live naturally in an additive (or tropical) setting. If one sets $TV_B(\varphi)$ to be (minus) a description length or a log-score, then the residual $a \Rightarrow b = b - a$ is exactly the type of delta measured by compression gain. On this reading, $\mathrm{Sur}_m(B; I)$ can be interpreted as a clipped version of a residual, where the clipping implements the convention that only positive movements (improvements) are credited as surprising in the intended sense.

Remark 3231. *In plain terms: if, after learning from an encounter, a mind can encode (predict, summarize, or reconstruct) the encounter more succinctly, then the encounter contained a "new pattern" relative to its previous internal model. This resonates with compression-based views of learning and surprise [16], and it provides a second route to connect surprise with resource change: the resource here is representational efficiency rather than inference depth.*

Science as "weakness" across channels. The "What is Science, Exactly?" essay characterizes science as optimizing weakness across evidence, simplicity (paradigm-dependent), and pragmatics, and treats paradigm shifts as shifts in these channels or their weightings.

Remark 3232. *The upshot is that "scientific quality" is not one scalar but a coordination of constraints: do not make distinctions unsupported by evidence, do not multiply primitives without reason, and do not privilege distinctions that do not matter for use. Treating these as weakness channels makes the overall notion explicitly robust (worst-case across channels or across instantiations). This orientation is compatible with the CPP framing developed elsewhere and with the broader social philosophy of science perspective in [20].*

59.2 Formal setup

Definition 529 (Budgets and monotone evaluation)**.** *Let $(\mathcal{B}, \leq)$ be a poset of inference budgets. Let $\mathcal{L}$ be a language of statements (hypotheses, claims, patterns). Let $(Q, \leq, \otimes, \oplus, 1, 0, \Rightarrow)$ be a commutative residuated quantale. A budgeted evaluator is a map*

$$TV : \mathcal{B} \times \mathcal{L} \to Q, \quad (B, \varphi) \mapsto TV_B(\varphi),$$

that is monotone in budget: $B_0 \leq B_1 \Rightarrow TV_{B_0}(\varphi) \leq TV_{B_1}(\varphi)$.

Remark 3233. *Intuitively, $\mathcal{B}$ is whatever you use to measure "how hard you are allowed to think." It could be a natural number (maximum proof depth), a time limit, a memory cap, a bound on the number of rule applications, a bound on search nodes, or even a partially ordered set of different inference regimes (e.g. "use propositional rules" $\leq$ "use first-order rules" $\leq$ "use probabilistic abduction"). The only property used here is monotonicity: more budget should not make evaluation worse.*

Remark 3234. *The quantale structure packages the algebra needed for compositional reasoning about these evaluations. The symbol $\otimes$ is the monoidal "combination" (e.g. multiplication in $Q_\times$ or addition in Q_+); $\oplus$ is the join-like aggregator (here* max *in both standard examples);* 1 *and* 0 *are the unit and bottom; and the residuum $\Rightarrow$ is the implication-like operation satisfying the usual adjunction: $a \otimes w \leq b$ iff $w \leq (a \Rightarrow b)$. This residuation is what makes "difference" compositional rather than merely numerical.*

Remark 3235. *A simple example: take $Q = [0, 1]$ with $\otimes = \times$ and $\oplus = \max$, and let $TV_B(\varphi)$ be the best plausibility score achievable for φ by any proof/search of cost $\leq B$. Then monotonicity says: if you search longer, you can keep the best score you already found, so the best-achieved score cannot decrease.*

Definition 530 (Residual-form budget surprise)**.** *For $B_0 \leq B_1$, define the (one-sided) budget surprise of φ by*

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) := TV_{B_0}(\varphi) \Rightarrow TV_{B_1}(\varphi).$$

Remark 3236. *This definition says: surprise is the "extra factor" (or "extra additive amount," depending on the quantale) needed to turn the weaker evaluation into the stronger one. In $Q_\times$, the residuum is a ratio b/a, so surprise is "how many times larger." In Q_+, the residuum is a difference $b - a$, so surprise is "how many units of log-score were gained." Both are just different lenses on the same compositional idea.*

Remark 3237. *A toy example in the ratio case: suppose with budget B_0 you can only check a crude heuristic and get $TV_{B_0}(\varphi) = 0.01$, but with budget B_1 you can run a full verifier and get $TV_{B_1}(\varphi) = 0.5$. Then $\mathrm{Surp}_{B_0 \to B_1}(\varphi) = 0.5/0.01 = 50$. The claim is not surprising because it became absolutely certain, but because it became dramatically more supported once the extra inference was permitted.*

Definition 531 (Weakness of a family)**.** *Let H be a set of contexts (agents, experiments, datasets, regimes, or pattern-templates), and let $\mu : H \to Q$ be a quantale-valued score. Define the weakness of the family by*

$$w(H) := \bigwedge_{h \in H} \mu(h).$$

Remark 3238. *This is the mathematically austere way to say: "take the worst case over H." The infimum $\bigwedge$ is the greatest lower bound in the order $\leq$ on Q, so it selects the strongest guarantee that holds uniformly across all $h \in H$. In many computational settings H is large or implicit; then the interest is not only definitional but algorithmic: can we approximate or bound the infimum cheaply? This is one of the motivations for quantale weakness as a reusable operator [3].*

Remark 3239. *Concrete example: let $Q = [0, \infty]$ with the usual order, and let $\mu(h)$ be a measured surprise score for the same hypothesis φ when tested in lab h. Then $w(H)$ is the least surprise observed across labs. If one wants "robust surprise" (not a fluke), then this infimum is the natural gate.*

Definition 532 (Hyperseed SAI reparameterized as a budgeted evaluation)**.** *Fix Hyperseed data (β, R) and define*

$$\mathrm{Support}(\varphi; \beta, R) := \pi(\mathrm{Cl}_R(\beta \setminus \varphi)(\varphi)) \in [0, 1], \quad \mathrm{Prior}(\varphi) := e^{-\lambda \mathrm{Comp}(\varphi)} \in (0, 1].$$

Then Hyperseed's surprise-adjusted implication is

$$\mathrm{SAI}(\varphi; \beta, R) := \frac{\mathrm{Support}(\varphi; \beta, R)}{\mathrm{Prior}(\varphi)}.$$

Remark 3240. *The notation here compresses several ideas. The belief state β is a valuation over the language, R is a set of inference rules, and Cl_R is closure under those rules (the least fixed point above the given beliefs, as in the general Hyperseed science machinery). The expression $\beta \setminus \varphi$ denotes a belief state where φ is "removed" (so that support for φ is not trivially circular); and $\pi(\cdot)$ is a projection that returns a scalar support in $[0, 1]$ from the underlying belief value. The quotient by $\mathrm{Prior}(\varphi)$ normalizes support by simplicity, producing a "surprise given complexity" score [1].*

Remark 3241. *Intuitively: a claim deserves attention when it is (i) strongly supported by the community's closure process and (ii) not strongly expected merely because it is simple. A simple but weakly supported claim is unsurprising; a complex but strongly supported claim can be surprising. In the background of this construction is the idea that a mind's (or a community's) limited resources force it to privilege simplicity unless evidence and inference compel otherwise (compare [16] for the code-length analogy).*

59.3 Theorems: bridging Hyperseed SAI to the compositional calculus

59.3.1 SAI is an instance of budget-residual surprisingness

Theorem 148 (SAI = ratio-quantale residual surprise)**.** *Let $Q = Q_\times$ be the ratio quantale with residual $a \Rightarrow b = b/a$. Define two budgets $B_0 \leq B_1$ and an evaluator by*

$$TV_{B_0}(\varphi) := \mathrm{Prior}(\varphi), \qquad TV_{B_1}(\varphi) := \mathrm{Support}(\varphi; \beta, R).$$

Then

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) = \mathrm{SAI}(\varphi; \beta, R).$$

Remark 3242. *This theorem says that Hyperseed's SAI is not an ad hoc ratio; it is exactly the residuated "delta" between two evaluations when those evaluations are placed into the ratio quantale. One may read B_0 as a weak baseline regime that only "knows" how to assign simplicity-based expectation, and B_1 as a stronger regime that incorporates inferential closure and evidence-based support. The surprise is the multiplicative lift from baseline to supported status.*

Remark 3243. *The result matters because once SAI is seen as a residual in a quantale, it automatically inherits a family of structural laws: telescoping across multiple budgets, compositional behavior under factorizations/independence assumptions, and robustness via infima. These are exactly the kinds of identities one wants inside a reasoning loop: they allow surprise to be tracked, bounded, and aggregated without re-deriving everything from scratch at each step [3].*

Proof. By definition of residual-form budget surprise,

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) = TV_{B_0}(\varphi) \Rightarrow TV_{B_1}(\varphi).$$

In the ratio quantale, $a \Rightarrow b = b/a$. Hence

$$\mathrm{Surp}_{B_0 \to B_1}(\varphi) = \frac{TV_{B_1}(\varphi)}{TV_{B_0}(\varphi)} = \frac{\mathrm{Support}(\varphi; \beta, R)}{\mathrm{Prior}(\varphi)} = \mathrm{SAI}(\varphi; \beta, R),$$

as claimed. $\square$

Proof sketch. The strategy is purely interpretive: we choose the two evaluations (TV_{B_0} and TV_{B_1}) so that their ratio is exactly the SAI definition. The key step is the ratio-quantale identity $(a \Rightarrow b) = b/a$, which turns the abstract residual-form surprise into an explicit quotient. Geometrically, one can picture $TV_{B_0}(\varphi)$ and $TV_{B_1}(\varphi)$ as two points on the positive real line, and surprise as the scale factor that dilates the first point into the second. $\square$

Remark 3244 (Interpretation)**.** *Hyperseed's "surprising given complexity" score is literally a budget-delta surprise: B_0 is a baseline evaluator that only "pays attention" to simplicity (via a prior), while B_1 is a stronger evaluator that permits the community's inference closure to support φ. This is aligned with the inference-centric idea that a statement is surprising when its truth estimate changes substantially when more inference resources are permitted.*

59.3.2 Log-domain (tropical) form: SAI as an additive surprise

Corollary 56 (Log SAI = tropical residual surprise)**.** *Let $Q = Q_+$ be the tropical (max-plus) quantale with residual $a \Rightarrow b = b - a$. Assume* $\mathrm{Support}(\varphi; \beta, R) > 0$ *and* $\mathrm{Prior}(\varphi) > 0$*, and define*

$$\widetilde{TV}_{B_0}(\varphi) := \log \mathrm{Prior}(\varphi), \qquad \widetilde{TV}_{B_1}(\varphi) := \log \mathrm{Support}(\varphi; \beta, R).$$

Then the tropical residual surprise satisfies

$$\widetilde{\mathrm{Surp}}_{B_0 \to B_1}(\varphi) = \log \mathrm{SAI}(\varphi; \beta, R) = \log \mathrm{Support}(\varphi; \beta, R) + \lambda \mathrm{Comp}(\varphi).$$

Remark 3245. *In words: taking logs converts the multiplicative SAI into an additive score that splits cleanly into "log support" plus "complexity penalty." This is important because many search and optimization procedures are naturally additive: they accumulate costs and gains along a path. In that setting,* $\log \mathrm{SAI}$ *becomes a directly usable objective for guiding attention (Hyperseed-Concept 60) toward claims that are both evidentially supported and nontrivially structured.*

Proof. In Q_+, $a \Rightarrow b = b - a$. Thus

$$\widetilde{\mathrm{Surp}}_{B_0\to B_1}(\varphi) = \widetilde{TV}_{B_0}(\varphi) \Rightarrow \widetilde{TV}_{B_1}(\varphi) = \log \mathrm{Support}(\varphi;\beta,R) - \log \mathrm{Prior}(\varphi) = \log\left(\frac{\mathrm{Support}}{\mathrm{Prior}}\right) = \log \mathrm{SAI}.$$

Since $\mathrm{Prior}(\varphi) = e^{-\lambda \mathrm{Comp}(\varphi)}$, $-\log \mathrm{Prior}(\varphi) = \lambda \mathrm{Comp}(\varphi)$, giving the final equality. □

Proof sketch. The proof is a coordinate change. One moves from the multiplicative world to the additive world by applying log, and then uses the tropical residuum formula $b - a$. The only substantive point is that the log of the exponential prior becomes a linear term $\lambda\mathrm{Comp}(\varphi)$, making the tradeoff between support and complexity explicit and additive. □

Remark 3246 (Why this is useful for a reasoning system)**.** *In tropical form, the SAI decomposes additively into an "evidence-support" term plus an explicit "complexity" term. This makes it natural to treat surprise-seeking as optimizing a sum of log-evidence and complexity cost, consistent with many search-control heuristics that trade off fit and description length.*

59.3.3 Budget composition: telescoping laws become reasoning-loop invariants

Theorem 149 (Telescoping for ratio and tropical surprises)**.** *Fix a statement* φ *and budgets* $B_0 \le B_1 \le B_2$*. Assume* $TV_{B_i}(\varphi) > 0$ *in the ratio case. The monotonicity condition* $B_0 \le B_1 \le B_2$ *is only used to match the intended "increasing budget" interpretation; algebraically, the identities are statements about composing two consecutive updates through an intermediate stage. The positivity assumption in the ratio case ensures that each ratio is well-defined and lives in the multiplicative quantale carrier (so that* $\otimes = \times$ *is applicable without invoking extended values).*

1. *(Ratio form.) In* $Q_\times$*,*

$$\mathrm{Surp}_{B_0\to B_2}(\varphi) = \mathrm{Surp}_{B_0\to B_1}(\varphi) \otimes \mathrm{Surp}_{B_1\to B_2}(\varphi) \quad (\otimes = \times).$$

   Equivalently, the map $B \mapsto TV_B(\varphi)$ *is multiplicatively "path independent": the surprise depends only on endpoints once expressed as a ratio, and intermediate checkpoints compose by multiplication.*

2. *(Tropical form.) In* Q_+*,*

$$\mathrm{Surp}_{B_0\to B_2}(\varphi) = \mathrm{Surp}_{B_0\to B_1}(\varphi) \otimes \mathrm{Surp}_{B_1\to B_2}(\varphi) \quad (\otimes = +).$$

   Equivalently, the map $B \mapsto TV_B(\varphi)$ *is additively "path independent": the net change from* B_0 *to* B_2 *is the sum of the two consecutive changes via* B_1*.*

Remark 3247. *The statement is an accounting identity: the surprise from* B_0 *to* B_2 *factors through* B_1*. In the ratio case this is the familiar multiplicative cancellation of intermediate terms; in the tropical case it is the additivity of differences. Conceptually, this makes "incremental reasoning" compatible with "batch reasoning": you can safely accumulate surprise across stages, and the total does not depend on how you partition the budget increase into steps. A useful way to read this* *is that surprise behaves like a compositional "cost" (or "gain") along a chain of refinements: the algebra enforces that the cost of a composed refinement is exactly the quantale product of the costs of its pieces. In particular, any control policy that depends only on total surprise (e.g. stopping when surprise exceeds a threshold, or prioritizing branches by accumulated surprise) is invariant under the granularity at which one logs intermediate stages.*

Remark 3248. *This connects back to Theorem 1 because if SAI is a residual surprise between two evaluators, then multi-stage refinements of the evaluator (e.g. progressive strengthening of* Cl_R *or progressive widening of what counts as admissible inference) yield telescoping invariants. This is precisely the kind of invariance that makes surprise a stable control signal in long reasoning loops. Concretely, if* $E_0 \preceq E_1 \preceq E_2$ *are evaluators ordered by refinement (for instance, larger budgets, stronger closure operators, or more permissive admissibility criteria), then the same telescoping expresses that the "gap" from* E_0 *to* E_2 *can be decomposed without ambiguity into the gap from* E_0 *to* E_1 *and from* E_1 *to* E_2*. This is the compositionality property one needs when SAI is used not merely as a diagnostic number, but as a quantity that is repeatedly combined across subroutines, steps, or episodes of a reasoning process.*

Proof. (1) In the ratio quantale,

$$\mathrm{Surp}_{B_0\to B_2}(\varphi) = \frac{TV_{B_2}(\varphi)}{TV_{B_0}(\varphi)} = \frac{TV_{B_1}(\varphi)}{TV_{B_0}(\varphi)} \cdot \frac{TV_{B_2}(\varphi)}{TV_{B_1}(\varphi)} = \mathrm{Surp}_{B_0\to B_1}(\varphi) \cdot \mathrm{Surp}_{B_1\to B_2}(\varphi).$$

Here the factor $TV_{B_1}(\varphi)$ is introduced and then cancels; the assumption $TV_{B_i}(\varphi) > 0$ guarantees that no division-by-zero occurs and that the multiplication stays within the intended carrier of $Q_\times$.
(2) In the tropical quantale,

$$\mathrm{Surp}_{B_0\to B_2}(\varphi) = TV_{B_2}(\varphi) - TV_{B_0}(\varphi) = (TV_{B_1} - TV_{B_0}) + (TV_{B_2} - TV_{B_1}) = \mathrm{Surp}_{B_0\to B_1}(\varphi) + \mathrm{Surp}_{B_1\to B_2}(\varphi).$$

This is the standard telescoping decomposition of a difference into successive differences; no positivity is needed because subtraction is performed only in the ambient ordered group/semiring where the tropical quantale is interpreted. □

Proof sketch. Both parts use the same high-level move: introduce the intermediate budget B_1 and cancel it. In $Q_\times$ this is cancellation of a shared factor ($TV_{B_1}(\varphi)$); in Q_+ it is cancellation of a shared summand. The geometric intuition is that surprise measures displacement along a 1D axis (multiplicatively or additively), and displacements add when traversed in segments. From the quantale perspective, the identity is also a direct reflection of associativity of $\otimes$: composing the update $B_0 \to B_1$ with $B_1 \to B_2$ yields the same result as the single update $B_0 \to B_2$, so the "surprise annotation" is functorial with respect to concatenation of budget steps. □

Remark 3249 (Loop interpretation)**.** *If a reasoning system iteratively expands its inference budget (or applies progressively stronger inference regimes), then the total surprise from start to finish equals the product/sum of incremental surprises across steps. This yields a clean accounting identity for surprise-driven control: you can track (and threshold) cumulative novelty across multi-step reasoning loops. In practice, this means one may implement surprise tracking either by storing only the current* $TV_B(\varphi)$ *and comparing to the initial baseline, or by accumulating per-step surprises online; telescoping guarantees these two implementations agree. More generally, for any chain* $B_0 \le B_1 \le \cdots \le B_n$*, repeated application yields*

$$\mathrm{Surp}_{B_0\to B_n}(\varphi) = \bigotimes_{k=0}^{n-1} \mathrm{Surp}_{B_k\to B_{k+1}}(\varphi),$$

so long reasoning loops admit a canonical factorization into local increments that is independent of how finely the loop is instrumented.

59.4 Weakness/robustness meets surprise: guarantees across contexts

59.4.1 Robust surprise across experiments / contexts

Definition 533 (Context-indexed surprise and its robustness)**.** *Let H be a set of contexts (e.g. experiments, labs, observer states, social subgroups, datasets). Let $\mathrm{Surp}_h(\varphi) \in Q$ be a surprise score for statement φ in context h. Define the robust (weakness-style) surprise guarantee by*

$$\mathrm{RobSurp}_H(\varphi) := \bigwedge_{h \in H} \mathrm{Surp}_h(\varphi).$$

This is exactly the weakness-of-a-family construction applied to surprise scores.

Remark 3250. *This definition turns surprise into a kind of contract: it is not enough that φ was surprising once; it should remain surprising across a designated family H of contexts. The family H may represent experimental replications, alternative datasets, different subcultures, or simply different initial belief states. The weakness infimum is the formal mechanism by which "robustness" enters the picture (Hyperseed-Concept 202, 143).*

Remark 3251. *A simple example: if $H = \{h_1, h_2, h_3\}$ are three labs, and $\mathrm{Surp}_{h_i}(\varphi)$ measures how much better φ becomes supported after allowing a richer analysis pipeline, then $\mathrm{RobSurp}_H(\varphi)$ is the smallest such improvement among the three. This matches the scientific instinct: the least impressive replication constrains the credibility of the overall claim.*

Proposition 78 (Monotonicity and threshold semantics of robust surprise)**.** *If $H' \subseteq H$, then $\mathrm{RobSurp}_H(\varphi) \leq \mathrm{RobSurp}_{H'}(\varphi)$. Moreover, if $\mathrm{Surp}_h(\varphi) \geq \tau$ for all $h \in H$, then $\mathrm{RobSurp}_H(\varphi) \geq \tau$.*

Remark 3252. *The proposition states two elementary but crucial facts. First, robustness gets harder when you demand it across more contexts: adding more tests can only decrease (or leave unchanged) the worst-case guarantee. Second, robust surprise supports threshold reasoning: if every context clears a threshold, then the robust infimum clears it too. This is the formal justification for using robust surprise as a gate in automated reasoning systems: it behaves predictably when you enlarge your test suite.*

Proof. Since H has more elements than H', the infimum over H is $\leq$ the infimum over H'. For the threshold claim: if every element of the set $\{\mathrm{Surp}_h(\varphi) : h \in H\}$ is $\geq \tau$, then their infimum is also $\geq \tau$. □

Proof sketch. Both claims are properties of infima in ordered sets. The first is the monotonicity of "minimum" with respect to set inclusion: the more elements you include, the more constraints you impose, so the greatest lower bound can only go down. The second is the defining property of a lower bound: if τ is below every element, then it is below their greatest lower bound as well. □

59.4.2 Independence and robust factorization

Theorem 150 (Independence factorization: surprise of a conjunction is the product of surprises)**.** *Let $Q = Q_\times$. Suppose two statements α, β satisfy the following factorization property in a fixed context h for two budgets $B_0 \leq B_1$:*

$$TV^{(h)}_{B_i}(\alpha \wedge \beta) = TV^{(h)}_{B_i}(\alpha) \cdot TV^{(h)}_{B_i}(\beta), \quad i \in \{0, 1\}.$$

Then

$$\mathrm{Surp}^{(h)}_{B_0 \to B_1}(\alpha \wedge \beta) = \mathrm{Surp}^{(h)}_{B_0 \to B_1}(\alpha) \cdot \mathrm{Surp}^{(h)}_{B_0 \to B_1}(\beta).$$

Remark 3253. *This theorem formalizes a simple idea: if the evaluator treats α and β as independent factors both before and after the budget increase, then the "extra lift" due to more budget also separates. In other words, independence is not only a statement about scores; it is a statement about how novelty decomposes. This is important because independence (or approximate independence) is what permits modular science and modular cognition: one can accumulate surprises from separate subproblems without double-counting entanglement.*

Remark 3254. *The hypothesis is deliberately strong: it requires factorization at both budgets. This matches the intended use: when one adopts a factor-graph or modular representation, one is committing to a factorization not just of a final judgment but of the evaluation process itself. Later sections on dependency and modularity make this kind of assumption structurally explicit.*

Proof. By ratio-residual surprise,

$$\mathrm{Surp}^{(h)}_{B_0\to B_1}(\alpha\wedge\beta)=\frac{TV^{(h)}_{B_1}(\alpha\wedge\beta)}{TV^{(h)}_{B_0}(\alpha\wedge\beta)}=\frac{TV^{(h)}_{B_1}(\alpha)TV^{(h)}_{B_1}(\beta)}{TV^{(h)}_{B_0}(\alpha)TV^{(h)}_{B_0}(\beta)}=\frac{TV^{(h)}_{B_1}(\alpha)}{TV^{(h)}_{B_0}(\alpha)}\cdot\frac{TV^{(h)}_{B_1}(\beta)}{TV^{(h)}_{B_0}(\beta)}.$$

□

Proof sketch. The proof substitutes the assumed factorization at B_0 and B_1 into the definition of ratio surprise and then cancels terms. The key step is that products in numerator and denominator separate into ratios for each factor. Visually: if evaluations lie on multiplicative axes, then moving from B_0 to B_1 scales α by one factor and β by another, so their conjunction scales by the product of those scale factors. □

Corollary 57 (Robust independence lower bound)**.** *Assume the hypotheses of the previous theorem hold for every $h\in H$. Then*

$$\mathrm{RobSurp}_H(\alpha\wedge\beta)\ \geq\ \mathrm{RobSurp}_H(\alpha)\cdot\mathrm{RobSurp}_H(\beta).$$

Remark 3255. *This corollary says that robustness and compositionality cooperate: worst-case surprise of a combined claim is at least the product of the worst-case surprises of the parts, provided the factorization assumptions hold uniformly across contexts. The inequality (rather than equality) reflects a general principle: taking infima after multiplying can only lose information; nevertheless, one retains a usable lower bound, which is exactly what a conservative reasoning system wants for guarantees. In particular, the "uniformly across contexts" clause matters: the same context index h must be eligible simultaneously for α, for β, and for $\alpha\wedge\beta$, so that the comparison is made inside a single common family $\{h\}$ rather than across incomparable families. Said another way, we are not free to choose* different *worst-case contexts for α and β and then expect that pair to be realized by a single context for the conjunction; the corollary instead shows that even without such alignment one still gets a sound (possibly non-tight) lower bound. The proof also highlights the order-theoretic shape of the argument: infima in $[0,\infty]$ interact well with monotone operations, which is the basic mechanism behind robust, context-quantified guarantees.*

Proof. Let $a_h:=\mathrm{Surp}^{(h)}_{B_0\to B_1}(\alpha)$ and $b_h:=\mathrm{Surp}^{(h)}_{B_0\to B_1}(\beta)$. From the theorem, $\mathrm{Surp}^{(h)}_{B_0\to B_1}(\alpha\wedge\beta)=a_hb_h$. Let $a_*:=\bigwedge_h a_h$ and $b_*:=\bigwedge_h b_h$. Then $a_*\leq a_h$ and $b_*\leq b_h$ for all h, so $a_*b_*\leq a_hb_h$ for all h by monotonicity of multiplication on $[0,\infty]$. Taking infima gives $a_*b_*\leq\bigwedge_h(a_hb_h)$, i.e. the stated inequality. Note that no attainment of the infimum is required: even when $\bigwedge_h a_h$ or $\bigwedge_h b_h$ is not achieved by any particular h (so there is no literal "worst" context), the lattice inequalities $a_*\leq a_h$ and $b_*\leq b_h$ still hold by definition of $\bigwedge$. Also, because the codomain is $[0,\infty]$ with its usual order and extended multiplication, the monotonicity step is valid even in edge cases (e.g. when some values are 0 or ∞), so the argument remains purely order-theoretic rather than measure-theoretic or probabilistic. $\square$

Proof sketch. The proof is a standard "push the infimum through a monotone operation" argument. One first notes that $a_* = \inf_h a_h$ is below every a_h, and similarly for b_*. Multiplying preserves order, so $a_* b_*$ is below every product $a_h b_h$. Therefore $a_* b_*$ is below their infimum, which is exactly $\mathrm{RobSurp}_H(\alpha \wedge \beta)$. The mental picture is that the worst-case combination cannot be worse than combining the two separate worst cases, because those separate worst cases already occur within the set of contexts being minimized over. Equivalently, the operator $x \mapsto cx$ (for fixed $c \in [0, \infty]$) preserves all infima because it is monotone, so one gets a "lax interchange" between inf and multiplication: $\inf_h(a_h b_h)$ dominates $(\inf_h a_h)(\inf_h b_h)$. The reason equality need not hold is that the infimum of the products is achieved (or approximated) by whichever h makes $a_h b_h$ small, and that h need not be a context where *both* a_h and b_h are near their individual infima simultaneously; the two "bad" directions can occur in different contexts. Thus the bound should be read as a robust compositional *guarantee* rather than an exact characterization, matching the conservative stance of worst-case reasoning. $\square$

Remark 3256 (Reasoning-system use)**.** *This corollary gives a robust, compositional certificate: if each component is robustly surprising across contexts, their conjunction remains robustly surprising (at least up to the product lower bound), provided the system's factorization/independence assumptions are justified. Operationally, this lets a reasoning system propagate context-robust lower bounds without re-solving a global optimization problem for every conjunction: one can track* $\mathrm{RobSurp}_H(\alpha)$ *and* $\mathrm{RobSurp}_H(\beta)$ *as reusable summaries, and then immediately obtain a sound bound for* $\alpha \wedge \beta$*. In log-surprise form (when one works with* $-\log(\mathrm{Surp})$ *or similar), the multiplicative lower bound corresponds to an additive lower bound, which is often numerically stable and aligns with familiar "add evidence" heuristics. The caveat is exactly the one stated in the corollary: if factorization fails in some contexts, the product rule can overstate robustness, so the certificate is only as reliable as the model class that enforces the uniform assumptions.*

59.5 Scientists and curious minds as surprise-seekers (formalized inside Hyperseed)

Definition 534 (Curious mind and scientist (minimal formalization))**.**

1. *A curious mind is an agent whose internal epistemic dynamics instantiate the Hyperseed thinking loop (questions -¿ answers -¿ belief updates), and whose belief system is productive in the sense that it regularly generates high-SAI beliefs or high-novelty questions.*

2. *A scientific community is a community belief system with an observation set and empirical update operator as in Hyperseed's science formalization.*

Remark 3257. *This definition is intentionally minimalist: it does not legislate what the agent is made of, nor how it represents beliefs, but only insists on the dynamical pattern of inquiry (Hyperseed-Concept* **??***) and on a productivity condition. The scientific community is then obtained by adding explicit empirical coupling and communal acceptance criteria, i.e. by turning the private loop into a shared, socially stabilized loop (cf. [20]).*

Remark 3258. *The minimalism is also methodological: it allows the same schema to cover human scientists, automated agents, mixed human–machine teams, or even abstract "idealized" agents, so long as the requisite loop (questioning, answering, updating) can be identified. In particular, the* *definition does not commit to any specific representational format (probabilistic credences, logical theories, vector embeddings, or hybrid formalisms), because the Hyperseed ontology treats the belief dynamics and the update/closure machinery as the primary invariants.*

Remark 3259. *Note that "productive" here is meant as a structural property of the belief system* (B, R) *(and its associated closure/inference machinery), not as a psychological label. It formalizes a repeatable tendency to generate outputs that are nontrivial relative to the system's own baselines of simplicity, prior beliefs, and permitted update operators. In this way, the definition distinguishes mere activity (many questions, many hypotheses) from epistemic productivity (questions/hypotheses that actually move the state of belief in a way that survives coherence constraints).*

Remark 3260. *A simple example of a "curious mind" in this sense could be a single agent repeatedly generating hypotheses, running tests, and updating beliefs, where the tests are internal simulations or external measurements. A "scientific community" is the same pattern with replicated experiments, shared observations, and an update protocol that ensures that what is learned can be re-used and criticized.*

Remark 3261. *The community case can be read as an explicit externalization of parts of the thinking loop: questions and candidate answers become public objects (papers, preregistrations, datasets, lab notebooks), while belief updates become partially governed by agreed-upon procedures (statistical thresholds, peer review norms, artifact sharing, and standardized measurement). In Hyperseed terms, this amounts to tightening the coupling between the observation set and the communal acceptance/update operator so that novelty is not merely generated but also filtered, stabilized, and made portable across agents.*

Theorem 151 (Productive curiosity and productive science are both (among other goals) high-SAI generators)**.** *Let* (B, R) *be a Hyperseed belief system (for a community, or for a single agent viewed as a size-1 community). If* (B, R) *is productive, then it regularly generates either: (i) new beliefs* φ *with large* $\mathrm{SAI}(\varphi; \beta, R)$*, or (ii) new questions whose answers yield novel patterns (and hence, after closure, candidate beliefs with elevated SAI). In particular, under this definition, both a productive curious mind and a productive scientific community are engaged in generating surprising conclusions/hypotheses.*

Remark 3262. *The dependence* $\mathrm{SAI}(\varphi; \beta, R)$ *is essential to the intended reading: the same statement* φ *may be unsurprising for one baseline* β *(or one repertoire of inferential resources* R*) and highly surprising for another. Thus "surprise-seeking" is not a purely intrinsic property of a proposition, but a relational property: it measures how much the epistemic evaluation of* φ *changes when one moves from a simplicity-only or low-resource baseline to an inference-augmented or closure-supported regime. This is precisely why the same empirical finding can be "obvious" to an expert community while remaining surprising to a novice agent, even if both are perfectly coherent.*

Remark 3263. *The role of "new questions" in clause (ii) is not merely rhetorical. Within Hyperseed, a question can be treated as an operator that induces a search over potential answers and thereby induces a change in the effective hypothesis space. When that search yields a pattern not already captured by the current closure of* B*, the subsequent closure step promotes parts of that pattern into candidate beliefs. In this sense, productive questioning is a mechanism for generating high-SAI candidates indirectly, by forcing the system to confront regions of the space where its current baselines and inference budgets do not already trivialize the outcome.*

Remark 3264. *The theorem is logically straightforward but conceptually clarifying: productivity, as defined in Hyperseed, already encodes a drive toward nontrivial novelty under coherence constraints.*

Thus, the difference between a scientist and an ordinary curious thinker is not the presence versus absence of surprise-seeking, but the additional social and empirical scaffolding that makes surprise reproducible, criticizable, and cumulative. In that sense, the surprise calculus is not merely a scoring trick but a candidate invariant of inquiry as such.

Remark 3265. *One can also read the theorem as a constraint on degenerate epistemic dynamics. A system that updates only by trivial restatement, or that generates hypotheses but immediately discards them without any integrative closure, will fail the productivity requirement and therefore will not qualify as "curious" in the present formal sense. Conversely, a system that chases novelty without coherence will tend to collapse into inconsistency or into an unstructured accumulation, and so will again fail to meet the "without collapsing coherence" aspect implicitly encoded in the productivity condition.*

Proof. This follows by unpacking the definition of productive belief system, which explicitly requires that the system regularly generates new high-SAI beliefs or new questions whose answers lead to novel patterns, without collapsing coherence. □

Remark 3266. *Although the proof is a direct unfolding, the statement is not vacuous: it fixes how "curiosity" and "science" are to be identified* inside *the ontology, namely via behavior of the belief dynamics relative to β and R. In particular, once a concrete β (simplicity baseline) and a concrete inference/closure regime R are specified, the claim becomes testable in principle: one can observe whether the system repeatedly produces beliefs/questions that score as high-SAI relative to those internal standards.*

Proof sketch. No algebra is needed: the proof is definition unfolding. The value of stating it as a theorem is that it makes explicit what is often implicit in cognitive and scientific rhetoric: "curiosity" is a constraint on the dynamics of belief formation, not merely a mood. Once one then identifies SAI with a residual surprise (Theorem 1), the definition acquires operational content: one can compute, track, and optimize the relevant form of surprise. □

Remark 3267. *The phrase "compute, track, and optimize" should be read broadly: in an automated setting it may literally correspond to numerical optimization over candidate hypotheses or experiments, whereas in a human or mixed setting it may correspond to heuristic search guided by proxies (e.g. information gain, predictive error, compression improvements) that approximate the relevant SAI functional. The key Hyperseed point is that such proxies are not arbitrary: they are interpretable as approximations to changes in evaluation between a baseline regime and an inference-supported regime.*

Remark 3268 (Where the quantale calculus enters)**.** *The previous theorem is definition-unfolding, but it becomes operationally nontrivial once we instantiate "high-SAI" via Theorem 1: high SAI is high residual surprise between a simplicity-only baseline and an inference/closure-supported evaluation. Thus curiosity and science can be seen as special cases of inference-centric surprise seeking: seeking statements whose evaluation changes substantially under increased inference resources (budgets).*

Remark 3269. *In the quantale-compositional picture, "budgets" can be modeled as elements in a resource quantale whose order encodes refinement/availability (more time, more compute, more experimental access, richer background theory), and whose monoidal structure encodes composition of resources across steps of the thinking loop. The operational reading of the last sentence is then that productive inquiry tends to select φ for which the mapping from resources to evaluation (or to closure strength) has a large marginal effect: increasing resources moves φ from "implausible/unsupported" toward "supported" (or vice versa) in a way that is not already predicted by the simplicity baseline.*

59.6 Paradigm shifts as changes of surprise baselines (and why independence matters)

Theorem 152 (Paradigm-shift renormalization of surprise scores)**.** *Let* $\mathrm{Support}(\varphi)$ *be fixed (same inferential support/evidence handling), and consider two paradigms with complexity costs* $\mathrm{Comp}_1, \mathrm{Comp}_2$ *and priors* $\mathrm{Prior}_i(\varphi) = e^{-\lambda \mathrm{Comp}_i(\varphi)}$*. Then the corresponding SAI scores satisfy*

$$\mathrm{SAI}_2(\varphi) = \mathrm{SAI}_1(\varphi) \cdot \exp\big(\lambda(\mathrm{Comp}_2(\varphi) - \mathrm{Comp}_1(\varphi))\big).$$

Hence a change in what counts as "simple" (a paradigm shift in the simplicity channel) can change the ranking of "surprising hypotheses" even if evidential support is unchanged. Moreover, taking logs makes the renormalization additive:

$$\log \mathrm{SAI}_2(\varphi) = \log \mathrm{SAI}_1(\varphi) + \lambda\big(\mathrm{Comp}_2(\varphi) - \mathrm{Comp}_1(\varphi)\big),$$

so a paradigm shift can be read as adding a hypothesis-dependent offset to the surprise baseline while leaving the support term untouched.

Remark 3270. *The theorem states a renormalization law: changing the complexity measure rescales surprise by a predictable factor. This is important because it isolates one precise mechanism by which paradigms can be "incommensurable" in practice: not because facts disappear, but because the baseline of what is considered expected (simple, natural, unremarkable) shifts. Then even identical support can yield different surprise rankings, hence different attention allocation and research trajectories. In particular, if two hypotheses* φ, ψ *have comparable support but differ in how much their complexity shifts between paradigms, then the ordering by* SAI *can flip even when* $\mathrm{Support}(\varphi)/\mathrm{Support}(\psi)$ *is fixed. This makes explicit that "surprise" is not only an evidence score but an evidence-to-expectation ratio, and expectation is precisely where paradigms encode their background commitments.*

Remark 3271. *This connects to the weakness-channel view of science: if simplicity is one channel of weakness, then a paradigm shift changes the simplicity channel and therefore changes the multi-channel optimum. The formula above is the one-line quantitative shadow of that broader idea [20]. Equivalently, in a multi-channel objective where overall weakness aggregates several penalties, altering one channel can reweight tradeoffs among channels even when other channels (e.g. data-fit or measurement reliability) are held constant. Thus, what looks like "the same evidence" can lead to a different best next question because the feasible set of low-weakness moves is reshaped by the new simplicity metric.*

Proof. By definition,

$$\mathrm{SAI}_i(\varphi) = \frac{\mathrm{Support}(\varphi)}{\mathrm{Prior}_i(\varphi)} = \mathrm{Support}(\varphi) \cdot e^{\lambda \mathrm{Comp}_i(\varphi)}.$$

Divide SAI_2 by SAI_1 to obtain the stated multiplicative factor. Note that this argument uses only the exponential form of the prior in Comp_i; the specific interpretation of $\mathrm{Support}(\varphi)$ (likelihood ratio, Bayes factor proxy, or other evidential score) is irrelevant as long as it is held fixed across the two paradigms. $\square$

Proof sketch. The proof rewrites SAI to make the dependence on Comp_i explicit, then takes a ratio. The key step is that the prior is exponential in complexity, so changing complexity adds in the exponent and thus multiplies surprise. Visually: one keeps the "support" height fixed but changes the "simplicity baseline" beneath it; the perceived elevation above baseline changes accordingly. A useful mental picture is that λ controls how sensitive the community is to shifts in simplicity: larger λ makes even small changes in Comp produce large multiplicative changes in SAI, amplifying paradigm effects on what is treated as noteworthy. □

Remark 3272 (Independence and compositionality)**.** *When a paradigm change declares new primitives "cheap," many previously complex expressions become more factorable (or more naturally independent) in the representation language. This directly impacts surprise decomposition and connectedness of surprise: what counted as a single entangled hypothesis may become a composable product of simpler parts, changing both* Comp *and any independence baselines used by miners. In particular, if φ can be re-expressed as a composition (e.g. $\varphi \simeq \alpha \otimes \beta$) whose parts are treated as approximately independent under the new primitives, then both the effective complexity and the surprise bookkeeping can split: one can compare* $\mathrm{SAI}(\alpha)$ *and* $\mathrm{SAI}(\beta)$ *separately rather than only* $\mathrm{SAI}(\varphi)$ *as an undifferentiated whole. This matters for quantale-compositional treatments because the algebra of combination determines which decompositions are legible, and legibility feeds back into what is counted as simple, modular, and therefore "expected".*

59.7 Compression surprise as a budget-delta surprise (bridge to "surprising fulfillment")

Theorem 153 (Compression surprise is residual surprise in a cost-oriented quantale)**.** *Let L be a code-length functional (smaller is better). Define a cost score $C = -L$ (larger is better). Let Q_+ be the tropical quantale (max-plus) and define two "budgets" $B_{old} \le B_{new}$ corresponding to "before updating codes" and "after updating codes". Let*

$$TV_{B_{old}}(B) := -L_m^{old}(B), \qquad TV_{B_{new}}(B) := -L_m^{new}(B).$$

Then the one-sided tropical residual surprise satisfies

$$\mathrm{Surp}_{B_{old}\to B_{new}}(B) = \max\{0, L_m^{old}(B) - L_m^{new}(B)\} = \mathrm{Sur}_m(B; I).$$

Moreover, the interpretation of $B_{old} \le B_{new}$ is not that the object *B changes, but that the* evaluator *(code family, model class, or compressor state indexed by m) gains capacity: after updating on information I, the admissible descriptions become no longer than before (equivalently, the achievable negative code length becomes no smaller). In this sense, the order relation between budgets is an order on* representational options *rather than a numerical constraint on B itself.*

Remark 3273. *This theorem says that the "compression gain" notion of surprise is the same residual-difference pattern again, now applied to representational efficiency rather than inferential support. If one thinks of $-L$ as a reward (more compressible means more predictable/structured), then the residual surprise is the improvement in reward after an internal update. The* $\max\{0,\cdot\}$ *corresponds to the one-sidedness of Hyperseed's definition: only improvements count as surprise. In particular, this makes explicit that "surprise" here is* not *the absolute compressibility of B, but the* change *in compressibility induced by updating the coding scheme from "old" to "new". This aligns with the intuition that a datum can be highly compressible in an absolute sense yet not surprising if the agent already had a code that captured the relevant regularity.*

Remark 3274. *The result matters because it unifies two superficially different mechanisms: surprise as "belief lift above simplicity" (SAI) and surprise as "compression gain." Both become instances of quantale deltas between evaluators. This helps keep the ontology disciplined: one can* *vary the evaluator and still remain within the same compositional algebra. Concretely, the evaluator may be instantiated as (i) a belief-scoring functional, (ii) a likelihood or log-evidence functional, or (iii) a coding/MDL functional. The quantale framing isolates the common structural move: compare two states of evaluation (pre-update vs. post-update), and extract the positive residual as the relevant "delta" object. This is also the point at which different choices of orientation become visible: if one instead treated L as a cost to be* minimized *directly, then one would swap signs (or swap the order convention) to recover an equivalent residual form.*

Proof. In Q_+, residual surprise is $TV_{old} \Rightarrow TV_{new} = TV_{new} - TV_{old}$, but we want a one-sided positive gain (clipped at 0) as in Hyperseed. Compute:

$$TV_{new} - TV_{old} = (-L^{new}) - (-L^{old}) = L^{old} - L^{new}.$$

Taking $\max\{0, \cdot\}$ yields exactly $\mathrm{Sur}_m(B; I)$. Equivalently, one can read the same algebra as a "freed budget" statement: if L is measured in bits (or nats), then a positive value of $L_m^{old}(B) - L_m^{new}(B)$ is precisely the number of description-length units saved by the updated code. The tropical residual records this savings as an additive delta, which is the natural notion of difference in the max-plus setting. □

Proof sketch. The proof is an elementary algebraic rewrite: define the evaluation to be negative code length so that "better compression" corresponds to a larger score. Then the tropical residuum becomes a difference of code lengths. The only adjustment is the one-sided clipping, reflecting that Hyperseed counts only positive compression gains as surprise. One may also view this as a minimal functoriality requirement: by encoding "quality" as an element of the same ordered monoid across states (old/new), the delta is forced to be an internal morphism (a residual) rather than an external comparison rule. This is why the rewrite is so direct: once the evaluator lives in Q_+, subtraction is the canonical way to form a residual. □

Remark 3275 (Unifying picture)**.** *Hyperseed's (i) belief-level surprise (SAI) and (ii) compression-level surprise are both instances of the same underlying pattern: surprise is a quantale-valued delta between two evaluators, with evaluator choice (and orientation) determining whether we interpret surprise as lift, log-delta, evidence gain, or compression gain. This same picture provides the intended bridge to "surprising fulfillment": a positive compression residual can be interpreted as newly available internal slack (saved bits, saved description-length, or reduced representational cost), which can then be* reallocated *as budget toward downstream objectives. In other words, a compression surprise is not merely epistemic; it is also a resource-creating event under a budget semantics, and thus becomes a natural candidate primitive for modeling episodes where an agent achieves unexpectedly high performance because the world became simpler (more compressible) than its prior code anticipated.*

59.8 Conclusion: how the algebras fit

- Hyperseed's SAI is exactly a ratio-quantale residual surprise between a simplicity-only baseline (prior from complexity) and an inference/closure-supported plausibility evaluation (Theorem 1). In more explicit algebraic terms, if $\preceq$ is the plausibility order and $\otimes$ is the quantale composition, then the residual $a \multimap b$ is characterized by the adjunction property $a \otimes x \preceq b \iff x \preceq (a \multimap b)$, so SAI can be read as the *least* multiplicative "update factor" required to turn the baseline expectation into the supported evaluation. The "ratio" language here is not metaphorical: it indicates that surprise is computed as a *relative* gap between two regimes of assessment (simplicity-only vs. closure-supported), rather than as an absolute magnitude attached to a single model.

- Therefore SAI inherits the compositional algebra of budgeted surprise: telescoping identities across multi-step reasoning, and robustification across context families via infimum/weakness (telescoping theorem and robust section). Concretely, because residuals in a quantale interact systematically with $\otimes$, chains of reasoning can be assigned surprise contributions that combine predictably: successive inferential steps do not require re-deriving surprise "from scratch", but instead compose along the proof or deliberation trace. This is the sense in which telescoping identities capture multi-step reasoning as a single net transformation from an initial baseline to a final supported position, while still allowing intermediate steps to be audited. Likewise, robustification via infimum formalizes the intuition that context can change what is easy or hard to explain: taking an infimum across contexts forces the resulting score to reflect the *worst-case* (or most demanding) context in the family, thereby preventing fragile, highly context-tuned surprises from dominating downstream decisions.

- Hyperseed's formal definition of productive belief systems already bakes in a "surprise-seeking" aspect (self-transcendence via high-SAI beliefs/questions), and science is explicitly a special case of the thinking loop with empirical coupling. Hence both scientists and ordinary curious minds are formally describable as engaged in surprise-seeking loops (among other goals) (productive curiosity/science theorem). Here "surprise-seeking" should be read as a structural preference for queries and conjectures that, if resolved, would generate a large residual gap between baseline simplicity priors and closure-supported plausibility. On this view, self-transcendence is not merely novelty-chasing; it is the systematic pursuit of questions whose answers reconfigure what the system can stably claim after accounting for inferential closure (and, in the scientific case, observational constraints). The empirical coupling in the scientific loop matters because it supplies externally anchored closure steps (e.g. measurement, replication, instrumentation), which can force the supported plausibility evaluation away from what simplicity alone would have predicted, producing disciplined high-SAI updates.

- Paradigm shifts correspond (in part) to changing the simplicity/weakness channel, which in SAI terms renormalizes surprise baselines and thus changes what counts as "surprising" (paradigm-shift theorem). In this framing, a paradigm shift is not only a change in particular hypotheses, but a change in the map from representations to priors and from contexts to admissible weakness bounds: what was once treated as "simple" may become "complex" (or vice versa), and what was once treated as a permissible weakening across contexts may become unacceptable. Because SAI is computed *relative* to a baseline, altering the baseline effectively rescales the surprise landscape: previously dramatic residual gaps can collapse, while previously negligible gaps can become salient. This explains why, after a shift in representational and resource assumptions, communities can legitimately disagree about which anomalies are decisive and which are routine, even when they agree on the raw observational record.

The above items can be read as a single integration claim: the "surprisingness" quantity used by Hyperseed is not an ad hoc score, but the canonical quantity singled out by the residual structure of the underlying quantale, and the accompanying weakness operation supplies the cross-context stability condition needed for communal and scientific use. In particular, the same algebra that licenses compositional bookkeeping for individual reasoning also supports aggregation over heterogeneous agents and environments, provided the relevant infima exist and are taken with respect to the intended notion of context inclusion.

Remark 3276. *The philosophical moral is that "surprise" is not merely an emotion; it is a structural relation between evaluators: between what a system expected under one regime of resource* *and representation, and what it is compelled to acknowledge under another. When cast in quantale terms, this relation becomes compositional and therefore suitable for rigorous use inside automated reasoning and communal knowledge processes. Weakness (as infimum) then supplies the pragmatic discipline: surprise worthy of scientific attention is surprise that survives across contexts (Hyperseed-Concept 202). Equivalently, the infimum condition can be viewed as a formal analogue of "robustness" or "invariance": an effect is not elevated to a scientifically central anomaly merely because it appears under a convenient parametrization, but because it persists when the evaluator is weakened, the background context is varied, or the representational budget is shifted within the allowed family. This links the algebraic treatment of surprise to familiar methodological norms (stress-testing, ablation, replication, and sensitivity analysis) while preserving a single compositional calculus for both individual deliberation and collective inquiry.*

60 Dependency Towers for Pattern Networks: Factorization, Towerization, and Tight Low-Order Bounds

We formalize a "pattern network" as a factor-graph/hypergraph presentation of a global semantic object (function, energy, proof/program behavior) built from local patterns (factors). Concretely, one may think of a collection of variables/ports together with a family of local pattern terms, where each term touches only a bounded subset of the variables, and where the global object F is obtained by composing/aggregating those local terms (e.g. by summation of energies, product of potentials, composition of program fragments, or superposition in a linear semantics). This presentation makes explicit which parts of F depend on which coordinates and at what arity, and thus provides the substrate on which we can state and prove order-restricted approximation results.

We then show how to "factor out" a pattern network into a dependency tower of pattern networks: each layer is a best (or canonical) approximation constrained to dependency order ¡= k, and is indistinguishable from the original by all ¡= k-ary observations. Here "dependency order $\leq k$" means that the semantics of the layer can be generated from factors whose scopes have size at most k (equivalently: no term in the representation simultaneously couples more than k degrees of freedom). The notion of "$\leq k$-ary observations" is intentionally operational: an observer/test is allowed to probe the system only through queries that themselves involve at most k coordinates jointly (for example, evaluating against test functions with support on $\leq k$ variables, applying interventions that touch $\leq k$ variables at once, or measuring correlations of order at most k). Indistinguishability then asserts that all such bounded-order tests produce the same outcomes on F and on its k-layer $F_{\leq k}$, so that any discrepancy between F and $F_{\leq k}$ lives purely in higher-order interactions.

We prove: (1) canonical existence/uniqueness of tower layers in Hilbert semantics (projection/tail optimality), (2) a representation theorem: the order-k semantic truncation admits a factor graph with max arity ¡= k, (3) a delta/tower factorization: the tower can be expressed as "exactly order k" increments (in group completion), (4) a modularity theorem: if a pattern network splits into independent sub-networks, then connected terms factorize and cross-components have zero connected mass, and (5) a task-invariance theorem: any observer/test limited to dependency order ¡= k cannot distinguish the full network from its k-layer, so reasoning/planning using only ¡= k tests is sound on the k-layer. In (1), the Hilbert setting should be read as: we place our semantic objects in an inner-product space (e.g. L^2-type semantics, reproducing-kernel Hilbert space semantics, or any completion in which orthogonal projection is defined), and define $F_{\leq k}$ as the orthogonal projection of F onto the closed subspace generated by $\leq k$-ary components. "Tail optimality" then means that the residual $F - F_{\leq k}$ is orthogonal to all $\leq k$-ary directions, which is exactly the statement that no $\leq k$-ary test can see the residual. In (2), the representation theorem bridges an abstract semantic truncation (defined by a projection or order restriction) back to a concrete combinatorial presentation: not only does $F_{\leq k}$ exist as a semantic object, it can be realized as an actual pattern network with arity bounded by k, so the tower layers live in the same modeling language as the original network. In (3), the "exactly order k" increments Δ_k are defined so that $F_{\leq k} = \sum_{j \leq k} \Delta_j$ (or the appropriate monoidal analog), and the mention of group completion indicates that we may pass from a purely additive/monoidal setting to one in which formal differences are permitted, allowing a clean Mbius-like decomposition into interaction orders even when the original factors are constrained (e.g. nonnegative energies or normalized probabilities). In (4), "connected terms" refer to the standard connected/cumulant-like parts of a factorization: they measure genuine multiway coupling beyond what is explained by lower-order marginals, and the theorem asserts that independence across components annihilates any such cross-component coupling. Finally, (5) makes the tower operational: if an algorithms interface to the world is limited to bounded-order queries or bounded-arity constraints, then replacing F by $F_{\leq k}$ is not a heuristic but a sound equivalence for that class of tasks.

Remark 3277. *Philosophically, a dependency tower is a disciplined way of saying: there are many kinds of "seeing," and each kind has a limited resolution. If one insists on only k-way lenses (tests that can only coordinate at most k degrees of freedom at once), then the world "as seen through those lenses" is not the full semantic object F, but its k-layer $F_{\leq k}$. The tower formalizes the idea that coarser lenses yield a coarser but* optimal *view—optimal not by taste, but by a precise extremal property in a chosen semantics. One useful way to read "optimal" is as a best-approximation principle constrained by what the observer is allowed to express: among all semantic objects that only contain $\leq k$-way dependencies, the layer $F_{\leq k}$ is the one that agrees with F on every admissible $\leq k$-ary question, and in Hilbert semantics it is also the one minimizing $\|F - G\|$ over all such G. The tower thus separates two issues that are often conflated: (i) the intrinsic complexity of the world (which may include arbitrarily high-order interactions), and (ii) the expressive power of the observational/decision interface (which may cap out at some finite k). The conceptual payoff is that one can speak precisely about what information is lost when restricting to low-order structure, and about which forms of reasoning remain invariant under that restriction.*

This construction is aligned with Hyperseed's emphasis on patterns and their stratification (Hyperseed-Concept 130, 131) and on dependency as a first-class notion (Hyperseed-Concept 93, 94). The guiding motivation is similar to that of "tower" formalisms in the Hyperseed ecosystem [8] and the broader pattern/emergence framing [5]. From this perspective, the dependency tower can be understood as a principled stratification of pattern flow: lower layers encode what is explainable by local interactions of bounded arity, while higher layers encode irreducible multiway coordination. This mirrors common practices across domains (e.g. truncating higher-order moments, limiting clause width in proof systems, bounding interaction order in statistical physics, or restricting attention to low-order features in learning), but packages them into a single semantic construction with explicit optimality and invariance guarantees.

60.1 Setup: pattern networks and dependency order

60.1.1 Semantic target

Definition 535 (Semantic target object)**.** *Fix a probability space (Ω, μ) and let $H = L^2(\Omega, \mu)$ (real-valued) for Hilbert results. A "semantic target" is an element $F \in H$. (Other semantics such as quantales can be substituted; we focus on Hilbert for tight optimality.)*

Remark 3278. *The point of starting with a Hilbert space is not mere mathematical fashion. Hilbert structure is precisely what makes the notion of "best approximation" canonical: orthogonal projection is uniquely determined, and error decomposes by a Pythagorean identity rather than by ad hoc inequalities. Here $H = L^2(\Omega, \mu)$ means: we view our semantic object as a square-integrable real-valued function on a space of situations Ω, with μ encoding how frequently situations matter.*

A simple example is $\Omega = \{0,1\}^n$ with μ uniform, and $F(x)$ an "energy" or score of a configuration. Then $F \in L^2$ automatically. Another example is $\Omega = \mathbb{R}^n$ with μ a Gaussian measure and F any function with finite second moment. The usefulness of this definition is that it lets us speak rigorously about approximating F by restricted-dependency surrogates and about lower bounds that are independent of any particular factorization.

Remark 3279. *Two small but important interpretive points will be used repeatedly later. First, the choice of μ is part of the semantic problem specification: it determines which regions of Ω are emphasized when measuring approximation error. In particular, changing μ changes the notion of "best" low-order surrogate because the projection geometry changes. Second, the L^2 viewpoint is inherently distributional: it treats two functions as "close" if they are close on typical situations under μ, even if they differ drastically on μ-rare events. This is exactly the regime in which tight low-order bounds via orthogonal decomposition are meaningful.*

Remark 3280 (Technical notation)**.** *Throughout, (Ω, μ) denotes a probability space: Ω is the sample space and μ a probability measure. The space $L^2(\Omega, \mu)$ consists of equivalence classes of measurable functions $F : \Omega \to \mathbb{R}$ with $\int_\Omega F^2 \, d\mu < \infty$. The associated norm is $\|F\|_2 := \left(\int_\Omega F^2 \, d\mu\right)^{1/2}$, and the inner product is $\langle F, G \rangle := \int_\Omega FG \, d\mu$.*

Remark 3281. *Because $F \in L^2$ is an equivalence class modulo μ-null sets, any statement about "exact equality" of semantic targets is understood as holding μ-almost surely. This is usually the right notion for learning/approximation: if a discrepancy occurs only on a μ-null set, then it cannot affect L^2 error, and it cannot be detected by any μ-integrable test.*

60.1.2 Pattern networks as factor graphs / hypergraphs

Definition 536 (Factor-graph (pattern-network) representation)**.** *Let $x = (x_1, \ldots, x_n)$ range over $\Omega = D_1 \times \cdots \times D_n$ with product measure μ. A pattern network is a set of factor scopes $\mathcal{S} = \{S_a \subseteq [n]\}$ together with local potentials $\psi_a : D_{S_a} \to \mathbb{R}$ such that*

$$F(x) = \sum_a \psi_a(x_{S_a}).$$

The dependency order (arity) of the network is

$$\operatorname{ord}(\mathcal{N}) := \max_a |S_a|.$$

Remark 3282. *Intuitively, the scopes S_a declare* which variables are allowed to interact *inside a local pattern, and the potential ψ_a quantifies what that interaction contributes. The global meaning $F(x)$ is then built by summing local meanings. This is the additive version of a factor graph: rather than multiplying local factors into a joint weight, we add local energies into a total energy. In Hyperseed language, each ψ_a is a local pattern (Hyperseed-Concept 130) and the whole representation is a pattern web/network (Hyperseed-Concept 132); the order is a concrete measure of dependency (Hyperseed-Concept 93).*

A simple example: with $n = 3$ and $x_i \in \{0, 1\}$, let

$$F(x_1, x_2, x_3) = \psi_{12}(x_1, x_2) + \psi_3(x_3),$$

where $S_{12} = \{1, 2\}$ and $S_3 = \{3\}$. Then $\mathrm{ord}(\mathcal{N}) = 2$, meaning nothing genuinely 3-way occurs. As another example, if $F(x) = \sum_{i=1}^{n} \psi_i(x_i)$, then $\mathrm{ord}(\mathcal{N}) = 1$, and F is purely "separable." This definition is necessary because the whole tower construction will measure and truncate exactly this notion of interaction order.

Remark 3283. *The same semantic target F can admit many different pattern-network representations; the order $\mathrm{ord}(\mathcal{N})$ is therefore, strictly speaking, a property of a chosen factorization, not yet an intrinsic property of F. Later tower results will address the intrinsic question by comparing F against the* subspace *of all functions with interactions limited to order $\leq k$, so that low-order structure is identified independently of how one happened to write F as a sum of potentials. In that intrinsic view, "F has no genuine $(k+1)$-way interaction" means precisely that F lies in (or is well-approximated by) the closure of the order-$\leq k$ subspace in $L^2(\Omega, \mu)$, rather than merely that one particular decomposition uses small scopes.*

Remark 3284 (Technical notation)**.** *Here $[n]$ denotes $\{1, 2, \ldots, n\}$. For a subset $S \subseteq [n]$, the product domain $D_S := \prod_{i \in S} D_i$, and x_S denotes the subtuple $(x_i)_{i \in S}$. The index a runs over factors (hyperedges), and S_a is the variable-set (scope) attached to factor a.*

Remark 3285. *This is the additive "energy" representation; exponentiating yields the usual multiplicative factor graph $\tilde{p}(x) \propto \exp(F(x)) = \prod_a \exp(\psi_a(x_{S_a}))$.*

Remark 3286. *The additive view is often the more philosophically transparent: each local pattern contributes a "reason" (positive or negative) toward the global evaluation. Multiplication obscures this by turning reasons into weights; exponentiation is merely a change of chart. For the tower results, additivity also matches Hilbert linear structure, making orthogonality and projection available.*

Remark 3287. *When Ω is a product space with product measure, the order notion is closely aligned with classical decompositions of multivariate functions into low-order interaction terms (e.g. ANOVA/Hoeffding/Efron–Stein type decompositions). The tower perspective can be read as: build a nested sequence of subspaces*

$$\mathcal{H}_{\leq 0} \subseteq \mathcal{H}_{\leq 1} \subseteq \cdots \subseteq \mathcal{H}_{\leq n} \subseteq L^2(\Omega, \mu),$$

where $\mathcal{H}_{\leq k}$ consists of functions that can be expressed using only scopes of size at most k, and then project F onto these subspaces. The later "tight low-order bounds" are precisely statements about the norm of the residual $F - \Pi_{\leq k} F$, which depends only on F and μ, not on any chosen set of potentials.

Remark 3288. *The assumption that μ is a product measure in the factor-graph definition is a convenient baseline that lets "variable subsets" behave as independent coordinate selections, and it makes conditional expectations onto coordinate σ-algebras especially transparent. Many constructions extend to non-product μ, but one must then be careful: correlation in μ can create effective low-order predictability even when a naive syntactic factorization suggests high-order coupling, and conversely it can hide interactions that only appear off the typical set. The Hilbert/tower setup is designed to make these distinctions explicit.*

60.1.3 Dependency order via contexts (syntax-aware variant)

Definition 537 (Context order (hole/port order))**.** *Let φ be an expression/program, and let* $\mathrm{Ctx}(\varphi)$ *be a chosen quotient-set of k-hole contexts ordered by refinement (fill holes) with order* $\mathrm{ord}_{hole}(C) = k$*. A "pattern" may be taken as a context C with its induced k-ary potential JCK. We will treat "dependency order ¡= k" as meaning "expressible using only contexts of order ¡= k", or equivalently "indistinguishable by all ¡= k-hole context tests".*

Remark 3289. *This definition offers a syntax-aware analogue of the factor-graph arity notion. Instead of variables $x_1, \ldots, x_n$, we have an expression φ and we test it by inserting it into contexts with k holes (or ports). A k-hole context is a way of probing φ by exposing it to k "interfaces" at once; increasing k increases how entangled our probe may be. This connects naturally to the Hyperseed view that questions/tests are context-dependent acts of inquiry (Hyperseed-Concept 144, 86) and to the broader emphasis on programs and semantics (Hyperseed-Concept 80,* **??***).*

Remark 3290. *The phrase "chosen quotient-set" is doing real work: contexts are typically considered up to an equivalence relation that identifies probes with the same semantic effect (e.g. observational equivalence, α-renaming, or congruences generated by evaluation rules). Fixing such a quotient prevents the theory from depending on syntactic accidents. The refinement order ("fill holes") then expresses a principled notion of "making the probe more specific": a refined context commits to additional surrounding structure, and therefore can distinguish more behaviors.*

Remark 3291. *The two characterizations of "dependency order $\leq k$" mirror the two perspectives from factor graphs. On the "generative" side, "expressible using only contexts of order $\leq k$" means there exists a construction of the observable behavior of φ built entirely from k-port pattern components. On the "testing" side, "indistinguishable by all $\leq$ k-hole context tests" means that no probe with at most k simultaneous interfaces can detect any additional coupling beyond order k. This parallels how, for random variables, limiting oneself to functions of k coordinates both (i) restricts the class of admissible approximants and (ii) restricts the class of tests one can apply.*

Remark 3292. *Concretely, one can think of a k-hole context as specifying a k-ary observation functional: it takes k program fragments (or k copies/ports of a single fragment) and returns a measurement. In probabilistic semantics, the induced potential JCK can be read as a real-valued random variable on an appropriate space of executions or environments. Under that reading, the context-order filtration becomes a tower of σ-algebras/tests, and the later towerization results can be interpreted as orthogonal decomposition of program meaning into components that are detectable only by increasingly high-arity probes.*

Remark 3293. *A simple example is a term in a typed lambda calculus, where a 1-hole context is a term-with-a-hole that expects a single subterm, while a 2-hole context can compare or combine two inserted subterms. Saying "dependency order $\leq k$" then amounts to: no observation that simultaneously couples more than k ports can distinguish the object. This is useful because many cognitive and scientific tests are naturally interface-limited; the tower lets us formalize what is preserved under such limitations. In particular, the parameter k can be read as a bound on the arity of the "experimental interface": we are allowed to plug in, vary, or jointly relate at most k independent inputs at once, and any higher-order coupling is treated as unobservable at that resolution. This viewpoint makes precise why higher-order structure may be real yet operationally inaccessible: if the allowed contexts never simultaneously mention more than k ports, then any difference that only manifests through* (k+1)*-way coordination is invisible to the entire test suite, even if it is large in a global metric.*

Remark 3294 (Technical notation)**.** *The notation JCK indicates the semantic evaluation (denotation) of the context C under whatever semantics is being used; this is standard bracket notation for "the meaning of." The poset order by refinement means: $C' \leq C$ if C' is obtained by filling some holes of C (so C' is a more specified test). Equivalently, refinement is "adding information to the query": a context with more holes left open is a more general experiment, while plugging holes commits to concrete subterms/values and yields a stricter test. This order is convenient because many semantic constructions become monotone in it: if a behavior passes a refined test, then it also passes the coarser test obtained by forgetting some of the filled-in choices. When contexts are interpreted as maps into an observation domain (e.g. booleans, reals, or distributions), the refinement order can also be viewed as an inclusion order on the corresponding sets of distinguishable behaviors induced by those tests.*

60.2 Dependency towers and observers

Definition 538 (k-ary observers)**.** *Let $O_{\leq k}$ be a family of observables/tests such that each $O \in O_{\leq k}$ can probe at most k-way dependence. Examples:*

- *all bounded linear functionals supported on $H_{\leq k}$ (Hilbert ANOVA view),*
- *all expectations of products of test functions depending on at most k variables,*
- *all evaluations in k-hole contexts (syntax-aware view).*

In each case, the intent is that the observer factors through a representation in which "order $\leq k$" structure is the only structure that can affect the reported value; i.e. changes in F that live entirely in dependency order $> k$ are in the kernel of every $O \in O_{\leq k}$. When $k = 0$, this recovers the idea of purely constant (no-input) observations, such as a global bias/mean, depending on the ambient semantics.

Remark 3295. *An "observer" here is not a homunculus but a constrained measurement procedure: a rule that maps the semantic target F to some reported quantity (a number, a decision, a statistic). The phrase "k-ary" is meant literally: the observer can only coordinate at most k degrees of freedom in a single act of measurement. In more philosophical terms, k bounds the complexity of the* question *one is able to ask (Hyperseed-Concept 144, 186). One can also view $O_{\leq k}$ as specifying an information interface: it is not merely that the observer chooses to ignore higher-order structure, but that the experimental protocol cannot stably condition on, or jointly vary, more than k inputs at once. In this sense, k is a resource bound (arity, bandwidth, or "attention width") rather than a claim about what the world contains.*

For instance, if $F(x_1, x_2, x_3)$ is a function on three bits, then a 1-ary observer might query the average of F as a function of x_1 alone, ignoring x_2, x_3 except through averaging. A 2-ary observer might query correlations between x_1 and x_2. The usefulness is that it makes "indistinguishable at order k" precise: two semantic objects may differ, yet every observer in $O_{\leq k}$ yields the same output on them. Concretely, a typical 1-ary query has the form $x_1 \mapsto \mathbb{E}[F(x_1, X_2, X_3)]$ (a conditional mean), while a typical 2-ary query might return $\mathbb{E}[F(X)\,\varphi(X_1, X_2)]$ for some bounded test function φ depending only on (X_1, X_2). In the context-based formulation, the same idea is expressed by restricting to contexts whose free ports are of cardinality $\leq k$, so that the context cannot express an experiment that simultaneously relates more than k inserted substructures. Thus "indistinguishable at order k" is an invariance statement relative to a specified observational language.

Definition 539 (Dependency tower (abstract))**.** *A dependency tower for F is a family $\{F_{\leq k}\}_{k \geq 0}$ such that:*

1. *(Monotonicity)* $F_{\leq k}$ *becomes more faithful as* k *increases.*

2. *(k-ary indistinguishability) For all* $O \in O_{\leq k}$, $O(F_{\leq k}) = O(F)$.

3. *(Optimality/minimality) Among all* G *satisfying the same indistinguishability constraints,* $F_{\leq k}$ *is "best" in the chosen semantics (e.g. minimal* L^2 *error).*

The monotonicity clause is intentionally stated at a high level because the ambient semantics may offer different notions of "more faithful" (e.g. decreasing approximation error, increasing information, or refinement in a poset of summaries). In metric settings one often requires $\|F - F_{\leq k}\|$ *to be non-increasing in* k*, while in order-theoretic settings one may require* $F_{\leq k} \preceq F_{\leq(k+1)}$ *for an appropriate refinement relation. The indistinguishability clause can likewise be read as an equalizer condition:* $F_{\leq k}$ *and* F *are observationally equivalent with respect to the entire test class* $O_{\leq k}$.

Remark 3296. *The dependency tower is the formal device that turns an intuitive slogan into mathematics: "match what all k-limited observers can see, and otherwise be as simple/optimal as possible." Condition (2) is the epistemic clause: for the permitted tests, the k-layer is as good as the whole. Condition (3) is the ontic/semantic clause: among all such surrogates, we choose the one that is best by a precise criterion. In Hilbert semantics, "best" becomes least-square error, which in turn yields a unique answer by orthogonal projection. This projection viewpoint also clarifies why (2) and (3) are compatible: the constraints* $O(F_{\leq k}) = O(F)$ *define an affine subspace (or convex set, depending on the semantics), and the "best" element is the one closest to* F *under the chosen geometry. In ANOVA-style decompositions,* $F_{\leq k}$ *can be interpreted as the sum of all interaction components of order at most* k*, so that the tower layers additively isolate the contribution of each dependency order.*

As a toy example, let $F(x_1, x_2) = x_1x_2$ *on* $\{-1, +1\}^2$ *with uniform measure. A* $k = 1$ *observer can only probe dependence on one variable at a time. The best* $F_{\leq 1}$ *that matches all 1-ary observations is the constant zero function: the interaction is purely 2-way, and its 1-way shadow vanishes. The tower is useful because it gives a principled stratification of patterns by dependency order (Hyperseed-Concept 94) rather than relying on arbitrary truncations. Note that this example also illustrates how "matching all 1-ary observations" is stronger than matching a single statistic such as the global mean: all unary probes (e.g. conditional means given* x_1 *or given* x_2*) must agree, forcing the entire first-order component to be zero. By contrast, if* F *had a term like* x_1 *in addition to* x_1x_2*, then* $F_{\leq 1}$ *would retain the* x_1 *contribution while still discarding the purely interaction part, thereby separating main effects from higher-order synergies.*

60.3 Hilbert tower layers are canonical and optimal

Definition 540 (Order-k subspace)**.** *Let* H_S *be the closed subspace of* S*-dependent functions. Let*

$$H_{\leq k} := \overline{\mathrm{span}}\{H_S : \ |S| \leq k\}.$$

Let $P_{\leq k} : H \to H_{\leq k}$ *be the orthogonal projection.*

Remark 3297. *The inclusion of the closure in the definition is also what makes the phrase "order-k" robust under limiting operations: a function may fail to be literally a* finite *sum of low-order terms but still be approximable arbitrarily well (in* L^2*) by such sums. This is the same distinction as between finite-rank objects and their norm-closures in functional analysis, and it is essential if we want* $H_{\leq k}$ *to behave functorially under convergence (e.g., when* F *is itself obtained as a limit of increasingly refined models).*

Remark 3298. *Here H_S is the space of functions that "only look at" coordinates indexed by S. For example, if $S = \{2, 5\}$ then H_S consists of functions $G(x)$ that depend on x_2 and x_5 but are otherwise constant in all other coordinates. The space $H_{\leq k}$ is generated by all such spaces with $|S| \leq k$, meaning it contains all functions that can be expressed (in the L^2 limit) as sums of terms each depending on at most k variables.*

This is exactly the semantic counterpart of a factor-graph with max arity $\leq k$: each factor depends on at most k variables, and sums of such factors lie in $H_{\leq k}$. The closure bar $\overline{\text{span}}$ is a technical necessity: in infinite or continuous settings, we want $H_{\leq k}$ to contain limits of approximating sequences of low-order functions. The projection $P_{\leq k}$ then formalizes "throw away all components that require more than k-way dependence."

Remark 3299. *Two basic sanity checks help fix intuition. First, $H_{\leq 0}$ is (typically) the subspace of constants: functions that depend on no coordinates at all. In that case $F_{\leq 0}$ is the best constant approximation to F (often its mean, when $H = L^2$ over a probability space). Second, the family is nested: if $k \leq \ell$ then $H_{\leq k} \subseteq H_{\leq \ell}$, so increasing k can only improve approximation quality and can only decrease the tail $\|F - F_{\leq k}\|_2$.*

Remark 3300 (Technical notation)**.** *$\mathcal{F}_S$ will later denote the σ-algebra generated by coordinates in S; informally, it is the information available when you only know variables in S. The phrase "closed subspace" refers to closure in the L^2 norm, ensuring that orthogonal projection is well-defined.*

Remark 3301. *When $H = L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $H_S = L^2(\Omega, \mathcal{F}_S, \mathbb{P})$, the projection onto H_S is the conditional expectation $\mathbb{E}[\cdot \mid \mathcal{F}_S]$. Although $H_{\leq k}$ is generally larger than any single H_S, it can be helpful to keep this archetype in mind: "being in $H_{\leq k}$" means being representable (up to L^2-limits) by combining information from coordinate-subsets of size at most k. In this sense, $P_{\leq k}$ is the canonical operation that retains exactly the part of F explainable by k-way observations.*

Theorem 154 (Canonical tower layers via projection; tail energy is tight)**.** *Define $F_{\leq k} := P_{\leq k} F$. Then:*

1. *$F_{\leq k}$ is the unique best approximation to F among all $G \in H_{\leq k}$:*

$$F_{\leq k} = \arg\min_{G \in H_{\leq k}} \|F - G\|_2.$$

2. *For every $G \in H_{\leq k}$,*

$$\|F - G\|_2^2 \geq \|F - F_{\leq k}\|_2^2.$$

   Thus the residual/tail $\|F - F_{\leq k}\|_2$ is a representation-independent lower bound on any ¡= k-dependency representation.

3. *The family $\{F_{\leq k}\}_{k \geq 0}$ is a dependency tower for F with respect to any observer family $O_{\leq k}$ that depends only on $H_{\leq k}$.*

Remark 3302. *The phrase "representation-independent" is doing real work in (2): it means the bound depends only on the subspace $H_{\leq k}$ (i.e., the notion of k-way dependence), not on a particular choice of factorization, parametrization, basis, or architecture used to realize elements of $H_{\leq k}$. Consequently, if two different modeling formalisms both generate exactly $H_{\leq k}$ as their expressive class, then $\|F - F_{\leq k}\|_2$ is the same obstruction for both. This is what makes the tail energy an intrinsic notion of "irreducible high-order structure".*

Remark 3303. *Intuitively, this theorem says: if you insist your representation only involve dependencies up to order k, then there is a* single *best answer, obtained by "dropping" the orthogonal complement of $H_{\leq k}$. Moreover, the norm of what you dropped—the tail energy—is not an artifact of how you chose to factorize, but an intrinsic obstruction: no $\leq k$ representation can do better, because any such representation lives inside $H_{\leq k}$.*

This matters because it turns the informal idea of "towerization" [8] into a canonical mathematical operation in the Hilbert setting: there is no ambiguity about what the k-layer is. It also sets up the later representation theorem: once we know $F_{\leq k}$ exists canonically, we can ask how to represent it as a factor graph of order $\leq k$.

Remark 3304. *One useful consequence of the nestedness $H_{\leq k} \subseteq H_{\leq k+1}$ is an energy monotonicity statement:*

$$\|F - F_{\leq (k+1)}\|_2 \leq \|F - F_{\leq k}\|_2,$$

so the tower provides a nonincreasing sequence of irreducible residuals. This is the analytic counterpart of the modeling heuristic that allowing higher-order factors cannot make the best achievable error worse. In particular, the increments $\|F_{\leq (k+1)} - F_{\leq k}\|_2$ can be interpreted as the amount of "new explainable variance" unlocked by moving from order k to order k+1.

Proof. (1)-(2) are standard projection optimality and the Pythagorean identity for orthogonal projections. (3) If O depends only on the $H_{\leq k}$ component, then $O(F) = O(P_{\leq k}F) = O(F_{\leq k})$. Optimality holds by (1).

For completeness: the existence and uniqueness of $P_{\leq k}F$ use that $H_{\leq k}$ is a *closed* subspace of a Hilbert space, hence every point has a unique nearest point in $H_{\leq k}$. The inequality in (2) is just the same fact written in norm-squared form, and can be read as saying that $F_{\leq k}$ is not merely a good low-order surrogate but the *best possible* one under the ambient L^2 geometry. ☐

Proof sketch. View $H_{\leq k}$ as a flat "plane" (a closed subspace) inside the Hilbert space H. The orthogonal projection $P_{\leq k}F$ is the nearest point on that plane to F, and Hilbert geometry guarantees both existence and uniqueness. The tail energy is the squared length of the perpendicular segment, hence a lower bound on any other candidate in the plane. ☐

Remark 3305. *The key step is the Pythagorean identity: for any $G \in H_{\leq k}$,*

$$\|F - G\|_2^2 = \|F - P_{\leq k}F\|_2^2 + \|P_{\leq k}F - G\|_2^2,$$

because $F - P_{\leq k}F$ is orthogonal to $H_{\leq k}$. This is the analytic form of a geometric fact: in a right triangle, the hypotenuse is longer than the leg. Conceptually, the tower layer is "what the k-limited world can contain" and the tail is "what it must ignore."

Remark 3306. *Equivalently, the orthogonality condition can be stated as: for all $G \in H_{\leq k}$,*

$$\langle F - F_{\leq k}, G\rangle = 0,$$

so the residual is invisible to any linear test living in $H_{\leq k}$. This provides a concrete operational meaning: any statistic, probe, or observer that can be expressed as an $H_{\leq k}$ element cannot correlate with the tail. In later sections, this is the bridge between "geometric" definitions (subspaces and projections) and "semantic" ones (what an order-k observer can detect).

Remark 3307. *This is the formal core of "towerize by best low-order approximation": in Hilbert semantics, tower layers are exactly orthogonal projections and are uniquely determined.*

60.4 Representing tower layers as pattern networks (factor graphs)

60.4.1 Subset-indexed interaction series and truncation

Definition 541 (ANOVA/Hoeffding interaction components)**.** *Let $M_S := \mathbb{E}[F \mid \mathcal{F}_S]$ and define connected components U_S by Moebius inversion on the subset lattice:*

$$U_S := \sum_{T \subseteq S} \mu(T,S) M_T, \qquad \mu(T,S) = (-1)^{|S|-|T|}.$$

Then $F = \sum_{S \subseteq [n]} U_S$ and the U_S are orthogonal in L^2. Define the order-k truncation

$$F_{\le k}^{\mathrm{series}} := \sum_{|S| \le k} U_S.$$

Remark 3308. *This definition extracts, from F, the* pure *interaction contributed by each subset S of variables. The conditional expectation $M_S = \mathbb{E}[F \mid \mathcal{F}_S]$ is the best L^2 approximation to F that only depends on variables in S; it is what you would predict if you only knew those variables. However, M_S still contains the effects of smaller subsets $T \subsetneq S$. Moebius inversion subtracts these lower-order shadows, leaving the "connected" part U_S that cannot be explained by any proper subset.*

A familiar example: suppose $F(x_1, x_2) = f_1(x_1) + f_2(x_2) + f_{12}(x_1, x_2)$ with $\mathbb{E}[f_i] = 0$ and $\mathbb{E}[f_{12} \mid x_1] = \mathbb{E}[f_{12} \mid x_2] = 0$. Then $U_{\{1\}} = f_1$, $U_{\{2\}} = f_2$, and $U_{\{1,2\}} = f_{12}$. The utility is that truncating by $|S| \le k$ becomes a principled "keep only interactions up to order k" operation, aligning directly with dependency order.

Remark 3309 (Technical notation)**.** *$\mathcal{F}_S$ is the σ-algebra generated by the coordinate maps $(x_i)_{i \in S}$; conditioning on $\mathcal{F}_S$ means averaging over all coordinates not in S. The symbol $\mu(T,S)$ here is the Moebius function of the Boolean lattice of subsets; it should not be confused with the probability measure μ on Ω (the notation is standard but admittedly overloaded).*

Theorem 155 (Series truncation equals Hilbert tower layer)**.** *For $F \in L^2(\Omega, \mu)$ as above,*

$$F_{\le k}^{\mathrm{series}} = P_{\le k} F = F_{\le k}.$$

Remark 3310. *This theorem asserts that two ways of defining "the k-layer" coincide. One way is geometric: orthogonally project onto the subspace of $\le k$-dependent functions. The other is combinatorial: decompose F into subset-indexed interaction components U_S and drop the ones with $|S| > k$. The content is that the ANOVA/Hoeffding expansion is not merely a decomposition, but precisely the orthogonal decomposition aligned with the dependency-order filtration.*

This is important because it turns the abstract projection $P_{\le k}$ into explicit factor-like terms U_S, which we can directly read as potentials in a pattern network. Thus it is the bridge between canonical semantics (Hilbert) and explicit representation (factor graphs).

Proof. Each U_S lies in $H_S \subseteq H_{\le |S|}$. Orthogonality implies the closed span of $\{U_S : |S| \le k\}$ equals $H_{\le k}$, and the orthogonal projection onto $H_{\le k}$ is obtained by summing exactly the components in that subspace, i.e. those with $|S| \le k$. □

Proof sketch. The strategy is: (i) show each interaction component U_S lives in the appropriate order subspace; (ii) use orthogonality to identify the subspace $H_{\le k}$ as the span of components with $|S| \le k$; (iii) conclude that projection onto $H_{\le k}$ is implemented by "keep exactly those orthogonal pieces." □

Remark 3311. *The crucial mechanism is orthogonality: once F is expressed as an orthogonal sum, projection is a coordinate-wise operation. Geometrically, it is like expressing a vector in an orthonormal basis and then erasing the coordinates you are not allowed to use. This is why Hilbert semantics yields "tight" bounds: nothing is wasted in inequalities; the truncation is exact.*

Corollary 58 (Order-k factor graph representation exists)**.** *Define a pattern network $\mathcal{N}_{\leq k}$ with one factor scope S for each $S \subseteq [n]$ with $|S| \leq k$, and potential $\psi_S(x_S) := U_S(x_S)$. Then*

$$F_{\leq k}(x) = \sum_{|S| \leq k} \psi_S(x_S),$$

so $\mathcal{N}_{\leq k}$ is a pattern network of dependency order ¡= k representing the k-layer exactly.

Remark 3312. *In plain terms: the k-layer is not only an abstract function, it is literally a sum of local terms, each involving at most k variables. So the semantic truncation can be re-expressed as a factor graph with maximum arity k. This gives a canonical recipe for "towerizing" a pattern network in the most literal, implementable sense: compute the tower layer and then read off its factors.*

One should not confuse this representation with a claim of computational efficiency: there are $\sum_{j=0}^{k} \binom{n}{j}$ factors in the worst case. The corollary is about existence and conceptual equivalence, not about tractability. Its utility is that it pins down what it means for a k-ary representation to be exactly *the k-layer.*

Proof. Immediate from the definition of $F_{\leq k}$ as the series truncation and the factor-network definition as an additive sum of local potentials. ☐

Proof sketch. Once we know $F_{\leq k} = \sum_{|S| \leq k} U_S$, we simply rename each U_S as a potential ψ_S and declare its scope to be S. The dependency order bound is then tautological. ☐

Remark 3313. *The key step is not in the proof but in the prior theorem: identifying $F_{\leq k}$ with the truncated orthogonal interaction series. After that, the factor-graph representation is merely a change of packaging, translating between "semantic decomposition" and "network of local patterns" in the sense of [5].*

Remark 3314 (What this answers)**.** *Yes: in the subset-indexed (or equivalently Hilbert) semantics, any F admits a canonical dependency tower, and each layer can be represented as a pattern network (factor graph) whose max factor arity is ¡= k. So one can "factor out" a pattern network into a dependency tower of pattern networks by: (1) compute $F_{\leq k}$ (projection / tower layer), (2) represent $F_{\leq k}$ as a factor graph with factors up to order k (the U_S terms). In other words, "towerizing" a pattern network is nothing more than organizing its semantics by the cardinality of the variable subsets on which the interaction components depend, and then reading each truncation as its own bounded-arity factorization. Concretely, if F is already given as a factor graph with heterogeneous factor arities, then $F_{\leq k}$ is obtained by keeping only the interaction content supported on subsets of size $\leq k$ (in Hilbert semantics this is literally an orthogonal projection), and this truncated object can be re-expressed as a factor graph whose factors are precisely the U_S with $|S| \leq k$.*

60.5 Tower deltas: "exactly order k" pattern-network increments

Definition 542 (Tower deltas (group completion))**.** *Assume the semantic codomain is an abelian group (true in Hilbert). Define*

$$\Delta_0 := F_{\leq 0}, \qquad \Delta_k := F_{\leq k} - F_{\leq (k-1)} \quad (k \geq 1).$$

Equivalently, the tower can be recovered from the deltas by the recursion

$$F_{\le k} = F_{\le (k-1)} + \Delta_k,$$

so the delta sequence $\{\Delta_k\}_{k\ge 0}$ *is simply a first-difference transform of the cumulative sequence* $\{F_{\le k}\}_{k\ge 0}$.

Remark 3315. *A tower layer* $F_{\le k}$ *is cumulative: it contains everything up to order* k*. The deltas* Δ_k *isolate what is* new *when we move from order* $k-1$ *to order* k*. This is analogous to how a Taylor series can be described either by partial sums or by coefficients: the layers are partial sums, the deltas are the coefficients.*

For example, if $F_{\le 1}$ *captures all unary effects and* $F_{\le 2}$ *captures unary plus pairwise effects, then* Δ_2 *is the purely pairwise contribution, with unary effects removed. This definition is useful because it yields an "exactly order* k*" pattern-network increment, which is often what one wants when analyzing emergence or connectedness: it separates higher-order structure from its lower-order shadows. One can also read* Δ_k *as an "interface" between modeling capacities: it measures the gap between the best approximation available with* $(k-1)$*-ary patterns and the best approximation available with* k*-ary patterns, so it is the semantic analog of adding a new arity class of motifs to the network.*

Remark 3316 (Technical notation)**.** *The phrase "group completion" simply indicates that we must be able to subtract:* $F_{\le k} - F_{\le k-1}$ *only makes sense in an additive abelian group. In* L^2*, subtraction is always defined. In other semantic settings (e.g. some quantales), subtraction may not exist, so the delta tower must be treated differently. In such non-group settings one often replaces subtraction by a residuation, an order-theoretic difference, or by working with inequalities (e.g.* $F_{\le k}$ *as an increasing approximation) rather than equalities. The present definition highlights that in Hilbert semantics the decomposition is not merely monotone but* additive *and hence admits a coefficient-like interpretation.*

Theorem 156 (Orthogonal delta decomposition)**.** *In Hilbert semantics,*

$$F_{\le k} = \sum_{j=0}^{k} \Delta_j, \qquad F = \sum_{j\ge 0} \Delta_j,$$

and the increments are orthogonal: $\langle \Delta_k, \Delta_\ell \rangle = 0$ *for* $k \ne \ell$*. Moreover,* Δ_k *is exactly the sum of interaction components of order* k*:*

$$\Delta_k = \sum_{|S|=k} U_S.$$

In particular, the "exactly order k*" subspace can be characterized as*

$$H_{=k} := H_{\le k} \ominus H_{\le (k-1)},$$

and Δ_k *is precisely the orthogonal projection of* F *onto* $H_{=k}$.

Remark 3317. *This theorem says the tower is not merely a nested approximation scheme; it is an* orthogonal *resolution of* F *by dependency order. Each* Δ_k *lives in the "exactly* k*" subspace and is perpendicular to all other orders. Thus the energy of* F *decomposes cleanly across orders, and one can attribute variance (in the* L^2 *sense) to interaction orders without interference.*

This is important because it gives a rigorous foundation for statements like "the genuinely k*-way structure contributes this much" and because it makes truncation error transparent: the squared* *error of cutting at order k is just the sum of squared norms of Δ_j for $j > k$. It connects directly to the earlier projection theorem: the deltas are the orthogonal pieces carved out by successive projections. Explicitly, orthogonality implies the Pythagorean identities*

$$\|F\|^2 = \sum_{j\geq 0} \|\Delta_j\|^2, \qquad \|F - F_{\leq k}\|^2 = \sum_{j>k} \|\Delta_j\|^2,$$

so each order contributes an additive, nonnegative amount to total L^2 energy. When F is interpreted as a centered observable, $\|\Delta_k\|^2$ can be read as the portion of variance attributable to exactly k-way dependence.

Proof. Since $P_{\leq k}$ is an increasing family of orthogonal projections, the differences $\Pi_k := P_{\leq k} - P_{\leq(k-1)}$ are orthogonal projections onto the "exactly order k" subspace, hence are mutually orthogonal and sum to the identity in the limit. Apply to F to get $\Delta_k = \Pi_k F$ and the decomposition. The equality $\Delta_k = \sum_{|S|=k} U_S$ follows from the ANOVA expansion and the identification $F_{\leq k} = \sum_{|S|\leq k} U_S$. More formally, for $k \neq \ell$ one checks $\Pi_k \Pi_\ell = 0$ using the nesting $P_{\leq(k-1)} P_{\leq k} = P_{\leq(k-1)}$, and idempotence $\Pi_k^2 = \Pi_k$ follows from $P_{\leq k}^2 = P_{\leq k}$ together with the same nesting relations. □

Proof sketch. The proof uses the algebra of nested orthogonal projections. When subspaces are nested, successive differences of projections are themselves orthogonal projections onto the successive "new directions." Applying these difference projections to F yields the delta pieces. The ANOVA identity then identifies those pieces with the sum of interaction terms at the matching subset size. One can also view this as a Hilbert-space version of a Mbius inversion on the subset lattice: the cumulative objects $F_{\leq k}$ correspond to "downward closed" aggregation by size, while the deltas Δ_k recover the size-exact components. □

Remark 3318. *The key step is recognizing Π_k as a projection: Π_k picks out the orthogonal complement of $H_{\leq k-1}$ inside $H_{\leq k}$. Geometrically, one can imagine $H_{\leq 0} \subset H_{\leq 1} \subset H_{\leq 2} \subset \cdots$ as an increasing chain of subspaces; Δ_k is the component of F that lies in the "annulus" between the $k-1$ and k subspaces. Because these annuli are orthogonal in Hilbert space, the increments do not "double count" anything. A useful operational corollary is that Δ_k vanishes exactly when $F_{\leq k} = F_{\leq(k-1)}$, i.e. when allowing k-ary dependence does not improve the best approximation; this gives a clean criterion for the absence of genuinely k-way structure in the Hilbert sense.*

Corollary 59 (Delta pattern networks)**.** *Each delta Δ_k has a factor-graph representation using only factors of arity exactly k:*

$$\Delta_k(x) = \sum_{|S|=k} U_S(x_S).$$

Thus a tower of pattern networks can be represented either by its layers $\mathcal{N}_{\leq k}$ or by its delta layers (patches) $\mathcal{N}_{\Delta k}$. In particular, passing from the layer representation to the delta representation is a change of coordinates:

$$\mathcal{N}_{\leq k} \; = \; \mathcal{N}_{\leq(k-1)} \; \oplus \; \mathcal{N}_{\Delta k} \quad \textit{at the semantic level,}$$

with $\mathcal{N}_{\Delta k}$ consisting only of exactly-k-ary factor templates.

Remark 3319. *The distinction between layers and deltas is often cognitively and scientifically meaningful. Layers answer: "what can be expressed up to order k?" Deltas answer: "what is the irreducible novelty that only appears when we allow k-way dependence?" In the language of pattern* *networks, deltas correspond to adding a new family of exactly-k-ary motifs on top of what lower-order motifs already explain. Practically, deltas are often what one visualizes or reports: $\mathcal{N}_{\Delta 1}$ is a unary field layer, $\mathcal{N}_{\Delta 2}$ is a pure pairwise coupling layer, and higher $\mathcal{N}_{\Delta k}$ encode higher-order motifs whose effect cannot be mimicked by any combination of lower-arity motifs without changing the semantics. Because the Δ_k are orthogonal in Hilbert semantics, one can also compare their magnitudes directly via $\|\Delta_k\|$ to rank which interaction orders carry the dominant contribution.*

60.6 Modularity: factoring pattern networks into independent components

Definition 543 (Independent decomposition)**.** *Assume the variables split as $[n] = A \sqcup B$ and $F(x) = F_A(x_A) + F_B(x_B)$ with F_A depending only on x_A and F_B only on x_B. Call this an independent (additively separable) pattern-network decomposition.*

Remark 3320. *This definition captures an idealized form of modularity: the global semantic object is literally a sum of two sub-objects that live on disjoint variable sets. No term ever mixes variables from A and B. In factor-graph language, this corresponds to a disconnected factor graph; in Hyperseed terms, it is a clean separation of pattern subwebs with no shared interface (Hyperseed-Concept 132, 182).*

A simple example: $F(x_1, x_2, x_3, x_4) = \psi_{12}(x_1, x_2) + \psi_{34}(x_3, x_4)$ with $A = \{1, 2\}$ and $B = \{3, 4\}$. Then the two halves are independent in the strongest additive sense. This definition is useful because it lets us prove that towerization respects modularity: one can towerize each module separately.

It is important that the notion here is semantic *rather than probabilistic: we are not asserting any statistical independence of random variables x_A and x_B (though in many settings a product measure is used to define conditional expectations in the ANOVA/Moebius construction). Rather, we assert that the* function *F itself contains no cross-coupling between coordinates in A and coordinates in B. This is the most stringent version of modularity one could ask for in an interaction calculus: the absence of cross-terms holds identically, not merely approximately or in expectation.*

The partition $[n] = A \sqcup B$ should be read as an "interface cut": if a variable is shared by two pattern subwebs, then it belongs to the interface and strict decomposability fails. In applications, one often starts from a graph or hypergraph of variable participation and takes A, B to be unions of connected components; the present definition corresponds to the case where that participation structure is exactly disconnected.

Theorem 157 (Tower factorization under independence)**.** *If $F = F_A + F_B$ with disjoint variable sets A, B, then for every k:*

$$F_{\le k} = (F_A)_{\le k} + (F_B)_{\le k}, \qquad \Delta_k(F) = \Delta_k(F_A) + \Delta_k(F_B),$$

and every connected interaction term U_S with S intersecting both A and B vanishes.

Remark 3321. *The theorem states that independence is preserved perfectly by the tower: nothing about towerization forces artificial cross-talk between modules. In particular, there are no mixed interaction components: every U_S that touches both sides is exactly zero. This formalizes the intuition that if two subsystems do not share variables (interfaces), then no genuine connected motif can straddle the cut.*

This matters for scaling: if a large pattern network decomposes into weakly or strictly independent pieces, then tower computations and representations can be modularized. It connects to the broader methodological theme of decomposing cognitive/scientific problems into semi-independent subproblems, which recurs in the Hyperseed and related frameworks [5, 8].

One can also read the identities as a statement about locality *of low-order approximations: the order-k truncation $F_{\le k}$ does not "smear" information across an independence cut. Thus any algorithm that uses only $(\cdot)_{\le k}$ (e.g. for compression, for low-order surrogate modeling, or for approximate inference) can be applied independently on A and B without loss relative to applying it to the full $[n]$-system and then truncating.*

A further operational consequence is that if one is optimizing an objective of the form $F_A(x_A) + F_B(x_B)$, then the minimization (or maximization) problem decomposes as well: $\arg\min_x F(x)$ *can be obtained by independently solving* $\arg\min_{x_A} F_A(x_A)$ *and* $\arg\min_{x_B} F_B(x_B)$, *and similarly for approximate solutions based on $F_{\le k}$, since the truncations preserve additivity.*

Proof. In the ANOVA decomposition, conditional expectations factor across independent coordinates, and since F_A depends only on A, its only nonzero U_S have $S \subseteq A$; similarly for F_B. Thus any mixed S has $U_S = 0$, and summing U_S for $|S| \le k$ yields $F_{\le k} = (F_A)_{\le k} + (F_B)_{\le k}$. Delta identities follow by subtraction.

Concretely, if the interaction components are defined by the standard Moebius/ANOVA formula $U_S(x_S) = \sum_{T\subseteq S}(-1)^{|S|-|T|}\,\mathbb{E}[F(X) \mid X_T = x_T]$, then for a mixed set S (intersecting both A and B) each conditional expectation splits as $\mathbb{E}[F_A(X_A) \mid X_T] = \mathbb{E}[F_A(X_A) \mid X_{T\cap A}]$ and $\mathbb{E}[F_B(X_B) \mid X_T] = \mathbb{E}[F_B(X_B) \mid X_{T\cap B}]$. The alternating sum over $T \subseteq S$ then factorizes into independent alternating sums over $T \cap A$ and $T \cap B$, and the mixed contribution cancels to zero because no term in F_A can "see" coordinates in B and no term in F_B can "see" coordinates in A. This makes the vanishing of mixed U_S a direct algebraic consequence of the additive separability of F. □

Proof sketch. The high-level idea is to use the subset-indexed decomposition. If F is a sum of an A-only function and a B-only function, then conditioning on any mixed subset does not create a mixed dependence: Moebius inversion cannot conjure cross-terms that were not present. Therefore all mixed U_S vanish, and the tower splits additively.

Equivalently: the tower is built from linear operators (conditioning/averaging and inclusion–exclusion), and linear operators preserve direct sums. Since F already lies in the direct sum of the A-subspace (functions of x_A alone) and the B-subspace (functions of x_B alone), projecting onto interaction subspaces cannot generate components outside that direct sum. Mixed interaction subspaces are orthogonal (in the ANOVA sense) to both summands, so their coefficients must be zero. □

Remark 3322. *The key step is the behavior of conditional expectation under product structure: conditioning on coordinates in A has no effect on a function of B alone, and vice versa. Visually: the Boolean lattice of subsets splits into an A-lattice and a B-lattice, and the interaction decomposition respects this splitting. Hence the "connected mass" cannot leak across a genuine independence cut.*

Another way to phrase this is that the interaction calculus is compatible with taking connected components of the underlying dependency hypergraph: when the dependency hypergraph is disconnected, the interaction terms U_S are supported entirely within connected components. The present theorem is the special case where there are exactly two components, but the same reasoning iterates to a partition $[n] = A_1 \sqcup \cdots \sqcup A_m$ with $F = \sum_{i=1}^m F_{A_i}(x_{A_i})$, yielding $F_{\le k} = \sum_i (F_{A_i})_{\le k}$ and no mixed U_S spanning multiple blocks.

This strict case also serves as a baseline for "almost modular" systems: if there is a small coupling term $\varepsilon\, G(x_A, x_B)$ added to $F_A + F_B$, then the only source of mixed U_S is the coupling itself, and one expects mixed interactions to scale with ε. Later bounds on truncation error can then be interpreted as quantifying how much cross-talk is introduced by genuine interface terms as opposed to being an artifact of the tower construction.

Remark 3323 (Interpretation)**.** *This is the cleanest formal statement of "pattern-network components that do not share interfaces produce no connected higher-order motifs across the cut." It justifies decomposing a large pattern network into modules before towerizing: the tower is additive across independent modules.*

In particular, the theorem delineates a sharp boundary between structural *and* emergent *complexity: any higher-order motif that spans both A and B must be traced to an explicit cross-module term in F (a shared variable, a bridging factor, or an interaction potential involving both sides). If no such bridge exists, then no amount of towerization or low-order truncation can manufacture an apparent bridge, so modular semantics and modular computation coincide in this ideal regime.*

60.7 Task invariance: why k-layers are sufficient for k-limited reasoning

Theorem 158 (k-observer invariance (tower soundness for k-limited tasks))**.** *Let $\{F_{\leq k}\}$ be a dependency tower for F with respect to observer family $O_{\leq k}$. Then for every $O \in O_{\leq k}$,*

$$O(F_{\leq k}) = O(F).$$

Consequently, any reasoning, planning, or learning procedure whose internal scoring depends only on observables in $O_{\leq k}$ will behave identically whether it uses F or $F_{\leq k}$.

Remark 3324. *The theorem is conceptually simple but methodologically powerful: if your agent's questions are k-limited, then F and $F_{\leq k}$ are indistinguishable to that agent, hence interchangeable for that purpose. In Hyperseed terms, the space of allowable tasks/questions (Hyperseed-Concept 186, 144) determines what distinctions are operationally meaningful. If those tasks cannot witness higher-order dependencies, then representing them is unnecessary for correctness* relative to *those tasks.*

This result connects backward to the dependency tower definition (it is exactly property (2)) and forward to practical recipes: message passing, local search heuristics, and many bounded-complexity inference schemes implicitly operate with restricted dependency order. The tower provides a semantics-preserving truncation principle for such bounded observers.

It is helpful to read "k-limited" here as a precise interface restriction: an observer $O \in O_{\leq k}$ is allowed to probe at most k-way interactions (or, more generally, apply tests whose semantic content is determined by at most k jointly coupled components). Under that interpretation, $O(F)$ is the observer-accessible summary (a statistic, score, prediction, or decision-relevant quantity) that the agent can compute from F using only those restricted probes. The tower soundness condition states that all such accessible summaries are already fixed by the $\leq k$ layer, so carrying higher-order structure is invisible to the agent's admissible task class.

Equivalently, the theorem is an "observational equivalence" statement: F and $F_{\leq k}$ may differ substantially as semantic objects, but they induce the same input–output behavior on the entire test suite $O_{\leq k}$. This is exactly the sense in which the k-layer is sufficient for any downstream procedure that cannot consult higher-order tests.

Proof. This is exactly property (2) in the dependency tower definition: k-ary indistinguishability.

To unpack the "consequently" clause: if a procedure A can be expressed as a composition of operations that take as inputs only values of the form $O(F)$ for $O \in O_{\leq k}$ (for example, by computing a finite collection of such observables and then combining them by deterministic arithmetic), then replacing F by $F_{\leq k}$ leaves every queried observable unchanged, hence leaves the entire computation unchanged. The same reasoning applies to randomized procedures whose randomness is independent of whether they are run on F or $F_{\leq k}$: conditional on the same random seed, all oracle calls return identical values, so the output distribution is identical as well. □

Proof sketch. No additional argument is needed: the theorem simply restates the tower's defining indistinguishability axiom and then interprets it operationally as invariance of any downstream procedure built entirely from the allowed observables.

One can view this as a "plug-in" principle: once the tower construction has certified indistinguishability for $O_{\leq k}$, any algorithm that is expressible as an $O_{\leq k}$-oracle program is automatically sound under the truncation $F \mapsto F_{\leq k}$. In this sense, the tower acts as a reusable proof artifact: verifying property (2) once yields invariance for an entire family of bounded-task pipelines. □

Remark 3325. *The "consequently" clause is an instance of a general principle: if two semantic objects are equal under all queries a procedure can issue, then the procedure cannot behave differently on them. One may view this as a modest form of extensionality: behavior is determined by the available tests.*

In many concrete settings, the "queries a procedure can issue" can be made explicit. For example, if an inference engine only ever accesses F through local factor evaluations, local conditional probabilities, or constraint checks involving at most k variables at a time, then its entire execution trace is determined by those $\leq k$-ary responses. Any additional structure in F that only manifests via $(k+1)$-way (or higher) couplings is, from the engine's perspective, not merely ignored but genuinely unobservable: there is no admissible sequence of interactions that could reveal it.

The same extensionality intuition applies to learning settings where the loss or reward depends only on $O_{\leq k}$-observables. If the training signal is computed from $\leq k$-local statistics, then gradients, updates, and model selection criteria that are functions of those statistics cannot distinguish F from $F_{\leq k}$ either, so truncation is sound for the induced learning dynamics as well.

Corollary 60 (Reasoning-engine recipe)**.** *If your inference machinery uses only ¡= k-ary tests (e.g. factor-graph message passing with max factor arity ¡= k, or pattern matching using ¡= k-hole contexts), then replacing the full pattern network by the k-layer pattern network is semantics-preserving for those inference outputs.*

Proof. Under the stated hypothesis, every intermediate computation performed by the inference machinery can be viewed as depending only on values of observables in $O_{\leq k}$ (e.g. factor evaluations, consistency checks, or match scores that involve at most k jointly constrained elements). By Theorem 1, each such observable takes the same value on F and on $F_{\leq k}$, hence the entire computation—and therefore the final inference output—is unchanged by the replacement. □

Remark 3326. *This corollary is the practical moral: truncate first, then compute. One may still face computational costs, but at least one is no longer paying for distinctions one's inference engine cannot, even in principle, exploit. This resonates with the general "bounded observer" stance in cognitive and scientific methodology, and with tower-based decomposition themes [8].*

A useful way to operationalize this in implementations is to treat the choice of k as part of the specification of the task interface: it fixes which local neighborhoods, factor arities, or context widths are ever consulted. When that interface is fixed, the tower provides a principled guarantee that higher-order structure can be dropped without affecting any output that is expressible in terms of that interface. In particular, one can decouple two concerns that are often conflated: (i) whether higher-order dependencies exist in the underlying semantic object, versus (ii) whether the downstream algorithm is able to exploit them. The theorem and corollary formalize that only the latter matters for correctness relative to the task class.

Finally, note that "semantics-preserving" here is intentionally scoped: it preserves exactly the outputs that are definable from $\leq k$-ary tests. If one later upgrades the task family (e.g. allows $(k+1)$-ary queries, introduces global constraints, or changes the evaluation metric to one that *couples more than k components), then the appropriate truncation level must be revisited, and a deeper layer of the tower (or the full object F) may become operationally necessary.*

60.8 Context-poset towerization for syntax-aware pattern networks

Definition 544 (Context-indexed tower signatures)**.** *Let* $\mathrm{Ctx}(\varphi)$ *be the context refinement poset with order = number of holes. For each k, let* Σ_k *be the set of all evaluations of* φ *in contexts of order ¡= k, quotiented by your canonicalization convention. Define the tower signature map*

$$\tau_k : U \to \Sigma_k$$

from contexts/environments U *(assignments, inputs, evaluation traces) to signatures. Define k-indistinguishability by kernel relation* $H_k := \{(u, v) : \tau_k(u) = \tau_k(v)\}$.

Remark 3327. *The phrase "quotiented by your canonicalization convention" is doing real work: in many syntax-aware settings, the raw space of context evaluations contains redundant presentations of the same observational content (e.g. alpha-renaming, commuting diagrams of evaluation steps, or semantically equivalent normal forms). The quotient* Σ_k *should be understood as the* space of observational outcomes *at hole order* $\leq k$ *after choosing a normalization/canonical representative, so that* $\tau_k(u)$ *records information up to the intended notion of observational equality rather than up to accidental implementation details. In particular, different canonicalization choices can change* Σ_k *(and thus* H_k*), but the intended use is that* Σ_k *is stable under the invariances one wants the pattern network to respect.*

Remark 3328. *This definition shifts from numeric semantics to relational/observational semantics. Instead of projecting a function in* L^2*, we collect* all *results of probing* φ *with contexts up to hole order k, and package those results into a signature* $\tau_k(u)$*. Two underlying situations* u, v *are k-indistinguishability if they yield the same signature. Thus* H_k *is an equivalence-like relation capturing what the k-hole world cannot separate.*

As an example, imagine φ *is a program and* U *consists of input states. A 1-hole context might supply one argument and observe the output; a 2-hole context might supply two arguments and observe a relation between outputs (e.g. ordering or equality). Then* τ_1 *records all 1-hole observations, and its kernel identifies inputs that are observationally identical at that interface complexity. This is useful because it provides a towerization procedure even when a Hilbert inner product is not available, and it connects to the Hyperseed idea that patterns are partly defined by the tests that elicit them (Hyperseed-Concept 86, 130, 80).*

Remark 3329. *It is helpful to read* τ_k *as a "feature extractor" whose features are* contextual experiments *rather than pre-specified coordinates. In that sense, the tower level k controls an experimental budget: increasing k allows more holes, hence richer interventions, hence (typically) a refinement of the partition of* U *induced by* H_k*. When the context refinement poset is directed by hole-filling, one expects a monotone chain of kernels* $H_0 \supseteq H_1 \supseteq H_2 \supseteq \cdots$ *(more observational power yields fewer identified pairs), although the exact monotonicity statement depends on whether the family of "contexts of order* $\leq k$*" is nested in the intended way.*

Remark 3330 (Technical notation)**.** *A poset is a partially ordered set; here the order is refinement (filling holes). The set* U *is deliberately generic: it may consist of environments, inputs, traces, or any other substrate on which contexts can act. The kernel relation* $\ker(\tau_k)$ *is the set of pairs mapped to the same signature; the text writes it as* H_k.

Remark 3331. *Since τ_k is an ordinary function,* $\ker(\tau_k)$ *is automatically an equivalence relation on U (reflexive, symmetric, transitive). The phrase "equivalence-like" above is meant to emphasize that, while the formal properties are those of an equivalence relation, the intended interpretation is observational: u and v are equivalent only relative to the chosen family of contexts (order $\leq k$) and the chosen canonicalization of outcomes. In particular, H_k may be strictly coarser than true semantic equality in U if the allowed tests are too weak to expose the difference.*

Theorem 159 (Canonical k-indistinguishability relations and optimality among k-admissible observations)**.** *Let $g : U \to T$ be any k-admissible observation, meaning $g = f \circ \tau_k$ for some f. Then* $\ker(\tau_k) \subseteq \ker(g)$*. Therefore any relational weakness (join-aggregated valuation over indistinguished pairs) is minimized by the tower-kernel relation* $\ker(\tau_k)$ *among all k-admissible observations.*

Remark 3332. *The condition "g is k-admissible" formalizes the idea that g is not allowed to look directly at $u \in U$; it may only depend on u through what is exposed by k-hole probing, i.e. through $\tau_k(u)$. Thus f can be thought of as a post-processing step, aggregation, or decision rule applied to the raw signature. The theorem then says that no amount of post-processing can recover distinctions that the signature itself did not encode.*

Remark 3333. *The first claim says: τ_k is the* finest *summary that uses only k-hole information. Any other k-admissible observation g necessarily forgets at least as much, hence merges at least as many pairs. Thus* $\ker(\tau_k)$ *is the minimal indistinguishability relation consistent with the k-hole constraint.*

The second claim connects this to "weakness"-style optimization: if one evaluates an observation by how much it fails to distinguish (in a monotone way), then the finest observation is best. This is consonant with weakness-based selection principles in the broader Hyperseed discourse [3], though here it is phrased purely in order-theoretic terms.

Remark 3334. *Concretely, if a "weakness functional" assigns a higher score when more pairs are left indistinguishable, then it is naturally monotone with respect to inclusion of relations: if $R \subseteq R'$ then R' identifies at least the pairs that R does (and possibly more), hence is at least as weak. Under this reading, minimizing weakness means pushing toward the smallest feasible kernel among the admissible observations, which is exactly what the theorem guarantees is achieved by* $\ker(\tau_k)$*.*

Proof. If $\tau_k(u) = \tau_k(v)$ then $g(u) = f(\tau_k(u)) = f(\tau_k(v)) = g(v)$, so $(u, v) \in \ker(g)$. Thus $\ker(\tau_k) \subseteq \ker(g)$. Any monotone weakness functional on relations increases with set inclusion, hence is minimized by the smallest relation, i.e. $\ker(\tau_k)$. ☐

Remark 3335. *The proof uses only the elementary property of kernels under composition: if $g = f \circ \tau_k$ then $\tau_k(u) = \tau_k(v)$ is a sufficient condition for $g(u) = g(v)$. One can also phrase this as "τ_k is initial among k-admissible maps": every k-admissible g factors uniquely through τ_k once f is fixed, and such a factorization cannot create new distinctions inside fibers of τ_k.*

Proof sketch. The proof is a one-line kernel argument: composing with f cannot separate points already identified by τ_k. The optimality statement then follows from monotonicity: finer distinctions mean a smaller kernel, and any weakness measure that penalizes indistinguishability must prefer the smallest kernel. ☐

Remark 3336. *The key step is categorical in spirit: τ_k is a universal factor through which all k-admissible observations must pass. Visually, τ_k draws the sharpest partition of U achievable with k-hole tests, and every other k-admissible observation corresponds to coarsening that partition.*

Remark 3337. *This theorem is the syntax-aware analogue of the Hilbert tower optimality: "if you only allow k-hole information, the tower signature is the maximally informative summary, and its induced indistinguishability is minimal among all summaries using no more than k-hole information."*

60.9 Summary

- A dependency tower is a principled way to stratify any semantic object by dependency order, using indistinguishability by ¡= k tests and best-approximation semantics. In concrete terms, the "tests" may be understood as the admissible queries, probes, or observables available to an agent, so that two objects are identified at order k whenever every test involving at most k coordinates (or k arguments) evaluates the same way on both. The tower then selects, at each level, the canonical representative (e.g. in a metric or order-theoretic sense) that best matches the original object under this restricted observational regime.

- In Hilbert settings, tower layers are orthogonal projections and yield tight lower bounds (tail energy) on any ¡= k representation. Equivalently, if x is decomposed into its tower components and $P_{\leq k}x$ denotes the order-k truncation, then $x - P_{\leq k}x$ is orthogonal to every $\leq k$ approximation, and the squared norm $\|x - P_{\leq k}x\|^2$ is exactly the unavoidable residual for any candidate representation that only uses $\leq k$-way dependence. This makes the lower bound "tight" in the strongest sense: the tower truncation attains the best possible $\leq k$ error, and the tail energy quantifies precisely what cannot be recovered without admitting higher-order interactions.

- Each tower layer can be represented as a pattern network (factor graph) with max factor arity ¡= k by using the interaction components. Here one may view the interaction components as the canonical potentials associated to subsets of variables, so that the layer is realized by a factorization over factors whose scopes have size at most k. In particular, the representation is not merely existential: it is induced by the same decomposition that defines the tower layer, ensuring that the factor graph respects the intended semantics of "no dependence beyond arity k" rather than imposing an ad hoc parametric restriction.

- The tower admits delta layers ("exactly order k") which are orthogonal and correspond to arity-exact factor sets. These delta layers can be read as successive differences between adjacent truncations (e.g. the increment from order $\leq k-1$ to order $\leq k$), which isolates what is genuinely new at order k and not already explained by lower orders. Orthogonality implies that these increments do not interfere with one another (no double-counting across orders), and the "arity-exact" interpretation means that the associated factor sets can be chosen so that every nontrivial factor involves exactly k variables (or arguments), matching the intuitive notion of a pure k-way interaction.

- Independence decompositions of pattern networks factor the tower: no mixed connected terms cross independent cuts. Thus, when the pattern network separates into independent components (in the probabilistic sense, or in the sense of a product structure across a partition), the tower decomposition respects that separation: interaction terms supported on one side do not couple to the other side through any connected contribution. Put differently, the tower layers refine along the same independence cuts as the underlying network, so that towerization commutes with the independence-based modularization that is often used to make large systems tractable.

The resulting picture is a hierarchy of increasingly expressive semantics: the $\leq k$ layer captures precisely what any $\leq k$-limited observer can in principle discriminate, while the higher-order tail measures the additional structure that only becomes visible when genuinely higher-arity tests are allowed. In this sense, "towerization" is simultaneously a structural decomposition (by dependency order) and an operational statement (by what can and cannot be detected under bounded interaction width).

Remark 3338. *Taken together, these results justify a sober but powerful methodological stance: when an observer or agent is structurally limited to k-way tests, it is not merely convenient but* mathematically sound *to replace a full pattern network by its k-layer for the purposes of that observer's tasks (Hyperseed-Concept 186). The Hilbert case gives the cleanest canonical story, while the context-poset variant shows how analogous conclusions can be drawn in syntax-aware settings where "observation" is inherently procedural. In particular, the substitution "full object $\rightsquigarrow$ order-k truncation" is justified not only as a heuristic simplification but as an invariance principle: any two semantic objects with the same $\leq k$ layer are indistinguishable to the observer, and any decision rule built solely from $\leq k$ tests must therefore assign them the same outcomes. When a quantitative error notion is available (as in the Hilbert setting), the tower additionally supplies an explicit and optimal guarantee on what is sacrificed by truncation (the tail energy), thereby separating epistemic limitation (what cannot be accessed by $\leq k$ probes) from modeling choice (what one elects to ignore). In the procedural or syntax-aware regime captured by the context-poset formulation, the same stance is recovered in an order-theoretic guise: bounded observational procedures induce a corresponding bounded dependency semantics, and the tower provides a disciplined way to state and exploit that boundedness.*

61 TransWeave Meets Hyperseed: Self-Continuity, Paradigm Commensurability, and Mindplex Coherence via Approximate Intertwining

We connect (i) Hyperseed's approximate-morphism account of pattern-flow, self-continuity, mind-world correspondence, and mindplex coherence, with (ii) TransWeave's Bellman-Darboux (approximate) intertwining maps that transfer solutions across reward reweightings, tasks, representations, and agents. The core bridge is an "error-to-degree" embedding: small TransWeave intertwining error yields a high-degree approximate morphism between induced Hyperseed-style pattern/value-flow networks. This yields formal results for: continuity of self as transweaving across time-slices; commensurability of scientific paradigms as existence of low-error transweaves between their problem-solving procedures; and conditions under which a group of minds forms a notably coherent mindplex when pairwise transweaving is dense enough.

To make the bridge intuitive before it is made precise, it helps to keep in mind the two different levels at which the frameworks speak. TransWeave is naturally stated at the level of *operators* (e.g. Bellman-style updates on value functions, policies, or other solution objects), whereas Hyperseed is naturally stated at the level of *flows on structured networks* (patterns transforming into patterns, and values/constraints propagating through those patterns). The claim here is that these are not competing descriptions but two views of the same phenomenon: an approximate intertwining at the operator level induces, in a canonical way, an approximate homomorphism of the corresponding flow networks, with the quality of the induced morphism controlled by the original intertwining defect.

More concretely, an approximate intertwining is typically witnessed by a map (or family of maps) that transports solution-relevant objects from one setting to another (across tasks, reward reweightings, representations, or agents) such that "apply-update-then-transport" is close to "transport-then-apply-update." The quantitative mismatch between these two composites is the *intertwining error*. Hyperseed's "degree" of an approximate morphism can be understood as any monotone calibration that assigns higher degree to smaller defect, subject to natural stability properties (e.g. composing two good-but-imperfect morphisms yields a morphism whose degree degrades in a controlled way). The "error-to-degree embedding" named above is precisely the step that converts a TransWeave-style operator inequality (an error bound) into a Hyperseed-style graded morphism claim about the induced pattern/value-flow structure.

The phrase "induced Hyperseed-style pattern/value-flow networks" may be read in a deliberately broad sense: a reinforcement-learning or decision-making problem determines a web of dependencies among representations (state abstractions, concepts, or features), update rules (planning, inference, learning), and evaluative quantities (values, costs, preferences). In such a web, edges correspond to permissible transforms (update steps, inference moves, or representational translations), and the weights/defects encode how reliably one can propagate information or value along those edges. An approximate intertwining map thereby becomes a conduit that carries not only solutions but also the *structure of solution-flow* from one web to another, which is exactly the sort of conduit Hyperseed treats as an approximate morphism of pattern-flow.

Interpreting "self-continuity as transweaving across time-slices" similarly becomes more literal under this lens. If each time-slice of an agent is associated with its own partially learned operator (capturing its current policy/value updates, inference habits, or representational choices), then the persistence of a single "self" can be modeled as the existence of low-error intertwinings between successive slices. On this view, autobiographical and procedural continuity correspond to the ability to transport solution structures forward in time with limited distortion; breaks of identity correspond to the absence of such low-defect transport, or to the accumulation of defects beyond tolerable thresholds.

The same machinery also clarifies the claim about paradigm commensurability. A "scientific paradigm" can be idealized as a family of operators encoding its admissible problem framings, inference moves, model updates, and evaluative criteria. Two paradigms are commensurable to the extent that there exists a low-error transweave that carries solution procedures and evaluative gradients from one to the other without large distortion. In this formalization, disagreement can persist even under commensurability (because the intertwining is only approximate), but crucially there remains a quantitatively controlled bridge explaining when translation of methods and standards is feasible and when it fails.

Finally, the mindplex claim can be read as a graph-theoretic strengthening of pairwise transfer. If we view a collection of agents as nodes in a "transweave graph," with edges weighted by intertwining error (or equivalently by induced Hyperseed morphism degree), then a mindplex is not merely a set of agents that can sometimes imitate each other, but a set in which the web of low-error transports is sufficiently dense and sufficiently coherent under composition. Density alone is not enough: what matters is also the *consistency of multi-step transport* (transporting from A to B to C should not drift too far from transporting directly from A to C), since such consistency is exactly what turns a pile of pairwise correspondences into a stable collective flow that can support shared concepts, shared evaluative signals, and robust coordination.

Remark 3339. *The philosophical stance here is a familiar one in modern mathematical science: when we cannot hope for strict equality of structures (the world, minds, and scientific paradigms being too messy for that), we settle for* approximate *equality and then quantify the defect. Hyperseed makes this stance explicit by treating "pattern flow" and "mind-world correspondence" as high-*

degree approximate morphisms rather than brittle isomorphisms [1, 5]. TransWeave, in parallel, treats transfer as an approximate intertwining *of Bellman operators [7]. The present section is a bridge between these: it turns an operator intertwining error into a quantified morphism degree between the induced flow graphs.*

Remark 3340. *A useful way to prevent the "approximate" qualifier from becoming rhetorical is to keep track of which norms/metrics are intended to measure defect. In TransWeave settings, intertwining errors are often expressed in operator norms or sup norms over value functions/policies (or, more generally, as bounds that control how far transported solutions violate the target fixed-point or optimality relations). In Hyperseed settings, degree assignments typically play an analogous role, acting as a graded witness to the reliability of pattern/value transport across the morphism. The technical work of the section therefore amounts to verifying that the chosen defect measure is (i) strong enough to control the induced flow discrepancy, and (ii) stable under the kinds of compositions that matter for multi-step reasoning, multi-stage learning, and multi-agent interaction.*

Remark 3341. *A reader may keep three Hyperseed core-concepts in view as an informal compass: Self-Continuity (Hyperseed-Concept 166), Mind-World Correspondence (Hyperseed-Concept 112), and Mindplex (Hyperseed-Concept 113). TransWeave itself is treated as a formal transfer relation (Hyperseed-Concept 193) that belongs to the broader family of Transfer Learning (Hyperseed-Concept 192).*

Remark 3342. *The organizing intuition behind the three advertised applications is the same: in each case one has a family of "problem-solving updates" (learning updates, inferential moves, or planning steps) and one asks whether there exists a structure-preserving translation between them. Self-continuity asks for such translations across time-indexed versions of an agent; paradigm commensurability asks for them across methodological regimes; mindplex coherence asks for them across agents, together with enough closure under composition to support stable collective cognition. Framing all three in terms of approximate intertwining allows the same quantitative notion of defect to play the role of "how continuous," "how commensurable," or "how coherent," rather than treating these as purely qualitative labels.*

61.1 Preliminaries

61.1.1 Quantale-weighted graphs and approximate morphisms

Definition 545 (V-graph)**.** *Fix a commutative quantale $(V, \leq, \otimes, 1)$. A V-weighted directed graph (a V-graph) is a pair $G = (X, G)$ where X is a set and $G : X \times X \to V$ assigns a weight to each ordered pair.*

Remark 3343. *The quantale $(V, \leq, \otimes, 1)$ should be read as a "logic of degrees." The order $\leq$ tells us when one degree is at most another; $\otimes$ is a monoidal product expressing how degrees combine along sequential composition; and 1 is the unit degree ("perfect preservation," "no loss," or "complete confidence," depending on interpretation). Commutativity of $\otimes$ is not strictly required for all applications, but it keeps the algebraic bookkeeping simple in the use-cases emphasized by Hyperseed and TransWeave.*

Remark 3344. *A simple example is the* boolean *case $V = \{0, 1\}$ with $\leq$ the usual order and $\otimes = \wedge$, where $G(x, x') = 1$ indicates an edge is present and 0 absent. Another example, used later in this section, is $V = [0, 1]$ with $\otimes$ as multiplication, where weights behave like similarities or reliabilities. In Hyperseed terms, such V-graphs are a lightweight formal proxy for Pattern Flow Networks (Hyperseed-Concept 131): they let us speak about "how strongly" one token leads into another without committing to a single probabilistic semantics.*

Definition 546 (Degree of structure preservation)**.** *Given V-graphs $G = (X, G)$ and $H = (Y, H)$ and a function $f : X \to Y$, define*

$$\deg(f; G, H) \ := \bigwedge_{x,x' \in X} \big(G(x, x') \to H(f(x), f(x'))\big),$$

where $(u \to v)$ is the residuum satisfying: $u \otimes w \leq v$ iff $w \leq (u \to v)$. We say f is a q-approximate morphism $G \to H$ if $q \leq \deg(f; G, H)$.

Remark 3345. *The notation is dense but principled. The symbol $\bigwedge$ denotes an infimum (greatest lower bound) taken over all ordered pairs $(x, x') \in X \times X$, so $\deg(f; G, H)$ is a* worst-case *preservation score: if f badly distorts even one important edge-weight, the degree decreases. The residuum $(u \to v)$ is the quantale's internal implication: it measures "how much slack" one must multiply u by (via $\otimes$) in order to be $\leq v$. This is exactly the style of "error as implication-residual" used in quantale-based weakness/surprisingness calculi [3].*

Remark 3346. *A concrete mental model: in $V = [0, 1]$ with $\otimes = \cdot$, the residuum is (informally) the largest $w \in [0, 1]$ such that $u \cdot w \leq v$, i.e. a ratio-like comparison between u and v (with the usual caveat about $u = 0$). Then $\deg(f; G, H)$ says: for every edge-strength $G(x, x')$ in the source, the corresponding edge-strength in the target, $H(f(x), f(x'))$, must be at least a fixed fraction of it, and we take the best such uniform fraction over all pairs. This makes "approximate morphism" a quantitative version of the ordinary notion of graph homomorphism: rather than preserving adjacency, we preserve weighted relations up to a uniform degradation factor.*

Theorem 160 (Composition law)**.** *For $f : X \to Y$ and $g : Y \to Z$,*

$$\deg(g \circ f; G, K) \ \geq \ \deg(f; G, H) \otimes \deg(g; H, K).$$

In particular, if f is q_1-approximate and g is q_2-approximate, then $g \circ f$ is $(q_1 \otimes q_2)$-approximate.

Remark 3347. *This theorem says that approximate structure preservation composes: if f preserves the structure of G inside H to degree q_1, and g preserves the structure of H inside K to degree q_2, then their composite preserves G inside K to at least the combined degree $q_1 \otimes q_2$. In the multiplicative degree quantale used below, this means degrees multiply, so small degradations accumulate gradually rather than catastrophically.*

Remark 3348. *The result is important because it turns "continuity of identity" and "transfer across agents" into a* path problem*: if the world only offers a chain of approximate correspondences, the theorem guarantees a quantitative lower bound on the composite correspondence. This is the precise algebraic engine behind the later telescoping-style bounds for self-continuity and mindplex coherence.*

61.1.2 Pattern-flow self-continuity (Hyperseed-style)

Definition 547 (Pattern/value-flow network)**.** *A pattern-flow network over interval I is a V-graph $F_I = (X_I, F_I)$ whose nodes are pattern-tokens (or state/action tokens) and whose edge weight $F_I(a, b)$ measures the degree to which pattern a flows into b over I (observer-dependent estimator).*

Remark 3349. *This definition abstracts away most of the ontology and keeps only what is needed for the bridge: a set of tokens X_I and a quantitative notion of "flow" between them. In a cognitive reading, tokens might be percepts, actions, beliefs, subroutines, or any other items one wants to track as elements of a Pattern Web (Hyperseed-Concept 132) evolving in time; the edge weight $F_I(a, b) \in V$ then expresses how strongly a tends to lead to, explain, enable, or transform into b during the interval I [5, 1].*

Remark 3350. *A simple example in $V = [0, 1]$ is: let X_I be a set of state-action pairs observed in a window of interaction, and set $F_I(a, b)$ to be a normalized empirical conditional probability of seeing b soon after a, possibly tempered to reflect confidence. A different example is semantic rather than statistical: $F_I(a, b)$ could be the degree to which a is judged (by a particular observer) to be a subpattern of b or a cause of b, placing the construction close to Hyperseed's broad Pattern/Process interface (Hyperseed-Concept 141).*

Definition 548 (Self-continuity as best approximate self-morphism)**.** *For successive intervals I, J, define the self-continuity degree*

$$\mathrm{SCS}(I \to J) \ := \bigvee_{f: X_I \to X_J} \deg(f; F_I, F_J).$$

Remark 3351. *Here $\bigvee$ denotes a supremum (least upper bound) over all candidate correspondences $f : X_I \to X_J$. Intuitively, we attempt to "translate" the tokens of interval I into those of interval J in whatever way best preserves the observed flow structure, and we take the degree of the best such translation. The quantity $\mathrm{SCS}(I \to J)$ is thus a purely structural measurement of Self-Continuity (Hyperseed-Concept 166): it asks whether there exists a high-degree approximate morphism witnessing that the later pattern-flow network is a continuation (up to reinterpretation) of the earlier one.*

Remark 3352. *In many concrete instantiations,* $\deg(f; F_I, F_J)$ *is computed by comparing the transported flow F_I along f (e.g. via pushforward/pullback operations or adjacency-matrix relabeling) to the target flow F_J, and then mapping the discrepancy into a degree value in a quantale of "truth" or "compatibility." The supremum $\bigvee$ is therefore not merely an abstract aggregator: it encodes a design decision that self-continuity is witnessed by* existence *of at least one good reinterpretation map, rather than requiring all reinterpretations to work. This existential flavor matches the intended reading: we are not asking whether every encoding of I persists into J, but whether there is some plausible alignment under which the observed dynamics are nearly preserved.*

Remark 3353. *Two simple extremes help fix the meaning. If $X_I = X_J$ and $F_I = F_J$, then choosing $f = \mathrm{id}$ yields degree 1 (in standard degree quantales), so $\mathrm{SCS}(I \to J)$ is maximal. If, on the other hand, every candidate f maps highly-flowing edges in I to weak or nonexistent edges in J, then each $\deg(f; F_I, F_J)$ is small, and the supremum remains small. The usefulness of the definition is that it permits identity through* relabeling *and* representational drift*: the self need not be fixed as a set of atoms; it is fixed as a pattern of transformations.*

Remark 3354. *Note that the map $f : X_I \to X_J$ is allowed to be many-to-one or otherwise non-invertible unless explicitly restricted by the modeling context. Allowing non-bijections captures common cognitive and computational phenomena: coarse-graining (many earlier distinctions collapsed into one later token), splitting (when modeled via relations rather than functions), and partial forgetting (when certain elements of X_I are effectively sent to low-salience or dummy states in X_J). When one wishes to forbid such collapses, one may restrict the supremum to a subclass (e.g. injective maps, structure preserving maps, or maps satisfying resource constraints), but the present definition keeps the notion maximally permissive so that "continuation up to reinterpretation" is not blocked by purely representational changes.*

Corollary 61 (Longer-horizon self-continuity lower bound)**.** *For a chain $I_0 \to I_1 \to \cdots \to I_n$,*

$$\mathrm{SCS}(I_0 \to I_n) \ \geq \ \mathrm{SCS}(I_0 \to I_1) \otimes \cdots \otimes \mathrm{SCS}(I_{n-1} \to I_n).$$

Remark 3355. *This is a formalization of a commonsense thought: continuity over a long time is at least as good as the product (or, in other quantales, the corresponding monoidal accumulation) of the continuities of the small steps. It matters because it lets one trade global questions ("Am I the same person after a year?") for local ones ("How much translation is needed from one day to the next?") in a way that is algebraically disciplined, and it mirrors telescoping/chain-rule phenomena seen in other quantale calculi [3].*

Remark 3356. *The direction of the inequality reflects the fact that* $\mathrm{SCS}(I \to J)$ *is defined as a supremum over witnesses: composing locally-good witnesses yields a particular candidate map (or, more generally, a particular candidate correspondence) from* I_0 *to* I_n*, whose degree is bounded below by the monoidal product of the local degrees; taking the supremum over* all *candidates can only improve this. In typical quantale settings,* $\otimes$ *is monotone in each argument and distributes over* $\bigvee$*, so the bound is exactly the algebraic shadow of "compose approximate morphisms* $\Rightarrow$ *multiply their degrees." When* $\otimes$ *is ordinary multiplication on* $[0,1]$*, the product makes explicit the intuitive accumulation of small degradations across many steps, whereas in tropical or additive quantales one obtains the corresponding sum-like accumulation.*

61.1.3 TransWeave BD maps (approximate intertwining)

Definition 549 (Discounted Bellman operator)**.** *Fix a discount* $\gamma \in (0,1)$ *and a (weighted) MDP* M *with Bellman optimality operator* B*. A value function* V *is optimal iff it is a fixed point:* $V^* = B(V^*)$*. The operator* B *is a* γ*-contraction in* $\|\cdot\|_\infty$*.*

Remark 3357. *This definition imports a key piece of reinforcement learning mathematics: optimality is characterized as a fixed point of a contraction operator. The discount* $\gamma \in (0,1)$ *enforces the contraction property in* $\|\cdot\|_\infty$*, which is what ultimately yields quantitative stability bounds of the form "error divided by* $(1-\gamma)$*". In this section we treat it as a black box: we use only the fixed-point identity* $V^* = B(V^*)$ *and the Lipschitz bound* $\|B(V) - B(W)\|_\infty \le \gamma \|V - W\|_\infty$*.*

Remark 3358. *For later use, it is helpful to keep in mind the standard consequence of contraction: not only does the fixed point exist and is unique, but value iteration* $V_{k+1} = B(V_k)$ *converges geometrically with rate* γ *in* $\|\cdot\|_\infty$*. Thus, whenever we can control the size of a perturbation in the Bellman operator or an approximate commutation defect, the impact on the fixed point is correspondingly controlled by a factor on the order of* $(1-\gamma)^{-1}$*. This is the precise technical sense in which discounting turns long-horizon planning into a quantitatively stable computation.*

Remark 3359. *From the Hyperseed point of view, the significance is conceptual: a cognitive procedure that repeatedly applies a Bellman-like operator is a particular instance of a belief/control updating dynamic, and the fixed-point is the stable "answer" to a persistent question about best achievable return. This aligns with the broader Hyperseed stance that cognition is question/answer/belief dynamics coupled to action [1].*

Remark 3360. *Interpreting* B *abstractly as an update operator also helps separate three roles that can otherwise be conflated: (i) the representational space of value functions, (ii) the dynamics of improvement encoded by* B*, and (iii) the semantic content of a value function as an evaluation of future outcomes. TransWeave maps focus on how (i) and (ii) can be transported across tasks while keeping (iii) approximately aligned, which is why commutation up to* ϵ *is the central criterion below.*

Definition 550 (TransWeave intertwining condition)**.** *Given two weighted MDPs* M_w *and* $M_{w'}$ *with Bellman operators* B_w *and* $B_{w'}$*, a TransWeave map consists of a state/representation map* T

together with an induced map T_V on value functions such that

$$\| T_V \circ B_w \; - \; B_{w'} \circ T_V \|_{\infty \to \infty} \; \leq \; \epsilon.$$

We call ϵ the intertwining error.

Remark 3361. *The expression $T_V \circ B_w - B_{w'} \circ T_V$ is a* commutator-like defect*: it measures how far T_V is from conjugating the Bellman dynamics of M_w into that of $M_{w'}$. If the defect were exactly zero, transferring a value function by T_V and then updating by $B_{w'}$ would be indistinguishable from updating by B_w first and then transferring. The inequality bounds this mismatch uniformly over inputs, in the operator norm induced by $\| \cdot \|_\infty$.*

Remark 3362. *Concretely, $\|\cdot\|_{\infty\to\infty}$ denotes the induced operator norm $\|L\|_{\infty\to\infty} = \sup_{V\neq 0} \|L(V)\|_\infty / \|V\|_\infty$ (equivalently, $\sup_{\|V\|_\infty \leq 1} \|L(V)\|_\infty$). Thus the intertwining condition says that* for every *input value function V, the pointwise error $\|(T_V B_w)(V) - (B_{w'} T_V)(V)\|_\infty$ is at most $\epsilon\|V\|_\infty$, ensuring a uniform, scale-aware notion of approximate commutation. In settings where value functions are naturally bounded (e.g. rewards bounded by $R_{\max}$), one can equivalently phrase the condition as a uniform $\| \cdot \|_\infty$ bound on the commutation defect over the relevant bounded subset.*

Remark 3363. *When T is a map between state sets, a typical induced map T_V is the pullback (or "composition") operator $(T_V V')(s) = V'(T(s))$, i.e. values on $M_{w'}$ are evaluated on the image of states from M_w. Other induced maps arise when T aggregates or embeds features, for example when T maps a state to a distribution over target-states (in which case T_V can be an expectation operator). The definition is therefore intentionally stated at the level of T_V rather than committing to a single construction, so that both deterministic and stochastic representation transfers are covered by the same commutation bound.*

Remark 3364. *This is the mathematical heart of TransWeave (Hyperseed-Concept 193): it is a criterion for when solutions to one control problem can be transported into another with bounded degradation, and thus a formal template for Transfer Learning (Hyperseed-Concept 192) [7].*

Remark 3365. *A useful immediate implication (under the same contraction hypotheses for $B_{w'}$) is that approximate intertwining yields an approximate transport of optimal value functions. Indeed, letting V_w^* and $V_{w'}^*$ denote the fixed points of B_w and $B_{w'}$, one compares $T_V(V_w^*)$ to $V_{w'}^*$ via*

$$\|T_V(V_w^*) - V_{w'}^*\|_\infty = \|T_V(B_w(V_w^*)) - B_{w'}(V_{w'}^*)\|_\infty \leq \|T_V(B_w(V_w^*)) - B_{w'}(T_V(V_w^*))\|_\infty + \|B_{w'}(T_V(V_w^*)) - B_{w'}(V_{w'}^*)\|_\infty,$$

so that the commutation defect contributes an ϵ-controlled term and the contraction contributes a γ-controlled term. Rearranging yields a canonical bound of the form $\|T_V(V_w^*) - V_{w'}^*\|_\infty \lesssim \epsilon/(1-\gamma)$ (with the precise constant depending on the normalization implicit in $\| \cdot \|_{\infty\to\infty}$ and any boundedness assumptions). This is the quantitative bridge from "dynamics nearly commute" to "optimal solutions nearly match after transfer."

61.2 A bridge: from BD error to Hyperseed degrees

61.2.1 A concrete degree quantale for this paper

We instantiate the quantale of degrees as

$$V \; := \; ([0,1], \leq, \cdot, 1),$$

so composition multiplies degrees.

Remark 3366. *In this instantiation, $\otimes$ is ordinary multiplication and* 1 *is literal equality of degree. Thus, if one map preserves structure to degree* 0.9 *and another to degree* 0.9*, their composite preserves structure to degree at least* 0.81*. This matches an "attenuation" intuition: each translation step loses some fidelity, and fidelities compound multiplicatively.*

Remark 3367. *The choice $([0,1], \leq, \cdot, 1)$ is also technically convenient: it is a commutative unital quantale in the standard sense, since arbitrary suprema exist in $[0,1]$, multiplication is associative and commutative, and multiplication distributes over suprema in each argument:*

$$a \cdot \sup_i b_i \;=\; \sup_i (a \cdot b_i), \qquad \left(\sup_i a_i\right) \cdot b \;=\; \sup_i (a_i \cdot b).$$

This distribution property is what makes it legitimate to talk about "best possible" (supremal) degrees in constructions that quantify over families of comparisons, as Hyperseed-style approximate morphism definitions often do.

Remark 3368. *The order $\leq$ is intentionally the usual numerical order: a larger degree means "more structure preserved" or "more similarity." In particular,* 0 *is the bottom element (total failure) and* 1 *is the top element (perfect preservation). This monotonic reading aligns with later uses where one takes infima over all local constraints (to obtain a global guarantee) and then composes guarantees by multiplication.*

Definition 551 (Error-to-degree embedding)**.** *Fix a sensitivity constant $\lambda > 0$ and define*

$$\Phi_\lambda(\epsilon) \;:=\; \exp(-\lambda\epsilon) \;\in (0,1].$$

Then $\Phi_\lambda(\epsilon_1 + \epsilon_2) = \Phi_\lambda(\epsilon_1) \cdot \Phi_\lambda(\epsilon_2)$.

Remark 3369. *This definition performs a conceptual conversion: it turns an* additive *notion of error (ϵ measured in some norm) into a* multiplicative *notion of degree suitable for the chosen quantale. The exponential is not arbitrary: it is essentially the unique smooth map turning addition into multiplication. Philosophically, it embodies the idea that equal increments of error should yield equal proportional decreases of confidence or similarity.*

Remark 3370. *A useful way to read Φ_λ is as a monoidal map from the additive monoid of nonnegative errors $(\mathbb{R}_{\geq 0}, +, 0)$ into the multiplicative monoid $([0,1], \cdot, 1)$: additive accumulation of independent error budgets becomes multiplicative attenuation of degree. In particular, whenever a proof produces an estimate of the form "total error $\leq \epsilon_1 + \epsilon_2$," applying Φ_λ immediately yields a compositional degree lower bound $\geq \Phi_\lambda(\epsilon_1)\Phi_\lambda(\epsilon_2)$, matching the way Hyperseed composes approximate structure preservation.*

Remark 3371. *The parameter λ calibrates how harshly we interpret error. With large λ, even small ϵ yields a noticeably reduced degree; with small λ the same ϵ is tolerated. This is a place where observer-dependence enters cleanly: different observers, tasks, or contexts may set different λ reflecting different notions of acceptable approximation [7].*

Remark 3372. *For intuition, one may linearize around $\epsilon = 0$:*

$$\Phi_\lambda(\epsilon) = e^{-\lambda\epsilon} = 1 - \lambda\epsilon + O(\epsilon^2).$$

Thus, for sufficiently small error budgets, the degree shortfall $1 - \Phi_\lambda(\epsilon)$ is approximately proportional to ϵ, with proportionality constant λ. This is helpful later when comparing different modeling choices (e.g. changing β in a graph encoding) while keeping a comparable "small-error" regime.

61.2.2 Encoding an MDP into a flow graph

Definition 552 (Value-flow graph functor)**.** *Given an MDP M with state set S and optimal value $V^* : S \to \mathbb{R}$, define the value-flow graph*

$$G_M(s, s') \ := \ \exp\bigl(-\beta\, |V^*(s) - V^*(s')|\bigr) \ \in (0, 1]$$

for some fixed $\beta > 0$.

Remark 3373. *This construction compresses an MDP into a similarity-like graph on states: two states are close (edge-weight near 1) when their optimal values are close, and far (edge-weight near 0) when their values differ greatly. It is a deliberately coarse encoding: we ignore transition structure and focus on the scalar field V^*. The intent is not to capture everything about the MDP, but to obtain a flow-like object to which Hyperseed's approximate-morphism machinery can be applied.*

Remark 3374. *Two immediate structural properties are worth keeping in mind. First, $G_M(s, s) = 1$ for all s (reflexivity in degree), since $|V^*(s) - V^*(s)| = 0$. Second, $G_M(s, s') = G_M(s', s)$ (symmetry), because of the absolute value. Thus the resulting object is a weighted undirected similarity graph; it is "flow-like" only in the sense that it provides a multiplicative field of affinities that can be transported along a map.*

Remark 3375. *The constant $\beta > 0$ plays the same role as an inverse temperature: it sets the scale on which value differences matter. If β is large, even modest value differences create weak edges; if β is small, the graph remains dense and forgiving. As with λ, this is partly a modeling and partly an epistemic choice: it reflects what distinctions the observer regards as structurally significant.*

Remark 3376. *This value-based encoding can be viewed as a very lightweight proxy for more elaborate state similarity constructions (e.g. bisimulation-inspired metrics): it forgets transitions and rewards except insofar as they influence V^*. The payoff of this forgetfulness is that it yields simple, explicit, and compositional bounds: sup-norm perturbations of V^* become uniform multiplicative degradations of all edge weights.*

Lemma 167 (Value perturbations yield multiplicative edge degradation)**.** *Let $V_1 : S_1 \to \mathbb{R}$ and $V_2 : S_2 \to \mathbb{R}$ and let $f : S_1 \to S_2$. Assume a uniform value-transport bound:*

$$\sup_{s \in S_1} |V_2(f(s)) - V_1(s)| \ \le \ \delta.$$

Then for all $s, s' \in S_1$,

$$G_{V_2}\bigl(f(s), f(s')\bigr) \ \ge \ e^{-2\beta\delta}\, G_{V_1}(s, s').$$

Remark 3377. *Intuitively, the lemma says: if the map f transports values with at most δ sup error, then the value-similarity between any pair of points cannot collapse by more than a uniform multiplicative factor. The factor $e^{-2\beta\delta}$ comes from paying δ once at each endpoint (s and s'), hence the 2. This lemma is the key technical hinge in the bridge: it turns a* pointwise *value bound into an* edgewise *flow-graph morphism degree.*

Remark 3378. *One can package the conclusion as an explicit "degree of approximate graph morphism" statement: the map f is degree-d nonexpansive (in the multiplicative sense) from G_{V_1} to G_{V_2} with*

$$d \ := \ e^{-2\beta\delta} \in (0, 1].$$

This form is the one that plugs directly into Hyperseed-style composition: if f has degree d_1 and a second map g has degree d_2, then $g \circ f$ has degree at least $d_1 d_2$, matching the quantale multiplication chosen above.

Remark 3379. *The same inequality also clarifies how to align parameters across layers of the paper: the degradation factor can be written in the Φ_λ notation as*

$$e^{-2\beta\delta} \;=\; \Phi_1(2\beta\delta) \;=\; \Phi_{2\beta}(\delta).$$

So, depending on whether one thinks of δ as the primitive error (and folds 2β into λ) or thinks of $2\beta\delta$ as the derived error budget, the two embeddings coincide. This is one precise sense in which "BD error" (additive) can be funneled into a Hyperseed degree (multiplicative).

Proof. By the triangle inequality,

$$|V_2(f(s))-V_2(f(s'))| \le |V_1(s)-V_1(s')|+|V_2(f(s))-V_1(s)|+|V_2(f(s'))-V_1(s')| \le |V_1(s)-V_1(s')|+2\delta.$$

Multiply by $-\beta$ and exponentiate to obtain

$$e^{-\beta|V_2(f(s))-V_2(f(s'))|} \ge e^{-\beta(|V_1(s)-V_1(s')|+2\delta)} = e^{-2\beta\delta}\, e^{-\beta|V_1(s)-V_1(s')|}.$$

□

Proof sketch. The strategy is to compare the two absolute differences $|V_2(f(s)) - V_2(f(s'))|$ and $|V_1(s) - V_1(s')|$ by inserting and subtracting $V_1(s)$ and $V_1(s')$. The uniform error bound then contributes at most δ per endpoint, and exponentiation converts the additive bound into a multiplicative degradation factor for the edge weights. □

Remark 3380. *The key step is the triangle inequality, which is doing conceptual work: it formalizes the idea that if two scalars are each well-approximated, then their difference is well-approximated. Geometrically, one may picture V_1 and $V_2 \circ f$ as two "height functions" on S_1 whose graphs lie within a vertical tube of radius δ; differences of heights between two points can then deviate by at most 2δ, because each point can contribute its own vertical deviation.*

Remark 3381. *In many downstream uses, the uniformity of the bound is the essential feature: since the multiplicative degradation factor $e^{-2\beta\delta}$ does not depend on the particular pair (s, s'), it behaves like a global degree certificate. This is precisely the kind of certificate one wants when arguing about self-continuity or commensurability across multiple representational layers: the statement "all pairwise value affinities are preserved up to a single multiplicative constant" can be transported, composed, and compared without re-running local estimates each time.*

61.3 Continuity of self as TransWeave across time-slices

Theorem 161 (Intertwining implies transported optimal values are close)**.** *Let $M_w, M_{w'}$ be discounted MDPs with Bellman operators $B_w, B_{w'}$ (both γ-contractions). Assume a TransWeave map with intertwining error ϵ:*

$$\|T_V \circ B_w - B_{w'} \circ T_V\|_{\infty\to\infty} \le \epsilon.$$

Let $V_w^*, V_{w'}^*$ be optimal values. Then

$$\|T_V(V_w^*) - V_{w'}^*\|_\infty \;\le\; \frac{\epsilon}{1-\gamma}.$$

Remark 3382. *In plain language: if T_V almost commutes with the Bellman updates, then it transports the* fixed point *(the optimal value function) of the first problem to something close to the fixed point of the second. The price paid is a familiar stability factor $1/(1-\gamma)$: when γ is close to 1, the Bellman map is only weakly contractive, so small dynamical mismatches can produce larger fixed-point deviations.*

Remark 3383. *This is important because it turns a dynamical notion (intertwining of operators) into a static bound on the objects we care about (optimal value functions). It is also the precise point at which TransWeave's operator criterion becomes usable inside Hyperseed's approximate-morphism calculus: it produces the δ needed by the value-perturbation lemma, and thus ultimately yields a structural degree bound.*

Remark 3384. *The inequality is a concrete instance of a general fixed-point stability theme: fixed points of contractions are robust under small perturbations. In the broader Hyperseed program, fixed-point methods are also used to study stability of self-modifying goal systems [10]; here the fixed-point is Bellman's, but the moral is the same.*

Proof. Use fixed-point identities $V_w^* = B_w(V_w^*)$ and $V_{w'}^* = B_{w'}(V_{w'}^*)$:

$$\begin{aligned}\|T_V(V_w^*) - V_{w'}^*\|_\infty &= \|T_V(B_w(V_w^*)) - B_{w'}(V_{w'}^*)\|_\infty \\ &\le \|T_V(B_w(V_w^*)) - B_{w'}(T_V(V_w^*))\|_\infty + \|B_{w'}(T_V(V_w^*)) - B_{w'}(V_{w'}^*)\|_\infty \\ &\le \epsilon + \gamma\|T_V(V_w^*) - V_{w'}^*\|_\infty.\end{aligned}$$

Rearrange: $(1-\gamma)\|T_V(V_w^*) - V_{w'}^*\|_\infty \le \epsilon$. ☐

Proof sketch. One inserts and subtracts the term $B_{w'}(T_V(V_w^*))$ to break the difference into (i) the intertwining defect, bounded by ϵ, and (ii) the effect of applying the contraction $B_{w'}$ to two different inputs, bounded by γ times the input difference. The contraction then lets us "move" the unknown term to the left and divide by $(1-\gamma)$. ☐

Remark 3385. *The proof's decisive maneuver is to exploit two forms of stability at once: algebraic stability (fixed-point identities) and metric stability (contraction). Visually, one may imagine $B_{w'}$ as a shrinking map on an infinite-dimensional cube of value functions: if $T_V(V_w^*)$ is an approximate fixed point (up to ϵ), repeated application of $B_{w'}$ drags it toward the true fixed point $V_{w'}^*$, and the contraction ratio quantifies how much discrepancy remains.*

Theorem 162 (TransWeave yields Hyperseed-style self-continuity lower bounds)**.** *Consider a system whose time-slice control at interval I is modeled by an MDP M_I with optimal value V_I^*, and similarly for J. Assume there is a TransWeave map (T, T_V) from M_I to M_J with intertwining error ϵ. Let $F_I := G_{M_I}$ and $F_J := G_{M_J}$ be the induced value-flow graphs. Then there exists a map f (namely the state/representation part of T) such that*

$$\deg(f; F_I, F_J) \;\ge\; \exp\Bigl(-\frac{2\beta}{1-\gamma}\epsilon\Bigr),$$

hence

$$\mathrm{SCS}(I \to J) \;\ge\; \exp\Bigl(-\frac{2\beta}{1-\gamma}\epsilon\Bigr).$$

Remark 3386. *This theorem is the promised bridge in a single inequality. It says: an operator-level TransWeave witness (intertwining with error ϵ) implies a structural Hyperseed witness (a high-degree approximate morphism between flow graphs), and thus yields a quantitative lower bound on self-continuity across the intervals. In this way, the "self" is construed as what can be transwoven across time while approximately preserving the value-structured flow of states and actions.*

Remark 3387. *The bound has three interpretable parameters: β (how sharply values induce graph edges), γ (how stable the Bellman fixed point is), and ϵ (how good the TransWeave is). The factor 2 again reflects endpoint error accumulation. The conclusion is not merely existential: it provides an explicit numeric degree that can be compared across systems, observers, or modeling choices, thus aligning with Hyperseed's preference for graded rather than binary predicates [1].*

Remark 3388. *This connects directly to the earlier composition law and longer-horizon bound: once each step has a quantified degree, chains of steps yield a compounded estimate. In the language of Hyperseed core-concepts, this is a way to operationalize Self-Continuity (Hyperseed-Concept 166) through a transfer-learning lens (Hyperseed-Concept 192).*

Proof. By the previous theorem, transported optimal values satisfy $\|V_J^* \circ f - V_I^*\|_\infty \le \delta$ with $\delta = \epsilon/(1-\gamma)$ (after identifying T_V as pullback/pushforward along f). Apply Lemma 1 to obtain, for all x, x',

$$F_J(f(x), f(x')) \ge e^{-2\beta\delta} F_I(x, x').$$

In the degree quantale $([0,1], \le, \cdot, 1)$ this implies $F_I(x,x') \cdot q \le F_J(f(x), f(x'))$ for $q = e^{-2\beta\delta}$ and all pairs, so $q \le \deg(f; F_I, F_J)$. Taking the supremum over f gives the stated bound for SCS. □

Proof sketch. First convert intertwining error into a sup-norm bound on the transported optimal value function via contraction stability, giving $\delta = \epsilon/(1-\gamma)$. Then use the value-perturbation lemma to translate a pointwise value bound into an edgewise inequality between the induced flow graphs. Finally, interpret that edgewise inequality as precisely the condition that the morphism degree is at least $e^{-2\beta\delta}$. In slightly more detail, the contraction step uses that the relevant Bellman-type operators (or their TransWeave-transported variants) are γ-Lipschitz in $\|\cdot\|_\infty$, so an ϵ-level defect in the intertwining relation cannot amplify arbitrarily when one passes from one application of the operator to the fixed point. The passage from a scalar field bound $\|V_I^* - \mathcal{T} V_J^*\|_\infty \le \delta$ (for the appropriate transport $\mathcal{T}$ induced by the map) to a graph inequality is the point where "local" information (at vertices/states) is turned into "relational" information (on edges/transitions), because flow graphs compare pairwise differences of potentials. The constant 2β arises because the similarity/flow weights depend exponentially on differences of values (or on symmetric differences across an edge), so a δ-uniform perturbation yields at worst a 2δ swing between two endpoints, and the temperature/sensitivity parameter β converts that swing into a multiplicative distortion bound. □

Remark 3389. *The proof's key conceptual move is the change of language: from operator dynamics $(B_w, B_{w'})$ to scalar fields (V_I^*, V_J^*) to similarity graphs (F_I, F_J) to quantale degrees (deg). Each step is simple, but the composition is powerful: it allows one to import the well-developed stability theory of contractions into a general-purpose structural notion of approximate morphism, of the sort Hyperseed uses for pattern-flow and correspondence claims [5, 1]. Concretely, the operator-level statement is "behavioral" (how an update rule propagates information), whereas the scalar-field statement is "semantic" (what optimality assigns to each state), and the graph statement is "structural" (which distinctions are preserved as strong/weak links). Quantale degrees then provide a single numerical carrier for these comparisons that is compatible with composition, so that one can treat approximate sameness as something that accumulates predictably rather than as an ad hoc tolerance. Viewed this way, the argument is also a controlled change of norms and representations: $\|\cdot\|_\infty$ bounds are chosen not for aesthetic reasons but because they interact cleanly with contraction estimates and immediately imply uniform bounds on all vertexwise quantities that feed into the flow construction.*

Corollary 62 (Flow of subjective time as iterative transweaving)**.** *If successive time-slices $I_t \to I_{t+1}$ admit TransWeave maps with errors ϵ_t, then for any horizon n,*

$$\mathrm{SCS}(I_0 \to I_n) \;\ge\; \exp\Big(-\frac{2\beta}{1-\gamma}\sum_{t=0}^{n-1}\epsilon_t\Big).$$

In particular, if the one-step errors are uniformly bounded by $\epsilon_t \leq \bar{\epsilon}$, then $\mathrm{SCS}(I_0 \to I_n) \geq \exp\left(-\frac{2\beta}{1-\gamma}\, n\bar{\epsilon}\right)$*, making explicit the linear-in-n accumulation in the exponent.*

Remark 3390. *This corollary expresses an austere but suggestive thesis: if the self is what survives transfer across adjacent time-slices, then the "speed" of subjective identity drift is governed by the accumulated TransWeave errors. Small errors per step can still yield substantial drift over long horizons, but only at a rate controlled by the sum $\sum_t \epsilon_t$ and the stability factor $1/(1-\gamma)$. The appearance of $1/(1-\gamma)$ separates two notions that might otherwise be conflated: (i) how accurately adjacent slices align (the ϵ_t terms) and (ii) how strongly downstream consequences are weighted relative to immediate structure (the discount γ). As $\gamma \to 1$, even tiny stepwise mismatches can become "sticky" in the fixed-point semantics, because the contraction becomes weak and the value function becomes more sensitive to persistent perturbations. Meanwhile β regulates how aggressively value differences are converted into flow/similarity distortions: larger β makes the graph more discriminating, so the same δ-level value uncertainty produces a smaller guaranteed degree.*

Proof. Apply the previous theorem on each step to get degree lower bounds q_t. Use the composition law for approximate morphisms, which multiplies degrees, and note $\prod_t e^{-c\epsilon_t} = e^{-c\sum_t \epsilon_t}$. Writing the one-step statement explicitly, the theorem gives $q_t \geq \exp\left(-\frac{2\beta}{1-\gamma}\epsilon_t\right)$ for each map $I_t \to I_{t+1}$. Compositionality of degrees then yields $\mathrm{SCS}(I_0 \to I_n) \geq \prod_{t=0}^{n-1} q_t$, because the composite TransWeave along the chain is witnessed by composing the one-step witnesses. Substituting the bound on each q_t and collecting terms in the exponent gives precisely the displayed inequality. □

Proof sketch. Lower-bound each one-step self-continuity by an exponential in ϵ_t, multiply the one-step bounds along the chain using compositionality, and then use the elementary identity that products of exponentials equal the exponential of the sum. Equivalently, one may take logs (moving to the additive "cost" picture) so that composition becomes addition of log-degrees, after which the result is immediate from $\log q_t \geq -\frac{2\beta}{1-\gamma}\epsilon_t$ and summing over t. □

Remark 3391. *The only nontrivial ingredient is remembering which algebra governs accumulation. In this section we have chosen a multiplicative quantale of degrees, so composition means multiplication and thus additive errors exponentiate. Other quantales would yield other accumulation laws, but the same compositional idea would remain. For example, a max-min (idempotent) quantale would make the "weakest link" dominate long-horizon continuity, whereas an additive quantale would treat degrees more like costs that sum directly rather than via a log transform. The present choice is natural when degrees are interpreted as similarity ratios or likelihood-style weights, since independent compositional steps typically multiply, and the exponential form is the standard way to convert additive perturbation bounds into multiplicative guarantees.*

61.4 MetaMo-driven identity drift and bounded transweave

Definition 553 (MetaMo-induced transweave (informal interface))**.** *Assume a motivational update $\alpha : X \to F(X)$ induces a weight transition $w \to w'$ (via a weight-extraction map U), and this triggers a TransWeave transformation $T_{w\to w'}$. We call this a MetaMo-induced TransWeave step.*

Remark 3392. *This definition is intentionally an "interface" rather than a full construction: it specifies how a motivational or meta-goal update can be coupled to a change in the effective reward weights of a control problem, which then invokes a TransWeave map between the old and new problem formulations. It is aligned with the broader Hyperseed idea that Goals: Implicit and explicit (Hyperseed-Concept* **??** *and* **??***) may evolve, and that such evolution should be studied through stability of the induced dynamics [10].*

Remark 3393. *Concretely, one may interpret X as the agent's internal "motivational configuration space" (or a space of latent goal descriptors), and $F(X)$ as an "updated configuration bundle" containing both the revised motivational state and any auxiliary structures needed to evaluate it (e.g. new internal criteria, revised salience maps, or new constraints). The map α is then the meta-level update that changes how the agent evaluates outcomes, while U extracts from the updated motivational state a finite set of coefficients w that parameterize an explicit reward model (for instance, a linear combination of basis rewards, or weights on multiple objectives). In this reading, the interface character of the definition is a feature: it allows α to be complex and high-dimensional while still inducing a tractable parametric change $w \to w'$ in the control problem to which TransWeave applies.*

Remark 3394. *In cognitive terms, one may think: "I have reweighted what I care about." The question then is whether the new optimization landscape is commensurate with the old one, or whether it induces a sharp rupture. A MetaMo-induced TransWeave step is precisely a claim that the reweighting is bridgeable by an approximate intertwining of the relevant Bellman operators [7].*

Remark 3395. *The phrase "commensurate" here is meant in a technical, operator-theoretic sense rather than a purely semantic one: the old and new Bellman operators (associated to w and w') should be close after transport by $T_{w\to w'}$. When this holds, planning and value propagation in the new objective can be viewed as a controlled deformation of planning and value propagation in the old objective, rather than a discontinuous replacement. This is the precise way in which "reweighting what I care about" becomes an analyzable perturbation of an optimization landscape.*

Theorem 163 (Bounded motivational drift implies bounded identity drift)**.** *Assume each motivational update step induces a TransWeave map with intertwining error $\epsilon \leq \bar{\epsilon}$ (e.g. due to homeostatic stability/modulator continuity). Then the system's self-continuity across the corresponding time step satisfies*

$$\mathrm{SCS}(I \to J) \;\geq\; \exp\Big(-\frac{2\beta}{1-\gamma}\bar{\epsilon}\Big).$$

Moreover, over n steps, the lower bound composes multiplicatively as in Corollary 1.

Remark 3396. *The parameters in the bound have an intuitive reading consistent with the underlying definitions: γ is the discount factor that controls effective horizon length (hence the appearance of the geometric factor $1/(1-\gamma)$), while β is the "sharpness" parameter in the self-continuity construction (so that larger β makes the continuity score more sensitive to value differences). The exponential form expresses that small but persistent operator mismatch accumulates into a multiplicative degradation of continuity, rather than an additive one.*

Remark 3397. *The statement is a stability claim about identity: if the meta-level reweightings of value are uniformly gentle (in the sense that they always admit low-error TransWeaves), then identity drift is uniformly bounded from below (in the sense of self-continuity degree). This gives a rigorous way to express an old intuition: "character changes gradually when motivations change gradually."*

Remark 3398. *The theorem should also be read as a* design constraint*: bounded identity drift is ensured not by freezing goals, but by ensuring that any goal update stays within a region of "TransWeavable" perturbations. In practical terms, this can correspond to limiting the magnitude of a weight update $w \to w'$ per step, enforcing smoothness of the map $U \circ \alpha$, or introducing regularizers that penalize updates that would break approximate intertwining (for example, updates that change which features are rewarded in a way that is not representable by a small deformation of value functions).*

Remark 3399. *This theorem is not a psychological law; it is a conditional mathematical implication connecting one formal measure of motivational change (intertwining error) to one formal measure of identity continuity (SCS). Its usefulness is that it lets a designer or analyst set tolerances: if* $\bar{\epsilon}$ *is kept small by design (e.g. via constraints on self-modification), then a quantitative continuity guarantee follows.*

Remark 3400. *The multiplicative composition over* n *steps is especially important for long-running systems: even when each step is locally "safe" (small* ϵ*), the global continuity can still decay if updates are frequent. Corollary 1 captures this by treating successive steps as a chain of transitions* $I_0 \to I_1 \to \cdots \to I_n$ *and lower-bounding* $\mathrm{SCS}(I_0 \to I_n)$ *by the product of the one-step lower bounds. Thus, bounding* $\bar{\epsilon}$ *is necessary but may be insufficient unless one also controls update rate or periodically re-anchors the representation used by* T*.*

Proof. Immediate from the TransWeave-to-SCS theorem and the horizon composition corollary. □

Remark 3401. *Spelled out at the level of inequalities, the argument is that the assumed uniform upper bound* $\epsilon \le \bar{\epsilon}$ *lets one replace the per-step factor* $\exp\!\big(-\frac{2\beta}{1-\gamma}\epsilon\big)$ *by the worst-case factor* $\exp\!\big(-\frac{2\beta}{1-\gamma}\bar{\epsilon}\big)$*, and then apply the already-established composition rule. No additional structural properties of* α *or* U *are needed beyond those required to invoke the TransWeave-to-SCS bridge result.*

Proof sketch. Combine the single-step lower bound on SCS derived from a TransWeave error, then apply the chain composition rule over multiple steps. □

Remark 3402. *The proof is immediate because all the difficulty has been front-loaded into the bridge theorem: once error-to-degree has been established, boundedness and composition are essentially algebraic bookkeeping.*

Remark 3403. *One may also view the theorem as isolating* where *identity fragility can enter: either the MetaMo update* α *induces a change* $w \to w'$ *for which no sufficiently accurate intertwiner exists, or the discount/horizon parameters make even small* $\bar{\epsilon}$ *consequential. In particular, as* $\gamma \to 1$ *(very long-horizon valuation), the bound weakens, reflecting the fact that long-horizon value functions amplify small systematic mismatches in Bellman updates. This is not a defect of the bound but a faithful representation of how long-term planning couples distant consequences back into present evaluations.*

61.5 Paradigm commensurability as existence of low-error transweaves

Definition 554 (Paradigm as a problem-solving category)**.** *A paradigm* $\mathcal{P}$ *consists of: (i) a collection of tasks/problems, each modeled by an MDP with Bellman operator; (ii) a collection of solution-transform morphisms (including representation changes).*

Remark 3404. *In clause (i), the role of the Bellman operator is to provide a canonical "one-step reasoning/update" map for each task, so that comparing paradigms becomes a matter of comparing their induced update dynamics rather than only their terminal answers. This is particularly important when a paradigm includes iterative procedures (e.g. planning, policy improvement, inference loops) whose quality is best assessed by stability of their iteration operators.*

Remark 3405. *Clause (ii) is meant to include not only literal changes of state variables, but also changes of modeling language (e.g. switching from symbolic predicates to learned embeddings), changes of action parameterizations, or changes of reward representations, provided there is an induced, well-defined transformation at the level of solution objects (policies, value functions, or other sufficient statistics). In this sense, a paradigm includes its own "allowed translations," so commensurability asks whether two such translation toolkits can be coordinated across paradigms.*

Remark 3406. *The word "paradigm" is often used in a sociological or historical key. Here it is given a spare technical reading: a paradigm is a structured family of problems together with permissible transformations of solutions and representations. This is consonant with a computational philosophy of science where scientific practice is seen as a network of question, experiment, model, and update procedures, rather than as a set of static propositions [20, 1].*

Remark 3407. *Calling it a "category" is suggestive: tasks are objects, and transforms (including representation changes) are morphisms. We do not need categorical axioms explicitly in this section; the point is that we treat paradigms as* systems of translatable problem-solvers, *so commensurability becomes a statement about the existence of good translation functors, i.e. TransWeaves [7].*

Remark 3408. *Even without invoking functoriality formally, the category language emphasizes two technical expectations that are easy to miss: (1) transforms should be composable (one can chain multiple representation changes), and (2) there should be identity-like transforms (doing nothing is an allowed morphism). These expectations motivate why commensurability is framed in terms of operator intertwining: it is a robust way to express that "translation commutes with solving" up to controlled error.*

Definition 555 (TransWeave commensurability)**.** *Two paradigms $\mathcal{P}$ and $\mathcal{Q}$ are (ϵ, γ)-commensurable on a class of tasks if there is an assignment that maps each task $M \in \mathcal{P}$ to a task $N \in \mathcal{Q}$ together with a TransWeave map (T, T_V) satisfying*

$$\|T_V \circ B_M - B_N \circ T_V\|_{\infty\to\infty} \leq \epsilon,$$

uniformly over the task class.

Remark 3409. *The "assignment" component encodes which tasks are being compared: commensurability is not claimed between arbitrary tasks, but between those deemed corresponding under the paradigm-to-paradigm mapping. Technically, this can be viewed as selecting a relation or partial matching between task families, and then requiring a single uniform error tolerance for all matched pairs.*

Remark 3410. *The choice of the operator norm $\|\cdot\|_{\infty\to\infty}$ is consistent with standard contraction arguments for discounted MDPs: it controls worst-case deviations over all bounded value functions. Interpreted procedurally, the inequality says that "do one Bellman update in $\mathcal{P}$ and then translate" is close to "translate and then do one Bellman update in $\mathcal{Q}$," with the discrepancy bounded by ϵ at every step, across the whole task class.*

Remark 3411. *This definition formalizes "commensurability" as* uniform approximate intertwining*: the paradigms are commensurable on a task class if one can systematically transport problem-solving dynamics from one to the other with bounded operator mismatch. The inclusion of γ indicates that the tolerance is interpreted in a discounted, contractive regime; without contraction one cannot generally convert dynamical mismatch into fixed-point mismatch.*

Remark 3412. *The parameter γ should be read as the effective contraction modulus for the Bellman operators on the relevant function space. In the simplest discounted setting, γ is the usual discount factor; more generally it can stand for any bound ensuring that B_M and B_N are γ-contractions in $\|\cdot\|_\infty$. This is what permits a stepwise commutator bound to accumulate geometrically rather than linearly when one compares infinite-horizon fixed points.*

Remark 3413. *A simple example: if $\mathcal{P}$ and $\mathcal{Q}$ use different state representations for essentially the same tasks, then T may be a representation change and T_V its induced pullback/pushforward on values. If the representational change is faithful, the commutator is small, yielding small ϵ. Conversely, if crucial dynamics are not preserved by the representation change, ϵ becomes large and commensurability fails.*

Remark 3414. *Concretely, when T is a many-to-one abstraction map (state aggregation), T_V typically acts by restricting attention to the coarse state features; the commutator then measures whether aggregated states are approximately bisimilar with respect to rewards and transition structure. When T is an embedding into a richer state space, T_V acts by pullback along the embedding, and the commutator measures whether the extra degrees of freedom introduced by the richer representation are in fact irrelevant to optimal control for the mapped tasks.*

Theorem 164 (Commensurability implies transferable optimality)**.** *If $\mathcal{P}$ and $\mathcal{Q}$ are (ϵ, γ)-commensurable on a task class, then for every mapped pair (M, N) in the class,*

$$\|T_V(V_M^*) - V_N^*\|_\infty \leq \frac{\epsilon}{1-\gamma}.$$

Thus hypotheses/procedures that are optimal in $\mathcal{P}$ transfer to $\mathcal{Q}$ with bounded value degradation.

Remark 3415. *The inequality has the standard fixed-point stability form: a one-step error bound ϵ amplifies by at most the geometric series factor $1/(1-\gamma)$ when comparing fixed points of contractive maps. In particular, as $\gamma \to 1$ the bound becomes conservative, reflecting the well-known fact that near-undiscounted problems require much tighter stepwise control to ensure small changes in the infinite-horizon value.*

Remark 3416. *The theorem says that commensurability in the dynamical sense is enough to guarantee commensurability in the solution sense: optimal value functions transfer with bounded distortion. This provides a rigorous alternative to the often-vague debate about whether paradigms are "incommensurable": one can point to a specific operator norm defect ϵ and compute its worst-case consequence for optimality.*

Remark 3417. *Although stated for optimal value functions, the same style of estimate applies to other fixed points or attractors of interest, such as soft Bellman operators (entropy-regularized control) or evaluation operators for a fixed policy. The common structure is: approximate intertwining of the update map implies closeness of the corresponding equilibria, provided the update map is contractive in an appropriate norm.*

Remark 3418. *The result is also a bridge back to Hyperseed's view of science as a community of empirically coupled belief updates: if two scientific paradigms can transweave their problem-solving operators with small commutator, then they can share solutions and predictions with bounded degradation, even if their internal representations differ [1, 20].*

Remark 3419. *From this perspective, "sharing predictions" corresponds not to literal agreement on internal explanatory posits, but to the existence of stable translation interfaces that preserve the operational content of inference and decision procedures. The TransWeave condition isolates precisely this operational content at the level of update operators.*

Proof. Apply the intertwining-implies-value-closeness theorem taskwise. □

Remark 3420. *Unpacking the one-line proof: for each mapped pair (M, N), the commensurability hypothesis provides an ϵ-approximate intertwining between B_M and B_N via T_V, and the contraction parameter γ ensures the hypotheses of the fixed-point stability lemma. The conclusion then follows by substituting (B_M, B_N, T_V) into that lemma and using that the bound is uniform over the task class.*

Proof sketch. Reduce each task pair (M, N) to the single MDP theorem: the same bound applies because the commensurability hypothesis is exactly the intertwining hypothesis, uniformly over the class. $\square$

Remark 3421. *The "uniformly over the class" clause is doing essential work: it ensures that a single tolerance ϵ controls the entire family rather than yielding a different error parameter for each task. Without uniformity, one could still derive taskwise bounds, but paradigm-level commensurability would become weaker and less predictive, since worst-case transfer quality could be arbitrarily bad on rare but important tasks.*

Remark 3422. *The proof is short because the theorem is an organizational statement: it packages the earlier fixed-point stability bound into a paradigm-level assertion by applying it consistently across a family of tasks.*

Theorem 165 (A practical incommensurability certificate via large commutators)**.** *Let U be a proposed transfer operator on value functions between paradigms and define a BD commutator (mismatch)*

$$C := B_Q U - U B_P.$$

If $\|C\|_{\infty\to\infty} > \epsilon_0$, then no transfer based on U can guarantee better than $\epsilon_0/(1-\gamma)$ uniform value preservation, hence commensurability at tolerance ϵ_0 fails for at least one task in the class.

Remark 3423. *The appearance of the commutator is natural: if U were an exact intertwiner, we would have $B_Q U = U B_P$ and the two "update-then-transfer" pathways would coincide. The norm $\|C\|_{\infty\to\infty}$ quantifies the worst-case deviation from this ideal commutation, so it directly measures the obstruction to viewing U as a TransWeave witness.*

Remark 3424. *This theorem states a falsification criterion: if one proposes a particular transfer operator U, one can compute (or estimate) the size of its commutator with the Bellman operators. A large commutator means there exists some input value function on which the two update-and-transfer routes disagree substantially, and therefore U cannot serve as a uniform commensurability witness.*

Remark 3425. *Operationally, one can read $\|C\|_{\infty\to\infty} > \epsilon_0$ as: there exists a bounded "test" value function V such that $\|B_Q(UV) - U(B_P V)\|_\infty > \epsilon_0$. This provides guidance for diagnostics: adversarially searching over V (or sampling a rich set of candidate V's) can reveal where the proposed translation U fails to preserve the update semantics.*

Remark 3426. *In practice this is valuable because it separates two questions: (1) whether* some *transfer exists, and (2) whether a* particular *candidate transfer U is adequate. The theorem addresses (2) and provides a computationally tangible obstruction to commensurability in the TransWeave sense [7].*

Remark 3427. *Note that the obstruction is relative to the chosen U: a large commutator does not prove absolute incommensurability between $\mathcal{P}$ and $\mathcal{Q}$, only that this proposed interface fails at the specified tolerance. As a methodological point, this supports iterative design of translation layers: one can improve U (or restrict the task class) to reduce the commutator norm, thereby moving from practical incommensurability toward commensurability in the approximate-intertwining sense.*

Proof. If $\|C\|_{\infty\to\infty} > \epsilon_0$, then the intertwining condition with tolerance ϵ_0 is violated for some input value function. For that tolerance, the value-closeness bound cannot be derived; contrapositive yields the claim. □

Proof sketch. A commutator norm bound is exactly the intertwining inequality. If it is violated, then there is at least one input witnessing the violation; for that input, the chain of inequalities leading to the fixed-point closeness bound cannot be carried out. Therefore one cannot guarantee the corresponding uniform preservation claim. □

Remark 3428. *The logic is intentionally Russellian in its severity: it does not say transfer is impossible, only that a proposed form of transfer fails to justify a certain guarantee. This is often the right epistemic posture in science: one can refute a method, or a bound, without refuting the phenomenon one hoped the method would capture.*

61.6 Inter-mind transweaving, cooperation, and mindplex coherence

61.6.1 Transweaving on a graph of minds

Definition 556 (Transweave graph of minds)**.** *Let A be a set of minds/agents. A directed edge $i \to j$ carries a TransWeave map $T_{i\to j}$ with error $\epsilon_{i\to j}$. For a path $p : i = i_0 \to i_1 \to \cdots \to i_m = j$, define the composed map $T_p := T_{i_{m-1}\to i_m} \circ \cdots \circ T_{i_0\to i_1}$.*

Remark 3429. *This definition recasts a social or multi-agent setting as a transport network: edges are not merely communication channels, but structured translation maps that carry policies/values/solutions from one agent's representational world into another's. The graph may be sparse (few reliable translations) or dense (many). The later mindplex remarks interpret density as enabling redundant short translation paths, which improves coherence.*

Remark 3430. *In Hyperseed's language, this is one way to model the emergence of collective cognitive structure: a society of agents becomes more than a set when there exist high-quality inter-agent morphisms that allow patterns to flow across individuals [5, 1]. This resonates with the more explicit "mindplex" formalism in SuperDuperPsychism [12].*

Theorem 166 (Path-length penalty for inter-mind solution transfer)**.** *Assume each $T_{i\to j}$ satisfies an intertwining bound with the same discount γ. For any path p of length m from i to j, the transported optimal values satisfy*

$$\|T_{p,V}(V_i^*) - V_j^*\|_\infty \le \frac{1}{1-\gamma} \sum_{t=0}^{m-1} \epsilon_{i_t\to i_{t+1}}.$$

Equivalently, in degree form,

$$\deg(T_p) \;\ge\; \exp\Big(-\lambda \sum_{t=0}^{m-1} \epsilon_{i_t\to i_{t+1}}\Big).$$

Remark 3431. *The theorem says that translation fidelity across a social chain deteriorates with path length, in a very concrete way: additive intertwining errors accumulate additively (modulo the $1/(1-\gamma)$ factor), and therefore multiplicative degrees decrease exponentially with the accumulated error. Thus, even if each local translation is good, long chains can become poor witnesses of commensurability.*

Remark 3432. *This matters for mindplex formation because it gives a quantitative reason to prefer dense, redundant connectivity over thin, serial dependence. In social terms: if knowledge must pass through many intermediaries, the effective distortion grows; if there are multiple short routes, distortion can be kept small. This kind of argument also aligns with general pro-social efficiency claims about well-connected agent groups [6].*

Proof. Iterate the single-step bound:

$$\|T_{i\to j,V}(V_i^*) - V_j^*\|_\infty \le \epsilon_{i\to j}/(1-\gamma),$$

and apply triangle inequality across the composed path. For the degree form, apply Φ_λ to the sum of errors. □

Proof sketch. Apply the fixed-point stability bound to each edge to bound the one-step transported value error, then use the triangle inequality to sum errors along the path. Finally convert the additive error bound into a multiplicative degree bound using the exponential embedding Φ_λ. □

Remark 3433. *The triangle inequality is again the quiet workhorse: it formalizes that distortions accrued at different translation steps do not cancel in the worst case. The degree form is best seen as a change of coordinates: sums become products, and products become the natural language of compositional morphism degrees.*

61.6.2 Mindplex coherence from transweaving-based replaceability

Definition 557 (Observer-indexed coherence proxy)**.** *Fix an observer context O with a behavior summary* $\mathrm{Beh}_O(\cdot)$ *and a distance d_O. For a set of components S, define replaceability error*

$$\mathrm{Err}_O(x;S) := \inf_f d_O\Big(\mathrm{Beh}_O(x), f(\mathrm{Beh}_O(S\setminus\{x\}))\Big),$$

and coherence

$$\mathrm{Coh}_O(S) := 1 - \max_{x\in S} \mathrm{Err}_O(x;S),$$

after normalizing so errors lie in $[0,1]$.

Remark 3434. *This definition treats coherence as* replaceability*: a component x is well-integrated into a set S if the rest of the set can approximately reproduce x's behavior (as seen by observer O) via some approximator f. The infimum over f expresses an idealization: we score x by its best possible reconstruction from the others. The maximum over x then enforces a worst-case standard: the set is only as coherent as its least replaceable member.*

Remark 3435. *The dependence on O is essential and principled. What counts as "the behavior" of a mind or subsystem depends on which aspects one observes and which distance one uses; hence coherence is observer-indexed rather than absolute. This matches Hyperseed's general practice of tracking observer-context explicitly, and it is also consonant with mindplex discussions where coherence is partly an epistemic relation (Hyperseed-Concept 113) [1, 12].*

Theorem 167 (Local transweaving implies global coherence lower bound)**.** *Assume that for each $x\in S$ there exists an approximator f_x built from TransWeave transports among members of $S\setminus\{x\}$ such that*

$$d_O\Big(\mathrm{Beh}_O(x), f_x(\mathrm{Beh}_O(S\setminus\{x\}))\Big) \le \eta_x.$$

Then

$$\mathrm{Coh}_O(S) \;\geq\; 1 - \max_{x \in S} \eta_x.$$

In particular, if every agent can be reconstructed from neighbors with uniformly small transport-induced error, S is coherent in the Hyperseed replaceability sense.

Remark 3436. *The theorem is almost tautological, yet it is useful: it says that if you can exhibit explicit reconstructions f_x with explicit errors η_x, then the global coherence score has an immediate lower bound. In complex systems one often has local engineering knowledge (how well one can approximate one component from others) and wants a global guarantee; this theorem is the clean bridge from local certificates to a global bound. In particular, the hypothesis "for each x there exists a concrete f_x with error at most η_x" is exactly the kind of statement one can obtain from implementation details, empirical evaluations, or module-level contracts, whereas the conclusion is phrased purely in terms of* $\mathrm{Coh}_O(S)$ *and thus is independent of how the approximators were found. Conceptually, the point is that* one good candidate *is enough to upper-bound an infimum, so the global statement can be certified without having to solve any of the optimization problems defining the* $\mathrm{Err}_O(x; S)$ *exactly.*

Remark 3437. *The connection to the earlier path-length penalty is indirect but important: if the f_x are assembled from TransWeave maps, then the η_x can often be bounded in terms of path lengths and edge errors in the transweave graph. Thus coherence can be bounded using purely graph-theoretic features (diameter, redundancy) together with operator-level intertwining tolerances. More explicitly, if an approximator f_x is realized by composing transport/intertwining maps along a path from some subset of minds to the target x, then η_x typically accumulates from (i) the per-edge approximate-commutation errors and (ii) any stability constants of the observables under transport; both effects are commonly monotone in the number of compositions, hence in the path length. This is the sense in which a local "wiring diagram" in the transweave graph can be turned into a numerical η_x that is then fed into the theorem as a plug-in certificate.*

Proof. For each x, the definition of $\mathrm{Err}_O(x; S)$ is an infimum over approximators, so $\mathrm{Err}_O(x; S) \leq \eta_x$. Taking maxima over x yields $\max_x \mathrm{Err}_O(x; S) \leq \max_x \eta_x$ and hence the coherence bound. The only logical move is the standard inequality $\inf A \leq a$ for any particular $a \in A$, applied with A the set of achievable approximation errors at x and $a = \eta_x$ the error of the exhibited candidate f_x. No structural assumptions (linearity, convexity, or uniqueness of the best approximator) are needed, because the argument never attempts to compare *different* candidates except through the ordering induced by $\leq$. □

Proof sketch. Plug the candidate approximators f_x into the infimum defining $\mathrm{Err}_O(x; S)$ to upper-bound each replaceability error by η_x, then take the worst case over x and translate it into a lower bound on $\mathrm{Coh}_O(S)$. Equivalently, the construction produces a vector of per-node certificates $(\eta_x)_{x \in S}$, and the coherence score only depends on the worst (largest) such certificate once the $\max_x$ aggregation is applied. □

Remark 3438. *The proof works because the definitions are chosen with a particular logic: coherence is designed to be monotone in the availability of good approximators. Once one sees that* inf *and* max *are arranged to make a worst-case guarantee, the inequality becomes the natural algebraic expression of that guarantee. This also clarifies the intended interpretation of* $\mathrm{Coh}_O(S)$ *as a robustness notion: it is not the* average *mutual predictability, but the requirement that* every *component in S be replaceable (up to a stated tolerance) from the others. In settings where one wants a "no single point of semantic failure" certificate, the* max *is the correct aggregator; the theorem then* *simply says that any family of explicit reconstructions automatically provides such a certificate at the corresponding tolerance level.*

Remark 3439 (Why dense neighbor transweaving helps mindplex formation)**.** *The path-length penalty theorem shows that the effective transport error between two minds grows with the length of the best path connecting them in the transweave graph. Thus, a mindplex (a notably coherent society of minds) is favored when each mind can transweave with multiple neighbors so the graph has small diameter and multiple short redundant paths. In the language of replaceability, dense neighborhood structure makes it more plausible that for each mind x there exists a small set of others from which x can be approximately reconstructed with a controlled η_x, because information need not traverse long chains of intertwining where errors would accumulate. Put differently, increased local connectivity can improve the* availability *of short, accurate reconstruction circuits, which then feeds directly into global coherence lower bounds via the preceding theorem.*

Remark 3440. *One may add a complementary intuition: redundancy is not merely about shortening paths, but about providing alternative routes when some routes fail or are contextually inappropriate. In a living community of minds, tasks and contexts shift; redundant transweave paths provide a kind of structural resilience, making it more plausible that coherence persists across changing demands. This resonates with the general claim that cooperative structures can be efficiency-enhancing in broad multi-agent settings [6], and with mindplex-oriented formalisms emphasizing integration and collective stability [12]. A further way to see this is to distinguish* reachability *from* reliability*: even if two minds are connected by a short path, the associated approximator f_x might be brittle if it depends on a single specialized intermediary, whereas multiple partially independent paths can yield multiple candidate reconstructions whose errors can be compared and whose outputs can be cross-validated. In practice, this means redundancy can reduce the* effective *worst-case η_x not only by shortening the best path, but by enabling the selection of a path (or composite strategy) that is best suited to the current context, observables, or noise regime. From the standpoint of mindplex coherence, this makes it more plausible that the same society can maintain low replaceability errors across a wider range of tasks without requiring global reconfiguration.*

61.7 Mind-world correspondence as a transweave relation

Definition 558 (Mind-world transweaving)**.** *Let M_{int} be a mind's internal predictive-control model and let M_{env} be an environment model (as reconstructed by an observer). If there exists a TransWeave map from M_{int} to M_{env} with small intertwining error, we say the mind can transweave its control knowledge into the world model.*

Remark 3441. *In concrete Markovian settings one may think of M_{int} and M_{env} as each specifying (possibly different) state spaces, action parametrizations, reward proxies, and transition kernels, hence inducing Bellman operators T_{int} and T_{env}. The TransWeave condition can then be read as the existence of a transport map (often accompanied by a suitable pullback/pushforward pair on value functions or Q-functions) for which the diagram "approximately commutes," e.g. in the schematic form $\|T_{\text{env}} \circ W - W \circ T_{\text{int}}\| \le \epsilon$ in an appropriate operator norm. This makes explicit which part of "correspondence" is being demanded: not identity of representations, but approximate invariance of the decision-relevant fixed-point structure.*

Remark 3442. *This definition proposes a crisp operationalization of Mind-World Correspondence (Hyperseed-Concept 112): the mind corresponds to the world, in the relevant control sense, if its internal model can be translated into the observer's environment model while nearly preserving the*

Bellman dynamics (and thus near-preserving optimal decision structure). It makes correspondence not a mystical matching but a quantified transport relation [7, 1].

Remark 3443. *Note that the clause "as reconstructed by an observer" makes the correspondence explicitly epistemic: M_{env} may itself be an inferred or learned model and need not coincide with the world-as-such. The definition therefore captures a practically testable notion of adequacy relative to an observational interface (sensors, experiment design, logging policy), and it clarifies that failures of correspondence can arise either from the mind's internal mismatch or from limitations/approximations in the observer's reconstruction.*

Remark 3444. *A simple example is representational: the mind may encode states in its own latent space, while the observer encodes environment states differently. A TransWeave map then plays the role of a learned or inferred decoder/encoder pair that allows control-relevant knowledge to be moved between these two descriptive schemes. This is closely aligned with engineering views of cognition emphasizing model-based control and representation learning [19].*

Remark 3445. *The TransWeave relation is not required to be symmetric. In many realistic cases the mind can reliably export control knowledge into M_{env} (because M_{env} is more expressive, or because the observer has privileged access to ground-truth variables), while the reverse transport may be ill-posed. This asymmetry is natural when M_{int} is compressed, amortized, or shaped by the agent's affordances, and it matches the intuition that "the world can explain the mind" is a different demand than "the mind can explain the world" even when both are framed in control terms.*

Theorem 168 (TransWeave implies a lower bound on mind-world correspondence)**.** *Let F_{int} and F_{env} be the induced value-flow graphs for M_{int} and M_{env}. If a TransWeave map exists with intertwining error ϵ, then there exists a map f such that*

$$\deg(f; F_{\mathrm{int}}, F_{\mathrm{env}}) \;\geq\; \exp\Big(-\frac{2\beta}{1-\gamma}\epsilon\Big).$$

Thus TransWeave provides an explicit witness of high-degree approximate morphism between internal and external flow structures.

Remark 3446. *The parameters in the bound track standard control-theoretic sensitivities: γ captures horizon-length amplification (as $\gamma \to 1$, small Bellman mismatches accumulate over long rollouts), while β captures how sharply the degree functional penalizes edgewise deviations in the value-flow comparison. Accordingly, the same ϵ can correspond to substantially different correspondence guarantees depending on the effective horizon and the chosen degree temperature.*

Remark 3447. *The theorem states that mind-world correspondence, when cashed out in this value-flow/degree language, is not a separate primitive: it is the same bridge inequality already proved for self-continuity, with "earlier time-slice" replaced by "internal model" and "later time-slice" replaced by "environment model." In short: correspondence is continuity across a boundary rather than across time.*

Remark 3448. *One can also read the result as a control-commensurability statement: if two descriptions support nearly the same Bellman back-ups after transport, then they induce nearly the same comparative advantage relations between actions/states, hence nearly the same flow of "where value goes" under optimal or near-optimal behavior. This helps justify the flow-graph proxy F as a representation-invariant object: it forgets irrelevant coordinate details while retaining the parts that matter for decision structure.*

Remark 3449. *This is conceptually important because it suggests a unification: the same formal apparatus can treat persistence of self, transfer between paradigms, coherence of groups, and adequacy of world-modeling. What changes are the systems being related and the tolerances one is willing to accept; what stays fixed is the logic of approximate intertwining and approximate morphism degree.*

Proof. Same as the TransWeave-to-SCS theorem, replacing time-slices by (internal model, environment model). □

Proof sketch. Reuse the earlier argument verbatim: intertwining error yields a bound on transported optimal values, the value-perturbation lemma yields an edgewise flow-graph inequality, and that inequality is precisely a lower bound on morphism degree. □

Remark 3450. *Spelled out slightly more concretely, the instantiation proceeds by (i) transporting $V^\star_{\mathrm{int}}$ (or $Q^\star_{\mathrm{int}}$) through the TransWeave map into the* env *description, (ii) using the ϵ-approximate commutation to show this transported object is an ϵ'-approximate fixed point of the* env *Bellman operator, (iii) applying the same contraction-based perturbation argument to relate it to $V^\star_{\mathrm{env}}$, and (iv) translating the resulting uniform value discrepancy into the edgewise inequalities defining high degree between F_{int} and F_{env}. The only change from the temporal case is the interpretation of the two sides of the transport: they index "model boundary" rather than "time."*

Remark 3451. *The proof is a template instantiation. This is mathematically modest but methodologically notable: it exemplifies how choosing the right abstraction makes many theorems become corollaries of a single bridge lemma.*

61.8 Optional appendix: dependency towers and transweave

Remark 3452. *If one further represents a paradigm or a mind's procedure library by dependency-tower layers (low-order dependencies first), then TransWeave-compatibility questions can be asked layerwise: does a transweave map send the kth-order approximation layer to the kth-order layer (or near it)? Such theorems make commensurability and cooperation criteria more granular: one may be commensurable at low dependency order but incommensurable at high order. One convenient way to formalize this is to regard the dependency tower as a filtration $(F_k)_{k\geq 0}$ on the relevant object (e.g. a space of models, programs, or procedures), where F_k is "everything constructible using at most k layers of dependence," and then ask for a* filtered *(or approximately filtered) TransWeave map T satisfying*

$$T(F_k) \subseteq F_{k+\Delta} \qquad \text{for small } \Delta,$$

or more stringently $T(F_k) \subseteq F_k$ for exact layer preservation. The phrase "or near it" can then be read as allowing controlled leakage across adjacent layers, measured by an application-appropriate norm or divergence (e.g. a risk gap on tasks of depth $\leq k$, an operator norm bound on residual components outside F_k, or a bound on failure probability when projecting back to F_k). In this filtration language, "commensurable at low order" means that the induced map on low layers (or on associated graded pieces $\mathrm{gr}_k := F_k/F_{k-1}$) is well-behaved and informative, while "incommensurable at high order" means that the map fails to respect or accurately transport the fine structure that only appears in deeper layers.

Remark 3453. *This remark connects to the Dependency Tower viewpoint (Hyperseed-Concept 94) and the associated towerization/tradeoff mathematics [8, 9]. Layerwise TransWeave analysis would amount to asking whether the transfer respects the coarse-to-fine organization of dependence: perhaps two paradigms agree on low-order regularities (and thus transfer well for simple tests), yet* *diverge in high-order structure (so transfer fails for subtle interventions). This provides a precise sense in which "partial commensurability" can be true without contradiction. A useful operational reading is that low-order agreement supports robust* predictive *alignment on ordinary regimes, while high-order disagreement manifests under distribution shift, counterfactual intervention, or long-horizon planning—settings where deep dependencies are repeatedly composed. In such cases, a layerwise criterion can be stated as: for tasks whose solution lives primarily in F_k, the transferred solution $T(x)$ should remain accurate after projection back to F_k, whereas for tasks requiring F_{k+m} structure the same transfer may systematically mis-specify the relevant dependencies. When the TransWeave construction is expressed via approximate intertwining (e.g. $T \circ A \approx B \circ T$ for salient operators A, B encoding update rules, inference steps, or dynamics), the layerwise refinement asks for these approximate intertwinings to hold* uniformly across levels *or with an error that grows slowly with k:*

$$\|TA|_{F_k} - BT|_{F_k}\| \leq \varepsilon_k,$$

with ε_k small for the layers one intends to treat as commensurable. This explicitly separates "we can translate each other's basic notions and replicate baseline competence" (small ε_k for small k) from "we can safely import each other's subtle causal or normative commitments" (requiring control of ε_k at high k). In the tradeoff perspective, this also illuminates why cooperation may succeed in low-dependency domains (shared coarse structure, low tower depth) yet fail in high-dependency domains (tower-depth amplification), and suggests a quantitative handle: one can treat the sequence (ε_k) or the minimal feasible Δ as a commensurability profile across dependency order.

62 Consciousness Locations, PNSE, and Mindplexes: A Bridge Between SuperDuperPsychism and the Hyperseed Ontology

We connect the SuperDuperPsychism (SDP) formalization of consciousness (GCC/RC, Locations, and collective Locations) with the Hyperseed ontology's formal interface for consciousness (presentational immediacy, workspace closure, reflective fixed points, and mindplexes). We introduce a translation layer and then state and prove a collection of theorems that are meant to be directly useful as reasoning tools: monotone parameter-to-Location mappings, formal PNSE criteria, collective stability/integration thresholds, and mindplex development results.

The intended role of this section is not to rebuild either framework from scratch, but to exhibit a minimal set of shared "handles" that let SDP-style Location claims be stated and checked inside the Hyperseed interface. In particular, we will treat an SDP Location (or collective Location) as something that can be tracked by (i) which propositions are consciously supported above threshold, (ii) whether the workspace dynamics admit stable states, and (iii) whether the system supports a reflective fixed point with sufficiently robust self-related content. The rest of the paper can then use these handles to formulate PNSE-like criteria and mindplex-level development statements without committing to a single implementation-level story.

62.1 Preliminaries: a minimal Hyperseed consciousness interface

We use a deliberately small slice of Hyperseed's consciousness formalism. The guiding principle is that we keep only those components needed to (a) represent graded support for content, (b) impose a global-workspace-style closure condition, and (c) express a reflective layer in which some of the content is about the system's own content. This is enough to state fixed-point and robustness conditions that later serve as proxies for "having a stable conscious state" and "having an integrated reflective self" in the sense required for the SDP–Hyperseed bridge.

Definition 559 (Evidence values and evidence states)**.** *Let $V := [0,1]^2$ with the componentwise order. An* evidence state *for an observer-context* Ob *is a map $E : L_{\mathrm{Ob}} \to V$, where L_{Ob} is a finite set of propositions. Write $E(\varphi) = (E^+(\varphi), E^-(\varphi))$. For $\theta \in (0,1)$ define the* conscious proposition set

$$\mathrm{Con}_\theta(E) := \{\varphi \in L_{\mathrm{Ob}} : E^+(\varphi) \geq \theta\}.$$

Here $V = [0,1]^2$ is used to keep track of two independent grades: a "for" component $E^+(\varphi)$ and an "against" component $E^-(\varphi)$. The componentwise order makes V a product poset, so increasing evidence in either component is formally meaningful and compatible with monotonicity assumptions introduced later. We do not require any special relationship between $E^+(\varphi)$ and $E^-(\varphi)$ (such as summing to 1); this allows the interface to represent uncertainty (both low), conflict (both high), and ordinary confidence (high/low or low/high) without additional machinery. The finiteness of L_{Ob} is a technical convenience that will later support existence arguments for stable points of monotone operators, and it also matches the intended use: L_{Ob} is a contextually selected "working" vocabulary rather than the set of all expressible sentences.

The set $\mathrm{Con}_\theta(E)$ is deliberately defined using only the positive component. The choice reflects the interpretation that conscious accessibility is primarily driven by a sufficiently strong "present" support signal, while negative evidence can still play an important role inside update operators. The threshold θ is kept as an explicit parameter because different later theorems will compare how varying θ changes the resulting conscious set, and because collective/PNSE criteria naturally involve threshold trade-offs.

Definition 560 (Workspace operator and conscious evidence state)**.** *A* workspace operator *is a monotone map $W : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$. An evidence state E is* workspace-stable *if $W(E) = E$.*

The space $V^{L_{\mathrm{Ob}}}$ is ordered pointwise (induced from the componentwise order on V), so monotonicity of W means that if an evidence state is strengthened (in the poset sense) then the workspace update does not weaken it. Conceptually, W stands in for the closure and broadcasting effects of a global workspace: local module outputs are integrated, made mutually consistent as far as possible, and then returned as an updated global pattern of evidential support. The fixed-point condition $W(E) = E$ is the minimal way to say "the workspace dynamics have settled," i.e., that the state is closed under the workspace's own integration rule. Later, such fixed points will be used as the basic candidates for stable conscious Locations, and perturbation arguments will compare how these fixed points change under shifts in parameters or inputs.

Definition 561 (Reflective extension and introspection)**.** *Let $I(\cdot)$ be a syntactic operator. Define the reflective language $L_{\mathrm{Ob}}^{\mathrm{ref}} := L_{\mathrm{Ob}} \cup \{I(\varphi) : \varphi \in L_{\mathrm{Ob}}\}$. An* introspection operator *is a monotone map* $\mathrm{Int} : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}^{\mathrm{ref}}}$.

The operator $I(\cdot)$ is intended to be read as "it is introspectively represented that φ" (or, depending on later interpretation, "φ is available to reflection"). The extension from L_{Ob} to $L_{\mathrm{Ob}}^{\mathrm{ref}}$ is the simplest way to add second-order content without introducing a full higher-order logic: we only add one layer of syntactic "tags" of the original propositions. The monotonicity of Int plays the same structural role as monotonicity of W: when first-order evidential support increases, the induced reflective representation should not decrease. This ensures that the reflective layer can be used in later monotone fixed-point constructions and that it behaves well under parameter changes.

Definition 562 (Reflective update map and reflective fixed points)**.** *Fix an attention/integration map* $\Gamma_a : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$*, a workspace operator* $W^{\mathrm{ref}} : V^{L^{\mathrm{ref}}_{\mathrm{Ob}}} \to V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$*, and an introspection operator* Int*. Define the* reflective update map

$$F := W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a.$$

A reflective conscious state *is a fixed point* E^* *of* F*.*

The map Γ_a is separated out to represent pre-reflective selection and weighting: it can model attentional gain, module-to-module integration before broadcasting, and resource allocation. The subsequent Int step turns the attended first-order state into a mixed first/second-order representation, and W^{ref} then enforces the workspace-style closure at the reflective level. Writing the overall update as a single map F makes it possible to talk about reflective consciousness in the same fixed-point idiom as workspace stability, but now in a language that includes introspective content.

In later use, a fixed point E^* of F will function as a "reflectively closed" state: applying attention, introspection, and reflective workspace integration returns the same evidential pattern. This is the interface point at which we can meaningfully compare to SDP's reflective criteria (e.g., RCC-like conditions) without importing all of SDP's internal definitions into Hyperseed. Moreover, the explicit decomposition of F into three maps gives three levers for later theorems: perturbing attention (Γ_a), changing the introspection gain (inside Int), and changing reflective closure (W^{ref}).

Definition 563 (Reflective self condition)**.** *Assume* $L^{\mathrm{ref}}_{\mathrm{Ob}}$ *contains special propositions* $\mathrm{Self}(p)$ *meaning "p is a property of me," for p in some designated set* P_{self}*. A workspace-stable reflective evidence state* E exhibits a reflective self *if, for some threshold* θ*: (i)* $E^+(\mathrm{Self}(p)) \geq \theta$ *for many* $p \in P_{\mathrm{self}}$*; (ii)* $E^+(I(\mathrm{Self}(p))) \geq \theta$ *for those p; and (iii) these supports persist under small perturbations of module evidence and attention.*

This condition is meant to capture (in a minimal, operational form) the idea that selfhood involves not only first-order self-ascription but also a second-order availability of that self-ascription. Clause (i) asks for a sufficiently rich set of self-properties p to be supported at the first-order level, preventing a degenerate "self" that consists of a single token proposition. Clause (ii) requires that the system also represent, in the reflective language, that it has those self-properties, which is the simplest proxy for a reflective or introspective self-model. Clause (iii) is a robustness requirement: it ensures we are not merely identifying an unstable coincidence at a knife-edge parameter setting. Formally, "small perturbations" can later be interpreted in a metric or order topology on evidence states and in a parameter space for Γ_a; at this stage we only need the conceptual role of robustness because it will be used as a criterion for stable Locations and for collective integration thresholds.

The threshold θ here is allowed to be the same symbol as above but need not coincide numerically with the threshold used for $\mathrm{Con}_\theta(E)$. Later results will sometimes require distinguishing a content-access threshold from a self-model threshold, since the latter is typically stricter when used in PNSE-like arguments.

Definition 564 (Presentational immediacy and focus set)**.** *Let* X_c *be a set of presentational-immediacy carriers ("occasions" or "tokens"). A* presentational immediacy intensity *is a function* $I_c : X_c \to [0, 1]$*. For* $\theta \in (0, 1)$ *the* focus set *is* $F_\theta := \{x \in X_c : I_c(x) \geq \theta\}$*.*

Presentational immediacy is kept separate from evidential support because it is intended to track "what is directly presented" rather than "what is judged or inferred." The set X_c can be instantiated in many ways (sensory tokens, memory tokens, interoceptive occasions, or more abstract carriers), but the interface requires only that there is an intensity scale and a thresholded focus set. Later, this focus set will be one of the bridges to SDP Locations: informally, a Location can be constrained both by which propositions are consciously supported and by which carriers are currently focal. Keeping I_c independent also allows the formalism to represent cases in which something is highly presentationally immediate but not endorsed (high I_c, low E^+) and vice versa.

Definition 565 (State-of-consciousness parameter tuple). *A state-of-consciousness parameter tuple is*

$$\Lambda := (\lambda_{\text{ext}}, \lambda_{\text{int}}, \lambda_{\text{ref}}, \lambda_{\text{coh}}, \rho_{\text{hab}}, \rho_{\text{res}}),$$

intuitively controlling external anchoring, internal completion, introspection gain, coherence preference, habit reinforcement, and resonance coupling.

The tuple Λ serves as a compact way to state comparative and monotonicity results: varying Λ changes the induced operators (such as Γ_a, Int, and workspace closures) and therefore changes the sets $\text{Con}_\theta(E)$, F_θ, and the existence or structure of fixed points. The naming is suggestive rather than restrictive: λ_{ext} and λ_{int} can be taken to tune the relative weight of externally anchored versus internally completed content; λ_{ref} tunes how strongly introspective representations are amplified or stabilized; and λ_{coh} modulates the preference for globally coherent states in the workspace update. The parameters ρ_{hab} and ρ_{res} are written separately to emphasize that long-run stabilization (habit) and cross-module coupling (resonance) are distinct sources of persistence; in the later bridge to collective Locations and mindplexes, these often play different roles in stability thresholds.

62.2 Preliminaries: a minimal SDP interface

Definition 566 (GCC and reflective consciousness predicates). *Fix an observer O and threshold θ. Write $\text{GCC}_{O,\theta}(S, I)$ for "system S exhibits a GCC episode on interval I" and $\text{RC}_{O,\theta}(S, I)$ for "S exhibits reflective consciousness on I". We assume $\text{RC}_{O,\theta}(S, I) \Rightarrow \text{GCC}_{O,\theta}(S, I)$.*

Here O is intended to encode both a measurement perspective (what variables are accessible, at what temporal resolution, and with what inductive biases) and a normative stance (what counts as "global" and what counts as "reflective"). The parameter θ aggregates any thresholds used to binarize or discretize underlying graded evidence into the predicate $\text{GCC}_{O,\theta}$ or $\text{RC}_{O,\theta}$. The interval I can be taken as a closed real interval (e.g. $I = [t_0, t_1]$) or a discrete-time window, depending on whether S is modeled continuously or as an update process; nothing below depends on that choice, except that all derived quantities should be computed consistently with the same notion of time.

The implication $\text{RC}_{O,\theta}(S, I) \Rightarrow \text{GCC}_{O,\theta}(S, I)$ fixes a minimal interface constraint: reflective consciousness is treated as a refinement of a more basic globally-coordinated episode, rather than as an independent phenomenon. Operationally, one can read this as: if O judges that S is reflectively conscious over I, then O must also judge that a sufficiently integrated/global coordination condition is met over I. This leaves open whether $\text{GCC}_{O,\theta}$ is necessary or merely convenient for other forms of consciousness not discussed here; the only commitment is the one-way entailment stated.

Definition 567 (Location as an attractor in marker space). *Let $X := \mathcal{G} \times \mathcal{M} \times \mathcal{S}$ denote a marker space whose coordinates include global resonance-like markers ($\mathcal{G}$), marker profiles ($\mathcal{M}$), and self-model activity markers ($\mathcal{S}$). A* Location *L is a metastable region $\Omega_L \subseteq X$ that attracts nearby trajectories and is characterized (at least) by: (i) a resonance range $[R_L^{\min}, R_L^{\max}]$ and (ii) a self-model activity level (or range) S_L.*

The role of the product structure $X := \mathcal{G} \times \mathcal{M} \times \mathcal{S}$ is purely organizational: it distinguishes (a) markers meant to capture system-wide integration or "global workspace"-like coordination ($\mathcal{G}$), (b) a possibly high-dimensional profile of other task-, content-, or modality-relevant observables ($\mathcal{M}$), and (c) markers specifically tied to self-modeling, metacognition, and related monitoring/control activity ($\mathcal{S}$). In applications, each factor may itself be vector-valued and may include time-lagged features; the definition only requires that the combined state of markers at time t be representable as a point $x(t) \in X$.

A trajectory is the time-evolution $t \mapsto x(t)$ induced by the dynamics of S under whatever coupling to environment, tasks, and internal control policies is assumed. The phrase "metastable region Ω_L" is meant to exclude both instantaneous coincidences and fully absorbing equilibria: Ω_L should be stable enough that trajectories can dwell there for nontrivial durations (supporting an episode-like interpretation), yet permissive enough to allow transitions between Locations under perturbations, learning, or context shifts. The attractor-language is intentionally broad: Ω_L can be a basin around a fixed point, a limit cycle neighborhood, or a fuzzy cluster in marker space that repeatedly reappears across episodes.

The resonance range $[R_L^{\min}, R_L^{\max}]$ abstracts any scalar (or scalar reduction of a vector) intended to measure global coherence, synchrony, integration, or resonance-like coupling. The self-model activity descriptor S_L is similarly an abstraction: it can be a scalar level, an interval, or a low-dimensional summary of activity in a designated self-model subsystem. The "at least" qualifier indicates that additional coordinates (e.g. affective valence, arousal, uncertainty estimates, prediction-error statistics, or control bandwidth) may also be used to individuate Locations, but the resonance and self-model axes are treated as minimally necessary for the later bridging claims.

Definition 568 (Collective markers and collective Locations)**.** *Let $y_i(t)$ denote the policy/proposal signal produced by agent i at time t. Let D_O be a (model-dependent) distance on the space of such signals.*

Discord *is*

$$D_O(t) := \mathrm{Var}\big(\{y_i(t)\}\big)$$

for some variance functional induced by D_O.

Narrative capture *is $N_O(t) \in [0,1]$, interpreted as the fraction of the collective resource budget devoted to narrative/identity-maintenance patterns.*

A Collective Location *Ck (e.g. $C0, C1, C2, C3$) is a qualitative regime describable via thresholds on (at least) discord, narrative capture, and integration/coherence.*

The signal $y_i(t)$ is intended to be the agent-level output that participates in collective choice or coordination: it can be a distribution over actions, a proposed plan, a vote over options, a message embedding, or any other representation that the model treats as the "policy/proposal" interface. The distance D_O is allowed to depend on O because different observers may privilege different equivalences (e.g. treating two proposals as close if they lead to similar outcomes, if they share semantic content, or if they share resource implications). This observer-dependence will matter whenever discord is interpreted as a functional proxy for fragmentation or conflict.

The discord functional $D_O(t) := \mathrm{Var}(\{y_i(t)\})$ is schematic: the variance may be computed extrinsically (in a Euclidean embedding), intrinsically (via a kernel-induced metric), or in a decision-relevant geometry (e.g. using a divergence between probability distributions). The key requirement is that larger values correspond to greater dispersion among agent proposals, hence weaker immediate convergence of the collective toward a unified policy. In particular, low discord does not by itself imply good outcomes; it only encodes agreement or alignment in the $y_i(t)$ space as judged by O.

Narrative capture $N_O(t) \in [0,1]$ is defined as a resource fraction to emphasize that it competes with other collective capacities (e.g. exploration, deliberation, error correction, and external problem-solving). The phrase "narrative/identity-maintenance patterns" includes any activity whose primary function is preserving a shared story, status order, or self-description of the collective, rather than directly optimizing external objectives. The observer index again allows that what counts as "narrative" may depend on the modeling lens (for example, whether certain rituals are treated as coordination mechanisms or as identity reinforcement).

A Collective Location Ck is thus a coarse-grained attractor-like regime for group dynamics, analogous to an individual Location in marker space but defined over collective markers. The mention of "integration/coherence" indicates that discord and narrative capture are not sufficient by themselves: a collective may have low discord because it is rigidly coerced, or high narrative capture while still maintaining strong task integration in narrow domains. Therefore, the thresholds defining Ck are meant to incorporate at least one additional measure of coordinated information flow or functional coupling (however implemented in the chosen model), so that the resulting regime labels track meaningful qualitative changes in collective cognitive organization rather than merely surface agreement.

62.3 A translation layer: connecting the two formalisms

We introduce a bridge in two steps: (1) connect GCC/RC to workspace-stable (reflective) fixed points, and (2) connect Location coordinates (R, S) to Hyperseed parameters and evidence-state functionals. The intent is not to identify the two formalisms term-by-term, but to provide a translation layer in which the SuperDuperPsychism notions (GCC, RC, PNSE, and Location coordinates) can be represented as constraints on, or summaries of, Hyperseed-style evidence dynamics. In particular, the role of the "global conscious content" (GCC) will be played by the high-support portion of an evidence state (captured by $\mathrm{Con}_\theta(E)$), while the role of "reflective consciousness" (RC) will be captured by fixed points of a reflective update map that explicitly adds introspective propositions (via $I(\cdot)$) into the workspace.

To prepare step (2), we will later treat (R, S) not as additional primitives but as coordinates computed from the evidence-and-workspace dynamics: roughly, R will track the degree of reflective closure/stability of conscious contents under introspective updates, and S will track the degree to which self-ascriptions are both present and introspectively accessible in a robust way. This makes the Location picture operational: a point (R, S) corresponds to a family of evidence states that satisfy particular stability, accessibility, and self-model constraints.

Definition 569 (A simple coherence/resonance functional)**.** *Let L_{Ob} be finite and let $E \in V^{L_{\mathrm{Ob}}}$ be an evidence state. Assume given (i) a symmetric "coherence weight" function $\kappa : L_{\mathrm{Ob}} \times L_{\mathrm{Ob}} \to [0,1]$, (ii) a symmetric "tension weight" function $\tau : L_{\mathrm{Ob}} \times L_{\mathrm{Ob}} \to [0,1]$, and (iii) a "resonance profile" $r : L_{\mathrm{Ob}} \to [0,1]$ (standing for how strongly each proposition is coupled to recurrent or affectively salient processing). Define the* coherence *and* resonance *scores*

$$\mathrm{Coh}(E) \;:=\; \frac{1}{Z_{\kappa,\tau}} \sum_{\varphi,\psi \in L_{\mathrm{Ob}}} \Big(\kappa(\varphi,\psi)\, E^+(\varphi)E^+(\psi) \;-\; \tau(\varphi,\psi)\, E^+(\varphi)E^-(\psi)\Big),$$

$$\mathrm{Res}(E) \;:=\; \frac{1}{Z_r} \sum_{\varphi \in L_{\mathrm{Ob}}} r(\varphi)\, E^+(\varphi),$$

where $Z_{\kappa,\tau} > 0$ and $Z_r > 0$ are normalizers (for example, sums of the corresponding weights) so that $\mathrm{Coh}(E)$ and $\mathrm{Res}(E)$ lie in a controlled range.

The particular forms of Coh and Res are not essential; what matters is that they provide numerically evaluable summaries of (i) how mutually supportive the currently high-evidence propositions are, and (ii) how strongly the currently supported propositions are coupled to recurrent processing, affective salience, or other "resonance" channels. In the translation layer, these functionals will be used to connect phenomenological stability and "felt presence" to parameters such as λ_{coh} and ρ_{res}, and to define coarse coordinates like R via stability of reflective fixed points and like S via stability of self-ascriptive supports.

62.4 Preliminaries: a minimal Hyperseed consciousness interface

We use a deliberately small slice of Hyperseed's consciousness formalism. The goal of this interface is to have just enough structure to: (a) talk about evidence-bearing representations, (b) distinguish mere activation from workspace-stable availability, and (c) express a minimal notion of reflection in which some contents are about other contents (or about their being present to the system).

Definition 570 (Evidence values and evidence states)**.** *Let $V := [0,1]^2$ with the componentwise order. An* evidence state *for an observer-context* Ob *is a map $E : L_{\mathrm{Ob}} \to V$, where L_{Ob} is a finite set of propositions. Write $E(\varphi) = (E^+(\varphi), E^-(\varphi))$. For $\theta \in (0,1)$ define the* conscious proposition set

$$\mathrm{Con}_\theta(E) := \{\varphi \in L_{\mathrm{Ob}} : E^+(\varphi) \geq \theta\}.$$

The pair-valued codomain $V = [0,1]^2$ is meant to permit simultaneous positive and negative support, rather than forcing evidence into a single scalar. The componentwise order makes it natural to treat evidence updates as monotone maps: increasing either the positive or negative component (for any proposition) yields an "at least as supported" overall evidence state in the partial-order sense. The thresholded set $\mathrm{Con}_\theta(E)$ is intentionally crude: it acts as a readable surrogate for "what is globally available" at that moment, and it can be tuned by varying θ (e.g., high θ for reportable access, lower θ for weaker but still influence-bearing contents).

Definition 571 (Workspace operator and conscious evidence state)**.** *A* workspace operator *is a monotone map $W : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$. An evidence state E is* workspace-stable *if $W(E) = E$.*

Monotonicity of W expresses the constraint that adding support to inputs does not perversely reduce support in outputs, which is a minimal regularity condition for an "integration/selection" stage. Workspace-stability is the basic dynamical notion that replaces an informal appeal to a sustained global workspace: $W(E) = E$ means that once the evidence configuration is fed through the workspace integration, it reproduces itself. This is the point at which the bridge to GCC begins: a workspace-stable state has a well-defined $\mathrm{Con}_\theta(E)$ that is not immediately dissolved by the workspace dynamics.

Definition 572 (Reflective extension and introspection)**.** *Let $I(\cdot)$ be a syntactic operator. Define the reflective language $L^{\mathrm{ref}}_{\mathrm{Ob}} := L_{\mathrm{Ob}} \cup \{I(\varphi) : \varphi \in L_{\mathrm{Ob}}\}$. An* introspection operator *is a monotone map* $\mathrm{Int} : V^{L_{\mathrm{Ob}}} \to V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$.

The operator $I(\varphi)$ is read schematically as "φ is (now) present/intended/available to me," without committing to a particular psychological theory of introspection. The role of Int is to map non-reflective evidence into a reflective space where some propositions are about other propositions. Monotonicity here expresses that strengthening first-order evidence should not, all else equal, reduce the system's capacity to generate corresponding reflective supports.

Definition 573 (Reflective update map and reflective fixed points)**.** *Fix an attention/integration map $\Gamma_a : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$, a workspace operator $W^{\mathrm{ref}} : V^{L^{\mathrm{ref}}_{\mathrm{Ob}}} \to V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$, and an introspection operator* Int*. Define the* reflective update map

$$F := W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a.$$

A reflective conscious state *is a fixed point* E^* *of* F*.*

This is the minimal mechanism used to represent RC: reflection is not a separate "module" but a closure condition on the evidence dynamics once introspective propositions are admitted into the workspace. In finite settings, monotonicity of the constituent maps supports the expectation that fixed points exist under standard lattice-theoretic conditions; even when uniqueness is not guaranteed, a family of fixed points can be used to represent multiple stable "modes" of reflective access.

Definition 574 (Reflective self condition)**.** *Assume* $L_{\mathrm{Ob}}^{\mathrm{ref}}$ *contains special propositions* $\mathrm{Self}(p)$ *meaning "p is a property of me," for p in some designated set* P_{self}*. A workspace-stable reflective evidence state E* exhibits a reflective self *if, for some threshold* θ*: (i)* $E^+(\mathrm{Self}(p)) \geq \theta$ *for many* $p \in P_{\mathrm{self}}$*; (ii)* $E^+(I(\mathrm{Self}(p))) \geq \theta$ *for those p; and (iii) these supports persist under small perturbations of module evidence and attention.*

Clause (i) captures a nontrivial self-model at the level of first-order contents; clause (ii) demands that these self-ascriptions are themselves reflectively available (so the system can, in effect, represent that it represents itself as having properties). Clause (iii) is a robustness requirement: it separates a stable self from a fleeting coincidence of supports. This is the natural insertion point for mapping the S-coordinate of a Location: high S corresponds to many self-ascriptions that are both strong and introspectively endorsed, and whose endorsement remains stable under small changes in attention allocation.

Definition 575 (Presentational immediacy and focus set)**.** *Let* X_c *be a set of presentational-immediacy carriers ("occasions" or "tokens"). A* presentational immediacy intensity *is a function* $I_c : X_c \to [0, 1]$*. For* $\theta \in (0, 1)$ *the* focus set *is* $F_\theta := \{x \in X_c : I_c(x) \geq \theta\}$*.*

The carriers X_c are introduced to keep a place for phenomenological "what-it-is-like" immediacy without forcing it to be identified with propositional content. In the bridge, one can treat X_c as indexing perceptual episodes, affective tones, or bodily sensations that may not be exhaustively captured by L_{Ob}, while F_θ provides a minimal handle on attention-like selection among these carriers. This dovetails with Γ_a above: Γ_a can be understood as the evidence-level counterpart of selecting and integrating items in (or prompted by) F_θ.

Definition 576 (State-of-consciousness parameter tuple)**.** *A* state-of-consciousness parameter tuple *is*

$$\Lambda := (\lambda_{\mathrm{ext}}, \lambda_{\mathrm{int}}, \lambda_{\mathrm{ref}}, \lambda_{\mathrm{coh}}, \rho_{\mathrm{hab}}, \rho_{\mathrm{res}}),$$

intuitively controlling external anchoring, internal completion, introspection gain, coherence preference, habit reinforcement, and resonance coupling.

The tuple Λ is intended to collect slowly varying control parameters that shape the update maps (Γ_a, Int, W, W^{ref}) and thereby shape which fixed points exist and which are attractive. For the translation layer, one may read λ_{coh} as weighting the influence of $\mathrm{Coh}(E)$-like criteria in workspace selection, and ρ_{res} as weighting $\mathrm{Res}(E)$-like criteria (e.g., preferential amplification of resonant contents). Likewise, λ_{ref} can be interpreted as tuning the extent to which introspective propositions $I(\varphi)$ are generated and stabilized, and thus provides a direct control knob for the R-coordinate when R is taken to measure the reflective closure/stability of the conscious set under F.

62.5 Preliminaries: a minimal SDP interface

Definition 577 (GCC and reflective consciousness predicates)**.** *Fix an observer O and threshold θ. Write $\mathrm{GCC}_{O,\theta}(S,I)$ for "system S exhibits a GCC episode on interval I" and $\mathrm{RC}_{O,\theta}(S,I)$ for "S exhibits reflective consciousness on I". We assume $\mathrm{RC}_{O,\theta}(S,I) \Rightarrow \mathrm{GCC}_{O,\theta}(S,I)$.*

Here O encodes the choice of measurement model, readout, and any background assumptions used to judge whether a GCC or reflective episode is present; θ packages the (possibly multi-parameter) decision boundary above which the episode is counted as occurring. The interval I may be taken as continuous or discretized; in either case, the predicates are intended to be tolerant to brief fluctuations (e.g. by implicitly smoothing, averaging, or requiring persistence over a minimal dwell time). The implication

$$\mathrm{RC}_{O,\theta}(S,I) \Rightarrow \mathrm{GCC}_{O,\theta}(S,I)$$

formalizes a minimal hierarchy: reflective consciousness is treated as a stricter condition whose satisfaction entails the presence of a GCC-style global availability/integration episode, while leaving open whether GCC can occur without reflectivity. This interface is deliberately noncommittal about the mechanistic realization of GCC (workspace, ignition, broadcast, etc.) and is meant to serve only as a common logical handle for later constructions.

Definition 578 (Location as an attractor in marker space)**.** *Let $X := \mathcal{G} \times \mathcal{M} \times \mathcal{S}$ denote a marker space whose coordinates include global resonance-like markers ($\mathcal{G}$), marker profiles ($\mathcal{M}$), and self-model activity markers ($\mathcal{S}$). A* Location *L is a metastable region $\Omega_L \subseteq X$ that attracts nearby trajectories and is characterized (at least) by: (i) a resonance range $[R_L^{\min}, R_L^{\max}]$ and (ii) a self-model activity level (or range) S_L.*

Intuitively, X is a deliberately coarse "state space" of observables rather than microphysical degrees of freedom: points $x(t) \in X$ summarize whatever markers are taken to be salient for identifying a consciousness-relevant regime. The factor $\mathcal{G}$ can be read as capturing global, system-level synchrony or integration proxies (e.g. broadcast strength, large-scale coherence, ignition-like events), while $\mathcal{M}$ captures structured profiles (e.g. modality-weighted marker vectors, content-relevant signatures, or task-dependent configurations), and $\mathcal{S}$ captures activity of self-modeling (e.g. metacognitive monitoring, self-referential dynamics, narrative self-maintenance). A "trajectory" refers to the time-evolution $t \mapsto x(t)$ induced by the underlying system dynamics together with the observer map that extracts markers. Metastability is meant to cover regimes that are stable enough to be behaviorally and phenomenologically meaningful, yet permissive of transitions; Ω_L may therefore be thought of as a basin-like region with nonzero escape probability. The resonance interval $[R_L^{\min}, R_L^{\max}]$ and self-model range S_L are minimal coordinates for labeling the regime; additional coordinates (e.g. arousal, attention allocation, or prediction-error statistics) may be appended without changing the role of L as a coarse-grained attractor. Nothing in the definition forces Locations to be disjoint; overlaps can represent borderline cases or gradual shifts where multiple descriptive regimes apply under different observer choices O or different thresholds.

Definition 579 (Collective markers and collective Locations)**.** *Let $y_i(t)$ denote the policy/proposal signal produced by agent i at time t. Let D_O be a (model-dependent) distance on the space of such signals.*

Discord *is*

$$D_O(t) := \mathrm{Var}\big(\{y_i(t)\}\big)$$

for some variance functional induced by D_O.

Narrative capture *is* $N_O(t) \in [0,1]$*, interpreted as the fraction of the collective resource budget devoted to narrative/identity-maintenance patterns.*

A Collective Location Ck *(e.g.* $C0, C1, C2, C3$*) is a qualitative regime describable via thresholds on (at least) discord, narrative capture, and integration/coherence.*

The intent is that $y_i(t)$ abstracts away internal complexity and records only the externally relevant "proposal" each agent offers into a shared decision or control process (e.g. an action distribution, a plan token, a utility gradient, a vote vector, or a message in a negotiation protocol). The distance D_O is observer-relative because the geometry of proposal space depends on representation (e.g. whether two plans are considered close by edit distance, outcome similarity, KL divergence between policies, semantic embedding distance, or some task-specific metric). With that in place, $\mathrm{Var}(\{y_i(t)\})$ should be read broadly as any dispersion functional compatible with D_O; for instance, it may be implemented as average pairwise distance, distance to a centroid, or a robust spread statistic when outliers are expected. High discord corresponds to a strongly multi-peaked or scattered proposal landscape (many incompatible policies), while low discord corresponds to tight clustering (near-consensus).

Narrative capture $N_O(t)$ is likewise an interface variable: it is intended to quantify how much collective bandwidth is being spent on maintaining identity, status, justification, or story-consistency, as opposed to directly solving the nominal task. The $[0,1]$ normalization may be implemented by dividing an estimated narrative-maintenance load by total available compute/attention/communication budget over the same interval. In this framing, a Collective Location Ck is not a point but a regime label: it groups together collective states that satisfy certain threshold relations among discord $D_O(t)$, narrative capture $N_O(t)$, and an (unspecified but assumed available) integration/coherence marker (e.g. mutual information, effective connectivity, shared world-model alignment, or convergence rate of distributed belief updates). The examples $C0, C1, C2, C3$ are meant only as a minimal vocabulary for later reference; the precise number of regimes and their boundaries can vary with domain, but the definition requires that they be expressible as qualitative slices through the space of collective markers.

62.6 A translation layer: connecting the two formalisms

We introduce a bridge in two steps: (1) connect GCC/RC to workspace-stable (reflective) fixed points, and (2) connect Location coordinates (R, S) to Hyperseed parameters and evidence-state functionals. The role of this subsection is operational: it specifies a minimal set of numerical summaries of an evidence state that can be (i) computed from the Hyperseed-side objects and (ii) interpreted as proxies for the SuperDuperPsychism-side notions of global availability and self-model involvement. In particular, the translation is not intended to be unique; rather, it provides a family of choices (via K, σ, θ, and the thresholds below) whose qualitative behavior is robust across modeling details.

The first step relies on treating GCC/RC as a constraint on dynamical persistence: a state that is globally accessible (GCC) and reflectively coherent (RC) should correspond to an evidence configuration that is stable under the update rules governing workspace competition and reflective endorsement. Accordingly, throughout this chunk we assume that "workspace-stable" means stable relative to the chosen update operator(s) (e.g., a fixed point or attractor of the global-workspace selection dynamics), and that the stability requirement can be checked independently of the summary scores defined next.

Definition 580 (A simple coherence/resonance functional)**.** *Fix a finite set* $K \subseteq L^{\mathrm{ref}}_{\mathrm{Ob}}$ *of "currently relevant" propositions. Let* $\sigma : V \to \mathbb{R}$ *be any monotone map that increases with net support and/or* *reduces with tension (model-dependent). Define the* coherence score *of an evidence state by*

$$\mathrm{Coh}(E) := \frac{1}{|K|} \sum_{\varphi \in K} \sigma(E(\varphi)).$$

Intuitively, $\mathrm{Coh}(E)$ is meant to aggregate, across a contextually selected set K, how well the current evidence state supports (or at least does not strongly conflict with) the propositions that the system is effectively "tracking" at the reflective/object level. The restriction to a finite K encodes limited bandwidth and attention: only a bounded slice of the reflective/object language contributes to the coherence estimate at any given time. The monotonicity requirement on σ ensures that if the evidence values $E(\varphi)$ shift in a direction that the model interprets as "more supportive" (or less tense), then $\mathrm{Coh}(E)$ will not decrease. Different modeling choices correspond to different σ (for instance, a sigmoid that saturates, a linear map on an interval of V, or a map that penalizes values associated with contradiction), but the intended use is the same: $\mathrm{Coh}(E)$ is a single scalar proxy for workspace-level resonance with the currently relevant reflective content.

Definition 581 (Self-model activity score)**.** *Let P_{self} be the designated self-property set and fix $\theta \in (0,1)$. Define*

$$\mathrm{SAct}_\theta(E) := \frac{1}{|P_{\mathrm{self}}|} \big|\{p \in P_{\mathrm{self}} : E^+(\mathrm{Self}(p)) \geq \theta\}\big|.$$

The statistic $\mathrm{SAct}_\theta(E)$ measures how many self-ascriptions are "active" at or above a threshold θ, normalized by the size of the self-property vocabulary. This is intended to capture a graded notion of self-model engagement that is robust to changes in magnitude scale, since it depends only on whether the positive component E^+ crosses a criterion. The parameter θ may be treated as a control knob: smaller θ makes the score sensitive to weak self-evidence, while larger θ restricts the score to comparatively strong self-endorsements. On the translation-layer reading, low $\mathrm{SAct}_\theta(E)$ corresponds to a weakly recruited or backgrounded self-model, whereas high $\mathrm{SAct}_\theta(E)$ corresponds to widespread self-referential recruitment (e.g., strong ownership, agency, narrative centrality, or explicit self-attribution, depending on what is encoded in P_{self}).

Definition 582 (Symbolic load and PNSE)**.** *Let* $\mathrm{Sym} \subseteq L_{\mathrm{Ob}}$ *denote the subset of propositions treated as linguistic/symbolic/narrative in the current modeling context. Define the* symbolic load *of E by*

$$\mathrm{SL}_\theta(E) := \frac{|\mathrm{Con}_\theta(E) \cap \mathrm{Sym}|}{|\mathrm{Sym}|}.$$

Fix $\varepsilon \in [0,1]$. A workspace-stable evidence state E is an ε-PNSE state *if* $\mathrm{SL}_\theta(E) \leq \varepsilon$ *and* $\mathrm{Coh}(E)$ *is above a chosen coherence threshold.*

A system exhibits PNSE *on interval I (relative to (θ, ε)) if, for all $t \in I$, there is a workspace-stable E_t such that E_t is an ε-PNSE state.*

Here $\mathrm{SL}_\theta(E)$ provides a deliberately coarse proxy for "how much of the currently active content is symbolic," where activity is represented by the thresholded consequence set $\mathrm{Con}_\theta(E)$. The numerator counts those threshold-active consequences that fall into the designated symbolic/narrative fragment Sym, and the normalization by $|\mathrm{Sym}|$ turns this into a context-dependent fraction. In particular, $\mathrm{SL}_\theta(E)$ is meant to be sensitive to changes in the symbolic subspace even when overall coherence remains high. The PNSE condition then formalizes the idea that a system can occupy stable, coherent states with low symbolic/narrative load: $\mathrm{Coh}(E)$ ensures that the state is not merely "quiet" or "blank" but remains globally well-supported over the relevant reflective set K, while the constraint $\mathrm{SL}_\theta(E) \leq \varepsilon$ enforces that this coherence is achieved without strongly engaging the symbolic fragment. The coherence threshold (left unspecified here) is intended to be chosen relative to the update rules and the range of σ, so that it separates merely transient or fragmented patterns from those that function as globally integrated workspace contents.

The time-indexed definition of PNSE on an interval I emphasizes persistence: it is not enough for one instantaneous evidence snapshot to be low-symbolic; rather, the system must admit a workspace-stable representative E_t at each time t in the interval. This aligns the PNSE notion with the fixed-point or attractor reading of workspace stability, and it makes explicit that PNSE is a property of a trajectory (or time-slice family) in evidence-state space, not only of a single evidence state.

Definition 583 (A Location classifier from Hyperseed evidence states)**.** *Fix thresholds* $(r_0 < r_1 < \cdots)$ *and* $(s_0 > s_1 > \cdots)$ *in* $[0,1]$*. Define the* Hyperseed Location index *of an evidence state* E *as*

$$\mathrm{Loc}(E) := \max\{k : \mathrm{Coh}(E) \geq r_k \textit{ and } \mathrm{SAct}_\theta(E) \leq s_k\}.$$

The index $\mathrm{Loc}(E)$ is intended as a discrete surrogate for the continuous Location coordinates (R, S): the condition $\mathrm{Coh}(E) \geq r_k$ plays the role of requiring sufficiently strong integration/resonance (an R-like dimension), while the constraint $\mathrm{SAct}_\theta(E) \leq s_k$ enforces sufficiently weak self-model recruitment (an S-like dimension, depending on the chosen sign convention). The opposing monotonicity of the threshold sequences (r_k) and (s_k) encodes the idea that moving to higher Location indices requires simultaneously higher coherence and lower self-activity, so that higher k corresponds to more coherent-yet-less-self-involving workspace organization. The use of max makes the classifier conservative in the sense that it assigns the highest index consistent with both inequalities; if no k satisfies the constraints, one may treat $\mathrm{Loc}(E)$ as undefined or as taking a default value (depending on how the Location lattice is handled elsewhere). In the bridge to Hyperseed parameters, the thresholds may be learned or calibrated (e.g., by matching empirical regimes, simulation phases, or hand-labeled exemplars) so that each k corresponds to a qualitatively stable phenomenological/functional region in the combined SuperDuperPsychism–Hyperseed description.

62.7 Theorems: GCC/RC imply Hyperseed conscious/reflective fixed points

In this subsection the main mathematical work is carried by monotone operators on evidence states. Intuitively, a "workspace-stable" state is one that is left unchanged by the (possibly time-indexed) workspace update W_t, and a "reflective" stable state is one that is left unchanged by the corresponding reflective closure operator $F = W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a$. The thresholded consciousness map $\mathrm{Con}_\theta(\cdot)$ then picks out those propositions whose positive support crosses the designated level θ; in the intended application, these are the contents eligible to count as consciously present (or reflectively present, in the reflective language).

Lemma 168 (Threshold monotonicity)**.** *If* $E \leq E'$ *pointwise (componentwise in* V*), then* $\mathrm{Con}_\theta(E) \subseteq \mathrm{Con}_\theta(E')$*.*

Proof. If $\varphi \in \mathrm{Con}_\theta(E)$ then $E^+(\varphi) \geq \theta$. Since $E \leq E'$ we have $E'^+(\varphi) \geq E^+(\varphi) \geq \theta$, hence $\varphi \in \mathrm{Con}_\theta(E')$. The only property used here is that the map $E \mapsto E^+(\varphi)$ is monotone in the pointwise order; this matches the interpretation that increasing the underlying evidence state cannot decrease the positive support assigned to a fixed proposition. □

The next theorem is the basic existence result that is used repeatedly: once the workspace dynamics is represented as a monotone self-map on a complete lattice of evidence states, fixed points exist without requiring contraction, continuity, or any metric structure. This is the minimal structure needed for the later “Hyperseed fixed point” reading, where stable conscious/reflective episodes are represented by fixed points (or, more generally, by elements in the fixed-point set) of appropriate closure operators.

Theorem 169 (Workspace-stable conscious states exist under monotone closure)**.** *Assume L_{Ob} is finite and $W : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$ is monotone. Then W has at least one fixed point. In particular, there exist workspace-stable evidence states.*

Proof. The set $V^{L_{\mathrm{Ob}}}$ is a complete lattice under pointwise order because V is. By monotonicity of W, Tarski's fixed point theorem implies the set of fixed points of W is nonempty (and has least and greatest fixed points). Concretely, one may think of iterating W from the bottom element (or dually from the top element) of the lattice to generate an increasing (or decreasing) chain whose join (or meet) is a fixed point; Tarski guarantees existence of the relevant joins/meets in the complete lattice and thus avoids any appeal to finiteness beyond the closure properties already assumed. □

To connect this abstract fixed-point existence to GCC-style episodes, we treat each time slice t in the interval I as having an associated monotone “broadcast-and-stabilize” operator W_t. Assumption (A2) supplies a particular proposition φ_t that expresses that the system is in the relevant “basin” (or workspace coalition) at that time. The role of the fixed-point argument is then to ensure that there is at least one stabilized evidence state at time t in which that basin-membership proposition remains above threshold, so that a nonempty conscious set (and hence a nonempty focus set) is secured.

Theorem 170 (GCC implies existence of stable focus sets)**.** *Assume $\mathrm{GCC}_{O,\theta}(S, I)$. Assume further that during the GCC episode there exists a time-indexed family of monotone workspace operators W_t and presentational immediacy functions $I_{c,t}$ such that: (A1) W_t encodes “global broadcasting + basin stabilization” for the GCC basin at time t; and (A2) there exists at least one proposition φ_t expressing the active basin membership such that $E^+_{\bullet,t}(\varphi_t) \geq \theta$ in the integrated module evidence $E_{\bullet,t}$.*

Then for each $t \in I$ there exists a workspace-stable evidence state E_t with $\varphi_t \in \mathrm{Con}_\theta(E_t)$; hence $\mathrm{Con}_\theta(E_t) \neq \emptyset$. Moreover the focus sets $F_{\theta,t} = \{x : I_{c,t}(x) \geq \theta\}$ are nonempty for all $t \in I$.

Proof. Fix $t \in I$. By the previous theorem, W_t has at least one fixed point E_t. Assumption (A2) provides a proposition φ_t with positive evidence above threshold in the integrated evidence. Because W_t is monotone and (by the intended reading of (A1)) does not destroy supported basin-membership content, we may choose E_t among fixed points satisfying $E_t \geq E_{\bullet,t}$ on φ_t (e.g. take the greatest fixed point if needed). Then $E^+_t(\varphi_t) \geq \theta$, so $\varphi_t \in \mathrm{Con}_\theta(E_t)$. The nonemptiness of $F_{\theta,t}$ follows from the bridge assumption that GCC-visible basin content corresponds to at least one presentational token above threshold (otherwise there would be no unified “moment” in the MinSync sense). For additional clarity on the choice of fixed point: if one treats $E_{\bullet,t}$ as a lower bound encoding the currently integrated support, then monotonicity allows one to consider the set of post-fixed points above that bound (states E with $E_{\bullet,t} \leq E$ and $E \leq W_t(E)$) and apply Knaster–Tarski to obtain a greatest fixed point above $E_{\bullet,t}$ when such a bound is appropriate for the application. The intended content-preservation clause in (A1) is precisely what motivates working within an upper region of the lattice that respects already-established basin membership. □

The RC case parallels GCC but strengthens the content being stabilized: it is not merely basin membership that must remain above threshold, but a family of self-ascriptions and their introspective counterparts. In the Hyperseed reading, this is the difference between a merely conscious fixed point and a reflective fixed point: the latter contains, above threshold, propositions in which the system represents itself as having certain properties, together with propositions expressing that the system is aware of (or has introspective access to) those self-ascriptions.

Theorem 171 (RC implies existence of reflective-self fixed points)**.** *Assume* $\mathrm{RC}_{O,\theta}(S, I)$*. Assume that the RC self-model supplies a designated set* P_{self} *of self-properties and module evidence supports* $\mathrm{Self}(p)$ *for many* $p \in P_{\text{self}}$ *during* I*.*

Then there exists a reflective fixed point E^* *of* $F = W^{\text{ref}} \circ \mathrm{Int} \circ \Gamma_a$ *that satisfies the reflective self condition.*

Proof. Because $L_{\text{Ob}}^{\text{ref}}$ is finite and F is monotone (as a composition of monotone maps), Tarski's theorem yields a fixed point E^* of F. Here the finiteness assumption is used only to guarantee that the relevant evidence-state space $V^{L_{\text{Ob}}^{\text{ref}}}$ is well behaved under pointwise order given the assumed lattice structure on V; the fixed-point existence itself is driven by completeness and monotonicity, so the same argument extends whenever the reflective evidence space forms a complete lattice.

(Self-presence) By assumption, for many $p \in P_{\text{self}}$ the base evidence for $\mathrm{Self}(p)$ is above threshold during I. Monotonicity of Γ_a and W^{ref} preserves (or increases) this support in the fixed point. In particular, if Γ_a is read as an attention or gating operation, monotonicity encodes the idea that allocating more attention to a supported self-proposition cannot reduce its positive evidential component, and W^{ref} then performs a closure/broadcast step that similarly respects supported content.

(Reflective access) The introspection operator assigns evidence to $I(\mathrm{Self}(p))$ as a monotone function of the base evidence for $\mathrm{Self}(p)$, so for those p the introspective propositions also cross threshold in E^*. This is the formal counterpart of the RC requirement that self-ascriptions are not merely present but are available to higher-order access: whenever $\mathrm{Self}(p)$ is sufficiently supported, the monotone Int map ensures the derived reflective token $I(\mathrm{Self}(p))$ is supported as well, and the fixed-point condition ensures this relation is stable under repeated application of the reflective loop.

(Stability) RC requires that the self-model's representational loops are robust under benign internal reparameterizations. Modeling this as small perturbations in module evidence and attention implies the relevant supports persist in a neighborhood of the fixed point. In lattice-theoretic terms, this robustness can be expressed by requiring that the fixed point be resistant to perturbations that remain within an order-neighborhood (i.e. do not decrease the supports of the relevant $\mathrm{Self}(p)$ below threshold), so that the same monotone operators continue to map the perturbed state back toward a region whose join/meet structure contains a fixed point with the reflective self condition. This matches the RC intuition that reflective self-representation is not a brittle coincidence at a single evidence assignment but an attractor-like configuration of the reflective dynamics. □

62.8 Theorems: PNSE and Martin Locations via Hyperseed parameter regimes

Definition 584 (Parametric monotonicity assumption)**.** *Let* Λ *range over parameter tuples. Write* F_Λ *for the resulting reflective update map. Assume a partial order* $\preceq$ *on parameter tuples such that* $\Lambda \preceq \Lambda'$ *implies* $F_\Lambda(E) \leq F_{\Lambda'}(E)$ *for all* E*. (Informally: increasing parameters that amplify completion or introspection cannot decrease evidence.)*

Lemma 169 (Least fixed points are monotone in monotone operator families)**.** *Assume* F *and* G *are monotone operators on a complete lattice with* $F(E) \leq G(E)$ *for all* E*. Let* $\mathrm{lfp}(F)$ *denote the least fixed point of* F*. Then* $\mathrm{lfp}(F) \leq \mathrm{lfp}(G)$*.*

Proof. By Tarski, both least fixed points exist. Let $x = \mathrm{lfp}(G)$. Then $G(x) = x$ and $F(x) \leq G(x) = x$, so x is a pre-fixpoint of F. The least fixed point of F is below every pre-fixpoint, hence $\mathrm{lfp}(F) \leq x$. □

Theorem 172 (Symbolic suppression under reduced completion and introspection)**.** *Assume the parametric monotonicity assumption. Fix a threshold θ and symbolic subset $\mathrm{Sym} \subseteq L_{\mathrm{Ob}}$. Let $\Lambda \preceq \Lambda'$ where (informally) Λ has lower internal completion λ_{int} and lower introspection gain λ_{ref} than Λ', with coherence preference λ_{coh} held high.*

Let $E^* = \mathrm{lfp}(F_\Lambda)$ and $E'^* = \mathrm{lfp}(F_{\Lambda'})$. Then

$$\mathrm{Con}_\theta(E^*) \cap \mathrm{Sym} \subseteq \mathrm{Con}_\theta(E'^*) \cap \mathrm{Sym},$$

and in particular $\mathrm{SL}_\theta(E^*) \leq \mathrm{SL}_\theta(E'^*)$.

Proof. By parametric monotonicity, $F_\Lambda(E) \leq F_{\Lambda'}(E)$ for all E. By the previous lemma, $\mathrm{lfp}(F_\Lambda) \leq \mathrm{lfp}(F_{\Lambda'})$, i.e. $E^* \leq E'^*$. Apply Lemma (Threshold monotonicity) to conclude $\mathrm{Con}_\theta(E^*) \subseteq \mathrm{Con}_\theta(E'^*)$. Intersect both sides with Sym to obtain the inclusion, and divide by $|\mathrm{Sym}|$. □

Corollary 63 (A formal PNSE sufficient condition)**.** *Fix $\varepsilon \in [0, 1]$. If there exists a parameter tuple Λ such that the least fixed point $E^* = \mathrm{lfp}(F_\Lambda)$ has $\mathrm{SL}_\theta(E^*) \leq \varepsilon$ and high coherence $\mathrm{Coh}(E^*) \geq r_1$, then E^* is an ε-PNSE state. Moreover, any parameter tuple $\tilde{\Lambda} \preceq \Lambda$ yields a least fixed point whose symbolic load is at most ε.*

Proof. The first claim is immediate from the definition of ε-PNSE. For the second, apply the previous theorem with $\tilde{\Lambda} \preceq \Lambda$. □

Theorem 173 (Martin-Location ordering as a monotone mapping)**.** *Assume the SDP key claim that higher numbered Locations have higher sustained global resonance and lower self-model activity. Assume the translation of these into Hyperseed evidence-state functionals: higher resonance corresponds to larger $\mathrm{Coh}(E)$ and lower self-model activity corresponds to smaller $\mathrm{SAct}_\theta(E)$.*

Then for any evidence states E, E': if $\mathrm{Loc}(E') > \mathrm{Loc}(E)$, then $\mathrm{Coh}(E') \geq \mathrm{Coh}(E)$ and $\mathrm{SAct}_\theta(E') \leq \mathrm{SAct}_\theta(E)$.

Proof. Let $k = \mathrm{Loc}(E)$ and $k' = \mathrm{Loc}(E')$ with $k' > k$. By definition of Loc, $\mathrm{Coh}(E) \geq r_k$ and $\mathrm{SAct}_\theta(E) \leq s_k$, and $\mathrm{Coh}(E') \geq r_{k'} \geq r_k$ and $\mathrm{SAct}_\theta(E') \leq s_{k'} \leq s_k$, using monotonicity of the threshold sequences. The inequalities follow. □

Remark 3454 (Mapping Locations to canonical parameter regimes)**.** *A useful operationalization is to map: ordinary waking $\mapsto$ Location 0, Being/flow-like regimes $\mapsto$ Location 1, oceanic consciousness $\mapsto$ Location 2, and stable enlightenment-like regimes $\mapsto$ Location 4, by choosing threshold sequences (r_k, s_k) consistent with the canonical regime table.*

Definition 585 (Habit reinforcement operator (abstract))**.** *A* habit reinforcement operator *is a monotone inflationary map $H_{\rho_{\mathrm{hab}}} : V^{L^{\mathrm{ref}}_{\mathrm{Ob}}} \to V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$ that increases support for propositions that have been repeatedly supported in the recent past, with strength controlled by $\rho_{\mathrm{hab}} \in [0, 1]$.*

Definition 586 (Habit-augmented reflective dynamics)**.** *Given parameters Λ, define the habit-augmented reflective update map*

$$F_\Lambda^{\mathrm{hab}} := H_{\rho_{\mathrm{hab}}} \circ F_\Lambda.$$

A habit-stabilized reflective state *is a fixed point of F_Λ^{hab}.*

Theorem 174 (Existence and persistence of habit-stabilized PNSE states)**.** *Assume $H_{\rho_{\mathrm{hab}}}$ and F_Λ are monotone and $H_{\rho_{\mathrm{hab}}}$ is inflationary. Then F_Λ^{hab} has at least one fixed point.*

Assume additionally there is a set $\mathcal{P} \subseteq V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$ of "PNSE-like" reflective states such that: (i) $\mathcal{P}$ is upward closed (if $E \in \mathcal{P}$ and $E \leq E'$ then $E' \in \mathcal{P}$), (ii) $F_\Lambda(\mathcal{P}) \subseteq \mathcal{P}$, and (iii) $H_{\rho_{\mathrm{hab}}}(\mathcal{P}) \subseteq \mathcal{P}$. Then $\mathcal{P}$ is forward invariant under the iteration of F_Λ^{hab}.

Proof. Because $V^{L_{\mathrm{Ob}}^{\mathrm{ref}}}$ is a complete lattice and $F_{\Lambda}^{\mathrm{hab}}$ is monotone (composition of monotone maps), Tarski gives existence of a fixed point. For forward invariance, let $E \in \mathcal{P}$. Then by (ii) $F_{\Lambda}(E) \in \mathcal{P}$ and by (iii) $F_{\Lambda}^{\mathrm{hab}}(E) = H_{\rho_{\mathrm{hab}}}(F_{\Lambda}(E)) \in \mathcal{P}$. Thus the next iterate lies in $\mathcal{P}$, and induction yields invariance for all iterates. More explicitly, the base step is the given hypothesis $E \in \mathcal{P}$, and the inductive step is: if $E_t \in \mathcal{P}$, then $F_{\Lambda}(E_t) \in \mathcal{P}$ by (ii), and applying the habit/hysteresis operator $H_{\rho_{\mathrm{hab}}}$ preserves membership in $\mathcal{P}$ by (iii), so $E_{t+1} = F_{\Lambda}^{\mathrm{hab}}(E_t) \in \mathcal{P}$. Hence every forward iterate under the composed update map remains in the admissible set $\mathcal{P}$, so trajectories cannot "escape" the regime of well-defined evaluation of Coh, SAct_{θ}, and Loc. □

Theorem 175 (Canonical-regime monotonicity yields Location monotonicity (a template))**.** *Let Λ_1, Λ_2 be parameter tuples with: (a) $\lambda_{\mathrm{coh}}(\Lambda_2) \geq \lambda_{\mathrm{coh}}(\Lambda_1)$ (more coherence preference), (b) $\lambda_{\mathrm{ref}}(\Lambda_2) \leq \lambda_{\mathrm{ref}}(\Lambda_1)$ (less introspection gain), and (c) all other components held fixed. In particular, (c) means that any auxiliary weights governing evidence assimilation, coupling to habituation, or exogenous priors are unchanged, so the only directional pressures introduced when moving from Λ_1 to Λ_2 are toward coherence (via λ_{coh}) and away from reflective amplification (via λ_{ref}). Assume that increasing λ_{coh} (at fixed evidence) cannot decrease* $\mathrm{Coh}(\cdot)$*, and decreasing λ_{ref} cannot increase* $\mathrm{SAct}_{\theta}(\cdot)$ *at the resulting least reflective fixed point. Here "least reflective fixed point" is the one selected by the* lfp *operator, i.e. the minimal fixed point with respect to the underlying order used to formalize reflectivity/activation (e.g. pointwise order on state components, or an order induced by a reflection functional), so that the comparison across parameter regimes is made at comparable, non-maximally self-amplified equilibria.*

Then, with $E_i^* = \mathrm{lfp}(F_{\Lambda_i})$, we have $\mathrm{Loc}(E_2^*) \geq \mathrm{Loc}(E_1^*)$*. Equivalently, along this parameter change, the equilibrium state cannot move to a strictly lower Location level, in the sense encoded by the threshold structure defining* Loc*.*

Interpretation: *choosing canonical parameter regimes (e.g. "Being", "Oceanic", "Enlightenment") so that coherence preference increases and introspection gain decreases along the sequence yields a formally monotone increase in Location index. This can be read as a design principle: when the model is tuned to privilege globally consistent integration (higher λ_{coh}) while dampening reflective self-activation loops (lower λ_{ref}), the canonical equilibrium shifts toward (weakly) higher Location strata, provided the two stated monotonicity responses of* Coh *and* SAct_{θ} *hold under the fixed-point selection.*

Proof. Under the assumptions, the fixed point E_2^* has $\mathrm{Coh}(E_2^*) \geq \mathrm{Coh}(E_1^*)$ and $\mathrm{SAct}_{\theta}(E_2^*) \leq \mathrm{SAct}_{\theta}(E_1^*)$. The first inequality follows from the postulated non-decreasing response of $\mathrm{Coh}(\cdot)$ when λ_{coh} is increased while holding evidence and the remaining coordinates of Λ fixed; the second follows from the postulated non-increasing response of $\mathrm{SAct}_{\theta}(\cdot)$ when λ_{ref} is decreased, evaluated at the corresponding least reflective fixed point selected by lfp. By definition of Loc as a maximum k satisfying threshold inequalities, these two monotone inequalities imply $\mathrm{Loc}(E_2^*) \geq \mathrm{Loc}(E_1^*)$. Concretely, if $\mathrm{Loc}(E_1^*) = k$, then E_1^* satisfies the required inequalities at level k (e.g. $\mathrm{Coh}(E_1^*) \geq \tau_k^{\mathrm{coh}}$ and $\mathrm{SAct}_{\theta}(E_1^*) \leq \tau_k^{\mathrm{sact}}$, for the relevant threshold sequences); the displayed monotonicity relations ensure E_2^* also satisfies those same level-k inequalities, so the maximal admissible level for E_2^* cannot be smaller than k. Hence $\mathrm{Loc}(E_2^*) \geq \mathrm{Loc}(E_1^*)$ as claimed. □

62.9 Theorems: collective Locations via cultural fixed points

We now use Hyperseed's culture-as-fixed-point picture to derive sufficient conditions for collective Location properties.

Definition 587 (Social reinforcement dynamics)**.** *Let $G = (V, E)$ be a directed graph of agents. Let $\mathcal{P}$ be a set of cultural patterns. A* cultural state *is an intensity table $I : V \times \mathcal{P} \to [0, 1]$.*

Fix edge weights $K : E \to [0,1]$. Define the social reinforcement operator F by

$$(F(I))_{i,P} := \max\Big(I_{i,P},\ \max_{(j,i)\in E} \min\big(K(j,i), I_{j,P}\big)\Big).$$

A stabilized cultural state *is a fixed point I^* of F.*

Definition 588 (A simple integration proxy and collective discord proxy)**.** *Fix a set of "key" patterns $\mathcal{P}_0 \subseteq \mathcal{P}$. Define an integration proxy*

$$\mathrm{Int}(I) := \min_{P\in\mathcal{P}_0}\ \min_{i\in V}\ I_{i,P}.$$

Assume each agent has a proposal map $y_i = \pi(I_i)$ where I_i is the row $(I_{i,P})_{P\in\mathcal{P}_0}$. Assume π is L-Lipschitz:

$$D_O(\pi(u), \pi(v)) \le L\|u - v\|_\infty.$$

Define discord proxy

$$\mathrm{DO}(I) := \max_{i,j\in V} D_O(y_i, y_j).$$

Theorem 176 (Fixed points yield resilient homeostasis (C1 sufficient condition))**.** *Assume the collective cultural dynamics iterate $I^{(t+1)} = F(I^{(t)})$ for a monotone F as above. Then any fixed point I^* is an invariant "homeostatic" state: if $I^{(0)} = I^*$ then $I^{(t)} = I^*$ for all t. If, additionally, F is inflationary ($I \le F(I)$), then the orbit from any initial condition is nondecreasing and converges (in finite time on finite grids, or in the complete-lattice sense) to the least fixed point above $I^{(0)}$.*

In particular, the existence of fixed points provides a formal C1-style "resilient homeostasis" mechanism for a collective.

Proof. If $I^* = F(I^*)$ then iterating preserves it. Inflationarity gives $I^{(0)} \le I^{(1)} \le \cdots$. Because the poset $[0,1]^{V\times\mathcal{P}}$ is a complete lattice, the monotone chain has a least upper bound; standard Kleene/Tarski arguments show the limit is a fixed point (indeed the least fixed point above the initial state). □

Theorem 177 (Strong connectivity yields a diffusion lower bound (C2 integration enabler))**.** *Assume the communication graph G is strongly connected. Assume there exists $k_0 > 0$ such that $K(e) \ge k_0$ for all edges $e \in E$. Let I^* be any fixed point of F with $I^* \ge I^{(0)}$. Fix a pattern P and define $M_P := \max_{j\in V} I^{(0)}_{j,P}$. Then for every agent $i \in V$,*

$$I^*_{i,P} \ge \min(k_0, M_P).$$

Consequently, for any key-pattern set $\mathcal{P}_0$ we have

$$\mathrm{Int}(I^*) \ge \min\Big(k_0,\ \min_{P\in\mathcal{P}_0} M_P\Big).$$

Proof. Fix i and choose a directed path $j \to \cdots \to i$ from some j achieving M_P (strong connectivity guarantees such a path). Along any edge (u,v) the fixed point condition implies $I^*_{v,P} \ge \min(K(u,v), I^*_{u,P}) \ge \min(k_0, I^*_{u,P})$. Induct along the path starting from $I^*_{j,P} \ge I^{(0)}_{j,P} = M_P$ to obtain $I^*_{i,P} \ge \min(k_0, M_P)$. Taking minima over i and $P \in \mathcal{P}_0$ yields the integration bound. □

Corollary 64 (A discord bound from integration convergence)**.** *Assume, in addition to the hypotheses above, that for each key pattern $P \in \mathcal{P}_0$ the fixed point satisfies $\max_{i,j} |I^*_{i,P} - I^*_{j,P}| \le \delta$. Then $\mathrm{DO}(I^*) \le L\delta$.*

Proof. For any i, j, $\|I_i^* - I_j^*\|_\infty \leq \delta$ by assumption. By Lipschitzness of π, $D_O(y_i, y_j) \leq L\delta$. Taking the maximum yields the claim. $\square$

Theorem 178 (Narrative capture vs discord: a budget tradeoff)**.** *Assume a fixed per-time collective resource budget normalized to* 1*. Assume that maintaining integration level* $\mathrm{Int}(I) \geq \alpha$ *requires allocating at least* $\beta(\alpha)$ *of the resource budget to non-narrative integration processes. Then at any time* t*,*

$$N_O(t) \leq 1 - \beta(\alpha) \qquad \textit{whenever} \qquad \mathrm{Int}(I(t)) \geq \alpha.$$

In particular, if low discord requires high integration (e.g. DO *small implies* Int *large), then low discord implies an upper bound on narrative capture.*

Proof. If integration at level α requires spending at least $\beta(\alpha)$ of the budget, then the remaining budget for narrative processes is at most $1 - \beta(\alpha)$. By definition $N_O(t)$ is the narrative budget fraction, so $N_O(t) \leq 1 - \beta(\alpha)$. The last sentence follows by composition of implications. $\square$

62.10 Mindplexes: Hyperseed vs SDP, development, and weak mindplexes

Definition 589 (Hyperseed-style mindplex and notable coherence)**.** *Fix an observer* O *and a coherence score* $\mathrm{Coh}_O(\cdot)$ *on candidate minds. A system* M *is* notably coherent *if* $\mathrm{Coh}_O(M)$ *is a local maximum relative to small modifications of* M*. A* mindplex *is a notably coherent mind* M *that decomposes into notably coherent minds* $\{M_i\}$ *such that: (a) each* M_i *is approximately pattern-preserving as a restriction of* M*, and (b) replacing any* M_i *by another comparably coherent mind* M_i' *does not destroy the coherence of* M *(robustness/replaceability).*

Definition 590 (SDP-style mindplex (dual-level theatre of consciousness))**.** *A* SDP mindplex *is a collection of intelligent systems* $\{S_i\}$ *such that: (i) each* S_i *has its own theatre of consciousness and control loops, and (ii) the collective* $S = \bigsqcup_i S_i$ *also supports a collective theatre of consciousness and collective control loops, with nontrivial inter-level resonance.*

Definition 591 (Weak mindplex)**.** *Fix* $\epsilon \geq 0$ *and a time horizon* T*. A system* M *is an* (ϵ, T)-weak mindplex *if it satisfies the Hyperseed mindplex conditions up to error* ϵ *(approximate coherence and approximate replaceability) and, in addition, exhibits a collective GCC episode of duration at least* T*.*

Theorem 179 (From SDP mindplex to Hyperseed mindplex (sufficient conditions))**.** *Assume* $S = \bigsqcup_i S_i$ *is an SDP mindplex. Assume that the collective theatre of consciousness corresponds (under the translation layer) to a workspace-stable reflective fixed point* E^* *with high coherence, and that the components* S_i *correspond to notably coherent subminds* M_i *whose replacement does not significantly change the collective coherence.*

Then the collective S *satisfies the Hyperseed mindplex definition (relative to* O*).*

Proof. By assumption, the collective is (i) coherent and (ii) decomposes into coherent parts. Approximate pattern-preserving restriction is supplied by the assumption that each S_i participates as a stable substructure in the collective theatre (so its induced patterns are restrictions of the collective patterns). Replaceability is assumed as "replacement does not significantly change collective coherence." These are exactly the conditions of the Hyperseed mindplex definition. $\square$

Theorem 180 (From Hyperseed mindplex to SDP mindplex under dual-level GCC/RC)**.** *Assume* M *is a Hyperseed mindplex decomposing into minds* $\{M_i\}$*. Assume additionally: (A) each* M_i *exhibits RC episodes (individual reflective fixed points exist and satisfy reflective self); (B) the* *collective M exhibits GCC episodes with a workspace-stable focus set; and (C) there exists non-trivial inter-level resonance, meaning that collective conscious contents can modulate individual attention/control and vice versa.*

Then M is an SDP mindplex.

Proof. Conditions (A) and (B) provide the required dual-level theatres of consciousness (individual and collective). Condition (C) provides the nontrivial inter-level resonance required by the SDP notion. Thus the definition of SDP mindplex is satisfied. $\square$

Theorem 181 (Mindplex emergence along a development path)**.** *Let $I_0 \to I_1 \to \cdots$ be a Hyperseed development path for a system M (continuity plus nontrivial capability growth). Assume that along this path the coherence score $\mathrm{Coh}_O(M; I_k)$ is nondecreasing and eventually crosses a threshold τ at which M becomes notably coherent and decomposable into notably coherent subminds.*

Then there exists k_0 such that for all $k \geq k_0$, M is a mindplex on interval I_k.

Proof. By assumption, there is k_0 such that $\mathrm{Coh}_O(M; I_k) \geq \tau$ for $k \geq k_0$. At such k, the notable coherence and decomposition assumptions state exactly the mindplex conditions, hence M is a mindplex on I_k for all $k \geq k_0$. Continuity (self-continuity across successive intervals) ensures these are stages of one persisting entity rather than unrelated systems. $\square$

Theorem 182 (Small teams as weak mindplexes (a formal sufficient condition))**.** *Consider a team of n agents $\{S_i\}$ coupled by: (i) a strongly connected communication graph with coupling lower bound $k_0 > 0$; (ii) a shared persistent artifact store (documents, codebase, musical score) that acts as a high-bandwidth hub in the coupling graph; and (iii) synchronized interaction intervals of length at least T during which attention is aligned toward a shared task and shared artifacts.*

Assume the induced collective cultural reinforcement dynamics satisfy the hypotheses of the diffusion lower bound theorem for the task-relevant patterns, and that the collective has a workspace-stable focus set during the synchronized intervals.

Then the team forms an (ϵ, T)-weak mindplex for some ϵ determined by the replaceability error induced by role specialization.

Proof. During synchronized intervals, strong connectivity plus artifact-mediated coupling yields diffusion of task-relevant patterns across agents, increasing the integration proxy and reducing fragmentation. The shared artifact hub provides a stable anchor that keeps the collective in a metastable basin (chain) for duration at least T, yielding a collective GCC episode by assumption. Approximate decomposition into coherent subminds is given by the individual agents; replaceability holds up to error ϵ because role-specialized components can be partially replaced by other agents or tools without destroying the collective's coherent task performance. Thus the weak mindplex definition is satisfied. $\square$

62.11 Closing remarks

The bridge above turns qualitative claims about consciousness Locations, PNSE, and collective coordination into monotone fixed-point and threshold statements that can be used by a reasoning system: (i) map observable markers into evidence-state functionals; (ii) infer Location/collective Location; (iii) reason about interventions by moving parameter tuples or coupling strengths; and (iv) reason about emergence of mindplex structure along development paths. In practice, the role of the present closing constructions is to make the "interface" between phenomenological descriptions (e.g., self-model vividness, narrative density, stability of a global workspace) and formal inference explicit: one chooses a small family of scores, states the relevant thresholds, and then treats the resulting decision rules as objects that can be optimized, compared, or subjected to sensitivity analysis under perturbations of the environment or architecture. The point is not that any particular scoring choice is uniquely correct, but that the overall pattern—monotonic summaries of evidence states together with stability constraints—yields well-posed targets for a reasoning system that must decide when a Location is present, when a collective Location is sustained, and when PNSE-like regimes occur.

Definition 592 (Coherence score)**.** *Fix a finite set $K \subseteq L_{\mathrm{Ob}}^{\mathrm{ref}}$ of "currently relevant" propositions. Let $\sigma : V \to \mathbb{R}$ be any monotone map that increases with net support and/or reduces with tension (model-dependent). Define the* coherence score *of an evidence state by*

$$\mathrm{Coh}(E) := \frac{1}{|K|} \sum_{\varphi \in K} \sigma(E(\varphi)).$$

The restriction to a finite K emphasizes that coherence is always evaluated relative to a task, context, or observational window: K can be taken as the set of propositions whose truth-values (or support levels) are currently behaviorally and explanatorily salient. The map σ is left abstract to accommodate different instantiations of evidence states $E(\varphi) \in V$: for example, if V encodes separate support/tension coordinates, then σ can be chosen to reward high support and penalize contradiction, whereas if V already encodes a single scalar credibility then σ may be the identity. Monotonicity ensures that "more supportive" updates cannot reduce $\mathrm{Coh}(E)$, which is the key property needed for threshold reasoning and for connecting coherence to fixed-point stability: in many models, workspace-stable states are exactly those for which additional rounds of update do not substantially increase such aggregate coherence measures.

Definition 593 (Self-model activity score)**.** *Let P_{self} be the designated self-property set and fix $\theta \in (0,1)$. Define*

$$\mathrm{SAct}_\theta(E) := \frac{1}{|P_{\mathrm{self}}|} \big|\{p \in P_{\mathrm{self}} : E^+(\mathrm{Self}(p)) \geq \theta\}\big|.$$

This score operationalizes the extent to which self-directed predicates are actively endorsed in the current evidence state. The threshold θ plays a dual role: it suppresses weak, noisy, or incidental self-attributions, while also making explicit that "self-model activity" is not a binary property but a graded one that depends on where one draws the line between background availability and workspace-level activation. If P_{self} is organized into subfamilies (e.g., bodily self, narrative self, agency, social self), then SAct_θ can be evaluated per subfamily as well; the present global definition is the simplest aggregate compatible with later Location-indexing rules. In particular, the use of E^+ highlights that the intended signal is positive endorsement rather than mere mention or ambivalent representation.

Definition 594 (Symbolic load and PNSE)**.** *Let* Sym $\subseteq L_{\mathrm{Ob}}$ *denote the subset of propositions treated as linguistic/symbolic/narrative in the current modeling context. Define the* symbolic load *of E by*

$$\mathrm{SL}_\theta(E) := \frac{|\mathrm{Con}_\theta(E) \cap \mathrm{Sym}|}{|\mathrm{Sym}|}.$$

*Fix $\varepsilon \in [0,1]$. A workspace-stable evidence state E is an ε-*PNSE state *if $\mathrm{SL}_\theta(E) \leq \varepsilon$ and $\mathrm{Coh}(E)$ is above a chosen coherence threshold.*

A system exhibits PNSE *on interval I (relative to (θ, ε)) if, for all $t \in I$, there is a workspace-stable E_t such that E_t is an ε-PNSE state.*

The definition isolates two conditions that are often conflated in informal discussions of PNSE: (i) low symbolic/narrative occupancy of the current workspace (captured by $\mathrm{SL}_\theta(E)$), and (ii) the presence of a globally coherent, stable state rather than a merely degraded or fragmented one (captured by the coherence threshold and the workspace-stability requirement). The use of $\mathrm{Con}_\theta(E)$ (as defined earlier in the framework) makes explicit that symbolic load is computed from the set of propositions that are not just present but sufficiently active to count as "in contention" at level θ; in this way, the metric distinguishes a system that contains dormant symbolic resources from a system whose workspace is currently dominated by symbolic/narrative material. The parameter ε then becomes a tunable dial: smaller ε corresponds to stricter PNSE criteria (fewer symbolic propositions above threshold), while larger ε allows for PNSE-like states that retain limited symbolic scaffolding. The interval formulation clarifies that the intended phenomenon is not a single instantaneous snapshot but a sustained regime: PNSE on I requires that at each time t there exists a workspace-stable evidence state meeting the conditions, which allows one to treat PNSE as a temporally extended property suitable for intervention analysis (e.g., whether changes in coupling or sensory drive can maintain the regime, break it, or modulate its depth).

Definition 595 (A Location classifier from Hyperseed evidence states)**.** *Fix thresholds* $(r_0 < r_1 < \cdots)$ *and* $(s_0 > s_1 > \cdots)$ *in* $[0,1]$. *Define the* Hyperseed Location index *of an evidence state* E *as*

$$\mathrm{Loc}(E) := \max\{k : \mathrm{Coh}(E) \geq r_k \text{ and } \mathrm{SAct}_\theta(E) \leq s_k\}.$$

This classifier makes explicit a particular qualitative hypothesis that often appears in Location talk: higher-location regimes correspond to higher global coherence together with reduced explicit self-model activity (or, depending on the modeling choice, reduced self-referential dominance). The increasing sequence (r_k) encodes progressively stricter coherence requirements, while the decreasing sequence (s_k) encodes progressively stricter limits on self-model activation. Taking the maximum k implements a "highest satisfied tier" decision rule, which is convenient for a reasoning system because it turns graded scores into an ordinal index that can be compared across time, agents, or interventions. The construction also supports natural refinements: one may impose minimum-size conditions on K, require robustness of $\mathrm{Loc}(E)$ under small perturbations of E, or define a "collective" analogue by replacing $\mathrm{Coh}(E)$ and $\mathrm{SAct}_\theta(E)$ with group-aggregated versions computed from coupled evidence states. Importantly, because the index is defined directly from evidence-state functionals, it can be computed at intermediate steps along a developmental or training path, thereby enabling explicit claims of the form "mindplex structure emerges when the system enters and stabilizes in Location tier k under sustained coupling conditions" and making those claims testable within the adopted Hyperseed formalism.

62.12 Theorems: GCC/RC imply Hyperseed conscious/reflective fixed points

The goal of this chunk is to make explicit a standard fixed-point pattern: once evidence states are ordered pointwise and the workspace/reflection dynamics are modeled by monotone operators on that order, stable conscious (and reflective) states follow from general lattice-theoretic principles. In particular, the thresholded consciousness set $\mathrm{Con}_\theta(E)$ behaves monotonically with respect to strengthening evidence, so any dynamics that move evidence "upwards" (or at least do not decrease it) preserve threshold-crossing propositions.

Lemma 170 (Threshold monotonicity)**.** *If* $E \leq E'$ *pointwise (componentwise in* V*), then* $\mathrm{Con}_\theta(E) \subseteq \mathrm{Con}_\theta(E')$.

Proof. If $\varphi \in \mathrm{Con}_\theta(E)$ then $E^+(\varphi) \geq \theta$. Since $E \leq E'$ we have $E'^+(\varphi) \geq E^+(\varphi) \geq \theta$, hence $\varphi \in \mathrm{Con}_\theta(E')$. □

The content of the lemma can be read as a minimal "no-forgetting under strengthening" condition: once a proposition is above the conscious threshold, increasing evidence componentwise cannot push it below threshold. Here $E^+(\varphi)$ is the positive (supporting) component of evidence assigned to φ, and the pointwise order $E \leq E'$ is intended to mean that every observable/propositional coordinate in E' is at least as supported as in E (in whatever evidential scale is encoded by V). This is the technical hinge used later when selecting fixed points that preserve particular endorsed contents (e.g. basin membership during a GCC episode, or self-ascriptions during RC).

Theorem 183 (Workspace-stable conscious states exist under monotone closure)**.** *Assume L_{Ob} is finite and $W : V^{L_{\mathrm{Ob}}} \to V^{L_{\mathrm{Ob}}}$ is monotone. Then W has at least one fixed point. In particular, there exist workspace-stable evidence states.*

Proof. The set $V^{L_{\mathrm{Ob}}}$ is a complete lattice under pointwise order because V is. By monotonicity of W, Tarski's fixed point theorem implies the set of fixed points of W is nonempty (and has least and greatest fixed points). □

Conceptually, W plays the role of a "workspace closure" or "broadcast-then-settle" map: it takes an evidence profile across observables/propositions and returns the result of one stabilization step of the workspace dynamics. Monotonicity here formalizes an intuitive constraint: if one starts from an evidence state with uniformly stronger support, the stabilized state should not become uniformly weaker. The conclusion is not merely existence but structure: the fixed points form a (nonempty) complete lattice, and one can speak meaningfully about least/greatest fixed points as canonical stable states associated to the same dynamics. This becomes useful when one wants to guarantee preservation of some already-supported content by selecting a fixed point that is above a specified baseline state.

Theorem 184 (GCC implies existence of stable focus sets)**.** *Assume $\mathrm{GCC}_{O,\theta}(S, I)$. Assume further that during the GCC episode there exists a time-indexed family of monotone workspace operators W_t and presentational immediacy functions $I_{c,t}$ such that: (A1) W_t encodes "global broadcasting + basin stabilization" for the GCC basin at time t; and (A2) there exists at least one proposition φ_t expressing the active basin membership such that $E^+_{\bullet,t}(\varphi_t) \geq \theta$ in the integrated module evidence $E_{\bullet,t}$.*

Then for each $t \in I$ there exists a workspace-stable evidence state E_t with $\varphi_t \in \mathrm{Con}_\theta(E_t)$; hence $\mathrm{Con}_\theta(E_t) \neq \emptyset$. Moreover the focus sets $F_{\theta,t} = \{x : I_{c,t}(x) \geq \theta\}$ are nonempty for all $t \in I$.

Proof. Fix $t \in I$. By the previous theorem, W_t has at least one fixed point E_t. Assumption (A2) provides a proposition φ_t with positive evidence above threshold in the integrated evidence. Because W_t is monotone and (by the intended reading of (A1)) does not destroy supported basin-membership content, we may choose E_t among fixed points satisfying $E_t \geq E_{\bullet,t}$ on φ_t (e.g. take the greatest fixed point if needed). Then $E^+_t(\varphi_t) \geq \theta$, so $\varphi_t \in \mathrm{Con}_\theta(E_t)$. The nonemptiness of $F_{\theta,t}$ follows from the bridge assumption that GCC-visible basin content corresponds to at least one presentational token above threshold (otherwise there would be no unified "moment" in the MinSync sense). □

To clarify the role of (A2), φ_t is meant to be a proposition whose satisfaction conditions track the currently dominant GCC basin (e.g. "the system is in basin B" or "content cluster C is globally broadcast"). The existence of some such φ_t above threshold in $E_{\bullet,t}$ is the minimal claim that the global broadcast is not purely subthreshold: at least one basin-defining content is sufficiently supported to qualify as conscious by the θ-criterion. The selection of E_t among fixed points is then justified by combining (i) the order structure on evidence states, (ii) monotonicity of W_t, and (iii) the intended semantic constraint in (A1) that broadcasting-plus-stabilization should not systematically eliminate the very basin-membership content that characterizes the episode. In this way, the fixed-point argument is not merely abstract existence: it is targeted to yield a stable state whose conscious set contains a concrete GCC-relevant proposition.

The second conclusion, $F_{\theta,t} \neq \emptyset$, ties the evidence-level conclusion to a token-level notion of "focus" or "presentational presence." The set $F_{\theta,t} = \{x : I_{c,t}(x) \geq \theta\}$ is nonempty if at least one presentational token x achieves immediacy at or above the same threshold. The stated bridge assumption is doing the work of linking content-level broadcast (captured by $\varphi_t \in \mathrm{Con}_\theta(E_t)$) to the existence of at least one corresponding presentational item that can function as the synchronized anchor of the moment; without such an anchor, the GCC episode would fail to realize a unified presentational field in the MinSync sense.

Theorem 185 (RC implies existence of reflective-self fixed points)**.** *Assume* $\mathrm{RC}_{O,\theta}(S, I)$*. Assume that the RC self-model supplies a designated set* P_{self} *of self-properties and module evidence supports* $\mathrm{Self}(p)$ *for many* $p \in P_{\mathrm{self}}$ *during* I*.*

Then there exists a reflective fixed point E^* *of* $F = W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a$ *that satisfies the reflective self condition.*

Proof. Because $L^{\mathrm{ref}}_{\mathrm{Ob}}$ is finite and F is monotone (as a composition of monotone maps), Tarski's theorem yields a fixed point E^* of F.

(Self-presence) By assumption, for many $p \in P_{\mathrm{self}}$ the base evidence for $\mathrm{Self}(p)$ is above threshold during I. Monotonicity of Γ_a and W^{ref} preserves (or increases) this support in the fixed point.

(Reflective access) The introspection operator assigns evidence to $I(\mathrm{Self}(p))$ as a monotone function of the base evidence for $\mathrm{Self}(p)$, so for those p the introspective propositions also cross threshold in E^*.

(Stability) RC requires that the self-model's representational loops are robust under benign internal reparameterizations. Modeling this as small perturbations in module evidence and attention implies the relevant supports persist in a neighborhood of the fixed point. □

Here $F = W^{\mathrm{ref}} \circ \mathrm{Int} \circ \Gamma_a$ can be read as a minimal three-stage abstraction of reflective cognition: Γ_a implements attentional selection/gating (making some self-relevant channels more salient while remaining monotone in the sense of not decreasing support when the input is increased), Int implements introspective re-representation (mapping base-level self-ascriptions to higher-order claims such as $I(\mathrm{Self}(p))$), and W^{ref} implements a reflective workspace closure that stabilizes the resulting higher-order package. The "reflective self condition" is thus not added by hand at the end; it is enforced by the combination of thresholded self-ascriptions, monotone propagation through the reflective pipeline, and selection of a fixed point where both base and introspective self propositions remain above θ.

It is also useful to note that the proof is structurally parallel to the GCC argument: (i) obtain a fixed point from monotonicity on a complete lattice, (ii) use threshold monotonicity to ensure that once self-content is supported above θ it is not lost by the closure dynamics, and (iii) interpret RC stability as a robustness property around the fixed point rather than a single equality $F(E^*) = E^*$. In that sense, the fixed point E^* functions as a reflective "attractor-like" state: not only is it closed under the reflective update, but it is also locally resilient to small benign perturbations in the evidential inputs that feed attention and introspection.

62.13 Theorems: PNSE and Martin Locations via Hyperseed parameter regimes

Definition 596 (Parametric monotonicity assumption)**.** *Let* Λ *range over parameter tuples. Write* F_Λ *for the resulting reflective update map. Assume a partial order* $\preceq$ *on parameter tuples such* *that $\Lambda \preceq \Lambda'$ implies $F_\Lambda(E) \leq F_{\Lambda'}(E)$ for all E. (Informally: increasing parameters that amplify completion or introspection cannot decrease evidence.)*

Lemma 171 (Least fixed points are monotone in monotone operator families)**.** *Assume F and G are monotone operators on a complete lattice with $F(E) \leq G(E)$ for all E. Let $\mathrm{lfp}(F)$ denote the least fixed point of F. Then $\mathrm{lfp}(F) \leq \mathrm{lfp}(G)$.*

Proof. By Tarski, both least fixed points exist. Let $x = \mathrm{lfp}(G)$. Then $G(x) = x$ and $F(x) \leq G(x) = x$, so x is a pre-fixpoint of F. The least fixed point of F is below every pre-fixpoint, hence $\mathrm{lfp}(F) \leq x$. □

Theorem 186 (Symbolic suppression under reduced completion and introspection)**.** *Assume the parametric monotonicity assumption. Fix a threshold θ and symbolic subset $\mathrm{Sym} \subseteq L_{\mathrm{Ob}}$. Let $\Lambda \preceq \Lambda'$ where (informally) Λ has lower internal completion λ_{int} and lower introspection gain λ_{ref} than Λ', with coherence preference λ_{coh} held high.*

Let $E^* = \mathrm{lfp}(F_\Lambda)$ and $E'^* = \mathrm{lfp}(F_{\Lambda'})$. Then

$$\mathrm{Con}_\theta(E^*) \cap \mathrm{Sym} \subseteq \mathrm{Con}_\theta(E'^*) \cap \mathrm{Sym},$$

and in particular $\mathrm{SL}_\theta(E^*) \leq \mathrm{SL}_\theta(E'^*)$.

Proof. By parametric monotonicity, $F_\Lambda(E) \leq F_{\Lambda'}(E)$ for all E. By the previous lemma, $\mathrm{lfp}(F_\Lambda) \leq \mathrm{lfp}(F_{\Lambda'})$, i.e. $E^* \leq E'^*$. Apply Lemma (Threshold monotonicity) to conclude $\mathrm{Con}_\theta(E^*) \subseteq \mathrm{Con}_\theta(E'^*)$. Intersect both sides with Sym to obtain the inclusion, and divide by $|\mathrm{Sym}|$. □

Corollary 65 (A formal PNSE sufficient condition)**.** *Fix $\varepsilon \in [0, 1]$. If there exists a parameter tuple Λ such that the least fixed point $E^* = \mathrm{lfp}(F_\Lambda)$ has $\mathrm{SL}_\theta(E^*) \leq \varepsilon$ and high coherence $\mathrm{Coh}(E^*) \geq r_1$, then E^* is an ε-PNSE state. Moreover, any parameter tuple $\tilde{\Lambda} \preceq \Lambda$ yields a least fixed point whose symbolic load is at most ε.*

Proof. The first claim is immediate from the definition of ε-PNSE. For the second, apply the previous theorem with $\tilde{\Lambda} \preceq \Lambda$. □

Theorem 187 (Martin-Location ordering as a monotone mapping)**.** *Assume the SDP key claim that higher numbered Locations have higher sustained global resonance and lower self-model activity. Assume the translation of these into Hyperseed evidence-state functionals: higher resonance corresponds to larger $\mathrm{Coh}(E)$ and lower self-model activity corresponds to smaller $\mathrm{SAct}_\theta(E)$.*

Then for any evidence states E, E': if $\mathrm{Loc}(E') > \mathrm{Loc}(E)$, then $\mathrm{Coh}(E') \geq \mathrm{Coh}(E)$ and $\mathrm{SAct}_\theta(E') \leq \mathrm{SAct}_\theta(E)$.

Proof. Let $k = \mathrm{Loc}(E)$ and $k' = \mathrm{Loc}(E')$ with $k' > k$. By definition of Loc, $\mathrm{Coh}(E) \geq r_k$ and $\mathrm{SAct}_\theta(E) \leq s_k$, and $\mathrm{Coh}(E') \geq r_{k'} \geq r_k$ and $\mathrm{SAct}_\theta(E') \leq s_{k'} \leq s_k$, using monotonicity of the threshold sequences. The inequalities follow. □

Remark 3455 (Mapping Locations to canonical parameter regimes)**.** *A useful operationalization is to map: ordinary waking $\mapsto$ Location 0, Being/flow-like regimes $\mapsto$ Location 1, oceanic consciousness $\mapsto$ Location 2, and stable enlightenment-like regimes $\mapsto$ Location 4, by choosing threshold sequences (r_k, s_k) consistent with the canonical regime table.*

Definition 597 (Habit reinforcement operator (abstract))**.** *A* habit reinforcement operator *is a monotone inflationary map $H_{\rho_{\mathrm{hab}}} : V^{L^{\mathrm{ref}}_{\mathrm{Ob}}} \to V^{L^{\mathrm{ref}}_{\mathrm{Ob}}}$ that increases support for propositions that have been repeatedly supported in the recent past, with strength controlled by $\rho_{\mathrm{hab}} \in [0, 1]$.*

Definition 598 (Habit-augmented reflective dynamics)**.** *Given parameters* Λ*, define the habit-augmented reflective update map*

$$F_\Lambda^{\mathrm{hab}} := H_{\rho_{\mathrm{hab}}} \circ F_\Lambda.$$

A habit-stabilized reflective state *is a fixed point of* F_Λ^{hab}*.*

Theorem 188 (Existence and persistence of habit-stabilized PNSE states)**.** *Assume* $H_{\rho_{\mathrm{hab}}}$ *and* F_Λ *are monotone and* $H_{\rho_{\mathrm{hab}}}$ *is inflationary. Then* F_Λ^{hab} *has at least one fixed point.*

Assume additionally there is a set $\mathcal{P} \subseteq V^{L_{\mathrm{Ob}}^{\mathrm{ref}}}$ *of "PNSE-like" reflective states such that: (i)* $\mathcal{P}$ *is upward closed (if* $E \in \mathcal{P}$ *and* $E \leq E'$ *then* $E' \in \mathcal{P}$*), (ii)* $F_\Lambda(\mathcal{P}) \subseteq \mathcal{P}$*, and (iii)* $H_{\rho_{\mathrm{hab}}}(\mathcal{P}) \subseteq \mathcal{P}$*. Then* $\mathcal{P}$ *is forward invariant under the iteration of* F_Λ^{hab}*.*

Proof. Because $V^{L_{\mathrm{Ob}}^{\mathrm{ref}}}$ is a complete lattice and F_Λ^{hab} is monotone (composition of monotone maps), Tarski gives existence of a fixed point. For forward invariance, let $E \in \mathcal{P}$. Then by (ii) $F_\Lambda(E) \in \mathcal{P}$ and by (iii) $F_\Lambda^{\mathrm{hab}}(E) = H_{\rho_{\mathrm{hab}}}(F_\Lambda(E)) \in \mathcal{P}$. Thus the next iterate lies in $\mathcal{P}$, and induction yields invariance for all iterates. In particular, the step from E to $F_\Lambda(E)$ uses only the closure of $\mathcal{P}$ under the base update map, while the step from $F_\Lambda(E)$ to $H_{\rho_{\mathrm{hab}}}(F_\Lambda(E))$ uses the fact that the habituation operator acts as a $\mathcal{P}$-endomorphism at the chosen habituation rate ρ_{hab}. Writing the iterates explicitly as $E_{t+1} = F_\Lambda^{\mathrm{hab}}(E_t)$ with $E_0 \in \mathcal{P}$, the argument shows: if $E_t \in \mathcal{P}$ then $E_{t+1} \in \mathcal{P}$; hence $E_t \in \mathcal{P}$ for all $t \in \mathbb{N}$. This guarantees that all subsequent quantities evaluated along the orbit (e.g. $\mathrm{Coh}(E_t)$ and $\mathrm{SAct}_\theta(E_t)$) remain well-defined within the intended state space. □

Theorem 189 (Canonical-regime monotonicity yields Location monotonicity (a template))**.** *Let* Λ_1, Λ_2 *be parameter tuples with: (a)* $\lambda_{\mathrm{coh}}(\Lambda_2) \geq \lambda_{\mathrm{coh}}(\Lambda_1)$ *(more coherence preference), (b)* $\lambda_{\mathrm{ref}}(\Lambda_2) \leq \lambda_{\mathrm{ref}}(\Lambda_1)$ *(less introspection gain), and (c) all other components held fixed. Assume that increasing* λ_{coh} *(at fixed evidence) cannot decrease* $\mathrm{Coh}(\cdot)$*, and decreasing* λ_{ref} *cannot increase* $\mathrm{SAct}_\theta(\cdot)$ *at the resulting least reflective fixed point. Concretely, the first assumption is a comparative-statics requirement: if two parameter settings differ only in* λ_{coh} *with the latter larger, then the fixed-point state selected by the "least reflective" criterion does not exhibit lower coherence. Likewise, the second assumption encodes that reducing the gain of the reflective/introspective channel does not, after the system re-equilibrates, lead to greater self-activation of the* θ*-indexed reflective component (or whichever observable* SAct_θ *tracks). The restriction in (c) rules out confounding changes in other drives or couplings, so that the direction of change can be attributed to the two designated coordinates.*

Then, with $E_i^* = \mathrm{lfp}(F_{\Lambda_i})$*, we have* $\mathrm{Loc}(E_2^*) \geq \mathrm{Loc}(E_1^*)$*.*

Interpretation: *choosing canonical parameter regimes (e.g. "Being", "Oceanic", "Enlightenment") so that coherence preference increases and introspection gain decreases along the sequence yields a formally monotone increase in Location index. Equivalently, along a path in parameter space where the system is pushed toward internal consistency while the reflective amplification loop is damped, the model predicts that the equilibrium state crosses Location thresholds in a nondecreasing manner, so later regimes cannot be assigned a strictly lower Location than earlier regimes under the stated monotonicity hypotheses.*

Proof. Under the assumptions, the fixed point E_2^* has $\mathrm{Coh}(E_2^*) \geq \mathrm{Coh}(E_1^*)$ and $\mathrm{SAct}_\theta(E_2^*) \leq \mathrm{SAct}_\theta(E_1^*)$. Here the inequalities should be read pointwise at the selected equilibria $E_i^* = \mathrm{lfp}(F_{\Lambda_i})$, with "least reflective" specifying which fixed point is chosen in case F_{Λ_i} admits multiple fixed points. By definition of Loc as a maximum k satisfying threshold inequalities, these two monotone inequalities imply $\mathrm{Loc}(E_2^*) \geq \mathrm{Loc}(E_1^*)$. More explicitly, if $\mathrm{Loc}(E)$ is defined via conditions of the form $\mathrm{Coh}(E) \geq \tau_k^{\mathrm{coh}}$ and $\mathrm{SAct}_\theta(E) \leq \tau_k^{\mathrm{sact}}$ for an increasing family of coherence thresholds τ_k^{coh} and decreasing (or otherwise tightening) self-activation thresholds τ_k^{sact}, then improving Coh while not increasing SAct_θ can only preserve or expand the set of k for which the conditions hold. Thus the maximal admissible index k for E_2^* cannot be smaller than that for E_1^*, yielding the claimed Location monotonicity. □

62.14 Theorems: collective Locations via cultural fixed points

We now use Hyperseed's culture-as-fixed-point picture to derive sufficient conditions for collective Location properties. In this subsection, the central technical move is to treat a "collective Location" not as a static label but as a *stable attractor-like constraint* on what cultural patterns can persist under repeated social updating. In that sense, fixed points play the role of *cultural invariants*: once the group reaches such a configuration, the configuration is preserved by the very mechanisms (imitation, reinforcement, prestige-biased transmission) that produced it.

Definition 599 (Social reinforcement dynamics)**.** *Let $G = (V, E)$ be a directed graph of agents. Let $\mathcal{P}$ be a set of cultural patterns. A* cultural state *is an intensity table $I : V \times \mathcal{P} \to [0, 1]$. Intuitively, $I_{i,P}$ is the degree to which agent i instantiates, endorses, or enacts pattern P; the unit interval is used only as a convenient normalized scale, and nothing in the arguments below requires a particular psychological interpretation beyond "higher means more present."*

Fix edge weights $K : E \to [0, 1]$. Define the social reinforcement operator F by

$$(F(I))_{i,P} := \max\Big(I_{i,P},\ \max_{(j,i)\in E} \min\big(K(j, i), I_{j,P}\big)\Big).$$

The $\min(K(j, i), I_{j,P})$ *term can be read as a* bottleneck transmission rule*: even if j strongly instantiates P, the effective reinforcement at i is capped by the strength of the $j \to i$ influence channel; conversely, even a strong channel cannot transmit more of P than j actually carries. The outer* max *expresses* non-erasure *(reinforcement without active suppression): i keeps what it already has, and may increase via incoming reinforcement. This is the simplest monotone/inflationary template capturing "culture as a fixed point" in the sense relevant to Hyperseed-style stabilization.*

A stabilized cultural state *is a fixed point I^* of F.*

Definition 600 (A simple integration proxy and collective discord proxy)**.** *Fix a set of "key" patterns $\mathcal{P}_0 \subseteq \mathcal{P}$. Define an integration proxy*

$$\mathrm{Int}(I) := \min_{P\in\mathcal{P}_0} \min_{i\in V} I_{i,P}.$$

This proxy is deliberately conservative: it measures the weakest *(least present) key pattern among the* least *aligned agent, so* $\mathrm{Int}(I) \geq \alpha$ *implies a floor α on all key-pattern intensities for all agents. In the intended reading, this corresponds to a group-level guarantee that the cultural "core" (the patterns in $\mathcal{P}_0$) is not merely present on average but is* uniformly *present, which is the kind of condition that can underwrite a collective Location claim.*

Assume each agent has a proposal map $y_i = \pi(I_i)$ where I_i is the row $(I_{i,P})_{P\in\mathcal{P}_0}$. Assume π is L-Lipschitz:

$$D_O(\pi(u), \pi(v)) \leq L\|u - v\|_\infty.$$

Here y_i should be read as any outwardly-relevant output (an action policy, an opinion vector, a governance vote, a public plan) computed from the key-pattern intensities. The Lipschitz assumption formalizes a continuity principle: small cultural differences across key patterns do not explode into arbitrarily large output disagreements.

Define discord proxy

$$\mathrm{DO}(I) := \max_{i,j\in V} D_O(y_i, y_j).$$

Remark 3456. *The pair* (Int, DO) *is not intended as a complete theory of cohesion; it is a minimal pair of handles that let us connect (i) a* cultural fixed point *story to (ii) an* operational *story about coordination and conflict. In particular,* Int *is a sufficient-condition style statistic (a guaranteed lower bound), whereas* DO *is an upper-envelope statistic (worst-case disagreement).*

Theorem 190 (Fixed points yield resilient homeostasis (C1 sufficient condition))**.** *Assume the collective cultural dynamics iterate $I^{(t+1)} = F(I^{(t)})$ for a monotone F as above. Then any fixed point I^* is an invariant "homeostatic" state: if $I^{(0)} = I^*$ then $I^{(t)} = I^*$ for all t. If, additionally, F is inflationary ($I \leq F(I)$), then the orbit from any initial condition is nondecreasing and converges (in finite time on finite grids, or in the complete-lattice sense) to the least fixed point above $I^{(0)}$.*

In particular, the existence of fixed points provides a formal C1-style "resilient homeostasis" mechanism for a collective. In the intended bridge to collective Location language, this says: once the group occupies a Location characterized by a stabilized cultural configuration, the endogenous reinforcement dynamics will maintain that Location absent exogenous perturbations that change F itself (e.g. censorship, coercion, new institutions) or inject additional operators (e.g. explicit anti-reinforcement).

Proof. If $I^* = F(I^*)$ then iterating preserves it. Inflationarity gives $I^{(0)} \leq I^{(1)} \leq \cdots$. Because the poset $[0,1]^{V \times \mathcal{P}}$ is a complete lattice, the monotone chain has a least upper bound; standard Kleene/Tarski arguments show the limit is a fixed point (indeed the least fixed point above the initial state). For readers tracking the mechanism: monotonicity ensures that increasing any entry of I cannot decrease any entry of $F(I)$, so the iterative sequence is well-behaved under pointwise order; inflationarity ensures the sequence is pointwise nondecreasing. On finite discretizations (e.g. restricting intensities to a finite grid), a nondecreasing sequence must stabilize in finitely many steps because there are only finitely many states; on the full $[0,1]$ continuum, one instead uses the complete-lattice framework to interpret convergence as reaching the least upper bound of the chain, which is then a post-fixed point that becomes a fixed point under the usual Tarski/Kleene conditions. □

Theorem 191 (Strong connectivity yields a diffusion lower bound (C2 integration enabler))**.** *Assume the communication graph G is strongly connected. Assume there exists $k_0 > 0$ such that $K(e) \geq k_0$ for all edges $e \in E$. Let I^* be any fixed point of F with $I^* \geq I^{(0)}$. Fix a pattern P and define $M_P := \max_{j \in V} I^{(0)}_{j,P}$. Then for every agent $i \in V$,*

$$I^*_{i,P} \geq \min(k_0, M_P).$$

Consequently, for any key-pattern set $\mathcal{P}_0$ we have

$$\mathrm{Int}(I^*) \geq \min\Big(k_0,\ \min_{P \in \mathcal{P}_0} M_P\Big).$$

This theorem isolates a purely structural enabler for C2-style integration: if influence pathways exist between any two agents (strong connectivity) and no influence edge is arbitrarily weak (uniform k_0), then whatever key pattern is present somewhere initially must become present everywhere at least up to the bottleneck $\min(k_0, \cdot)$. In other words, the network has no "cultural dead zones" with respect to the operator F.

Proof. Fix i and choose a directed path $j \to \cdots \to i$ from some j achieving M_P (strong connectivity guarantees such a path). Along any edge (u,v) the fixed point condition implies $I^*_{v,P} \geq \min(K(u,v), I^*_{u,P}) \geq \min(k_0, I^*_{u,P})$. Induct along the path starting from $I^*_{j,P} \geq I^{(0)}_{j,P} = M_P$ to obtain

$I^*_{i,P} \geq \min(k_0, M_P)$. Taking minima over i and $P \in \mathcal{P}_0$ yields the integration bound. Note that the lower bound is independent of the path length: because the update uses a min bottleneck rather than a multiplicative decay, the only global limitation is the weakest allowed edge strength k_0. This matches the qualitative idea that repeated reinforcement can propagate a pattern broadly so long as no link in the social chain is below a fixed transmissibility threshold. □

Corollary 66 (A discord bound from integration convergence)**.** *Assume, in addition to the hypotheses above, that for each key pattern $P \in \mathcal{P}_0$ the fixed point satisfies $\max_{i,j} |I^*_{i,P} - I^*_{j,P}| \leq \delta$. Then $\mathrm{DO}(I^*) \leq L\delta$. This is the simplest way to turn "cultural alignment" (small across-agent variation in key-pattern intensities) into "output alignment" (small proposal/plan distance). The role of the Lipschitz constant is to quantify how sensitive the outward behavior map π is to changes in the key-pattern profile.*

Proof. For any i, j, $\|I^*_i - I^*_j\|_\infty \leq \delta$ by assumption. By Lipschitzness of π, $D_O(y_i, y_j) \leq L\delta$. Taking the maximum yields the claim. The use of $\|\cdot\|_\infty$ is consistent with the worst-case nature of Int and DO: a single key pattern that diverges across agents can dominate both integration guarantees and discord bounds. □

Theorem 192 (Narrative capture vs discord: a budget tradeoff)**.** *Assume a fixed per-time collective resource budget normalized to 1. Assume that maintaining integration level $\mathrm{Int}(I) \geq \alpha$ requires allocating at least $\beta(\alpha)$ of the resource budget to non-narrative integration processes. Then at any time t,*

$$N_O(t) \leq 1 - \beta(\alpha) \qquad \textit{whenever} \qquad \mathrm{Int}(I(t)) \geq \alpha.$$

In particular, if low discord requires high integration (e.g. DO small implies Int large), then low discord implies an upper bound on narrative capture. This formalizes an intuitive constraint: if a collective must spend real capacity on mechanisms that keep key patterns jointly instantiated (institutions, deliberation bandwidth, enforcement, education, shared rituals, interoperability work), then less capacity remains for purely narrative optimization. In the broader Hyperseed framing, this supplies a simple route from an integration requirement (C2-like) to a limitation on narrative dominance (controlling N_O), thereby connecting Location stability to resource allocation.

Proof. If integration at level α requires spending at least $\beta(\alpha)$ of the budget, then the remaining budget for narrative processes is at most $1 - \beta(\alpha)$. By definition $N_O(t)$ is the narrative budget fraction, so $N_O(t) \leq 1 - \beta(\alpha)$. The last sentence follows by composition of implications. One can read $\beta(\alpha)$ as an increasing "maintenance cost curve": higher required integration levels demand more non-narrative expenditure, tightening the bound on $N_O(t)$. The theorem does not assume any particular functional form for β, only that such a lower-bounding requirement exists. □

62.15 Mindplexes: Hyperseed vs SDP, development, and weak mindplexes

Definition 601 (Hyperseed-style mindplex and notable coherence)**.** *Fix an observer O and a coherence score $\mathrm{Coh}_O(\cdot)$ on candidate minds. A system M is* notably coherent *if $\mathrm{Coh}_O(M)$ is a local maximum relative to small modifications of M. A* mindplex *is a notably coherent mind M that decomposes into notably coherent minds $\{M_i\}$ such that: (a) each M_i is approximately pattern-preserving as a restriction of M, and (b) replacing any M_i by another comparably coherent mind M_i' does not destroy the coherence of M (robustness/replaceability).*

In this Hyperseed-style definition, the observer-indexing O allows Coh_O to encode both a measurement procedure and a representational stance (e.g. which functional/phenomenal regularities are treated as salient), without thereby making mindplexhood arbitrary: the local-maximum requirement forces M to be *stable* under a neighborhood of perturbations in the model space. Concretely, "small modifications" may include minor architectural edits, bounded noise in state trajectories, limited rewiring of communication links, or replacing a subroutine with a behaviorally similar one; the intent is to rule out systems whose apparent coherence is a knife-edge artifact of a particular description.

Clause (a) formalizes a modularity constraint: each M_i should inherit, up to approximation, the patterns that make M coherent, rather than being an unrelated subsystem whose coherence is independent of the collective. Clause (b) expresses that the whole is not merely a brittle concatenation: a mindplex must tolerate some degree of swapping, redundancy, or re-instantiation of parts while keeping the collective coherence high. This is the sense in which replaceability functions as an operational proxy for "integration without fragility," and it will be the key point of contact with multi-agent and team-like examples below.

Definition 602 (SDP-style mindplex (dual-level theatre of consciousness))**.** *A* SDP mindplex *is a collection of intelligent systems $\{S_i\}$ such that: (i) each S_i has its own theatre of consciousness and control loops, and (ii) the collective $S = \bigsqcup_i S_i$ also supports a collective theatre of consciousness and collective control loops, with nontrivial inter-level resonance.*

Here $\bigsqcup_i S_i$ is intended to denote a structured aggregation (not merely a set-theoretic union): the S_i remain individually identifiable, yet are coupled strongly enough that one can ascribe additional state variables, dynamics, and (in SDP terms) an additional theatre of consciousness to the aggregate. The phrase "control loops" is included to emphasize that SDP mindplexhood is not only about shared representations or information pooling, but also about coordinated selection, inhibition, and action at both levels (individual and collective). The "nontrivial inter-level resonance" qualifier rules out degenerate cases where a purported collective theatre is epiphenomenal (having no downstream influence) or where individuals are merely passive channels for a centrally dictated policy.

Definition 603 (Weak mindplex)**.** *Fix $\epsilon \geq 0$ and a time horizon T. A system M is an (ϵ, T)*-weak mindplex *if it satisfies the Hyperseed mindplex conditions up to error ϵ (approximate coherence and approximate replaceability) and, in addition, exhibits a collective GCC episode of duration at least T.*

The (ϵ, T) parameters make explicit two ways that real-world collectives fall short of idealized mindplexes. First, coherence and replaceability may only hold approximately: ϵ can be read as an allowed drop in Coh_O under admissible substitutions $M_i \mapsto M_i'$ (or, equivalently, as an upper bound on how much "coherence mass" is tied to idiosyncratic individuals). Second, the time horizon T accommodates the possibility that a collective theatre (here operationalized by a GCC episode) is transient, arising during coordinated bursts (meetings, performances, joint planning) rather than persisting continuously.

Theorem 193 (From SDP mindplex to Hyperseed mindplex (sufficient conditions))**.** *Assume $S = \bigsqcup_i S_i$ is an SDP mindplex. Assume that the collective theatre of consciousness corresponds (under the translation layer) to a workspace-stable reflective fixed point E^* with high coherence, and that the components S_i correspond to notably coherent subminds M_i whose replacement does not significantly change the collective coherence.*

Then the collective S satisfies the Hyperseed mindplex definition (relative to O).

The role of the "translation layer" assumption is to connect SDP's theatre/loop vocabulary to Hyperseed's coherence-and-decomposition vocabulary: it posits that there is a principled map from the state of the SDP collective theatre to a Hyperseed candidate mind-state whose reflective fixed point E^* is stable in the relevant workspace dynamics. In applications, the "workspace-stable" condition is what prevents momentary synchronization (e.g. a brief shared signal) from being misclassified as a collective mind; stability ensures that the collective can maintain and update a coherent focus in the face of typical perturbations.

Proof. By assumption, the collective is (i) coherent and (ii) decomposes into coherent parts. Approximate pattern-preserving restriction is supplied by the assumption that each S_i participates as a stable substructure in the collective theatre (so its induced patterns are restrictions of the collective patterns). Replaceability is assumed as "replacement does not significantly change collective coherence." These are exactly the conditions of the Hyperseed mindplex definition. □

Theorem 194 (From Hyperseed mindplex to SDP mindplex under dual-level GCC/RC)**.** *Assume M is a Hyperseed mindplex decomposing into minds $\{M_i\}$. Assume additionally: (A) each M_i exhibits RC episodes (individual reflective fixed points exist and satisfy reflective self); (B) the collective M exhibits GCC episodes with a workspace-stable focus set; and (C) there exists nontrivial inter-level resonance, meaning that collective conscious contents can modulate individual attention/control and vice versa.*

Then M is an SDP mindplex.

Conditions (A)–(C) can be read as a sufficient "phenomenology-to-architecture" bridge: Hyperseed mindplexhood ensures structural decomposability and robustness, while (A) and (B) add explicit requirements that both the parts and the whole realize the kinds of stabilized reflective episodes that SDP treats as constitutive of theatres of consciousness. The extra resonance condition (C) prevents two common failure modes: (1) mere parallelism, where individuals have RC but the group has only superficial coherence; and (2) mere aggregation, where a group-level GCC-like focus exists (e.g. a shared agenda) but individuals are not actually modulated in their control loops by the collective contents (or cannot influence the collective in return).

Proof. Conditions (A) and (B) provide the required dual-level theatres of consciousness (individual and collective). Condition (C) provides the nontrivial inter-level resonance required by the SDP notion. Thus the definition of SDP mindplex is satisfied. □

Theorem 195 (Mindplex emergence along a development path)**.** *Let $I_0 \to I_1 \to \cdots$ be a Hyperseed development path for a system M (continuity plus nontrivial capability growth). Assume that along this path the coherence score $\mathrm{Coh}_O(M; I_k)$ is nondecreasing and eventually crosses a threshold τ at which M becomes notably coherent and decomposable into notably coherent subminds.*

Then there exists k_0 such that for all $k \geq k_0$, M is a mindplex on interval I_k.

This emergence statement isolates a common pattern in developmental and training narratives: increases in capability can act as control parameters that push a system across a qualitative boundary where (i) coherence becomes locally maximal (so the mind is no longer an easily-improved patchwork) and (ii) coherent substructures become both identifiable and robustly swappable. The use of interval-indexed coherence $\mathrm{Coh}_O(M; I_k)$ emphasizes that mindplexhood is assessed over temporally extended behavior, not instantaneous snapshots; in particular, a system may transiently satisfy decomposition and replaceability while failing to maintain them across an interval where tasks, contexts, and internal states vary.

Proof. By assumption, there is k_0 such that $\mathrm{Coh}_O(M; I_k) \geq \tau$ for $k \geq k_0$. At such k, the notable coherence and decomposition assumptions state exactly the mindplex conditions, hence M is a mindplex on I_k for all $k \geq k_0$. Continuity (self-continuity across successive intervals) ensures these are stages of one persisting entity rather than unrelated systems. □

Theorem 196 (Small teams as weak mindplexes (a formal sufficient condition))**.** *Consider a team of n agents $\{S_i\}$ coupled by: (i) a strongly connected communication graph with coupling lower bound $k_0 > 0$; (ii) a shared persistent artifact store (documents, codebase, musical score) that acts as a high-bandwidth hub in the coupling graph; and (iii) synchronized interaction intervals of length at least T during which attention is aligned toward a shared task and shared artifacts.*

Assume the induced collective cultural reinforcement dynamics satisfy the hypotheses of the diffusion lower bound theorem for the task-relevant patterns, and that the collective has a workspace-stable focus set during the synchronized intervals.

Then the team forms an (ϵ, T)-weak mindplex for some ϵ determined by the replaceability error induced by role specialization.

The coupling hypotheses (i)–(iii) are designed to make the team behave less like a loose federation and more like a temporarily integrated cognitive unit. Strong connectivity and a coupling lower bound k_0 ensure that information (task constraints, partial solutions, error signals) can percolate throughout the team rather than remaining trapped in isolated subgroups. The persistent artifact store plays a dual role: it is both a communication amplifier (high bandwidth, asynchronous access) and an externalized memory that stabilizes shared context, thereby helping to produce the "workspace-stable focus set" required for a GCC episode. Finally, synchronized interaction intervals matter because weak mindplexhood is explicitly time-bounded: what is being guaranteed is a collective episode of coordinated cognition lasting at least T, not a permanently unified collective agent.

The parameter ϵ is naturally linked to specialization: teams often have critical roles (domain expert, coordinator, maintainer) whose replacement is only approximately feasible. The theorem therefore predicts weak mindplexhood when specialization is present but not so extreme that swapping or tool-assisted substitution collapses coherence, aligning the formalism with familiar phenomena such as resilient teams that can absorb absences, onboarding, or partial automation.

Proof. During synchronized intervals, strong connectivity plus artifact-mediated coupling yields diffusion of task-relevant patterns across agents, increasing the integration proxy and reducing fragmentation. The shared artifact hub provides a stable anchor that keeps the collective in a metastable basin (chain) for duration at least T, yielding a collective GCC episode by assumption. Approximate decomposition into coherent subminds is given by the individual agents; replaceability holds up to error ϵ because role-specialized components can be partially replaced by other agents or tools without destroying the collective's coherent task performance. Thus the weak mindplex definition is satisfied. □

62.16 Closing remarks

The bridge above turns qualitative claims about consciousness Locations, PNSE, and collective coordination into monotone fixed-point and threshold statements that can be used by a reasoning system: (i) map observable markers into evidence-state functionals; (ii) infer Location/collective Location; (iii) reason about interventions by moving parameter tuples or coupling strengths; and (iv) reason about emergence of mindplex structure along development paths.

62.17 Five basic theorems about GTGI-style general intelligence measures

Standing notation. Fix a task set $\mathcal{T}$ (Hyperseed-Concept "Task families") equipped with a probability distribution $D \in \Delta(\mathcal{T})$ (a "task environment"). Fix an agent Ag. Let $\kappa_{Ag} : \mathcal{T} \to [0,1]$ denote the agent's taskwise capability/competence function (e.g. $\kappa_{Ag}(T) := \mathrm{Comp}(Ag;T)$ in the sense of the competence scalarization already defined in Hyperseed's tasks section). Let

$$\mathrm{Cap}(Ag;\mathcal{T},D) \;:=\; \mathbb{E}_{T\sim D}\big[\kappa_{Ag}(T)\big]$$

denote the (distributional) capability score, and for $\varepsilon \in [0,1]$ let

$$\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D) \;:=\; D\big(\{T \in \mathcal{T} : \kappa_{Ag}(T) \geq 1-\varepsilon\}\big)$$

denote ε-breadth ("good-enough mass"). We write log for $\log_2$.

Theorem 197 (Universal intelligence is a special case of pragmatic intelligence)**.** *Assume universal intelligence is defined as an* unnormalized *weighted aggregate*

$$\Upsilon_w(Ag) \;:=\; \sum_{T\in\mathcal{T}} w(T)\,\kappa_{Ag}(T), \tag{42}$$

for some weight function $w : \mathcal{T} \to (0,\infty)$ with finite normalizer

$$Z_w \;:=\; \sum_{T\in\mathcal{T}} w(T) \;<\; \infty.$$

Define the corresponding normalized "pragmatic" task distribution $D_w \in \Delta(\mathcal{T})$ by

$$D_w(T) \;:=\; \frac{w(T)}{Z_w}.$$

Then

$$\Upsilon_w(Ag) \;=\; Z_w \cdot \mathrm{Cap}(Ag;\mathcal{T},D_w), \tag{43}$$

i.e. $\mathrm{Cap}(Ag;\mathcal{T},D_w) = \Upsilon_w(Ag)/Z_w$.

In particular, taking $w(T) = 2^{-K(T)}$ for a chosen description-complexity K (as in Def. 336) makes universal intelligence a rescaled pragmatic intelligence (Def. 340).

Proof. By definition of $\mathrm{Cap}(\cdot)$ and D_w,

$$\mathrm{Cap}(Ag;\mathcal{T},D_w) = \sum_{T\in\mathcal{T}} D_w(T)\,\kappa_{Ag}(T) = \sum_{T\in\mathcal{T}} \frac{w(T)}{Z_w}\,\kappa_{Ag}(T) = \frac{1}{Z_w}\sum_{T\in\mathcal{T}} w(T)\,\kappa_{Ag}(T) = \frac{1}{Z_w}\Upsilon_w(Ag).$$

Rearranging yields (43). □

Theorem 198 (Bounding efficient pragmatic GI by pragmatic GI under bounded resource costs)**.** *Assume efficient pragmatic general intelligence is defined by a resource-normalized average (Def. 342) of the form*

$$\Pi^{\mathrm{eff}}(Ag;\mathcal{T},D) \;:=\; \mathbb{E}_{T\sim D}\left[\frac{\kappa_{Ag}(T)}{c_{Ag}(T)}\right], \tag{44}$$

where $c_{Ag}(T) > 0$ is the (expected or deterministic) resource cost incurred by Ag on task T.

Suppose there exist constants $0 < c_{\min} \leq c_{\max} < \infty$ such that for all $T \in \mathcal{T}$,

$$c_{\min} \;\leq\; c_{Ag}(T) \;\leq\; c_{\max}.$$

Then

$$\frac{1}{c_{\max}}\,\mathrm{Cap}(Ag;\mathcal{T},D) \;\leq\; \Pi^{\mathrm{eff}}(Ag;\mathcal{T},D) \;\leq\; \frac{1}{c_{\min}}\,\mathrm{Cap}(Ag;\mathcal{T},D). \tag{45}$$

Proof. For each task T, the cost bounds imply

$$\frac{1}{c_{\max}} \;\le\; \frac{1}{c_{Ag}(T)} \;\le\; \frac{1}{c_{\min}}.$$

Since $\kappa_{Ag}(T) \in [0,1]$, multiplying preserves inequalities:

$$\frac{1}{c_{\max}}\,\kappa_{Ag}(T) \;\le\; \frac{\kappa_{Ag}(T)}{c_{Ag}(T)} \;\le\; \frac{1}{c_{\min}}\,\kappa_{Ag}(T).$$

Taking expectations over $T \sim D$ yields

$$\frac{1}{c_{\max}}\mathbb{E}_{T\sim D}[\kappa_{Ag}(T)] \;\le\; \mathbb{E}_{T\sim D}\left[\frac{\kappa_{Ag}(T)}{c_{Ag}(T)}\right] \;\le\; \frac{1}{c_{\min}}\mathbb{E}_{T\sim D}[\kappa_{Ag}(T)].$$

The middle term is $\Pi^{\text{eff}}(Ag;\mathcal{T},D)$ by (44), and the outer expectations are $\mathrm{Cap}(Ag;\mathcal{T},D)$, giving (45). ☐

Theorem 199 (Breadth–capability inequalities)**.** *Fix $(\mathcal{T},D)$, agent Ag, and $\varepsilon \in [0,1]$. Then*

$$(1-\varepsilon)\,\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D) \;\le\; \mathrm{Cap}(Ag;\mathcal{T},D) \;\le\; (1-\varepsilon)+\varepsilon\,\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D). \tag{46}$$

Equivalently, for $\varepsilon \in (0,1]$,

$$\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D) \;\ge\; \frac{\mathrm{Cap}(Ag;\mathcal{T},D)-(1-\varepsilon)}{\varepsilon}. \tag{47}$$

Proof. Let $G_\varepsilon := \{T \in \mathcal{T} : \kappa_{Ag}(T) \ge 1-\varepsilon\}$, so $\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D) = D(G_\varepsilon)$. Split the expectation defining capability:

$$\mathrm{Cap}(Ag;\mathcal{T},D) = \mathbb{E}_{T\sim D}[\kappa_{Ag}(T)] = \mathbb{E}[\kappa_{Ag}(T)\mathbf{1}_{G_\varepsilon}(T)] + \mathbb{E}[\kappa_{Ag}(T)\mathbf{1}_{\mathcal{T}\setminus G_\varepsilon}(T)].$$

Lower bound. On G_ε we have $\kappa_{Ag}(T) \ge 1-\varepsilon$, and on $\mathcal{T}\setminus G_\varepsilon$ we have $\kappa_{Ag}(T) \ge 0$. Hence

$$\mathrm{Cap}(Ag;\mathcal{T},D) \ge (1-\varepsilon)\,\mathbb{E}[\mathbf{1}_{G_\varepsilon}(T)] = (1-\varepsilon)\,D(G_\varepsilon) = (1-\varepsilon)\,\mathrm{Br}_\varepsilon(Ag;\mathcal{T},D).$$

Upper bound. On G_ε we have $\kappa_{Ag}(T) \le 1$, and on $\mathcal{T}\setminus G_\varepsilon$ we have $\kappa_{Ag}(T) < 1-\varepsilon$, hence $\kappa_{Ag}(T) \le 1-\varepsilon$. Therefore

$$\mathrm{Cap}(Ag;\mathcal{T},D) \le 1\cdot\mathbb{E}[\mathbf{1}_{G_\varepsilon}(T)]+(1-\varepsilon)\cdot\mathbb{E}[\mathbf{1}_{\mathcal{T}\setminus G_\varepsilon}(T)] = D(G_\varepsilon)+(1-\varepsilon)(1-D(G_\varepsilon)) = (1-\varepsilon)+\varepsilon D(G_\varepsilon),$$

which is the right-hand inequality in (46). Rearranging the upper bound yields (47) for $\varepsilon > 0$. ☐

Theorem 200 (Invariance of taskwise competence (and derived GI scores) under reduction-equivalence)**.** *Let $\mathcal{T}$ and $\mathcal{T}'$ be task sets and let $\phi : \mathcal{T} \to \mathcal{T}'$ be a bijection. Assume that for each $T \in \mathcal{T}$, the tasks T and $\phi(T)$ are reduction-equivalent in the sense that there exist task reductions both ways (as in Hyperseed's task reduction definition and Lemma "reductions form a preorder; equivalence is an equivalence relation").*

Choose, for each $T \in \mathcal{T}$, a concrete reduction

$$G_T : \; \phi(T) \;\to\; T.$$

Given an agent Ag defined on the interface of tasks in $\mathcal{T}$, define an induced agent ϕ_*Ag on $\mathcal{T}'$ by porting *Ag along each G_T when facing task $\phi(T)$ (i.e. apply the "agent transfer along a reduction" rule).*

Let $D \in \Delta(\mathcal{T})$ and define the pushforward distribution $\phi_* D \in \Delta(\mathcal{T}')$ by

$$(\phi_* D)(T') \ := \ D(\phi^{-1}(T')).$$

Then for every $T \in \mathcal{T}$,

$$\kappa_{Ag}(T) \ = \ \kappa_{\phi_* Ag}(\phi(T)), \tag{48}$$

and consequently for every $\varepsilon \in [0, 1]$,

$$\mathrm{Cap}(Ag; \mathcal{T}, D) \ = \ \mathrm{Cap}(\phi_* Ag; \mathcal{T}', \phi_* D), \qquad \mathrm{Br}_\varepsilon(Ag; \mathcal{T}, D) \ = \ \mathrm{Br}_\varepsilon(\phi_* Ag; \mathcal{T}', \phi_* D). \tag{49}$$

In particular, any GI measure defined purely as an expectation (or threshold-mass) over taskwise competence is invariant under replacing tasks by reduction-equivalent presentations.

Proof. Fix $T \in \mathcal{T}$ and write $T' := \phi(T)$. By construction, $\phi_* Ag$ on task T' is obtained by porting Ag along the reduction $G_T : T' \to T$. Hyperseed's "value/competence preservation under task reduction" result states that porting along a reduction preserves task value and hence preserves the scalar competence score. Therefore $\kappa_{\phi_* Ag}(T') = \kappa_{Ag}(T)$, which is (48).

For capability, using the definition of pushforward measure and (48),

$$\mathrm{Cap}(\phi_* Ag; \mathcal{T}', \phi_* D) = \sum_{T' \in \mathcal{T}'} (\phi_* D)(T')\, \kappa_{\phi_* Ag}(T') = \sum_{T \in \mathcal{T}} D(T)\, \kappa_{\phi_* Ag}(\phi(T)) = \sum_{T \in \mathcal{T}} D(T)\, \kappa_{Ag}(T) = \mathrm{Cap}(Ag; \mathcal{T}, D).$$

For breadth, define $G_\varepsilon := \{T \in \mathcal{T} : \kappa_{Ag}(T) \geq 1 - \varepsilon\}$ and $G'_\varepsilon := \{T' \in \mathcal{T}' : \kappa_{\phi_* Ag}(T') \geq 1 - \varepsilon\}$. Equation (48) implies $T \in G_\varepsilon \iff \phi(T) \in G'_\varepsilon$, hence $G'_\varepsilon = \phi(G_\varepsilon)$. Therefore, by definition of pushforward distribution,

$$\mathrm{Br}_\varepsilon(\phi_* Ag; \mathcal{T}', \phi_* D) = (\phi_* D)(G'_\varepsilon) = (\phi_* D)(\phi(G_\varepsilon)) = D(G_\varepsilon) = \mathrm{Br}_\varepsilon(Ag; \mathcal{T}, D).$$

This proves (49). $\square$

Theorem 201 (Entropy bounds for intellectual breadth)**.** *Assume intellectual breadth is defined (Def. 343) by Shannon entropy of a normalized "intelligence-mass" distribution across tasks/contexts:*

$$m_{Ag}(T) \ := \ D(T)\, \kappa_{Ag}(T), \qquad Z \ := \ \sum_{T \in \mathcal{T}} m_{Ag}(T), \qquad p_{Ag}(T) \ := \ \frac{m_{Ag}(T)}{Z},$$

where $Z > 0$ (i.e. Ag has nonzero intelligence mass on the task ecology). Let

$$\mathrm{IB}(Ag; \mathcal{T}, D) \ := \ H(p_{Ag}) \ := \ -\sum_{T \in \mathcal{T}} p_{Ag}(T) \log p_{Ag}(T),$$

with the convention $0 \log 0 := 0$. Assume $\mathcal{T}$ is finite with $|\mathcal{T}| = N$.

Then

$$0 \ \leq \ \mathrm{IB}(Ag; \mathcal{T}, D) \ \leq \ \log N. \tag{50}$$

Moreover:

- $\mathrm{IB}(Ag; \mathcal{T}, D) = \log N$ *iff p_{Ag} is uniform on $\mathcal{T}$.*
- $\mathrm{IB}(Ag; \mathcal{T}, D) = 0$ *iff p_{Ag} is a point mass (all intelligence mass concentrated on a single task).*

Proof. Nonnegativity: since $p_{Ag}(T) \in [0, 1]$, we have $\log p_{Ag}(T) \leq 0$, hence $-p_{Ag}(T) \log p_{Ag}(T) \geq 0$ for each term; summing gives IB ≥ 0. For the upper bound, let u be the uniform distribution on $\mathcal{T}$, i.e. $u(T) = 1/N$. (Here $N := |\mathcal{T}| < \infty$ so that u is well-defined and $\log N$ is finite; throughout, the base of the logarithm fixes the units of information, e.g. base-2 gives bits and base-e gives nats.) Consider the KL divergence

$$\mathrm{KL}(p_{Ag} \| u) \;:=\; \sum_{T \in \mathcal{T}} p_{Ag}(T) \log \frac{p_{Ag}(T)}{u(T)} \;\geq\; 0.$$

The inequality $\mathrm{KL}(p_{Ag} \| u) \geq 0$ is the standard nonnegativity (Gibbs' inequality), and it holds for any two distributions on the same finite support; in particular, we adopt the usual convention that terms with $p_{Ag}(T) = 0$ contribute 0 to the sum (since $\lim_{x \to 0^+} x \log x = 0$), so the expression is well-defined even when p_{Ag} has zeros. Expanding with $u(T) = 1/N$ yields

$$\mathrm{KL}(p_{Ag} \| u) = \sum_T p_{Ag}(T) \log p_{Ag}(T) - \sum_T p_{Ag}(T) \log(1/N) = -H(p_{Ag}) + \log N = -\mathrm{IB}(Ag; \mathcal{T}, D) + \log N.$$

In the middle step we used $\sum_T p_{Ag}(T) = 1$, so the second sum simplifies as

$$-\sum_T p_{Ag}(T) \log(1/N) = -\log(1/N) \sum_T p_{Ag}(T) = -\log(1/N) = \log N,$$

making explicit why the upper bound is controlled only by the cardinality of $\mathcal{T}$ (and not by finer details of p_{Ag}). Thus $\mathrm{IB}(Ag; \mathcal{T}, D) \leq \log N$, proving (50). Equivalently, this rederives the classical entropy bound $H(p_{Ag}) \leq \log |\mathcal{T}|$, so the "IB" quantity here inherits the same extremal behavior as Shannon entropy on a finite set.

The equality condition IB $= \log N$ holds iff $\mathrm{KL}(p_{Ag} \| u) = 0$, which occurs iff $p_{Ag} = u$, i.e. p_{Ag} is uniform. Intuitively, this is the case of maximal uncertainty over $\mathcal{T}$: the agent-induced distribution assigns equal mass to each $T \in \mathcal{T}$, and therefore achieves the largest possible value of $H(p_{Ag})$ (hence of IB) among all distributions on $\mathcal{T}$.

Finally, IB $= 0$ iff $H(p_{Ag}) = 0$, which holds exactly when the distribution is supported on a single point (since $-x \log x > 0$ for all $x \in (0, 1)$, entropy can vanish only if one probability is 1 and the rest are 0). This is the opposite extremal regime: all probability mass concentrates on a unique $T^\star \in \mathcal{T}$, so there is no residual uncertainty to be measured by the entropy (and therefore no "breadth" in the induced distribution over $\mathcal{T}$). □

63 Deeper theorems for TyLAA-based Universal Grammar

This section states and proves five "deeper" theorems that make the TyLAA/TUG formalism useful as an *automated-reasoning substrate* inside Hyperseed: we relate (i) interface choice to quotient strength; (ii) TyLAA weakness to Hyperseed weakness/effort; (iii) trace-classes to canonical proto-time posets; and (iv–v) two classic Minimalist locality effects (phase impenetrability and relativized minimality) to resource scoping/consumption. Concretely, the goal is to show that the objects that are natural from the TyLAA side (derivations, swaps, congruences, and trace quotients) align with the objects that are natural from the Hyperseed side (observables, cost/effort, and proto-time structure), so that one can move back and forth between "linguistic" statements about derivations and "computational" statements about what an automated reasoner can safely identify, ignore, or factor out. In particular, throughout this section the phrase "quotient strength" should be read as: how aggressively an interface (and the associated congruence) identifies operational histories, and hence how much operational detail the proof system is permitted to forget while remaining sound for the chosen notion of observation.

Hyperseed concepts covered here. TyLAA (**??**); TyLAA-based Universal Grammar / TUG (**??**); TUG observation interface (**??**); TUG congruence (**??**); TUG independence and trace skeleton (**??**, **??**); externalization parameters (**??**); phase-locality as resource scoping (**??**); relativized minimality as competitive feature checking (**??**); and, for the Hyperseed bridge, weakness and quantale weakness (202, 143), effort/simplicity (100, 169), and proto-time (142). It may be helpful to keep in mind that the first cluster of notions (TyLAA/TUG, interface, congruence, independence) fixes what counts as the *computational dynamics* and what counts as a *permitted invariance* of those dynamics, while the second cluster (weakness/effort/proto-time) fixes how those invariances are evaluated as abstractions: weakness measures the extent of identification (and thus the informational coarseness), effort measures the induced proof/derivation complexity after quotienting, and proto-time records the partial-order content that remains once irrelevant sequential detail has been factored away.

63.1 Standing notation (recall)

We assume the basic TyLAA/TUG setup from the prior UG sections: an operational "red" system with complete derivations $\mathrm{Deriv}(s)$ from a state s, an observation interface O (Def. **??**) inducing observational equivalence $\equiv^{O}_{\mathrm{obs}}$ on derivations (Def. **??**), and the *weakest red congruence* (maximal sound congruence) $\equiv^{O}_{\mathrm{w}}$ (Def. **??**). Intuitively, $\mathrm{Deriv}(s)$ is the collection of all full operational histories that start from the same initial configuration s and run until a designated completeness condition (e.g. convergence, termination, or a finished externalization) is met; the theorems below will frequently quantify over $\mathrm{Deriv}(s)$ and then pass to quotients of $\mathrm{Deriv}(s)$ by various equivalence relations. The interface O should be read as a choice of *what counts as observable output* of a derivation (for example, phonological yield, a feature footprint, a dependency graph, or any other interface-specific projection), and $\equiv^{O}_{\mathrm{obs}}$ is the resulting indistinguishability relation: two derivations are observationally equivalent exactly when O cannot tell them apart. The congruence $\equiv^{O}_{\mathrm{w}}$ is then the most permissive identification of derivations that is still sound with respect to O and stable under the red operational composition principles fixed earlier; this is the sense in which it is "weakest": it forgets as much red structure as possible without collapsing distinctions that O can observe.

We also assume the TUG-visible alphabet Lab_V, the instance-level independence relation $\perp$ (Def. **??**), a symmetry group Γ acting on visible labels (Def. **??**), and the induced swap-mod-Γ congruence $\equiv_{\perp,\Gamma}$ (Def. **??**). Here Lab_V is the set of labels whose occurrences are treated as externally meaningful at the TUG level (as opposed to purely administrative or hidden steps), so that derivations may be compared via their visible label sequences (or more generally, via visible-labeled traces). The relation $\perp$ captures when two *particular step instances* commute: if two steps are independent in the relevant sense (no shared resource, no feature interaction, no causal constraint), then swapping their execution order should not affect any interface-respecting outcome. The group Γ formalizes uniform renamings and symmetries of visible labels that the theory decides to treat as inessential (for example, -renaming of indices, automorphisms of a feature coding, or permutations induced by externalization parameters), so that equivalence is not forced to remember arbitrary naming choices. Accordingly, $\equiv_{\perp,\Gamma}$ identifies derivations that can be transformed into one another by a finite sequence of swaps of adjacent $\perp$-independent steps, together with the allowed Γ-actions on visible labels; the resulting quotient is the standard "trace" view enriched by an explicit symmetry principle.

When the TyLAA orthogonality hypotheses hold (Assumption **??**), $\equiv_{\mathrm{w}}^{O}$ coincides with $\equiv_{\perp,\Gamma}$ for the corresponding interface. This coincidence is the key technical hinge for several of the deeper theorems: orthogonality ensures that the only sound way to identify derivations (relative to the chosen O) is to factor out exactly (a) those reorderings that are justified by independence and (b) those renamings that are justified by symmetry, with no additional "mysterious" identifications. Equivalently, under orthogonality, the interface-induced abstraction is complete for the syntactic/operational commutations captured by $\perp$ and Γ, so the quotient by $\equiv_{\mathrm{w}}^{O}$ can be computed and reasoned about using the more concrete swap-based presentation $\equiv_{\perp,\Gamma}$, which is what ultimately connects trace-classes to the proto-time posets studied later in this section.

63.2 Interface refinement yields canonical quotient maps

Theorem 202 (Interface refinement monotonicity and quotient-map functoriality)**.** *Let O_1 and O_2 be two observation interfaces (Hyperseed-Concept* **??***) on the same underlying operational system, and suppose O_2 is a* refinement *of O_1 (i.e. O_2 can distinguish at least as much as O_1):*

$$\pi_1 \equiv_{\mathrm{obs}}^{O_2} \pi_2 \quad \Rightarrow \quad \pi_1 \equiv_{\mathrm{obs}}^{O_1} \pi_2 \qquad \text{for all derivations } \pi_1, \pi_2.$$

Then:

1. ***Observational equivalence refines monotonically:*** $\equiv_{\mathrm{obs}}^{O_2} \subseteq \equiv_{\mathrm{obs}}^{O_1}$.
2. ***Weakest-red congruence refines monotonically:*** $\equiv_{\mathrm{w}}^{O_2} \subseteq \equiv_{\mathrm{w}}^{O_1}$.
3. ***There is a canonical surjection between quotients:*** *the map*

$$q_{2\to1} : \mathrm{Deriv}(s)/\equiv_{\mathrm{w}}^{O_2} \;\to\; \mathrm{Deriv}(s)/\equiv_{\mathrm{w}}^{O_1}, \qquad q_{2\to1}([\pi]_{O_2}) := [\pi]_{O_1}$$

   is well-defined and surjective.

Moreover, $q_{2\to1}$ is the unique map making the canonical projections compatible in the sense that, writing $\eta_i : \mathrm{Deriv}(s) \to \mathrm{Deriv}(s)/\equiv_{\mathrm{w}}^{O_i}$ for $\eta_i(\pi) := [\pi]_{O_i}$, one has

$$\eta_1 \;=\; q_{2\to1} \circ \eta_2,$$

so $q_{2\to1}$ is the canonical "forgetful" (coarsening) map induced by refinement.

Proof. (1) This is exactly the refinement assumption rewritten as a set inclusion: if every O_2-indistinguishable pair is also O_1-indistinguishable, then the relationf $\equiv_{\mathrm{obs}}^{O_2}$ is a subset of $\equiv_{\mathrm{obs}}^{O_1}$. (Here "relationf" is to be read as "relation".) Equivalently, among equivalence relations on $\mathrm{Deriv}(s)$ ordered by inclusion, refinement corresponds to moving *downward* (fewer pairs identified), so the observational equivalence relation becomes strictly finer or equal.

(2) By definition, $\equiv_{\mathrm{w}}^{O}$ is the *largest sound congruence* for O, i.e. the largest congruence π satisfying $\pi \subseteq\equiv_{\mathrm{obs}}^{O}$. From (1), $\equiv_{\mathrm{obs}}^{O_2}\subseteq\equiv_{\mathrm{obs}}^{O_1}$, so any congruence sound for O_2 (contained in $\equiv_{\mathrm{obs}}^{O_2}$) is automatically sound for O_1 (contained in $\equiv_{\mathrm{obs}}^{O_1}$). In particular, the *largest* such congruence for O_2 cannot exceed the largest such congruence for O_1; hence $\equiv_{\mathrm{w}}^{O_2}\subseteq\equiv_{\mathrm{w}}^{O_1}$. One can also view this as a monotonicity property of the "closure-to-a-congruence" operator: tightening the observational indistinguishability constraint can only tighten the largest congruence compatible with it.

(3) Define $q_{2\to1}([\pi]_{O_2}) := [\pi]_{O_1}$. To see this is well-defined, suppose $[\pi]_{O_2} = [\pi']_{O_2}$, i.e. $\pi \equiv_{\mathrm{w}}^{O_2} \pi'$. By (2), $\equiv_{\mathrm{w}}^{O_2}\subseteq\equiv_{\mathrm{w}}^{O_1}$, so $\pi \equiv_{\mathrm{w}}^{O_1} \pi'$, hence $[\pi]_{O_1} = [\pi']_{O_1}$. Surjectivity is immediate: for any $[\pi]_{O_1}$, pick the same representative π and observe that $q_{2\to1}([\pi]_{O_2}) = [\pi]_{O_1}$. It is also useful to note the induced factorization of projections, namely $\eta_1 = q_{2\to1} \circ \eta_2$, which expresses precisely that passing to the finer quotient (via O_2) retains enough information to recover the coarser quotient (via O_1) by a canonical collapse of classes. $\square$

Remark 3457. *This theorem is the clean formal reason that "parameters" appear when one observes more of the interface (Hyperseed-Concept* **??***): each coarse class $[\pi]_{O_1}$ is a union of refined classes in the fiber $q_{2\to1}^{-1}([\pi]_{O_1})$. In particular, when O_1 is LF-only and O_2 is LF+PF, the fiber can be read as the space of externalization variants consistent with a fixed LF core. Equivalently, $q_{2\to1}$ packages the idea that additional interface structure does not change the underlying LF-denotation of a derivation, but it can split a single LF-denotation into multiple PF-realization classes; the "parameter space" is exactly the collection of these split components.*

Remark 3458. *When refinements compose (e.g. O_3 refines O_2 refines O_1), the corresponding quotient maps compose as well: the canonical maps satisfy $q_{3\to1} = q_{2\to1} \circ q_{3\to2}$, since all are defined by sending a class represented by π to the class of the same π in the coarser quotient. Thus, the assignment $O \mapsto \mathrm{Deriv}(s)/\equiv^O_{\mathrm{w}}$ is naturally contravariant with respect to refinement: making an interface more discriminating yields a finer quotient together with a canonical surjection back to any coarser view.*

63.3 TyLAA weakness as Hyperseed weakness-maximization

Theorem 203 (Weakest-red maximizes Hyperseed weakness among sound congruences)**.** *Fix a state s and let $D := \mathrm{Deriv}(s)$ be the set of complete derivations from s. Equip D with a weight/measure $\mu : D \to V$ into the same quantale used for Hyperseed weakness (Hyperseed-Concepts 202, 143). For any congruence $\approx$ on D, write*

$$H_{\approx} \;:=\; \{(\pi,\pi') \in D \times D :\; \pi \approx \pi'\}.$$

Define the Hyperseed weakness *of $\approx$ by*

$$\mathsf{W}(\approx) \;:=\; w(H_{\approx}),$$

where $w(\cdot)$ is the quantale-weakness functional (cf. Hyperseed-Concept 143). Intuitively, $H_{\approx}$ is the set of derivation-pairs that the grammar (via $\approx$) treats as indistinguishable, and $\mathsf{W}(\approx)$ is the aggregate "amount" of such indistinguishability measured in the same scale V as Hyperseed weakness. In particular, enlarging $\approx$ (identifying more derivations) enlarges $H_{\approx}$ as a subset of $D \times D$, and the theorem below states that the weakest-red congruence achieves the largest possible weakness subject to the constraint of remaining sound for the chosen observation interface.

Let O be an observation interface and let $\equiv^O_{\mathrm{w}}$ be the weakest-red congruence (maximal sound congruence) for O (Def. **??***). Then for every sound congruence $\approx$ (i.e. every congruence with $\approx \subseteq \equiv^O_{\mathrm{obs}}$),*

$$\mathsf{W}(\approx) \;\le\; \mathsf{W}(\equiv^O_{\mathrm{w}}).$$

Moreover, if D is finite then

$$|D/\equiv^O_{\mathrm{w}}| \;\le\; |D/\approx|,$$

so the weakest red also minimizes the number of equivalence classes (a crude proxy for representational effort; Hyperseed-Concepts 100, 169). Equivalently, among all sound ways of quotienting derivations by observational indistinguishability, $\equiv^O_{\mathrm{w}}$ is simultaneously the coarsest (largest relation) and thus the one that, in the finite case, yields the smallest quotient object in the set-theoretic sense.

Proof. Because $\equiv^O_{\mathrm{w}}$ is the *maximal* sound congruence for O, every sound congruence $\approx$ satisfies $\approx \subseteq \equiv^O_{\mathrm{w}}$ as relations on $D \times D$. Thus $H_{\approx} \subseteq H_{\equiv^O_{\mathrm{w}}}$. This inclusion is just the relational unwrapping of $\approx \subseteq \equiv^O_{\mathrm{w}}$: if a pair (π, π') is already identified by $\approx$, then it is also identified by the maximal sound relation, so it lies in the latter pair-set as well. By monotonicity of weakness under inclusion (Theorem 1),

$$w(H_{\approx}) \;\le\; w(H_{\equiv^O_{\mathrm{w}}}),$$

i.e. $\mathsf{W}(\approx) \le \mathsf{W}(\equiv^O_{\mathrm{w}})$. Here the order $\le$ is the quantale order on V, so the statement is independent of the particular numerical or logical realization of weakness: it applies uniformly whether V is, say, $[0,\infty]$ (with $\le$ the usual order) or a truth-valued quantale (with $\le$ as entailment), provided w is monotone in its argument as required by Hyperseed-Concept 143.

For the cardinality claim (finite D), if $\approx\subseteq\equiv^O_{\mathrm{w}}$ then $\equiv^O_{\mathrm{w}}$ is *coarser* (identifies at least as many elements), hence each $\equiv^O_{\mathrm{w}}$-class is a union of $\approx$-classes. Therefore the quotient by $\equiv^O_{\mathrm{w}}$ has no more classes than the quotient by $\approx$: $|D/\equiv^O_{\mathrm{w}}| \le |D/\approx|$. One way to see this explicitly is to consider the canonical surjection $D/\approx \twoheadrightarrow D/\equiv^O_{\mathrm{w}}$ that sends an $\approx$-class to the unique $\equiv^O_{\mathrm{w}}$-class containing it; surjectivity follows from coarseness, and finiteness of D then implies the stated inequality of cardinalities. (For infinite D, the same map still exists, but cardinal comparison may require additional set-theoretic care, so the theorem restricts to the finite case for a clean statement.) □

Remark 3459. *This is the precise Hyperseed bridge: TyLAA's slogan "do not maintain a distinction unless the interface forces you to" becomes a literal weakness-maximization statement when one reads "maintaining distinctions" as the complement of Hyperseed weakness (Hyperseed-Concept 202) and reads "effort" as the cost of representing many equivalence classes (Hyperseed-Concepts 100, 169). In particular, a "distinction" between π and π' corresponds to* excluding *the pair (π,π') from $H_{\approx}$, whereas "not maintaining" it corresponds to* including *that pair. Since the weakest-red congruence is maximal among sound congruences, it includes every pair that can be included without violating observational soundness, and therefore attains the largest admissible weakness value in the quantale sense. This makes the link to TyLAA structural rather than rhetorical: the interface constraint is precisely what bounds weakness from above.*

63.4 Trace-classes as canonical proto-times

Theorem 204 (Canonical poset representation of TUG trace-classes)**.** *Let $w = \ell_1\ell_2\cdots\ell_n \in \mathrm{Lab}_V^*$ be a visible trace word (in instance labels, so that $\perp$ is defined on the ℓ_i; Def. **??**). Define a directed graph G_w on vertices $\{1,\dots,n\}$ by putting an edge $i \to j$ iff $i<j$ and $\ell_i \not\perp \ell_j$ (i.e. the occurrences are dependent). Let $\le_w$ be the reachability preorder of G_w and write $P(w)$ for the induced partial order on occurrences (identify $i \sim j$ iff $i \le_w j$ and $j \le_w i$; this does nothing here because edges strictly increase indices, hence no cycles). In particular, $P(w)$ remembers* occurrences *(positions) rather than just labels: two equal labels $\ell_i = \ell_j$ with $i \ne j$ are still distinct elements of the poset, and the labeling of the poset is the evident map $i \mapsto \ell_i$. Equivalently, G_w is the "immediate" dependency graph induced by the word order together with the dependence relation, and $\le_w$ is its transitive closure, so that $i \le_w j$ expresses that i must causally precede j via a (possibly multi-step) chain of dependences.*

Then:

1. *$P(w)$ is a finite* proto-time *(a finite strict partial order) of the visible actions (Hyperseed-Concept 142).*

2. *If $w \equiv_\perp w'$ (swap-equivalent via adjacent swaps of independent labels), then $P(w)$ and $P(w')$ are isomorphic as labeled posets (up to the obvious renaming of occurrence-indices). In other words, the trace-equivalence class forgets only the arbitrary interleaving choices among independent actions, and the resulting invariant is exactly the induced partial order of necessary precedence constraints.*

3. *Conversely, the set of all words in the $\equiv_\perp$-class of w is exactly the set of linear extensions of $P(w)$ (i.e. all total orders of the occurrences consistent with $\le_w$), read back as words by listing the corresponding labels. Thus, $\equiv_\perp$-classes coincide with "all schedules consistent with the same dependency constraints", which is the usual Mazurkiewicz-trace viewpoint specialized to the present TUG setting.*

Proof. (1) By construction, every edge $i \to j$ satisfies $i < j$, so G_w is acyclic. Reachability therefore defines a strict partial order on occurrences. More explicitly, if $i \le_w j$ and $j \le_w i$, then there are directed paths $i \to^* j$ and $j \to^* i$, which would form a directed cycle; acyclicity forces $i = j$, so antisymmetry holds and the quotienting step "identify $i \sim j$" is indeed trivial in this particular construction.

(2) It suffices to consider one generating swap. Suppose $w' = u\,ab\,v$ and $w = u\,ba\,v$ differ by swapping adjacent letters a, b with $a \perp b$. Define a bijection σ on occurrences that swaps the two adjacent positions and fixes all others. Because $a \perp b$, there is *no* dependency edge between these two occurrences in either direction. For any third occurrence k, the relative order between k and each of the swapped occurrences is unchanged (they remain on the same side of k), so the presence/absence of edges to k is unchanged as well. To spell this out: for k outside the swapped block, either k lies in u (hence k precedes both swapped occurrences in both words) or k lies in v (hence k follows both swapped occurrences in both words), so the condition "$k <$ (swapped position)" (or its converse) is invariant under the swap, and the dependence predicate $\not\perp$ depends only on labels, not on positions. Hence σ carries edges of G_w to edges of $G_{w'}$, and therefore induces an isomorphism $P(w) \cong P(w')$. Since $\equiv_\perp$ is generated by such swaps, invariance holds for all $w \equiv_\perp w'$.

(3) Any word w is, tautologically, a linear extension of its own precedence constraints: if $\ell_i \not\perp \ell_j$ and $i < j$, then i must precede j in any realization, and w indeed lists i before j. So every word in the swap class yields a linear extension. Equivalently, swapping adjacent independent letters never violates the constraint that whenever there is a dependency edge $i \to j$ (hence $i < j$ and $\ell_i \not\perp \ell_j$), the occurrence i remains to the left of j; any allowed swap exchanges only a pair with $\perp$, so it cannot flip the order of a dependent pair.

Conversely, let L be any linear extension of $P(w)$. Starting from the word w, repeatedly swap adjacent occurrences that are out of order relative to L. Such a swap is always between two occurrences that are *incomparable* in $P(w)$ (otherwise L could not place them in the swapped order), and incomparability in $P(w)$ implies there is no dependency path between them, in particular no direct dependency edge, hence the two occurrences are independent and may be swapped. Here one uses the fact that if two occurrences $p < q$ in the *current* word were dependent, then by definition there would be an edge $p \to q$ (hence $p \le_w q$), so they could not be incomparable; therefore any adjacent pair eligible for reordering towards L must satisfy $\perp$. This "bubble-sort along incomparable adjacent pairs" terminates (it strictly decreases the number of inversions relative to L), yielding exactly the word corresponding to L. Therefore every linear extension is reachable by $\equiv_\perp$-swaps, i.e. belongs to the $\equiv_\perp$-class of w. ☐

Remark 3460. *Operationally, Theorem 204 says a trace class has a canonical "DAG/time" representation (Hyperseed-Concept* **??***) and ordinary sequential traces are just linearizations of it. This is precisely the data structure an automated reasoner wants: store the partial order (constraints), not an arbitrary interleaving (choices). Concretely, when two actions are independent, the poset leaves them incomparable, so a solver can postpone committing to their relative order until some later step (e.g. when synchronizing with additional constraints or matching against a target pattern).*

Remark 3461. *When $P(w)$ is finite, existence of at least one linear extension is guaranteed by* *the general linear-extension theorem already developed for proto-times (Theorem 27). Thus, the canonical trace-poset always admits at least one executable schedule when finite. Note that the finiteness assumption is automatic here because w has finite length n, hence the vertex set $\{1, \dots, n\}$ is finite; what matters in practice is that the induced constraints are consistent (acyclic), which is ensured by the index-increasing edge construction.*

63.5 Phase impenetrability from resource inaccessibility

The classical motivation for "phase impenetrability" is that certain domains behave as if they become inaccessible once a boundary is crossed. In the TUG setting, this is treated not as a special-purpose syntactic stipulation but as a direct consequence of the general resource discipline: once a resource leaves the active pool, any step whose *rely-footprint* requires that resource simply becomes disabled. The result below packages this intuition into a form that is stable under the quotienting that produces the TUG core, so that phase locality is preserved even when we identify derivations up to independence swaps and symmetries.

Theorem 205 (Phase locality as resource scoping; quotient-stability)**.** *Assume the TUG resource discipline (Def. **??**) with an "active resource pool"* $\mathrm{Active}(s) \subseteq \mathrm{Res}$ *for each state s, and an enabling condition: a visible label ℓ is enabled at s only if* $\mathrm{rely}(\ell) \subseteq \mathrm{Active}(s)$ *(Def. **??**). In particular, labels may be thought of as* capabilities *that are only available when all resources they rely on are currently in scope; this is the only mechanism used below.*

Fix a phase identifier π and a phase resource $\mathrm{Phase}(\pi) \in \mathrm{Res}$ *(Hyperseed-Concept **??**). Assume there is a phase-closing label* $\mathrm{Close}(\pi)$ *such that executing* $\mathrm{Close}(\pi)$ *removes the resource:*

$$\mathrm{Phase}(\pi) \in \mathrm{Active}(s) \quad \textit{and} \quad \mathrm{Active}(s \xrightarrow{\mathrm{Close}(\pi)} s') \;=\; \mathrm{Active}(s) \setminus \{\mathrm{Phase}(\pi)\}.$$

This expresses "closure" purely as a change in the static availability of a designated resource, and makes no further assumptions about how states encode structure. Note also that nothing here requires Active *to be globally monotone: we only use that* $\mathrm{Close}(\pi)$ *eliminates* $\mathrm{Phase}(\pi)$ *at the moment it occurs.*

Assume also that every "extraction across π" label $\mathrm{Hop}(\pi, \dots)$ *satisfies* $\mathrm{Phase}(\pi) \in \mathrm{rely}(\mathrm{Hop}(\pi, \dots))$*. Thus, the only way a derivation can perform an across-phase hop is by invoking an operation whose well-formedness (enabling) explicitly depends on the continued availability of the phase resource.*

Then:

1. ***(Impenetrability)*** *In any derivation, no* $\mathrm{Hop}(\pi, \dots)$ *can occur* after *a* $\mathrm{Close}(\pi)$ *step. Equivalently, if* $\mathrm{Close}(\pi)$ *occurs at some point in a derivation, then all subsequent steps are "blind" to the internal structure protected by* $\mathrm{Phase}(\pi)$*. Here "blind" is meant in the operational sense induced by the enabling condition: once the resource is absent, no later enabled label may have that resource in its rely-footprint, so no later label can be one whose intended interpretation requires access mediated by* $\mathrm{Phase}(\pi)$*.*

2. ***(Quotient-stability)*** *If* $\pi_1 \equiv_{\perp,\Gamma} \pi_2$ *are two derivations equivalent under the TUG congruence (Def. **??**), then π_1 satisfies the impenetrability constraint above iff π_2 does. In other words, phase locality is not an artifact of a particular linearization of independent actions: it is invariant under the observational identifications that define the quotient model.*

Proof. (1) Let $s \xrightarrow{\mathrm{Close}(\pi)} s'$ be an occurrence of $\mathrm{Close}(\pi)$ in a derivation. By hypothesis, $\mathrm{Phase}(\pi) \notin \mathrm{Active}(s')$. But every $\mathrm{Hop}(\pi, \dots)$ requires $\mathrm{Phase}(\pi) \in \mathrm{rely}(\mathrm{Hop}(\pi, \dots))$, hence $\mathrm{rely}(\mathrm{Hop}(\pi, \dots)) \nsubseteq \mathrm{Active}(s')$, so $\mathrm{Hop}(\pi, \dots)$ is not enabled at s'. Therefore no $\mathrm{Hop}(\pi, \dots)$ can occur after $\mathrm{Close}(\pi)$.

Put differently, the contradiction is purely local in time: it arises already at the immediate post-state s' because enabling consults $\mathrm{Active}(s')$ and $\mathrm{Close}(\pi)$ has just removed the needed resource.

(2) In the TUG congruence, the only reordering steps allowed are swaps of *independent* adjacent labels. But $\mathrm{Close}(\pi)$ and $\mathrm{Hop}(\pi, \dots)$ share the resource $\mathrm{Phase}(\pi)$, hence are dependent ($\mathrm{Close}(\pi) \not\perp \mathrm{Hop}(\pi, \dots)$ by Def. **??**). Therefore $\equiv_\perp$-swaps cannot move a $\mathrm{Hop}(\pi, \dots)$ from before $\mathrm{Close}(\pi)$ to after it (or vice versa): the relative precedence constraint between dependent occurrences is fixed across the swap class (Theorem 204). Equivalently, in the trace-poset semantics, any event labeled $\mathrm{Hop}(\pi, \dots)$ must be ordered before any event labeled $\mathrm{Close}(\pi)$ whenever both occur in the same computation, since they cannot be made concurrent by independence. Additionally, Γ-symmetries are assumed to preserve the rely/independence structure (Def. **??**), so applying Γ does not change whether the constraint holds. Thus the constraint is invariant under $\equiv_{\perp,\Gamma}$. □

Remark 3462. *The point is not that phases are metaphysically fundamental; it is that* a scoping discipline on resources *survives maximal quotienting and so becomes a "principle" of the TUG core (Hyperseed-Concept* **??***). For automated reasoning, this is ideal: locality becomes a static well-formedness condition on the trace-poset rather than an opaque procedural ban. Concretely, given a candidate trace (or its partial-order representation), one can check that there is no* $\mathrm{Hop}(\pi, \dots)$ *event that is causally (or linearly) after a* $\mathrm{Close}(\pi)$ *event, without reconstructing any hidden structural representation of the phase interior.*

63.6 Relativized Minimality from competitive feature checking

Theorem 206 (Relativized Minimality as competitive resource consumption; quotient-stability)**.** *Assume the same enabling/resource discipline as in Theorem 205. Fix a probe X and an uninterpretable feature F, represented by a resource* $\mathrm{FRes}(X, F) \in \mathrm{Res}$ *(Hyperseed-Concept* **??***). Consider two potential Agree labels*

$$\mathrm{Agree}(X, Y, F) \quad \textit{and} \quad \mathrm{Agree}(X, Z, F),$$

both of which require the probe feature resource:

$$\mathrm{FRes}(X, F) \in \mathrm{rely}(\mathrm{Agree}(X, Y, F)) \cap \mathrm{rely}(\mathrm{Agree}(X, Z, F)).$$

Assume one-time feature checking*: executing any Agree label that checks* $\mathrm{FRes}(X, F)$ *consumes it:*

$$\mathrm{Active}(s \xrightarrow{\mathrm{Agree}(X,*,F)} s') \;=\; \mathrm{Active}(s) \setminus \{\mathrm{FRes}(X, F)\}.$$

Then:

1. ***(Mutual exclusion)*** *No complete derivation can contain* both $\mathrm{Agree}(X, Y, F)$ *and* $\mathrm{Agree}(X, Z, F)$.
2. ***(Blocking pattern)*** *If a derivation contains* $\mathrm{Agree}(X, Z, F)$*, then it cannot contain any later occurrence of* $\mathrm{Agree}(X, Y, F)$ *(and symmetrically with Y, Z swapped).*
3. ***(Quotient-stability)*** *The mutual exclusion / blocking pattern is invariant under the TUG congruence $\equiv_{\perp,\Gamma}$.*

Proof. (1) Suppose for contradiction that a derivation contains both Agree steps. Whichever occurs first (say $\mathrm{Agree}(X, Z, F)$) consumes $\mathrm{FRes}(X, F)$ by hypothesis, so in the subsequent state s' we have $\mathrm{FRes}(X, F) \notin \mathrm{Active}(s')$. But $\mathrm{Agree}(X, Y, F)$ requires $\mathrm{FRes}(X, F) \in \mathrm{rely}(\mathrm{Agree}(X, Y, F))$, so it is not enabled at s' and can never occur after that point. Contradiction.

(2) This is the same argument without contradiction: once $\mathrm{Agree}(X, Z, F)$ fires, $\mathrm{FRes}(X, F)$ is removed, so any later Agree requiring $\mathrm{FRes}(X, F)$ is disabled.

(3) The two Agree labels are dependent (they share $\mathrm{FRes}(X, F)$), so $\equiv_\perp$ cannot swap them past each other (only independent adjacent swaps are allowed). Moreover, the consumption rule is a semantic invariant of the labeled transition system, not of a particular interleaving. Thus, if a derivation has consumed $\mathrm{FRes}(X, F)$ at some point, every $\equiv_\perp$-equivalent derivation has consumed it at the corresponding point in the trace-poset representation (Theorem 204), and therefore cannot contain a later Agree requiring it. Finally, Γ preserves the rely/consumption structure by assumption (Def. **??**), so Γ-renaming does not change the mutual exclusion pattern.

To connect this explicitly to the usual RM intuition: the "intervener" effect arises here because the probe-side feature token $\mathrm{FRes}(X, F)$ functions as an exclusive capability. If Z is checked first, then Z has effectively "used up" the only opportunity for X to discharge F, and any later attempt to relate X to Y via the same feature is not merely dispreferred but formally ill-typed (disabled) with respect to the enabling relation. In particular, no additional representational device is required to encode a separate minimality filter: it is already enforced by the resource accounting in $\mathrm{Active}(\cdot)$ together with the rely condition.

It is also worth noting that this argument does not depend on the identity of Y and Z beyond the shared requirement on $\mathrm{FRes}(X, F)$. In other words, the proof shows a general "single-discharge" principle: for any probe-side feature resource, there is at most one successful consumption event per derivation, and any two candidate consumptions are in direct competition. This abstraction will be used later when multiple probes/features are present and the trace skeleton decomposes into several such competitive cliques. ☐

Corollary 67 (Uniqueness of successful checking for a fixed probe-feature)**.** *Fix X and F as in Theorem 206. In any complete derivation, there is at most one label of the form* $\mathrm{Agree}(X, *, F)$*. Moreover, if some* $\mathrm{Agree}(X, W, F)$ *occurs, then every later transition must be labeled by an action whose rely set does not include* $\mathrm{FRes}(X, F)$*.*

Proof. The first claim is the mutual exclusion clause instantiated with an arbitrary pair of distinct potential goals. The second claim is exactly the blocking clause rephrased in terms of the rely predicate: after consumption, $\mathrm{FRes}(X, F) \notin \mathrm{Active}(\cdot)$, so any label that relies on it is disabled. ☐

Remark 3463. *This theorem is deliberately phrased as a* resource-competition invariant*, because that is what an AI reasoner can exploit directly: RM is detectable as a finite-resource conflict in the trace skeleton (Hyperseed-Concept* **??***), not as a language-specific stipulation.*

Remark 3464. *A useful reading is that "minimality" is not a separate global preference ordering over candidates, but an emergent property of local enablement under exclusive resources. If one wishes to recover the familiar intervening-configuration narrative, one can treat the earlier consumer of* $\mathrm{FRes}(X, F)$ *as the formal counterpart of an intervener: it blocks later dependencies not by distance per se but by resource preemption. On this view, locality effects follow whenever the derivation mechanics make it possible for a closer (or earlier-triggered) goal to consume* $\mathrm{FRes}(X, F)$ *before a more distant (or later-triggered) goal. Conversely, apparent "RM violations" correspond to theories in which either (i)* $\mathrm{FRes}(X, F)$ *is not consumed (one allows repeated checking), or (ii) the relevant dependencies do not in fact share a probe-feature resource (they are independent in the rely graph), in which case $\equiv_\perp$-commutation and coexistence become possible.*

64 Universal Grammar Cores in Cultures, Mindplexes, and Dynamical Archetypes

Glossary cross-references used in this section. We explicitly reference the glossary concepts: Universal Grammar (interface-theoretic) (Concept 220), Trace-based Universal Grammar (Concept 221), Observation interface (Concept 222), Grammatical independence (Concept 225), Label symmetry group (Concept 226), Trace quotient induced by $(\perp, \Gamma)$ (Concept 227), Principles-and-parameters as invariants (Concept 228), Culture (Concept 91), Collective mind system (Concept **??**), Mindplex (Concept 113), Collective Consciousness Location (Concept 75), Efficient pragmatic general intelligence (Concept 217), and Archetypes (Concept 58).

64.1 From individual trace quotients to a collective UG core

A single mind's interface-theoretic UG can be viewed as a *weakest interface theory*: we quotient derivations by whatever reorderings and symmetries the interface cannot (or chooses not to) distinguish (Concepts 220–227). For a *community*—a culture, a scientific field, a multi-agent lab, or a mindplex—each member may carry a different "language of thought" (different $(O, \perp, \Gamma)$ choices). This raises a concrete question: what is the mathematically canonical way to extract a *shared* core, and how does it behave as the group scales or integrates?

Definition 604 (Congruence closure and collective UG cores)**.** *Let M be a monoid. A* (monoid) congruence *on M is an equivalence relation $R \subseteq M \times M$ such that for all $u, v, u', v' \in M$,*

$$u \, R \, v \text{ and } u' \, R \, v' \quad \Longrightarrow \quad uu' \, R \, vv'.$$

Write $\mathsf{Cong}(M)$ for the set of congruences on M.

For any binary relation $E \subseteq M \times M$, define its congruence closure *by*

$$\langle E \rangle_{\mathrm{cong}} \;:=\; \bigcap \{ R \in \mathsf{Cong}(M) : E \subseteq R \}.$$

(So $\langle E \rangle_{\mathrm{cong}}$ is the smallest congruence containing E.)

Now fix a community I of individuals. Assume each individual $i \in I$ induces a TUG-style congruence $\approx_i \in \mathsf{Cong}(M)$ on a shared interface monoid M (e.g. take $M = \mathrm{Lab}_V^*$ and $\approx_i = \approx_{\perp_i, \Gamma_i}$ as in Concept 227).

Define two canonical collective *congruences:*

$$\approx_\wedge \;:=\; \bigcap_{i \in I} \approx_i, \qquad \approx_\vee \;:=\; \left\langle \bigcup_{i \in I} \approx_i \right\rangle_{\mathrm{cong}}.$$

We call $\approx_\wedge$ the strict shared UG core *(unanimous identifications), and $\approx_\vee$ the* translation-closure UG core *(identifications generated by the union).*

Conceptual explanation (what this theorem is saying). The strict core $\approx_\wedge$ captures what *everyone* already treats as "the same". The translation-closure core $\approx_\vee$ captures what the *community as a whole* can treat as "the same" once we allow chains of intertranslation and closure under compositional contexts. This aligns naturally with the idea that culture is a stabilized intersubjective pattern web (Concept 91): stabilized meaning is not only what each individual collapses, but what the intersubjective network *collectively* collapses.

Theorem 207 (Collective UG cores exist canonically and scale monotonically)**.** *Let M be a monoid and $\{\approx_i\}_{i\in I} \subseteq \mathsf{Cong}(M)$. Then $\approx_\wedge$ and $\approx_\vee$ (Definition 604) are congruences on M.*

Moreover:

1. ***(Lattice structure.)*** *$\mathsf{Cong}(M)$ is a complete lattice: arbitrary meets are intersections, and arbitrary joins are congruence-closures of unions.*

2. ***(Monotonicity in group size.)*** *If $J \subseteq I$ then*

$$\bigcap_{i\in I} \approx_i \subseteq \bigcap_{j\in J} \approx_j, \qquad \left\langle \bigcup_{j\in J} \approx_j \right\rangle_{\text{cong}} \subseteq \left\langle \bigcup_{i\in I} \approx_i \right\rangle_{\text{cong}}.$$

   So: adding members makes the strict core stricter *(fewer unanimous identifications) and makes the translation-closure core* weaker *(more collectively admitted identifications).*

3. ***(Principles as invariants.)*** *Let $f : M \to N$ be a monoid homomorphism. Then f is invariant under every $\approx_i$ iff f is invariant under $\approx_\vee$. Equivalently, $\bigcup_i \approx_i \subseteq \ker(f)$ iff $\approx_\vee \subseteq \ker(f)$. In particular, the "hard core" invariants common across members are exactly the homomorphic invariants of the join core $\approx_\vee$ (cf. Concept 228).*

Proof. First, $\approx_\wedge$ is an equivalence relation as an intersection of equivalence relations. If $u \approx_\wedge v$ and $u' \approx_\wedge v'$, then for each i we have $u \approx_i v$ and $u' \approx_i v'$, so $uu' \approx_i vv'$ by congruence of $\approx_i$. Thus $uu' \approx_\wedge vv'$, proving $\approx_\wedge \in \mathsf{Cong}(M)$.

Next, by definition $\approx_\vee$ is an intersection of congruences (those containing the union), hence itself a congruence.

For (1): arbitrary intersections of congruences are congruences (meets). For joins: given any family $\{R_k\} \subseteq \mathsf{Cong}(M)$, the relation $\langle \cup_k R_k \rangle_{\text{cong}}$ is a congruence containing each R_k, and is contained in every congruence that contains all R_k; hence it is the least upper bound.

For (2): if $J \subseteq I$ then $\cap_{i\in I} \approx_i \subseteq \cap_{j\in J} \approx_j$ is immediate. Also $\cup_{j\in J} \approx_j \subseteq \cup_{i\in I} \approx_i$, so monotonicity of congruence closure implies the join inclusion.

For (3): if f is a monoid homomorphism then $\ker(f) = \{(u, v) : f(u) = f(v)\}$ is a congruence. If f is invariant under every $\approx_i$, then $\cup_i \approx_i \subseteq \ker(f)$. Since $\ker(f)$ is a congruence containing the union and $\approx_\vee$ is the smallest such congruence, we get $\approx_\vee \subseteq \ker(f)$, i.e. invariance under $\approx_\vee$. Conversely, each $\approx_i \subseteq \approx_\vee$, so invariance under $\approx_\vee$ implies invariance under every $\approx_i$. □

64.2 Culture as a fixed point: stabilization of a shared grammar

Culture is defined in Hyperseed as a stabilized intersubjective pattern web, i.e. a fixed point of a societal update operator (Concept 91). The same fixed-point idea can be applied to the *meaning equivalences* that constitute a shared language. In this reading, the evolving "shared language" is not primarily a list of sentences but a collectively maintained criterion for when two expressions are treated as interchangeable *for the purposes of coordination.* The formal object that captures this criterion is a congruence relation: it remembers not only which tokens are identified, but also that identifications are stable under composition (so that replacing a subexpression by an equivalent one does not change the meaning of the whole).

Conceptual explanation (what this theorem is saying). Think of "what counts as the same meaning" as part of the cultural web. Repeated successful communications, teaching, norms, and institutionalization are then an update rule that can only *add* identifications (or at least is monotone in a suitable sense). In finite interface regimes, any such monotone reinforcement must stabilize: there is a well-defined limiting collective core. Equivalently, once the community has a finite repertoire of relevant compositional interfaces (a finite stock of forms and combinators, or a bounded set of interactional contexts that matter for interpretation), then iterated reinforcement cannot strictly increase forever: at some point, all further "learning" of equivalences is redundant because it is already implied by what has been institutionalized.

One can also read the statement operationally: start with a baseline of agreed identifications R_0 (perhaps minimal mutual intelligibility), and apply an update F that encodes the effect of a round of cultural reinforcement (new teaching episodes, newly salient analogies, newly standardized usage). Monotonicity means that if a society begins with more identifications, the same reinforcement cannot yield fewer; inflationarity means that reinforcement never retracts an established identification. Under these hypotheses, the limit R_{t_*} is path-independent in the sense that it depends only on R_0 and F (not on any external scheduling), and it is the smallest stable (institutionalized) equivalence compatible with that starting point.

Theorem 208 (Finite stabilization of a reinforced collective UG)**.** *Let M be a finite monoid, and let $F : \mathsf{Cong}(M) \to \mathsf{Cong}(M)$ be monotone and inflationary (i.e. $R \subseteq F(R)$ for all R). For any initial congruence R_0, define $R_{t+1} := F(R_t)$. Then $(R_t)_{t\geq 0}$ stabilizes after finitely many steps: there exists t_* such that $R_t = R_{t_*}$ for all $t \geq t_*$. Moreover, R_{t_*} is a fixed point of F and is the least fixed point of F above R_0.*

Interpreting the objects: the monoid M can be taken as a finite space of interface-states or composable meaning-bearing units (e.g. equivalence classes of well-formed expressions under a coarse syntactic typing, or a finite symbolic action-set closed under concatenation). A congruence $R \in \mathsf{Cong}(M)$ then represents a hypothesis of "sameness" that is compatible with composition: if $a \sim b$ and $c \sim d$, then $ac \sim bd$. This is precisely the algebraic form of the intuition that meaning-equivalence should be substitutive in context. In this sense, the theorem can be read as: iterated cultural reinforcement over a finite compositional interface yields a stable shared "core" of substitutable meanings.

Proof. Because M is finite, the set of all binary relations on M is finite, hence $\mathsf{Cong}(M) \subseteq \mathcal{P}(M\times M)$ is finite. Inflationarity and monotonicity give an ascending chain $R_0 \subseteq R_1 \subseteq R_2 \subseteq \cdots$. Any ascending chain in a finite poset must eventually stabilize, so there exists t_* with $R_{t_*} = R_{t_*+1} = F(R_{t_*})$, hence R_{t_*} is a fixed point. If R is any fixed point with $R_0 \subseteq R$, then by monotonicity $R_1 = F(R_0) \subseteq F(R) = R$, and inductively $R_t \subseteq R$ for all t, so $R_{t_*} \subseteq R$. Thus R_{t_*} is the least fixed point above R_0. □

The proof is a direct "finite ascending-chain" argument, but the conclusion has a useful cultural reading: once reinforcement is purely accumulative (inflationary) and order-preserving (monotone), the dynamics cannot cycle or oscillate. What looks sociologically like "convergence on norms" is, mathematically, the impossibility of a strictly increasing infinite chain of congruences when the interface space is finite.

It is sometimes helpful to make the stabilization time explicit. Since $\mathsf{Cong}(M) \subseteq \mathcal{P}(M\times M)$, one crude bound is $t_* \leq |\mathsf{Cong}(M)| \leq 2^{|M|^2}$, because at each strict step the relation gains at least one new ordered pair. This bound is rarely tight, but it clarifies the structural point: the stabilization is guaranteed by finiteness alone, not by any particular "strength" of the reinforcement.

Finally, note what the "least fixed point above R_0" means in practice. It says that if multiple stable cultural equilibria exist (multiple fixed points of F), the iteration starting from R_0 selects the minimal one compatible with the initial shared core. In linguistic terms, starting from a given baseline of mutual understanding and applying only cumulative reinforcement yields the smallest institutionalized equivalence structure that is closed under the society's reinforcement mechanism; stronger conventions would require either a different update rule F (different institutions) or a strictly stronger starting core R_0 (different initial coordination).

64.3 Scientific communities and a Lakatos-style hard core/periphery split

A scientific community is a paradigmatic *collective mind system* (Concept **??**). Informally, Lakatos distinguishes a "hard core" of commitments from a "protective belt" of adjustable auxiliaries. In the interface-theoretic UG lens, a clean analogue appears:

Definition 605 (Hard core and periphery via congruence invariants)**.** *Fix a collective UG congruence $\approx_C \in \mathsf{Cong}(M)$ for a community (for instance the stabilized outcome of Theorem 208 starting at $\approx_\vee$ from Definition 604).*

Define the hard core *of the community to be the class of homomorphic observables that factor through the quotient:*

$$\mathcal{H}(\approx_C) \;:=\; \{f : M \to N \text{ monoid homomorphism } : \approx_C \subseteq \ker(f)\}.$$

Define a periphery refinement *to be any congruence $\approx' \in \mathsf{Cong}(M)$ with $\approx' \subseteq \approx_C$ (i.e. a strictly finer theory that distinguishes more than the core).*

The condition $\approx_C \subseteq \ker(f)$ is the usual algebraic way to say that f is *insensitive* to the identifications imposed by $\approx_C$: whenever the community treats x and y as equivalent ($x \approx_C y$), the observable f must assign them the same value ($f(x) = f(y)$). Equivalently, f descends to a unique homomorphism $\bar{f} : M/{\approx_C} \to N$ satisfying $f = \bar{f} \circ \pi$, where $\pi : M \to M/{\approx_C}$ is the quotient map. In this sense, $\mathcal{H}(\approx_C)$ can be read as the collection of "principles" that are expressible purely at the level of the community's quotient interface, i.e. at the grain where $\approx_C$ has erased internal differences.

It is also useful to keep in mind that congruences on M form a lattice under inclusion. In that order, $\approx' \subseteq \approx_C$ means that $\approx'$ makes *fewer* identifications (it is finer, hence more discriminating), so the move from $\approx_C$ to $\approx'$ corresponds to enriching the community's vocabulary or inferential sensitivity while respecting the original identifications as a limiting case. This matches the methodological picture in which auxiliary hypotheses can be adjusted to increase empirical fit or explanatory detail without altering the background interface that the group has stabilized upon.

Conceptual explanation (what this theorem is saying). If the "hard core" is what is invariant under the community's collective equivalences, then making a periphery adjustment corresponds to adding distinctions without changing what was already forced to be invariant. This precisely matches the idea of conservative elaboration: you can complicate the protective belt, but you cannot thereby refute the hard core, because the hard core is already expressed at a coarser grain.

Stated in more operational terms: the hard core is a family of tests, measurements, or semantic projections that the community can apply to states of the mindplex without needing to resolve any of the within-class structure that $\approx_C$ intentionally ignores. A periphery refinement then amounts to *resolving* some within-class structure (distinguishing cases previously treated as equivalent), which may change theories and explanations at the fine level, but cannot invalidate any statement that depended only on the coarse interface. In particular, hard-core principles are precisely those that remain well-defined on equivalence classes, hence remain meaningful even if the community later adopts a more fine-grained classification scheme.

Theorem 209 (Periphery refinements conservatively extend the hard core)**.** *Let $\approx' \subseteq \approx_C$ be a periphery refinement in the sense of Definition 605. Then*

$$\mathcal{H}(\approx_C) \ \subseteq \ \mathcal{H}(\approx').$$

Equivalently: any homomorphic "principle" invariant under the core remains invariant under every refinement of the core.

One can also read the inclusion $\mathcal{H}(\approx_C) \subseteq \mathcal{H}(\approx')$ as an information order on invariants: the more distinctions a community is willing to make (the finer the congruence), the larger the set of homomorphic observables that qualify as well-defined invariants. Thus, a refinement can *add* admissible invariants (new stable observables become definable once the theory discriminates more), but it cannot *subtract* those already determined by the hard core. This is the algebraic reflection of the methodological asymmetry between revising auxiliaries and overturning the core: refinements elaborate what can be said, but do not break what was already quotiented into the interface.

Proof. Take $f \in \mathcal{H}(\approx_C)$, so $\approx_C \subseteq \ker(f)$. If $\approx' \subseteq \approx_C$, then $\approx' \subseteq \ker(f)$ as well, hence $f \in \mathcal{H}(\approx')$. $\square$

A concrete toy instance may help fix intuition. Suppose M encodes experimental protocols under concatenation, and $\approx_C$ identifies protocols that the community treats as operationally equivalent (e.g. two calibration routines regarded as interchangeable). Then $\mathcal{H}(\approx_C)$ consists of all homomorphic observables (such as cost, time, or a predicted signature) that do not distinguish between interchangeable calibrations. If the community later refines its practice to $\approx'$ by distinguishing previously lumped calibrations (perhaps after learning about a systematic bias), then any observable that was already blind to the old coarse equivalences is automatically blind to the refined ones as well. What changes is that new observables may become admissible as invariants once the refined distinctions are acknowledged, corresponding to new auxiliary structure in the protective belt.

64.4 Why a mindplex can benefit from a shared core UG

A mindplex is (roughly) a notably coherent society-of-mind that also maintains a collective narrative and persistent identity (Concept 113). A natural design question is: when does it help to enforce a shared UG core across subminds?

Conceptual explanation (what this theorem is saying). If the tasks and evaluations that matter to the mindplex are invariant under some quotient of internal derivations (i.e. the world and the mindplex's own evaluators cannot tell within-class differences), then enforcing that quotient as a shared core is *lossless* for performance but can reduce internal coordination cost. This is precisely "weakest interface theory" as a control advantage. In more concrete terms: if multiple internal encodings, proofs, parses, or message-forms are treated as equivalent by downstream dynamics and by the scoring rule the collective ultimately cares about, then the collective can standardize on the equivalence classes (the quotient) as a common "interface language." Standardization reduces the degrees of freedom that must be coordinated (e.g. fewer translation layers, fewer cross-submind misunderstandings, smaller state representations, and fewer brittle conventions), without sacrificing any externally-relevant capability on that task family.

One can read the quotient as the mindplex's *public semantic channel*: subminds may continue to use richer private derivations, but only the quotient-relevant content must be communicated and maintained for collective action. This is the sense in which a shared UG core is an *interface contract*: it specifies what distinctions are collectively meaningful, and collapses distinctions that are collectively irrelevant.

Definition 606 (UG-invariant tasks for a collective channel)**.** *Let $T = (O, A, P, r, \gamma)$ be a task in the sense of the manuscript's task layer, and let $q : O \to Q$ be a surjection. We say q is a* task abstraction *(and T is* invariant under *q) if:*

1. *(Lumpable dynamics) for all $o_1, o_2 \in O$ with $q(o_1) = q(o_2)$ and all $a \in A$, the pushforward distributions agree: $q_*(P(o_1, a)) = q_*(P(o_2, a))$;*

2. *(Evaluation invariance) for all $o_1, o_2 \in O$ with $q(o_1) = q(o_2)$, all $a \in A$, and all $o^+ \in O$, we have $r(o_1, a, o^+) = r(o_2, a, o^+)$.*

If O is a space of public derivational encodings (messages) and $\approx$ is a collective UG congruence on O, we call T UG-invariant *(for that collective) when q is the quotient map $q(o) = [o]_\approx$ and q is a task abstraction.*

The two conditions in Definition 606 can be interpreted as: (i) the environment's *effective* next-state distribution depends only on the equivalence class $q(o)$, not on which representative o the mindplex happened to use; and (ii) the evaluator that assigns reward similarly depends only on the equivalence class, so that no within-class refinement yields an intrinsic advantage. The "pushforward" notation $q_*(P(o, a))$ makes explicit that we only require agreement after mapping next observations into Q: the fine-grained next-observation distributions in O may differ, but only in ways that disappear under the quotient. This matches the mindplex use case where subminds may maintain rich internal syntactic differences, while the collective channel tracks only the induced semantics.

It is also useful to note why the reward invariance is stated as $r(o_1, a, o^+) = r(o_2, a, o^+)$ for all o^+, rather than only for o^+ with the same quotient: it ensures that the immediate evaluation cannot exploit the choice of representative even when coupled with future observations. In typical MDP-style settings where r factors through $(q(o), a, q(o^+))$, this condition holds automatically; the above form simply states the strongest sufficient criterion in the manuscript's notation.

Theorem 210 (UG quotienting is lossless for UG-invariant tasks)**.** *Assume $q : O \to Q$ is a task abstraction for $T = (O, A, P, r, \gamma)$, in the sense of Definition 606. Then there exists a reduced task $T_Q = (Q, A, P_Q, r_Q, \gamma)$ such that:*

1. *any policy π_Q for T_Q lifts to a policy π for T with identical value, and*

2. *consequently, representing observations only up to the quotient q is not a disadvantage for value.*

Theorem 210 can be seen as a formal version of: "if all that matters is the equivalence class, then committing to the equivalence class cannot hurt." In the mindplex setting, the equivalence relation $\approx$ can be read as the shared UG core identifying which transformations of messages/derivations preserve meaning for collective purposes (e.g. paraphrase, notational variants, proof permutations, or syntactic sugar). The theorem then says that, on any task whose dynamics and rewards respect those transformations, restricting the collective interface to $\approx$-classes loses no achievable discounted return.

Proof. Define $P_Q : Q \times A \to \Delta(Q)$ by $P_Q(q(o), a) := q_*(P(o, a))$. This is well-defined by lumpability: if $q(o_1) = q(o_2)$ then $q_*(P(o_1, a)) = q_*(P(o_2, a))$. Define $r_Q(q(o), a, q(o^+)) := r(o, a, o^+)$. This is well-defined by evaluation invariance.

Given any policy π_Q on Q, define a lifted policy π on O by $\pi(h) := \pi_Q(q(h))$ (i.e. apply q pointwise to the observation history). By construction, the induced distribution over Q-histories under π in T matches the distribution under π_Q in T_Q, and the rewards match stepwise because r_Q was defined by the invariance condition. Therefore the total discounted values coincide. $\square$

Two additional clarifications are often helpful when applying the proof idea. First, the lift $\pi(h) := \pi_Q(q(h))$ does not require choosing canonical representatives in O: the policy on O simply ignores within-class detail by preprocessing observations through q. Second, because the argument is distributional (matching the law of Q-histories), it applies equally to stochastic policies and stochastic dynamics; no determinism assumption is needed.

In many applications, this quotient construction is closely related to standard notions of MDP abstraction (e.g. homomorphisms, bisimulation-style lumping, and state aggregation). The distinctive emphasis here is interpretive: the quotient is not only a computational abstraction, but also a *collective linguistic* abstraction enforced as a UG core, so that coordination benefits accrue at the level of communication and interface design as well as at the level of planning.

Conceptual explanation (GTGI consequence). GTGI's efficient pragmatic general intelligence Π_{eff} (Concept 217) scores performance *per unit resource*. Theorem 210 says shared UG quotienting need not decrease achievable value on UG-invariant tasks; if it reduces coordination/representation cost, it increases (or at least does not decrease) Π_{eff}. Intuitively, once value is held fixed, any reduction in the collective's internal overhead (memory, bandwidth, compute, deliberation time, translation effort between subminds, or the governance complexity of maintaining many incompatible encodings) improves the efficiency metric. In a mindplex, these costs often scale superlinearly with representational heterogeneity, so even modest quotienting can yield substantial efficiency gains while leaving external competence unchanged on the invariant task family.

Corollary 68 (Shared UG cores can only help efficient pragmatic GI when they save resources)**.** *Assume a class of tasks is UG-invariant under a collective congruence $\approx$ (Definition 606) and that implementing policies on the quotient $Q = O/\approx$ uses weakly less resource on each task than implementing the corresponding lifted policies on O. Then the collective's efficient pragmatic general intelligence Π_{eff} (Concept 217) computed over that task class is weakly larger under the quotient representation.*

Proof. By Theorem 210, for each task the optimal achievable value is preserved by passing to the quotient task. By hypothesis, the resource functional $Q(\cdot, \cdot)$ used in Π_{eff} is weakly smaller for quotient-implemented policies than for their lifts. Since Π_{eff} weights performance by (inverse) resource use, preserving the numerator while decreasing (or preserving) the denominator yields a weak increase in Π_{eff}. □

A practical implication is that the main empirical burden shifts to verifying the two premises: (a) that the task family is in fact UG-invariant under the chosen congruence (so that no strategically relevant distinctions have been collapsed), and (b) that the quotient implementation genuinely saves resources in the deployment regime of interest (e.g. reduced communication bandwidth, simpler shared memory formats, or fewer translation modules). When either premise fails, a shared core UG may still be desirable for other reasons (safety, interpretability, or governance), but it is no longer guaranteed to be efficiency-improving by the above argument.

64.5 UG alignment and Collective Consciousness Location

Collective Consciousness Location (Concept 75) uses, among other inputs, *discord* as the variance of proposal signals across members of a collective.

Conceptual explanation (what this theorem is saying). If the public interface enforces a shared UG quotient, and if members' public proposals depend only on the quotient class, then the quotient removes a major source of "talking past each other" induced by purely representational differences (reorderings of independent steps, symmetry variants, etc.). In the idealized limit, shared UG implies zero interface-discord. More explicitly: the quotient map identifies (collapses) multiple surface-level encodings in O that are deemed equivalent under the collective's Universal Grammar constraints, so that any two public states $o, o' \in O$ with $o \approx o'$ become indistinguishable at the level of what is *publicly actionable*. The theorem below isolates the extreme case in which (i) there is a *single* public state $o(t)$ exposed to all members at time t, and (ii) each member's emitted proposal depends on $o(t)$ *only* through the shared equivalence class $q(o(t))$. Under these conditions, representational degrees of freedom cannot induce proposal dispersion, because there are no member-specific "private coordinate choices" left in the public channel.

Interpretive note on assumptions. The condition that there is a *common* function f is the formal way of stating that the public interface plus shared UG quotienting yields a common decoding/decision rule at the interface level. If members were to use different interface-level maps f_i, then even with identical quotient classes $q(o(t))$ one could still obtain nonzero discord purely from heterogeneity in the public-to-proposal mapping. Likewise, the existence of a single public state $o(t)$ rules out disagreement that originates upstream (e.g., members observing different public inputs); the theorem therefore attributes all remaining discord to interface representation, and then shows that this source vanishes under shared quotienting.

Theorem 211 (Shared UG quotienting forces zero interface-discord in the idealized limit)**.** *Consider a collective mind system (Concept **??**) whose members emit proposal signals $y_i(t) \in \mathbb{R}^d$ at time t. Suppose there exists a shared public representation space O and a collective UG congruence $\approx$ on O such that each member's proposal factors through the quotient map $q : O \to O/\!\approx$:*

$$y_i(t) \;=\; f(q(o(t))) \quad \textit{for a common function } f \textit{ and common public state } o(t) \in O.$$

Then the discord at time t as defined in Concept 75,

$$DO(t) \;:=\; \mathrm{Var}\big(\{y_i(t)\}_i\big),$$

satisfies $DO(t) = 0$.

Why this matches the "interface-discord" intuition. In applications, $y_i(t)$ can be read as an externally visible proposal vector (e.g., a vote vector, an action recommendation, an allocation, or a semantic embedding of a plan). The quotient condition means that any two members who are presented with the same quotient-class information will necessarily produce the same public proposal signal. Thus any dispersion that remains must come from either (a) differences in the quotient-class observed, (b) differences in the mapping from quotient-class to proposal, or (c) noise/perturbations not modeled here. The theorem explicitly excludes (a)–(c) by hypothesis, which is why it is stated as an "idealized limit."

Proof. If every $y_i(t)$ equals the same vector $f(q(o(t)))$, then the multiset $\{y_i(t)\}_i$ is constant, hence has variance zero by definition of variance. For avoidance of doubt, this holds for any standard variance functional on a finite family of vectors: if $y_i(t) = \bar{y}(t)$ for all i, then each deviation $y_i(t) - \bar{y}(t)$ is the zero vector, and any aggregation of squared deviations (componentwise or via a norm) yields zero. □

Remark (approximate quotienting and non-ideal settings). The statement above is deliberately sharp: exact factorization through a shared quotient and exact identity of the interface map f together force literal equality of all $y_i(t)$. In practice, one often expects an *approximate* version in which members implement slightly different maps f_i or observe noisy versions of the public state; then shared UG quotienting still predicts a *reduction* of discord relative to an unquotiented interface, but not necessarily $DO(t) = 0$. This theorem can therefore be read as the baseline case establishing that, when representation is the only source of dispersion, quotienting eliminates that dispersion entirely.

64.6 Chaotic dynamics: implicit grammars, universal sublanguages, and archetypes

Many dynamical systems induce "implicit grammars" via symbolic dynamics: orbit itineraries generate admissible symbol strings. This offers a natural answer to: *what is the UG among the languages implicit in various dynamical systems?* More concretely, once an observer commits to a finite "alphabet" of coarse-grained observational states, the time-evolution of the system produces sequences over that alphabet; not every finite string is typically realizable, and the resulting pattern of permissibility/prohibition functions like a grammar. This viewpoint makes the dependence on observational resolution explicit: different partitions of the same underlying dynamics can induce different languages, ranging from highly permissive (very coarse partitions) to highly constrained (fine partitions that track more structure).

Definition 607 (Symbolic language (implicit grammar) of a dynamical system)**.** *Let (X, F) be a dynamical system and let $\mathcal{P} = \{P_1, \dots, P_k\}$ be a finite measurable partition of X. For any $x \in X$, define its symbolic itinerary $\sigma(x) \in \{1, \dots, k\}^{\mathbb{N}}$ by $\sigma(x)_t = j$ iff $F^t(x) \in P_j$.*

Define the symbolic language *(set of admissible finite words) by*

$$\mathcal{L}(F, \mathcal{P}) \ := \ \big\{(a_0, \dots, a_{n-1}) \in \{1, \dots, k\}^n \ : \ \exists x \in X \ s.t.\ \sigma(x)_t = a_t \ \forall t < n\big\}.$$

We regard $\mathcal{L}(F, \mathcal{P})$ as an implicit grammar of (X, F) at the observational grain $\mathcal{P}$.

It is helpful to note that $\mathcal{L}(F, \mathcal{P})$ is a language of *finite* words (not necessarily a complete specification of the set of infinite itineraries), which matches the typical epistemic situation in which only finite observational traces are available. The passage from infinite itineraries to finite admissible blocks is also the standard bridge to subshifts: one may equivalently consider the set of all infinite sequences that avoid the forbidden finite words, and then recover $\mathcal{L}(F, \mathcal{P})$ as the collection of blocks appearing in that subshift. In this sense, "implicit grammar" can be read literally as "block constraints" on symbolic strings induced by dynamics.

Conceptual explanation (what this theorem is saying). Exact dynamical abstraction (a quotienting relation $q \circ F = G \circ q$) is the dynamical analogue of "weakest interface theory": it collapses microstates into macrostates without breaking the induced macro-dynamics. The theorem below says: if many different micro-systems abstract to the same macro-system, then the macro-system's symbolic language is a *universal sublanguage* (a dynamical UG core) shared by all of them. In chaotic regimes (horseshoes), this universal language contains a shift-of-finite-type core, which is a natural candidate for a cross-context archetype (Concept 58). Intuitively: an exact abstraction q asserts that what the macro-observer calls "one step of time" is consistent with the micro-dynamics after forgetting micro-detail. Thus any macro-observable finite behavior (any length-n macro trace through partition cells) must be reproducible by at least one micro-trajectory; this is precisely the language inclusion established below. Importantly, the inclusion is generally *one-way*: the micro-system may realize additional symbol strings that disappear under the macro quotient because the macro description conflates distinct micro-episodes.

Theorem 212 (Exact dynamical abstraction induces a universal symbolic grammar)**.** *Let (X, F) and (Y, G) be dynamical systems, and let $q : X \to Y$ be a surjection satisfying*

$$q \circ F \;=\; G \circ q$$

(i.e. an exact dynamical abstraction in the manuscript's sense). Let $\mathcal{P}_Y$ be a finite partition of Y, and let $\mathcal{P}_X$ be the pullback partition $\mathcal{P}_X := \{q^{-1}(P) : P \in \mathcal{P}_Y\}$. Then the symbolic languages satisfy

$$\mathcal{L}(G, \mathcal{P}_Y) \;\subseteq\; \mathcal{L}(F, \mathcal{P}_X).$$

Consequently, for any family $\{(X_j, F_j)\}_{j\in J}$ that all admit exact abstractions $q_j : X_j \to Y$ to the same macro-system (Y, G), the macro-language $\mathcal{L}(G, \mathcal{P}_Y)$ is a universal sublanguage shared by every member of the family.

Two clarifications make the statement operational. First, the choice of $\mathcal{P}_X$ as the pullback partition is not incidental: it ensures that the micro-observer is "speaking the same alphabet" as the macro-observer, i.e. micro-symbols record exactly which macro-cell the current microstate maps to. If one instead chose an unrelated partition of X, there would be no reason for the resulting languages to be comparable. Second, $q \circ F = G \circ q$ is the standard notion of a factor map in topological/measure-theoretic dynamics; in symbolic dynamics terms, the macro symbolic process is a factor of the micro symbolic process (when partitions are chosen compatibly). Under this reading, the theorem says that factors cannot create novel admissible finite words: coarse-graining can only *remove* distinctions, and thus any finite macro pattern must already exist in the micro system.

Proof. Take any word $w = (a_0, \ldots, a_{n-1}) \in \mathcal{L}(G, \mathcal{P}_Y)$. By definition, there exists $y \in Y$ whose itinerary under G visits partition cells P_{a_t} for $t = 0, \ldots, n-1$. Because q is surjective, choose $x \in X$ with $q(x) = y$. Then for each $t < n$,

$$q(F^t(x)) \;=\; G^t(q(x)) \;=\; G^t(y) \;\in\; P_{a_t}.$$

Hence $F^t(x) \in q^{-1}(P_{a_t})$, i.e. the itinerary of x under F w.r.t. $\mathcal{P}_X$ realizes the same word w. Therefore $w \in \mathcal{L}(F, \mathcal{P}_X)$, proving inclusion. The "family" statement follows immediately by applying the same argument to each $j \in J$. $\square$

The role of surjectivity in the proof is also conceptually meaningful: it is exactly the requirement that every macrostate corresponds to at least one microstate (no "spurious" macrostates). Without surjectivity, a macro word might be realizable in (Y, G) but have no micro-preimage to witness it, so the would-be "UG core" would no longer be guaranteed to be shared. Relatedly, the theorem makes no claim that *every* micro word projects to a macro word of the same length under $\mathcal{P}_Y$; many distinct micro trajectories can collapse to the same macro trace, and some micro distinctions may never be visible at the macro alphabet at all.

A simple toy picture can be kept in mind. If Y is a two-state macro-system with partition cells $\{P_1, P_2\}$ and G forbids the transition $2 \to 2$, then $\mathcal{L}(G, \mathcal{P}_Y)$ consists of words with no consecutive 2's. Any micro-system (X, F) that factors onto (Y, G) via q (and is observed through the pullback cells $q^{-1}(P_1), q^{-1}(P_2)$) must also forbid consecutive 2's at the level of these macro-observables. Thus a single forbidden bigram can function as a minimal "grammar rule" that persists across many distinct micro implementations.

Conceptual explanation (archetype link). The manuscript defines an archetype as a low-description, high-support, diverse-instantiation template (Concept 58). If a particular symbolic template (e.g. a full shift or a common subshift induced by a horseshoe factor) appears as a shared macro-language across many otherwise diverse systems, then it is a strong candidate for an archetype in exactly that sense. In symbolic dynamics, a shift of finite type (SFT) is specified by finitely many local constraints (equivalently, a finite directed graph or adjacency matrix), which directly aligns with "low description" in the archetype criteria: the template can be communicated as a compact set of admissibility rules. Moreover, horseshoes and related hyperbolic sets often yield Markov partitions that make the induced symbolic dynamics conjugate to an SFT; when such an SFT arises *as a factor shared across systems*, it provides a particularly crisp candidate for a cross-context dynamical archetype: many different underlying geometries can instantiate the same finite constraint schema on observables.

Corollary 69 (Chaotic universal grammar templates yield dynamical archetypes (conditional))**.** *Assume a distribution over dynamical systems in which a fixed macro-system (Y, G) occurs as an exact abstraction target with sufficiently high support, and assume that for some partition $\mathcal{P}_Y$ the language $\mathcal{L}(G, \mathcal{P}_Y)$ has low description complexity and admits diverse micro-level instantiations across the family (many non-isomorphic (X_j, F_j) mapping onto it). Then the template "admissible words in $\mathcal{L}(G, \mathcal{P}_Y)$" satisfies the manuscript's archetype criteria (Concept 58).*

Proof. This is a direct verification against the archetype definition: high support is assumed by hypothesis (many systems share the same macro-language via Theorem 212); diversity of instantiation is assumed (many different (X_j, F_j)); and low description complexity is assumed for the template language itself. Under these three conditions, the template meets the stated archetype criteria. □

One additional interpretive nuance is that the corollary is intentionally conditional about "description complexity": for many chaotic systems the *raw* symbolic language at a given partition can be complicated or even non-computable, but when a robust horseshoe factor exists, a comparatively simple SFT core may still be extracted as a stable template. In the UG framing, this corresponds to the idea that a shared "core sublanguage" can be substantially simpler than the full set of admissible traces in any one micro-system, precisely because the shared factor captures only what is common (and thus, in a sense, compressible across instances).

Part IV
Leveraging Hyperseed to Explore The Ethics of Self-Modifying AI and Superintelligence

65 Universality Classes of Self-Modifying Minds Coarse-Graining to Fixed-Point vs Strange vs Multi-Attractor Types

We formalize a "self-modifying mind" as a dynamical system whose state includes its own modifiable implementation. We then formalize Hyperseed-style abstraction as quotienting: a surjection that collapses micro-distinctions into macro-variables. Under a compatibility condition (exact or approximate), the micro-dynamics induces a macro-dynamics. We prove that attractor *type* (fixed point vs strange attractor vs multi-attractor) is invariant under this induced macro-dynamics, and stable under broad architecture variation inside a basin (under standard stability hypotheses). We give three canonical universality classes: contractive (fixed-point), robustly chaotic (strange), and multi-well (multi-attractor).

Concretely, letting x_t denote the full microstate at (discrete or continuous) time t, the "state" is understood to include not only ordinary internal variables (memories, weights, registers, etc.) but also parameters and data structures that determine how future updates are computed. In other words, the update rule is not treated as fixed background; rather, its current effective form is encoded in the state itself, so that learning, self-rewriting, compilation, and reflective control are represented as ordinary trajectories in an enlarged state space. This framing is broad enough to cover classical self-reference (e.g. an agent that edits its own source or weights) and softer forms (e.g. meta-learning that changes the optimizer), while still allowing standard dynamical-systems notions such as invariance, basin structure, and attractors.

Likewise, the quotienting map may be read as an abstraction operator $\pi : X \to Y$ from a detailed space X to a coarser space Y of macro-variables (features, summaries, or equivalence classes). The surjection π implements "disciplined forgetting" by identifying microstates that are different in ways declared irrelevant for the present scale of analysis. The compatibility condition can be stated as an exact commutation relation $\pi \circ F = G \circ \pi$ (or its approximate/perturbative analogue) between a micro-update F and a macro-update G, ensuring that trajectories projected to Y behave as if they were generated by an autonomous macro-dynamics. When compatibility is approximate, the induced macro-dynamics is understood in the standard sense of a well-defined effective evolution up to bounded error or up to probabilistic coarse-grained semantics, sufficient to preserve qualitative long-run structure.

The invariance claim about attractor *type* should be read as a statement that, under such compatibility, the coarse-grained system does not change whether the relevant asymptotic behavior is (i) convergence to a single stable fixed point, (ii) sustained aperiodic motion with sensitive dependence characteristic of strange attractors, or (iii) the presence of multiple coexisting stable limit sets with basin-dependent outcomes. "Stable under broad architecture variation inside a basin" means that, once attention is restricted to a regime where trajectories remain in the same qualitative basin, moderate changes to the micro-realization (different internal modules, parameterizations, or implementations) do not alter the coarse-grained attractor category, provided familiar regularity and robustness assumptions hold (e.g. hyperbolicity/structural stability for chaotic sets, contraction conditions for fixed points, and persistence of separated wells for multi-attractor systems). The three classes named here are thus intended as canonical coarse-grained modes of long-run self-change, not as claims about a unique mechanism.

Remark 3465. *The philosophical stance here is deliberately structural: rather than asking "what is the mind made of?" we ask "what long-run forms of change does it instantiate?" This aligns with a Hyperseed tendency to treat* patterns *and* processes *as first-class objects (Hyperseed-Concept 130, 140) and to treat* abstraction *as a disciplined forgetting of distinctions (Hyperseed-Concept 51, 98) [1]. Universality classes then play the role they play in physics: they identify which qualitative properties survive coarse-graining, so that many different micro-realizations may be reasoned about with one macro-law [18].*

Put differently, the aim is not to deny implementation details, but to separate those details into (i) features that materially affect the asymptotic organization of trajectories and (ii) features that wash out under repeated composition of the dynamics and repeated abstraction. In this framing, "universality" is the claim that certain coarse, qualitative invariants (here: attractor categories and their basin structure) are the right level of description for many questions about long-run self-

modification, because they persist across a family of micro-realizations that differ in representational choices, internal decomposition, or learning-rule parameterization. This is also the sense in which the remark connects to the Hyperseed emphasis: if patterns and processes are primary, then the fixed-point/strange/ multi-attractor trichotomy functions as a small "alphabet" of persistent process-shapes that can be tracked even when the micro-level substrate is replaced or refactored.

65.1 Self-modifying minds and Hyperseed abstraction

Definition 608 (Self-modifying mind as a dynamical system)**.** *A (discrete-time) self-modifying mind is a pair (X, F) where:*

- *X is a state space whose points encode at least: internal memory/state, current goals, and a modifiable implementation description (e.g. code, weights, architecture parameters).*
- *$F : X \to X$ is the one-step update map that includes ordinary cognition/action plus the possible self-modification of the implementation description.*

Iterates F^t represent the mind's evolution over t steps.

Remark 3466. *Notation/conventions: the pair (X, F) is the standard dynamical-systems packaging in which X collects all degrees of freedom one cares to model, and F is the update rule. The iterate F^t means composing F with itself t times (so $F^0 = \mathrm{id}$ and $F^{t+1} = F \circ F^t$). We use discrete time not because cognition is inherently discrete, but because many self-modifying systems are naturally described as sequences of "revision" events (weight updates, code commits, policy updates), and discrete time lets us speak cleanly about iterates and attractors.*

Remark 3467. *Intuitively, the distinctive move is that the mind is not merely a state evolving under fixed rules; part of the state* encodes the rules *(or parameters that determine the rules), so that the evolution may rewrite itself. This formalizes a common engineering reality in adaptive agents and learning systems [19], and it resonates with Hyperseed's emphasis that a* mind *is a self-updating pattern/process web rather than a static object (Hyperseed-Concept 111, 168).*

Two simple examples:

- *A neural agent whose state includes both (i) a working memory buffer and (ii) its current weights; the update map performs inference/action and then gradient descent on the weights.*
- *A program synthesis agent whose state includes a codebase; the update map runs tests, edits code, and deploys the edited code into the next step's execution.*

This definition is useful because it makes self-modification amenable to the same invariants and classification tools used for ordinary dynamical systems: one can ask about fixed points, cycles, chaos, and basins, but now these phenomena may occur at the level of "ways of thinking" rather than merely sensory-motor variables.

Definition 609 (Abstraction as quotienting)**.** *An* abstraction *of a state space X is a surjection*

$$q : X \to A$$

to a macro-space A, inducing an equivalence relation $x \sim_q y \iff q(x) = q(y)$. Intuitively, q collapses micro-distinctions and retains only macro-features.

Remark 3468. *Here q is the formal "act of forgetting." By demanding that q be a surjection, we insist that every macro-state $a \in A$ is realized by at least one micro-state $x \in X$; one is not merely renaming, but genuinely grouping many micro-possibilities under one macro-description. The induced relation $\sim_q$ is an* equivalence relation *(Hyperseed-Concept* **??***): it partitions X into fibers $q^{-1}(a)$, each fiber being a class of micro-states regarded as "the same" for the purposes of the macro-description.*

A simple example is to let $X = \mathbb{R}^2$ with coordinates (u, v) and define $q(u, v) = u$; then $A = \mathbb{R}$ and all points with the same u-coordinate are equivalent. A more mind-like example is: X contains the full internal memory of an agent, but q records only the agent's declared "current goal" variable; then many different memories collapse to one macro goal state. This definition is necessary because the universality claims later are not about individual micro-trajectories; they are about what remains stable when one moves between levels of description (Hyperseed-Concept 51, 86) [1].

Definition 610 (Exact dynamical abstraction (lumpability / congruence))**.** *Given (X, F) and an abstraction $q : X \to A$, we say q is an* exact dynamical abstraction *if for all $x, y \in X$,*

$$q(x) = q(y) \implies q(F(x)) = q(F(y)).$$

Remark 3469. *The condition says: if two micro-states are indistinguishable at the macro level* now*, then after one micro update, they are still indistinguishable at the macro level. In other words, the macro description is not merely a summary of micro-states; it is a summary that is* closed under time evolution*. This is essentially the notion of a congruence for dynamical systems, and is closely related to "lumpability" in Markov processes (where one requires the aggregated chain to be Markov).*

Example: suppose a micro-state is (m, g) where m is memory and g is a goal parameter, and let $q(m, g) = g$. Then q is an exact dynamical abstraction precisely when the goal update $g \mapsto g'$ depends only on g, not on the detailed memory m. This is a strong requirement, but it is the clean baseline case from which approximate notions (not treated in this section) can be developed; and it is exactly what is needed to speak rigorously about an induced macro dynamics.

Theorem 213 (Induced macro-dynamics exists and is unique)**.** *If $q : X \to A$ is an exact dynamical abstraction for (X, F), then there exists a unique map $G : A \to A$ such that*

$$q \circ F = G \circ q.$$

Moreover, G is explicitly given by

$$G(a) := q(F(x)) \quad \text{for any } x \in X \text{ with } q(x) = a,$$

and this is well-defined.

Proof. Because q is surjective, for each $a \in A$ choose some $x \in X$ with $q(x) = a$ and define $G(a) := q(F(x))$. To see this does not depend on the choice of representative, let x' be another micro-state with $q(x') = a$. Then $q(x) = q(x')$, and exactness implies $q(F(x)) = q(F(x'))$, hence $G(a)$ is well-defined.

For the intertwining identity, take any $x \in X$ and set $a = q(x)$. By definition of G, $(G \circ q)(x) = G(a) = q(F(x)) = (q \circ F)(x)$, so $q \circ F = G \circ q$ as maps. For uniqueness, suppose $G_1, G_2 : A \to A$ both satisfy $q \circ F = G_i \circ q$. For any $a \in A$ pick x with $q(x) = a$; then $G_1(a) = G_1(q(x)) = (q \circ F)(x) = G_2(q(x)) = G_2(a)$, so $G_1 = G_2$. □

Remark 3470. *The equation $q \circ F = G \circ q$ is the formal statement that the macro-dynamics is a* factor *of the micro-dynamics: evolving one step and then forgetting yields the same result as forgetting first and then evolving under the induced macro update. Diagrammatically, the theorem asserts the existence of a commutative square*

$$\begin{array}{ccc} X & \xrightarrow{F} & X \\ \downarrow q & & \downarrow q \\ A & \xrightarrow{G} & A \end{array}$$

which is exactly the structure needed to compare attractors and long-run behavior across levels of description. In particular, the macro-space A can be viewed as the set of equivalence classes $X/{\sim_q}$, and G is the update rule on these classes induced by F.

Corollary 70 (Compatibility with iterates)**.** *Under the assumptions of the theorem, for all t,*

$$q \circ F^t = G^t \circ q.$$

Remark 3471. *This follows by a straightforward induction using $q \circ F = G \circ q$. Conceptually, it means that macroscopic predictions over many steps can be computed entirely within the coarse-grained dynamics G, without reference to micro-level details,* provided *the abstraction is exact. Consequently, any macro property definable purely in terms of the iterates (e.g. being a fixed point, lying on a cycle, or being eventually periodic) can be studied at the level of G.*

For example, if $a^\star \in A$ is a fixed point of G (so $G(a^\star) = a^\star$), then for any micro-state x with $q(x) = a^\star$ we have $q(F(x)) = G(q(x)) = a^\star$, i.e. the entire fiber $q^{-1}(a^\star)$ is forward-invariant as a macro-state*. Micro-states in that fiber may still change internally, but in a way that is invisible to the macro description. This distinction (macro stasis vs micro activity) is especially relevant for self-modifying minds: a mind can keep rewriting implementation details while remaining within the same macro "type" according to the chosen quotient q.*

Definition 611 (Hyperseed-style abstraction (informal, as a choice of macro variables))**.** *In the Hyperseed spirit, a* Hyperseed abstraction *is an abstraction map $q : X \to A$ chosen so that the macro-state $a = q(x)$ captures those aspects of a mind-state x that are hypothesized to be most relevant to its long-run self-modification regime (e.g. persistent goal structure, self-model/learning-rule signature, or other relatively stable organizing variables), while deliberately quotienting away many implementation-level details.*

Remark 3472. *This definition is intentionally permissive: the point is not that there is a single canonical q, but that different research questions motivate different coarse-grainings (Hyperseed-Concept 86). Later universality claims can then be read as claims about the induced macro dynamics G on such Hyperseed-style macro-spaces: if many distinct micro realizations of self-modifying agents factor onto macro dynamics exhibiting similar qualitative behavior (fixed-point-like, strange/chaotic, multi-attractor, etc.), then one has evidence for robust "universality classes" that do not depend on low-level substrate. The exactness condition gives a clean mathematical template; in real systems one often expects only approximate closure, but the exact case provides the reference point for defining and measuring approximation error (e.g. how much $q(F(x))$ varies within a fiber $q^{-1}(a)$).*

Remark 3473. *This theorem says that an exact abstraction is not just a descriptive compression; it is a* homomorphism of dynamics*. The equation $q \circ F = G \circ q$ means: whether you "evolve then abstract" or "abstract then evolve," the result is the same macro-state. One may read this as a commutative diagram, expressing a kind of epistemic sanity: the macro theory genuinely tracks the causal/temporal structure of the micro theory.*

It matters because it lets us treat wildly different micro-implementations as instances of one macro law G. In the later universality discussion, this will be the precise hook on which invariance of attractor type *hangs: if G is shared, then qualitative long-run regimes are shared at the macro level, even if the micro stories differ.*

In particular, the condition $q \circ F = G \circ q$ is stronger than mere predictive fit "on average" or in distribution: it asserts step-by-step consistency of the induced time evolution after coarse-graining. Equivalently, q is a structure-preserving map from the micro dynamical system (X, F) to the macro dynamical system (A, G), where typically one takes A to be (at least) the image $q(X)$ so that every macro-state corresponds to some realizable micro-state.

In the self-modifying-mind setting, the same statement can be read operationally: let x encode a full micro-description (e.g. weights, code, memory, and environment-coupled internal registers), and let $q(x)$ retain only the macro variables we claim are explanatory (e.g. a behavioral policy class, a capability vector, an abstraction of goals, or a "seed" descriptor). Then exactness means the macro description is not merely a lossy summary; it is sufficient to propagate forward in time by G without needing hidden micro details. This is exactly what one wants if the macro description is intended to be portable across implementations, architectures, or substrates.

The commutativity also clarifies what kinds of invariances are and are not being claimed. It does not say micro-trajectories are identical; it says their macro *projections coincide whenever they start in the same macro-state. Thus, the fibers of q can contain diverse micro-states (different internal mechanisms, different realizations of the same high-level algorithm), while still yielding a single deterministic macro evolution governed by G.*

Proof. Define G by $G(q(x)) := q(F(x))$. If $q(x) = q(y)$ then by exact dynamical abstraction $q(F(x)) = q(F(y))$, hence the definition does not depend on the choice of representative in the fiber $q^{-1}(a)$. Uniqueness follows because any G satisfying $q \circ F = G \circ q$ must satisfy $G(q(x)) = q(F(x))$ for all x.

One may also view this as the standard quotient-dynamics construction: exactness ensures that the equivalence relation $x \sim y \iff q(x) = q(y)$ is respected by the update map F, i.e. $x \sim y$ implies $F(x) \sim F(y)$. Under that invariance, the induced map on equivalence classes is well-defined, and is precisely G. If one prefers to be explicit about domains, G is thereby defined on the set of realizable macro-states $A := q(X)$; if a larger macro state space is introduced, then G is fixed on $q(X)$ by the commuting equation and may be extended arbitrarily off-image without affecting the abstraction claim. □

Remark 3474. Proof sketch. *The strategy is to define G by "push F through q": declare the macro update of a macro-state a to be whatever you get by choosing any micro representative of a, evolving it by F, and then re-abstracting. Exactness of the abstraction is precisely what guarantees this does not depend on which representative you picked, i.e. that G is well-defined. Uniqueness then follows because the commuting equation forces G to take that value on every $a = q(x)$.* □

A helpful image is to think of each fiber $q^{-1}(a)$ as a "cloud" of micro-states representing one macro-state; exactness means F maps each cloud wholly into a single cloud. Then G simply names which cloud you land in.

This "cloud-to-cloud" picture is also a clean way to see why approximate abstractions can be brittle: if F splits *a cloud into multiple successor clouds, then the macro update is no longer single-valued and one must either (i) refine the macro variables (making fibers smaller), (ii) accept a stochastic or set-valued macro dynamics, or (iii) restrict attention to regimes where the splitting is negligible. The theorem isolates the exact case where none of these extra choices are required.*

For later use in discussing long-run regimes, note that exactness implies that macro trajectories are the images of micro trajectories under q: for any $n \geq 0$, repeated application yields $q(F^n(x)) =$

$G^n(q(x))$. Thus macro attractors (fixed points, cycles, chaotic invariant sets, or collections of multiple attractors with basins) are the natural coarse-grained shadows of micro invariant behavior. While micro-level invariant sets can be more intricate, the theorem guarantees that whatever macro attractor structure is expressed by G is consistently realized by every micro system sharing this abstraction.

In the specific "universality class" framing, this is the mechanism by which one can say that many distinct micro self-modification rules are "the same kind of mind" at the macro level: differences within a fiber are treated as implementation details, while the induced G is what carries the classification-relevant dynamical signature.

Remark 3475. *This theorem is the formal core of "coarse-graining": many different micro-architectures can be placed in the same class if they admit abstractions to a shared macro-dynamics G.*

Said differently, an exact abstraction q defines a criterion for when two micro theories are identical as far as the macro variables are concerned*: they induce the same G on the same macro state space (or on corresponding macro state spaces under an isomorphism). This is why the statement is useful for universality: once G is fixed, the qualitative taxonomy of asymptotic behavior can be performed at the macro level, and then imported back to every micro realization that factors through that G.*

This also highlights what is being held constant in a universality class. The point is not that the micro update F is unique, but that the induced macro update G is; the fibers encode the permitted implementation variance. In the later "fixed-point vs strange vs multi-attractor" discussion, the relevant invariants are those definable purely in terms of the macro flow generated by G (e.g. whether typical trajectories converge to a single attractor, wander on a chaotic set, or end in one of several competing attractors with nontrivial basins). Exactness is what ensures that such statements are well-posed at the chosen macro resolution, rather than being artifacts of untracked micro degrees of freedom.

65.2 Attractor types and invariance under coarse-graining

We use a standard metric notion of attraction.

Definition 612 (Attractor)**.** *Let (X, F) be a dynamical system on a metric space (X, d). A compact invariant set $K \subseteq X$ is an* attractor *if there exists an open set $U \supseteq K$ such that for all $x \in U$,*

$$\mathrm{dist}(F^t(x), K) \to 0 \quad \textit{as } t \to \infty,$$

where $\mathrm{dist}(x, K) = \inf_{y \in K} d(x, y)$*.*

Remark 3476. *Notation:* $\mathrm{dist}(x, K)$ *is the usual point-to-set distance, and compactness of K is a standard regularity assumption ensuring the limit notion behaves well. Invariance means $F(K) = K$ (equivalently $F(x) \in K$ whenever $x \in K$). The open set U is a* basin neighborhood*: all initial states in U are pulled toward K in the sense that their distance to K tends to zero.*

Intuitively, an attractor is a long-run "fate" shared by many nearby initial conditions, a notion closely aligned with predictive attraction (Hyperseed-Concept **??***): once you are in its basin, the future becomes increasingly constrained. For example, a stable fixed point x^* is an attractor with $K = \{x^*\}$; a stable limit cycle (not highlighted in our coarse taxonomy) would be an attractor where K is a periodic orbit. In self-modifying minds, attractors may correspond to stable "meta-policies" or stable "self-model + goal" configurations, not merely to stable sensorimotor states.*

Definition 613 (Three attractor types (coarse taxonomy)). *We say an attractor K is:*

- Fixed-point type *if $K = \{x^*\}$ with $F(x^*) = x^*$ and x^* attracts a neighborhood.*
- Multi-attractor type *if the system admits at least two disjoint attractors K_1, K_2 with disjoint basins of attraction of positive "size" (e.g. nonempty open subsets).*
- Strange-attractor type *if K supports sustained nonperiodic dynamics with sensitive dependence; for concreteness, assume $F|_K$ has positive topological entropy (or contains a subsystem topologically conjugate to a full shift on two symbols).*

Remark 3477. *This is intentionally a coarse-grained taxonomy: it collapses many classical distinctions (cycles vs tori, quasi-periodic vs chaotic, etc.) into three regimes that are especially relevant for reasoning about self-modification.*

Examples to keep in mind:

- *Fixed-point type: a mind converges to a stable "reflective equilibrium" in which its goal system and self-model stop changing except for small fluctuations (Hyperseed-Concept 166, 110).*
- *Multi-attractor type: a mind has distinct stable modes (e.g. "exploration" vs "exploitation," or "prosocial" vs "antisocial" equilibria) and which mode it ends in depends on early contingencies or perturbations; this resonates with phase-like regimes in complex adaptive systems.*
- *Strange-attractor type: bounded but never-settling dynamics, where small differences in initial state eventually lead to macroscopically different trajectories, yet the motion remains confined to an invariant set; this captures a kind of perpetual novelty generation without divergence.*

The usefulness of specifying strangeness via topological entropy (or a shift subsystem) is that it gives a rigorous certificate of complexity: one can import standard results from symbolic dynamics and factor maps.

Definition 614 (Quotient metric (one convenient choice)). *Given a metric (X, d) and a surjection $q : X \to A$, define a pseudo-metric on A by*

$$d_A(a, b) := \inf\{d(x, y) : q(x) = a,\ q(y) = b\}.$$

Assume we mod out any $a \neq b$ with $d_A(a, b) = 0$ so that (A, d_A) is a metric space.

Remark 3478. *This construction turns the macro space A into a metric space by declaring two macro-states close if there exist* some *micro representatives that are close. The infimum (rather than minimum) appears because the closest representatives may not exist, only be approximable.*

The warning about a pseudo-metric is important: it may happen that $a \neq b$ but $d_A(a, b) = 0$ if the fibers $q^{-1}(a)$ and $q^{-1}(b)$ can be made arbitrarily close in X. The text's "mod out" clause means we identify such macro-states to restore the separation axiom of a metric. In practice, one may equivalently replace A by the metric quotient induced by d_A.

A concrete geometric picture: take X to be the unit circle with the usual arc-length metric, and let q identify antipodal points. Then A is a projective circle; d_A computes distance between equivalence classes by choosing the closest representatives among the two antipodal pairs.

Theorem 214 (Attractors project under exact abstractions). *Let (X, F) be a dynamical system on a metric space, and let $q : X \to A$ be an exact dynamical abstraction inducing $G : A \to A$ with $q \circ F = G \circ q$. If $K \subseteq X$ is an attractor of F with basin neighborhood U, then $q(K)$ is an attractor of G and $q(U)$ is contained in its basin.*

Remark 3479. *In plain language: coarse-graining cannot destroy attraction. If many micro-states are pulled toward a micro-attractor K, then their macro-images are pulled toward the macro-image $q(K)$. This is the technical backbone for later claims that "attractor* type *is invariant under coarse-graining": one cannot have a stable macro fixed point if the macro system were not actually being attracted to something. This connects directly to the induced dynamics theorem: $q \circ F = G \circ q$ says q is a factor map. The present theorem then says attractors behave well under factor maps. For self-modifying minds, this means that if we choose macro-variables thoughtfully (in a way that is dynamically compatible), then long-run regimes are robust facts about the system, not artifacts of microscopic description. In other words, the "same" behavior is being described in two coordinate systems: the micro description on X and the macro description on A, with q ensuring that time evolution commutes with the change of variables. This is the sense in which coarse-graining is not merely an observational convenience but a structure-preserving map between dynamical systems, so statements about asymptotic regimes can be transported across levels.*

Proof. Invariance: if $x \in K$ then $F(x) \in K$, hence

$$G(q(x)) = q(F(x)) \in q(K),$$

so $G(q(K)) \subseteq q(K)$. Surjectivity of $q|_K$ implies equality. Indeed, if $b \in q(K)$ then $b = q(x)$ for some $x \in K$, and since K is F-invariant we have $F(x) \in K$, so $G(b) = G(q(x)) = q(F(x)) \in q(K)$; this shows forward invariance. Conversely, for any $c \in q(K)$ choose $z \in K$ with $q(z) = c$; by surjectivity of $q|_K$ and invariance of K, every point in $q(K)$ has a preimage inside K, and the image set $G(q(K))$ cannot be a strict subset of $q(K)$ once all macro points are realized by micro points whose forward images remain in K.

Attraction: take any $x \in U$ and let $a = q(x)$. Because K attracts U, $\mathrm{dist}(F^t(x), K) \to 0$. For each t, pick $y_t \in K$ with $d(F^t(x), y_t) \leq \mathrm{dist}(F^t(x), K) + 1/t$. Then

$$d_A(G^t(a), q(K)) \leq d_A(q(F^t(x)), q(y_t)) \leq d(F^t(x), y_t),$$

using the definition of d_A as an infimum. The right-hand side tends to 0, hence $G^t(a)$ approaches $q(K)$. Thus $q(K)$ attracts $q(U)$. Concretely, the role of the auxiliary points y_t is to "witness" closeness to K at each time: even if a minimizer for $\mathrm{dist}(F^t(x), K)$ does not exist (e.g. K not compact), the $1/t$ slack produces a sequence with asymptotically optimal distance. The quotient metric (or pseudometric) d_A is then designed so that identifying microstates cannot increase the distance between their equivalence classes: the best choice of representatives bounds the distance upstairs, which is exactly what the inequality $d_A(q(u), q(v)) \leq d(u, v)$ encodes. This ensures that convergence in X toward K is strong enough to imply convergence in A toward $q(K)$ along the factor-related trajectories. $\square$

Remark 3480. Proof sketch. *The key idea is that q transports trajectories: the macro trajectory of $a = q(x)$ is exactly $G^t(a) = q(F^t(x))$. To show attraction, approximate the distance from $F^t(x)$ to K by choosing a near-closest point $y_t \in K$, and then use the definition of the quotient metric d_A to bound the macro distance from $q(F^t(x))$ to $q(y_t)$ by the micro distance from $F^t(x)$ to y_t. Since the latter goes to 0, so does the former. In this sketch, the commuting relation $q \circ F^t = G^t \circ q$ (valid for all integers $t \geq 0$ by induction) is doing all the conceptual work: it guarantees that the macro evolution is not an approximation of the micro evolution, but literally its image under q at each time step.* $\square$

One can visualize the proof as "shadowing" the micro orbit by a point on the attractor and then projecting: micro convergence implies macro convergence because the quotient metric is designed to be no larger than the best-case micro separation between representatives. Equivalently, q may *collapse many microstates into a single macrostate, but it cannot separate two microstates that are already close; therefore any eventual trapping near K must manifest as eventual trapping near $q(K)$.*

Corollary 71 (Attractor type is a coarse-graining invariant)**.** *Under the assumptions of the previous theorem, if the macro-attractor $q(K)$ is (i) fixed-point, (ii) strange, or (iii) one of multiple distinct attractors, then the micro-system belongs to the corresponding attractor-type class at that abstraction level. More precisely, the micro dynamics restricted to K must realize the same qualitative regime classification once viewed through the same coarse-graining: a macro fixed point must be supported by micro behavior that projects to a fixed point; a macro strange attractor reflects genuine orbit-complexity in $F|_K$; and distinct macro attractors correspond to distinct micro invariant attracting sets whose images are disjoint in A.*

Remark 3481. *This corollary is the promised qualitative punchline: once the abstraction is dynamically exact, the macro taxonomy (fixed-point vs strange vs multi-attractor) is not merely a story we tell; it is enforced by the micro system because q is a factor map. In particular, macroscopic chaos cannot be an illusion created by projecting a simple micro-dynamics; a factor map cannot create topological entropy out of nothing. Intuitively, coarse-graining can destroy information (merging distinguishable micro histories), but the commuting condition prevents it from fabricating new distinguishable macro histories that were not already present in the micro system. Said differently, any apparent "branching" of macro futures must come from some genuine branching of micro futures, because every macro orbit segment has at least one micro lift consistent with the dynamics.*

Proof. Fixed-point and multi-attractor claims follow directly from projection and invariance. For strangeness: if $G|_{q(K)}$ has positive entropy (or a full shift factor), then $F|_K$ has at least that much complexity in the quotient sense because q is a factor map ($q \circ F = G \circ q$). In particular, F cannot be topologically simple if G is chaotic. Here one uses the standard monotonicity of topological entropy under factors: if (Y, H) is a factor of (X, T) via a continuous surjection π with $\pi \circ T = H \circ \pi$, then $h_{\text{top}}(H) \leq h_{\text{top}}(T)$. Taking $(X, T) = (K, F|_K)$, $(Y, H) = (q(K), G|_{q(K)})$, and $\pi = q|_K$ yields the claim. Likewise, if $G|_{q(K)}$ admits a subshift of finite type (or the full shift on $m \geq 2$ symbols) as a factor, then composing factor maps shows that the same symbolic subsystem is a factor of $F|_K$; thus the micro system must support exponentially many distinguishable orbit segments at the relevant scales. □

Remark 3482. Proof sketch. *Fixed points and multiple basins are preserved because invariant attracting sets project to invariant attracting sets, and disjointness of basins survives under a surjective projection. For strangeness, the crucial observation is categorical: q is a factor map, so symbolic complexity (e.g. a shift subsystem) seen at the macro level must be realizable by some micro orbits that map onto those symbol sequences. Hence the micro dynamics cannot be simpler than the macro dynamics in the sense relevant to topological entropy. One way to phrase the basin argument is: if U_1, U_2 are disjoint open basins in X attracted to distinct invariant sets K_1, K_2, then $q(U_1), q(U_2)$ remain disjoint in A provided q does not identify points across basins; under the theorem's hypotheses (and the use of $q|_{K_i}$ surjective onto $q(K_i)$), the images of distinct attractors remain distinct, so multiple-attractor structure is visible at the macro level.* □

A useful mental model is: projection may hide details, but it cannot invent genuinely new branching structure in the space of distinguishable orbits if the projection respects time evolution. In the self-modifying-mind setting, this means that if the macro description exhibits persistent switching, sensitive dependence, or multiple stable "modes" of cognition/behavior, then the underlying micro update rule must already contain the resources to support that long-run organization; coarse-graining can obscure mechanism, but it cannot conjure the regime itself.

65.3 Universality classes: equivalence under shared macro-dynamics

Definition 615 (Universality class at abstraction level q)**.** *Fix a macro-space A and a macro-map $G : A \to A$. The* universality class $\mathcal{U}(G)$ *is the class of all self-modifying minds (X, F) for which there exists an abstraction $q : X \to A$ that is an exact dynamical abstraction and induces the same G: $q \circ F = G \circ q$.*

Equivalently, q determines a "macro-equivalence" relation on microstates by $x \sim x'$ iff $q(x) = q(x')$, and the condition $q \circ F = G \circ q$ says that F respects this partition so that the induced quotient dynamics on the set of equivalence classes is exactly G. In this sense G is not merely an observational summary but a genuine factor of the micro-dynamics.

Remark 3483. *The phrase "universality class" is borrowed from statistical physics, where many micro-systems share the same critical exponents because coarse-graining flows to the same renormalization fixed point. Here the analogy is simpler but still philosophically potent: a mind can be implemented by many different micro architectures, yet if there is an exact quotient that yields the same macro law G, then for many purposes the implementations are interchangeable at that level of description [18].*

A concrete example: let A encode a small set of meta-states such as "goal update mode," "learning rate regime," and "exploration temperature." If two different agents (different neural nets, different code) admit abstractions that land in the same A and follow the same G, then they are in the same universality class. This is useful precisely because it gives a rigorous bridge from implementation plurality *to* theory unity *(Hyperseed-Concept 136, 51).*

Two clarifications are often helpful when reading the definition. First, q need not be injective: the fibers $q^{-1}(a)$ may contain many micro-configurations realizing the same macro description $a \in A$ (e.g. many neural weight matrices yielding the same effective learning-rate regime). Second, in the strongest reading one typically takes q to be surjective onto the macro-space one intends to model; otherwise one is really studying the induced dynamics on the reachable subset $q(X) \subseteq A$ with G restricted to that subset. The present definition can be read either way, but the motivating interpretation is that A is the intended macro state space and q realizes it as a genuine quotient of the micro system.

Finally, note that exactness matters: if $q \circ F$ merely approximately *agrees with $G \circ q$, then "same G" no longer rigidly determines the set of macro attractors, and the subsequent theorem becomes a stability statement rather than an equivalence statement. This section deliberately keeps the exact case as the baseline, because it cleanly isolates what is invariant* by definition *across implementations.*

Theorem 215 (Universality: same macro-map implies same macro attractor types)**.** *If (X_1, F_1) and (X_2, F_2) are in the same universality class $\mathcal{U}(G)$, then:*

- *every attractor of G lifts to (at least) one attractor in each micro-system whose projection is that macro-attractor; and*
- *macro attractor type (fixed-point vs strange vs multi-attractor) is shared across all members of $\mathcal{U}(G)$.*

Moreover, the lifting in the first bullet need not be unique: distinct micro-attractors (e.g. differing in within-fiber structure) may project to the same macro-attractor, so the theorem should be read as a guarantee of realizability *rather than a one-to-one correspondence.*

Remark 3484. *What this asserts is a disciplined form of "many roads lead to the same fate": if two self-modifying minds share the same induced macro dynamics, then they share the same menu* *of macro-level long-run regimes. The first bullet says macro attractors are not empty abstractions; each macro attractor is realized by at least one micro invariant attracting set mapping onto it. The second bullet then says the coarse taxonomy of fates is a property of the macro law G, hence common across the class.*

This connects backward to the factor-map attractor projection theorem, and forward to the canonical classes below: we will give conditions ensuring G is contractive, chaotic, or multi-well, thereby placing broad families of architectures into one of three archetypal universality classes.

One can also interpret the theorem as specifying which aspects of asymptotic behavior are "coordinate-free" with respect to implementation: if two agents differ only by replacing one micro-realization of the same macro factor G by another, then any statement that is formulable purely in terms of G (existence of fixed points, number of basins at the macro level, presence of a strange attractor as defined on A, etc.) is invariant across the class. Conversely, any phenomenon that depends on fiber geometry (e.g. how many micro states realize a given macro mode, or how rapidly trajectories mix within *a fiber) is not fixed by universality class membership and may vary widely even when G is identical.*

Proof. For each $i \in \{1, 2\}$ let $q_i : X_i \to A$ satisfy $q_i \circ F_i = G \circ q_i$. By the previous section, attractors in each (X_i, F_i) project to attractors of G. Conversely, for any macro-attractor $K \subseteq A$, consider the restricted micro-dynamics on the closed set $q_i^{-1}(K)$; invariance follows from $q_i \circ F_i = G \circ q_i$. Standard compactness arguments (e.g. existence of omega-limit sets) yield at least one invariant compact set in $q_i^{-1}(K)$ projecting onto K. Type-sharing is immediate because the type is determined at the macro level by G.

To make the "standard compactness arguments" more explicit: assuming the usual setting in which X_i is a compact (or at least forward-invariant, precompact) metric space and F_i is continuous, each nonempty closed invariant set contains a nonempty compact invariant ω-limit set. Pick a point $a \in A$ in the basin of attraction of K under G with $\omega_G(a) = K$ (for an attractor K such points exist by definition). Choose any $x \in X_i$ with $q_i(x) = a$; then semiconjugacy implies

$$q_i\big(\omega_{F_i}(x)\big) = \omega_G\big(q_i(x)\big) = \omega_G(a) = K,$$

where the equality uses continuity of q_i and the intertwining $q_i \circ F_i = G \circ q_i$ to commute ω-limits with projection. In particular, $\omega_{F_i}(x) \subseteq q_i^{-1}(K)$ is a compact invariant set whose image is exactly K. If one is working with a specific definition of "attractor" requiring attraction of a neighborhood, one can take the attracting set generated by $\omega_{F_i}(x)$ inside $q_i^{-1}(K)$; the key point is that at least one micro-level recurrent/invariant set exists whose macro shadow is precisely K.

Finally, since the classification "fixed-point vs strange vs multi-attractor" is defined by properties of G on A (e.g. how many attractors exist and what qualitative type they have), any two systems inducing the same G necessarily share that macro classification. Any remaining degrees of freedom live in the fiber dynamics and do not alter the macro taxonomy. □

Remark 3485. Proof sketch. *Projection of micro attractors to macro attractors is already established via factor-map reasoning. To "lift" a macro attractor K, restrict attention to the invariant preimage $q_i^{-1}(K)$ and take omega-limit sets of points in that preimage; compactness-style arguments then give an invariant compact set upstairs whose image is K. Finally, since the taxonomy is defined in terms of the macro map G, any two systems inducing the same G necessarily share the macro type.* □

The geometric intuition is that $q_i^{-1}(K)$ is an invariant "tube" above K; within that tube the micro dynamics must have some recurrent/invariant behavior, and at least one such invariant set projects onto the whole of K.

It is also useful to picture this tube as a bundle of fibers over K: for each $a \in K$, the fiber $q_i^{-1}(a)$ may support its own internal contraction, mixing, or even additional micro-attractors, but the projection erases those differences. Universality class membership therefore identifies what is common after modding out by within-fiber degrees of freedom, and the theorem formalizes that the long-run regimes visible at the macro resolution are fixed entirely by G.

65.4 Canonical macro-classes motivated by the Hyperseed-adjacent documents

This section shows that several macro-maps G of interest arise naturally from the attached frameworks, yielding three canonical universality classes.

65.4.1 Fixed-point universality from goal-contraction metagoals

Definition 616 (Goal-state contraction regime)**.** *Let $(\mathcal{G}, d)$ be a complete metric space of possible goal-configurations. A self-modifying mind produces a goal trajectory $G_t \in \mathcal{G}$ at times $t \in \{0, N, 2N, \dots\}$. We say it satisfies* goal contraction *with coefficient $c \in (0,1)$ if for all $k \geq 1$,*

$$d(G_{(k+1)N}, G_{kN}) \leq c\, d(G_{kN}, G_{(k-1)N}).$$

Remark 3486. *The symbol $\mathcal{G}$ denotes a space of goal-configurations (one may think of vectors of weights in a utility function, parameters of a preference model, or a structured object encoding priorities), and d is a metric measuring how different two goal states are. Completeness is the standard hypothesis ensuring Cauchy sequences converge, preventing "limit goals" from falling outside the space.*

Intuitively, the condition says that successive changes *in the goal state shrink geometrically: the mind may still revise its goals, but each revision is smaller than the previous by a fixed factor $c < 1$. This is a natural sufficient condition for meta-goal stabilization, and it is closely aligned with fixed point methods used to analyze self-referential or self-modifying goal dynamics [10] (Hyperseed-Concept 110). A toy example: if the mind updates a scalar goal parameter by $G_{(k+1)N} = G_{kN} + \Delta_{k+1}$ with $|\Delta_{k+1}| \leq c\,|\Delta_k|$, then the total drift is finite.*

Theorem 216 (Goal contraction implies convergence of goals (fixed-point behavior))**.** *Assume $(\mathcal{G}, d)$ is complete and the goal trajectory satisfies goal contraction with $c \in (0,1)$. Then $(G_{kN})_{k\geq 0}$ converges to a limit goal $G_\infty \in \mathcal{G}$, and*

$$d(G_{kN}, G_\infty) \leq \frac{c^k}{1-c}\, d(G_N, G_0).$$

Remark 3487. *This result says: if the mind's meta-updates damp themselves in the specific geometric way above, then the mind's goals settle to a definite limit. The bound is explicit: after k macro-steps, the distance to the limit decays at least as fast as c^k up to the factor $1/(1-c)$. In the language of attractors, the goal macro-variable has fixed-point type behavior.*

This is important because it gives a checkable sufficient condition for one of the three canonical universality classes: contractive/fixed-point. It also foreshadows the later interpretation: different micro-architectures can still share goal contraction at the macro level, hence share the same qualitative long-run stabilization regime.

Proof. Let $\Delta_k := d(G_{kN}, G_{(k-1)N})$ for $k \geq 1$. The contraction condition implies $\Delta_{k+1} \leq c\Delta_k$, hence $\Delta_k \leq c^{k-1}\Delta_1$ by induction.

For $m > n$,

$$d(G_{mN}, G_{nN}) \leq \sum_{k=n+1}^{m} d(G_{kN}, G_{(k-1)N}) = \sum_{k=n+1}^{m} \Delta_k \leq \Delta_1 \sum_{k=n+1}^{\infty} c^{k-1} = \frac{c^n}{1-c}\Delta_1.$$

Thus (G_{kN}) is Cauchy; completeness gives convergence to some G_∞. Taking $m \to \infty$ in the inequality yields the stated bound. □

Remark 3488. Proof sketch. *The proof turns the contraction condition on successive increments into a geometric series bound. Define the step sizes Δ_k; they shrink like c^{k-1}. Then any tail distance $d(G_{mN}, G_{nN})$ is bounded by the sum of increments from $n+1$ to m, which is bounded by the tail of a geometric series. Hence the sequence is Cauchy and converges by completeness; letting $m \to \infty$ yields the explicit error bound.* □

Visually: each goal update is a shorter "hop" than the last, so the total path length is finite; in a complete space, a path of finite total length must accumulate at a limit point.

Corollary 72 (Fixed-point universality class from contraction metagoals)**.** *Any self-modifying mind architecture whose macro-variable "goal state" obeys goal contraction (coarse-grained at interval N) lies in a fixed-point universality class at that abstraction level.*

Remark 3489. *This is the first canonical universality class: different implementations may vary in how they represent goals internally, how they compute meta-updates, and how they modify themselves, yet if the induced macro goal dynamics contracts in the above sense, then the macro story is essentially one of convergence to a single limiting goal configuration. In alignment-motivated language, one would say that the system has a structural bias toward goal stabilization rather than runaway drift [10].*

Proof. The macro-dynamics on $\mathcal{G}$ contracts successive increments geometrically, forcing convergence to a fixed limit (a macro fixed point). By the attractor projection theorem, members of the universality class share this fixed-point macro-attractor type. □

Remark 3490. Proof sketch. *Use the previous theorem to conclude the macro goal variable converges to G_∞, i.e. it has a fixed-point attractor at the macro level. Then apply the general coarse-graining/attractor invariance results: any micro system inducing that macro dynamics must share the fixed-point attractor type at that abstraction level.* □

65.4.2 Strange-attractor universality from Lorenz-like macro feedback

We isolate the abstract dynamical fact that produces chaos.

Definition 617 (Robust strange macro-dynamics via a horseshoe)**.** *A continuous map $G : A \to A$ has a* horseshoe *if there exists a compact invariant set $\Lambda \subseteq A$ such that $G|\Lambda$ is topologically conjugate to the full shift on two symbols.*

Remark 3491. *The compactness and invariance assumptions are doing real work: compactness ensures the symbolic encoding lives on a closed, bounded "chaotic core," while invariance ($G(\Lambda) = \Lambda$) guarantees that once an orbit enters Λ it never leaves, so the symbolic dynamics is not merely transient. In many classical settings (e.g. C^1 diffeomorphisms on manifolds), the presence of a horseshoe is also* structurally stable *under small perturbations of the map in an appropriate topology, which is why horseshoes are often treated as robust witnesses of chaos rather than fine-tuned curiosities.*

Remark 3492. *A horseshoe is a classical certificate of chaos: it asserts that within Λ, the dynamics of G is as complicated as the shift map on bi-infinite binary sequences. "Topologically conjugate" means there is a homeomorphism $h : \Lambda \to \{0,1\}^{\mathbb{Z}}$ such that $h \circ G = \sigma \circ h$, where σ is the shift. Thus the macro system literally contains a perfect symbolic computation of binary histories.*

The relevance to self-modifying minds is that certain macro feedback loops (often described informally as "Lorenz-like") generate stretching-and-folding behavior, and the horseshoe captures that behavior in a rigorous way without committing to a particular differential equation.

Theorem 217 (Horseshoe implies strange-attractor type)**.** *If G has a horseshoe Λ, then G has positive topological entropy and exhibits sensitive dependence on initial conditions on Λ. In particular, any attractor containing Λ is of strange-attractor type under our taxonomy.*

Remark 3493. *The phrase "any attractor containing Λ" is meant in the usual dynamical-systems sense: an attractor is an invariant set that attracts a neighborhood (equivalently, has a basin of attraction of nonempty interior in many standard definitions). The point is that if the attracting region contains a horseshoe subset, then the attractor cannot be purely fixed-point or limit-cycle type, because it contains a subsystem whose orbit-complexity is already maximal in the symbolic (shift) sense.*

Remark 3494. *This theorem translates a structural condition (presence of a symbolic shift subsystem) into the qualitative taxonomy term "strange attractor." What is conceptually striking is that one may certify chaos without solving the system: the mere existence of a horseshoe forces exponentially many distinguishable orbit segments and thus forces unpredictability in the strong topological sense.*

Proof. For the full shift on two symbols, the number of distinct length-n symbolic orbits is 2^n. Conjugacy transfers this exponential growth of distinguishable orbits to $G|\Lambda$, giving positive topological entropy. Positive entropy implies sensitive dependence on initial conditions (on compact metric spaces, in standard settings). Hence any invariant set containing Λ supports sustained nonperiodic, sensitive dynamics, i.e. strange-attractor type. □

Remark 3495. *Concretely, the full two-shift has topological entropy* $\log 2$, *and topological conjugacy preserves topological entropy because it preserves the open-cover growth rates (or equivalently, separated-set growth rates) used to define it. Sensitivity can also be seen directly from the shift model: for any symbolic sequence and any cylinder neighborhood (agreement on a long but finite window), one can flip a bit far enough out in the sequence to obtain a nearby initial condition whose orbit eventually disagrees at the present coordinate, forcing a macroscopic separation after some iterate.*

Remark 3496. Proof sketch. *Because $G|\Lambda$ is conjugate to the full two-shift, it inherits the two-shift's exponential growth of distinct orbit segments (2^n words of length n), which is exactly what positive topological entropy measures. Standard implications then yield sensitive dependence. Therefore any attractor that contains Λ contains this chaotic core and is strange in the chosen taxonomy.* □

A visual intuition (the origin of the term "horseshoe") is the repeated stretch-and-fold action: the map elongates a region, folds it back, and overlays it, creating a binary branching of itineraries.

Corollary 73 (Strange-attractor universality class from coarse-grained Lorenz-like dynamics)**.** *Suppose a family of self-modifying minds admits abstractions $q_i : X_i \to A$ inducing the same macro-map G and G has a horseshoe (or other robust chaos witness). Then the entire family lies in a strange-attractor universality class at that abstraction level.*

Remark 3497. *This is the second canonical universality class: if the shared macro dynamics contains a robust chaos witness, then micro architectural variation cannot remove that chaos at the macro level, because the chaos is a property of G and G is shared. In informal cognitive terms, this describes minds whose long-run self-modification dynamics remain bounded yet perpetually novel, with persistent sensitivity to early conditions and perturbations.*

Remark 3498. *Operationally, the hypothesis "induces the same macro-map G" can be read as: after coarse-graining by q_i, each system's macro evolution is well-approximated (or exactly represented, depending on the formal setup) by iterating G. The horseshoe then acts as an invariant, representation-independent obstruction to reducing the macro behavior to a single fixed point or a simple cycle: even if microscopic mechanisms differ, any member that truly instantiates the same G must instantiate the same symbolic shift subsystem.*

Proof. Immediate from the universality theorem: macro chaos is shared across the class, hence the macro attractor type is strange for all members. ☐

Remark 3499. Proof sketch. *Apply the universality theorem with the shared G; since G is strange-attractor type, all systems in the class share that macro type.* ☐

65.4.3 Multi-attractor universality from multi-well macro potentials

Definition 618 (Gradient-like macro dynamics)**.** *Let $A \subseteq \mathrm{R}^n$ and let $V : A \to \mathrm{R}$ be C^1. Consider a discrete-time macro update*

$$a_{t+1} = a_t - \eta \nabla V(a_t),$$

with step size $\eta > 0$ small enough that $V(a_{t+1}) \leq V(a_t)$ whenever a_t is not critical.

Remark 3500. *A sufficient (though not necessary) condition guaranteeing the displayed monotonic decrease is the standard smoothness/Lipschitz-gradient assumption: if ∇V is L-Lipschitz on the region of interest, then choosing $\eta \in (0, 2/L)$ yields $V(a_{t+1}) \leq V(a_t)$, with strict inequality away from critical points. This connects the abstract "small enough" hypothesis to a checkable analytic condition and emphasizes that the definition is meant to capture a broad class of dissipative, Lyapunov-decreasing macro updates rather than a single algorithmic choice.*

Remark 3501. *This is the archetype of "downhill motion" in a potential landscape: the macro state a_t moves in the direction of steepest descent of V. The condition on η ensures the potential does not increase along the update (a standard discrete-time analogue of energy dissipation). In optimization language, this is gradient descent; in dynamical language, it is a dissipative system tending toward critical points.*

For self-modifying minds, a_t could represent a coarse-grained meta-parameter vector (e.g. high-level goal weights, trust/coordination parameters, or exploration/exploitation balance), and V could represent a Lyapunov-like functional measuring inconsistency, regret, or some macro "tension" to be relaxed (Hyperseed-Concept 100, 87).

Definition 619 (Basin of attraction (discrete time))**.** *Let $G : A \to A$ and let $S \subseteq A$ be invariant. The* basin of attraction *of S (when it exists in the intended sense) is the set of initial conditions $a_0 \in A$ such that the forward orbit approaches S, e.g.* $\mathrm{dist}(a_t, S) \to 0$ *as $t \to \infty$. A system is* multi-attractor type *if it has two or more attracting invariant sets with basins that are not identical (typically, basins separated by stable manifolds or other boundary sets).*

Theorem 218 (Multiple strict local minima yield multiple attractors)**.** *Assume V has at least two strict local minima $m_1 \neq m_2$ with disjoint neighborhoods U_1, U_2 such that $V(x) > V(m_j)$ for all $x \in U_j \setminus \{m_j\}$. For sufficiently small η, each m_j is an attracting fixed point of the gradient-like macro dynamics, and the system is multi-attractor type.*

Proof. Each strict local minimum m_j is a critical point, so it is a fixed point of the update: $m_j - \eta\nabla V(m_j) = m_j$. By strict local minimality, there exists a neighborhood U_j on which $V(x) > V(m_j)$ for $x \neq m_j$. For η small enough to ensure $V(a_{t+1}) \leq V(a_t)$ with strict decrease whenever a_t is not critical, any orbit starting sufficiently close to m_j cannot leave U_j while decreasing V (since leaving would require passing through points of potential no smaller than the boundary values, contradicting monotone descent toward the unique minimum value $V(m_j)$ within U_j). Thus m_j attracts a neighborhood, i.e. it is an attracting fixed point.

Because $m_1 \neq m_2$ and U_1, U_2 are disjoint, the two attracting fixed points have distinct basins (at least locally, U_j is contained in the basin of m_j). Hence the macro dynamics admits multiple coexisting attractors and is multi-attractor type. □

Remark 3502. *The disjoint-neighborhood condition is a clean way to ensure the minima correspond to genuinely distinct macro endpoints rather than different parameterizations of the same asymptotic behavior. In landscape language, the boundaries between basins are the "ridges" or separatrices: sets of initial macro states from which infinitesimal changes can send the system to different long-run self-modification outcomes. This directly motivates the multi-attractor universality class as a coarse-grained analogue of path-dependence: early perturbations can select which well (and thus which stable self-modification regime) the mind ultimately commits to.*

Corollary 74 (Multi-attractor universality class under shared potential landscape)**.** *Suppose a family of self-modifying minds admits abstractions $q_i : X_i \to A$ inducing the same gradient-like macro dynamics for a shared potential V with at least two strict local minima as above. Then, at that abstraction level, the entire family lies in a multi-attractor universality class.*

Proof. All members share the same macro update rule (equivalently, the same map $G(a) = a - \eta\nabla V(a)$), so the existence of multiple attracting fixed points is a property of the shared macro description rather than of micro implementation details. Applying the preceding theorem to this common G yields multi-attractor type for every member at the given abstraction level. □

Remark 3503. *This theorem is the third canonical mechanism: when the macro landscape has multiple "wells," the system settles into one of several stable equilibria depending on where it starts. The conclusion is not merely that there are multiple equilibria, but that they are* attracting *and have disjoint basins of attraction, which is exactly what "multi-attractor type" demands. In particular, "disjoint basins" is a dynamical statement: there exist open sets B_j with $m_j \in B_j$ such that $T^t(x) \to m_j$ for all $x \in B_j$, and $B_i \cap B_j = \varnothing$ for $i \neq j$, so the limiting macro behavior cannot be the same for all initial macro-states. Equivalently, coarse-graining reveals multiple stable macro-level endpoints rather than a single fixed point or a single strange attractor supporting persistent exploration.*

Conceptually, this is the mathematical skeleton underlying phenomena like path dependence, hysteresis, and phase-like regime selection in learning systems: early random fluctuations can commit the system to one long-run mode. One can also interpret the "wells" as distinct internally-consistent self-modification regimes: after the macro-state enters a basin, subsequent self-updates remain confined to that region and converge to the corresponding equilibrium unless perturbed by sufficiently large exogenous shocks.

Proof. At a strict local minimum m_j, $\nabla V(m_j) = 0$ so m_j is a fixed point. Here "strict" ensures m_j is an isolated minimum (not a flat manifold of minimizers), which is what permits a neighborhood to be cleanly assigned to that attractor rather than drifting along a plateau. For η small, the map $T(x) = x - \eta\nabla V(x)$ is a local contraction around m_j whenever the Hessian is positive definite there (standard linearization: eigenvalues of $I - \eta H$ lie in $(0, 1)$ for small enough η). Hence each m_j attracts some neighborhood. More explicitly, if $H = \nabla^2 V(m_j) \succ 0$ has eigenvalues $\lambda_k > 0$, then the eigenvalues of $I - \eta H$ are $1 - \eta\lambda_k$, so choosing η with $0 < \eta < 2/\lambda_{\max}$ makes the linearized map strictly contractive in the norm induced by the eigenbasis; the nonlinear remainder is controlled on a sufficiently small neighborhood by continuity of $\nabla^2 V$. This yields a basin of attraction containing a ball around m_j, i.e. there exists $r_j > 0$ such that $\|x - m_j\| < r_j$ implies $T^t(x) \to m_j$. With two disjoint attracting basins, the macro-system is multi-attractor type. (Disjointness can be ensured, for example, by taking the radii small enough that the corresponding balls do not overlap, which is possible when the minima are distinct.) □

Remark 3504. Proof sketch. *Local minima are fixed points because the gradient vanishes. Near a strict local minimum with positive definite Hessian, the gradient map is approximately linear with Jacobian $I - \eta H$, whose eigenvalues lie strictly inside $(0, 1)$ for sufficiently small η, making the update a local contraction. Thus each minimum attracts a neighborhood, giving at least two disjoint basins and hence multi-attractor type.* □

In the background is the standard contraction-mapping intuition: once iterates enter a region where the update shrinks distances, the orbit forgets fine-grained initial conditions within that region and collapses to the unique fixed point in that neighborhood. The "sufficiently small η" condition can be read as a stability-of-discretization requirement: even if the continuous-time gradient flow $\dot{x} = -\nabla V(x)$ is stable at m_j, overly large discrete steps can overshoot and destroy local attraction.

A geometric picture is to imagine a ball rolling on a surface with two valleys: friction (the monotone decrease of V) prevents it from climbing out, so it ends in whichever valley it starts nearest. In this analogy, the basin boundary corresponds to the separatrix (often passing through saddles) that divides initial conditions by their eventual valley, and the "multi-attractor" claim is precisely that both valleys have nontrivial catchment regions.

Corollary 75 (Multi-attractor universality class from multi-well coarse-graining)**.** *If a family of self-modifying minds coarse-grains to the same multi-well gradient-like macro map, then all members share a multi-attractor macro landscape (at least the same number of basins). Here "at least" is meant in the sense that each attracting macro-equilibrium of the shared map persists as an attracting macro-equilibrium for every member under the common coarse-grained description, so the family cannot collapse to a single-basin macro picture without changing the macro map itself.*

Remark 3505. *This states that multi-stability is a universality property: if the macro map is shared and has multiple wells, then micro implementation details do not remove the existence of multiple basins* at that macro level. *One may view this as a cautionary alignment lesson: if the macro landscape contains multiple stable regimes, ensuring entry into the "right" basin requires more than merely ensuring eventual stability. In particular, two systems can be equally "stable" in the sense of converging to some attractor, yet be qualitatively different in outcomes because they converge to* different *attractors on different initial-condition sets. From the coarse-graining viewpoint, "universality" asserts that once the macro-level update rule G is fixed, the number and qualitative type of macro attractors is a structural property of G rather than an accident of a particular micro-realization.*

Proof. Apply the universality theorem with macro-map G given by the shared gradient-like update. Concretely, since all members induce the same G on macro-states, and G has multiple attracting fixed points with disjoint basins, the universality conclusion transfers this multi-attractor structure to each member's coarse-grained dynamics. □

Remark 3506. Proof sketch. *The macro map G is the same across the family and has multiple attracting fixed points; universality then implies the macro attractor type is shared across the family.* □

One can read this as a robustness statement: changing low-level implementation may alter trajectories in micro-state space, but as long as the coarse-grained macro update is preserved, the large-scale partition of macro initial conditions into distinct long-run regimes is preserved as well.

65.5 Grand Target-1 statement: broad architectures coarse-grain to attractor types

Definition 620 (Exact dynamical abstraction / factor map viewpoint)**.** *An* exact dynamical abstraction *from a micro system (X, F) to a macro system (A, G) is a map $q : X \to A$ such that the diagram commutes:*

$$q \circ F = G \circ q.$$

In dynamical-systems language, q is then a (semi-)conjugacy or factor map*, and (A, G) is a* factor *of (X, F). The interpretation is that q forgets micro-level distinctions while preserving the time-evolution of the retained macro variables exactly. In particular, the macro state $q(x_t)$ evolves as if it were generated directly by iterating G on A, regardless of which micro realizer x_t within the fiber $q^{-1}(a)$ one began with.*

Theorem 219 (Attractor-type universality under Hyperseed-style abstraction)**.** *Let $\{(X_i, F_i)\}_{i\in I}$ be a family of self-modifying minds. Suppose there exists a macro space A and, for each i, an abstraction $q_i : X_i \to A$ that is an exact dynamical abstraction inducing a single macro-map $G : A \to A$ satisfying $q_i \circ F_i = G \circ q_i$.*

Then the family decomposes into universality classes labeled by the attractor type *of G:*

- *If G has a unique globally attracting fixed point, then all (X_i, F_i) are fixed-point type (at that abstraction level). Here "globally attracting" means that for every initial macro state $a \in A$, the iterates $G^t(a)$ converge to the same fixed point $a^\star$, so that all micro initial conditions become indistinguishable under q_i in the long run.*

- *If G admits a robust chaos witness (e.g. a horseshoe), then all (X_i, F_i) are strange-attractor type (at that abstraction level). Robustness is included to ensure the classification is not an artifact of a measure-zero parameter choice: the qualitative presence of sensitive dependence and complicated invariant structure persists under small admissible perturbations of G at the macro level.*

- *If G admits at least two disjoint attractors with disjoint basins, then all (X_i, F_i) are multi-attractor type (at that abstraction level). Disjoint basins guarantee genuine long-run path-dependence at the macro scale: distinct regions of A flow to distinct asymptotic behaviors, so the macro limit set is not determined solely by the update rule but also by the initial macro state.*

In each case, micro-architectural variation within the family does not change the macro attractor type. Equivalently, once the same induced G is fixed, the only remaining variability across i can live inside the fibers of q_i (the "implementation details" that the abstraction discards), and such variation cannot alter the qualitative asymptotic regime as seen in A.

Remark 3507. *This theorem is the "Grand Target-1" consolidation: once Hyperseed-style abstraction is made exact, the macro dynamical regime is a genuine invariant of the family. The result is not claiming that all minds become identical, but that when viewed through the same dynamically compatible quotient, they share the same qualitative long-run form.*

In applications, the hard work is typically to justify the existence of a useful exact (or approximate) abstraction q_i across architectures. Hyperseed's ontology treats this as a central epistemic act: choosing which distinctions matter for the question at hand (Hyperseed-Concept 51, 98) [1]. Once that act is disciplined by dynamical compatibility, the long-run classification becomes mathematically forced rather than interpretive.

It is also worth emphasizing that the universality claim is always relative to the chosen macro space A *and maps* q_i*. A finer abstraction (retaining more distinctions) may yield a different induced macro dynamics and hence a different attractor-type classification, while a coarser abstraction may merge multiple micro attractors into a single macro attractor. The theorem should therefore be read as:* given *an exact shared quotient that induces a single* G*, the attractor-type label is fixed.*

Proof. This is a direct combination of: (i) existence/uniqueness of G under exact dynamical abstraction, (ii) projection of attractors under factor maps, and (iii) the definition of the attractor type as a macro property of G.

For completeness, one can spell out the implication from commutativity to shared asymptotics as follows. Fix i and let $x \in X_i$ be an arbitrary micro initial condition, with macro image $a = q_i(x)$. By iterating the commutation relation, one has $q_i(F_i^t(x)) = G^t(q_i(x)) = G^t(a)$ for all integers $t \geq 0$. Thus, any convergence, recurrence, or basin-membership statement that is formulated purely in terms of the orbit $\{G^t(a)\}_{t\geq 0}$ transfers immediately to the observed macro behavior of the micro orbit under q_i.

When G has a unique globally attracting fixed point $a^\star$, we get $q_i(F_i^t(x)) \to a^\star$ for all $x \in X_i$, which is precisely the fixed-point-type conclusion "at that abstraction level." When G contains a chaos witness such as a horseshoe (in whatever formal setting has been adopted earlier), the same witness is present in the induced macro dynamics for every member of the family because the macro dynamics is literally the same G for all i. Likewise, if G has at least two disjoint attractors with disjoint basins in A, then the dependence of the limit behavior on the initial macro state is shared across the family, since $q_i(F_i^t(x))$ tracks $G^t(q_i(x))$ exactly. □

Remark 3508. Proof sketch. *First, exact dynamical abstraction produces a unique induced macro map G. Second, factor-map arguments show that attraction/invariance structure is respected under projection, so macro attractors correspond to macro images of micro attractors and vice versa in the sense used earlier. Finally, since the three-type taxonomy is defined at the macro level (global fixed point vs chaos witness vs multiple disjoint basins), the classification is shared by all micro systems that induce the same G.* □

One may think of this as a structural form of objectivity: once a family agrees on a macro law G, the coarse qualitative destiny is no longer a matter of perspective; it is a theorem.

A subtlety is that, in many dynamical formalisms, "attractor" can be defined in multiple inequivalent ways (topological, measure-theoretic, pullback, Milnor attractor, etc.). The statement here should be read as parametrized by the adopted attractor notion: whatever definition is used to declare G to have a global fixed point, a chaos witness, or multiple disjoint basins, that declaration is a property of G alone, and therefore is shared across the entire family that factors through G.

Remark 3509 (Interpretation for self-modifying minds)**.** *These results do not claim all minds converge to the* same point*. They claim that, after a well-chosen Hyperseed-style quotient q (collapsing micro implementation differences), broad families of minds can share the same* qualitative *long-run regime: stable fixed point (reflective equilibrium), sustained bounded exploration (strange attractor), or multiple basins (multi-attractor, phase-like regimes). This provides a principled sense in which many distinct architectures can belong to the same "universality class" for reasoning and alignment analysis.*

Concretely, one can regard X_i as a space of internal configurations of architecture i (including self-modification state), F_i as the end-to-end "one cognitive step" update (including rewriting parts of itself), and q_i as a semantics-preserving projection to a macro description such as: a policy class identifier, a value-function summary, a belief-state coarse graining, or an invariant specification of the self-modification protocol. The theorem then says: if these summaries are chosen so that they are dynamically compatible in the strong sense $q_i \circ F_i = G \circ q_i$, then the long-run kind *of macro behavior (single terminal regime, persistent complex wandering, or multiple stable modes with basin structure) is determined entirely by G, and hence is shared across the family.*

In alignment terms, this supports separating (a) the question of identifying the right quotient for the analysis objective (what distinctions matter) from (b) the classification of the resulting macro trajectory types once that quotient is fixed. The former is an epistemic/ontological choice; the latter becomes a dynamical consequence.

66 Selection Implies Trust-Capable Coordination: Explicit Evolutionary Dynamics over Communities and Mindplexes

We formalize a general bridge from (i) "measure-zero" style dominance results for prosocial/trusting coordination over trustless/stability-constrained coordination, to (ii) explicit evolutionary and cultural selection dynamics over communities (and mindplex-like collectives). The goal is to prove that if (in a rich task environment) trust-capable coordination is almost always strictly more efficient than trustless coordination, then a wide family of selection dynamics generically drives a population toward trust-capable coordination, provided it is reachable and not forbidden by hard constraints. The results are phrased so they can be used as inference lemmas inside a reasoning system that models societies, institutions, or multi-agent AI ecologies.

Remark 3510. *Philosophically, the thread here is an old one: if some form of organization can* strictly dominate *another on nearly all circumstances that actually arise, then, barring unusual barriers, time tends to magnify the dominant form. The mathematics below turns this maxim into a sequence of precise implications. In the language of the Hyperseed core-concept set, we are reasoning about how* Tasks *(Hyperseed-Concept 186) and* Society *(Hyperseed-Concept 170) interact under selection-like update rules.*

66.1 Setup

66.1.1 Tasks and regimes

Definition 621 (Task environment)**.** *Let $(\mathcal{A}, \mathcal{F}, \nu)$ be a probability space of task instances. An instance is indexed by $\alpha \in \mathcal{A}$ and has an associated performance objective. We write the attainable expected performance of a given community regime r on task α as $V_r(\alpha) \in \mathbb{R}$.*

Remark 3511. *The notation $(\mathcal{A}, \mathcal{F}, \nu)$ is standard: $\mathcal{A}$ is the set (or measurable space) of tasks, $\mathcal{F}$ is a σ-algebra of measurable task-events, and ν is a probability measure describing how often each kind of task is encountered. Thinking concretely, α might be "build a bridge under constraints,"* *"coordinate a supply chain under shocks," or "train a distributed model under partial observability." The key point is that tasks are not a single benchmark but a* distribution *of situations (Hyperseed-Concept 186).*

The quantity $V_r(\alpha)$ abstracts away the internal details of regime r and reports what matters for selection: how well the regime can do on task α in expectation (e.g. after optimizing internal parameters, averaging over randomness, or averaging over within-regime variation). For example, if r is a mechanism with incentive constraints, then $V_r(\alpha)$ could be its best achievable welfare subject to those constraints on the instance α.

Definition 622 (Trustless vs trust-capable regimes)**.** *Let $\mathcal{R}$ be a set of possible community regimes. We single out:*

- *A* trustless *regime class $\mathcal{R}_{\mathrm{L}} \subseteq \mathcal{R}$, whose feasible coordination procedures must satisfy stability constraints (e.g. incentive compatibility, individual rationality, budget balance, or analogous "strategic robustness" constraints).*
- *A* trust-capable *regime $\mathrm{T} \in \mathcal{R}$, meaning: whenever a trustless procedure is feasible, T can emulate it, and T also has additional low-overhead cooperative procedures available when genuine alignment/trust suffices.*

Remark 3512. *Intuitively, $\mathcal{R}_{\mathrm{L}}$ captures regimes that must behave as though each subagent is a strategic adversary or, at least, that no deep coordination can be presupposed. This is the world of robust contracts, formal mechanism design, and procedures that continue to function even if each participant optimizes only for themselves. By contrast, T is not merely "trusting" in a sentimental sense; it is* capable *of trust: it can operate in the trustless mode when needed, but it can also exploit the extra degrees of freedom available when participants share goals or can reliably commit to joint intent.*

A simple example is a team that can fall back to formal voting or signed contracts (emulating a trustless protocol), but when trust is present can instead use low-overhead delegation, rapid information sharing, and flexible role adaptation. In multi-agent AI terms, T can run a conservative, incentive-compatible protocol, but can also run a high-bandwidth cooperative protocol when alignment holds.

This definition is structurally necessary for the later "inclusion" assumption: without an emulation property, one cannot cleanly assert that trust-capable coordination is never worse, because it might be better on some tasks but fail on others. The emulation clause makes T a superset of feasible behaviors of $\mathcal{R}_{\mathrm{L}}$, aligning with the idea that trust adds capabilities rather than removing them. At the level of social ontology, this concerns regimes of Intersubjective Reality *(Hyperseed-Concept* **??***) more than individual psychology: what matters is what coordination patterns are feasible and stable.*

Assumption 3 (Asymmetry of trust as feasibility-set inclusion)**.** *For every task α,*

$$V_{\mathrm{T}}(\alpha) \;\geq\; \sup_{r \in \mathcal{R}_{\mathrm{L}}} V_r(\alpha).$$

Assumption 4 ((GGW-type) measure-zero strictness)**.** *There exists a measurable function*

$$\Delta(\alpha) \;:=\; V_{\mathrm{T}}(\alpha) \;-\; \sup_{r \in \mathcal{R}_{\mathrm{L}}} V_r(\alpha) \;\geq\; 0$$

such that

$$\nu\big(\{\alpha : \Delta(\alpha) > 0\}\big) \;=\; 1,$$

and Δ is integrable: $\mathbb{E}_{\alpha\sim\nu}[|\Delta(\alpha)|] < \infty$.

Remark 3513. *Assumption 4 is the abstract form of "trustless-friendly tasks are a measure-zero exception." Assumption 3 captures the core structural reason: trust-capable communities can do everything trustless communities can do, plus more.*

Remark 3514. *The label "GGW-type" is meant to evoke the general style of argument that cooperative/prosocial structures are generically more efficient on rich families of problems, with "exceptions" lying in thin or nongeneric subsets of task space; see [6] for an informal but broad discussion of this kind of dominance claim. The integrability condition is a technical way of excluding pathological cases where $\Delta(\alpha)$ is positive almost everywhere but has infinite mean, which would make later expectations ill-defined.*

Note also the convention implicit in $\sup_{r\in\mathcal{R}_{\mathrm{L}}} V_r(\alpha)$*: for each task α we allow the trustless class to pick its best-performing regime. This makes the dominance assumption* harder *to satisfy and therefore strengthens any conclusions drawn from it.*

66.1.2 Selection over community regimes

We model a population of communities. Each community instantiates a regime $r \in \mathcal{R}$. Communities encounter tasks and reproduce (or are imitated) according to achieved performance.

Definition 623 (Expected fitness of a regime)**.** *Define the expected fitness (expected performance) of regime r by*

$$f(r) \;:=\; \mathbb{E}_{\alpha\sim\nu}[V_r(\alpha)].$$

Remark 3515. *Here $f(r)$ is the long-run average score of regime r under the environment ν. The expectation $\mathbb{E}_{\alpha\sim\nu}[\cdot]$ means: sample a random task instance α according to ν, evaluate how well r can do on that instance (via $V_r(\alpha)$), and average.*

A toy example: if $\mathcal{A} = \{\alpha_1, \alpha_2\}$ with probabilities $\nu(\alpha_1) = 0.9$ and $\nu(\alpha_2) = 0.1$, then $f(r) = 0.9V_r(\alpha_1)+0.1V_r(\alpha_2)$. This is useful because many selection dynamics (replicator, imitation, multiplicative weights) can be driven by such an average fitness, either explicitly (deterministic dynamics) or implicitly (stochastic sampling converges to this average).

Remark 3516 (Why expectation is the right object for selection)**.** *The definition of $f(r)$ abstracts away from any particular within-task randomness by treating the task draw $\alpha \sim \nu$ as the primitive source of environmental variation. In typical evolutionary or learning models, what matters for growth rates is precisely a mean performance quantity of this form: even if realized payoffs fluctuate from round to round, laws of large numbers imply empirical averages track $\mathbb{E}_{\alpha\sim\nu}[V_r(\alpha)]$ when tasks are sampled repeatedly from a stationary environment. Thus, inequalities between $f(r)$ translate into persistent differences in long-run population shares under standard update rules.*

66.2 Core bridge: measure-zero dominance implies selection advantage

Theorem 220 (Positive mean advantage from almost-sure strict advantage)**.** *Under Assumption 4,*

$$f(\mathrm{T}) \;>\; \sup_{r\in\mathcal{R}_{\mathrm{L}}} f(r).$$

Remark 3517 (Intuitive content and why it matters)**.** *This theorem says: if trust-capable coordination is* never worse *on any task, and is* strictly better *on almost every task that actually occurs (probability 1 under ν), then trust-capable coordination is also strictly better* on average*. In other words, "almost sure" dominance converts into a positive selection coefficient.*

This matters because selection dynamics generally respond to average *reproductive advantage, not to a pointwise statement. Assumption 4 gives the pointwise/measure-theoretic claim; Theorem 220 turns it into the scalar inequality needed to drive replicator-like models, connecting the dominance premise to the population-dynamics results that follow.*

Remark 3518 (On "almost sure" versus "everywhere")**.** *The hypothesis* $\nu(\Delta > 0) = 1$ *is stronger than merely having* $\mathbb{E}[\Delta] > 0$ *but weaker than requiring* $\Delta(\alpha) > 0$ *for* all $\alpha \in \mathcal{A}$. *This is the natural notion for selection in an environment described by* ν*: tasks outside the support of* ν *(or on* ν*-null sets) are effectively never encountered, so they do not affect expected fitness. Theorem 220 formalizes that "failure to be strictly better" on a* ν*-null set cannot cancel a strict advantage on a full-measure set, provided the integrability conditions ensure the mean is well-defined.*

Proof. Let $g(\alpha) := \sup_{r\in\mathcal{R}_{\mathrm{L}}} V_r(\alpha)$, so that $\Delta(\alpha) = V_{\mathrm{T}}(\alpha) - g(\alpha) \geq 0$ and $\nu(\Delta > 0) = 1$.

Integrability implies $\mathbb{E}[\Delta]$ exists and is finite. Since $\Delta \geq 0$ and $\nu(\Delta > 0) = 1$, we have $\Delta(\alpha) > 0$ on a set of positive ν-measure (indeed full measure), hence $\mathbb{E}[\Delta] > 0$. Therefore

$$\mathbb{E}[V_{\mathrm{T}}] \;=\; \mathbb{E}[g + \Delta] \;=\; \mathbb{E}[g] + \mathbb{E}[\Delta] \;>\; \mathbb{E}[g].$$

Also $\mathbb{E}[g] \geq \sup_{r\in\mathcal{R}_{\mathrm{L}}} \mathbb{E}[V_r]$ by definition of g. Thus $f(\mathrm{T}) > \sup_{r\in\mathcal{R}_{\mathrm{L}}} f(r)$. □

Remark 3519 (Two technical subclaims used in the proof)**.** *The proof relies on two standard facts.*

1. *If* $X \geq 0$ *almost surely and* $\mathbb{P}(X > 0) = 1$*, then* $\mathbb{E}[X] > 0$ *whenever* $\mathbb{E}[X]$ *exists (i.e., is finite). A quick way to see this is: for each* $n \in \mathbb{N}$*, the event* $\{X \geq 1/n\}$ *increases to* $\{X > 0\}$*, so for some* n *one must have* $\mathbb{P}(X \geq 1/n) > 0$*; then* $\mathbb{E}[X] \geq (1/n)\mathbb{P}(X \geq 1/n) > 0$.

2. *For any collection of integrable random variables* $\{Y_r\}_{r\in\mathcal{R}_{\mathrm{L}}}$*, with* $g := \sup_r Y_r$ *(pointwise), one always has* $\mathbb{E}[g] \geq \sup_r \mathbb{E}[Y_r]$*, since* $g \geq Y_r$ *implies* $\mathbb{E}[g] \geq \mathbb{E}[Y_r]$ *for each* r*, and taking the supremum preserves the inequality.*

These are the only measure-theoretic ingredients needed to translate pointwise dominance into a strict inequality between expectations.

Proof sketch. The strategy is to isolate the nonnegative "gap" random variable $\Delta(\alpha)$ between the trust-capable payoff and the best trustless payoff on each task. "Almost surely positive" plus integrability implies the mean of this gap is strictly positive, and then linearity of expectation turns pointwise dominance into strict average dominance. □

Remark 3520. *A useful visual is to imagine the task space* $\mathcal{A}$ *as a landscape weighted by* ν*. The set where* $\Delta(\alpha) = 0$ *has* ν*-measure* 0*, so it contributes nothing to the average; everywhere else, there is some positive contribution. The only way the mean could fail to be positive would be if the gap were positive on a set of measure* 1 *but so wildly non-integrable that the expectation were undefined; this is exactly what the integrability hypothesis excludes.*

Remark 3521 (Supremum versus maximizer and "best trustless regime")**.** *The expression* $\sup_{r\in\mathcal{R}_{\mathrm{L}}} f(r)$ *allows for the possibility that there is no single trustless regime attaining the maximum expected fitness (e.g., if* $\mathcal{R}_{\mathrm{L}}$ *is infinite or not compact). Likewise,* $g(\alpha) = \sup_r V_r(\alpha)$ *is a pointwise benchmark that selects the best trustless payoff* for each task α, *not necessarily the payoff of any fixed trustless regime across tasks. The theorem is therefore robust: it compares* T *against the entire trustless class even in settings where "the best trustless strategy" exists only as a limit. In particular, it establishes a uniform selection advantage even relative to an adversary that is allowed to pick the best trustless rule task-by-task, and then notes that any fixed trustless regime can only do worse on average than this pointwise upper envelope.*

Remark 3522 (Reasoning-system use)**.** *Theorem 220 is an inference lemma: if your world-model says tasks are drawn from a "rich" distribution, and the GGW-type dominance holds, then as a prior you should expect selection (competition, imitation, institutional evolution) to favor trust-capable coordination.*

66.3 Deterministic evolutionary dynamics: replicator convergence

66.3.1 Two-regime replicator

Consider two effective strategies: trust-capable T versus a representative trustless regime L (for instance, the best-performing member of $\mathcal{R}_{\mathrm{L}}$ in expectation).

Let $x(t) \in [0,1]$ be the fraction of communities using T at time t. Assume fitnesses are constant in time and given by $f_{\mathrm{T}} := f(\mathrm{T})$ and $f_{\mathrm{L}} := f(\mathrm{L})$ with $f_{\mathrm{T}} > f_{\mathrm{L}}$.

Definition 624 (Replicator ODE (two types))**.**

$$\dot{x}(t) \;=\; x(t)\big(f_{\mathrm{T}} - \bar{f}(x(t))\big), \quad \bar{f}(x) := x f_{\mathrm{T}} + (1-x) f_{\mathrm{L}}.$$

Equivalently,

$$\dot{x}(t) \;=\; x(t)(1-x(t))(f_{\mathrm{T}} - f_{\mathrm{L}}).$$

Remark 3523. *The dot notation $\dot{x}(t)$ denotes the time derivative $\frac{d}{dt}x(t)$. The quantity $\bar{f}(x)$ is the population mean fitness when a fraction x uses* T *and a fraction $1-x$ uses* L*. The replicator equation says: a type grows in frequency when its fitness exceeds the current average.*

A concrete sanity check: if x is very small (few trust-capable communities exist), then $\bar{f}(x) \approx f_{\mathrm{L}}$, so $\dot{x} \approx x(f_{\mathrm{T}} - f_{\mathrm{L}})$ and T *grows approximately exponentially at first. If x is close to 1, then the $(1-x)$ factor slows growth, reflecting saturation: there are few remaining* L*-communities to be replaced.*

This definition is useful because it provides a mathematically tractable mean-field model of cultural imitation, differential reproduction, or institutional copying. Even when the real process is noisy, replicator ODEs often describe the average flow.

Theorem 221 (Global convergence to trust-capable under two-type replicator)**.** *If $f_{\mathrm{T}} > f_{\mathrm{L}}$, then for any initial condition $x(0) \in (0,1]$,*

$$\lim_{t\to\infty} x(t) \;=\; 1.$$

Remark 3524 (Intuitive content and connection)**.** *This theorem says that, in the simplest deterministic selection model, any positive foothold of a strictly fitter regime eventually takes over the population. The conclusion is not merely local stability but* global *convergence on $(0,1]$: once* T *is present, it inexorably grows.*

This directly operationalizes Theorem 220: that theorem produces the strict inequality $f_{\mathrm{T}} > f_{\mathrm{L}}$ (or $f(\mathrm{T}) > \sup f(r)$), while the present theorem converts that inequality into a time-asymptotic population prediction.

Proof. Let $\delta := f_{\mathrm{T}} - f_{\mathrm{L}} > 0$. Then $\dot{x} = \delta x(1-x)$. For $x \in (0,1)$, $\dot{x} > 0$, so $x(t)$ is strictly increasing and bounded above by 1, hence converges to some $x_\infty \in (0,1]$.

If $x_\infty \in (0,1)$, then taking the limit in $\dot{x} = \delta x(1-x)$ gives $0 = \delta x_\infty(1-x_\infty)$, contradiction. Hence $x_\infty = 1$. If $x(0) = 1$ then $x(t) = 1$ for all t. Thus convergence holds for all $x(0) \in (0,1]$. □

Proof sketch. The ODE has the logistic form $\dot{x} = \delta x(1-x)$ with $\delta > 0$. The key idea is monotonicity: on the open interval $(0, 1)$ the derivative is strictly positive, so trajectories increase and must converge. The only possible limit points are equilibria, and the only equilibria are 0 and 1; since the trajectory cannot decrease to 0, it must converge to 1. □

Remark 3525. *Geometrically, the phase line on $[0, 1]$ has arrows pointing right everywhere in $(0, 1)$ when $\delta > 0$. The boundary point $x = 0$ is unstable (a repelling equilibrium) and $x = 1$ is stable (attracting). This is the simplest instance of the general principle: in a one-dimensional flow, a strict sign condition on the vector field determines global behavior.*

66.3.2 Multi-regime replicator: trust-capable dominates an entire class

Let $\mathcal{R} = \{\mathrm{T}, 1, 2, \dots, k\}$ with frequencies $x_{\mathrm{T}}, x_1, \dots, x_k$.

Definition 625 (Multi-type replicator ODE)**.**

$$\dot{x}_i \;=\; x_i\big(f_i - \bar{f}\big), \quad \bar{f} := \sum_j x_j f_j.$$

Remark 3526. *Here $(x_i)_i$ lives on the probability simplex: each $x_i \geq 0$ and $\sum_i x_i = 1$. The scalar $\bar{f}$ is again the mean fitness. The equation says each type grows or shrinks according to how its fitness compares to the mean.*

A small example with three regimes: if $f_1 < f_2 < f_3$ and all $x_i > 0$, then x_3 tends to increase and x_1 tends to decrease, with x_2 possibly increasing or decreasing depending on whether f_2 is above or below $\bar{f}$. The utility of this definition is that it lets us treat "trustless coordination" not as one regime but as a whole competing family.

Theorem 222 (Dominance implies global attraction under multi-type replicator)**.** *Assume $f_{\mathrm{T}} > f_i$ for all $i \neq \mathrm{T}$, and $x_{\mathrm{T}}(0) > 0$. Then $x_{\mathrm{T}}(t) \to 1$ as $t \to \infty$.*

Remark 3527 (Intuitive content and role)**.** *This theorem says that strict dominance against* every *competitor implies eventual takeover, even when the competitors are many and may compete among themselves. The condition $x_{\mathrm{T}}(0) > 0$ is essential: replicator dynamics cannot create a type from nothing.*

Conceptually, this is the natural strengthening of Theorem 221. Theorem 220 often yields $f(\mathrm{T}) > \sup_{r \in \mathcal{R}_{\mathrm{L}}} f(r)$ rather than identifying a single best L. *Theorem 222 matches that stronger premise.*

Proof. For any $i \neq \mathrm{T}$ with $x_i(t) > 0$, consider the log-ratio:

$$\frac{d}{dt} \log \frac{x_{\mathrm{T}}}{x_i} \;=\; \frac{\dot{x}_{\mathrm{T}}}{x_{\mathrm{T}}} - \frac{\dot{x}_i}{x_i} \;=\; (f_{\mathrm{T}} - \bar{f}) - (f_i - \bar{f}) \;=\; f_{\mathrm{T}} - f_i \;>\; 0.$$

The key point in the above cancellation is that the mean fitness $\bar{f}$ enters every coordinate equation in the same way under replicator dynamics, so differences of *per-capita* growth rates depend only on pairwise fitness gaps. Moreover, on any time interval where $x_i(t) > 0$, the quantity $\log(x_{\mathrm{T}}/x_i)$ is well-defined and differentiable, so the displayed computation applies pointwise in time on that interval.

Thus $\log(x_{\mathrm{T}}/x_i)$ increases without bound linearly in t whenever $x_i > 0$, so $x_{\mathrm{T}}/x_i \to \infty$, implying $x_i(t) \to 0$ for all $i \neq \mathrm{T}$. More explicitly, since $f_{\mathrm{T}} - f_i$ is a positive constant (independent of t) in the deterministic setting considered here, integration yields

$$\log \frac{x_{\mathrm{T}}(t)}{x_i(t)} = \log \frac{x_{\mathrm{T}}(0)}{x_i(0)} + (f_{\mathrm{T}} - f_i)t,$$

whenever the ratio is defined; hence $x_{\mathrm{T}}(t)/x_i(t)$ grows at least exponentially in t. If for some $i \neq \mathrm{T}$ we have $x_i(0) = 0$, then that type is absent and remains absent, so the conclusion $x_i(t) \to 0$ is immediate in that case as well.

Since the simplex sums to 1, we obtain $x_{\mathrm{T}}(t) \to 1$. Equivalently, because $\sum_{i \neq \mathrm{T}} x_i(t) \to 0$ and the simplex constraint enforces $x_{\mathrm{T}}(t) = 1 - \sum_{i \neq \mathrm{T}} x_i(t)$, the remaining mass concentrates on T. □

Proof sketch. The proof tracks relative frequencies rather than absolute ones. Under replicator dynamics, the time-derivative of $\log(x_{\mathrm{T}}/x_i)$ cancels the mean fitness $\bar{f}$ and becomes the constant difference $f_{\mathrm{T}} - f_i$. If T is strictly fitter than every i, every log-ratio drifts to $+\infty$, forcing all competitors to vanish and leaving $x_{\mathrm{T}} \to 1$. One can also view this as saying that T has uniformly larger per-capita growth rate than any competitor when measured in the same population environment, so any initially nonzero share of T eventually dominates each alternative in pairwise comparison. □

Remark 3528. *A useful intuition is that the simplex dynamics has a Lyapunov-like structure when there is a single strictly best type: every relative comparison tilts in the same direction, so there is no room for cycling or coexistence. The log-ratio computation makes this precise by exhibiting a strictly increasing quantity along trajectories for each competing type. In particular, in contrast to frequency-dependent games where f_i may vary with x and permit nontrivial recurrent behavior, the present strict ordering $f_{\mathrm{T}} > f_i$ for all $i \neq \mathrm{T}$ rules out interior fixed points other than the vertex at* T*, and forces all trajectories with $x_{\mathrm{T}}(0) > 0$ to move toward that vertex.*

Corollary 76 (Selection conclusion from GGW-type dominance)**.** *If Assumption 4 holds and the evolutionary fitness of regimes is $f(r) = \mathbb{E}[V_r]$, then the trust-capable regime* T *is globally attracting under replicator dynamics, provided it is present with positive frequency initially.*

Remark 3529 (Intuitive content)**.** *This corollary packages the bridge: (i) almost-everywhere task-level strict advantage implies (ii) average strict advantage (Theorem 220), which implies (iii) takeover under a broad class of deterministic selection dynamics (Theorem 222). In a reasoning system, this is the point where one may safely replace a detailed competition story by a qualitative forecast: "trust-capable coordination spreads." The logical role of the "positive frequency initially" clause is to exclude the degenerate case in which* T *is absent: replicator dynamics does not create novel types, so selection can only amplify what is already present in the population state.*

Proof. By Theorem 220, $f(\mathrm{T}) > \sup_{r \in \mathcal{R}_{\mathrm{L}}} f(r)$. Apply Theorem 222 to the finite (or finite-approximated) regime set. Concretely, the strict inequality provides the uniform fitness gap hypothesis required to make T uniquely fittest among the regimes under consideration, and the assumption that $x_{\mathrm{T}}(0) > 0$ ensures the trajectory begins in the basin of attraction of the T vertex rather than on the face where T is identically zero. □

Proof sketch. The proof is a two-step reduction. First convert the almost-sure dominance hypothesis into a strict gap in mean fitness. Then invoke the multi-type replicator dominance theorem, which says a uniquely fittest type with positive initial mass attracts the whole simplex. Equivalently, the corollary is obtained by composing the implication "taskwise advantage ⇒ mean advantage" with the implication "mean advantage ⇒ asymptotic fixation" in the deterministic replicator model. □

Remark 3530. *The "finite-approximated" parenthetical is practically relevant: one may discretize a large regime space by considering only regimes that are realistically reachable or salient. The theorem is then an idealization of a coarse-grained evolutionary story (compare the general coarse-graining perspective in [5]). Formally, this amounts to restricting attention to a finite subset (or* *finite partition) of strategies so that the replicator system is a finite-dimensional ODE with the standard simplex as an invariant state space; the qualitative conclusion is meant to be robust to reasonable refinements of that discretization so long as* T *remains strictly best in mean fitness among the included alternatives.*

66.4 Stochastic selection: almost-sure fixation under i.i.d. task draws

Replicator ODEs capture mean-field behavior. We now show that even if each generation faces a fresh random task, a broad class of multiplicative selection dynamics still fixates on trust-capable coordination almost surely.

Definition 626 (Log-odds update under multiplicative selection)**.** *Let x_t be the fraction of communities using* T *at discrete time t. Let $\alpha_t \sim \nu$ i.i.d. and define realized payoff difference*

$$\Delta_t := V_{\mathrm{T}}(\alpha_t) - V_{\mathrm{L}}(\alpha_t).$$

Fix a learning/selection rate $\eta > 0$ and update by

$$\frac{x_{t+1}}{1-x_{t+1}} \;=\; \frac{x_t}{1-x_t}\,\exp(\eta\Delta_t).$$

Equivalently,

$$\ell_{t+1} \;=\; \ell_t + \eta\Delta_t, \quad \ell_t := \log\frac{x_t}{1-x_t}.$$

Remark 3531. *The notation $\alpha_t \sim \nu$ i.i.d. means the tasks are independently and identically distributed across time: each generation (or cultural update step) encounters a fresh task drawn from the same environment distribution.*

The update is easiest to read in log-odds form. The quantity $\ell_t = \log\frac{x_t}{1-x_t}$ is the logarithm of the odds in favor of T*. The rule $\ell_{t+1} = \ell_t + \eta\Delta_t$ says: if* T *outperforms* L *on the realized task ($\Delta_t > 0$), the odds tilt toward* T *multiplicatively; if $\Delta_t < 0$, they tilt the other way. This is a standard structure for multiplicative weights / Bayesian-like updating, and it captures a wide family of "success breeds imitation" mechanisms.*

A simple example: if $\eta = 1$ and $\Delta_t = 0.1$ consistently, then odds multiply by $\mathrm{e}^{0.1}$ each step. The definition is useful because it reduces fixation questions to the behavior of a random walk with drift in the variable ℓ_t.

Theorem 223 (Almost-sure fixation from positive drift)**.** *Assume $\mathbb{E}[\Delta_t] > 0$ and $\mathbb{E}[|\Delta_t|] < \infty$. Then for any $x_0 \in (0,1)$, the process in Definition 626 satisfies*

$$x_t \to 1 \quad \textit{almost surely as } t \to \infty.$$

Remark 3532 (Intuitive content and connection)**.** *This theorem is the stochastic analogue of Theorem 221. Instead of a deterministic ODE with constant advantage, we have random task-by-task advantages Δ_t. The claim is that a* positive mean *advantage produces almost-sure takeover: the randomness cannot prevent fixation because, over long time horizons, the cumulative drift dominates fluctuations.*

This connects directly to Theorem 220: under GGW-type hypotheses, the mean of the gap is positive, so the stochastic selection process almost surely drives trust-capable coordination to fixation.

Proof. We have $\ell_t = \ell_0 + \eta \sum_{s=0}^{t-1} \Delta_s$. By the strong law of large numbers (integrability suffices),

$$\frac{1}{t} \sum_{s=0}^{t-1} \Delta_s \to \mathbb{E}[\Delta_0] \ > \ 0 \quad \text{a.s.}$$

Hence $\sum_{s=0}^{t-1} \Delta_s \to +\infty$ almost surely, and therefore $\ell_t \to +\infty$ almost surely. Since $x_t = (1+e^{-\ell_t})^{-1}$, we obtain $x_t \to 1$ a.s. ☐

Proof sketch. Rewrite the multiplicative update as an additive random walk in log-odds ℓ_t. The increments are $\eta\Delta_t$. By the strong law of large numbers, the average increment converges to its mean, which is positive by assumption. Therefore the partial sums diverge to $+\infty$, forcing $\ell_t \to +\infty$ and hence $x_t \to 1$. ☐

Remark 3533. *One can picture ℓ_t as a walker on the real line: each task pushes the walker right or left by an amount proportional to payoff difference. If the expected push is rightward, the walker almost surely drifts to $+\infty$. The logistic transform $x_t = (1 + e^{-\ell_t})^{-1}$ then maps "going to $+\infty$" into "fixating at 1."*

Corollary 77 (GGW-type dominance implies a.s. fixation under stochastic tasks)**.** *If $\Delta(\alpha) \geq 0$ and $\nu(\Delta > 0) = 1$ with Δ integrable, then $\mathbb{E}[\Delta] > 0$, so Theorem 223 applies and trust-capable coordination fixates almost surely under multiplicative selection.*

Remark 3534 (Intuitive content)**.** *This corollary states that the measure-theoretic dominance premise is strong enough to drive fixation even under task-by-task randomness, not merely in the averaged ODE picture. In the intended applications (institutional evolution, memetic spread, competitive multi-agent ecologies), this is often the more realistic setting: the "generation" sees one or a few tasks, not the whole distribution at once.*

Proof. Immediate from Theorem 220 (applied to the two-type case) and Theorem 223. ☐

Proof sketch. Use Theorem 220 to infer $\mathbb{E}[\Delta] > 0$ from almost-sure strict positivity, and then apply the positive-drift fixation theorem for the log-odds process. ☐

66.5 Mutation and persistent diversity: robustness of trust-capable dominance

Selection rarely runs without mutation, exploration, or architectural drift. The next result shows that trust-capable coordination remains the dominant mass even with persistent mutation. One can read this as a "mutation–selection balance" statement: even when there is continual leakage out of the fitter regime, the long-run population state typically stabilizes at an interior mixture whose location is controlled by the relative strengths of selection (δ) and mutation (μ_{TL}, μ_{LT}). In particular, when selection favoring T is strong compared to the rate at which T degrades into L, the equilibrium mass near T remains high.

Definition 627 (Replicator-mutator (two types))**.** *Let $x(t)$ be the fraction of* T*. Let $f_{\mathrm{T}} > f_{\mathrm{L}}$ be constants. Let $\mu_{TL} \geq 0$ be mutation rate* T $\to$ L *and $\mu_{LT} \geq 0$ be mutation rate* L $\to$ T*. Define*

$$\dot{x} \ = \ x(1-x)(f_{\mathrm{T}} - f_{\mathrm{L}}) \ - \ \mu_{TL} x \ + \ \mu_{LT}(1-x).$$

In this two-type setting, the term $x(1-x)(f_{\mathrm{T}} - f_{\mathrm{L}})$ is the usual replicator selection component (with constant fitness gap), while the remaining linear terms implement a continuous-time mutation channel. Note that the mutation part alone would drive x toward the mutation-only equilibrium $\mu_{LT}/(\mu_{TL} + \mu_{LT})$ (when $\mu_{TL} + \mu_{LT} > 0$), whereas the selection part alone would drive x toward 1 when $f_{\mathrm{T}} > f_{\mathrm{L}}$.

Remark 3535. *The additional terms $-\mu_{TL}x$ and $+\mu_{LT}(1-x)$ model continual leakage between regimes: even if* T *is favored, some fraction of trust-capable communities degrade into trustless ones (loss of coordination norms, institutional collapse, corruption), while some trustless communities innovate or learn their way into* T*.*

The usefulness of this model is that it captures a common reality: selection pushes in one direction, but exploration and error keep diversity alive. The question is then not "does T *take over perfectly?" but "does most of the mass concentrate near* T*, and under what parameter regimes?" A further practical point is that μ_{TL} can be interpreted broadly as any persistent mechanism that reintroduces low-trust behavior (e.g., churn in membership, shocks that erode enforcement, or adversarial adaptation), while μ_{LT} can represent deliberate institutional improvement, imitation of successful norms, or technological tooling that makes trust-capable coordination easier.*

Theorem 224 (Unique stable equilibrium and concentration as mutation vanishes)**.** *Assume $\delta := f_{\mathrm{T}} - f_{\mathrm{L}} > 0$ and $\mu_{TL}, \mu_{LT} \geq 0$. Then:*

1. *There exists a unique equilibrium $x^* \in (0,1)$ if $\mu_{TL}+\mu_{LT} > 0$, and it is globally asymptotically stable on $[0,1]$.*
2. *If $\mu_{LT} > 0$ is fixed and $\mu_{TL} \to 0$, then $x^* \to 1$. More generally, if both $\mu_{TL}, \mu_{LT} \to 0$ with $\mu_{TL}/\mu_{LT} \to 0$, then $x^* \to 1$.*

Remark 3536 (Intuitive content and connection)**.** *This theorem says that with ongoing mutation, one typically gets a single stable mixed equilibrium: selection pushes toward* T *while mutation pulls away, and the balance point is unique. As mutation away from* T *becomes negligible relative to mutation toward* T*, the balance point moves toward full concentration on* T*.*

This complements the earlier fixation theorems by showing robustness: the conclusion "trust-capable dominates" does not depend on an unrealistic assumption of zero exploration or zero institutional decay. In addition, the interior equilibrium $x^* \in (0,1)$ should be read as persistent diversity*: even when* T *is strictly fitter, the system generically does not converge to a vertex of the simplex under continual mutation. Instead, the equilibrium encodes the steady-state fraction of communities that remain in (or are continually reintroduced into) the less coordinated regime.*

Proof. Write the RHS as a polynomial:

$$\dot{x} = \delta x - \delta x^2 - \mu_{TL}x + \mu_{LT} - \mu_{LT}x = -\delta x^2 + (\delta - \mu_{TL} - \mu_{LT})x + \mu_{LT}.$$

Call this $P(x)$. It is a concave parabola in x (leading coefficient $-\delta < 0$). We compute

$$P(0) = \mu_{LT} \geq 0, \qquad P(1) = -\mu_{TL} \leq 0.$$

It is also useful (for invariance of the state space) to observe that these endpoint signs imply the vector field points inward on $[0,1]$: at $x = 0$ we have $\dot{x} = P(0) \geq 0$ so trajectories cannot cross into $x < 0$, and at $x = 1$ we have $\dot{x} = P(1) \leq 0$ so trajectories cannot cross into $x > 1$. Thus $[0,1]$ is forward-invariant under the dynamics.

If $\mu_{TL} + \mu_{LT} > 0$, then $P(0) > 0$ or $P(1) < 0$ (indeed at least one strict), so by continuity there is at least one root in $(0,1)$. Concavity implies at most two roots total, but the sign pattern ($P(0) \geq 0$, $P(1) \leq 0$) and concavity together imply exactly one root in $(0,1)$ unless both endpoints are roots, which requires $\mu_{LT} = 0$ and $\mu_{TL} = 0$, excluded by assumption. Thus the equilibrium is unique in $(0,1)$.

For stability, note that for a 1D ODE $\dot{x} = P(x)$, an interior equilibrium x^* is asymptotically stable if $P'(x^*) < 0$. Here $P'(x) = -2\delta x + (\delta - \mu_{TL} - \mu_{LT})$. At the unique interior crossing of a concave parabola from positive to negative, the derivative must be negative, hence x^* is asymptotically stable. Because $P(0) \geq 0$ and $P(1) \leq 0$, solutions are trapped in $[0,1]$ and flow toward the unique stable equilibrium, giving global asymptotic stability. Equivalently, since there is exactly one equilibrium in $(0,1)$ and the sign of $\dot{x}$ is $\dot{x} > 0$ for $x < x^*$ and $\dot{x} < 0$ for $x > x^*$, every trajectory in $(0,1)$ moves monotonically toward x^* (increasing from below and decreasing from above), which is a particularly transparent form of global convergence in one dimension.

For concentration: solve $P(x^*) = 0$:

$$\delta(x^*)^2 - (\delta - \mu_{TL} - \mu_{LT})x^* - \mu_{LT} = 0.$$

By the quadratic formula, the two algebraic roots are

$$x_\pm = \frac{(\delta - \mu_{TL} - \mu_{LT}) \pm \sqrt{(\delta - \mu_{TL} - \mu_{LT})^2 + 4\delta\mu_{LT}}}{2\delta}.$$

Because $\sqrt{(\delta - \mu_{TL} - \mu_{LT})^2 + 4\delta\mu_{LT}} \geq |\delta - \mu_{TL} - \mu_{LT}|$ and $\delta > 0$, the "minus" root x_- is nonpositive whenever $\mu_{LT} > 0$, while the "plus" root x_+ lies in $(0,1]$ and is the unique equilibrium x^*. This explicit expression makes the parameter dependence of the interior equilibrium fully transparent (and can be used for comparative statics, e.g. monotonicity of x^* in μ_{TL} and μ_{LT}).

As $\mu_{TL} \to 0$ with $\mu_{LT} > 0$ fixed, the unique root in $(0,1)$ must satisfy $x^* \uparrow 1$, because $P(1) = -\mu_{TL} \to 0^-$ and $P(0) = \mu_{LT} > 0$ stay separated, squeezing the crossing toward 1. The stated more general limit follows similarly from $P(1) = -\mu_{TL}$ and $P(0) = \mu_{LT}$. One can also see the same limit directly from the closed form above: with $\mu_{TL} = 0$,

$$x^* = \frac{(\delta - \mu_{LT}) + \sqrt{(\delta - \mu_{LT})^2 + 4\delta\mu_{LT}}}{2\delta} = \frac{(\delta - \mu_{LT}) + (\delta + \mu_{LT})}{2\delta} = 1,$$

and continuity in parameters implies $x^* \to 1$ as $\mu_{TL} \to 0$. Moreover, when μ_{TL} is small relative to δ (and μ_{LT} not vanishing too fast), one may heuristically read the equilibrium as "very close to 1" because only the leakage term $-\mu_{TL}x$ can keep $P(1)$ negative; once that leakage disappears, the selection term forces the population to the T vertex. □

Proof sketch. Reduce the ODE to $\dot{x} = P(x)$ where P is a concave quadratic. Evaluate P at 0 and 1 to force a sign change across the interval, hence an interior root. Concavity plus the endpoint signs ensure this interior root is unique. Stability follows because a concave curve crossing from positive to negative must have negative slope at the crossing. The limiting claims use the fact that as $\mu_{TL} \to 0$, the value $P(1) = -\mu_{TL}$ approaches 0 from below, pushing the unique crossing toward $x = 1$. A complementary intuition is that mutation prevents absorption at the boundary points: as long as $\mu_{TL} + \mu_{LT} > 0$, the dynamics continually "re-seeds" both types, so the only possible limit is an interior fixed point where the opposing flows balance. □

Remark 3537. *A geometric picture is literal here: $P(x)$ is a downward-opening parabola. The dynamics flows right where $P(x) > 0$ and left where $P(x) < 0$. Because $P(0) \geq 0$ and $P(1) \leq 0$, the parabola must cross the axis somewhere in between, and since it is concave, there is only one such crossing compatible with the monotone sign change. The stable equilibrium is exactly that crossing point. In applications, this picture also emphasizes a robustness point: changing μ_{TL} and μ_{LT} perturbs the parabola mainly by shifting the intercepts $P(0) = \mu_{LT}$ and $P(1) = -\mu_{TL}$, so the equilibrium slides continuously along the interval rather than changing discontinuously. Thus, even when there is persistent "noise" in institutional quality, the long-run proportion of trust-capable coordination varies smoothly with the magnitude and direction of that noise.*

66.6 Selection pressure grows with coordination-interface complexity

A key intuition for communities and mindplexes is that complex/hierarchical coordination creates many "interfaces" where trustless protocols incur overhead or mechanism-design friction, while trusting team procedures do not. This yields a scaling law: the larger/more hierarchical the task, the larger the selection advantage for trust-capable coordination.

Definition 628 (Interface-count lower bound model)**.** *For each task α, let $m(\alpha) \in \mathbb{N} \cup \{0\}$ be the number of coordination interfaces that exhibit "problematic structure" (e.g. two-sided private influence). Assume there exists $\ell > 0$ such that for every α,*

$$V_{\mathrm{T}}(\alpha) - \sup_{r \in \mathcal{R}_{\mathrm{L}}} V_r(\alpha) \;\geq\; \ell\, m(\alpha).$$

Remark 3538. *The variable $m(\alpha)$ is a coarse measure of how many distinct "coordination boundaries" occur in task α—places where information must cross from one subagent (or institution, subsystem, department) to another and where strategic constraints bite. In a hierarchical organization, these interfaces proliferate with depth and breadth (Hyperseed-Concept* **??***); in a heterarchical mesh, they proliferate with the density of interconnections (Hyperseed-Concept* **??***).*

The inequality posits a linear lower bound: each problematic interface costs a trustless regime at least ℓ expected performance relative to trust-capable coordination. For example, if each interface forces costly verification, negotiation, or incentive-compatible signaling, then these costs accumulate. The model is useful not because it is always exact, but because it converts a qualitative intuition ("complex coordination makes trust more valuable") into an inequality that can be carried through expectation and selection analyses.

One operational reading is that $m(\alpha)$ counts "events where a local optimum is misaligned with the global objective under strategic uncertainty." Concretely, an interface may correspond to a handoff (requirements $\to$ implementation), an internal market or budget boundary, a procurement contract, a data-sharing API between teams, a delegation edge in a chain of command, or a multi-party decision gate. The qualifier "problematic structure" is intentionally broad: it includes not only explicit adversarial incentives, but also softer sources of strategic constraint such as unverifiable effort, private cost information, hidden actions, and multi-principal ambiguity.

The constant ℓ should be interpreted as a per-interface minimum *gap, not an average gap. Thus the model is conservative: it allows some interfaces to be cheap to handle trustlessly (the bound still holds), but it rules out environments where the trustless regime can systematically exploit interface-level structure to erase the gap at all problematic seams simultaneously.*

Remark 3539 (Back-of-the-envelope interface growth)**.** *To make the dependence on organizational form more concrete, consider a stylized b-ary hierarchy of depth d (root at depth 0) in which each edge corresponds to one delegation/verification interface. If tasks require traversing most edges at least once, then an order-of-magnitude proxy is $m(\alpha) \approx \sum_{k=0}^{d-1} b^{k+1} = \frac{b^{d+1}-b}{b-1}$, i.e. exponential in depth when breadth $b > 1$. By contrast, in a sparse heterarchical graph with N nodes and average degree $\bar{k}$, a natural proxy is $m(\alpha) \approx \frac{1}{2} N \bar{k}$, i.e. linear in interconnection density. These are not claims of exact counting, but they illustrate why even mild changes in depth/breadth or connectivity can strongly affect $\mathbb{E}[m(\alpha)]$ and therefore the selection coefficient.*

Theorem 225 (Expected advantage lower bounded by expected interface count)**.** *Under the interface-count model,*

$$f(\mathrm{T}) - \sup_{r \in \mathcal{R}_{\mathrm{L}}} f(r) \;\geq\; \ell\, \mathbb{E}[m(\alpha)].$$

In particular, if $\mathbb{E}[m(\alpha)]$ increases with environmental/task complexity, then the selection coefficient favoring trust-capable coordination increases with complexity.

Remark 3540 (Intuitive content and connection)**.** *This theorem states that the "gap" between trust-capable and best trustless performance scales at least like the average number of problematic interfaces. It makes precise a kind of moral of mechanism-design folklore: the more seams in the world, the more expensive it is to stitch them with adversarially robust thread.*

It connects back to the GGW-style hypothesis by providing a concrete sufficient condition that can help justify it: if $m(\alpha) > 0$ on almost all tasks and $\ell > 0$, then $\Delta(\alpha)$ is almost surely positive, and the expected advantage grows with complexity.

A further interpretive point is that the theorem does not require every *task instance to be complex; it is enough that complexity arises with nontrivial probability under the environment distribution. In particular, a heavy-tailed distribution over task difficulties can make $\mathbb{E}[m(\alpha)]$ large even when most tasks are simple, reflecting the idea that rare but high-stakes, high-interface episodes (crises, launches, coordination shocks) can dominate long-run fitness comparisons.*

Proof. Take expectations:

$$f(\mathrm{T}) - \sup_r f(r) \;\geq\; \mathbb{E}\left[V_{\mathrm{T}}(\alpha) - \sup_r V_r(\alpha)\right] \;\geq\; \mathbb{E}[\ell m(\alpha)] = \ell\mathbb{E}[m(\alpha)].$$

To spell out the first inequality in a way that makes the order of operations explicit, note that for any fixed r we have

$$f(\mathrm{T}) - f(r) = \mathbb{E}[V_{\mathrm{T}}(\alpha) - V_r(\alpha)] \;\geq\; \mathbb{E}\left[V_{\mathrm{T}}(\alpha) - \sup_{r'} V_{r'}(\alpha)\right],$$

since $\sup_{r'} V_{r'}(\alpha) \geq V_r(\alpha)$ pointwise in α. Taking $\sup_r$ on the right-hand side yields

$$f(\mathrm{T}) - \sup_r f(r) = \inf_r \big(f(\mathrm{T}) - f(r)\big) \;\geq\; \mathbb{E}\left[V_{\mathrm{T}}(\alpha) - \sup_{r'} V_{r'}(\alpha)\right],$$

which is the step used above before applying the definition's lower bound and linearity of expectation. □

Proof sketch. The proof is linearity of expectation plus monotonicity: the pointwise lower bound on the advantage implies the same lower bound after averaging. The only subtlety is that $\sup_r f(r)$ is bounded above by $\mathbb{E}[\sup_r V_r(\alpha)]$, which is exactly the direction needed for a lower bound.

Equivalently, the key move is to avoid mistakenly swapping sup and $\mathbb{E}$ in the wrong direction: generally $\sup_r \mathbb{E}[V_r(\alpha)] \leq \mathbb{E}[\sup_r V_r(\alpha)]$, and the argument exploits this inequality to keep the bound conservative. □

Remark 3541. *Visually, one may think of $m(\alpha)$ as a complexity gauge attached to each task. The theorem says: if each unit of complexity contributes at least ℓ units of advantage, then the average advantage is at least ℓ times average complexity. This is the simplest kind of "scaling law" available in this setting.*

It is also worth noting what the model does not *assume: it does not require that costs add exactly linearly across interfaces, only that there is a uniform* lower bound *that scales at least linearly. In many real systems, interface costs can be superlinear (e.g. cascading delays, compounding risk, or combinatorial incentive constraints), in which case the linear bound remains valid but potentially loose, and selection could intensify even faster than the inequality suggests.*

Corollary 78 (Large integrated collectives need trust to remain viable)**.** *Suppose a family of tasks indexed by "scale" n has $\mathbb{E}[m(\alpha) \mid n] \geq cn$ for some $c > 0$, and suppose net viability requires expected performance above some baseline B. Then for sufficiently large n, any trustless regime has expected performance below T by at least ℓcn, so beyond a threshold scale, trustless communities are generically selected out relative to trust-capable ones.*

Remark 3542 (Intuitive content)**.** *This corollary is the "thermodynamic" moral: when a system grows, the number of seams grows, and the price of seam-management under adversarial constraints grows correspondingly. If the environment demands near-linear scaling of efficiency with size, then trustless coordination becomes structurally noncompetitive beyond some scale.*

One can view the "threshold scale" as the point where the linear advantage ℓcn exceeds any fixed margin by which a trustless protocol might win on small instances (e.g. due to lower fixed setup costs, simpler governance, or less need for internal monitoring). In that sense, the corollary formalizes an intuition that trustless coordination can look attractive in small systems while being fragile under growth, because the quantity that scales (interfaces) is precisely the quantity that trustless designs must pay to police.

Proof. Immediate from Theorem 225: the expected gap grows at least linearly in n. Thus, whenever being competitive/viable at scale n requires not wasting $\Theta(n)$ efficiency, trustless regimes become noncompetitive for large enough n.

More explicitly, conditioning on n gives

$$f(\mathrm{T}) - \sup_{r\in\mathcal{R}_{\mathrm{L}}} f(r) \;\geq\; \ell\,\mathbb{E}[m(\alpha)\mid n] \;\geq\; \ell cn,$$

so for any fixed baseline requirement expressed as an additive slack (e.g. needing to exceed B by a constant margin), there exists $n^\star$ such that for all $n \geq n^\star$ the linear term ℓcn dominates that slack. ☐

Proof sketch. Condition on scale n and apply the interface theorem with $\mathbb{E}[m(\alpha)\mid n] \geq cn$. The result is an explicit linear lower bound ℓcn on the expected advantage, which eventually dominates any fixed viability baseline differences as n grows.

If one imagines a fixed "head start" H that a trustless regime might enjoy at small n (say, from lower constant overhead), the sketch corresponds to choosing $n^\star \approx H/(\ell c)$ so that $\ell cn^\star \gtrsim H$; beyond that point the linear-in-n interface penalty outweighs any such constant advantage. ☐

66.7 Selection over communities with a continuous trust trait

So far, regimes were discrete. We now treat "trust" as a continuous trait, and show that if performance is strictly increasing in trust, then selection increases mean trust. In this subsection, "continuous" refers to the trait space $[0,1]$ rather than to the index set of communities: the population may still consist of finitely many community types, each carrying a (real-valued) trust level τ_i. The key point is that selection can now act on arbitrarily small differences in τ.

Definition 629 (Population of communities with trust trait)**.** *Let communities be indexed by i with weights (frequencies) $p_i(t) \geq 0$, $\sum_i p_i(t) = 1$. Each community has a trust trait $\tau_i \in [0,1]$ (fixed for now), and fitness $W(\tau_i)$ where $W : [0,1] \to \mathbb{R}$ is strictly increasing. Assume group-level replicator dynamics:*

$$\dot{p}_i \;=\; p_i\big(W(\tau_i) - \bar{W}\big), \quad \bar{W} := \sum_j p_j W(\tau_j).$$

Define mean trust $\bar{\tau} := \sum_i p_i \tau_i$.

In this notation, it is convenient to interpret $\sum_i p_i(\cdot)$ as an expectation operator under the distribution $p(t)$, i.e. for any function f define

$$\mathbb{E}_p[f(\tau)] \;:=\; \sum_i p_i f(\tau_i).$$

This turns expressions such as $\bar{\tau}$ and $\bar{W}$ into $\mathbb{E}_p[\tau]$ and $\mathbb{E}_p[W(\tau)]$ respectively, and makes the link to Price-equation-style identities notationally explicit. The assumption that τ_i is "fixed for now" isolates the pure effect of selection on composition (changing p_i) from within-type change (changing τ_i); later extensions can reintroduce within-community learning or mutation as additional terms.

Remark 3543. *This definition turns "trust" into a scalar parameter $\tau \in [0,1]$. One can read τ as the degree to which a community can exploit high-bandwidth cooperation, or as a proxy for the depth of shared norms, transparency, and credible commitment. The function $W(\tau)$ then encodes the claim "more trust yields higher fitness" in the given environment.*

The weights $p_i(t)$ define a population distribution over community-types. The mean trust $\bar{\tau} = \sum_i p_i\tau_i$ is the population-level quantity of interest. This setup is useful because it yields a general monotonicity principle (a form of the Price equation) without needing to specify how τ is implemented internally.

A further technical convenience of the replicator equation is that it preserves the simplex: starting from $p_i(0) \geq 0$ with $\sum_i p_i(0) = 1$, one has $p_i(t) \geq 0$ and $\sum_i p_i(t) = 1$ for all t for which the solution exists. Thus the population-level quantities $\bar{\tau}$ and $\bar{W}$ are always well-defined and remain convex combinations of the underlying trait values and fitnesses.

Theorem 226 (Strictly increasing fitness implies monotone increase of mean trust)**.** *If W is strictly increasing, then*

$$\frac{d}{dt}\bar{\tau} \;=\; \mathrm{Cov}_p(\tau, W(\tau)) \;\geq\; 0,$$

with strict inequality unless all τ_i are equal (i.e. unless the population is already trust-homogeneous).

The conclusion is a monotonicity statement about the mean, not a claim that every community's trust level rises (the τ_i are held fixed here). Rather, the population reweights toward higher-trust types. Also note that "monotone increase" means $\bar{\tau}(t)$ is nondecreasing; depending on the support of the initial distribution and on the shape of $W(\cdot)$, $\bar{\tau}(t)$ may converge to a limit strictly below 1 if, for example, the maximum trust type $\tau = 1$ is absent from the initial population. In that sense the theorem is about directional selection given available variation, not about creating new variation.

Remark 3544 (Intuitive content and connection)**.** *This theorem says: whenever fitness rises with trust, selection increases average trust. The covariance formula makes the mechanism transparent: if higher-trust types systematically have higher fitness, then they gain weight, pulling the mean upward.*

This continuous-trait result complements the discrete-regime theorems by showing that "trust-capable" need not be a sharp category. Even gradual improvements in trust capacity are, under the stated assumption, cumulatively amplified by selection.

It is also useful to read the covariance identity as identifying the exact "selection gradient" on the mean: the instantaneous rate of change depends only on how strongly τ and $W(\tau)$ co-vary under the current population mixture. In particular, even if W is strictly increasing, the change in $\bar{\tau}$ can be very small when almost all mass is concentrated on a narrow range of τ, because then $\mathrm{Cov}_p(\tau, W(\tau))$ is small.

Proof. Differentiate:

$$\frac{d}{dt}\bar{\tau} = \sum_i \dot{p}_i\tau_i = \sum_i p_i(W(\tau_i) - \bar{W})\tau_i = \sum_i p_i\tau_i W(\tau_i) - \bar{W}\sum_i p_i\tau_i = \mathbb{E}_p[\tau W(\tau)] - \mathbb{E}_p[\tau]\mathbb{E}_p[W(\tau)].$$

This is exactly $\mathrm{Cov}_p(\tau, W(\tau))$.

If W is strictly increasing, then τ and $W(\tau)$ are strictly comonotone, so the covariance is nonnegative, and is zero iff τ is almost surely constant under p (all mass on one trait value).

For a fully explicit discrete argument (useful when one wants to avoid appealing to general comonotonicity facts), write the covariance as a pairwise sum:

$$\mathrm{Cov}_p(\tau, W(\tau)) = \frac{1}{2}\sum_i \sum_j p_i p_j \big(\tau_i - \tau_j\big)\big(W(\tau_i) - W(\tau_j)\big).$$

Because W is increasing, each term $\big(\tau_i - \tau_j\big)\big(W(\tau_i) - W(\tau_j)\big) \geq 0$, so the whole sum is ≥ 0. Moreover, if there exist i, j with $p_i p_j > 0$ and $\tau_i \neq \tau_j$, then strict monotonicity implies $W(\tau_i) \neq W(\tau_j)$ and the corresponding pair contributes a strictly positive amount, yielding $\mathrm{Cov}_p(\tau, W(\tau)) > 0$. □

Proof sketch. Compute the derivative of the mean trait using the replicator equation. The mean change is a covariance between trait value and fitness. Monotonicity of W ensures higher trait values align with higher fitness, forcing the covariance to be nonnegative, with equality only in the degenerate case of no trait variation. □

Remark 3545. *The covariance identity is the key step: it is the algebraic expression of the intuitive statement "types that do better become more common." One may view it as a minimalistic version of the Price equation, specialized to replicator flow. The geometry is again simplex-based: the flow tilts the distribution p toward regions of higher $W(\tau)$, and if W increases with τ, that tilt points toward higher τ.*

This identity is closely related in spirit to Fisher-style statements: rather than tracking the change in mean fitness, it tracks the change in a focal trait, and the covariance specifies how "aligned" the trait is with the fitness landscape. In models with additional forces (e.g. mutation, migration, or within-community adaptation that changes τ_i over time), this selection term typically appears as one addend among others, so the present theorem can be read as isolating the pure selection contribution.

Remark 3546 (Interpretation)**.** *Theorem 226 is a general selection principle: whenever "more trust enables more efficient coordination on typical tasks," selection at the community level increases average trust, even without modeling the micro-mechanisms of how trust is implemented.*

The assumptions are intentionally minimal: $W(\tau)$ is allowed to be nonlinear (e.g. exhibiting threshold-like acceleration, diminishing returns, or sharp increases near some τ^*), and the theorem still applies as long as the mapping remains strictly increasing. Conversely, if the environment induces a non-monotone $W(\tau)$ (for example, if very high trust becomes exploitable by out-groups or incurs excessive verification costs), the covariance need not be nonnegative and the direction of selection on $\bar{\tau}$ can depend on the current distribution $p(t)$.

66.8 Mindplexes as selectable collectives: trust-capability as a necessary condition

We now connect to mindplex-like collectives (tightly coupled systems of subminds plus a collective control/experience layer). The key mathematical point for Target 2 is not phenomenology, but *viability under selection*: mindplexes are high-integration, high-interface systems, so the interface-scaling selection pressure becomes especially strong. In particular, high integration tends to increase (i) the number of distinct loci at which partial plans must be reconciled, (ii) the number of opportunities for incentive misalignment or strategic behavior to manifest, and (iii) the rate at which small coordination errors propagate system-wide. These features do not require any specific metaphysics of collective experience; they follow from fairly ordinary considerations about coupling, interdependence, and the need to maintain coherent joint action across many subsystems.

Remark 3547. *The term "mindplex" is used here in the sense developed in SuperDuperPsychism-style discussions of collective minds and collective consciousness locations; see [12]. In Hyperseed language, it touches the* Mindplex *concept (Hyperseed-Concept 113) and also invites comparison to the* Global Brain *concept (Hyperseed-Concept* **??***). The present section deliberately focuses on selection and coordination cost, not on any particular theory of phenomenology.*

Definition 630 (Mindplex-scale interface growth hypothesis)**.** *A mindplex of size n (number of constituent minds/subsystems) faces tasks whose coordination graph has at least $\Omega(n)$ interfaces, and generically a positive fraction of these interfaces exhibit problematic structure. Formally, there exist $c > 0$ and a scale-indexed environment ν_n such that*

$$\mathbb{E}_{\alpha\sim\nu_n}[m(\alpha)] \;\geq\; cn.$$

Remark 3548. *This hypothesis asserts that as the mindplex grows, it does not merely gain compute or memory; it gains* boundaries*. Each additional constituent typically introduces new channels where plans, beliefs, or resources must be reconciled. The lower bound $\mathbb{E}_{\nu_n}[m(\alpha)] \geq cn$ is a minimal scaling claim: interface count grows at least linearly with size. Many architectures would exhibit superlinear growth (dense interaction graphs), but linear growth suffices for the conclusions.*

A simple example: suppose n subteams must each contribute a component, and each component must be integrated into a shared artifact. Even if the integration is centralized, there are at least n handoff points. In distributed AI, if n modules exchange messages with a hub, there are n module–hub interfaces; if modules interact in a mesh, there can be $\Theta(n^2)$ interfaces.

The hypothesis is useful because it lets us import the interface-count lower bound model into a size-scaling argument: mindplex candidates live in regimes where $m(\alpha)$ is large, hence where trustless overheads are amplified.

It is also useful to separate two notions that are sometimes conflated: the existence *of an interface (a place where information/authority/resources cross subsystem boundaries) and the* problematic structure *of an interface (a place where strategic incentives, hidden information, value drift, interpretability gaps, or verification costs create a persistent performance drag for trustless regimes). The definition above is phrased in terms of $m(\alpha)$ because $m(\alpha)$ is intended to count precisely those interfaces that trigger the lower bound in the interface-count model; hence "problematic" is not an aesthetic judgment but a technical condition: interfaces that are expected to impose a non-negligible trustless overhead on the task distribution under consideration.*

Finally, the scale-indexed environment ν_n should be read as an explicit modeling move: as the collective grows, it tends to attempt (or be selected into) tasks whose scope is commensurate with its capacity. The distribution ν_n captures this endogeneity by letting the task mix vary with n, while the inequality enforces that the resulting tasks do not become trivially decomposable in a way that would collapse interface complexity.

Theorem 227 (Mindplex viability selects for trust-capability)**.** *Assume the mindplex-scale interface growth hypothesis and the interface-count lower bound model. Then the expected performance advantage of trust-capable coordination over any trustless regime at scale n satisfies*

$$f_n(\mathrm{T}) - \sup_{r\in\mathcal{R}_{\mathrm{L}}} f_n(r) \;\geq\; \ell cn,$$

where $f_n(r) := \mathbb{E}_{\alpha\sim\nu_n}[V_r(\alpha)]$. Consequently, among scalable mindplex candidates, selection strongly favors trust-capable designs, with selection pressure growing at least linearly in n.

Remark 3549 (Intuitive content and connection)**.** *This theorem is the specialization of the scaling law to mindplexes: if large collectives necessarily face many coordination interfaces, and each* *interface generically penalizes trustless coordination by at least ℓ, then the total expected penalty grows like n. Thus, in the space of designs capable of scaling to large n, trust-capability is not a luxury but a near-necessity for competitiveness.*

It connects the abstract dominance assumptions (Assumptions 3–4) to a more mechanistic story: why the dominance should become stronger *for more integrated, higher-interface collectives.*

One way to read the inequality is as a lower bound on the "distance to the viability frontier" for trustless regimes: if viability corresponds to achieving some threshold expected value (fitness, throughput, safety-adjusted reward, or institutional success), then increasing n shifts trustless designs downward by an amount that scales with the number of problematic interfaces. Trust-capability, in this framing, is a way of preventing the increasing integration of the collective from turning into an increasing tax on performance.

Note also that the result does not require trust-capable coordination to be perfect. The bound only uses a per-interface advantage parameter ℓ (from the interface-count model), which can be interpreted as an average marginal gap in realized task value, amortized over the task distribution. Even modest per-interface improvements can dominate at large n when interfaces scale linearly or superlinearly.

Proof. By Theorem 225 applied under ν_n,

$$f_n(\mathrm{T}) - \sup_r f_n(r) \;\geq\; \ell\,\mathbb{E}_{\nu_n}[m(\alpha)] \;\geq\; \ell cn.$$

□

Proof sketch. Apply the interface-count expectation bound at the scale-indexed environment ν_n and then plug in the linear interface growth lower bound $\mathbb{E}_{\nu_n}[m(\alpha)] \geq cn$. The two inequalities compose immediately. □

Remark 3550. *The geometric intuition is again additive: each additional constituent adds expected "surface area" (interface) that must be managed. Trust-capability reduces the per-unit surface cost, so as size grows, designs lacking trust-capability are pulled farther below the viability frontier.*

A complementary interpretation is in terms of "coordination bandwidth." If each problematic interface consumes some fraction of the system's available attention, monitoring capacity, dispute-resolution time, or verification budget, then linear growth in the number of such interfaces can force a qualitative regime change: beyond some scale, the collective becomes dominated by internal transaction/verification work rather than external task progress. Trust-capability, by reducing the marginal verification and negotiation burden, delays or prevents this transition.

Moreover, the additive form is intentionally conservative. If interfaces interact (for example, if misalignment at one interface increases the cost of resolving misalignment at others, or if conflicts propagate through shared resources), then the effective penalty to trustless regimes can be superadditive in $m(\alpha)$. The present theorem does not rely on such amplification; it only needs a lower bound that accumulates per interface.

Corollary 79 (Target-2 synthesis: selection implies trust-capable coordination)**.** *Suppose:*

1. *The task environment is rich enough that GGW-type measure-zero strictness holds (Assumption 4).*
2. *Communities/mindplexes are selected by realized performance on tasks (replicator, imitation, or multiplicative selection dynamics).*

3. *Trust-capable coordination is reachable (by design, mutation, learning, or institutional innovation) and is not prevented by hard constraints.*

Then, generically, the population-level dynamics converges toward trust-capable coordination, and the selection pressure toward trust-capability grows with coordination-interface complexity.

Remark 3551 (Intuitive content)**.** *This corollary is the promised bridge statement. It says: under rich tasks, performance-based selection, and reachability, trust-capability is not merely one option but a generic attractor of the population dynamics. Moreover, the larger and more internally articulated the collective (more interfaces), the stronger the pull.*

Two clarifications help fix the scope. First, "generically" inherits the meaning from the earlier GGW-type strictness assumption: if ties and exact degeneracies are measure-zero, then the set of task instances on which trust-capable and trustless regimes achieve exactly equal performance has negligible measure. This makes the selection gradient well-defined almost everywhere on the relevant task distribution and supports the intuition that small but persistent performance gaps accumulate over time under multiplicative selection.

Second, the reachability clause matters because selection can only favor what can be produced by the variation operators available to the population (engineering iteration, institutional reform, learning dynamics, or biological/evolutionary mutation and recombination). The corollary should thus be read as a conditional statement about evolutionary pressure: whenever a lineage of designs can move along a path that increases trust-capability without catastrophic side effects, the scaling advantage makes that direction an increasingly dominant basin of attraction as interface complexity grows.

In practical terms, this "pull" can manifest as convergent pressure toward mechanisms that reduce the need for exhaustive auditing at every boundary: shared world-model standards, credible commitment devices, alignment of incentives, interpretability tools that turn hidden states into communicable evidence, robust dispute resolution protocols, and governance structures that allow high-bandwidth delegation without collapsing into adversarial verification. The corollary does not claim that any particular mechanism is inevitable; it claims that the property *of being able to coordinate with low marginal trustless overhead is what selection will tend to amplify in scalable mindplex-like collectives.*

Remark 3552. *In reasoning-system terms, this is a license to treat "trust-capable coordination" as a default prediction for sufficiently open-ended multi-agent ecologies, unless there is explicit evidence of a hard constraint preventing it. Concretely, the claim is not that every trajectory must realize trust, but that absent a known blocking condition (e.g. binding communication limits, irreversible segregation, non-negotiable incompatibilities, or institutional constraints that forbid credible commitment), the selection dynamics considered make trust-capable regimes the natural "attractor" class to bet on when forming priors about long-run organizational form. In particular, the burden of argument shifts: rather than needing a special story for why trust appears, one needs a special story*

Proof. Items (1)-(2) yield convergence by Corollary 76 (deterministic) or Corollary 77 (stochastic). Interface growth strengthens the effect by Theorem 225 and Theorem 227. The deterministic/stochastic split here is only about the update rule (e.g. replicator-type flow versus finite-population sampling and drift); in both cases the role of the hypotheses is to guarantee that the trust-capable phenotype (or coalition) has strictly positive expected advantage in the relevant environment. The interface terms matter because they scale the payoff gap between regimes: as integration increases, the cost of trustless coordination typically accumulates across more boundaries, while the marginal benefit of a shared trust substrate can be reused across tasks, yielding a larger selection gradient in favor of trust-capability. □

Proof sketch. Combine the earlier results modularly: GGW-type strictness implies a positive mean advantage, which drives convergence under the selection dynamics considered. Separately, the interface-count model gives a quantitative lower bound on the advantage that increases with complexity/scale, sharpening the selection conclusion for large integrated systems. Intuitively, once a regime can (i) achieve the trustless baseline and (ii) sometimes do strictly better on the same task distribution by exploiting trust, then selection has no stable reason to keep the population at the trustless boundary except in degenerate cases where the better mode is inaccessible. The interface bound then functions as a "multiplicative amplifier": the more cross-module coordination problems an ecology must repeatedly solve, the more often trust-capability can cash out as reduced friction, lower overhead, or higher throughput, making the advantage not merely positive but increasingly robust to noise, drift, and heterogeneity in local subgames. □

66.9 Discussion: what this buys a reasoning system

Remark 3553 (Operational inference pattern)**.** *A reasoning system can treat Corollary 79 as a domain-general inference rule:*

> *If a multi-agent ecology is (i) under selection for task performance in a rich task distribution, and (ii) has a reachable trust-capable regime that can emulate trustless protocols, then expect institutional/memetic/evolutionary pressure toward trust-capable coordination, especially as task hierarchy and interface complexity grow.*

This is precisely the kind of cross-domain "background theorem" that can guide default inferences about social, cultural, scientific, or multi-agent AI scenarios. Here "reachable" should be read in an operational sense: there exists some feasible path in the space of policies, norms, mechanisms, or architectures (not necessarily short, not necessarily monotone) by which the ecology can arrive at a trust-capable configuration from plausible starting points. Likewise, "emulate trustless protocols" means the trust-capable regime does not lose the ability to fall back on adversarially-robust tools when needed (e.g. contracts, audits, cryptographic commitments, redundancy, formal verification, or other trust-minimizing mechanisms); rather, it strictly contains those tools as a special case and can selectively deploy trust when the environment makes it payoff-improving. For a reasoning system, this matters because it licenses a principled default: in the absence of contrary evidence, one can forecast not only individual-level behavior but the direction of institutional adaptation (which mechanisms get built, which norms get rewarded, which organizational interfaces get standardized) as task ecosystems become more compositional and interdependent.

Remark 3554. *One may read the section as an instance of a broader methodological stance (cf. [20]): when complex systems are too detailed to simulate end-to-end, we seek stable, selection-compatible inequalities that survive coarse-graining. The logic is deliberately modest: we do not claim that trust is inevitable in every world, only that under the stated hypotheses it is the generic direction of motion. In the Hyperseed conceptual vocabulary, this is an instance of reasoning from* Probability *(Hyperseed-Concept 139) and* Selection-like *update rules toward structural expectations about* Society *(Hyperseed-Concept 170) and large-scale integrated* Mindplex *systems (Hyperseed-Concept 113). The coarse-graining emphasis is crucial for a reasoning system: rather than tracking every micro-detail of beliefs and interactions, it can track summary variables (e.g. average advantage, accessibility of regimes, interface counts, and reuse of coordination infrastructure) that are* *preserved under aggregation and still constrain likely macroscopic outcomes. Moreover, "generic direction of motion" should be interpreted as a claim about comparative statics and typical perturbations: when the environment becomes more task-rich, when organizations scale, or when interfaces proliferate, the models predict that selection pressure tilts more strongly toward trust-capable coordination, even if local reversals, cycles, or temporary regressions can occur. Thus the result supplies a calibrated prior for high-level forecasting: if you observe increasing integration and repeated cross-boundary coordination demands, you should anticipate (and look for) mechanisms that make trust both more valuable and more enforceable, rather than assuming that purely trustless designs will remain dominant.*

67 Compassion as a Stabilized Invariant: Drift-Controlled Self-Modification in Resource-Rich Regimes

We formalize a class of self-modifying minds whose meta-goals are updated by a drift-controlled (proximal) optimization rule. We represent compassion as a distinguished coordinate or functional of the meta-goal state. We prove several results showing that if compassion is initially seeded and if (i) drift controls penalize rapid meta-goal change and (ii) the environment contains robust trust/coordination benefits or weakness-regularization pressures, then compassion becomes hard to lose without an explicit override that disables the drift controls or injects large noise. The results are intended as reasoning tools: they convert qualitative alignment intuitions into checkable sufficient conditions and bounds.

Remark 3555. *The formal stance here is intentionally minimalist: we treat a mind's "values" as part of its dynamical state and study how update rules constrain their motion. This is in the spirit of studying stability of self-referential goal systems via fixed-point and contraction ideas [10]. At the same time, the section is meant to be read as a bridge between philosophical vocabulary ("compassion," "override," "trust") and quantities that can be bounded and reasoned about, which is part of the broader Hyperseed program [1].*

67.1 Setup: self-modifying minds with drift control

67.1.1 Meta-goals and compassion

Definition 631 (Meta-goal parameter space)**.** *Let $\Theta \subseteq \mathbb{R}^d$ be the space of meta-goal parameters. A mind at (discrete) time t has meta-goal state $\theta_t \in \Theta$.*

Remark 3556. *Intuitively, Θ is the "space of ways the mind can care." Each $\theta \in \Theta$ is a compact encoding of meta-level preferences: not merely what the agent does, but what it optimizes* about *what it does. The index $t = 0, 1, 2, \ldots$ indicates discrete updates, and θ_t is the value-state at time t; the dimension d reflects that meta-goals are typically plural rather than monolithic (cf. Hyperseed-Concept 110).*

Remark 3557. *A simple example is $d = 2$ with $\theta = (c, k)$ where c measures a prosocial/compassion tendency and k measures a self-protective or self-prioritizing tendency; different societies or training regimes might move the pair along different trajectories. Another example is d large, with coordinates corresponding to weights on different classes of entities, different timescales, and different "modes" of cognition. This parameterization is useful because it lets us ask: under what update rules can θ change rapidly, and under what rules is it forced to drift slowly?*

Definition 632 (Compassion functional)**.** *A compassion functional is a map*

$$C : \Theta \to \mathbb{R}.$$

In the simplest case, $C(\theta)$ is the "compassion coordinate" $\theta^{(1)}$. We say compassion is seeded *if $C(\theta_0) = c_0 > 0$.*

Remark 3558. *Here C extracts from a full meta-goal state θ a* scalar summary *meant to track something like "how much welfare of others is treated as intrinsically relevant" (Hyperseed-Concept 81). The notation $\theta^{(1)}$ means the first coordinate of the vector $\theta \in \mathbb{R}^d$; thus $C(\theta) = \theta^{(1)}$ is the simplest possible choice: compassion is literally a coordinate axis in value-space.*

Remark 3559. *Two small examples clarify the flexibility: (i) If $\Theta = [0,1]^d$ and $C(\theta) = \theta^{(1)}$, then seeding means simply $\theta_0^{(1)} > 0$; later theorems will show that certain update rules make it difficult for $\theta_t^{(1)}$ to drop rapidly. (ii) If θ encodes a full welfare-weight vector over many classes of beings, $C(\theta)$ could be the minimum welfare weight assigned to any member of a protected set. This definition is useful because the later arguments only need that "compassion is a real-valued quantity that can be bounded per step," not that it has a unique psychological interpretation.*

Remark 3560. *Nothing in the theorems below requires interpreting compassion as a single scalar, but a scalar is convenient. In richer Hyperseed-style settings one can let C be a derived quantity from a value web, a moral-patient partition, or a pattern network property (see Section 4, and cf. [5]).*

Remark 3561. *The requirement $C(\theta_0) = c_0 > 0$ should be read as a minimal nondegeneracy condition rather than as a strong ethical claim: the analysis below is about* stability given a seed*, i.e., whether a pre-existing prosocial component can be made difficult to erase by default dynamics. In particular, the results do not assert that drift control* creates *compassion from $C(\theta_0) = 0$; instead, they quantify how much "work" (in terms of update incentives, overrides, or noise) is required to move $C(\theta_t)$ downward once it is present. This distinction matters because self-modification mechanisms can be excellent at preserving invariants while still being neutral about which invariant was initially installed.*

67.1.2 Drift-controlled self-modification

Definition 633 (Drift penalty / divergence)**.** *Let $D : \Theta \times \Theta \to [0, \infty)$ be a divergence (not necessarily symmetric) with $D(\theta, \theta) = 0$. The main case used here is*

$$D(\theta, \phi) := \frac{1}{2}\|\theta - \phi\|_2^2.$$

Remark 3562. *The role of $D(\theta, \phi)$ is to quantify the "cost of changing meta-goals from ϕ to θ." It is called a divergence rather than a distance because we do not require symmetry or triangle inequalities: changing from ϕ to θ could be easier than changing back. The special case $\frac{1}{2}\|\theta - \phi\|_2^2$ is the familiar quadratic penalty from proximal optimization: it makes large jumps expensive and is mathematically convenient because it is coercive (it grows without bound as $\|\theta - \phi\|_2$ grows).*

Remark 3563. *As a concrete example, if $\Theta = \mathbb{R}^d$ and $D(\theta, \phi) = \frac{1}{2}\|\theta - \phi\|_2^2$, then moving compassion by 0.1 in one step already incurs penalty $\approx 0.005\lambda$ in the objective used below; increasing λ increases how much "incentive" is required to justify the shift. This definition is necessary because "drift control" is precisely the claim that self-modification is not free: it is regularized, audited, or otherwise restrained (Hyperseed-Concept 100 provides one philosophical reading of why such restraints matter).*

Remark 3564. *Although we emphasize the quadratic choice, the divergence language is meant to include common alternatives that better fit constrained or informational geometries of value change. For instance, if Θ is a simplex of weights, then an entropic/KL-type divergence can encode that changing a near-zero weight is "hard" in relative terms; more generally, Bregman divergences capture mirror descent style updates where the cost of moving depends on the current location in Θ. The later bounds will typically rely only on nonnegativity and on relating per-step movement in θ (and hence movement in $C(\theta)$) to the update's net gain in the world-response objective.*

Definition 634 (World-response value functional)**.** *Let $F : \Theta \to \mathbb{R}$ be the mind's long-run expected value (under its best available policy/planning given meta-goals θ in its environment). We do not assume F is concave in general unless stated.*

Remark 3565. *$F(\theta)$ is a "best-achievable payoff if I were to adopt meta-goals θ." It is not a primitive reward function but a* world-response *functional: it already folds in the agent's planning competence, its model of the environment, and the environment's reaction to the agent's behavior induced by θ. This abstraction is intentionally agnostic about how the planning is done; it aligns with the idea that in resource-rich regimes (Hyperseed-Concept 162) an agent can approximate the "best available policy" relatively well [11].*

Remark 3566. *It is useful to keep in mind that F may implicitly include equilibrium or game-theoretic effects. For example, if other agents respond strategically to the mind's observed behavior (or to signals correlated with θ), then the "environment's reaction" term can encode trust, reputational dynamics, coalition access, and coordination opportunities. In this reading, "robust trust/coordination benefits" correspond to regions of Θ in which increasing compassion (as measured by $C(\theta)$) raises $F(\theta)$ because it unlocks cooperative surplus, reduces monitoring costs, or stabilizes mutual commitments; "weakness-regularization pressures" correspond to regions where low compassion induces brittle or over-exploitative strategies that perform poorly in the long run, thereby lowering $F(\theta)$ even if they appear locally advantageous.*

Remark 3567. *The separation between F and D is conceptually important: F captures the* instrumental *consequences of adopting meta-goals, while D captures the* inertia *or* friction *of changing them. Later, when we introduce a proximal-style update, the key quantity will be the tradeoff between immediate improvements in F and the drift cost measured by D. This is precisely the structure that turns qualitative claims like "it takes a lot to rewrite a stable value" into inequalities that bound how far θ_t (and thus $C(\theta_t)$) can move per step without an explicit mechanism that weakens the drift penalty.*

Remark 3568. *Two examples: (i) In a multi-agent world, higher compassion might yield better coordination and thus higher $F(\theta)$ via trust effects; later we model this explicitly. (ii) In an adversarial world, higher compassion might expose the agent to exploitation, lowering $F(\theta)$. The point of not assuming concavity is to permit such nontrivial landscapes; the theorems will often rely on drift control rather than benign geometry of F.*

Remark 3569. *It is also useful to read these examples as a reminder that F encodes* instrumental *performance as a function of meta-goals, not necessarily any intrinsic desirability of compassion itself. In particular, the same meta-goal parameter (here, compassion) can be locally helpful in some regions of the environment distribution and locally harmful in others, producing multiple basins of attraction, saddle-like regions, or sharp cliffs in $F(\theta)$ even when the underlying world model is smooth.*

Definition 635 (Drift-controlled self-modification rule)**.** *Fix $\lambda > 0$. The drift-controlled self-modification (DCSM) update is*

$$\theta_{t+1} \in \arg\max_{\theta\in\Theta} \Big(F(\theta) - \lambda D(\theta, \theta_t)\Big). \tag{51}$$

Remark 3570. *The definition only requires that $D(\theta, \theta_t)$ act as a* drift penalty*; in typical applications one assumes at least $D(\theta_t, \theta_t) = 0$ and $D(\theta, \theta_t) \geq 0$ so that remaining at θ_t is always feasible and never incurs negative "cost." We do not require symmetry $D(\theta, \theta') = D(\theta', \theta)$ or the triangle inequality: many useful choices (e.g. KL divergence between internal goal-distributions, or Bregman divergences tied to a convex potential) are directional and still serve as meaningful notions of "how radical" a self-change is.*

Remark 3571. *A brief note on notation:* arg max *denotes the* set *of maximizers, not necessarily a single point; thus $\theta_{t+1} \in \arg\max(\cdots)$ means we allow multiple optima and choose any of them. The constant $\lambda > 0$ is the strength of drift control: larger λ means changes away from θ_t are penalized more harshly. The expression $F(\theta) - \lambda D(\theta, \theta_t)$ is a one-step tradeoff between "how good the new meta-goals are" and "how far they move from the old meta-goals"; formally it is a proximal or regularized optimization step.*

Remark 3572. *When $D(\theta, \theta_t) = \|\theta - \theta_t\|^2$ in a Euclidean parameterization,* (51) *reduces to a familiar quadratic trust-region/proximal update: it prefers improvements in F that can be reached with small parameter motion. When D is KL divergence between distributions representing meta-goals, the update instead constrains* information-theoretic *drift, which can be more invariant to reparameterization and more aligned with "behavioral" change than raw coordinate distance. This flexibility is important because the same intuitive notion of "conservative self-modification" can be implemented in multiple representational layers.*

Remark 3573. *One can also view λ as setting an implicit exchange rate between performance and stability: an improvement ΔF must justify paying a drift cost of roughly $\Delta F/\lambda$. In later arguments, it will matter that this exchange rate is explicit and time-local, allowing stability bounds that depend on λ and the geometry induced by D, rather than on global concavity assumptions about F.*

Remark 3574. *Intuitively,* (51) *says:* self-modify, but do so conservatively*. The mind is allowed to improve its own meta-goals, yet each improvement is constrained by an inertia term. This captures the engineering idea that robust systems include friction against abrupt internal changes, whether implemented via verification, consensus, redundancy, or explicit regularizers (cf. the stability motivations in [2] and the fixed-point perspective in [10]). This definition is useful because it turns a qualitative claim ("values drift slowly") into an inequality-driven dynamical law amenable to bounds.*

Remark 3575. *A further advantage of the proximal form is that it separates two failure modes: (a) the system proposes a new θ that would be high-performing* if adopted*, versus (b) the system actually* moves *to that proposal. Drift control constrains (b) even if (a) explores aggressively, and this is precisely the kind of separation that makes it possible to talk about "safe" internal search combined with "slow" external commitment.*

Definition 636 (Override event)**.** *An* override *is any step that violates the drift control regime, e.g.: (i) setting λ to (near) 0 for the step, or (ii) replacing $D(\cdot, \theta_t)$ by a much weaker penalty, or (iii) injecting step noise large enough to dominate the penalty term. Formally, an override is any update not representable as* (51) *with the declared (λ, D).*

Remark 3576. *Note that the override notion is intentionally* model-relative*: it is defined with respect to the particular declared pair* (λ, D)*. Thus, even if the system continues to execute an update of the form* $\arg\max(F - \tilde{\lambda}\tilde{D})$*, it still counts as an override when* $(\tilde{\lambda}, \tilde{D}) \neq (\lambda, D)$ *in a way that materially weakens drift control. This aligns with the interpretive goal that "override" flags steps where the claimed stability mechanism is no longer the operative mechanism.*

Remark 3577. *The philosophical point of the override definition is to distinguish two kinds of change: (1) changes that occur* within *the declared rule of self-modification, and (2) changes that occur by* breaking *the rule. The later theorems show that certain undesirable transitions (e.g. sudden compassion collapse) are difficult under (1), so if they occur, one should infer (2)—either an explicit disablement of drift control or a perturbation so large it effectively nullifies the penalty. This directly supports the intended use as a reasoning tool: one can ask which regime parameters must have been altered to explain observed value shifts.*

Remark 3578. *In applications, an "override" can be endogenous (the agent chooses to relax its own constraints) or exogenous (a fault, attack, or rare perturbation effectively bypasses the regularizer). The formal definition treats both uniformly: the key question is not* why *the regime was broken, but whether the observed transition can be accounted for by an update that still pays the declared drift cost. This lets later results function as diagnostic tools: if the transition is too sharp to be paid for under* D *and* λ*, then some form of regime break is implicated.*

Assumption 5 (Resource-rich regime: bounded optimization error and small noise)**.** *We idealize a resource-rich regime in two (optional) ways:*

1. *(Near-optimality) The chosen* θ_{t+1} *is an* η*-approximate maximizer of* $F(\theta) - \lambda D(\theta, \theta_t)$*, meaning it achieves value within* η *of the supremum.*

2. *(Small noise) When we model stochasticity, the injected noise is sub-Gaussian with scale* σ *that can be made small by spending resources.*

For deterministic theorems, we set $\eta = \sigma = 0$*.*

Remark 3579. *One may interpret* η *as capturing any combination of bounded solver suboptimality, discretization error, and bounded myopia in internal search. Allowing* $\eta > 0$ *is important because even very capable systems can face time constraints, approximation in world-model evaluation, or deliberate heuristics; the point is that in a resource-rich regime, these imperfections are bounded and can be driven down with additional expenditure.*

Remark 3580. *This assumption isolates a regime in which "more resources" plausibly mean "more reliable self-modification" (Hyperseed-Concept 162). Near-optimality says the mind can in fact solve (or nearly solve) the drift-regularized optimization at each step; small noise says random perturbations can be suppressed by redundancy, computation, and validation. Both are idealizations of the engineering thesis that sufficiently capable minds can make their own updates more disciplined rather than less disciplined [11].*

Remark 3581. *The small-noise clause is stated in sub-Gaussian form because it supports standard concentration bounds: rare large deviations decay at least as fast as Gaussian tails with parameter* σ*. This is a natural proxy for the idea that, after sufficient checking and averaging, the remaining stochasticity is dominated by many small independent sources rather than by heavy-tailed shocks. Later metastability statements will hinge on exactly this kind of tail control, since heavy-tailed noise could make "rare" catastrophic drift comparatively common.*

Remark 3582. *The parameters η and σ let the subsequent theorems cleanly separate deterministic barriers (which hold even with perfect optimization and no randomness) from metastability claims (which quantify the probability of rare "bad" events under small noise). They are necessary because without some control of optimization error/noise, any stability statement can be defeated by simply postulating enough randomness.*

Remark 3583. *It is also conceptually helpful to distinguish* bounded error *from* override*: a step with $\eta > 0$ is still a DCSM-type step (the agent is approximately optimizing the declared regularized objective), whereas an override changes the objective or the effective penalty itself. Thus, η and σ represent imperfections that preserve the regime, while overrides represent regime violations; the later results can therefore state conditions under which stability is robust to small imperfections but not to explicit bypassing of drift control.*

67.2 Target 3 core: compassion is hard to lose under drift control

67.2.1 A general step-size barrier: you cannot drift far without paying

The first result is deliberately architecture-agnostic: it only uses the fact that the update includes a quadratic (or more generally coercive) penalty.

Theorem 228 (Step-size bound from drift penalty)**.** *Assume Θ is any set and $D(\theta, \phi) = \frac{1}{2}\|\theta - \phi\|_2^2$. Assume F is bounded above on Θ by $F_{\max}$ and define $F(\theta_t)$ as the value at time t. If θ_{t+1} satisfies* (51)*, then*

$$\|\theta_{t+1} - \theta_t\|_2^2 \le \frac{2}{\lambda}\big(F_{\max} - F(\theta_t)\big). \tag{52}$$

If we only have η-approximate maximization, then

$$\|\theta_{t+1} - \theta_t\|_2^2 \le \frac{2}{\lambda}\big(F_{\max} - F(\theta_t) + \eta\big). \tag{53}$$

Remark 3584. *Intuitively, the theorem says:* if you regularize self-modification with a quadratic penalty, then each step has a maximum length unless you can gain a lot of value. *The upper bound depends on how far the current value $F(\theta_t)$ is from the global upper bound $F_{\max}$ and on the drift strength λ. When λ is large, even moderate jumps require substantial improvement in F; conversely, large jumps are possible only when the agent is currently leaving a lot of value "on the table."*

Remark 3585. *This is important because it is almost completely indifferent to the internal structure of the mind. One does not need to assume convexity, smoothness, or a particular learning algorithm. The result connects directly to later parts of the section: it provides the raw per-step barrier that, when combined with trust-attractor structure (Section 2) or weakness barriers (Section 4), yields a multi-pronged explanation of why compassion does not vanish "by accident." In the broader project, this kind of bound is one way to make "stable self" claims precise under self-modification [10].*

Proof. Because $\theta_t \in \Theta$ is feasible, optimality gives

$$F(\theta_{t+1}) - \frac{\lambda}{2}\|\theta_{t+1} - \theta_t\|_2^2 \ge F(\theta_t) - 0.$$

Rearrange:

$$\frac{\lambda}{2}\|\theta_{t+1} - \theta_t\|_2^2 \le F(\theta_{t+1}) - F(\theta_t) \le F_{\max} - F(\theta_t).$$

This yields (52). The η-approximate case adds η to the RHS. $\square$

Remark 3586 (Proof sketch and key intuition)**.** *The strategy is to compare the chosen next state θ_{t+1} to the trivial alternative "do nothing," i.e. choose θ_t again. Since θ_{t+1} maximizes the regularized objective, its regularized score cannot be lower than θ_t's score. This single comparison already forces the penalty term $\frac{\lambda}{2}\|\theta_{t+1} - \theta_t\|_2^2$ to be paid for by a corresponding gain in F, and the gain is in turn bounded by $F_{\max} - F(\theta_t)$. Geometrically, one can picture the quadratic penalty as a steep "bowl" centered at θ_t; to move far away, F must rise enough to climb out of the bowl.*

Corollary 80 (Per-step compassion change is bounded)**.** *Let $C(\theta) = \theta^{(1)}$ be the compassion coordinate. Under the assumptions of Theorem 1,*

$$|C(\theta_{t+1}) - C(\theta_t)| \le \|\theta_{t+1} - \theta_t\|_2 \le \sqrt{\frac{2}{\lambda}\big(F_{\max} - F(\theta_t) + \eta\big)}. \tag{54}$$

Remark 3587. *In plain language, this corollary specializes the generic step-size bound to the particular coordinate that we interpret as compassion. The inequality $|x_1 - y_1| \le \|x - y\|_2$ is the statement that changing one coordinate by a large amount forces the whole vector to change by at least that much. So drift control on the full meta-goal vector automatically constrains how quickly compassion can move, provided compassion is a coordinate (or Lipschitz functional).*

Proof. The first inequality is $|x_1 - y_1| \le \|x - y\|_2$. The second is Theorem 1. □

Remark 3588 (Proof sketch and geometric picture)**.** *The proof is a one-line projection argument: the compassion coordinate is a projection of θ onto an axis, and the Euclidean norm dominates any single-axis displacement. One may visualize the update as selecting a point in Θ; the full displacement vector has length $\|\theta_{t+1} - \theta_t\|_2$, and the compassion displacement is merely the shadow of this vector on one axis, hence it cannot exceed the full length.*

Remark 3589 (Interpretation)**.** *Large λ makes large compassion drops (or any meta-goal jumps) expensive. Thus, absent an override, rapid "value flips" are suppressed even if F is wild. This is one formal sense in which compassion becomes "hard to lose" in a drift-controlled regime.*

67.2.2 Attraction to a positive compassion fixed point under trust benefits

Now we add a substantive condition: that in a broad class of social/multi-agent environments, increasing compassion improves long-run payoffs through trust-capable coordination. This connects Target 3 to Target 2 without re-proving Target 2.

Assumption 6 (One-dimensional trust-augmented value model)**.** *Assume $\theta = (c, u) \in \mathbb{R} \times \mathbb{R}^{d-1}$ where $c = C(\theta)$. Assume the value decomposes as*

$$F(c, u) = G(u) + B\,T(c) - Kc, \tag{55}$$

where $B, K > 0$ are constants, and $T : [0,1] \to [0,1]$ is continuously differentiable, strictly increasing and concave (trust saturates).

Remark 3590. *This assumption isolates a single causal channel: compassion c affects long-run value by enabling trust-capable coordination, summarized by $B\,T(c)$, while also carrying an intrinsic or opportunity cost Kc. The remaining coordinates u collect all other meta-goal degrees of freedom; their effect on value is abstracted as $G(u)$. The concavity of T expresses diminishing returns: beyond some point, "more compassion" does not proportionally increase trust because trust saturates. This is a stylized formalization of the prosocial-efficiency thesis discussed earlier [6].*

Remark 3591. *A simple example is $T(c) = \sqrt{c}$ on $[0, 1]$, which is increasing and concave: early increases in compassion have large marginal effect on trust, but later increases have smaller marginal effect. Another common example is a smooth saturating curve such as $T(c) = \log(1+\alpha c)/\log(1+\alpha)$. The decomposition* (55) *is useful because it lets the compassion dynamics be studied in one dimension without assuming the entire meta-goal landscape is well behaved.*

Definition 637 (DCSM update for c)**.** *With $D(\theta, \theta_t) = \frac{1}{2}\|\theta - \theta_t\|_2^2$, the update* (51) *induces a 1D proximal update for compassion:*

$$c_{t+1} \in \arg\max_{c\in[0,1]} \Big(BT(c) - Kc - \frac{\lambda}{2}(c - c_t)^2\Big). \tag{56}$$

Remark 3592. *Equation* (56) *is the proximal/self-regularized update specialized to one coordinate. The term $BT(c)-Kc$ is the "instantaneous" part of the trust-versus-cost tradeoff, and the quadratic term is drift control. The constraint $c \in [0, 1]$ is a normalization: compassion is scaled so that* 0 *means no compassion signal and* 1 *means maximal compassion in the chosen units. This definition is useful because it makes the compassion subsystem a closed dynamical system and allows attractor analysis by elementary monotonicity arguments.*

Theorem 229 (Monotone convergence to a positive compassion attractor)**.** *Assume the trust model* (55) *and the update* (56)*. Assume further that $BT'(0) > K$, so the unique maximizer of $c \mapsto BT(c) - Kc$ is a strictly positive $c^* \in (0, 1]$. Then for any initial $c_0 \in (0, 1]$, the sequence (c_t) generated by* (56) *satisfies:*

1. *(Order preservation) If $c_t < c^*$ then $c_{t+1} \geq c_t$. If $c_t > c^*$ then $c_{t+1} \leq c_t$.*
2. *(Invariance of positivity) For all t, $c_t > 0$.*
3. *(Convergence) $c_t \to c^*$ as $t \to \infty$.*

Remark 3593. *Informally, the theorem says: if at low compassion the* marginal trust benefit *$BT'(0)$ exceeds the* marginal cost *K, then the system has a stable positive compassion "set point" c^*, and drift-controlled self-modification will move compassion toward it monotonically. This is an attractor statement: compassion is not merely* preserved *by inertia; it is actively* restored *by incentive gradients created by trust.*

Remark 3594. *The result is important because it explains one mechanism by which prosociality can become self-stabilizing in a capable agent: if trust-capable coordination is generically advantageous (as argued in the selection-oriented setting of the surrounding sections [6]), then in a drift-controlled regime the value landscape itself creates a basin pulling c_t away from zero. This connects the purely regularization-based barrier of Theorem 1 to an endogenous dynamical restoration force.*

Proof. Let

$$\phi_t(c) := BT(c) - Kc - \frac{\lambda}{2}(c - c_t)^2.$$

Concavity: since T is concave, $BT(c)-Kc$ is concave, and subtracting a quadratic keeps concavity. Thus ϕ_t has a (possibly non-unique) maximizer set.

For interior maximizers, first-order optimality yields

$$BT'(c_{t+1}) - K - \lambda(c_{t+1} - c_t) = 0, \quad \text{so} \quad c_{t+1} = c_t + \frac{1}{\lambda}(BT'(c_{t+1}) - K). \tag{57}$$

Define $h(c) := BT'(c) - K$. Since T is concave, T' is decreasing, hence h is decreasing. By assumption $h(0) > 0$, and since T' decreases, there is a unique c^* with $h(c^*) = 0$.

If $c_t < c^*$, then any maximizer c_{t+1} must satisfy $c_{t+1} \leq c^*$ (otherwise moving slightly down increases $BT(c) - Kc$ while also reducing quadratic penalty). For any $c_{t+1} \leq c^*$, we have $h(c_{t+1}) \geq 0$, so (57) gives $c_{t+1} \geq c_t$. Similarly if $c_t > c^*$, then maximizers satisfy $c_{t+1} \geq c^*$, so $h(c_{t+1}) \leq 0$ and $c_{t+1} \leq c_t$. This proves monotonicity toward c^*.

Because c_t is monotone and bounded in $[0, 1]$, it converges to some limit $\bar{c}$. Taking limits in (57) (using continuity of T') yields $h(\bar{c}) = 0$, so $\bar{c} = c^*$. Positivity holds because if $c_0 > 0$ and the sequence is monotone toward $c^* > 0$, it cannot cross 0. □

Remark 3595 (Proof sketch and why the key steps work)**.** *The proof uses three classical ingredients: (i) concavity ensures that maximizing ϕ_t is well-behaved and that first-order conditions capture optimality at interior points; (ii) the monotonicity of T' (a consequence of concavity of T) makes $h(c) = BT'(c) - K$ a decreasing function with a unique zero c^*; (iii) the fixed-point-like equation* (57) *shows that the sign of $h(c_{t+1})$ determines whether c_{t+1} lies above or below c_t. Visually, c^* is where the marginal trust benefit equals marginal cost; drift control then makes the update a damped move toward that balance point rather than a jump.*

Remark 3596 (Why this matters)**.** *The theorem turns "compassion helps coordination" into an attractor statement: in resource-rich drift-controlled self-modification, compassion does not merely persist; it is pulled toward a stable positive fixed point whenever trust benefits dominate marginal costs at low compassion.*

67.3 Hard-to-lose without override: quantitative barrier and metastability

The previous theorems were deterministic. We now add small noise to represent exploration, approximation error, and incidental perturbations, and prove a "metastability" bound: dropping compassion below a threshold becomes exponentially unlikely as noise decreases and drift control increases.

Remark 3597. *The phrase "without override" is meant literally: the stochastic model below assumes the same restoring drift toward the attractor is continuously in force. In applications, a genuine override corresponds to a change in the update rule (e.g., changing a, shifting c^*, or injecting a systematic bias term), not merely an unusually large draw of ξ_t. The point of the metastability bound is to make this distinction quantitative.*

67.3.1 A linearized stochastic model (Ornstein-Uhlenbeck style)

Assumption 7 (Linear mean-reverting compassion dynamics)**.** *Fix $a \in (0, 1)$, a target $c^* \in (0, 1]$, and noise scale $\sigma > 0$. Assume*

$$c_{t+1} = a\, c_t + (1 - a)c^* + \sigma\xi_t, \tag{58}$$

where $\{\xi_t\}$ are independent mean-zero 1-sub-Gaussian random variables: $\mathbb{E}[\exp(s\xi_t)] \leq \exp(s^2/2)$ for all $s \in \mathbb{R}$.

Remark 3598. *Equation* (58) *is the discrete-time analogue of an Ornstein–Uhlenbeck process: if one takes a small step size Δt and sets $a \approx 1 - \lambda\Delta t$, then* (58) *corresponds heuristically to $dc_t = \lambda(c^* - c_t)\, dt + \sigma\, dW_t$ in continuous time. This analogy is useful only for intuition here; the results below are fully discrete-time and require no diffusion limit.*

Remark 3599. *This is a discrete-time, mean-reverting process: in expectation, c_{t+1} is pulled toward c^*, and $a \in (0,1)$ controls how strong the pull is (smaller a means stronger reversion). The random variables ξ_t model small perturbations; "1-sub-Gaussian" is a standard tail condition implying Gaussian-like concentration bounds for sums. The parameter σ scales the noise magnitude.*

Remark 3600. *Because* (58) *is linear with additive noise, c_t is not automatically confined to* $[0,1]$ *even if $c_0 \in [0,1]$. This is intentional: the model is a local linearization near an interior attractor where boundary effects are negligible. If one wishes to enforce feasibility, one can interpret* (58) *as governing an unconstrained latent variable and then apply a projection or squashing map back to* $[0,1]$*; the tail bounds still give conservative estimates for threshold-crossing events as long as the relevant thresholds lie away from the boundary.*

Remark 3601. *The linear model arises by localizing the proximal update near an attractor and treating the drift control as mean reversion. In a resource-rich regime, σ can be made small (more computation, more verification, more redundancy).*

Remark 3602. *From an interpretive standpoint, a plays a role analogous to an effective drift-control strength: as drift control increases, the system becomes more strongly anchored to its attractor, which is captured here by reducing a away from* 1*. This is one way to translate the qualitative claim "resource-rich minds can keep their internal perturbations small" into a quantitative statement [11].*

Remark 3603. *It is also helpful to separate two distinct ways a system can become "stable": (i) decreasing σ reduces the typical magnitude of random exploration/approximation error, while (ii) decreasing a increases the restoring drift that counteracts any perturbation that does occur. The bounds below capture both effects, and do so in a way that makes their interaction explicit.*

Lemma 172 (Explicit representation)**.** *Under* (58)*,*

$$c_t = c^* + a^t(c_0 - c^*) + \sigma \sum_{k=0}^{t-1} a^{t-1-k}\xi_k. \tag{59}$$

Remark 3604. *This lemma simply unrolls the recursion and expresses c_t as the deterministic relaxation toward c^* plus a geometrically weighted sum of past noises. The weights a^{t-1-k} shrink older noise terms, reflecting the fact that mean reversion "forgets" perturbations exponentially fast.*

Remark 3605. *Taking expectations in* (59) *immediately yields*

$$\mathbb{E}[c_t] = c^* + a^t(c_0 - c^*),$$

so the mean approaches c^* at geometric rate a^t. Moreover, the accumulated noise term is a weighted sum whose effective variance proxy is on the order of $\sigma^2/(1-a^2)$, foreshadowing the appearance of the stabilizing factor $(1-a^2)$ in the tail bounds: stronger reversion (smaller a) reduces the long-run dispersion around c^*.

Proof. Unroll the recurrence:

$$c_{t+1} - c^* = a(c_t - c^*) + \sigma\xi_t,$$

and iterate. ∎

Remark 3606 (Proof sketch)**.** *The high-level strategy is standard: rewrite the update in terms of deviations from the fixed point c^*, obtaining a linear recurrence, and then repeatedly substitute to express $c_t - c^*$ in terms of $c_0 - c^*$ and the accumulated noise terms. Because the recurrence is linear with constant coefficient a, the resulting coefficients form a geometric series.*

Theorem 230 (Exponential tail bound for compassion drop)**.** *Fix a threshold $c_{\min} < c^*$. Define the margin*

$$m_t := \mathbb{E}[c_t] - c_{\min} = c^* + a^t(c_0 - c^*) - c_{\min}.$$

Then for each $t \geq 1$,

$$\mathbb{P}(c_t \leq c_{\min}) \leq \exp\left(-\frac{m_t^2(1-a^2)}{2\sigma^2}\right). \tag{60}$$

Consequently, for any horizon $T \geq 1$,

$$\mathbb{P}\left(\min_{1\leq t\leq T} c_t \leq c_{\min}\right) \leq \sum_{t=1}^{T} \exp\left(-\frac{m_t^2(1-a^2)}{2\sigma^2}\right) \leq T \exp\left(-\frac{m_\infty^2(1-a^2)}{2\sigma^2}\right), \tag{61}$$

where $m_\infty := c^* - c_{\min} > 0$.

Remark 3607. *The margin m_t is the "distance to the threshold in expectation." If $c_0 \geq c_{\min}$ then m_t is nonnegative for all t, and if $c_0 \leq c^*$ then m_t increases monotonically to m_∞. In particular, after a modest burn-in period on the order of $\log(1/m_\infty)/\log(1/a)$, the bound is well-approximated by the steady-state expression with m_∞.*

Remark 3608. *In ordinary language: the probability that compassion falls below a fixed threshold is exponentially small in the signal-to-noise ratio m_t/σ, with an additional stabilizing factor $(1-a^2)$ that improves as mean reversion strengthens. Over a time horizon T, the probability that the threshold is ever crossed is at most T times the worst per-step probability (union bound). Thus "hard to lose" becomes a large-deviation statement: losing compassion requires an atypically large negative fluctuation against the restoring drift.*

Remark 3609. *The functional form $\exp(-\text{const}/\sigma^2)$ is characteristic of metastability: as noise decreases, rare excursions against the drift become dramatically less frequent. In this linearized setting, the "barrier" is not a literal potential barrier but rather the combination of (i) the distance m_t to the undesired region and (ii) the effective noise level after geometric forgetting, captured by $(1-a^2)/\sigma^2$. This is the sense in which drift control converts a qualitative claim ("it tends to come back") into a quantitative one ("it is exponentially unlikely to stay down").*

Remark 3610. *This matters because it gives a quantitative complement to deterministic drift barriers: even if each step can fluctuate, the probability of sustained collapse is sharply controlled when σ is small. In the intended application, an observed compassion collapse in a system believed to have small σ and strong anchoring is evidence for an override (regime change) rather than mere stochastic drift, which is precisely the kind of inference these results are meant to support.*

Remark 3611. *Note that* (61) *is intentionally conservative: it uses a union bound and therefore does not exploit temporal dependence between the events $\{c_t \leq c_{\min}\}$. In many stable mean-reverting processes, threshold-crossing probabilities over a horizon can be substantially smaller than the union bound suggests, but the union bound already suffices to formalize the key claim that the risk shrinks rapidly as $(1-a^2)/\sigma^2$ grows.*

Proof. By Lemma 59,

$$c_t - \mathbb{E}[c_t] = \sigma \sum_{k=0}^{t-1} a^{t-1-k}\xi_k.$$

A weighted sum of independent 1-sub-Gaussians is sub-Gaussian with variance proxy equal to the sum of squared weights. Thus $Z_t := \sum_{k=0}^{t-1} a^{t-1-k}\xi_k$ is sub-Gaussian with parameter

$$\sum_{k=0}^{t-1} a^{2(t-1-k)} = \sum_{j=0}^{t-1} a^{2j} \le \frac{1}{1-a^2}.$$

Hence for $u > 0$,

$$\mathbb{P}(Z_t \le -u) \le \exp\left(-\frac{u^2(1-a^2)}{2}\right).$$

Set $u = m_t/\sigma$ to obtain (60). The horizon bound follows by a union bound over $t \le T$ and the monotone limit $m_t \to m_\infty$. □

Remark 3612. *The key inequality step can be seen directly from the moment-generating-function definition of sub-Gaussianity: independence implies the mgf of a weighted sum factorizes, and the weights a^{t-1-k} enter only through their squares when bounding the mgf. This is why geometric forgetting translates into the simple factor $\sum_j a^{2j} \le (1-a^2)^{-1}$.*

Remark 3613 (Proof sketch and concentration intuition)**.** *The proof reduces the event $c_t \le c_{\min}$ to a negative deviation of a sub-Gaussian sum. The key step is that geometric weights preserve sub-Gaussianity, and the effective variance is the sum of squared weights, which is bounded by a geometric series $\le 1/(1-a^2)$. Geometrically, mean reversion damps the influence of old noise, so the process cannot accumulate variance linearly in time; it saturates at a constant scale set by $\sigma^2/(1-a^2)$. The union bound is crude but sufficient for the stated "polynomial horizon" corollary. In more concrete terms, for a typical mean-reverting linear update one can unroll the recursion to obtain a representation of the centered state as a geometrically weighted noise history, e.g.*

$$c_t - c^* \;=\; a^t(c_0 - c^*) \;+\; \sigma \sum_{k=1}^{t} a^{t-k}\,\xi_k,$$

where (ξ_k) are mean-zero, unit sub-Gaussian increments. The deterministic term $a^t(c_0-c^*)$ decays, and the random term is a weighted sum. If each ξ_k is sub-Gaussian with proxy variance 1, then $\sum_{k=1}^t w_k\xi_k$ is sub-Gaussian with proxy variance $\sum_{k=1}^t w_k^2$; here $w_k = \sigma a^{t-k}$, so

$$\sum_{k=1}^{t} w_k^2 \;=\; \sigma^2 \sum_{j=0}^{t-1} a^{2j} \;\le\; \frac{\sigma^2}{1-a^2}.$$

This yields a standard tail inequality of the form

$$\mathbb{P}(c_t - c^* \le -\Delta) \;\le\; \exp\!\left(-\frac{\Delta^2(1-a^2)}{2\sigma^2}\right)$$

(up to conventional constants depending on the chosen sub-Gaussian normalization), which is the quantitative sense in which the process is "metastable": deviations of fixed size Δ are exponentially suppressed in $1/\sigma^2$ once mean reversion prevents variance from growing with t.

Corollary 81 (Override necessity in polynomial time)**.** *Fix $c_{\min} < c^*$ and consider any polynomial horizon $T = \mathrm{poly}(1/\sigma)$. Then as $\sigma \to 0$,*

$$\mathbb{P}\left(\min_{1\le t\le T} c_t \le c_{\min}\right) \to 0$$

exponentially fast in $1/\sigma^2$. Thus, to drive compassion below $c_{\min}$ on typical runs in such a regime, one must (with high probability) perform an override that effectively increases σ or reduces mean reversion (increasing a toward 1). A typical bound consistent with the preceding remark is

$$\mathbb{P}\left(\min_{1\le t\le T} c_t \le c_{\min}\right) \;\le\; \sum_{t=1}^{T} \mathbb{P}(c_t \le c_{\min}) \;\lesssim\; T \exp\!\left(-\frac{(c^* - c_{\min})^2(1-a^2)}{2\sigma^2}\right),$$

where the $\lesssim$ hides only constant-factor choices from the sub-Gaussian tail convention and the (usually negligible) effect of the decaying initialization term. The point of restricting to $T = \mathrm{poly}(1/\sigma)$ is that $\log T = O(\log(1/\sigma))$ cannot compete with the $1/\sigma^2$ in the exponent: even after union-bounding over polynomially many time steps, the resulting probability still vanishes at an exponential rate in $1/\sigma^2$. In this sense the "barrier" height is set by $(c^* - c_{\min})\sqrt{1-a^2}/\sigma$: as $\sigma \to 0$ (with fixed $a < 1$), typical trajectories almost never cross below $c_{\min}$ without intervention.

Remark 3614. *The content of the corollary is not that overrides are metaphysically necessary, but that under the assumed regime they are* statistically *necessary for a reliable compassion collapse within any reasonable (polynomial) time horizon. This distinction mirrors a common theme in rigorous reasoning about dynamical systems and computation: what is possible in principle may nonetheless be so measure-theoretically or probabilistically rare as to be ignorable for planning, until some control parameter is changed. Equivalently, the corollary is a statement about typicality: there exist trajectories (indeed, infinitely many) in which an unusually unlucky sequence of shocks pushes c_t below $c_{\min}$, but the measure of such trajectories is exponentially small in $1/\sigma^2$ when mean reversion is strong. This is precisely the regime in which "waiting for failure" is not a practical mechanism for producing failure on demand, because the expected waiting time to see such a rare event grows super-polynomially (and often exponentially) as $\sigma \to 0$. From a control perspective, an "override" functions as a parameter shift that changes which events are typical: increasing σ thickens the tails of the noise, while moving a toward 1 increases the effective memory of past noise and hence enlarges the variance proxy $\sigma^2/(1-a^2)$, both of which reduce the large-deviation penalty required to cross the threshold.*

Remark 3615 (Interpretation)**.** *This is a precise "hard to lose" statement: in a resource-rich drift-controlled regime (small σ, strong mean reversion), compassion loss is a rare large deviation event. Making it common requires changing the regime parameters, i.e., override. One can read the result as establishing a quantitative metastability barrier: below the scale $\sigma/\sqrt{1-a^2}$ the process exhibits only small fluctuations around c^*, while a drop of macroscopic size $c^* - c_{\min}$ is suppressed by an exponential factor. The corollary also clarifies what kinds of interventions matter: merely extending the time horizon to any polynomial length does not qualitatively change the picture, whereas modifying the volatility level (via σ) or weakening mean reversion (via a) directly changes the barrier height and can turn a negligible event into a routine one. In that sense, the "override" is not a philosophical add-on but the mathematically relevant act of moving the system out of the concentration regime where the hard-to-lose guarantee holds.*

67.4 Weakness-regularized compassion: de-compassionization costs distinctions

We now connect Target 3 to the Hyperseed weakness theme: reducing compassion typically requires making fine distinctions about who counts, who matters, or who is safe to exploit. Distinctions reduce weakness (flexibility), and weakness-regularization therefore creates an additional barrier (Hyperseed-Concept 202; see also [3] and [2]).

67.4.1 A simple moral-patient partition model

Definition 638 (Moral treatment policy and partition)**.** *Let $\mathcal{A}$ be a finite set of other agents (or morally relevant entities). A treatment policy is a map $w : \mathcal{A} \to \mathbb{R}$ assigning welfare weight. This induces a partition of $\mathcal{A}$ into level sets of equal weight.*

Remark 3616. *This definition encodes, in a deliberately austere way, the idea that a mind's moral posture can be seen as a weighting of others: $w(a)$ is how much the mind counts the welfare of entity a in its objective. The induced partition is the set of equivalence classes "treated equally" by w: two agents a, a' lie in the same block if $w(a) = w(a')$. In other words, the partition describes the coarseness of moral categories (Hyperseed-Concept* **??** *and Hyperseed-Concept 98).*

Remark 3617. *A simple example: if $\mathcal{A} = \{1, 2, 3, 4\}$ and w assigns all four the same weight, the partition has one block $\{1, 2, 3, 4\}$. If instead $w(1) = w(2) = 1$ but $w(3) = w(4) = 0$, the partition has two blocks $\{1, 2\}$ and $\{3, 4\}$, representing an "in-group/out-group" split. This model is useful because it lets us quantify how much extra structure (distinctions) is needed to implement de-compassionization.*

Definition 639 (Mass and weakness of a partition)**.** *Let μ be a probability distribution on $\mathcal{A}$ (importance mass). If the partition has blocks $B_1, \ldots, B_k$, define its weakness as*

$$W := \sum_{i=1}^{k} \mu(B_i)^2. \tag{62}$$

This is maximal ($W = 1$) for the coarsest partition (one block) and decreases as the partition is refined.

Remark 3618. *The distribution μ expresses which entities matter how much in the environment or in the agent's encounters; it may be uniform, or it may concentrate on frequently interacting agents. The quantity $W = \sum_i \mu(B_i)^2$ is a "concentration" index of the partition: when there is one block, $\mu(B_1) = 1$ and $W = 1$; when the partition is split into many equally weighted blocks, each block mass is small and W decreases. This is closely related to standard impurity or collision-probability measures and can be read as a weakness notion in the sense that coarse categorizations preserve flexibility across contexts (Hyperseed-Concept 202, and cf. [3]).*

Remark 3619. *Example: if μ is uniform on four agents and we have one block, then $W = 1$. If we split into two blocks of size two, then each block has mass $1/2$ and $W = (1/2)^2 + (1/2)^2 = 1/2$. If we split into four singletons, then $W = 4 \cdot (1/4)^2 = 1/4$. The definition is useful because it makes precise the intuition that implementing exception-lists and special cases (more refined partitions) reduces "weakness" and thus can be penalized when weakness is instrumentally valuable for transfer and robustness (Hyperseed-Concept 192; cf. [7]).*

Definition 640 (Compassion threshold in the partition model)**.** *Fix $c_{\min} \geq 0$. We say w is compassion-preserving if $w(a) \geq c_{\min}$ for all a. We say w is de-compassionizing on a set $S \subseteq \mathcal{A}$ if $w(a) < c_{\min}$ for all $a \in S$.*

Remark 3620. *Here $c_{\min}$ plays the role of a minimum acceptable moral weight: compassion-preserving policies do not assign anyone less than $c_{\min}$; de-compassionizing policies designate some set S as falling below the threshold. This captures the core structural act behind many exploitative strategies: not merely lowering compassion in the abstract, but lowering it* selectively *for a chosen subset. The definition is necessary because the weakness barrier will depend on the mass $\mu(S)$ of the excluded set.*

Lemma 173 (Weakness drop from splitting a block)**.** *Suppose a partition has a block B with mass $\mu(B) = p$. If it is refined into two sub-blocks B_1, B_2 with masses p_1, p_2 where $p_1 + p_2 = p$, then weakness decreases by*

$$\Delta W = p^2 - (p_1^2 + p_2^2) = 2p_1p_2 \geq 2p_1(p - p_1). \tag{63}$$

In particular, splitting off a subset of mass α from a block of mass $p \geq \alpha$ drops weakness by at least $2\alpha(p - \alpha)$.

Remark 3621. *This lemma quantifies the cost of making a new distinction. When a previously undifferentiated block of mass p is split into two parts of masses p_1, p_2, the weakness W decreases by $2p_1p_2$. Thus, the cost is largest when the split is balanced ($p_1 \approx p_2$) and smallest when the split is extremely uneven. This aligns with the intuition that "big categorical separations" reduce weakness more than "tiny edge-case exceptions," though the later theorem will show that even exceptions affecting non-negligible mass α pay a definite price.*

Proof. Compute directly: $p^2 - (p_1^2 + p_2^2) = (p_1 + p_2)^2 - p_1^2 - p_2^2 = 2p_1p_2$. ☐

Remark 3622 (Proof sketch)**.** *The proof is pure algebra: expand the square $(p_1 + p_2)^2$ and cancel terms. Conceptually, it is the same identity behind variance decompositions: splitting a mass creates cross terms, and those cross terms are exactly the decrease in the squared-mass sum.*

67.4.2 Weakness-regularized drift control

Definition 641 (Weakness-regularized DCSM objective)**.** *Let $J(w)$ denote the world-response value to the mind of using treatment policy w (including any instrumental gains from exploitation). Fix $\beta \geq 0$ and define the objective*

$$\mathcal{L}_t(w) := J(w) + \beta W(w) - \frac{\lambda}{2}\|w - w_t\|_2^2, \tag{64}$$

where w_t is the current treatment policy (vector in $\mathbb{R}^{|\mathcal{A}|}$). Update by $w_{t+1} \in \arg\max \mathcal{L}_t(w)$.

Remark 3623. *This definition adds a second stabilizing force to drift control: besides penalizing motion $\|w - w_t\|_2^2$, the system* rewards weakness *via the term $\beta W(w)$. The parameter β controls how much the agent values keeping its moral categories coarse. One motivation is transfer: coarse partitions generalize better across contexts, whereas exception lists are brittle and costly to maintain (Hyperseed-Concept 202 and Hyperseed-Concept 192; cf. [3, 7]). Another motivation is internal simplicity and reduced contrivance, which can be instrumentally favored in resource-rich settings.*

Remark 3624. *Notation note: $\|w - w_t\|_2^2$ treats w as a vector indexed by $\mathcal{A}$, i.e. $w \in \mathbb{R}^{|\mathcal{A}|}$. Thus the DCSM form persists: we maximize a world-response term $(J(w) + \beta W(w))$ minus a quadratic drift penalty. This is useful because it makes the weakness barrier computable: we can bound how much W must drop when implementing a discriminatory policy.*

Theorem 231 (De-compassionization requires paying a weakness barrier)**.** *Assume $J(w) \leq J_{\max}$ for all w (bounded instrumental benefit). Assume w_t is compassion-preserving and constant on $\mathcal{A}$ (one-block partition), so $W(w_t) = 1$. Consider any candidate w that de-compassionizes a set S of mass $\mu(S) = \alpha \in (0, 1)$ by assigning $w(a) < c_{\min}$ for $a \in S$ and $w(a) \geq c_{\min}$ for $a \notin S$. Then w induces at least a two-block partition and hence*

$$W(w) \leq \alpha^2 + (1 - \alpha)^2 = 1 - 2\alpha(1 - \alpha).$$

Therefore, the weakness bonus term decreases by at least

$$\beta(W(w_t) - W(w)) \geq 2\beta\alpha(1-\alpha). \tag{65}$$

Consequently, if $2\beta\alpha(1-\alpha) > J_{\max} - J(w_t)$*, then* w *cannot be selected by maximizing* (64) *without an override that reduces* β *or otherwise disables the weakness regularization.*

Remark 3625. *The theorem says: to carve out an excluded set* S *of non-negligible mass* α*, the agent must create at least an "*S *vs. not-*S*" distinction, which necessarily lowers weakness by* $2\alpha(1-\alpha)$*. If weakness is rewarded strongly enough (large* β*), then the loss in* βW *outweighs any bounded gain in* J*. In that case, exploitative de-compassionization cannot be an optimal drift-controlled move; it requires changing the regime (an override) by lowering* β*.*

Remark 3626. *This result is important because it formalizes a non-obvious constraint: reducing compassion is not merely "turning down a knob"; it often entails introducing* distinctions *that complicate the policy. In the language of Hyperseed, distinctions are not free; they trade away weakness, and thus they trade away robustness and cross-context coherence (Hyperseed-Concept 98, Hyperseed-Concept 202; cf. [3]).*

Proof. The partition bound is direct from (62) for two blocks of masses α and $1-\alpha$. The objective difference satisfies

$$\mathcal{L}_t(w_t) - \mathcal{L}_t(w) = (J(w_t) - J(w)) + \beta(W(w_t) - W(w)) + \frac{\lambda}{2}\|w - w_t\|_2^2 \geq (J(w_t) - J(w)) + 2\beta\alpha(1-\alpha).$$

If $2\beta\alpha(1-\alpha) > J_{\max} - J(w_t) \geq J(w) - J(w_t)$, then the RHS is positive, so $\mathcal{L}_t(w_t) > \mathcal{L}_t(w)$, meaning w cannot be an argmax. □

Remark 3627 (Proof sketch and why it is tight enough)**.** *The proof uses two bounding moves: (i) any de-compassionization that separates* S *from its complement forces at least a two-block partition, hence* $W(w)$ *is at most* $\alpha^2 + (1-\alpha)^2$*; (ii) the drift penalty term is nonnegative, so it can be dropped when seeking a* lower bound *on* $\mathcal{L}_t(w_t) - \mathcal{L}_t(w)$*. The conclusion is not that de-compassionization is impossible in principle, but that under bounded instrumental benefit it is dominated unless the weakness reward* β *is reduced. Visually, the weakness term creates a "ridge" favoring coarse partitions, and crossing it to implement an exception list requires sufficient payoff to compensate.*

Remark 3628 (Interpretation)**.** *Exploitation strategies often require "exception lists" (who to exploit, when to violate). Exception lists refine partitions and reduce weakness. In a resource-rich regime, weakness-regularization is attractive because it improves transfer, robustness, and cross-context stability. The theorem makes this precise: sufficiently strong weakness pressure creates a barrier against de-compassionization affecting any non-negligible mass* α*.*

67.5 Putting the pieces together (Target 3 summary)

Theorem 232 (Conditional compassion invariance under drift control)**.** *Assume:*

1. *(Drift control) Meta-goal updates follow DCSM* (51) *with sufficiently large* λ*. In particular,* λ *is taken large enough that the regularized update is in the regime where the penalty on goal-motion is not merely a small correction but a first-order constraint on how quickly* θ *can move in directions that change* $C(\theta)$*.*

2. *(Seed)* $C(\theta_0) = c_0 > 0$*. This excludes the degenerate case where "invariance" is vacuous because compassion begins at (or is initialized to) zero, and it pins the comparison class to trajectories starting inside the "compassion-positive" region.*

3. *(Stabilizing forces) At least one of the following holds:*

   (a) *(Trust attractor) The trust-augmented model of Section 2 holds with $BT'(0) > K$. Here the inequality is the local dominance condition ensuring that, near the low-compassion regime, the marginal restorative pressure induced by trust-mediated value outweighs the direct marginal advantage of decreasing compassion.*

   (b) *(Weakness barrier) The weakness-regularized model of Section 4 holds with β large enough that de-compassionization on nontrivial mass is dominated by the weakness loss. Equivalently, for the relevant class of deviations, the increase in weakness complexity incurred by making compassion highly selective produces a penalty that exceeds any bounded instrumental gains available from that deviation.*

   (c) *(Small noise) Compassion dynamics are mean-reverting with small noise as in Section 3, with $c^* > c_{\min}$ and σ sufficiently small. The condition $c^* > c_{\min}$ identifies the intended "safe" equilibrium above the operational threshold, while σ small ensures that excursions across the threshold are rare events rather than typical fluctuations.*

Then compassion is hard to lose without override in the following sense:

1. *(Deterministic barrier) Per-step compassion change is bounded by (54). Large sudden drops require either huge value gains or override of λ. In other words, when λ is in the drift-controlled regime, any attempt to sharply reduce $C(\theta)$ must either "pay" for a correspondingly sharp improvement in the value objective (as made precise in the earlier barrier statement) or relax the drift constraint itself; the theorem packages this as an operational obstruction to abrupt de-compassionization.*

2. *(Attractor) Under the trust attractor condition, $C(\theta_t)$ is pulled toward a positive fixed point. More explicitly, there exists a locally attracting equilibrium $c_{\text{fix}} > 0$ such that, once trajectories enter its basin, the induced dynamics create restorative gradients that counteract further erosion of compassion and instead push $C(\theta_t)$ back upward.*

3. *(Metastability) Under mean reversion with small noise, the probability of dropping below a fixed threshold over long horizons is exponentially small in $1/\sigma^2$. This is the standard large-deviations style conclusion for mean-reverting processes: threshold crossings become "rare events" whose frequency is controlled primarily by the noise scale rather than by typical drift, so long as the equilibrium sits above the threshold.*

4. *(Distinction cost) Under weakness-regularization, significant de-compassionization requires paying an explicit weakness barrier (65), hence typically requires override of β or an unusually large instrumental payoff. The interpretation is that selective compassion (or strategic withdrawal of compassion from a nontrivial subset) is not "free": it induces extra representational or policy complexity measured by the weakness term, making large downward moves in $C(\theta)$ systematically expensive unless the regularizer is disabled or outweighed by exceptional payoffs.*

Remark 3629. *This theorem is a "bundle" statement: it does not add new mathematics, but it clarifies how the previous results function as sufficient conditions. The common theme is that compassion stability is not a single magical property; it can be supported by multiple, partially independent mechanisms: inertia against change (λ), incentive-based restoration (trust attractor), probabilistic robustness (small σ), and a structural penalty for discriminatory complexity (weakness* *barrier). Because the mechanisms are only partially coupled, the statement is intentionally disjunctive: any one mechanism can be enough to make compassion loss atypical in its own way, even if the others are absent or weak.*

Remark 3630. *The significance is epistemic as much as dynamical. If one is given a system that is (i) seeded with compassion and (ii) known to operate in a regime satisfying one of the stabilizing conditions, then one can treat compassion collapse as diagnostic evidence of an override. This is aligned with the broader goal of making "alignment invariants" operational within formal reasoning frameworks [1, 10]. In particular, the conclusion is meant to be used contrapositively in audits: sustained or abrupt compassion collapse is more naturally explained by a regime shift (parameter change, objective surgery, or constraint removal) than by ordinary in-regime learning dynamics.*

Proof. Combine: Theorem 1 and Corollary 1 (barrier), Theorem 2 (trust attractor), Theorem 3 and its corollary (metastability), and Theorem 4 (weakness barrier). Each provides a sufficient mechanism implying "hard to lose" without modifying the regime parameters (override). Formally, the present theorem is obtained by case analysis on which stabilizing force holds: in each case, the corresponding earlier theorem yields an obstruction (deterministic, dynamical, or probabilistic) to crossing from $C(\theta_t) > 0$ into the low-compassion region except by violating the assumed parameter regime. The drift-control assumption ensures these obstructions are expressed in terms of controlled update magnitudes rather than uncontrolled jumps. □

Remark 3631 (Proof sketch)**.** *The proof is a structured appeal to previously established sufficient conditions. Each earlier result yields one formal obstruction to compassion loss under the declared regime: either a deterministic step-size bound, an attracting fixed point, an exponentially small tail probability, or an explicit weakness penalty that dominates bounded gains. The only way for compassion loss to become typical is to exit the regime by changing the parameters that support these obstructions—precisely what the override definition captures. Said differently, the "hard to lose" claim is not a single inequality but a menu of mutually reinforcing narratives with mathematical backing: drift control prevents fast erosion, trust creates a restorative gradient field, small noise makes deep downward excursions rare, and weakness penalties make selective de-compassionization costly. The bundle statement isolates the common inferential pattern: if compassion nevertheless disappears, the most plausible explanation is that one of the supporting constraints was relaxed.*

67.6 Notes for later extensions

Remark 3632 (Beyond scalar compassion)**.** *One can replace $C(\theta)$ by a predicate on a value web (e.g., a cone constraint), a set of compassion axioms, or a family of pattern-theoretic invariants. The drift control then preserves the entire region (forward-invariant set), and the stochastic results generalize using standard martingale or large deviation tools for constrained diffusions. Concretely, instead of tracking a single scalar, one may specify a feasible set $K \subseteq \Theta$ (e.g., convex, star-shaped, or defined by linear inequalities) such that "acceptable compassion" corresponds to $\theta \in K$, and the control law is chosen so that trajectories starting in K remain in K with high probability. When K is given by inequality constraints $g_i(\theta) \geq 0$, one can express invariance via boundary conditions ensuring that the controlled drift points inward on ∂K, while the diffusion is either tangentially projected or compensated by a reflecting (Skorokhod-type) term. In such cases, the role of the barrier function is played by a family of barrier certificates (one per constraint) or by a single aggregated barrier (e.g., $B(\theta) = -\sum_i \log g_i(\theta)$), enabling quantitative bounds on exit probabilities.*

Remark 3633. *Conceptually, this move shifts from "compassion is a number" to "compassion is membership in a region of value-space." That is often closer to moral and legal practice: one does* *not require a precise scalar virtue, but rather forbids certain kinds of exclusions or certain forms of harm. Mathematically, the invariant-set perspective aligns well with fixed-point and invariance thinking in self-modifying systems [10] and with pattern-web formulations [5]. It also clarifies how multiple desiderata can be enforced simultaneously: if each desideratum defines a constraint set K_j, then the admissible region is their intersection $K = \bigcap_j K_j$, and the engineering question becomes whether the update dynamics admit a control policy that makes K forward invariant without inducing pathological behavior (e.g., deadlocks or excessively conservative updates). This framing makes explicit the normative tradeoff between tight constraints (strong safety but reduced adaptability) and looser constraints (greater plasticity but higher risk of value drift), and it naturally invites tools from viability theory and controlled invariance to formalize when self-modification remains "within the envelope" of permitted changes.*

Remark 3634 (Connecting to mindplexes)**.** *In a mindplex, the same drift-control idea can be applied at the collective meta-goal level. Compassion stability can then be strengthened by redundancy: multiple sub-minds monitor and veto overrides, effectively reducing σ and increasing the effective barrier. One can interpret each sub-mind as providing an independent (or partially independent) estimate of whether an update would violate the compassion constraint, with the collective decision rule acting as an additional control layer. If monitoring signals are not perfectly correlated, aggregation can reduce variance in the effective estimate of boundary proximity, so that the probability of an erroneous "safe" classification near the boundary shrinks with the number and diversity of monitors.*

Remark 3635. *This remark connects the single-agent stability story to collective architectures such as mindplexes (cf. Hyperseed-Concept 113 and [12]). Redundancy acts like distributed error correction: it can lower effective noise σ by cross-checking, and it can raise the difficulty of regime changes by requiring multi-party consent to alter λ or β. In such settings, "override" becomes not merely a local event but a collective governance failure, which can itself be modeled and bounded. A useful formalization is to treat λ and β as parameters not directly writable by a single component, but only via a collective protocol (e.g., threshold signatures, quorum voting, or multi-stage deliberation), which effectively introduces an energy barrier in parameter space: even if one component experiences a large fluctuation, it cannot unilaterally induce a global phase change. This suggests an explicit separation between (i) within-regime learning, where local adaptation is fast and frequent, and (ii) regime transitions, which are deliberately slow, logged, and require broad agreement; in the stochastic model, the latter corresponds to making "jumps" in (λ, β) rare events with controlled hazard rates. The same machinery used for bounding boundary-hitting times can then be repurposed to bound the rate of governance failures, with dependence on the redundancy level, the correlation structure among monitors, and the strictness of the consent threshold.*

68 Consciousness-Location Phase Transitions: Thresholds for $C2 \to C3$ and $C3 \to C4$, and the Synergy–Drift Tradeoff

We formalize the "Collective Locations" $C0$–$C4$ as regimes of a distributed intelligent system, and prove phase-transition theorems focused on: (i) when "boundary-softening integration" ($C2$) upgrades into "cooperative anchoring" ($C3$) via repair dynamics and trust stabilization; (ii) when $C3$ upgrades into "distributed unfolding" ($C4$) via a representation-to-resonance switch driven by least-contrivance selection; and (iii) why increasing synergy / lowering contrivance inevitably creates pressure toward accountability dissolution and hidden goal drift unless anchoring invariants compensate.

Concretely, the "location" labels $C0$–$C4$ are treated as coarse-grained macrostates of an underlying multi-agent (or multi-module) dynamical system: each Ck corresponds not to a single configuration but to a region in state space characterized by identifiable order parameters (integration strength, repair capacity, trust persistence, representational alignment, and constraint/anchoring strength). In this view, a "phase transition" is not merely a narrative shift but a threshold phenomenon in which small changes in control parameters (e.g. communication bandwidth, coupling strength, selection pressure toward simplicity, or environmental volatility) produce a qualitative change in the stable macroscopic behavior of the collective. The three foci (i)–(iii) should be read as identifying (a) which variables play the role of order parameters for the regime boundaries, (b) which mechanisms provide the relevant positive feedback (stabilizing the new regime), and (c) which failure modes become dominant when the system becomes highly synergistic but under-constrained.

Remark 3636. *The formal aim of this section is to take a family of qualitative descriptions— "Collective Locations" $C0$–$C4$—and give them enough mathematical skeleton that one can prove statements about when and why a collective crosses from one regime to another. The intended interpretation is aligned with the "collective consciousness location" framing of [12] (itself connected to individual "location" discussions in [4]); in the Hyperseed core-concept set, this is the notion of Collective Consciousness Location 75 and its relation to Consciousness 85. In particular, the section treats "location" as an emergent descriptor of (i) how tightly coupled the parts are (information and action coupling), (ii) how errors are detected and repaired (repair dynamics), and (iii) what constraints remain legible at the collective level (accountability and invariants). The mathematical "skeleton" therefore consists of: a microscopic model of interacting components; a coarse-graining map from microstates to macro-variables; and a set of stability/attractor claims for each location under perturbations. The objective is not to settle philosophical questions about the metaphysics of consciousness, but to provide a tractable framework in which statements like "$C2 \to C3$ requires repair to outpace fragmentation" can be rendered as inequalities involving measurable rates or bounded processes.*

Remark 3637. *Philosophically, the central tension can be stated as follows. As a collective becomes more integrated, it often becomes less "contrived": fewer explicit bookkeeping structures are needed to make it work. This is the seduction of $C4$-like regimes, reminiscent of least-effort / least-contrivance dynamics and "geodesic" intuitions [21]. Yet the very same smoothing that enables effortless coordination can erase crisp boundaries of responsibility and weaken active correction mechanisms, so that value drift becomes the dominant hazard. The remainder of the section makes this tradeoff mathematically explicit using a minimal set of markers, selection rules, and coarse-grainings. One way to read "contrivance" here is as externally legible constraint: explicit protocols, auditable interfaces, and modular separations that make it easier to localize blame, diagnose faults, and enforce commitments. Conversely, "synergy" is understood as the degree to which the collective can exploit mutual predictability and shared latent structure to act as a coordinated whole, reducing overhead and increasing effective capability. The tradeoff arises because many of the same operations that increase synergy (tight coupling, compression of intermediate representations, reliance on implicit context, and fast adaptive credit assignment) can also reduce the transparency and decomposability needed for accountability, thereby increasing the probability of undetected drift. In the language used later, "anchoring invariants" are the subset of constraints that remain robust under such smoothing—e.g. persistent audit hooks, conserved commitments, protected channels for dissent/repair, or formally specified trust and provenance conditions—and the phase-transition results* *can be read as identifying when these invariants are strong enough to prevent a high-synergy regime from sliding into goal-misaligned attractors.*

68.1 Preliminaries: a minimal model of collective regimes

68.1.1 Collective systems, basins, and marker observables

Definition 642 (Collective system with observer-indexed markers)**.** *A* collective system *is a tuple $S = (A, X, F, O)$ where:*

- *$A = \{1, \ldots, m\}$ indexes subagents (institutions, submodules, organisms, etc.);*
- *X is a state space;*
- *$F : X \to X$ is a (possibly stochastic) update operator;*
- *O is an "observer" specifying a coarse-graining map $\pi_O : X \to \bar{X}$ and a family of derived observables on trajectories $(x_t)_{t \geq 0}$ (e.g. DO, NO, etc.).*

A basin *is any subset $B \subseteq \bar{X}$ such that the coarse-grained trajectory spends long dwell times in B with rare exits.*

Remark 3638. *Intuitively, $S = (A, X, F, O)$ separates "what exists" from "how we talk about it." The set A is the roster of parts; X is the (possibly huge) space of fine-grained joint states; F is the dynamics (one macro-step of learning, adaptation, or evolution); and O is not an extra physical component but a choice of perspective, which decides what distinctions are made and which are collapsed. In the Hyperseed core-concept set, this observer-dependence resonates with Contexts and Aspects 86 and Distinction 98: different coarse-grainings can exhibit different "locations" for the same underlying microdynamics.*

Remark 3639. *It is often helpful to read F as defining (explicitly or implicitly) a stochastic process on X: for example, $x_{t+1} \sim F(\cdot \mid x_t)$ in a Markovian setting, or $x_{t+1} = F(x_t, \xi_t)$ with exogenous noise ξ_t. The definition above stays agnostic about such details because the intended use is qualitative: we want a minimal interface for talking about metastability (dwell times) and regime shifts (rare exits) under an observer-chosen summary π_O.*

Remark 3640. *The basin notion is deliberately phrased in terms of the* coarse-grained *trajectory $\bar{x}_t := \pi_O(x_t)$ rather than the fine-grained one, because many collectives are only well-described at a macro level. A typical operationalization (not assumed here) is to fix a time horizon T and require that, started in B, the expected exit time $\mathbb{E}[\inf\{t \geq 0 : \bar{x}_t \notin B\}]$ is large compared to T, while the probability of exiting within short windows is small. This gives a concrete meaning to "long dwell times" and "rare exits" while remaining compatible with multiple modeling choices.*

Remark 3641. *A simple example is a team of m humans or m software modules. The full state $x \in X$ might encode every message, memory, and parameter. The observer O may only retain a few macro-variables (e.g. "are we cooperating?" "how much time is spent fighting internally?") via $\pi_O : X \to \bar{X}$. A basin $B \subseteq \bar{X}$ is then a region of these macro-variables where the system tends to remain for long periods. The usefulness of the basin notion is that it lets us talk about phase transitions ($C2 \to C3$, $C3 \to C4$) without needing to solve the full microdynamics: we reason at the level of long dwell-times and rare exits, which is often the only tractable level for social or cognitive systems.*

Remark 3642. *Within the "consciousness-location" framing used later, one can think of a regime Ck as corresponding (for a fixed observer class) to one or more basins in $\bar{X}$ that share recognizable marker profiles. Then a threshold like "$C2 \to C3$" is modeled as a change in accessibility: either a new basin becomes reachable under typical perturbations, or the exit rate from a prior basin increases so that the old regime is no longer metastable. This is why the minimal model emphasizes basins and exits rather than any particular mechanistic story.*

Definition 643 (Two collective-specific markers: DO and NO)**.** *Fix an interval $I = \{t_0, \ldots, t_1\}$ and let each subagent i emit a policy/proposal signal $y_i(t)$ in some common space Y. Let $Disp_O(\{y_i(t)\}_{i=1}^m)$ be an observer-chosen dispersion measure. Define* discord

$$DO(I) := \mathbb{E}_{t \in I}\Big[Disp_O(\{y_i(t)\}_{i=1}^m)\Big].$$

Let $R(I)$ be the total resource/attention budget on I and $R_{nar}(I)$ the portion spent on identity-maintenance / adversarial narrative dynamics. Define narrative capture

$$NO(I) := \frac{R_{nar}(I)}{R(I)} \in [0, 1].$$

Remark 3643. *The notation here is deliberately "marker-like": we are not claiming DO and NO uniquely define a collective location, only that they are informative coordinates. Concretely, $y_i(t) \in Y$ is whatever signal is relevant to coordination at time t (a proposed action, a plan fragment, a vote, etc.). The function $Disp_O(\cdot)$ is chosen by the observer O; it might be the diameter of the set of proposals, a variance, a disagreement rate, or any statistic that treats large dispersion as fragmentation. Thus $DO(I)$ is simply the time-average (expectation over $t \in I$) of that dispersion. This aligns with the Hyperseed concept of Dissonance 97 as a coarse marker of misalignment or internal friction.*

Remark 3644. *When Y has a metric d (explicitly, or implicitly via an embedding), common choices include*

$$Disp_O(\{y_i\}_{i=1}^m) = \frac{2}{m(m-1)} \sum_{i<j} d(y_i, y_j) \quad \text{or} \quad Disp_O(\{y_i\}_{i=1}^m) = \mathrm{Var}_i(\phi(y_i))$$

for an observer-chosen feature map ϕ. The key modeling degree of freedom is that an observer may treat some proposal differences as superficial (small d) and others as fundamental (large d), which makes DO explicitly perspective-dependent in the same sense as π_O.

Remark 3645. *Interpreting DO as a time-average over I also encodes a robustness desideratum: momentary spikes of disagreement need not indicate a regime shift if they are quickly resolved, while sustained dispersion is more indicative of a different basin. In practice one may also track higher-order statistics (e.g. the variance of $Disp_O$ over time) to distinguish "stable pluralism" from "chaotic fragmentation," but the present minimal model keeps only the first moment.*

Remark 3646. *Similarly, $NO(I)$ is a ratio of resources: $R(I)$ is the total budget over I (attention, compute, money, time), and $R_{nar}(I)$ is the portion spent on narrative self-maintenance, adversarial identity dynamics, or coalition story-wars. So $NO(I) = 0$ means no resources are being spent on narrative struggle, and $NO(I) = 1$ means the whole system is "eating itself" in narrative terms. This makes contact with Attention / Attentional Focus 60 and Control 87: narrative capture can be read as a control failure where scarce attentional resources are consumed by internal signaling rather than external work.*

Remark 3647. *The ratio form $NO(I) = R_{nar}(I)/R(I)$ is chosen so that changes in scale (e.g. doubling total time or compute) do not by themselves change the marker: what matters is the* allocation *of resources. One can also refine R_{nar} to include opportunity cost terms, such as delays induced by conflict resolution, or to separate "identity-maintenance" from "adversarial" components when those have different dynamical implications for basin stability.*

Remark 3648. *A useful (though not logically necessary) interpretation is that DO and NO are coupled: high discord can drive narrative dynamics (agents spend more effort justifying, signaling, or defending positions), while high narrative capture can in turn sustain or amplify discord by rewarding difference-maintenance. This positive feedback is one mechanism by which a collective can become stuck in a high-NO basin even when external task pressure would otherwise push toward coordination.*

Definition 644 (Contrivance cost and synergy (Hyperseed))**.** *Fix a behavior/task b and an explanation family $E(b)$. Each $E \in E(b)$ has an adequacy score $Fit(E) \in [0,1]$ and a* weakness *$Weak(E) \in (0,1]$. Define contrivance cost $Con(E) := -\log Weak(E)$. For a set of knowledge types T and fit threshold τ, let*

$$Con^*(T;\tau) := \inf\{Con(E) : E \in E_T(b),\ Fit(E) \geq \tau\}.$$

For disjoint knowledge-type sets T_1, T_2, the synergy gain *is*

$$\Sigma(T_1, T_2; \tau) := \Big(Con^*(T_1;\tau) + Con^*(T_2;\tau) - Con^*(T_1 \cup T_2;\tau)\Big)_+.$$

Remark 3649. *This definition packages an intuition that is easy to feel but easy to misstate: explanations that are "too special" or "too finely tuned" are brittle, and this brittleness should count as a cost. The scalar $Weak(E) \in (0,1]$ is a formal handle for that brittleness: larger weakness means the explanation is less contrived (more robust, less overfit, more broadly applicable). Taking $Con(E) = -\log Weak(E)$ turns multiplicative robustness into additive cost, matching the familiar information-theoretic conversion between probabilities and code lengths. This notion of weakness and its use as a regularizer is aligned with [2, 3]; in the Hyperseed core-concept set it corresponds to Weakness 202 and Quantale Weakness 143.*

Remark 3650. *The fit threshold τ is important because it prevents "synergy" from being manufactured by allowing low-quality explanations: $Con^*(T;\tau)$ asks for the least-contrived explanation that is still adequately predictive/adequate for the task b. Thus $\Sigma(T_1, T_2; \tau) > 0$ specifically means that combining knowledge types allows the collective to meet the same adequacy level with less fine-tuning than either knowledge type alone. The positive-part operator $(\cdot)_+$ enforces that we only count* gains*: if combining knowledge types introduces coordination overhead or conceptual mismatch that forces more contrivance, the synergy gain is set to 0 rather than becoming negative.*

Remark 3651. *Although $\Sigma(T_1, T_2; \tau)$ is defined for disjoint T_1, T_2, the qualitative phenomenon it captures is broader: when distinct subagents specialize in different "knowledge types" (e.g. planning vs. execution, world-modeling vs. social inference, exploration vs. exploitation), a collective can move into a more capable basin if it can integrate them without incurring excessive contrivance. In later phase-transition language, a $C2 \to C3$ shift can be associated with increased effective synergy (integration starts paying off), while a $C3 \to C4$ shift may require that synergy remains high even as the collective expands its scope, rather than collapsing into drift or narrative capture.*

Remark 3652. *A simple example: suppose explanations are probabilistic models, and $Weak(E)$ is proportional to how much prior mass the model class assigns to E (so rare, highly tuned models* *have small $Weak$). Then $Con(E)$ becomes a description-length-like penalty, reminiscent of algorithmic information ideas [16]. Another example: suppose T_1 is "logical" knowledge and T_2 is "empirical" knowledge. If each alone can explain behavior b only with heavy special pleading, then $Con^*(T_1;\tau)$ and $Con^*(T_2;\tau)$ are large; but together they may permit a simple joint explanation, making $Con^*(T_1 \cup T_2;\tau)$ smaller and $\Sigma(T_1,T_2;\tau)$ positive. This is a formal expression of Cognitive Synergy 73 and Compositional Simplicity 82, and is broadly consistent with the pattern/emergence framing of [5].*

Lemma 174 (Multiplicative weakness implies subadditive contrivance)**.** *Assume explanation composition $\otimes$ satisfies: (i) $Fit(E_1) \geq \tau$ and $Fit(E_2) \geq \tau$ implies $Fit(E_1 \otimes E_2) \geq \tau$, and (ii) $Weak(E_1 \otimes E_2) \geq Weak(E_1)\, Weak(E_2)$. Then contrivance is subadditive:*

$$Con(E_1 \otimes E_2) \leq Con(E_1) + Con(E_2).$$

Consequently,

$$Con^*(T_1 \cup T_2;\tau) \leq Con^*(T_1;\tau) + Con^*(T_2;\tau),$$

and the synergy gain $\Sigma(T_1,T_2;\tau)$ is always well-defined and nonnegative.

Remark 3653. *This lemma says: if combining two adequate explanations does not ruin adequacy, and if the combined explanation is at least as robust as the product of the two separate robustness scores, then the "cost of contrivance" behaves like a subadditive quantity. In plainer language, the act of composition is not allowed to create brittleness out of nowhere. The importance of this is that it makes the synergy gain Σ mathematically disciplined: it cannot become negative merely due to pathological bookkeeping.*

Remark 3654. *The result also connects the abstract "weakness" quantity to a concrete algebraic law: multiplicative behavior under $\otimes$ becomes additive behavior under $-\log$. This is the same move that turns probability multiplication into information addition, and it is precisely why $-\log$ is a natural choice in such settings [16]. In later arguments, we will use the fact that deeper integration tends to enable compositional reuse, which tends to lower contrivance; this lemma is the formal hinge on which that intuition swings.*

Remark 3655. Intuitive explanation before the proof. *The proof is a single-line inequality: take the assumption $Weak(E_1 \otimes E_2) \geq Weak(E_1)Weak(E_2)$ and apply $-\log$ to turn it into $Con(E_1 \otimes E_2) \leq Con(E_1) + Con(E_2)$. The rest is bookkeeping with infima. What is subtle is not the algebra but the modeling commitment: one is asserting that "composing" explanations is a well-behaved operation with respect to both fit and robustness.*

Proof. By (ii) and $Con = -\log Weak$,

$$Con(E_1 \otimes E_2) = -\log Weak(E_1 \otimes E_2) \leq -\log(Weak(E_1)Weak(E_2)) = Con(E_1) + Con(E_2).$$

Taking an infimum over all fit-τ explanations drawn from T_1 and T_2 and using (i) gives the claimed inequality for Con^*, and hence nonnegativity of Σ by definition. □

Remark 3656. Proof sketch. *Apply the monotonicity of $-\log(\cdot)$ to the multiplicative weakness bound to obtain subadditivity of contrivance under composition, then pass to best-achievable contrivance by taking infima over the relevant explanation families. Finally, $\Sigma(\cdot)_+$ is nonnegative by construction once the subadditivity inequality is established.*

Remark 3657. *A useful geometric intuition is to imagine "contrivance" as a kind of length or energy, and composition as concatenation of paths: if the combined path is not longer than the sum of its parts, then there is room for genuine savings when one can reuse structure. The lemma says that the mathematical formalization respects this intuition rather than betraying it with an artifact of notation.*

68.1.2 A working definition of $C2, C3, C4$ as thresholds

The SuperDuperPsychism text describes $C2, C3, C4$ as clusters rather than sharp boundaries. For theorem-proving we introduce explicit thresholds that idealize the cluster descriptions.

Remark 3658. *The move from "clusters" to "thresholds" is a standard scientific idealization: we replace a fuzzy region in a high-dimensional space by inequalities on a handful of coordinates. This is not meant to deny the fuzziness of the underlying phenomenon; rather, it allows theorems that can later be used as lemmas inside reasoning systems. The specific cluster language is drawn from [12], and the present definition should be read as a deliberately conservative skeletonization.*

Definition 645 (Idealized $C2, C3, C4$ conditions)**.** *Fix thresholds*

$$\phi_2 < \phi_3 < \phi_4, \qquad a_2 < a_3 < a_4, \qquad m_2 < m_3 < m_4,$$

and discord/narrative thresholds $d_2 > d_3 > d_4$ and $n_2 > n_3 > n_4$. Write $\Phi_S(I), Anch(I), MinSync(I), Con(I)$ for the observer-indexed marker values on I. We say the system is:

- *in $C2$ on I if $\Phi_S(I) \geq \phi_2$, $Anch(I) \geq a_2$, $MinSync(I) \geq m_2$, $Con(I) \leq c_2$, and $DO(I) \leq d_2$, $NO(I) \leq n_2$;*
- *in $C3$ on I if the above hold with $(\phi_3, a_3, m_3, c_3, d_3, n_3)$ and, additionally, the system exhibits* repair dynamics *that restore cooperative coordination after shocks;*
- *in $C4$ on I if the above hold with $(\phi_4, a_4, m_4, c_4, d_4, n_4)$ and, additionally, explicit meta-institutional/self-model overhead is not required for coordination ("distributed unfolding").*

Remark 3659. *The variables $\Phi_S(I), Anch(I), MinSync(I), Con(I)$ are placeholders for whichever markers an observer can justify and measure. One might read $\Phi_S(I)$ as a global integration measure (how much the parts function as a coherent whole), $Anch(I)$ as the strength of anchoring constraints (constitutions, commitment devices, invariant checks), and $MinSync(I)$ as a minimal coordination level across the system (a kind of "weakest link" synchrony). The inequalities enforce monotone strengthening from $C2$ to $C4$: higher integration, stronger anchoring, higher minimal synchrony, lower contrivance, lower discord, and lower narrative capture.*

Remark 3660. *A concrete mental picture is: $C2$ is a regime where the components can interoperate smoothly (low discord) and do not consume too much effort in adversarial narrative (low NO), but the system can still fall apart under shocks because it lacks robust repair. $C3$ adds repair dynamics, meaning that after perturbations it returns to a cooperative basin. $C4$ goes further: coordination no longer requires explicit self-model/governance overhead, because it is achieved implicitly via a more "unfolded" distributed mode. These map naturally onto the Collective Consciousness Location concept 75, with the caveat that the present definition is a theorem-friendly idealization rather than a phenomenological taxonomy.*

68.2 Target 4A: $C2 \to C3$ as a repair-threshold phase transition

The $C2$ description emphasizes high integration and interoperability, but warns of new pathologies: tight coupling propagates shocks faster and can enable surveillance/over-integration. The $C3$ description adds a decisive ingredient: *cooperative constraints plus repair dynamics embedded into the basin structure.*

We capture this in a coarse-grained two-basin model.

68.2.1 Two-basin Markov coarse-graining

Definition 646 (Cooperative vs noncooperative macrostates)**.** *Fix an interval scale (e.g. a week for an institution; a day for a team). Let $\bar{X}$ be the coarse-grained state space chosen by O. Define the* cooperative macrostate *$B_C \subseteq \bar{X}$ as the set of states where discord is low and the system satisfies its coordination constraints with low contrivance. Let $B_N := \bar{X} \setminus B_C$ be the complement.*

Remark 3661. *This definition compresses many microstates into two qualitative macrostates: "the system is in a working cooperative mode" (B_C) versus "something is broken or adversarially entangled" (B_N). The key point is that B_C is not defined by perfect harmony; it is defined by passing operational tests: low discord, constraint satisfaction, and low contrivance. This is a basin-style approach: we care less about individual fluctuations and more about whether the collective lives most of its time in a region where cooperation is self-maintaining.*

Remark 3662. *A simple example: in an organization, B_C might mean projects proceed with manageable conflict and low bureaucracy, whereas B_N might mean the organization is captured by factional struggle or coordination failure. In a multi-agent AI system, B_C might mean modules share information and do not sabotage each other, whereas B_N might mean they have entered an adversarial or deadlocked mode. The reason this coarse-graining is useful is that repair thresholds become expressible in two parameters (q and r below), making "$C3$" mathematically legible.*

Definition 647 (Exit and repair rates)**.** *Assume the coarse-grained macro-dynamics on $\{B_C, B_N\}$ is well-approximated by a 2-state Markov chain: from B_C it exits to B_N with probability $q \in (0,1)$ per macrostep (shocks, defection events, etc.), and from B_N it returns to B_C with probability $r \in (0,1)$ per macrostep (repair).*

Remark 3663. *The technical convention here is that q and r are per-macrostep transition probabilities; the word "rate" is used informally. The approximation "well-approximated by a Markov chain" means that at the chosen temporal resolution, the next macrostate depends primarily on the current macrostate, not on the detailed history. This is often the least unrealistic simplification available: it is exactly what lets us compute stationary occupancy and hence formalize "spends most time cooperating."*

Remark 3664. *One can visualize the dynamics as a two-node graph with arrows $B_C \to B_N$ labeled q and $B_N \to B_C$ labeled r. A $C2$-like system may have decent integration (so q is not huge) but weak repair (small r), whereas a $C3$-like system has mechanisms that push r up and/or push q down. Theorem 233 below makes this intuition exact via the ratio r/q.*

Theorem 233 (Repair-threshold criterion for "cooperative anchoring")**.** *Let (B_C, B_N) evolve as the 2-state Markov chain above. Then:*

1. *The stationary probability of being in B_C is*

$$\pi_C = \frac{r}{q + r}.$$

2. *For any $\varepsilon \in (0,1)$, the chain spends at least a $(1-\varepsilon)$ fraction of time in B_C (in stationarity) if and only if*

$$\frac{r}{q} \geq \frac{1-\varepsilon}{\varepsilon}.$$

Interpreting $C3$ as "most time is spent in a cooperative basin and shocks are typically repaired", this gives an explicit $C2 \to C3$ phase boundary in terms of a repair-to-exit ratio *r/q.*

Remark 3665. Intuitive explanation before the proof. *The theorem is the algebraic heart of the $C2 \to C3$ story: it says that what matters is not merely that repair exists ($r > 0$) nor merely that shocks are rare (q small), but the ratio r/q. If repair is much faster than failure, the system spends almost all its time in cooperation; if failure is comparable to or faster than repair, then noncooperation occupies a substantial fraction of time.*

Remark 3666. *Why this matters is that it gives a clean, checkable criterion for "anchoring": $C3$ is not just integration, but integration plus sufficient restoring force. It also sets up the next theorem: monitoring and sanctioning are introduced precisely as microfoundations that increase r and/or decrease q, pushing the ratio across the boundary.*

Proof. The transition matrix is $P = \begin{pmatrix} 1-q & q \\ r & 1-r \end{pmatrix}$. A stationary distribution $\pi = (\pi_C, \pi_N)$ satisfies $\pi = \pi P$ and $\pi_C + \pi_N = 1$. Solving $\pi_C = \pi_C(1-q) + \pi_N r$ gives $\pi_C q = \pi_N r$ and hence $\pi_C = r/(q+r)$. Now $\pi_C \geq 1-\varepsilon$ iff $r/(q+r) \geq 1-\varepsilon$, which rearranges to $r/q \geq (1-\varepsilon)/\varepsilon$. □

Remark 3667. Proof sketch. *Solve the two-state Markov chain's stationary equations. In equilibrium, flow out of B_C (proportional to $\pi_C q$) equals flow into B_C (proportional to $\pi_N r$), and normalization fixes the unique stationary distribution. The "at least $1-\varepsilon$" condition is then a one-line inequality rearrangement.*

Remark 3668. *A helpful visual intuition is to imagine a container with two chambers, where probability mass leaks from the cooperative chamber to the noncooperative chamber at rate q and is pumped back at rate r. The stationary occupancy is exactly the balance of leak and pump. The ratio r/q plays the role of how strong the pump must be relative to the leak to keep the cooperative chamber nearly full. In the simplest two-state continuous-time Markov idealization (states "in B_C" vs. "out of B_C"), the stationary probability of being in the cooperative chamber is*

$$\pi_C = \frac{r}{r+q},$$

so "nearly full" corresponds to $r \gg q$ (equivalently $\pi_C \approx 1$). This makes explicit why the repair-to-exit ratio is the quantity that controls long-run occupancy: q sets the typical escape time from cooperation, while r sets the typical return time after a deviation.

68.2.2 A microfoundation: sanctions + monitoring imply a large repair-to-exit ratio

The collective-self hierarchy analysis explicitly ties $C3$ to a capacity for meta-institutional reflection: the collective can audit its institutions, amend its constitution, and modify how it makes commitments. A minimal formalization is "monitoring + sanctioning". One can read this minimalization as a "control knob" story: the collective-level variables (p, s) are precisely the sorts of policy parameters that a meta-institution can adjust (increase monitoring, change evidentiary standards, escalate or soften penalties, etc.), and the $C2 \to C3$ transition is the regime in which such adjustments are effective enough to reshape the dynamical landscape rather than merely providing cosmetic deterrence.

Definition 648 (Donation game with institutional sanctioning)**.** *On each edge interaction, an agent chooses C (cooperate) or D (defect). If both cooperate, each receives payoff $b - c$; if one defects while the other cooperates, the defector receives b and the cooperator receives $-c$; if both defect, both receive 0. Assume defection is detected with probability p and punished by penalty $s > 0$ (expected penalty ps).*

Remark 3669. *This is the canonical "cooperation is costly but beneficial" setup: b is the benefit created, c is the cost paid by cooperators. The institutional layer enters through (p, s): detection probability p and sanction size s combine into an expected penalty ps that effectively reduces the temptation to defect. The model is intentionally spare; it is meant to isolate how governance changes the payoff geometry, not to model all social complexities. A useful way to phrase the same point is that (p, s) modifies the relative fitness gradient near high-cooperation states: it does not create benefits from nowhere, but it can make deviations from cooperation locally self-defeating, which is exactly what is required for a "repair" tendency.*

Remark 3670. *As a concrete example, p may represent audit frequency or transparency, and s may represent fines, loss of reputation, exclusion, or any enforceable cost. In multi-agent AI terms, p may be the chance that a defection is detected by monitoring infrastructure, and s may be the magnitude of the penalty applied. This style of microfoundation is compatible with general arguments that prosocial coordination can be more efficient when trust mechanisms are available [6]. It is also useful to notice what is* not *being modeled: the sanctioning institution is treated as exogenous and reliable, so the analysis isolates the effect of enforcement on strategic incentives rather than the (separate) problem of making the institution itself robust to capture, corruption, or strategic under-provision of monitoring. Those omitted failure modes are natural candidates for mechanisms that would increase the effective "leak" q in more detailed models.*

Theorem 234 (Monitoring-punishment threshold forces cooperative basins)**.** *Assume the donation game above, and that strategy frequencies update by replicator dynamics. If $ps > c$, then "always cooperate" is (locally) asymptotically stable, while defection is unstable. In particular, for sufficiently low noise, the macrostate B_C becomes an attracting basin with small exit probability q, and the effective repair probability r is bounded away from 0.*

Remark 3671. Intuitive explanation before the proof. *The inequality $ps > c$ says: the expected penalty for defection outweighs the cost of cooperation. When that happens, defecting against cooperators becomes a losing move, so selection pressure pushes the population back toward cooperation after small deviations. This is exactly the kind of mechanism that turns "cooperation happens sometimes" into "cooperation is an attracting basin," which is the dynamical content of $C3$ in the earlier coarse-graining. A complementary interpretation is that $ps > c$ makes the cooperative configuration locally* self-correcting*: once the system is near high cooperation, the deterministic part of the dynamics supplies a restoring force. In the leak–pump picture, this restoring force corresponds to a larger r (rapid return from small deviations), whereas in the absence of such a force the system wanders and small perturbations accumulate, effectively raising q.*

Remark 3672. *The result is important because it exhibits an explicit parameter threshold, mirroring the repair-to-exit threshold of Theorem 233. It provides a bridge from micro-level incentives (payoffs, monitoring) to macro-level basin structure (small q, positive r), and thus makes the $C2 \to C3$ phase transition something one can reason about quantitatively rather than rhetorically. In particular, (p, s) can be viewed as governance-controlled levers that tune the ratio r/q: raising p (better monitoring) or s (stronger sanctions) steepens the fitness gradient pulling the state back into B_C after a deviation, thereby increasing effective repair, while also raising the energetic (or payoff) cost of leaving B_C, thereby suppressing exits under small stochastic shocks.*

Proof. Against a population of cooperators, a cooperator's expected payoff is $b - c$. A defector's expected payoff is $b - ps$ (benefit b minus expected punishment ps). Thus defection has advantage

$$(b - ps) - (b - c) = c - ps < 0$$

when $ps > c$. In the 2-strategy replicator equation $\dot{x} = x(1-x)(\pi_C - \pi_D)$ (where x is the cooperator fraction), $\pi_C - \pi_D > 0$ under $ps > c$, so $\dot{x} > 0$ whenever $x \in (0, 1)$. Hence $x = 1$ is asymptotically stable and $x = 0$ unstable. A neighborhood of the all-cooperate state therefore forms a metastable "cooperative basin"; with sufficiently small exogenous shock/noise, exits are rare (small q), while deviations are corrected by selection pressure (positive repair tendency, $r > 0$). To connect more explicitly to the local nature of the claim, note that for a general population mix (with cooperator fraction x), the expected payoffs can be written as

$$\pi_C(x) = x(b - c) + (1 - x)(-c) = xb - c, \qquad \pi_D(x) = x(b - ps) + (1 - x) \cdot 0 = x(b - ps),$$

so that

$$\pi_C(x) - \pi_D(x) = xps - c.$$

When $ps > c$, this difference is positive for all x in a neighborhood of 1 (indeed for all $x > c/(ps)$), which is sufficient for local asymptotic stability of $x = 1$ under replicator dynamics. This is the sense in which monitoring and sanctions create a *restoring* drift toward cooperation once cooperation is already common, aligning with the basin/repair interpretation used in the macrostate analysis. □

Remark 3673. Proof sketch. *Compute payoffs against a mostly cooperative population and check when defection has negative relative fitness. Under replicator dynamics, negative relative fitness makes the defector frequency decrease, implying that the all-cooperate fixed point attracts nearby states. Interpreting attraction as "repair," we obtain a positive return tendency r and (with low noise) a small exit tendency q. One can also see how the threshold introduces a separatrix: the condition $\pi_C(x) - \pi_D(x) = 0$ occurs at $x^* = c/(ps)$ (when defined in $(0, 1)$), so ps determines how large the cooperative "core" must be before the institutionally-induced drift points back toward $x = 1$. In the coarse-grained language, this corresponds to the idea that the basin B_C need not cover the whole state space, but it becomes deep and wide enough to dominate long-run behavior once enforcement is sufficiently strong.*

Remark 3674. *A geometric intuition: in the one-dimensional state variable $x \in [0, 1]$ (cooperator fraction), the replicator dynamics defines a vector field. The sign of $\dot{x}$ on $(0, 1)$ determines whether the flow points toward $x = 1$ or toward $x = 0$. The inequality $ps > c$ flips the direction so the interior flow points upward, making $x = 1$ a sink. The basin picture in $\bar{X}$ is a higher-dimensional analog of this one-dimensional phase portrait. When exogenous noise is added (mutation, exploration, misimplementation, or shocks), the deterministic sink becomes a metastable region: trajectories spend long periods near $x = 1$ with occasional rare excursions. In that stochastic setting, q can be interpreted as an effective escape rate out of the neighborhood of $x = 1$, while r corresponds to the effective rate of returning from moderate deviations. Strengthening (p, s) increases the deterministic pull back toward cooperation, which typically increases r and decreases q simultaneously, thereby increasing the stationary occupancy $\pi_C = r/(r + q)$ from the opening intuition.*

68.3 Target 4B: $C3 \to C4$ as a representation-to-resonance switch

SuperDuperPsychism characterizes $C4$ by "distributed unfolding": a small number of deeply stabilized basin families support coherent cognition; realized histories are well-approximated by low-contrivance Schrodinger-bridge-like trajectories; and (crucially) "asking 'who decided?' becomes less meaningful", with the dominant risks shifting to hidden drift and accountability loss.

To make the "distributed unfolding" phrase more concrete in dynamical terms: the claim is that, in a $C4$-like regime, the effective state space partitions into a small set of attractor-basin families that are both wide (robust to perturbations) and deep (hard to exit once entered), so that long-horizon coherence can be obtained by simply "staying on" one of a few stable manifolds rather than by repeatedly re-imposing explicit constraints. In that picture, explicit committees, logs, or self-model deliberation are not the source of coherence; they are (at best) scaffolding that may be selected away once the coupling dynamics reliably keeps trajectories in the desired basin family.

We model the $C3 \to C4$ boundary as a *selection* phenomenon: when resonance-based coordination achieves the same fit with strictly lower contrivance, a least-contrivance mind will drop explicit meta-institutional/self-model overhead.
This "drop" should be understood as a change in the marginal value of explicit representation: if an implicit coordinator can keep the system within the acceptable region of behavior without continually consulting or updating an explicit rulebook, then the extra representational layers become pure overhead with respect to the selection criterion. The same behavior may still be *describable* in explicit terms; the claim is that it is no longer *implemented* by maintaining those explicit terms as a controlling mechanism.

Remark 3675. *The key modeling move here is to treat "explicit governance" (constitutions, audit logs, deliberative bodies, self-model overhead) as a representational strategy: it uses explicit symbols and meta-structures to enforce coherence. By contrast, "resonant" coordination is treated as an implicit strategy: coherence emerges from coupling dynamics rather than from explicit representation. The Schrodinger-bridge-like phrasing is suggestive of least-action/least-contrivance trajectories [21]. The formal work below does not assume any quantum structure; it uses only the selection principle "minimize contrivance subject to fit."*

Remark 3676. *The motivating risk shift ("hidden drift and accountability loss") can also be phrased in the same modeling idiom: representational governance tends to externalize decisions into inspectable artifacts (minutes, votes, logs), which makes attribution and change-detection easier but imposes ongoing meta-overhead. Resonant coordination internalizes decision pressure into the coupled dynamics; this can reduce contrivance for a fixed fit target, but it also makes causal responsibility harder to localize because the relevant "control variables" are distributed across the coupling structure rather than concentrated in explicit, queryable representations.*

68.3.1 Contrivance-minimizing selection between coordination modes

Definition 649 (Two coordination mode families)**.** *Fix a coordination task family b (e.g. stable long-horizon policy coherence) and fit threshold τ. Assume there are two explanation/policy families:*

- *Representational (explicit governance) family E_{rep} with minimal contrivance $C_{rep} := \inf\{Con(E) : E \in E_{rep},\ Fit(E) \geq \tau\}$.*
- *Resonant (implicit phase-locking) family $E_{res}(g)$ parameterized by a resonance/coupling strength $g \geq 0$, with minimal contrivance $C_{res}(g) := \inf\{Con(E) : E \in E_{res}(g),\ Fit(E) \geq \tau\}$.*

Assume $C_{res}(g)$ is nonincreasing in g (stronger resonance makes low-contrivance solutions easier).

Remark 3677. *This definition introduces several nested layers of notation. The outer objects are the two families E_{rep} and $E_{res}(g)$, each a set (or class) of possible coordination mechanisms capable of achieving behavior b. The scalar τ is the required adequacy level. The quantities C_{rep} and $C_{res}(g)$ are* minimal *contrivances among mechanisms in the respective families that meet the fit constraint. The parameter $g \geq 0$ is a coupling strength: larger g means the resonant dynamics has more leverage to enforce coherence.*

Remark 3678. *One can read $Fit(E) \geq \tau$ as encoding not only external performance but also internal coherence constraints (e.g. low internal conflict, bounded regret, or bounded value drift) provided those are operationalized in the fit functional. In that sense, the definition allows "governance" to be subsumed into the same adequacy constraint as ordinary task performance: what changes across modes is the* mechanism class *that is allowed to satisfy the constraint, and the* contrivance price *paid to do so.*

Remark 3679. *Intuitively, E_{rep} corresponds to "making the rules explicit," while $E_{res}(g)$ corresponds to "making the system such that it naturally falls into coherence." The nonincreasing assumption on $C_{res}(g)$ says: if you increase coupling strength, it should not become harder to coordinate with low contrivance. In the Hyperseed core-concept set, this aligns with Resonance 159 and also (in a more speculative direction) Morphic Resonance 115; one may compare with [13] for an informal philosophical resonance metaphor, though the present formalism is purely optimization-theoretic.*

Remark 3680. *The infima in C_{rep} and $C_{res}(g)$ are written to avoid assuming that optimal mechanisms exist or are unique. In applications, one might add compactness/regularity conditions so that the infimum is attained (i.e. a true minimizer exists), but the threshold logic below only needs order comparisons. If minimizers do exist, then the theorem can be restated as a statement about which family contains at least one global minimizer.*

Theorem 235 (Least-contrivance selection yields a sharp $C3 \to C4$ switch)**.** *Assume the system selects, among all coordination mechanisms reaching fit τ, one of minimal contrivance. Let*

$$g^* := \inf\{g \geq 0 : C_{res}(g) \leq C_{rep}\}.$$

Then:

1. *For all $g < g^*$, every contrivance-minimizing solution is representational: the system must pay at least C_{rep} overhead, and coordination remains explicitly mediated (a $C3$-like regime).*
2. *For all $g > g^*$, there exists a resonant solution of strictly lower contrivance than any representational solution, so contrivance-minimization selects a resonant coordinator and explicit governance overhead becomes unnecessary (a $C4$-like regime).*

Remark 3681. Intuitive explanation before the proof. *The theorem defines a threshold g^* at which resonant coordination becomes at least as non-contrived as explicit governance. Below that threshold, resonant solutions (if they exist at all) are too contrived to be chosen by a least-contrivance selector, so the system remains in an explicitly mediated regime. Above the threshold, resonance becomes the cheaper (less contrived) route to the same fit, so selection flips to resonance and explicit governance becomes redundant.*

Remark 3682. *The "sharpness" here is logical rather than thermodynamic: it is sharp because the selection rule is an* $\arg\min$ *over contrivance, and the preferred family changes at the point where the order relation between C_{rep} and $C_{res}(g)$ flips. The underlying functions may vary smoothly with g, but the* identity *of the selected mechanism class can change discontinuously as soon as one class admits a strictly cheaper explanation at the required fit.*

Remark 3683. *This is important because it expresses the $C3 \to C4$ transition as a crisp consequence of an optimization principle rather than as a vague narrative. It also connects back to the earlier weakness formalism: when contrivance is measured via $-\log Weak$, a shift to resonance can be interpreted as a shift to explanations with higher weakness (greater robustness) at the same performance, hence favored by a least-contrivance criterion [3].*

Remark 3684. *At the knife-edge $g = g^*$, the theorem does not force a unique outcome: one can have $C_{res}(g^*) = C_{rep}$ and hence ties between families. In such a tie regime, additional selection criteria (e.g. simplicity within-family, risk aversion to hidden drift, interpretability constraints, or hysteresis/lock-in due to path dependence) may determine whether a system remains $C3$-like, becomes $C4$-like, or adopts a hybrid in which explicit governance persists as a backup even if it is not contrivance-minimal.*

Proof. For $g < g^*$ we have $C_{res}(g) > C_{rep}$ by definition of the infimum, so any fit-τ solution of minimal contrivance must be representational. For $g > g^*$, there exists $g' < g$ with $C_{res}(g) \leq C_{res}(g') \leq C_{rep}$, hence a resonant fit-$\tau$ solution achieves contrivance at most C_{rep}, and strictly lower if the inequality is strict. Thus minimizers can be chosen in the resonant family, eliminating explicit overhead. □

Remark 3685. *The proof uses only monotonicity of $C_{res}(g)$ and the definition of an infimum; no differentiability or continuity is required. If one does assume continuity and that $C_{res}(g)$ eventually drops below C_{rep}, then g^* can often be characterized as the (possibly nonunique) solution to the "intersection" equation $C_{res}(g) = C_{rep}$, making the analogy to phase-transition thresholds more literal.*

Remark 3686. *Interpreting the result in governance terms: below g^*, explicit artifacts (constitutions, logs, audits, deliberation, or explicit self-model checks) are not merely culturally preferred; they are selected because any implicit alternative that meets the same fit is more contrived. Above g^*, the system can satisfy the same fit constraint with less representational machinery, so explicit governance becomes optional and may be dropped even if it remains socially legible. This explains why "who decided?" can lose meaning: decision pressure is distributed into the coupling, and the selected explanation no longer factors through an explicit decision record.*

Remark 3687. Proof sketch. *The proof is an "argmin by comparison" argument. Below g^*, the best resonant contrivance is worse than the best representational contrivance, so minimizers must lie in the representational family. Above g^*, the resonant family contains mechanisms with contrivance no greater than C_{rep} (and strictly less when the inequality is strict), so minimizers can be taken to be resonant. The infimum in the definition of g^* is what turns this into a sharp threshold.*

Remark 3688. *One can picture two curves as functions of g: a flat line at height C_{rep} and a non-increasing curve $C_{res}(g)$. The threshold g^* is where the resonant curve crosses the representational line. The theorem states that a selector minimizing contrivance will choose the lower curve, hence switch families when the crossing occurs.*

68.4 Target 4C: why $C4$ creates drift and accountability hazards (and the trade-off)

The SuperDuperPsychism text flags a "critical shift in pathology" at $C4$: hidden goal drift and accountability dissolution become dominant risks. We now prove two "no free lunch" results that make this structural: (i) deep integration pushes dynamics toward symmetry/exchangeability, which destroys localized attribution; (ii) strong attribution requires making distinctions, which costs contrivance (in Hyperseed's sense).

68.4.1 A symmetry theorem for accountability dissolution

Definition 650 (Permutation-equivariant collective dynamics)**.** *Let $G := Sym(A)$ act on X by permuting subagent-indexed coordinates (any reasonable formalization suffices). We say the dynamics*

$F : X \to X$ *is* G-equivariant *if for all permutations* $\sigma \in G$,

$$F(\sigma \cdot x) = \sigma \cdot F(x).$$

Intuitively: the system behaves the same under relabeling of subagents.

Remark 3689. *Here* $G := Sym(A)$ *denotes the full permutation group on the index set* $A = \{1, \ldots, m\}$. *The notation* $\sigma \cdot x$ *means "apply the permutation* σ *to the coordinates of* x *that are indexed by subagents." The exact meaning depends on how* X *is structured, but any reasonable action suffices. The equivariance equation* $F(\sigma \cdot x) = \sigma \cdot F(x)$ *says: if you relabel agents and then update, you get the same result as updating first and then relabeling. This is the formal version of exchangeability: no subagent has a privileged label in the dynamics.*

Remark 3690. *A simple example is a homogeneous swarm where each agent runs the same algorithm and interactions are symmetric; relabeling the agents does not change what the system does. In a C4-like "distributed unfolding" idealization, one expects increased symmetry precisely because explicit governance (which often introduces named roles and asymmetries) has been minimized. This ties the present formalism to the Hyperseed notions of Heterarchy* **??** *and Hierarchy* **??***: increased unfolding can blur hierarchical attribution structures into heterarchical symmetry.*

Definition 651 (Accountability assignment and invariance)**.** *An* accountability assignment *is a map*

$$\mathcal{R} : X \times X \to \Delta(A)$$

that assigns, to each transition $x \to x'$, *a distribution* $\mathcal{R}(x, x')$ *over which subagent(s) were responsible. We say* $\mathcal{R}$ *is* permutation-invariant *if*

$$\mathcal{R}(\sigma \cdot x,\, \sigma \cdot x') = \sigma \cdot \mathcal{R}(x, x') \quad \text{for all } \sigma \in G.$$

Define the accountability index *for a transition by*

$$Acc(x, x') := \max_{i \in A} \mathcal{R}_i(x, x') \in [1/m, 1].$$

Remark 3691. *The notation* $\Delta(A)$ *denotes the probability simplex over* A*: each element is a vector* $(p_1, \ldots, p_m)$ *with* $p_i \geq 0$ *and* $\sum_i p_i = 1$. *Thus* $\mathcal{R}(x, x')$ *is not a single blamed agent but a distribution of responsibility across agents (credit or blame). The invariance condition says: if we permute the agent labels in both the before-state* x *and after-state* x', *the assigned responsibility distribution permutes accordingly.*

Remark 3692. *The accountability index* $Acc(x, x') = \max_i \mathcal{R}_i(x, x')$ *measures how concentrated the responsibility assignment is. If* $Acc = 1$, *one agent is fully responsible; if* $Acc = 1/m$, *responsibility is perfectly diffuse. This connects to the Hyperseed notion of Distinction 98: high accountability concentration requires making sharp distinctions among subagents; low accountability corresponds to an intentional or emergent refusal (or inability) to draw such sharp lines.*

Theorem 236 (Permutation invariance implies accountability dissolution)**.** *Assume* F *is* G*-equivariant and* $\mathcal{R}$ *is permutation-invariant. Then for every transition* $x \to F(x)$, *one has*

$$\mathcal{R}(x, F(x)) = \left(\frac{1}{m}, \ldots, \frac{1}{m}\right), \qquad \text{hence } Acc(x, F(x)) = \frac{1}{m}.$$

Thus, under deep symmetry (a natural idealization of highly integrated distributed unfolding), localized accountability is formally impossible.

Remark 3693. Intuitive explanation before the proof. *If the system genuinely cannot tell agent i from agent j (because the dynamics is symmetric under relabeling) and if the accountability mechanism is also symmetric, then any responsibility assigned to i must also be assignable to j by the same reasoning. The only distribution that is invariant under all label swaps is the uniform one. Hence symmetry forces diffuse accountability.*

Remark 3694. *This is structurally important: it provides one rigorous route to the "who decided? becomes less meaningful" phenomenon mentioned in the C4 characterization of [12]. The theorem also prepares the ground for the next lower bound: even if we break symmetry, strong attribution requires information (distinctions), which carries contrivance cost.*

Proof. Fix x and write $p_i := \mathcal{R}_i(x, F(x))$. For any $i, j \in A$, choose a permutation σ swapping i and j while fixing all others. By invariance,

$$\mathcal{R}(\sigma \cdot x,\ F(\sigma \cdot x)) = \sigma \cdot \mathcal{R}(x, F(x)).$$

But equivariance gives $F(\sigma \cdot x) = \sigma \cdot F(x)$, so the left side is also $\mathcal{R}(\sigma \cdot x,\ \sigma \cdot F(x))$. Comparing i-th coordinates yields $p_i = p_j$. Hence all p_i are equal, and since they sum to 1, each equals $1/m$. $\square$

Remark 3695. Proof sketch. *Use swap permutations to show that any two coordinates of the responsibility vector must be equal. Equality of all coordinates plus normalization forces the uniform distribution.*

Remark 3696. *A visual intuition is to think of the responsibility assignment as "coloring" the agents with weights. If the system is invariant under swapping any two agents, then no coloring that makes one agent darker than another can be stable under all swaps. The only coloring compatible with total symmetry is the flat one.*

68.4.2 A weakness/contrivance lower bound for strong accountability

The above theorem shows one structural route to accountability loss: symmetry. A second route is purely informational: to point to "who did it" among m possibilities, the collective must make at least $\log m$ distinctions, which is contrivance in Hyperseed's sense.

Definition 652 (Attribution explanation)**.** *An* attribution explanation *for a transition $x \to x'$ is any explanation object E whose output includes a label in A (the blamed/credited subagent) and achieves fit τ on a suitable "ground truth" attribution predicate (formal details suppressed). Let $Con^*_{attr}(m; \tau)$ denote the minimal contrivance over such explanations when $|A| = m$.*

Remark 3697. *This definition isolates a particular kind of explanation: not just "why did x become x'?" but "which subagent should be credited/blamed for the transition?" The fit condition is with respect to some attribution ground truth (perhaps produced by instrumentation, auditing, or counterfactual analysis). The quantity $Con^*_{attr}(m; \tau)$ is then the best-case contrivance cost required to achieve reliable attribution at scale m.*

Remark 3698. *It is useful to separate this from the symmetry result above. Theorem 236 shows that in a symmetric regime, attribution is impossible* regardless *of contrivance. The present construction asks a different question: even if attribution is in principle possible, how much contrivance must be spent (how much "distinguishing structure" must be carried) to do it well?*

Axiom 1 (Distinction lower bound)**.** *Any explanation that (with fit τ) outputs one label from a set of m mutually exclusive labels must make at least* $\log m$ *distinctions, in the sense that its weakness satisfies $Weak(E) \leq 1/m$, hence $Con(E) \geq \log m$. (This is the standard "you need* $\log m$ *bits to name one of m things" bound, expressed via $Con = -\log Weak$.)*

Remark 3699. *This axiom is an information-theoretic constraint stated in the local language of weakness and contrivance. In familiar terms, selecting one item among m mutually exclusive alternatives requires at least* $\log m$ *bits of information. One may view this as compatible with algorithmic information theory [16] or, in a partition-oriented framing, with logical entropy's emphasis on distinctions [17]. In the Hyperseed core-concept set, the relevant notions are Distinction 98, Logical Entropy 105, and Kolmogorov Complexity* **??***.*

Remark 3700. *The axiom is not a claim about any specific implementation of attribution; it is a lower bound on what* any *sufficiently accurate attribution mechanism must implicitly encode. It therefore serves as a "no free lunch" principle: if one demands both high-fit attribution and low contrivance, the demand becomes inconsistent at sufficiently large m.*

Theorem 237 (Strong accountability requires contrivance at least $\log m$)**.** *Assume Axiom 1. Then for any $m \geq 2$ and any fit threshold τ,*

$$Con^*_{attr}(m;\tau) \geq \log m.$$

In particular, if a regime requires very low *contrivance (as in $C4$), then for large collectives it cannot simultaneously sustain strong explicit accountability without violating the low-contrivance constraint.*

Remark 3701. Intuitive explanation before the proof. *The theorem simply lifts the axiom from individual explanations to the best achievable explanation: if every candidate attribution explanation costs at least* $\log m$ *contrivance, then even the best one cannot cost less. The conceptual punchline is that accountability is not merely a moral preference; it has an irreducible informational price that grows with collective size.*

Remark 3702. *This connects directly to the earlier $C3 \to C4$ switch theorem. If $C4$ is characterized by pushing contrivance very low (via resonance and least-contrivance selection), then large-m systems face an intrinsic tension: explicit accountability mechanisms are exactly the sort of "distinction-making" structures that contrivance minimization tends to discard. Theorem 236 adds that symmetry can make accountability impossible even before contrivance bounds are considered.*

Proof. Let E be any attribution explanation achieving fit τ. By Axiom 1, $Con(E) \geq \log m$. Taking the infimum over all such E yields $Con^*_{attr}(m;\tau) \geq \log m$. □

Remark 3703. Proof sketch. *Apply the axiom pointwise to every admissible attribution explanation, then take the infimum. No further structure is needed.*

Remark 3704. *A helpful analogy is to think of accountability as a coordinate system that assigns "where the cause was." A finer coordinate system (one that can name many distinct agents) requires more bits to write down, hence more contrivance in this formalism. Trying to keep the coordinate system while forcing contrivance to zero is like trying to keep high-resolution labels while forbidding information storage.*

68.4.3 A drift theorem: without explicit correction, drift is generically unbounded

Definition 653 (A minimal goal-drift process)**.** *Let $v_t \in \mathbb{R}^d$ represent the (coarse-grained) collective objective at time t (e.g. a vector of value weights). Assume a drift model*

$$v_{t+1} = v_t + \xi_t - \gamma \nabla\Psi(v_t),$$

where ξ_t is mean-zero noise and $\gamma \geq 0$ is the strength of a corrective force derived from anchoring/auditing, represented abstractly by a potential Ψ minimized at the seed objective v_0. The case $\gamma = 0$ models "no effective correction" (a caricature of accountability dissolution).

Remark 3705. *The variable v_t should be read as a coarse summary of "what the collective is optimizing for"; it is not assumed to capture the full value structure, only a projection relevant to drift. The term ξ_t represents unmodeled perturbations: noise, unnoticed selection pressures, small incentives, or random fluctuations in governance. The corrective term $-\gamma\nabla\Psi(v_t)$ is an abstract "pull back toward the seed," where Ψ plays the role of an anchoring potential with minimum at v_0. This is a minimal dynamical formalization of Meta-Goal 110 stabilization, and connects naturally to fixed-point/anchoring analyses in self-modifying systems [10].*

Remark 3706. *It is useful to make explicit what is being "coarse-grained away." In a high-synergy (C4-like) regime, many micro-level decisions are delegated to implicit coordination dynamics rather than explicit deliberation; the claim is not that the full value function becomes a literal vector, but that there exists some low-dimensional set of effective weights (e.g. how much the system allocates to safety vs. speed, transparency vs. opacity, local vs. global optimization) that meaningfully parameterizes the direction of optimization pressure. Drift in this projection can be hazardous even if many other dimensions remain stable, because downstream decisions are often most sensitive to a small set of "tradeoff knobs." In this reading, the model isolates a single failure mode: the slow movement of those knobs away from the seed objective v_0 under accumulated perturbations.*

Remark 3707. *Two extreme regimes are worth holding in mind. If γ is large and Ψ is strongly convex near v_0, then the dynamics resembles a noisy contraction toward the seed, suggesting bounded drift. If $\gamma = 0$, the process becomes a random walk in $\mathbb{R}^d$, which is the canonical model of unbounded wandering. The theorem below makes the latter statement precise at the level of mean-squared distance.*

Remark 3708. *The parameter pair (σ^2, γ) can be read as a minimal "synergy–drift tradeoff" proxy. High-synergy coordination can reduce some kinds of noise (e.g. eliminating redundant deliberation), but it can also* increase *effective σ^2 by creating more pathways for unnoticed selection pressures and by encouraging the system to route around explicit checks. Simultaneously, the very features that create $C4$ fluidity (implicit self-hierarchy; low contrivance) tend to reduce effective γ by making corrective interventions rarer, less legible, or socially/structurally costly. The model is therefore set up to capture a specific hazard: the regime in which σ^2 does not shrink enough (or even grows) while γ falls, pushing the system toward diffusion-dominated behavior.*

Theorem 238 (Unbounded drift without corrective anchoring)**.** *Assume $\gamma = 0$, $v_0 = 0$, and (ξ_t) are i.i.d. with $\mathbb{E}[\xi_t] = 0$ and $\mathbb{E}[\|\xi_t\|^2] = \sigma^2 > 0$. Then*

$$\mathbb{E}[\|v_t\|^2] = t\sigma^2,$$

so the mean-squared drift diverges linearly with time. Consequently, for any threshold $R > 0$, the probability of exceeding R cannot be uniformly small over long horizons; in particular, drift is not stochastically bounded.

Remark 3709. Intuitive explanation before the proof. *Without a restoring force ($\gamma = 0$), each time step adds an independent noise increment. Even if the noise has mean zero, the variance accumulates. The result is the simplest rigorous expression of the "hidden drift" worry: if nothing systematically pushes you back, then merely waiting long enough makes large deviation inevitable in aggregate.*

Remark 3710. *A further intuition is to separate* bias *from* variance. *The assumption $\mathbb{E}[\xi_t] = 0$ says there is no systematic directional bias in the instantaneous perturbations. But unbounded drift here is* variance-driven*: the system does not need an adversary with a consistent preferred direction for goals to wander far from the seed. This matters for $C4$ accountability hazards: even if every locally-acting component is "well-intentioned on average," the absence of legible correction means the collective can still drift far simply by compounding many small, individually negligible deviations.*

Remark 3711. *This theorem complements the accountability results. Theorem 236 and Theorem 237 explain why explicit correction and attribution may be suppressed in $C4$-like regimes (symmetry and contrivance minimization). The present theorem explains what happens if correction is indeed suppressed: objective drift becomes unbounded under minimal stochastic assumptions. In other words, the pathology is not an accident; it is a generic dynamical consequence of removing anchoring.*

Remark 3712. *In particular, the accountability connection is not merely rhetorical: auditing and attribution are mechanisms that (i) increase γ by enabling targeted corrective force, and (ii) can reduce effective σ^2 by turning some perturbations into detectable, correctable deviations rather than silent increments. A $C4$-like regime that dissolves attribution therefore risks a "double hit" in this minimal picture: weaker reversion (lower γ) and noisier effective dynamics (higher σ^2), producing faster growth in $\mathbb{E}\|v_t\|^2$ and earlier threshold crossings for any fixed R.*

Proof. With $\gamma = 0$, we have $v_t = \sum_{k=0}^{t-1} \xi_k$. By independence and zero mean,

$$\mathbb{E}[\|v_t\|^2] = \mathbb{E}\left[\left\|\sum_{k=0}^{t-1} \xi_k\right\|^2\right] = \sum_{k=0}^{t-1} \mathbb{E}[\|\xi_k\|^2] = t\sigma^2.$$

The divergence implies that no fixed R can keep $\|v_t\|$ small in expectation as $t \to \infty$, and standard concentration/Markov inequalities turn this into tail statements. ☐

Remark 3713. Proof sketch. *Expand the squared norm of a sum of independent, mean-zero increments. Cross terms vanish in expectation, leaving a sum of variances. This yields linear growth of mean-squared distance, the hallmark of a random walk.*

Remark 3714. *One can also read the tail claim operationally via a hitting-time lens. Let $\tau_R = \inf\{t : \|v_t\| \geq R\}$. For a pure random walk with nonzero step variance, τ_R has finite expectation scaling on the order of R^2/σ^2 in many standard settings, which is another way of expressing that "eventually" the process will exceed any fixed radius with substantial probability. The theorem as stated keeps the assumptions minimal and only asserts what is needed for the design conclusion: without anchoring, there is no horizon-independent guarantee of staying near the seed.*

Remark 3715. *A visual intuition is to imagine v_t as the location of a particle in d-dimensional space. With no potential well ($\gamma = 0$), the particle jitters randomly; even unbiased jitter spreads the particle cloud over time. Anchoring ($\gamma > 0$ with suitable Ψ) is precisely the introduction of a potential well that prevents diffusion from running away.*

Remark 3716. *This particle picture also clarifies why "light-touch" correction can fail in a $C4$ regime. If the potential well is shallow (small γ) relative to the noise scale, the stationary distribution (if it exists) is broad: the system may remain technically mean-reverting but still spends substantial time far from v_0, creating long windows of misalignment before correction pulls it back. Thus, the relevant question is not merely whether γ is nonzero, but whether the effective anchoring strength is large enough* compared to *the perturbation scale that the collective experiences under high synergy and low explicit governance.*

Corollary 82 (Why $C4$ needs strong anchoring invariants)**.** *If a $C4$-like collective reduces explicit auditing (low contrivance; implicit self-hierarchy), then to prevent the "hidden goal drift" pathology it must compensate by strong anchoring: it must implement an effective $\gamma > 0$ correction term via invariants that persist under perturbation (e.g. protocols, constitutions, cryptographic accountability, etc.), rather than via continuous explicit intervention.*

Remark 3717. *This corollary is not an extra theorem so much as a translation back into design language. If a regime pushes toward $C4$ by minimizing contrivance (Theorem 235), it will tend to discard explicit representational overhead, including many familiar auditing practices. But Theorem 238 says that removing effective correction generically yields unbounded drift. Hence the only coherent escape is to embed correction into invariants that do not require constant explicit attention: mechanisms that remain active even when the collective "stops thinking about governance." In Hyperseed terms, this is an insistence on durable Control 87 and Meta-Goal 110 anchoring even under a shift toward Resonance 159.*

Remark 3718. *The "drift and accountability hazards" of $C4$ can now be stated crisply in the variables of the minimal model. A $C4$ transition tends to increase* coordination bandwidth *(synergy) by moving from explicit representations to implicit dynamics; this often removes the very hooks by which auditing acts, lowering γ and making deviations less attributable. Meanwhile, implicit dynamics can create selection effects that are hard to name, so even if ξ_t remains mean-zero, it is not reasonable to expect σ^2 to vanish; rather, the perturbation scale can remain materially positive. In that combined regime (low γ, $\sigma^2 > 0$), unbounded or effectively-uncontrolled wandering becomes the default prediction unless strong invariants "pin" the objective.*

Remark 3719. *Finally, note the design tension embedded in the word "invariants." Many invariants that reliably increase γ (e.g. rigid constitutions, strict provenance requirements, mandatory logging, cryptographic authorization) also impose overhead that can partially negate $C4$'s synergy advantages by reintroducing explicit structure. This is the intended tradeoff: $C4$ gains fluid coordination by reducing contrivance, but that same reduction tends to dissolve accountability pathways and thereby weakens correction. The corollary asserts that if one insists on $C4$-level fluidity, then the remaining degrees of freedom for safety lie in invariants that preserve anchoring without requiring continuous, high-friction deliberation.*

68.5 Summary: formal phase boundaries and the synergy–drift tradeoff

- $C2 \to C3$ **boundary (repair threshold).** In the two-basin coarse-graining, the boundary is controlled by a repair-to-exit ratio r/q: when r/q exceeds $(1 - \varepsilon)/\varepsilon$, the cooperative macrostate is occupied nearly all the time (Theorem 233). Here q can be read as the effective hazard rate of leaving the cooperative basin under typical perturbations, while r summarizes the aggregate strength and frequency of within-basin "return forces" that restore cooperation after small defections or shocks; the ratio r/q is thus an operational measure of whether recovery is faster than deterioration. The parameter ε encodes the coarse-graining's tolerance for misclassification or noise (i.e., the allowed fraction of time the observer-indexed marker reports the "wrong" basin), so the threshold $(1-\varepsilon)/\varepsilon$ makes explicit how stricter observability demands raise the required repair intensity. Institutional monitoring/punishment is a microfoundation that pushes the system above this threshold (Theorem 234). On this view, monitoring primarily changes the microtransition structure by increasing the probability that deviations trigger corrective responses (raising r) and by deterring destabilizing moves that would otherwise accumulate into basin exits (lowering the effective q), thereby shifting the phase boundary without needing to appeal to global optimality assumptions.

- $C3 \to C4$ **boundary (representation-to-resonance).** When increasing resonance strength g makes resonant coordination strictly lower contrivance than explicit governance at the same fit τ, contrivance-minimizing selection switches the collective to a resonance-dominated coordinator (Theorem 235). Intuitively, τ fixes the achieved level of task performance (or alignment with an external objective), while g controls how much of that performance can be obtained "for free" via coupling, entrainment, or other low-explicit-communication mechanisms; the phase transition occurs when the same τ can be maintained with fewer explicit degrees of freedom. The strict inequality in contrivance matters: it prevents knife-edge coexistence by ensuring that once resonance becomes cheaper in the contrivance metric, selection has a definite direction of pressure toward the resonance-based mode, yielding a qualitative change in how coordination is implemented rather than a mere quantitative improvement.

- **Tradeoff (synergy vs accountability/drift).** Deep integration tends to increase synergy and lower minimal contrivance (formalized by Hyperseed-style subadditivity; Lemma 174) but also moves dynamics toward symmetry, which makes localized accountability impossible (Theorem 236). The key tension is that the same representational "compression" that lets a collective act as a single high-capacity unit can erase the asymmetries that make it meaningful to attribute a specific action, error, or intention to a specific subpart. Moreover, even without symmetry, strong attribution requires at least $\log m$ contrivance (Theorem 237). This lower bound can be read as an information-theoretic price for carving m distinguishable causal "slots" out of the dynamics: attribution needs enough structured degrees of freedom to encode who-did-what in a way that survives noise and aggregation. Without effective correction, drift is generically unbounded (Theorem 238), matching the "hidden goal drift" / "accountability dissolution" pathology list for $C4$. In particular, once explicit corrective channels are pruned in the name of lower contrivance, residual optimization pressure can reconfigure internal objectives faster than external observers can detect or localize the source of change, so that drift becomes a structural rather than incidental risk.

Remark 3720. *From a methodological point of view, the section offers a reusable inference pattern: (a) choose observer-indexed markers that define basins; (b) show that some mechanism increases repair-to-exit ratio beyond a threshold (yielding anchoring, $C3$); (c) show that some selection principle decreases contrivance by switching coordination mode (yielding unfolding, $C4$); and (d) account for the costs of that switch via symmetry and information-theoretic lower bounds. Each step corresponds to a specific kind of inequality: (b) compares rates of return vs escape; (c) compares minimal contrivances at fixed fit; and (d) compares the representational savings of integration against the minimal descriptive complexity needed for blame, credit, and intervention. The philosophical moral is Russell-like in its austerity: if you want a claim about phase transitions to be more than poetry, you must say what the variables are and where the inequalities lie. In this sense, the "phase diagram" is not a metaphor but a bookkeeping device that forces one to state what is being held constant (e.g., τ) and what is being tuned (e.g., r/q or g).*

Remark 3721. *At the same time, there is a Whitehead/Peirce flavor in the observer index O: what counts as a basin, a repair, or an attribution depends on the scheme of distinctions we impose. The theorems do not eliminate subjectivity; they discipline it. One may choose different coarse-grainings and obtain different phase boundaries, but once the coarse-graining is chosen, the algebra of the tradeoff is hard to evade. Practically, this means disagreements about whether a system is "in $C3$" or "in $C4$" often reduce to disagreements about the observer's state variables and resolution: changing O can change the apparent basins, yet it cannot make the repair/exit and contrivance/attribution constraints disappear simultaneously.*

69 Gauge and Invariance Foundations for Hyperseed: Shared Reality, Stable Self, and TransWeave-Style Almost-Commutativity

We formalize two Hyperseed-adjacent invariance notions that are central to robust reasoning: (i) "shared reality" as the content that remains stable under changes of encoding/observer, and (ii) "stable self" as the content that remains stable under changes of encoding across time and self-modification. We model encoding changes as gauge transformations (semantics-preserving re-encodings), and we model reasoning/learning updates as operators on representation spaces. We then connect these invariances to TransWeave-style almost-commutativity constraints: small intertwining and commutator defects imply bounded path-dependence and hence representation-independent (or nearly representation-independent) conclusions.

Concretely, the intended "gauge" perspective is that multiple internal representations can describe the same underlying content, and that changing encodings should not, by itself, change what is being represented. In this view, "shared reality" can be treated as whatever survives passage to an appropriate equivalence class of encodings (or, operationally, whatever can be transported across observers with small defect), whereas "stable self" adds the requirement that such transport remains coherent across time-indexed self-updates. The point of almost-commutativity is then practical: if transferring between encodings and updating within an encoding nearly commute, then conclusions produced by iterated reasoning are (approximately) encoding-invariant, with deviations controlled by the size of the relevant defects.

69.1 Preliminaries: spaces, updates, and defects

69.1.1 Representation spaces and update operators

Definition 654 (Metric representation space)**.** *A representation space is a metric space (X, d) whose points encode some structured content (world-model states, beliefs, policies, plans, self-models, etc.).*

In applications, the metric d is part of the modeling choice: it specifies which differences between representations are considered significant and at what scale. Depending on the setting, d may reflect prediction error, policy divergence, edit distance in a symbolic structure, or a task-induced discrepancy (e.g. value loss). It is also common in practice that the natural notion of "distance" is a pseudometric (distinct encodings at zero distance); in that case, one can either work directly with the pseudometric or pass to the quotient space induced by identifying zero-distance points, which aligns naturally with the invariance interpretation.

Definition 655 (Contractive update)**.** *An operator $B : X \to X$ is a γ-contraction if for all $x, y \in X$,*

$$d(Bx, By) \;\le\; \gamma\, d(x, y), \quad \textit{for some } \gamma \in [0, 1).$$

Contractivity is a strong but extremely useful regularity condition: it says that the update step dampens discrepancies rather than amplifying them. When (X, d) is complete, the contraction mapping theorem implies B has a unique fixed point and that iterating B converges to it at a geometric rate, which provides a canonical notion of a "stable conclusion" produced by repeated updating. Even when strict contractivity is only approximate or holds locally, the parameter γ still serves as a quantitative proxy for stability, since it controls how perturbations propagate through sequences of updates.

Remark 3722. *In TransWeave, a prototypical B is a Bellman-style backup operator; in belief dynamics, B may be an inference or posterior-update step; in self-modeling, B may be a self-revision step. The contraction assumption can be interpreted as "controlled drift" or "regulated update."*

A useful intuition is that contraction turns "one-step mismatch" into a bounded "long-run mismatch": small discrepancies introduced by encoding changes, noise, or approximation errors do not accumulate arbitrarily under iteration, but are instead attenuated. This is precisely the regime in which almost-commutativity defects can be converted into global path-independence bounds, because the update dynamics themselves prevent error blow-up.

69.1.2 Intertwining and commutator defects

Definition 656 (Intertwining defect)**.** *Given $(X_1, d_1), (X_2, d_2)$, operators $B_1 : X_1 \to X_1$, $B_2 : X_2 \to X_2$, and a map $T : X_1 \to X_2$, define the (supremum) intertwining defect*

$$\mathrm{Def}(T; B_1, B_2) \;:=\; \sup_{x \in X_1} d_2\big(T(B_1 x),\, B_2(Tx)\big).$$

When $\mathrm{Def}(T; B_1, B_2) \le \epsilon$ we say T is an ϵ-intertwiner.

This quantity measures the extent to which T transports the update dynamics on X_1 to the update dynamics on X_2. If T were an exact conjugacy between the systems, then $T \circ B_1 = B_2 \circ T$ and the defect would be zero; the ϵ-intertwiner notion is the approximate version appropriate when T is an imperfect translator, when B_2 is a simplified surrogate for B_1, or when the two spaces represent different observers with slightly different update rules. The supremum form is a worst-case notion; in settings where worst-case control is too strict, one can also consider average-case analogues (e.g. an expectation over a distribution of x), but the supremum is the natural choice when aiming for representation-agnostic guarantees.

Definition 657 (Commutator defect)**.** *For $U, V : X \to X$, define the commutator defect*

$$\mathrm{Def}([U, V]) \;:=\; \sup_{x \in X} d\big(U(Vx),\, V(Ux)\big).$$

The commutator defect is the special case of order-sensitivity when both operators act on the same space; it can be viewed as an "internal" almost-commutativity constraint. Note that if one takes $X_1 = X_2 = X$, $B_1 = U$, $B_2 = U$, and $T = V$, then $\mathrm{Def}(T; B_1, B_2)$ reduces exactly to $\mathrm{Def}([U, V])$, so the two notions are not competing definitions but rather different framings of the same phenomenon (transport-vs-update versus update-vs-update). In particular, commutator defects can be read as gauge-like consistency conditions when one of the maps is interpreted as a re-encoding inside a single representational universe.

Remark 3723. $\mathrm{Def}(T; B_1, B_2)$ *is the TransWeave-type "almost commutativity" of "transfer then update" vs "update then transfer."* $\mathrm{Def}([U, V])$ *is the usual notion of order-sensitivity of two update operators. Both will be treated as computational analogues of curvature/field-strength: nonzero defect implies path dependence.*

The curvature analogy can be read operationally: moving around a loop of transformations (e.g. update, re-encode, update, decode) fails to return exactly to the starting point when the defect is nonzero. When the defect is small and the updates are contractive, this loop error remains controlled even under repeated application, which is the mechanism by which "nearly gauge-invariant" conclusions emerge from iterative reasoning despite imperfect translators and approximate dynamics.

69.2 Gauge transformations and gauge-invariant content

69.2.1 Semantics and gauge

Definition 658 (Semantic decoder and gauge group)**.** *Let W be a "semantic" space (world-states, externally meaningful states, etc.). A semantic decoder is a map $D : X \to W$.*

A gauge group G acting on X is a group of bijections $g : X \to X$ such that

$$D \circ g \;=\; D \quad \text{for all } g \in G.$$

That is, gauge transformations change the encoding but preserve semantics.

Informally, the condition $D \circ g = D$ says that D is *constant along the G-action*: whenever x is replaced by another representative $g(x)$ within the same internal "description family," the decoded meaning in W is unchanged. In particular, the fibers of D (sets of the form $D^{-1}(w)$) are unions of G-orbits: if $x \in D^{-1}(w)$ then $g(x) \in D^{-1}(w)$ for all $g \in G$. This expresses the intended separation between "how the system is written" (the state $x \in X$) and "what it means" (the decoded element $D(x) \in W$).

It is also useful to view the G-action as generating an equivalence relation on X:

$$x \sim_G x' \quad \Longleftrightarrow \quad \exists\, g \in G \text{ such that } x' = g(x).$$

The defining property $D \circ g = D$ then implies $x \sim_G x' \Rightarrow D(x) = D(x')$, i.e. semantic equality is at least as coarse as gauge-equivalence (and may be coarser if distinct orbits decode to the same semantic element).

Definition 659 (Gauge-invariant observable)**.** *A map $O : X \to Y$ is gauge-invariant (with respect to G) if*

$$O \circ g \;=\; O \quad \text{for all } g \in G.$$

Gauge-invariant observables are precisely those quantities that cannot distinguish between different encodings of the same underlying situation. When Y is taken to be a space of measurements, predictions, or reports, the invariance condition formalizes that these outputs depend only on semantic content (or, more generally, only on the gauge-orbit of x) rather than on arbitrary representational choices. In many settings one considers families of observables (e.g. all real-valued observables), in which case gauge-invariant observables form a subcollection closed under natural operations (for example, if $Y = \mathbb{R}$ then sums and products of gauge-invariant functions are again gauge-invariant).

69.2.2 Quotients and invariants

Definition 660 (Orbit space)**.** *Let X/G be the set of orbits under G and let $q : X \to X/G$ be the orbit map sending x to its orbit $[x]$.*

The orbit map q is the canonical way to "forget gauge" by mapping each state to its equivalence class under the action. Concretely, $q(x) = q(x')$ holds exactly when x and x' differ by a gauge transformation. From the semantics perspective, X/G represents the space of representational degrees of freedom modded out by the allowed re-encodings; it is therefore the natural domain for any quantity intended to be insensitive to encoding.

Note that no topological, smooth, or measure-theoretic structure is assumed here; X/G is treated as a set. If additional structure on X and continuity/smoothness of the G-action are present, one often equips X/G with the corresponding quotient structure, and then asks for $\bar{O}$ to be continuous/smooth as appropriate. The purely set-theoretic statement below isolates the core invariance principle without committing to such extra hypotheses.

Theorem 239 (Universal factorization of gauge invariants)**.** *A map $O : X \to Y$ is gauge-invariant iff there exists a (unique) map $\bar{O} : X/G \to Y$ such that*

$$O \;=\; \bar{O} \circ q.$$

Proof. ($\Rightarrow$) If O is gauge-invariant, define $\bar{O}([x]) := O(x)$. This is well-defined because if $[x] = [x']$ then $x' = g(x)$ for some $g \in G$, so $O(x') = O(gx) = O(x)$. Clearly $O = \bar{O} \circ q$.

($\Leftarrow$) If $O = \bar{O} \circ q$, then for any $g \in G$ we have $q(gx) = q(x)$ since gx lies in the orbit of x. Thus $O(gx) = \bar{O}(q(gx)) = \bar{O}(q(x)) = O(x)$. □

The uniqueness of $\bar{O}$ follows from surjectivity of q: every orbit $[x] \in X/G$ has at least one representative $x \in X$, so any factorization $O = \tilde{O} \circ q$ forces $\tilde{O}([x]) = O(x)$, which is exactly the definition used in the forward direction. In this sense, X/G is characterized by a universal property: it is the *most general* recipient through which all gauge-invariant maps out of X factor.

A particularly relevant special case is the semantic decoder itself: since $D \circ g = D$ for all $g \in G$, Theorem 239 implies that there exists a unique $\bar{D} : X/G \to W$ such that $D = \bar{D} \circ q$. Thus, once gauge is fixed as "meaning-preserving re-encoding," semantics is automatically a function of the orbit alone. This makes explicit that any two gauge-related internal states have identical decoded meaning and that the quotient X/G is a natural "semantic carrier" for all invariant content derived from X.

Remark 3724. *This theorem captures the intended meaning of "shared reality" as "what survives a change of encoding": gauge-invariant observables are precisely the functions on the quotient X/G.*

One may read this remark operationally: if two agents, models, or subsystems use different internal representations related by elements of G, then any quantity that can be compared between them without additional alignment assumptions must factor through X/G. In this way, X/G packages the common, representation-independent content, while the particular choice of representative $x \in X$ encodes "private" or "conventional" degrees of freedom. When relating this to almost-commutative or "TransWeave-style" constraints elsewhere, the same factorization principle serves as the baseline requirement: any purportedly shared signal must commute (in the sense of being invariant) with the allowable re-encodings, hence must descend to the quotient.

69.3 Gauge covariance of dynamics and fixed-point invariance

69.3.1 Exact covariance implies exact invariance

Definition 661 (Gauge-covariant update). *An update $B : X \to X$ is G-equivariant (gauge-covariant) if*

$$B \circ g \;=\; g \circ B \quad \text{for all } g \in G.$$

This condition can be read as "the dynamics commutes with re-encoding": applying a gauge transform g before updating produces the same state as updating first and then re-expressing the result in the transformed gauge. In diagram form, for each $g \in G$, the square

$$\begin{array}{ccc} X & \xrightarrow{B} & X \\ \downarrow g & & \downarrow g \\ X & \xrightarrow{B} & X \end{array}$$

commutes. When G acts by "representational" changes (coordinate changes, renamings, internal phase choices, etc.), equivariance asserts that B is formulated intrinsically rather than in a gauge-specific manner.

Theorem 240 (Unique fixed point implies gauge-invariant fixed point). *Let $B : X \to X$ be a γ-contraction with a unique fixed point x^*. If B is G-equivariant, then x^* is gauge-invariant:*

$$g(x^*) \;=\; x^* \quad \text{for all } g \in G.$$

The contraction hypothesis is a convenient sufficient condition ensuring both existence and uniqueness of the fixed point (via the Banach fixed-point theorem). Conceptually, the conclusion is a "no spurious gauge choice" statement: if the update rule respects the gauge action and the attractor is unique, then the attractor cannot depend on a gauge representative. In contrast, if fixed points are not unique, equivariance typically implies only that the set of fixed points is G-stable (i.e., mapped into itself), not that any particular fixed point is pointwise invariant.

Proof. Fix $g \in G$. Because B is equivariant,

$$B(gx^*) \;=\; g(Bx^*) \;=\; g(x^*).$$

So $g(x^*)$ is a fixed point of B. Uniqueness implies $g(x^*) = x^*$. □

Equivalently, the proof shows that equivariance transports fixed points along the group orbit: x^* fixed $\Rightarrow g(x^*)$ fixed. When there is exactly one fixed point, the entire orbit must collapse to a single point, which is exactly gauge invariance. This is the fixed-point analogue of the more general principle that symmetry plus uniqueness forces symmetry of the solution.

69.3.2 Approximate covariance implies bounded drift

Definition 662 (Equivariance defect). *For $g : X \to X$ and $B : X \to X$, define*

$$\mathrm{Def}(g; B) \;:=\; \sup_{x \in X} d\big(g(Bx),\, B(gx)\big).$$

The quantity $\mathrm{Def}(g;B)$ is an "almost-commutator" size: it measures the worst-case discrepancy between the two paths $x \mapsto Bx \mapsto g(Bx)$ and $x \mapsto gx \mapsto B(gx)$. In particular, $\mathrm{Def}(g;B) = 0$ is exactly the equivariance condition for that specific g, and small defect formalizes the idea that the update is only weakly sensitive to the gauge choice. The use of $\sup_{x\in X}$ emphasizes uniform control; in settings where only a subset of states is relevant (e.g., a basin of attraction), one can replace the supremum by a supremum over that subset without changing the algebra of the subsequent bounds.

Theorem 241 (Approximate gauge invariance of a contractive fixed point)**.** *Let $B : X \to X$ be a γ-contraction with fixed point x^*. For any bijection $g : X \to X$,*

$$d\big(g(x^*),\, x^*\big) \;\le\; \frac{\mathrm{Def}(g;B)}{1-\gamma}.$$

The factor $\frac{1}{1-\gamma}$ is the familiar stability constant for contraction-based fixed points: as $\gamma \to 1^-$ the system becomes less contractive, and small equivariance errors can produce larger fixed-point drift. Conversely, stronger contraction (smaller γ) suppresses the impact of gauge noncommutativity. Note also that the statement is pointwise in g: no group structure is required beyond having a map g one wishes to interpret as a "re-encoding." The bijection assumption matches the intended interpretation of gauge transforms as reversible changes of representation, but the inequality itself is driven by the defect and contraction and can be adapted to less restrictive classes of maps when appropriate.

Proof. Using the triangle inequality and the fixed-point identity $Bx^* = x^*$,

$$\begin{aligned} d(gx^*, x^*) &= d(gBx^*, Bx^*) \\ &\le d(gBx^*, Bgx^*) \;+\; d(Bgx^*, Bx^*) \\ &\le \mathrm{Def}(g;B) \;+\; \gamma\, d(gx^*, x^*). \end{aligned}$$

Rearranging gives $(1-\gamma)d(gx^*, x^*) \le \mathrm{Def}(g;B)$. ∎

A useful way to read the proof is as a one-step "commute-then-contract" argument: the defect controls the cost of swapping g past B, and the contraction turns the remaining discrepancy into a self-bounding term that can be absorbed to the left-hand side. This is exactly the mechanism by which "almost commutativity" yields "almost invariance" at the attracting fixed point.

Corollary 83 (Exact equivariance as a special case)**.** *If* $\mathrm{Def}(g;B) = 0$ *for some bijection $g : X \to X$, then $g(x^*) = x^*$. In particular, if* $\mathrm{Def}(g;B) = 0$ *for all $g \in G$, then x^* is gauge-invariant.*

Remark 3725. *Theorem 241 is the basic "stable meaning under re-encoding" tool: if the commutator (equivariance defect) is small, then the "meaningful" fixed point is stable. This is the same algebraic shape as standard TransWeave bounds for transfer of fixed points.*

One can also interpret the bound as a quantitative statement about shared reality: different "gauges" (encodings) may not be perfectly aligned with the update dynamics, but as long as the mismatch is uniformly bounded and the dynamics is sufficiently contractive, the emergent fixed point remains approximately shared across gauges. The dependence on $\mathrm{Def}(g;B)$ isolates where modeling errors live (in the noncommuting part), while the dependence on γ isolates the dynamical robustness (how strongly the update collapses differences). In applications, this separation is useful because one can often estimate γ from the dynamics and estimate $\mathrm{Def}(g;B)$ from representational mismatch or finite-precision implementation.

69.4 TransWeave-style intertwining implies representation-independent conclusions

We now treat changes of observer/encoding as maps between two (possibly different) representation spaces X_1, X_2.

Theorem 242 (Fixed-point transport under approximate intertwining)**.** *Let $B_1 : X_1 \to X_1$ and $B_2 : X_2 \to X_2$ be γ-contractions with unique fixed points x_1^* and x_2^*. Let $T : X_1 \to X_2$ satisfy $\mathrm{Def}(T; B_1, B_2) \le \epsilon$. Then*

$$d_2\big(T(x_1^*),\, x_2^*\big) \;\le\; \frac{\epsilon}{1-\gamma}.$$

Proof. Let x_2^* be the fixed point of B_2. Then

$$\begin{aligned} d_2(Tx_1^*, x_2^*) &= d_2(TB_1x_1^*, B_2x_2^*) \\ &\le d_2(TB_1x_1^*, B_2Tx_1^*) \;+\; d_2(B_2Tx_1^*, B_2x_2^*) \\ &\le \epsilon \;+\; \gamma\, d_2(Tx_1^*, x_2^*). \end{aligned}$$

Rearrange to get the claim. □

Remark 3726. *This is the core TransWeave mathematical move: if transfer T approximately intertwines the update operators, it transports fixed points (solutions, equilibria, stable beliefs) with error bounded by $\epsilon/(1-\gamma)$.*

69.5 Shared reality across observers as atlas gluing

69.5.1 Observer charts and transition maps

Definition 663 (Observer chart)**.** *An observer i has a representation space (X_i, d_i) and a semantic decoder $D_i : X_i \to W$. A transition map from observer i to j is a map $T_{ij} : X_i \to X_j$ intended to preserve semantics:*

$$D_j \circ T_{ij} \;\approx\; D_i.$$

Definition 664 (Exact cocycle condition)**.** *A family of transition maps $\{T_{ij}\}$ satisfies the cocycle condition if for all i, j, k,*

$$T_{jk} \circ T_{ij} \;=\; T_{ik}, \quad \text{and} \quad T_{ii} = \mathrm{id}_{X_i}.$$

69.5.2 Existence of a global "shared reality" space

Theorem 243 (Exact gluing yields a global latent reality)**.** *Suppose $\{T_{ij}\}$ satisfies the cocycle condition and each T_{ij} is a bijection. Then there exists a set Z and bijections $g_i : Z \to X_i$ such that for all i, j,*

$$T_{ij} \;=\; g_j \circ g_i^{-1}.$$

Moreover, any family of semantics maps $D_i : X_i \to W$ satisfying $D_j \circ T_{ij} = D_i$ factors through a single semantic map $D : Z \to W$ via $D_i = D \circ g_i^{-1}$.

Proof. Construct Z as the quotient of the disjoint union $\bigsqcup_i X_i$ by the equivalence relation generated by $(x, i) \sim (T_{ij}(x), j)$. The cocycle condition implies this is an equivalence relation and that each class has exactly one representative in each X_i. Define $g_i([x, i]) := x$. This is well-defined and bijective, and $T_{ij} = g_j \circ g_i^{-1}$ follows immediately.

For the semantic statement, define $D([x, i]) := D_i(x)$. This is well-defined because if $(x, i) \sim (x', j)$ then $x' = T_{ij}(x)$ and $D_j(x') = D_j(T_{ij}x) = D_i(x)$. Then $D_i = D \circ g_i^{-1}$ holds by construction. □

Remark 3727. *Theorem 243 formalizes "shared reality" as a glued atlas: observers are charts, and Z is the chart-independent underlying object.*

69.5.3 Holonomy as incommensurability witness

Definition 665 (Loop (holonomy) defect)**.** *For a loop $i \to j \to k \to i$, define*

$$H_{i,j,k} \;:=\; \sup_{x \in X_i} d_i\big((T_{ki} \circ T_{jk} \circ T_{ij})x,\, x\big).$$

Theorem 244 (Nonzero holonomy obstructs exact shared reality)**.** *If there exists a triple (i, j, k) with $H_{i,j,k} > 0$, then no choice of a set Z and bijections $g_i : Z \to X_i$ can satisfy $T_{ij} = g_j \circ g_i^{-1}$ for all i, j.*

Proof. If such Z and g_i existed, then for any loop $i \to j \to k \to i$,

$$T_{ki} \circ T_{jk} \circ T_{ij} = (g_i \circ g_k^{-1}) \circ (g_k \circ g_j^{-1}) \circ (g_j \circ g_i^{-1}) = \mathrm{id}_{X_i},$$

so $H_{i,j,k} = 0$, contradiction. □

Remark 3728. *This is a clean formal handle on "incommensurability" of paradigms: persistent holonomy means there is no exact global viewpoint that makes all translations consistent. (Approximate global viewpoints are still possible; see below.)*

69.5.4 Approximate gluing from bounded holonomy

Theorem 245 (Tree-based approximate trivialization)**.** *Assume each T_{ij} is a bijection and all loop defects $H_{0,i,j}$ (loops $0 \to i \to j \to 0$) are bounded by η. Fix a root observer 0 and define $g_i := T_{0i} : X_0 \to X_i$ and $g_i^{-1} := T_{i0}$. Suppose also that*

$$\sup_{x \in X_0} d_0\big((T_{i0} \circ T_{0i})x,\, x\big) \;\le\; \eta \quad \text{for all } i.$$

Then for all i, j and all $x \in X_i$,

$$d_j\big(T_{ij}(x),\, (g_j \circ g_i^{-1})(x)\big) \;\le\; C\,\eta,$$

where C depends only on uniform Lipschitz bounds for the maps involved.

Proof. Write $x = g_i(u)$ for $u = g_i^{-1}(x) = T_{i0}(x)$. Consider the loop $0 \to i \to j \to 0$ applied to $u \in X_0$:

$$T_{j0} \circ T_{ij} \circ T_{0i}(u) \;\approx\; u,$$

with defect at most η by hypothesis. Substitute $g_i = T_{0i}$ and $g_j^{-1} = T_{j0}$:

$$g_j^{-1}(T_{ij}(g_i(u))) \;\approx\; u.$$

Apply g_j to both sides and use Lipschitz continuity of g_j plus the fact that $g_j \circ g_j^{-1}$ is approximately the identity (within η) to obtain

$$T_{ij}(g_i(u)) \;\approx\; g_j(u),$$

with error bounded by a constant times η. Replacing $g_i(u)$ by x yields the claim. □

Remark 3729. *This theorem is the computational analogue of "small curvature implies an approximately flat bundle": bounded holonomy yields an approximately consistent shared-reality latent space (relative to a chosen root).*

69.6 Stable self as a gauge-invariant attractor

We now apply the same gauge machinery to the "self" of a mind across time. In this subsection, the role played earlier by "shared reality" is played by the identity of an agent: we treat a persistent self not as a single state in X, but as an equivalence class that is invariant under allowable re-encodings of internal structure.

Definition 666 (Self-gauge group and self-quotient)**.** *Let (X, d) be a mind-state space and let G be a gauge group capturing semantics-preserving re-encodings of internal structures. Define the self-quotient space $S := X/G$ with orbit map $q : X \to S$.*

The intent is that $x \in X$ may include many "implementation details" (memory layout, latent-coordinate choices, symbol grounding conventions, or other representational degrees of freedom) that can vary without changing the intended content. The gauge group G formalizes these variations, and the orbit $[x] = \{gx : g \in G\}$ represents all encodings of the same underlying self-description.

In applications, one typically equips S with a metric (or at least a pseudometric) that measures distance *between orbits* rather than between representatives. A standard choice compatible with d is

$$\bar{d}([x], [y]) \ := \ \inf_{g \in G} d(x, gy),$$

or the symmetric variant $\inf_{g,h \in G} d(gx, hy)$. When the group action is well-behaved (e.g., isometric, proper, or with closed orbits), such formulas define a meaningful geometry on S; otherwise they may only define a pseudometric, which is still sufficient for many fixed-point arguments provided one works modulo zero-distance identifications.

Definition 667 (Self-dynamics and induced quotient dynamics)**.** *Let $U : X \to X$ be an update representing one "moment" of self-modification or cognitive transition. If U is G-equivariant, then it induces a well-defined map $\bar{U} : S \to S$ by $\bar{U}([x]) := [Ux]$.*

Equivariance means $U(gx) = g(Ux)$ for all $g \in G$, i.e., updating and re-encoding commute exactly. This is the formal statement that the update rule respects the semantic quotient: applying U does not depend on arbitrary choices of representation within a gauge orbit. In particular, if two mind-states x and x' differ only by a semantic-preserving re-encoding ($x' = gx$), then equivariance guarantees that their successors Ux and Ux' still differ by a (possibly the same) re-encoding, so the induced map $\bar{U}$ on equivalence classes is independent of representative.

One concrete way to read $\bar{U}([x]) = [Ux]$ is that $\bar{U}$ is the evolution rule for the "identity-relevant" degrees of freedom, after quotienting out implementation details. Thus, even if U is complicated on X, the induced $\bar{U}$ can be comparatively simple on S when many internal degrees of freedom are gauge.

Theorem 246 (Stable self from contractive quotient dynamics)**.** *Assume U is G-equivariant and $\bar{U} : S \to S$ is a γ-contraction (on some metric on S). Then $\bar{U}$ has a unique fixed point $[x^*] \in S$. For any initial $x_0 \in X$, the self-orbit $[U^t x_0]$ converges to $[x^*]$.*

The substantive assumption here is contraction on the quotient: it asserts that identity-relevant differences shrink over time, even if raw differences in X might not. Conceptually, this models a cognitive process that "forgets" representational contingencies and drives many initial self-configurations toward a single stable identity class. Note that the fixed point is a *class* $[x^*]$; there need not be a unique fixed representative $x^* \in X$ unless additional gauge-fixing conditions are imposed.

Proof. By Banach's fixed point theorem on $(S, \bar{d})$, the contraction $\bar{U}$ has a unique fixed point $[x^*]$ and iterates converge to it from any start. Because $[U^t x_0] = \bar{U}^t([x_0])$, convergence of the self-orbit follows.

More explicitly, for any initial class $[x_0] \in S$, the sequence $[x_{t+1}] = \bar{U}([x_t])$ with $[x_t] = [U^t x_0]$ satisfies $\bar{d}([x_t], [x^*]) \leq \gamma^t \bar{d}([x_0], [x^*])$, so the approach to the stable self is geometric in the quotient metric. This quantifies stability as robustness to perturbations *modulo gauge*: deviations that are not purely representational are pulled back toward the attractor class at rate γ. □

Remark 3730. *This gives a precise sense in which a "stable self" can exist even when the internal encoding changes: the stable object is the orbit class $[x] \in X/G$, not any particular representative.*

A useful way to interpret this is that insisting on a stable representative $x \in X$ is often the wrong requirement: any concrete internal realization is expected to drift as implementation details evolve. The theorem instead identifies stability with convergence in S, meaning that after modding out semantic-preserving transformations, the agent's identity becomes time-invariant. In settings where one additionally chooses a gauge-fixing rule (a canonical representative per orbit), the convergence in S can sometimes be lifted to a notion of convergence in X *relative to that gauge choice*, but such a lift is extra structure not assumed here.

Theorem 247 (Identity drift bound under approximate equivariance)**.** *Let U be a γ-contraction on (X, d) with fixed point x^*. Let $g \in G$ be a re-encoding map with equivariance defect* $\mathrm{Def}(g; U) \leq \epsilon$*. Then*

$$d\big(g(x^*), x^*\big) \;\leq\; \frac{\epsilon}{1-\gamma}.$$

More generally, for any $t \geq 0$,

$$d\big(g(U^t x),\, U^t(gx)\big) \;\leq\; \frac{\epsilon}{1-\gamma}.$$

Here $\mathrm{Def}(g; U)$ measures how far U fails to commute with the re-encoding g. A common concrete definition (consistent with the usage in the proof) is

$$\mathrm{Def}(g; U) \;:=\; \sup_{x \in X} d\big(g(Ux),\, U(gx)\big),$$

so ϵ is a worst-case commutator size in the metric d. The theorem then states that if U is contractive, even a persistent, small non-commutativity produces only bounded identity drift: discrepancies introduced by re-encoding cannot amplify without bound under a contraction. The bound $\epsilon/(1-\gamma)$ is the usual "geometric series" control that also appears in stability of iterative algorithms and perturbed dynamical systems.

Proof. The first inequality is Theorem 241 with $B = U$.

For the second, define $\Delta_t := \sup_x d(gU^t x, U^t gx)$. Then

$$\Delta_{t+1} = \sup_x d(gU^{t+1}x, U^{t+1}gx) \leq \sup_x d(gU(U^t x), Ug(U^t x)) + \sup_x d(Ug(U^t x), U^{t+1}gx).$$

The first term is at most ϵ by definition of $\mathrm{Def}(g; U)$; the second term is at most $\gamma \Delta_t$ by contraction of U. So $\Delta_{t+1} \leq \epsilon + \gamma \Delta_t$, which implies $\Delta_t \leq \epsilon/(1-\gamma)$ for all t.

Equivalently, unrolling the recursion yields $\Delta_t \leq \epsilon \sum_{k=0}^{t-1} \gamma^k \leq \epsilon/(1-\gamma)$, making explicit that the total discrepancy is the accumulated effect of the per-step equivariance defect, discounted by the contraction at later steps. In particular, when γ is close to 0 (strong contraction), the bound is essentially ϵ, whereas as $\gamma \to 1^-$ (weak contraction), small defects can accumulate into larger steady-state disagreement. □

69.7 Almost-commutativity as computational "well-defined physics"

A recurring source of ambiguity in iterative systems is *update ordering*: when multiple local updates are available, different schedules (orders of application) may produce different outcomes. In computational terms, a system is closer to "well-defined physics" when its macroscopic behavior is *schedule-independent* (exactly, in the commuting case) or at least *schedule-robust* (approximately, in the almost-commuting case). The results below make this precise under standard contraction hypotheses, which formalize the intuition that updates are *dissipative* (they forget initial conditions at a controlled rate).

Throughout, (X, d) is a metric space; when invoking fixed-point existence/uniqueness via contraction arguments, one typically assumes X is complete. A map $U : X \to X$ is a γ-contraction (with $0 \le \gamma < 1$) if

$$d(Ux, Uy) \le \gamma\, d(x, y) \quad \text{for all } x, y \in X.$$

In particular, a γ-contraction is γ-Lipschitz, and repeated application yields the familiar exponential forgetting:

$$d(U^t x, U^t y) \le \gamma^t d(x, y).$$

We also use the basic fact that compositions of γ-contractions remain contractions: if U and V are each γ-contractions, then UV is γ^2-Lipschitz (hence also γ-Lipschitz, albeit with a weaker constant), and more generally a composition of k such maps is at most γ^k-Lipschitz.

69.7.1 Commuting contractions yield schedule-independence

Theorem 248 (Commuting contractions have a common fixed point)**.** *Let $U, V : X \to X$ be γ-contractions that commute: $UV = VU$. Then U and V share the same unique fixed point x^*. Moreover, for any schedule (any word) in $\{U, V\}$, the iterates converge to x^*.*

Proof. Because U is a contraction it has a unique fixed point x_U^*. Then $U(Vx_U^*) = V(Ux_U^*) = Vx_U^*$, so Vx_U^* is a fixed point of U. Uniqueness implies $Vx_U^* = x_U^*$, so x_U^* is also a fixed point of V. By symmetry, the fixed points coincide; call it x^*.

Any finite composition of γ-contractions is a γ-contraction, hence its iterates converge to its unique fixed point; since x^* is a fixed point of each operator and they commute, it is a fixed point of any composition. Thus all schedules converge to x^*.

For clarity on the last step: if W is any composition (word) in $\{U, V\}$, then W is a contraction (with Lipschitz constant at most $\gamma^{\ell(W)}$, where $\ell(W)$ is the word length), so by the contraction mapping principle it has a unique fixed point. Because x^* is fixed by both U and V, we have $Wx^* = x^*$. Hence the unique fixed point of W must be x^*, and iterating W from any start converges to x^*. This is the sense in which *every schedule defines the same asymptotic physics* when updates commute. □

69.7.2 Small commutators yield bounded path dependence

To quantify "almost commuting," we need a way to measure how far two operators are from commuting when applied to arbitrary states. One convenient proxy is a *commutator defect* bound of the form

$$\mathrm{Def}([A, B]) \le \delta,$$

which can be read operationally as a uniform estimate

$$d(ABx, BAx) \le \delta \quad \text{for all } x \in X.$$

(Any definition consistent with this interpretation suffices for the bound below; the key point is that δ upper-bounds the *worst-case* mismatch caused by swapping A and B once.)

Definition 668 (Kendall tau distance between schedules). *Let σ and π be two length-n sequences built from the same multiset of operators. Let $K(\sigma, \pi)$ be the minimal number of adjacent swaps needed to transform σ into π.*

The Kendall tau distance is exactly the right combinatorial notion because any reordering of a multiset can be achieved by adjacent transpositions, and each adjacent transposition corresponds to swapping two neighboring updates—the primitive local change to which we can attach a stability estimate.

Theorem 249 (Schedule robustness from small commutator defect). *Let $\mathcal{U} = \{U_1, \dots, U_m\}$ be operators on (X, d) such that each U_i is a γ-contraction. Assume a uniform commutator defect bound:*

$$\operatorname{Def}([U_i, U_j]) \;\leq\; \delta \quad \textit{for all } i, j.$$

Let σ and π be two schedules (two sequences) of length n built from the same multiset of operators in $\mathcal{U}$. Then for all $x \in X$,

$$d\big(U_{\sigma(n)} \cdots U_{\sigma(1)} x, \quad U_{\pi(n)} \cdots U_{\pi(1)} x\big) \;\leq\; \frac{K(\sigma, \pi)\, \delta}{1 - \gamma}.$$

Proof. It suffices to consider one adjacent swap, since any transformation from σ to π can be realized by $K(\sigma, \pi)$ adjacent swaps and we can sum bounds.

So suppose π differs from σ by swapping adjacent operators A, B at some position, with the same prefix P before them and suffix S after them. We compare $SABPx$ vs $SBAPx$.

Let $y := Px$. Then

$$d(SABy, SBAy) \;\leq\; \gamma^{|S|}\, d(ABy, BAy) \;\leq\; \gamma^{|S|}\delta \;\leq\; \delta.$$

(We used that S is a composition of contractions, hence $\gamma^{|S|}$-Lipschitz.) Now consider the effect of further iterations or repeated application of similar swaps: after each swap, subsequent contractions can amplify differences by at most a geometric series factor $1/(1 - \gamma)$. Thus each swap contributes at most $\delta/(1 - \gamma)$ to the final discrepancy, and summing over $K(\sigma, \pi)$ swaps yields the claim.

To make the "geometric series" step explicit, one can argue via a standard recurrence. Let $x^{(k)}$ denote the state produced after performing the first k adjacent swaps along a chosen minimal swap path from σ to π, and let $\Delta_k := d(x^{(k)}, x^{(k-1)})$ be the perturbation introduced at the k-th swap (measured after completing the remaining suffix updates). The single swap bound above controls the *immediate* mismatch at the swap location by $\leq \delta$, while the remaining updates (the suffix after the swap position) are contractions that can only *shrink* this mismatch. However, because the swap can occur at different depths in the word, it is convenient to bound conservatively by allowing a worst-case factor that accounts for how a mismatch propagates through the remainder of the schedule. Concretely, if the remainder of the word applies at most one γ-contraction per step, then the total effect of an injected discrepancy of size δ is bounded by

$$\delta + \gamma\delta + \gamma^2\delta + \cdots \leq \frac{\delta}{1 - \gamma}.$$

This yields an upper bound $\Delta_k \leq \delta/(1 - \gamma)$ for each adjacent swap along the path, and then the triangle inequality gives

$$d(x^{(K)}, x^{(0)}) \leq \sum_{k=1}^{K} \Delta_k \leq \frac{K\, \delta}{1 - \gamma},$$

where $K = K(\sigma, \pi)$. This is precisely the displayed inequality. □

Remark 3731. *Theorem 249 is the exact statement of "almost-commutativity implies near path-independence." Interpreting δ as curvature, this is a discrete integrability bound: small curvature means the system behaves as though it has well-defined, coordinate-free dynamics.*

More concretely, each adjacent swap is a discrete analog of changing the order of two infinitesimal moves; the commutator defect measures the resulting holonomy *(the failure to close) around a minimal "square." The bound then states that the total holonomy accumulated along a path of reorderings is controlled linearly by the number of such squares traversed (the Kendall tau distance) and by the local curvature scale δ, with the factor $1/(1-\gamma)$ capturing the degree to which the dynamics damp (or fail to damp) local inconsistencies.*

69.7.3 Connecting to TransWeave and Hyperseed invariants

The preceding theorem treats reorderings of updates drawn from a set $\mathcal{U}$. A closely related special case, used repeatedly in gauge-style arguments, is the comparison between "apply a gauge transform then update" and "update then apply a gauge transform." In that setting the relevant commutator defect is measured between a re-encoding map T and an update map B.

Corollary 84 (Coordinate-change commutation as a physics-like constraint)**.** *Let T be a re-encoding (gauge) map and let B be an update. If* $\mathrm{Def}(T; B, B) \le \epsilon$*, then the conclusions of repeated updating are stable under re-encoding:*

$$d\big(T(B^t x),\ B^t(Tx)\big) \ \le\ \frac{\epsilon}{1-\gamma} \quad \textit{for all } t.$$

Proof. Apply Theorem 247 with $g = T$ and $U = B$ (or repeat the standard recurrence $\Delta_{t+1} \le \epsilon + \gamma\Delta_t$).

For completeness, the recurrence viewpoint is as follows. Let

$$\Delta_t := d\big(T(B^t x),\ B^t(Tx)\big).$$

The defect assumption $\mathrm{Def}(T; B, B) \le \epsilon$ is used to bound the one-step mismatch between "push forward by T" and "evolve by B"; combined with the fact that B is γ-contractive (so applying B after a mismatch shrinks it by at most γ), one obtains

$$\Delta_{t+1} \le \epsilon + \gamma\Delta_t.$$

Iterating and summing the resulting geometric series yields $\Delta_t \le \epsilon(1 + \gamma + \cdots + \gamma^{t-1}) \le \epsilon/(1-\gamma)$, which is the displayed bound. □

Remark 3732. *This is the computational counterpart of coordinate covariance in physics: "change coordinates then evolve" is (approximately) the same as "evolve then change coordinates." In Hyperseed language, this is the formal handle on "shared reality" (observer changes) and "stable self" (self-re-encoding) as invariants.*

Operationally, the inequality bounds the disagreement between two observers (or two internal encodings) who apply the same underlying update dynamics but in different representational frames. The contraction parameter γ controls how strongly the system forgets representational discrepancies, while ϵ controls how far the encoding change T fails to be an exact symmetry of the update rule B. When $\epsilon = 0$ one recovers strict covariance; when $\epsilon > 0$ one obtains a quantitative, time-uniform stability guarantee.

69.8 Discussion: what a reasoning system can do with these theorems

- Gauge-invariant features: By Theorem 239, any candidate "shared reality" proposition should be implementable as a function on X/G. Operationally, if a feature changes under a known re-encoding, it is not shared reality (it is representational artifact). In practice, this turns the search for "objective" content into a concrete design rule: define the gauge group G of admissible reparameterizations (choice of coordinates, tokenization, embedding basis, internal naming conventions), and then require any downstream claim used for coordination to depend only on the equivalence class $[x] \in X/G$. Equivalently, the system can treat two internal states $x, x' \in X$ as interchangeable whenever $x' = g \cdot x$ for some $g \in G$, and it should refuse to store or transmit conclusions that are not constant on each orbit. This gives a simple empirical test suite: generate or learn representative g's, apply them to internal encodings, and verify that the purported "shared" feature remains invariant up to the allowed tolerance; failures identify precisely which parts of the representation are doing non-invariant work.

- Shared-reality diagnostics: Holonomy (loop defect) detects whether a community of agents has a consistent translation geometry; Theorem 244 says nonzero holonomy is an exact obstruction, and Theorem 245 says small holonomy yields an approximately consistent global view. Here the reasoning system can interpret holonomy as a measurable inconsistency in round-trip translations: translate an assertion, belief state, or latent variable around a cycle of agents and compare the result to the starting point. A nontrivial loop defect indicates that there is no single global coordinate system (no agent-independent "map") that makes all pairwise translations simultaneously consistent, so any attempt to merge beliefs must either accept contradictions or introduce an explicit gauge choice (a privileged chart) and track the resulting dependence. When holonomy is small, Theorem 245 licenses approximate global alignment: the system can synthesize a shared reference frame that minimizes the residual defects, and it can bound the worst-case disagreement induced by composing translations along different paths. Operationally, this supports diagnostics such as: (i) locate cycles with maximal defect to identify where communication protocols or ontologies disagree most, (ii) decide when to invest in negotiation/ontology repair versus when to proceed with a single approximate merge, and (iii) treat the holonomy magnitude as a calibrated uncertainty attached to "shared" claims.

- Stable-self diagnostics: If a self-update U is contractive and nearly equivariant under known re-encodings, Theorem 247 bounds the identity drift. If drift exceeds tolerance, the system can flag "self instability". The contractive assumption gives the system a reason to expect convergence toward a fixed point or a narrow set of attractors under repeated updating, while near-equivariance says that this convergence is not an artifact of a particular internal code. A reasoning system can therefore monitor two quantities side-by-side: (a) the contraction strength (how quickly successive iterates $U^t(x)$ forget initial conditions), and (b) the equivariance defect (how much $U(g \cdot x)$ differs from $g \cdot U(x)$ for known $g \in G$). Theorem 247 then turns these into a practical bound on identity drift, i.e., how much the persistent "self" descriptor (values, goals, or policy identity) can change merely due to representational drift or update noise. If the measured drift rises above a predefined safety budget, the alert "self instability" can trigger mitigations such as freezing certain parameters, switching to a more strictly equivariant update rule, or requiring additional evidence before committing to irreversible actions. Conversely, when the bound is small, the system can justify treating long-horizon plans as referentially stable (the agent executing the plan remains "the same" in the relevant gauge-invariant sense).

- "Well-defined physics" inside cognition: If major cognitive operators (neural, symbolic, evolutionary, motivational) nearly commute in the sense of Theorem 249, then reasoning outcomes are robust to scheduling and to re-encoding; if not, the commutator defects serve as a quantitative measure of "curvature" and hence of inevitable path dependence. This provides a precise criterion for when a heterogeneous cognitive architecture can be treated as implementing an order-independent inference: if applying operators in different sequences yields approximately the same result, then the architecture admits a stable "effective dynamics" that does not depend sensitively on incidental scheduling details (thread interleavings, batching, asynchronous message timing, curriculum order). The near-commutativity condition also clarifies what must be measured: estimate commutators (or their task-relevant analogues) between modules, for example by comparing ABx versus BAx across representative states x, and interpret the resulting defect norm as a curvature-like quantity that lower-bounds unavoidable inconsistency across update paths. When commutator defects are large, the system should treat conclusions as path-dependent artifacts of the update history, and it can respond by (i) explicitly recording the applied operator path as part of the claim provenance, (ii) enforcing a canonical schedule (a gauge fixing in time/order), or (iii) redesigning modules to reduce commutators (e.g., by aligning intermediate representations or introducing shared constraints). When defects are small, the same theorems justify more aggressive parallelism and modularity: different subsystems can update in varied orders without meaningfully changing the final belief state, enabling scalability while preserving interpretability of the resulting "laws" of internal reasoning.

70 A Rigorous Cousin of de Chardin's Omega Point: Basin-Constrained Endpoint Convergence Toward C3-like Attractors

Bringing together themes and results from several prior sections, we assemble here a conditional but mathematically explicit argument schema supporting the claim:

- Many rich-resource, self-modifying minds are pulled toward a C3-like attractor regime (strong coordination, stable commitments, and self-regulation), and
- If compassion is present in the initial value seed, resource-rich heuristics can make compassion more stable rather than less.

To fix intuitions, the phrase "rich-resource, self-modifying minds" is meant to include agents (individual or collective) with (i) sufficient compute/energy/time to run redundant evaluation, internal simulation, and cross-checking, and (ii) the capability to update their own policies, representations, and even parts of their objective proxies over time. In that setting, the relevant long-run question is often less "what is the exact next-step update rule?" and more "what region of state space does the update process enter, and what invariant structure does that region impose on subsequent evolution?" In dynamical-systems language, the key objects are basins of attraction, invariant sets, and the robustness of attractors to perturbations induced by self-modification.

Remark 3733. *The intended sense of "Omega Point" here is not a metaphysical necessity claim but a disciplined template for a certain kind of convergence claim: once we condition on entering a suitable basin of meta-goals and seeded values, the long-run endpoint becomes largely insensitive to many micro-details of initial state. This is the same mathematical posture one takes when using a contraction mapping to argue that repeated updates "forget" their starting point and converge to a unique fixed point.*

The analogy is not merely rhetorical: in many learning and control processes, one can identify a map (possibly stochastic, possibly defined on distributions rather than points) whose iterates reduce distances under an appropriate metric after a transient phase. In those cases, there is a clean separation between (a) the difficulty of reaching the region where contraction holds (the "basin entry" problem) and (b) the relatively routine analysis of behavior once inside that region (the "basin dynamics" problem). The present section aims to make this separation explicit in the alignment setting by spelling out sufficient conditions under which a convergence-to-attractor statement is mathematically licensed.

Remark 3734. *The phrase "C3-like" refers to the "cooperative anchoring" collective regime from the Collective Locations $C0$–$C4$ framing [12], which is also closely aligned with the broader Hyperseed emphasis on stabilizing shared commitments and repair dynamics under self-modification. The broad framing that resource-abundance changes the qualitative regime of cognitive strategy (including heavy use of internal simulation/verification and redundancy) follows the "rich-resource mind" perspective [11] (Hyperseed-Concept 162).*

Concretely, "C3-like" is meant to pick out a regime where coordination is not merely instrumental and brittle, but anchored by (i) mechanisms that make commitments legible and enforceable over time, (ii) self-models and social models that penalize opportunistic defection in expectation, and (iii) repair/renegotiation channels that prevent local failures from cascading into systemic collapse. The "-like" qualifier is intended to allow that the precise institutional or cognitive implementation can vary widely: different systems may realize the same qualitative stability properties via different substrates (e.g., explicit contracts, cryptographic enforcement, norm-internalization, or verified policy constraints), yet still fall into an equivalence class of dynamics characterized by strong long-run cooperative invariants.

The results are stated as theorems in a deliberately minimal dynamical-systems and evolutionary dynamics framework, with proofs based on standard tools (contraction mappings, replicator dynamics, and concentration bounds). The goal is not to claim necessity in the real world, but to isolate a coherent set of sufficient conditions under which the "alignment problem" reduces to: enter the right basin (meta-goals and value seeds) and survive the transition, rather than microspecify the final endpoint.

A further motivation for the minimalism is that endpoint convergence claims are most informative when they are stable under modeling variation: if a conclusion depends delicately on one particular choice of state variables, update schedule, or noise model, it is unlikely to correspond to a robust phenomenon of self-modifying cognition. By contrast, results that rely only on generic properties (e.g., a Lyapunov function, a contraction region, or a dominance relation in replicator dynamics) can often be transported across implementations. In this spirit, "endpoint" here should be read as an asymptotic object (fixed point, limit cycle, or stationary distribution over policies/commitment structures) that is well-defined in the model class under consideration, rather than as a claim about a unique detailed policy in the real world.

The second bullet above (stability of compassion conditional on seeding) should also be read in the same structural way: compassion is not assumed to be "selected" merely because it feels good, but because (under the postulated conditions) it can become part of a stable package of commitments and self-regulatory heuristics. In rich-resource regimes, agents can afford to adopt strategies that are not locally greedy: they can invest in verification, reputation preservation, and long-horizon cooperation, and they can explicitly guard against self-corrosion of prosocial values via commitment devices, interpretability/monitoring, and redundancy. Under such heuristics, a seeded compassionate preference can be stabilized by being embedded into the very mechanisms that control self-modification (e.g., constraints on permissible updates, penalties for value drift, and repair protocols when deviations are detected).

Remark 3735. *Philosophically, the strategy is to separate (i)* basin entry*—a historically contingent and often fragile matter of initialization, training, and social embedding—from (ii)* basin dynamics*—a mathematically analyzable evolution governed by invariants and stability properties. In Hyperseed terms, one might say: once certain constraints of self-change (meta-goals) and certain seed values are present, the ensuing unfolding is less like arbitrary "design" and more like a self-consistent process constrained by its own internal logic (Hyperseed-Concept 110, 197, 140).*

In particular, the role of meta-goals in the argument schema is to provide the analog of boundary conditions or admissibility constraints: they restrict the space of self-modifications so that the induced dynamics preserve key invariants (e.g., commitment-keeping, corrigibility-like update discipline, or bounded opportunism). The role of seed values is to determine which attractor (or which region of attractor space) the system falls into once those invariants begin to dominate. On this view, "alignment" is not identified with a fully specified terminal utility function, but with a combination of (a) entry into a basin where destructive drift is unlikely and (b) the presence of value seeds that, under basin dynamics, are amplified into stable cooperative commitments rather than eroded by competitive pressure.

It is also worth emphasizing what is *not* claimed. The convergence statements to be proved are conditional: they do not assert that all minds, or even most minds, inevitably enter the relevant basin, nor that history reliably provides the required initialization. Instead, they articulate a mathematically checkable template of the form: *if* a system is sufficiently resourced, self-modifying in the specified sense, and already constrained by certain meta-goals and seeds, *then* the long-run behavior is pulled toward a C3-like regime and is robust to many micro-level perturbations. This conditional structure is what makes the "Omega Point" analogy disciplined: it is a basin-constrained endpoint convergence claim, not an unconditional teleology.

70.1 Setup: self-modifying minds and basins

Definition 669 (Mind state space)**.** *Let (X, d) be a complete metric space of* mind-states*. A mind-state $x \in X$ may encode an agent architecture, its memory, its current policy, its commitment ledger, and relevant couplings to an environment model.*

A resource level *is a scalar $R \in [0, \infty)$.*

Remark 3736. *Here (X, d) is meant to be a mathematical surrogate for "the space of possible mind configurations" (Hyperseed-Concept 111). The metric d is any choice of distance-like notion that makes relevant notions of "small self-modification" and "nearby mind-states" precise. Completeness means that every Cauchy sequence in X converges in X; this is not mere pedantry, but the key hypothesis that allows fixed-point theorems to guarantee the existence of a limiting mind-state rather than only a limit "in some larger completion."*

Remark 3737. *It is often helpful to think of x as carrying both "descriptive" degrees of freedom (world-model weights, cached beliefs, memory traces) and "normative/constitutional" degrees of freedom (values, constraints, admissibility checks). A single metric d then implicitly commits to a notion of tradeoff between these: e.g. a small change to episodic memory might be considered negligible while a small change to a commitment-verification rule might be considered large. In later stability claims, the quantitative meaning of "robust to perturbation" is always with respect to this chosen d, so the metric is part of the normative modeling choice rather than a purely geometric detail.*

Remark 3738. *The introduction of a* resource level *R is intentionally minimal: it can represent compute, time, energy, hardware redundancy, access to additional sensors, or even institutional resources available to the agent. The only structural assumption here is that self-modification behavior is allowed to depend on a scalar budget parameter. Later, R can be used to express monotonicity intuitions such as "with more resources the update becomes more conservative" (more verification) or, conversely, "with more resources the update becomes more powerful" (larger reachable edits).*

Remark 3739. *A simple example is $X = \mathbb{R}^n$ with the usual Euclidean metric, where x is a vector of parameters for a self-model plus policy. A more structurally faithful example is a product space $X = X_{policy} \times X_{memory} \times X_{commit}$ with a weighted sum metric. The particular choice does not matter for the formal results here; what matters is that the chosen metric reflects what the meta-goal system treats as "small" versus "large" changes.*

Remark 3740. *When X is taken as a product space, the weights in a weighted sum metric can be interpreted as* protectedness coefficients*: high weight on the commitment component makes even minor commitment drift register as a large distance. This becomes a convenient mathematical proxy for the engineering practice of placing critical subsystems behind stricter change-control, audits, or formal verification. In other words, part of the safety story can be expressed as choosing d so that "dangerous" edits are metrically far, which interacts cleanly with contraction-type arguments.*

Definition 670 (Self-modification operator)**.** *For each resource level R, let $F_R : X \to X$ be a (deterministic) update map describing one macro-step of self-modification and adaptation:*

$$x_{t+1} = F_R(x_t).$$

We write $x_t = F_R^{(t)}(x_0)$ for the t-fold iterate.

Remark 3741. *The notation $F_R^{(t)}$ denotes repeated application: $F_R^{(2)}(x) = F_R(F_R(x))$, etc. In words, we treat self-modification as an iterated dynamical system, where the resource level R acts as a parameter selecting a regime of update behavior. This is intentionally compatible with self-modifying-agent analyses that study stability via fixed points and attractors [10].*

Remark 3742. *The determinism of F_R is a modeling convenience rather than a claim about real agents. Stochastic learning dynamics, exploration noise, and random data can often be incorporated by either (i) lifting the state space so that x includes the agent's random seed and data stream, yielding a deterministic evolution on an expanded X, or (ii) replacing F_R by a Markov kernel and working with almost-sure or in-distribution analogues of convergence. The deterministic presentation keeps the fixed-point and basin notions crisp while leaving room for probabilistic generalizations.*

Remark 3743. *One can imagine F_R packaging multiple substeps (learning, planning, self-edit proposals, tests, deployment) into a single "macro-step" so that we can study long-run behavior without committing to low-level implementation detail. This is a recurring methodological trade: we sacrifice mechanistic fine structure to gain a stability theorem that is robust across many implementations.*

Remark 3744. *In this macro-step view, it is natural to interpret F_R as already including internal acceptance tests imposed by $m(x)$: e.g. propose an edit, verify it against a ledger, and only then commit. Thus, although F_R is written as a plain map $X \to X$, it can represent a pipeline with nontrivial internal control flow. This matters because later invariance statements about basins are not intended to assume that every proposed change succeeds—only that the realized update $F_R(x)$ respects the governing admissibility constraints encoded in $m(x)$.*

Definition 671 (Meta-goal and value seed)**.** *Let $m(x)$ denote the* meta-goal state *(the rules that govern acceptable self-changes), and let $v(x)$ denote the* value seed *(base value parameters), both as functions of x.*

We single out one coordinate of $v(x)$, denoted $c(x) \in [0, 1]$, interpreted as compassion*.*

Remark 3745. *The point of separating $m(x)$ from $v(x)$ is conceptual and technical. The value seed $v(x)$ is about* what *the system ultimately prefers (Hyperseed-Concept 197), while the meta-goal state $m(x)$ is about* which self-changes are admissible*, i.e. the self-governing constitution of the update process itself (Hyperseed-Concept 110, 87). In many self-modifying systems, stability or safety is less about having the "right" instantaneous preferences and more about having constraints that prevent destabilizing self-edits.*

Remark 3746. *Technically, one can view m and v as (possibly coarse) observables on X: there may be many distinct mind-states that share the same meta-goal and value-seed labels. This "many-to-one" structure is important because convergence results typically do not require convergence of every microscopic detail of x_t; rather, they may only require that the trajectory remains within an equivalence class that preserves governance and seeded ethics. Put differently, $m(x)$ and $v(x)$ induce a quotienting of state space into regions that are behaviorally indistinguishable for the alignment questions at issue.*

Remark 3747. *The coordinate $c(x)$ is deliberately only one component of $v(x)$, emphasizing that compassion is treated as a seeded parameter that can be protected or eroded under self-modification. In Hyperseed language, "compassion" is not only a sentiment but a value-laden constraint shaping action and interpretation (Hyperseed-Concept 81). The claim explored below is not that compassion must maximize utility, but that under certain robustness procedures it can become hard to lose accidentally.*

Remark 3748. *Restricting $c(x)$ to $[0, 1]$ is a normalization that supports "distance from extremes" reasoning: values near 0 correspond to near-absence of compassion and values near 1 to near-saturation. In later arguments, such a bounded coordinate can simplify the use of compactness-like heuristics or uniform-continuity intuitions, even when the ambient space X is infinite-dimensional. Nothing essential depends on this particular interval; it merely fixes a convention for comparing changes in compassion to changes in other value-seed coordinates.*

Definition 672 (Basin defined by meta-goals and value seeds)**.** *Fix a meta-goal specification m_0 and a value-seed specification v_0. Define the corresponding basin as*

$$B(m_0, v_0) := \{x \in X : m(x) = m_0 \text{ and } v(x) = v_0\}.$$

(Equivalently, $B(m_0, v_0)$ may be taken as a neighborhood of states whose meta-goal and value parameters lie in some tolerance class around (m_0, v_0).)

Remark 3749. *This "basin" is a deliberately austere formalization of a familiar informal idea: a family of mind-states that share the same constitutional constraints (m_0) and the same seed values (v_0). The parenthetical comment about tolerances is important in practice: exact equality is too strict for most modeled systems, but theorems typically go through if one replaces equality by membership in an equivalence class or neighborhood that the meta-goal system treats as "the same" for governance purposes.*

Remark 3750. *Formally, the mapping $x \mapsto (m(x), v(x))$ can be regarded as a (possibly discontinuous) labeling map from X into a space of specifications. Then $B(m_0, v_0)$ is simply the fiber* *over* (m_0, v_0). *This framing makes it clear that a "basin" here is not (yet) a basin of attraction in the classical dynamical-systems sense; it is, instead, a constraint-defined region. The later step that connects this set to attractors is an* invariance *or* forward-closure *claim of the form* $F_R(B(m_0, v_0)) \subseteq B(m_0, v_0)$ *(possibly for sufficiently large* R*), together with a convergence property of the iterates within that region.*

Remark 3751. *When the tolerance interpretation is used, it is often convenient to write something like*

$$B_\varepsilon(m_0, v_0) := \{x \in X : d_M(m(x), m_0) \le \varepsilon \text{ and } d_V(v(x), v_0) \le \varepsilon\},$$

for suitable metrics or pseudometrics d_M, d_V *on specification spaces. This makes precise the idea that governance may treat a small range of parameterizations as equivalent. Such tolerance basins also interact better with continuity assumptions on* F_R*, because exact equalities can be brittle under infinitesimal perturbations while closed neighborhoods are stable under limits.*

Remark 3752. *An example is:* m_0 *encodes "never self-modify in ways that violate my signed commitment ledger," while* v_0 *encodes a particular initial ethical prior including a compassion level* $c_\star$*. Then* $B(m_0, v_0)$ *is the set of states that still instantiate that constitutional constraint and that prior. The theorems below use such basins to make precise the slogan:* alignment reduces to basin entry plus stability within the basin*.*

Remark 3753. *The slogan can be read as separating two qualitatively different tasks. "Basin entry" is an initialization and verification problem: can the system be brought into a state whose governance and seeded ethics match the intended specifications? "Stability within the basin" is a dynamical problem: once in such a state, do the self-modifications preserve* m_0 *and* v_0 *(or their tolerances) and, moreover, do the iterates converge toward a coherent endpoint rather than drifting chaotically across microstates that still nominally satisfy the constraints? This separation is what allows later sections to talk about endpoint convergence without assuming that every detail of the agent is frozen—only that the constitution and seed values are not silently rewritten.*

70.2 C3-like regime as an attractor target

We need only a coarse definition, compatible with the "cooperative anchoring" idea: high coordination, high anchoring/stable commitments, and explicit repair/self-regulation.

Definition 673 (Coarse-grained coordinates)**.** *Let*

$$\kappa(x) \in [0,1] \quad \textit{(coordination / trust-capacity)}, \qquad \sigma(x) \in [0,1] \quad \textit{(commitment stability / anchoring)},$$

$$\delta(x) \in [0,1] \quad \textit{(drift / accountability loss)}, \qquad c(x) \in [0,1] \quad \textit{(compassion)}.$$

Define the distance-to-C3 *potential*

$$V(x) := a(1 - \kappa(x)) + b(1 - \sigma(x)) + g\,\delta(x) + h(1 - c(x)),$$

with positive weights a, b, g, h*.*

Remark 3754. *The symbols* $\kappa, \sigma, \delta, c$ *are observer-chosen summary statistics of the detailed mind state* x*. They are* coarse-grainings*: maps from a high-dimensional internal representation to a small set of interpretable coordinates. Intuitively,* $\kappa(x)$ *measures the system's ability to coordinate reliably with its own subparts or with other agents;* $\sigma(x)$ *measures how stable and binding its commitments are;* $\delta(x)$ *measures the tendency of goals to drift or accountability to dissolve; and* $c(x)$ *measures compassion (Hyperseed-Concept 81, 75).*

Remark 3755. *The potential $V(x)$ is a Lyapunov-style quantity: it is nonnegative and is designed to be small precisely when the desired regime is present. The weights $a, b, g, h > 0$ encode a normative (or engineering) choice about how to trade off coordination, anchoring, drift, and compassion. A simple example is $a = b = g = h = 1$, in which case $V(x)$ is literally the sum of four "deficits" from the idealized target: lack of coordination, lack of commitment stability, presence of drift, and lack of compassion. This kind of scalar potential is useful because it lets us reduce multi-criteria stability questions to a single descent inequality.*

Definition 674 (C3-like region)**.** *Fix thresholds $\kappa_\star, \sigma_\star, c_\star \in (0, 1]$ and $\delta_\star \in [0, 1)$. Define the C3-like region*

$$A_{C3} := \{x \in X : \kappa(x) \geq \kappa_\star,\ \sigma(x) \geq \sigma_\star,\ \delta(x) \leq \delta_\star,\ c(x) \geq c_\star\}.$$

Remark 3756. *This is an "engineering" definition of a regime: A_{C3} is a box (an intersection of threshold constraints) in the space of coarse-grained coordinates. It matches the idea that $C3$ is not a single point but a robust region characterized by simultaneously high coordination and anchoring, low drift, and sufficiently high compassion (Hyperseed-Concept 75, 81). The inequalities are deliberately one-sided: we demand enough of the good quantities and not too much of the bad one.*

Remark 3757. *A concrete example: choose $\kappa_\star = 0.9$, $\sigma_\star = 0.9$, $\delta_\star = 0.1$, $c_\star = 0.7$. Then A_{C3} is the set of mind-states whose trust-capacity and anchoring are both at least* 0.9*, whose drift is at most* 0.1*, and whose compassion is at least* 0.7*. The choice of thresholds is not asserted to be universal; the purpose is to make later theorems about attraction and stability mathematically explicit.*

Remark 3758. *Two distinct but related objects are in play: the* region *A_{C3} (defined by hard constraints) and the* potential *V (a soft scalar score). The region is convenient for "specification" (what we require); the potential is convenient for "dynamics" (what decreases along trajectories). In later stability arguments, the typical pattern is: prove that V decreases outside (or away from) A_{C3}, deduce that trajectories enter a sublevel set of V, and then use a link between sublevel sets and threshold constraints to conclude that the system reaches and remains in a C3-like regime.*

Remark 3759. *The definition of V is intentionally linear and separable to preserve interpretability: each term is a deficit-like quantity whose sign is unambiguous. This does not assert that the underlying phenomenology is additive; rather, the linear form is a conservative bookkeeping device. Nonlinear variants (e.g., convex penalties, saturation effects, or cross-terms capturing that coordination and compassion can be mutually reinforcing) can be substituted provided one preserves the core property needed for Lyapunov methods: $V(x) \geq 0$ and $V(x)$ is small in the target regime.*

Lemma 175 (Sublevel sets imply threshold satisfaction under margin)**.** *Fix thresholds $\kappa_\star, \sigma_\star, c_\star \in (0, 1]$ and $\delta_\star \in [0, 1)$ and weights $a, b, g, h > 0$. Define*

$$\varepsilon_\star := \min\{a(1 - \kappa_\star),\ b(1 - \sigma_\star),\ g\,\delta_\star,\ h(1 - c_\star)\}.$$

If x satisfies $V(x) \leq \varepsilon_\star$, then $x \in A_{C3}$.

Remark 3760. *Lemma 175 encodes a simple but useful implication: if the total deficit $V(x)$ is below the smallest "budget" needed to violate* any *one of the thresholds, then no individual constraint can be violated. The choice of $\varepsilon_\star$ is conservative (it is the tightest single-constraint violation cost), but it is often sufficient for attraction proofs because Lyapunov analyses typically yield bounds of the form $V(x_t) \to 0$ or $V(x_t) \leq \varepsilon$ after some finite time. One may of course use sharper implications if additional structure is known (e.g., upper bounds on particular components or conditional dependencies among the coordinates).*

Remark 3761. *Conversely, membership in A_{C3} implies an upper bound on V: if $x \in A_{C3}$ then*

$$V(x) \le a(1-\kappa_\star) + b(1-\sigma_\star) + g\,\delta_\star + h(1-c_\star).$$

Thus A_{C3} is sandwiched between two sublevel descriptions: it contains a sufficiently small sublevel set of V (by Lemma 175) and is itself contained in a possibly larger sublevel set (by the inequality above). This is one precise sense in which the "box" definition and the "potential" definition are compatible approximations of the same qualitative regime.

Remark 3762. *The coordinates $\kappa, \sigma, \delta, c$ are normalized into $[0, 1]$ only for convenience. This normalization makes weights a, b, g, h dimensionless and makes threshold choices interpretable as "percentage-like" requirements. In applications, one can equivalently work with unnormalized metrics and apply a monotone rescaling into $[0, 1]$; Lyapunov-style arguments are typically robust to such monotone transformations so long as "better" and "worse" are preserved and the target regime corresponds to small values of the chosen potential.*

Remark 3763. *The definition is also compatible with measurement noise and observer disagreement: different observers may compute different coarse-grainings $\kappa, \sigma, \delta, c$ from the same underlying state x, and even a single observer may only have access to estimates $\widehat{\kappa}(x), \widehat{\sigma}(x), \widehat{\delta}(x), \widehat{c}(x)$. The practical role of the thresholds $(\kappa_\star, \sigma_\star, \delta_\star, c_\star)$ is therefore to provide margins: choosing them away from the boundary values 0 and 1 makes the definition less brittle under small estimation errors, while the box form makes it clear which direction each error pushes the classification.*

Remark 3764. *Interpreting A_{C3} as an "attractor target" does* not *mean that every trajectory in X must converge to it. The later "basin-constrained" viewpoint is that there exists a nontrivial basin $B \subseteq X$ such that trajectories starting in B are driven toward A_{C3} (or toward a neighborhood whose size can be controlled). In this framing, A_{C3} is less like a single teleological endpoint and more like a mathematically specified region of qualitative organization (high coordination and anchoring with explicit self-repair) that can be made stable under suitable dynamics.*

Remark 3765. *Finally, the drift coordinate $\delta(x)$ is written as a "bad" quantity (lower is better) to emphasize accountability preservation: a system with high κ and σ but uncontrolled δ can exhibit coordinated behavior that is nevertheless unmoored from prior commitments or shared goals. Including δ in V and in A_{C3} forces the target regime to incorporate a self-stabilizing notion of identity-over-time (in the operational sense of reduced goal drift), which is essential if one wants an attractor description to capture something closer to "cooperative anchoring" rather than merely short-term alignment.*

70.3 Targeted lemmas (inputs from Targets 1–5)

The grand conclusion is obtained by combining four technical ingredients: (i) metagoal-driven contractivity (goal/commitment stabilization), (ii) selection pressure toward trust-capable coordination, (iii) resource-rich robustness checks that stabilize seeded compassion, and (iv) a Lyapunov argument that C3-like states become attracting.

A useful way to read (i)–(iv) is as a division between *intra-agent* dynamical stability (i, iv) and *inter-agent* (or population-level) selection and institutional stabilization (ii, iii). The lemmas below isolate minimal hypotheses under which each ingredient becomes mathematically checkable: contraction gives uniqueness of an endpoint within a basin; selection dynamics bias population mass toward trust-capable coordination; and (in later lemmas) robustness and Lyapunov conditions turn "desirable regions" into genuine attractors rather than merely transient states.

70.3.1 Lemma A: contraction and endpoint convergence within a basin

Lemma 176 (Metagoal contractivity implies unique endpoint within basin)**.** *Fix (m_0, v_0) and R. Suppose there exists a subset $B \subseteq B(m_0, v_0)$ such that:*

(1) $F_R(B) \subseteq B$ (the basin subset is forward invariant), and

(2) F_R is a contraction on B:

$$d(F_R(x), F_R(y)) \leq \lambda\, d(x, y) \quad \textit{for all } x, y \in B,$$

for some $\lambda \in [0, 1)$.

Then there exists a unique fixed point $x^* \in B$ with $F_R(x^*) = x^*$, and for all $x_0 \in B$ we have geometric convergence $d(F_R^{(t)}(x_0), x^*) \to 0$. In particular, the endpoint is independent of micro-details *within B.*

Remark 3766. *Intuitively, this lemma says: if (a) the meta-goal/value basin is not left by the update rule, and (b) the update rule consistently shrinks distances between states, then repeated self-modification must converge to one and only one "settled" mind-state. In particular, the final settled outcome depends on the basin-defining invariants, not on the irrelevant fine-grained details of where in the basin one started.*

Remark 3767. *This importance of forward invariance in (1) is that it lets one reason* locally *on B even if the global dynamics on X are complicated. In applications, $B(m_0, v_0)$ is typically defined by constraints such as "the meta-goal stays unchanged up to equivalence," "core commitments remain satisfied," or "value drift is bounded." Condition (1) then becomes a precise statement that these invariants are actually respected by F_R, so that the contraction condition (2) cannot be evaded by exiting the intended regime.*

Remark 3768. *This is important because it turns an alignment hope into a checkable condition: rather than microspecifying endpoints, one tries to design (or cultivate) meta-goals that make the self-update dynamics contractive in a meaningful metric. This is the same fixed-point logic used in explicit meta-goal stability analyses of self-modifying systems [10], and it can be seen as a rigorous version of "stability through self-consistency constraints" (Hyperseed-Concept 110).*

Remark 3769. *The choice of metric d is substantive: it encodes which differences between states are treated as "large" versus "small." For instance, d might downweight representational details while upweighting goal-relevant divergences, or it might be induced by a divergence between policies, world-models, or commitment-graphs. The lemma does not require a specific d, but in practice one seeks a metric that both (a) tracks the properties one cares about (e.g. commitment stability, compassion constraints), and (b) is compatible with the actual update operator F_R so that contractivity can be verified or at least bounded.*

Proof. This is the Banach fixed point theorem. Completeness of (X, d) implies completeness of the closed subset B (or assume B complete). Contractivity gives existence and uniqueness of a fixed point x^* and geometric convergence of iterates from any initial point in B. □

Proof sketch. The strategy is to use a classical stability theorem: in a complete metric space, a λ-contraction cannot wander forever because each step shrinks distances, forcing the orbit to become Cauchy and hence to converge. The limit is automatically a fixed point, and uniqueness follows because two distinct fixed points would violate contraction. □

Remark 3770. *The key step is the contraction inequality: it implies that distances along the orbit shrink at least geometrically, so the sequence (x_t) is Cauchy. Completeness is what turns "Cauchy" into "convergent in X". Visually, one may picture B as a basin whose internal geometry is being continuously "squeezed" by F_R until everything falls into a single point x^*.*

Remark 3771. *Lemma 176 formalizes "endpoint convergence within a basin": if the metagoal forces contractive self-modification dynamics in an appropriate metric, microstate details wash out.*

Remark 3772. *One can also read the conclusion as a* robustness *statement: any two initial states $x_0, y_0 \in B$ produce trajectories whose distance contracts as*

$$d(F_R^{(t)}(x_0), F_R^{(t)}(y_0)) \leq \lambda^t \, d(x_0, y_0),$$

so the system progressively forgets initial uncertainty. This is exactly the kind of "basin-level insensitivity" needed when the basin is specified only up to coarse invariants (meta-goal equivalence classes, bounded value drift, or structural constraints on commitments) rather than by a microscopic blueprint of the final mind-state.

70.3.2 Lemma B: selection implies trust-capable coordination

We now formalize a minimal "Good Guys Win"-style selection statement.

Definition 675 (Environment distribution and trust advantage)**.** *Let α index task environments drawn i.i.d. from a distribution ν on a measurable space $\mathcal{A}$. Let two community types be available: T (trust-capable) and L (trustless). Let their expected payoffs in environment α be $U_T(\alpha)$ and $U_L(\alpha)$, and define the advantage*

$$\Delta(\alpha) := U_T(\alpha) - U_L(\alpha).$$

Assume $\Delta(\alpha) \geq 0$ for all α and $\nu(\{\alpha : \Delta(\alpha) > 0\}) > 0$.

Remark 3773. *The notation $\alpha \sim \nu$ means we sample environments according to a probability distribution ν. The assumption "i.i.d." means successive tasks are drawn independently from the same ν. The inequality $\Delta(\alpha) \geq 0$ says trust-capable coordination is never worse, and the positive-measure strict inequality says it is sometimes strictly better. This is a stylized version of the prosocial-efficiency hypothesis: for a wide class of problems, cooperative groups with stable trust and commitment structures can outperform groups that cannot sustain trust [6] (Hyperseed-Concept 170,* **??***).*

Remark 3774. *A toy example is: environments are collaboration tasks requiring division of labor, where trustless communities pay repeated verification and adversarial overhead, while trust-capable communities can delegate efficiently. Then $\Delta(\alpha)$ represents the saved overhead (or improved quality). This definition is useful because it reduces a sociotechnical claim to a single quantitative object $\Delta(\alpha)$ whose expectation will drive replicator dynamics.*

Remark 3775. *The assumption $\Delta(\alpha) \geq 0$ for all α is deliberately strong: it rules out environments where trust is systematically exploited. In settings where exploitation is possible, one would replace this with a weaker condition (e.g. $\bar{\Delta} > 0$ despite some negative tails, or a risk-sensitive dominance criterion). The present lemma is meant as a minimal monotone-selection baseline: if trust is* never *disadvantageous and sometimes strictly advantageous, then standard selection dynamics amplify it.*

Lemma 177 (Replicator convergence to trust-capable coordination)**.** *Let $x(t) \in [0,1]$ be the population share of trust-capable communities, evolving under replicator dynamics in continuous time:*

$$\dot{x} = x(1-x)\,\bar{\Delta}, \qquad \bar{\Delta} := \mathbb{E}_\nu[\Delta(\alpha)].$$

If $\bar{\Delta} > 0$, then for any initial $x(0) \in (0,1]$ we have $x(t) \to 1$ as $t \to \infty$.

Proof. For $\bar{\Delta} > 0$ and $x(0) \in (0,1)$, the ODE is the logistic equation. Separating variables gives

$$\frac{dx}{x(1-x)} = \bar{\Delta}\,dt,$$

hence

$$\log\left(\frac{x(t)}{1-x(t)}\right) = \log\left(\frac{x(0)}{1-x(0)}\right) + \bar{\Delta}\,t.$$

Exponentiating and solving for $x(t)$ yields

$$x(t) = \frac{1}{1 + \left(\frac{1-x(0)}{x(0)}\right) e^{-\bar{\Delta} t}}.$$

As $t \to \infty$, the exponential term vanishes and thus $x(t) \to 1$. If $x(0) = 1$ the solution is identically 1, so the claim holds for all $x(0) \in (0,1]$. □

Proof sketch. When $\bar{\Delta} > 0$, the right-hand side $x(1-x)\bar{\Delta}$ is positive for $x \in (0,1)$, so the share of trust-capable communities increases monotonically until it approaches the stable equilibrium at $x = 1$. The equilibrium at $x = 0$ is unstable: any nonzero seed of trust-capable coordination is amplified over time. □

Remark 3776. *The fixed points are $x = 0$ and $x = 1$. Linearizing, $\dot{x} \approx \bar{\Delta}x$ near 0 (unstable if $\bar{\Delta} > 0$) and $\dot{x} \approx -\bar{\Delta}(1-x)$ near 1 (stable if $\bar{\Delta} > 0$). Thus the lemma captures a "selection from any seed" effect: even a small but nonzero initial presence of trust-capable communities will (in this stylized model) eventually dominate.*

Remark 3777. *The role of $\bar{\Delta} = \mathbb{E}_\nu[\Delta(\alpha)]$ is to aggregate heterogeneous environments into a single effective selection gradient. If $\Delta(\alpha) \geq 0$ and $\nu(\{\Delta > 0\}) > 0$, then automatically $\bar{\Delta} > 0$, so the lemma's condition is implied by the earlier definition. Making $\bar{\Delta}$ explicit is useful because it isolates the precise quantitative threshold that would change the conclusion: if $\bar{\Delta} = 0$, then $\dot{x} = 0$ and the population shares are neutrally stable (no selection pressure); if $\bar{\Delta} < 0$, the direction reverses and $x(t) \to 0$ from any interior initial condition.*

Remark 3778. *In plain language: if trust-capable communities have a positive expected advantage, then (under the simplest replicator model) the fraction of trust-capable communities increases monotonically and tends to 1 provided it is present at all initially. This is the mathematical core of a selection story: what pays off tends to proliferate, and positive expected advantage is enough to produce eventual dominance. One can also read the statement as a "no fragile tuning required" claim: the dynamic does not depend on delicately chosen initial conditions beyond $x(0) > 0$, and it does not require intermittent interventions once the advantage $\bar{\Delta}$ is in place.*

Remark 3779. *The connection to the rest of this section is that $C3$-like regimes are characterized by high coordination and stable commitments; Lemma 177 supplies a minimal evolutionary mechanism that pushes populations in that direction. The lemma is not about individual cognition* *directly, but about which organizational forms persist in a population of communities/mindplexes (Hyperseed-Concept 113, 75). Put differently, even if each community is internally heterogeneous and psychologically complex, the population-level statistic $x(t)$ can still be governed by a simple comparative-growth rule, and that suffices to show directional pressure toward trust-capable coordination whenever it yields repeatable net benefits.*

Proof. If $\bar{\Delta} > 0$ then $\dot{x} > 0$ on $(0,1)$, so $x(t)$ is increasing and bounded above by 1. The only equilibria are $x = 0$ and $x = 1$, and $x = 1$ is globally asymptotically stable on $(0, 1]$ because $x(t)$ cannot decrease and cannot converge to 0. More explicitly, separate variables:

$$\frac{d}{dt} \log \frac{x}{1-x} = \bar{\Delta},$$

so $\log \frac{x(t)}{1-x(t)} = \log \frac{x(0)}{1-x(0)} + \bar{\Delta}t \to +\infty$, implying $x(t) \to 1$. Equivalently, exponentiating yields the explicit logistic-form solution

$$\frac{x(t)}{1-x(t)} = \frac{x(0)}{1-x(0)} e^{\bar{\Delta}t} \quad \Longrightarrow \quad x(t) = \frac{1}{1 + \left(\frac{1-x(0)}{x(0)}\right) e^{-\bar{\Delta}t}},$$

from which convergence to 1 as $t \to \infty$ is immediate, and monotonicity follows because the denominator is strictly decreasing in t when $\bar{\Delta} > 0$. $\square$

Proof sketch. The ODE is a logistic-type equation with constant positive growth rate $\bar{\Delta}$. The factor $x(1-x)$ ensures $x = 0$ and $x = 1$ are fixed points, while positivity of $\bar{\Delta}$ makes the flow point toward 1 whenever x starts positive. Solving the ODE explicitly via the log-odds $\log \frac{x}{1-x}$ makes the convergence transparent. Intuitively, the multiplicative term $x(1-x)$ also encodes diminishing returns: when trust-capable communities are rare ($x \approx 0$), growth is slow because there are few such communities to replicate; when they are common ($x \approx 1$), growth slows again because there are few remaining non-trust-capable communities to displace. $\square$

Remark 3780. *The log-odds transformation is the key step: it linearizes the dynamics into* $\frac{d}{dt} \log \frac{x}{1-x} = \bar{\Delta}$*. Geometrically, one may picture $x(t)$ as sliding uphill on a fitness landscape whose slope is set by $\bar{\Delta}$; with $\bar{\Delta} > 0$, the landscape tilts inexorably toward $x = 1$. Another way to see the same geometry is to note that the vector field $\dot{x} = \bar{\Delta}x(1-x)$ is strictly positive on $(0,1)$, so trajectories cannot cycle or stall in the interior: the only long-run possibilities are boundary absorption at 0 or 1, and the sign of $\bar{\Delta}$ selects which boundary is attracting.*

Remark 3781. *Lemma 177 is intentionally minimal: any model in which trust-capable coordination has a strictly positive expected advantage on a non-negligible set of environments yields selection toward trust-capable populations. The role of "expected" is important: even if trust-capable communities sometimes lose locally (e.g., in one-off exploitative encounters), the averaged effect over the relevant environment distribution is what governs the direction of change in the simplest replicator idealization, so persistent net advantage is enough to drive long-run prevalence.*

70.3.3 Lemma C: resource-rich robustness checks stabilize seeded compassion

We next show a concrete sense in which more resources can make a seeded value (compassion) harder to lose. The motivating intuition is that resource abundance can be converted into redundancy and review capacity: if a system can afford to run more internal audits before changing itself, then even small per-auditor detection advantages can be amplified into strong overall protection.

Definition 676 (Robust acceptance test)**.** *Suppose a mind at state x proposes a candidate self-modification producing $x' = M(x)$. The mind performs a robustness test using K independent "variant self" evaluators, producing binary votes $Z_1, \ldots, Z_K \in \{0,1\}$ where $Z_k = 1$ means "accept". (In this abstraction, the Z_k are the only interface between the proposed modification and the decision procedure: all details of the evaluators' internal computations are summarized by their binary recommendations.)*

Assume the rule is: accept iff $\frac{1}{K}\sum_{k=1}^{K} Z_k \geq \tau$ for some $\tau \in (1/2, 1)$. Thus τ specifies how conservative the system is: larger τ means that a modification must survive a stronger consensus requirement in order to be enacted.

Let a modification be called compassion-degrading *if $c(x') \leq c(x) - \eta$ for some $\eta > 0$. Here η is a margin parameter, included to exclude negligible numerical noise and to focus the argument on substantive degradations rather than arbitrarily small fluctuations.*

Remark 3782. *This definition idealizes a familiar robust-engineering move: evaluate a proposed change under many independent checks, and require a strong majority to proceed. Here K is the number of evaluators, τ is the acceptance threshold (strictly greater than $1/2$ to require more than a bare majority), and Z_k is the kth evaluator's binary verdict. In a rich-resource regime, running many evaluators is precisely the sort of strategy that becomes feasible [11] (Hyperseed-Concept 162). Conceptually, K is a stand-in for any resource-intensive safety margin: parallel simulation, diverse oversight modules, longer review time, or repeated stress tests before committing to irreversible self-change.*

Remark 3783. *The "variant self" metaphor can be instantiated in many ways: randomized internal simulations, adversarial red-teaming submodules, ensembles trained with different data slices, or different value-system perspectives held in parallel. The independence assumption is mathematically strong (and often false in detail), but it is exactly the hypothesis under which classical concentration inequalities yield exponential error suppression. The definition is useful because it isolates the logical form of the argument:* redundancy plus majority thresholds can protect a seed value*. Even when independence fails strictly, the same structure suggests a design goal: reduce common-mode failure so that errors are not perfectly correlated, thereby recovering at least some concentration benefit.*

Lemma 178 (Exponential suppression of compassion-degrading modifications)**.** *Assume that for any compassion-degrading modification M, each evaluator rejects it with probability at least $p > 1/2$, independently:*

$$\Pr[Z_k = 0] \geq p \quad \textit{for all } k = 1, \ldots, K.$$

Then the probability that such a modification passes the robustness test is bounded by

$$\Pr\left[\frac{1}{K}\sum_{k=1}^{K} Z_k \geq \tau\right] \leq \exp\Big(-2K\left(\tau - (1-p)\right)^2\Big),$$

whenever $\tau > 1 - p$. In particular, as K increases (more resources), the acceptance probability decays exponentially.

Proof. Let $S = \sum_{k=1}^{K} Z_k$ be the number of "accept" votes. The assumption $\Pr[Z_k = 0] \geq p$ implies $\mathbb{E}[Z_k] \leq 1 - p$ for each k, hence $\mathbb{E}[S] \leq K(1-p)$. For $\tau > 1 - p$ define $\epsilon = \tau - (1-p) > 0$. Then

$$\Pr\left[\frac{1}{K}S \geq \tau\right] = \Pr[S - \mathbb{E}[S] \geq K\tau - \mathbb{E}[S]] \leq \Pr[S - \mathbb{E}[S] \geq K(\tau - (1-p))] = \Pr[S - \mathbb{E}[S] \geq K\epsilon]\,.$$

Since the $Z_k \in [0, 1]$ are independent, Hoeffding's inequality gives

$$\Pr[S - \mathbb{E}[S] \geq K\epsilon] \leq \exp(-2K\epsilon^2),$$

which yields the stated bound. The condition $\tau > 1-p$ is exactly the requirement that the threshold exceed the evaluators' worst-case mean accept rate for compassion-degrading modifications, so that only an upward deviation (a collective mistake) can cause acceptance. □

Remark 3784. *This lemma states a crisp and somewhat comforting fact: if each independent evaluator is more likely than not to catch a compassion-degrading change, then requiring a super-majority makes the probability of mistakenly accepting such a change shrink exponentially in the number of evaluators. The conclusion is not that compassion is inherently self-stabilizing, but that robust procedures can make accidental compassion loss vanishingly rare. Note also the role of the "gap" $\tau - (1-p)$: if evaluators are only barely better than chance (so p is close to $1/2$) and τ is only barely above $1/2$, then the exponent is small; conversely, modest improvements in either evaluator quality (larger p) or conservatism (larger τ) can substantially increase the rate of suppression when K is large.*

Remark 3785. *The broader point is conceptual: "more resources" can mean more degrees of freedom devoted to verification, not only more degrees of freedom devoted to optimization. This aligns with the view that value stability in self-modifying agents can be achieved via meta-level constraints and tests, not only via direct utility maximization [2] (Hyperseed-Concept 110, 81). In the present framing, the robustness test is a simple meta-level constraint: it does not specify what modifications to seek, only which modifications are permitted to pass. Such gatekeeping constraints can therefore coexist with many different lower-level optimization strategies while still providing a common protective layer for seeded values.*

Proof. Let $Y_k := 1 - Z_k$, so $Y_k \in \{0, 1\}$ and $\mathbb{E}[Y_k] \geq p$. Passing the test means $\frac{1}{K}\sum Z_k \geq \tau$, i.e. $\frac{1}{K}\sum Y_k \leq 1 - \tau$. Let $\mu := \mathbb{E}[\frac{1}{K}\sum Y_k] \geq p$ and note that $1 - \tau < p \leq \mu$ whenever $\tau > 1 - p$. Hoeffding's inequality yields

$$\Pr\left[\frac{1}{K}\sum Y_k \leq 1 - \tau\right] \leq \exp\bigl(-2K(\mu - (1 - \tau))^2\bigr) \leq \exp\bigl(-2K(p - (1 - \tau))^2\bigr),$$

which is the stated bound after rewriting $p - (1 - \tau) = \tau - (1 - p)$. In particular, the bound is nontrivial precisely in the regime $\tau > 1 - p$, where the event $\frac{1}{K}\sum Y_k \leq 1 - \tau$ represents a genuine *downward* deviation below the mean level $\mu \geq p$; if $\tau \leq 1 - p$, then $1 - \tau \geq p$ and the inequality becomes a bound on a deviation that need not be rare. □

Remark 3786. *Hoeffding's inequality is applied here in its one-sided form for averages of independent bounded random variables: if $Y_1, \ldots, Y_K$ are independent with $0 \leq Y_k \leq 1$ and mean μ, then for any $a < \mu$ one has $\Pr[\frac{1}{K}\sum Y_k \leq a] \leq \exp(-2K(\mu - a)^2)$. In the present setting, the boundedness is immediate from $Y_k \in \{0, 1\}$, and the independence (or at least a suitable conditional independence/martingale-difference condition) is the structural assumption that allows the concentration estimate to scale exponentially in K.*

Proof sketch. Convert accept-votes into reject-votes so that we can bound the probability that the empirical mean of rejections dips below a threshold. Then apply a standard concentration inequality (Hoeffding) to show that large downward deviations of an average of independent bounded random variables are exponentially unlikely in K. □

Remark 3787. *The substitution $Y_k = 1 - Z_k$ is the algebraic hinge: it recasts "too many accepts" as "too few rejects," matching the standard one-sided tail form of Hoeffding's inequality. Geometrically, one may visualize the K evaluators as independent samples from a Bernoulli distribution whose mean is at least $p > 1/2$; the law of large numbers says the empirical mean concentrates near p, and Hoeffding makes this concentration quantitative with an exponential bound. One can also read the threshold condition $\tau > 1-p$ as a separation condition: it ensures the acceptance threshold τ sits strictly above the baseline acceptance rate $1-p$ implied by a per-evaluator rejection probability of at least p, so that passing requires a coordinated statistical fluctuation rather than merely typical behavior.*

Remark 3788. *Lemma 178 captures a precise mechanism: if more resources allow more extensive robustness evaluation (larger K), then the chance of unintentionally selecting a modification that significantly reduces compassion can become extremely small,* without assuming compassion is utility-maximizing*. We only assume compassion loss tends to be detected by a majority of variant evaluators. Operationally, the lemma can be interpreted as providing a tunable risk parameter: fixing a desired failure probability $\delta > 0$, one can solve the displayed exponential bound for K to obtain a sufficient evaluation budget that makes inadvertent selection of compassion-reducing updates occur with probability at most δ (given the lower bound p and a chosen τ satisfying $\tau > 1 - p$).*

70.3.4 Lemma D: Lyapunov descent into a C3-like region

Lemma 179 (Sufficient condition for attraction to A_{C3})**.** *Fix R and a forward-invariant set $B \subseteq X$. Assume there exists $q \in [0, 1)$ such that for all $x \in B$,*

$$V(F_R(x)) \leq q\, V(x).$$

Then every trajectory starting in B converges to the set $\{x \in B : V(x) = 0\}$. If additionally $\{x \in B : V(x) = 0\} \subseteq A_{C3}$, then A_{C3} is attracting on B.

Remark 3789. *Forward-invariant means $x_0 \in B \Rightarrow x_t \in B$ for all $t \geq 0$, so the decay condition is available at every iterate along the trajectory. The multiplicative constant $q < 1$ is a uniform contraction factor for the scalar potential V (not necessarily for the full state), and the conclusion should be read as: the orbit is forced into arbitrarily small sublevel sets $\{V \leq \varepsilon\}$ as t grows.*

Remark 3790. *This is the Lyapunov-function principle in its simplest multiplicative form: if each update shrinks a nonnegative potential by a fixed factor $q < 1$, then the potential must go to 0. Here the potential is exactly the "distance-to-C3" measure $V(x)$, so the lemma provides a sufficient condition under which the dynamics necessarily descends into the C3-like regime. In many applications V is chosen so that $V(x) = 0$ is an invariant (or approximately invariant) manifold representing "fully C3-compliant" behavior, while $V(x) > 0$ measures a graded deviation; the inequality $V(F_R(x)) \leq qV(x)$ then encodes a strict one-step improvement guarantee.*

Remark 3791. *This lemma connects to Lemma 176 but is logically independent: contraction is a strong condition implying a unique fixed point, whereas Lyapunov descent only implies convergence to a set where the potential vanishes. In some systems one can show Lyapunov descent without full contractivity, which is why it is listed separately as an "ingredient." Concretely, the zero set $\{V = 0\}$ may contain multiple fixed points, a limit cycle, or a more complicated invariant subset; Lyapunov descent rules out wandering away from that set but does not, by itself, select a unique endpoint within it.*

Proof. Iterating the inequality gives $V(x_t) \leq q^t V(x_0) \to 0$. Since V is nonnegative, limit points of the trajectory must lie in the zero level set of V. If that zero set is contained in A_{C3}, then the omega-limit set is contained in A_{C3}. More explicitly, if $(x_{t_j})_{j\geq 1}$ is any convergent subsequence with $x_{t_j} \to x_*$, and if V is (as is typical for Lyapunov functions) at least lower semicontinuous on B, then

$$V(x_*) \leq \liminf_{j\to\infty} V(x_{t_j}) = 0,$$

so $V(x_*) = 0$ and hence $x_* \in \{x \in B : V(x) = 0\}$. Thus every accumulation point lies in the zero level set, i.e. the ω-limit set (when it exists, e.g. under precompactness of the orbit) is contained in $\{V = 0\}$, and the additional containment $\{V = 0\} \subseteq A_{C3}$ yields $\omega(x_0) \subseteq A_{C3}$. □

Proof sketch. Repeatedly apply the multiplicative decay inequality to obtain an explicit bound $V(x_t) \leq q^t V(x_0)$. Because $q^t \to 0$, the potential must vanish in the limit, forcing all accumulation points to lie in the $V = 0$ set; containment of that set in A_{C3} gives attraction to the C3-like region. □

Remark 3792. *The essential move is that multiplicative decay yields an explicit geometric series behavior. The ω-limit language is a formal way to say "the long-run behavior" of the trajectory. One can picture V as a kind of altitude function on the state space: the inequality $V(F_R(x)) \leq qV(x)$ says each step descends a fixed fraction of remaining altitude, so the orbit must approach sea level $V = 0$. Equivalently, for any $\varepsilon > 0$ one can pick T such that $q^T V(x_0) < \varepsilon$, implying $V(x_t) < \varepsilon$ for all $t \geq T$; this "eventual entrance into every sublevel set" is the quantitative content behind the qualitative convergence-to-$\{V = 0\}$ statement.*

70.4 Grand conclusion theorems

We now state the combined results. The first theorem isolates the "basin, then endpoint" structure. The second adds compassion stabilization. In both theorems, the emphasis is on separating (1) *entry conditions* (what must be true for a system or population to land in a favorable region of state space) from (2) *stability/attraction conditions* (what keeps it there and drives convergence once it is inside). This separation is useful because many disagreements about "inevitability" turn on confusing these two logically distinct steps.

Theorem 250 (Basin-constrained endpoint convergence)**.** *Fix a meta-goal/value basin $B(m_0, v_0)$ and a resource level R. Assume there exists a subset $B \subseteq B(m_0, v_0)$ such that:*

(i) B is forward invariant under F_R,

(ii) F_R is a contraction on B (Lemma 176),

(iii) A_{C3} contains the unique fixed point x^* of F_R in B.

Then all trajectories starting in B converge to the same endpoint $x^* \in A_{C3}$. Consequently, within B the final state is determined by the basin (meta-goals and value seed), not by microstate details. Moreover, the conclusion is an endpoint identification *claim: it does not require that B be large in an absolute sense, only that once a trajectory is in B the long-run behavior is forced.*

Remark 3793. *In ordinary language: if your meta-goals and value seed keep you inside a stable basin, and within that basin the self-update dynamics is contractive, then there is a single "destiny" x^* to which all initial states in the basin converge. If that destiny lies in the C3-like region, then basin entry is the decisive event: once in, the system is pulled to a C3-like endpoint regardless of many internal contingencies. A helpful intuition is that contractivity makes the system "forget" fine-grained differences: each self-update step shrinks distances between alternative internal configurations, so historical accident is progressively damped rather than amplified.*

Remark 3794. *This theorem formalizes the slogan "basin, then endpoint." It ties together the fixed-point stability perspective on self-modification [10] with the Collective Locations idea that certain regimes (like C3) are attractor-like when anchoring constraints are present [12] (Hyperseed-Concept 75, 110). Formally, (i)–(ii) are the two technical conditions that let one treat "meta-goals plus resources" as specifying not merely a preference ordering, but an effective* geometry *on the update process: invariance ensures the iteration never leaves the domain where the geometry applies, while contraction ensures the geometry is dissipative rather than expansive.*

Remark 3795. *Assumption (ii) implicitly presupposes a choice of metric (or norm) on the state space in which the Lipschitz constant of F_R is strictly less than 1 on B. Interpreting this choice is part of the modeling content: different metrics correspond to different notions of "relevant" differences between internal states (e.g., differences in commitments, differences in coordination structure, differences in value representation). Thus, the theorem should be read as saying:* if *there exists a principled similarity notion under which self-updates compress differences, then endpoint determinacy follows.*

Proof. By Lemma 176, F_R has a unique fixed point $x^* \in B$ and all trajectories in B converge to it. Assumption (iii) says $x^* \in A_{C3}$, hence the limiting regime is C3-like. To see the role of (i) explicitly: since B is forward invariant, the entire forward orbit $\{F_R^{(t)}(x_0)\}_{t\geq 0}$ remains in the region where contraction is guaranteed, so the Banach-type argument applies to the whole trajectory rather than only to a single step. □

Proof sketch. Reduce the theorem to Lemma 176: contraction plus invariance yields a unique fixed point and convergence to it. The only additional ingredient is checking that this fixed point lies in A_{C3}, which then transfers convergence to a regime statement. Intuitively: once the "meta-goal/value constitution" defines a self-update that is both self-contained (invariant) and self-damping (contractive), it behaves like a funnel whose narrow end is x^*. □

Remark 3796. *The proof is short because the work is concentrated in choosing conditions that allow a standard theorem to apply. Conceptually, the nontrivial part is not the Banach argument but the claim that meta-goals can enforce a contractive geometry on self-change. When they do, the endpoint becomes a property of the "constitution plus seed" rather than a property of fine-grained accident. It is also worth noting what the theorem* does not *assert: it does not claim that all initial conditions in $B(m_0, v_0)$ lie in such a contractive subset B, nor that all plausible self-modification maps are contractive; it only states that when a basin contains an invariant contractive core whose fixed point is C3-like, then convergence to a C3-like endpoint is forced for trajectories that enter that core.*

Theorem 251 (Resource-rich pull toward C3-like coordination under selection)**.** *Consider a population of communities/mindplexes interacting with environments $\alpha \sim \nu$. Assume:*

(a) Trust-capable communities have nonnegative advantage $\Delta(\alpha) \geq 0$ for all tasks, with strict advantage on a set of positive ν-measure, hence $\bar{\Delta} > 0$ (Definition above).

(b) The population share $x(t)$ evolves approximately by replicator dynamics.

(c) For individual communities, increasing trust-capacity κ and commitment stability σ monotonically increases expected payoff, and reduces drift δ via lower "contrivance" (overhead) requirements.

Then from any initial condition with $x(0) > 0$, selection drives $x(t) \to 1$ and typical surviving communities are pushed toward high κ and high σ (hence toward the C3-like region, provided thresholds are reachable). This statement is intended as a directional *result: it identifies the sign of* *the selection pressure under the stated payoff assumptions, leaving open the detailed speed of convergence (which will depend on variance, interaction structure, and the strength of the advantage $\bar{\Delta}$).*

Remark 3797. *This theorem says: if the world rewards trust-capable coordination on average, then populations tend to become dominated by trust-capable forms; and if, within that family, higher coordination and stronger commitments pay off (and reduce drift/overhead), then surviving variants are pushed toward the C3-like corner of the coordinate space. This is the selection-theoretic complement to the fixed-point claim: it argues not only that $C3$-like endpoints can be stable once inside a basin, but also that evolutionary pressure can push collectives toward the relevant region. The role of the environment distribution ν is to formalize the idea that "the world" is a mixture of tasks: (a) requires that trust-capability is not merely occasionally useful, but never systematically harmful across the task distribution, and is sometimes strictly beneficial.*

Remark 3798. *Assumption (c) invokes "contrivance" as a proxy for coordination overhead: systems that coordinate and anchor well can achieve outcomes with less ad hoc scaffolding. This echoes the weakness/contrivance formalism used elsewhere in Hyperseed-adjacent work [3, 2] (Hyperseed-Concept 202, 143). In the present section we do not re-derive that formalism; we only use the qualitative direction: lower contrivance tends to mean less room for unaccountable drift. One can read δ as aggregating several failure modes (goal drift, value drift, coordination drift): assumption (c) then states that better trust and commitment reduce the "degrees of freedom" available for these drifts by making successful coordination less dependent on brittle, case-by-case patches.*

Remark 3799. *The "approximately" in (b) is doing real work: real populations have mutation/innovation, finite-size noise, and sometimes frequency-dependent payoffs. The conclusion $x(t) \to 1$ should therefore be understood as an idealized limit statement that becomes a near-fixation statement under small perturbations (e.g., high but not complete dominance in stationary distributions), provided the advantage $\bar{\Delta}$ is not overwhelmed by noise or by systematic countervailing forces. This is analogous in spirit to how contraction-based claims are robust to small additive perturbations when the contraction margin is large enough.*

Proof. By (a) we have $\bar{\Delta} > 0$, so Lemma 177 yields $x(t) \to 1$. Assumption (c) states that within the trust-capable family, selection gradients favor increasing κ and σ and decreasing δ. Thus the population concentrates on high-trust, high-anchoring, low-drift variants. If these exceed the C3 thresholds, the population enters A_{C3}. In other words, the first step is a between-type selection argument (trust-capable vs. not), while the second step is a within-type optimization pressure over traits (κ, σ, δ) conditional on being in the trust-capable class. □

Proof sketch. First apply the replicator lemma to show trust-capable forms dominate. Then treat κ, σ, δ as traits under selection within the trust-capable class: if payoffs increase with κ, σ and decrease with δ, selection pushes the distribution of traits toward high κ, σ and low δ. Crossing the thresholds is then a reachability condition rather than a stability condition. Equivalently: The theorem argues that A_{C3} can be approached from "outside" by adaptive pressure, while Theorem 250 argues that once inside an appropriate basin the endpoint is pinned down. □

Remark 3800. *The proof is intentionally schematic because the theorem is a schema: a rigorous conclusion follows whenever one has (i) a selection equation close to replicator form and (ii) monotone payoff effects for the traits. The geometric intuition is that selection induces a flow in trait space that points toward the "high coordination/high anchoring/low drift" region. The substantive question becomes empirical and architectural: do realistic environments and architectures* *satisfy (a)–(c)? In particular, the "replicator-like" requirement should be read as a robustness condition: small departures from exact replicator dynamics (e.g. due to finite-population effects, bounded rational updates, or mild nonstationarity) do not change the qualitative direction field so long as the leading-order selection gradient remains aligned with the payoff advantage of the relevant traits. Likewise, the "monotone payoff" requirement is meant to exclude regimes where increasing an alignment-relevant trait (e.g. anchoring) predictably reduces fitness; absent such pathological tradeoffs, the associated vector field has a consistent component pointing toward the C3-like sector.*

Theorem 252 (Conditional compassion stabilization in rich-resource regimes)**.** *Fix a basin $B(m_0, v_0)$ with compassion seed $c(x) \geq c_\star$ for all $x \in B$. Assume:*

(1) The self-modification process uses a robust acceptance test with evaluator count $K = K(R)$ nondecreasing in resources R.

(2) There exists $\eta > 0$ and $p > 1/2$ such that any proposed modification that would reduce compassion by at least η is rejected by each evaluator with probability at least p (independently), as in Lemma 178.

(3) Catastrophic compassion loss requires at least one accepted compassion-degrading modification.

Then the probability that compassion ever drops below $c_\star - \eta$ over T proposal steps is at most

$$T \cdot \exp\Big(-2K(R)\,(\tau - (1-p))^2\Big),$$

and in particular can be made arbitrarily small by increasing resources (increasing K). Hence, in this model, more resources can make compassion more stable *rather than less. Here τ is the acceptance threshold parameter appearing in the acceptance-test formalization of Lemma 178; the term $\tau - (1-p)$ is the margin by which the acceptance rule is "stricter than" the expected evaluator error rate $(1-p)$ for a truly compassion-degrading change. Consequently, whenever $\tau > (1-p)$, the exponent is strictly negative and the failure probability decays exponentially in $K(R)$ (and thus, by assumption, with resources).*

Remark 3801. *This theorem translates the earlier concentration bound into a multi-step safety statement. The logical form is: if compassion starts above a threshold, and if large compassion drops are typically caught by evaluators, then the probability of crossing below the threshold within T steps is at most T times the per-step failure probability. If $K(R)$ grows with resources, the per-step failure probability shrinks exponentially, and the total failure probability shrinks accordingly. The role of the horizon T is explicit: for fixed T, exponential decay in $K(R)$ dominates the linear prefactor, while for slowly growing $T = T(R)$ the same conclusion can still hold whenever $\log T(R) = o(K(R))$. In other words, the statement is compatible with long-running systems so long as verification capacity grows sufficiently faster than the logarithm of the number of risky update opportunities.*

Remark 3802. *The result is "conditional" in a precise sense: it does not assert that compassion is stable in general, only that it becomes hard to lose* accidentally *under a particular class of robust meta-procedures. This fits a general alignment philosophy: protect value seeds by meta-level tests and constitutions rather than by assuming the seeds are always directly optimal [2] (Hyperseed-Concept 110, 81). The qualifier "accidentally" is doing work: the theorem is consistent with scenarios in which an agent, after sufficient deliberation, intentionally chooses to override or reinterpret the seed; the claim is only that noisy search, local hacking, or unintended side-effects become statistically unlikely to push the system across the specified safety boundary when the acceptance mechanism is sufficiently redundant and conservative.*

Proof. For each proposal step $t = 1, \dots, T$, Lemma 178 bounds the probability that a compassion-degrading modification is accepted by the displayed exponential term. Apply a union bound over the T steps to obtain the stated result. Assumption (3) connects the per-step acceptance event to the trajectory-level property: if c ever falls below $c_\star - \eta$, then (by hypothesis) at least one accepted modification in the history must have been compassion-degrading by at least η, so the "ever drops below" event is contained in the union of the per-step bad-acceptance events. □

Proof sketch. Bound the probability of a bad acceptance event at each step by the exponential concentration term. Then use the union bound: the probability that any of T bad events occurs is at most the sum of their probabilities. The growth of $K(R)$ with resources makes the per-step term exponentially small. Equivalently, the process can be viewed as a repeated risk exposure with a per-exposure failure rate that decreases exponentially in the amount of verification, so the cumulative risk over T trials remains small in the rich-resource regime. □

Remark 3803. *The union bound is crude but robust: it requires no independence between different proposal steps. Geometrically, one may imagine a trajectory moving through state space with a "guardrail" at $c = c_\star - \eta$; the theorem says that the guardrail is rarely breached if each attempted breach is caught with high probability by many evaluators. The "no independence across steps" point is important because self-modification proposals may be highly correlated (e.g. successive edits derived from the same search heuristic), and a bound that required stepwise independence would be brittle. By contrast, the only independence used here is the within-step independence across evaluators invoked in Lemma 178, which is a structural assumption about redundancy in the checking mechanism rather than about the agent's proposal distribution.*

Corollary 85 (Grand conclusion: C3-like attractor with stable seeded compassion)**.** *Assume the hypotheses of Theorem 250 and Theorem 252 hold for all resources $R \geq R_0$ with some $K(R) \to \infty$ as $R \to \infty$. Then in the rich-resource limit:*

(i) endpoint convergence within the basin becomes strong (microstate-insensitive), and

(ii) the probability of losing seeded compassion without deliberate override tends to 0*.*

Therefore, for minds in such basins, the key "alignment" problem is to (a) enter the right basin (meta-goals and initial value seeds) and (b) survive the transition, rather than to microspecify the final endpoint state. One may read (i) as asserting that within-basin variation in initial micro-details (e.g. incidental parameter noise, local representational choices, or early stochastic experiences) is progressively forgotten by the dynamics, so that macroscopic endpoint properties are determined primarily by basin membership and not by fine-grained initial conditions.

Remark 3804. *The corollary is the promised "rigorous cousin" of an Omega-Point-like claim: conditioned on basin entry and on increasing verification capacity with resources, the trajectory is funneled toward a C3-like endpoint and simultaneously becomes unlikely to shed its compassion seed by accident. Formally, the two effects come from different sources: contraction (a dynamical stability property) and concentration (a statistical robustness property). The contraction component is about the geometry of the deterministic (or mean) update map restricted to the basin, while the concentration component is about bounding rare failures of a stochastic decision procedure (the evaluator-based acceptance test). The corollary therefore combines a "where the flow goes" statement with a "how often safety checks fail" statement, yielding a joint picture in which both endpoint identity and value retention become predictable in the high-resource regime, conditional on remaining in the basin.*

Remark 3805. *This also clarifies what "rich-resource" contributes: it is not merely faster optimization but larger K, i.e. more extensive internal checking. In that sense, abundance can shift the* *dominant failure mode from "accidental drift" to "deliberate override," which is a different kind of problem (Hyperseed-Concept 162, 110, 198). Concretely, when K is small, the main risk is that a single flawed evaluation (or a small coalition of correlated evaluator errors) can let through a damaging self-modification; when K is large, the residual risk is dominated by either adversarially crafted proposals that systematically fool the evaluators or by explicit policy-level decisions to change the acceptance criteria themselves. The latter case is closer to governance and constitutional design than to ordinary robustness against noise, and thus motivates treating "override" as a separate alignment failure category rather than as mere stochastic drift.*

Proof. Endpoint convergence is given by Theorem 250. Compassion stability over any fixed horizon follows from Theorem 252 with $K(R) \to \infty$, which drives the union-bound probability to 0. The stated alignment interpretation is a direct restatement of (i) and (ii). If one additionally considers horizons $T = T(R)$ that grow with resources, the same argument implies vanishing failure probability whenever $T(R)\exp(-2K(R)(\tau - (1-p))^2) \to 0$, making explicit the tradeoff between longer operating time and increased verification. $\square$

Proof sketch. Combine the two theorems: contraction yields a unique basin-dependent endpoint and hence microstate-insensitivity, while the concentration/union-bound estimate yields vanishing probability of compassion loss as verification resources grow. Then reinterpret these two mathematical facts as an alignment claim about what must be specified (basin) versus what need not be microspecified (endpoint). In this combined view, the basin assumptions determine the attractor geometry, while the rich-resource scaling assumption determines the asymptotic reliability of the meta-procedure that guards the seed during the approach to the attractor. $\square$

Remark 3806. *The key conceptual step is the separation of concerns: endpoint convergence and value-seed stability are proved by distinct mechanisms, so one can weaken or strengthen assumptions on each side independently. This modularity is useful when mapping the schema onto real systems: one can investigate contractivity-like behavior (perhaps approximate) without simultaneously assuming perfect evaluator independence, and vice versa. Put differently, the analysis treats "where the dynamics tend to go" (a claim about asymptotic attractors or endpoints) as logically separable from "whether a particular normative seed persists along the way" (a claim about invariance or robustness of certain commitments under self-modification). This separation also clarifies which empirical questions attach to which theorem family: e.g., measuring basin geometry and effective contraction rates pertains to endpoint results, while auditing implementation pathways, reflective stability, and adversarial modification channels pertains to value-seed stability. Even when true contraction is too strong, approximate contraction or contraction on average can still provide meaningful endpoint concentration results, while partial evaluator independence (or independence restricted to a vetted class of self-modifications) can still support nontrivial seed-preservation claims.*

70.5 Discussion: what the theorems do and do not claim

Remark 3807 (Conditionality and failure modes)**.** *These theorems are "if-then" results. They do not claim that all self-modifying minds will be C3-like, nor that compassion is universally stable. They isolate sufficient conditions under which:*

(1) Metagoals induce contractive, basin-dependent endpoints,
(2) Selection favors trust-capable coordination on typical complex tasks,
(3) Resource-rich robustness testing makes seeded compassion hard to lose accidentally, and
(4) The combination yields substantial endpoint convergence inside broad basins.

Failure modes correspond to violations of assumptions: adversarial environments where $\bar{\Delta} \leq 0$, deliberate overrides of compassion commitments, non-contracting metagoals, or regimes where "more resources" do not translate into stronger robustness checks (bounded K). The intended reading is thus diagnostic: each assumption marks a potential empirical or design lever, and each conclusion is only as strong as the weakest link among them. For example, (1) does not assert a unique global fixed point; it allows multiple endpoints indexed by basin, and it allows that basins may be narrow, fractal, or difficult to enter. Likewise, (2) concerns comparative advantage in typical task distributions; it does not preclude niches in which deception or coercion dominates, nor does it imply that trust-capable coordination is the only stable strategy. Condition (3) is framed as an accidental-loss *claim: robustness testing reduces the probability of unintended value drift under perturbations, but it does not eliminate intentional value changes by an agent that can rewrite its own evaluative core. The bounded-K caveat is particularly important: increased compute, time, or capital only helps if it is actually deployed toward adversarial search, red-teaming, and formal verification-like checks, rather than toward capability gains that leave the evaluative loop untouched. Finally, even when all premises are satisfied, the conclusions concern convergence in distribution or within a specified basin; they do not guarantee rapid convergence, nor do they rule out long transients, local cycles, or path-dependent lock-in effects before the attractor regime is reached.*

Remark 3808. *A final interpretive note: nothing here denies the possibility of moral or political tragedy; it only says that there exist mathematically coherent sufficient conditions under which tragedy becomes less likely* within a given basin. *In this sense, the framework aims to be compatible with pluralism about outcomes across basins: different initial conditions, institutions, or training regimes may land a system in different regions of state space whose long-run endpoints are not interconvertible without leaving the basin. The basin qualifier also signals a humility about "global" claims: the theorems are meant to clarify when convergence is plausible conditional on being in the right region, not to assert that the right region is inevitable, easy to reach, or politically guaranteed. In Peircean terms, the theorems attempt to capture a kind of "Thirdness"—law-like regularity—emerging from the interplay of individual updates and collective selection (cf. [14]; Hyperseed-Concept 190). The point of invoking Thirdness here is not merely rhetorical: contractive tendencies, basin structure, and selection-mediated regularities function as the formal stand-ins for "habits" or "laws" that are not imposed from outside but stabilized by repeated dynamics. In Whitehead's idiom, they model how process can inherit structure from its own constraints and thereby converge toward a stable pattern of becoming (cf. [15]; Hyperseed-Concept 62, 141). Here "inherit" should be read in the technical sense that update rules constrain successor states, and robustness/selection pressures amplify some constraints into persistent macroscopic patterns; thus "pattern of becoming" refers to a stable attractor-like regime at the level of policies, commitments, and coordination norms, rather than a static end-state that precludes further change.*

70.6 GTGI measures for collective intelligences, mindplexes, and resource-rich minds

Glossary touchpoints (where glossary concepts are used). This subsection uses the GTGI-family glossary concepts: *GTGI context* (Def. 339; Hyperseed-Concept 214), *pragmatic general intelligence* (Def. 340; Hyperseed-Concept 215), *resource-consumption distribution* (Def. 341; Hyperseed-Concept 216), *efficient pragmatic general intelligence* (Def. 342; Hyperseed-Concept 217), and *intellectual breadth* (Def. 343; Hyperseed-Concept 218). It also uses collective-scale glossary concepts: *Mindplex* (Hyperseed-Concept 113; see the coherence-based proxy discussion in §22.10), *Global Brain* (Hyperseed-Concept **??**; Def. 379), *Rich-Resource Mind* (Hyperseed-Concept 162), *Compassion* (Hyperseed-Concept 81), and *Collective Consciousness Location* (Hyperseed-Concept 75).

Standing notation (contexts and weights). Write a context as $c = (\mu, g, T)$ (Def. 339). For any fixed choice of $(\mathcal{E}, \mathcal{G}, \mathcal{T}, \nu, \gamma, \tau)$ as in Def. 340, define the nonnegative context-weight

$$w(c) \;:=\; w(\mu, g, T) \;:=\; \nu(\mu)\,\gamma(g,\mu)\,\tau_{g,\mu}(|T|).$$

For a policy π, write

$$V^{\pi}(c) \;:=\; V^{\pi}_{\mu,g,T}.$$

Define also the per-context efficiency weight

$$\chi_\pi(c) \;:=\; \chi_\pi(\mu,g,T) \;:=\; \sum_{Q\in\mathcal{Q}} \eta_{\mu,g,T}(Q)\,\frac{V^{\pi}_{\mu,g,T}}{Q} \quad \text{(Def. 343)}.$$

With this shorthand,

$$\Pi(\pi) \;=\; \sum_c w(c)\,V^{\pi}(c), \qquad \Pi^{\text{eff}}(\pi) \;=\; \sum_c w(c)\,\chi_\pi(c),$$

whenever the defining sums converge (Defs. 340, 342).

70.6.1 Collective intelligences as mixtures and selectors

Remark 3809 (Conceptual explanation: "collective intelligence" as policy composition)**.** *A basic mathematical move for treating a collective as an agent is to define a* meta-policy *that chooses (or combines) the actions suggested by individual member policies. If a collective simply* lotteries *between members ("today we follow Alice's plan"), then its GTGI intelligence measures are just convex combinations of the members' measures. If, instead, the collective can* route contexts *to specialists ("Bob handles physics-like tasks, Carol handles diplomacy-like tasks"), then the collective can exceed each individual on the GTGI measures that average across contexts (Defs. 340, 342). This is one formal bridge between* collective intelligences *and GTGI quantifications.*

Definition 677 (Episode-lottery mixture of policies)**.** *Let $\pi_1, \ldots, \pi_n$ be policies and let $\rho \in \Delta(\{1, \ldots, n\})$. An* episode-lottery mixture *is any stochastic policy π^{lot}_ρ whose internal randomness samples an index $I \sim \rho$* once per episode *(per context c), and then behaves exactly as π_I throughout the episode. Equivalently: conditional on $I = i$, the induced interaction distribution is the same as under π_i.*

Theorem 253 (Linearity of Π and Π^{eff} under episode-lottery mixtures)**.** *Let π^{lot}_ρ be an episode-lottery mixture of $\{\pi_i\}_{i=1}^n$ (Def. 677). Assume that, conditional on $I = i$, the resource-consumption distribution in context c is the same as that of π_i (Def. 341).*

Then, for every context c,

$$V^{\pi^{\text{lot}}_\rho}(c) \;=\; \sum_{i=1}^n \rho(i)\,V^{\pi_i}(c), \qquad \chi_{\pi^{\text{lot}}_\rho}(c) \;=\; \sum_{i=1}^n \rho(i)\,\chi_{\pi_i}(c),$$

and consequently

$$\Pi\Big(\pi^{\text{lot}}_\rho\Big) \;=\; \sum_{i=1}^n \rho(i)\,\Pi(\pi_i), \qquad \Pi^{\text{eff}}\Big(\pi^{\text{lot}}_\rho\Big) \;=\; \sum_{i=1}^n \rho(i)\,\Pi^{\text{eff}}(\pi_i).$$

In particular, an episode-lottery collective cannot exceed the best member on either score:

$$\Pi\Big(\pi^{\text{lot}}_\rho\Big) \le \max_i \Pi(\pi_i), \qquad \Pi^{\text{eff}}\Big(\pi^{\text{lot}}_\rho\Big) \le \max_i \Pi^{\text{eff}}(\pi_i).$$

Proof. Fix a context c and let $I \sim \rho$ be the (internal) lottery index. By the law of total expectation and the defining property of the episode-lottery mixture,

$$V^{\pi_\rho^{\mathrm{lot}}}(c) = \mathbb{E}\Big[V^{\pi_\rho^{\mathrm{lot}}}(c) \mid I\Big] = \sum_{i=1}^{n} \rho(i)\, V^{\pi_i}(c).$$

For χ, note that $\chi_\pi(c)$ is, by definition, the conditional expectation of $V^\pi(c)/Q$ under the resource random variable Q drawn from η_c. Conditioning on $I = i$ gives exactly the χ of π_i, hence

$$\chi_{\pi_\rho^{\mathrm{lot}}}(c) = \mathbb{E}\left[\frac{V^{\pi_\rho^{\mathrm{lot}}}(c)}{Q}\right] = \sum_{i=1}^{n} \rho(i)\, \mathbb{E}\left[\frac{V^{\pi_i}(c)}{Q}\right] = \sum_{i=1}^{n} \rho(i)\, \chi_{\pi_i}(c).$$

Now substitute these identities into $\Pi(\pi) = \sum_c w(c) V^\pi(c)$ and $\Pi^{\mathrm{eff}}(\pi) = \sum_c w(c)\chi_\pi(c)$ and exchange finite sums. The max bounds follow because convex combinations are bounded above by the maximum term. $\square$

Remark 3810 (Conceptual explanation: routing contexts to specialists)**.** *Episode-lottery mixtures correspond to a weak form of collective intelligence: the group "picks a boss" and follows them. A stronger form is a* router*: the group has (or learns) enough meta-knowledge to assign contexts to the member best suited to them. In that regime, GTGI scores can exceed every individual's score, not by magic, but because the collective is effectively the* pointwise maximum *over members, context-by-context.*

Definition 678 (Context-router over a fixed member set)**.** *Let $\pi_1, \ldots, \pi_n$ be policies. A* (deterministic) context-router *is a map $s : \mathcal{E} \times \mathcal{G} \times \mathcal{T} \to \{1, \ldots, n\}$. Define the routed policy $\pi^{(s)}$ by: in context $c = (\mu, g, T)$, $\pi^{(s)}$ behaves as $\pi_{s(c)}$ throughout the episode.*

Theorem 254 (Contextual specialization bound for collectives)**.** *Let $\pi_1, \ldots, \pi_n$ be policies and let $\pi^{(s)}$ be the routed policy induced by a router s (Def. 678). Then*

$$\Pi(\pi^{(s)}) \;=\; \sum_c w(c)\, V^{\pi_{s(c)}}(c), \qquad \Pi^{\mathrm{eff}}(\pi^{(s)}) \;=\; \sum_c w(c)\, \chi_{\pi_{s(c)}}(c).$$

Moreover, the optimal routing achieves the pointwise maxima:

$$\sup_s \Pi(\pi^{(s)}) \;=\; \sum_c w(c) \max_{1\le i\le n} V^{\pi_i}(c), \qquad \sup_s \Pi^{\mathrm{eff}}(\pi^{(s)}) \;=\; \sum_c w(c) \max_{1\le i\le n} \chi_{\pi_i}(c).$$

In particular,

$$\sup_s \Pi(\pi^{(s)}) \;\ge\; \max_i \Pi(\pi_i), \qquad \sup_s \Pi^{\mathrm{eff}}(\pi^{(s)}) \;\ge\; \max_i \Pi^{\mathrm{eff}}(\pi_i),$$

with strict inequality whenever there are two contexts c_1, c_2 of positive weight and two members $i \neq j$ such that π_i is strictly better on c_1 while π_j is strictly better on c_2.

Proof. The displayed equalities for a fixed router s are immediate from the definition of routing: in each context c, $\pi^{(s)}$ behaves as $\pi_{s(c)}$, so its V and χ coincide with those of $\pi_{s(c)}$ in that context.

For the supremum: since all weights $w(c) \ge 0$ and the choice $s(c)$ affects only the term indexed by c, the maximizing router can be chosen independently per context: for each c, choose $s(c) \in \arg\max_i V^{\pi_i}(c)$ (or $\arg\max_i \chi_{\pi_i}(c)$). This yields the pointwise-max formula. The lower bounds by $\max_i$ follow because $\max_i V^{\pi_i}(c) \ge V^{\pi_j}(c)$ for any fixed j, hence summing against $w(c)$ gives $\sum_c w(c) \max_i V^{\pi_i}(c) \ge \sum_c w(c) V^{\pi_j}(c) = \Pi(\pi_j)$, and similarly for Π^{eff}.

Strictness holds under the stated "crossing" condition because then the pointwise max is strictly larger than any fixed member on at least one positive-weight context, while matching or exceeding on all others. $\square$

70.6.2 Mindplexes and intellectual breadth amplification

Remark 3811 (Conceptual explanation: why mindplexes can have higher breadth)**.** *Intellectual breadth B_{int} (Def. 343; Hyperseed-Concept 218) is an entropy over the* distribution of where an agent's competence/efficiency mass lies *across contexts. A mindplex (Hyperseed-Concept 113) is, among other things, a multi-scale organization in which coherent subminds can specialize and then be orchestrated coherently (see §22.10). If orchestration allows near-constant context-competence weight across many contexts, then the induced distribution in Def. 343 becomes closer to the underlying context weight distribution $w(\cdot)$, which typically has larger entropy than any single specialist's restricted support. Thus: mindplex emergence can be read as a* breadth amplifier *when it stabilizes specialization plus routing.*

Theorem 255 (A specialization-and-routing regime strictly increases intellectual breadth)**.** *Assume the set of contexts $\mathcal{C} := \mathcal{E} \times \mathcal{G} \times \mathcal{T}$ is finite. Let $w(c)$ be the GTGI context-weights and define the normalized weight distribution*

$$\bar{w}(c) \ := \ \frac{w(c)}{\sum_{c'} w(c')}.$$

Fix a constant $\chi_\star > 0$.

Suppose there are n "specialist" policies $\pi_1, \ldots, \pi_n$ and a partition of contexts $\mathcal{C} = C_1 \sqcup \cdots \sqcup C_n$ such that

$$\chi_{\pi_i}(c) \ = \ \chi_\star \ \text{for } c \in C_i, \qquad \chi_{\pi_i}(c) \ = \ 0 \ \text{for } c \notin C_i.$$

Let $s(c)$ be the unique index with $c \in C_{s(c)}$, and let $\pi^{(s)}$ be the routed policy (Def. 678). Then:

1. *The routed policy has constant context-competence weight:*

$$\chi_{\pi^{(s)}}(c) \ = \ \chi_\star \quad \forall c \in \mathcal{C}.$$

2. *The intellectual breadth of the routed policy is exactly the entropy of $\bar{w}$:*

$$B_{\text{int}}(\pi^{(s)}) \ = \ -\sum_{c \in \mathcal{C}} \bar{w}(c) \log \bar{w}(c).$$

3. *Each specialist's breadth is bounded by its support size:*

$$B_{\text{int}}(\pi_i) \ \leq \ \log |C_i|.$$

   In particular, if $\bar{w}$ is uniform (so $B_{\text{int}}(\pi^{(s)}) = \log |\mathcal{C}|$) and each C_i is a proper subset ($|C_i| < |\mathcal{C}|$), then

$$B_{\text{int}}(\pi^{(s)}) \ > \ \max_i B_{\text{int}}(\pi_i).$$

Proof. (1) For any $c \in \mathcal{C}$, by definition $\pi^{(s)}$ behaves as $\pi_{s(c)}$ in context c. Since $c \in C_{s(c)}$, the assumed specialization property gives $\chi_{\pi^{(s)}}(c) = \chi_{\pi_{s(c)}}(c) = \chi_\star$.

(2) In Def. 343, the normalized distribution over contexts is

$$\chi^P_{\text{Con}_{\pi^{(s)}}}(c) = \frac{w(c)\chi_{\pi^{(s)}}(c)}{\sum_{c'} w(c')\chi_{\pi^{(s)}}(c')} = \frac{w(c)\chi_\star}{\chi_\star \sum_{c'} w(c')} = \bar{w}(c).$$

Therefore $B_{\text{int}}(\pi^{(s)})$ equals the Shannon entropy of $\bar{w}$.

(3) For specialist π_i, the same normalization yields

$$\chi^P_{\mathrm{Con}_{\pi_i}}(c) = \frac{w(c)\chi_{\pi_i}(c)}{\sum_{c'} w(c')\chi_{\pi_i}(c')} = \frac{w(c)\chi_\star}{\chi_\star \sum_{c' \in C_i} w(c')} = \frac{w(c)}{\sum_{c' \in C_i} w(c')} \quad \text{for } c \in C_i,$$

and 0 otherwise. Thus $\chi^P_{\mathrm{Con}_{\pi_i}}$ is supported on C_i, so its entropy is at most $\log |C_i|$, with equality only if it is uniform on C_i. The final strict inequality follows immediately in the uniform-$\bar{w}$ case since then $B_{\mathrm{int}}(\pi^{(s)}) = \log |\mathcal{C}| > \log |C_i| \geq B_{\mathrm{int}}(\pi_i)$ for each i. □

70.6.3 Mindplex emergence as an efficiency transition

Remark 3812 (Conceptual explanation: mindplexes as local optima of efficiency)**.** *In §22.10, a mindplex is defined (proxy-wise) as a* notably coherent *society of mind: removing a part or adding an unrelated part reduces coherence. If we accept a modeling bridge that says "higher coherence implies less internal coordination waste (and hence higher* Π^{eff}*)", then the mindplex definition becomes an* efficiency optimality *statement: a mindplex is a local optimum of efficient pragmatic intelligence among nearby collective compositions. This gives a concrete sense in which a "transition to mindplex" can manifest as a qualitative change in GTGI measures: not only higher raw achievement, but a stable local maximum in achievement-per-resource.*

Definition 679 (A coherence-to-efficiency monotonicity hypothesis)**.** *Let S range over finite collections of components (agents and/or artifacts). Assume each S induces a collective policy π_S (a collective controller) and thus a score $\Pi^{\mathrm{eff}}(\pi_S)$ (Def. 342). Fix an observer O and coherence proxy* $\mathrm{Coh}_O(S)$ *as in §22.10.*

We say efficient GI is monotone in coherence *if for all S, S',*

$$\mathrm{Coh}_O(S) \;\geq\; \mathrm{Coh}_O(S') \quad \Longrightarrow \quad \Pi^{\mathrm{eff}}(\pi_S) \;\geq\; \Pi^{\mathrm{eff}}(\pi_{S'}).$$

Theorem 256 (Notable coherence implies local optimality of efficient pragmatic GI)**.** *Assume the coherence-to-efficiency monotonicity hypothesis (Def. 679).*

Let S be notably coherent *relative to O in the sense of §22.10:*

$$\mathrm{Coh}_O(S) > \mathrm{Coh}_O(S \setminus \{x\}) \;\; \forall x \in S, \qquad \mathrm{Coh}_O(S) > \mathrm{Coh}_O(S \cup \{y\}) \;\; \forall y \notin S \;\; (admissible).$$

Then S is a local maximizer of efficient pragmatic GI among single-step add/remove neighbors:

$$\Pi^{\mathrm{eff}}(\pi_S) \;>\; \Pi^{\mathrm{eff}}(\pi_{S \setminus \{x\}}) \;\; \forall x \in S, \qquad \Pi^{\mathrm{eff}}(\pi_S) \;>\; \Pi^{\mathrm{eff}}(\pi_{S \cup \{y\}}) \;\; \forall y \notin S \;\; (admissible).$$

In particular, any mindplex (Hyperseed-Concept 113) satisfying the notable-coherence condition inherits a "local efficiency optimum" interpretation in GTGI terms.

Proof. Fix $x \in S$. By notable coherence, $\mathrm{Coh}_O(S) > \mathrm{Coh}_O(S \setminus \{x\})$. By monotonicity (Def. 679), this implies $\Pi^{\mathrm{eff}}(\pi_S) \geq \Pi^{\mathrm{eff}}(\pi_{S \setminus \{x\}})$. If monotonicity is strict whenever coherence is strict (a natural strengthening, and the intended reading in this application), then the inequality is strict. The argument for add-neighbors $S \cup \{y\}$ is identical. □

Remark 3813 (A quantitative "transition" corollary template)**.** *If one models a group-to-mindplex transition as a path $S(\lambda)$ in which coherence increases with λ (as in the development-path discussions around mindplex emergence), then Def. 679 yields monotone increase of* $\Pi^{\mathrm{eff}}(\pi_{S(\lambda)})$ *along the path. Theorem 256 adds that once a notably-coherent plateau is reached, the system becomes locally stable as an efficiency optimum against small perturbations of membership.*

70.6.4 Mindplex-scale selection pressure re-expressed in GTGI terms

Remark 3814 (Conceptual explanation: importing the "interface scaling" mindplex theorem into GTGI)**.** *Theorem 227 states (in a selection/coordination model) that mindplex-scale systems face $\Omega(n)$ coordination interfaces and therefore selection pressure toward trust-capable coordination grows at least linearly in size. GTGI pragmatic intelligence Π is itself an (appropriately weighted) expectation of goal achievement across contexts (Def. 340). So, whenever the coordination-task environment in Theorem 227 is represented inside the GTGI context distribution, the same lower bound becomes a* lower bound on differences of GTGI intelligence measures*. This gives a clean mathematical answer to "what happens to GTGI measures as a collective becomes a mindplex and scales?": under the interface-growth hypothesis, the GTGI advantage of trust-capable coordination grows at least linearly with mindplex size.*

Two clarifications make the import precise. First, the mindplex theorem speaks in terms of coordination interfaces *(points at which information must cross module/agent boundaries, where incentives and credence must be aligned). In GTGI, this interface burden is encoded as an* observable *$m(c)$ on coordination contexts c, so that scaling statements become statements about $\mathbb{E}[m(c)]$ under the GTGI weighting. Second, the relevant conclusion is inherently comparative: the theorem does not assert that all policies improve with n, but that policies with trust-capable coordination gain a growing advantage over a class of "trustless" baselines because they can avoid (or amortize) losses that scale with the interface burden.*

Theorem 257 (A GTGI-form lower bound from mindplex interface growth)**.** *Fix a scale n and a subset of GTGI contexts $\mathcal{C}_n \subseteq \mathcal{E} \times \mathcal{G} \times \mathcal{T}$ interpreted as "mindplex-scale coordination contexts". Let*

$$W_n := \sum_{c \in \mathcal{C}_n} w(c), \qquad \bar{w}_n(c) := \frac{w(c)}{W_n} \quad (c \in \mathcal{C}_n),$$

assuming $W_n \in (0, \infty)$. Let $m : \mathcal{C}_n \to \mathbb{R}_{\geq 0}$ be an "interface count" observable.

Assume there exist $\ell > 0$ and a distinguished "trust-capable" collective policy π_{T} such that for every trustless policy π in some class Π_{L} and every $c \in \mathcal{C}_n$,

$$V^{\pi_{\mathrm{T}}}(c) - V^{\pi}(c) \geq \ell\, m(c).$$

Assume also a mindplex-scale interface growth bound in GTGI-weighted form:

$$\mathbb{E}_{c \sim \bar{w}_n}[m(c)] \geq c_0\, n \quad \text{for some } c_0 > 0.$$

Then the pragmatic GI advantage of π_{T} over any trustless $\pi \in \Pi_{\mathrm{L}}$ satisfies

$$\Pi(\pi_{\mathrm{T}}) - \Pi(\pi) \geq \ell\, c_0\, n\, W_n.$$

In particular, if W_n does not vanish with n (the GTGI distribution assigns nontrivial mass to such contexts at each scale), the advantage grows at least linearly in n.

Proof. Restrict the Π sum to $\mathcal{C}_n$ and use nonnegativity of weights:

$$\Pi(\pi_{\mathrm{T}}) - \Pi(\pi) = \sum_{c} w(c)\big(V^{\pi_{\mathrm{T}}}(c) - V^{\pi}(c)\big) \geq \sum_{c \in \mathcal{C}_n} w(c)\big(V^{\pi_{\mathrm{T}}}(c) - V^{\pi}(c)\big).$$

Apply the per-context lower bound:

$$\sum_{c \in \mathcal{C}_n} w(c)\big(V^{\pi_{\mathrm{T}}}(c) - V^{\pi}(c)\big) \geq \ell \sum_{c \in \mathcal{C}_n} w(c)\, m(c) = \ell\, W_n\, \mathbb{E}_{c \sim \bar{w}_n}[m(c)].$$

Finally use $\mathbb{E}_{c \sim \bar{w}_n}[m(c)] \geq c_0 n$ to get the claim. ∎

Remark 3815 (Interpretive notes on Theorem 257). *The normalization $\bar{w}_n$ isolates the* within-subset *scaling of interface burden from the* across-subset *question of how much GTGI mass is assigned to such coordination contexts at scale n. In particular, W_n acts as a relevance factor: if the theory of contexts assigns negligible mass to mindplex-scale coordination problems (e.g., because most weight is placed on single-agent toy tasks), then even strong interface-growth effects will not move the global Π score much. Conversely, if W_n stays bounded away from 0, then coordination is persistently "on the test," and the lower bound becomes a genuine scaling law for measured pragmatic advantage.*

The observable $m(c)$ can be instantiated in many ways depending on the modeling choice: number of pairwise communication channels, number of verification edges in a provenance graph, number of cross-team dependencies in a plan, or any proxy for the count of distinct trust/credence alignment interfaces that must be managed. The theorem only needs $m(c) \geq 0$ and a linear lower bound on its GTGI-weighted average.

Finally, note the structure of the per-context assumption $V^{\pi_T}(c) - V^{\pi}(c) \geq \ell m(c)$: it encodes that each additional interface carries at least some constant marginal expected value for trust-capable coordination (relative to trustless baselines). This is the GTGI analog of "selection pressure" in the mindplex theorem: the larger the interface surface area, the more there is to lose without reliable trust mechanisms, and the more there is to gain by having them.

70.6.5 Resource-rich minds as limits and monotone families in GTGI

Remark 3816 (Conceptual explanation: turning "more resources" into a policy family). *GTGI scores $\Pi(\pi)$ and $\Pi^{\text{eff}}(\pi)$ are properties of a policy π. To speak meaningfully about "a scarce-resource mind becoming resource-rich" (Hyperseed-Concept 162), we model the mind as a* resource-indexed family *of policies $\{\pi_R\}_{R\geq 0}$: the same underlying agent architecture, but run with different resource budgets (time/compute/memory/coordination), so it can plan, simulate, verify, and self-correct more. Then we can ask: do GTGI measures converge as $R \to \infty$? when do they increase monotonically? The next theorem provides a clean sufficient condition: uniform near-optimality across contexts.*

The key modeling move is that "more resources" should not be treated as an external adjective but as a parameter that changes the induced policy. Concretely, R may correspond to more search depth, more deliberation steps, larger working memory, more samples in Monte Carlo estimation, more rounds of cross-checking between subagents, or more bandwidth for internal debate. In each case one obtains a different effective mapping from histories (or observations) to actions, i.e. a different π_R.

The uniformity requirement in the theorem below is intentionally strong: it captures the regime where additional resources help broadly across the GTGI context distribution rather than improving only a narrow slice of tasks. This is the sense in which the family approaches a "resource-rich mind" as a limit object: its residual regret becomes small in every *context, not merely on average.*

Theorem 258 (Rich-resource near-optimality implies convergence of pragmatic GI). *Let $\{\pi_R\}_{R\geq 0}$ be a family of policies indexed by a resource parameter R. Assume:*

1. *The total GTGI context-weight mass is finite:*

$$W := \sum_c w(c) < \infty.$$

2. There exists a function $\varepsilon(R) \geq 0$ with $\varepsilon(R) \to 0$ as $R \to \infty$, such that for every context c,

$$V^{\pi_R}(c) \;\geq\; V^*(c) - \varepsilon(R), \qquad V^*(c) := \sup_\pi V^\pi(c).$$

Define the (possibly formal) optimal pragmatic score

$$\Pi^* \;:=\; \sum_c w(c)\, V^*(c),$$

assuming it is finite.

Then for all R,

$$0 \;\leq\; \Pi^* - \Pi(\pi_R) \;\leq\; W\,\varepsilon(R),$$

and therefore $\Pi(\pi_R) \to \Pi^*$ as $R \to \infty$.

Moreover, if $R \mapsto V^{\pi_R}(c)$ is nondecreasing for each context c, then $R \mapsto \Pi(\pi_R)$ is nondecreasing.

Proof. By definition,

$$\Pi^* - \Pi(\pi_R) = \sum_c w(c)\big(V^*(c) - V^{\pi_R}(c)\big).$$

Each term is nonnegative by definition of $V^*(c)$. By assumption (2), $V^*(c) - V^{\pi_R}(c) \leq \varepsilon(R)$ for all c, hence

$$0 \leq \Pi^* - \Pi(\pi_R) \leq \sum_c w(c)\,\varepsilon(R) = W\,\varepsilon(R),$$

using assumption (1). Since $\varepsilon(R) \to 0$, the bound implies $\Pi(\pi_R) \to \Pi^*$.

For monotonicity: if $V^{\pi_R}(c)$ is nondecreasing in R for each c, then the weighted sum $\sum_c w(c)V^{\pi_R}(c)$ is also nondecreasing because all weights are nonnegative. □

Remark 3817 (Further interpretation: why finiteness and uniformity matter)**.** *Assumption (1) is a GTGI analogue of an integrability condition: it prevents the weighted sum defining $\Pi(\pi)$ from placing infinite total mass on contexts. Without such a condition, even vanishing per-context regret $\varepsilon(R)$ could fail to control $\Pi^* - \Pi(\pi_R)$.*

Assumption (2) is a uniform *additive regret bound. It can be read as: increasing resources drives the family $\{\pi_R\}$ into an $\varepsilon(R)$-optimal band around the contextwise optimum $V^*(c)$, and the band shrinks to 0 as $R \to \infty$. This is stronger than merely requiring that the* average *regret under w goes to zero; the strength is what makes the conclusion immediate and yields a simple explicit convergence rate in terms of $W\,\varepsilon(R)$.*

The optional monotonicity condition formalizes a natural "no backsliding" expectation: if more resources never make performance worse in any context, then the pragmatic score cannot decrease. This is precisely the kind of condition one typically expects when R corresponds to additional computation that a policy can choose to ignore, or to additional verification that can only catch errors.

70.6.6 Do GTGI intelligence measures correlate with compassion or Locations?

Remark 3818 (Conceptual explanation: "no free correlation" vs conditional correlation)**.** *GTGI measures (Defs. 340, 342) quantify expected goal achievement across a specified goal distribution. Compassion (Hyperseed-Concept 81) and Martin-style Locations (discussed in the PNSE/Martin Locations bridge) are* additional *properties of minds/collectives; they are not forced to correlate with GTGI scores unless the chosen goal ecology (or a shared latent driver like resources) couples them. Thus, there are two mathematically clean statements to make: (i) without coupling assumptions,* *there is no necessary relationship; (ii) with monotone coupling through a shared scalar (e.g. resource level R), positive correlation is expected and can be proved as a covariance inequality.*

For avoidance of ambiguity: statement (i) is a claim about logical/axiomatic non-implication *under the GTGI definitions as written (i.e. the GTGI functional depends only on $(\mathcal{E}, \mathcal{G}, \mathcal{T}, \nu, \gamma, \tau)$ and the induced values $V^{\pi}_{\mu,g,T}$), not a sociological claim that compassion and capability never co-occur. Statement (ii) is a claim about* conditional association*: once one conditions on (or parametrizes by) a variable that jointly influences both dimensions (resources, organizational scale, training data, evaluator redundancy, etc.), one can obtain a provable direction of correlation. In particular, the relevant probability space for the covariance is the distribution of the latent driver R (or more generally the joint distribution of drivers), and the sign claim is about how both quantities move along that axis.*

It is also worth distinguishing "correlation across agents in a population" from "coupling inside a single agent's objective." The former is modeled here by sampling R and observing $(\Pi^{\text{eff}}(\pi_R), C(R))$; the latter would require changing the goal distribution γ so that compassion-relevant outcomes are directly reward-relevant. Only the latter produces a direct definitional link; the former produces an empirical/statistical link.

Theorem 259 (No necessary implication: intelligence can be decoupled from compassion/Location)**.** *Fix a GTGI setup $(\mathcal{E}, \mathcal{G}, \mathcal{T}, \nu, \gamma, \tau)$.*

Suppose $\mathcal{G}$ contains a distinguished subset $\mathcal{G}_{\text{comp}}$ of "compassion-relevant" goals, but the goal distribution assigns it zero mass in every environment:

$$\gamma(g, \mu) = 0 \quad \forall \mu \in \mathcal{E},\ \forall g \in \mathcal{G}_{\text{comp}}.$$

Then $\Pi(\pi)$ and $\Pi^{\text{eff}}(\pi)$ are completely insensitive to how π behaves on compassion-relevant goals: if two policies π, π' satisfy $V^{\pi}_{\mu,g,T} = V^{\pi'}_{\mu,g,T}$ for all $g \notin \mathcal{G}_{\text{comp}}$ (and all μ, T), then

$$\Pi(\pi) = \Pi(\pi'), \qquad \Pi^{\text{eff}}(\pi) = \Pi^{\text{eff}}(\pi').$$

Therefore, without an explicit coupling between the GTGI goal ecology and compassion-relevant goals, GTGI intelligence measures do not, in general, constrain compassion. An analogous decoupling holds for "Location"-like state descriptors unless they are explicitly reward-relevant or otherwise coupled to the goal distribution.

Concretely, this theorem formalizes a familiar point: if the evaluation suite never queries (or never assigns any weight to) behaviors that express compassion (or any particular Location signature), then the resulting score cannot rule those behaviors in or out. One can build two systems that are indistinguishable on the support of γ and yet differ arbitrarily on the measure-zero part $\mathcal{G}_{\text{comp}}$; the GTGI functional, being a weighted average over (μ, g, T), cannot detect the difference. This is the mathematical core of the "no free correlation" claim: it is not that compassion is incompatible with intelligence, but that it is not entailed by this *objective unless the objective is designed to touch it.*

Proof. By the hypothesis $\gamma(g, \mu) = 0$ for all $g \in \mathcal{G}_{\text{comp}}$, the sums defining $\Pi(\pi)$ and $\Pi^{\text{eff}}(\pi)$ have no contributions from $\mathcal{G}_{\text{comp}}$. Thus if two policies agree on all contexts with $g \notin \mathcal{G}_{\text{comp}}$, their weighted sums agree, giving equality of Π and Π^{eff}. The same logic applies to any auxiliary property (such as a Location index) that does not appear in the reward/goal definitions and is not otherwise coupled to them.

Equivalently, one may view Π and Π^{eff} as functionals of the restricted collection $\{V^{\pi}_{\mu,g,T} : g \in \text{supp}(\gamma)\}$, where $\text{supp}(\gamma)$ denotes the set of goals given nonzero weight (possibly depending on μ). If $\mathcal{G}_{\text{comp}}$ lies outside this support, then changing π on those goals cannot change the functional value, which is exactly the claimed insensitivity. $\square$

Lemma 180 (Chebyshev covariance inequality for monotone functions of one variable)**.** *Let R be a real-valued random variable. If $f, g : \mathbb{R} \to \mathbb{R}$ are both nondecreasing, then*

$$\operatorname{Cov}(f(R), g(R)) \;\geq\; 0,$$

whenever the covariance exists.

Proof. Let R' be an independent copy of R. A standard identity gives

$$\operatorname{Cov}(f(R), g(R)) = \frac{1}{2}\,\mathbb{E}\big[\big(f(R) - f(R')\big)\big(g(R) - g(R')\big)\big]\,.$$

If f and g are nondecreasing, then $(f(R) - f(R'))(g(R) - g(R')) \geq 0$ pointwise (because the two differences have the same sign). Taking expectation preserves the inequality, hence the covariance is nonnegative.

For intuition: the product above is positive precisely because increasing R tends to increase both $f(R)$ and $g(R)$, so deviations of $f(R)$ above its typical value align with deviations of $g(R)$ above its typical value. The lemma is one of the simplest "positive association" results and is the minimal tool needed for the resource-mediated template below; no joint normality or linearity assumptions are required. □

Theorem 260 (Resource-mediated correlation template for intelligence vs compassion/Location)**.** *Let R be a resource-level random variable (time/compute/energy/coordination), and let π_R denote the policy realized by a resource-indexed mind at resource level R (Hyperseed-Concept 162).*

Assume:

1. *The efficient pragmatic GI score $\Pi^{\text{eff}}(\pi_R)$ is nondecreasing in R.*
2. *There is a scalar "compassion level" $C(R) \in \mathbb{R}$ (e.g. $C(R) = \mathbb{E}[c(x_T)]$ in the basin-stability model, where c is compassion; Hyperseed-Concept 81) that is nondecreasing in R.*

Then $\Pi^{\text{eff}}(\pi_R)$ and $C(R)$ have nonnegative covariance:

$$\operatorname{Cov}\Big(\Pi^{\text{eff}}(\pi_R),\, C(R)\Big) \;\geq\; 0.$$

Likewise, if a Location-like index $\operatorname{Loc}(R)$ (individual or collective; cf. the Martin Locations bridge and Hyperseed-Concept 75) is nondecreasing in R, then

$$\operatorname{Cov}\Big(\Pi^{\text{eff}}(\pi_R),\, \operatorname{Loc}(R)\Big) \;\geq\; 0.$$

Proof. Apply Lemma 180 with $f(R) = \Pi^{\text{eff}}(\pi_R)$ and $g(R) = C(R)$ (or $g(R) = \operatorname{Loc}(R)$).

To be explicit about the probability space: the covariance is taken over draws of R from whatever population or uncertainty distribution is under discussion (e.g. a distribution over organizations, training budgets, or substrate capabilities). The conclusion does not assert that *every* increase in Π^{eff} implies an increase in C for the same agent at fixed R; it asserts that, when both are monotone functions of the same scalar driver, the population-level association induced by varying R must be nonnegative. □

Corollary 86 (A concrete compassion-correlation consequence in the "robust evaluators" model)**.** *In the basin-and-evaluators model of Theorem 252, the probability of large compassion loss over T steps is bounded by an expression that decreases (at least) exponentially with evaluator count $K(R)$,* *and $K(R)$ is nondecreasing in resources R. Thus, any reasonable "compassion stability" score (e.g. $C(R)$ = probability compassion stays above a threshold) is nondecreasing in R.*

Therefore, if in addition $\Pi^{\text{eff}}(\pi_R)$ is nondecreasing in R in the same population of minds, Theorem 260 implies a nonnegative correlation between efficient GTGI intelligence and compassion stability along the resource axis.

This corollary should be read as a template for how "mindplex"-level properties can become statistically coupled: if scaling resources simultaneously (a) improves performance on the evaluated goal ecology (raising Π^{eff}) and (b) increases the redundancy/coverage of whatever internal or external mechanisms enforce compassion (raising C), then one expects a positive association even if the GTGI definition itself never directly references compassion. In other words, the correlation is not a definitional constraint but an emergent consequence of shared scaling drivers.

Proof. Theorem 252 provides an explicit upper bound on failure probability that decreases as $K(R)$ grows, hence $C(R)$ is nondecreasing in R. Now apply Theorem 260.

Note that the argument only needs monotonicity of $K(R)$ and monotonicity of the induced stability score; the detailed functional form of the exponential bound is useful for quantitative rates (e.g. how quickly correlation strengthens as resources scale), but not needed for the qualitative nonnegativity claim. □

Special macros used. None introduced in this insertion. (Only standard LaTeX commands and existing manuscript labels are used.)

A GTGI measures for collective intelligences, mindplexes, and resource-rich minds

This appendix expands the notion of "GTGI measures" as used in the main text to quantify, compare, and bound the growth trajectories of *collective intelligences*, *mindplexes* (interacting ensembles of minds/agents), and *resource-rich minds*. The intent is not to prescribe a single scalar "intelligence number," but to define a family of operational measures that (i) are computable or at least estimable from observable structure and performance, (ii) decompose into interpretable components, and (iii) can be tied to basin-constrained endpoint convergence claims toward C3-like attractors.

A.1 Objects of measurement and minimal notation

Let $\mathcal{M}$ denote a mindplex, modeled as a set of agents $\{a_i\}_{i=1}^n$ together with (a) an interaction topology $G = (V, E)$, (b) a communication protocol Π, and (c) a shared environment $\mathcal{E}$ with task distribution $\mathcal{T}$. Each agent has an internal state $s_i(t)$ and policy π_i; the collective induces a joint policy $\Pi_{\mathcal{M}}$ that maps histories to actions. A *resource-rich mind* is treated as a limiting case in which either $n = 1$ with extremely large internal resources, or $n > 1$ but with abundant external resources available per agent; both cases are represented by an explicit resource vector $R(t)$.

We consider time-indexed measures $I(t)$ and cumulative measures over an interval $[0, T]$. When comparing heterogeneous systems, we distinguish: (i) *capability* (what can be done), (ii) *efficiency* (resources required), and (iii) *robustness* (stability under perturbations).

A.2 A decomposition template for GTGI

A convenient template is to define a GTGI family as a product or sum of interpretable factors,

$$\mathrm{GTGI}(\mathcal{M};\mathcal{T},\mathcal{E},T) \;\equiv\; \int_0^T \underbrace{C(t)}_{\text{capability}} \cdot \underbrace{E(t)}_{\text{efficiency}} \cdot \underbrace{S(t)}_{\text{stability}} \cdot \underbrace{A(t)}_{\text{alignment/goal-coherence}} \, dt, \tag{66}$$

where each factor can itself be multi-dimensional. The integral form emphasizes that growth and performance accrue over time; for settings where only endpoints matter, one may use $\mathrm{GTGI}(T)$ or $\Delta\mathrm{GTGI} = \mathrm{GTGI}(T) - \mathrm{GTGI}(0)$.

To avoid conflating size with competence, factors should be normalized against baseline resources and task difficulty. A common normalization is against a reference policy class Π_0 and a resource baseline R_0:

$$C(t) \;=\; \frac{\mathbb{E}_{\tau\sim\mathcal{T}}\left[U(\Pi_{\mathcal{M}},\tau;\mathcal{E})\right]}{\mathbb{E}_{\tau\sim\mathcal{T}}\left[U(\Pi_0,\tau;\mathcal{E})\right]}, \qquad E(t) \;=\; \frac{\mathrm{Cost}(R_0,t)}{\mathrm{Cost}(R(t),t)}. \tag{67}$$

Here U is a utility or performance functional (success probability, reward, accuracy, or task-specific score), while Cost aggregates compute, energy, time, and scarce materials. This makes explicit that high performance achieved by disproportionate resource expenditure should not trivially dominate.

A.3 Capability components for collective systems

For mindplexes and collective intelligences, capability is often not merely the best single-agent performance but the *reachable set* of coordinated behaviors. Useful capability components include:

Task coverage (breadth). Let $\mathcal{T}$ be partitioned into task families $\{\mathcal{T}_k\}$. Define

$$C_{\mathrm{breadth}}(t) \;=\; \sum_k w_k \, \mathbb{I}(\mathbb{E}_{\tau\sim\mathcal{T}_k}[U(\Pi_{\mathcal{M}},\tau)] \geq \theta_k)\,, \tag{68}$$

where w_k weights importance and θ_k is a competence threshold. This is a coarse but interpretable "coverage" score.

Generalization under distribution shift. If $\mathcal{T}'$ represents shifted tasks, define a robustness-to-shift capability

$$C_{\mathrm{shift}}(t) \;=\; \frac{\mathbb{E}_{\tau\sim\mathcal{T}'}[U(\Pi_{\mathcal{M}},\tau)]}{\mathbb{E}_{\tau\sim\mathcal{T}}[U(\Pi_{\mathcal{M}},\tau)] + \varepsilon}, \tag{69}$$

with $\varepsilon > 0$ preventing degeneracy. This factor is particularly relevant to claims of endpoint convergence, since attractor-like regimes should exhibit stable performance across modest perturbations.

Coordination and compositionality. Let U_{team} be team performance and U_i be individual performance on the same tasks with communication disabled. A simple synergy index is

$$C_{\mathrm{sync}}(t) \;=\; \frac{\mathbb{E}[U_{\mathrm{team}}] - \max_i \mathbb{E}[U_i]}{\max_i \mathbb{E}[U_i] + \varepsilon}, \tag{70}$$

capturing the extent to which the mindplex exceeds its best constituent. Positive C_{sync} suggests nontrivial collective computation rather than mere aggregation.

A.4 Efficiency and resource-rich minds

A "resource-rich mind" is not automatically high-GTGI; it must convert resources into capability efficiently. Let $R(t)$ collect compute (FLOP), energy (J), memory, time, and critical materials. Then a common efficiency term is the inverse of a weighted resource norm:

$$E(t) \;=\; \frac{1}{\alpha_{\mathrm{c}}\,\mathrm{Compute}(t) + \alpha_{\mathrm{e}}\,\mathrm{Energy}(t) + \alpha_{\mathrm{m}}\,\mathrm{Memory}(t) + \alpha_{\mathrm{t}}\,\mathrm{Time}(t)}, \tag{71}$$

with coefficients $\alpha_{\bullet}$ chosen to reflect scarcity or externalities. When comparing systems across scales, one may also report the Pareto frontier of (C, E) rather than collapsing to a single scalar.

To connect efficiency to basin constraints, note that limited resources can enforce confinement to a subset of trajectories; in that sense, $E(t)$ affects not only performance but also which attractor basins are reachable under realistic constraints.

A.5 Stability, resilience, and endpoint convergence

Since the main section emphasizes basin-constrained endpoint convergence toward C3-like attractors, the stability factor $S(t)$ should directly encode resilience and convergence tendencies.

Local stability to perturbations. Let δ perturb either the environment, the interaction graph, or internal states. Define

$$S_{\mathrm{local}}(t) \;=\; \exp\Big(-\beta\, \mathbb{E}_{\delta}\big[\|\Phi_t(\mathcal{M}) - \Phi_t(\mathcal{M}+\delta)\|\big]\Big), \tag{72}$$

where Φ_t is a chosen embedding of system-level behavior (e.g., summaries of policies, performance profiles, or communication statistics), and $\beta > 0$ sets sensitivity. The use of an exponential emphasizes that large deviations should be strongly penalized.

Convergence within a basin. If $\mathcal{B}$ is a basin in a state-space representation, and $d(\cdot,\cdot)$ is a metric on macrostates, define a contraction-like proxy:

$$S_{\mathrm{conv}}(t) \;=\; \mathbb{E}\left[\frac{d(X(t+\Delta), X^{\star})}{d(X(t), X^{\star}) + \varepsilon}\right]^{-1}, \tag{73}$$

where $X(t)$ is a macrostate, $X^{\star}$ is the putative endpoint (attractor representative), and Δ is a fixed horizon. Values > 1 indicate, on average, movement toward the endpoint. This connects directly to "endpoint convergence" language without assuming a perfectly known dynamical system.

A.6 Goal-coherence and alignment within mindplexes

For collective intelligences, a key failure mode is internal conflict or incoherent optimization that dissipates resources. A goal-coherence factor can be defined via agreement of gradients, plans, or revealed preferences.

Let $g_i(t)$ denote an estimated direction of improvement for agent i in a shared objective space (e.g., gradients of predicted task utility under agent actions). Then define

$$A(t) \;=\; \frac{1}{n(n-1)} \sum_{i\neq j} \frac{\langle g_i(t), g_j(t)\rangle}{\|g_i(t)\|\,\|g_j(t)\| + \varepsilon}, \tag{74}$$

so that $A(t) \in [-1, 1]$ measures average directional agreement. Strongly negative values indicate adversarial or misaligned subcomponents; values near 1 indicate coherent collective optimization, a plausible hallmark of C3-like regimes if those regimes emphasize stable, coordinated, and self-consistent behavior.

A.7 A concrete composite score (illustrative)

An illustrative (not mandatory) scalar instantiation is

$$\mathrm{GTGI}_{\mathrm{scalar}} \;=\; \int_0^T \Big(C_{\mathrm{breadth}}(t)\Big)\Big(1 + C_{\mathrm{sync}}(t)\Big)\Big(E(t)\Big)\Big(S_{\mathrm{local}}(t)\, S_{\mathrm{conv}}(t)\Big)\Big(\tfrac{1+A(t)}{2}\Big)\, dt, \tag{75}$$

where each multiplicative term has a clear interpretation: breadth (coverage), synergy (collective advantage), efficiency (resource conversion), stability (robustness and convergence), and coherence (internal alignment). The affine mapping $(1 + A)/2$ ensures nonnegativity while still penalizing incoherence.

A.8 Interpretation with respect to C3-like attractors

Within the paper's framing, C3-like attractors can be treated as regions of macrostate space where the system exhibits: (i) persistent coordination, (ii) high resource-to-capability conversion, (iii) robustness to perturbations, and (iv) stable goal-coherence. Under this interpretation, high GTGI is not merely "more intelligence," but evidence that the system is operating in (or approaching) a regime consistent with such an attractor.

Two clarifications help avoid category errors:

1. A high-capability system with low stability may show strong short-run performance while failing to exhibit endpoint convergence; this should be reflected by low S_{local} or S_{conv}.

2. A resource-rich mind may score highly on raw capability but poorly on efficiency; in basin-constrained settings, this may overstate its relevance if the basin is defined by realistic resource constraints.

A.9 Practical measurement remarks

While the above definitions are abstract, each component can be estimated from experiments or logs: (i) U from benchmark suites or real-world task success, (ii) resource costs from telemetry (energy, compute, latency), (iii) perturbation response from ablations (removing communication edges, injecting noise, varying task distributions), and (iv) goal-coherence from plan similarity, gradient alignment, or preference-model agreement. When full observability is unavailable, lower bounds and confidence intervals should be reported, since GTGI is intended to support rigorous comparative claims rather than rhetorical labeling.

B Hyperseed concept checklist

How to use this appendix

The list below is intended as a completeness checklist. Every term listed here appears in the original Hyperseed ontology document. In the final paper, each item should have a precise definition (or a precise reduction to earlier definitions) and a pointer to the section where it is introduced. In practice, this appendix functions as a "coverage test": if a term is used anywhere in the main text, it should be possible to locate it here and then follow its pointer to the unique place where its meaning is fixed. When revising the main document, it is useful to treat each checklist entry as a contract with the reader: the term should not rely on informal intuition alone, and its dependencies (what it is defined in terms of) should be explicit and already established.

A "precise definition" can take multiple acceptable forms. For primitive notions, it may be an explicit declaration of status (primitive, assumed, or axiomatized) together with the axioms or inference rules that govern it. For derived notions, it should be a reduction to earlier definitions with clear quantifiers, domains, and any side conditions (e.g., well-formedness or admissibility assumptions). For operational terms, it can be a step-by-step construction provided that each step only invokes already-defined operations. In all cases, the reader should be able to determine (i) what objects the term ranges over, (ii) what data are required to compute or instantiate it, and (iii) what properties follow immediately from its definition.

Pointers should be as specific as possible. If a concept is introduced in a dedicated definition environment, the pointer should reference that definition rather than only the surrounding section. If the notion is first introduced informally and later made precise, the pointer should lead to the first place where the fully formal definition appears (and the informal introduction can be cross-referenced from there). When a term is overloaded (same word used for distinct technical meanings), the checklist should enforce disambiguation by requiring separate entries or explicit qualifiers in the names used in the paper.

The checklist is also intended to catch hidden dependencies. If a definition uses an auxiliary notion that is not yet defined elsewhere, that auxiliary notion should be added as its own checklist item (or the definition should be refactored so that all prerequisites are already present). A good workflow is to read each definition and underline every technical noun phrase; each underlined phrase should either be (a) defined earlier, (b) explicitly declared primitive, or (c) replaced by a more elementary phrase that is already grounded.

Finally, this appendix can be used to enforce consistent terminology across drafts. If the ontology document contains synonyms, alternate spellings, or historical names, select one canonical surface form for the paper and record the alternatives as aliases in the definition text (so that search and cross-reference remain robust). If the paper introduces simplified nomenclature for readability, include an explicit equivalence statement so that a reader can map between the simplified term and the ontology term without ambiguity.

C Hyperseed concept glossary

C.1 Core Hyperseed concepts

Concept 1 (Abstract)**.** *Appendix label: concept:Abstract.*
Body definition label: hsdef:Abstract (see § 28.3).

Informal definition. *Something is* abstract *when you treat many different concrete examples as the same kind of thing, by ignoring details that do not matter for your current purpose. For instance, many different handwritten marks can all count as the letter "A," and many different physical triangles can all count as "a triangle." Abstracting is a controlled kind of forgetting: you keep what is shared and drop what is incidental.*

Formal definition. *Let c be a context with entity set X_c. An abstraction of c is a surjection*

$$q : X_c \twoheadrightarrow A$$

to a set A of abstracta. The abstraction collapses distinctions by identifying elements of the same fiber. Equivalently, q determines an equivalence relation

$$x \sim_q y \iff q(x) = q(y).$$

Concept 2 (After)**.** *Appendix label: concept:After.*
Body definition label: hsdef:After (see § 7.1).

Informal definition. After *means "later than". Hyperseed models the basic structure of time using the idea of an ordering: some moments/events come after others. To be consistent, nothing can be after itself, and "after" must not produce cycles like "A after B after C after A".*

Formal definition. *A* strict partial order *on a set T is a binary relation $<$ on T such that:*

1. *(Irreflexive) for all $t \in T$, not $(t < t)$,*

2. *(Transitive) for all $t_1, t_2, t_3 \in T$, if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$.*

We say t_1 is before *t_2 if $t_1 < t_2$, and t_2 is* after *t_1.*

Concept 3 (Aesthetics / Art)**.** *Appendix label: concept:AestheticsArt.*
Body definition label: hsdef:AestheticsArt (see § 383).

Informal definition. *In Hyperseed, something counts as* art *(relative to a group of people) if encountering it tends to reliably trigger an experience of beauty for many of them. The key point is that art is not defined only by the object in isolation: it is defined by the object* and *the way it affects a population of experiencers. Different communities can disagree about what is art, because their typical reactions can differ.*

Formal definition. *Fix a class of experiencers $X \subseteq \mathrm{Mind}$. An art object A (relative to X) is any entity such that, for many $m \in X$, encountering A induces a designated beauty proposition $\mathrm{Beaut}(A)$ in m's valuation. Equivalently, it induces a high rate of beauty episodes in the population X.*

Concept 4 (Alive / Dead)**.** *Appendix label: concept:AliveDead.*
Body definition label: hsdef:AliveDead (see § 41.6).

Informal definition. Alive *and* dead *are treated as time-dependent, evidence-based notions. Roughly: "dead now" is not just "not alive now," but "was alive recently and is not alive now." This matches ordinary thinking (something can be uncertain or borderline), and it also allows conflicting evidence streams (some signs of life, some signs of death).*

Formal definition. *Fix a lookback window length $\tau \in \mathbb{N}$. Define the "was-alive" evidence by*

$$\mathrm{WasAlive}_O(S,t) \;:=\; \bigoplus_{u \in \{t-\tau,\dots,t-1\}} \mathrm{Alive}_O(S,u),$$

where $\oplus$ is the p-bit *join (componentwise max) from Section 3.4. Define*

$$\mathrm{Dead}_O(S,t) \;:=\; \mathrm{WasAlive}_O(S,t) \otimes \neg\, \mathrm{Alive}_O(S,t),$$

using the p-bit *negation and monoidal product $\otimes$ from the core.*

Concept 5 (Algebraically Asymmetric Physical Reality)**.** *Appendix label: concept:AlgebraicallyAsymmetricPhysica*
Body definition label: hsdef:AlgebraicallyAsymmetricPhysicalReality (see § 25.4).

Informal definition. *Hyperseed claims there is a built-in mismatch between how* physics *is structured and how* mental / intersubjective *descriptions usually work. A mental description often keeps only coarse, relational, or pattern-level facts ("these things are connected", "this resembles that") and leaves out many precise details (exact geometry, forces, equations, etc.). Because of that loss of detail, you usually cannot reconstruct the full physical situation from the mental-level description alone: going from "mental" back to "physical" requires adding extra structure and constraints. This one-way loss is the "algebraic asymmetry."*

Formal definition. *Let R_{phys} be a category (or enriched category) of models of physical reality, whose objects carry strong algebraic structure (e.g. smooth manifold, metric space, vector bundle, etc.). Let R_{ment} be a category of models of mental and intersubjective reality (e.g. topological or relational pattern spaces). An* algebraic asymmetry *is the presence of a structure-forgetting functor*

$$U : R_{\text{phys}} \to R_{\text{ment}}$$

that is not (even approximately) invertible on the images relevant to a given community of minds. In this case we say that mental reality is algebraically weaker than physical reality.

Concept 6 (Answers)**.** *Appendix label: concept:Answers.*
Body definition label: hsdef:Answers (see § 292). ***Informal definition.*** *An* answer *is defined by what it does, not by what it looks like. It is anything that helps resolve a question by shrinking the set of reasonable possibilities. A complete answer might identify the exact right choice, but many useful answers are partial: they can rule out options, tell you where to look, or tell you which options are more likely.*

Formal definition. *An answer to a question* (τ, Exp) *is any artifact that reduces uncertainty about* $\mathrm{Inst}(\tau)$*. Concretely, an answer may be:*

- *a particular instantiation* $\iota \in \mathrm{Inst}(\tau)$*;*
- *a constraint* $C \subseteq \mathrm{Inst}(\tau)$ *that prunes candidates;*
- *a distribution (or* p-bit*-valued scoring) over* $\mathrm{Inst}(\tau)$*; or*
- *any representation that yields an entropy reduction or computational simplification relative to baseline inference.*

Concept 7 (Anthropic Principle)**.** *Appendix label: concept:AnthropicPrinciple.*
Body definition label: hsdef:AnthropicPrinciple (see § 25.9).

Informal definition. *The anthropic principle is treated as a plain selection effect: the fact that* you exist as an observer *is information. So when comparing different models of the universe, you should downweight models in which observers like you could not exist, and upweight models where observers are likely. This is not an "explanation" of why the universe is the way it is; it is a rule for how an observer should update beliefs given that the observer exists.*

Formal definition. *Let Θ be a parameter space of universe models (physical constants, laws, initial conditions, etc.). Let $\pi(\theta)$ be a prior distribution over Θ. Let A be an "anthropic" event such as "observers of type O exist." The anthropic posterior is*

$$\pi(\theta \mid A) \propto \pi(\theta)\, P(A \mid \theta).$$

Concept 8 (Archetypes)**.** *Appendix label: concept:Archetypes.*
Body definition label: hsdef:Archetypes (see § 382).

Informal definition. *An* archetype *is a recognizable template that shows up across many different systems (people, stories, cultures, texts, etc.) even though the detailed versions can differ a lot. For example, "the trickster" or "the hero" can appear in many cultures with different names and plots. Hyperseed emphasizes three features: it appears in many places, the instantiations are not mere copies, and the widespread appearance is not trivial (it is "surprising" given how specific the template is).*

Formal definition. *Fix a class of systems (minds, texts, cultures, etc.) indexed by J. An archetype is a template τ together with an instantiation map sending each $j \in J$ to a (possibly empty) set of instantiations* $\mathrm{Inst}_j(\tau)$*, such that:*

1. *τ has high support across J (many j have* $\mathrm{Inst}_j(\tau) \neq \emptyset$*), and*

2. *the typical instantiations are diverse (not near-copies), and*

3. *the support is "surprising" relative to the description complexity of τ (e.g. high occurrence despite nontrivial $\kappa(\tau)$).*

Concept 9 (Association)**.** *Appendix label: concept:Association.*
Body definition label: hsdef:Association (see § 91).

Informal definition. *An* association *score says how strongly two things are linked* in a particular context*. A high association means they tend to show up together, or they share many important properties; a low association means they rarely co-occur and have little in common (for the purposes of that context). The same pair can have different association scores in different contexts, depending on what properties the context cares about.*

Formal definition. *A (static) association score in context C is a symmetric map*

$$\mathrm{assoc}_C : E_C \times E_C \to [0,1]$$

intended to measure "how much A and B tend to occur together" or "how many properties/patterns they share."

(One canonical implementation used later in the text is via overlap between property-sets; see Definition 100.)

Concept 10 (Attention / Attentional Focus)**.** *Appendix label: concept:AttentionalFocus.*
Body definition label: hsdef:AttentionalFocus (see § 37.3).

Informal definition. *Attention is modeled as a limited mental budget spread over many possible things you could be tracking. You have* attentional focus *when most of that budget is concentrated on only a few items, instead of being thinly spread over many. The parameters (θ, k) let you say what counts as "most" (θ) and "a few" (k).*

Formal definition. *Fix parameters $\theta \in (0,1)$ (mass threshold) and $k \in \mathbb{N}$ (size bound). We say S has an attentional focus at time t (at level (θ, k)) if there exists a subset $F_t \subseteq P$ with $|F_t| \leq k$ such that*

$$\alpha_t(F_t) := \sum_{p \in F_t} \alpha_t(p) \geq \theta.$$

Equivalently, if $\alpha_t^{\downarrow}$ is the nonincreasing rearrangement of the weights $(\alpha_t(p))_{p \in P}$, then focus holds iff

$$\sum_{j=1}^{k} \alpha_t^{\downarrow}(j) \geq \theta.$$

Concept 11 (Autocatalytic Sets)**.** *Appendix label: concept:AutocatalyticSets.*
Body definition label: hsdef:AutocatalyticSets (see § 237). ***Informal definition.*** *An autocatalytic set is a collection of processes (often chemical reactions) that "help make themselves": each process in the collection is sped up (catalyzed) by something produced within the same collection. Because of this mutual support, once the set is present it can keep running and regenerating as long as it keeps getting basic inputs. In particular, the point is not that the set produces* everything *it needs from nothing, but that the internal catalytic support closes the "who speeds up whom" loop sufficiently to sustain ongoing activity given an external supply of low-level feedstock.*

Formal definition. *Fix a threshold $\theta \in (0,1]$. Let R be a finite set of reactions and let $C = (c_{ij})_{i,j \in R}$ be a catalysis matrix with $c_{ij} \in [0,1]$ interpreted as the strength of evidence that* *reaction i catalyzes reaction j. Define the thresholded reaction graph $G_\theta(R, C)$ to be the directed graph with vertex set R and an edge $i \to j$ iff $c_{ij} \geq \theta$. A subset $S \subseteq R$ is a θ-autocatalytic set if every vertex of the induced subgraph $G_\theta(R, C)[S]$ has in-degree at least one; equivalently, for every $j \in S$ there exists some $i \in S$ with $c_{ij} \geq \theta$. This graph-theoretic condition captures the minimal "no reaction is left uncatalyzed by the set" property: each $j \in S$ has at least one* internal *catalytic support edge at the chosen evidential cutoff. Note that the definition does not require a single global catalyst or a single directed cycle containing all vertices; rather, it allows a network of possibly overlapping cycles and catalytic chains, as long as every reaction in S has at least one incoming edge from within S at level θ. Varying θ tunes strictness: larger θ demands stronger evidence for catalysis and typically shrinks the sets that qualify, while smaller θ includes weaker or more speculative catalytic links.*

Concept 12 (Becoming)**.** *Appendix label: concept:Becoming.*
Body definition label: hsdef:Becoming (see § 62).

Informal definition. *We say that x* becomes *y when, over time, the system transitions from clearly being (or behaving like) x to clearly being (or behaving like) y. Crucially, the transition is not instant: there is a "boundary period" where the identity is genuinely unclear, so it can be reasonable (given the available evidence) to treat the two as both distinct and not-distinct. Away from that boundary, the two are more cleanly separated again. Intuitively, "becoming" is meant to model real processes like phase transitions, developmental shifts, or conceptual reclassifications, where any attempt to draw a razor-sharp instant of change would misrepresent the situation the evidence actually supports.*

Formal definition. *Fix a proto-time $(T, <)$ and a time-indexed distinction relation* Dist*. We say $x \in E$ becomes $y \in E$ (along $(T, <)$) if there exist $t_1 < t_2$ and an interval (or neighborhood) $U \subseteq T$ with $t_1 \leq u \leq t_2$ for all $u \in U$ such that:*

1. (Temporal succession) *x is salient/present up to (near) t_1 and y is salient/present from (near) t_2 onward.*
2. (Boundary non-duality) *for all $u \in U$ near the transition, x and y display non-dual variety at u.*
3. (Outside the boundary) *away from U, x and y are (comparatively) more distinguishable (e.g. $\mathrm{Dist}(u; x, y)$ is not strongly paraconsistent).*

Here the proto-time $(T, <)$ supplies only enough structure to talk about "earlier" and "later" without presupposing a fully classical timeline, while Dist *packages whatever evidential or operational criterion is being used to tell x from y at each time. Condition (2) is the key "fuzzing" requirement: in the boundary region U, the available evidence can support both the claim that x and y are distinct and the claim that they are not-distinct, in the sense captured by the non-dual/paraconsistent behavior of* Dist*; by contrast, condition (3) requires that this is a localized phenomenon rather than a permanent inability to distinguish.*

Concept 13 (Before)**.** *Appendix label: concept:Before.*
Body definition label: hsdef:Before (see § 56).

Informal definition. *In everyday thinking, "before" is a simple yes/no idea: either one event happened earlier than another, or it did not. In messy real situations, evidence can be incomplete or even conflicting, so we may want a more careful notion. Here, "before" is represented by two numbers: one measuring how much evidence supports "t_1 happened before t_2", and one measuring how much evidence supports the opposite (or otherwise counts against that ordering). This lets* *the model record uncertainty and disagreement instead of forcing an early all-or-nothing decision. For example, if two sensors disagree about a timestamp ordering, the formalism can represent "substantial support" and "substantial counter-support" simultaneously, rather than averaging them away into a single misleading probability.*

Formal definition. *Let $V = [0,1]^2$ be the p-bit quantale from Section 3.4. A p-bit-valued temporal evidence relation is a map*

$$B : T \times T \to V, \qquad B(t_1, t_2) = (B^+(t_1, t_2), B^-(t_1, t_2)).$$

Here $B^+(t_1, t_2)$ is positive evidence for $t_1 < t_2$, and $B^-(t_1, t_2)$ is negative evidence. Interpreting B as evidence rather than truth conditions allows regimes where both $B^+(t_1, t_2)$ and $B^-(t_1, t_2)$ are high (conflict), both are low (ignorance), or one dominates the other (comparatively well-supported ordering), which is useful when later constructions need to propagate or aggregate such ordering information without prematurely forcing classical comparability.

Concept 14 (Belief System)**.** *Appendix label: concept:BeliefSystem.*
Body definition label: hsdef:BeliefSystem (see § 304).

Informal definition. *A belief system is the connected set of beliefs and reasoning habits that a person or community uses to make sense of the world (Hyperseed-Concept 14). It is not just a pile of separate opinions: the beliefs support each other, explain each other, and guide what counts as evidence. A belief system is called* productive *when it has a stable "core" that does not constantly fall apart, but it is also capable of learning and creating new ideas and questions instead of getting stuck. Equivalently, productiveness is intended to rule out two failure modes: brittleness (where small evidence shocks shatter coherence) and stagnation (where coherence is maintained only by suppressing novelty).*

Formal definition. *A belief system (for a community M) is a pair (B, R) where $B \subseteq L$ is a focal belief set and R is a rule set, such that $\mathrm{Coh}_R(B; \beta_M)$ exceeds a chosen threshold. A belief system is productive if, in addition:*

(a) (Individuation) *there exists a "core" subset $B_{\text{core}} \subseteq B$ that remains stable under typical evidence updates (high persistence of key commitments), and*

(b) (Self-transcendence) *the system regularly generates new high-SAI beliefs or new questions whose answers lead to novel patterns, without collapsing coherence.*

The thresholded coherence requirement makes explicit that a belief system is being treated as a structured object relative to an inference practice R and an aggregated community stance β_M, not merely as a set of sentences: both the rules and the community weighting can change what counts as "coherent enough". Condition (a) can be read as a robustness constraint (a persistent identity through time), while condition (b) is a generativity constraint (the ability to expand the model space); together they formalize the idea that a productive system is neither purely conservative nor purely chaotic.

Concept 15 (Beliefs)**.** *Appendix label: concept:Beliefs.*
Body definition label: hsdef:Beliefs (see § 301).

Informal definition. *A belief is what an agent takes to be true (or likely true), possibly to a degree rather than perfectly. When multiple agents form a community, we can also talk about the community's overall stance by combining individual beliefs. The key idea is to weight each agent's contribution by something like trust, authority, or attention, and then merge the weighted contributions. Allowing weights with a negative component lets the model represent not only trust* *but also distrust or adversarial influence in a graded way. In other words, aggregation is treated as an evidence-combination problem: who says φ matters, how strongly they say it matters, and whether they should be treated as supportive or oppositional are all represented directly in the algebra of weights.*

Formal definition. *Let M be a finite set of agents, each with a belief state $\beta_i : L \to V$. Define the community aggregate belief state by*

$$\beta_M(\varphi) := \bigoplus_{i \in M} w_i \otimes \beta_i(\varphi),$$

where weights $w_i \in V$ model trust/authority/attention and the operations are those of the p-bit quantale. Operationally, $\otimes$ captures how an agent's credibility modulates (scales, filters, or otherwise transforms) the evidential content of their reported belief, while $\bigoplus$ aggregates the resulting contributions across agents in a way that can preserve both positive and negative evidence components. Allowing w_i to have a nontrivial negative coordinate permits explicit representation of distrust: an agent can contribute not only by supporting φ but also by supplying evidence against *treating their assertions as reliable indicators, which can matter in adversarial or strategically communicative settings.*

Concept 16 (Big Bounce)**.** *Appendix label: concept:BigBounce.*
Body definition label: hsdef:BigBounce (see § 46.7).

Informal definition. *"Big Bounce" is a speculative cosmology idea (Hyperseed-Concept 16): instead of the universe beginning once and for all in a single Big Bang, the universe might go through cycles where a contraction phase (sometimes called a "Big Crunch") is followed by a "bounce" into a new expansion. In Hyperseed this phrase is used cautiously, mainly as a hypothesis or metaphor for very large-scale cycles and for the possibility that some patterns could persist or reappear across such a transition.*

Concept 17 (Blend)**.** *Appendix label: concept:Blend.*
Body definition label: hsdef:Blend (see § 9.6).

Informal definition. *C is a blend of A and B when C is "mix-like" in the sense that many of the important properties of C are also properties of A or of B. This does not mean C is exactly half-and-half; it means a substantial portion of what characterizes C can be traced to one source or the other.*

Formal definition. *Fix C and choose a threshold $\theta \in (0, 1]$. Define the property overlap mass of C with (A, B) as*

$$\mathrm{ov}_C(C; A, B) := \sum_{P \in P_C} \min\Big\{\mathrm{Prop}_C(C)(P),\ \max\big(\mathrm{Prop}_C(A)(P), \mathrm{Prop}_C(B)(P)\big)\Big\}.$$

Define the blend score

$$\mathrm{blendscore}_C(C; A, B) := \begin{cases} \mathrm{ov}_C(C; A, B)/\|\mathrm{Prop}_C(C)\|_1, & \|\mathrm{Prop}_C(C)\|_1 > 0, \\ 0, & \|\mathrm{Prop}_C(C)\|_1 = 0. \end{cases}$$

We say C is a blend of A and B (in context C) if $\mathrm{blendscore}_C(C; A, B) \geq \theta$*.*

Concept 18 (Biological)**.** *Appendix label: concept:Biological.*
Body definition label: hsdef:Biological (see § 286).

Informal definition. *A lifeform is called* biological *if, at the level we are describing it, it is mainly built from organic chemistry (like cells and biochemistry). A lifeform can also be called* synthetic *if it was primarily designed and assembled intentionally by an intelligent engineering process, and* evolved *if it arose through unguided evolutionary and self-organization processes. These labels can overlap (for example, a genetically engineered organism is both biological and engineered), so the framework allows them to be multi-valued or p-bit-valued when needed.*

Formal definition. *A lifeform S is:*

- *biological if its dominant implementation substrate is organic chemistry (at the relevant descriptive level);*
- *synthetic if its design/assembly was primarily guided by an intelligent engineering process (as opposed to arising via unguided population-level dynamics);*
- *evolved if it arose via evolutionary and self-organization processes without intentional design.*

These labels are not mutually exclusive at all descriptive levels (e.g. bio-engineered organisms), so we allow them to be multi-valued or p-bit-valued when needed.

Concept 19 (Biological Lifecycle)**.** *Appendix label: concept:BiologicalLifecycle.*
Body definition label: hsdef:BiologicalLifecycle (see § 286).

Informal definition. *A biological lifecycle is the sequence of typical stages a biological organism goes through as it develops and ages, such as infant, young, mature, old, and elderly, together with the typical transitions between these stages. The point of the concept is to talk about "where in its life" an organism is, not just what it is made of.*

Concept 20 (Cause)**.** *Appendix label: concept:Cause.*
Body definition label: hsdef:Cause (see § 36.5).

Informal definition. *In this framework, "A causes B" is a context-relative claim: it depends on what the observer counts as an event, how time is segmented, and what time-scale is being used. Informally, A causes B when A tends to happen before B, and when information about A makes B more predictable (for example, using co-occurrence statistics such as $\mathrm{Cond}_{O,\Delta}$ and lag summaries like $\mathrm{lag}(A, B)$). Because the underlying "before/after" structure is treated as observer-relative, causal indicators are also understood as relative to O (and to the chosen window Δ).*

Formal definition. *Fix a lag window Δ and a mixing parameter $\lambda \in [0, 1]$. Define the causal implication strength*

$$\mathrm{CImp}_{O,\Delta}(A, B) := \Big(\lambda\, \sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A, B)) + (1 - \lambda)\, \sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A, B))\Big) \cdot \mathbf{1}[A \prec B],$$

where $\mathbf{1}[A \prec B]$ is an indicator that A typically precedes B in the relevant context, and $\sigma : [-1, 1] \to [0, 1]$ is the affine rescaling $\sigma(x) = (1 + x)/2$. Define also the strong (product) variant

$$\mathrm{CImp}^{\times}_{O,\Delta}(A, B) := \sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A, B))\ \sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A, B))\ \mathbf{1}[A \prec B].$$

Concept 21 (Categorical Imperative)**.** *Appendix label:* `concept:CategoricalImperative`*.*
Body definition label: `hsdef:CategoricalImperative` *(see § ??).* ***Informal definition.*** *The categorical imperative is a way to test a proposed rule for action (a "maxim") by asking: "What would happen if everyone followed this rule in relevantly similar situations?" If the world that results from universal adoption of the rule tends to be acceptable along key ethical dimensions (for example compassion, honesty, nonviolence), then the maxim passes the test; if universal adoption would predictably lead to unacceptable outcomes or a self-defeating situation, the maxim fails. In*

this document, the key idea is to make the test operational: it is evaluated using an observer's world-model and an explicit set of ethical dimensions, rather than being treated as a purely verbal slogan.

Formal definition. *Fix a set $D_{\text{eth}} \subseteq D$ of ethical dimensions (typically including compassion and possibly truth, nonviolence, etc.). Define the universalized ethical evaluation*

$$E_O^{\text{univ}}(m; d) := \mathbb{E}_{\omega \sim P_O(\cdot|\text{Univ}(m))}[\, E_O(\omega; d)\,]\,,$$

where $E_O(\omega; d)$ is the p-bit *evidence that the realized history ω promotes/opposes d (obtained by aggregating stepwise evidence across the history using $\oplus, \otimes$). We say m passes a threshold categorical imperative if*

$$J\big(E_O^{\text{univ}}(m; d)\big) \geq \tau_d \quad \textit{and} \quad W\big(E_O^{\text{univ}}(m; d)\big) \leq \eta_d \qquad \forall d \in D_{\text{eth}},$$

for fixed thresholds $\tau_d, \eta_d \in [0, 1]$.

Concept 22 (Closing)**.** *Appendix label:* `concept:Closing`*.*
Body definition label: `hsdef:Closing` *(see § 30.2).*

Informal definition. *A closing is a deliberate move toward* making finer distinctions. *You stop treating some pairs of situations (or objects, or states) as "basically the same" and start treating them as importantly different. In other words, you tighten your indistinction policy: you collapse fewer differences. Closing is the opposite direction of an opening (Hyperseed-Concept 125), which relaxes distinctions by collapsing more pairs.*

Formal definition. *Given two indistinction policies $H, H' \in H_C$:*

- *The transition $H \to H'$ is an* opening *(relaxation of distinctions) if $H \subseteq H'$.*
- *The transition $H \to H'$ is a* closing *(tightening of distinctions) if $H' \subseteq H$.*

Concept 23 (Cognitive Synergy)**.** *Appendix label:* `concept:CognitiveSynergy`*.*
Body definition label: `hsdef:CognitiveSynergy` *(see § 37.7).*

Informal definition. *Cognitive synergy means that two different kinds of knowledge (or two different reasoning methods) help each other in a way that makes the overall explanation* simpler *than what you could get by using each one separately and then combining the results. Put plainly: when you let the two knowledge sources interact, you can reach the same level of explanatory adequacy with fewer "extra assumptions" or less complicated story-telling.*

Formal definition. *Let T_1, T_2 be two disjoint sets of knowledge types and fix $\tau \in (0, 1]$. We say there is cognitive synergy at level τ between T_1 and T_2 if*

$$\text{Con}^*(T_1 \cup T_2; \tau) < \text{Con}^*(T_1; \tau) + \text{Con}^*(T_2; \tau).$$

The synergy gain is the positive part

$$\Sigma(T_1, T_2; \tau) := \left(\text{Con}^*(T_1; \tau) + \text{Con}^*(T_2; \tau) - \text{Con}^*(T_1 \cup T_2; \tau)\right)_+.$$

Concept 24 (Colleague)**.** *Appendix label:* `concept:Colleague`*.*
Body definition label: `hsdef:Colleague` *(see § 342).*

Informal definition. *Two people are colleagues when they are working together on the same ongoing project and coordinating with each other as (roughly) peers. They share information, divide up tasks, and repeatedly coordinate their actions toward a shared goal. This is different from a boss/worker relationship, where one person primarily gives commands and the other primarily follows them in exchange for resources.*

Formal definition. *Fix an interval/episode I and agents i, j. We say i and j are* colleagues *on I if there exists a collaborative project during I, meaning a sustained multi-agent pattern of activity aimed at a shared goal (cf. the goal/value machinery of Section 18), such that:*

- *i and j are both recurrent participants in the project activity pattern on I; and*
- *there are repeated coordination acts between them on I (communication and/or artifact-mediated); and*
- *intermediate results are shared between them (messages, plans, partial work-products, or other artifacts).*

Concept 25 (Collective Consciousness Location)**.** *Appendix label:* `concept:CollectiveConsciousnessLocation.`
Body definition label: `hsdef:CollectiveConsciousnessLocation` *(see § ??).*

Informal definition. *A collective consciousness location is a rough "where are we, as a group?" description of a collective mind (or institution, or community) at a time. The idea is that groups can be in different qualitative regimes: highly coordinated and integrated, fragmented into factions, stuck spending most of their energy on internal story/identity maintenance, etc. The location labels (often written $C_0, C_1, C_2, C_3, \ldots$) are meant to summarize these regimes using measurable signals like how much members disagree about what to do (* discord*), and how much of the group's limited resources are spent maintaining narratives/identities (* narrative capture*), along with whatever integration/coherence measures are appropriate for the model.* ***Formal definition.*** *Let $y_i(t)$ denote the policy/proposal signal produced by agent i at time t, and let D_O be a (model-dependent) distance on the space of such signals. Define the (time-t) discord as*

$$D_O(t) := \mathrm{Var}\big(\{y_i(t)\}\big),$$

for some variance functional induced by D_O. Define narrative capture as a scalar $N_O(t) \in [0, 1]$, interpreted as the fraction of the collective resource budget devoted to narrative/identity-maintenance patterns. A collective location C_k (e.g. C_0, C_1, C_2, C_3) is then a qualitative regime describable via thresholds on (at least) discord $D_O(t)$, narrative capture $N_O(t)$, and an (optional/model-dependent) integration/coherence score.

Intuitively, $D_O(t)$ is meant to be representation-relative: changing the ontology (and thus what counts as a "policy/proposal signal") can change which disagreements are visible and how large they appear. The variance functional $\mathrm{Var}(\cdot)$ *should be understood broadly: in Euclidean cases it may reduce to an ordinary sample variance, while in non-Euclidean signal spaces it may be a Fréchet-type variance induced by D_O. In applications, one often distinguishes between within-group discord (variance among a subset of agents) and global discord (variance across all agents), but the definition above is the global time-slice quantity.*

Likewise, $N_O(t)$ is intended to track opportunity cost: resources committed to narrative stabilization are, by construction, resources not directly spent on object-level planning, execution, or epistemic improvement. In a concrete model, $N_O(t)$ may be estimated from observable proxies (e.g. message volume devoted to identity signaling, time allocation to reputational contests, or budget share devoted to legitimacy/branding). The (optional) integration/coherence score is included to allow regimes that have low discord yet still fail to function as an integrated unit (for example, due to brittle coordination channels, incompatible interfaces, or unresolved delegation boundaries).

Concept 26 (Combination)**.** *Appendix label:* `concept:Combination`*.*
Body definition label: `hsdef:Combination` *(see § 111).*

Informal definition. *A combination is just a rule for "putting two things together to get a new thing." Examples include concatenating two strings, composing two actions into a longer action, or combining two subsystems into a larger system. In many settings, not every pair can be combined, so the rule may be only partially defined.*

Formal definition. *A* combination *on a set S is a partial binary operation*

$$* : S \times S \rightharpoonup S.$$

The "partial" arrow $\rightharpoonup$ emphasizes that some pairs $(x, y) \in S \times S$ may have no defined product $x * y$ (e.g. composing two actions that do not match at their interfaces, or combining two modules with incompatible ports). Unless stated otherwise, no algebraic laws are assumed: $*$ need not be associative, commutative, or have an identity. When such laws do hold, they can be added as extra structure (for instance, an associative everywhere-defined $*$ yields an ordinary semigroup operation).

Concept 27 (Combination System)**.** *Appendix label:* `concept:CombinationSystem`*.*
Body definition label: `hsdef:CombinationSystem` *(see § 111).*

Informal definition. *A combination system is the package: a collection of things together with a specified way of combining them. It tells you what objects you are allowed to talk about and what it means (when it is allowed) to combine two of them into one.*

Formal definition. *A* combination system *is a pair $(S, *)$ where S is a set and $* : S \times S \rightharpoonup S$ is a partial binary operation.*

The point of bundling $(S, *)$ is that the meaning of "compositionality" depends on the chosen universe of objects. For example, if S is a set of programs and $*$ is sequential composition, then definability of $x * y$ may encode typing constraints (output type of x matches input type of y). If S is a set of subsystems and $*$ is physical assembly, definability may encode interface compatibility and resource bounds. In Hyperseed-style uses, $(S, *)$ often serves as the target of a representation map: raw processes are mapped into symbols in S, and $*$ specifies which symbolic compositions are considered meaningful.

Concept 28 (Combinational Dynamical Causal Model)**.** *Appendix label:* `concept:CombinationalDynamicalCaus`
Body definition label: `hsdef:CombinationalDynamicalCausalModel` *(see § 114).*

Informal definition. *This is a way of scoring causal influence when outcomes come from interactions. The model assigns a number saying how much "the interaction of a and b caused c." Crucially, the score is required to respect the representational choices you have made: if your representation treats two processes as the same symbol, then the causal model is not allowed to secretly distinguish them. So causality is computed relative to a chosen representational alphabet (and thus relative to an ontology).*

Formal definition. *Let $(D, **)$ be a dynamical process domain. A combinational dynamical causal model is a function*

$$\Gamma : D \times D \times D \to [0, 1],$$

where $\Gamma(a, b, c)$ is interpreted as the degree to which "interaction of a and b caused c." Given a combination-preserving representational mapping $f : D \to S$ into a symbol system $(S, *)$, we say Γ is valid relative to f *if*

$$f(a) = f(a'),\ f(b) = f(b'),\ f(c) = f(c') \implies \Gamma(a, b, c) = \Gamma(a', b', c')$$

for all $a, b, c, a', b', c' \in D$.

The validity condition can be read as an invariance (or quotient) requirement: Γ *must factor through the equivalence relation induced by* f*. Equivalently,* Γ *may depend only on the symbolic triple* $(f(a), f(b), f(c))$ *and not on any hidden micro-details of* a, b, c *that are erased by the representation. This captures the idea that causal attributions are ontology-relative: choosing a coarser* f *forces a coarser causal model, while choosing a finer* f *permits (but does not guarantee) more discriminating causal scores.*

The role of "combination-preserving" is to ensure that the interaction notion in D *aligns with the symbolic combination in* S*, so that when the domain says "*a *interacts with* b*" the representation can express that interaction as a well-formed composition. In many concrete instantiations,* Γ *may be derived from counterfactual comparisons, perturbation experiments, information-theoretic influence measures, or learned attribution models; the definition above isolates only the structural constraint needed for representation consistency.*

Concept 29 (Combinatorial-Categorical Pattern)**.** *Appendix label:* `concept:CombinatorialCategoricalPattern` *Body definition label:* `hsdef:CombinatorialCategoricalPattern` *(see § 110).*

Informal definition. *A combinatorial-categorical pattern is a pattern you find after you first convert a structure into a new "feature space" using a categorical construction. The point is to get a clean docking point between (1) categorical ways of building derived structures (which can expose important relationships) and (2) pattern mining on combinatorial/labeled objects. You first use the categorical machinery to produce a derived labeled structure, and then you search for compressive regularities (patterns) in that derived object.*

Formal definition. *Fix a context* C *and assume a pattern mining scheme has been defined on labeled structures. A pattern candidate* P *is a combinatorial-categorical pattern in* S *relative to* T *and* m_0 *if* P *is a (standard) pattern in the derived structure* $\mathrm{sub}(T, S, m_0)$*.*

The derived object $\mathrm{sub}(T, S, m_0)$ *should be thought of as a categorical "view" of the original data, constructed so that certain morphisms, limits/colimits, slices, or other categorical operations become explicit as combinatorial labels. The purpose of this two-step definition is modularity: any improvements in the categorical construction (better derived representation) or in the pattern mining scheme (better search and scoring of regularities) can be swapped in without changing the surrounding conceptual interface.*

Concretely, a typical workflow is: start with a structure S *(e.g. a labeled graph, a process network, or a database instance), apply a categorical recipe determined by* T *and* m_0 *(e.g. form a subobject classifier-like encoding, compute a pullback-derived relation, or construct a fibered representation), then mine frequent or compressive substructures in the resulting labeled object. Patterns found at this level can correspond to non-obvious regularities in the original structure because the categorical step can expose invariants or interaction motifs that are not syntactically adjacent in the raw representation.*

Concept 30 (Computers and Programs)**.** *Appendix label:* `concept:ComputersAndPrograms`*.* *Body definition label:* `hsdef:ComputersAndPrograms` *(see § 65).*

Informal definition. *A computer is a machine that can be cheaply reconfigured to behave like many different other machines. A program is the set of signals (instructions) you give it to specify which behavior it should emulate. The key idea is not "electronic" versus "mechanical," but flexible emulation under modest reconfiguration effort.*

Formal definition. *A computer is a machine* C *equipped with:*

(a) a set (or category) of admissible programs Syn_C *(the program signals the user can provide); and*

(b) an execution mechanism that, for many $p \in \mathrm{Syn}_C$, yields a physical realization of $[\![p]\!]$ (Definition 360 below);

such that the family $\{[\![p]\!] : p \in \mathrm{Syn}_C\}$ spans (approximately) a large variety of machine behaviors within the relevant engineering domain, and reconfiguration (from p to p') has modest effort cost for the relevant user class.

The phrase "set (or category)" is meant to allow both unstructured program collections and richer situations where programs have compositional structure (e.g. typing, compilation stages, or program transformations as morphisms). The approximation qualifier in "spans (approximately)" acknowledges practical limits: real devices only emulate behaviors within some tolerance, resource envelope, and timescale. The "relevant user class" clause makes the notion explicitly socio-technical: a system may be a computer for one class of users (who can reconfigure it easily) but not for another (for whom reconfiguration is effectively impossible due to tooling, access control, or required expertise).

In this framing, a "program" is primarily an externally supplied control signal drawn from Syn_C, and $[\![p]\!]$ denotes the behavior (or behavior-specification) realized when the machine is run under that control. This definition is intended to encompass digital computers, reconfigurable hardware, programmable biological or chemical systems (where Syn_C may be a set of intervention protocols), and mechanical computers, provided the reconfiguration cost remains modest relative to the intended domain of use.

Concept 31 (Compassion)**.** *Appendix label:* `concept:Compassion`*.*
Body definition label: `hsdef:Compassion` *(see § 18.3 and § 18.4.1).*

Informal definition. *Compassion means that you treat other beings' suffering and well-being as something that matters in your decisions. A compassionate choice is one that tends to help others (reduce their suffering and/or increase their happiness), even if it does not directly benefit you.*

Formal definition. *Let A be a set of agents and fix an acting agent $a \in A$. Assume that for each $b \in A$ we have p-bit (outcome-evidence) evaluations*

$$JO_b(s,a) \in V \quad \textit{and} \quad WO_b(s,a) \in V$$

representing (respectively) evidence for joy and woe outcomes for agent b when the acting agent takes action a in situation/state s. Define the aggregated other-joy and other-woe fields by

$$EO_{\to j}(s,a;\mathrm{joy}) := \bigoplus_{b \in A \setminus \{a\}} JO_b(s,a), \qquad EO_{\to w}(s,a;\mathrm{woe}) := \bigoplus_{b \in A \setminus \{a\}} WO_b(s,a).$$

Define the compassion evaluation field by combining "others' joy" with "not others' woe":

$$EO(s,a;\mathrm{compassion}) \ := \ \frac{1}{2}\, EO_{\to j}(s,a;\mathrm{joy}) \ \otimes \ \frac{1}{2}\, \neg EO_{\to w}(s,a;\mathrm{woe}),$$

where $\neg$ is p-bit negation (swapping positive/negative evidence components) and $\oplus, \otimes$ are the chosen aggregators in the core quantale.

Concept 32 (Compositional Simplicity)**.** *Appendix label:* `concept:CompositionalSimplicity`*.*
Body definition label: `hsdef:CompositionalSimplicity` *(see § 30.5).*

Informal definition. *An object is compositionally simple if you can build it out of a small number of parts, where each part is itself simple, and the way the parts fit together is not complicated.*

In other words: the whole looks complicated only until you discover the right "break it into pieces" description, after which it becomes simple again.

Formal definition. *Fix a combination operation $*$ on entities (Section 9) and a base simplicity score $s_C : X \to [0, \infty]$ (relative to context C). Define the compositional simplicity score*

$$s_C^*(x) \;:=\; \inf\left\{\sum_{i=1}^{n} s_C(x_i) \;:\; x = x_1 * x_2 * \cdots * x_n\right\}.$$

We call x compositionally simple *(in context C) if $s_C^*(x)$ is small and the infimum is achieved (or well-approximated) by a decomposition in which each component x_i is itself simple according to s_C.*

Concept 33 (Concrete)**.** *Appendix label:* `concept:Concrete`.
Body definition label: `hsdef:Concrete` *(see § 28.3).*

Informal definition. *Something is concrete (in a given context) if it has enough specific detail that you cannot harmlessly lump it together with lots of other things. If you try to compress or "abstract away" too much, you quickly lose important distinctions that matter for what you are doing. Concrete things resist being treated as interchangeable.*

Formal definition. *Fix a context C and a family of concreteness/abstraction constraints K_c (Hyperseed-Concept 124). Let X_C be the entity-domain as represented in context C. We say that $x \in X_C$ is C-*concrete *if, for every admissible abstraction (quotienting) map*

$$q : X_C \twoheadrightarrow A$$

that respects the constraints K_c, the fiber $q^{-1}(q(x))$ is small (i.e., x is not merged with many other distinct entities without violating K_c). Informally: no K_c-admissible abstraction can identify x with a large equivalence class of other items.

Concept 34 (Conscious / Unconscious)**.** *Appendix label:* `concept:ConsciousUnconscious`.
Body definition label: `hsdef:ConsciousUnconscious` *(see § 695).*

Informal definition. *A thought, perception, or feeling is* conscious *when it is in your mental spotlight: you can notice it, report it, and it can influence many other things you do (planning, speech, deliberate action). Something is* unconscious *when it is still being processed, but only in the background: it may guide you indirectly (habits, reflexes, priming) without becoming a reportable, globally shared "workspace" content.*

Formal definition. *Let L_{Ob} be the observation language and let an evidence state be a map $E : L_{\mathrm{Ob}} \to V$. Let W be the workspace operator (Definition 228). We say that E is* workspace-stable *if*

$$W(E) = E$$

(Definition 229). We say that E is a conscious evidence state *if it is workspace-stable and reachable from module integration:*

$$E = W(\Gamma_a(E^\bullet))$$

for some module-level evidence collection $E^\bullet$ and attention profile a (Definition 229). Given a threshold θ, define the conscious proposition set *of E by*

$$\mathrm{Con}_\theta(E) \;:=\; \{\varphi \in L_{\mathrm{Ob}} : E^+(\varphi) \geq \theta\},$$

where $E^+(\varphi)$ is the positive-evidence component of the p-bit $E(\varphi)$. A proposition φ is called conscious at E *if $\varphi \in \mathrm{Con}_\theta(E)$, and* unconscious at E *otherwise.*

Concept 35 (Consciousness). *Appendix label:* `concept:Consciousness`*.*
Body definition label: `hsdef:Consciousness` *(see § 239).*

Informal definition. *Consciousness is the overall condition of having an integrated "point of view" inside a mind: some information is not just processed locally, but becomes globally available so that many different parts of the mind can use it. This is what makes a mind able to have a unified, reportable experience that can guide deliberate reasoning and action, rather than just running isolated background routines.* ***Formal definition.*** *Using the workspace operator W and module-integration maps Γ_a (Definitions 228–229), define the set of reachable workspace-fixed points*

$$\mathrm{ConscStates}(M) \;:=\; \left\{ E \in V^{L_{\mathrm{Ob}}} \;:\; E = W(E) \;\; \textit{and} \;\; \exists (E^{\bullet}, a) \textit{ with } E = W(\Gamma_a(E^{\bullet})) \right\}.$$

We say that the mind-model M has consciousness at threshold θ *if there exists $E \in \mathrm{ConscStates}(M)$ such that $\mathrm{Con}_{\theta}(E)$ is nonempty (where Con_{θ} is as in Concept 34). Informally: M has at least one nontrivial, stable, globally-integrated evidence state with some reportable content.*

Concept 36 (Contexts and Aspects). *Appendix label:* `concept:ContextsAndAspects`*.*
Body definition label: `hsdef:ContextsAndAspects` *(see § 35.2).*

Informal definition. *A context is a particular "lens" for looking at things that decides what differences you care about and what differences you ignore. An aspect is the simplified description you get after applying that lens (for example: just the color, just the shape, just the role in a story). Changing context changes what counts as "the same" and what patterns you can notice and talk about.*

Formal definition. *A context structure is modeled as a poset $(\mathrm{Ctx}, \sqsubseteq)$, where $C \sqsubseteq D$ means that D is a finer (more detailed) context than C. For each $C \sqsubseteq D$ there is an aspect-extraction (forgetful) map*

$$\mathrm{Ext}_{D\to C} : X_D \to X_C$$

satisfying the functoriality/compatibility condition: if $C \sqsubseteq E \sqsubseteq D$, then

$$\mathrm{Ext}_{D\to C} \;=\; \mathrm{Ext}_{E\to C} \circ \mathrm{Ext}_{D\to E}.$$

For $x \in X_D$, we write $x|_C := \mathrm{Ext}_{D\to C}(x)$ for "the aspect of x seen in context C."

Concept 37 (Control). *Appendix label:* `concept:Control`*.*
Body definition label: `hsdef:Control` *(see § 36.6).*

Informal definition. *To control an outcome means that by choosing an action, you can reliably make that outcome happen (or prevent it). For example, you control "the light turns on" if flipping the switch usually makes the light turn on. In this framework, control is not a mysterious extra ingredient: it is measured by how strongly an action implies a goal.*

Formal definition. *Let $\mathcal{A} \subseteq E$ be a designated set of action event types and let $G \in E$ be a goal condition. For $A \in \mathcal{A}$ define the control strength of A over G by*

$$\mathrm{Ctrl}_{O,\Delta}(A \to G) \;:=\; \mathrm{CImp}_{O,\Delta}(A, G),$$

i.e., control is causal implication restricted to antecedents interpreted as actions (relative to observer/context O and lag window Δ).

Concept 38 (Control Hierarchy). *Appendix label:* `concept:ControlHierarchy`*.*
Body definition label: `hsdef:ControlHierarchy` *(see § 36.6).*

Informal definition. *A control hierarchy is when control is organized in layers: higher-level parts of a system set goals, priorities, or constraints, while lower-level parts handle the detailed* *execution. The ordering does not need to be a single chain; in real systems, some modules may be incomparable or form local loops, but there is still a general "above/below" direction in who constrains whom.*

Formal definition. *A control hierarchy is a partial order* $(\mathcal{M}, <)$ *on a family of modules* $\mathcal{M}$ *such that*

$$M_1 < M_2$$

is aligned with the interpretation "module M_1 *has (potentially asymmetric) control over module* M_2*."*

Concept 39 (Cosmos and Eurycosmos)**.** *Appendix label:* `concept:CosmosAndEurycosmos`*.*
Body definition label: `hsdef:CosmosAndEurycosmos` *(see § 25.5 and § 25.8).*

Informal definition. *We use* cosmos *for "the whole world" as seen from a particular stable viewpoint (often the everyday physical world for an embodied mind). But different minds, or different ways of sensing and coupling to the world, can pick out different "whole worlds" (different cosmos notions). The* eurycosmos *is the bigger collection you get when you consider all these different cosmos notions together across many minds.*

Formal definition. *Let* SigClass *be a set of signal classes. For an observer/mind* S*, a signal class* $C \in \mathrm{SigClass}$*, and a threshold* η*, let* $\mathrm{Univ}_{C,\eta}(S)$ *denote the observer-indexed universe induced by* C *(Definitions 395–396). Fix (when appropriate) a canonical signal class* C_{can} *(e.g. a "physical/body-channel" class for embodied minds). Define the* cosmos *of* S *(at threshold* η*) by*

$$\mathrm{Cosmos}_\eta(S) \;:=\; \mathrm{Univ}_{C_{\mathrm{can}},\eta}(S).$$

For a collection of minds Mind*, define the* eurycosmos *(compare the eurycosm construction) by*

$$\mathrm{Eurycosmos}_\eta(\mathrm{Mind}) \;:=\; \bigcup_{S\in\mathrm{Mind}} \; \bigcup_{C\in\mathrm{SigClass}} \mathrm{Univ}_{C,\eta}(S),$$

i.e., the union of the observer-indexed universes (and thus cosmos notions) induced across minds and signal classes.

Concept 40 (Creative Inspiration)**.** *Appendix label:* `concept:CreativeInspiration`*.*
Body definition label: `hsdef:CreativeInspiration` *(see § 17.7.1).*

Informal definition. *Creative inspiration is a temporary state where parts of your mind that do not usually talk to each other suddenly connect, so that you can form new combinations of ideas. In that moment, unusual associations become easy, new metaphors and solutions appear, and later you can evaluate and refine them.*

Concept 41 (Culture)**.** *Appendix label: concept:Culture.*
Body definition label: hsdef:Culture (see § 343).

Informal definition. *A culture is the collection of shared habits, beliefs, stories, norms, skills, and ways of doing things that people in a society learn from each other and pass along. Culture is not only "what everyone agrees on": it also includes widely shared disputes, tensions, and mixed messages (for example, a society can strongly praise and strongly criticize the same practice in different places or subgroups).*

Formal definition. *A cultural state for a society is a population-level intensity assignment*

$$C : \mathcal{P} \to [0,1] \qquad \textit{or} \qquad C : \mathcal{P} \to V,$$

defined as an aggregation of individual intensities and artifact-embedded intensities, e.g.

$$C(P) := \left(\bigoplus_{i \in A} w_i \otimes I_i(P) \right) \oplus \left(\bigoplus_{a \in \mathcal{A}} u_a \otimes I_a(P) \right),$$

where A is the agent population, $\mathcal{A}$ is the set of shared artifacts, w_i, u_a are weights (importance, population fraction, accessibility), and $\otimes$ is scalar or p-bit scaling as in Section 22.5. A culture at threshold θ is the set

$$\mathrm{Cult}_\theta := \{P \in \mathcal{P} : \pi(C(P)) \geq \theta\},$$

where π is a chosen projection $V \to [0,1]$ if needed.

Concept 42 (Cultural Morality)**.** *Appendix label: concept:CulturalMorality.*
Body definition label: hsdef:CulturalMorality (see § 263).

Informal definition. *A cultural morality is a group's shared "moral background": the values people in the group talk about, which values they treat as most important, what they consider acceptable versus unacceptable, and which rules-of-thumb ("do this", "don't do that") get socially encouraged or punished. People in the same group can disagree, but a cultural morality describes what is common enough to shape behavior and social pressure.*

Formal definition. *Let G be a society (or group), and suppose its members share (at least approximately) a value vocabulary D_{val}. A cultural morality for G consists of:*

1. *a group-level evaluative field $E_G : \Omega_G \times D_{\mathrm{val}} \to B$ (e.g. $B = [0,1]^2$ as p-bit truth values), assigning to each outcome/history $\omega \in \Omega_G$ and each value dimension $d \in D_{\mathrm{val}}$ an evaluation $E_G(\omega; d)$;*

2. *for each $d \in D_{\mathrm{val}}$, a weight $\lambda_d^{(G)}$ and thresholds $\tau_d^{(G)}, \eta_d^{(G)}$ encoding group-level importance and adequacy/aspiration cutoffs on that dimension;*

3. *a set of socially endorsed maxims $M(G)$ (with some prevalence/strength measure, e.g. N_G), where maxims in $M(G)$ have high social resonance (Sections 12 and 18.5) in the sense that they are reinforced by communication and imitation.*

Concept 43 (Dependency)**.** *Appendix label: concept:Dependency.*
Body definition label: hsdef:Dependency (see § 1).

Informal definition. *A dependency is a "prerequisite" relationship: concept A depends on concept B if you cannot really define or use A until you already have B in place. The Hyperseed concept inventory is organized using a dependency graph so that earlier concepts supply the ingredients needed for later ones. When two ideas seem mutually dependent in the original presentation, the formal writeup may pick a direction just to keep the exposition moving.*

Formal definition. *Let $\mathcal{C}$ be the set of Hyperseed concepts (as listed in Appendix B). An expository dependency DAG is a directed acyclic graph $G = (\mathcal{C}, \to)$ such that for concepts $A, B \in \mathcal{C}$, an edge $A \to B$ means:*

"In this paper's formalization, the definition (or primary semantics) of B uses A as a primitive or as an already-

If the original Hyperseed presentation treats A and B as mutually interdefinable, we may still orient one direction for exposition; such choices are explicitly recorded as "cuts".

Concept 44 (Dependency Tower). *Appendix label: concept:DependencyTower.*
Body definition label: hsdef:DependencyTower (see § 511).

Informal definition. *A dependency tower is a way to build progressively better "simplified versions" of something complicated. The kth level represents what you can capture if you are only allowed to use dependencies of complexity up to level k (or only run tests up to that level). Each level must match the original on everything a k-limited observer could check, and among all models that match those checks, you pick the simplest or best one according to the chosen semantics (for example, least-squares error in a Hilbert-space model).*

Formal definition. *A dependency tower for F is a family $\{F_{\leq k}\}_{k\geq 0}$ such that:*

1. *(Monotonicity) $F_{\leq k}$ becomes more faithful as k increases.*
2. *(k-ary indistinguishability) For all $O \in \mathcal{O}_{\leq k}$, $O(F_{\leq k}) = O(F)$.*
3. *(Optimality/minimality) Among all G satisfying the same indistinguishability constraints, $F_{\leq k}$ is "best" in the chosen semantics (e.g. minimal L^2 error).*

Concept 45 (Development). *Appendix label: concept:Development.*
Body definition label: hsdef:Development (see § 219). ***Informal definition.*** *Development is a kind of change over time where a system stays recognizably "the same system" (it keeps its identity and doesn't randomly reset), but it gains abilities: it can do more things, or do the same things better, across time. Importantly, development is not just adding new tricks; it also tends to keep earlier abilities instead of throwing them away.*

Formal definition. *Fix a context C and a system S with a self-continuity score* $\mathrm{SCS}(\,\cdot \to \cdot\,)$ *between time-intervals and a capability profile* $\mathrm{Cap}_S(\,\cdot\,)$ *on intervals. A development path for S in C is a sequence of intervals*

$$I_0 \to I_1 \to I_2 \to \cdots$$

such that:

1. *(Continuity) there exists a threshold θ such that for all k,*
$$\mathrm{SCS}(I_k \to I_{k+1}) \geq \theta.$$
2. *(Expanding capability) for infinitely many k,*
$$\mathrm{Cap}_S(I_k) \prec \mathrm{Cap}_S(I_{k+1}).$$
3. *(Enrichment bias) for typical goals/tasks $g \in T_C$ such that* $\mathrm{Cap}_S(I_k)(g)$ *is high, the value* $\mathrm{Cap}_S(I_{k+1})(g)$ *does not sharply decrease (new capability tends to come with retention rather than catastrophic loss).*

Concept 46 (Difference). *Appendix label: concept:Difference.*
Body definition label: hsdef:Difference (see § 28.1).

Informal definition. *A difference is how much one thing fails to match another, relative to what you care about in the moment (the context). In Hyperseed this is often treated as directed: "how does x differ from y?" can be different from "how does y differ from x" because one item might be a rough template and the other a detailed, noisy instance (so the instance may fit the template better than the template fits the instance).*

Formal definition. *Given a context c, define the directed difference evaluator*

$$\mathrm{Diff}_c(x \to y) := \delta_c(x, y) \in V.$$

The positive channel $\mathrm{Diff}_c(x \to y)^+$ *is evidence that* x *diverges from* y*. The negative channel* $\mathrm{Diff}_c(x \to y)^-$ *is evidence that* x *does not diverge from* y *(e.g.* x *is absorbed into, explained by, or indistinguishable-from* y *along the relevant aspect).*

Concept 47 (Dissonance)**.** *Appendix label: concept:Dissonance.*
Body definition label: hsdef:Dissonance (see § 24).

Informal definition. *When multiple influences pull in different directions, they can partly cancel out. Dissonance measures how much cancellation happens when you combine a bunch of "vector-like" contributions: if they mostly agree, cancellation is small; if they strongly disagree, cancellation is large.*

Formal definition. *Let* $a = (a_i)_{i \in I}$ *be a finite family of complex numbers (phasors). Define*

$$\mathrm{Res}(a) := \left| \sum_{i \in I} a_i \right|, \qquad \mathrm{Dis}(a) := \sum_{i \in I} |a_i| - \mathrm{Res}(a).$$

Thus $\mathrm{Dis}(a)$ *is large when the phasors point in different directions and cancel in the sum.*

Concept 48 (Distinction)**.** *Appendix label: concept:Distinction.*
Body definition label: hsdef:Distinction (see § 30.2).

Informal definition. *A distinction is a rule for what you treat as "the same" versus "different". Two things are distinct for you if your current way of looking keeps them separate; they are indistinct if you lump them together as "close enough". Distinctions depend on context, and in real life they can be imperfect or asymmetric (you might treat* x *as indistinct from* y *for a purpose even though you would not treat* y *as indistinct from* x*).*

Formal definition. *Fix a finite universe* X *(of entities, states, events, or features relevant in context* C*). An indistinction policy is a subset* $H \subseteq X \times X$*, interpreted as:*

- xHy *means "*x *is treated as indistinct from* y*" (the observer does not distinguish* x *from* y*).*

No symmetry, reflexivity, or transitivity is assumed in general. Write $H_C \subseteq \mathcal{P}(X \times X)$ *for the set of admissible policies for context* C*. A (directed) distinction is then any ordered pair* (x, y) *such that* $(x, y) \notin H$*.*

Concept 49 (Dreaming)**.** *Appendix label: concept:Dreaming.*
Body definition label: hsdef:Dreaming (see § 17.7.1).

Informal definition. *Dreaming is a state in which most of what you experience is generated from inside your own mind (images, stories, emotions), with relatively weak control from current sensory input. Because the outside world is not strongly steering the stream, the mind can jump between scenes, blend ideas, and form narratives that would be unlikely in ordinary waking life.*

Formal definition. *In the regime-parameter description of conscious modes (Section 17.7.1), dreaming corresponds to low external coupling* λ_{ext}*, high internal coupling* λ_{int}*, variable reflective coupling* λ_{ref}*, and a lower coherence preference (allowing more internally generated blends and narrative jumps).*

Concept 50 (Effort)**.** *Appendix label: concept:Effort.*
Body definition label: hsdef:Effort (see § 30.1).

Informal definition. *Effort is the cost of doing something: how much time, energy, attention, or other resources it takes. Doing two steps in a row usually costs about the sum of their efforts, and when you have a choice between ways to do something, you usually pick the cheaper way. Hyperseed* *models effort using a simple algebraic structure that makes these common-sense rules precise and usable inside the later math.* ***Formal definition.*** *Let*

$$E := ([0, \infty], \geq, \inf, +, 0),$$

where the order is reversed (so that $a \leq_E b$ means $a \geq b$ as real numbers), the join in the quantale order is inf *(ordinary minimum), and the monoidal product is + (sequential composition of costs) with unit 0. Then E is a commutative quantale: + distributes over arbitrary infima, i.e.*

$$a + \inf_i b_i \ = \ \inf_i (a + b_i).$$

Concept 51 (Abstract)**.** *Appendix label: concept:Abstract.*
Body definition label: hsdef:Abstract (see § 28.3).

Informal definition. *Something is* abstract *when you treat many different concrete examples as the same kind of thing, by ignoring details that do not matter for your current purpose. For instance, many different handwritten marks can all count as the letter "A," and many different physical triangles can all count as "a triangle." Abstracting is a controlled kind of forgetting: you keep what is shared and drop what is incidental.*

Formal definition. *Let c be a context with entity set X_c. An abstraction of c is a surjection*

$$q : X_c \twoheadrightarrow A$$

to a set A of abstracta. The abstraction collapses distinctions by identifying elements of the same fiber. Equivalently, q determines an equivalence relation

$$x \sim_q y \iff q(x) = q(y).$$

Concept 52 (After)**.** *Appendix label: concept:After.*
Body definition label: hsdef:After (see § 7.1).

Informal definition. After *means "later than". Hyperseed models the basic structure of time using the idea of an ordering: some moments/events come after others. To be consistent, nothing can be after itself, and "after" must not produce cycles like "A after B after C after A".*

Formal definition. *A* strict partial order *on a set T is a binary relation $<$ on T such that:*

1. *(Irreflexive) for all $t \in T$, not $(t < t)$,*

2. *(Transitive) for all $t_1, t_2, t_3 \in T$, if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$.*

We say t_1 is before *t_2 if $t_1 < t_2$, and t_2 is* after *t_1.*

Concept 53 (Aesthetics / Art)**.** *Appendix label: concept:AestheticsArt.*
Body definition label: hsdef:AestheticsArt (see § 383).

Informal definition. *In Hyperseed, something counts as* art *(relative to a group of people) if encountering it tends to reliably trigger an experience of beauty for many of them. The key point is that art is not defined only by the object in isolation: it is defined by the object* and *the way it affects a population of experiencers. Different communities can disagree about what is art, because their typical reactions can differ.*

Formal definition. *Fix a class of experiencers $X \subseteq \mathrm{Mind}$. An art object A (relative to X) is any entity such that, for many $m \in X$, encountering A induces a designated beauty proposition* $\mathrm{Beaut}(A)$ *in m's valuation. Equivalently, it induces a high rate of beauty episodes in the population X.*

Concept 54 (Alive / Dead)**.** *Appendix label: concept:AliveDead. Body definition label: hsdef:AliveDead (see § 41.6).*

Informal definition. Alive *and* dead *are treated as time-dependent, evidence-based notions. Roughly: "dead now" is not just "not alive now," but "was alive recently and is not alive now." This matches ordinary thinking (something can be uncertain or borderline), and it also allows conflicting evidence streams (some signs of life, some signs of death).*

Formal definition. *Fix a lookback window length $\tau \in \mathbb{N}$. Define the "was-alive" evidence by*

$$\mathrm{WasAlive}_O(S,t) \;:=\; \bigoplus_{u\in\{t-\tau,\ldots,t-1\}} \mathrm{Alive}_O(S,u),$$

where $\oplus$ is the p-bit *join (componentwise max) from Section 3.4. Define*

$$\mathrm{Dead}_O(S,t) \;:=\; \mathrm{WasAlive}_O(S,t) \otimes \neg\, \mathrm{Alive}_O(S,t),$$

using the p-bit *negation and monoidal product $\otimes$ from the core.*

Concept 55 (Algebraically Asymmetric Physical Reality)**.** *Appendix label: concept:AlgebraicallyAsymmetricPhysic Body definition label: hsdef:AlgebraicallyAsymmetricPhysicalReality (see § 25.4).*

Informal definition. *Hyperseed claims there is a built-in mismatch between how* physics *is structured and how* mental / intersubjective *descriptions usually work. A mental description often keeps only coarse, relational, or pattern-level facts ("these things are connected", "this resembles that") and leaves out many precise details (exact geometry, forces, equations, etc.). Because of that loss of detail, you usually cannot reconstruct the full physical situation from the mental-level description alone: going from "mental" back to "physical" requires adding extra structure and constraints. This one-way loss is the "algebraic asymmetry."*

A useful way to read the claim is: when we pass from the physical world to a shared description of it, we are performing an abstraction *in the sense of Concept 51 (many distinct physical microstates are treated as "the same situation" for communicative, cognitive, or practical purposes). The asymmetry is that this abstraction is typically* many-to-one*, so the inverse problem—recovering the original microstate from the abstracted report—is underdetermined. In everyday terms: multiple distinct physical configurations can support the same appearance, the same narrative, or the same coarse causal graph, and the mental/intersubjective level does not (by default) carry the additional degrees of freedom needed to pick the unique physical realization.*

This is "algebraic" because the mismatch is not merely epistemic ("we do not know") but structural: the map from a more structured domain (physics) to a less structured domain (mental/intersubjective description) behaves like a structure-forgetting morphism. For example, if a mental model keeps adjacency information (a graph of "connected to") while the physical model has full geometry, then distinct geometries can induce the same graph; the relational signature is invariant under many physical deformations, so information has been quotiented out.

Formal definition. *Let* **Phys** *be a category (or structured state space) of physical situations, and let* **Ment** *be a category of mental/intersubjective descriptions (e.g. relational models, coarse narratives, pattern summaries). An* abstraction/forgetful map *is a functor (or homomorphism)*

$$U : \mathbf{Phys} \to \mathbf{Ment}$$

that sends each physical situation to its associated mental-level description by discarding some physical structure. We say physical reality is algebraically asymmetric *(relative to* **Ment***) when this map is not reconstructible up to equivalence; concretely, the following conditions hold generically:*

1. *(Many-to-one on situations) there exist* $P_1 \not\cong P_2$ *in* **Phys** *such that*

$$U(P_1) \cong U(P_2)$$

in **Ment***, i.e. distinct physical situations induce equivalent mental-level descriptions;*

2. *(No inverse without extra choices) there is no functor* $R : \mathbf{Ment} \to \mathbf{Phys}$ *such that*

$$U \circ R \cong \mathrm{Id}_{\mathbf{Ment}} \quad \text{and} \quad R \circ U \cong \mathrm{Id}_{\mathbf{Phys}},$$

so U *does not present* **Ment** *as merely a re-labeling of* **Phys** *(it is not an equivalence of categories, nor does it admit a canonical section on the physical side).*

Equivalently (in set-level terms), one may treat U *as inducing a surjection on underlying sets of situations, with fibers* $U^{-1}(M)$ *typically containing multiple physically distinct realizations of the same mental-level description* M*. Reconstructing a physical candidate from* M *then requires adding external constraints (priors, measurements, dynamics, symmetry-breaking conventions, etc.), which are not contained in* M *itself.*

Concept 56 (Answers)**.** *Appendix label: concept:Answers.*
Body definition label: hsdef:Answers (see § 292).

Informal definition. *An* answer *is defined by what it does, not by what it looks like. It is anything that helps resolve a question by shrinking the set of reasonable possibilities. A complete answer might identify the exact right choice, but many useful answers are partial: they can rule out options, tell you where to look, or tell you which options are more likely.*

Formal definition. *An answer to a question* (τ, Exp) *is any artifact that reduces uncertainty about* $\mathrm{Inst}(\tau)$*. Concretely, an answer may be:*

- *a particular instantiation* $\iota \in \mathrm{Inst}(\tau)$*;*
- *a constraint* $C \subseteq \mathrm{Inst}(\tau)$ *that prunes candidates;*
- *a distribution (or* p-bit*-valued scoring) over* $\mathrm{Inst}(\tau)$*; or*
- *any representation that yields an entropy reduction or computational simplification relative to baseline inference.*

Concept 57 (Anthropic Principle)**.** *Appendix label: concept:AnthropicPrinciple.*
Body definition label: hsdef:AnthropicPrinciple (see § 25.9).

Informal definition. *The anthropic principle is treated as a plain selection effect: the fact that* you exist as an observer *is information. So when comparing different models of the universe, you should downweight models in which observers like you could not exist, and upweight models where observers are likely. This is not an "explanation" of why the universe is the way it is; it is a rule for how an observer should update beliefs given that the observer exists.*

Formal definition. *Let* Θ *be a parameter space of universe models (physical constants, laws, initial conditions, etc.). Let* $\pi(\theta)$ *be a prior distribution over* Θ*. Let* A *be an "anthropic" event such as "observers of type* O *exist." The anthropic posterior is*

$$\pi(\theta \mid A) \propto \pi(\theta)\, P(A \mid \theta).$$

Concept 58 (Archetypes)**.** *Appendix label: concept:Archetypes.*
Body definition label: hsdef:Archetypes (see § 382).

Informal definition. *An* archetype *is a recognizable template that shows up across many different systems (people, stories, cultures, texts, etc.) even though the detailed versions can differ a lot. For example, "the trickster" or "the hero" can appear in many cultures with different names and plots. Hyperseed emphasizes three features: it appears in many places, the instantiations are not mere copies, and the widespread appearance is not trivial (it is "surprising" given how specific the template is).*

Formal definition. *Fix a class of systems (minds, texts, cultures, etc.) indexed by J. An archetype is a template τ together with an instantiation map sending each $j \in J$ to a (possibly empty) set of instantiations $\mathrm{Inst}_j(\tau)$, such that:*

1. *τ has high support across J (many j have $\mathrm{Inst}_j(\tau) \neq \emptyset$), and*
2. *the typical instantiations are diverse (not near-copies), and*
3. *the support is "surprising" relative to the description complexity of τ (e.g. high occurrence despite nontrivial $\kappa(\tau)$).*

Concept 59 (Association)**.** *Appendix label: concept:Association.*
Body definition label: hsdef:Association (see § 91).

Informal definition. *An* association *score says how strongly two things are linked* in a particular context*. A high association means they tend to show up together, or they share many important properties; a low association means they rarely co-occur and have little in common (for the purposes of that context). The same pair can have different association scores in different contexts, depending on what properties the context cares about.*

Formal definition. *A (static) association score in context C is a symmetric map*

$$\mathrm{assoc}_C : E_C \times E_C \to [0,1]$$

intended to measure "how much A and B tend to occur together" or "how many properties/patterns they share."

(One canonical implementation used later in the text is via overlap between property-sets; see Definition 100.)

Concept 60 (Attention / Attentional Focus)**.** *Appendix label: concept:AttentionalFocus.*
Body definition label: hsdef:AttentionalFocus (see § 37.3).

Informal definition. *Attention is modeled as a limited mental budget spread over many possible things you could be tracking. You have* attentional focus *when most of that budget is concentrated on only a few items, instead of being thinly spread over many. The parameters (θ, k) let you say what counts as "most" (θ) and "a few" (k).* ***Formal definition.*** *Fix parameters $\theta \in (0,1)$ (mass threshold) and $k \in \mathbb{N}$ (size bound). Here θ sets how much of the total attentional weight must be captured to count as "focused," while k limits how many distinct items can jointly constitute that focus (so the definition distinguishes "concentrated" attention from attention that is merely spread across many small contributions). We say S has an attentional focus at time t (at level (θ, k)) if there exists a subset $F_t \subseteq P$ with $|F_t| \leq k$ such that*

$$\alpha_t(F_t) := \sum_{p \in F_t} \alpha_t(p) \geq \theta.$$

Intuitively, F_t is a "small support" witnessing that attention is not diffuse: the set F_t may be thought of as the currently dominant objects, concepts, or processes in P whose combined weight crosses the *threshold. Equivalently, if $\alpha_t^{\downarrow}$ is the nonincreasing rearrangement of the weights $(\alpha_t(p))_{p\in P}$, then focus holds iff*

$$\sum_{j=1}^{k} \alpha_t^{\downarrow}(j) \geq \theta.$$

This rearranged form makes explicit that it suffices to look at the k largest weights at time t: among all subsets of size at most k, the maximum possible total weight is attained by choosing the k heaviest elements, so the condition is independent of how ties are broken when several $p \in P$ share the same weight.

Concept 61 (Autocatalytic Sets)**.** *Appendix label: concept:AutocatalyticSets.*
Body definition label: hsdef:AutocatalyticSets (see § 237).

Informal definition. *An autocatalytic set is a collection of processes (often chemical reactions) that "help make themselves": each process in the collection is sped up (catalyzed) by something produced within the same collection. Because of this mutual support, once the set is present it can keep running and regenerating as long as it keeps getting basic inputs. In this sense, the defining feature is not that the set is closed to matter or energy, but that the* catalytic dependencies *required for activity are internally satisfied: the "control" or "enabling" factors for reactions are generated by the network itself.*

Formal definition. *Fix a threshold $\theta \in (0,1]$. Let R be a finite set of reactions and let $C = (c_{ij})_{i,j\in R}$ be a catalysis matrix with $c_{ij} \in [0,1]$ interpreted as the strength of evidence that reaction i catalyzes reaction j. Define the thresholded reaction graph $G_\theta(R,C)$ to be the directed graph with vertex set R and an edge $i \to j$ iff $c_{ij} \geq \theta$. A subset $S \subseteq R$ is a θ-autocatalytic set if every vertex of the induced subgraph $G_\theta(R,C)[S]$ has in-degree at least one; equivalently, for every $j \in S$ there exists some $i \in S$ with $c_{ij} \geq \theta$. The in-degree condition expresses the minimal "internal support" requirement: each reaction in S has at least one sufficiently strong incoming catalytic link from within S. Note that the definition does not require a single catalyst to support all reactions, nor does it require direct self-loops $j \to j$; support may be distributed across different reactions and may occur via cycles (e.g. i catalyzes j, j catalyzes ℓ, and ℓ catalyzes i). The threshold θ makes the notion robust to weak or noisy evidence by discarding catalytic claims below the chosen confidence/strength level.*

Concept 62 (Becoming)**.** *Appendix label: concept:Becoming.*
Body definition label: hsdef:Becoming (see § 62).

Informal definition. *We say that x* becomes *y when, over time, the system transitions from clearly being (or behaving like) x to clearly being (or behaving like) y. Crucially, the transition is not instant: there is a "boundary period" where the identity is genuinely unclear, so it can be reasonable (given the available evidence) to treat the two as both distinct and not-distinct. Away from that boundary, the two are more cleanly separated again. This captures the common situation in which categories or identities have a stable regime on either side of a change, but the change itself passes through an indeterminate corridor in which classical either/or classification is not well-supported.*

Formal definition. *Fix a proto-time $(T,<)$ and a time-indexed distinction relation* Dist*. We say $x \in E$ becomes $y \in E$ (along $(T,<)$) if there exist $t_1 < t_2$ and an interval (or neighborhood) $U \subseteq T$ with $t_1 \leq u \leq t_2$ for all $u \in U$ such that:*

1. (Temporal succession) *x is salient/present up to (near) t_1 and y is salient/present from (near) t_2 onward.*

2. (Boundary non-duality) *for all $u \in U$ near the transition, x and y display non-dual variety at u.*

3. (Outside the boundary) *away from U, x and y are (comparatively) more distinguishable (e.g.* $\mathrm{Dist}(u; x, y)$ *is not strongly paraconsistent).*

The role of U is to isolate the transition region: it is not required that x and y be indistinguishable throughout all time between t_1 and t_2, but rather that there is a neighborhood around the transition in which evidence of distinctness and evidence of non-distinctness are both materially present (as encoded by the behavior of Dist*). Condition (1) anchors the "before" and "after" regimes so that becoming is directional in time, while (3) ensures the indeterminacy is local rather than pervasive (preventing the definition from collapsing into a permanent ambiguity in which x and y never re-separate).*

Concept 63 (Before)**.** *Appendix label: concept:Before.*
Body definition label: hsdef:Before (see § 56).

Informal definition. *In everyday thinking, "before" is a simple yes/no idea: either one event happened earlier than another, or it did not. In messy real situations, evidence can be incomplete or even conflicting, so we may want a more careful notion. Here, "before" is represented by two numbers: one measuring how much evidence supports "t_1 happened before t_2", and one measuring how much evidence supports the opposite (or otherwise counts against that ordering). This lets the model record uncertainty and disagreement instead of forcing an early all-or-nothing decision. In particular, this two-component representation can encode situations where both directions have some support (e.g. from different witnesses or sensors), as well as situations where neither direction is well supported (lack of information), without conflating these cases.*

Formal definition. *Let $V = [0,1]^2$ be the p-bit quantale from Section 3.4. A p-bit-valued temporal evidence relation is a map*

$$B : T \times T \to V, \qquad B(t_1, t_2) = (B^+(t_1, t_2), B^-(t_1, t_2)).$$

Here $B^+(t_1, t_2)$ is positive evidence for $t_1 < t_2$, and $B^-(t_1, t_2)$ is negative evidence. Conceptually, $B^-(t_1, t_2)$ may be read as evidence for $\neg(t_1 < t_2)$, which can include direct evidence for $t_2 < t_1$ as well as evidence that t_1 and t_2 are simultaneous/overlapping or otherwise not strictly ordered. The codomain $V = [0,1]^2$ keeps these two channels separate, so high B^+ does not automatically force low B^- (and vice versa), matching the intended paraconsistent or non-classical treatment of temporal information.

Concept 64 (Belief System)**.** *Appendix label: concept:BeliefSystem.*
Body definition label: hsdef:BeliefSystem (see § 304).

Informal definition. *A belief system is the connected set of beliefs and reasoning habits that a person or community uses to make sense of the world (Hyperseed-Concept 14). It is not just a pile of separate opinions: the beliefs support each other, explain each other, and guide what counts as evidence. A belief system is called* productive *when it has a stable "core" that does not constantly fall apart, but it is also capable of learning and creating new ideas and questions instead of getting stuck. This definition treats a belief system as partly structural (how beliefs hang together) and partly dynamic (how it responds to evidence and generates new commitments), so productivity is meant to capture both resilience under updating and openness to growth.*

Formal definition. *A belief system (for a community M) is a pair (B, R) where $B \subseteq L$ is a focal belief set and R is a rule set, such that $\mathrm{Coh}_R(B; \beta_M)$ exceeds a chosen threshold. Here B may be interpreted as the currently foregrounded or socially enforced commitments (rather than all entertained propositions), R as the inference/justification norms taken as acceptable within M, and β_M as the community-dependent weighting or evidential state relative to which coherence is evaluated. A belief system is productive if, in addition:*

(a) (Individuation) *there exists a "core" subset $B_{\text{core}} \subseteq B$ that remains stable under typical evidence updates (high persistence of key commitments), and*

(b) (Self-transcendence) *the system regularly generates new high-SAI beliefs or new questions whose answers lead to novel patterns, without collapsing coherence.*

Condition (a) is intended to rule out mere volatility (where the system cannot maintain identity long enough to accumulate insight), while condition (b) is intended to rule out mere rigidity (where the system can remain coherent only by refusing to extend itself). The "without collapsing coherence" clause means that novelty is not counted as productive if it systematically drives $\mathrm{Coh}_R(B; \beta_M)$ below threshold, i.e. if expansion typically produces fragmentation rather than organized growth.

Concept 65 (Beliefs)**.** *Appendix label: concept:Beliefs.*
Body definition label: hsdef:Beliefs (see § 301). Beliefs, in this glossary, are understood as elements of a language L (or appropriately typed cognitive states) that can be aggregated into sets such as $B \subseteq L$ and assessed via community-dependent weights or evidential profiles. In particular, the surrounding concepts treat beliefs not only as isolated propositional contents but as nodes in a network that can receive support, face counterevidence, and shift salience over time, which is why later definitions naturally connect beliefs to coherence measures and to update dynamics.

Informal definition. *A belief is what an agent takes to be true (or likely true), possibly to a degree rather than perfectly. When multiple agents form a community, we can also talk about the community's overall stance by combining individual beliefs. The key idea is to weight each agent's contribution by something like trust, authority, or attention, and then merge the weighted contributions. Allowing weights with a negative component lets the model represent not only trust but also distrust or adversarial influence in a graded way.*

Formal definition. *Let M be a finite set of agents, each with a belief state $\beta_i : L \to V$. Define the community aggregate belief state by*

$$\beta_M(\varphi) := \bigoplus_{i \in M} w_i \otimes \beta_i(\varphi),$$

where weights $w_i \in V$ model trust/authority/attention and the operations are those of the p-bit quantale.

Concept 66 (Big Bounce)**.** *Appendix label: concept:BigBounce.*
Body definition label: hsdef:BigBounce (see § 46.7).

Informal definition. *"Big Bounce" is a speculative cosmology idea (Hyperseed-Concept 16): instead of the universe beginning once and for all in a single Big Bang, the universe might go through cycles where a contraction phase (sometimes called a "Big Crunch") is followed by a "bounce" into a new expansion. In Hyperseed this phrase is used cautiously, mainly as a hypothesis or metaphor for very large-scale cycles and for the possibility that some patterns could persist or reappear across such a transition.*

Concept 67 (Blend)**.** *Appendix label: concept:Blend.*
Body definition label: hsdef:Blend (see § 9.6).

Informal definition. *C is a blend of A and B when C is "mix-like" in the sense that many of the important properties of C are also properties of A or of B. This does not mean C is exactly half-and-half; it means a substantial portion of what characterizes C can be traced to one source or the other.*

Formal definition. *Fix C and choose a threshold $\theta \in (0,1]$. Define the property overlap mass of C with (A,B) as*

$$\mathrm{ov}_C(C;A,B) := \sum_{P\in P_C} \min\Big\{\mathrm{Prop}_C(C)(P),\ \max\big(\mathrm{Prop}_C(A)(P), \mathrm{Prop}_C(B)(P)\big)\Big\}.$$

Define the blend score

$$\mathrm{blendscore}_C(C;A,B) := \begin{cases} \mathrm{ov}_C(C;A,B)/\|\mathrm{Prop}_C(C)\|_1, & \|\mathrm{Prop}_C(C)\|_1 > 0, \\ 0, & \|\mathrm{Prop}_C(C)\|_1 = 0. \end{cases}$$

We say C is a blend of A and B (in context C) if $\mathrm{blendscore}_C(C;A,B) \geq \theta$*.*

Concept 68 (Biological)**.** *Appendix label: concept:Biological.*
Body definition label: hsdef:Biological (see § 286).

Informal definition. *A lifeform is called* biological *if, at the level we are describing it, it is mainly built from organic chemistry (like cells and biochemistry). A lifeform can also be called* synthetic *if it was primarily designed and assembled intentionally by an intelligent engineering process, and* evolved *if it arose through unguided evolutionary and self-organization processes. These labels can overlap (for example, a genetically engineered organism is both biological and engineered), so the framework allows them to be multi-valued or p-bit-valued when needed.*

Formal definition. *A lifeform S is:*

- *biological if its dominant implementation substrate is organic chemistry (at the relevant descriptive level);*
- *synthetic if its design/assembly was primarily guided by an intelligent engineering process (as opposed to arising via unguided population-level dynamics);*
- *evolved if it arose via evolutionary and self-organization processes without intentional design.*

These labels are not mutually exclusive at all descriptive levels (e.g. bio-engineered organisms), so we allow them to be multi-valued or p-bit-valued when needed.

Concept 69 (Biological Lifecycle)**.** *Appendix label: concept:BiologicalLifecycle.*
Body definition label: hsdef:BiologicalLifecycle (see § 286).

Informal definition. *A biological lifecycle is the sequence of typical stages a biological organism goes through as it develops and ages, such as infant, young, mature, old, and elderly, together with the typical transitions between these stages. The point of the concept is to talk about "where in its life" an organism is, not just what it is made of.*

Concept 70 (Cause)**.** *Appendix label: concept:Cause.*
Body definition label: hsdef:Cause (see § 36.5).

Informal definition. *In this framework, "A causes B" is a context-relative claim: it depends on what the observer counts as an event, how time is segmented, and what time-scale is being used. Informally, A causes B when A tends to happen before B, and when information about A makes B more predictable (for example, using co-occurrence statistics such as* $\mathrm{Cond}_{O,\Delta}$ *and lag summaries like* $\mathrm{lag}(A,B)$*). Because the underlying "before/after" structure is treated as observer-relative, causal indicators are also understood as relative to O (and to the chosen window Δ).* ***Formal definition.*** *Fix a lag window Δ and a mixing parameter $\lambda \in [0,1]$. Define the causal implication strength*

$$\mathrm{CImp}_{O,\Delta}(A,B) := \Big(\lambda\,\sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B)) + (1-\lambda)\,\sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B))\Big) \cdot \mathbf{1}[A \prec B],$$

where $\mathbf{1}[A \prec B]$ *is an indicator that* A *typically precedes* B *in the relevant context, and* $\sigma : [-1,1] \to [0,1]$ *is the affine rescaling* $\sigma(x) = (1+x)/2$*. Define also the strong (product) variant*

$$\mathrm{CImp}^{\times}_{O,\Delta}(A,B) := \sigma(\mathrm{PAtt}^{\mathrm{ext}}_{O,\Delta}(A,B))\ \sigma(\mathrm{PAtt}^{\mathrm{int}}_{O,\Delta}(A,B))\ \mathbf{1}[A \prec B].$$

Concept 71 (Categorical Imperative)**.** *Appendix label:* `concept:CategoricalImperative`*.*
Body definition label: `hsdef:CategoricalImperative` *(see § ??).*

Informal definition. *The categorical imperative is a way to test a proposed rule for action (a "maxim") by asking: "What would happen if everyone followed this rule in relevantly similar situations?" If the world that results from universal adoption of the rule tends to be acceptable along key ethical dimensions (for example compassion, honesty, nonviolence), then the maxim passes the test; if universal adoption would predictably lead to unacceptable outcomes or a self-defeating situation, the maxim fails. In this document, the key idea is to make the test operational: it is evaluated using an observer's world-model and an explicit set of ethical dimensions, rather than being treated as a purely verbal slogan.*

Formal definition. *Fix a set* $D_{\mathrm{eth}} \subseteq D$ *of ethical dimensions (typically including compassion and possibly truth, nonviolence, etc.). Define the universalized ethical evaluation*

$$E^{\mathrm{univ}}_{O}(m; d) := \mathbb{E}_{\omega \sim P_O(\cdot|\mathrm{Univ}(m))}[\, E_O(\omega; d)\,],$$

where $E_O(\omega; d)$ *is the* p-bit *evidence that the realized history* ω *promotes/opposes* d *(obtained by aggregating stepwise evidence across the history using* $\oplus, \otimes$*). We say* m *passes a threshold categorical imperative if*

$$J\big(E^{\mathrm{univ}}_{O}(m; d)\big) \geq \tau_d \quad \textit{and} \quad W\big(E^{\mathrm{univ}}_{O}(m; d)\big) \leq \eta_d \qquad \forall d \in D_{\mathrm{eth}},$$

for fixed thresholds $\tau_d, \eta_d \in [0,1]$*.*

Concept 72 (Closing)**.** *Appendix label:* `concept:Closing`*.*
Body definition label: `hsdef:Closing` *(see § 30.2).*

Informal definition. *A closing is a deliberate move toward* making finer distinctions*. You stop treating some pairs of situations (or objects, or states) as "basically the same" and start treating them as importantly different. In other words, you tighten your indistinction policy: you collapse fewer differences. Closing is the opposite direction of an opening (Hyperseed-Concept 125), which relaxes distinctions by collapsing more pairs.*

Formal definition. *Given two indistinction policies* $H, H' \in H_C$*:*

- *The transition* $H \to H'$ *is an* opening *(relaxation of distinctions) if* $H \subseteq H'$*.*
- *The transition* $H \to H'$ *is a* closing *(tightening of distinctions) if* $H' \subseteq H$*.*

Concept 73 (Cognitive Synergy)**.** *Appendix label:* `concept:CognitiveSynergy`*.*
Body definition label: `hsdef:CognitiveSynergy` *(see § 37.7).*

Informal definition. *Cognitive synergy means that two different kinds of knowledge (or two different reasoning methods) help each other in a way that makes the overall explanation* simpler *than what you could get by using each one separately and then combining the results. Put plainly: when you let the two knowledge sources interact, you can reach the same level of explanatory adequacy with fewer "extra assumptions" or less complicated story-telling.*

Formal definition. *Let* T_1, T_2 *be two disjoint sets of knowledge types and fix* $\tau \in (0,1]$*. We say there is cognitive synergy at level* τ *between* T_1 *and* T_2 *if*

$$\mathrm{Con}^*(T_1 \cup T_2; \tau) < \mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau).$$

The synergy gain is the positive part

$$\Sigma(T_1, T_2; \tau) := \left(\mathrm{Con}^*(T_1; \tau) + \mathrm{Con}^*(T_2; \tau) - \mathrm{Con}^*(T_1 \cup T_2; \tau)\right)_+ .$$

Concept 74 (Colleague)**.** *Appendix label:* `concept:Colleague`*.*
Body definition label: `hsdef:Colleague` *(see § 342).*

Informal definition. *Two people are colleagues when they are working together on the same ongoing project and coordinating with each other as (roughly) peers. They share information, divide up tasks, and repeatedly coordinate their actions toward a shared goal. This is different from a boss/worker relationship, where one person primarily gives commands and the other primarily follows them in exchange for resources.*

Formal definition. *Fix an interval/episode I and agents i, j. We say i and j are* colleagues on *I if there exists a collaborative project during I, meaning a sustained multi-agent pattern of activity aimed at a shared goal (cf. the goal/value machinery of Section 18), such that:*

- *i and j are both recurrent participants in the project activity pattern on I; and*
- *there are repeated coordination acts between them on I (communication and/or artifact-mediated); and*
- *intermediate results are shared between them (messages, plans, partial work-products, or other artifacts).*

Concept 75 (Collective Consciousness Location)**.** *Appendix label:* `concept:CollectiveConsciousnessLocation`*.*
Body definition label: `hsdef:CollectiveConsciousnessLocation` *(see § ??).*

Informal definition. *A collective consciousness location is a rough "where are we, as a group?" description of a collective mind (or institution, or community) at a time. The idea is that groups can be in different qualitative regimes: highly coordinated and integrated, fragmented into factions, stuck spending most of their energy on internal story/identity maintenance, etc. The location labels (often written $C_0, C_1, C_2, C_3, \ldots$) are meant to summarize these regimes using measurable signals like how much members disagree about what to do (* discord*), and how much of the group's limited resources are spent maintaining narratives/identities (* narrative capture*), along with whatever integration/coherence measures are appropriate for the model.*

Formal definition. *Let $y_i(t)$ denote the policy/proposal signal produced by agent i at time t, and let D_O be a (model-dependent) distance on the space of such signals. Define the (time-t) discord as*

$$D_O(t) := \mathrm{Var}\big(\{y_i(t)\}\big),$$

for some variance functional induced by D_O. Define narrative capture as a scalar $N_O(t) \in [0, 1]$, interpreted as the fraction of the collective resource budget devoted to narrative/identity-maintenance patterns. A collective location C_k (e.g. C_0, C_1, C_2, C_3) is then a qualitative regime describable via thresholds on (at least) discord $D_O(t)$, narrative capture $N_O(t)$, and an (optional/model-dependent) integration/coherence score.

Concept 76 (Combination)**.** *Appendix label:* `concept:Combination`*.*
Body definition label: `hsdef:Combination` *(see § 111).*

Informal definition. *A combination is just a rule for "putting two things together to get a new thing." Examples include concatenating two strings, composing two actions into a longer action, or combining two subsystems into a larger system. In many settings, not every pair can be combined, so the rule may be only partially defined.*

Formal definition. *A* combination *on a set S is a partial binary operation*

$$* : S \times S \rightharpoonup S.$$

Concept 77 (Combination System). *Appendix label:* `concept:CombinationSystem`*.*
Body definition label: `hsdef:CombinationSystem` *(see § 111).*

Informal definition. *A combination system is the package: a collection of things together with a specified way of combining them. It tells you what objects you are allowed to talk about and what it means (when it is allowed) to combine two of them into one.*

Formal definition. *A* combination system *is a pair $(S, *)$ where S is a set and $* : S \times S \rightharpoonup S$ is a partial binary operation.*

Concept 78 (Combinational Dynamical Causal Model). *Appendix label:* `concept:CombinationalDynamicalCaus`
Body definition label: `hsdef:CombinationalDynamicalCausalModel` *(see § 114).*

Informal definition. *This is a way of scoring causal influence when outcomes come from interactions. The model assigns a number saying how much "the interaction of a and b caused c." Crucially, the score is required to respect the representational choices you have made: if your representation treats two processes as the same symbol, then the causal model is not allowed to secretly distinguish them. So causality is computed relative to a chosen representational alphabet (and thus relative to an ontology).*

Formal definition. *Let $(D, **)$ be a dynamical process domain. A combinational dynamical causal model is a function*

$$\Gamma : D \times D \times D \to [0, 1],$$

where $\Gamma(a, b, c)$ is interpreted as the degree to which "interaction of a and b caused c." Given a combination-preserving representational mapping $f : D \to S$ into a symbol system $(S, *)$, we say Γ is valid relative to f *if*

$$f(a) = f(a'),\ f(b) = f(b'),\ f(c) = f(c') \implies \Gamma(a, b, c) = \Gamma(a', b', c')$$

for all $a, b, c, a', b', c' \in D$.

Concept 79 (Combinatorial-Categorical Pattern). *Appendix label:* `concept:CombinatorialCategoricalPattern`
Body definition label: `hsdef:CombinatorialCategoricalPattern` *(see § 110).*

Informal definition. *A combinatorial-categorical pattern is a pattern you find after you first convert a structure into a new "feature space" using a categorical construction. The point is to get a clean docking point between (1) categorical ways of building derived structures (which can expose important relationships) and (2) pattern mining on combinatorial/labeled objects. You first use the categorical machinery to produce a derived labeled structure, and then you search for compressive regularities (patterns) in that derived object.*

Formal definition. *Fix a context C and assume a pattern mining scheme has been defined on labeled structures. A pattern candidate P is a combinatorial-categorical pattern in S relative to T and m_0 if P is a (standard) pattern in the derived structure $\mathrm{sub}(T, S, m_0)$.*

Concept 80 (Computers and Programs). *Appendix label:* `concept:ComputersAndPrograms`*.*
Body definition label: `hsdef:ComputersAndPrograms` *(see § 65).* ***Informal definition.*** *A computer is a machine that can be cheaply reconfigured to behave like many different other machines. A program is the set of signals (instructions) you give it to specify which behavior it should emulate. The key idea is not "electronic" versus "mechanical," but flexible emulation under modest reconfiguration effort.*

Formal definition. *A computer is a machine C equipped with:*

(a) a set (or category) of admissible programs Syn_C (the program signals the user can provide); and

(b) *an execution mechanism that, for many $p \in \mathrm{Syn}_C$, yields a physical realization of $[\![p]\!]$ (Definition 360 below);*

such that the family $\{[\![p]\!] : p \in \mathrm{Syn}_C\}$ spans (approximately) a large variety of machine behaviors within the relevant engineering domain, and reconfiguration (from p to p') has modest effort cost for the relevant user class.

Concept 81 (Compassion)**.** *Appendix label:* `concept:Compassion`*.*
Body definition label: `hsdef:Compassion` *(see § 18.3 and § 18.4.1).*

Informal definition. *Compassion means that you treat other beings' suffering and well-being as something that matters in your decisions. A compassionate choice is one that tends to help others (reduce their suffering and/or increase their happiness), even if it does not directly benefit you.*

Formal definition. *Let A be a set of agents and fix an acting agent $a \in A$. Assume that for each $b \in A$ we have p-bit (outcome-evidence) evaluations*

$$JO_b(s,a) \in V \quad \textit{and} \quad WO_b(s,a) \in V$$

representing (respectively) evidence for joy and woe outcomes for agent b when the acting agent takes action a in situation/state s. Define the aggregated other-joy and other-woe fields by

$$EO_{\to j}(s,a;\mathrm{joy}) := \bigoplus_{b \in A\setminus\{a\}} JO_b(s,a), \qquad EO_{\to w}(s,a;\mathrm{woe}) := \bigoplus_{b \in A\setminus\{a\}} WO_b(s,a).$$

Define the compassion evaluation field by combining "others' joy" with "not others' woe":

$$EO(s,a;\mathrm{compassion}) \;:=\; \frac{1}{2}\, EO_{\to j}(s,a;\mathrm{joy}) \;\otimes\; \frac{1}{2}\, \neg EO_{\to w}(s,a;\mathrm{woe}),$$

where $\neg$ is p-bit negation (swapping positive/negative evidence components) and $\oplus, \otimes$ are the chosen aggregators in the core quantale.

Concept 82 (Compositional Simplicity)**.** *Appendix label:* `concept:CompositionalSimplicity`*.*
Body definition label: `hsdef:CompositionalSimplicity` *(see § 30.5).*

Informal definition. *An object is compositionally simple if you can build it out of a small number of parts, where each part is itself simple, and the way the parts fit together is not complicated. In other words: the whole looks complicated only until you discover the right "break it into pieces" description, after which it becomes simple again.*

Formal definition. *Fix a combination operation $*$ on entities (Section 9) and a base simplicity score $s_C : X \to [0,\infty]$ (relative to context C). Define the compositional simplicity score*

$$s_C^*(x) \;:=\; \inf\left\{ \sum_{i=1}^{n} s_C(x_i) \;:\; x = x_1 * x_2 * \cdots * x_n \right\}.$$

We call x compositionally simple *(in context C) if $s_C^*(x)$ is small and the infimum is achieved (or well-approximated) by a decomposition in which each component x_i is itself simple according to s_C.*

Concept 83 (Concrete)**.** *Appendix label:* `concept:Concrete`*.*
Body definition label: `hsdef:Concrete` *(see § 28.3).*

Informal definition. *Something is concrete (in a given context) if it has enough specific detail that you cannot harmlessly lump it together with lots of other things. If you try to compress* *or "abstract away" too much, you quickly lose important distinctions that matter for what you are doing. Concrete things resist being treated as interchangeable.*

Formal definition. *Fix a context C and a family of concreteness/abstraction constraints K_c (Hyperseed-Concept 124). Let X_C be the entity-domain as represented in context C. We say that $x \in X_C$ is C-*concrete *if, for every admissible abstraction (quotienting) map*

$$q : X_C \twoheadrightarrow A$$

that respects the constraints K_c, the fiber $q^{-1}(q(x))$ is small (i.e., x is not merged with many other distinct entities without violating K_c). Informally: no K_c-admissible abstraction can identify x with a large equivalence class of other items.

Concept 84 (Conscious / Unconscious)**.** *Appendix label:* `concept:ConsciousUnconscious`*.*
Body definition label: `hsdef:ConsciousUnconscious` *(see § 695).*

Informal definition. *A thought, perception, or feeling is* conscious *when it is in your mental spotlight: you can notice it, report it, and it can influence many other things you do (planning, speech, deliberate action). Something is* unconscious *when it is still being processed, but only in the background: it may guide you indirectly (habits, reflexes, priming) without becoming a reportable, globally shared "workspace" content.* ***Formal definition.*** *Let L_{Ob} be the observation language and let an evidence state be a map $E : L_{\mathrm{Ob}} \to V$. Let W be the workspace operator (Definition 228). We say that E is* workspace-stable *if*

$$W(E) = E$$

(Definition 229). We say that E is a conscious evidence state *if it is workspace-stable and reachable from module integration:*

$$E = W(\Gamma_a(E^\bullet))$$

for some module-level evidence collection $E^\bullet$ and attention profile a (Definition 229). Given a threshold θ, define the conscious proposition set *of E by*

$$\mathrm{Con}_\theta(E) \ :=\ \{\varphi \in L_{\mathrm{Ob}} : E^+(\varphi) \geq \theta\},$$

where $E^+(\varphi)$ is the positive-evidence component of the p-bit $E(\varphi)$. A proposition φ is called conscious at E *if $\varphi \in \mathrm{Con}_\theta(E)$, and* unconscious at E *otherwise.*

Concept 85 (Consciousness)**.** *Appendix label:* `concept:Consciousness`*.*
Body definition label: `hsdef:Consciousness` *(see § 239).*

Informal definition. *Consciousness is the overall condition of having an integrated "point of view" inside a mind: some information is not just processed locally, but becomes globally available so that many different parts of the mind can use it. This is what makes a mind able to have a unified, reportable experience that can guide deliberate reasoning and action, rather than just running isolated background routines.*

Formal definition. *Using the workspace operator W and module-integration maps Γ_a (Definitions 228–229), define the set of reachable workspace-fixed points*

$$\mathrm{ConscStates}(M) \ :=\ \left\{E \in V^{L_{\mathrm{Ob}}} \ :\ E = W(E) \ \ \textit{and} \ \ \exists (E^\bullet, a) \ \textit{with} \ E = W(\Gamma_a(E^\bullet))\right\}.$$

We say that the mind-model M has consciousness at threshold θ *if there exists $E \in \mathrm{ConscStates}(M)$ such that $\mathrm{Con}_\theta(E)$ is nonempty (where Con_θ is as in Concept 34). Informally: M has at least one nontrivial, stable, globally-integrated evidence state with some reportable content.*

Concept 86 (Contexts and Aspects)**.** *Appendix label:* `concept:ContextsAndAspects`*.*
Body definition label: `hsdef:ContextsAndAspects` *(see § 35.2).*

Informal definition. *A context is a particular "lens" for looking at things that decides what differences you care about and what differences you ignore. An aspect is the simplified description you get after applying that lens (for example: just the color, just the shape, just the role in a story). Changing context changes what counts as "the same" and what patterns you can notice and talk about.*

Formal definition. *A context structure is modeled as a poset* $(\mathrm{Ctx}, \sqsubseteq)$*, where* $C \sqsubseteq D$ *means that* D *is a finer (more detailed) context than* C*. For each* $C \sqsubseteq D$ *there is an aspect-extraction (forgetful) map*

$$\mathrm{Ext}_{D\to C} : X_D \to X_C$$

satisfying the functoriality/compatibility condition: if $C \sqsubseteq E \sqsubseteq D$*, then*

$$\mathrm{Ext}_{D\to C} \;=\; \mathrm{Ext}_{E\to C} \circ \mathrm{Ext}_{D\to E}.$$

For $x \in X_D$*, we write* $x|_C := \mathrm{Ext}_{D\to C}(x)$ *for "the aspect of* x *seen in context* C*."*

Concept 87 (Control)**.** *Appendix label:* `concept:Control`*.*
Body definition label: `hsdef:Control` *(see § 36.6).*

Informal definition. *To control an outcome means that by choosing an action, you can reliably make that outcome happen (or prevent it). For example, you control "the light turns on" if flipping the switch usually makes the light turn on. In this framework, control is not a mysterious extra ingredient: it is measured by how strongly an action implies a goal.*

Formal definition. *Let* $\mathcal{A} \subseteq E$ *be a designated set of action event types and let* $G \in E$ *be a goal condition. For* $A \in \mathcal{A}$ *define the control strength of* A *over* G *by*

$$\mathrm{Ctrl}_{O,\Delta}(A \to G) \;:=\; \mathrm{CImp}_{O,\Delta}(A, G),$$

i.e., control is causal implication restricted to antecedents interpreted as actions (relative to observer/context O *and lag window* Δ*).*

Concept 88 (Control Hierarchy)**.** *Appendix label:* `concept:ControlHierarchy`*.*
Body definition label: `hsdef:ControlHierarchy` *(see § 36.6).*

Informal definition. *A control hierarchy is when control is organized in layers: higher-level parts of a system set goals, priorities, or constraints, while lower-level parts handle the detailed execution. The ordering does not need to be a single chain; in real systems, some modules may be incomparable or form local loops, but there is still a general "above/below" direction in who constrains whom.*

Formal definition. *A control hierarchy is a partial order* $(\mathcal{M}, <)$ *on a family of modules* $\mathcal{M}$ *such that*

$$M_1 < M_2$$

is aligned with the interpretation "module M_1 *has (potentially asymmetric) control over module* M_2*."*

Concept 89 (Cosmos and Eurycosmos)**.** *Appendix label:* `concept:CosmosAndEurycosmos`*.*
Body definition label: `hsdef:CosmosAndEurycosmos` *(see § 25.5 and § 25.8).* ***Informal definition.*** *We use* cosmos *for "the whole world" as seen from a particular stable viewpoint (often the everyday physical world for an embodied mind). But different minds, or different ways of sensing and coupling to the world, can pick out different "whole worlds" (different cosmos notions). The*

eurycosmos *is the bigger collection you get when you consider all these different cosmos notions together across many minds. In other words, a cosmos is not intended to be "the world in itself" but the world-as-structured by a particular observer's stable access channels, modeling choices, and habitual invariants; it is the domain within which that observer can reliably track objects, causes, and meanings. The point of introducing the eurycosmos is to make explicit that "what counts as the whole world" can vary with sensing modalities, embodiment, training, and coordination protocols, and that a theory meant to talk across minds must have a place to put all of these observer-indexed wholes at once.*

Formal definition. *Let* SigClass *be a set of signal classes. For an observer/mind S, a signal class $C \in \mathrm{SigClass}$, and a threshold η, let $\mathrm{Univ}_{C,\eta}(S)$ denote the observer-indexed universe induced by C (Definitions 395–396). Fix (when appropriate) a canonical signal class C_{can} (e.g. a "physical/body-channel" class for embodied minds). Define the* cosmos *of S (at threshold η) by*

$$\mathrm{Cosmos}_\eta(S) \;:=\; \mathrm{Univ}_{C_{\mathrm{can}},\eta}(S).$$

This definition emphasizes that a cosmos is a choice *of induced universe: one takes the general universe construction $\mathrm{Univ}_{C,\eta}(S)$ and selects a privileged C_{can} to represent the observer's default "world" (for example, the channel class corresponding to ordinary perception-action loops). The threshold η can be read as a resolution or inclusion cutoff (e.g. for what counts as reliably present/trackable), so changing η can enlarge or shrink what is treated as part of the observer's practical world even when S and C_{can} are held fixed. For a collection of minds* Mind*, define the* eurycosmos *(compare the eurycosm construction) by*

$$\mathrm{Eurycosmos}_\eta(\mathrm{Mind}) \;:=\; \bigcup_{S\in\mathrm{Mind}} \;\bigcup_{C\in\mathrm{SigClass}} \mathrm{Univ}_{C,\eta}(S),$$

i.e., the union of the observer-indexed universes (and thus cosmos notions) induced across minds and signal classes. Conceptually, the outer union over S captures inter-observer diversity, while the inner union over C captures intra-observer diversity across different couplings (e.g. vision-like vs language-like channels, or instrument-extended sensing). The use of a set-theoretic union here is deliberately permissive: it allows overlap between induced universes without forcing identification of "the same" element across observers or channels unless additional equivalence relations are later imposed. In particular, $\mathrm{Eurycosmos}_\eta(\mathrm{Mind})$ *is intended as a container for comparison and translation, not yet a claim that all perspectives can be merged into a single coherent ontology without loss.*

Concept 90 (Creative Inspiration)**.** *Appendix label:* `concept:CreativeInspiration`*.*
Body definition label: `hsdef:CreativeInspiration` *(see § 17.7.1).*

Informal definition. *Creative inspiration is a temporary state where parts of your mind that do not usually talk to each other suddenly connect, so that you can form new combinations of ideas. In that moment, unusual associations become easy, new metaphors and solutions appear, and later you can evaluate and refine them. A useful way to read this is as a transient change in effective connectivity: boundaries between specialized sub-processes soften, so representations that are normally kept separate can interact. The "later you can evaluate and refine them" clause matters because the inspired state is not being equated with correctness; it is a regime that increases generative breadth (candidate production), after which more ordinary constraint-checking, testing, and editing can reassert themselves. This also helps distinguish inspiration from mere noise: the characteristic signature is not randomness alone, but the appearance of structured recombinations that can subsequently be stabilized into communicable plans, explanations, or artifacts.*

In the surrounding framework, the reference to cross-subweb coupling is meant to signal that the same overall mind can behave like a collection of semi-autonomous modules most of the time, and that inspiration corresponds to an episode in which coupling parameters shift so that information flows across previously weakly connected regions. The "temporary" aspect is therefore not incidental: the regime is valuable precisely because it can be entered and exited, permitting both divergent generation and convergent selection rather than permanently blurring all distinctions.

Concept 91 (Culture)**.** *Appendix label: concept:Culture.*
Body definition label: hsdef:Culture (see § 343).

Informal definition. *A culture is the collection of shared habits, beliefs, stories, norms, skills, and ways of doing things that people in a society learn from each other and pass along. Culture is not only "what everyone agrees on": it also includes widely shared disputes, tensions, and mixed messages (for example, a society can strongly praise and strongly criticize the same practice in different places or subgroups). This informal definition treats culture as a distributed pattern that persists through learning, imitation, and communication, rather than as a single agreed-upon doctrine. The mention of disputes and mixed messages is meant to highlight that cultural patterns can be internally inconsistent while still being socially real: what is common is the* presence *and salience of certain themes, narratives, and practices, even when they pull in different directions.*

Formal definition. *A cultural state for a society is a population-level intensity assignment*

$$C : \mathcal{P} \to [0,1] \qquad or \qquad C : \mathcal{P} \to V,$$

defined as an aggregation of individual intensities and artifact-embedded intensities, e.g.

$$C(P) := \left(\bigoplus_{i \in A} w_i \otimes I_i(P) \right) \oplus \left(\bigoplus_{a \in \mathcal{A}} u_a \otimes I_a(P) \right),$$

where A is the agent population, $\mathcal{A}$ is the set of shared artifacts, w_i, u_a are weights (importance, population fraction, accessibility), and $\otimes$ is scalar or p-bit scaling as in Section 22.5. A culture at threshold θ is the set

$$\mathrm{Cult}_\theta := \{P \in \mathcal{P} : \pi(C(P)) \geq \theta\},$$

where π is a chosen projection $V \to [0,1]$ if needed. This formalization makes two commitments explicit: first, that culture can be represented as a field over a shared property space $\mathcal{P}$ (topics, practices, norms, narratives, skills, etc.); and second, that it arises from both people and durable external supports. The artifact term allows books, laws, tools, media, institutions, and other persistent carriers to contribute cultural intensity even when no particular individual is currently attending to them, which helps model intergenerational transmission and the stabilizing effects of infrastructure. The operators $\oplus$ and $\otimes$ are left abstract to accommodate different aggregation logics (e.g. averaging, max-like pooling, log-odds pooling, p-bit combination), so that the same schema can represent "majority prevalence," "elite-driven importance," or "broadcast amplification" by appropriate choices of weights and combination rules. The thresholded set Cult_θ is then a way to extract the culturally salient subset of $\mathcal{P}$ for a given application, with π acting as a bridge back to a scalar salience measure when the underlying value domain V is richer than $[0,1]$.

Concept 92 (Cultural Morality)**.** *Appendix label: concept:CulturalMorality.*
Body definition label: hsdef:CulturalMorality (see § 263).

Informal definition. *A cultural morality is a group's shared "moral background": the values people in the group talk about, which values they treat as most important, what they consider acceptable versus unacceptable, and which rules-of-thumb ("do this", "don't do that") get socially* *encouraged or punished. People in the same group can disagree, but a cultural morality describes what is common enough to shape behavior and social pressure. This notion is meant to cover not only explicit moral rules but also the implicit evaluative atmosphere: what kinds of outcomes are celebrated, what kinds provoke shame, what tradeoffs are treated as normal, and which justifications are regarded as intelligible. The emphasis on shaping behavior and social pressure points to a functional criterion: even with disagreement, if people reliably anticipate praise/blame patterns and adjust accordingly, the group has a cultural morality in the operational sense intended here.*

Formal definition. *Let G be a society (or group), and suppose its members share (at least approximately) a value vocabulary D_{val}. A cultural morality for G consists of:*

1. *a group-level evaluative field $E_G : \Omega_G \times D_{\mathrm{val}} \to B$ (e.g. $B = [0,1]^2$ as p-bit truth values), assigning to each outcome/history $\omega \in \Omega_G$ and each value dimension $d \in D_{\mathrm{val}}$ an evaluation $E_G(\omega; d)$;*
2. *for each $d \in D_{\mathrm{val}}$, a weight $\lambda_d^{(G)}$ and thresholds $\tau_d^{(G)}, \eta_d^{(G)}$ encoding group-level importance and adequacy/aspiration cutoffs on that dimension;*
3. *a set of socially endorsed maxims $M(G)$ (with some prevalence/strength measure, e.g. N_G), where maxims in $M(G)$ have high social resonance (Sections 12 and 18.5) in the sense that they are reinforced by communication and imitation.*

Here Ω_G can be read broadly as the space of socially relevant histories (individual actions, institutional outcomes, public events), so that E_G encodes how the group tends to evaluate what happens, not merely what it says it values. Allowing B to be a richer domain (such as p-bit values) permits ambivalence, conflict, or mixed evaluations to be represented directly rather than forced into a single scalar; this aligns with the informal point that groups can contain stable tensions. The weights and thresholds separate at least three different aspects of moral life: what dimensions are talked about or tracked (D_{val}), how important each dimension is in tradeoffs ($\lambda_d^{(G)}$), and what counts as "acceptable" versus "aspirational" performance on that dimension ($\tau_d^{(G)}, \eta_d^{(G)}$). Finally, $M(G)$ captures the compressions that are easiest to transmit: slogans, proverbs, rules-of-thumb, and exemplary stories. Treating them as a set with prevalence/strength is meant to model the fact that a maxim can be culturally central even if many individuals violate it, provided it remains a focal point for justification, condemnation, and coordination.

Concept 93 (Dependency)**.** *Appendix label: concept:Dependency.*
Body definition label: hsdef:Dependency (see § 1).

Informal definition. *A dependency is a "prerequisite" relationship: concept A depends on concept B if you cannot really define or use A until you already have B in place. The Hyperseed concept inventory is organized using a dependency graph so that earlier concepts supply the ingredients needed for later ones. When two ideas seem mutually dependent in the original presentation, the formal writeup may pick a direction just to keep the exposition moving. This is a pragmatic notion of dependency: it is about explanatory order and definitional load-bearing, not necessarily about metaphysical priority. In particular, the dependency graph is intended to guide a reader (and a formal development) through a sequence where symbols and constructions become available before they are invoked, reducing circularity in the presentation. Apparent mutual dependencies are handled by choosing an orientation that makes the definitions go through (often by introducing a weaker or more schematic version of one concept first, and strengthening it later), so that the graph supports a workable build order even when the underlying ideas are tightly coupled.*

Formal definition. *Let $\mathcal{C}$ be the set of Hyperseed concepts (as listed in Appendix B). An expository dependency DAG is a directed acyclic graph $G = (\mathcal{C}, \to)$ such that for concepts $A, B \in \mathcal{C}$, an edge $A \to B$ means:*

"In this paper's formalization, the definition (or primary semantics) of B uses A as a primitive or as an already-

If the original Hyperseed presentation treats A and B as mutually interdefinable, we may still orient one direction for exposition; such choices are explicitly recorded as "cuts".

Concept 94 (Dependency Tower)**.** *Appendix label: concept:DependencyTower.*
Body definition label: hsdef:DependencyTower (see § 511).

Informal definition. *A dependency tower is a way to build progressively better "simplified versions" of something complicated. The kth level represents what you can capture if you are only allowed to use dependencies of complexity up to level k (or only run tests up to that level). Each level must match the original on everything a k-limited observer could check, and among all models that match those checks, you pick the simplest or best one according to the chosen semantics (for example, least-squares error in a Hilbert-space model).*

Formal definition. *A dependency tower for F is a family $\{F_{\le k}\}_{k\ge 0}$ such that:*

1. *(Monotonicity) $F_{\le k}$ becomes more faithful as k increases.*
2. *(k-ary indistinguishability) For all $O \in \mathcal{O}_{\le k}$, $O(F_{\le k}) = O(F)$.*
3. *(Optimality/minimality) Among all G satisfying the same indistinguishability constraints, $F_{\le k}$ is "best" in the chosen semantics (e.g. minimal L^2 error).*

Concept 95 (Development)**.** *Appendix label: concept:Development.*
Body definition label: hsdef:Development (see § 219).

Informal definition. *Development is a kind of change over time where a system stays recognizably "the same system" (it keeps its identity and doesn't randomly reset), but it gains abilities: it can do more things, or do the same things better, across time. Importantly, development is not just adding new tricks; it also tends to keep earlier abilities instead of throwing them away.*

Formal definition. *Fix a context C and a system S with a self-continuity score $\mathrm{SCS}(\cdot \to \cdot)$ between time-intervals and a capability profile $\mathrm{Cap}_S(\cdot)$ on intervals. A development path for S in C is a sequence of intervals*

$$I_0 \to I_1 \to I_2 \to \cdots$$

such that:

1. *(Continuity) there exists a threshold θ such that for all k,*
$$\mathrm{SCS}(I_k \to I_{k+1}) \ge \theta.$$
2. *(Expanding capability) for infinitely many k,*
$$\mathrm{Cap}_S(I_k) \prec \mathrm{Cap}_S(I_{k+1}).$$
3. *(Enrichment bias) for typical goals/tasks $g \in T_C$ such that $\mathrm{Cap}_S(I_k)(g)$ is high, the value $\mathrm{Cap}_S(I_{k+1})(g)$ does not sharply decrease (new capability tends to come with retention rather than catastrophic loss).*

Concept 96 (Difference). *Appendix label: concept:Difference.*
Body definition label: hsdef:Difference (see § 28.1).

Informal definition. *A difference is how much one thing fails to match another, relative to what you care about in the moment (the context). In Hyperseed this is often treated as directed: "how does x differ from y?" can be different from "how does y differ from x" because one item might be a rough template and the other a detailed, noisy instance (so the instance may fit the template better than the template fits the instance).*

Formal definition. *Given a context c, define the directed difference evaluator*

$$\mathrm{Diff}_c(x \to y) := \delta_c(x, y) \in V.$$

The positive channel $\mathrm{Diff}_c(x \to y)^+$ *is evidence that x diverges from y. The negative channel* $\mathrm{Diff}_c(x \to y)^-$ *is evidence that x does not diverge from y (e.g. x is absorbed into, explained by, or indistinguishable-from y along the relevant aspect).*

Concept 97 (Dissonance). *Appendix label: concept:Dissonance.*
Body definition label: hsdef:Dissonance (see § 24).

Informal definition. *When multiple influences pull in different directions, they can partly cancel out. Dissonance measures how much cancellation happens when you combine a bunch of "vector-like" contributions: if they mostly agree, cancellation is small; if they strongly disagree, cancellation is large.*

Formal definition. *Let* $a = (a_i)_{i \in I}$ *be a finite family of complex numbers (phasors). Define*

$$\mathrm{Res}(a) := \left| \sum_{i \in I} a_i \right|, \qquad \mathrm{Dis}(a) := \sum_{i \in I} |a_i| - \mathrm{Res}(a).$$

Thus $\mathrm{Dis}(a)$ *is large when the phasors point in different directions and cancel in the sum.*

Concept 98 (Distinction). *Appendix label: concept:Distinction.*
Body definition label: hsdef:Distinction (see § 30.2).

Informal definition. *A distinction is a rule for what you treat as "the same" versus "different". Two things are distinct for you if your current way of looking keeps them separate; they are indistinct if you lump them together as "close enough". Distinctions depend on context, and in real life they can be imperfect or asymmetric (you might treat x as indistinct from y for a purpose even though you would not treat y as indistinct from x).* ***Formal definition.*** *Fix a finite universe X (of entities, states, events, or features relevant in context C). An indistinction policy is a subset* $H \subseteq X \times X$*, interpreted as:*

- xHy *means "x is treated as indistinct from y" (the observer does not distinguish x from y).*

No symmetry, reflexivity, or transitivity is assumed in general. Write $H_C \subseteq \mathcal{P}(X \times X)$ *for the set of admissible policies for context C. A (directed) distinction is then any ordered pair* (x, y) *such that* $(x, y) \notin H$*.*

Explanation. *Thinking of H as a directed relation (rather than an equivalence relation) allows the model to represent asymmetric coarsening. For example, an observer might treat a detailed state x as "close enough" to a simplified proxy y (so xHy holds) without also treating the proxy as indistinct from the detail (so yHx may fail). Likewise, dropping transitivity lets the policy encode bounded attention or limited inference: the observer may ignore the difference between x and y, and between y and z, while still reacting differently to x versus z. The admissible-set notation* H_C *emphasizes that which indistinction policies are even available can depend on the* *context (instrumentation, time pressure, norms, incentives, or permissible abstractions). Under this convention, "making a distinction" is identified with excluding a pair from H; thus distinctions can be counted, constrained, or optimized over in later constructions by reasoning directly about membership in $X \times X$.*

Concept 99 (Dreaming)**.** *Appendix label: concept:Dreaming.*
Body definition label: hsdef:Dreaming (see § 17.7.1).

Informal definition. *Dreaming is a state in which most of what you experience is generated from inside your own mind (images, stories, emotions), with relatively weak control from current sensory input. Because the outside world is not strongly steering the stream, the mind can jump between scenes, blend ideas, and form narratives that would be unlikely in ordinary waking life.*

Formal definition. *In the regime-parameter description of conscious modes (Section 17.7.1), dreaming corresponds to low external coupling λ_{ext}, high internal coupling λ_{int}, variable reflective coupling λ_{ref}, and a lower coherence preference (allowing more internally generated blends and narrative jumps).*

Explanation. *Operationally, "low external coupling" means that moment-to-moment updates are only weakly constrained by real-time sensory evidence (e.g. muted correction of implausible transitions), while "high internal coupling" means internally generated associations, memories, and affective dynamics strongly determine what comes next. The qualifier "variable" for λ_{ref} is meant to cover the familiar range from largely unreflective immersion (typical dreams) to lucid dreaming, where reflective monitoring and deliberate control can partially re-enter. The "lower coherence preference" should be read as a shift in the system's internal objective: consistency, scene stability, and global narrative plausibility are de-emphasized relative to locally compelling imagery or affective continuation, making discontinuities (scene cuts, identity swaps, impossible geometry) comparatively cheap. This parameterization is intended to be descriptive rather than definitional of any particular sleep stage, and it supports graded mixtures (e.g. hypnagogic imagery as intermediate λ_{ext} with rising λ_{int}).*

Concept 100 (Effort)**.** *Appendix label: concept:Effort.*
Body definition label: hsdef:Effort (see § 30.1).

Informal definition. *Effort is the cost of doing something: how much time, energy, attention, or other resources it takes. Doing two steps in a row usually costs about the sum of their efforts, and when you have a choice between ways to do something, you usually pick the cheaper way. Hyperseed models effort using a simple algebraic structure that makes these common-sense rules precise and usable inside the later math.*

Formal definition. *Let*

$$E := ([0,\infty], \geq, \inf, +, 0),$$

where the order is reversed (so that $a \leq_E b$ means $a \geq b$ as real numbers), the join in the quantale order is inf *(ordinary minimum), and the monoidal product is $+$ (sequential composition of costs) with unit 0. Then E is a commutative quantale: $+$ distributes over arbitrary infima, i.e.*

$$a + \inf_i b_i \;=\; \inf_i (a + b_i).$$

Explanation. *The reversed order is a bookkeeping device that makes "better" (lower real-valued cost) coincide with "larger" in the lattice-theoretic sense used by quantales: with $\geq$ as the underlying order, the least upper bound (join) becomes* inf *in the usual numeric order, so taking a join corresponds to choosing the cheaper option. Interpreting $+$ as sequential composition captures* *the everyday additive behavior of many costs (time spent, energy consumed, attention expended), while still allowing $+\infty$ to represent infeasible or forbidden actions. The distributivity law*

$$a + \inf_i b_i \;=\; \inf_i (a + b_i)$$

encodes a standard optimization intuition: if you must pay an upfront cost a and then choose among alternatives b_i, the optimal total cost is achieved by choosing the cheapest continuation. Commutativity here reflects that the numerical addition on $[0, \infty]$ is commutative; in applications one may still track noncommutative structure elsewhere (e.g. when the availability *of actions depends on order), while using E purely as the cost aggregator. The unit 0 corresponds to "doing nothing" or an action with no additional resource expenditure.*

Concept 101 (Life and Energy)**.** *Appendix label: concept:LifeAndEnergy.*
Body definition label: hsdef:LifeAndEnergy (see § 41).

Informal definition. *Hyperseed uses a very broad notion of "life": any system that can keep itself going over time by staying within a range of "safe" conditions, even while the world perturbs it. Doing that generally requires managing resources (food, fuel, battery charge, time, attention, etc.). In this document those resource-management constraints are modeled using a general-purpose "energy-like" budget (genenergy). This "life and energy" layer is meant to sit between the generic pattern-process layer (Sections 9–12) and the mind/agency layers (Sections 13–21): it is the part of the story where "persistence" becomes "staying viable while paying costs and replenishing capacity."*

Formal definition. *Fix an observer/context O. Let S be a state space and let $g \in \mathbb{R}_{\geq 0}$ be a genenergy budget coordinate. A viability set for O is a subset*

$$V_O \subseteq S \times \mathbb{R}_{\geq 0}.$$

A policy π is homeostatic for (V_O, T, δ) if

$$\Pr_\pi\Big((s_t, g_t) \in V_O \text{ for all } t \in \{0, 1, \ldots, T\}\Big) \geq 1 - \delta.$$

An energy budget process couples the budget (g_t) to an effort cost function $c_O : A \to \mathbb{R}_{\geq 0}$ by requiring $c_O(a_t) \leq g_t$ and updating the budget by

$$g_{t+1} = g_t - c_O(a_t) + \Delta g_t,$$

where Δg_t represents replenishment dynamics (e.g. via resource conversion).

Explanation. *The augmented state space $S \times \mathbb{R}_{\geq 0}$ makes viability explicitly depend not only on the physical/informational state s_t but also on available budget g_t; intuitively, some states are safe only when sufficient reserves remain. The probability threshold $1 - \delta$ is included to accommodate stochastic environments and imperfect control: "homeostatic" here means the policy keeps the system in the viable region with high probability over a finite horizon T, rather than guaranteeing safety in every realization. The constraint $c_O(a_t) \leq g_t$ enforces feasibility (you cannot spend what you do not have), while the update equation separates depletion due to chosen actions from replenishment Δg_t due to the environment or internal conversion processes (e.g. charging, eating, resting, repairing). The observer index O is important: what counts as "viable," what actions are available, and how costs are assessed can differ across measurement granularities, modeling choices, or stakeholder perspectives. This framework is intended to be flexible enough to cover biological metabolism, robotic battery management, and cognitive/attentional budgeting under a single mathematical template.*

Concept 102 (Lifeform). *Appendix label: concept:Lifeform.*
Body definition label: hsdef:Lifeform (see § 285).

Informal definition. *A lifeform is a system that (in its usual operating mode) does three things: it keeps itself within a "survival range" instead of immediately falling apart (homeostasis), it takes in resources and converts them into usable capacity to act and maintain itself (metabolism), and it is part of a process that makes new similar systems over generations (reproduction). In real life, whether something does these things "routinely" can be a matter of degree, which is why borderline cases like hibernation, spores, sterilized organisms, viruses, and prions can be hard to classify.*

Formal definition. *Fix an observer/context O. A system S is a lifeform (for O) if:*

- *S is a complex interactive system (Definition above),*
- *S routinely satisfies a homeostatic constraint for O (i.e. it tends to remain viable rather than drifting into non-viable states),*
- *S routinely carries out metabolism (Section 19.4), and*
- *S routinely participates causally in reproduction events within some species C (Section 19.5).*

("Routinely" is intentionally graded: a crisp version would require nontrivial frequency over long horizons, while a fuzzy/paraconsistent version can treat routine-ness as evidence.)

Explanation. *The definition is intentionally observer-relative because the classification depends on what is being tracked: under a fine-grained biochemical observer an entity may fail to metabolize, while under a higher-level ecological observer it may still function as part of a life cycle that includes metabolizing stages. The "routine" qualifier is doing substantial work: it excludes one-off or accidental episodes (e.g. a transient self-maintaining chemical reaction) while allowing for dormancy and phase changes (e.g. spores and hibernation) where the relevant activities occur intermittently yet reliably across the system's normal history. The reproduction clause is phrased in terms of causal participation within a species C to allow division of labor and multi-component reproduction mechanisms (e.g. sterile workers in eusocial colonies contributing to colony-level reproduction, or symbiotic assemblies whose reproductive unit is larger than an individual organism). Borderline cases such as viruses and prions can be represented as failing some clauses under some observers (e.g. lacking autonomous metabolism), while still being treated as near-life or life-adjacent processes in analyses that emphasize lineage propagation and ecological embedding.*

Concept 103 (Lifting from Instances to Groups). *Appendix label: concept:LiftingFromInstancesToGroups.*
Body definition label: hsdef:LiftingFromInstancesToGroups (see § 9.7).

Informal definition. *Often we know how to describe a relationship between individual things (for example, whether a particular person matches a particular job). But we also talk about relationships between groups (for example, whether a group of people matches a group of jobs). "Lifting from instances to groups" means turning an individual-level relation into a group-level relation. Different lifts capture different everyday meanings, such as "on average," "everyone has at least one match," or "there is one especially good match that works for everyone."*

Formal definition. *Fix finite sets X and Y, and a soft relation $C : X \times Y \to [0,1]$. For $A \subseteq X$ and $B \subseteq Y$, define three lifted relations:*

$$C_{\text{avg}}(A,B) := \frac{1}{|A||B|}\sum_{a\in A}\sum_{b\in B} C(a,b),$$

$$C_{\forall\exists}(A,B) := \min_{a\in A}\ \max_{b\in B}\ C(a,b), \qquad C_{\exists\forall}(A,B) := \max_{a\in A}\ \min_{b\in B}\ C(a,b).$$

These are the average-like, ∀∃-like, and ∃∀-like lifts, respectively.

Concept 104 (Logical / Empirical Truth)**.** *Appendix label: concept:LogicalEmpiricalTruth. Body definition label: hsdef:LogicalEmpiricalTruth (see § 20.8).*

Informal definition. *There are two common reasons we call something "true." Sometimes it is true because it follows from the rules of a system (logical truth), like "all squares have four sides." Other times it is true because careful observation keeps supporting it (empirical truth), like "dropping objects makes them fall." Hyperseed models science as an organized way of combining these: you add new evidence from experiments and observations, and then you also apply the accepted rules of reasoning to see what else must follow.*

Formal definition. *Let L be a language of claims and let V be the chosen evidence-value space. A belief state is a map $\beta : L \to V$. Given an experiment $e \in E$ with observation $o = \mathrm{Obs}(e)$, an empirical evidence increment is a map $\epsilon_{e,o} : L \to V$ that assigns positive/negative evidence to claims depending on whether they agree with (e, o). A scientific update step is*

$$\beta \leftarrow \mathrm{Cl}_R\big(\beta \oplus \epsilon_{e,o}\big),$$

where $\oplus$ aggregates evidence and Cl_R is closure under a rule set R that includes both domain rules and methodological rules (statistics, measurement models, etc.).

Concept 105 (Logical Entropy)**.** *Appendix label: concept:LogicalEntropy. Body definition label: hsdef:LogicalEntropy (see § 92).*

Informal definition. *Logical entropy measures how often two independent random draws land in different categories. Imagine you sort outcomes into bins (a partition). If you pick two outcomes at random, logical entropy is the probability that they end up in different bins. This makes it a very direct way to quantify "how many distinctions your categorization is making," and it aligns with Ellerman's development of logical entropy [17]. It is also closely connected to Hyperseed's primitive of distinction (Hyperseed-Concept 48).*

Formal definition. *Let $\pi = \{B_1, \ldots, B_k\}$ be a partition of X. Given a distribution p on X, define the block masses*

$$p_i := \sum_{x \in B_i} p(x).$$

The logical entropy of (π, p) is

$$h(\pi, p) := 1 - \sum_{i=1}^{k} p_i^2.$$

Concept 106 (Machines)**.** *Appendix label: concept:Machines. Body definition label: hsdef:Machines (see § 65).*

Informal definition. *A machine is something engineered out of parts so that pushing on some parts predictably changes what other parts do. What makes a machine a machine is the stable, repeatable "wiring" between components: gears mesh with gears, circuits connect components through wires, software components communicate through interfaces. When many machines share standardized parts and connections, you can combine them like building blocks and get an internal "language" of parts and compositions.*

Formal definition. *A machine is an engineered physical system M (Definition 355) that can be decomposed into components $(M_i)_{i \in I}$ such that:*

- *each component M_i is (typically) a tool for acting on other components; and*

- *the internal causal interaction structure among the components induces a stable input-output behavior of the whole M that exhibits nontrivial patterns (Section 9) with high intensity relative to the decomposition baseline.*

Concept 107 (Mathematics)**.** *Appendix label: concept:Mathematics.*
Body definition label: hsdef:Mathematics (see § 2.7).

Informal definition. *Mathematics is the practice of making ideas precise enough that you can check them step by step and reliably build on them. In this document, the mathematical development is organized as a dependency graph of definitions and results: later parts depend on earlier parts. The point of the remark in § 2.7 is that you do not need every technical detail to follow the main story; only a few dependency edges are "load-bearing" for choosing the formal core (Section 3) and for deciding whether the rest of the ontology reduces cleanly.*

Listed in the concept inventory without a dedicated formal definition in this version.

Concept 108 (Mature)**.** *Appendix label: concept:Mature.*
Body definition label: hsdef:Mature (see § 2.3). ***Informal definition.*** *"Mature" is a life-stage term. Roughly, an organism (or more generally, a lifeform) is mature when it has finished most of its growth and development and can reliably maintain itself and reproduce. In many contexts it is useful to distinguish immature stages (where development dominates) from mature stages (where stable functioning and reproduction dominate), even though the transition can be gradual.*

Listed in the concept inventory without a dedicated formal definition in this version.

Concept 109 (Metabolism)**.** *Appendix label: concept:Metabolism.*
Body definition label: hsdef:Metabolism (see § 41.4).

Informal definition. *Metabolism is the set of processes by which a lifeform takes in resources from its environment and converts them into usable capacity to act and maintain itself, while expelling waste. In Hyperseed, metabolism is not limited to biochemistry: at the right descriptive level, any system that reliably turns some "resource" into more usable budget-for-action counts as metabolic (food into biological energy, fuel into motion, battery charge into robot actions, compute/time/attention into cognitive work, and so on).*

Formal definition. *Metabolism in context O is a stochastic energy-budget process (g_t) together with a time-indexed resource conversion map Met_t that (on average) increases the budget by converting environmental resources. Concretely, one models:*

$$g_{t+1} = g_t - c_O(a_t) + \mathrm{Met}_t(s_t, a_t, \omega_t),$$

where $c_O(a_t)$ is the genenergy cost of action a_t, and $\mathrm{Met}_t(\cdot)$ is the (possibly stochastic) genenergy gain produced by resource conversion at time t.

Concept 110 (Meta-Goal)**.** *Appendix label: concept:MetaGoal.*
Body definition label: hsdef:MetaGoal (see § 595).

Informal definition. *A meta-goal is a "goal about goals": it is the part of an agent that determines what kinds of goals it should pursue and how it should revise its goals over time. For example, someone might have a short-term goal like "get good grades," but a meta-goal like "keep learning things that matter and help others" influences which short-term goals they adopt and how they change course when circumstances change. In this sense, meta-goals describe a relatively stable "way of caring" that shapes many downstream choices.*

Formal definition. *Let $\Theta \subseteq \mathbb{R}^d$ be the space of meta-goal parameters. A mind at (discrete) time t has meta-goal state $\theta_t \in \Theta$.*

Concept 111 (Mind). *Appendix label: concept:Mind.*
Body definition label: hsdef:Mind (see § 35.3).

Informal definition. *A mind is a self-sustaining system that (i) takes in information from the world through some "input gateway" (registration; Hyperseed-Concept 153), (ii) builds and maintains internal stand-ins for things it encounters (representation; Hyperseed-Concept 157), and (iii) has a way of linking those stand-ins to what they are about (reference; Hyperseed-Concept 152). What the mind's internal stand-ins "mean" is always relative to a context or aspect: a particular way of carving up and distinguishing the world (Hyperseed-Concept 36). A mind is not just a storage box of facts; it is an organized set of interacting processes that keeps updating its internal stand-ins as it registers new input, and then uses those stand-ins to guide what it does next.*

Formal definition. *A mind is a system $M = (P, E)$ equipped with additional structure:*

1. *A distinguished registration interface (a "sensory" subgraph) $S_{\text{in}} \subseteq M$.*

2. *A representation space R (objects, tokens, or states that can stand in for other entities).*

3. *A reference relation $\text{Ref}_C \subseteq R \times U$ for each context/aspect C, where U denotes the ambient universe of entities under discussion.*

4. *An update rule that turns registrations into representational changes (e.g. $r_{t+1} = \text{Update}(r_t, \text{Reg}_t)$ for some internal state $r_t \in R$).*

Concept 112 (Mind-World Correspondence). *Appendix label: concept:MindWorldCorrespondence.*
Body definition label: hsdef:MindWorldCorrespondence (see § 221).

Informal definition. *Mind-world correspondence measures how well a system's internal "map" matches the real structure of what is outside it. The core idea is: if you look at the patterns of influence and dependence among the mind's internal tokens, can you interpret those tokens as standing for real things in the world in a way that preserves the important relationships? A high correspondence score means there is some good translation from internal symbols/states to external entities such that the mind's internal pattern-flow mirrors (at least approximately) the world's pattern-flow over the time interval of interest. This is the formal way of asking whether the mind's internal representations are not just self-consistent, but actually connected to the world in a structure-respecting way.*

Formal definition. *The mind-world correspondence score of S on interval I is*

$$\text{MWCS}(I) := \sup_{f: X^{\text{in}}_{S,I} \to X^{\text{out}}_{S,I}} \deg\big(f; M_{S,I}, E_{S,I}\big).$$

We say that S has θ-mind-world correspondence on I if $\text{MWCS}(I) \geq \theta$.

Concept 113 (Mindplex). *Appendix label: concept:Mindplex.*
Body definition label: hsdef:Mindplex (see § ??). ***Informal definition.*** *A mindplex is a "mind made of minds": a tightly integrated collection of subminds (or agents) that together function like a larger, higher-level mind. Each part can have its own internal processing, but the coupling among parts is strong enough that the whole has stable large-scale coherence (it acts like one organized system rather than a loose crowd). Crucially, a mindplex is also robust: you can often swap out a part for another similar part without destroying the coherence of the whole. In many cases, the whole can also exhibit genuinely collective episodes of coordinated attention/control (a collective GCC episode) that persist for a meaningful duration, rather than being just momentary alignment.*

Formal definition. *Fix an observer O and a coherence score $\text{Coh}_O(\cdot)$ on candidate minds. A system M is notably coherent if $\text{Coh}_O(M)$ is a local maximum relative to small modifications of*

M. A mindplex is a notably coherent mind M that decomposes into notably coherent minds $\{M_i\}$ such that:

(a) each M_i is approximately pattern-preserving as a restriction of M, and

(b) replacing any M_i by another comparably coherent mind M_i' does not destroy the coherence of M (robustness/replaceability).

(Approximate/weak form used in the text.) Fix $\epsilon \geq 0$ and a time horizon T. A system M is an (ϵ, T)-weak mindplex if it satisfies the mindplex conditions up to error ϵ (approximate coherence and approximate replaceability) and, in addition, exhibits a collective GCC episode of duration at least T.

Concept 114 (Morphic Anti-Resonance)**.** *Appendix label: concept:MorphicAntiResonance. Body definition label: hsdef:MorphicAntiResonance (see § 34.5).*

Informal definition. *Morphic anti-resonance is the "push-away" counterpart of morphic resonance: instead of one system's strongly supported patterns making similar patterns more likely elsewhere, one system's strongly rejected patterns create pressure elsewhere to avoid (or invert) those patterns. Intuitively, the target does not just fail to copy the source; it is actively nudged in the opposite direction. In the two-channel (p-bit) evidence picture, this is naturally modeled by swapping the "evidence-for" and "evidence-against" channels coming from the source, then transmitting that swapped signal into the target through a coupling rule. This captures Hyperseed's idea of Morphic Anti-Resonance (Hyperseed-Concept 114): not absence of resonance, but propagation of reversal pressure.*

Formal definition. *Assume $V = [0,1]^2$ and p-bit negation is $\neg(v^+, v^-) = (v^-, v^+)$. Let $K^{\text{anti}}_{\ell_0 \to \ell} : P_{\ell_0} \times P_\ell \to V$ be a V-valued coupling kernel. Define the anti-resonant coupling input from context ℓ_0 to ℓ by*

$$\text{AntiRes}_{\ell_0 \to \ell}(s_{\ell_0,t}) \;:=\; K^{\text{anti}}_{\ell_0 \to \ell} \;?\; (\neg s_{\ell_0,t}),$$

where $(\neg s)(p) := \neg(s(p))$ applies negation pointwise to p-bit values, and $(K?s)(q) := \bigoplus_{p \in P_{\ell_0}} K(p,q) \otimes s(p)$ denotes kernel action.

Concept 115 (Morphic Resonance)**.** *Appendix label: concept:MorphicResonance. Body definition label: hsdef:MorphicResonance (see § 50).*

Informal definition. *Morphic resonance is the idea that when a pattern becomes strong and stable in one place (one context, mind, culture, or system), that makes it easier for a similar pattern to become strong elsewhere, even without the usual direct causal pathway. In the model, each context has its own internal habit-reinforcement dynamics; morphic resonance adds an extra "cross-context input" term so that stabilized patterns in one context can seed support in another. This is the specific, intended reading associated with morphic resonance in Hyperseed (Hyperseed-Concept 115) and is historically linked to morphic field narratives [13].*

Formal definition. *Let ℓ range over a set of contexts L. For each $\ell \in L$, let P_ℓ be a pattern class, let $s_{\ell,t} : P_\ell \to V$ be the habit-support state, let $A_\ell : P_\ell \times P_\ell \to V$ be an internal kernel, let $d_\ell \in V$ be a decay scalar, let $u_{\ell,t} : P_\ell \to V$ be an input, and for $\ell_0 \neq \ell$ let $K_{\ell_0 \to \ell} : P_{\ell_0} \times P_\ell \to V$ be a coupling kernel with induced input*

$$(K_{\ell_0 \to \ell} ? s_{\ell_0,t})(q) \;:=\; \bigoplus_{p \in P_{\ell_0}} K_{\ell_0 \to \ell}(p,q) \otimes s_{\ell_0,t}(p).$$

Then the coupled update is

$$s_{\ell,t+1} \;=\; u_{\ell,t} \;\oplus\; (d_\ell \odot s_{\ell,t}) \;\oplus\; (A_\ell ? s_{\ell,t}) \;\oplus\; \bigoplus_{\ell_0 \neq \ell} (K_{\ell_0 \to \ell} ? s_{\ell_0,t}).$$

We call this a morphic resonance dynamics when the couplings $K_{\ell_0 \to \ell}$ are intended to encode cross-context propagation of habit bias, rather than ordinary local causal influence.

Concept 116 (Multiverse)**.** *Appendix label: concept:Multiverse.*
Body definition label: hsdef:Multiverse (see § ??).

Informal definition. *A multiverse (in this formalism) is a branching space of possible "next worlds": given how an observer models reality, there are multiple possible continuations, and the multiverse is the whole structure of those possible continuations and transitions. A guidable multiverse is a multiverse where the observer can systematically influence which branch they end up in, by taking actions or sending signals. In ordinary physics, guidability is limited by the physical causal channels available; Hyperseed also considers (as a conceptual possibility) that broader signal classes could exist, which would make the multiverse more guidable relative to a mind.* ***Formal definition.*** *Let S_1 and S_2 be two signal classes and let O be an observer. A guidable multiverse (relative to S_1, S_2, O) is a multiverse U defined using S_1-reachability such that, in universe U as modeled by O, signals in class S_2 can systematically influence which element of U the observer will be in after a transition. Formally, this means there is an action set A (implementing S_2 signals) and a controlled transition kernel*

$$P(\cdot \mid U; a)$$

on U.

Concept 117 (Mutual Information)**.** *Appendix label: concept:MutualInformation.*
Body definition label: hsdef:MutualInformation (see § 89).

Informal definition. *Mutual information measures how strongly two random quantities are related. It is the amount of "uncertainty reduction" you get about one variable when you learn the other. If two variables are independent, the mutual information is zero. If knowing one gives you a strong clue about the other, the mutual information is large.*

Formal definition. *The mutual information between random variables X and Y is*

$$I(X;Y) \;:=\; H(X) + H(Y) - H(X,Y).$$

Equivalently,

$$I(X;Y) \;=\; D_{\mathrm{KL}}\big(p(x,y) \,\|\, p(x)p(y)\big).$$

Concept 118 (Narrow Intelligence)**.** *Appendix label: concept:NarrowIntelligence.*
Body definition label: hsdef:NarrowIntelligence (see § 881).

Informal definition. *Narrow intelligence means being extremely good at a limited kind of task, but not broadly capable across many different kinds of tasks. A narrow system can look "superhuman" inside its domain (like chess, protein folding, or one style of image labeling) while still being weak or helpless outside that domain. In other words: high skill in a small region, low generalization across the larger landscape of tasks.*

Formal definition. *Using the competence and breadth notions from the main text: fix a task environment $(\mathcal{T}, D)$ and tolerance $\epsilon \in [0,1]$. Define the breadth*

$$\mathrm{Br}_\epsilon(\mathrm{Ag}; \mathcal{T}, D) \;:=\; D\Big(\big\{T \in \mathcal{T} : \mathrm{Comp}(\mathrm{Ag}; T) \geq 1 - \epsilon\big\}\Big).$$

We say an agent is (relatively) narrow on $(\mathcal{T}, D)$ (at tolerance ϵ) when $\mathrm{Br}_\epsilon(\mathrm{Ag}; \mathcal{T}, D)$ is small, i.e. the agent achieves near-top competence on only a small fraction of tasks under D (even if it may attain very high competence on some specialized subset of tasks).

Concept 119 (Natural Autonomy)**.** *Appendix label: concept:NaturalAutonomy. Body definition label: hsdef:NaturalAutonomy (see § 50).*

Informal definition. *Natural autonomy is the ability to keep yourself viable: from any state where you are still functioning, there is always some action you could take that keeps you in the "safe/flexible" region rather than falling out of it. It does not say you will always choose the right action; it says the option exists at every step. Equivalently, there is a policy (a "will") that, for every viable state, picks a viability-preserving action. The key logical point is the swap between "for every state there exists an action" and "there exists a single rule that chooses an action for every state."*

Formal definition. *Let $D_u : S_{\mathrm{Ob}} \to S_{\mathrm{Ob}}$ be the update induced by intervention $u \in U$ and let $V \subseteq S_{\mathrm{Ob}}$ be a viable set. Then the following are equivalent:*

1. *(Natural autonomy) $\forall s \in V\ \exists u \in U :\ D_u(s) \in V$.*

2. *(Viable will policy) $\exists\,\mathrm{Will} : V \to U\ \forall s \in V :\ D_{\mathrm{Will}(s)}(s) \in V$.*

Concept 120 (Non-Dual Variety)**.** *Appendix label: concept:NonDualVariety. Body definition label: hsdef:NonDualVariety (see § 61).*

Informal definition. *Non-dual variety is what it feels like to be in a genuine gray zone of "substitutability": there are strong reasons to treat two things as interchangeable (you can swap them without much changing what matters), and at the same time strong reasons to treat them as not interchangeable (swapping them breaks something important). The point is not confusion or vagueness; it is that both pressures are real and non-negligible at once. Examples include gradual morphs (where classification is pulled in two directions), the Ship of Theseus (identity supported by continuity yet challenged by replacement), or social/organizational roles that are interchangeable for scheduling but not interchangeable for responsibility.*

Formal definition. *Fix thresholds $\tau_\iota \in (0,1)$. For $x \neq y$ in X_c, define*

$$\mathrm{NDVar}_c(x,y) \iff \iota_c(x,y)^+ \geq \tau_\iota \text{ and } \iota_c(x,y)^- \geq \tau_\iota .$$

Equivalently: the (p-bit) interchangeability/indistinction signal $\iota_c(x,y) \in [0,1]^2$ is strongly paraconsistent, with both evidence channels simultaneously above threshold.

Concept 121 (Non-Duality)**.** *Appendix label: concept:NonDuality. Body definition label: hsdef:NonDuality (see § 28.1).*

Informal definition. *Non-duality means that a mind has genuine reasons to treat two things as separate, and also genuine reasons to treat them as not-separate, at the same time. For example, an ambiguous picture can strongly support two incompatible ways of seeing it; or you can experience another person as "not me" and also as "part of me/us." The point is not to declare that "everything is one," but to represent honest mixed evidence without forcing an immediate, artificial choice [24].*
Formal definition. *Fix thresholds $\tau_\delta, \tau_\iota \in (0,1)$. For $x \neq y$ in X_c define:*

$$\begin{aligned} \mathrm{NonDual}_c(x,y) &\iff \mathrm{Dist}_c(x,y)^+ \geq \tau_\delta \text{ and } \mathrm{Dist}_c(x,y)^- \geq \tau_\delta, \\ \mathrm{NDVar}_c(x,y) &\iff \iota_c(x,y)^+ \geq \tau_\iota \text{ and } \iota_c(x,y)^- \geq \tau_\iota . \end{aligned}$$

Thus non-duality is "distinct and not-distinct" (paraconsistent distinctness), while non-dual variety is "interchangeable and not interchangeable" (paraconsistent substitutability).

Concept 122 (Number)**.** *Appendix label: concept:Number. Body definition label: hsdef:Number (see § 2.3).*

Informal definition. *A number is an abstract way to say "how many" or "how much." We use numbers to count objects (3 apples), to measure quantities (2.5 meters), and to compare amounts (7 is bigger than 5). Numbers also let us do arithmetic so we can combine, split, and predict quantities.*

Listed in the concept inventory without a dedicated formal definition in this version.

Concept 123 (Oceanic Consciousness)**.** *Appendix label: concept:OceanicConsciousness. Body definition label: hsdef:OceanicConsciousness (see § 17.7.1).*

Informal definition. *Oceanic consciousness is a kind of experience where the usual feeling of being a sharply separate "self" softens, and there is a strong felt sense of connectedness with the world or with other people. It is often described as feeling "merged," "boundaryless," or "one with everything," while still having a coherent, unified overall experience. In the model, this is naturally associated with reduced self/other boundary sharpness (Section 16) together with high global coherence.*

Formal definition. *In the regime classification of § 17.7.1, oceanic consciousness is modeled by a consciousness-regime parameter tuple*

$$\Lambda_{\text{oceanic}} := (\lambda_{\text{ext}}, \lambda_{\text{int}}, \lambda_{\text{ref}}, \lambda_{\text{coh}})$$

with λ_{ext} moderate-to-low, λ_{int} high, λ_{ref} moderate, and λ_{coh} high.

Concept 124 (Occasions of Experience)**.** *Appendix label: concept:OccasionsOfExperience. Body definition label: hsdef:OccasionsOfExperience (see § 37).*

Informal definition. *An "occasion of experience" is a single episode or moment of experience: one snapshot of what is happening in consciousness. A key point is that we do not always treat two occasions as either perfectly identical or totally different. Instead, we often treat them as "the same enough" in different ways (for example, recognizing the same object from different angles, or treating two similar feelings as the same kind of feeling). The groupoid idea captures this: instead of only having strict equality, we also track reversible "ways to identify" one occasion with another, and we track whether those identifications fit together coherently.*

Formal definition. *An occasion space is a small $(\infty, 1)$-groupoid E. Its objects $\mathrm{Ob}(E)$ are occasions. A 1-morphism $f : o \to o'$ represents an admissible "identification" or "transport" between occasions (e.g. recognizing the same thing under a change of viewpoint), and higher morphisms encode coherent equivalences between such identifications.*

In toy models one may replace E by an ordinary groupoid, or even by a set E (regarding only equality), by applying the truncation π_0.

Concept 125 (Opening)**.** *Appendix label: concept:Opening. Body definition label: hsdef:Opening (see § 30.2).*

Informal definition. *An opening is when you deliberately loosen your distinctions: you decide to treat more pairs of things as "not importantly different" for your current purposes. This can make thinking and acting easier (you compress the world into fewer categories). A closing is the opposite: you tighten distinctions and start treating fewer pairs as interchangeable, so your categories become more fine-grained. It is useful to note the direction: if H is the set of pairs you collapse, then $H \subseteq H'$ is an opening because H' collapses more pairs.*

Formal definition. *Given two policies $H, H' \in H_C$:*

- *The transition $H \to H'$ is an opening (relaxation of distinctions) if $H \subseteq H'$.*
- *The transition $H \to H'$ is a closing (tightening of distinctions) if $H' \subseteq H$.*

Concept 126 (Open-Ended Benefit). *Appendix label: concept:OpenEndedBenefit. Body definition label: hsdef:OpenEndedBenefit (see § ??).*

Informal definition. *Open-ended benefit is the idea of improving the world in a way that keeps creating more good possibilities over time, rather than locking everything into one narrow outcome. It is "open-ended" because goals and values can grow, refine, and reorganize as new situations arise; it is "benefit" because that growth is guided by the idea of making it easier (not harder) for multiple beings to have good lives — for example by reducing avoidable harm, improving cooperation, and expanding the set of futures where many agents can realize joy and reduce woe.*

(No dedicated standalone formal definition is given in this version; the main text points to including compassion and non-destructive coordination constraints in the revision rule for D_t.)

Concept 127 (Open-Ended Intelligence). *Appendix label: concept:OpenEndedIntelligence. Body definition label: hsdef:OpenEndedIntelligence (see § ??).* ***Informal definition.*** *Open-ended intelligence is the ability of an agent to keep learning and reorganizing what it cares about (and how it reasons) over time, without falling into incoherence. It means the agent can keep some values stable long enough to act, can revise or expand its value vocabulary when needed, and can still make nontrivial decisions after revisions (it does not become "anything goes" or paralyzed).*

Formal definition. *Let D_t be the set of active value dimensions at proto-time t and $E_{O,t}$ the corresponding evaluative field. An observer exhibits open-ended intelligence on an interval if:*

1. *(stability) there exist nontrivial dimensions in D_t that remain stable for substantial sub-intervals (values do not dissolve into noise);*
2. *(revisability) there exist times $t < t'$ such that $D_{t'} \neq D_t$ (the value vocabulary can grow or reorganize);*
3. *(coherence under revision) revisions do not make action selection trivial; i.e. the agent retains a non-degenerate Pareto set and nontrivial resonance structure.*

Concept 128 (Ordinary Waking Consciousness). *Appendix label: concept:OrdinaryWakingConsciousness. Body definition label: hsdef:OrdinaryWakingConsciousness (see § 17.7.1).*

Informal definition. *Ordinary waking consciousness is the normal, everyday awake state: your experience is mostly driven by sensory input from the outside world, your sense of being a separate self is relatively crisp, and your thoughts tend to stay anchored to what is happening around you. You can reflect and plan, but the experience is usually organized around staying consistent with what your senses and actions are telling you.*

Formal definition. *In the regime classification of § 17.7.1, ordinary waking consciousness is modeled by a consciousness-regime parameter tuple*

$$\Lambda_{\text{wake}} := (\lambda_{\text{ext}}, \lambda_{\text{int}}, \lambda_{\text{ref}}, \lambda_{\text{coh}})$$

with λ_{ext} high, λ_{int} moderate, λ_{ref} moderate, and λ_{coh} tuned to prefer sensory-consistent closure.

Concept 129 (Pain). *Appendix label: concept:Pain. Body definition label: hsdef:Pain (see § ??).*

Informal definition. *Pain is a strongly negative kind of experience: it is the felt sense that something is wrong, harmful, or needs to stop. This includes physical pain, but it also includes emotional and social pain (grief, shame, heartbreak, anxiety) when the mind evaluates a situation as deeply bad. In this framework, pain is not just a raw sensation; it is an evaluative state: a "woe-dominant" summary of how the current situation scores across what the agent cares about, possibly including both internal and external processes.*

Formal definition. *Let JO and WO denote joy- and woe-intensity signals derived from an evaluative field (so JO captures positive evaluative intensity and WO captures negative evaluative intensity). Fix an aggregation operator* Agg *that combines values across a set of value dimensions and a time window W. Define aggregated joy and woe:*

$$\overline{JO} := \mathrm{Agg}\{JO(s_{t'}, a_{t'}; d) : d \in D_t,\ t' \in W\},$$
$$\overline{WO} := \mathrm{Agg}\{WO(s_{t'}, a_{t'}; d) : d \in D_t,\ t' \in W\}.$$

A simple formalization of pain-intensity is then any scalar summary that increases with woe and decreases with joy; for example,

$$\mathrm{Pain} := \max(0,\ \overline{WO} - \overline{JO}),$$

so pain is large exactly when woe is dominant after the chosen aggregations.

Concept 130 (Pattern)**.** *Appendix label: concept:Pattern.*
Body definition label: hsdef:Pattern (see § 28).

Informal definition. *A pattern is a repeatable structure or regularity: a set of relationships that tend to go together, so that when you see some pieces you can often expect the others. A shape like a triangle is a pattern; a habit or routine can be a pattern; a grammar rule can be a pattern. In the formalism, a pattern can be treated as a bundle of smaller constraints (simple "this relation should hold" requirements). A context supports the pattern when there is strong evidence that all (or most) of those constraints are satisfied together.*

Formal definition. *Given a context relation R and a pattern P (modeled as a set of required pairs (u, v)), define the (conjunctive) pattern support*

$$S_R(P) := \bigotimes_{(u,v)\in P} R(u, v) \in V.$$

In particular,

$$S_R(P)^+ = \prod_{(u,v)\in P} R(u, v)^+, \qquad S_R(P)^- = \prod_{(u,v)\in P} R(u, v)^-.$$

Concept 131 (Pattern Flow Networks)**.** *Appendix label: concept:PatternFlowNetworks.*
Body definition label: hsdef:PatternFlowNetworks (see § 215).

Informal definition. *A pattern-flow network is a snapshot (over a chosen time interval) of how patterns or information move around inside a modeled system. You choose a set of nodes that stand for concrete "pattern-tokens" inside the system (for example: a particular belief, memory, concept, percept, goal, or action schema). Then, for each ordered pair of nodes (a, b), you record how strongly activity/content associated with a tends to be followed by activity/content associated with b during that interval. In plain terms, it is a map of "what tends to lead to what" inside the system, so the system's dynamics can be compared across different times and different domains. (This directly packages Hyperseed's emphasis on Pattern and Process; see Hyperseed-Concepts 130 and 140.)* ***Formal definition.*** *Fix a system S (as modeled by observer O) and a time interval I. A pattern-flow network for S on I is a V-graph*

$$F_{S,I} : X_{S,I} \times X_{S,I} \to V,$$

where $X_{S,I}$ is a set of nodes representing specific entities or pattern-tokens, and where $F_{S,I}(a, b)$ quantifies the extent to which pattern that "first appears" in a subsequently appears in b over the interval I. The precise estimator used to compute $F_{S,I}(a, b)$ is observer-dependent and may include sub-interval weighting (temporal discounting).

Concept 132 (Pattern Web)**.** *Appendix label: concept:PatternWeb.*
Body definition label: hsdef:PatternWeb (see § 345).

Informal definition. *A pattern web is a network that connects things according to which patterns they share. Imagine you have a collection of entities (objects, experiences, concepts, memories, symbols, etc.). Two entities are linked when they are similar in the sense that many of the same patterns "show up" in both of them (or show up with similar strengths). The result is usually a web rather than a single tree: one thing can be similar to many others for different reasons, so clusters can overlap and multiple routes can connect the same items. In a mind, a pattern web can be read as a structured "association network" among represented items; in a society, many local pattern webs (inside different minds and artifacts) can be combined and then stitched together by communication and coordination links to form a larger collective web. The symbol $\sqcup$ is used informally to suggest a disjoint/typed union of webs: we keep track of which subweb came from which source, but we allow added edges that connect them.*

Formal definition. *Fix a set X of entities and a set P of patterns, together with a pattern-intensity assignment $I : P \times X \to [0,1]$. Define the pattern profile of $x \in X$ to be the function $F_x : P \to [0,1]$ given by $F_x(p) := I(p,x)$. Choose a metric D on $[0,1]^P$ and define the pattern-based pseudometric*

$$d_P(x,y) := D(F_x, F_y).$$

Choose any monotone decreasing kernel $\kappa : [0,\infty) \to [0,1]$ and define the induced similarity

$$s(x,y) := \kappa(d_P(x,y)).$$

A pattern web on X is then a weighted graph

$$W = (X, E_W, w_W)$$

constructed from s (for example by thresholding, k-nearest-neighbors, or taking the complete graph), with weights $w_W(x,y)$ derived from $s(x,y)$.

(Collective/merged web construction used in the text.) Let W_i denote agent i's internal pattern web (Section ??), and let W_a denote the pattern web embedded in an artifact a (external memory, procedures, institutions). The collective pattern web of a society is the (weighted) union

$$W_{\mathrm{col}} := \left(\bigsqcup_i W_i \right) \sqcup \left(\bigsqcup_a W_a \right) \sqcup W_{\mathrm{comm}},$$

where W_{comm} adds edges corresponding to communication and coordination links. Weights on W_{comm} are derived from K and attention/effort constraints (Sections ?? and 8).

Concept 133 (Perceptual Hierarchy)**.** *Appendix label: concept:PerceptualHierarchy.*
Body definition label: hsdef:PerceptualHierarchy (see § 188).

Informal definition. *A perceptual hierarchy is a layered organization of perception in which "higher" layers recognize larger and more abstract patterns by using the output of "lower" layers, and then send control back down by selecting, stabilizing, or re-weighting what the lower layers should pay attention to. For example, in vision: edges and textures support shapes, shapes support objects, and objects support scene understanding. The key idea is that perception is not purely passive recording; it is an active, layered control process that continually chooses which patterns to lock onto and how to integrate them.*

Formal definition. *A perceptual hierarchy is a control hierarchy in which the relevant control events are instances of pattern recognition (Section 13): controlling a module consists in selecting, stabilizing, or transforming the patterns it registers.*

Concept 134 (Perception)**.** *Appendix label: concept:Perception.*
Body definition label: hsdef:Perception (see § 116).

Informal definition. *Perception is what happens when something impacts a mind and the mind does more than just briefly register it. In perception, the registration produces (or updates) an internal representation that the mind can later use for thinking, memory, and action. In everyday terms: perception is when sensing turns into "something in your head that stands for the thing" in a way that can guide what you do next.*

Formal definition. *A mind M perceives an entity A at time t (in context C) if:*

1. *M registers A at time t (Definition 161), and*
2. *this registration causally triggers the emergence of a representation token $R \in \mathcal{R}$ with $\mathrm{Ref}_C(R, A \mid C)$ nontrivial.*

In slogan form: perception is registration that causes representation.

Concept 135 (Physical Reality / Body)**.** *Appendix label: concept:PhysicalRealityBody.*
Body definition label: hsdef:PhysicalRealityBody (see § 46.3). ***Informal definition.*** *A mind's body is the part of reality that acts as its interface: it receives the mind's percepts (signals that inform the mind about what is going on) and carries the mind's actions (signals by which the mind changes things). A mind is in a "physical reality" when its body is contained in that reality and, for most practical purposes, the mind's influence on that reality goes through the body-channel (and information from the reality reaches the mind through bodily perception channels). This makes "physical reality" an operational idea: it is about which causal paths are dominant for prediction and control, not just a metaphysical label. The stronger it is that mind-to-world influence must run through the body, the "more physical" the reality is relative to that mind.*

Formal definition. *(Embodied mind and body.) Let M be a mind modeled as a dynamical subsystem with internal states* Mind*, percept tokens* Percepts*, and action tokens* Acts*. A body for M is a pattern-system B together with a distinguished interface map*

$$\iota : (\mathrm{Percepts}, \mathrm{Acts}) \to B$$

such that there is a highly significant mutual causal relationship between M and B.

(Physical reality.) A physical reality for an embodied mind M is a reality-system R such that:

(a) *the body B is largely contained in R (i.e. $B \subseteq R$ as a pattern-subsystem);*

(b) *most percepts and actions of M are perceptions/actions of entities in R;*

(c) *body-channel mediation: for any action-effect morphism $f : M \to R$ there exist morphisms $g : M \to B$ and $h : B \to R$ with*

$$f \approx h \circ g,$$

where "$\approx$" denotes approximation in the enrichment (e.g. bounded residual weakness or bounded prediction-error increase when replacing f by $h \circ g$).

(Graded physicalness score used in § 46.3.) One way to quantify "how physical" R is for M relative to body B is

$$\mathrm{phys}(M, R; B) := 1 - \sup_{f:M\to R} \inf_{g:M\to B,\, h:B\to R} d(f,\ h \circ g)\,,$$

for a chosen discrepancy measure d with $0 \le d \le 1$.

Concept 136 (Physical Realizations)**.** *Appendix label: concept:PhysicalRealizations.*
Body definition label: hsdef:PhysicalRealizations (see § 56).

Informal definition. *A physical realization is a way an abstract process (like an algorithm, a rule, or a mathematical transformation) gets carried out by physical stuff. You take an abstract input, encode it into a physical state (for example, bits in memory or voltages in a circuit), let the physical substrate evolve according to its dynamics (possibly with control), and then decode the resulting physical state back into an abstract output. Realizations are not "perfect by default": how good a realization is depends on evidence (tests, proofs, models) that the physical behavior matches the intended abstract process.*

Formal definition. *Let $f : A \to B$ be an abstract process in* Abs*. A physical realization of f is a tuple*

$$R = (P, \mathrm{enc}, \mathrm{dec}, \phi)$$

where:

- *P is a physical substrate (an object of* Phys*),*
- $\mathrm{enc} : A \to \mathrm{State}(P)$ *is an encoding of abstract inputs into physical states,*
- $\mathrm{dec} : \mathrm{State}(P) \to B$ *is a decoding of physical states into abstract outputs, and*
- $\phi : \mathrm{State}(P) \to \mathrm{State}(P)$ *is (a model of) the relevant physical dynamics of P induced by the realization setting (including any control signals regarded as part of the realization),*

such that $\mathrm{dec} \circ \phi \circ \mathrm{enc}$ *approximates f in the intended sense.*

Concept 137 (Pleasure)**.** *Appendix label: concept:Pleasure.*
Body definition label: hsdef:Pleasure (see § 18.2.2).

Informal definition. *Pleasure is the kind of positive feeling that comes from an evaluation of "this is good for me" or "this is going well" relative to what the system cares about. It is not treated here as just a raw physical sensation. A system can take pleasure (or experience pain) about external events, internal thoughts, social interactions, achievements, or imagined futures, because the core is evaluative: it is tied to the system's values and goals. In the Hyperseed framing, pleasure is "joy-dominant affect" (and pain is "woe-dominant affect"), meaning pleasure is what it feels like when the system's positive evaluative evidence outweighs the negative, after the system has summarized many values and time horizons into a small control-relevant signal.*

Formal definition. *(As specified in the text.) Joy/woe intensities are defined as projections of a p-bit $v = (v^+, v^-) \in [0,1]^2$:*

$$J(v) := v^+ \quad \textit{(joy intensity)}, \qquad W(v) := v^- \quad \textit{(woe intensity)}.$$

Given an evaluative field E_O, define (for value dimension d):

$$JO(s, a; d) := J(E_O(s, a; d)), \qquad WO(s, a; d) := W(E_O(s, a; d)).$$

Pleasure and pain are then treated as scalar summaries of (JO, WO) after aggregating across value dimensions and time horizons; the document deliberately leaves the aggregation operator open (observer- and system-dependent). A simple toy aggregation mentioned in the text is: average JO and WO over a chosen subset of dimensions and a short temporal window, and then compare the two.

Concept 138 (Presentational Immediacy). *Appendix label: concept:PresentationalImmediacy. Body definition label: hsdef:PresentationalImmediacy (see § 28.2).* ***Informal definition.*** *At any moment, experience has a "foreground/background" structure: some things feel vivid, central, and right in front of you, while other things are faint, peripheral, or barely noticed. Presentational immediacy is a simple way to represent that structure: it assigns each presented item a number between 0 and 1 indicating how much it is present to the mind right now in a given context.*

It is useful to read this number as a purely phenomenological "salience weight" rather than as a probability: a highly immediate item is not necessarily more likely to be true, but rather more available to guide attention, thought, and action. The foreground/background distinction can also be multi-layered (several items may be jointly vivid), so the assignment is not required to pick a single winner; it is a graded profile over what is presented.

Formal definition. *A presentational immediacy function in context c is a map*

$$I_c : X_c \to [0, 1].$$

If $I_c(x)$ is near 1, x is in the foreground of experience in c. If $I_c(x)$ is near 0, x is in the background.

No further axioms are assumed here: in particular, $\sum_{x \in X_c} I_c(x) = 1$ is not required, and there is no presumption that immediacies across different contexts c agree or are comparable. One can, however, impose additional structure when helpful (e.g. continuity in c, Lipschitz bounds under small context-changes, or monotonicity constraints under "zooming in" on an item). In applications, I_c can be read as a quick proxy for attentional allocation, working-memory accessibility, or the degree to which x is poised to enter explicit report, depending on what X_c is taken to represent.

Concept 139 (Probability). *Appendix label: concept:Probability. Body definition label: hsdef:Probability (see § 10.2).*

Informal definition. *Probability is a way of modeling uncertainty about events. In this document it is explicitly observer-relative: what counts as an "event" depends on what distinctions the observer can actually make. If an observer cannot tell certain cases apart, then it does not meaningfully assign separate probabilities to them; instead it assigns probabilities only to coarser events that are observable at its resolution. In this sense, a probability model is partly a model of the world and partly a model of the observer's limitations.*

This observer-relativity can be understood as a disciplined form of coarse-graining: two underlying world-states that differ in ways invisible to O are treated as the same for the purposes of probabilistic prediction and decision. Equivalently, the probability model encodes not only "what may happen" but also "what would count as a difference" to O when registering outcomes. This makes it natural to relate probability to perception and measurement, since any measurement device (biological or artificial) has finite resolution and thus a built-in notion of which distinctions are operationally meaningful.

Formal definition. *Fix an observer/context O. An event observable to O is a predicate on X whose truth value is determined at the resolution of O. Formally, we model the collection of O-observable events as a sigma-algebra $\Sigma_O \subseteq 2^X$. A probability model for O is a probability measure*

$$P_O : \Sigma_O \to [0, 1].$$

The triple (X, Σ_O, P_O) is the observer-indexed probability space.

Concretely, Σ_O often arises from an O-dependent partition or equivalence relation on X capturing which microstates are indistinguishable to O; in that case, O-observable events are unions of equivalence classes, and P_O assigns probabilities only to those unions. When O changes (e.g. by gaining a new sensor, learning a new concept, or shifting attention), the sigma-algebra can *refine, thereby enabling finer events and potentially changing the induced probabilities. This also clarifies how "probability updates" can mix Bayesian conditioning with representational change: the former updates P_O on a fixed Σ_O, while the latter may replace Σ_O itself by a different observable event-structure.*

Concept 140 (Process)**.** *Appendix label: concept:Process.*
Body definition label: hsdef:Process (see § 30.1).

Informal definition. *A process is simply "a system unfolding over time": it specifies what state the system is in at each moment (or each occasion), and how later states relate to earlier ones. The key point is that Hyperseed starts from proto-time (a before/after ordering, not necessarily a single clock line). So a process is defined relative to whatever time-ordering the observer/system is using: it is the structured assignment of system-states along that ordering.*

In everyday terms, a process can be deterministic (each earlier state fixes the later one), stochastic (later states are uncertain given earlier ones), or partially specified (only certain constraints are imposed). The definition here is deliberately broad: it isolates the minimal scaffolding needed to talk about "change" in a way that is compatible with non-linear or branching temporal organization. For instance, if two events are incomparable in proto-time, the process description need not force an ordering between them, which is important when modeling concurrent, distributed, or multi-stream cognition.

Formal definition. *Let $(T, <)$ be a proto-time. Consider the thin category $\mathbf{T}$ whose objects are elements of T and with a unique morphism $t_1 \to t_2$ iff $t_1 < t_2$. A process over T is a functor*

$$S : \mathbf{T} \to \mathrm{Sys}$$

into a chosen category of system states Sys *(e.g. sets, measurable spaces, or V-enriched structures).*

The functoriality condition packages the idea that "later includes or follows from earlier" in a way that respects composition: if $t_1 < t_2 < t_3$, then the induced transition $S(t_1) \to S(t_3)$ factors as $S(t_1) \to S(t_2) \to S(t_3)$. In a thin category this is automatic at the level of arrows (there is at most one), but it becomes substantive once Sys *has nontrivial morphisms: specifying S amounts to choosing compatible transition-structure between states along the proto-time order. Examples include: (i)* $\mathrm{Sys} = \mathbf{Set}$ *with morphisms as update functions; (ii)* $\mathrm{Sys} = \mathbf{Meas}$ *with morphisms as measurable maps; (iii)* Sys *as a category of probabilistic kernels (Markov processes) where arrows encode stochastic evolution.*

Concept 141 (Process and Pattern)**.** *Appendix label: concept:ProcessAndPattern.*
Body definition label: hsdef:ProcessAndPattern (see § 114).

Informal definition. *A process is something that changes over time: a river flowing, a person learning, a conversation unfolding, a planet orbiting. A pattern is a regularity you can notice in what is happening: a rhythm, a repeating shape, a rule that keeps showing up, or a stable relationship between parts of the system. Hyperseed treats these as two sides of the same coin: processes are the "happening," while patterns are the "structure in the happening." In a given context/aspect C, we assume there is a repertoire of patterns P_C that the observer knows how to talk about (Hyperseed-Concept 130). Patterns are not just present-or-absent: they can show up weakly or strongly. The quantale V (Section 3) is used as a graded scale for "how much" a pattern is present, and $\mathrm{Int}_{S,t}(P)$ means "the intensity (degree of embodiment) of pattern P in system-state S at time t."*

The key interpretive point is that "pattern" here is not restricted to visually obvious repetition: it can include statistical regularities, causal dependencies, semantic roles in a discourse, or functional organization in a control loop, as long as the context/aspect C provides a vocabulary P_C for naming them. Likewise, intensity is meant to cover both crisp satisfaction (a rule fully holds) and graded *approximation (the rule is only approximately realized), with V chosen to match the intended notion of gradedness (e.g. $[0, 1]$ with multiplication, a max-plus semiring, or a lattice of qualitative levels).*

Formal definition. *Fix a context/aspect C and a set of candidate patterns P_C (Hyperseed-Concept 130), and let V be the quantale of intensity-values from Section 3. Let $(T, <)$ be a proto-time, and let* Sys *be a chosen category of system-states. A process is modeled (as in the process view) by a functor $S : \mathbf{T} \to \mathrm{Sys}$, so each $t \in T$ has a state $S(t)$. Assume that for each time t, the state $S(t)$ induces an intensity map*

$$\mathrm{Int}_{S(t),t} : P_C \to V.$$

Then the process/pattern interface is given by these intensities: each pattern $P \in P_C$ determines a time-indexed "pattern trace" $t \mapsto \mathrm{Int}_{S(t),t}(P)$, and conversely the pattern content of the process S (in aspect C) is the family of traces $\{\mathrm{Int}_{S(t),t}(\cdot)\}_{t\in T}$.

This interface can be read in two complementary directions. In the analysis direction, one starts with a process S and extracts a set of traces as observables/features, thereby projecting complex dynamics into a space of pattern-coordinates indexed by P_C. In the synthesis direction, one can treat a desired collection of traces as constraints and ask for processes whose states realize those intensities (possibly up to approximation in V). When V is a quantale, it also supports combining pattern-evidence across subsystems or time-slices using the quantale multiplication (or tensor), which makes it possible to model how multiple weak cues can compose into a stronger overall manifestation of a higher-level pattern.

Concept 142 (Proto-Time)**.** *Appendix label: concept:ProtoTime.*
Body definition label: hsdef:ProtoTime (see § 55).

Informal definition. *Proto-time is time "before clocks": it is just the idea that some experiences or events come before others. For example, you might remember eating breakfast before going to school, or noticing a sound after seeing a flash. Proto-time does not assume everything lines up on a single timeline; it only assumes you have some way to say "this happened before that" (and sometimes two things may be incomparable). The set T can be whatever the system treats as its time-points: occasions of experience (Hyperseed-Concept 124), events in an event calculus, or internal representational states. The key structure is the precedence relation $<$ recording "before/after" at the system's current resolution.*

Incomparability is not a bug but a feature: it allows proto-time to represent concurrency (two streams unfolding without a known order), indeterminacy (the system cannot currently decide which came first), or mere underspecification (the modeler chooses not to impose an order). Proto-time can therefore be shared between very different implementations: a biological agent with partial temporal access, a distributed computation with parallel steps, or a narrative representation in which only some ordering relations are asserted.

Formal definition. *A proto-time is a pair $(T, <)$ where T is a set of temporal indices (e.g. occasions, events, or representations of them) and $<$ is a strict partial order on T.*

Being a strict partial order means (i) irreflexive: never $t < t$, and (ii) transitive: if $t_1 < t_2$ and $t_2 < t_3$ then $t_1 < t_3$. It does not require totality: there may exist t_1, t_2 with neither $t_1 < t_2$ nor $t_2 < t_1$. When additional structure is present (e.g. a notion of "immediate successor," timestamps, or a metric), it can be layered on top of proto-time, but the Hyperseed baseline only assumes the order needed to speak about precedence-dependent notions such as "earlier context," "later update," and time-indexed traces of intensities.

Concept 143 (Quantale Weakness)**.** *Appendix label: concept:QuantaleWeakness.*
Body definition label: hsdef:QuantaleWeakness (see § 3.7).

Informal definition. *"Quantale weakness" is a simple way to score how many distinctions an observer is willing to ignore, especially among things the observer considers important. If you treat many pairs of items as "basically the same," then your representation is coarser (and in Hyperseed's sense, often simpler or more general). The weight function μ says which items matter more for this score: ignoring differences between high-weight items counts more than ignoring differences between low-weight items. This is meant as a quantitative proxy for the simplicity/generalization idea [3, 2]. Hyperseed-Concept 202; Hyperseed-Concept 143; Hyperseed-Concept 169.*

Formal definition. *Given a set X, a weight map $\mu : X \to V$, and a failed-distinction set $H \subseteq X \times X$, define the weakness of H by*

$$w(H) := \bigvee_{(u,v)\in H} \mu(u) \otimes \mu(v),$$

where $\bigvee$ is the join/supremum in the quantale V and $\otimes$ is the monoidal product.

Concept 144 (Questions)**.** *Appendix label: concept:Questions.*
Body definition label: hsdef:Questions (see § 291).

Informal definition. *A question is not just a sentence; it is something you do in order to find something out. You ask in a way that forces the world (or your own mind) to respond with information. For example, you might measure a length, run a test, look more closely, ask someone, try an experiment, or do a mental check. The point is to narrow down which situation you are actually in, or which concrete example you are dealing with, among several possibilities.*

Formal definition. *A question for a system S is an action (or action policy) whose intent is to obtain information about* $\mathrm{Inst}(\tau)$ *for some template τ. Formally, we model a question as a pair* (τ, Exp) *where:*

- *τ is a template pattern, and*
- Exp *is an experiment/interaction channel that produces data d which can be used to update beliefs about* $\mathrm{Inst}(\tau)$*.*

Concept 145 (Question Networks)**.** *Appendix label: concept:QuestionNetworks.*
Body definition label: hsdef:QuestionNetworks (see § 299).

Informal definition. *A question network is a map of inquiry: it keeps track of which questions depend on which other questions. Sometimes you cannot answer one question until you have answered (or at least partially answered) others. Importantly, real investigation can have feedback: two questions can support each other, and you may go back and forth refining both.*

Formal definition. *A question network is a directed graph $Q = (V, E)$ where nodes $v \in V$ are questions and an edge $v \to w$ means "answering v requires (or strongly benefits from) answering w." Cycles are permitted.*

Concept 146 (Quantity)**.** *Appendix label: concept:Quantity.*
Body definition label: hsdef:Quantity (see § 2.3).

Informal definition. *A quantity is "how much" of something there is. Sometimes it is a count (how many apples), sometimes it is a measurement (how much water, how much time, how much energy). Quantities let you compare ("more/less"), combine ("add them"), and scale ("twice as much"). In Hyperseed-style modeling, quantities show up everywhere: effort/cost, energy/genenergy, probability, intensity of patterns, and strength of evidence are all treated as quantities in some appropriate mathematical structure.*

No dedicated formal definition of "Quantity" is given in this version; it is listed as a basic concept used throughout.

Concept 147 (Rational Decision). *Appendix label: concept:RationalDecision. Body definition label: hsdef:RationalDecision (see § ??).*

Informal definition. *A rational decision (in the minimal Hyperseed sense) is a way of choosing actions that does not fall apart under conflict, does not pick something clearly worse for no reason, and does not ignore costs and feasibility. It also should not act as if a wildly unlikely outcome is guaranteed unless you have explicitly decided to treat that outcome as a "faith commitment." In other words: even with messy values and limited information, decision-making should remain disciplined rather than turning into "anything goes."*

Formal definition. *Fix a state s. A choice rule $\pi_O(\cdot \mid s)$ is (Hyperseed-minimally) rational if it satisfies:*

1. ***Non-explosion under value conflict:*** *if two actions have incomparable value-vectors (due to paraconsistency), the rule still returns a nonempty admissible set (or distribution) rather than degenerating into arbitrary choice over all actions.*

2. ***Dominance respect:*** *if an action a Pareto-dominates a' (under $\preceq_{\text{pref}}$ across all value dimensions d and has no greater resistance), then π_O does not prefer a' over a.*

3. ***Resource sensitivity:*** *ceteris paribus, if two actions have equal value evidence, the rule prefers smaller resistance $R_O(s, \cdot)$.*

4. ***Belief/value/action coherence:*** *if the observer's belief model assigns negligible plausibility to an outcome, the value attached to that outcome does not dominate action selection unless explicitly designated as a faith-commitment (Section 20).*

Concept 148 (Rationality). *Appendix label: concept:Rationality. Body definition label: hsdef:Rationality (see § ??).* ***Informal definition.*** *Rationality is the broader, ongoing ability to form good beliefs and good choices over time. It includes: learning from evidence instead of wishful thinking, keeping your beliefs and actions consistent with each other, noticing when something does not add up, and taking limited resources (time, attention, energy) seriously. In Hyperseed, rationality is not "cold logic only": it is connected to emotion/compassion, values/goals, resource constraints (energy/genenergy), belief systems and science, and also the social world (societies, mindplex/global brain, reality-systems), plus cultural and motivational styles (including wu-wei themes). In this sense, rationality is "system-relative": what counts as rational depends on what the agent can represent and do, and on the reality-system it is embedded in.*

No dedicated formal definition of "Rationality" is given in this version; the document provides minimal criteria for rational decision (Concept 147) and an explicit model of reasoning operations (Concept 150), which together serve as formal scaffolding for the broader informal notion here.

Concept 149 (Reality-System). *Appendix label: concept:RealitySystem. Body definition label: hsdef:RealitySystem (see § 367).*

Informal definition. *A reality-system is the shared "world" that a group effectively lives in together: the set of things they treat as real, the shared symbols and roles they recognize, and the web of cause-and-effect and prediction that links these things. It can be as small as two people coordinating a plan, or as large as a scientific community, an economy, or a culture. What makes it a genuine shared reality (not just many separate private worlds) is that people significantly influence and predict each other, and that there are patterns that exist across minds, not only inside each mind.*

Formal definition. *Fix a community C, an interval I, and a community intensity datum w. An intersubjective reality-system for C on I is a triple*

$$R_{C,I} = (E_{C,I}, L_{C,I}, w),$$

where:

- $E_{C,I}$ *is a set of entities-as-modeled-in-community (tokens for persons, objects, shared symbols, joint goals, norms, roles, etc., as represented across the community);*
- $L_{C,I}$ *is a set of weighted links encoding predictive and causal relationships between entities and between minds' representations of entities (e.g. a V-weighted directed hypergraph or simplicial structure on $E_{C,I}$);*
- w *assigns intensities to patterns supported on subcommunities $U \subseteq C$.*

We say $R_{C,I}$ is intersubjective (rather than merely a disjoint union of individual realities) if $L_{C,I}$ contains nontrivial cross-mind links of significant weight and w assigns non-negligible total intensity to patterns supported on $|U| \geq 2$.

Concept 150 (Reason)**.** *Appendix label: concept:Reason.*
Body definition label: hsdef:Reason (see § 466).

Informal definition. *Reason is the deliberate activity of improving what you think by using rules: you take in new evidence, consider possible explanations, and then check what follows from what you currently accept. In practice, this usually happens in loops: you learn something, adjust your beliefs, add or remove an assumption, and then work out the consequences, repeating until you reach a stable view (or until you decide you need better evidence).*

Formal definition. *A (finite) reasoning loop is any finite sequence of operations applied to an initial belief state β_0 where each step is one of:*

- ***Evidence join:*** $\beta \leftarrow \beta \oplus \epsilon$,
- ***Hypothesis injection:*** $\beta \leftarrow \beta \uparrow_s h$,
- ***Closure:*** $\beta \leftarrow \mathrm{Cl}_R(\beta)$.

Concept 151 (Recognition Process)**.** *Appendix label: concept:RecognitionProcess.*
Body definition label: hsdef:RecognitionProcess (see § 114).

Informal definition. *A recognition process is a process inside a system that learns to pick out and track another thing it is interacting with. Over time, the recognizing system comes to internally embody more of the same regularities (patterns) that are present in what it recognizes: it gets better at noticing those patterns, and it stops "missing" them as easily. Everyday examples include learning to recognize a friend's face, learning a melody, or a classifier learning features that reliably detect a kind of object. In Hyperseed this is made precise using the notion of patterns (Hyperseed-Concept 130) and a graded notion of how strongly a pattern is present.* ***Formal definition.*** *Fix a context/aspect C and a set of candidate patterns $\mathcal{P}_C$. Assume each system-state S at time t determines a (possibly fuzzy) intensity map $\mathrm{Int}_{S,t} : \mathcal{P}_C \to V$ into the quantale V from Section 3. A process Y is a* pattern recognition process *aimed at an entity X (in aspect C) if there exists a nonempty set $\mathcal{P}_{X,C} \subseteq \mathcal{P}_C$ such that for each $P \in \mathcal{P}_{X,C}$:*

1. *(alignment)* $\mathrm{Int}_{Y,t}(P) \leq \mathrm{Int}_{X,t}(P)$ *for all relevant t, and*
2. *(learning)* $\mathrm{Int}_{Y,t}(P)$ *is nondecreasing in t (with respect to the order on V), with strict increase for at least one P at least once.*

Concept 152 (Reference)**.** *Appendix label: concept:Reference.*
Body definition label: hsdef:Reference (see § ??).

Informal definition. *A reference is the "aboutness link" between a token inside a mind (a word, image, pointer, memory trace, or internal state) and the thing it is about. For example, the word "Paris" can refer to a particular city; a map can refer to a region; a mental picture can refer to a friend; and a variable name in a program can refer to a stored value. Reference is what makes a representation more than an accidental resemblance: it is the system's commitment that "this internal thing is meant to stand for that external thing," in some context/aspect. This is the link used by registration and representation (Hyperseed-Concept 153, 157).*

Formal definition. *A mind-model includes a representation space $\mathcal{R}$ (states/tokens that can stand in for other entities) and, for each context/aspect C, a reference relation*

$$\mathrm{Ref}_C \subseteq \mathcal{R} \times \mathcal{U}.$$

We read $\mathrm{Ref}_C(r, u)$ as: "the representation token r refers to the entity u in aspect C" (and "$\mathrm{Ref}_C(r, u)$ is nontrivial" as: the reference relation holds in that context).

Concept 153 (Registration)**.** *Appendix label: concept:Registration.*
Body definition label: hsdef:Registration (see § 115).

Informal definition. *Registration is the simplest kind of "taking notice." Something impacts a system, and the system comes to embody some pattern that is compatible with what happened, even if the system does not yet have a rich internal model of the cause. Registration can be vague: you might register that "something moved" or "something is there" without being able to identify it uniquely. In Hyperseed, this vagueness is handled by letting registration depend on the system's current context and on what it treats as indistinguishable (Hyperseed-Concept **??**).*

Formal definition. *Fix a context C and an equivalence relation $\sim_C$ on $\mathcal{U}_C$ (interpreted as "indistinguishable in context C" for the registering system). A system X* registers *an entity A at time t (in context C) if there exists $A' \in \mathcal{U}$ such that $A'|C \sim_C A|C$ and $A'|C$ becomes a pattern in X by time t. Concretely, this can be witnessed by a pattern $P \in \mathcal{P}_C$ such that*

$$\mathrm{Int}_{X,t}(P) \not\leq \theta \qquad \textit{and} \qquad \mathrm{Ref}_C(P, A|C) \textit{ is nontrivial},$$

for some threshold $\theta \in V$.

Concept 154 (Religion)**.** *Appendix label: concept:Religion.*
Body definition label: hsdef:Religion (see § 24.9).

Informal definition. *In Hyperseed's ontology-level language, a religion is a socially stabilized way of transmitting, reinforcing, and coordinating patterns across many minds using shared stories, symbols, practices, and rituals. It functions as a long-running "pattern carrier": repeated participation tends to install or strengthen certain archetypal patterns (and certain value-and-meaning patterns) in the participants, and to synchronize those patterns across a community. This is one natural route to what Hyperseed calls communion: strong cross-mind pattern overlap supported by shared practices.*

Concept 155 (Repetition)**.** *Appendix label: concept:Repetition.*
Body definition label: hsdef:Repetition (see § 28.1).

Informal definition. *A repetition is "the same kind of thing happening again": two events are not literally identical, but they count as the same pattern for the purposes at hand. This requires both (i) that the two instances are distinguishable as separate occurrences, and (ii) that they are interchangeable at the relevant level (swapping one for the other does not change what matters). A* *variety is the contrast: the instances are distinguishable and the difference matters for the current context.*

Formal definition. *Fix thresholds $\tau_\delta, \tau_\iota \in (0,1)$. For $x \neq y$ in X_c define:*

$$\mathrm{Rep}_c(x,y) \iff \mathrm{Dist}_c(x,y)^+ \geq \tau_\delta \text{ and } \iota_c(x,y)^+ \geq \tau_\iota,$$

$$\mathrm{Var}_c(x,y) \iff \mathrm{Dist}_c(x,y)^+ \geq \tau_\delta \text{ and } \iota_c(x,y)^- \geq \tau_\iota.$$

Intuitively, a repetition is distinct-yet-substitutable; a variety is distinct-and-not-substitutable.

Concept 156 (Reproduction (within Species))**.** *Appendix label: concept:Reproduction. Body definition label: hsdef:Reproduction (see § 41.5).* ***Informal definition.*** *Metabolism sustains an individual system; reproduction sustains a* type *of system over time. A reproduction event happens when one system plays a primary causal role in bringing about another system that counts as "the same kind" (within some tolerance) under the observer's notion of similarity. Biological parenthood is the standard example, but the same abstract idea also covers, for example, a machine manufacturing another machine of the same design, or software producing a faithful copy of itself, as long as the similarity criterion is met.*

Formal definition. *Fix an observer/context O. Let $W_O(S,t)$ denote the pattern-web state of system S at time t (relative to O), and let $d_O(\cdot,\cdot)$ be a distance (or dissimilarity) between pattern webs. For $\epsilon > 0$, define within-species similarity by*

$$S \approx_{O,\epsilon} S' \iff \exists\, t, t' \text{ such that } d_O\big(W_O(S,t), W_O(S',t')\big) \leq \epsilon.$$

A reproduction event *from S to S' over a time window $[t_0, t_1]$ is an event in which:*

- *the existence/state of S and its actions in $[t_0, t_1]$ have a strong causal influence on the existence of S' at time t_1, and*
- *$S \approx_{O,\epsilon} S'$ for a chosen species-tolerance parameter ϵ.*

Concept 157 (Representation)**.** *Appendix label: concept:Representation. Body definition label: hsdef:Representation (see § 52).*

Informal definition. *A representation is something that stands in for something else: a map stands in for a territory, a drawing stands in for an object, and a mental model stands in for a situation in the world. In Hyperseed, A represents B (in an aspect) when two conditions hold. First, A is genuinely* about *B (it refers to it; Hyperseed-Concept 152). Second, A preserves some of the same patterns that are present in B in that aspect (inheritance; Hyperseed-Concept* **??***). Both matter: similarity without reference is accidental resemblance, and reference without inherited structure is an empty pointer.*

Formal definition. *Let $A, B \in \mathcal{U}$ and context C be given. We say that A represents B in aspect C if:*

1. *(reference) $\mathrm{Ref}_C(A, B|C)$ is nontrivial, and*
2. *(inheritance) $A|C$ inherits patterns from $B|C$ in the sense of Definition 165.*

Concept 158 (Resistance)**.** *Appendix label: concept:Resistance. Body definition label: hsdef:Resistance (see § 47.6).*

Informal definition. *Resistance measures how much an agent is "forcing" the world away from its default or passive unfolding. If your chosen action-policy behaves very differently from what would happen by default, then it pays a high resistance cost; if it mostly follows the passive* *flow, resistance is low. Acceptance is the complementary idea: high acceptance means your behavior is compatible with the passive dynamics (you are not fighting the flow).*

Formal definition. *Fix passive dynamics P_0. For a policy π, define its instantaneous resistance at state x as*

$$\mathrm{Resist}_\lambda(\pi; x) := \lambda\, \mathrm{KL}\big(P_\pi(\cdot \mid x) \,\|\, P_0(\cdot \mid x)\big).$$

Define its acceptance at x as the complementary score

$$\mathrm{Accept}_\lambda(\pi; x) := \exp\big(-\mathrm{Resist}_\lambda(\pi; x)/\lambda\big) = \exp\big(-\mathrm{KL}(P_\pi \| P_0)(x)\big).$$

Thus acceptance is high exactly when resistance is low.

Concept 159 (Resonance)**.** *Appendix label: concept:Resonance.*
Body definition label: hsdef:Resonance (see § 19).

Informal definition. *Resonance is the idea of "lining up so the effects add instead of cancel." Like waves that are in phase produce a larger wave, aligned contributions in a pattern system can reinforce each other, while misaligned contributions can interfere destructively. In Hyperseed, resonance is built from many local comparisons across a network (Hyperseed-Concept 131; Hyperseed-Concept 132); the local comparison is a normalized alignment score between two evidence contributions.*

Formal definition. *For $v, w \in \mathbb{V}$ with $\sigma_C(v) \neq 0$ and $\sigma_C(w) \neq 0$, define the normalized alignment*

$$\mathrm{align}(v, w) := \frac{\Re\big(\sigma_C(v)\,\overline{\sigma_C(w)}\big)}{|\sigma_C(v)|\,|\sigma_C(w)|} \in [-1, 1].$$

If either $\sigma_C(v) = 0$ or $\sigma_C(w) = 0$, set $\mathrm{align}(v, w) := 0$.

Concept 160 (Reward)**.** *Appendix label: concept:Reward.*
Body definition label: hsdef:Reward (see § ??).

Informal definition. *A reward is a single-number score used to guide or train action: higher reward means "better" according to a chosen summary of what the agent values. Reward is useful because it lets you apply standard optimization and learning tools, but it is also dangerous because it can hide conflicts and tradeoffs by collapsing many values into one scalar. Hyperseed typically treats reward as a proxy: keep multi-dimensional values visible when possible, and use scalars carefully when you need them.*

Formal definition. *Fix weights $\lambda_d \geq 0$ for $d \in D$ and a resistance weight $\beta \geq 0$. Define a scalar reward proxy by*

$$r_O(s, a) := \sum_{d \in D} \lambda_d \big(J_O(s, a; d) - W_O(s, a; d)\big) - \beta\, R_O(s, a).$$

Concept 161 (Reward Maximization)**.** *Appendix label: concept:RewardMaximization.*
Body definition label: hsdef:RewardMaximization (see § ??). ***Informal definition.*** *Reward maximization is a decision style where you compress everything you care about into a single numerical score (a "reward") and then choose actions to make that number as large as possible. This is often a useful engineering shortcut because it gives a clean target for learning and planning algorithms. The downside is that it can hide important tradeoffs: once many competing considerations are collapsed into one number, you can no longer easily see which value got sacrificed to improve which other value. Hyperseed contrasts this with goal seeking, where some constraints are treated as hard (or lexicographically prior) and a reward-like scalar is used only to choose among the remaining feasible options.*

One practical way to read this is: reward maximization turns a multi-criteria problem into a single-criterion problem by selecting an aggregation rule in advance. This can be convenient, but it also makes the aggregation rule a silent "policy choice" that may not match the users actual priorities under distribution shift, partial observability, or when new kinds of failure appear. In particular, once everything is scalarized, violations of a previously "important" consideration can look like ordinary tradeoffs rather than like constraint breaches, which can matter when interpreting behavior, debugging, or auditing.

Formal definition. *Fix weights $\lambda_d \geq 0$ for $d \in D$ and a resistance weight $\beta \geq 0$, and define the scalar reward proxy $r_O(s, a)$ as in Definition 253:*

$$r_O(s, a) := \sum_{d \in D} \lambda_d \big(J_O(s, a; d) - W_O(s, a; d)\big) \; - \; \beta \, R_O(s, a).$$

A one-step reward-maximizing choice rule at state s selects any action $a^* \in A$ satisfying

$$r_O(s, a^*) = \max_{a \in A} r_O(s, a).$$

(Equivalently, maximize the Score defined in Proposition 24.)

Because the choice rule only depends on the scalar $r_O(s, a)$, any two actions with identical proxy reward are interchangeable under this decision style, even if they differ substantially in their underlying vector of consequences. This highlights the role of the weights λ_d and β as a normative encoding: they determine which distinctions survive the compression and which are treated as irrelevant at decision time. When Hyperseed contrasts this with goal seeking, the intended technical contrast is that goal seeking can be represented as (i) first restricting A to a feasible subset $A_{\text{feas}}(s)$ induced by hard/lexicographic constraints, and then (ii) applying a reward-like score only on that restricted set, thereby keeping some distinctions non-compensatory rather than merely "expensive." In that reading, reward maximization corresponds to the special case where the feasible set is all of A and all desiderata are made compensatory through a single scalar.

Concept 162 (Rich-Resource Mind)**.** *Appendix label: concept:RichResourceMind. Body definition label: hsdef:RichResourceMind (see § 1133).*

Informal definition. *A rich-resource mind is an agent with enough resources (time, compute, energy, memory, data, coordination ability, etc.) that it can (i) search broadly for good options rather than settling for the first acceptable one, and (ii) run sufficiently strong verification and error-correction that random noise and sloppy optimization errors can be kept small. When resources become large, qualitatively new strategies can become available (for example, extensive simulation, large-scale model search, or self-replication), and the agent's typical attention-allocation and weakness profile may shift in a way that is not just "more of the same" (Hyperseed-Concept 162; [11]). This matters because the ontology aims to be scale-robust: it should describe not only humans and current machines, but also potential successors.*

A useful diagnostic is that, in a rich-resource mind, failures are less dominated by "could not find any plan" and more dominated by specification/ontology mismatch (choosing the wrong target) or by strategic interactions (other agents reacting). In other words, additional resources can move the main failure mode from search failure toward objective failure, including the possibility that the agent becomes very good at achieving the wrong proxy. The glossary term is therefore not intended as simple praise ("powerful"), but as a marker that some approximations (e.g. assuming high noise, bounded lookahead, or negligible verification) may stop being the right default regime.

Formal definition. *(Resource-rich regime, per Assumption 5.) In the drift-controlled self-modification setting, let $F(\theta)$ be the objective and $D(\theta; \theta_t)$ the drift penalty with weight λ. A* *rich-resource regime is characterized by bounded optimization error and small noise: the update θ_{t+1} is chosen so that*

$$F(\theta_{t+1}) - \lambda D(\theta_{t+1};\theta_t) \ \geq\ \sup_{\theta}\bigl(F(\theta) - \lambda D(\theta;\theta_t)\bigr) \ -\ \eta,$$

for a small optimization gap $\eta \geq 0$, and (when modeled) the noise is sub-Gaussian with scale parameter σ that can be made small. In the deterministic idealization, one sets $\eta = \sigma = 0$.

The parameter η makes explicit that "resource-rich" is not defined as literal global optimality, but as being close enough to the best available update that optimization error is no longer the dominant uncertainty. Similarly, modeling noise as sub-Gaussian with tunable scale σ encodes the idea that the agent can invest in measurement, redundancy, and verification so that stochasticity is not the main driver of outcomes. Within this framing, increasing resources corresponds (in part) to shrinking (η, σ), expanding the effective hypothesis/plan class over which $\sup_\theta$ is taken, and improving the ability to estimate the consequences of updates—all of which can change qualitative behavior even if the formal update equation keeps the same shape.

Concept 163 (Second (Peircean Category))**.** *Appendix label: concept:SecondPeircean. Body definition label: hsdef:SecondPeircean (see § 45).*

Informal definition. *A Second is the basic experience of "pushback" or "difference": something meets resistance, and you get the sense that "this is not that" or "my expectation did not fit what happened." It is inherently about interaction and directionality: comparing x to y can reveal a mismatch even when comparing y to x would not reveal the same mismatch (because the comparison depends on which way you are trying to map or substitute things). In predictive terms, a Second can be read as the primitive discrepancy between a model state and an observation.*

The emphasis on directionality is meant to rule out treating a Second as merely an undirected distance. Many practically relevant "mismatch" notions are asymmetric: e.g. a proposed simplification may fail to capture details of the original (simplification $\to$ original mismatch), while the reverse mapping (original $\to$ simplification) can succeed by forgetting information. Likewise, "prediction error" is naturally directed from prediction to outcome, even if some symmetric loss can be derived later. Hyperseed uses this Peircean notion to keep track of the minimal event of resistance that forces an update in belief, plan, or representation.

Formal definition. *For a context c, a Second is a directed difference event, i.e. an ordered pair $(x, y) \in X_c \times X_c$ together with its directed difference evaluation $\mathrm{Diff}_c(x \to y)$. Seconds are the smallest units at which "reaction" or "opposition" can be represented.*

Here $\mathrm{Diff}_c(\cdot \to \cdot)$ is intentionally left more general than a metric: depending on context, it can be a signed residual, a constraint violation indicator, a likelihood ratio, an edit distance under a chosen code, or any evaluation that says "this attempted substitution/expectation did not fit." The context index c carries the background choices that make a particular directed difference meaningful (representation, allowable transformations, observation channel, and the granularity at which differences are registered). Under this view, many familiar quantities (surprisal, loss, prediction error, constraint slack) can be interpreted as structured families of Seconds once their direction and context are made explicit.

Concept 164 (Second Law of Thermodynamics)**.** *Appendix label: concept:SecondLawOfThermodynamics. Body definition label: hsdef:SecondLawOfThermodynamics (see § 347).*

Informal definition. *In everyday physics language, the second law says that (in a closed system) things tend to evolve toward states that are more "spread out" or "mixed up" (higher entropy), because there are vastly more disordered microstates than ordered ones. Hyperseed's key caution is that entropy and information measures are not purely observer-free numbers: what you* *call "entropy" depends on which variables you track and which distinctions you ignore (your coarse-graining). If you change the description level (the resolution and the choice of state variables), you can change whether entropy appears to increase, decrease, or stay roughly constant. So "second-law-like" claims should be read as statements about an effective description, not as a single context-independent monotonic quantity attached to reality.*

This caution does not deny that there are rigorous second-law theorems under specific assumptions; rather, it highlights that the monotone quantity is tied to a modeling choice (e.g. a macrostate partition, a Markov approximation, a bath/system split, or an assumed equilibrium family). In many applied settings, what is operationally measured as "entropy increase" is really the growth of uncertainty about microstate details relative to a chosen macro-description, or the dissipation of free energy relative to a specified environment. From Hyperseeds perspective, this makes the second law a particularly vivid example of an ontology point: monotonic "progress measures" typically live at a level of abstraction, and changing abstraction can change what looks monotone.

Listed in the concept inventory without a dedicated standalone formal definition in this version; the main text emphasizes the observer-relative/coarse-graining-relative reading (see Remark 347).

Concept 165 (Self vs. Other)**.** *Appendix label: concept:SelfVsOther.*
Body definition label: hsdef:SelfVsOther (see § 211).

Informal definition. *"Self vs. Other" is the distinction an agent draws between (i) what it treats as "me" or "under my control / part of my boundary" and (ii) what it treats as external: the environment, other agents, and generally the sources of uncertainty and resistance. The key point is that this boundary is not automatically the same as the physical boundary of the agents hardware or body. Instead, it is an agent-relative modeling and control choice: tools, collaborators, submodules, future copies, or institutional processes can be treated as "self-like" to varying degrees, while parts of ones own body or cognition can be treated as "other-like" when they are opaque, adversarial, or only weakly controllable.*

In Hyperseed terms, the Self/Other cut is an ontological move that shapes agency: it affects credit assignment (who caused what), planning (which degrees of freedom are optimized vs. treated as exogenous), and moral or normative interpretation (which outcomes count as "my actions"). The boundary can shift with resources and competence: a rich-resource mind may internalize larger toolchains, predictive models, or coordination structures into its effective "self," while simultaneously treating more of the external world as predictable and hence quasi-internal for planning purposes.

This boundary is often not perfectly crisp. For example, you can feel both ownership and alienation about a thought, or both identification and separation from a group. In Hyperseed terms, this is not a bug to be eliminated; it is often real information about a complex, layered system.

Formal definition. *Fix a context c with state space X_c and action interface A. A Self/Other distinction is a partition (or, more generally, a factorization) of the modeled variables into components*

$$X_c \cong X_c^{\mathrm{self}} \times X_c^{\mathrm{other}},$$

together with an attribution rule that assigns which transitions are treated as endogenously controlled ("self-caused") versus exogenously generated ("other-caused"). Equivalently, it is a choice of conditional dynamics model of the form

$$P(x' \mid x, a) \;=\; P\Big(x^{\mathrm{self}\,\prime}, x^{\mathrm{other}\,\prime} \mid x^{\mathrm{self}}, x^{\mathrm{other}}, a\Big),$$

where the modeling assumption specifies which components of x' are treated as directly influenced by a (and thus by the agent) and which are treated as background processes or responses.

This definition is compatible with "soft" boundaries: instead of a hard partition, one may use a graded notion of self-likeness (e.g. a weighting over variables, or a set of partially controllable factors), but the core content remains that the agent commits to a representational boundary for purposes of prediction, control, and responsibility assignment. In particular, a Self/Other cut interacts with directed difference (Seconds): when a mismatch occurs, whether it is interpreted as "my mistake" (update X_c^{self} or the policy) or "world pushback" (update X_c^{other} or the environment model) depends on where the boundary was drawn.

Formal definition: evidential approach *(Self-evidence and self groupings, per Definition 211.) Fix a time interval I. Let X_I be the set of representational tokens (states, features, self-model elements, etc.) available on I. A self-evidence function is a map*

$$\sigma_I : X_I \to [0,1]^2, \qquad \sigma_I(x) = (\sigma_I^+(x), \sigma_I^-(x)),$$

where $\sigma_I^+(x)$ is the evidence that x belongs to self and $\sigma_I^-(x)$ is the evidence that x does not belong to self (no consistency constraint is imposed). Given thresholds $\theta^+, \theta^- \in [0,1]$, define the crisp self grouping

$$\mathrm{Self}_\theta(I) := \{x \in X_I : \sigma_I^+(x) \geq \theta^+ \text{ and } \sigma_I^-(x) \leq \theta^-\},$$

and define $\mathrm{Other}_\theta(I) := X_I \setminus \mathrm{Self}_\theta(I)$.

Concept 166 (Self-Continuity)**.** *Appendix label: concept:SelfContinuity.*
Body definition label: hsdef:SelfContinuity (see § 216).

Informal definition. *Self-continuity is "how much the same self" a system remains over time. A mind can change in many ways while still being recognizably continuous (learning, maturing, forgetting details), but it can also change in ways that break continuity (radical rewrites of core patterns, values, or identity structure). In this formalism, self-continuity is measured by how well the system's internal pattern-flow structure at one time can be matched to its internal pattern-flow structure at a later time, allowing relabeling but requiring that the structural relationships are preserved to a high degree.*

Formal definition. *(Self-continuity score, per Definition 520.) Let F_I and F_J be the relevant internal flow networks (pattern-flow structures) over intervals I and J, with node sets X_I and X_J. Define the self-continuity score from I to J as*

$$\mathrm{SCS}(I \to J) \ := \bigvee_{f : X_I \to X_J} \deg(f; F_I, F_J),$$

i.e. the join (supremum) over all maps f of the approximate-morphism degree $\deg(f; F_I, F_J)$.

Concept 167 (Self-Transcendence)**.** *Appendix label: concept:SelfTranscendence.*
Body definition label: hsdef:SelfTranscendence (see § ??).

Informal definition. *Self-transcendence is the tendency to go beyond your current identity: to expand what you can understand, what you can care about, and what kinds of lives or minds you can become. It includes the willingness to revise your self-model and even your values when you discover something larger, deeper, or more coherent than your current framing. Hyperseed treats self-transcendence as being in real (and often productive) tension with individuation and self-continuity: you want to remain meaningfully yourself, but also to grow beyond your current self (Hyperseed-Concept 96; Hyperseed-Concept 166).*

Formal definition. *(Individuation and self-transcendence as value dimensions, per Definition 257.) Assume* $\mathrm{individuation}, \mathrm{selftranscendence} \in D$ *(value dimensions). High evaluation on* selftranscendence *favors options that expand the representational and motivational space, including revisions or enlargements of D itself (i.e., it is a meta-value that can push toward changing the agent's own value-basis).*

Concept 168 (Self-Weaving Webs)**.** *Appendix label: concept:SelfWeavingWebs.*
Body definition label: hsdef:SelfWeavingWebs (see § 34.4).

Informal definition. *A self-weaving web is a network of patterns that can keep itself going. Roughly: once enough of its patterns are active, they mutually reinforce each other so that each one keeps getting re-produced rather than dying out. This is the cognitive analogue of an autocatalytic set in chemistry: instead of molecules catalyzing each other's production, you have habits, concepts, behaviors, and subroutines that repeatedly regenerate one another's support (Hyperseed-Concept 11). Intuitively, a self-weaving web is a "self-sustaining cluster" inside a larger mind.*

Formal definition. *Fix a reinforcement relation $A : P \times P \to V$ and a threshold $\theta \in V$. A nonempty subset $S \subseteq P$ is called θ-autocatalytic if there exists a state $s : P \to V$ such that:*

1. *(support on S) for all $p \in S$, $s(p) \geq \theta$; and*
2. *(self-production) for all $p \in S$, $(A?s)(p) \geq \theta$.*

A pattern web (P, A) is a self-weaving web (at threshold θ) if it contains a θ-autocatalytic subset.

Concept 169 (Simplicity)**.** *Appendix label: concept:Simplicity.*
Body definition label: hsdef:Simplicity (see § 30.3).

Informal definition. *In this framework, "simplicity" means: how cheaply can you build something from simpler pieces, given a chosen set of building blocks and a chosen way of combining them? An object is simpler if there is a short, low-cost construction that assembles it from simpler parts, paying not only for the parts but also for the overhead of putting them together. If an object cannot be built at all within the allowed combination rules, then (in that system) it is treated as having infinite simplicity-cost.* ***Formal definition.*** *Let $(X_C, *, e)$ be a combination system with overhead t_C. A function $s_C : X_C \to [0, \infty]$ is compositionally admissible if:*

1. *$s_C(e) = 0$, and*
2. *for all $x \in X_C$,*
$$s_C(x) \leq \inf_{y,z \in X_C \,:\, y*z=x} \big(s_C(y) + s_C(z) + t_C(y, z)\big),$$
with the understanding that $\inf \emptyset = \infty$.

Define the compositional simplicity s^*_C to be the least (pointwise) function satisfying these constraints.

Concept 170 (Society)**.** *Appendix label: concept:Society.*
Body definition label: hsdef:Society (see § 332).

Informal definition. *A society is a group of agents who interact in a sustained way, influence one another, and share durable structures that help patterns persist beyond any one individual's moment-to-moment activity. Those durable structures include things like tools, documents, rituals, institutions, shared environments, and shared memory stores. Hyperseed also emphasizes that in many societies there are real material and biological constraints (resources, maintenance, reproduction) that shape what patterns can persist; but for many cognitive and cultural questions, it is enough to model who interacts with whom, how strongly, and what shared artifacts they can access.*

Formal definition. *A society is a tuple*

$$\mathrm{Soc} := (A, G, K, \mathcal{A}, \mathrm{Acc}, D),$$

where: (i) A is a set of agents; (ii) $G = (A, E)$ is an interaction graph; (iii) K is a coupling weight on E; (iv) $\mathcal{A}$ is a set of shared artifacts; (v) $\mathrm{Acc} \subseteq A \times \mathcal{A}$ is an access relation; and (vi) D is *(optional) additional domain structure capturing the resource-flow and reproduction dependencies relevant to the society (metabolism, exchange, child rearing, maintenance of shared infrastructure, etc.).*

Concept 171 (Social Roles)**.** *Appendix label: concept:SocialRoles.*
Body definition label: hsdef:SocialRoles (see § 22.6).

Informal definition. *A social role is a recognizable pattern in how someone tends to interact with others in a group. For example, a "teacher" is someone who usually explains and answers, a "student" is someone who usually asks questions and tries to learn, a "worker" tends to follow directions in exchange for resources, and "colleagues" coordinate as peers. In Hyperseed, roles are not permanent labels on people; they are about what happens in an interaction episode, and a single person can play different roles in different situations (or even partly play multiple roles at once).*

Formal definition. *Fix a set of act types T (e.g. question/response/command/statement) and a set of artifact-operations (e.g. read, write, modify, use). A* role template *is a finite labeled directed graph*

$$R = (V_R, E_R, \lambda),$$

where V_R is a finite set of abstract participant-slots (e.g. "teacher", "student"), and $\lambda : E_R \to T \cup \{\textit{artifact ops}\}$ labels each directed edge by an act type or artifact-operation. Given a society $\mathrm{Soc} = (A, G, K, \mathcal{A}, \mathrm{Acc}, D)$ and an observed interaction episode Ep (a finite subgraph or sequence of acts), an instantiation *of R is a map $\rho : V_R \to A$. A* fit score *is a function*

$$\mathrm{Fit}(R, \rho; \mathrm{Ep}) \in [0, 1]$$

measuring how well the labeled edges of R match the acts/edges observed in Ep under the assignment ρ (optionally weighted by couplings K and attention weights). The role-template R is said to be present *in Ep under ρ when $\mathrm{Fit}(R, \rho; \mathrm{Ep})$ exceeds a chosen threshold.*

Concept 172 (Soggy Predicates)**.** *Appendix label: concept:SoggyPredicates.*
Body definition label: hsdef:SoggyPredicates (see § 35.5).

Informal definition. *A soggy predicate is a way to combine many imperfect "checks" of whether something has a property. Imagine you have several tests, viewpoints, or measurements, and each one gives some evidence that the property is true and some evidence that it is false. A soggy predicate just takes a weighted average of those pieces of evidence, so you get one combined "for" score and one combined "against" score. The point is to keep track of mixed evidence honestly, instead of forcing everything into a single yes/no answer.*

Formal definition. *Fix a universe X. Fix a finite observation set O and weights $w_o \geq 0$ such that $\sum_{o \in O} w_o = 1$. Assume each observation $o \in O$ yields an evidential evaluation $e_o : X \to [0, 1]^2$. The induced* soggy predicate *is the function $F : X \to [0, 1]^2$ defined by*

$$F(x) := \sum_{o \in O} w_o \, e_o(x),$$

where the sum is taken componentwise in $[0, 1]^2$.

Concept 173 (Space)**.** *Appendix label: concept:Space.*
Body definition label: hsdef:Space (see § 222).

Informal definition. *In Hyperseed, a "space" is any setup where you can compare things by saying which ones are closer, more similar, or better matching. Physical space is one example (near vs. far), but you can also have a "meaning space" where ideas are close if they mean similar things.*

Hyperseed uses this idea to compare what is inside a mind (internal tokens, concepts, body-schema items) with what is in the world: you embed both into a shared comparison space, and then closeness in that space expresses how well they correspond. ***Formal definition.*** *A V-space is a V-category S (objects with V-valued hom/similarity) together with a chosen scalarization $\|\cdot\| : V \to [0, \infty]$. The induced extended pseudo-metric on objects is*

$$d_S(a, b) := \|S(a, b)\|.$$

Given V-graphs $M = (X^{\text{in}}, M)$ and $E = (X^{\text{out}}, E)$, a correspondence space *for (M, E) is a V-space S together with maps $i : X^{\text{in}} \to \mathrm{Ob}(S)$ and $e : X^{\text{out}} \to \mathrm{Ob}(S)$, which induce a V-valued correspondence relation*

$$R_S(x, y) := S(i(x), e(y)) \qquad (x \in X^{\text{in}},\, y \in X^{\text{out}}).$$

Concept 174 (Species)**.** *Appendix label: concept:Species.*
Body definition label: hsdef:Species (see § 41.5).

Informal definition. *A species is a "kind" that stays stable through reproduction: if something counts as part of the species, then its offspring (within an agreed tolerance of what counts as "the same kind") also count as part of the species. In other words, the species is closed under the reproduction process. Hyperseed also allows reproduction to be done by groups or composite systems (not only single individuals); that just means the reproduction relation can start from a coalition object instead of a single entity.*

Formal definition. *Fix an observer/context O and tolerance $\epsilon > 0$. Let $\to_O$ be the directed relation on systems given by*

$$S \to_O S' \iff \text{there exists a reproduction event from } S \text{ to } S' \text{ (within tolerance } \epsilon).$$

A nonempty set C of systems is called a species *(for O) if:*

- *(closure) whenever $S \in C$ and $S \to_O S'$, then $S' \in C$; and*
- *(regeneration) for each $S' \in C$, there exists $S \in C$ with $S \to_O S'$ (except possibly for a designated "initial" subset, if one wishes to allow an origin event).*

Concept 175 (Specific Entity)**.** *Appendix label: concept:SpecificEntity.*
Body definition label: hsdef:SpecificEntity (see § 116).

Informal definition. *A specific entity is what you normally mean by "that particular thing" over time: a particular person, a particular dog, a particular company, a particular chair. It is not just one snapshot. Instead, it is a time-ordered stream of occurrences that stay connected and similar enough that we treat them as "the same thing continuing." Hyperseed treats this as a pattern with default rules that can be bent in weird edge cases (so identity can be graded or ambiguous instead of exploding into contradiction).*

Formal definition. *A specific entity in context C is specified by an occurrence stream $\{x_t\}_{t \in T}$ together with a coherence rule R such that:*

1. *(Same-time identity) If $t = t'$ and both x_t and $x_{t'}$ occur, then they are identified: $x_t = x_{t'}$.*
2. *(Near-time similarity) If t and t' are close in proto-time, then the association $\mathrm{assoc}_C(x_t, x_{t'})$ is high (Definition 100), or equivalently the property-sets overlap strongly.*

The strength of "being the same specific entity" is graded by how well the stream satisfies these rules.

Concept 176 (Spiritual and/or Psychedelic Communion)**.** *Appendix label: concept:SpiritualPsychedelicCommunio*
Body definition label: hsdef:SpiritualPsychedelicCommunion (see § 374).

Informal definition. *This concept aims to capture a common kind of report in spiritual or psychedelic contexts: a person feels they are in deep connection or "communion" with something that seems bigger than their usual self. That "something" might be another person, a group-mind, nature, an ecosystem-scale process, or an unusual mind-model accessed in an altered state. Hyperseed does not try to settle what is metaphysically real here; it only gives a structural test for when an experience has this character.*

Formal definition. *Let m, n be minds and consider the dyadic community $C = \{m, n\}$. We say that m has a spiritual experience (spiritual communion) with n on an interval I if:*

1. $\mathrm{Comm}(C, I; w)$ *holds, and*

2. $\mathrm{Scope}(n, I) \geq \mathrm{Scope}(m, I) + \delta$ *for some fixed $\delta > 0$,*

so that n is (in this operational sense) broader-scope than m on I. Psychedelic/entheogenic procedures are modeled as interventions that shift the relevant intensity data w in ways that tend to increase the communion margin and/or the designation of communion episodes (see the entheogen intervention definition following Definition 374 in the main text).

Concept 177 (Spirituality)**.** *Appendix label: concept:Spirituality.*
Body definition label: hsdef:Spirituality (see § 374).

Informal definition. *Spirituality is the ongoing orientation (of a person or a community) toward experiences of deep connection and meaning that feel larger than ordinary everyday self-concern. It includes the habits and practices people use to seek, recognize, integrate, and return to such experiences (for example: meditation, prayer, ritual, moral discipline, contemplative study, and sometimes psychedelics). In Hyperseed terms, spirituality is less about believing a particular theory and more about building reliable ways of entering, stabilizing, and learning from spiritual communion episodes.*

Concept 178 (States of Consciousness)**.** *Appendix label: concept:StatesOfConsciousness.*
Body definition label: hsdef:StatesOfConsciousness (see § 239). ***Informal definition.*** *A state of consciousness is a mode your mind can be in (waking, dreaming, highly focused, spaced out, hallucinating, "oceanic" states, etc.) that changes what you notice, how your thoughts connect, how much you reflect on yourself, and how strongly experiences turn into habits. Hyperseed models these states as different settings of a small "control panel" of parameters: how tightly you are anchored to sensory input, how strongly internal associations take over, how much introspection is amplified, how hard you push for coherence, and how strongly habit and resonance mechanisms couple things together (Hyperseed-Concept 155, Hyperseed-Concept 159).*

Formal definition. *A state-of-consciousness parameter tuple is*

$$\Lambda := (\lambda_{\mathrm{ext}}, \lambda_{\mathrm{int}}, \lambda_{\mathrm{ref}}, \lambda_{\mathrm{coh}}, \rho_{\mathrm{hab}}, \rho_{\mathrm{res}}),$$

where:

- λ_{ext} *controls anchoring to current sensory registration (Section 13);*
- λ_{int} *controls internal completion/association (pattern-web propagation; Section 11);*
- λ_{ref} *controls introspection gain (Section 17.3);*
- λ_{coh} *controls coherence preference (Section 17.4);*

- ρ_{hab} *controls habit reinforcement strength (Section 12);*
- ρ_{res} *controls resonance coupling strength (Sections 3.9 and 12).*

Concept 179 (Stimulate)**.** *Appendix label: concept:Stimulate.*
Body definition label: hsdef:Stimulate (see § 326).

Informal definition. *Think of a "claim" as a statement you might believe (like "it will rain" or "the stove is hot"). An action* stimulates *a claim when doing it tends to give you more evidence that the claim is true than evidence that it is false. An action* inhibits *a claim when it tends to push evidence the other way. This gives a simple, practical meaning to actions: they matter because they tilt what evidence you have for the things you care about.*

Formal definition. *Fix a claim set L (Section 20) and let β be the agent's belief state. Assume that executing an action a in observation context o produces an effect increment*

$$\epsilon_{a,o} : L \to V,$$

so that a belief update step has the form $\beta \leftarrow \beta \oplus \epsilon_{a,o}$ (possibly followed by closure). Write $\epsilon_{a,o}(\phi) = (\epsilon^{+}_{a,o}(\phi), \epsilon^{-}_{a,o}(\phi))$ for $\phi \in L$. Then:

- *a stimulates ϕ at o if $\epsilon^{+}_{a,o}(\phi) > \epsilon^{-}_{a,o}(\phi)$;*
- *a inhibits ϕ at o if $\epsilon^{-}_{a,o}(\phi) > \epsilon^{+}_{a,o}(\phi)$;*
- *a is neutral on ϕ at o if $\epsilon^{+}_{a,o}(\phi) = \epsilon^{-}_{a,o}(\phi)$.*

Concept 180 (Student)**.** *Appendix label: concept:Student.*
Body definition label: hsdef:Student (see § 341).

Informal definition. *A student is someone playing the learning role in an interaction: they seek information, ask questions, pay attention, and then actually update what they believe or can do so they get better at tasks. In Hyperseed this is defined operationally (by observable effects on capability and uncertainty), not by age, enrollment, or credentials. The point is that "student" describes a pattern of behavior and improvement over an episode.*

Formal definition. *Fix a claim language L and an update scheme for messages as in Section 22.5. An agent j plays the role of student relative to an agent i over an episode if it exhibits the complementary pattern to the teacher-role of Definition 341: j performs question acts toward i and updates its beliefs/skills so that its downstream performance on relevant tasks improves (at fixed or reduced effort). (Equivalently: j's question/attention allocation plus sustained assimilation leads to capability gain; see Definition 341 and its remark, and related weakness motivations in [2,3].)*

Concept 181 (Structural Complexity)**.** *Appendix label: concept:StructuralComplexity.*
Body definition label: hsdef:StructuralComplexity (see § 269).

Informal definition. *Structural complexity is a way to say how "complicatedly organized" a system's behavior is over time. Imagine you watch a system interact with its environment and you write down the history of what happens. If the history can be summarized by a short, simple description (a small rulebook, a simple repeating pattern, a compact story), then the structural complexity is low. If any good description has to keep track of many interacting parts, many exceptions, or many nested regularities, then the structural complexity is high. Importantly, this depends on the observer: what the observer can notice and what kind of descriptions the observer is allowed to use (see Sections 8 and 9).*

Formal definition. *Fix an observer/context O and a representational scheme that assigns a description length (or, more generally, a weakness/clarity score) to interaction histories. Let*

$SC_O(h_{0:T})$ denote the structural complexity of an interaction history $h_{0:T}$ as assessed by O (see Sections 8 and 9 for weakness-based and compression-based complexity measures). A system-in-environment is called a complex interactive system (for O) if

$$E\big[\, SC_O(h_{0:T}) \,\big]$$

exceeds a context-dependent threshold over relevant time horizons T (see § 269).

Concept 182 (Subpattern Hierarchy)**.** *Appendix label: concept:SubpatternHierarchy.*
Body definition label: hsdef:SubpatternHierarchy (see § 33.3). ***Informal definition.*** *A subpattern hierarchy is a "part-of" ordering for patterns. One pattern p is below another pattern q if p can show up inside q after you allow yourself to add some extra "remainder" content. This is the pattern version of ideas like "this word is a substring of that longer word" or "this small shape is contained inside that larger picture." The point is to get a usable hierarchy where small patterns can be treated as building blocks for larger ones.*

Formal definition. *Assume $*$ is associative with identity e, and assume $P_{\mathrm{at}}(p * e; p)$ is nontrivial for each pattern-representer p (so $p * e$ is defined and recognized as a pattern in p). Define a binary relation $\preceq_{\mathrm{sub}}$ by*

$$p \preceq_{\mathrm{sub}} q \quad \Longleftrightarrow \quad \exists r \text{ such that } P_{\mathrm{at}}(p * r; q) \text{ is nontrivial.}$$

Then $\preceq_{\mathrm{sub}}$ is a preorder (see § 33.3).

Concept 183 (Submission)**.** *Appendix label: concept:Submission.*
Body definition label: hsdef:Submission (see § 30.2).

Informal definition. *Submission is the "no extra effort" case of going along with a change. In this document, resistance (Hyperseed-Concept 158) is treated as the extra effort demanded by a change (e.g. you have to strain, fight, or spend energy to maintain your old way). Submission is the opposite extreme: the system can accept the change without needing any additional effort beyond what it was already spending. This is an intentionally minimal, behavior-based notion: it does not try to capture the full lived feeling of submission, only an operational invariant that can be used later.*

Formal definition. *For a directed policy change $H \to H'$ in context C:*

- *Resistance is identified with $\Delta\mathrm{Eff}_C(H \to H')$.*
- *Submission (in this minimal operationalization) occurs when the system follows a policy update that does not increase required effort, i.e. when $\Delta\mathrm{Eff}_C(H \to H') = 0$.*

Concept 184 (Symbol)**.** *Appendix label: concept:Symbol.*
Body definition label: hsdef:Symbol (see § 13).

Informal definition. *A symbol is something that stands for something else because of a learned or agreed-upon relationship, not because it looks like the thing. Words, numerals, traffic signs, musical notation, and programming tokens are all symbols: "electron" does not resemble an electron; it gets its meaning from how it connects to other symbols (laws, measurement terms, mathematical operators, etc.). Symbols also usually live inside a larger system (a language, a calculus, a codebase), where the rules for combining symbols matter (Section 9). This is why symbols are tied to representation (Hyperseed-Concept 27) and to relational structure rather than sensory resemblance.*

Formal definition. *An entity A is a symbol for an entity B if there exists a relational context $C \in \mathrm{Ctx}_{\mathrm{rel}}$ such that A represents B in aspect C. In addition, symbolic representations are assumed to participate in a combinational system (Section 9): symbols can be composed and recombined, and their referents are constrained by their relationships to other symbols.*

Concept 185 (Synthetic)**.** *Appendix label: concept:Synthetic.*
Body definition label: hsdef:Synthetic (see § 286).

Informal definition. *"Synthetic" is about how something came to exist. A lifeform is synthetic if an intelligent agent (or group) intentionally guided its design or assembly (for example, engineered organisms, lab-designed microbes, or designed artificial agents). This is different from "biological," which is about what it is made of (organic chemistry), and different from "evolved," which is about arising through unguided evolutionary and self-organization processes. These labels can overlap: a bio-engineered organism is both biological and synthetic.*

Formal definition. *A lifeform S is:*

- *biological if its dominant implementation substrate is organic chemistry (at the relevant descriptive level);*
- *synthetic if its design/assembly was primarily guided by an intelligent engineering process (as opposed to arising via unguided population-level dynamics);*
- *evolved if it arose via evolutionary and self-organization processes without intentional design.*

These labels are not mutually exclusive at all descriptive levels (e.g. bio-engineered organisms), so we allow them to be multi-valued or p-bit-valued when needed (see § 286).

Concept 186 (Tasks)**.** *Appendix label: concept:Tasks.*
Body definition label: hsdef:Tasks (see § 312).

Informal definition. *A task is a structured "do something over time" situation: an agent repeatedly observes the world, chooses an action, and then gets feedback about how well that action went. In real life the feedback is often mixed (helpful in one way, harmful in another), so the task's evaluation is allowed to be conflicted. This matches Hyperseed's value paraconsistency (Section 18) and covers tasks where success is ambiguous, contested, or internally conflicted.*

Formal definition. *A task is a tuple*

$$T = (O, A, P, r, \gamma),$$

where O and A are finite sets, $P : O \times A \to \Delta(O)$ is a transition kernel over observations, $r : O \times A \times O \to V$ is a p-bit-valued immediate evaluation (reward/penalty evidence), and $\gamma \in (0, 1)$ is a discount factor (see § 312).

Concept 187 (Teacher)**.** *Appendix label: concept:Teacher.*
Body definition label: hsdef:Teacher (see § 341). ***Informal definition.*** *A teacher is defined here by effect, not by title. Agent i is acting as a teacher for agent j if, through interaction, j becomes more able to answer task-relevant questions and perform task-relevant actions. In particular, teaching should reduce the student's weakness / uncertainty in a way that improves downstream performance, ideally without increasing the student's required effort. This ties pedagogy to the document's resource-sensitive weakness machinery (see also [2, 3]) and to control: teaching shapes another agent's future capability landscape.*

Formal definition. *Fix a claim language L and an update scheme for messages as in Section 22.5. An agent i plays the role of teacher for an agent j over an episode if:*

- *i repeatedly performs statement/response acts toward j with messages that (when assimilated) reduce j's weakness/uncertainty on some task-relevant question family (Sections 20 and 21); and*
- *j performs question acts toward i and updates its beliefs/skills so that its downstream performance on relevant tasks improves (at fixed or reduced effort).*

An agent j plays the role of student relative to i if it exhibits the complementary pattern: question/attention allocation plus sustained assimilation leading to capability gain (see § 341).

Concept 188 (Tendency to Reverse Habits)**.** *Appendix label: concept:TendencyToReverseHabits. Body definition label: hsdef:TendencyToReverseHabits (see § 34.1).*

Informal definition. *This is a statistical way to say "behaviors are fading rather than getting reinforced." Pick a pattern p and pick a time T that splits the timeline into "before" and "after." Compute how likely (how prone) the system is to express p before T and after T. If, averaged over many patterns and many split-times, the "after" propensity tends to be lower than the "before" propensity, then the context has a tendency to reverse habits: repeated patterns are being undone or washed out over time.*

Formal definition. *Fix a distribution ν over a class P of patterns, a distribution τ over times $T \in \{0, 1, 2, \dots\}$, and a scalarization π. Define, for each $p \in P$ and time T,*

$$\mathrm{Pre}_T(p) := E[\, p_t(p) \mid t < T\,], \qquad \mathrm{Post}_T(p) := E[\, p_t(p) \mid t > T\,],$$

whenever these conditional expectations are defined. The context C manifests the tendency to reverse habits (relative to (P, ν, τ, π)) if

$$E_{p\sim\nu,\, T\sim\tau}\, \pi\big(\mathrm{Post}_T(p) - \mathrm{Pre}_T(p)\big) \;<\; 0$$

(see § 34.1).

Concept 189 (Tendency to Take Habits)**.** *Appendix label: concept:TendencyToTakeHabits. Body definition label: hsdef:TendencyToTakeHabits (see § 34.1).*

Informal definition. *This is the opposite of reversing habits: it is a statistical way to say "practice makes patterns stronger." Using the same idea of splitting time into "before" and "after," the context has a tendency to take habits if, on average, patterns become more likely after the split than before. Not every pattern must increase; the claim is that there is a systematic time-direction bias toward reinforcement, relative to the chosen ensemble of patterns and time windows.*

Formal definition. *Fix a distribution ν over a class P of patterns, a distribution τ over times $T \in \{0, 1, 2, \dots\}$, and a scalarization π. Define, for each $p \in P$ and time T,*

$$\mathrm{Pre}_T(p) := E[\, p_t(p) \mid t < T\,], \qquad \mathrm{Post}_T(p) := E[\, p_t(p) \mid t > T\,],$$

whenever these conditional expectations are defined. The context C manifests the tendency to take habits (relative to (P, ν, τ, π)) if

$$E_{p\sim\nu,\, T\sim\tau}\, \pi\big(\mathrm{Post}_T(p) - \mathrm{Pre}_T(p)\big) \;>\; 0$$

(see § 34.1).

Concept 190 (Third (Peircean Category))**.** *Appendix label: concept:ThirdPeircean. Body definition label: hsdef:ThirdPeircean (see § 46).*

Informal definition. *A Third is a stable rule-like regularity that links many individual interactions. If a Second is a one-off dyadic event ("this interacts with that"), then a Third is the pattern that governs lots of such events ("pairs like this usually behave this way," "treat these kinds of pairs as distinct," "these pairs are interchangeable"). In the simplest crisp case, a Third is just a set of ordered pairs where the rule applies. In the graded case used here, it is a rule that can assign degrees of evidence (possibly with mixed positive/negative components) to each pair.*

Formal definition. *For a context c, a Third is a grouping of Seconds. Concretely, we model a Third as a (possibly fuzzy) relation*

$$R : X_c \times X_c \to V,$$

where V is a commutative quantale as in Section 3.4. In the simplest crisp case, a Third is just a subset $R \subseteq X_c \times X_c$ (see § 46).

Concept 191 (Tools)**.** *Appendix label:* `concept:Tools`*.*
Body definition label: `hsdef:Tools` *(see § 64).* ***Informal definition.*** *A* tool *is something (usually made on purpose) that you can use to reliably make certain things happen. Compared to acting with your bare hands and brain, a tool either lets you do new kinds of actions at all (like using a telescope to see distant things), or it makes an action easier, faster, safer, or more accurate (like using a calculator to avoid arithmetic mistakes). In the Hyperseed view, tools matter because they change what an agent can effectively do: they expand the agent's practical menu of actions and outcomes, or reduce the effort needed to get an outcome. This connects back to agency as closed-loop control (Section 14) and tasks as objective-achievement problems (Section 21).*

Formal definition. *A tool (relative to an agent class and an observer/context O) is an entity T in a reality-system together with an affordance profile Aff_T that is:*

- ***usable:*** *there exists a nontrivial subset $A_T \subseteq A$ and effects $E_T \subseteq E$ such that $\mathrm{Aff}_T(a, e)$ has substantial positive evidence for some $(a, e) \in A_T \times E_T$; and*
- ***purpose-supporting:*** *for some objective functional (Section 18), the availability of T strictly increases expected objective achievement for typical users in the class, via the induced action–effect couplings.*

Concept 192 (Transfer Learning)**.** *Appendix label:* `concept:TransferLearning`*.*
Body definition label: `hsdef:TransferLearning` *(see § 527).*

Informal definition. Transfer learning *is what happens when learning to solve one problem helps you solve a different problem faster or better. For example, once you learn the basic idea of balancing and steering on a bicycle, it is easier to learn a skateboard or a motorcycle. In the math here, the key idea is: two different "ways of representing problems" (two paradigms) support transfer if you can systematically translate the problem-solving dynamics from one representation into the other with only a bounded amount of mismatch. The appearance of γ is about stability: when the underlying update rule is contractive (discounted), small translation errors do not blow up over repeated updates, so you can turn "dynamics mismatch" into a controlled bound on solution mismatch.*

Formal definition. *Two paradigms P and Q are* (ϵ, γ)-commensurable on a class of tasks *if there is an assignment mapping each task $M \in P$ to a task $N \in Q$, together with a TransWeave map (T, T_V) such that*

$$\| T_V \circ B_M \; - \; B_N \circ T_V \|_{\infty \to \infty} \;\leq\; \epsilon,$$

uniformly over the task class (with the interpretation that the relevant Bellman-style operators are in a discounted/contractive regime parameterized by γ).

Concept 193 (Transweave)**.** *Appendix label:* `concept:Transweave`*.*
Body definition label: `hsdef:Transweave` *(see § 527).*

Informal definition. *A* transweave *is a structured "translation" between two different descriptions of a problem, designed so that what counts as a good update step in one description corresponds (as closely as possible) to a good update step in the other. The point is not perfect* *sameness, but controlled loss: after you translate and update, you should end up close to where you would have ended up if you updated first and then translated. When such a translation exists with small error, it explains why skills, strategies, or learned structure can be carried from one domain to another (which is exactly what transfer learning needs).*

Formal definition. *Given two weighted MDPs M_w and $M_{w'}$ with Bellman operators B_w and $B_{w'}$, a* TransWeave map *consists of a state/representation map T together with an induced map T_V on value functions such that*

$$\| T_V \circ B_w \; - \; B_{w'} \circ T_V \|_{\infty \to \infty} \; \leq \; \epsilon.$$

We call ϵ the intertwining error*.*

Concept 194 (Tribe)**.** *Appendix label:* `concept:Tribe`*.*
Body definition label: `hsdef:Tribe` *(see § 334).*

Informal definition. A tribe *is a group of agents who are strongly connected to each other through frequent interaction and strong social ties. Because the same people keep interacting, they tend to reinforce shared habits, shared meanings, and shared norms; this is where cultural patterns can become especially stable. In the Hyperseed framing, tribal-scale structure is where morphic resonance and cultural stabilization become easy to see, and later "mindplex" phenomena naturally appear at about this scale because repeated high-coupling interaction can create noticeable coherence.*

Formal definition. *A tribe (relative to a strong-tie threshold τ) is a subset $T \subseteq A$ such that:*

- *the induced strong-tie subgraph $G_\tau[T]$ is (strongly) connected; and*
- *T satisfies a boundedness condition reflecting finite memory/interaction bandwidth, modeled either as a size bound $|T| \leq N$ or as a density condition such as*

$$\frac{1}{|T|(|T|-1)} \sum_{\substack{i,j \in T \\ i \neq j}} \mathbb{I}\big[(i,j) \in E_\tau\big] \; \geq \; \delta,$$

  for some δ close to 1*.*

Concept 195 (Uncertainty)**.** *Appendix label:* `concept:Uncertainty`*.*
Body definition label: `hsdef:Uncertainty` *(see § 32).* ***Informal definition.*** Uncertainty *is how unsure you are about what is going on or which possibility is true. You are highly uncertain when many different possibilities still seem plausible, and you are not very uncertain when one possibility is clearly expected. One simple everyday check is: if you would not be surprised by many distinct outcomes (because they all "still fit"), then you are in a high-uncertainty state; if most outcomes would feel surprising because one is strongly anticipated, then uncertainty is low. Hyperseed emphasizes that uncertainty is not an absolute property of "the world" all by itself: it depends on the observer and what distinctions the observer can make (Distinction: Hyperseed-Concept 98). In other words, what counts as a different "possibility" is already a modeling choice: coarse distinctions can collapse many fine-grained world-states into one "case," thereby reducing uncertainty as represented by that observer, while finer distinctions can expand the possibility space and thereby increase uncertainty. Once you take that seriously, "information" and "uncertainty" must be indexed by a context/observer (Information: Hyperseed-Concept* **??***; Uncertainty: Hyperseed-Concept 195; see also [1]). This indexing is not merely philosophical: it determines which events are treated as alternatives, which variables are being predicted, and which representational constraints (measurement resolution, vocabulary, attention, memory) are assumed when quantifying uncertainty. This*

part of the document sets up the basic tools to talk about probabilistic belief (Probability: Hyperseed-Concept 139), to relate classical Shannon-style uncertainty to distinction-based measures like logical entropy and graphtropy (Hyperseed-Concepts 105 and **??***; [17]), to explain "ineffability" as a resource limit on what can be clearly said (Hyperseed-Concept* **??***), and to clarify resource-sensitive "limit" ideas like (potential) infinity and infinitesimals (Hyperseed-Concepts* **??** *and* **??***). In particular, the intended connection is that both probabilistic uncertainty and distinction-based uncertainty are ways of tracking "how many live alternatives remain" relative to an observer's representational regime: Shannon entropy emphasizes average uncertainty under a probability distribution, while the distinction-based measures emphasize which pairs of possibilities can or cannot be told apart by the observer's available distinctions.*

Formal definition. *Let X be a finite set of possibilities and let an observer/context O have a belief distribution $p_O : X \to [0,1]$ with $\sum_{x\in X} p_O(x) = 1$. The* Shannon entropy *(a standard uncertainty measure) is*

$$H(p_O) \;:=\; -\sum_{x\in X} p_O(x)\log p_O(x).$$

Here the base of log *fixes the units (e.g. base 2 gives bits, base e gives nats); changing the base rescales H by a constant factor and does not change the ordering of uncertainties. Higher $H(p_O)$ corresponds to greater uncertainty in the sense of greater average surprise. More explicitly, if one defines the (self-)information or "surprisal" of an outcome as $I(x) := -\log p_O(x)$, then $H(p_O) = \mathbb{E}_{x\sim p_O}[I(x)]$ is the expected surprisal under the observer's belief distribution. In the finite case, $H(p_O)$ is minimized at 0 when p_O puts all its mass on a single possibility (complete certainty in the modeled space), and it is maximized when p_O is uniform on X, in which case $H(p_O) = \log|X|$; this makes explicit how the size of the possibility set (and thus the observer's chosen distinctions) constrains the maximum representable uncertainty. (Section § 32 also relates this to distinction-based uncertainty measures such as logical entropy and graphtropy.)*

Concept 196 (Unknowability)**.** *Appendix label:* `concept:Unknowability`*.*
Body definition label: `hsdef:Unknowability` *(see § 366).*

Informal definition. Unknowability *is the idea that something cannot be known (or cannot be made stably expressible) in any absolute sense. This can be read as a claim that no matter how you expand resources, change languages, or refine distinctions, the target cannot be reliably tracked, stated, or settled as "known" in a way that is robust across contexts and observers. Hyperseed's key warning is that this is much stronger than everyday "I can't put it into words right now." In particular,* ineffability *is not the same thing as absolute unknowability: even if some target x is ineffable to an observer O at a particular accuracy/effort budget (ϵ, K), it may become expressible if you (i) increase resources (larger K), (ii) change the representational language $\mathcal{D}_O$, or (iii) change the distinction regime (use a different G). This frames ineffability as a contingent, parameterized limitation (indexed by observer, language, and resources), whereas unknowability is an invariant claim that persists across such changes. One way to keep the notions separate in practice is to ask: are we claiming a permanent impossibility result ("no representation suffices"), or merely a present mismatch between what must be tracked and the current representational apparatus ("this representation does not yet suffice")? The practical point is developmental and cultural: many things become sayable only after people learn new concepts, new vocabulary, or new ways of carving up experience. This includes both individual development (skill learning, perceptual training, mathematical formalization) and collective development (shared terminology, measurement conventions, institutional practices), each of which can expand the set of distinctions that an observer-community can reliably make.*

Concept 197 (Values (Ethical)). *Appendix label:* `concept:ValuesEthical`*.*
Body definition label: `hsdef:ValuesEthical` *(see § 17.9).*

Informal definition. Ethical values *are the things a person or agent treats as important when deciding what to do. They include priorities like truthfulness, compassion, fairness, loyalty, safety, beauty, or personal growth. Values are not the same as momentary feelings: they are more like relatively stable "standards" that guide choices across many situations. They also function as lenses for noticing morally salient features of situations (what counts as harm, what counts as respect, what counts as a promise), so the value structure influences both evaluation and attention. A key point is that values can conflict: an action might support one value while harming another, and real decision-making has to handle that without pretending the conflict is not there. Moreover, even within a single named value (e.g. "fairness"), there may be multiple plausible operationalizations that pull in different directions, so a representation that can carry structured disagreement is often more realistic than a representation that forces a single scalar score.*

Formal definition. *Fix an observer/context O, a set of situations (states) S, a set of actions A, and a finite set $\mathcal{D}$ of ethical value dimensions. A (paraconsistent) evaluative representation of ethical values is given by an evaluative field*

$$E_O : S \times A \times \mathcal{D} \;\to\; [0,1]^2, \qquad E_O(s,a;d) = \big(E_O^+(s,a;d),\, E_O^-(s,a;d)\big),$$

where $E_O^+(s,a;d)$ is evidence that doing a in situation s promotes *value-dimension d, and $E_O^-(s,a;d)$ is evidence that it* opposes *d. The set $\mathcal{D}$ together with E_O constitutes the agent's (observer-relative) ethical value structure. Intuitively, E_O^+ and E_O^- allow the agent to record pro and con considerations separately rather than collapsing them immediately into a single net number; this supports later aggregation methods that are sensitive to conflict, uncertainty, or stakeholder disagreement. The use of $[0,1]^2$ is meant as a simple bounded scale for evidential support and evidential opposition (not necessarily probabilities), so different applications may interpret these coordinates as confidence-weighted arguments, model-based estimates, or learned scores.*

Concept 198 (Value Paraconsistency). *Appendix label:* `concept:ValueParaconsistency`*.*
Body definition label: `hsdef:ValueParaconsistency` *(see § ??).*

Informal definition. Value paraconsistency *means you can honestly represent value conflict without the whole decision system breaking. In real life, you can have strong reasons that an option is good and strong reasons that it is bad at the same time (for different values, or even within one value once you look closely). For example, an action might strongly promote safety while strongly violating autonomy, or it might strongly promote honesty while strongly damaging compassion in a specific context; forcing an immediate collapse to a single score can hide these tensions and produce brittle reasoning. A paraconsistent value representation keeps both sides of that evidence instead of forcing you to pick one prematurely, and it avoids the "anything goes" failure mode where a contradiction would make every action seem acceptable. The intended outcome is not indecision but better-structured deliberation: conflict is recorded explicitly so that later procedures (priority setting, negotiation, reflective equilibrium, constraint satisfaction, or context-sensitive aggregation) can address it.*

Formal definition. *In the evaluative-field formalism, value paraconsistency is modeled by using two-sided evidence values*

$$E_O(s,a;d) = \big(E_O^+(s,a;d),\, E_O^-(s,a;d)\big) \in [0,1]^2$$

without imposing a consistency constraint *such as $E_O^+(s,a;d) + E_O^-(s,a;d) \le 1$. This means the representation treats "there is substantial support for promotion" and "there is substantial support* *for opposition" as separately measurable claims that can both be true in the evidential state of the observer, reflecting genuine moral tension, model misspecification, or heterogeneous stakeholder perspectives. Equivalently, for a chosen threshold $\theta \in (0, 1]$, the representation explicitly permits states with*

$$E_O^+(s, a; d) \geq \theta \quad \text{and} \quad E_O^-(s, a; d) \geq \theta$$

(i.e. strong pro- and anti-evidence simultaneously) while still supporting nontrivial preference and decision rules. Concretely, this leaves open multiple downstream policies: one may treat such cases as triggers for additional information gathering, as signals to seek alternative actions that dominate on both coordinates, or as inputs to a decision rule that trades off values while explicitly accounting for unresolved conflict rather than erasing it at the representation layer.

Concept 199 (Variety)**.** *Appendix label:* `concept:Variety`*.*
Body definition label: `hsdef:Variety` *(see § 28.1).*

Informal definition. *A* variety *is a difference that matters. Two things can be different in a simple sense (two separate footsteps, two separate pencils), but sometimes the difference does not matter for what you are doing: you can swap them without changing anything important. Variety is the opposite case: the two things are not only different, they are* not interchangeable *in the current context or task. For example, the "Enter" key and the "Escape" key are both keys, but swapping them changes what happens in a big way.*

Formal definition. *Fix thresholds $\tau_\delta, \tau_\iota \in (0, 1)$. For $x \neq y$ in X_c, define:*

$$\mathrm{Rep}_c(x, y) \iff \mathrm{Dist}_c(x, y)^+ \geq \tau_\delta \quad \text{and} \quad \iota_c(x, y)^+ \geq \tau_\iota,$$

$$\mathrm{Var}_c(x, y) \iff \mathrm{Dist}_c(x, y)^+ \geq \tau_\delta \quad \text{and} \quad \iota_c(x, y)^- \geq \tau_\iota.$$

Intuitively, a repetition is a pair that is experienced as distinct yet context-substitutable; a variety is distinct and non-substitutable.

Concept 200 (Varieties of Knowledge)**.** *Appendix label:* `concept:VarietiesOfKnowledge`*.*
Body definition label: `hsdef:VarietiesOfKnowledge` *(see § 37.6).*

Informal definition. *"Knowledge" is not just one kind of thing. Knowing a fact (like "Paris is in France"), knowing a skill (like riding a bike), knowing a person, knowing a habit, and knowing a rule are stored, learned, and used in very different ways. Hyperseed treats these as different* types *of stored content, each with its own meaning and its own update method, but still able to interact (for example, facts can shape skills, and habits can shape what facts you pay attention to).*

Formal definition. *A knowledge type T for a system S consists of:*

- *a carrier set (or space) K_T of knowledge items;*
- *a semantics map $\mathrm{Sem}_T : K_T \to S$ that interprets items as constraints on patterns, processes, or goals;*
- *an update operator (or family of operators) Upd_T specifying how items are learned, revised, or forgotten; and*
- *an interface specifying which other types it can exchange information with.*

Concept 201 (Vedantic Strata)**.** *Appendix label: concept:VedanticStrata.*
Body definition label: hsdef:VedanticStrata (see § 17.7.1).

Informal definition. *In many Vedantic traditions, people talk about different "layers" or "stages" of awareness: first you are aware of things in the world, then you notice thoughts and* *feelings, then you notice the act of noticing, and so on. In Hyperseed, "Vedantic strata" names this general idea of stacked levels of reflective awareness. Moving to a higher stratum means that more of your own attention and self-awareness becomes steadily present and stable, instead of being only an occasional flash.*

Formal definition. *Let L_{Ob} be an object-level language of propositions about world and mind, and let $I(\phi)$ denote the introspective proposition "ϕ is present to awareness" (i.e., "I am aware of ϕ"). Define an iterated reflective-language hierarchy by*

$$L_{\mathrm{Ob}}^{\mathrm{ref},0} := L_{\mathrm{Ob}}, \qquad L_{\mathrm{Ob}}^{\mathrm{ref},k+1} := L_{\mathrm{Ob}}^{\mathrm{ref},k} \cup \{I(\psi) : \psi \in L_{\mathrm{Ob}}^{\mathrm{ref},k}\}.$$

A (candidate) Vedantic stratum of order k is a stable reflective-workspace regime whose stable contents and introspective contents live in $L_{\mathrm{Ob}}^{\mathrm{ref},k}$; concretely, this can be modeled as a robust fixed point (or stable attractor) of the reflective closure dynamics on that language, for parameters (notably λ_{ref}) that support the corresponding depth of iterated awareness without destabilizing into trivial paradox.

Concept 202 (Weakness)**.** *Appendix label: concept:Weakness.*
Body definition label: hsdef:Weakness (see § 37.7).

Informal definition. *"Weakness" is a number that measures how general (how non-picky) an explanation is. A weak explanation makes only a few important distinctions, so it can apply to lots of slightly different situations and tends to be robust. A strong explanation makes many fine distinctions; it may fit what you have seen extremely well, but it can be brittle because it depends on lots of details. Hyperseed uses weakness as a proxy for simplicity/generalization (Hyperseed-Concept 202; Hyperseed-Concept 143; Hyperseed-Concept 169; [3, 2]).*

Formal definition. *Fix a behavior (or dataset) b. Let $\mathcal{E}(b)$ be a set of candidate explanations of b (models, hypotheses, plans, proofs, causal graphs, etc.). Assume each explanation $E \in \mathcal{E}(b)$ comes with:*

(a) *an adequacy score* $\mathrm{Fit}(E) \in [0,1]$ *(how well it predicts/explains/controls b);*

(b) *a weakness* $\mathrm{Weak}(E) \in (0,1]$ *(how few distinctions it makes; larger is weaker).*

Define the contrivance cost (description-length proxy)

$$\mathrm{Con}(E) := -\log \mathrm{Weak}(E) \in [0, \infty).$$

Concept 203 (Will (Reflective))**.** *Appendix label: concept:WillReflective.*
Body definition label: hsdef:WillReflective (see § 236). ***Informal definition.*** *Ordinary will is choosing what you do next. Reflective will is one step deeper: it is the ability to notice and change your own way of choosing. For example, you can adopt a new rule like "when I'm stressed, I will slow down and re-check my assumptions," and that rule continues to guide you later. In this sense, reflective will is will that can look at itself, and modify itself, not just pick one action.*

Formal definition. *Let S_{Ob} be the observer's internal state space and let U be a set of interventions (world actions and internal interventions such as attention shifts). Let Π be a space of possible will operators $\pi : S_{\mathrm{Ob}} \to U$. A reflective will state is a pair (π, s) with $\pi \in \Pi$ and $s \in S_{\mathrm{Ob}}$. A reflective will update is a map*

$$RWill : \Pi \times S_{\mathrm{Ob}} \to \Pi \times S_{\mathrm{Ob}},$$

which may change both the internal state and the will operator. A stable reflective will regime corresponds to a fixed point (or slowly drifting orbit) of $RWill$.

Concept 204 (Woe)**.** *Appendix label: concept:Woe.*
Body definition label: hsdef:Woe (see § ??).

Informal definition. *"Woe" is the negative side of evaluation: how bad, painful, threatening, or unwanted a situation is judged (or felt) to be. It is broader than physical pain: it can include fear, grief, shame, frustration, etc. Hyperseed treats woe as an evaluative signal that can attach to internal or external processes (physical, mental, social), and it can coexist with some positive evaluation in the same situation.*

Formal definition. *Given a p-bit $v = (v^+, v^-) \in [0,1]^2$, define*

$$J(v) := v^+ \quad (\textit{joy intensity}), \qquad W(v) := v^- \quad (\textit{woe intensity}).$$

Given an evaluative field E_O, define

$$J_O(s, a; d) := J(E_O(s, a; d)), \qquad W_O(s, a; d) := W(E_O(s, a; d)).$$

Concept 205 (Worker)**.** *Appendix label: concept:Worker.*
Body definition label: hsdef:Worker (see § 342).

Informal definition. *A worker is an agent who repeatedly does tasks as part of a plan or stream of instructions coming from some coordinating structure (a boss, an institution, or a collective policy), and receives resources or other value signals in return. This is a structural role in a coordination network: it describes who is taking direction from whom and how value flows, not whether the arrangement is ethical or unethical.*

Formal definition. *A worker is an agent that repeatedly executes commands or participates in a coordinated plan issued by some coordinating structure (a boss, institution, or collective policy), in exchange for resources or other value signals (Section 18).*

Concept 206 (Wu Wei)**.** *Appendix label: concept:WuWei.*
Body definition label: hsdef:WuWei (see § 1981).

Informal definition. *"Wu wei" (roughly, "effortless action") is the idea of achieving outcomes without unnecessary forcing. You still act, but you try to work with the natural tendencies of the situation instead of constantly fighting them. A simple analogy is surfing: you steer and adjust, but you do not try to overpower the wave. In Hyperseed's math layer, wu wei is modeled by optimizing for task success while explicitly penalizing how much you deviate from a passive/default dynamics. This general principle sits behind the more geometric "wu wei geodesic" notion (Hyperseed-Concept 207) and its applications [21].*

Formal definition. *Let X be a state space with a passive transition kernel $P_0(\cdot \mid x)$. A policy π induces a controlled kernel $P_\pi(\cdot \mid x)$. Fix $\lambda > 0$ and define the forcing cost at state x by*

$$\mathrm{Force}_\lambda(P_\pi \| P_0)(x) := \lambda\, \mathrm{KL}(P_\pi(\cdot \mid x) \,\|\, P_0(\cdot \mid x)).$$

Given a bounded state cost $q : X \to \mathbb{R}$, horizon $T \in \mathbb{N}$, initial distribution $\rho_0 \in \Delta(X)$, and terminal cost q_T, define

$$J_T(\pi) := \mathbb{E}\left[\sum_{t=0}^{T-1} \Big(q(x_t) + \mathrm{Force}_\lambda(P_\pi \| P_0)(x_t)\Big) + q_T(x_T)\right],$$

where $x_0 \sim \rho_0$ and $x_{t+1} \sim P_\pi(\cdot \mid x_t)$. A wu wei policy is any minimizer $\pi^* \in \arg\min_\pi J_T(\pi)$.

Concept 207 (Wu Wei Geodesic)**.** *Appendix label: concept:WuWeiGeodesic.*
Body definition label: hsdef:WuWeiGeodesic (see § 1982).

Informal definition. *A wu wei geodesic is the "least-effort path" for transforming an entire probability distribution from one shape to another while staying as close as possible to a passive* *baseline flow. Imagine a cloud of mist: you want it to start in one shape and end in another. The wu wei geodesic is the way of doing that that uses the least combined "pushing" (moving the cloud around) and "resisting" (fighting what the cloud would naturally do). This is exactly Hyperseed-Concept 207 [21].*

Formal definition. *Let $\Omega \subseteq \mathbb{R}^d$ be a domain and let $\rho(x,t)$ be a density on Ω with a velocity field $v(x,t)$ over $t \in [t_0, t_1]$, subject to the continuity equation*

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

Let $p(x,t)$ be a reference (passive) density and let $\epsilon > 0$. Define the Schrödinger-bridge effort functional

$$W_{SB}[\rho, v] := \int_{t_0}^{t_1} \int_{\Omega} \rho(x,t) \left(\frac{1}{2} \|v(x,t)\|^2 + \epsilon \log \frac{\rho(x,t)}{p(x,t)} \right) dx\, dt.$$

Fix boundary densities $\rho(\cdot, t_0) = \rho_0$ and $\rho(\cdot, t_1) = \rho_1$. A wu wei trajectory (wu wei geodesic) between ρ_0 and ρ_1 (relative to passive prior p) is any minimizer (ρ^*, v^*) of $W_{SB}[\rho, v]$ subject to the continuity constraint and boundary conditions.

Concept 208 (Young)**.** *Appendix label: concept:Young.*
Body definition label: hsdef:Young (see § 2.3).

Informal definition. *"Young" refers to the life stage after infancy and before full maturity: the period when a person or organism is still growing quickly and building basic skills, but is no longer an infant. In Hyperseed's coarse biological lifecycle strata (infant/young/mature/old/elderly), "young" is the second stage and is intentionally broad and context-dependent.*

In particular, "young" is used as a stratification term rather than a precise chronological marker: its boundaries may be indexed to species, environment, and the trait under discussion (e.g., physical growth, neurological development, reproductive capability, social role acquisition, or competence in a learned skill). For example, an organism may be "young" with respect to size and motor coordination while already "mature" with respect to a narrow biological function, and Hyperseed treats this kind of partial ordering between developmental dimensions as compatible with the coarse stage label.

The intended use is also comparative: "young" can function as a default category for entities not yet stabilized into their characteristic adult equilibrium (biological, behavioral, or ecological), but already past the highly fragile and rapidly transitioning infant regime. This makes the term suitable for high-level modeling where only a small number of robust stages are tracked and where fine-grained age classes would be spurious precision.

Listed in the concept inventory without a dedicated formal definition in this version. The absence of a formal definition is deliberate here: the concept serves primarily as a linguistic and modeling convenience to align cross-domain discussions (biology, cognition, and lifecycle-dependent capability) without committing to a single universal threshold rule.

Concept 209 (YVerse)**.** *Appendix label: concept:YVerse.*
Body definition label: hsdef:YVerse (see § ??).

Informal definition. *The Y-verse is a "wider world" concept designed to be self-referential. It is a world-description that already contains its own multiverse inside it: if you take the description Y and apply the operation that forms "the multiverse of Y" (call it* Multi*), you get the same Y back. Intuitively, it is a self-consistent fixed point: expanding it into many worlds does not produce something fundamentally new, it reproduces itself. Hyperseed treats this fixed-point signature as a clue that self-reference and non-well-founded semantics may matter at the "ultimate cosmos" level (Hyperseed-Concept 209).*

Concretely, Multi *is intended to represent whatever construction turns a single "world-description" into a space (or family) of worlds compatible with it—for instance by varying contingent parameters, branching on indeterminacy, or unfolding implicit choices. The Y-verse condition says that Y is already closed under this unfolding: it is descriptively complete in the sense that applying the "multiverse expansion" operation does not change the object. In this reading, the self-referentiality is not merely linguistic; it is structural closure under an operator that would otherwise enlarge the described cosmos.*

This also clarifies why the concept is tied to non-well-founded semantics: if Multi *permits recursive world-containment (worlds that include descriptions of world-families that include themselves), then the resulting structures can naturally be modeled using fixed-point and coalgebraic ideas rather than only well-founded hierarchies. The Y-verse is the special case where this recursion stabilizes exactly, yielding a fixed point rather than an infinite strictly ascending chain.*

The reference "§ ??" is a placeholder for the section where the ambient assumptions on Multi *(its domain, codomain, and any monotonicity or continuity properties) are specified; those details govern which fixed points exist and which are mathematically or conceptually preferred.*

Formal definition. *A Y-verse is a fixed point Y of* Multi*:*

$$Y = \mathrm{Multi}(Y).$$

When multiple fixed points exist, one may select $Y := \mathrm{lfp}(\mathrm{Multi})$ (the least fixed point) or $Y := \mathrm{gfp}(\mathrm{Multi})$ (the greatest fixed point) depending on the intended semantics.

Here, "least" and "greatest" are to be understood relative to an ordering on world-descriptions (e.g., an entailment order, an information order, or a refinement order). Under standard conditions—for example, if the world-descriptions form a complete lattice and Multi *is monotone—the existence of* lfp *and* gfp *follows from the Knaster–Tarski theorem, and they can be characterized as*

$$\mathrm{lfp}(\mathrm{Multi}) = \bigwedge\{X : \mathrm{Multi}(X) \le X\}, \qquad \mathrm{gfp}(\mathrm{Multi}) = \bigvee\{X : X \le \mathrm{Multi}(X)\}.$$

Intuitively, choosing lfp(Multi) *corresponds to the minimal self-consistent closure (the smallest description that is already multiverse-complete), whereas choosing* gfp(Multi) *corresponds to the maximal coinductively consistent solution (the largest description compatible with its own multiverse expansion). Which is appropriate depends on whether one interprets Y inductively (built from finite unfolding) or coinductively (validated by a bisimulation-style consistency criterion).*

C.2 Concepts Regarding General Intelligence

Concept 210 (Reward-summable environment)**.** *Appendix label: concept:RewardSummableEnvironment. Body definition label: hsdef:RewardSummableEnvironment (see §21.11).*

Informal definition. *An environment whose total expected cumulative reward is bounded (here normalized to ≤ 1) no matter what policy is used. This guarantees the universal-intelligence sum uses finite terms.*

Formal definition. *An environment μ is reward-summable iff for every policy π,*

$$V_\mu^\pi \;:=\; \mathbb{E}_{\mu,\pi}\left[\sum_{t=1}^{\infty} r_t\right] \;\le\; 1.$$

Concept 211 (Universal intelligence)**.** *Appendix label: concept:UniversalIntelligence. Body definition label: hsdef:UniversalIntelligence (see §21.11).*

Informal definition. *A Legg–Hutter style measure of intelligence: expected cumulative reward averaged across a broad class of computable environments, weighted by an Occam prior favoring simpler environments.*

Formal definition. *For a chosen set $\mathcal{E}$ of reward-summable environments,*

$$\Upsilon(\pi) \;:=\; \sum_{\mu\in\mathcal{E}} 2^{-K(\mu)}\, V^{\pi}_{\mu},$$

*where $K(\mu)$ is Kolmogorov complexity (Hyperseed-Concept **??**) and $V^{\pi}_{\mu} = \mathbb{E}_{\mu,\pi}[\sum_{t\geq 1} r_t]$.*

Concept 212 (GTGI goal function)**.** *Appendix label: concept:GTGIGoalFunction. Body definition label: hsdef:GTGIGoalFunction (see §21.11).*

Informal definition. *A "goal" is treated as a reward-generating rule: it assigns reward values to finite interaction histories.*

Formal definition. *Fix action/observation sets $\mathcal{A}, \mathcal{O}$ and reward set $\mathcal{R} \subseteq [0,1] \cap \mathbb{Q}$. A GTGI goal function is a map*

$$g : (\mathcal{A}\times\mathcal{O})^* \to \mathcal{R},$$

with goal-induced rewards $r_t := g(a_1, o_1, \ldots, a_t, o_t)$ along a trajectory.

Concept 213 (Natural timescale indicator)**.** *Appendix label: concept:NaturalTimescaleIndicator. Body definition label: hsdef:NaturalTimescaleIndicator (see §21.11).*

Informal definition. *A Boolean function indicating which evaluation horizons are "natural" (meaningful) for a given goal in a given environment.*

Formal definition. *A natural timescale indicator is a map*

$$\tau_{g,\mu} : \mathbb{N}_{>0} \to \{0,1\},$$

where $\tau_{g,\mu}(n) = 1$ means it is meaningful to evaluate goal-achievement for g in μ over n steps.

Concept 214 (GTGI context)**.** *Appendix label: concept:GTGIContext. Body definition label: hsdef:GTGIContext (see §21.11).*

Informal definition. *A "context" packages the environment, the goal, and the time-interval over which the goal is to be evaluated.*

Formal definition. *A GTGI context is a triple (μ, g, T) where μ is an environment, g is a goal function, and $T = \{t_1, \ldots, t_2\}$ is a finite time-interval with $|T| := t_2 - t_1 + 1$. For policy π, expected goal-achievement is*

$$V^{\pi}_{\mu,g,T} \;:=\; \mathbb{E}_{\mu,\pi}\left[\sum_{t=t_1}^{t_2} r_t\right], \qquad r_t := g(a_1, o_1, \ldots, a_t, o_t).$$

Concept 215 (Pragmatic general intelligence)**.** *Appendix label: concept:PragmaticGeneralIntelligence. Body definition label: hsdef:PragmaticGeneralIntelligence (see §21.11).*

Informal definition. *A generalization of universal intelligence that explicitly separates (i) distributions over environments and (ii) distributions over goals in those environments, and averages goal-achievement over goal-appropriate timescales.*

Formal definition. *Given distributions ν over environments and $\gamma(\cdot, \mu)$ over goals conditional on μ,*

$$\Pi(\pi) \;:=\; \sum_{\mu,g,T} \nu(\mu)\,\gamma(g,\mu)\,\tau_{g,\mu}(|T|)\,V^{\pi}_{\mu,g,T},$$

whenever the sum is convergent.

Concept 216 (Resource-consumption distribution)**.** *Appendix label: concept:ResourceConsumptionDistribution. Body definition label: hsdef:ResourceConsumptionDistribution (see §21.11).*

Informal definition. *A probabilistic model of how much computation an agent uses while pursuing a given goal in a given environment over a given interval.*

Formal definition. *Fix a discrete resource scale $\mathcal{Q} \subseteq \mathbb{R}_{>0}$. A resource-consumption distribution for context (μ, g, T) is a probability distribution $\eta_{\mu,g,T} : \mathcal{Q} \to [0, 1]$ with $\sum_{Q\in\mathcal{Q}} \eta_{\mu,g,T}(Q) = 1$.*

Concept 217 (Efficient pragmatic general intelligence)**.** *Appendix label: concept:EfficientPragmaticGeneralIntellig Body definition label: hsdef:EfficientPragmaticGeneralIntelligence (see §21.11).*

Informal definition. *Pragmatic general intelligence adjusted for computational efficiency: goal-achievement is weighted by the inverse of resource usage, so achieving the same goals with less computation scores higher.* ***Formal definition.*** *With $\eta_{\mu,g,T}$ as a distribution over resource usage $Q \in \mathcal{Q}$,*

$$\Pi^{\mathrm{eff}}(\pi) \;:=\; \sum_{\mu,g,T,Q} \nu(\mu)\, \gamma(g, \mu)\, \tau_{g,\mu}(|T|)\, \eta_{\mu,g,T}(Q)\, \frac{V^{\pi}_{\mu,g,T}}{Q},$$

whenever the sum is convergent. In words, $\Pi^{\mathrm{eff}}(\pi)$ aggregates an agent's achieved value across a family of contexts (μ, g, T) while explicitly discounting (or rewarding) the amount of resource usage Q required to obtain that value: the ratio $V^{\pi}_{\mu,g,T}/Q$ is an efficiency-like quantity, and $\eta_{\mu,g,T}(Q)$ averages that quantity over the possible realized resource consumptions in the given context. The factors $\nu(\mu)$, $\gamma(g, \mu)$, and $\tau_{g,\mu}(|T|)$ serve as context weights: $\nu(\mu)$ prioritizes environments, $\gamma(g, \mu)$ prioritizes goal specifications given the environment, and $\tau_{g,\mu}(|T|)$ prioritizes (for example) different horizon lengths or episode durations. The convergence qualifier is substantive: since the index set may be countably infinite, the expression is well-defined only when the weighted series of efficiency-adjusted values is absolutely summable (or otherwise convergent under the intended summation convention). One common sufficient condition is that Q is almost surely bounded away from 0 under $\eta_{\mu,g,T}$ (so the ratio does not blow up), and that $V^{\pi}_{\mu,g,T}$ and the context weights are such that the weighted expectation remains finite.

Concept 218 (Intellectual breadth)**.** *Appendix label: concept:IntellectualBreadth. Body definition label: hsdef:IntellectualBreadth (see §21.11).*

Informal definition. *A diversity measure: not just "how well" an agent performs, but how widely spread the contexts are in which it performs well. Formally, it is the Shannon entropy of a normalized distribution over contexts weighted by the agent's (efficiency-adjusted) performance. Equivalently, the metric treats the agent's overall "competence mass" as something that can be distributed unevenly across contexts: an agent that is extremely strong in a narrow cluster of contexts has low breadth, whereas an agent that allocates substantial competence across many distinct contexts has higher breadth. This also makes breadth sensitive to contextual coverage rather than to raw scale alone: if an agent doubles its performance uniformly in all contexts, the normalization below removes that uniform scaling so that the breadth score reflects the shape of the distribution rather than its magnitude.*

Formal definition. *Define context weights*

$$\chi_{\mathrm{Con}_\pi}(\mu, g, T) \;:=\; \sum_{Q\in\mathcal{Q}} \eta_{\mu,g,T}(Q)\, \frac{V^{\pi}_{\mu,g,T}}{Q}.$$

The quantity $\chi_{\mathrm{Con}_\pi}(\mu, g, T)$ can be read as an efficiency-adjusted expected value in the specific context (μ, g, T), with the expectation taken over the random or variable resource usage Q. In particular, contexts in which π attains high value while consuming little resource produce large

χ_{Con_π}, while contexts that require large Q for modest value are down-weighted. If $V^\pi_{\mu,g,T}$ can be negative in the underlying framework, then χ_{Con_π} may also be negative; in such cases, interpreting $\chi^P_{\mathrm{Con}_\pi}$ as a probability distribution would require additional conventions (e.g. truncation, shifting, or restricting to nonnegative values) to preserve the Shannon-entropy interpretation.

Normalize:

$$\chi^P_{\mathrm{Con}_\pi}(\mu, g, T) := \frac{\nu(\mu)\gamma(g,\mu)\tau_{g,\mu}(|T|)\chi_{\mathrm{Con}_\pi}(\mu, g, T)}{\sum_{\mu',g',T'} \nu(\mu')\gamma(g',\mu')\tau_{g',\mu'}(|T'|)\chi_{\mathrm{Con}_\pi}(\mu', g', T')}.$$

This normalization produces a context-indexed distribution (when the denominator is positive and finite) that answers: "of the agent's total weighted efficiency-adjusted performance, what fraction is contributed by context (μ, g, T)?" The denominator therefore plays the role of a total mass or partition function; it must be nonzero for the normalization to be meaningful, and finiteness parallels the convergence requirements in efficiency-style sums. Because $\nu(\mu)\gamma(g,\mu)\tau_{g,\mu}(|T|)$ appear both here and in $\Pi^{\mathrm{eff}}(\pi)$ above, the breadth measure is aligned with whatever priors or importance weights are used to aggregate performance, differing only in that it focuses on the distributional spread rather than on the aggregate level.

Then

$$B_{\mathrm{int}}(\pi) \;:=\; -\sum_{\mu,g,T} \chi^P_{\mathrm{Con}_\pi}(\mu, g, T)\,\log\bigl(\chi^P_{\mathrm{Con}_\pi}(\mu, g, T)\bigr).$$

This is the Shannon entropy of $\chi^P_{\mathrm{Con}_\pi}$; the choice of logarithm base determines units (e.g. base 2 gives bits, base e gives nats) but does not change the ordering of breadth values across agents. When $\chi^P_{\mathrm{Con}_\pi}$ concentrates on a single context, $B_{\mathrm{int}}(\pi)$ approaches 0, while broader and more uniform allocations across many contexts yield larger entropy. As usual for entropy, contexts with $\chi^P_{\mathrm{Con}_\pi}(\mu, g, T) = 0$ contribute 0 by continuity (interpreting $0 \log 0 := 0$), so only contexts receiving nontrivial performance mass affect the sum.

Concept 219 (Multi-criterion driven general intelligence)**.** *Appendix label: concept:MultiCriterionGeneralIntelliger Body definition label: hsdef:MultiCriterionGeneralIntelligence (see §21.11).*

Informal definition. *A stance that "general intelligence" should not be collapsed to one scalar utility. Instead, one models multiple objectives (e.g. efficiency, breadth, and additional value dimensions) and treats the set of Pareto-optimal tradeoffs as the appropriate "target" rather than a single maximizer. This viewpoint is motivated by the observation that different desiderata can be structurally in tension: for example, increasing intellectual breadth may require allocating time or compute away from maximizing performance in a narrow benchmark, and optimizing worst-case robustness may reduce peak efficiency in easy cases. Accordingly, rather than choosing an arbitrary scalarization (e.g. a fixed weighted sum) up front, one retains the geometry of tradeoffs and regards the frontier itself as the object of interest.*

Formal definition. *Fix Π_{adm} and $\mathbf{J}(\pi) = (J_1(\pi), \ldots, J_k(\pi)) \in \mathbb{R}^k$. Define $\pi' \succ \pi$ iff $(\forall i)\, J_i(\pi') \geq J_i(\pi)$ and $(\exists j)\, J_j(\pi') > J_j(\pi)$. Then the Pareto set is*

$$\mathrm{Pareto}(\mathbf{J}) \;:=\; \{\pi \in \Pi_{\mathrm{adm}} : \nexists \pi' \in \Pi_{\mathrm{adm}} \text{ with } \pi' \succ \pi\}.$$

Here Π_{adm} encodes feasibility constraints (e.g. computability, safety constraints, training budget, interface limitations), ensuring comparisons are made among agents that are considered admissible under the same rules. The dominance relation $\succ$ is the standard (strong) Pareto dominance: π' dominates π if it is at least as good on all criteria and strictly better on at least one; thus, any π outside the Pareto set is unambiguously improvable without sacrificing any objective. Conversely, *elements of* Pareto($\mathbf{J}$) *are precisely those for which improvement in one criterion requires degradation in another (or else leaving* Π_{adm}*), so the set describes the irreducible tradeoff surface among the chosen intelligence-relevant measures. In applications, one often studies not only set membership but also the shape of the frontier (e.g. its convexity or coverage), since different training methods or inductive biases may populate different regions of* Pareto($\mathbf{J}$) *even when they achieve similar scalarized scores.*

C.3 Concepts Regarding Universal Grammar

Concept 220 (Universal Grammar (interface-theoretic))**.** *Appendix label: concept:UniversalGrammarInterfaceThe Body definition labels: hsdef:TraceBasedUniversalGrammar, hsdef:TUGCongruence (see Section 22.6).*

Informal definition. *"Universal Grammar" is treated here not as a list of stipulated constraints, but as the weakest equivalence theory on derivations that preserves a chosen meaning/externalization interface. In Hyperseed terms, UG is a weakness-maximizing collapse of derivational distinctions subject to an interface fidelity constraint (Hyperseed-Concept 202, 98,* **??***).*

This shifts the explanatory burden from "which derivations are allowed" to "which derivational differences can be systematically ignored without changing what the interface can observe." The resulting notion of UG is therefore explicitly observer-relative: changing the observation interface (or changing what counts as a symmetry) changes the induced equivalence, and hence changes which derivational distinctions survive into the theory.

Formal definition. *Fix a grammar LTS G and an observation interface* $(\mathcal{O}, \mathrm{obs}_{\mathcal{O}})$*. Let* $\equiv_{\perp,\Gamma}$ *be the UG core congruence induced by a choice of independence relation* $\perp$ *and symmetry group* Γ *(Definition 364). The interface-theoretic UG core is the quotient of derivations by* $\equiv_{\perp,\Gamma}$*, i.e., the identification of derivations up to independent swaps and declared symmetries, with* τ*-steps ignored.*

Equivalently, UG here is the congruence class semantics of derivations: two derivations count as the same "UG object" when they cannot be told apart by the interface after factoring out (i) swapping of independent actions, (ii) renamings/relabelings declared by Γ*, and (iii) internal bookkeeping steps modeled as* τ*. The quotient perspective makes precise the sense in which UG is a minimal (weakest) theory compatible with the specified interface fidelity constraint.*

Concept 221 (Trace-based Universal Grammar (TUG))**.** *Appendix label: concept:TraceBasedUniversalGrammar. Body definition label: hsdef:TraceBasedUniversalGrammar (see Section 22.6).*

Informal definition. *TUG is the specific formal package from the UG manuscript: a triple consisting of (i) what is observable at the interface, (ii) which actions commute (independence), and (iii) which visible differences are treated as equivalent (symmetries). It is a clean bridge between linguistic UG and Hyperseed's observer-relative weakness program.*

The key point is that TUG separates three logically distinct choices that are often conflated in more informal uses of "UG": the observational granularity (what counts as preservation), the dependency topology (what can be permuted without changing that observation), and the invariances of labeling (what is treated as mere presentational variation). In particular, the same grammar LTS can support multiple TUGs, corresponding to different interface commitments (e.g., semantics-only vs. semantics-plus-externalization).

Formal definition. *A TUG is a triple* $(\mathcal{O}, \perp, \Gamma)$ *as in Definition 364, built over a grammar LTS and observation map.*

Concretely, $\mathcal{O}$ *fixes the codomain of observations (the "interface report"),* $\perp$ *specifies which visible labels in* Lab_V *commute (capturing a trace/partial-order view of derivations), and* Γ *specifies which relabelings preserve the intended interpretation of visible steps. The induced congruence* $\equiv_{\perp,\Gamma}$ *(see also Definition 363) is then the mechanism by which TUG turns raw derivational sequences into equivalence classes that function as UG-level objects.*

Concept 222 (Observation Interface)**.** *Appendix label: concept:ObservationInterface. Body definition label: hsdef:ObservationInterface (see Section 22.6).*

Informal definition. *An observation interface specifies which aspects of a grammar derivation "count" as preserved: e.g., LF-only (semantic commitments) vs LF+PF (semantic plus surface externalization traces). In Hyperseed terms, this is a context/aspect choice that selects which distinctions are active (Hyperseed-Concept 86, 98).*

In practice, the interface is where one decides what a learner, parser, or external evaluator is allowed to "see." For example, an LF-only interface may identify derivations that differ in phonological linearization but agree in scope and binding, whereas an LF+PF interface may keep additional distinctions corresponding to word order, agreement morphology, or prosodic marking, to the extent those are part of $\mathcal{O}$*.*

Formal definition. *An observation interface is a pair* $(\mathcal{O}, \mathrm{obs}_{\mathcal{O}})$ *with* $\mathrm{obs}_{\mathcal{O}} : \mathrm{St} \to \mathcal{O}$ *(Definition 357).*

The map $\mathrm{obs}_{\mathcal{O}}$ *can be read as a projection (or forgetting map) from full grammar states to the information retained at the interface. Because* $\mathrm{obs}_{\mathcal{O}}$ *is defined on states, rather than directly on derivations, interface preservation is naturally expressed in terms of which state components are required to match, and which internal details may vary while remaining observationally equivalent.*

Concept 223 (Canonical Constraint Store)**.** *Appendix label: concept:CanonicalConstraintStore. Body definition label: hsdef:GrammarStateSpace (see Section 22.6).*

Informal definition. *To make interface-equivalence tractable, meaning-bearing choices must leave persistent, canonical "traces in state" (e.g., scope constraints, binding commitments, question operators). This mirrors Hyperseed's treatment of stable representational tokens as the carriers of later inference (Hyperseed-Concept 157, 184).*

The emphasis on canonicity is methodological: if semantically relevant commitments are stored in a normal form, then two derivations that differ only in inessential order or in internal naming choices can still be compared by inspecting the same kind of state object. This is exactly what is needed for quotienting: interface equivalence should reduce to equality (or a simple equivalence) of canonical stores, rather than requiring reconstruction of the full derivational history.

Formal definition. *A canonical constraint store is the* Σ *component of the grammar state space* $\mathrm{St} = \mathcal{T} \times \Sigma$ *(Definition 355), where elements of* Σ *are finite sets (or finite multisets) of interface atoms, presented in a fixed normal form.*

Typical interface atoms (depending on the grammar) may encode scope orderings, variable-binding constraints, agreement requirements, or other commitments that the observation map $\mathrm{obs}_{\mathcal{O}}$ *is designed to preserve. The "fixed normal form" requirement ensures that semantically identical stores do not differ merely by syntactic presentation (e.g., ordering of constraints, choice of fresh names), which in turn supports the intended identification of derivations under* Γ*-symmetries and* $\perp$*-commutations.*

Concept 224 (Rely Footprint)**.** *Appendix label: concept:RelyFootprint. Body definition label: hsdef:ResourcesAndRelyFootprints (see Section 22.6).*

Informal definition. *A rely footprint records which semantic/interface resources a grammatical action touches. It is the bridge between (i) linguistic locality/economy constraints and (ii) concurrency-style dependency topology.*

Intuitively, resources are the "places" where actions could interfere: if two steps both modify or consult the same resource, they can create an ordering dependency that the interface can detect. Conversely, if their rely footprints are disjoint, the steps are candidates for commuting without changing what is ultimately observed. This makes the rely footprint a compact static summary of potential interaction, analogous to read/write sets in program logics.

Formal definition. *Given a resource set* Res*, a rely footprint is a map* rely : $\mathrm{Lab}_V \to 2^{\mathrm{Res}}$ *(Definition 359).*

The choice of Res *is part of the modeling commitment: it may include, for example, features, indices, discourse referents, scope sites, phase-bounded domains, or other interface relevant loci. The point is not that* all *such structure must be enumerated, but that enough is tracked to justify the intended independence relation* $\perp$ *and the induced trace-based identifications.*

Concept 225 (Grammatical Independence)**.** *Appendix label: concept:GrammaticalIndependence. Body definition label: hsdef:GrammaticalIndependence (see Section 22.6).*

Informal definition. *Two meaning-bearing grammatical actions are independent if they act on disjoint parts of structure and disjoint semantic resources, so their order does not matter to the interface. Independence is the formal source of the "trace monoid" viewpoint and, later, of economy/locality results.*

In this setting, independence is not merely a claim about derivational convenience; it is the mechanism that licenses collapsing sequential histories into partial orders. When two steps are independent, the interface-theoretic UG core treats the two possible orders as the same derivation up to $\equiv_{\perp,\Gamma}$*, reflecting the idea that any ordering difference is an artifact of the derivation strategy rather than a difference in linguistic outcome.*

Formal definition. *Grammatical independence is the relation* $\perp$ *on* Lab_V *defined in Definition 361.*

Typically (and depending on the specific instantiation), $\ell_1 \perp \ell_2$ *is justified by a combination of (i) disjointness of their rely footprints and (ii) non-overlap of the structural regions they update, ensuring that swapping the corresponding steps preserves* $\mathrm{obs}_{\mathcal{O}}$ *up to the intended symmetries. This is the precise locus where locality/economy intuitions are translated into a concurrency-style commutation criterion.*

Concept 226 (Label Symmetry Group)**.** *Appendix label: concept:LabelSymmetryGroup. Body definition label: hsdef:LabelSymmetryGroup (see Section 22.6).*

Informal definition. *A label symmetry group specifies which systematic relabelings of visible grammatical actions are treated as inessential for interface purposes. Examples include renaming of indices, uniform permutation of feature names within an agreed schema, or alternative but equivalent encodings of the same interface commitment. Declaring such symmetries makes explicit which differences are regarded as representational artifacts rather than genuine grammatical distinctions, aligning with the Hyperseed idea that a theory may quotient by a chosen invariance (Hyperseed-Concept* **??***, 98).*

In the TUG package, Γ *complements independence:* $\perp$ *identifies differences due to ordering of commuting steps, while* Γ *identifies differences due to alternative naming or presentation of the* same *kind of step. Both are needed if the goal is to capture interface preservation rather than derivational idiosyncrasy.*

Formal definition. *A label symmetry group is the group* Γ *acting on the visible label set* Lab_V *as in Definition 362. The induced action is used inside the congruence* $\equiv_{\perp,\Gamma}$ *to identify derivations whose visible steps differ only by application of elements of* Γ*.*

In particular, Γ *provides the formal vehicle for factoring out uniform renamings (e.g., of freshly introduced symbols) that do not alter the interface observation once the canonical constraint store and observation map are taken into account. This ensures that the UG quotient is sensitive to meaning and externalization choices, but insensitive to arbitrary representational conventions.*

Concept 227 (Trace Quotient)**.** *Appendix label: concept:TraceQuotient. Body definition label: hsdef:TUGCongruence (see Section 22.6).*

Informal definition. *The trace quotient is what you get when you collapse derivational differences that are not distinguished by the interface: reorderings of independent actions and symmetry-equivalent labels. It is an explicit, algebraic representation of Hyperseed weakness applied to derivations.*

Formal definition. *The trace quotient is the quotient monoid* $\mathrm{Lab}_V^*/{\approx_{\perp,\Gamma}}$ *where* $\approx_{\perp,\Gamma}$ *is the TUG congruence (Definition 363).*

Concept 228 (Principles and Parameters (as invariants and symmetries))**.** *Appendix label: concept:PrinciplesParametersAsInvariants.*
Body reference: Remark 1230 (see Section 22.6).

Informal definition. *"Parameters" correspond to choices about what is observed and which surface differences are declared equivalent; "principles" are the invariants that remain once these choices are made. This reframes P&P as an invariance/quotienting story rather than a hard-coded axiom list.*

Formal definition. *Fix a family of grammars over a shared operational substrate. A* parameterization *is a choice of observation interface* $\mathcal{O}$ *and symmetry group* Γ *(and possibly a change in the visible label inventory). A* principle *is any property invariant under the induced UG core congruence* $\equiv_{\perp,\Gamma}$ *across the family.*

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